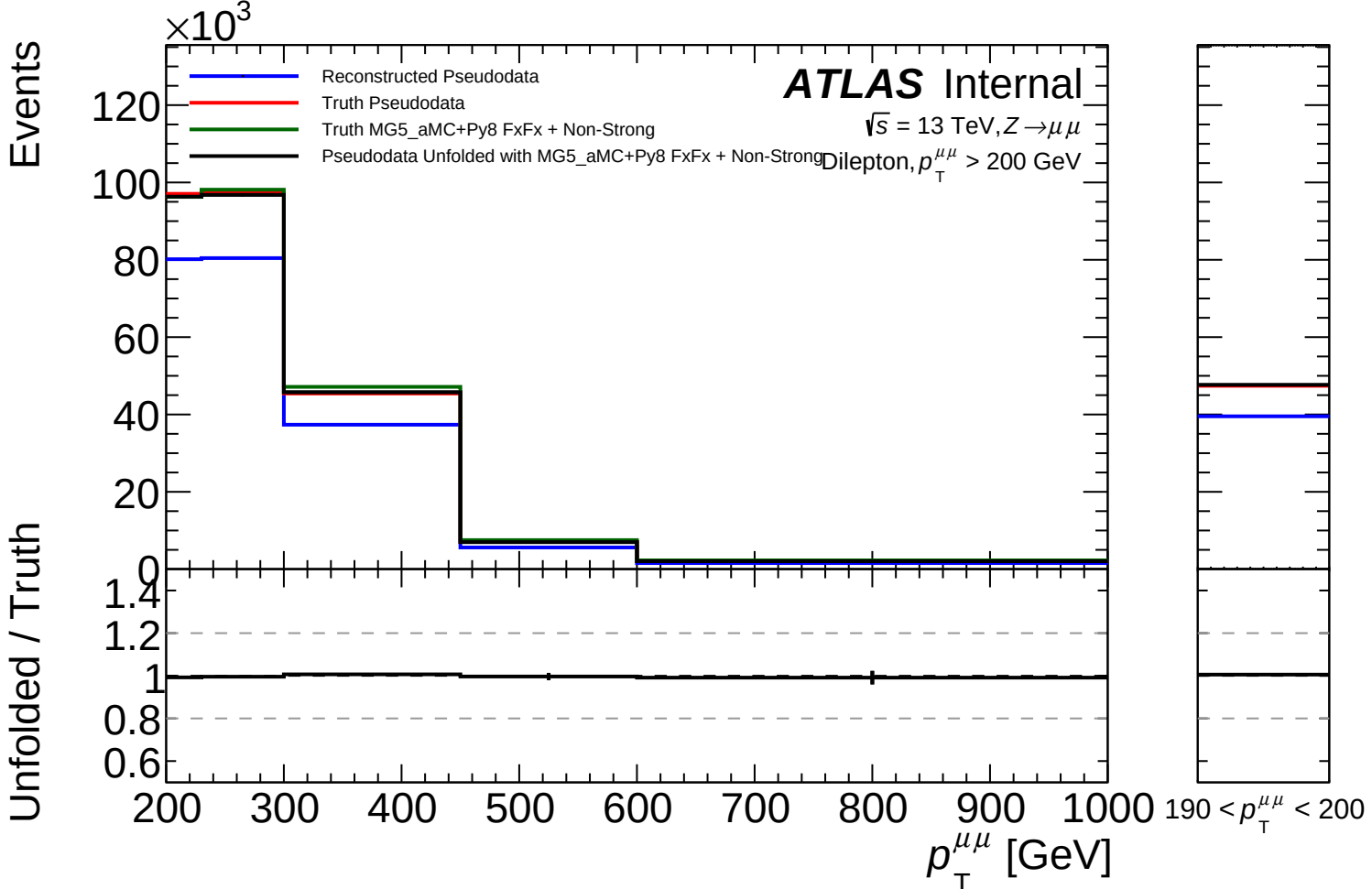
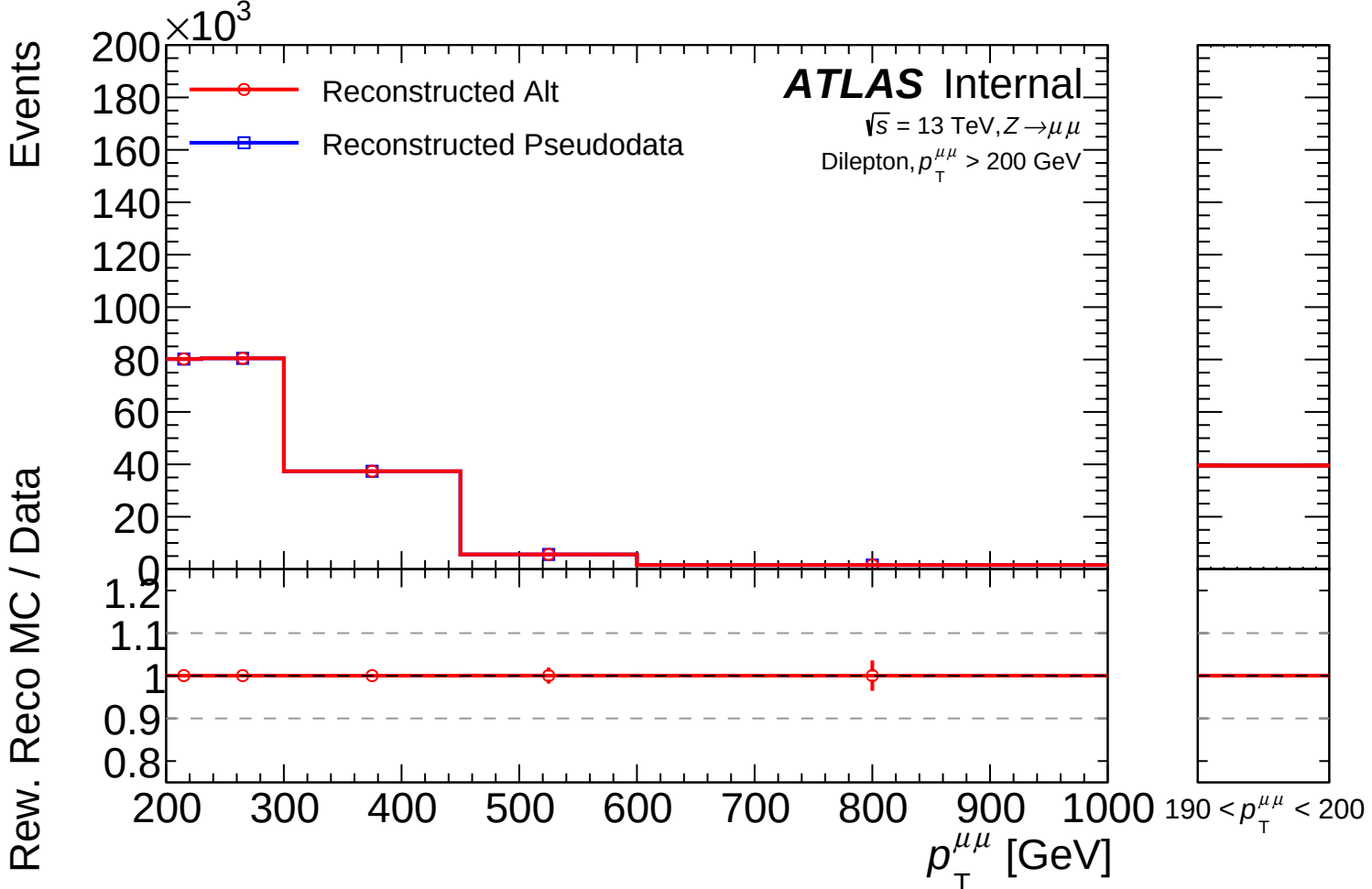
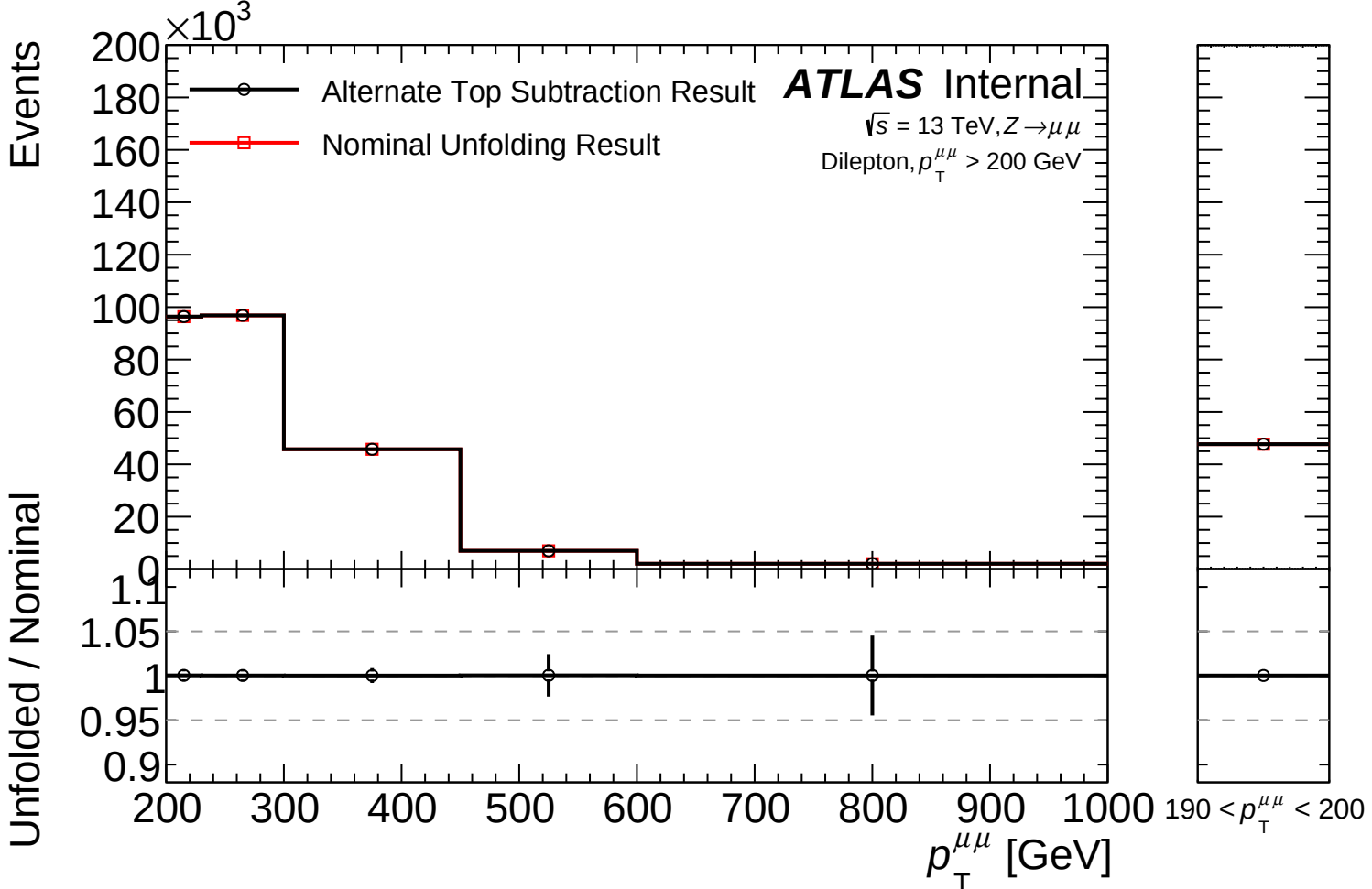








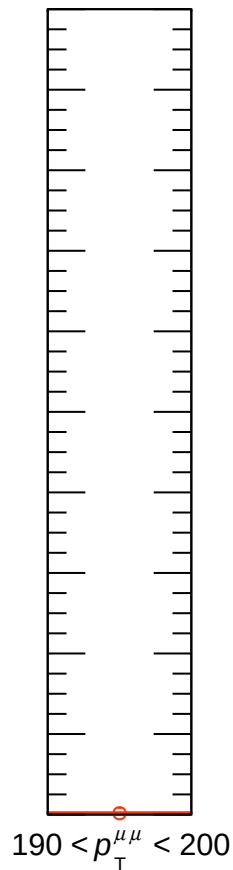
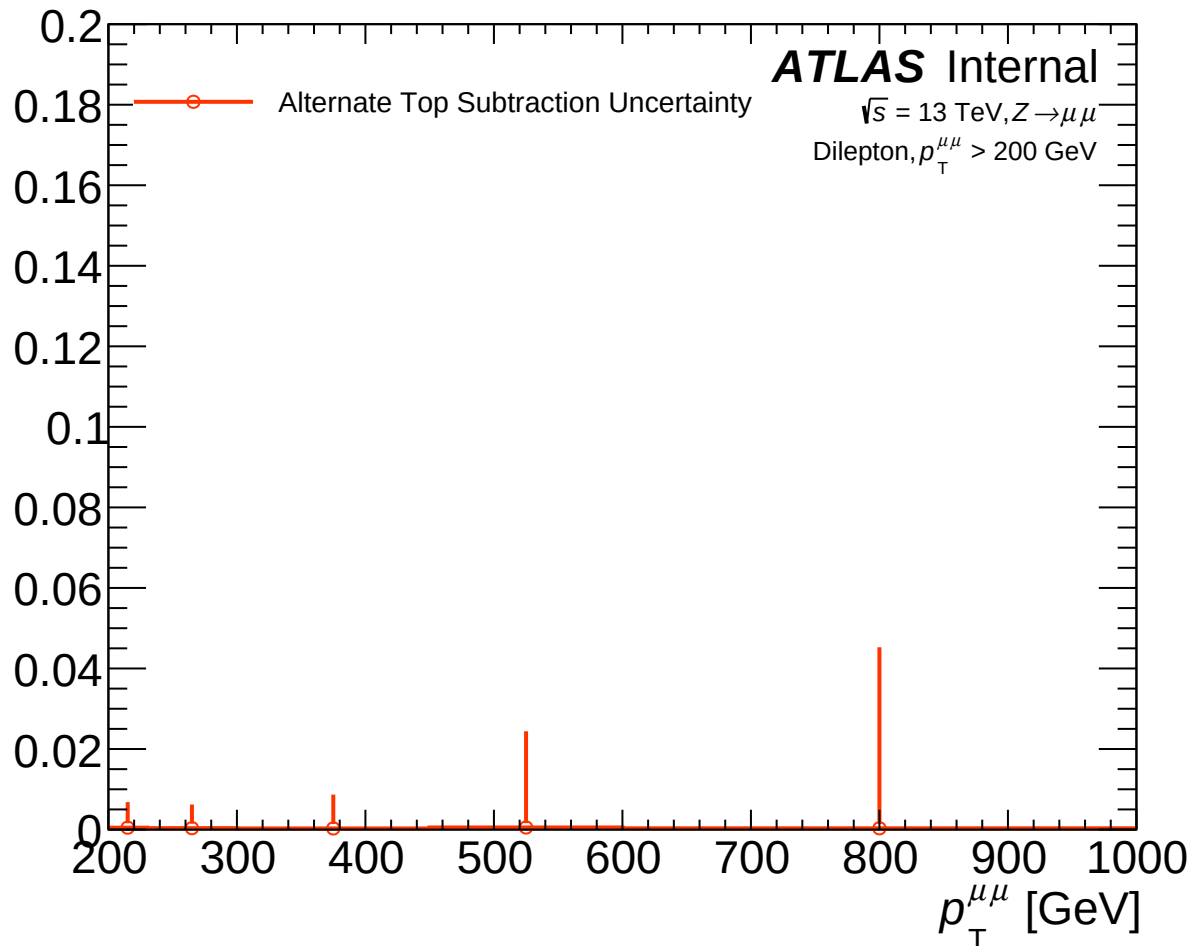
Fractional systematic impact

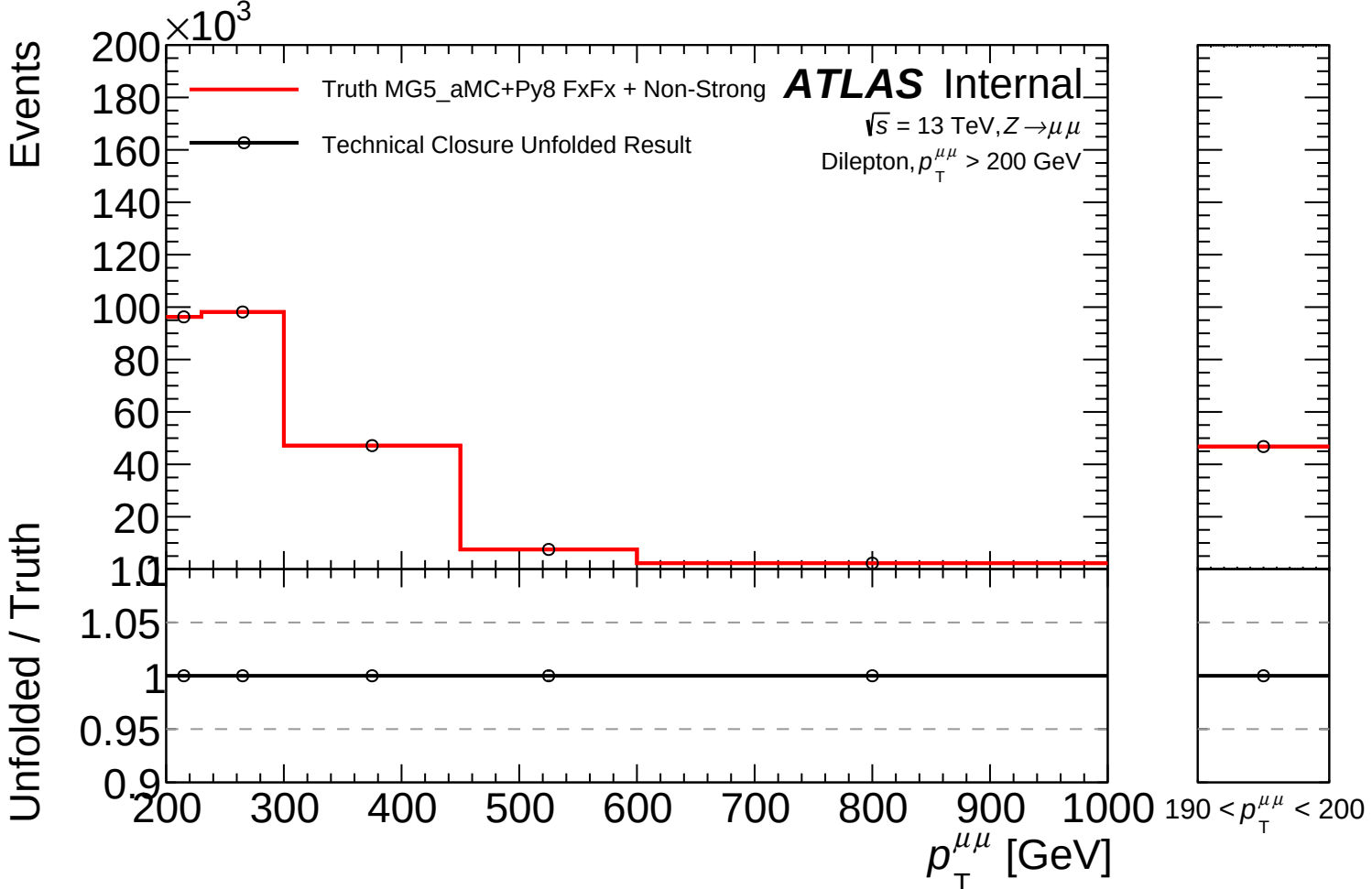
ATLAS Internal

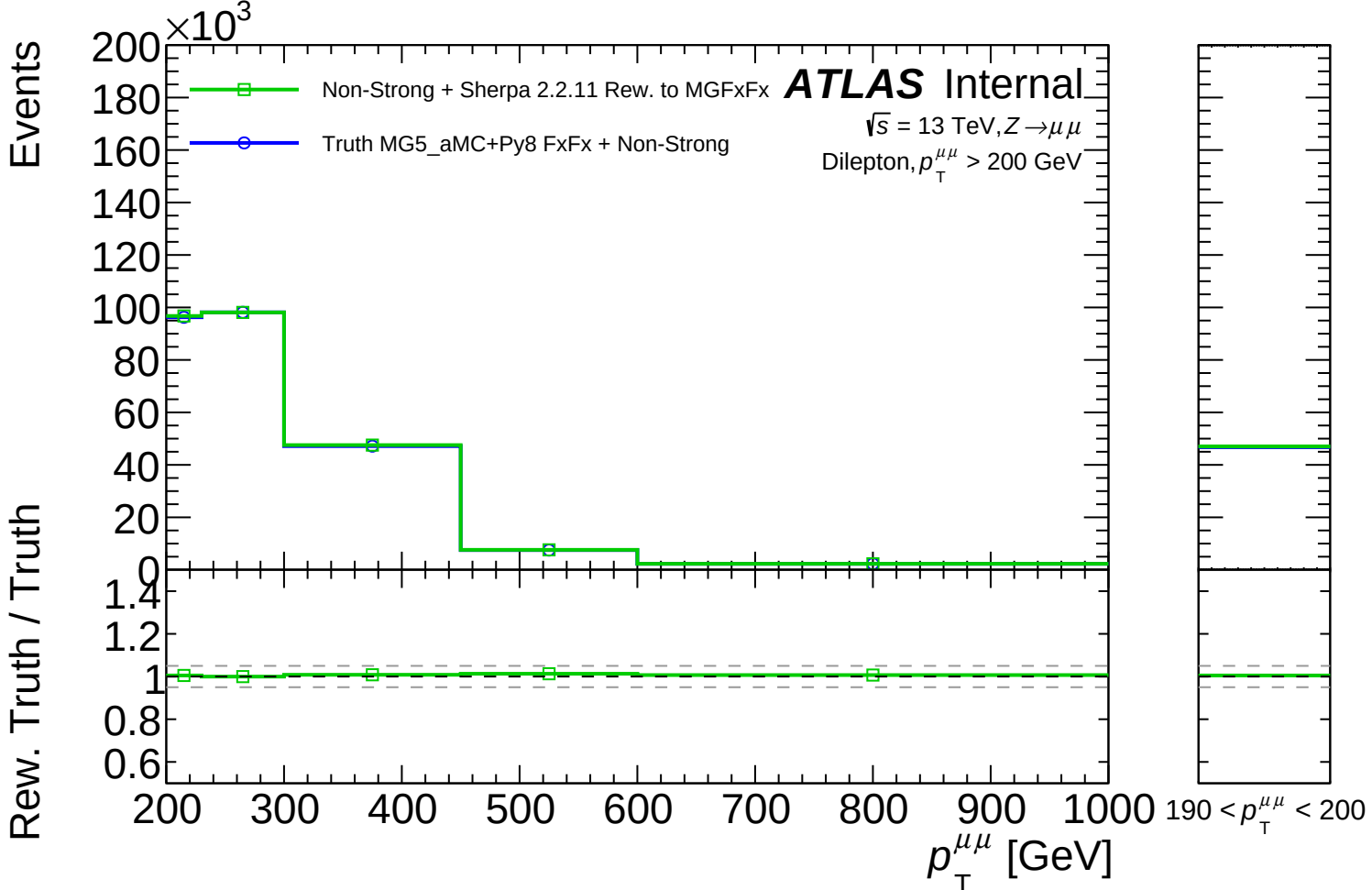
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

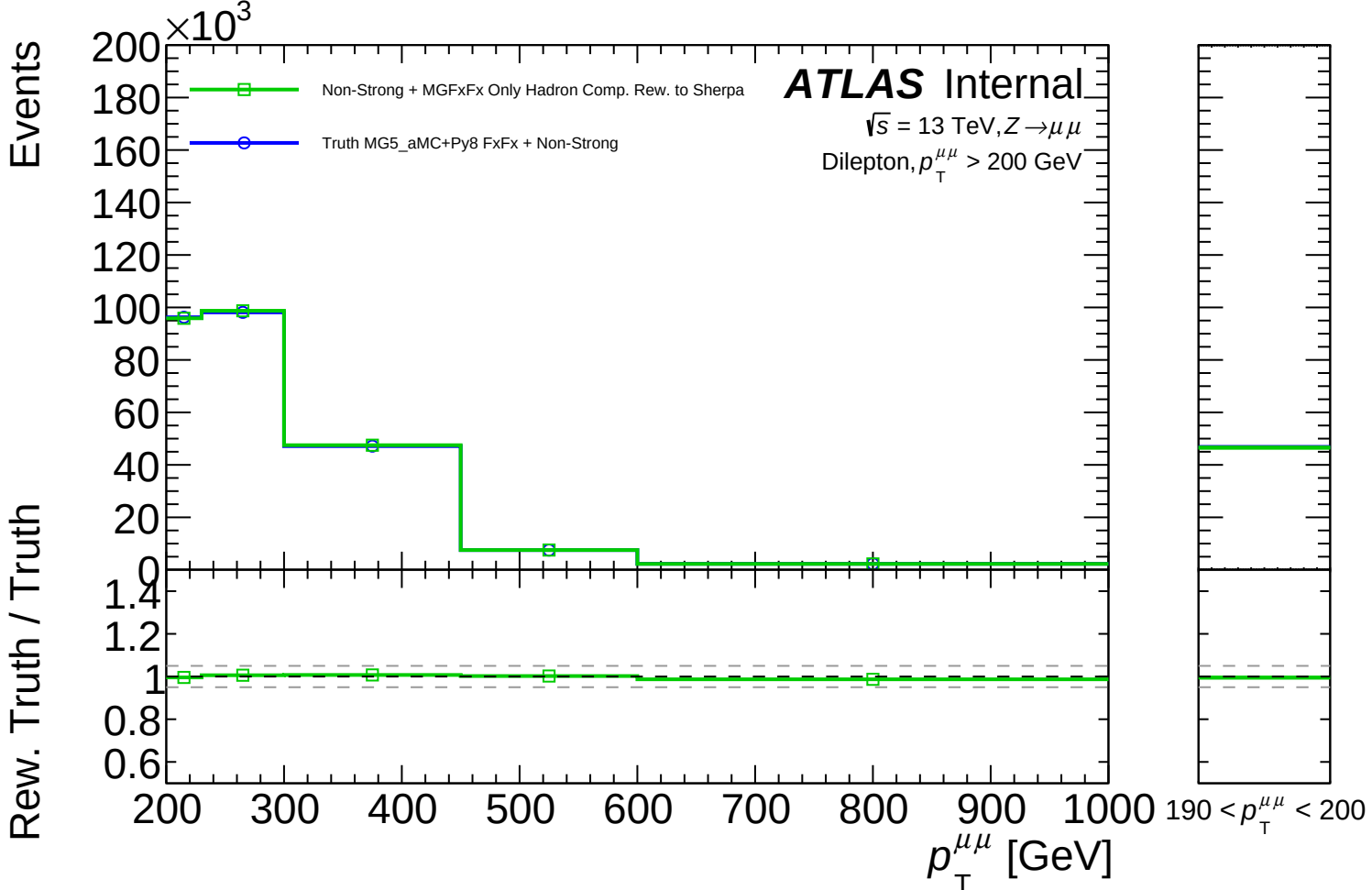
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

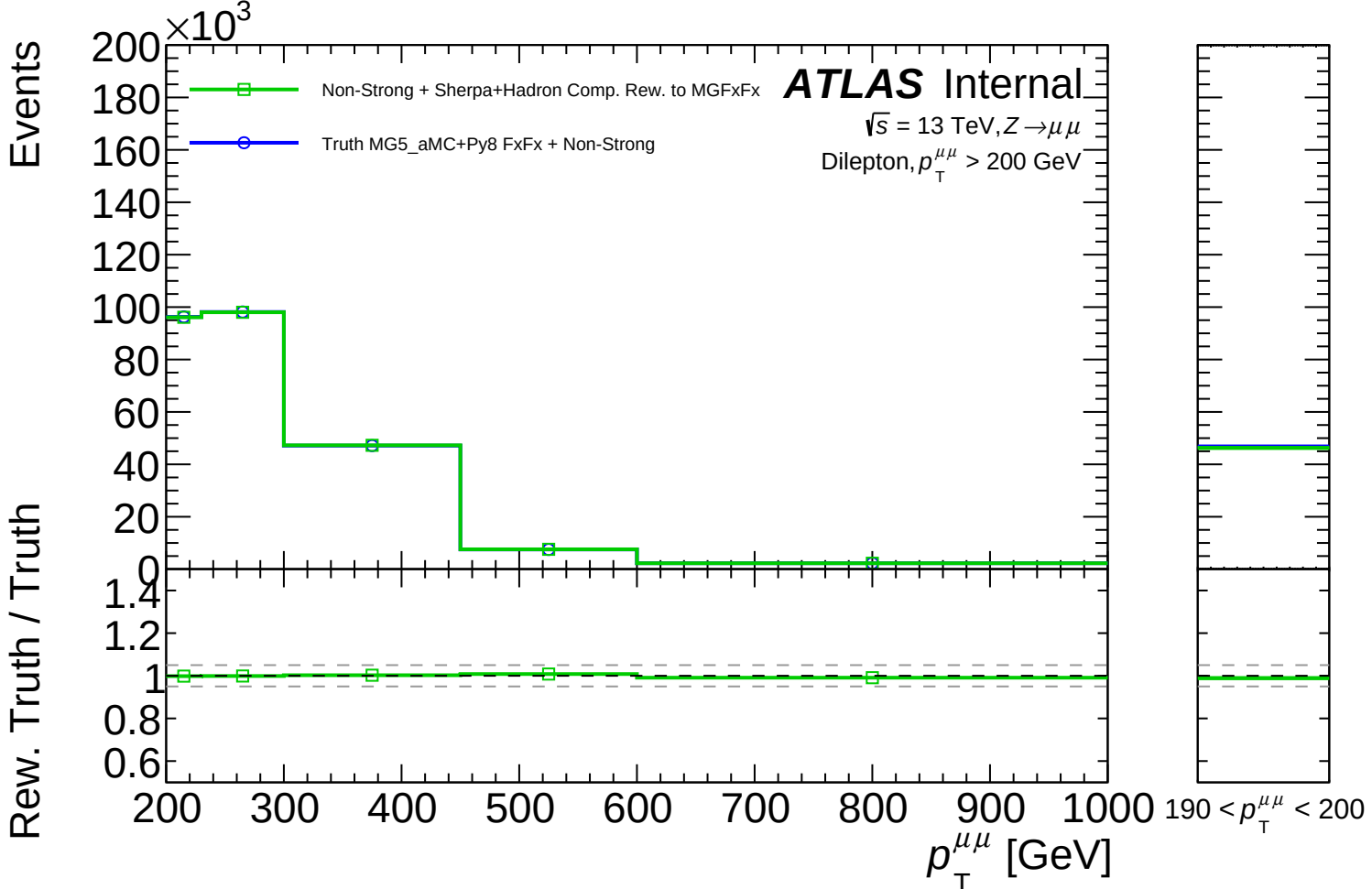
Alternate Top Subtraction Uncertainty

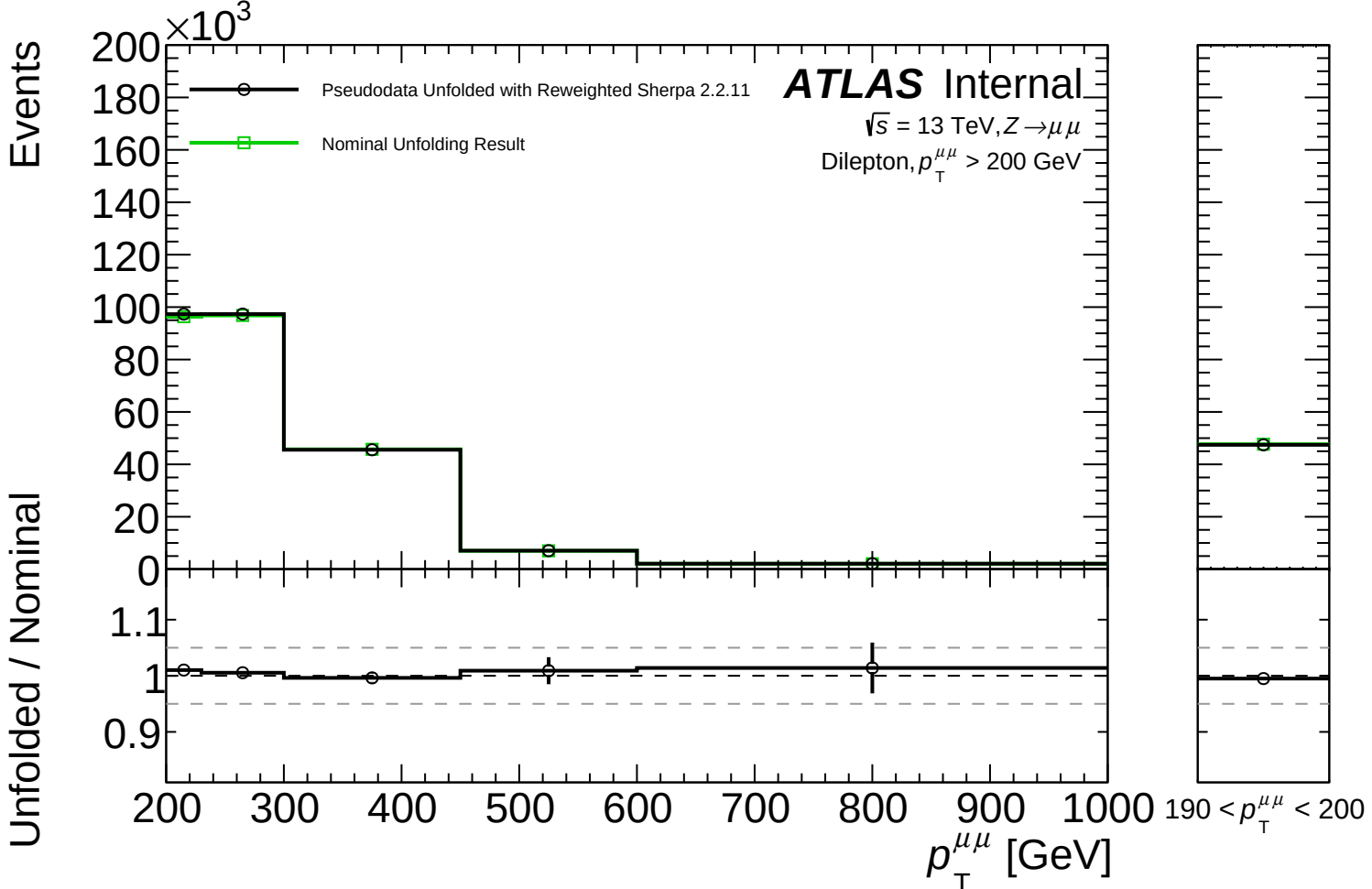


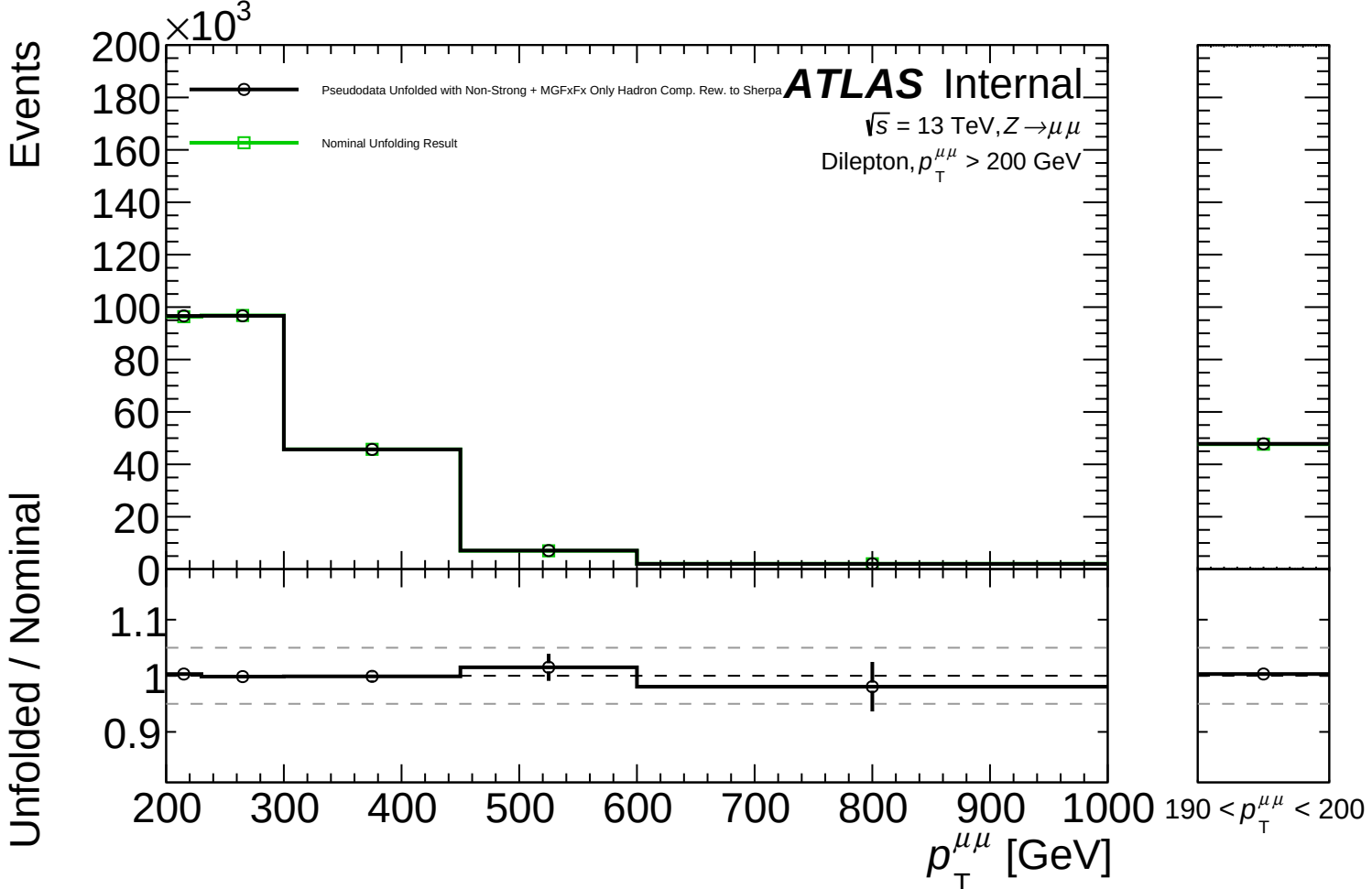


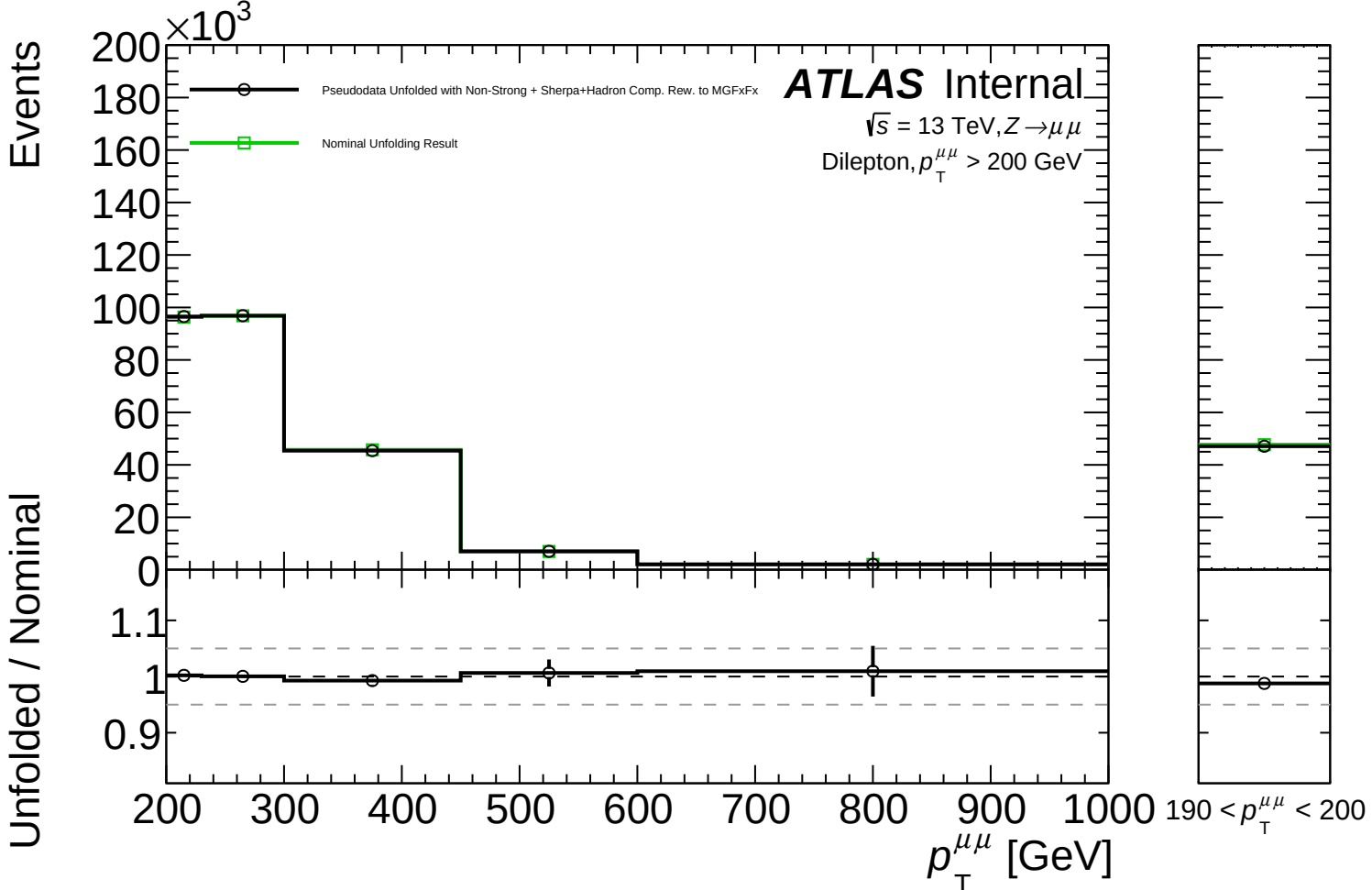


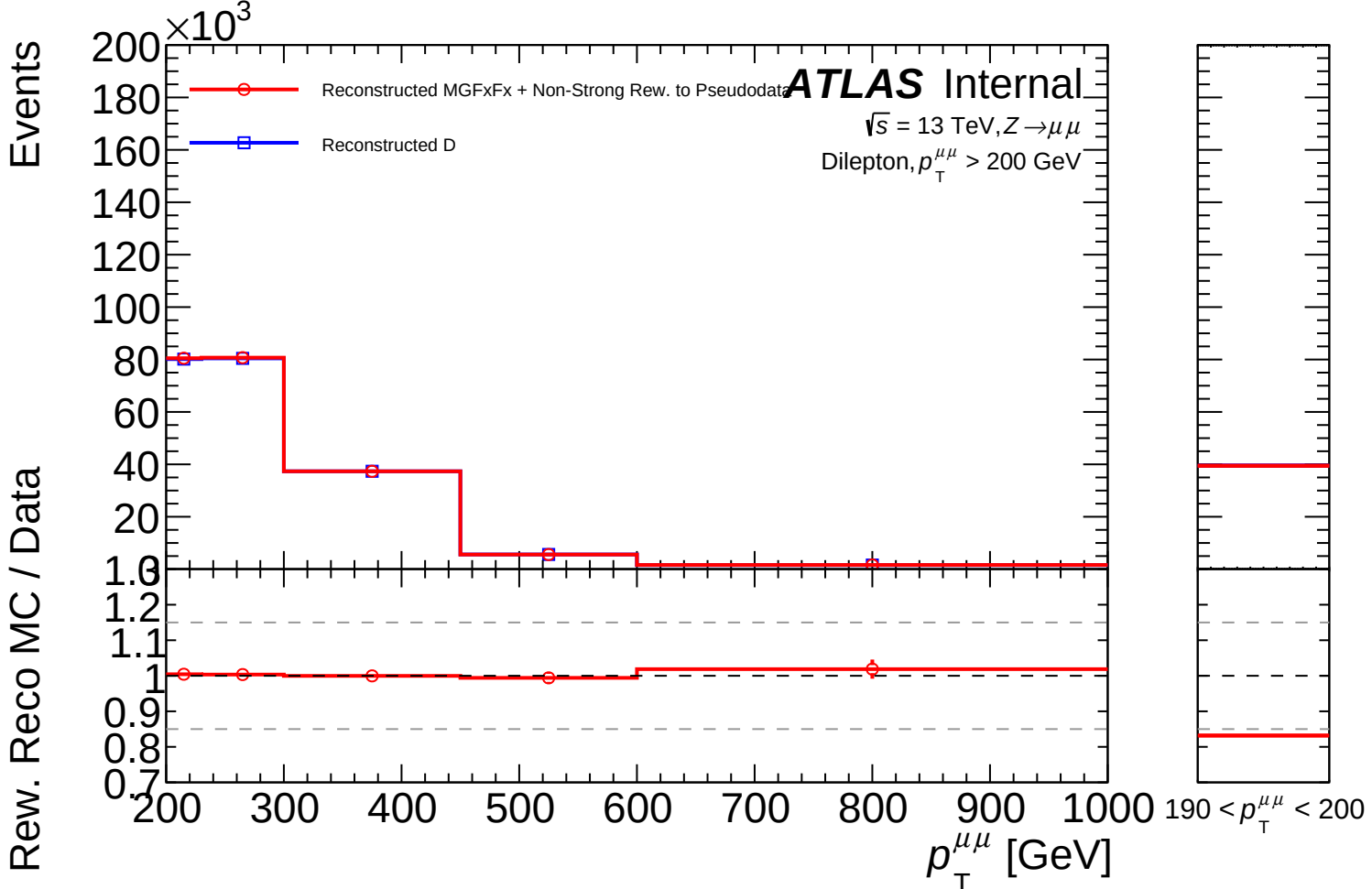


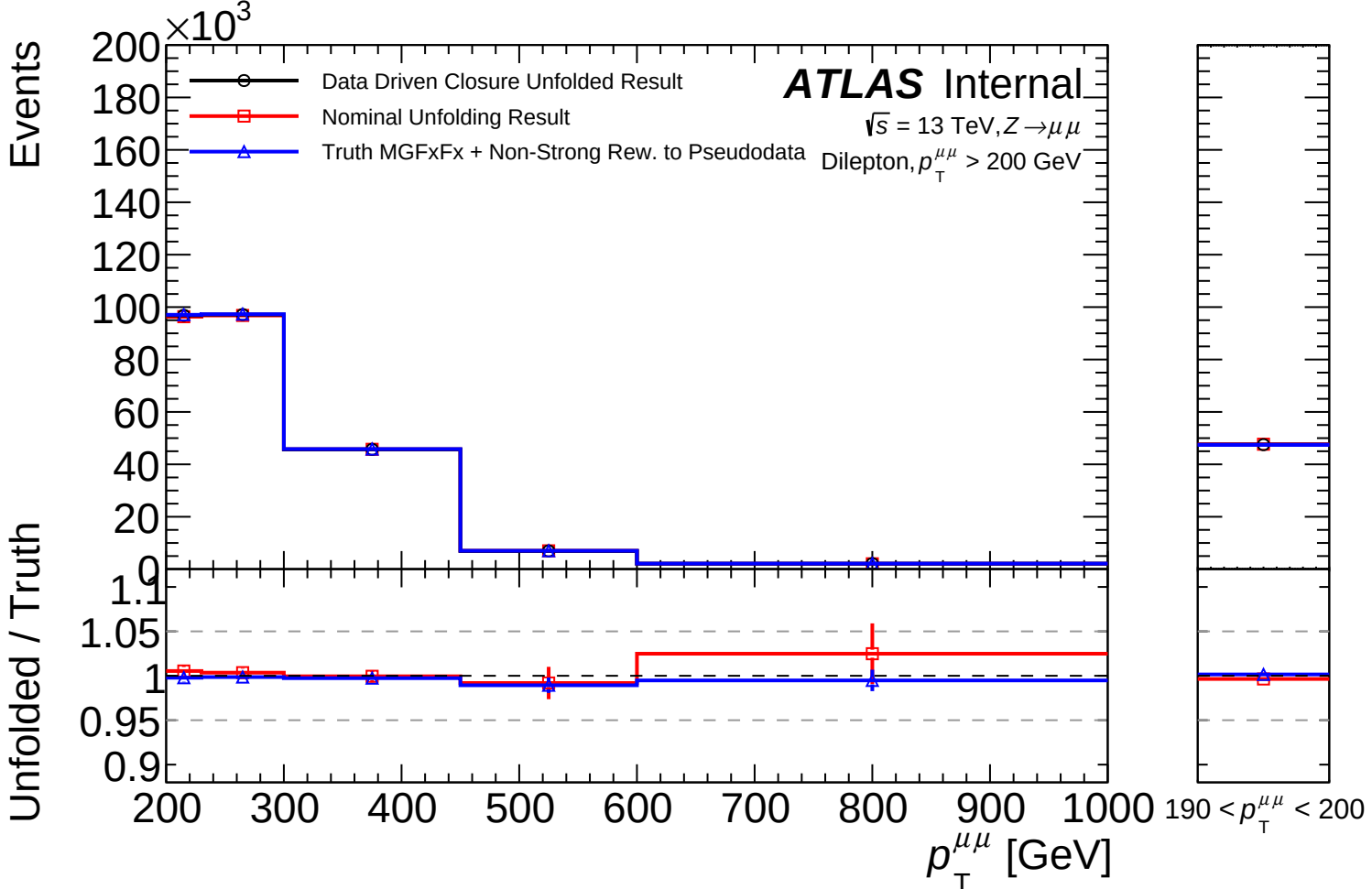




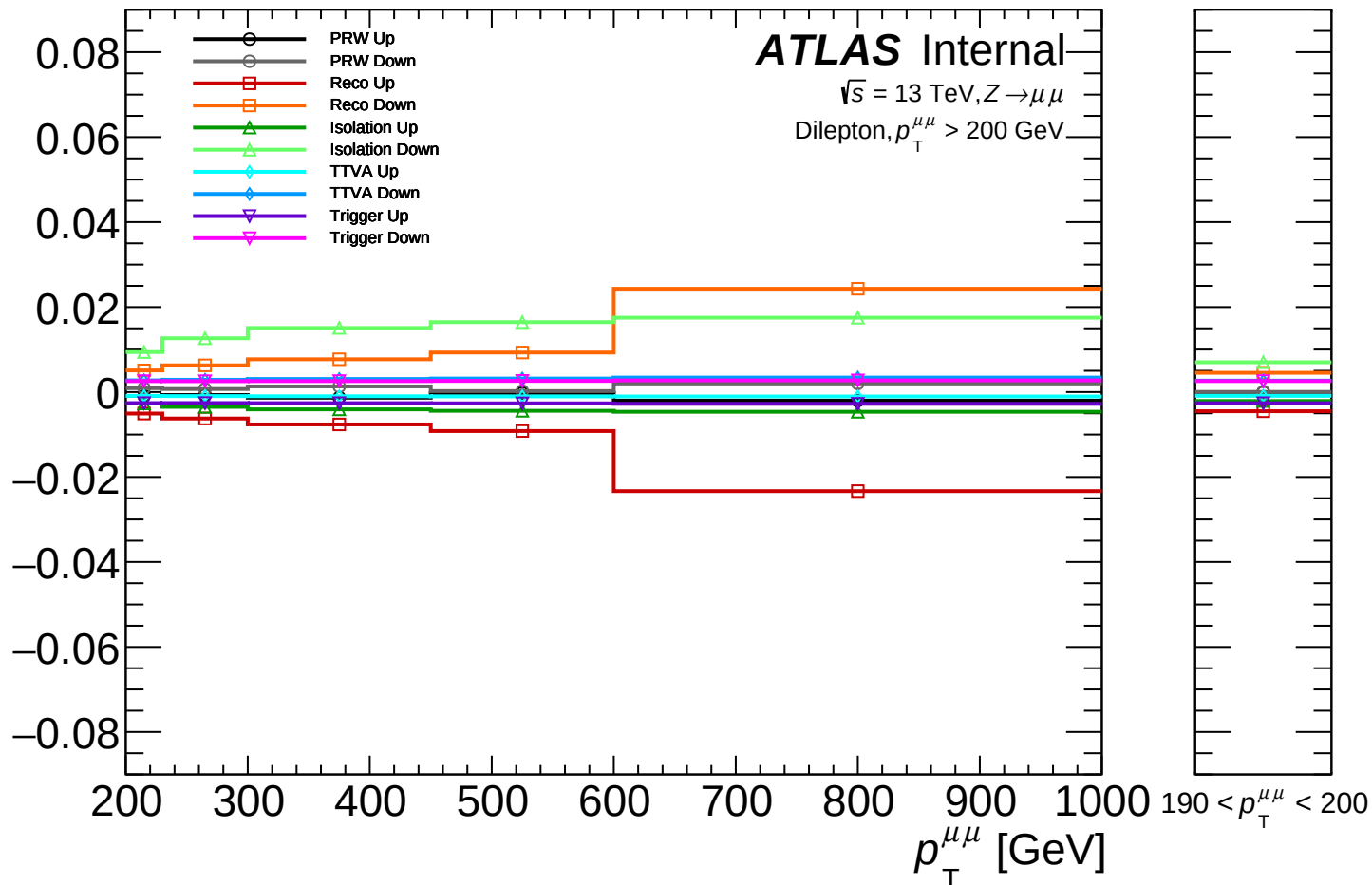




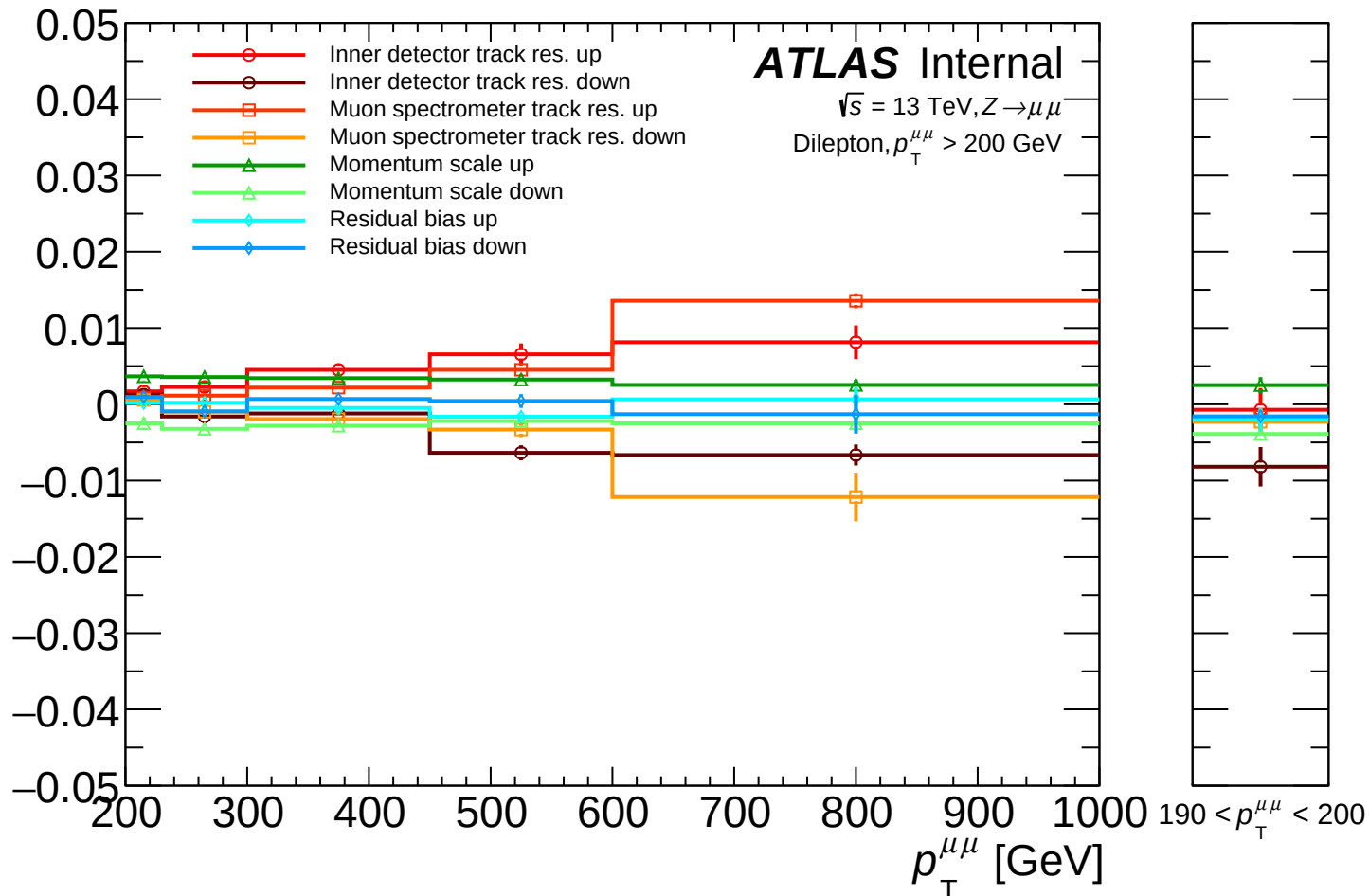




Fractional systematic impact



Fractional systematic impact



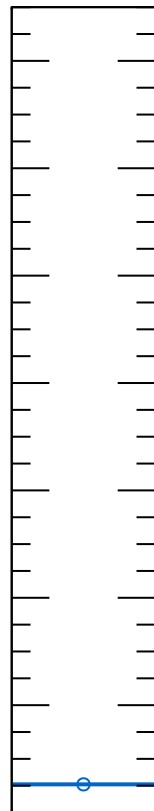
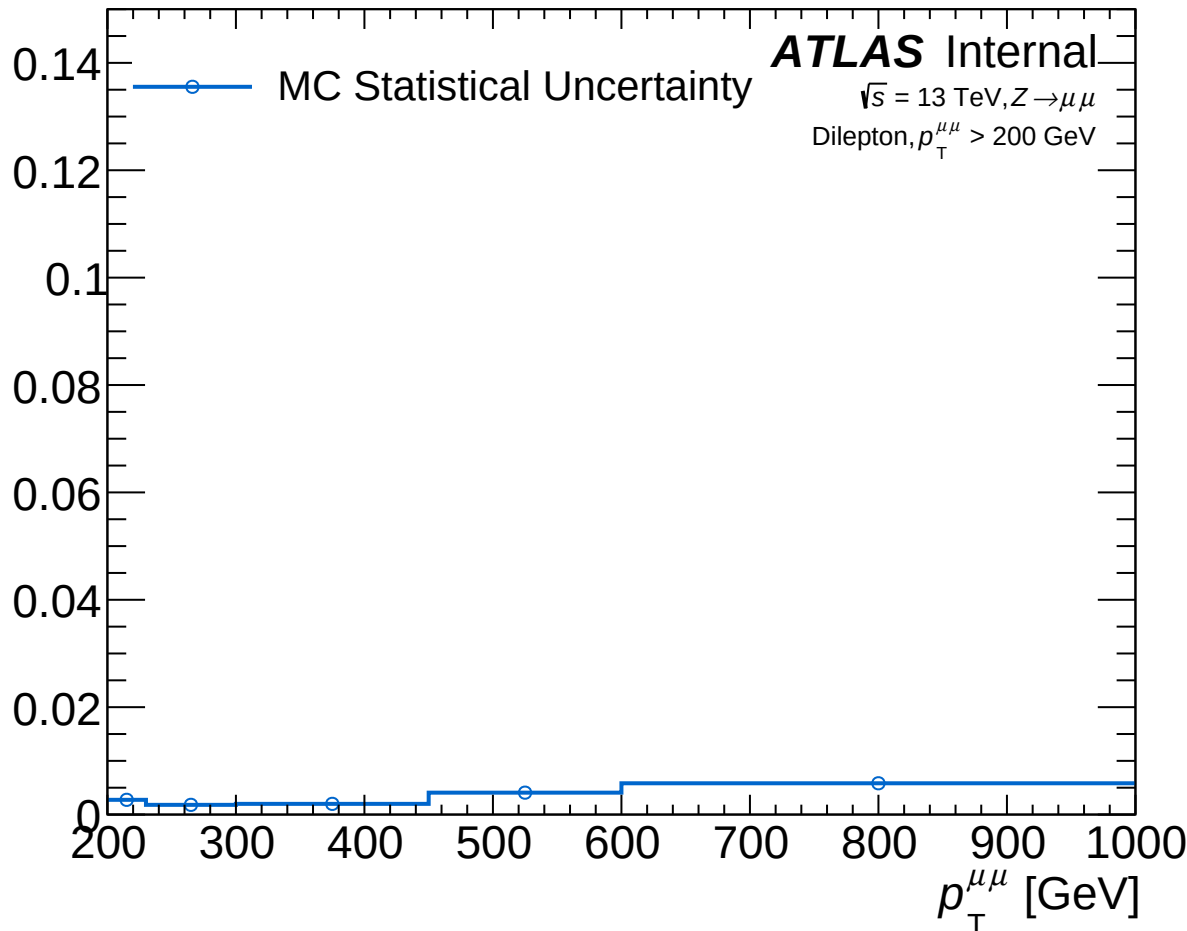
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



$190 < p_{\text{T}}^{\mu\mu} < 200$

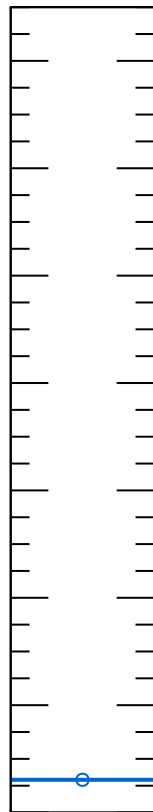
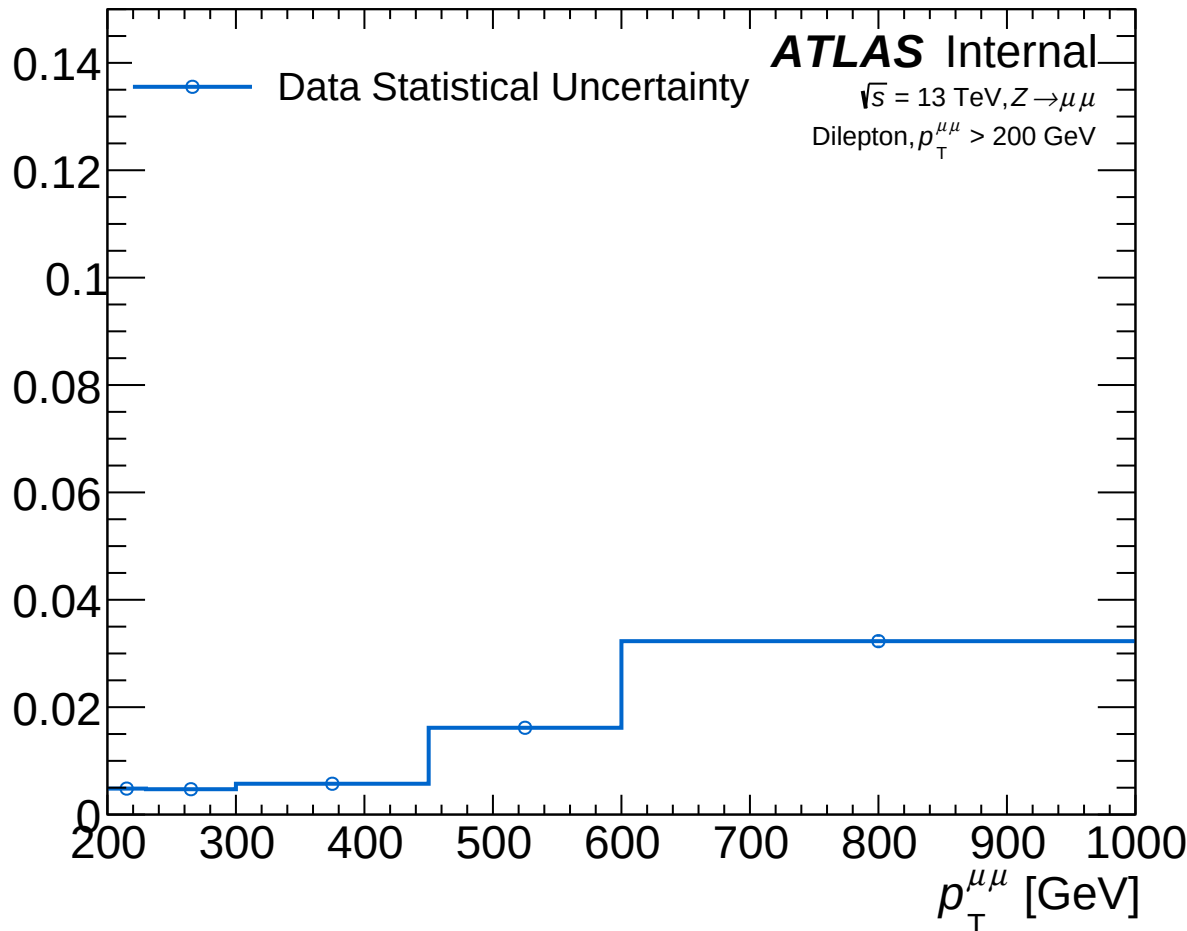
Fractional systematic impact

ATLAS Internal

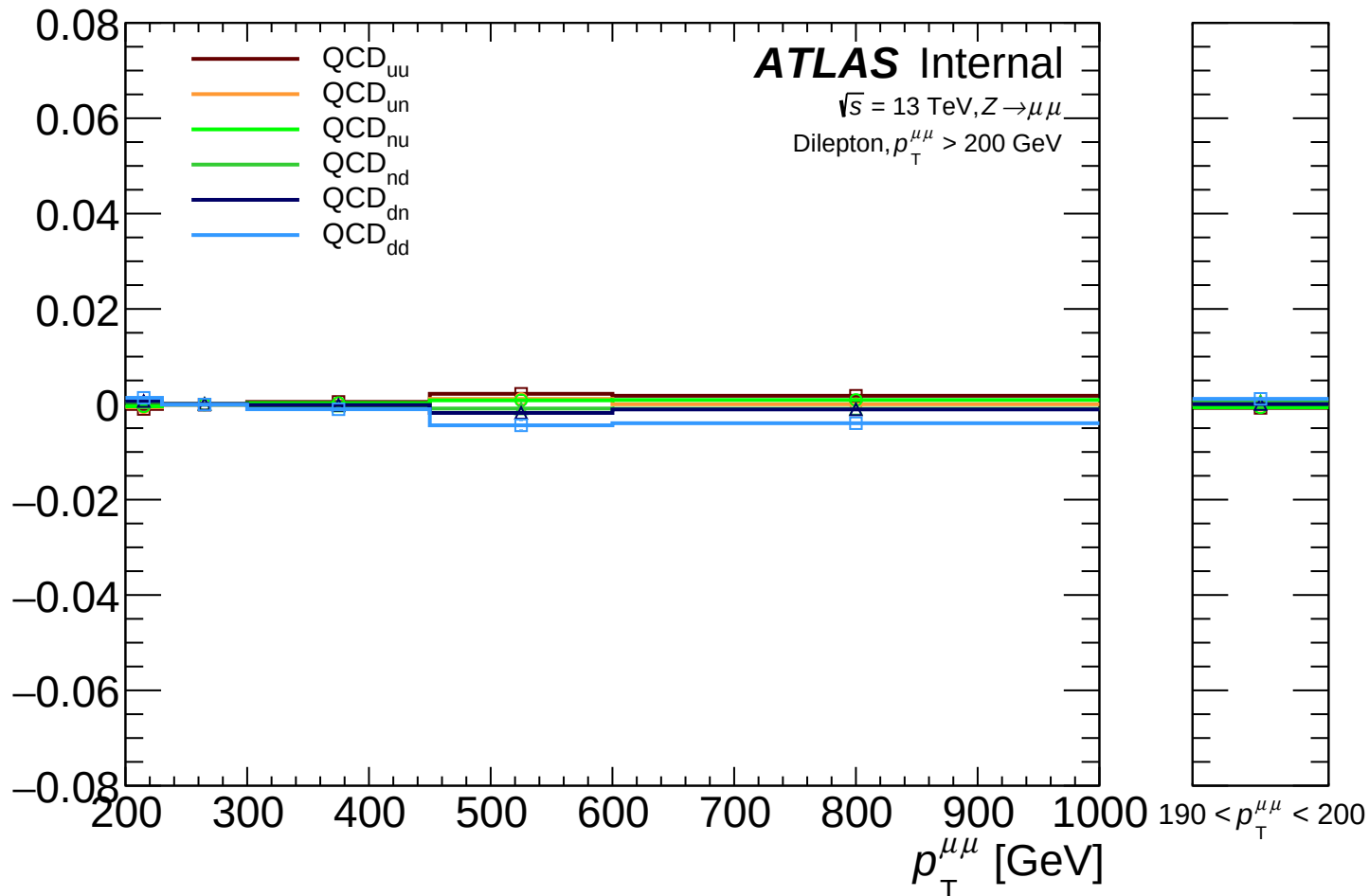
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200$ GeV

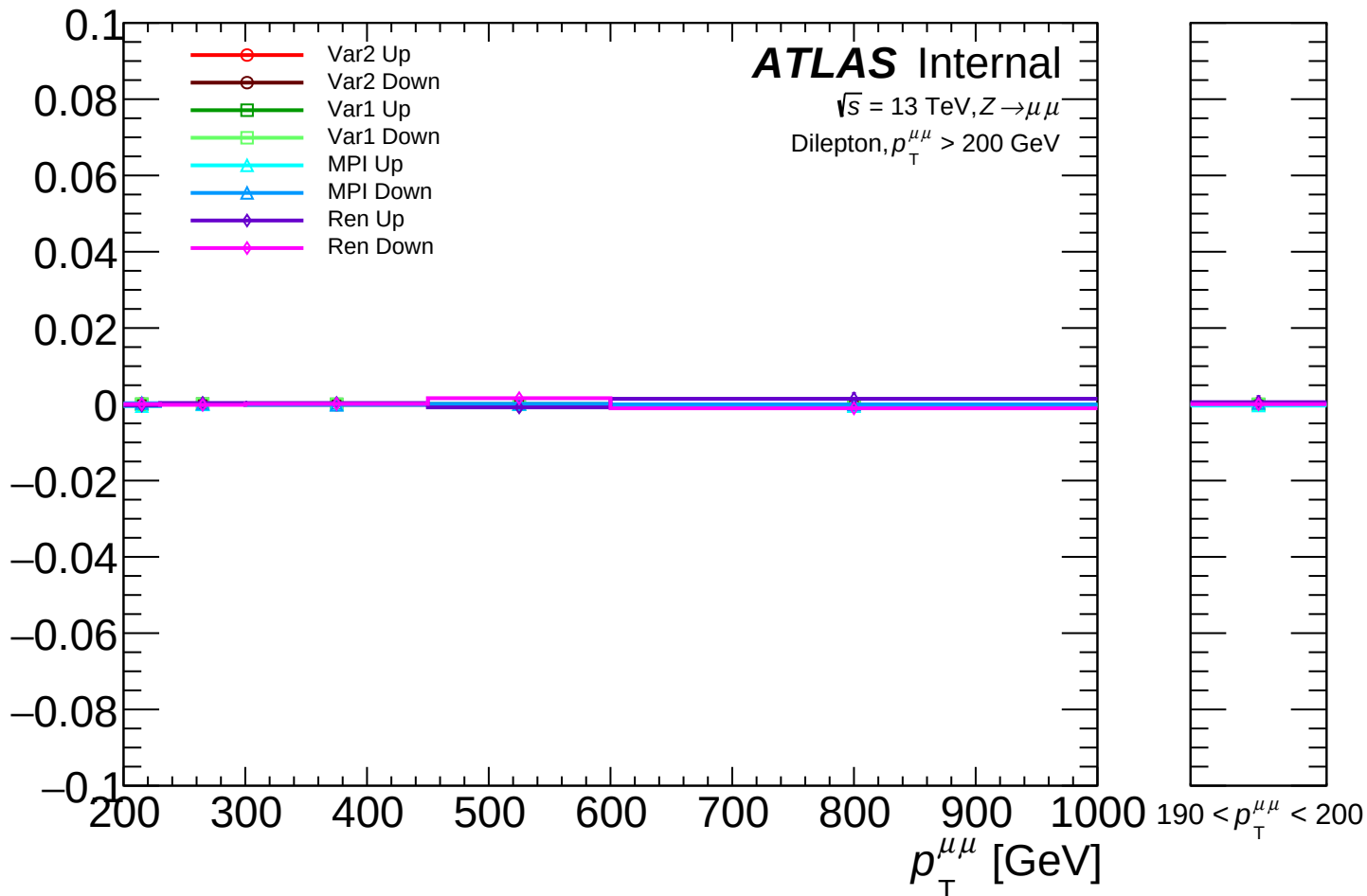
Data Statistical Uncertainty

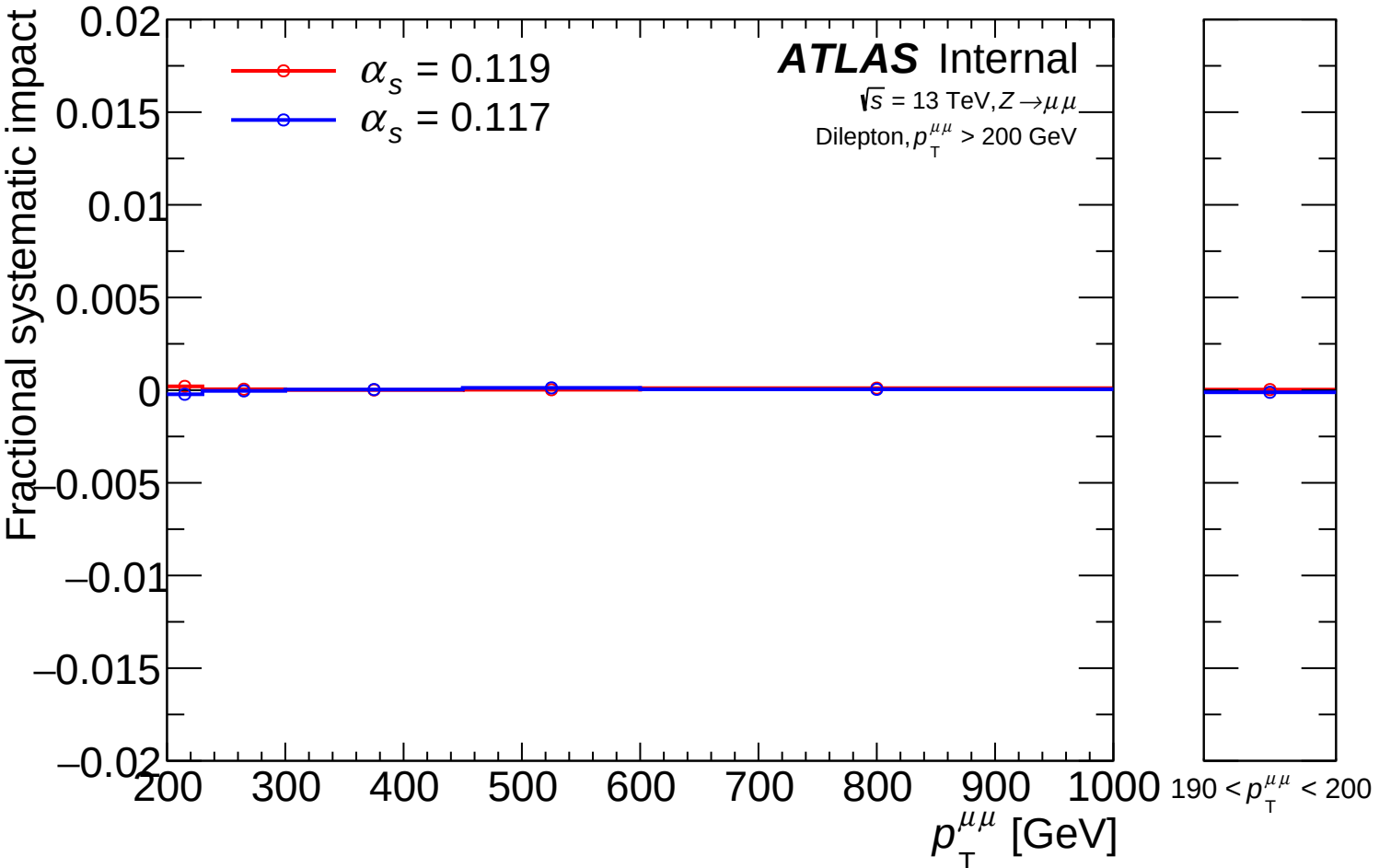


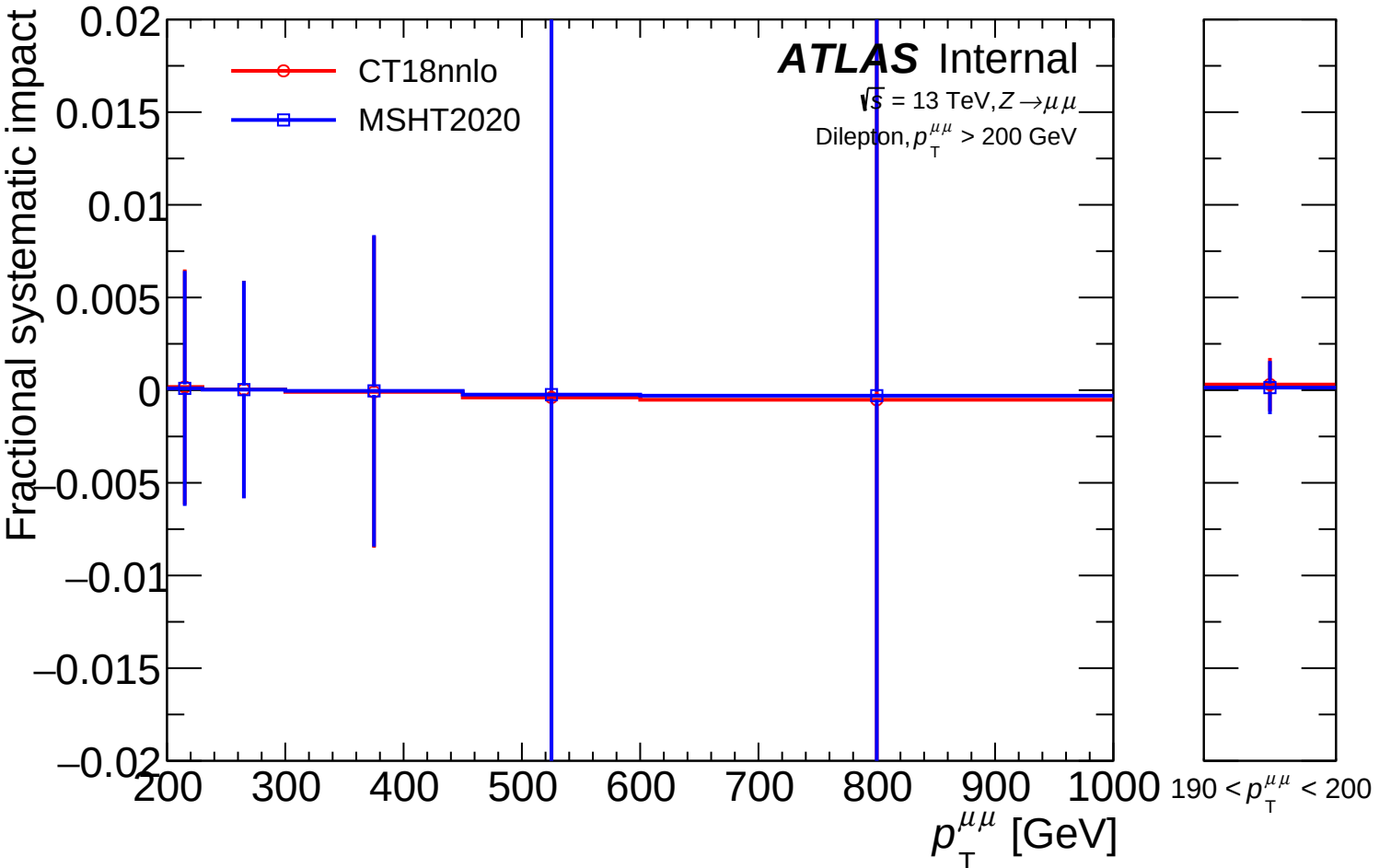
Fractional systematic impact



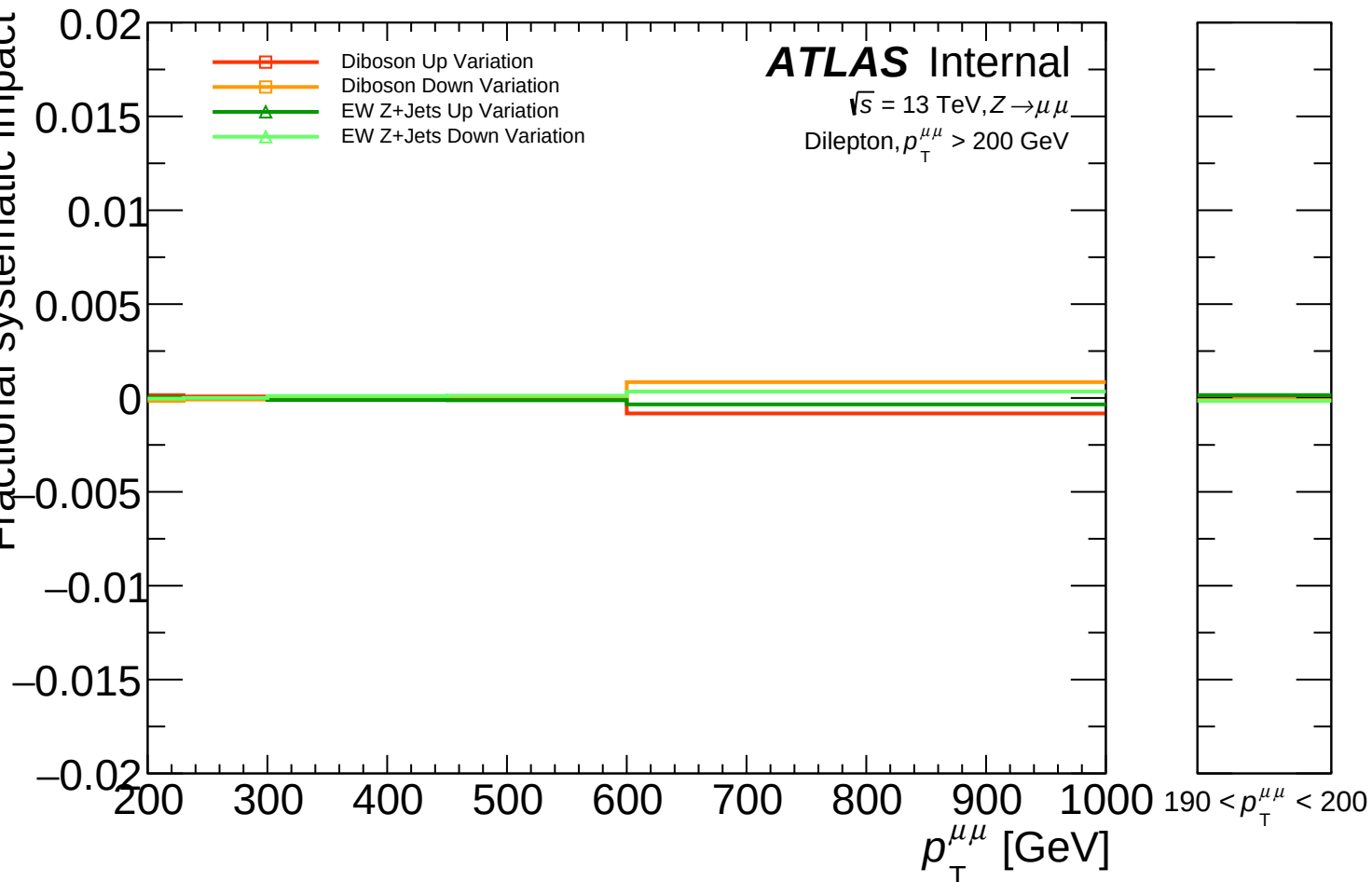
Fractional systematic impact







Fractional systematic impact

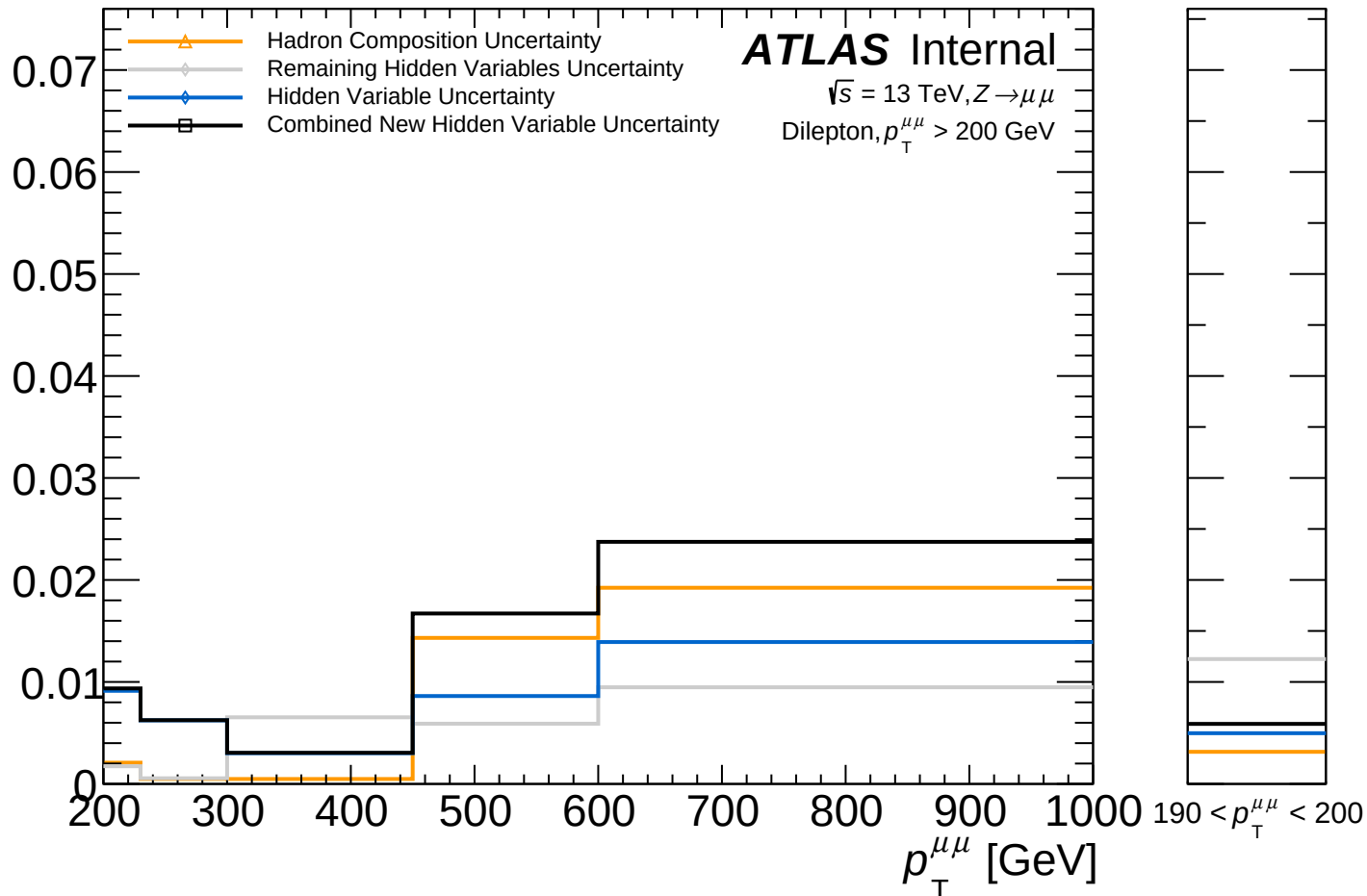


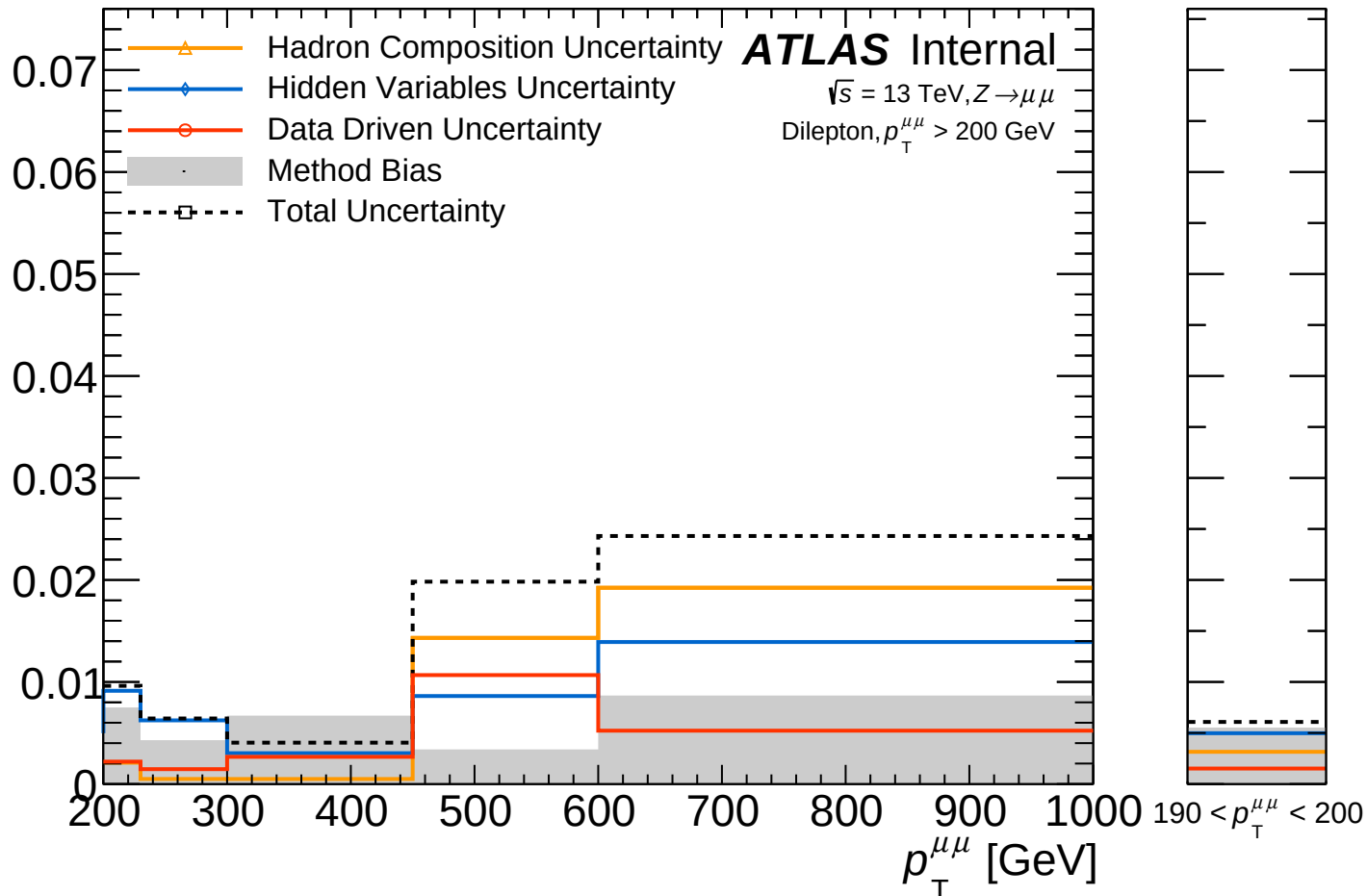
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

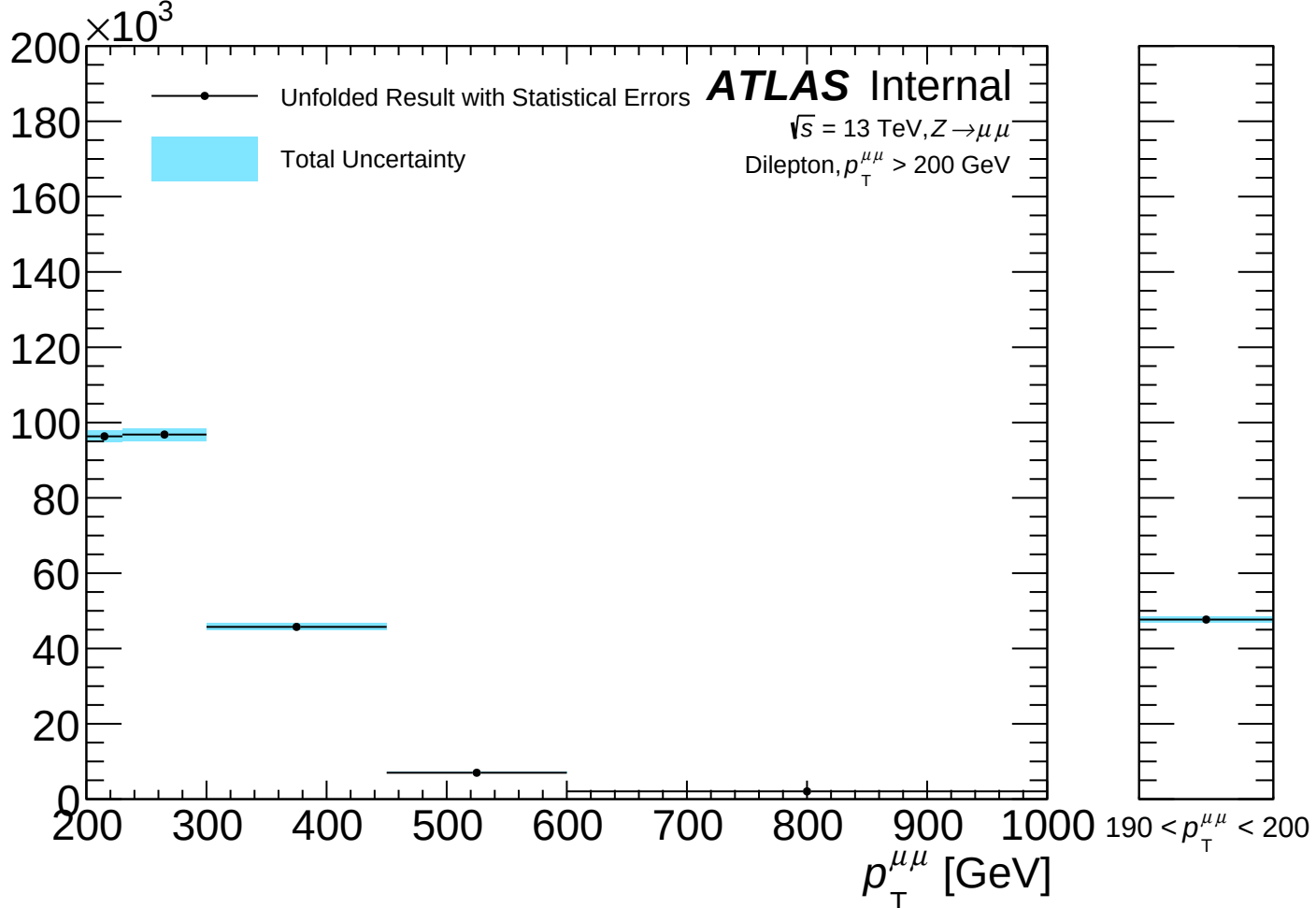
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

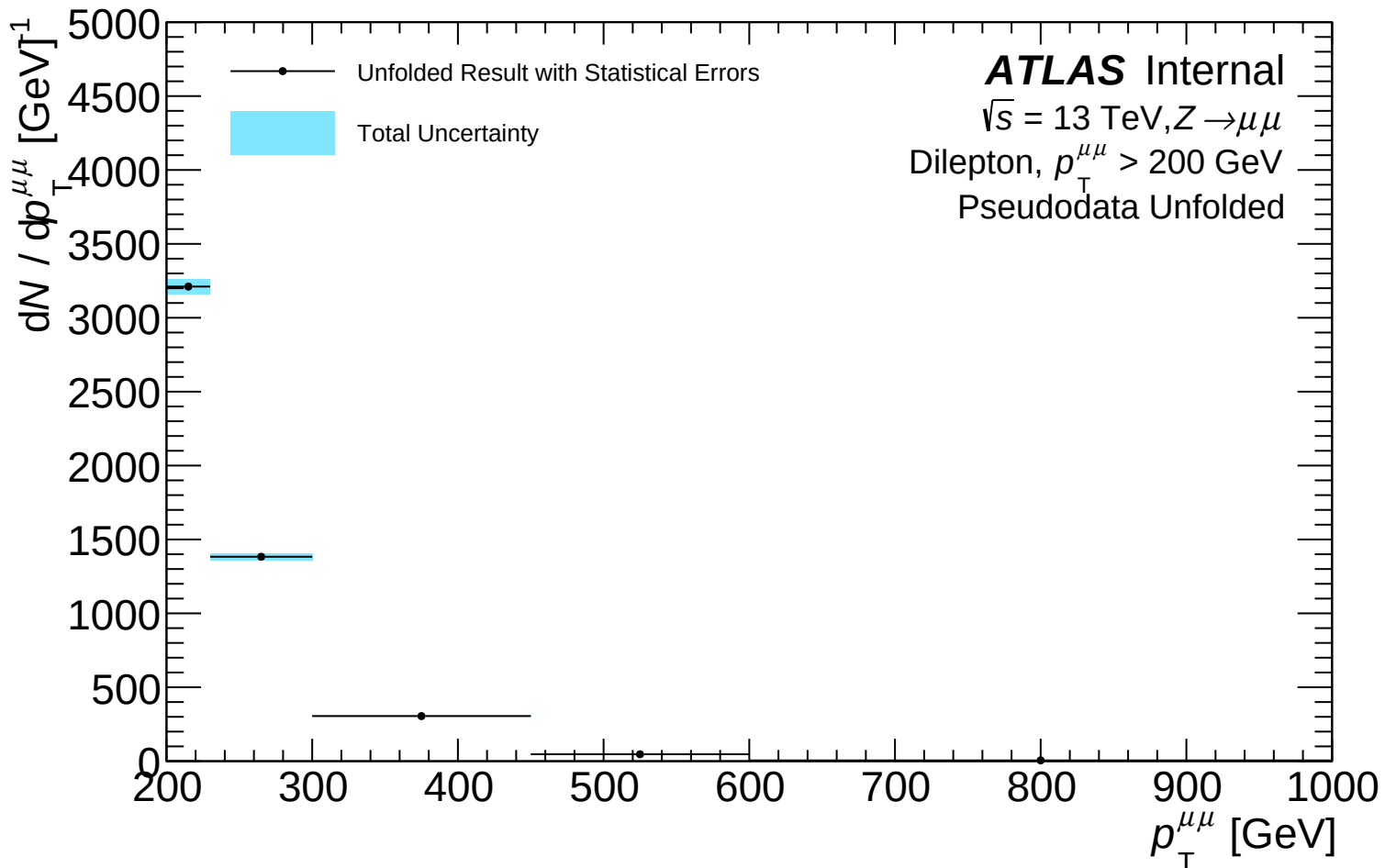
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty



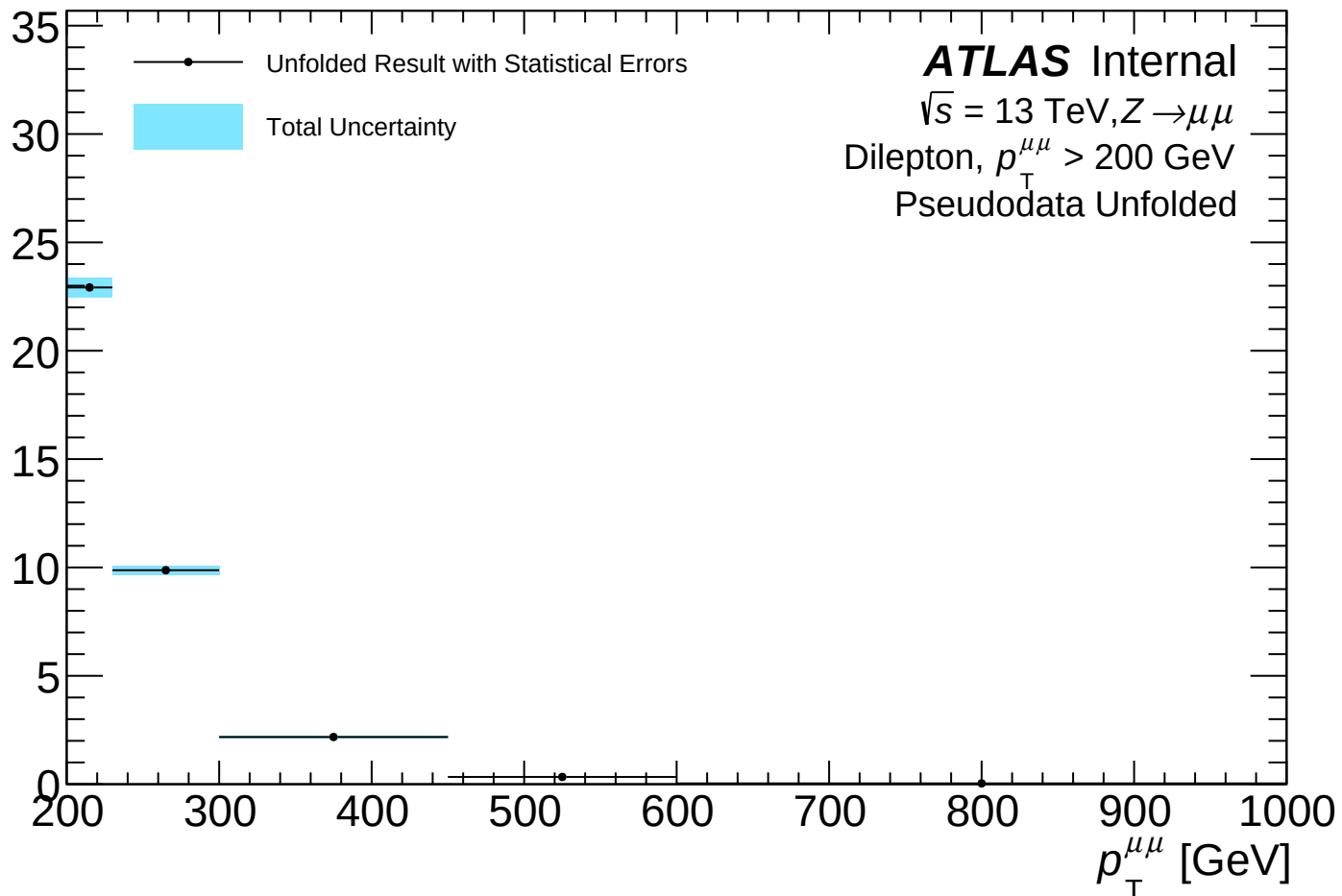


Events





$d\sigma / dp_{\text{T}}^{\mu\mu}$ [fb/GeV]



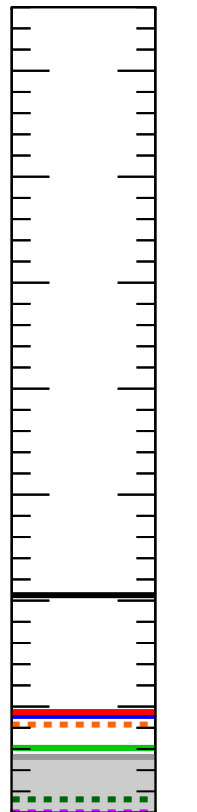
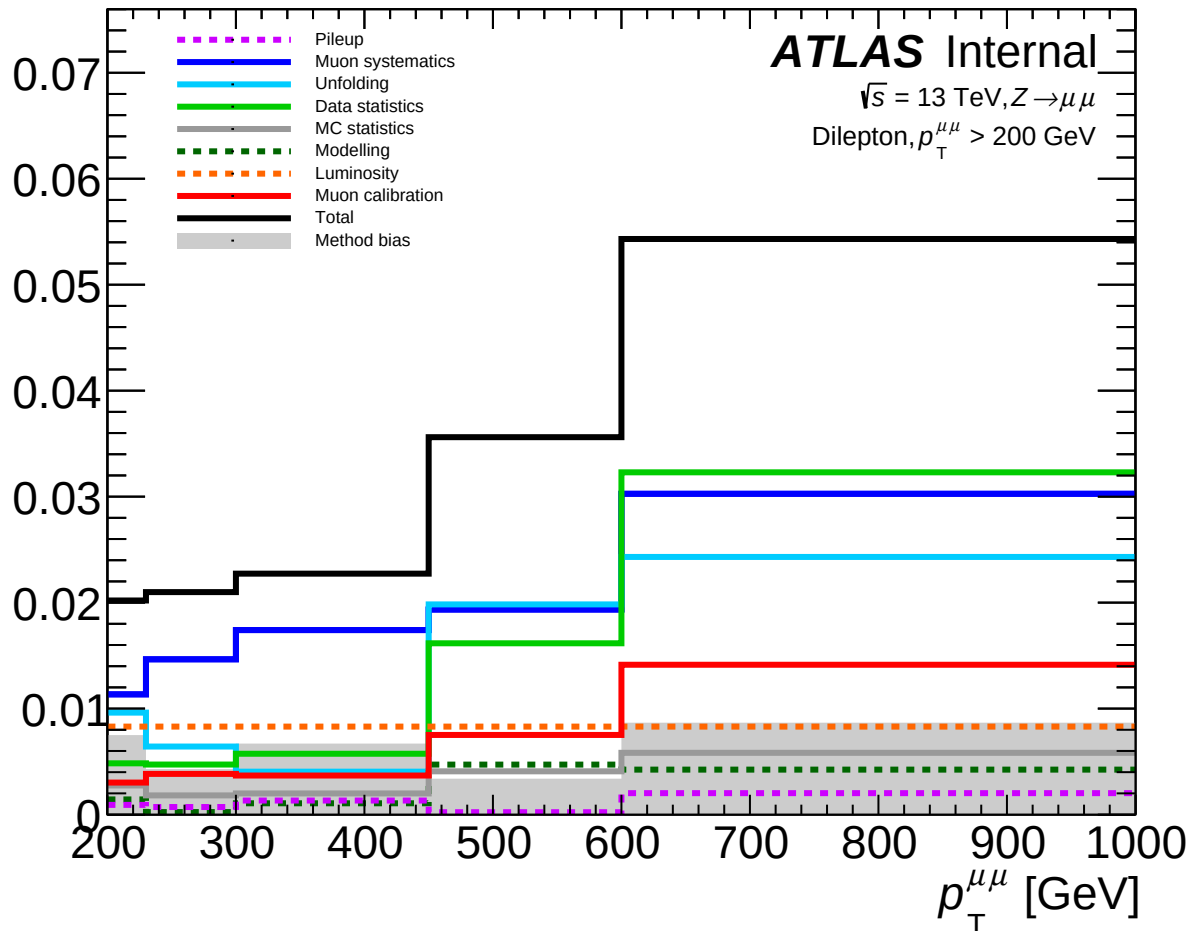
Fractional uncertainty

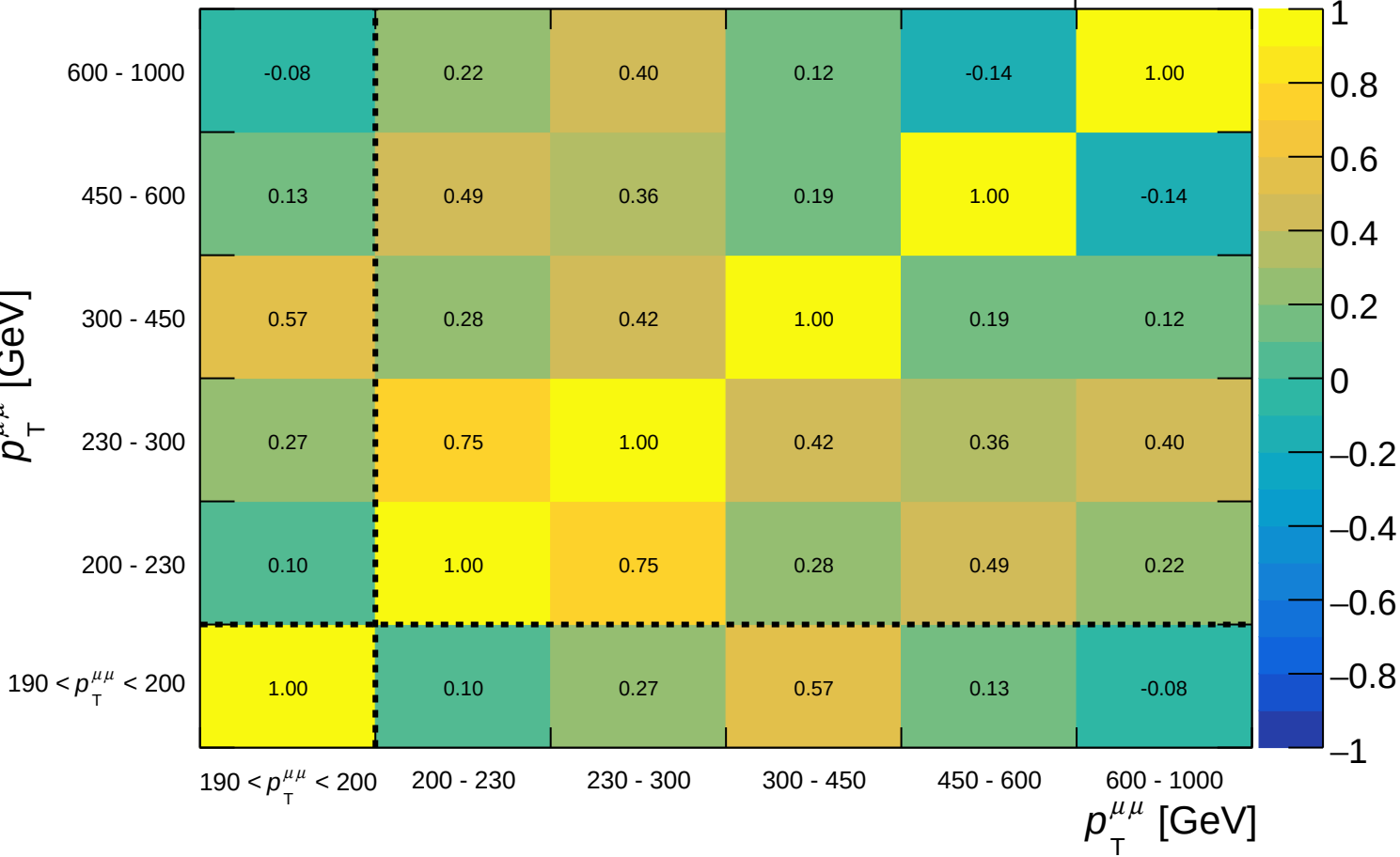
ATLAS Internal

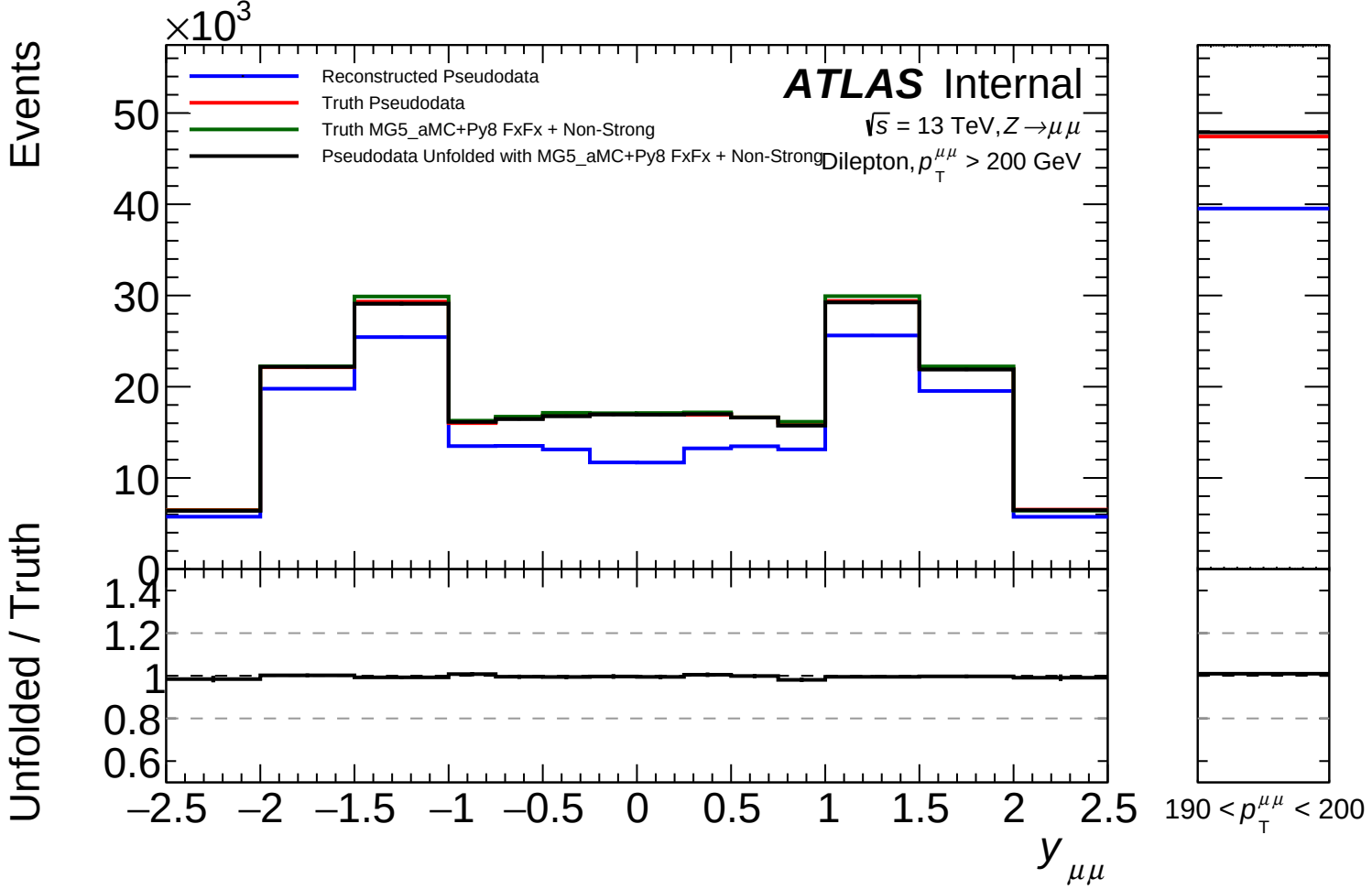
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

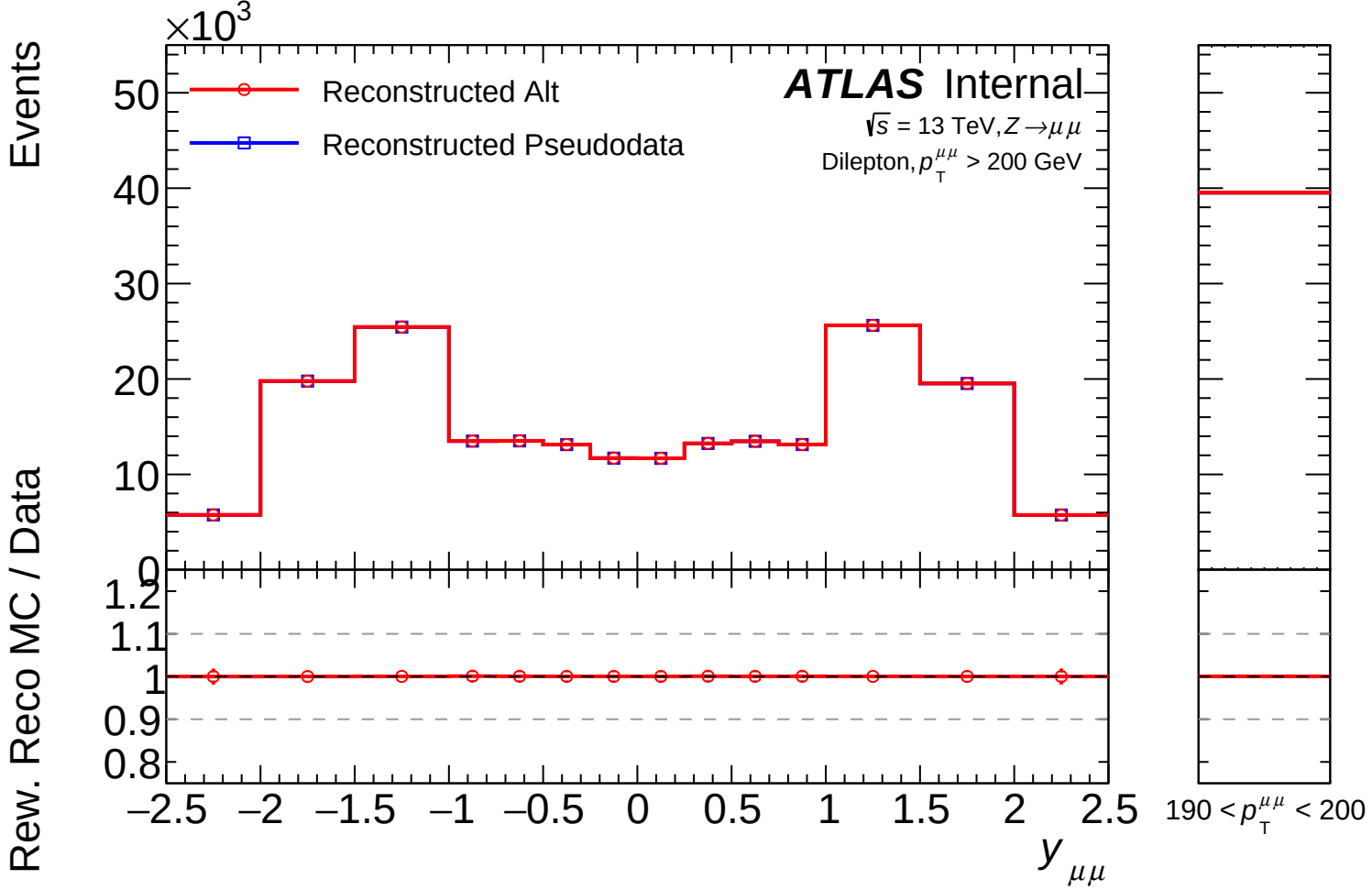
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

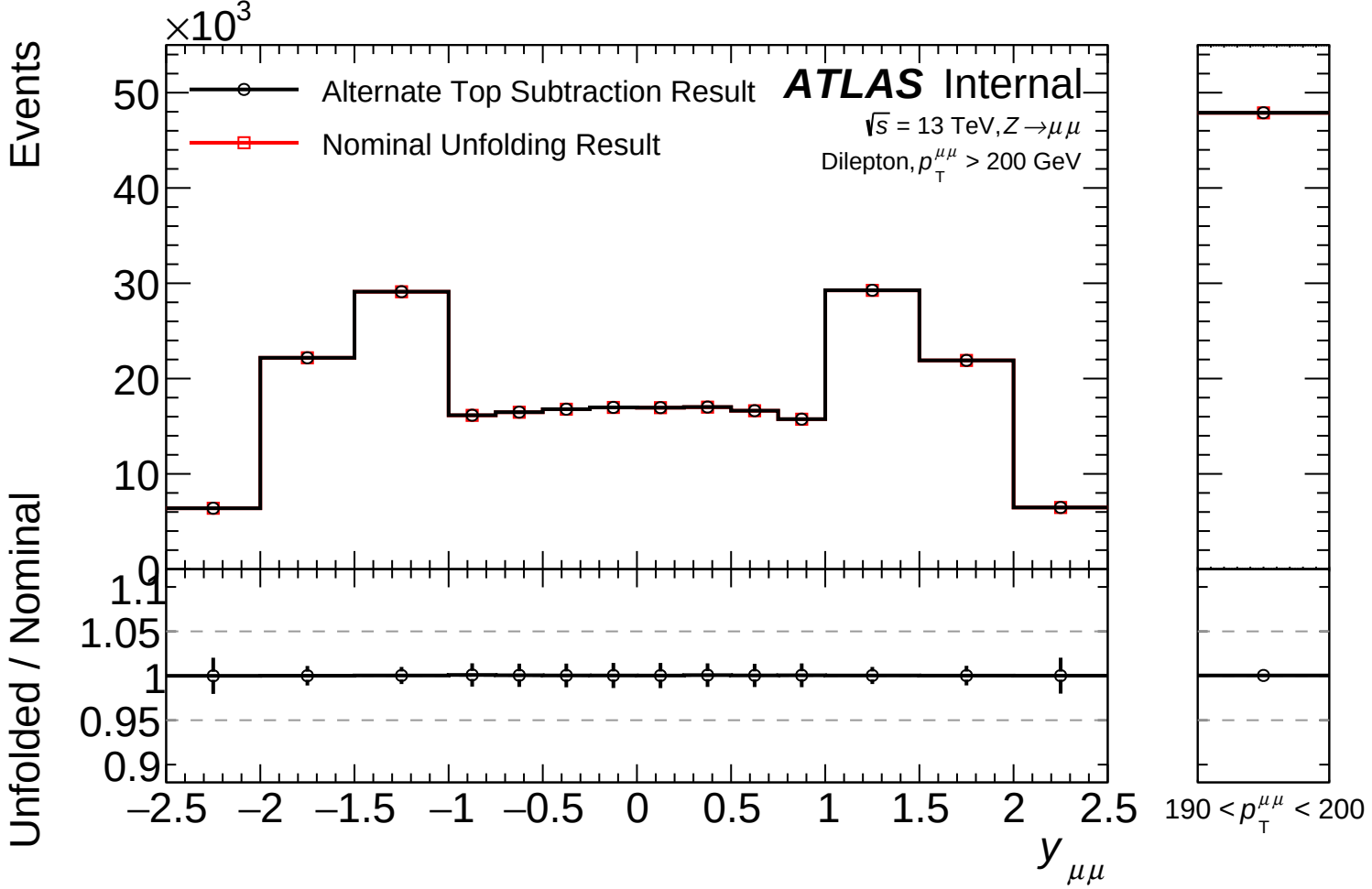
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Muon calibration
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $p_{\text{T}}^{\mu\mu} [\text{GeV}]$ 







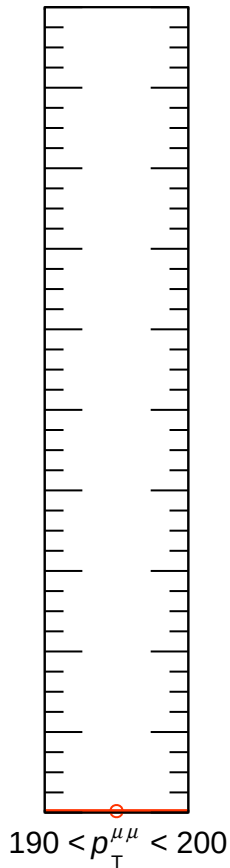
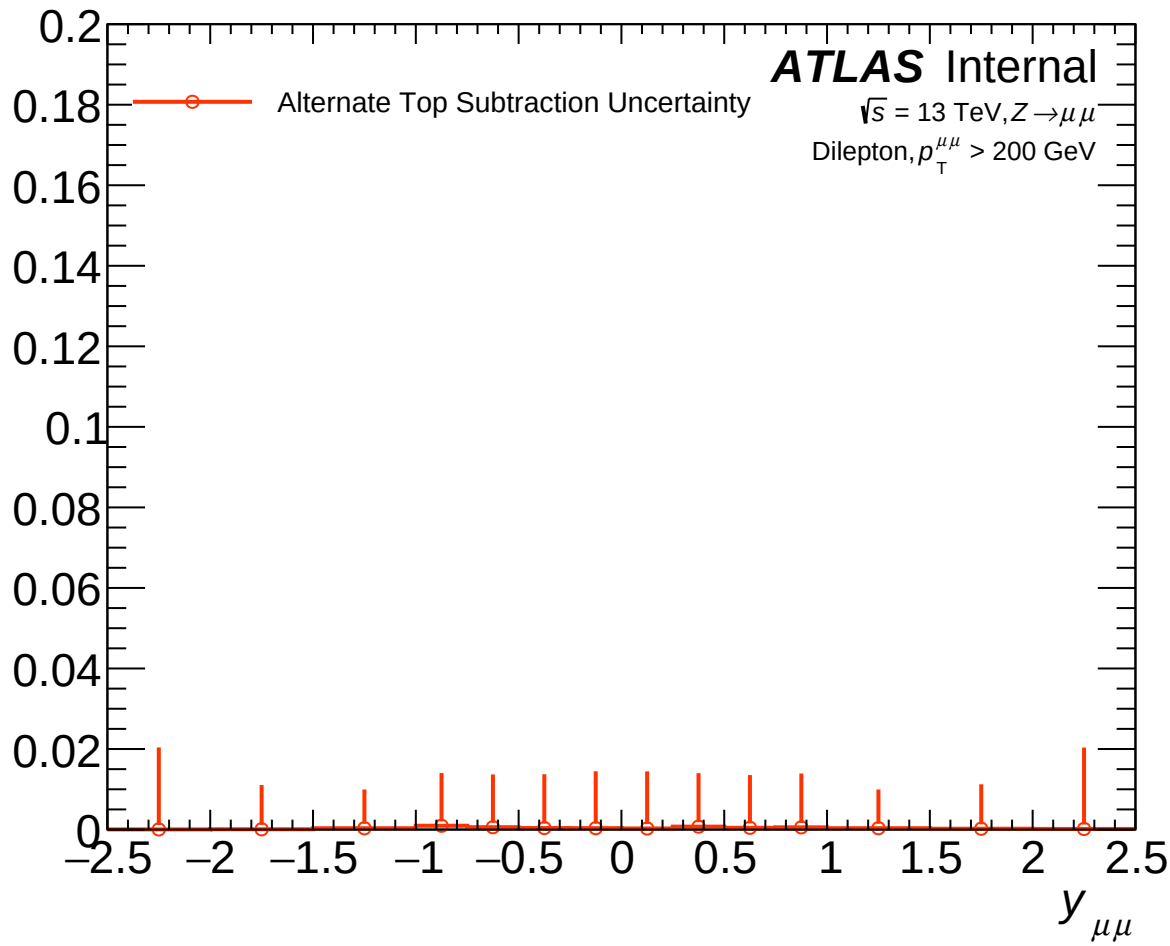
Fractional systematic impact

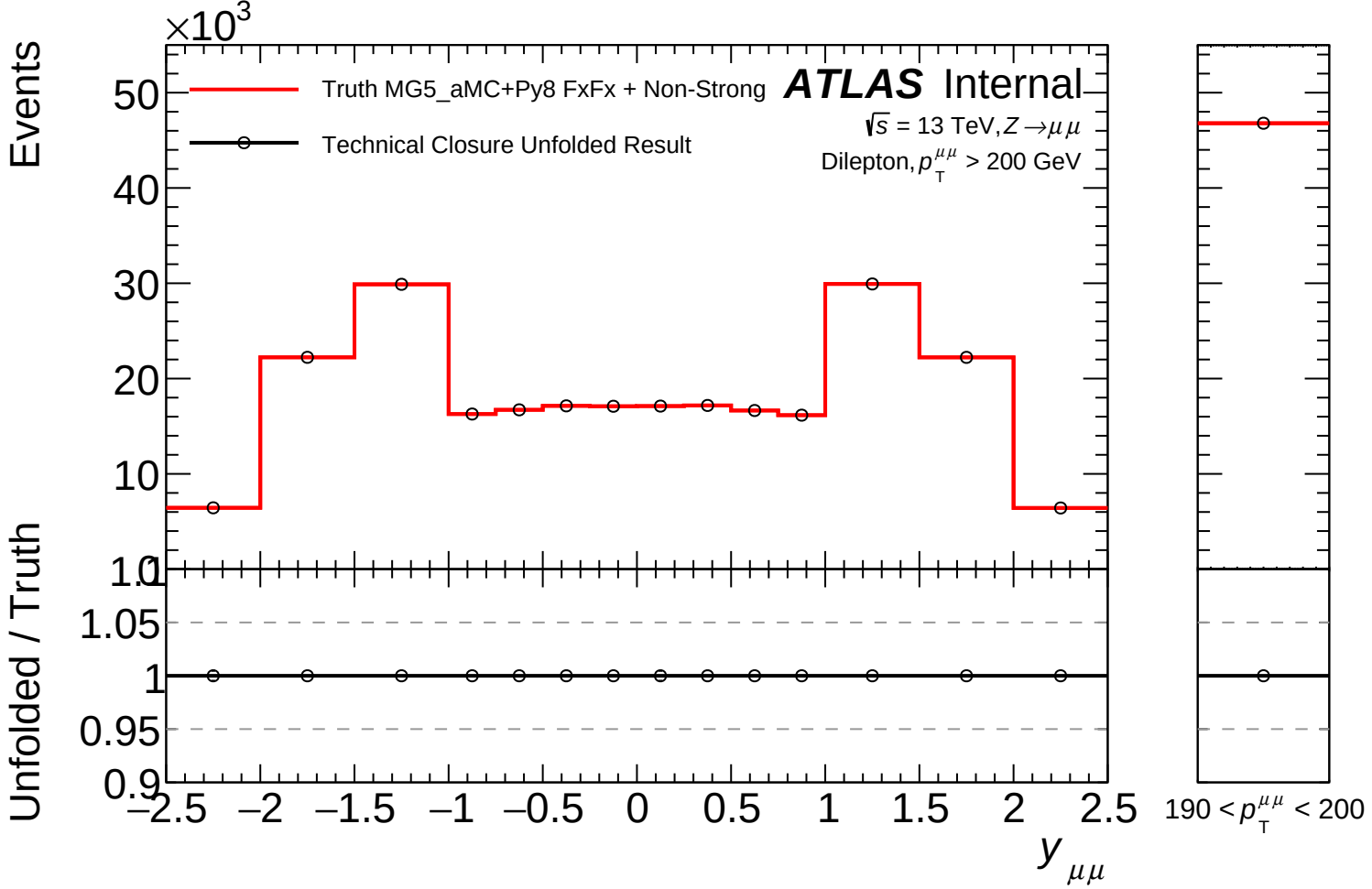
ATLAS Internal

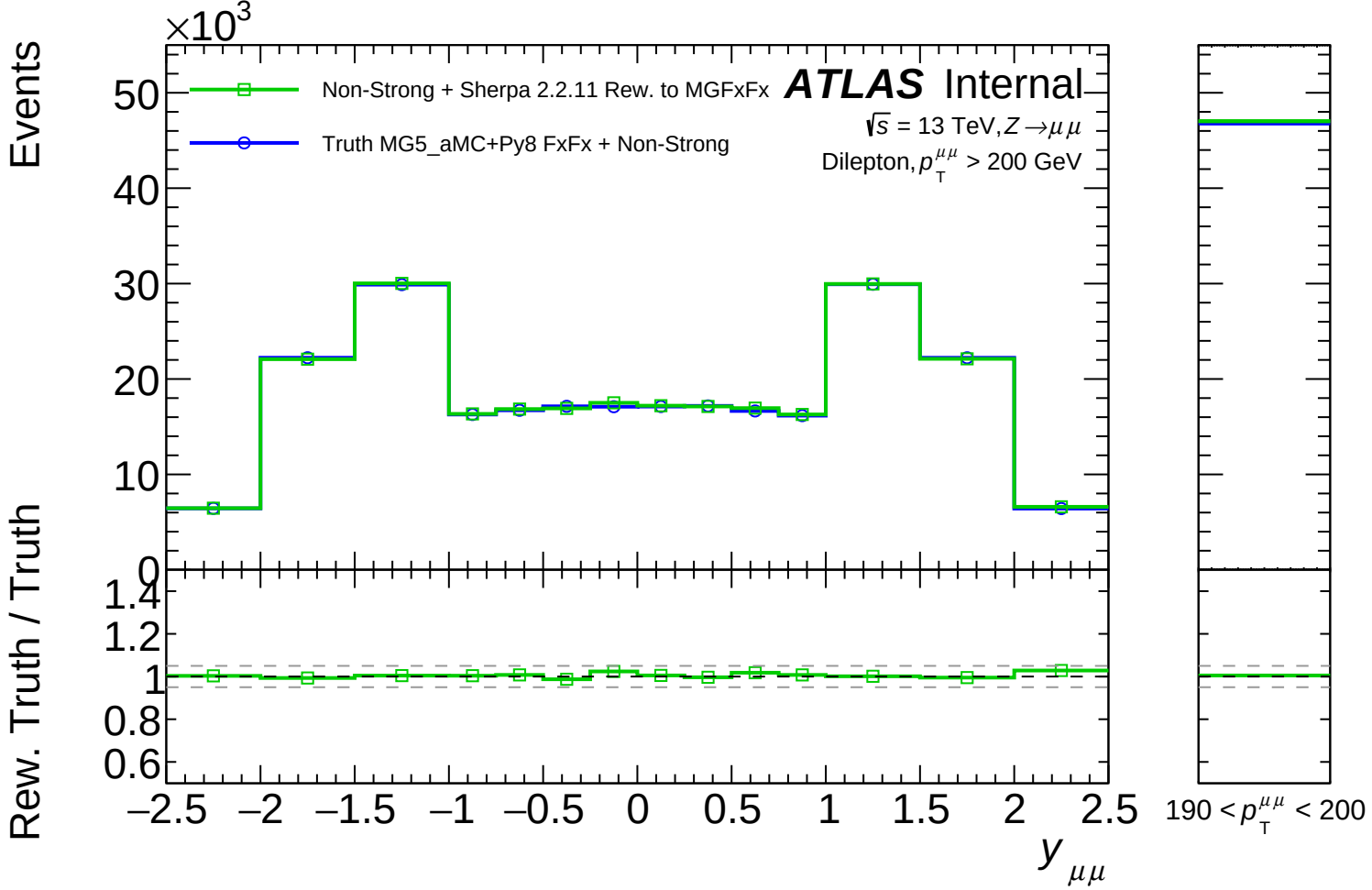
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

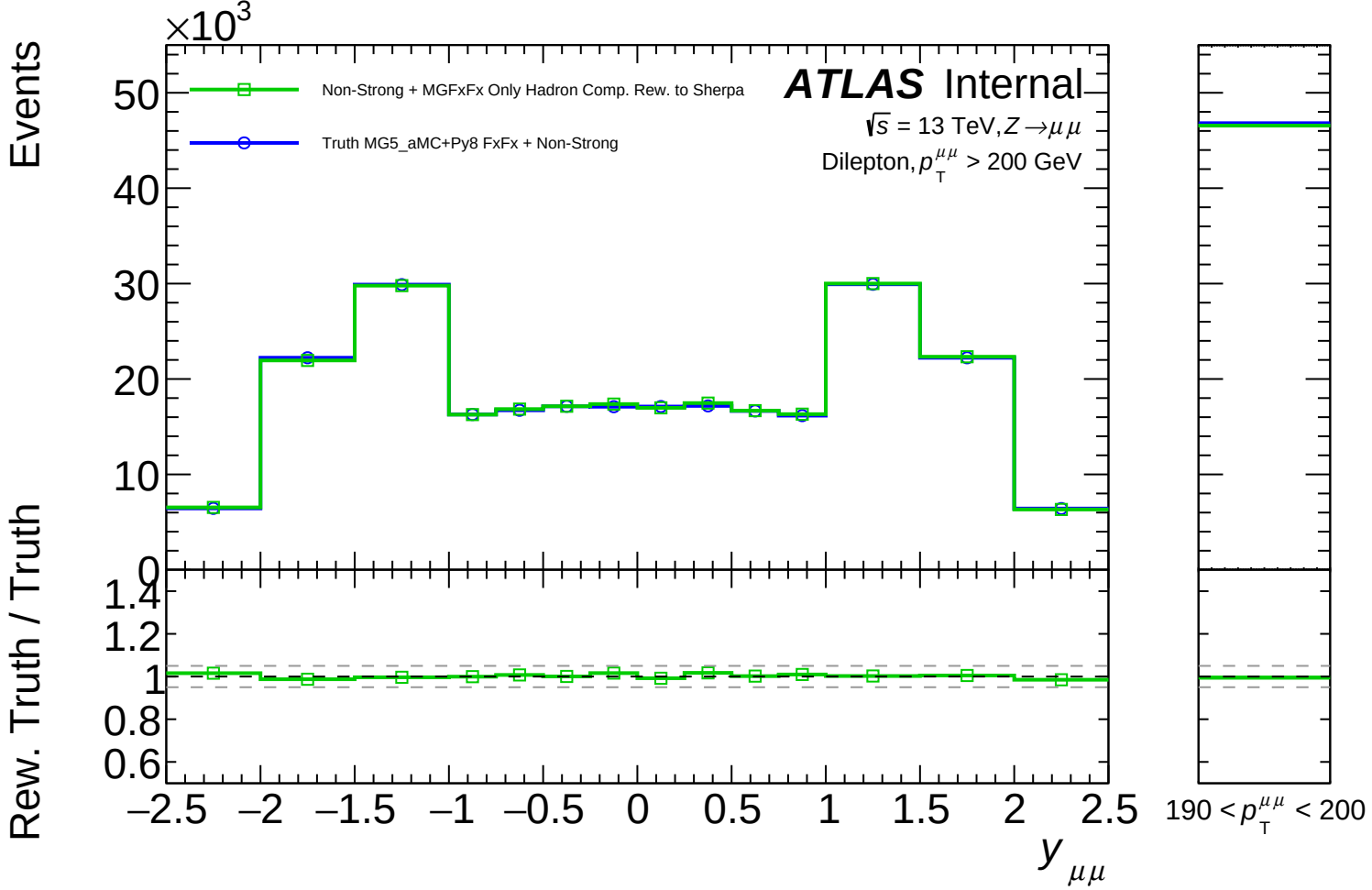
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

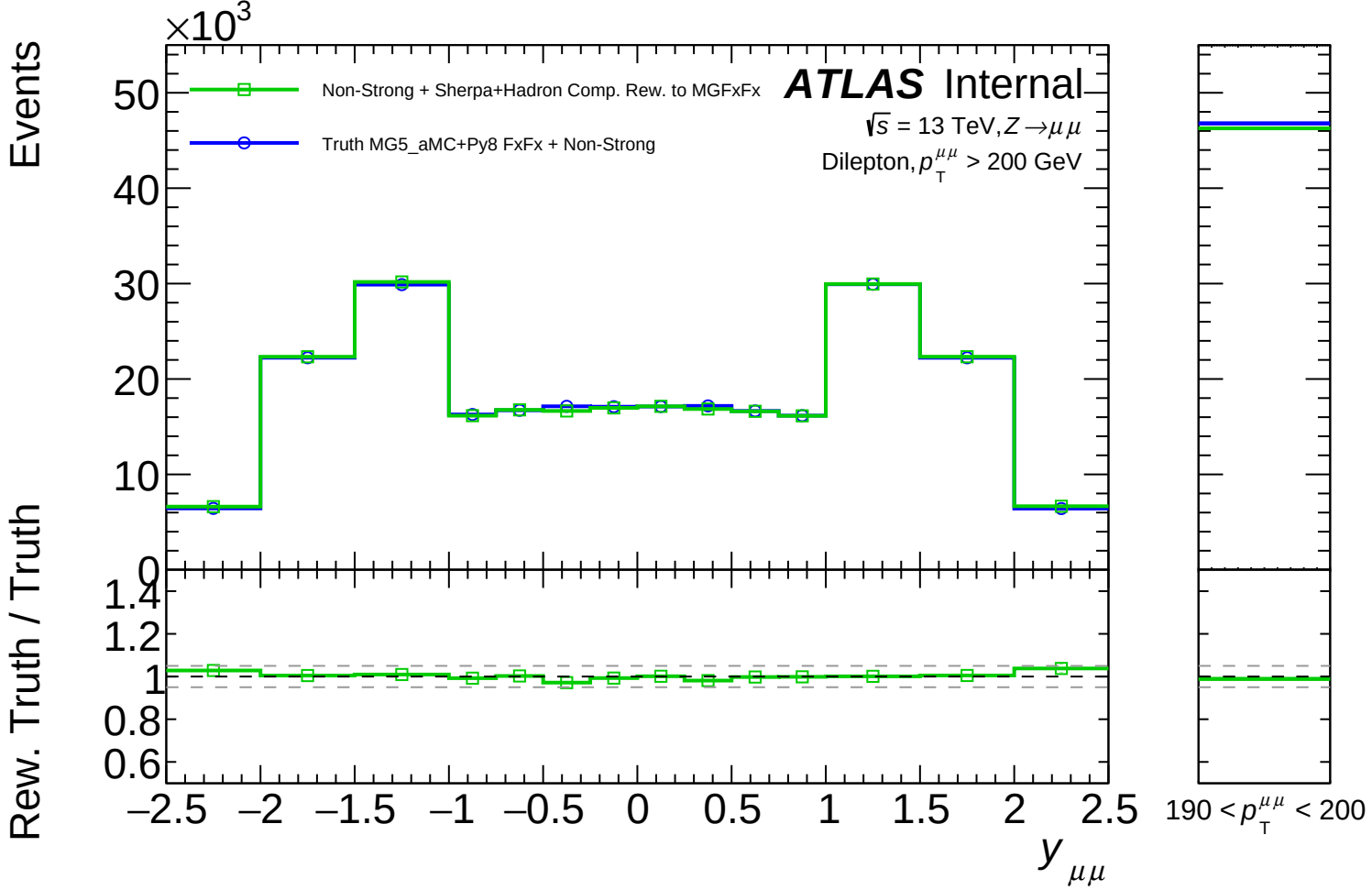
Alternate Top Subtraction Uncertainty

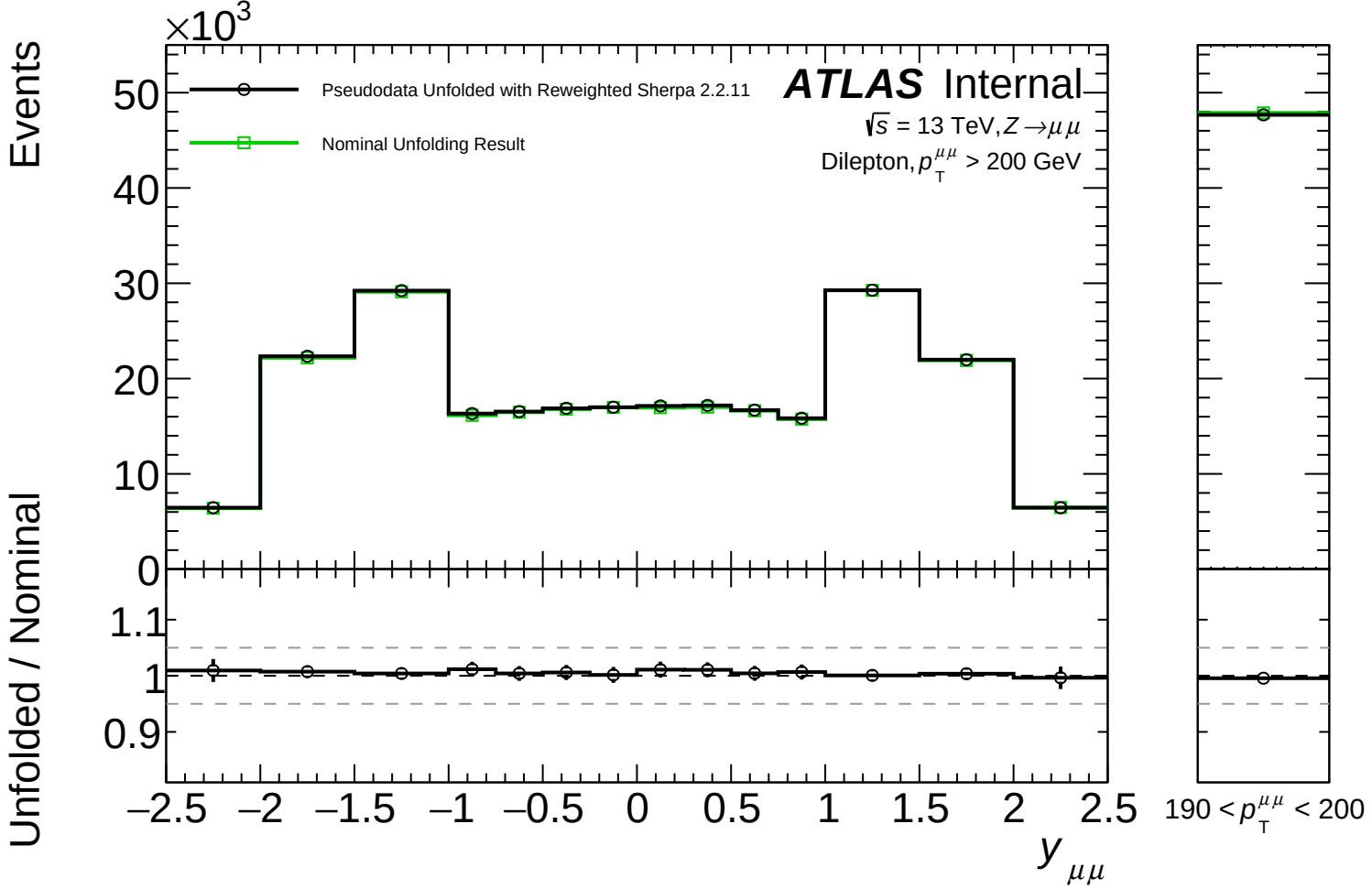


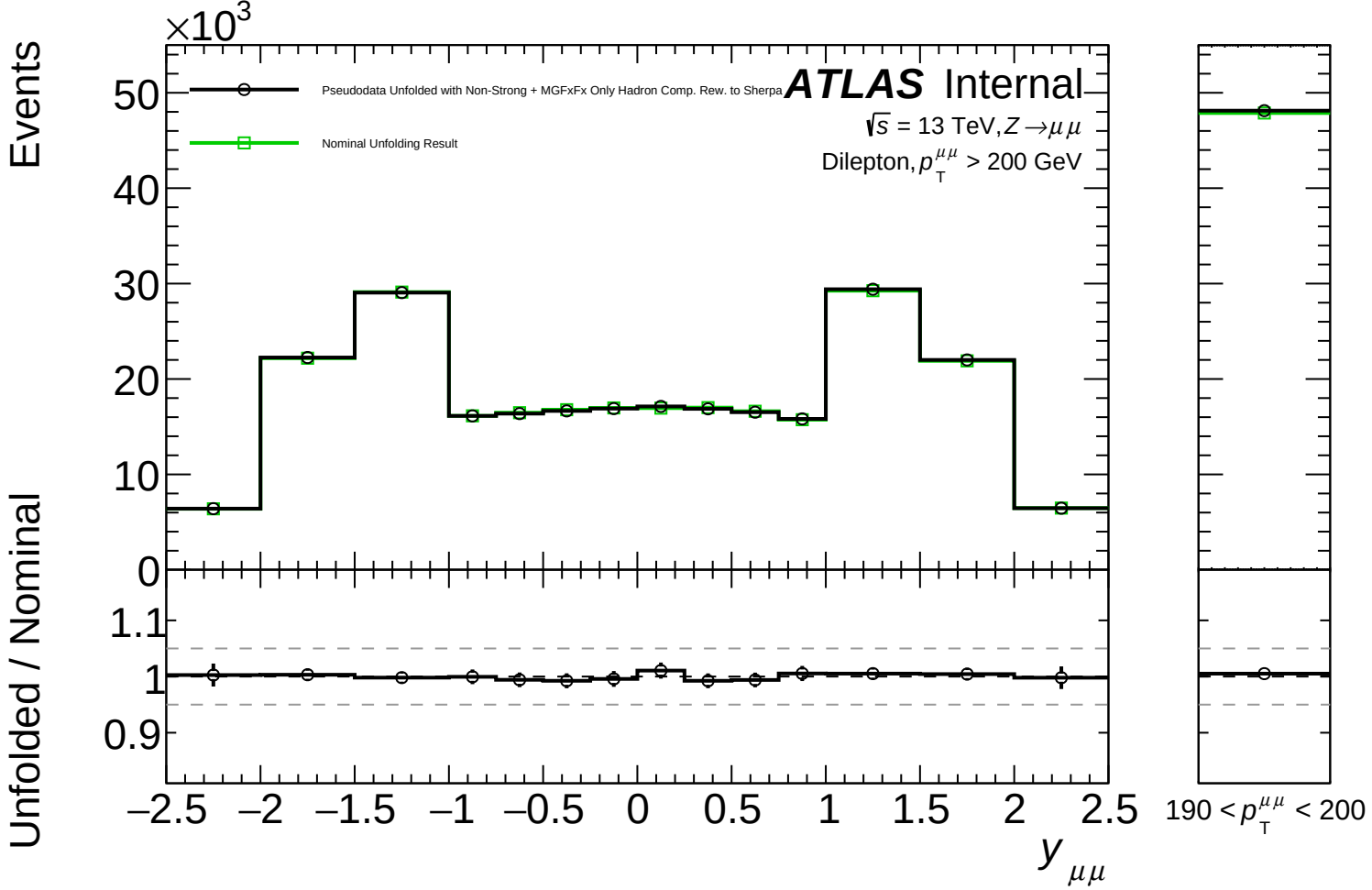


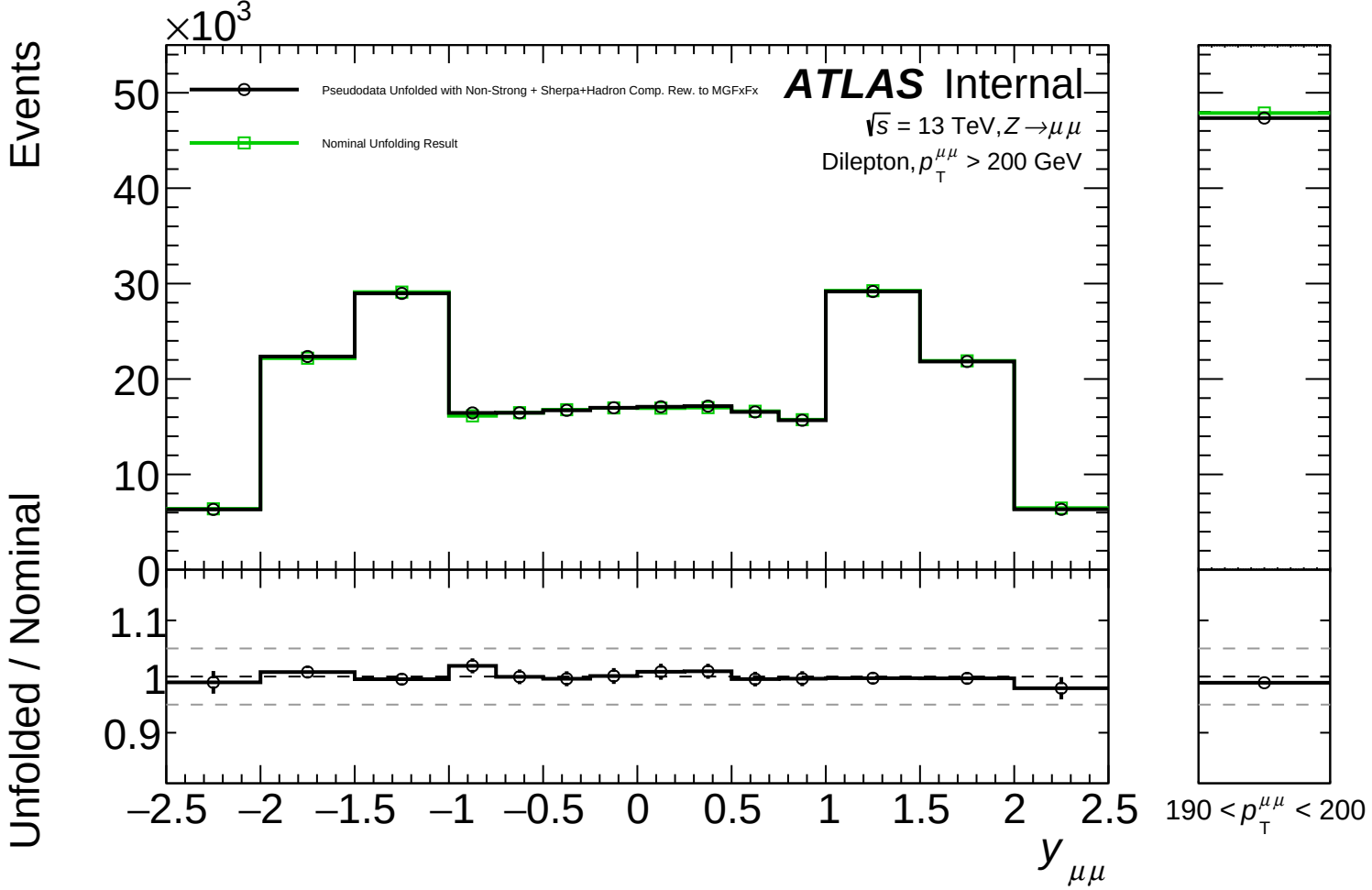






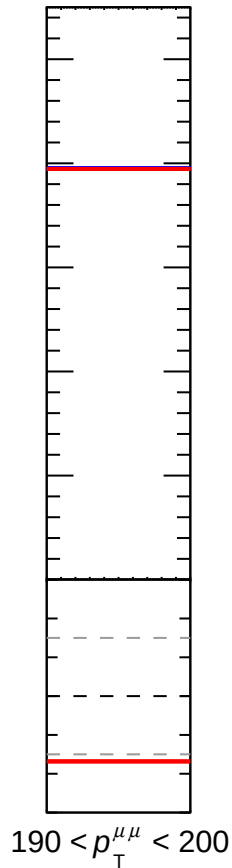
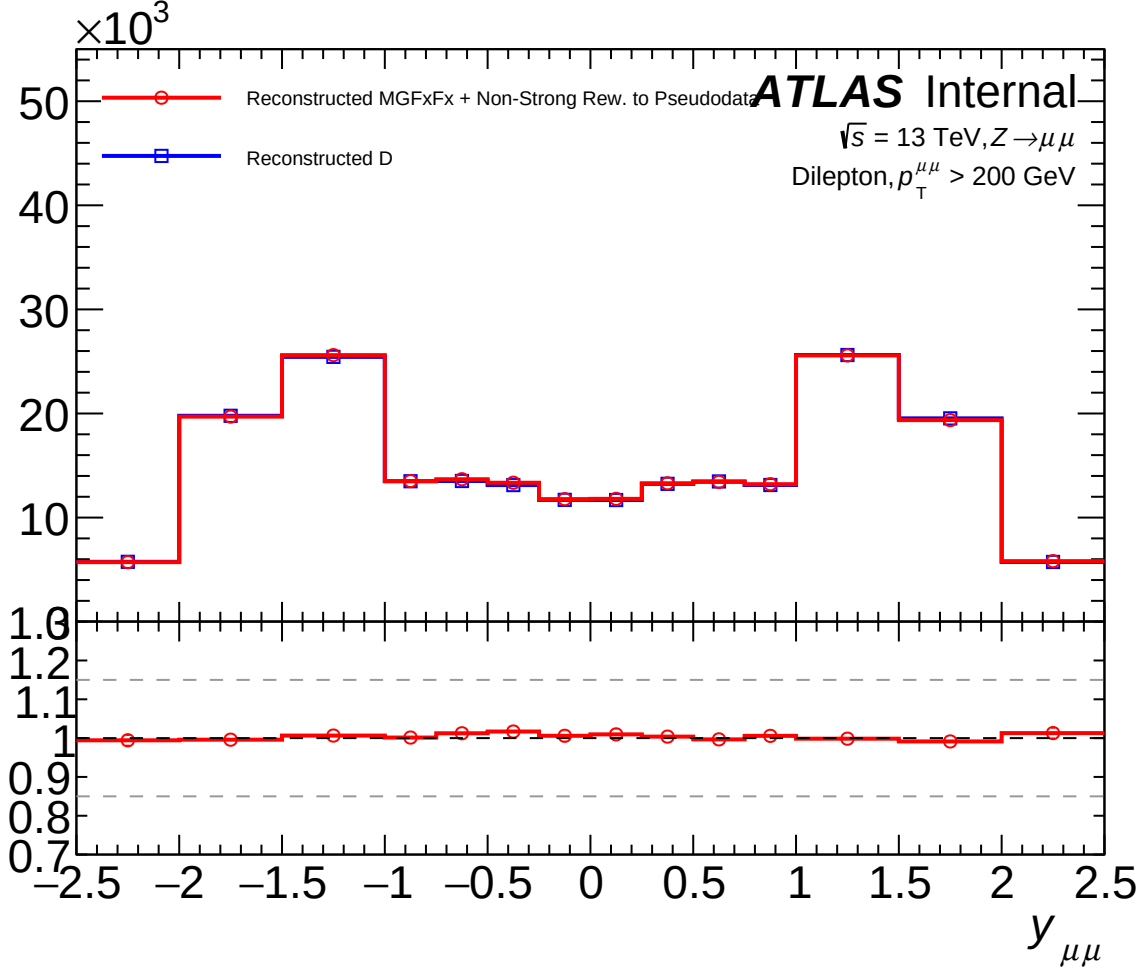


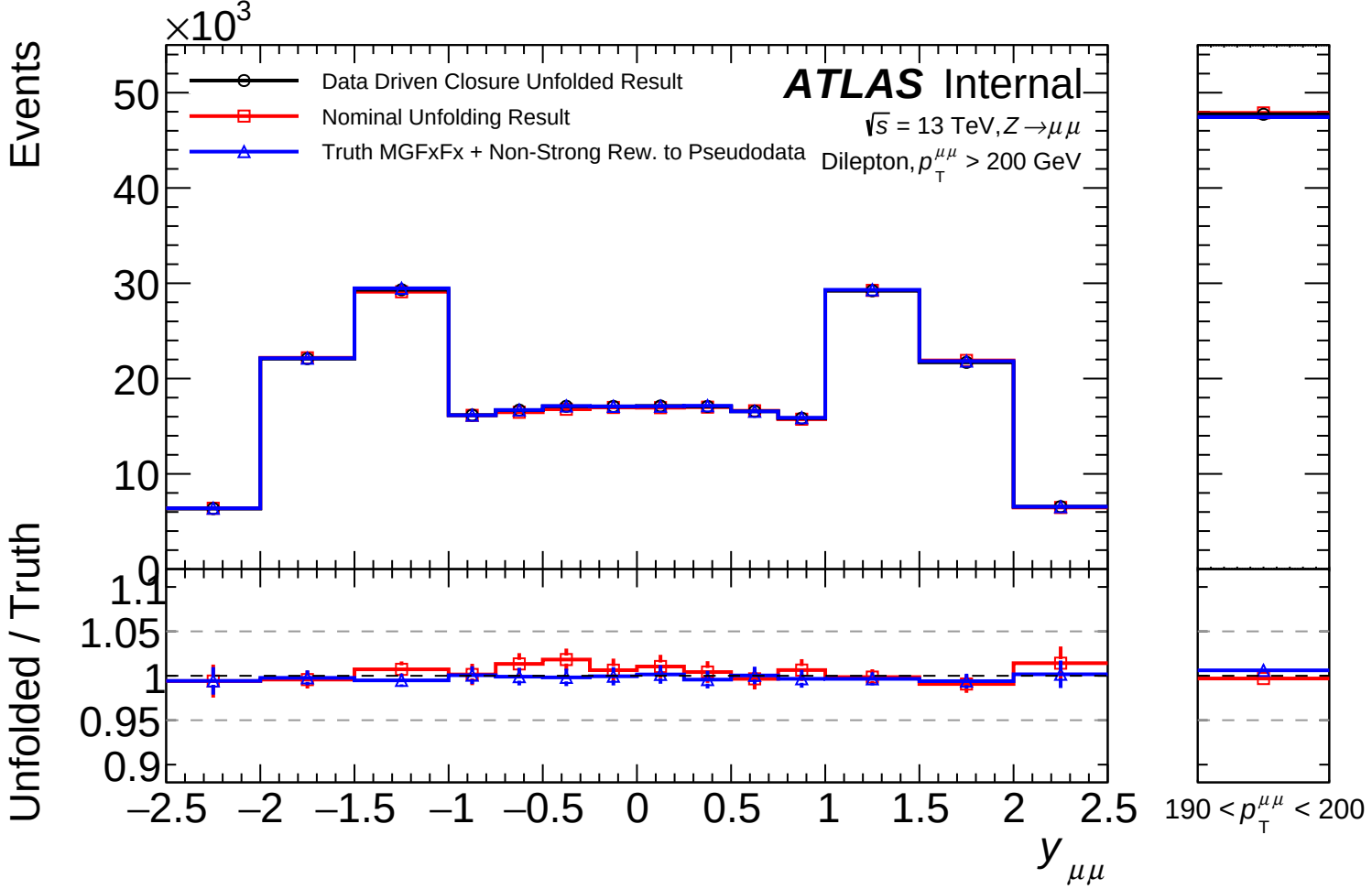




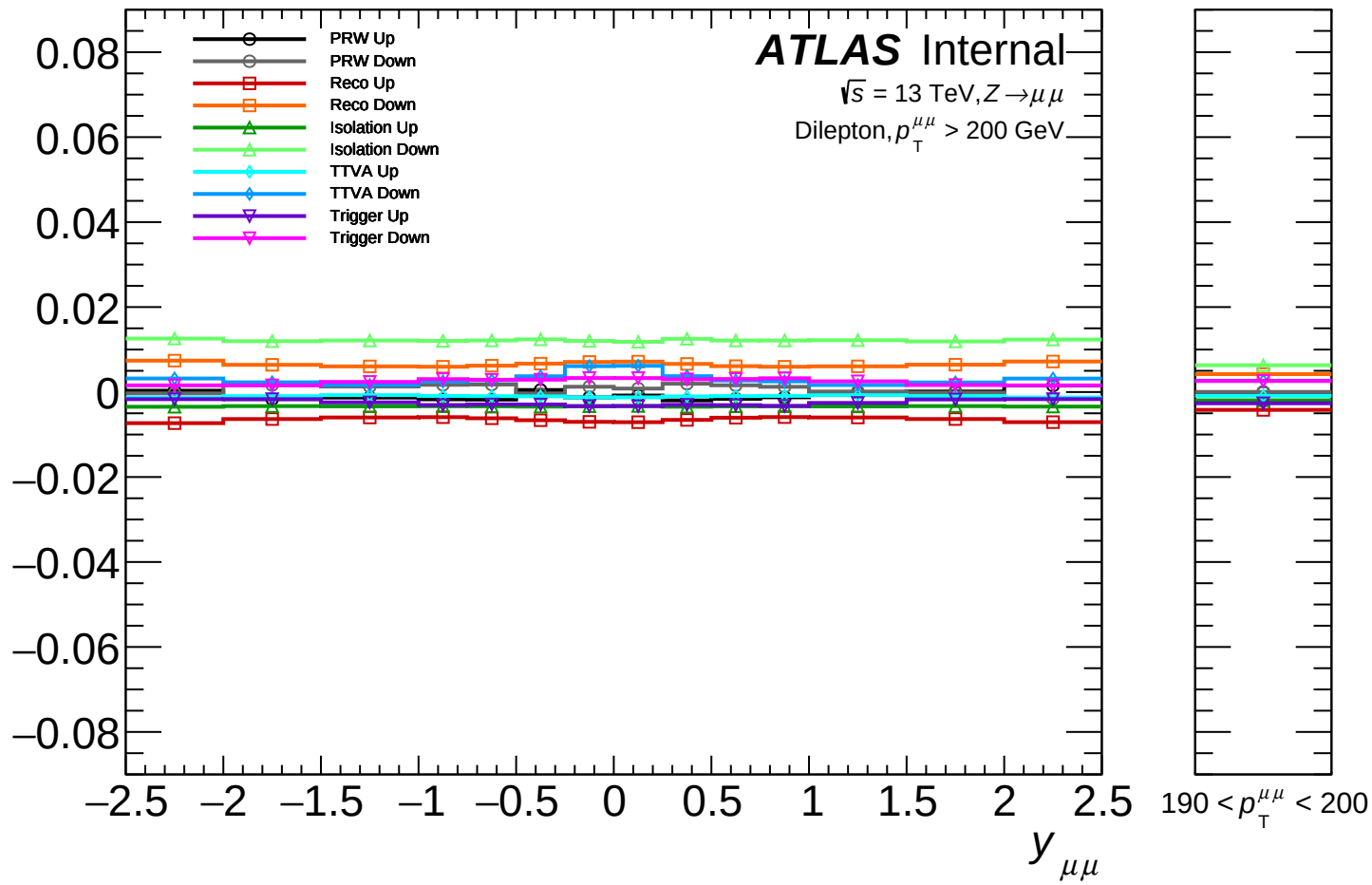
Events

Reco MC / Data

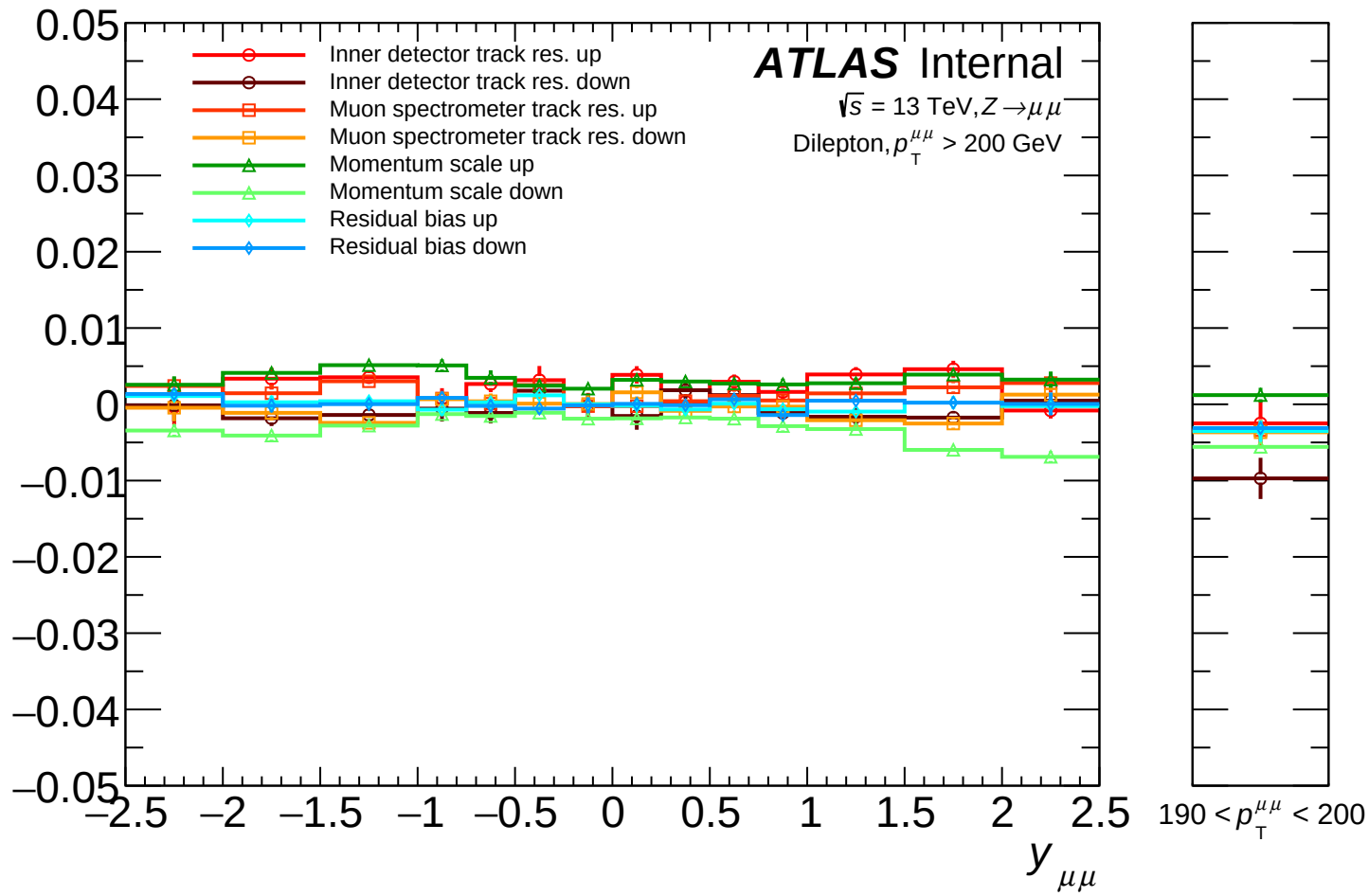




Fractional systematic impact



Fractional systematic impact



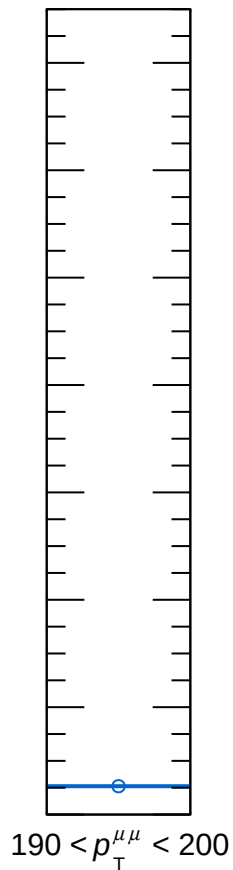
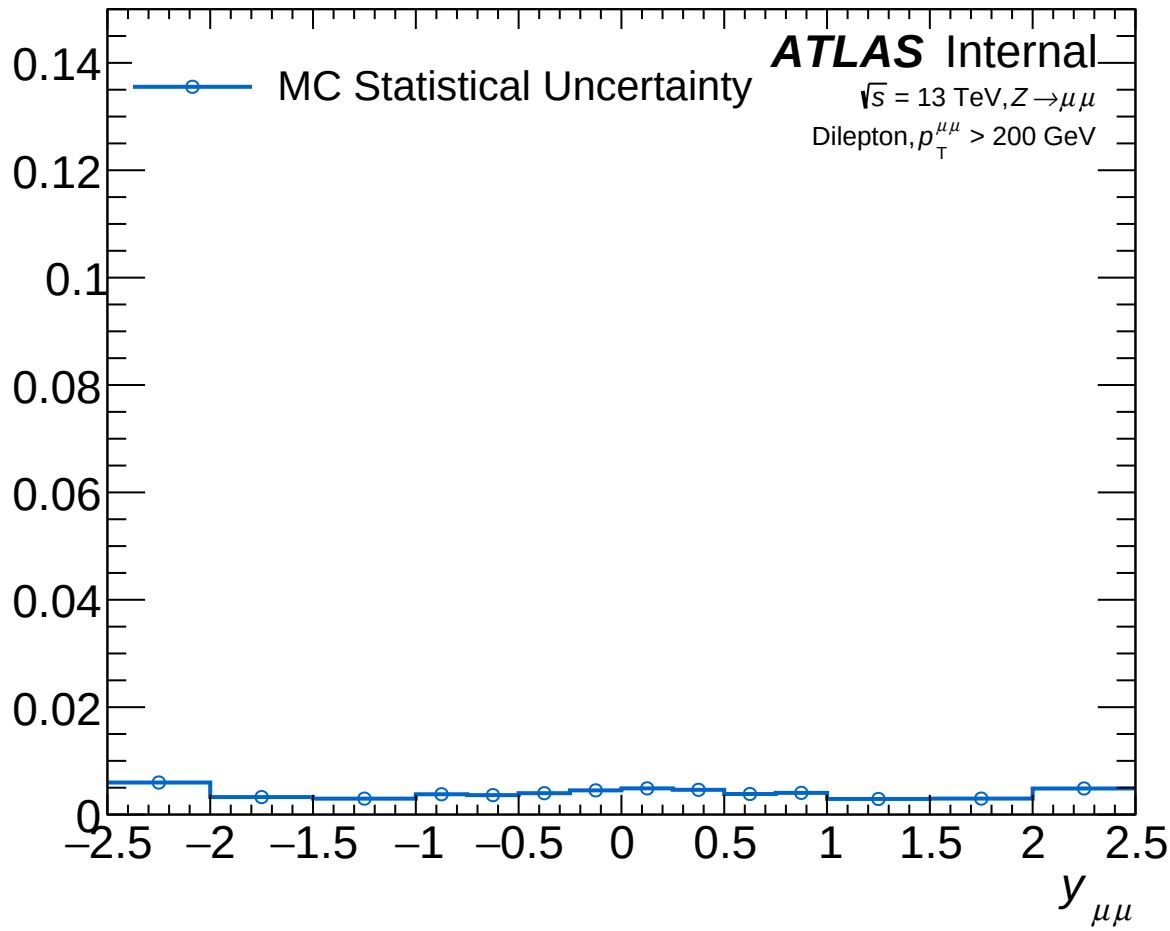
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



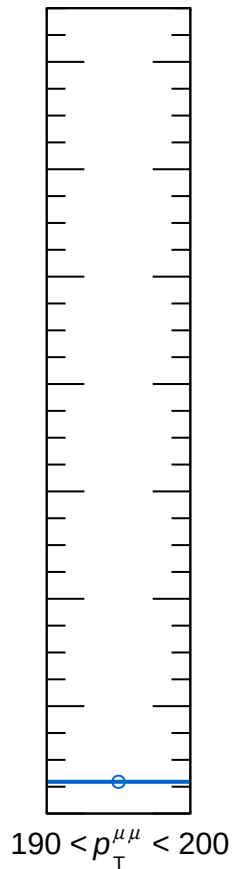
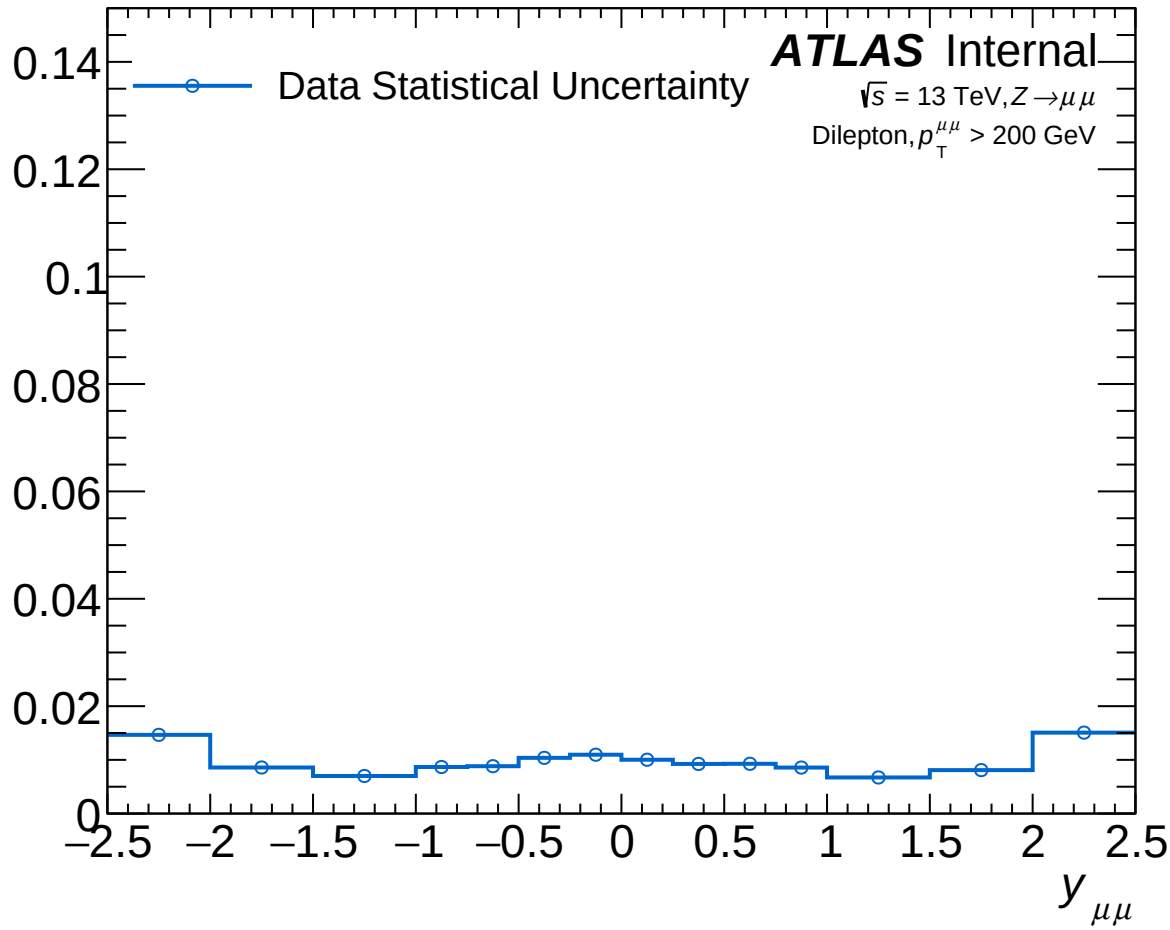
Fractional systematic impact

ATLAS Internal

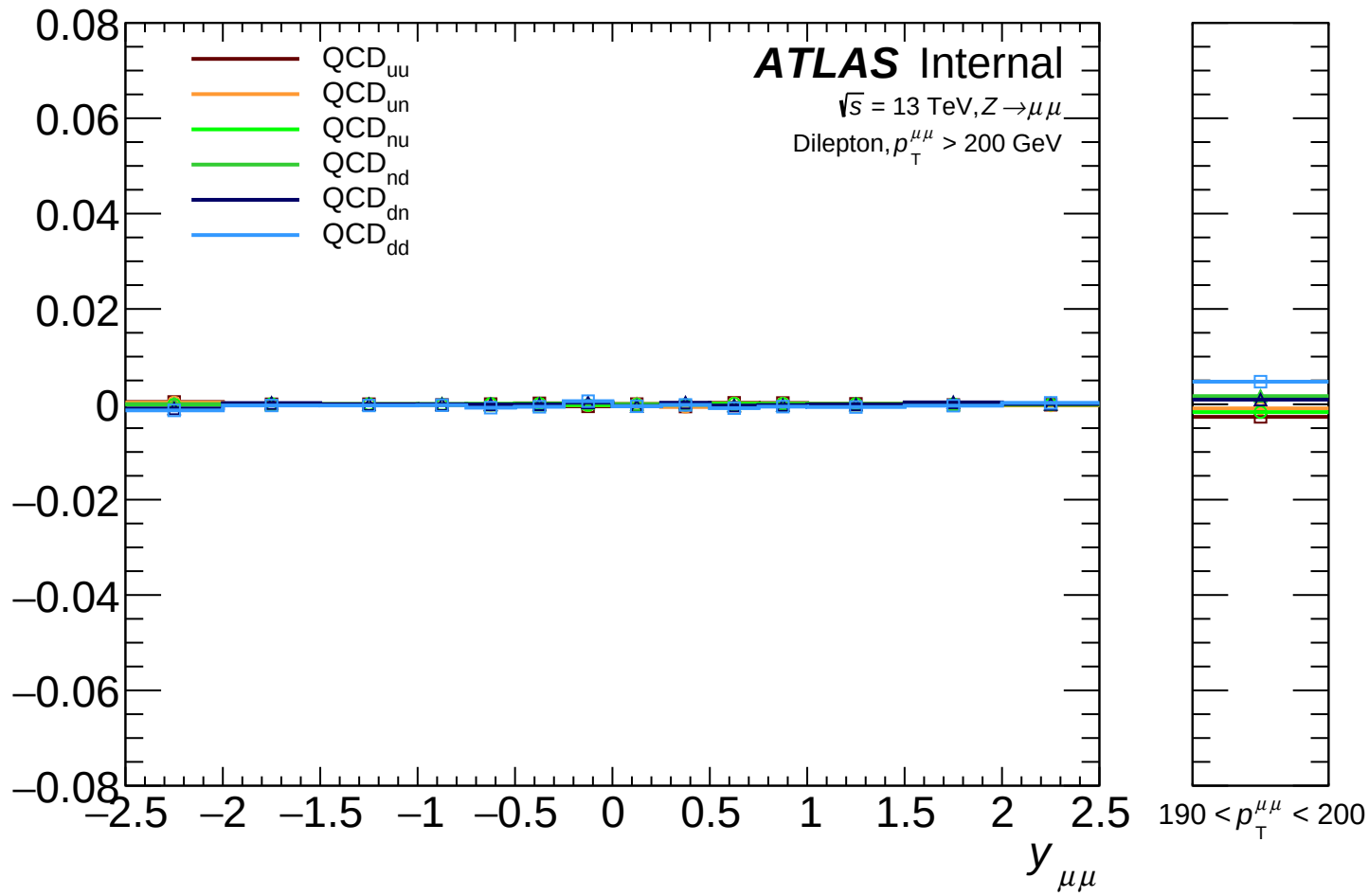
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

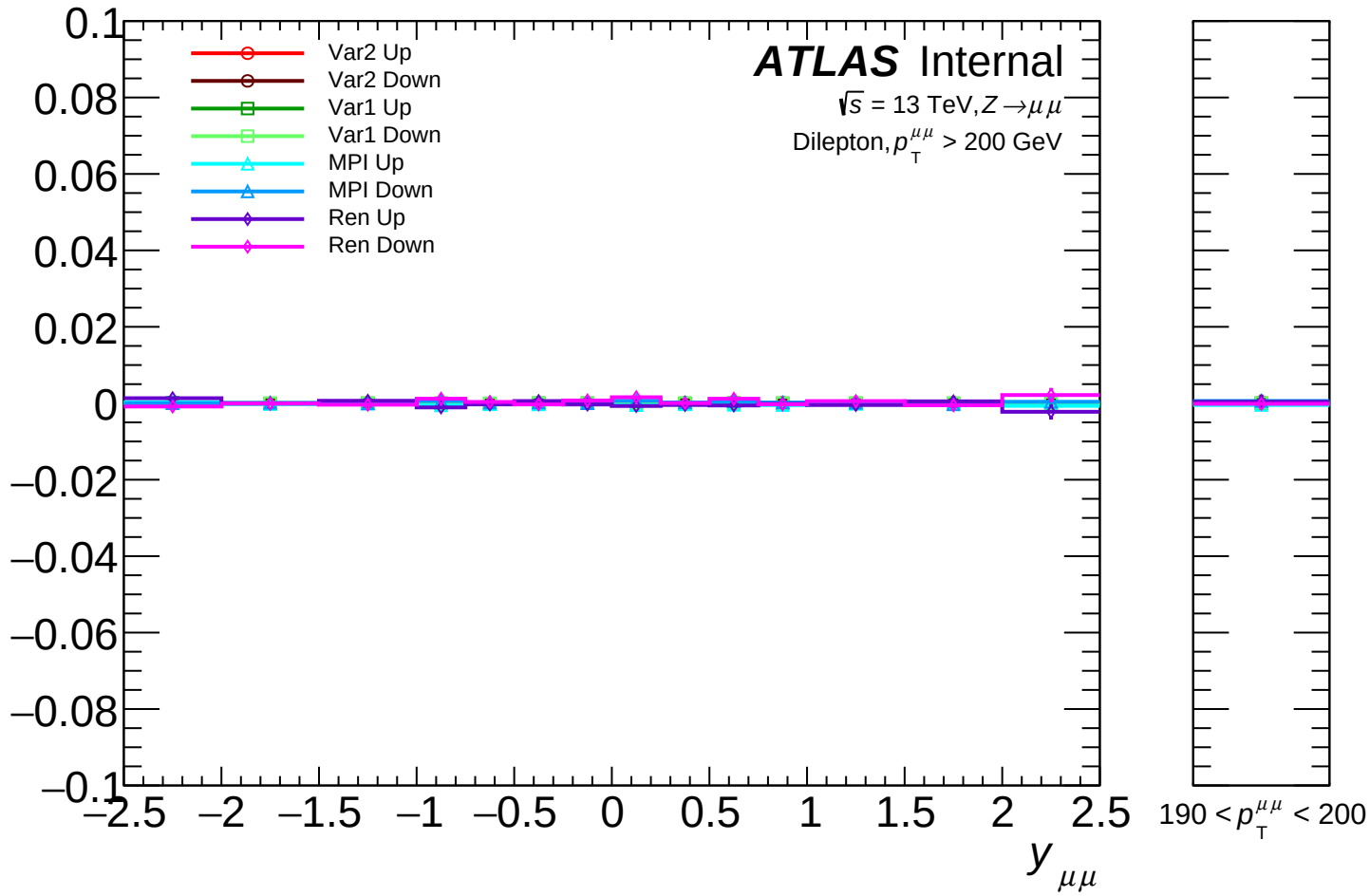
Data Statistical Uncertainty

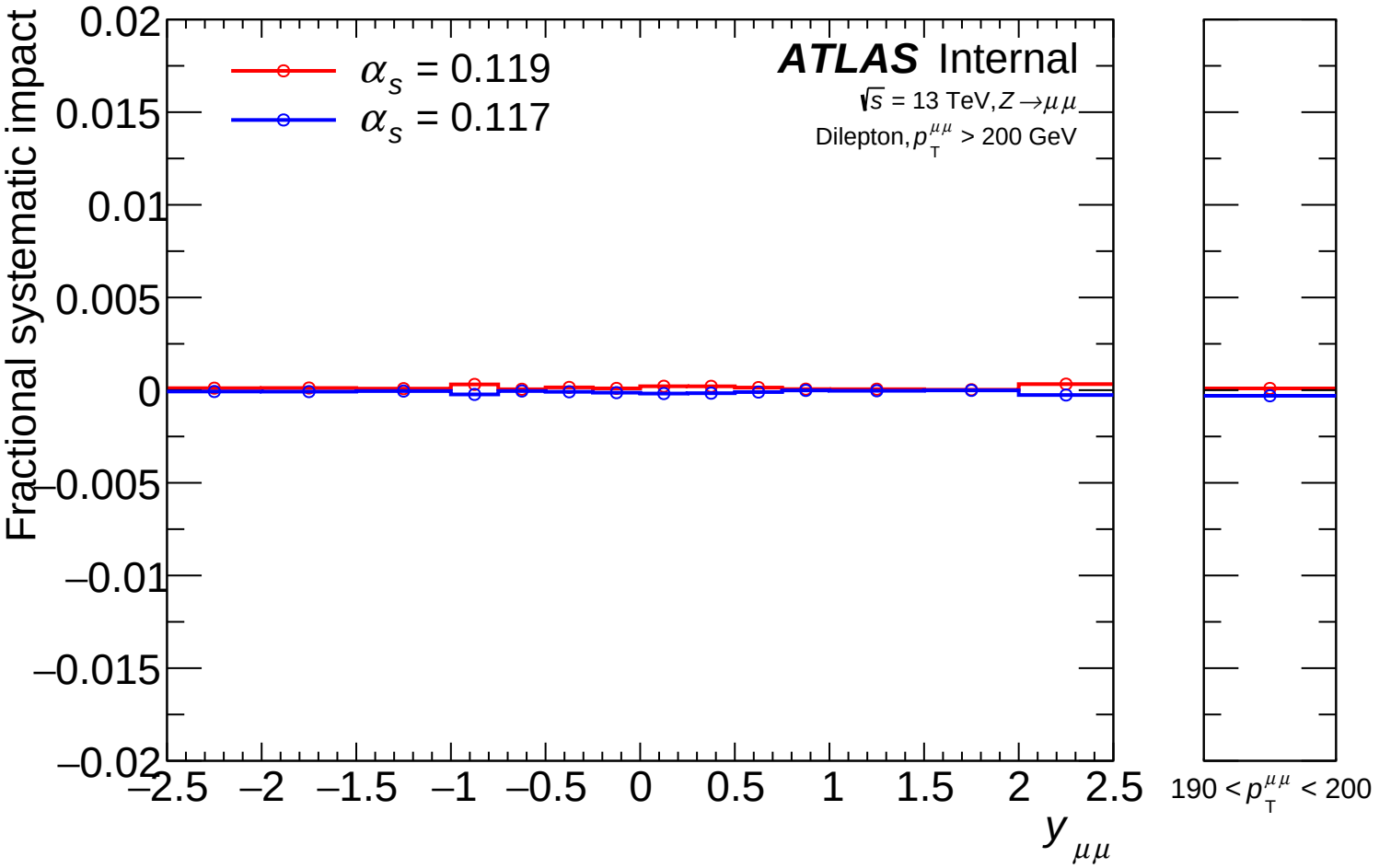


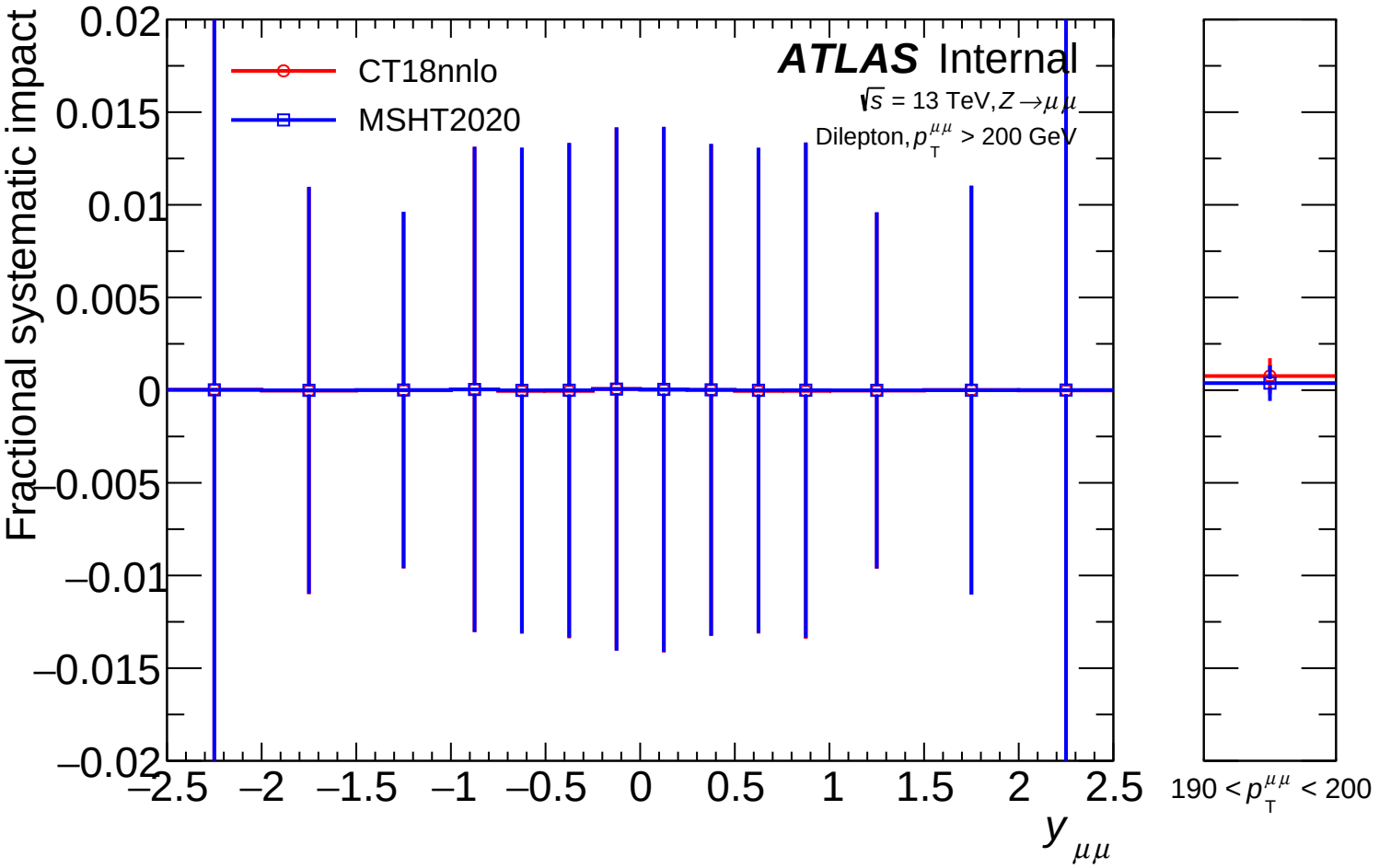
Fractional systematic impact



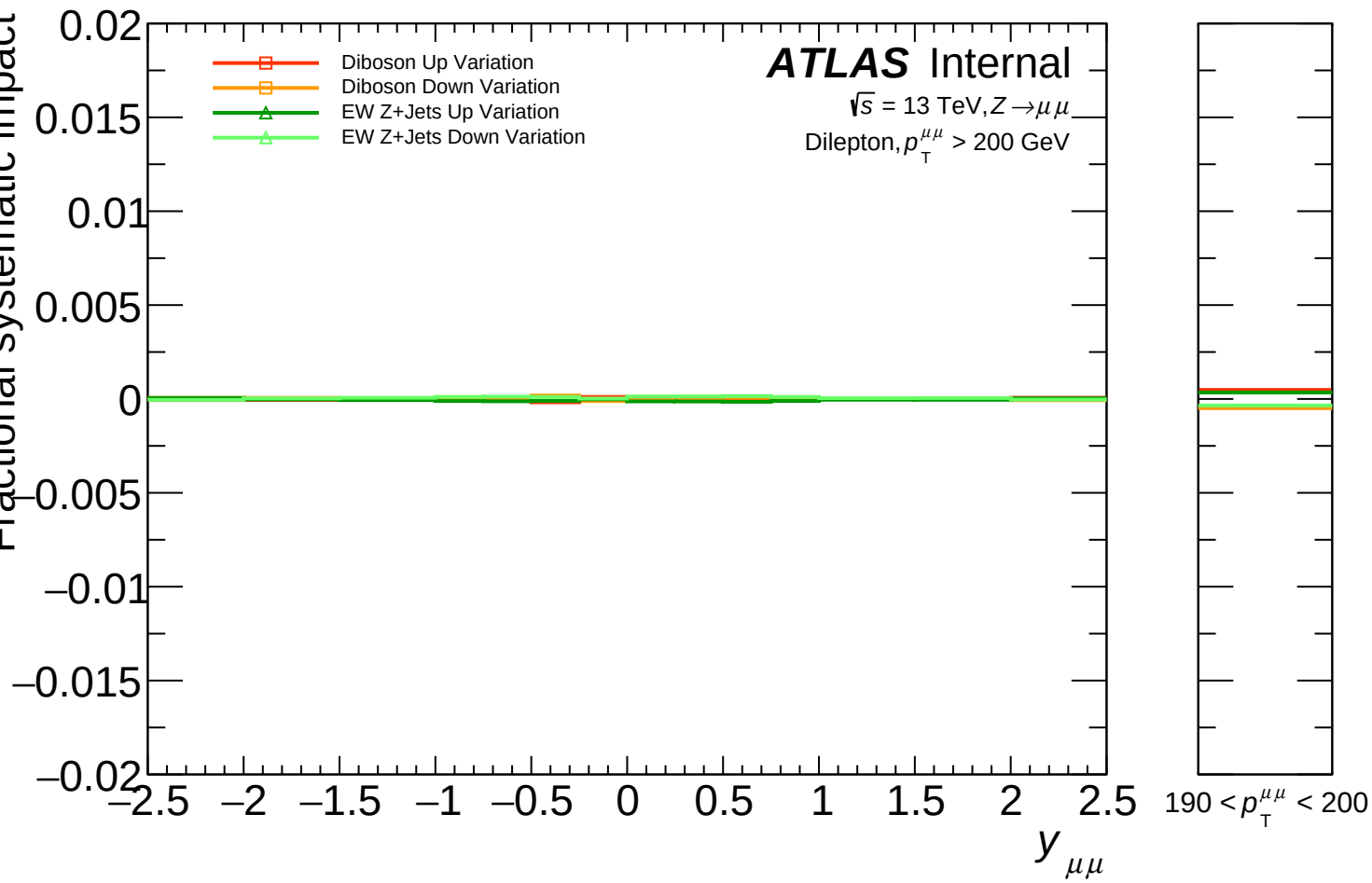
Fractional systematic impact







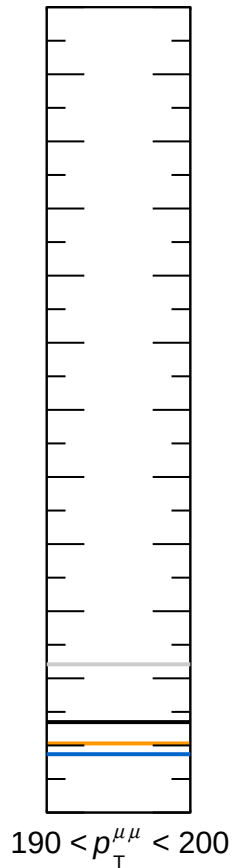
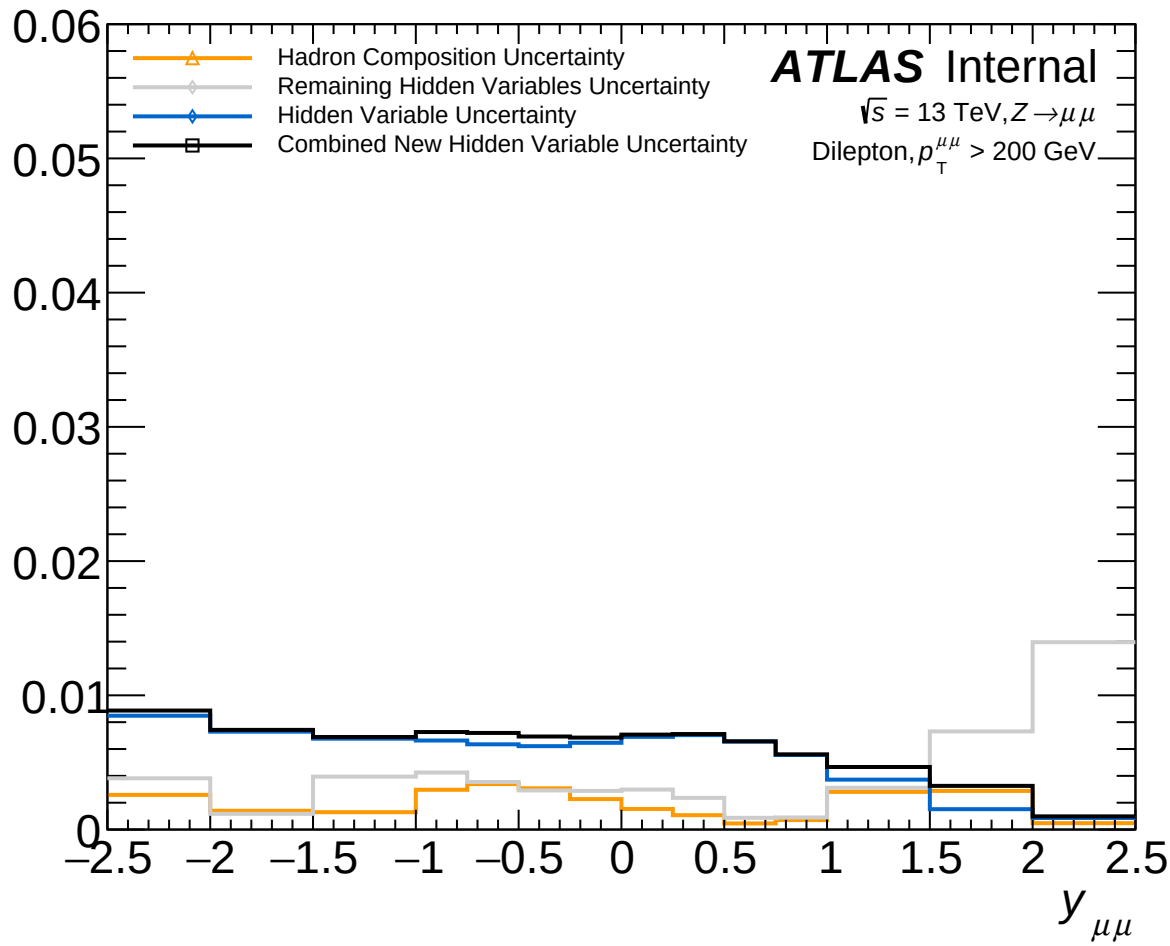
Fractional systematic impact

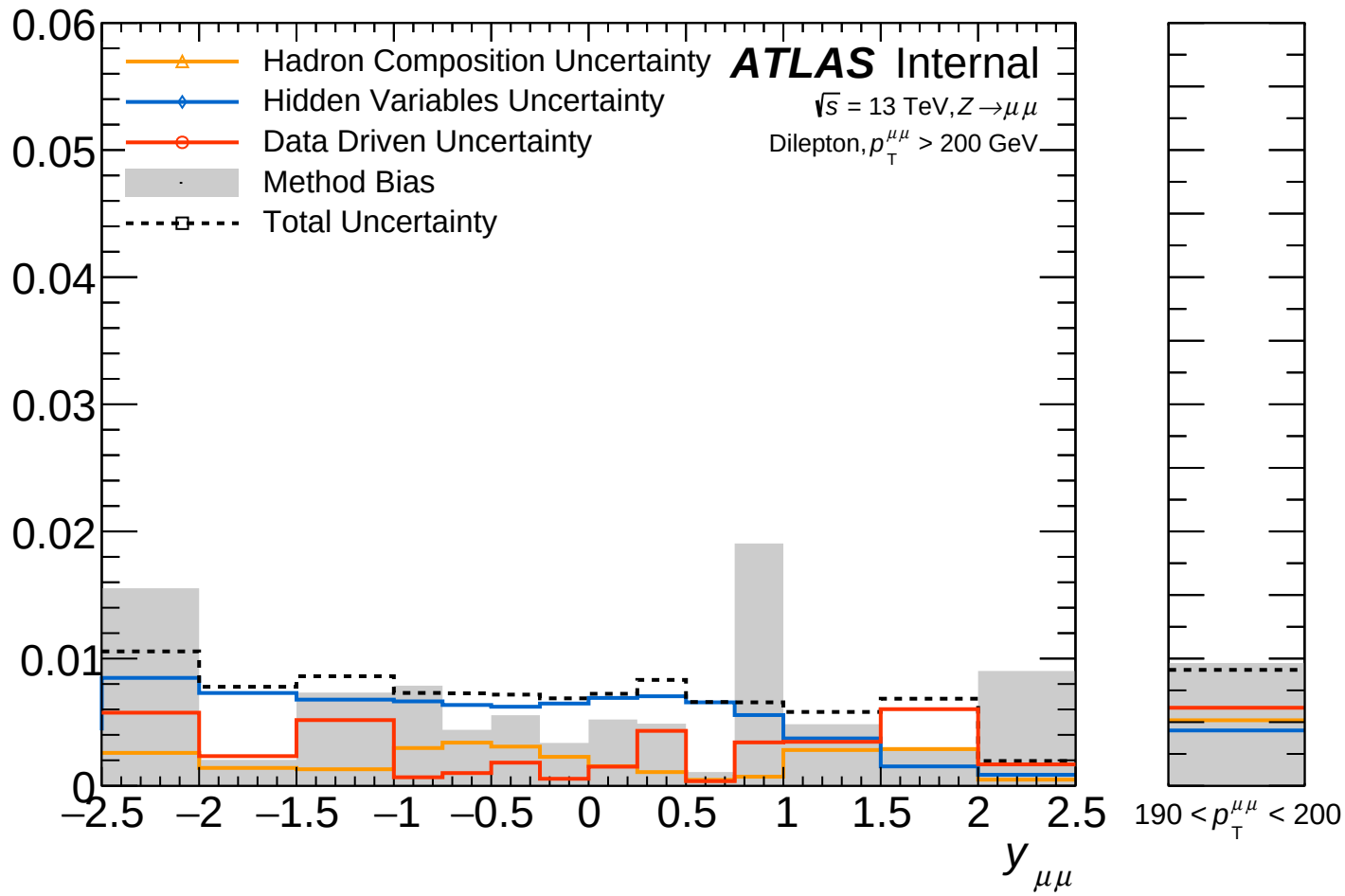


ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

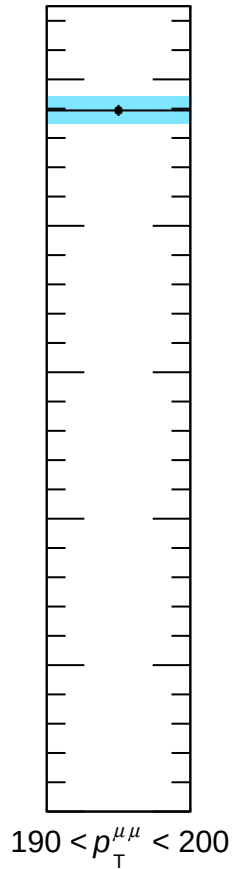
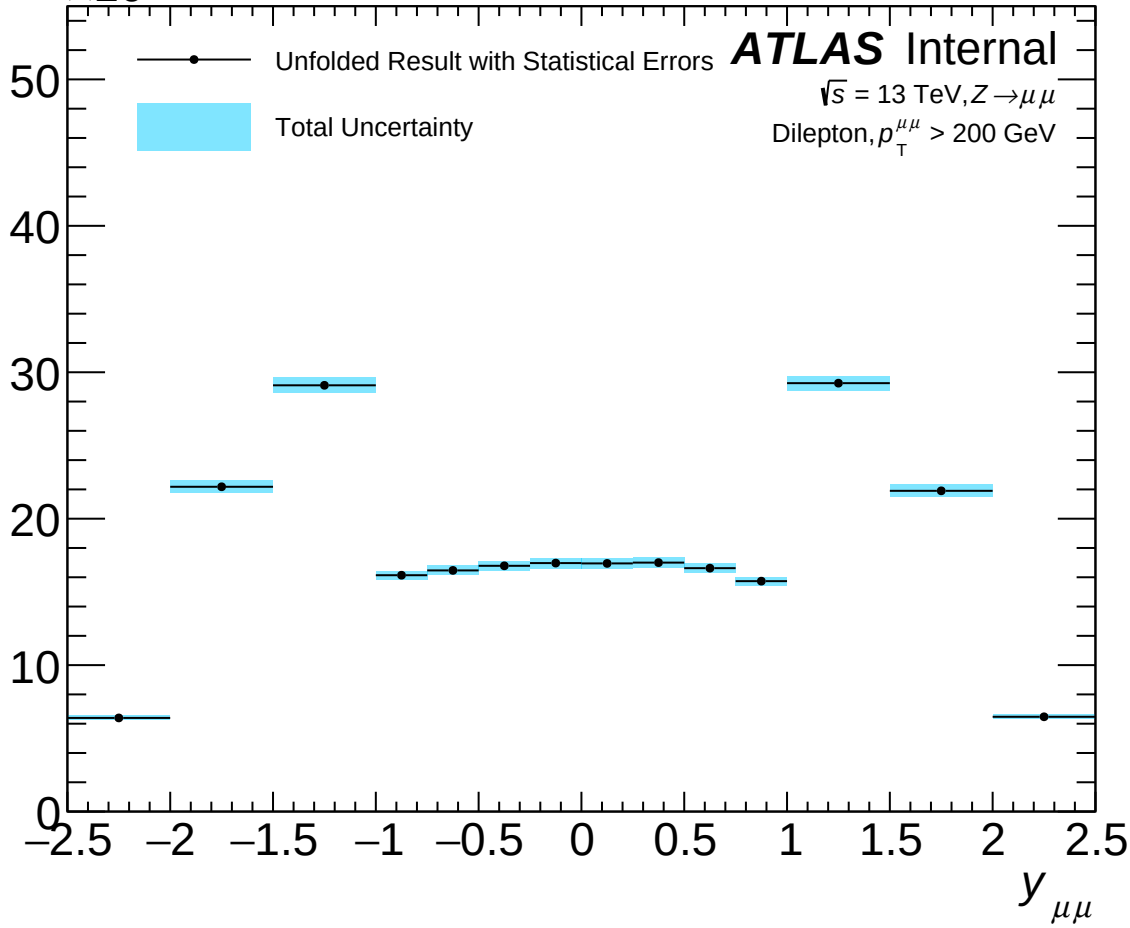
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

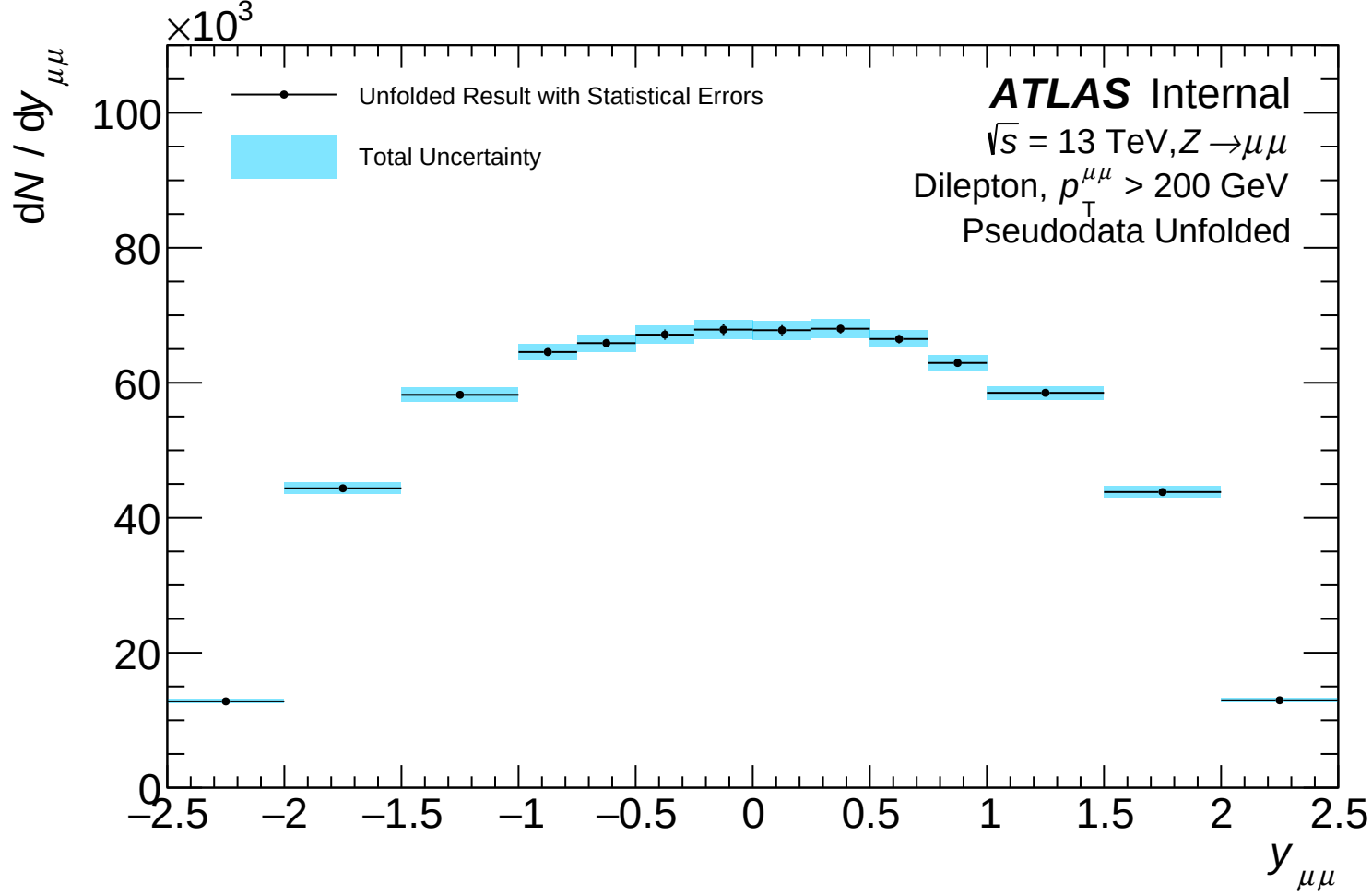


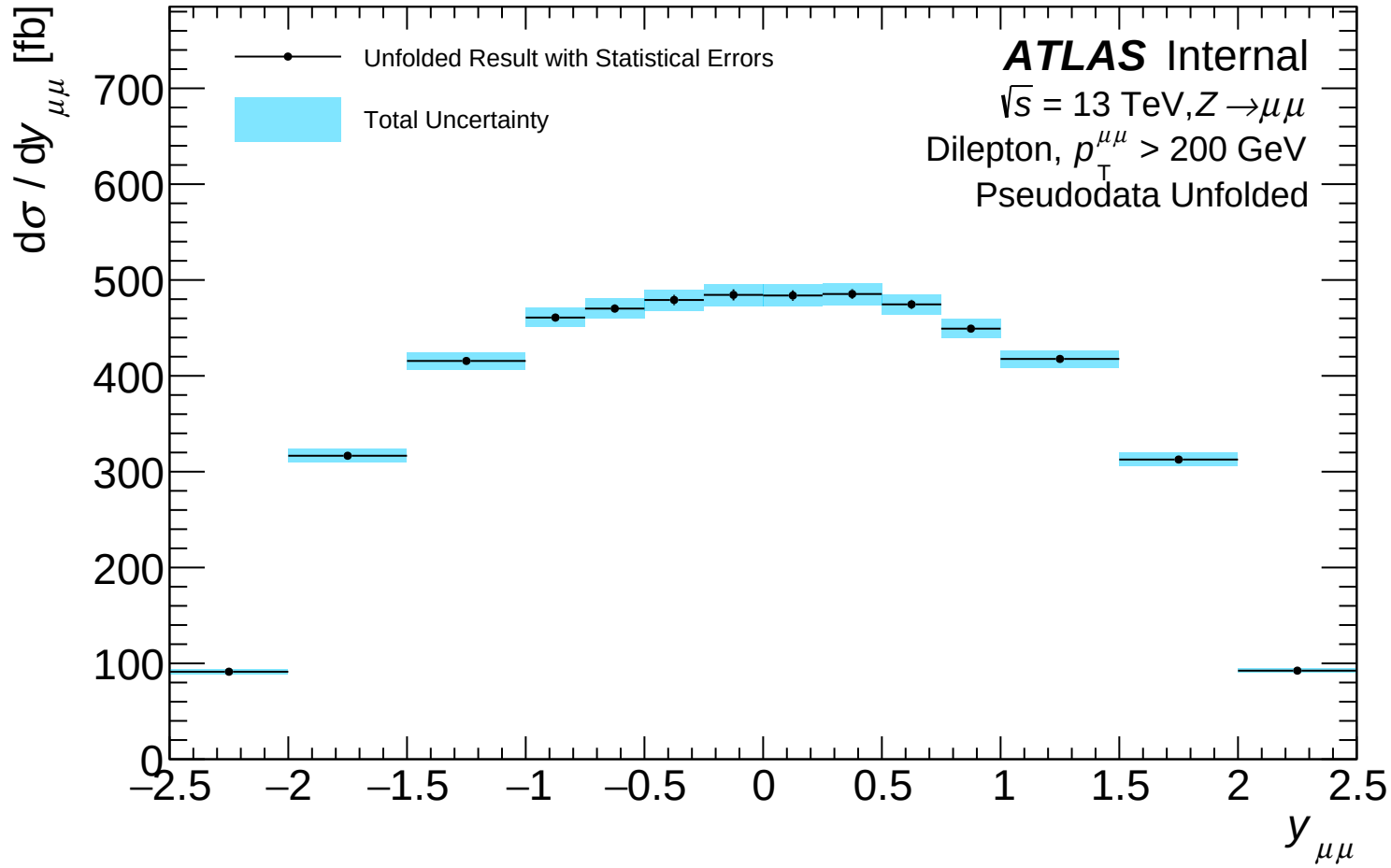


Events

$\times 10^3$







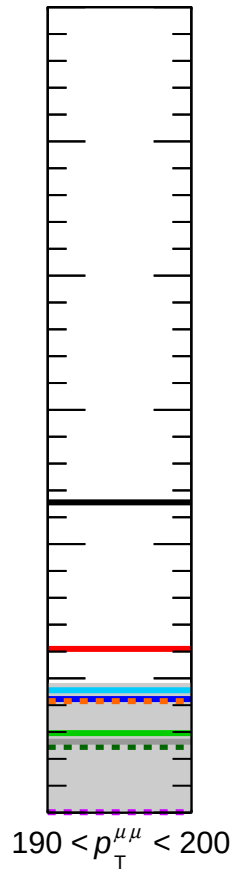
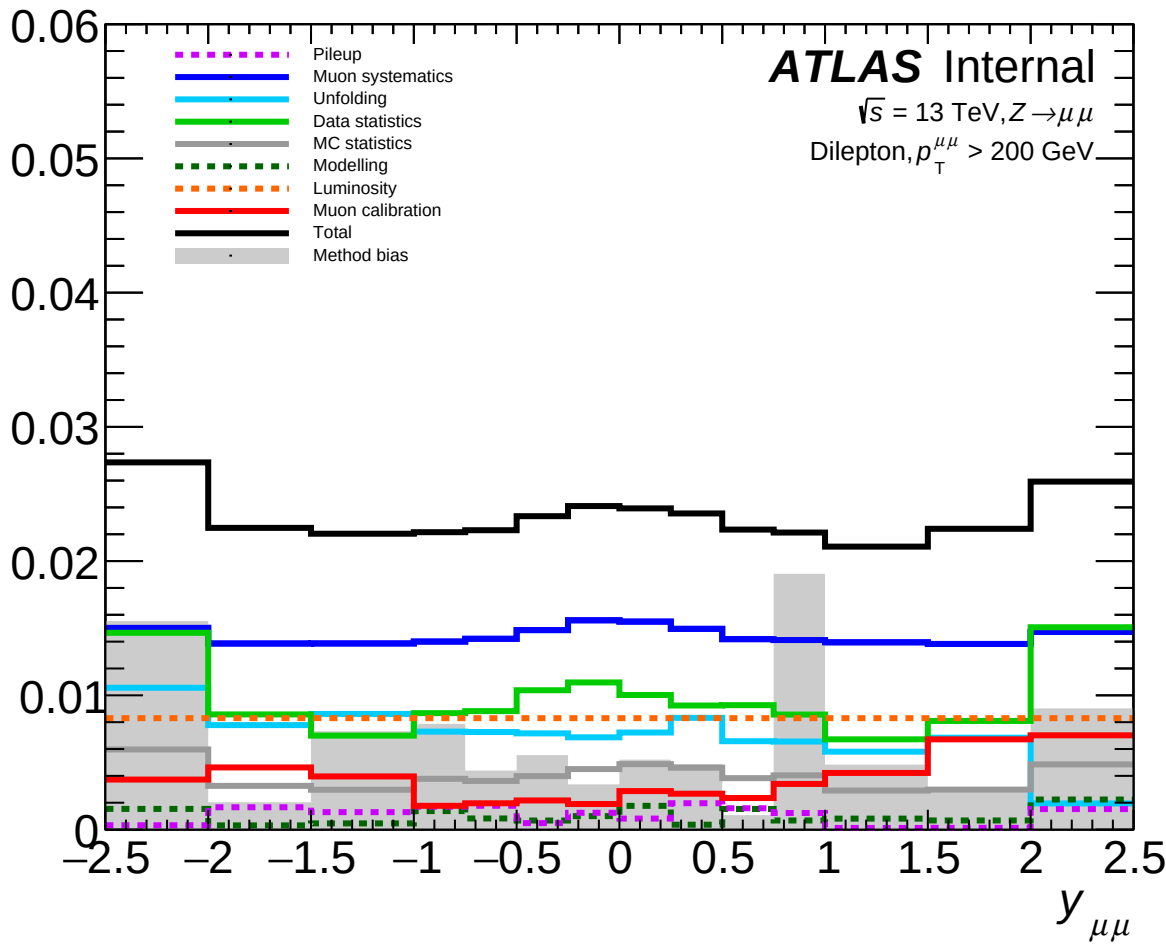
Fractional uncertainty

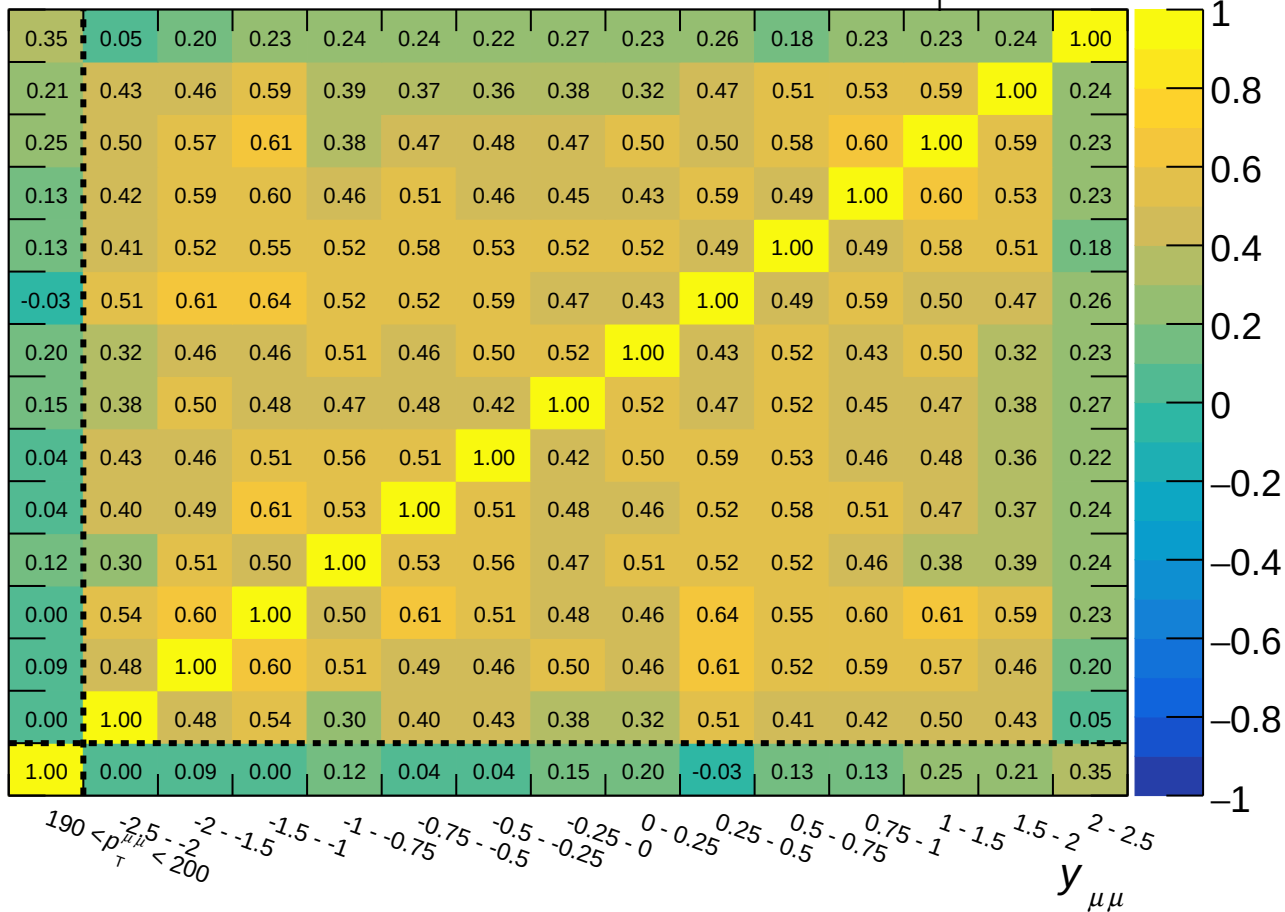
ATLAS Internal

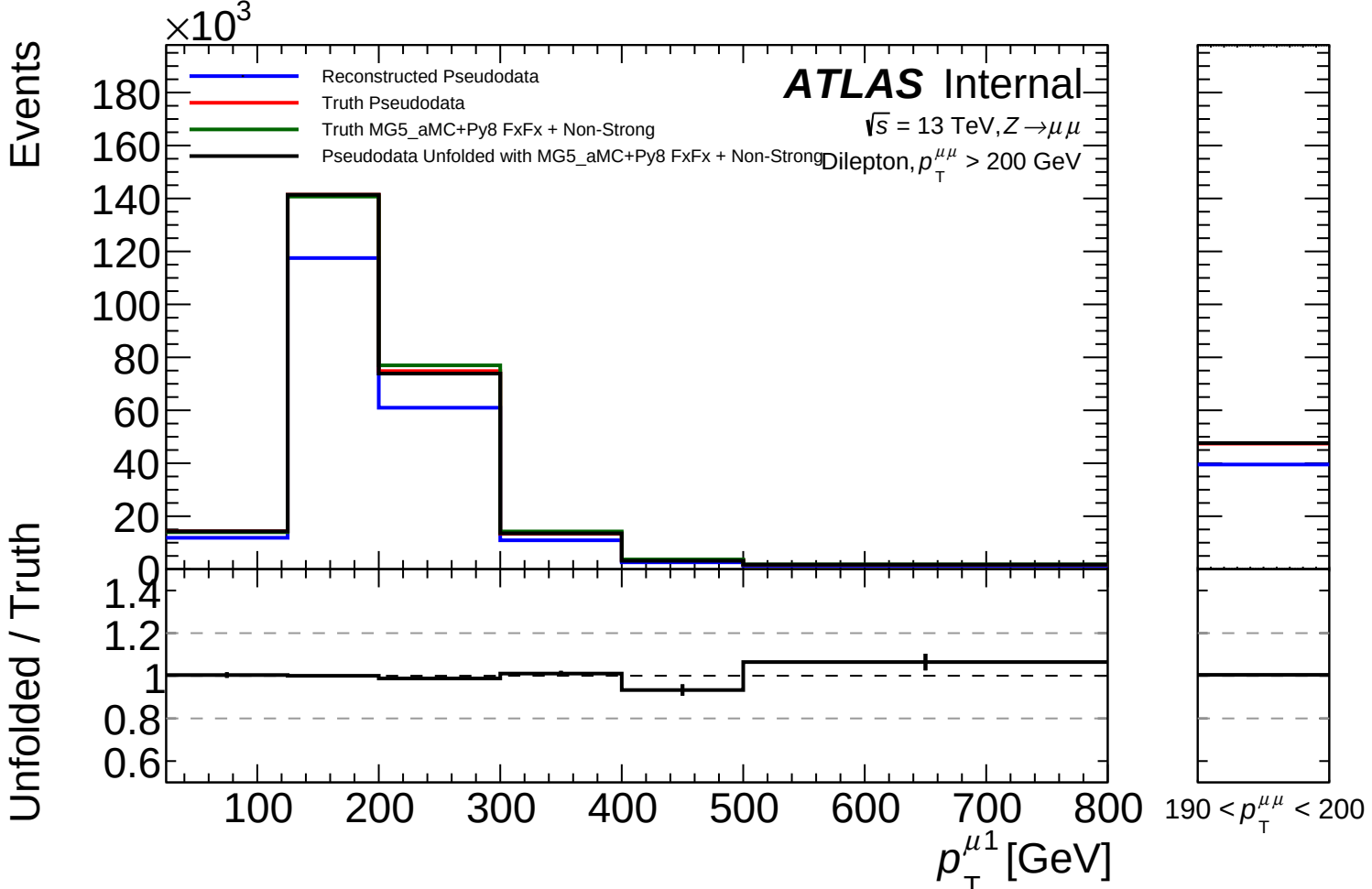
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

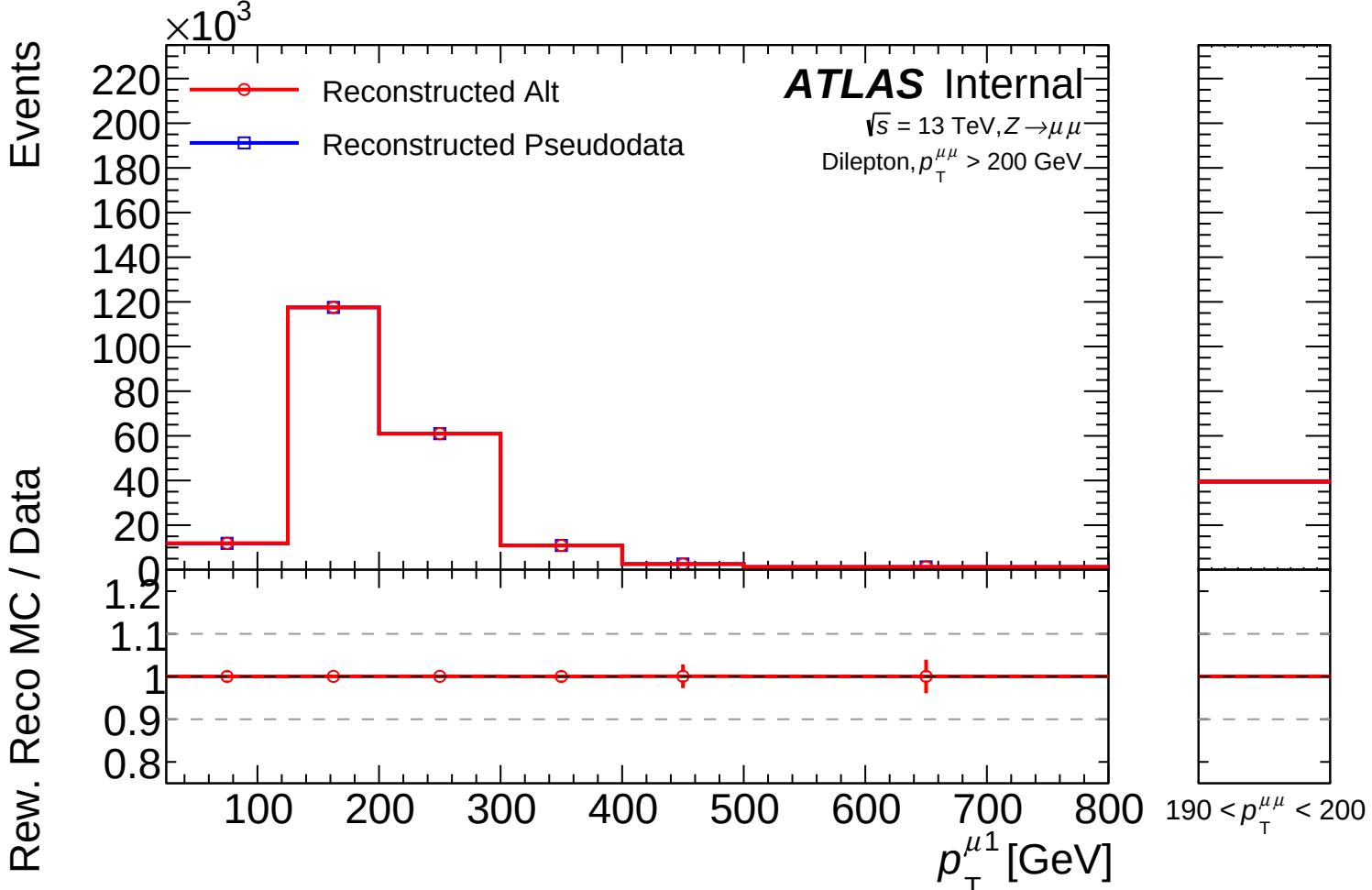
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

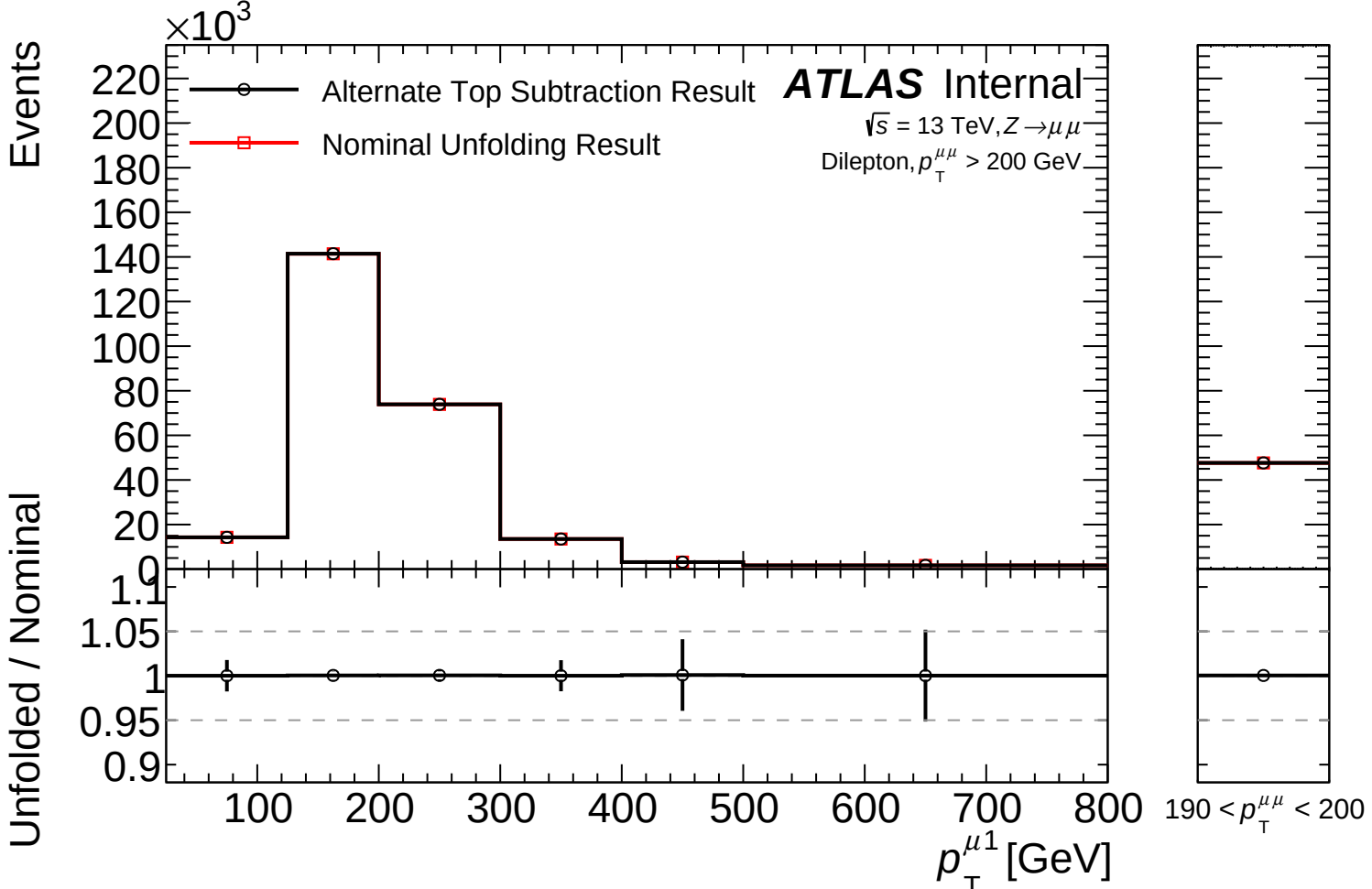
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Muon calibration
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $y_{\mu\mu}$ $190 < p_{\text{T}}^{\mu\mu} < 200$ 







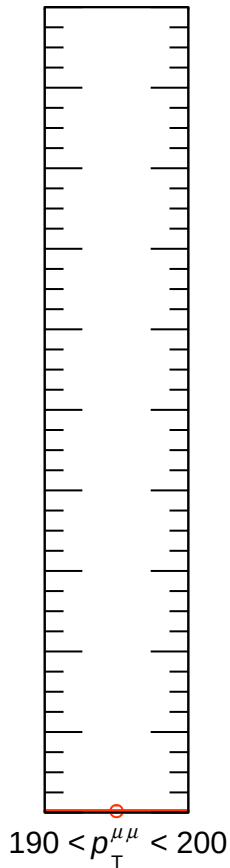
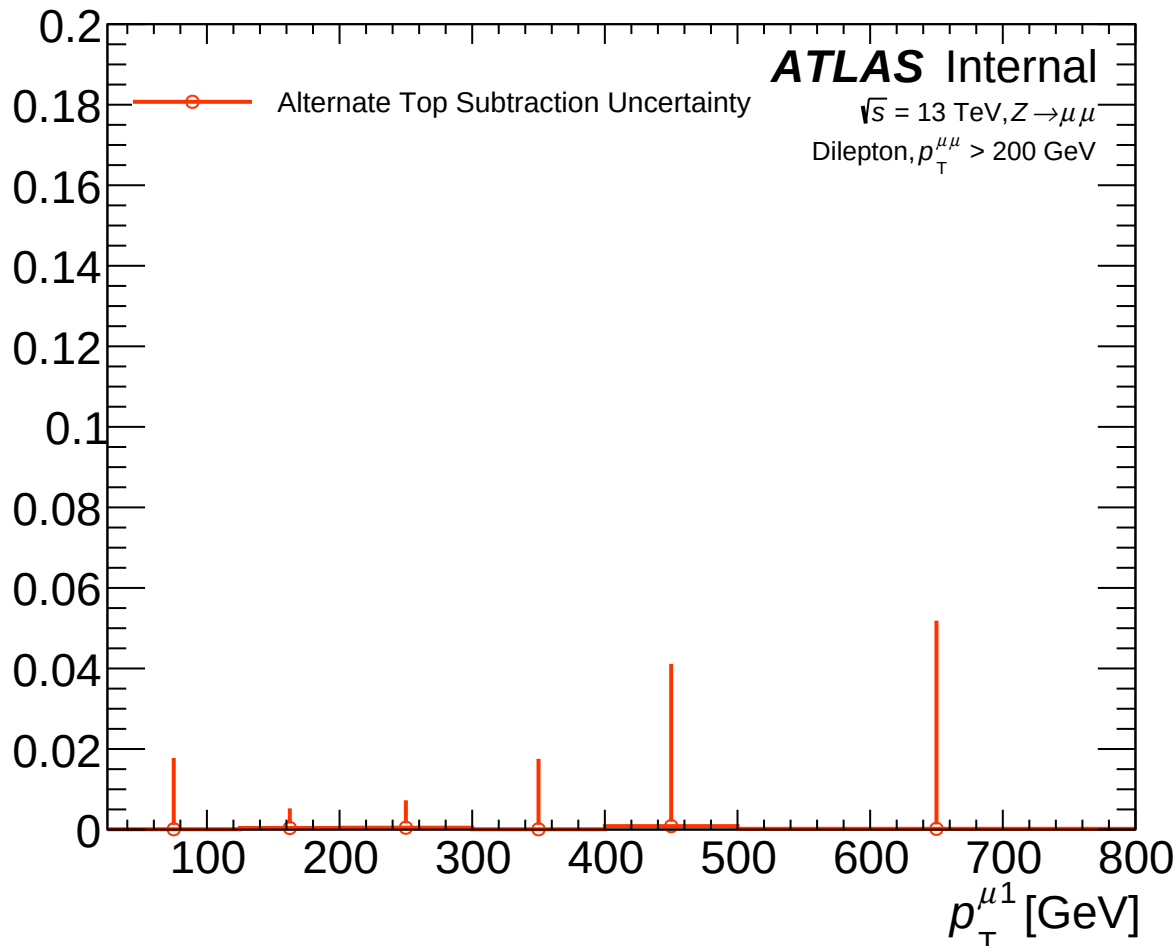
Fractional systematic impact

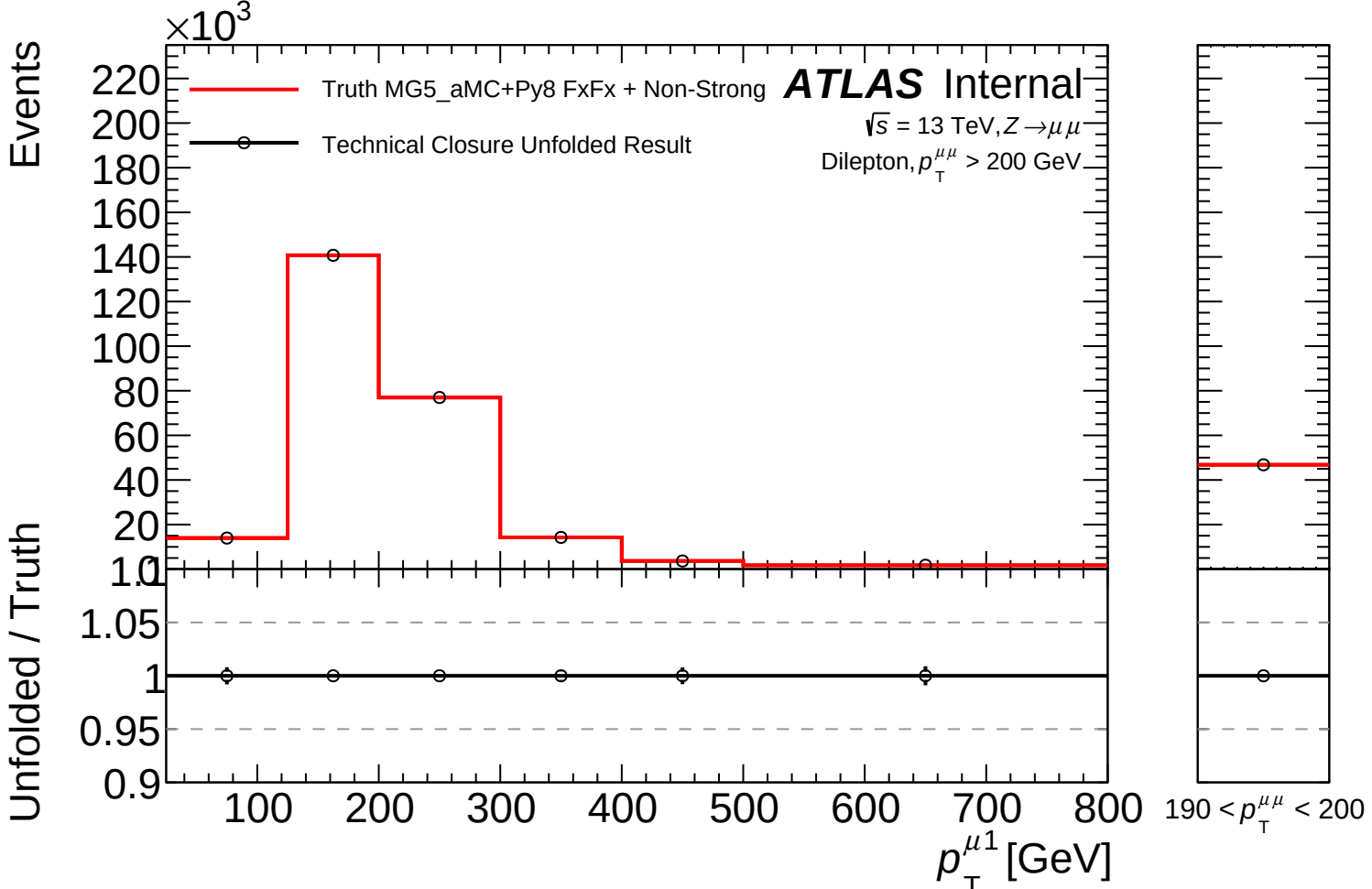
ATLAS Internal

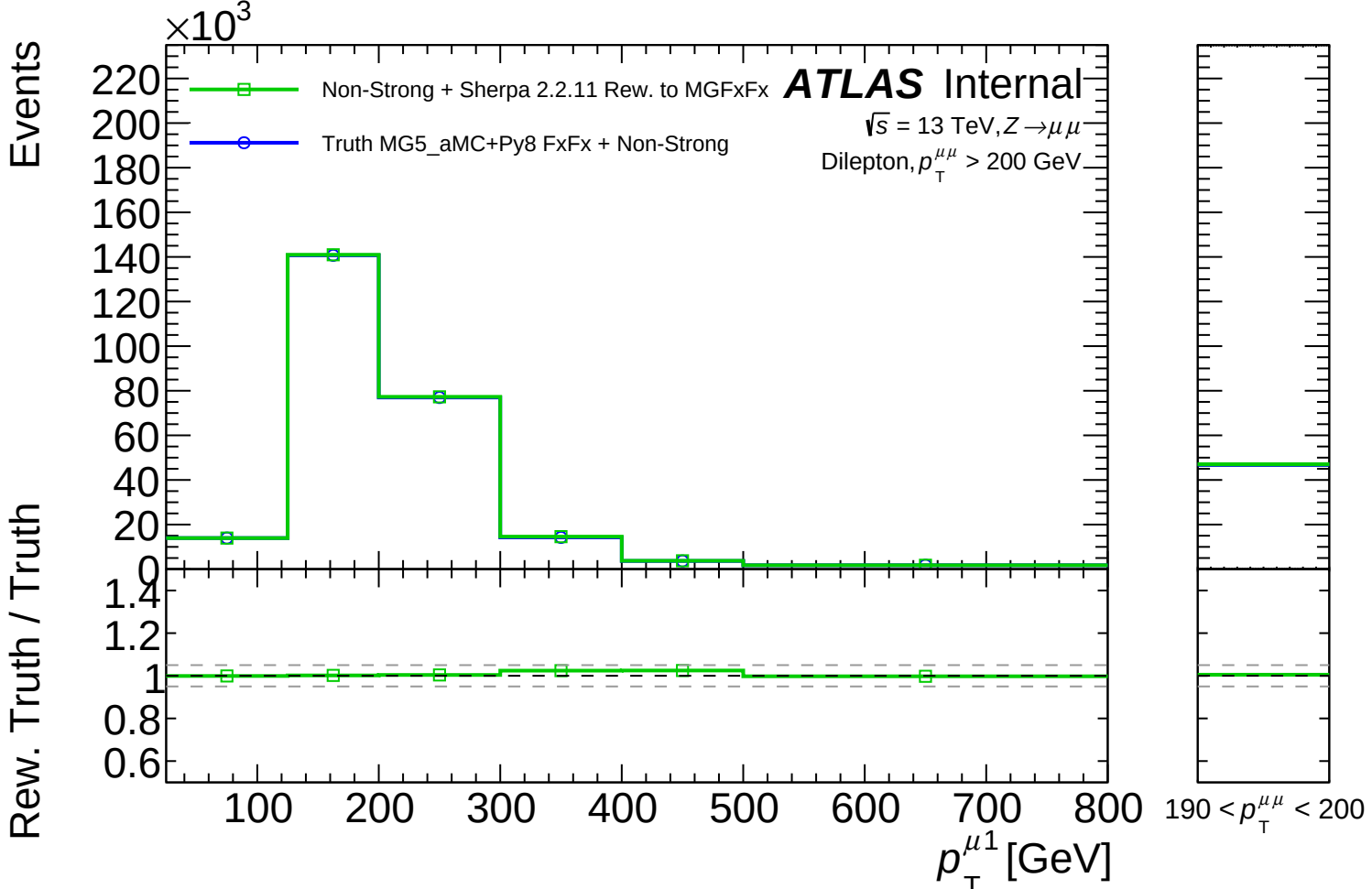
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

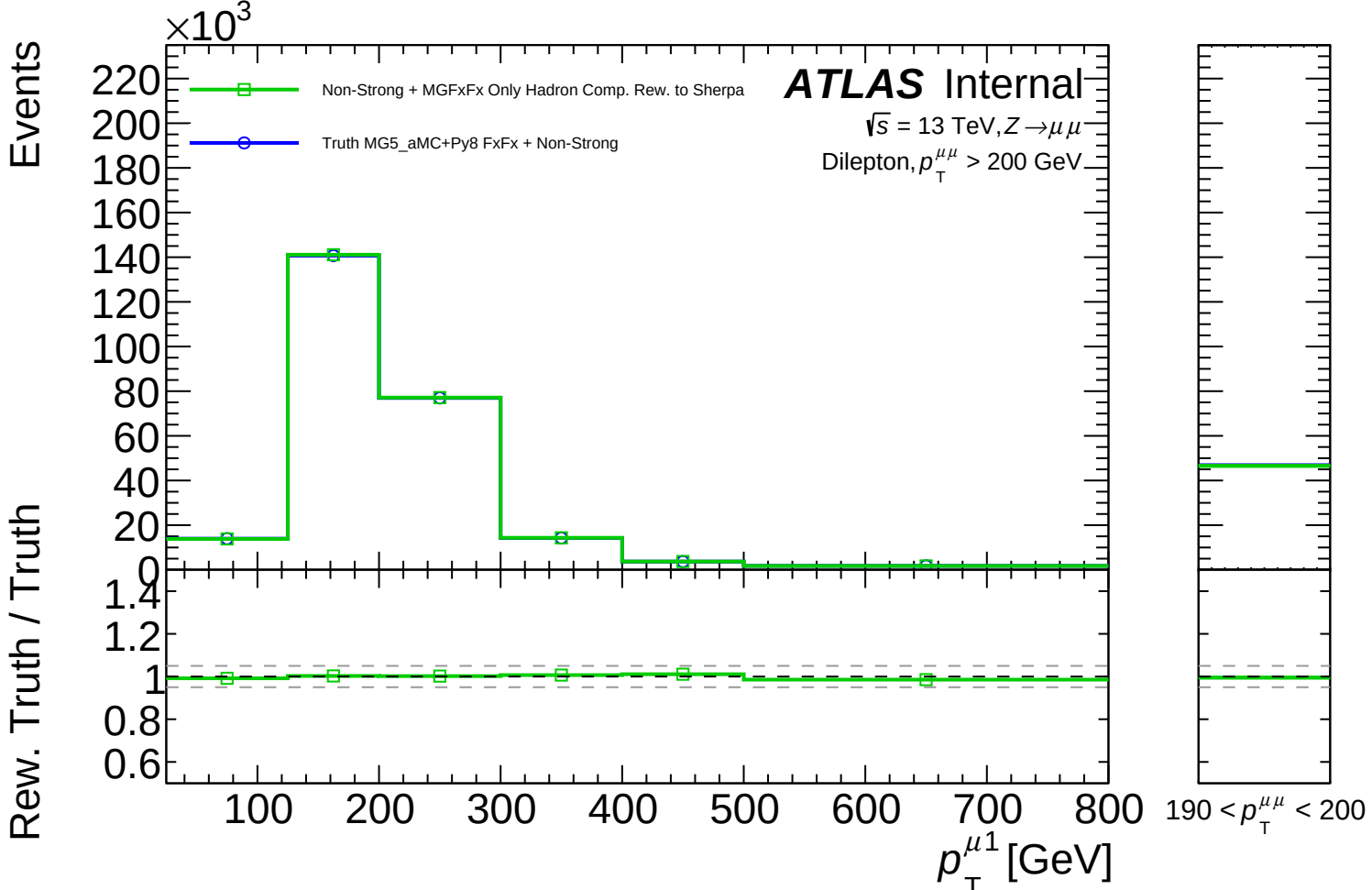
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

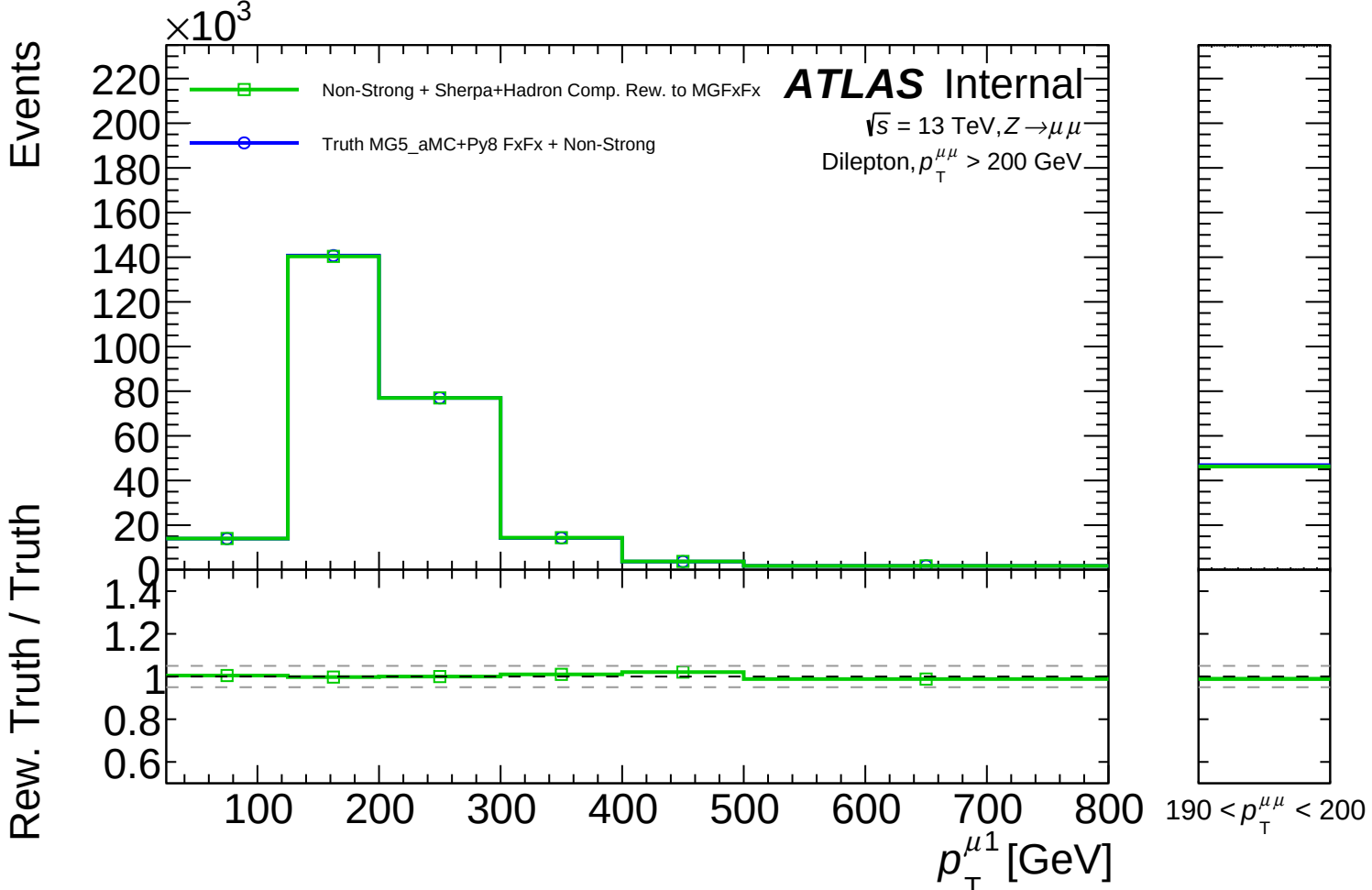
Alternate Top Subtraction Uncertainty

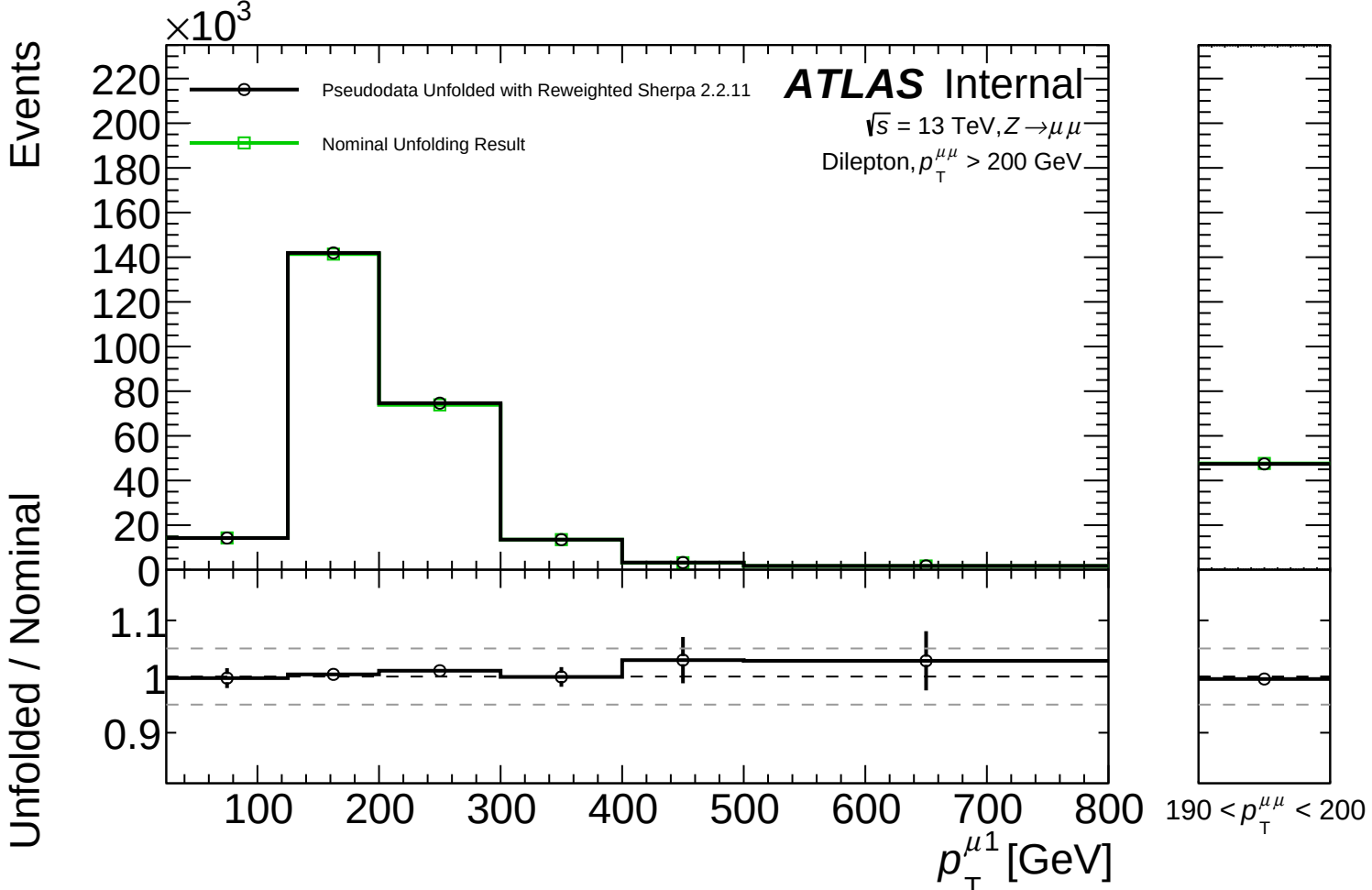


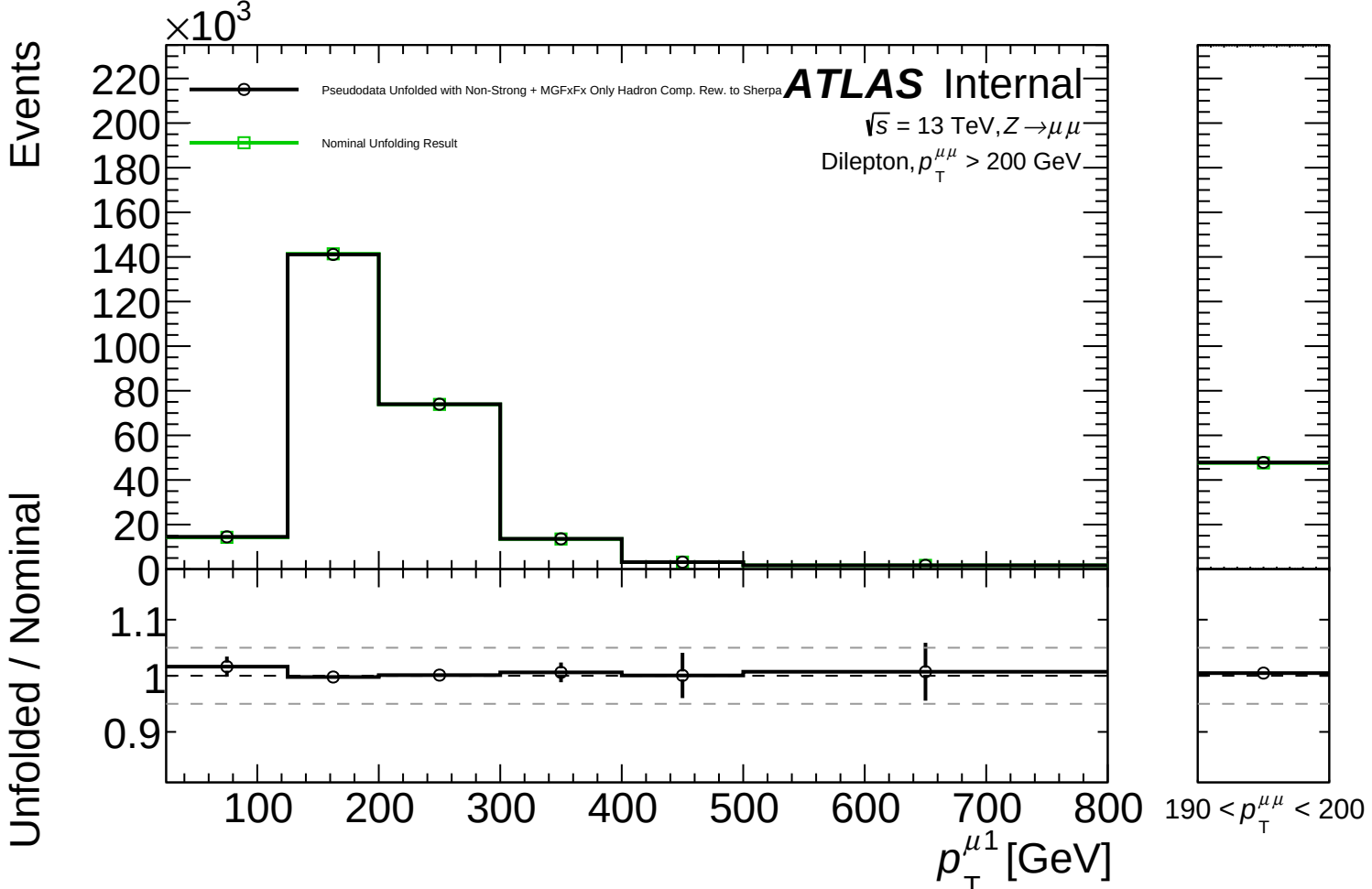


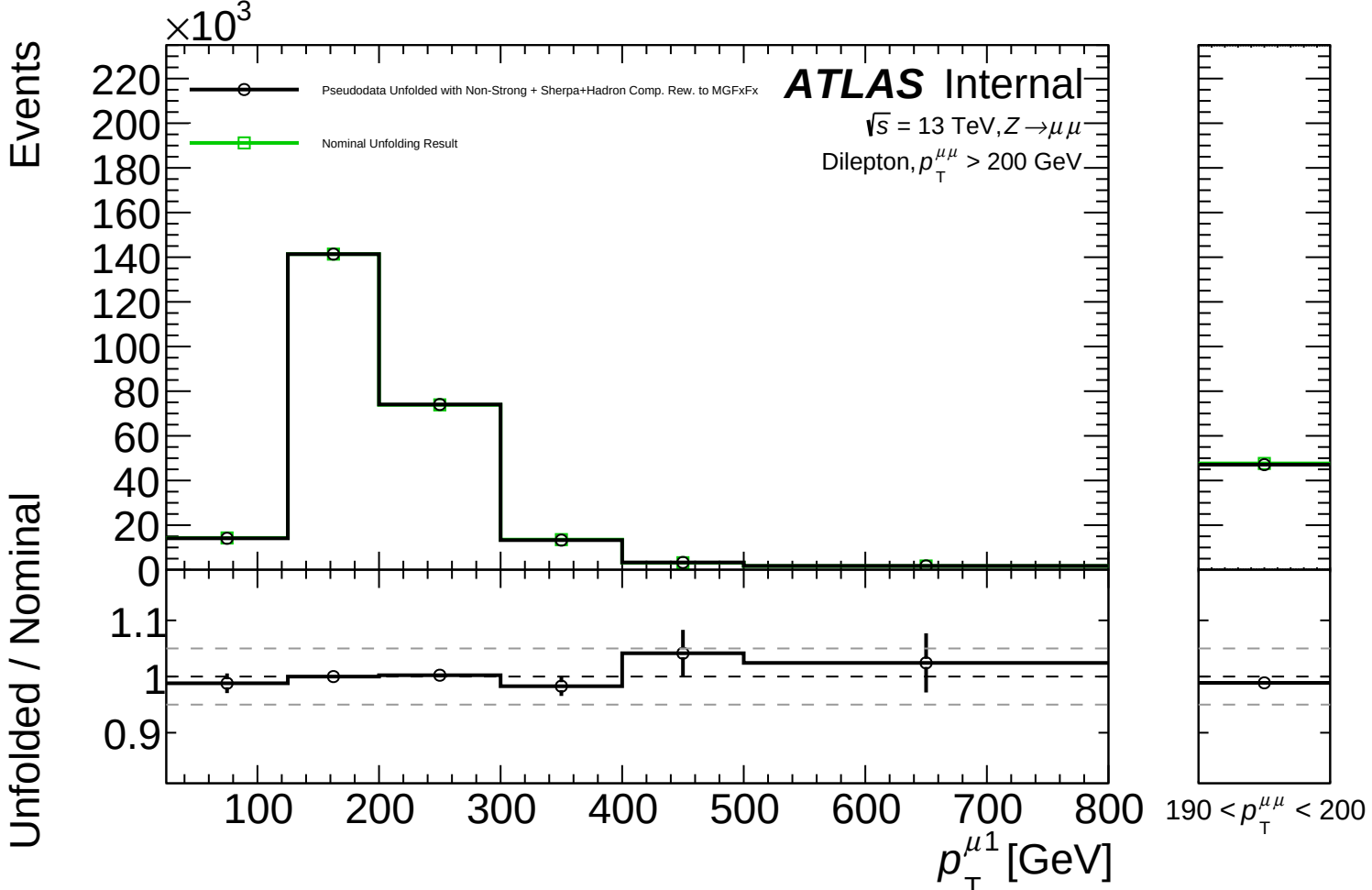


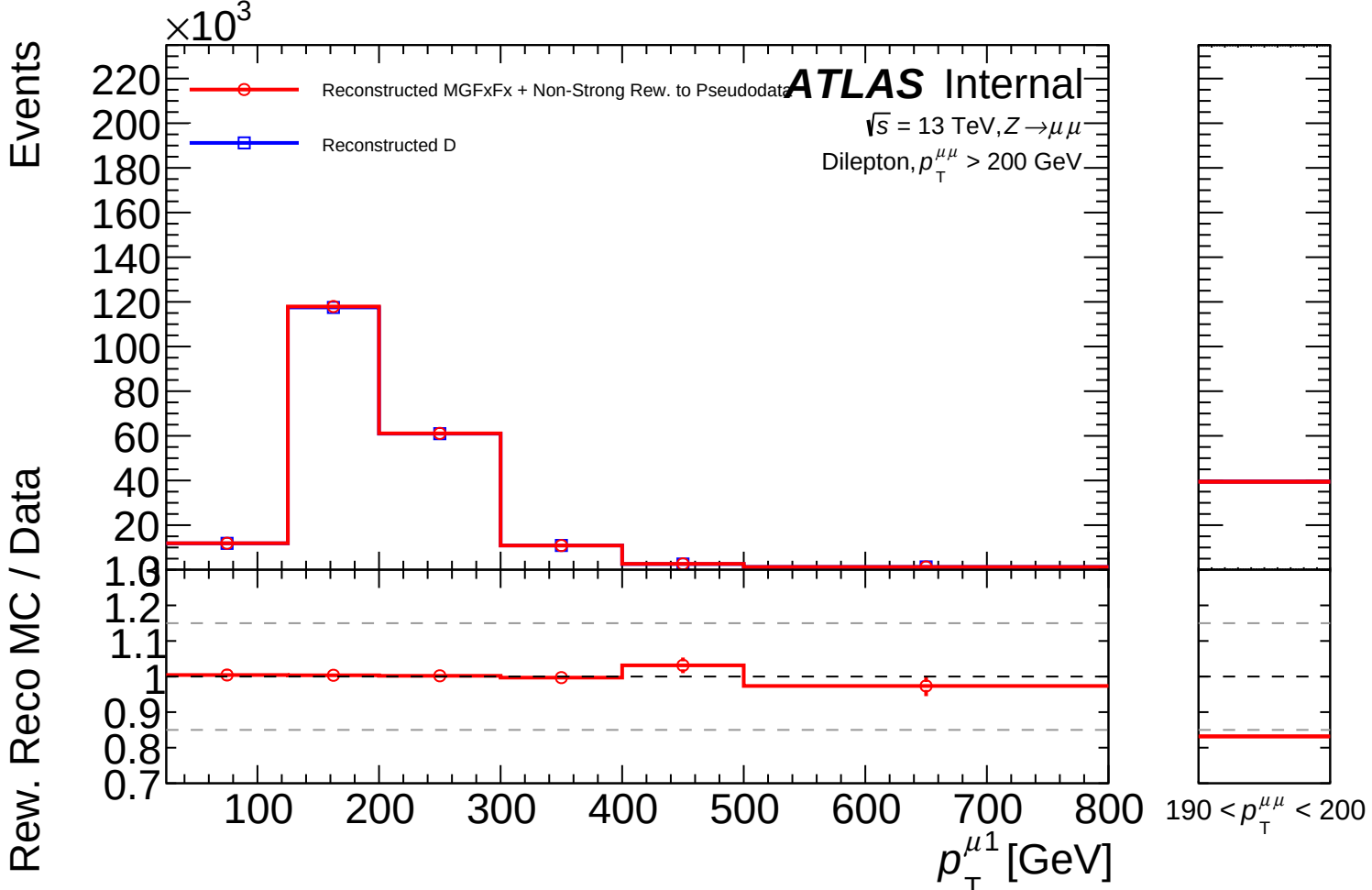


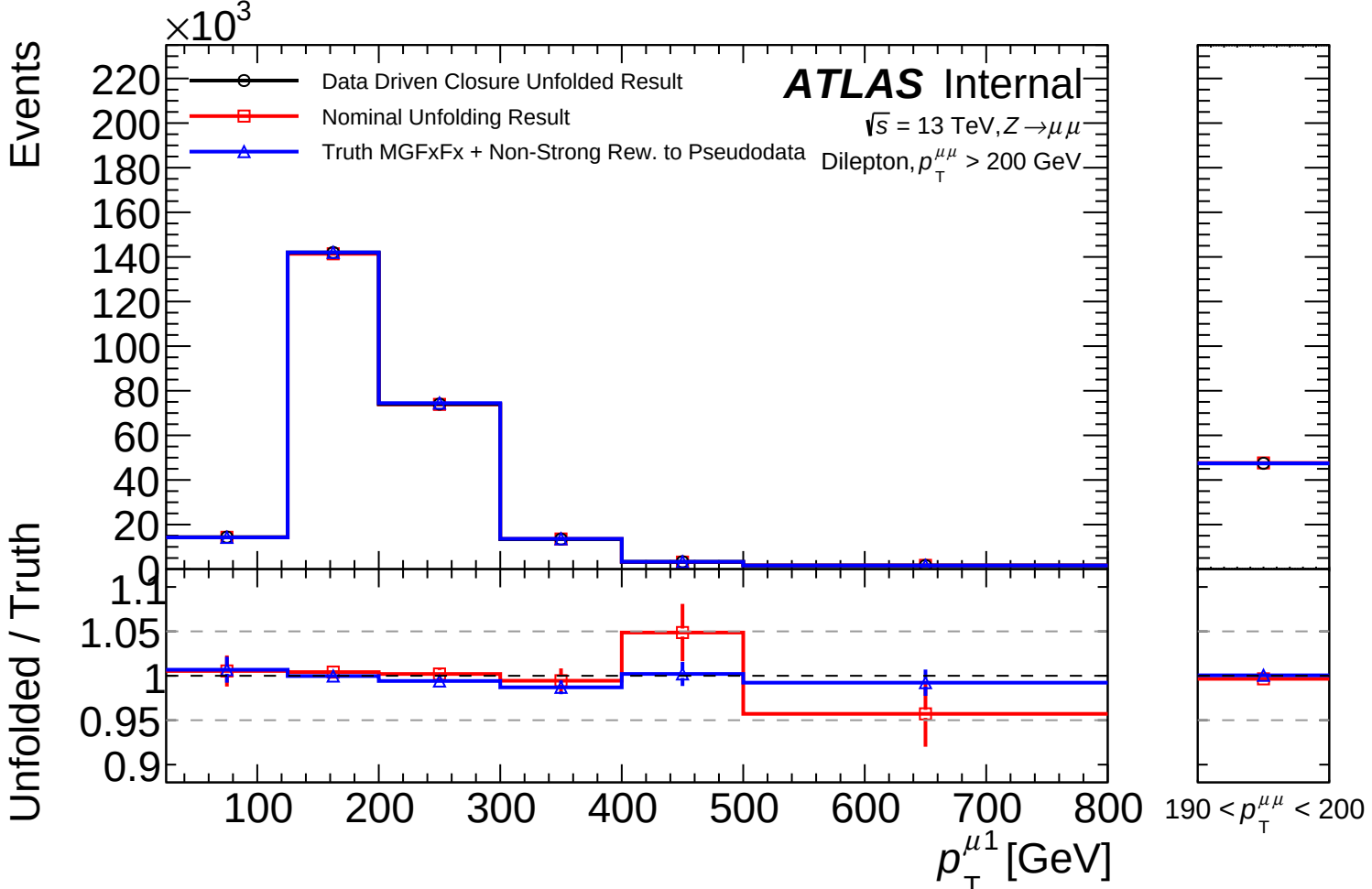




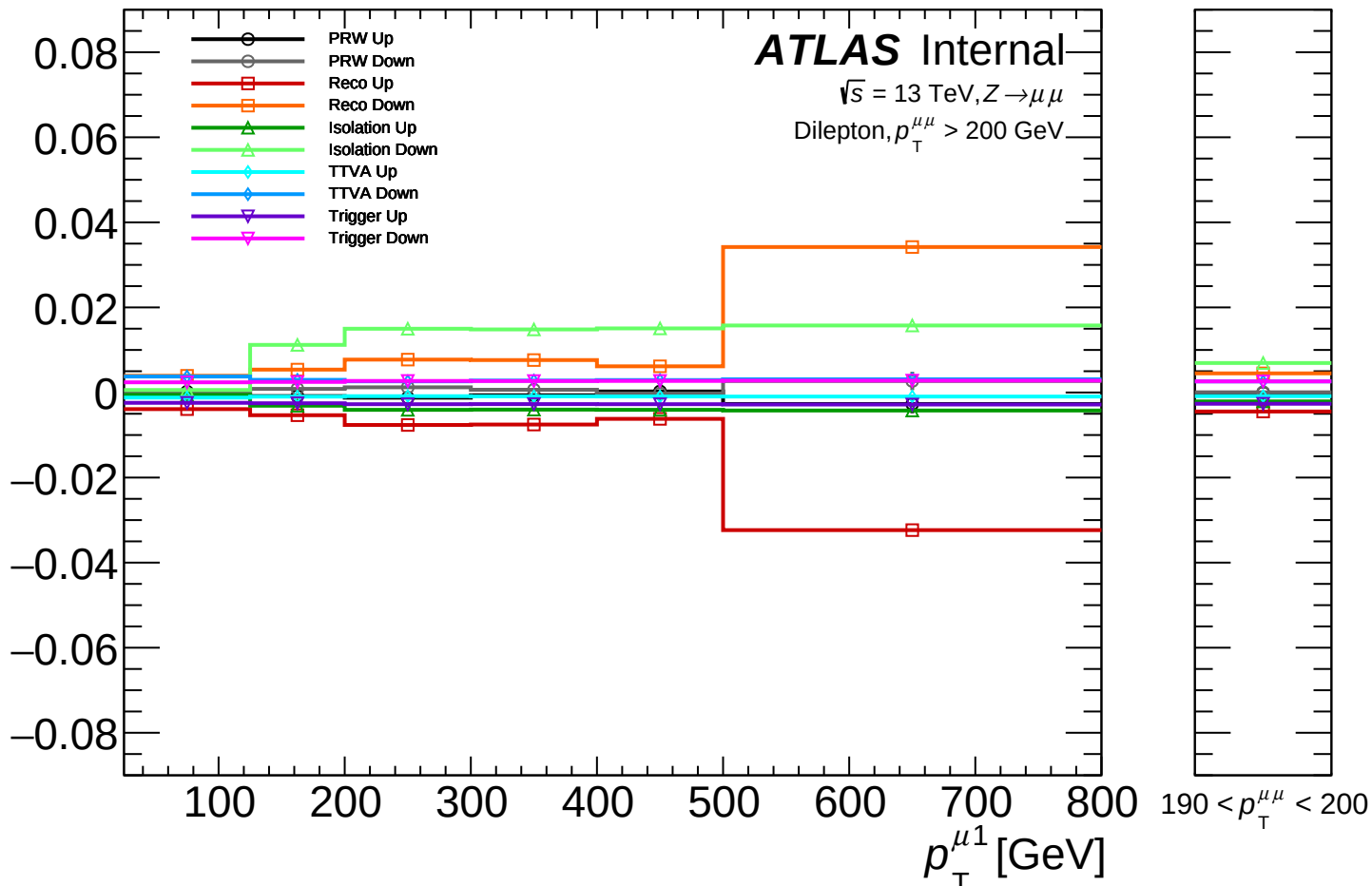




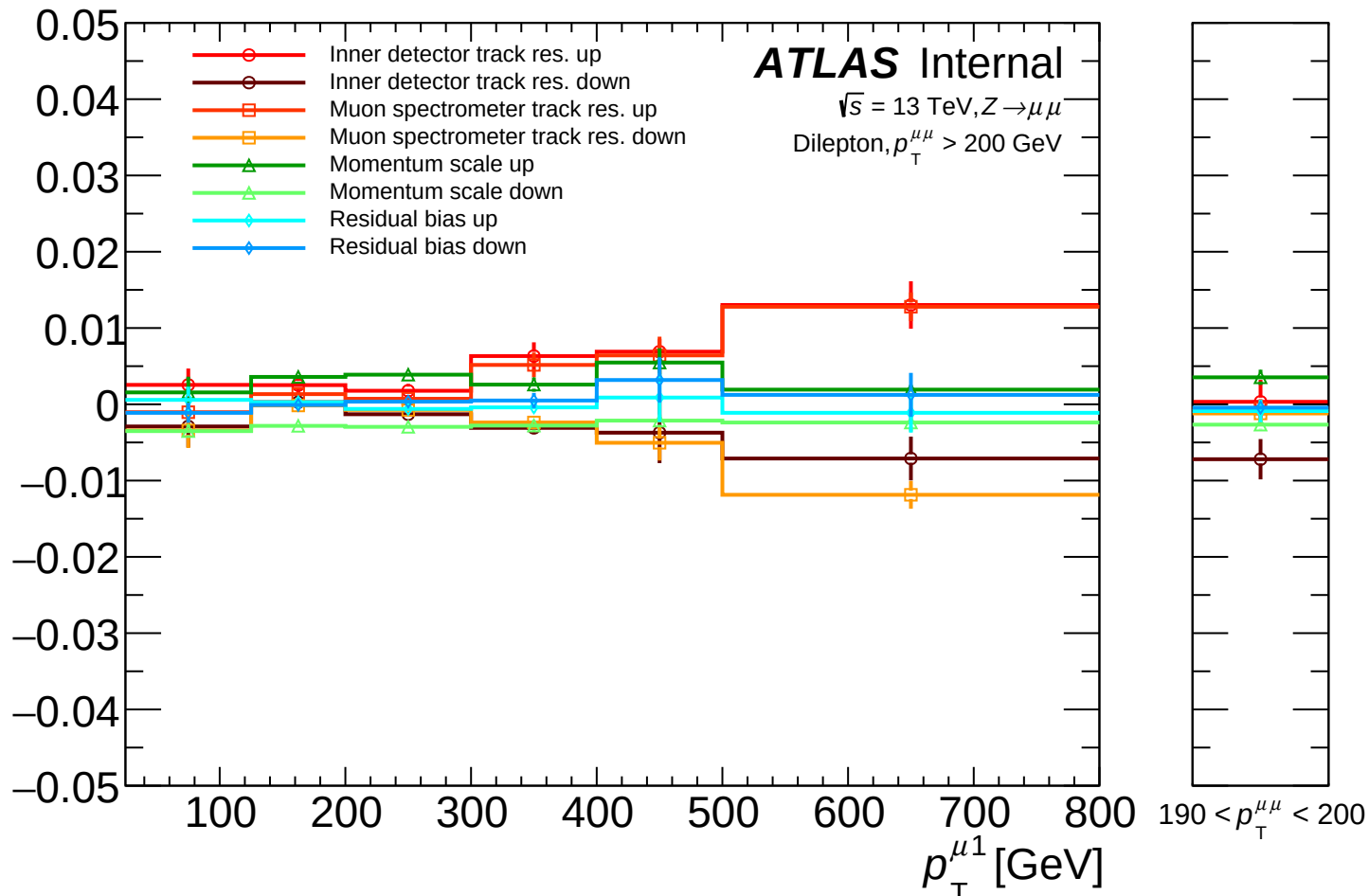




Fractional systematic impact



Fractional systematic impact



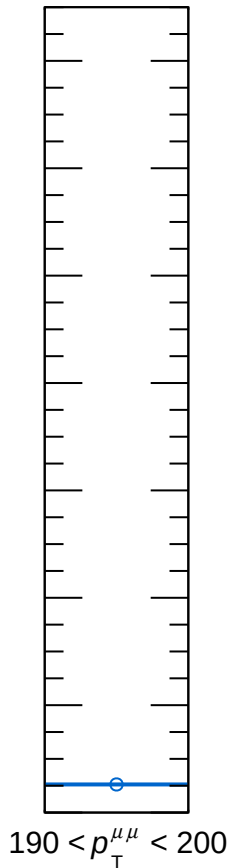
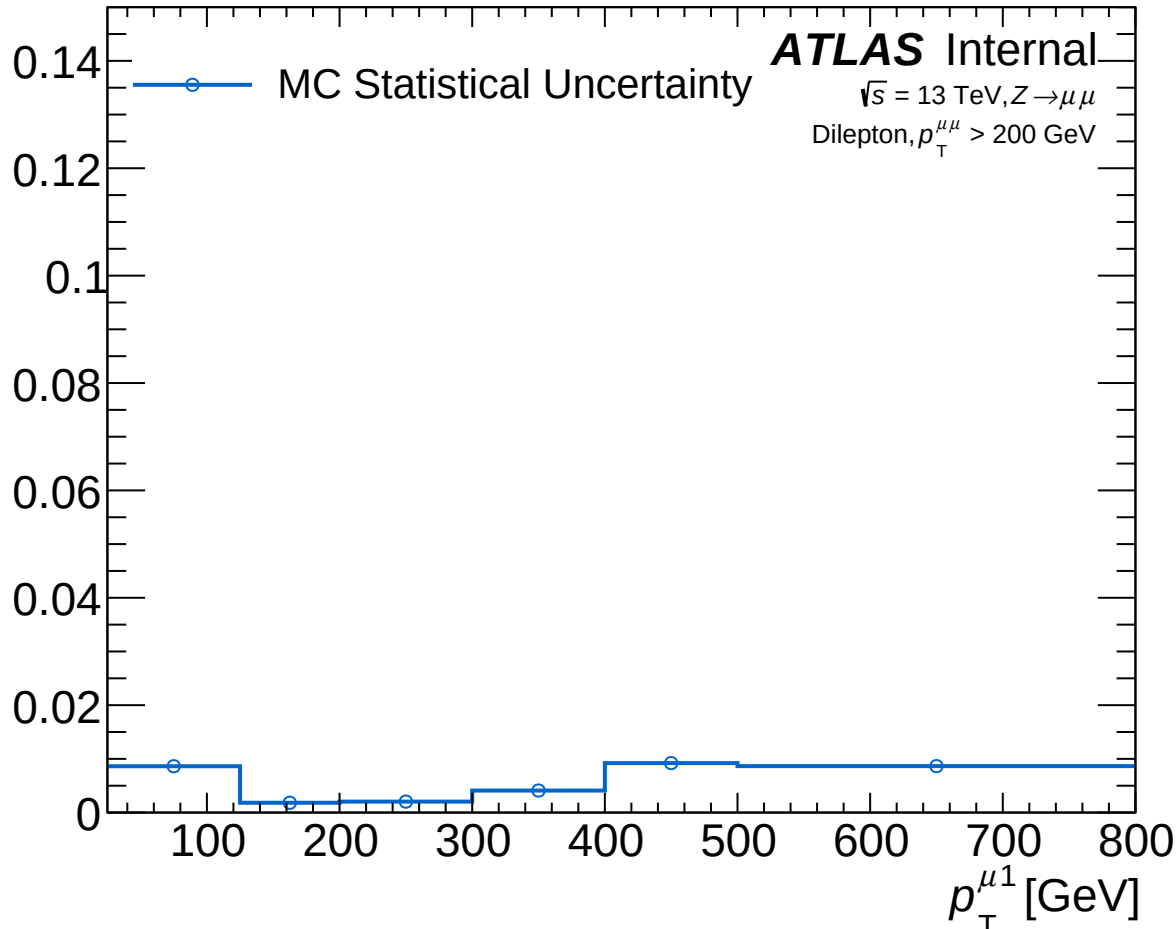
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



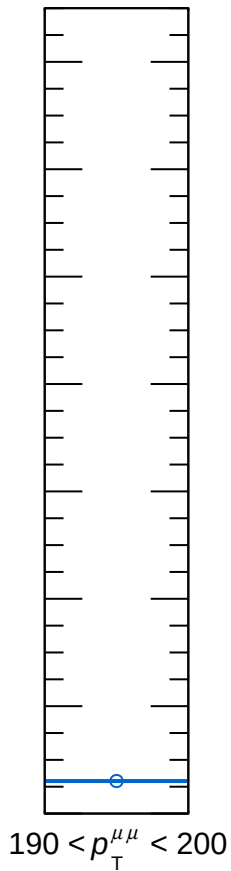
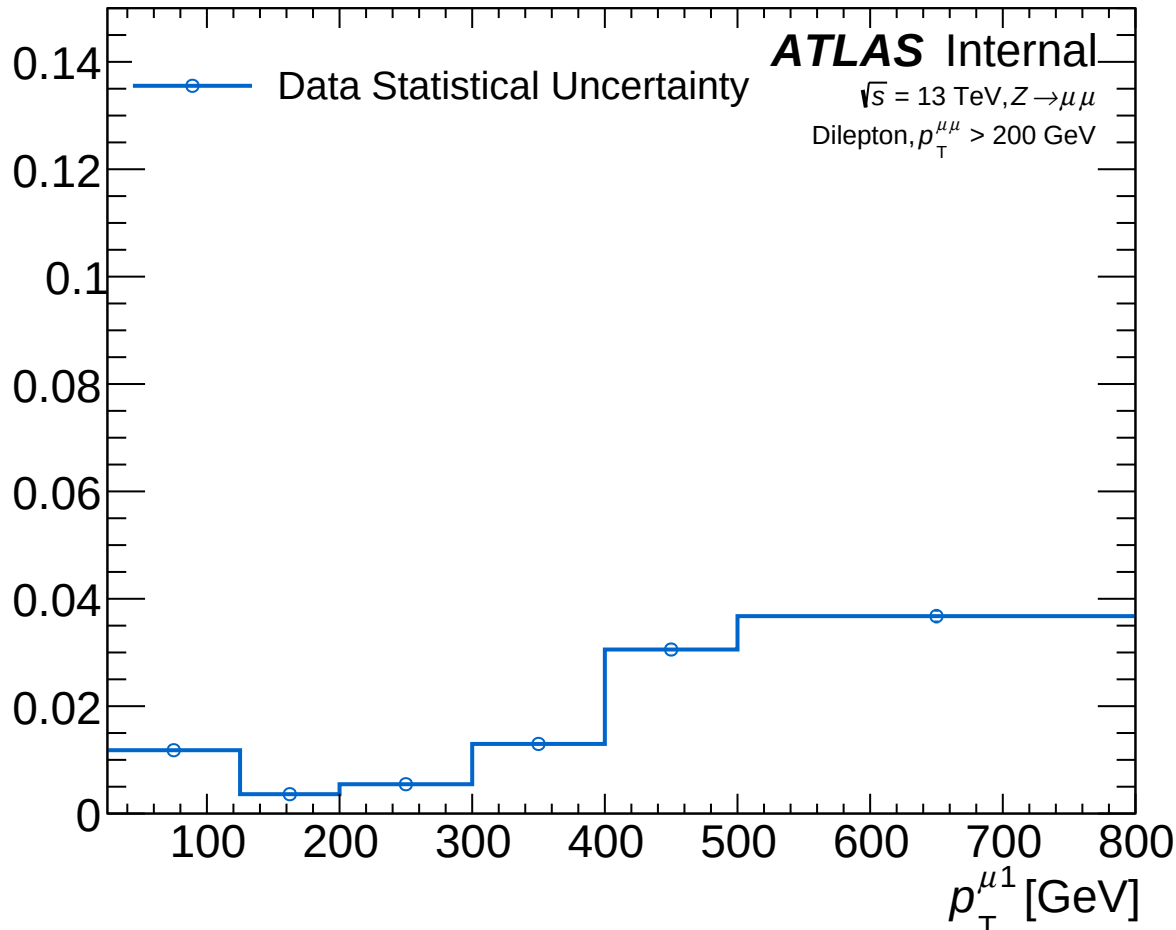
Fractional systematic impact

ATLAS Internal

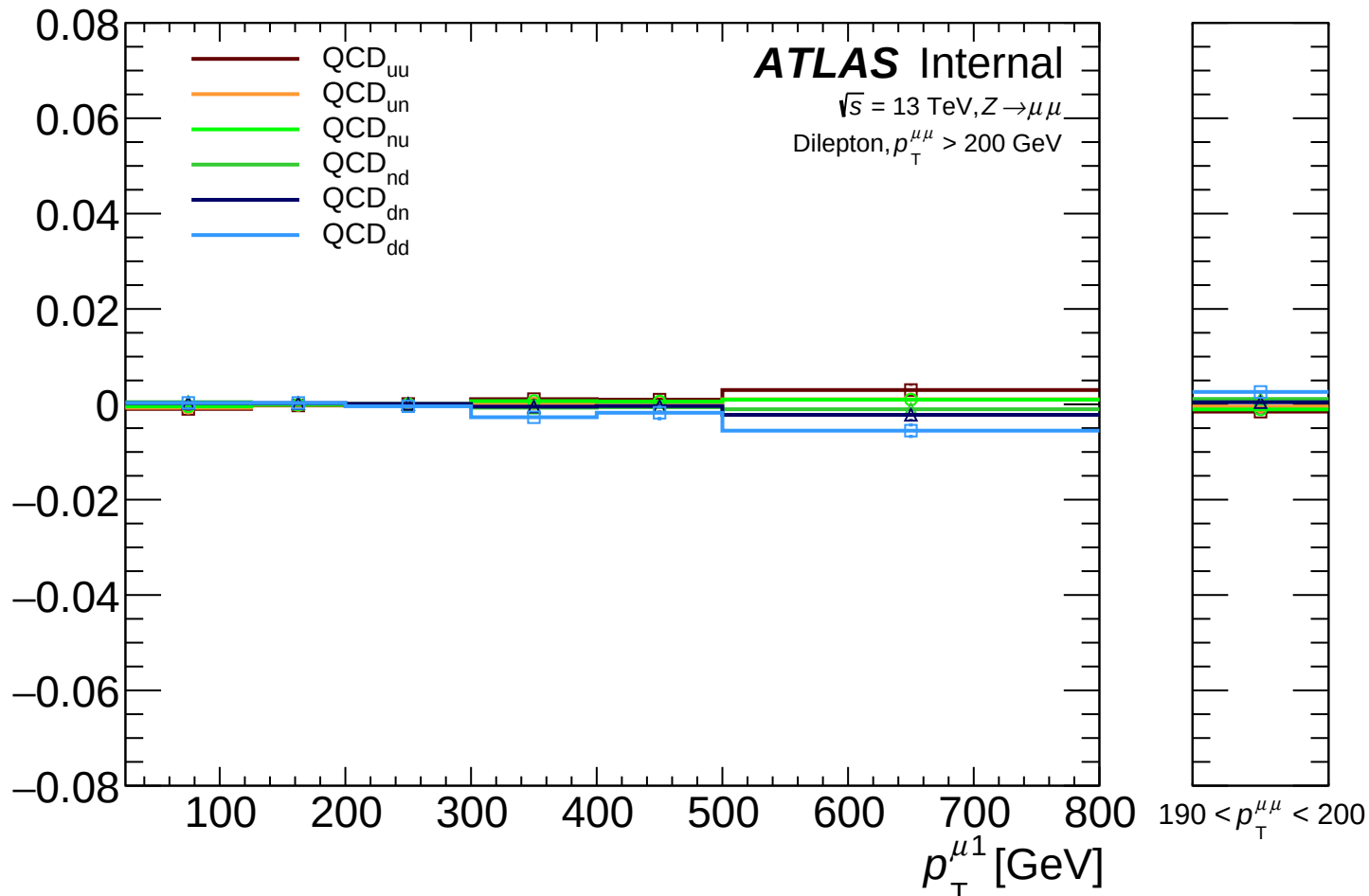
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

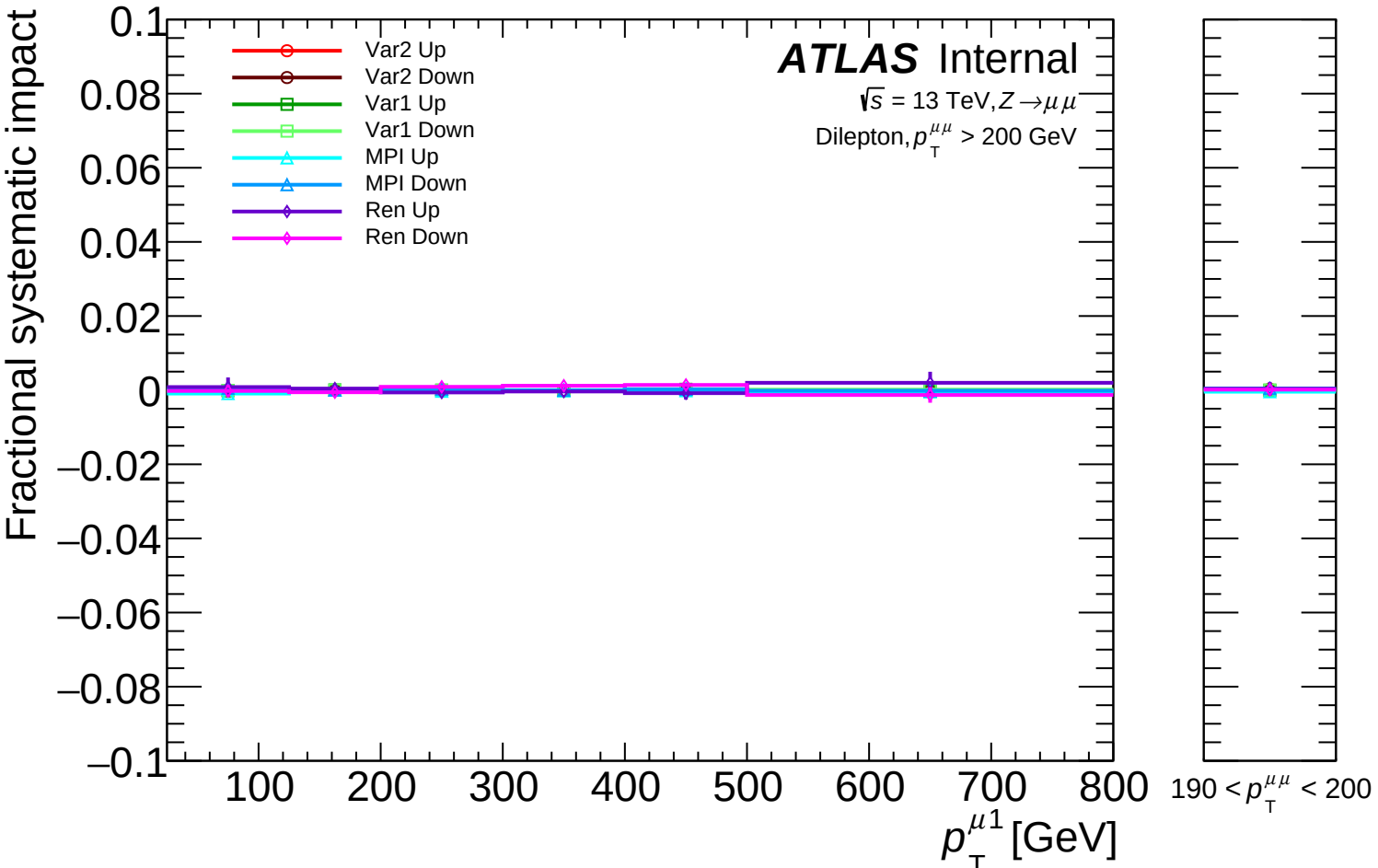
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

Data Statistical Uncertainty

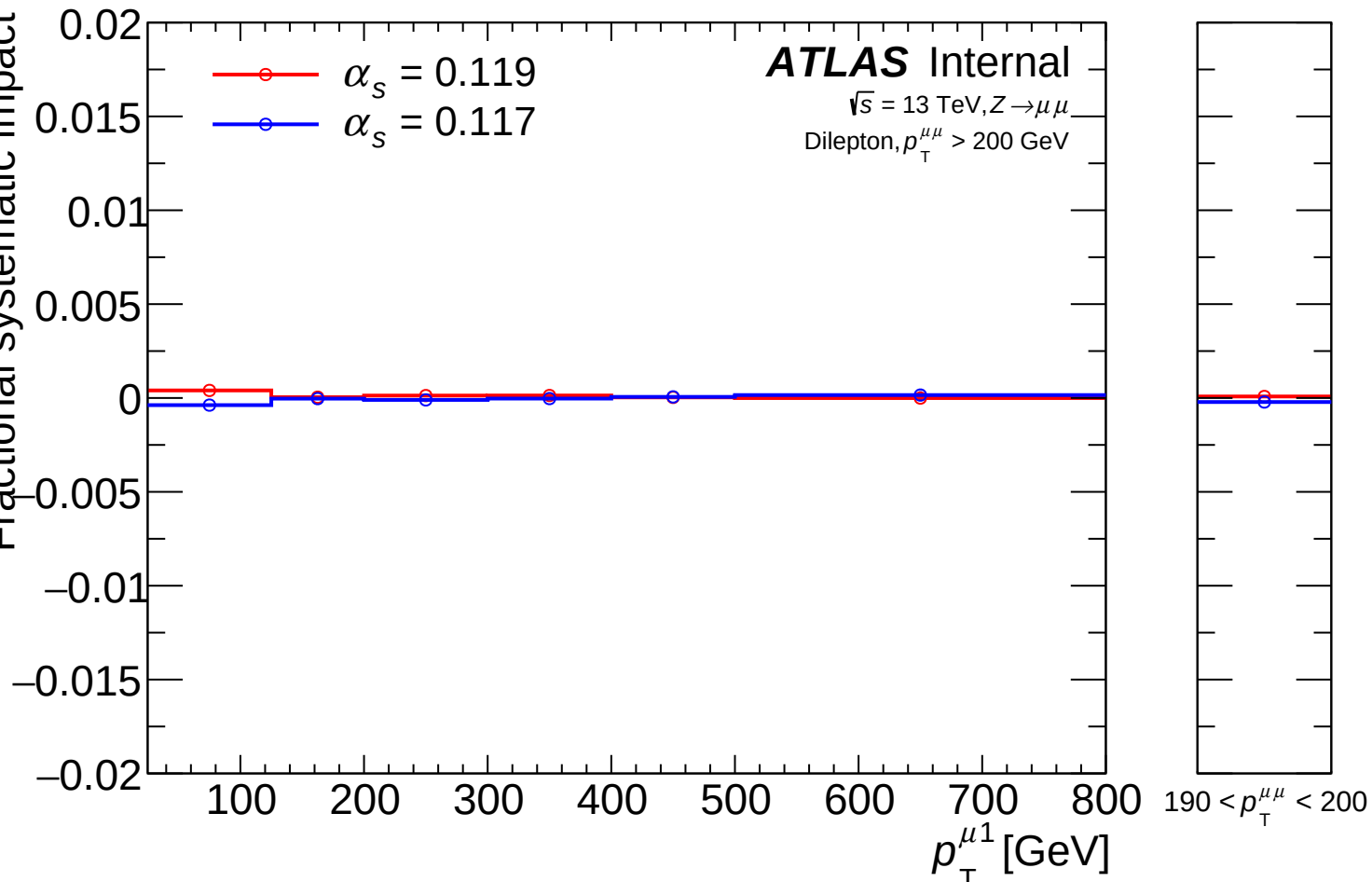


Fractional systematic impact

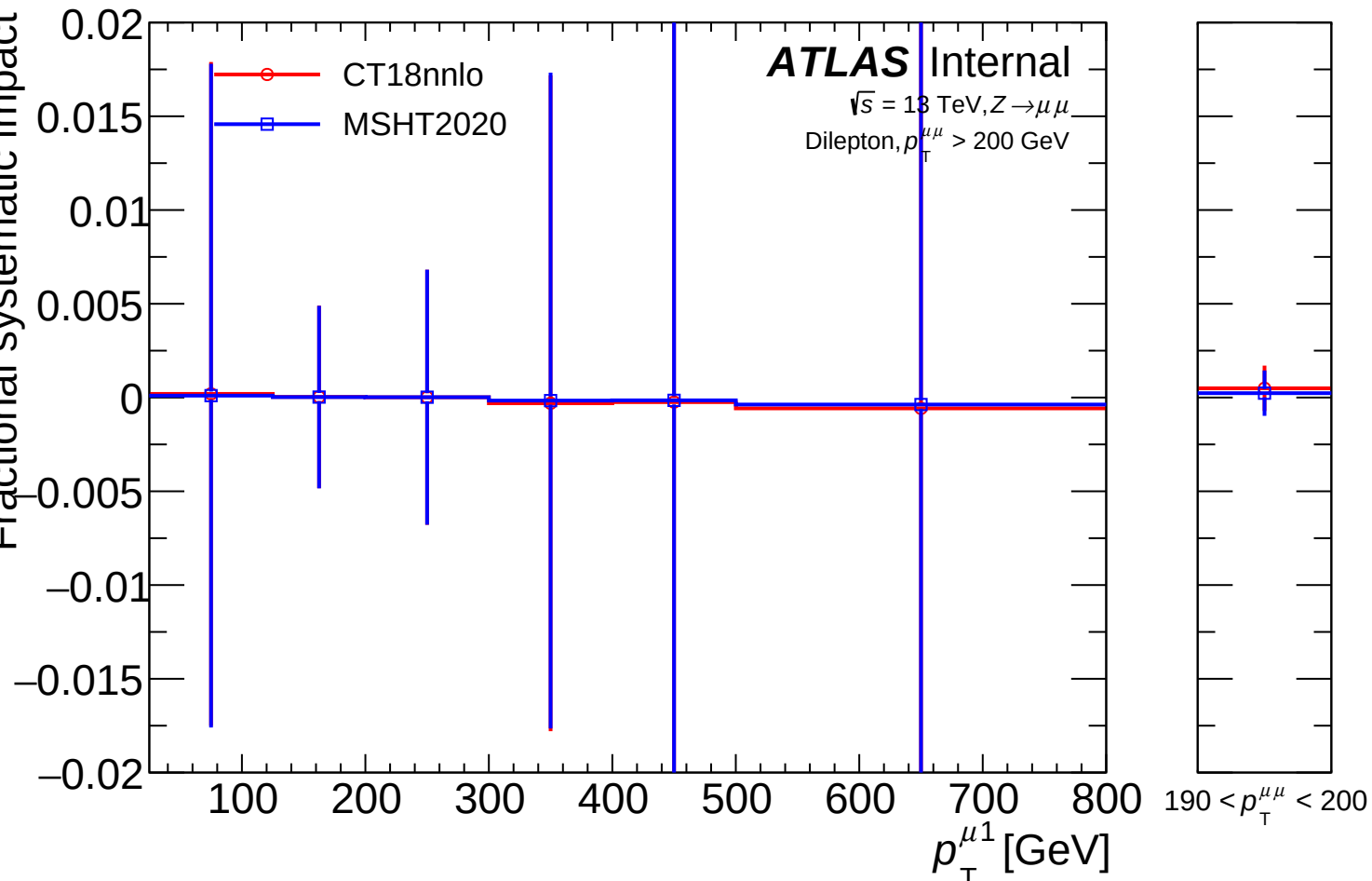




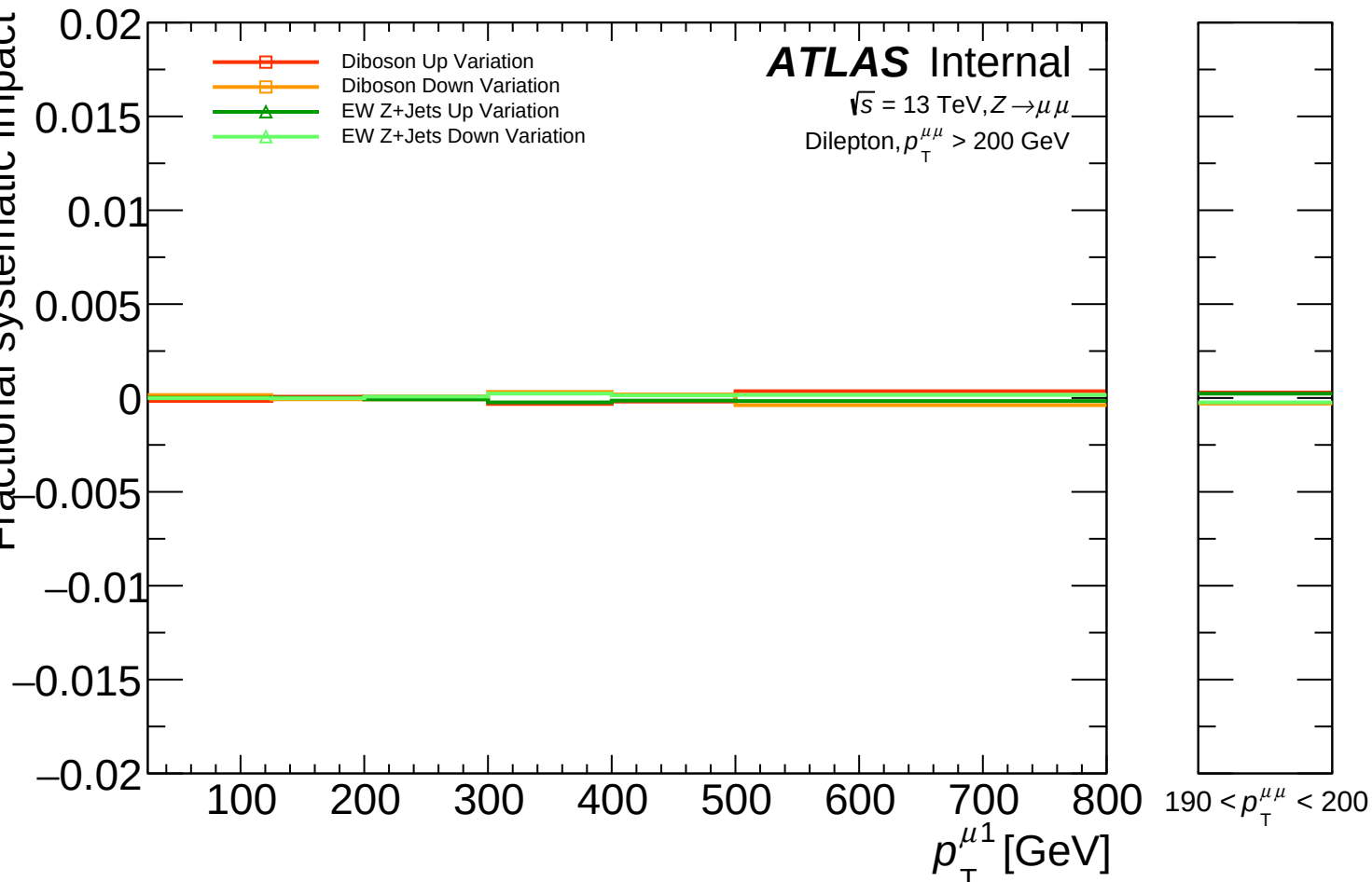
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact

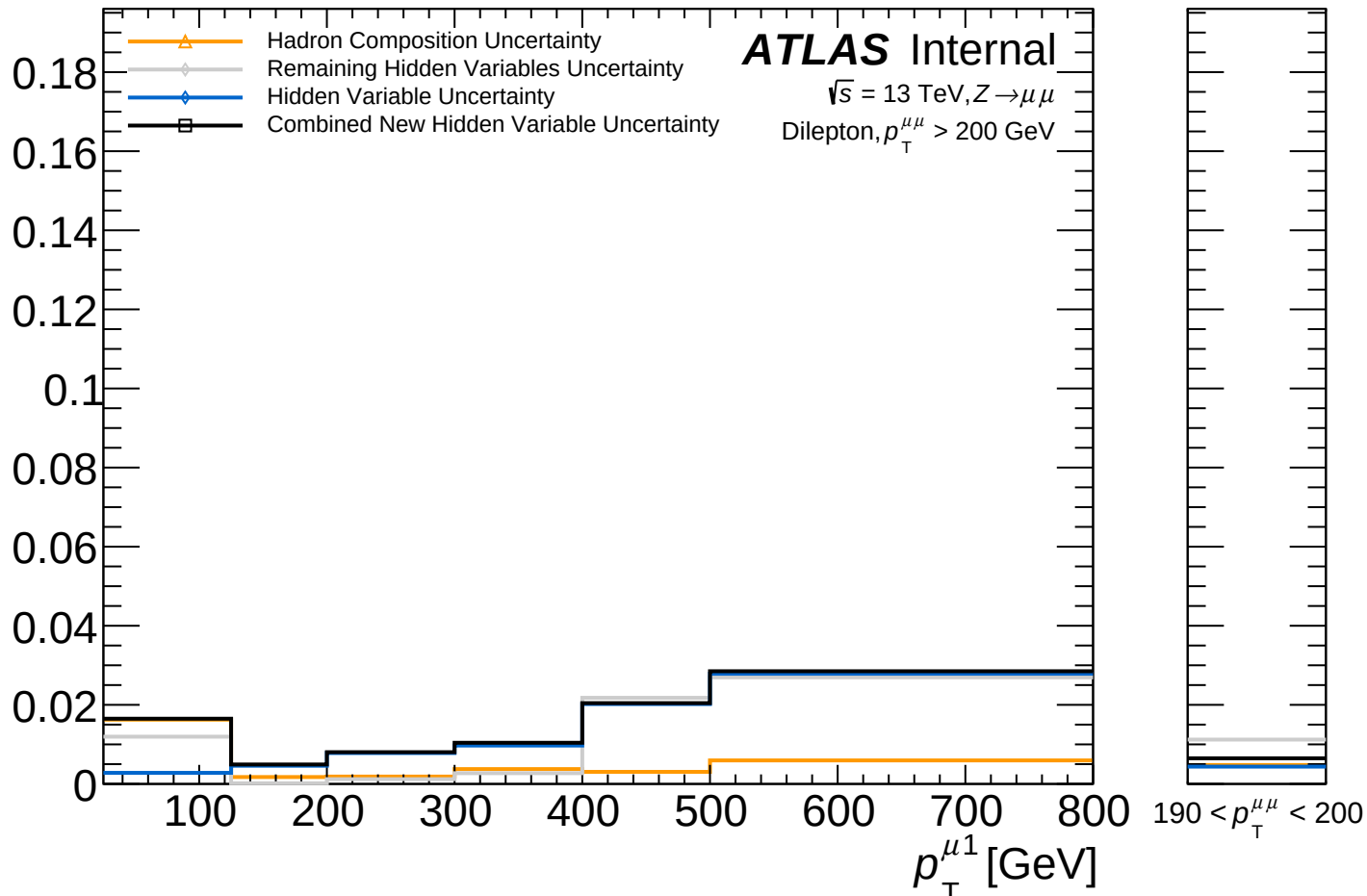


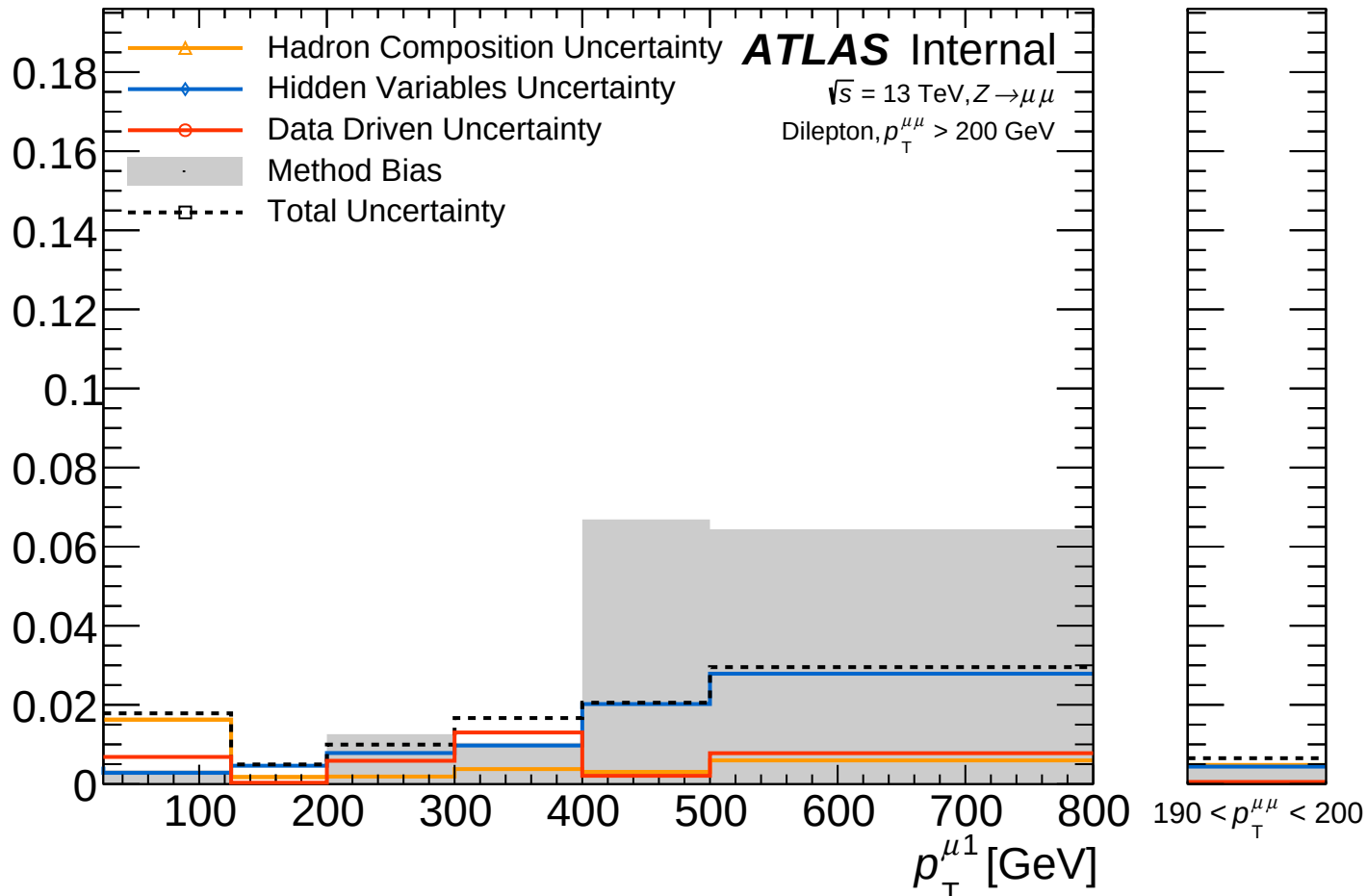
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

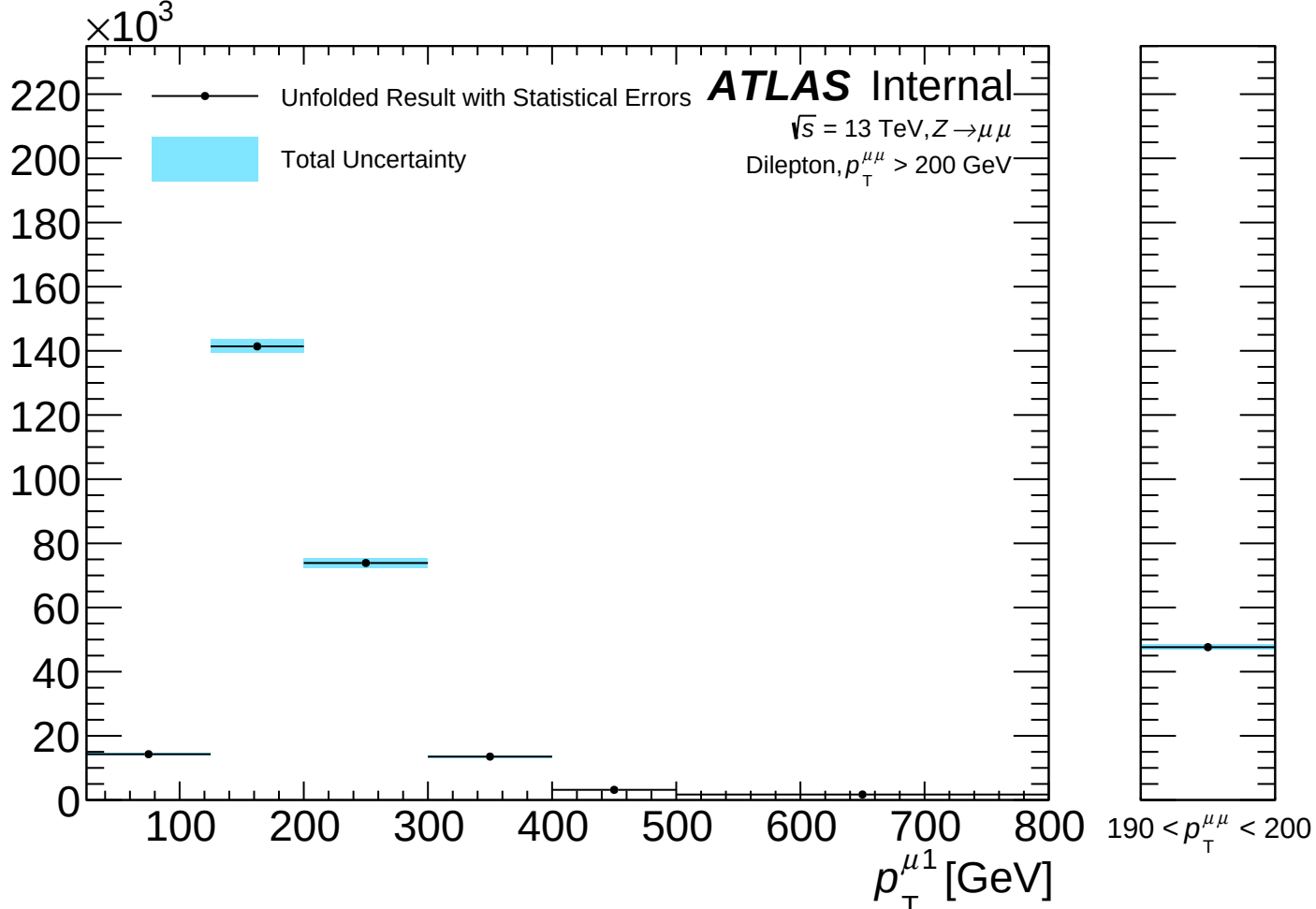
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

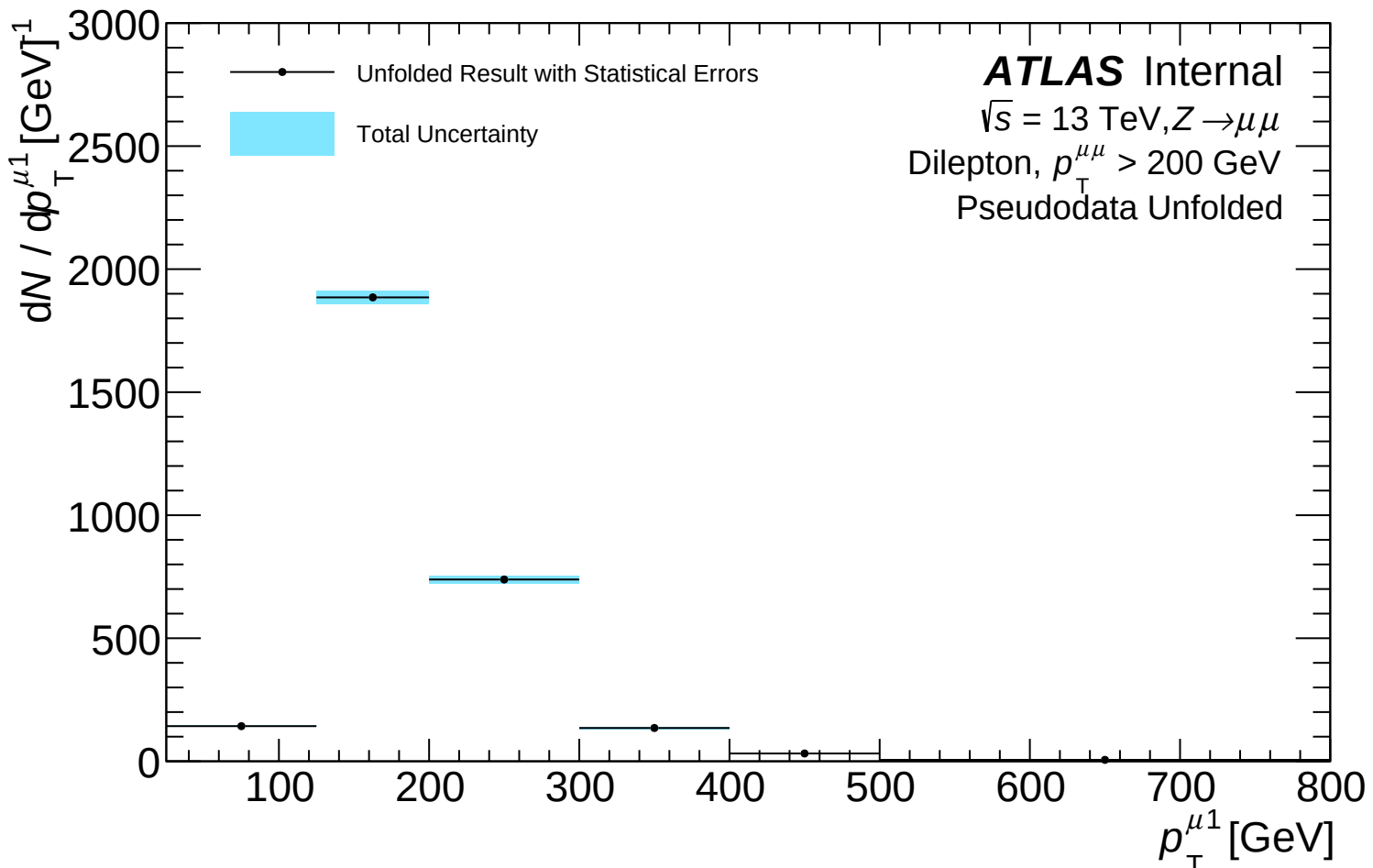
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty

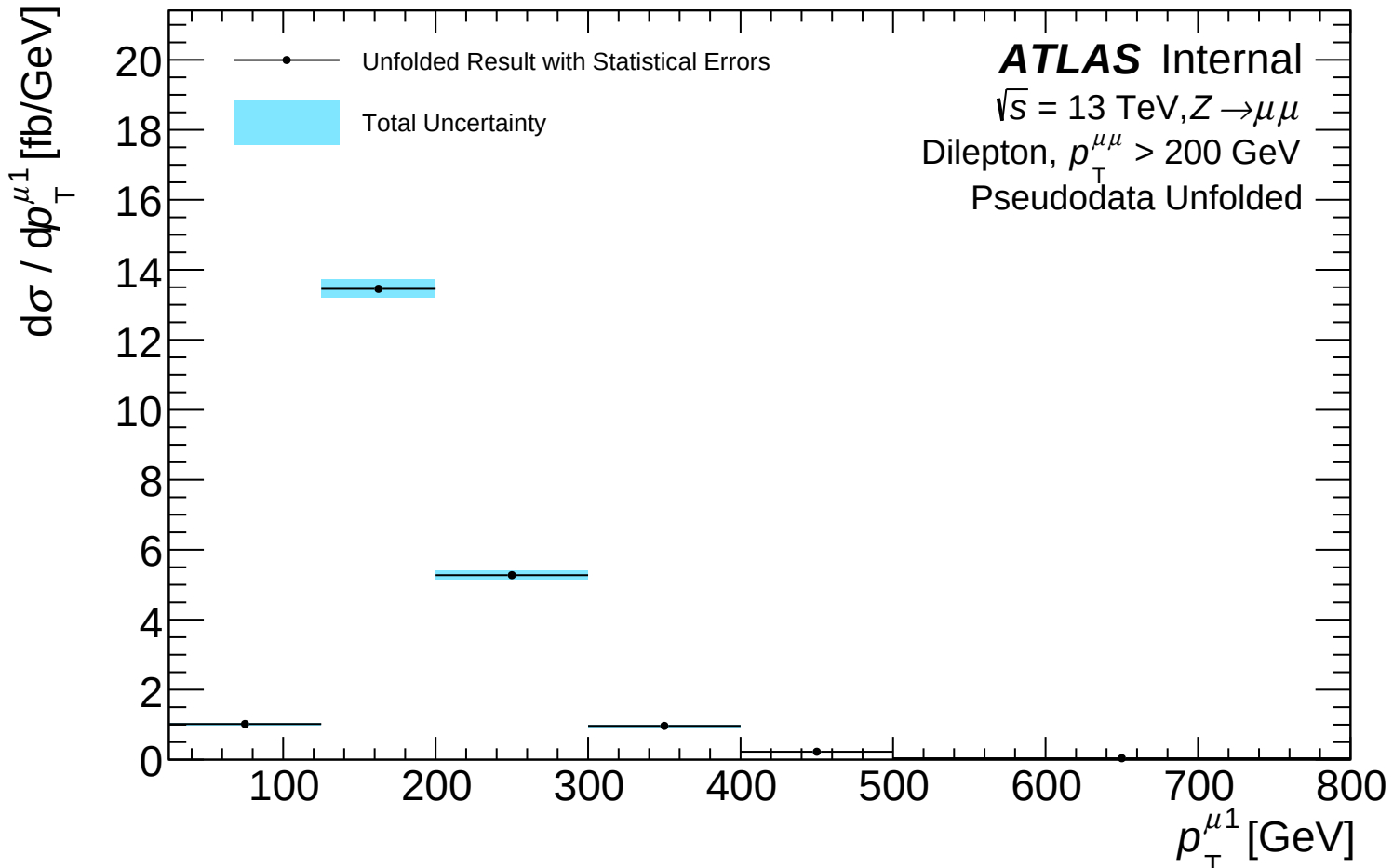


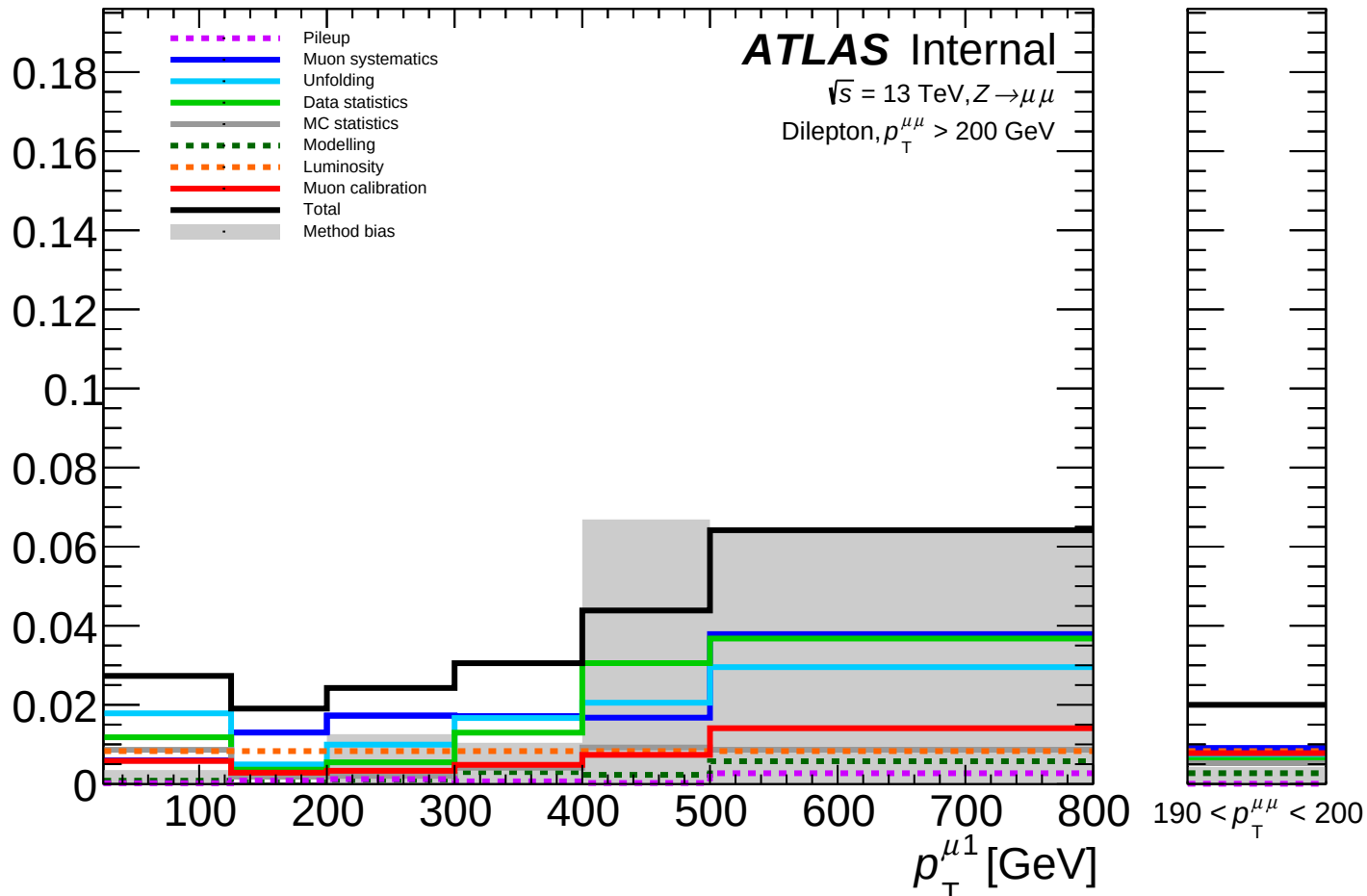


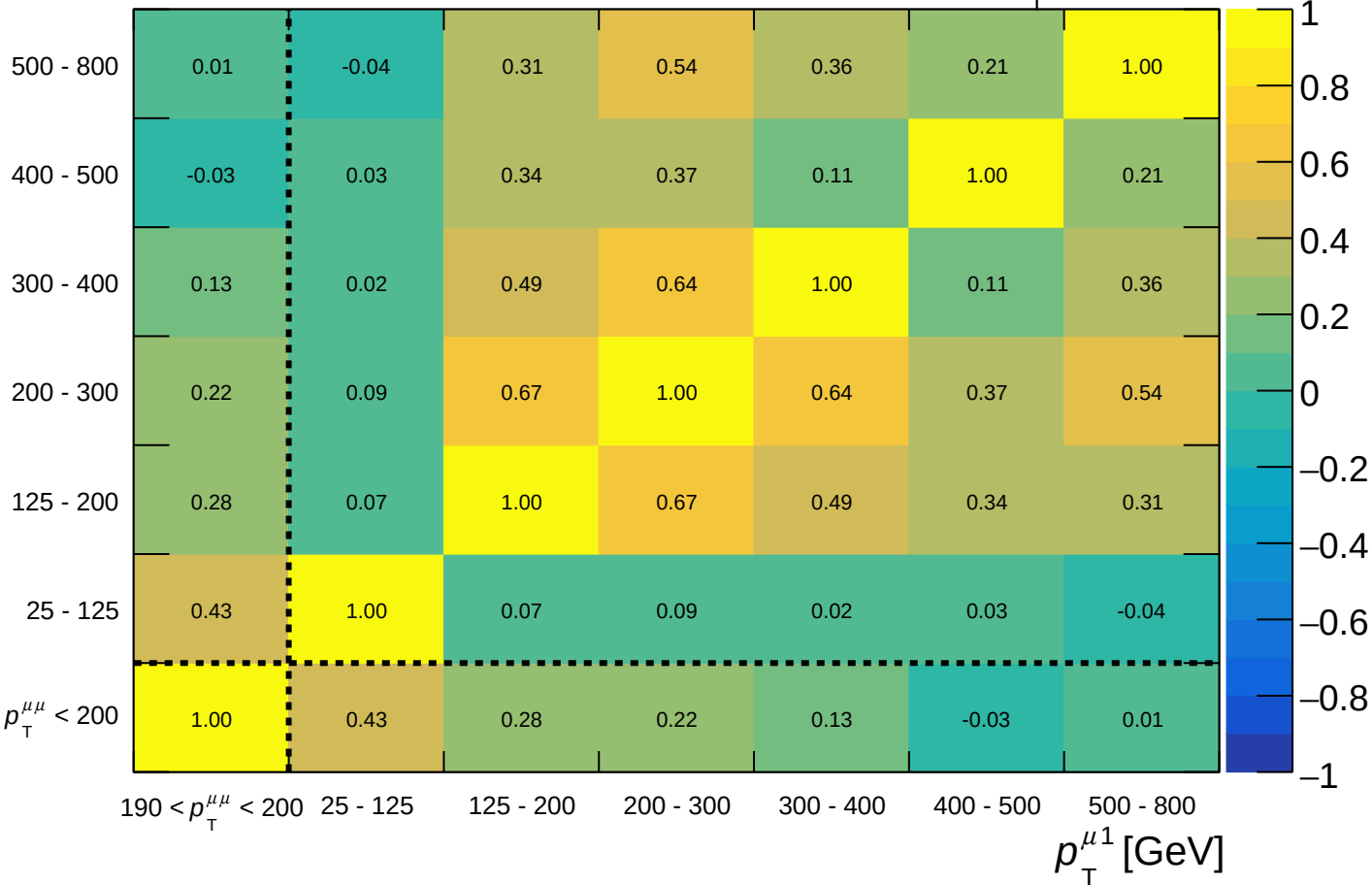
Events

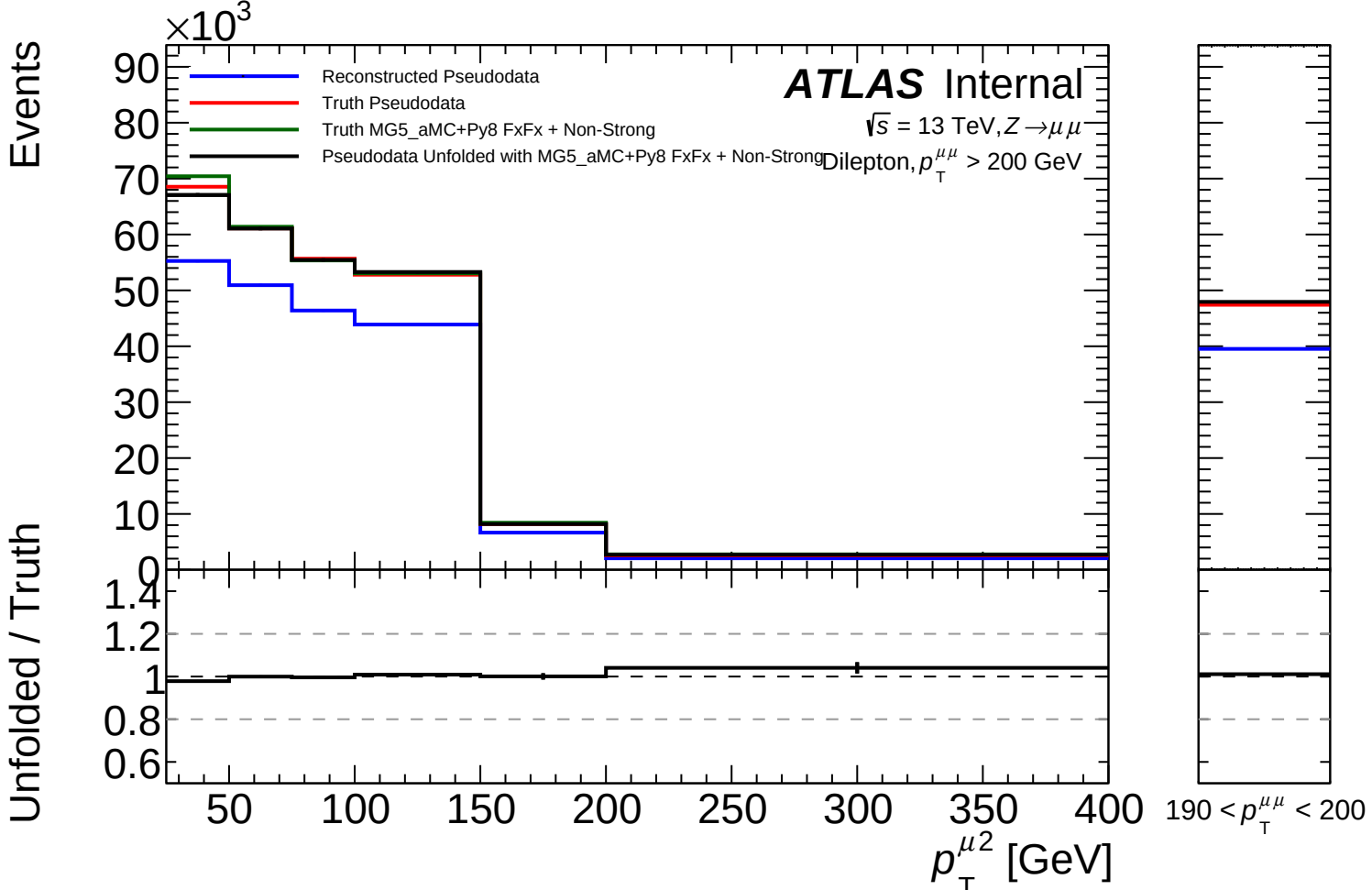


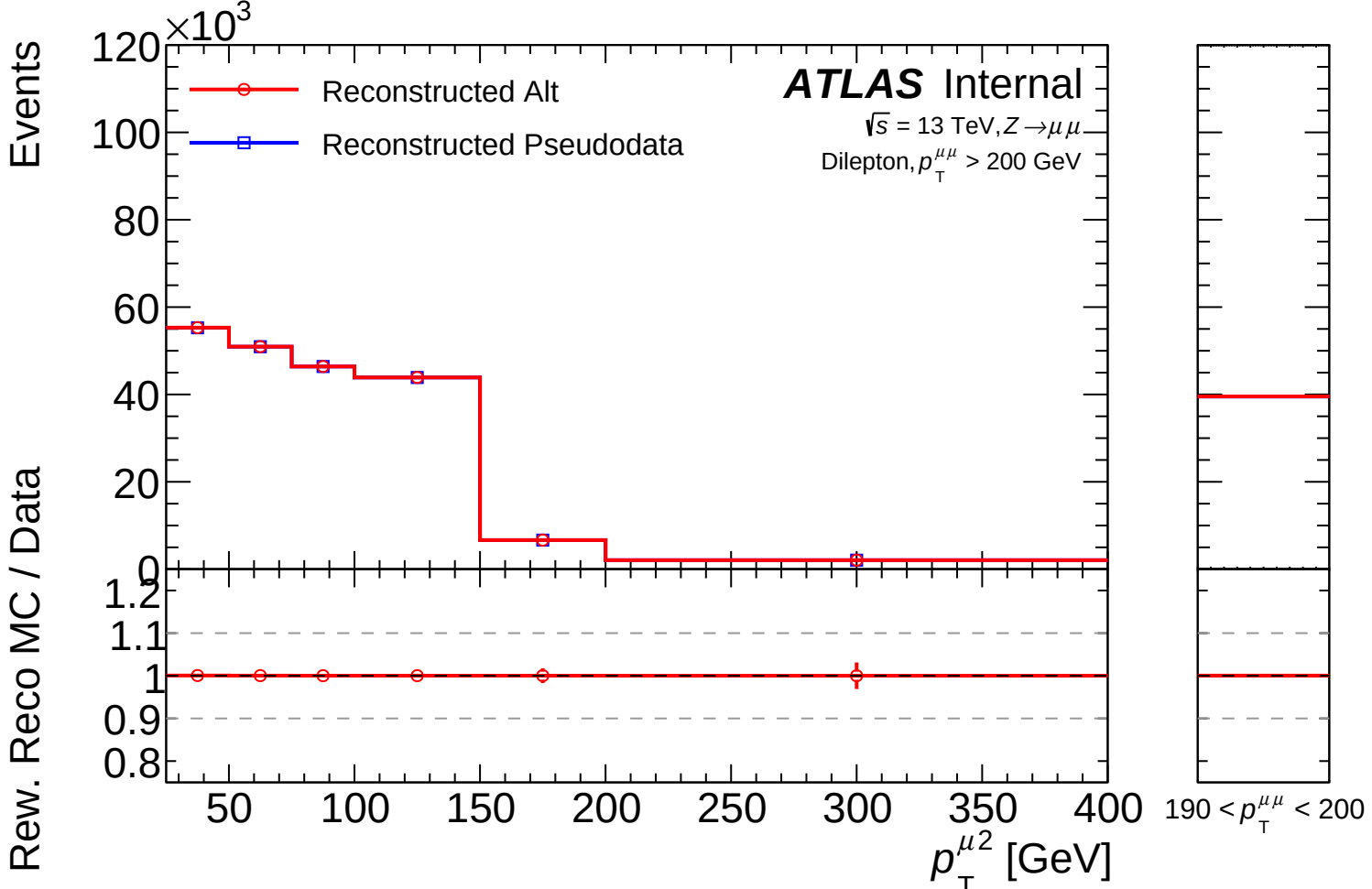


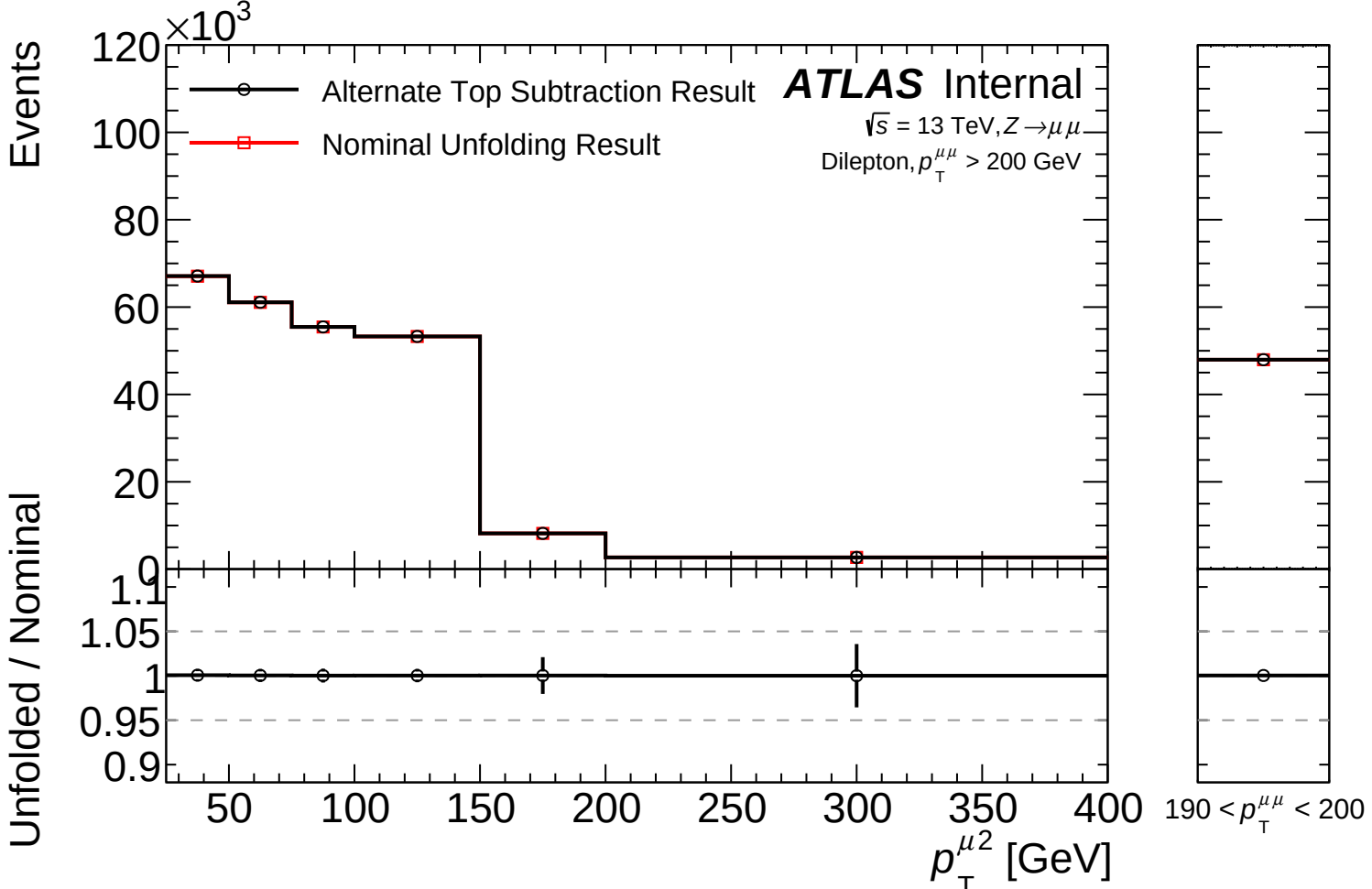




ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $p_{\text{T}}^{\mu 1} [\text{GeV}]$  $p_{\text{T}}^{\mu 1} [\text{GeV}]$







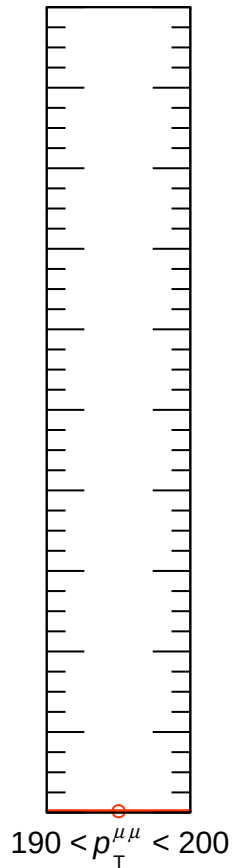
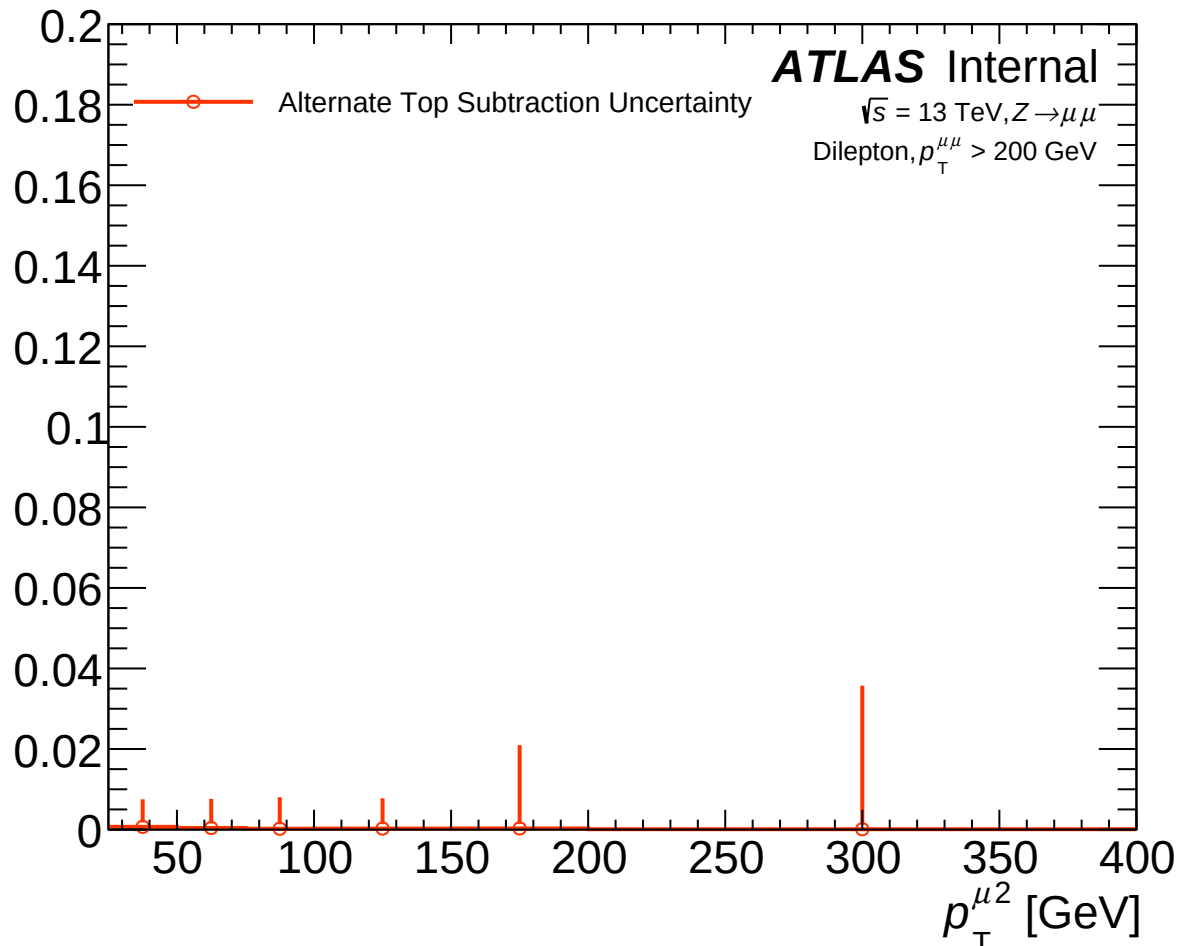
Fractional systematic impact

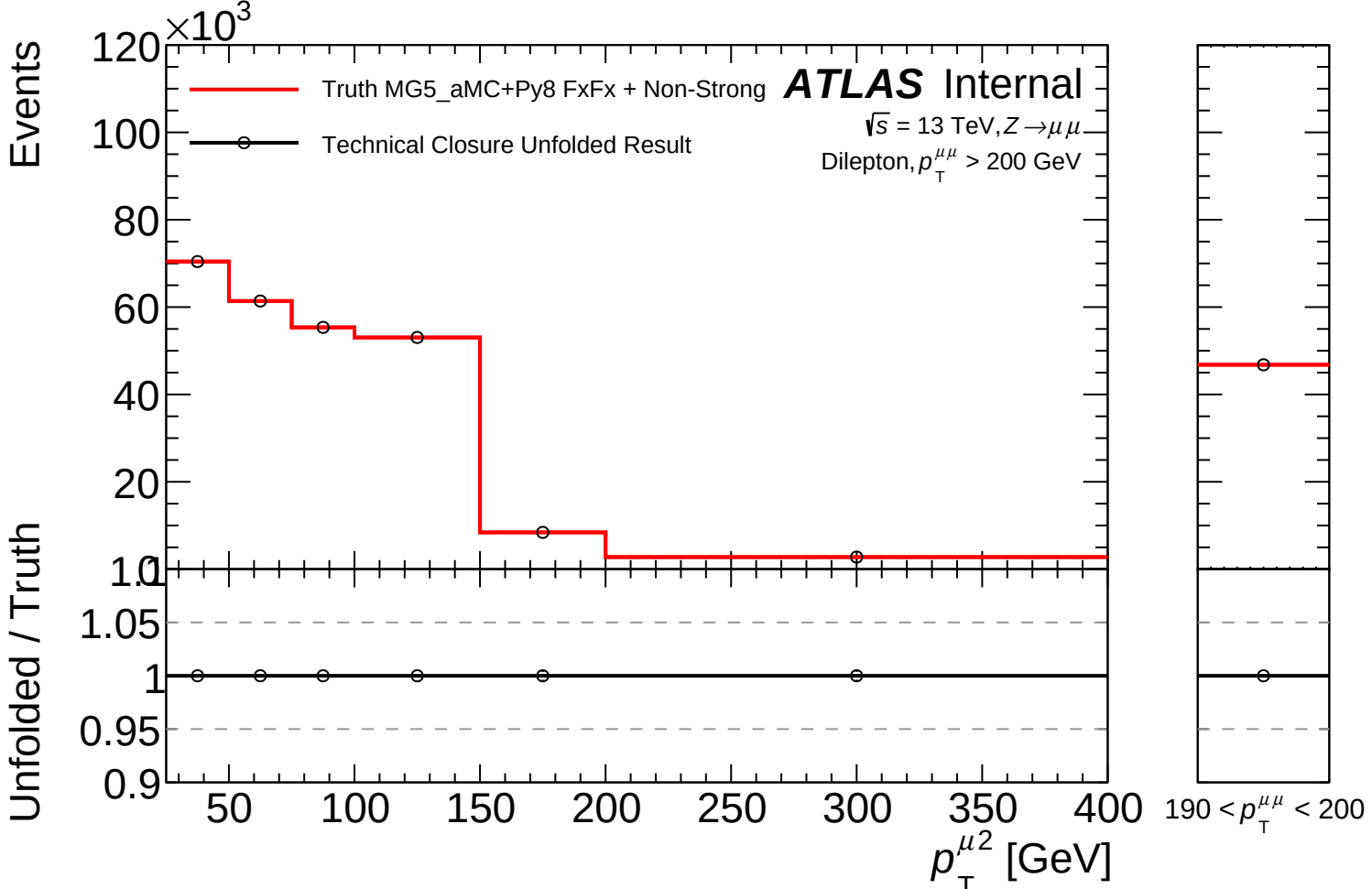
ATLAS Internal

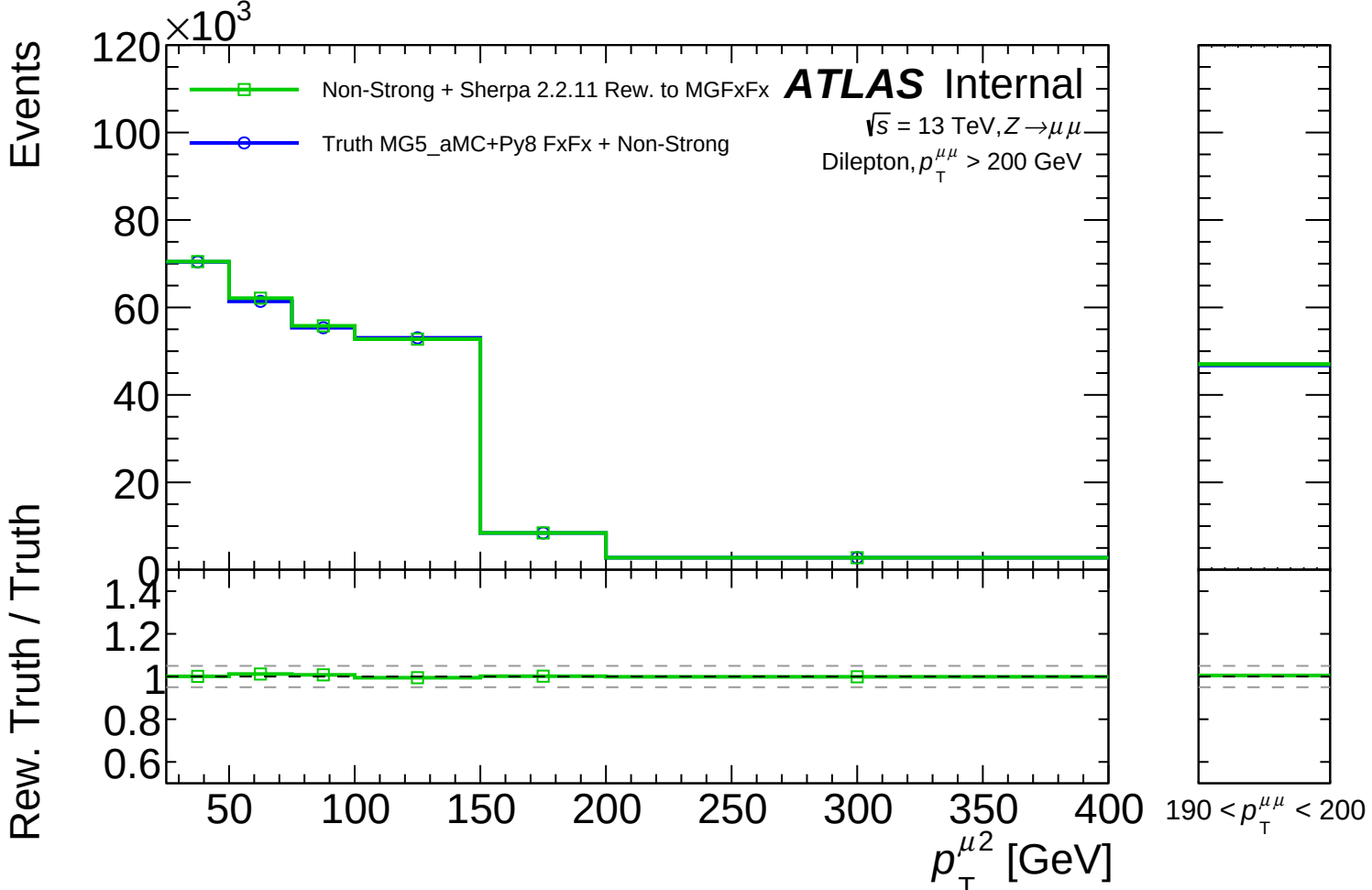
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

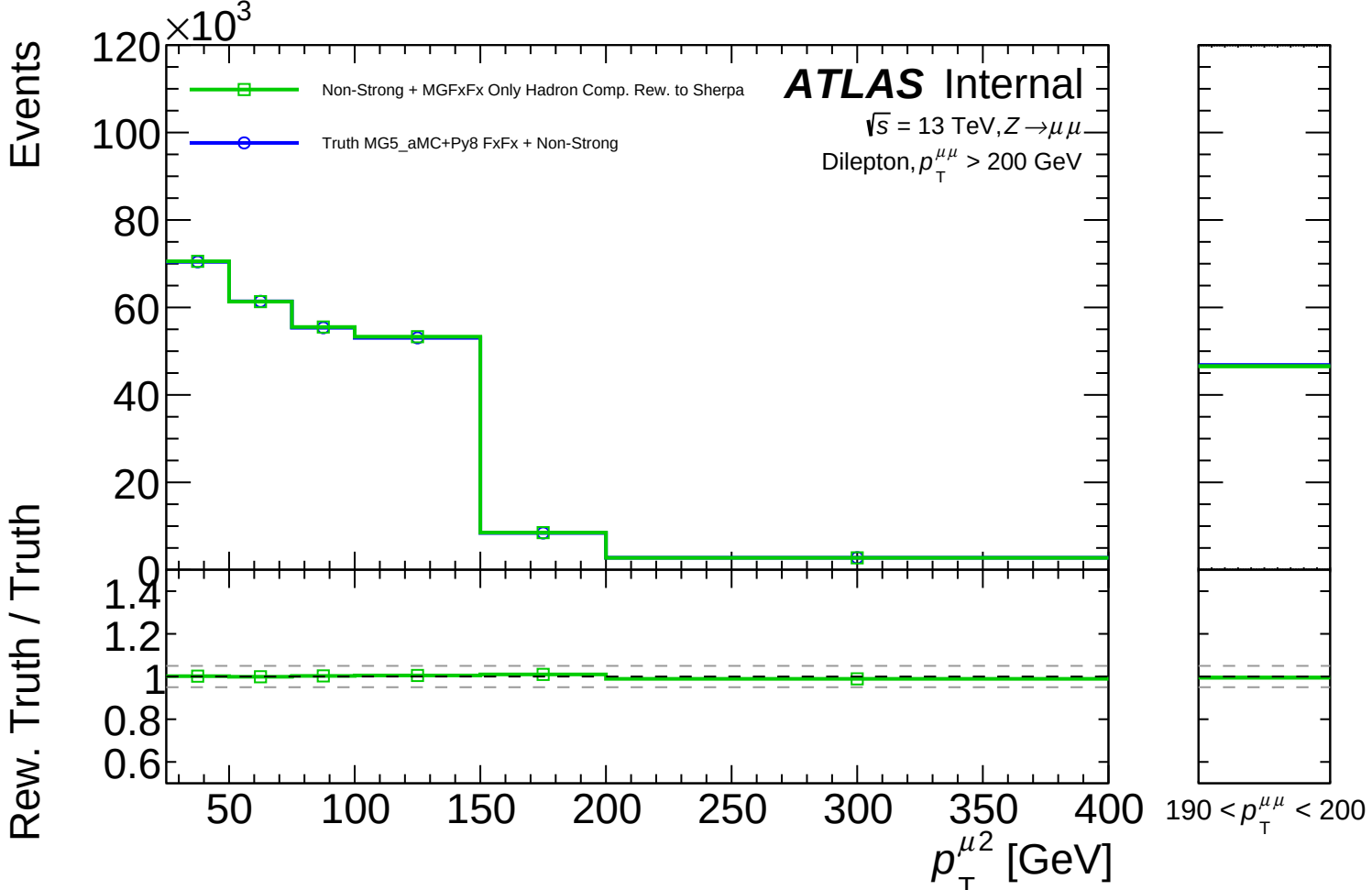
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

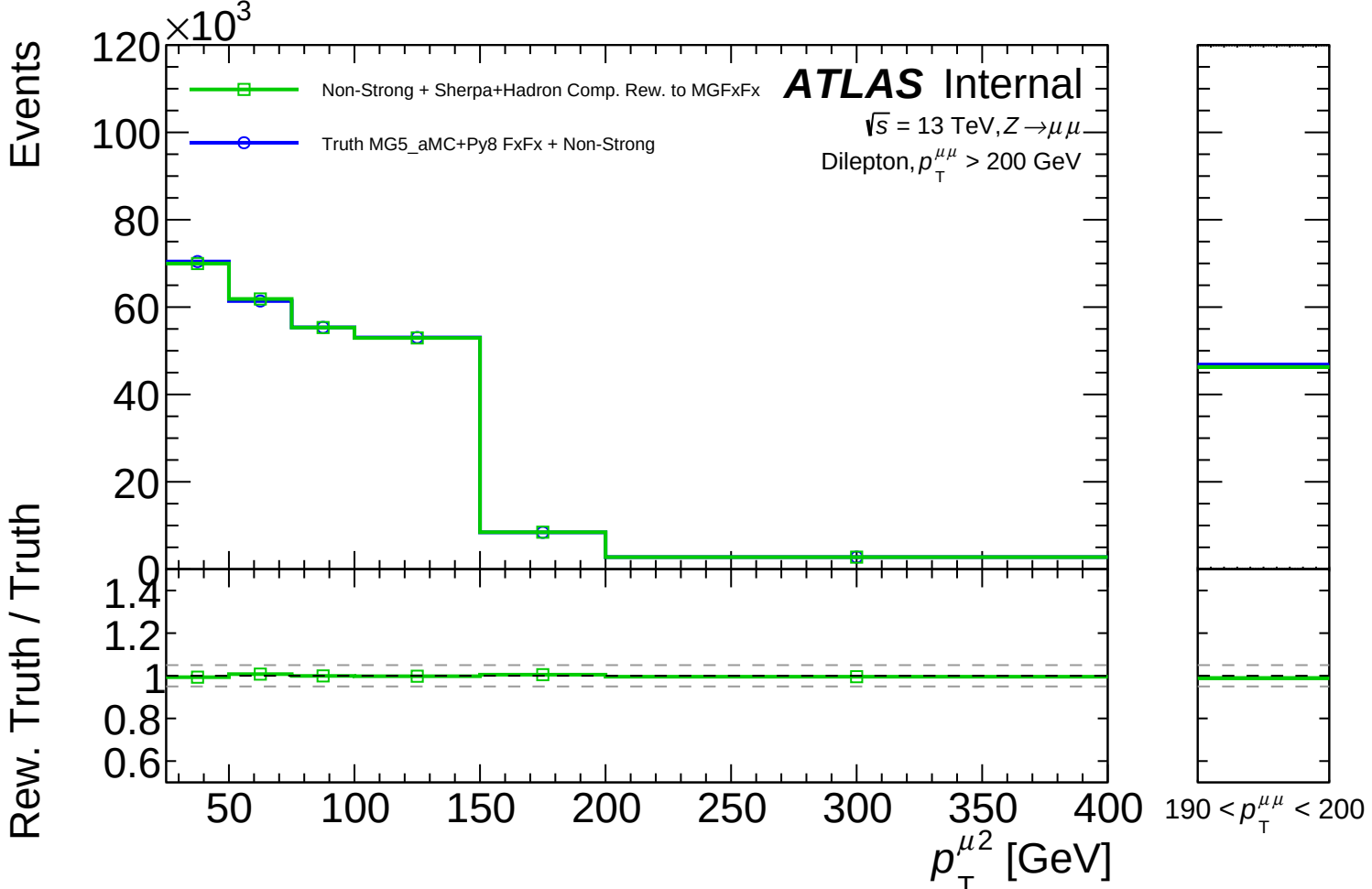
Alternate Top Subtraction Uncertainty

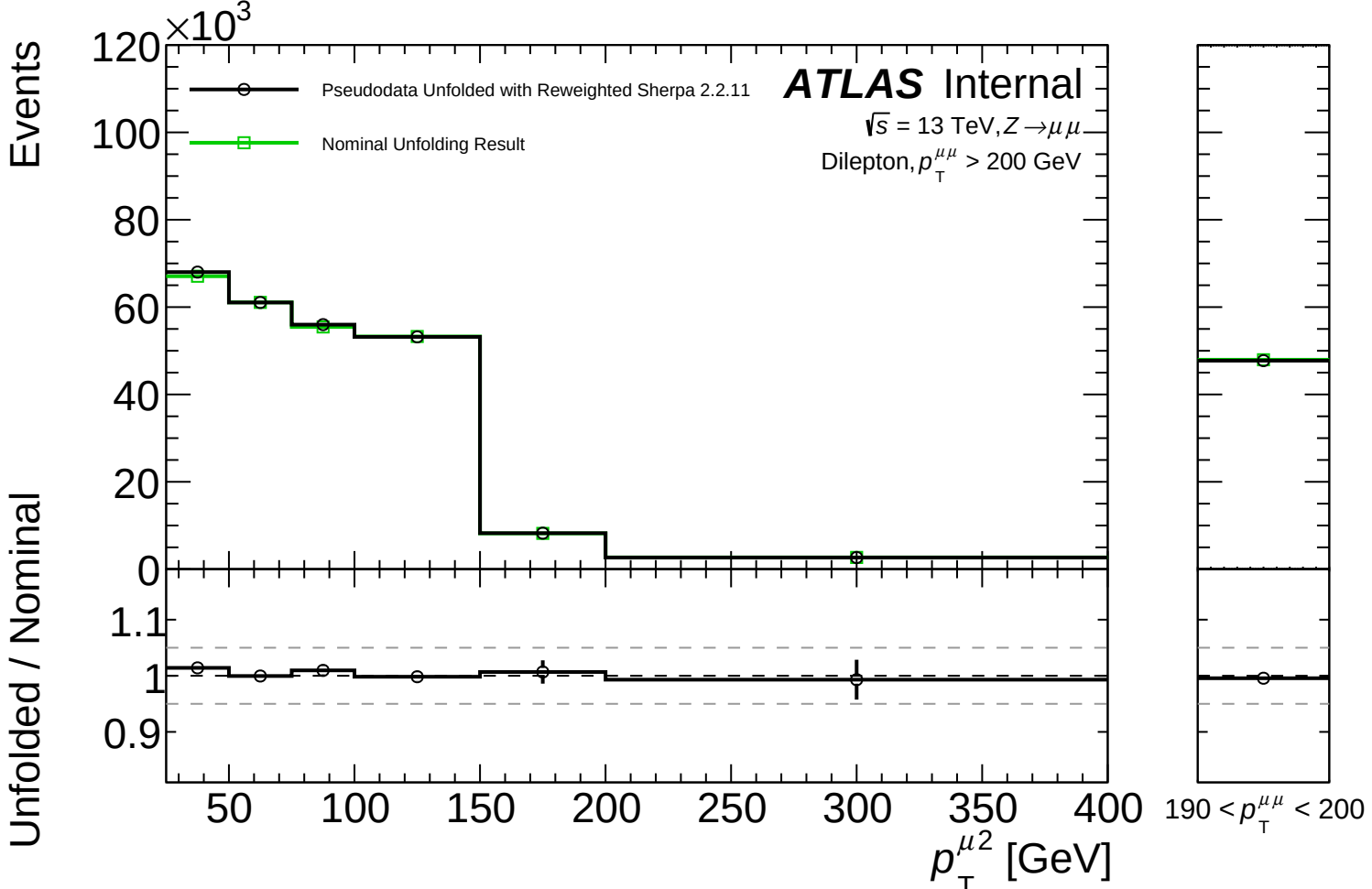


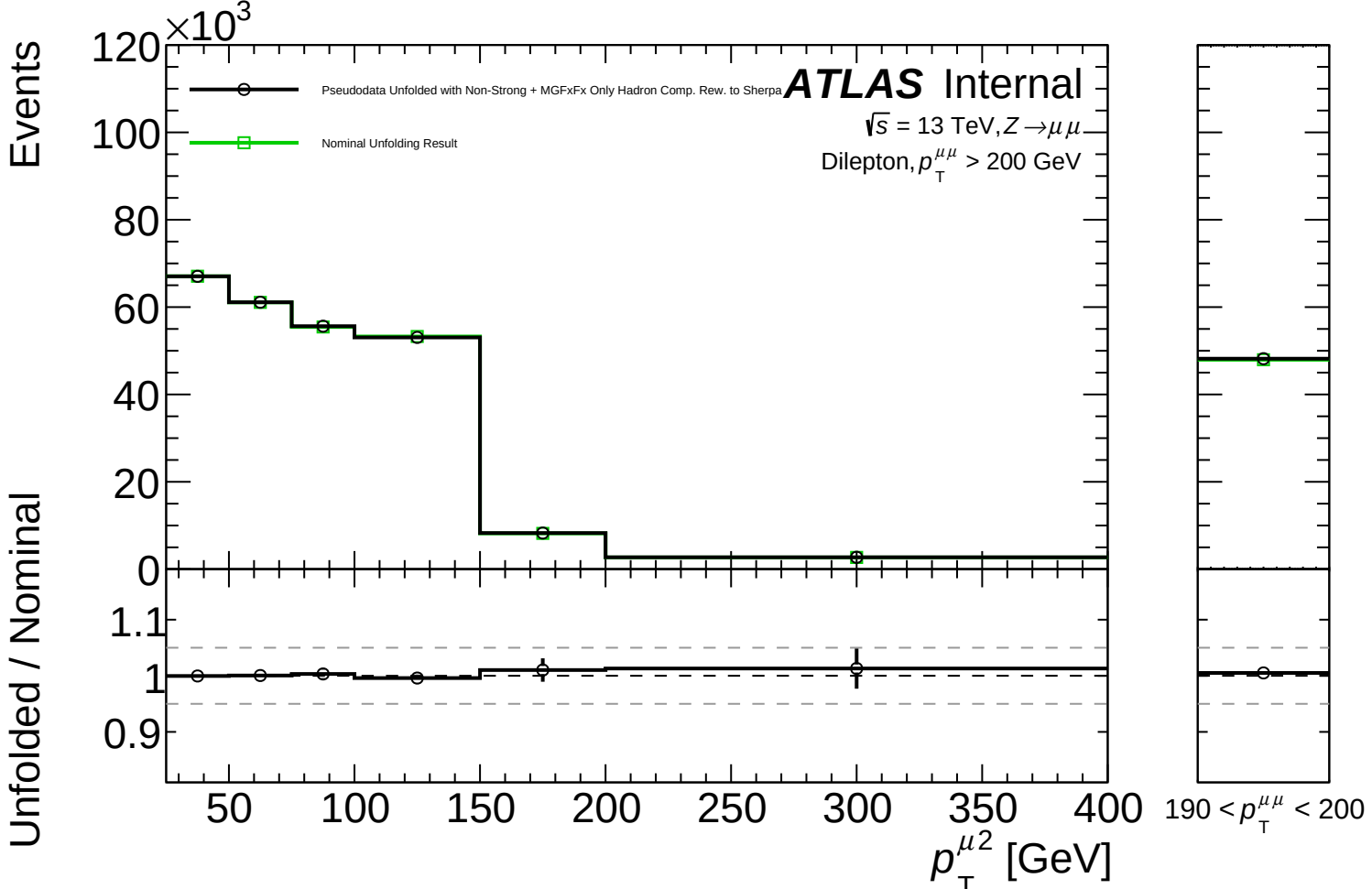


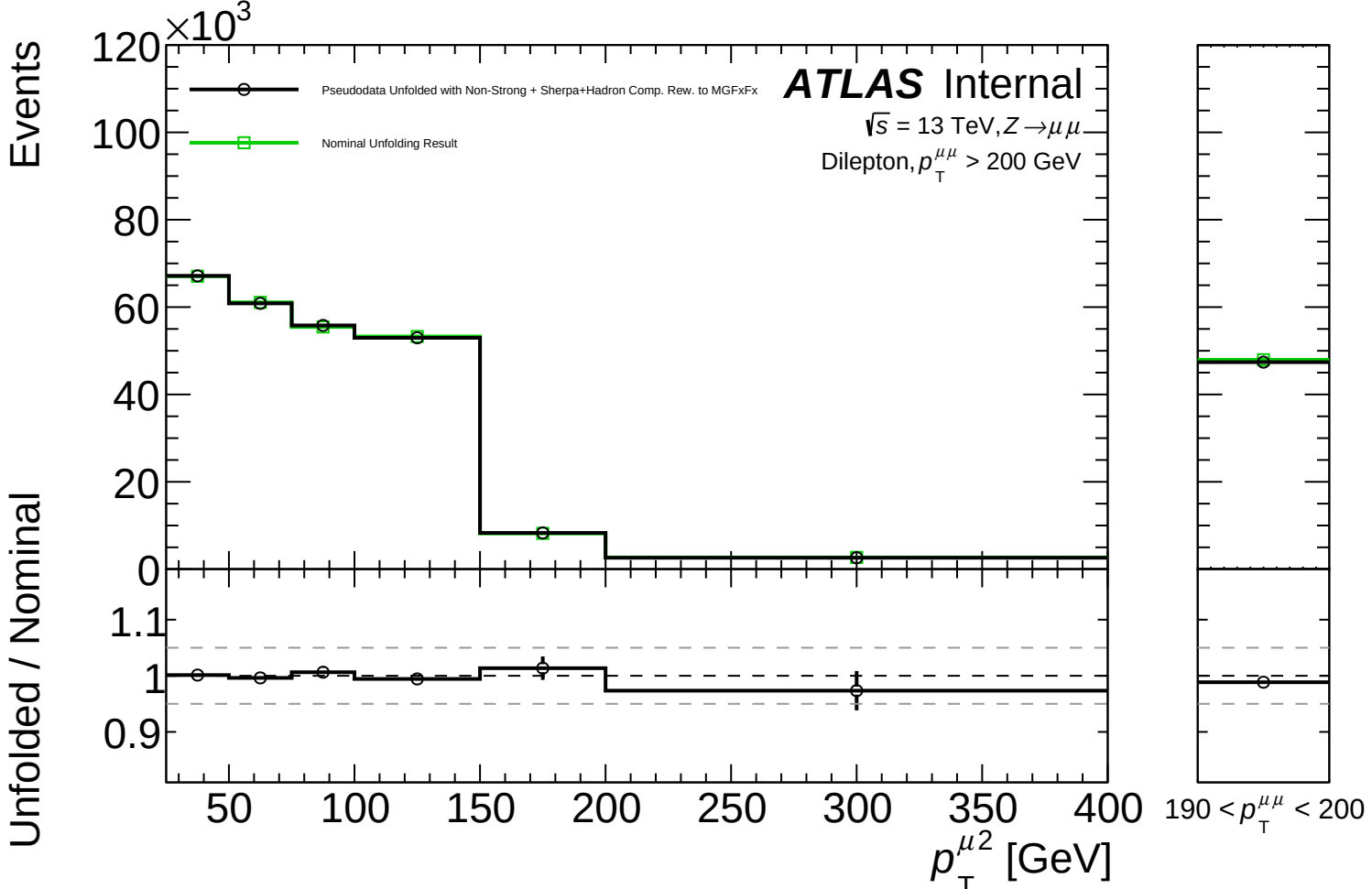


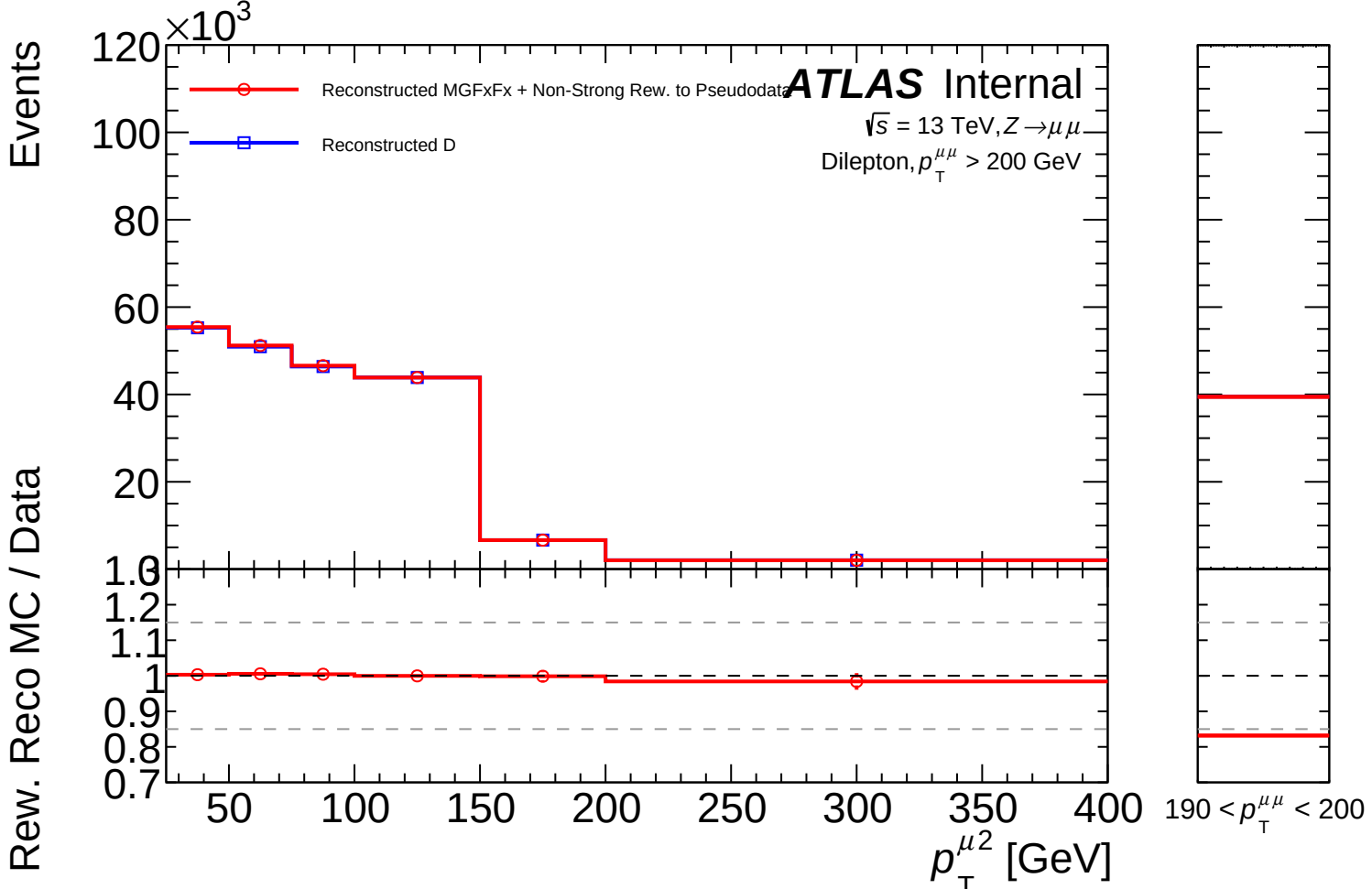


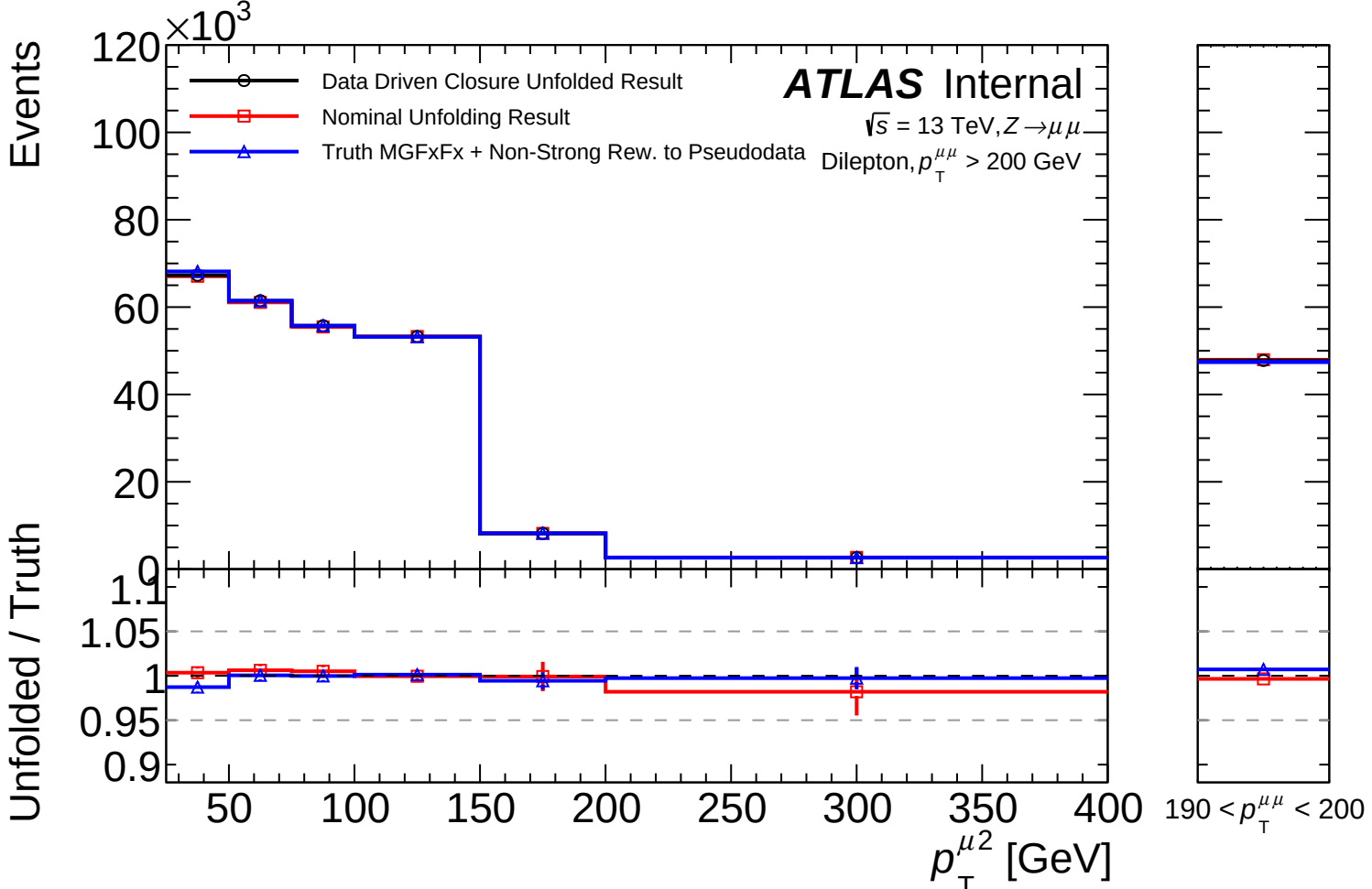




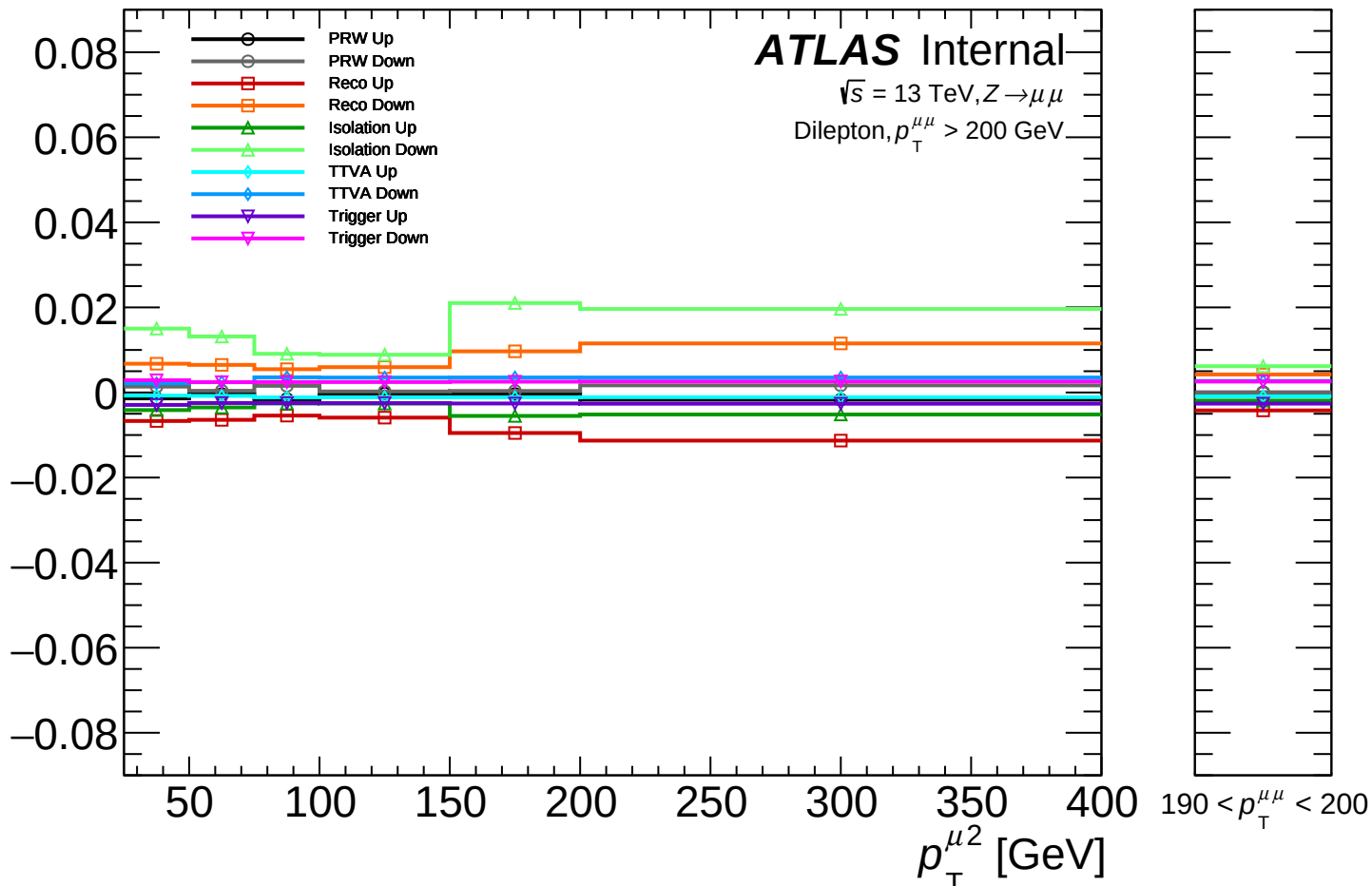




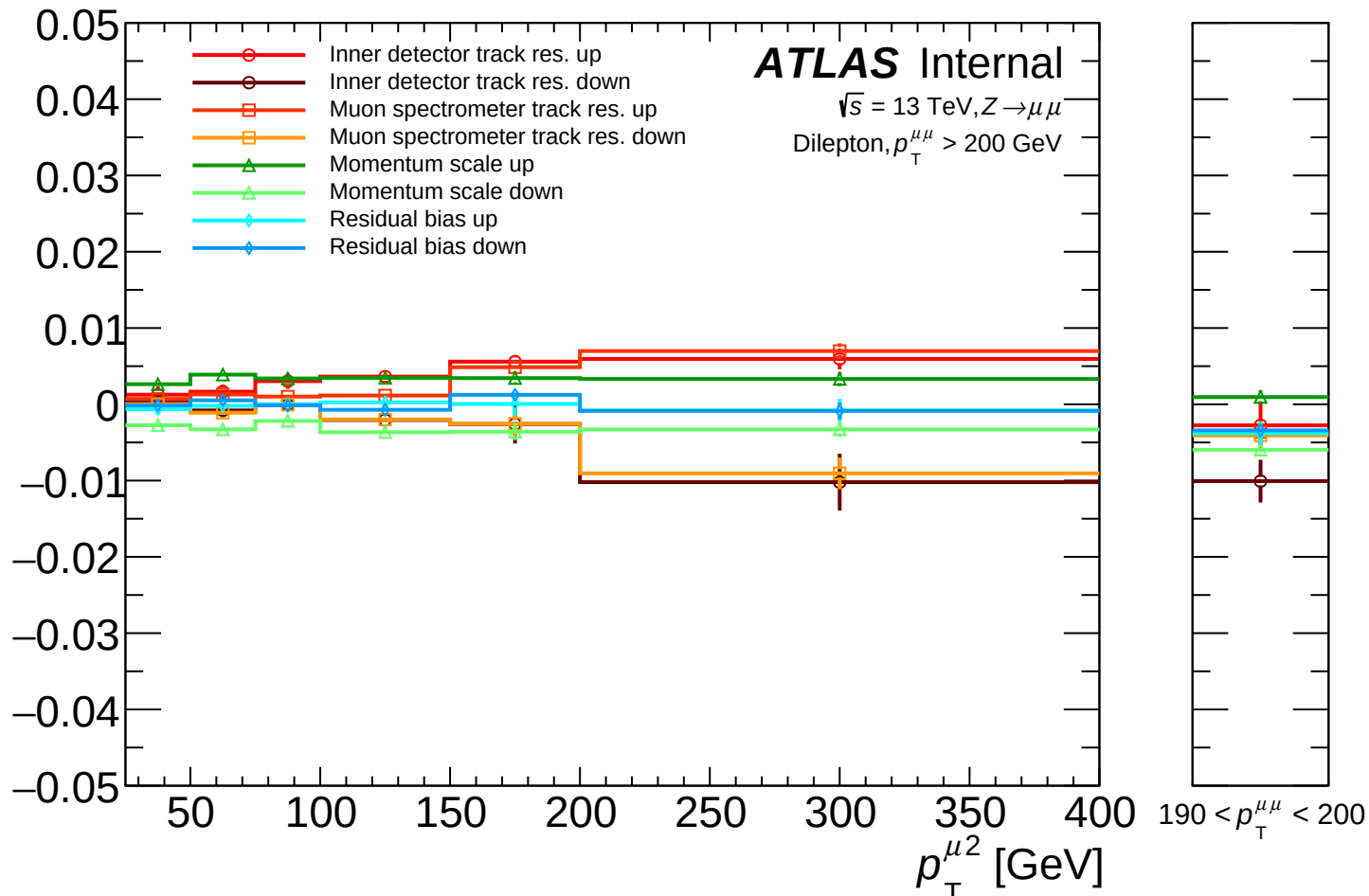




Fractional systematic impact



Fractional systematic impact



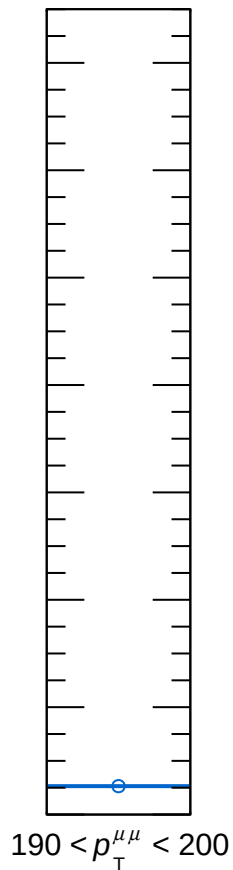
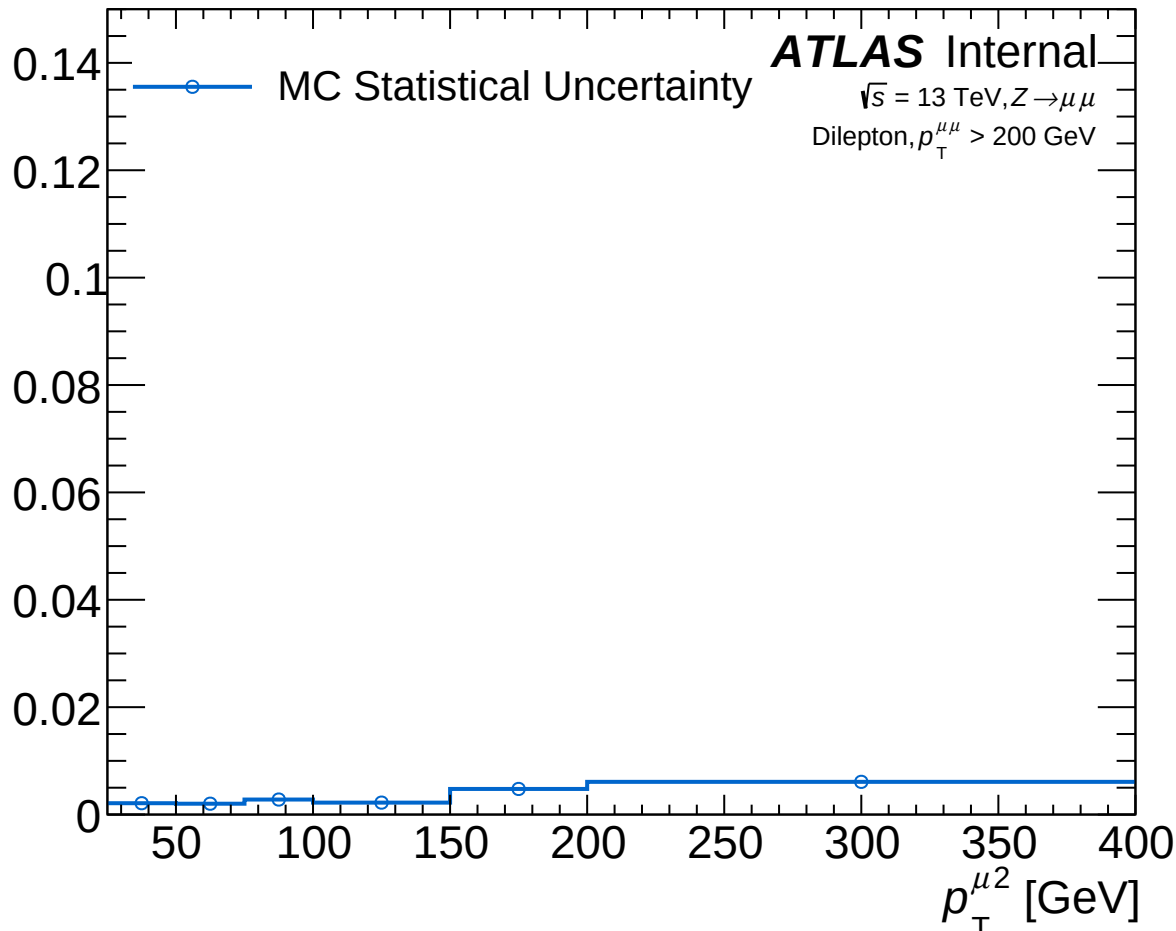
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



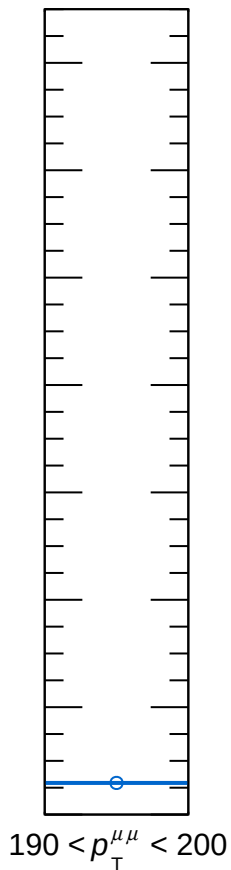
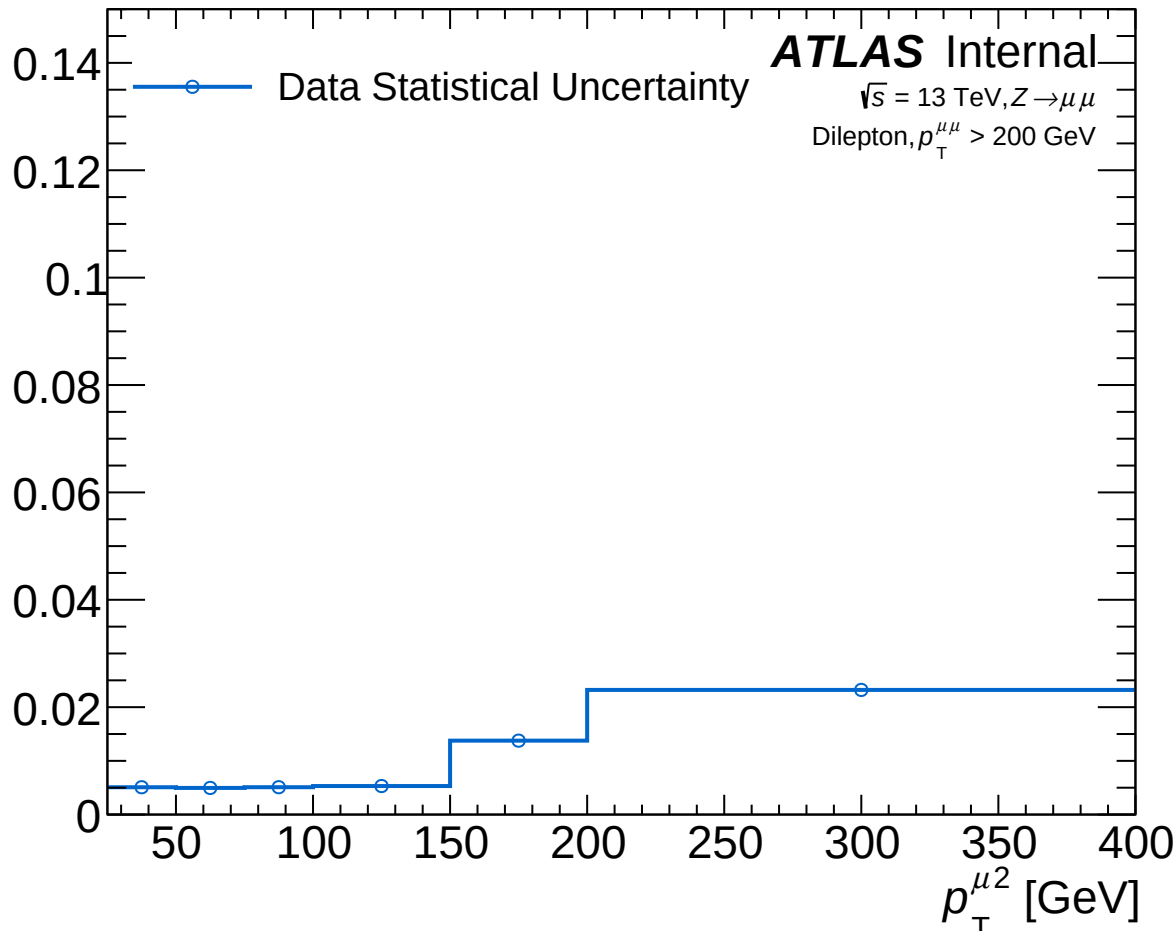
Fractional systematic impact

ATLAS Internal

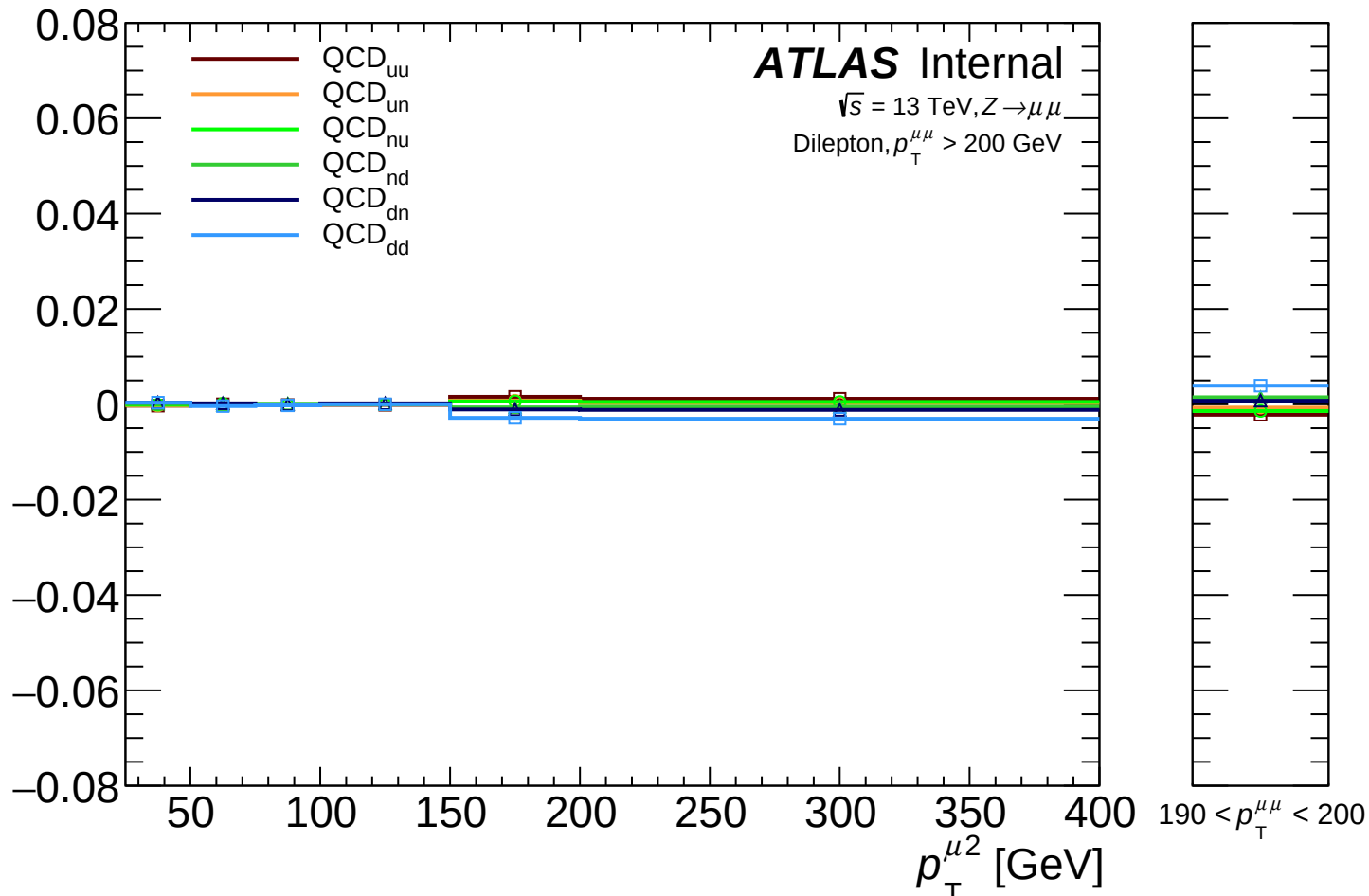
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

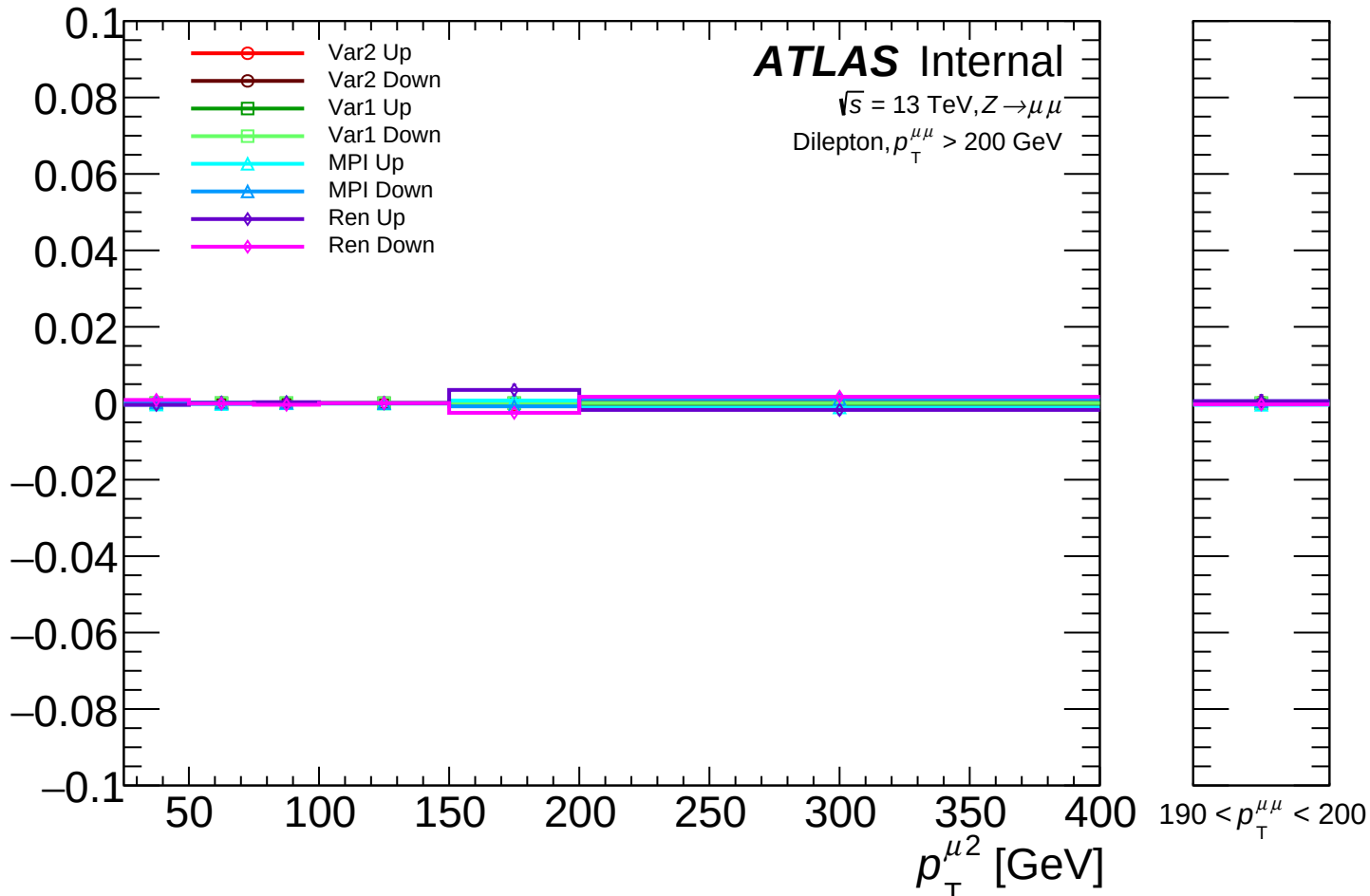
Data Statistical Uncertainty

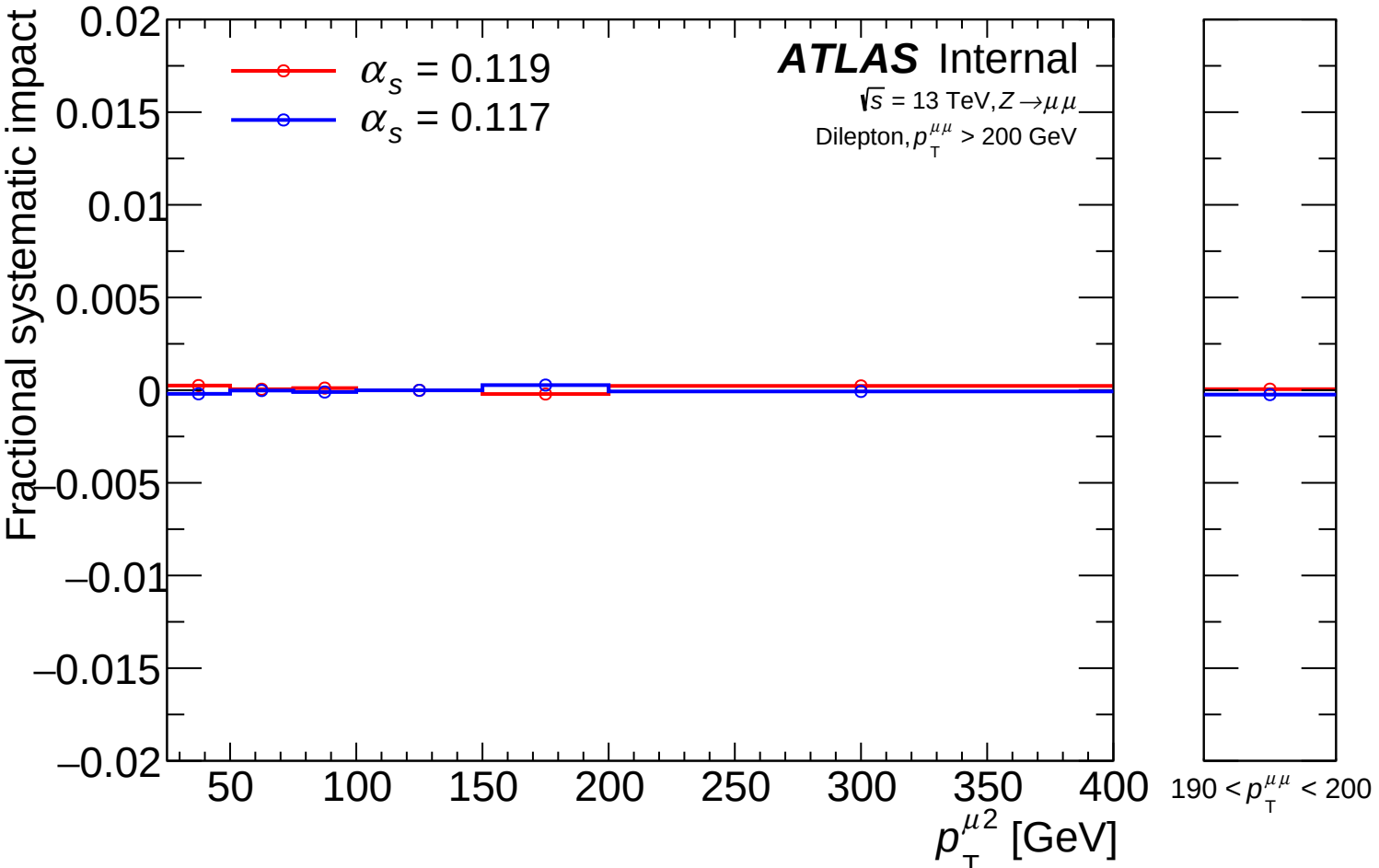


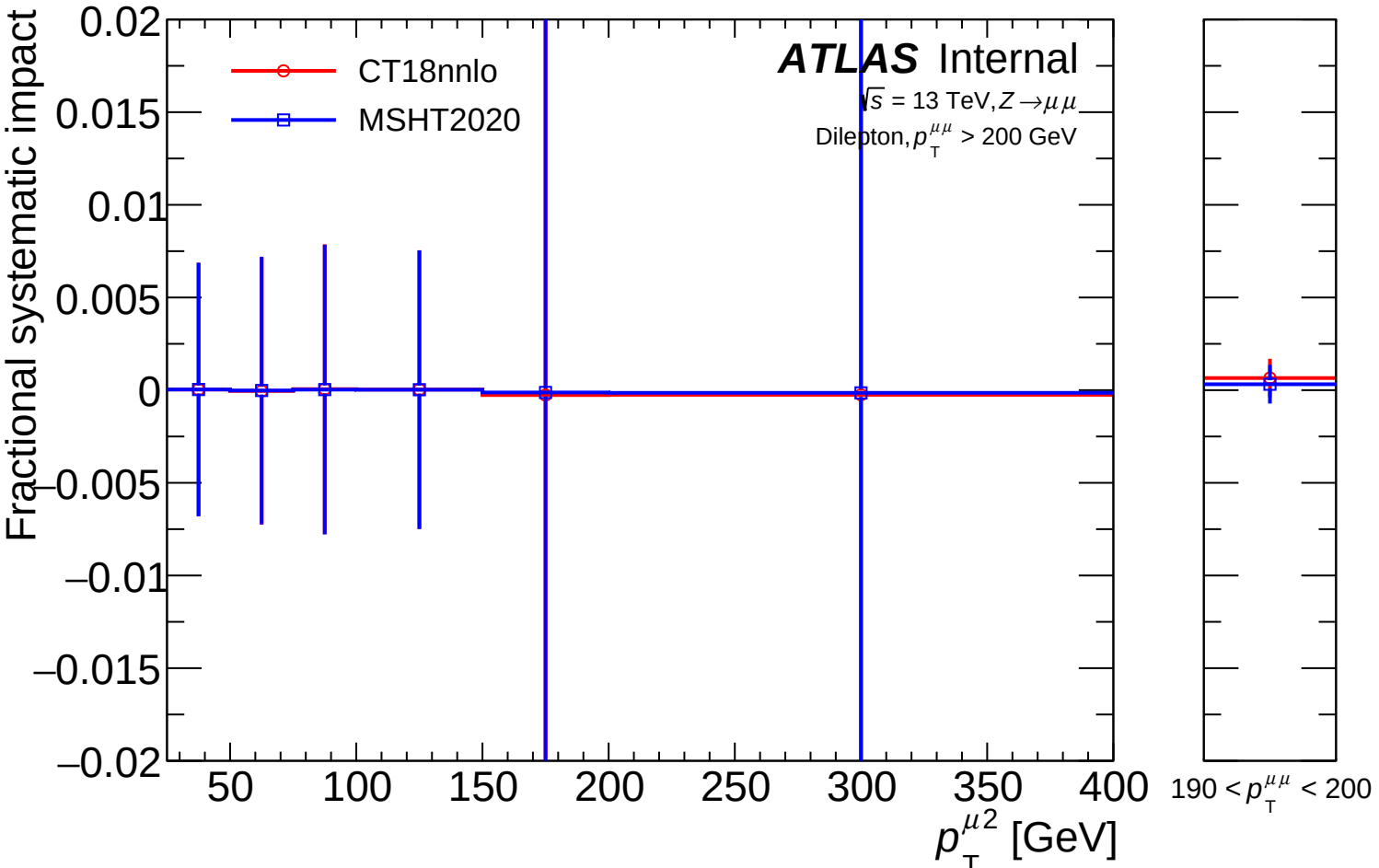
Fractional systematic impact



Fractional systematic impact







Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

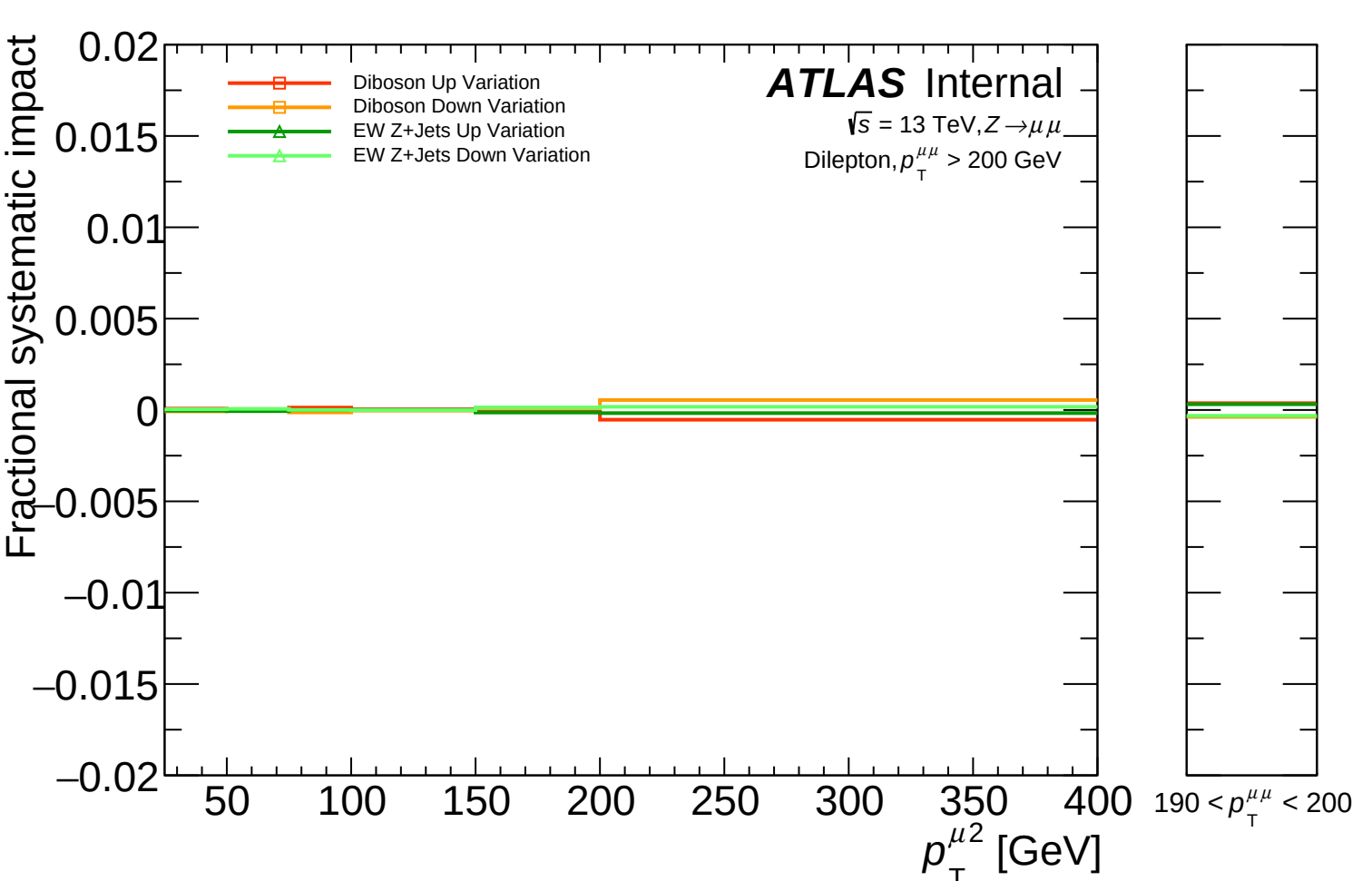
ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

$p_{\text{T}}^{\mu\mu}$ [GeV]

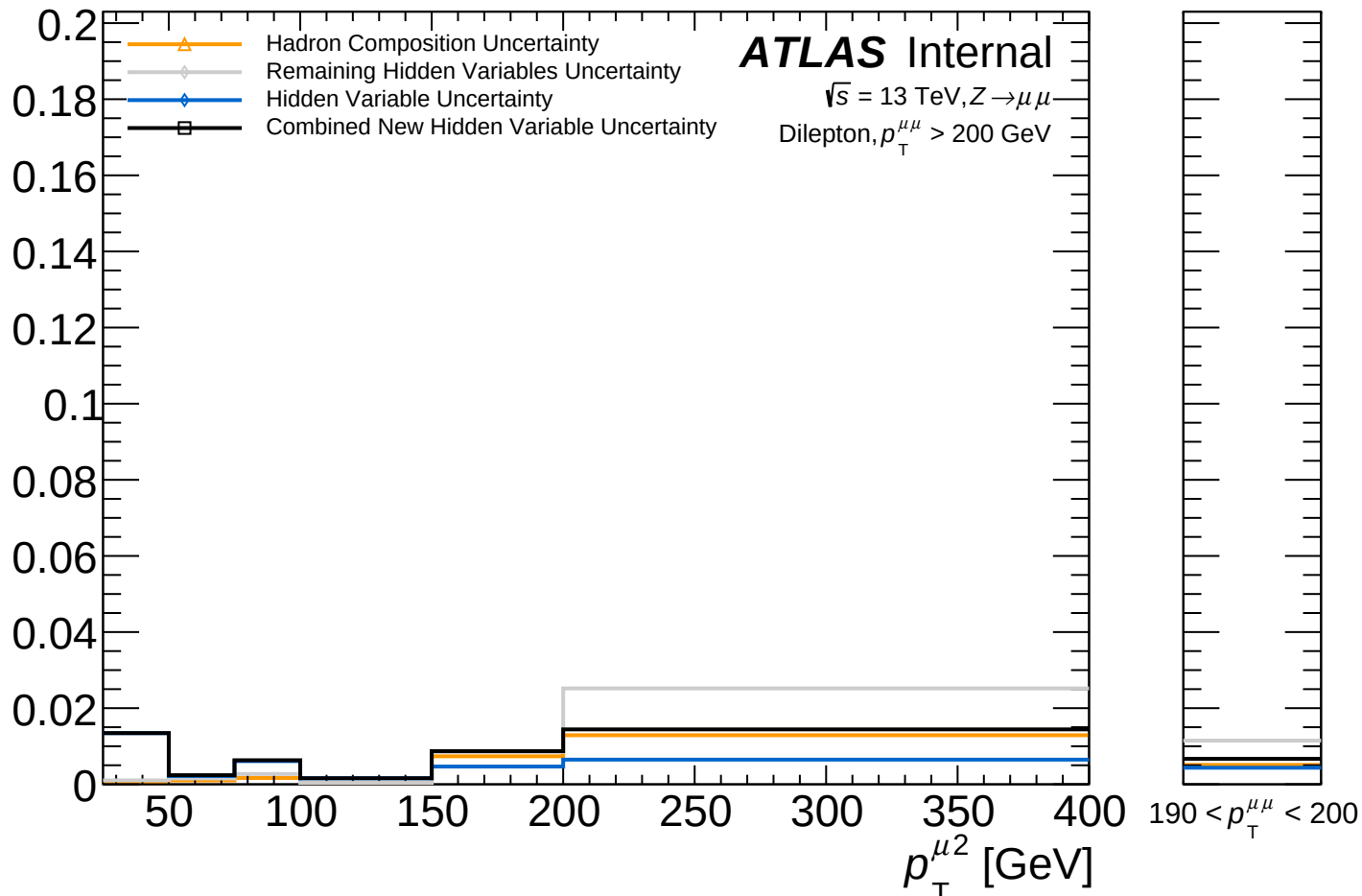
$190 < p_{\text{T}}^{\mu\mu} < 200$

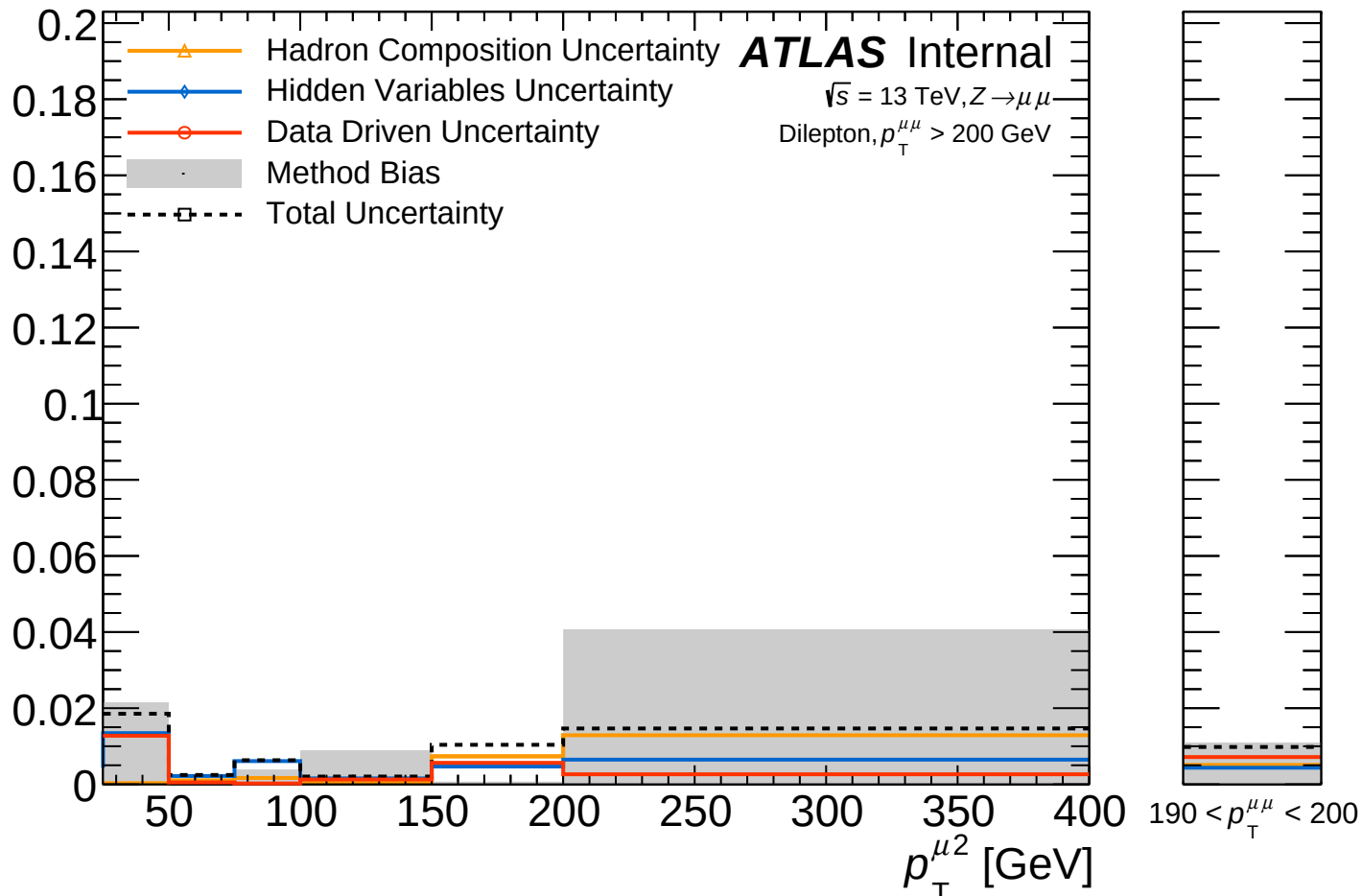


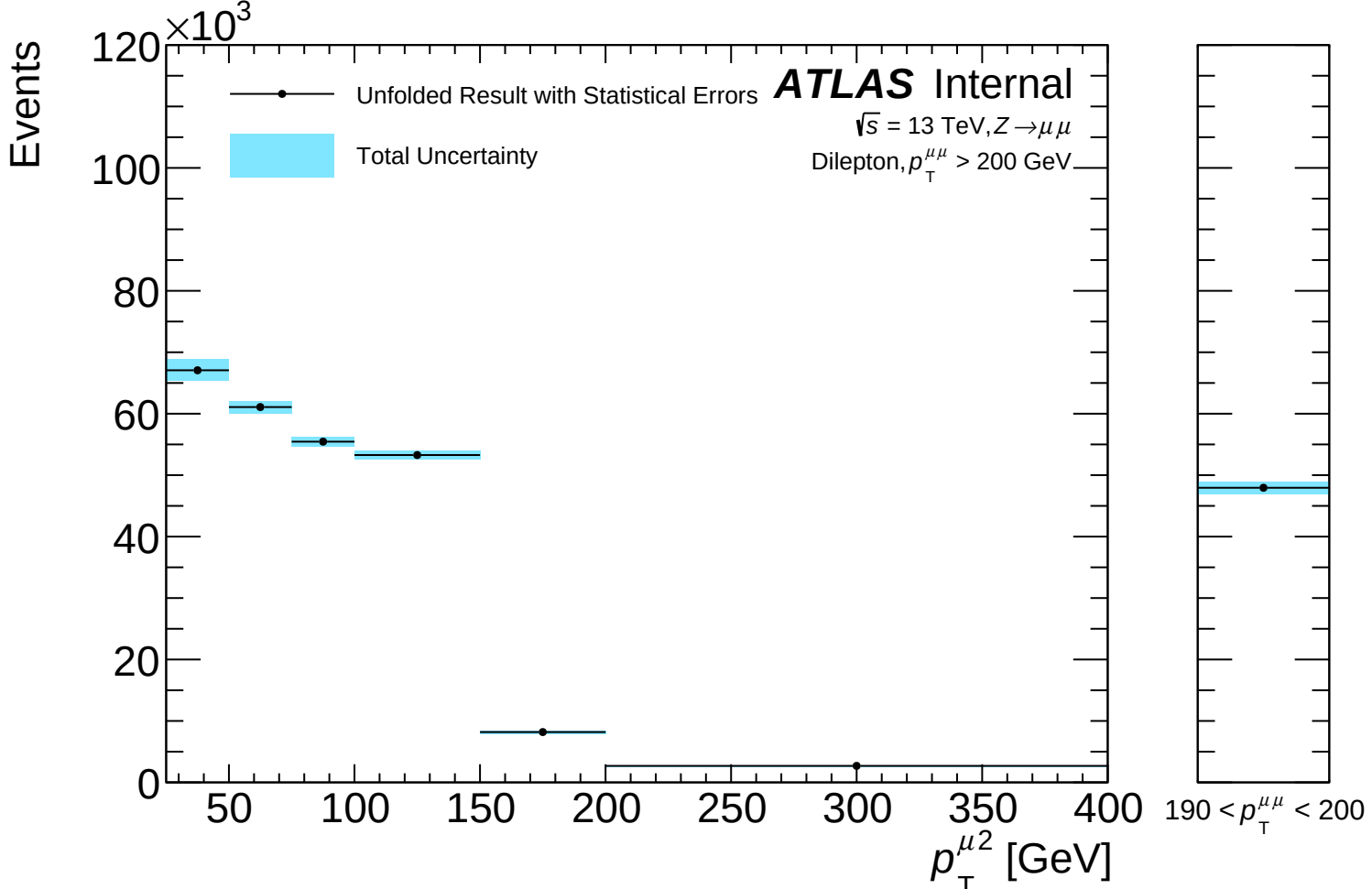
ATLAS Internal

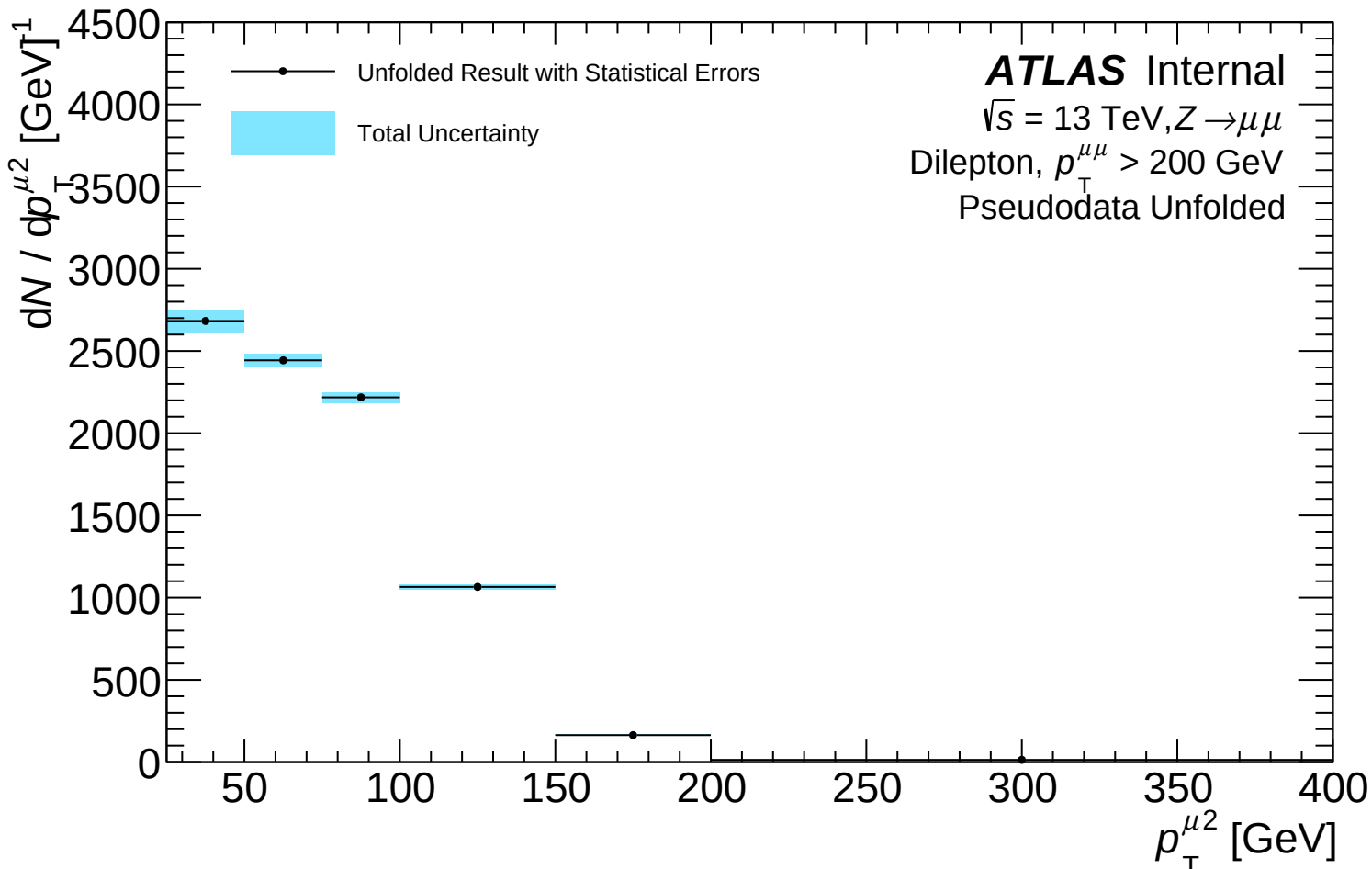
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

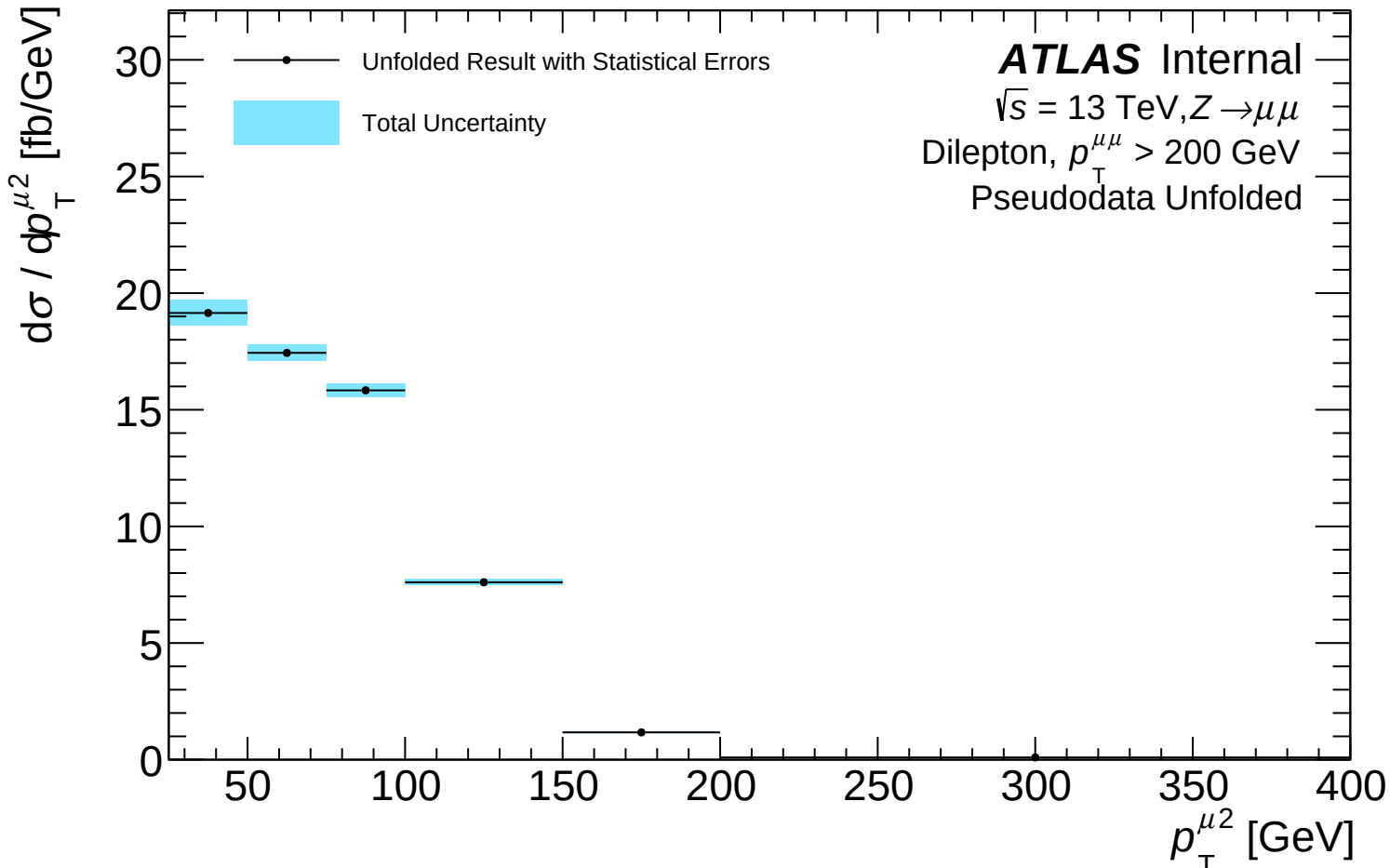
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$











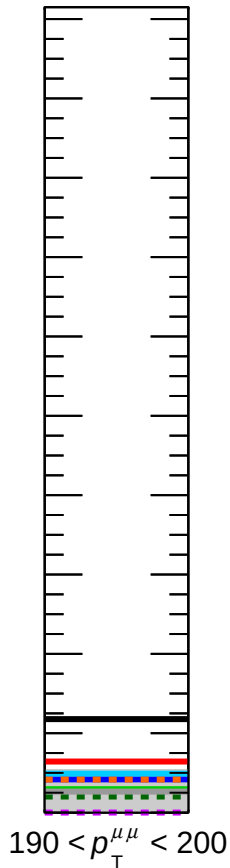
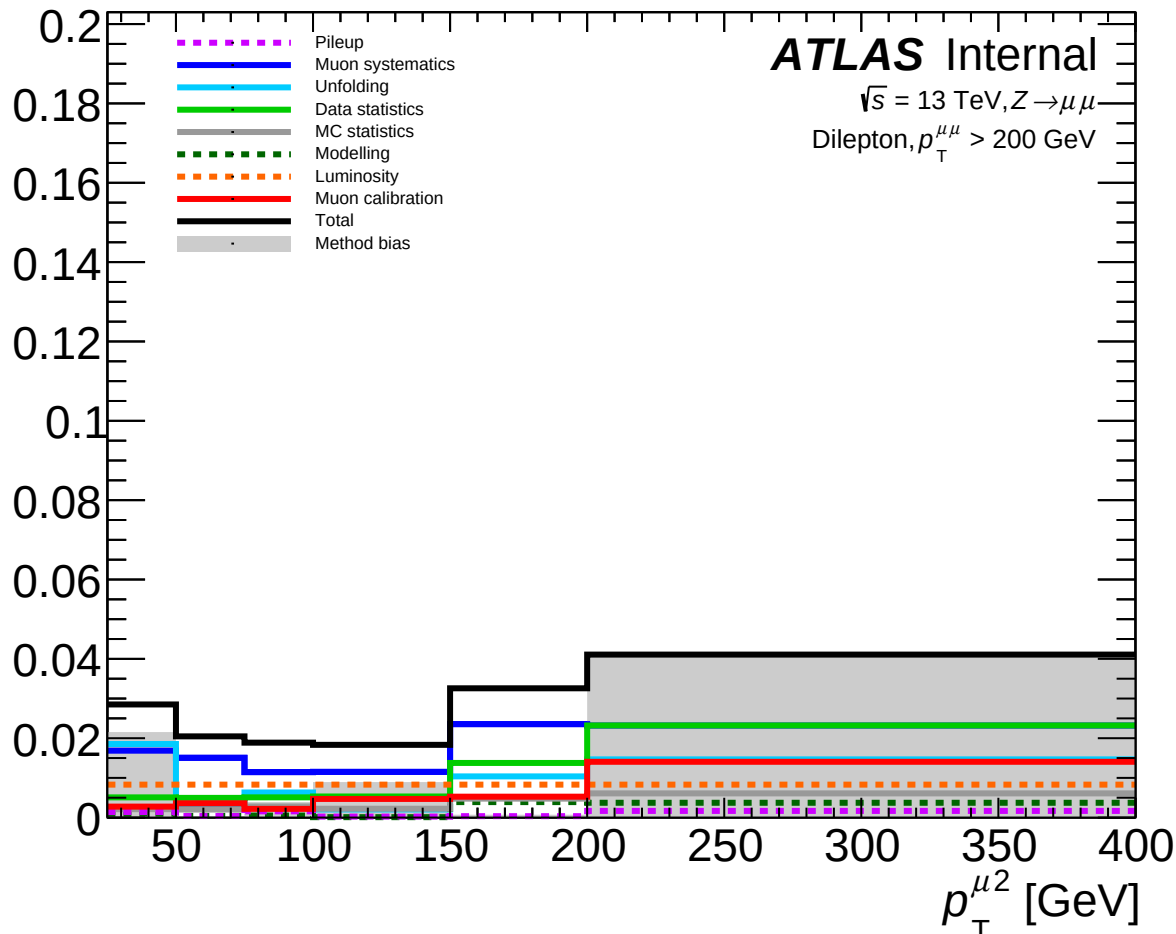
Fractional uncertainty

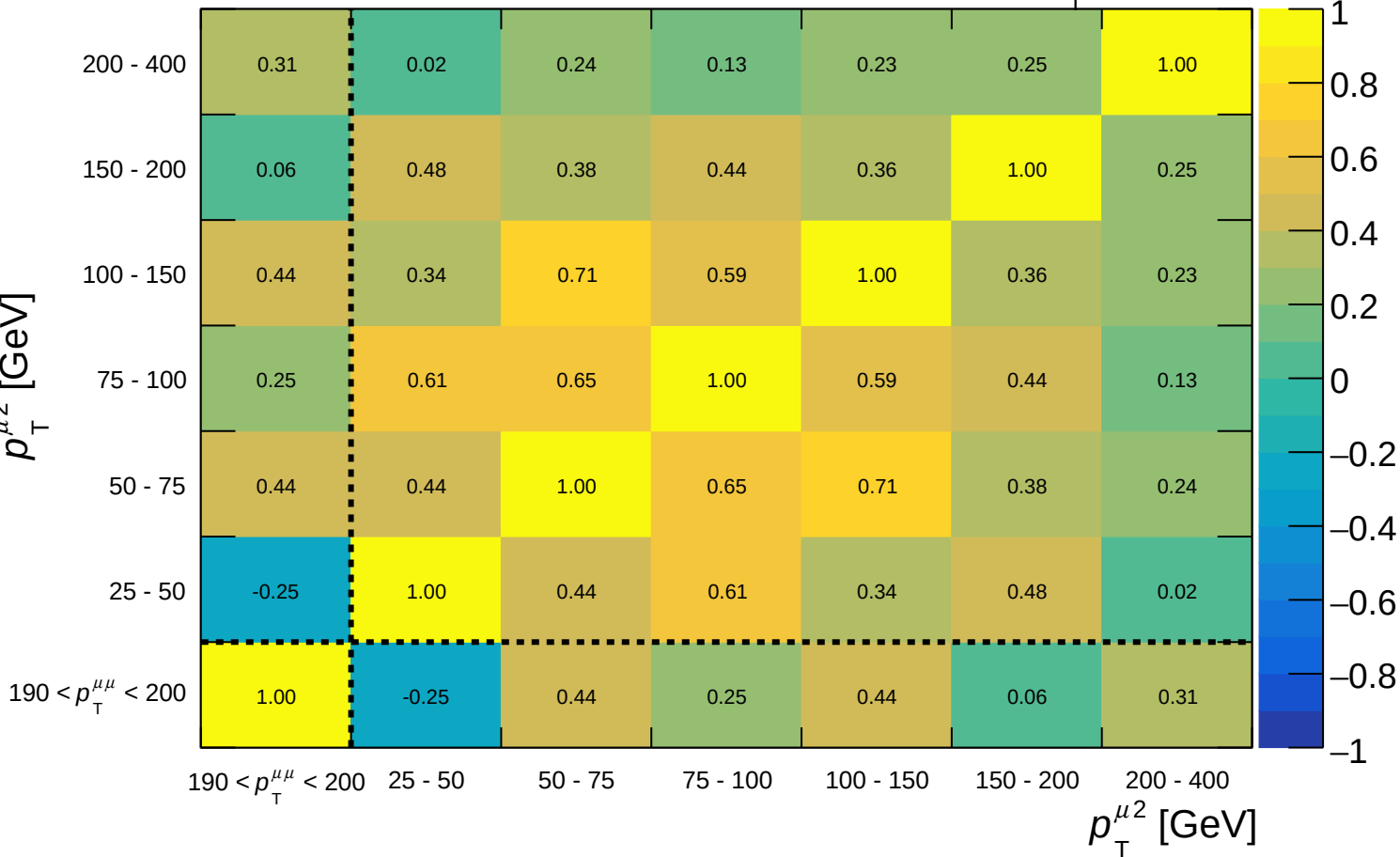
ATLAS Internal

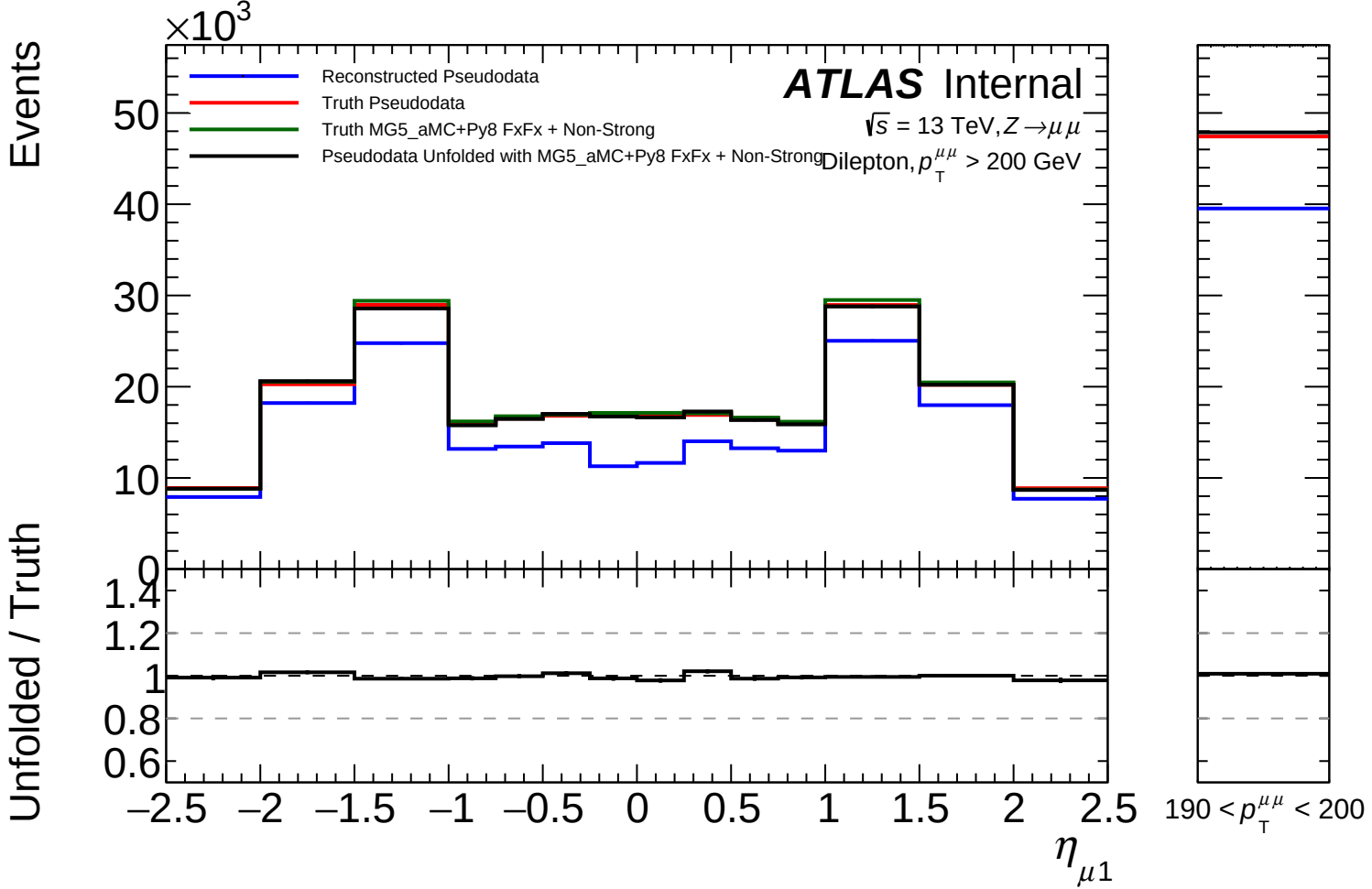
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

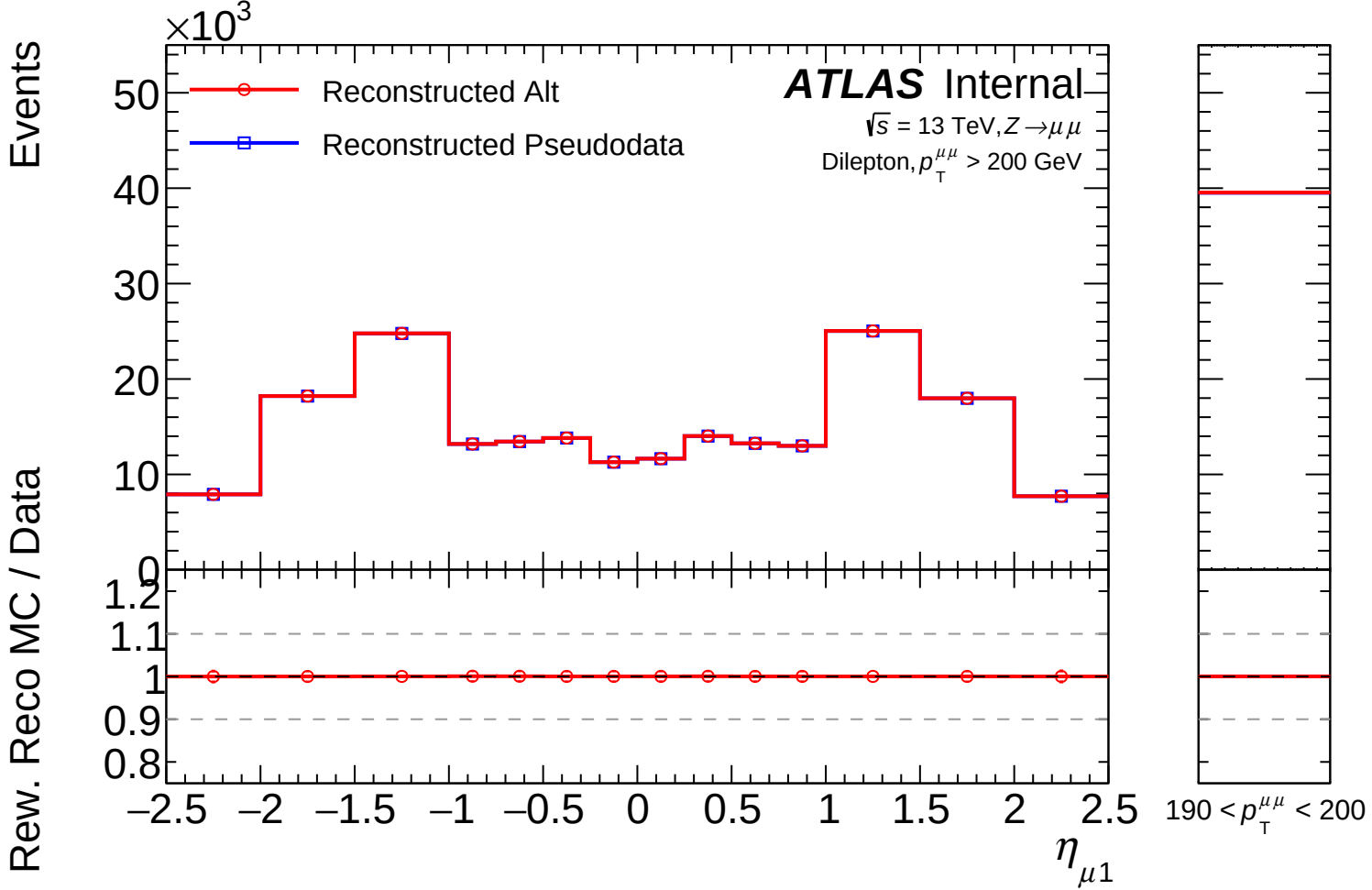
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

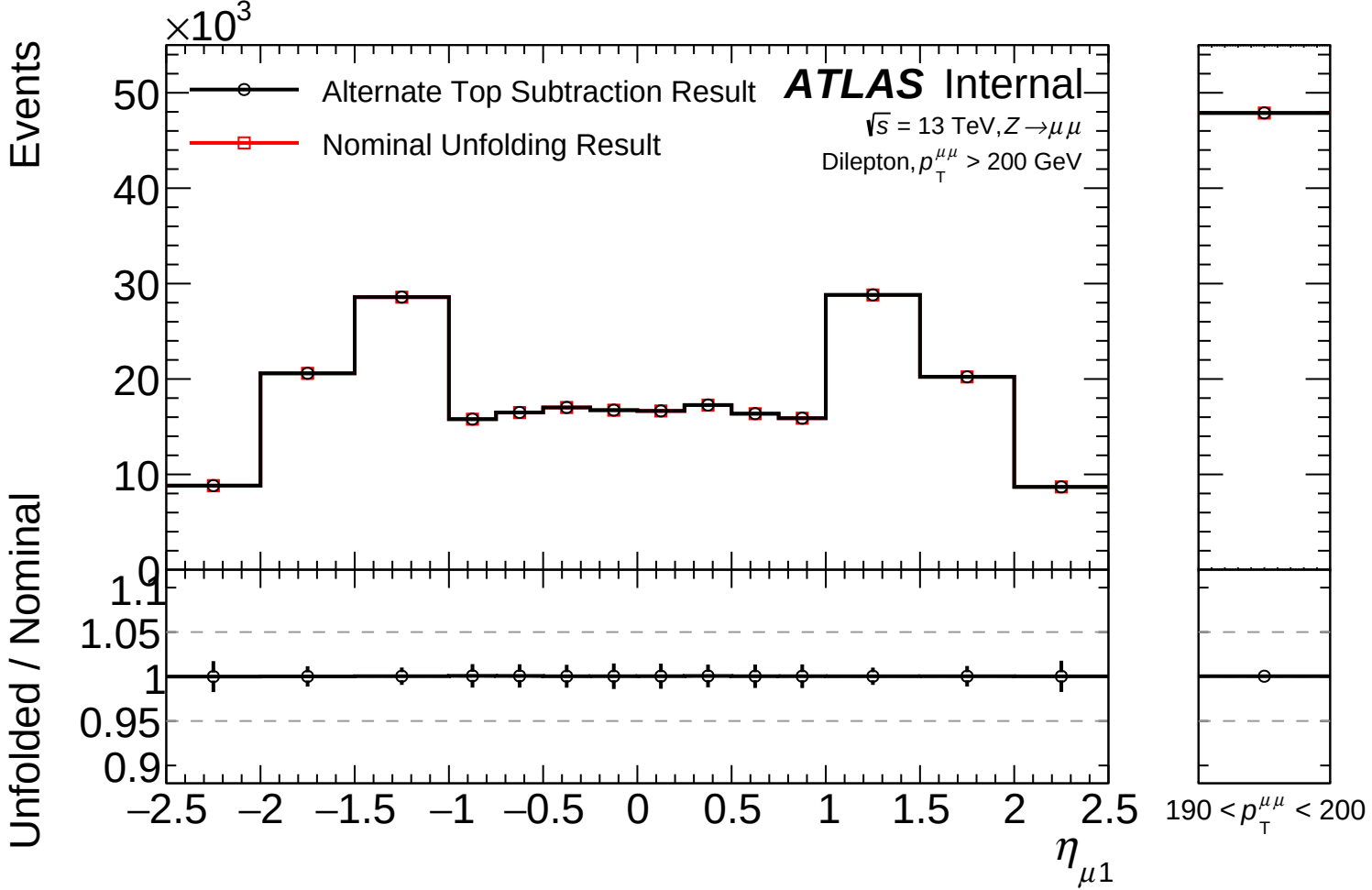
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Muon calibration
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $p_{\text{T}}^{\mu\mu} [\text{GeV}]$ 







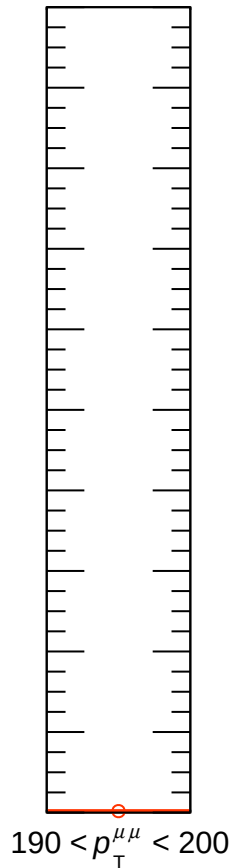
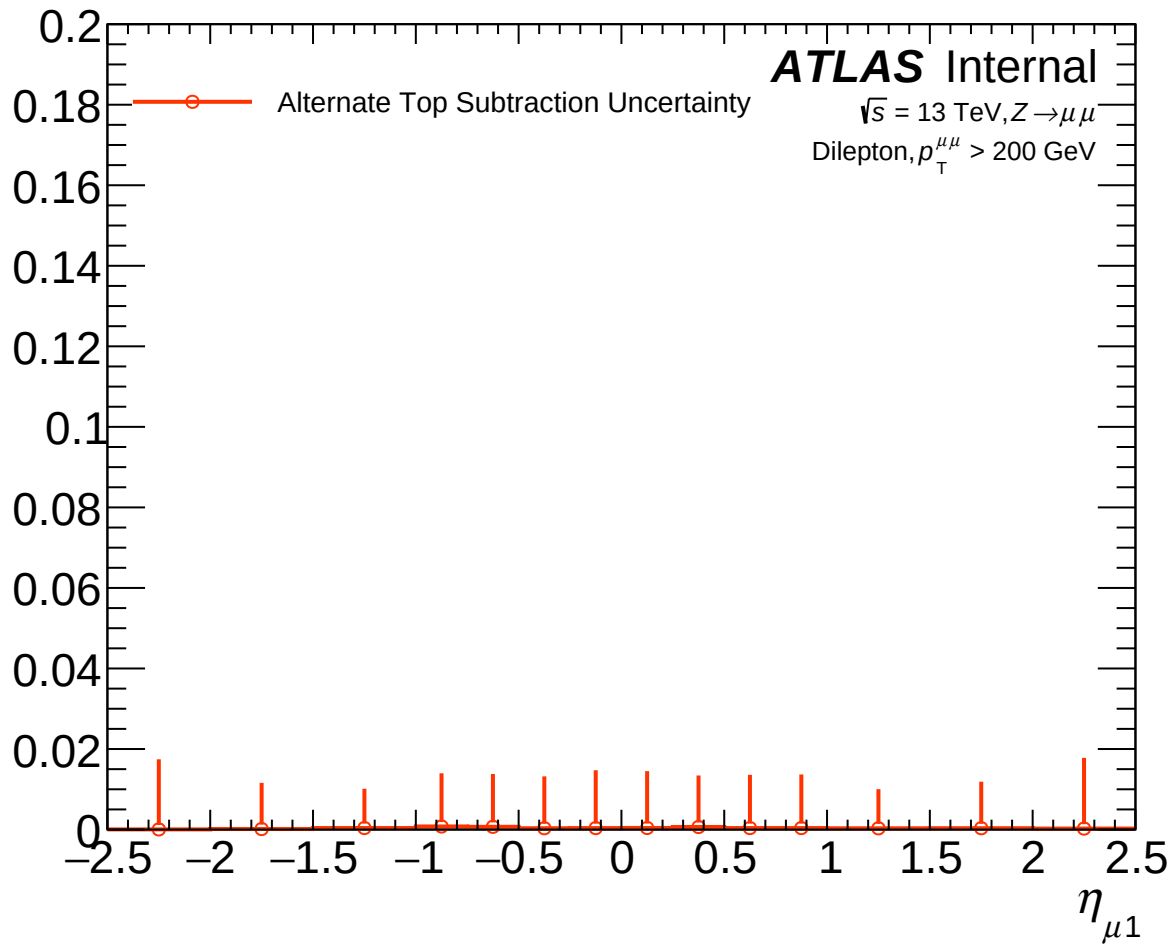
Fractional systematic impact

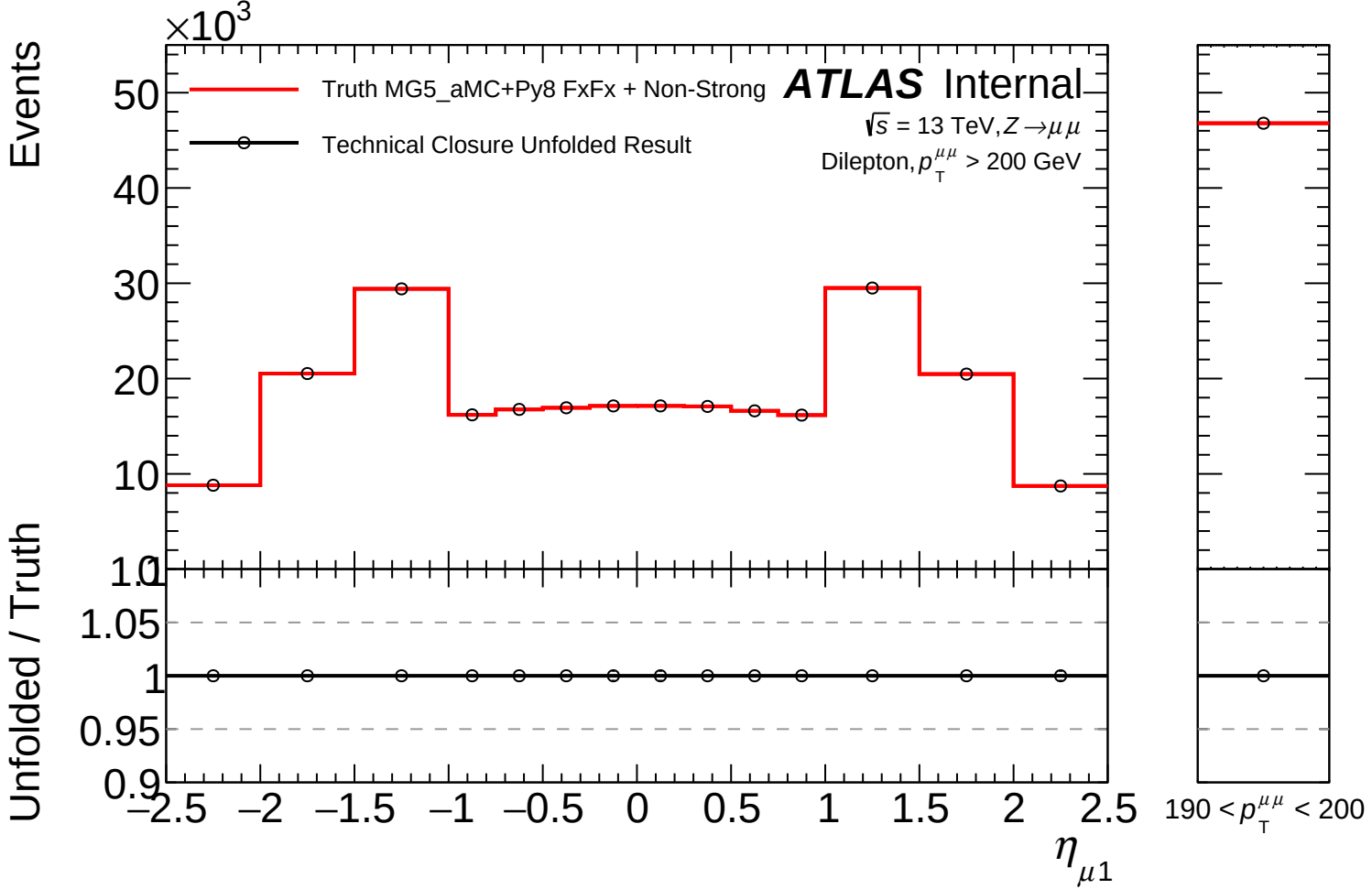
ATLAS Internal

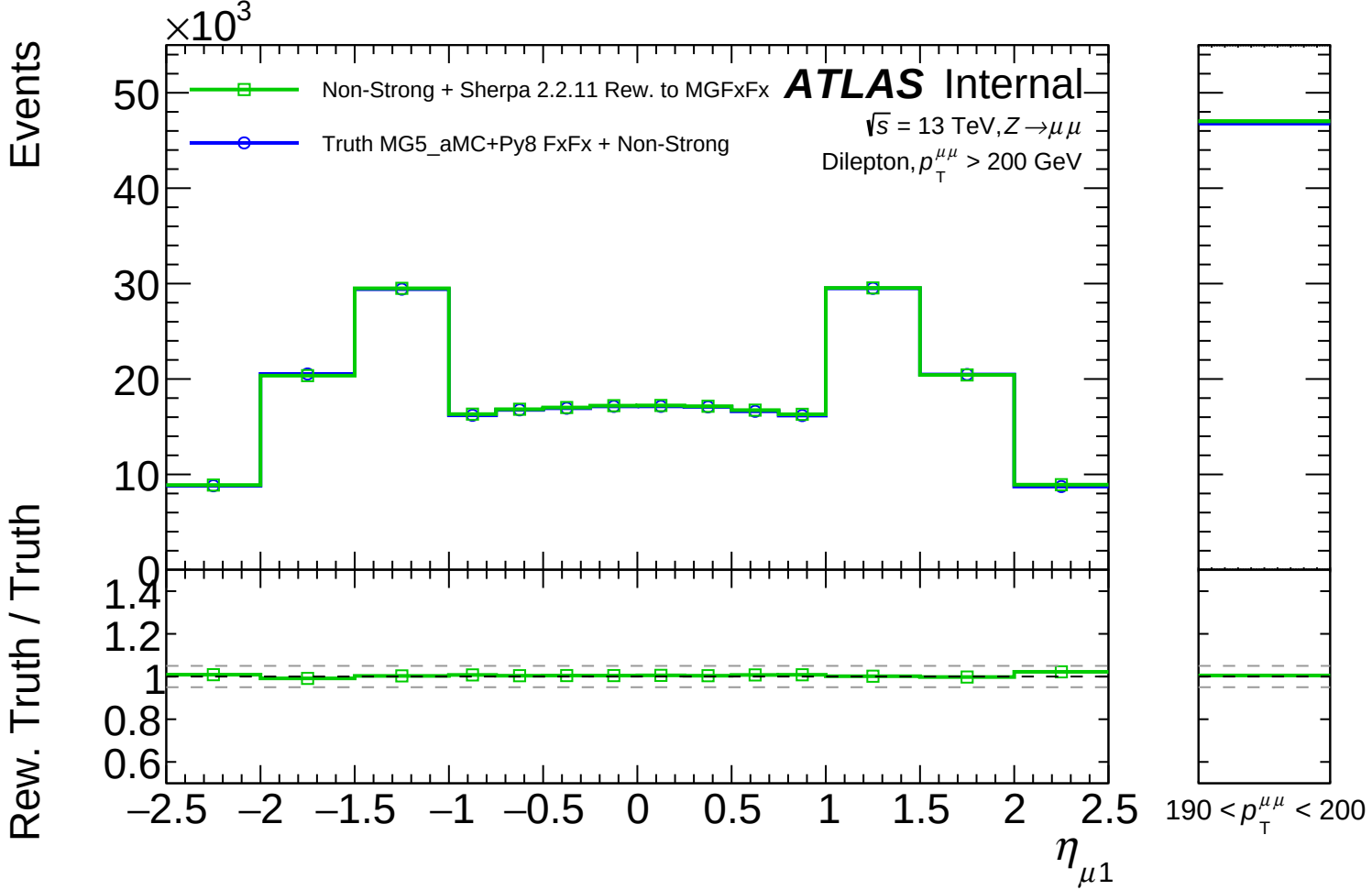
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

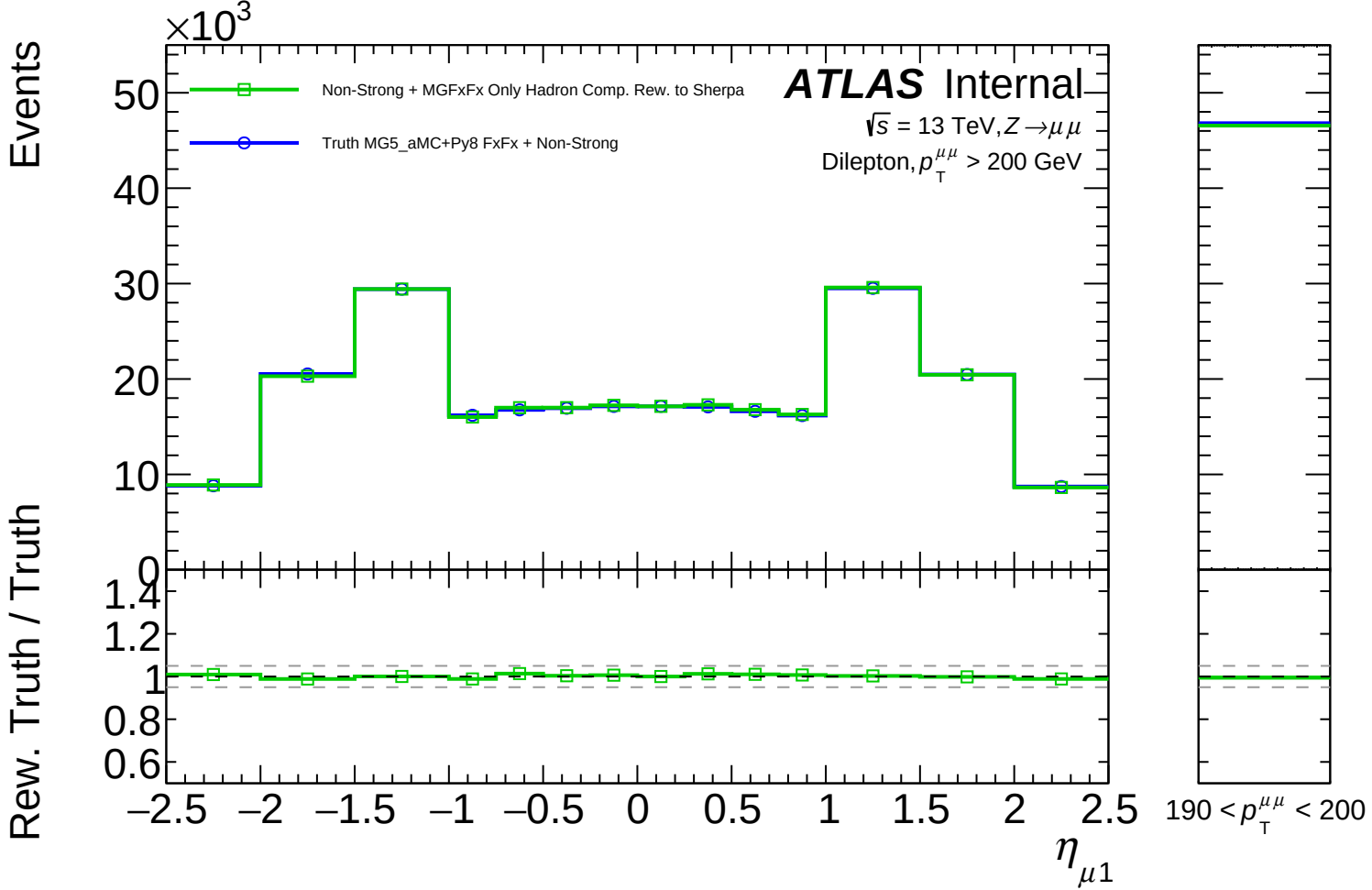
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

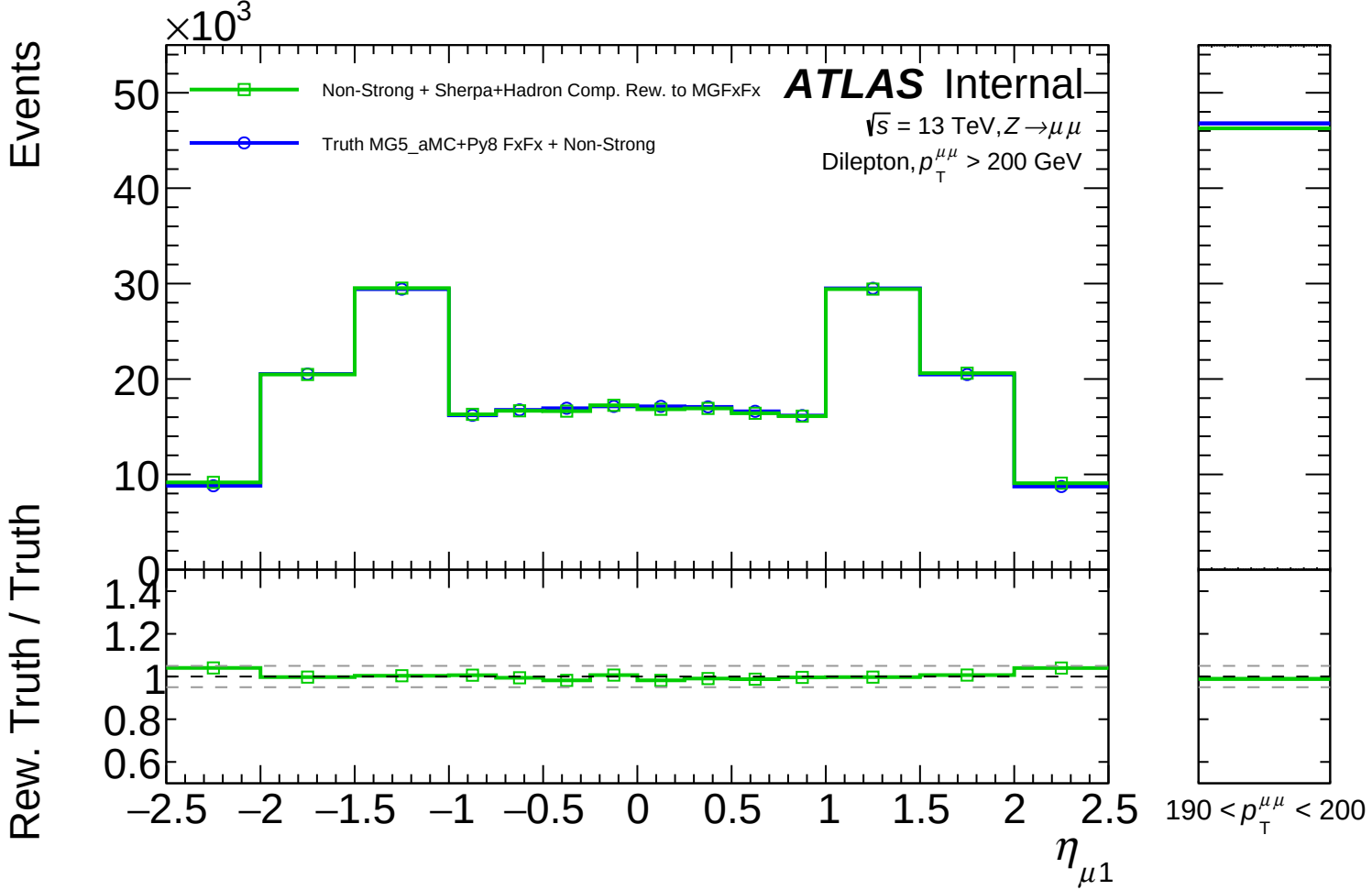
Alternate Top Subtraction Uncertainty

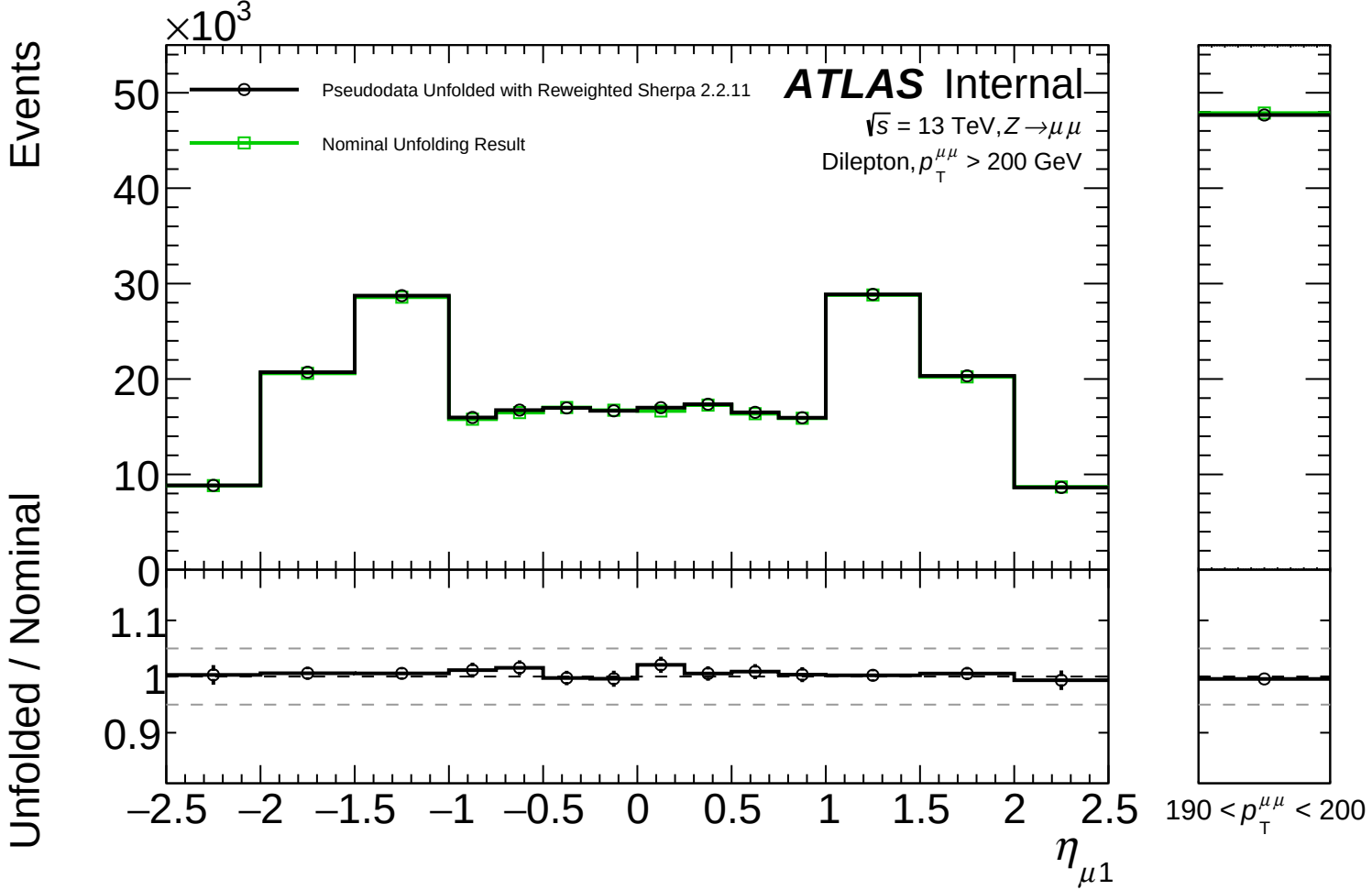


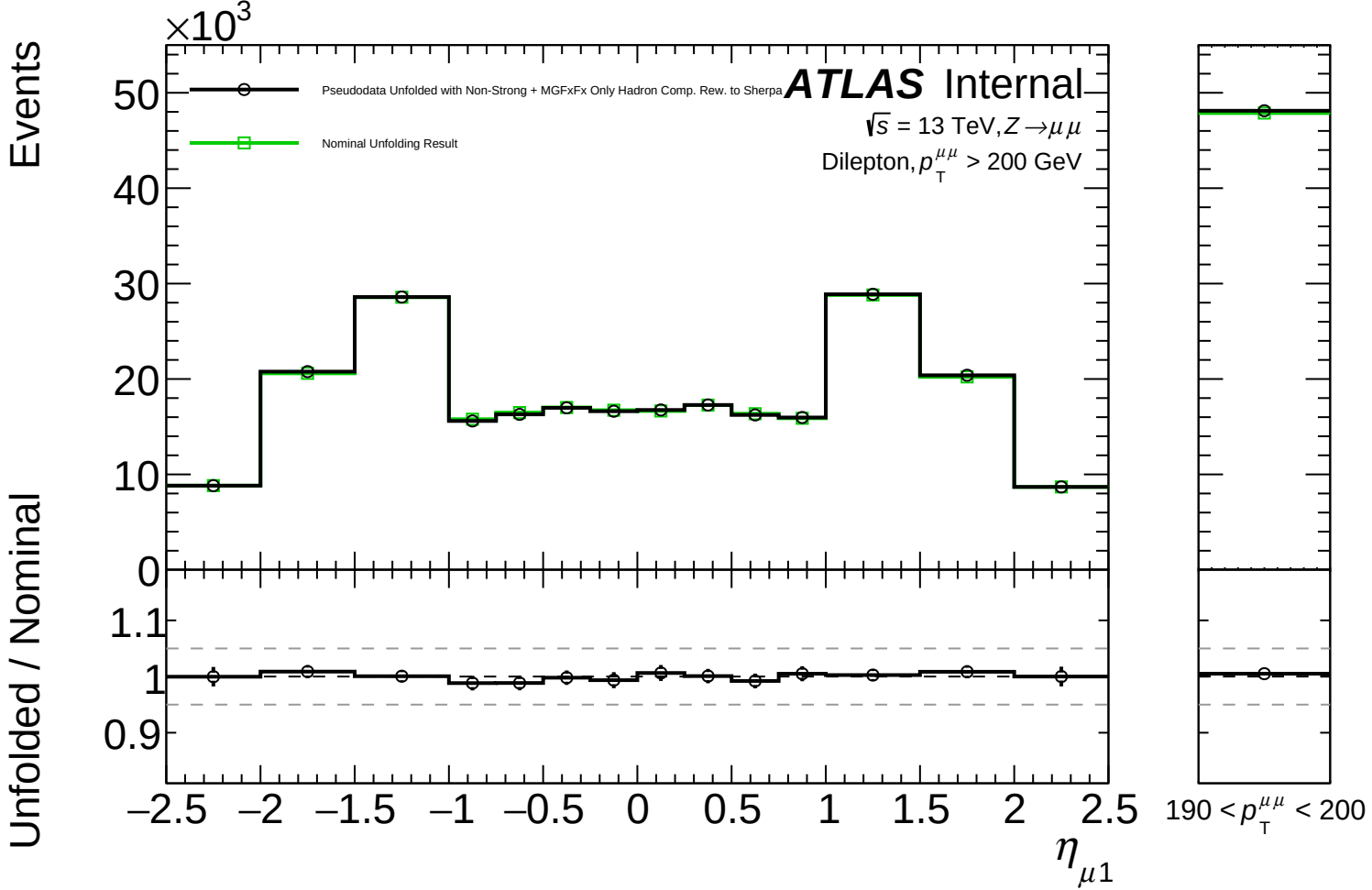


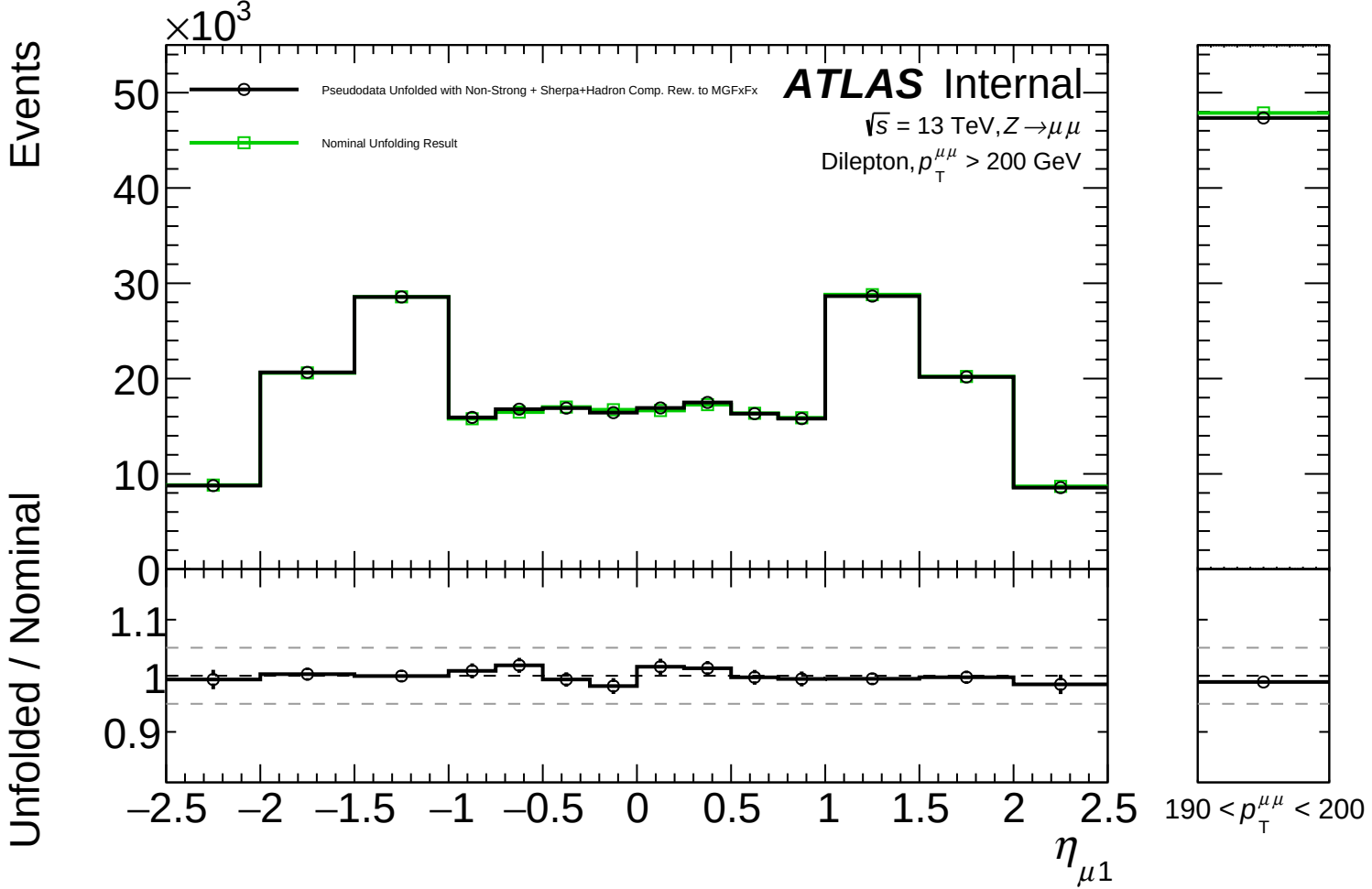


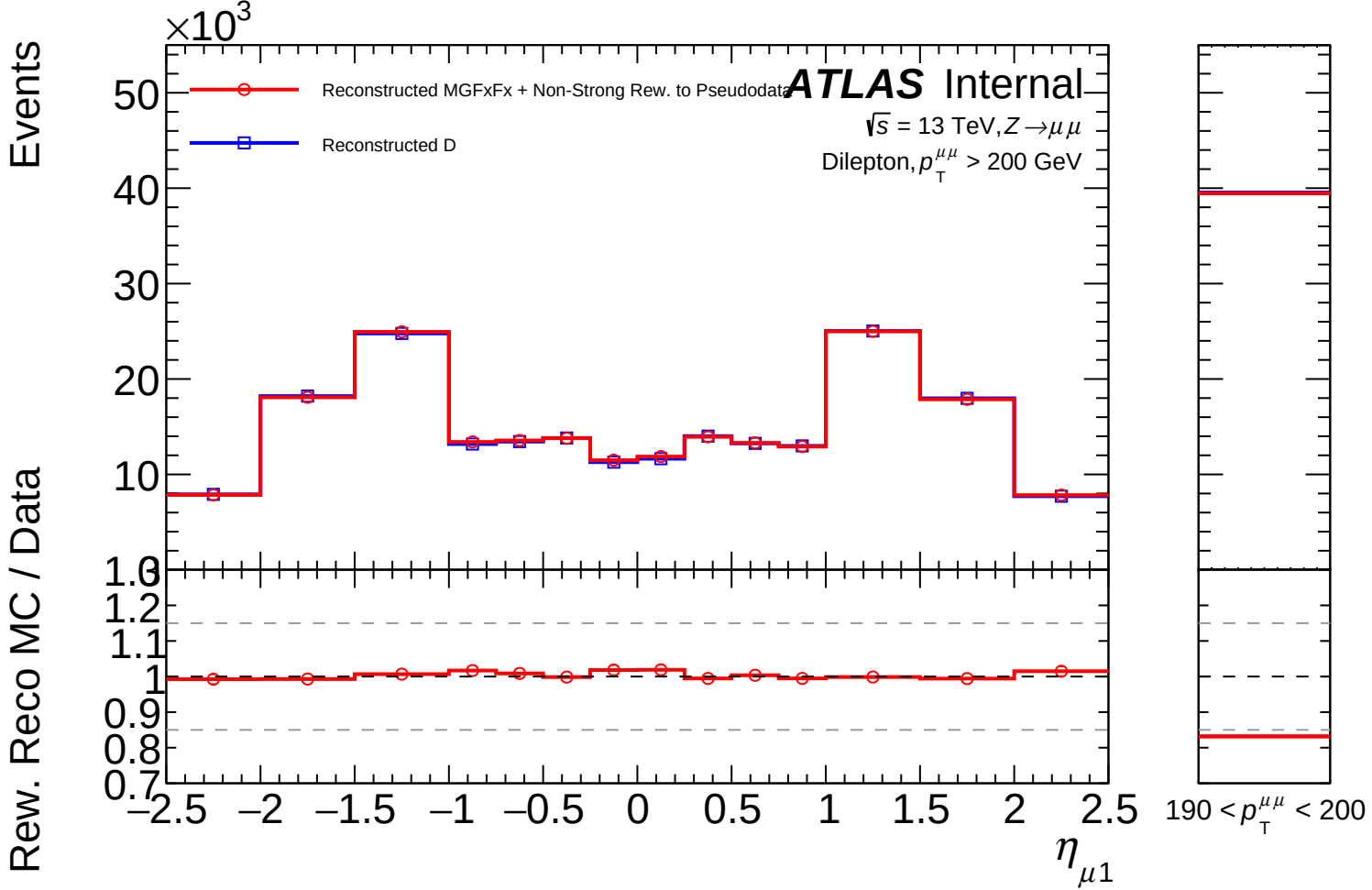


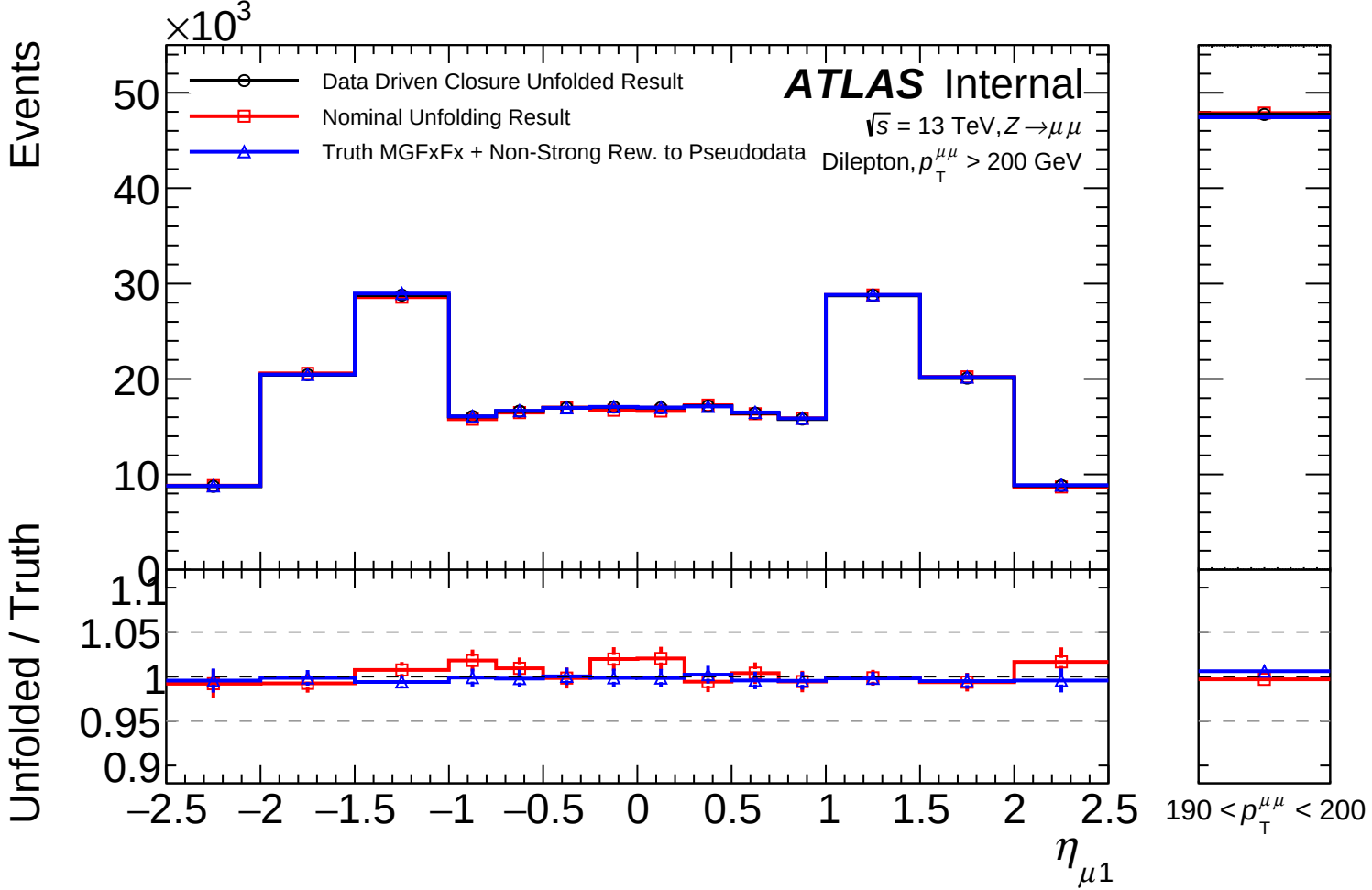




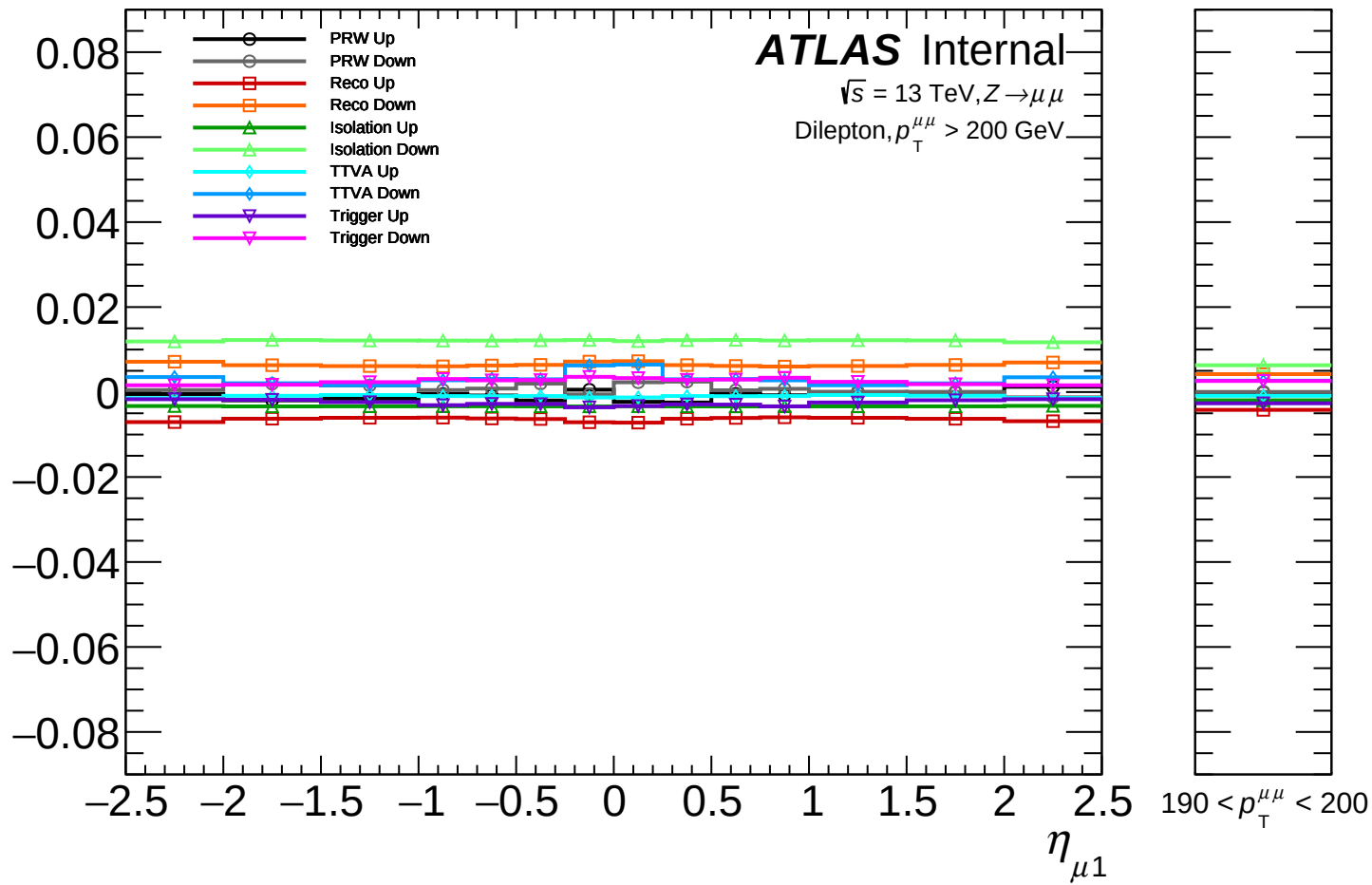








Fractional systematic impact



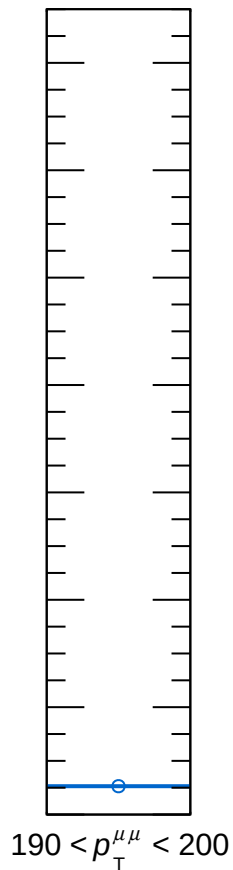
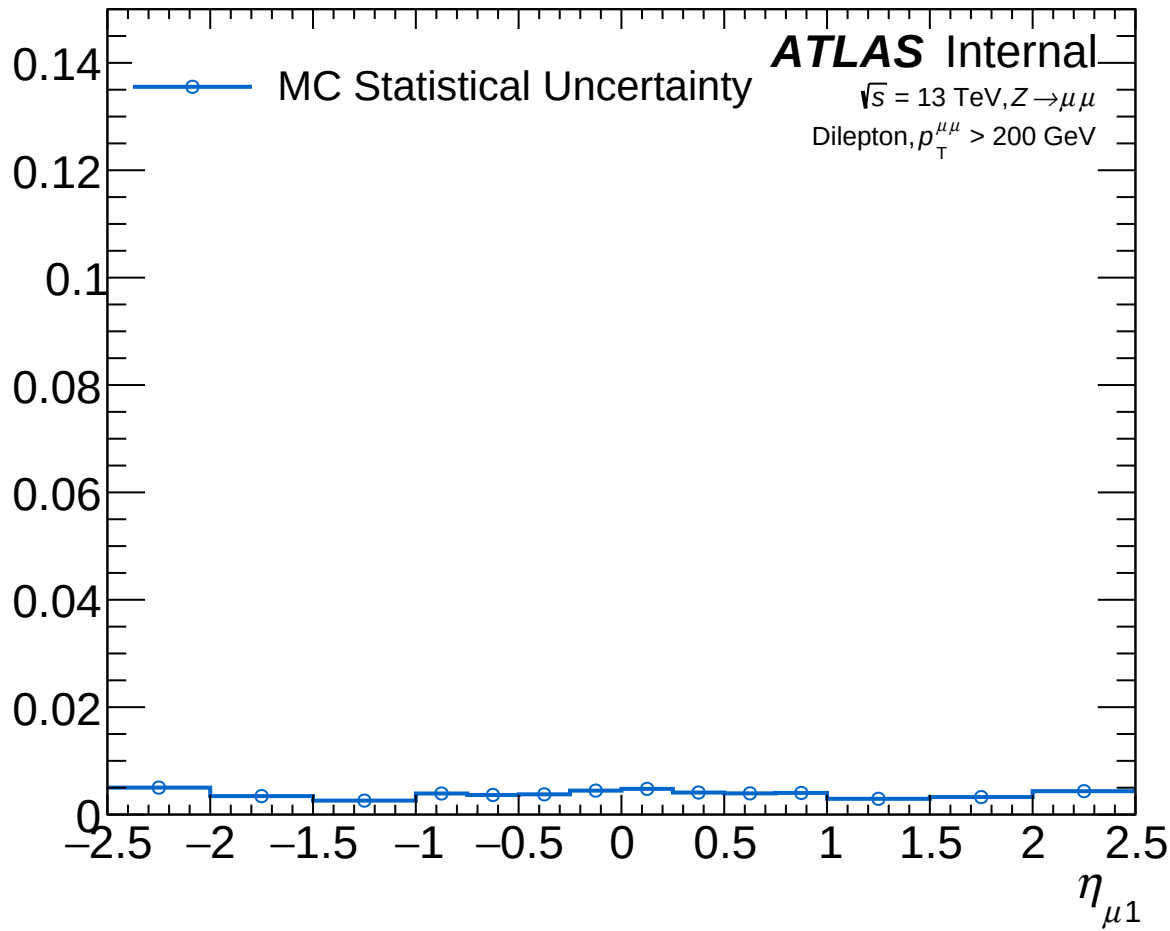
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



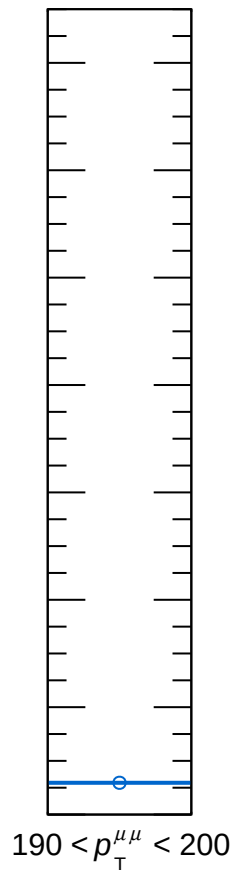
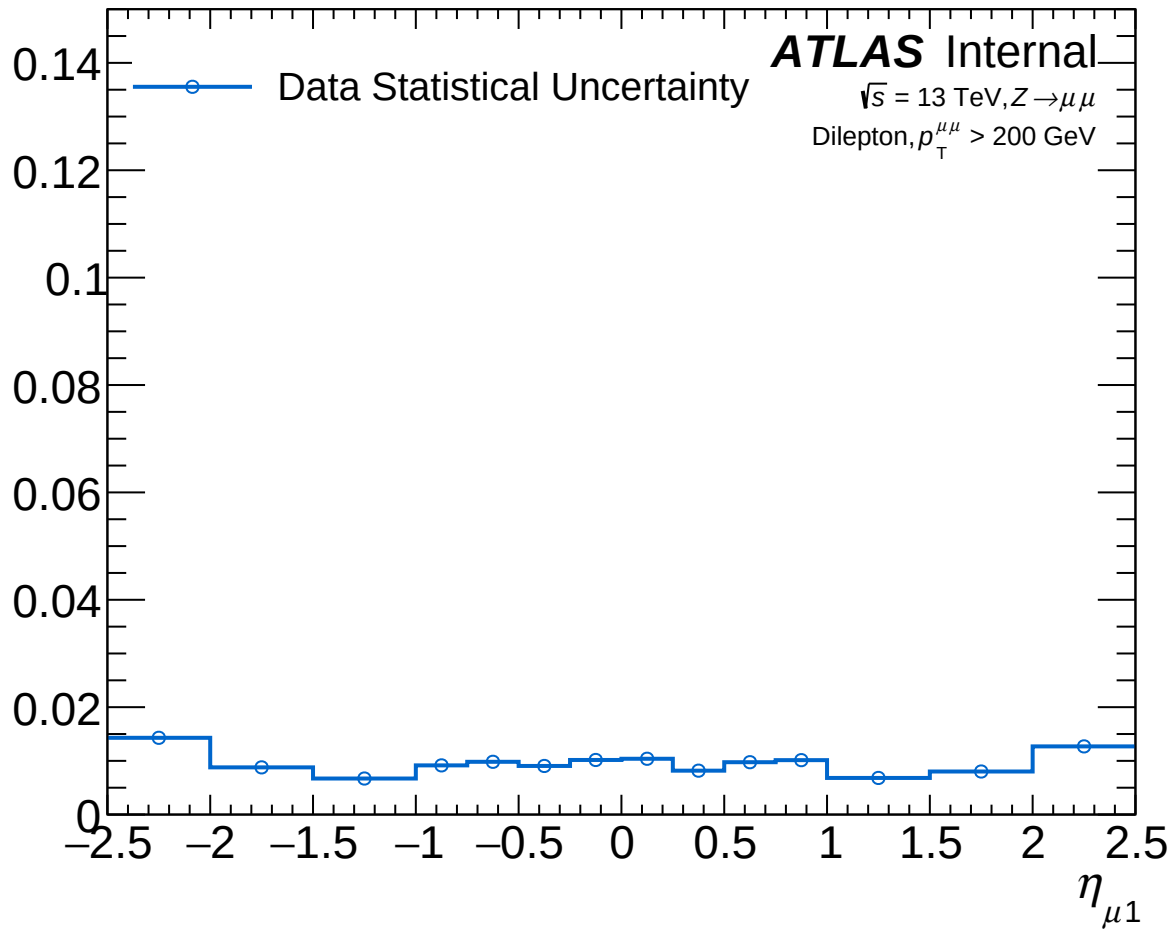
Fractional systematic impact

ATLAS Internal

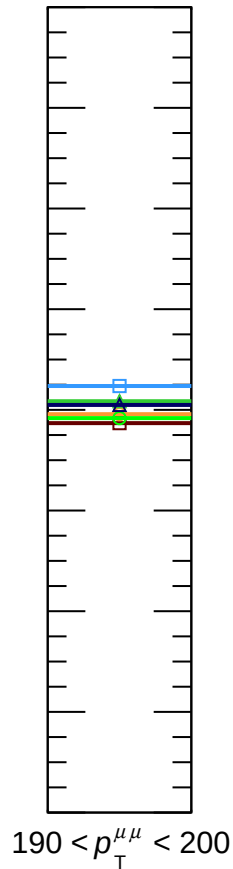
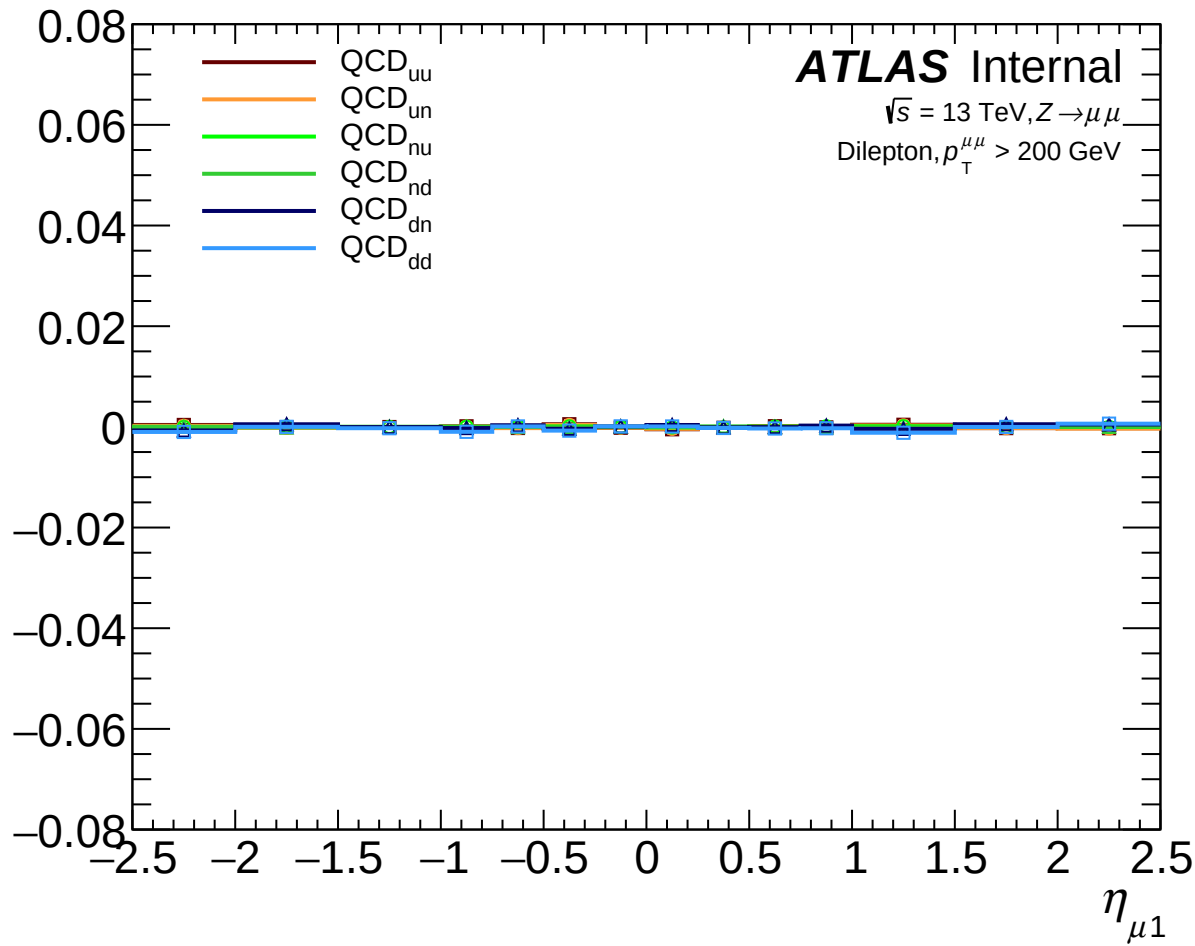
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

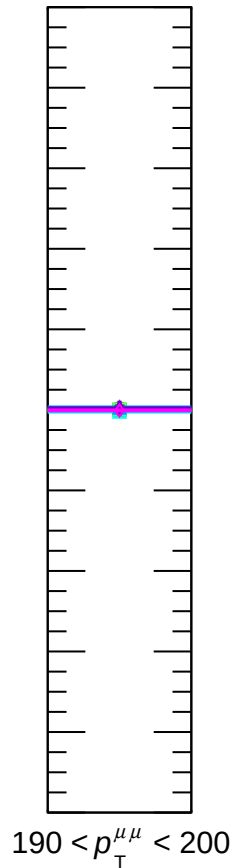
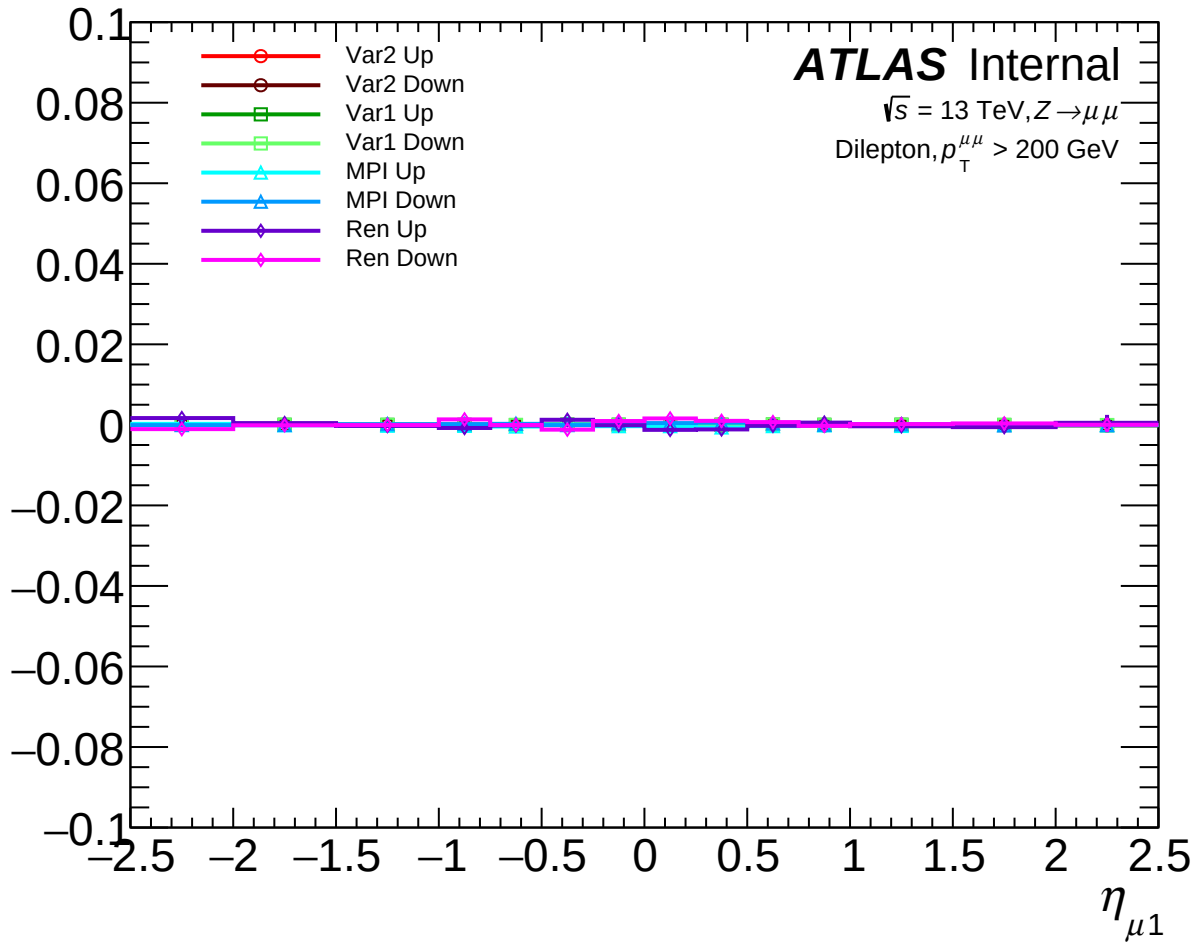
Data Statistical Uncertainty



Fractional systematic impact



Fractional systematic impact



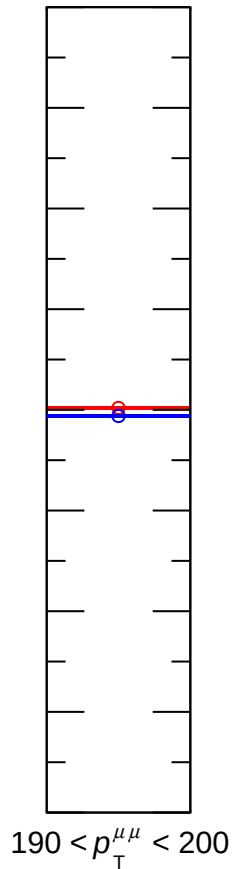
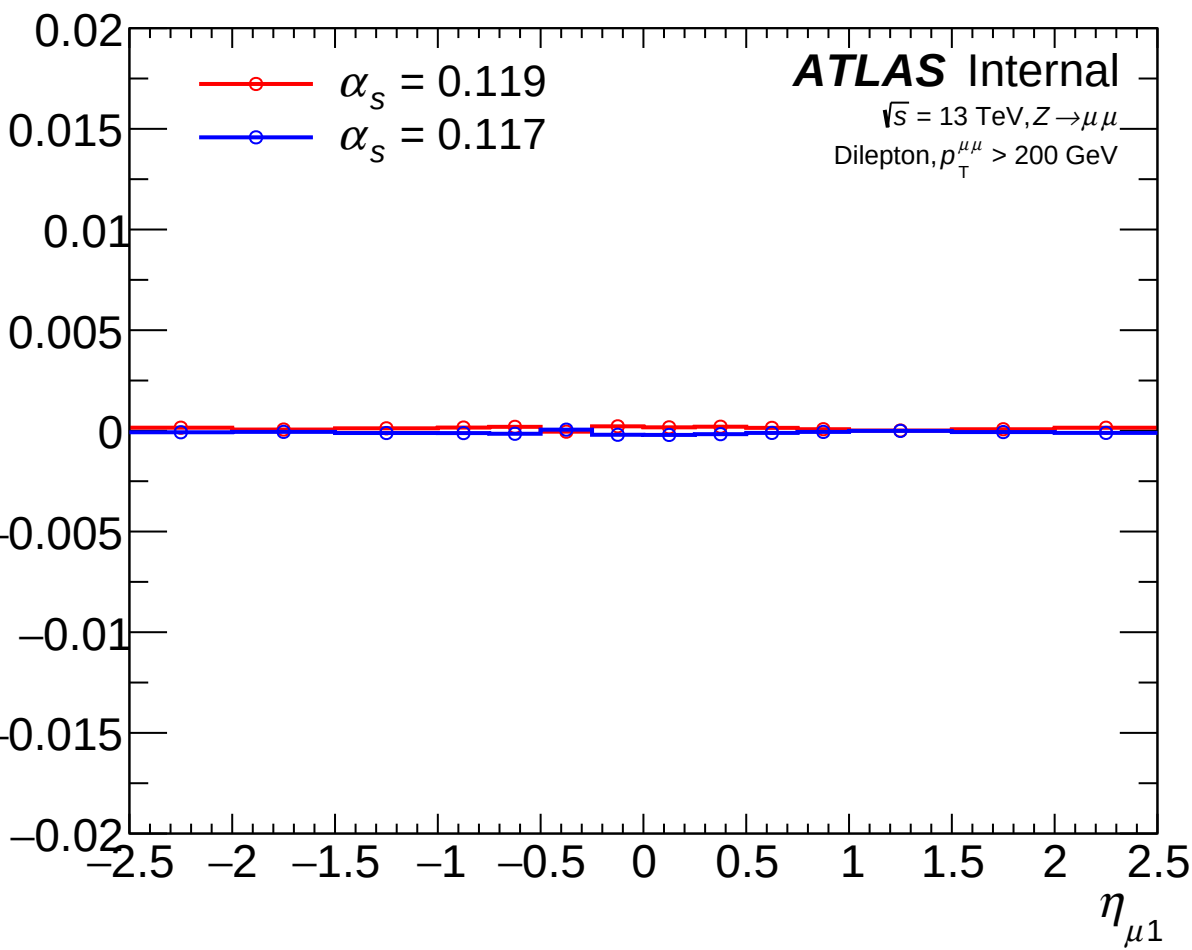
Fractional systematic impact

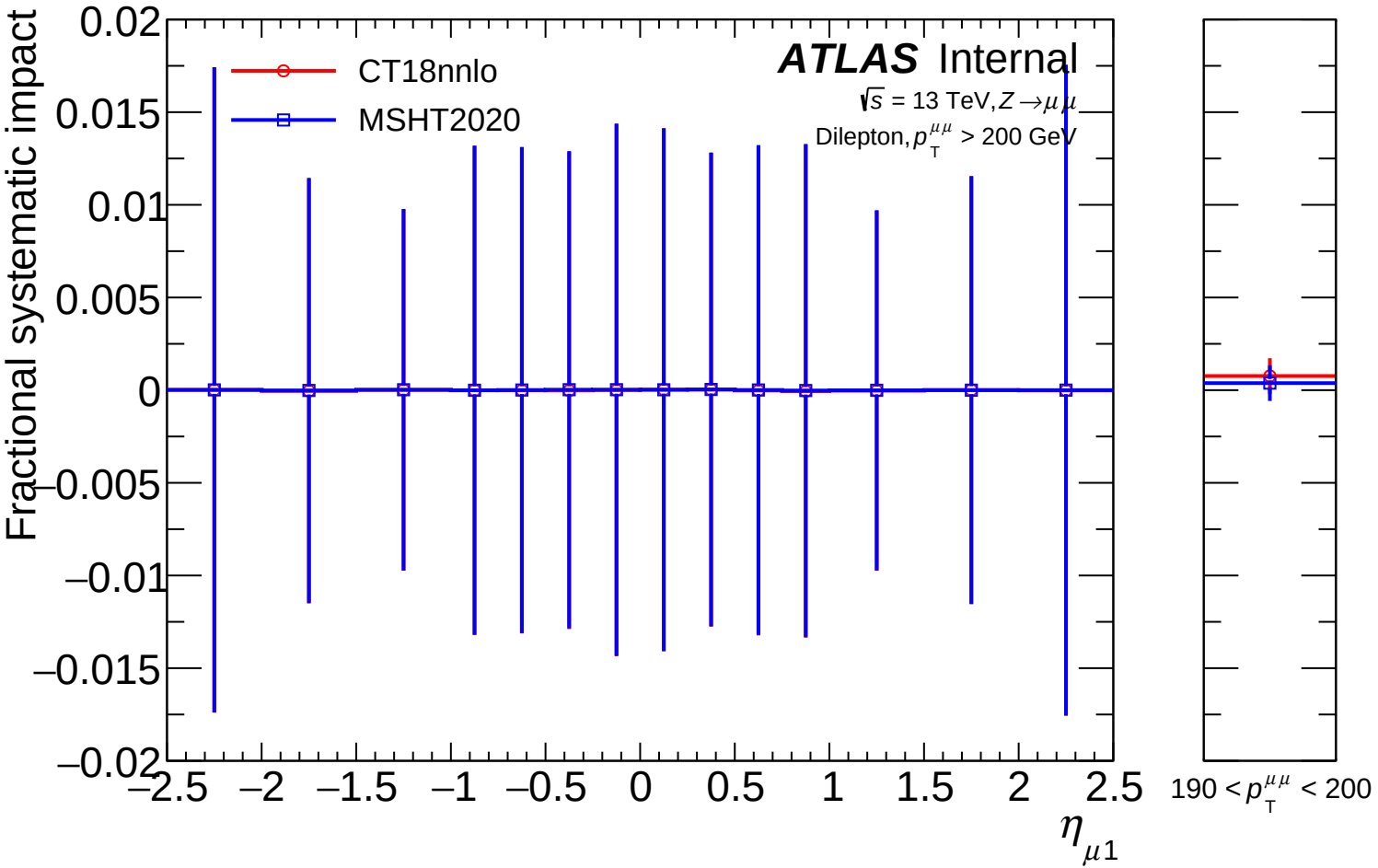
$\alpha_s = 0.119$
 $\alpha_s = 0.117$

ATLAS Internal

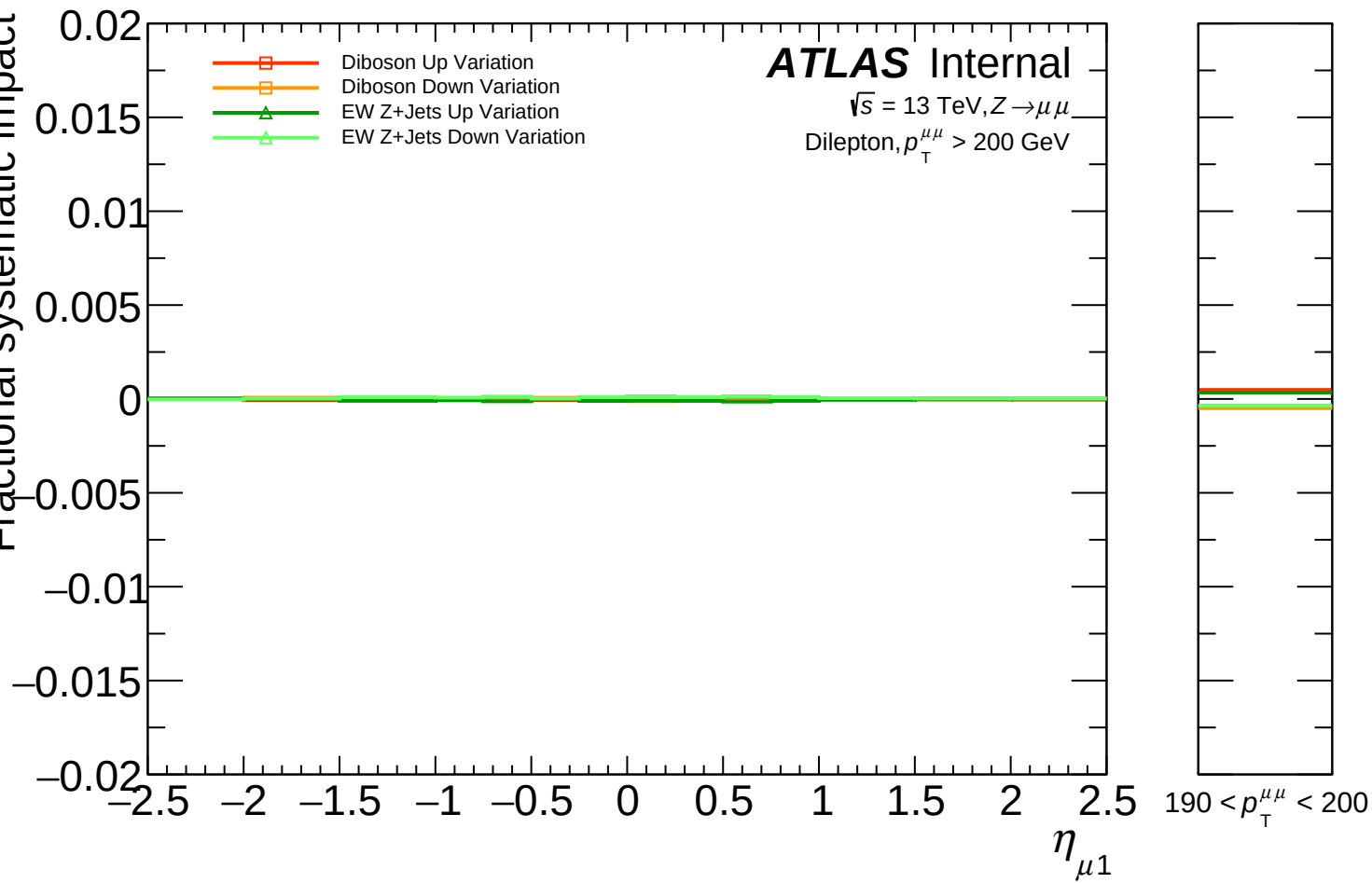
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200$ GeV





Fractional systematic impact



ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty

0.06

0.05

0.04

0.03

0.02

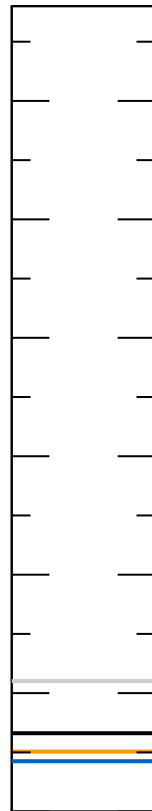
0.01

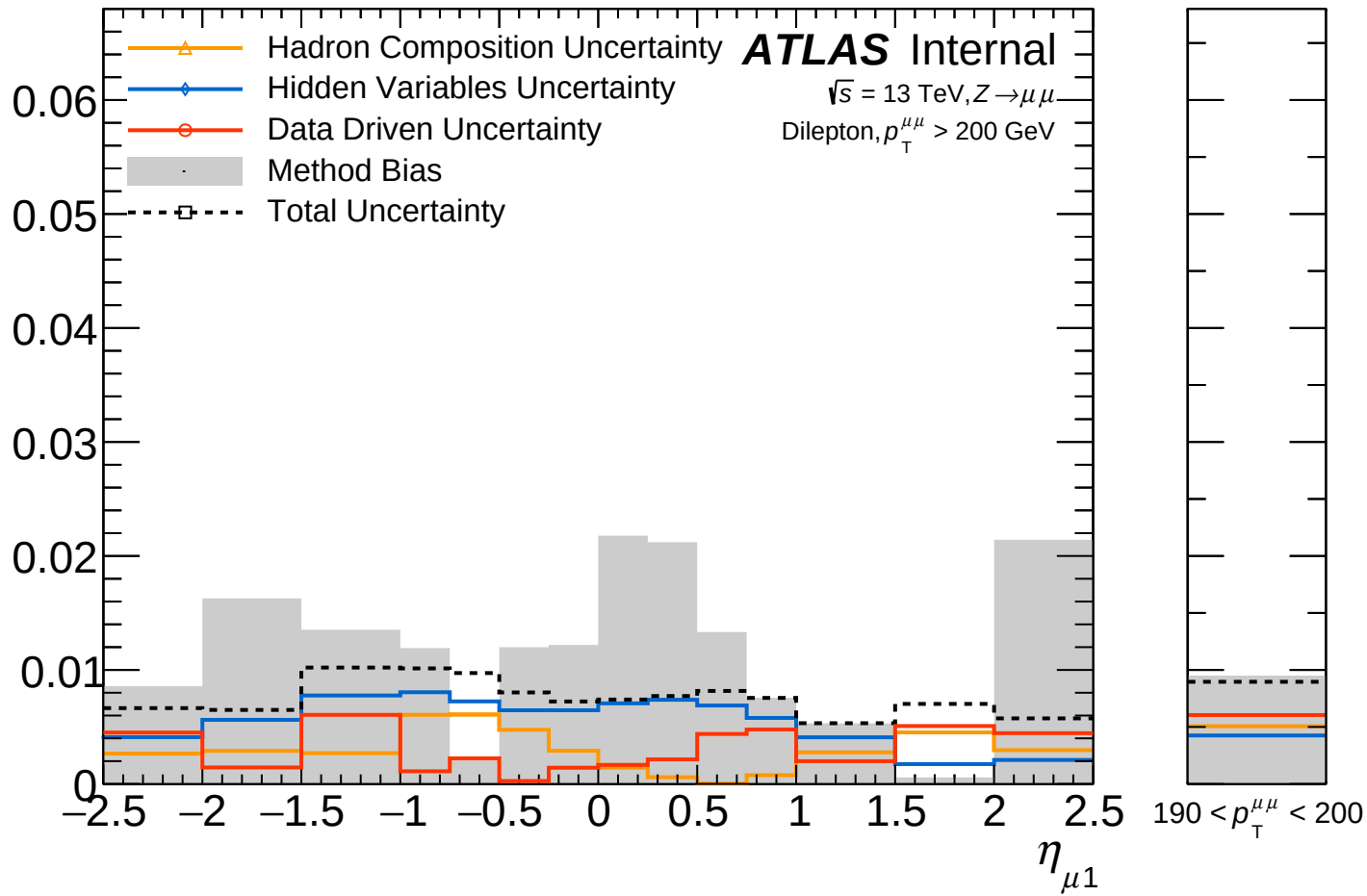
0

-2.5 -2 -1.5 -1 -0.5 0 0.5 1 1.5 2 2.5

$\eta_{\mu 1}$

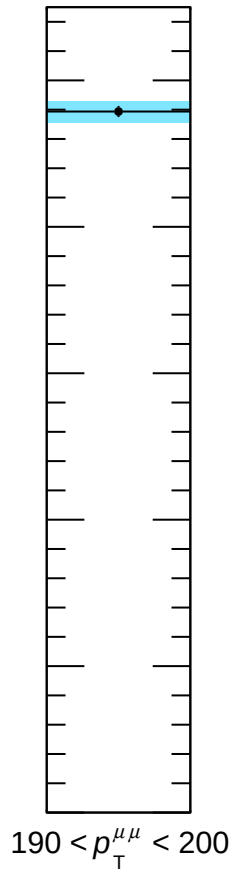
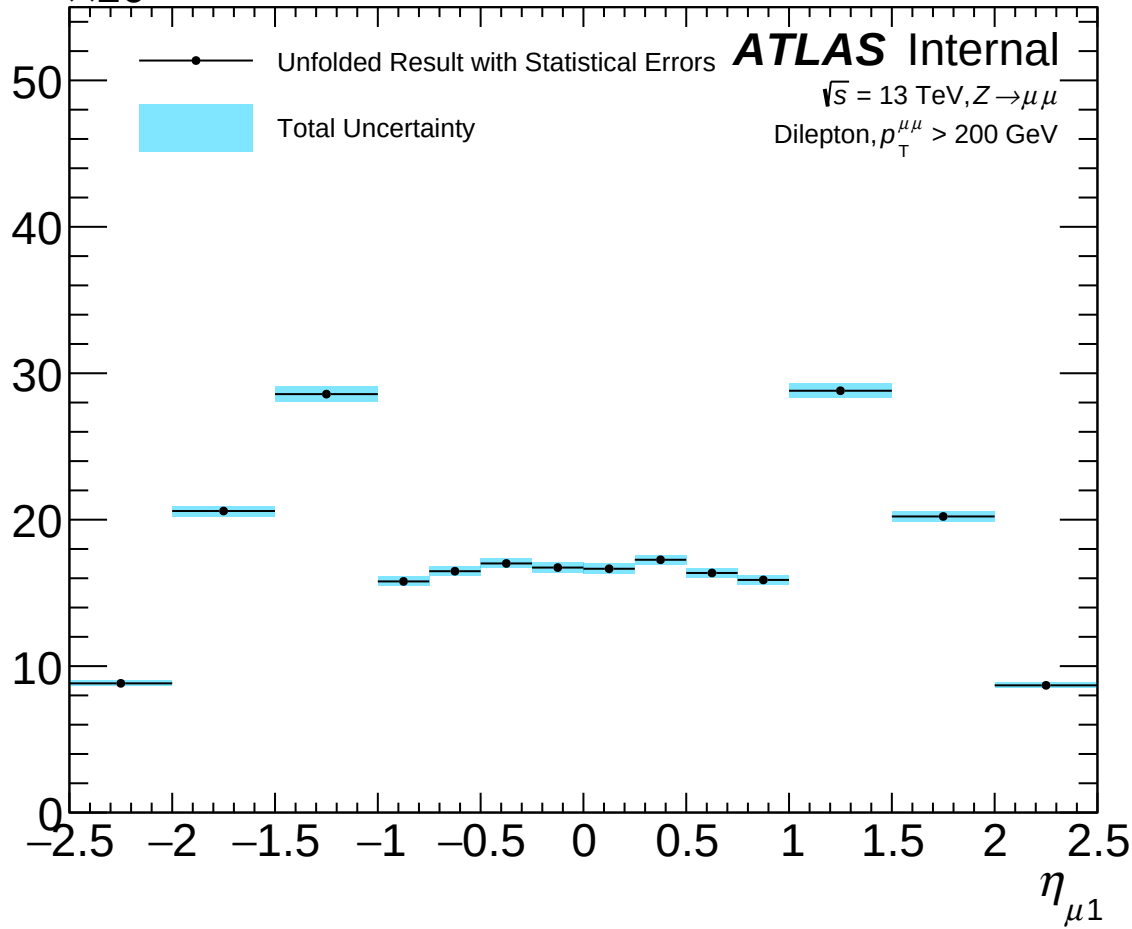
$190 < p_{\text{T}}^{\mu\mu} < 200$

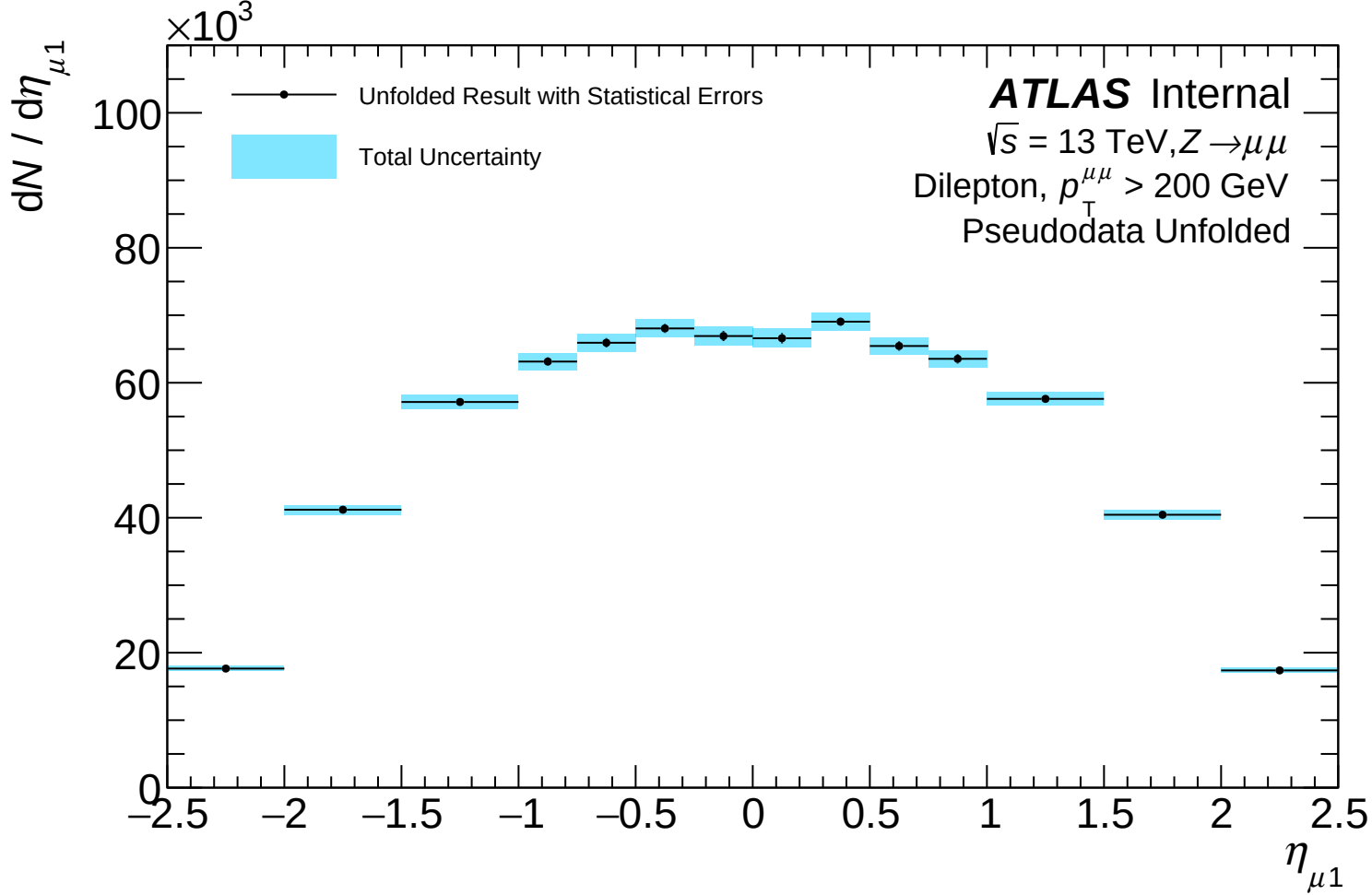


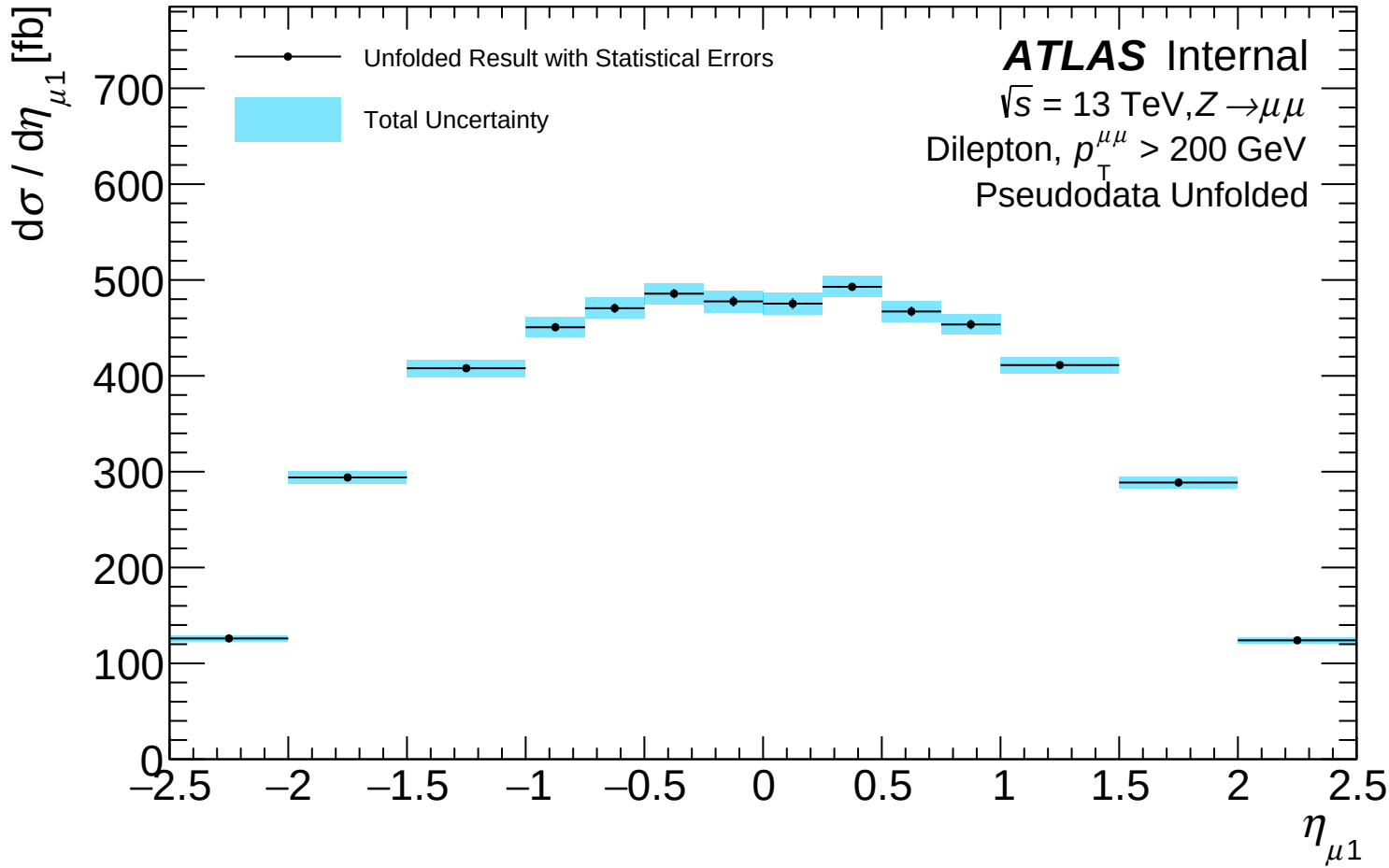


Events

$\times 10^3$







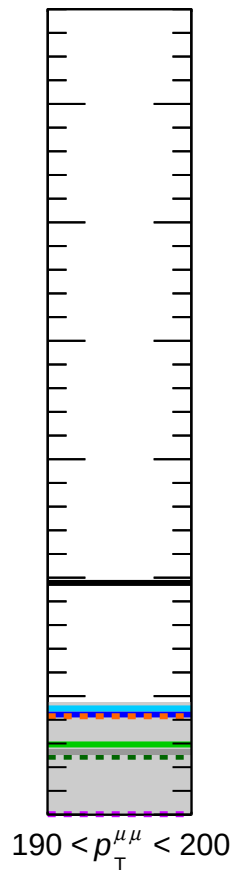
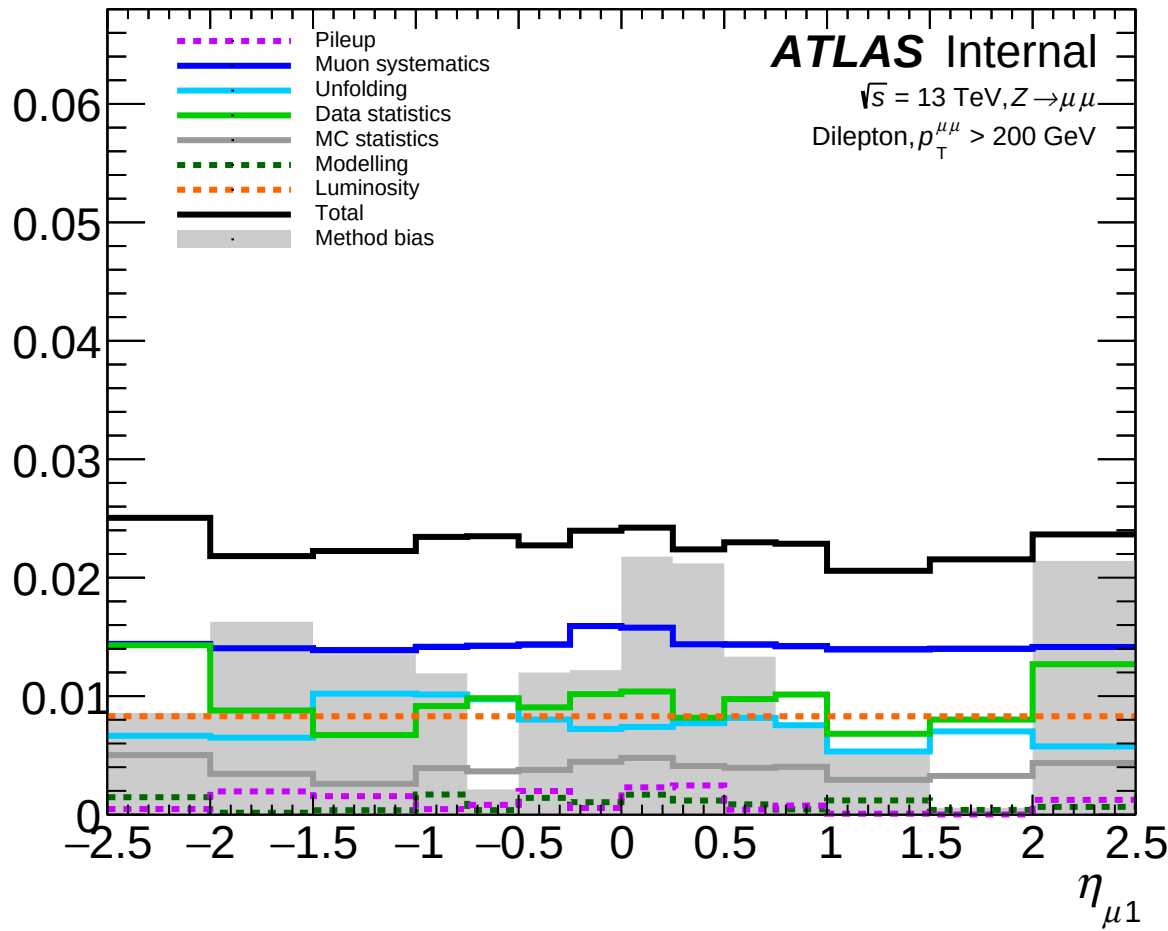
Fractional uncertainty

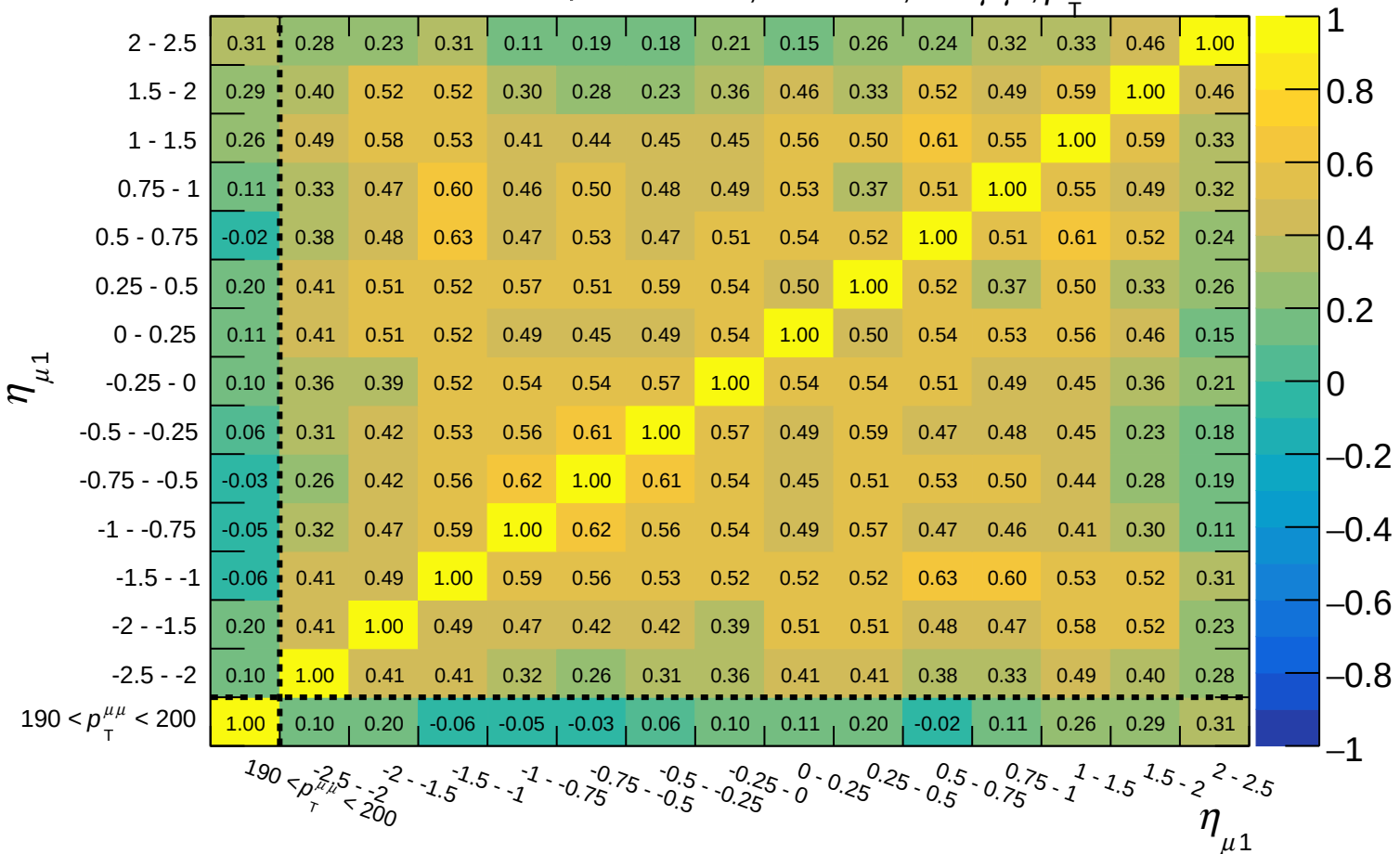
ATLAS Internal

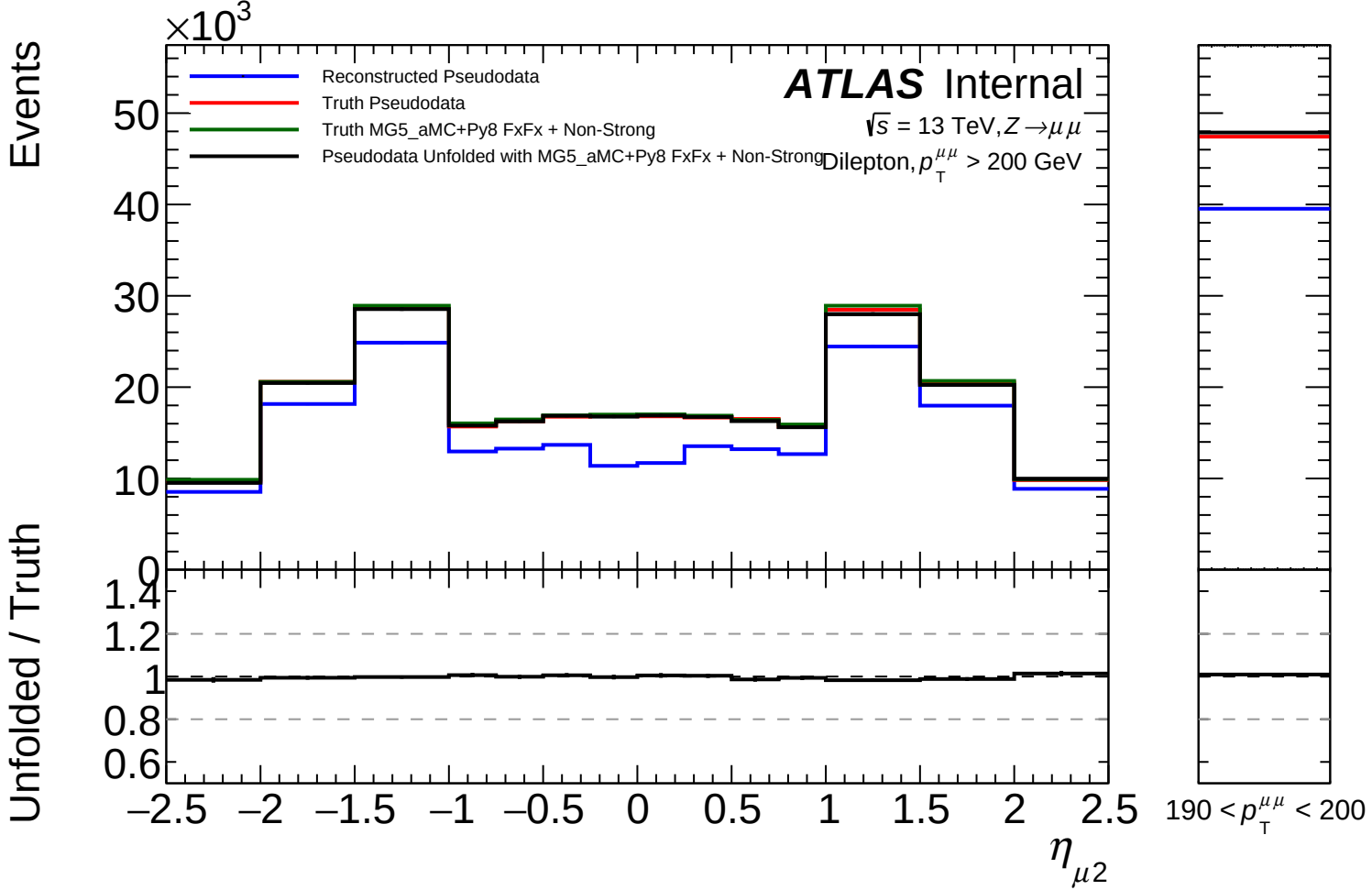
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

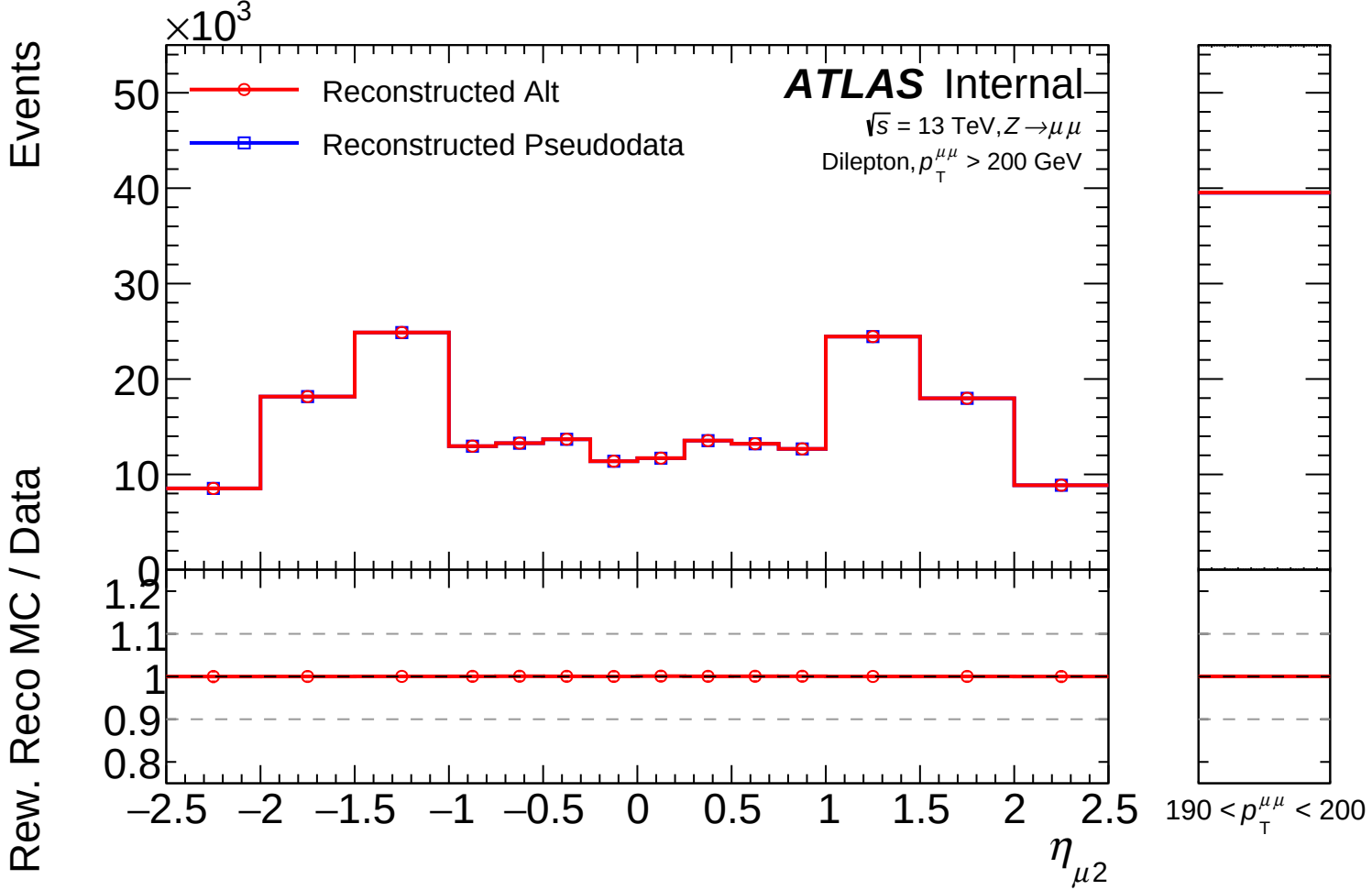
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

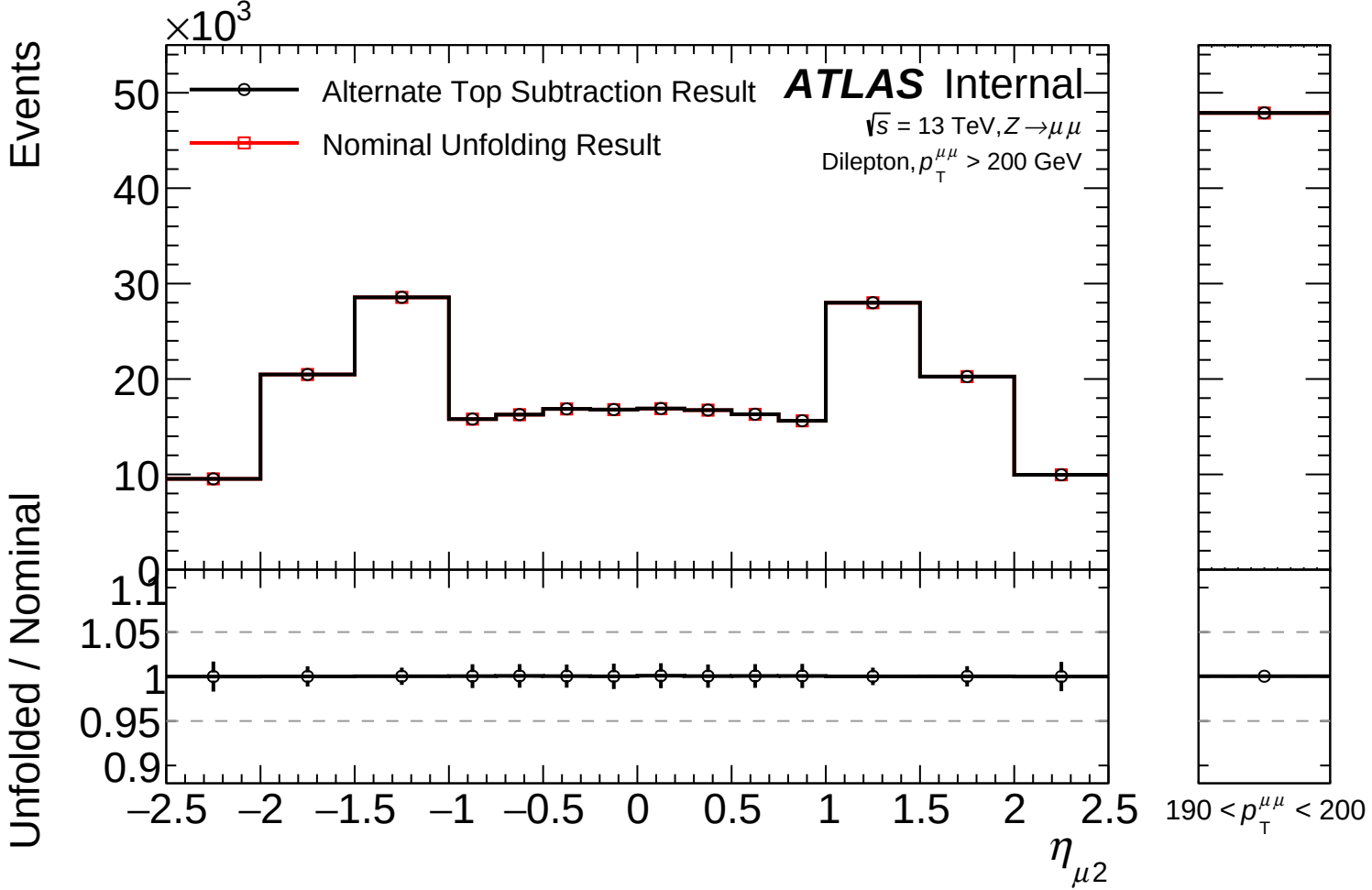
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ 







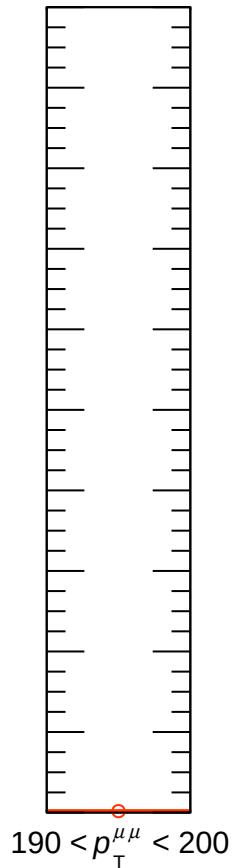
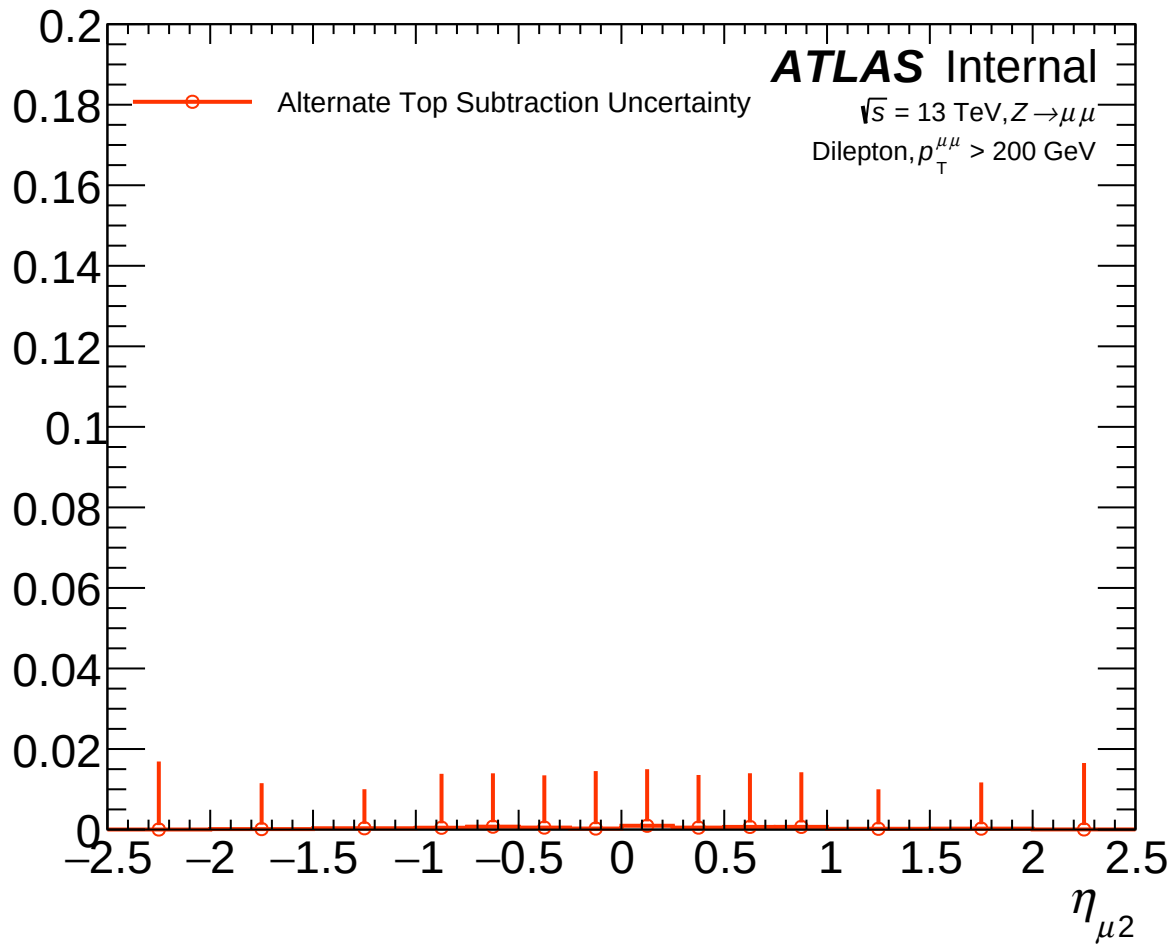
Fractional systematic impact

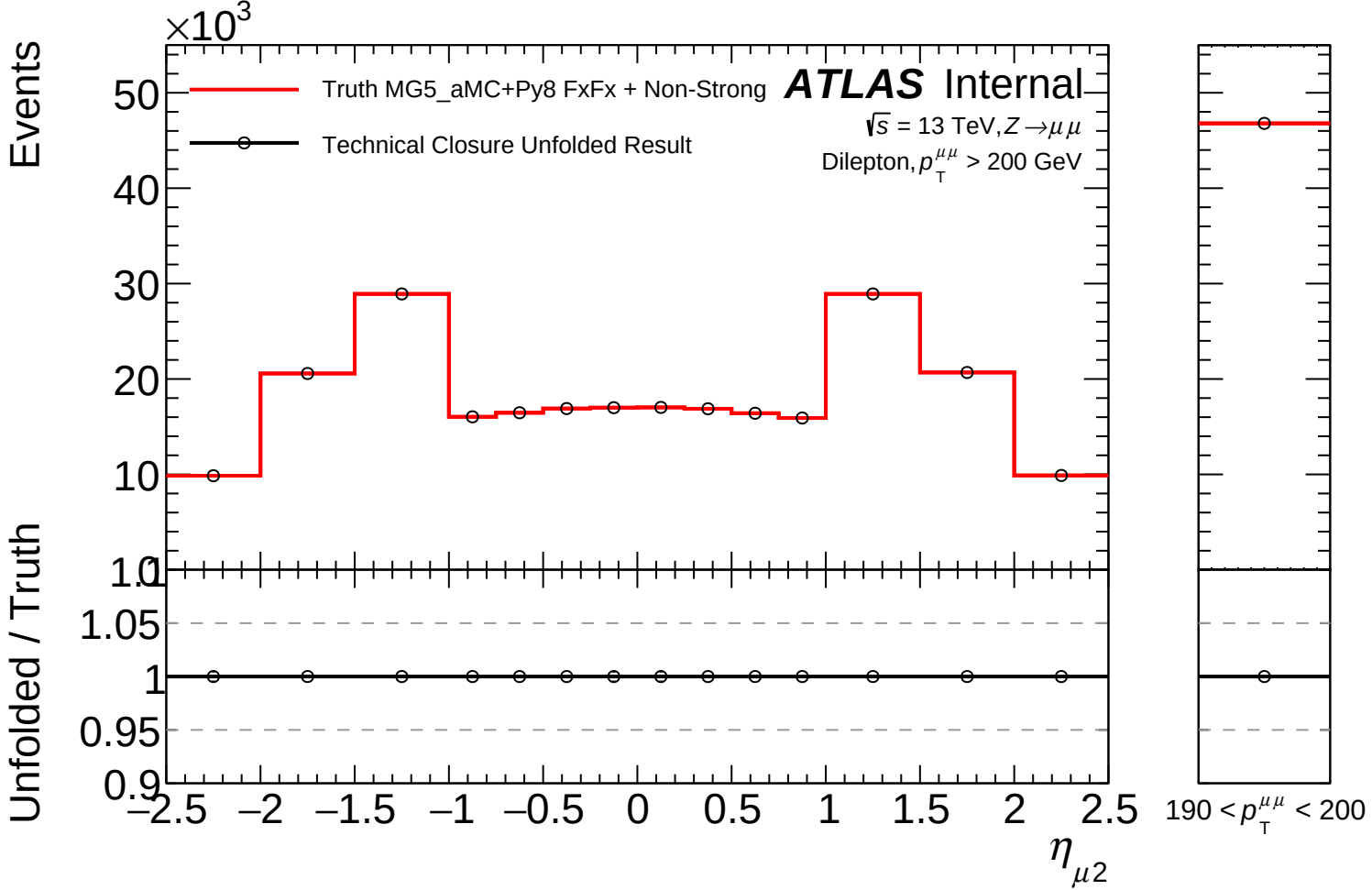
ATLAS Internal

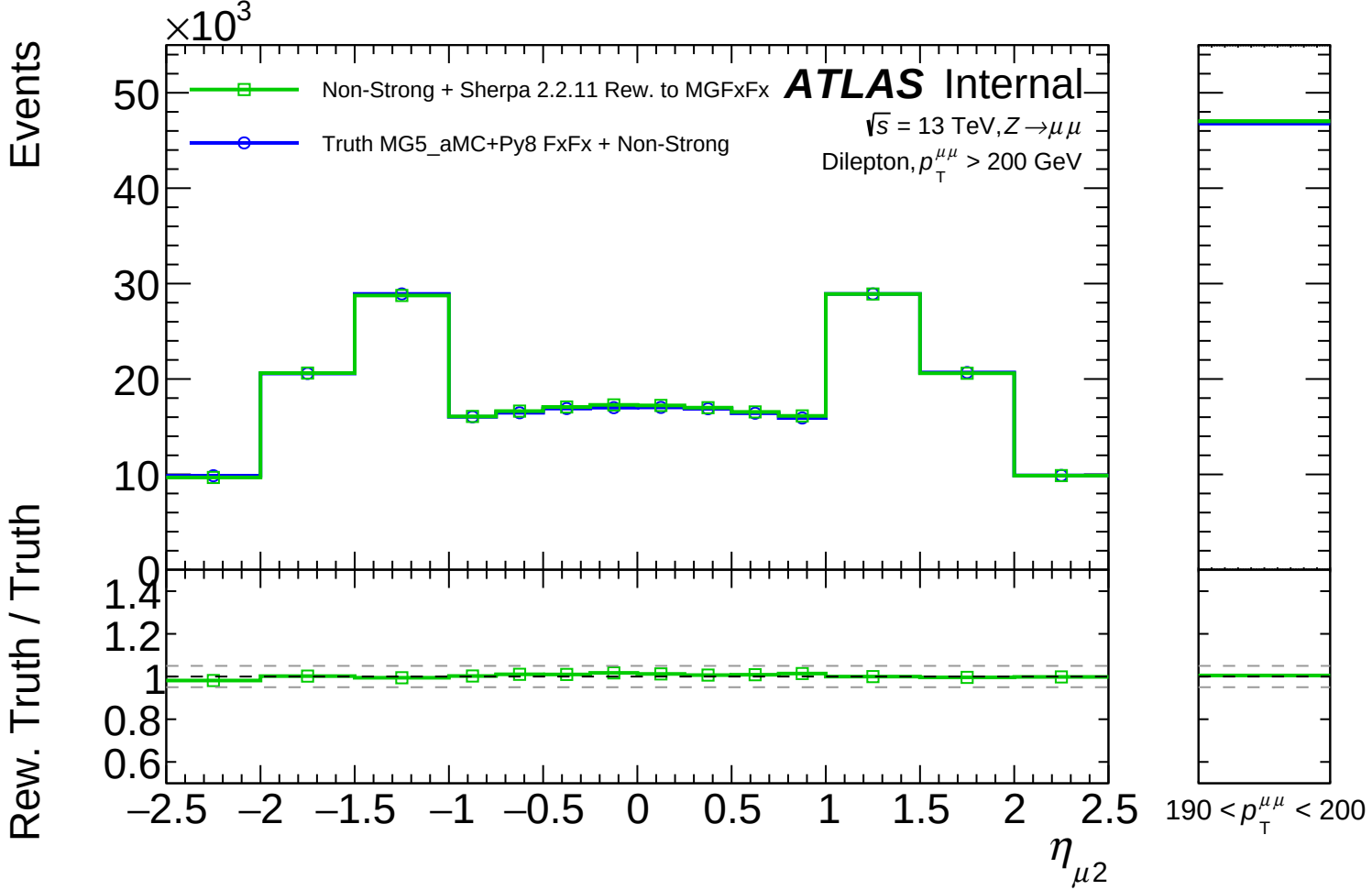
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

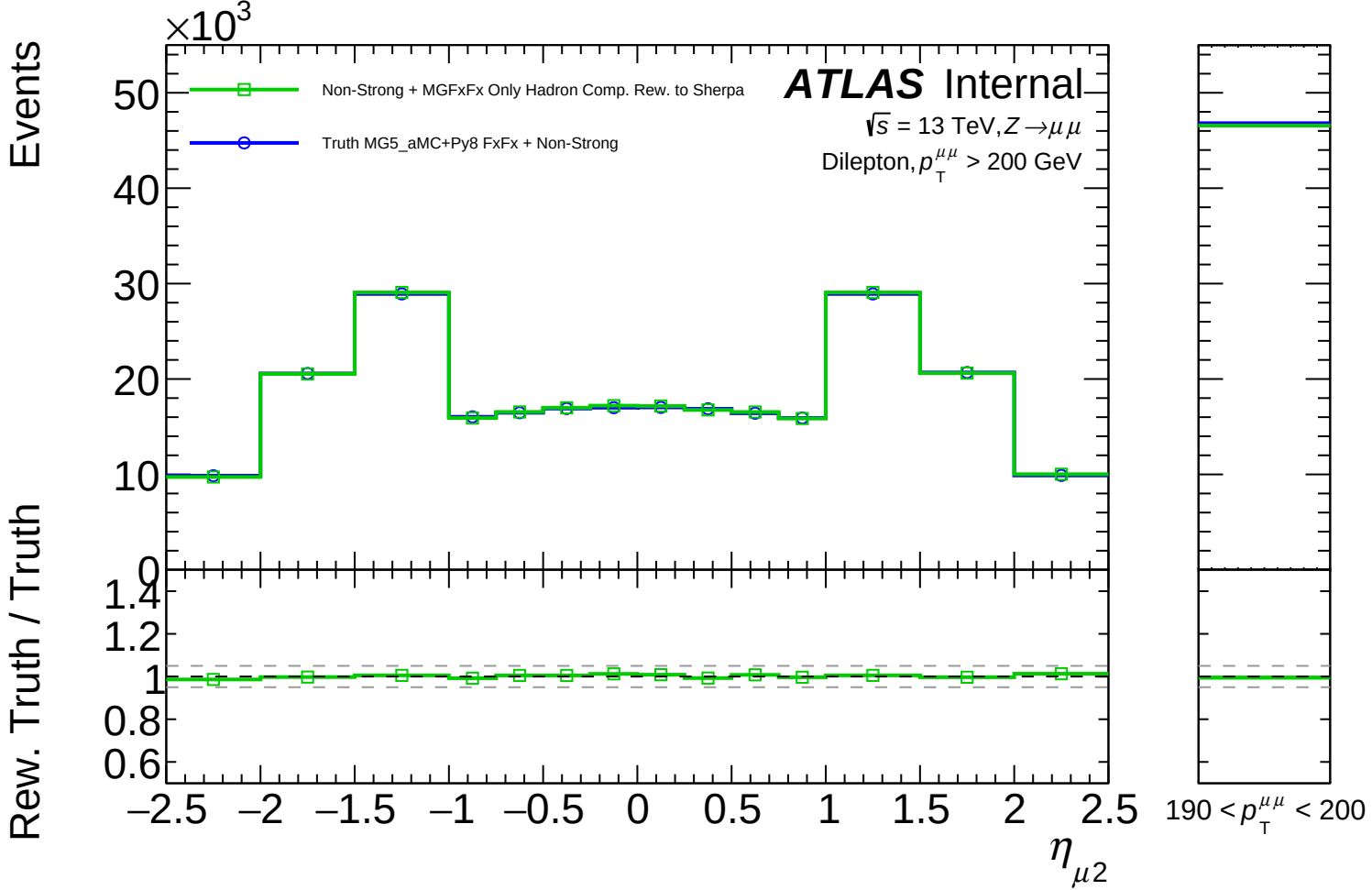
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

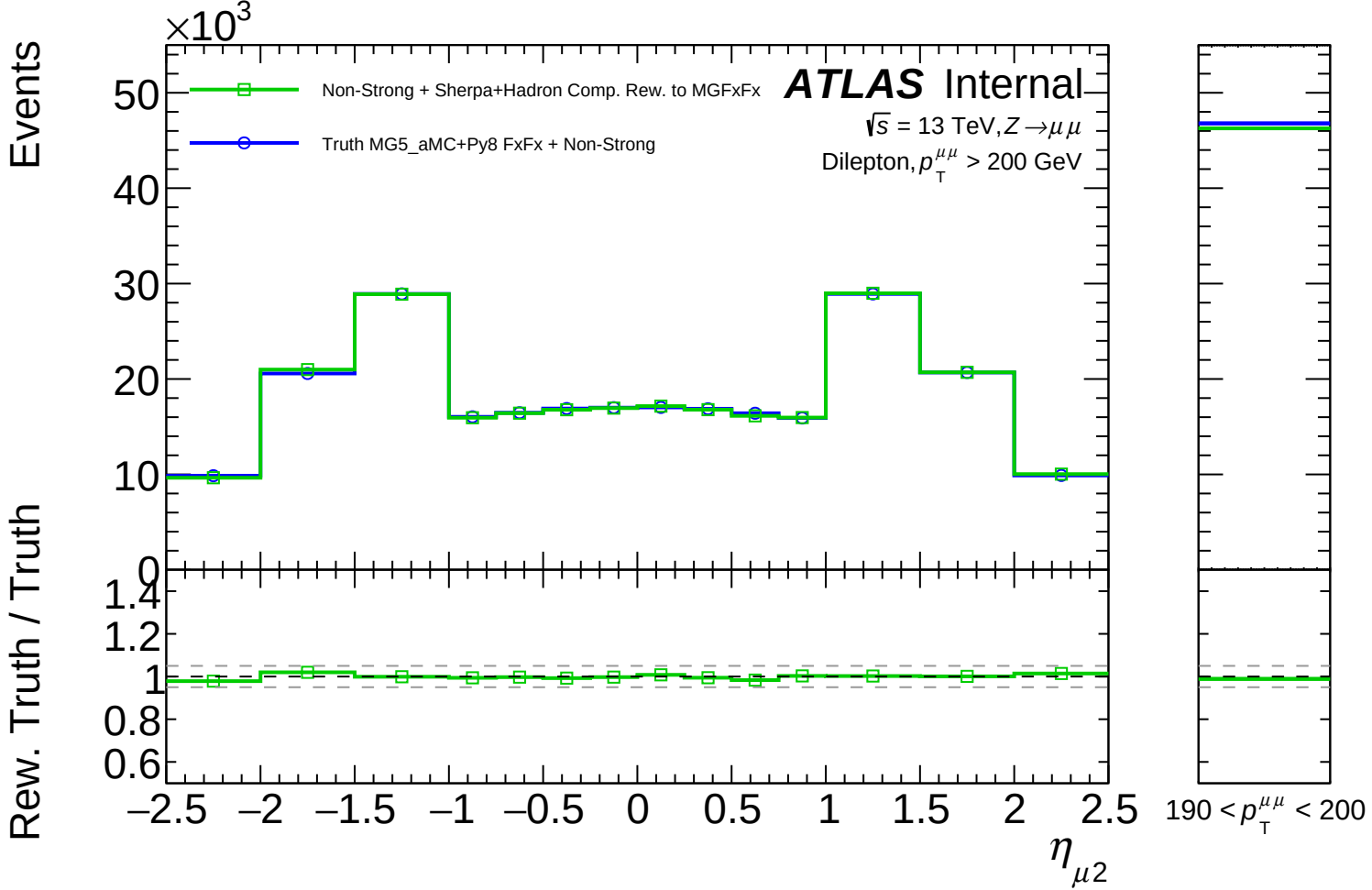
Alternate Top Subtraction Uncertainty

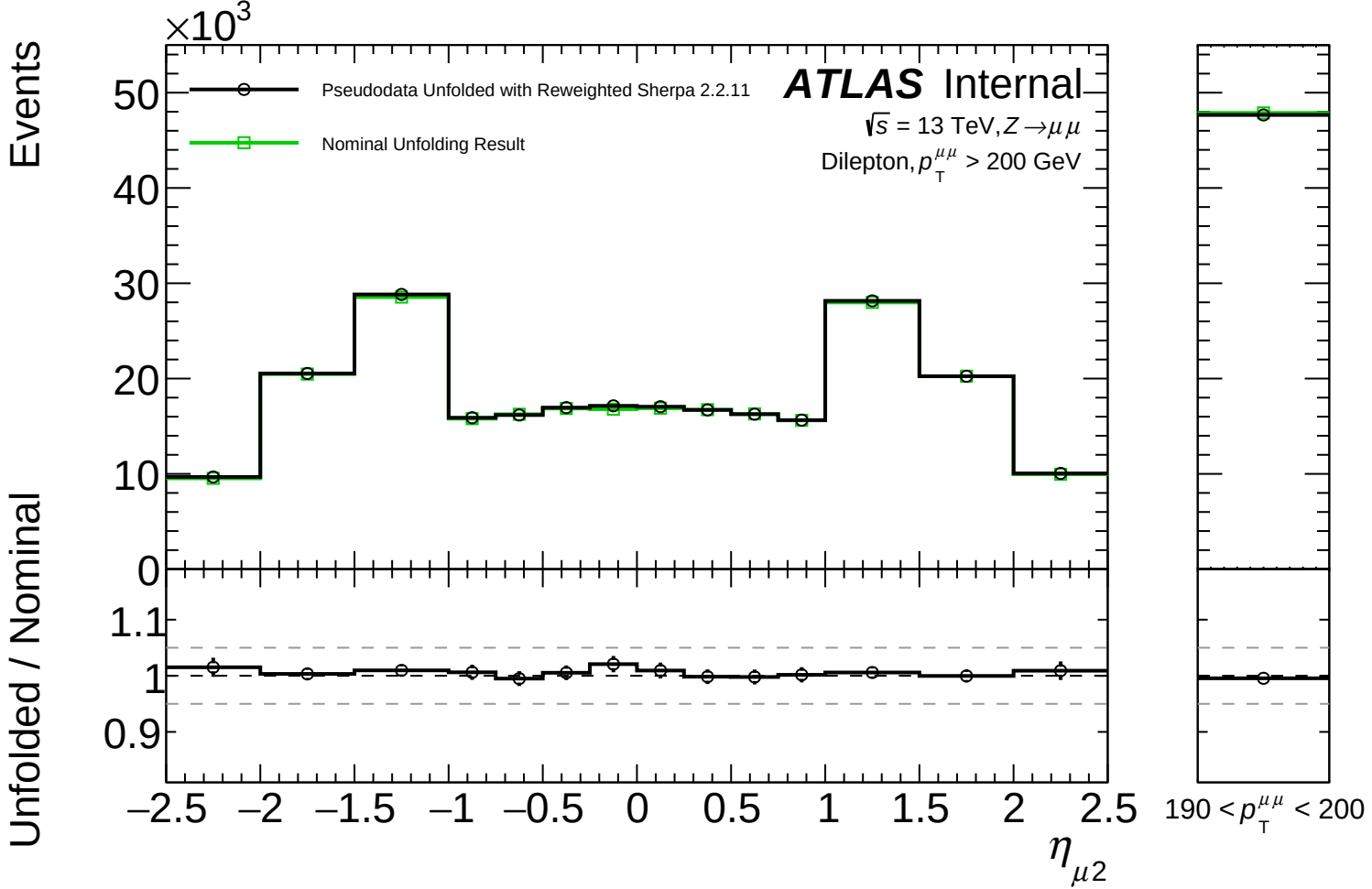


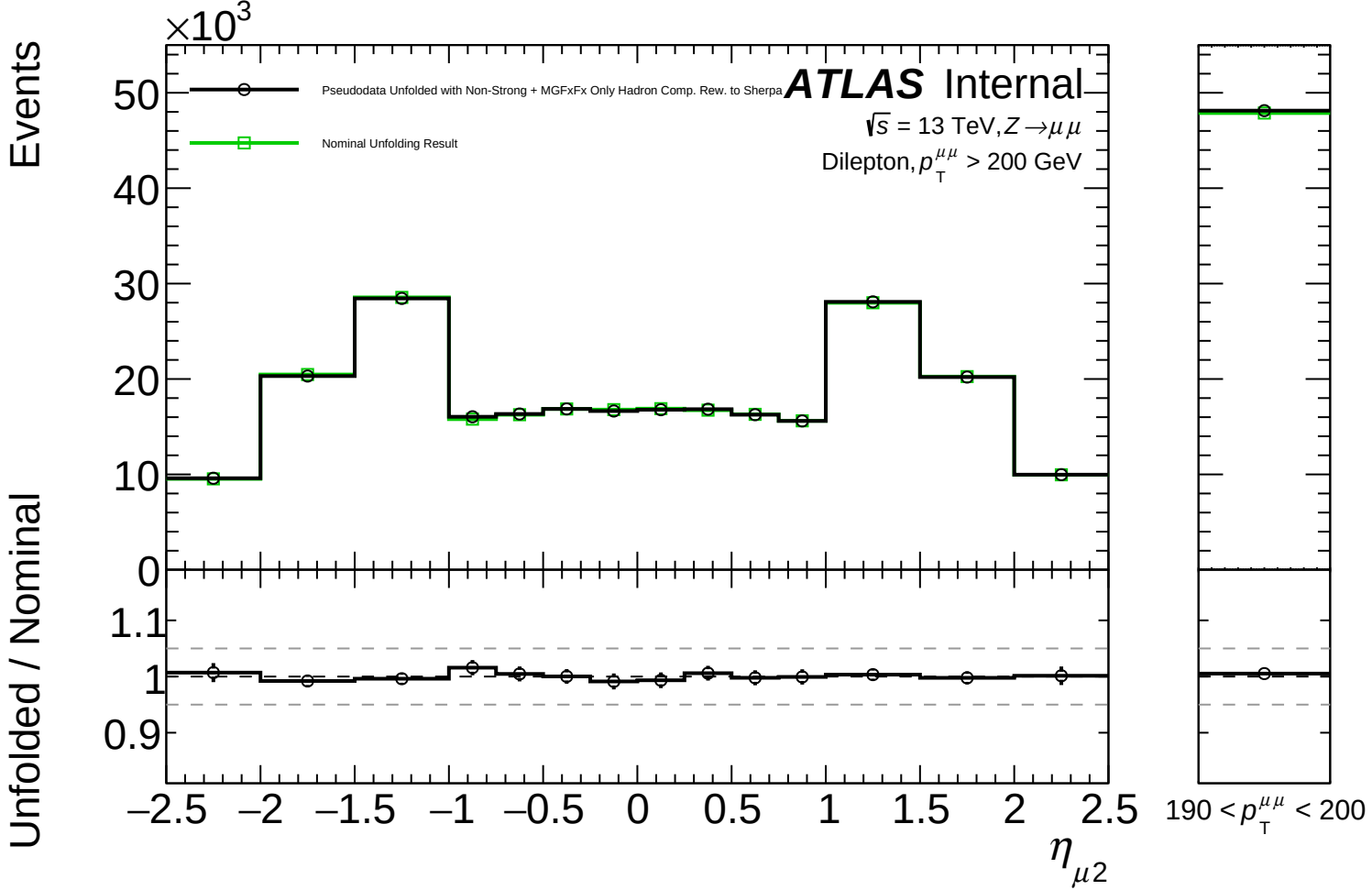


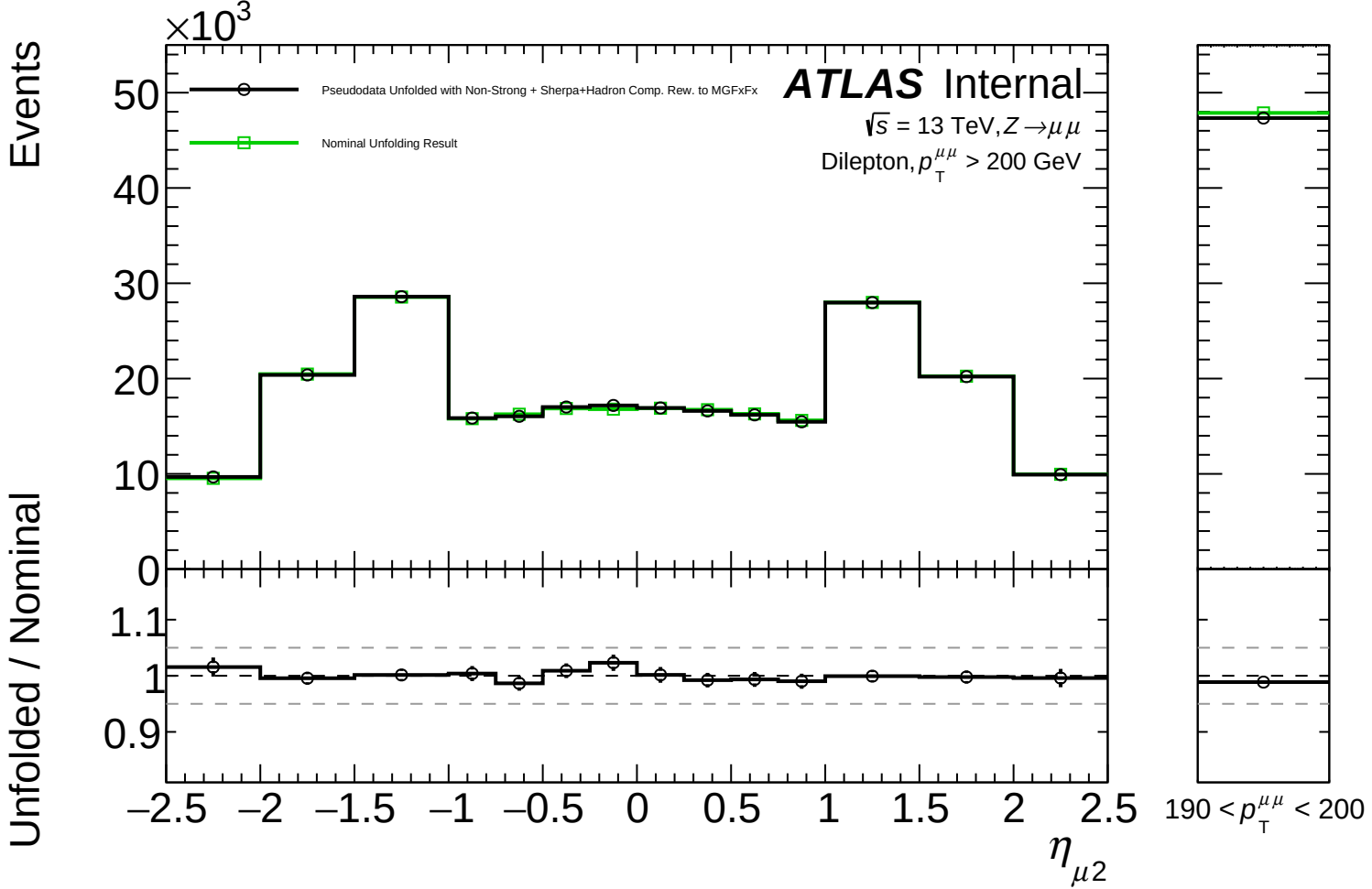


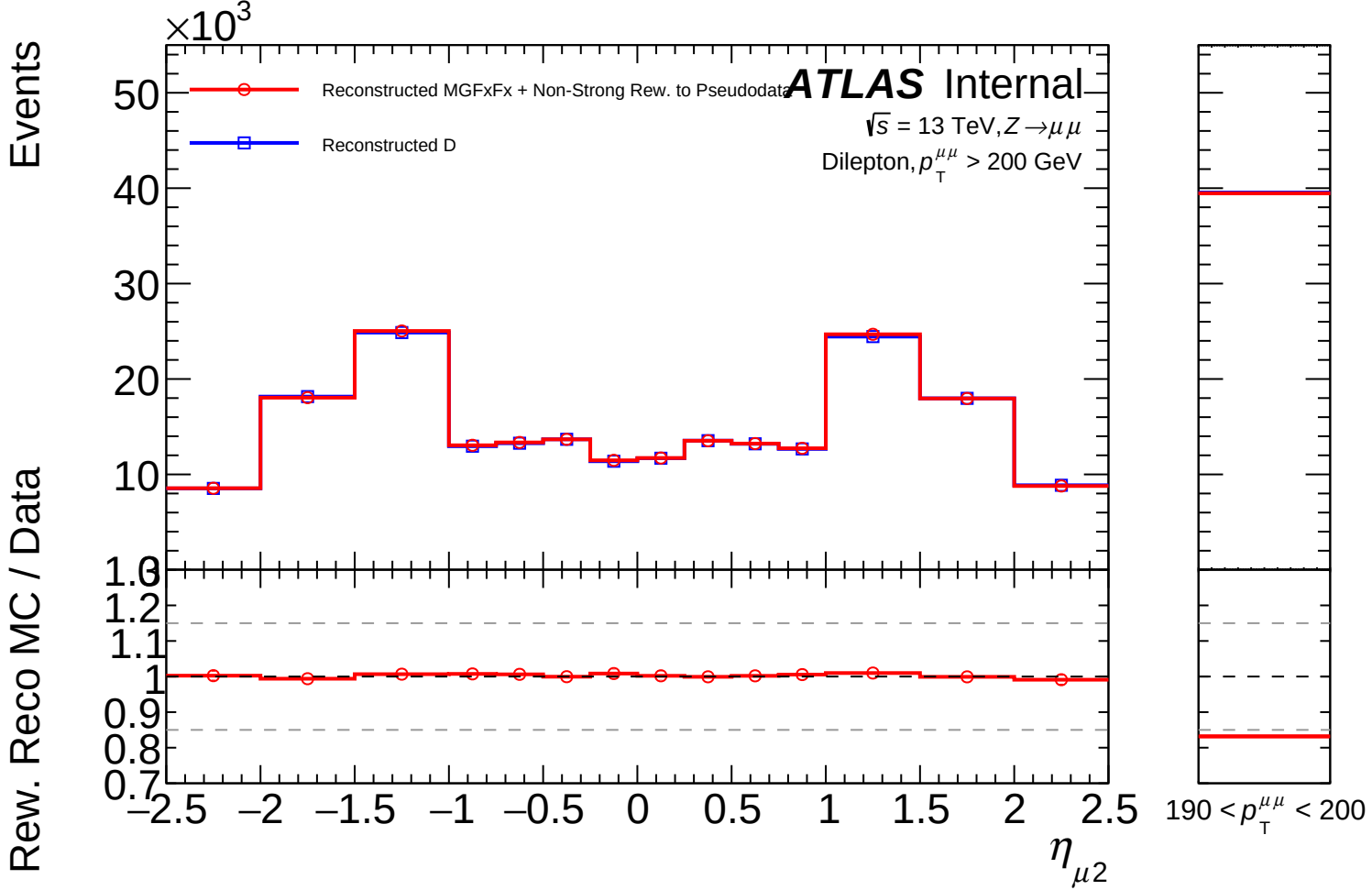


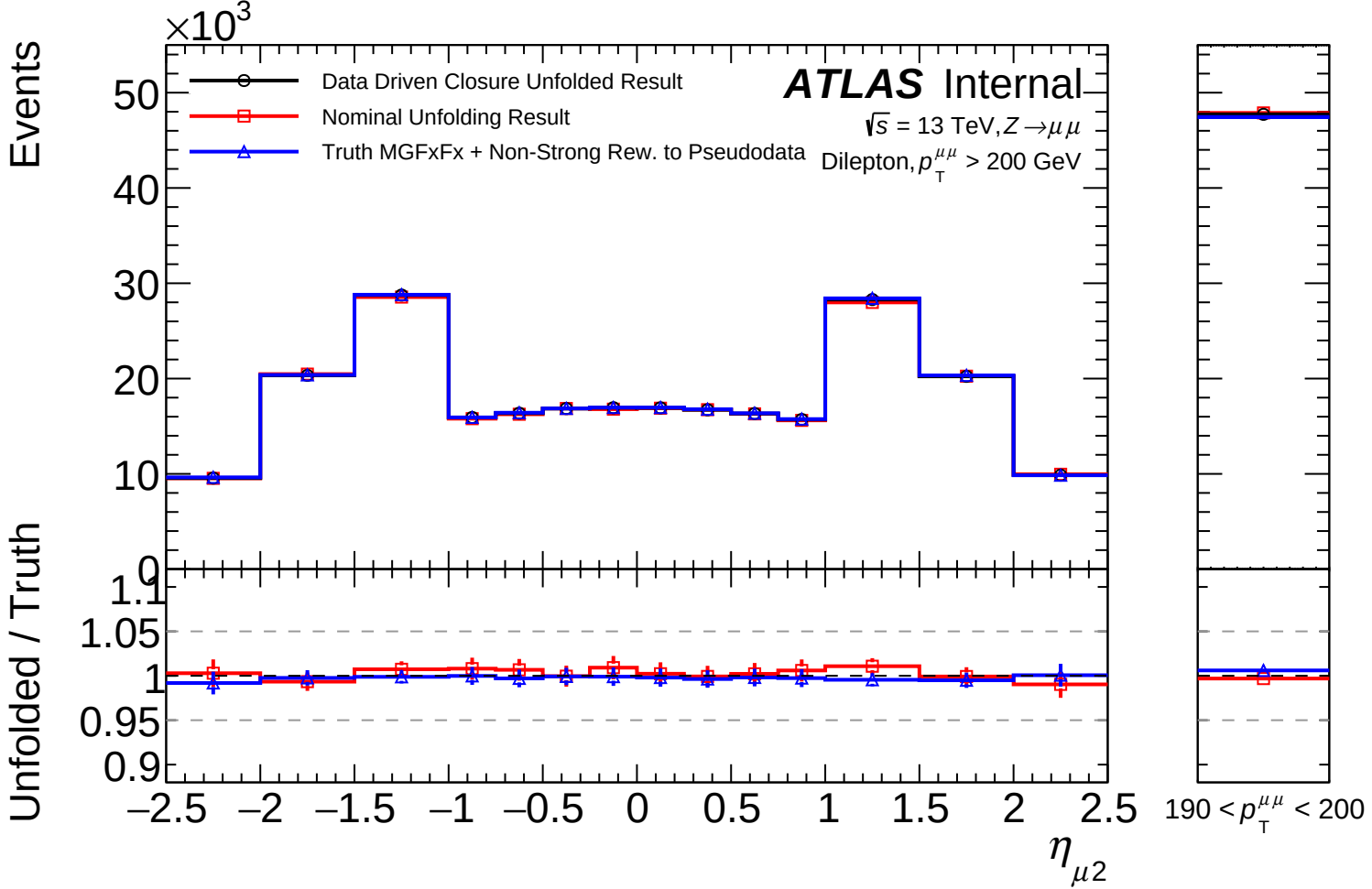




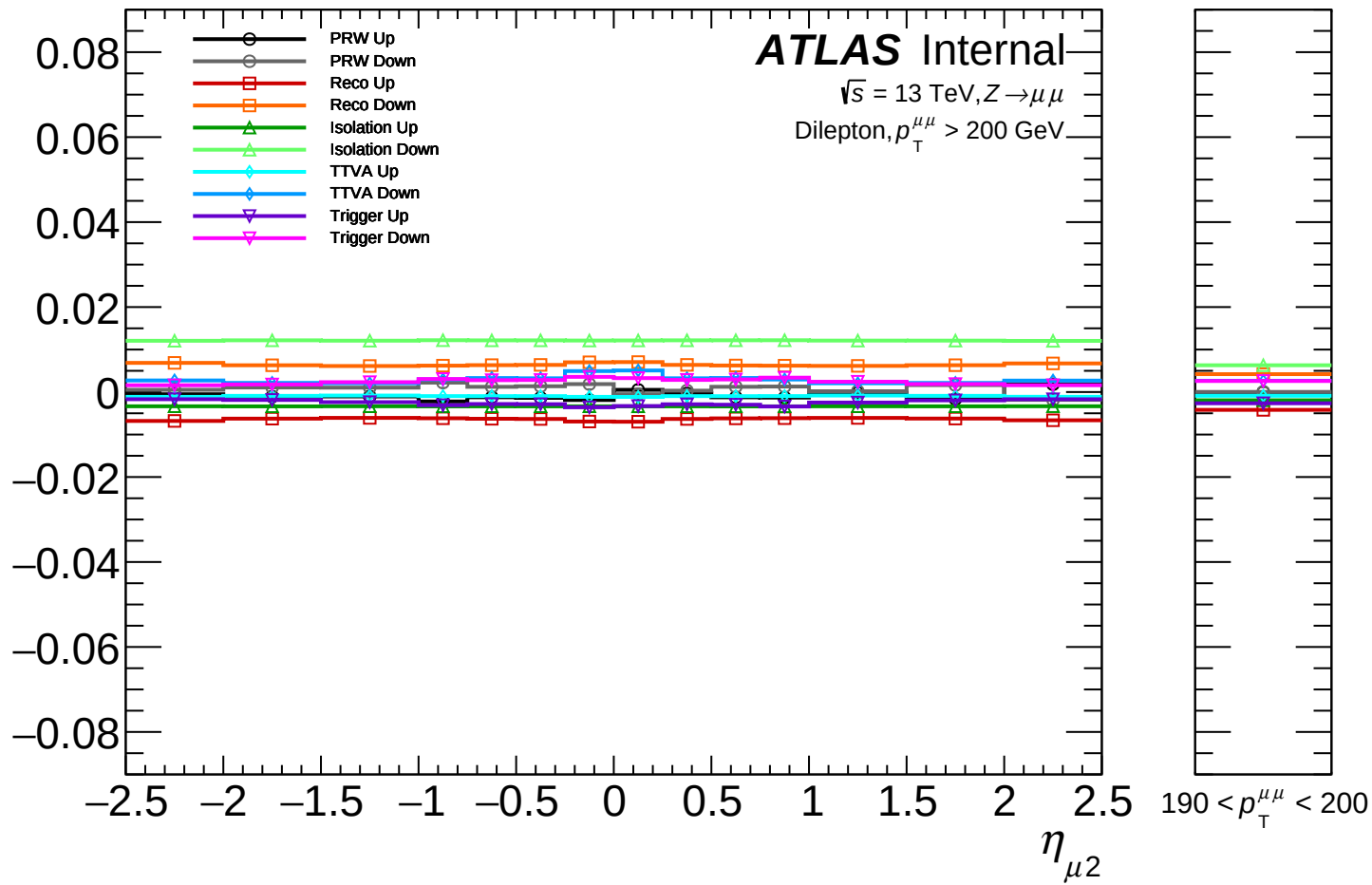




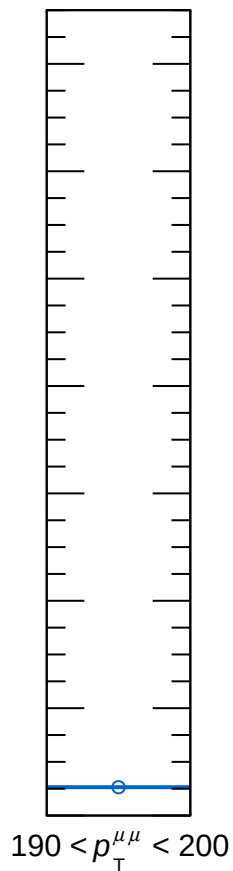
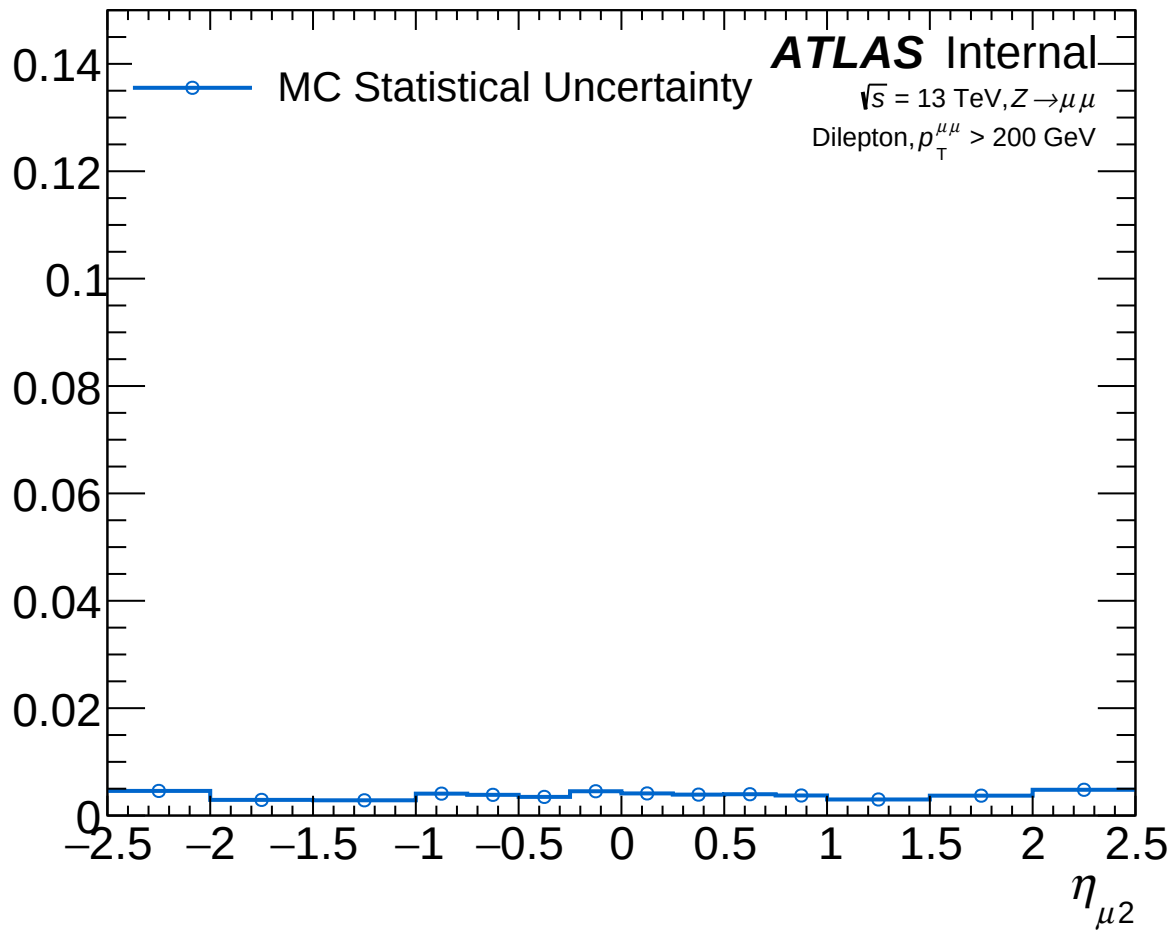




Fractional systematic impact



Fractional systematic impact



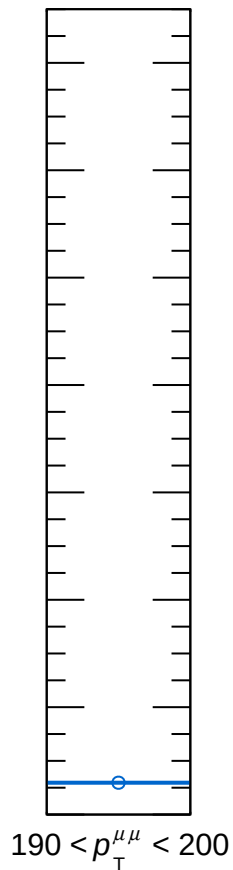
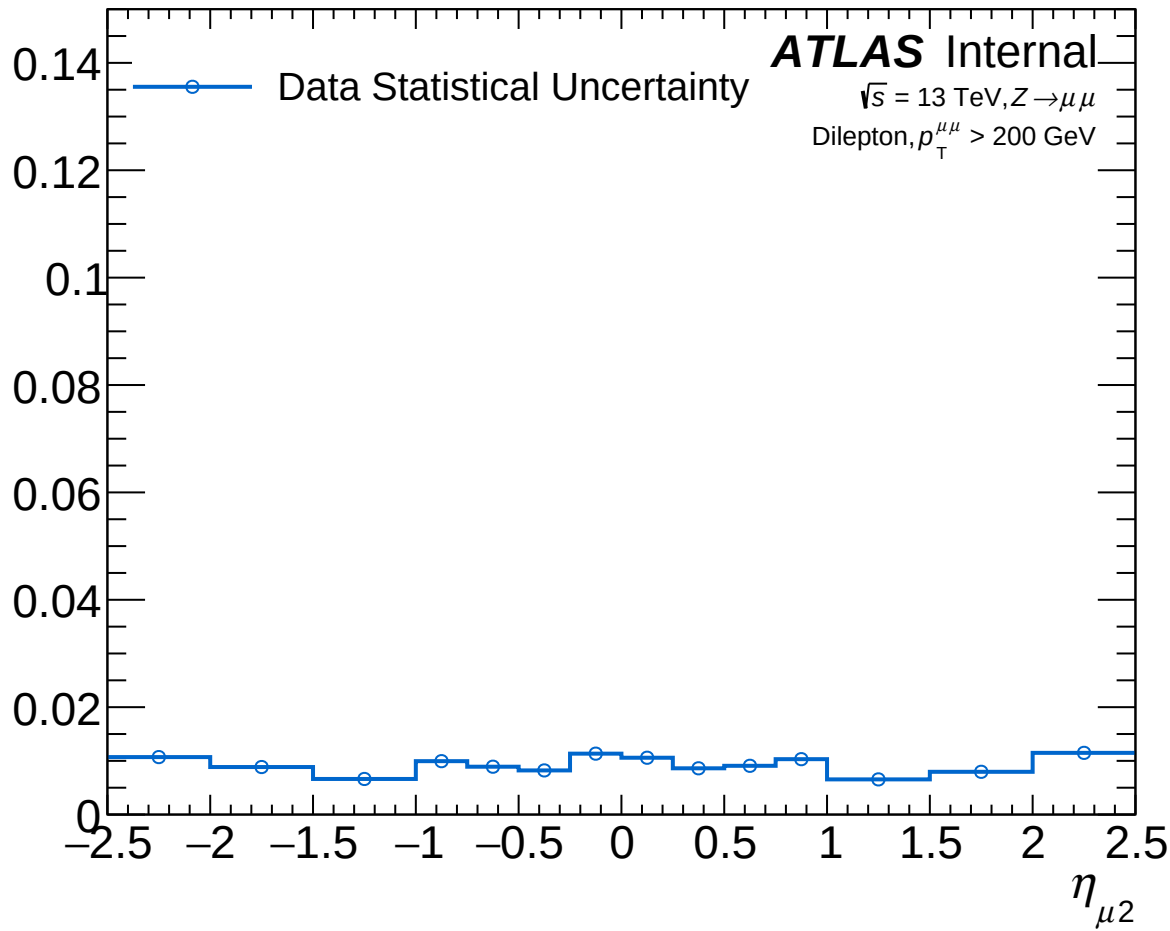
Fractional systematic impact

ATLAS Internal

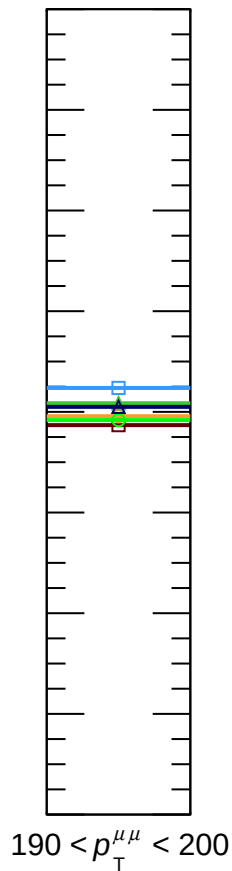
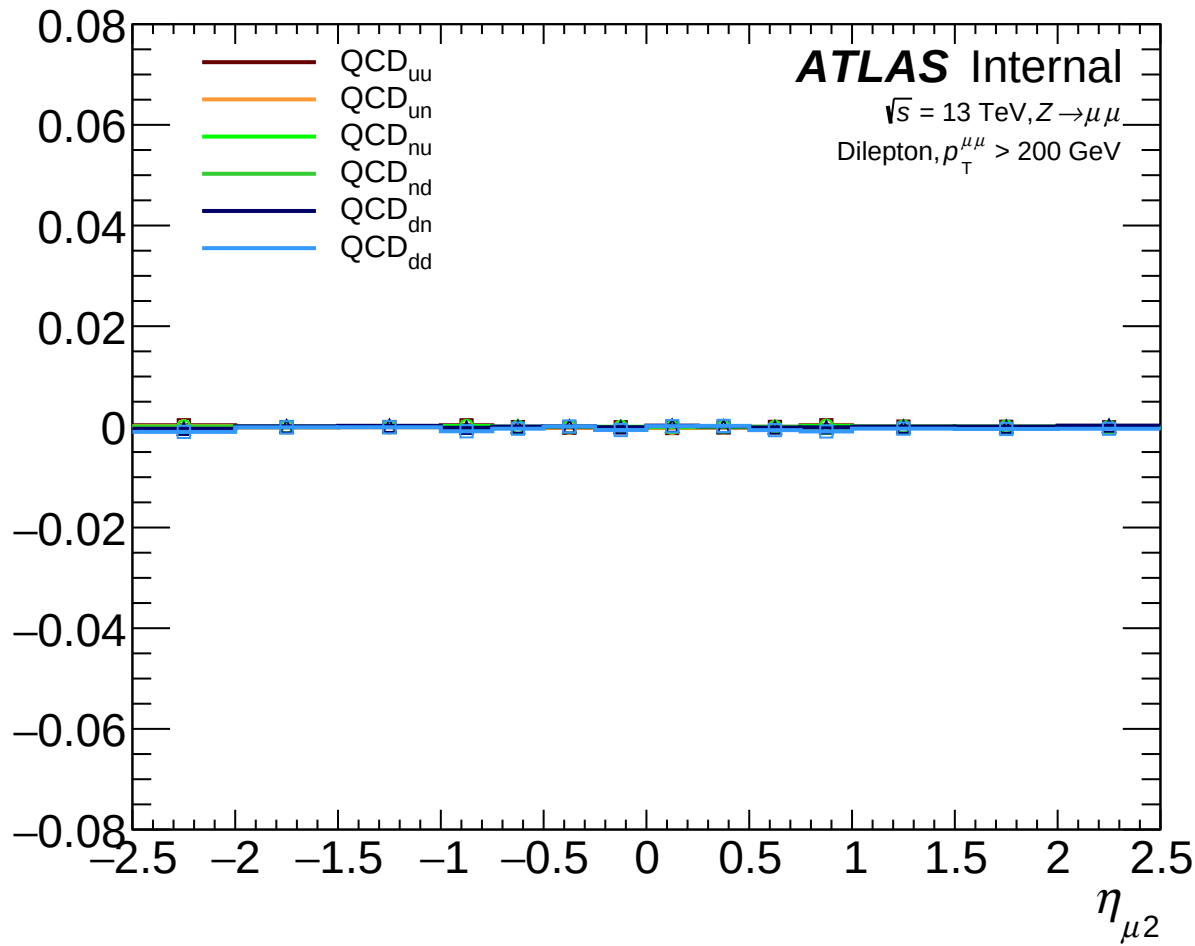
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

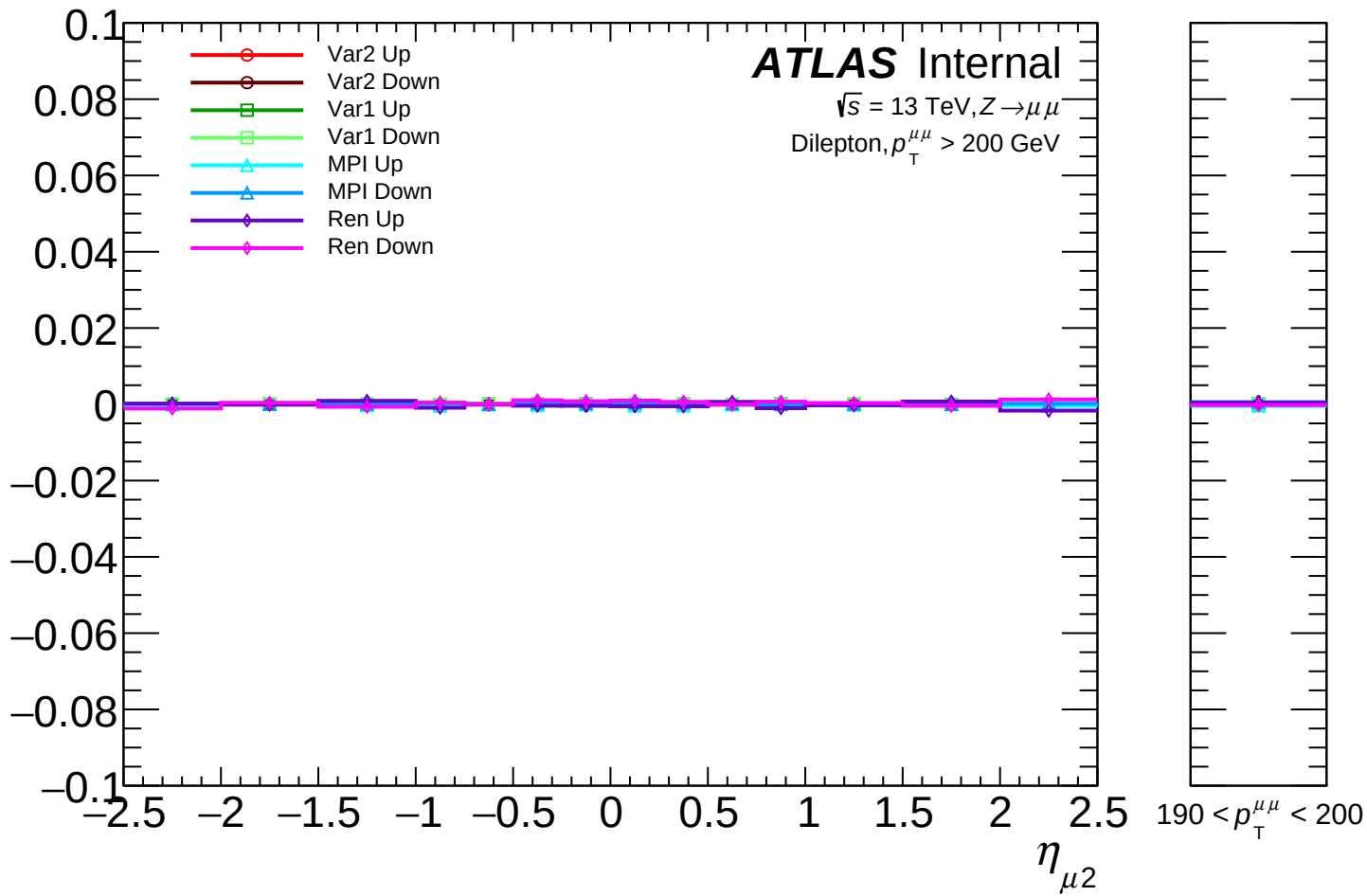
Data Statistical Uncertainty

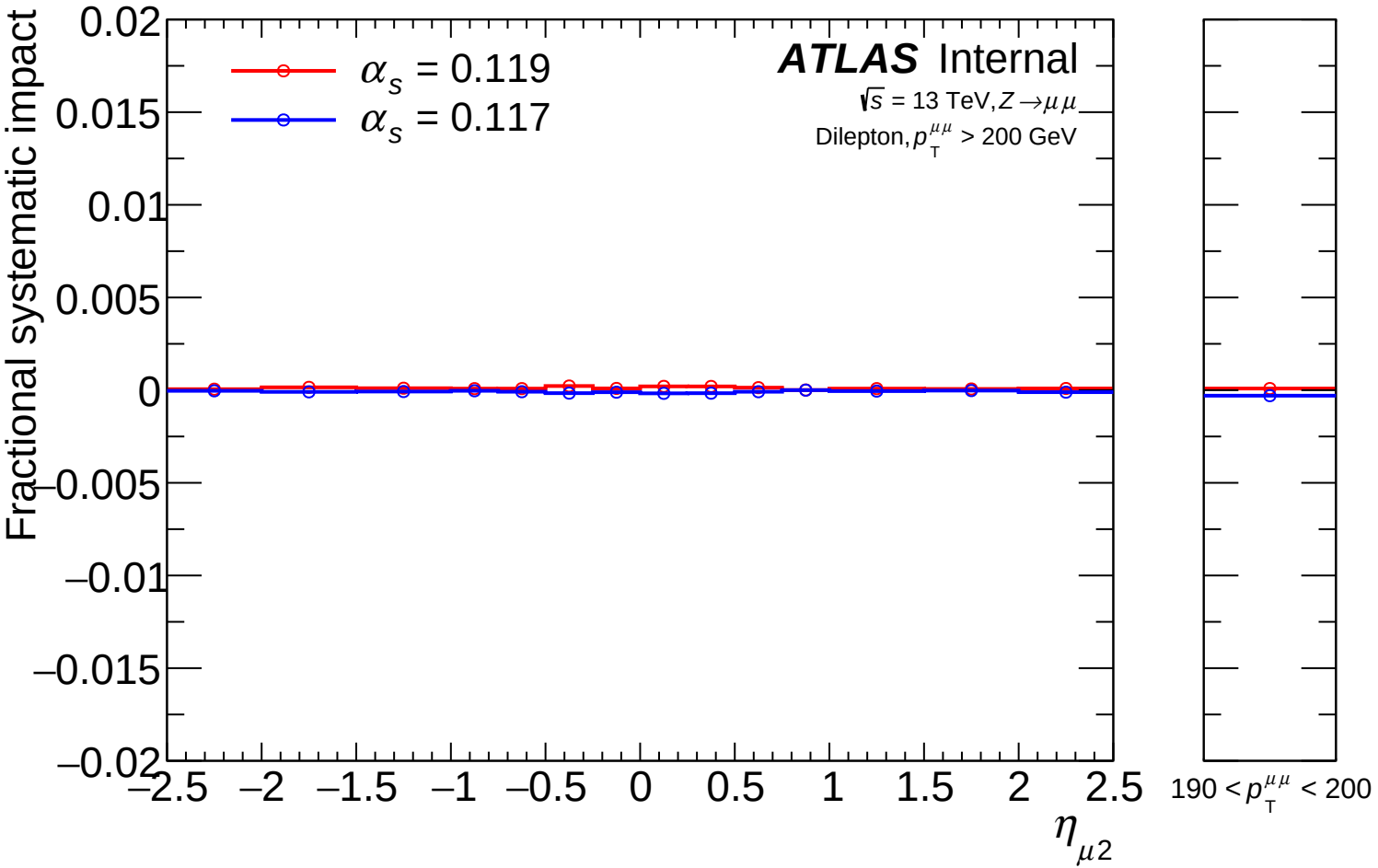


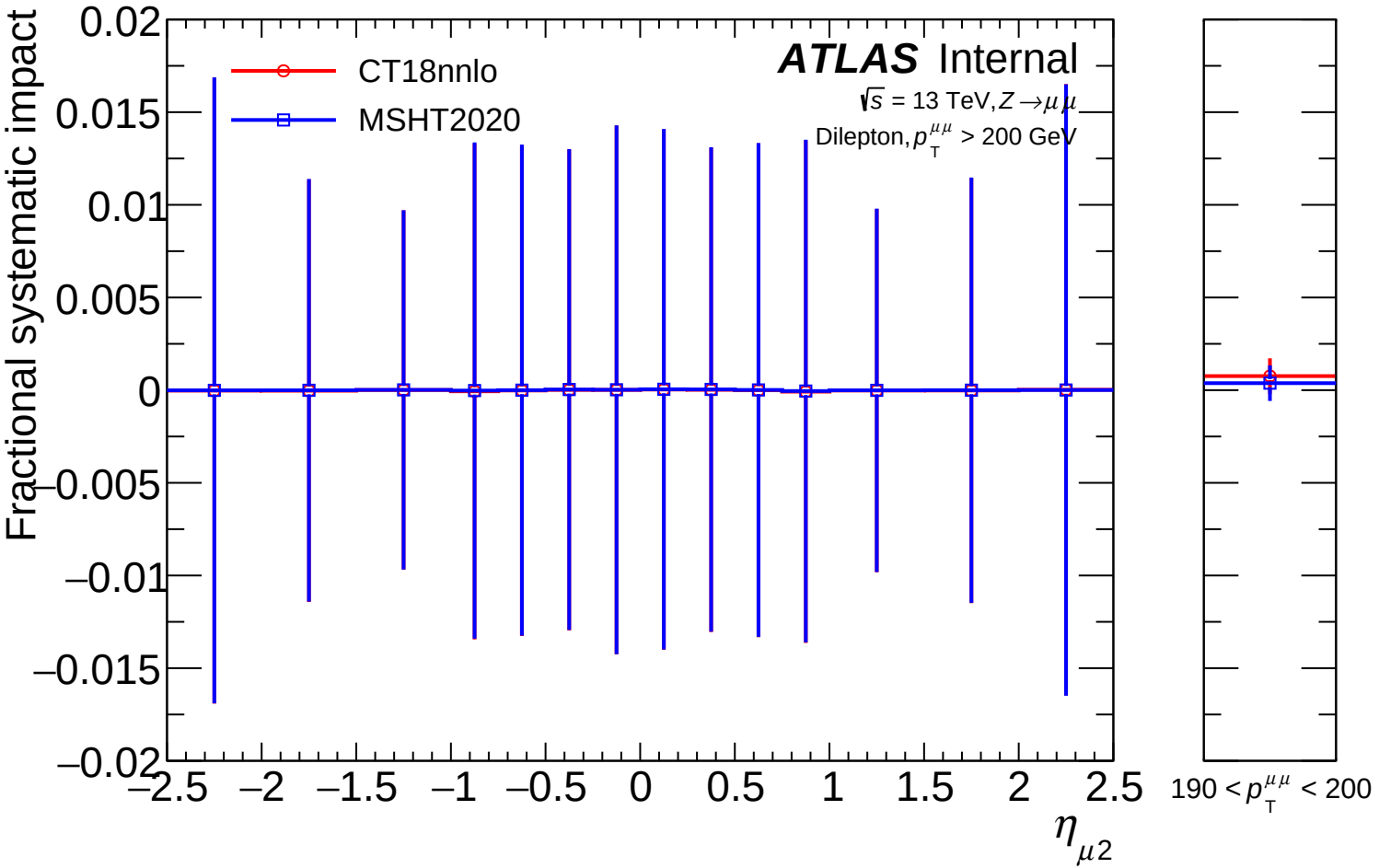
Fractional systematic impact



Fractional systematic impact







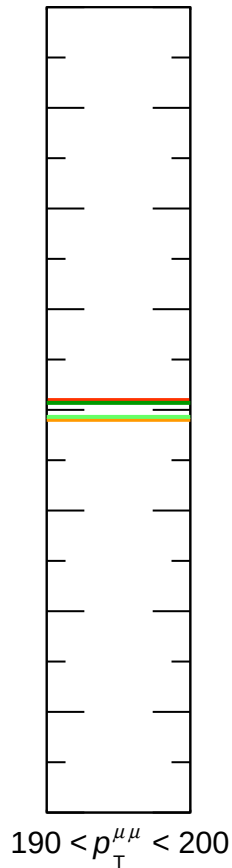
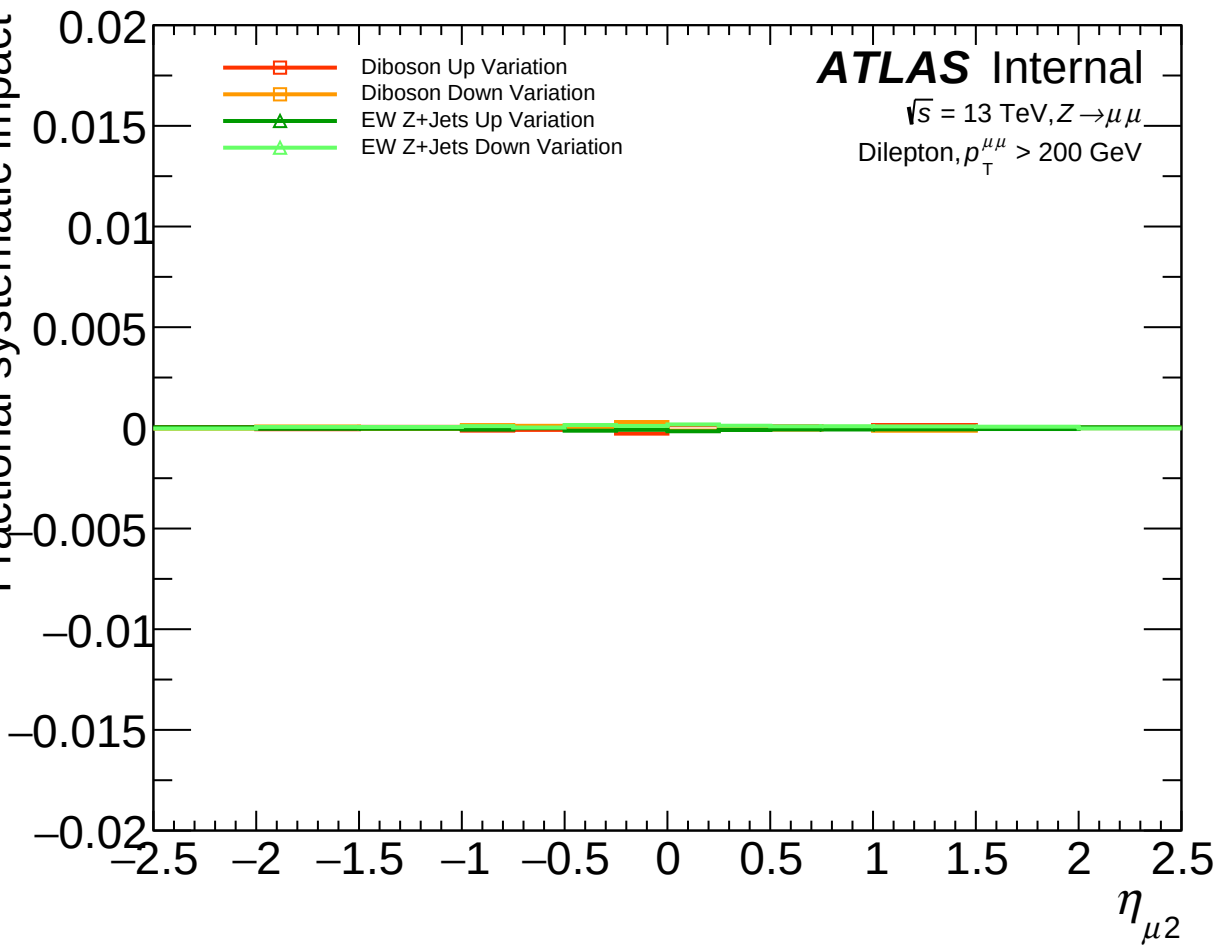
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$



ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty

0.06

0.05

0.04

0.03

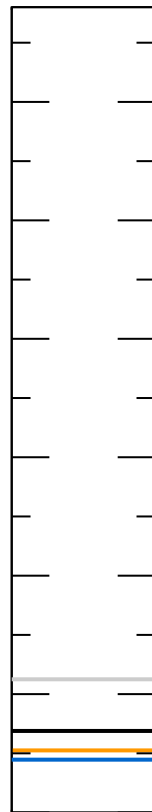
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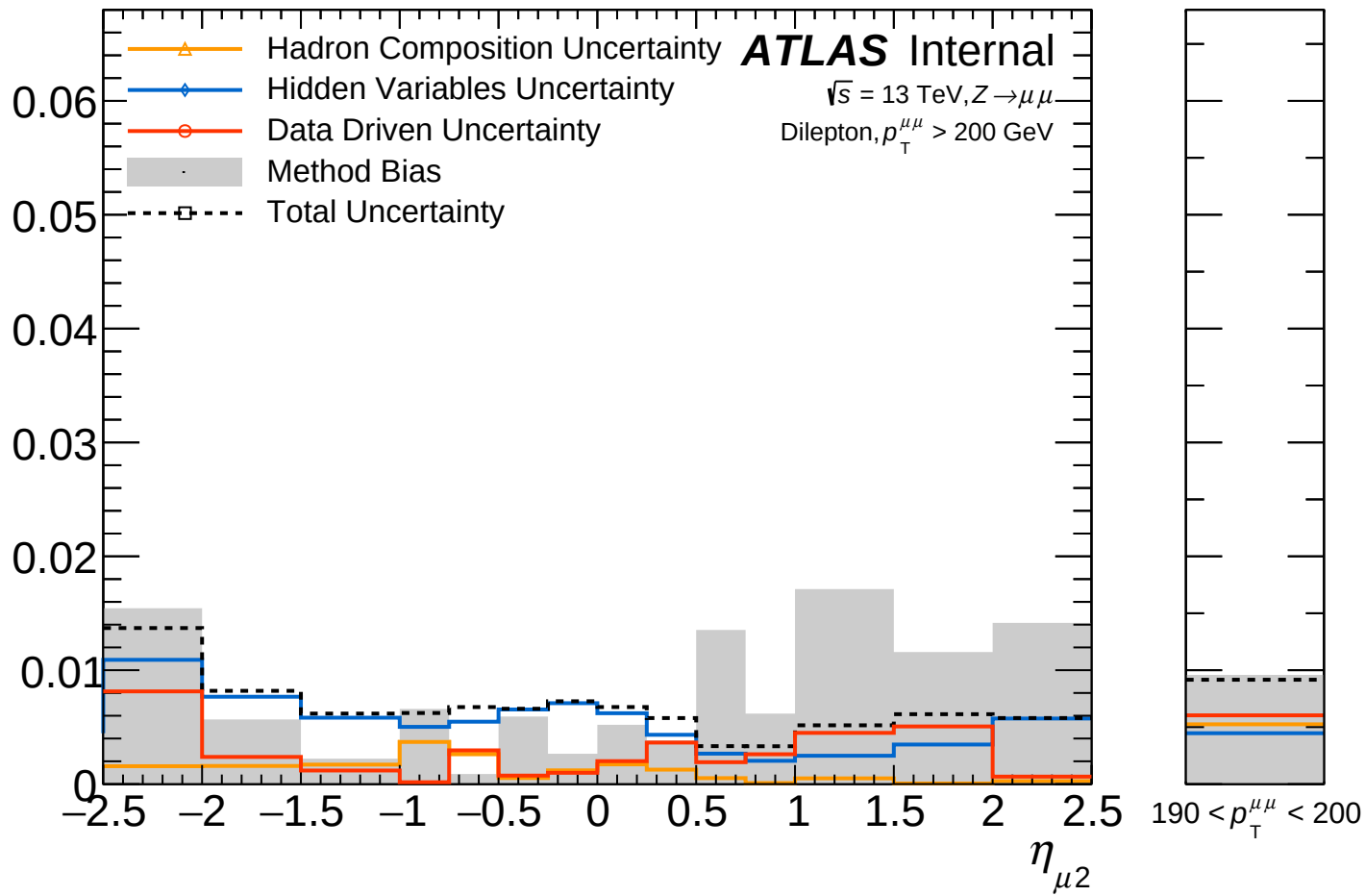
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0

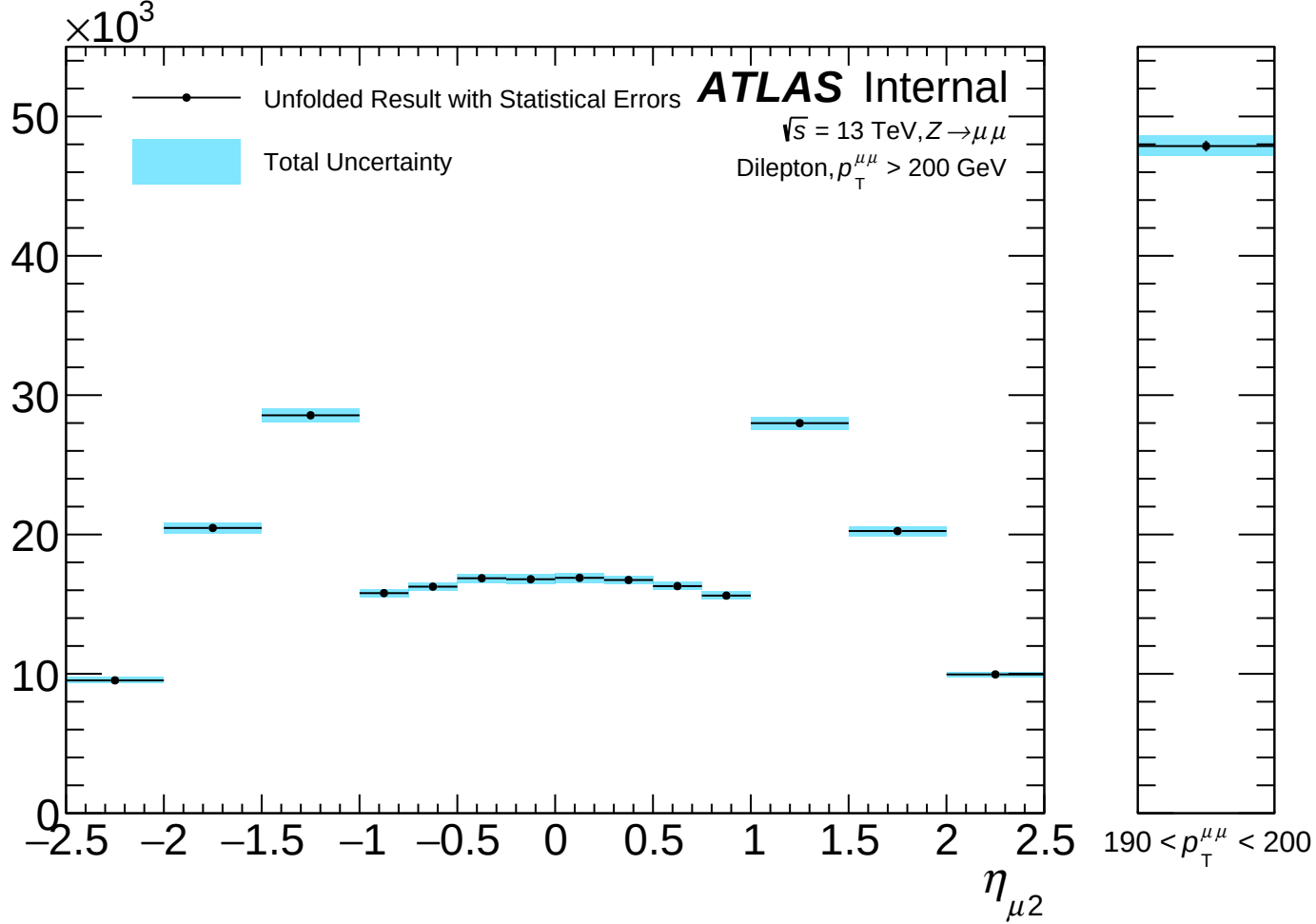
$\eta_{\mu 2}$

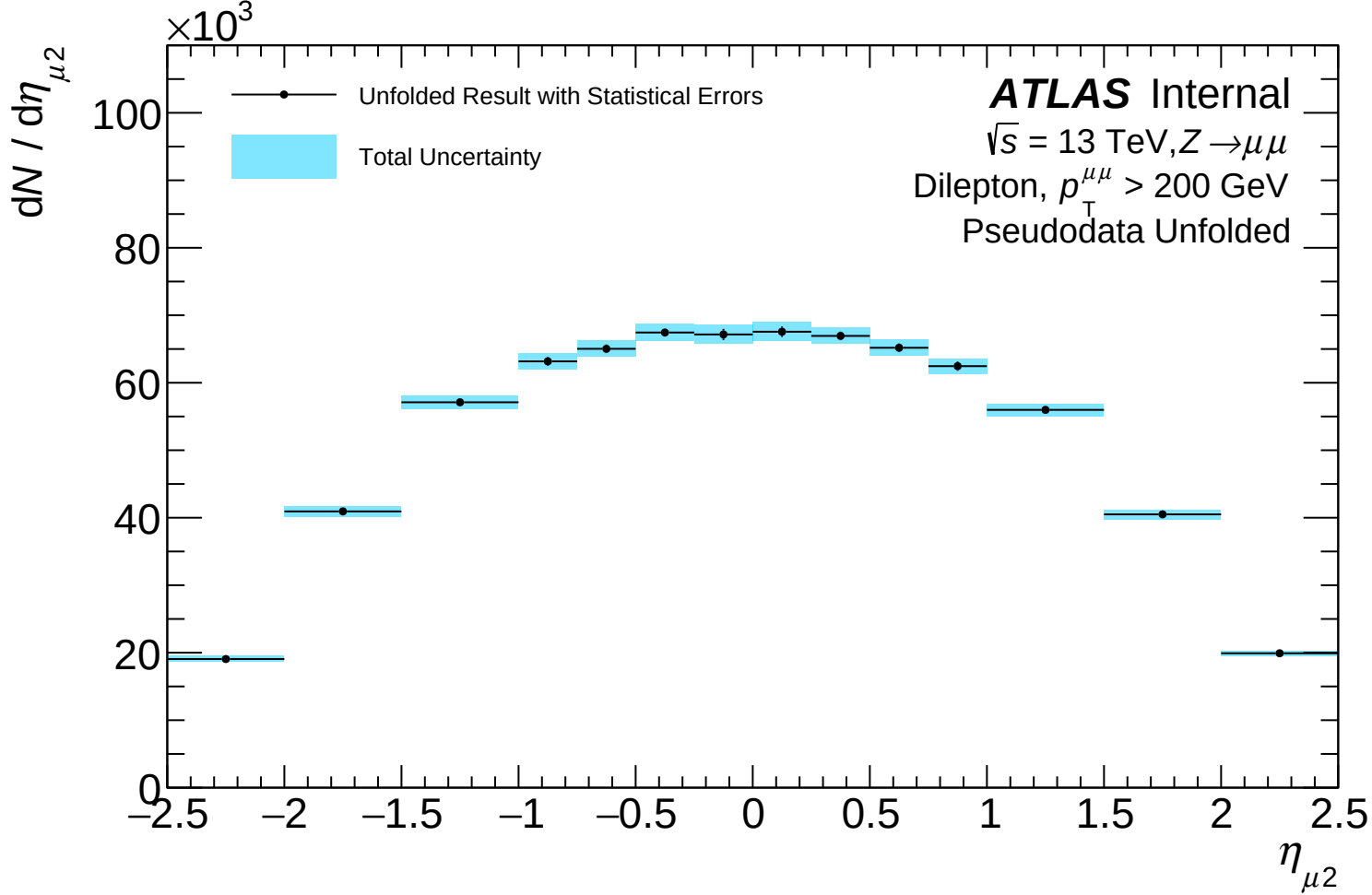
$190 < p_{\text{T}}^{\mu\mu} < 200$

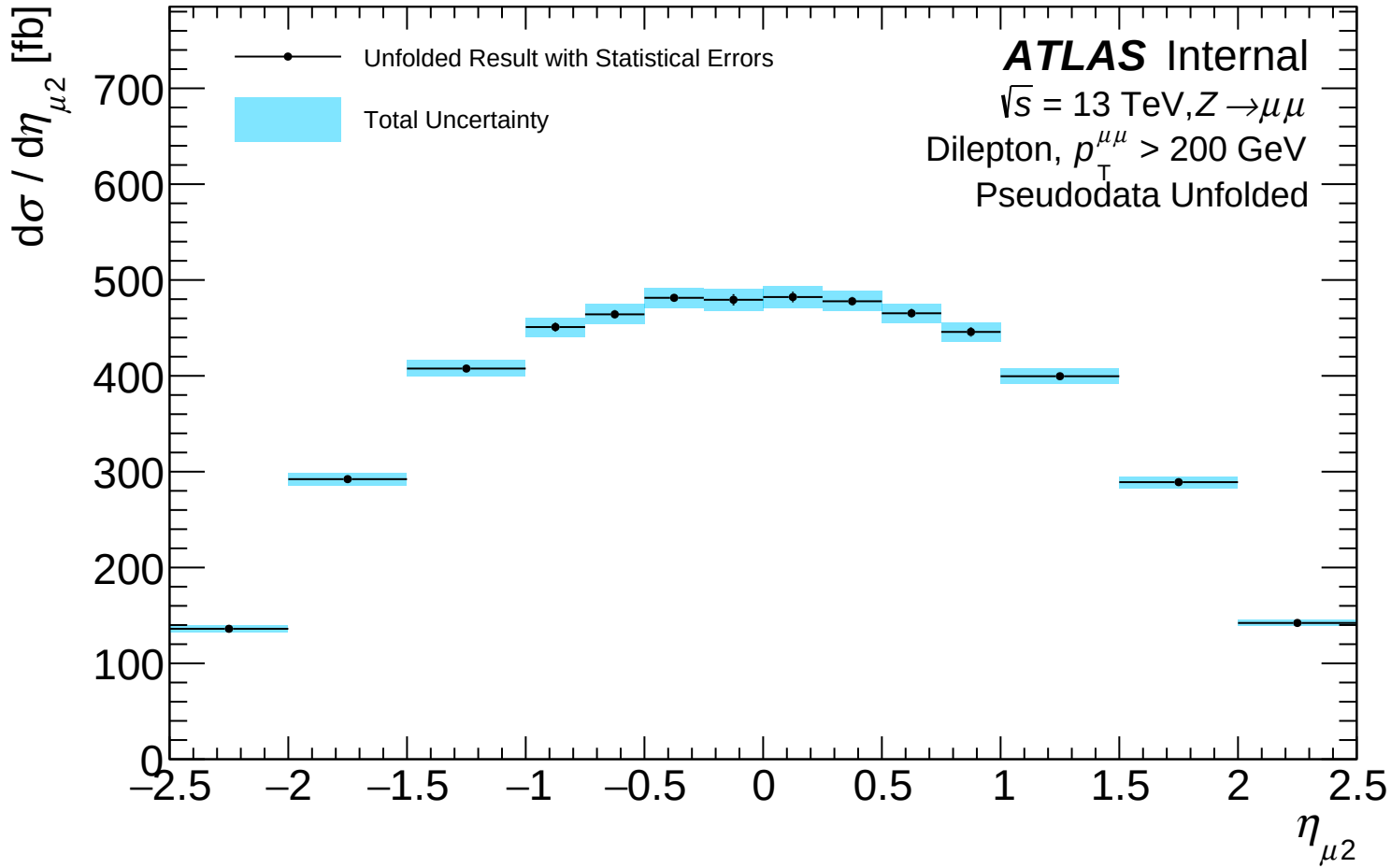




Events







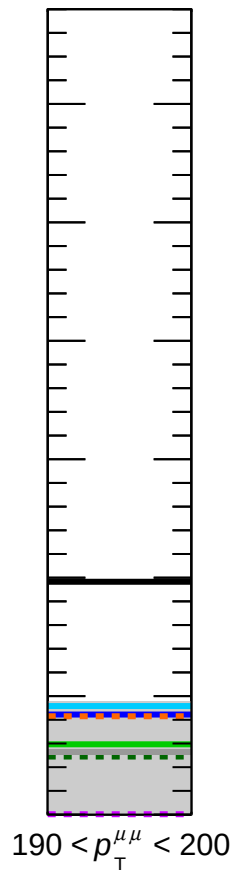
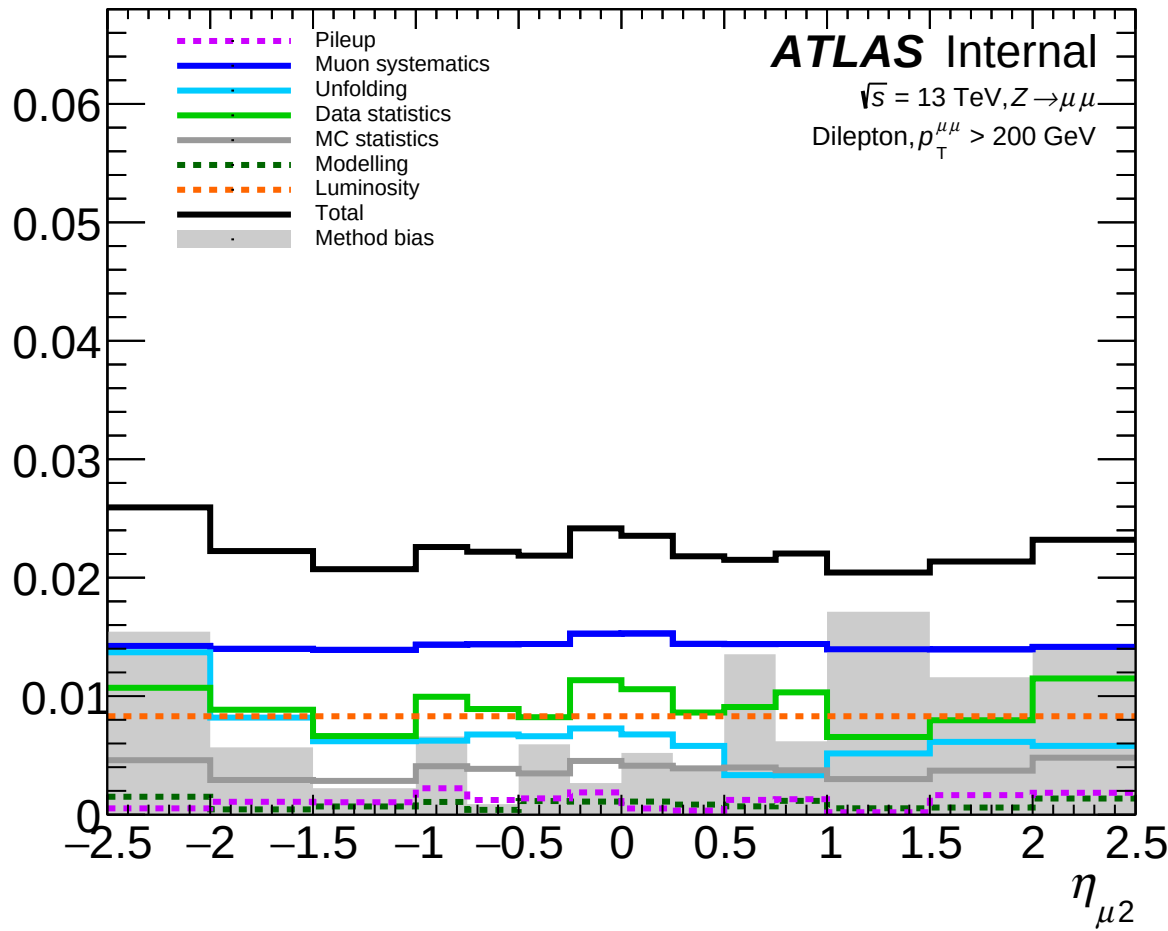
Fractional uncertainty

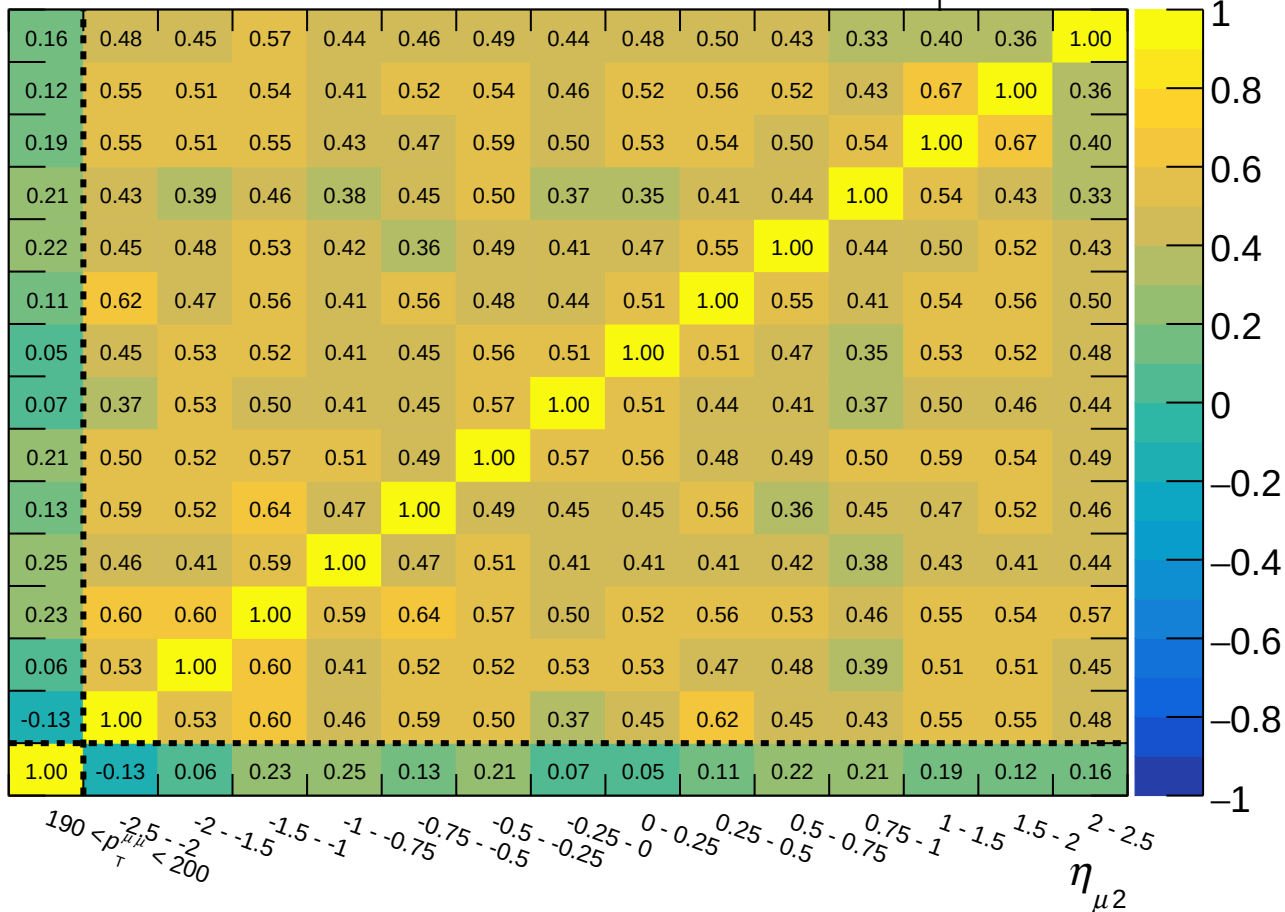
ATLAS Internal

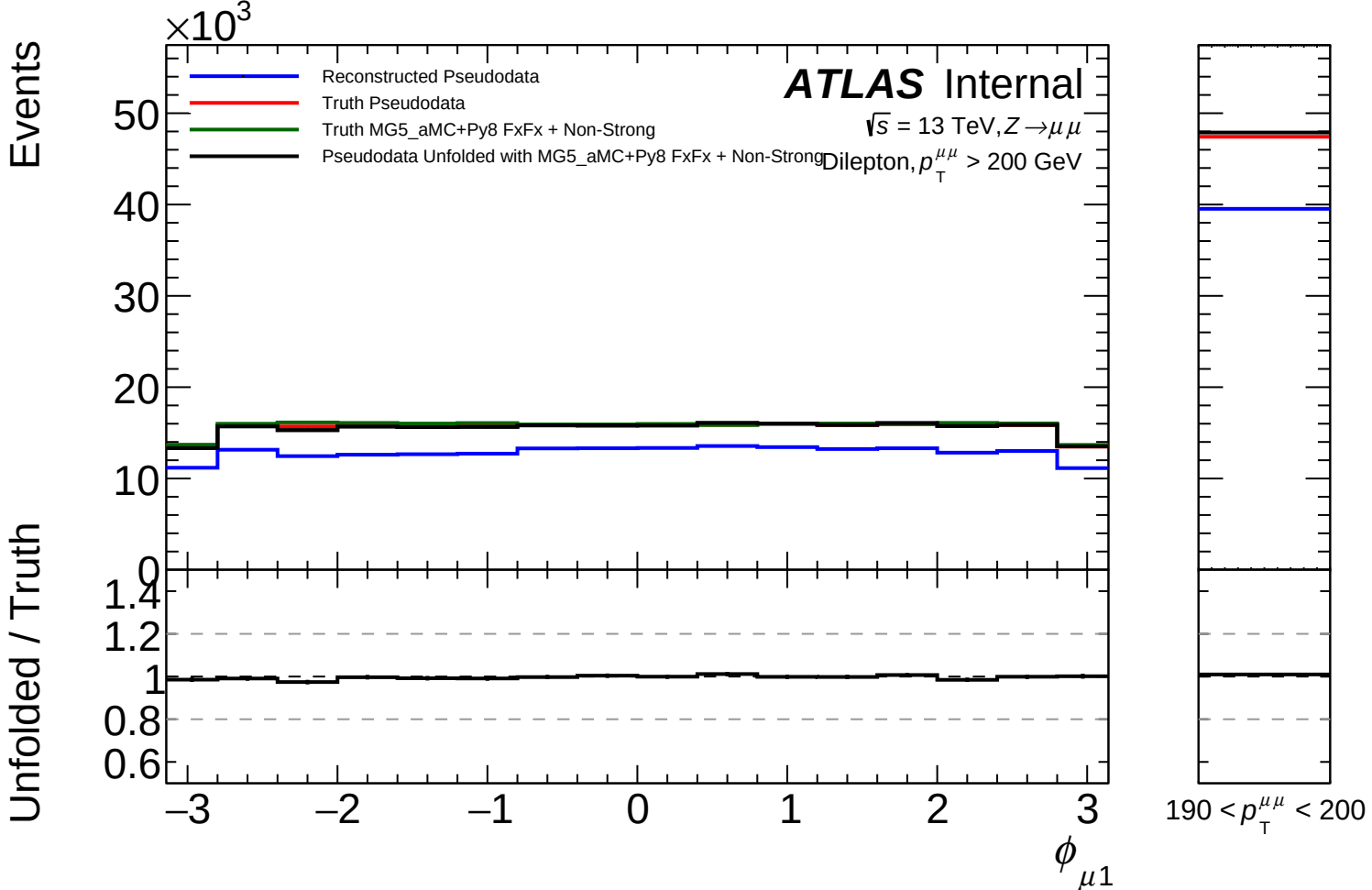
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

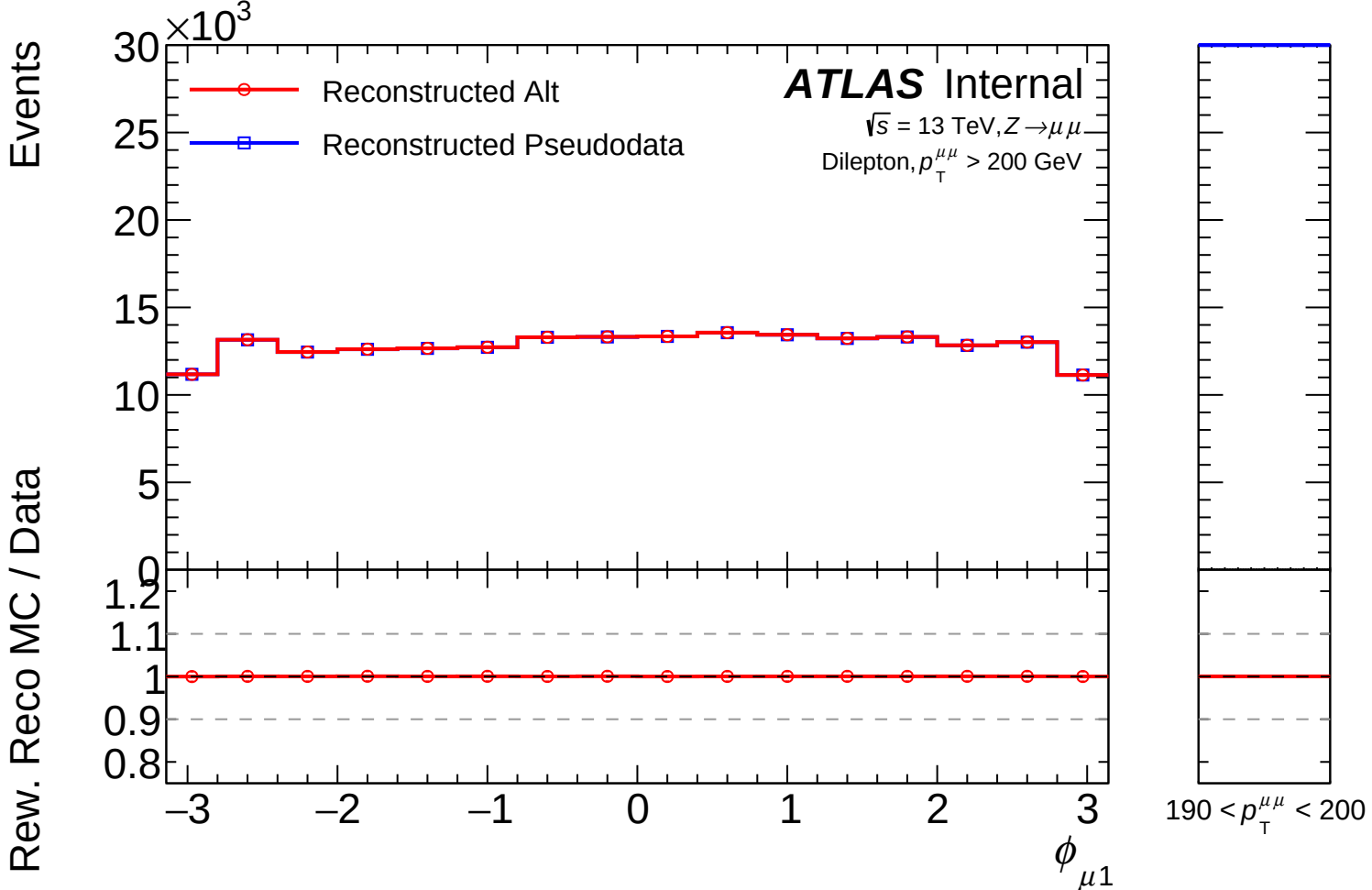
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

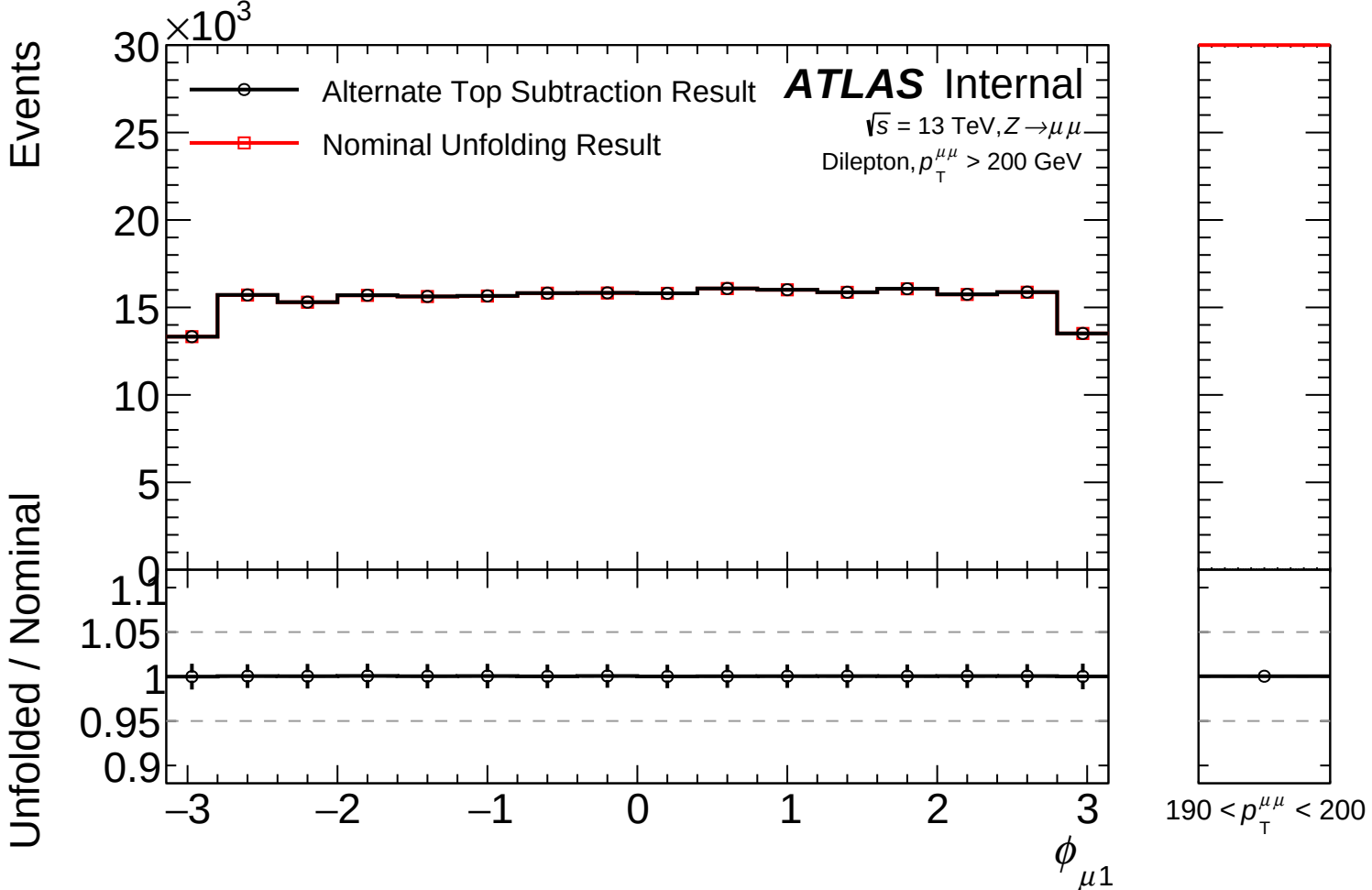
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $\eta_{\mu 2}$ $190 < p_{\text{T}}^{\mu\mu} < 200$ 







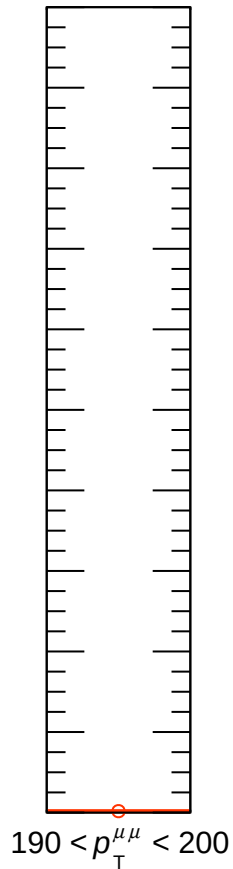
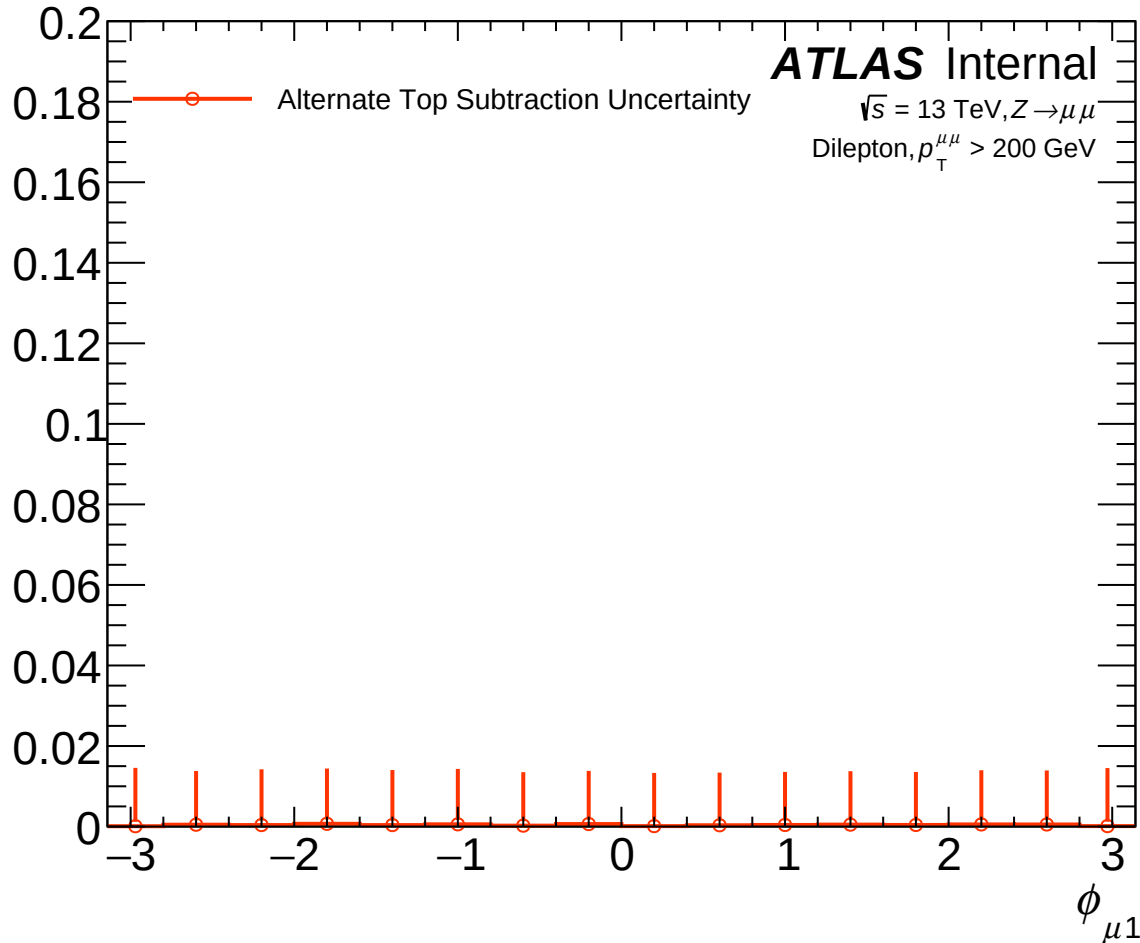
Fractional systematic impact

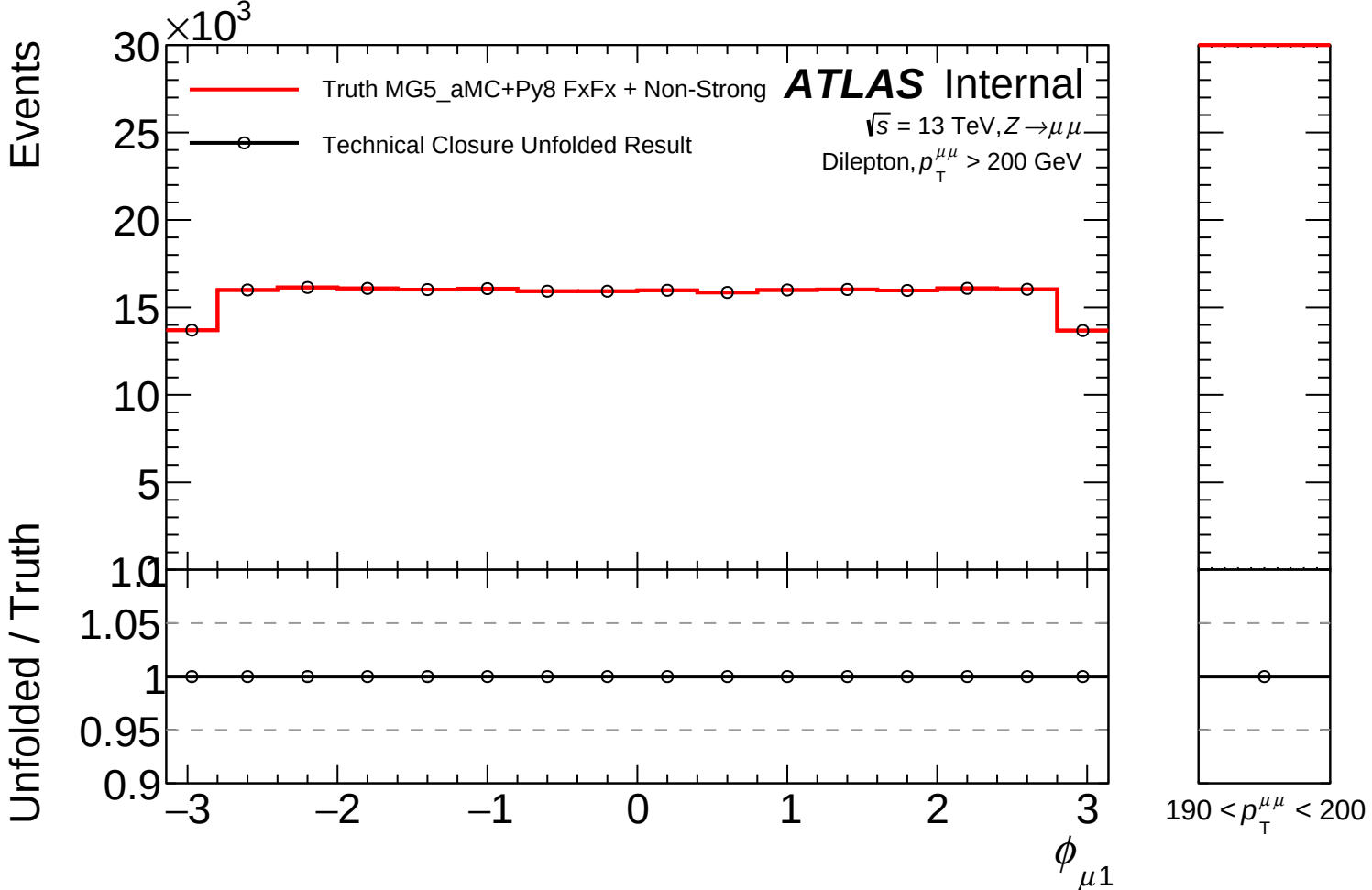
ATLAS Internal

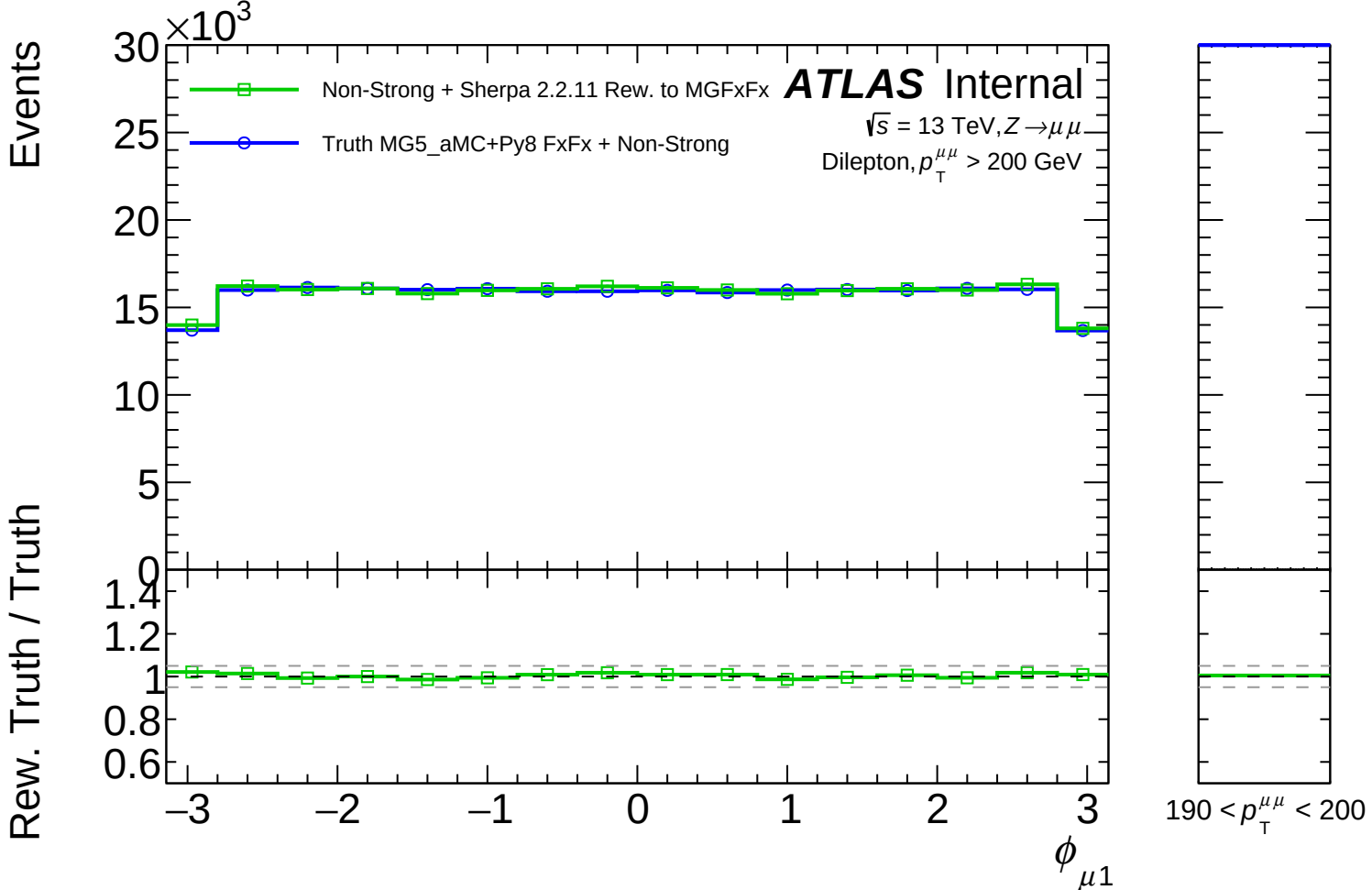
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

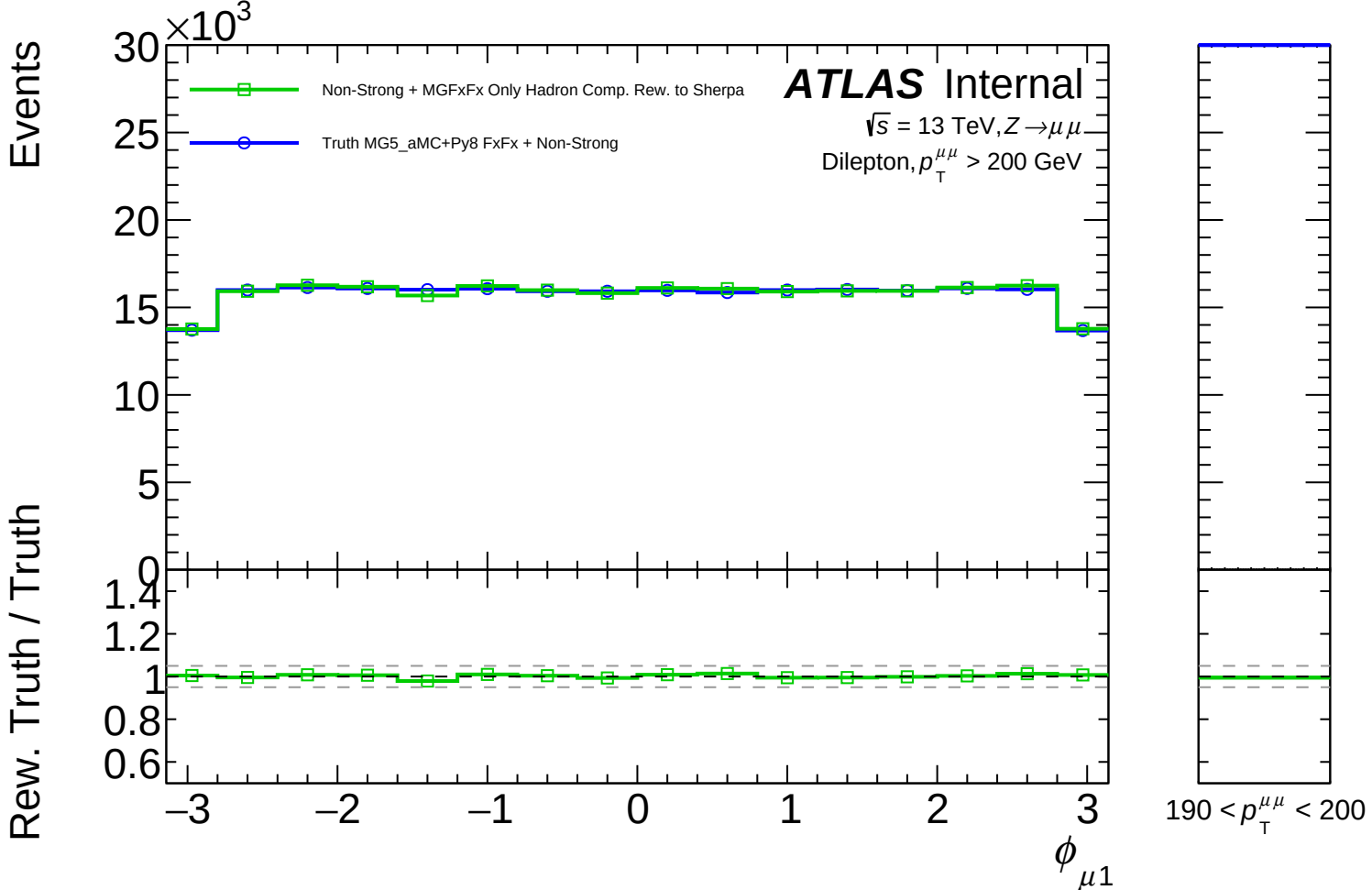
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

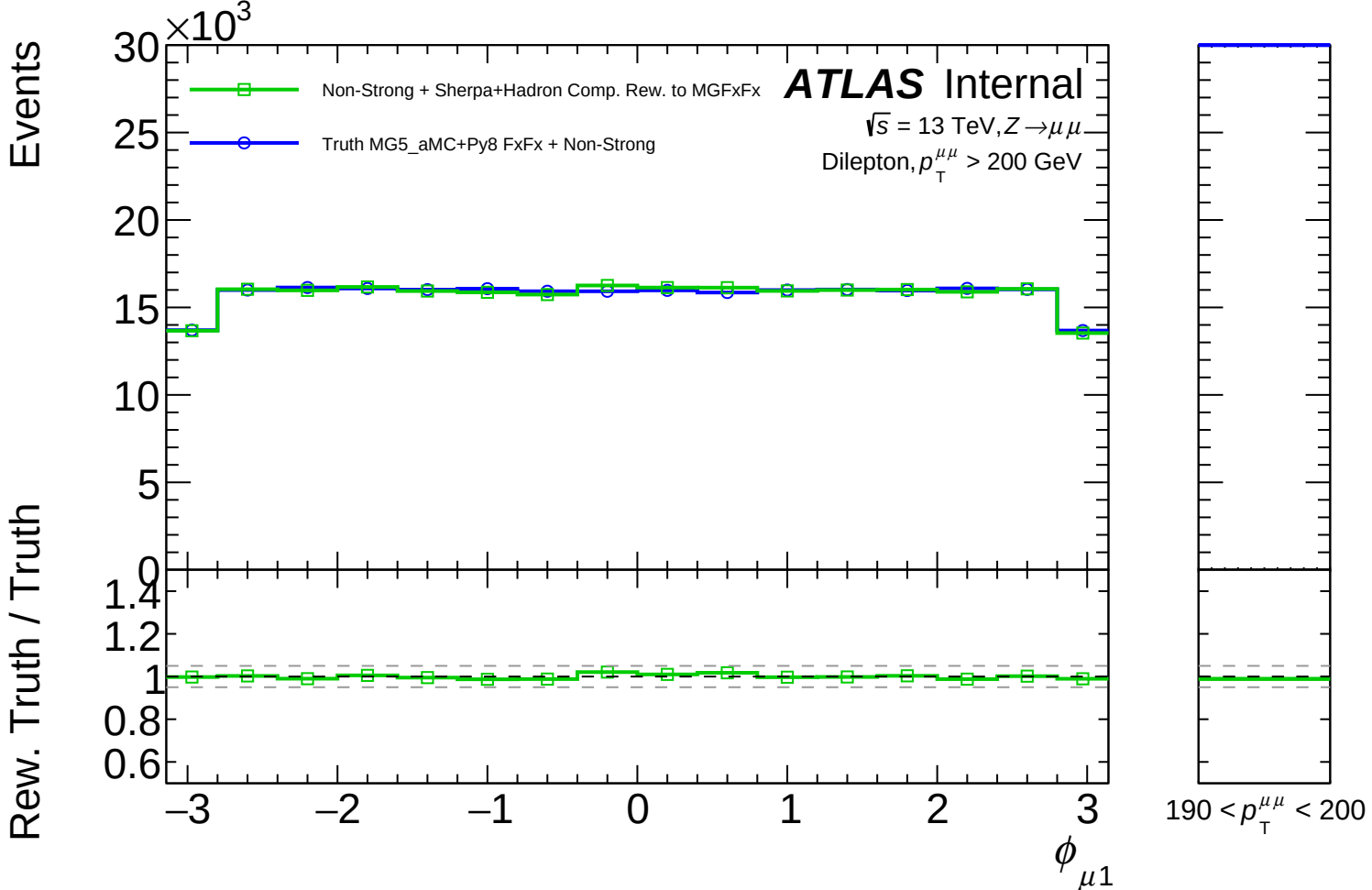
Alternate Top Subtraction Uncertainty

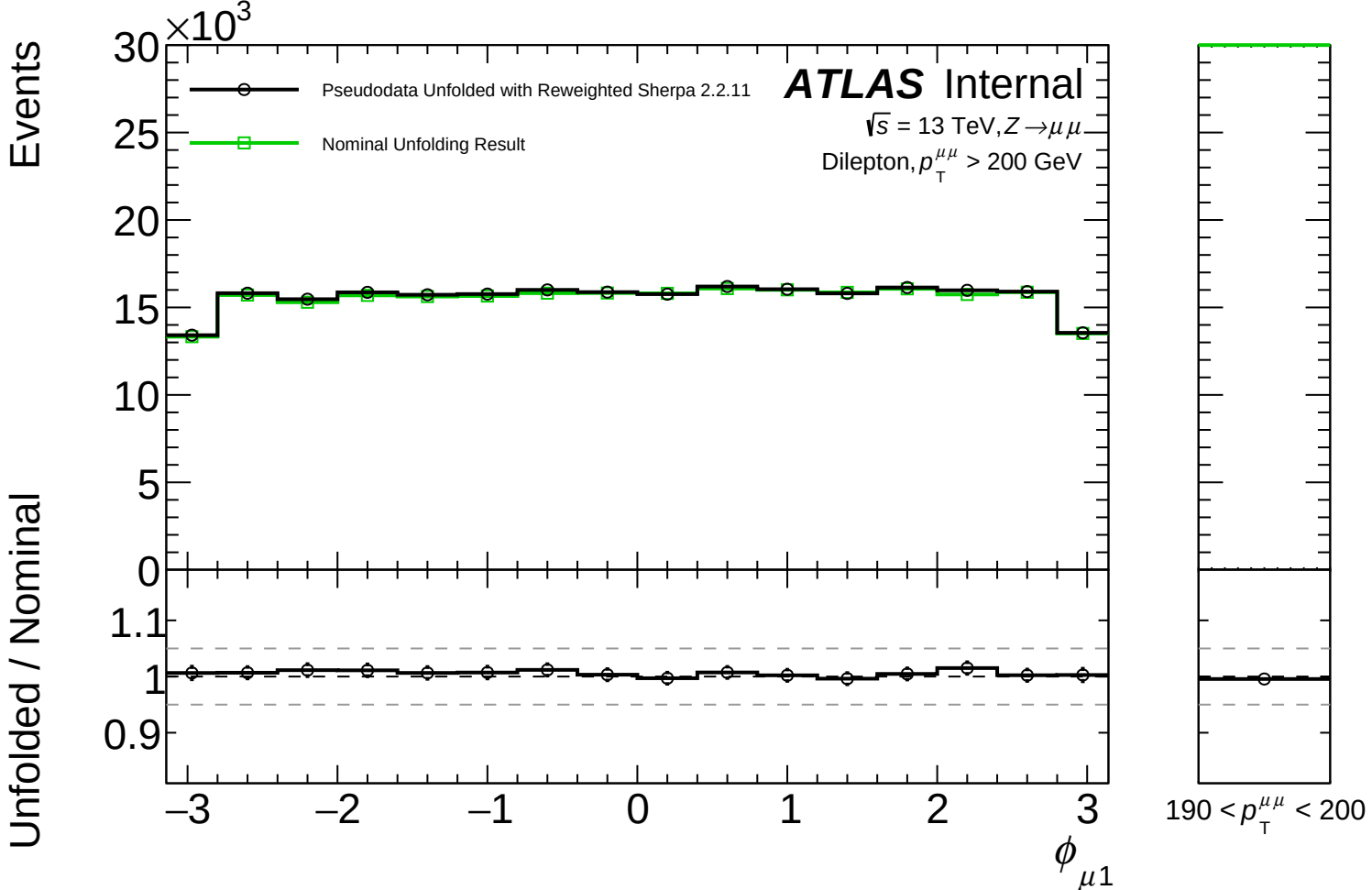


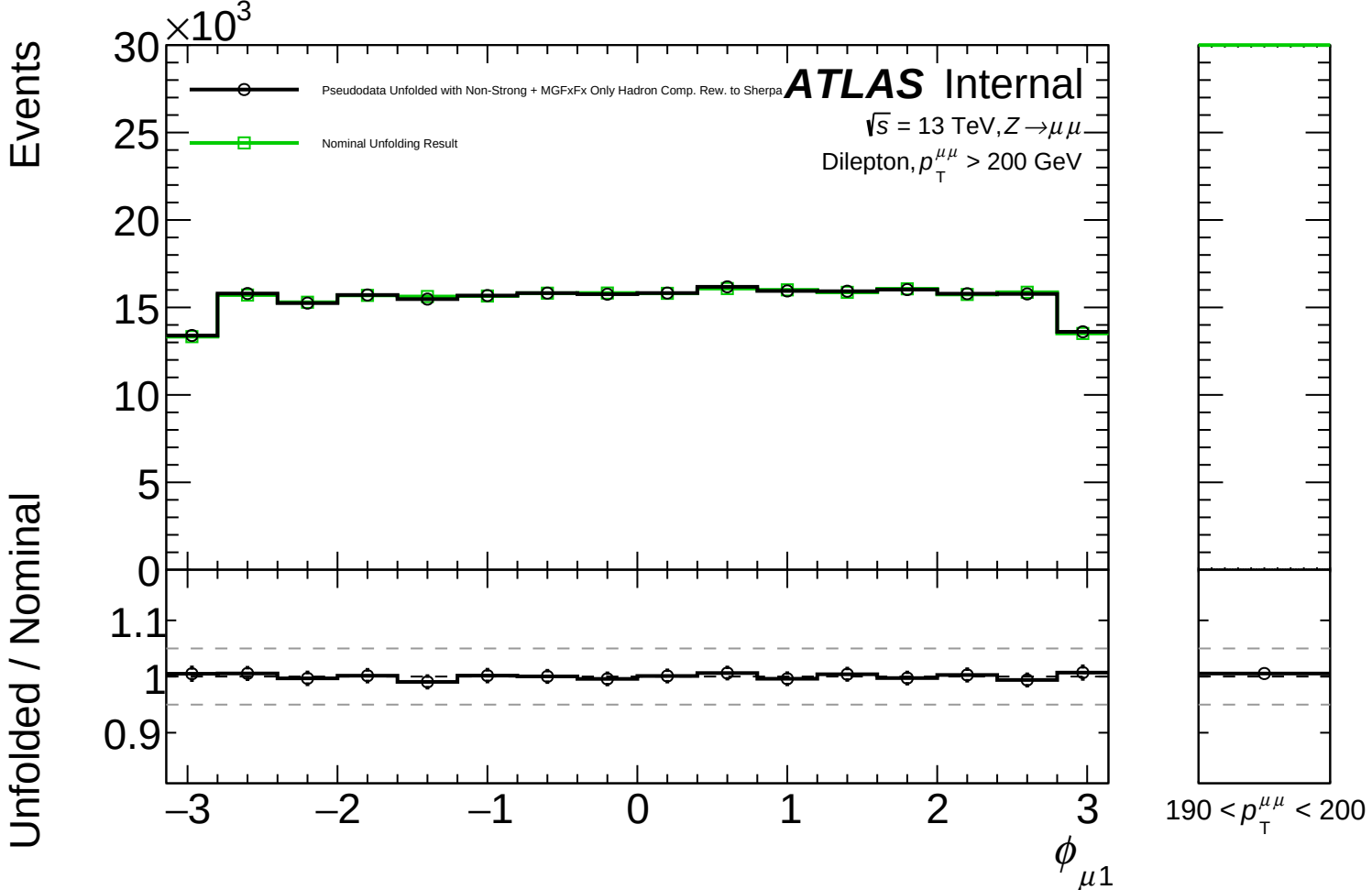


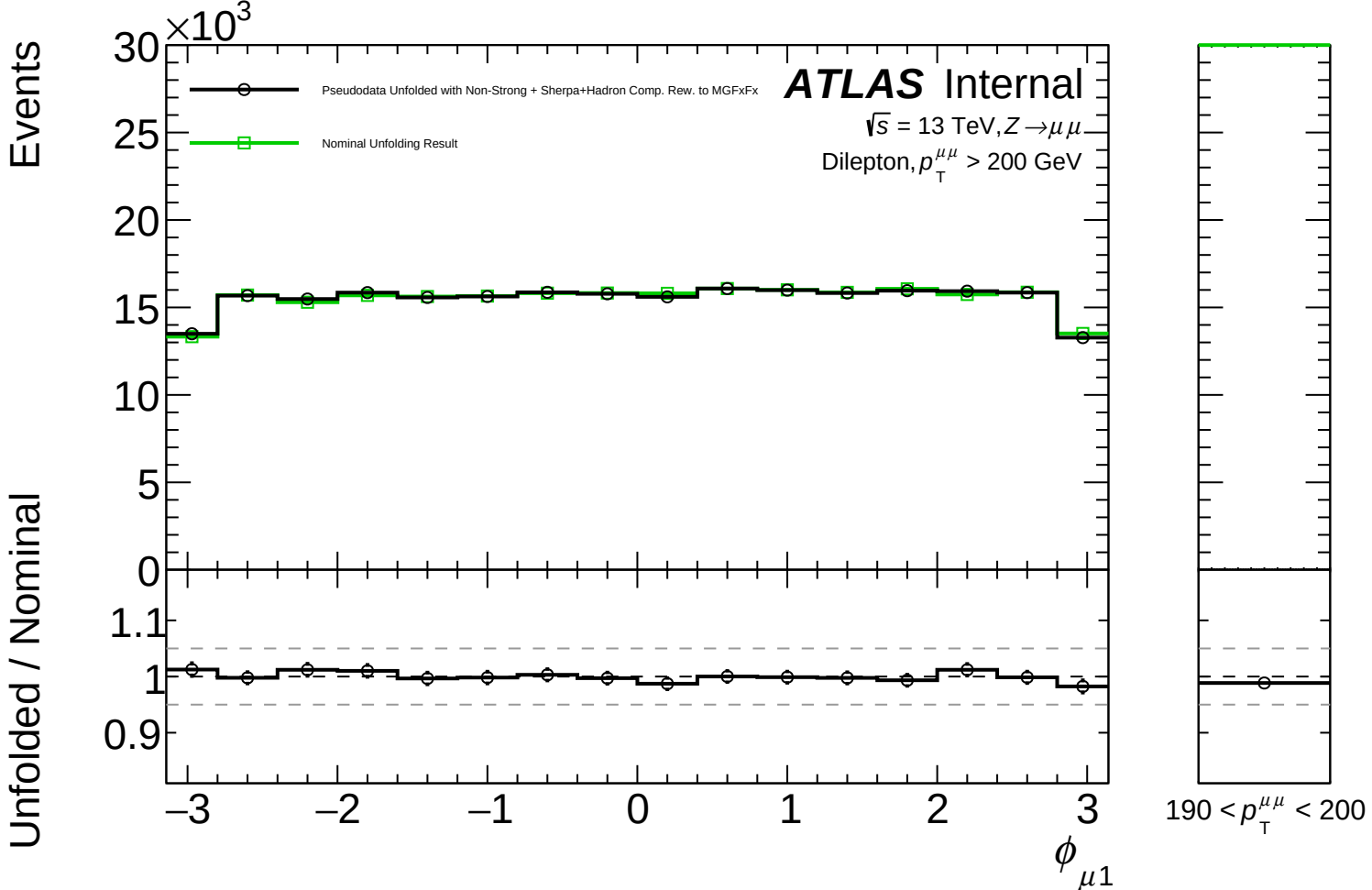


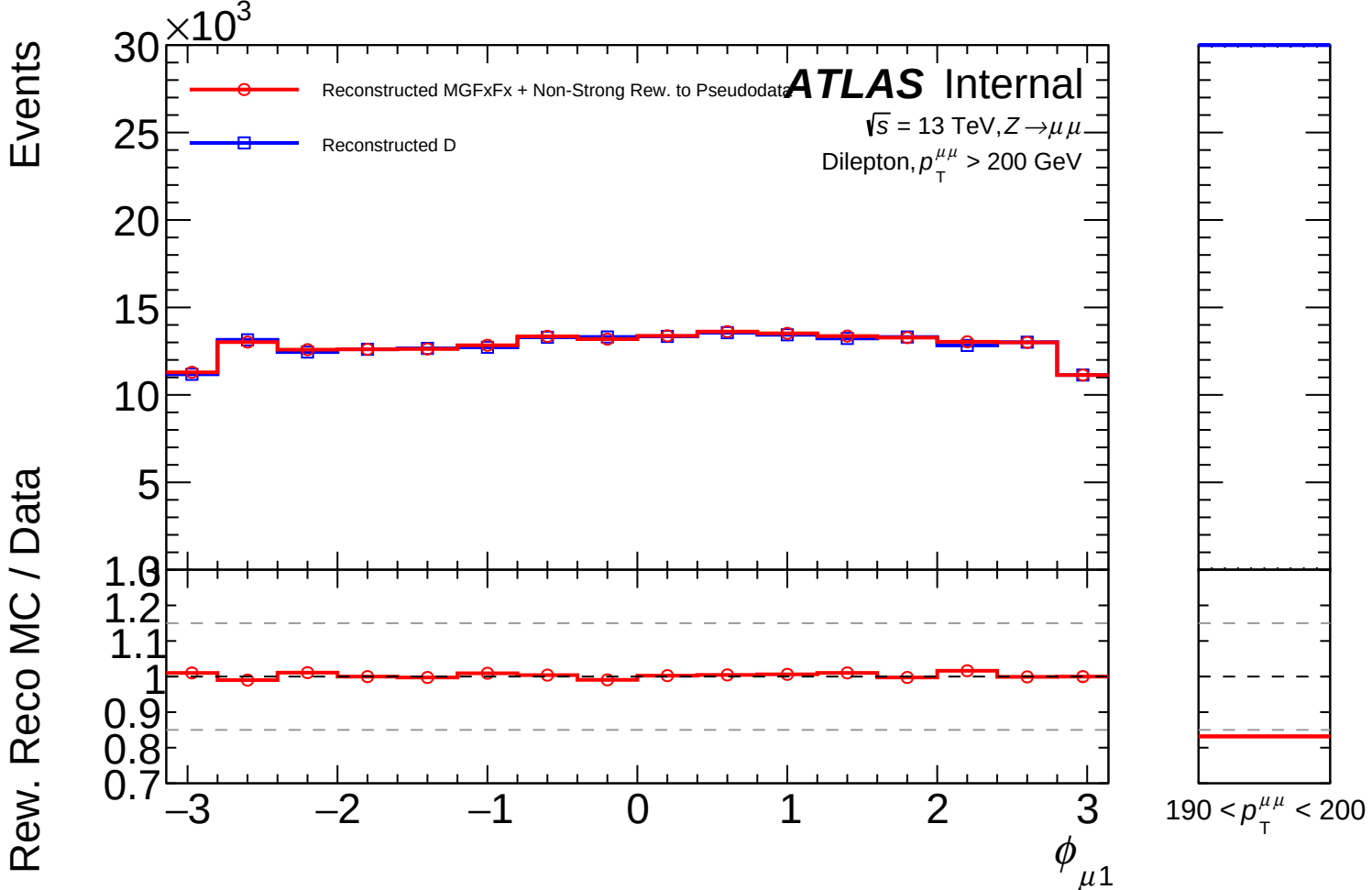


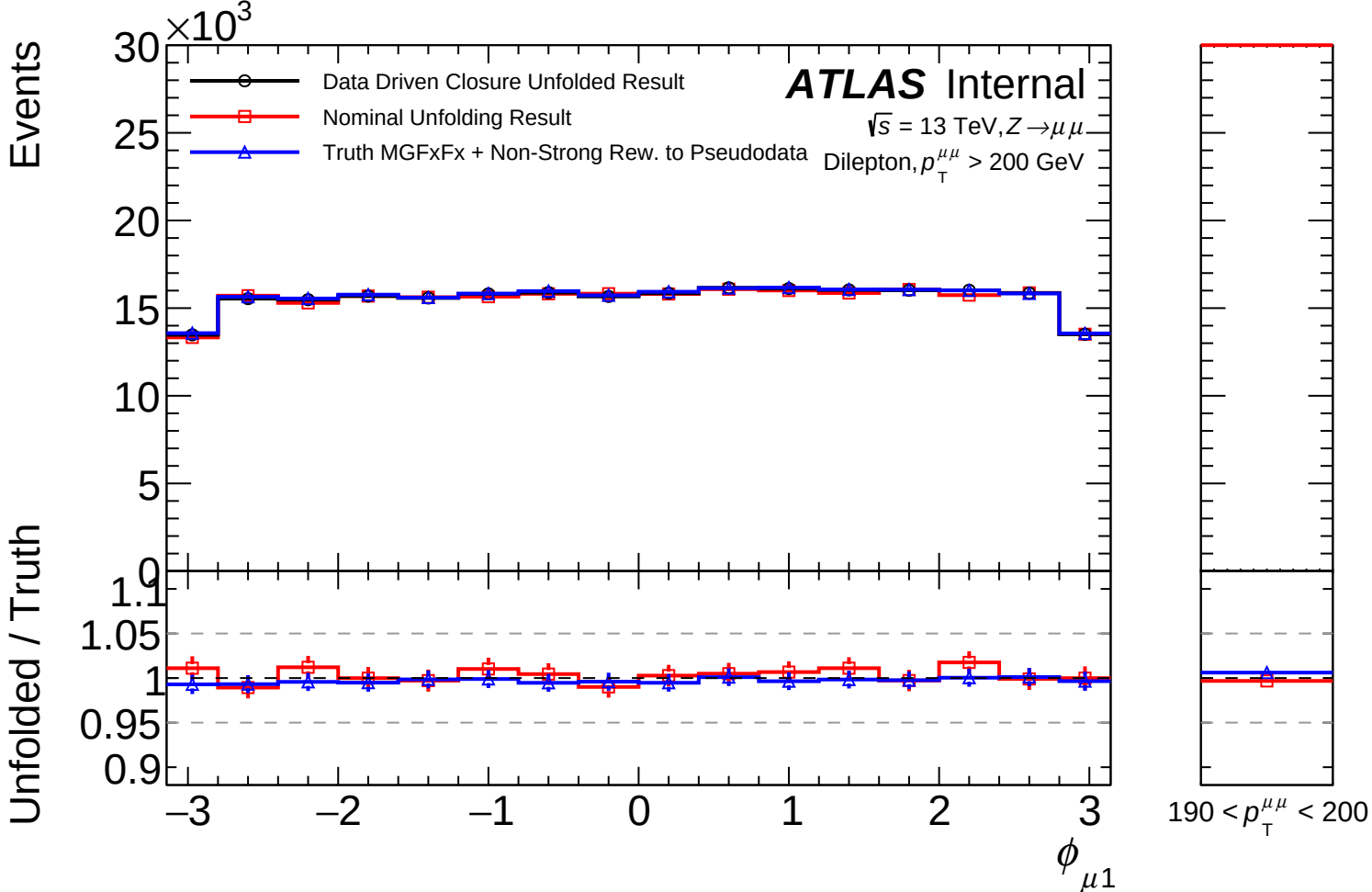




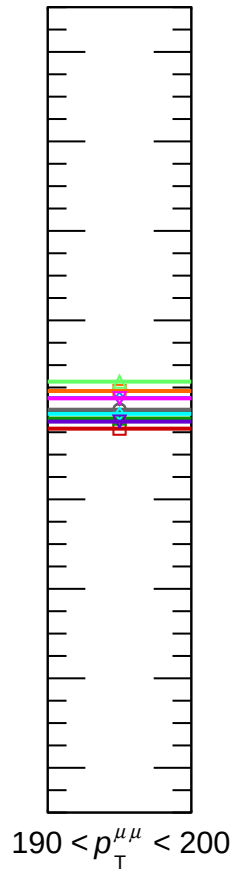
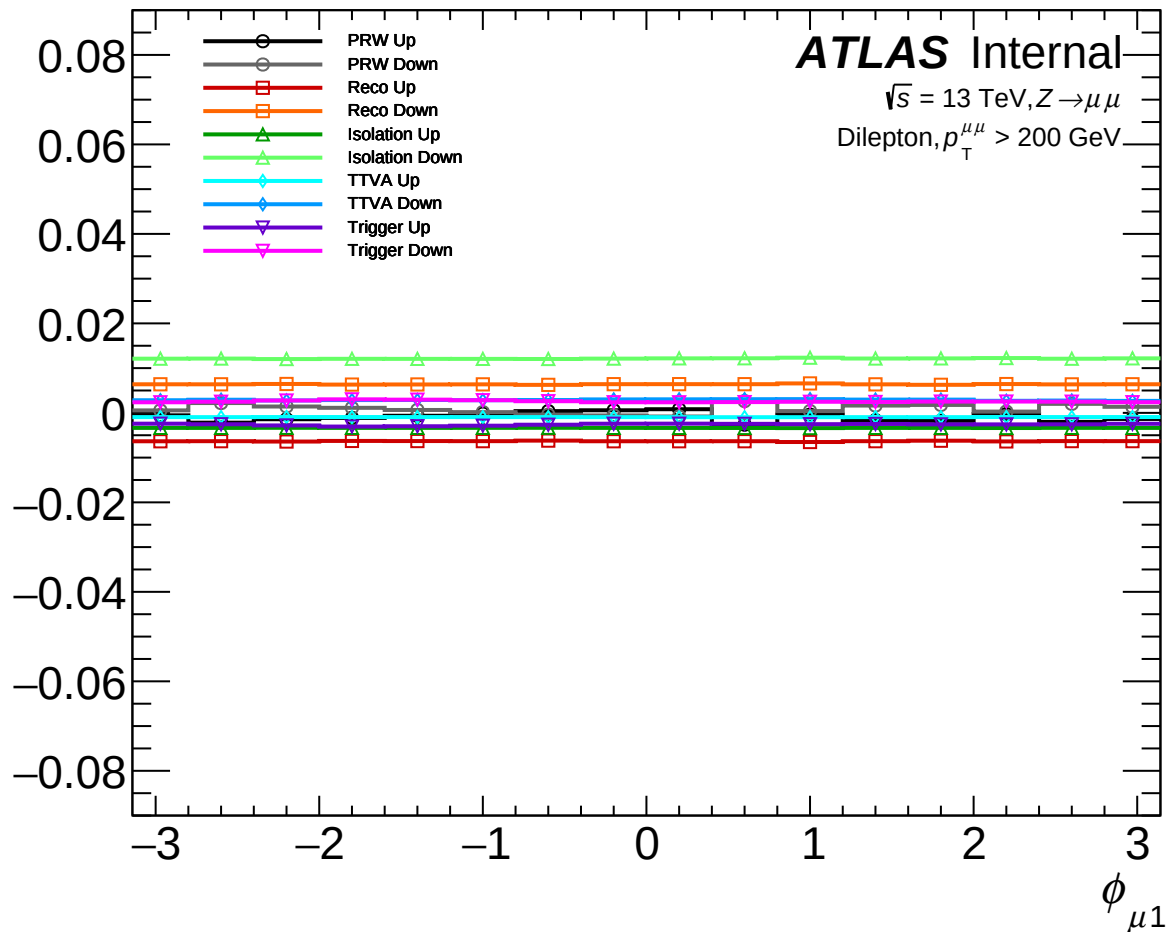




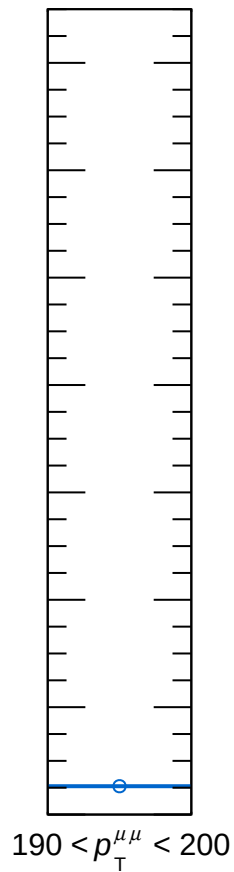
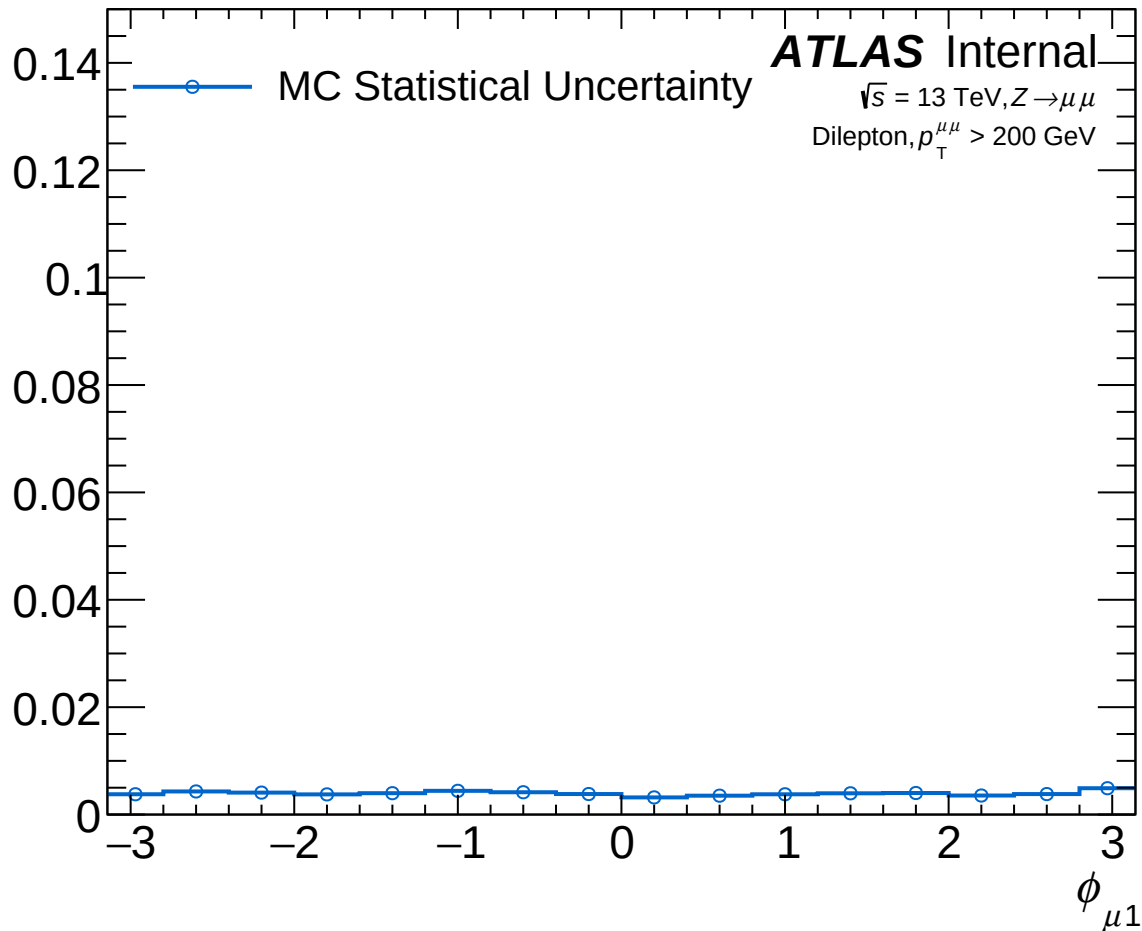




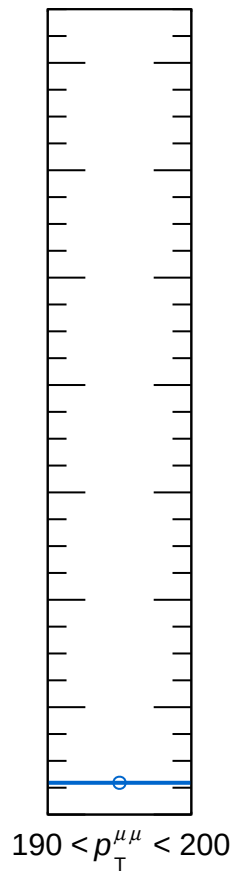
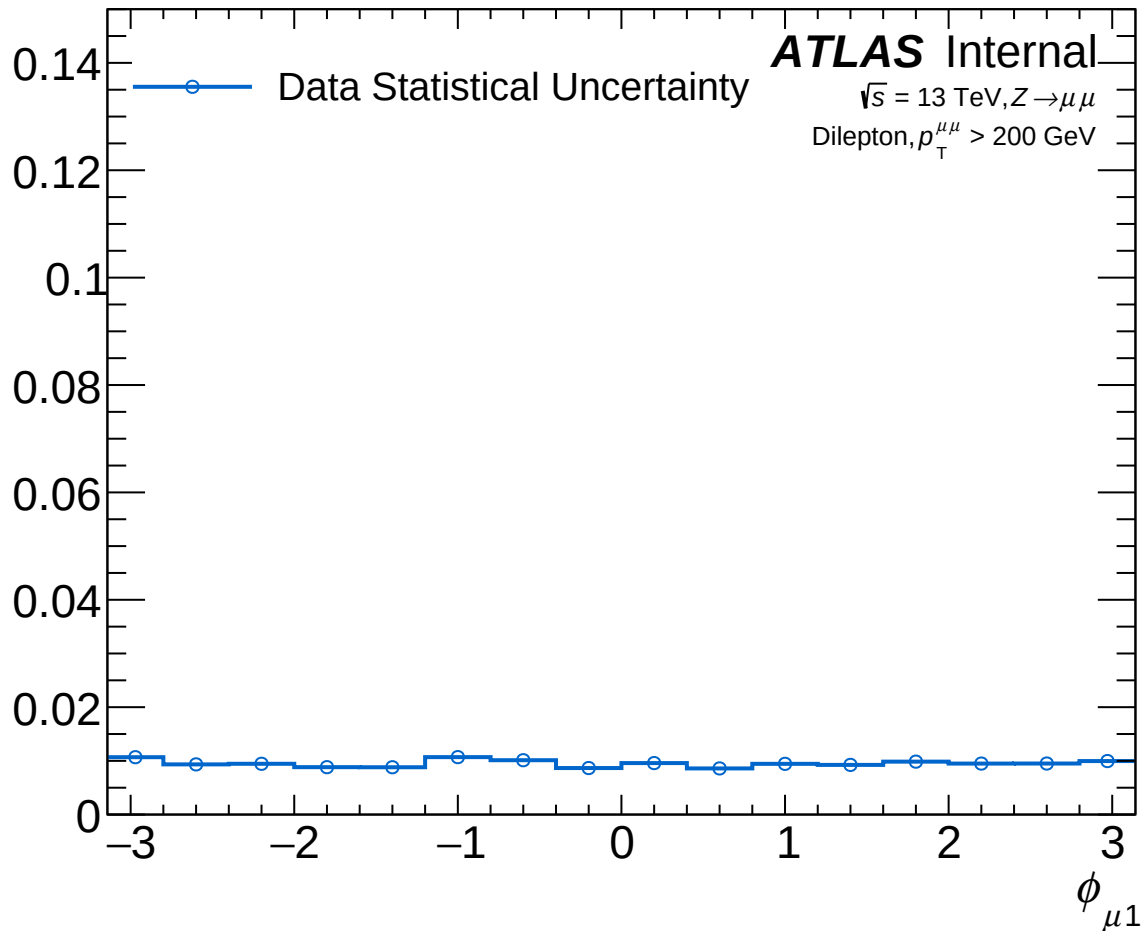
Fractional systematic impact



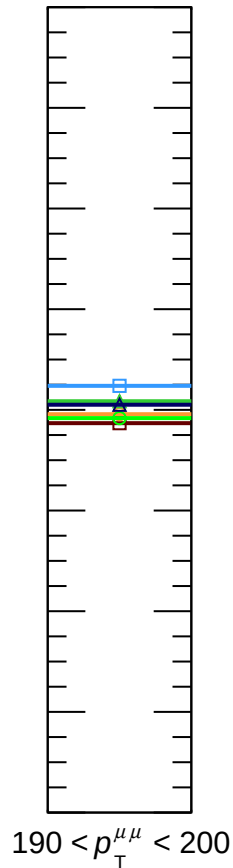
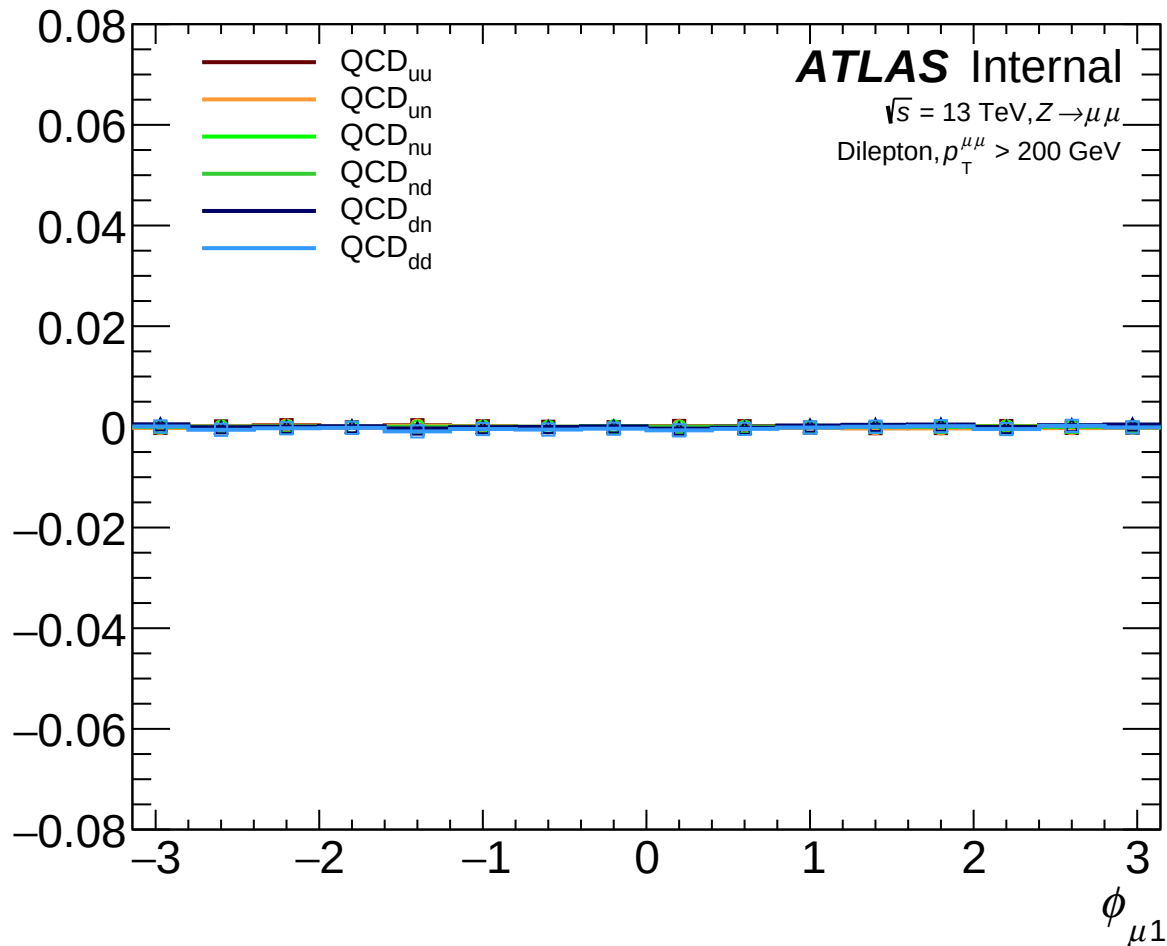
Fractional systematic impact



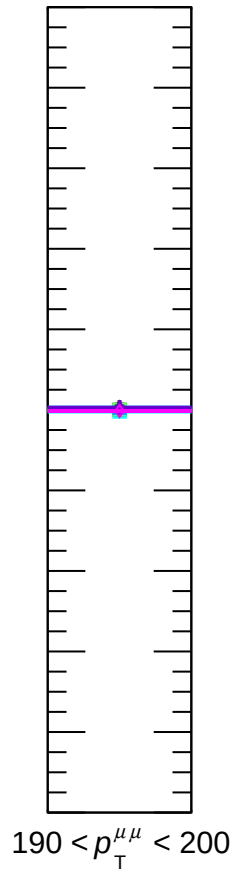
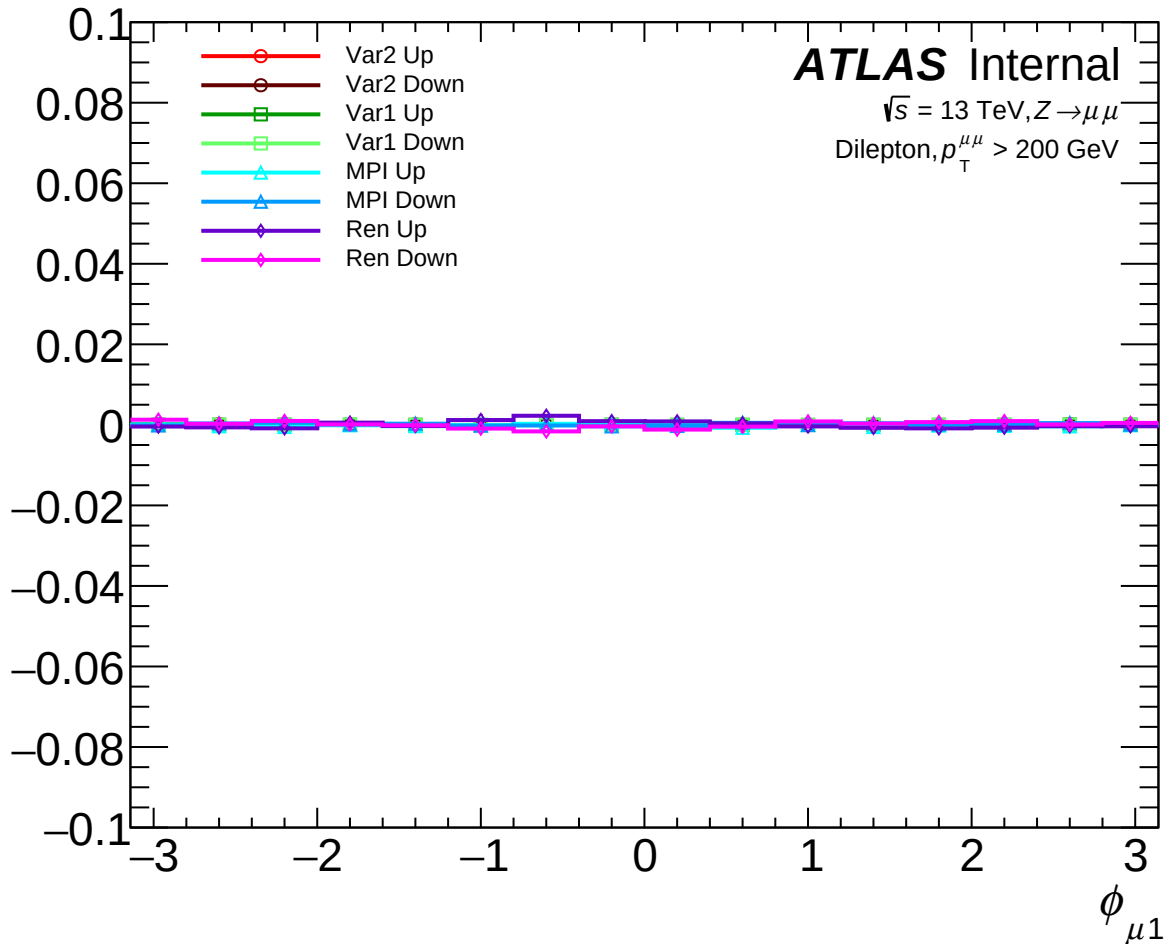
Fractional systematic impact



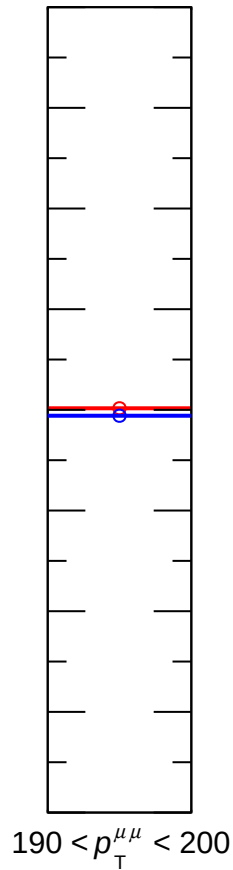
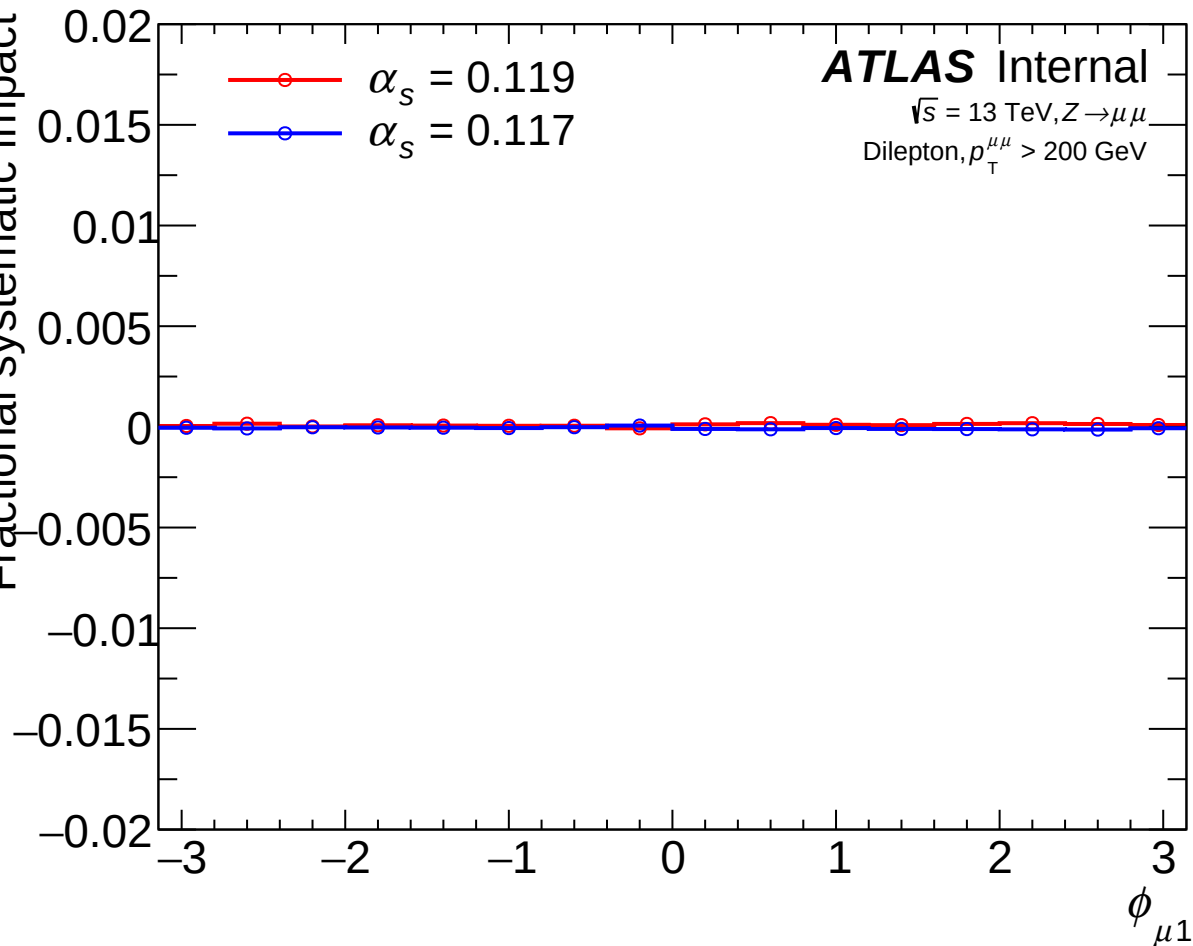
Fractional systematic impact

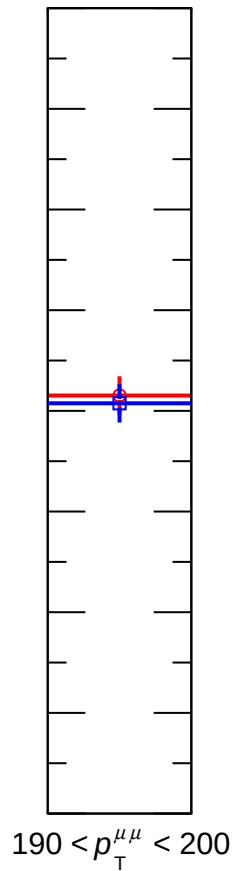
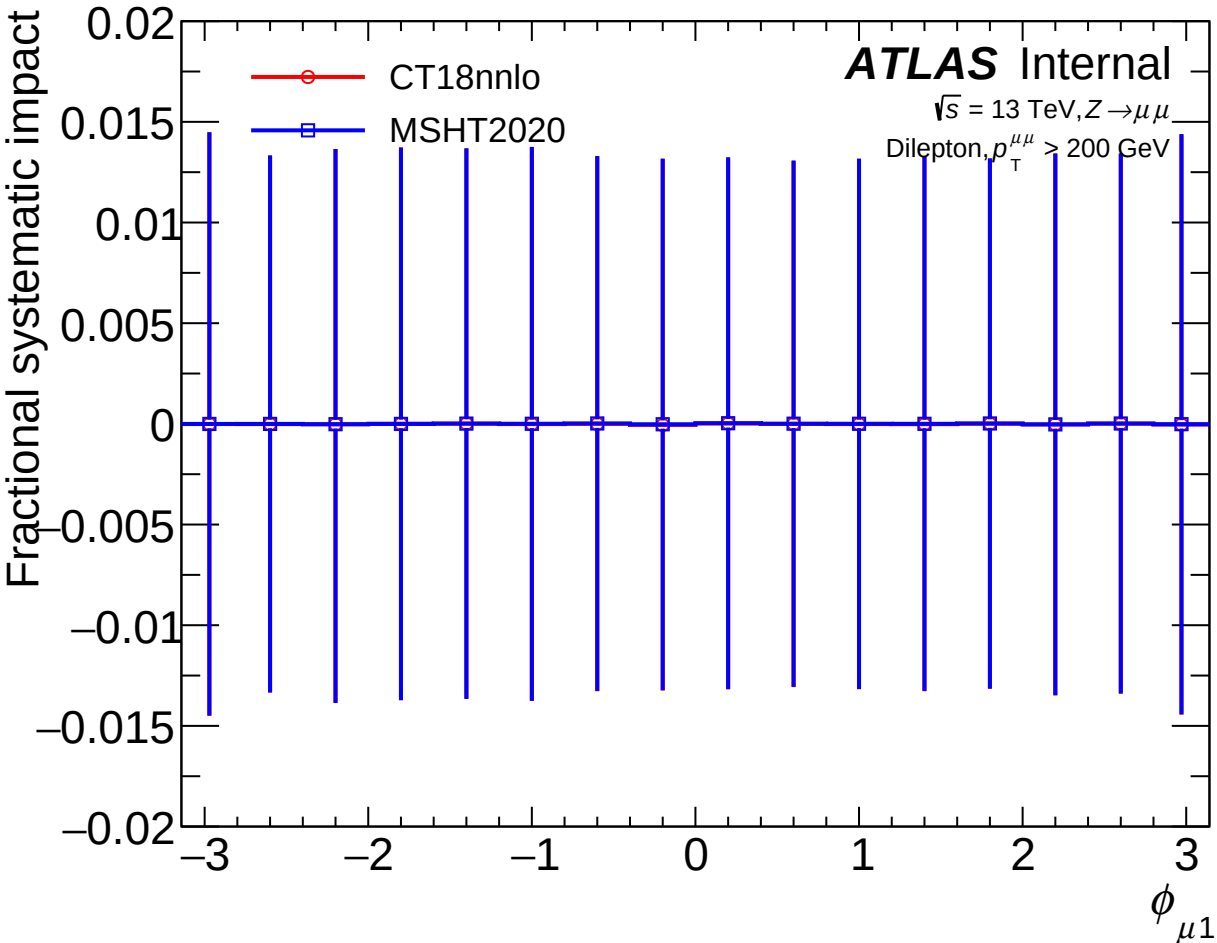


Fractional systematic impact

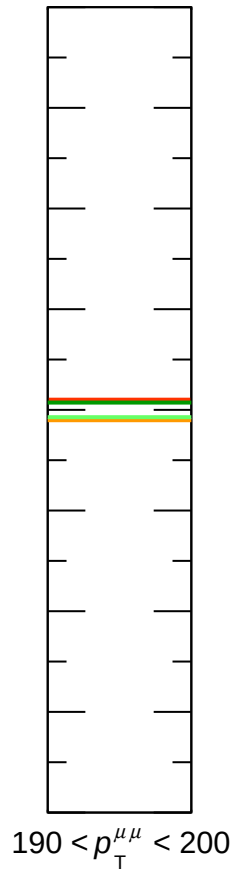
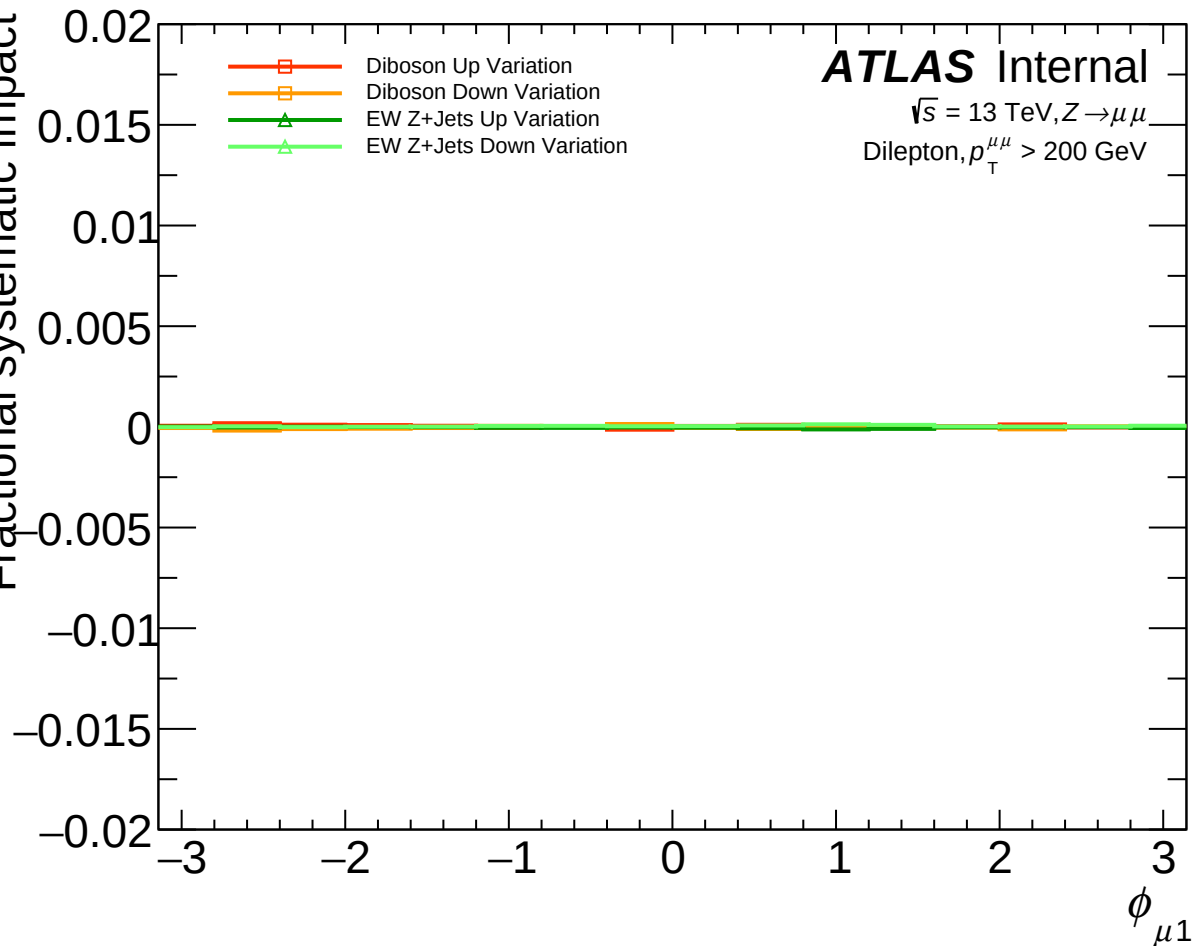


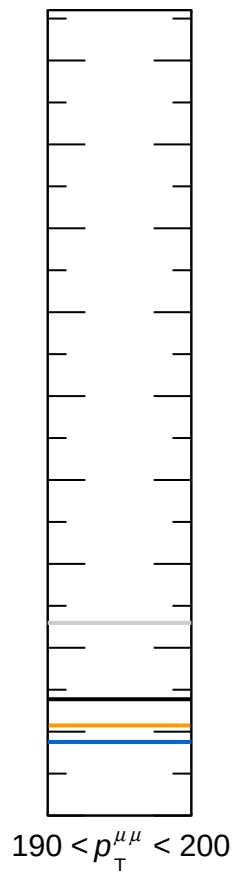
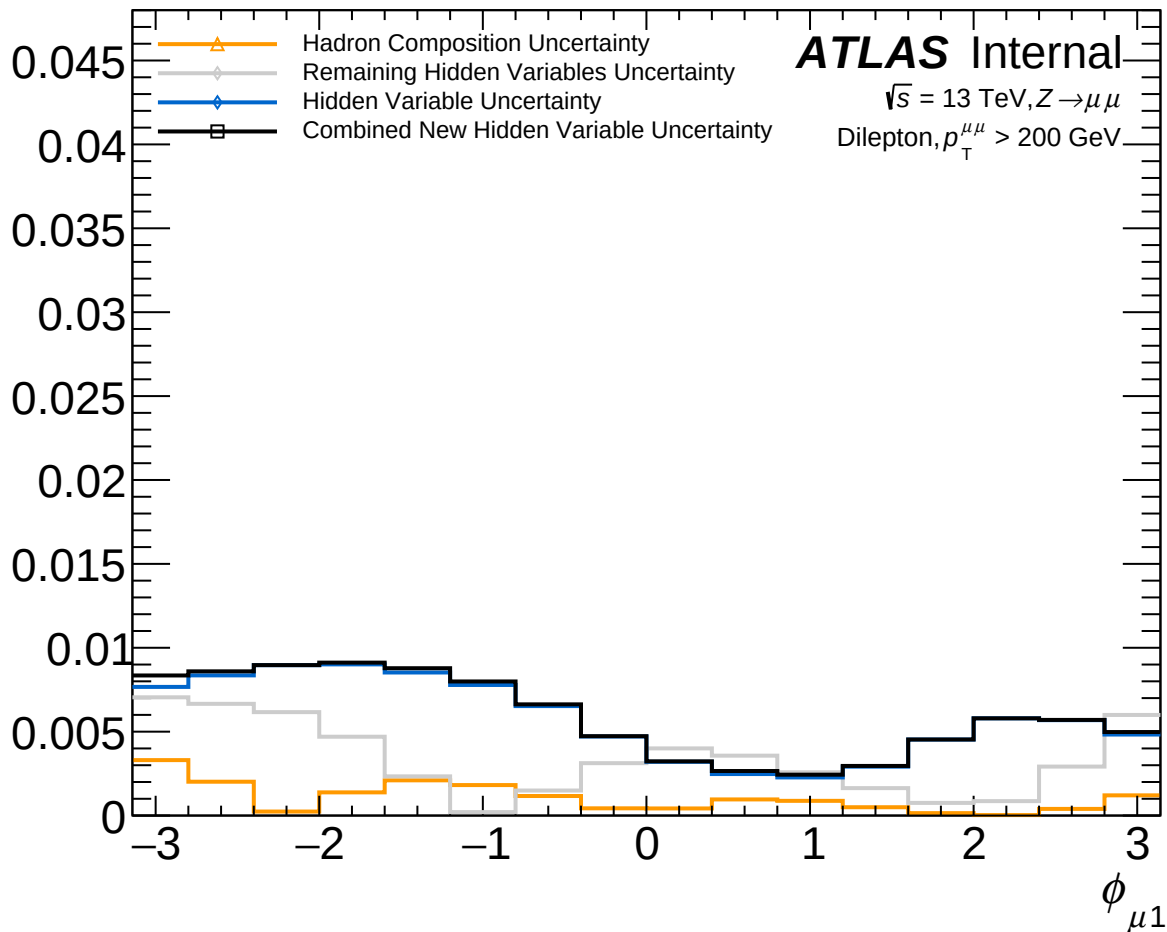
Fractional systematic impact

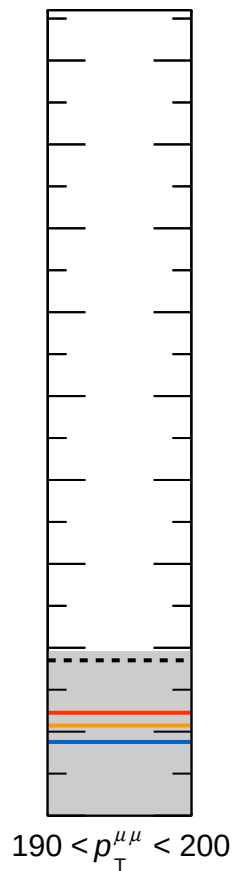
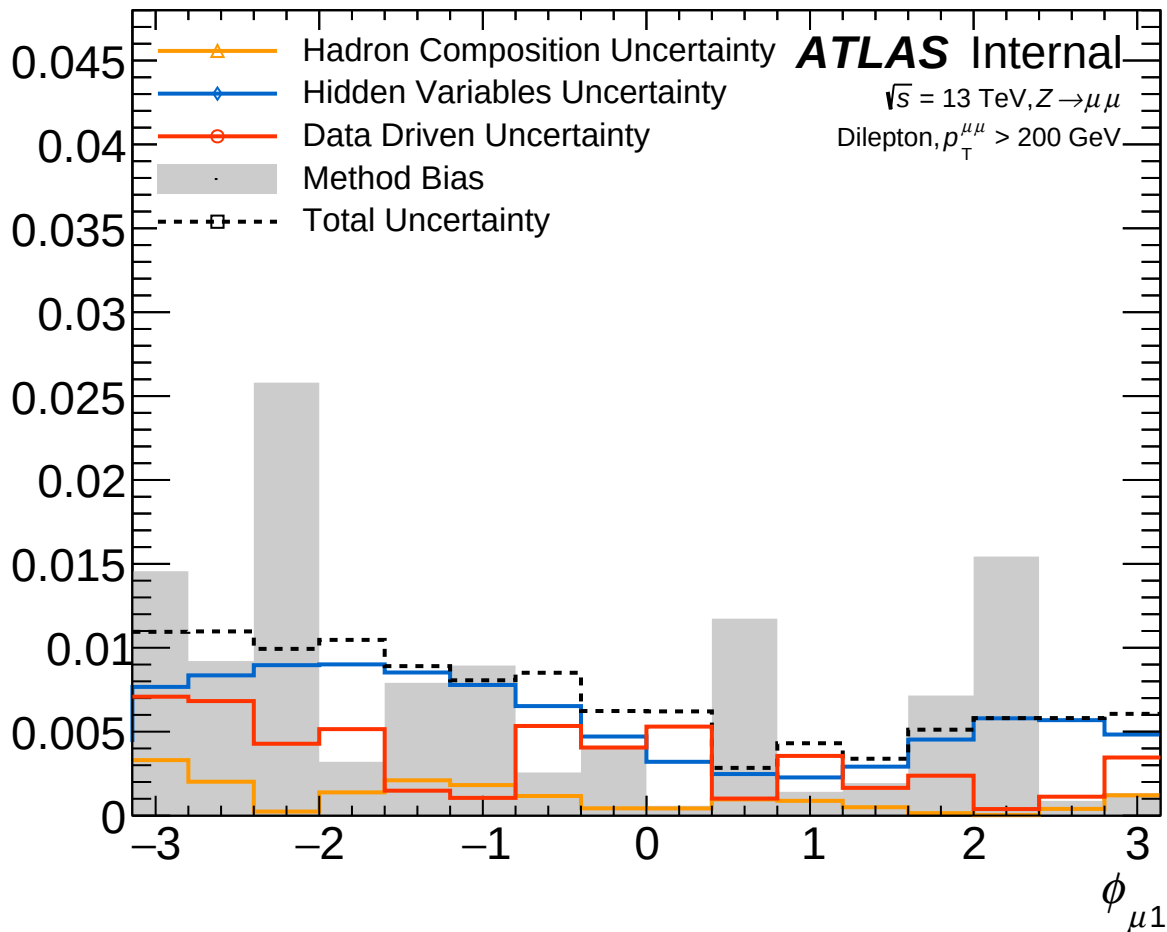




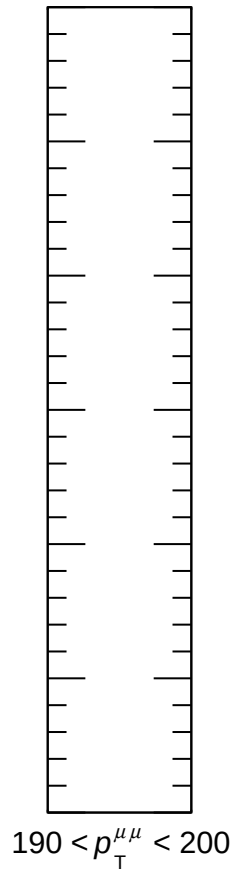
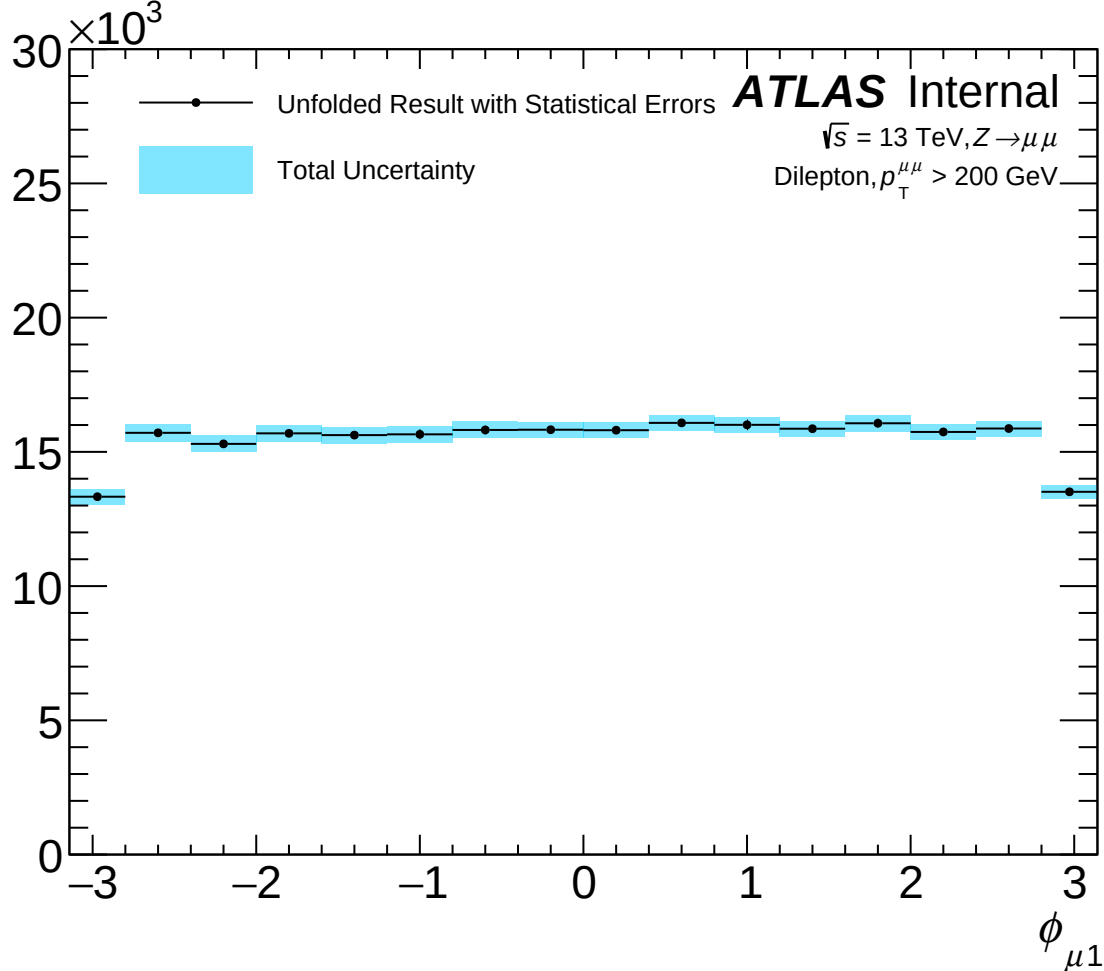
Fractional systematic impact

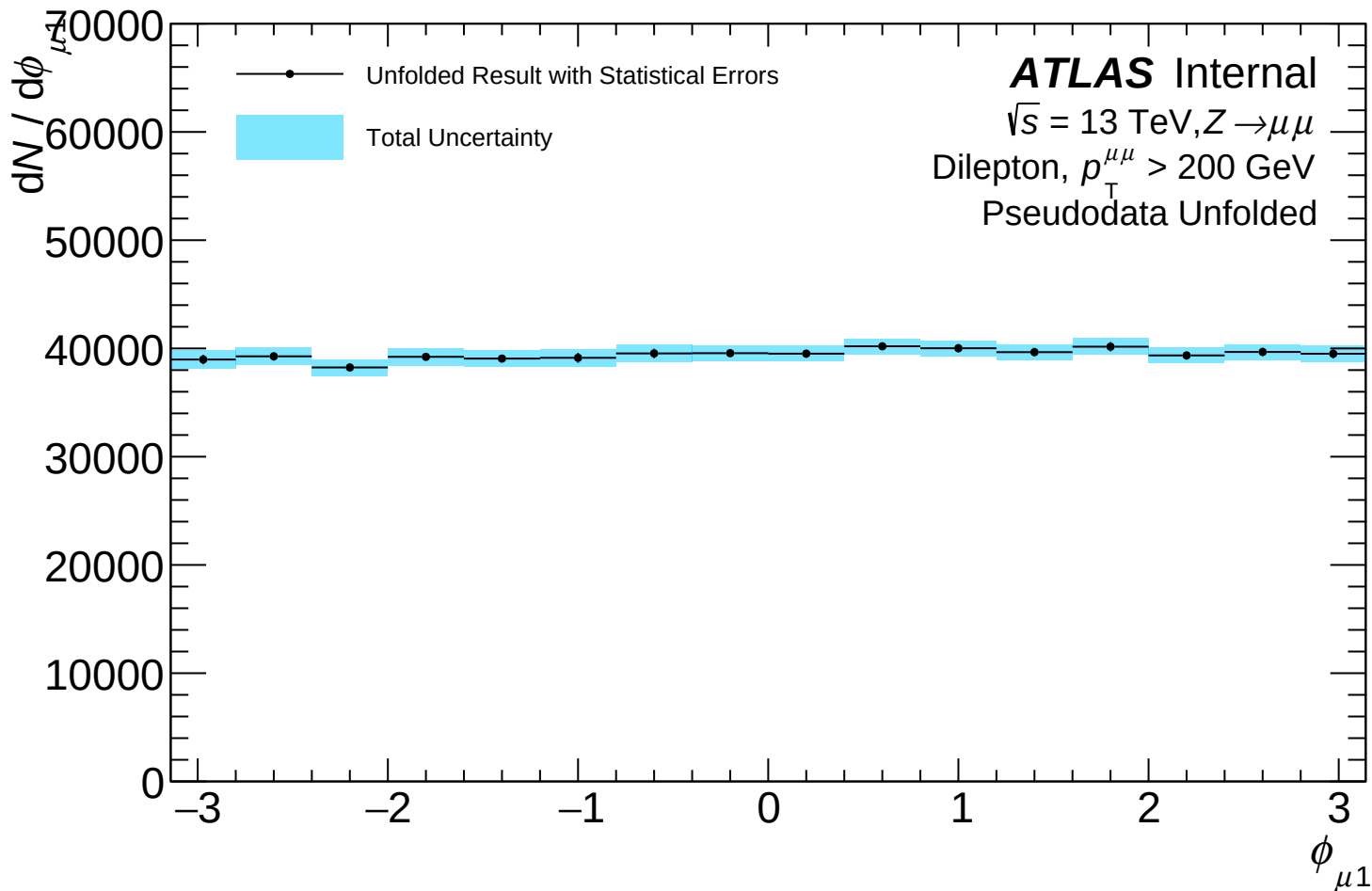


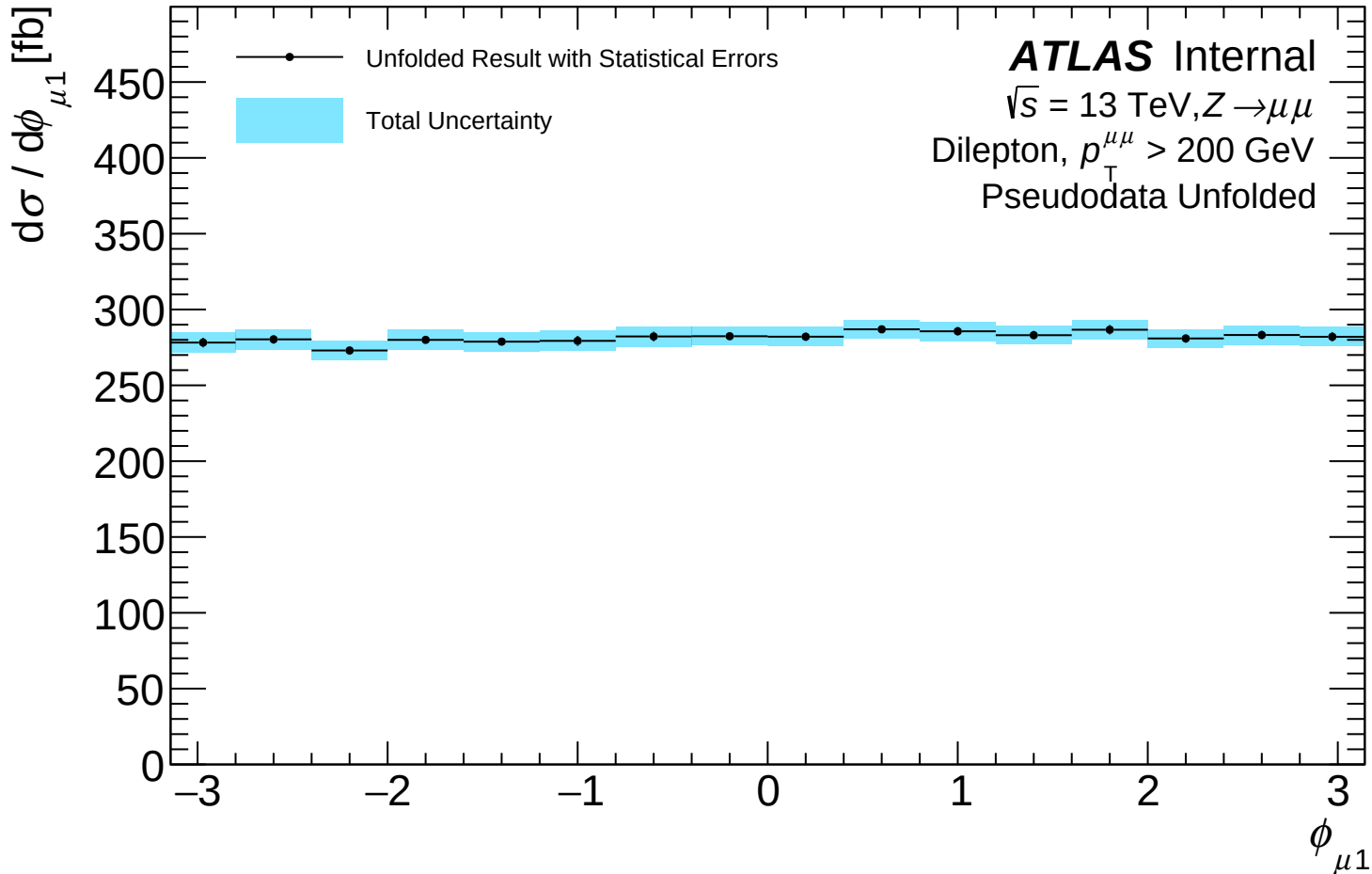




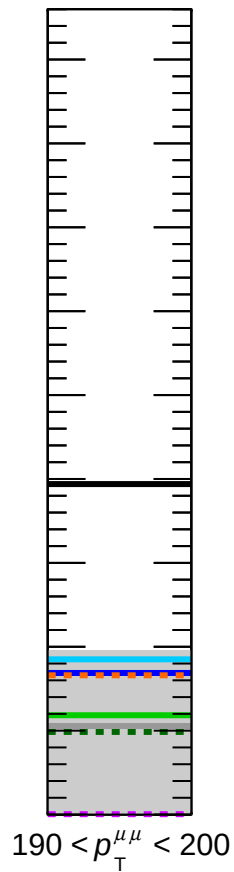
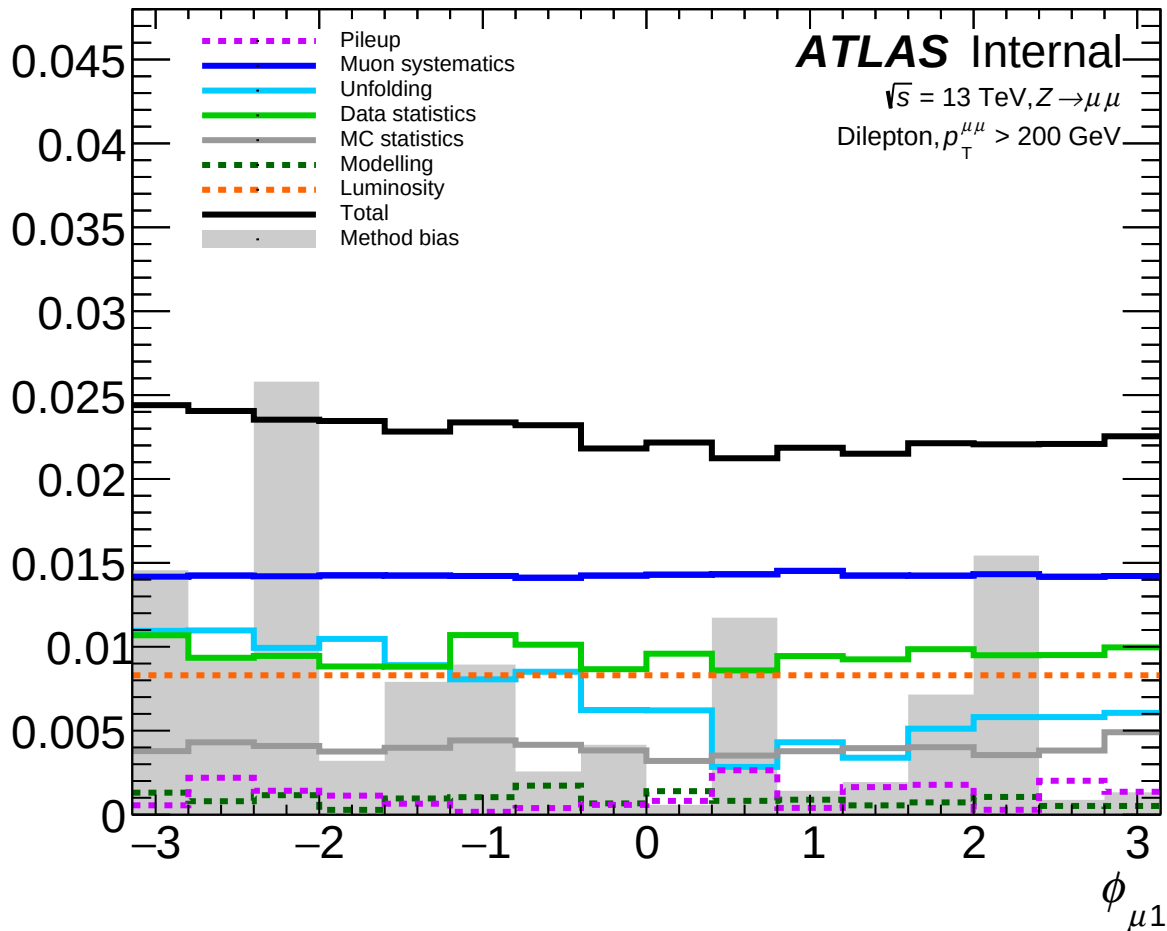
Events

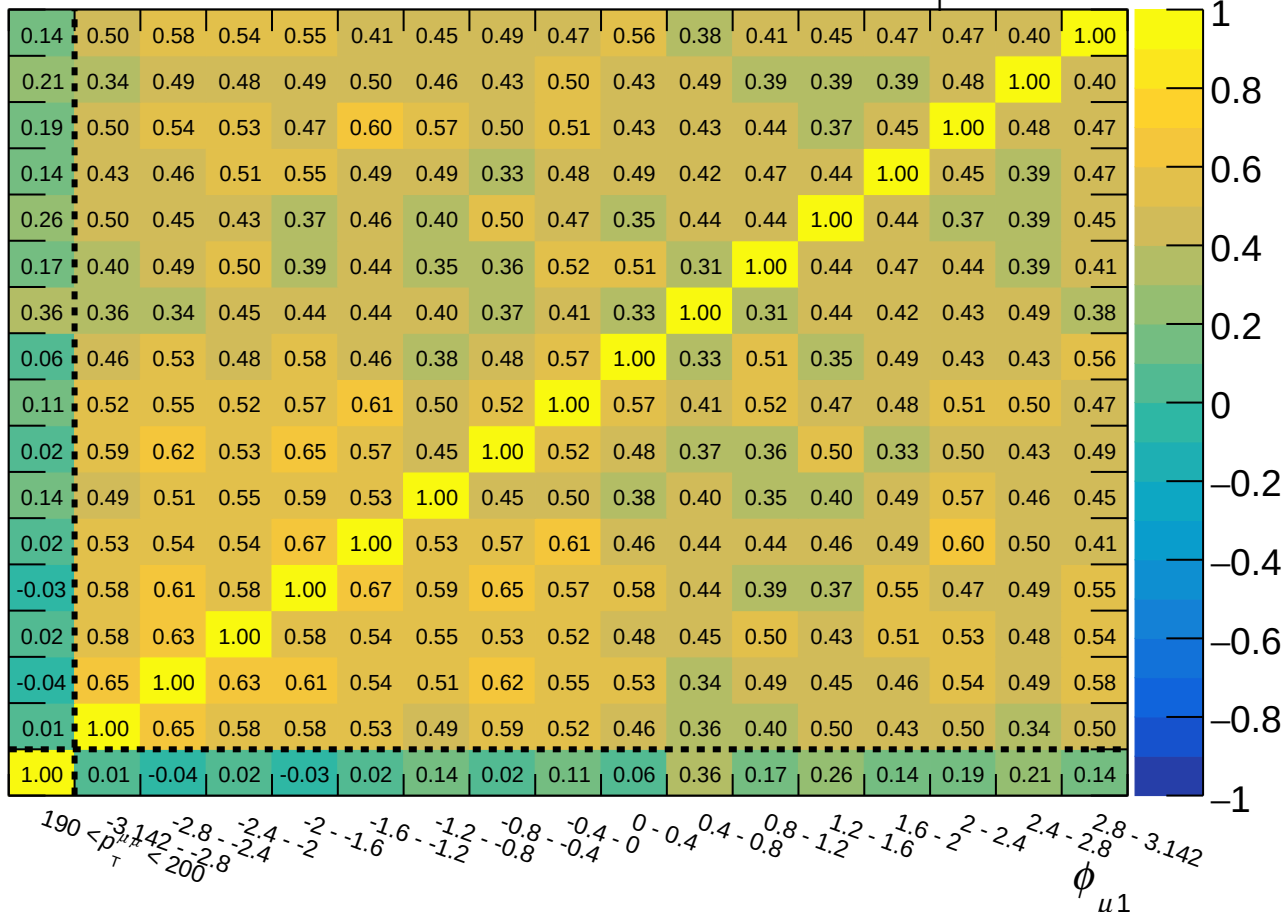


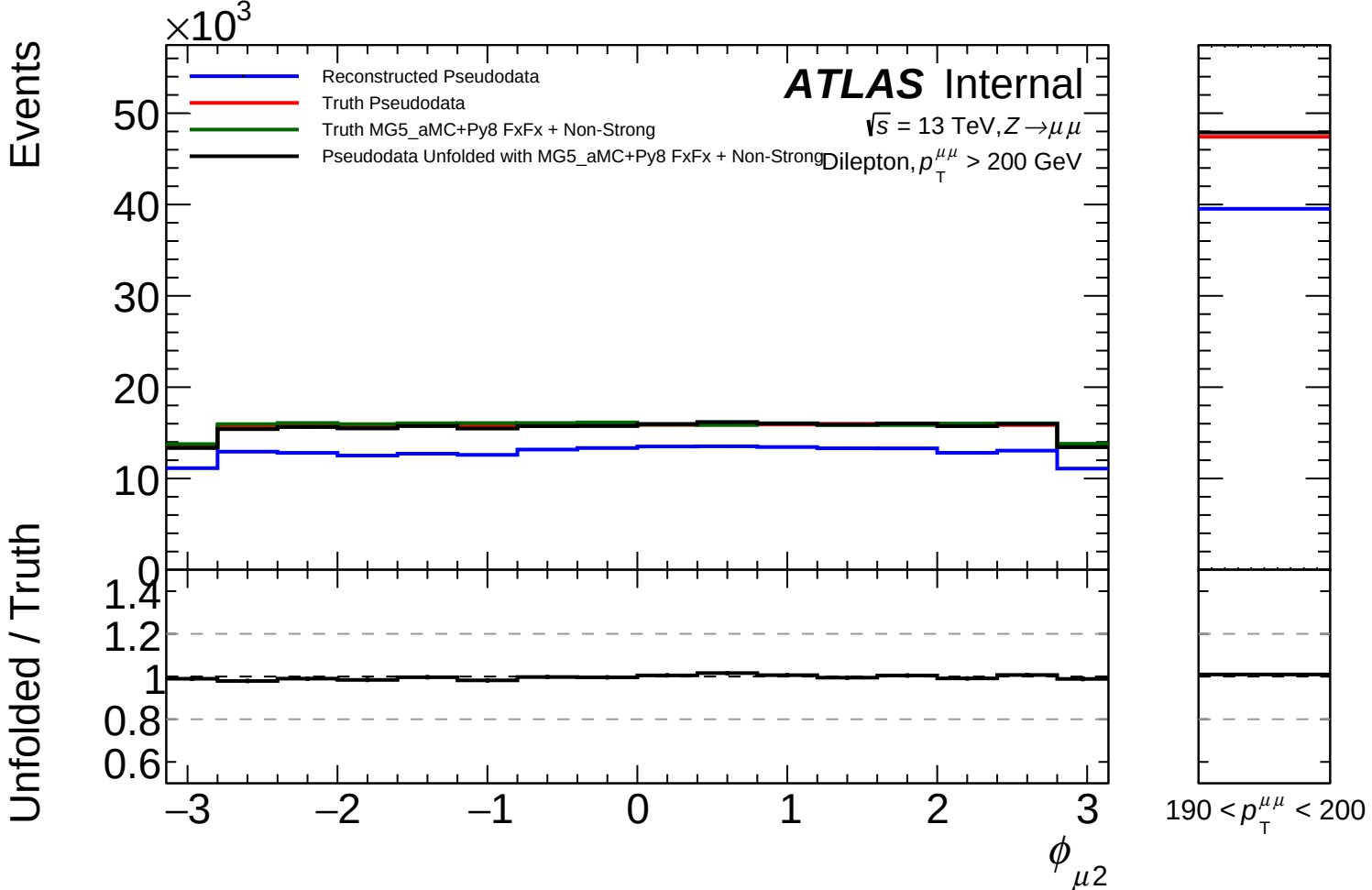


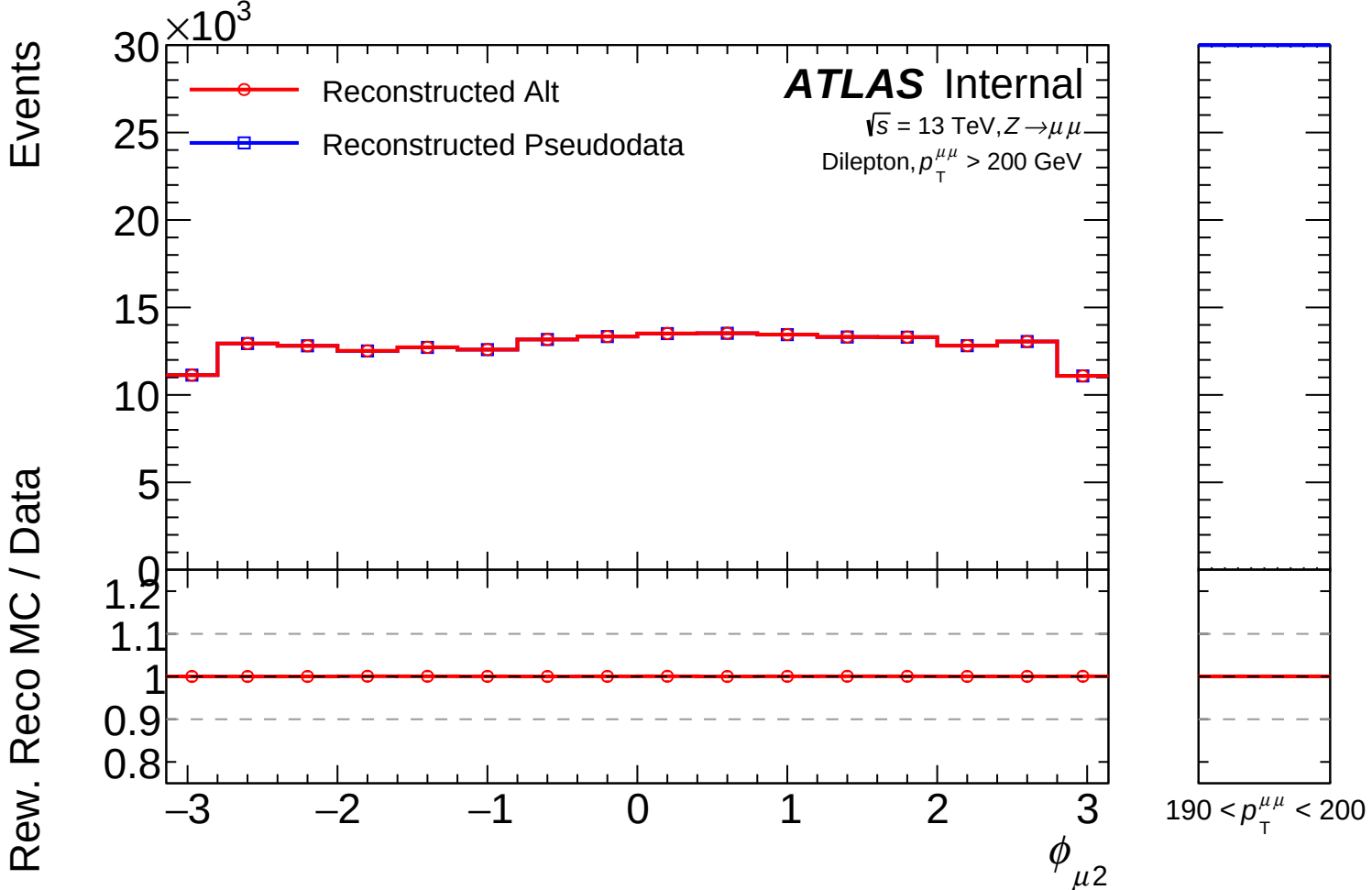


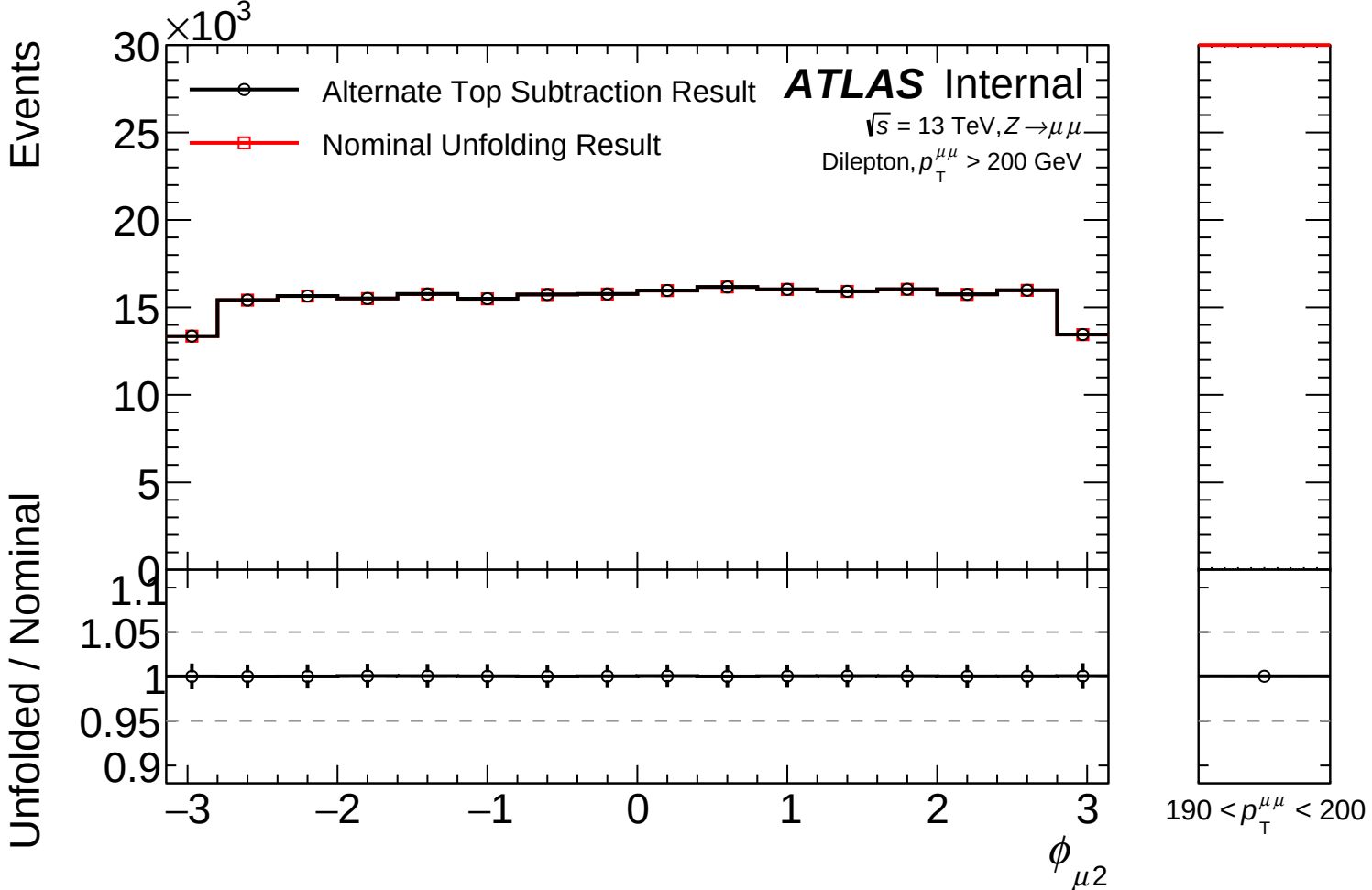
Fractional uncertainty



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $\phi_{\mu 1}$ $190 < p_{\text{T}}^{\mu\mu} < 200$ 







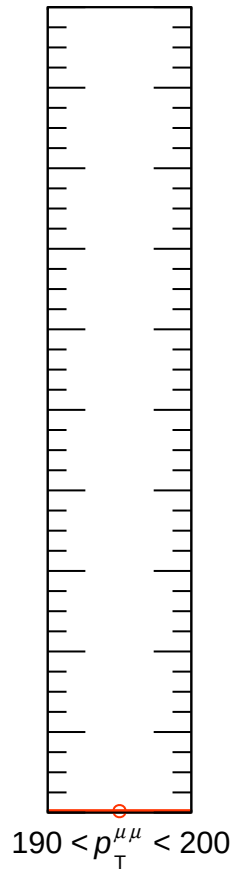
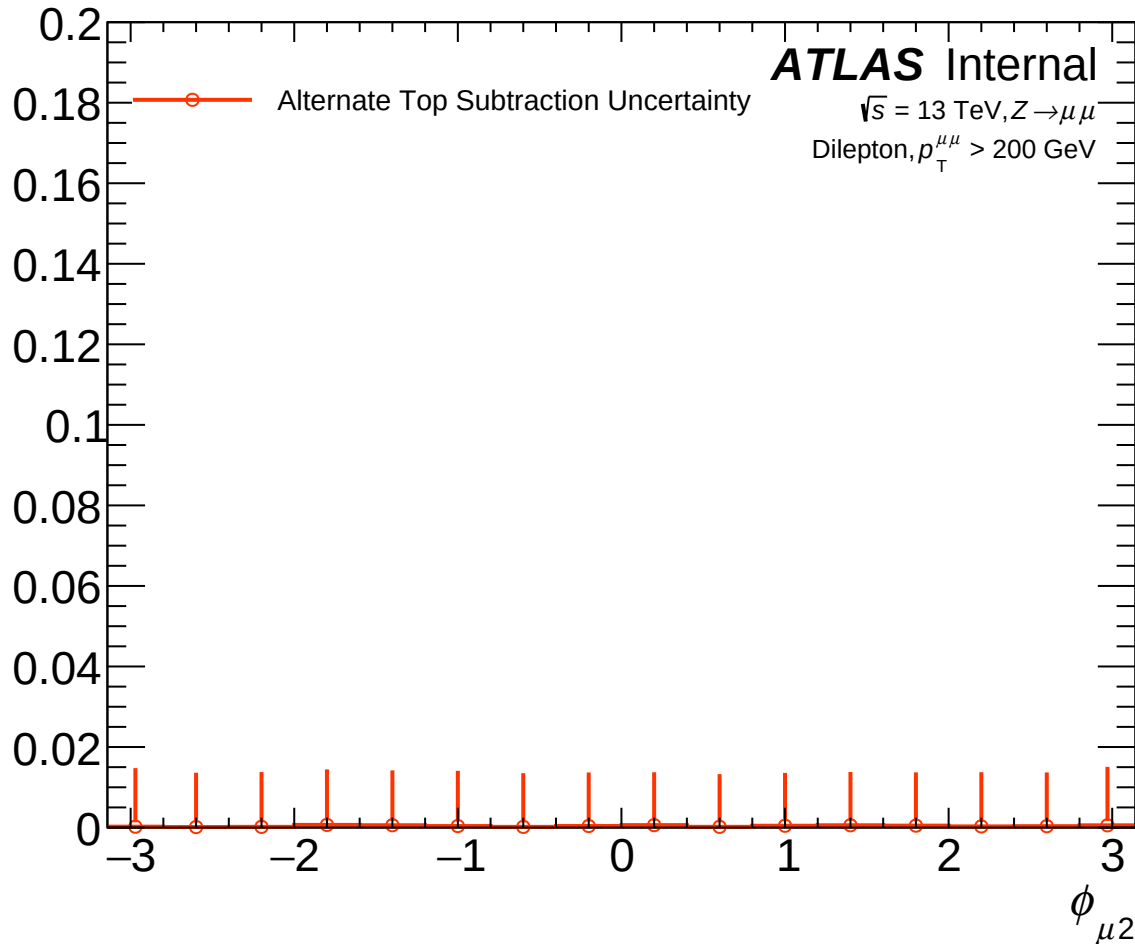
Fractional systematic impact

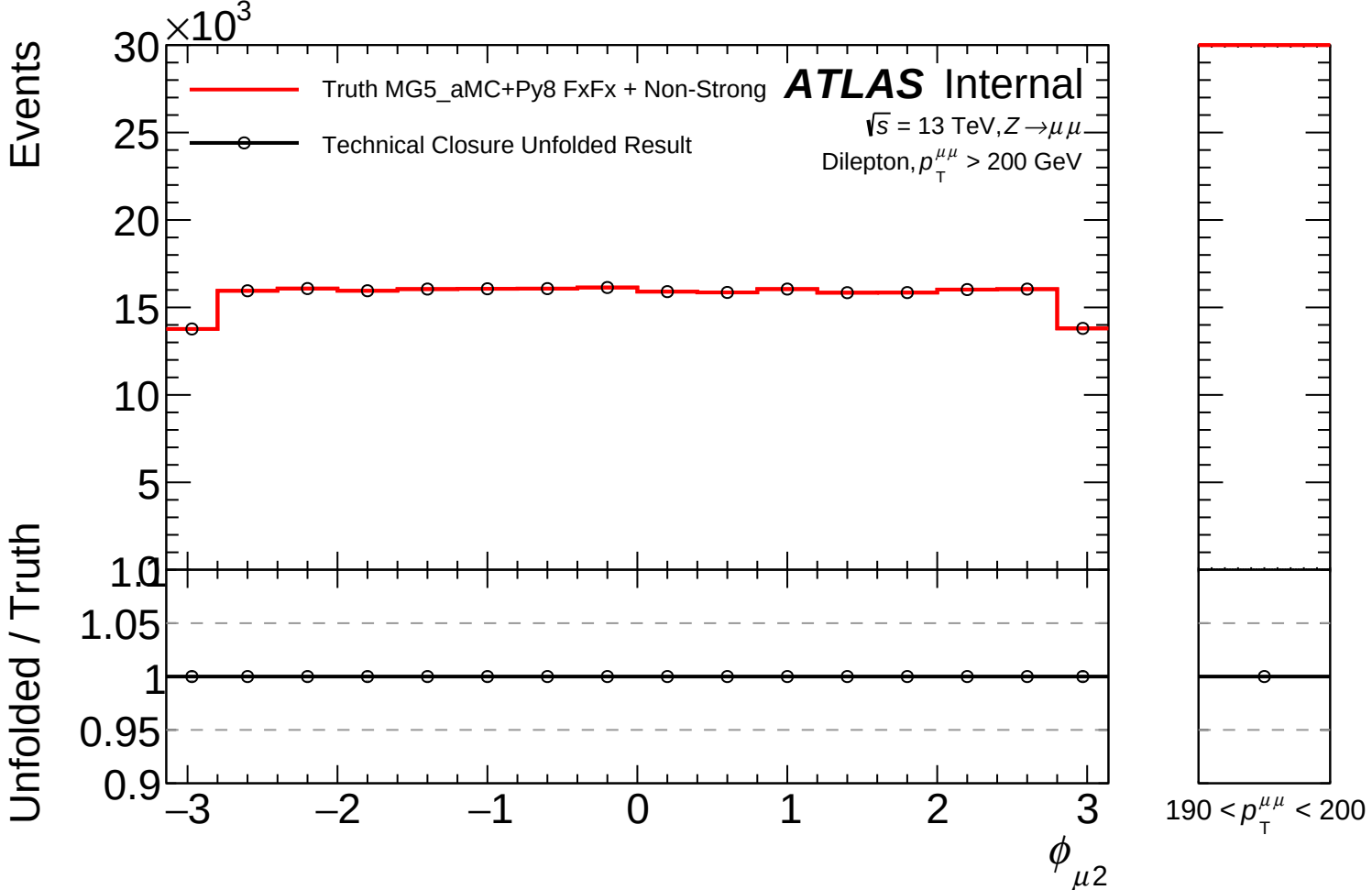
ATLAS Internal

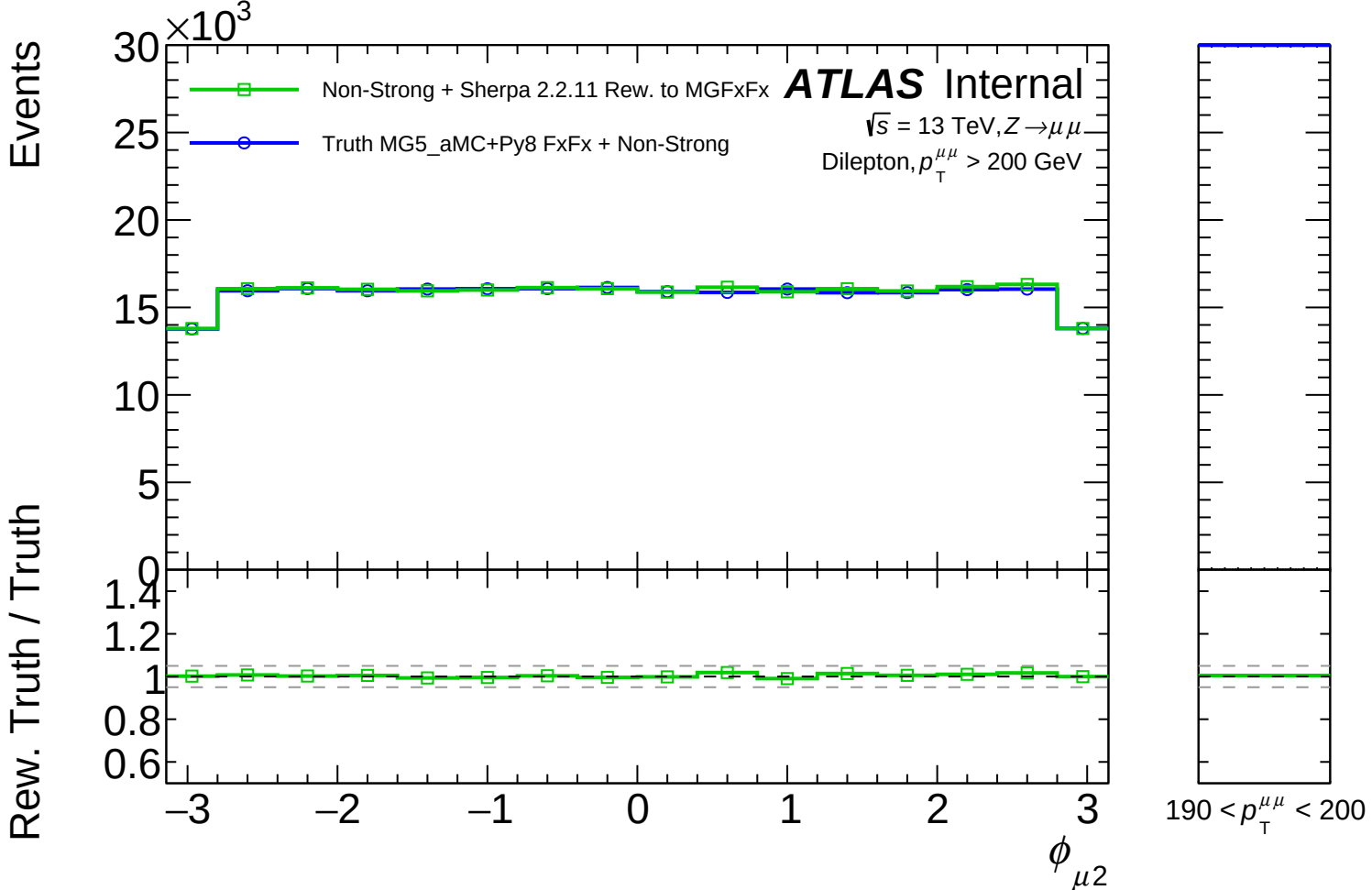
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

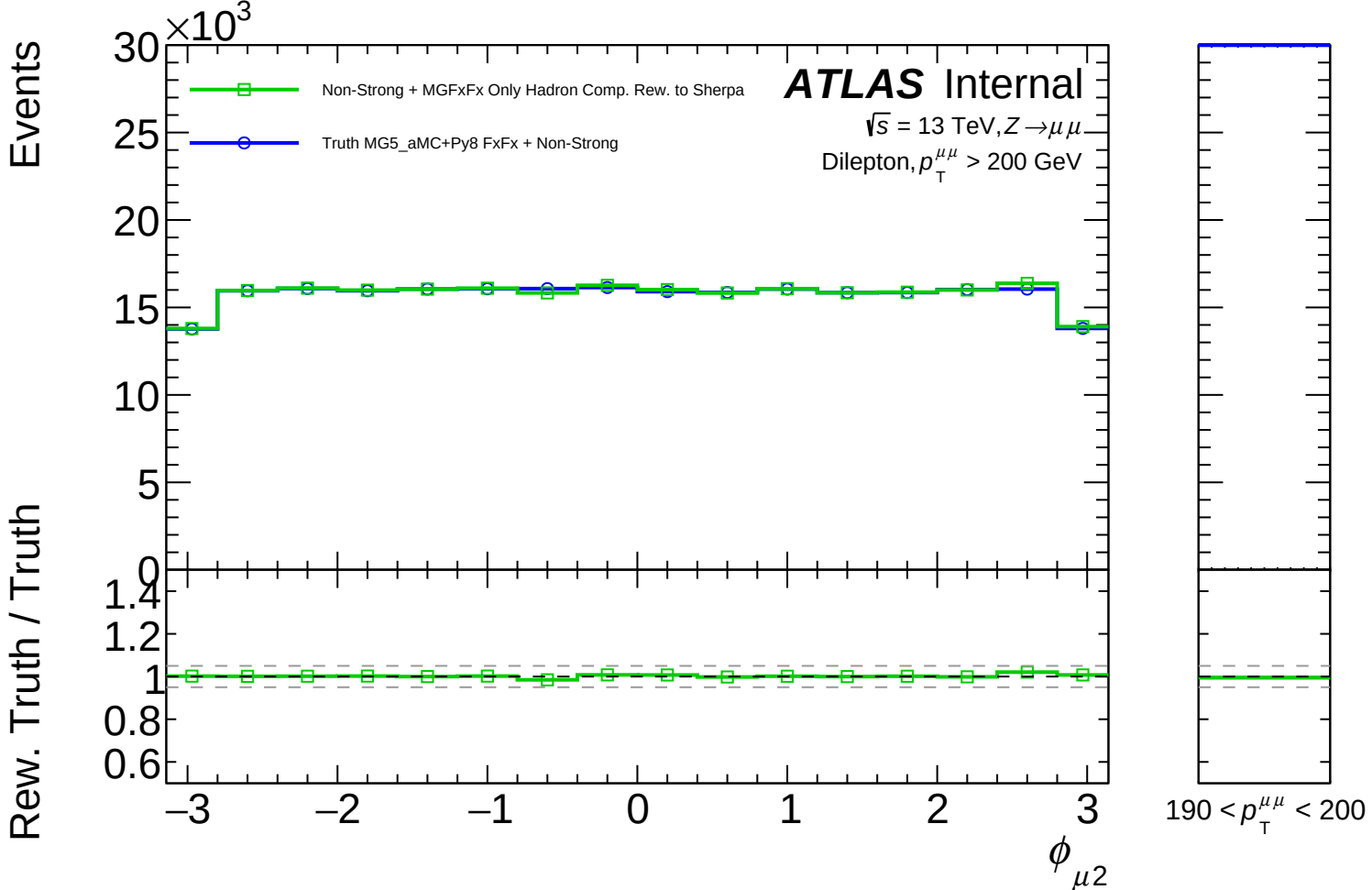
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

Alternate Top Subtraction Uncertainty



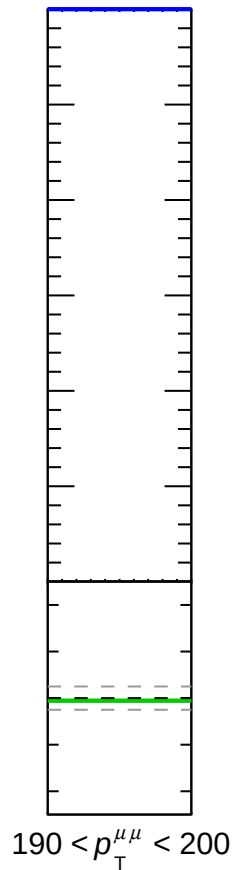
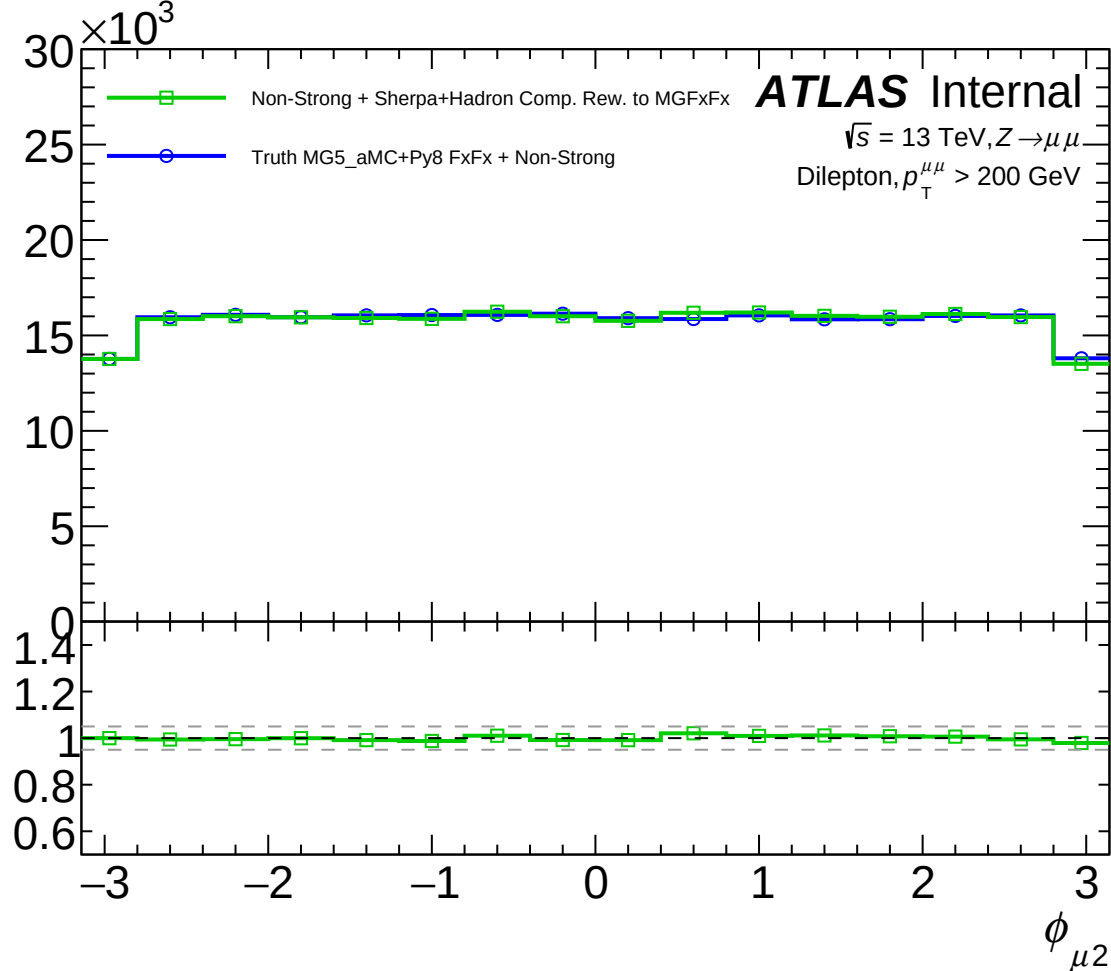


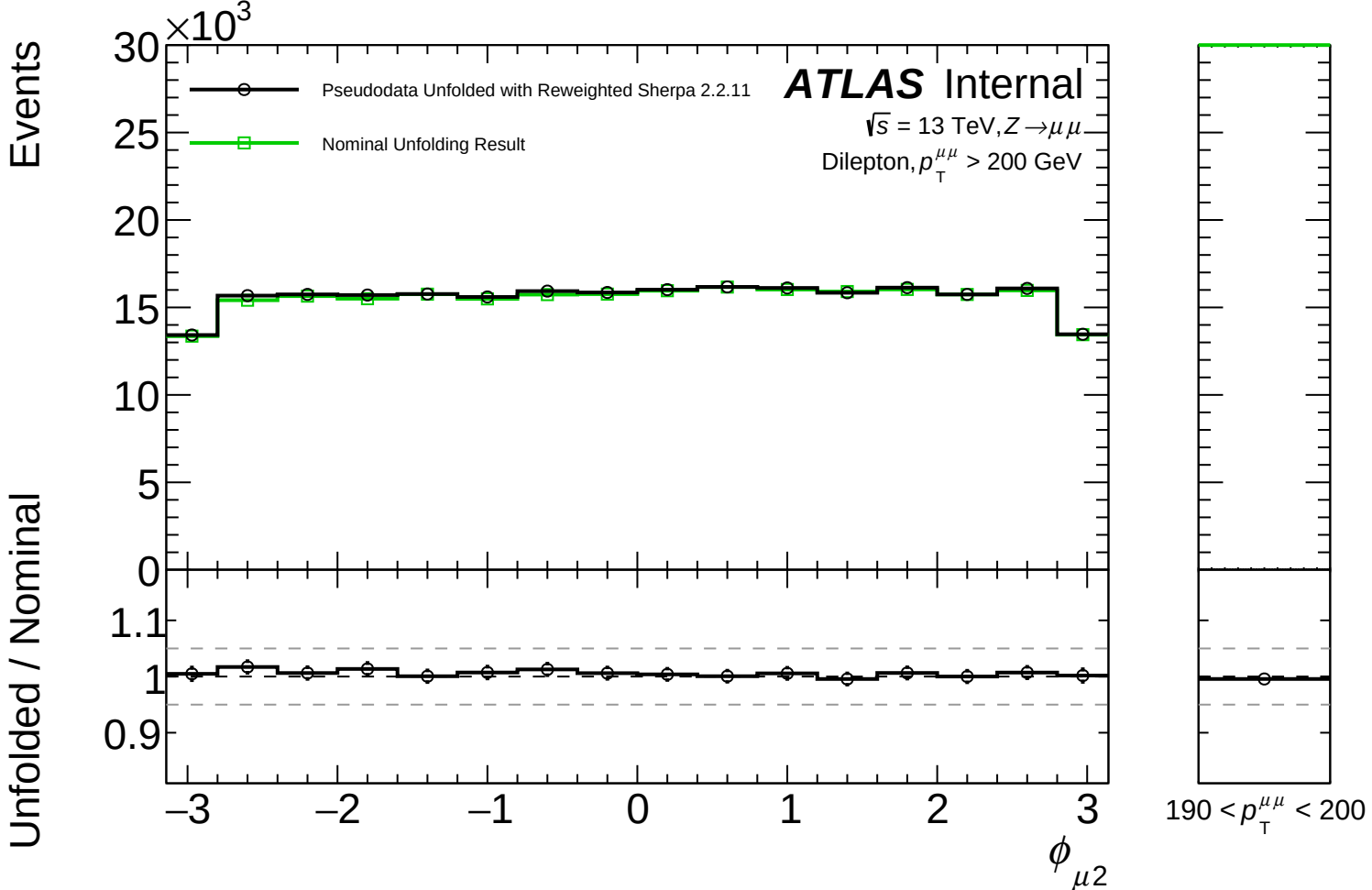


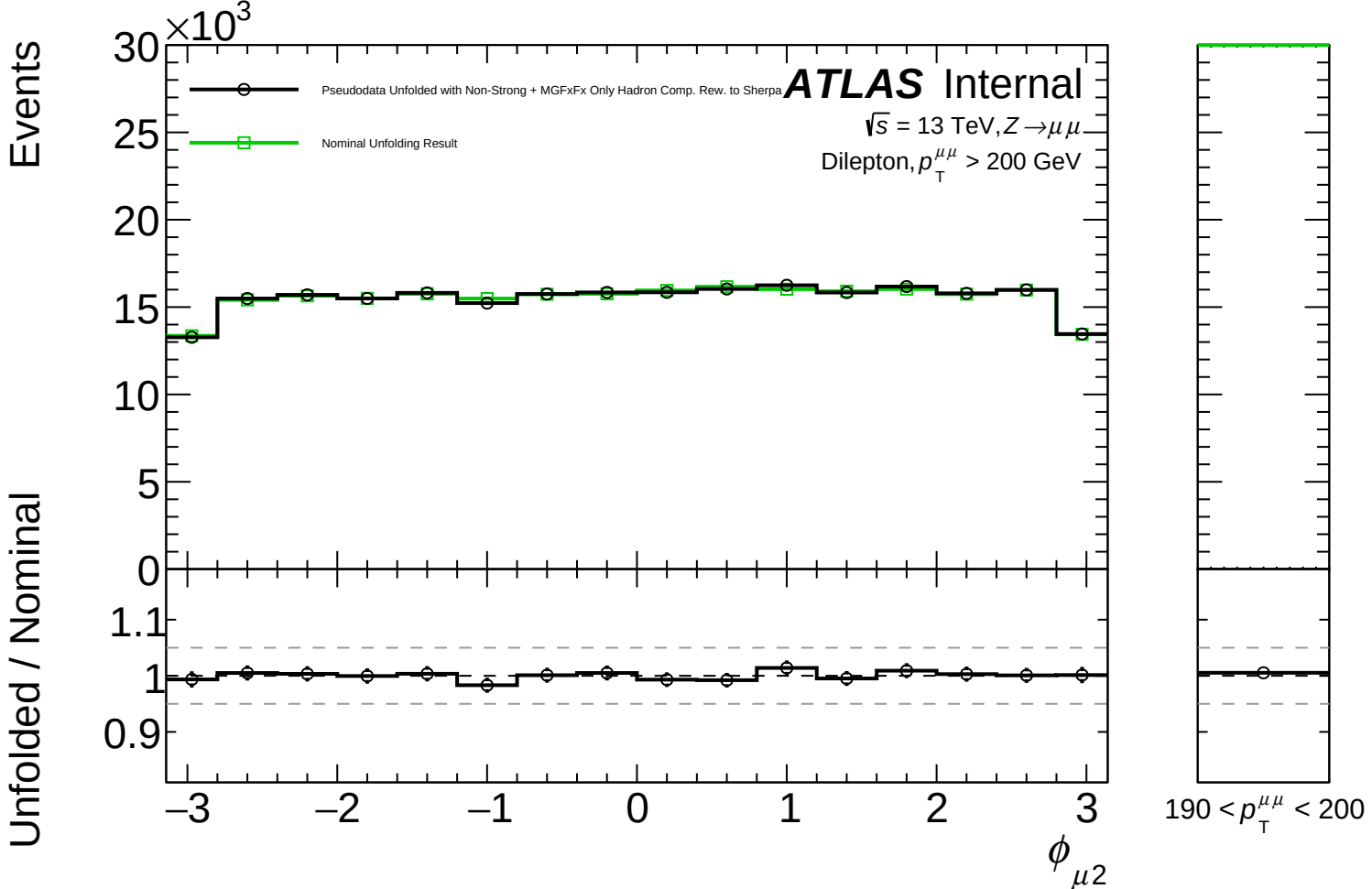


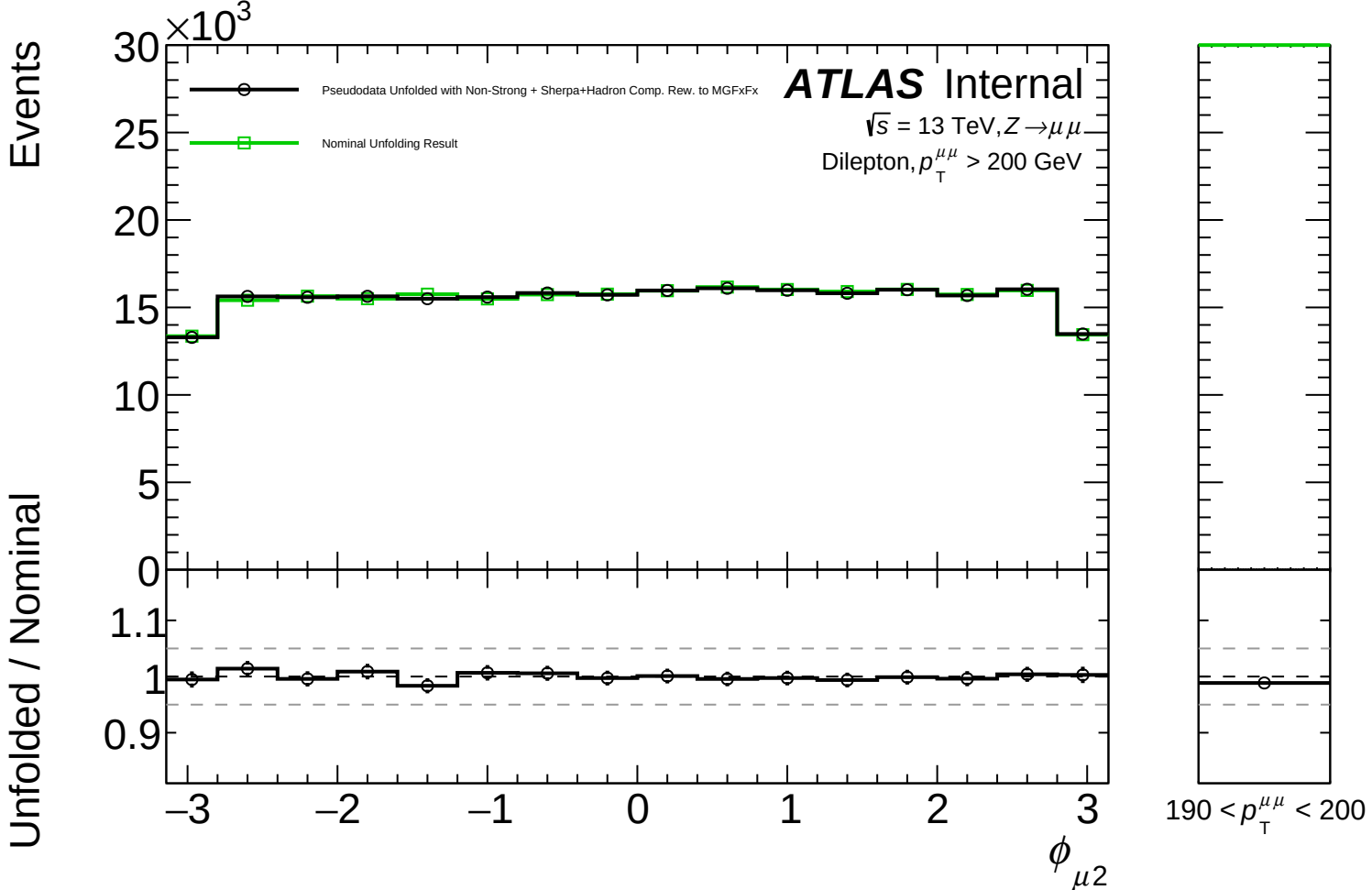
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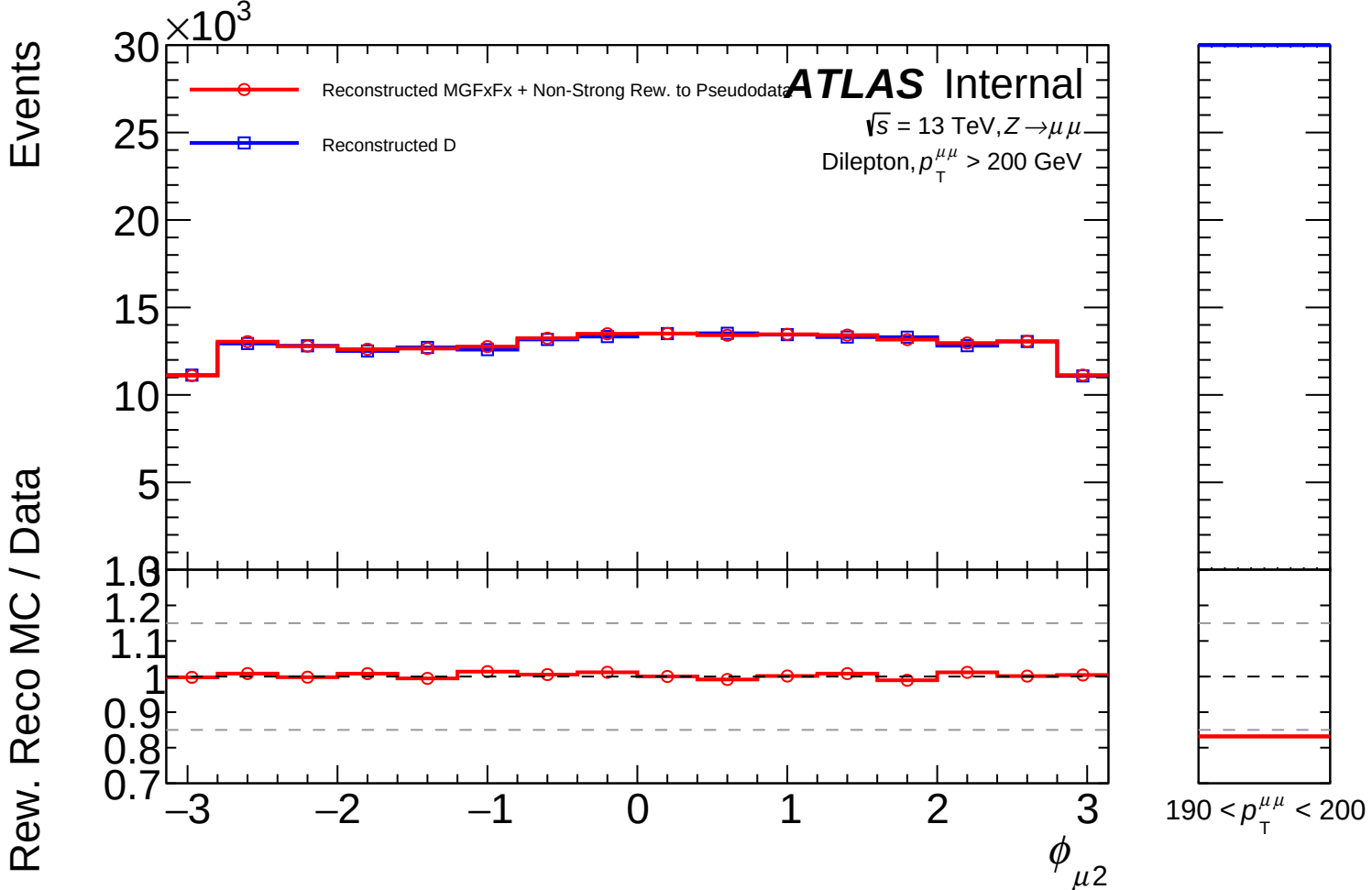
Rew. Truth / Truth

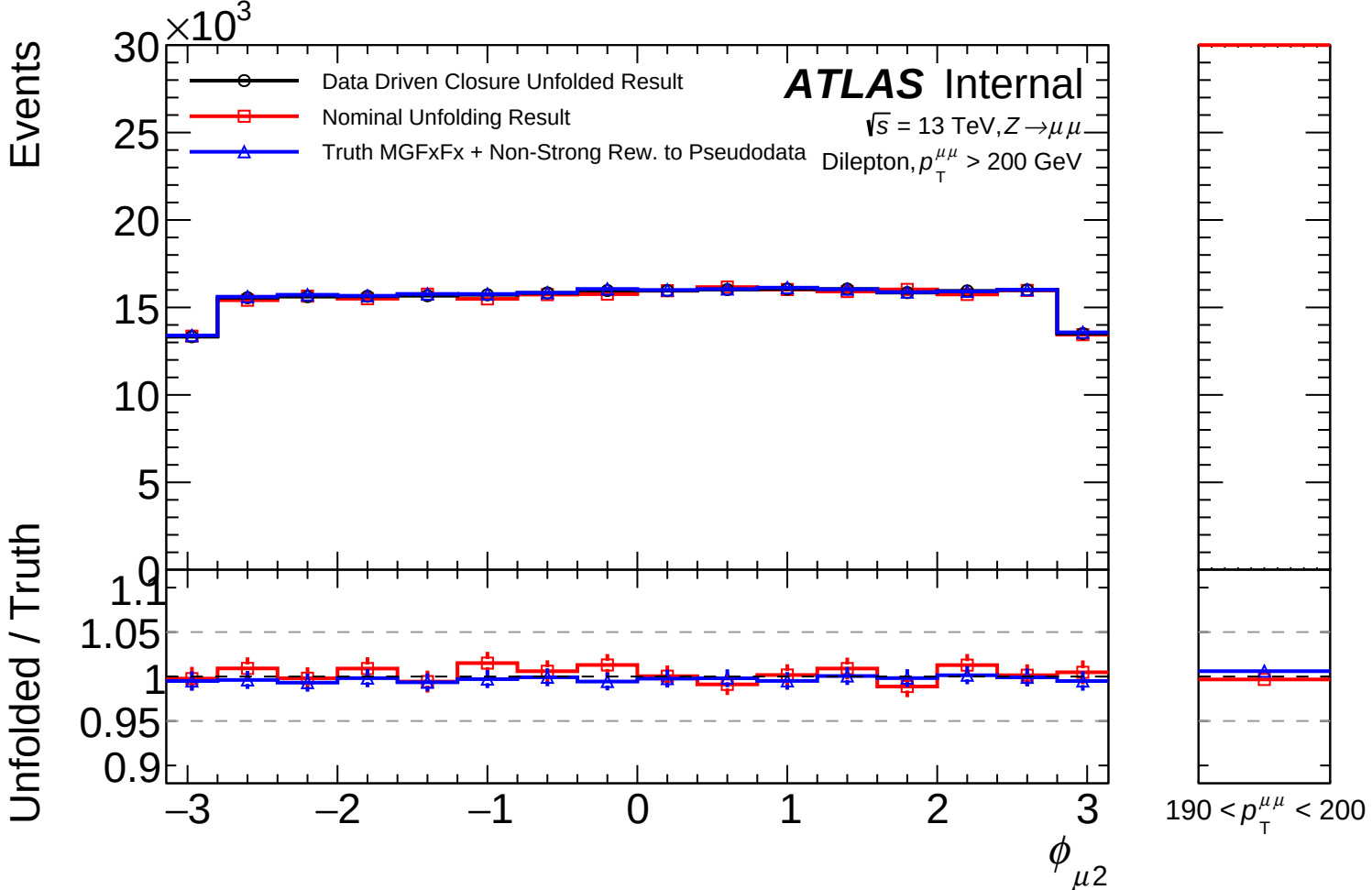




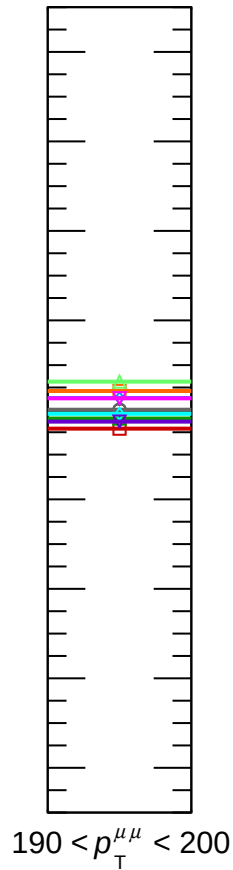
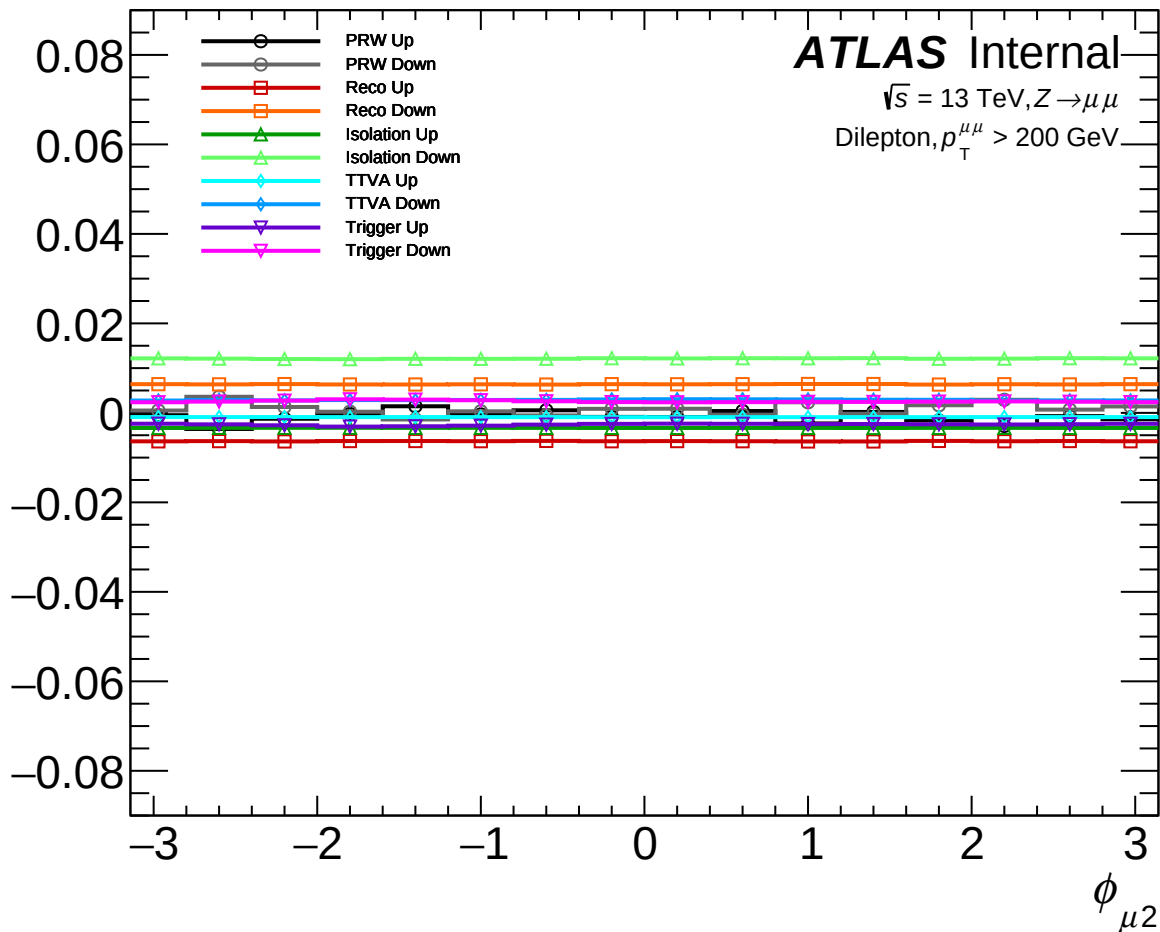




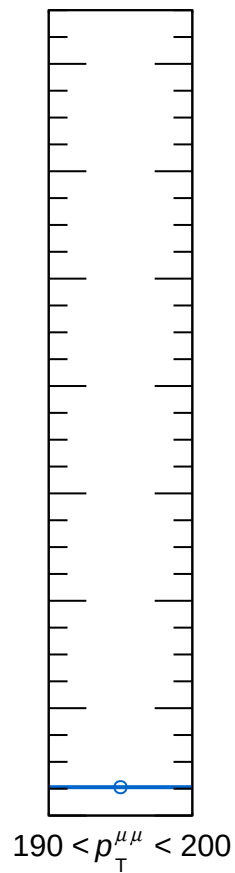
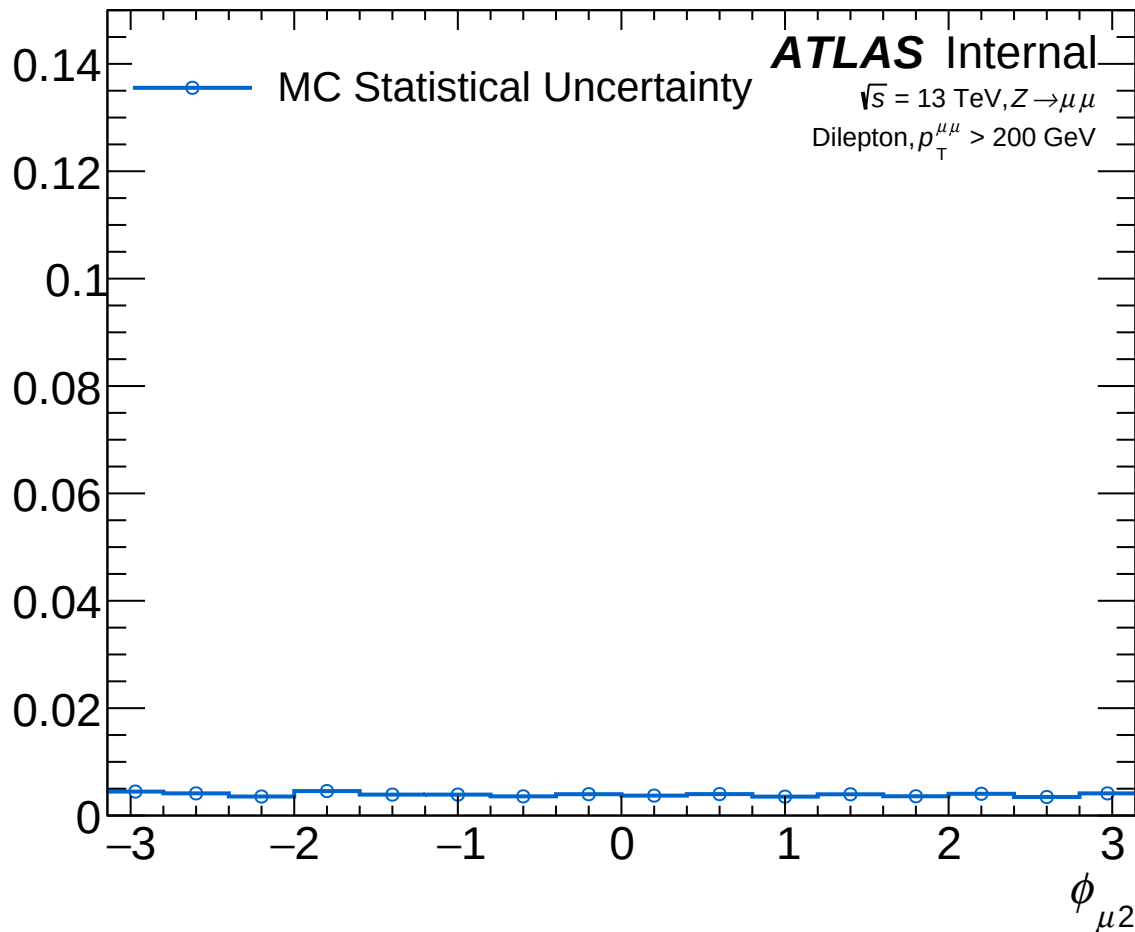




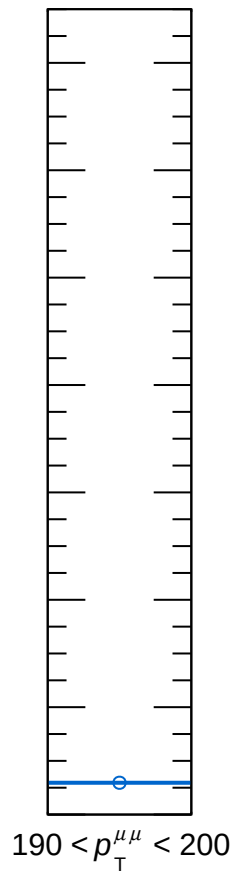
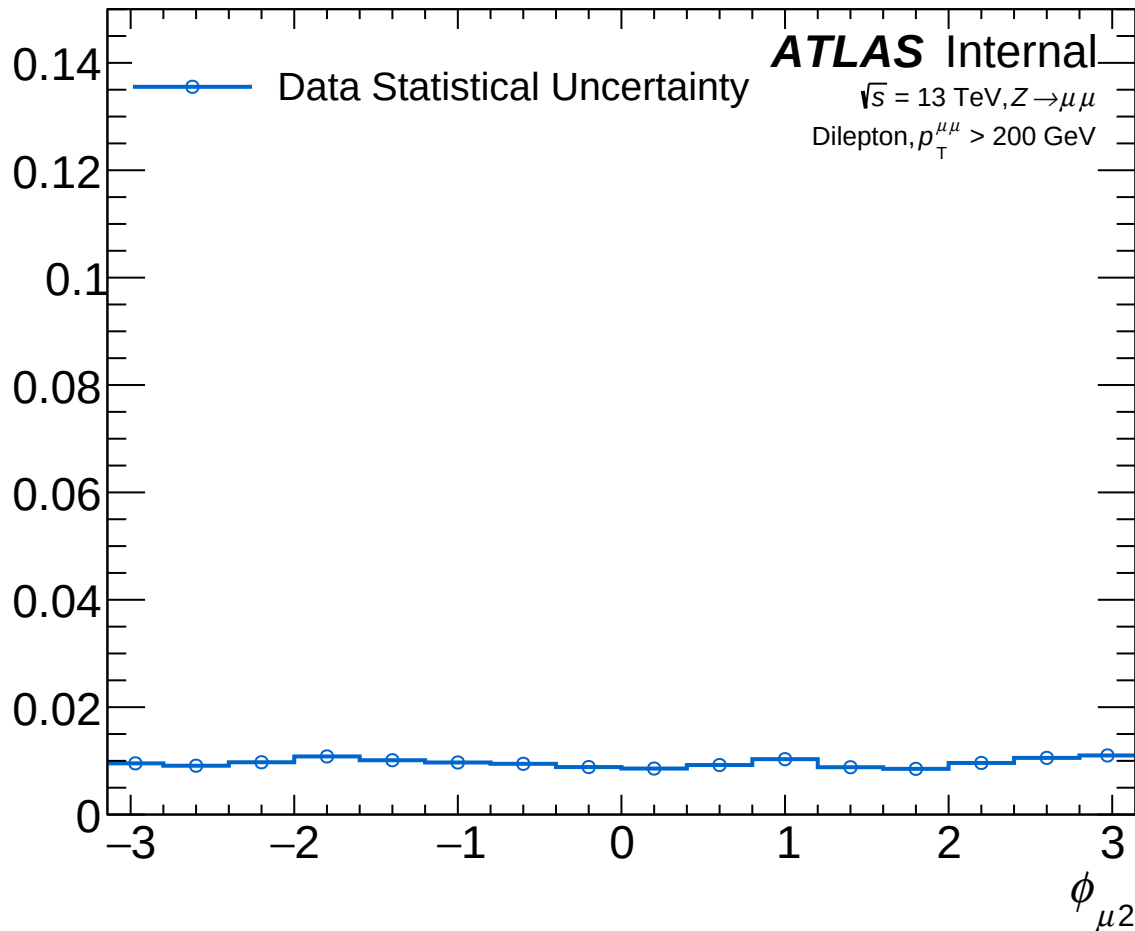
Fractional systematic impact



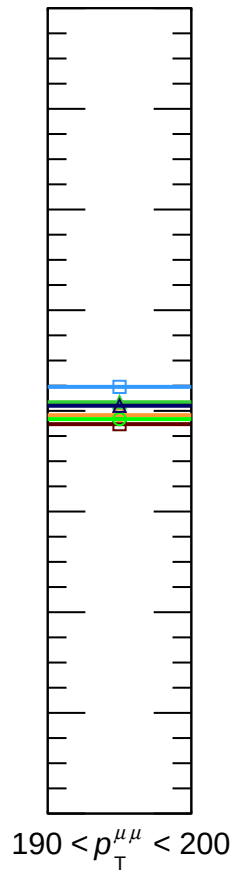
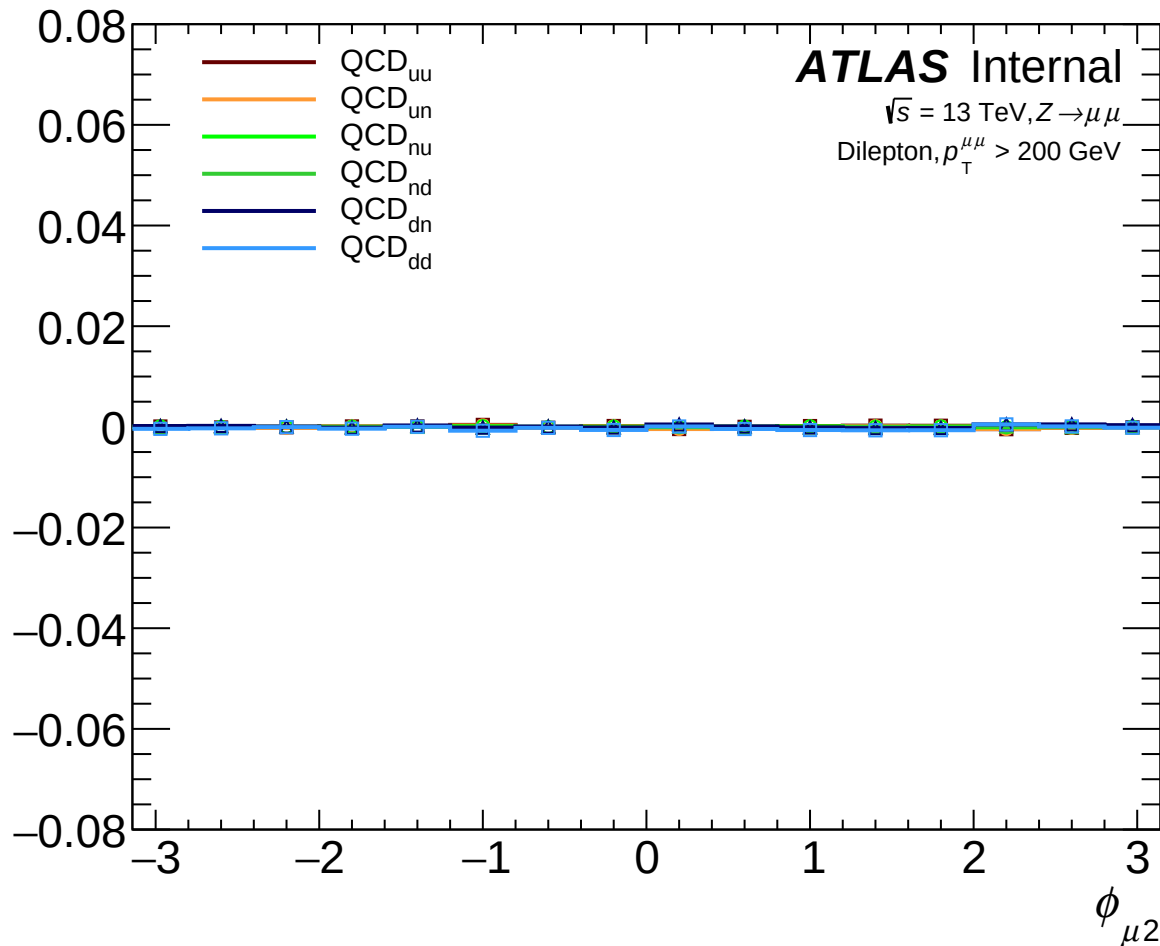
Fractional systematic impact



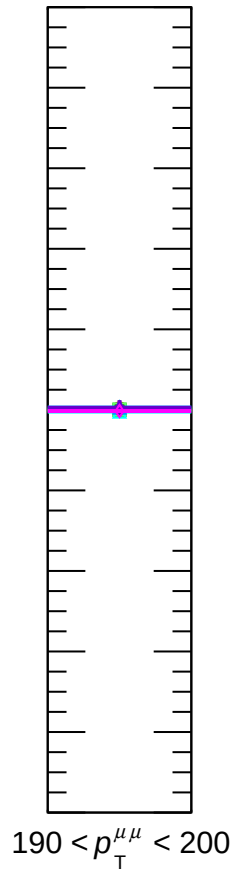
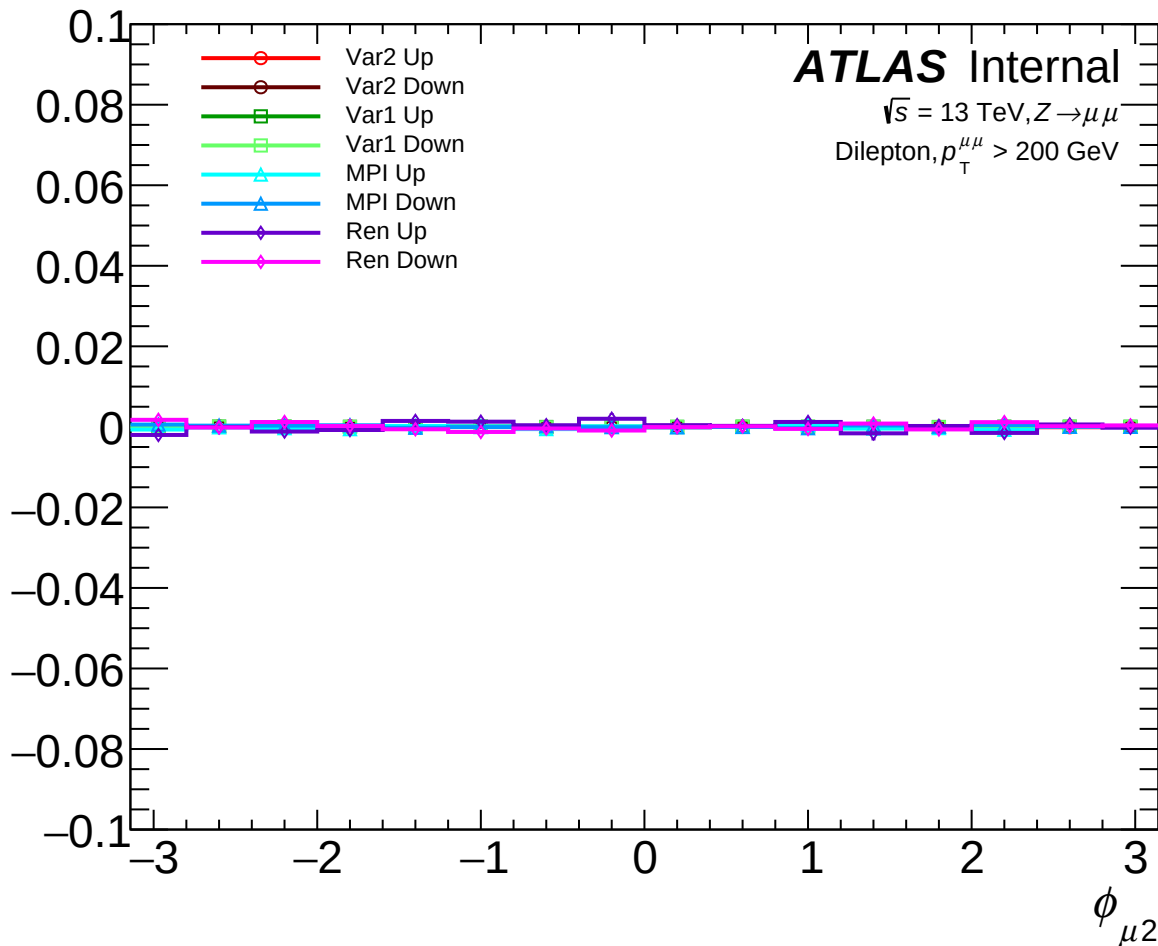
Fractional systematic impact



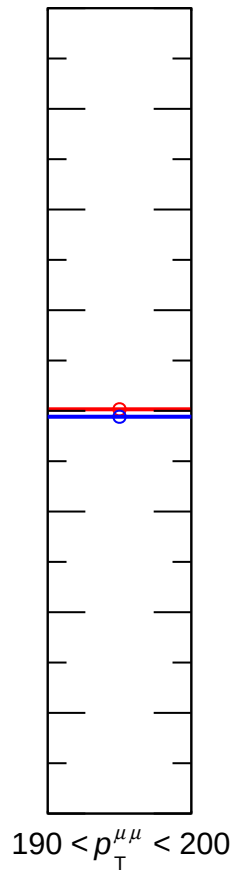
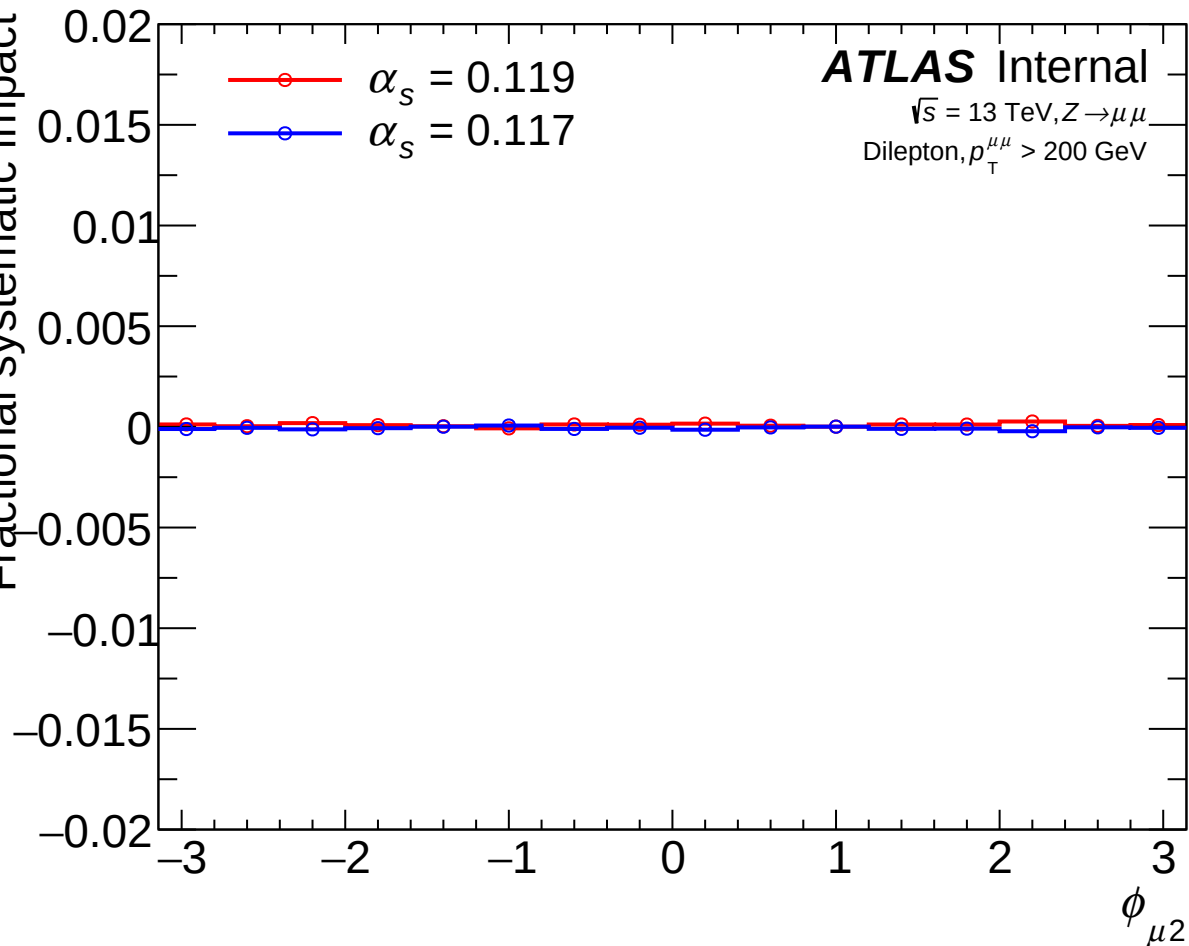
Fractional systematic impact

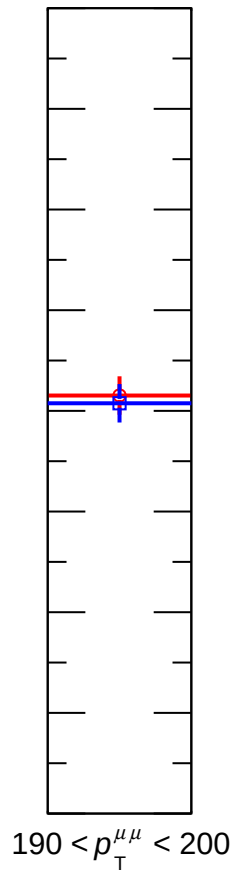
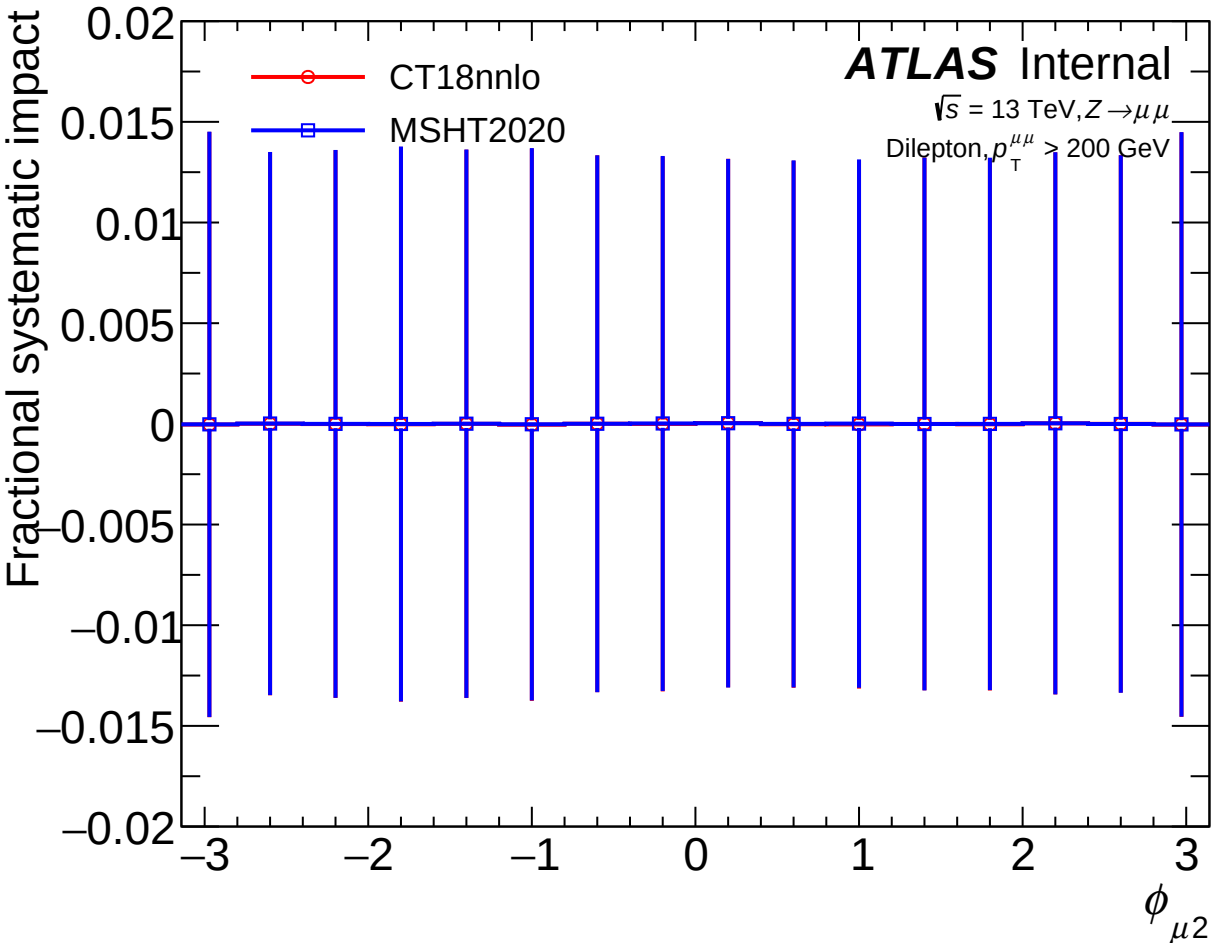


Fractional systematic impact

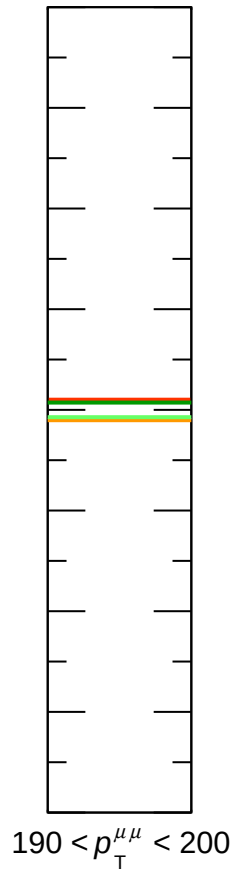
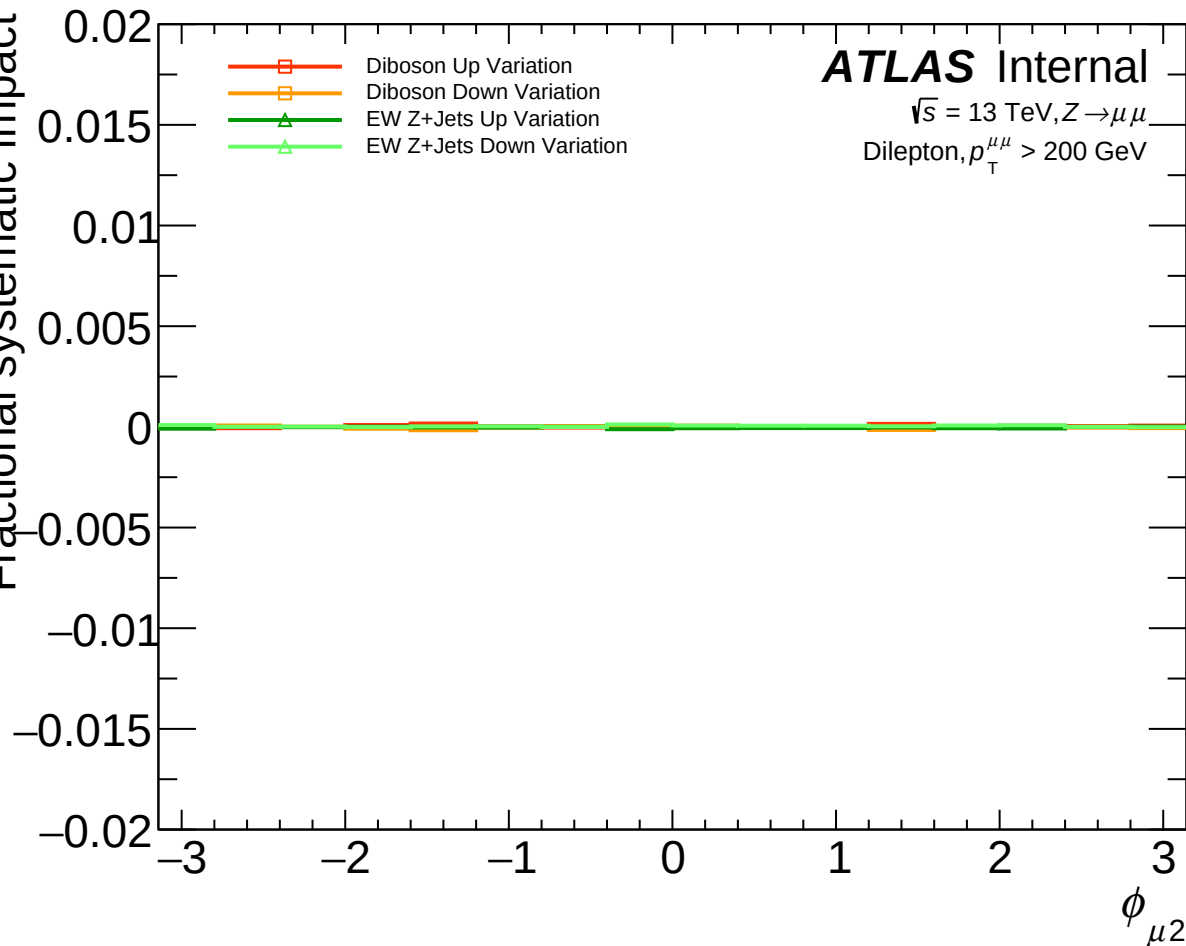


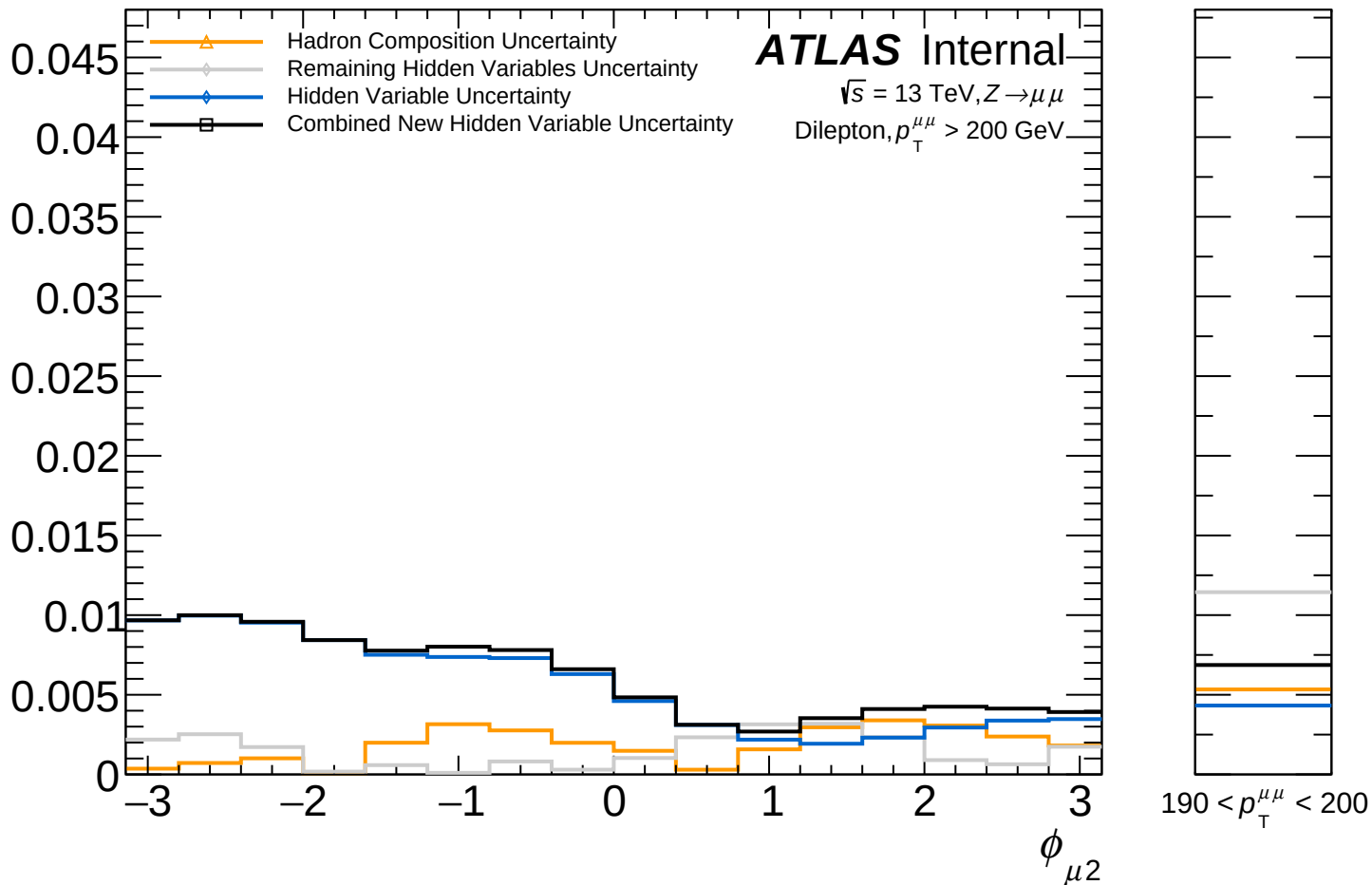
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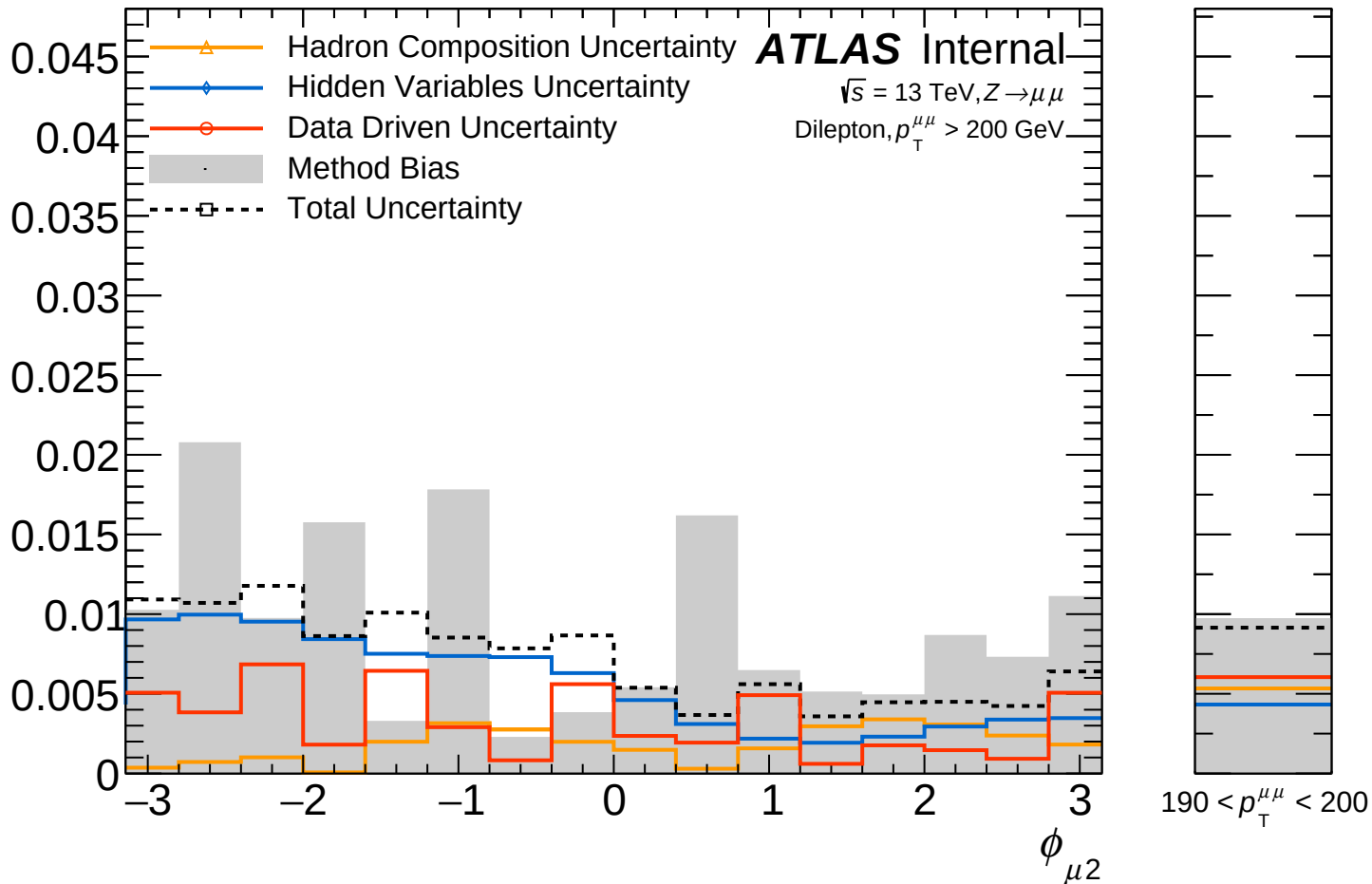


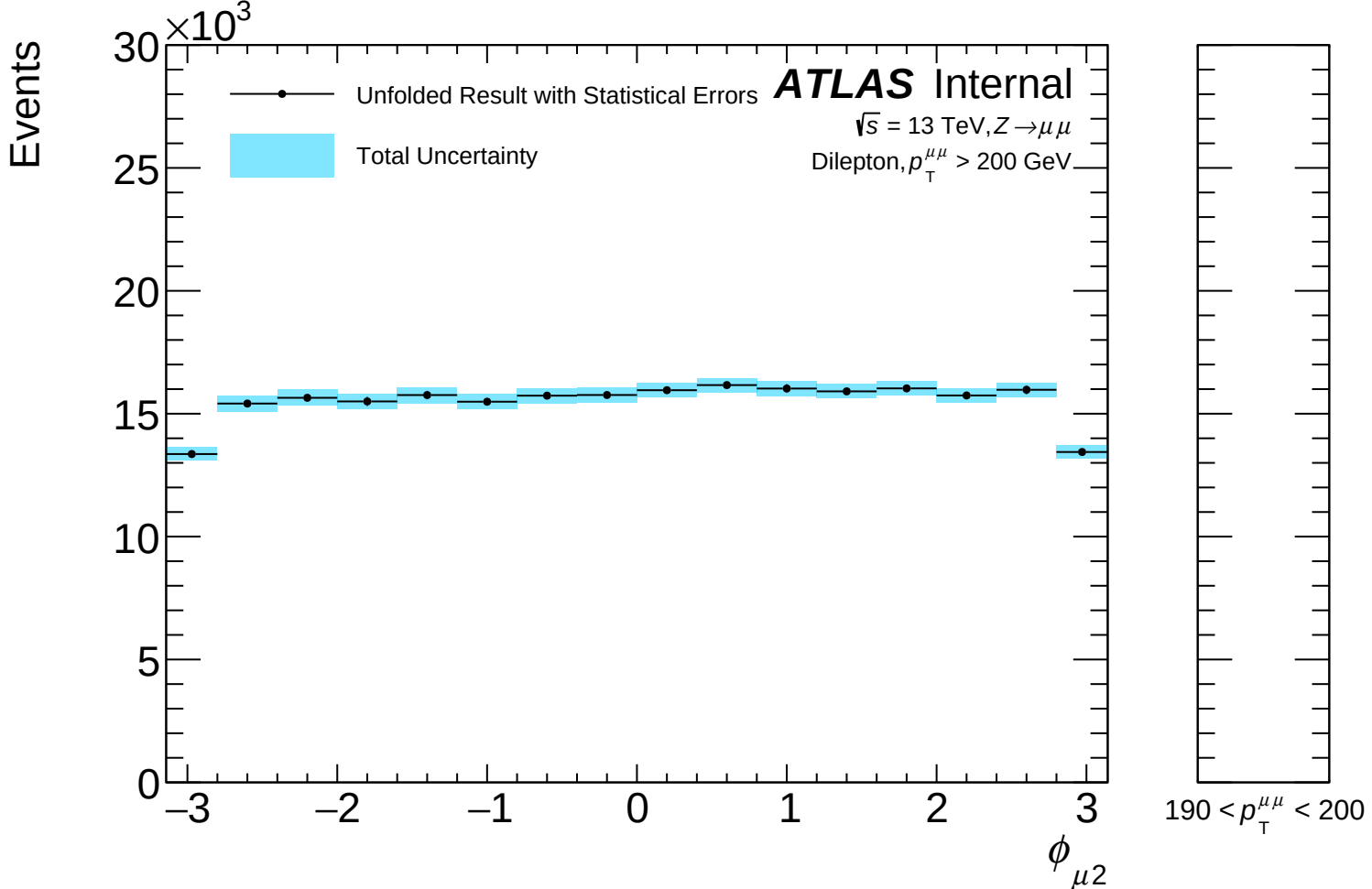


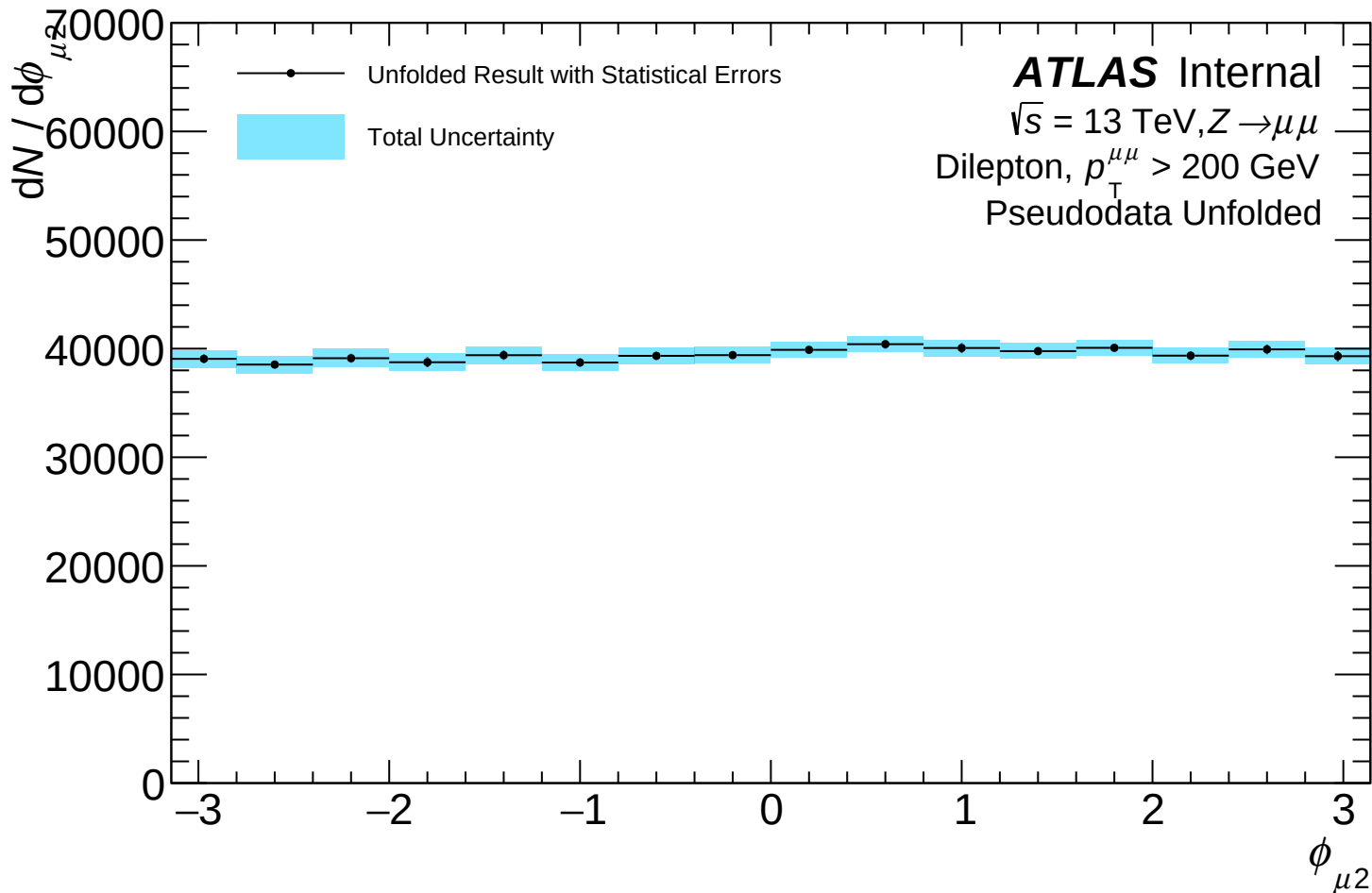
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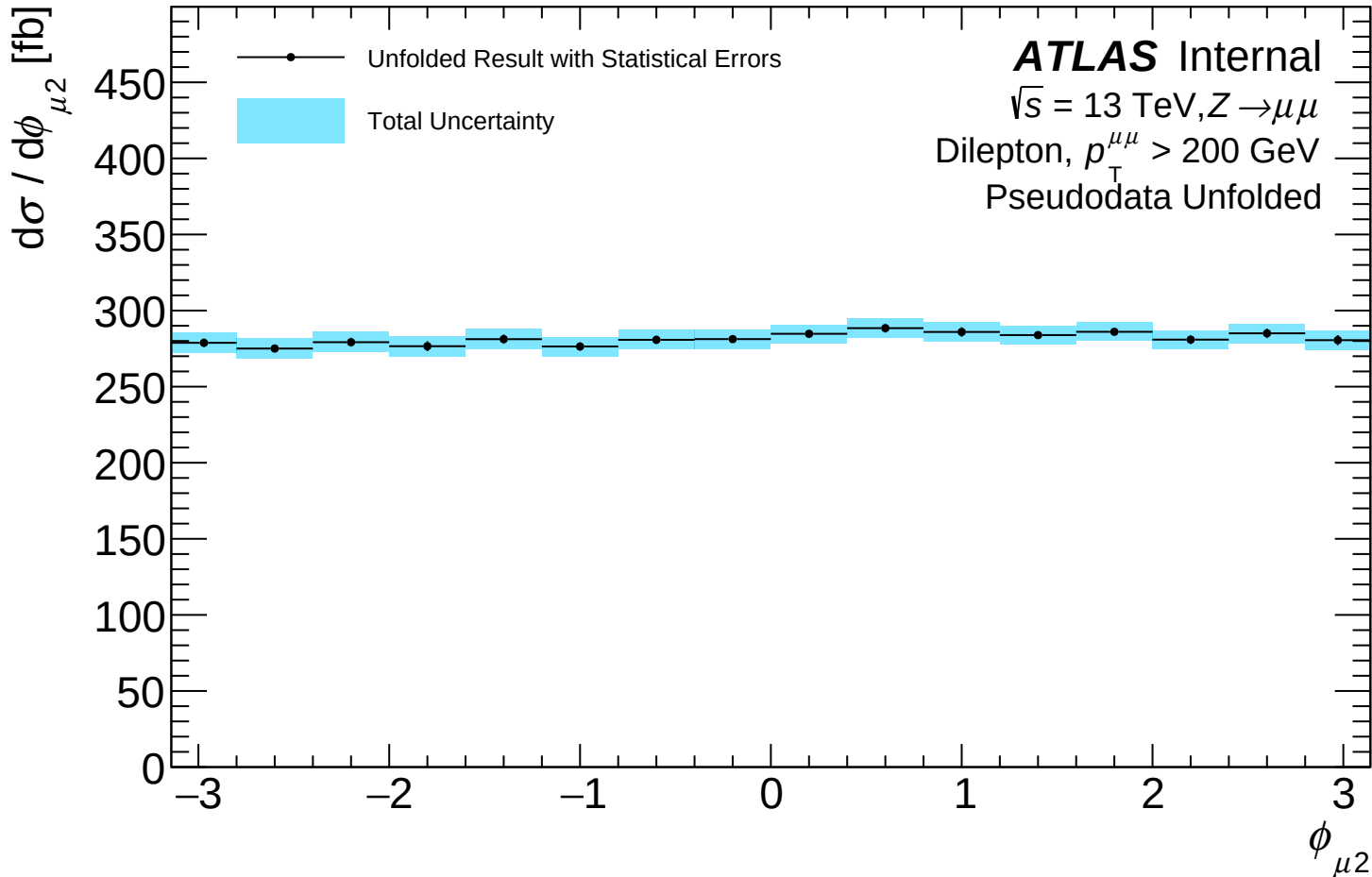




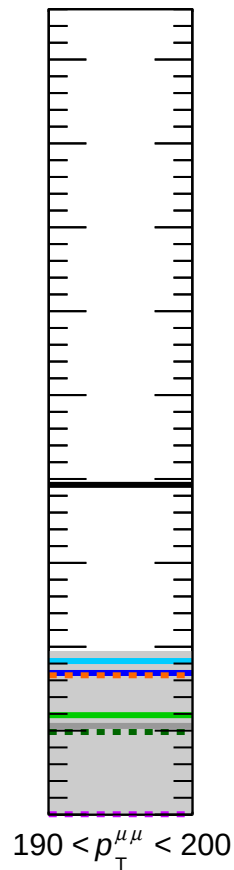
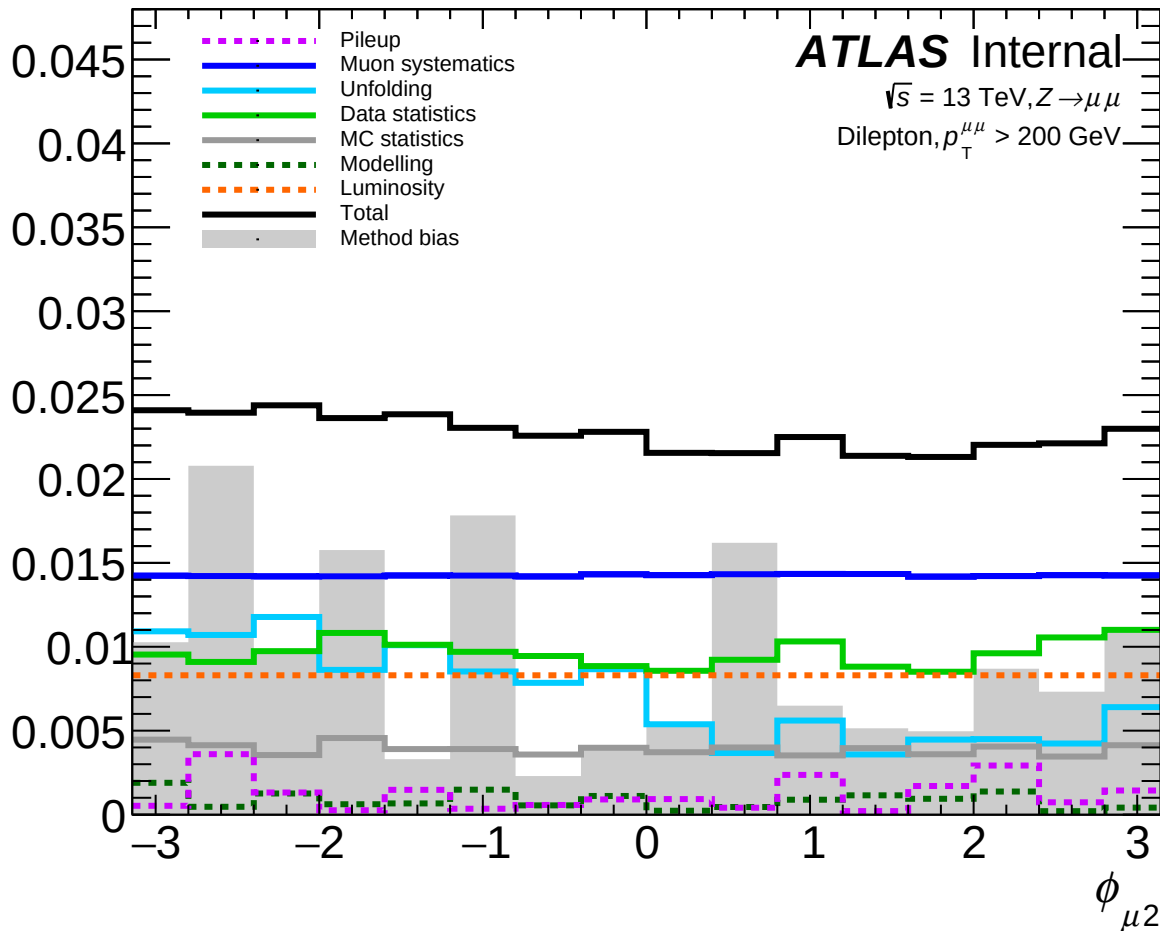


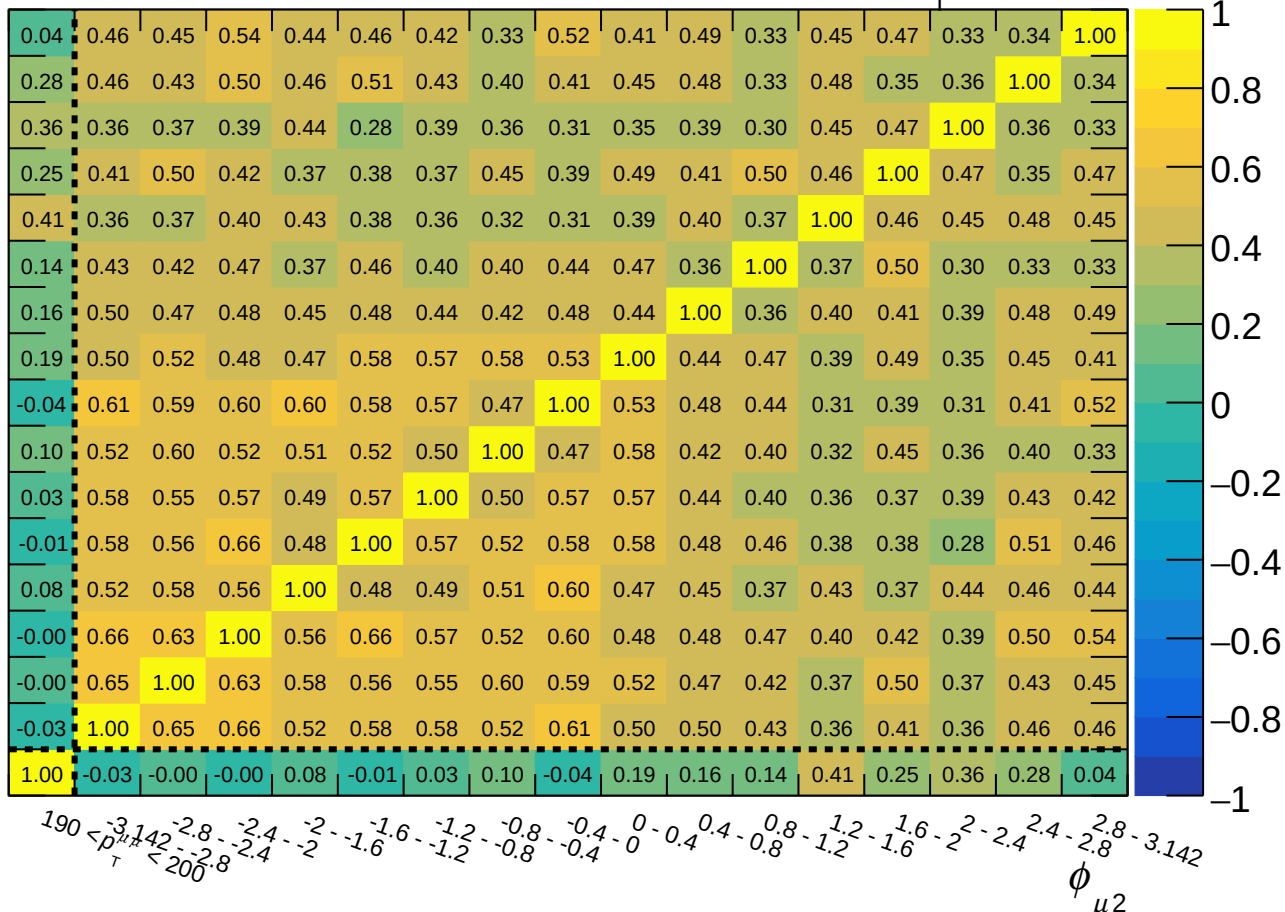


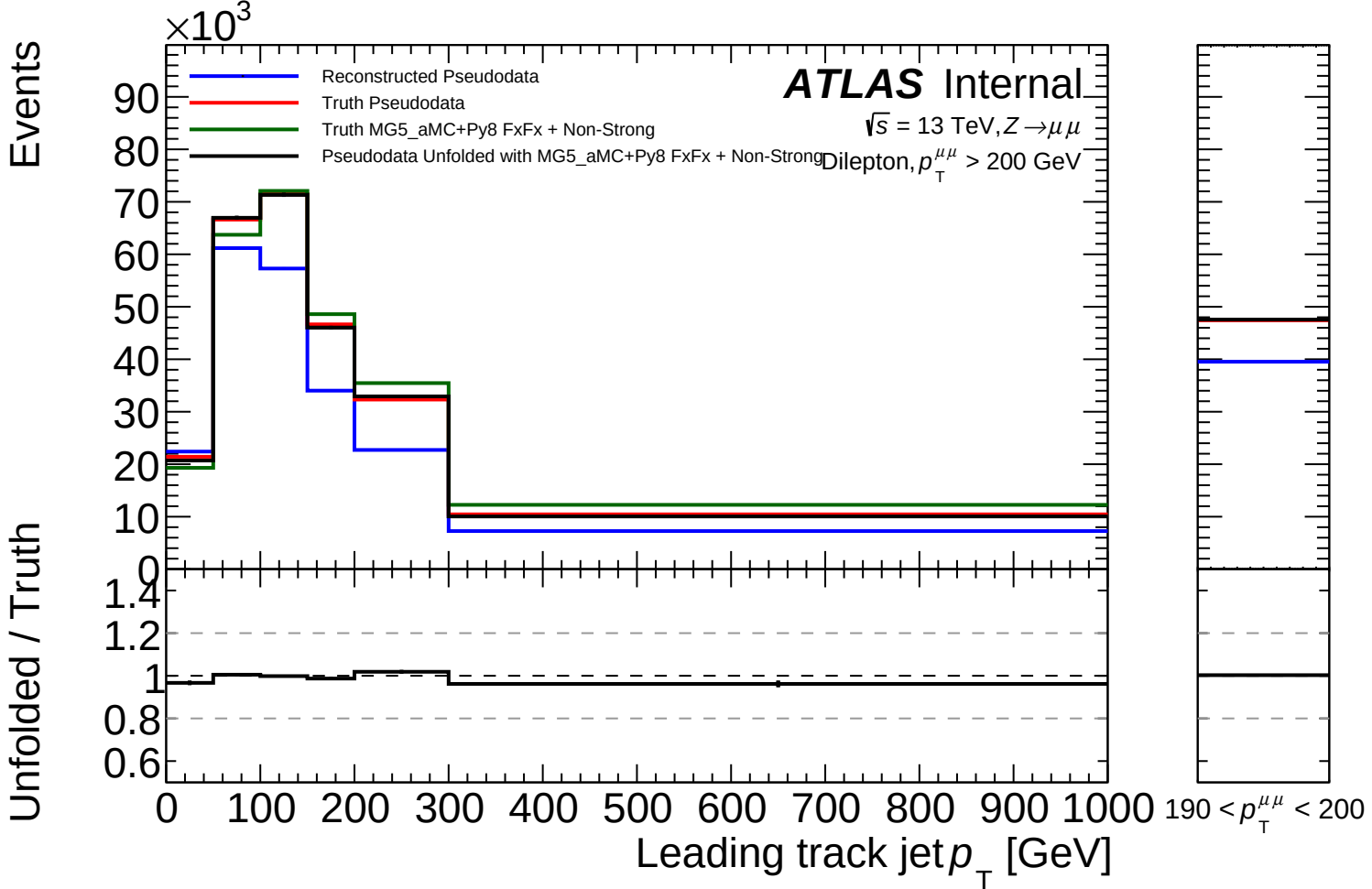


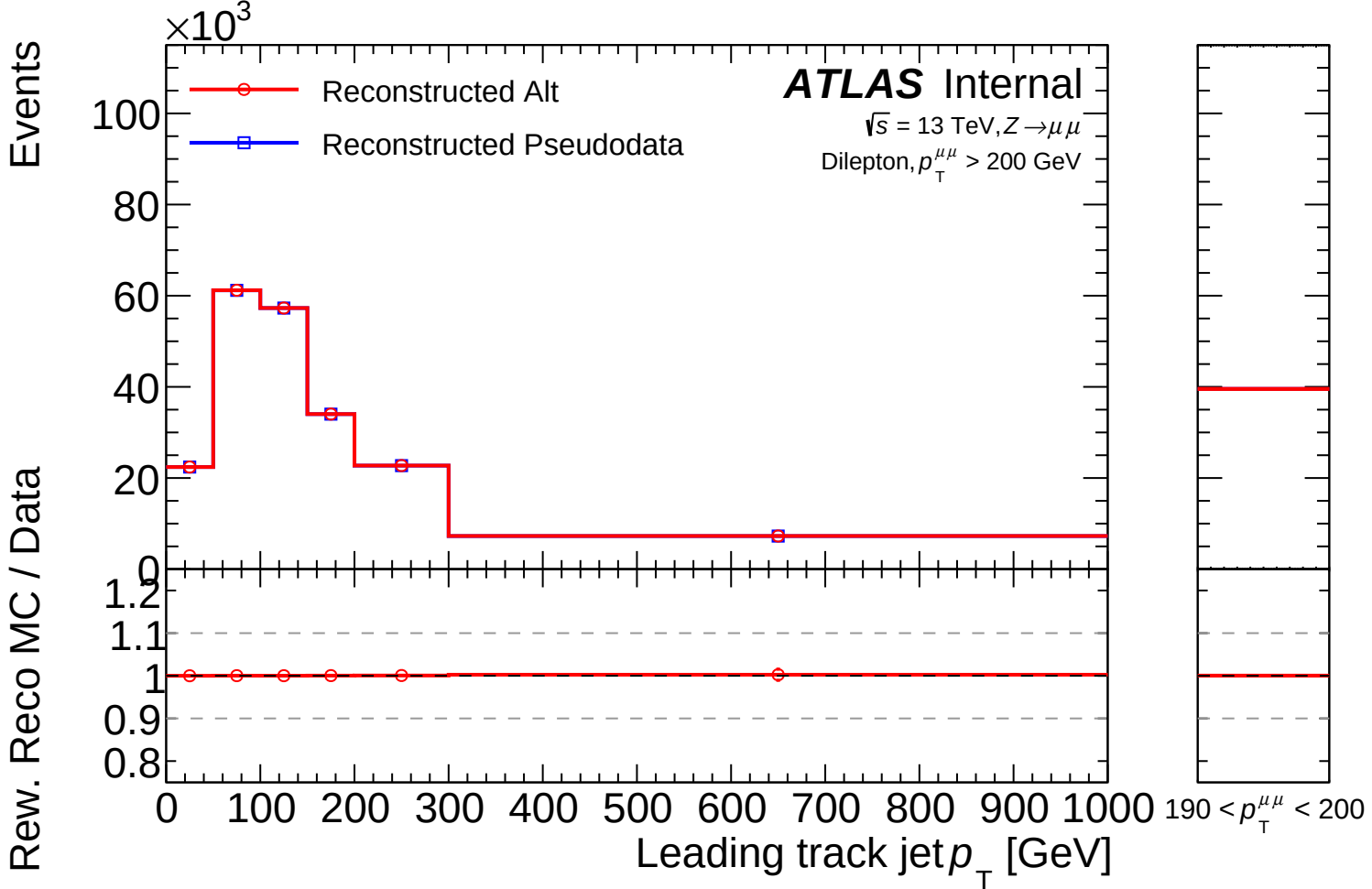


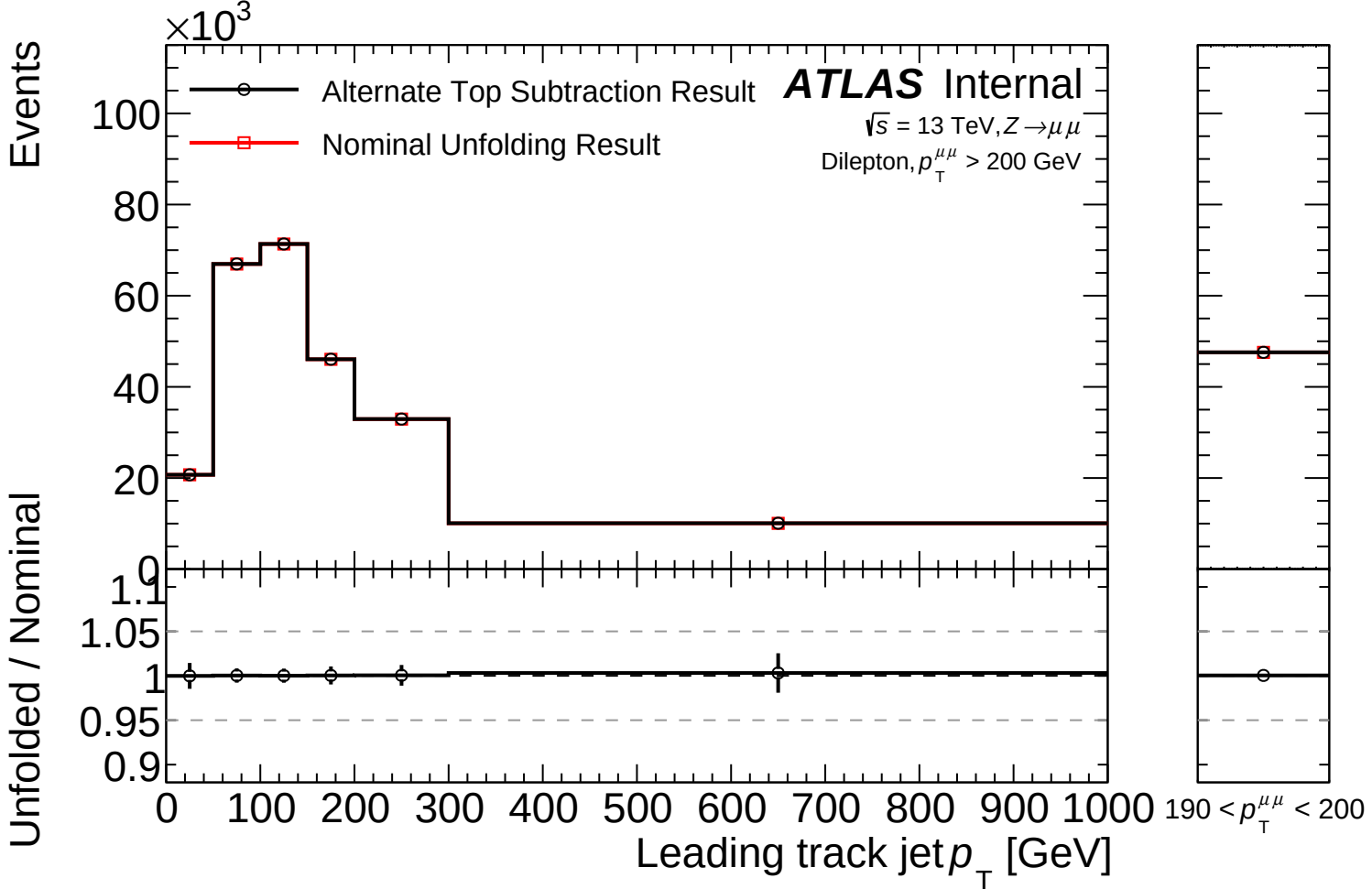
Fractional uncertainty



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $\phi_{\mu 2}$ $190 < p_{\text{T}}^{\mu\mu} < 200$ 







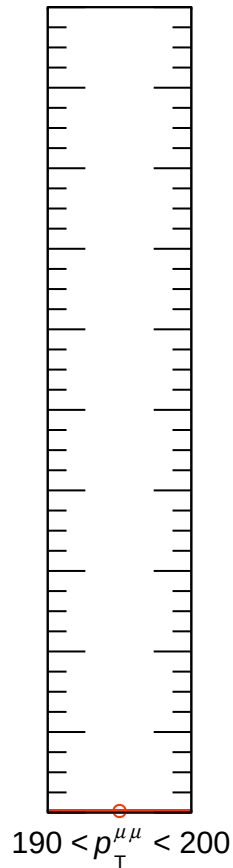
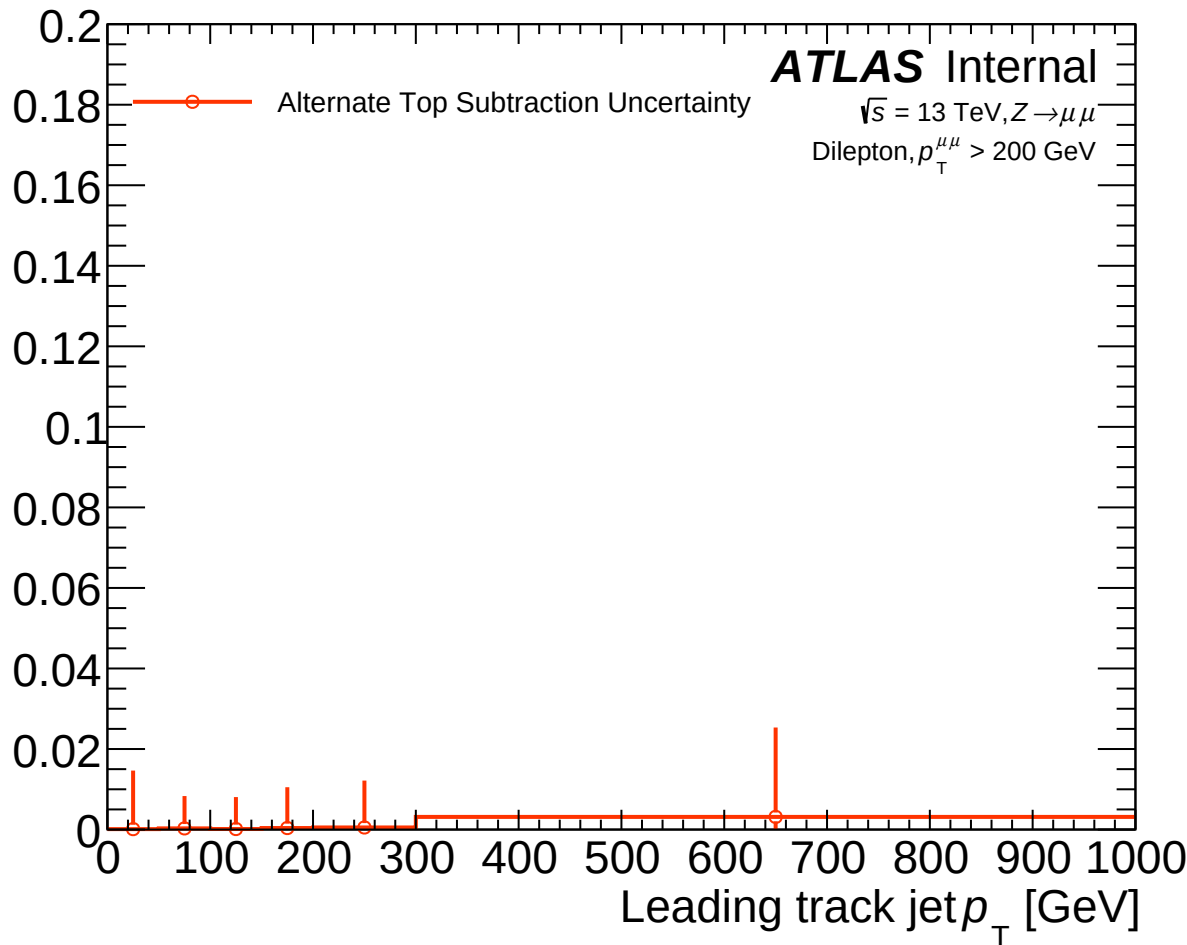
Fractional systematic impact

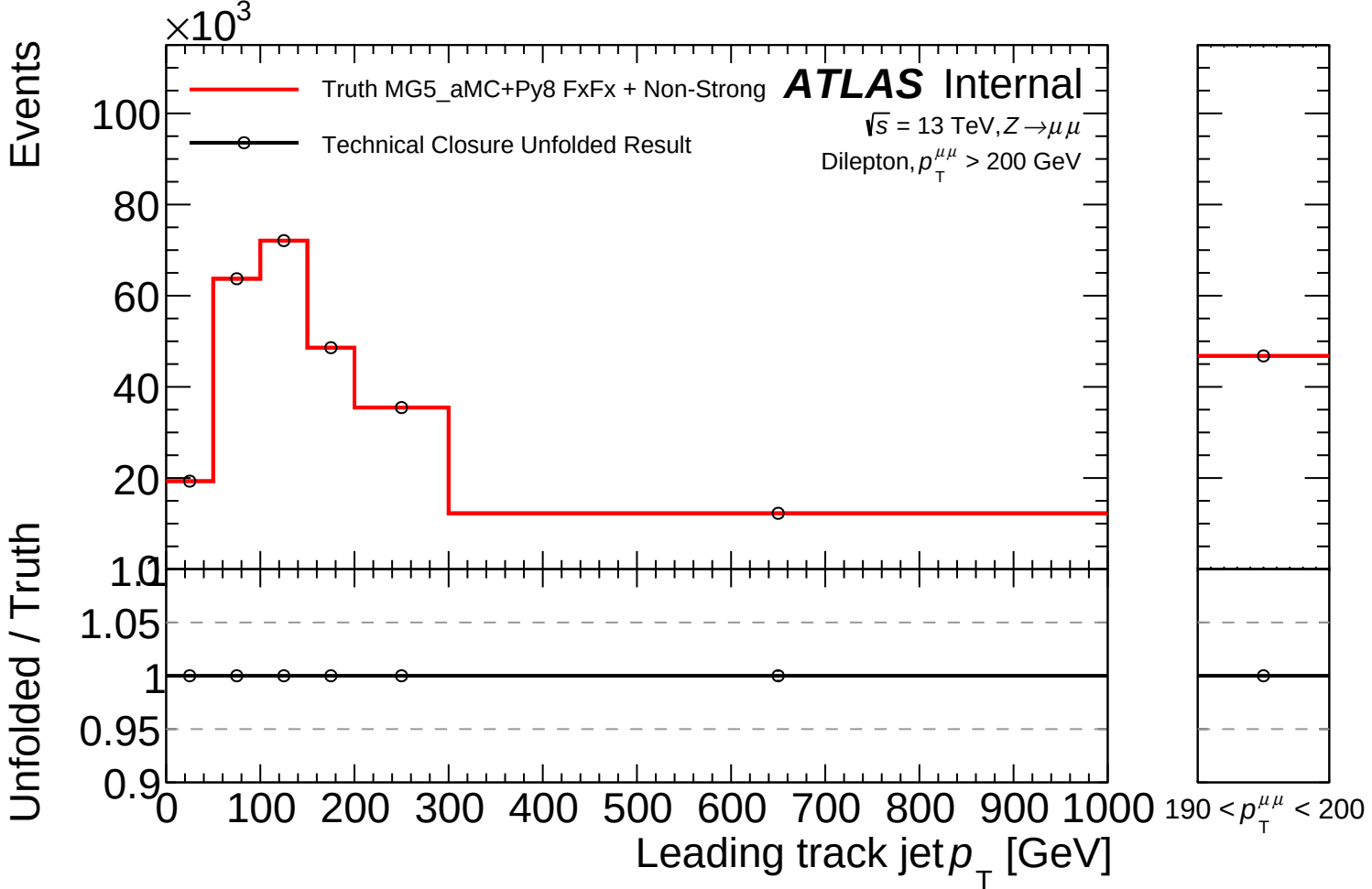
ATLAS Internal

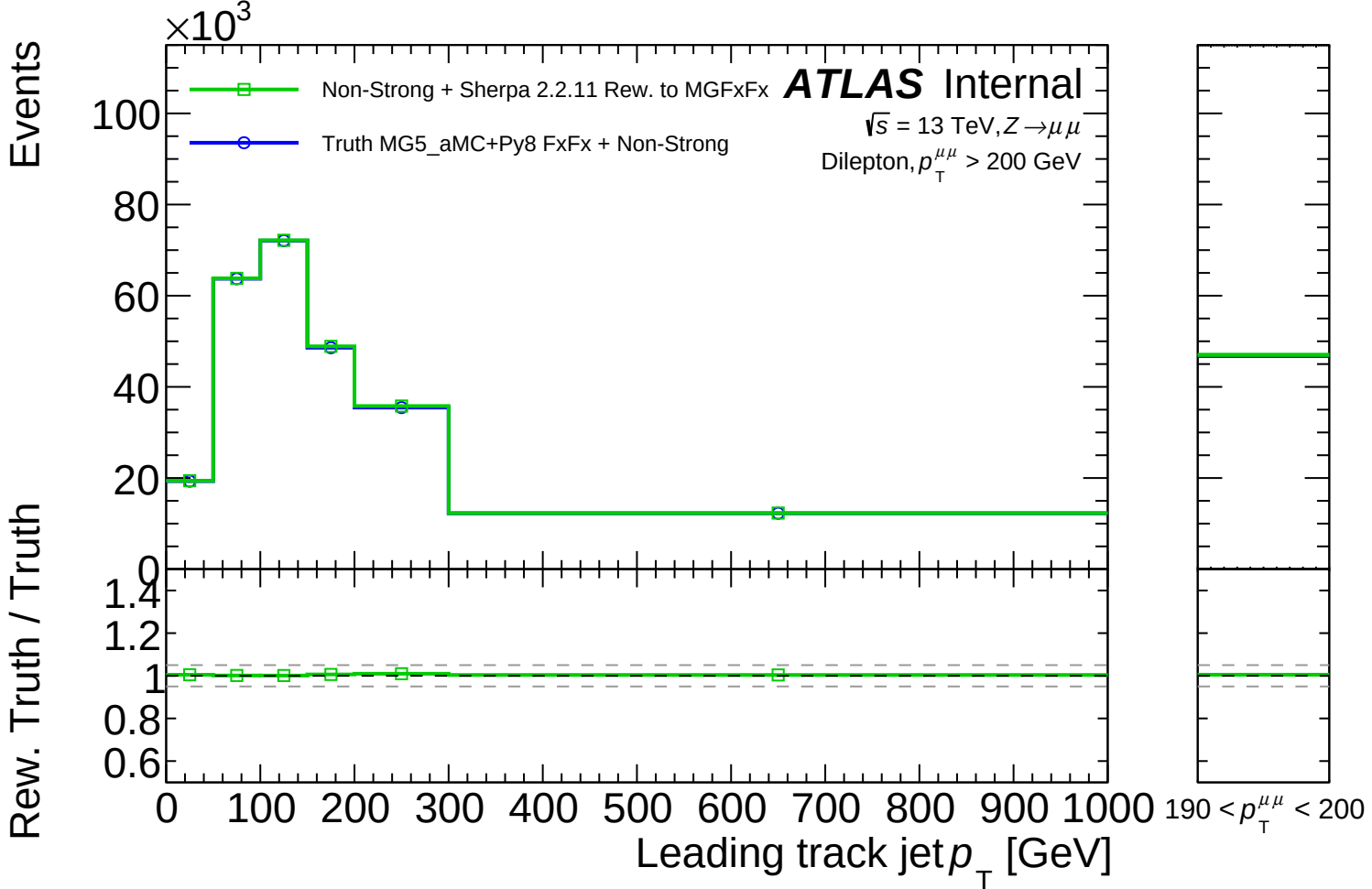
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

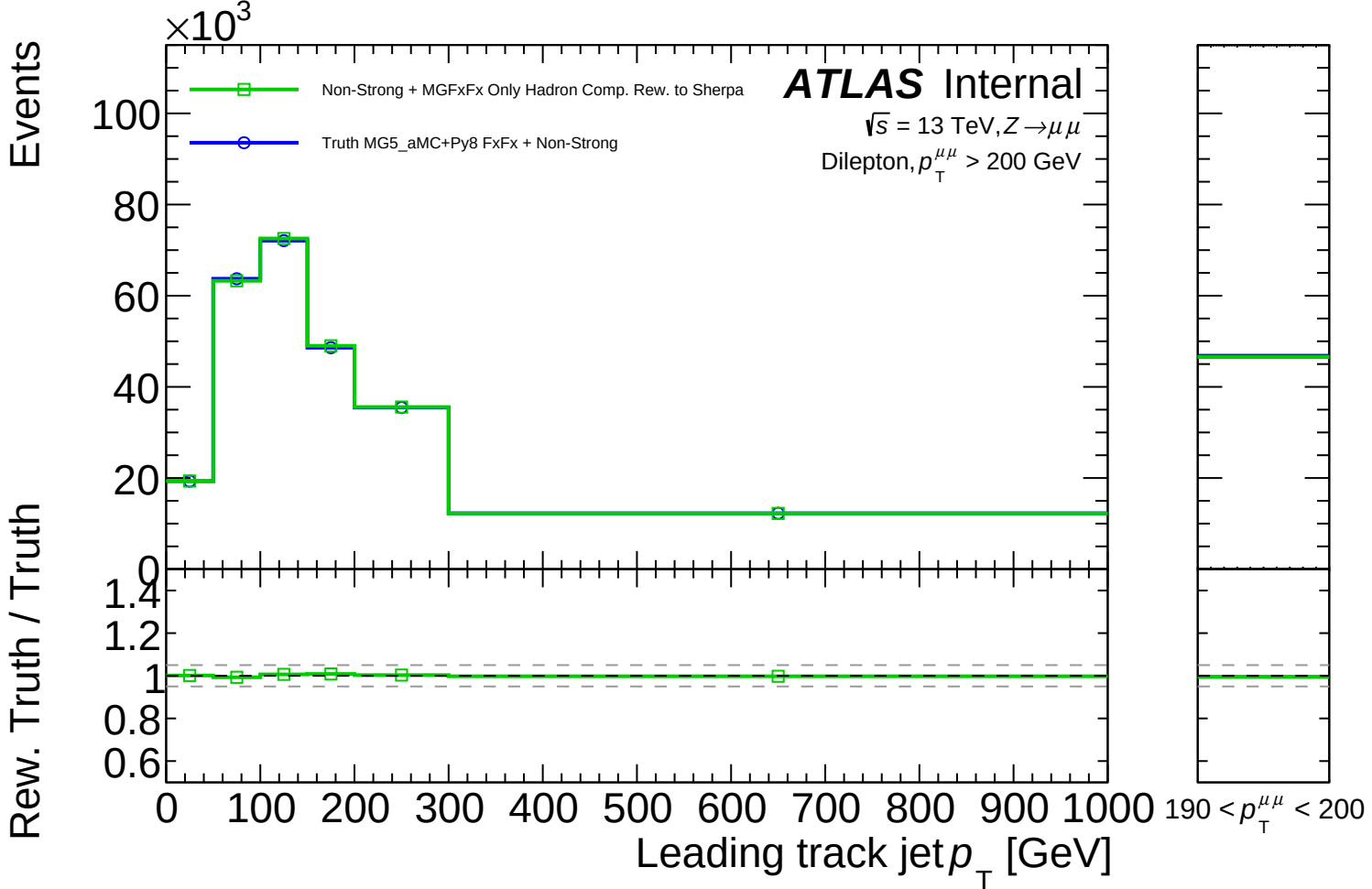
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

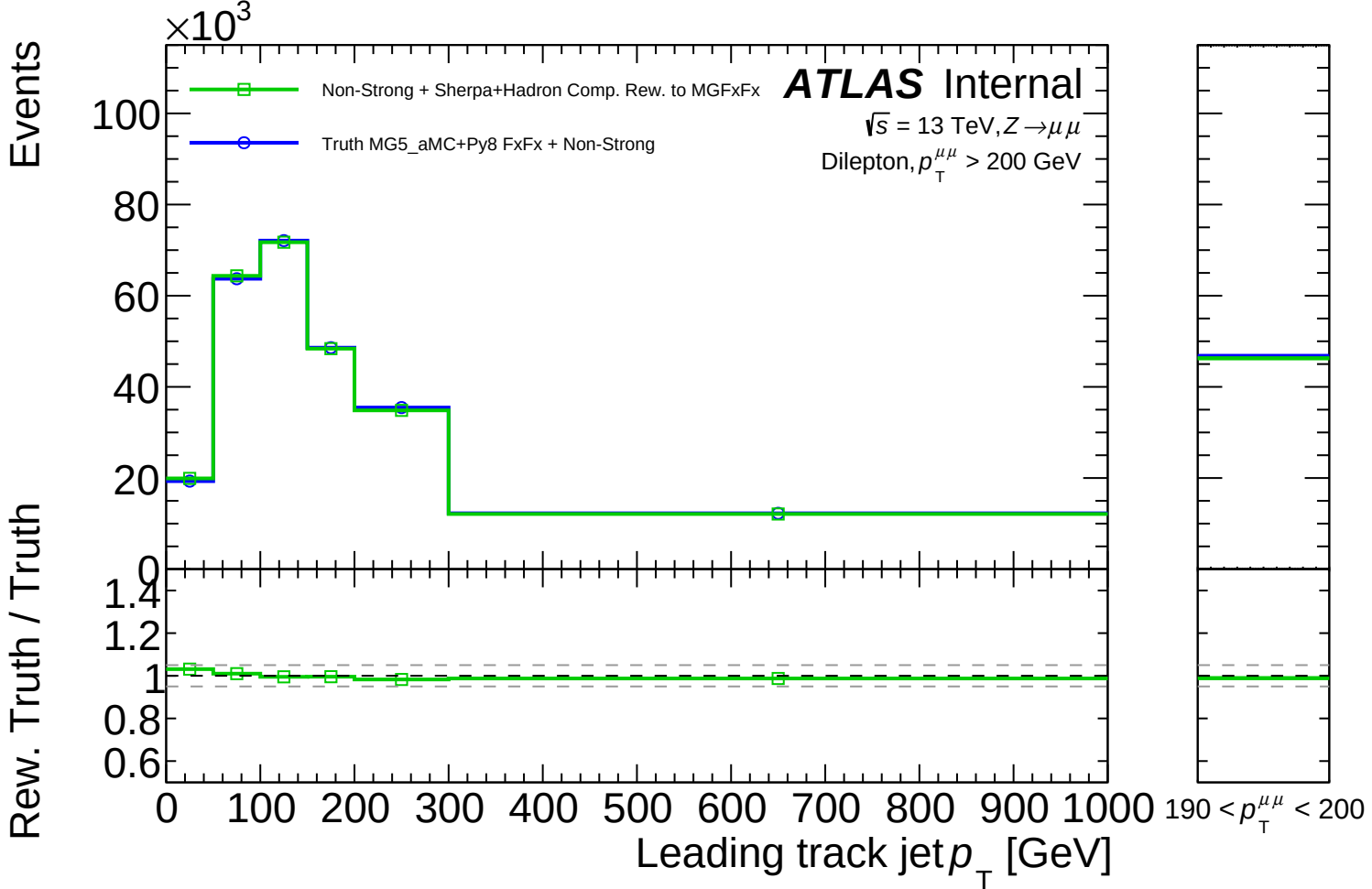
Alternate Top Subtraction Uncertainty

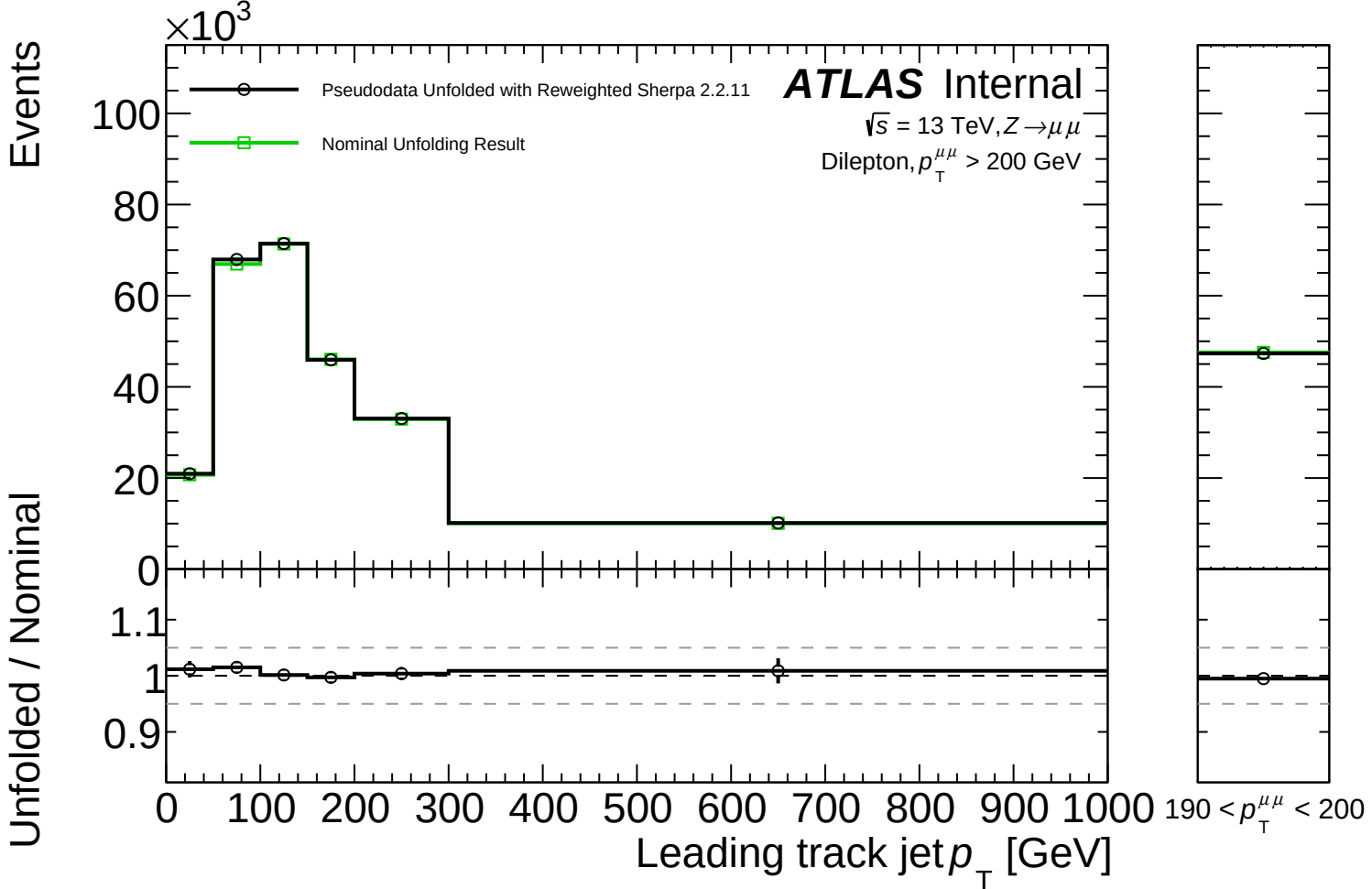


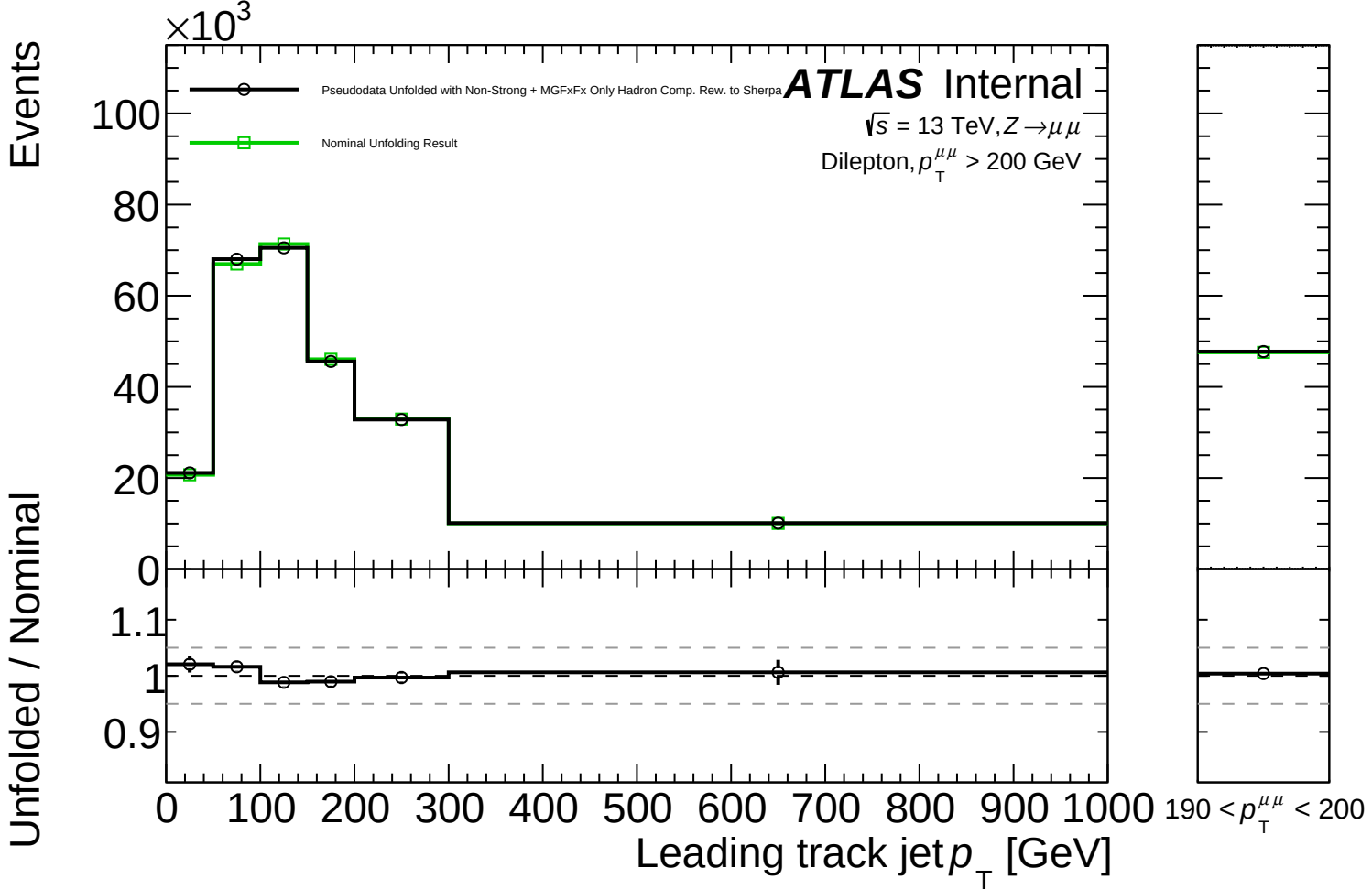


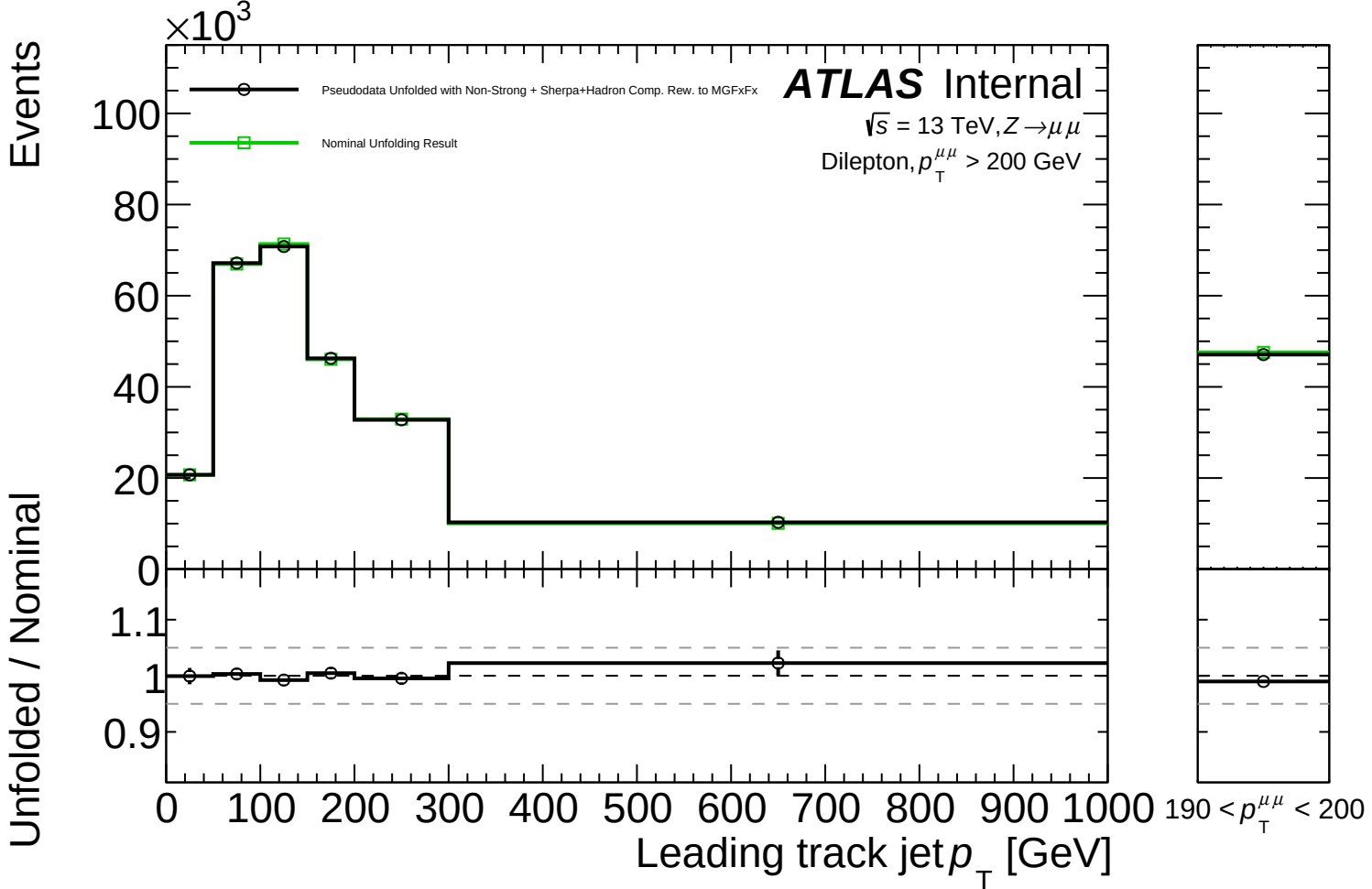


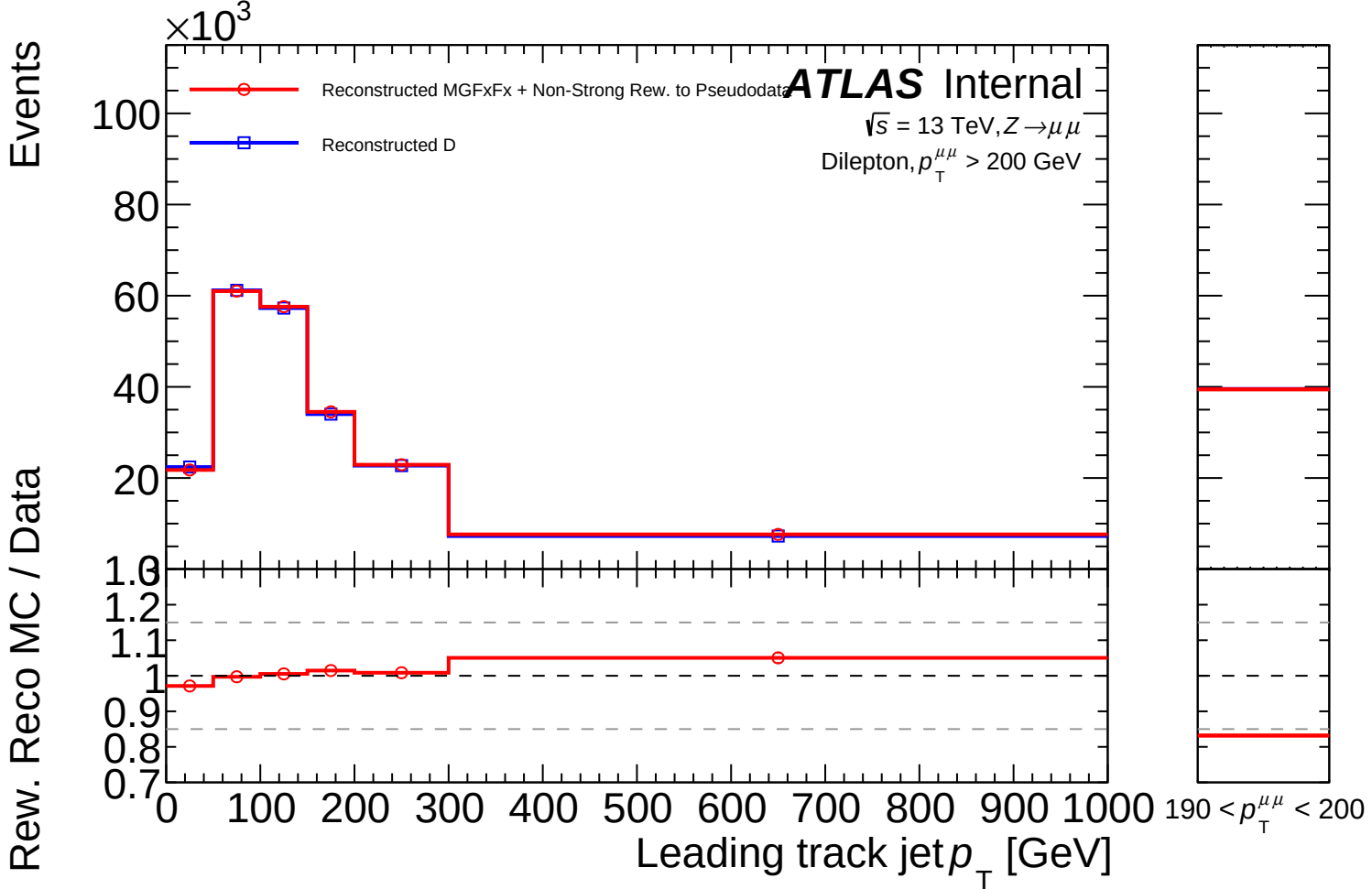


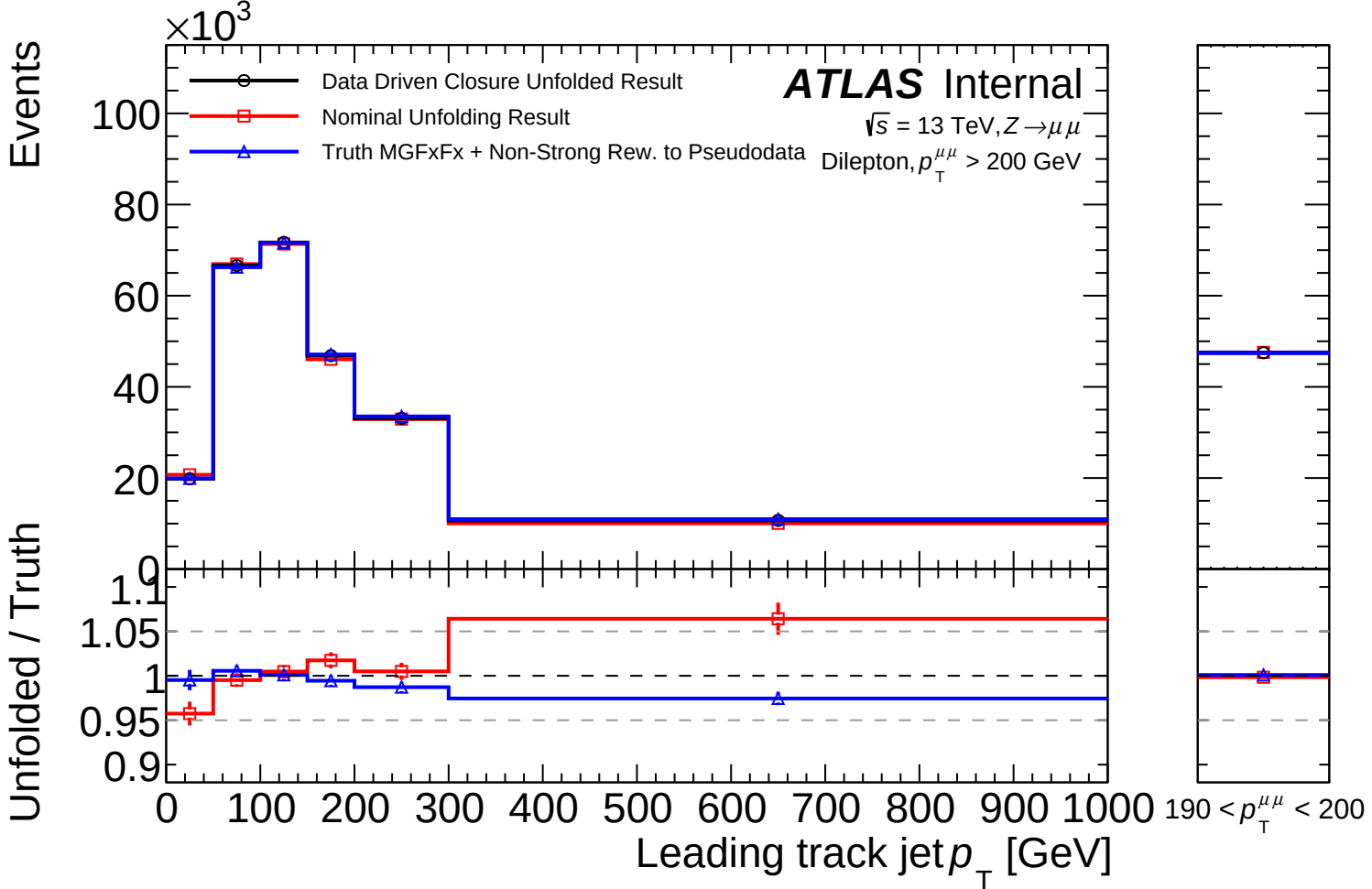




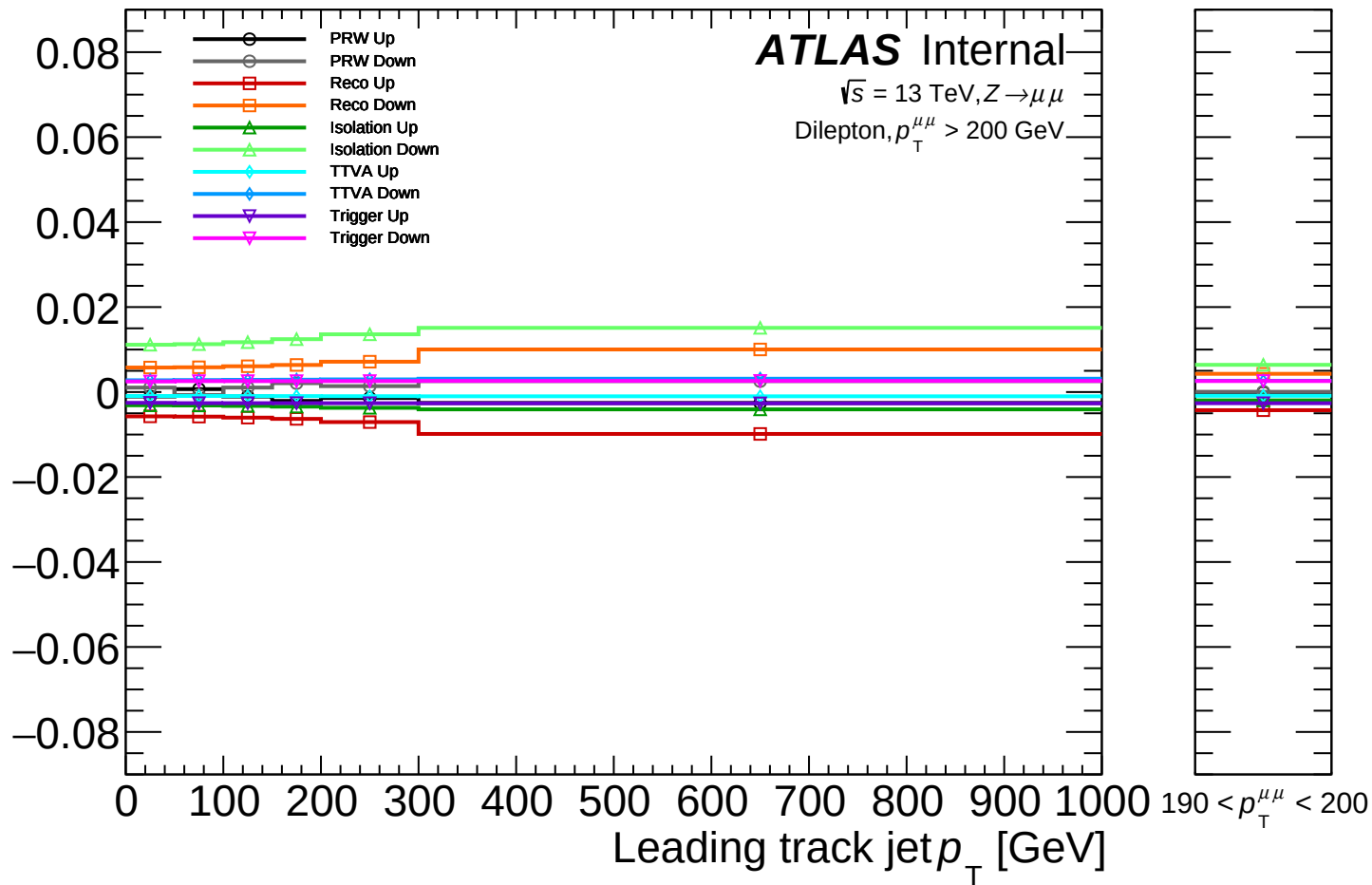




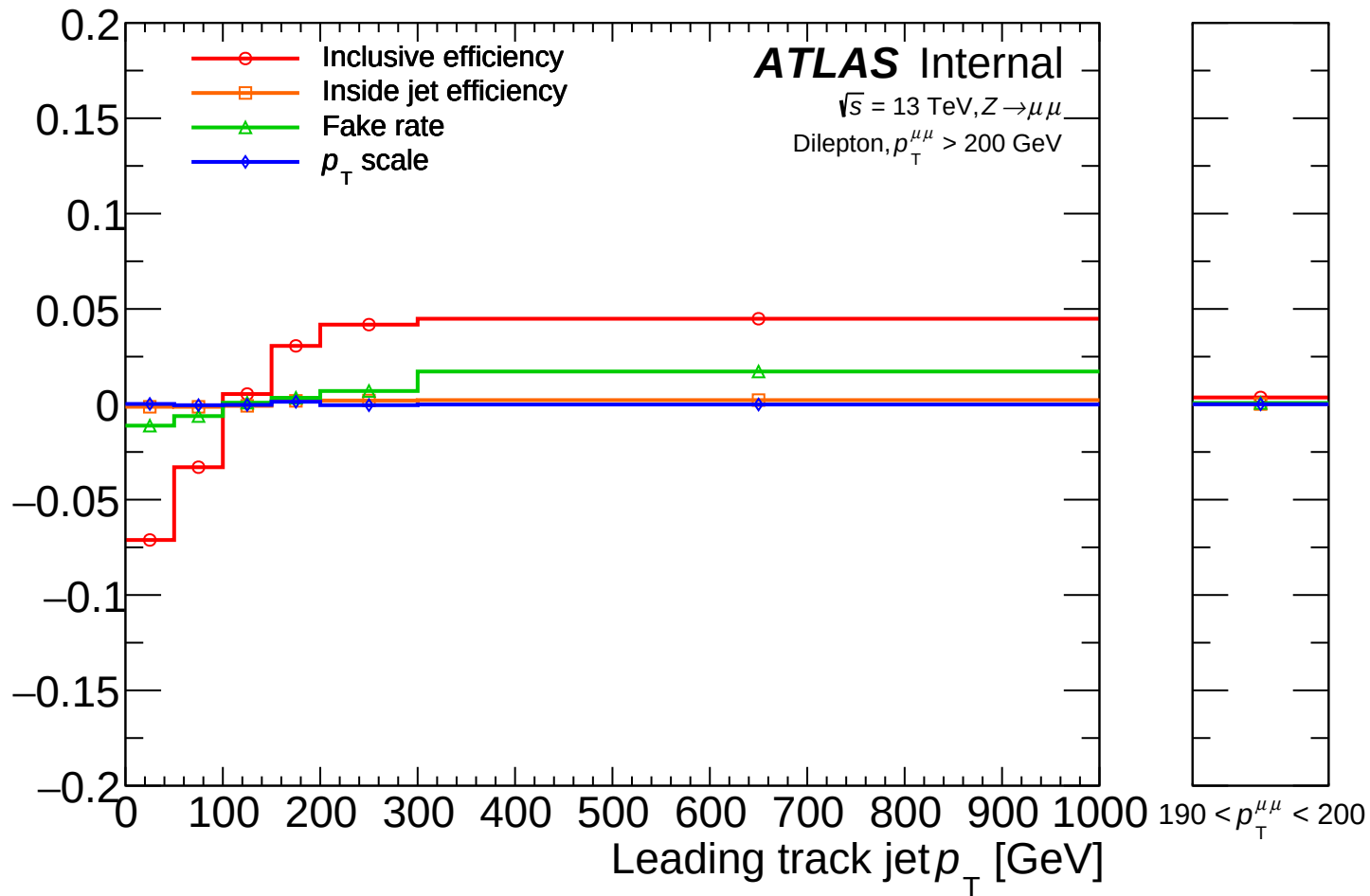




Fractional systematic impact



Fractional systematic impact



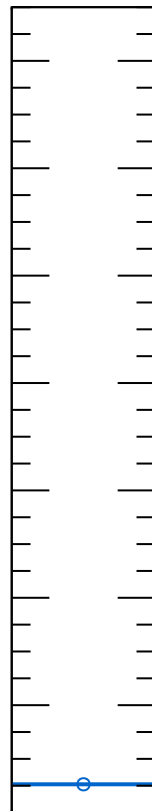
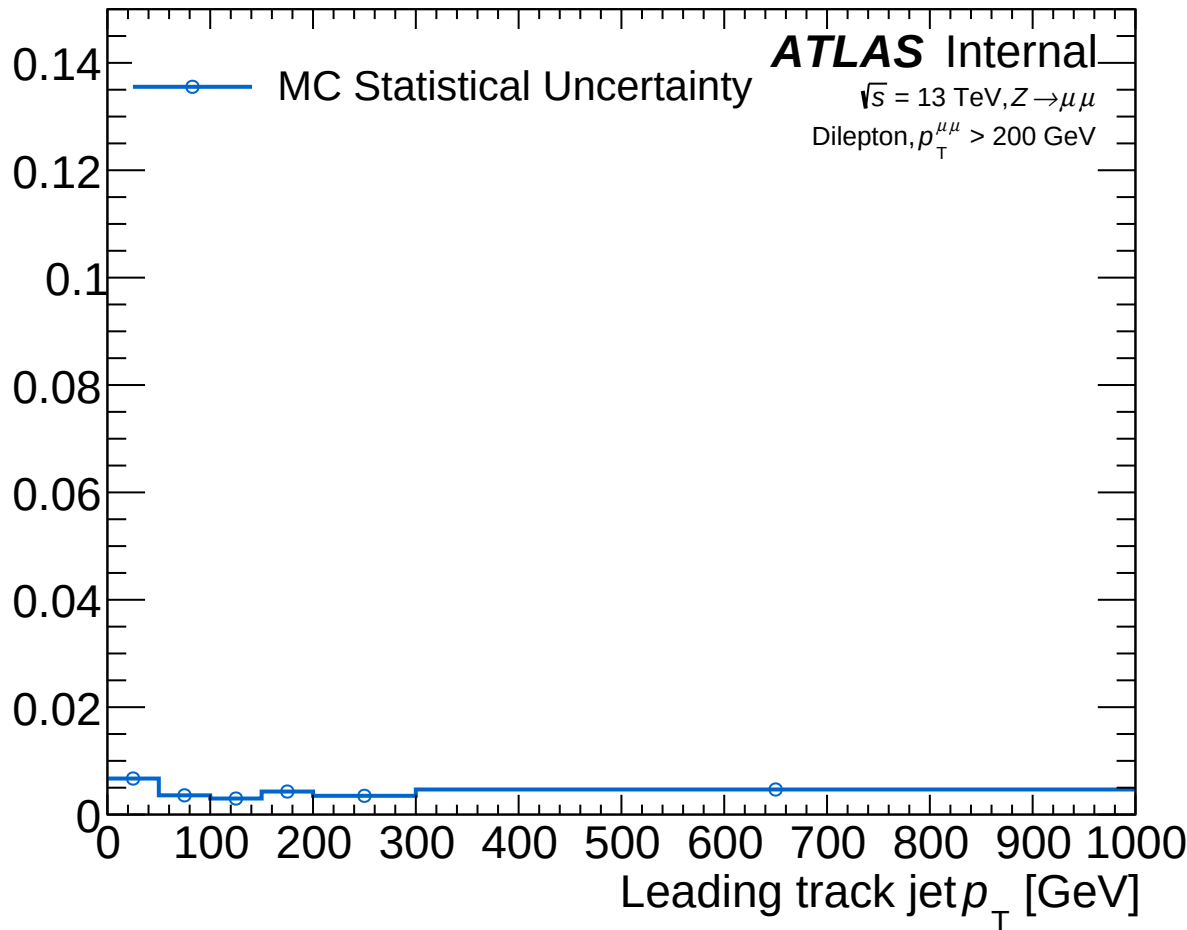
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

MC Statistical Uncertainty



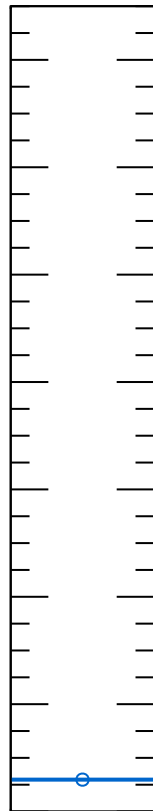
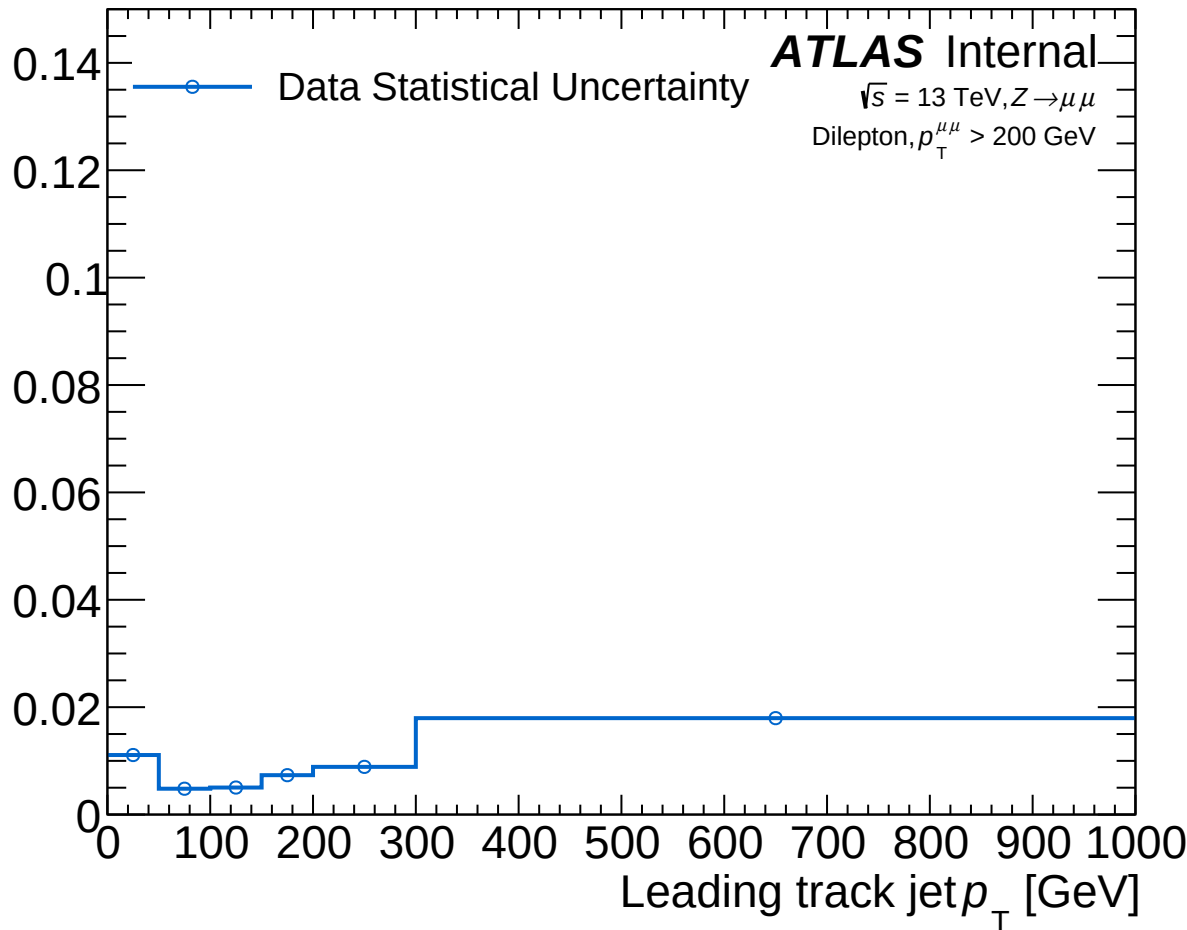
Fractional systematic impact

ATLAS Internal

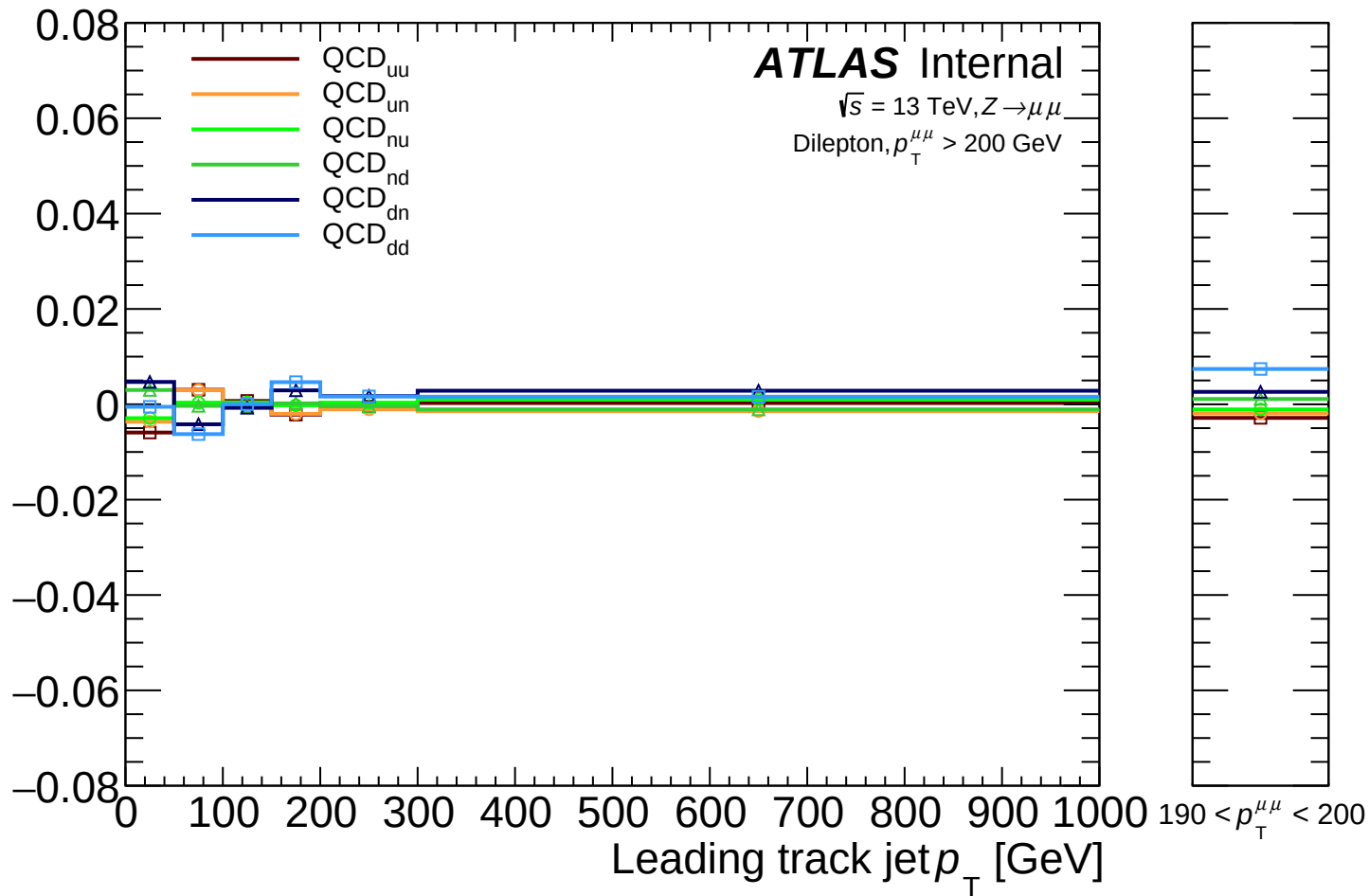
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

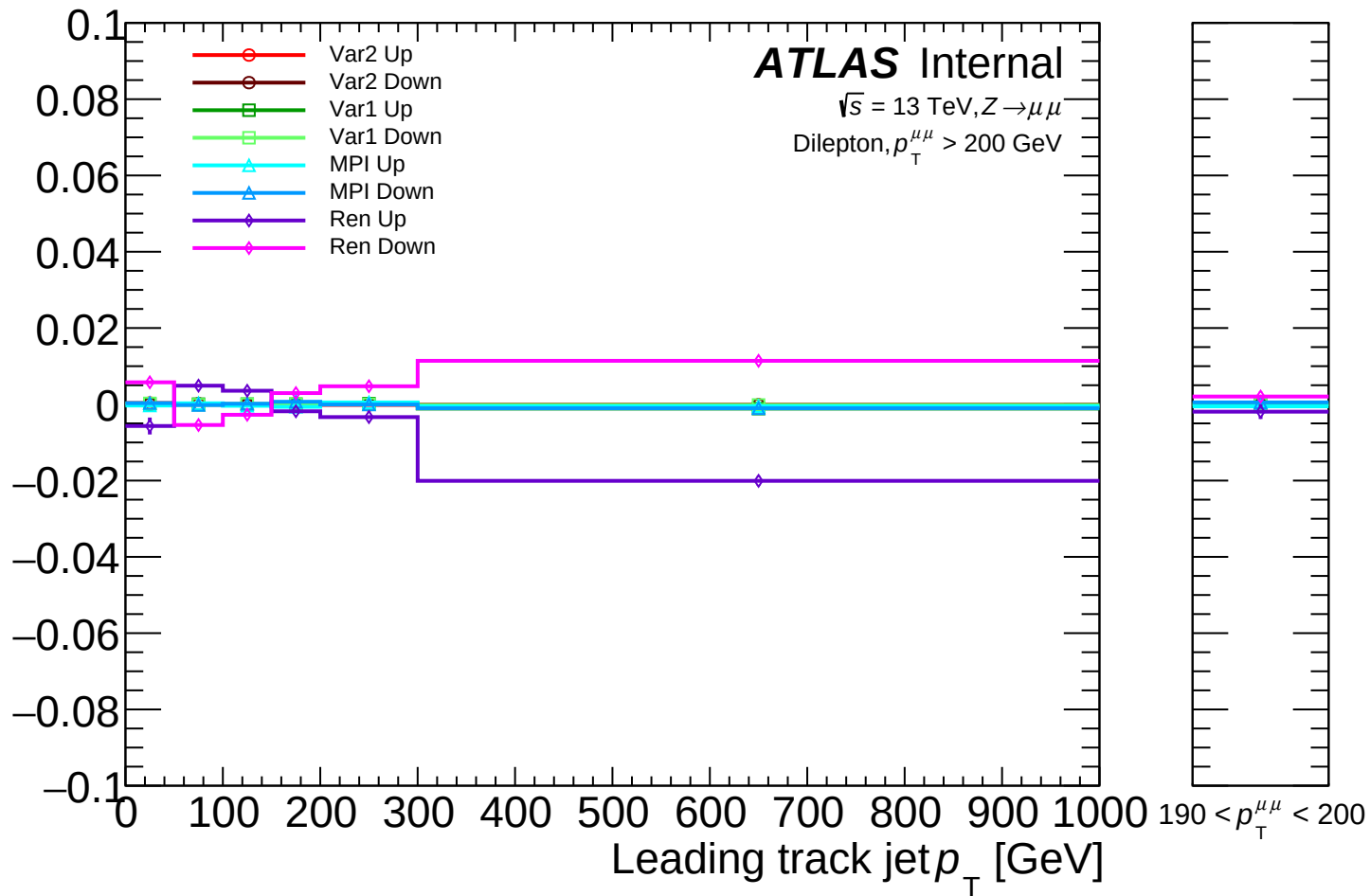
Data Statistical Uncertainty

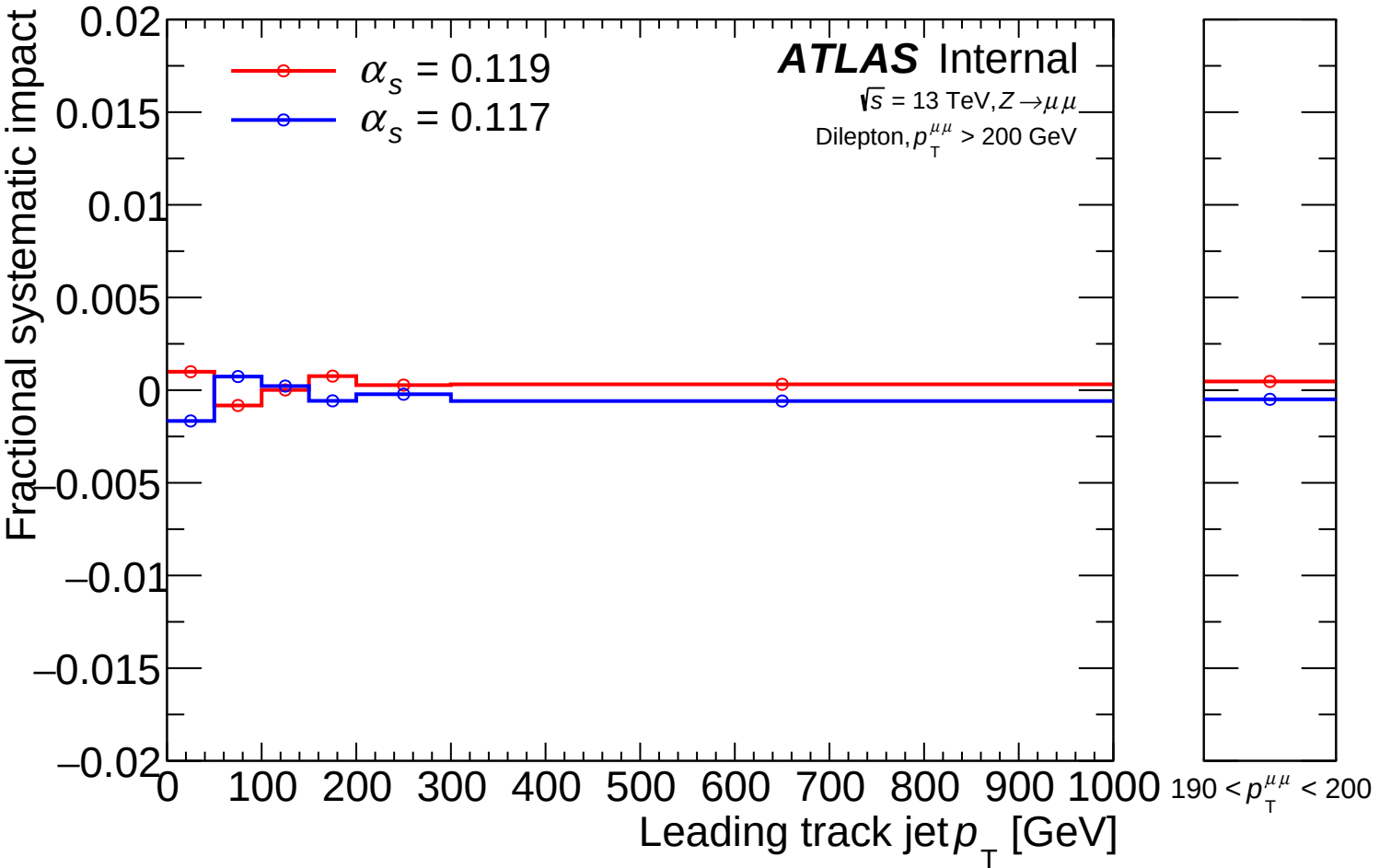


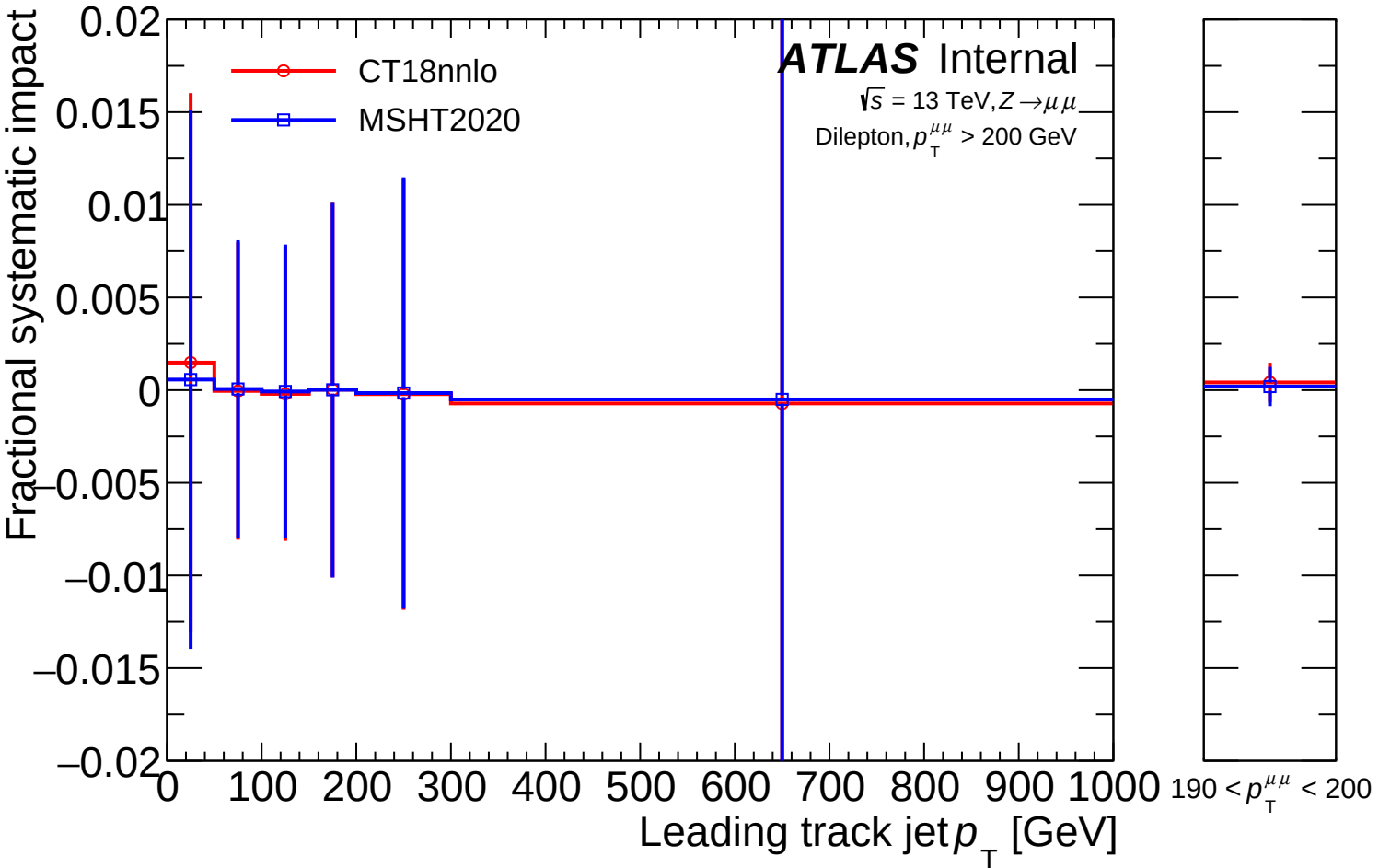
Fractional systematic impact



Fractional systematic impact







Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

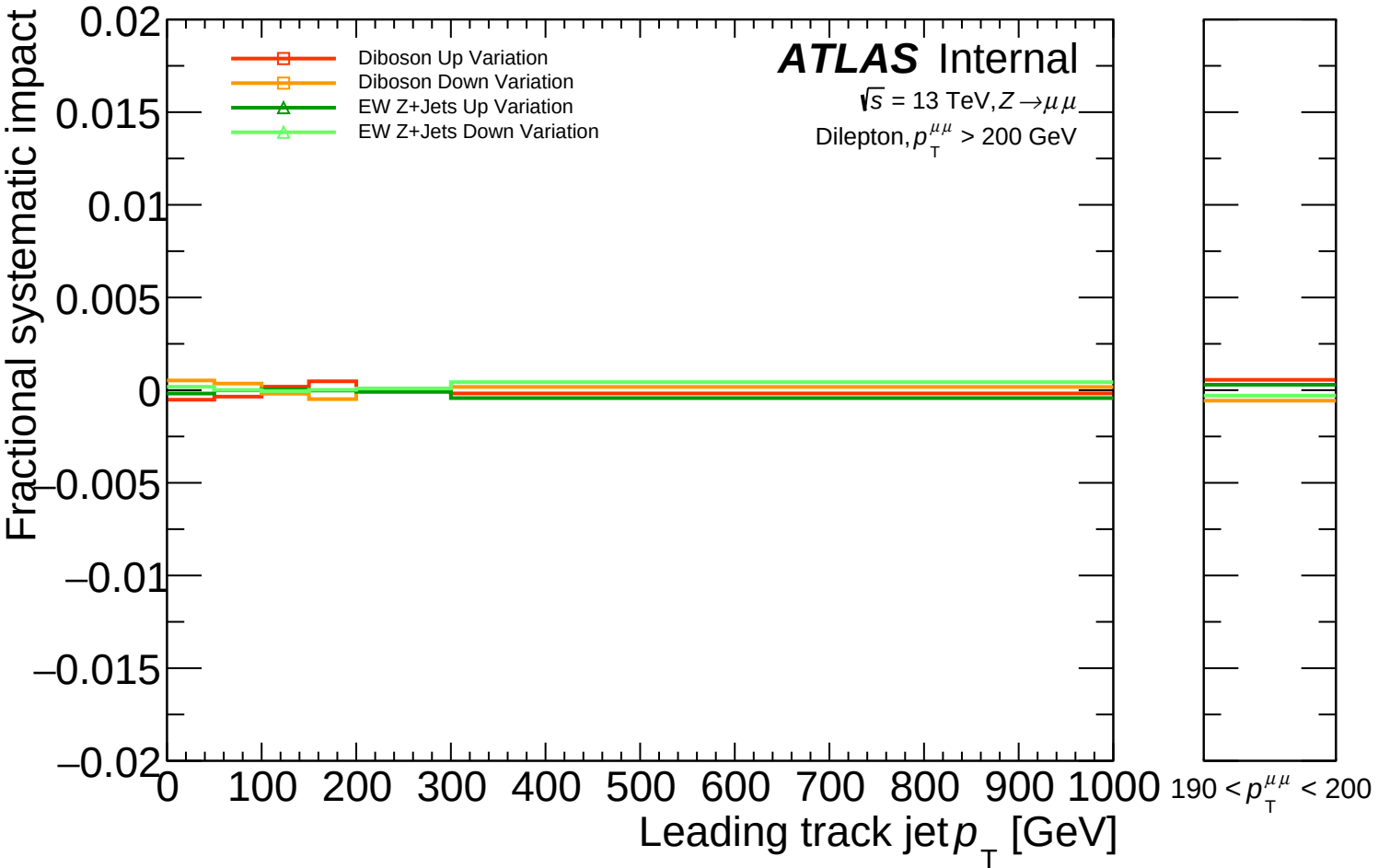
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

Leading track jet p_T [GeV]

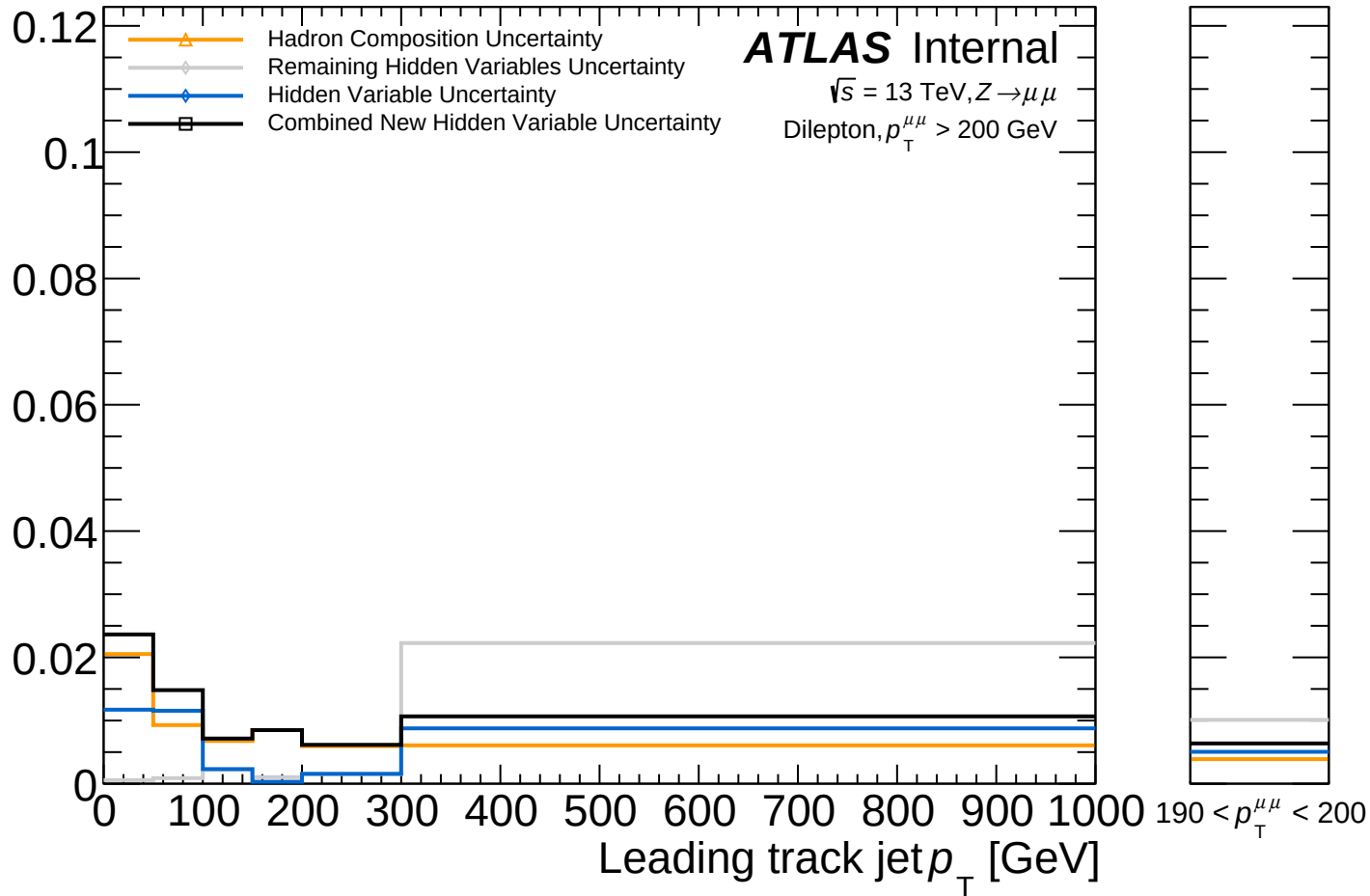
$190 < p_T^{\mu\mu} < 200$

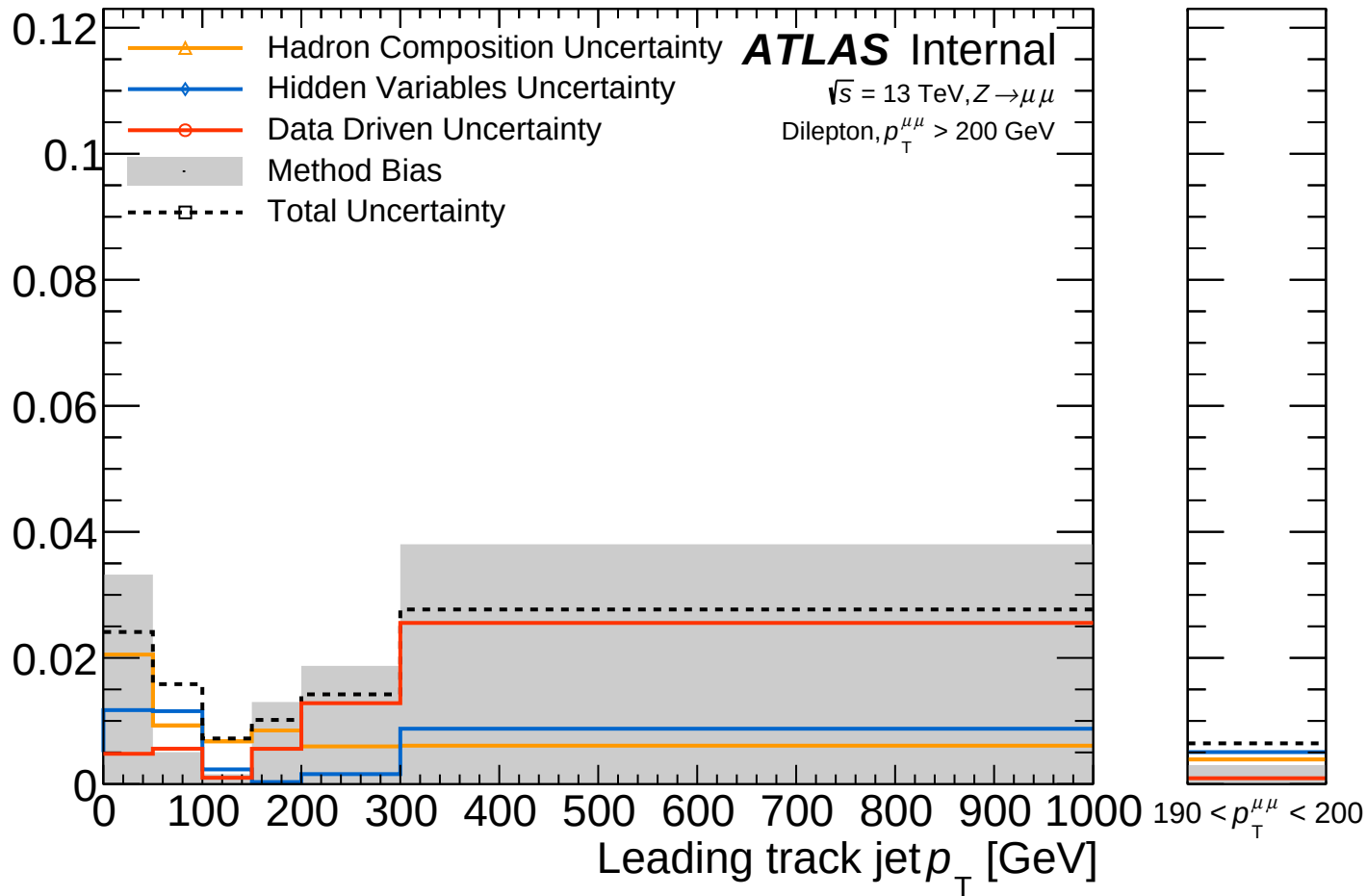


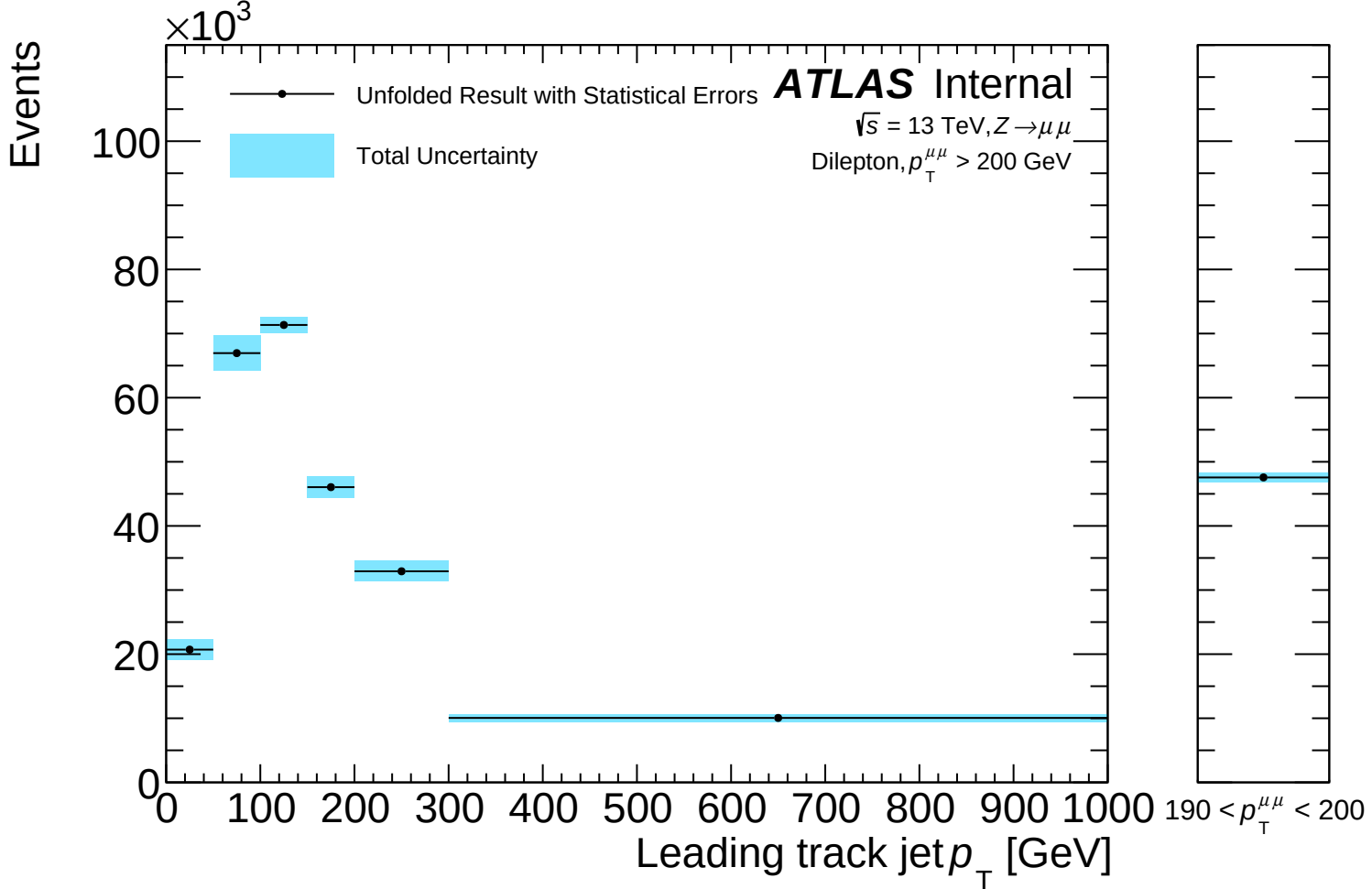
ATLAS Internal

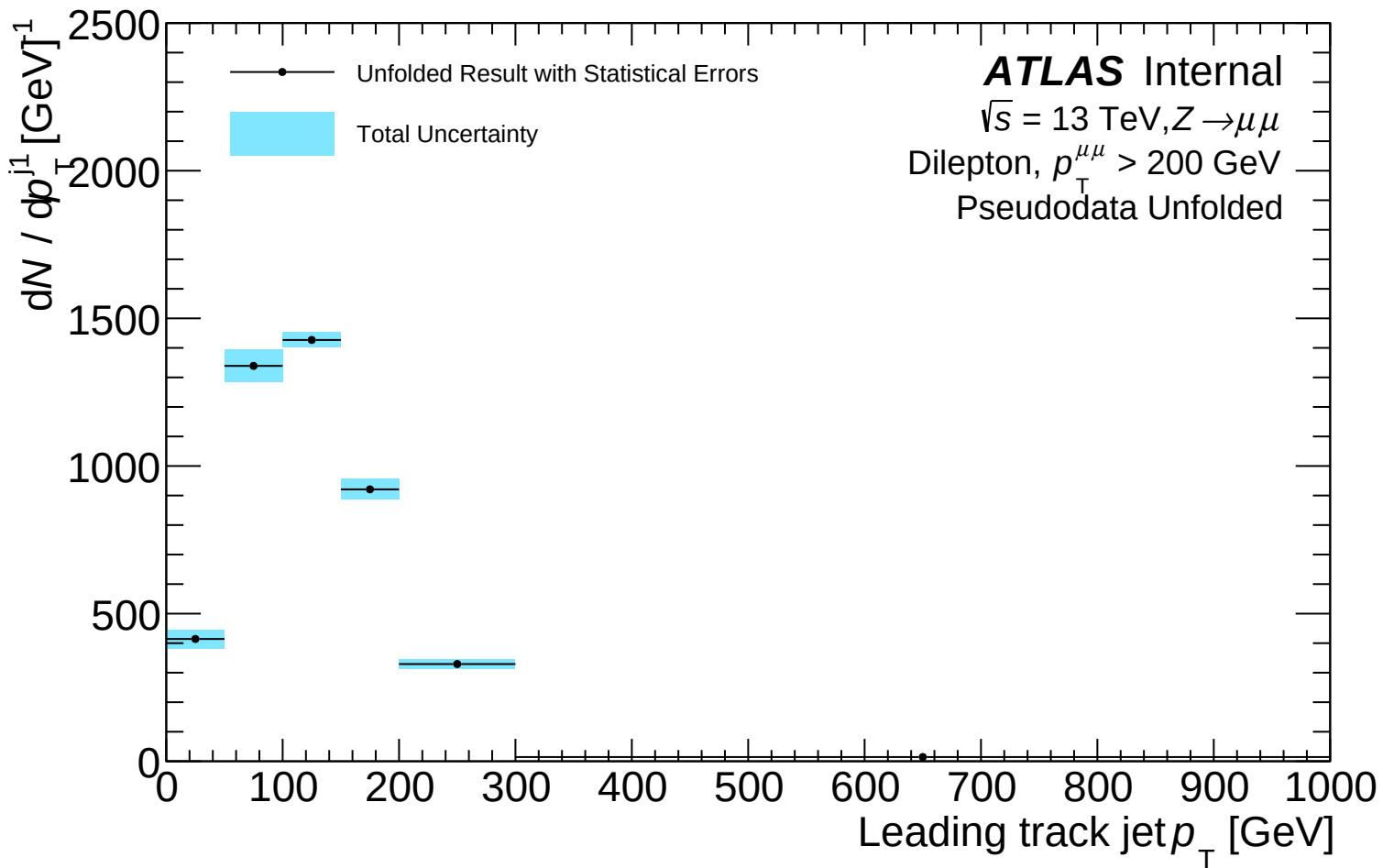
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

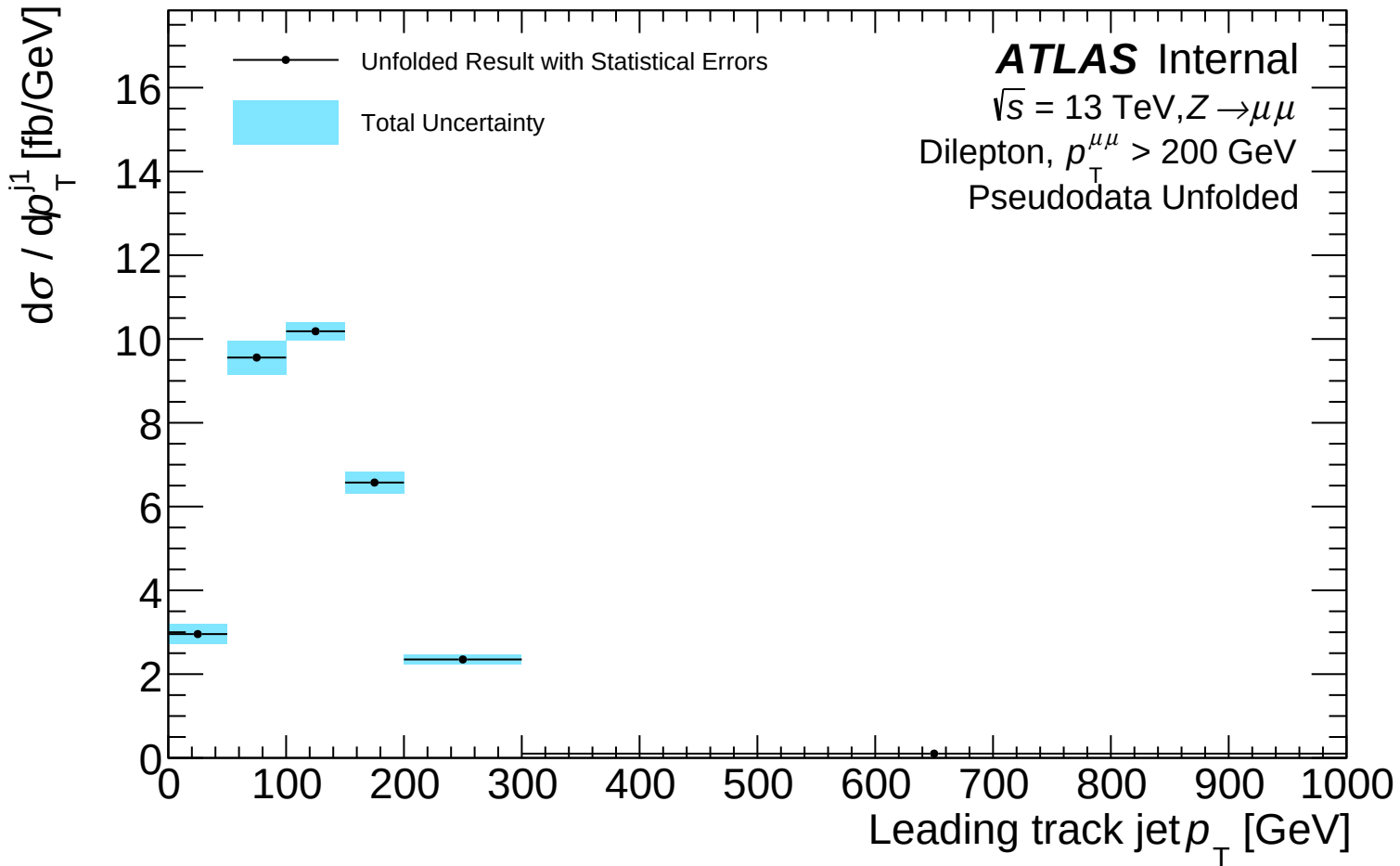
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$











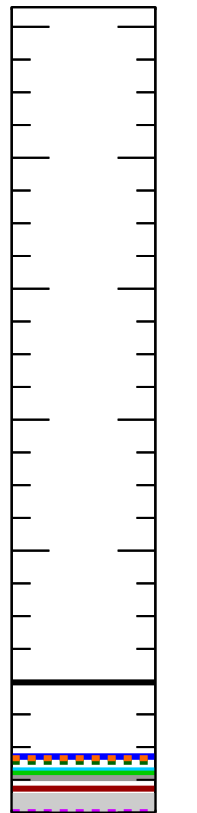
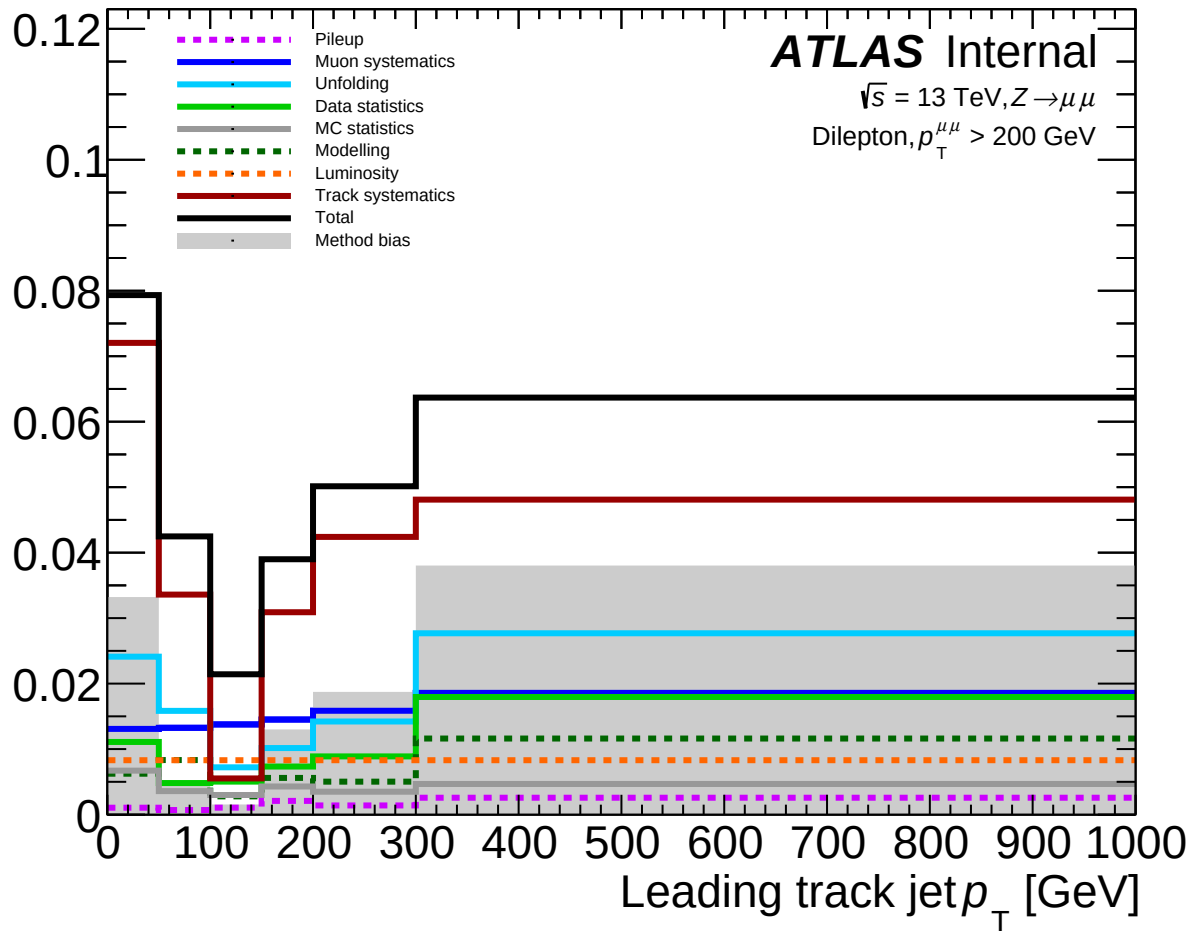
Fractional uncertainty

ATLAS Internal

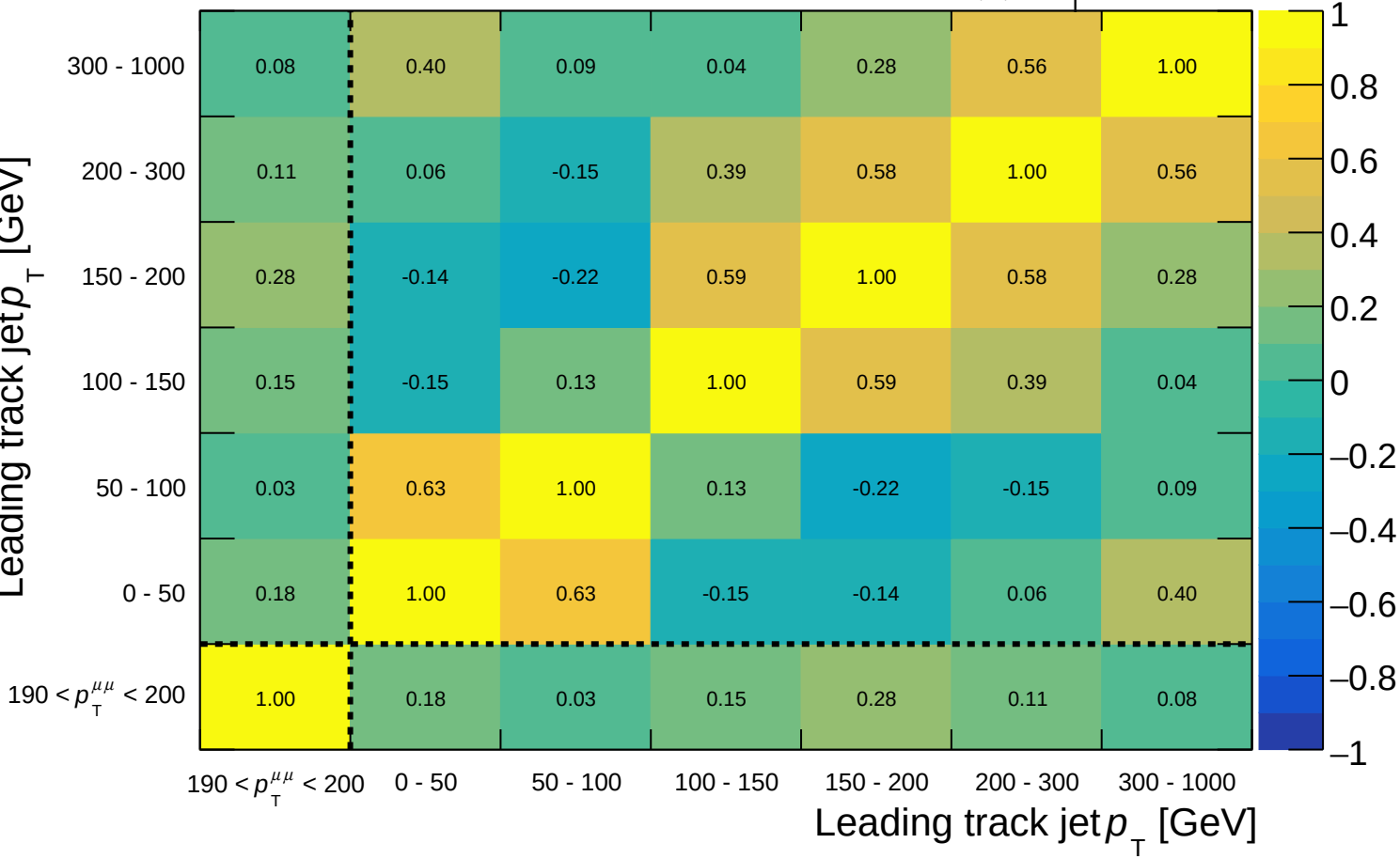
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

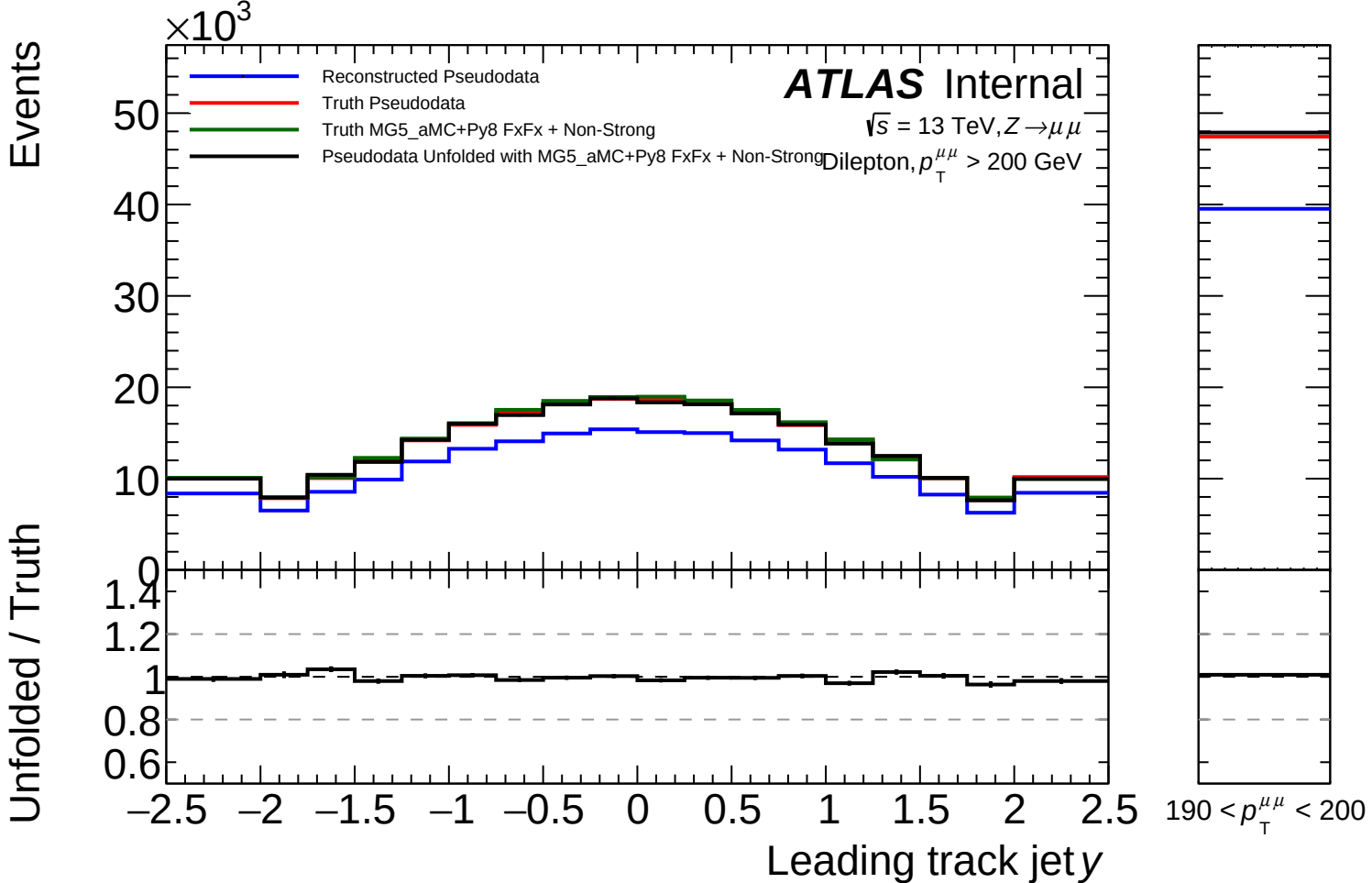
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

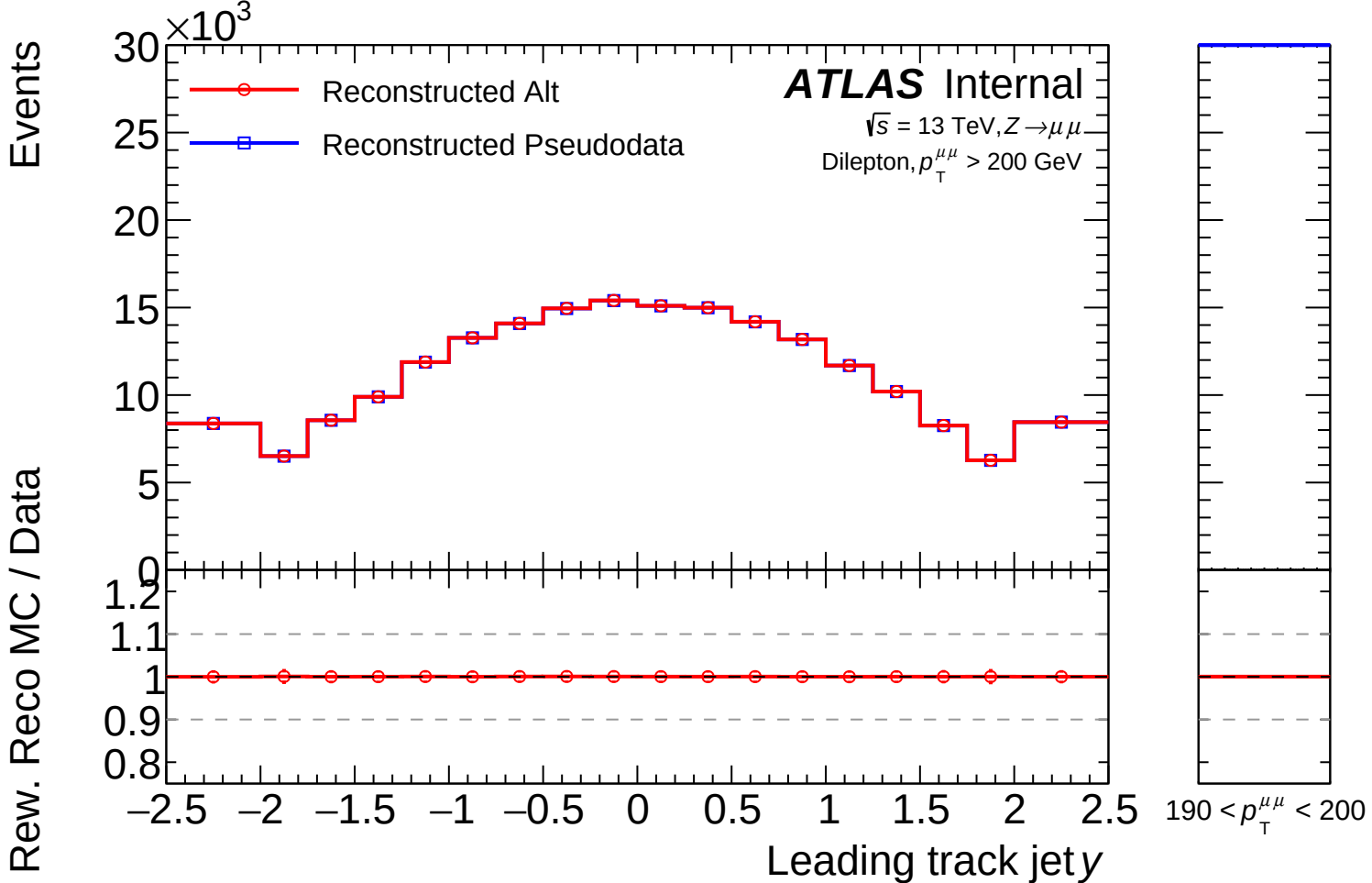
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

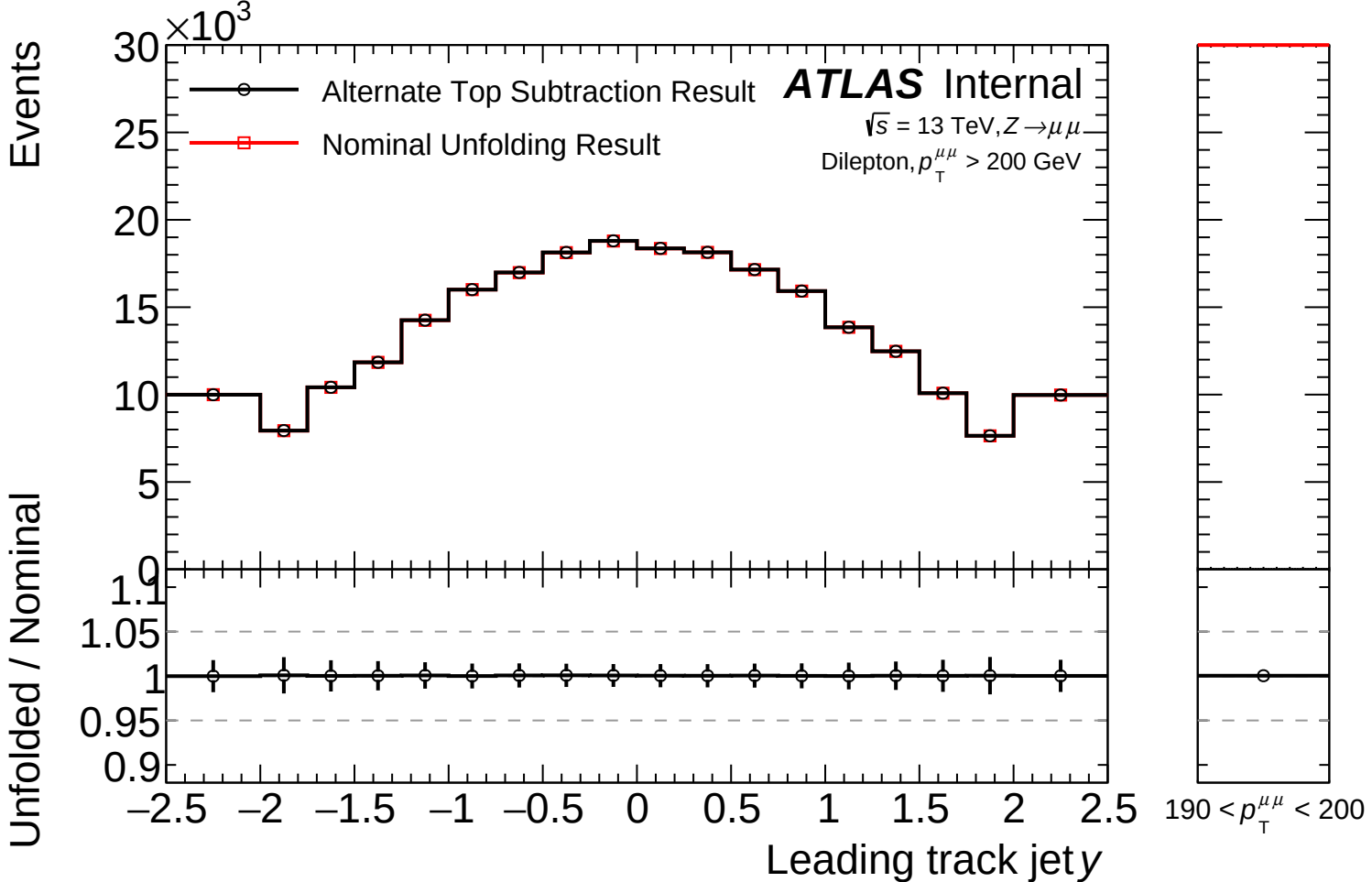


$190 < p_T^{\mu\mu} < 200$

ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet p_{T} [GeV]Leading track jet p_{T} [GeV]







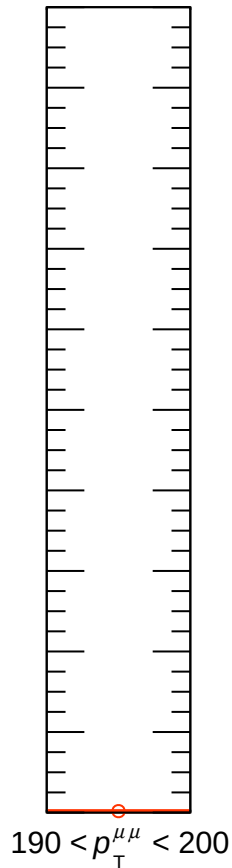
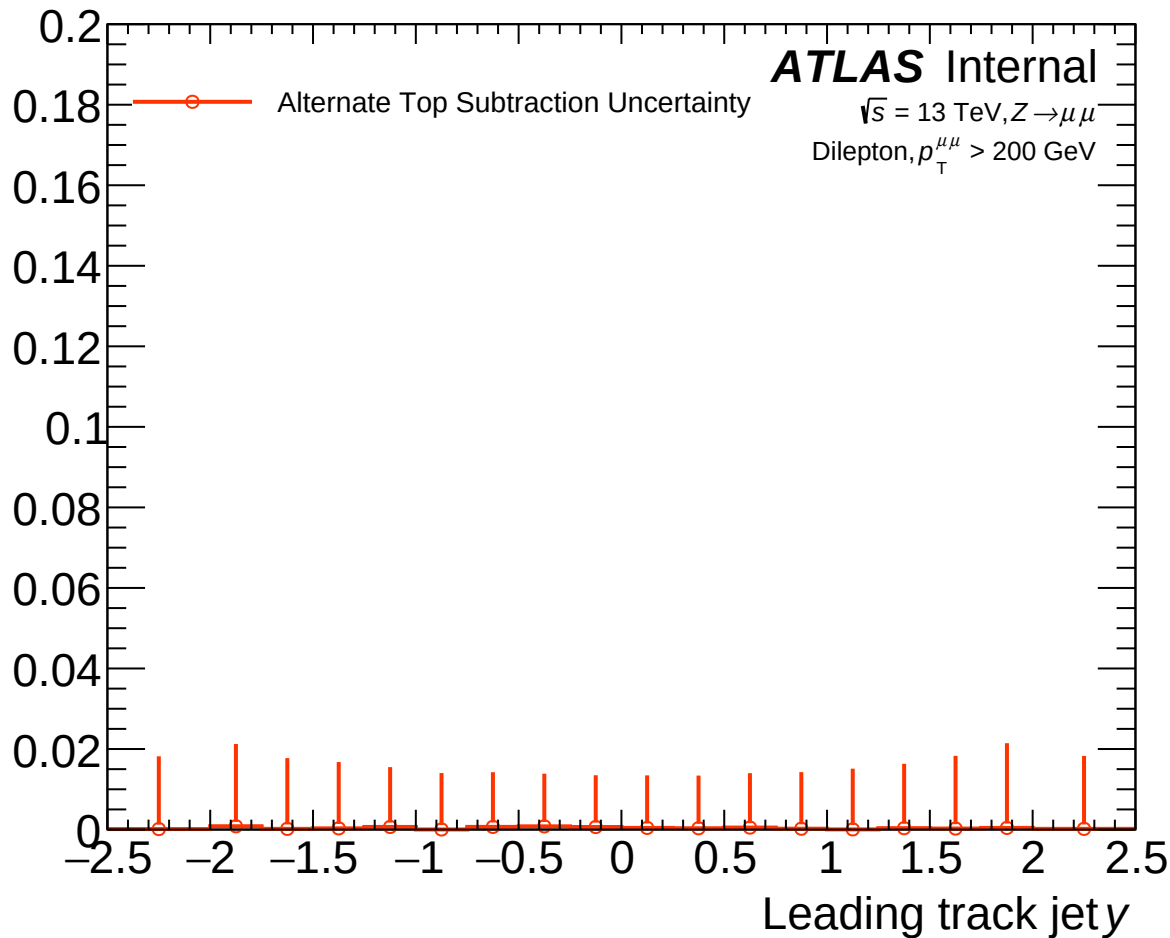
Fractional systematic impact

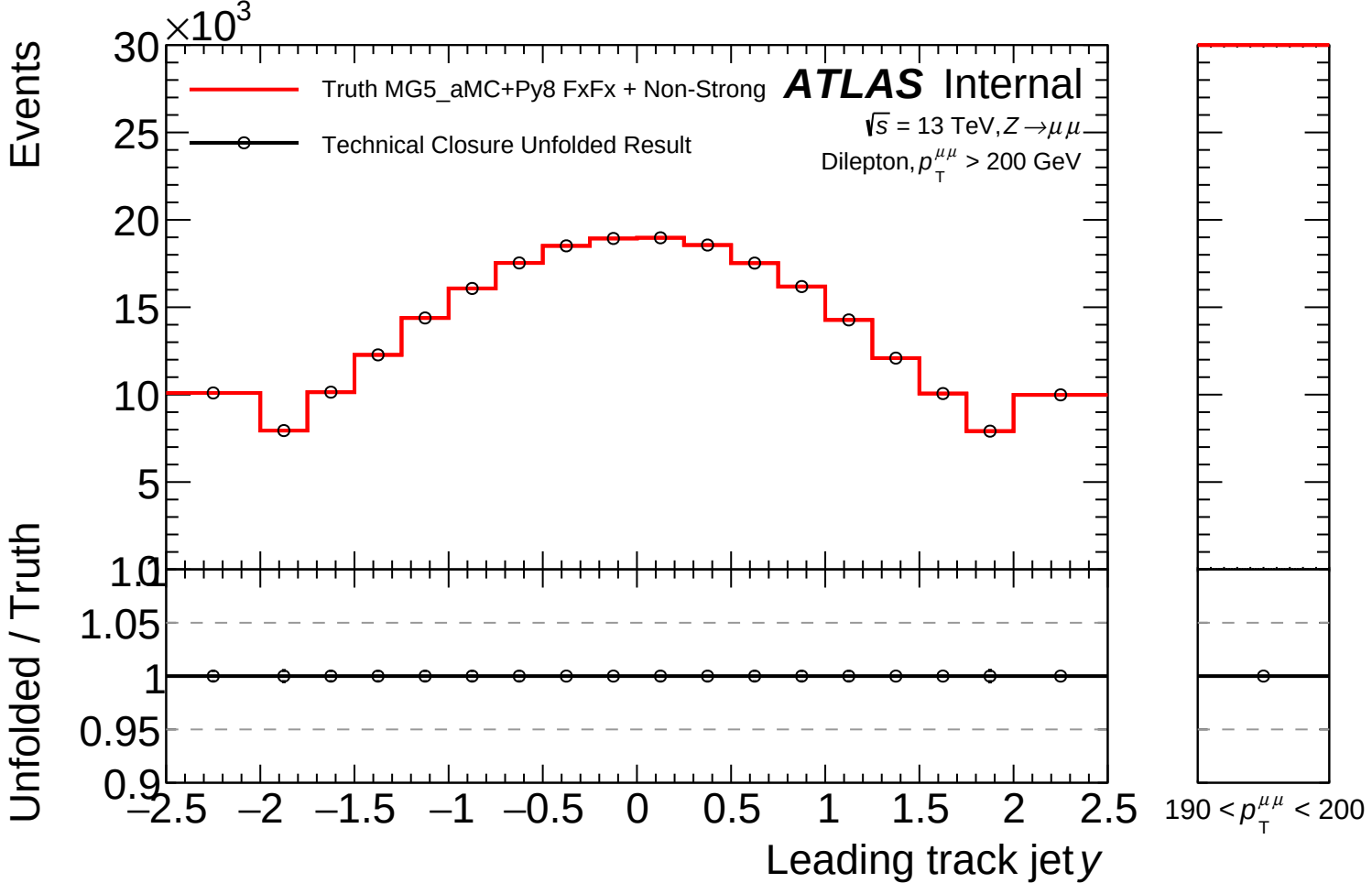
ATLAS Internal

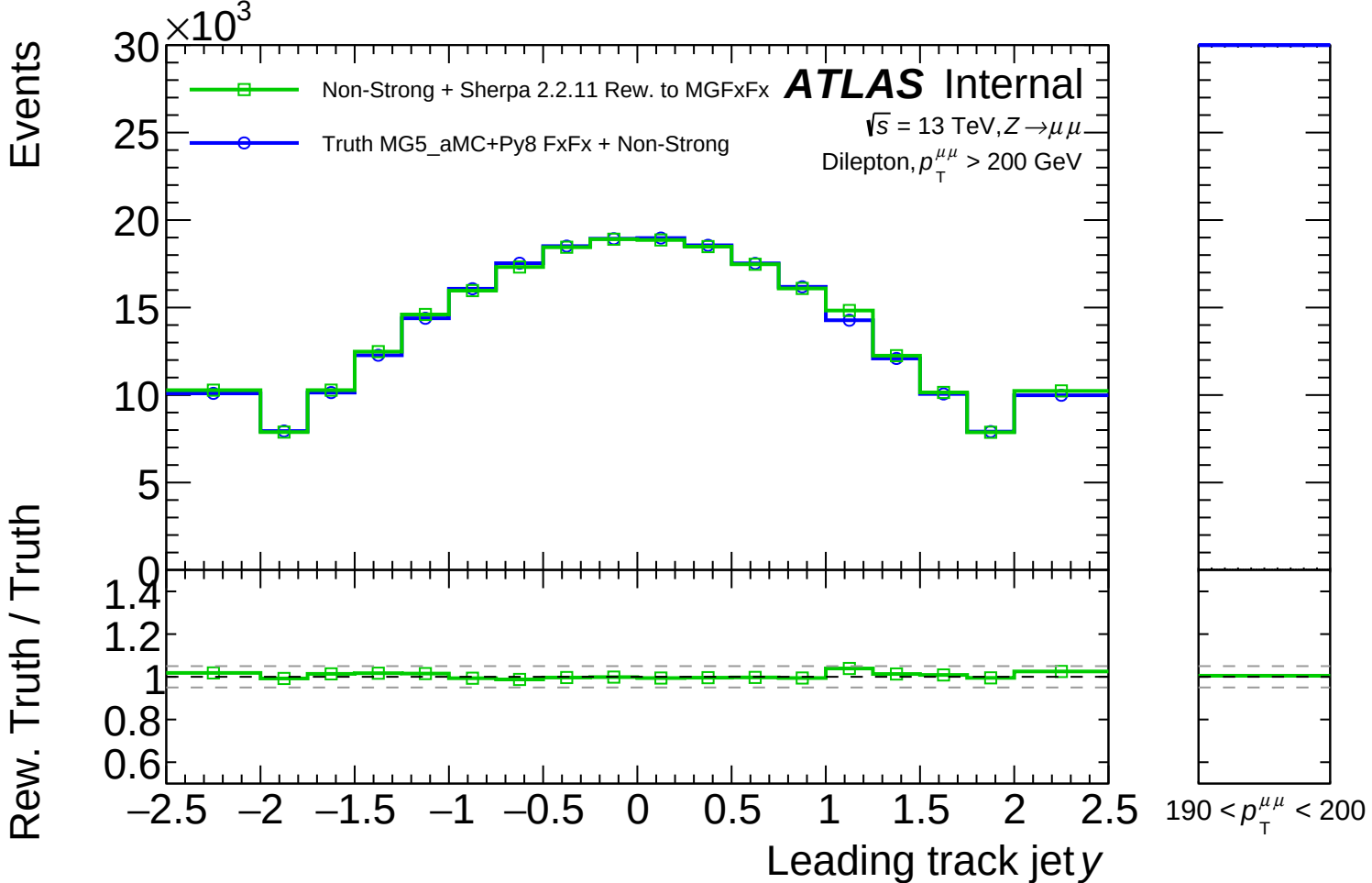
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

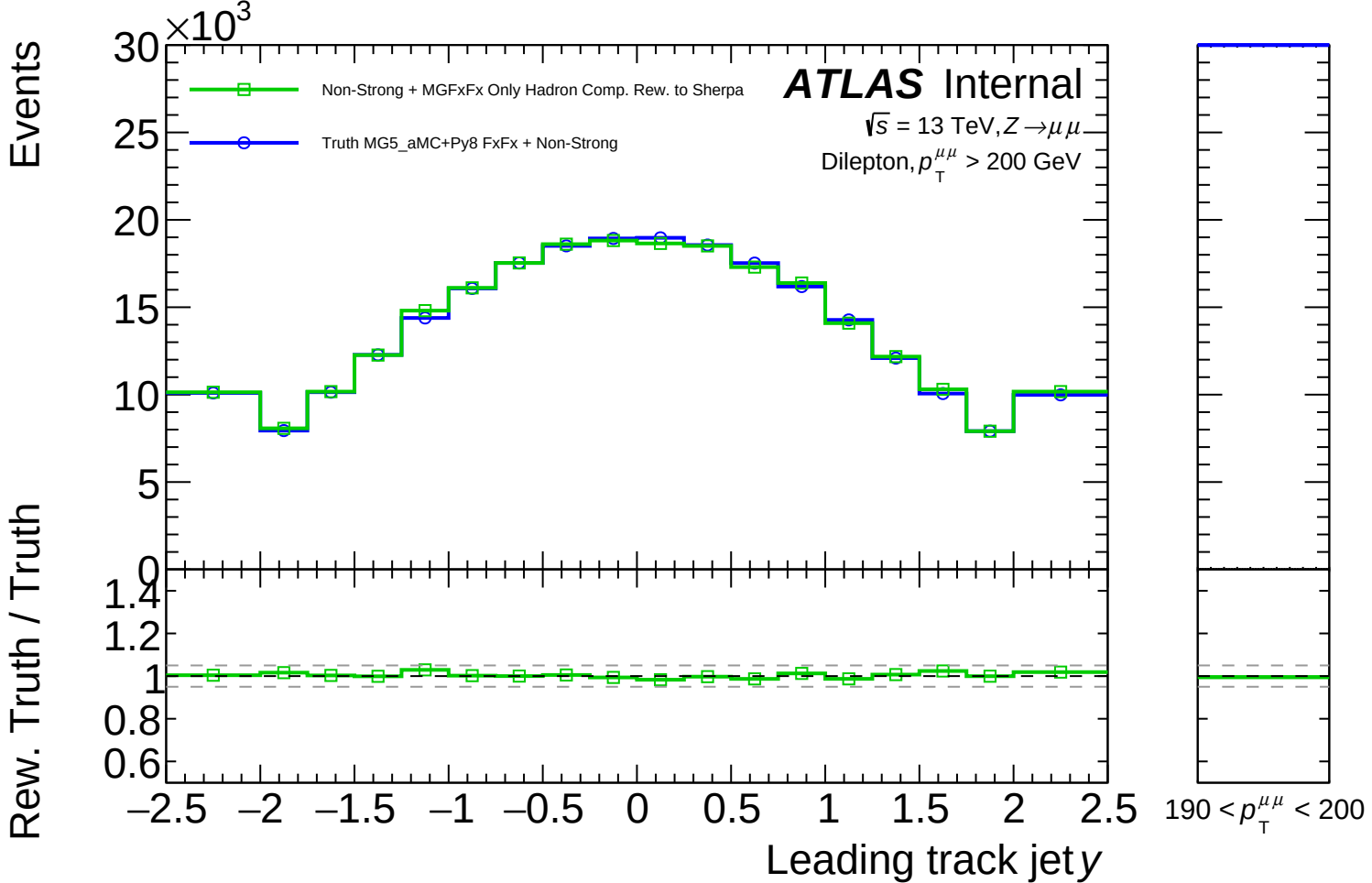
Dilepton, $p_T^{\mu\mu} > 200$ GeV

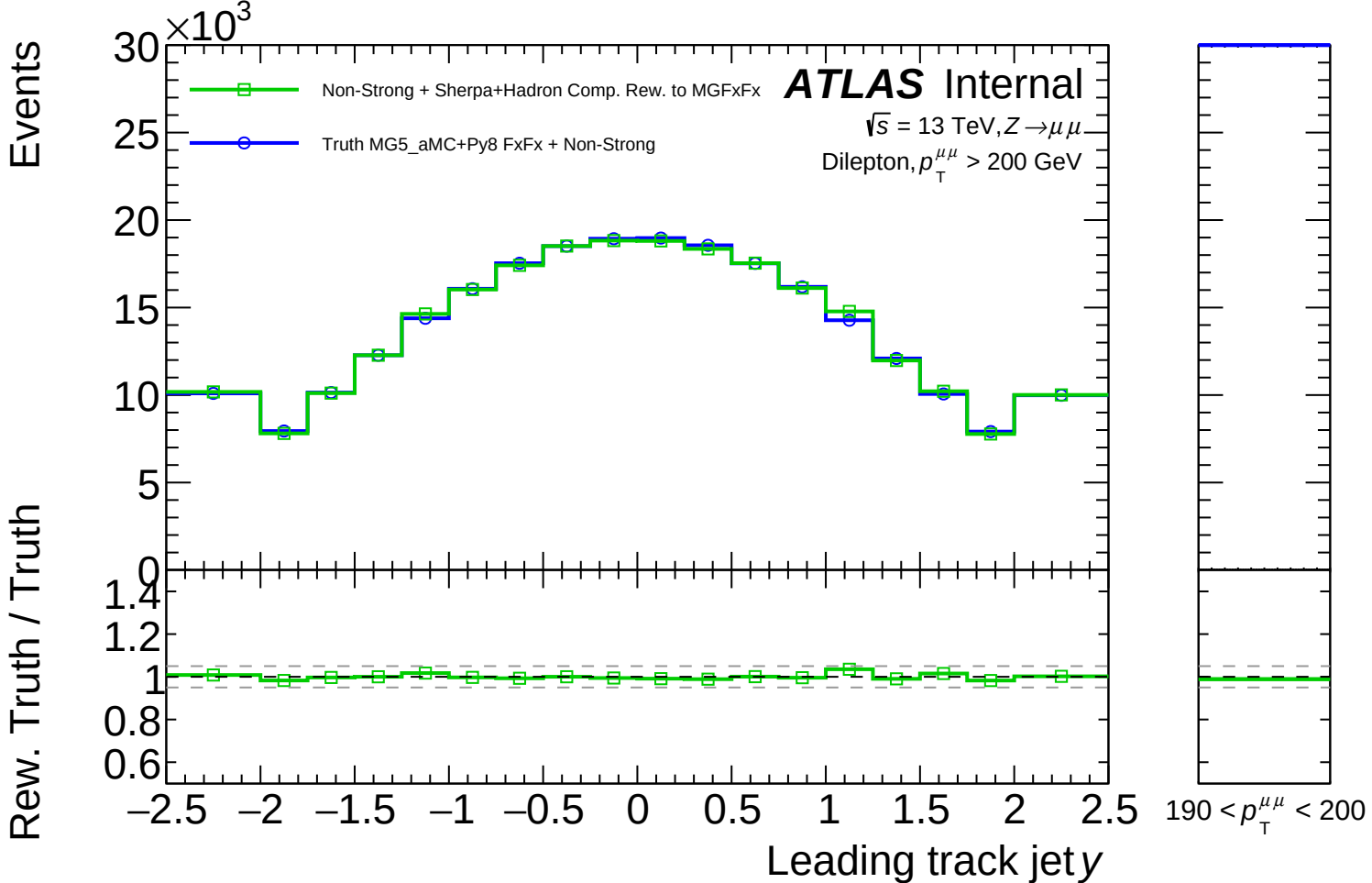
Alternate Top Subtraction Uncertainty

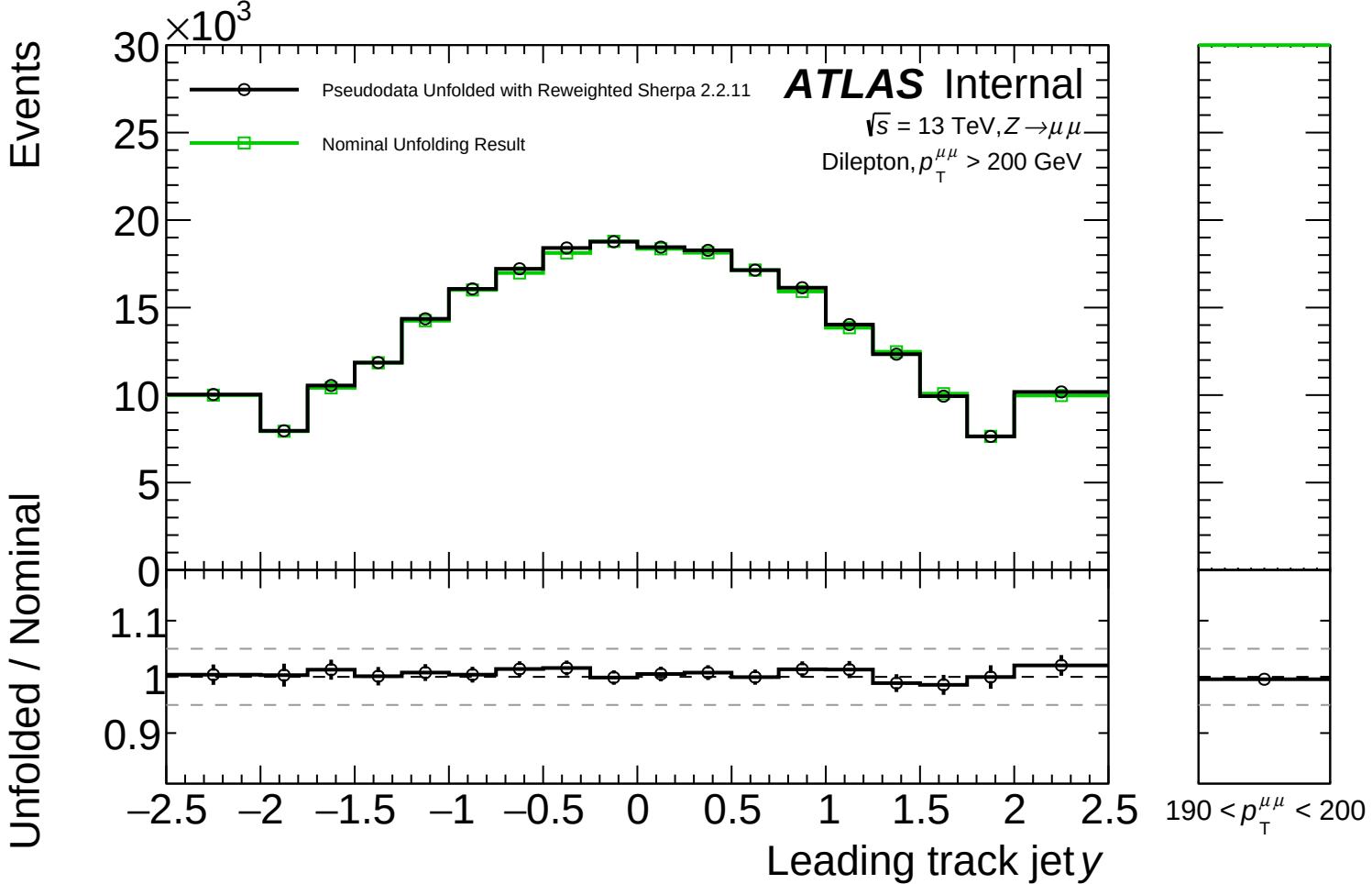


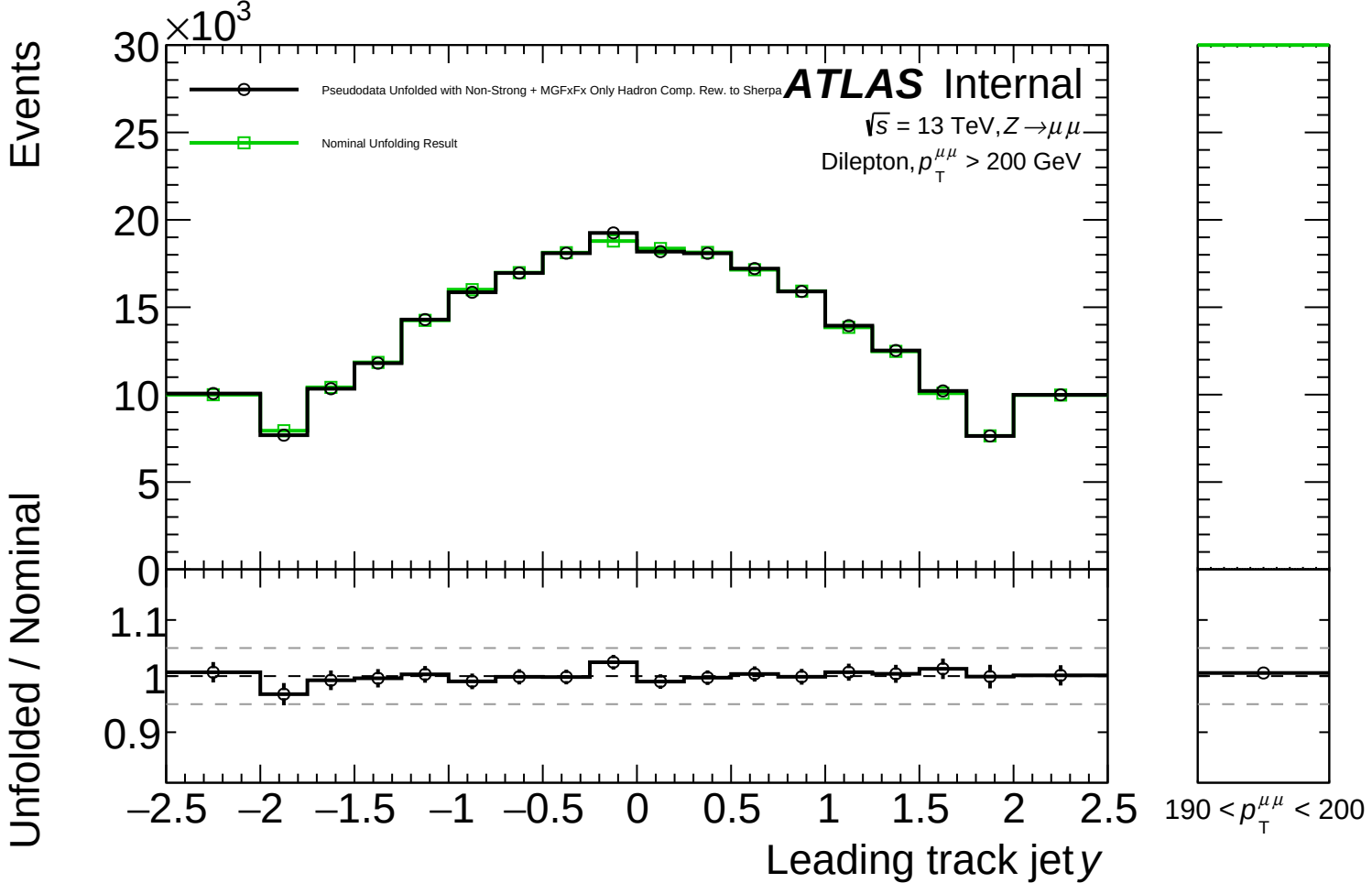


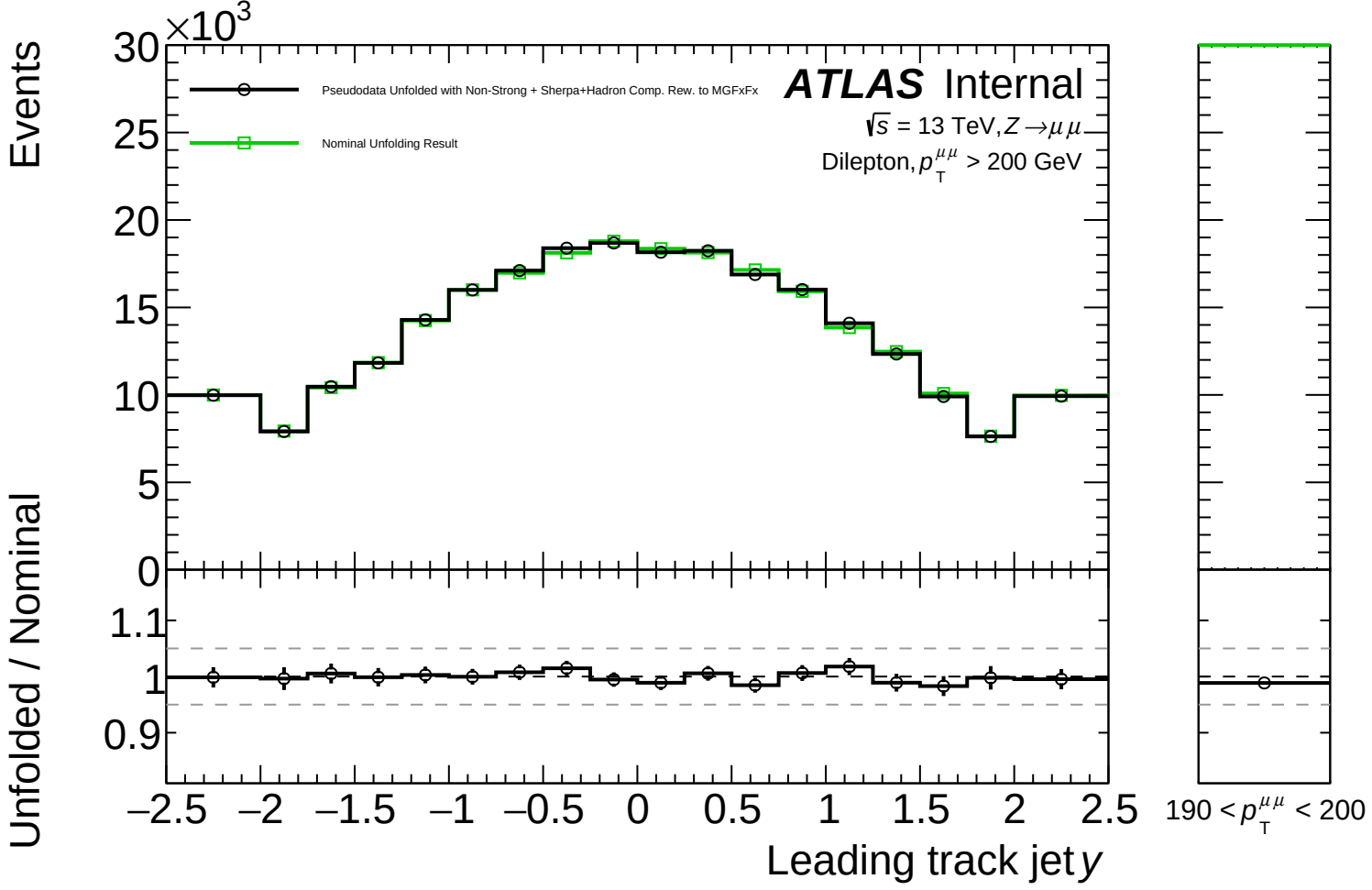


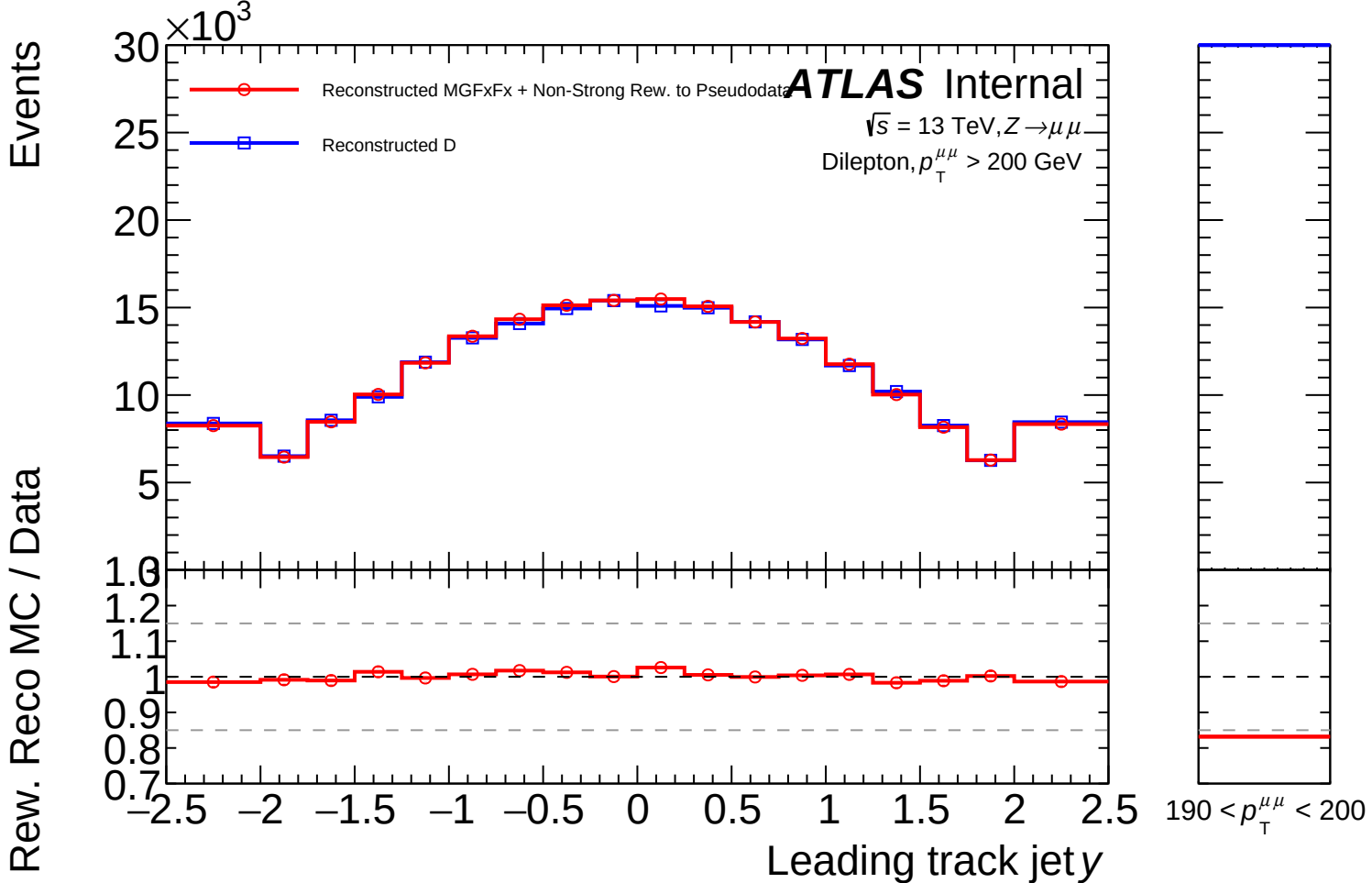


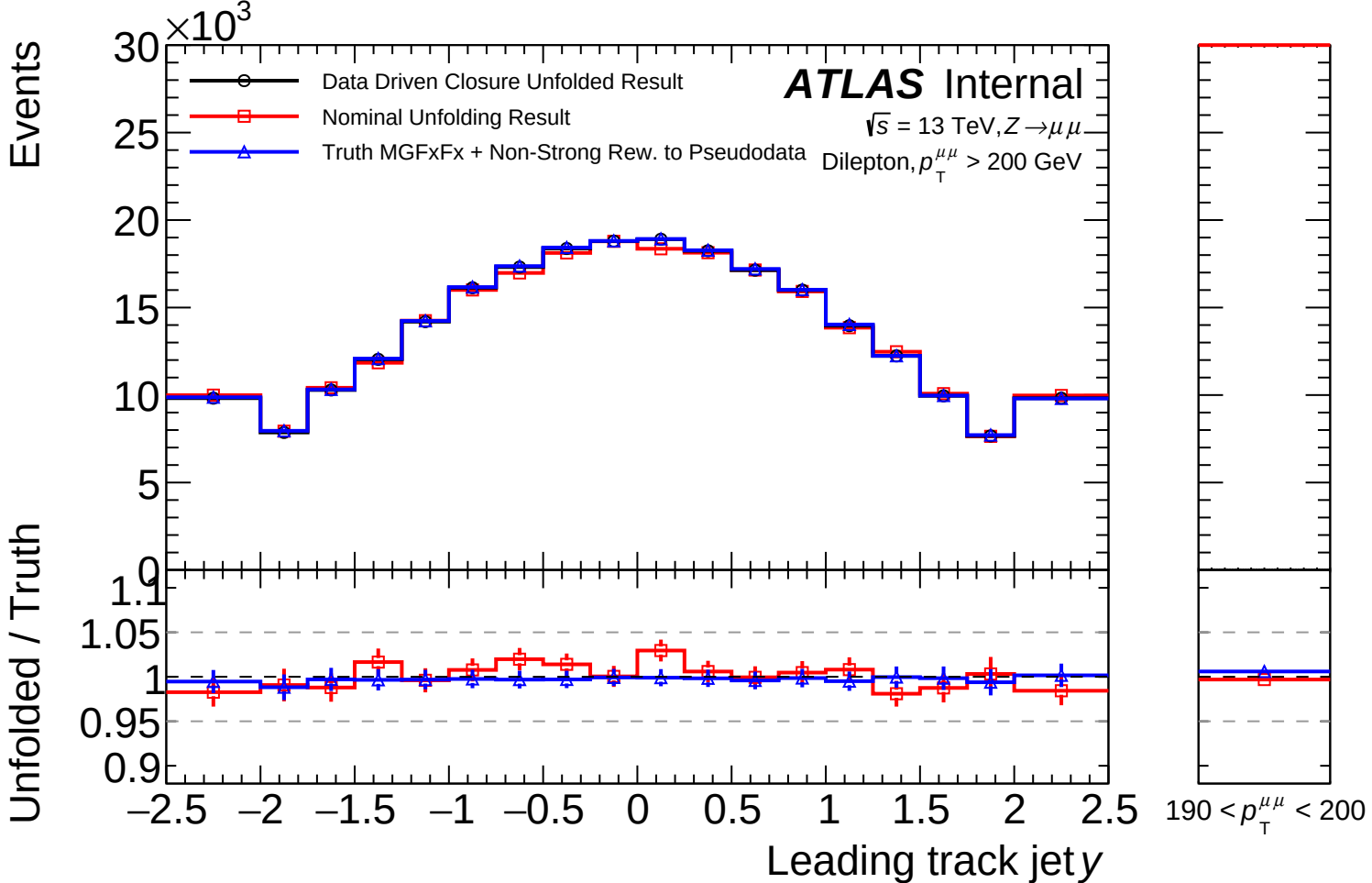




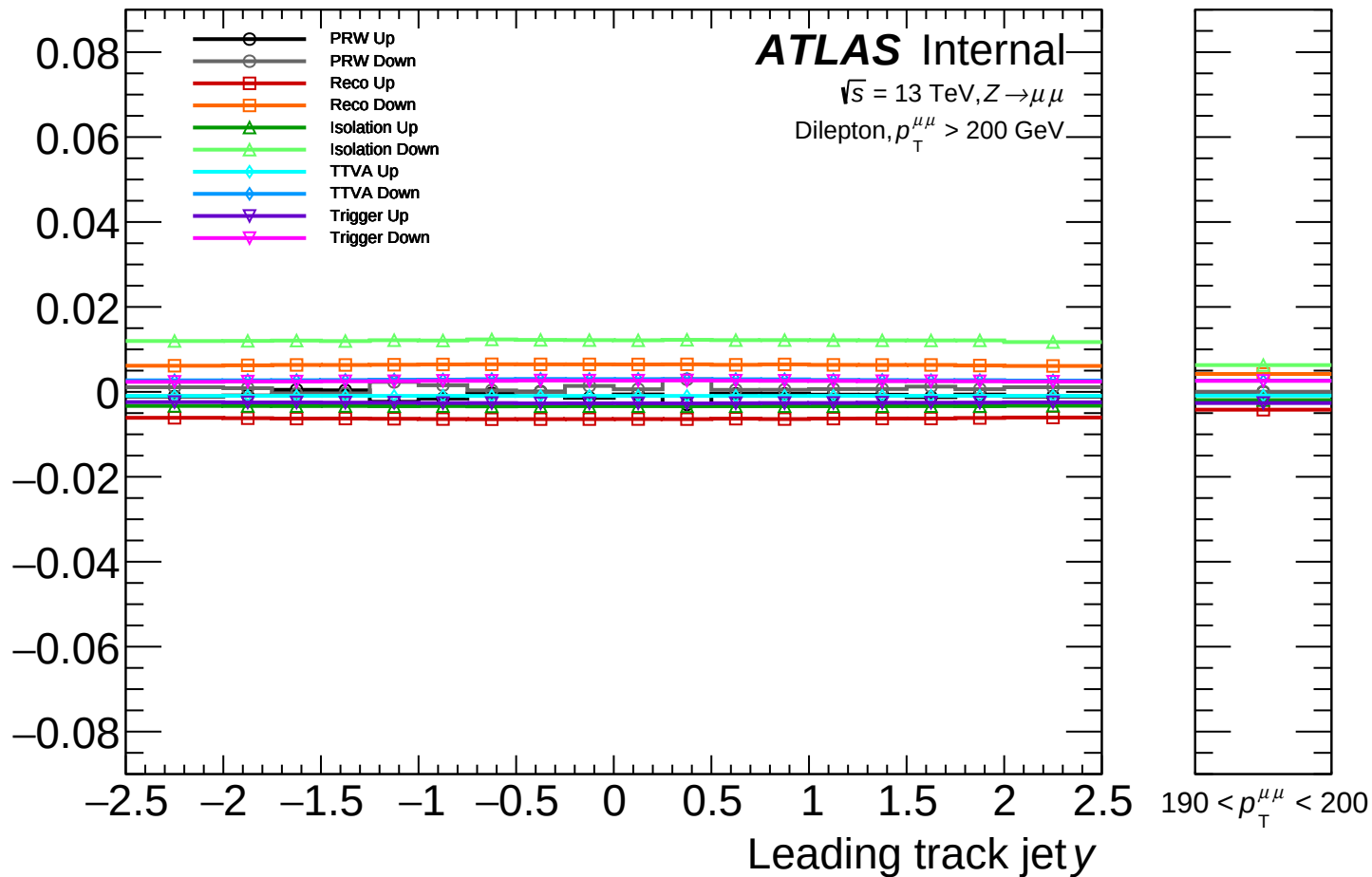




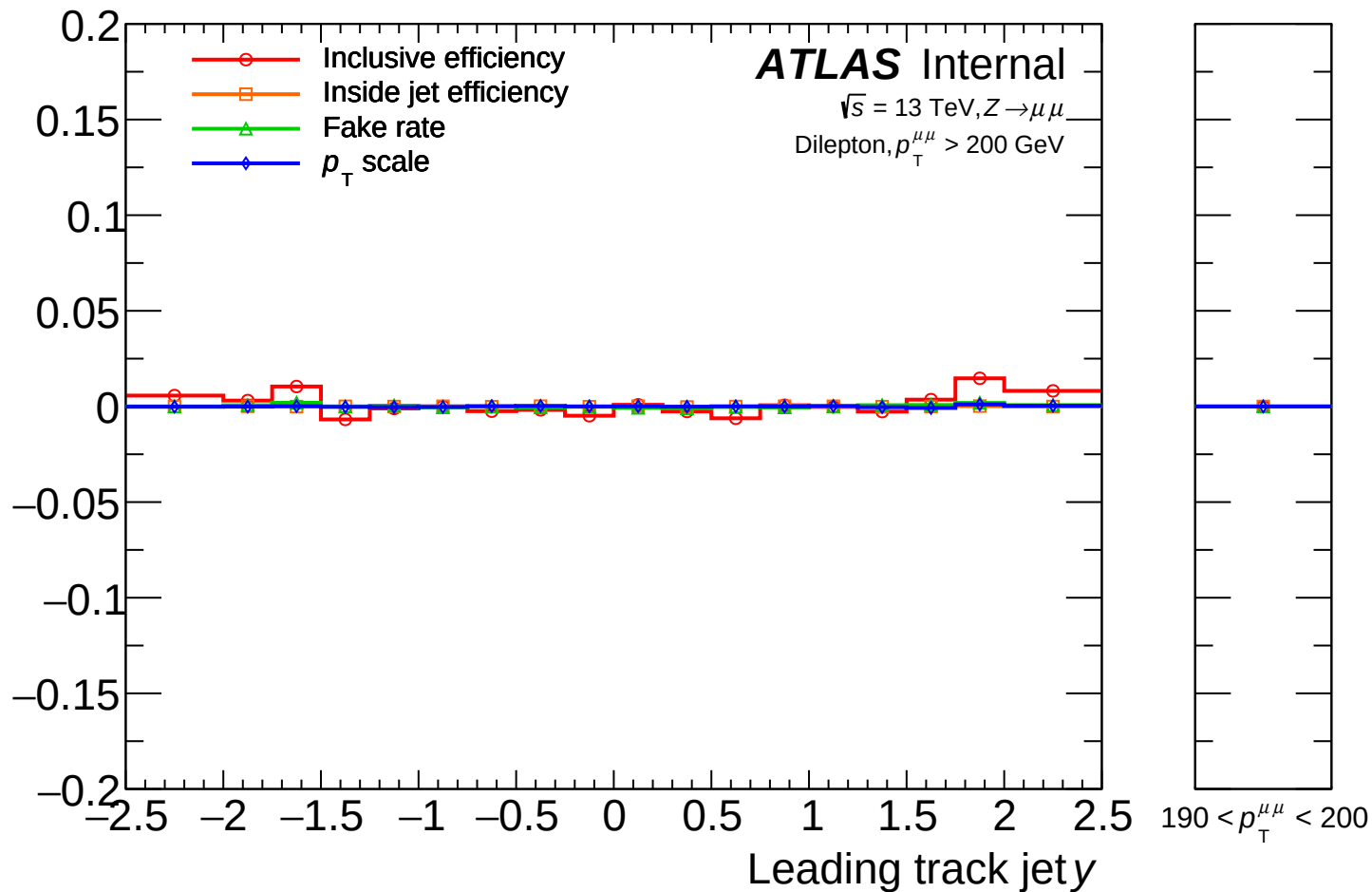




Fractional systematic impact



Fractional systematic impact



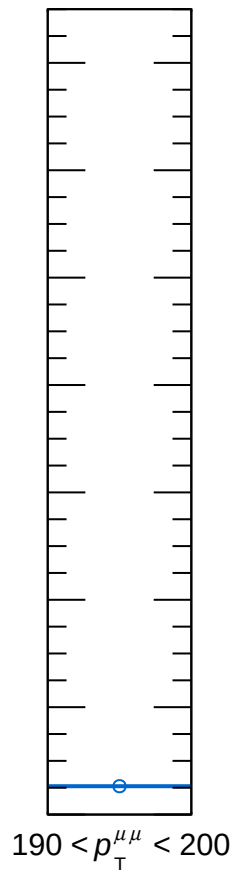
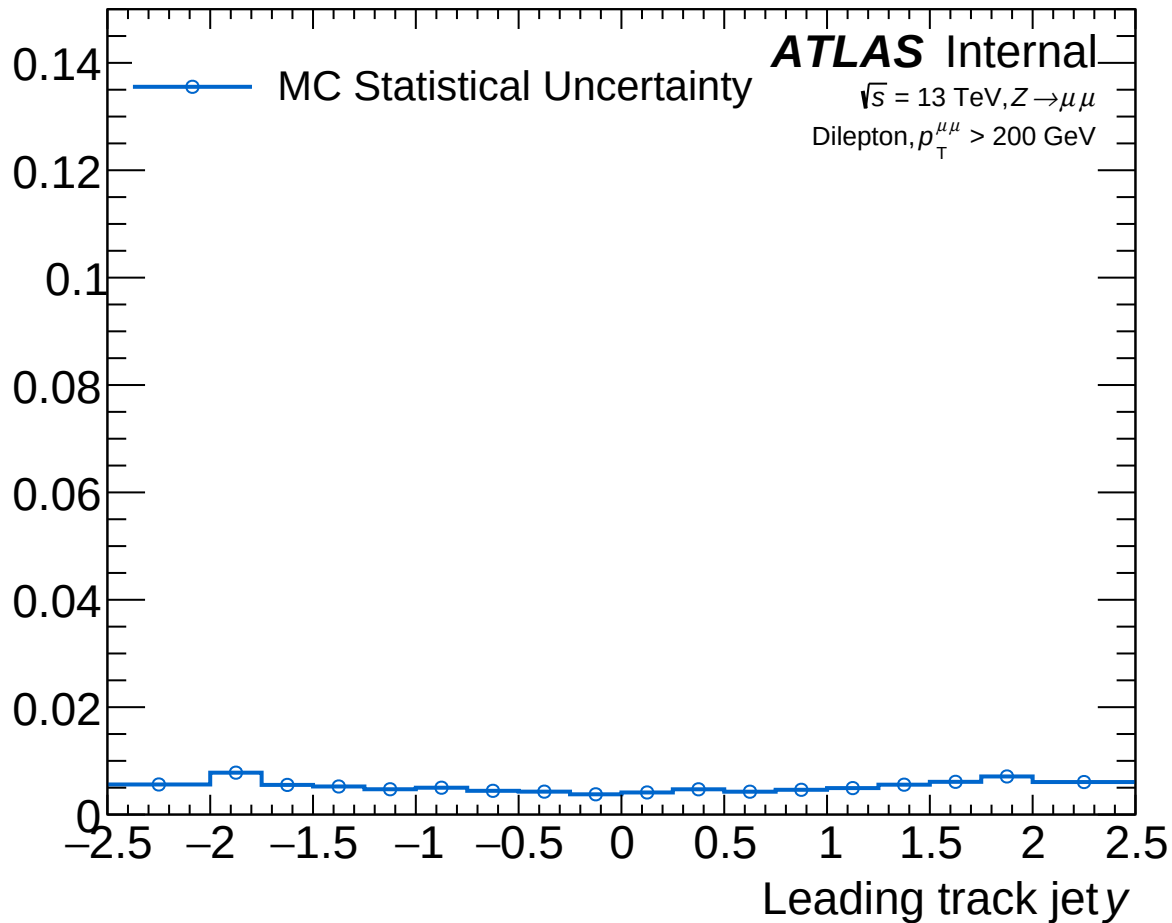
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



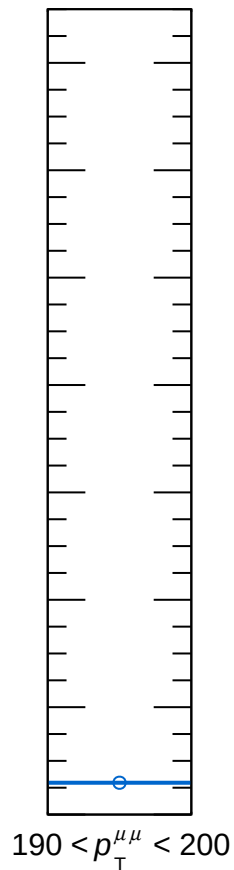
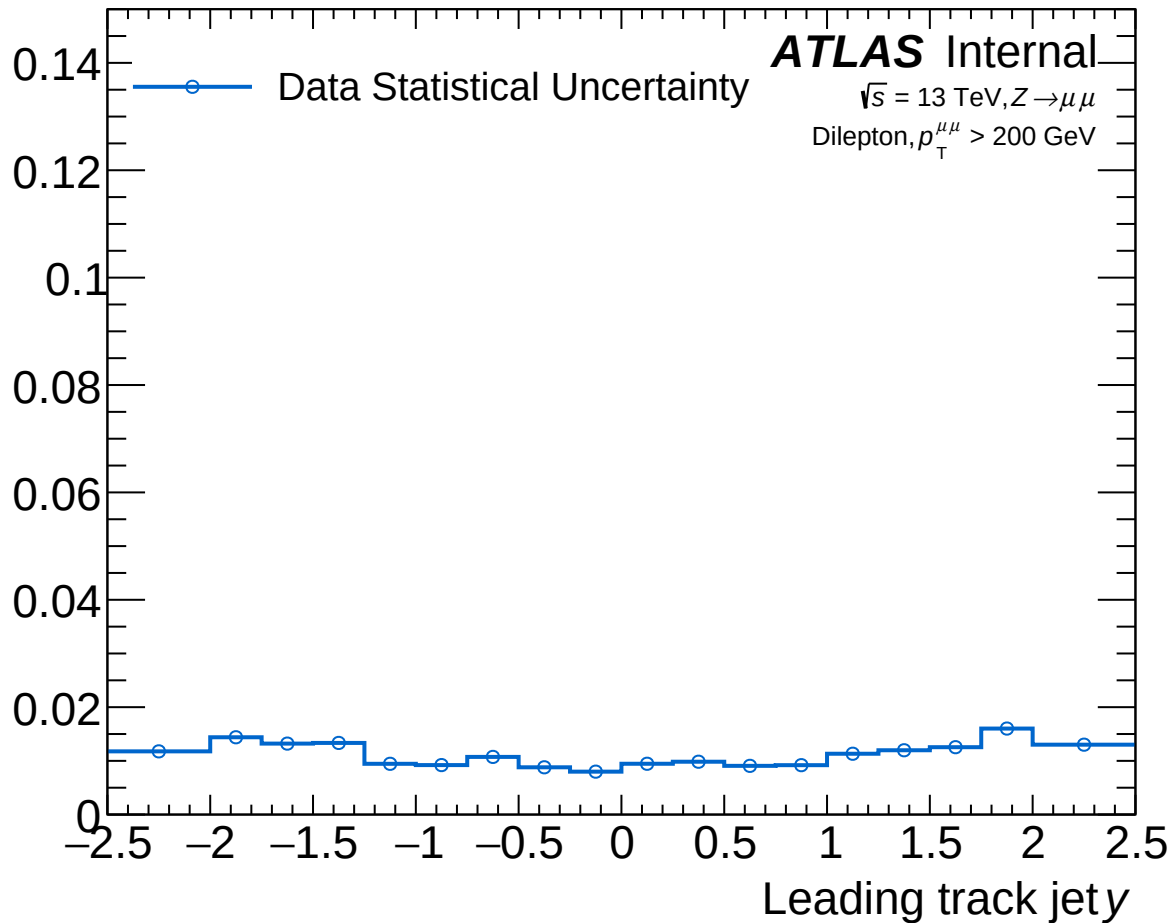
Fractional systematic impact

ATLAS Internal

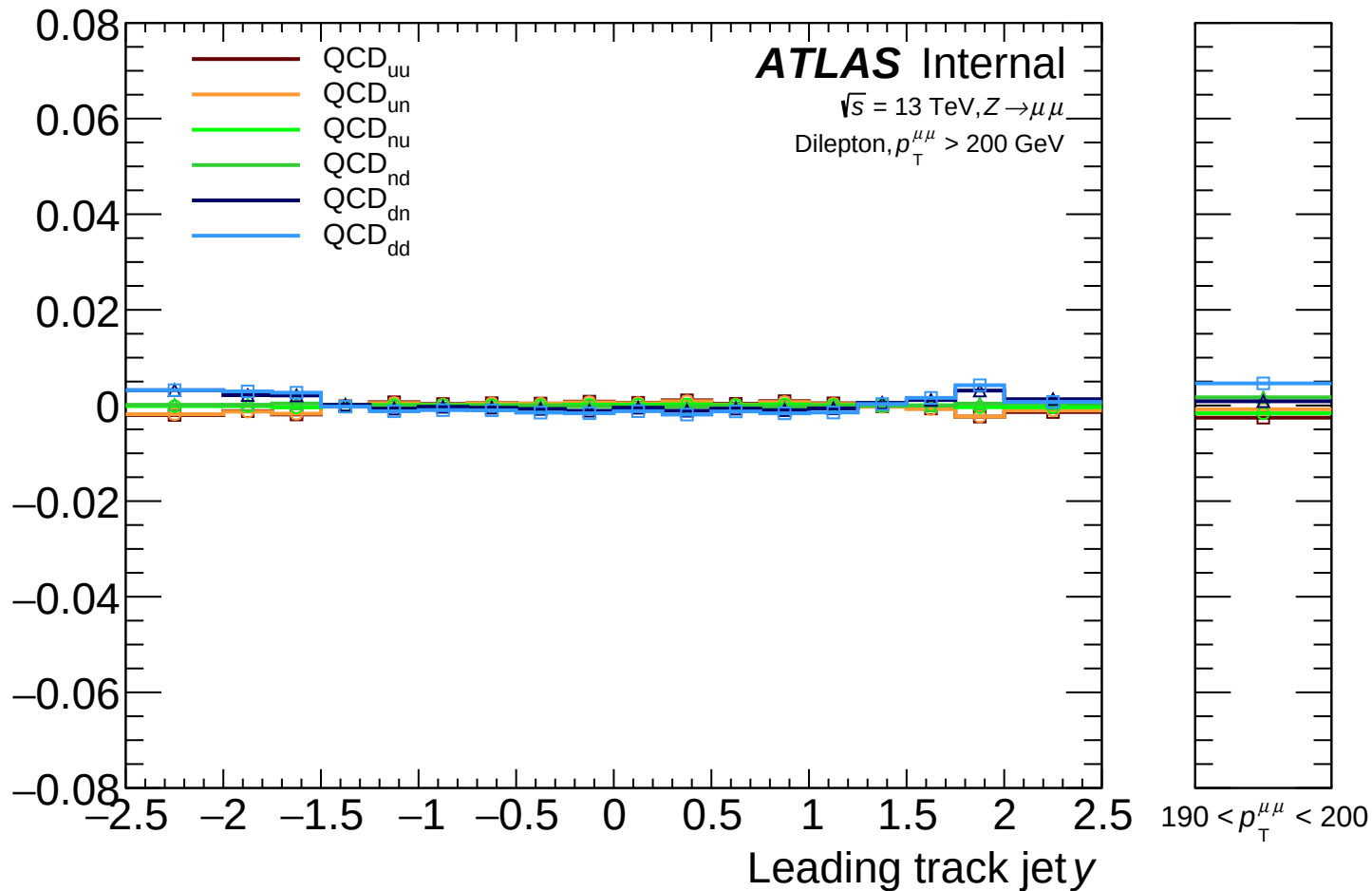
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

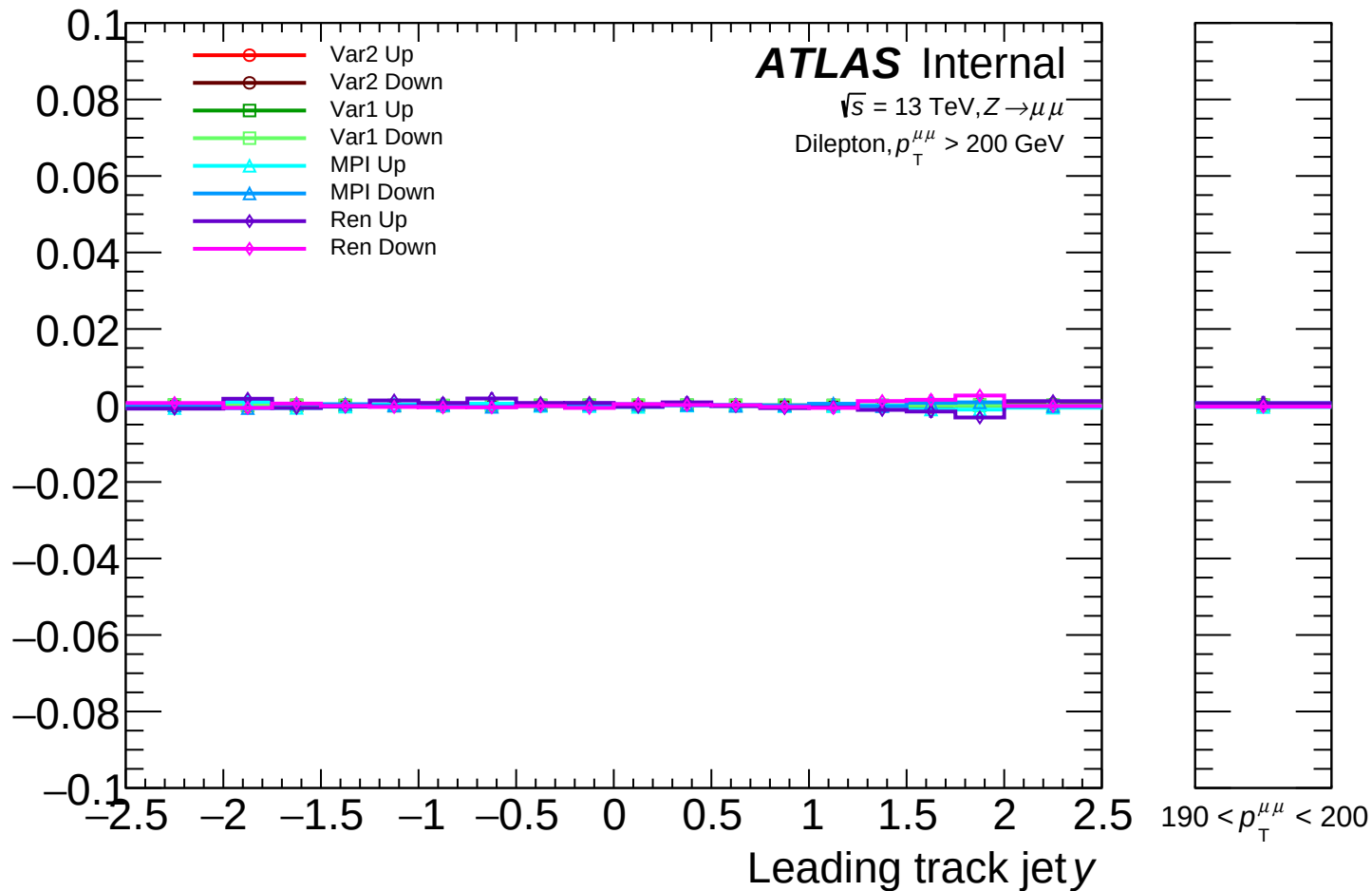
Data Statistical Uncertainty

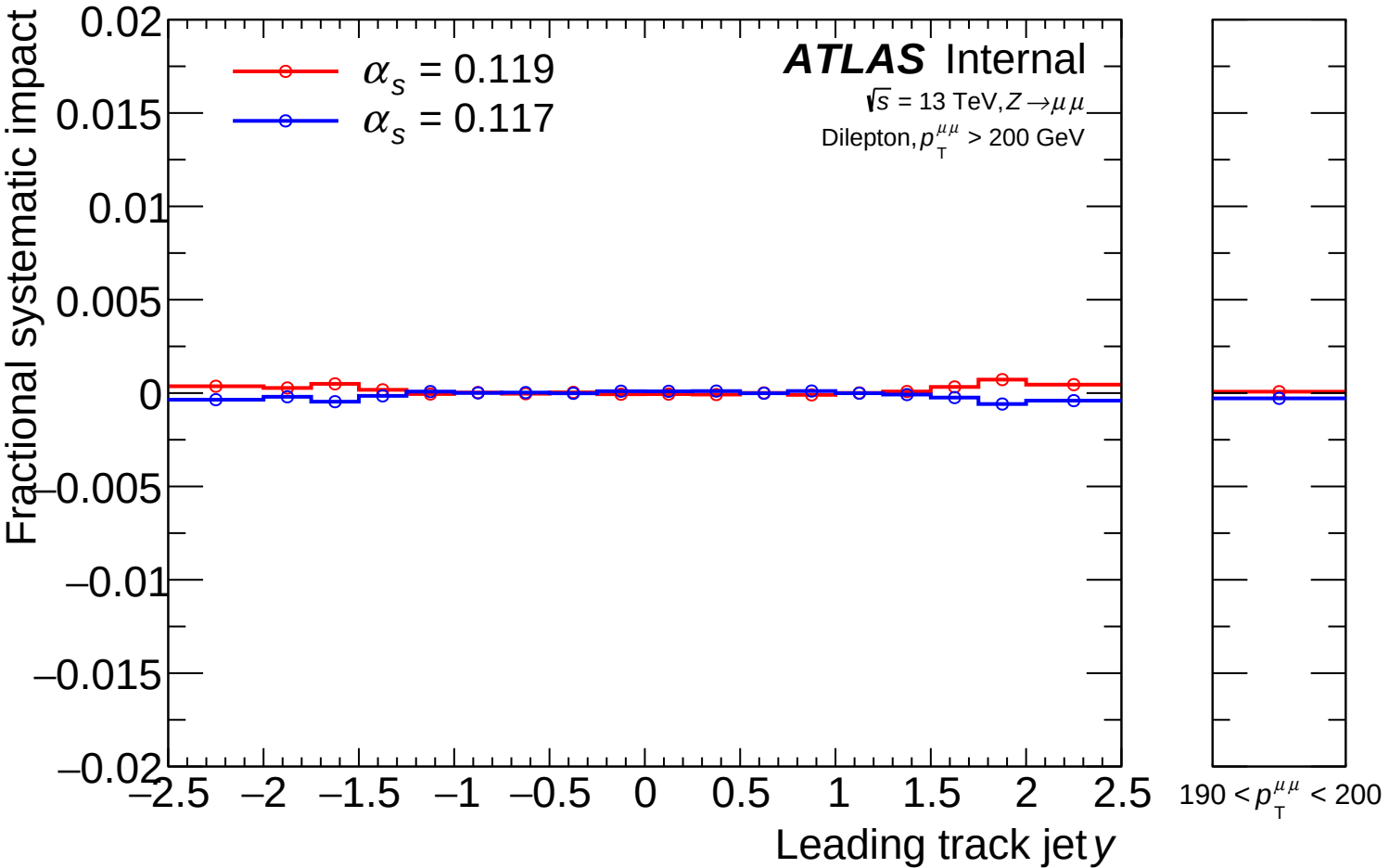


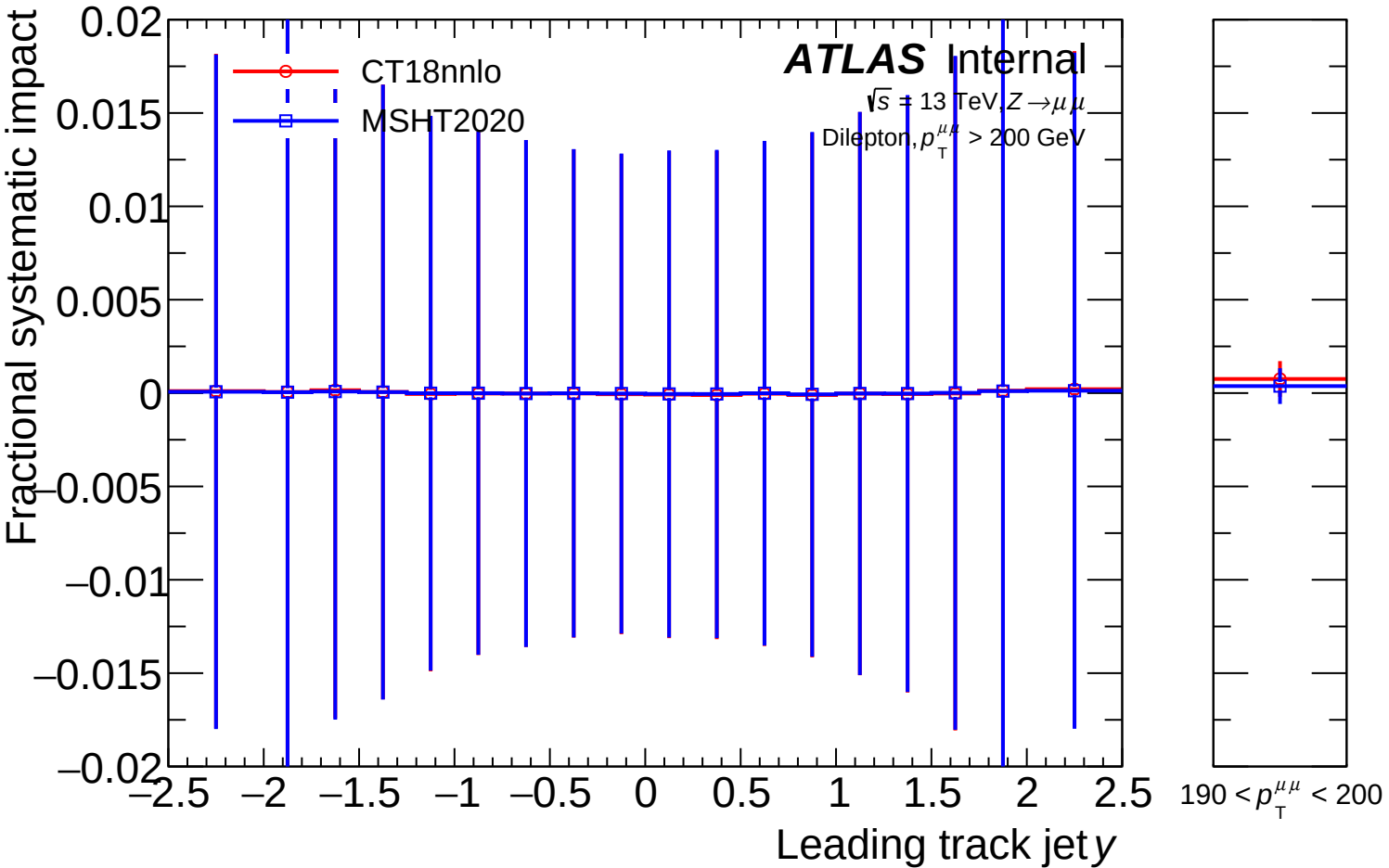
Fractional systematic impact

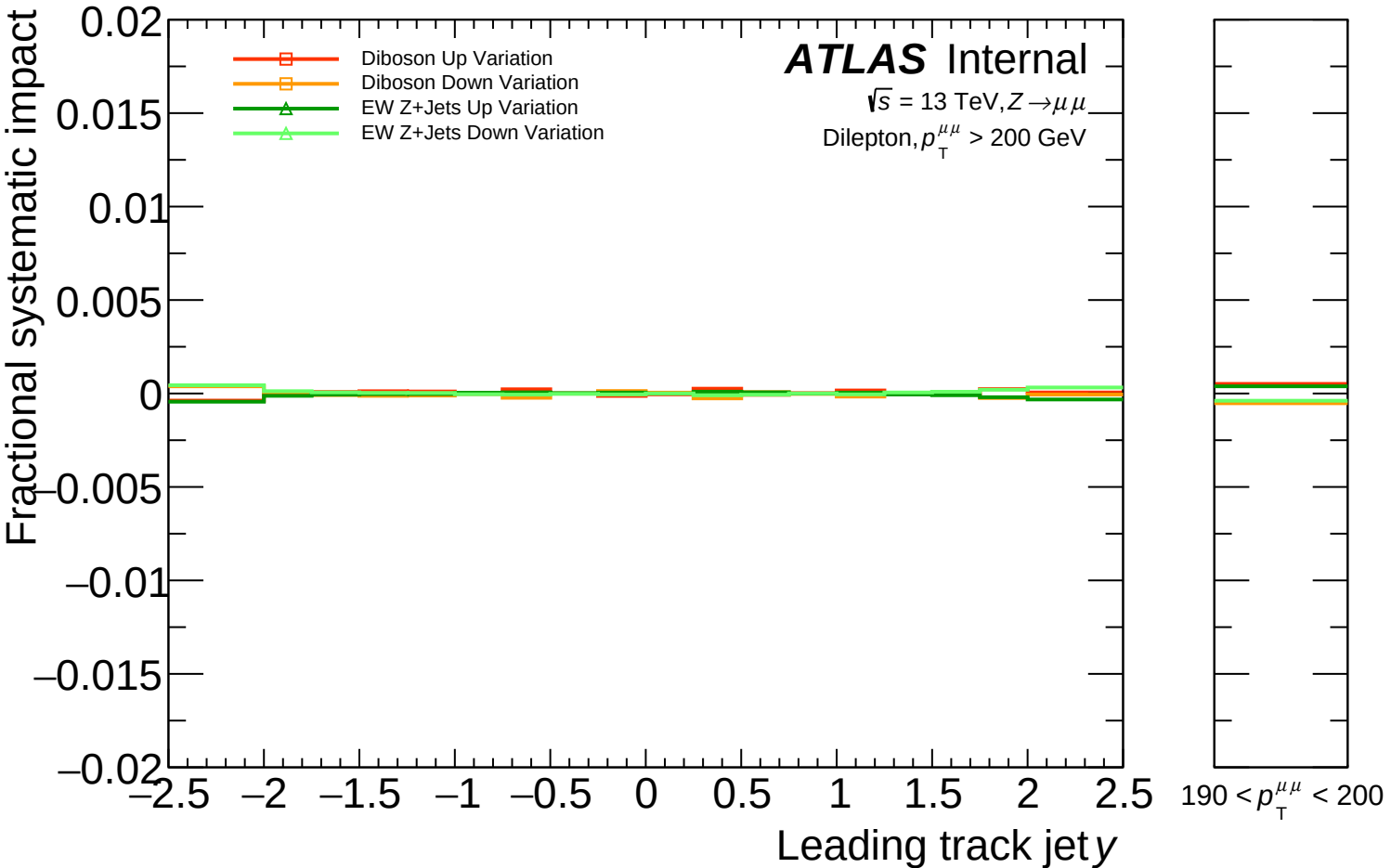


Fractional systematic impact







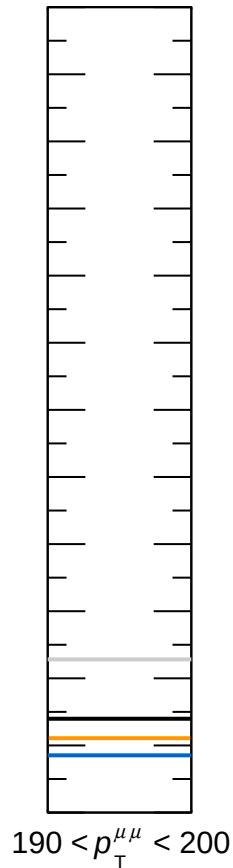
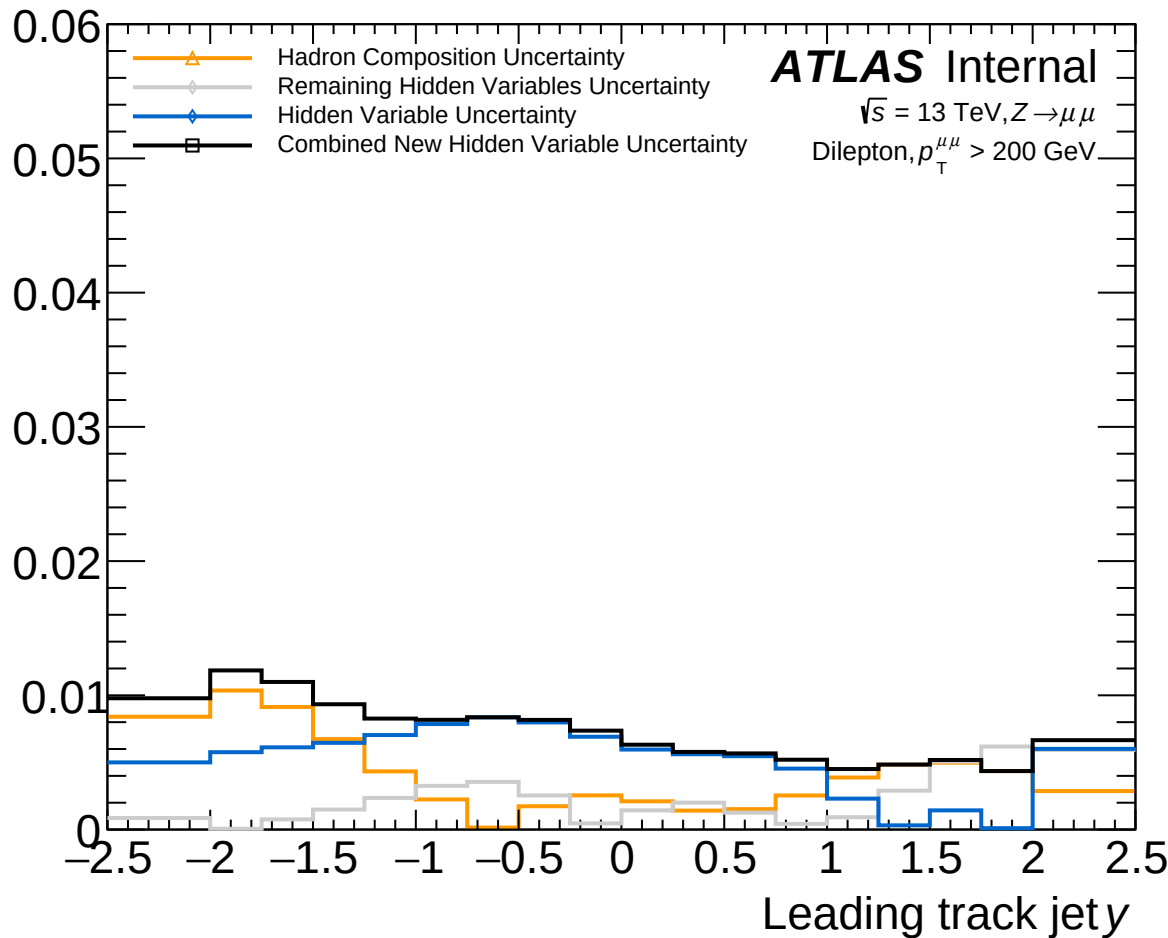


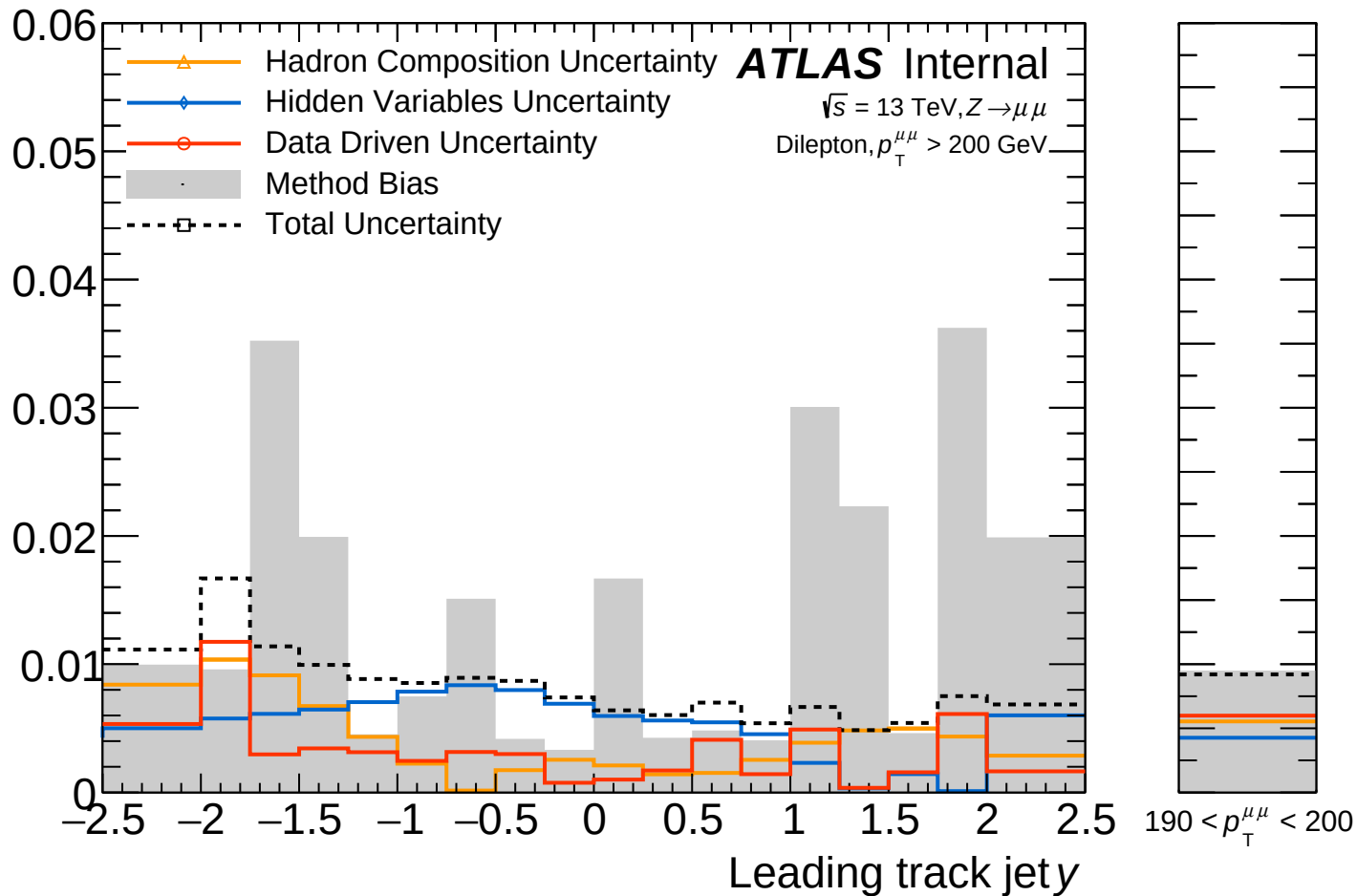
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

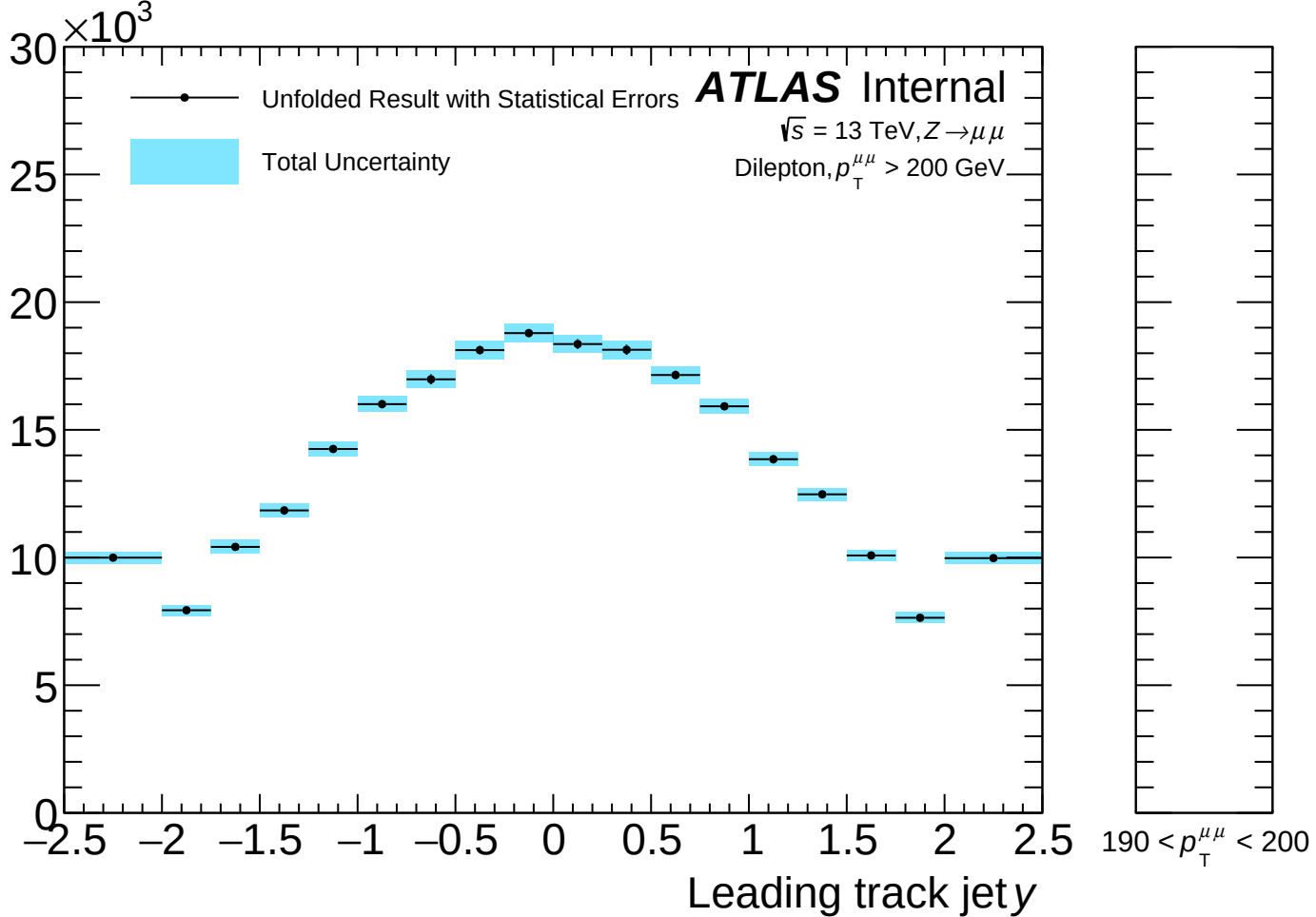
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

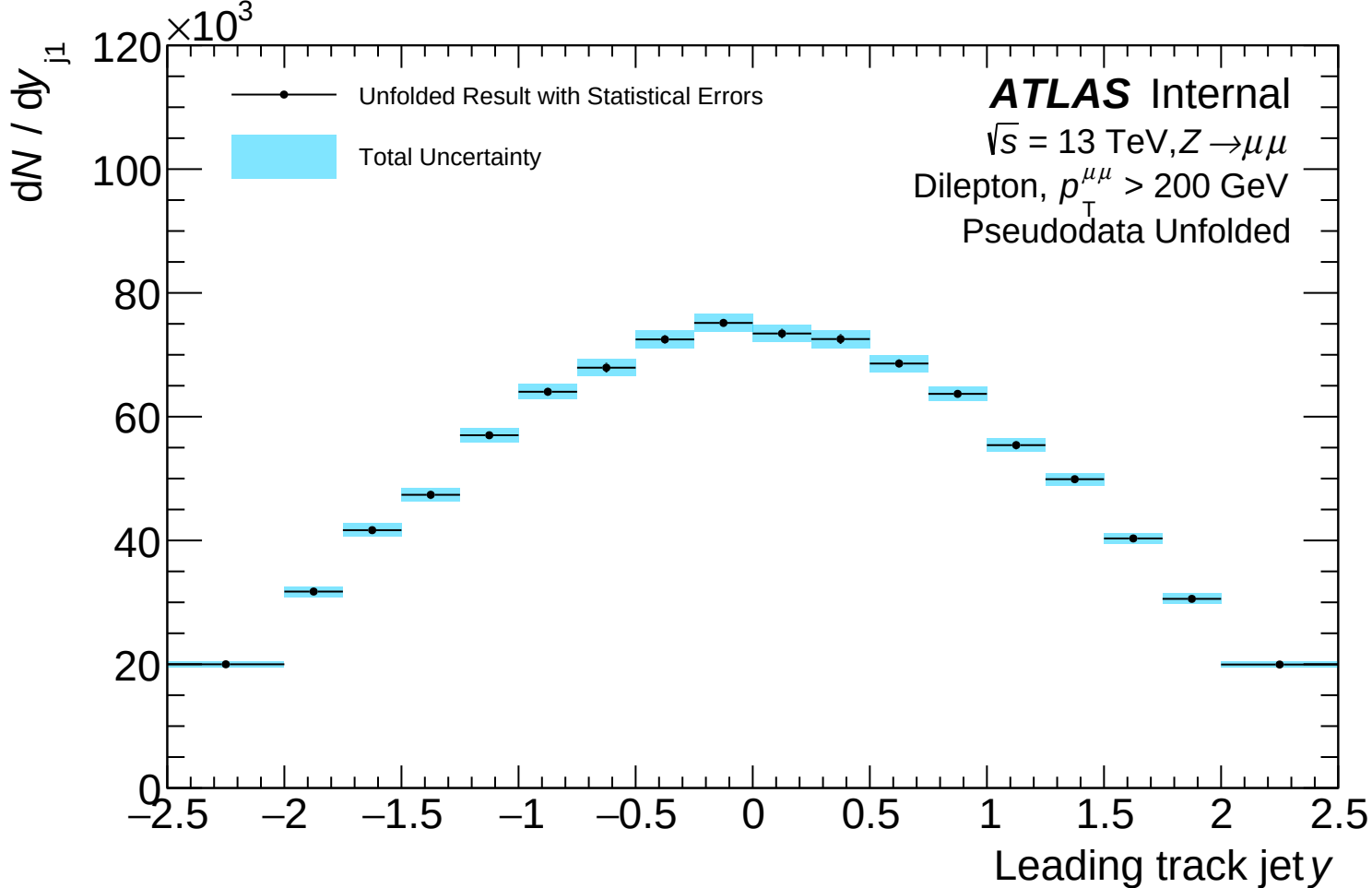
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty

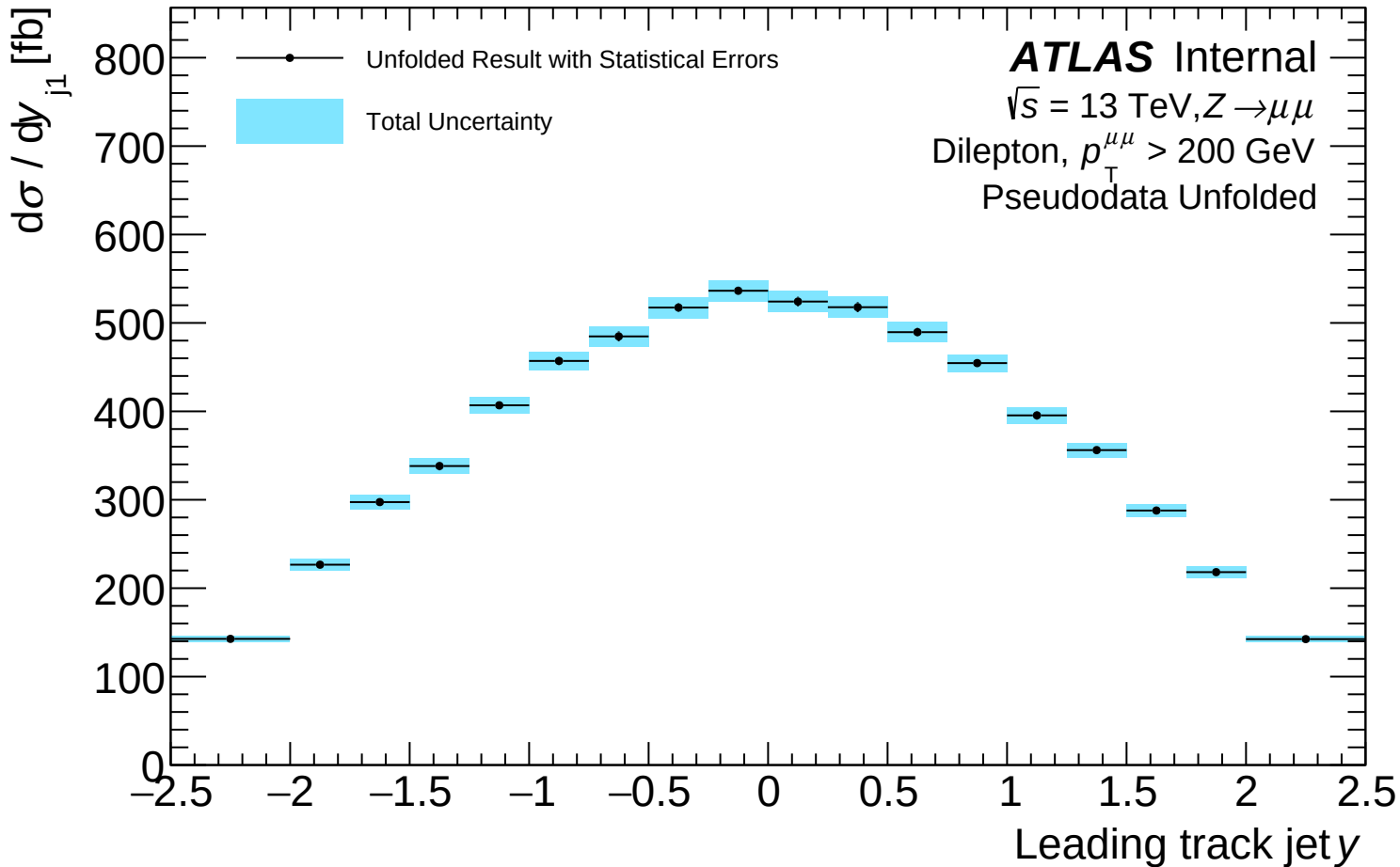




Events







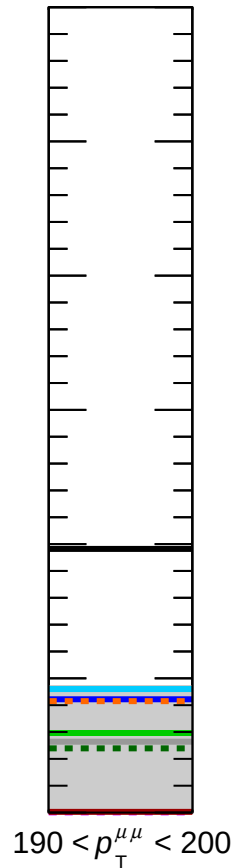
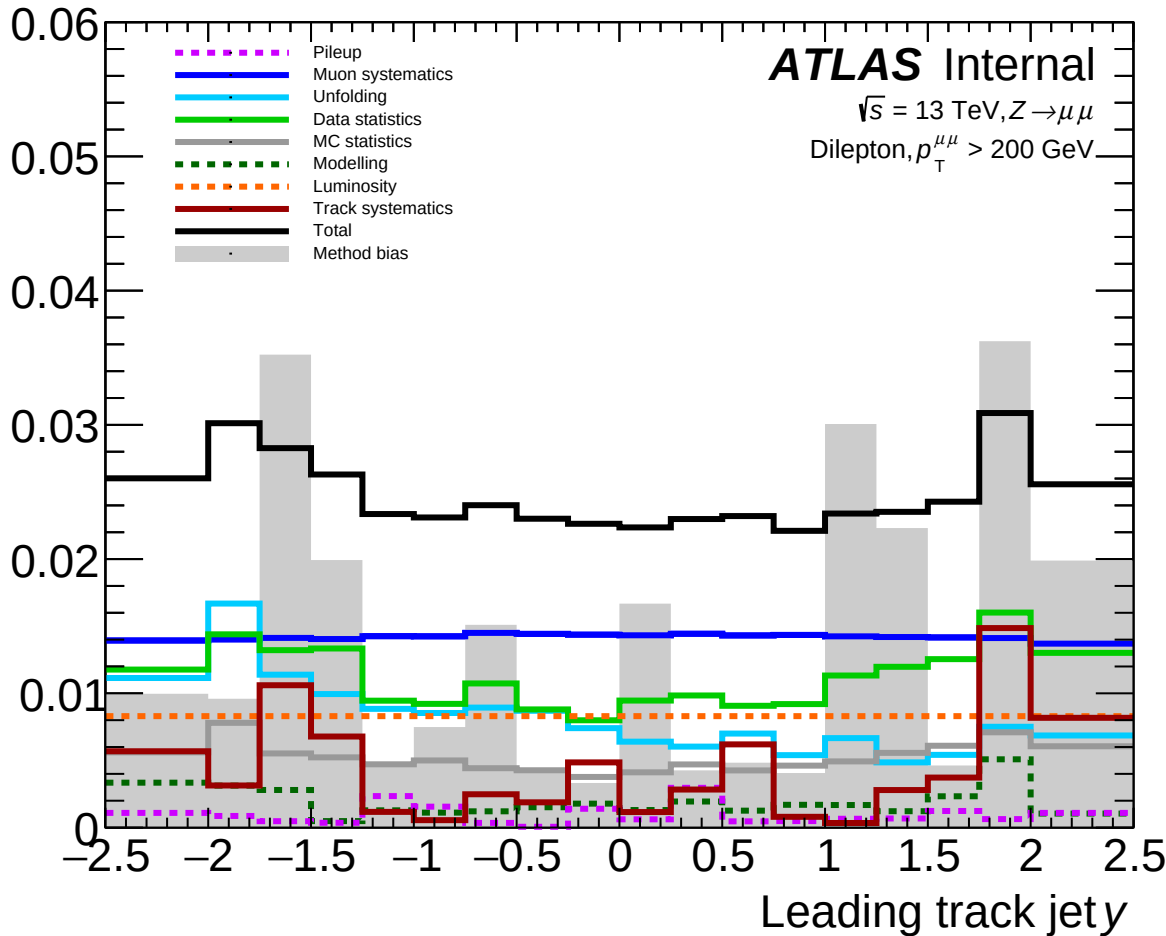
Fractional uncertainty

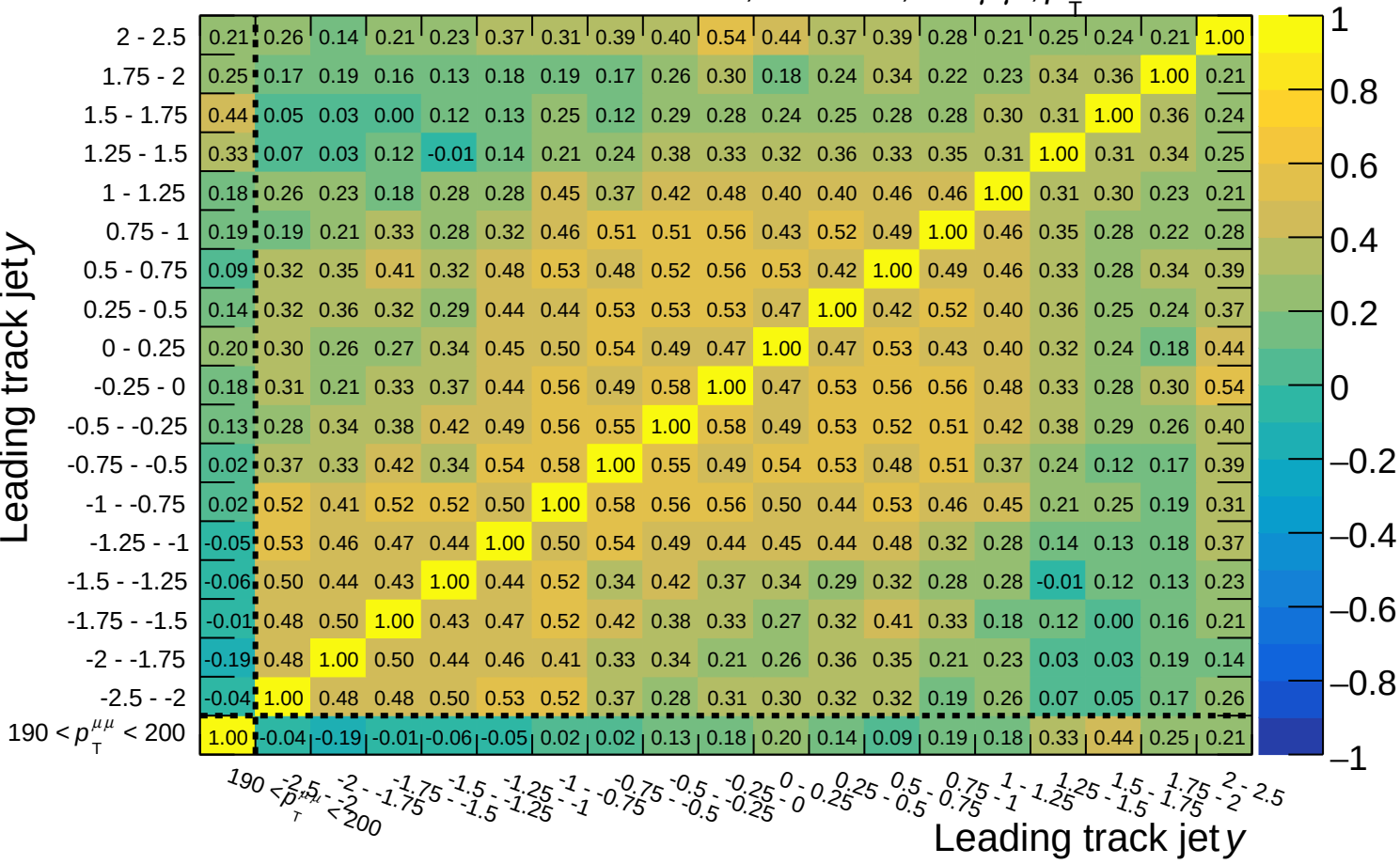
ATLAS Internal

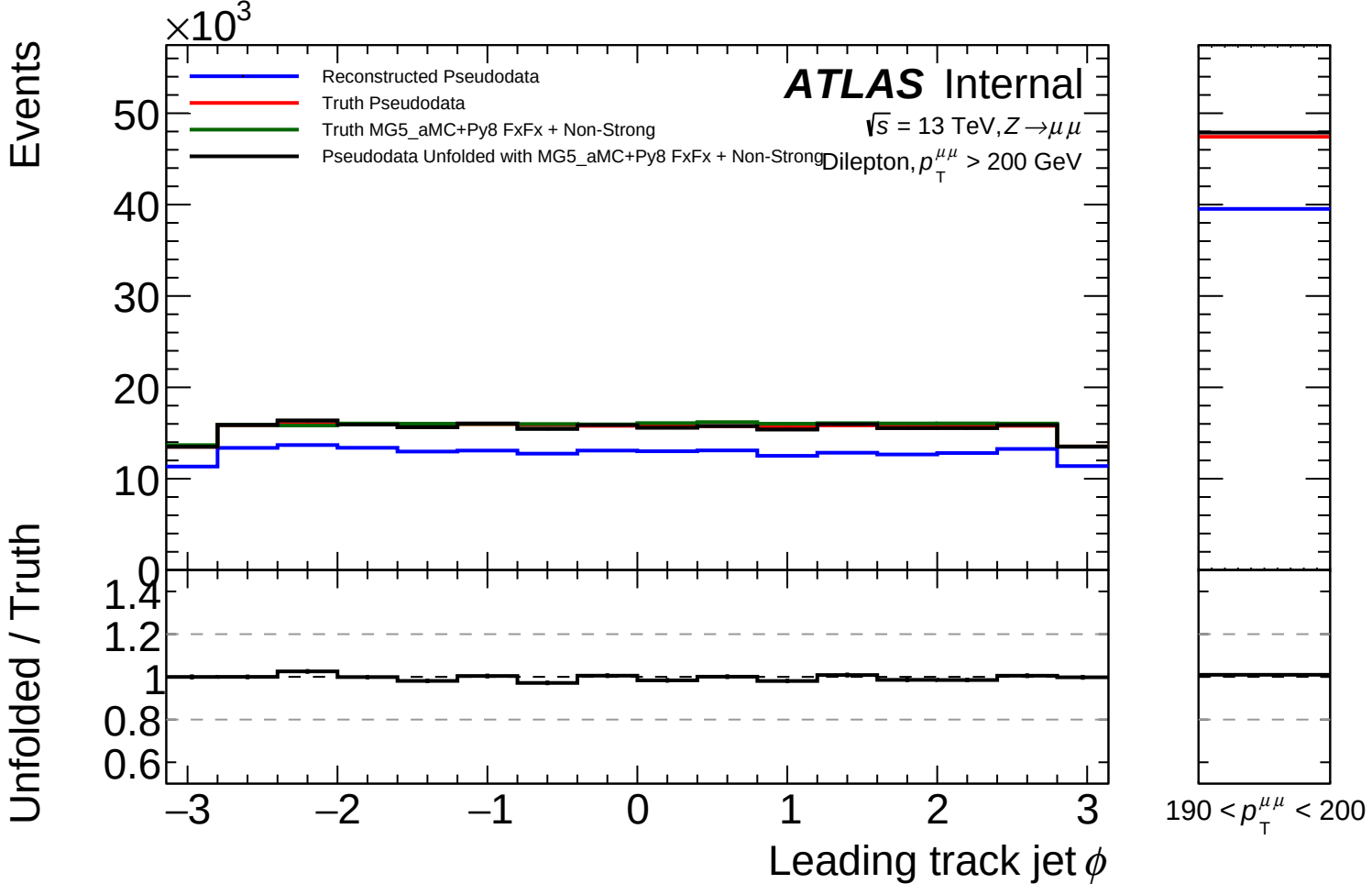
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

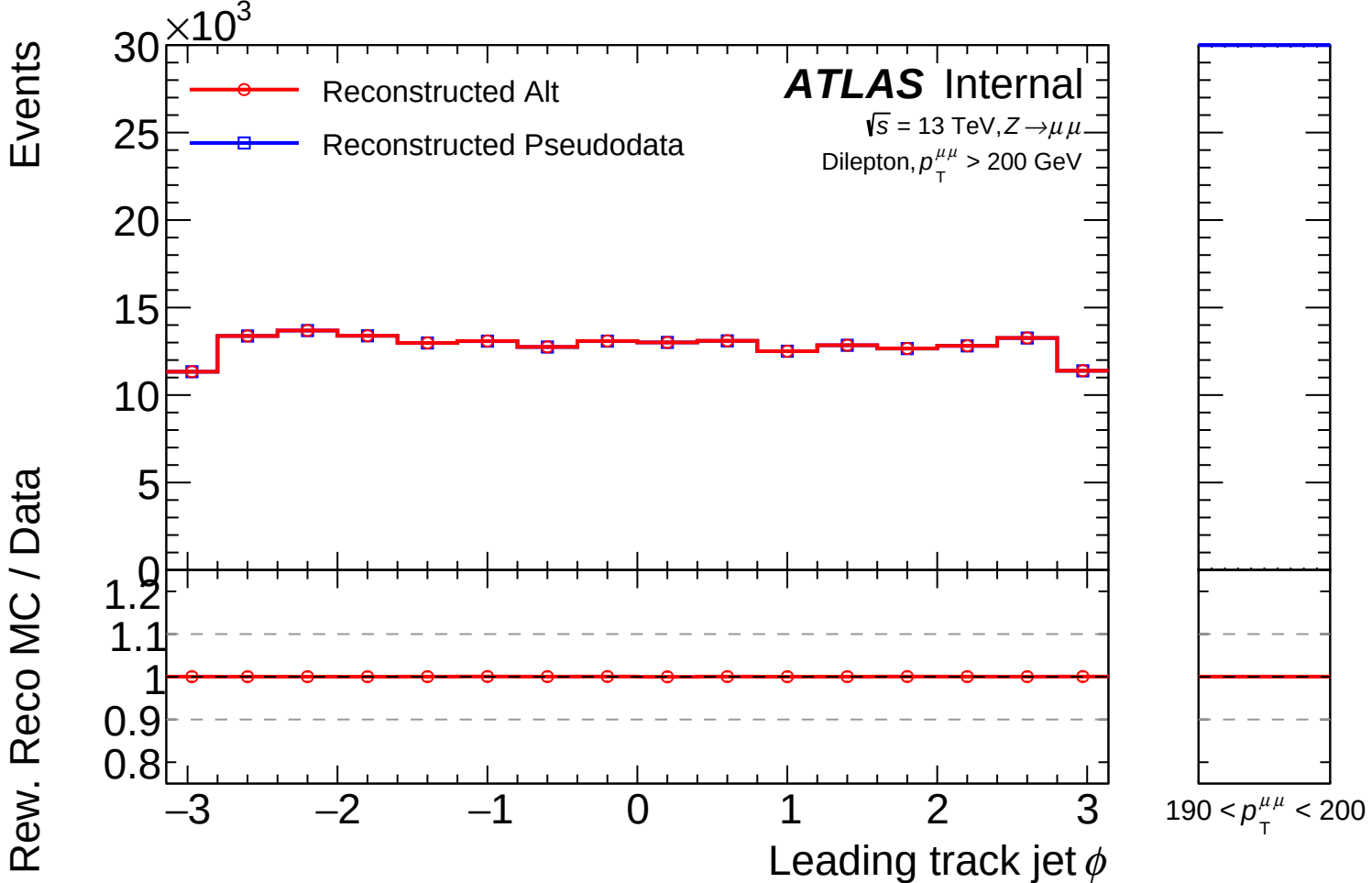
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

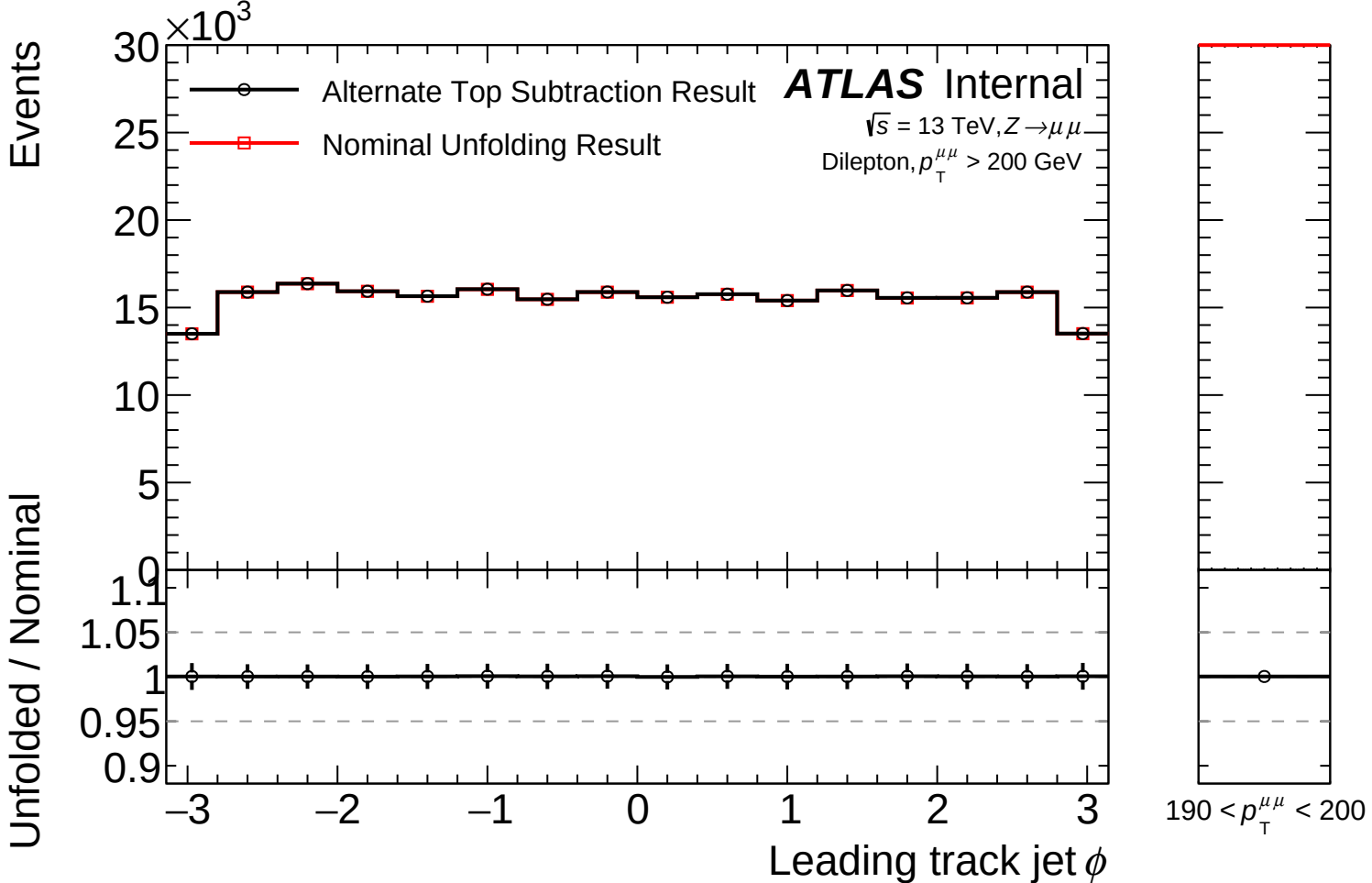
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet y 







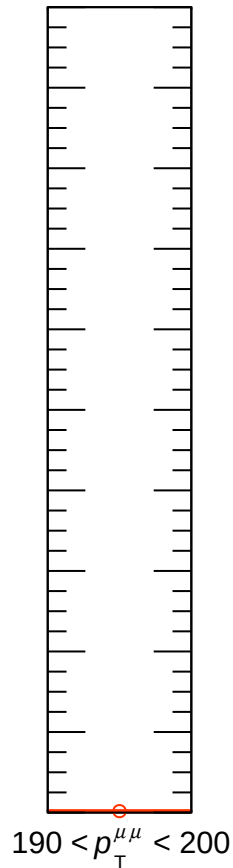
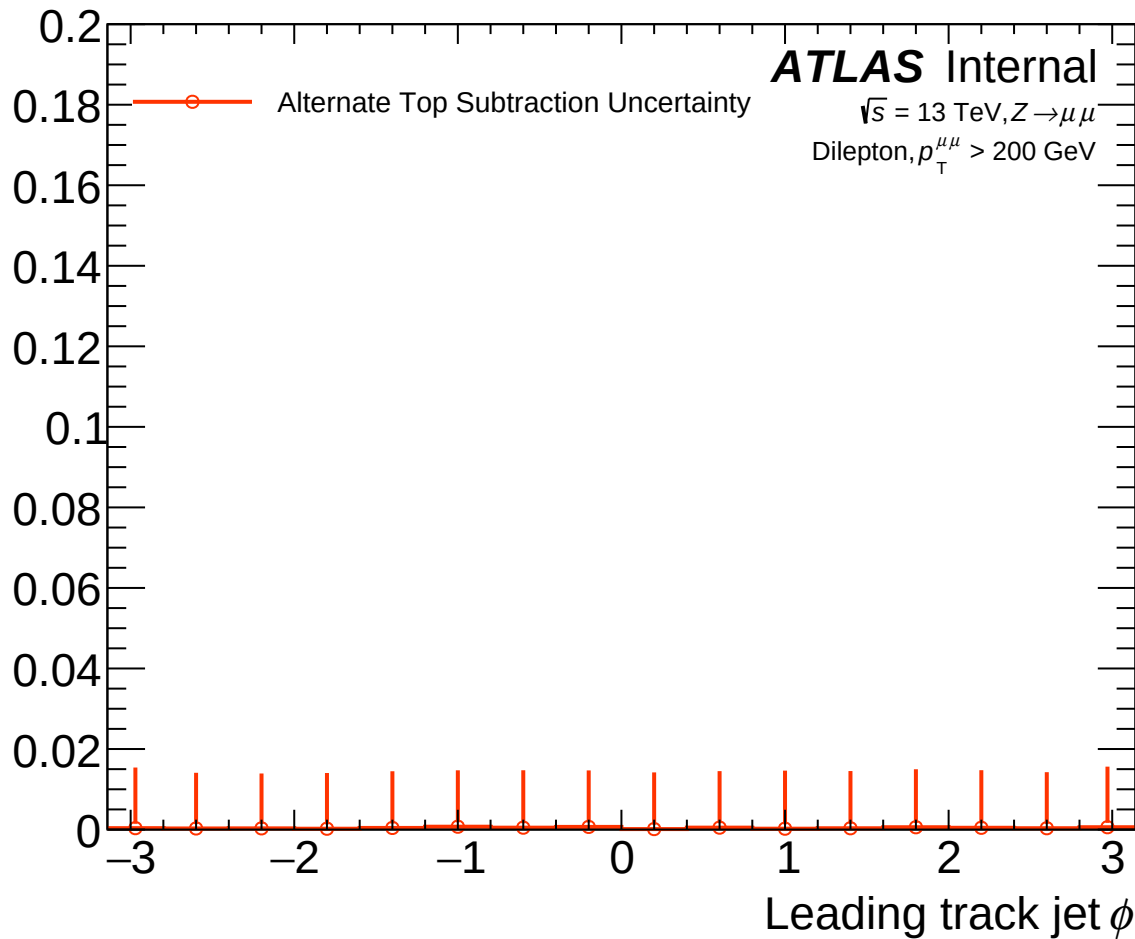
Fractional systematic impact

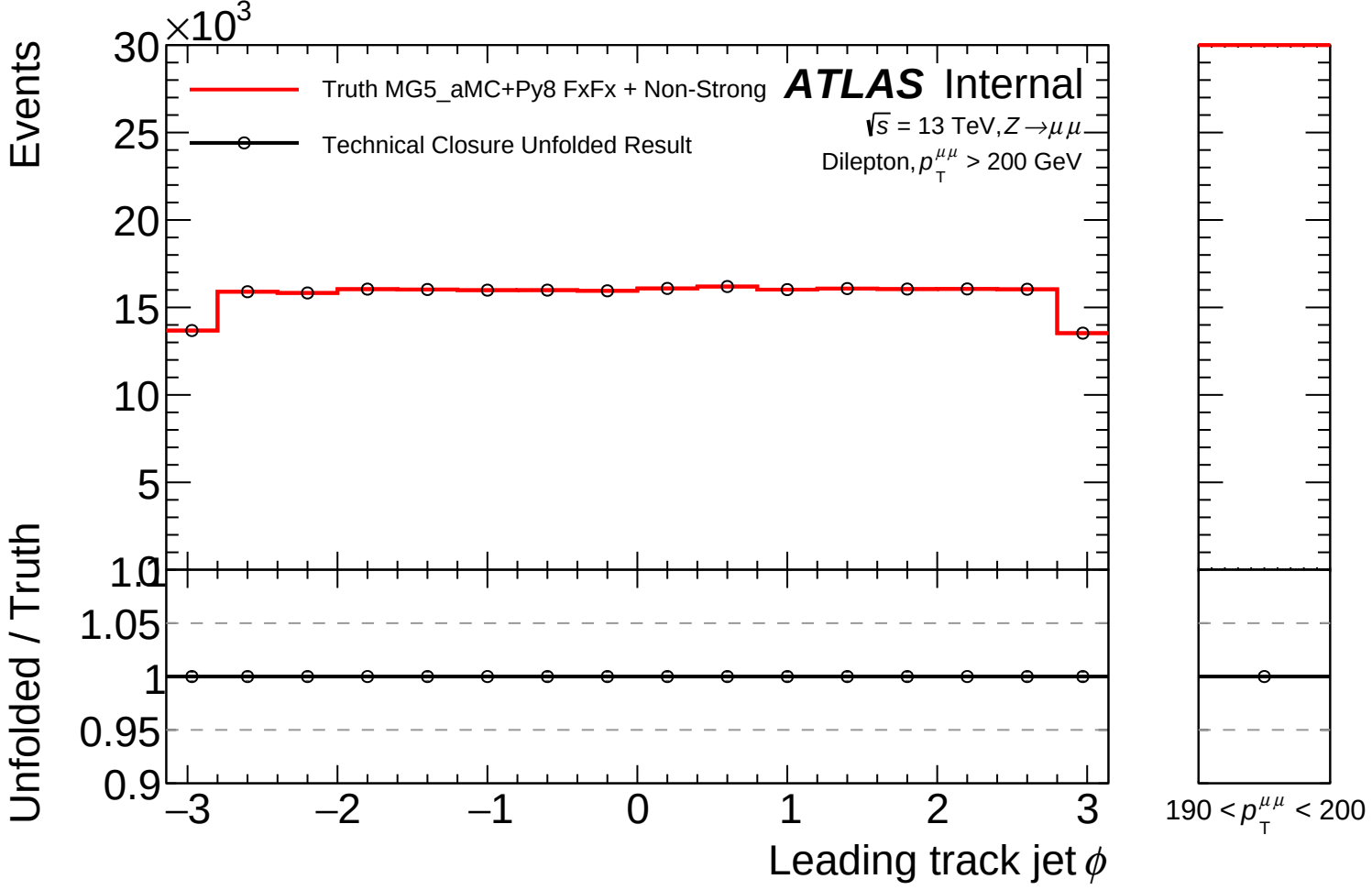
ATLAS Internal

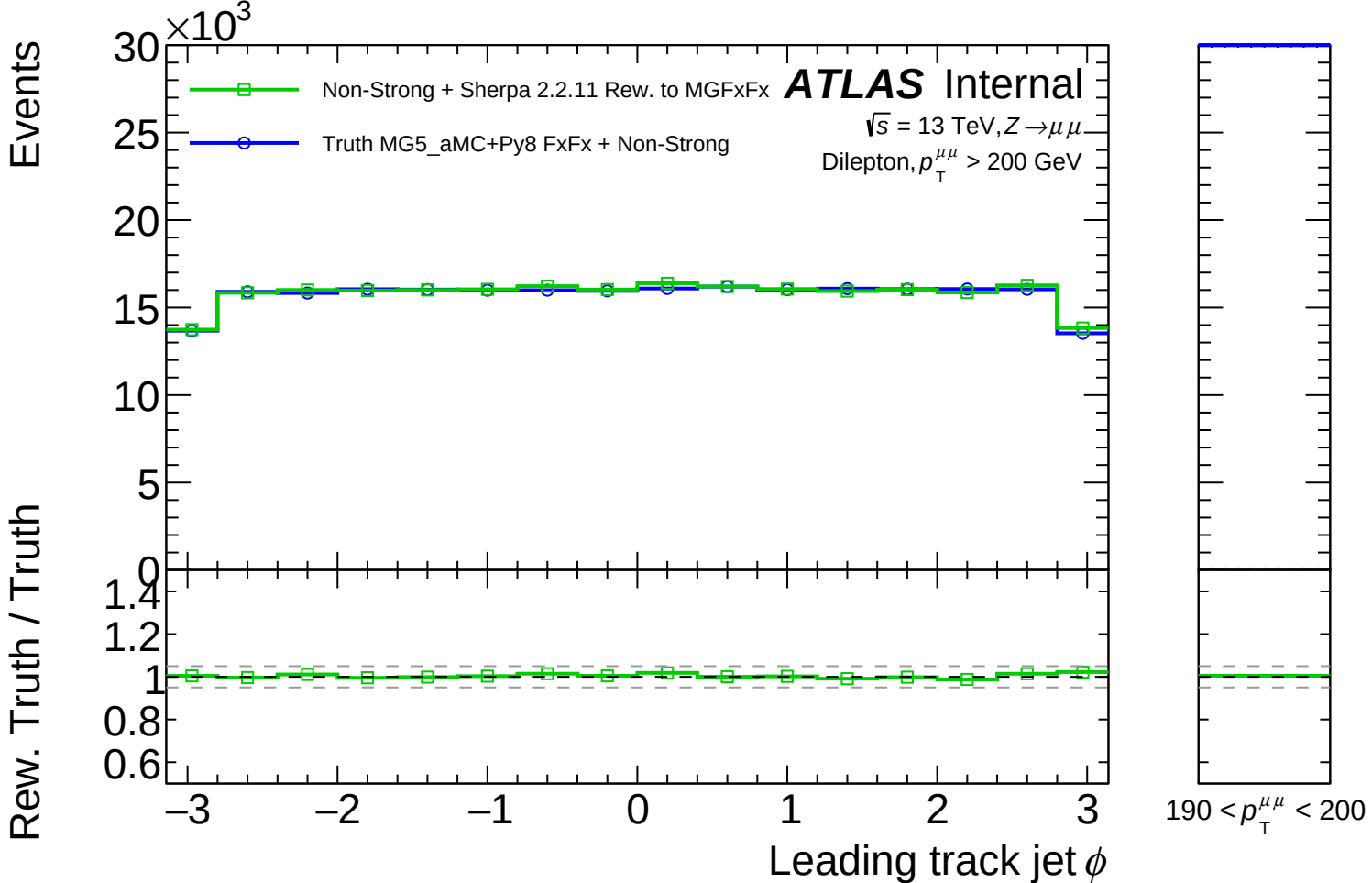
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

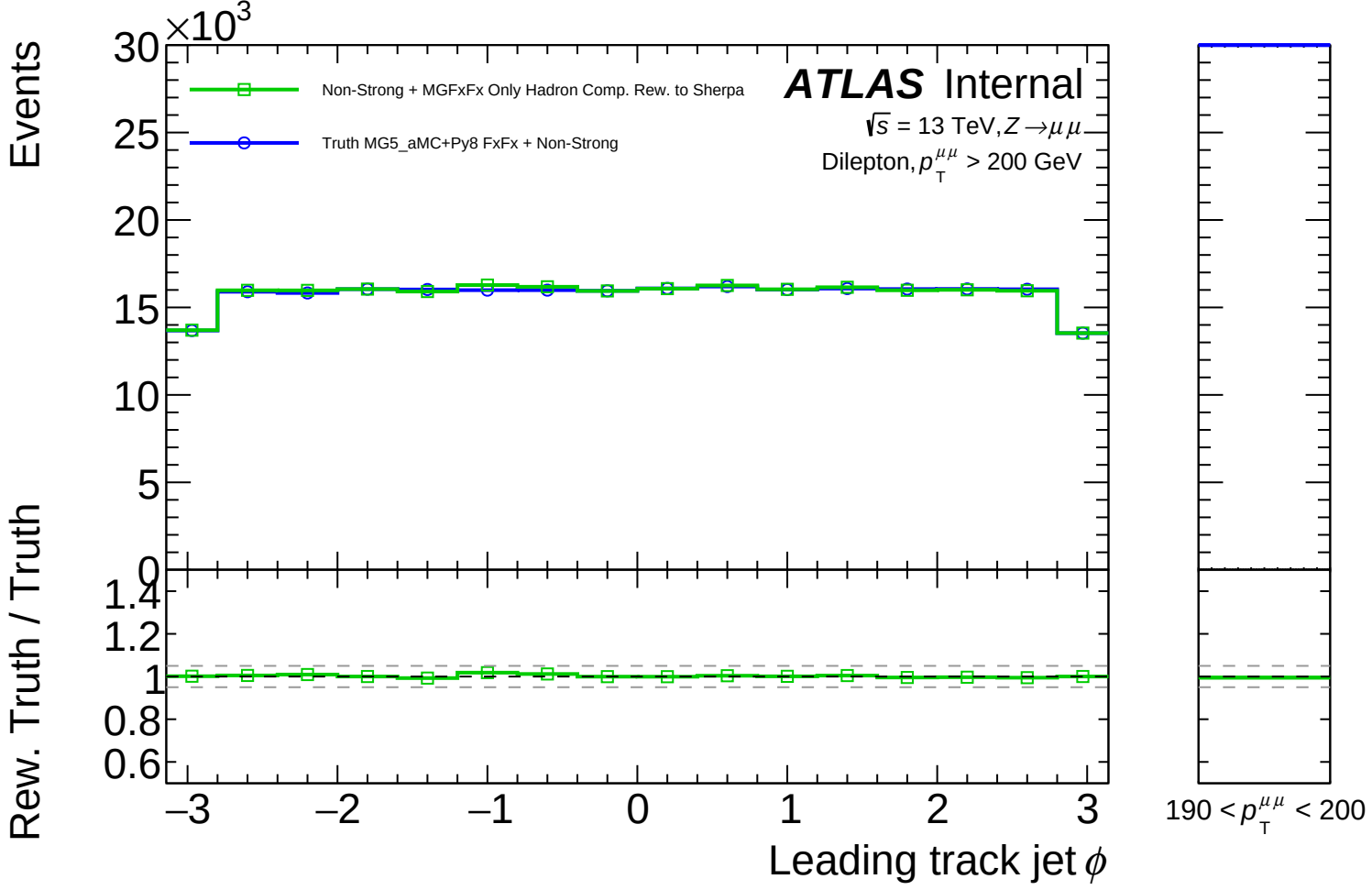
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

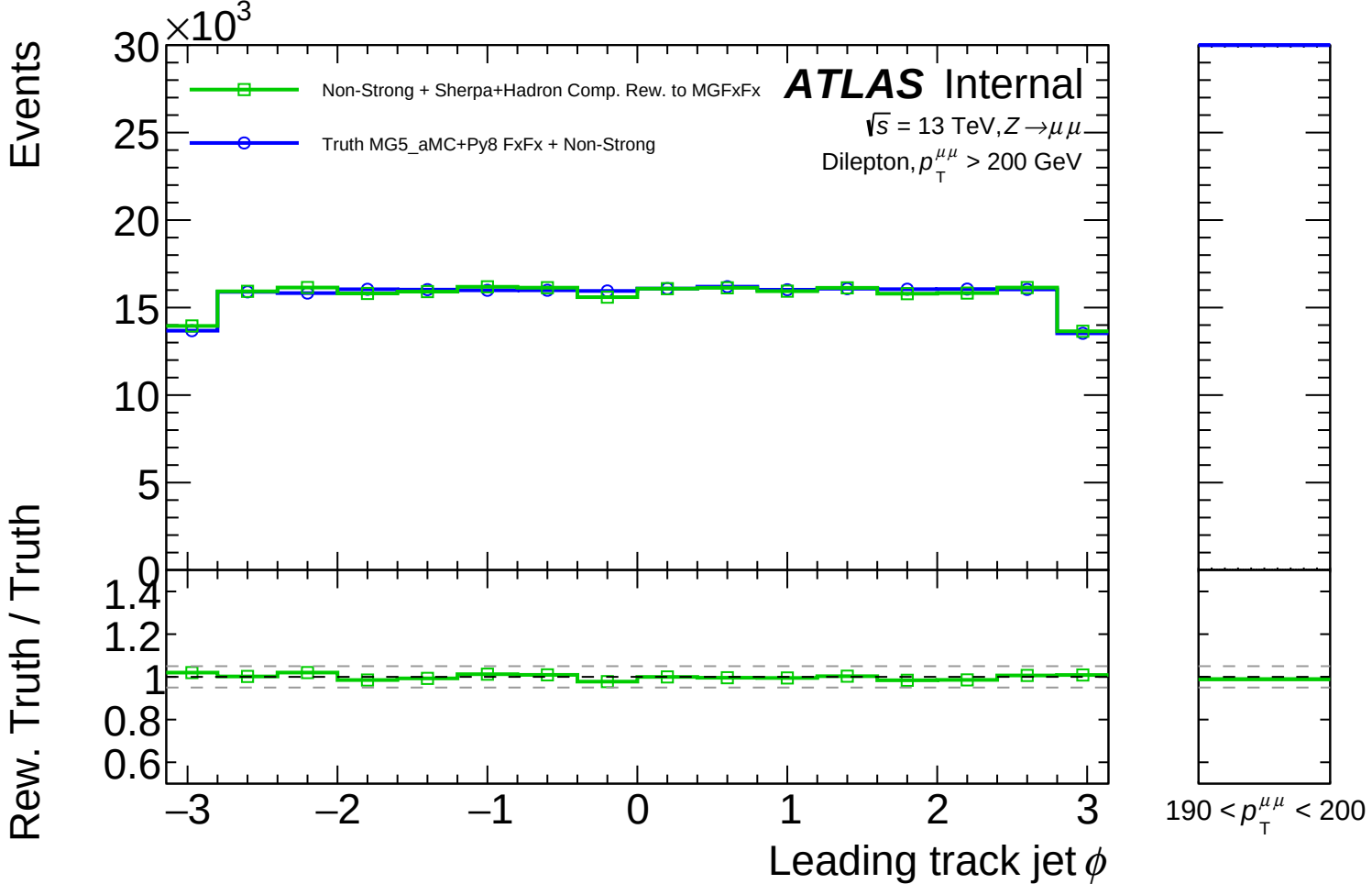
Alternate Top Subtraction Uncertainty

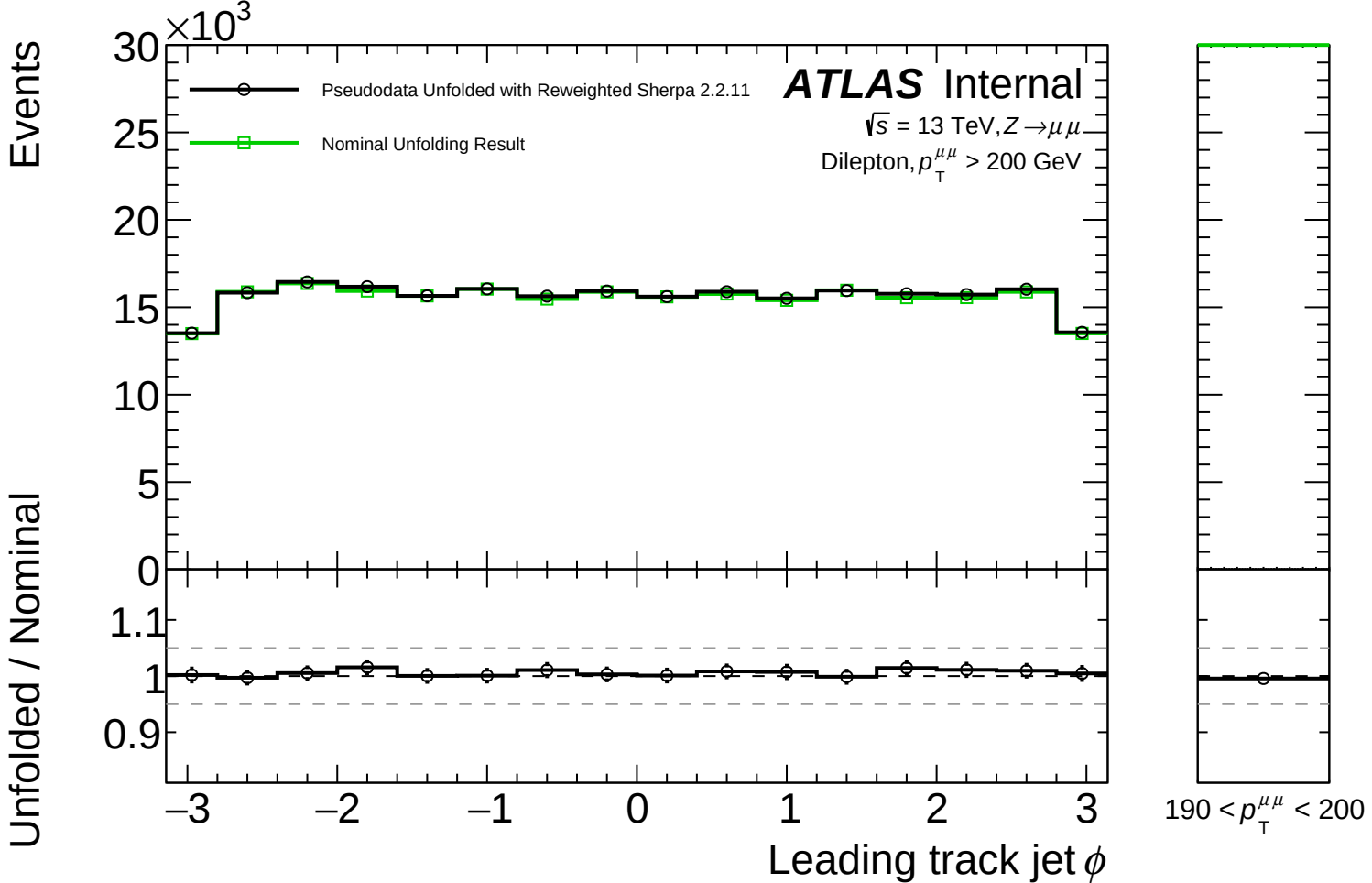


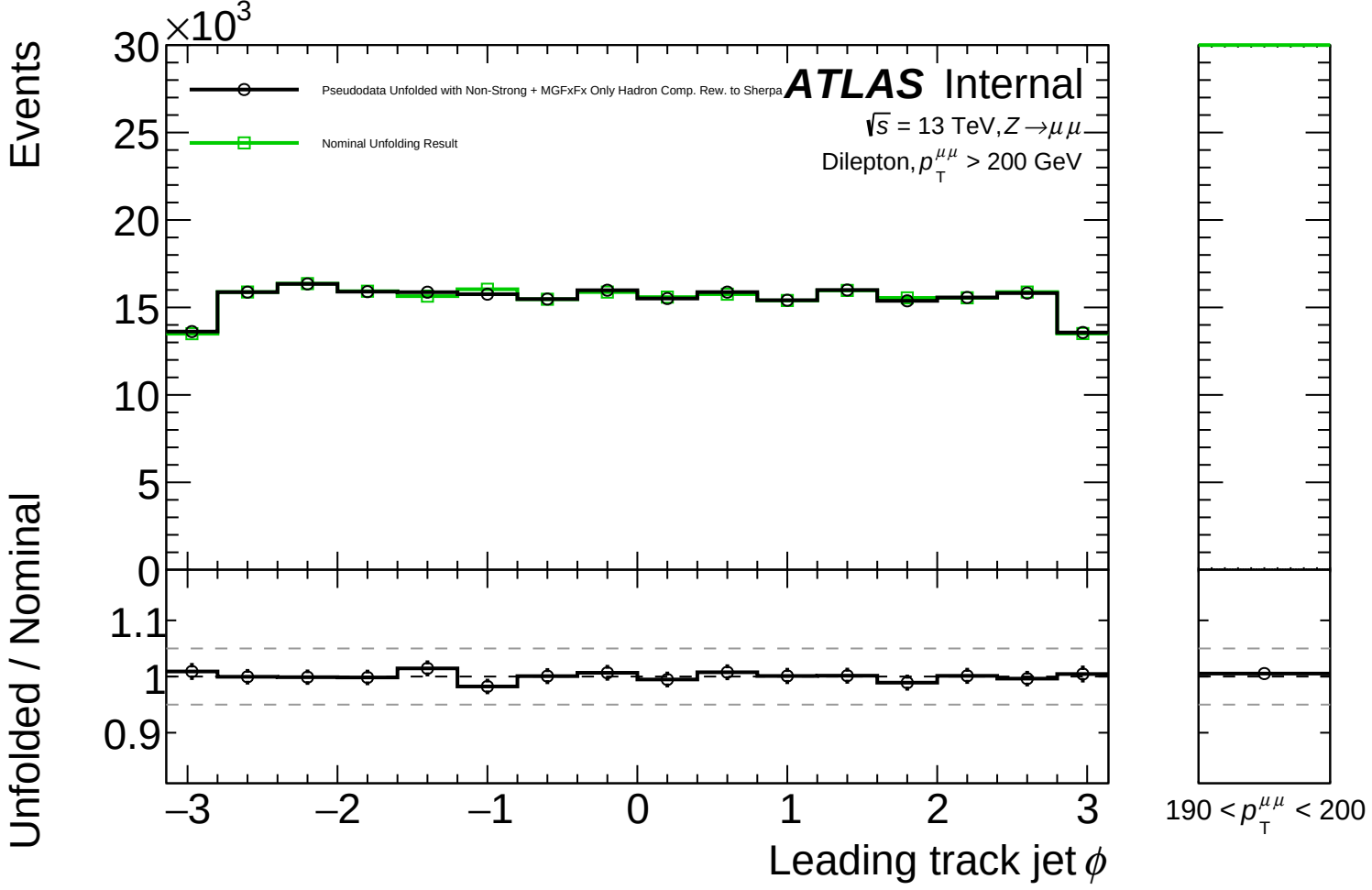


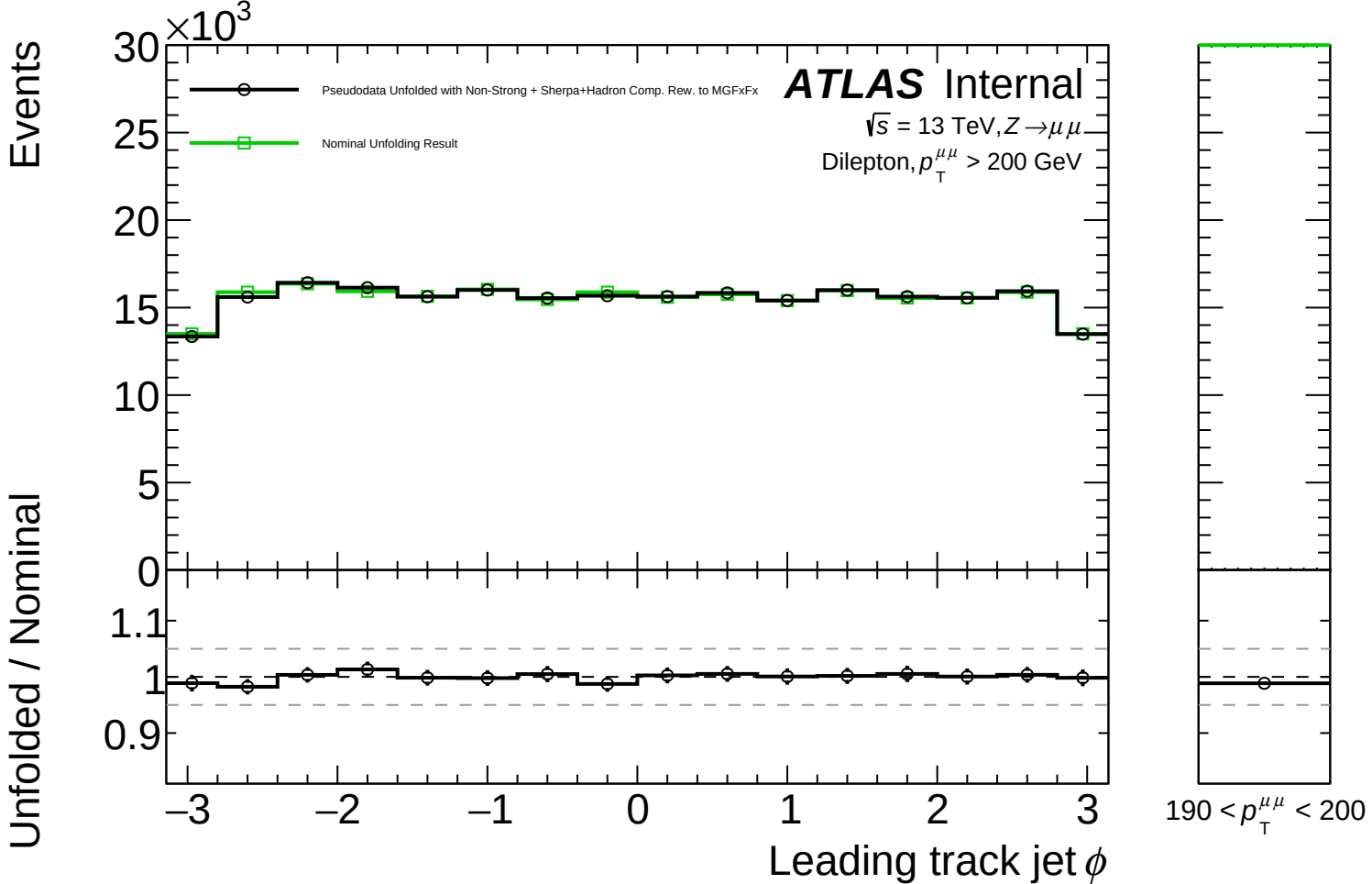


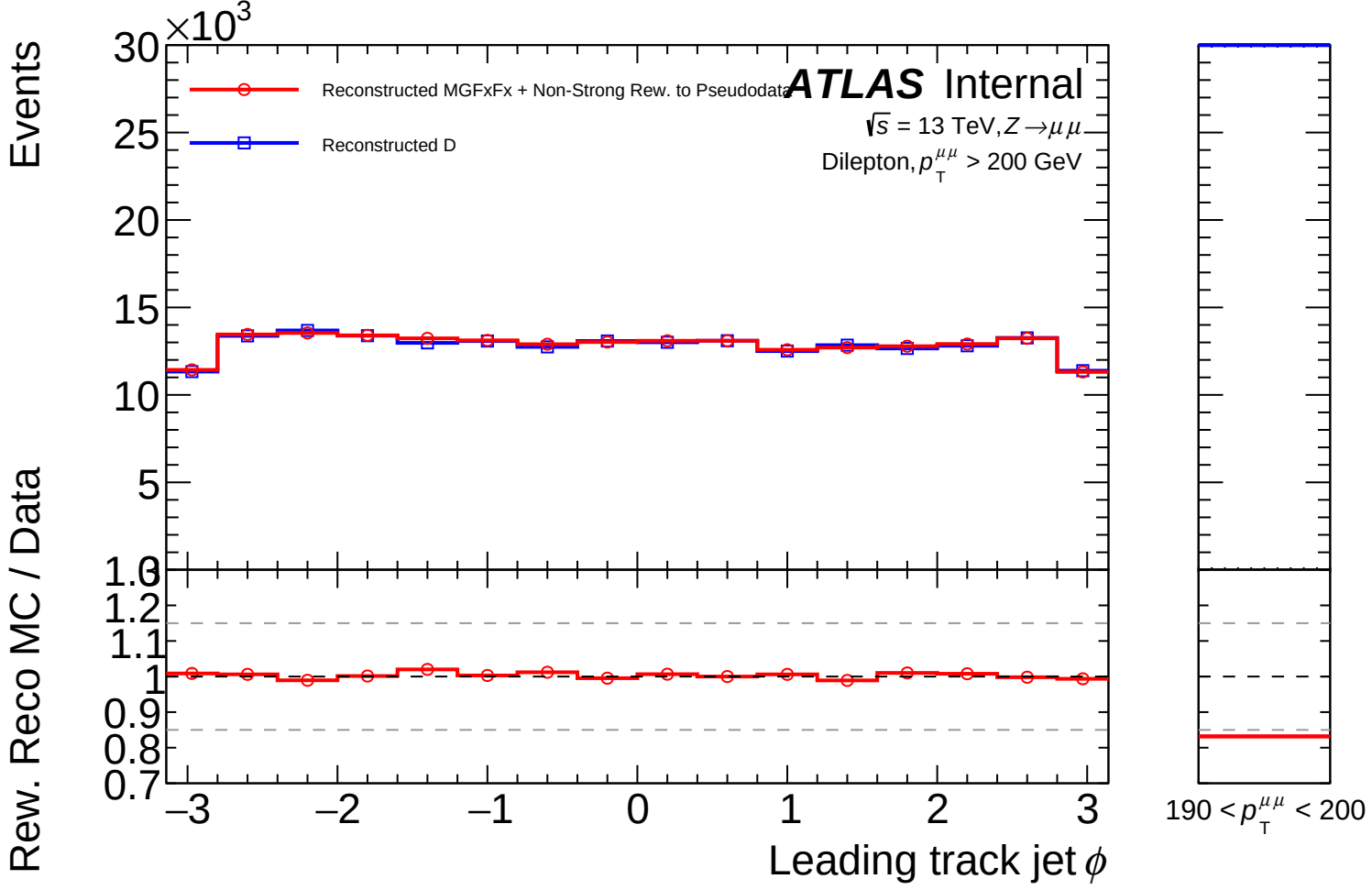


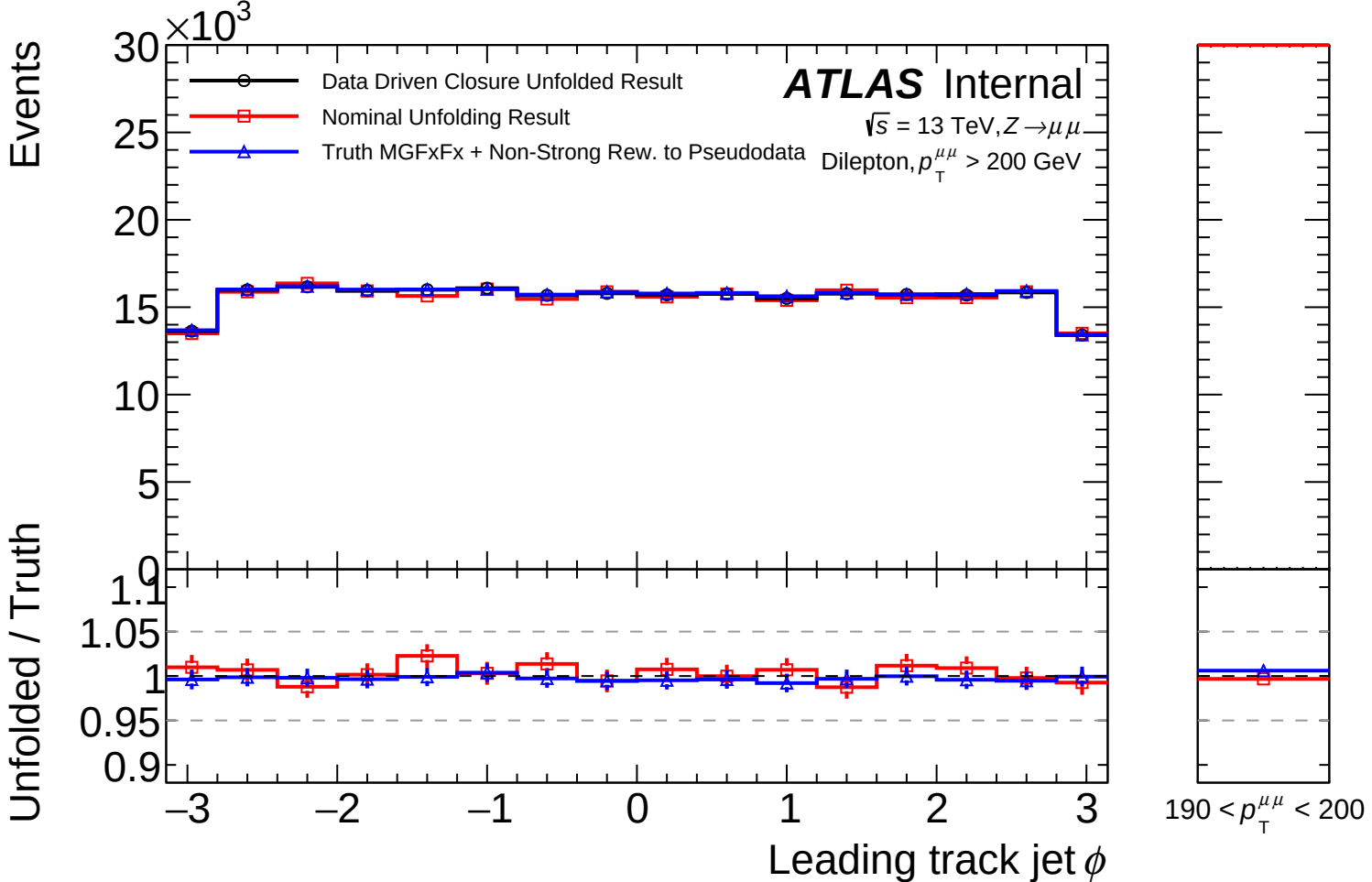




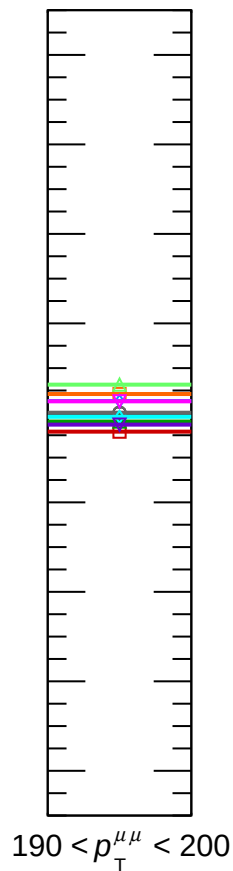
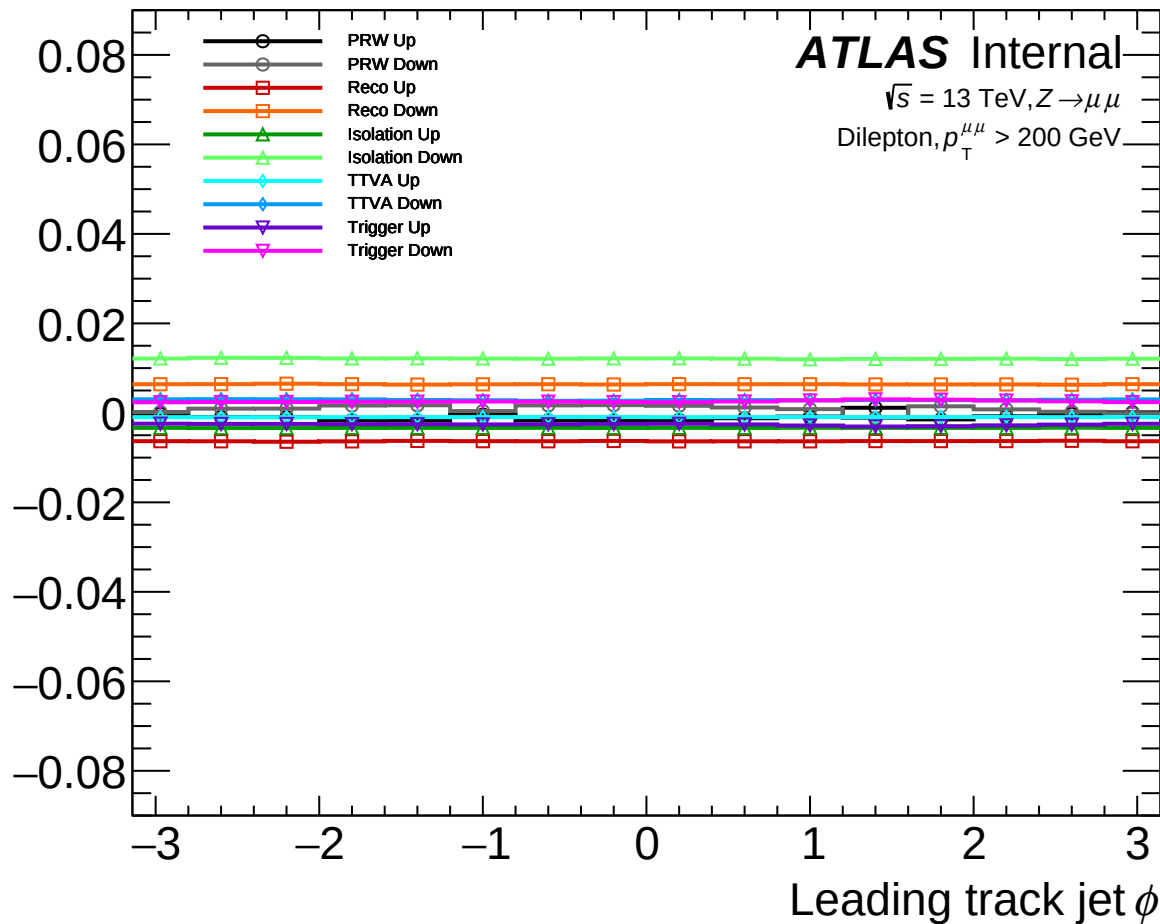




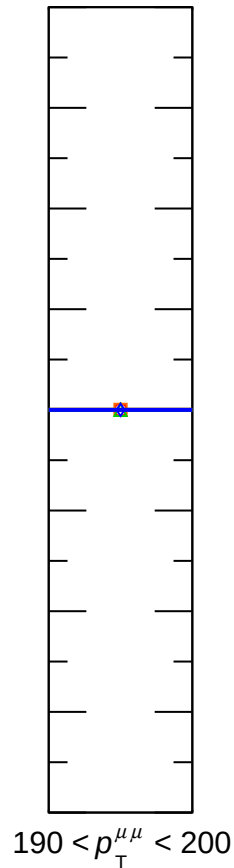
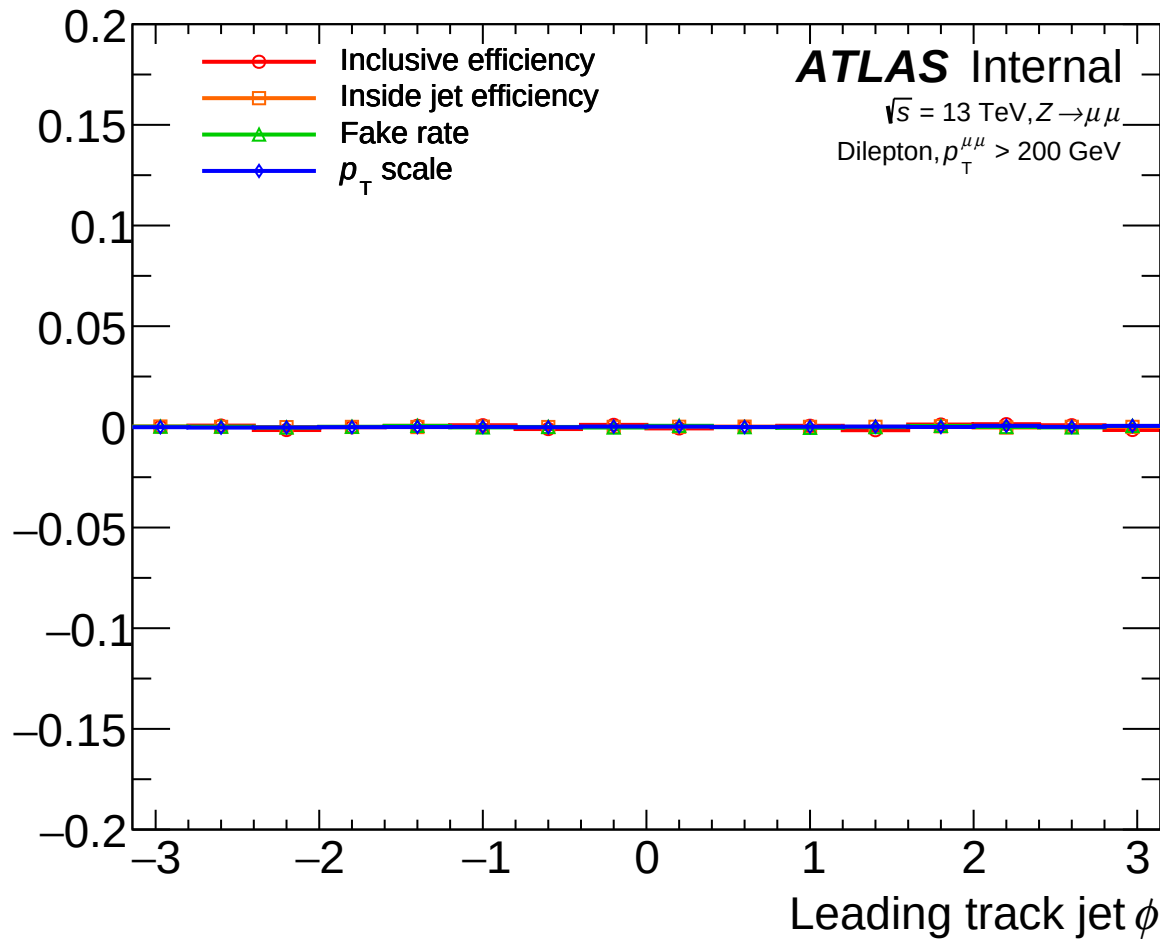




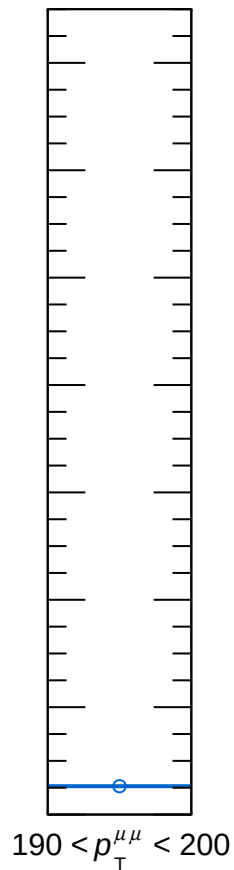
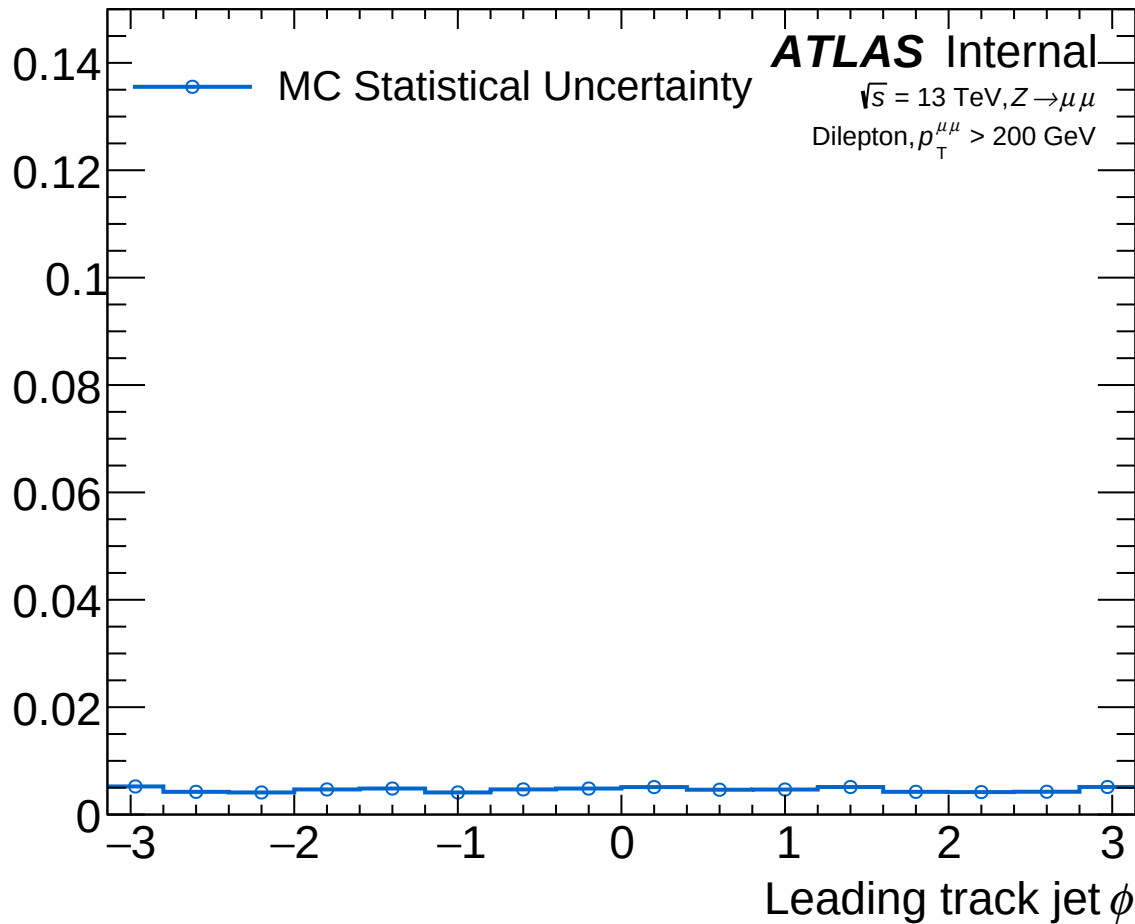
Fractional systematic impact



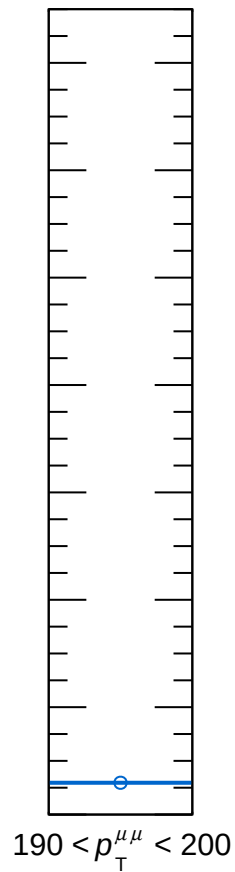
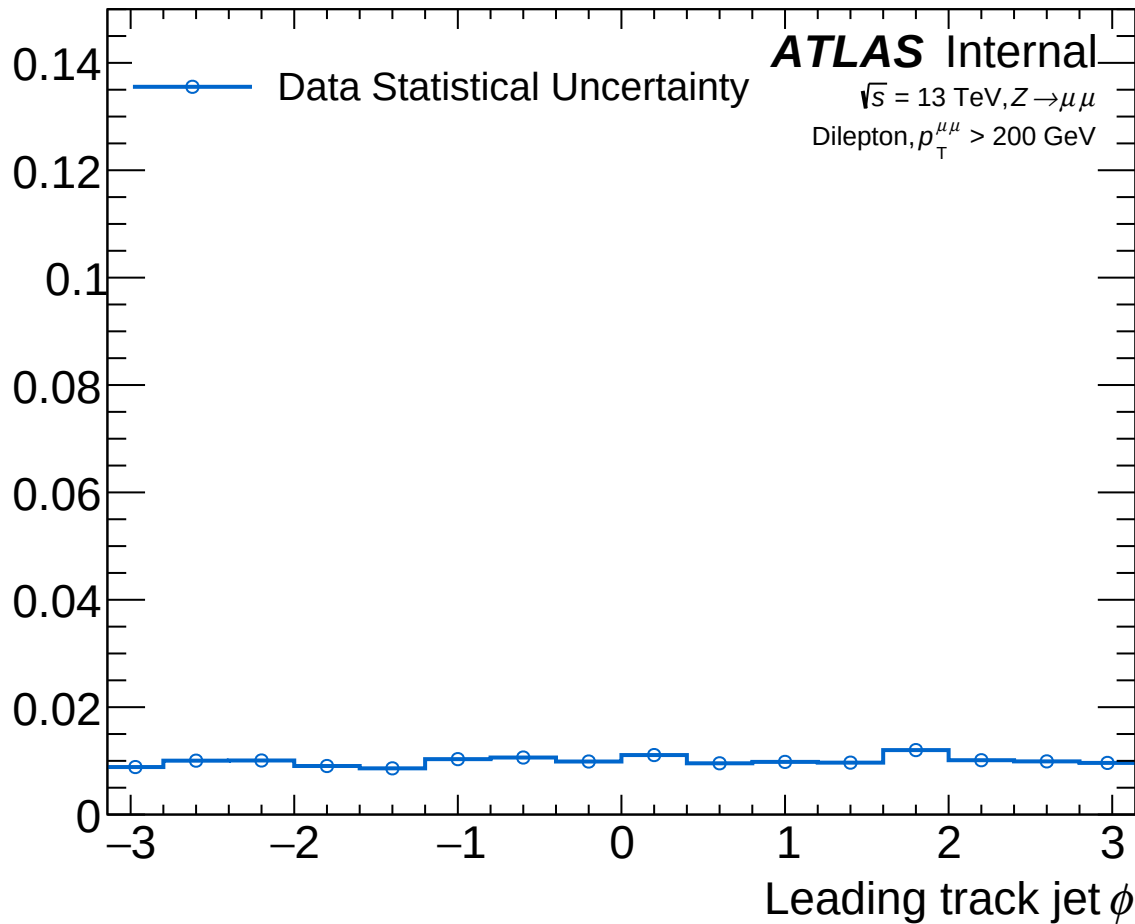
Fractional systematic impact



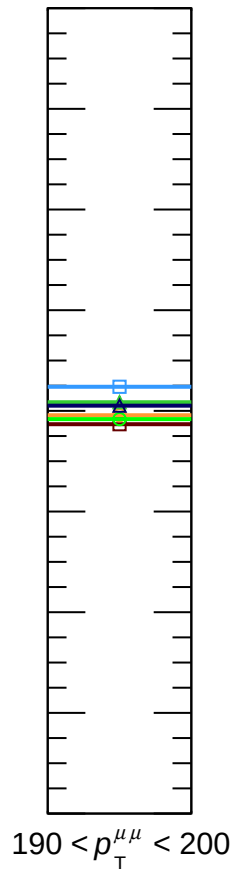
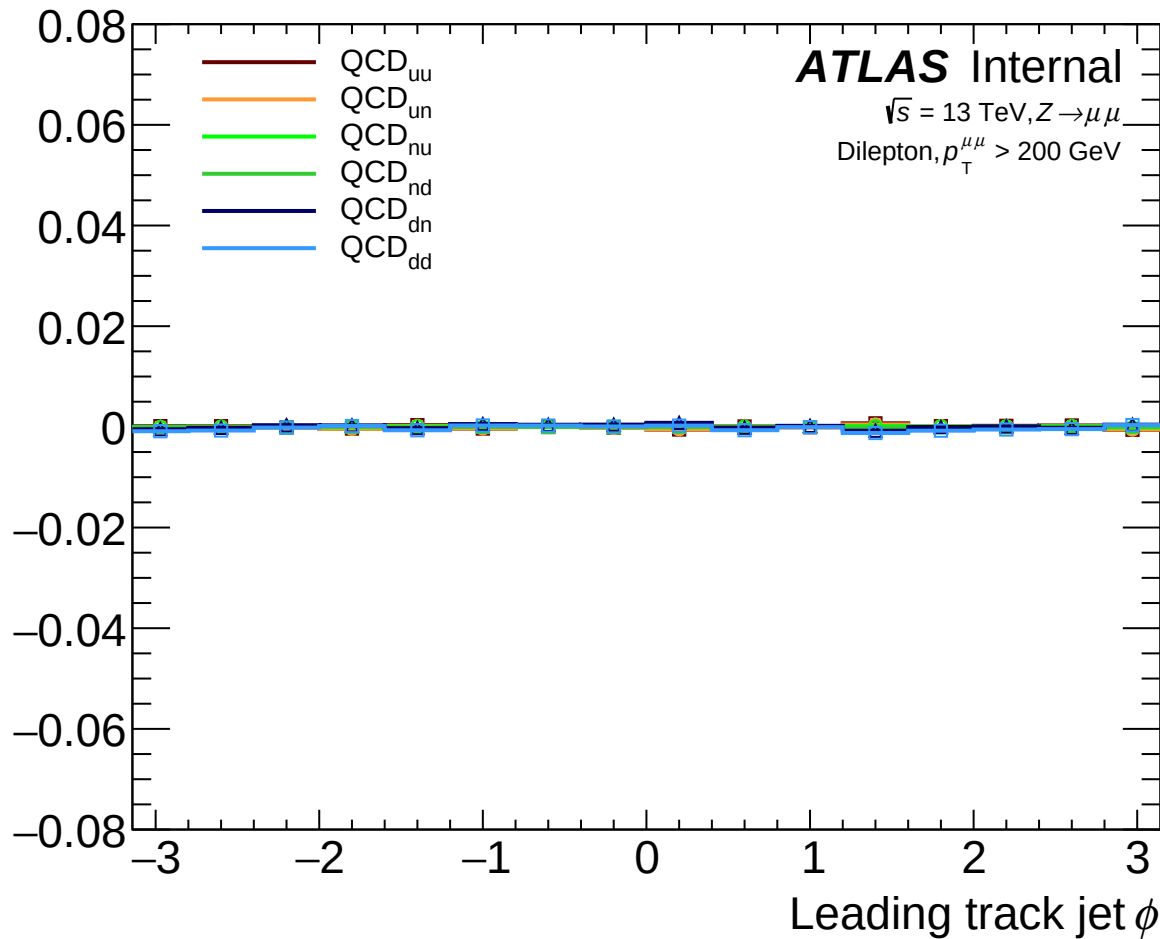
Fractional systematic impact



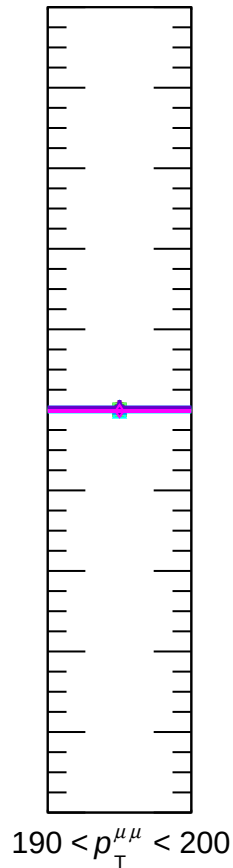
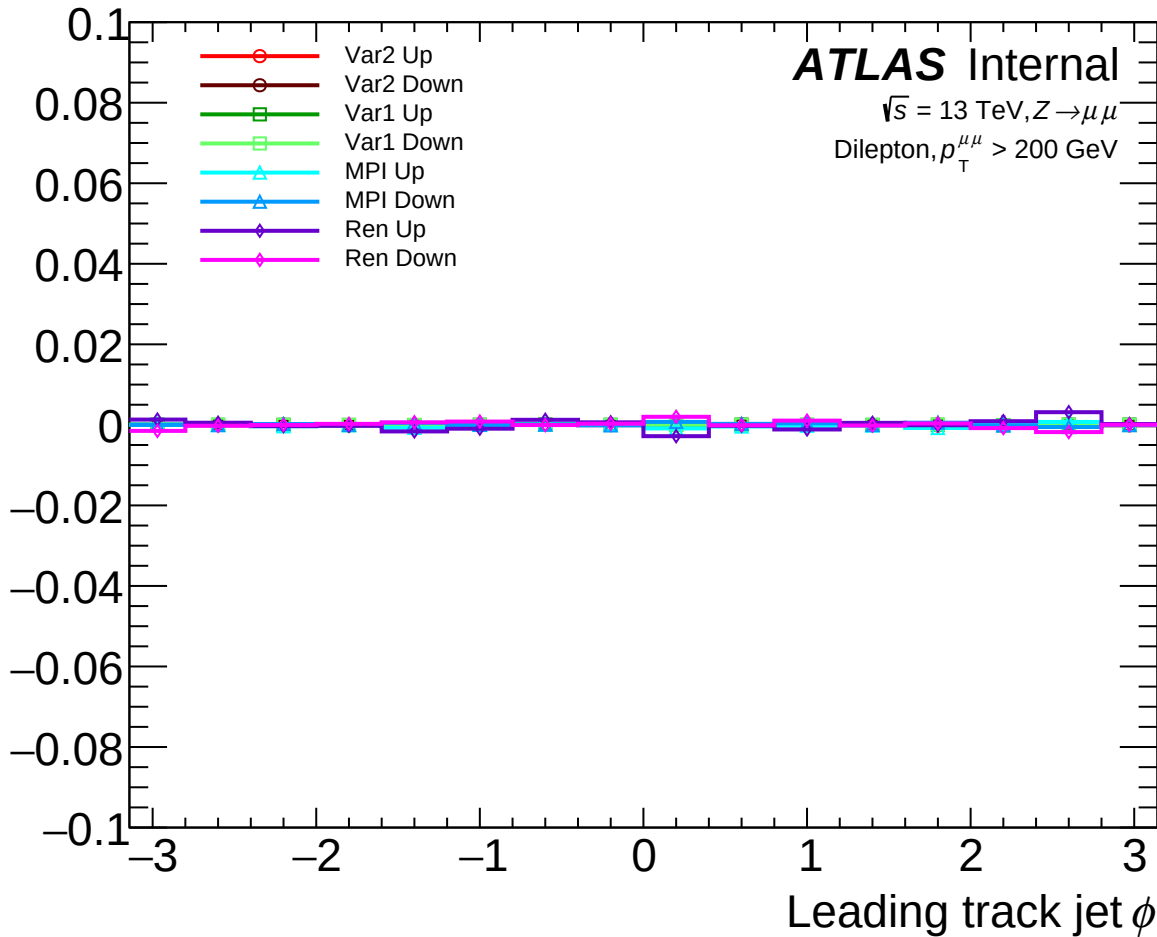
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



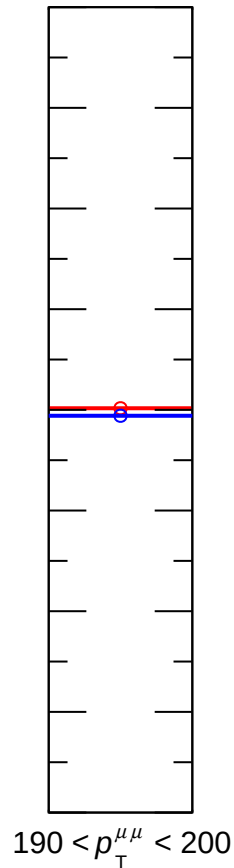
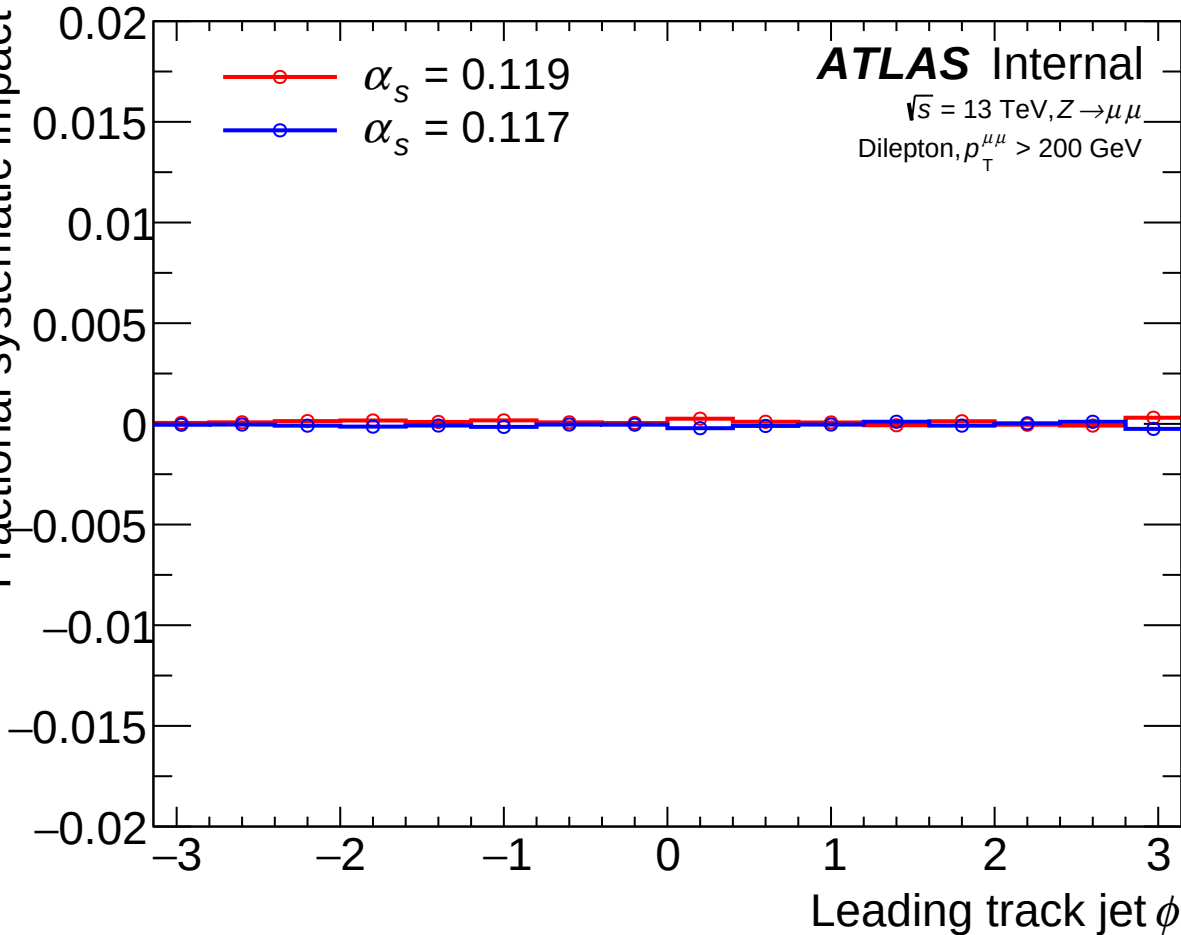
Fractional systematic impact

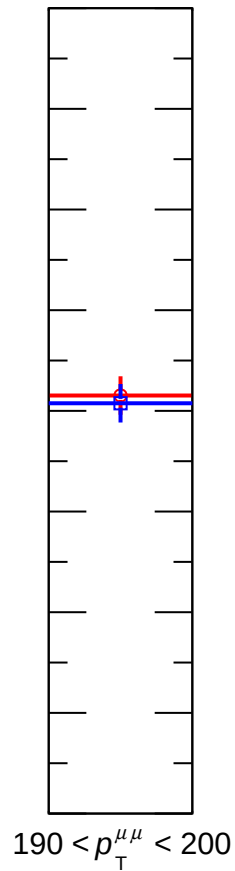
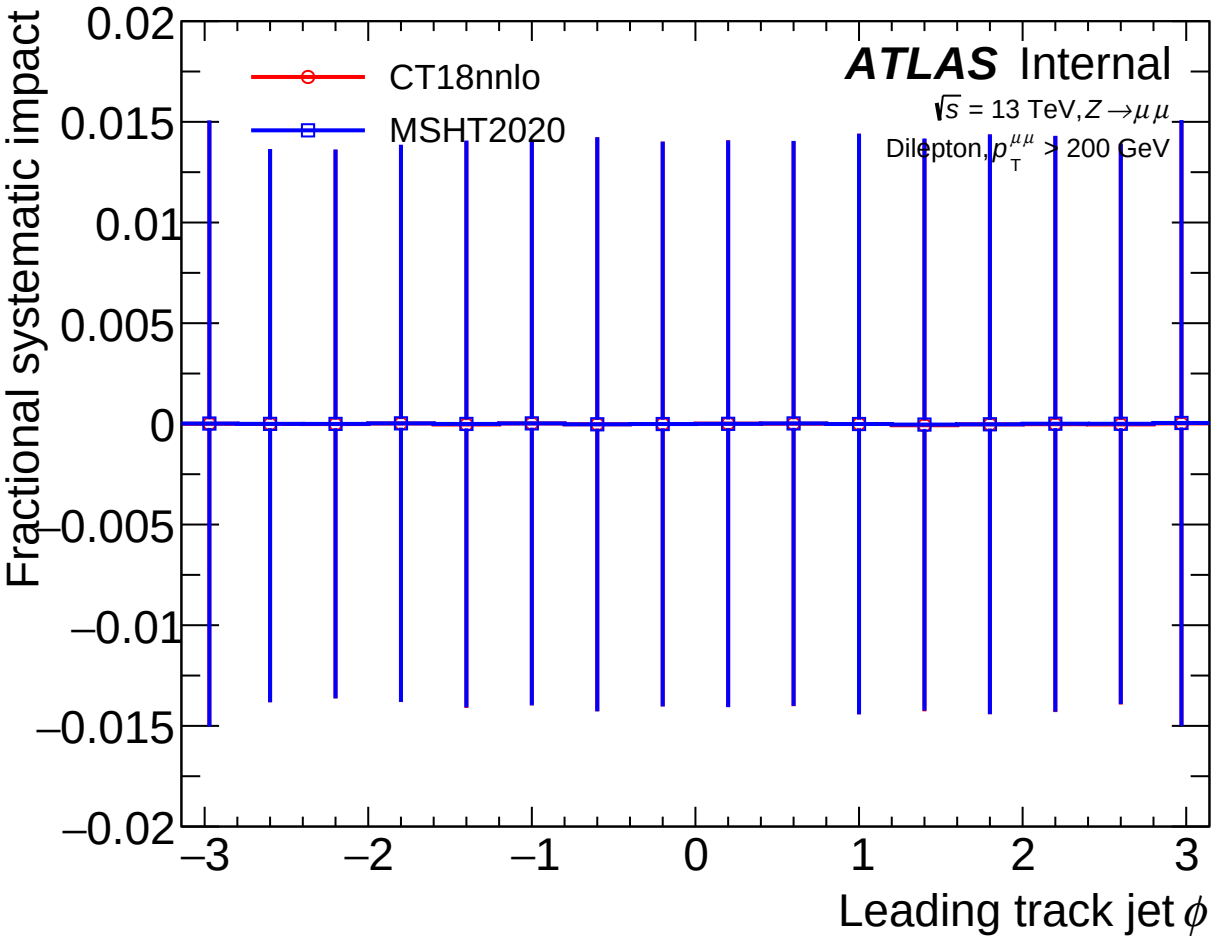
$\alpha_s = 0.119$
 $\alpha_s = 0.117$

ATLAS Internal

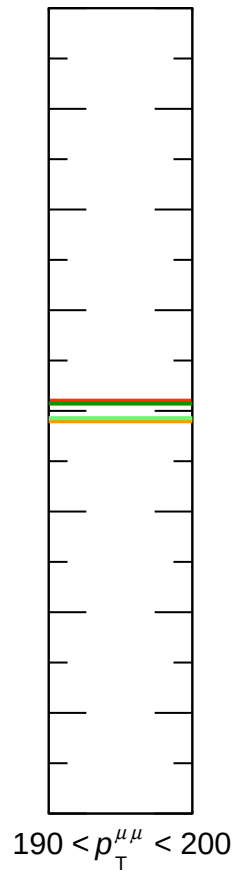
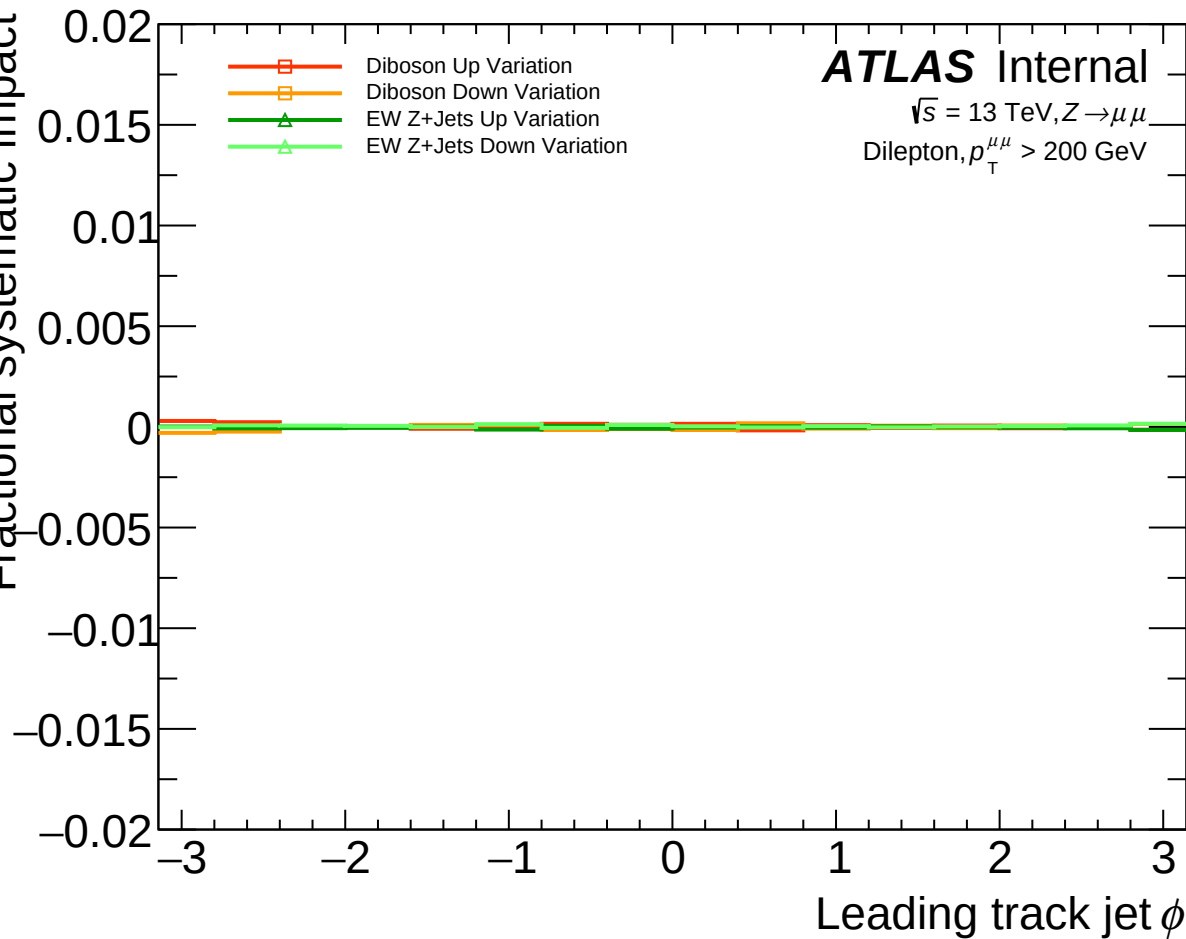
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

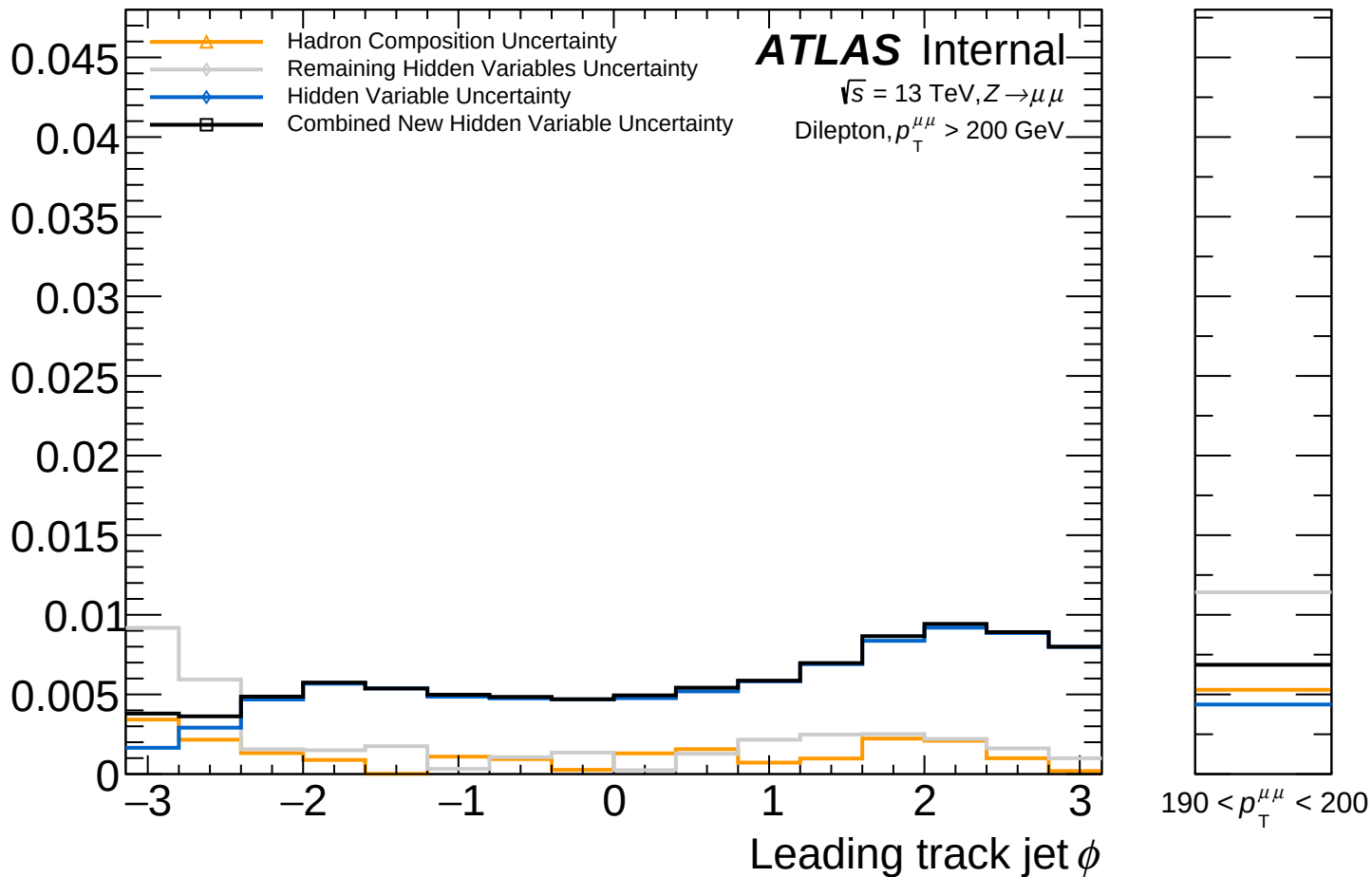
Dilepton, $p_T^{\mu\mu} > 200$ GeV

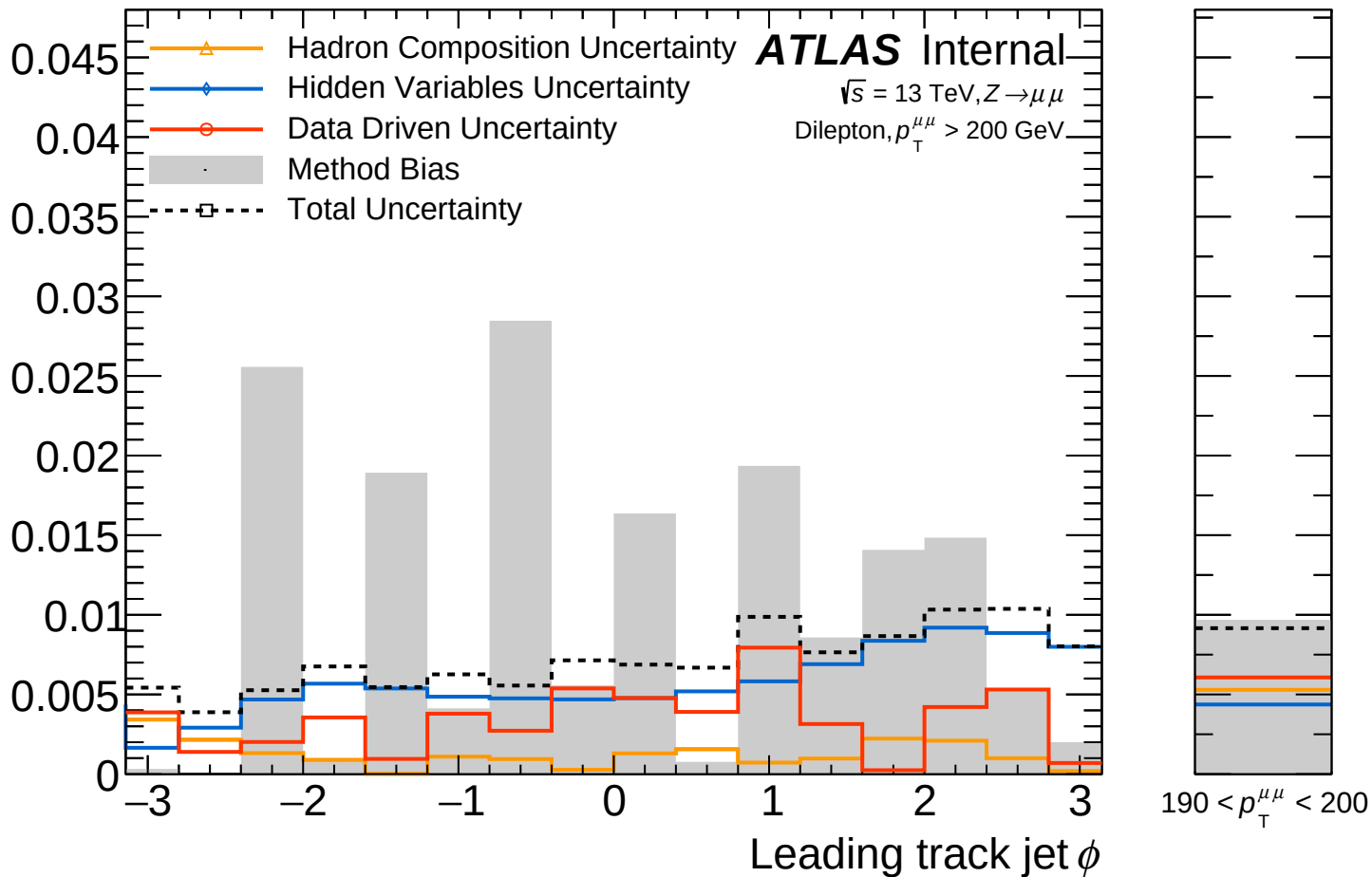




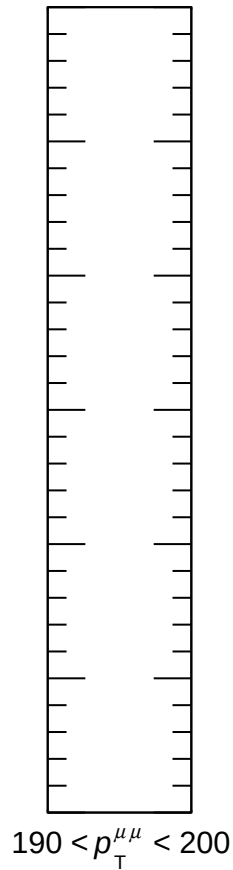
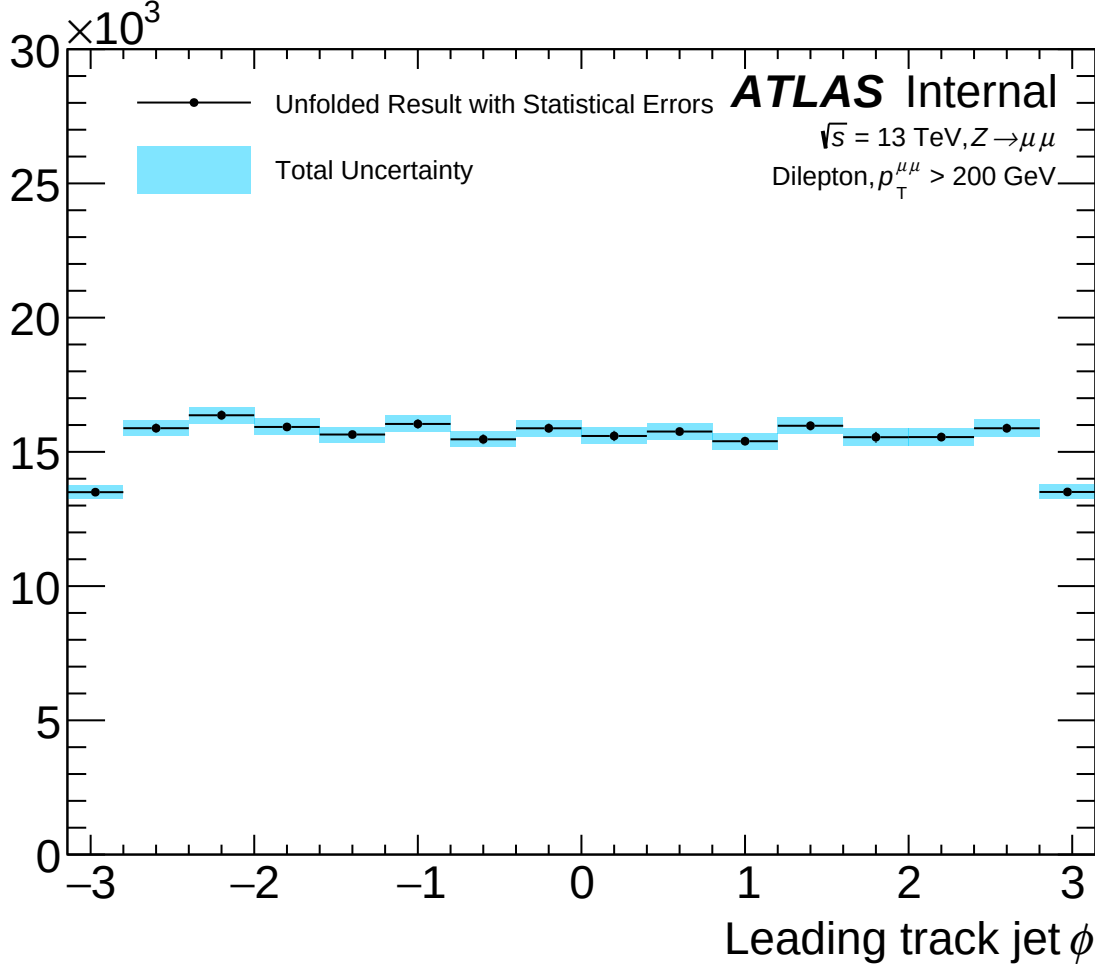
Fractional systematic impact

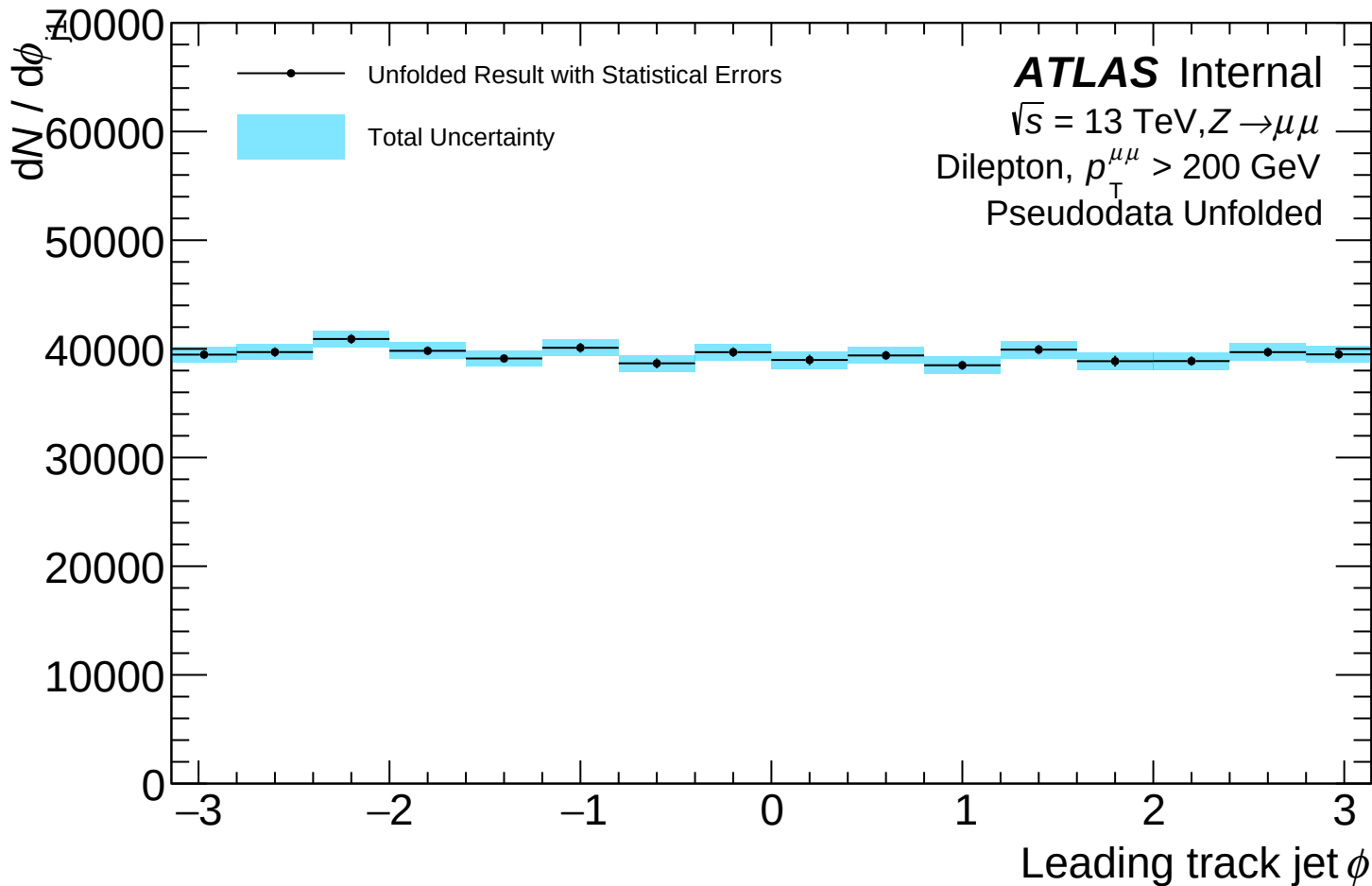


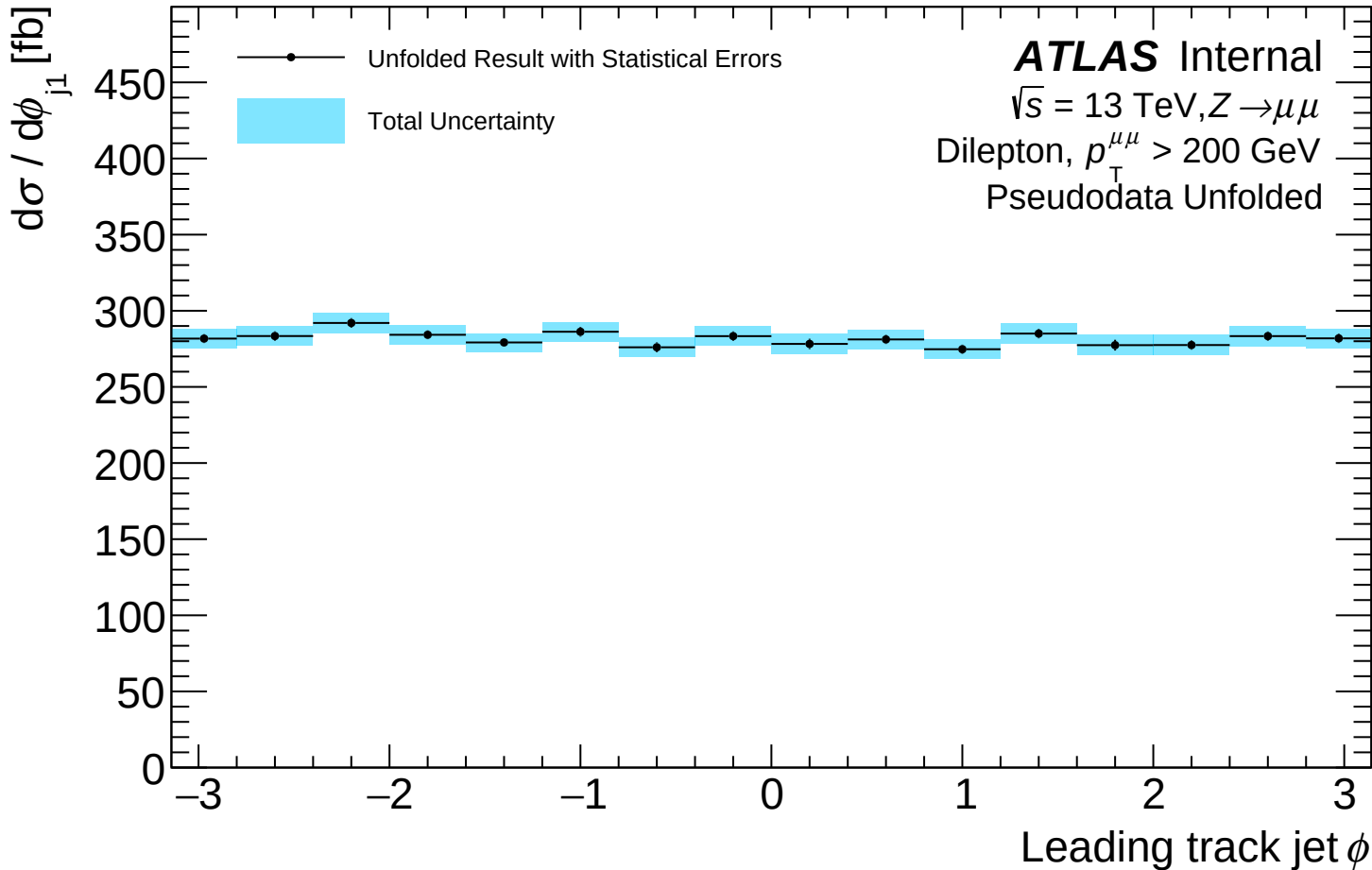




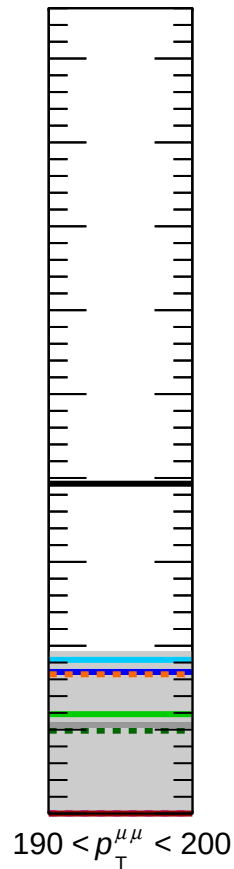
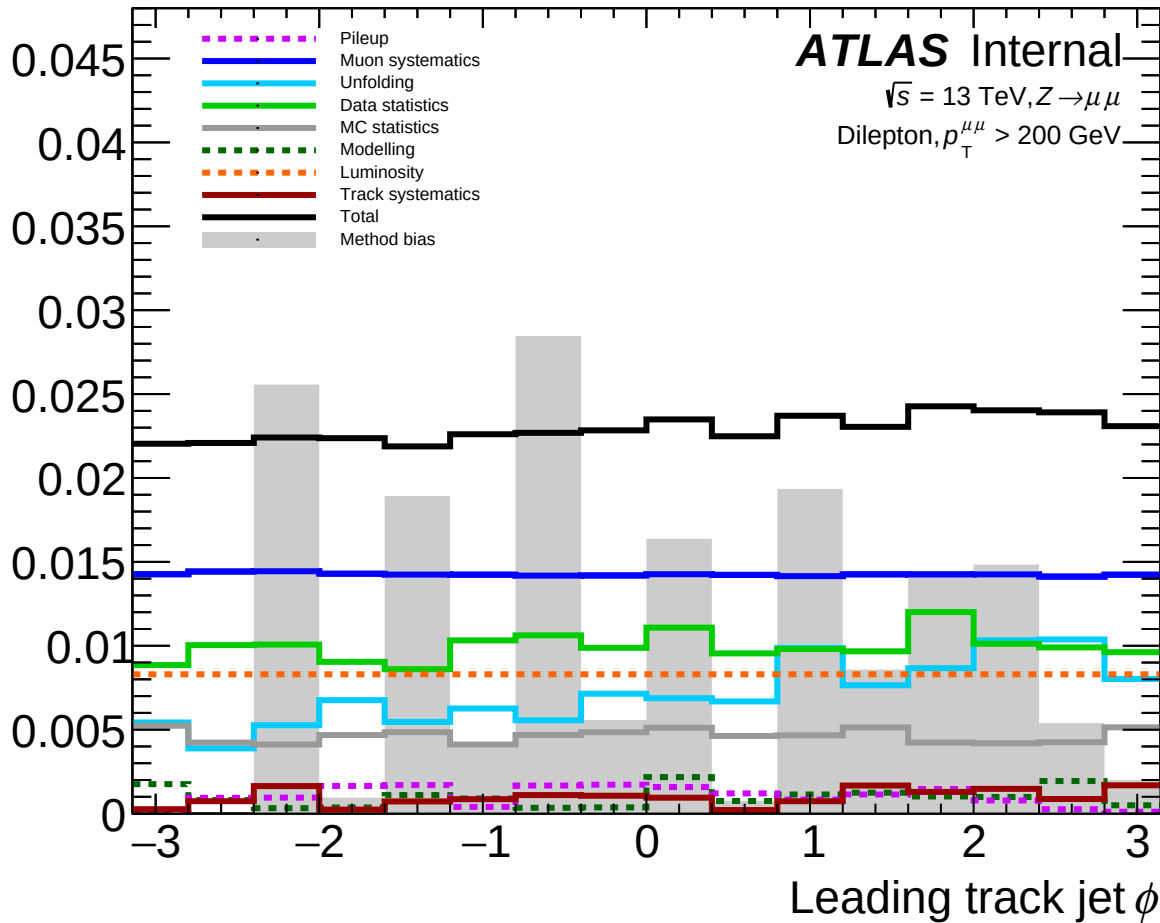
Events

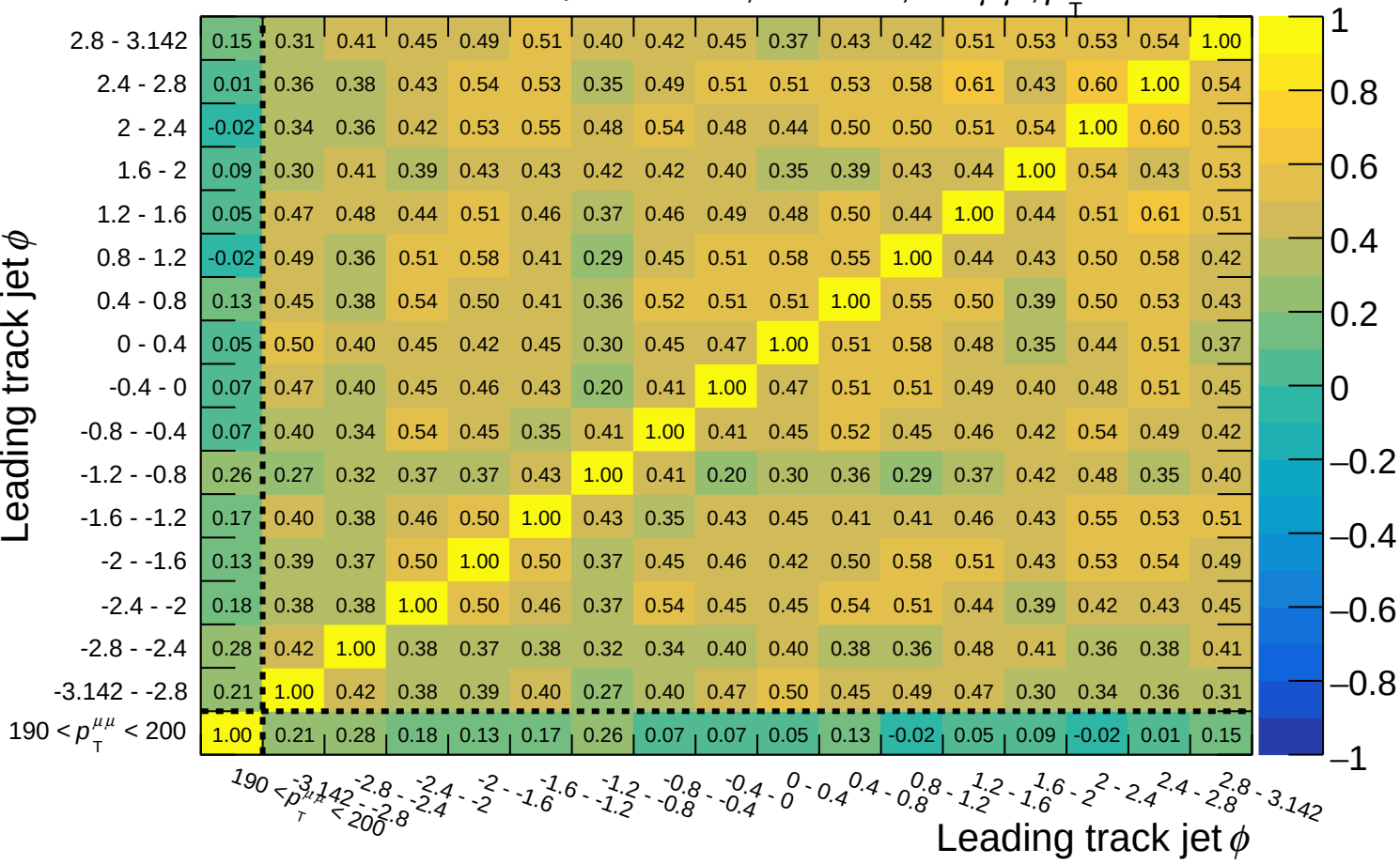


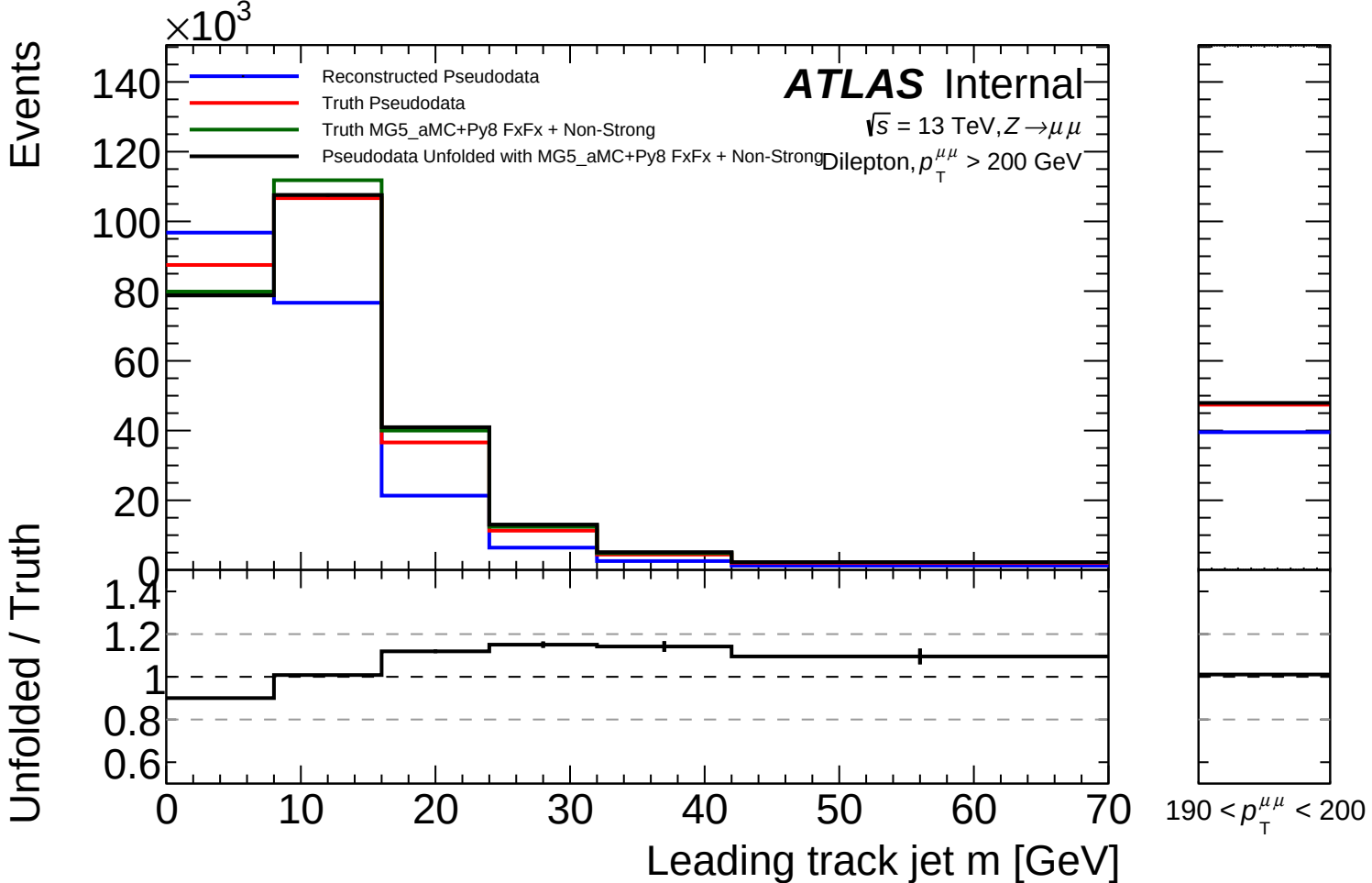


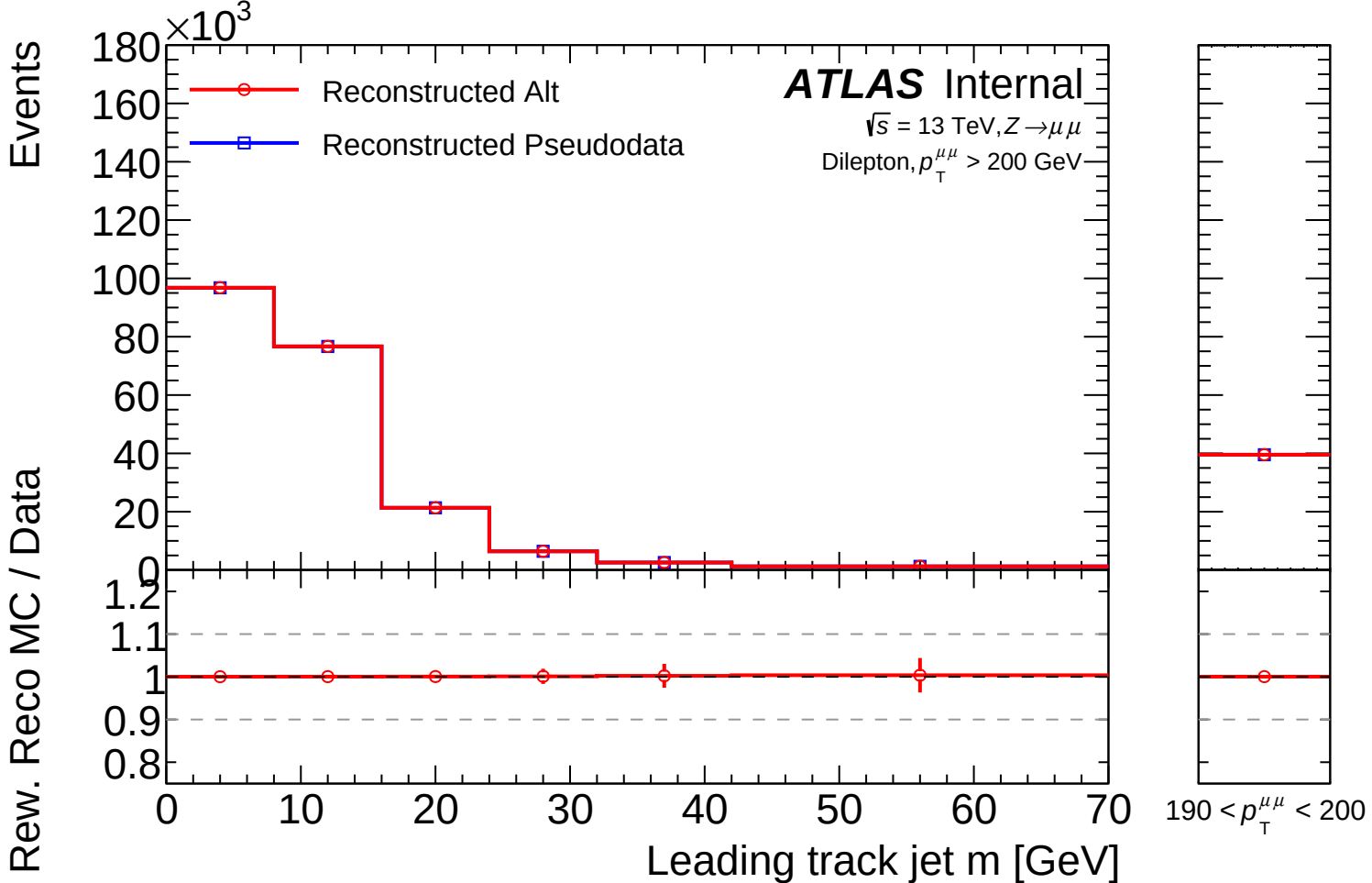


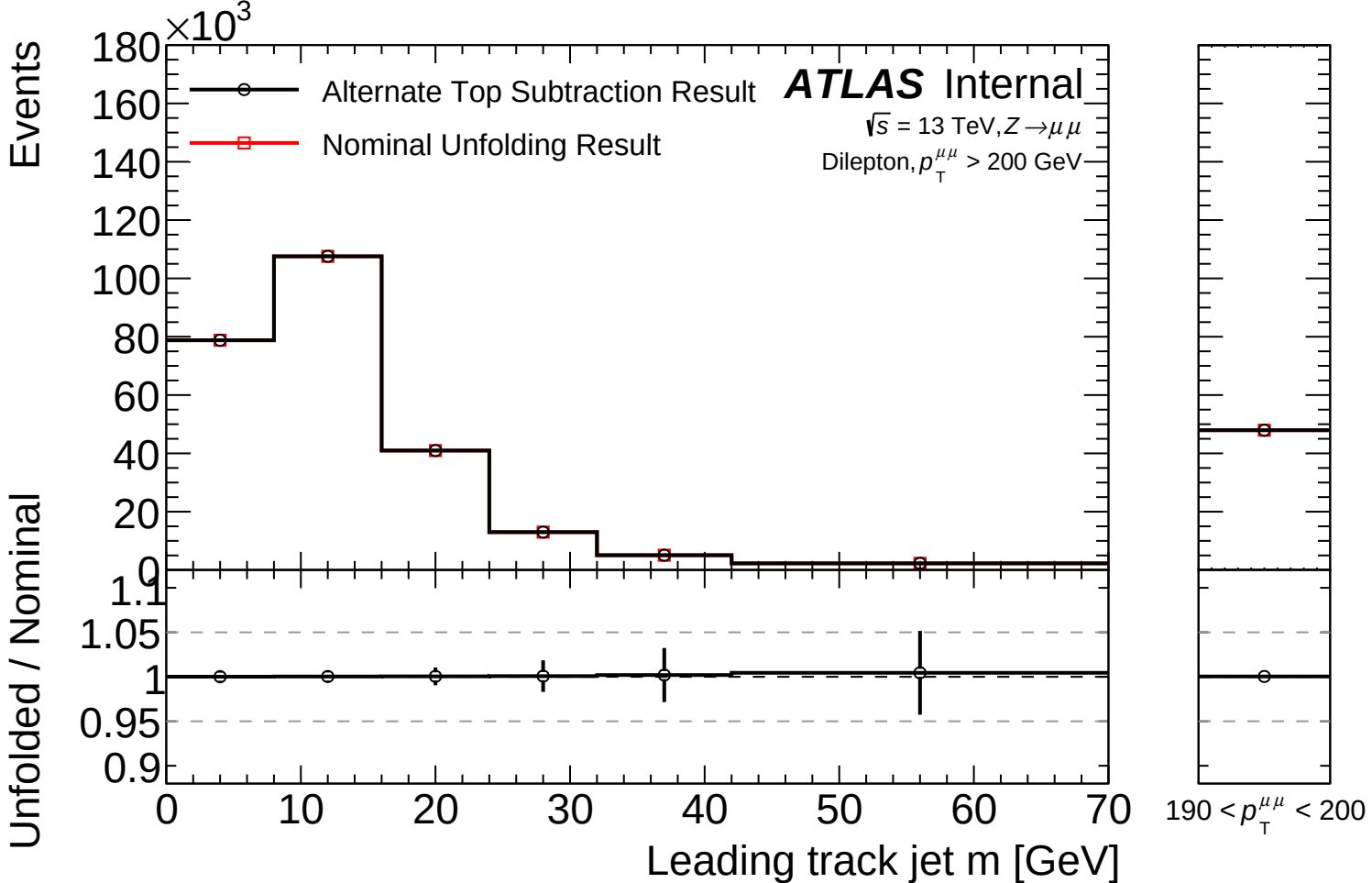
Fractional uncertainty



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet ϕ 







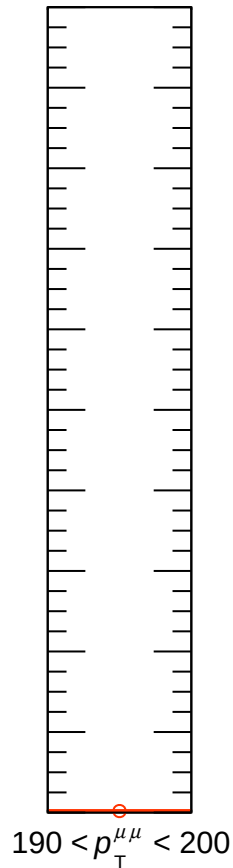
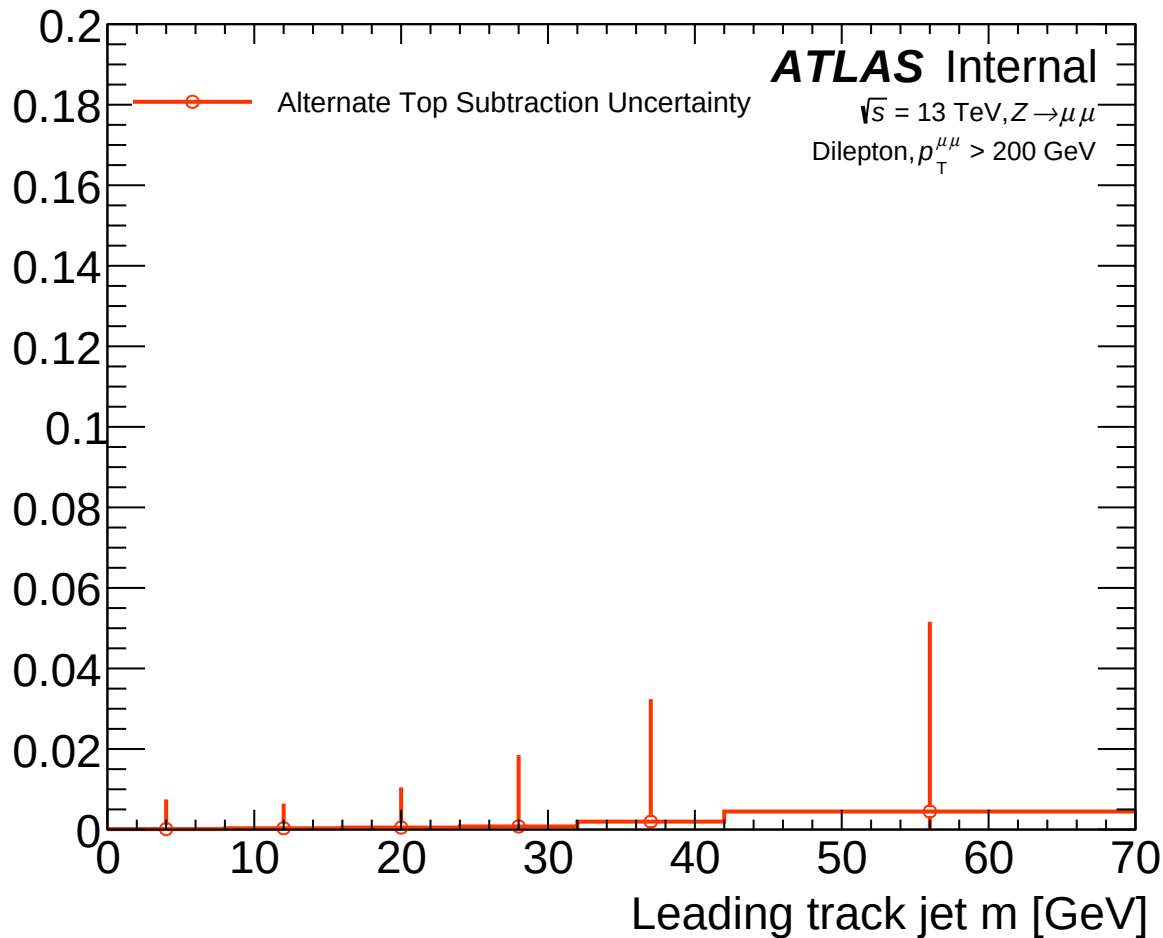
Fractional systematic impact

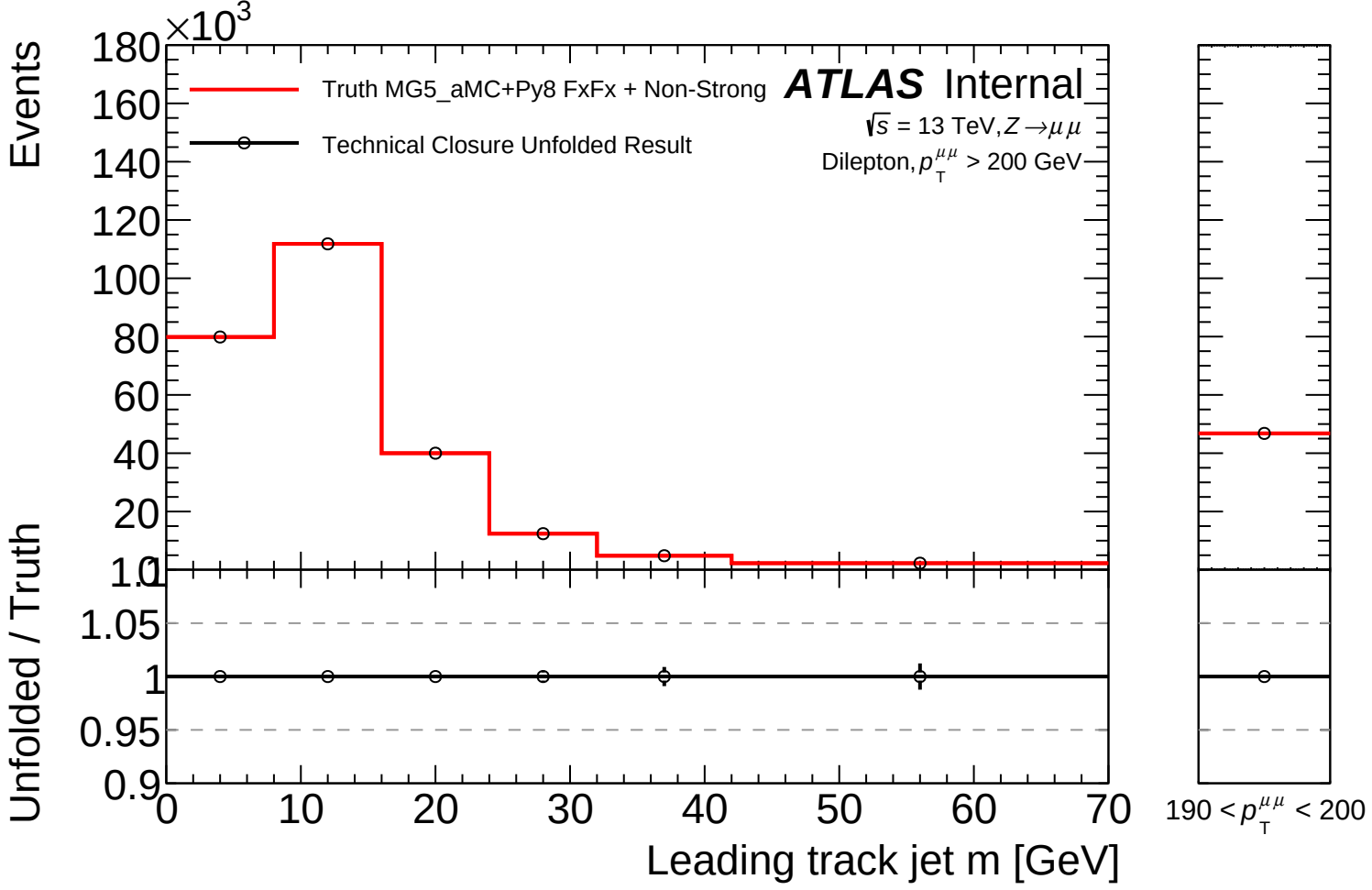
ATLAS Internal

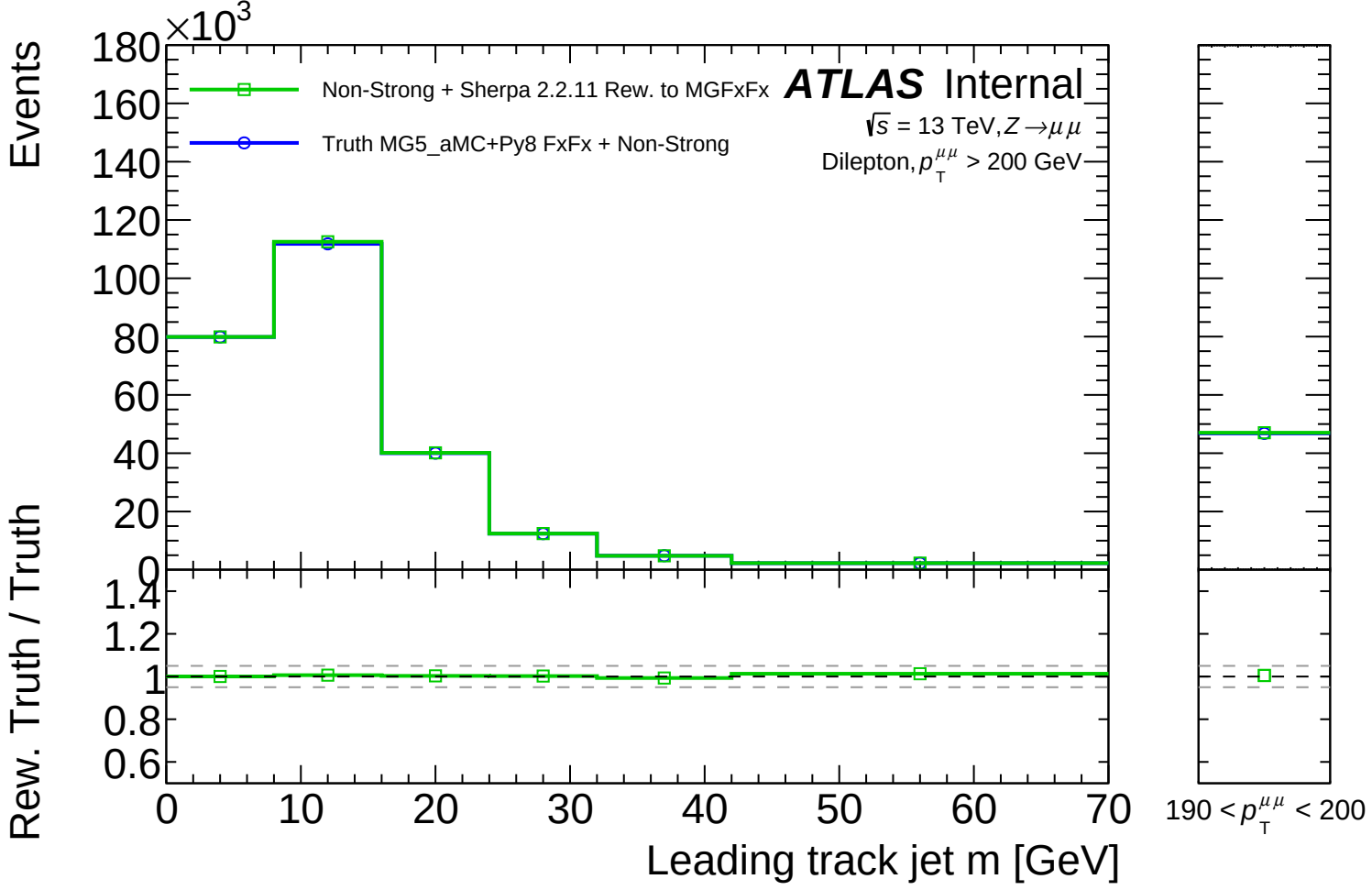
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

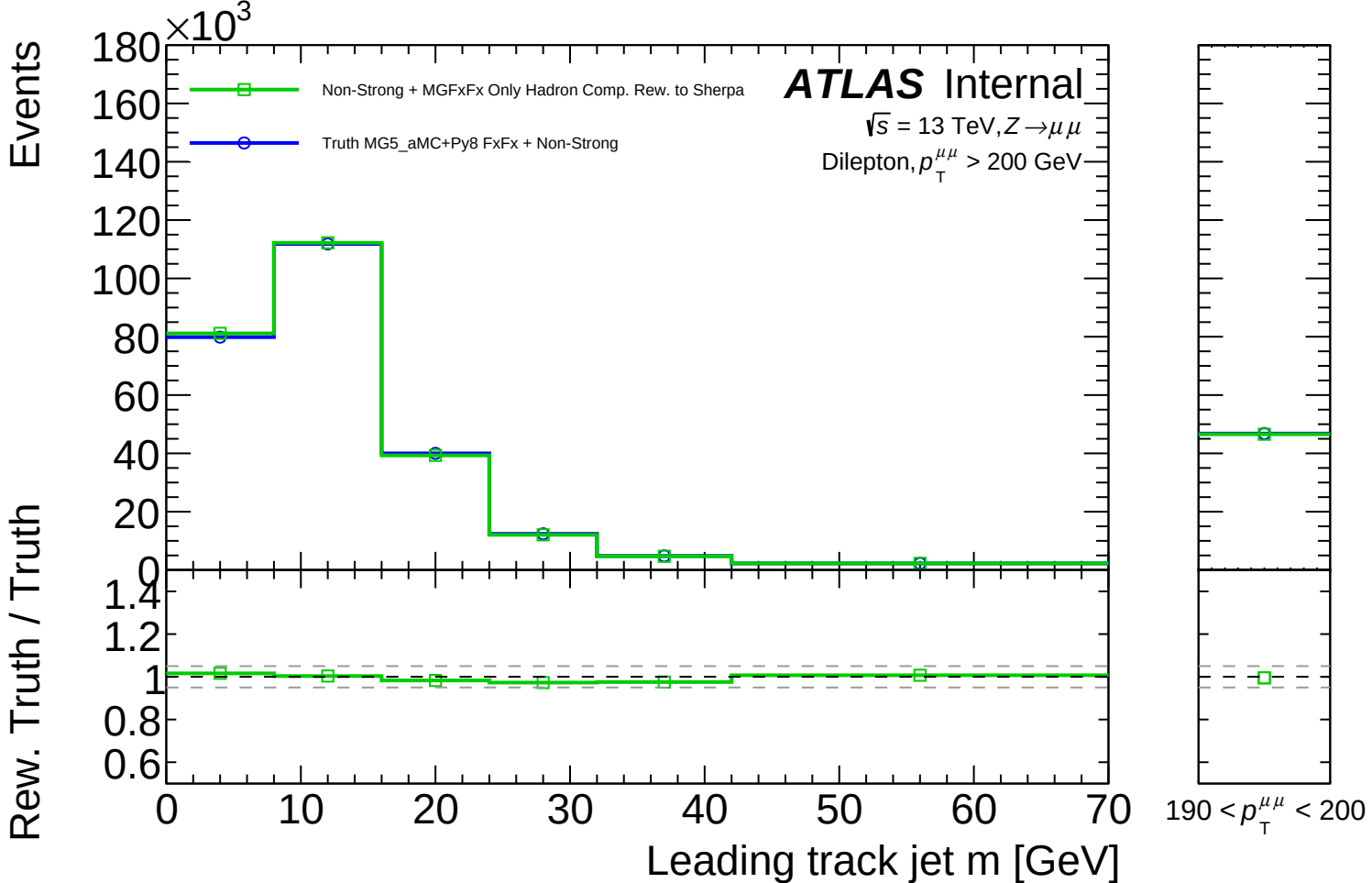
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

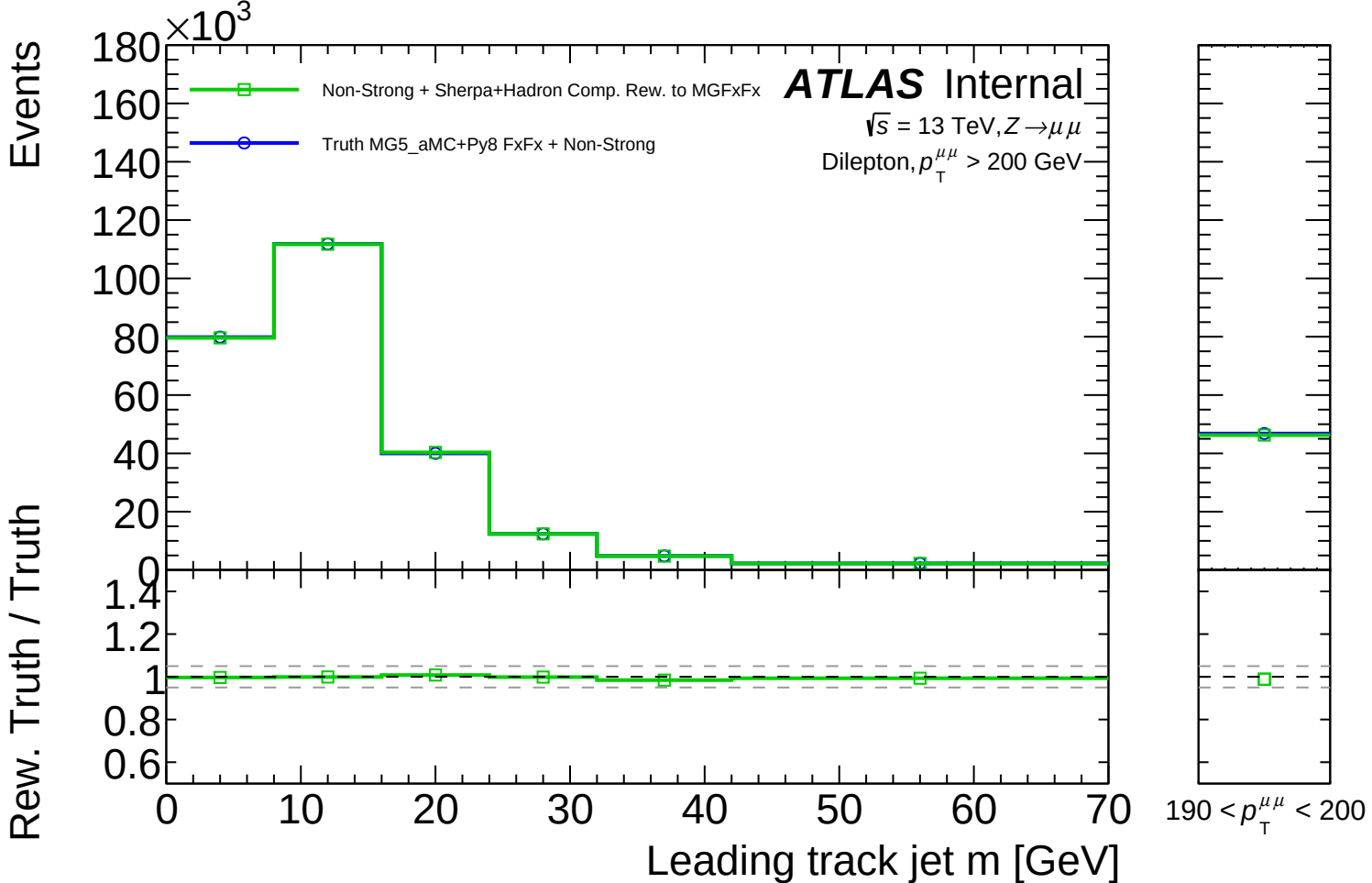
Alternate Top Subtraction Uncertainty

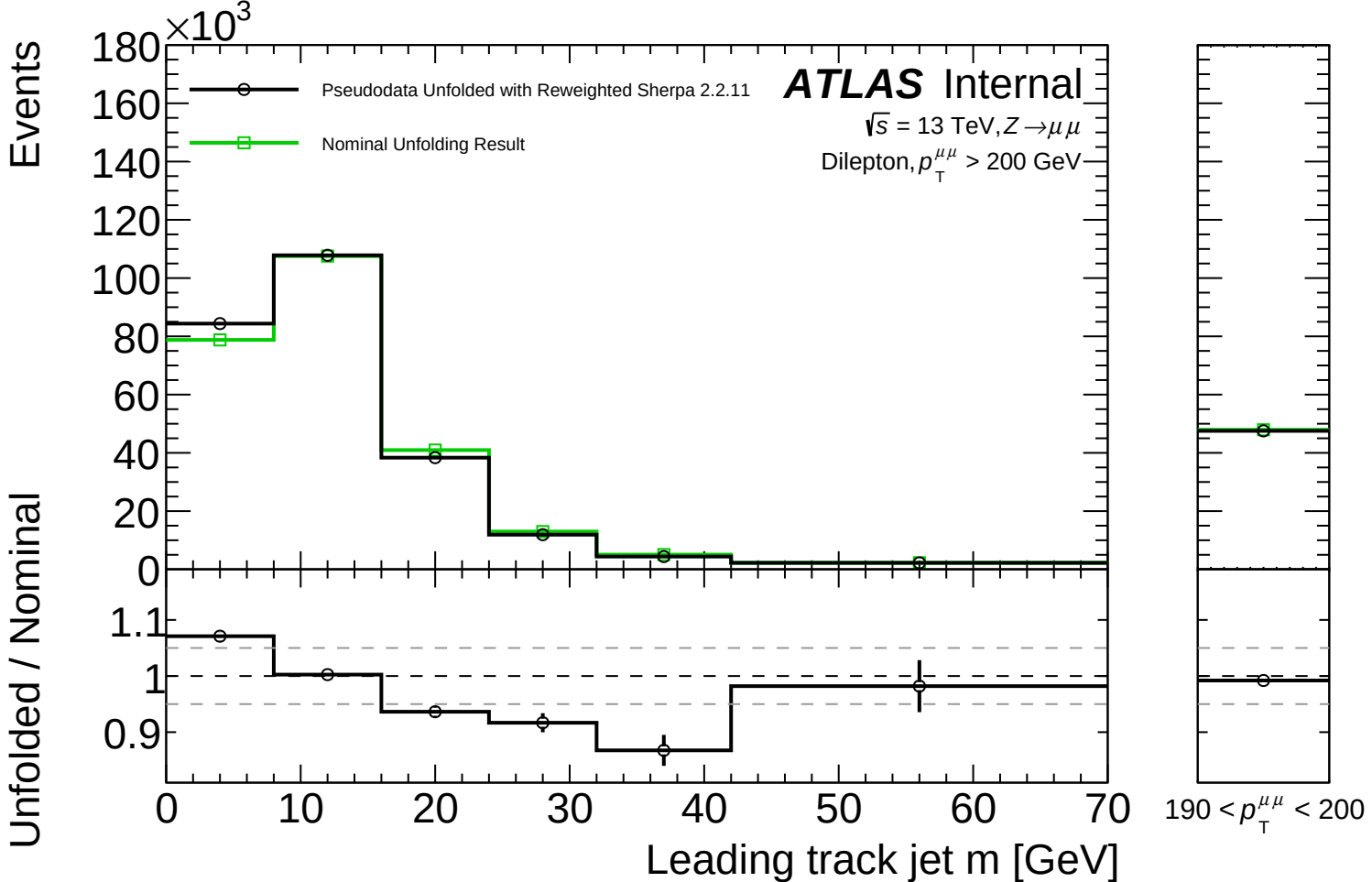


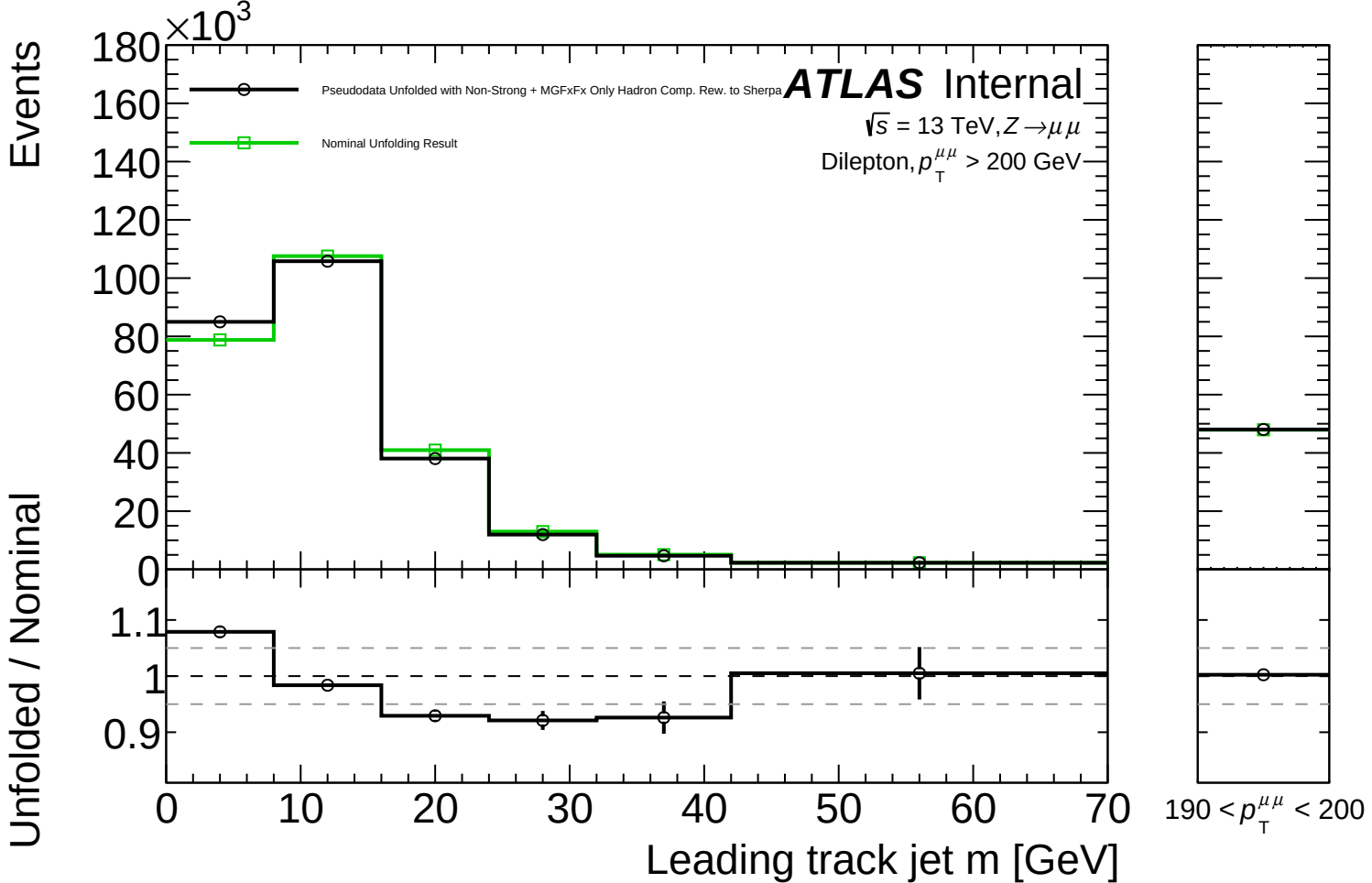


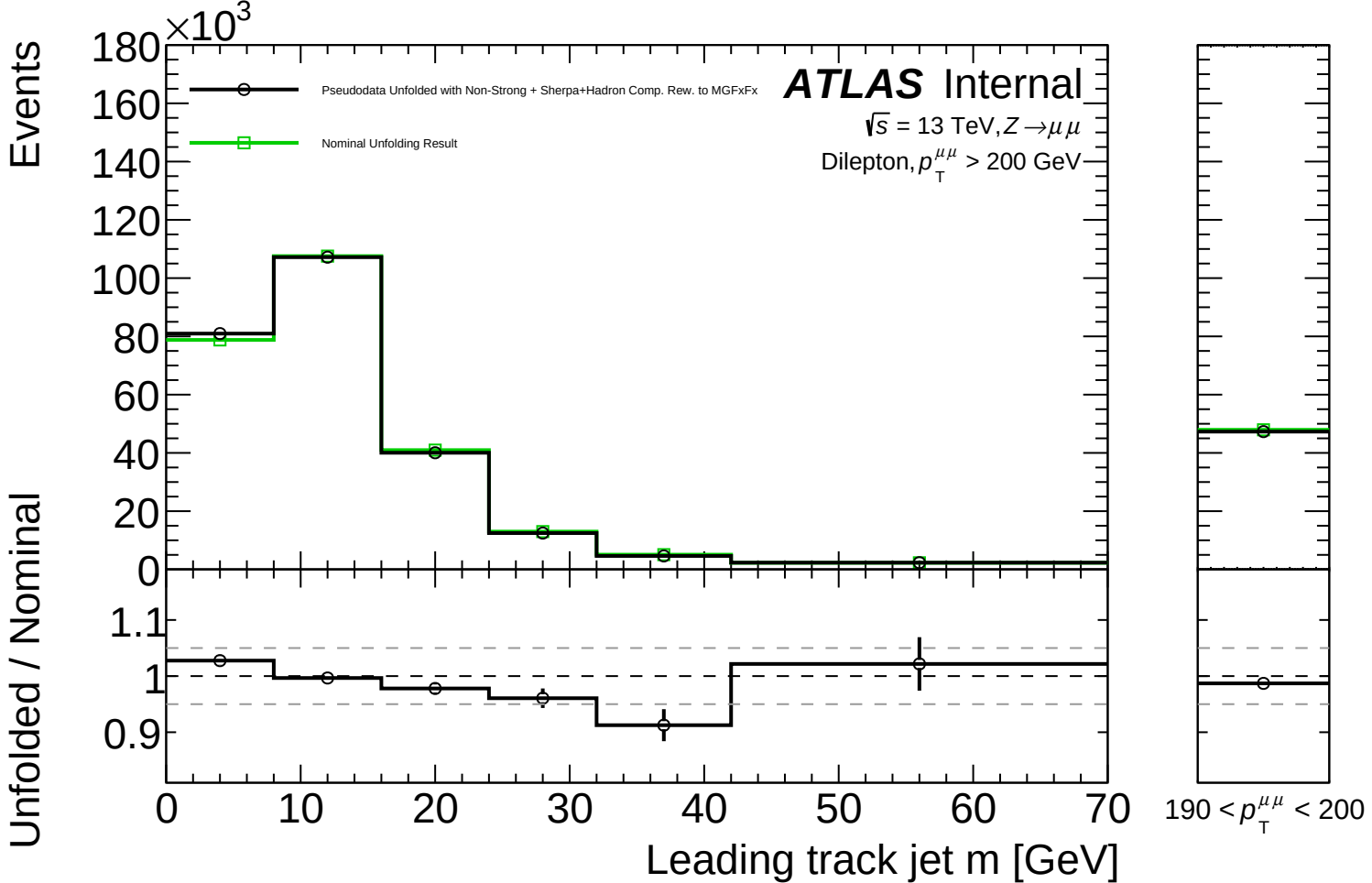


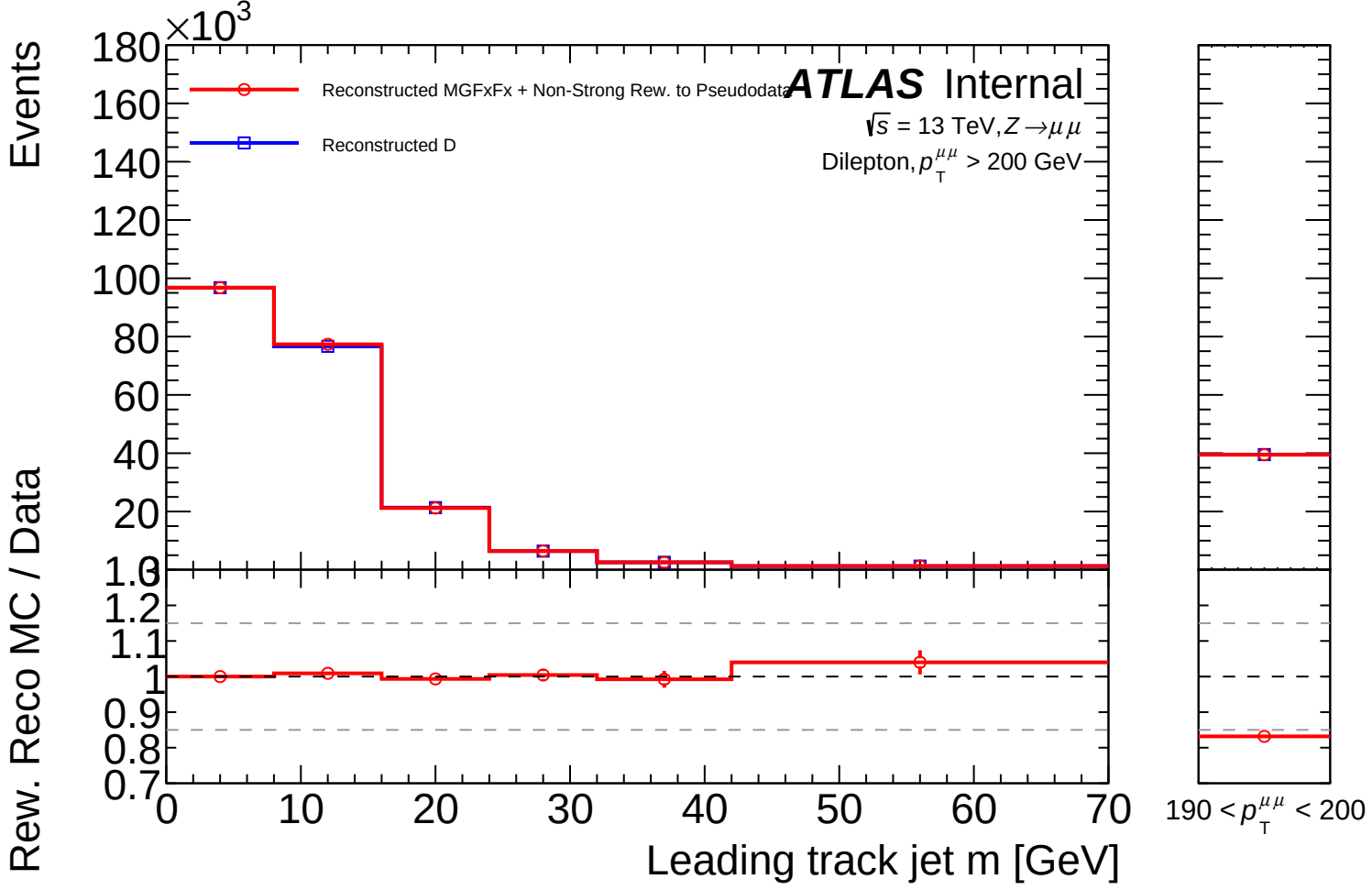


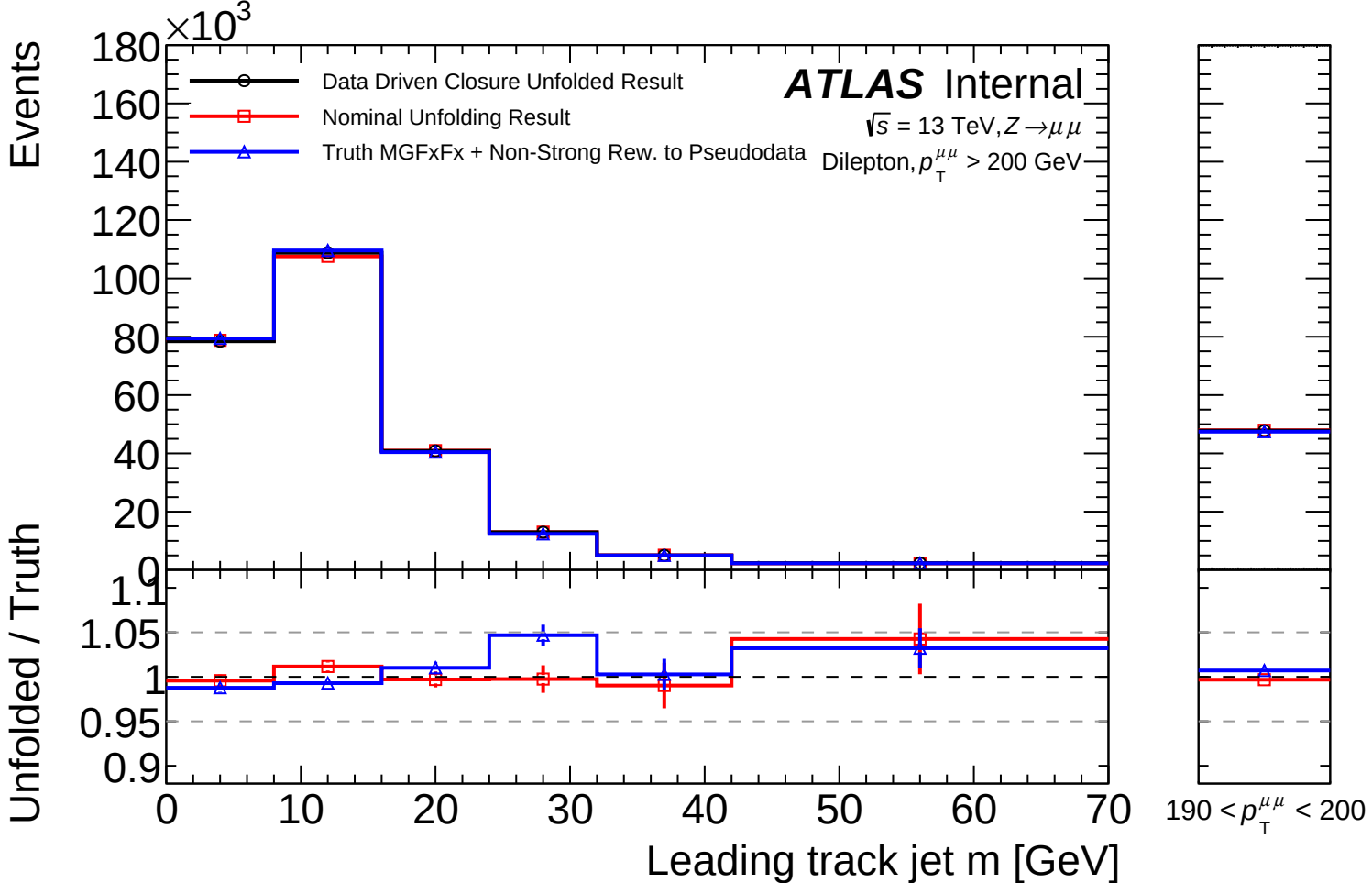




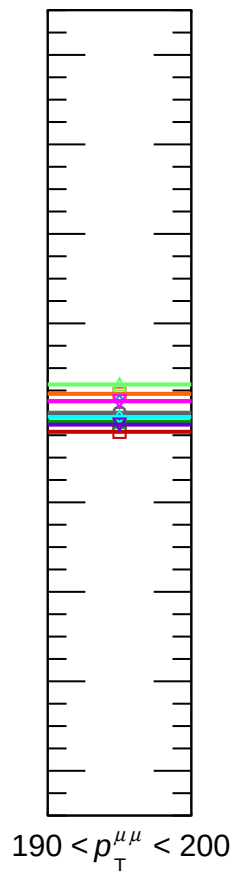
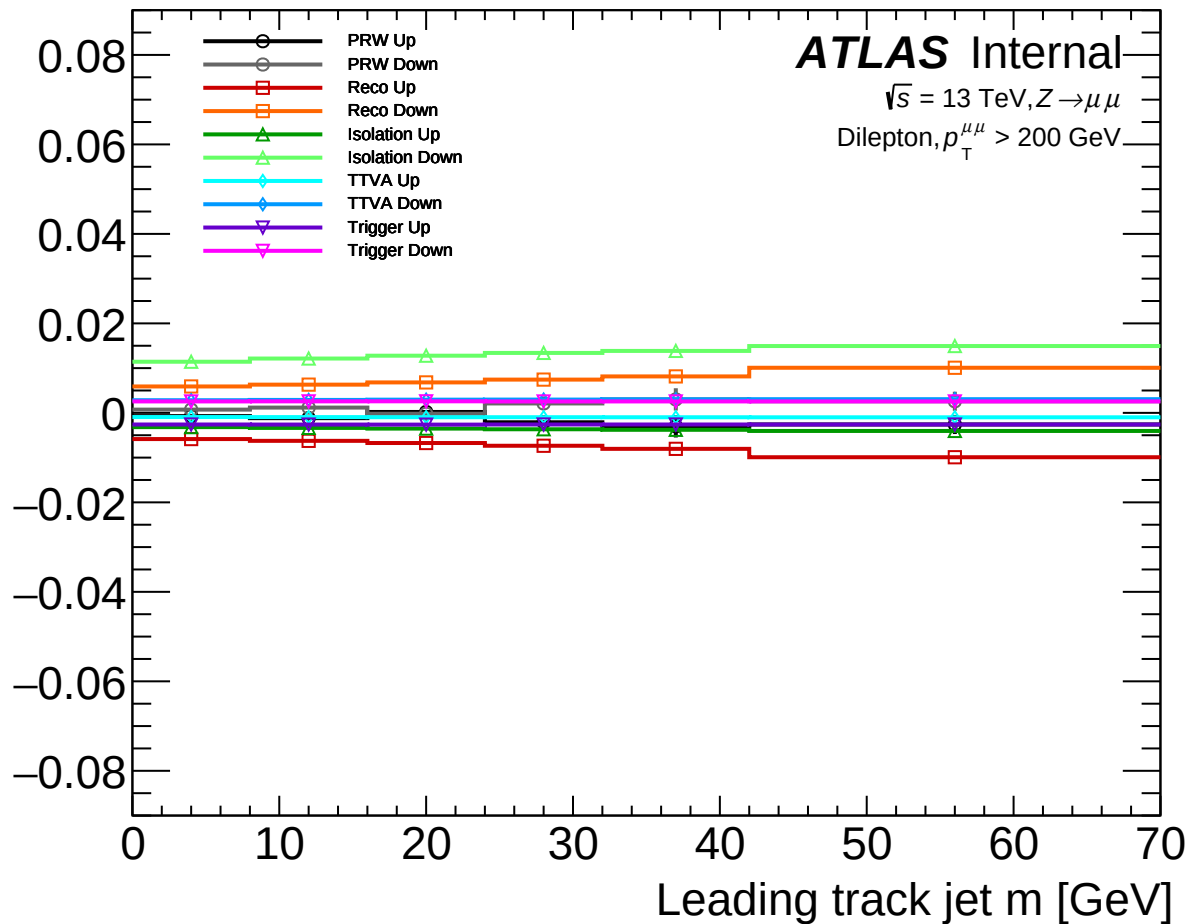




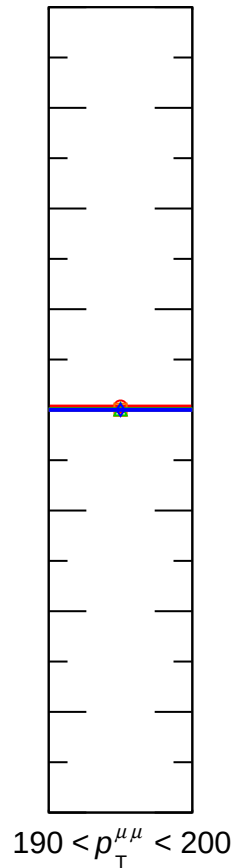
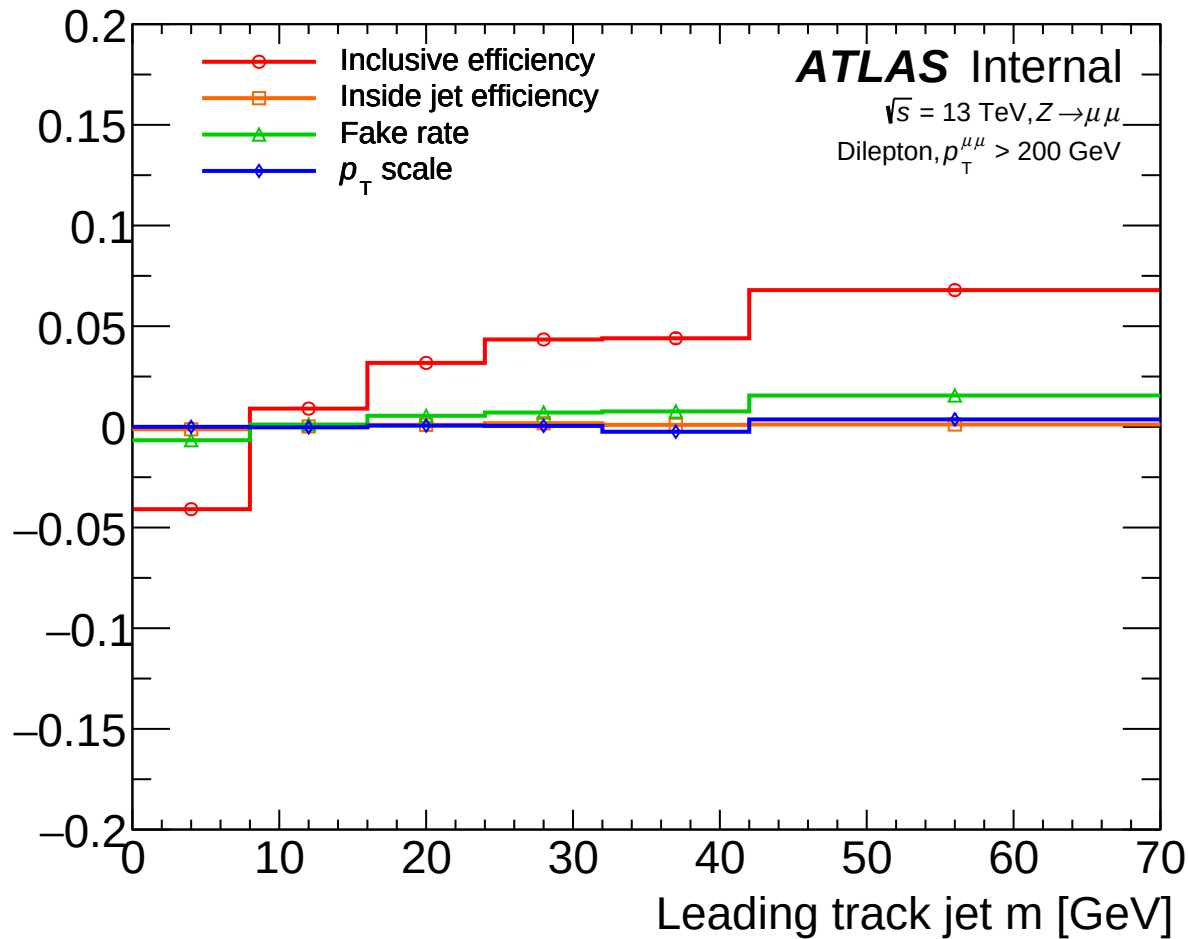




Fractional systematic impact



Fractional systematic impact



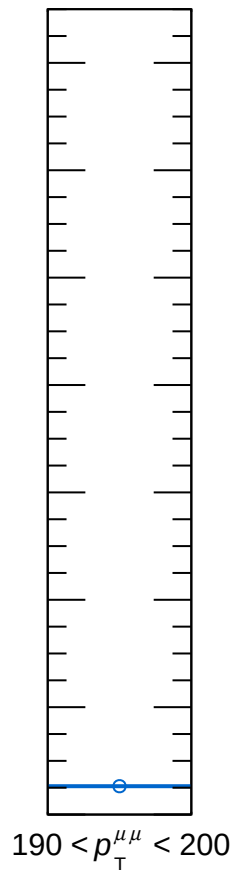
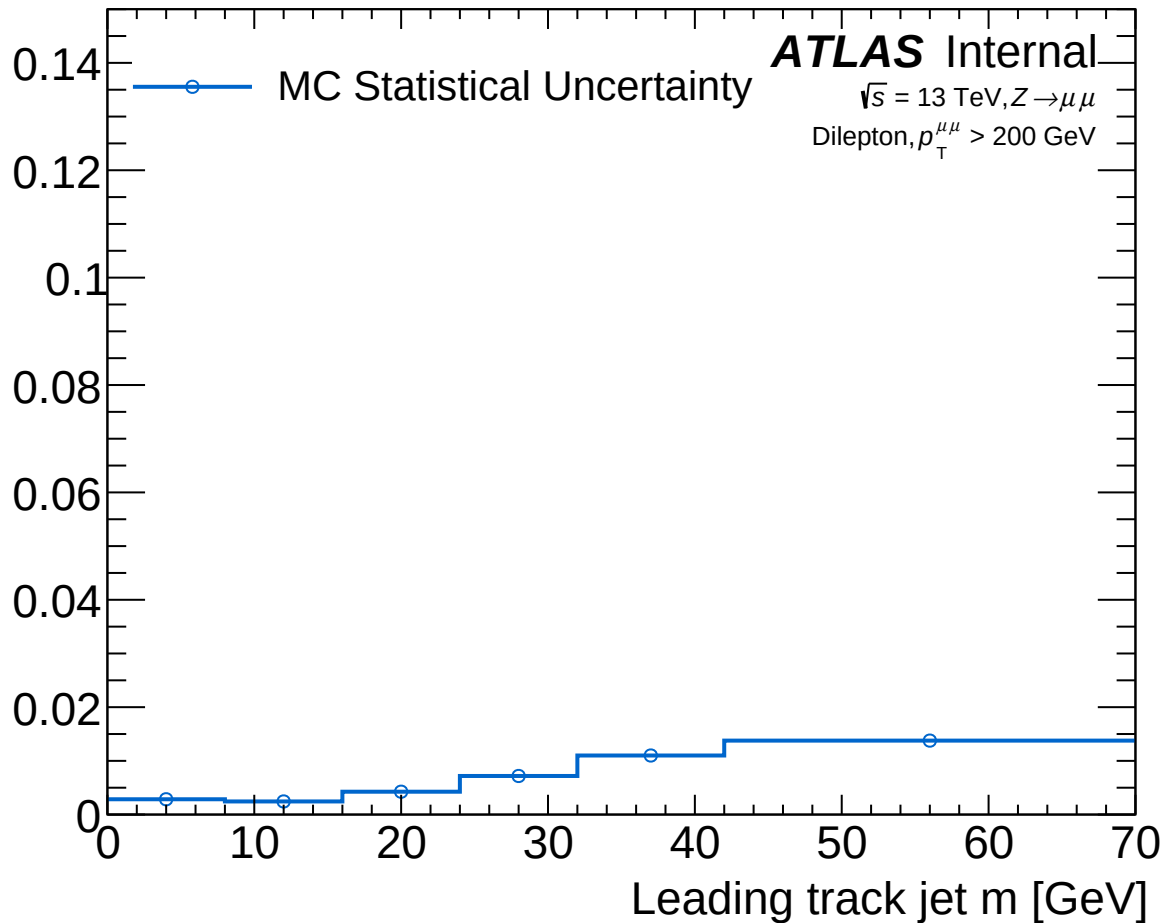
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



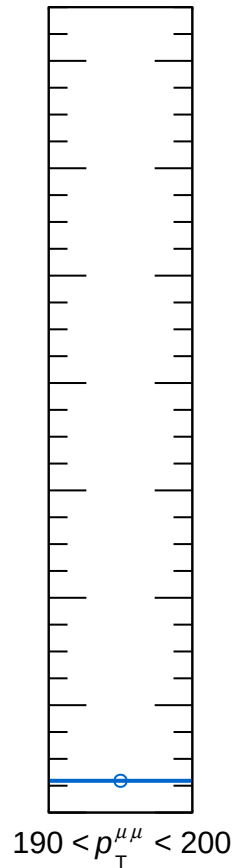
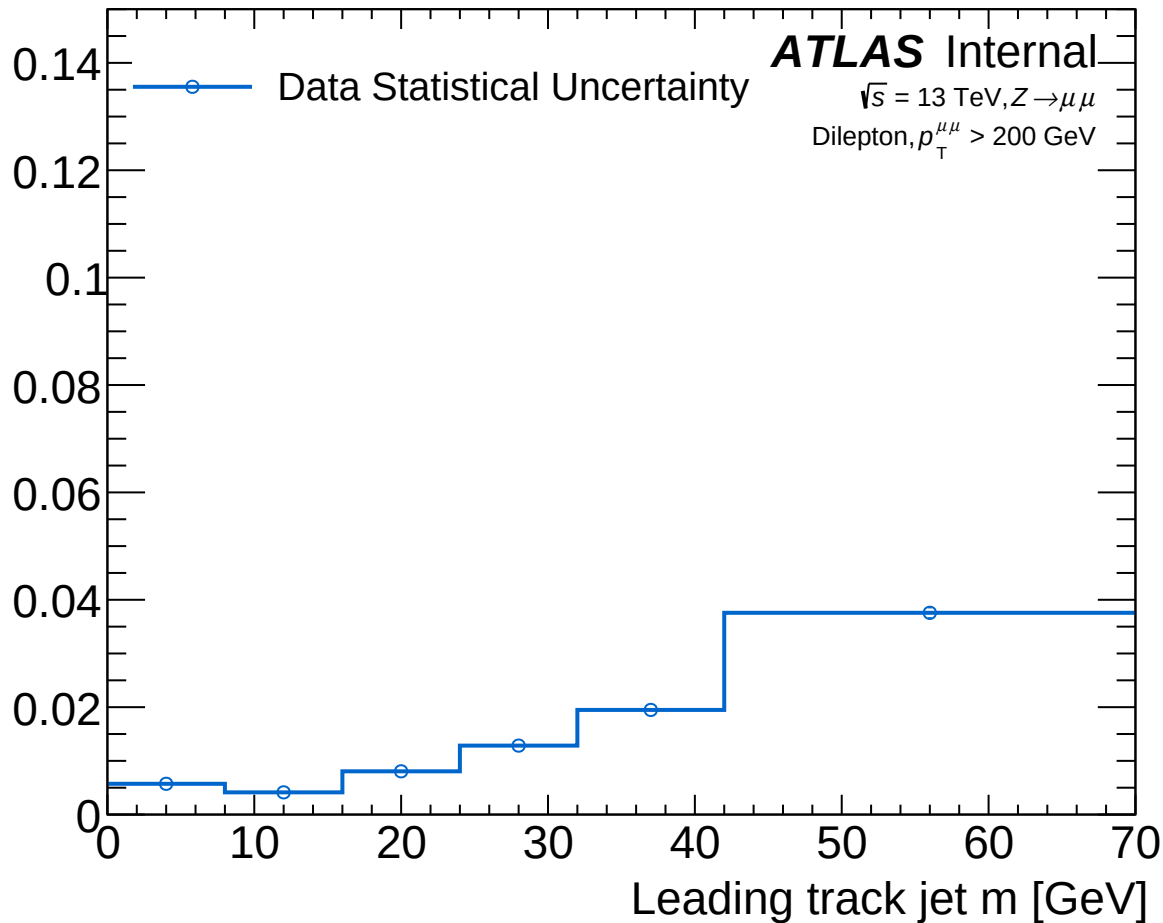
Fractional systematic impact

ATLAS Internal

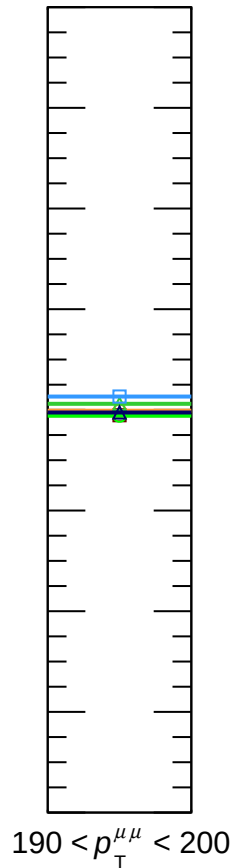
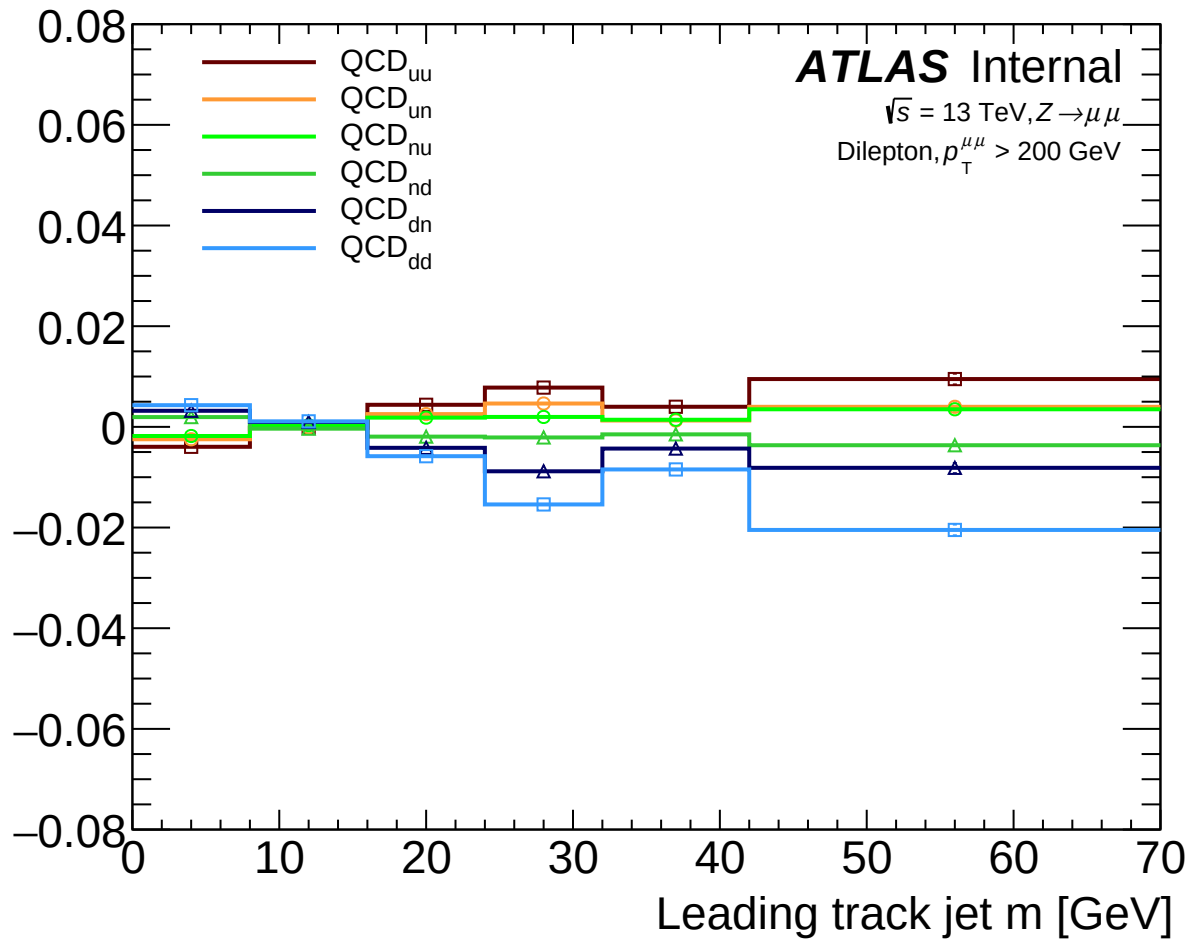
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

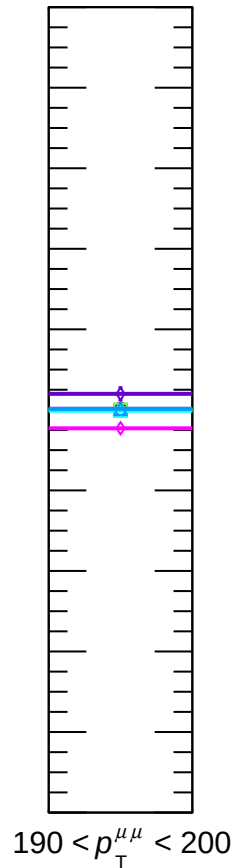
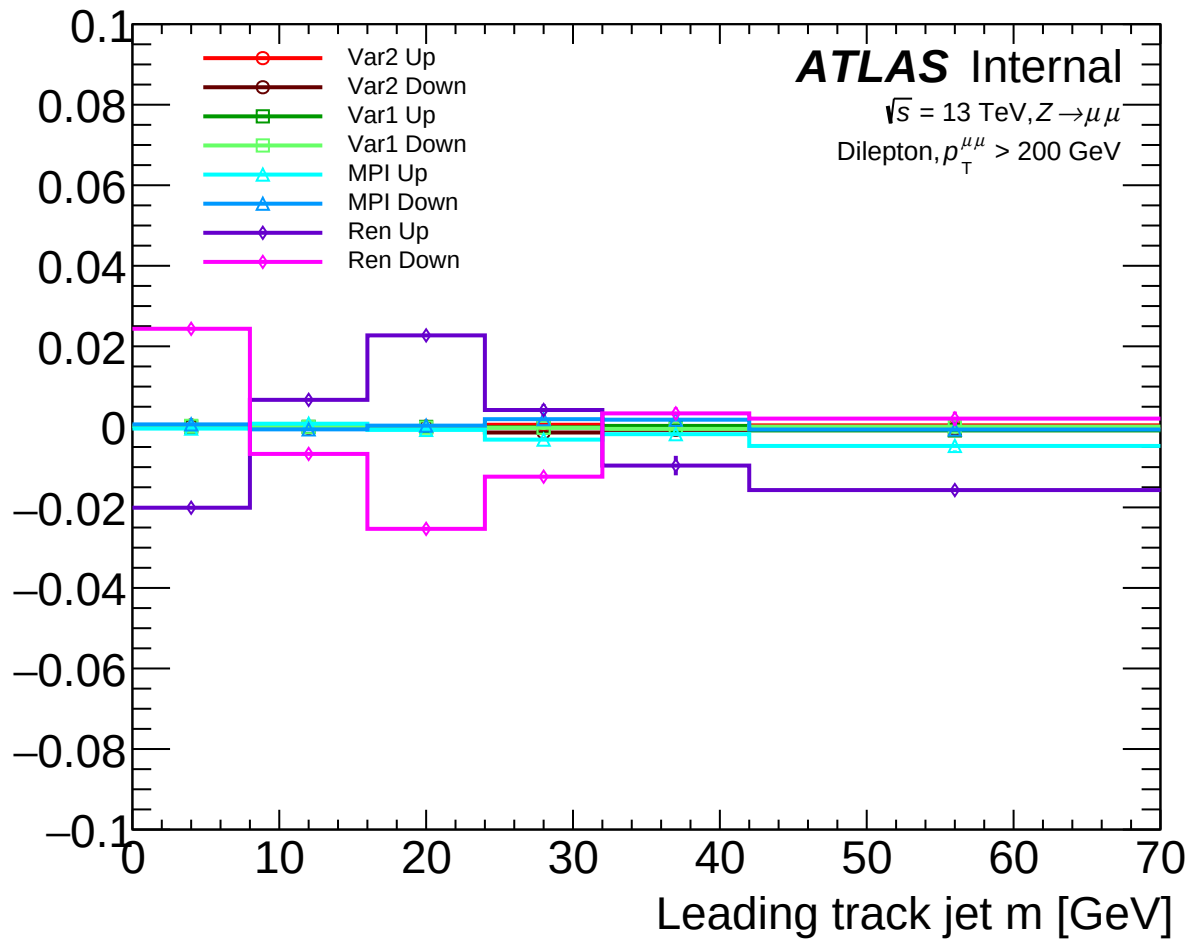
Data Statistical Uncertainty

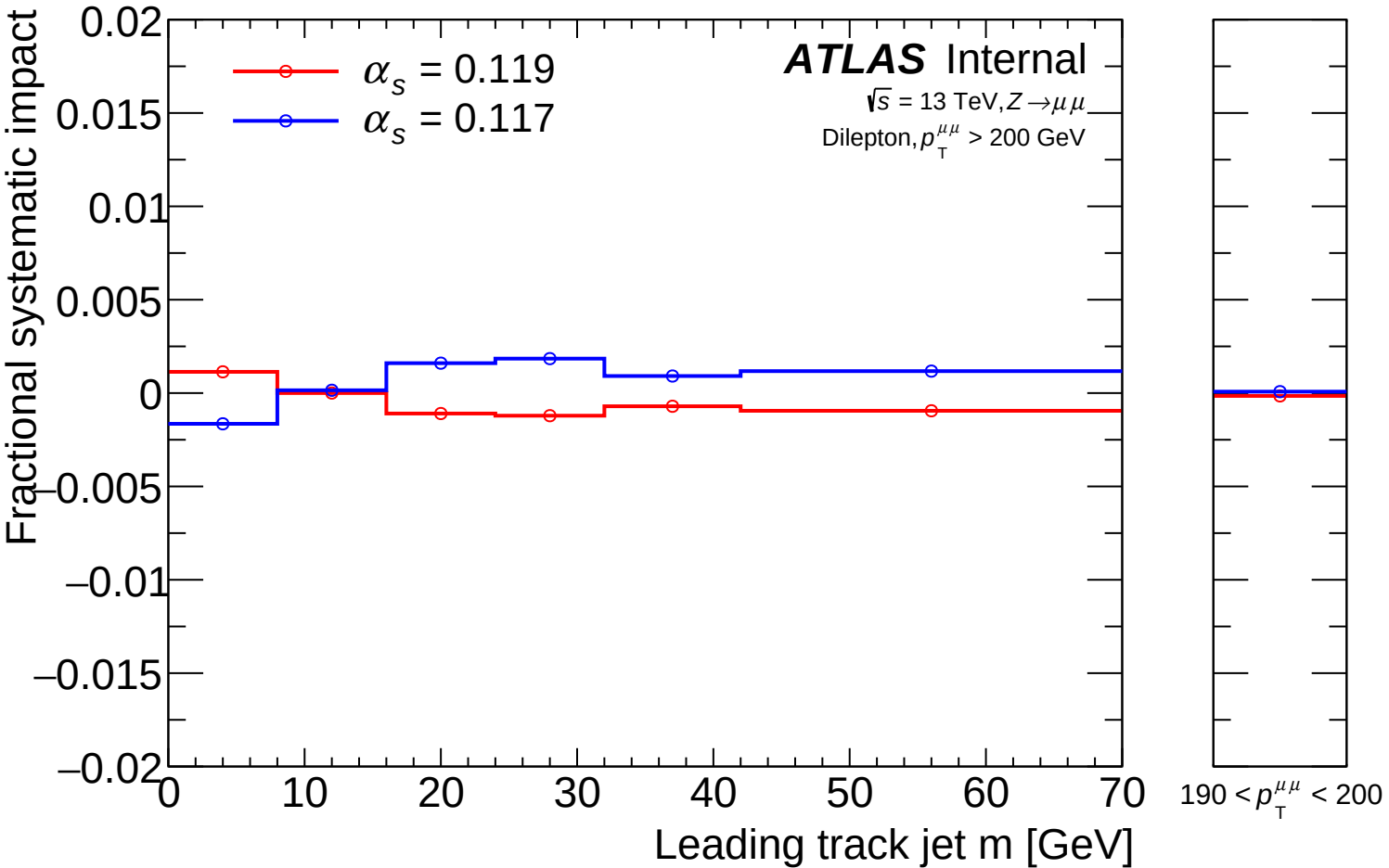


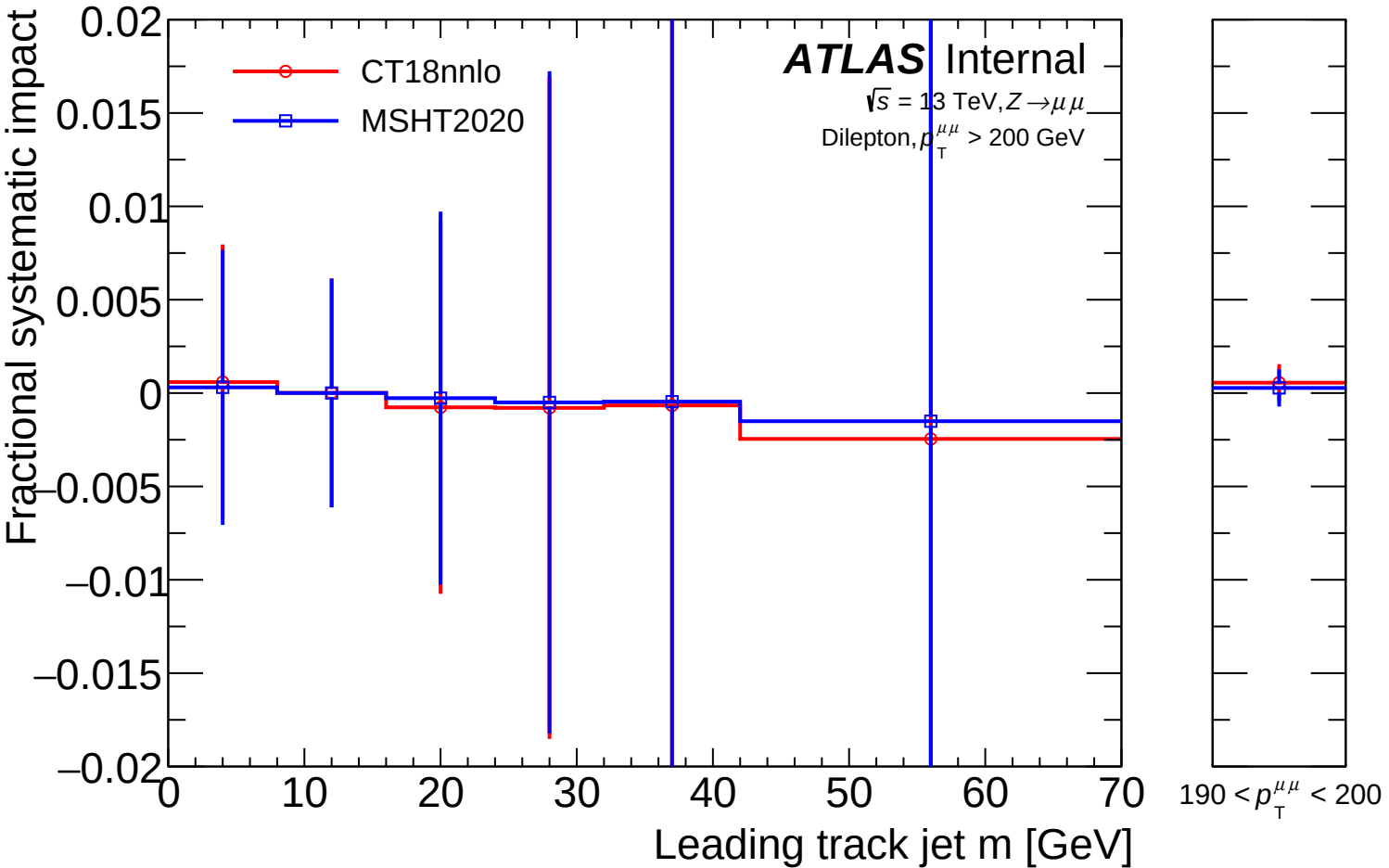
Fractional systematic impact



Fractional systematic impact







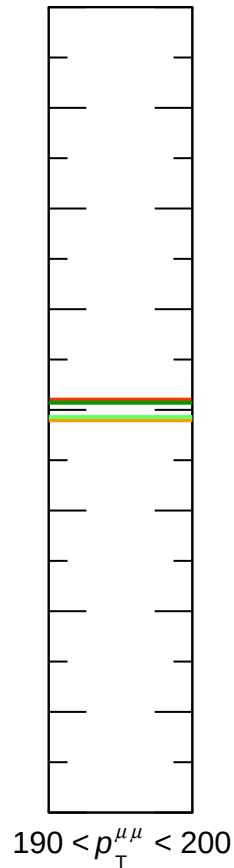
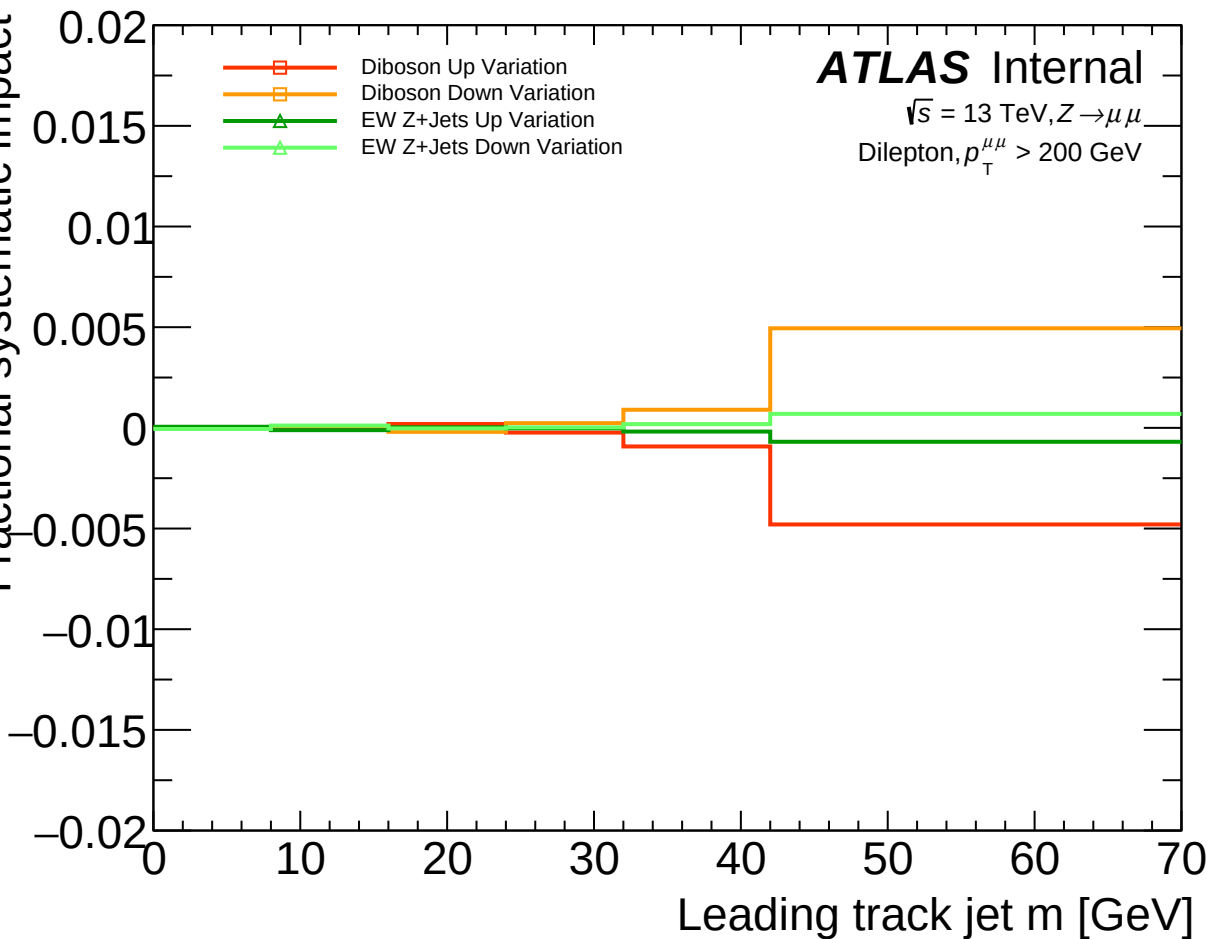
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

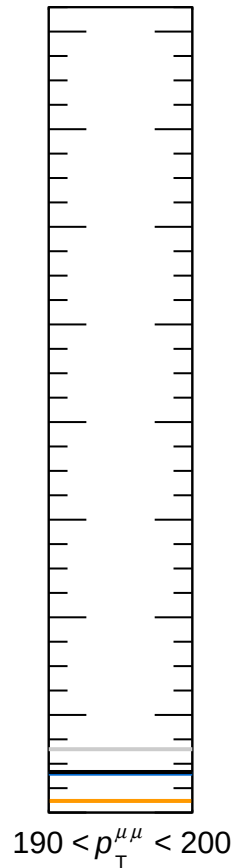
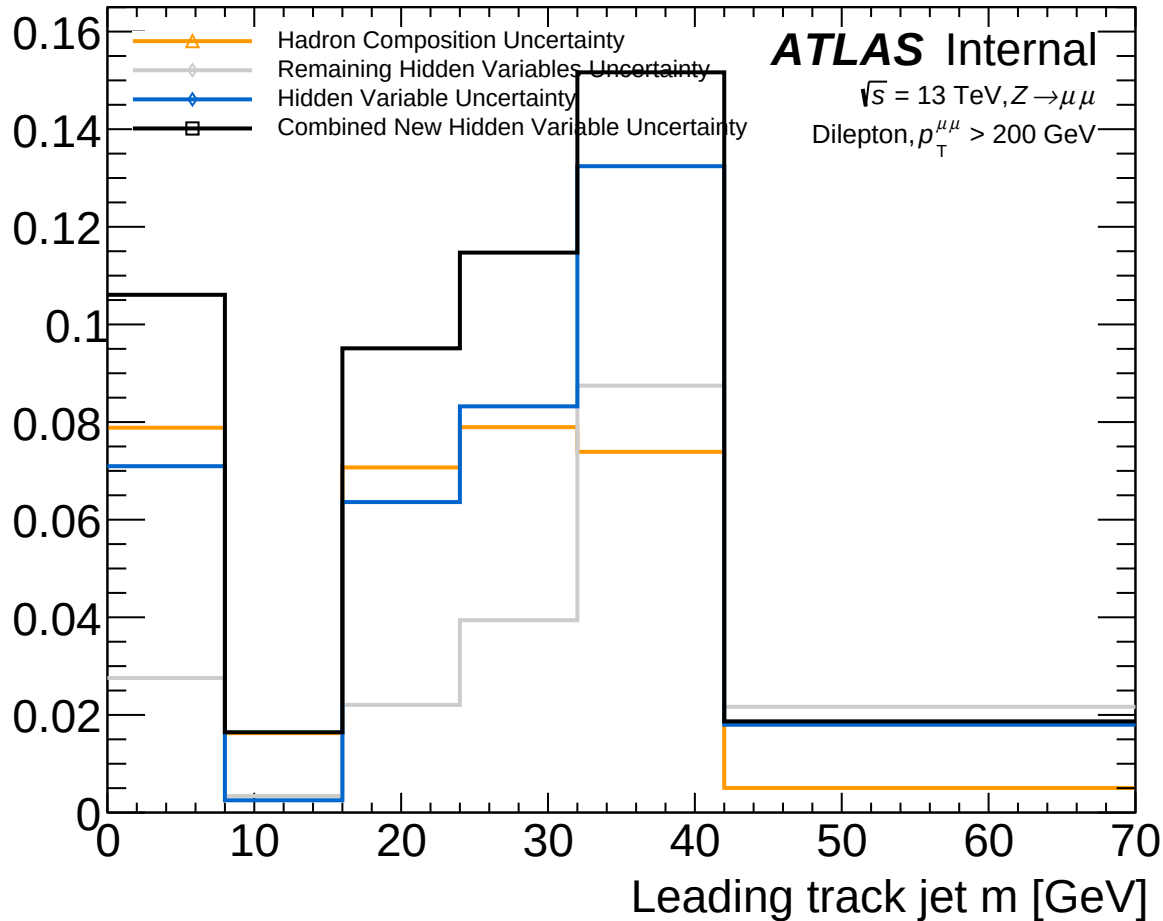
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

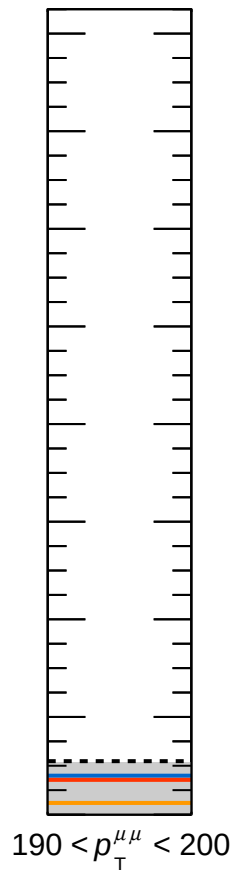
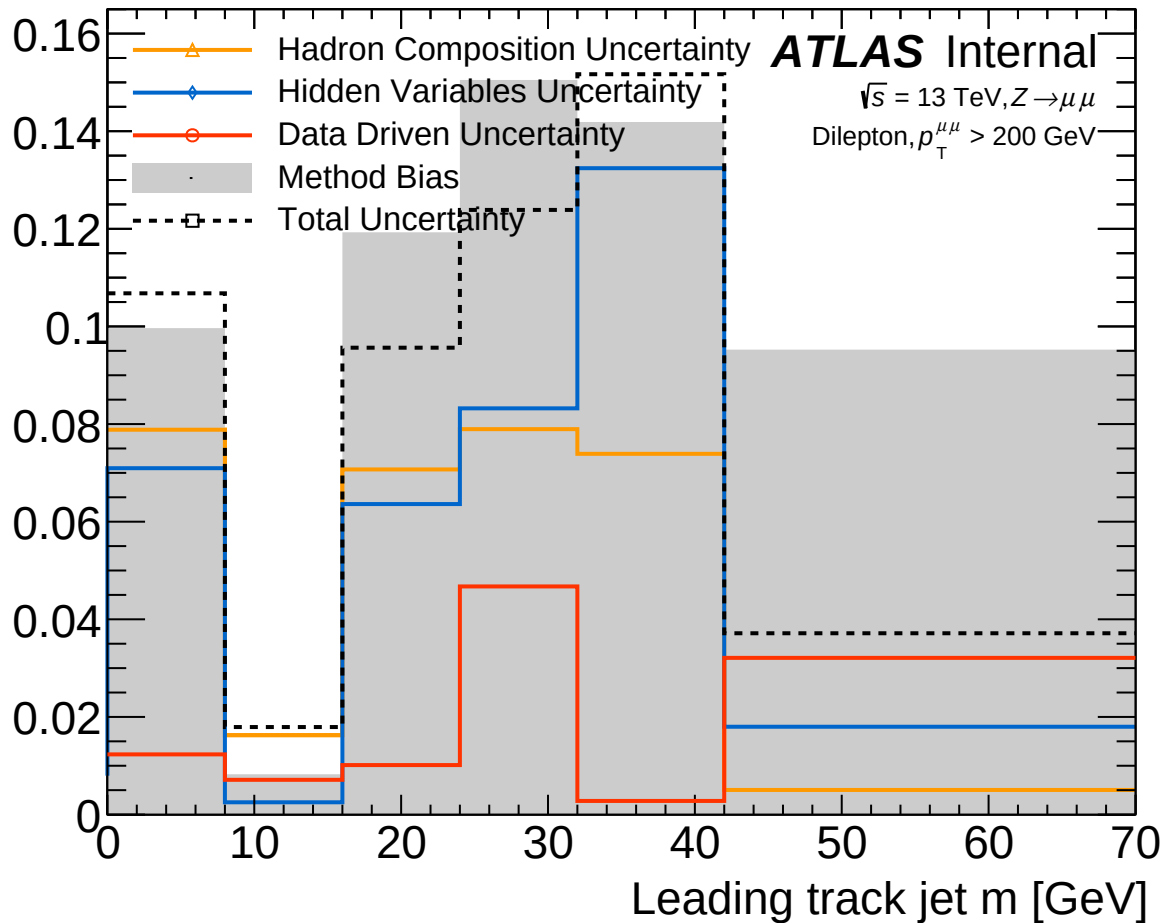


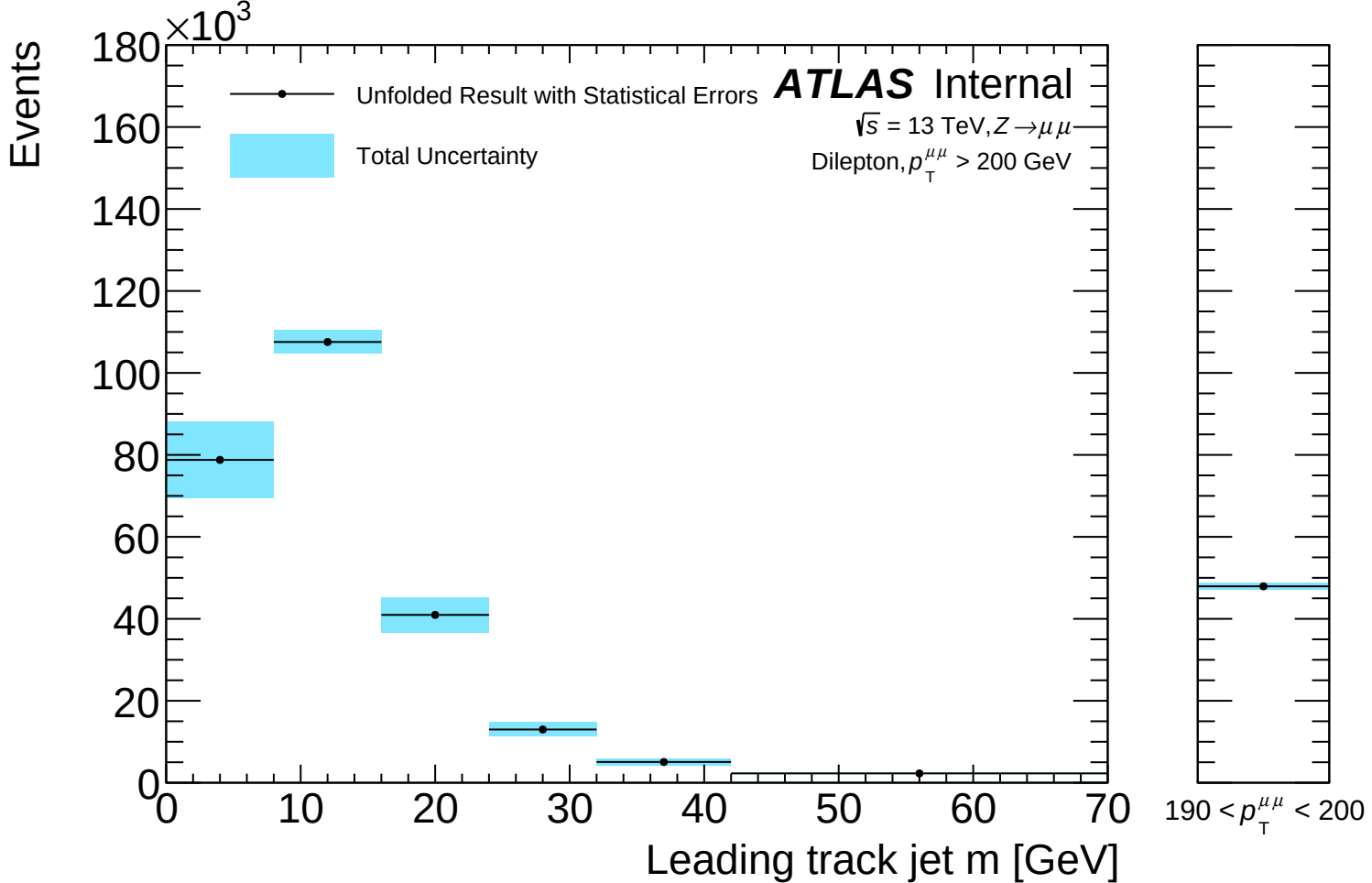
ATLAS Internal

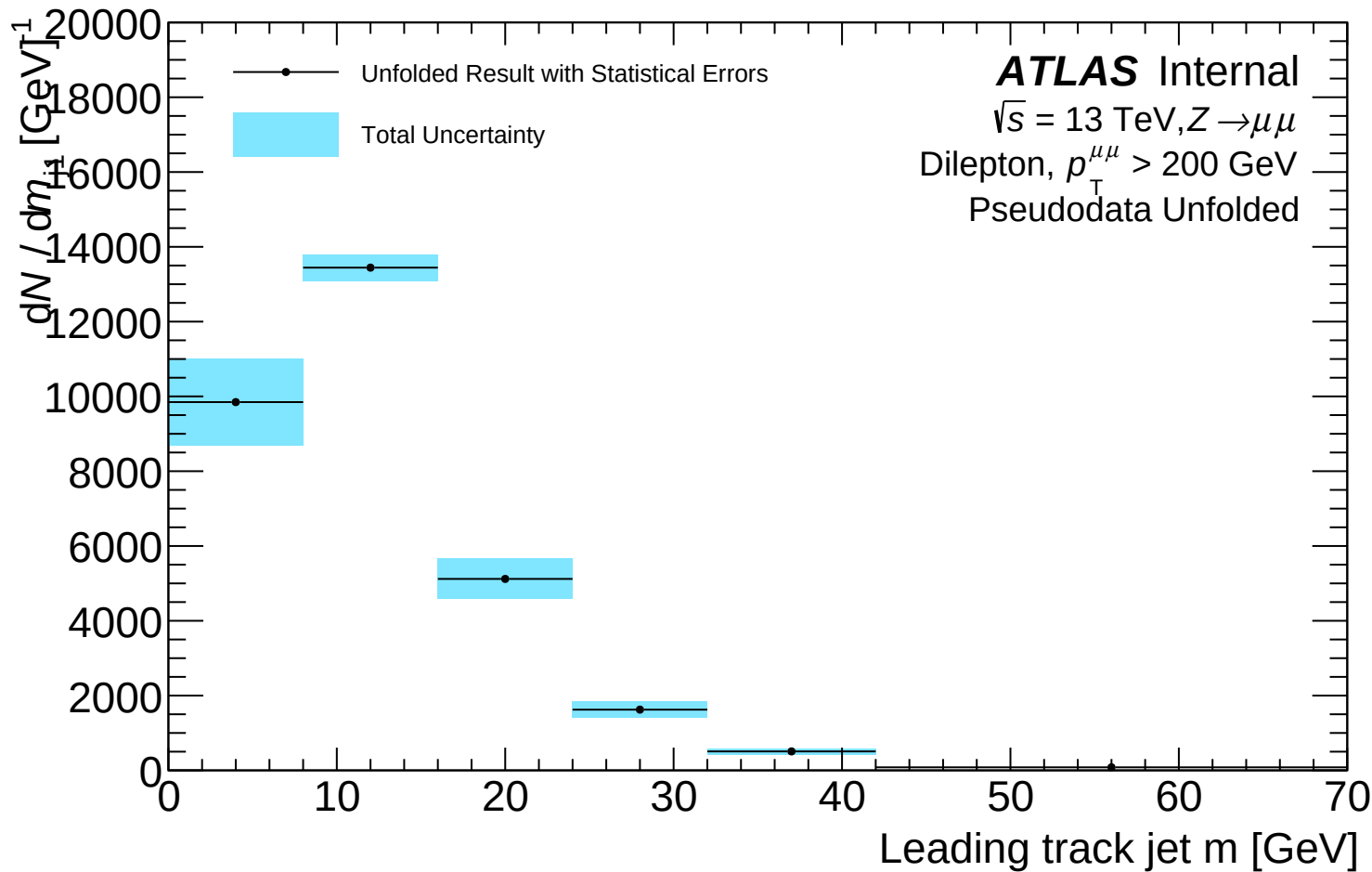
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

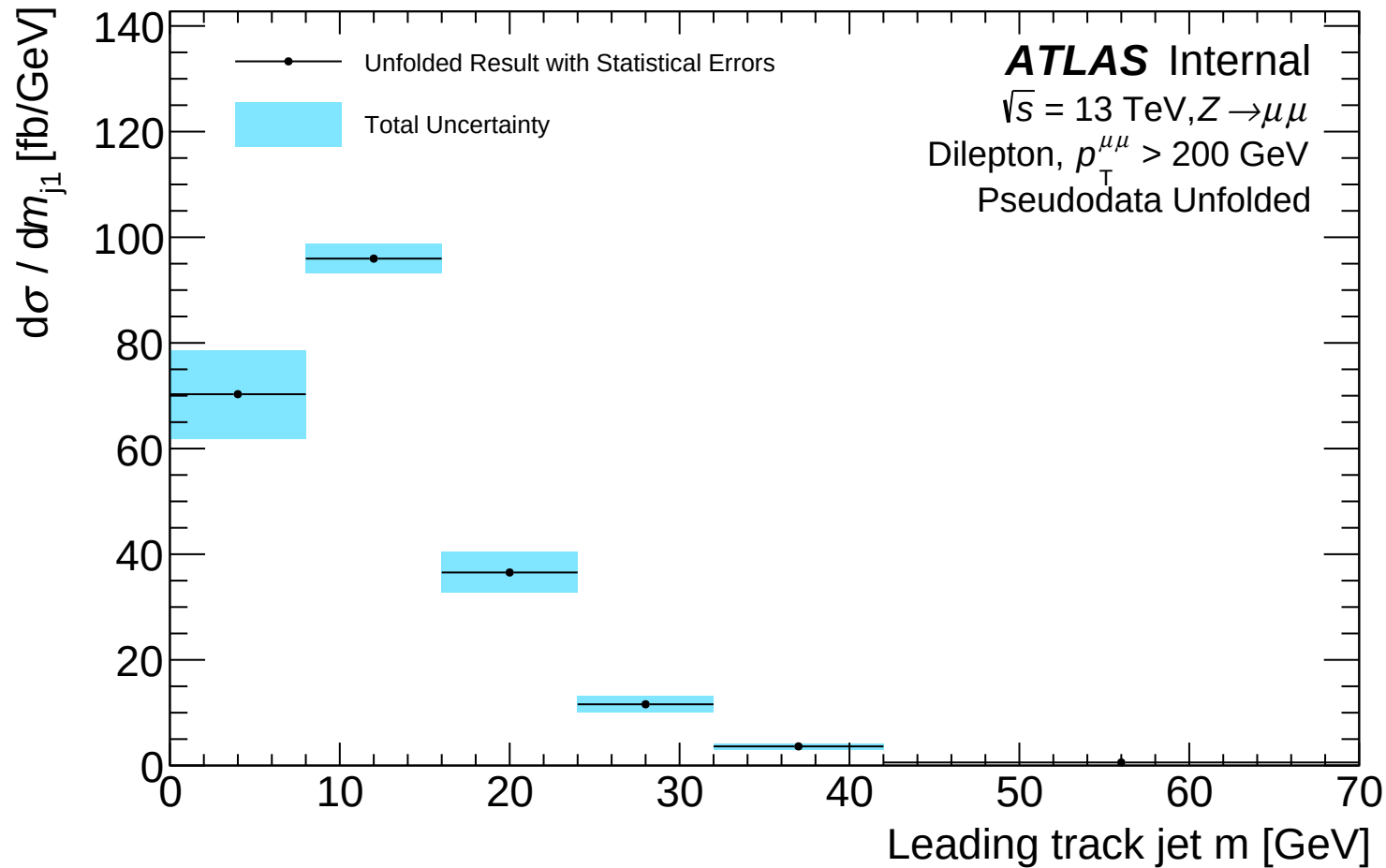
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$











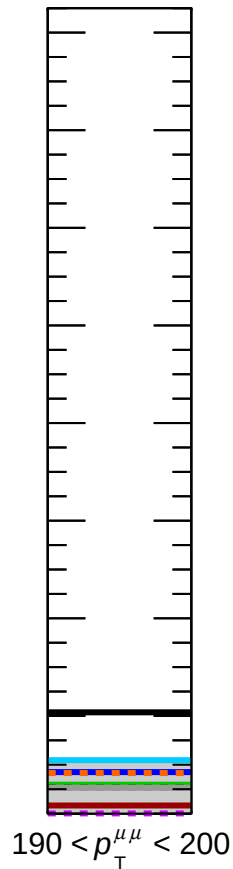
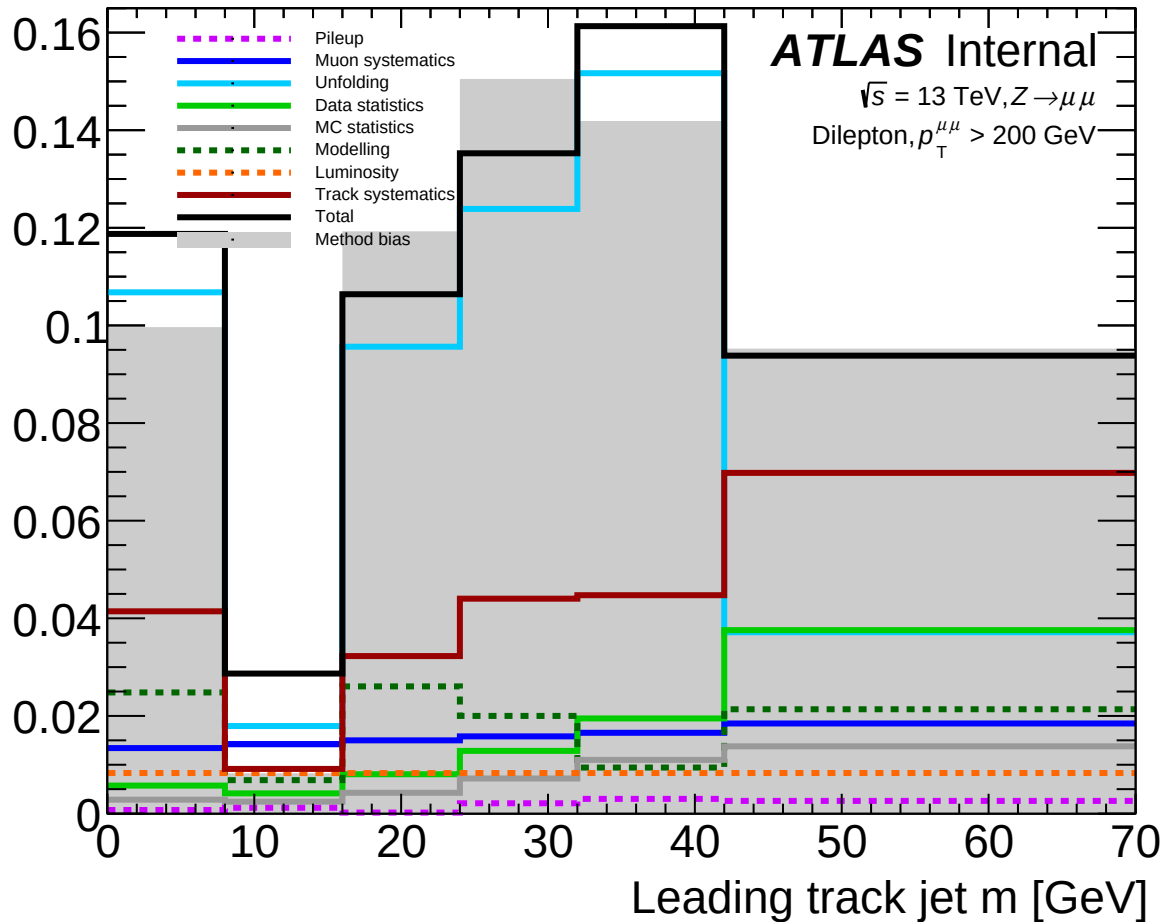
Fractional uncertainty

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

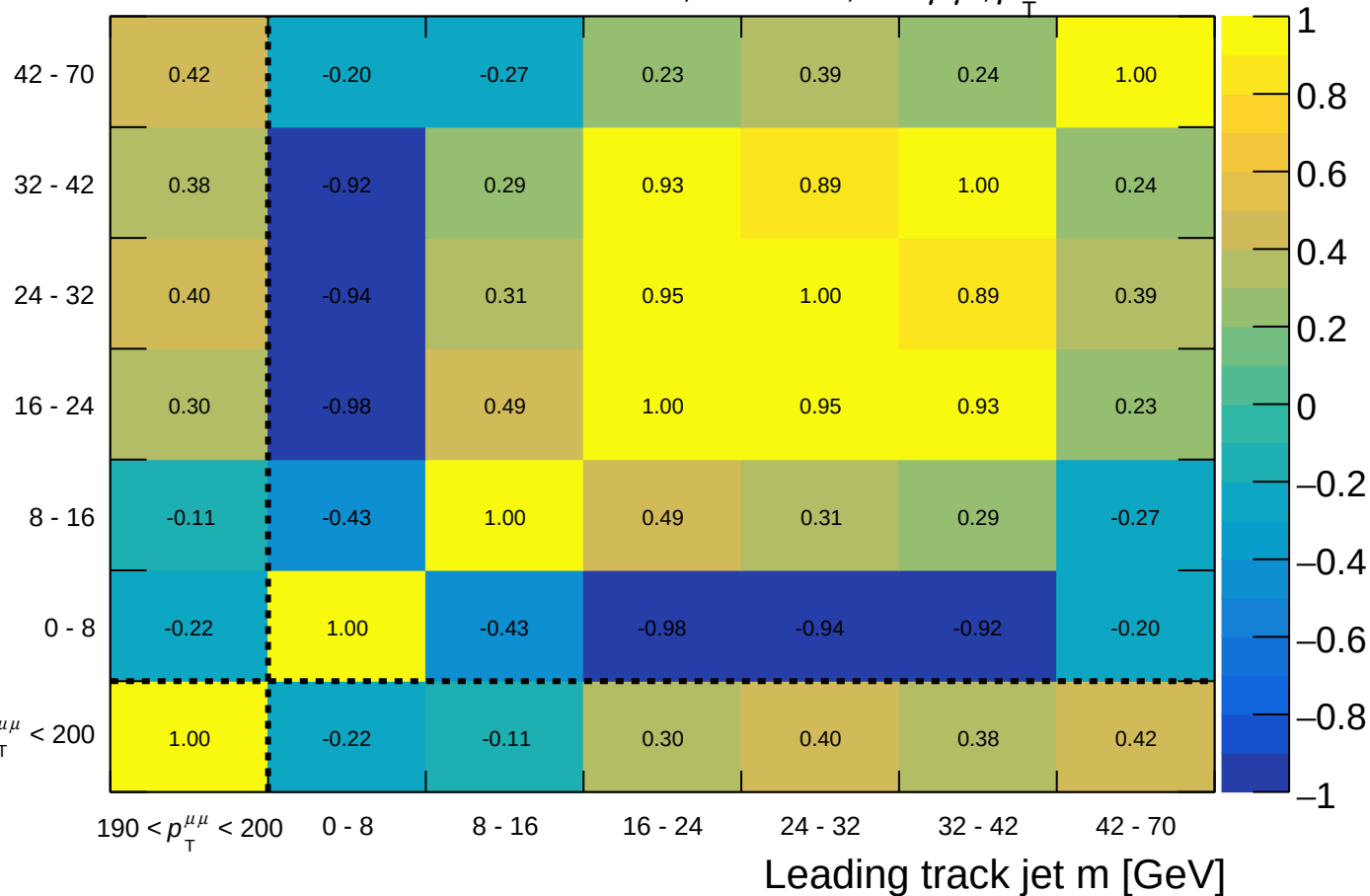
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

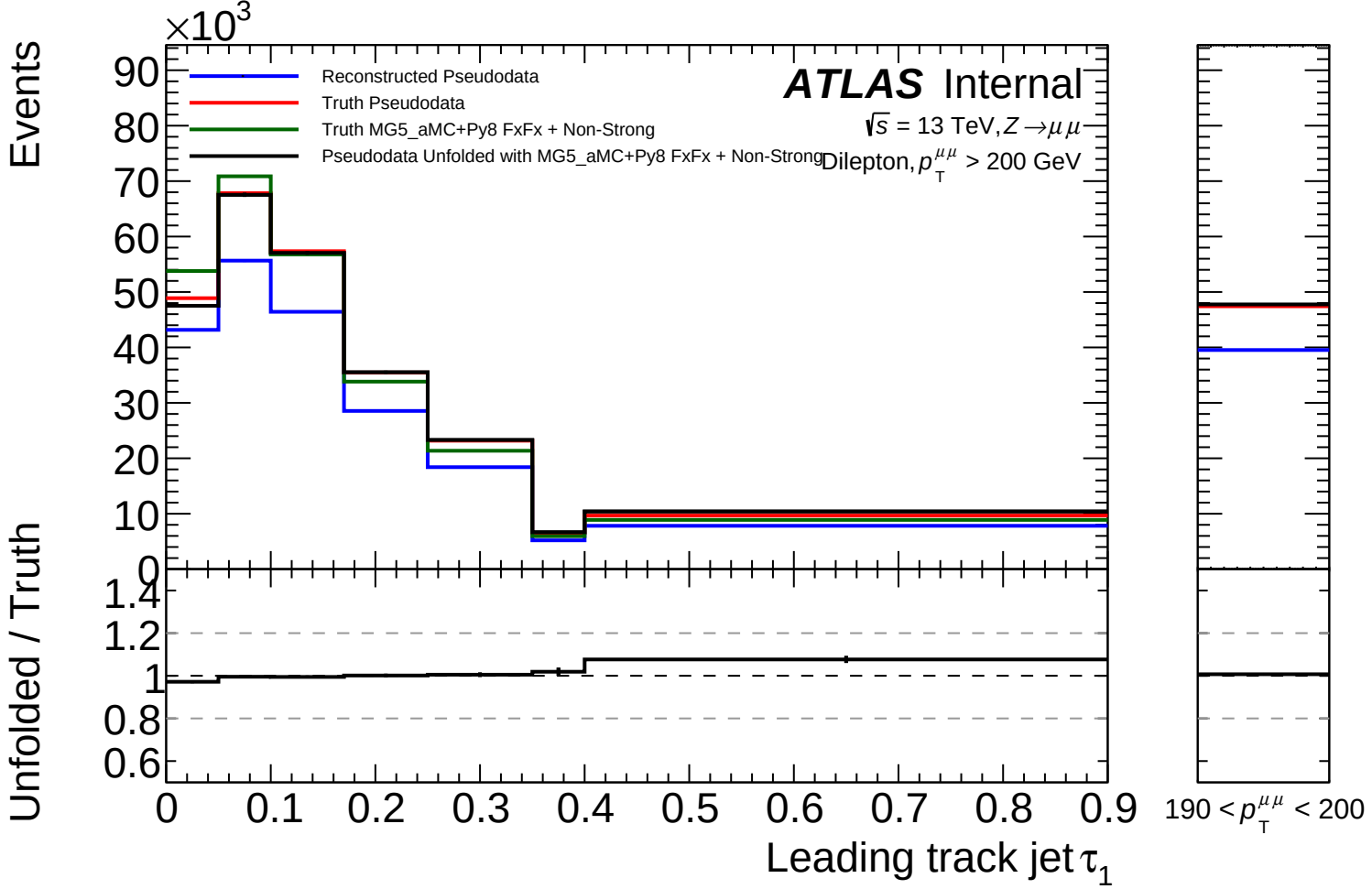
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

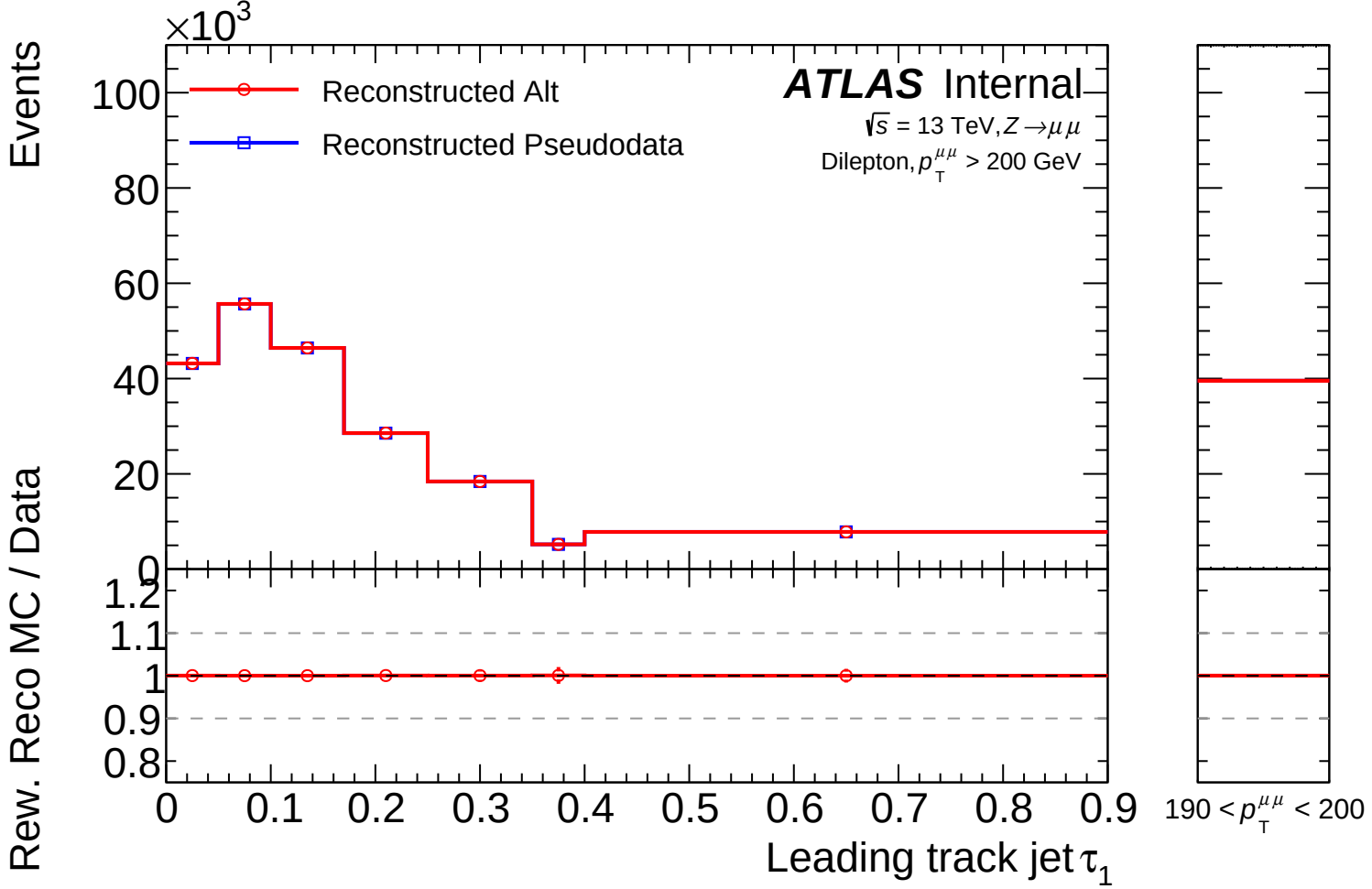


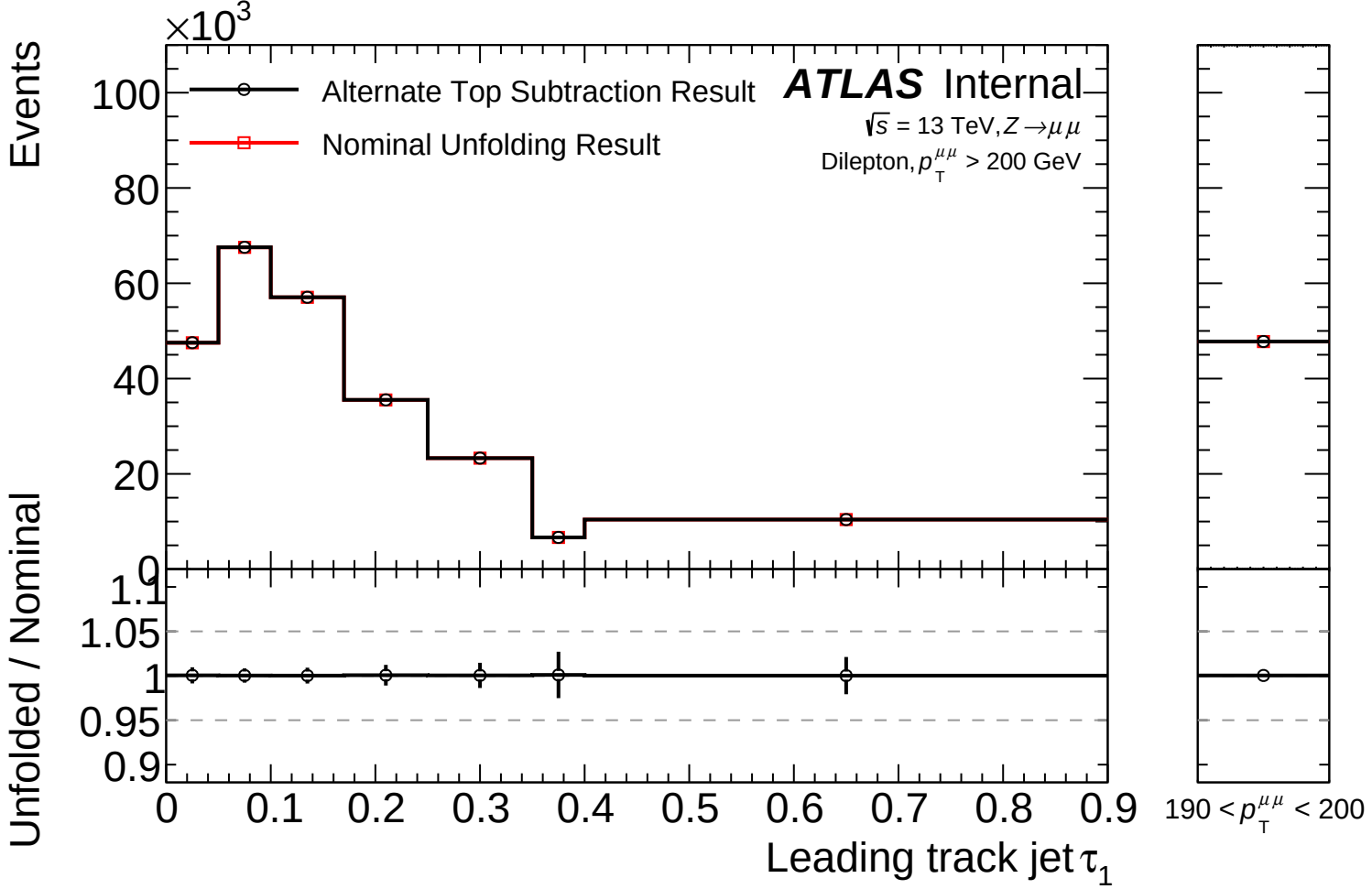
ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

Leading track jet m [GeV]









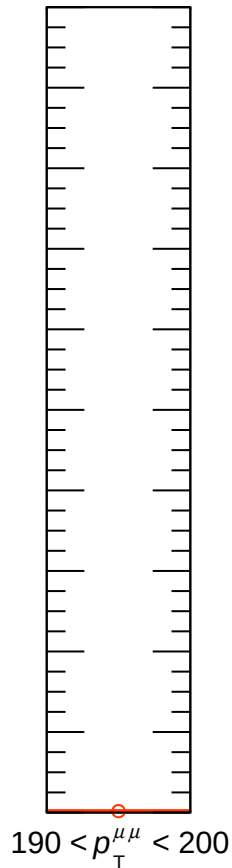
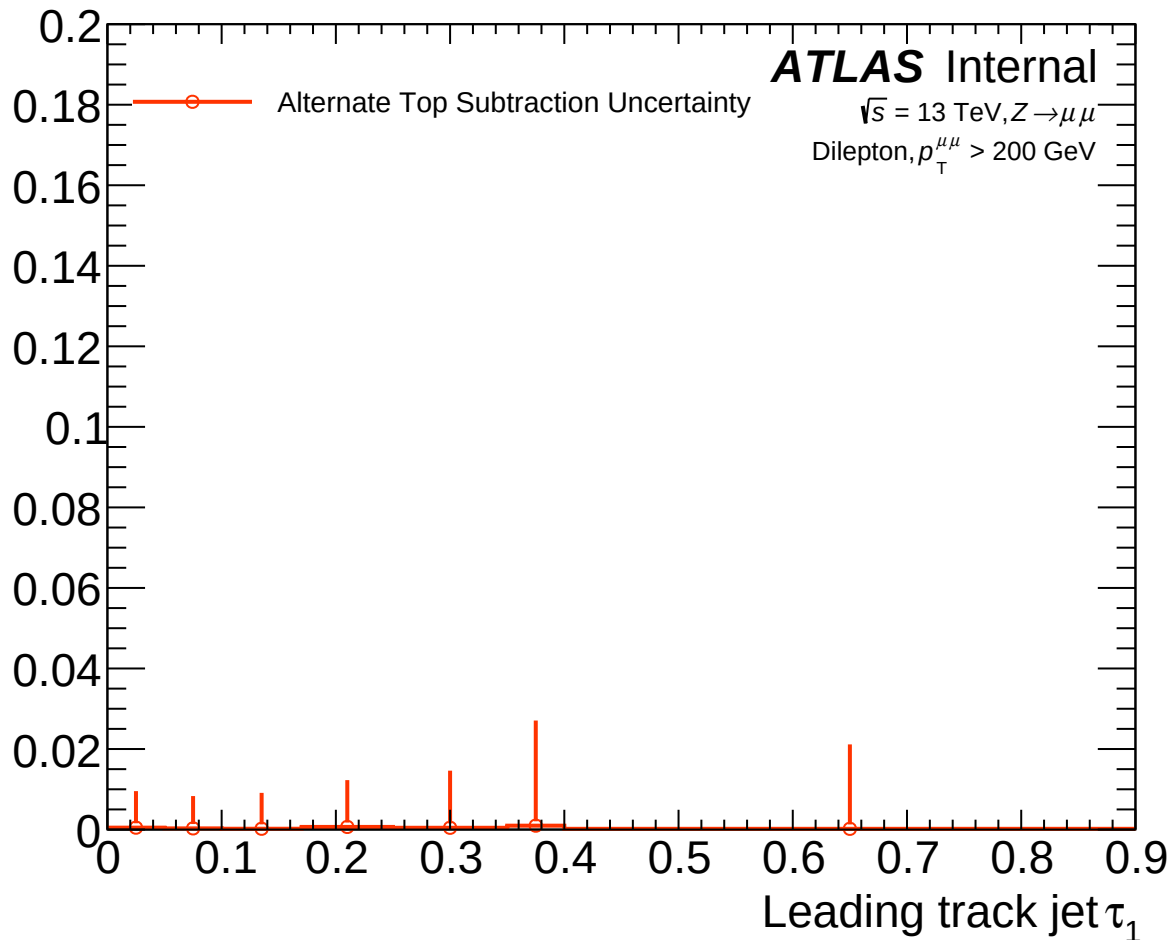
Fractional systematic impact

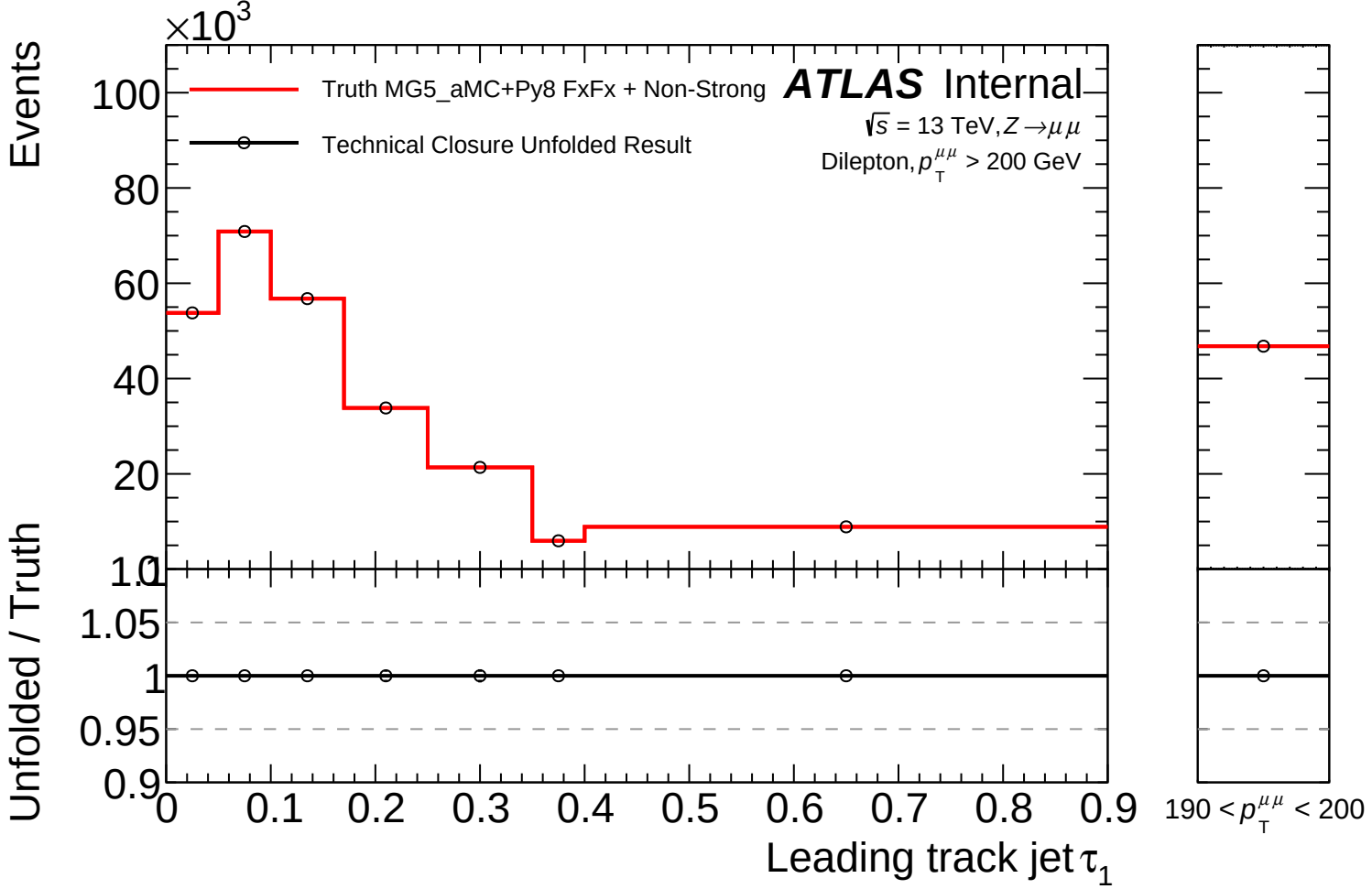
ATLAS Internal

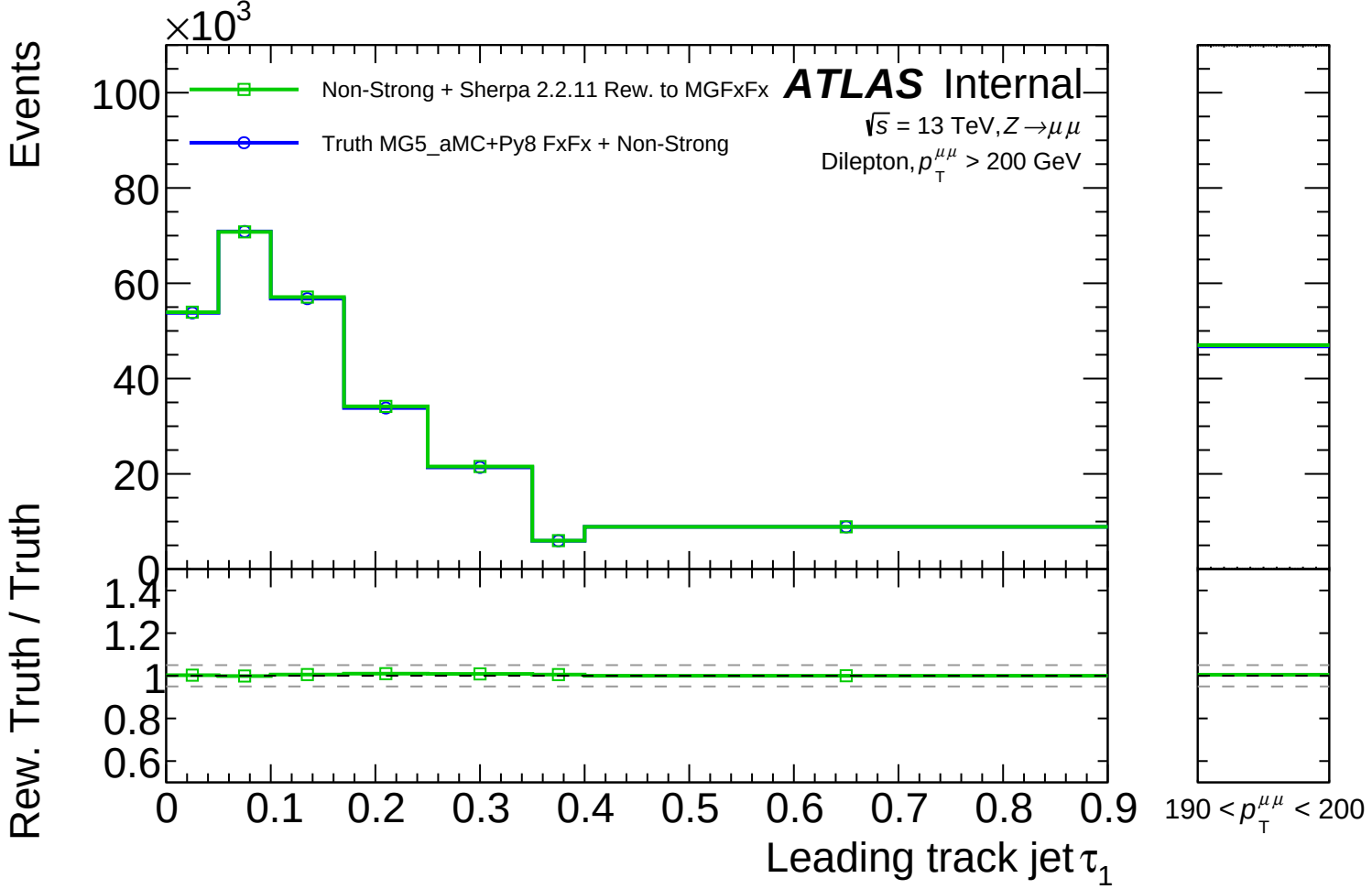
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

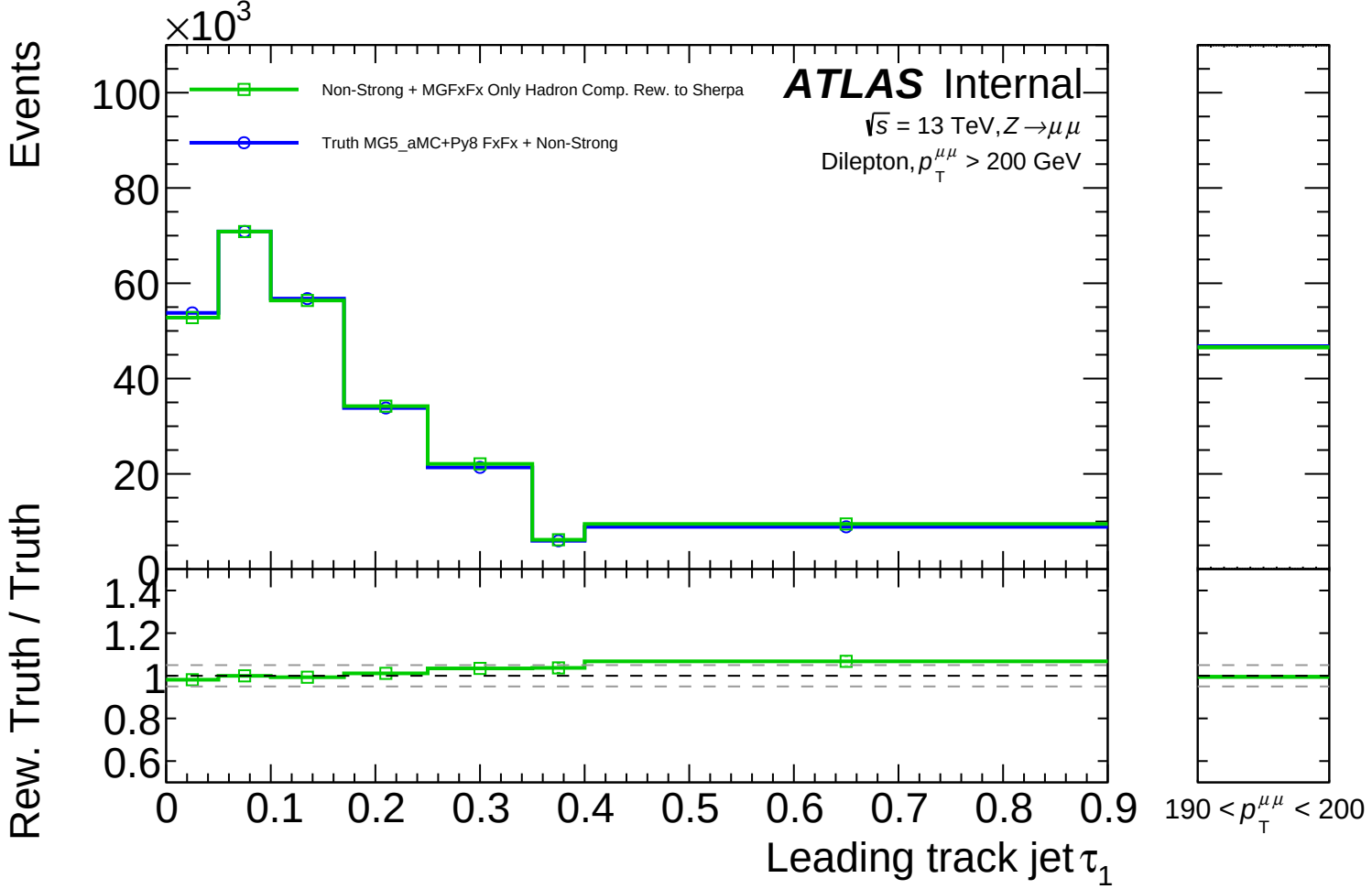
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

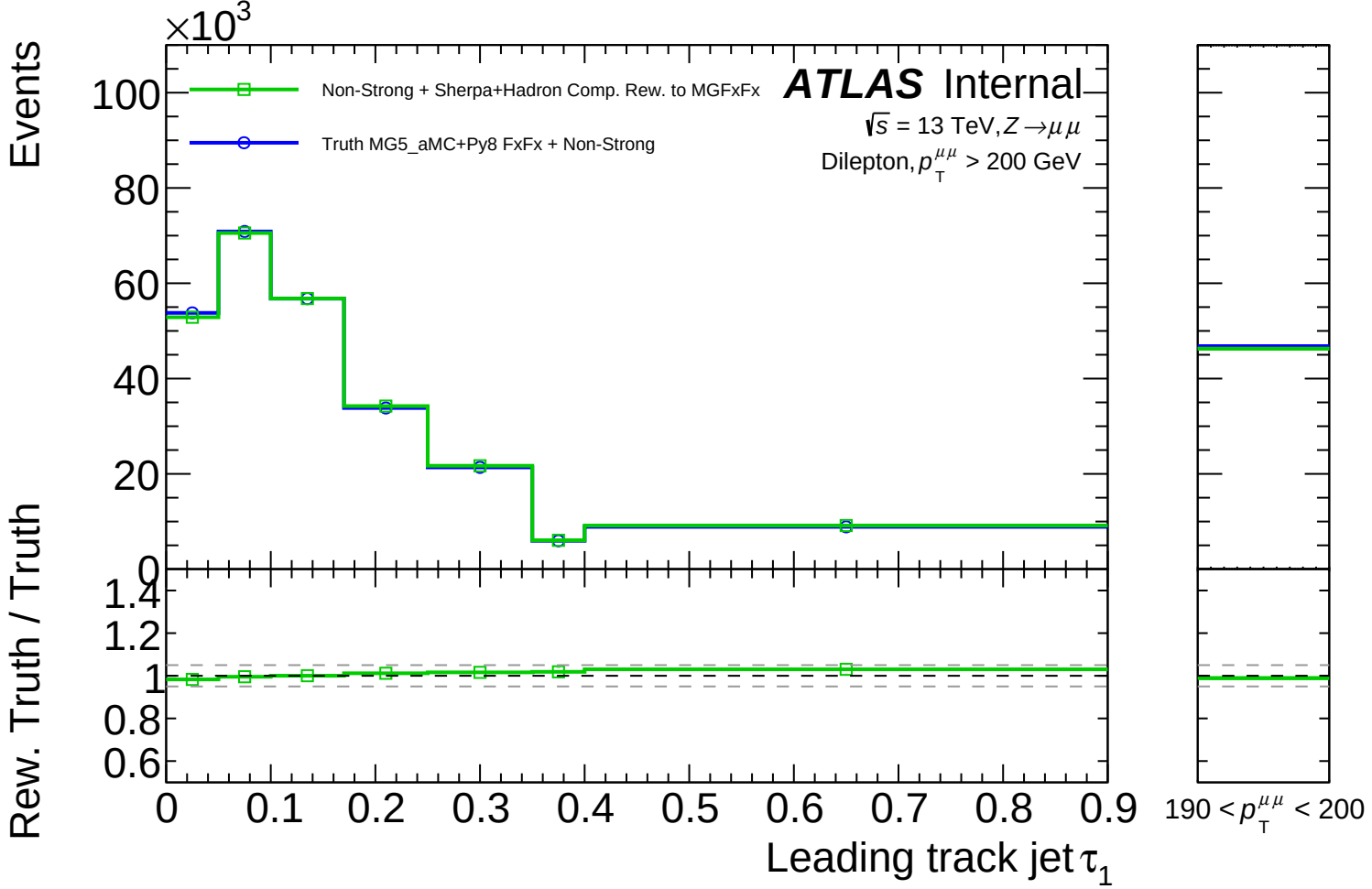
Alternate Top Subtraction Uncertainty

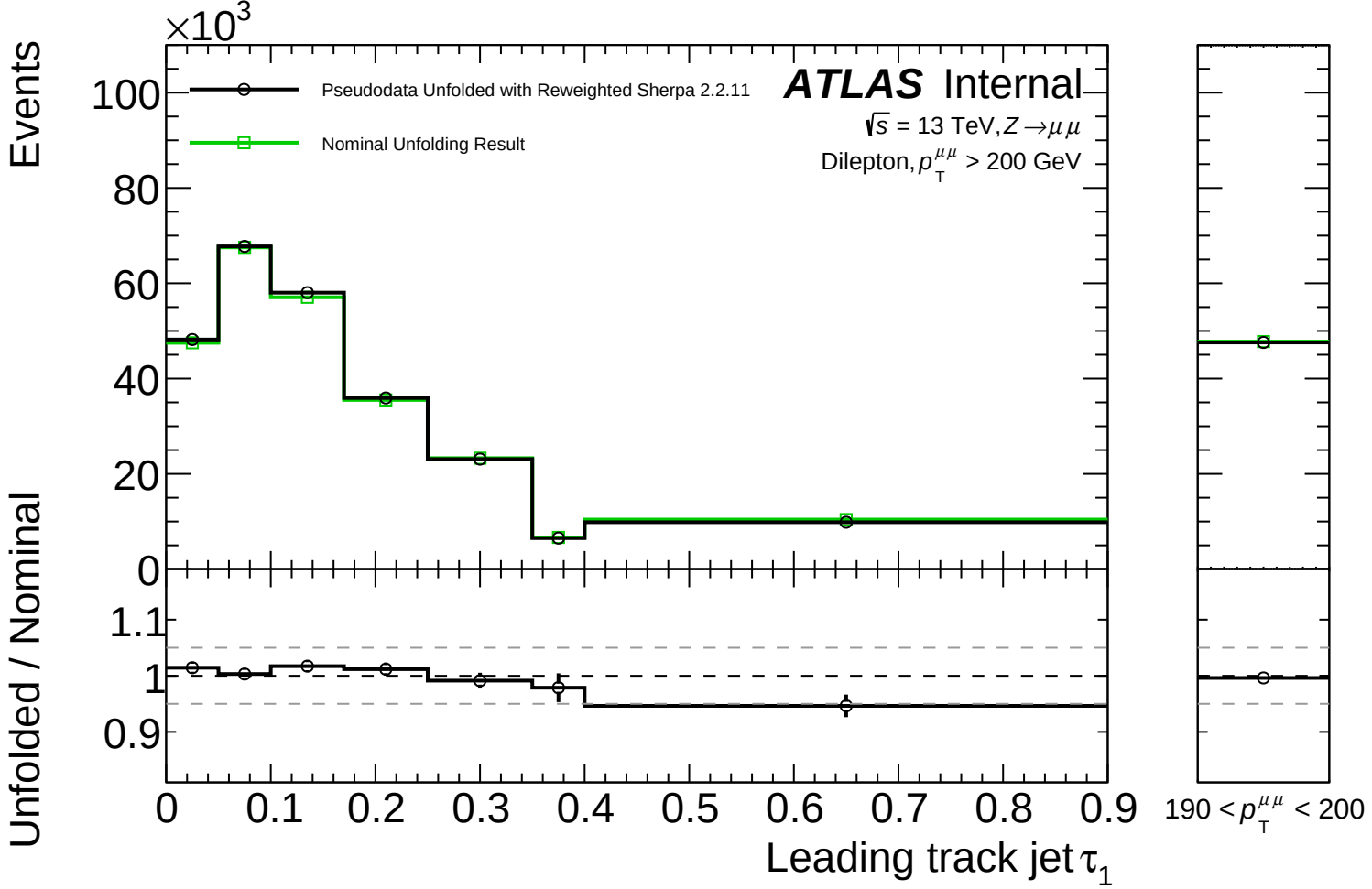


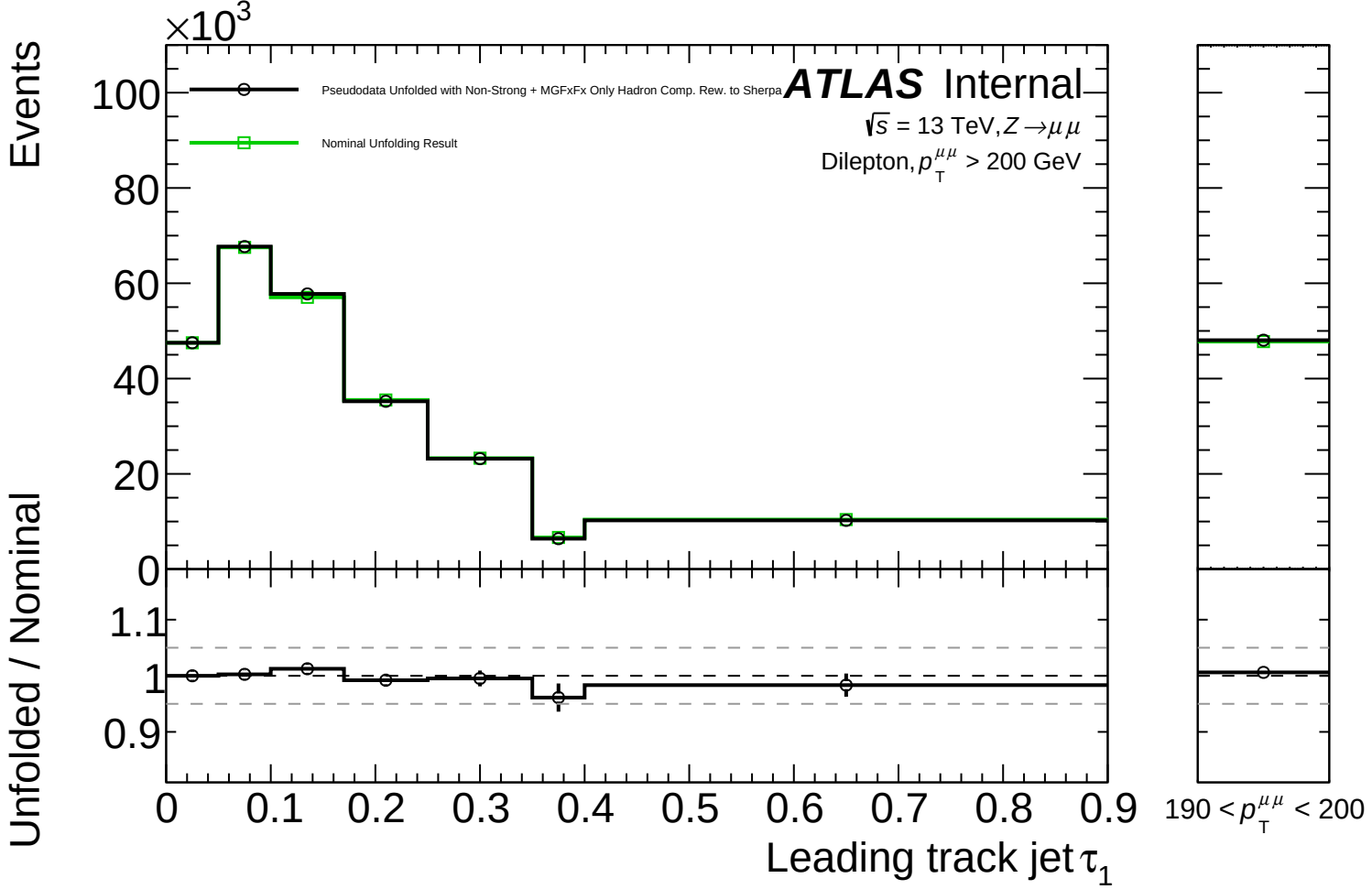


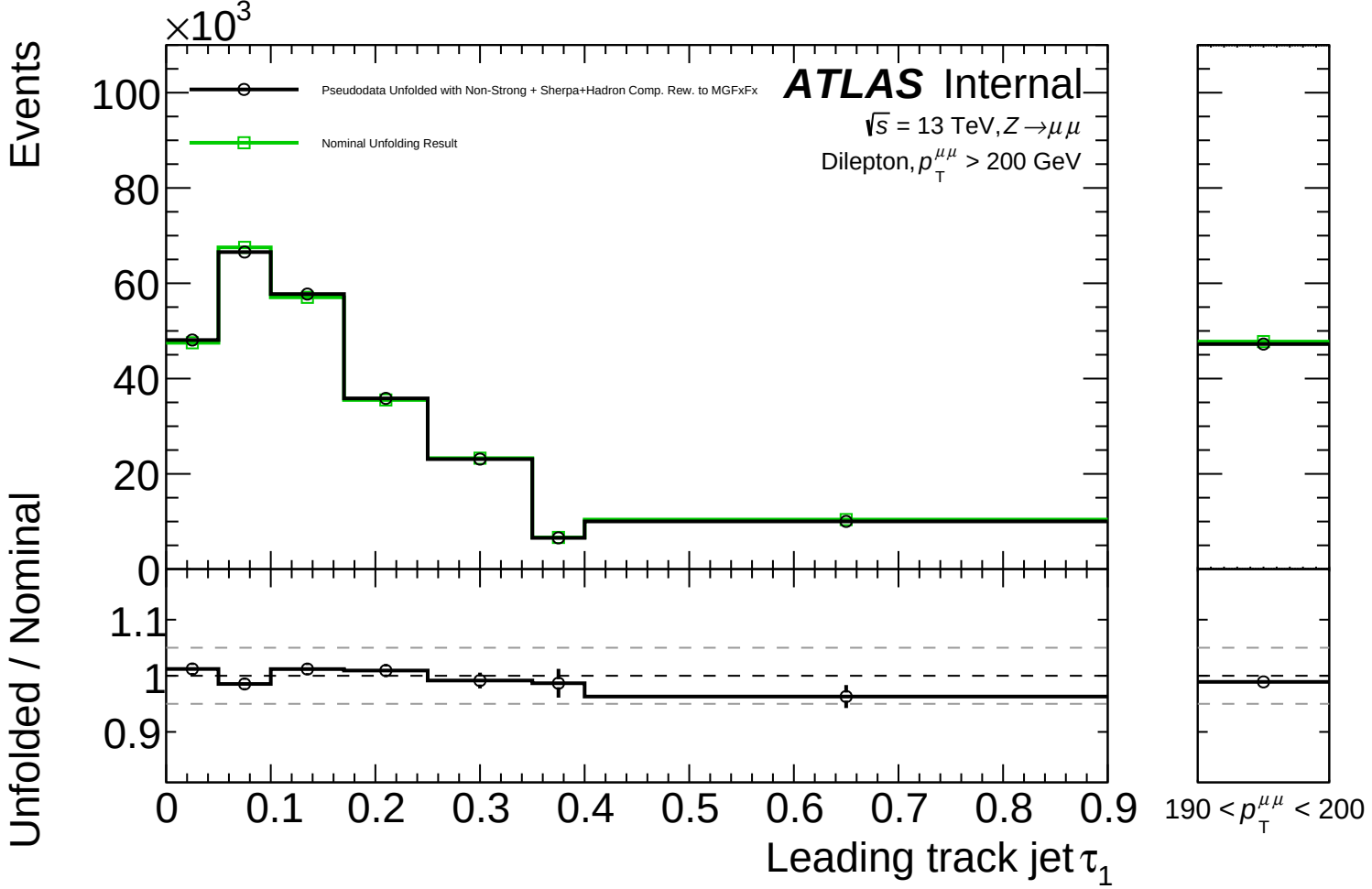


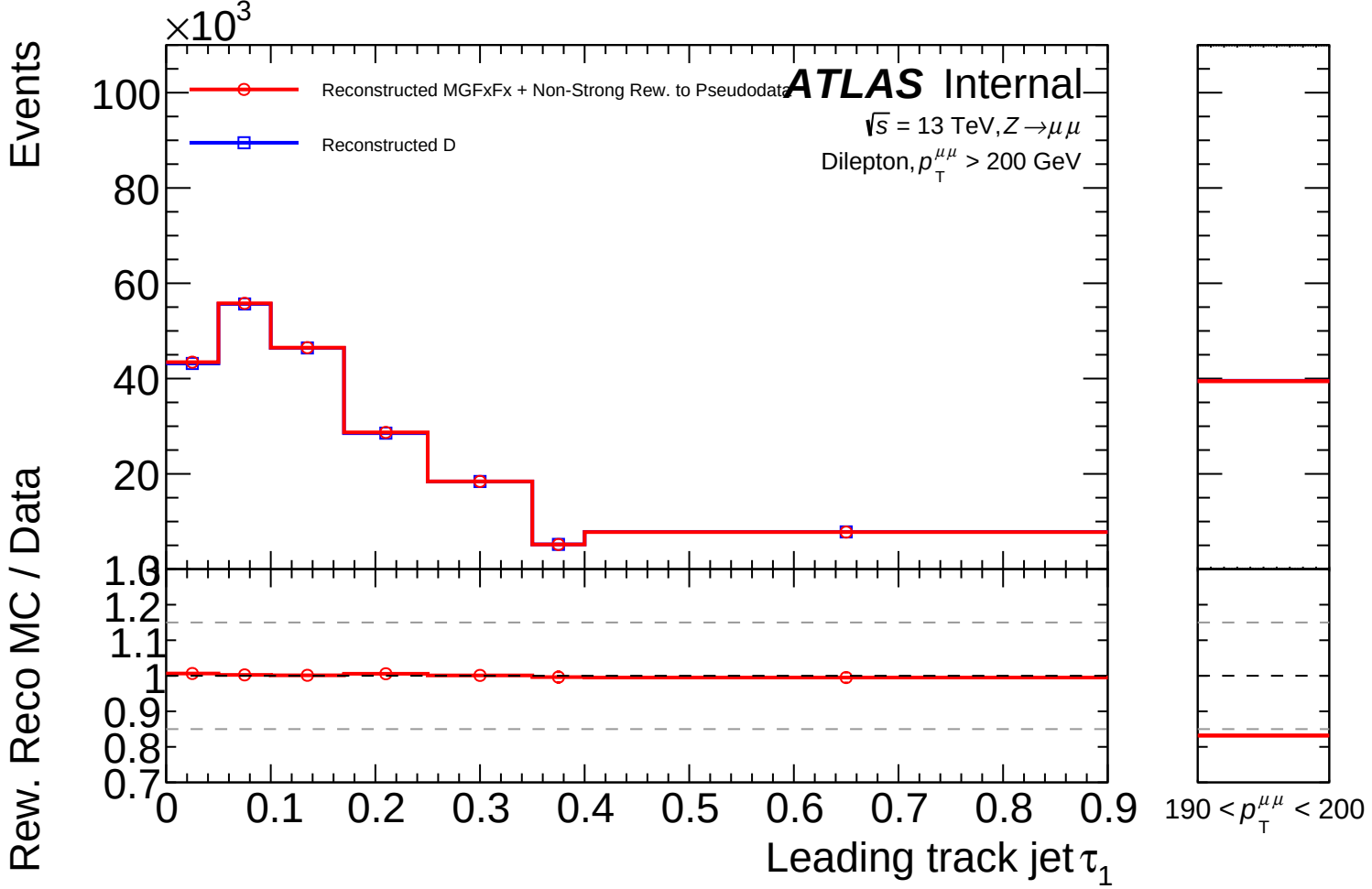


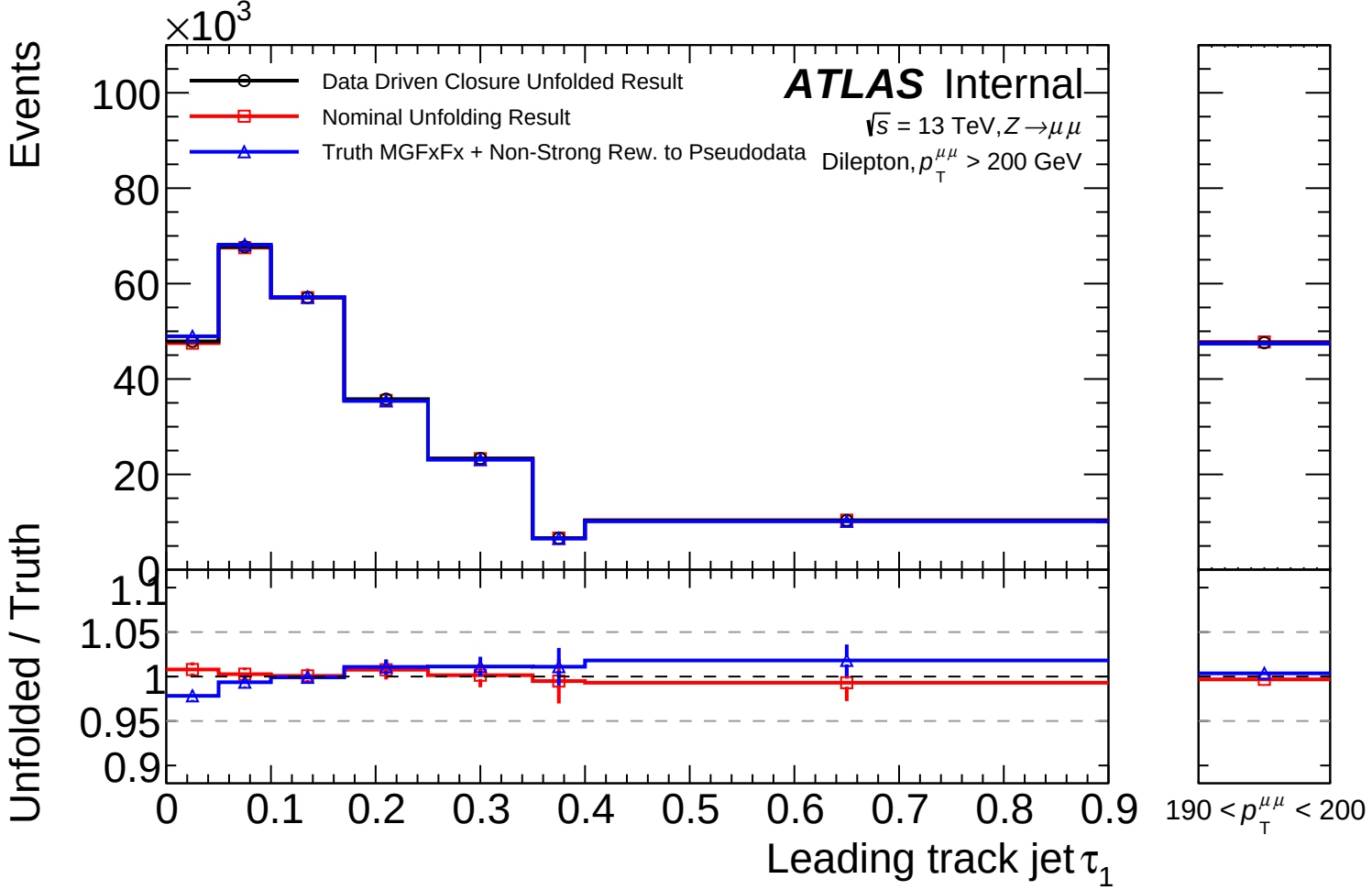




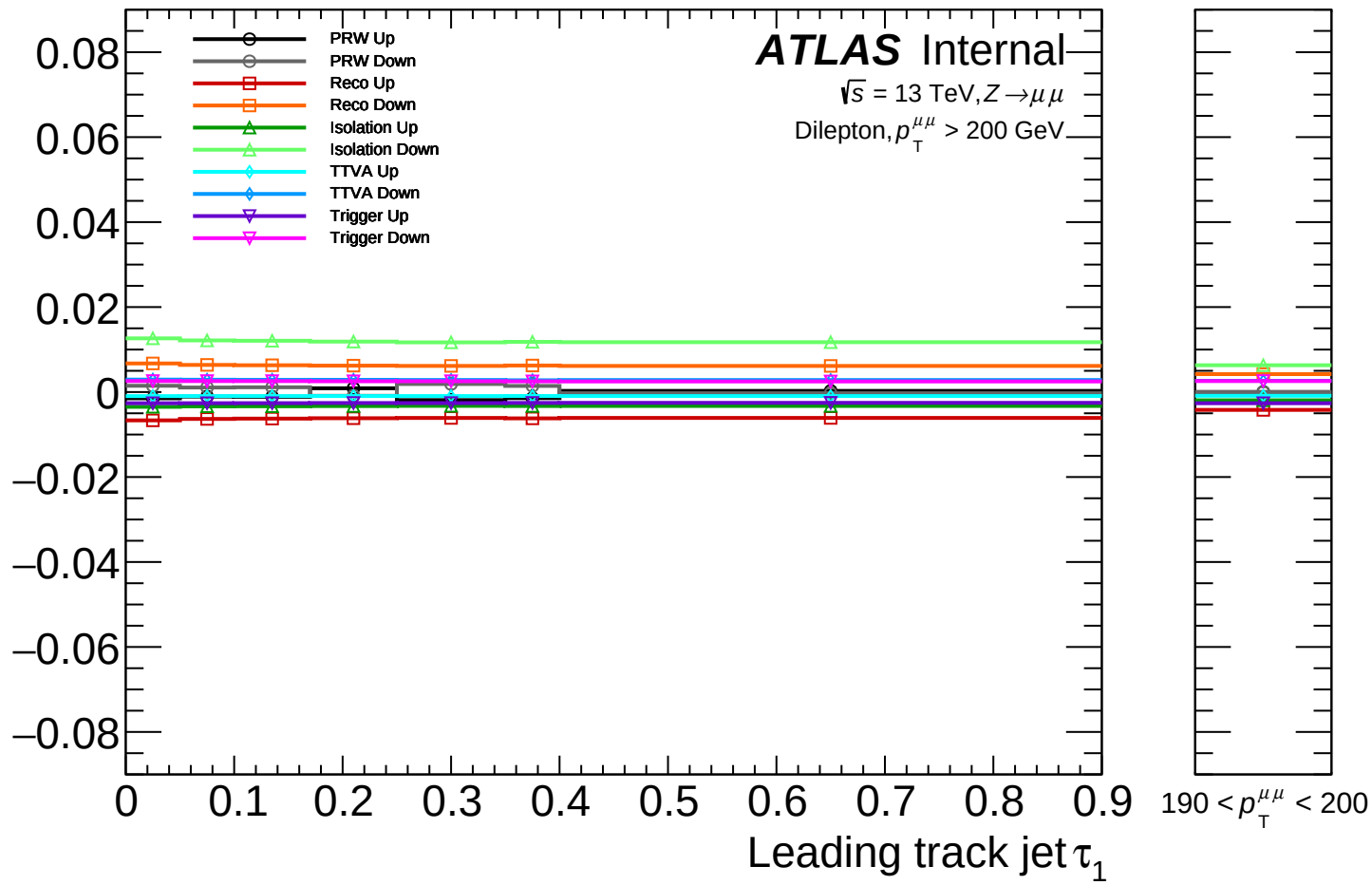




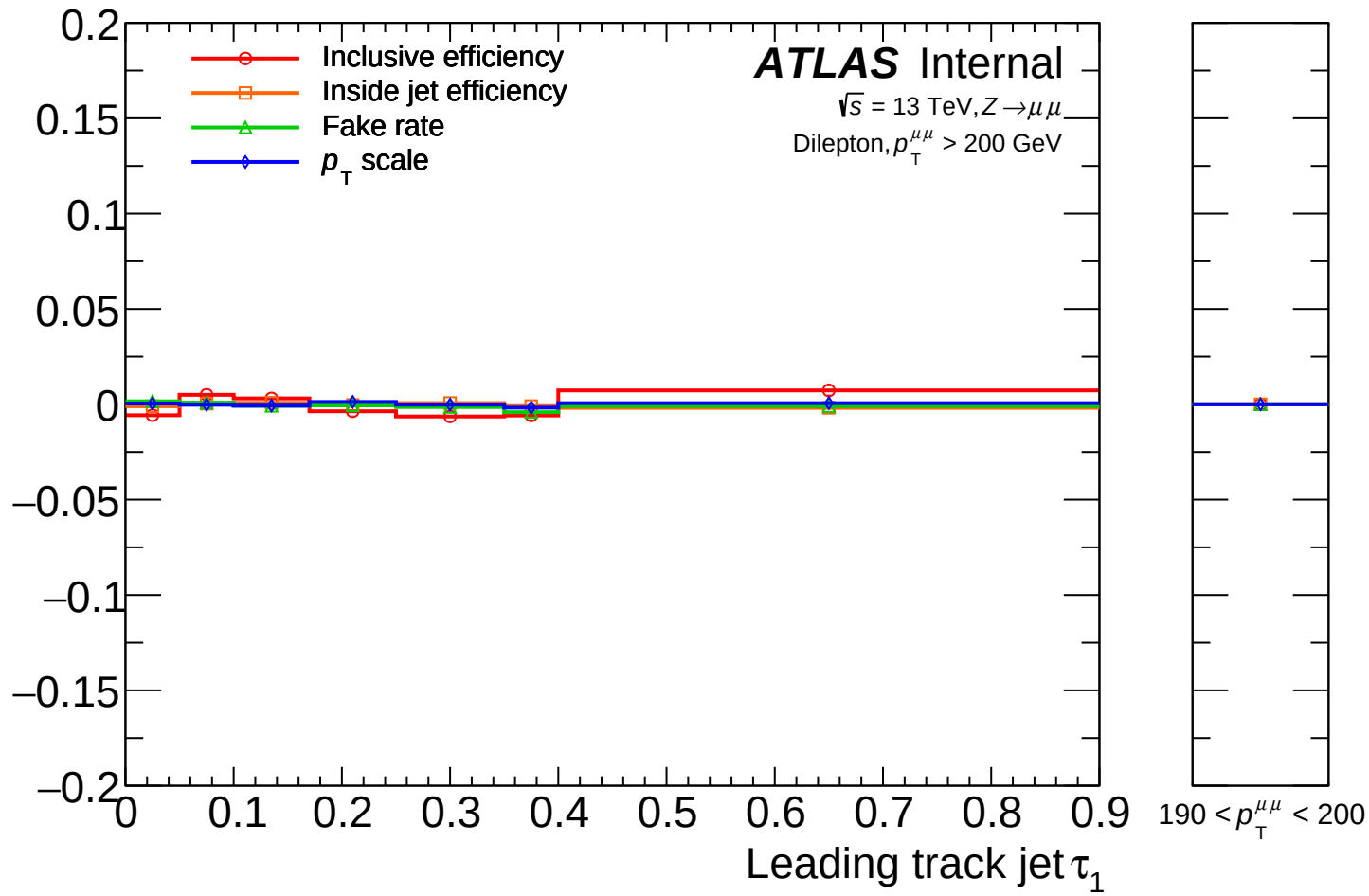




Fractional systematic impact



Fractional systematic impact



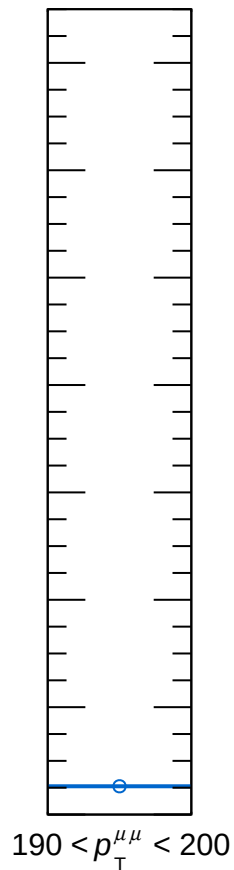
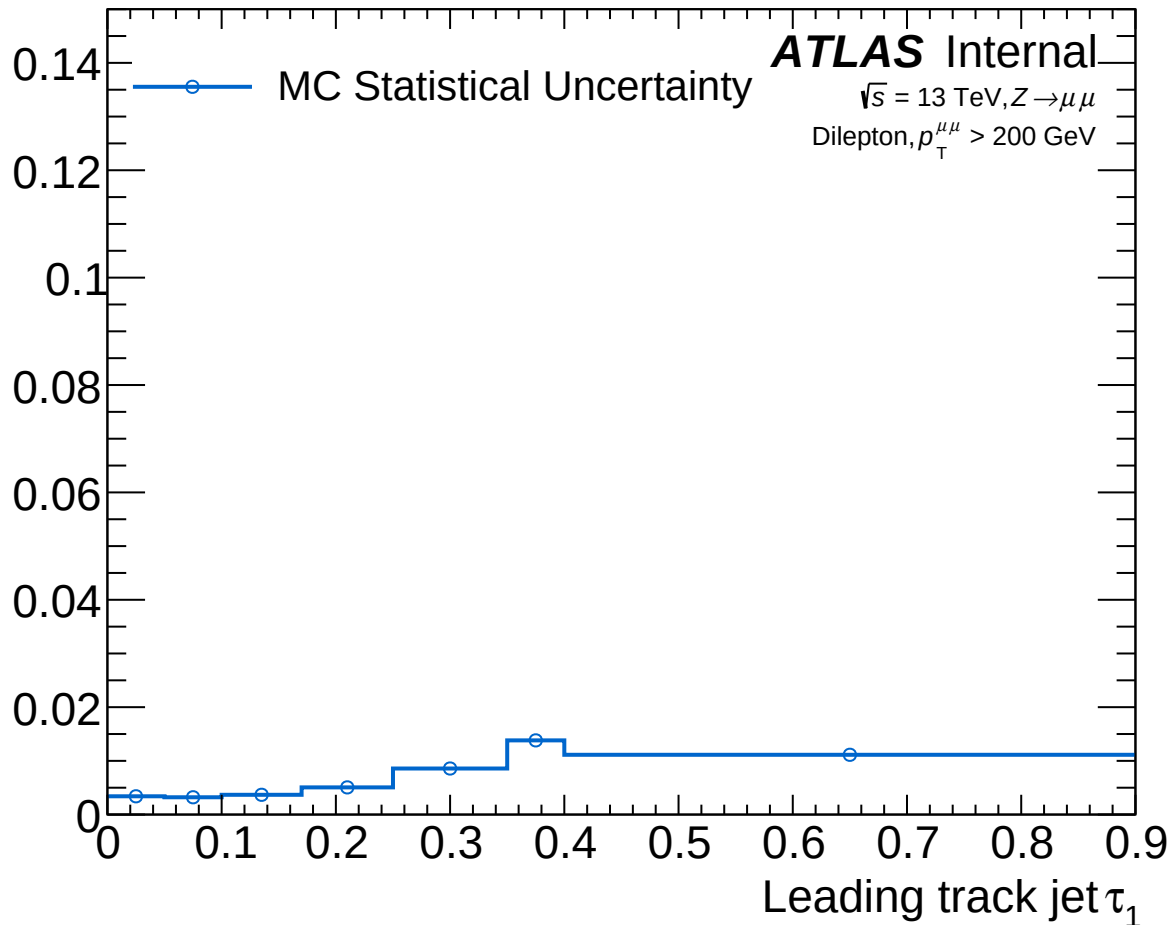
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



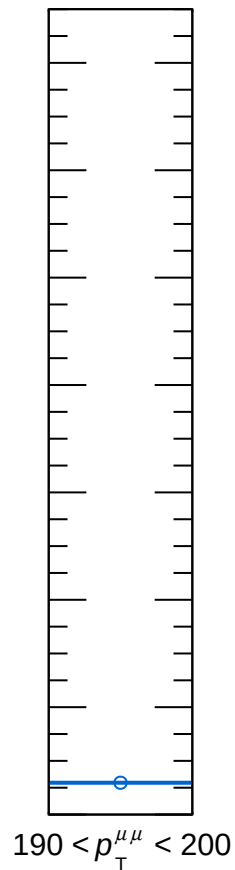
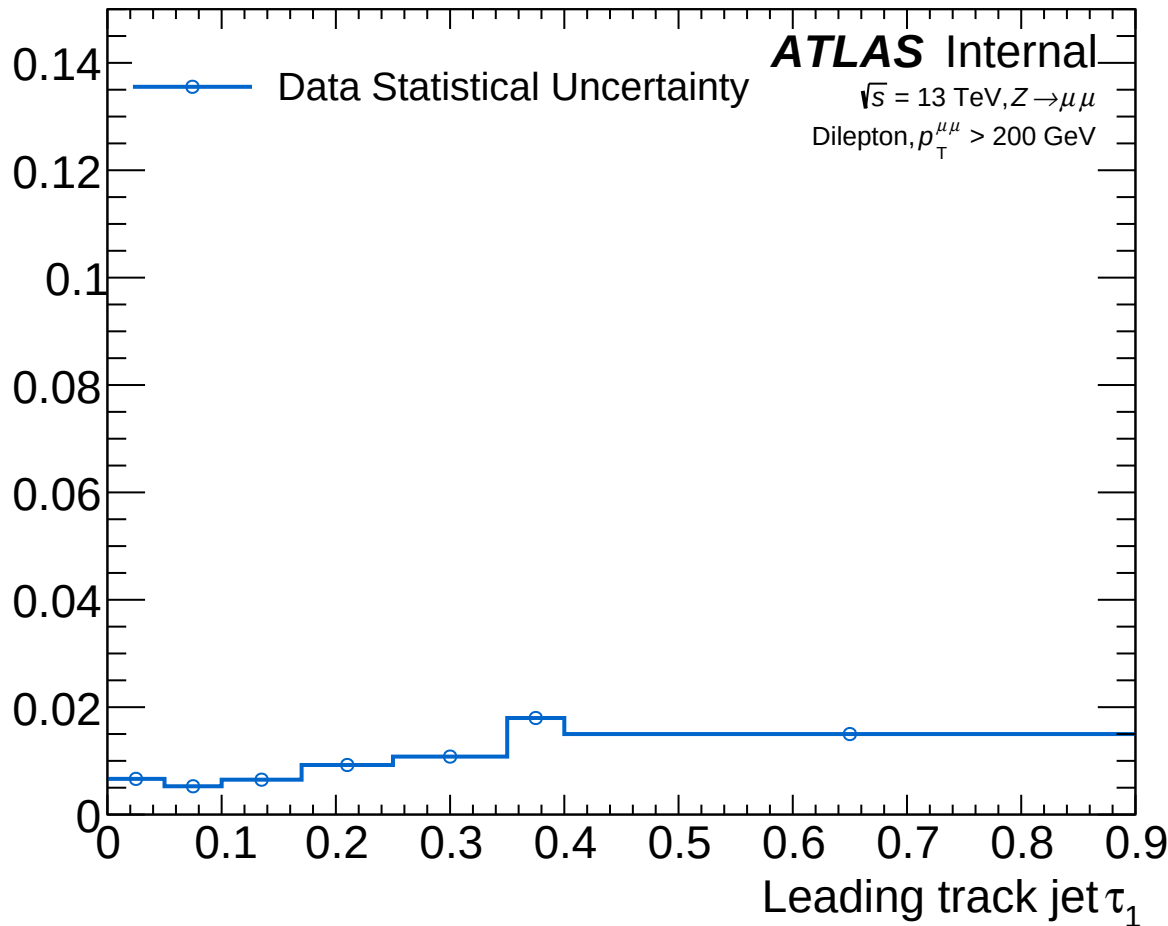
Fractional systematic impact

ATLAS Internal

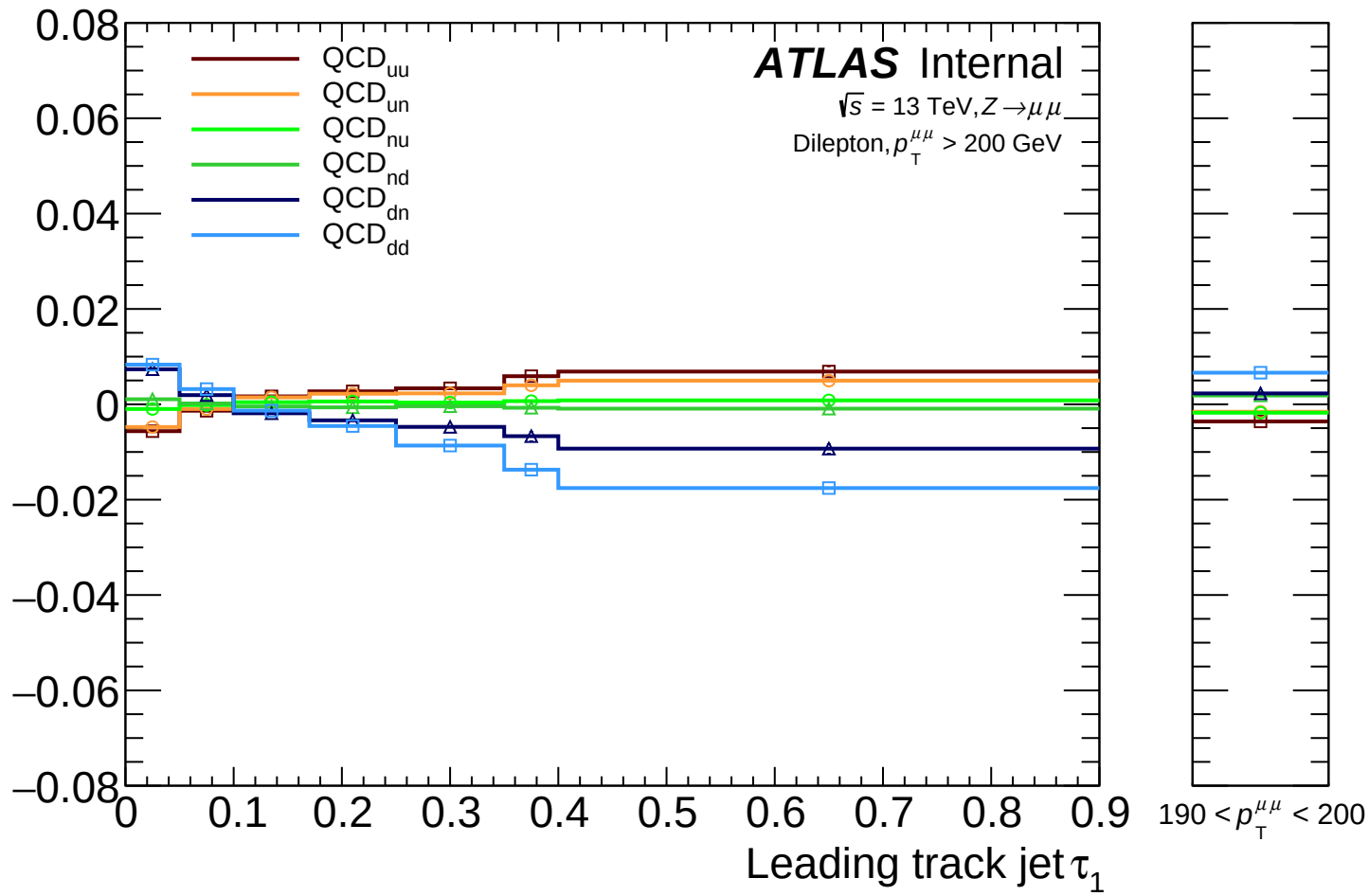
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

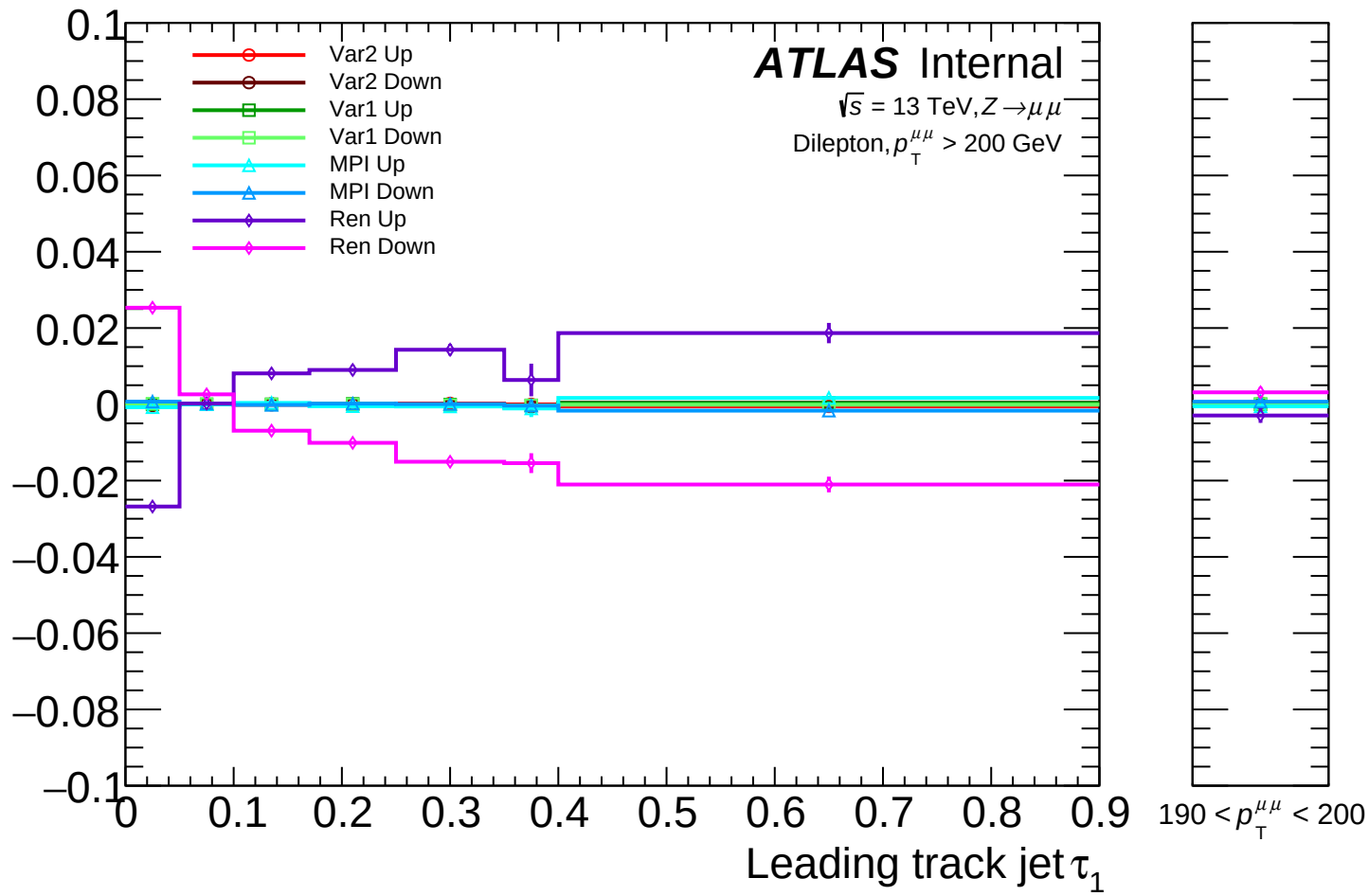
Data Statistical Uncertainty

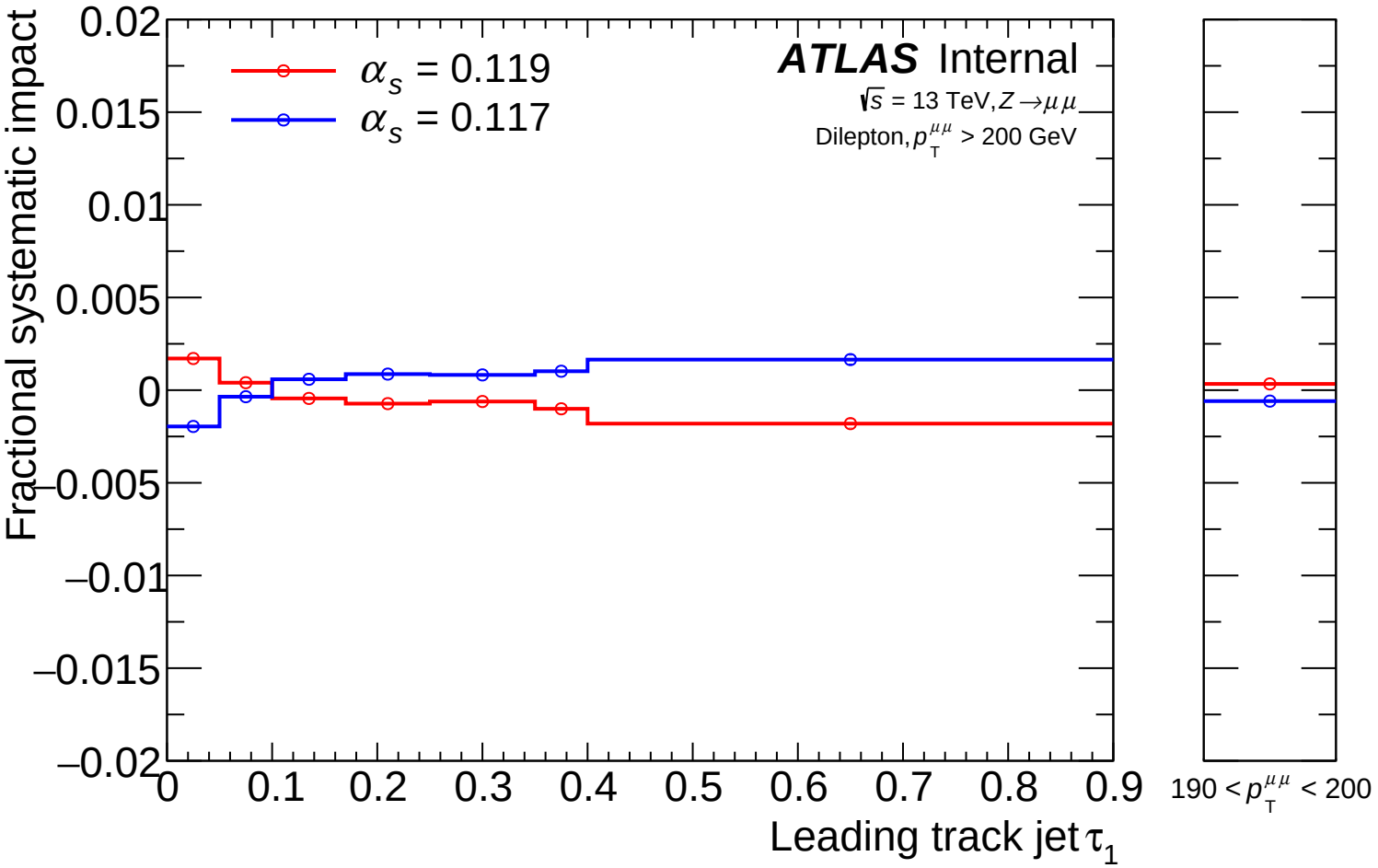


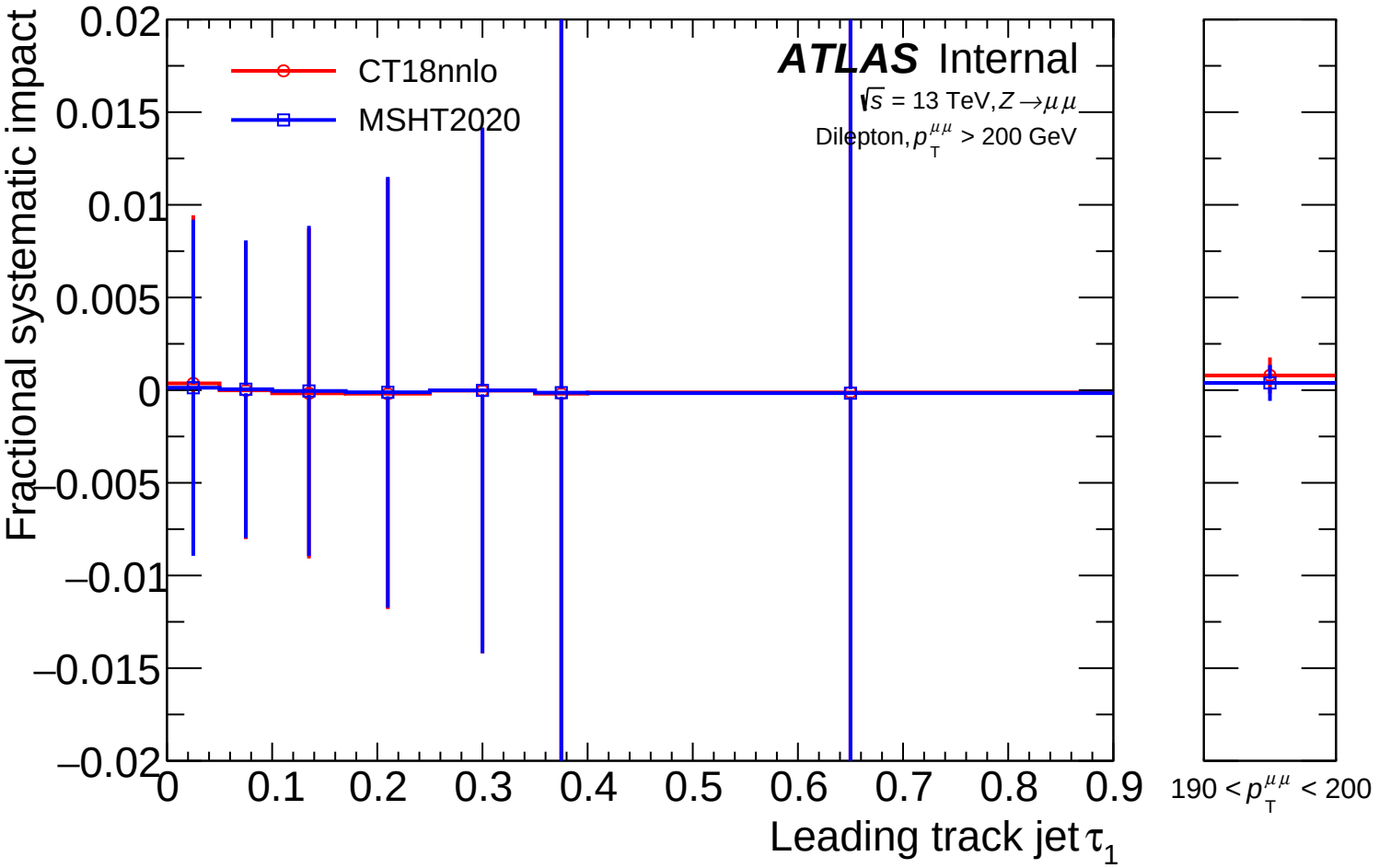
Fractional systematic impact



Fractional systematic impact







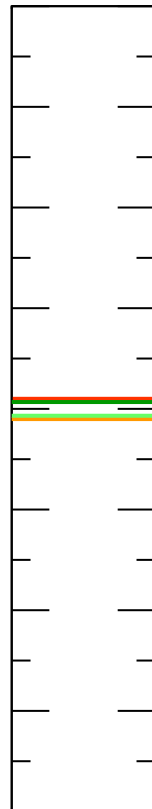
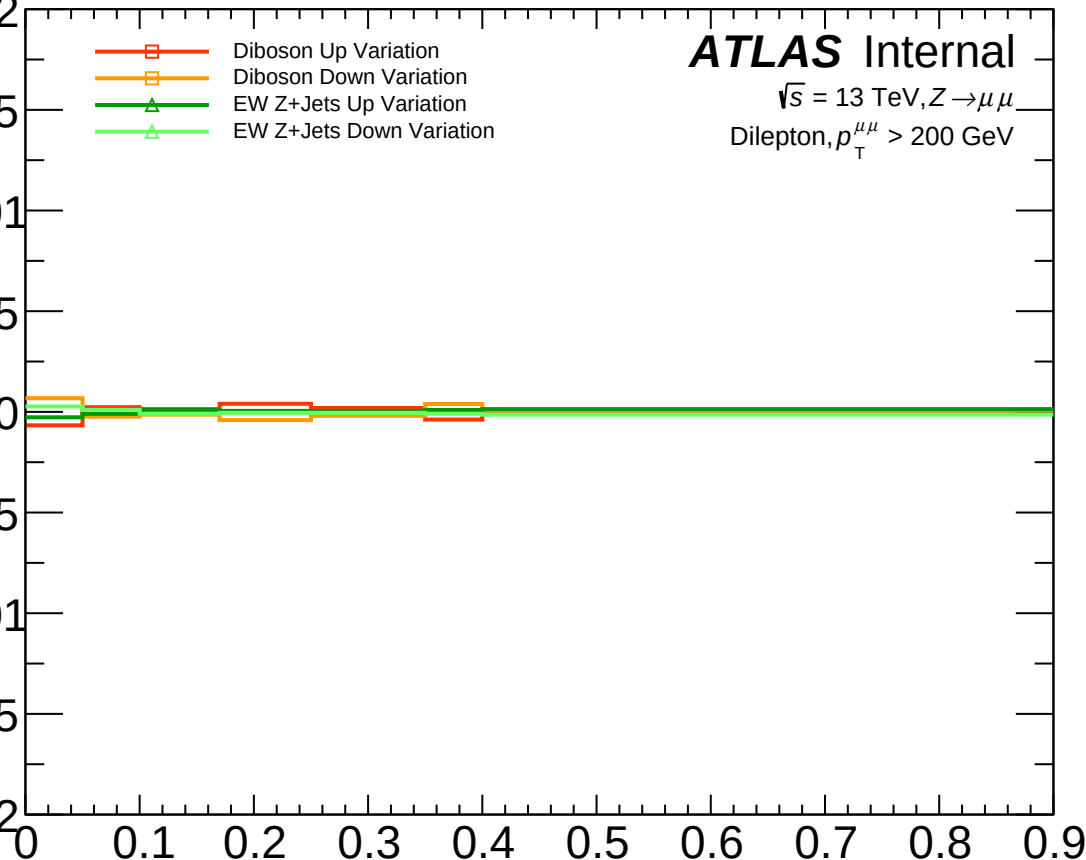
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200$ GeV



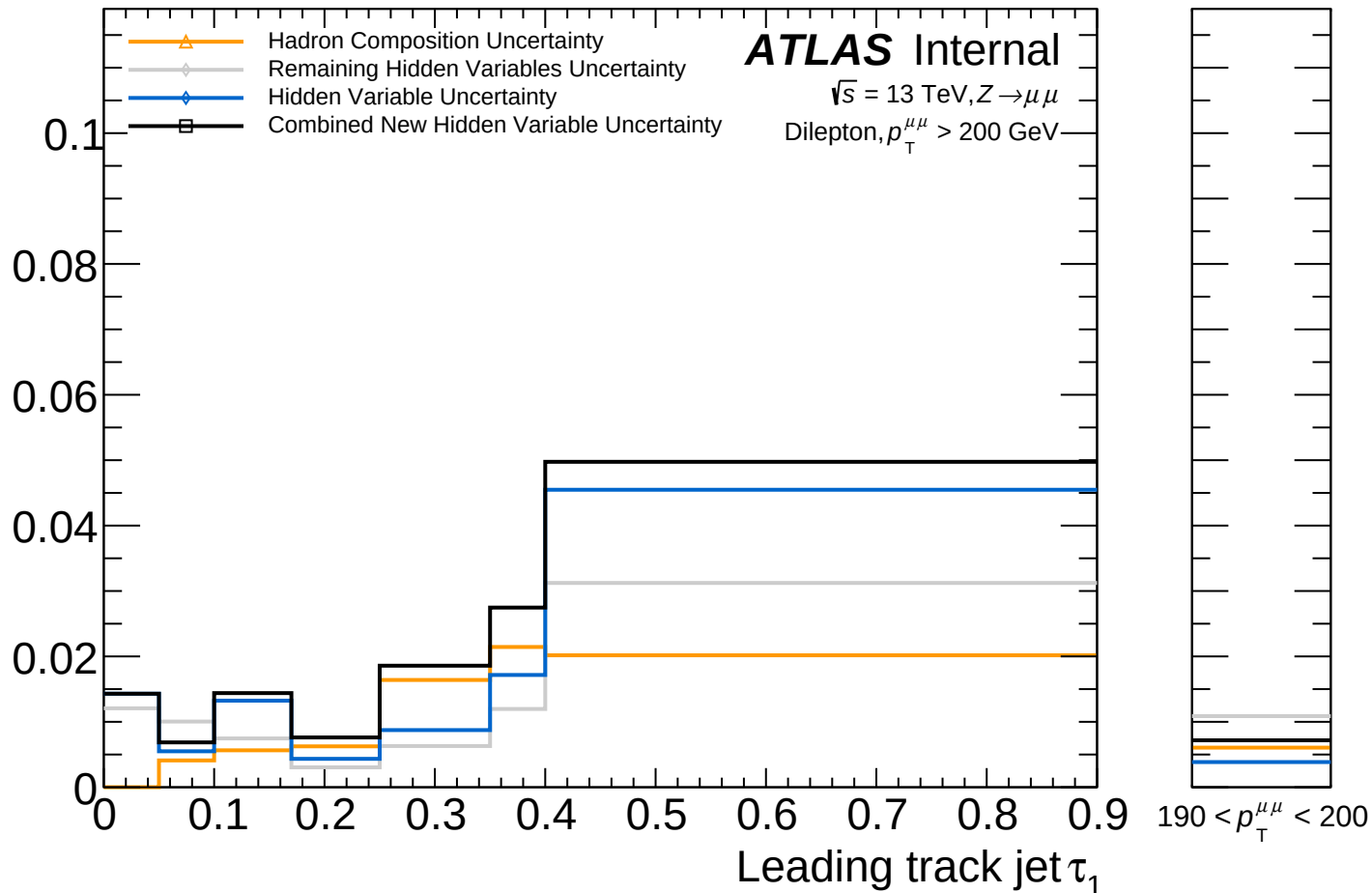
Leading track jet τ_1

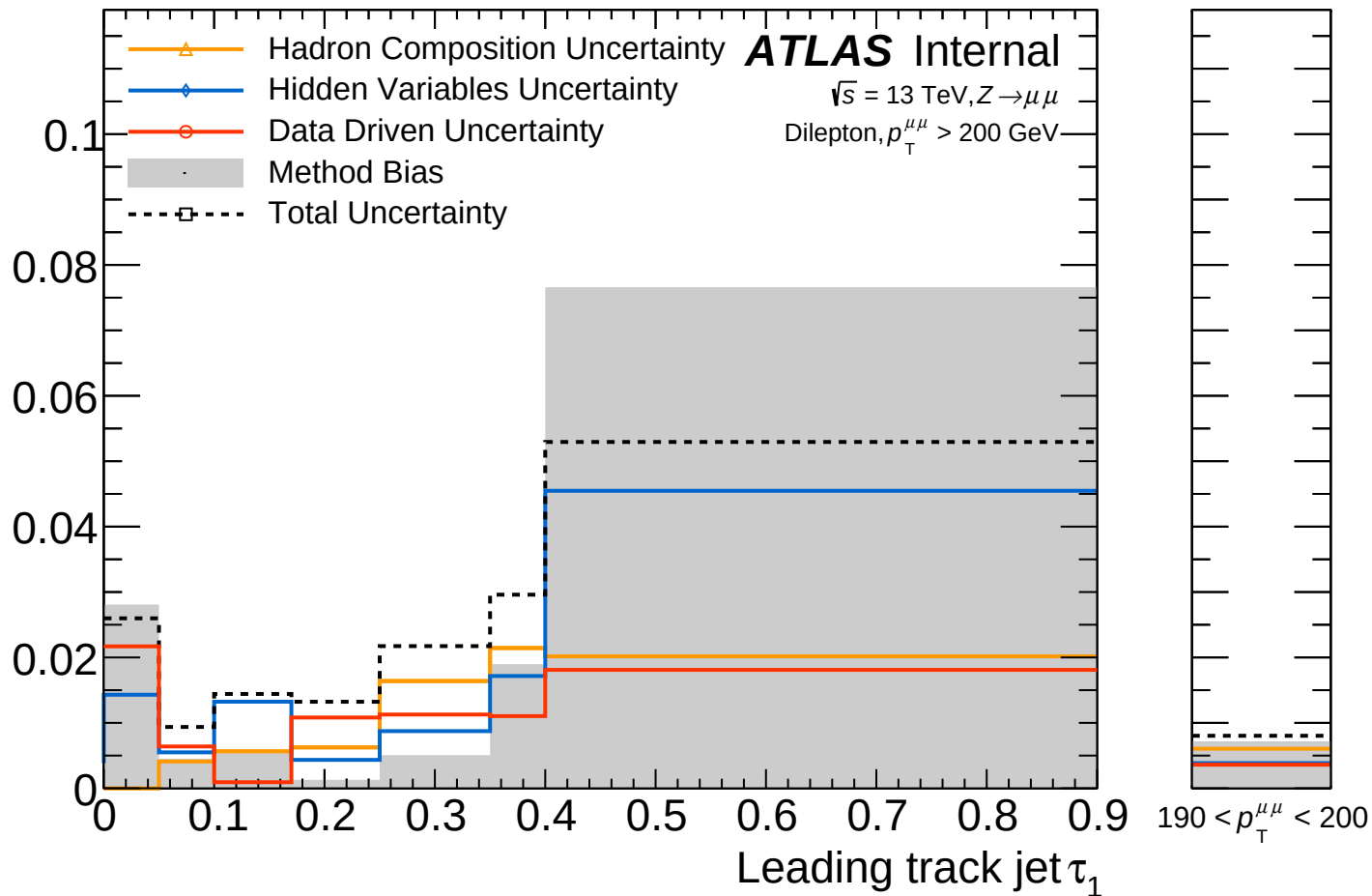
$190 < p_T^{\mu\mu} < 200$

ATLAS Internal

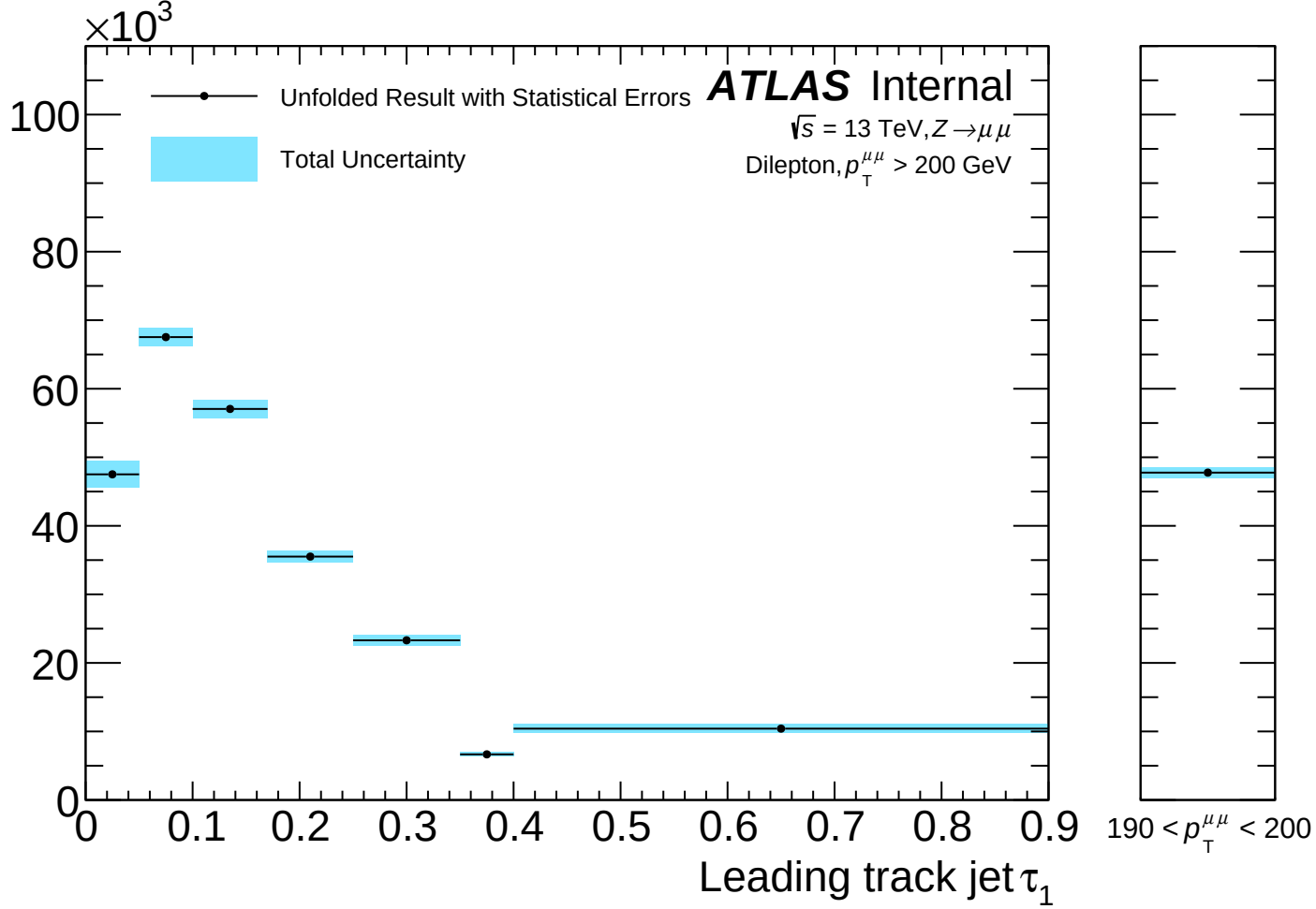
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

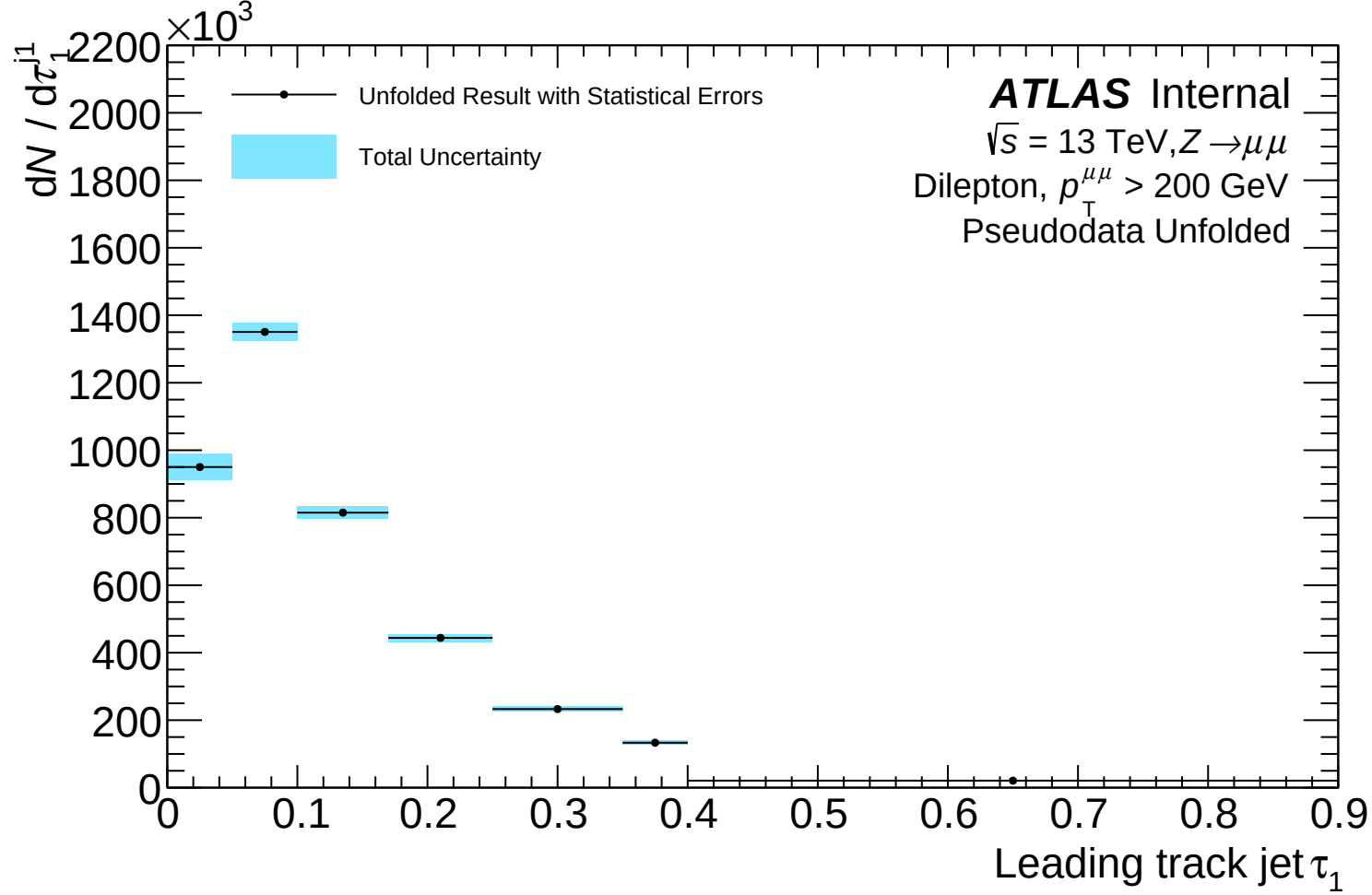
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

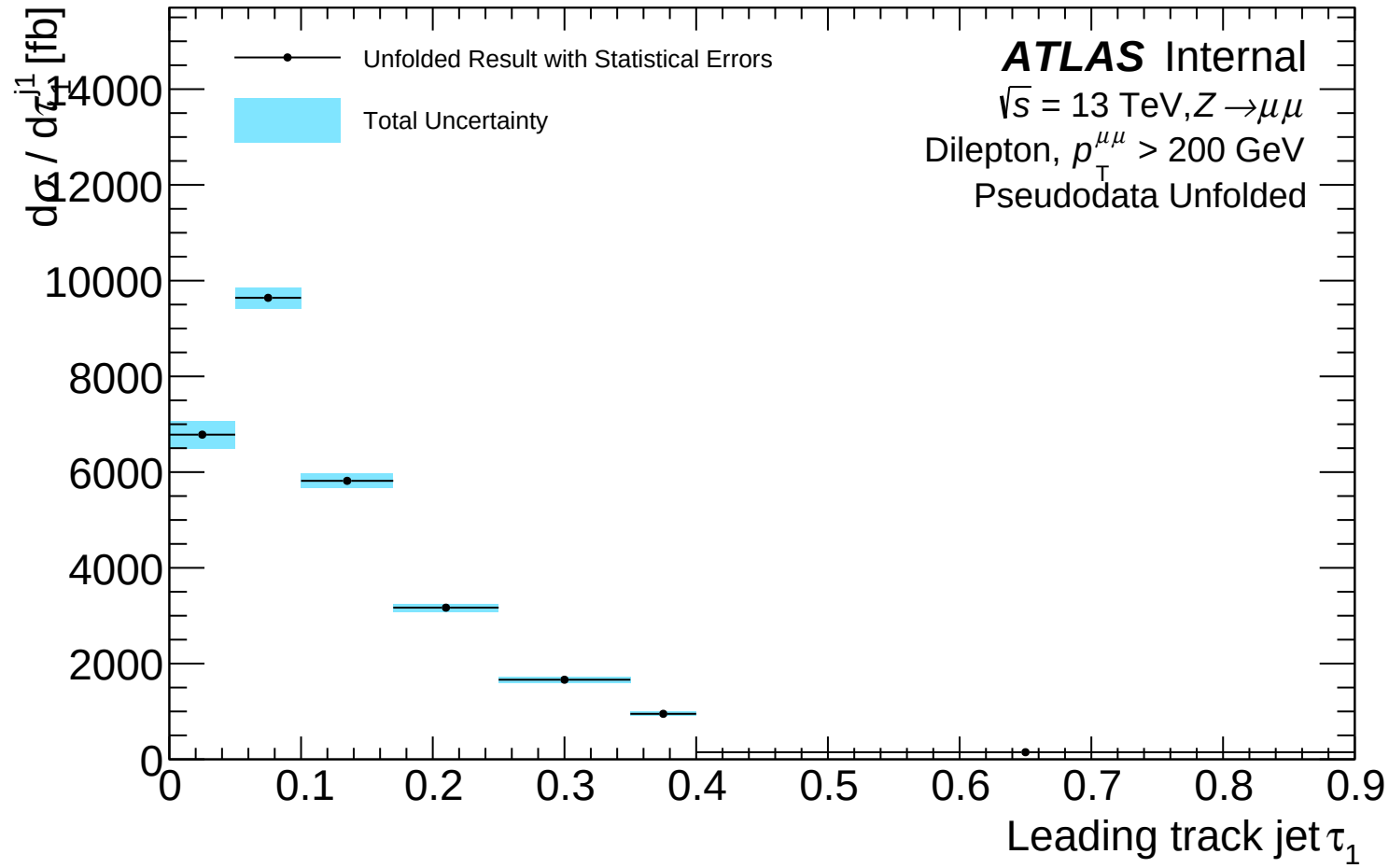




Events







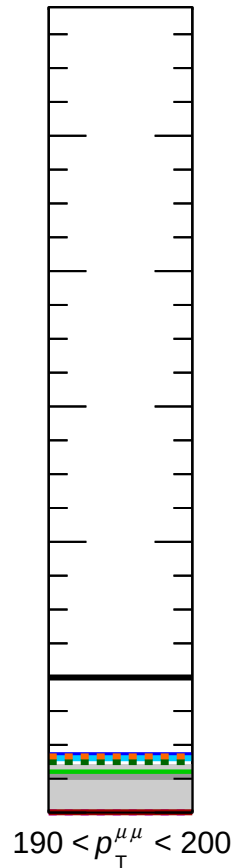
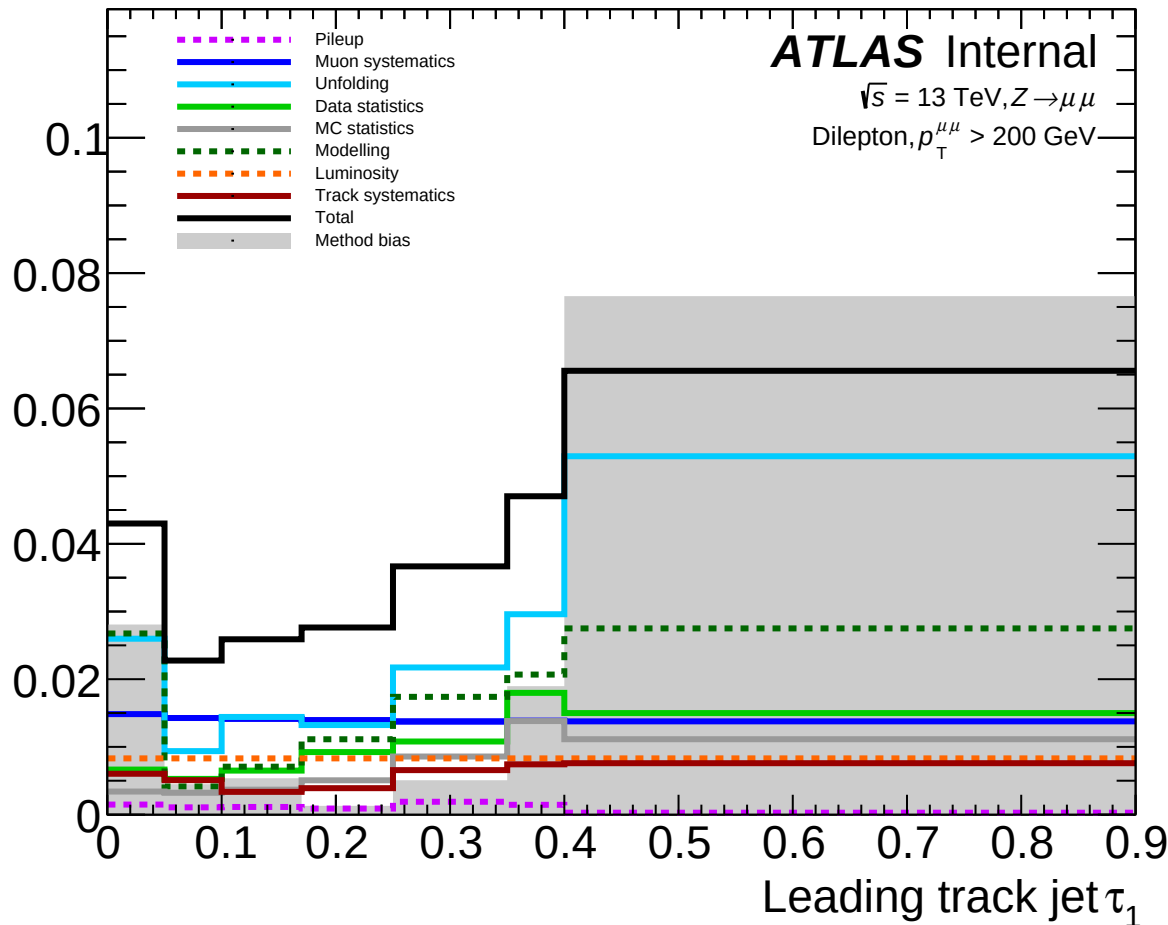
Fractional uncertainty

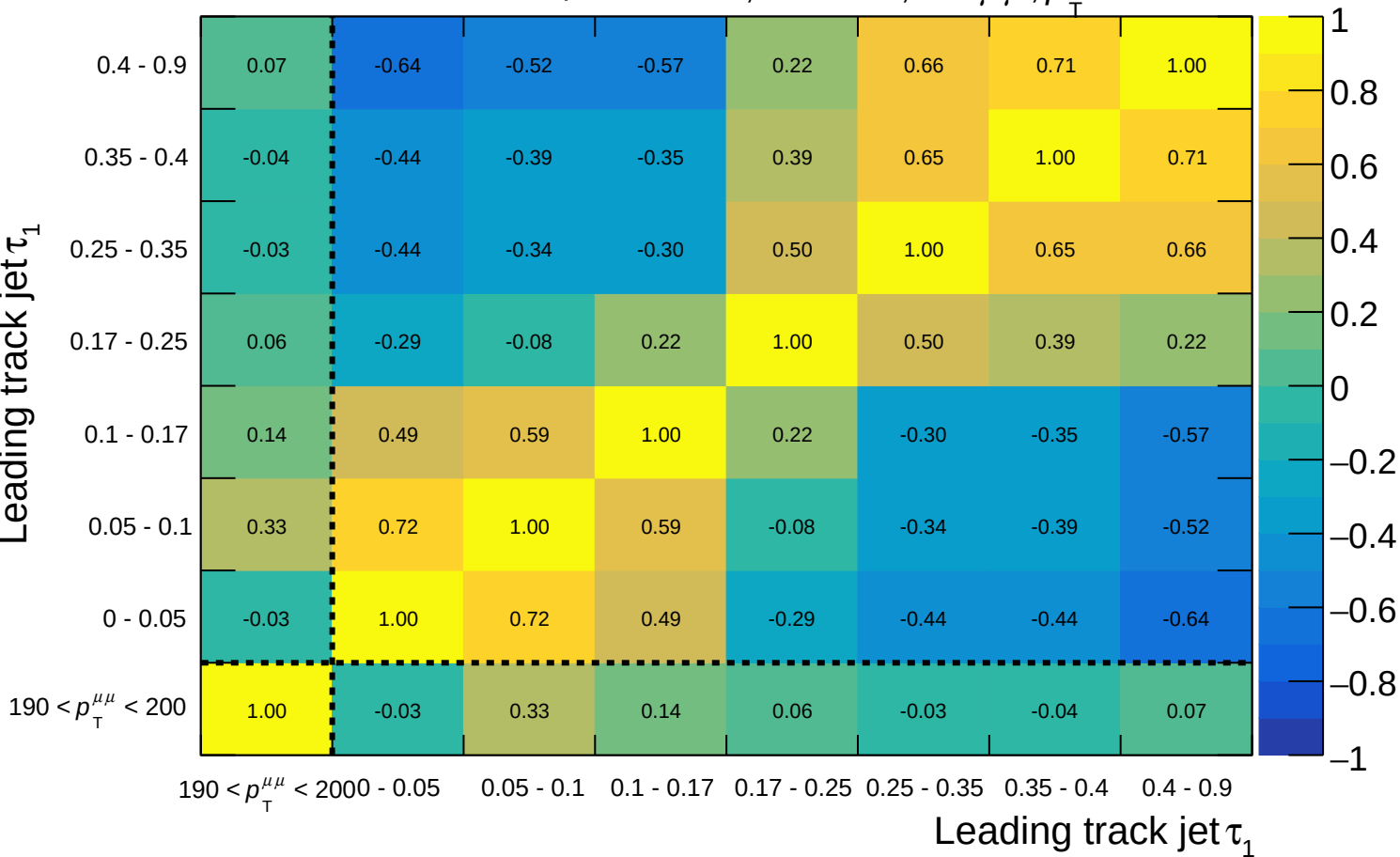
ATLAS Internal

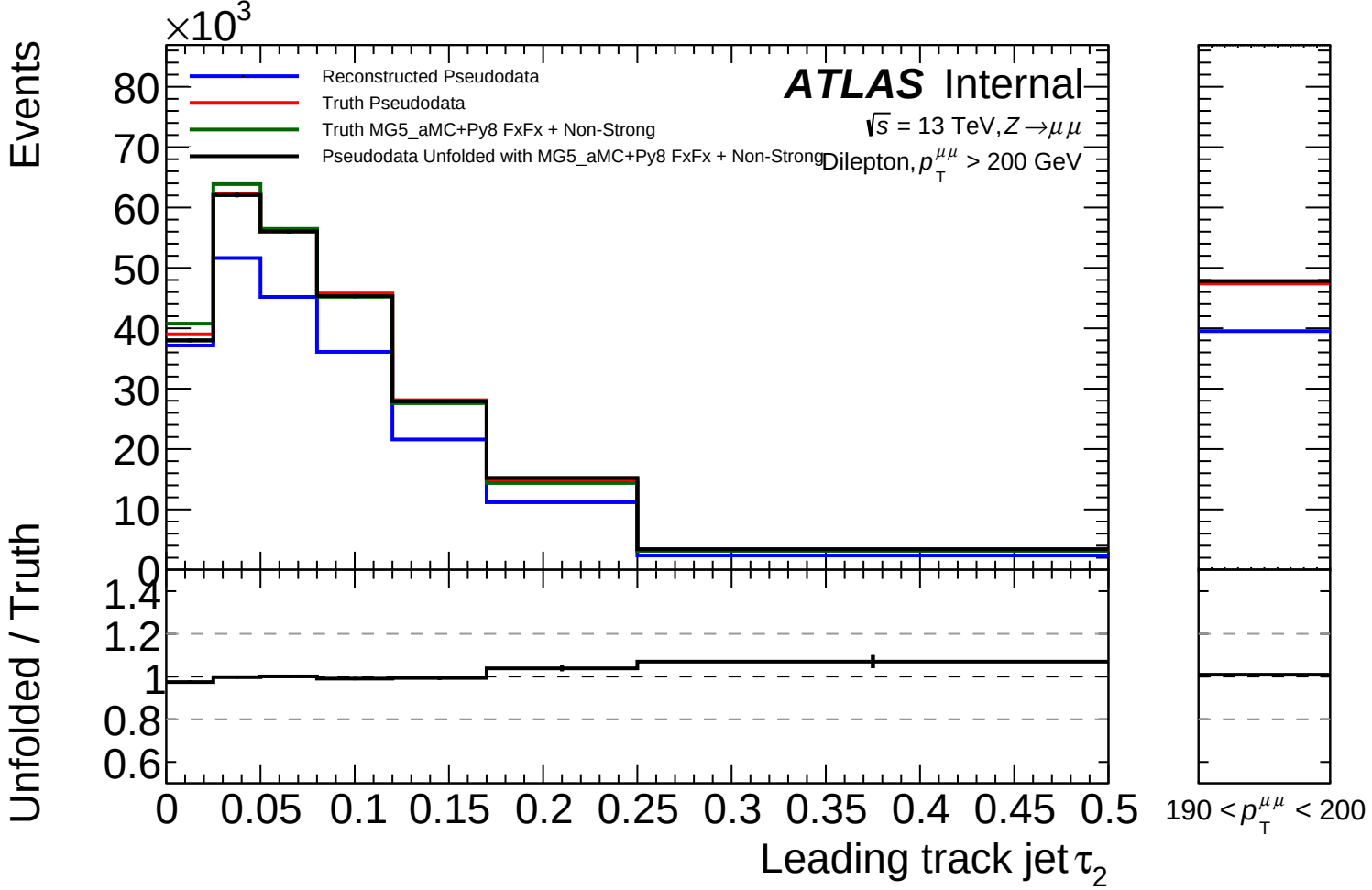
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

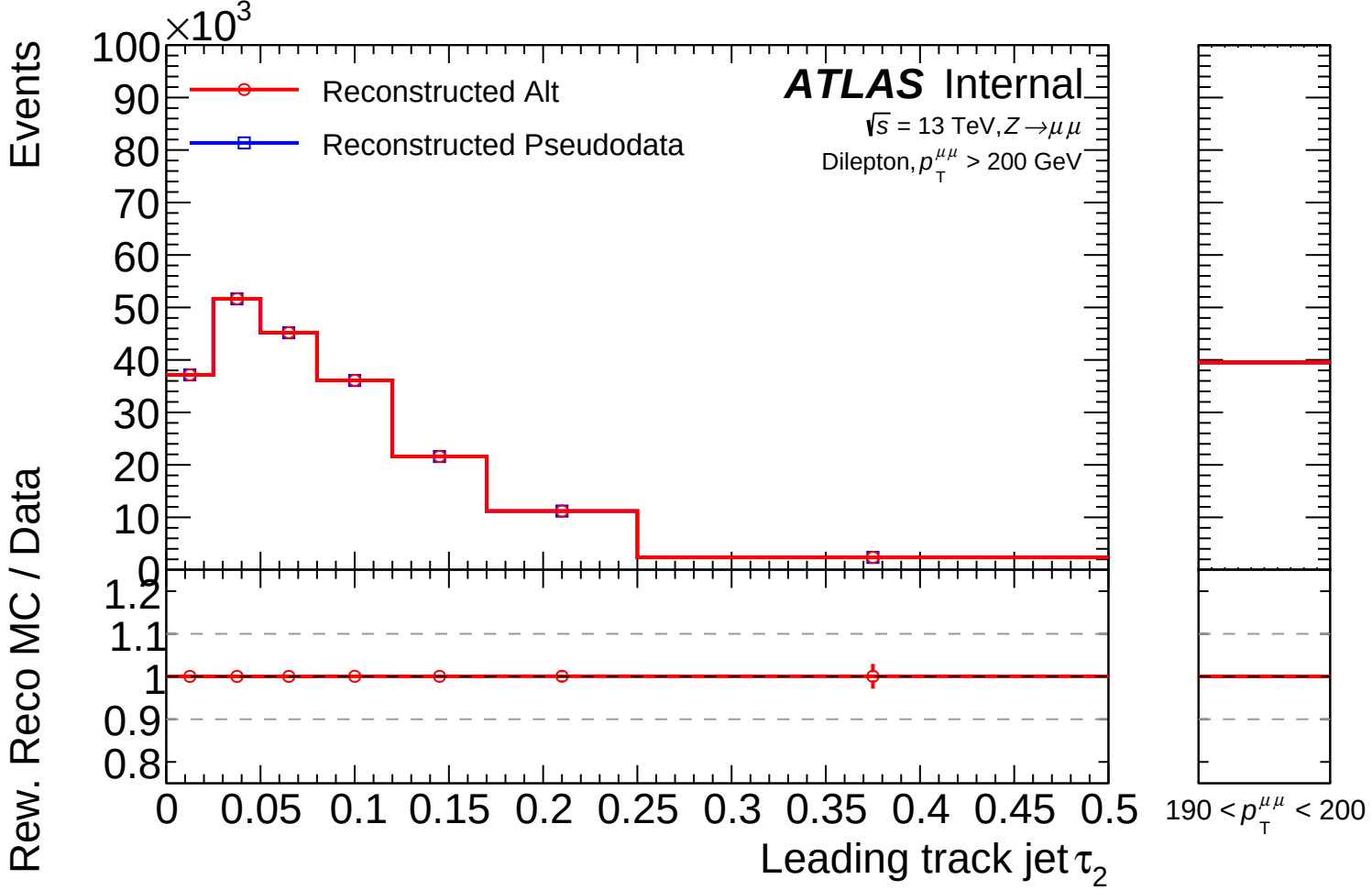
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

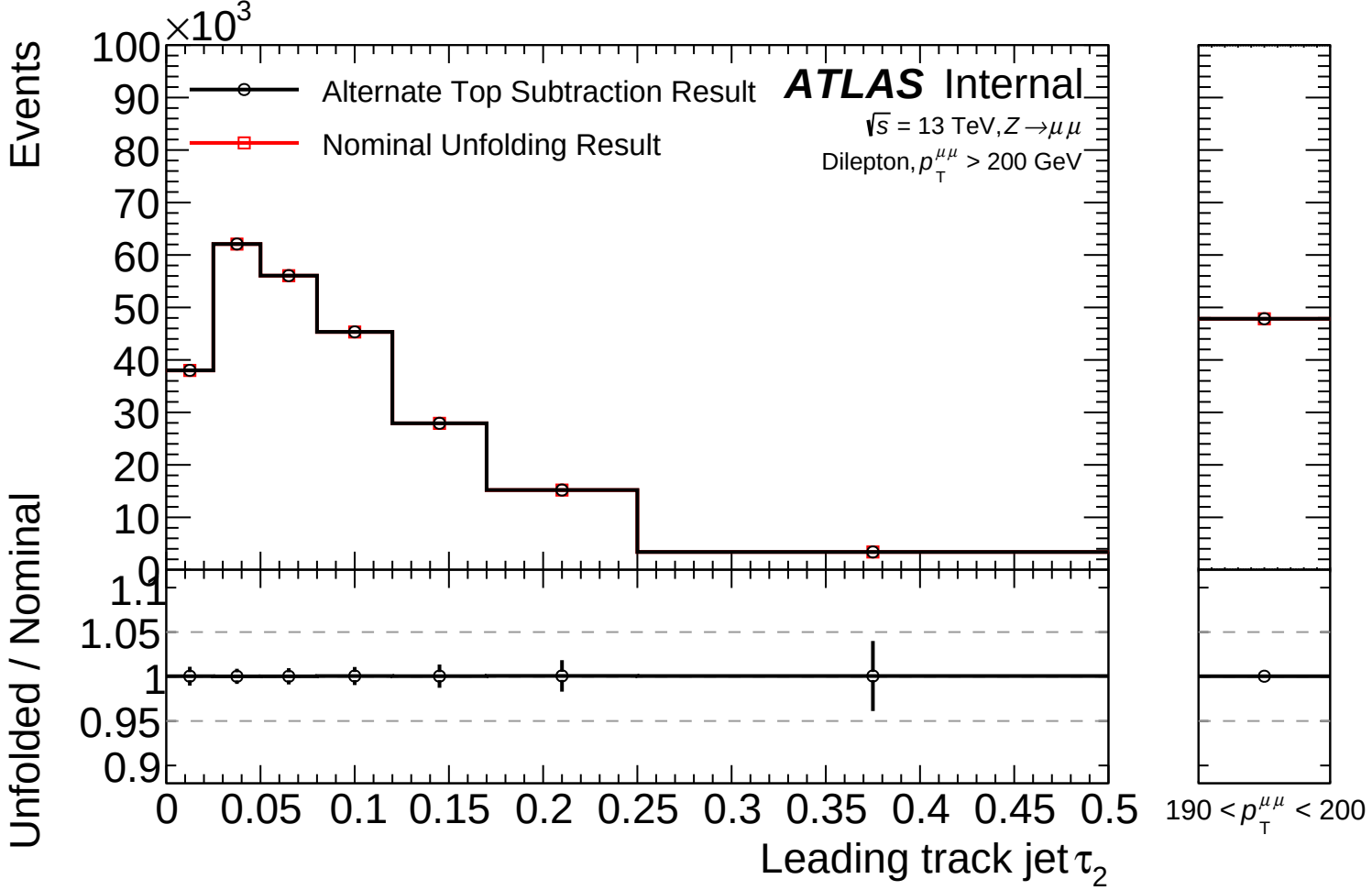
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet τ_1 







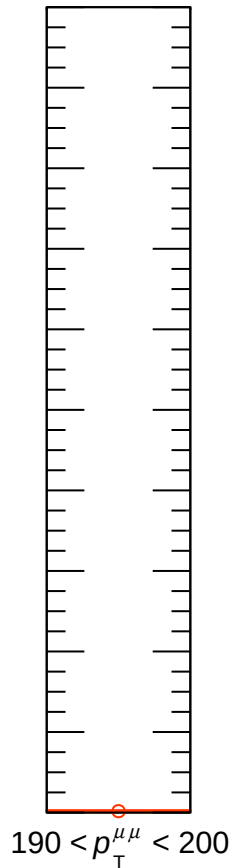
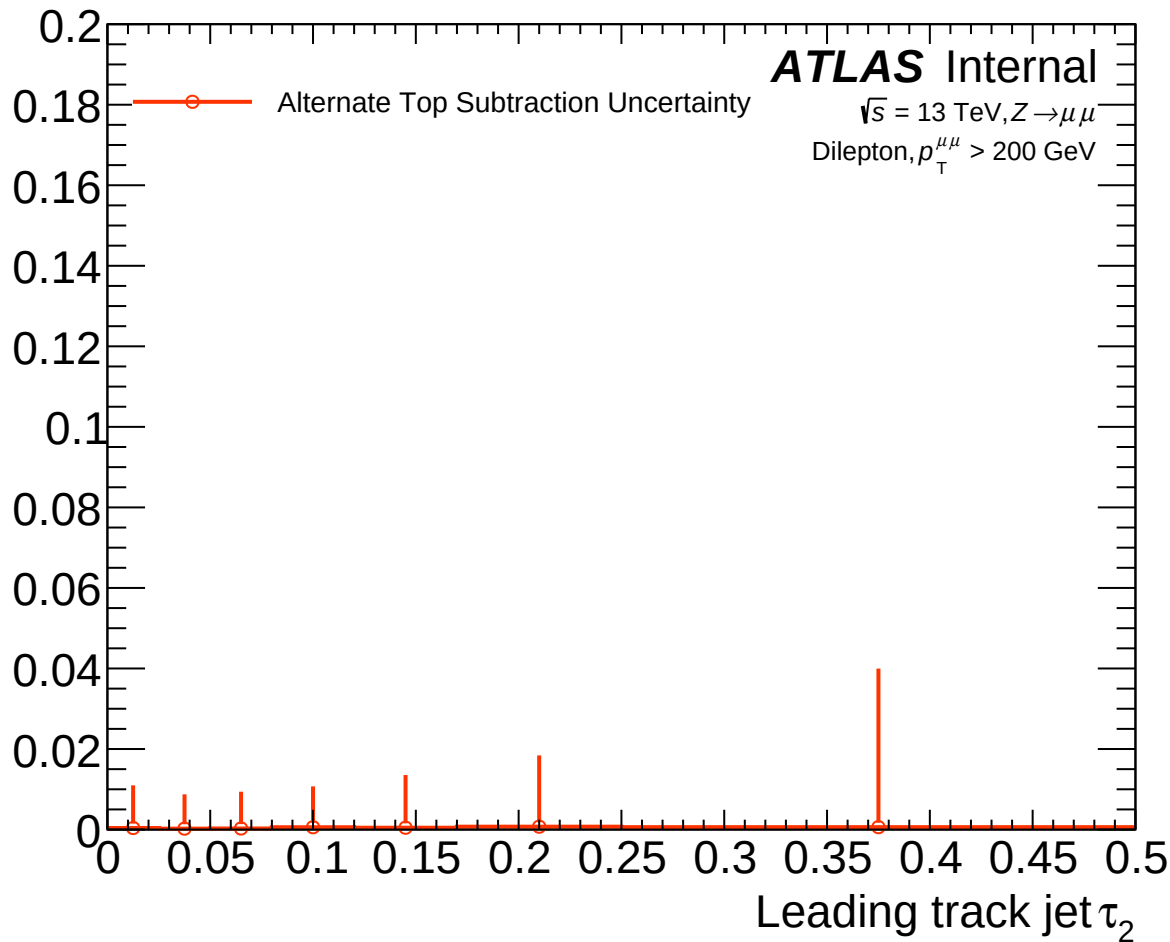
Fractional systematic impact

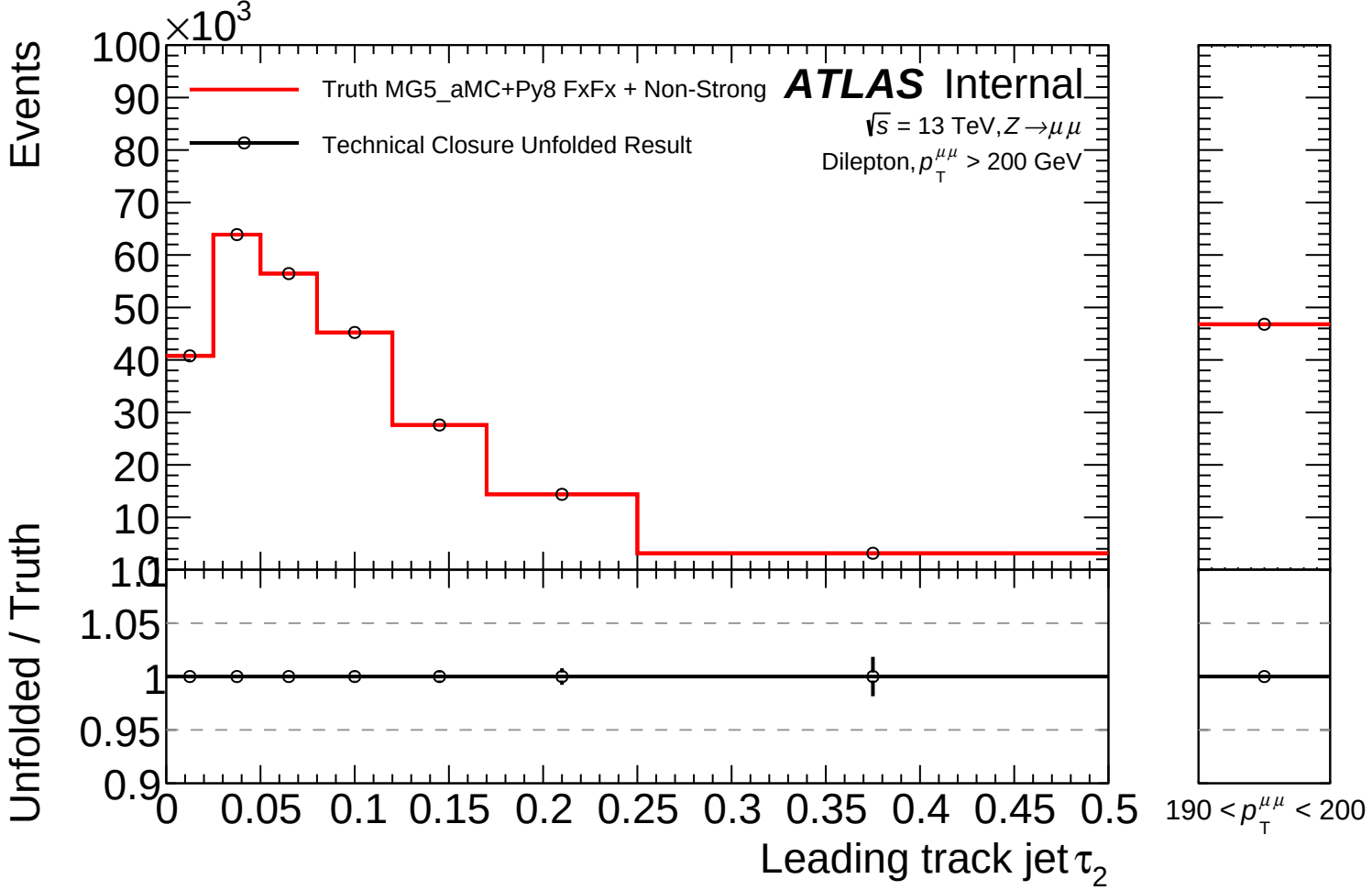
ATLAS Internal

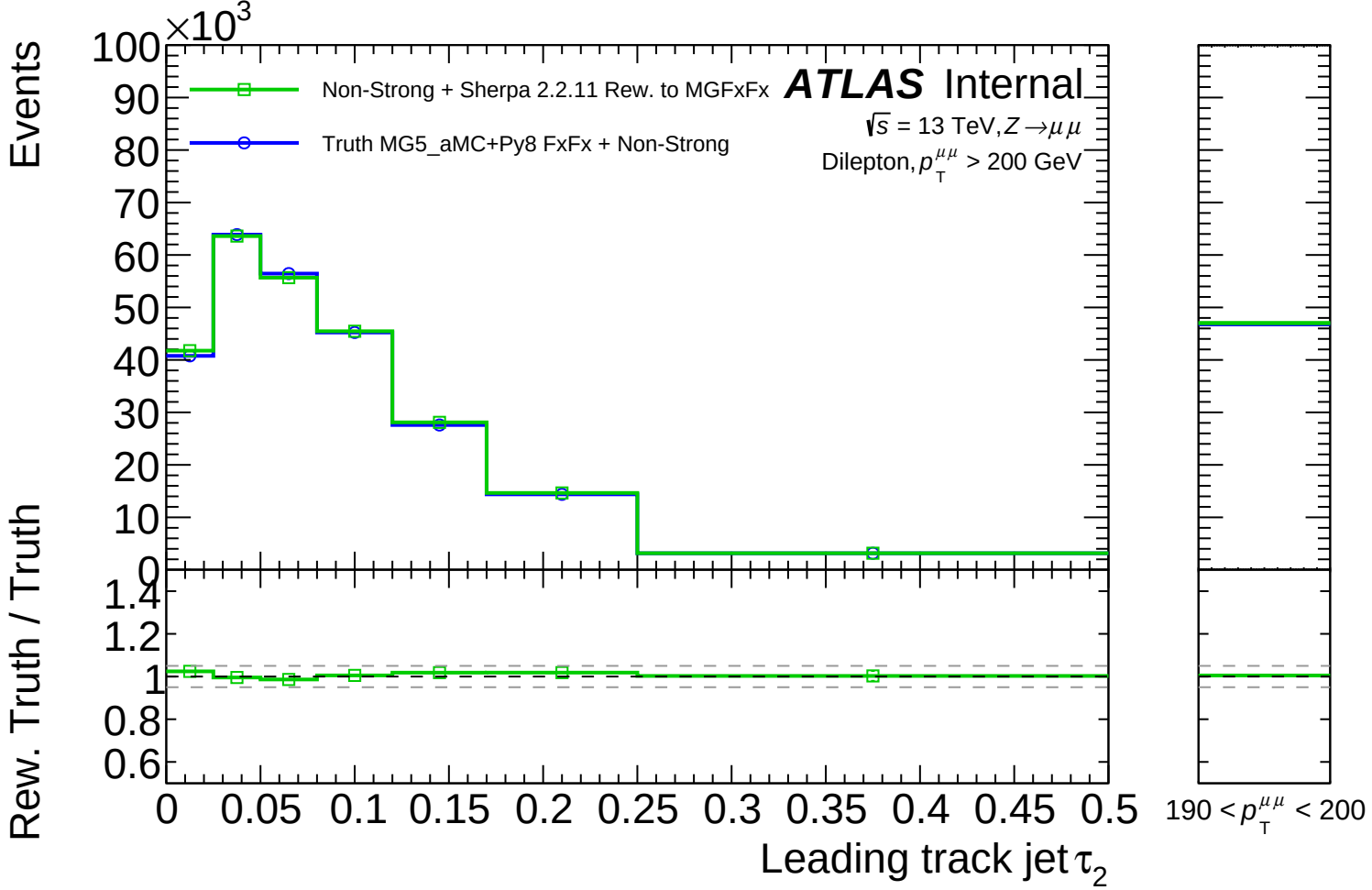
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

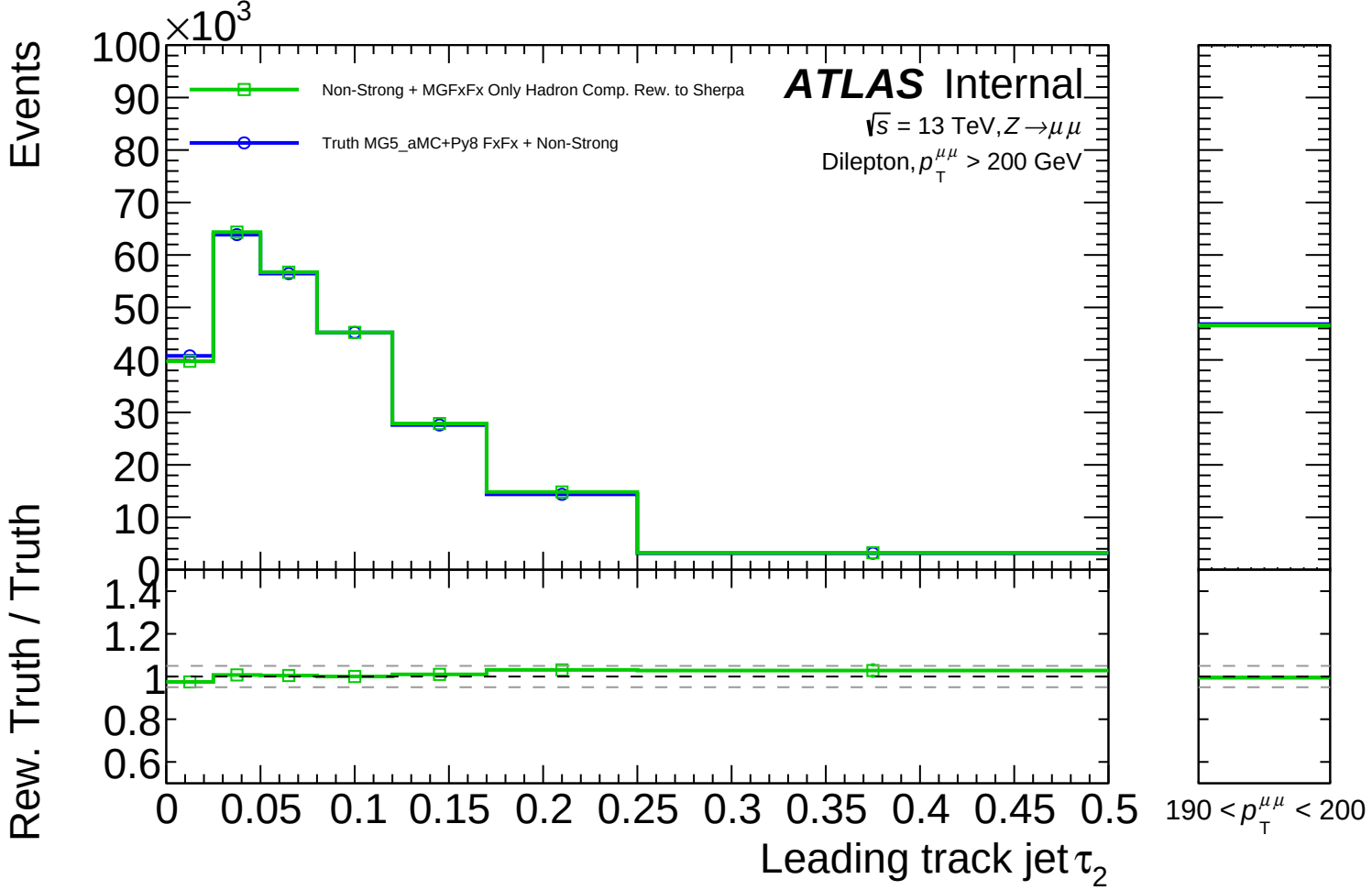
Dilepton, $p_T^{\mu\mu} > 200$ GeV

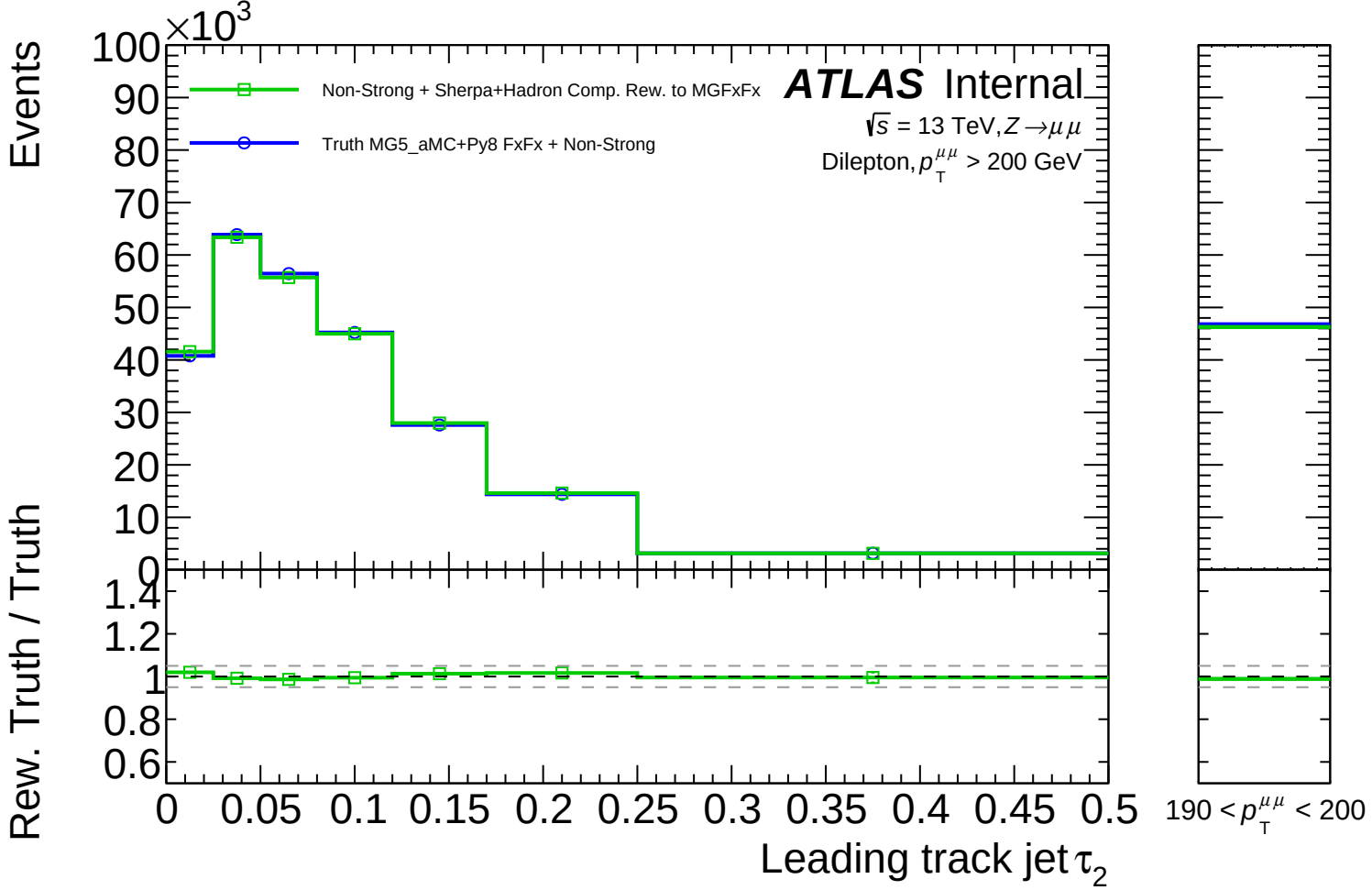
Alternate Top Subtraction Uncertainty

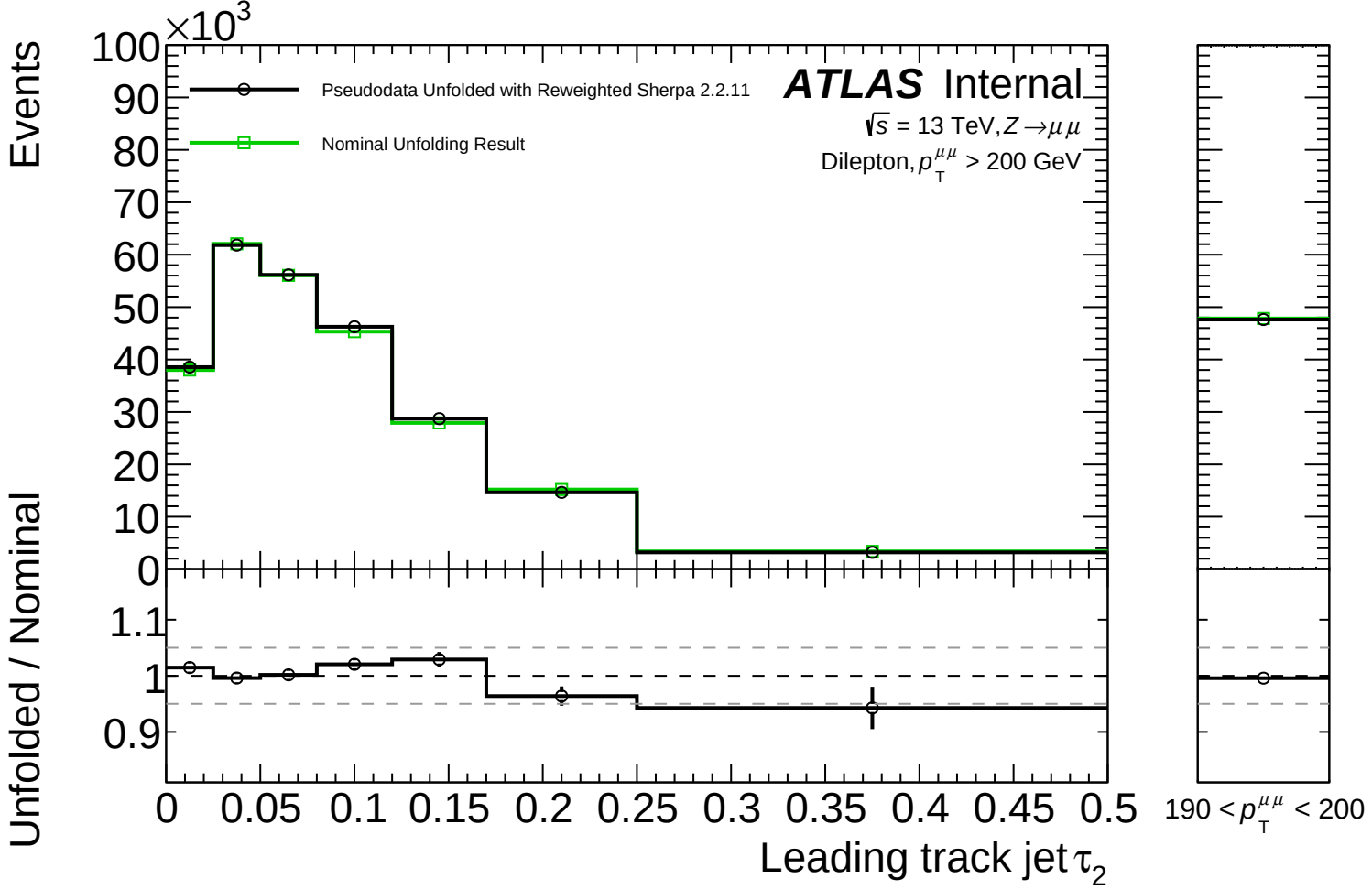


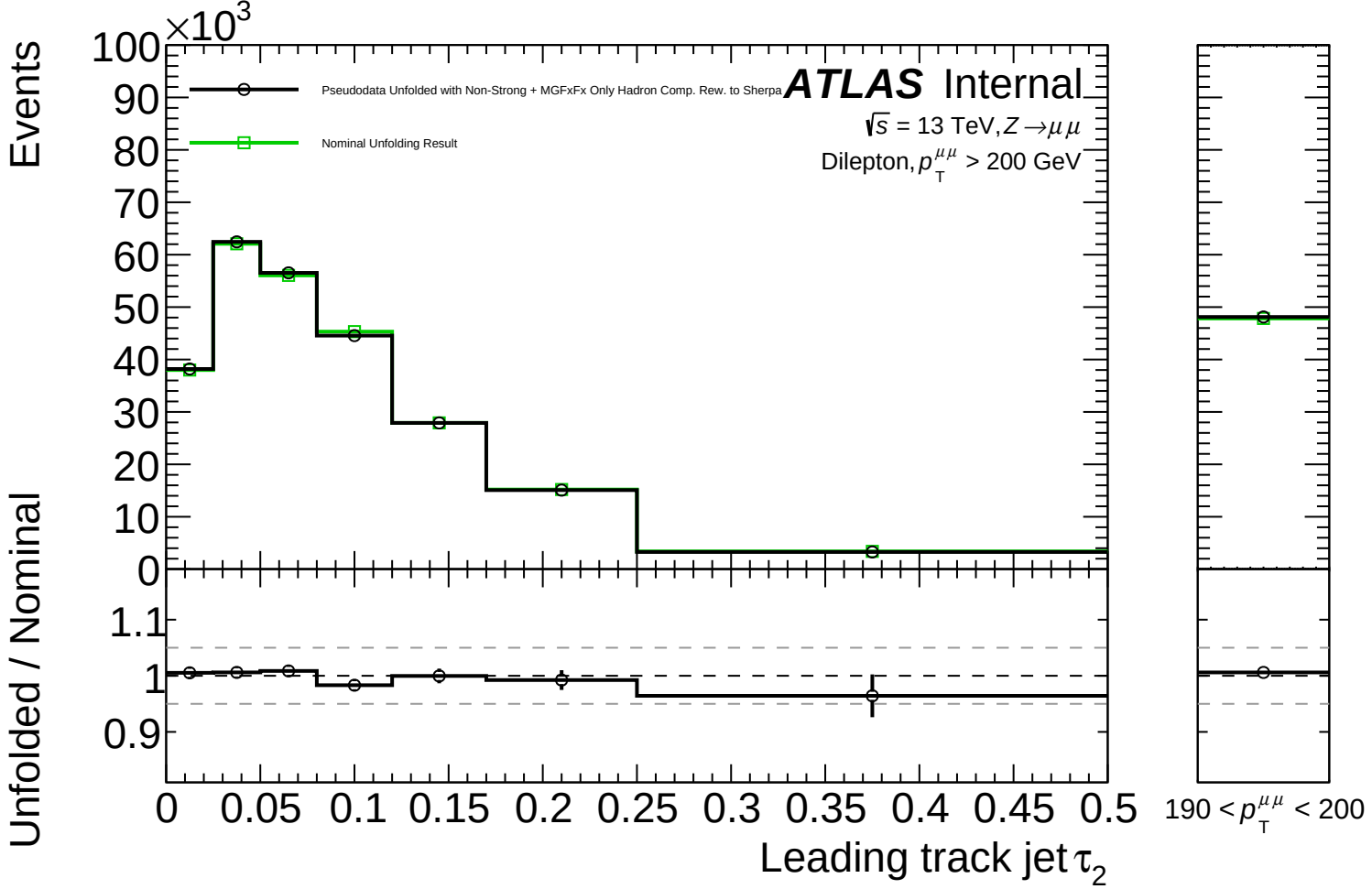


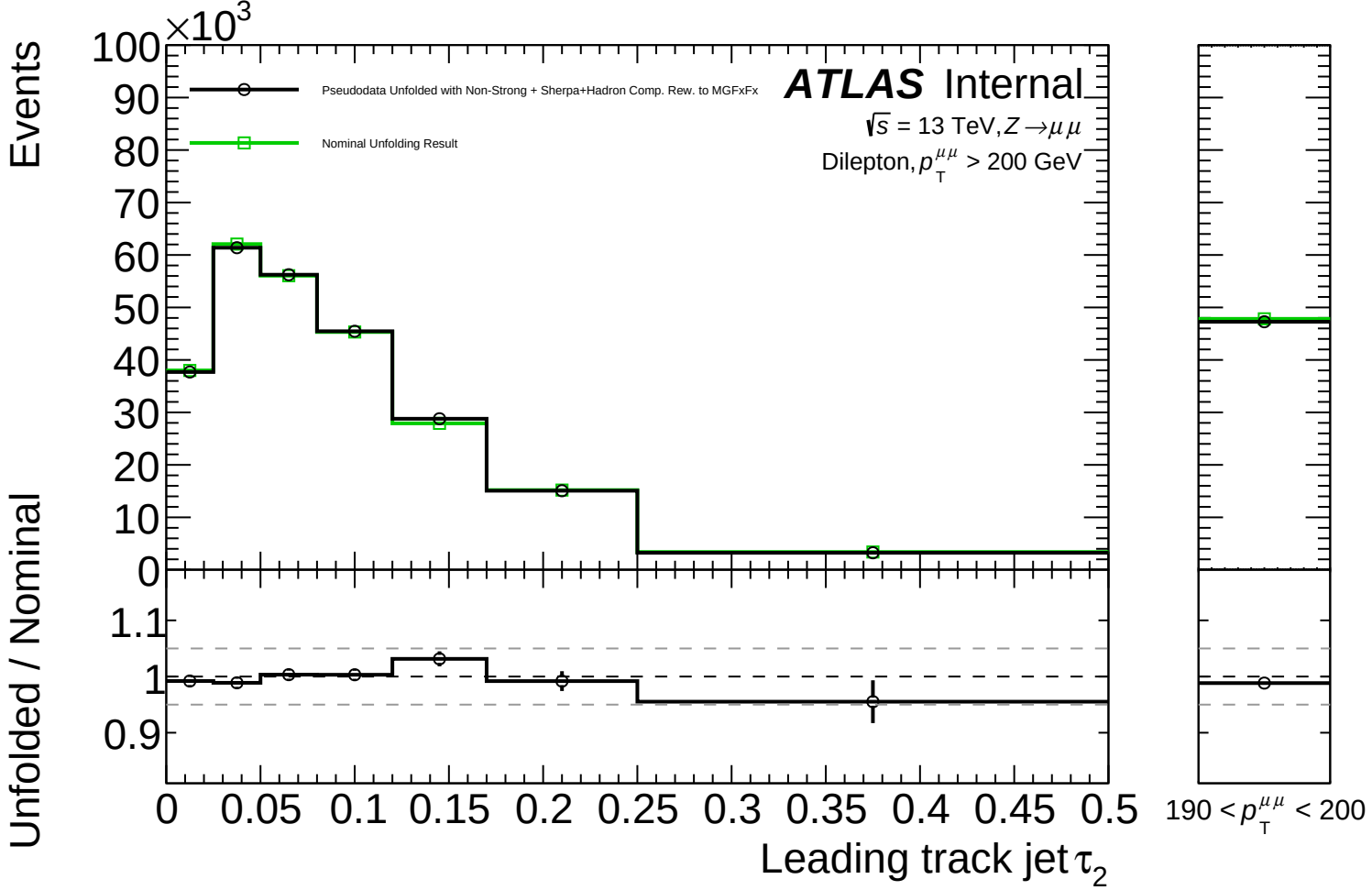


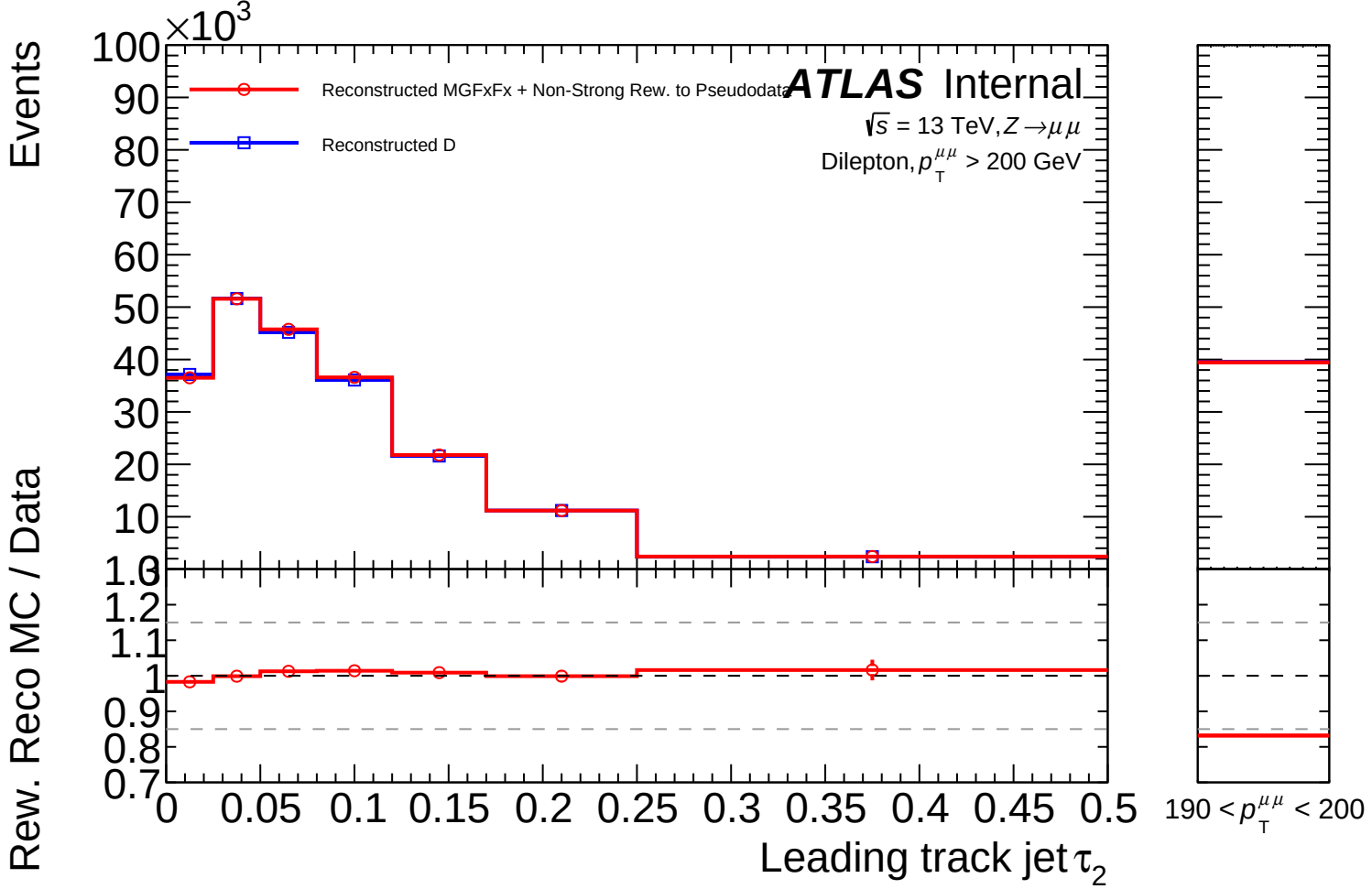


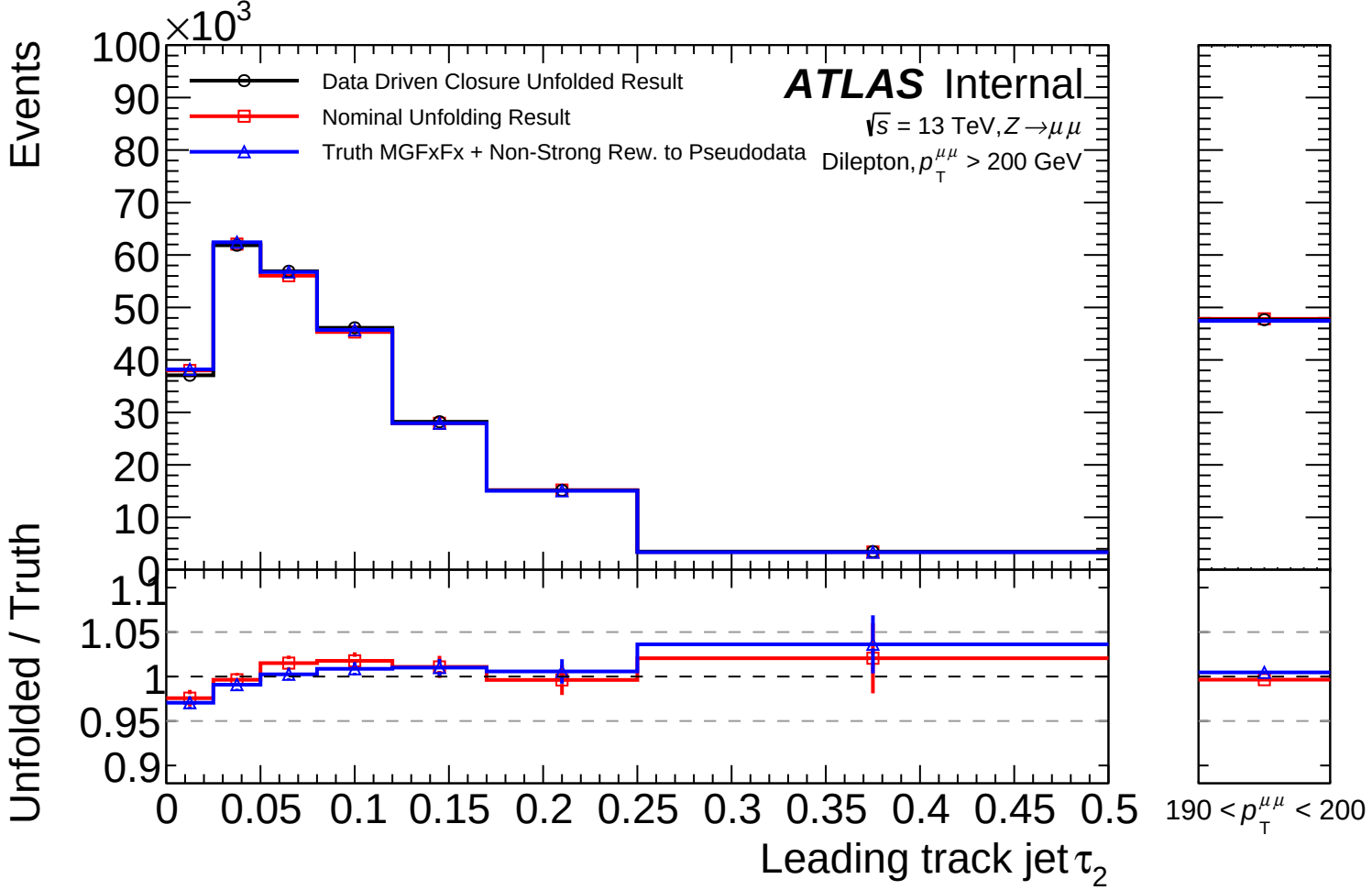




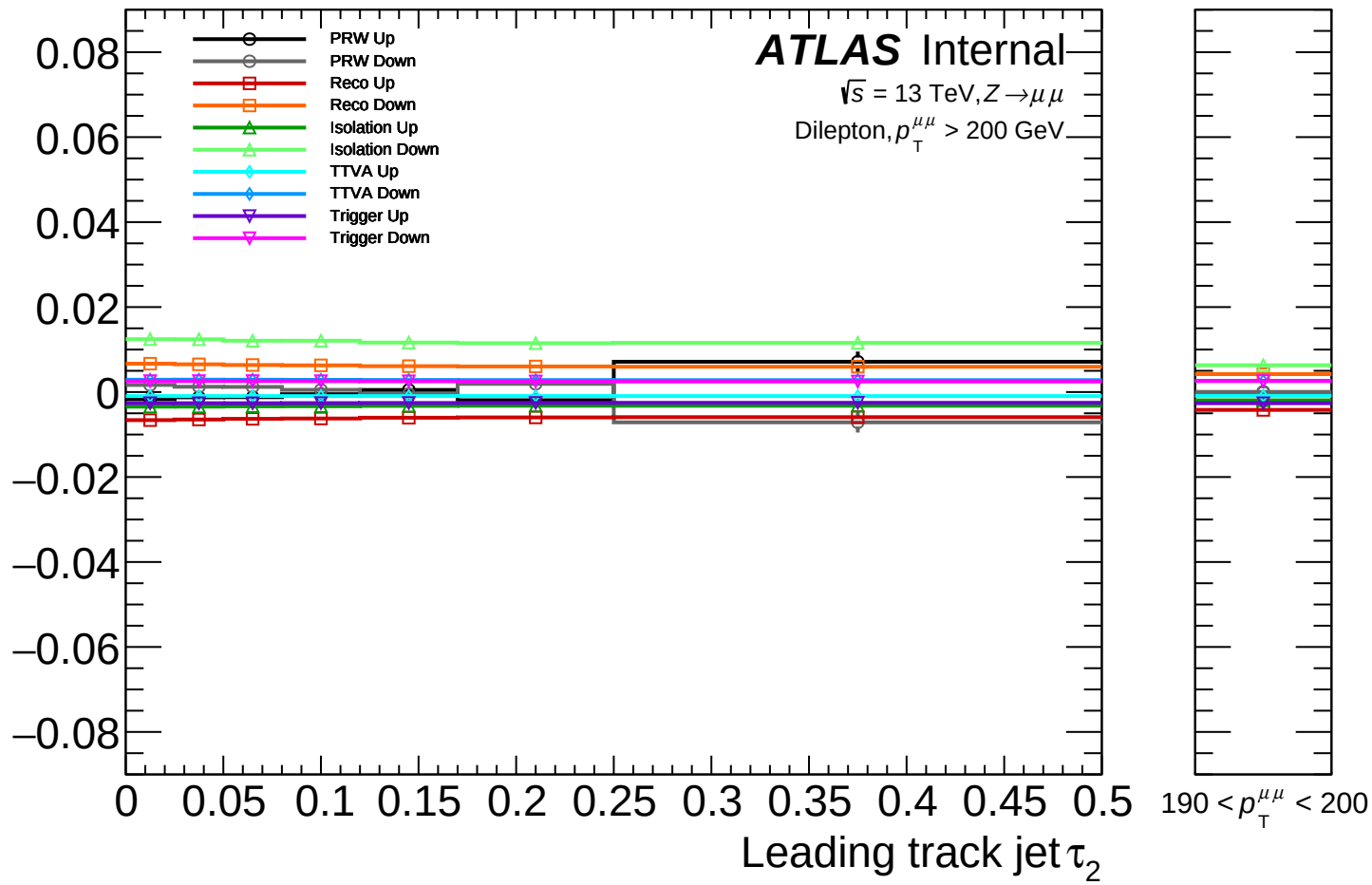




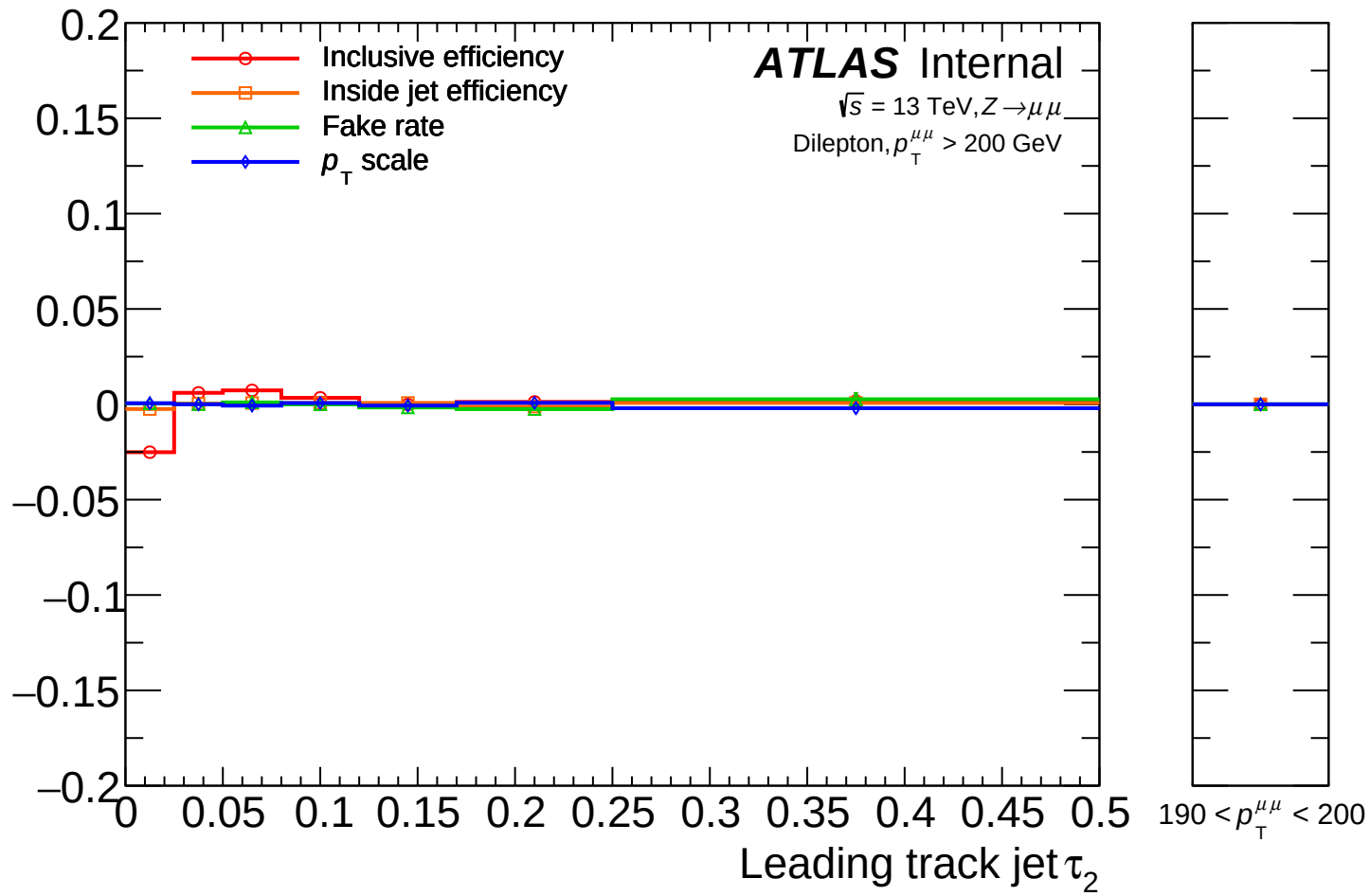




Fractional systematic impact



Fractional systematic impact



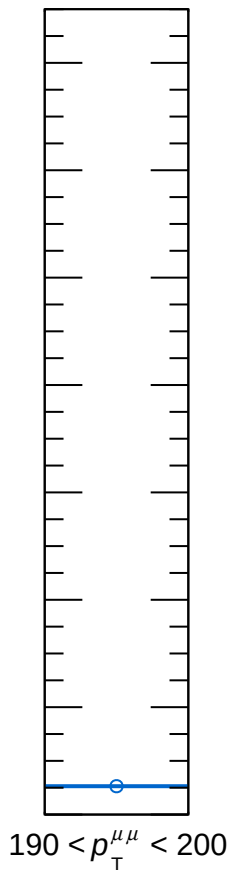
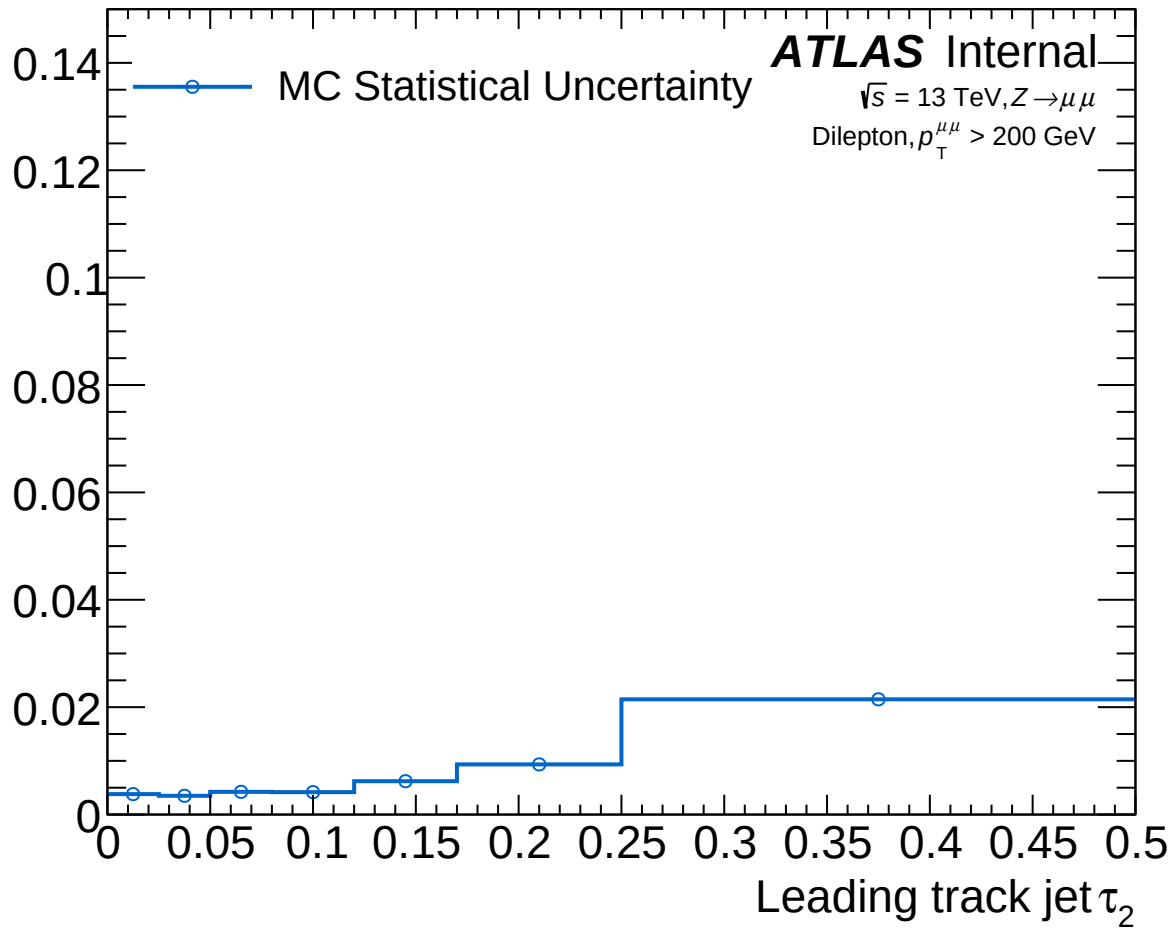
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



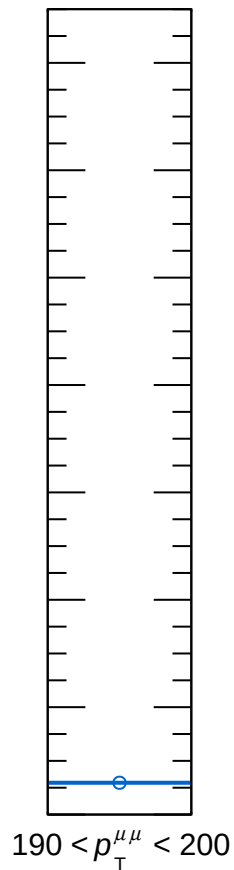
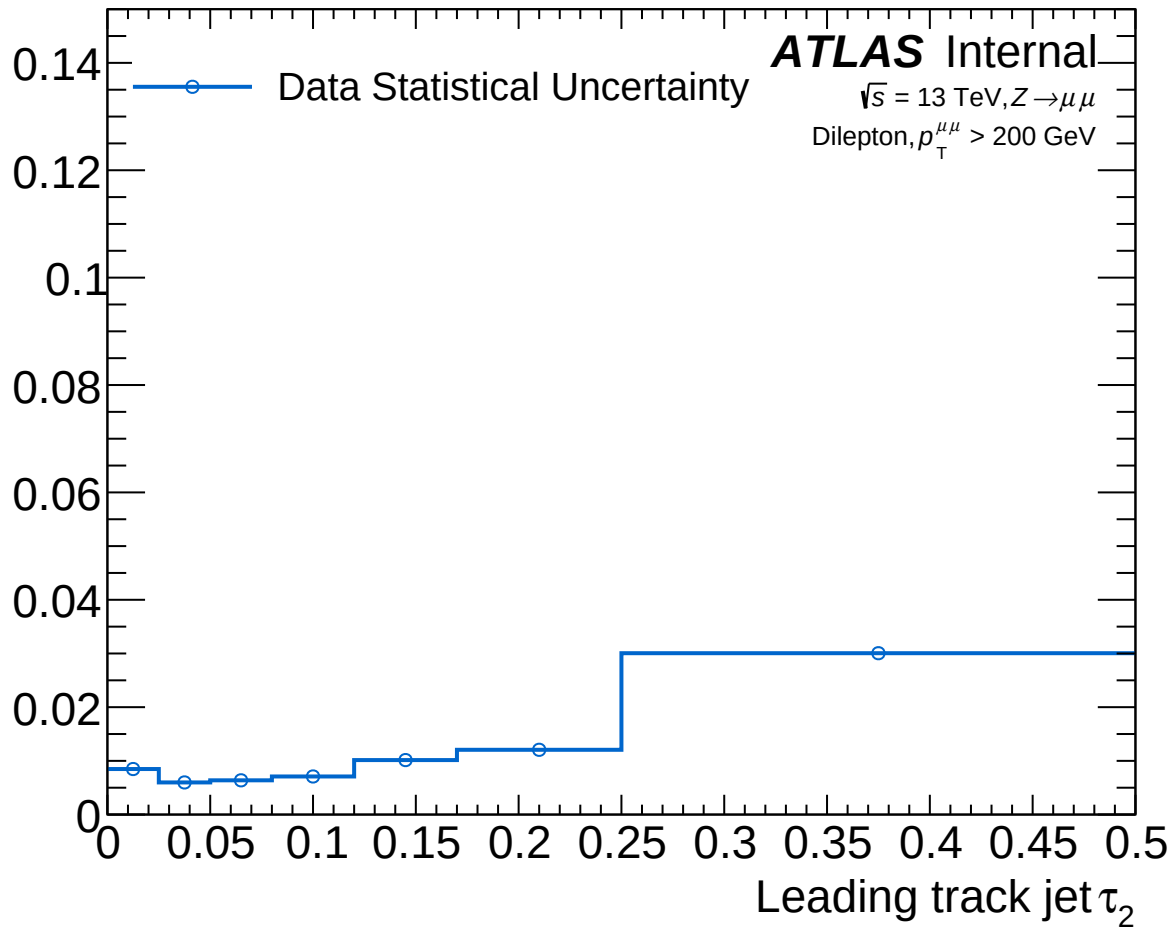
Fractional systematic impact

ATLAS Internal

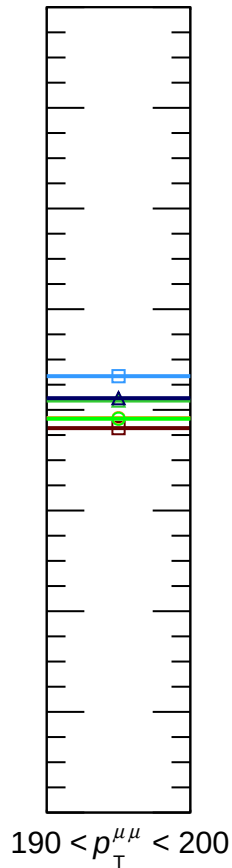
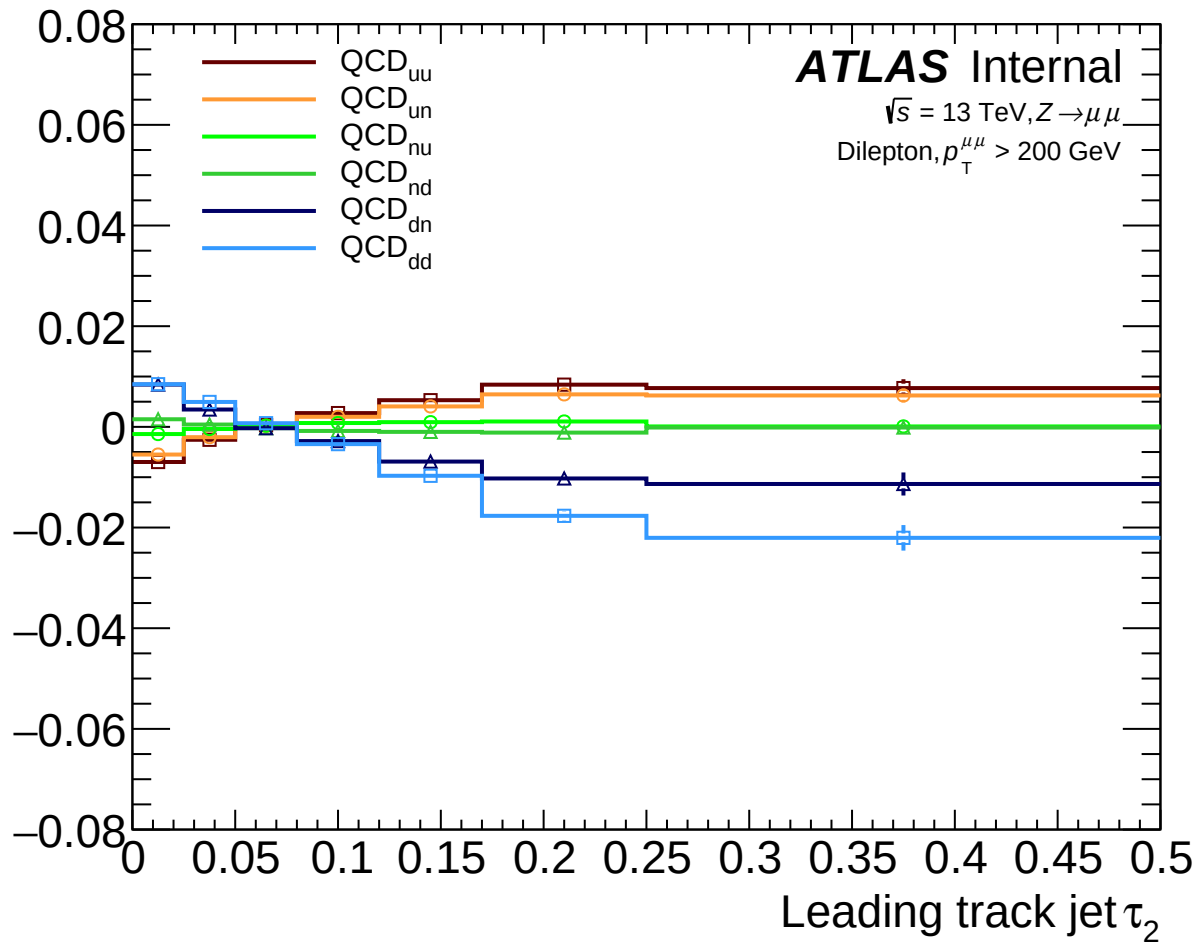
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

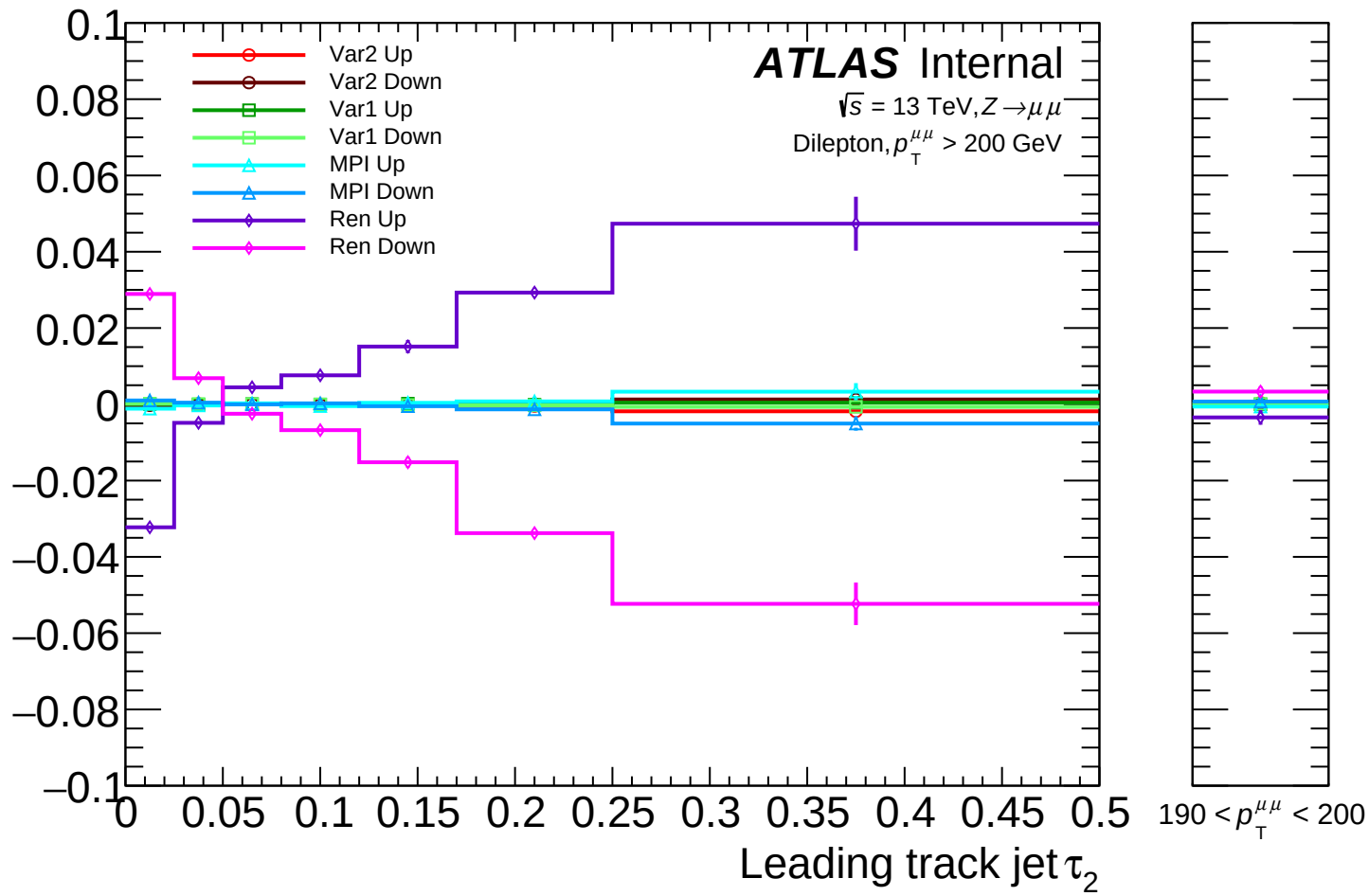
Data Statistical Uncertainty

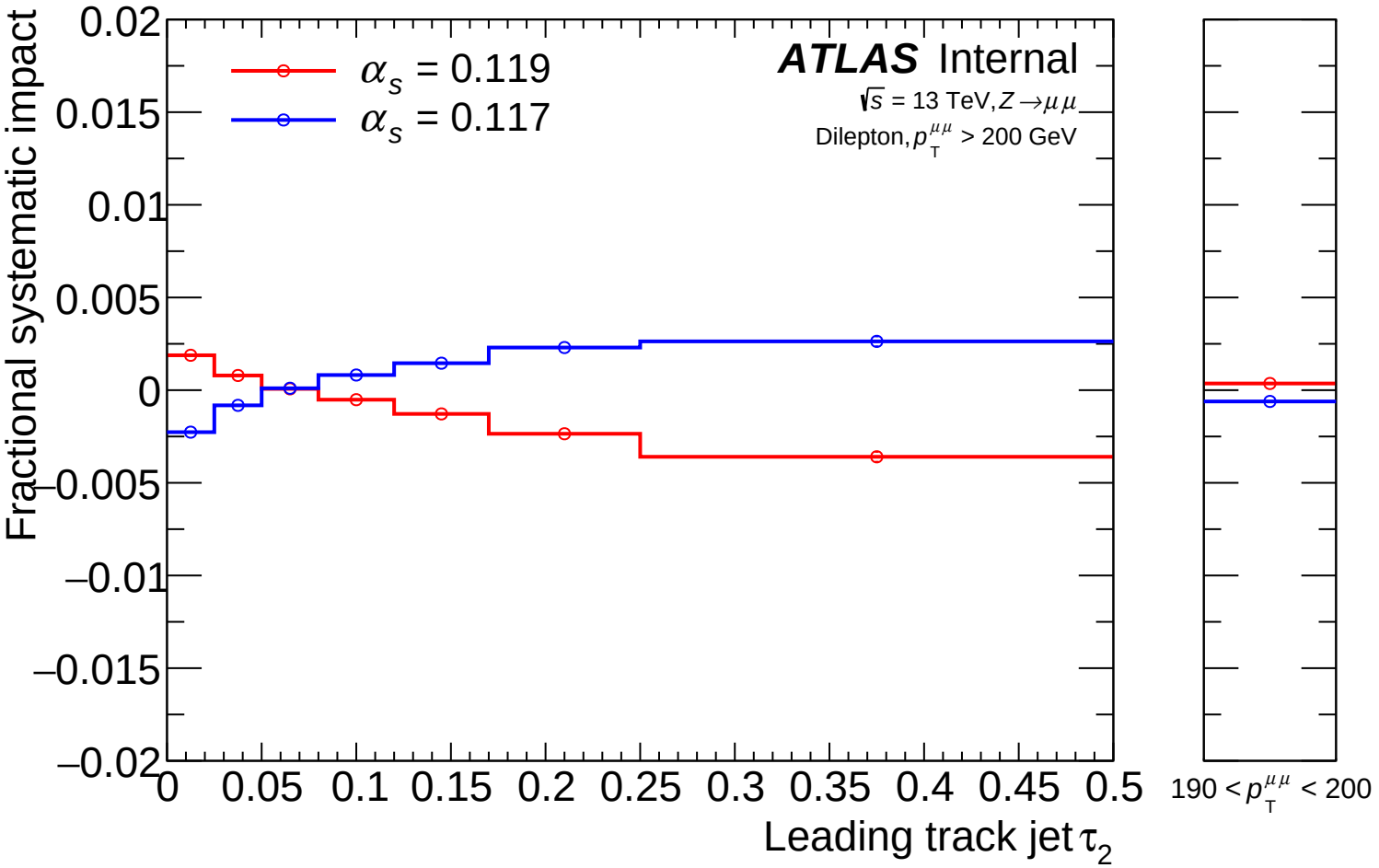


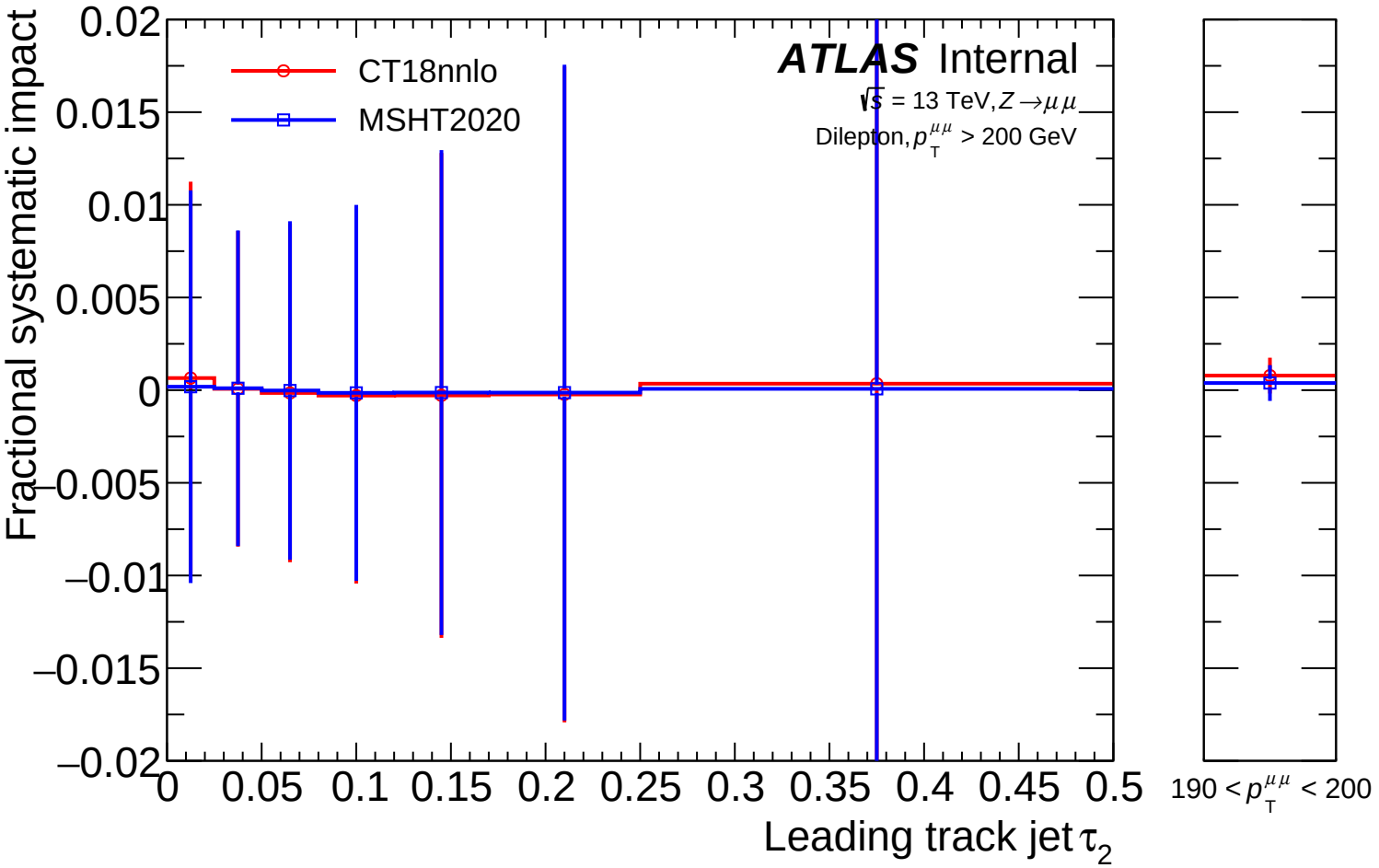
Fractional systematic impact

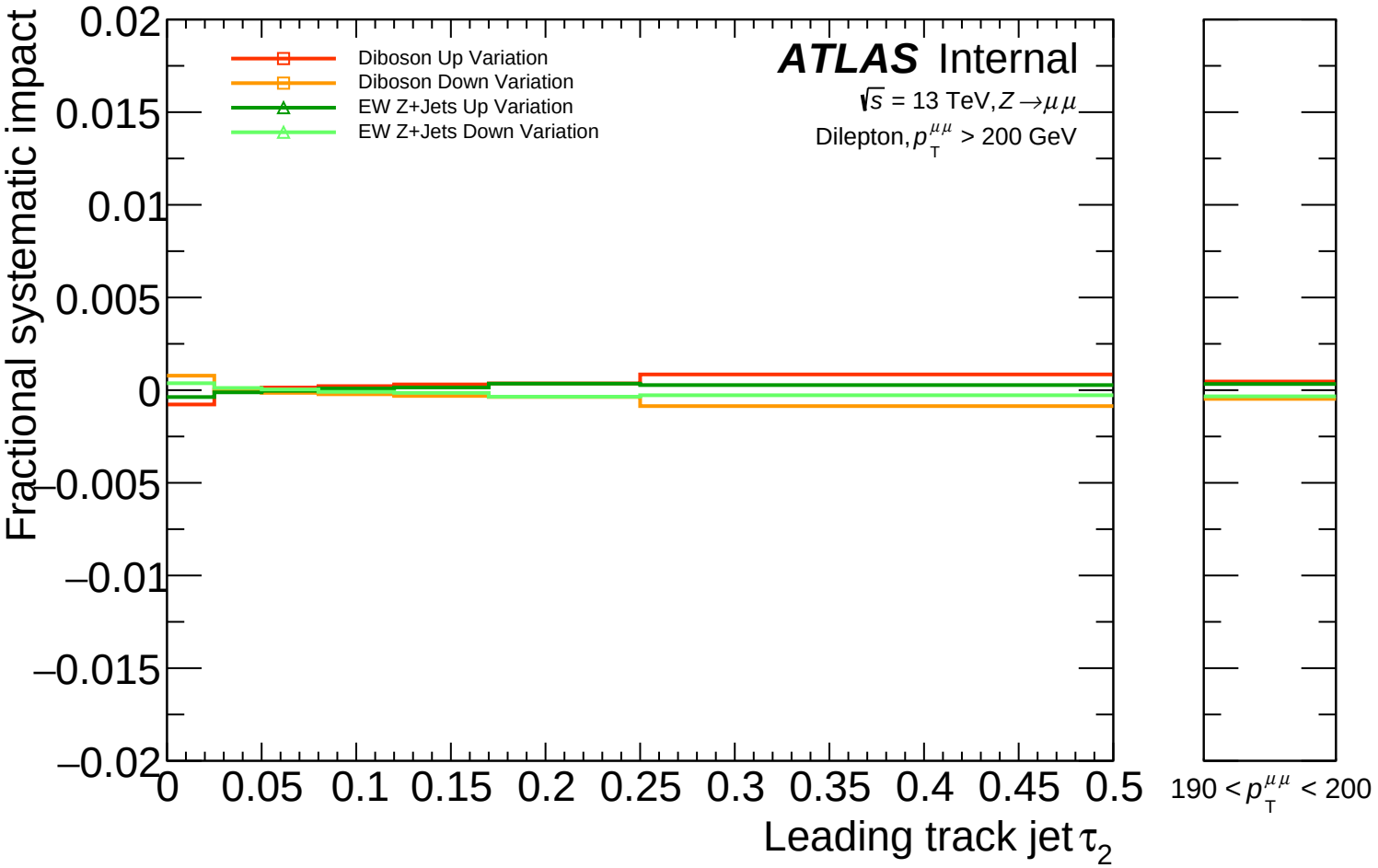


Fractional systematic impact





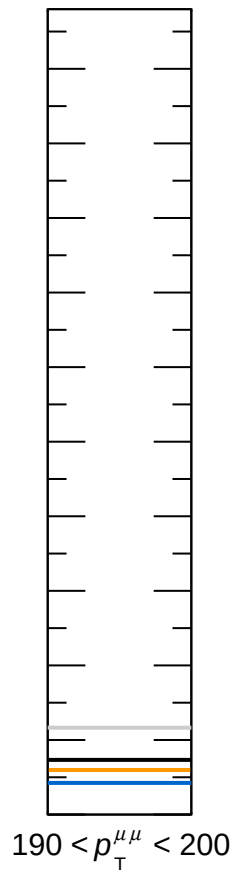
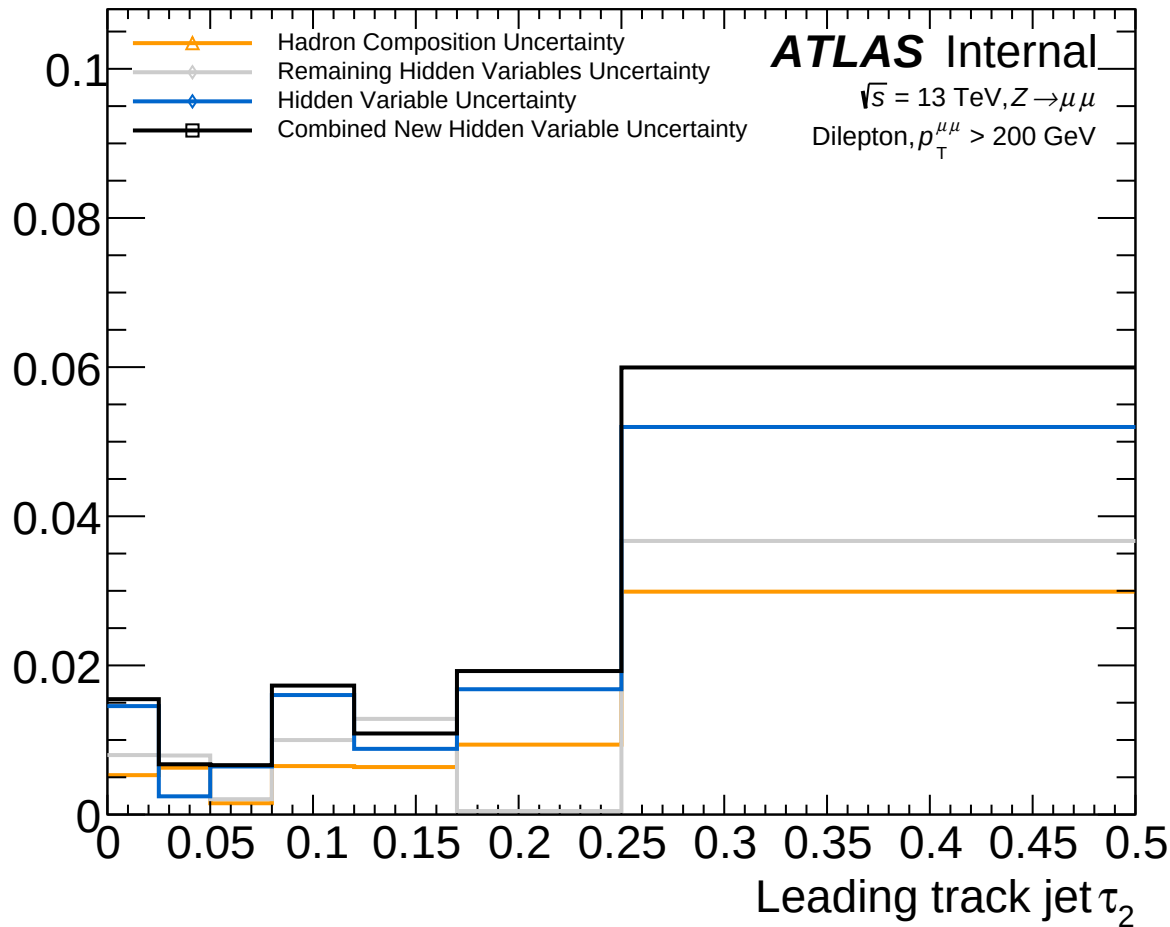


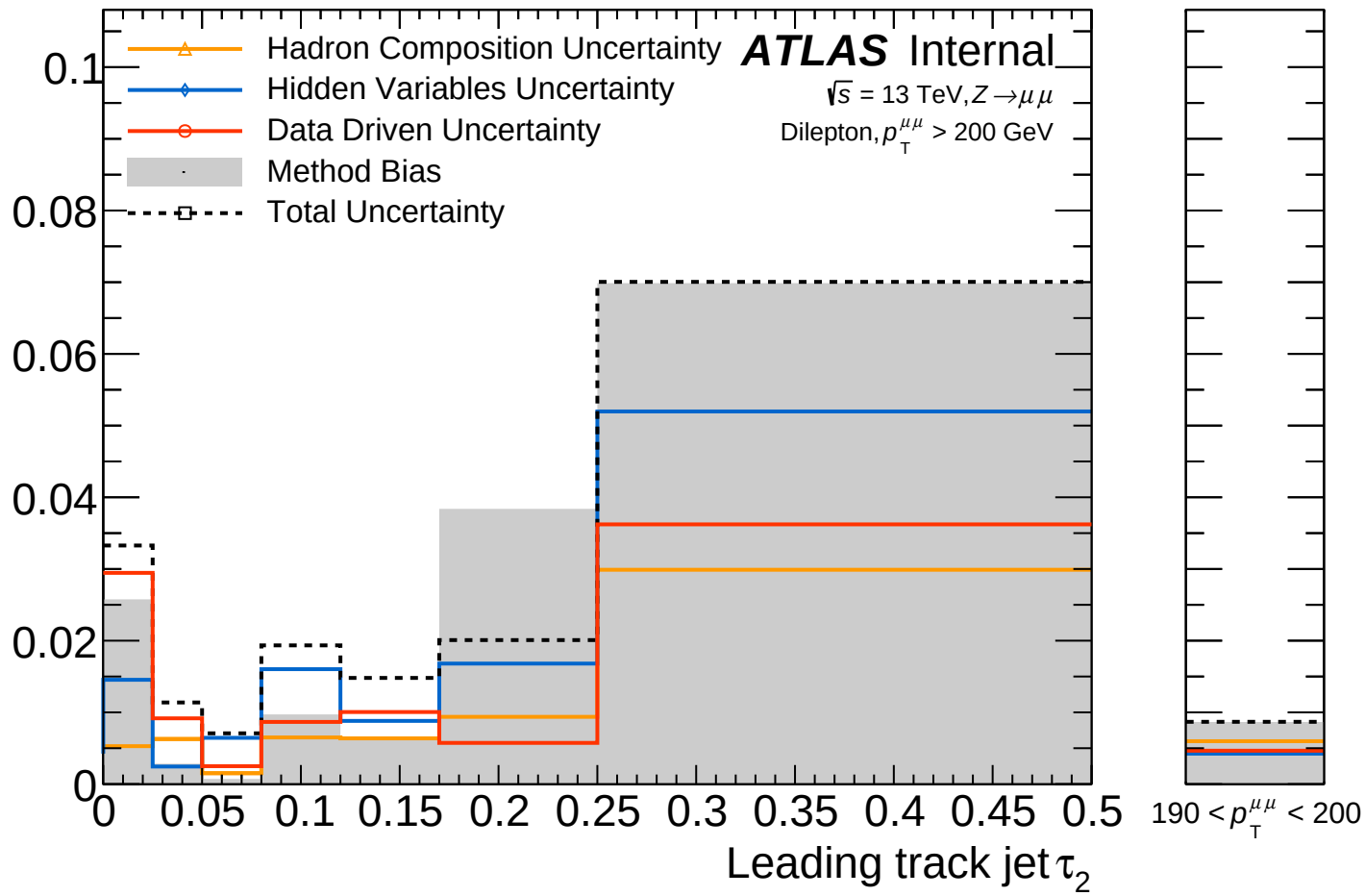


ATLAS Internal

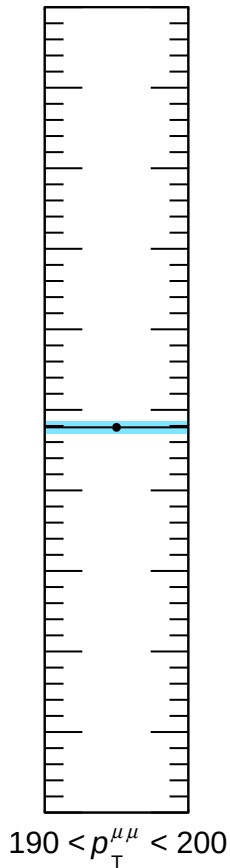
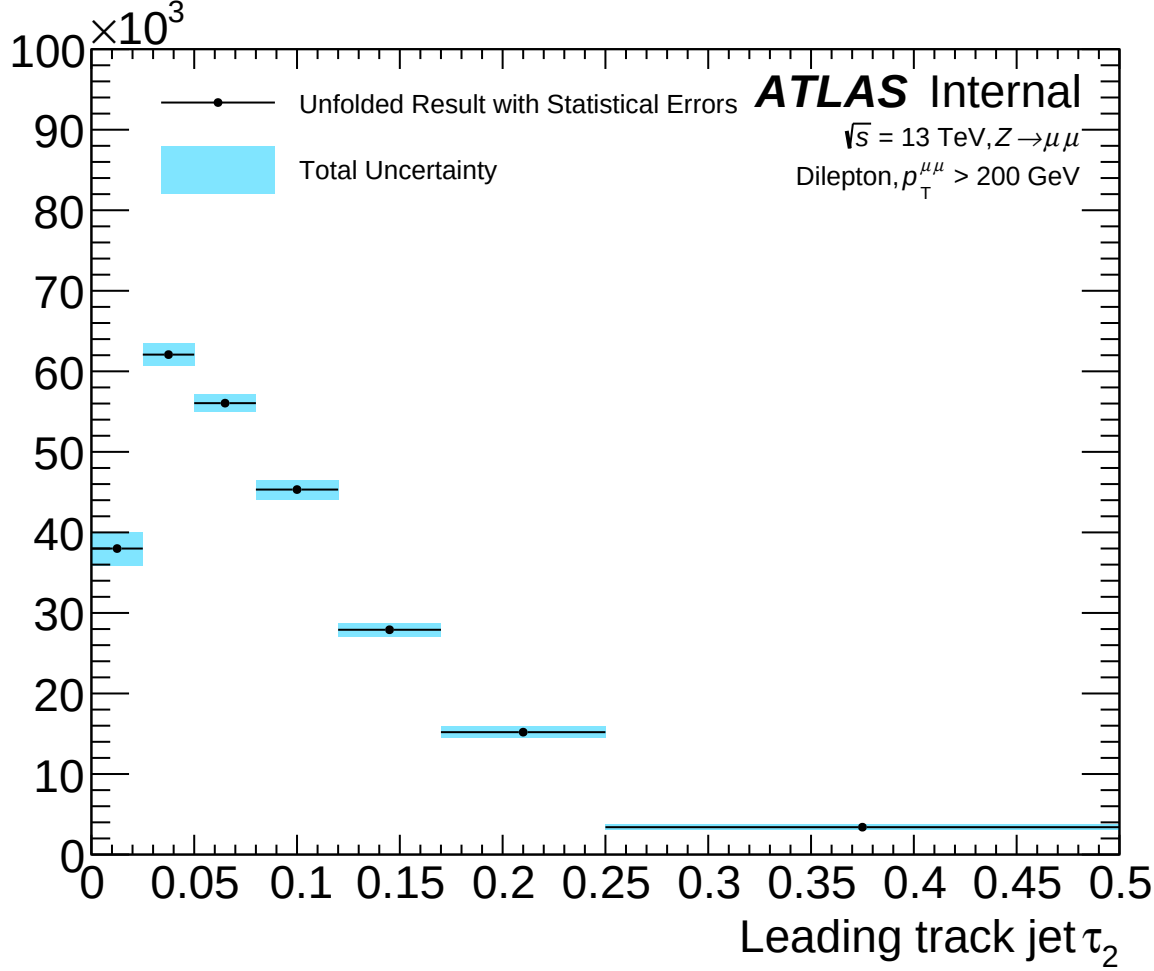
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

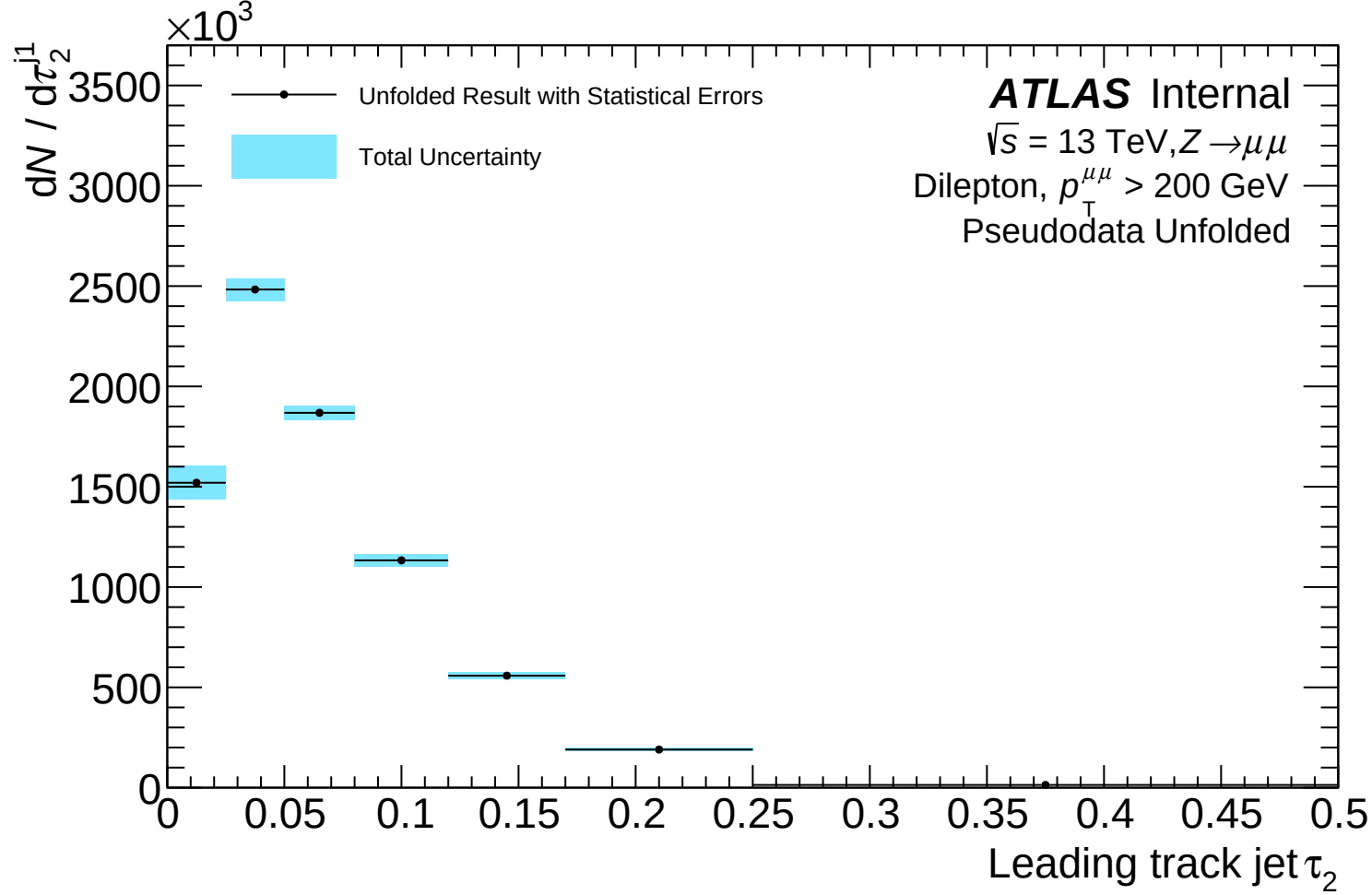
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

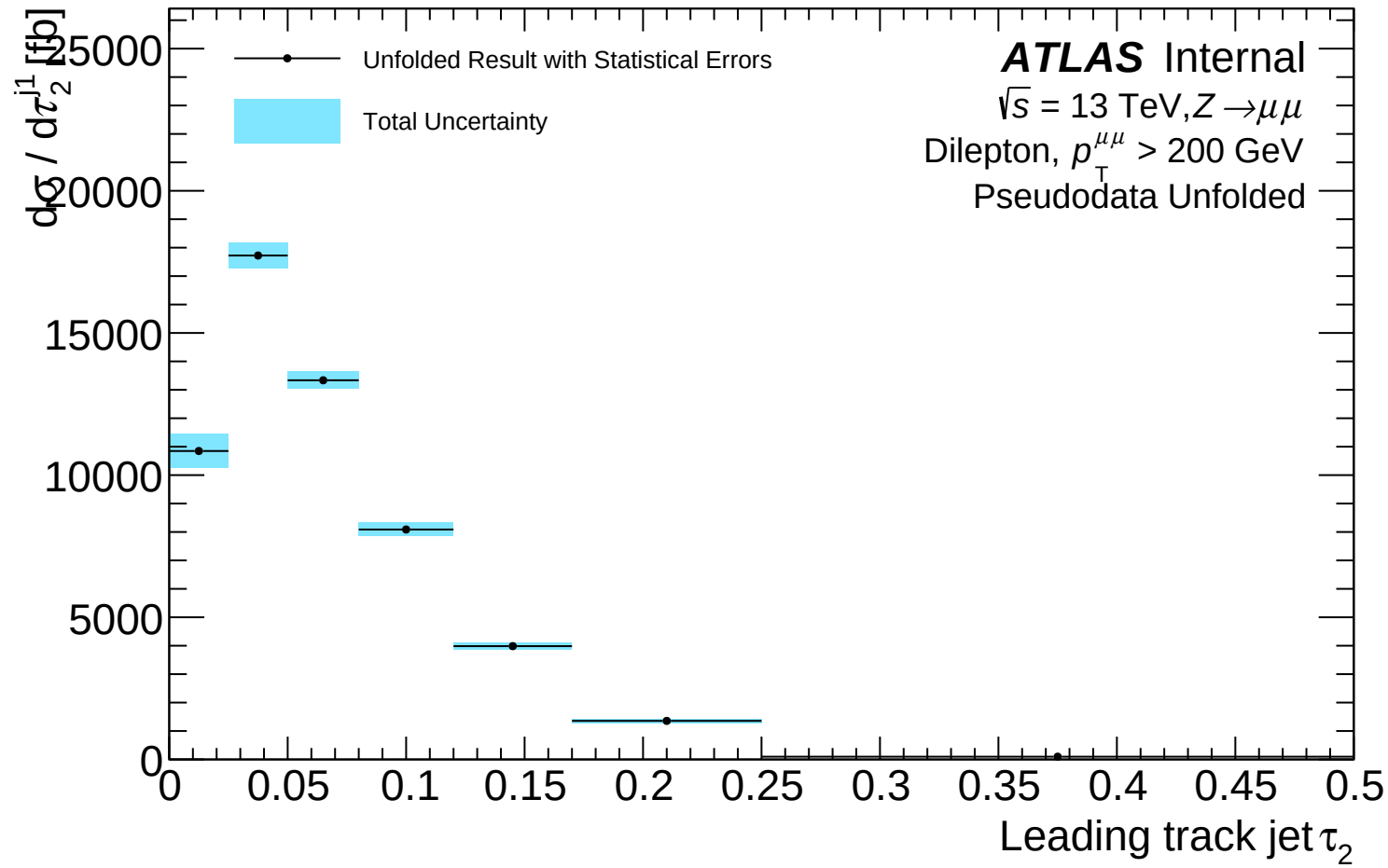




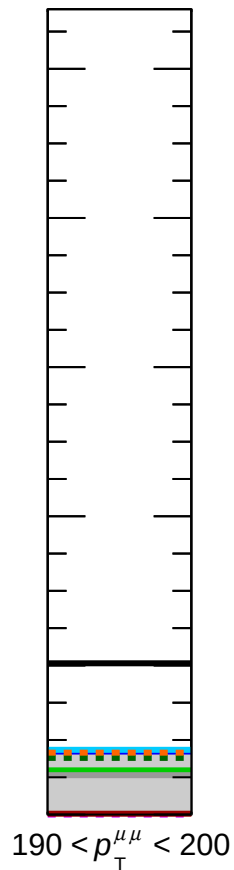
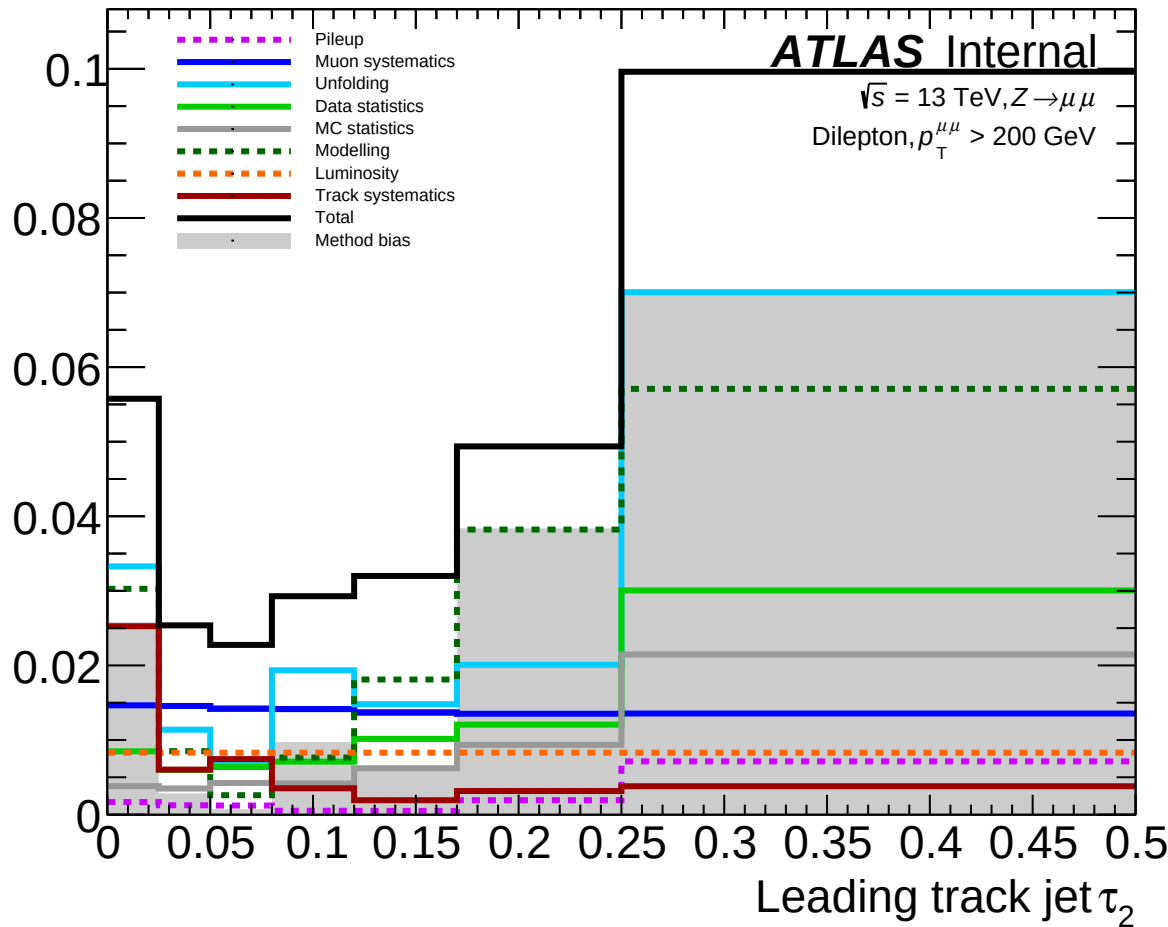
Events

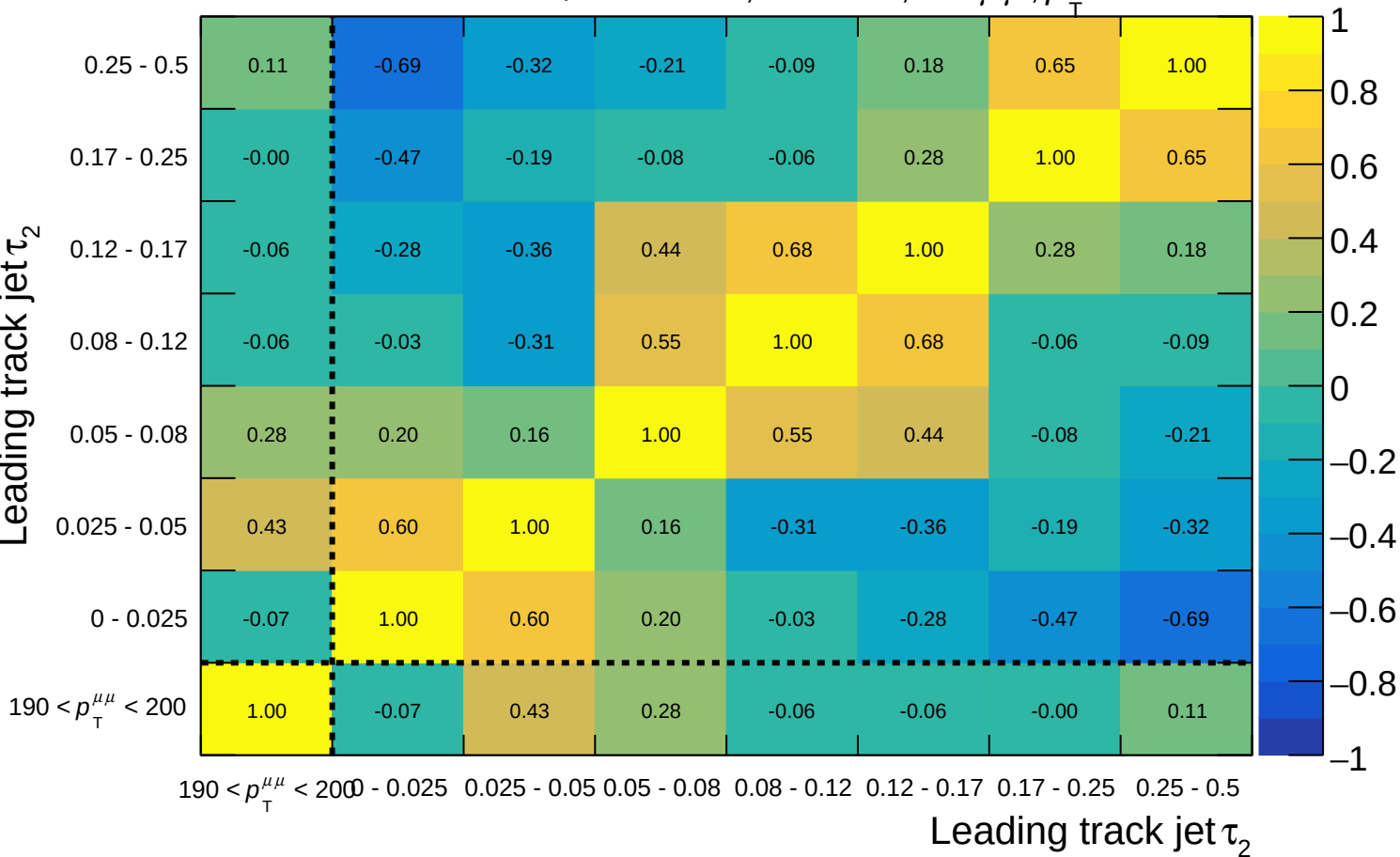


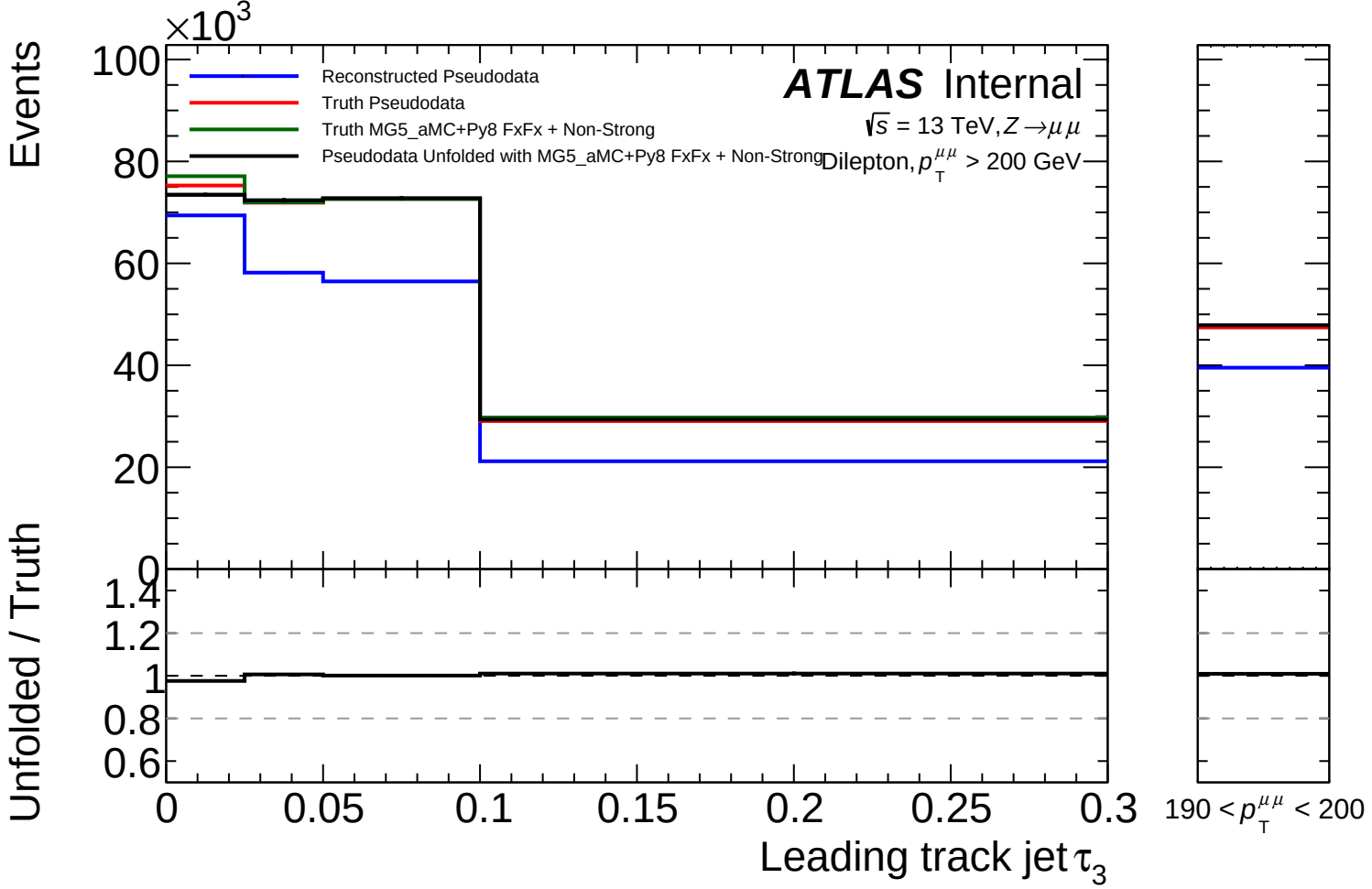


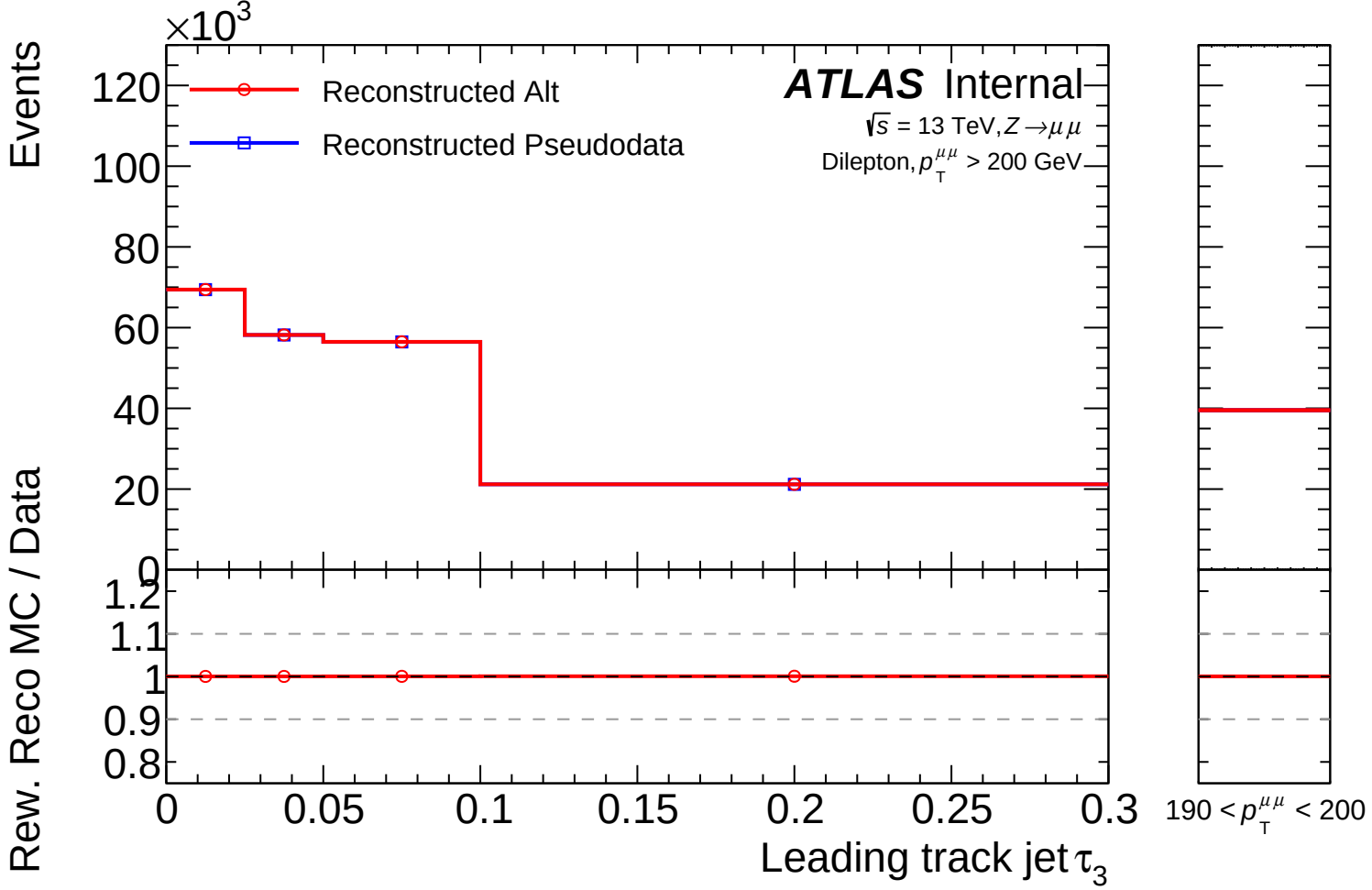


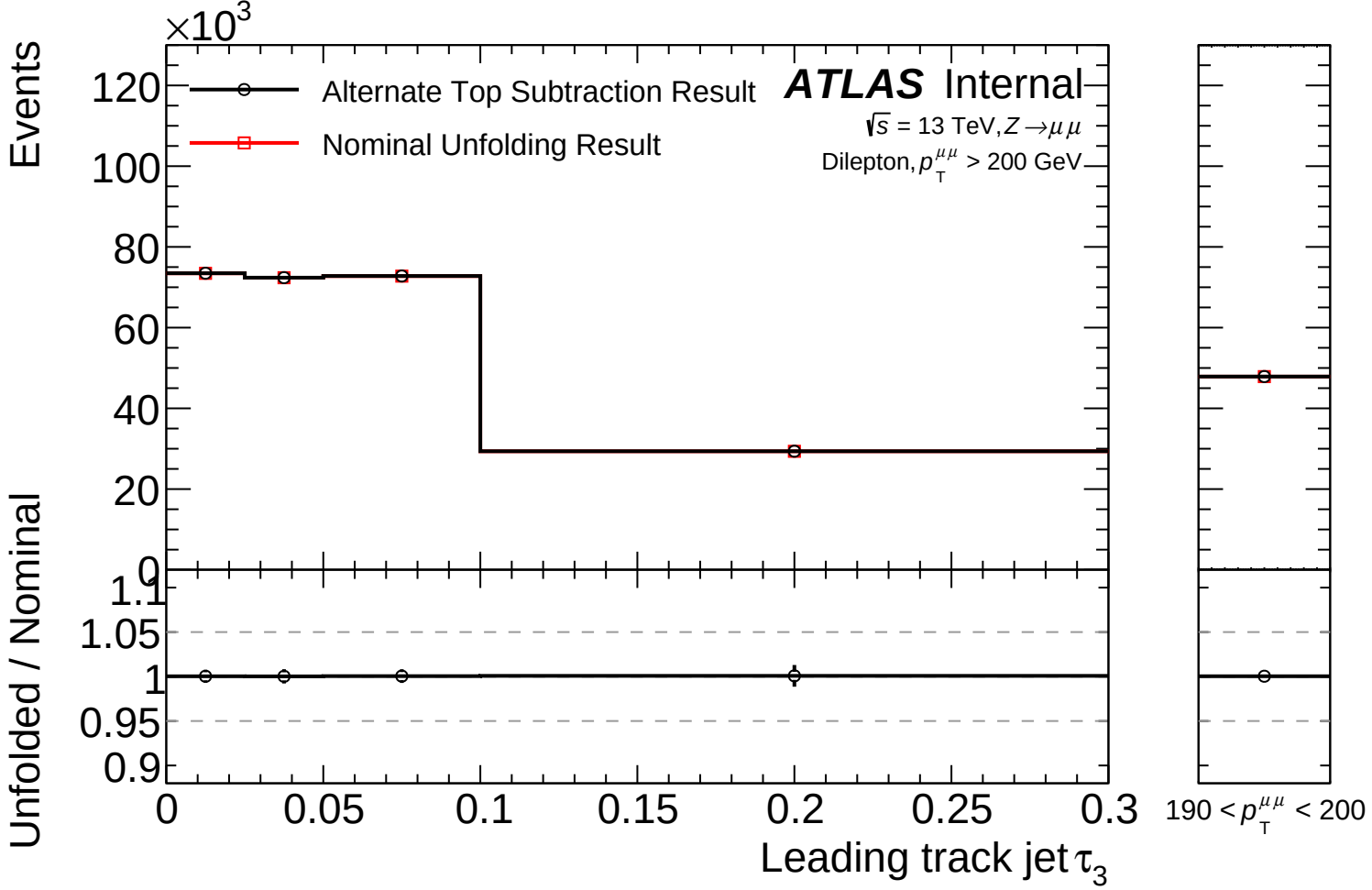
Fractional uncertainty



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet τ_2 Leading track jet τ_2







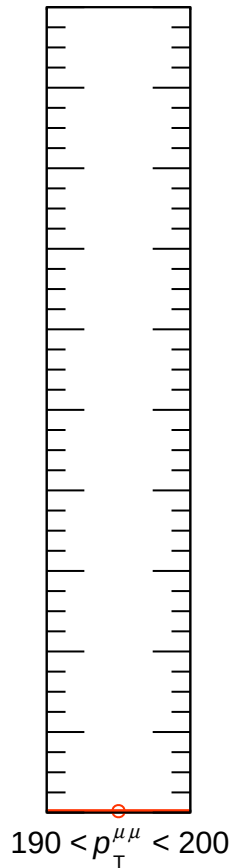
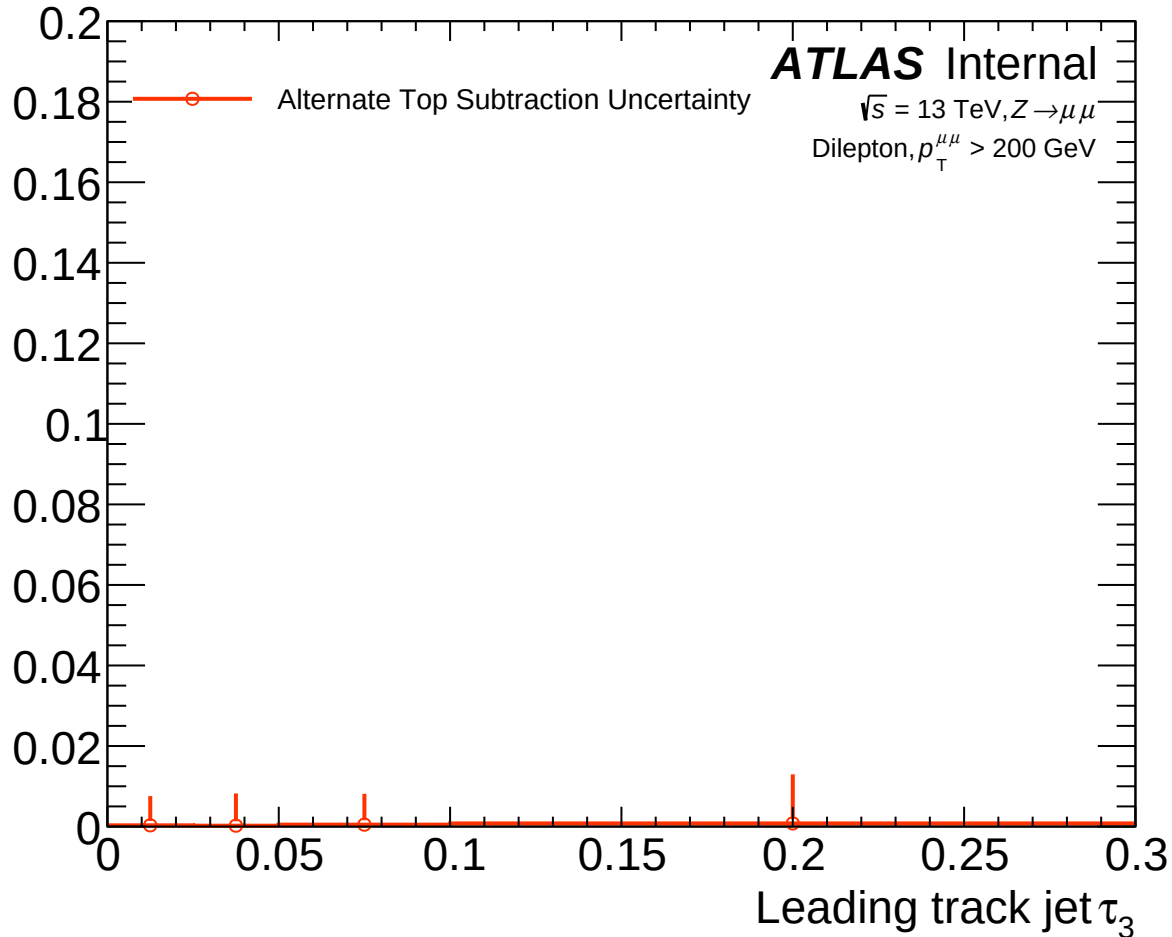
Fractional systematic impact

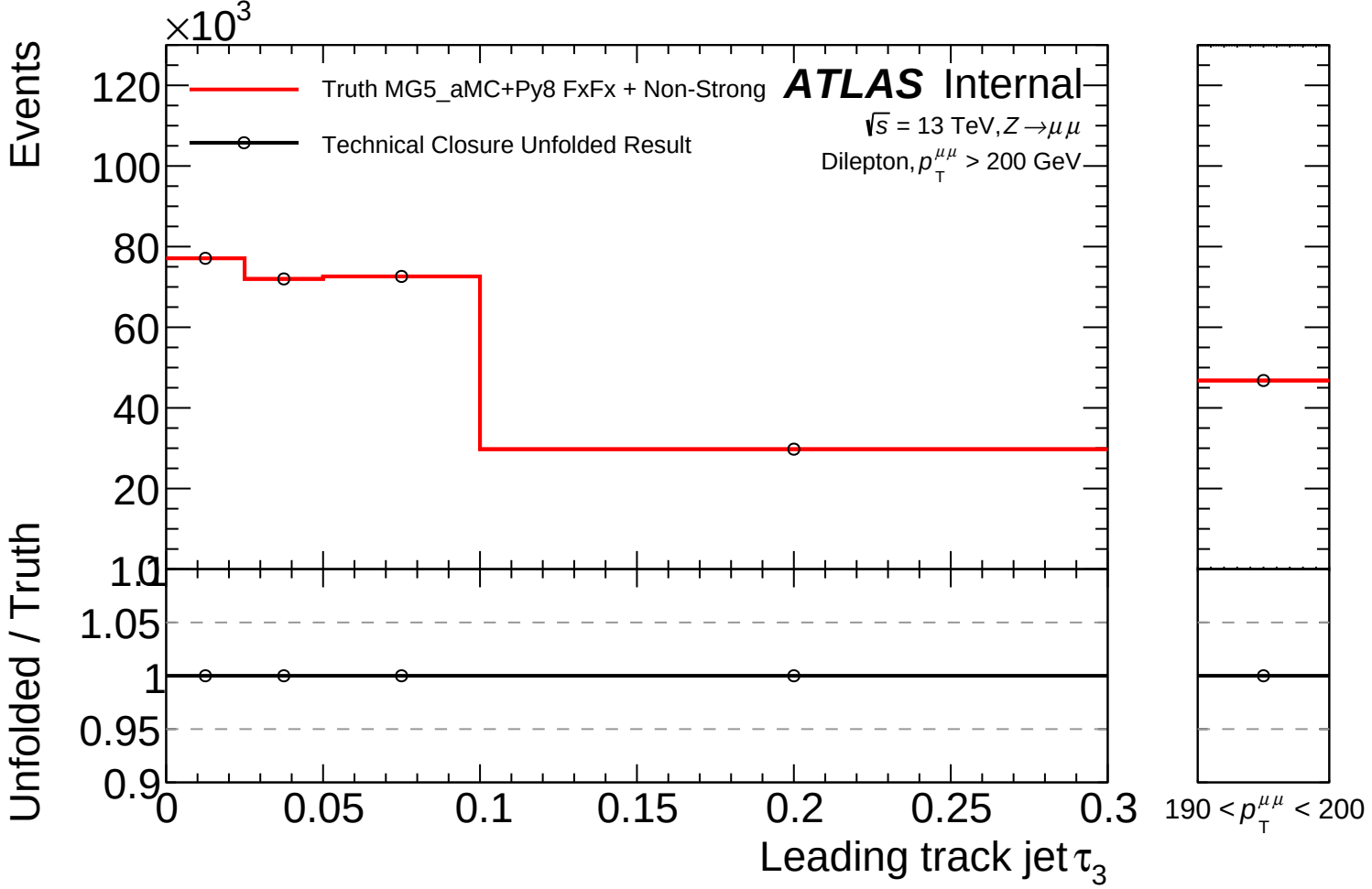
ATLAS Internal

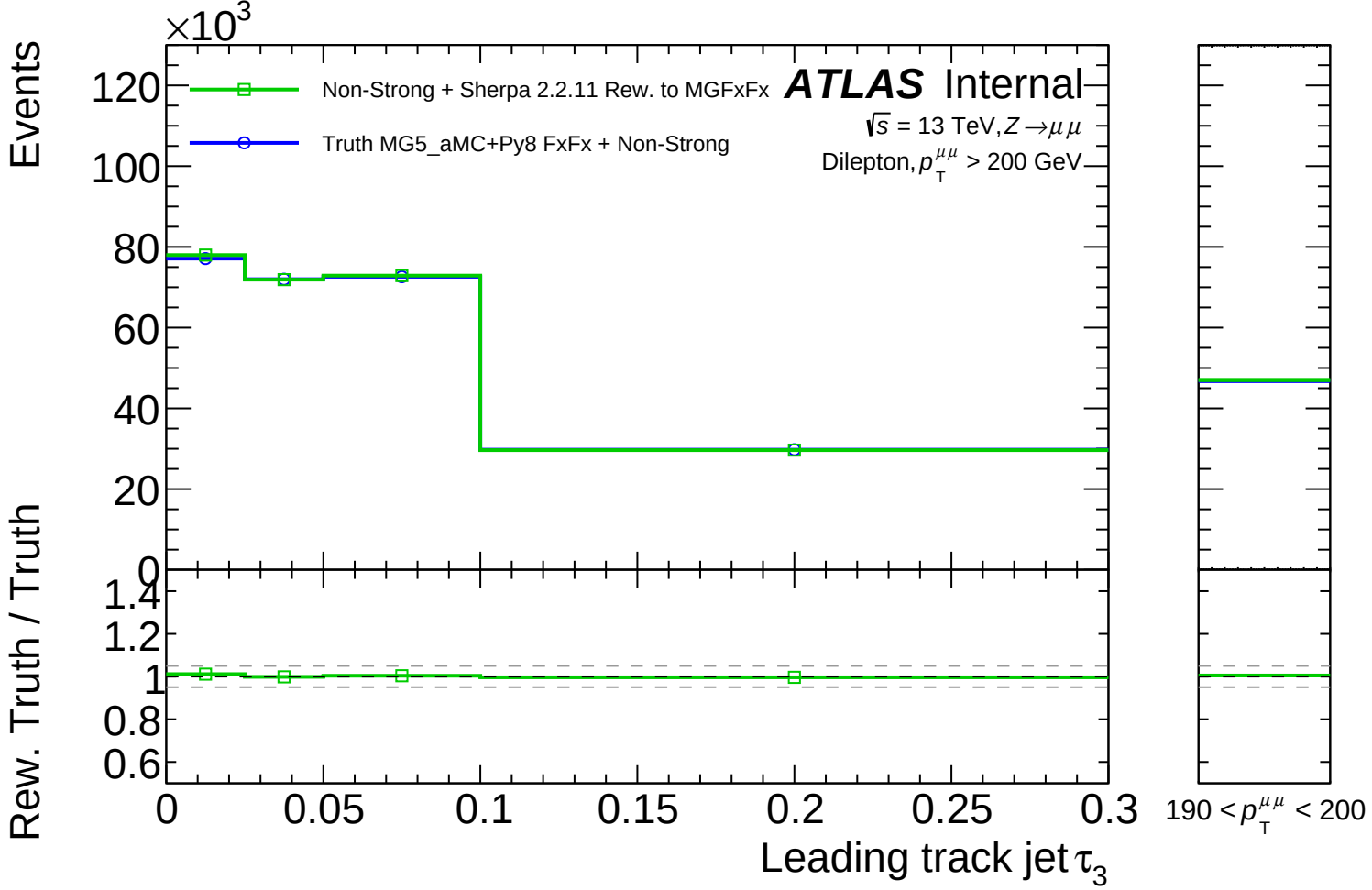
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

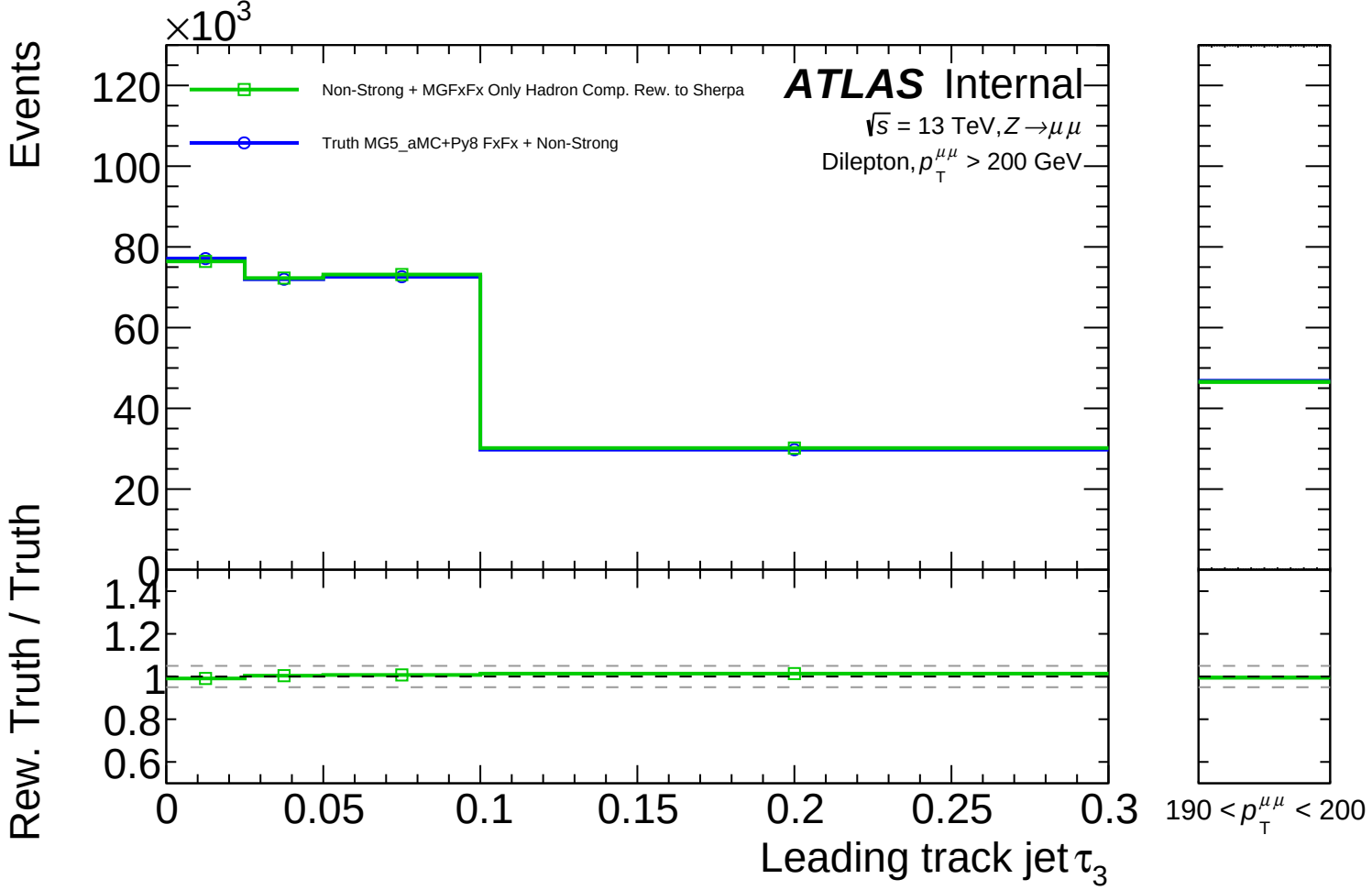
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

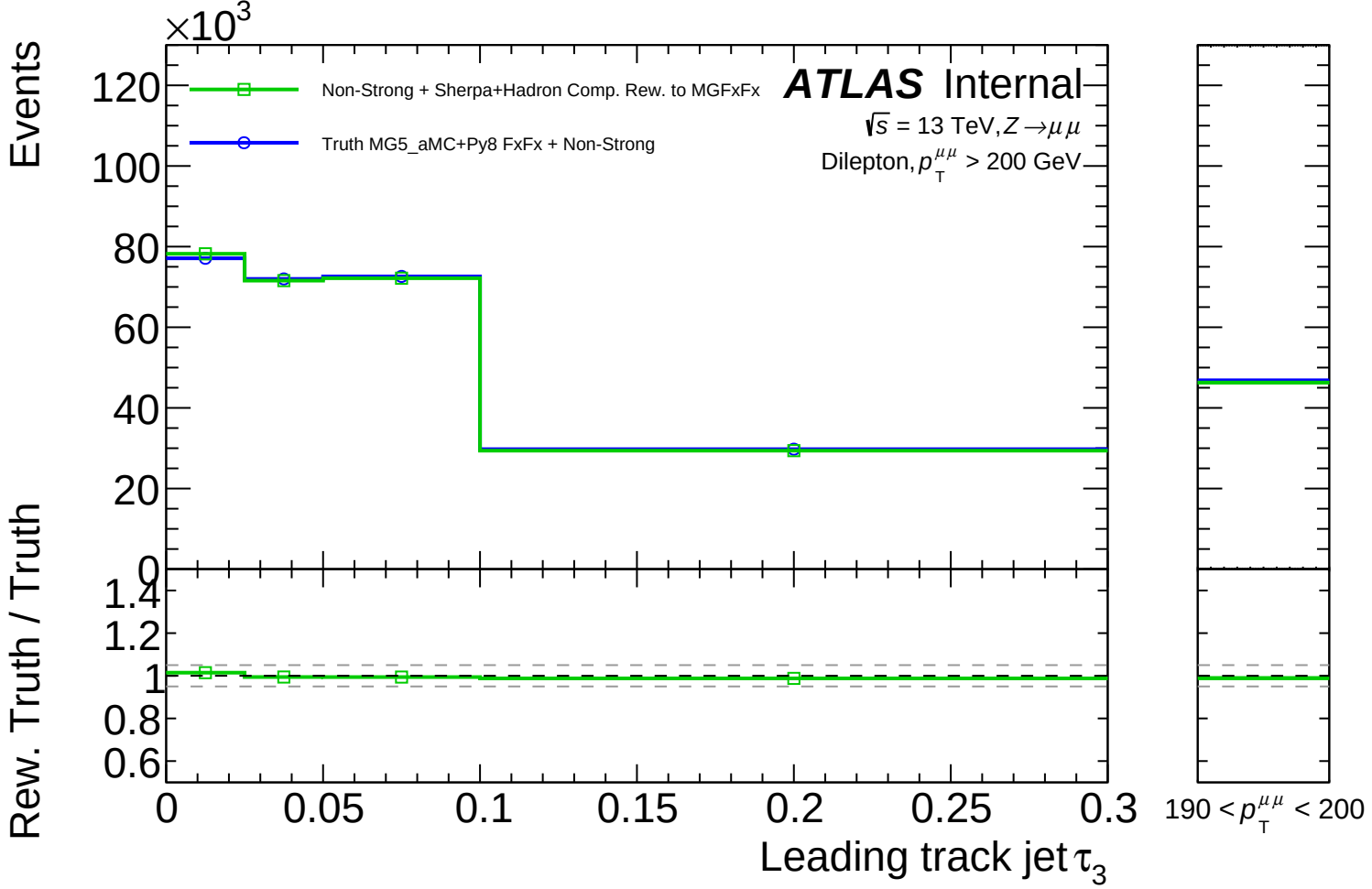
Alternate Top Subtraction Uncertainty

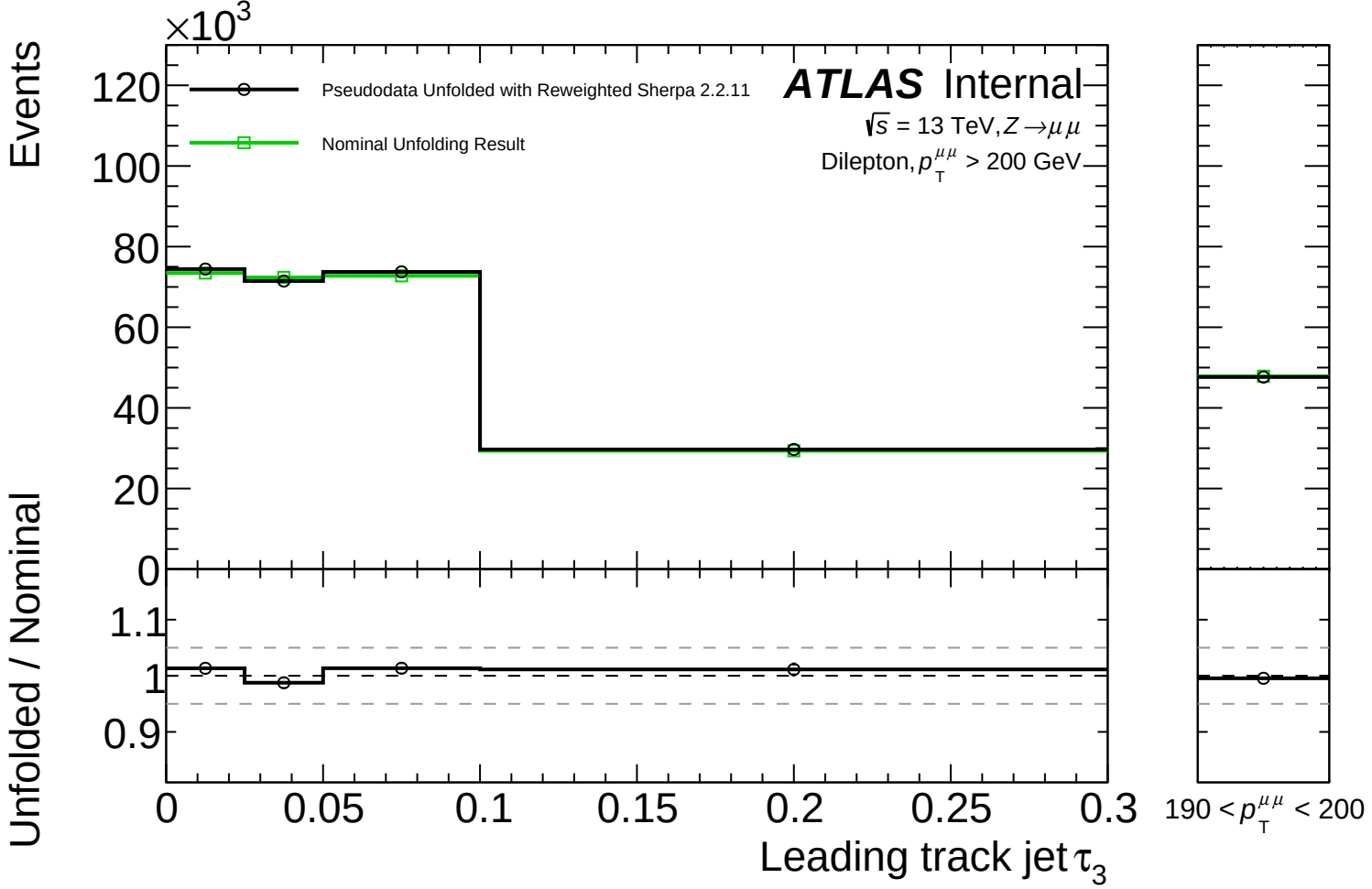


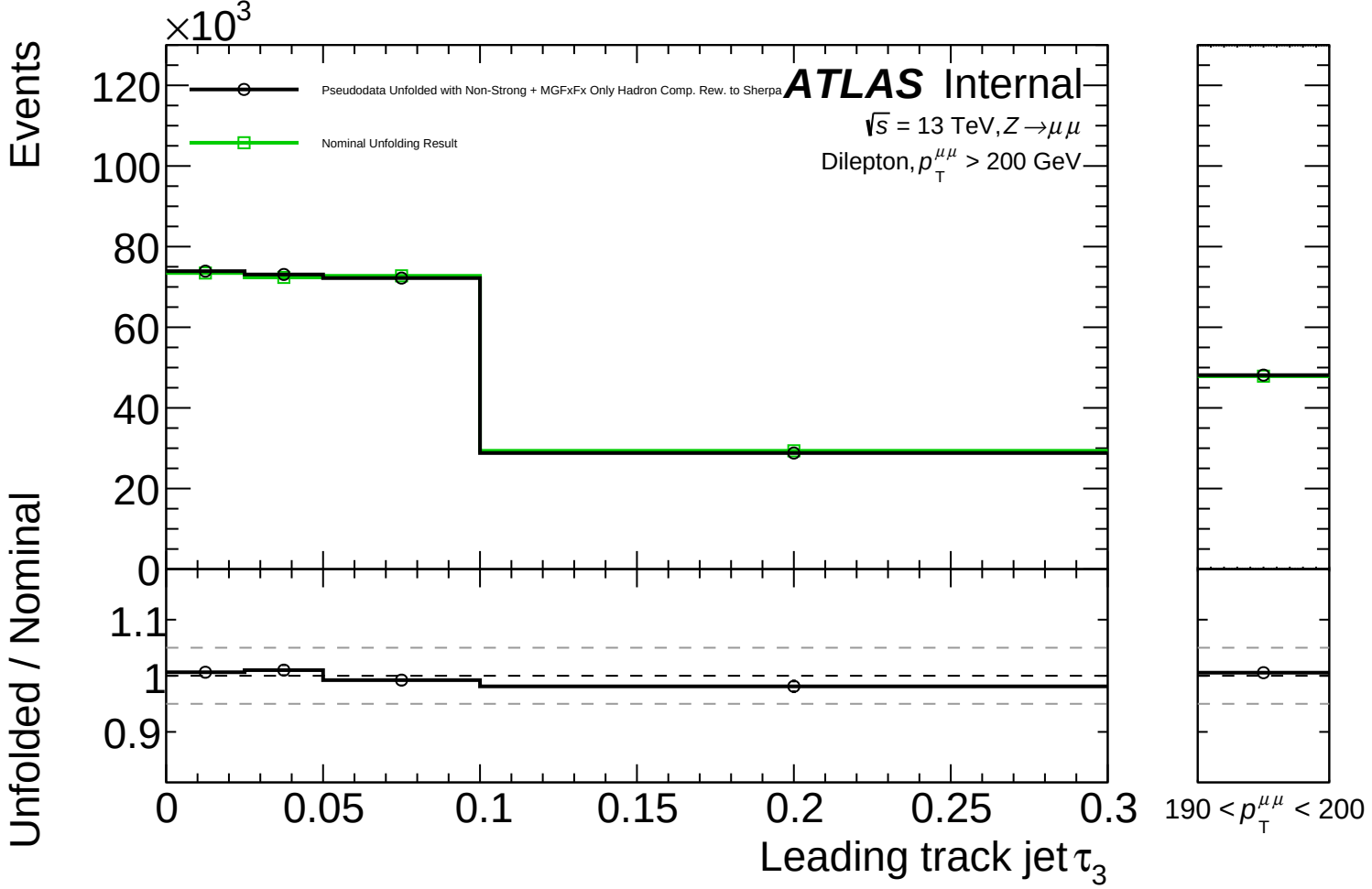


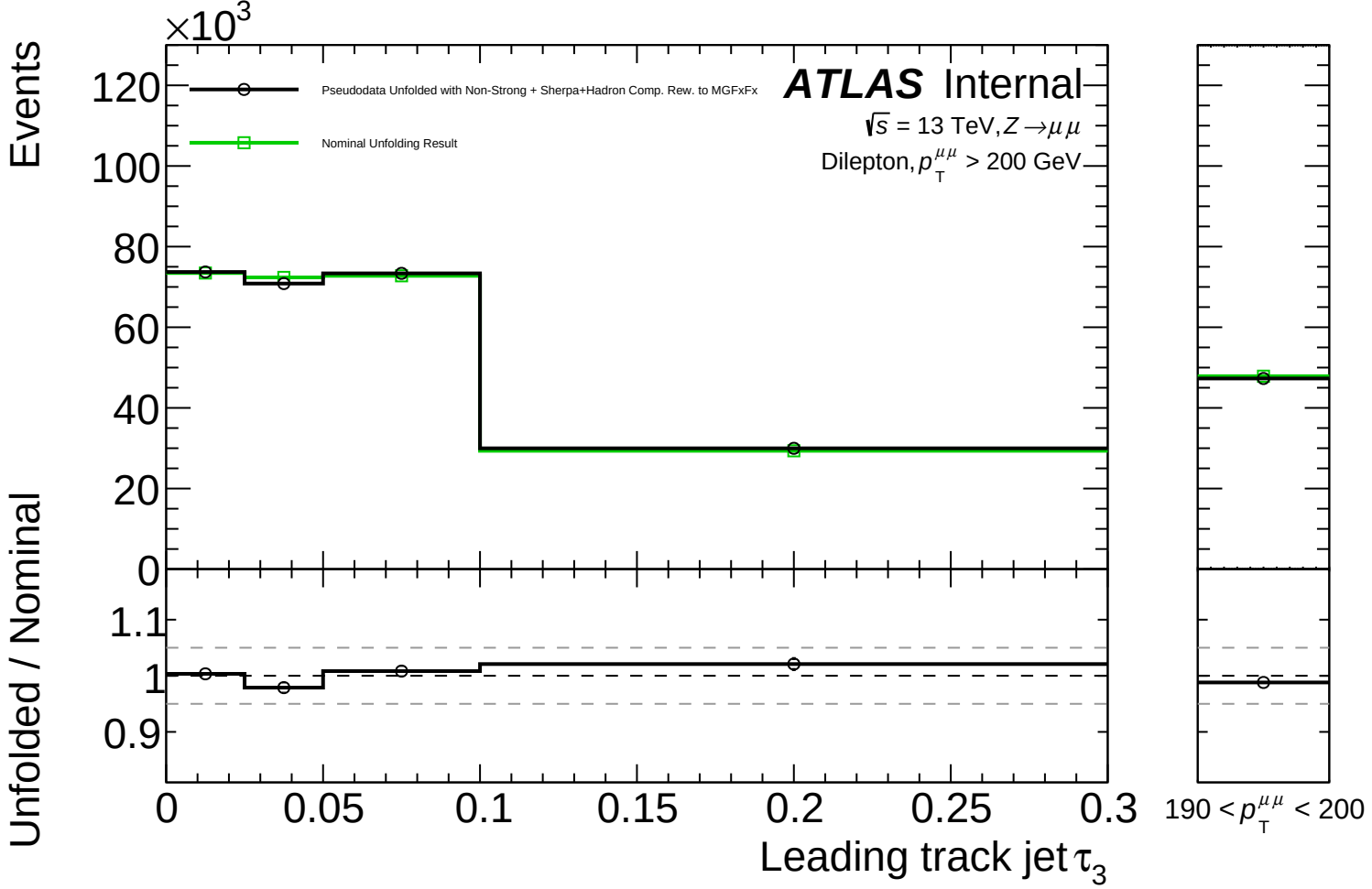


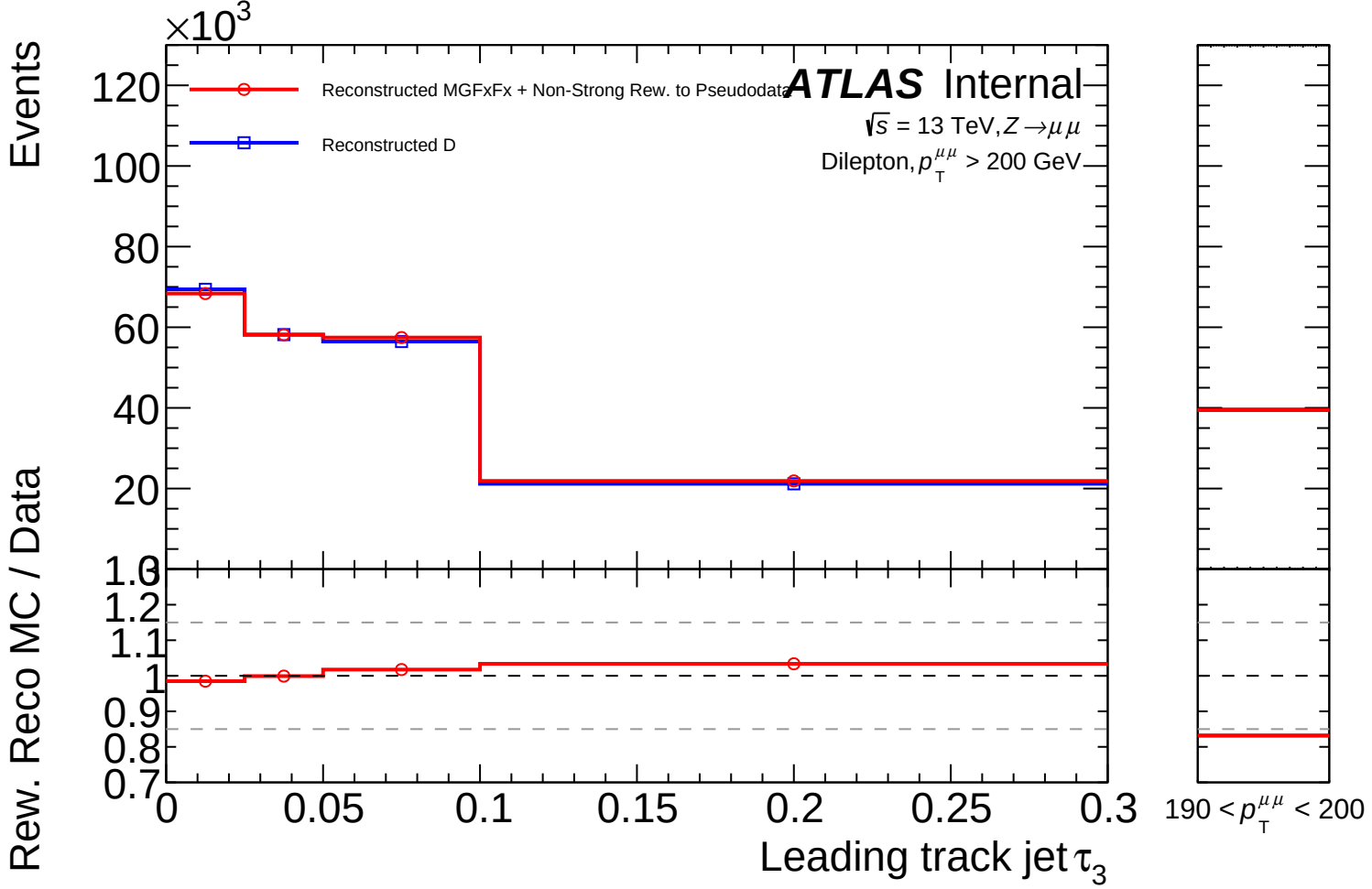


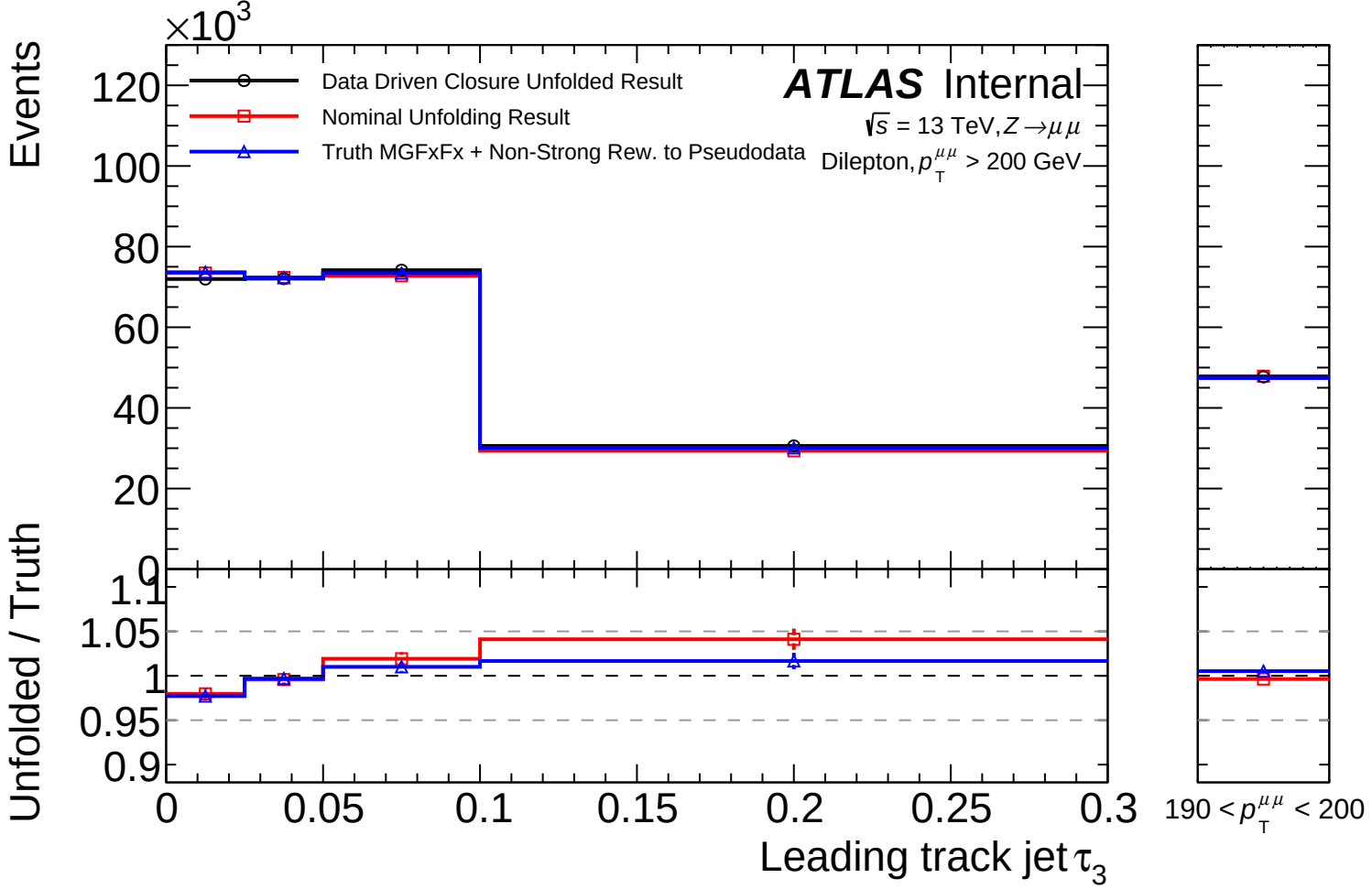




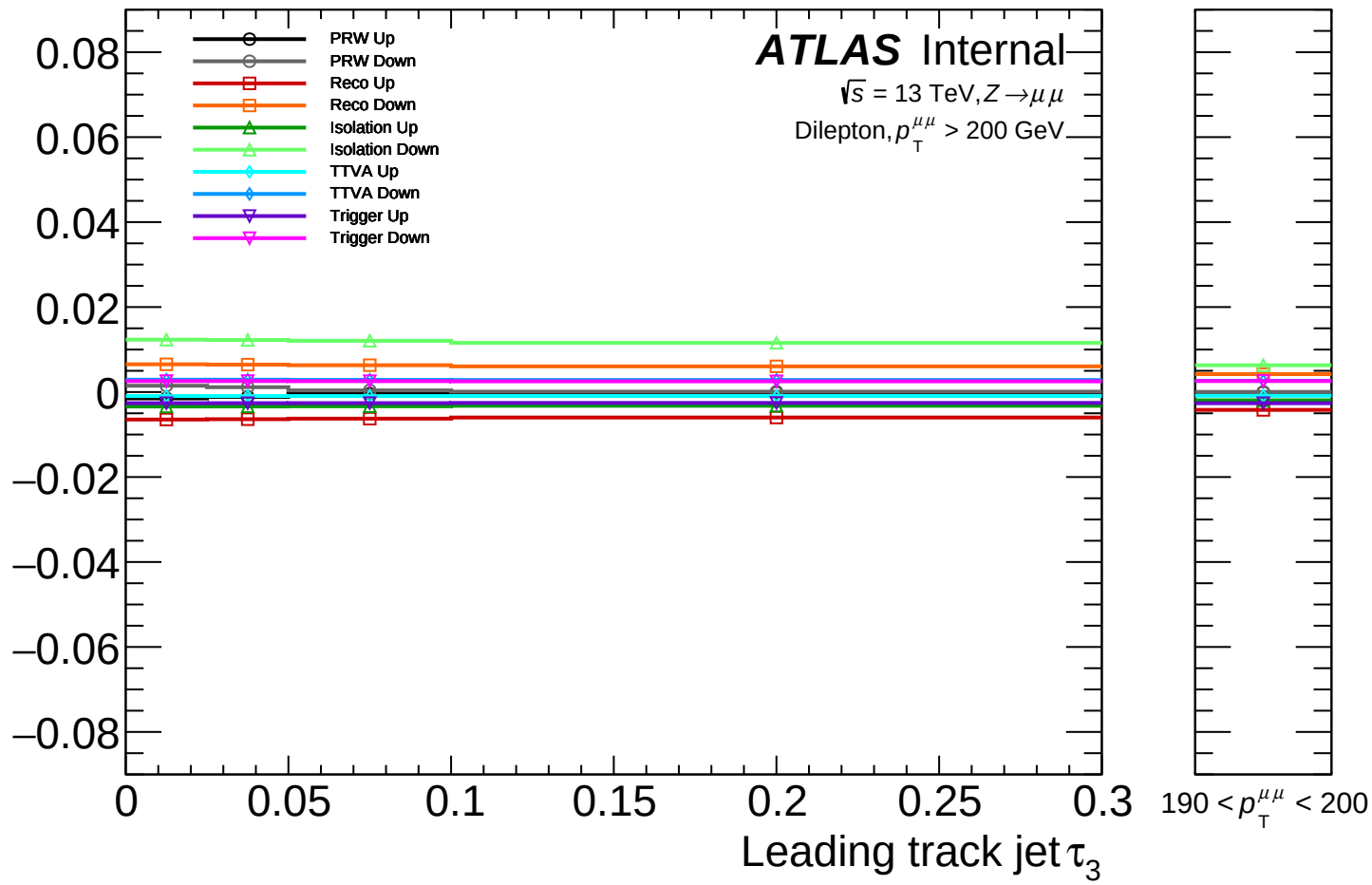




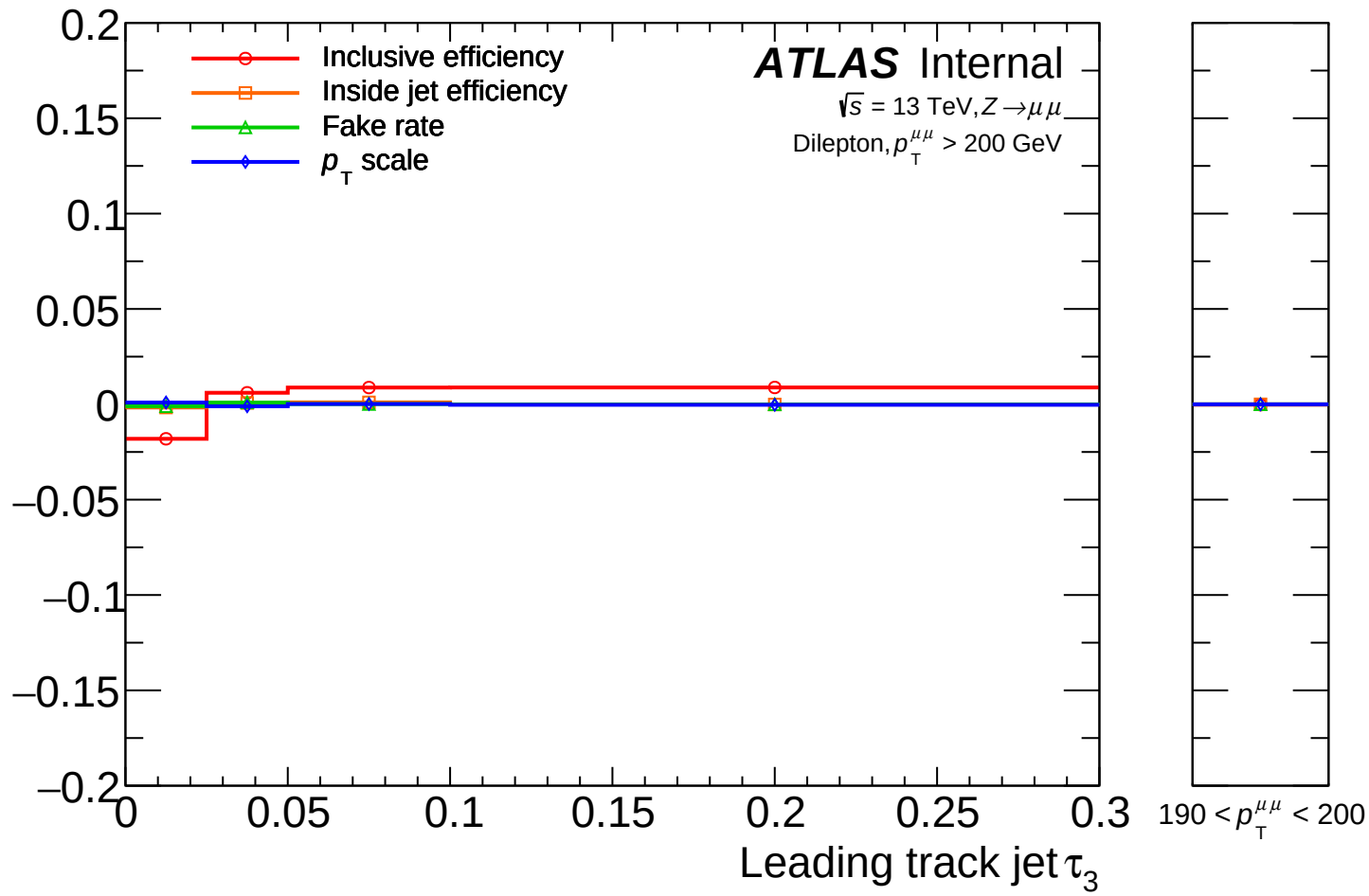




Fractional systematic impact



Fractional systematic impact



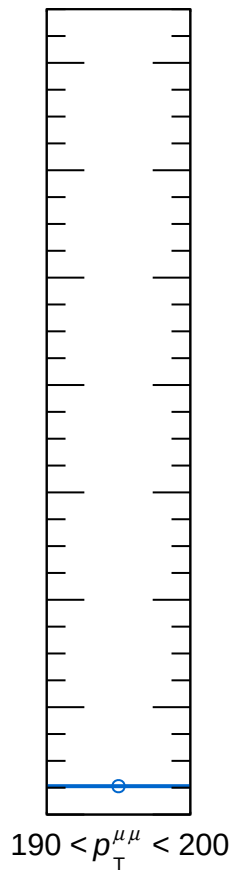
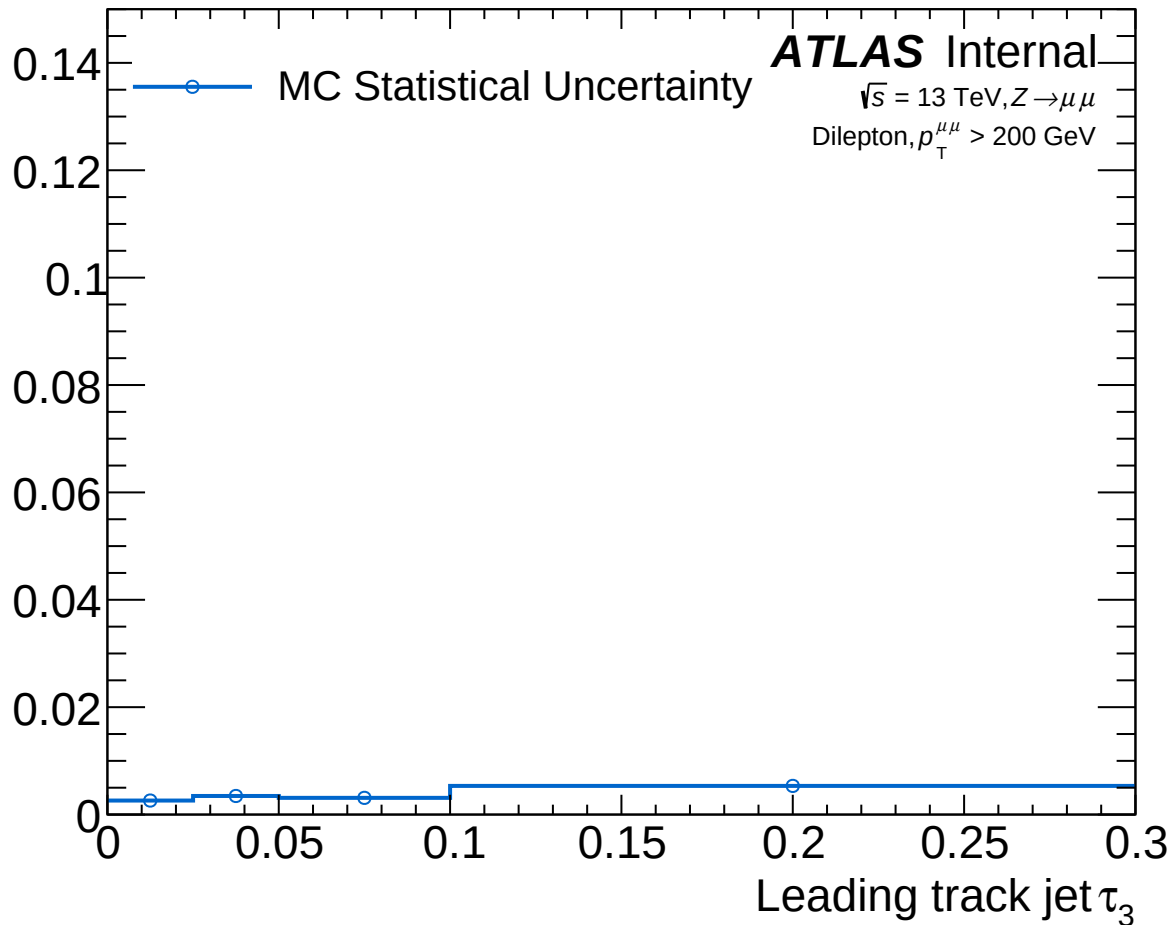
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



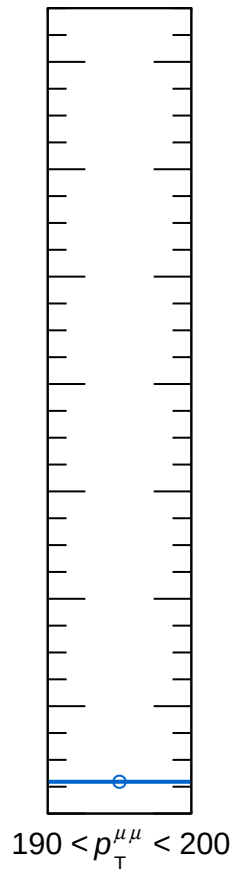
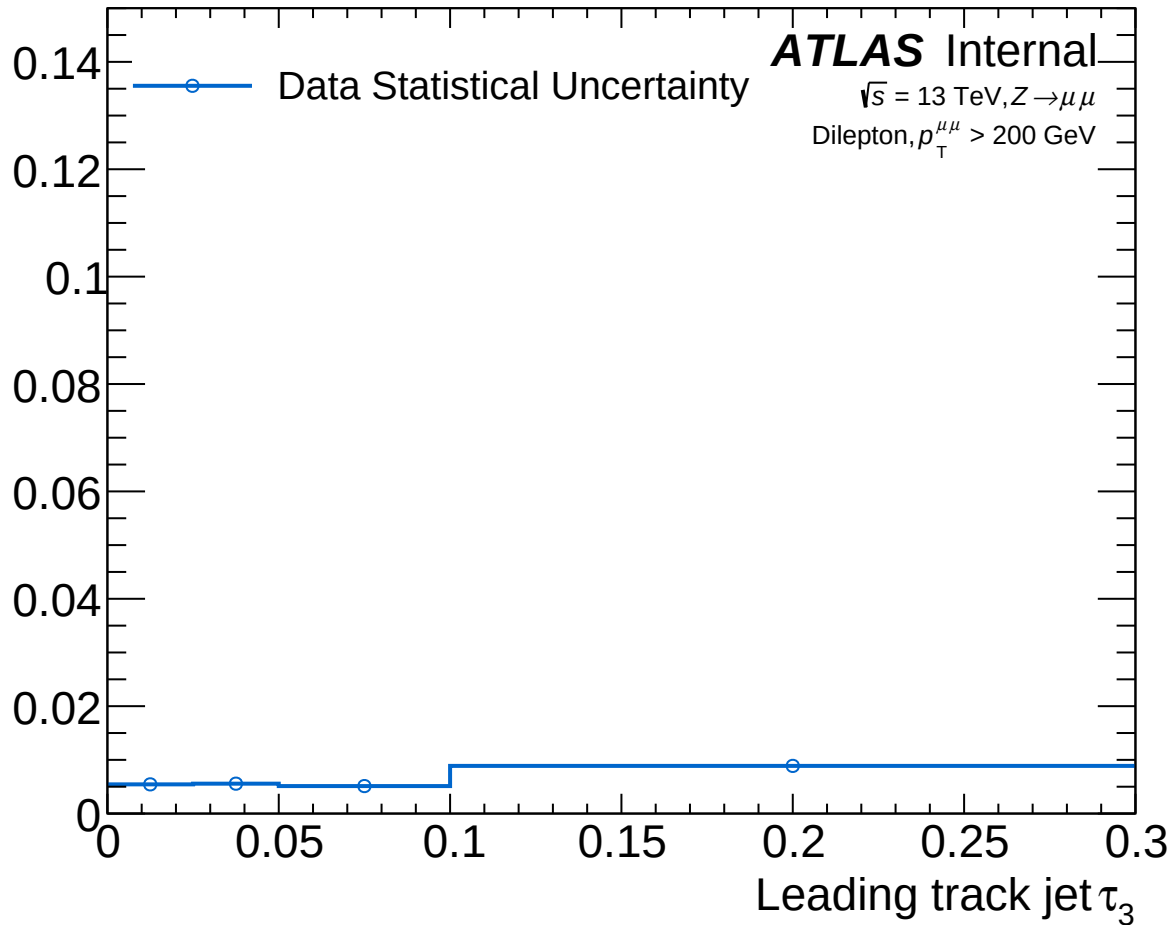
Fractional systematic impact

ATLAS Internal

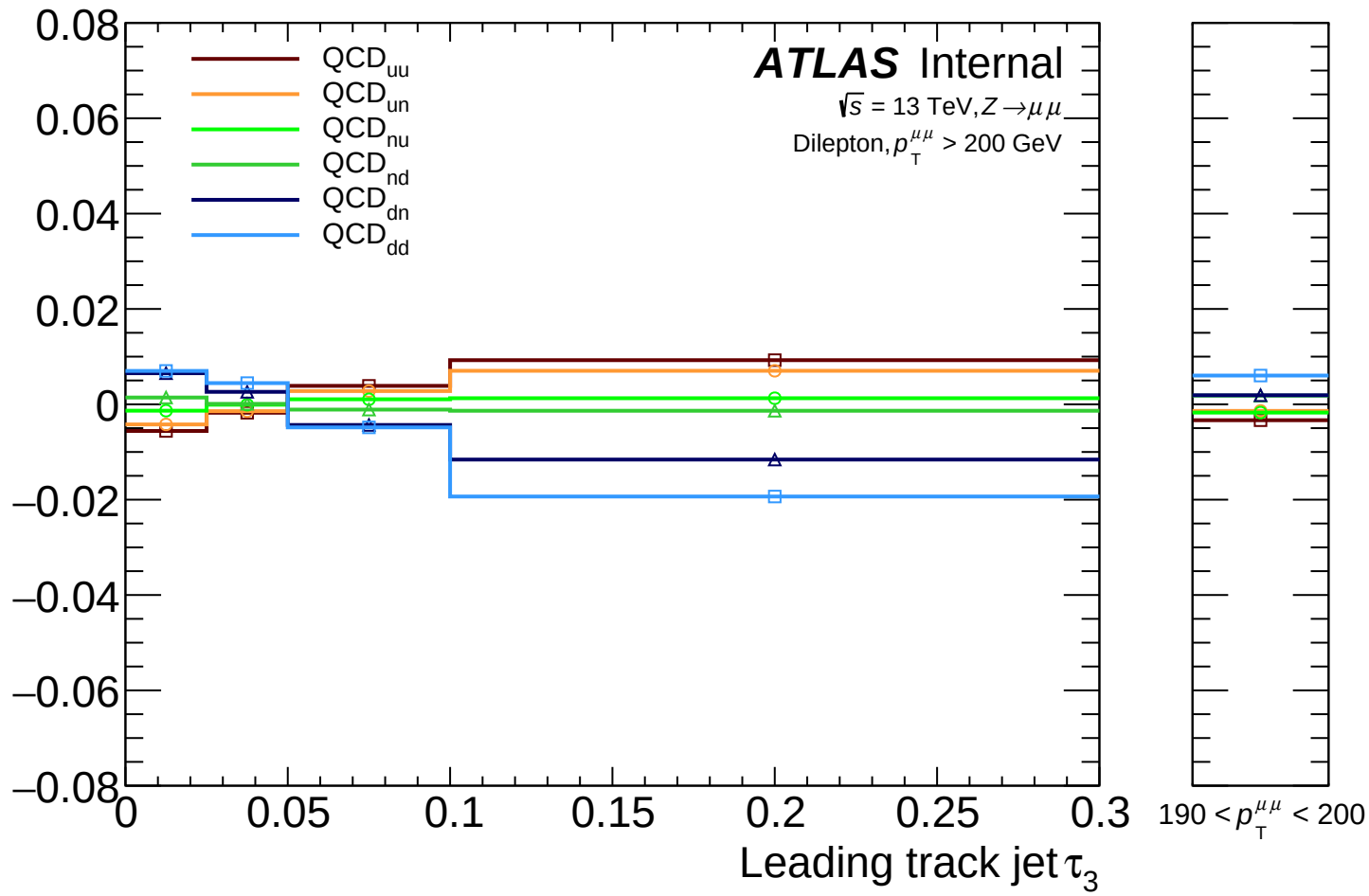
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

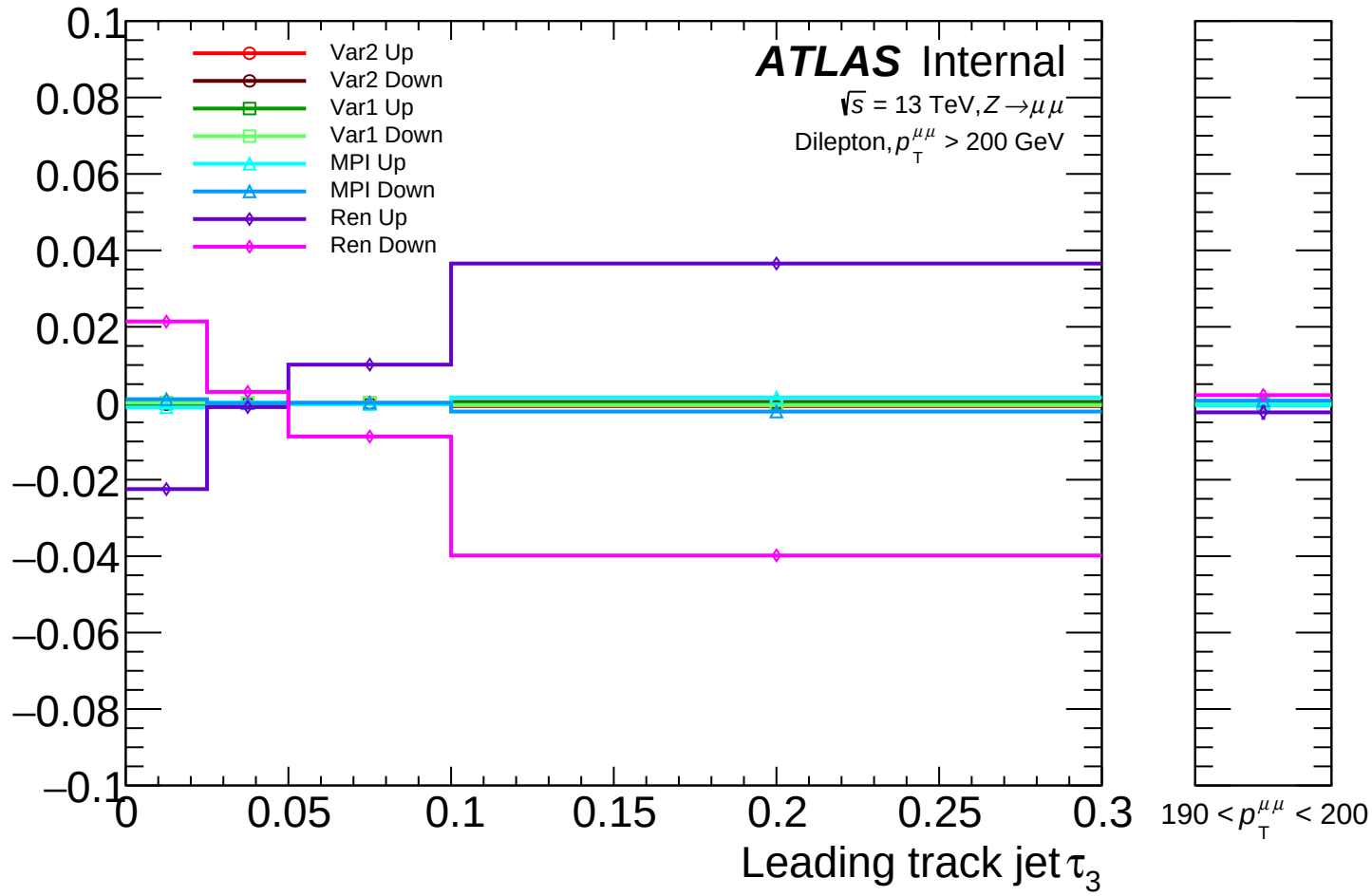
Data Statistical Uncertainty

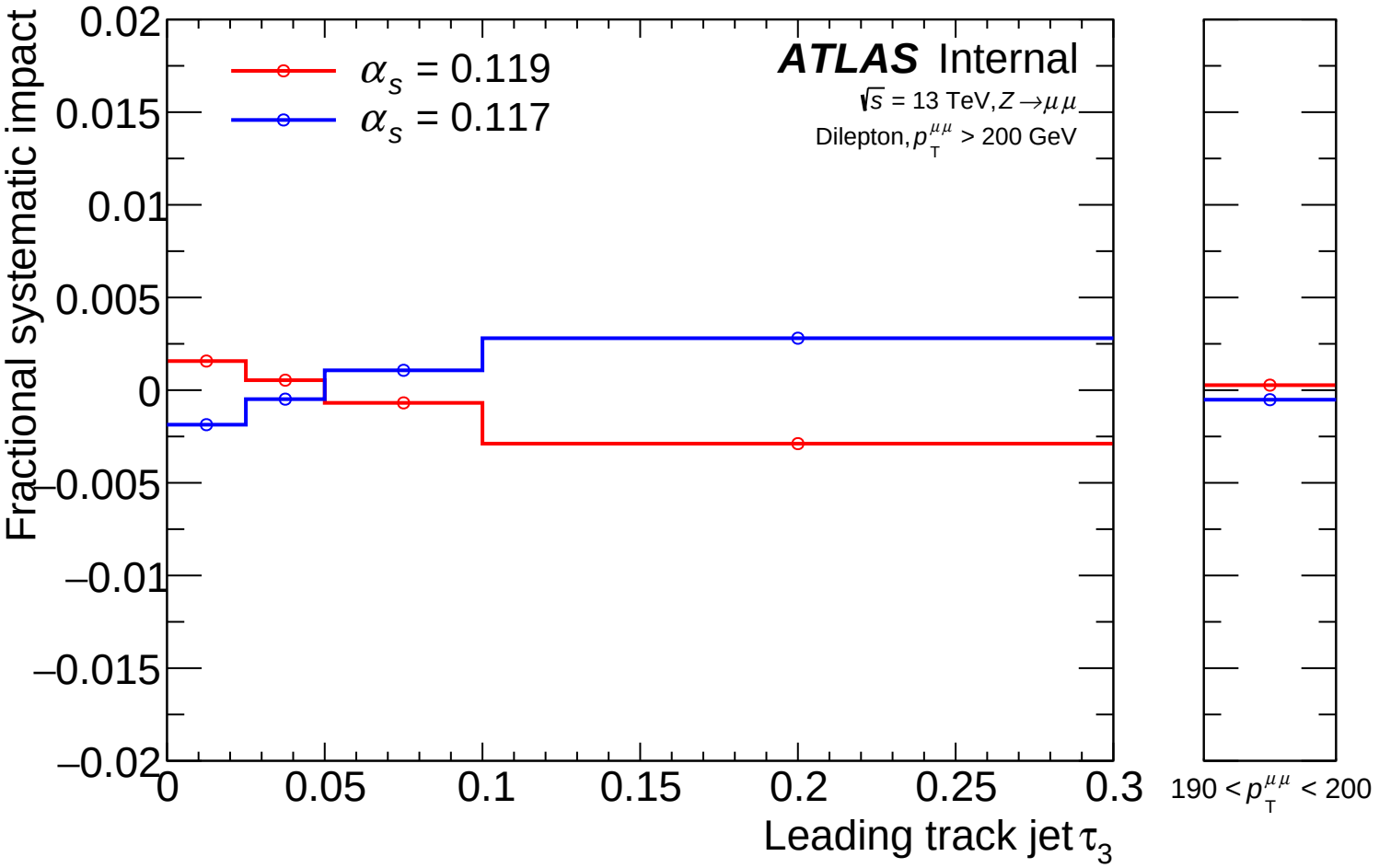


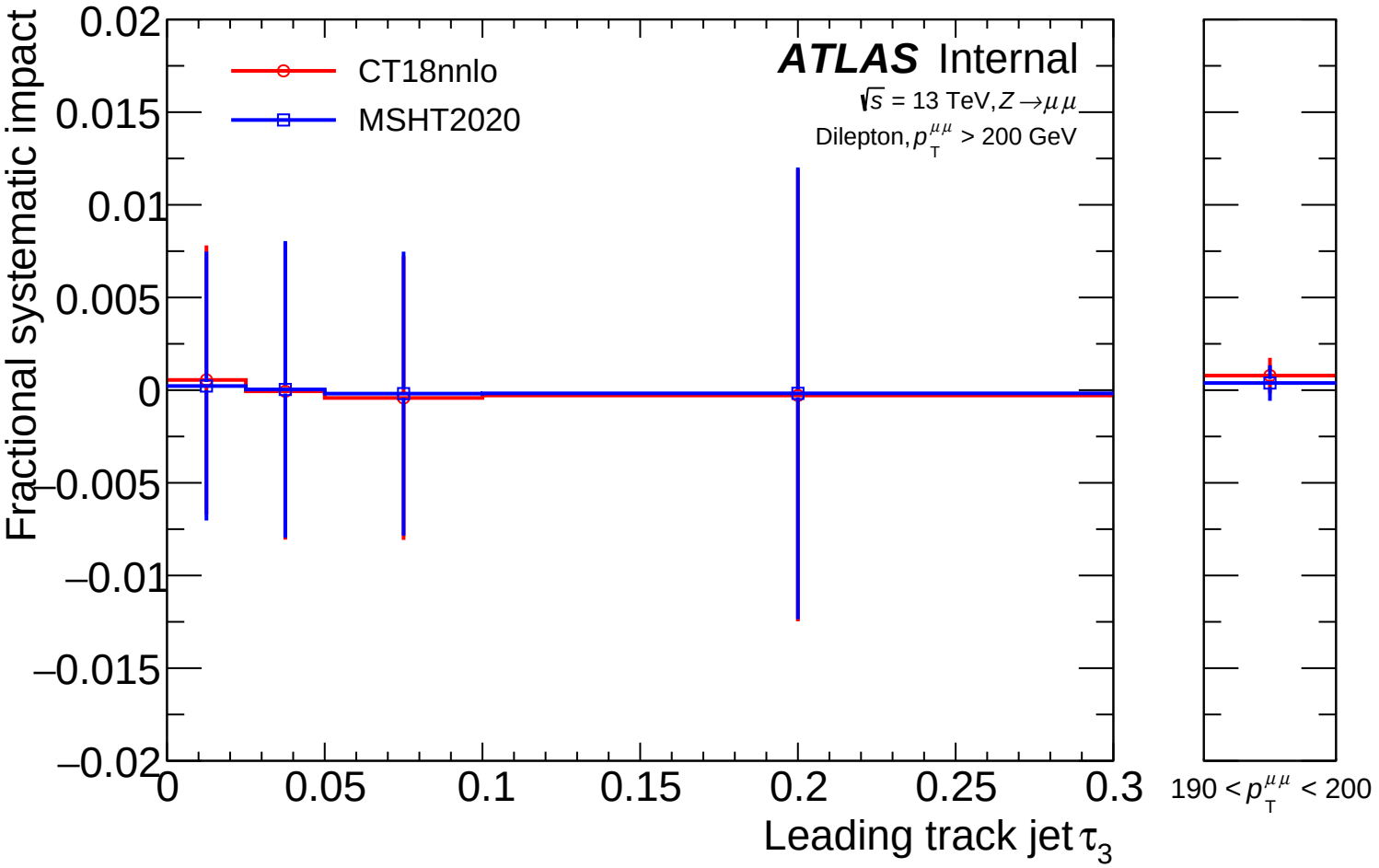
Fractional systematic impact



Fractional systematic impact







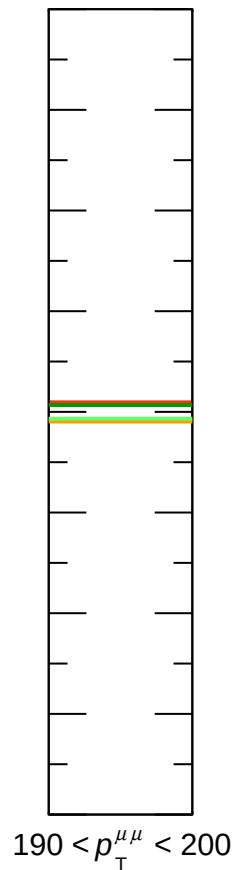
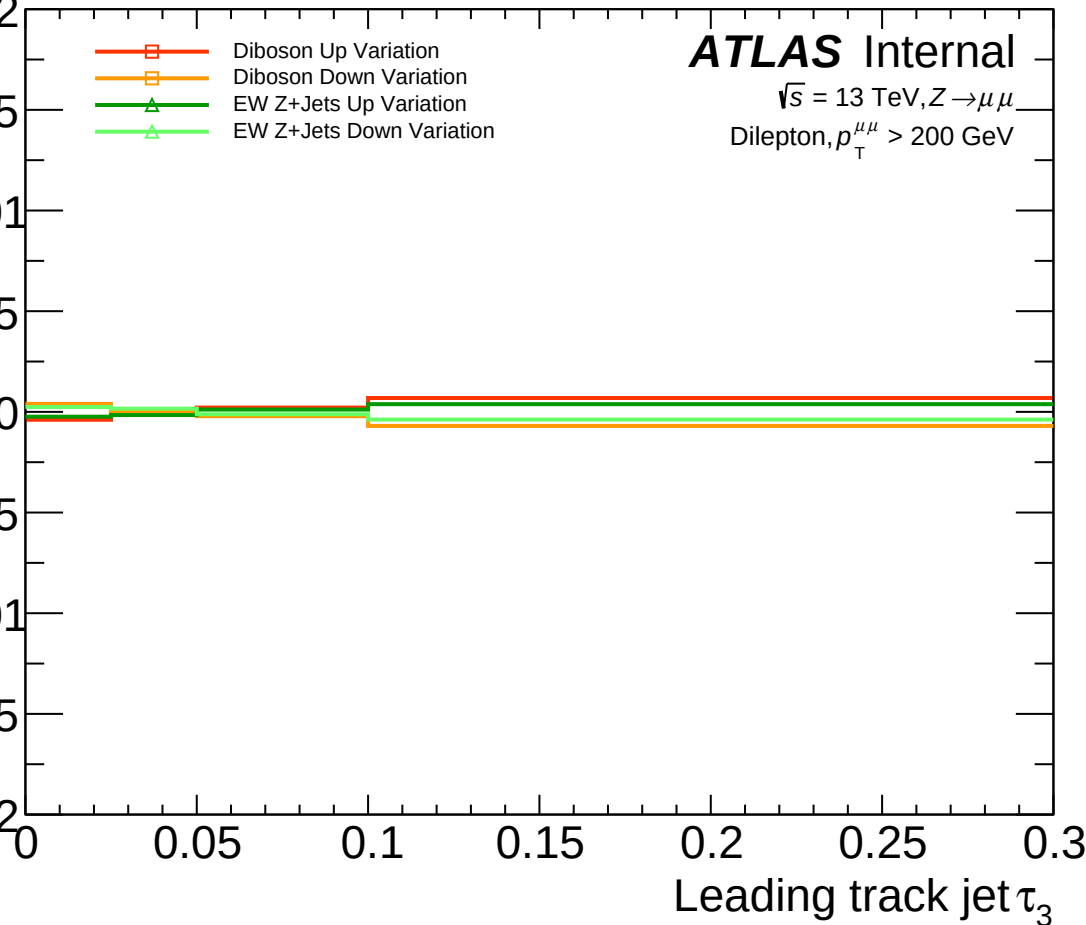
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

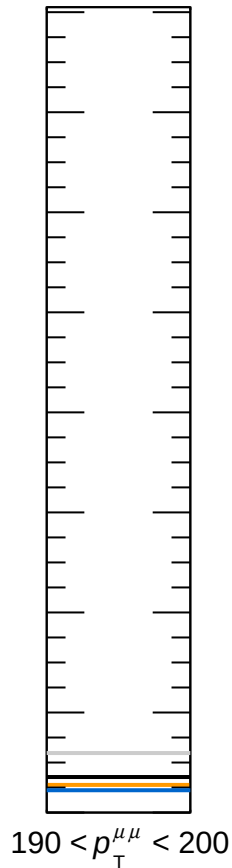
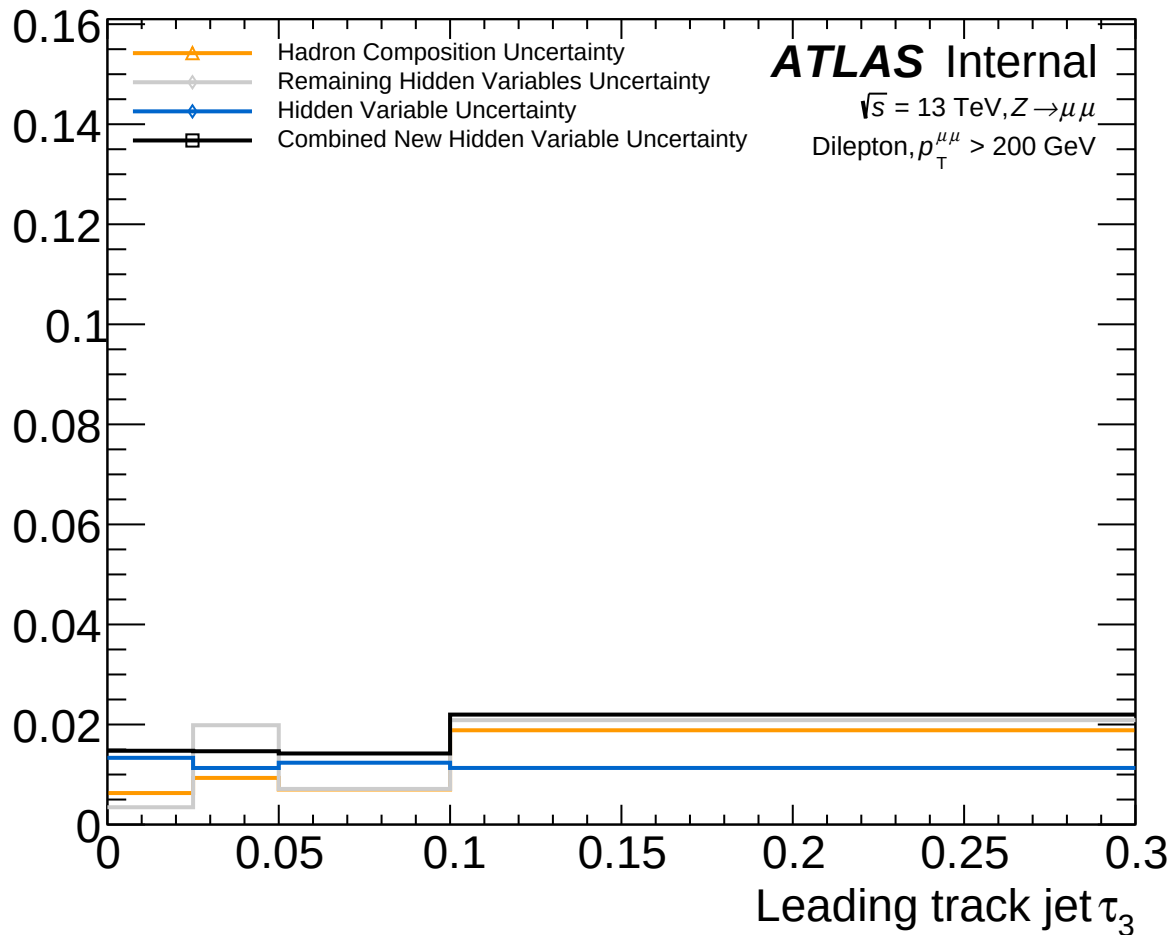


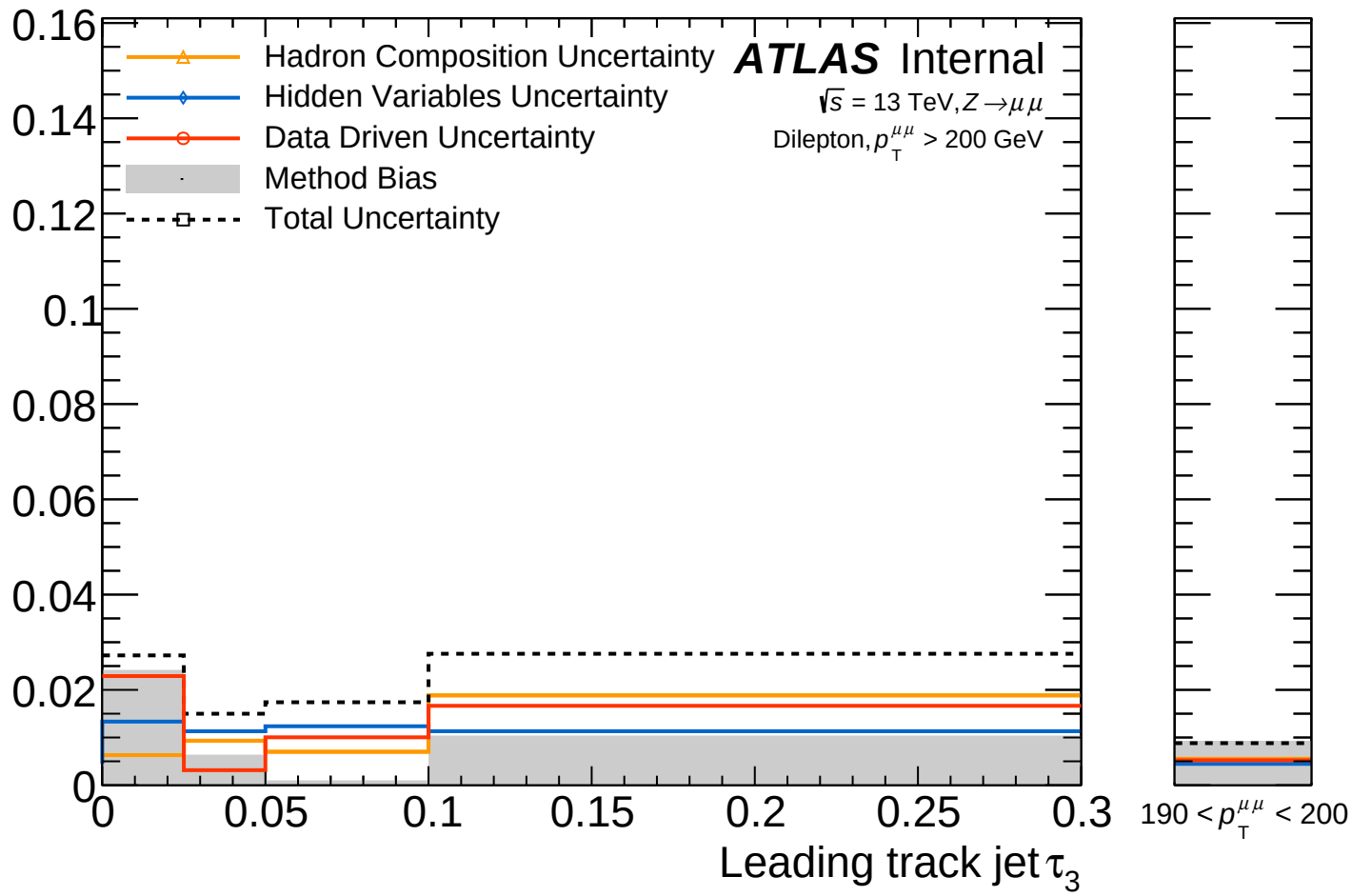
ATLAS Internal

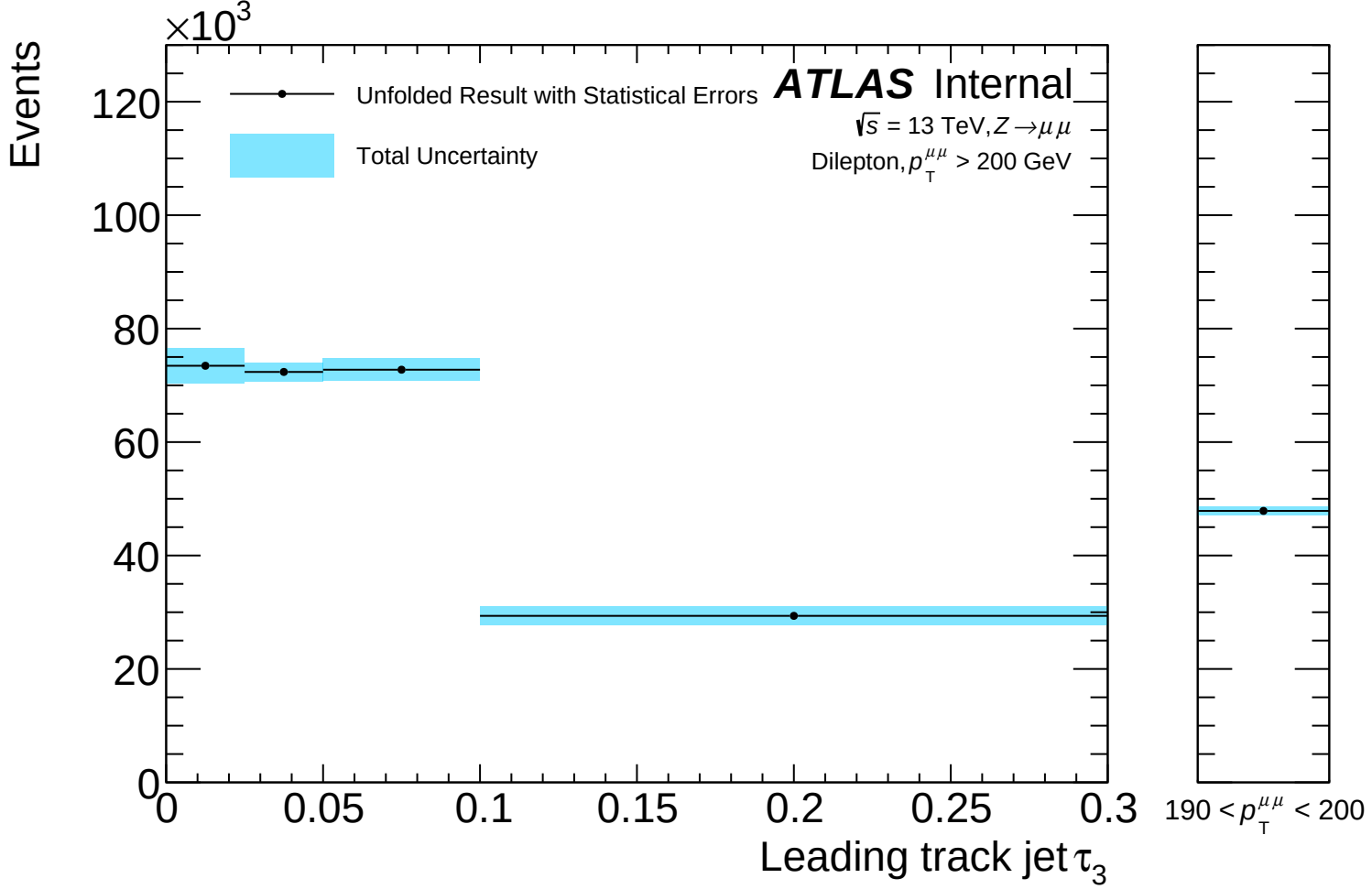
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

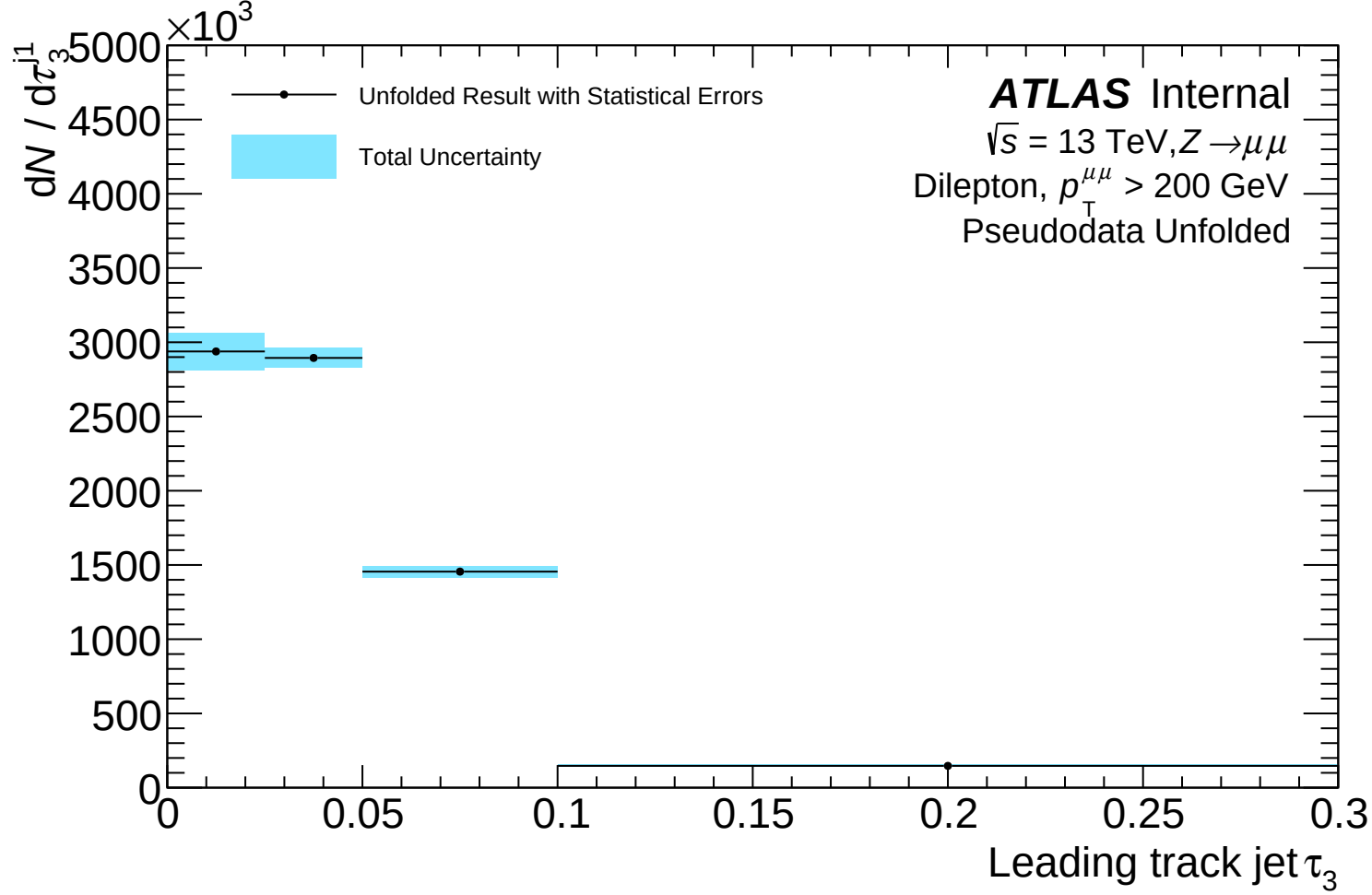
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

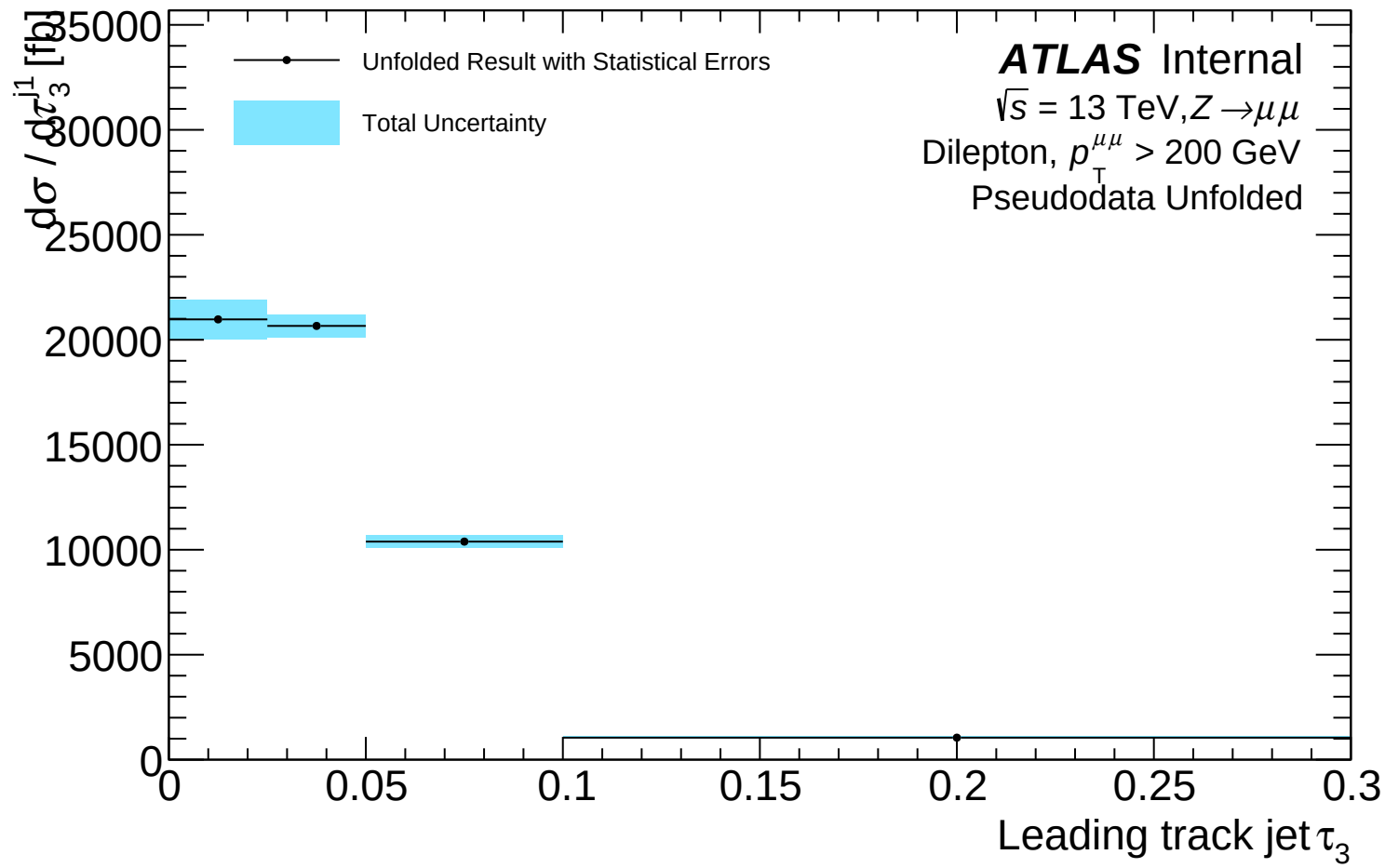
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty











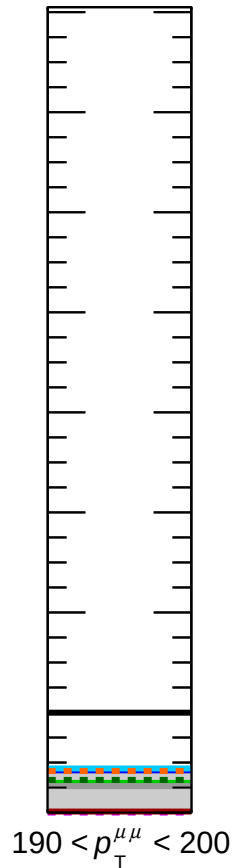
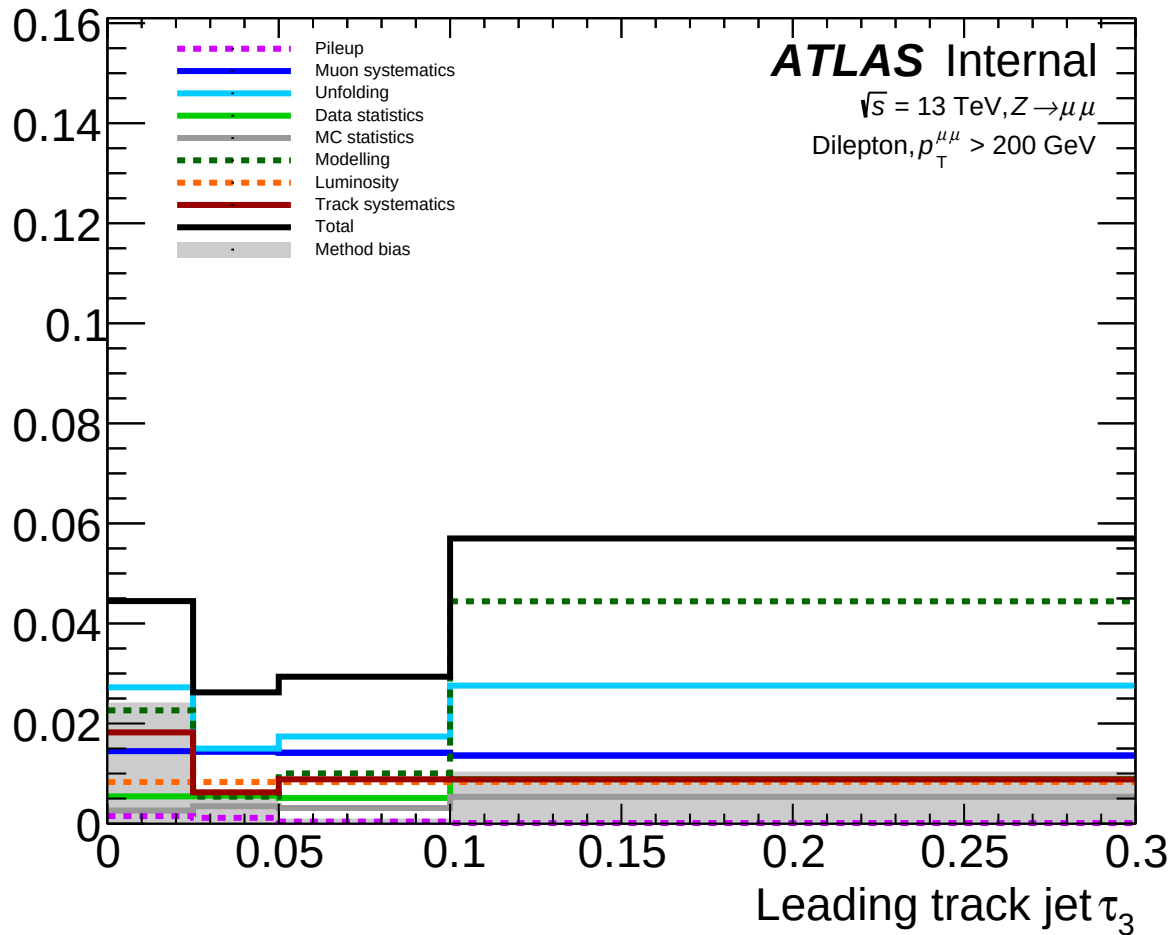
Fractional uncertainty

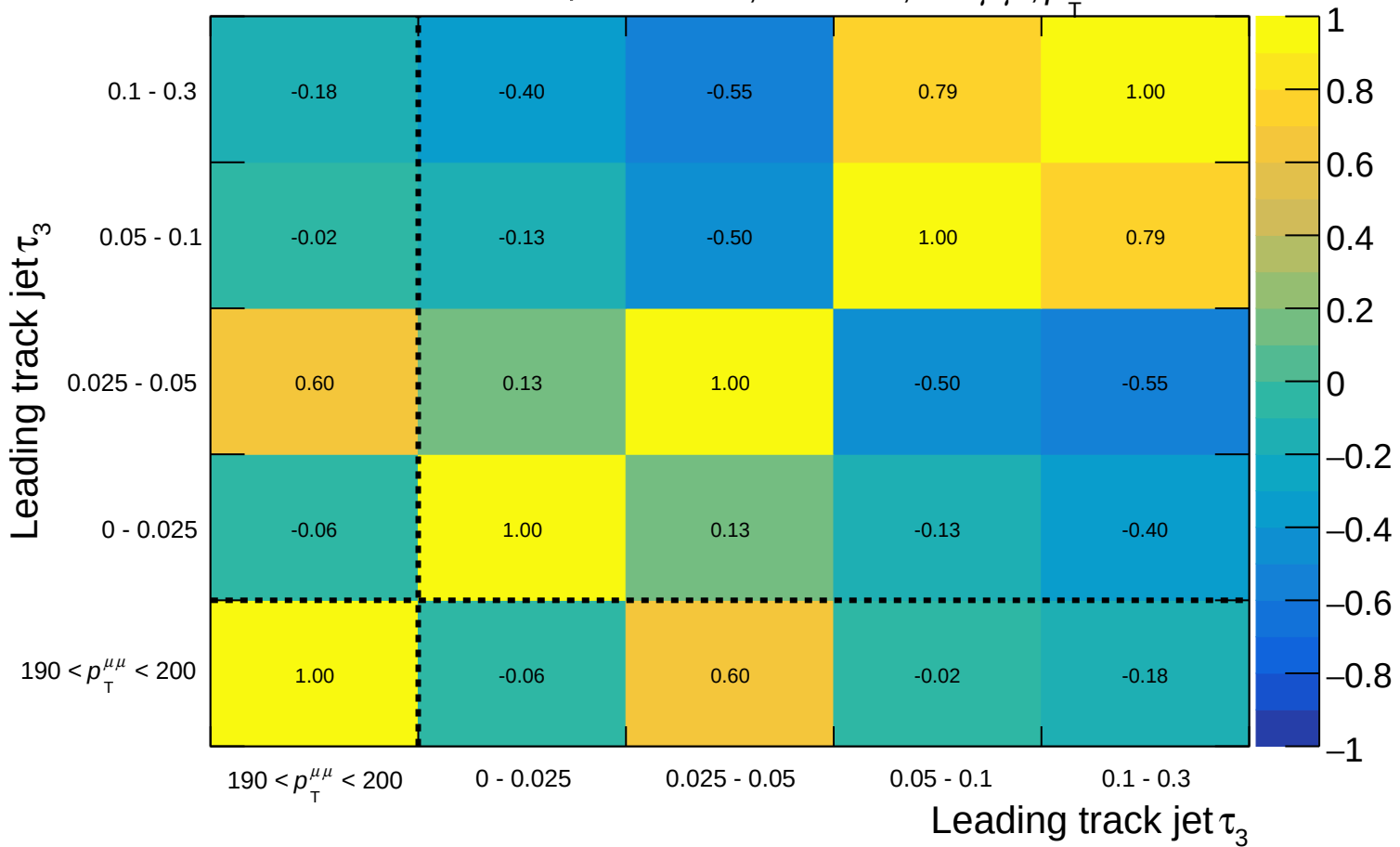
ATLAS Internal

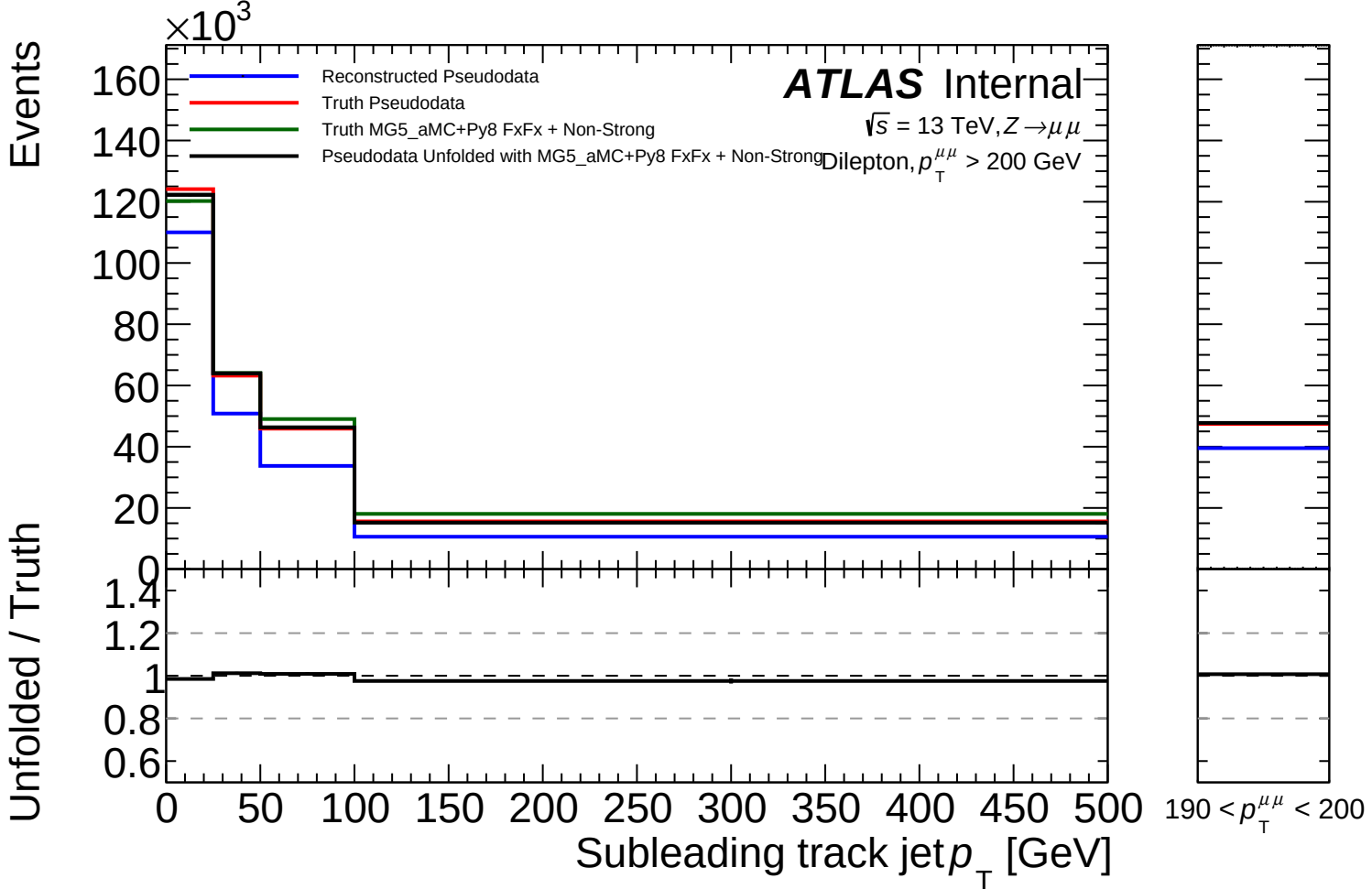
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

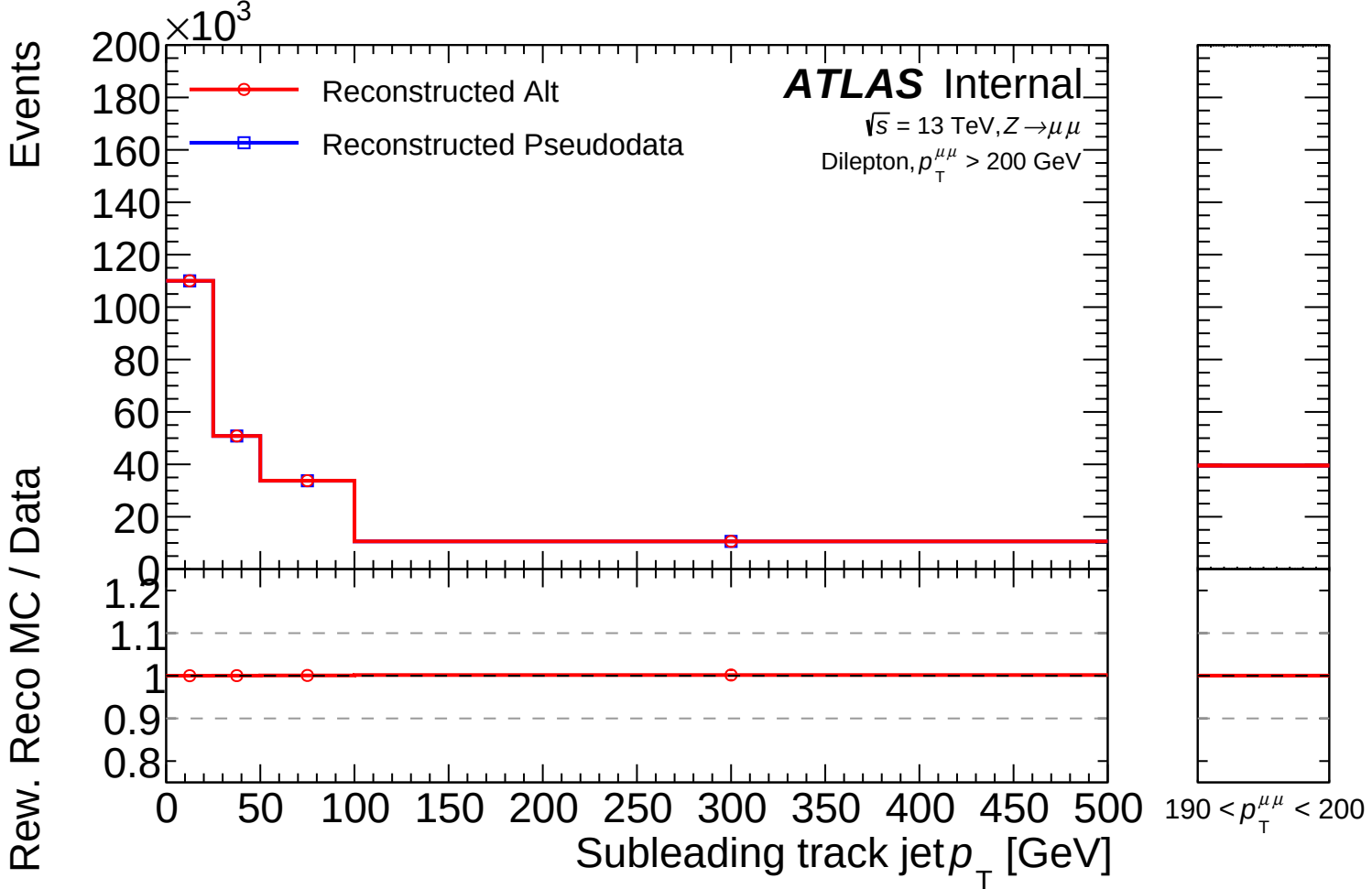
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

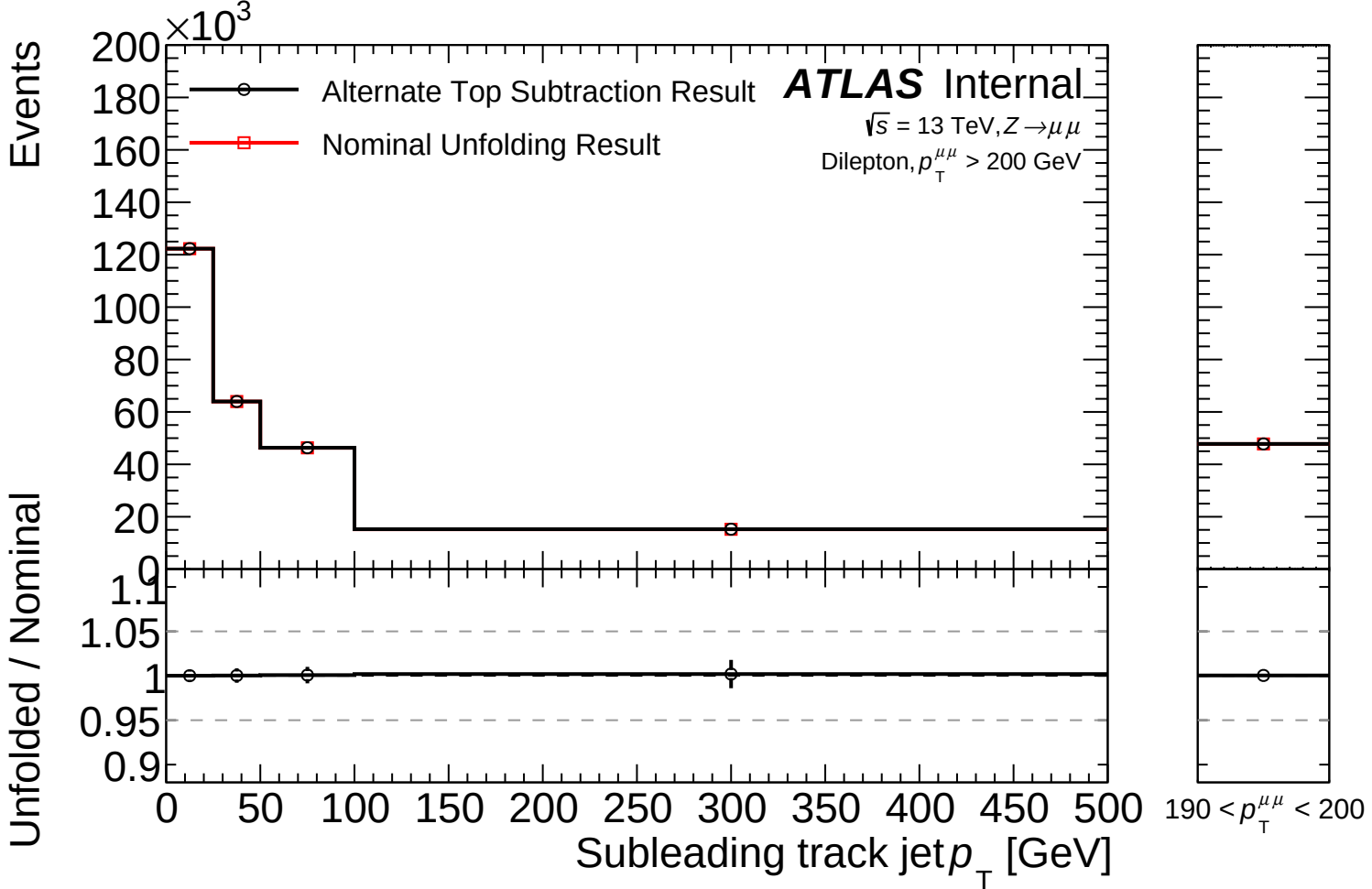
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ 







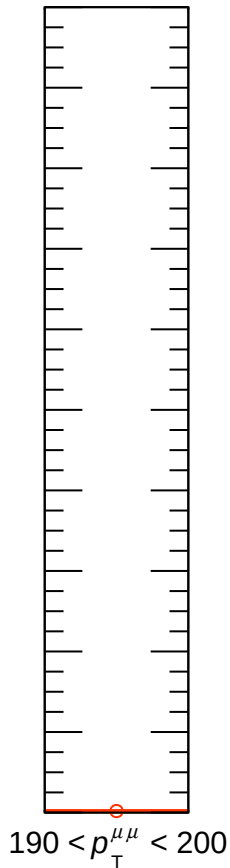
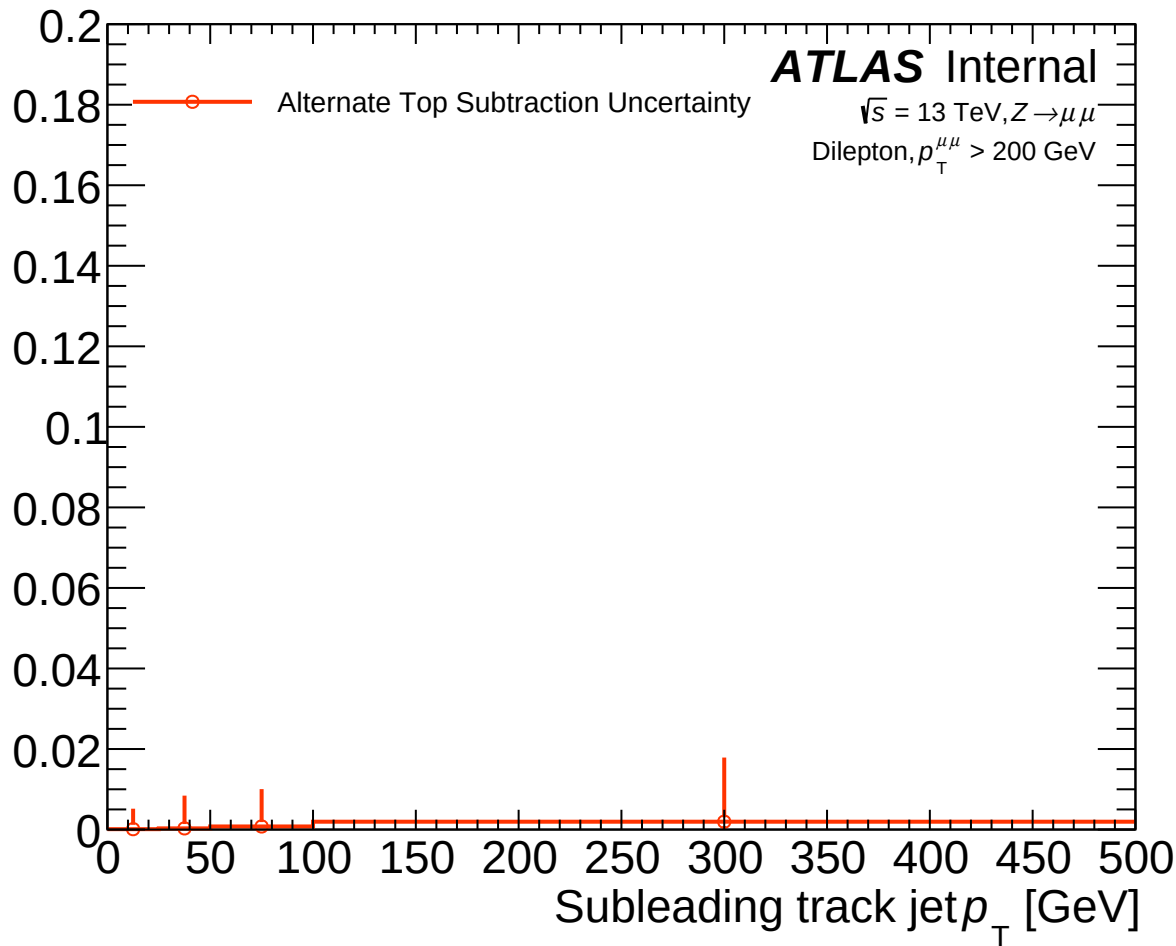
Fractional systematic impact

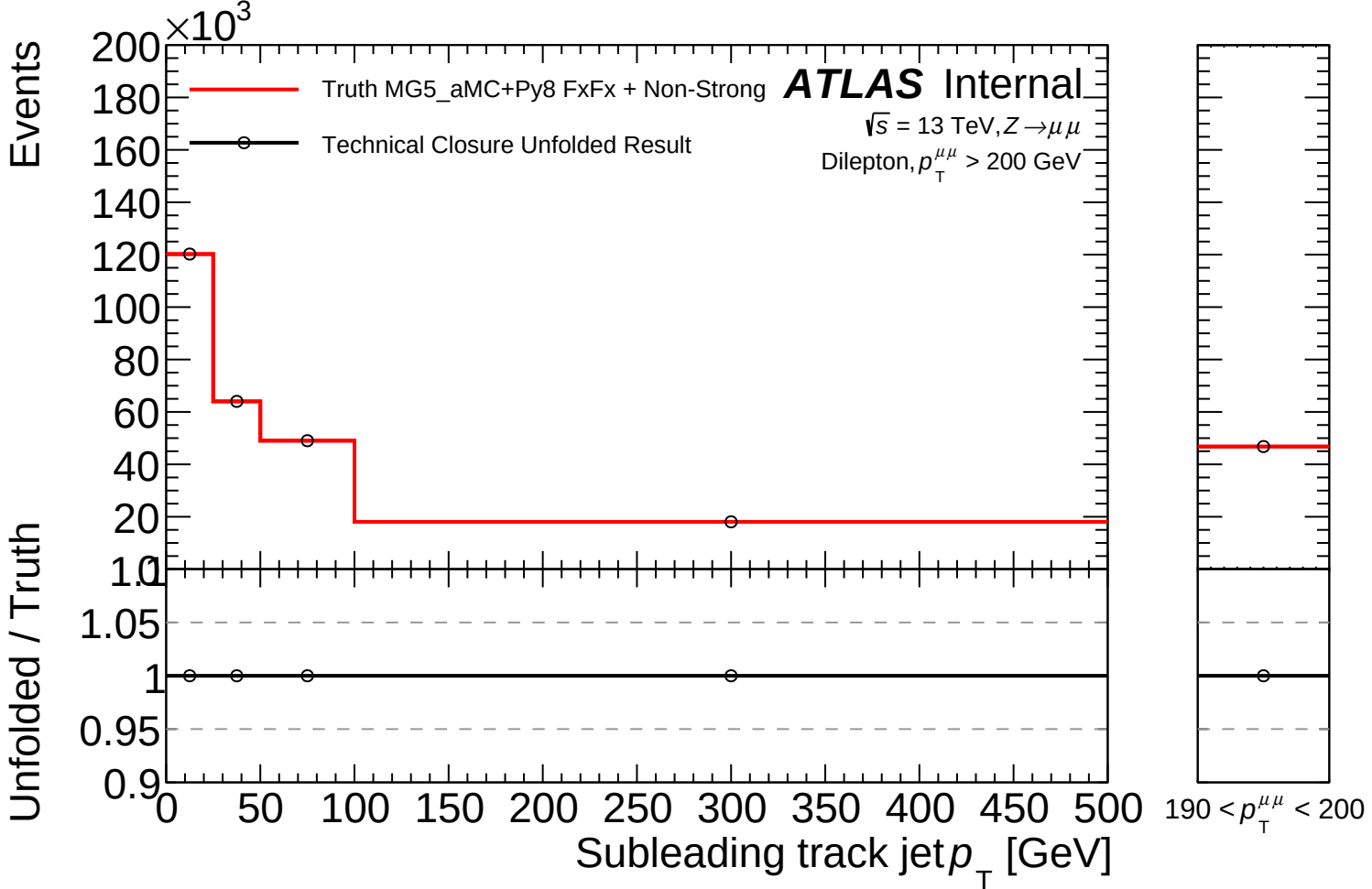
ATLAS Internal

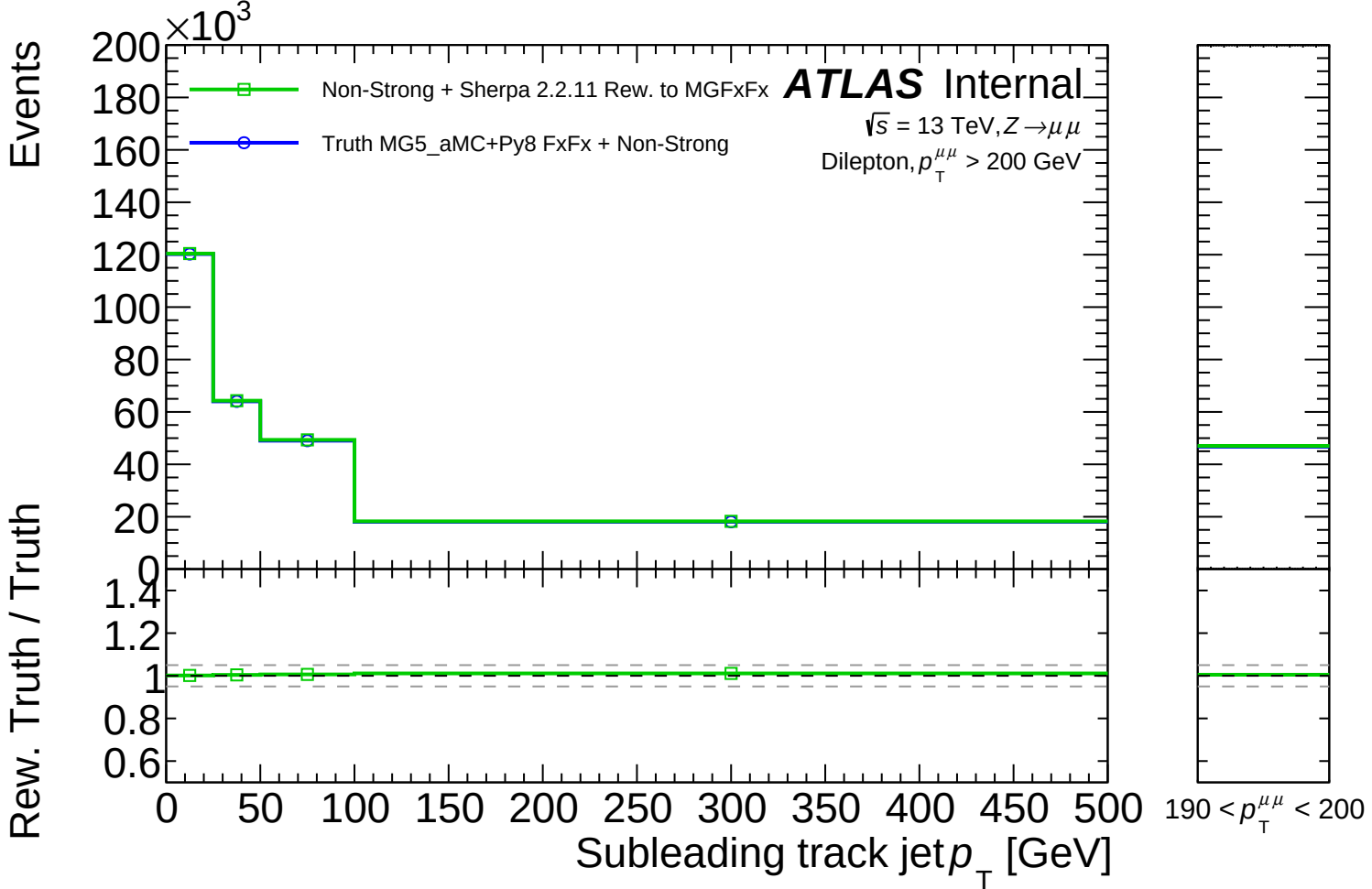
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

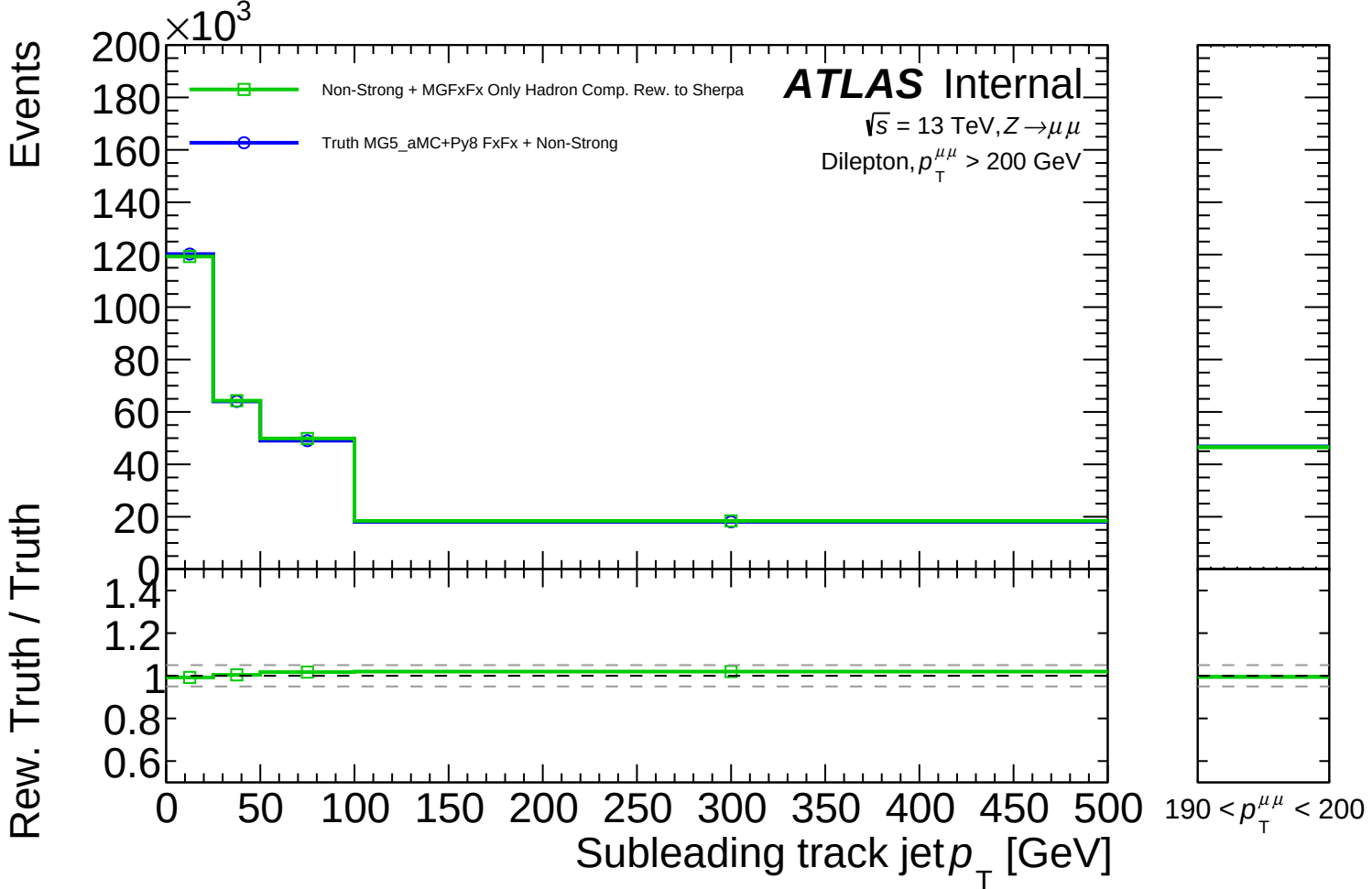
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

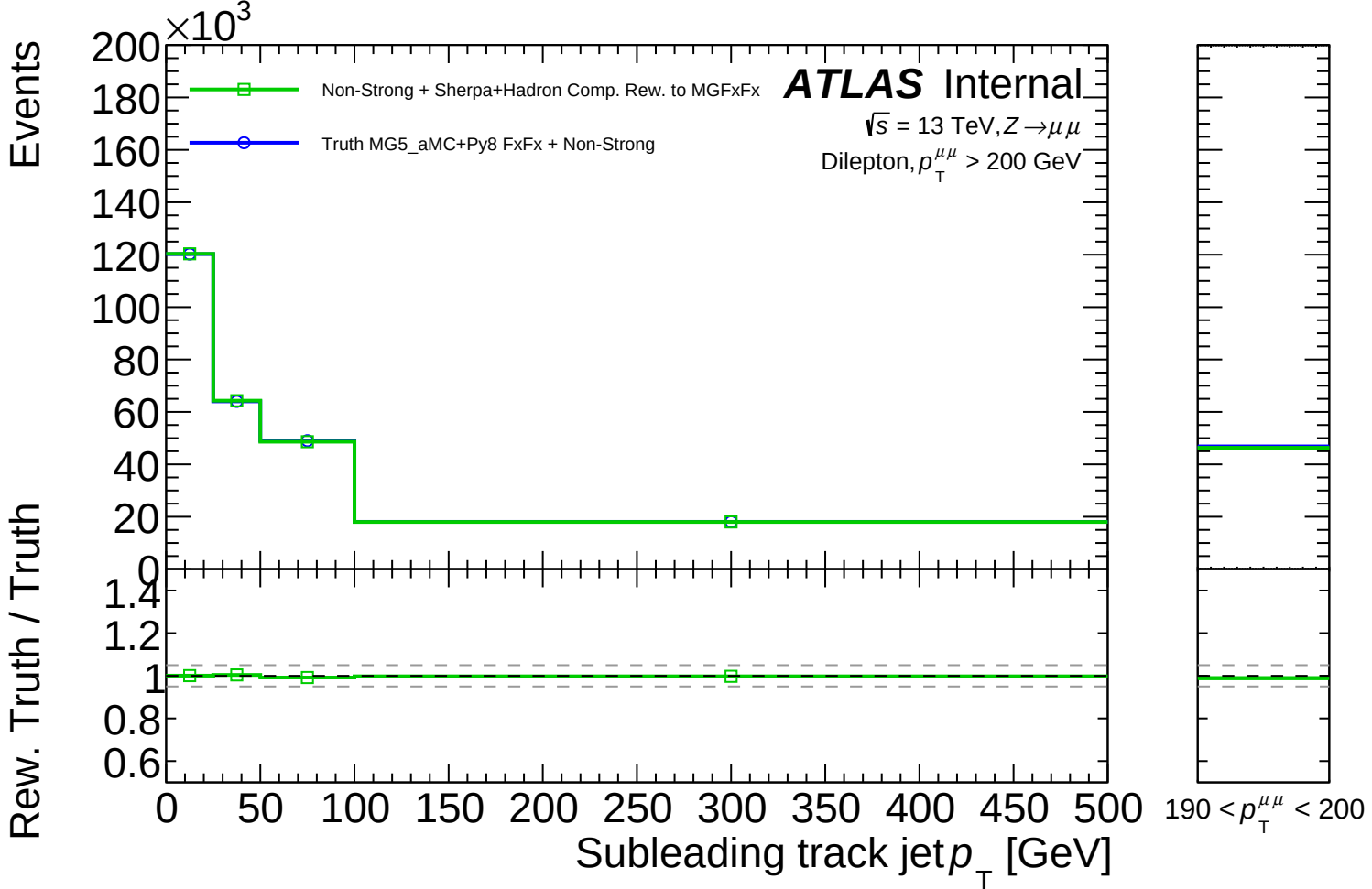
Alternate Top Subtraction Uncertainty

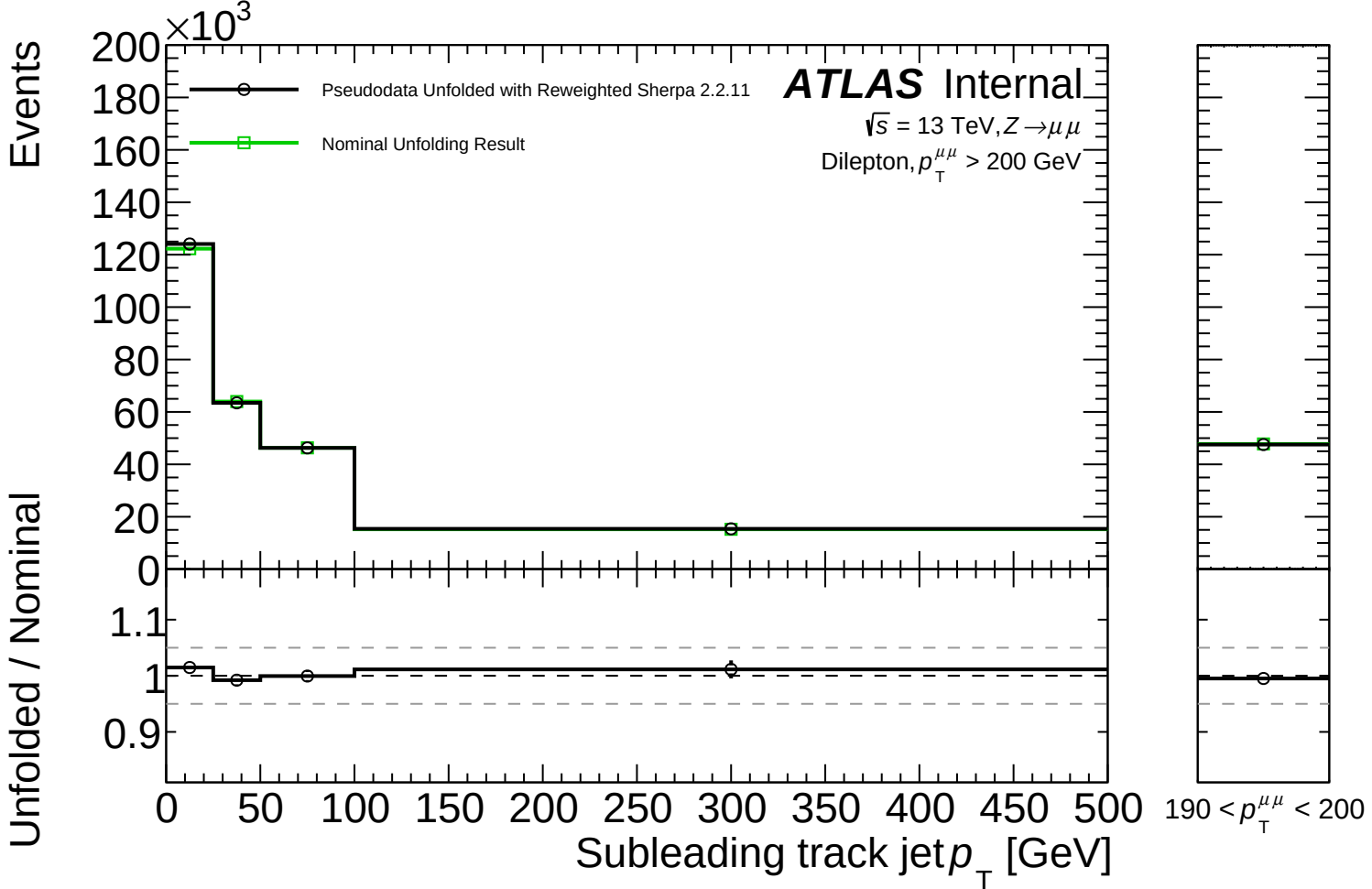


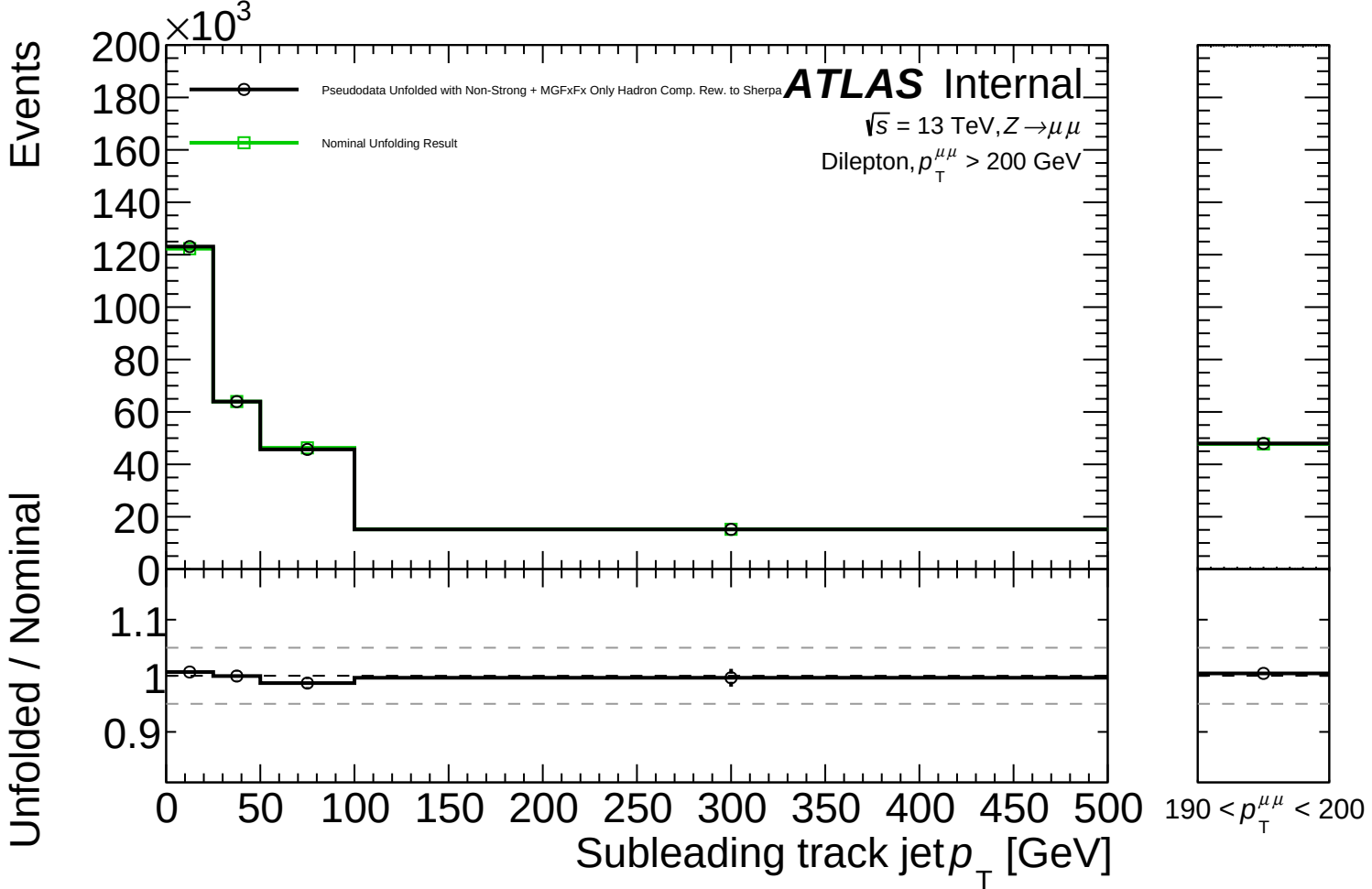


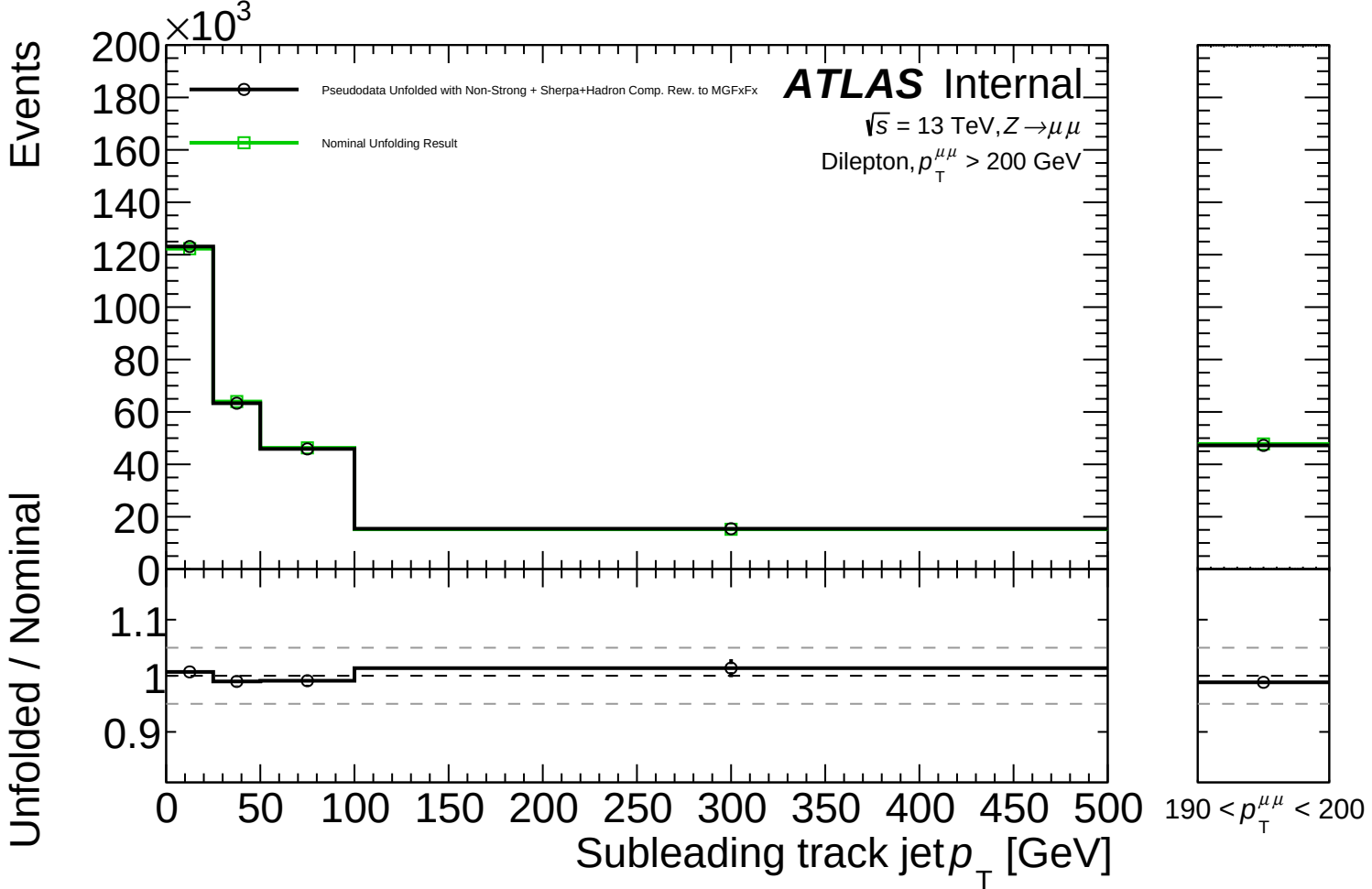


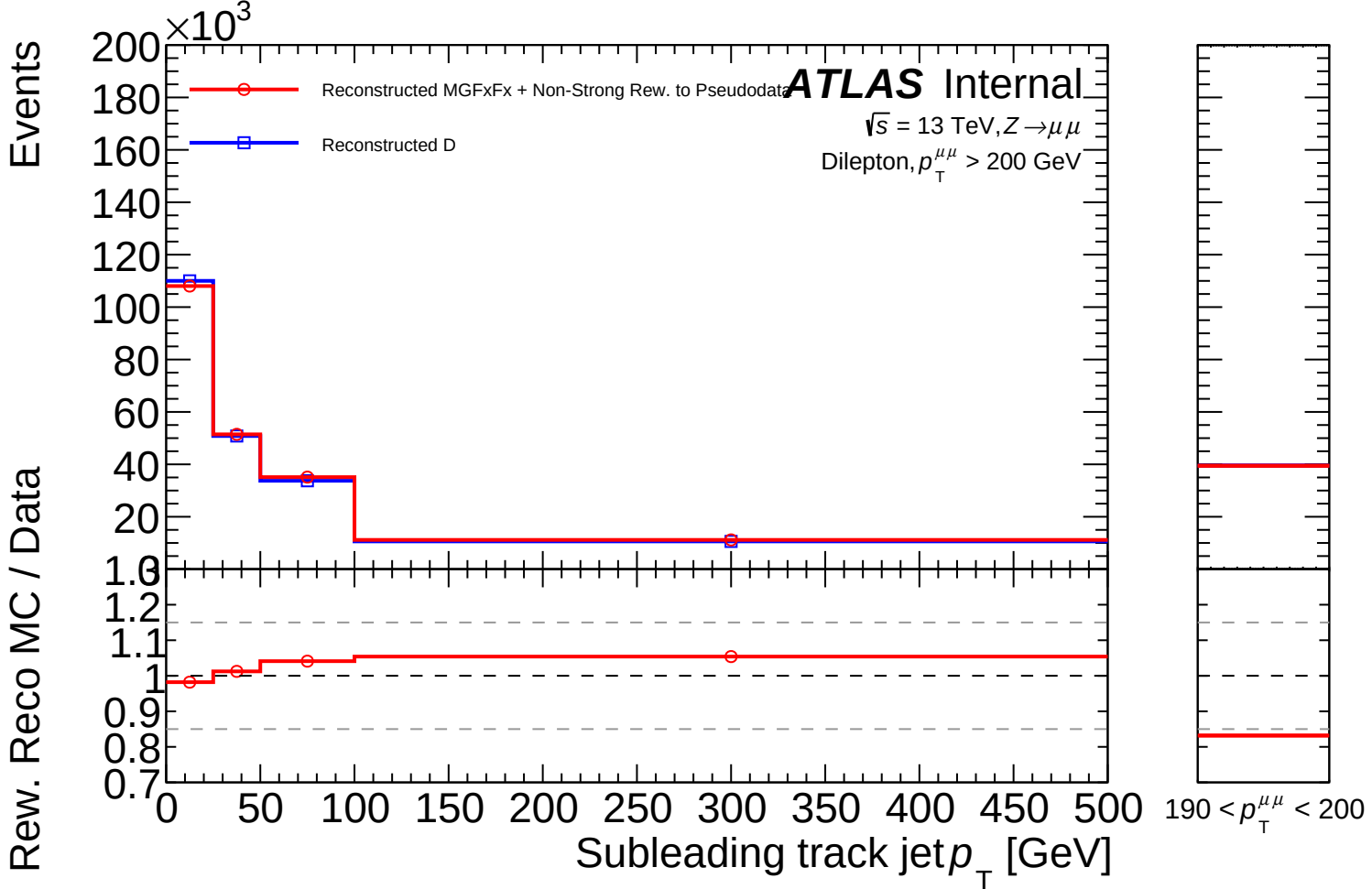


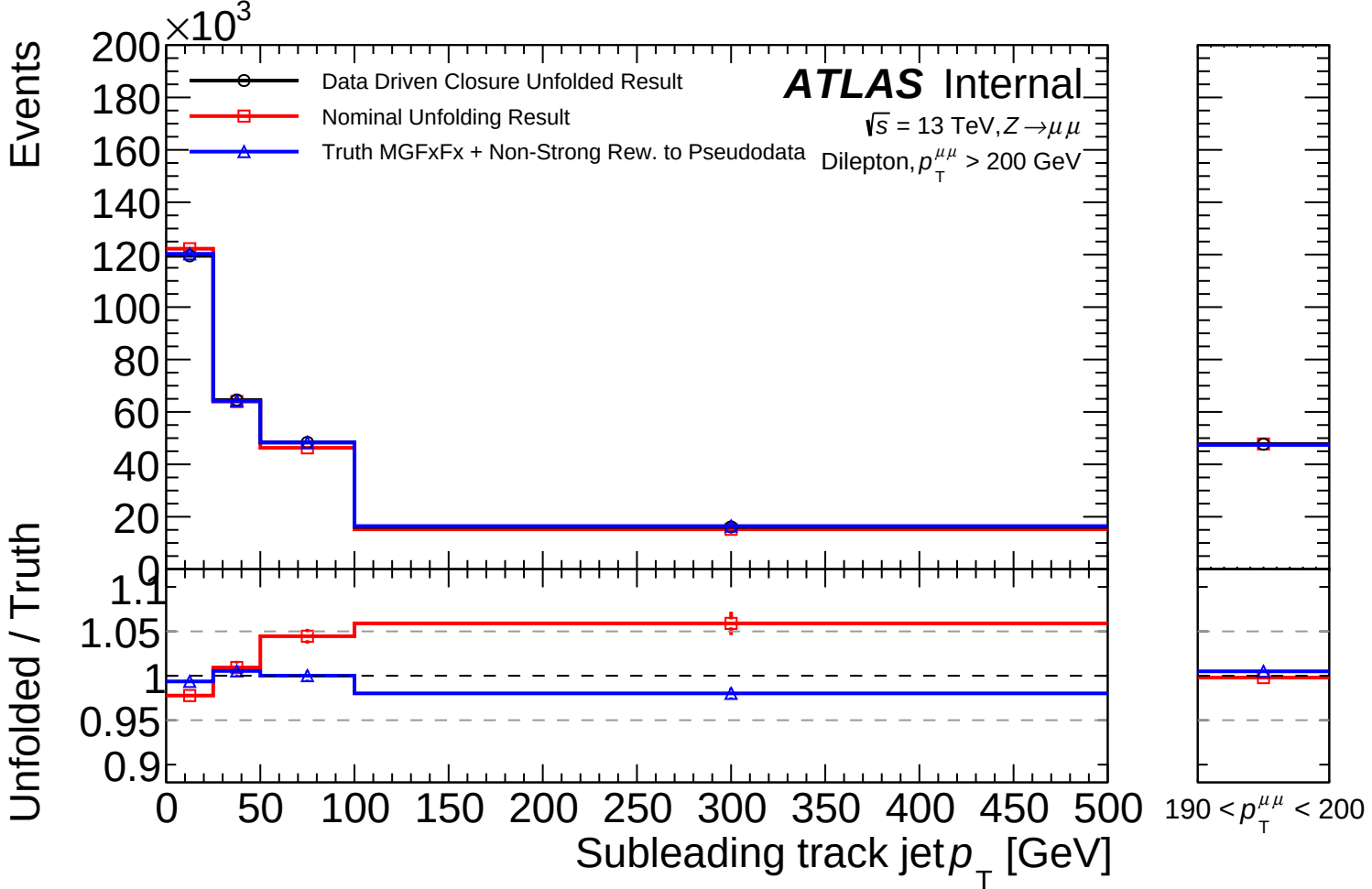




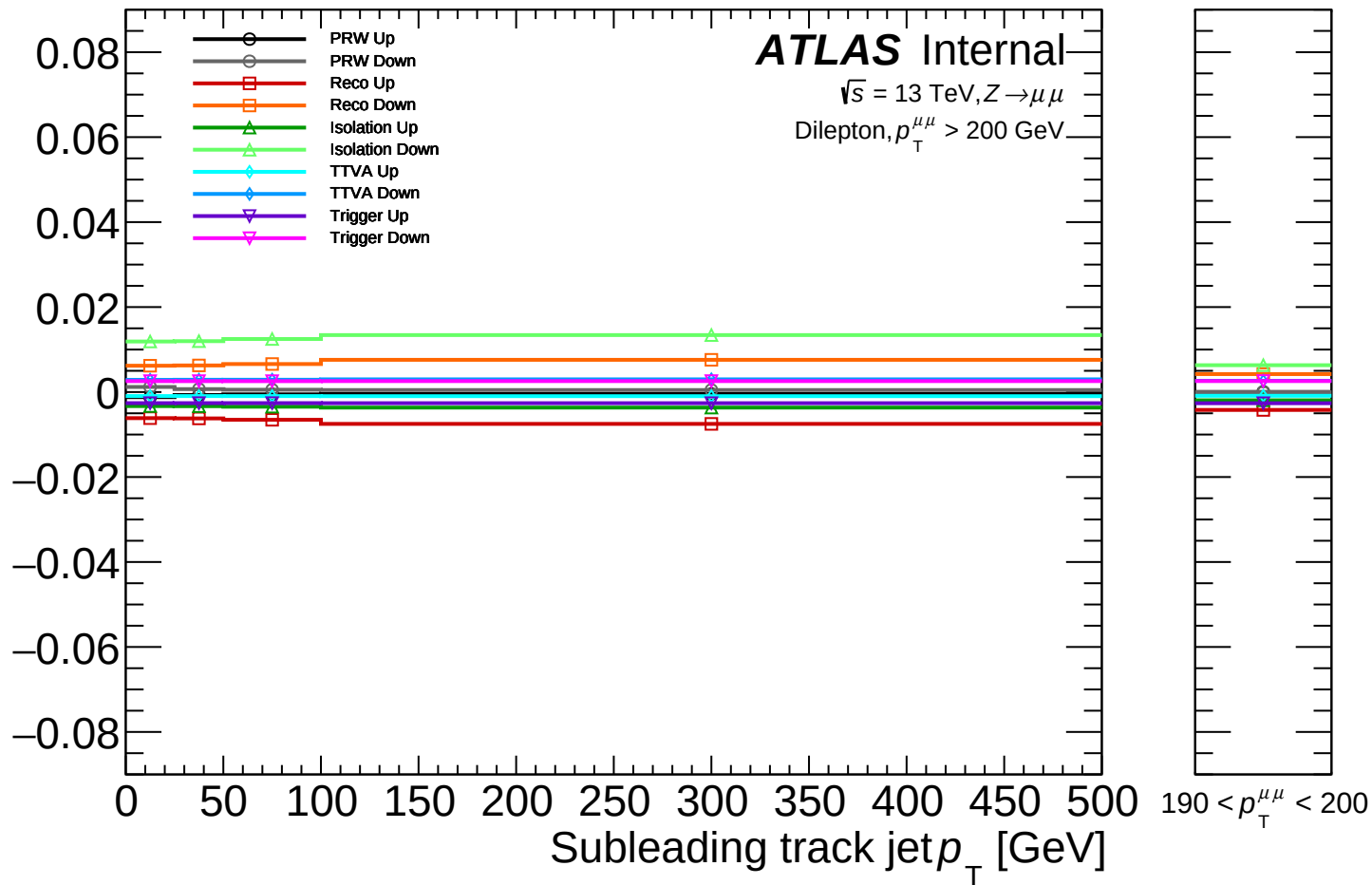




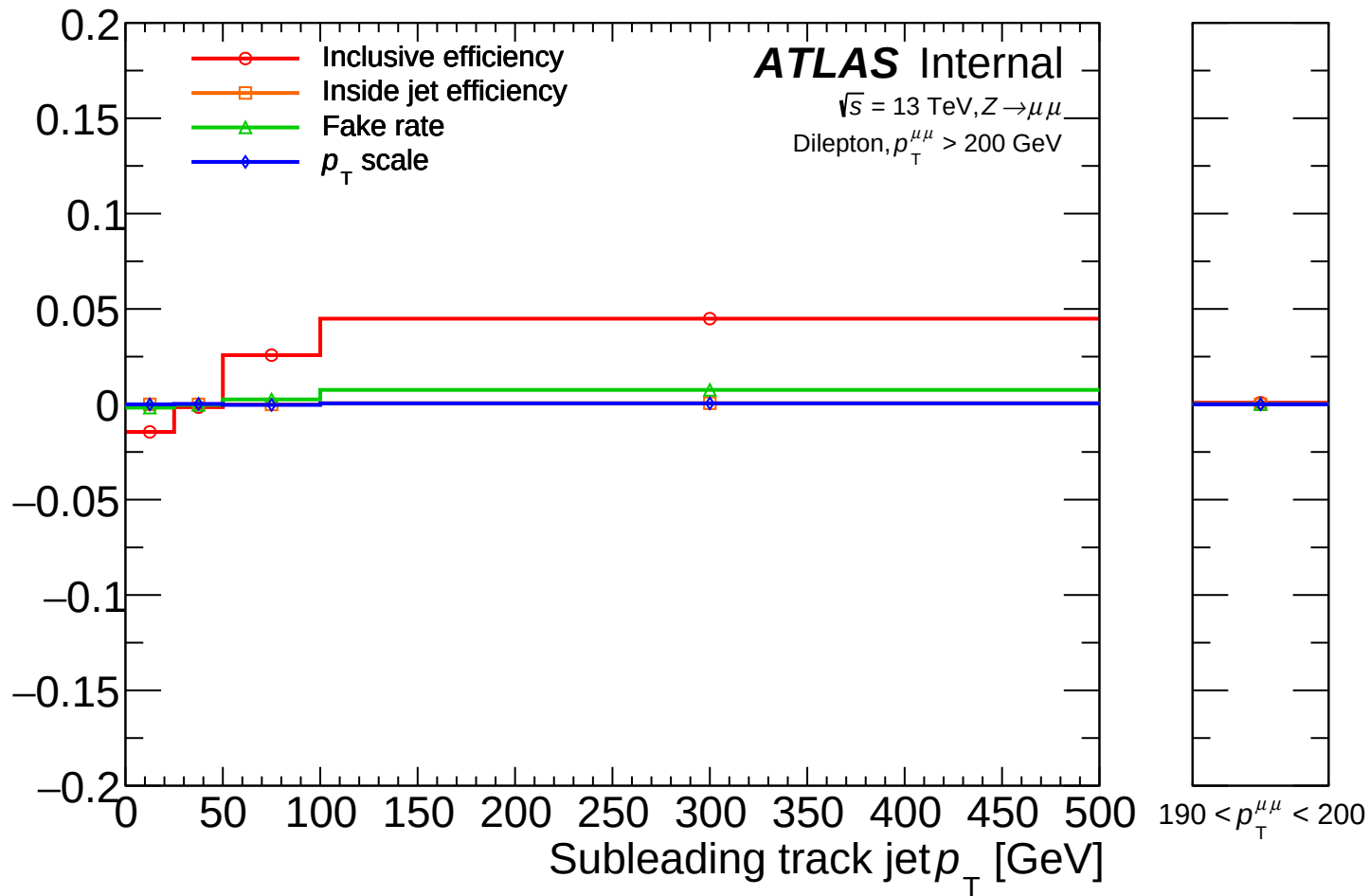




Fractional systematic impact



Fractional systematic impact



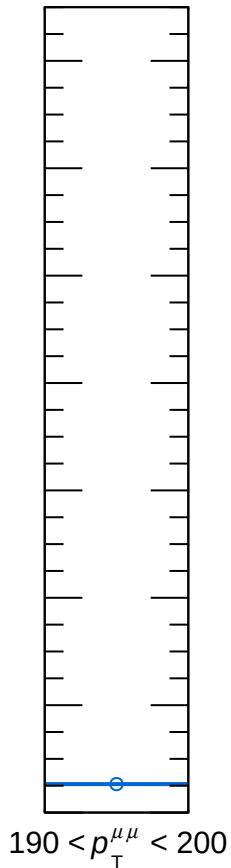
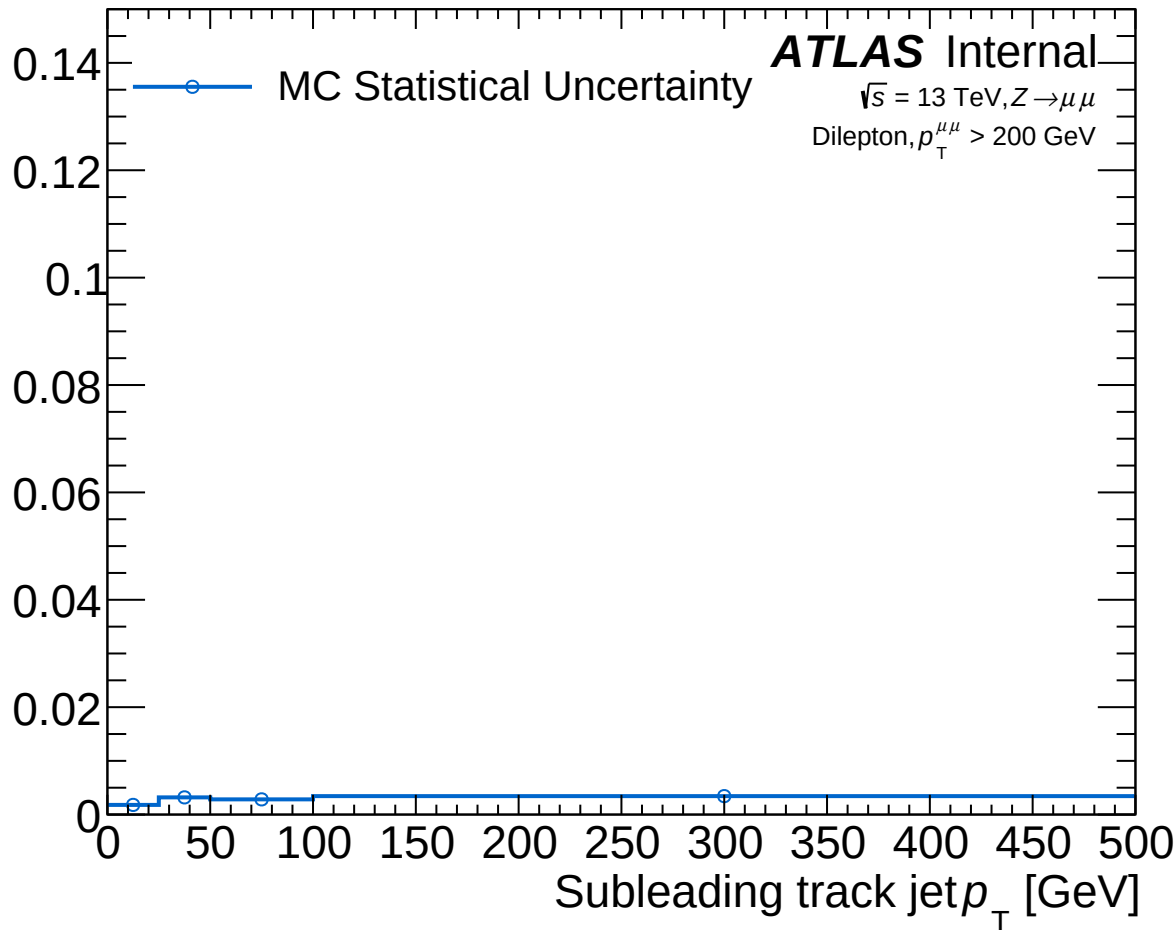
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



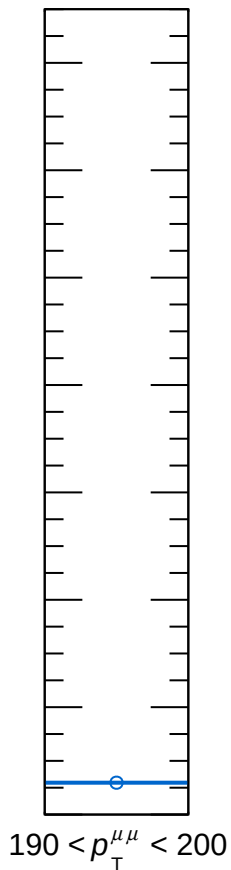
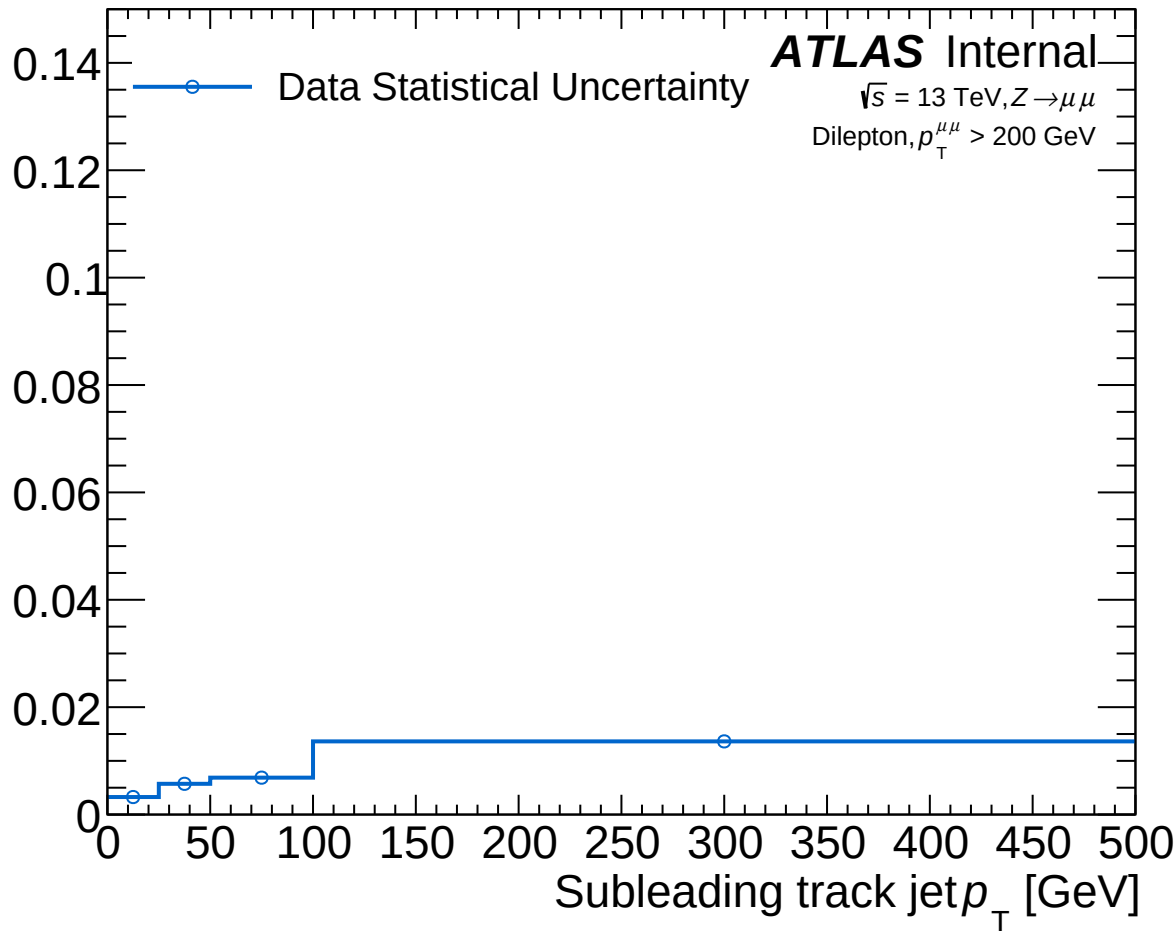
Fractional systematic impact

ATLAS Internal

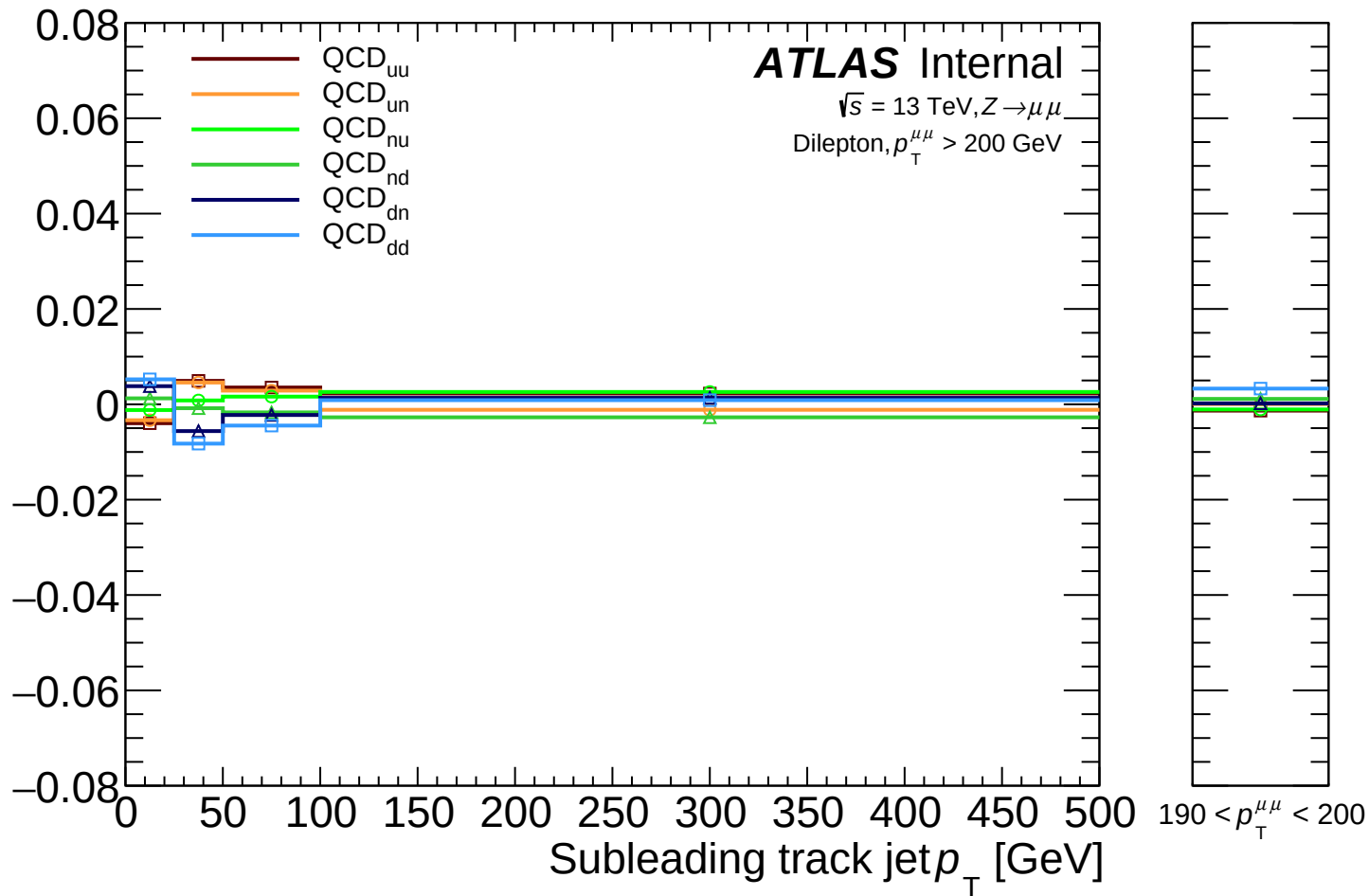
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

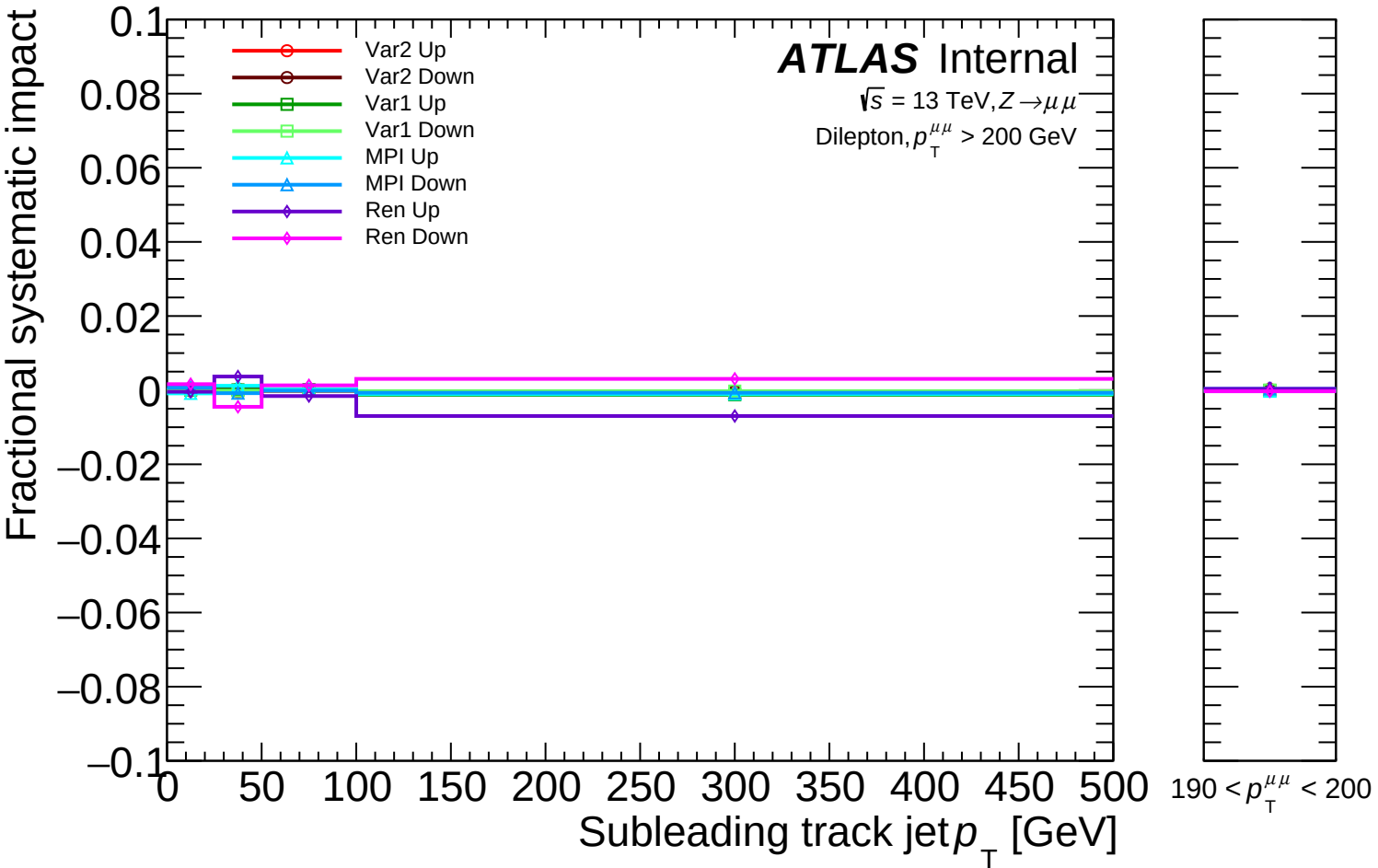
Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

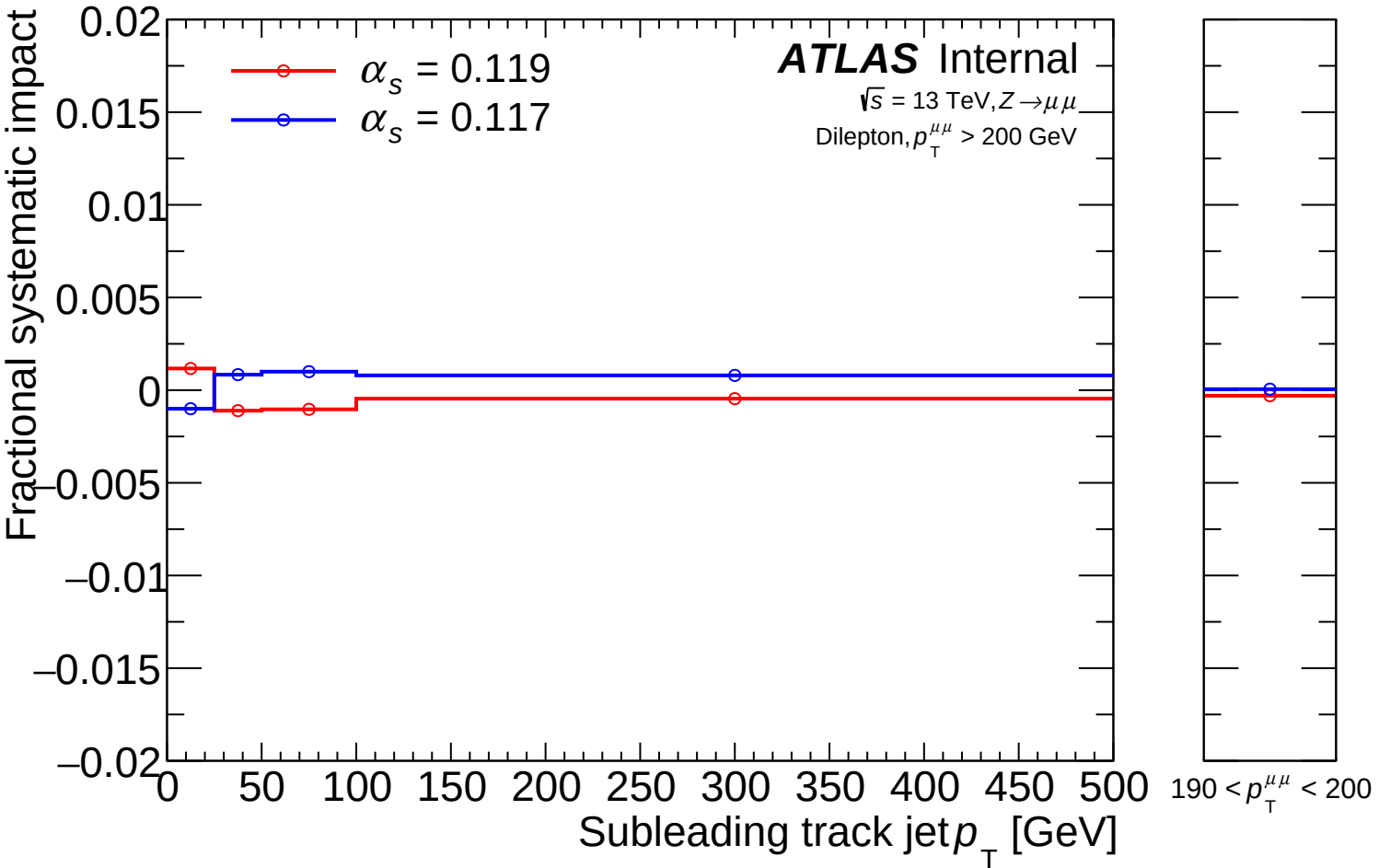
Data Statistical Uncertainty

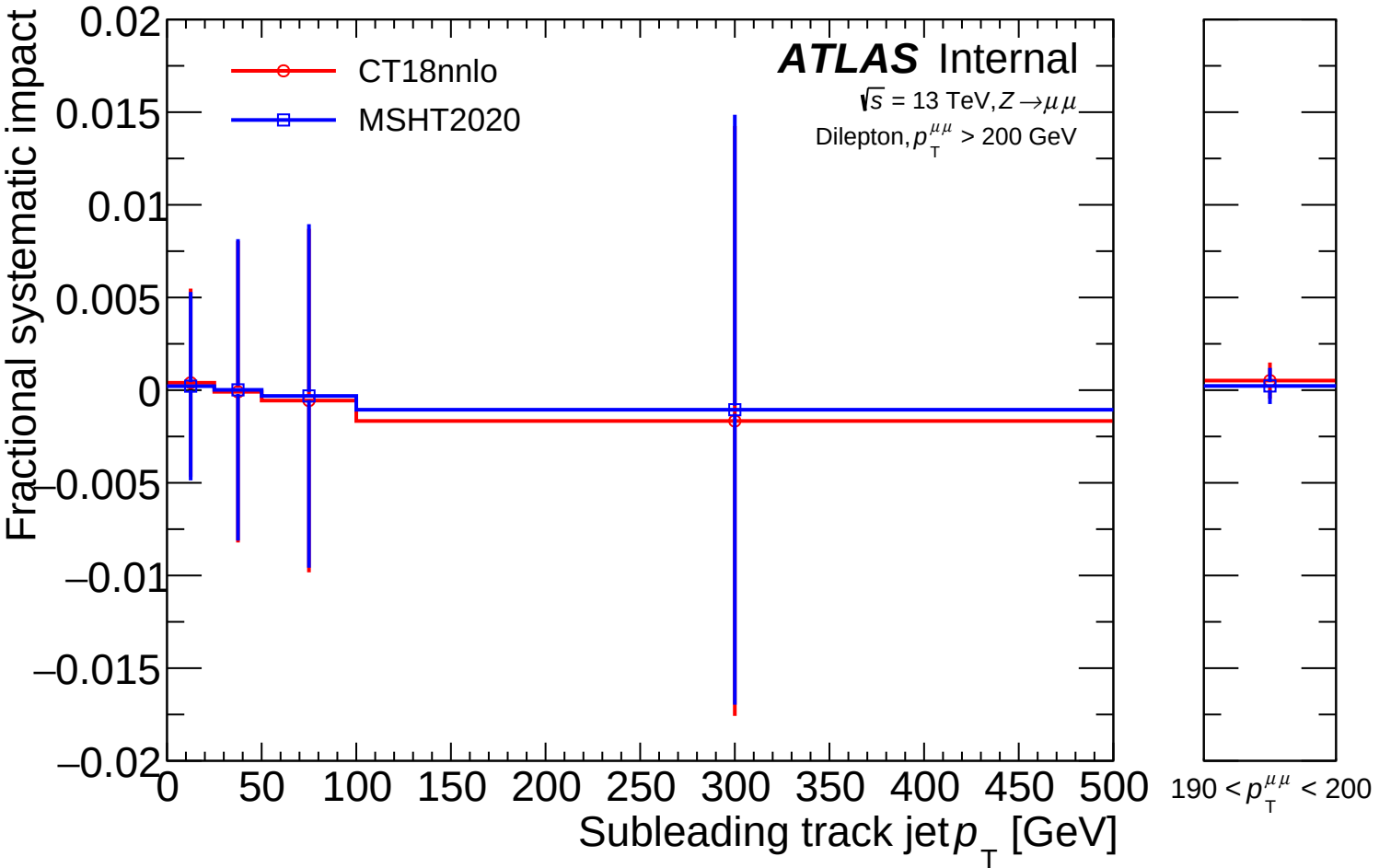


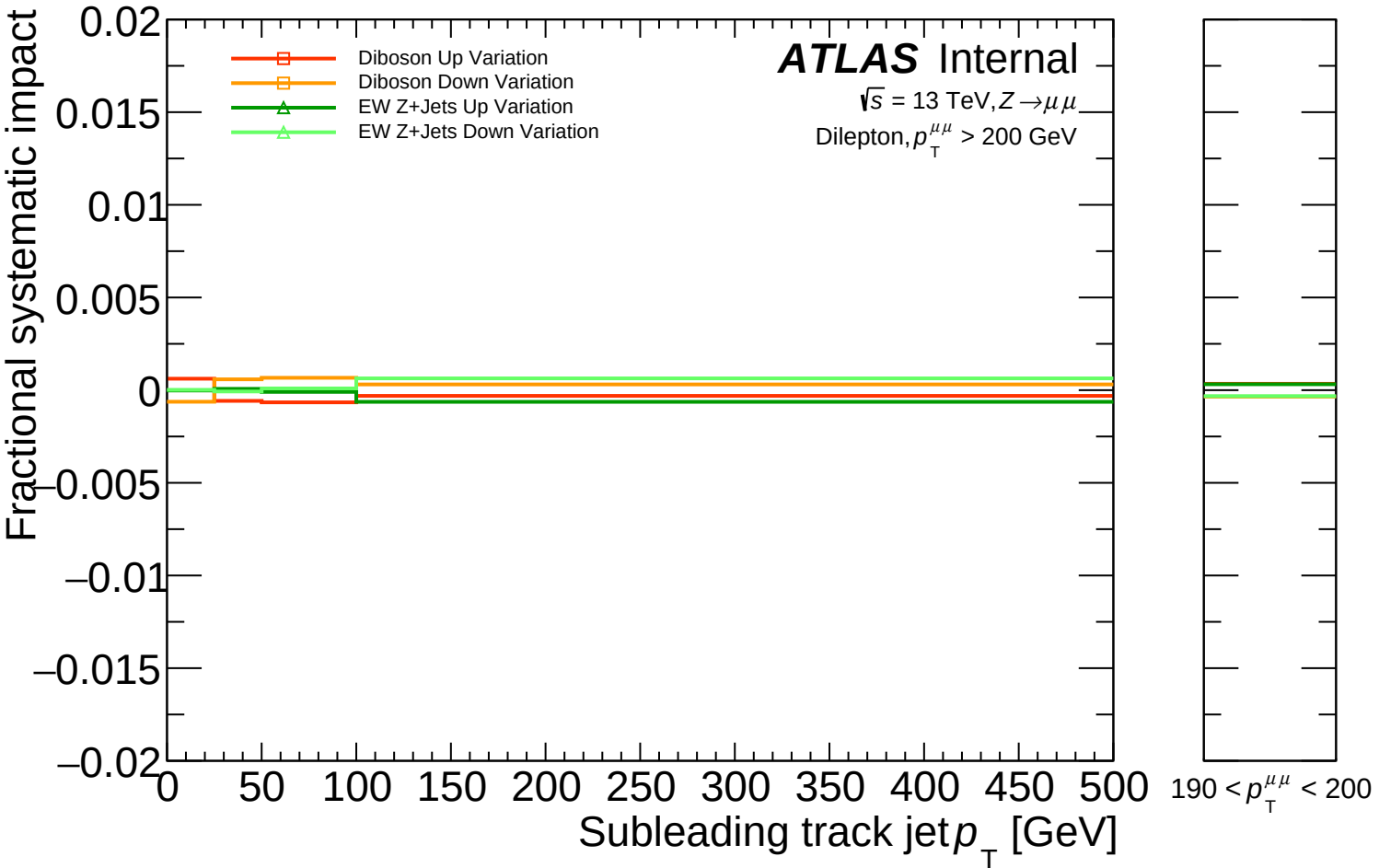
Fractional systematic impact









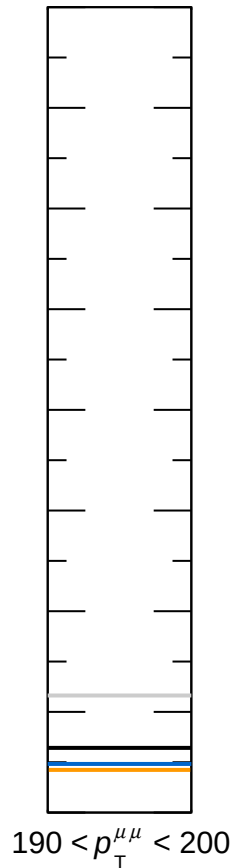
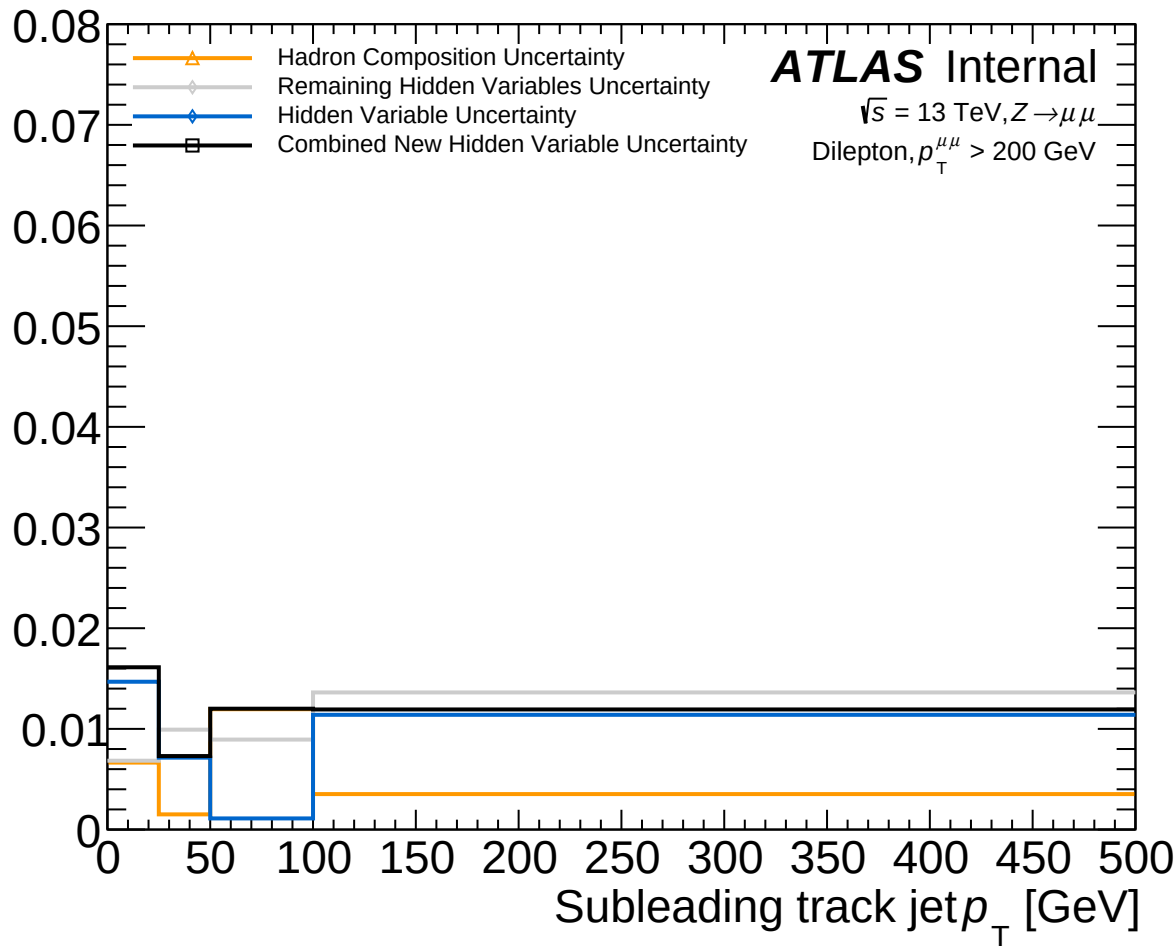


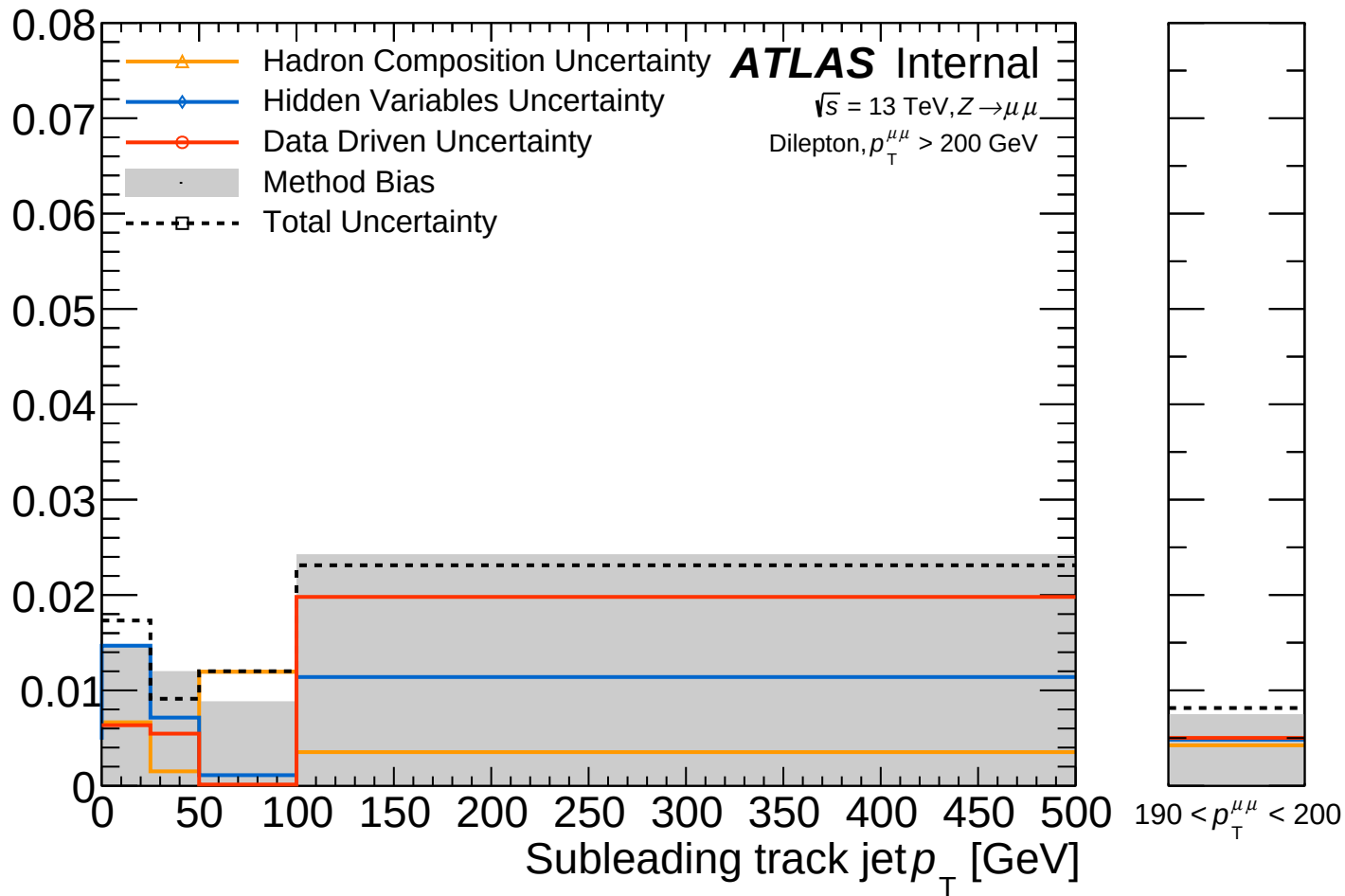
ATLAS Internal

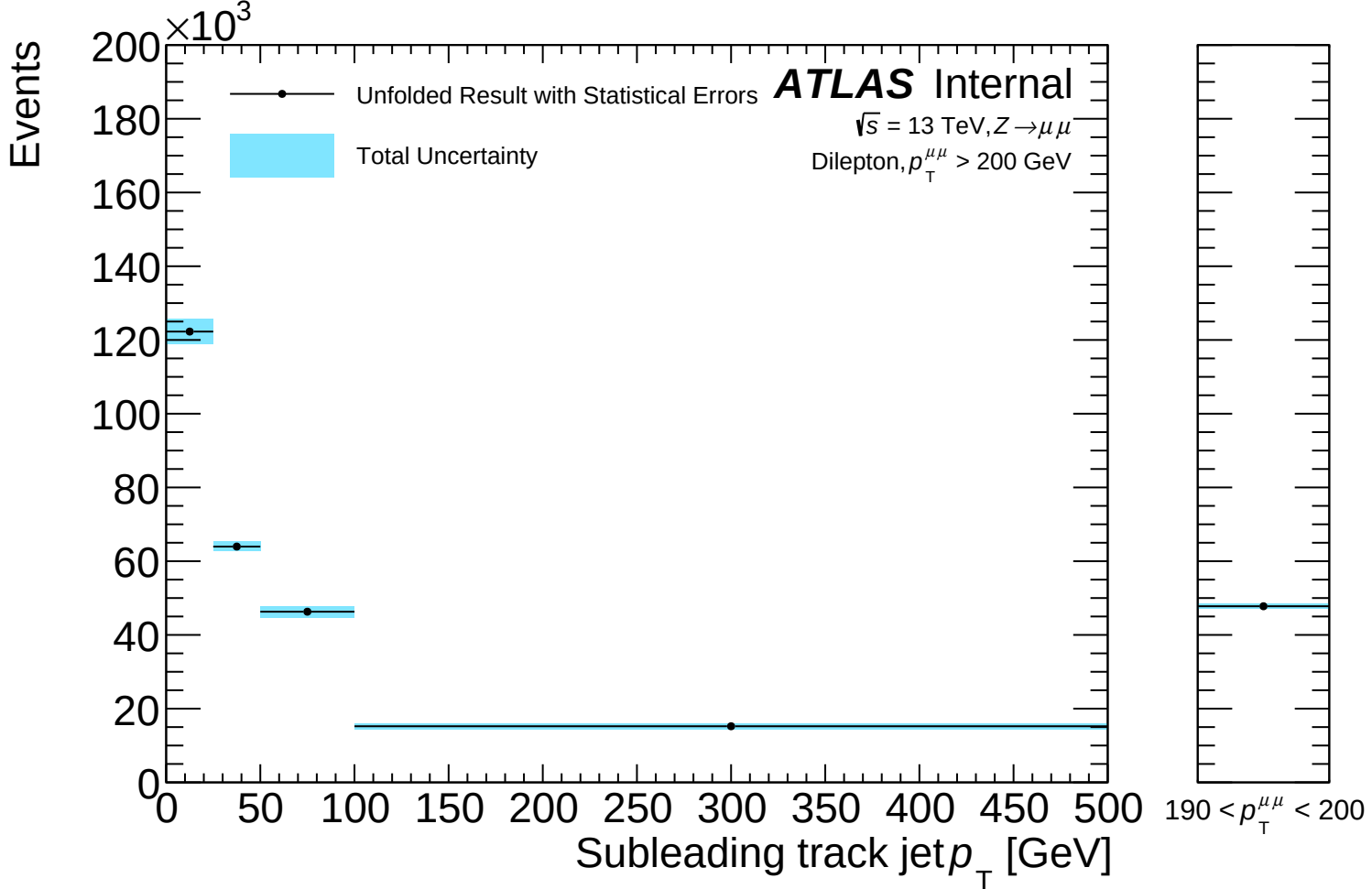
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

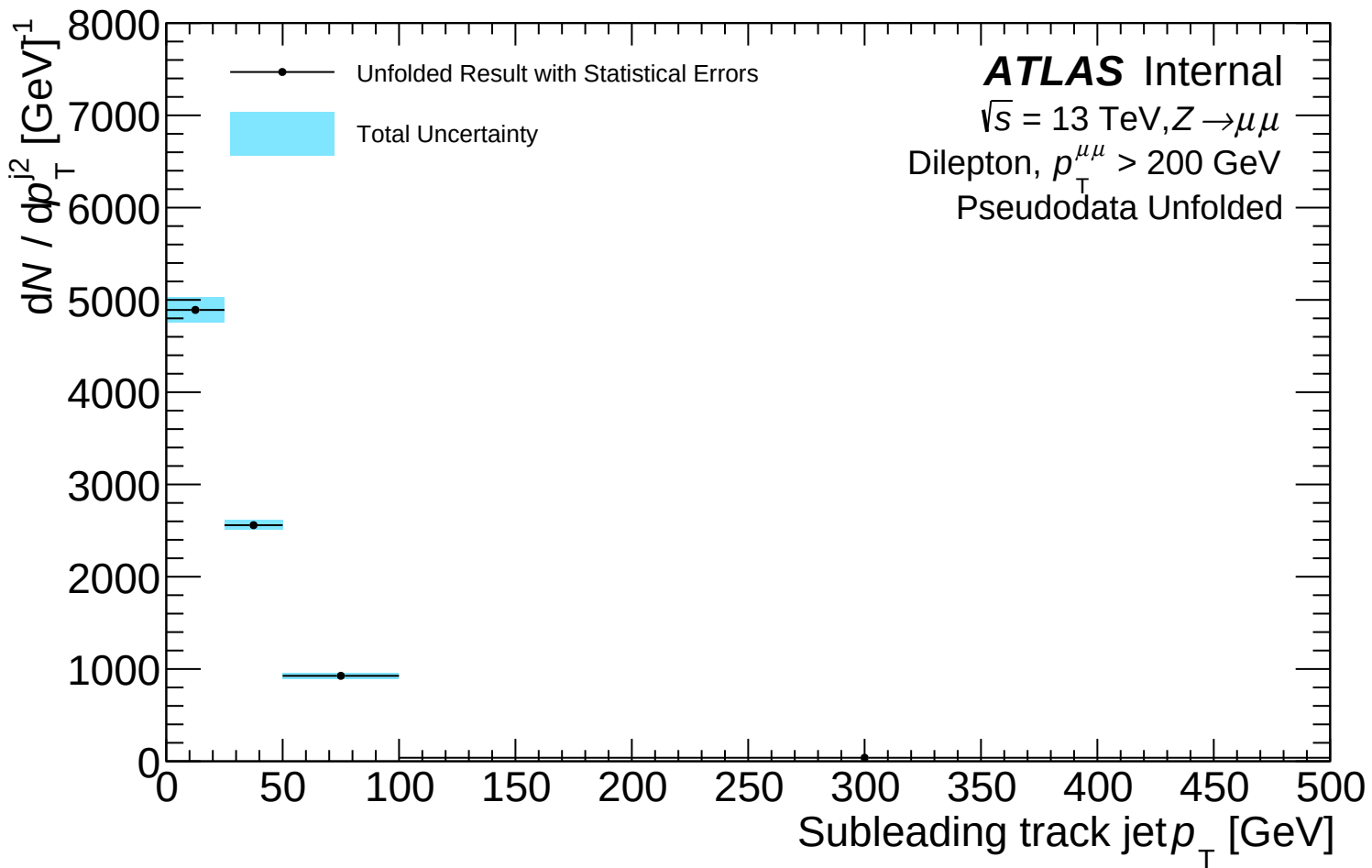
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

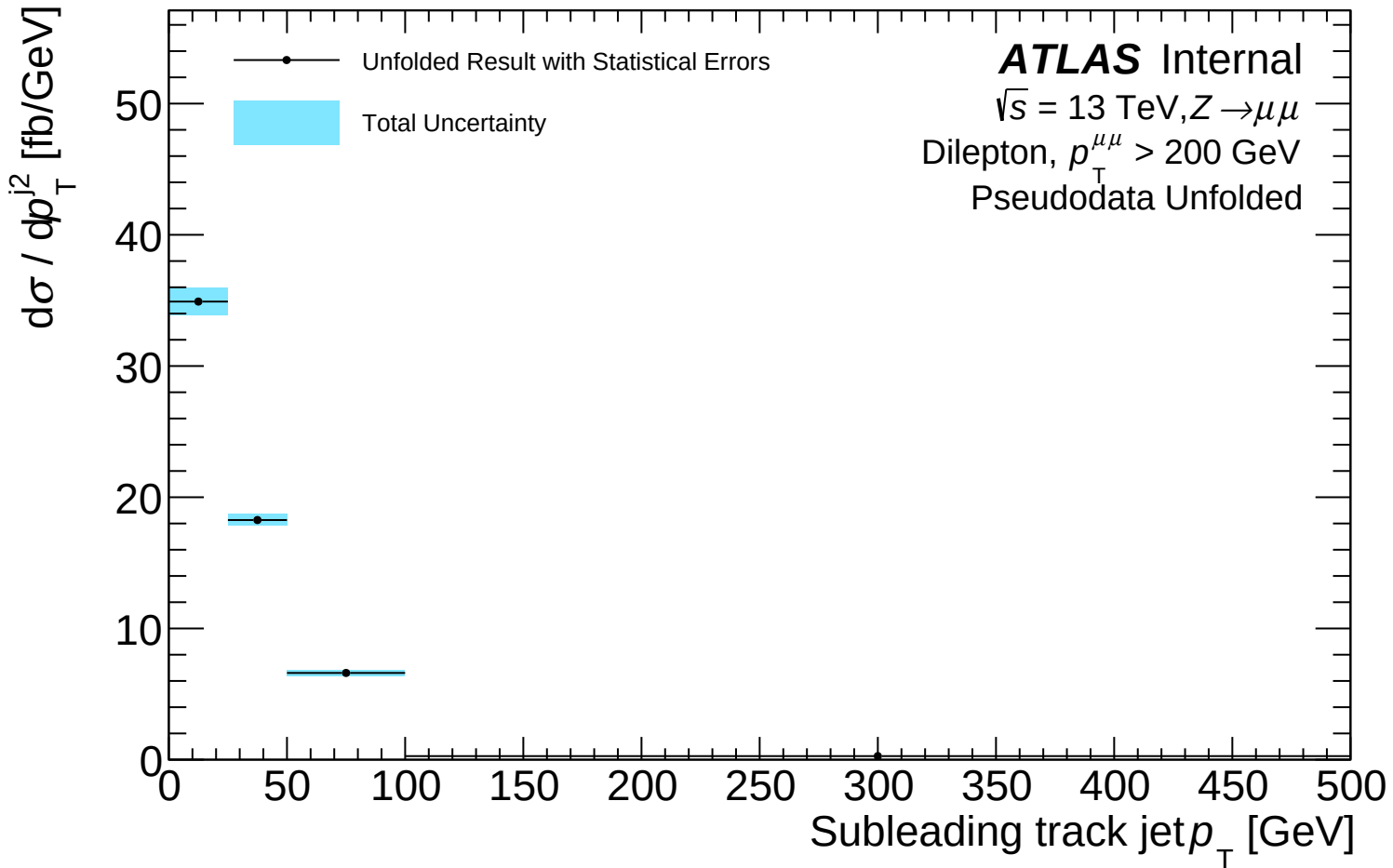
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty











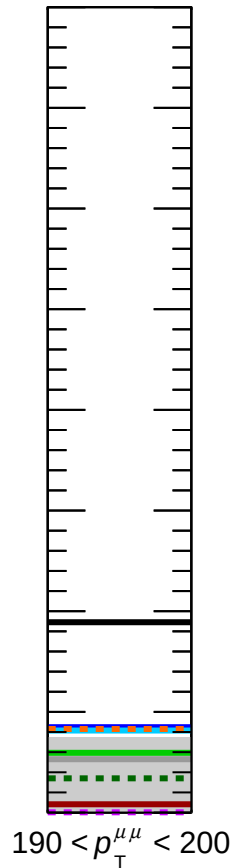
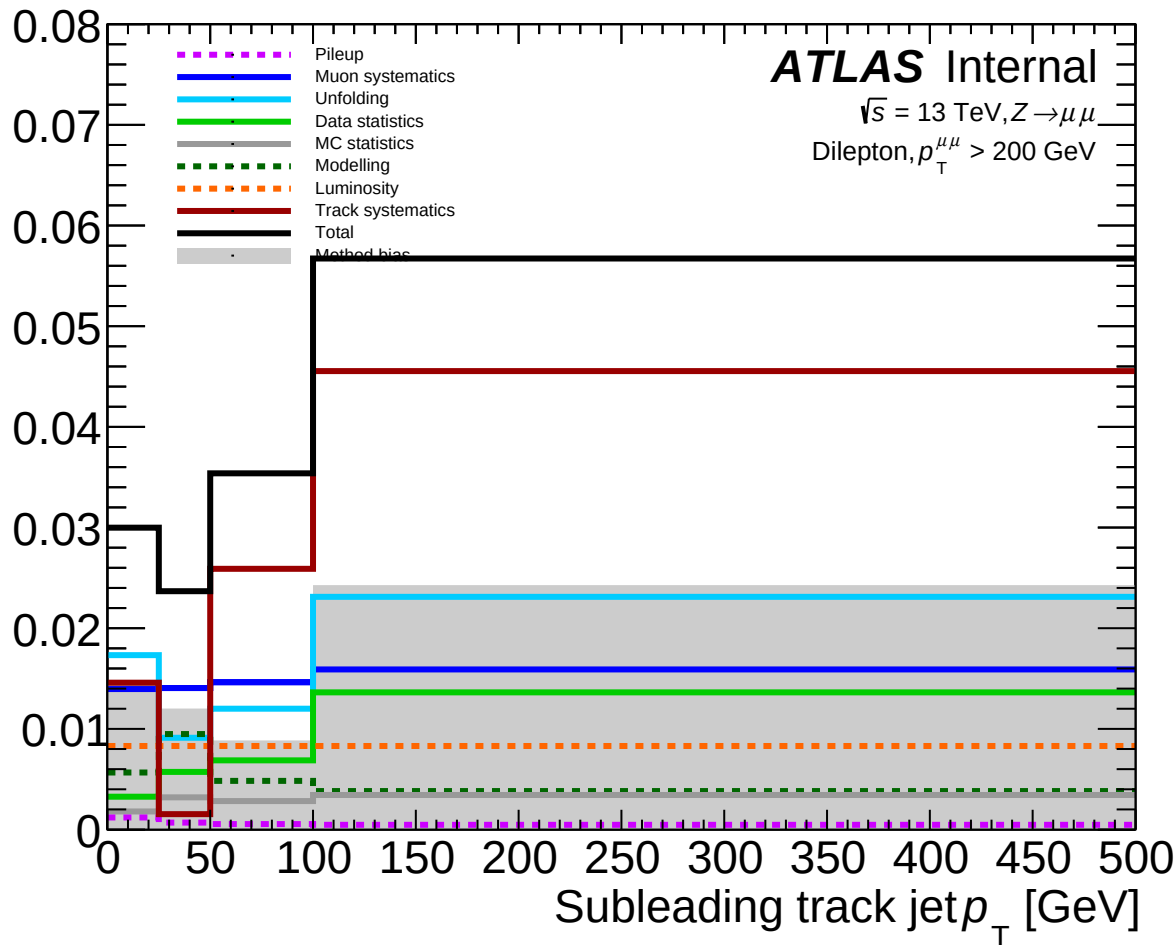
Fractional uncertainty

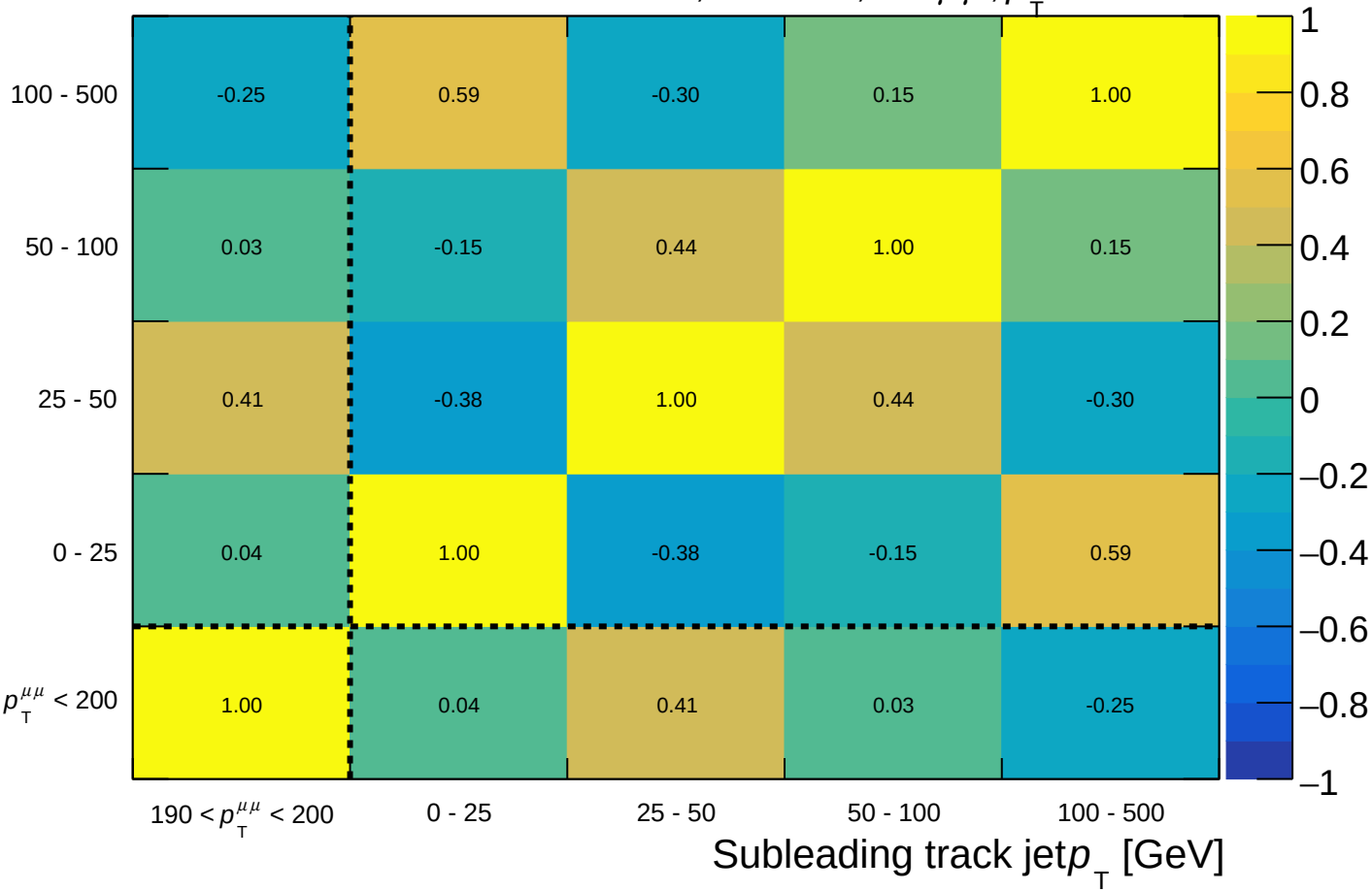
ATLAS Internal

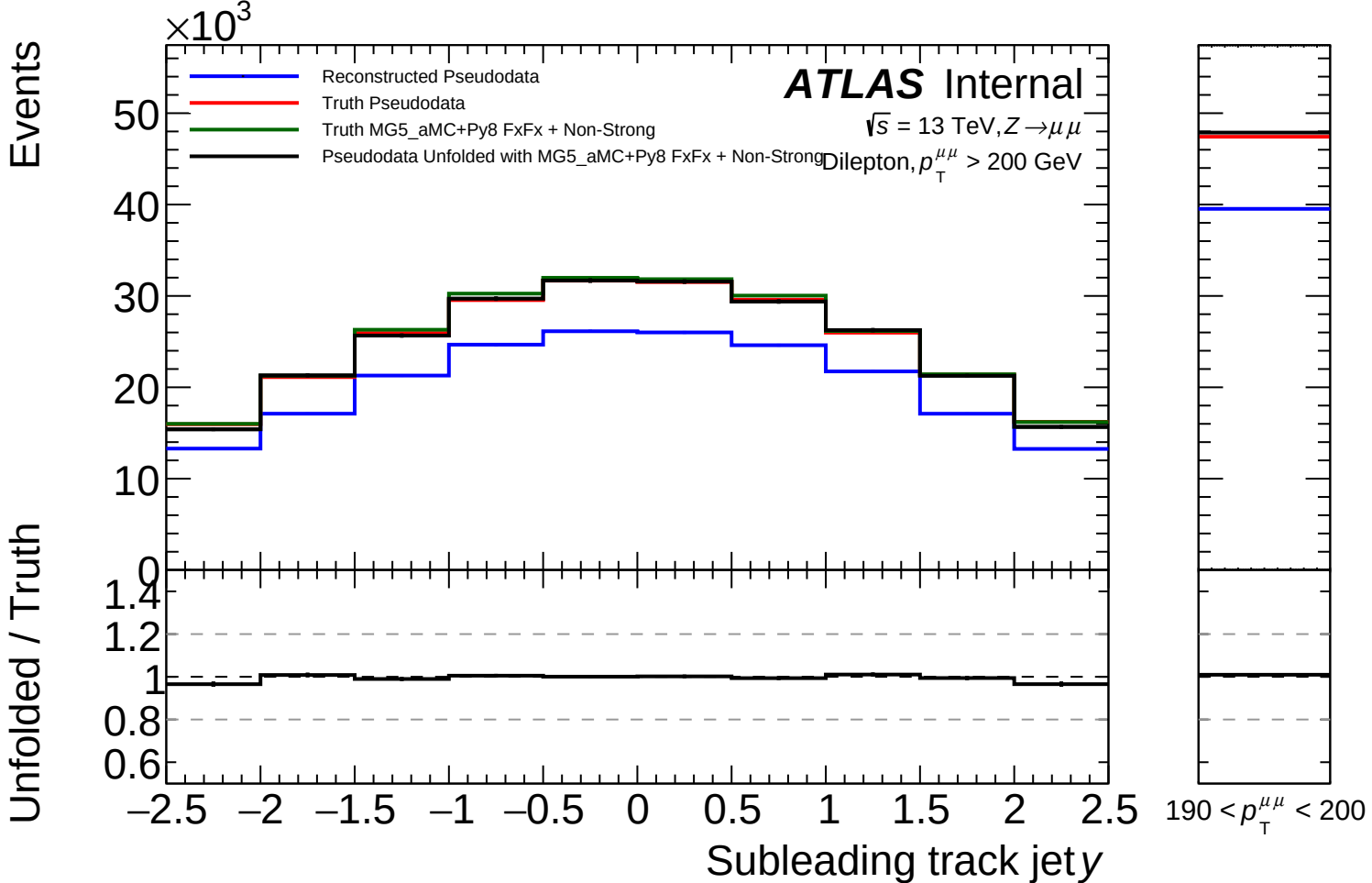
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

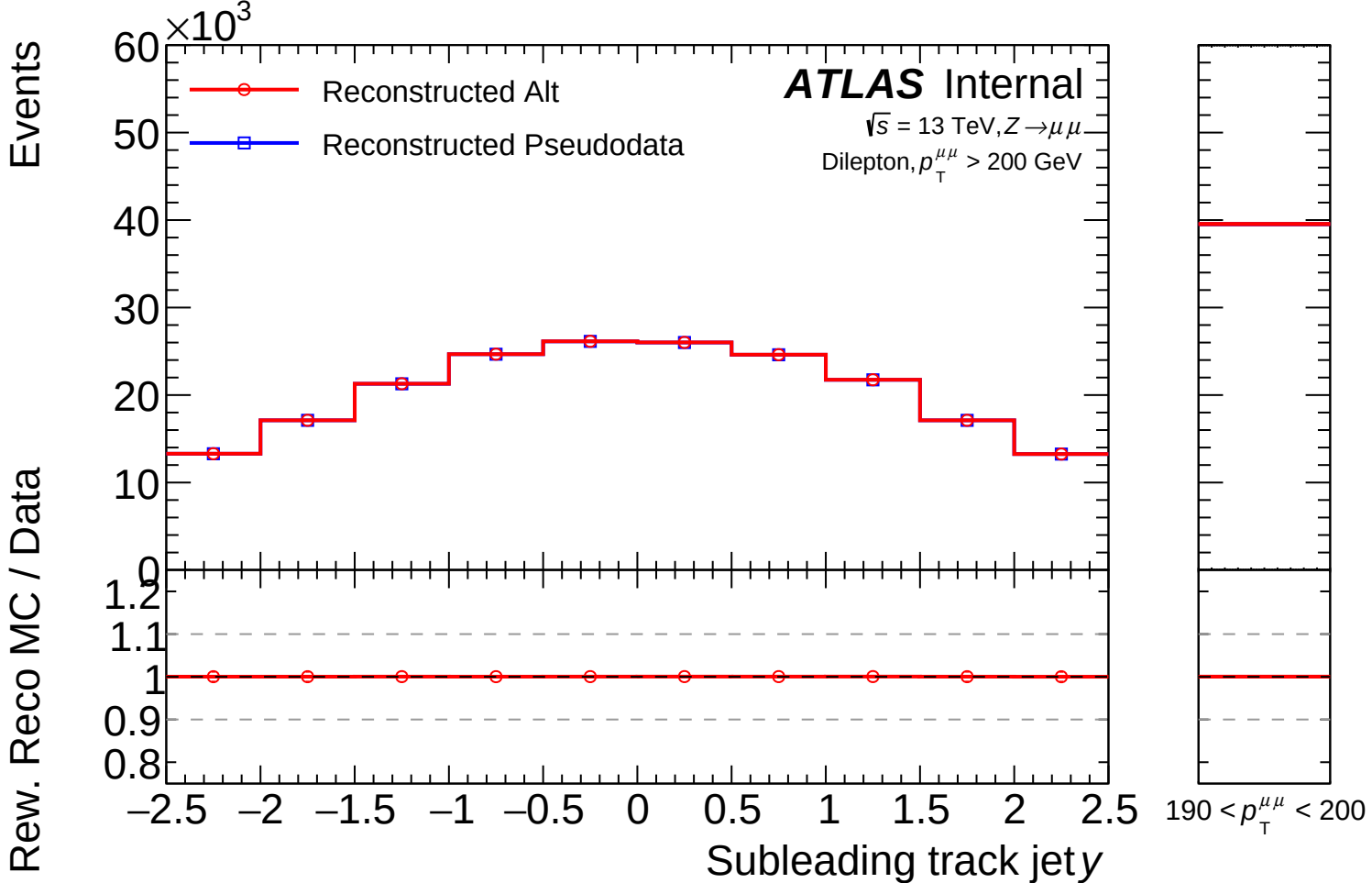
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

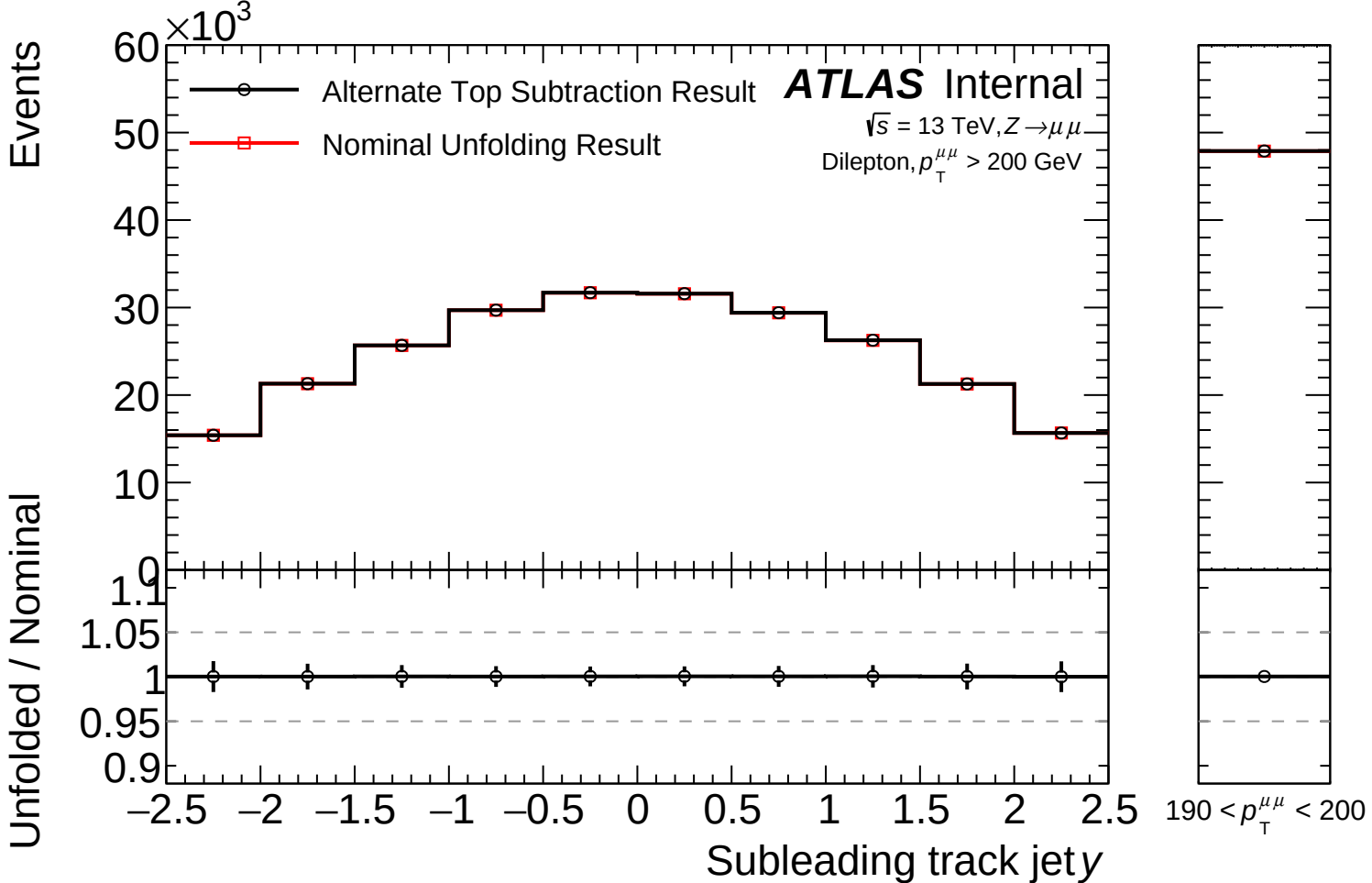
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Subleading track jet p_{T} [GeV]







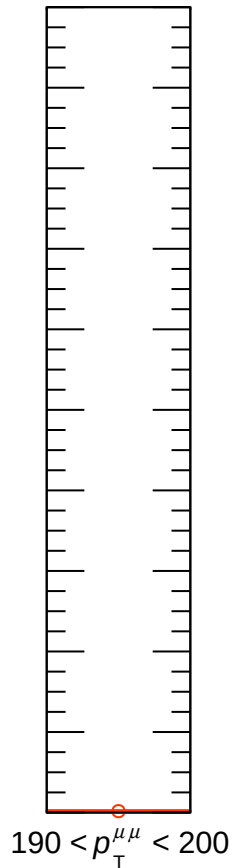
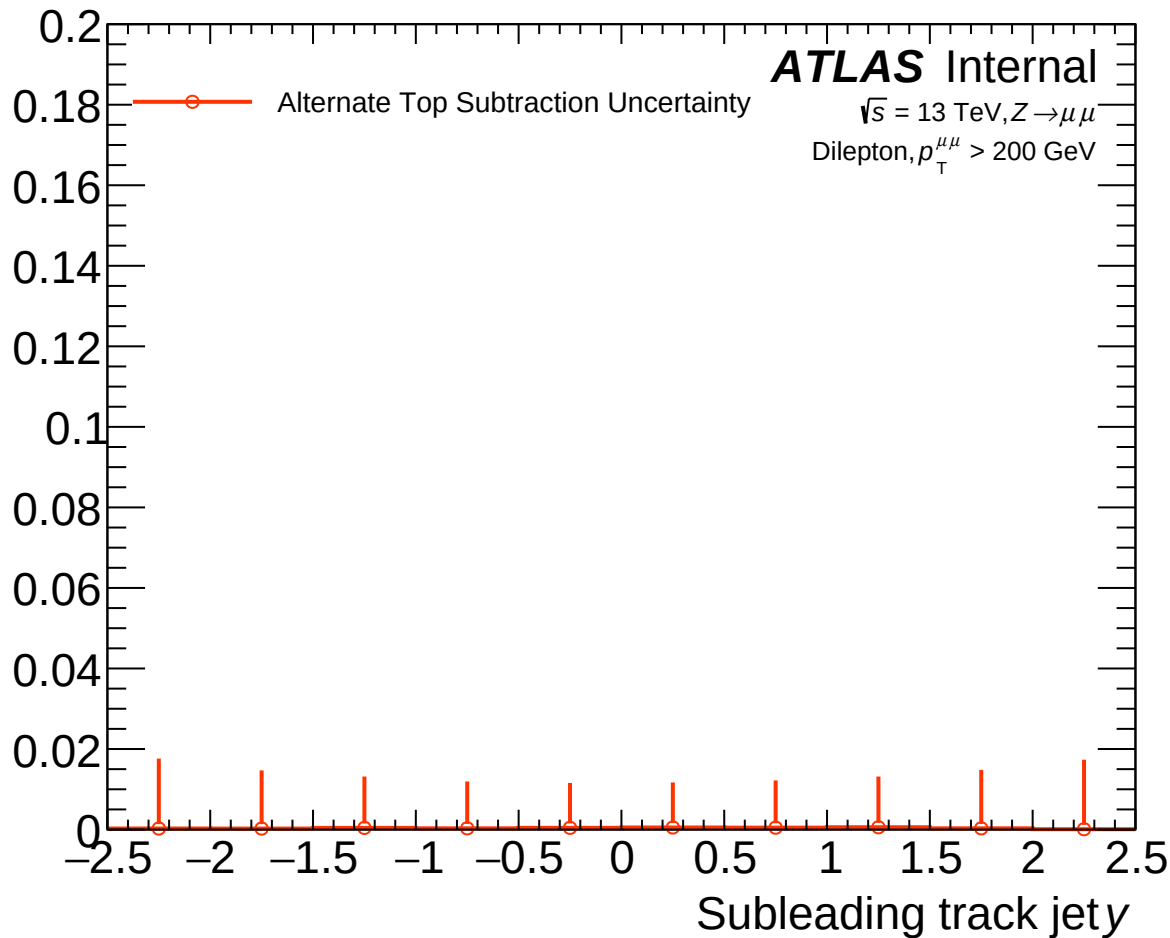
Fractional systematic impact

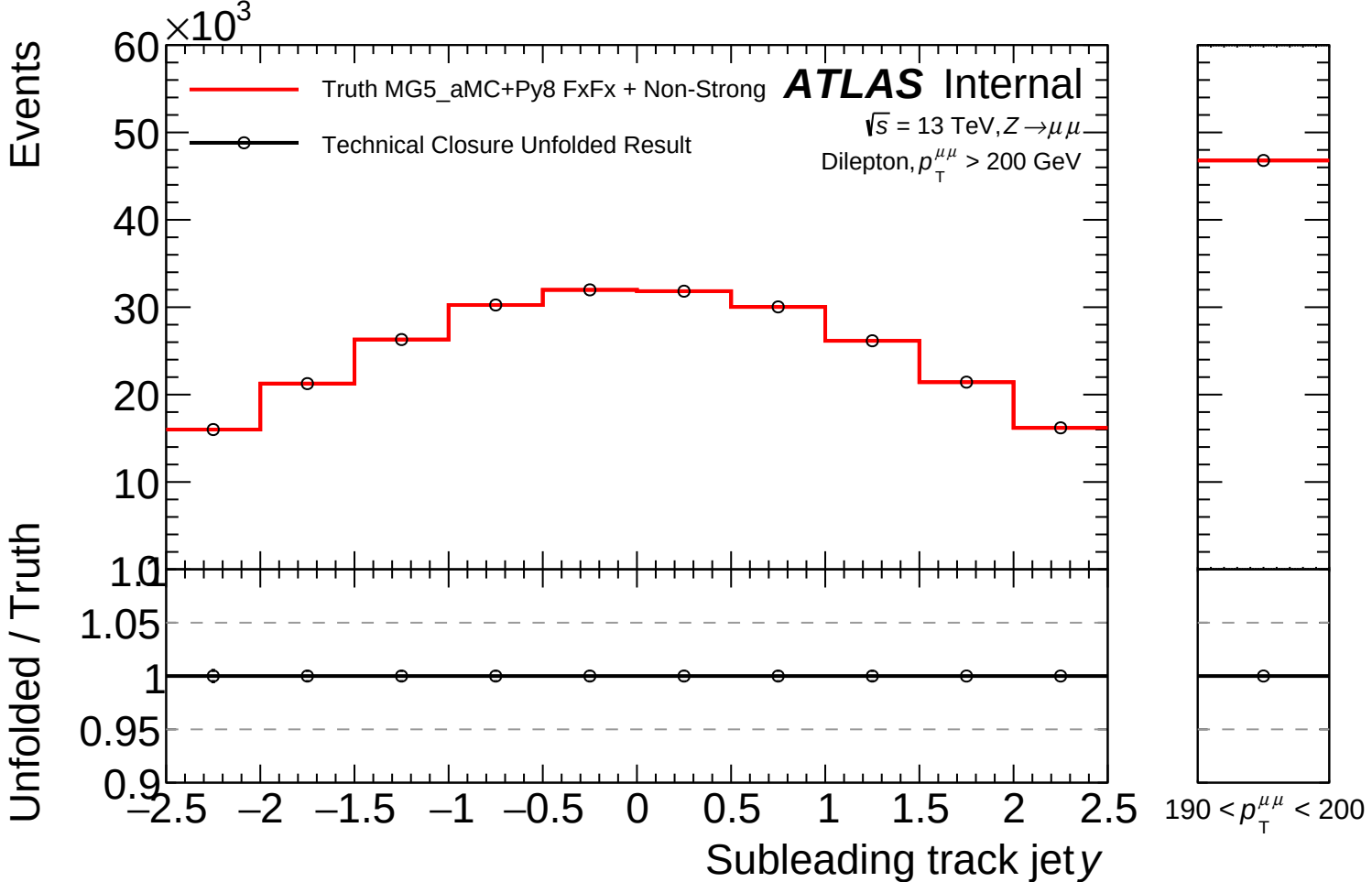
ATLAS Internal

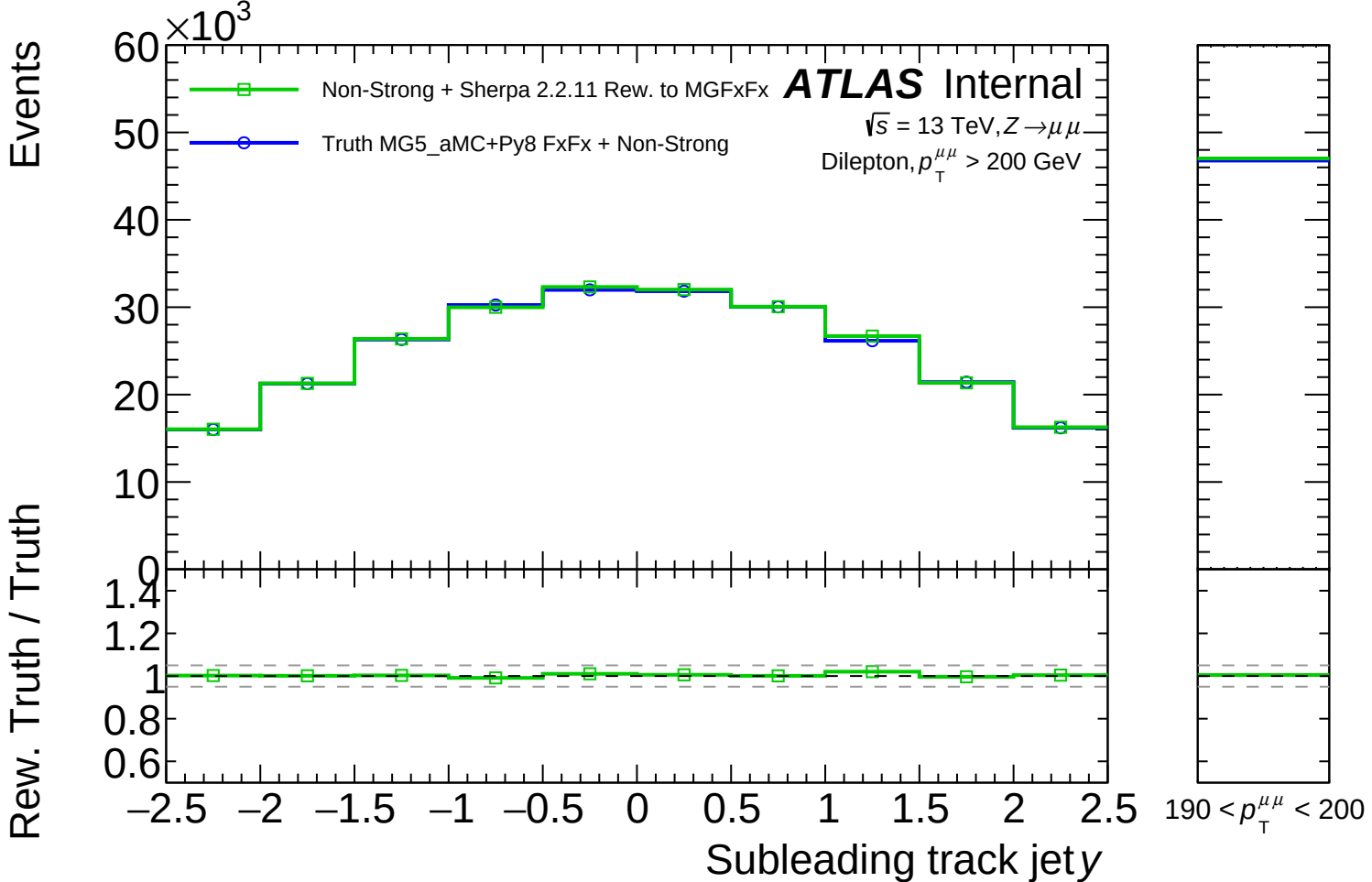
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

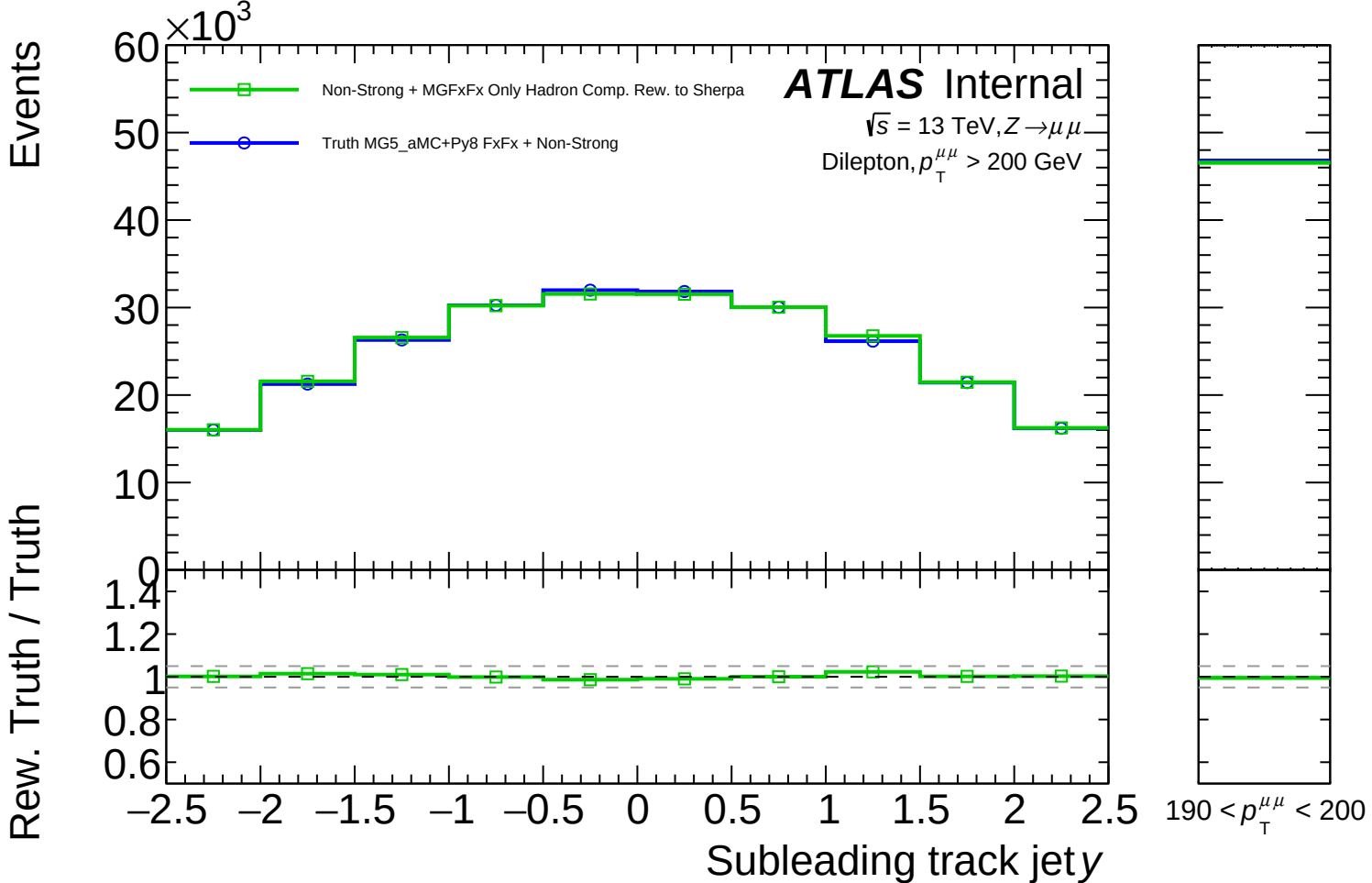
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

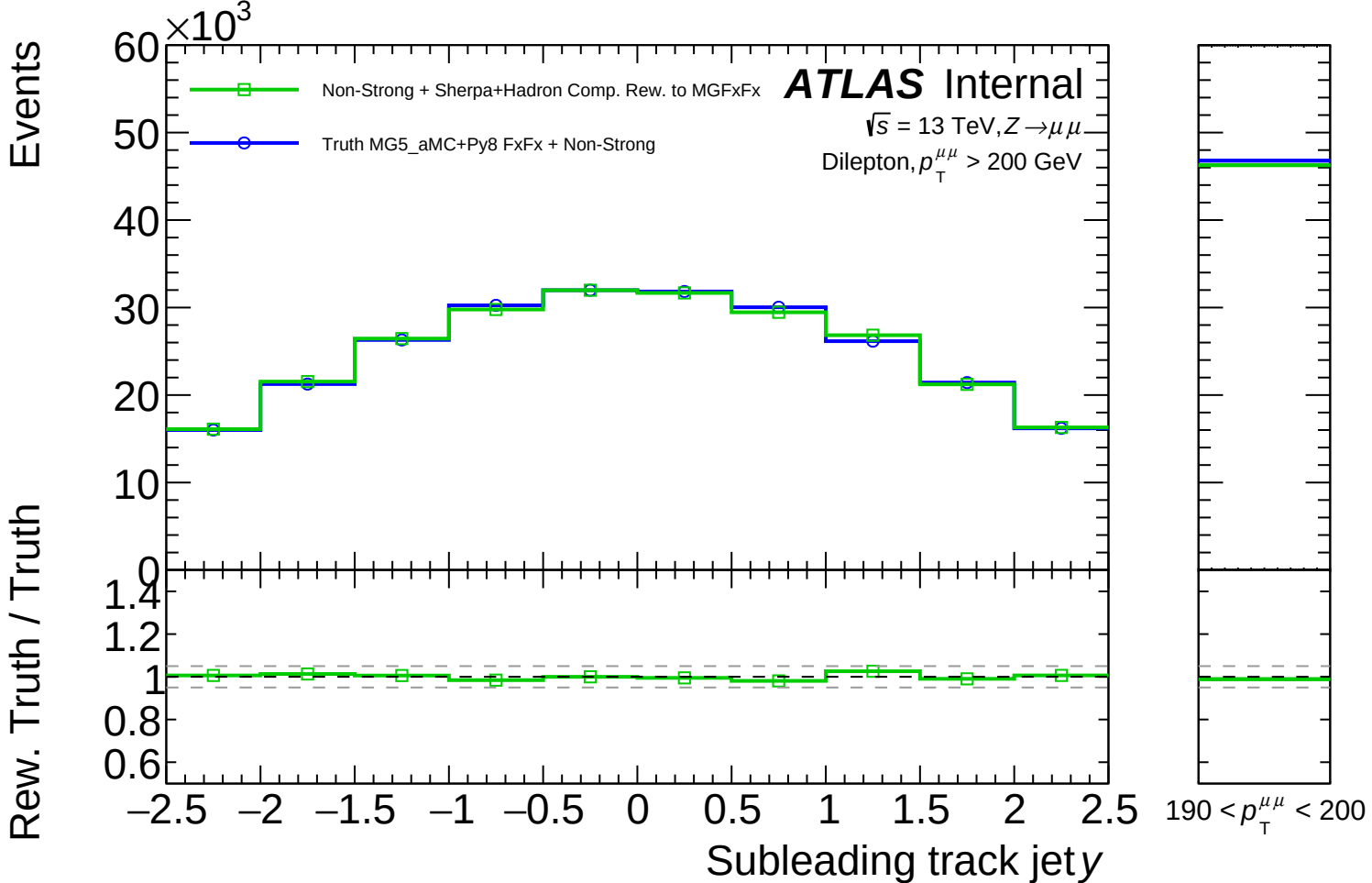
Alternate Top Subtraction Uncertainty

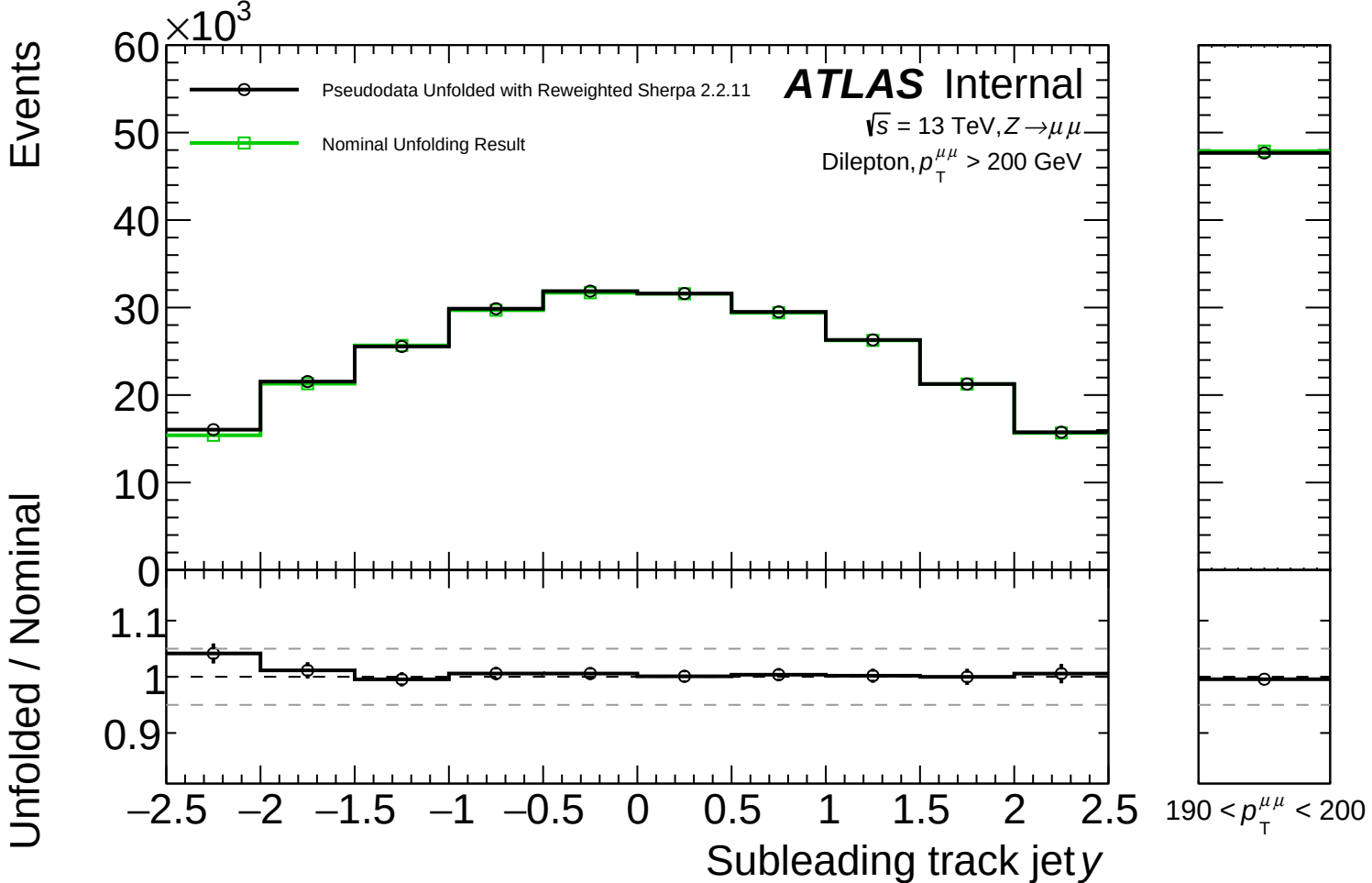


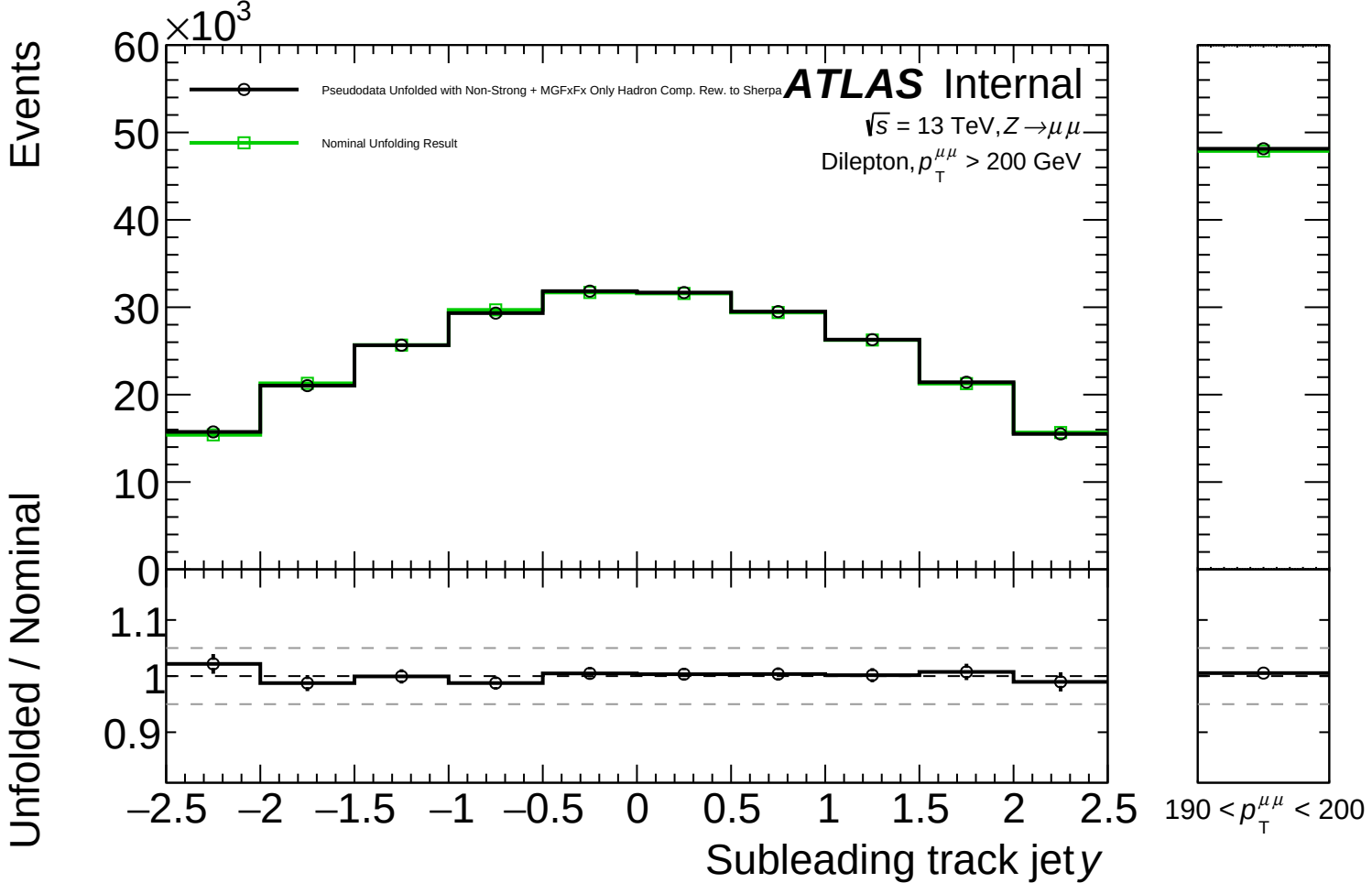


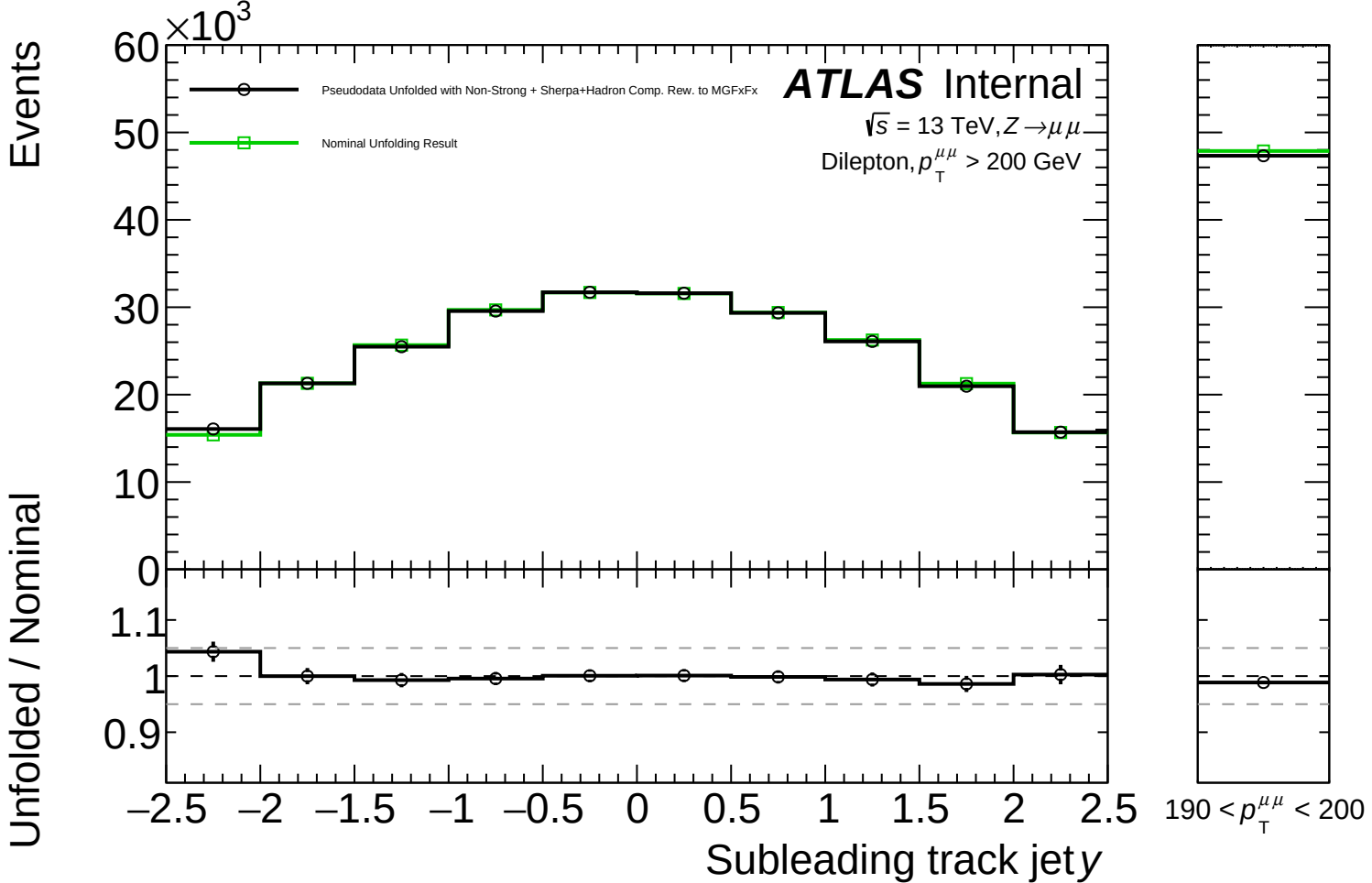


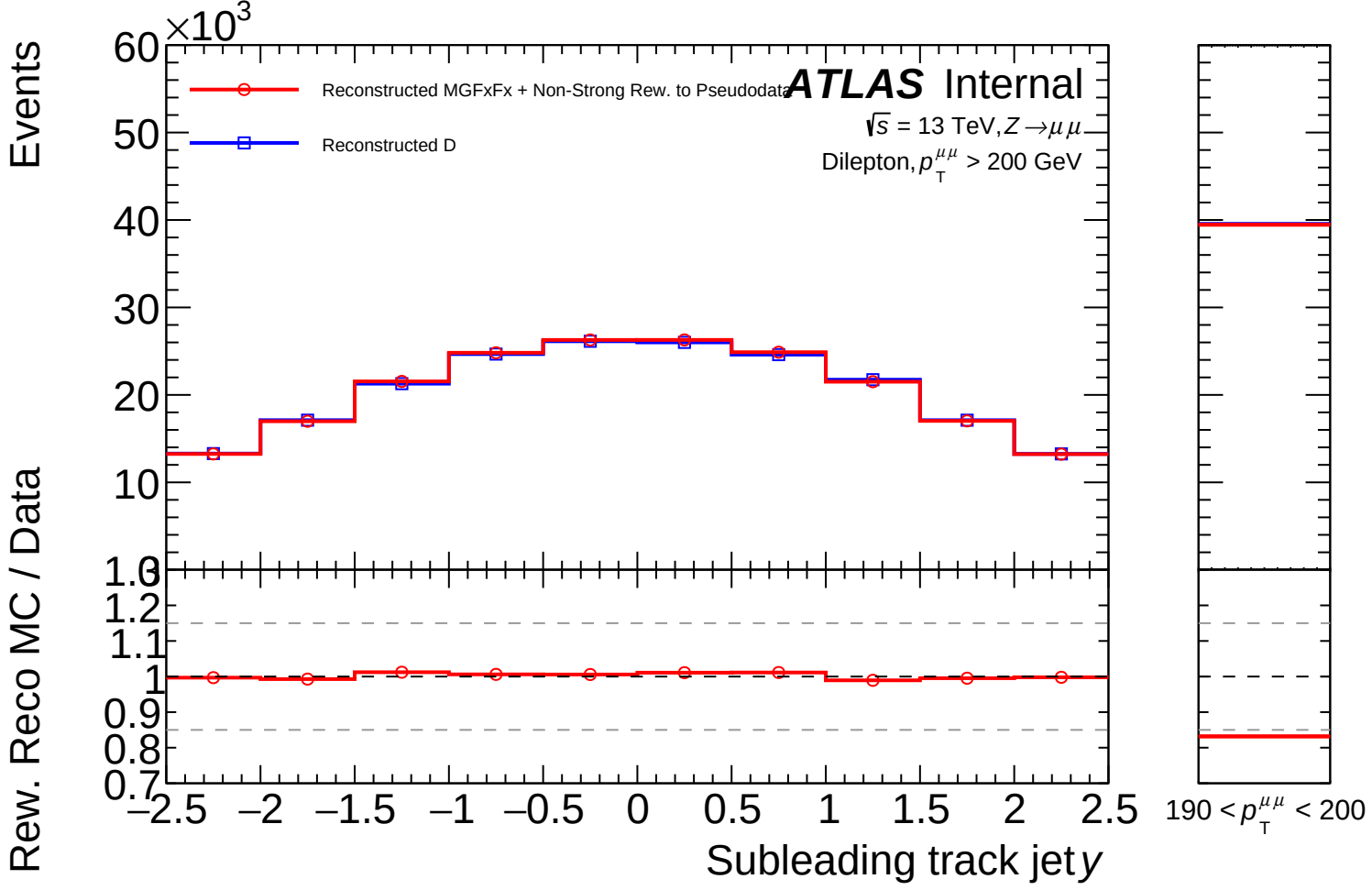


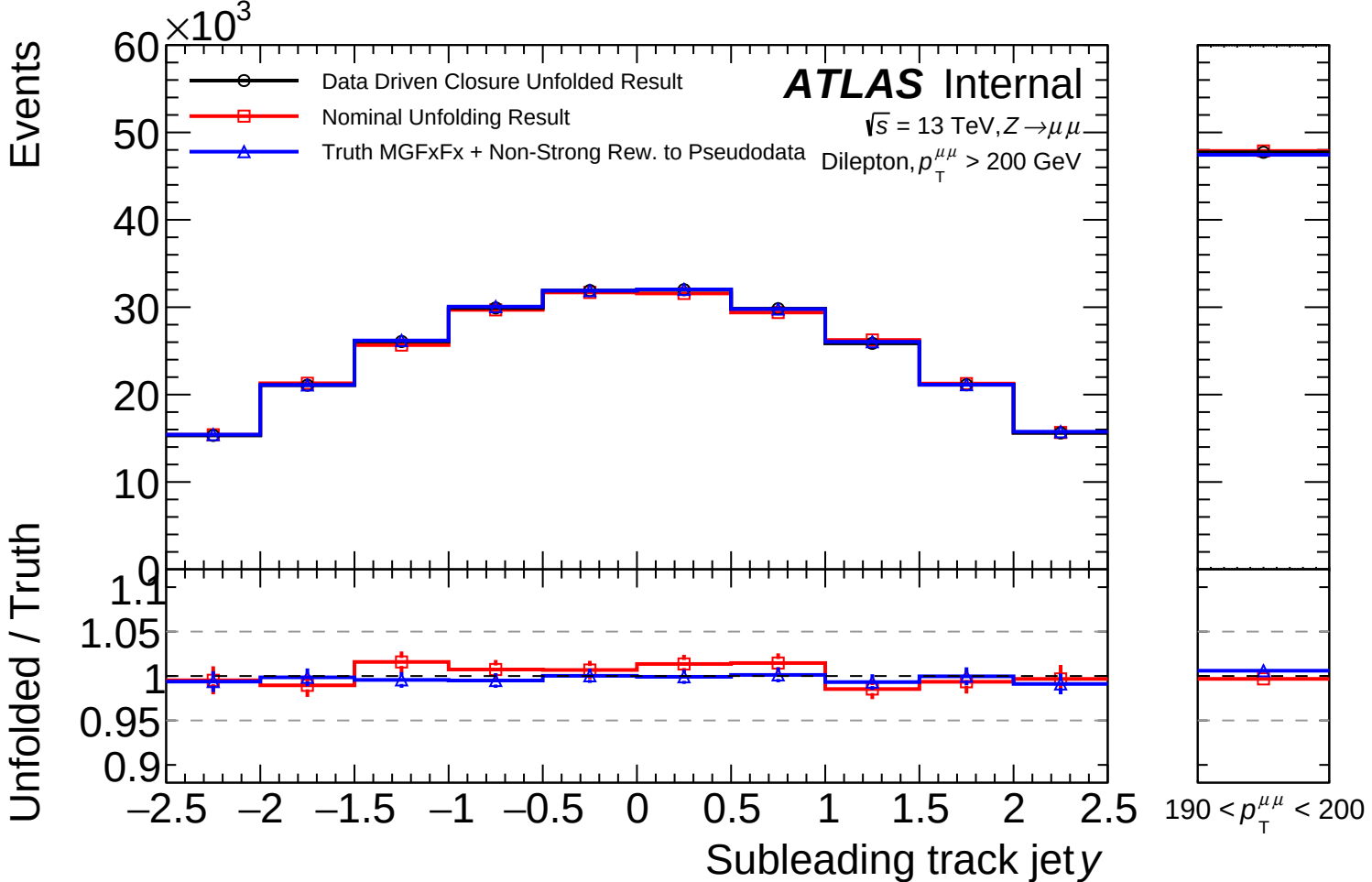




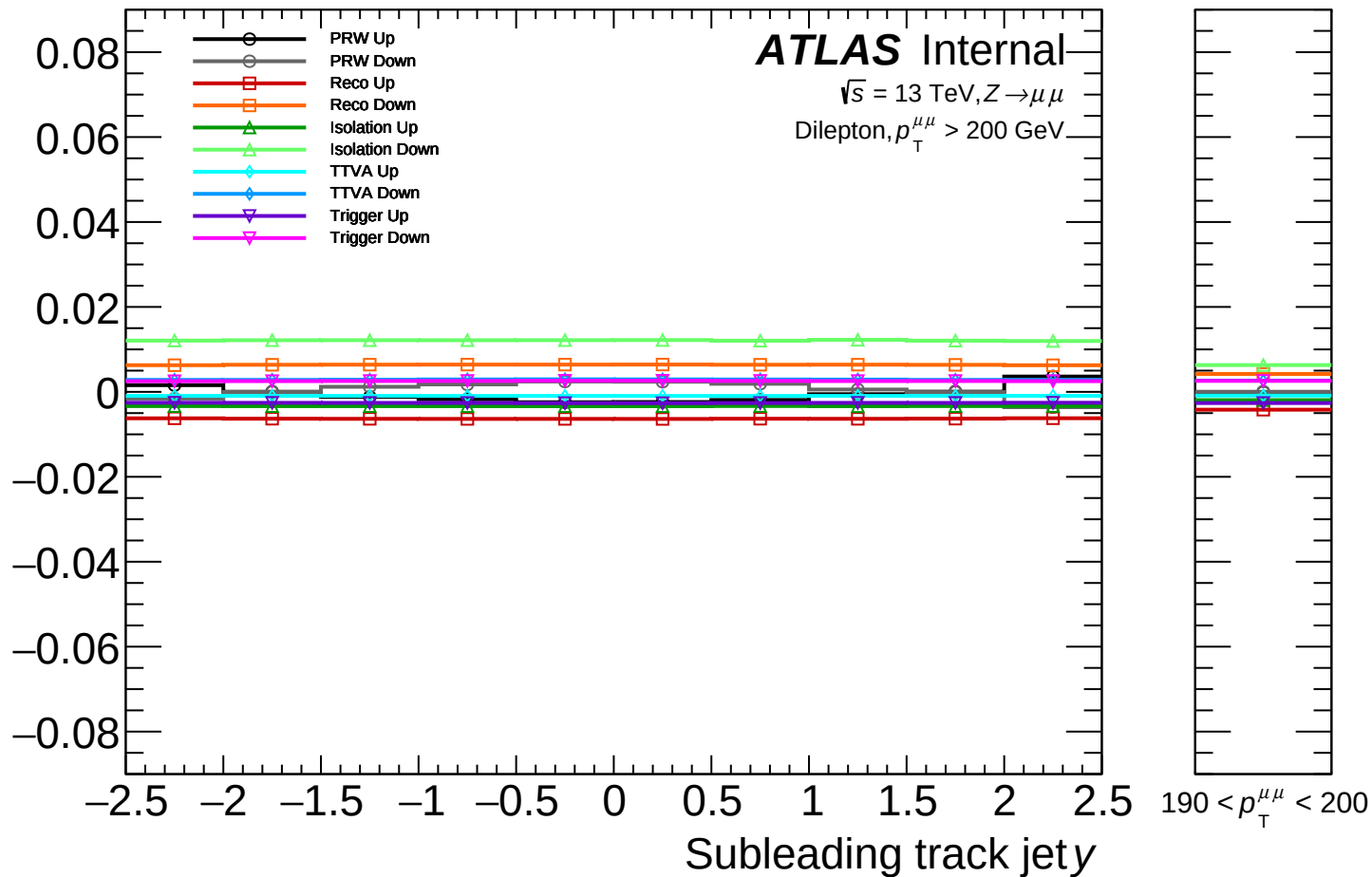




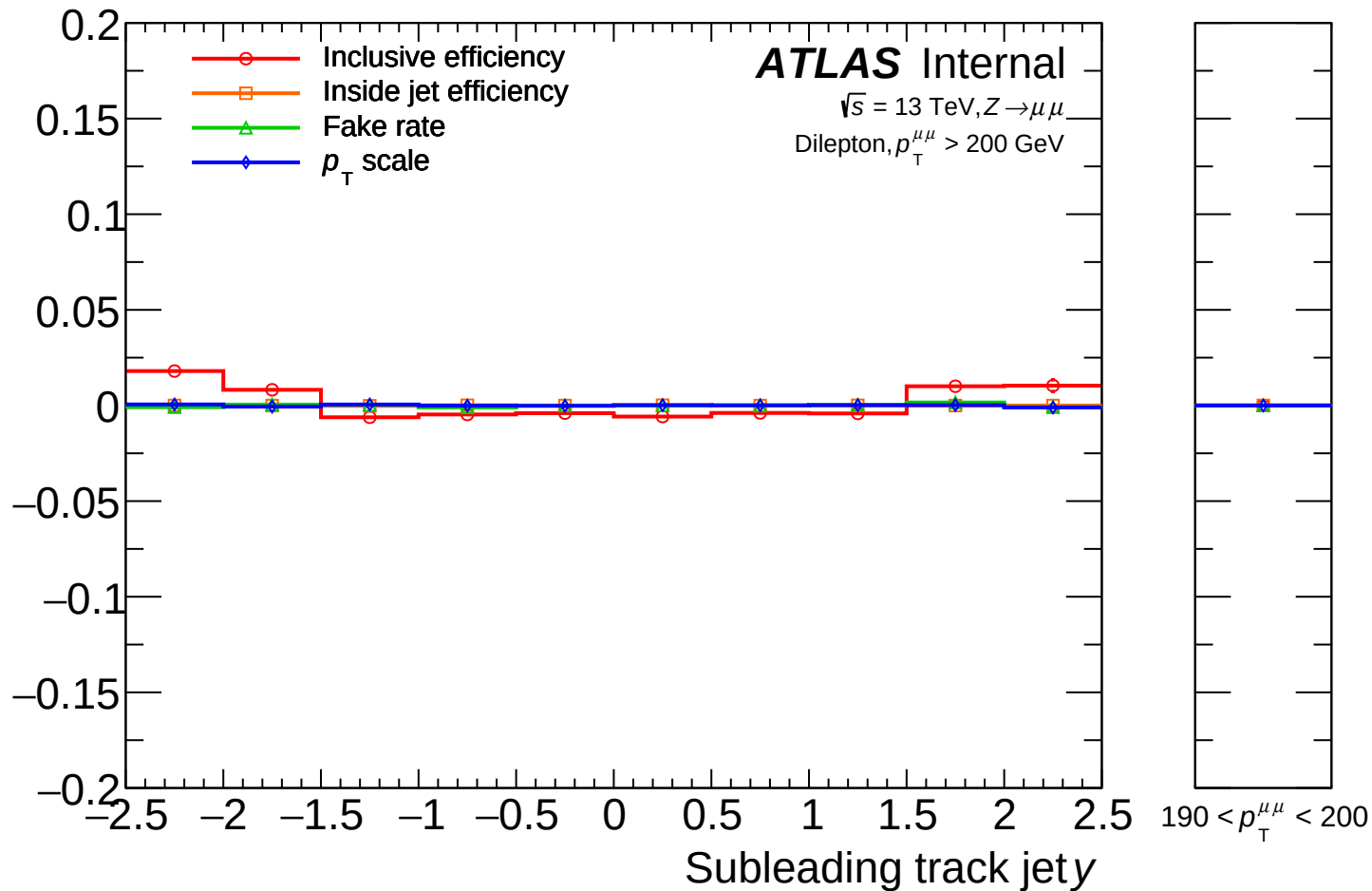




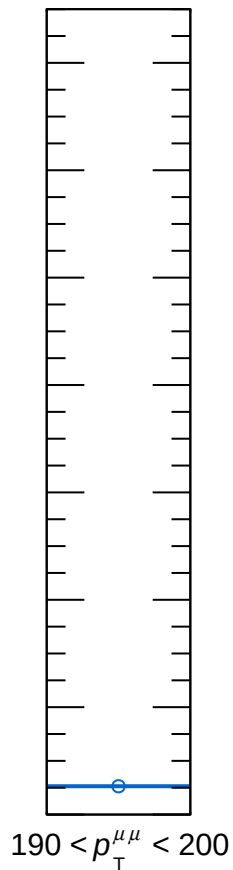
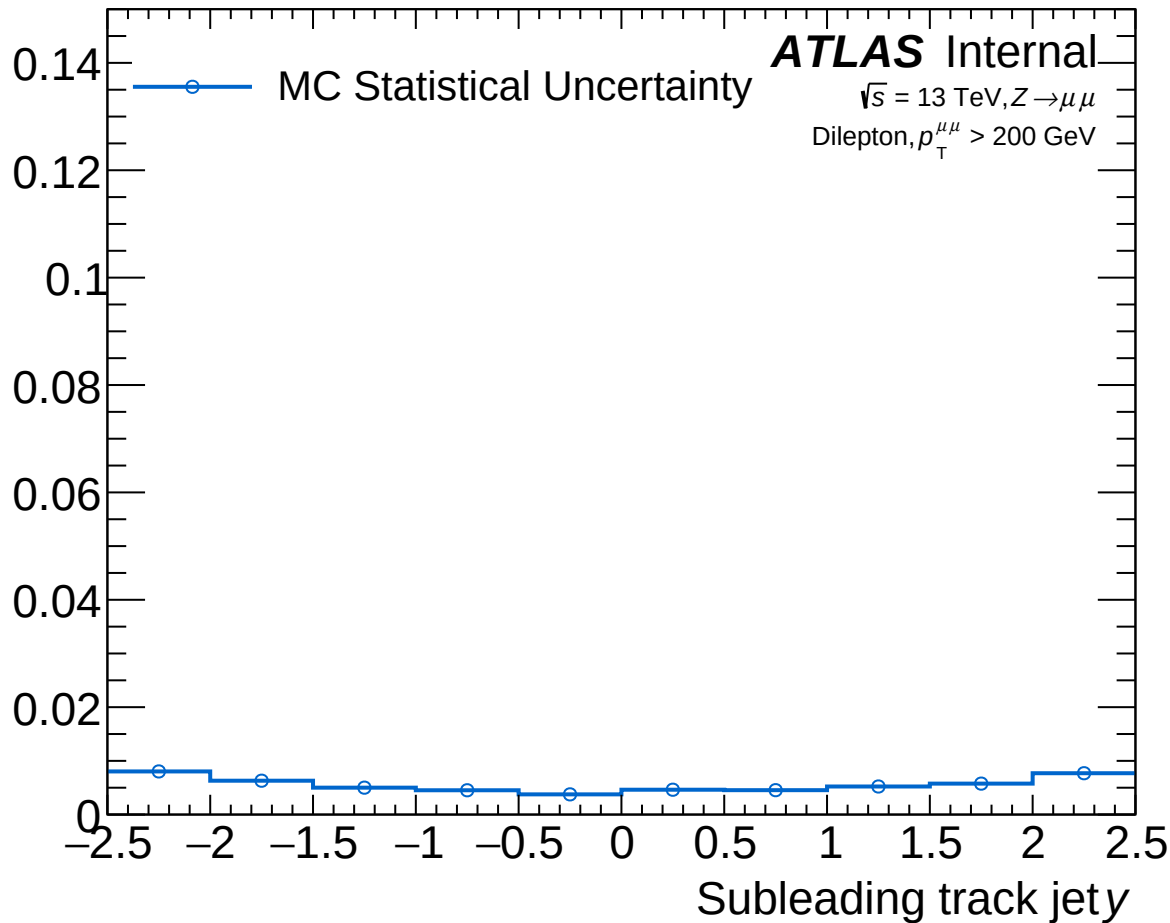
Fractional systematic impact



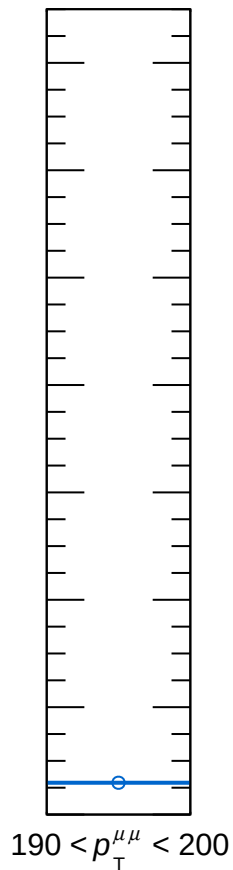
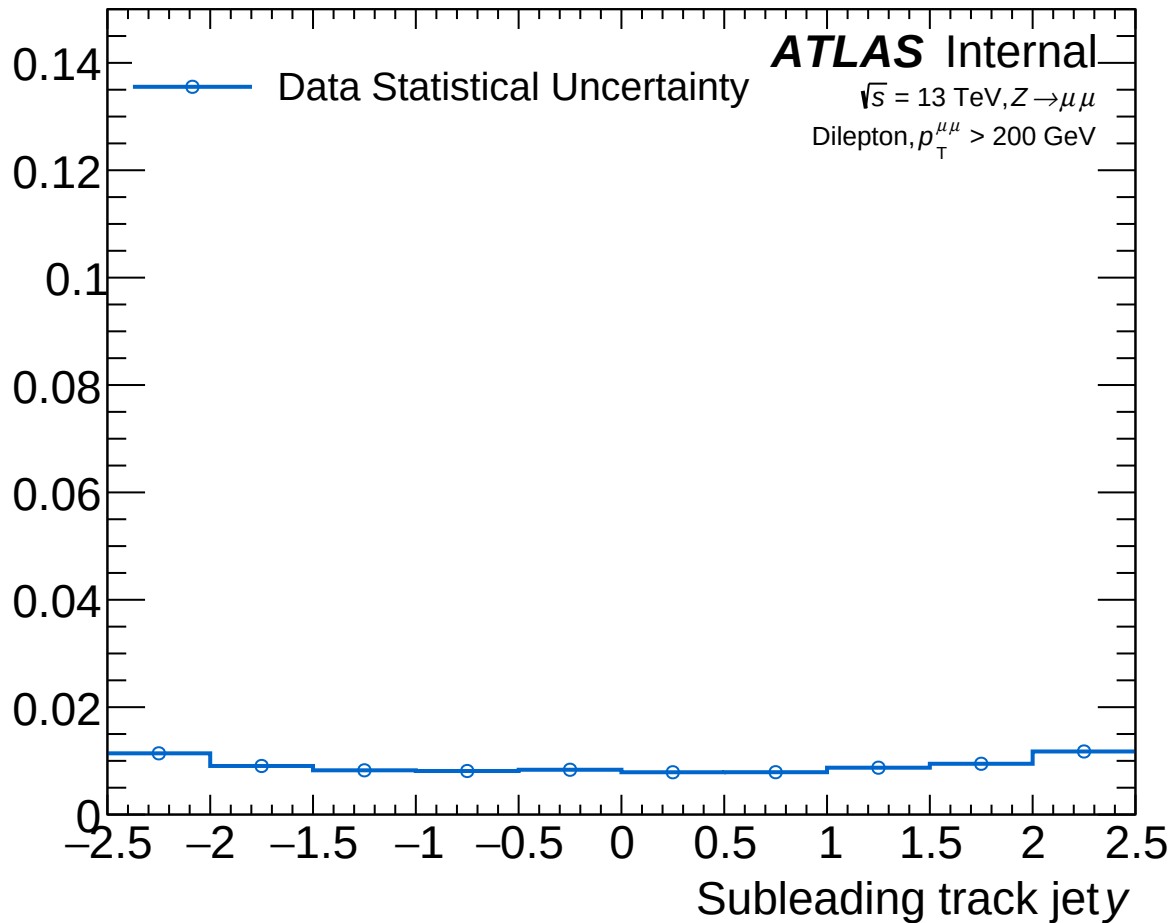
Fractional systematic impact



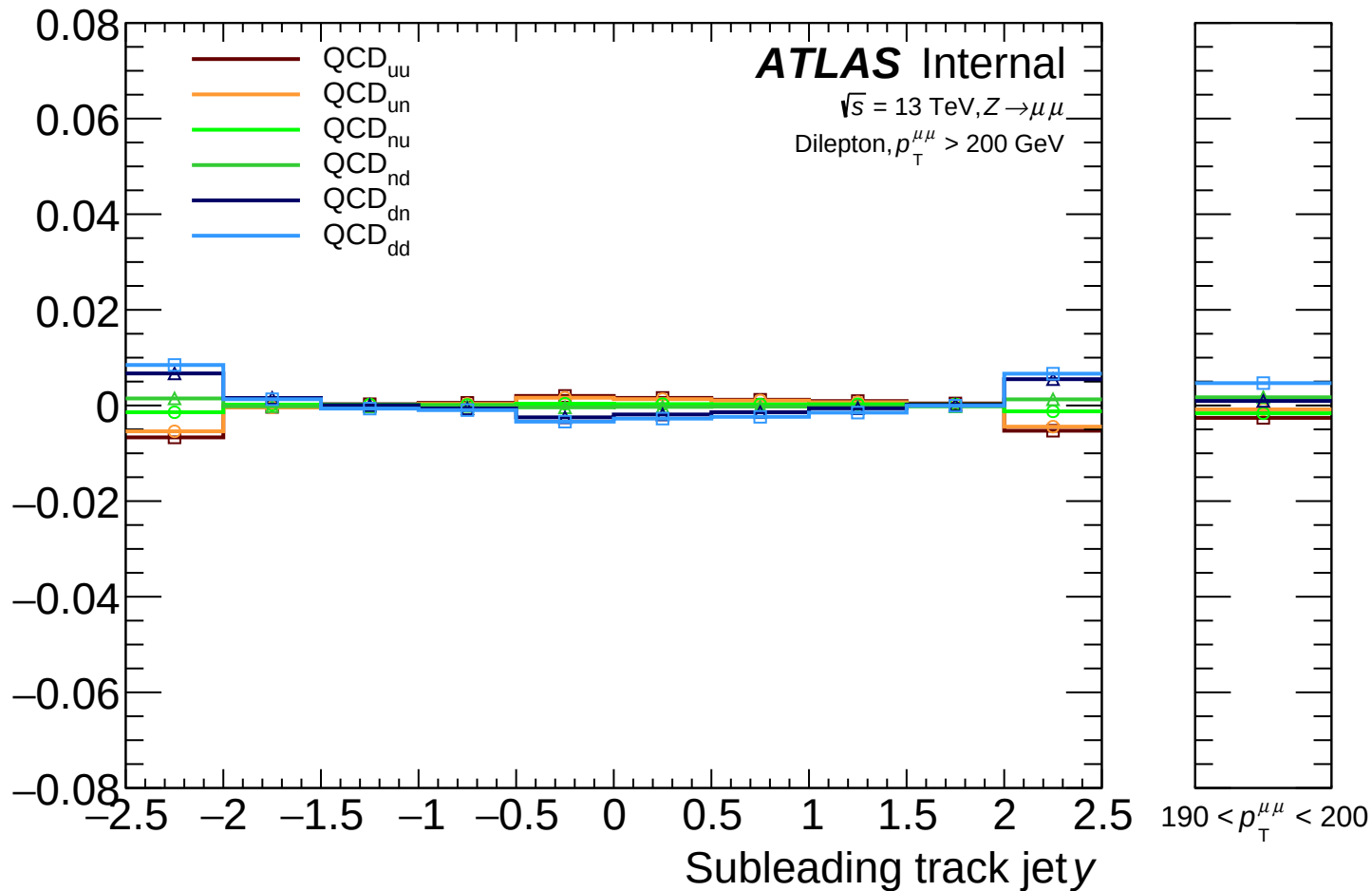
Fractional systematic impact



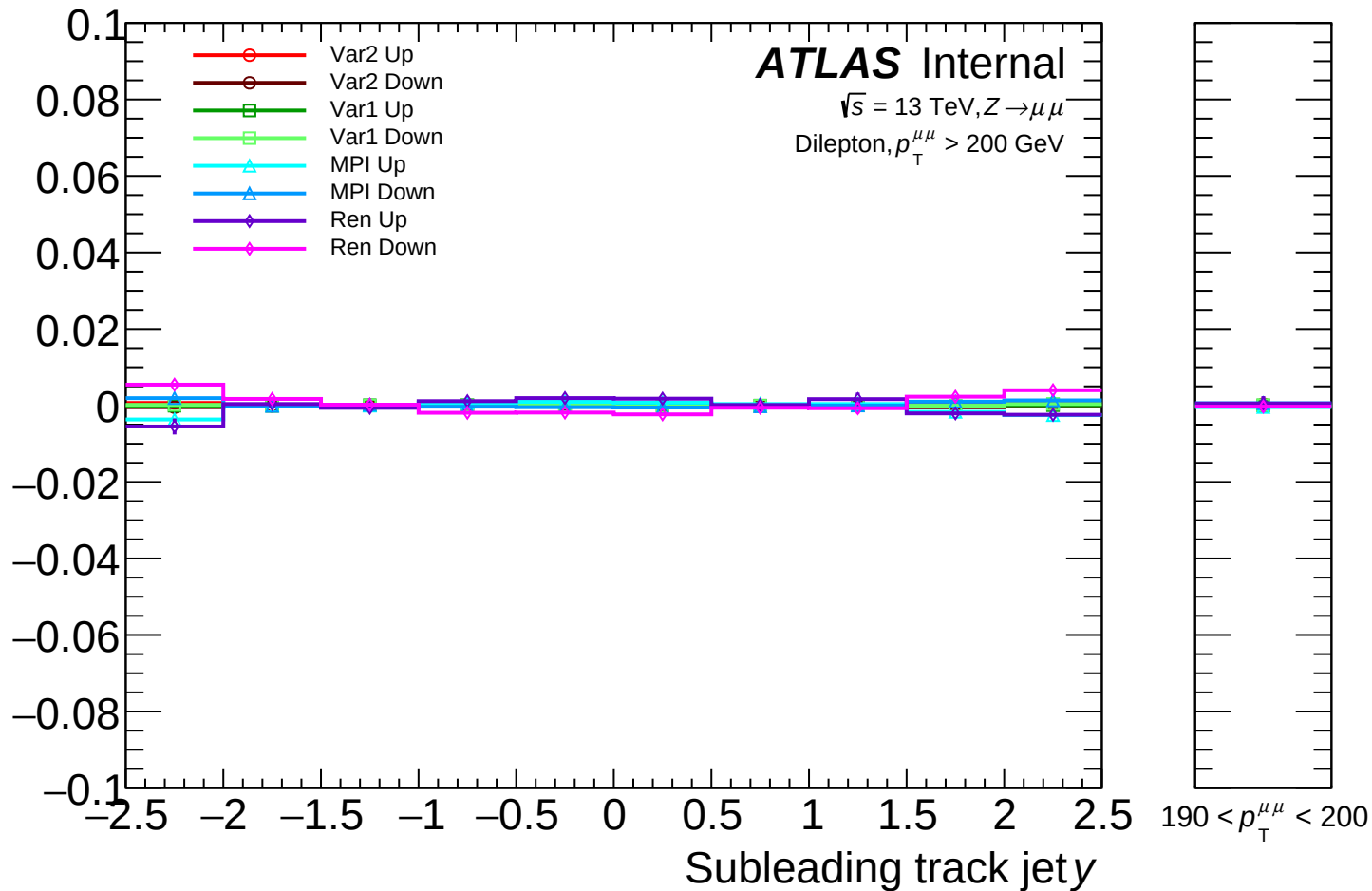
Fractional systematic impact

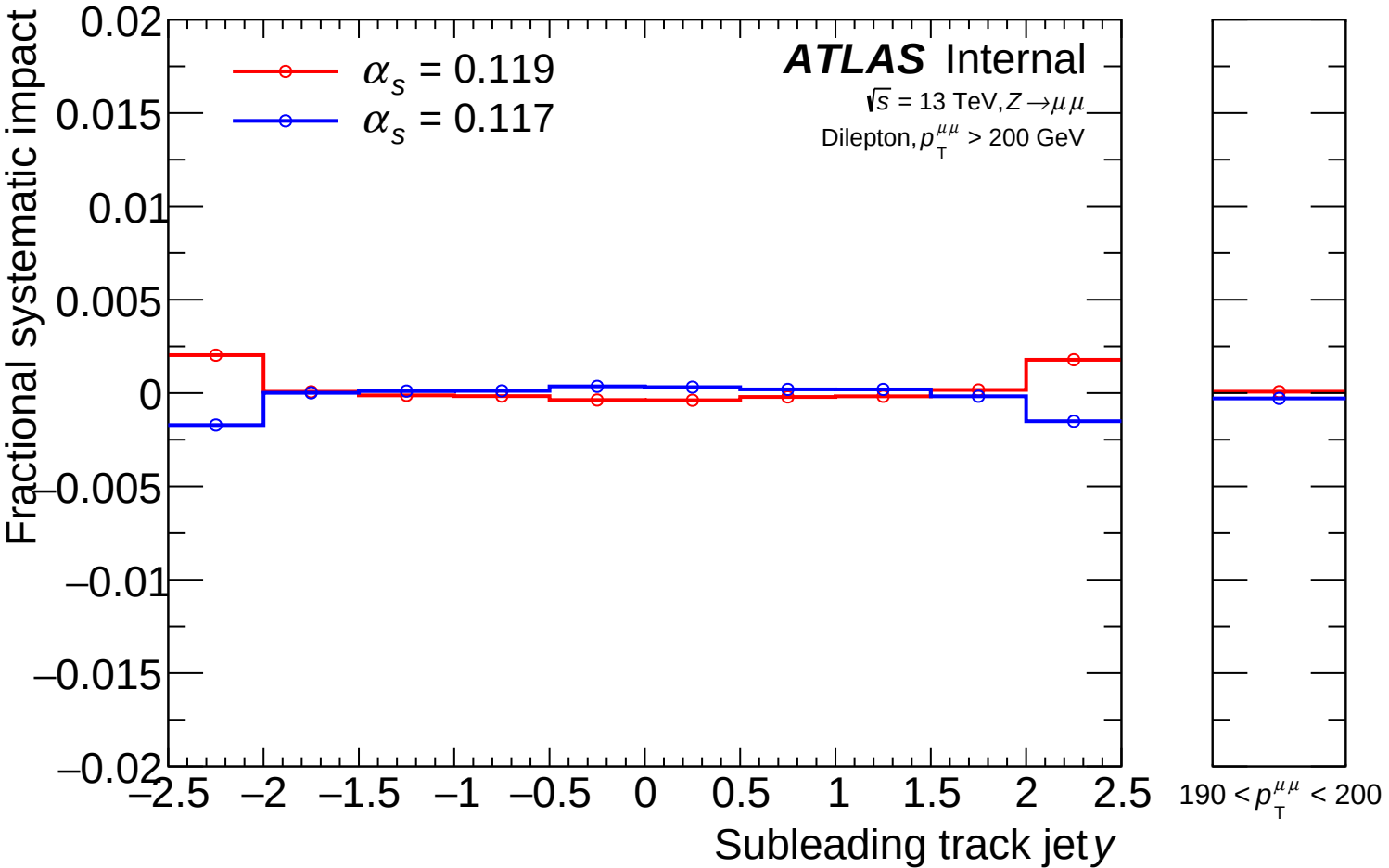


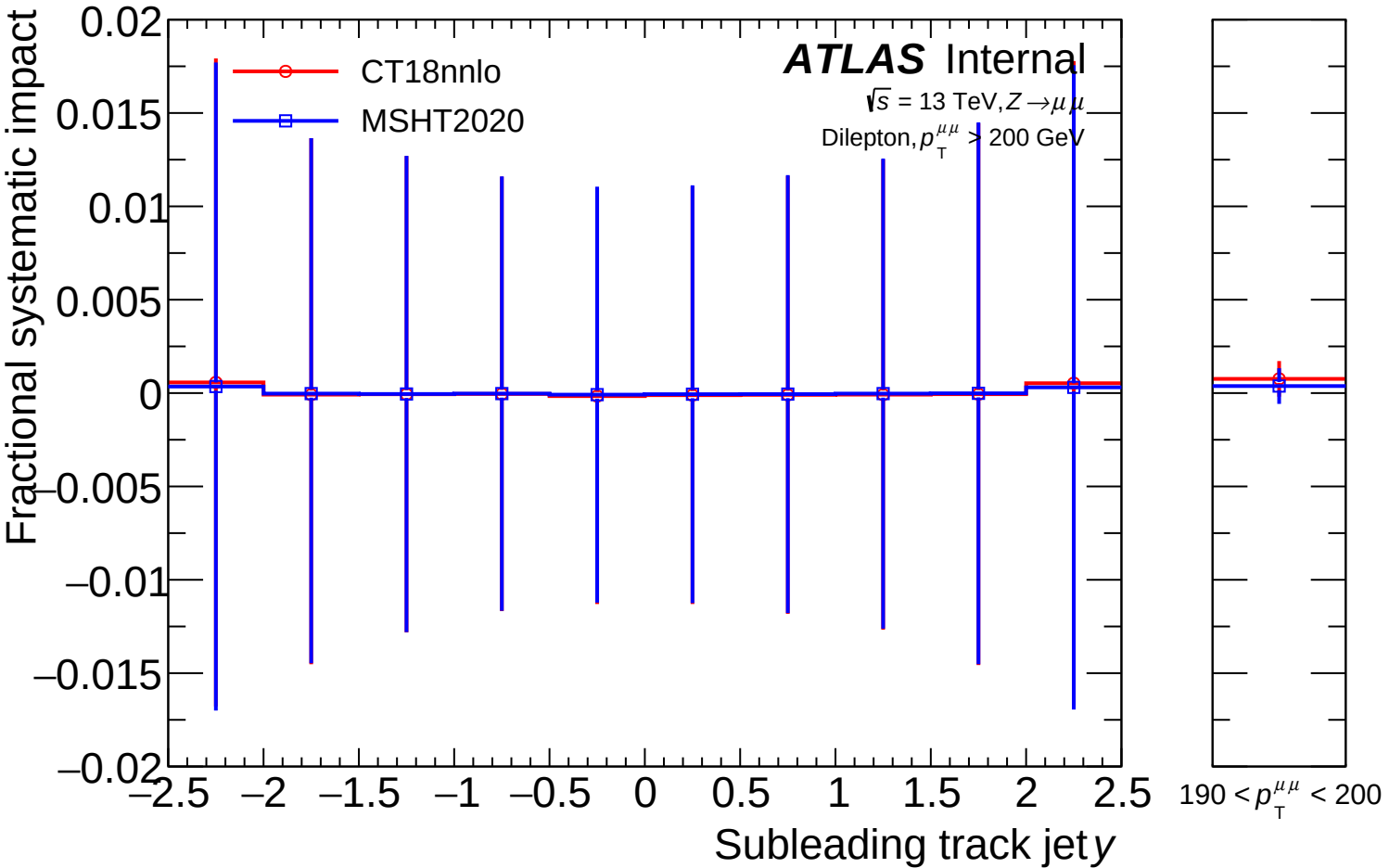
Fractional systematic impact

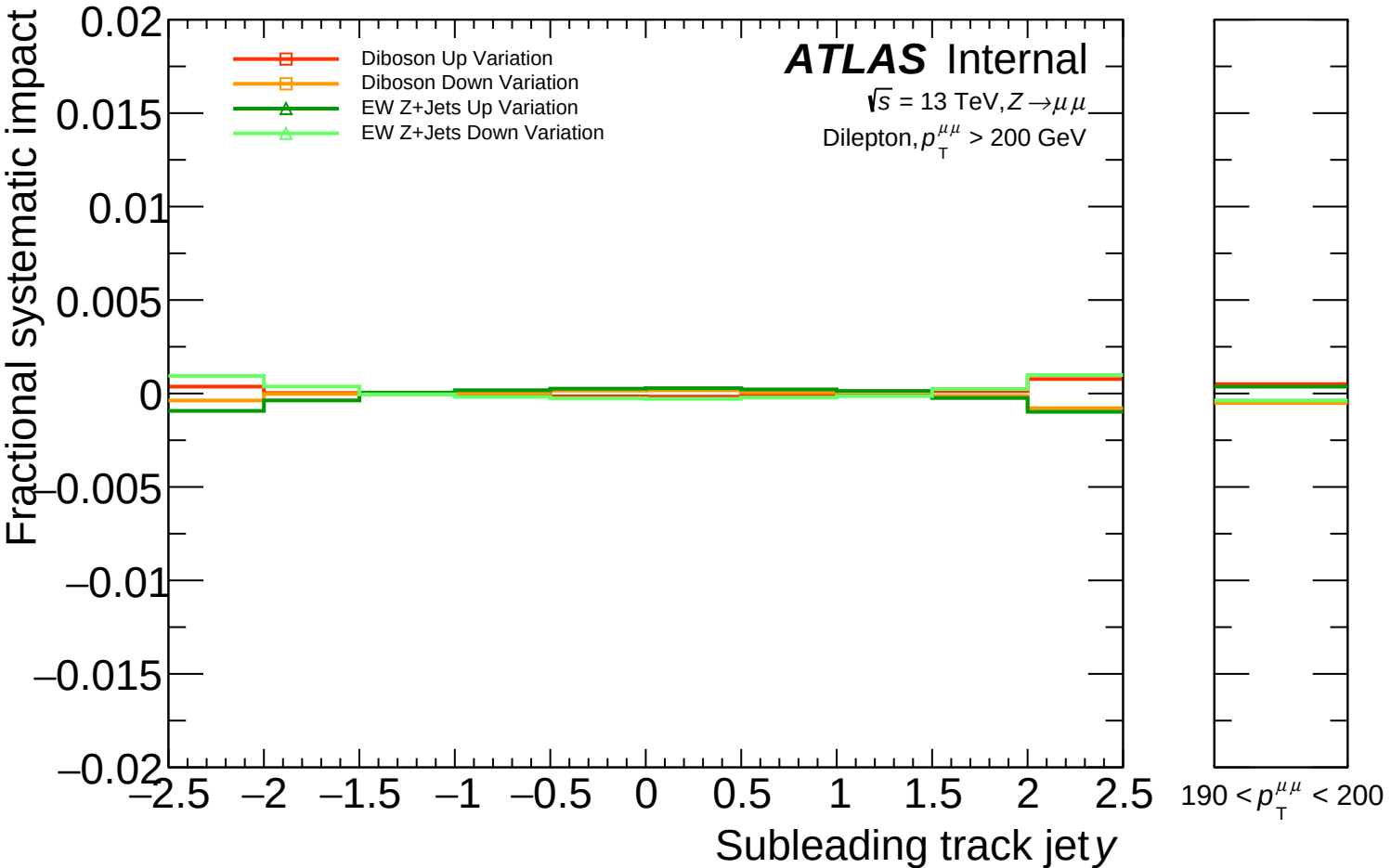


Fractional systematic impact





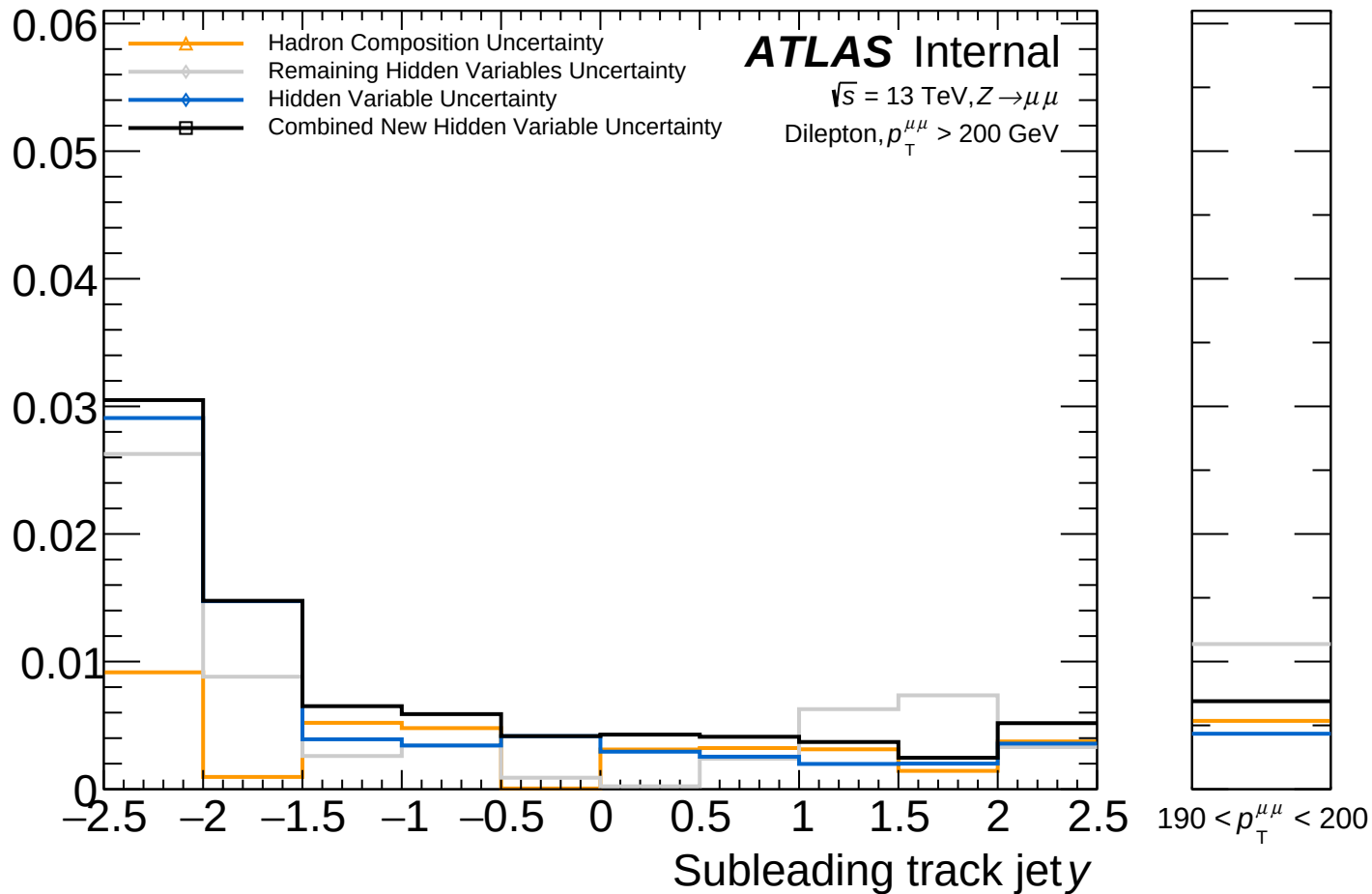


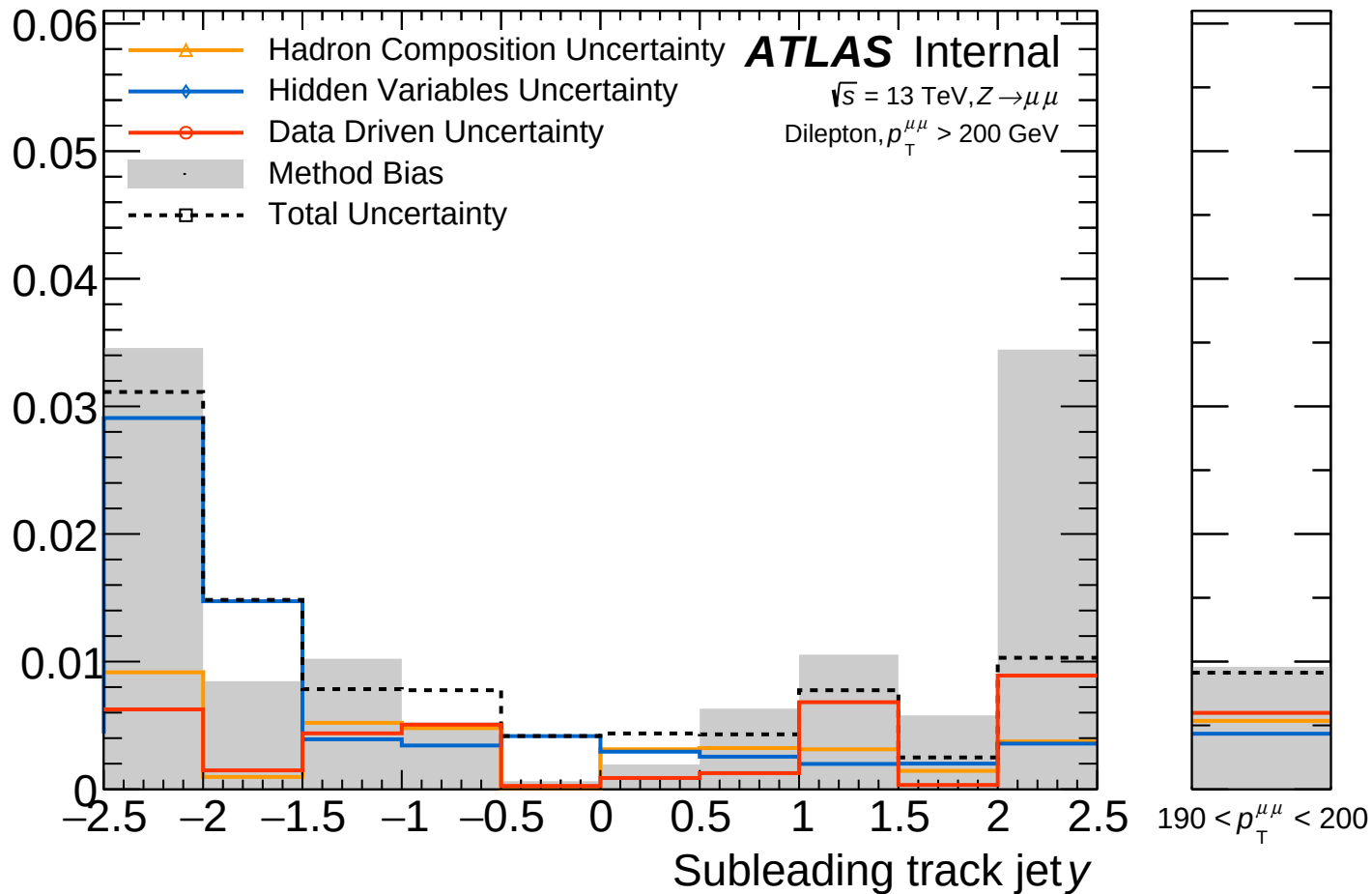


ATLAS Internal

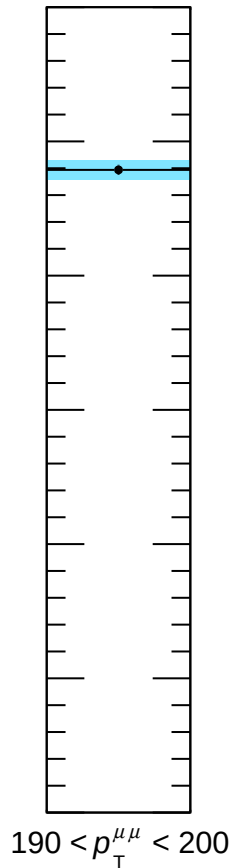
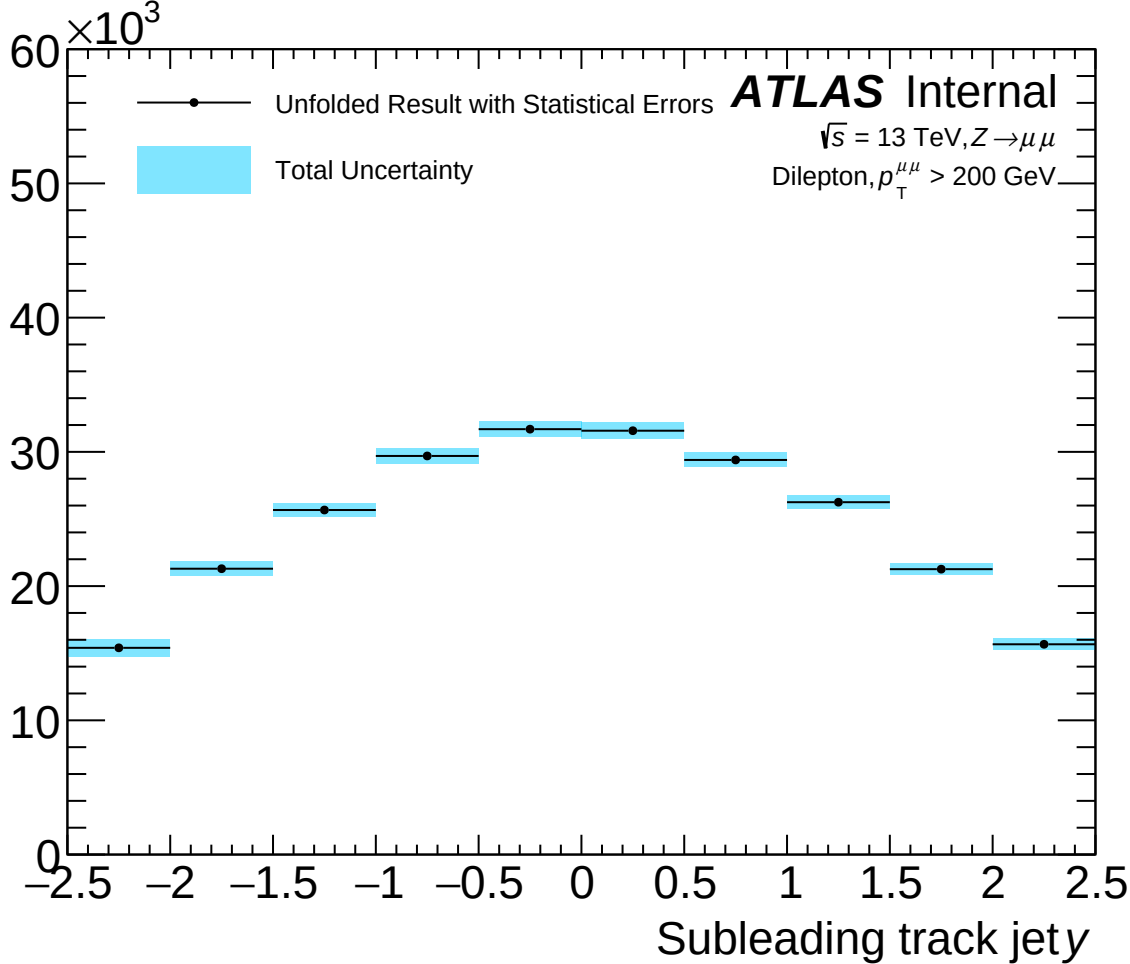
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

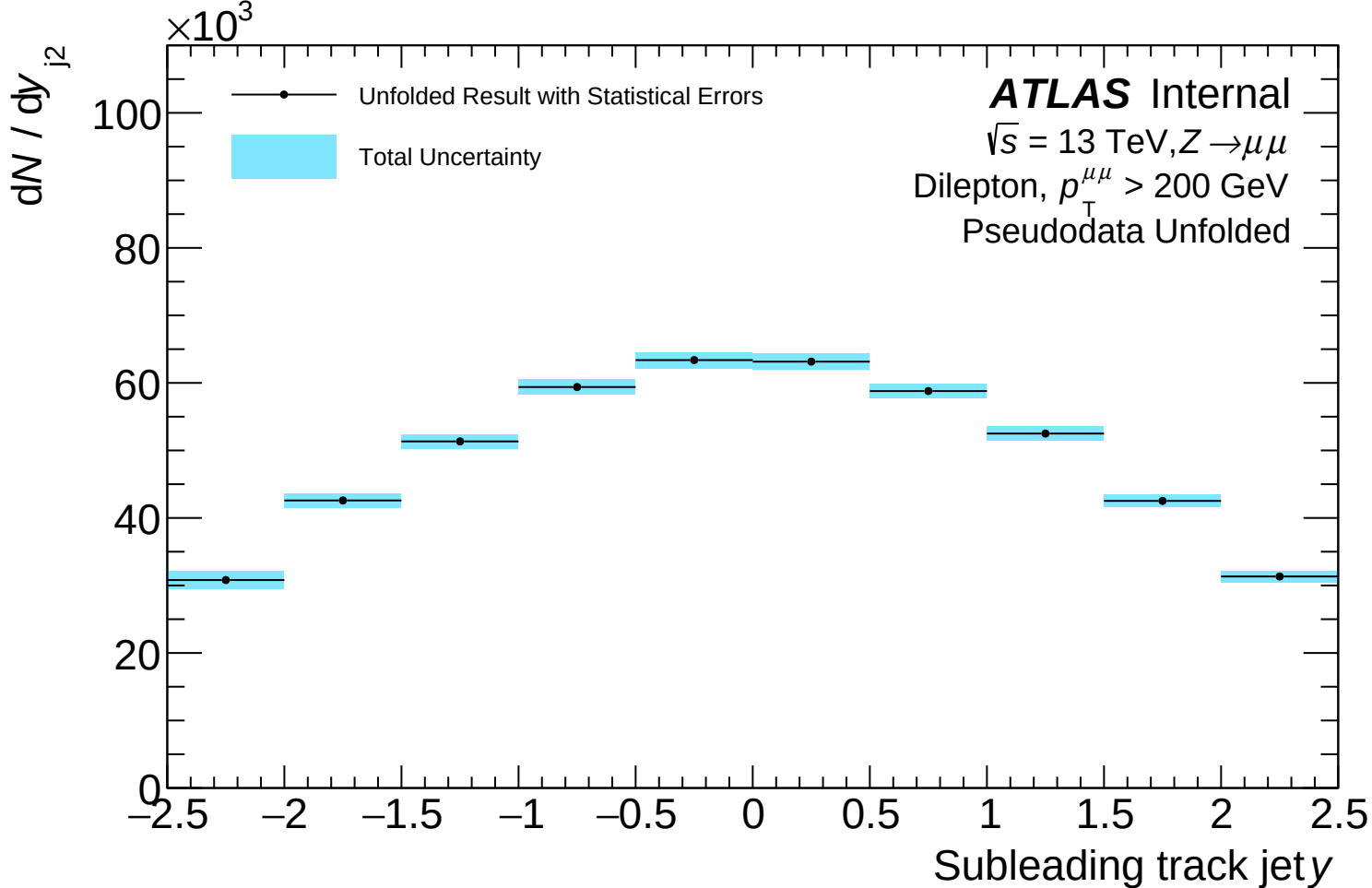
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

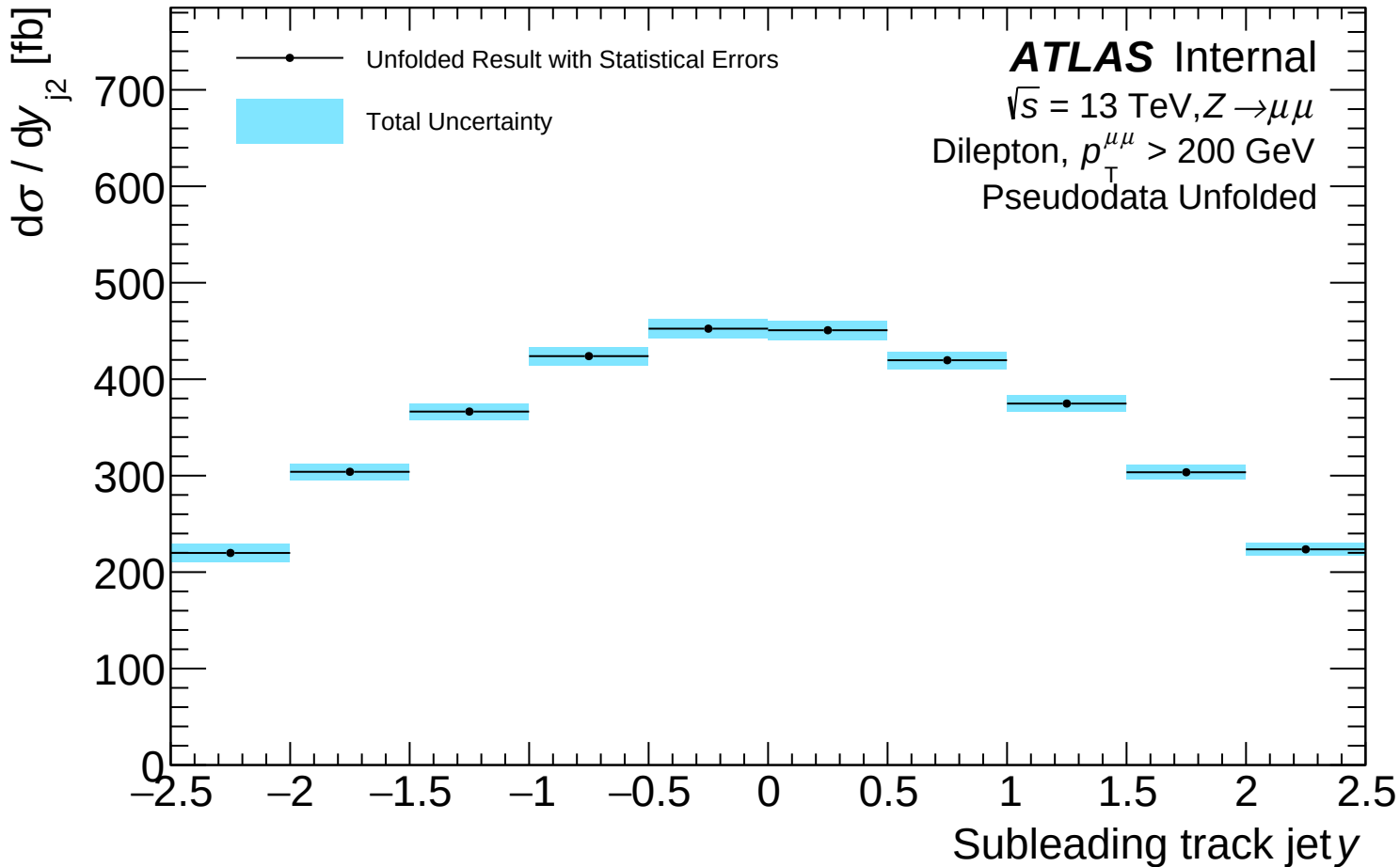




Events







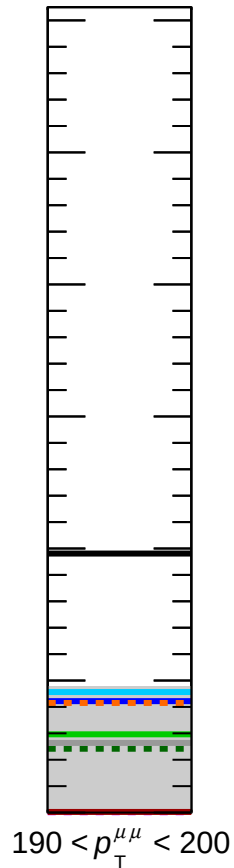
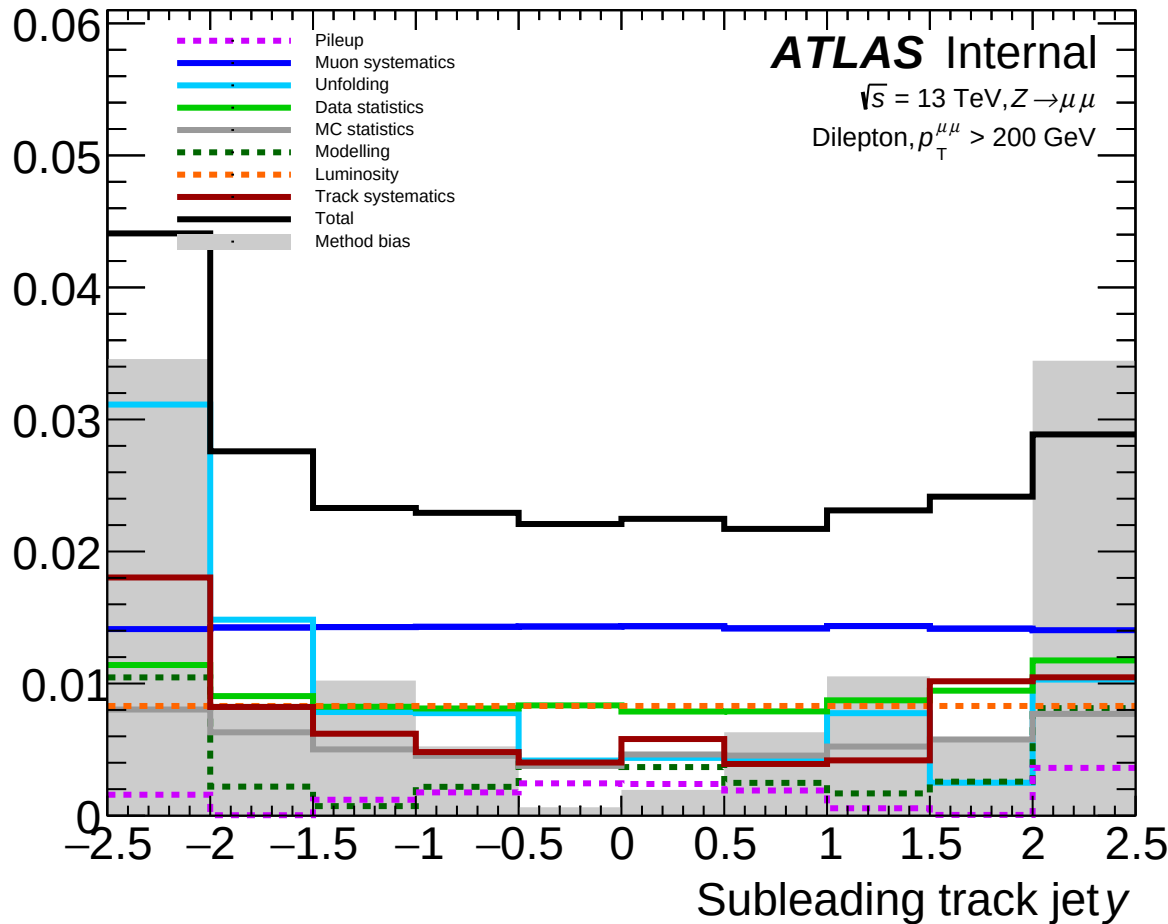
Fractional uncertainty

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

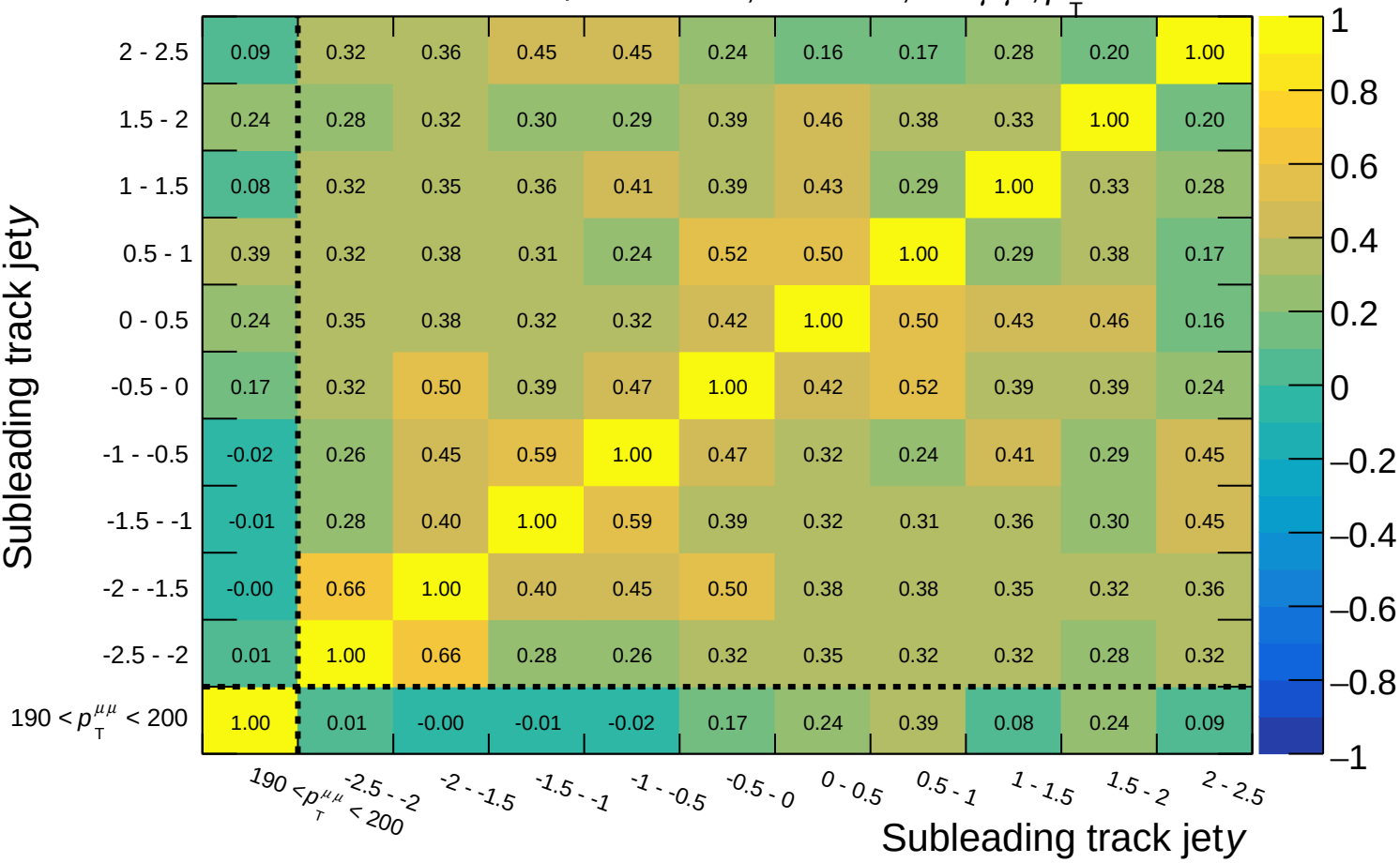
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

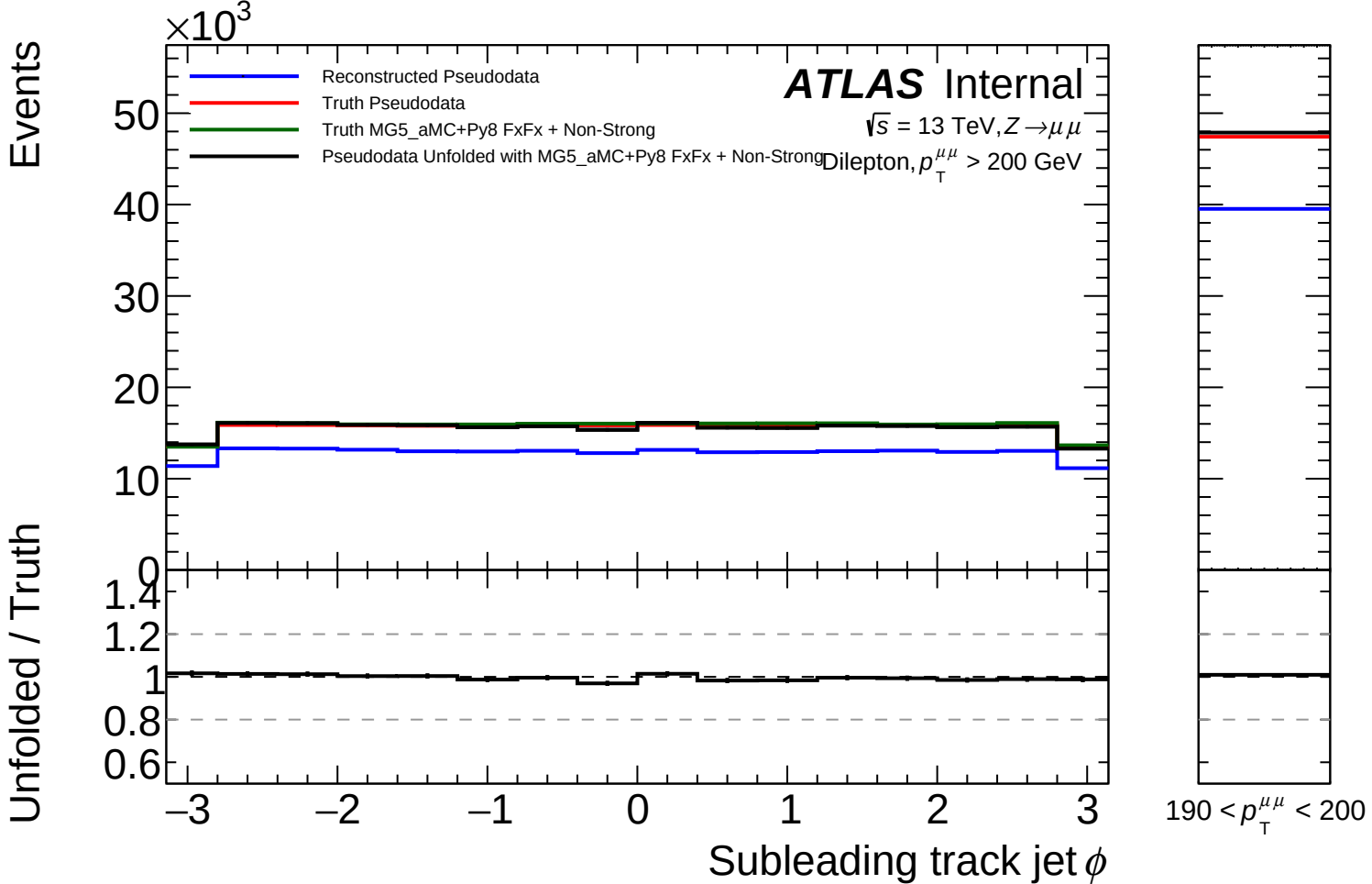
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

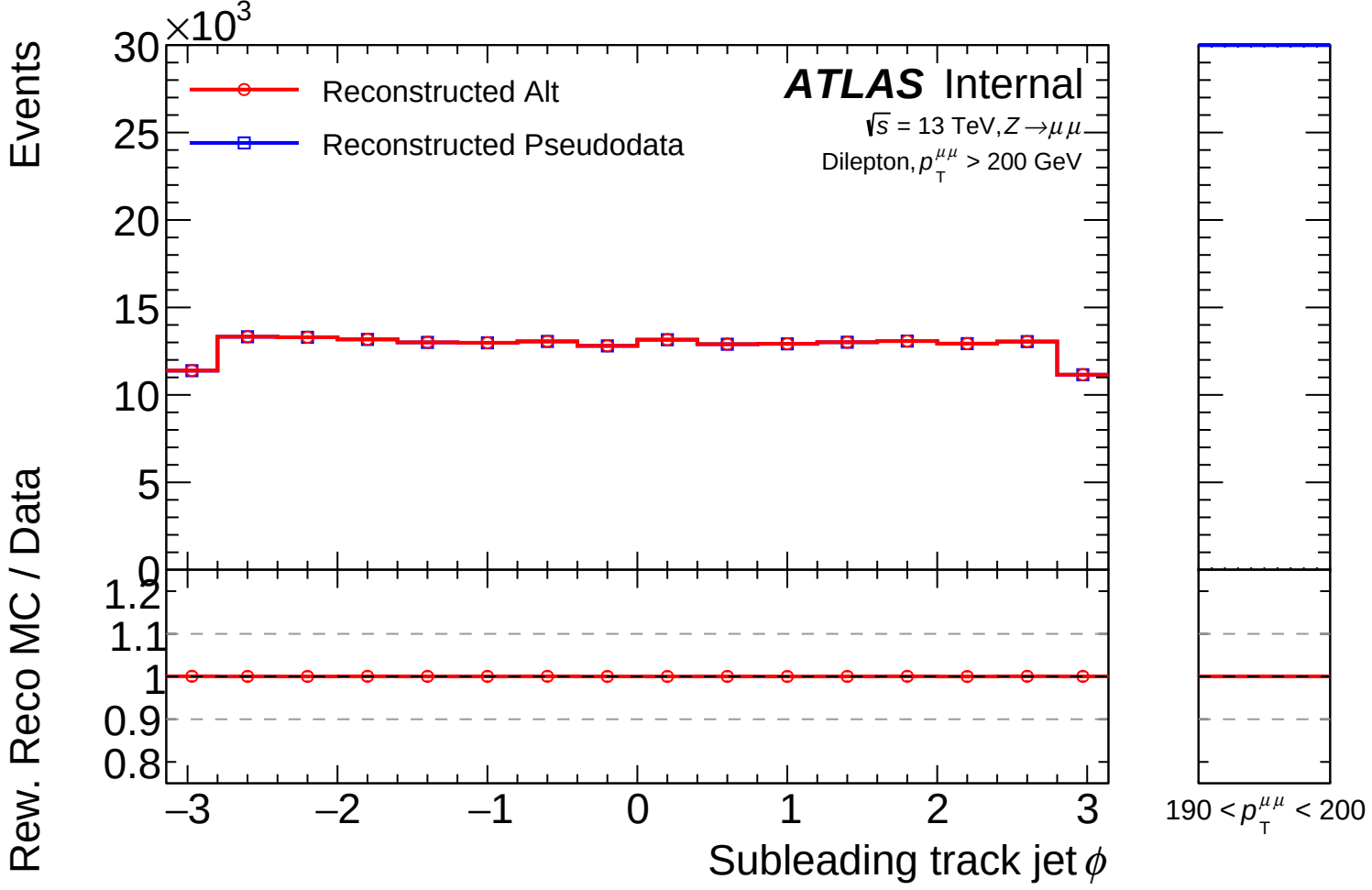


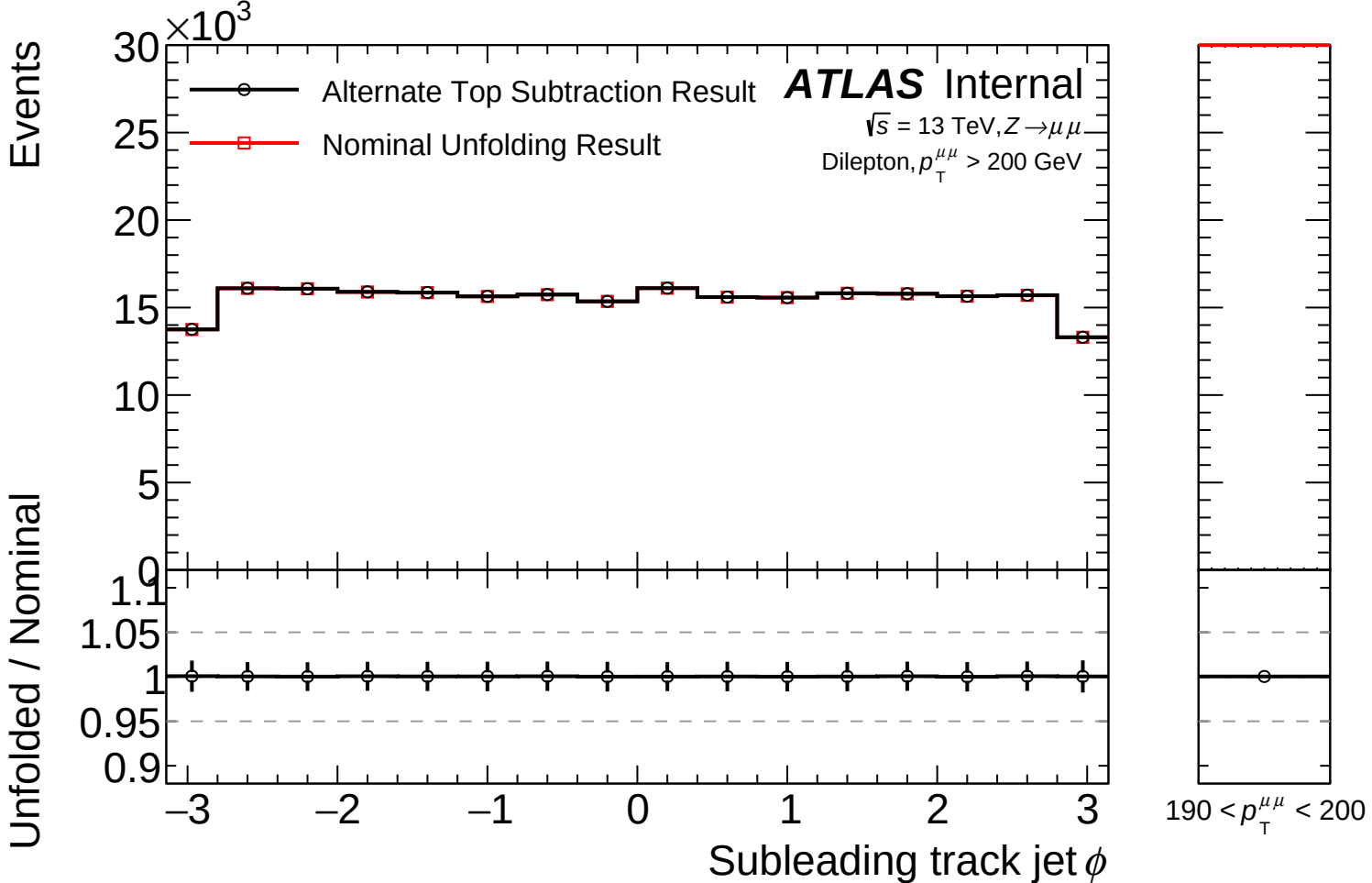
ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

Subleading track jety









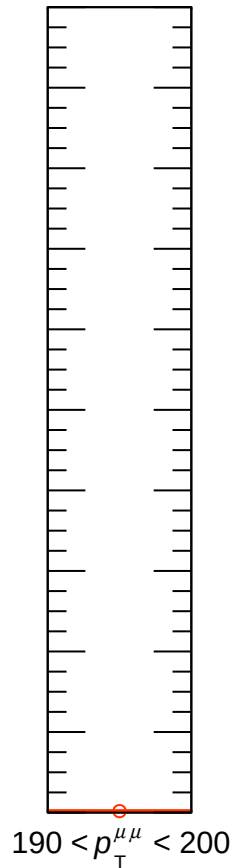
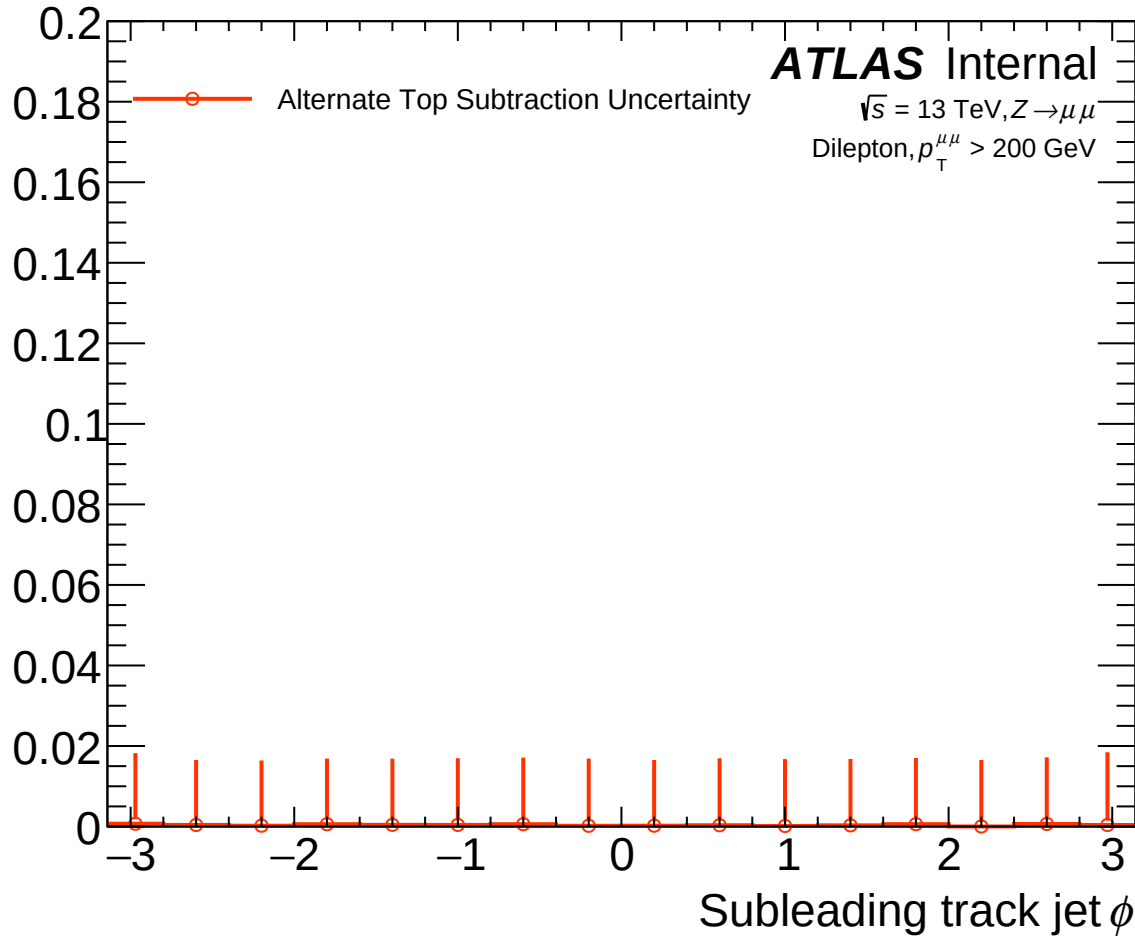
Fractional systematic impact

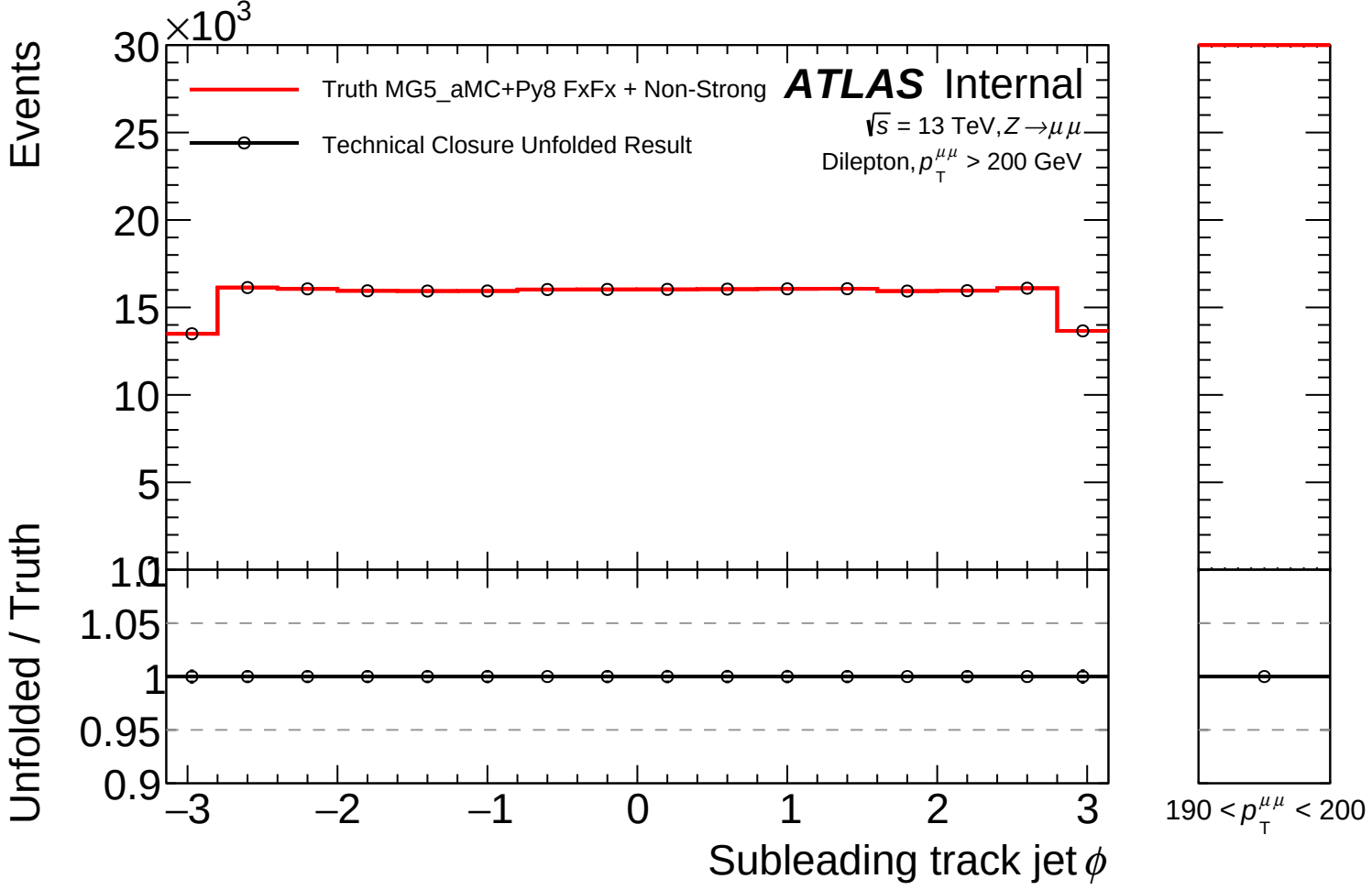
ATLAS Internal

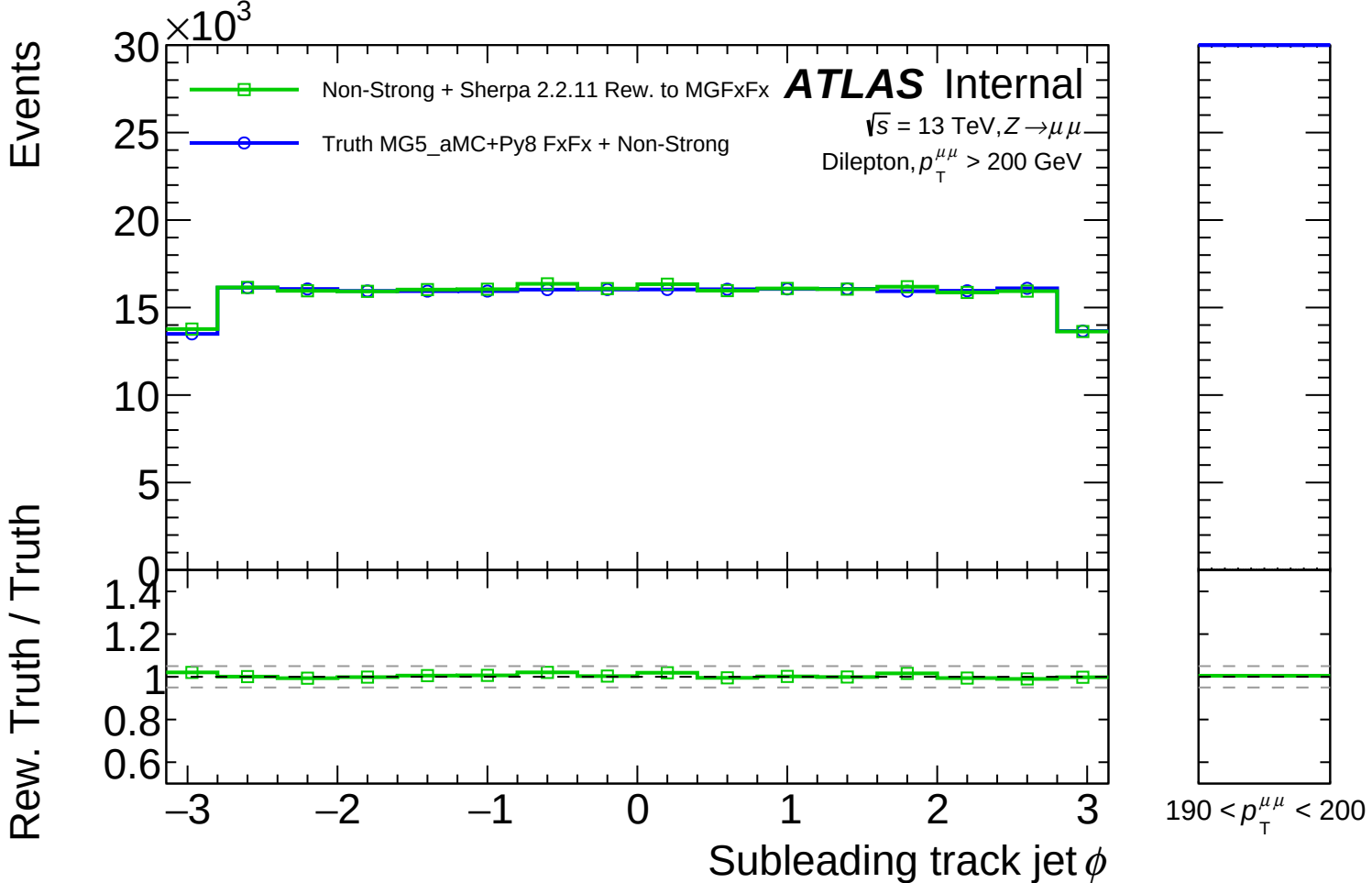
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

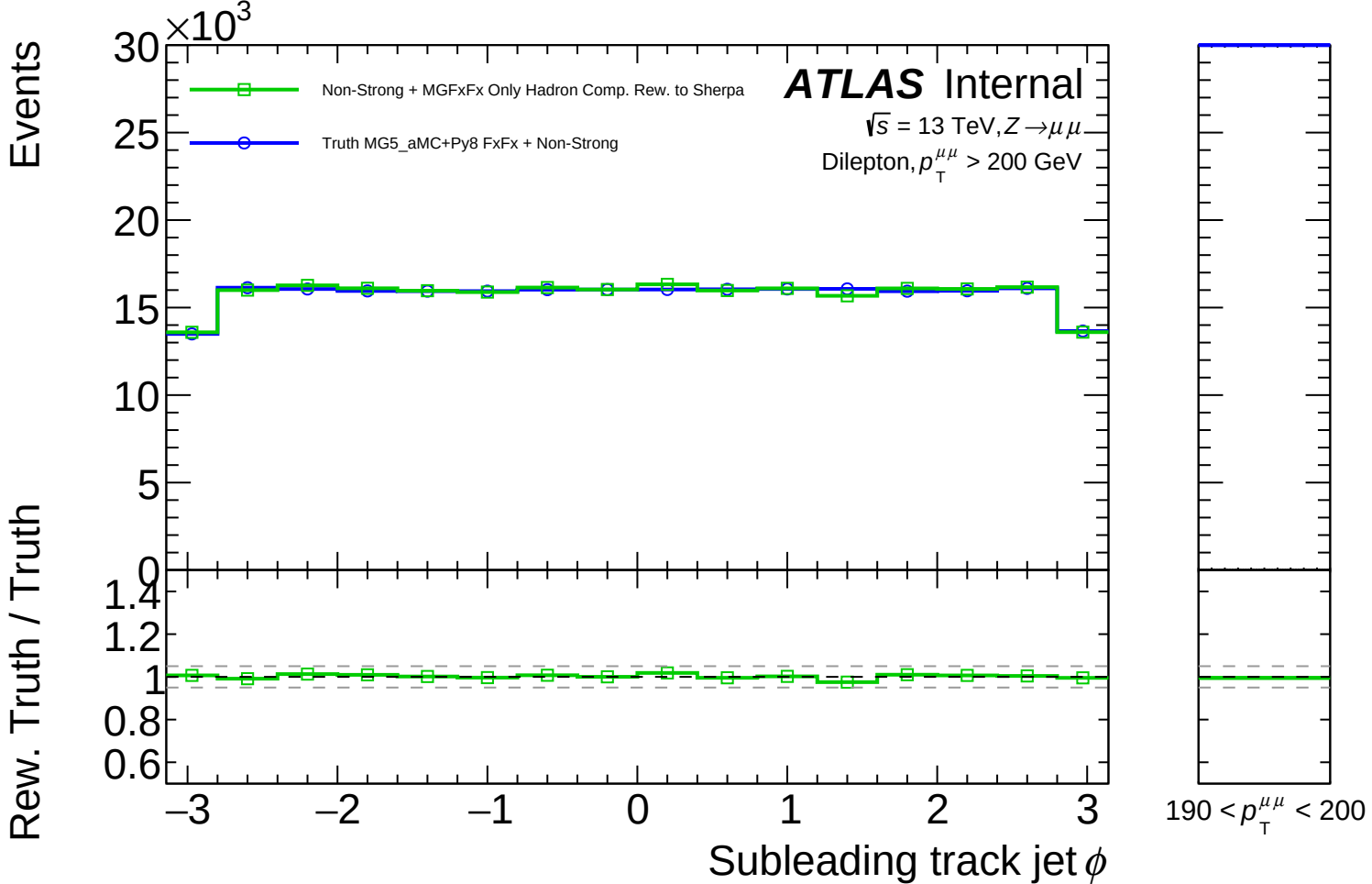
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

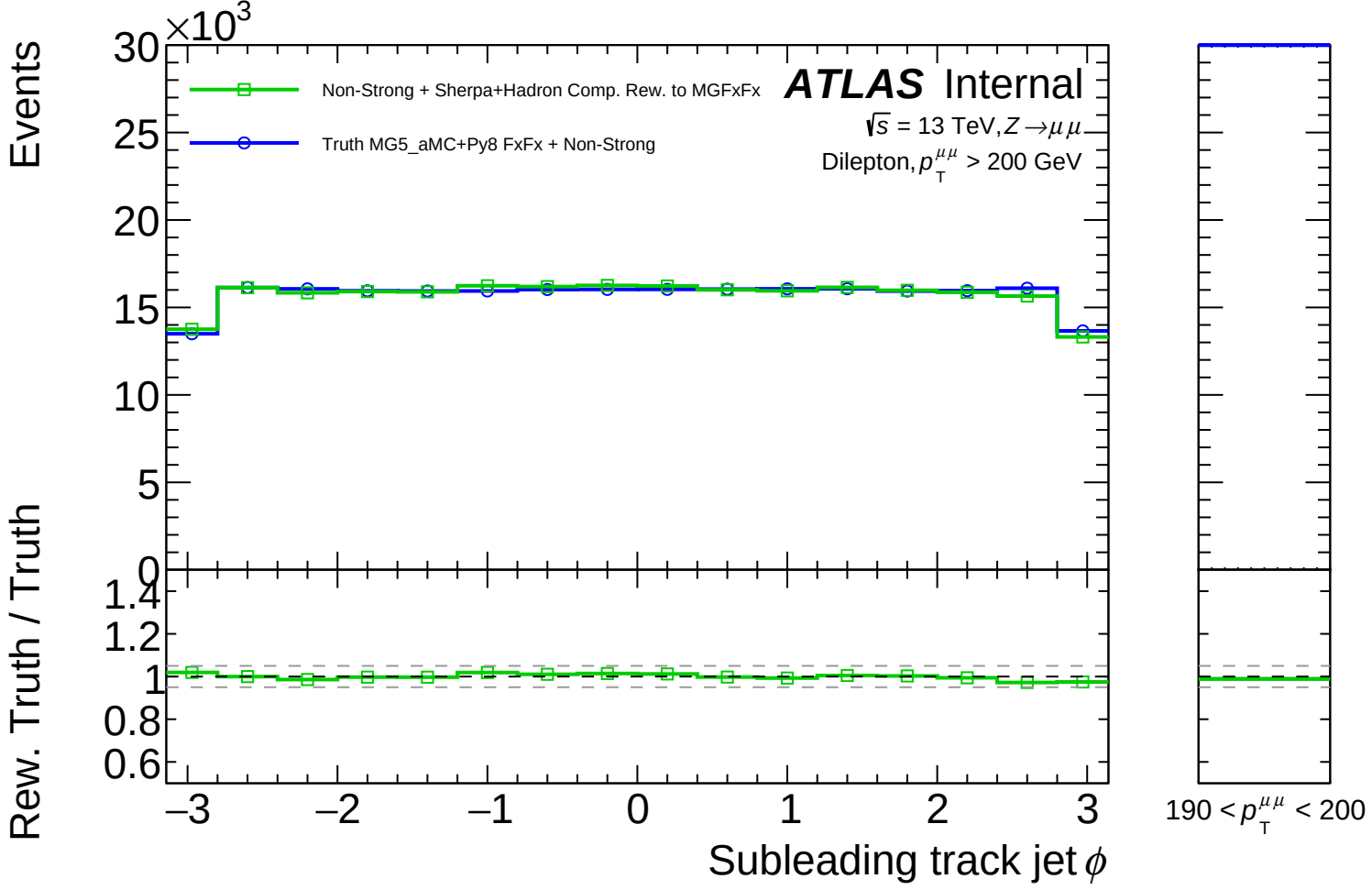
Alternate Top Subtraction Uncertainty

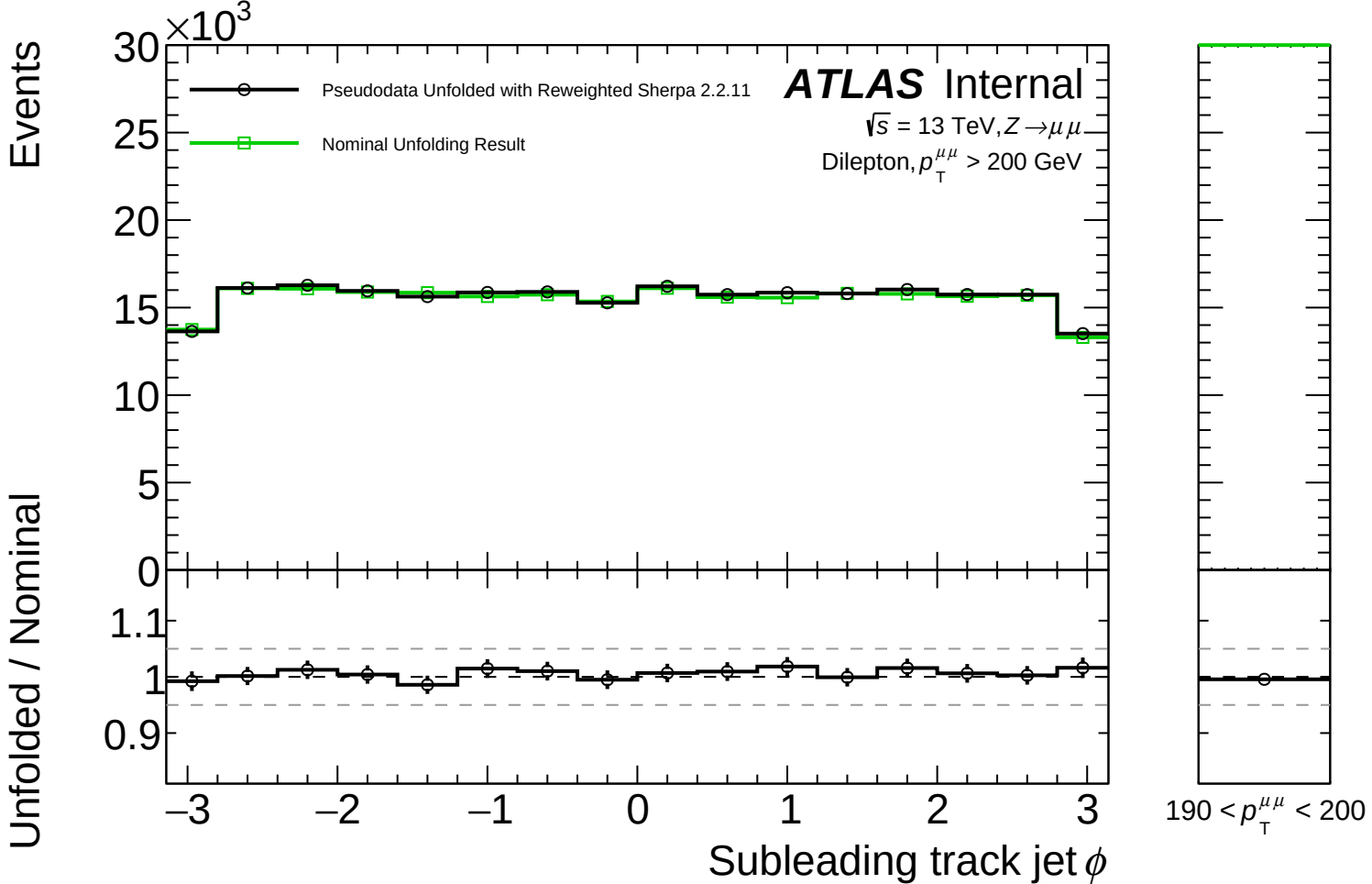


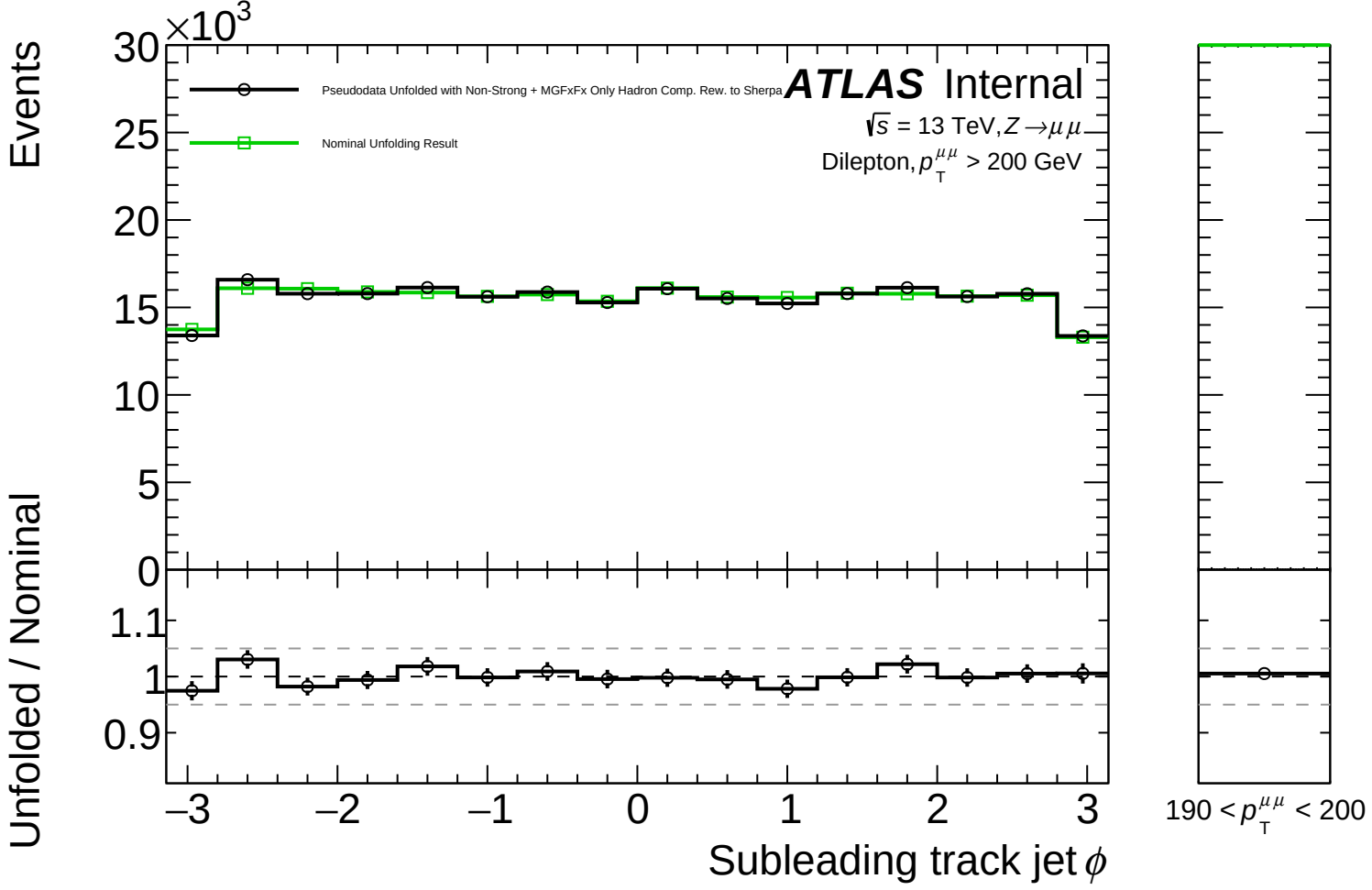


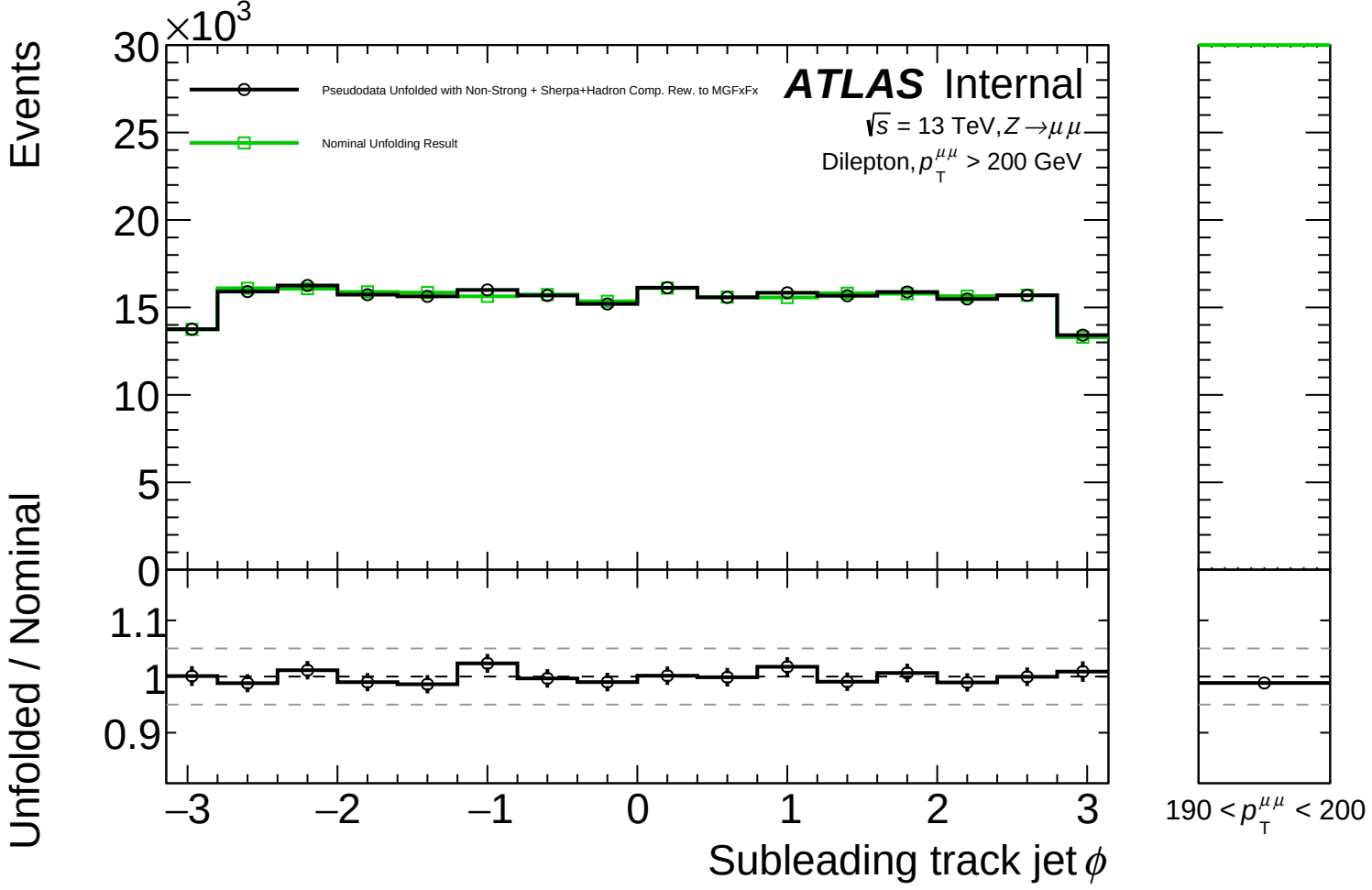


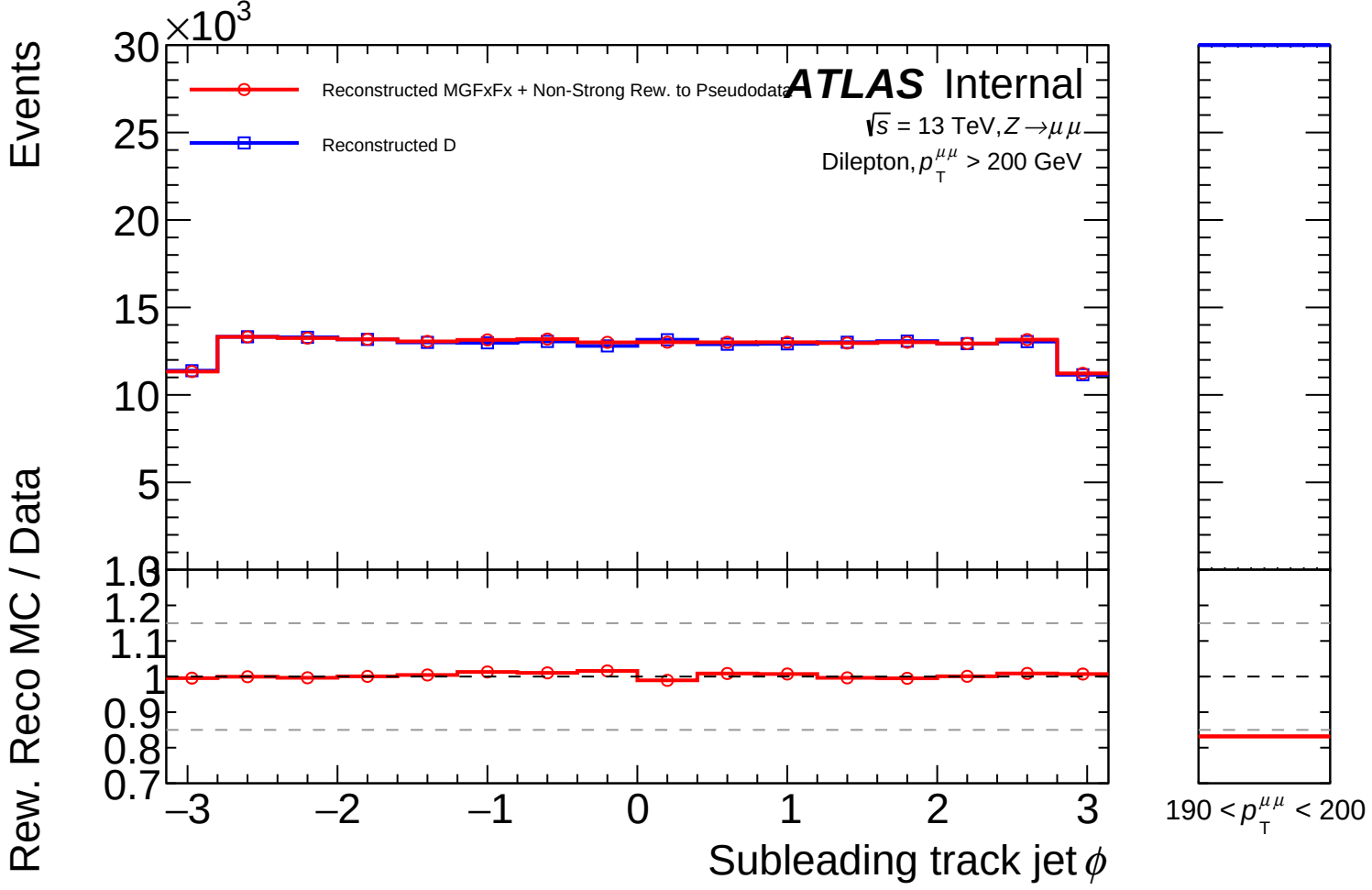


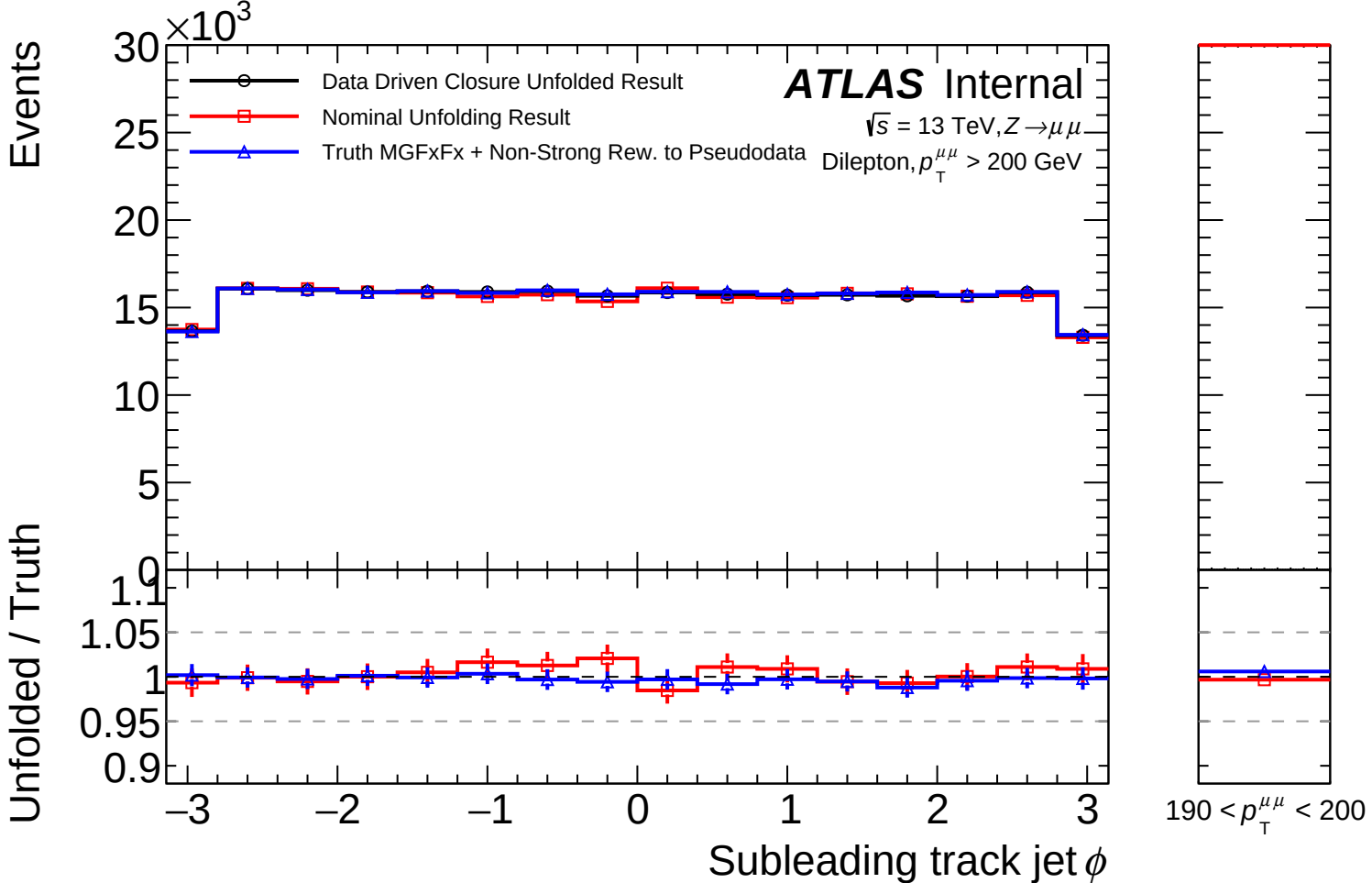




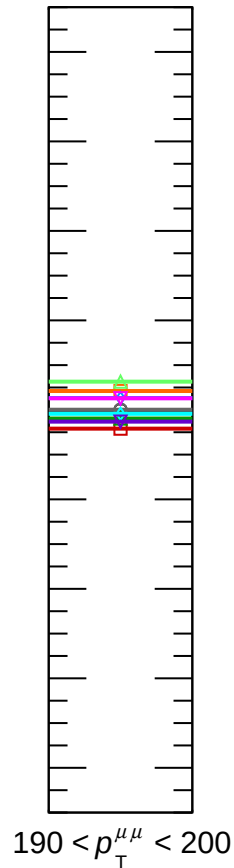
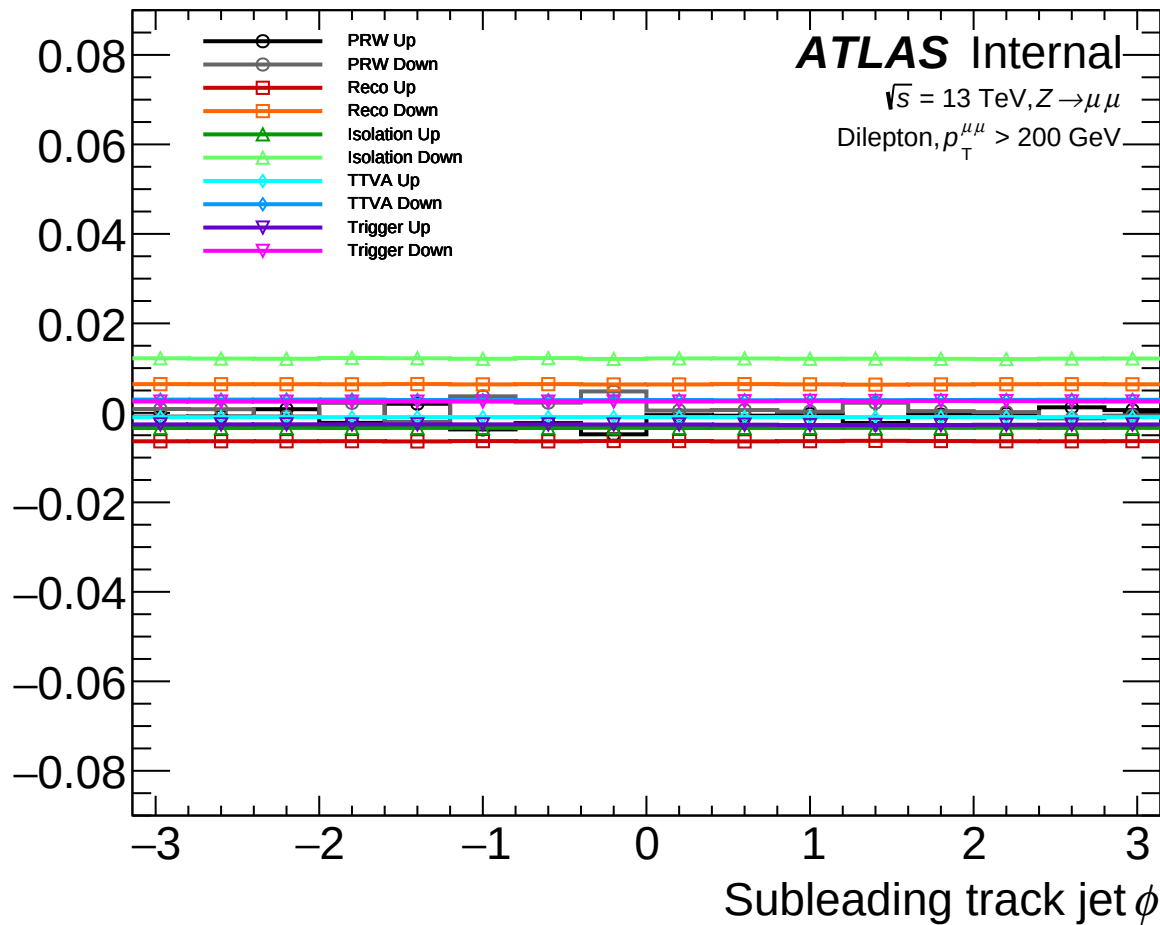




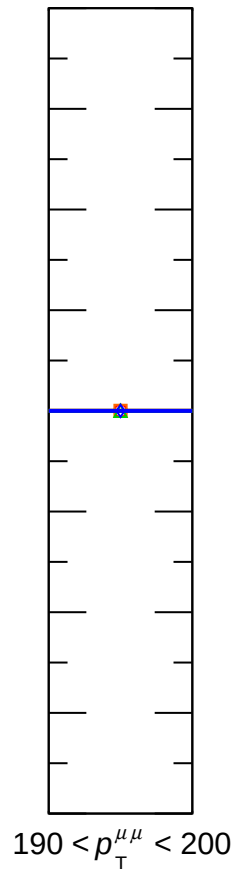
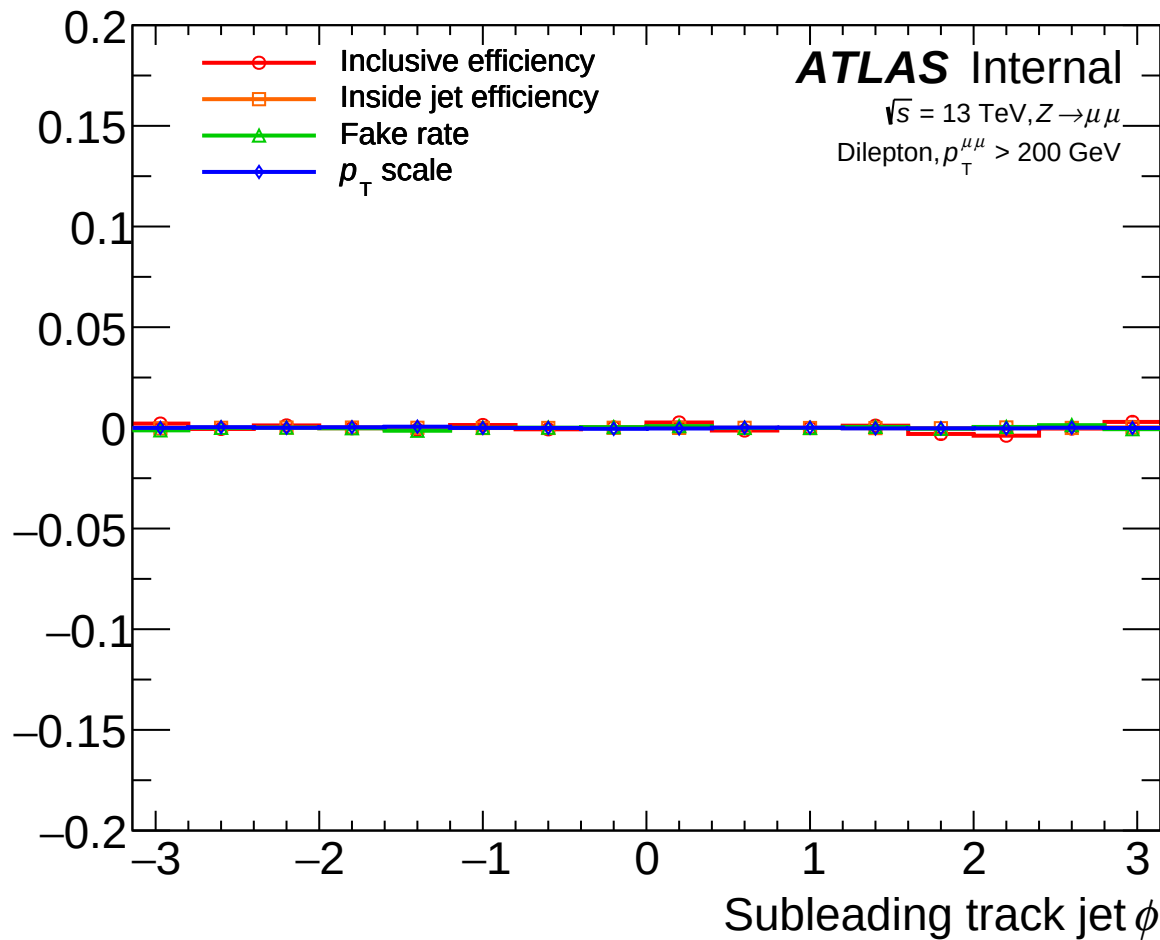




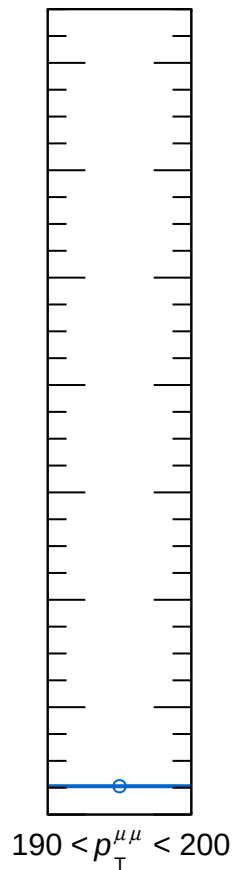
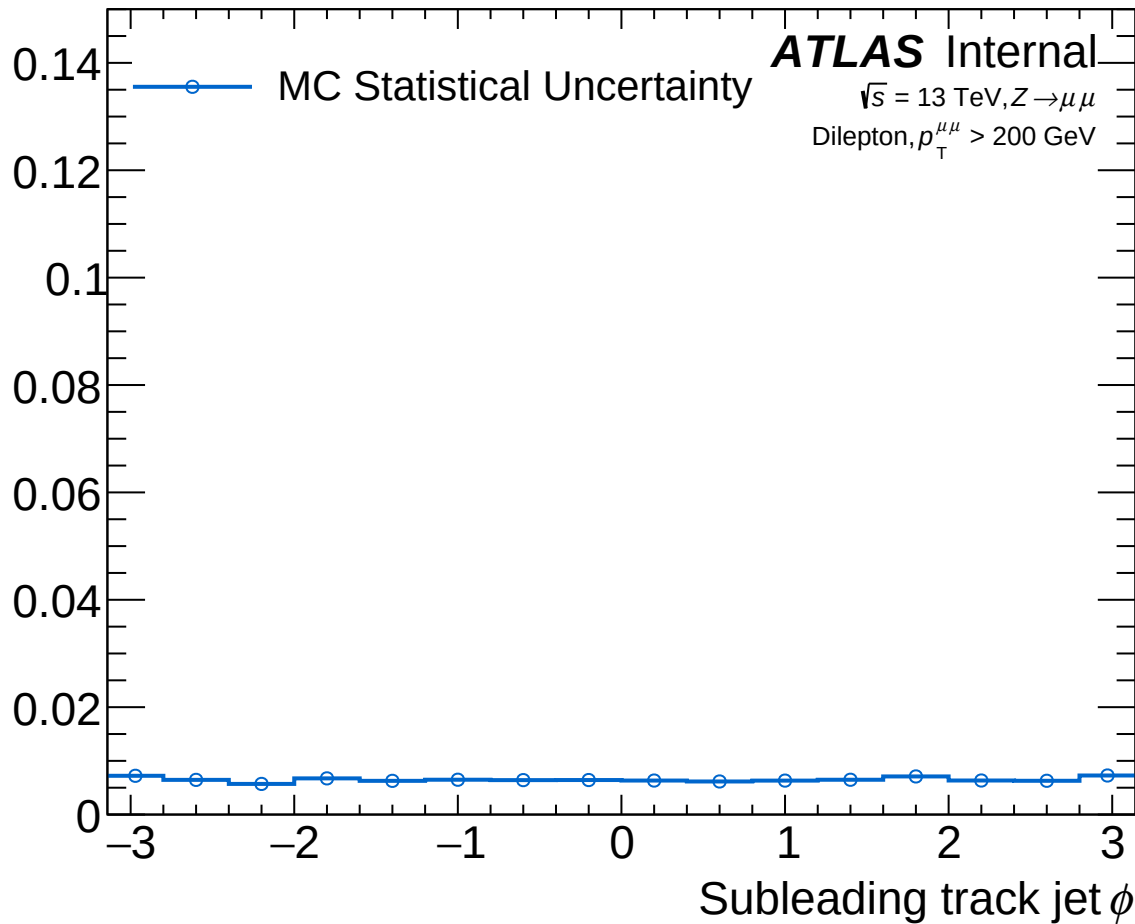
Fractional systematic impact



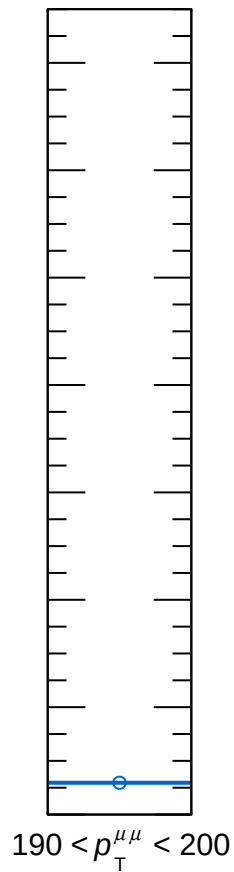
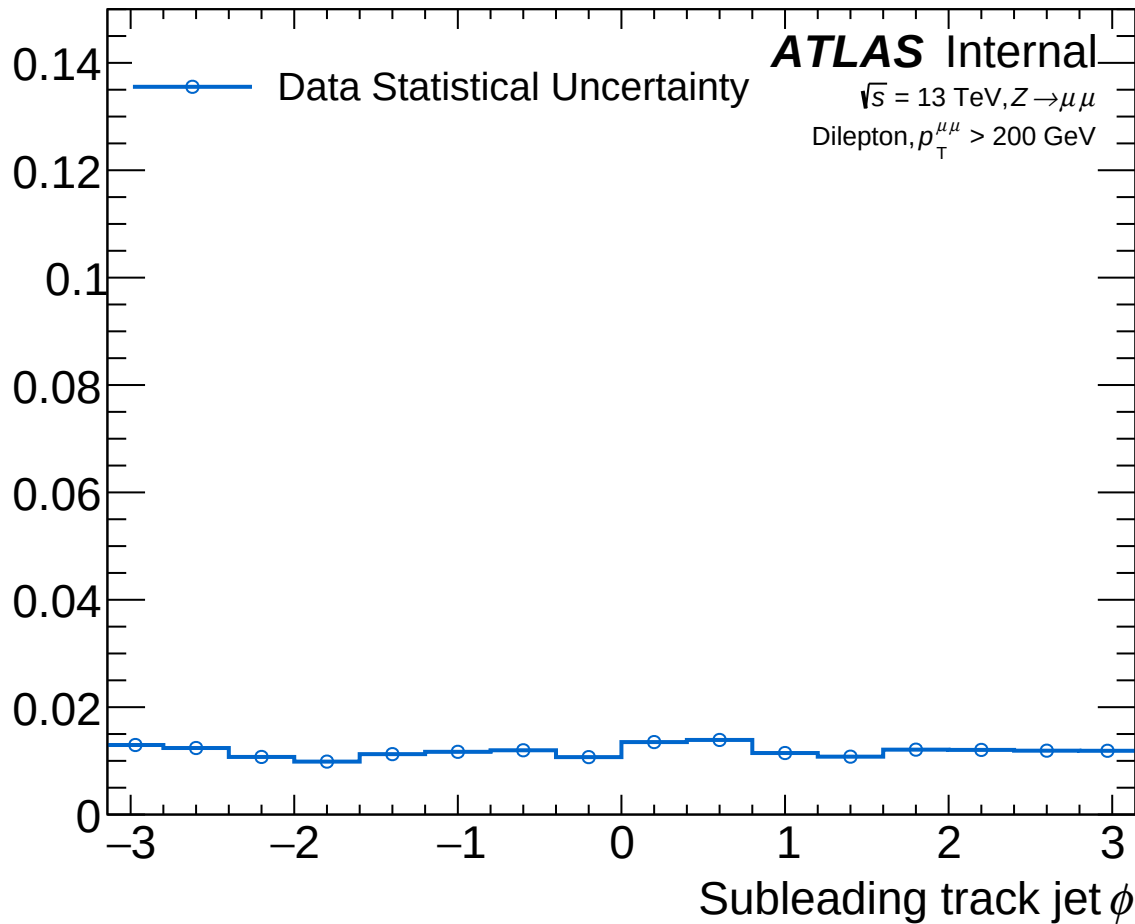
Fractional systematic impact



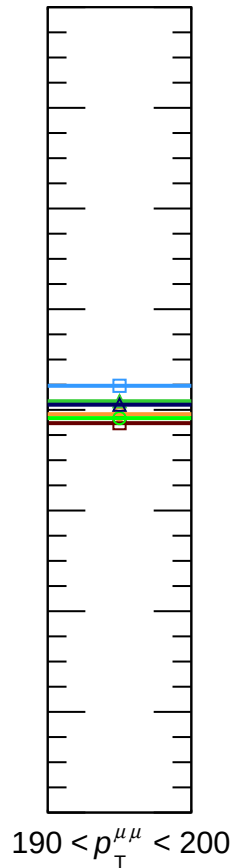
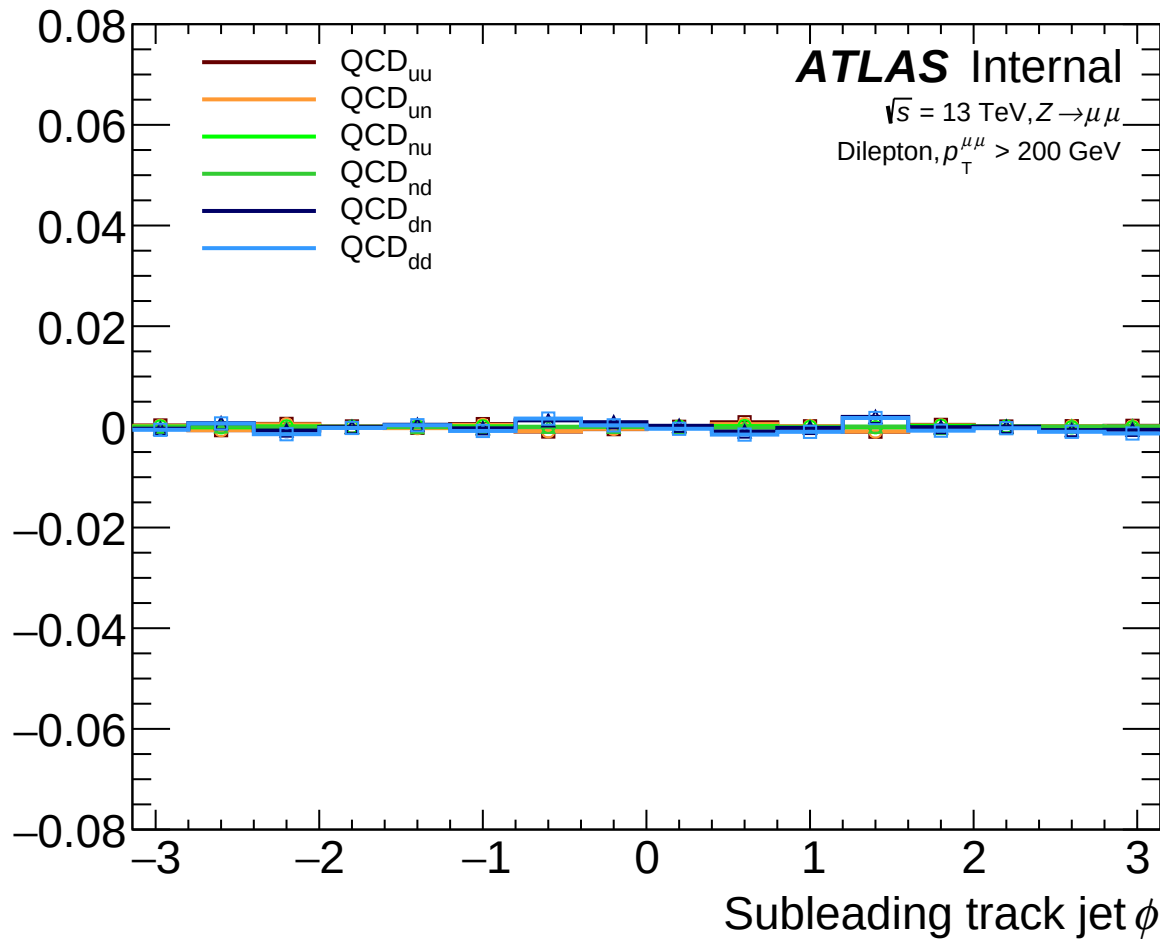
Fractional systematic impact



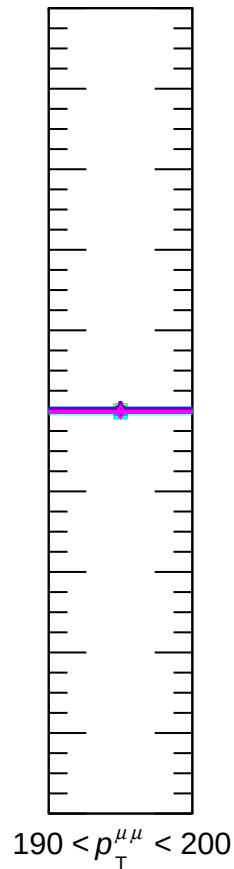
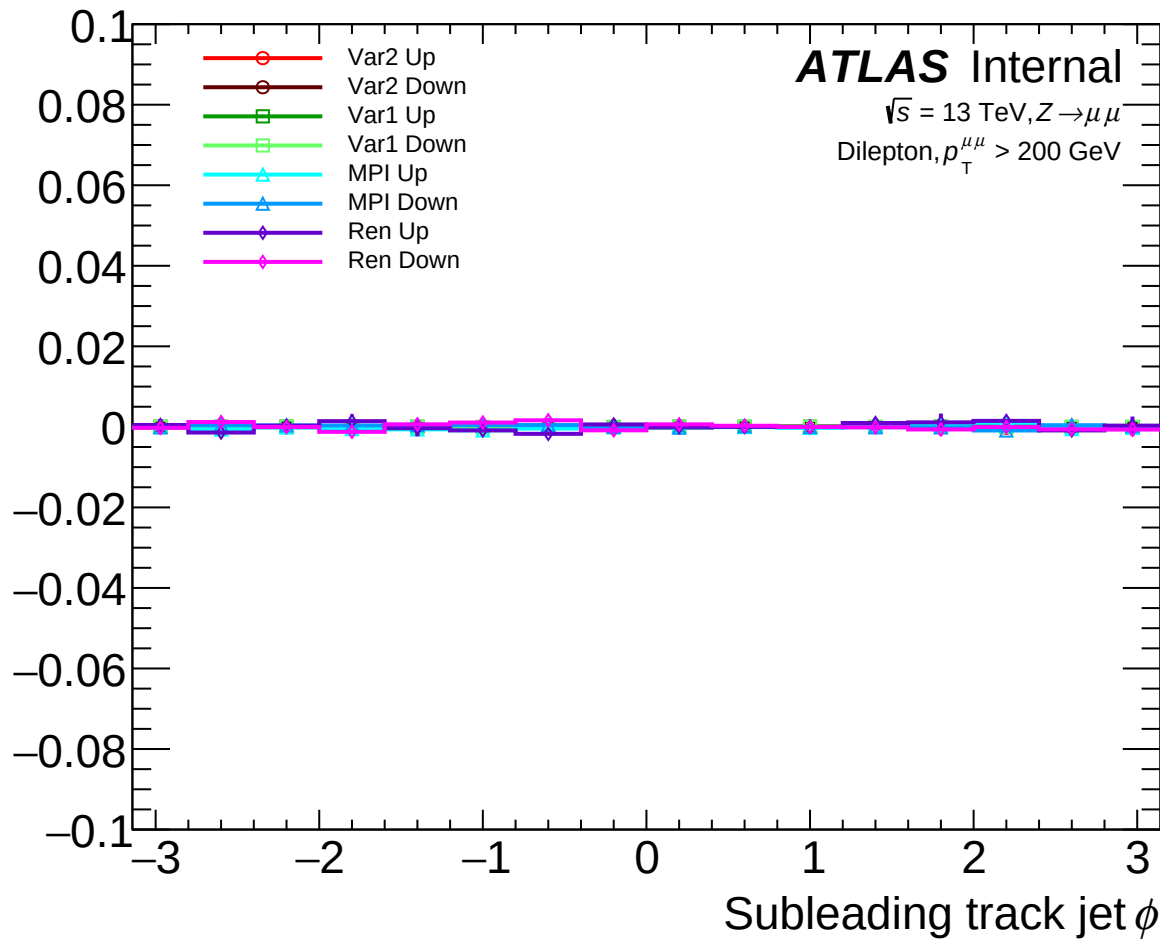
Fractional systematic impact

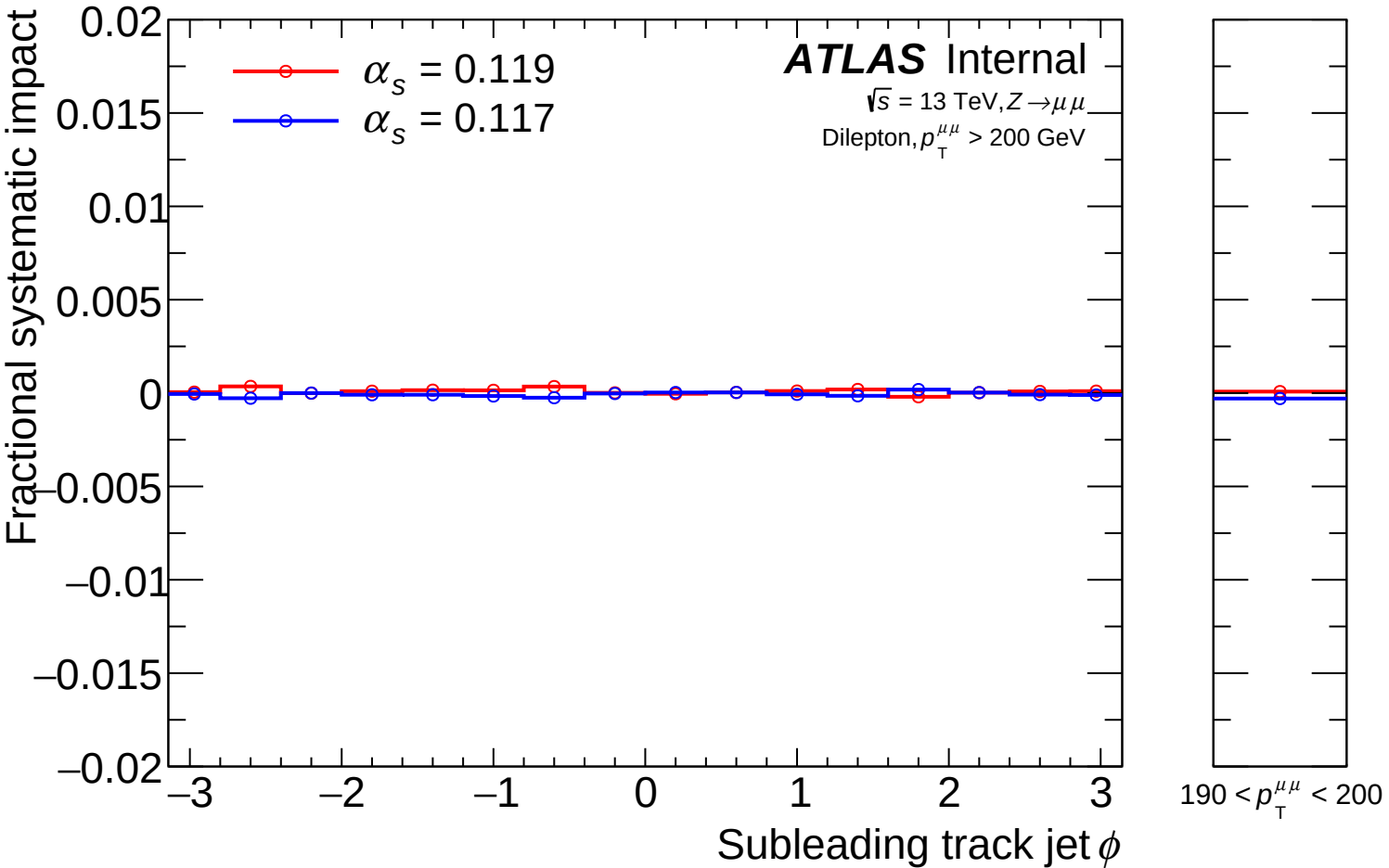


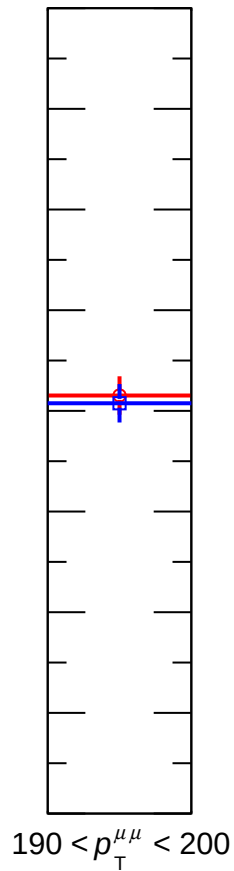
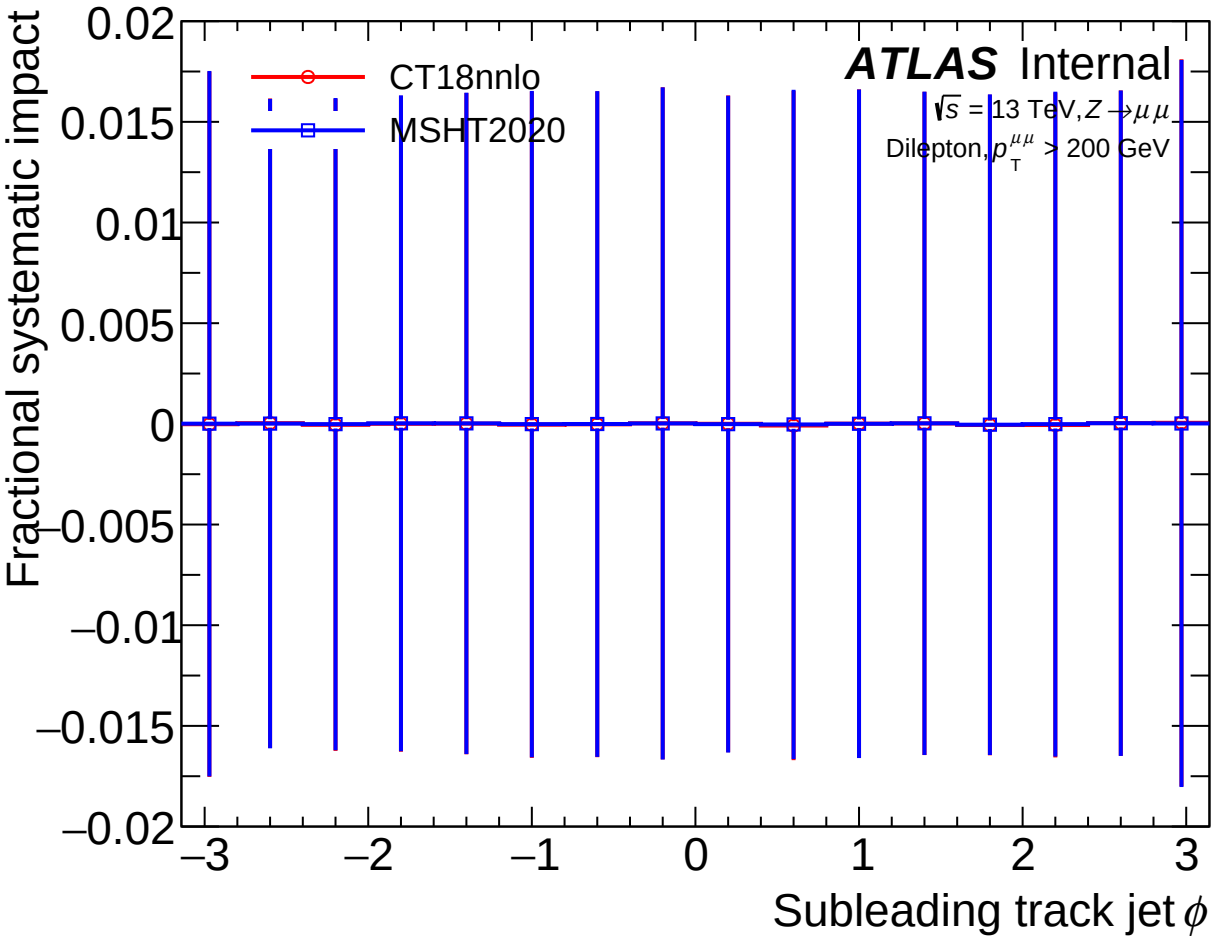
Fractional systematic impact



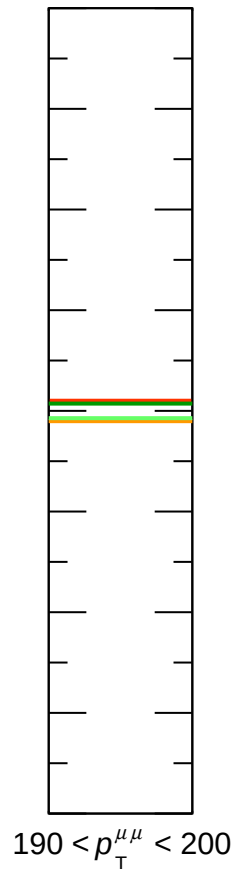
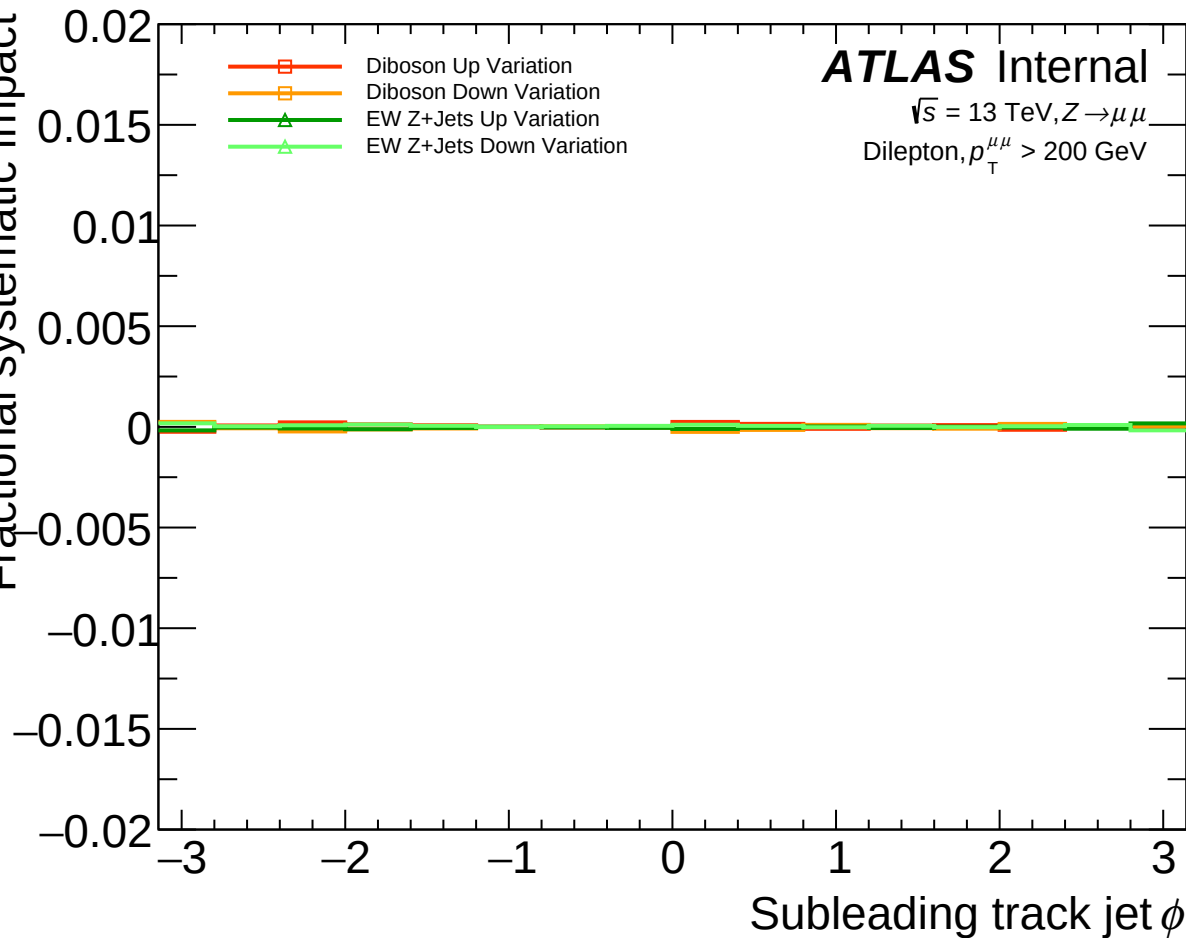
Fractional systematic impact

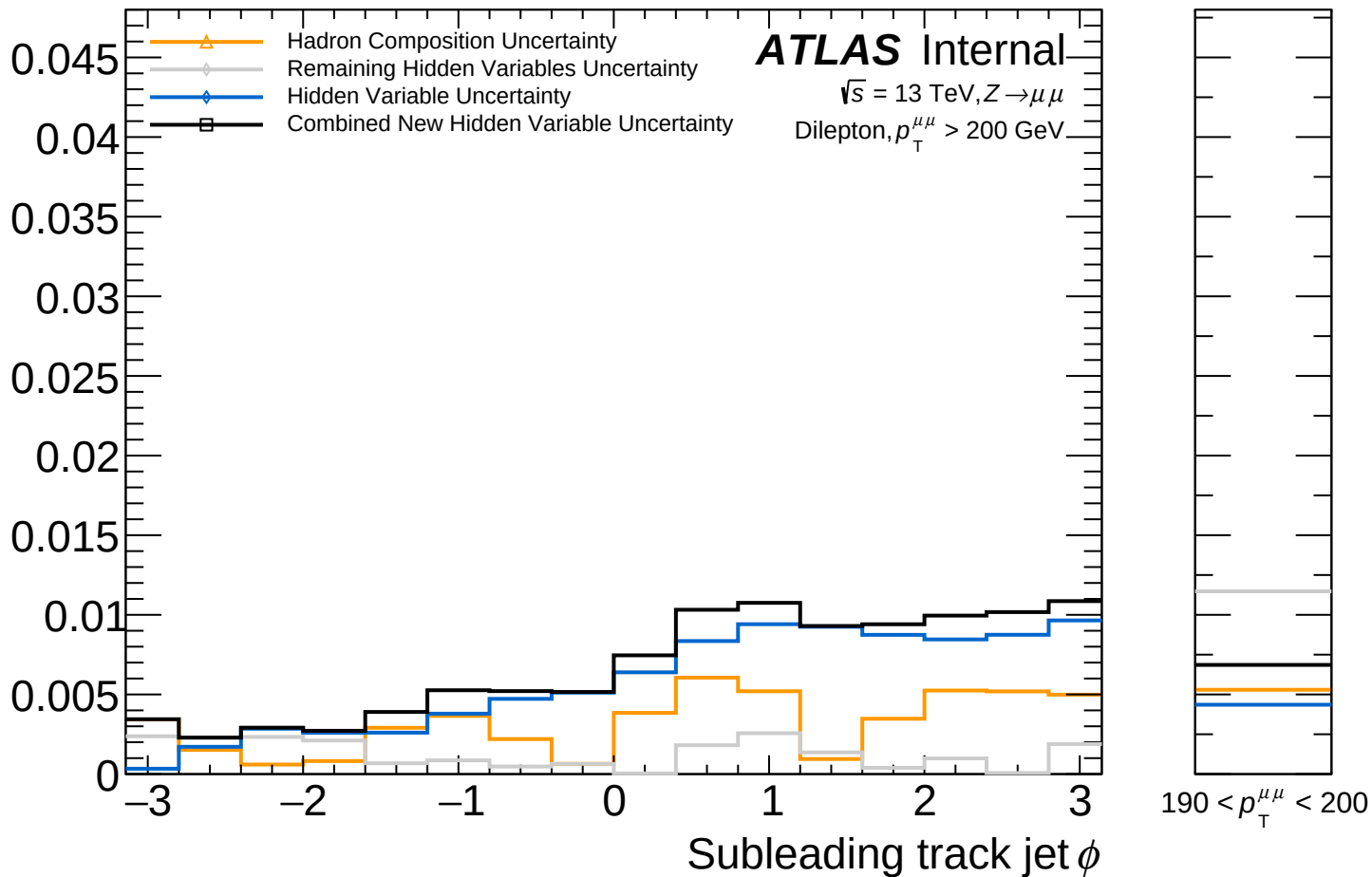


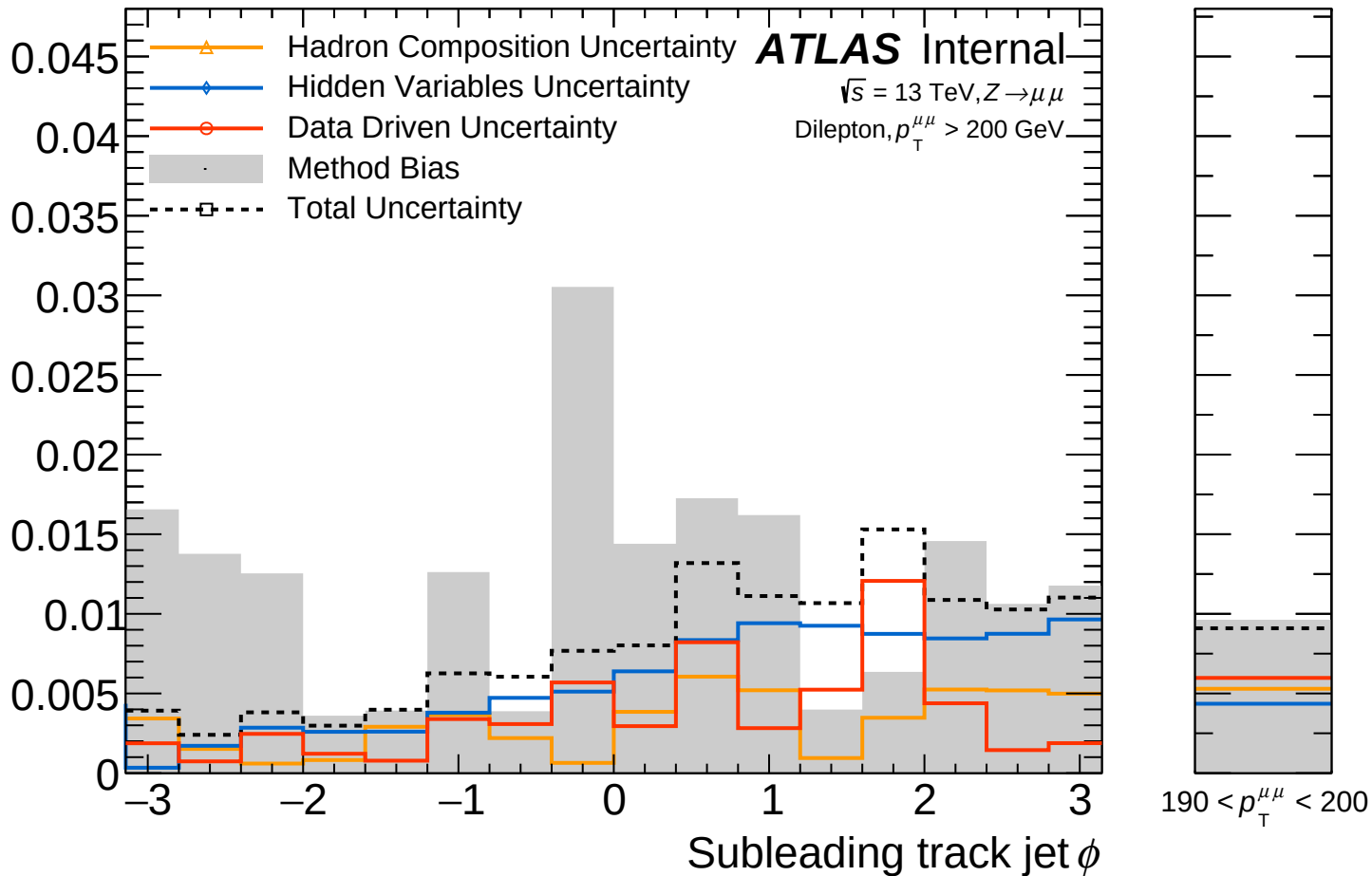


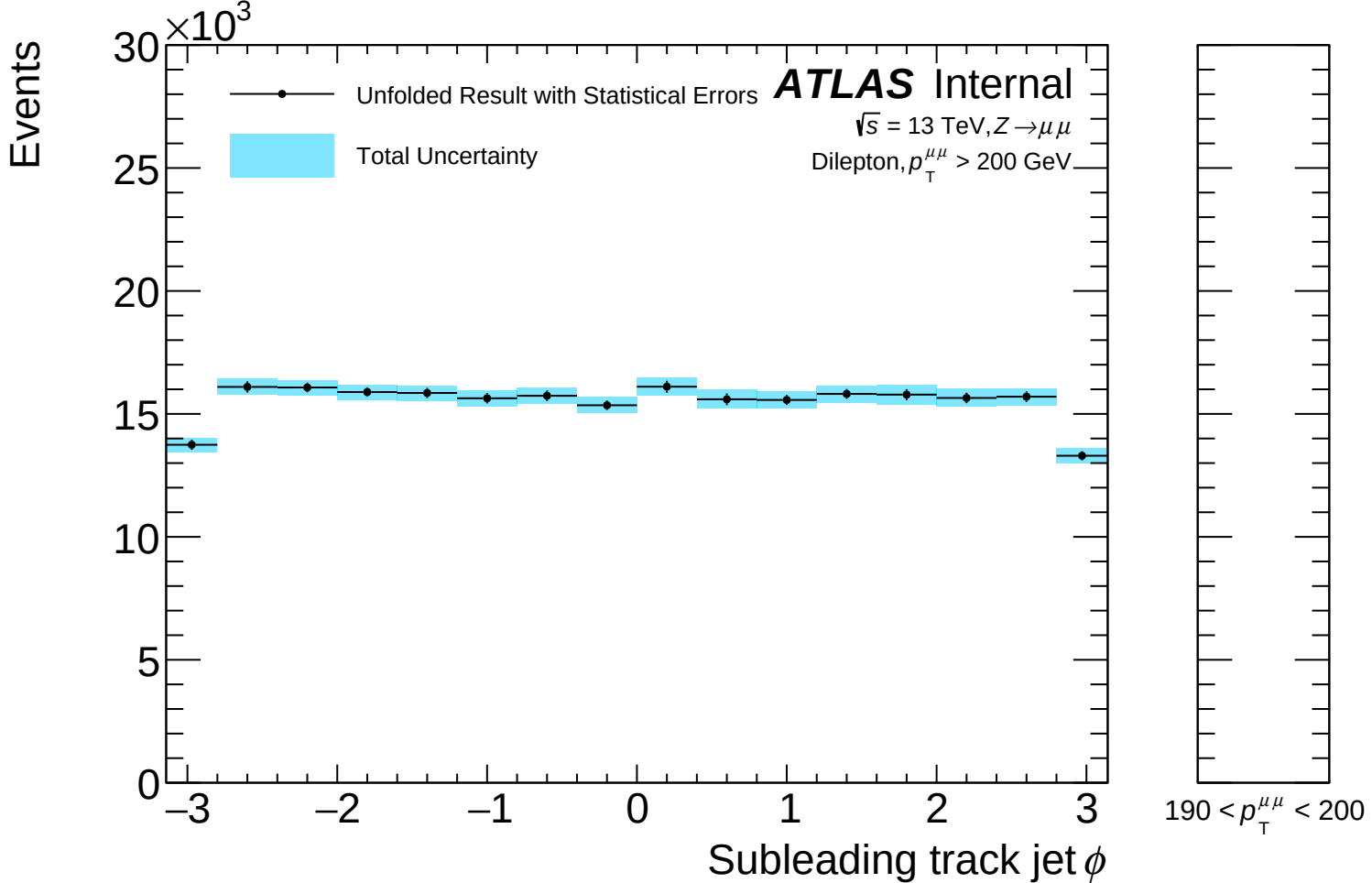


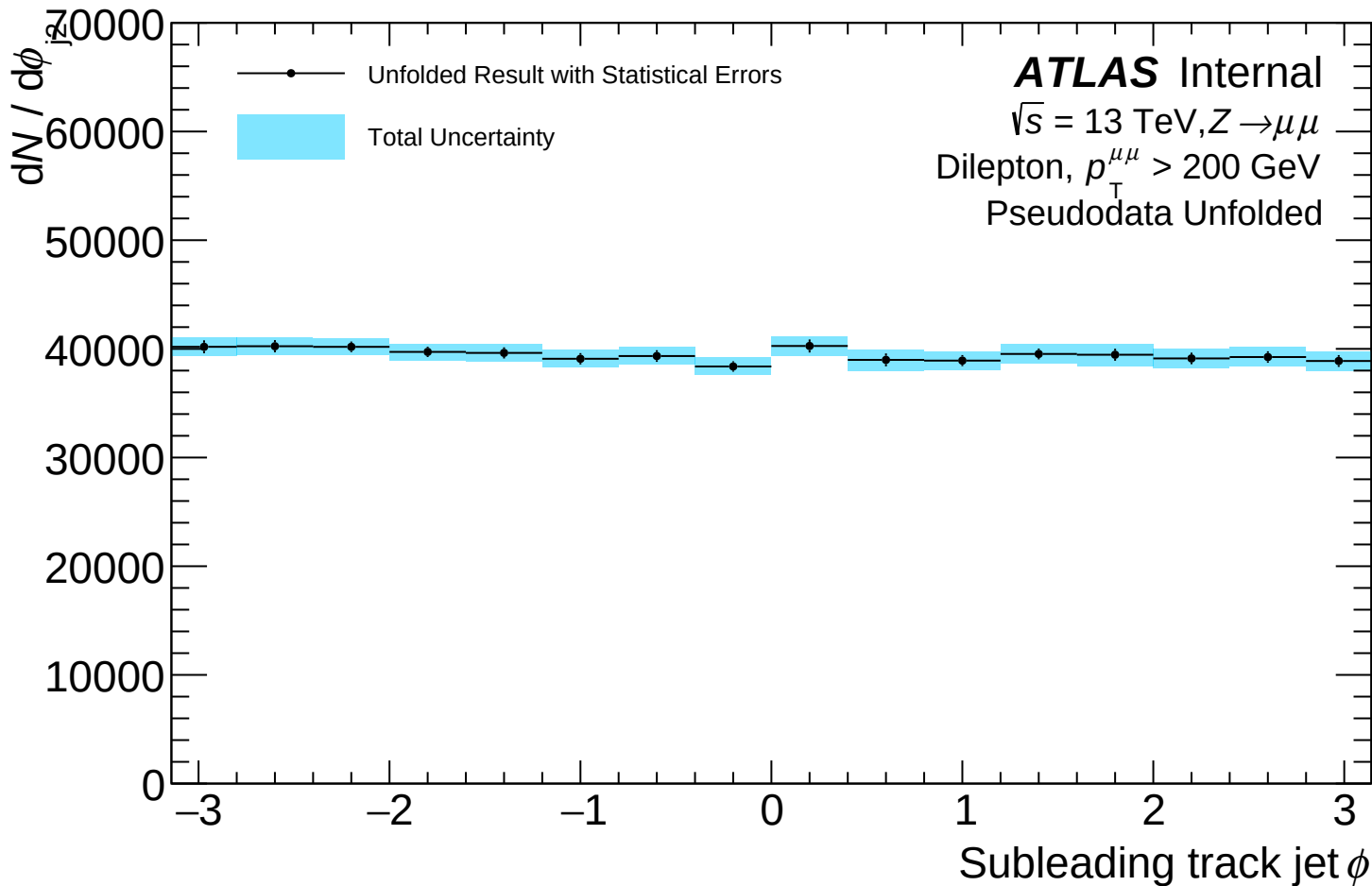
Fractional systematic impact

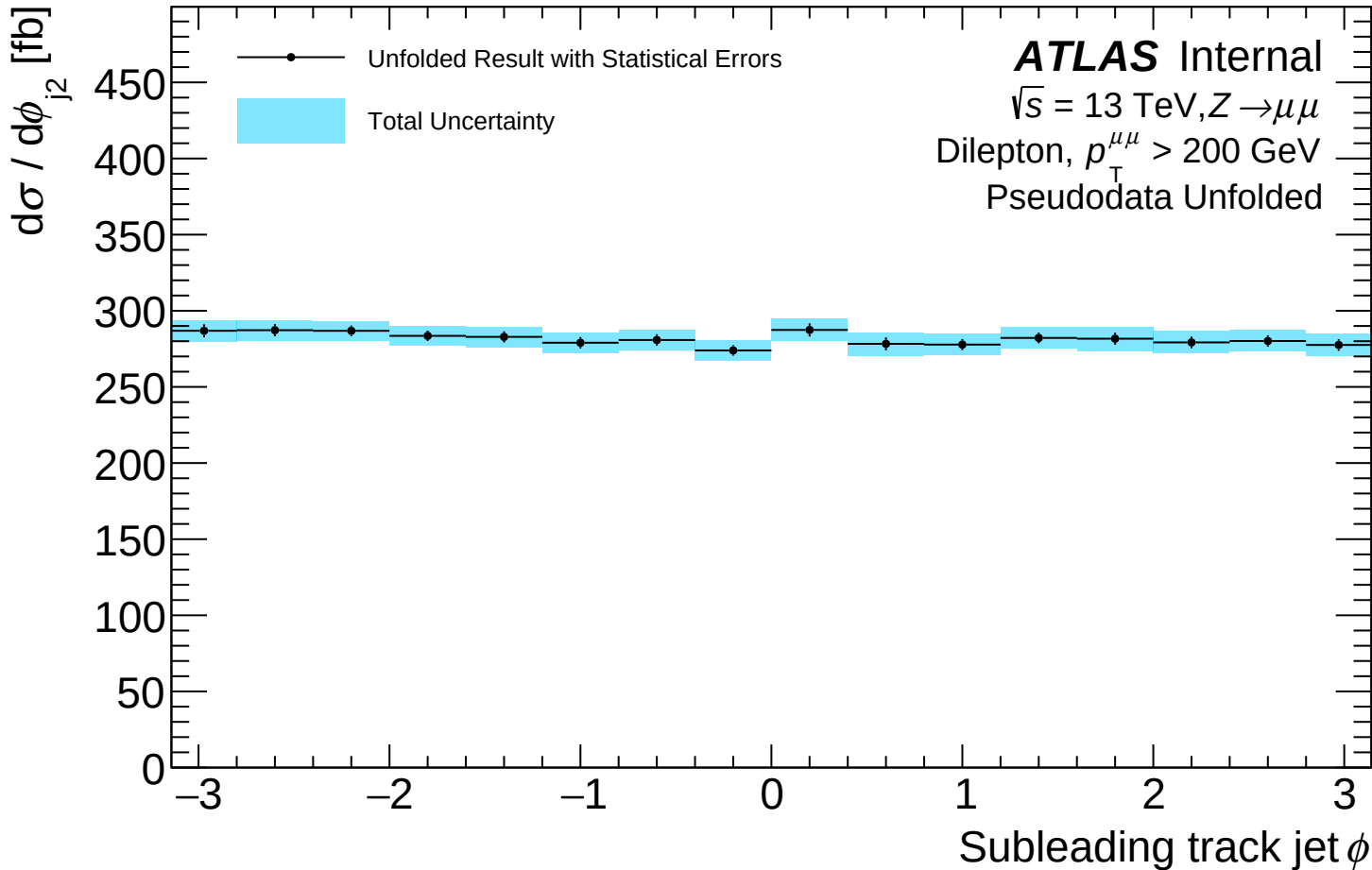




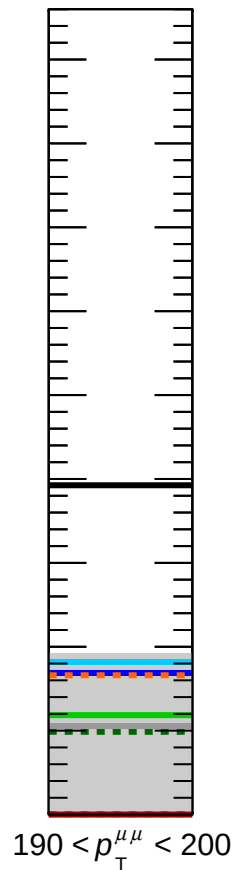
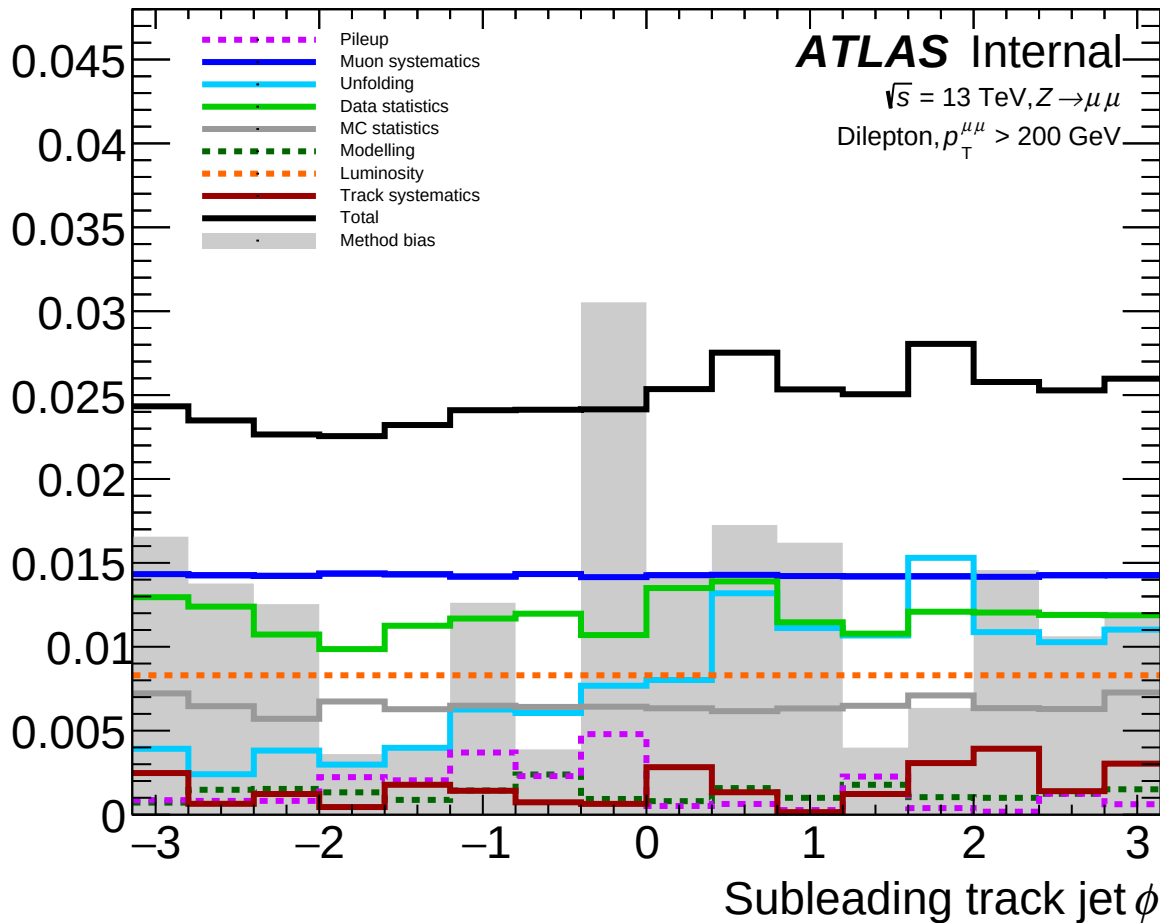


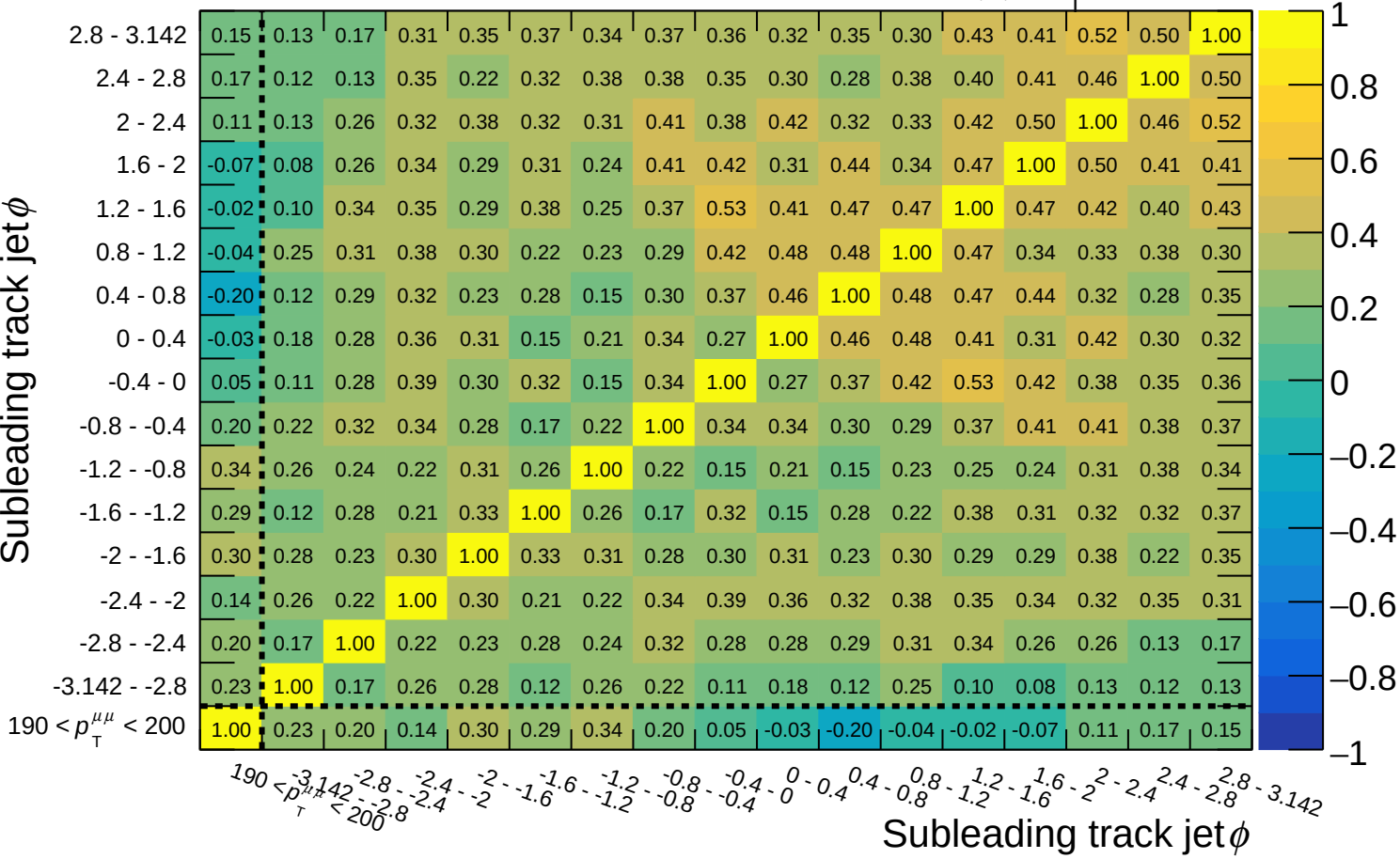


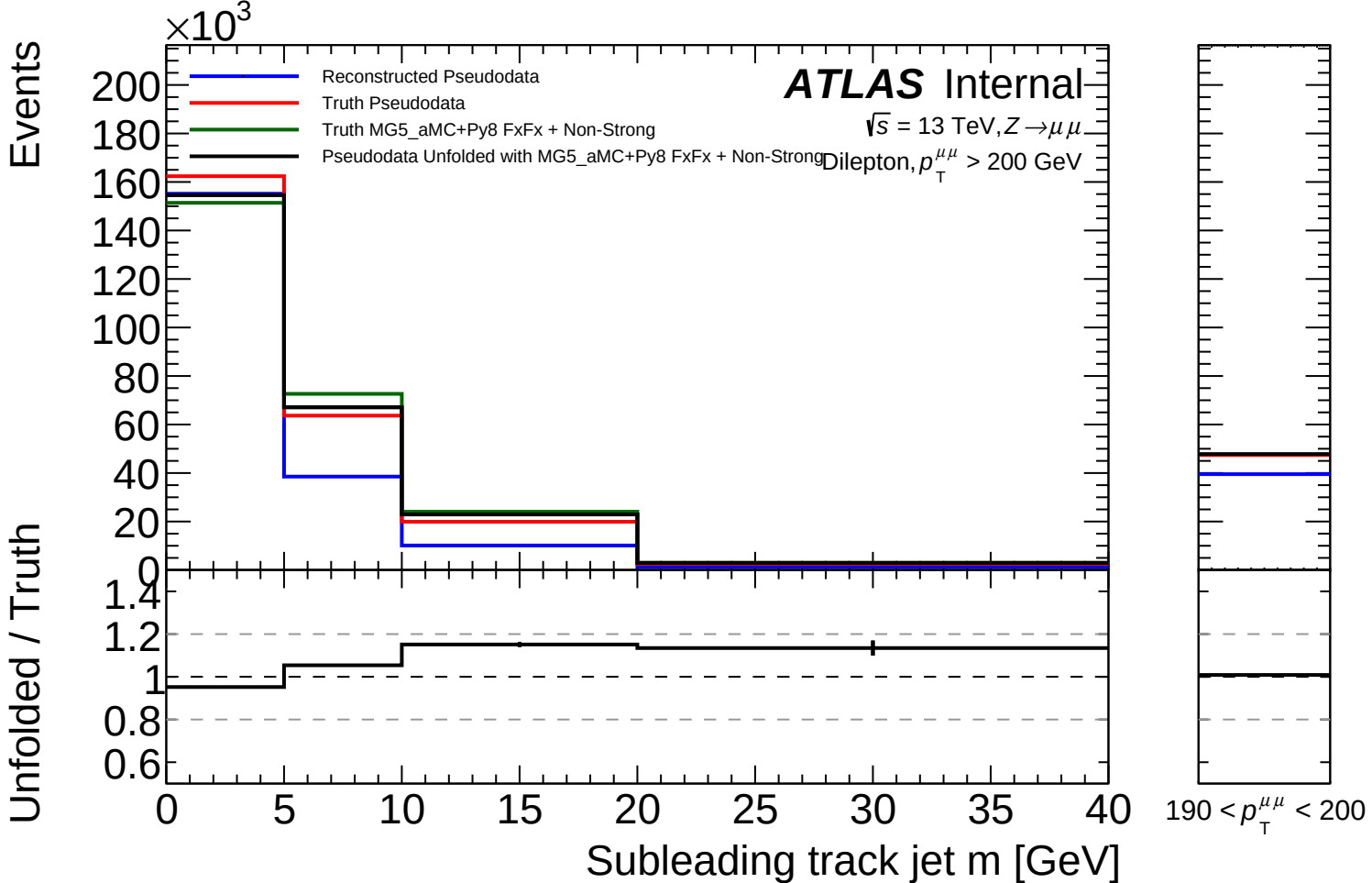


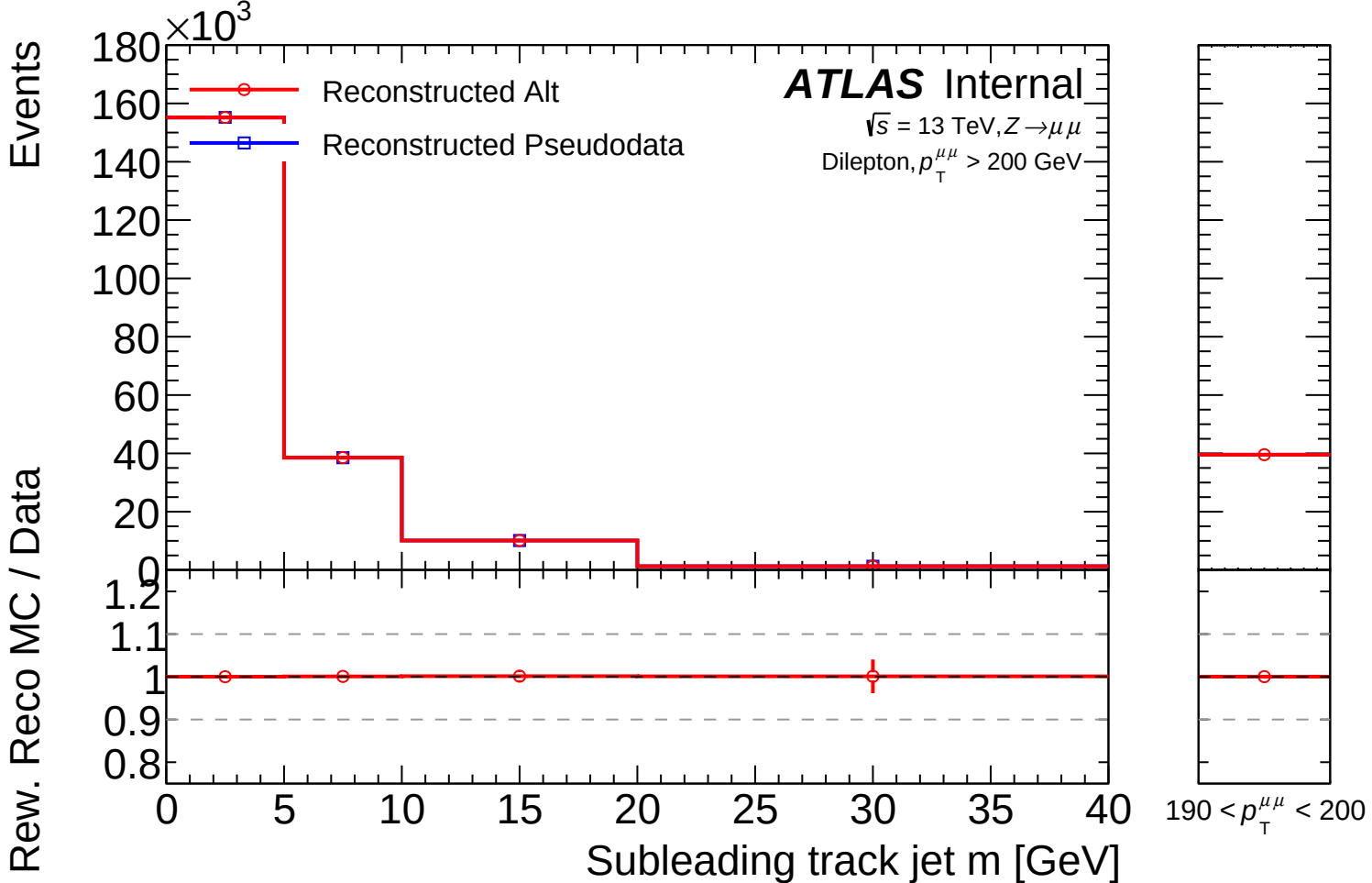


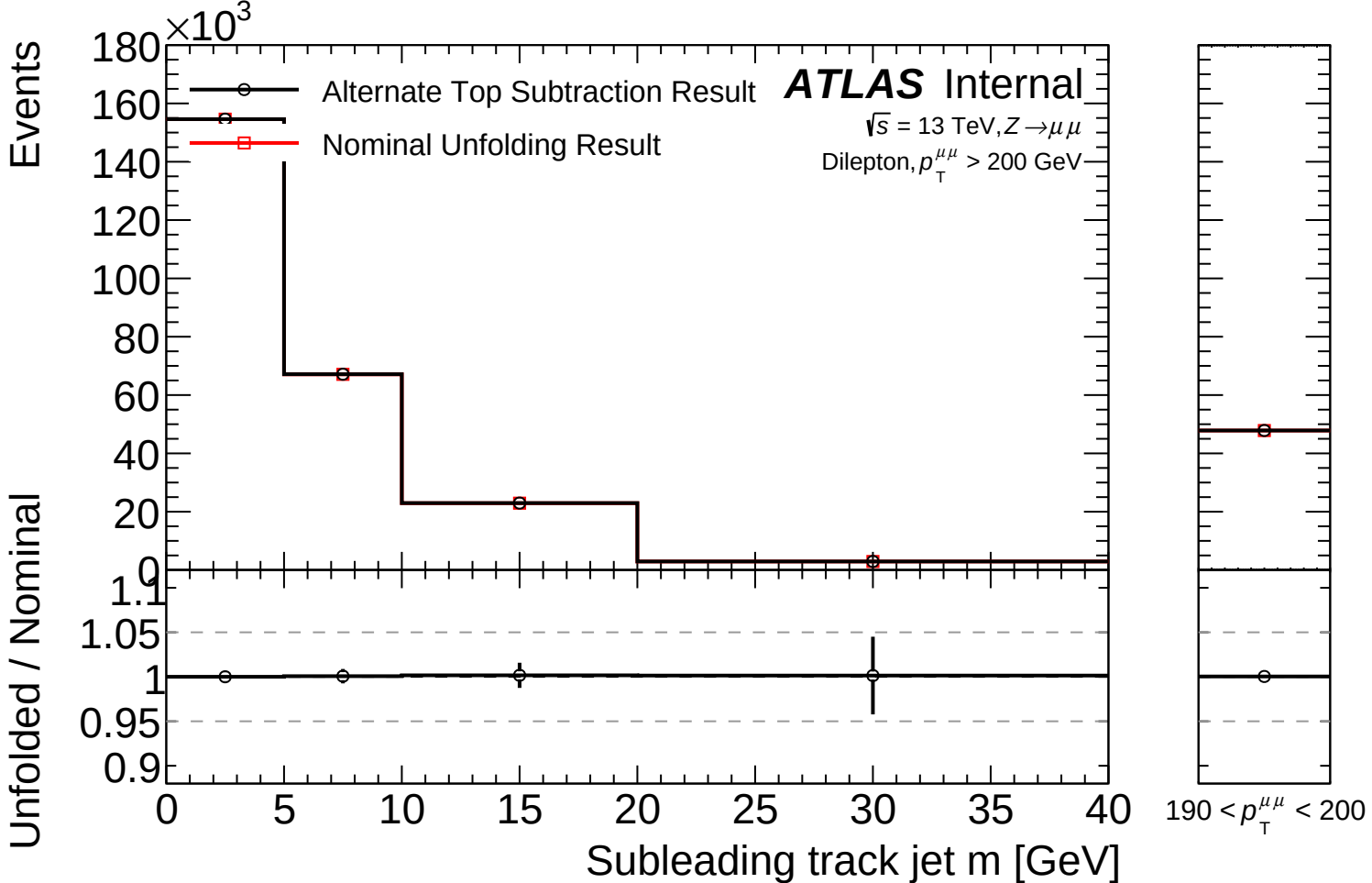
Fractional uncertainty



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Subleading track jet ϕ 







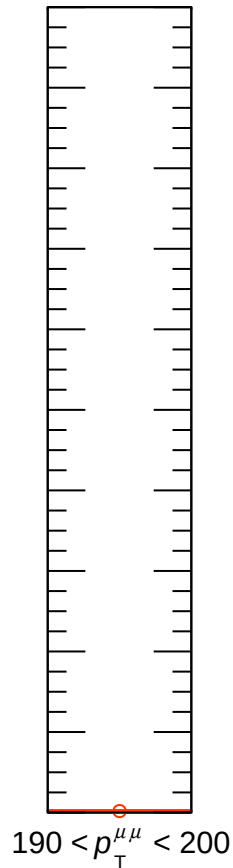
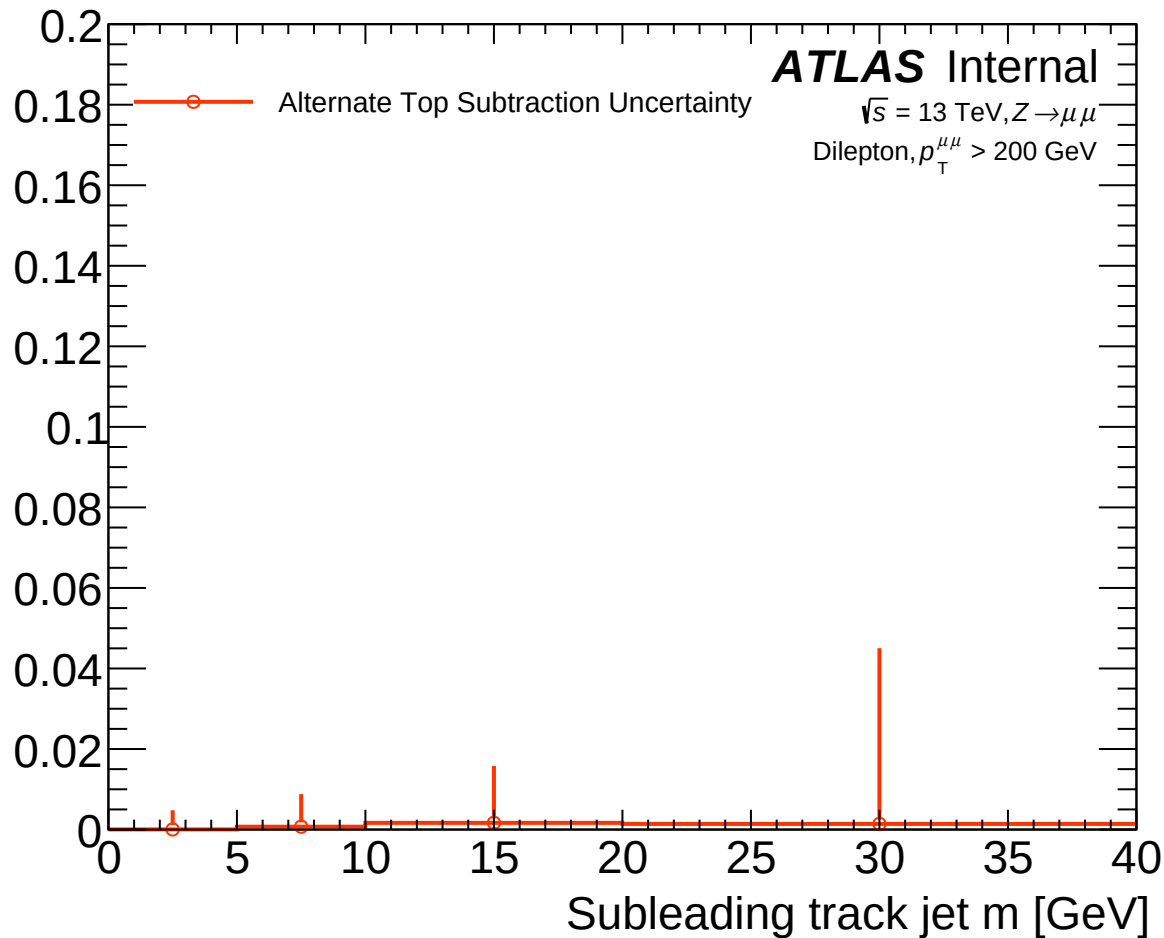
Fractional systematic impact

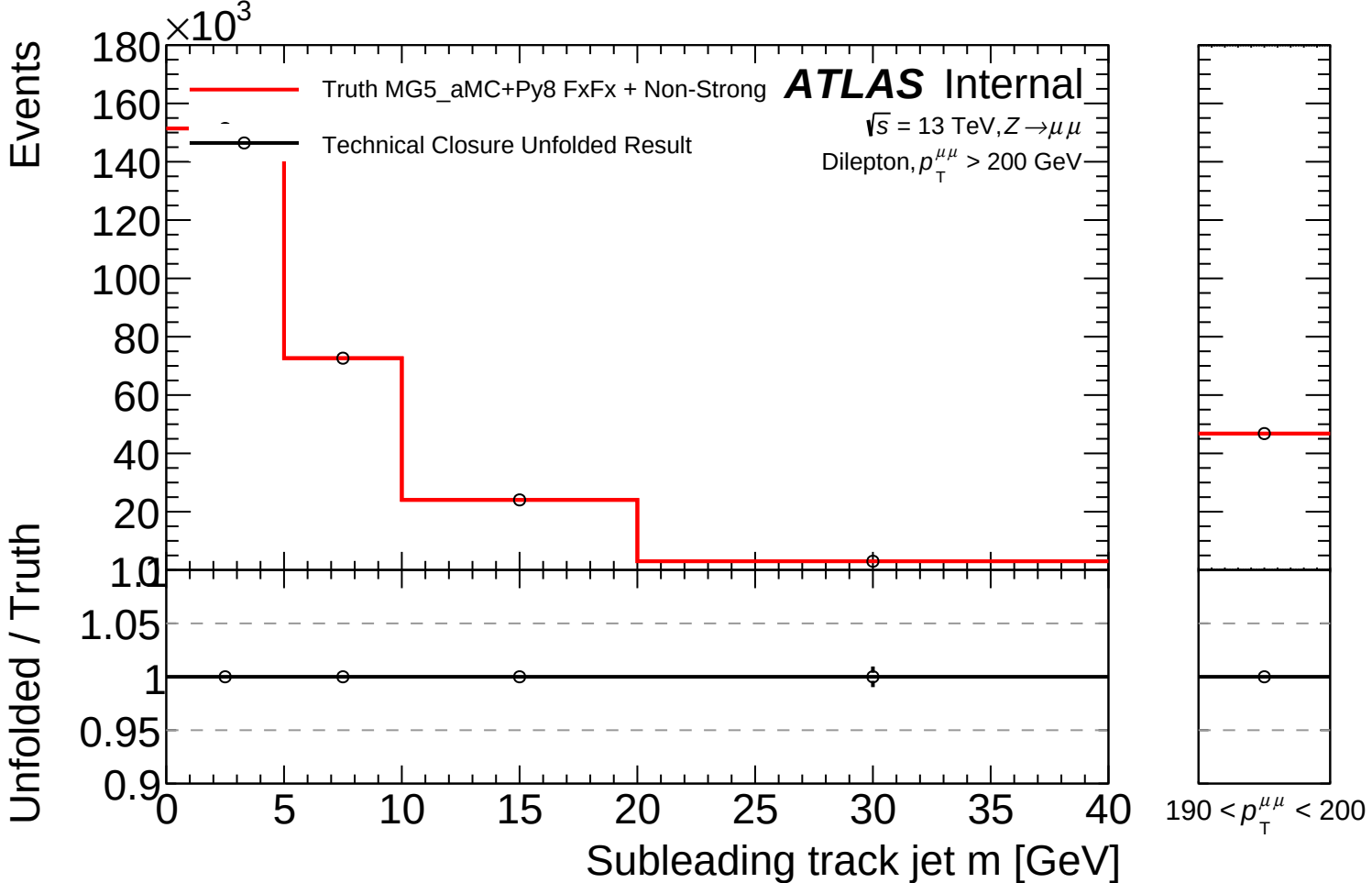
ATLAS Internal

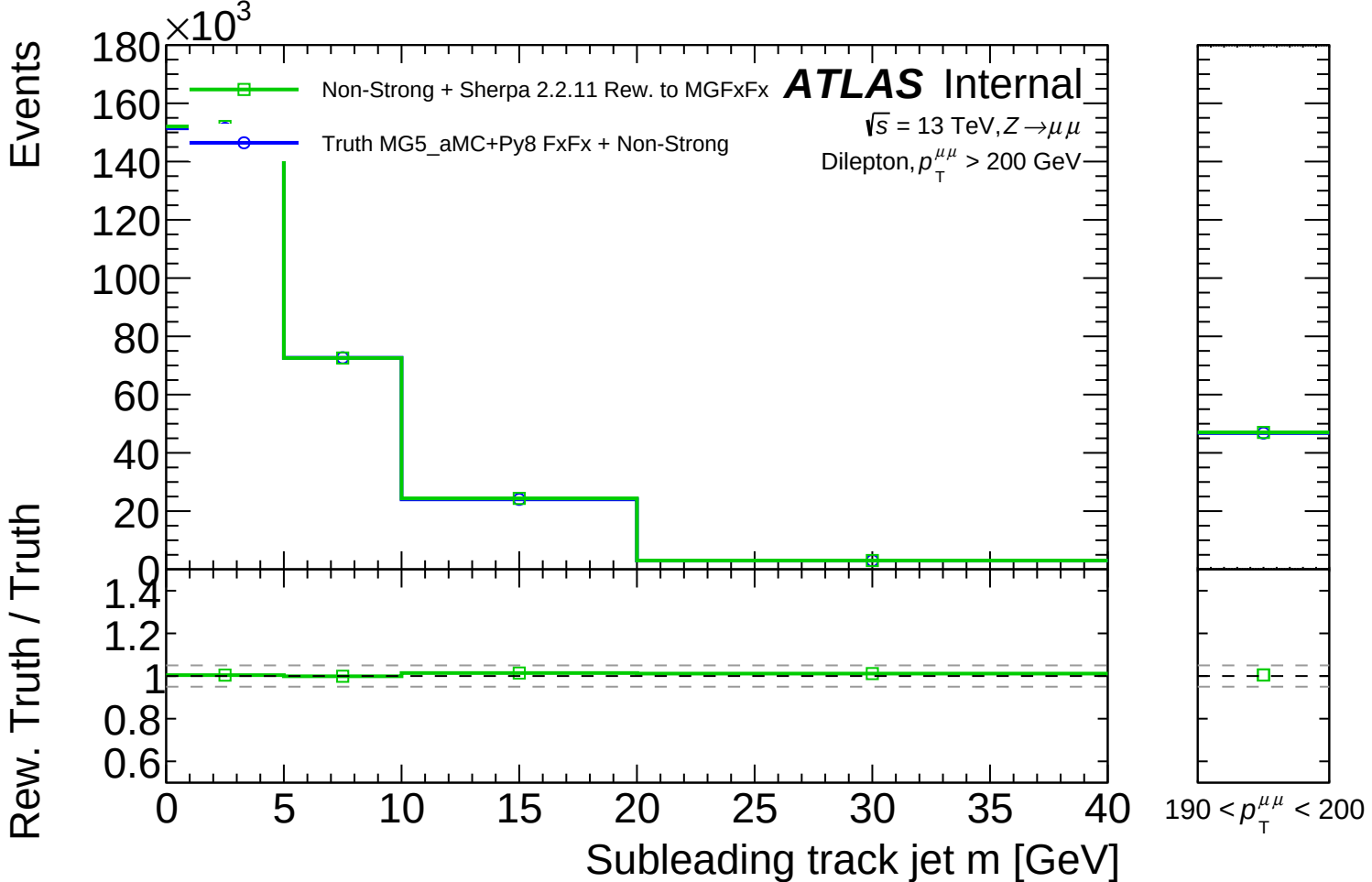
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

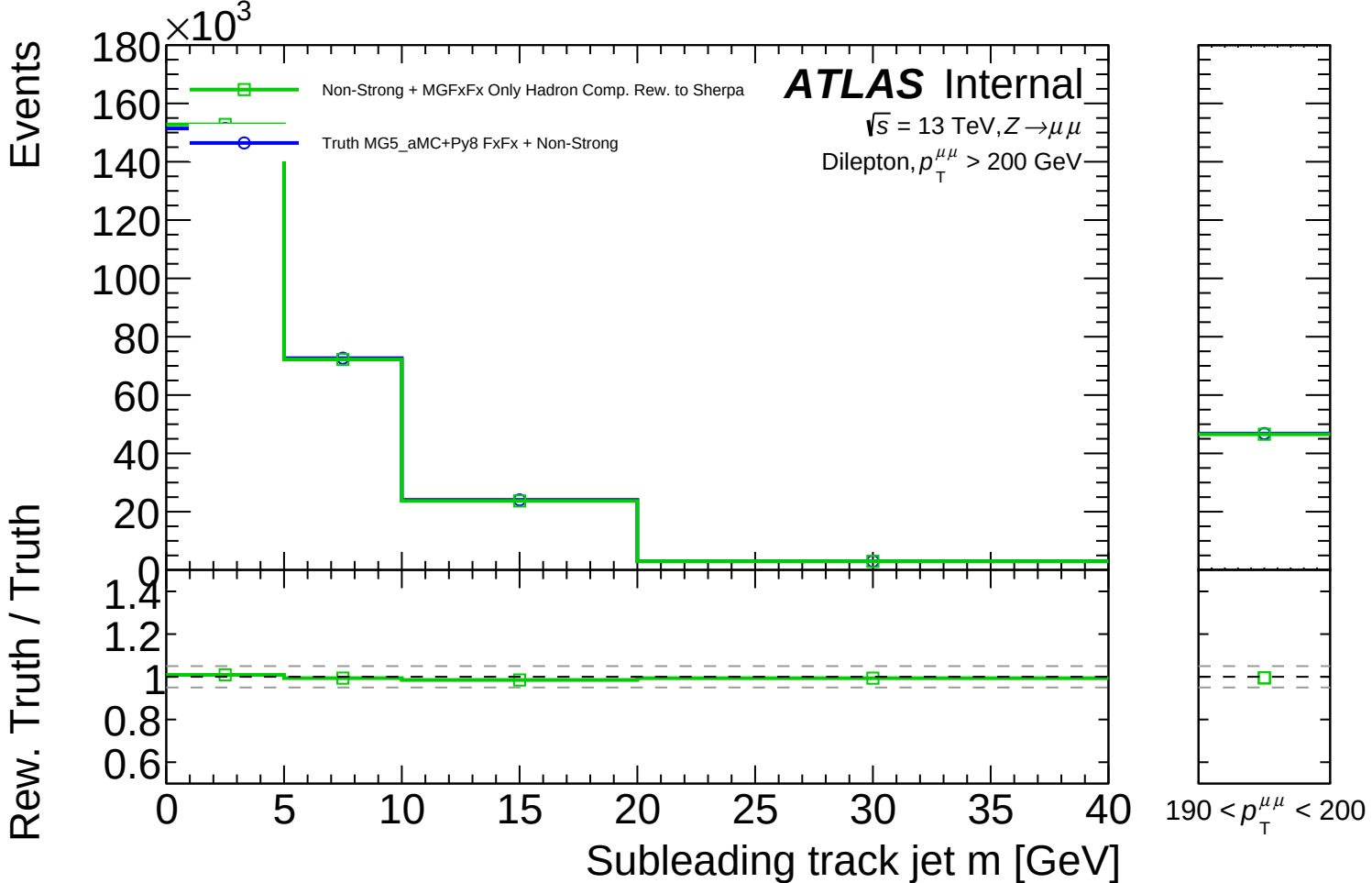
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

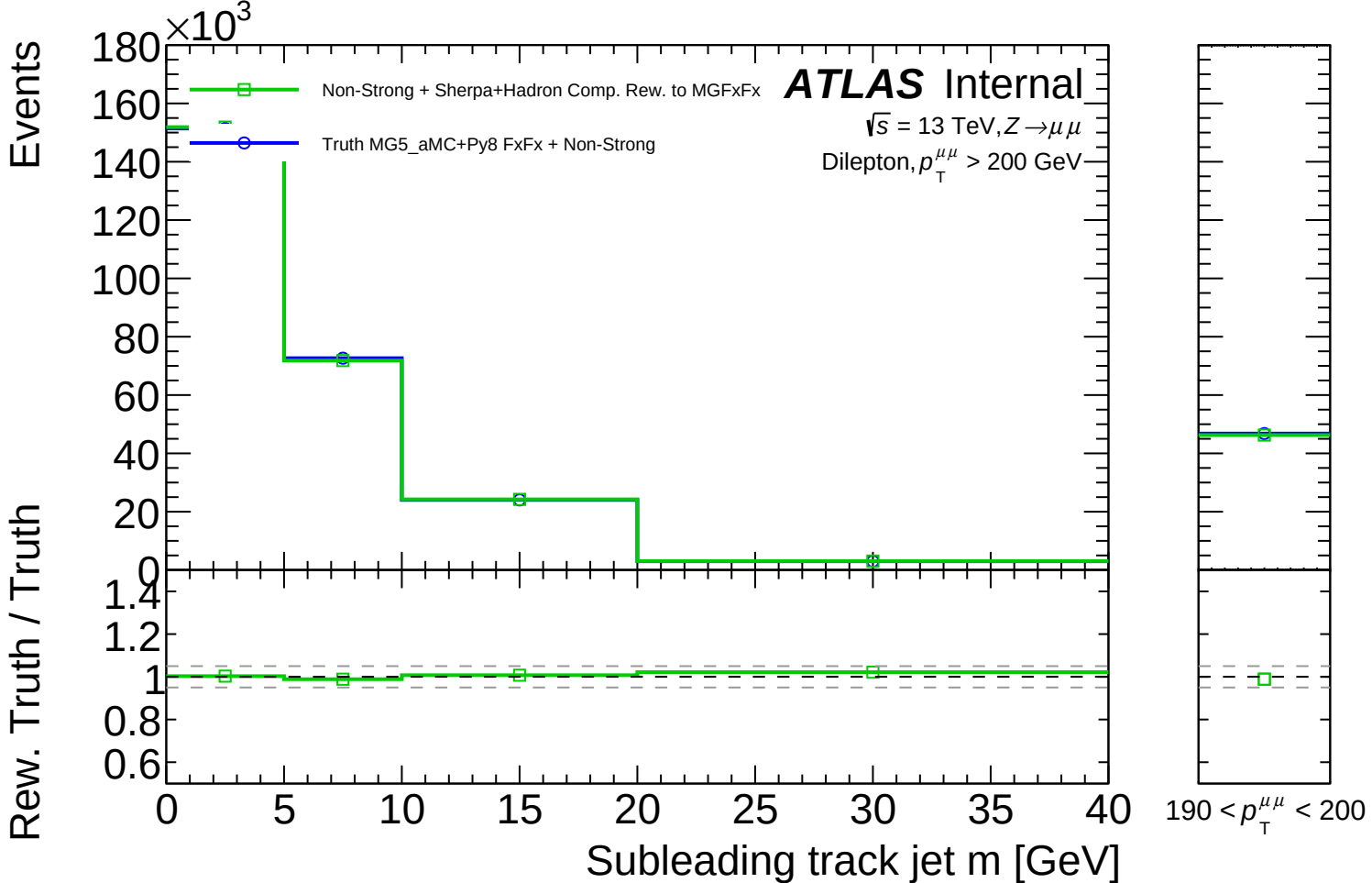
Alternate Top Subtraction Uncertainty

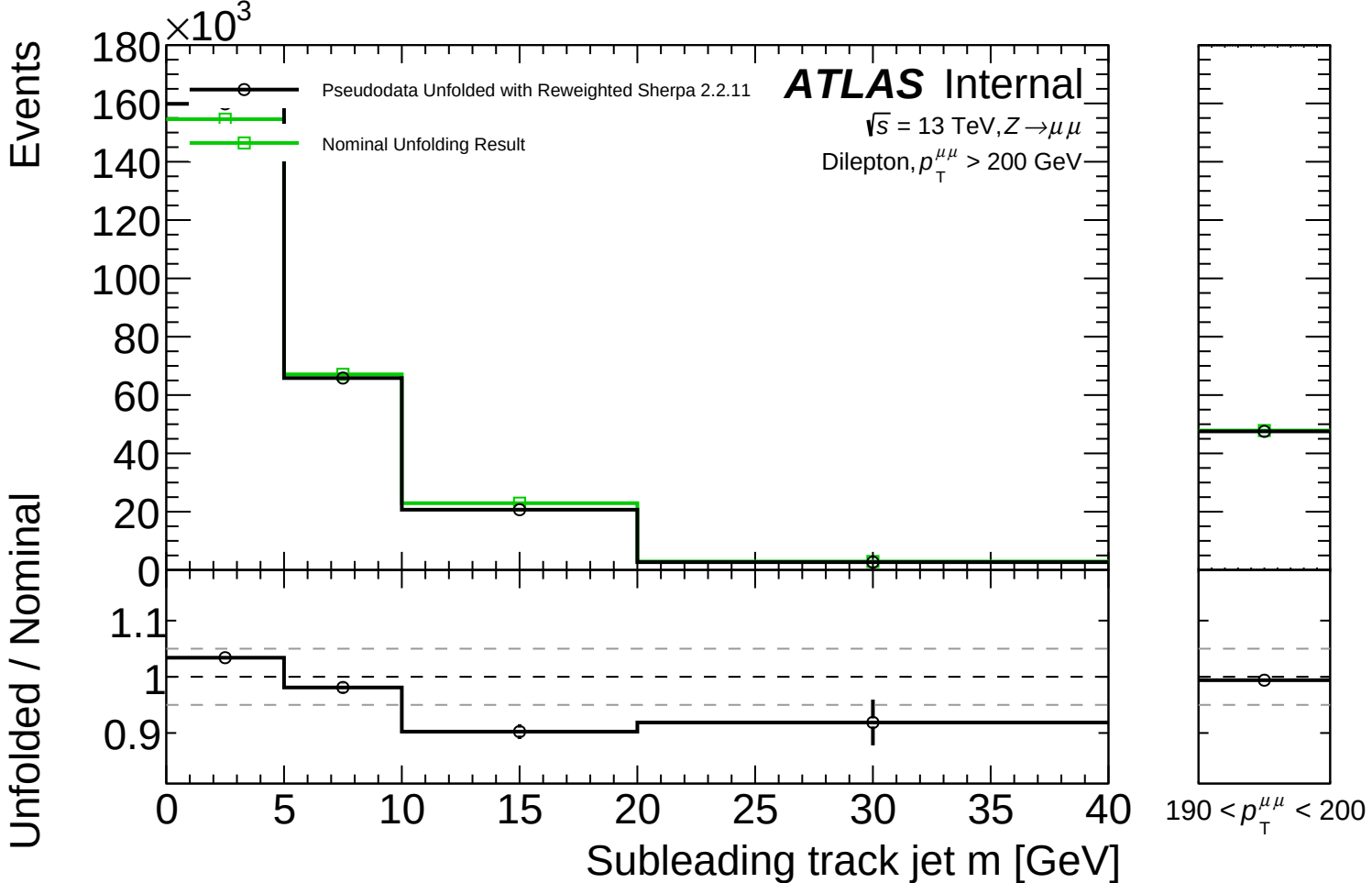


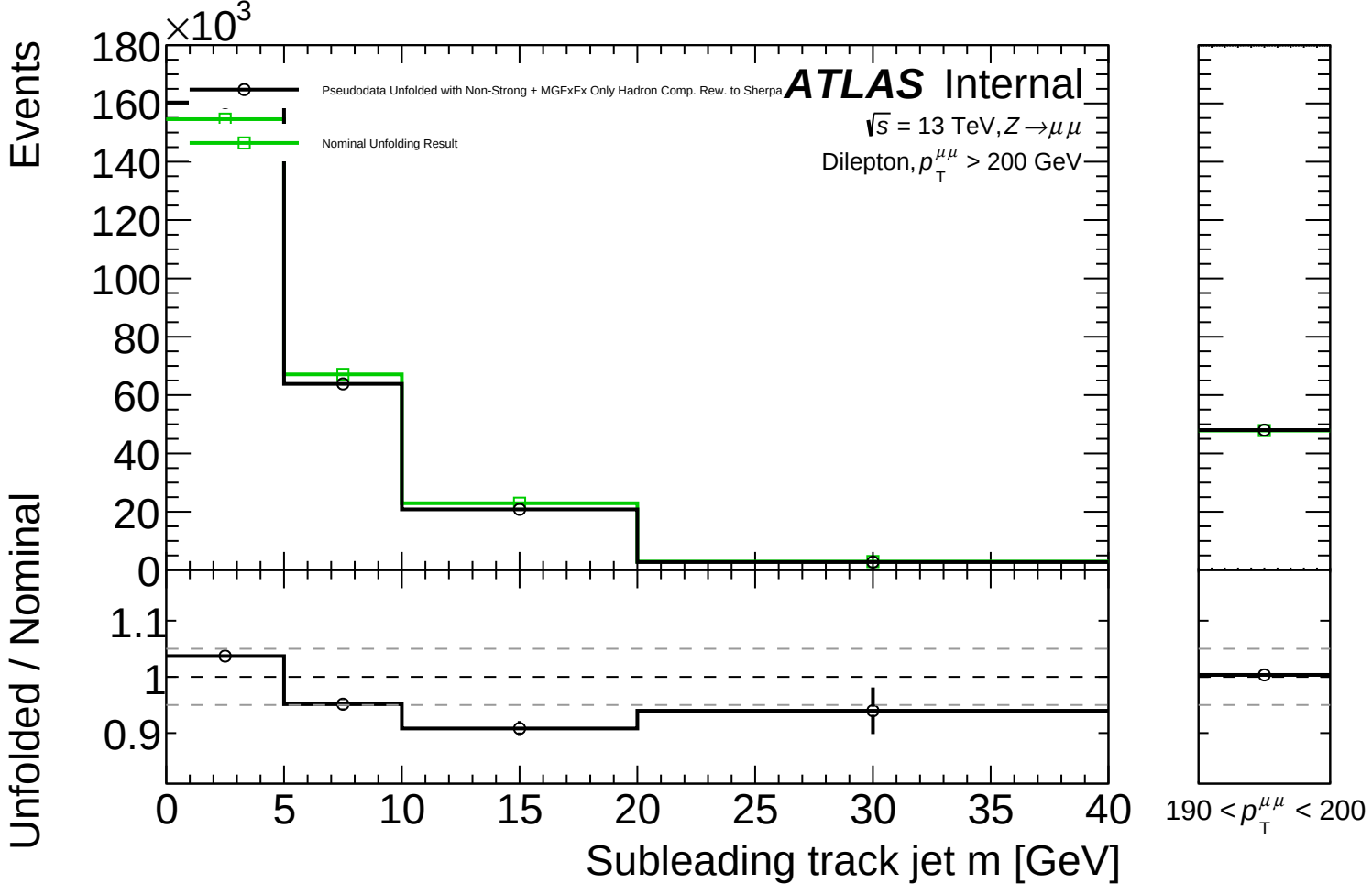


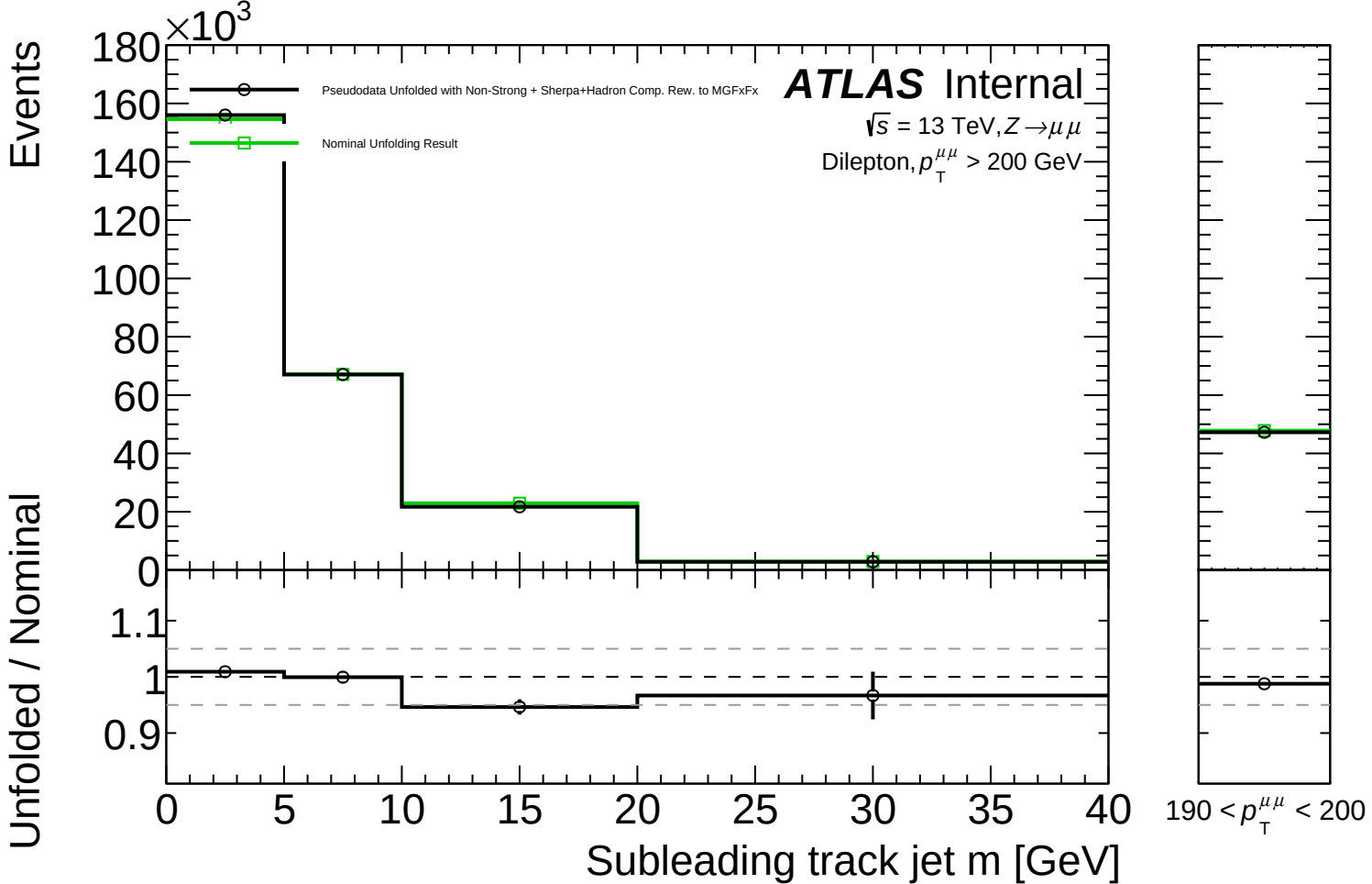


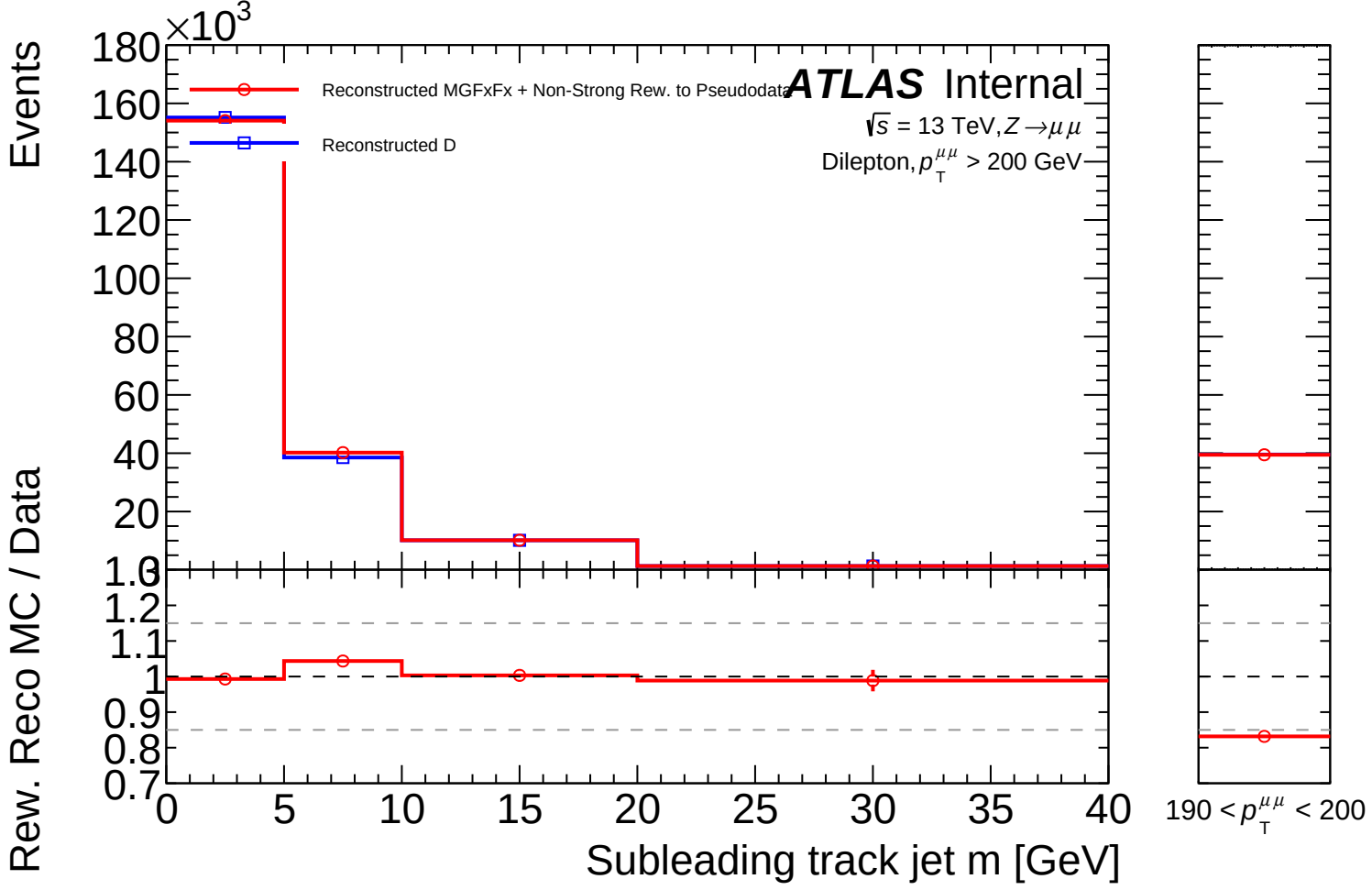


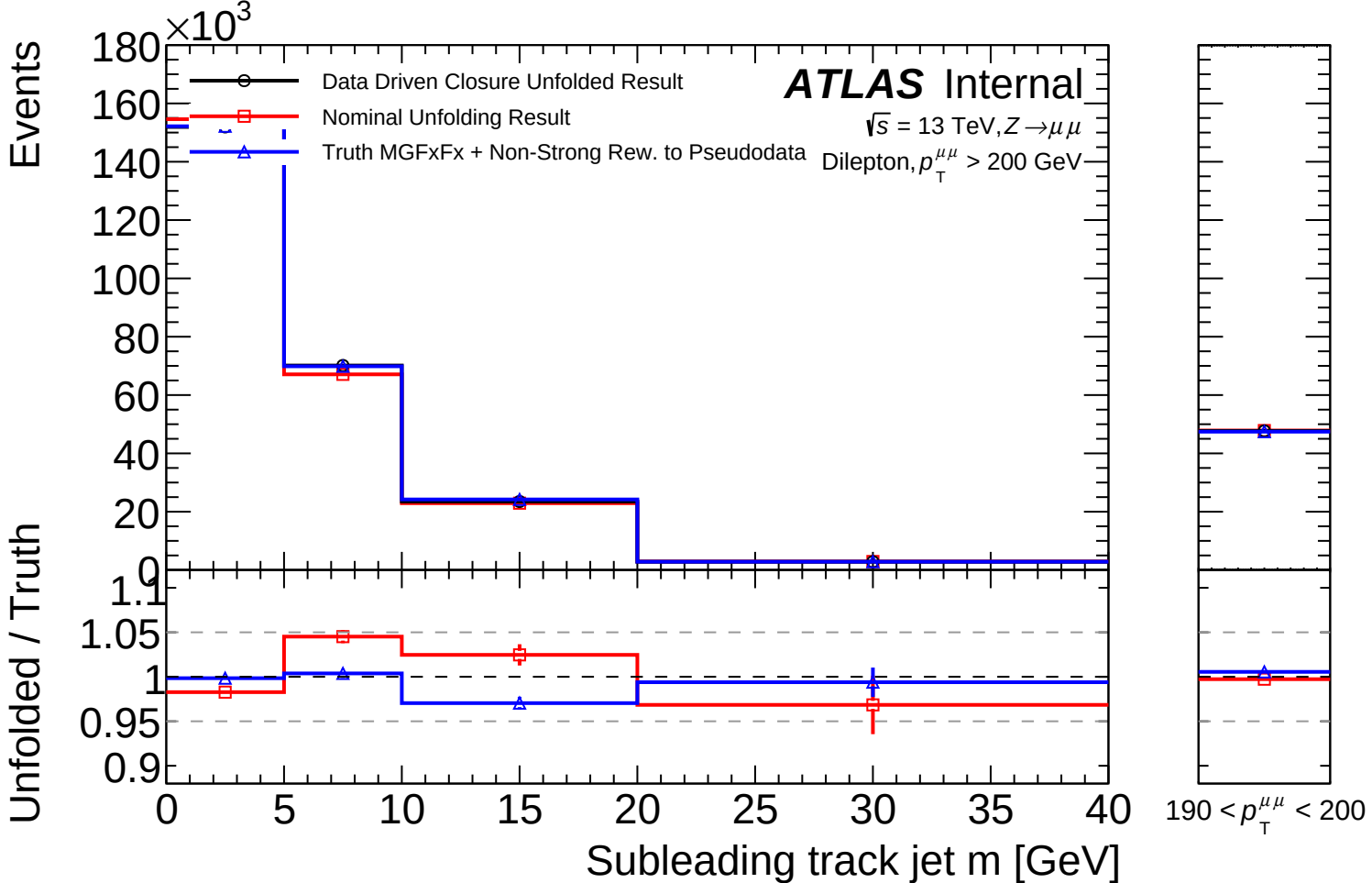




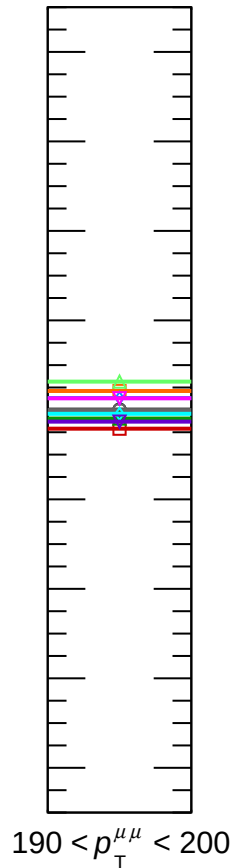
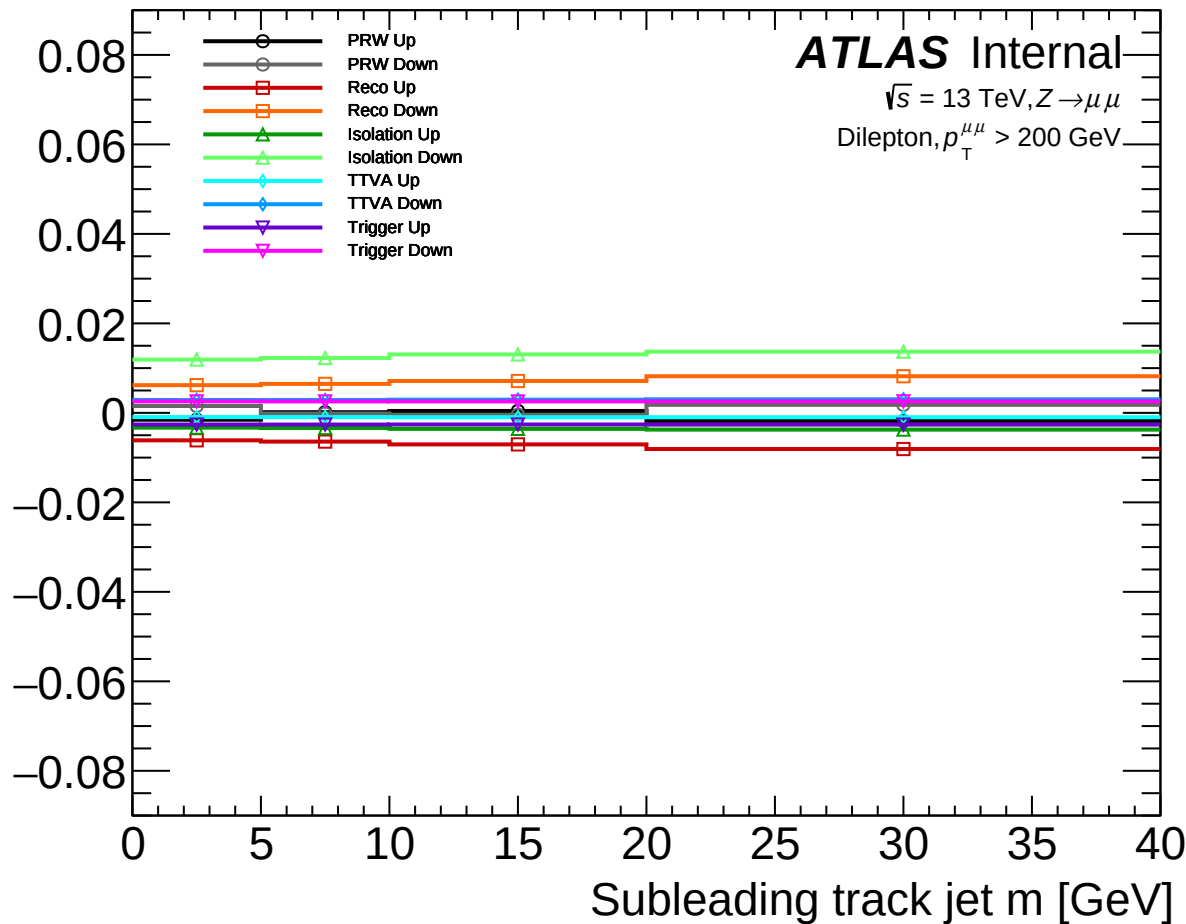




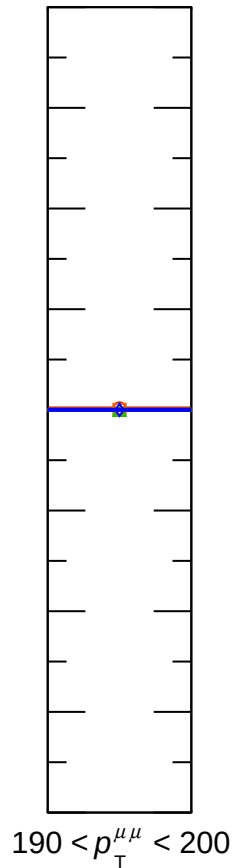
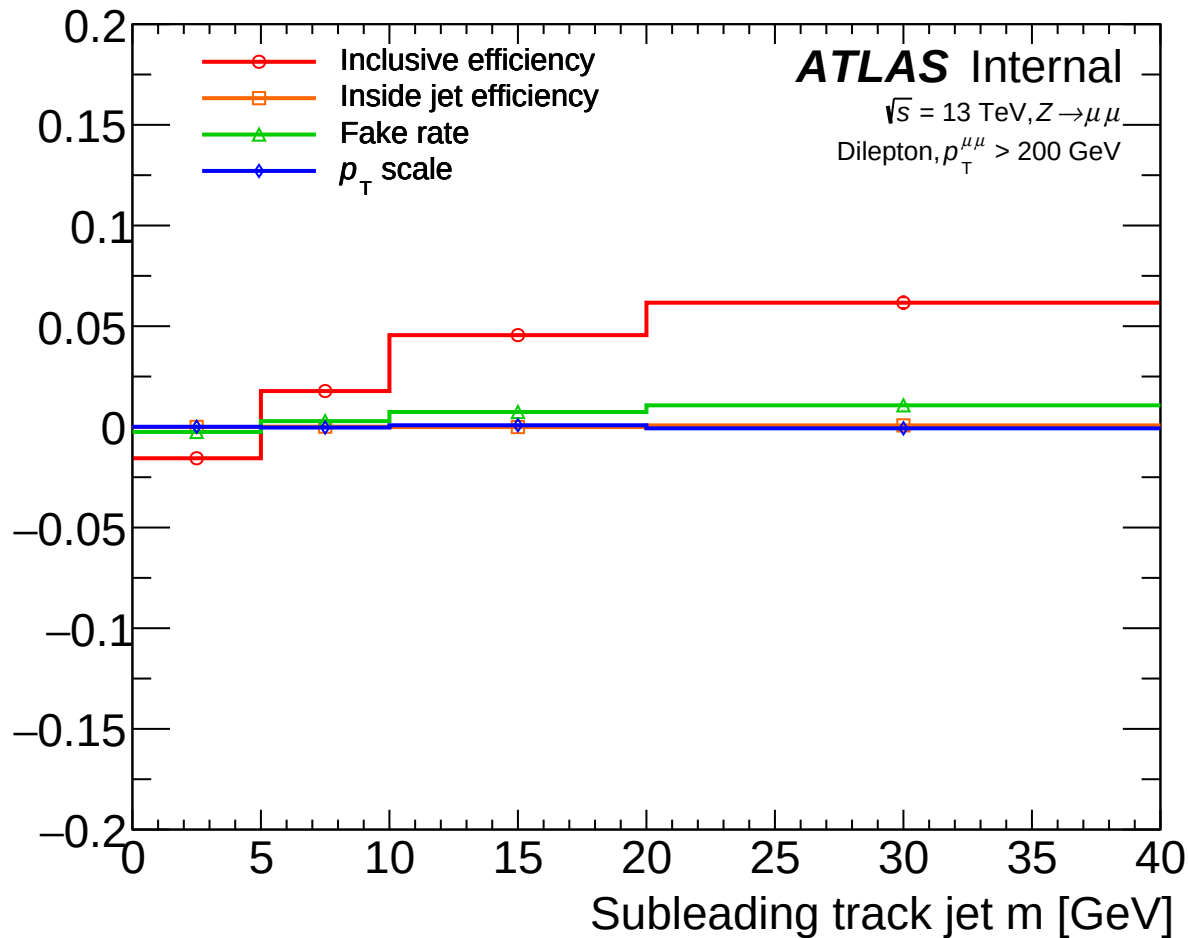




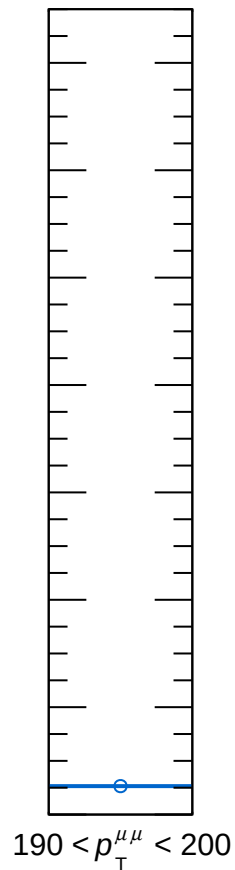
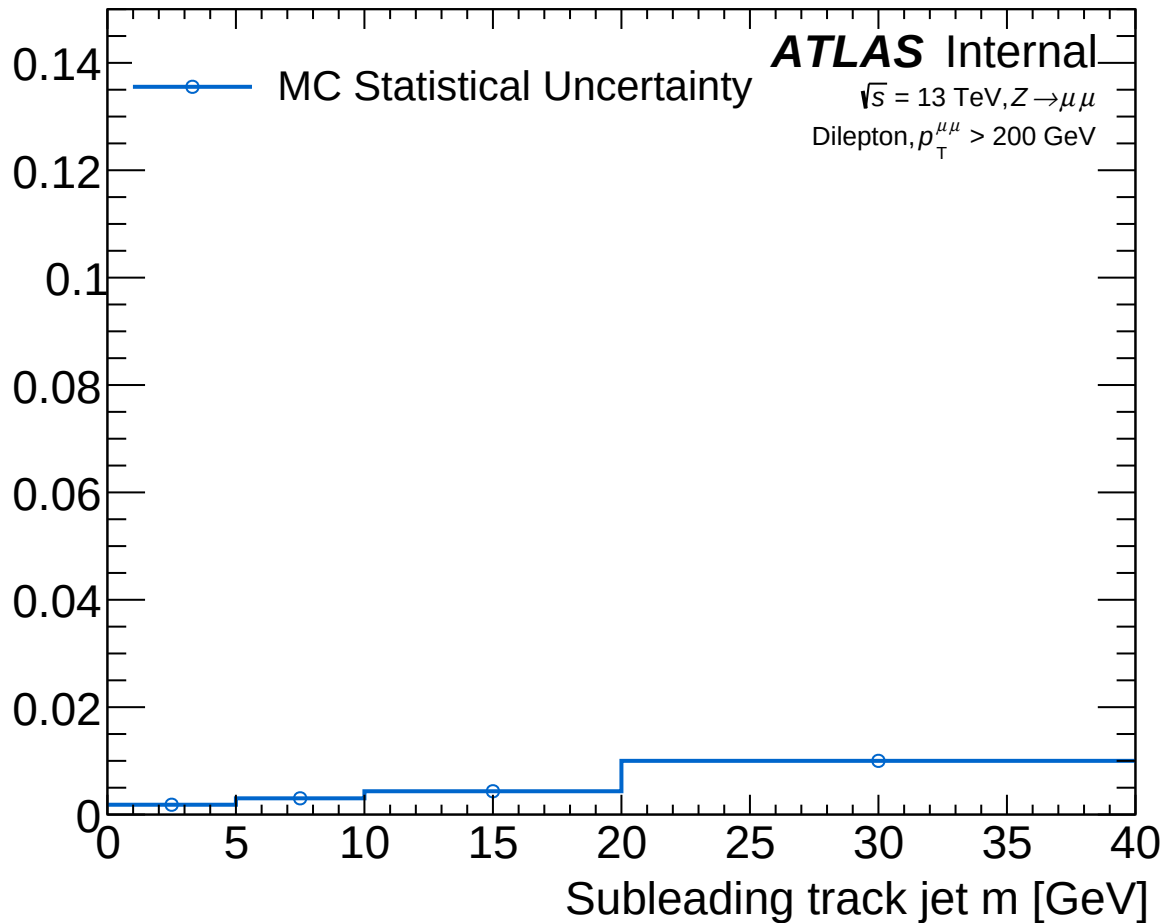
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



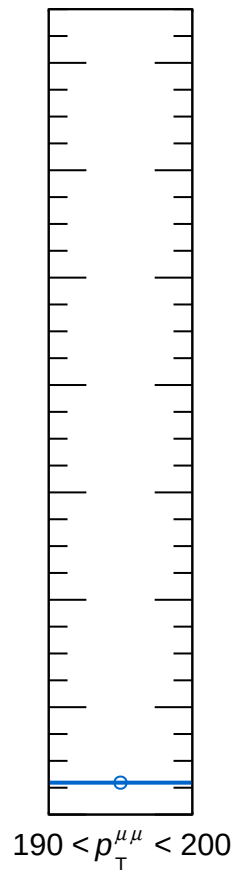
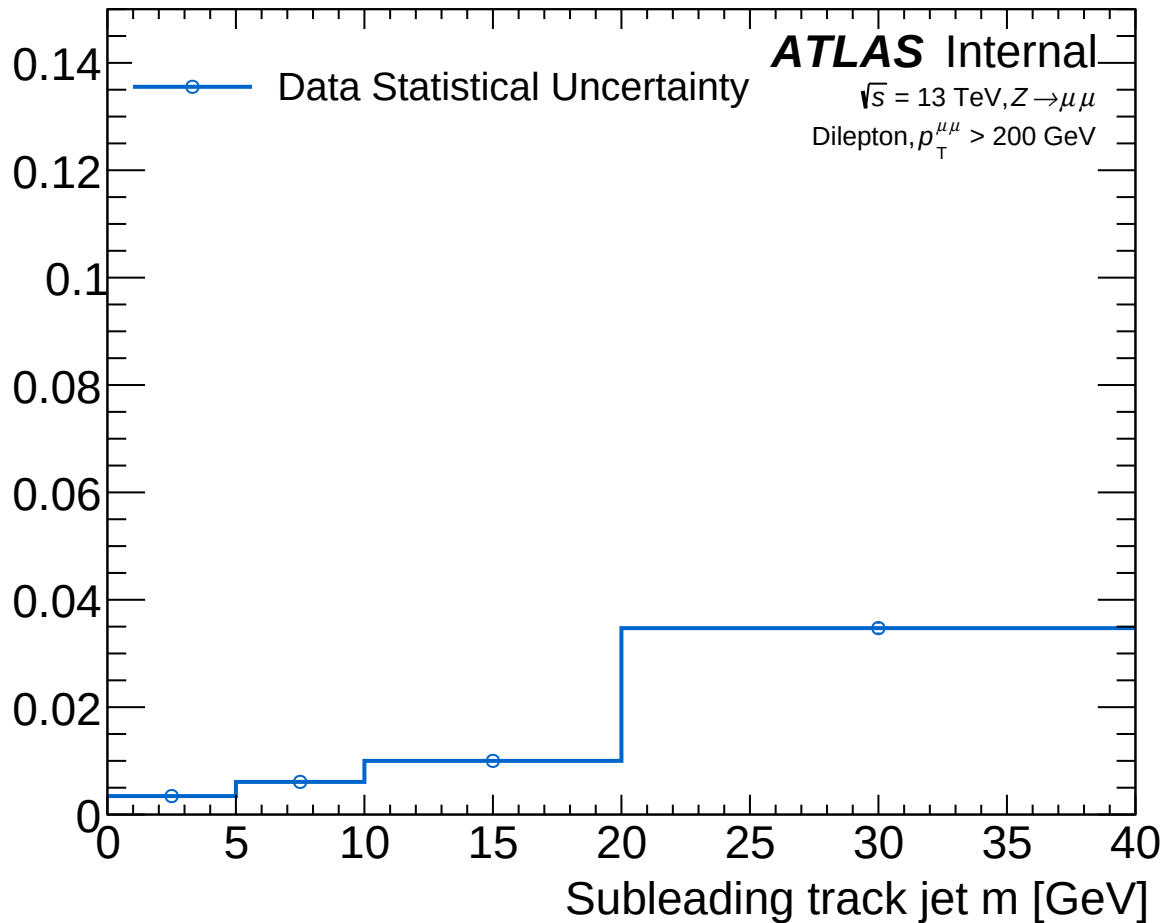
Fractional systematic impact

ATLAS Internal

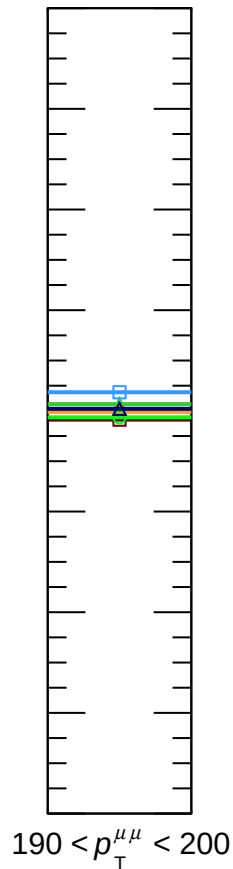
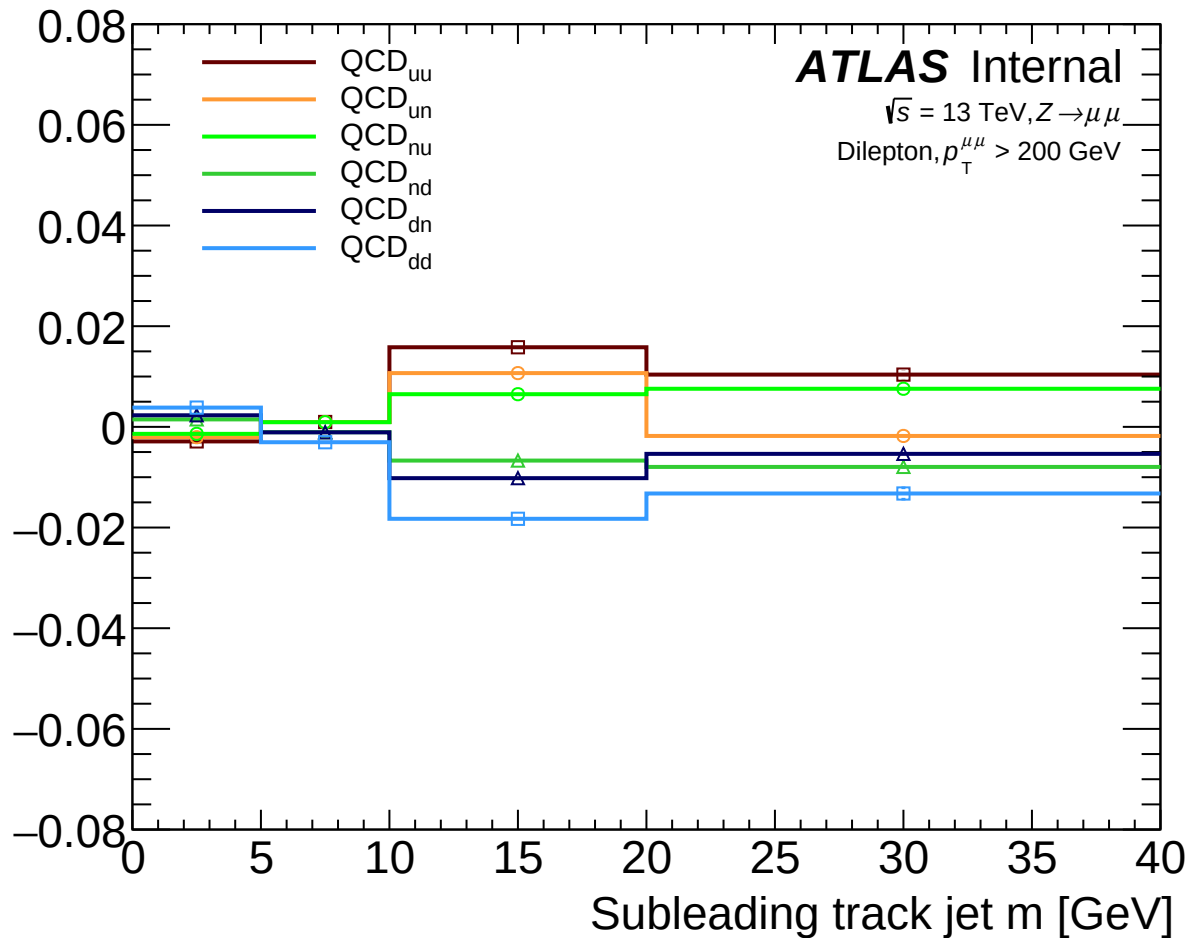
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

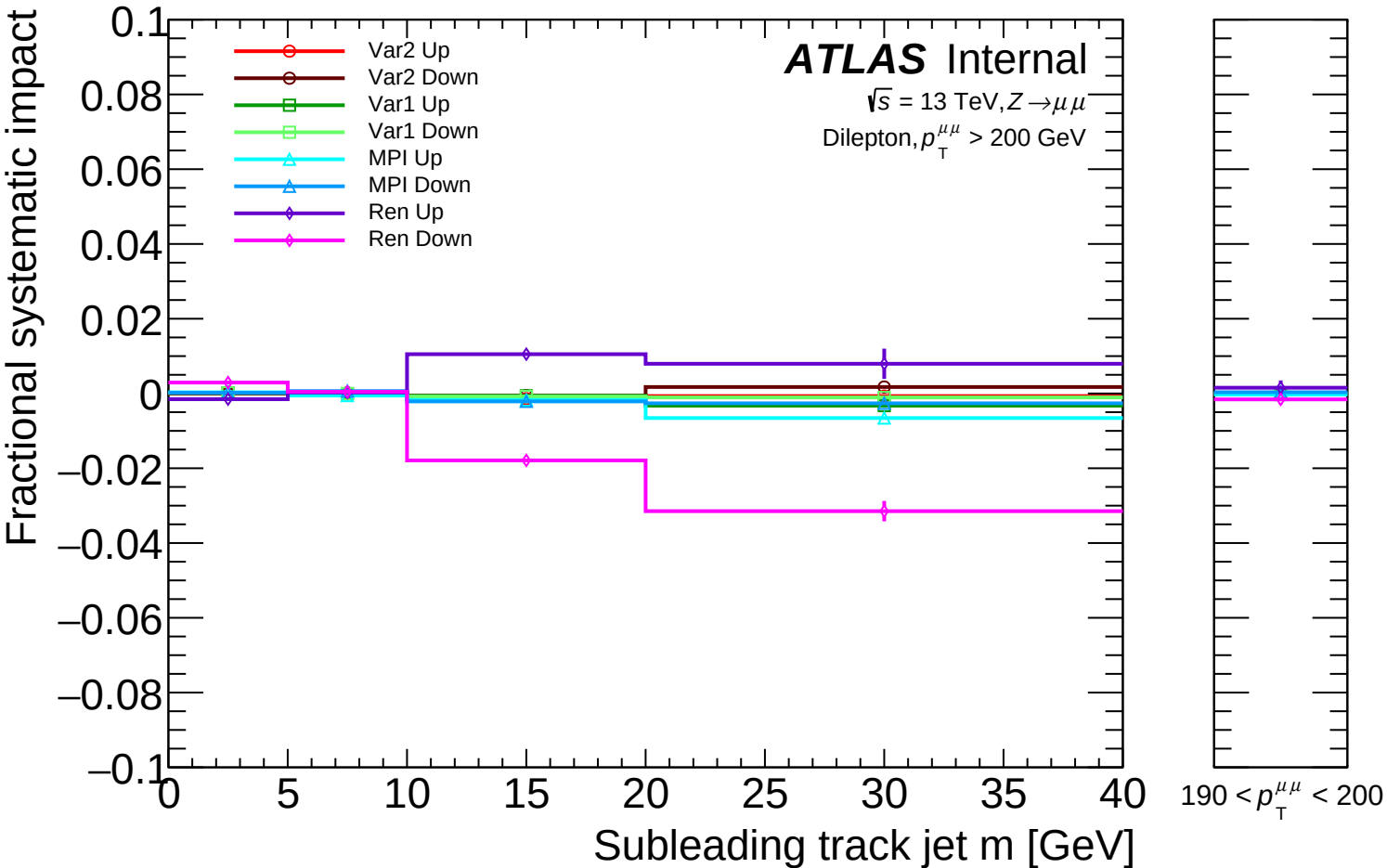
Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

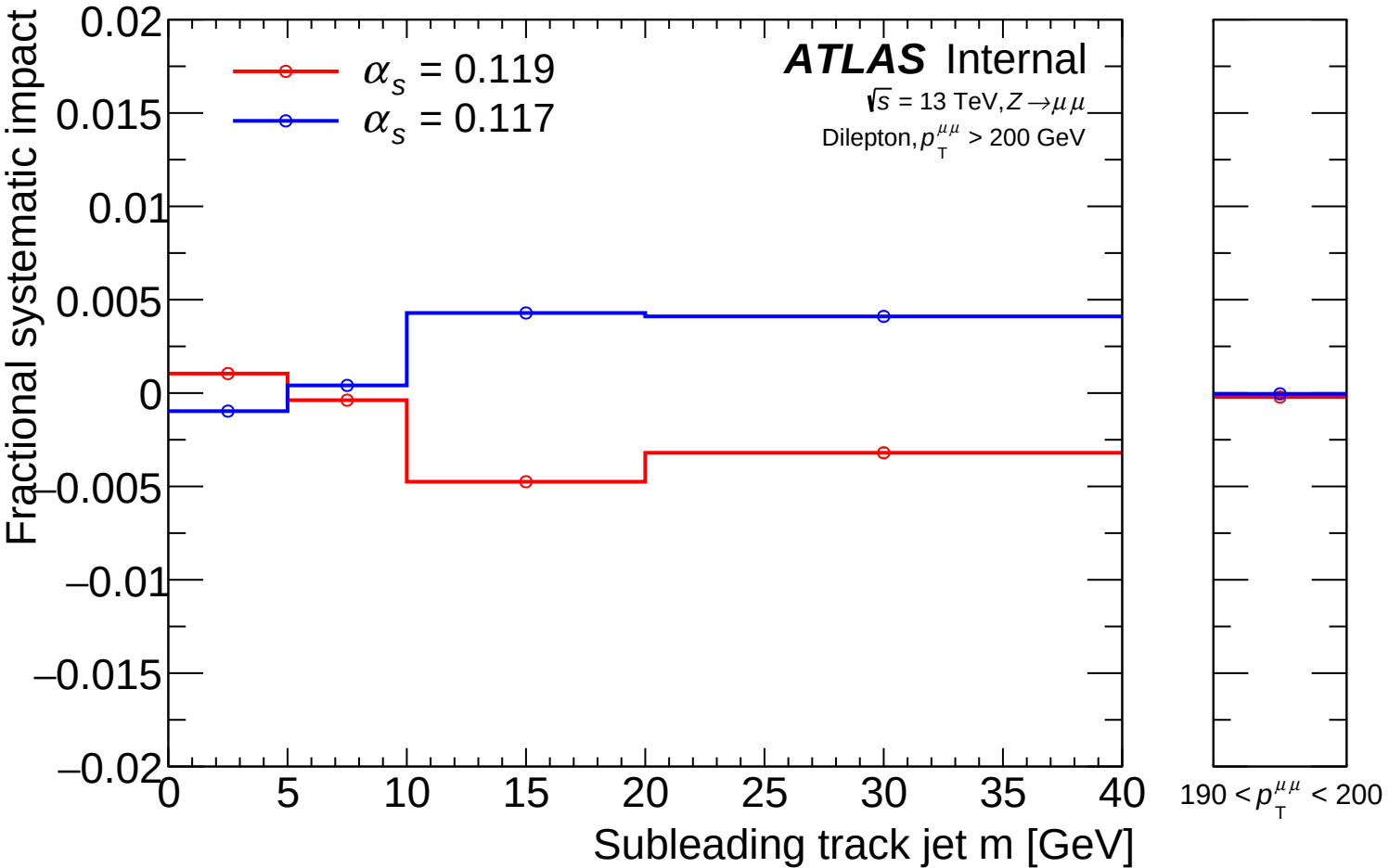
Data Statistical Uncertainty

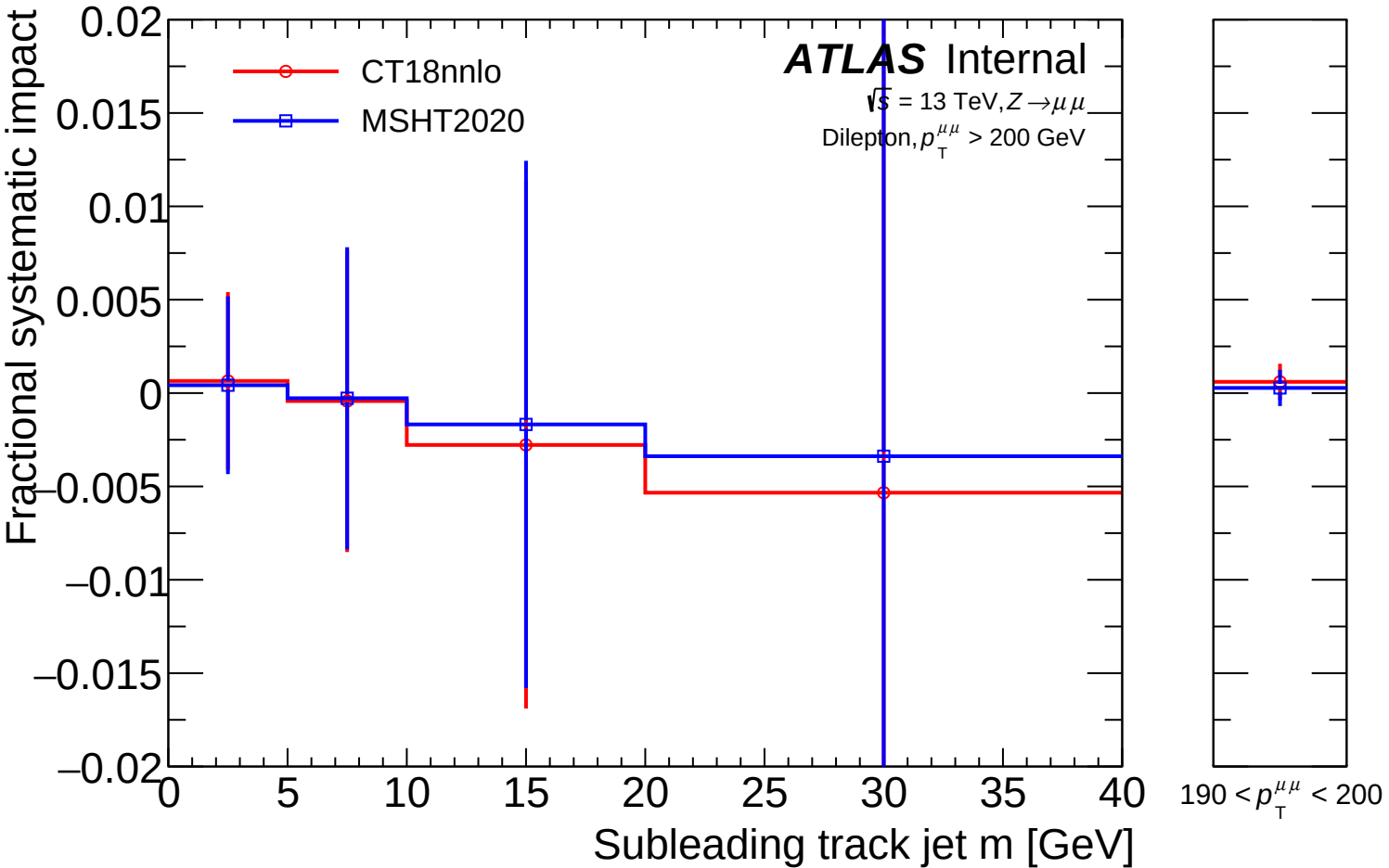


Fractional systematic impact









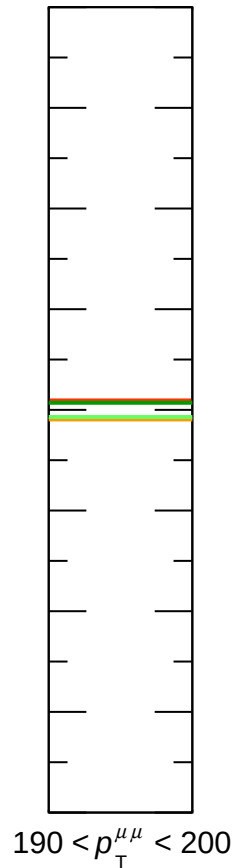
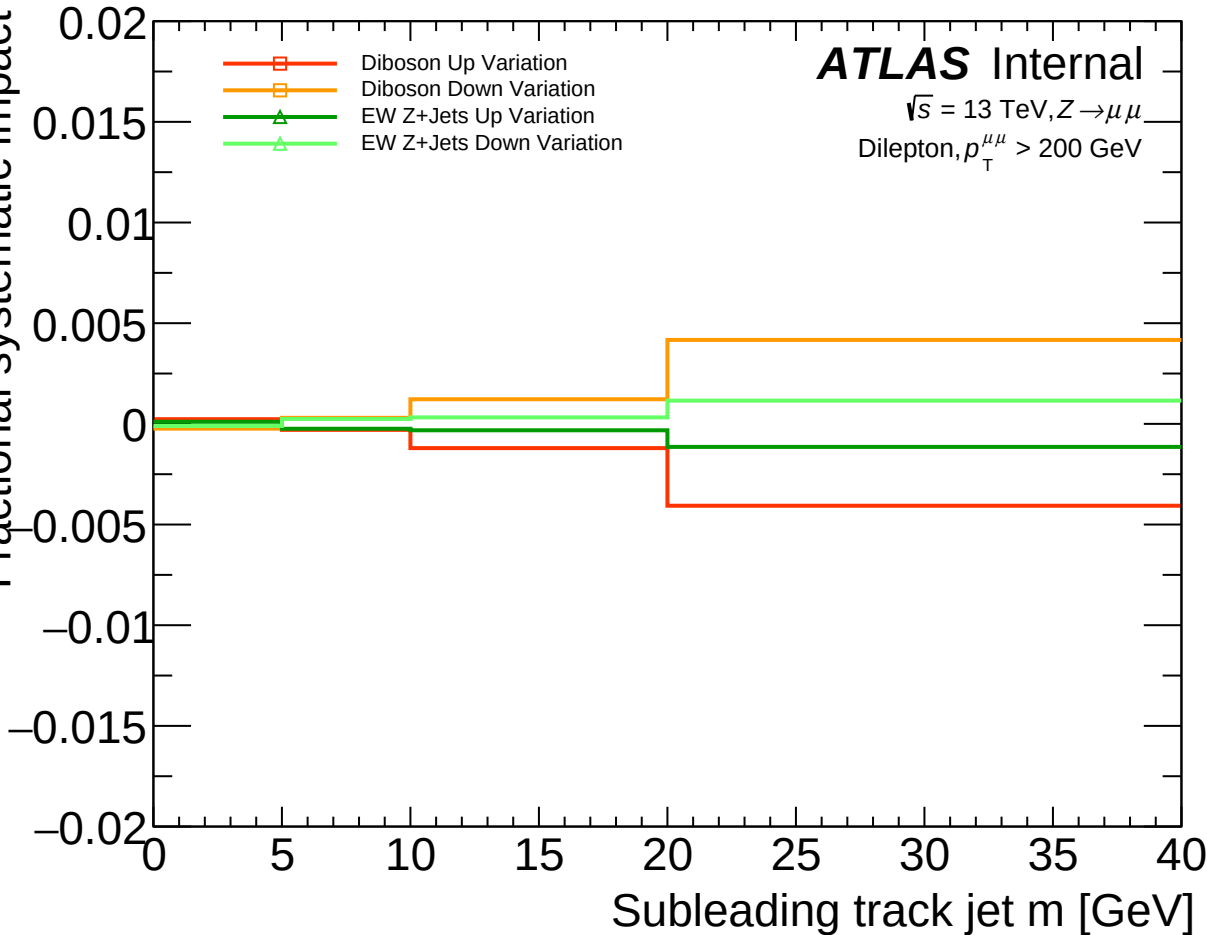
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

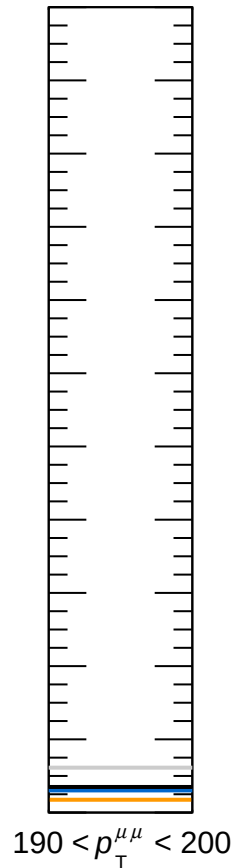
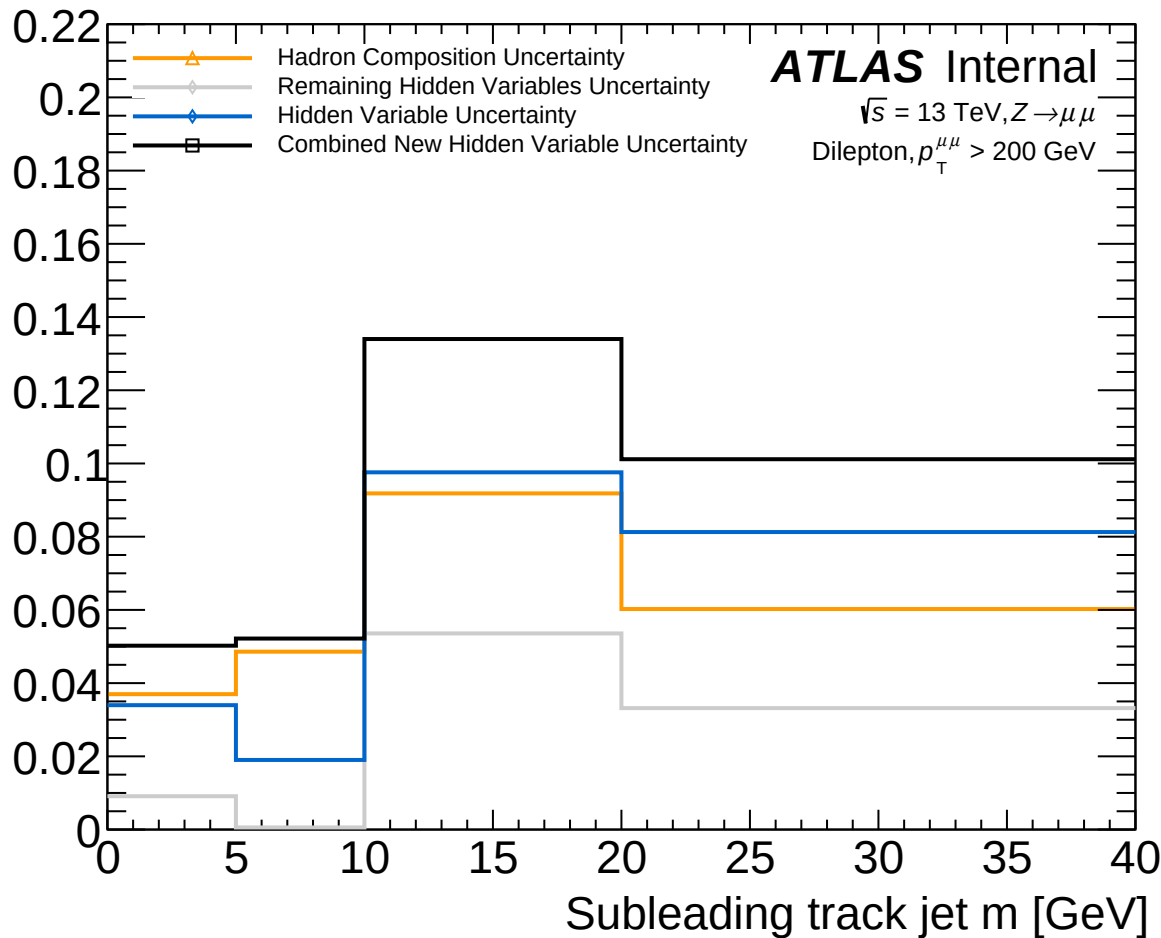
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

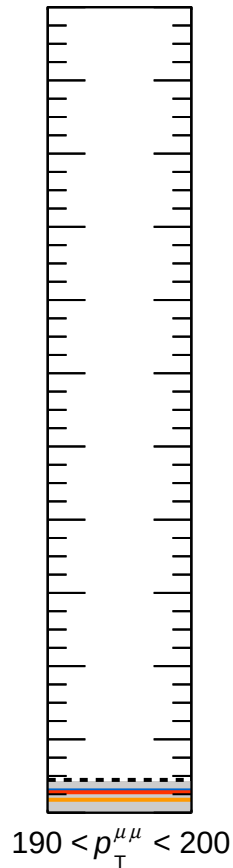
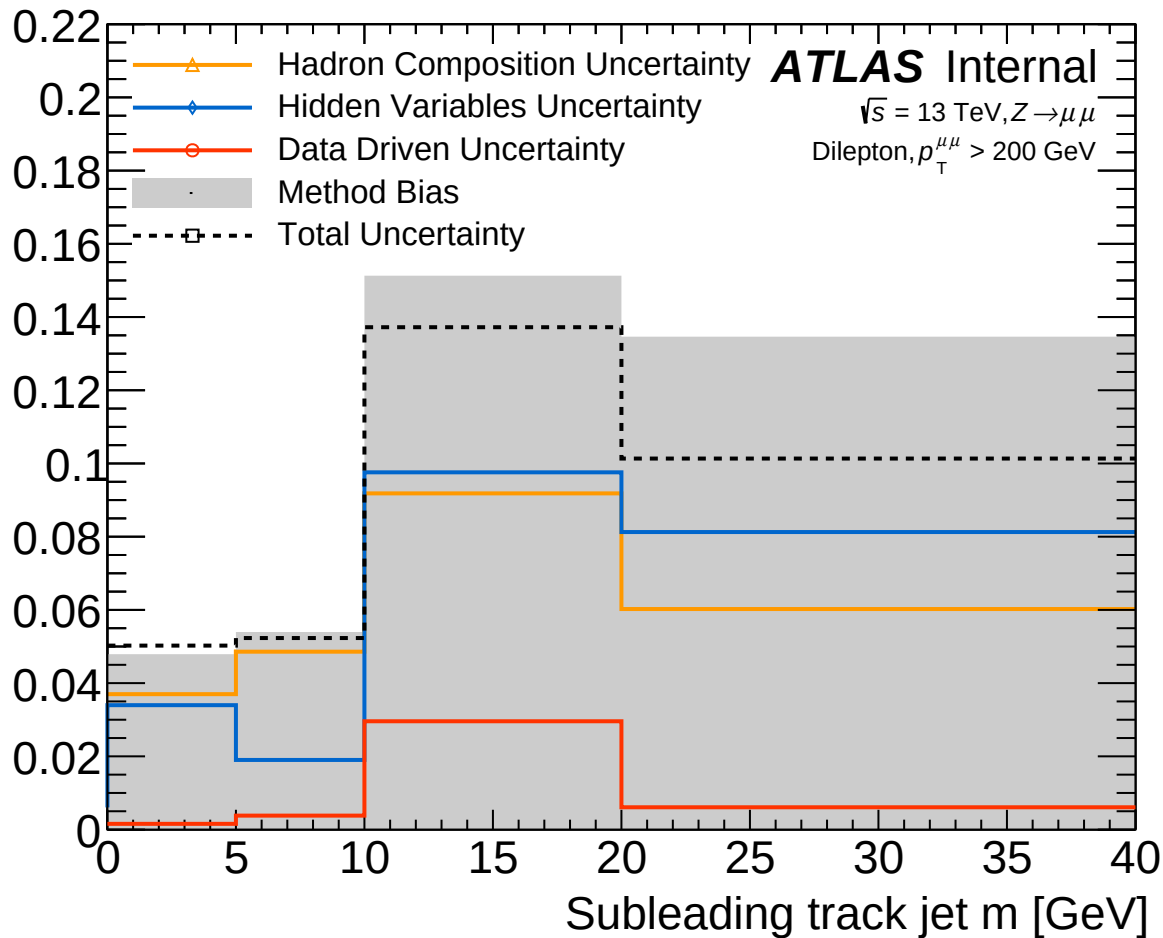


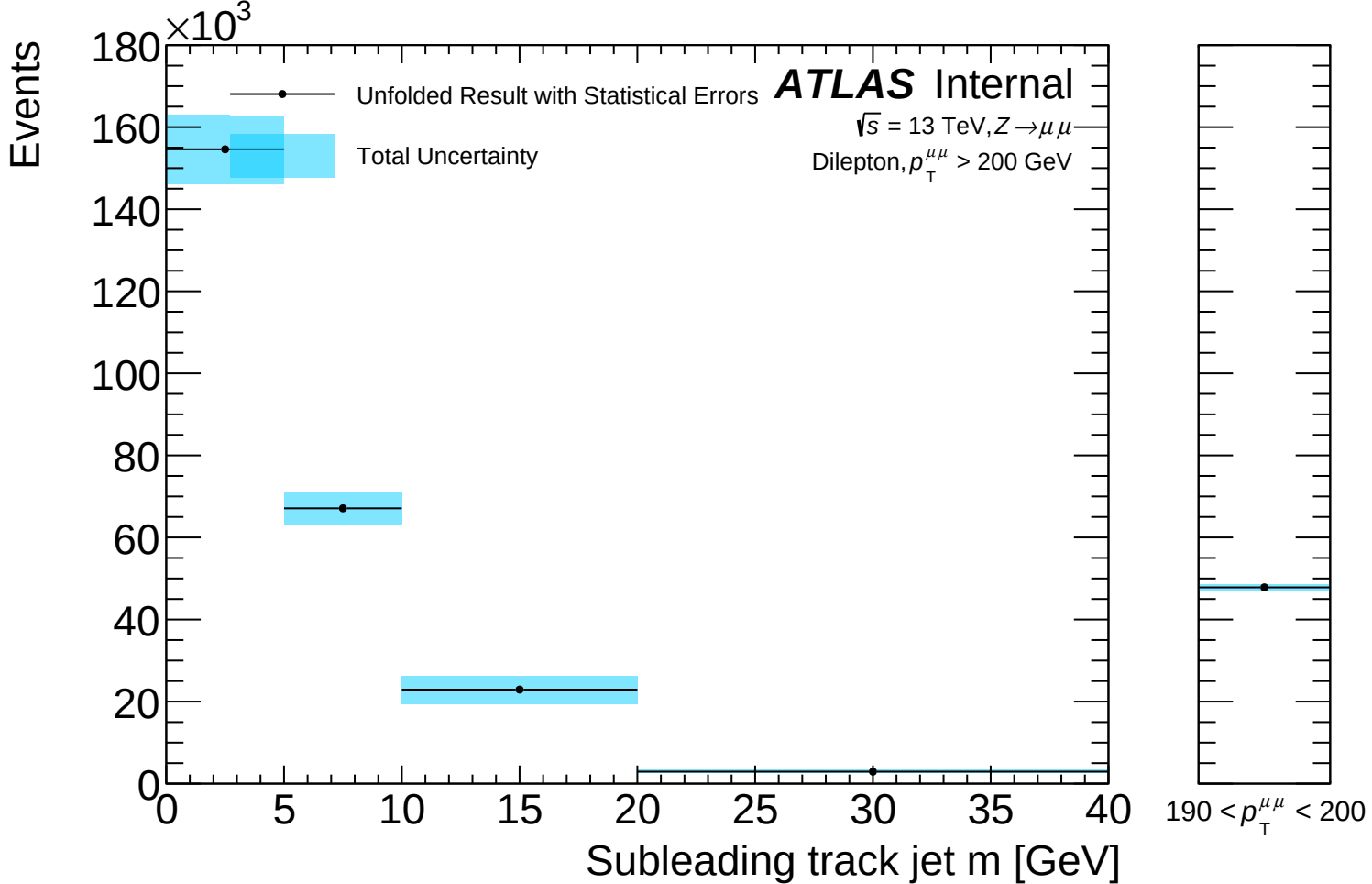
ATLAS Internal

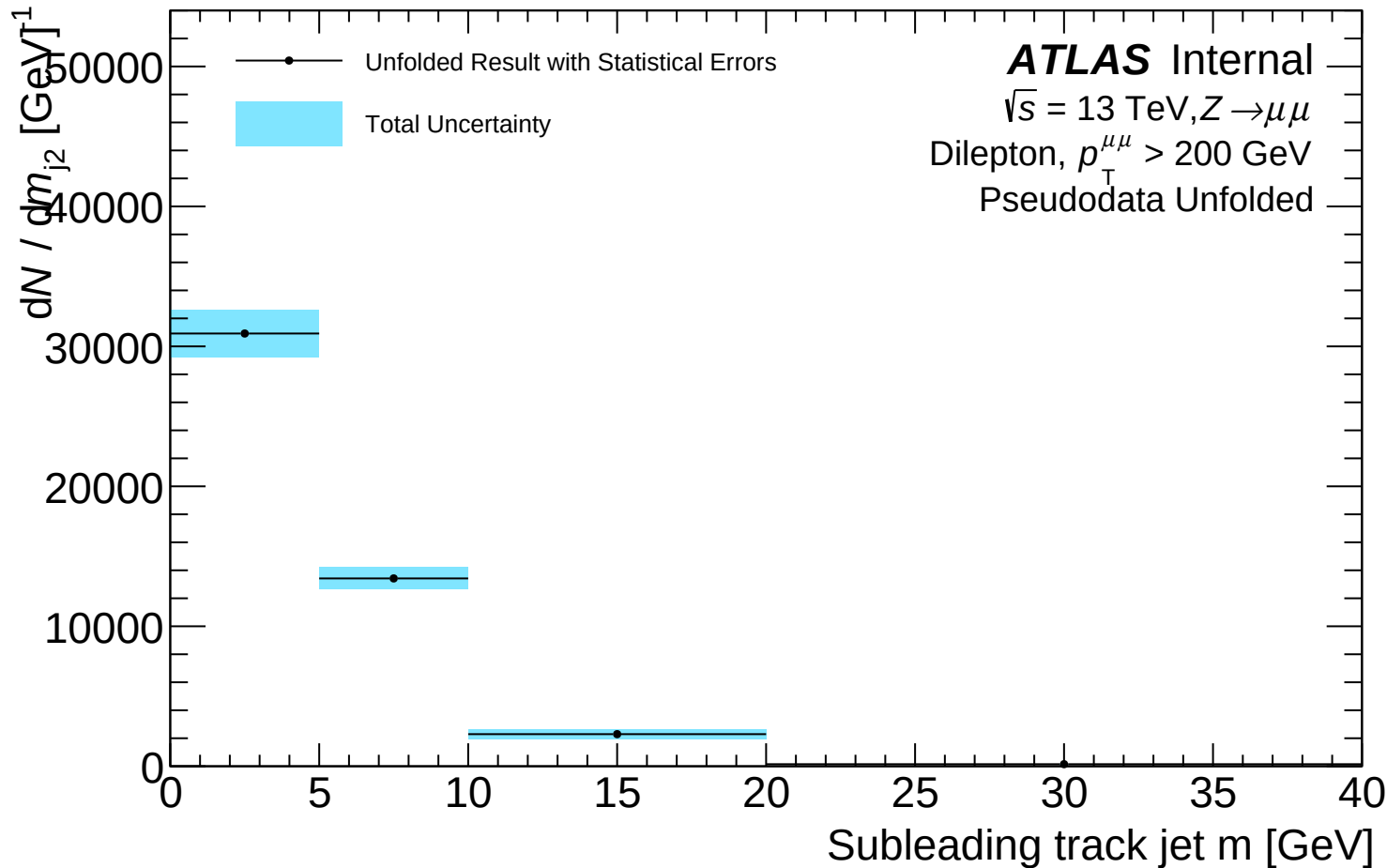
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

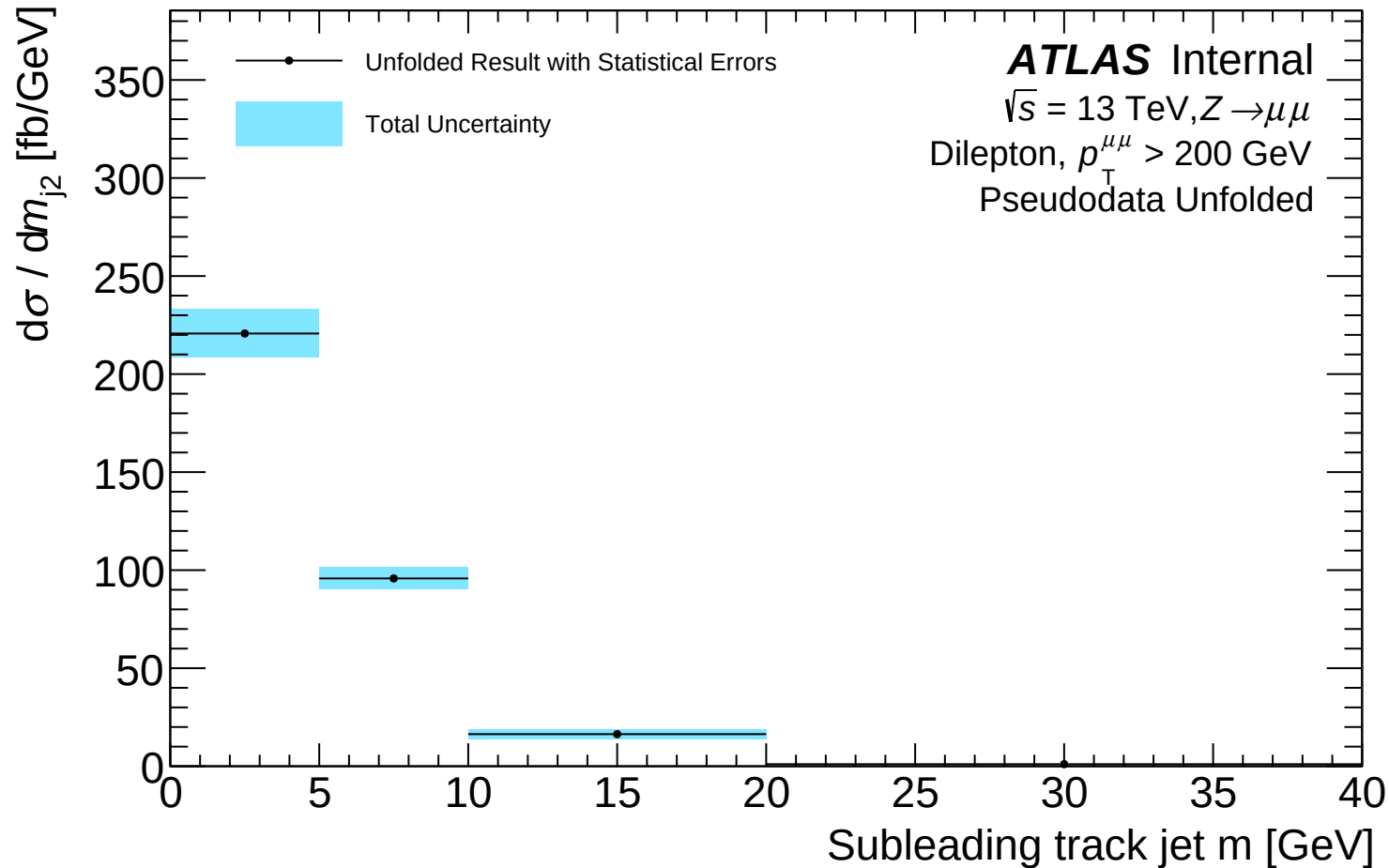
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$











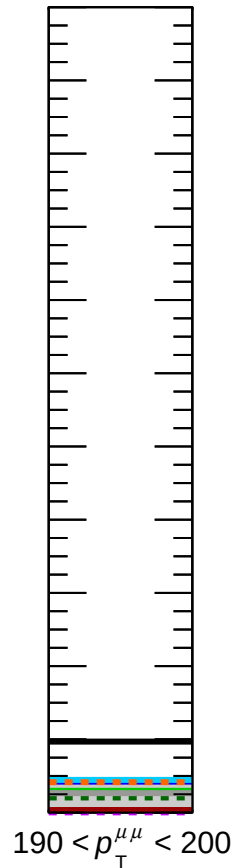
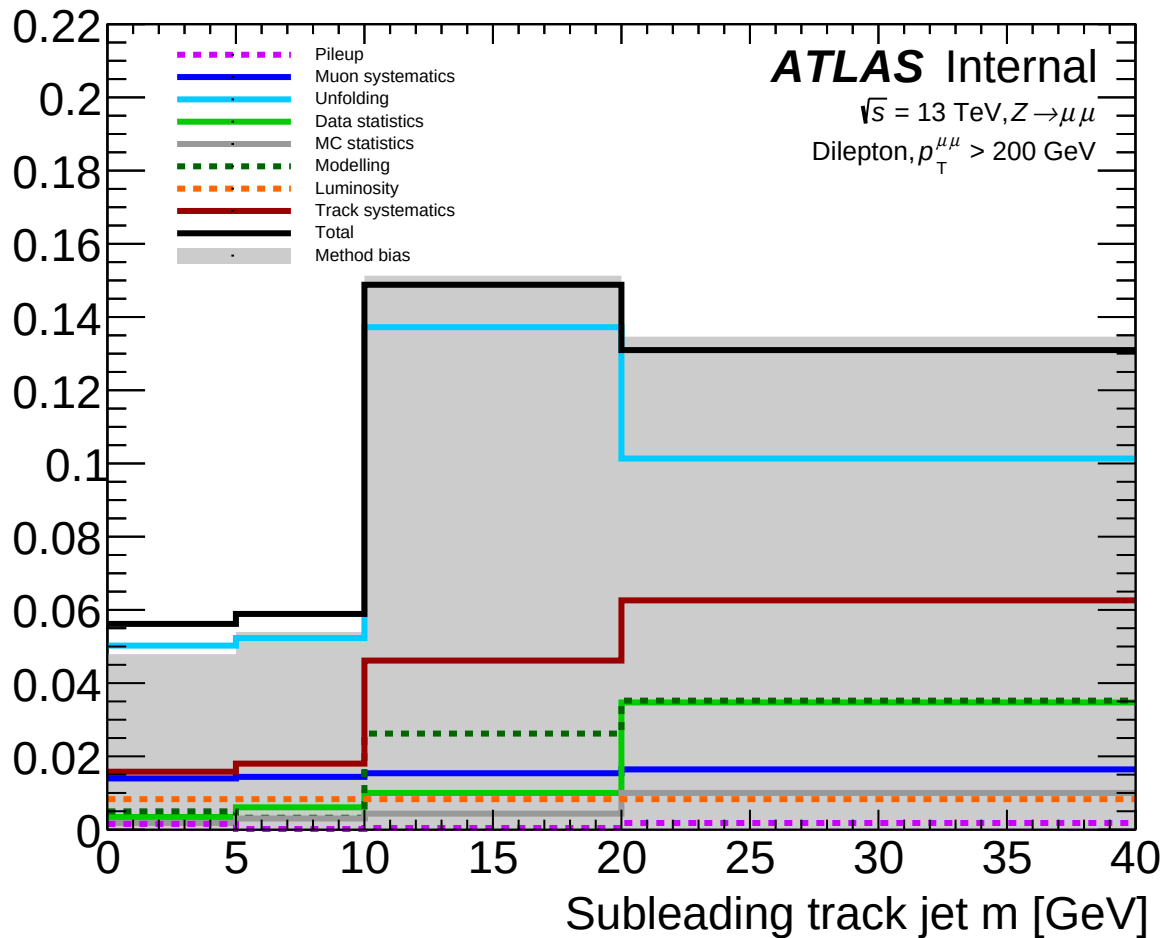
Fractional uncertainty

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

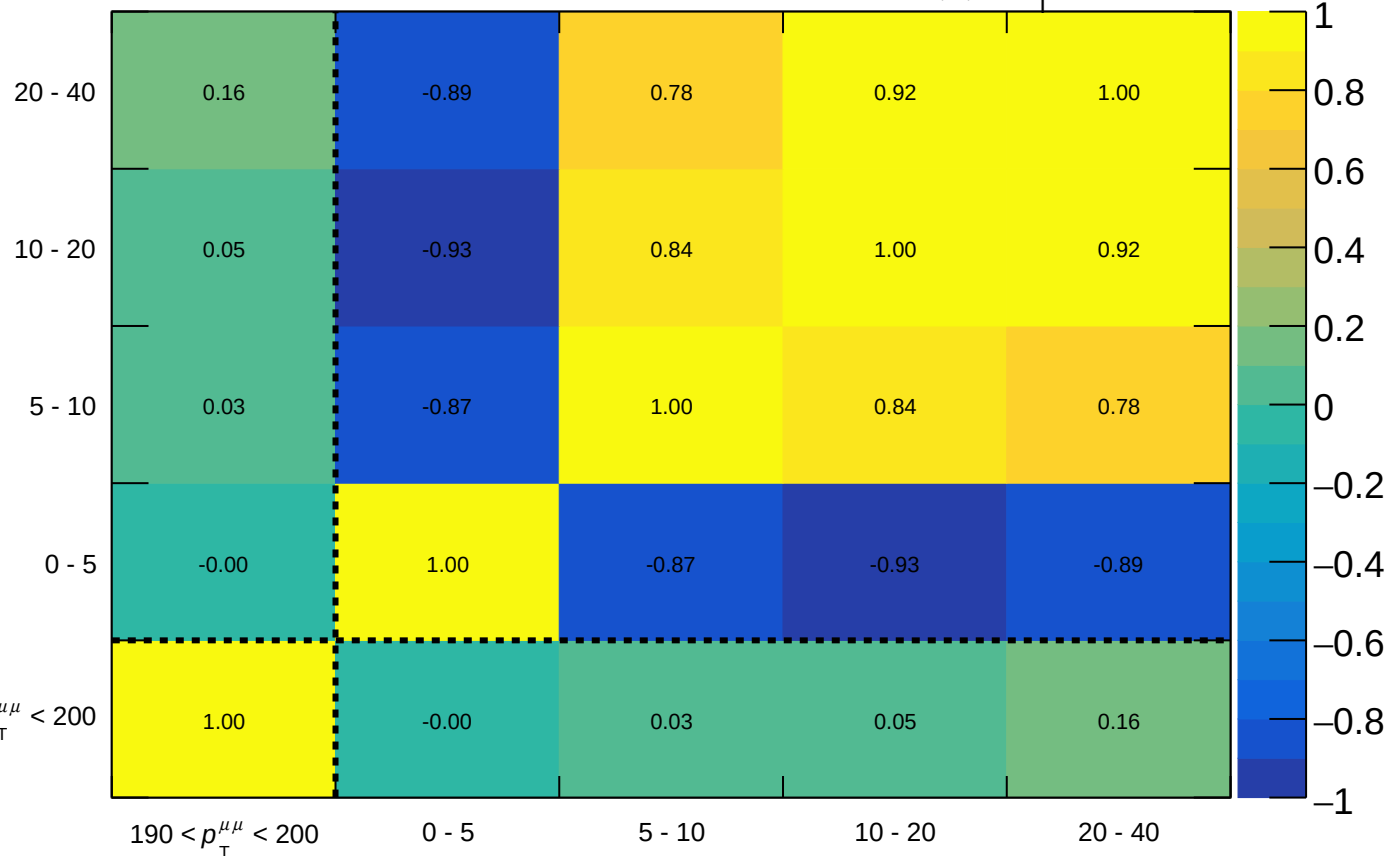
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

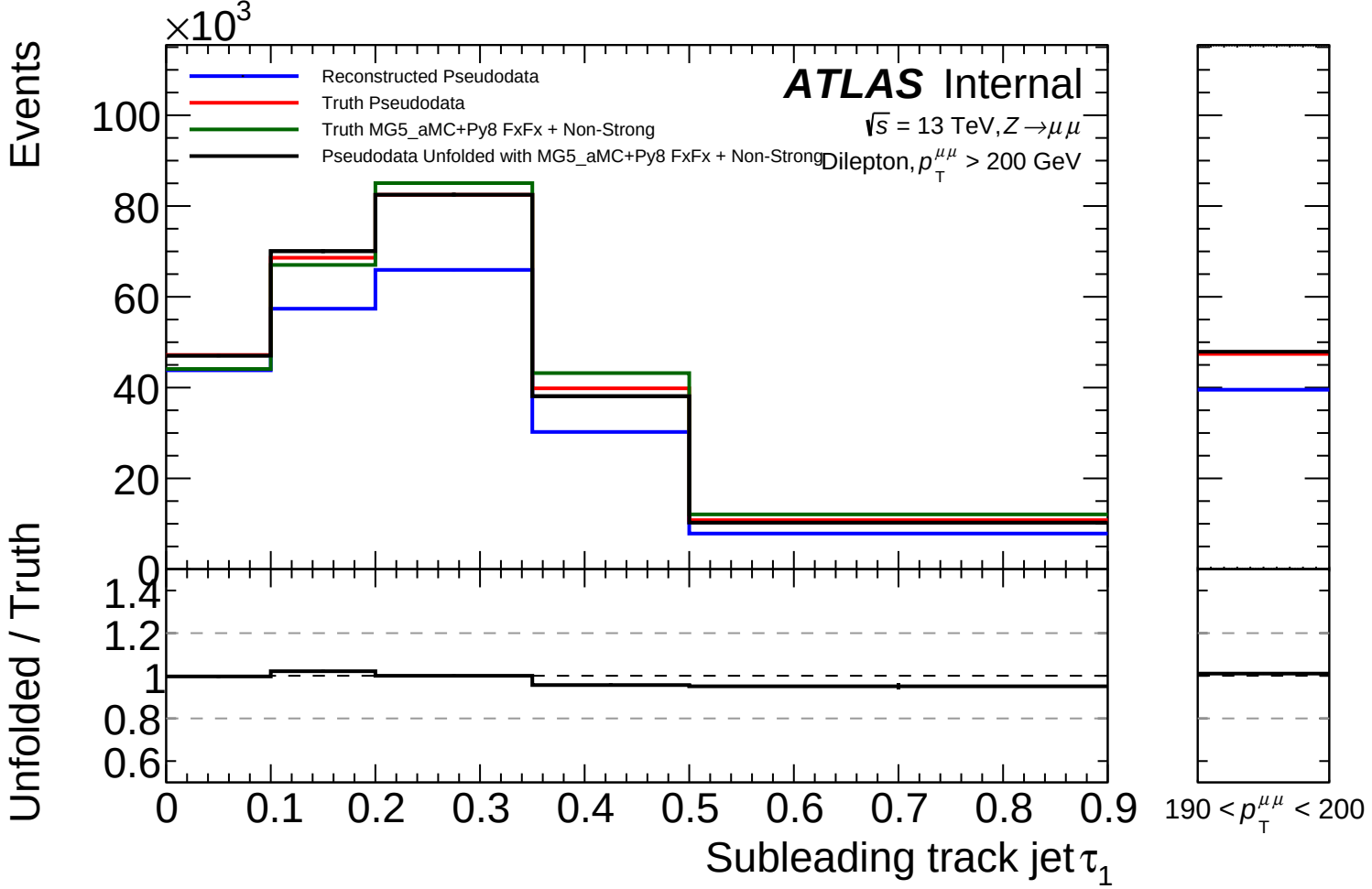


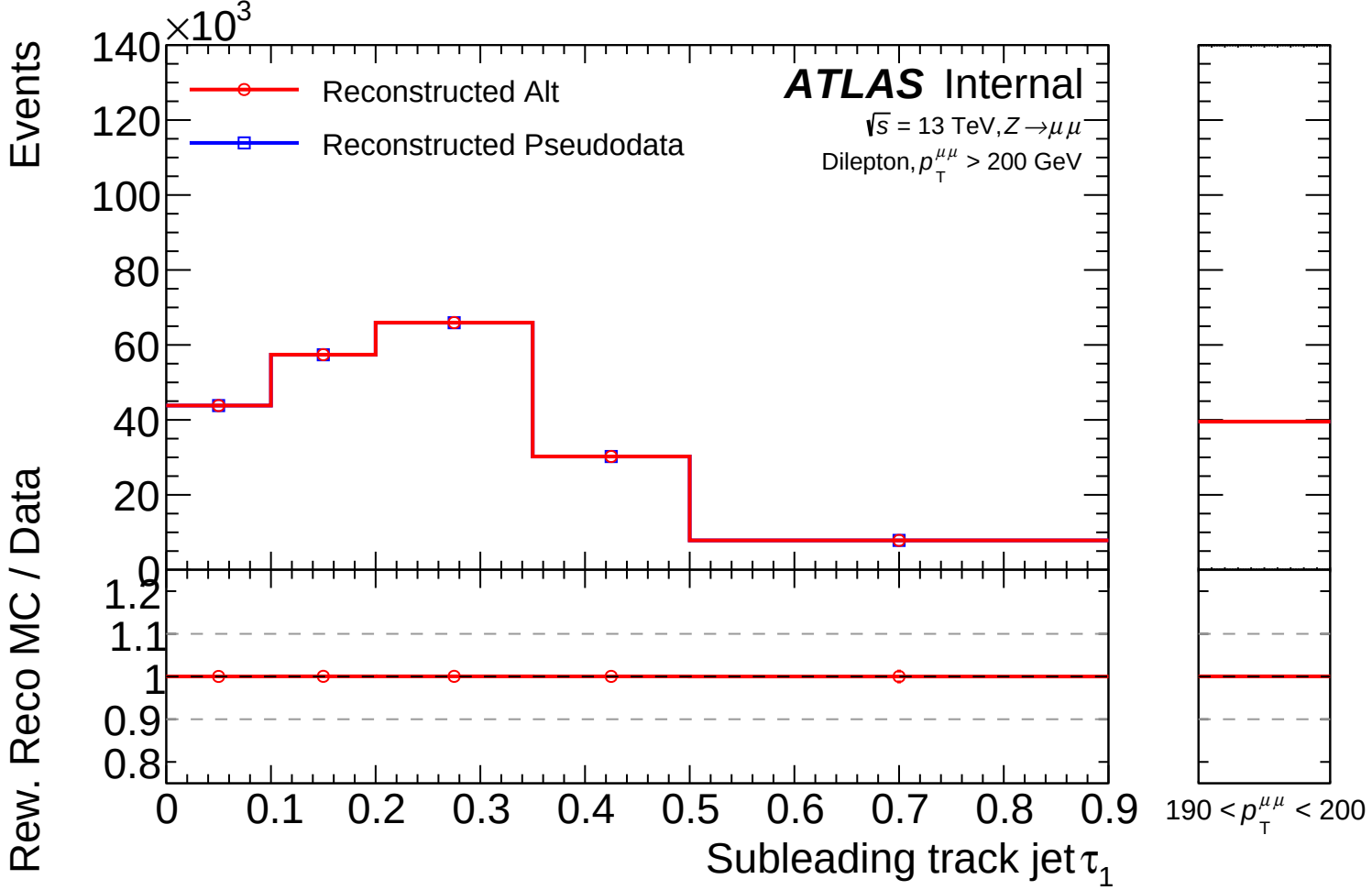
ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

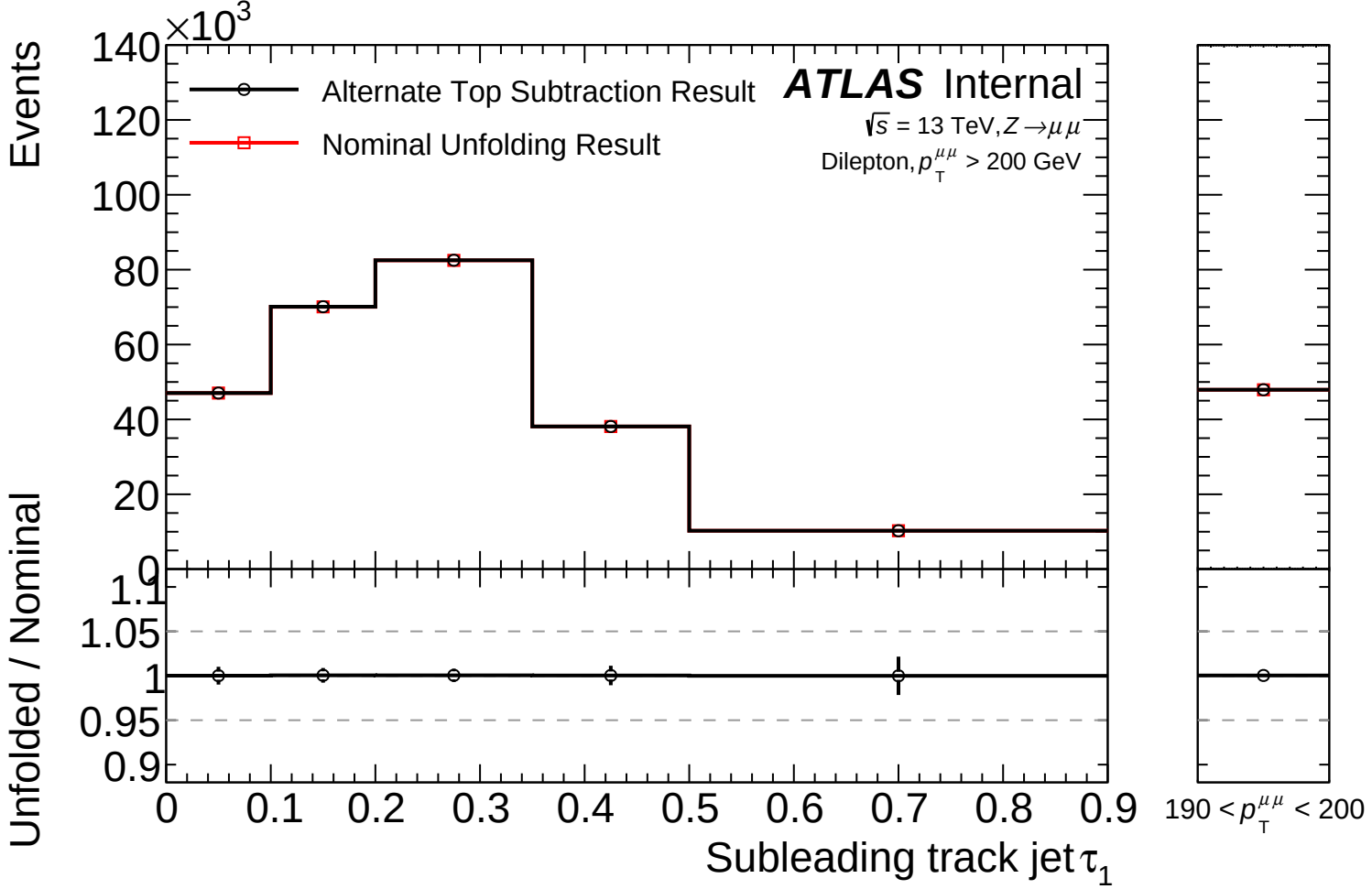
Subleading track jet m [GeV]



Subleading track jet m [GeV]







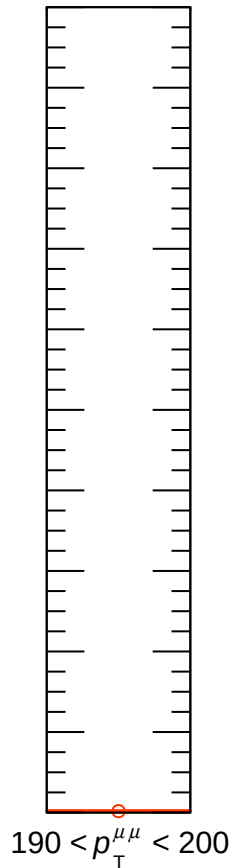
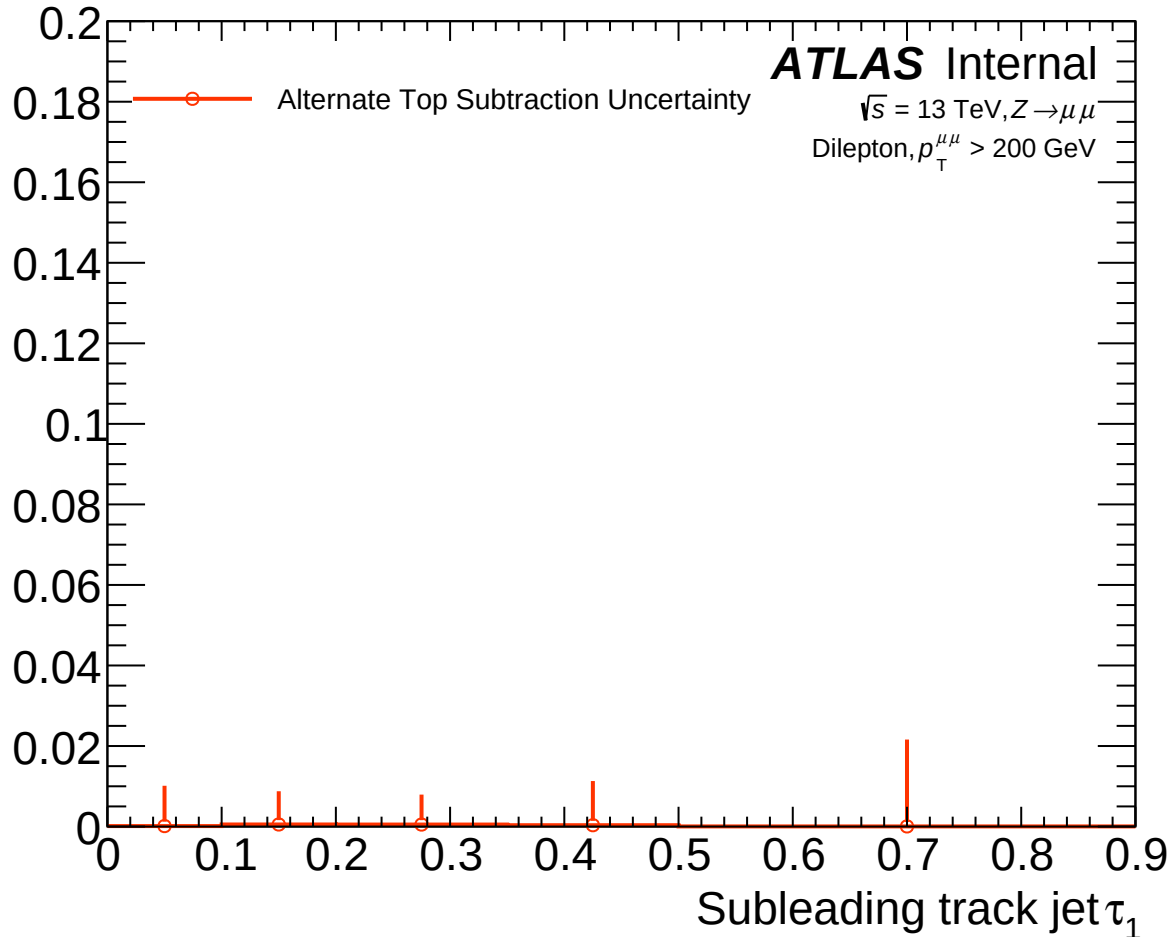
Fractional systematic impact

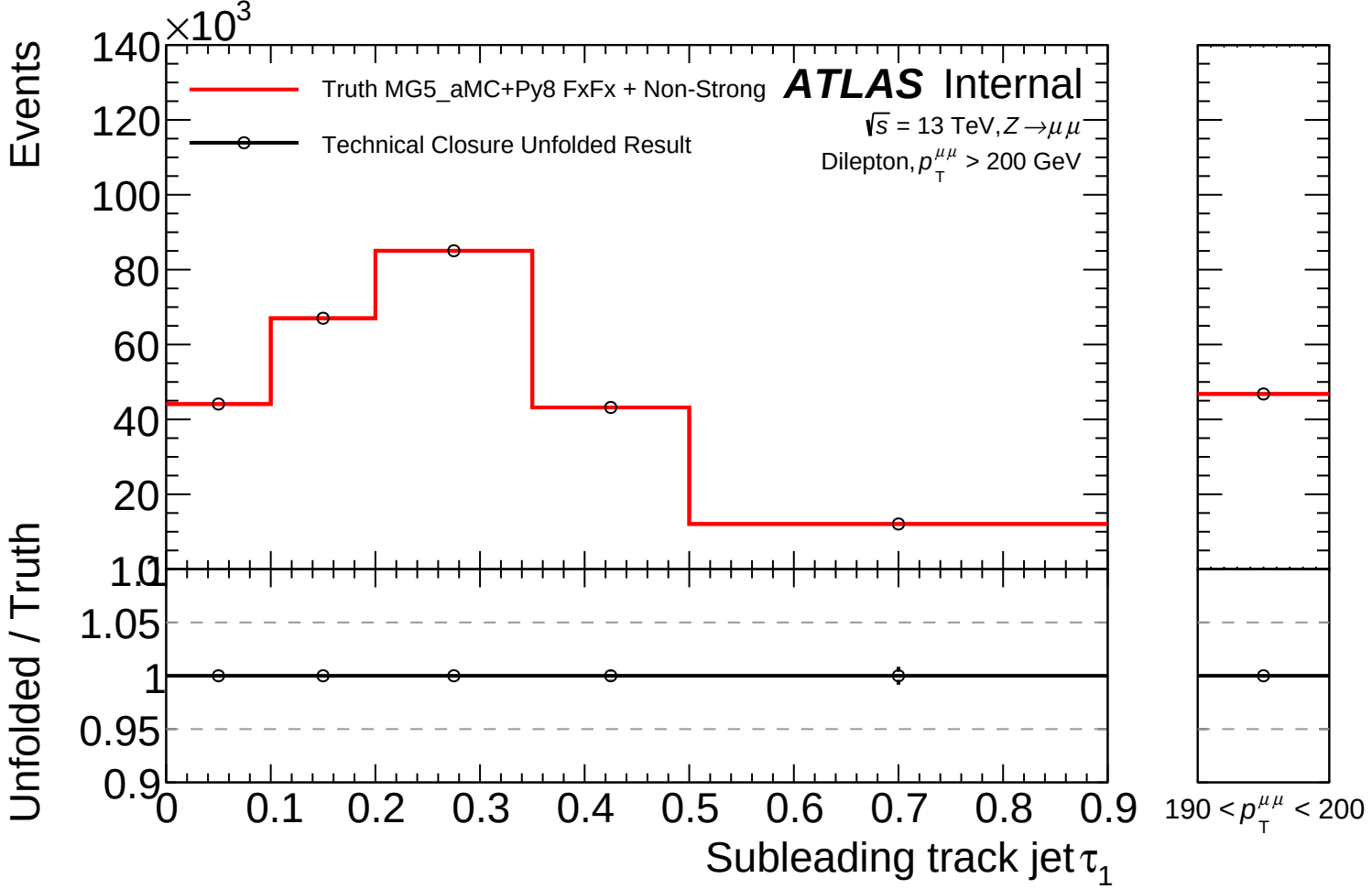
ATLAS Internal

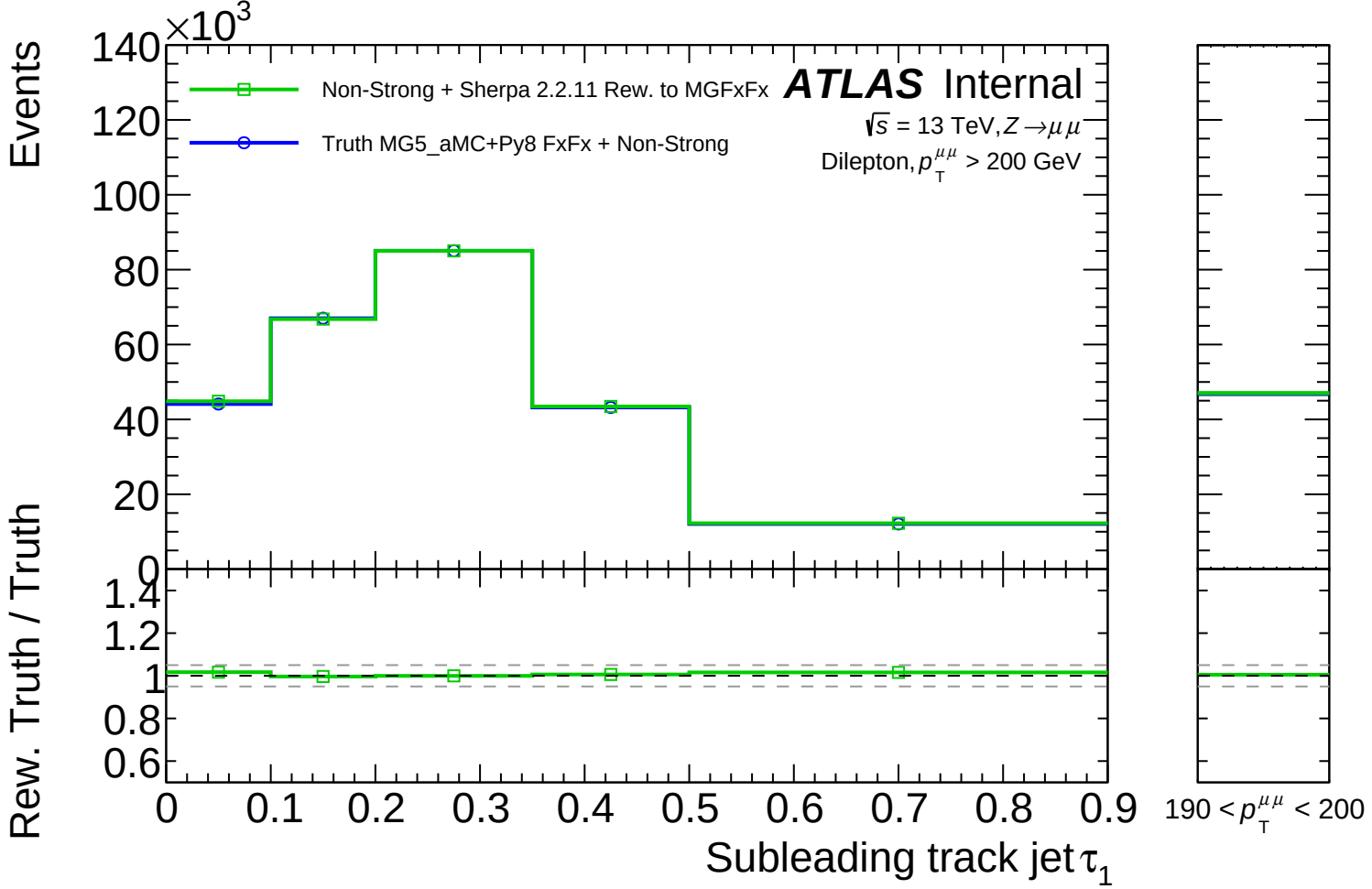
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

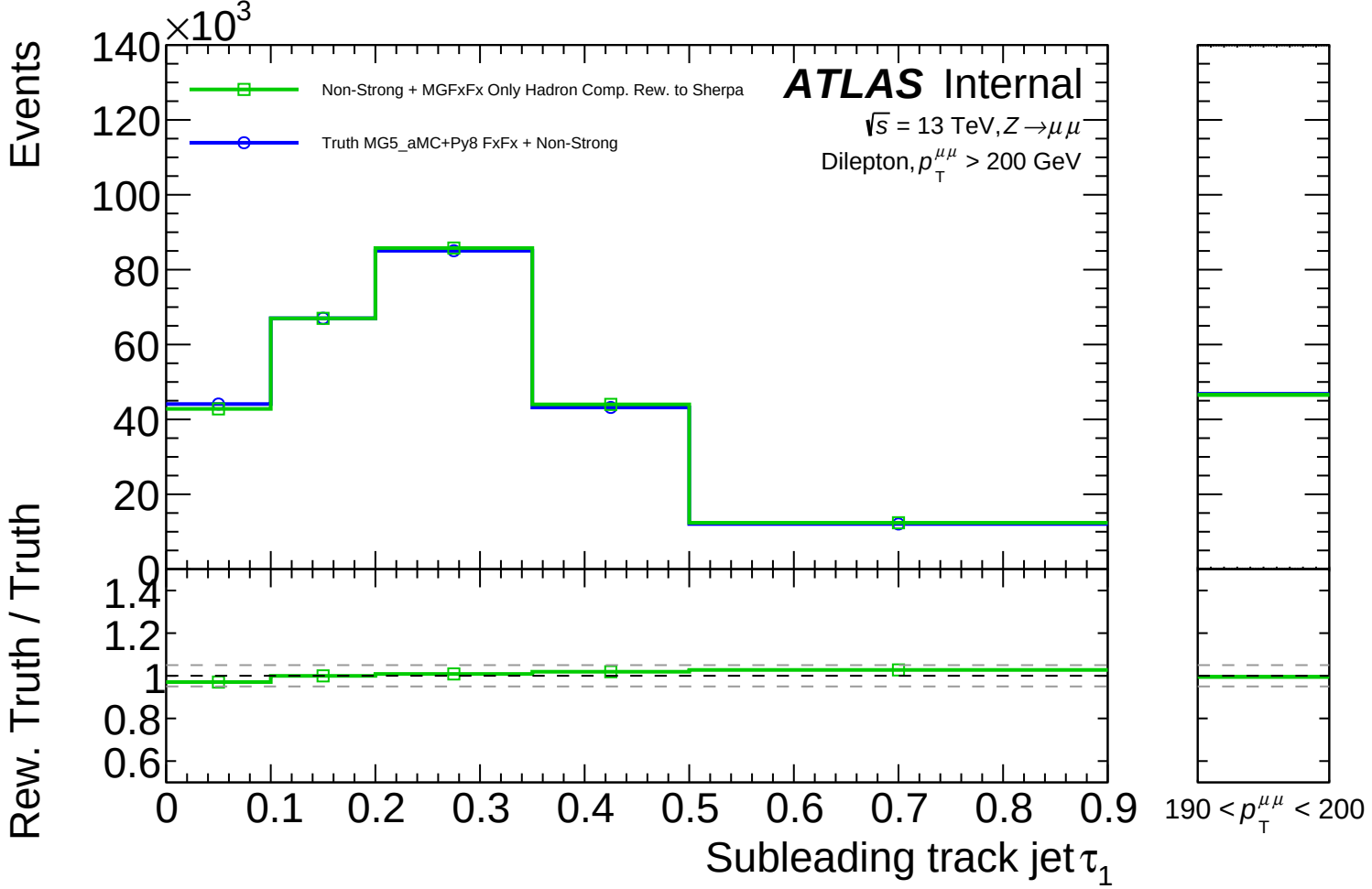
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

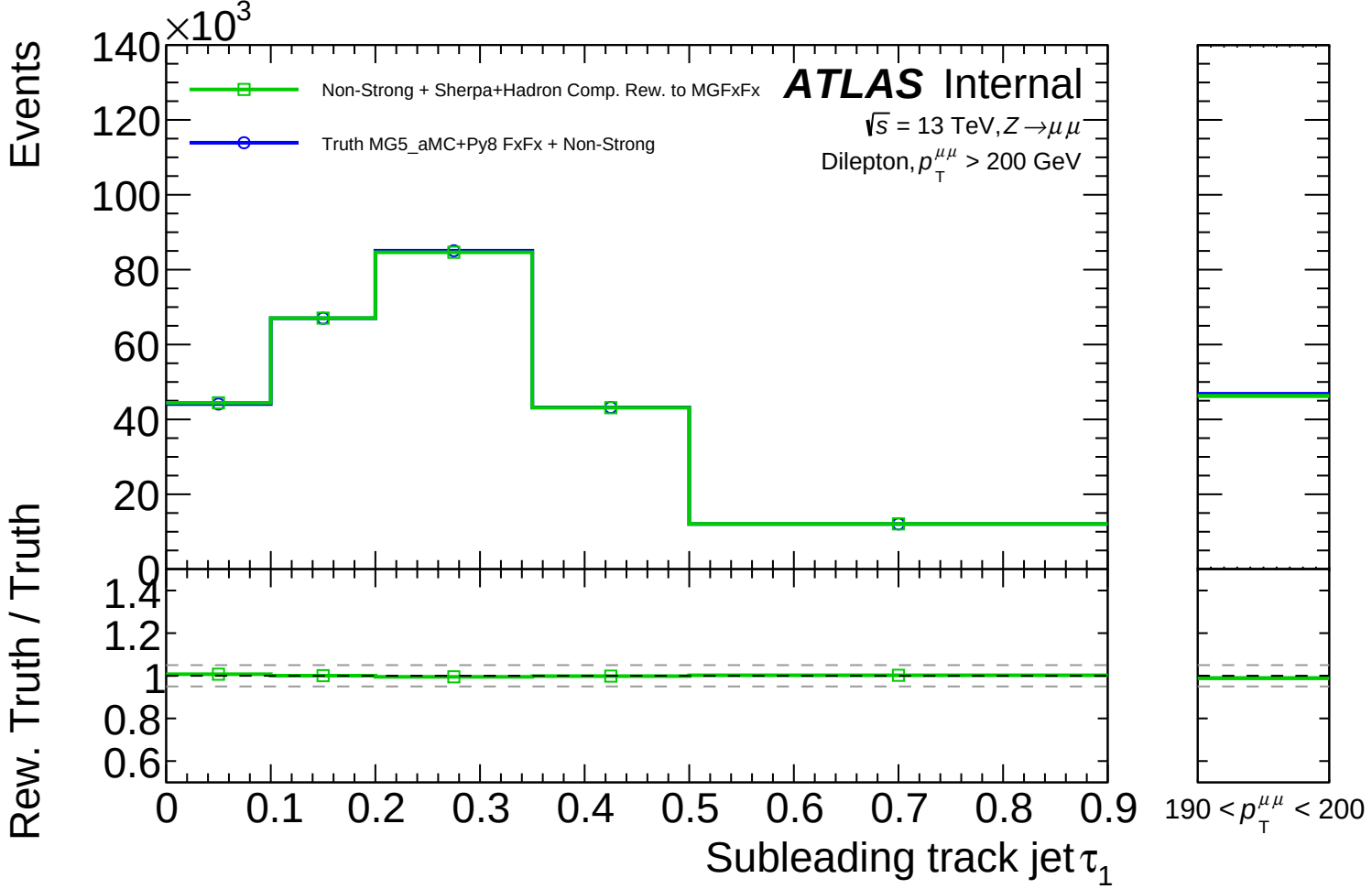
Alternate Top Subtraction Uncertainty

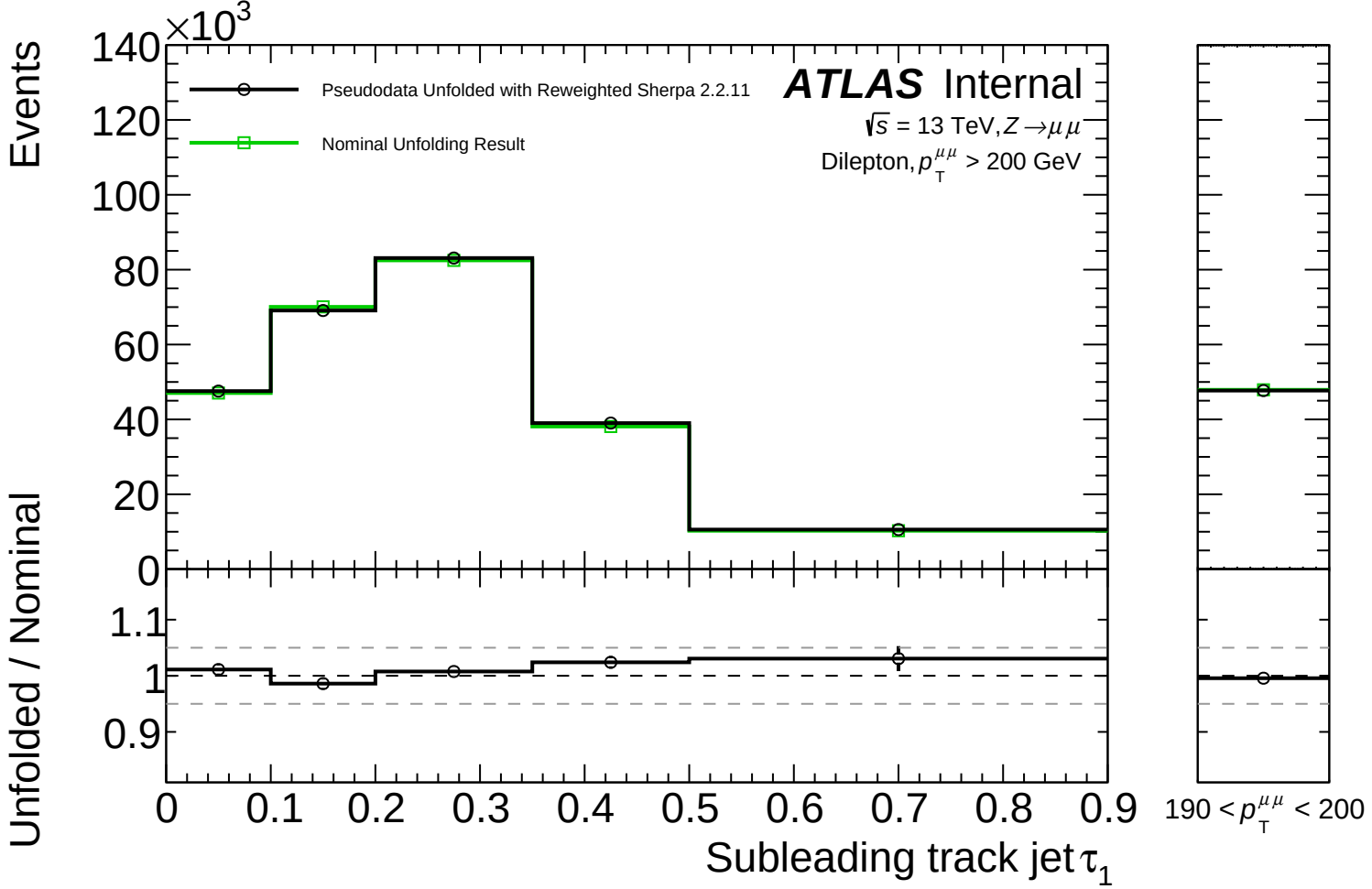


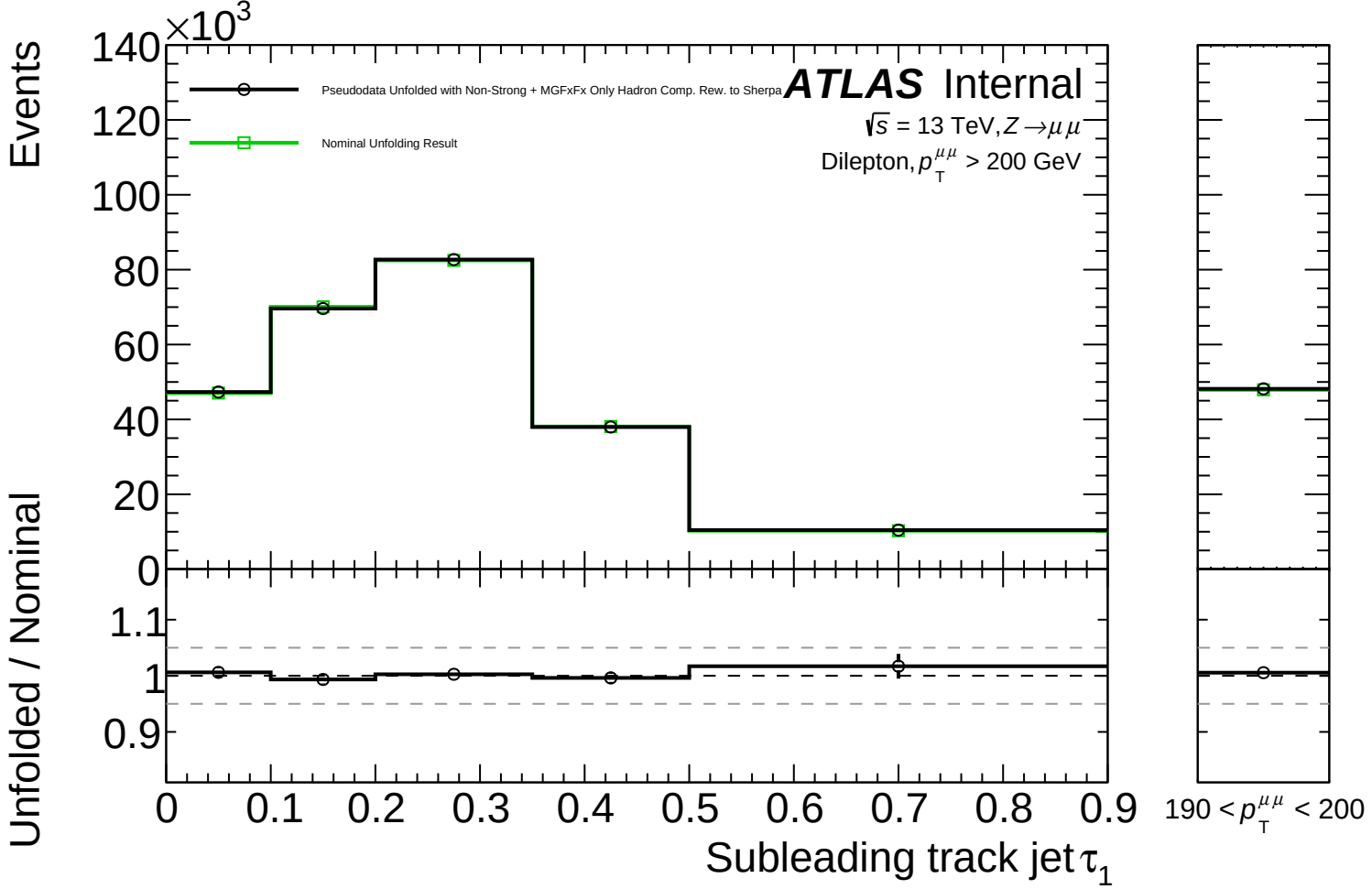


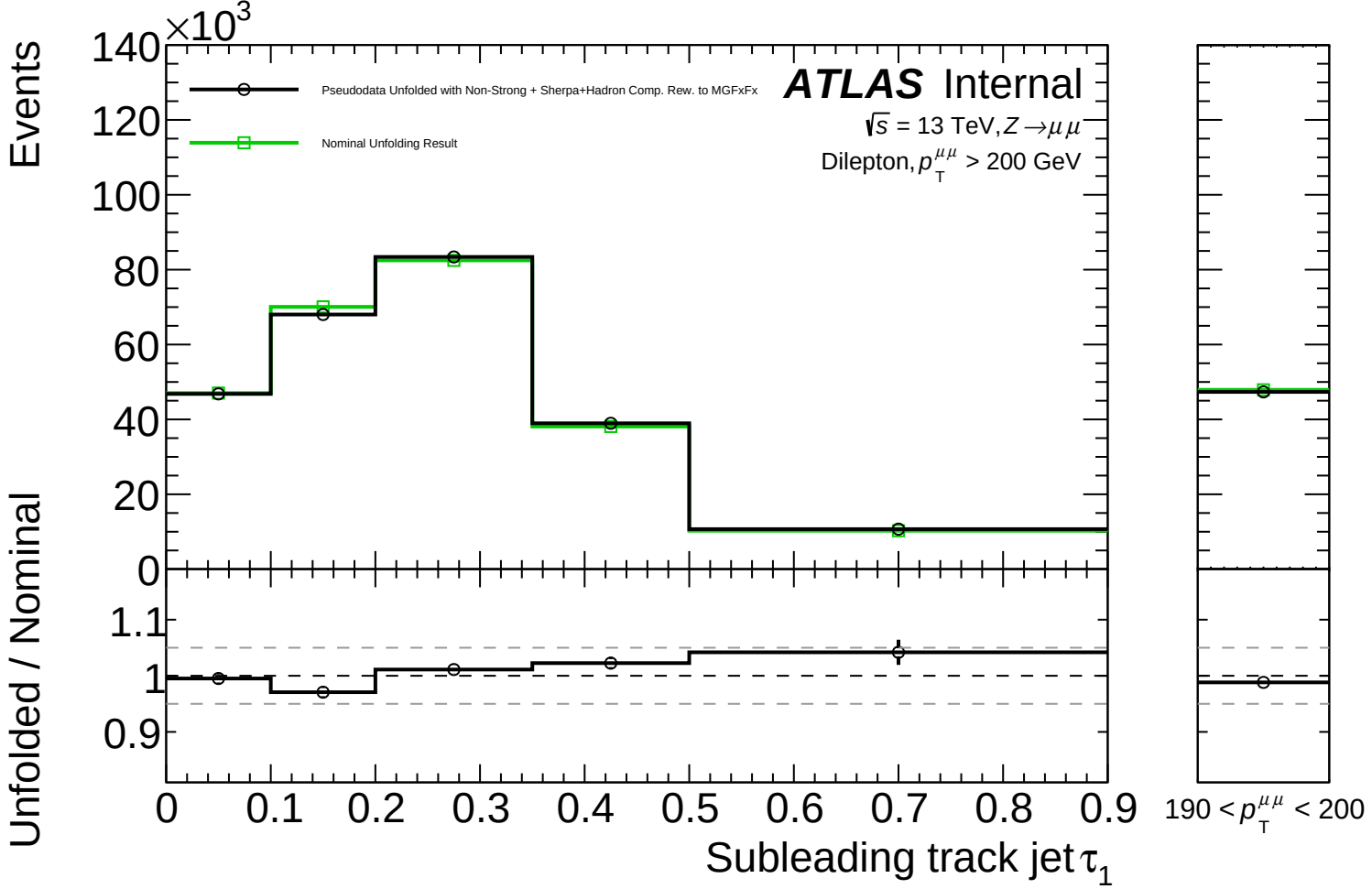


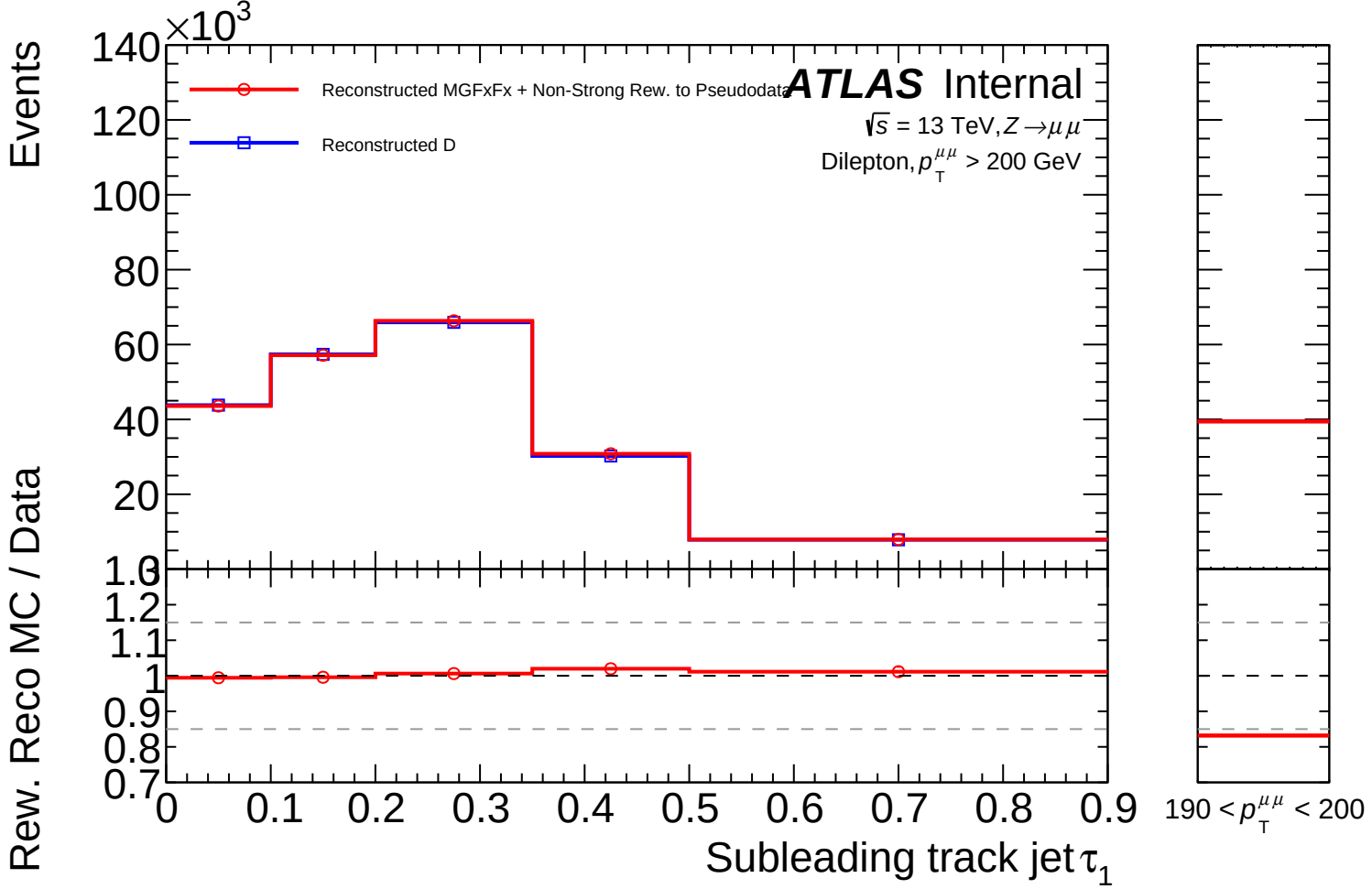


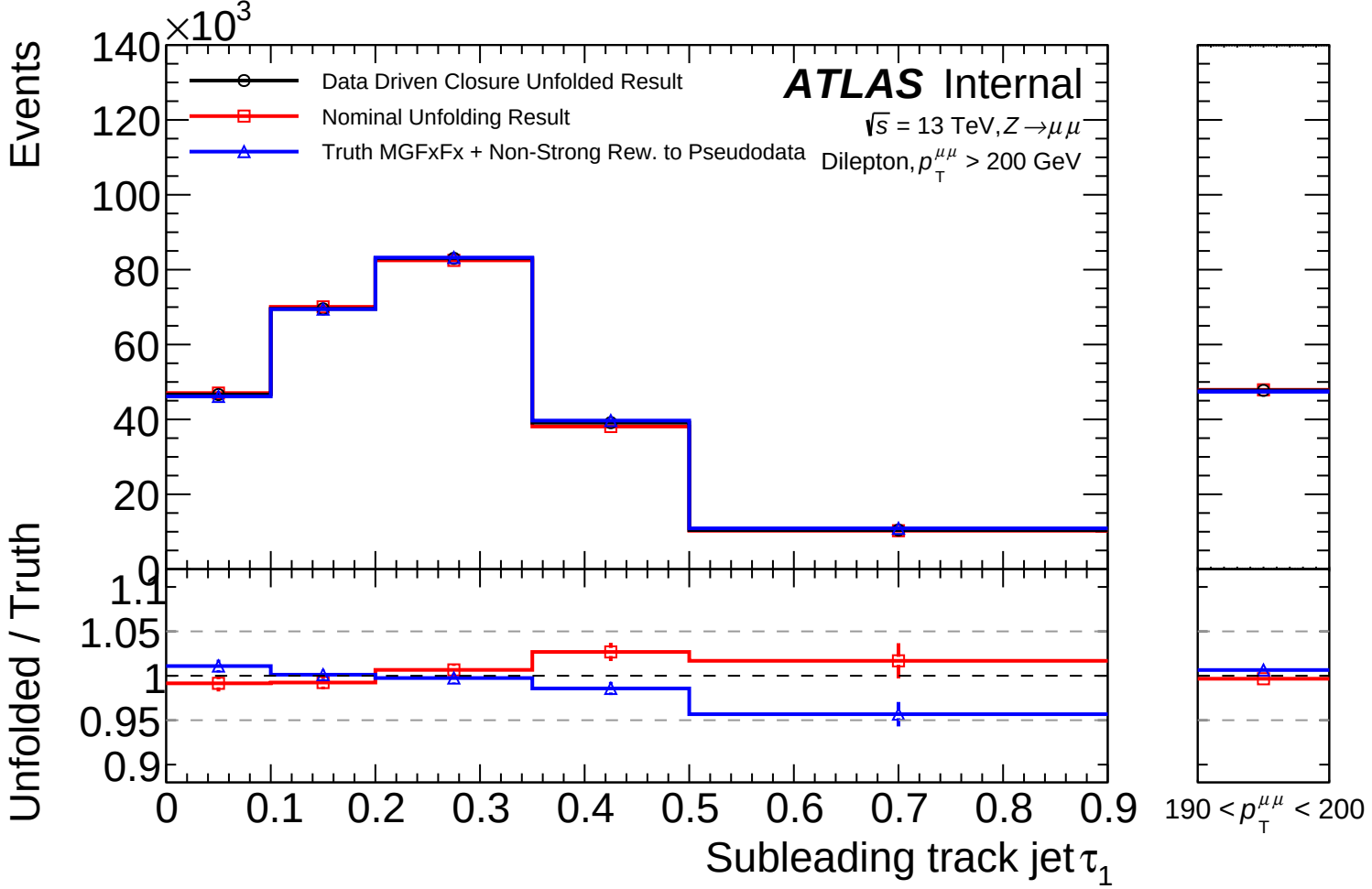




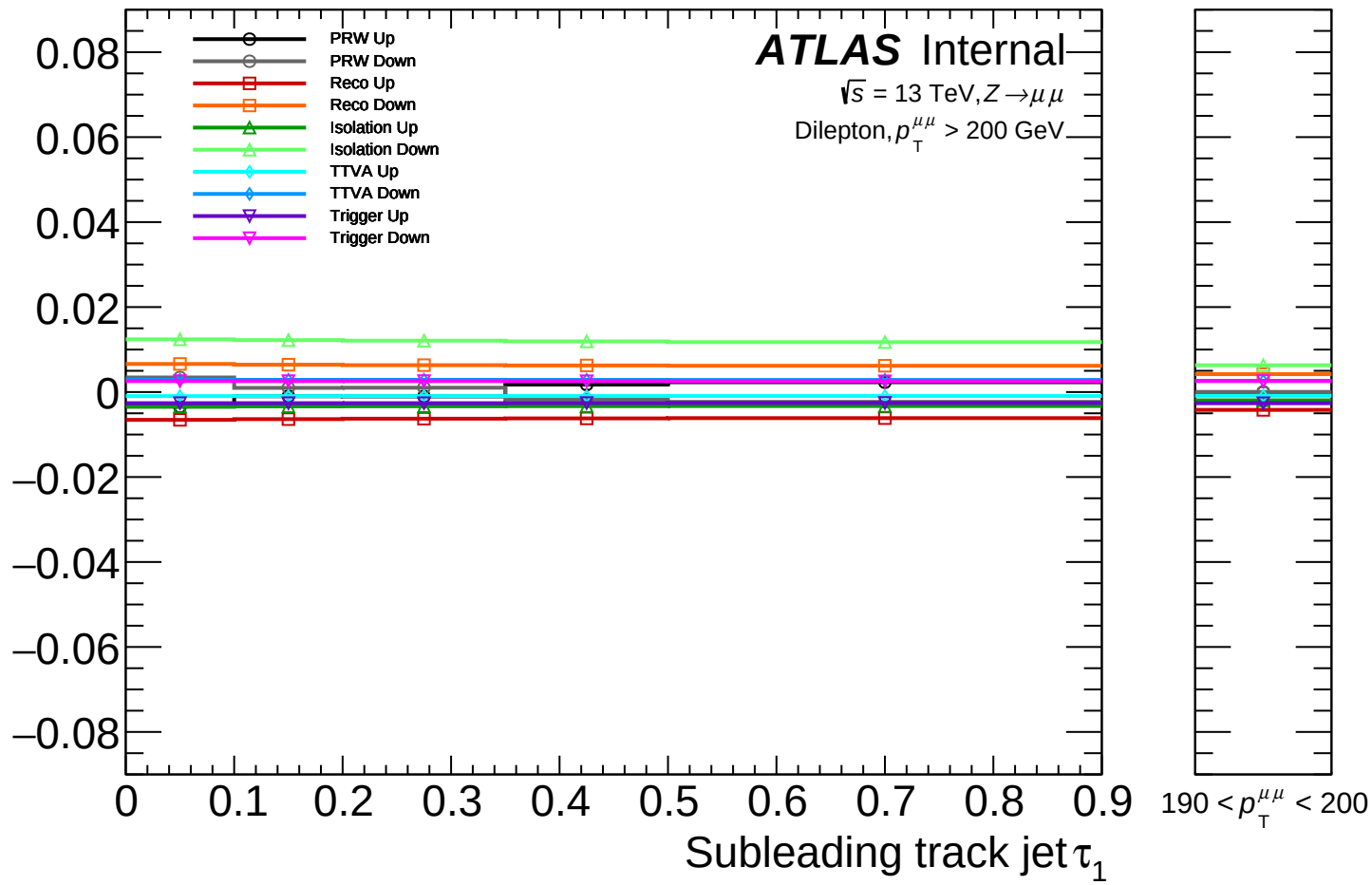




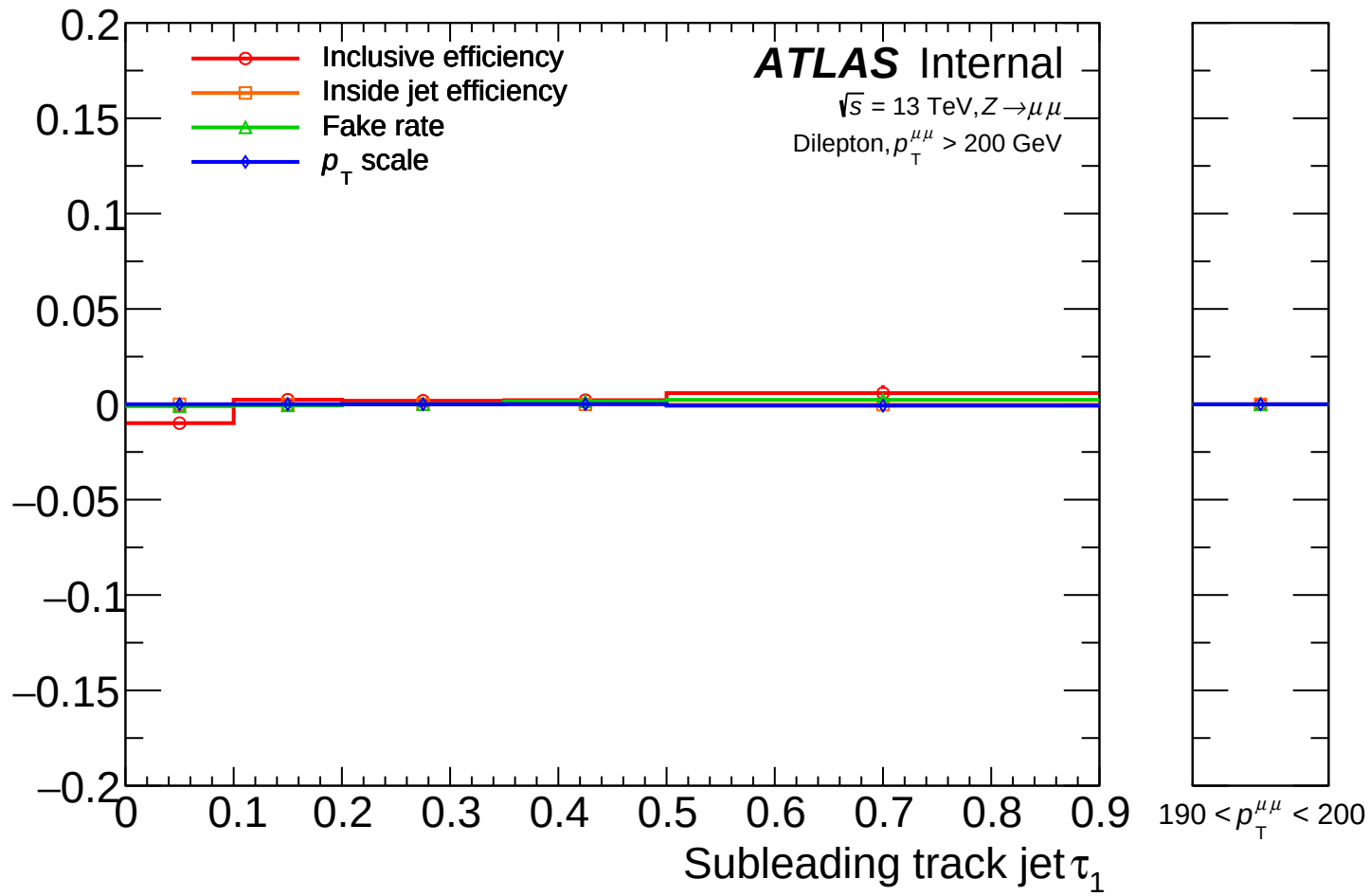




Fractional systematic impact



Fractional systematic impact



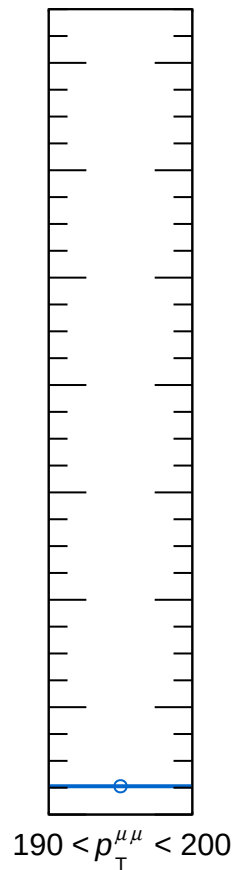
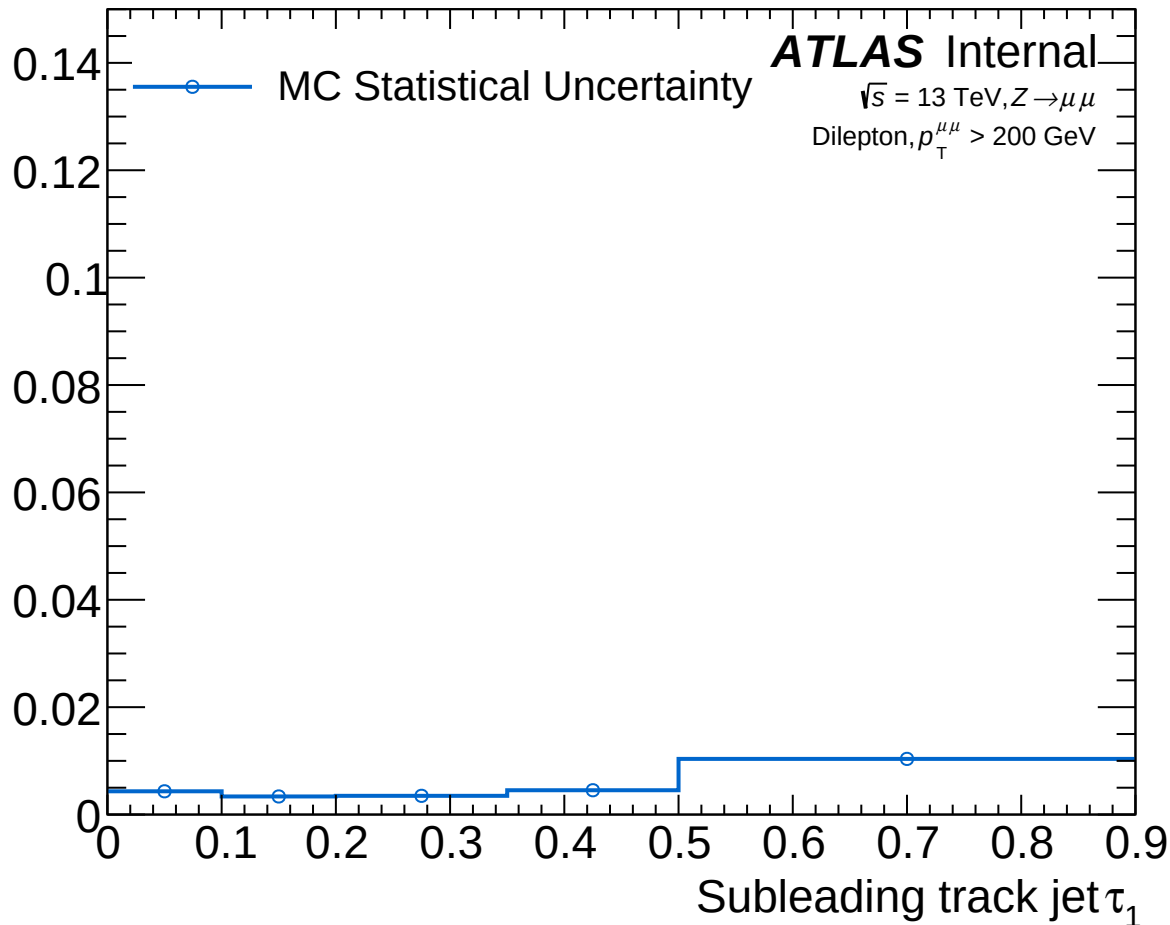
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



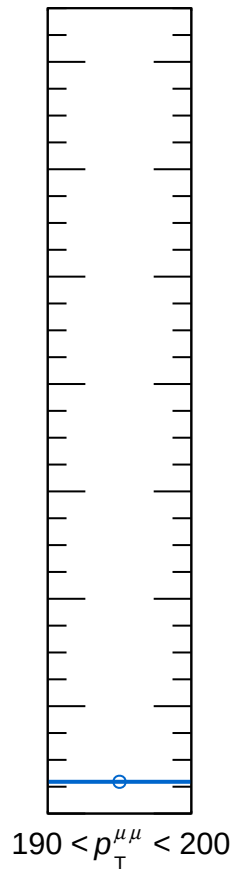
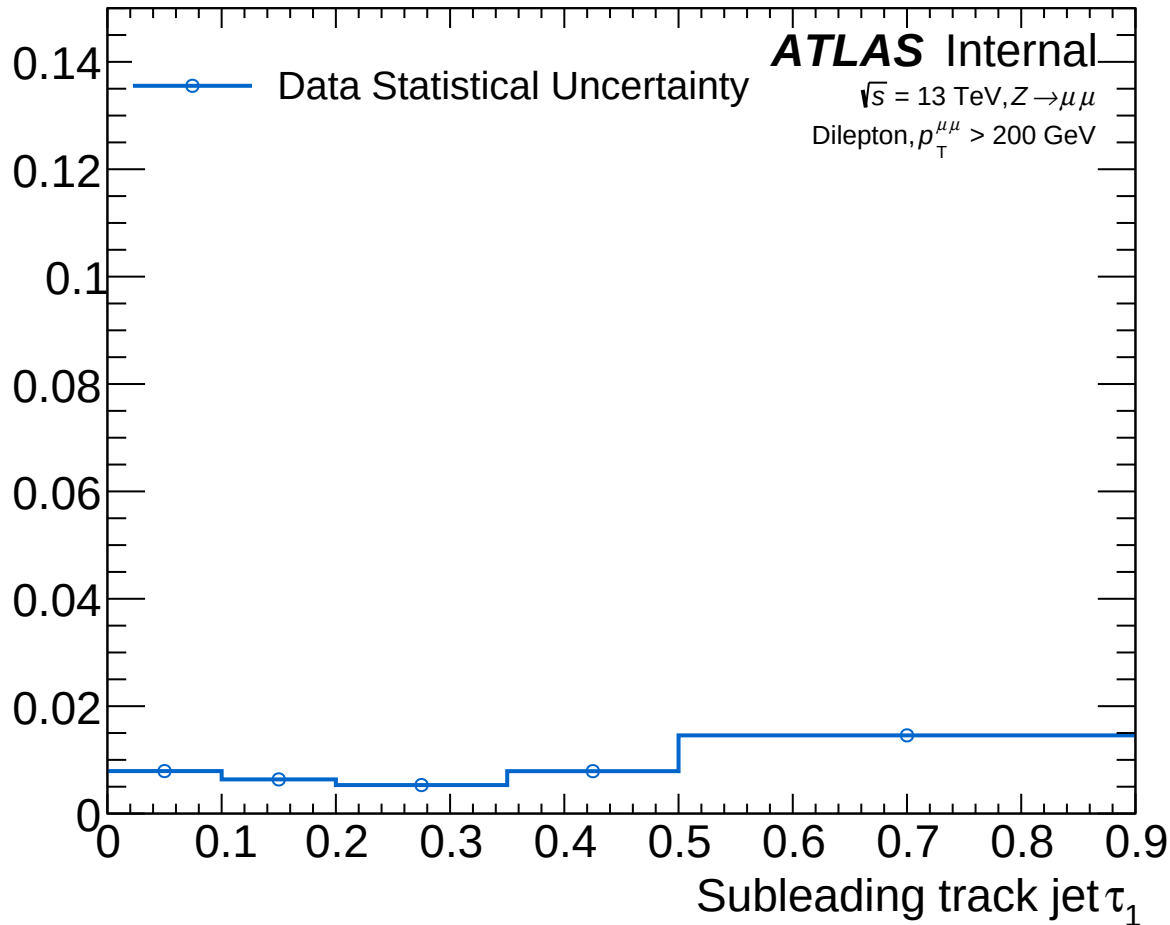
Fractional systematic impact

ATLAS Internal

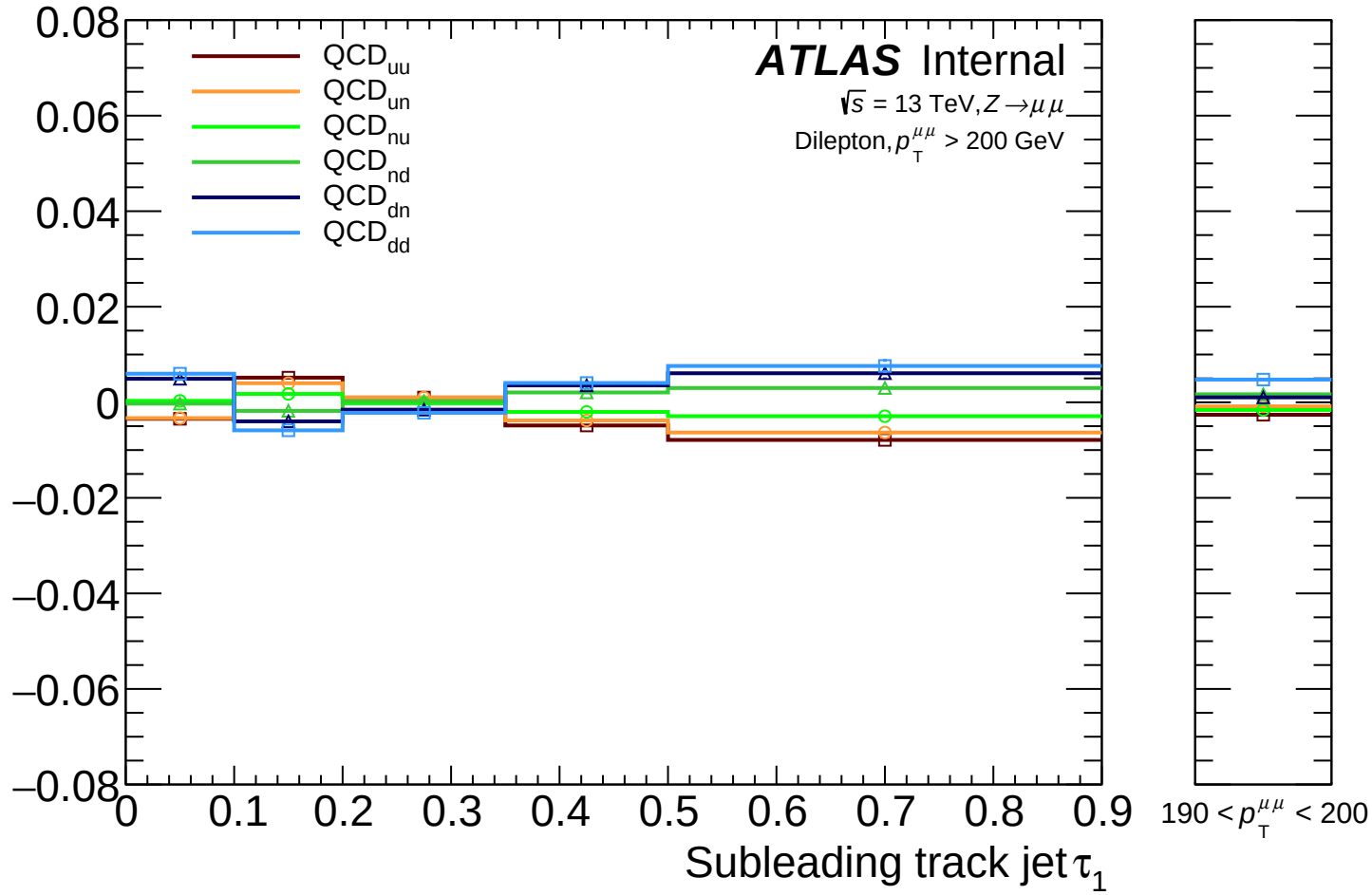
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

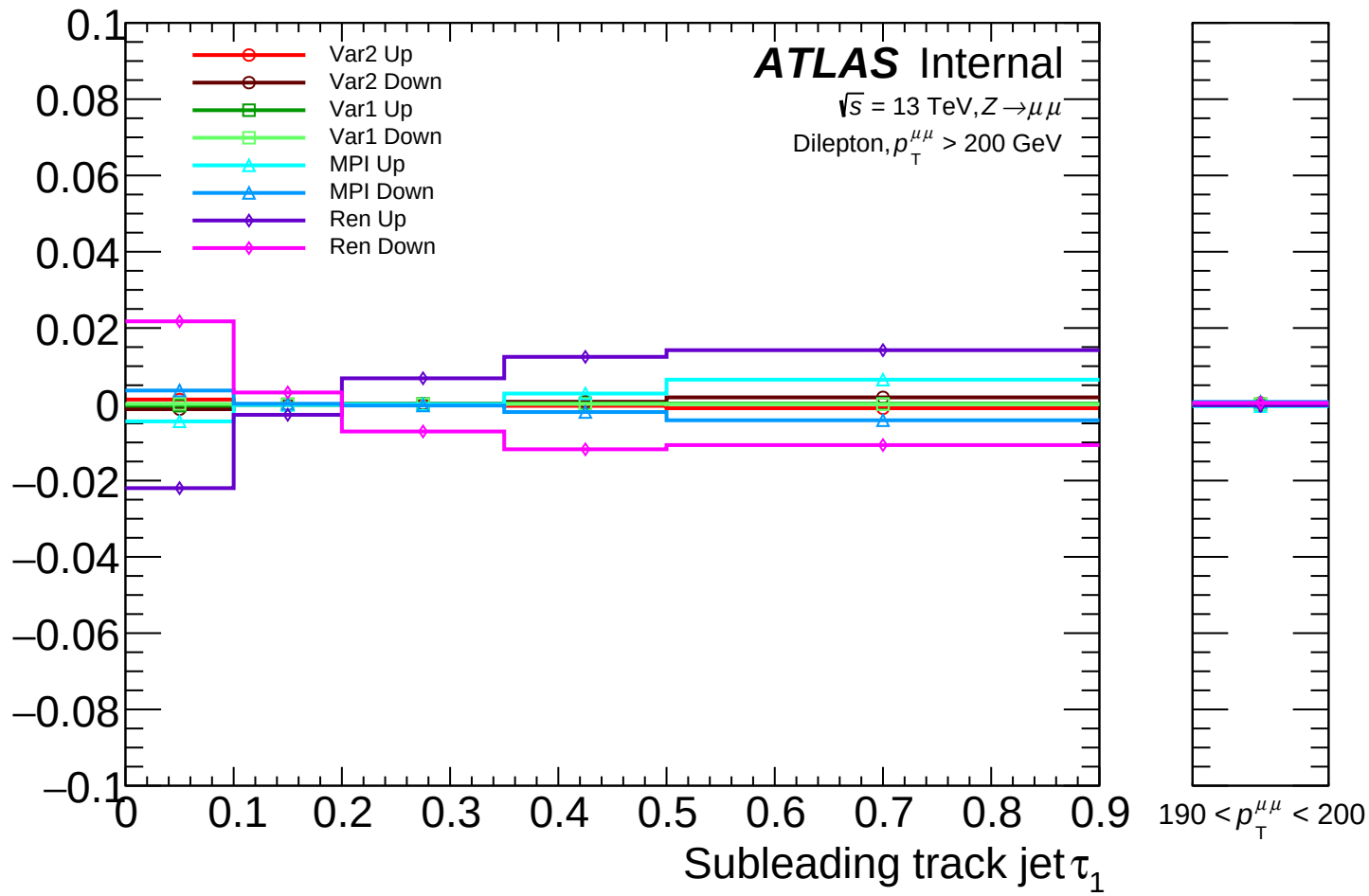
Data Statistical Uncertainty

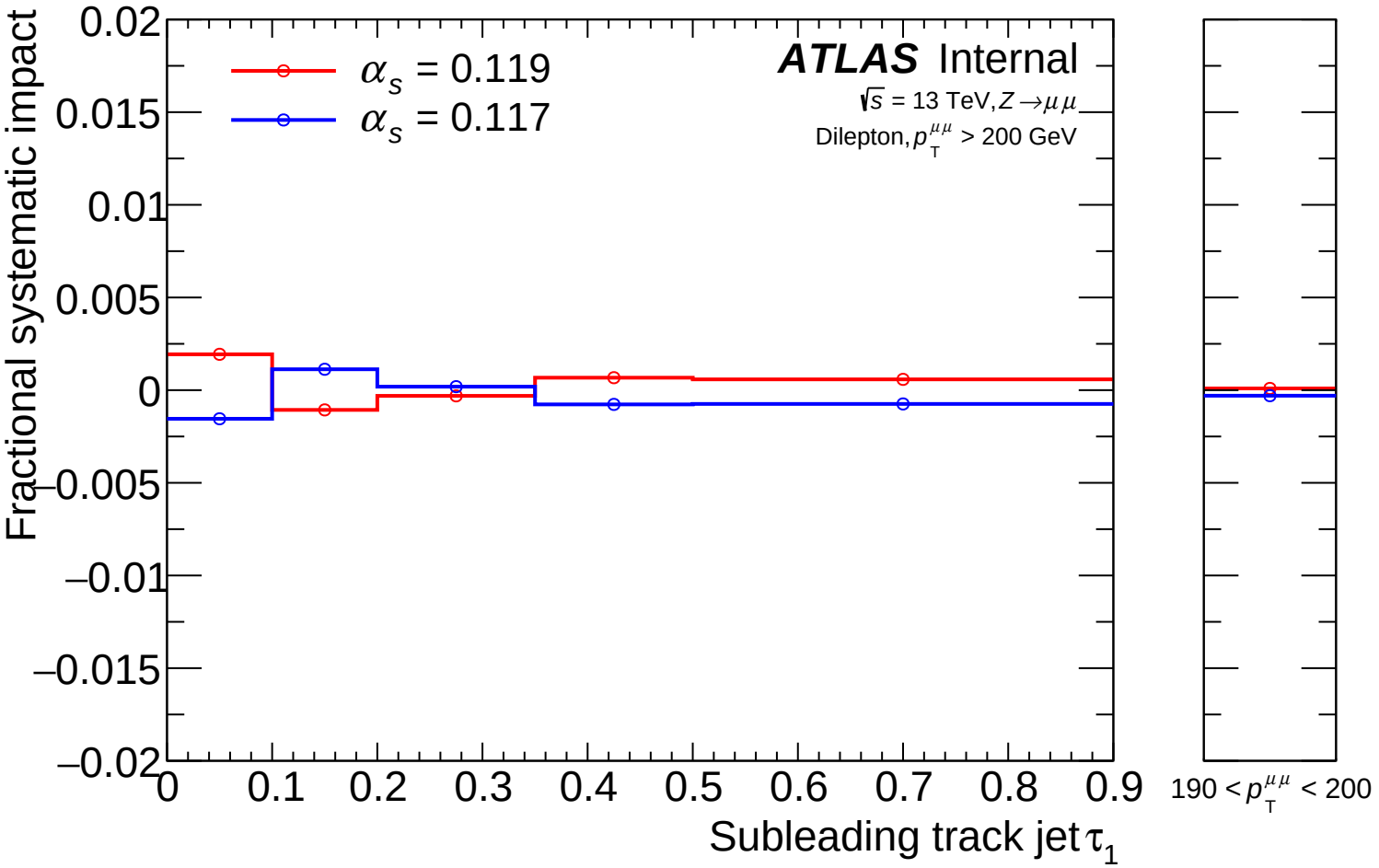


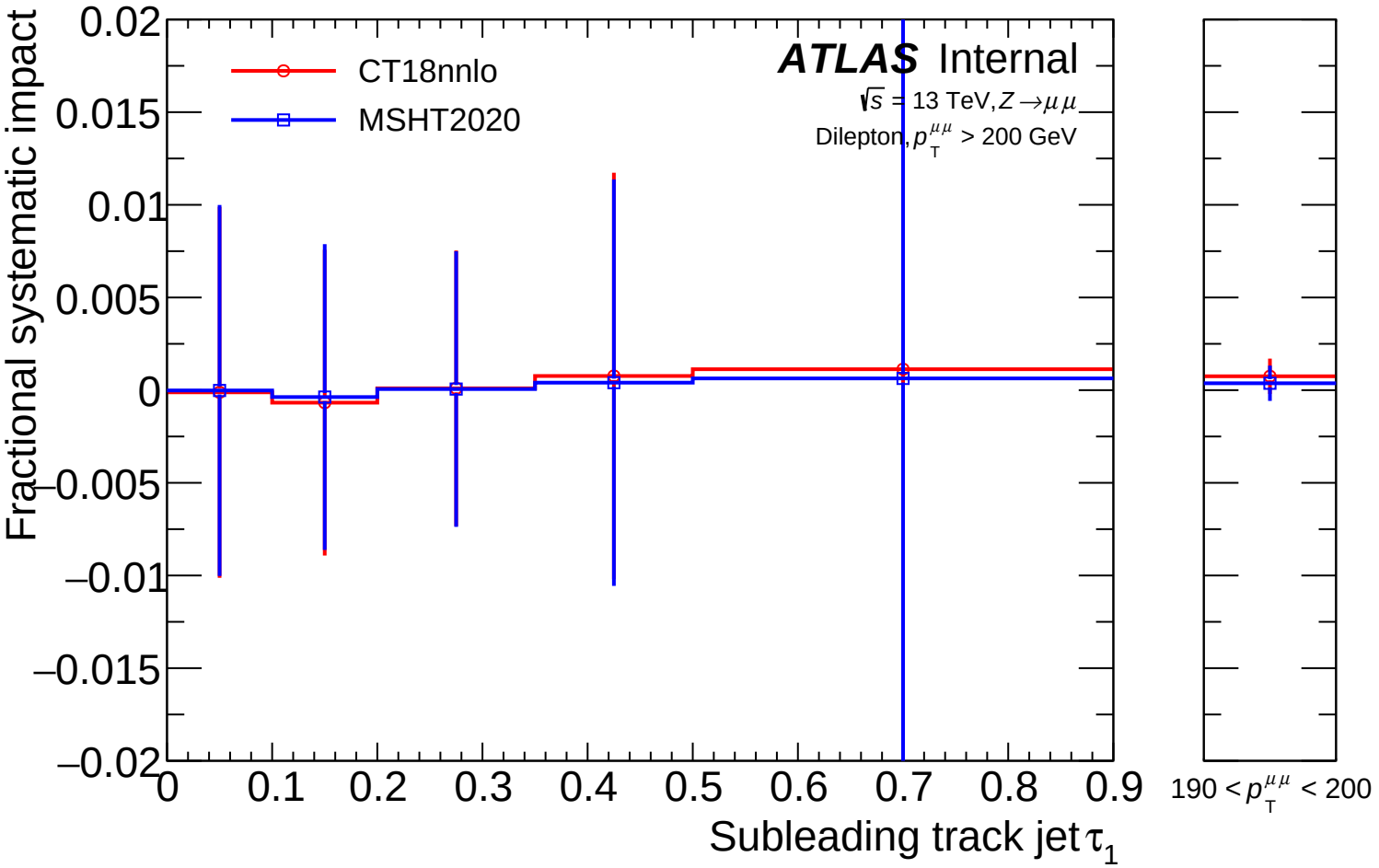
Fractional systematic impact



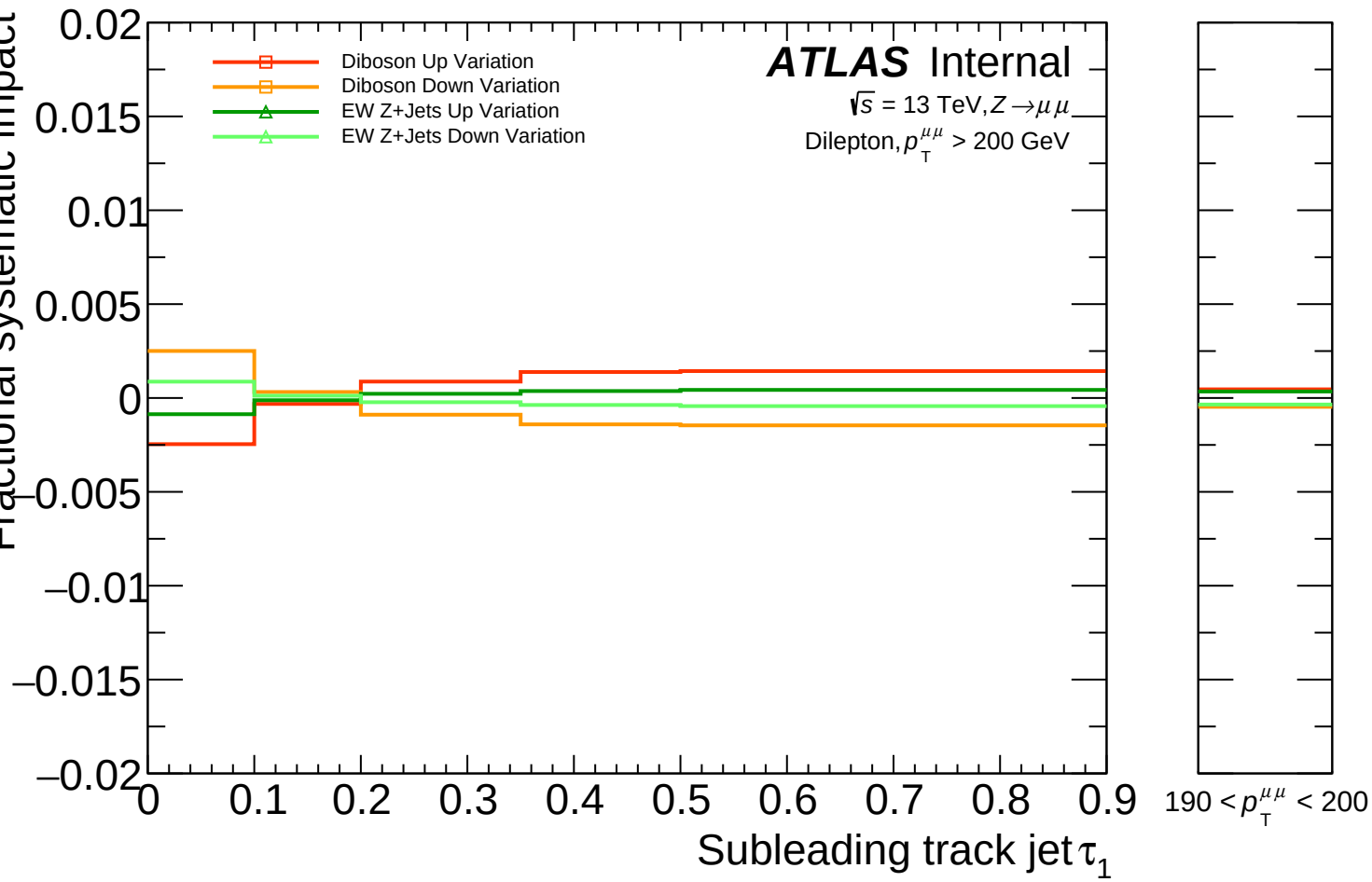
Fractional systematic impact







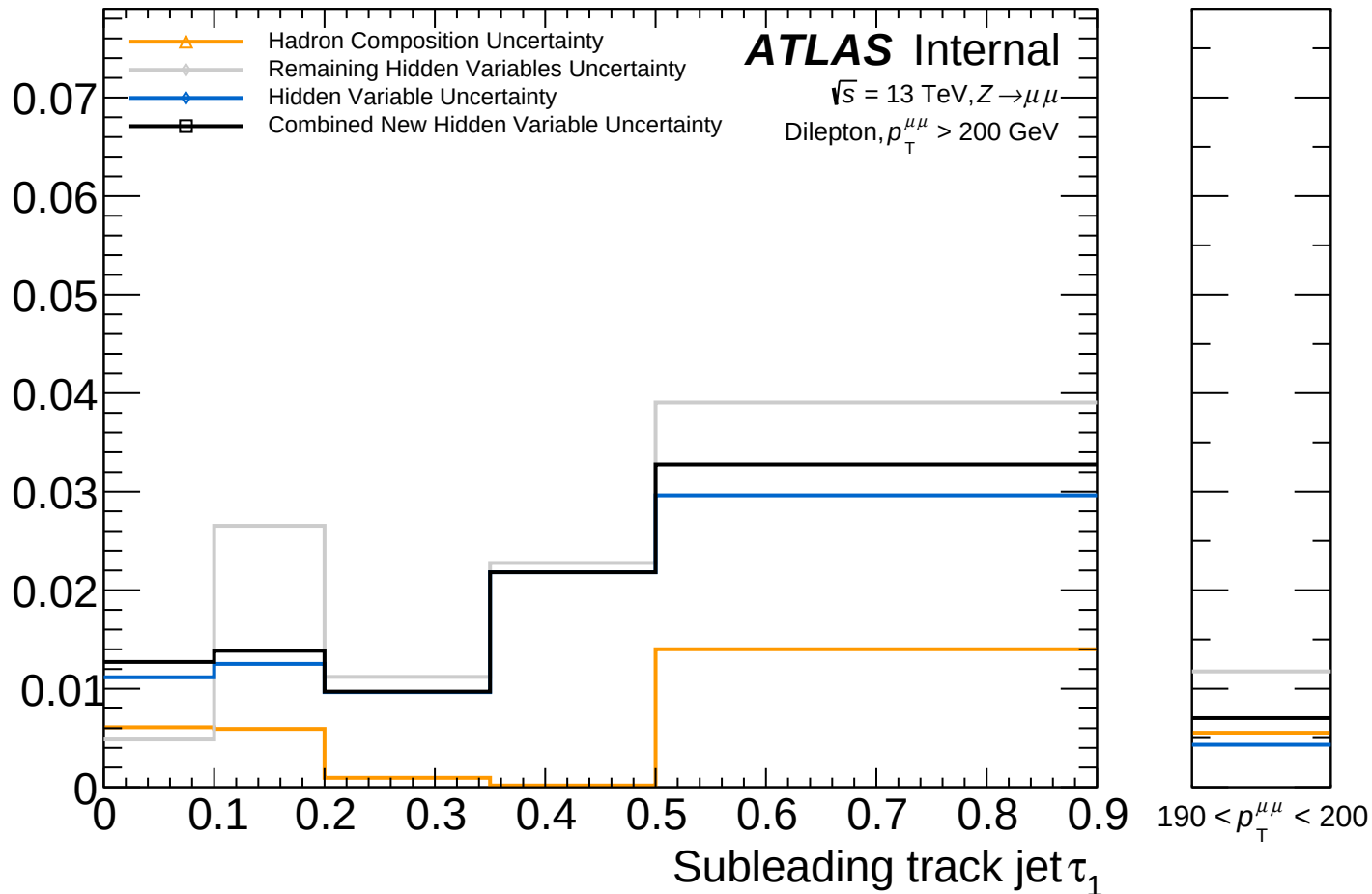
Fractional systematic impact

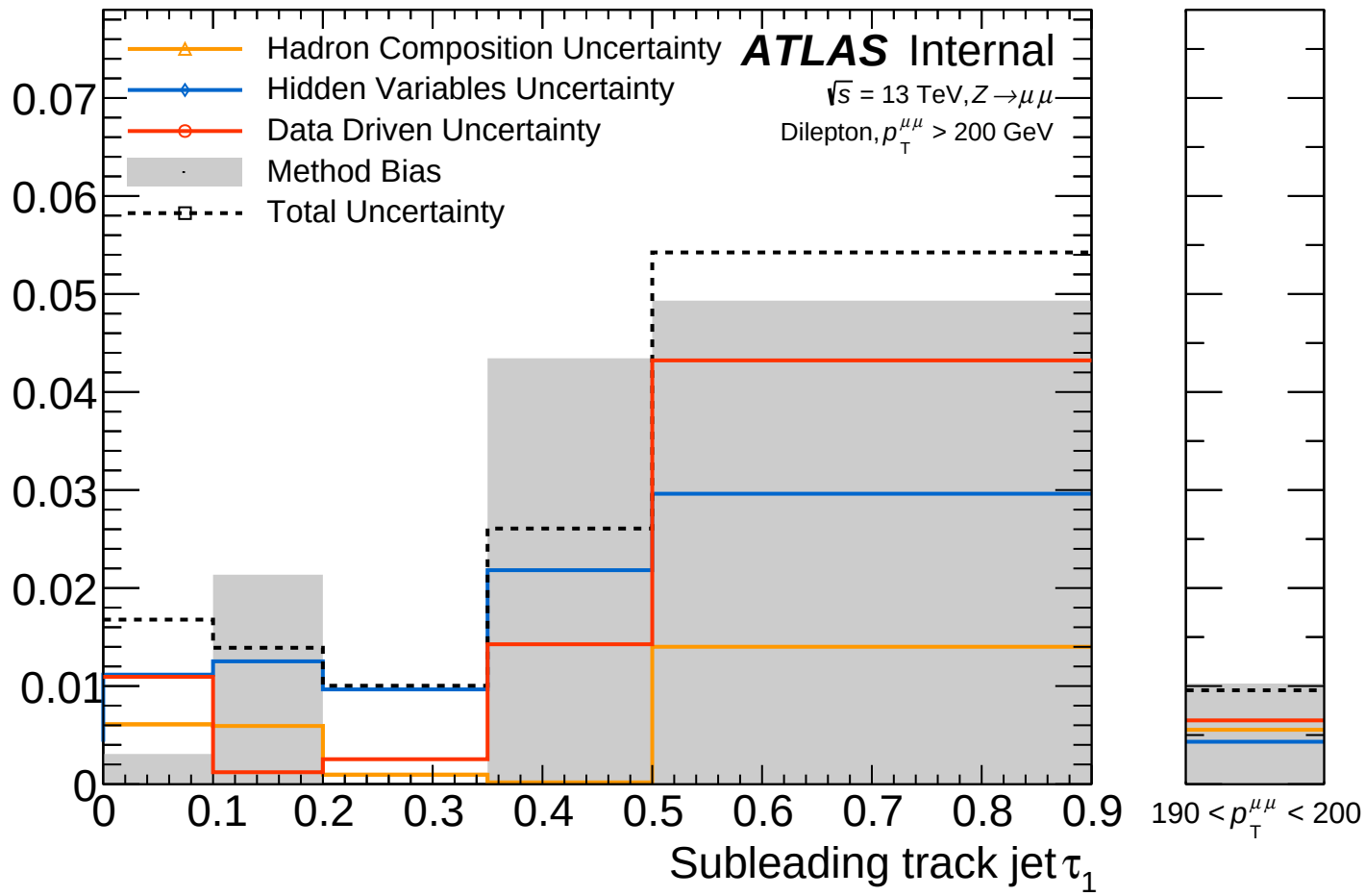


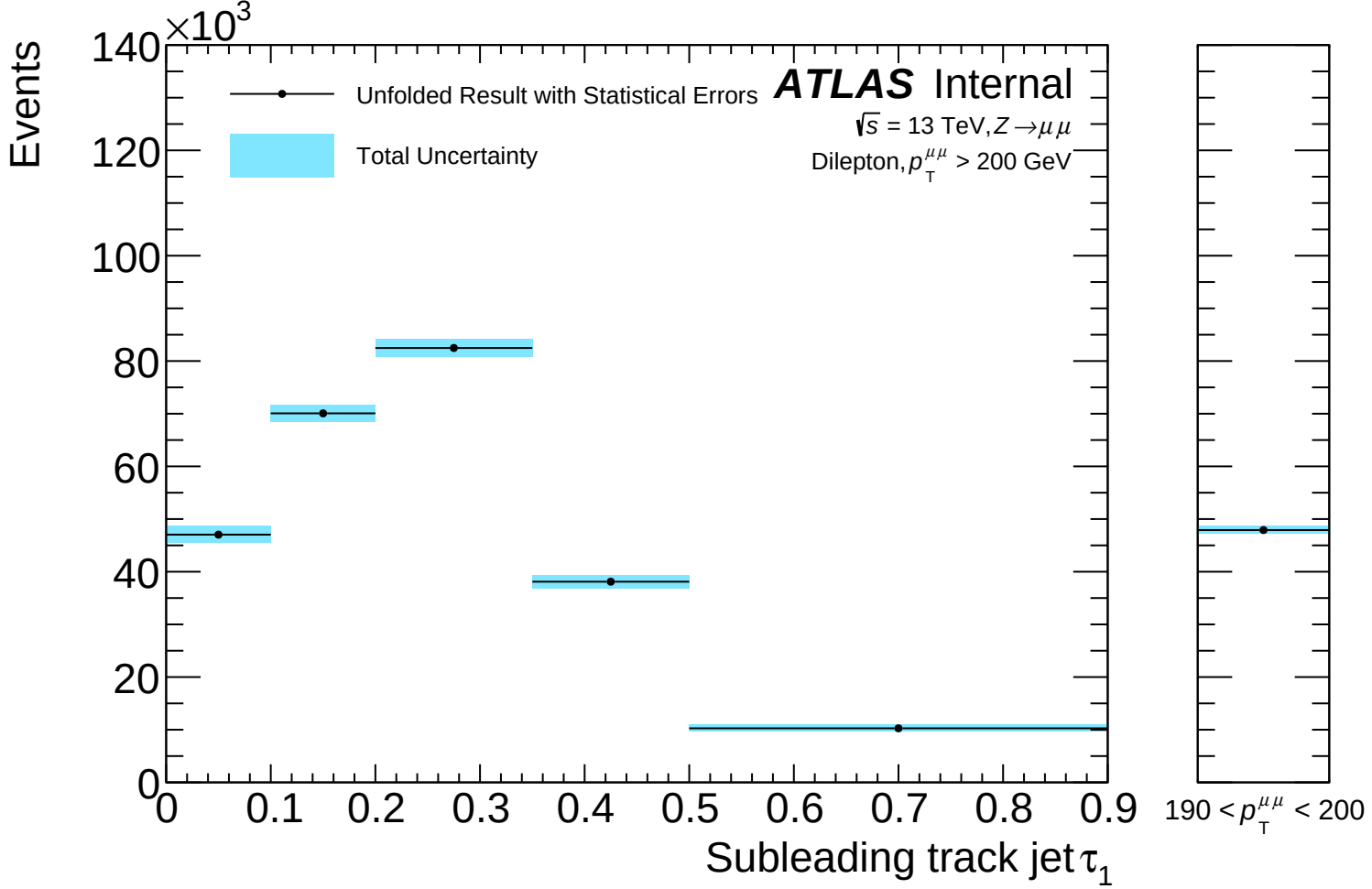
ATLAS Internal

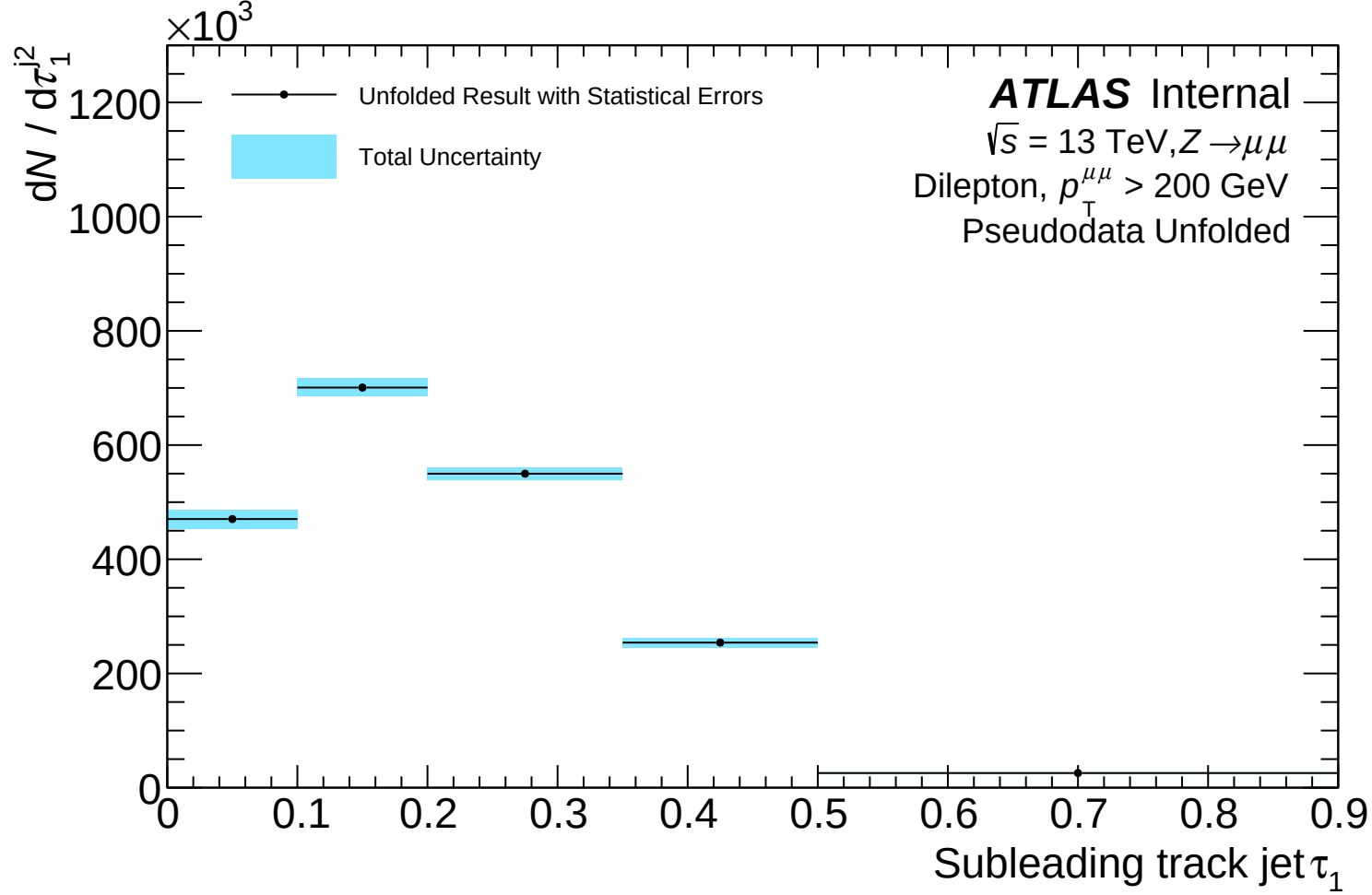
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

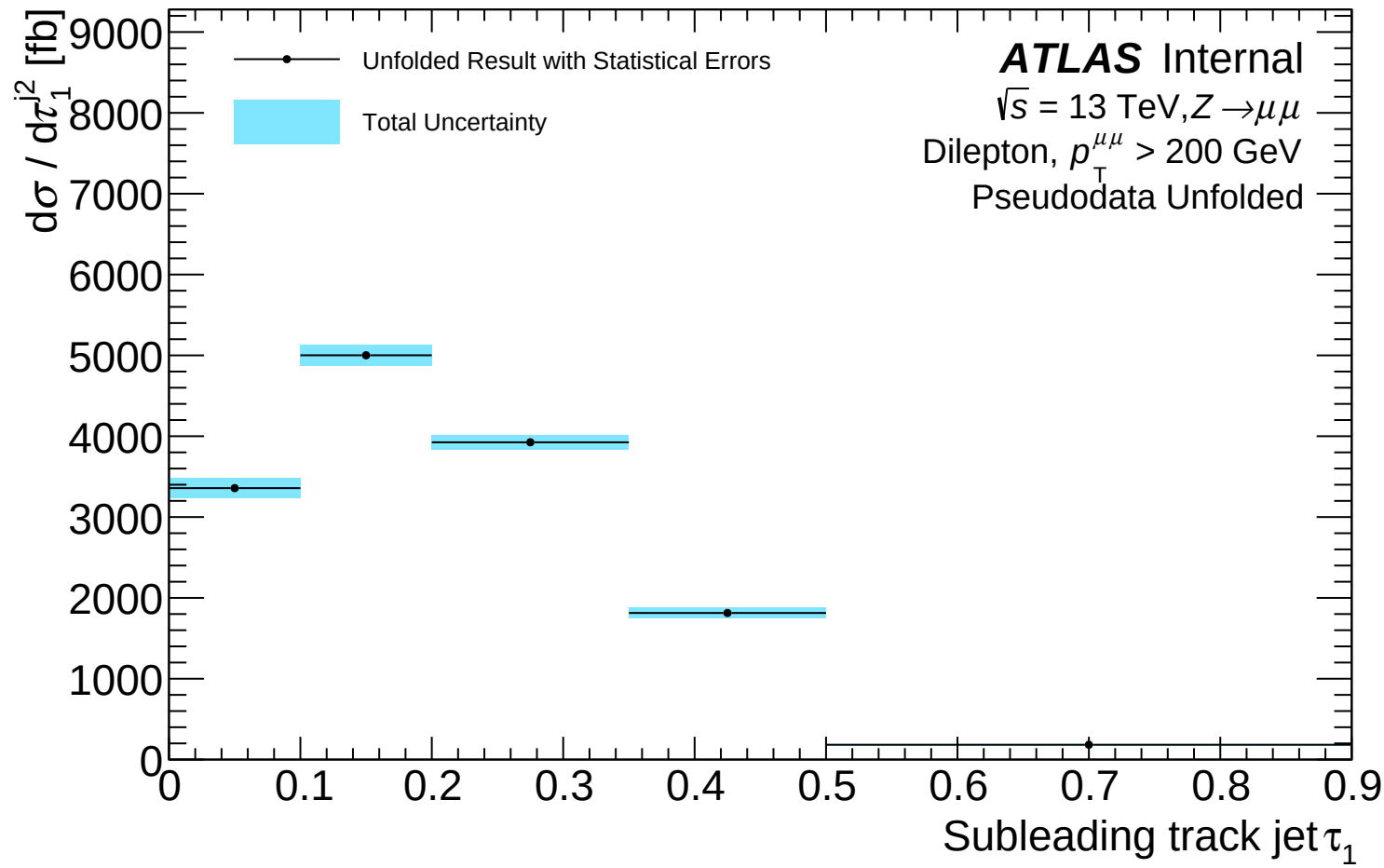
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

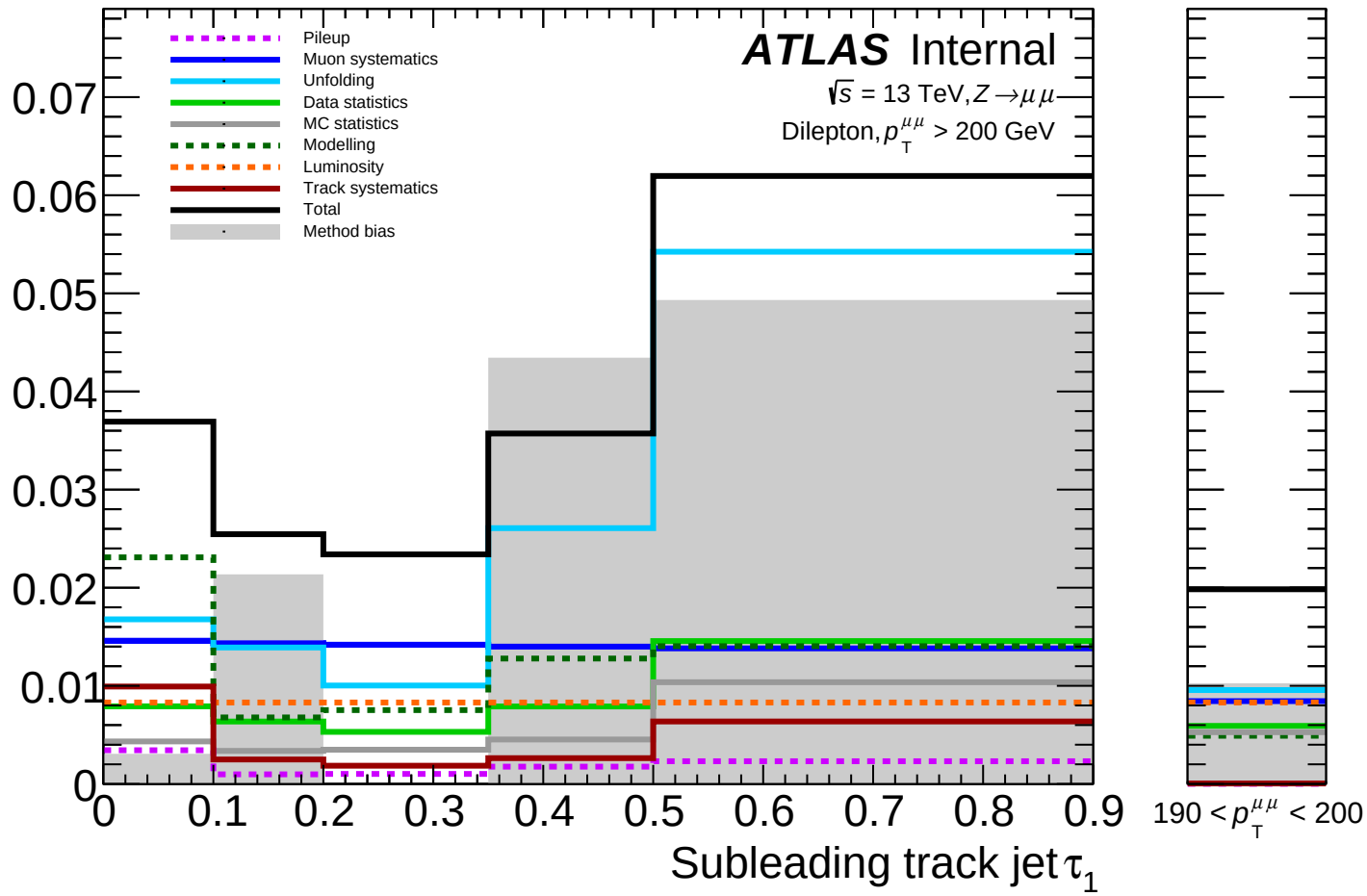


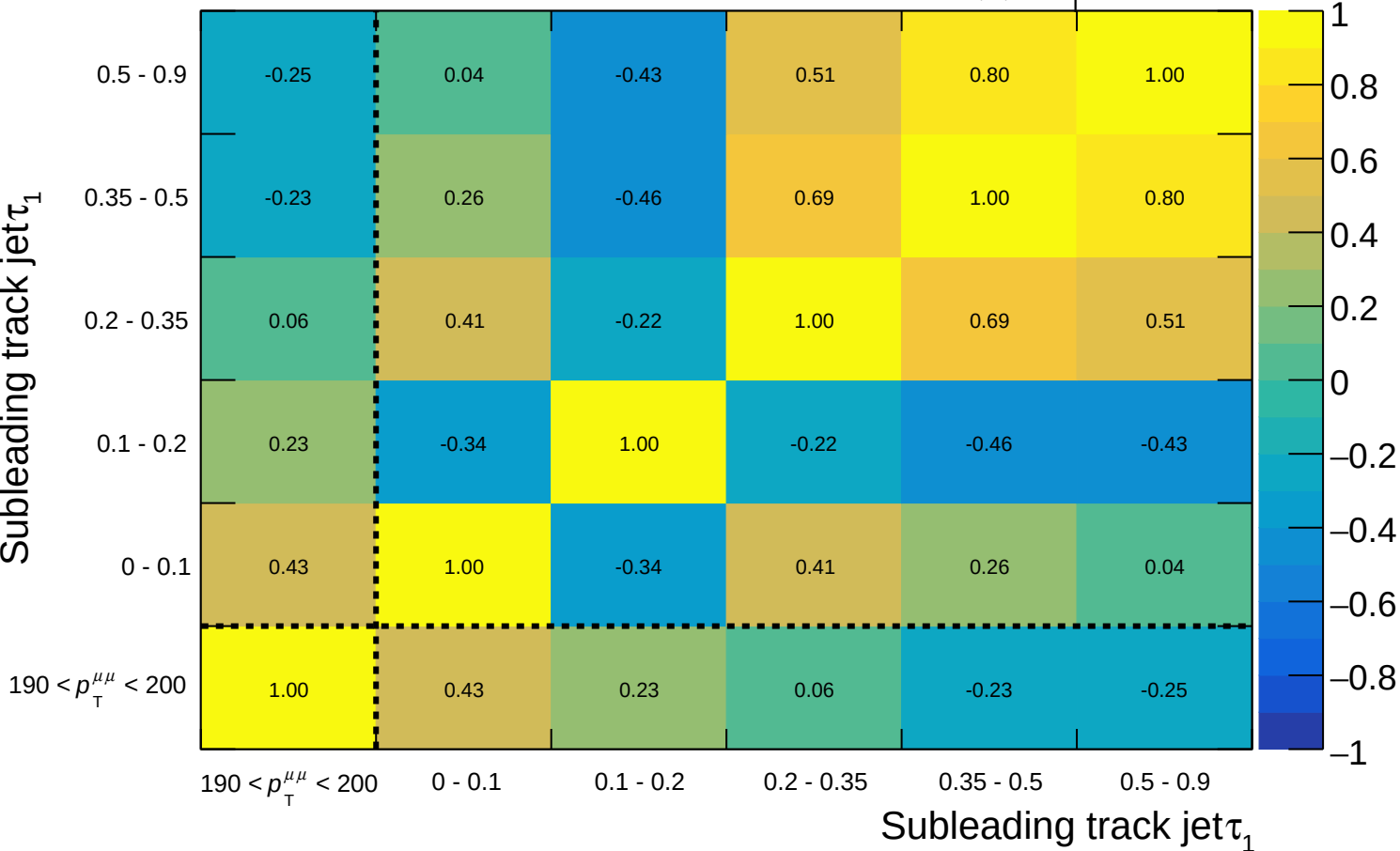


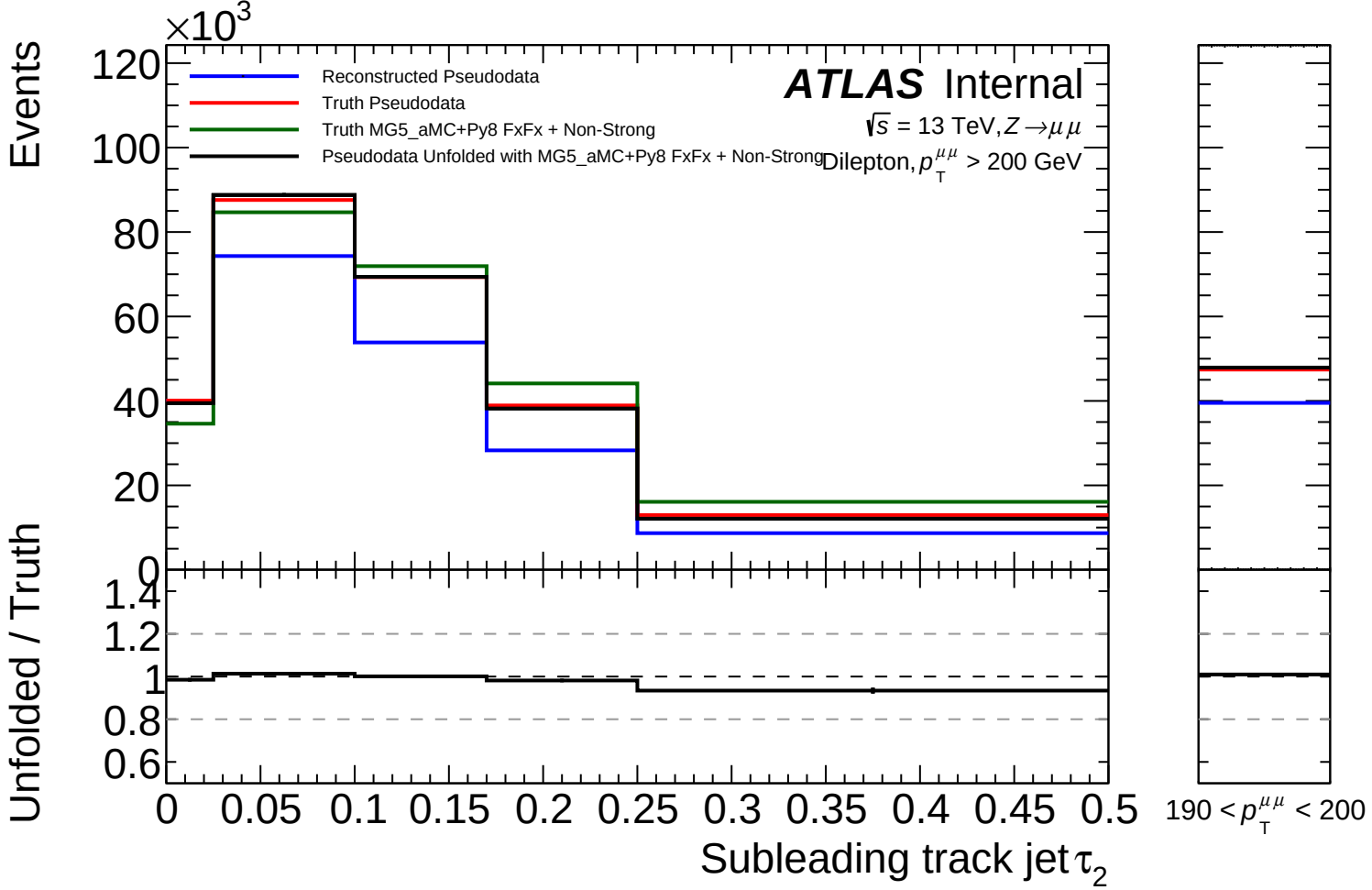


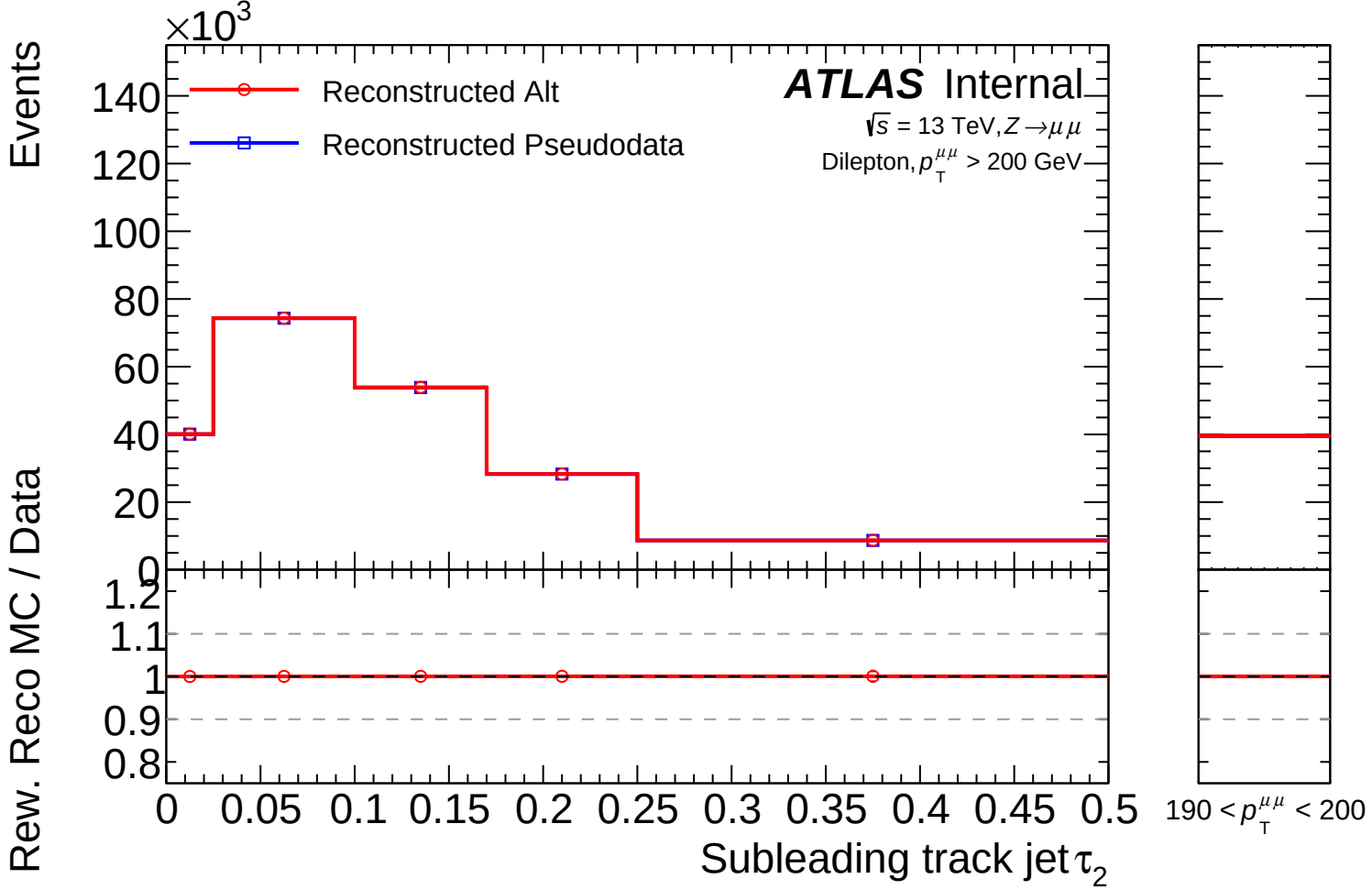


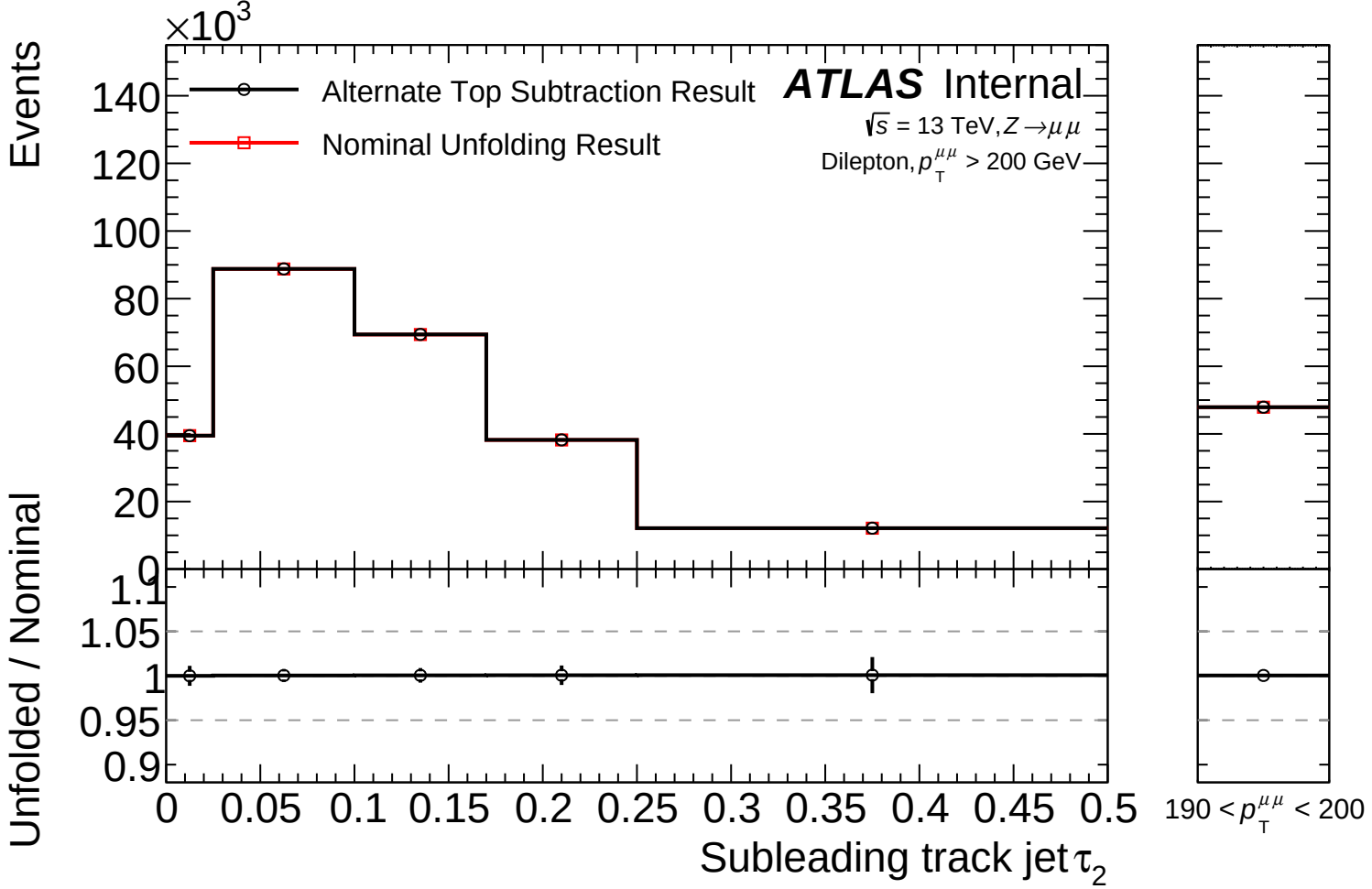




ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Subleading track jet τ_1 







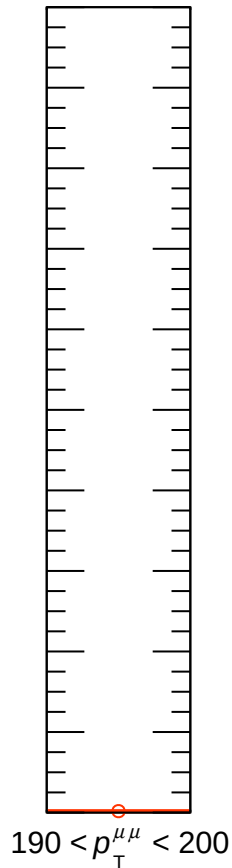
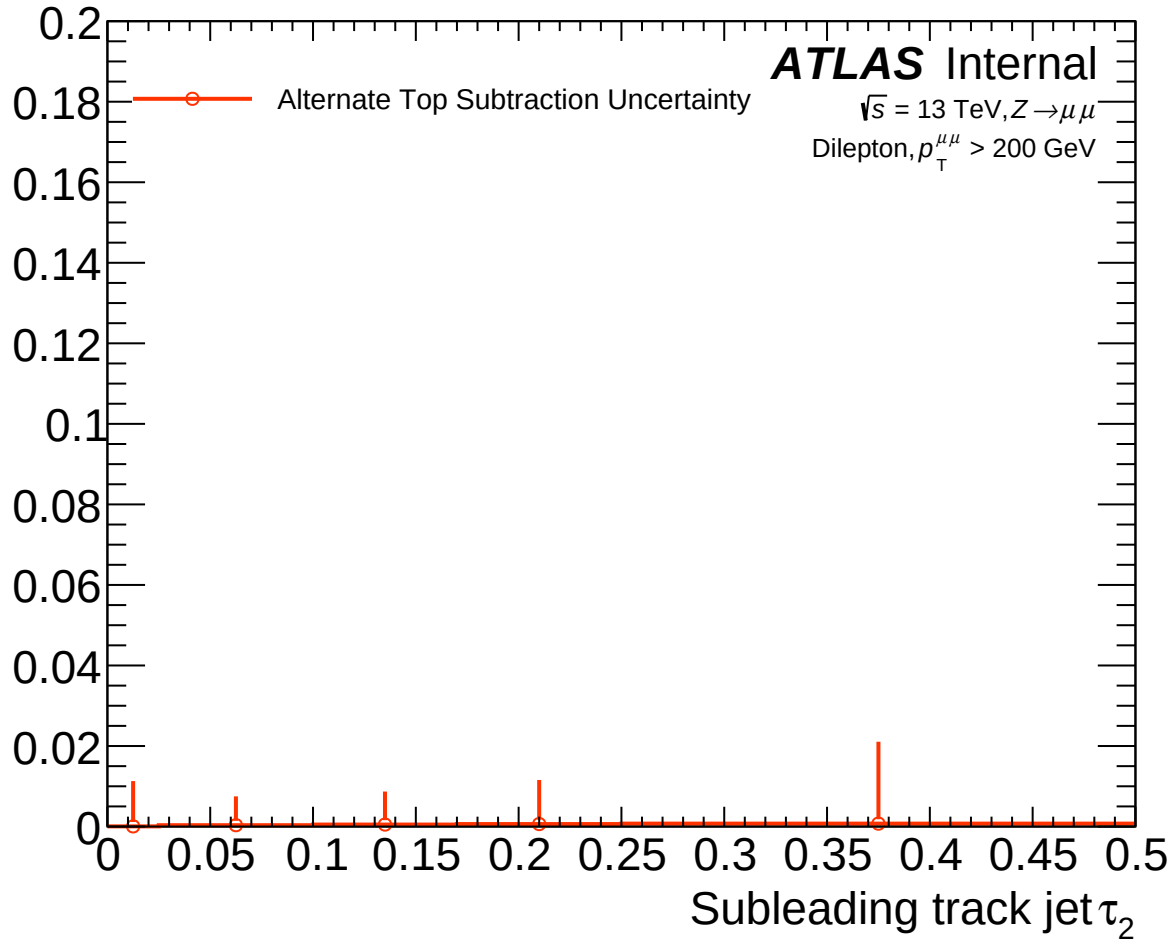
Fractional systematic impact

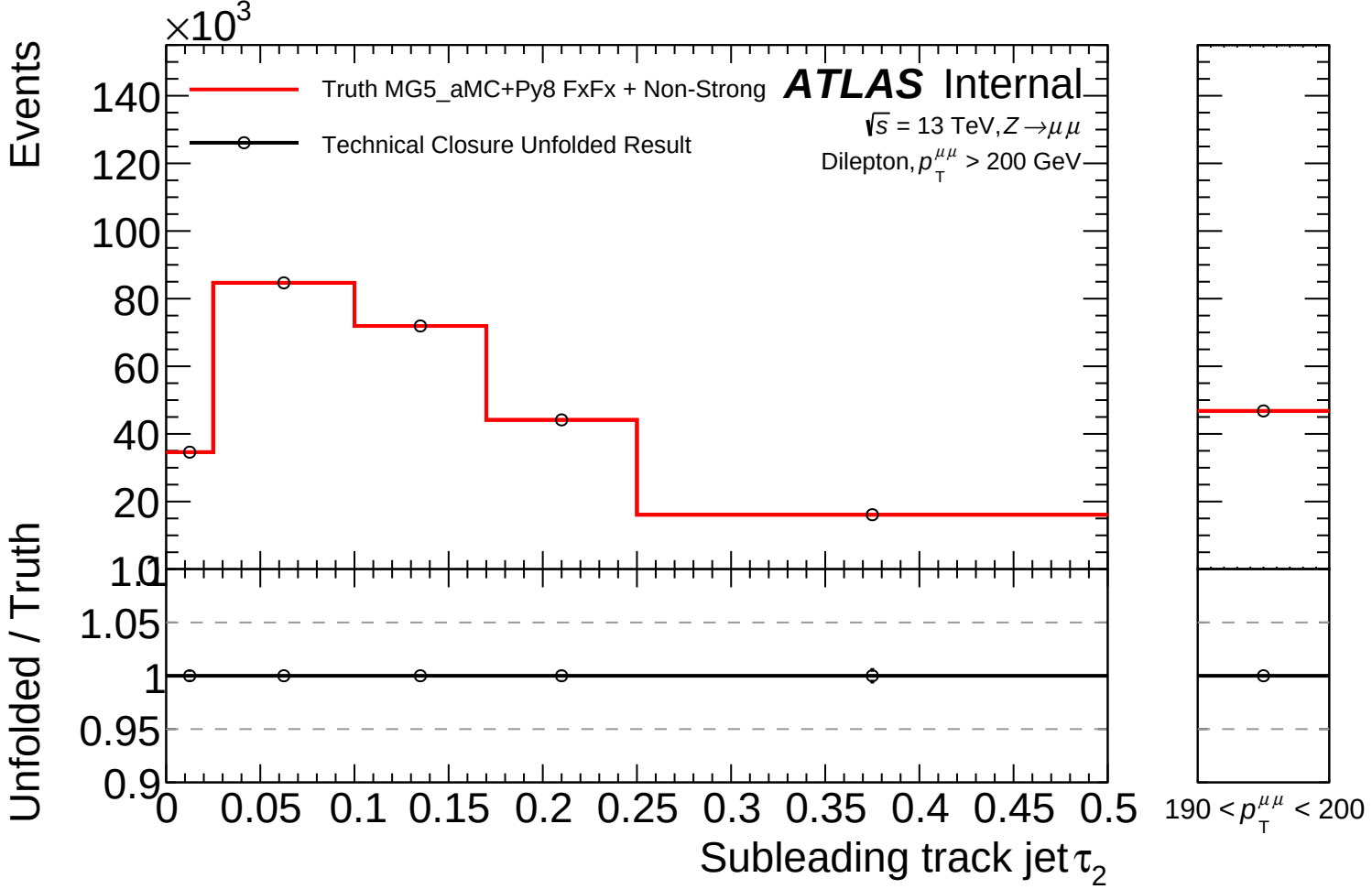
ATLAS Internal

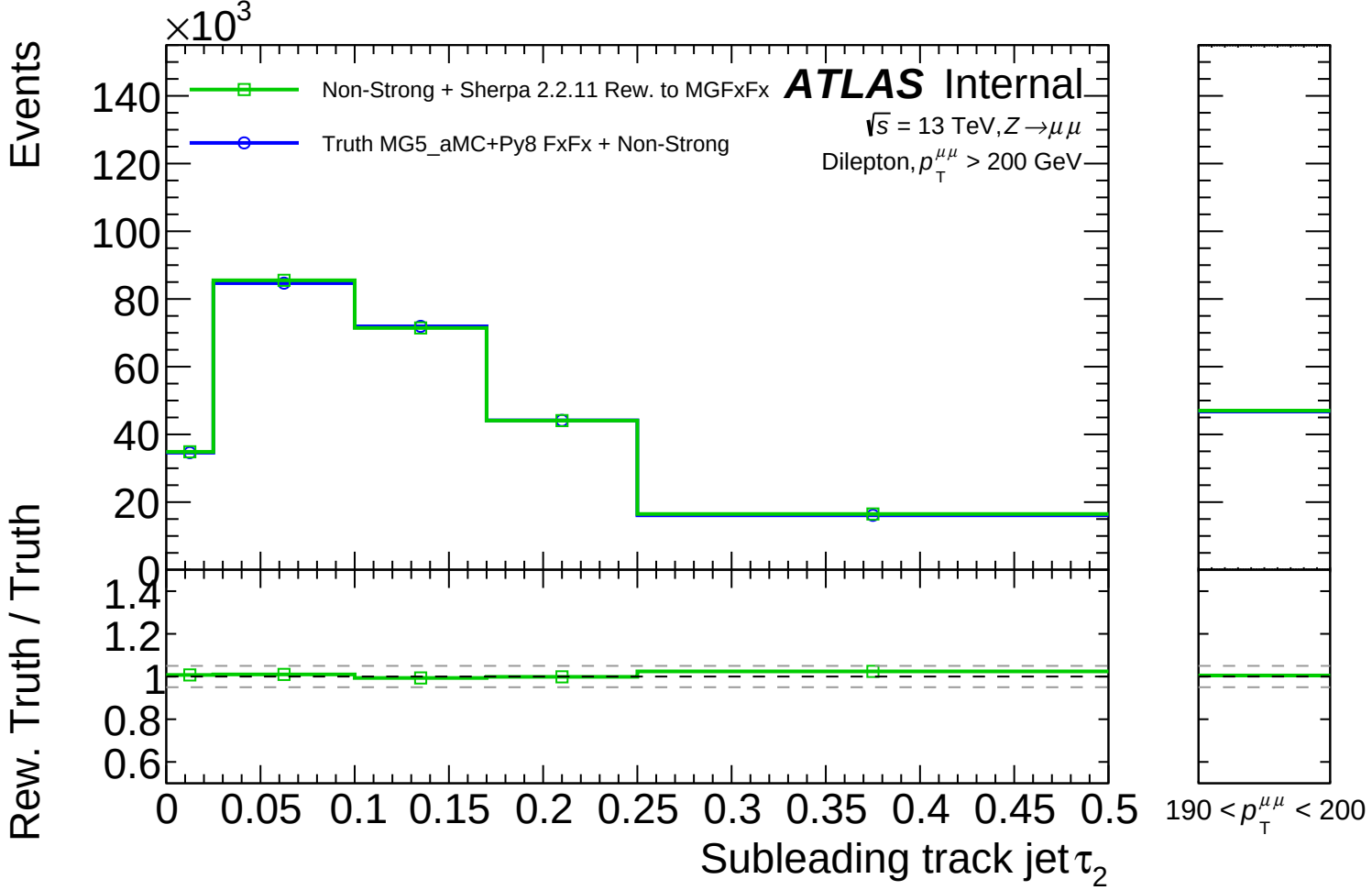
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

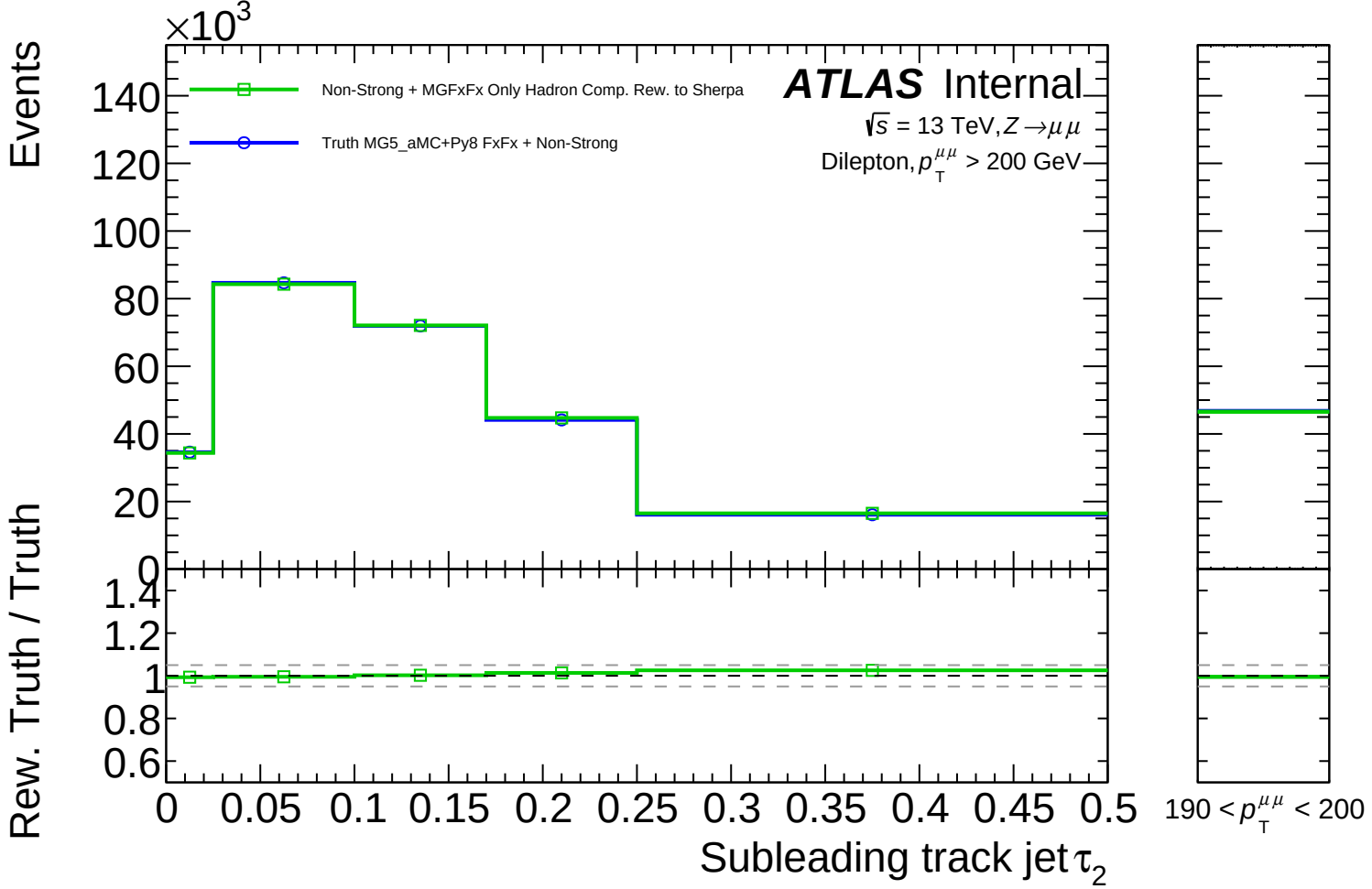
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

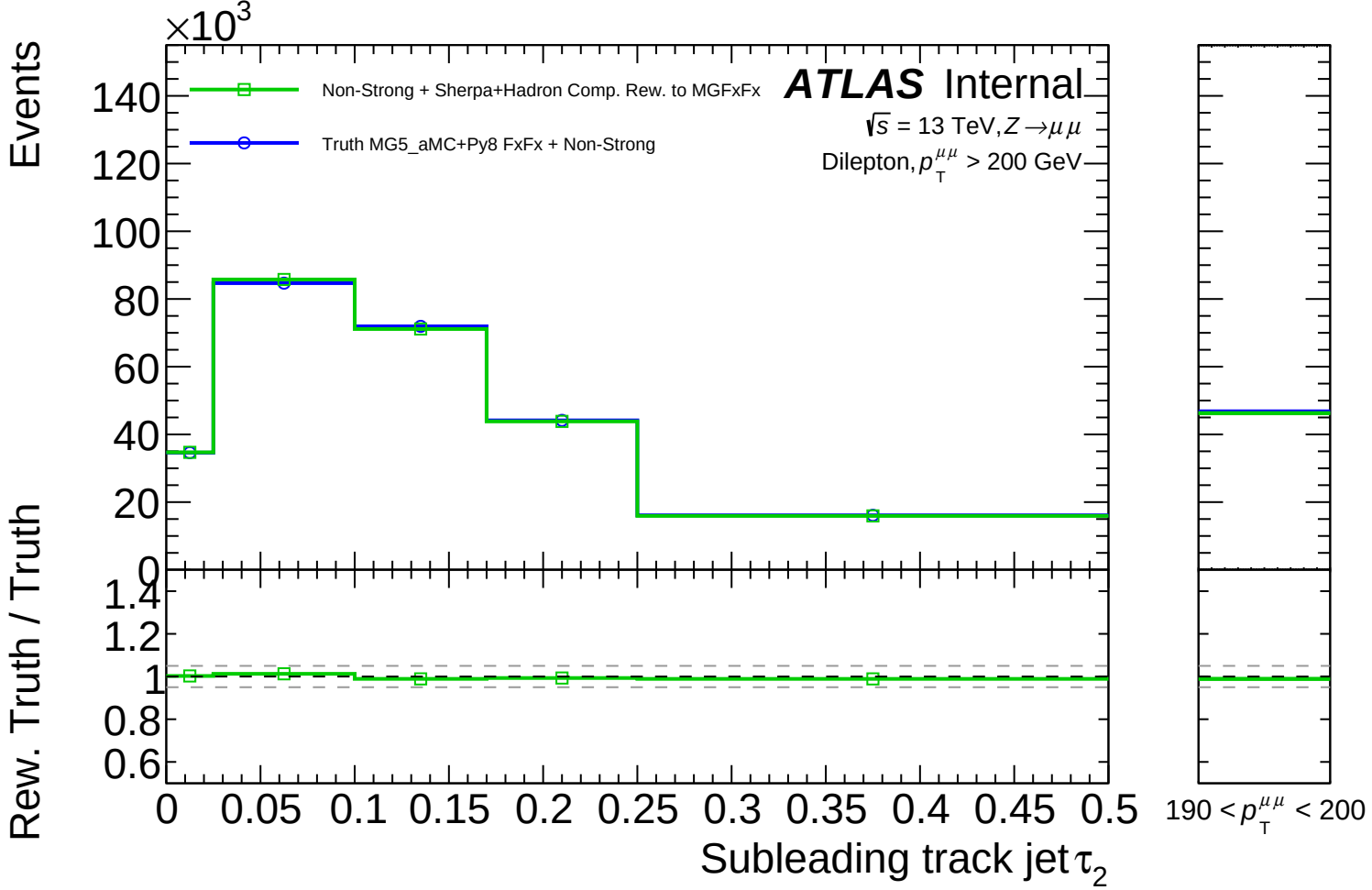
Alternate Top Subtraction Uncertainty

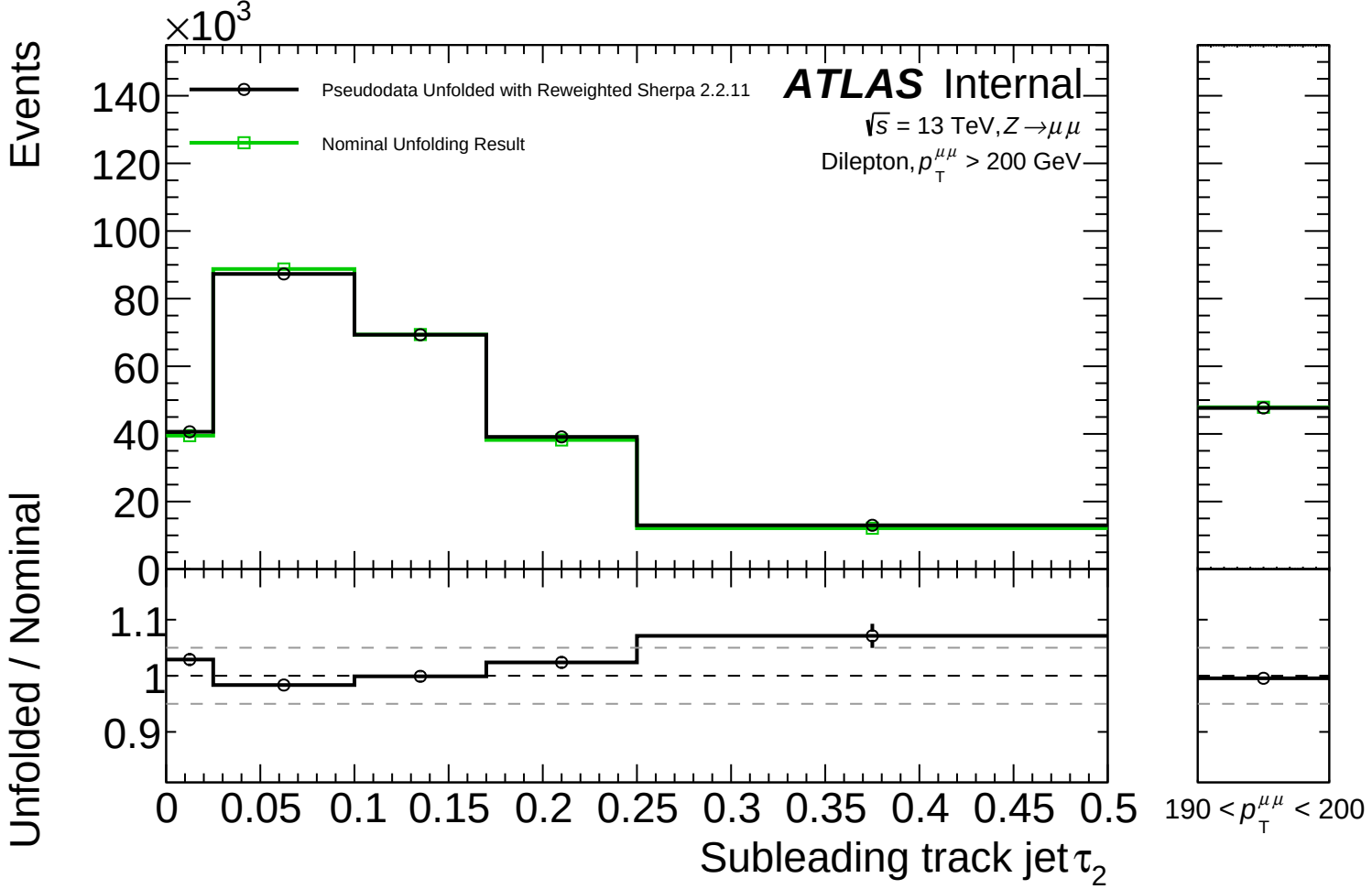


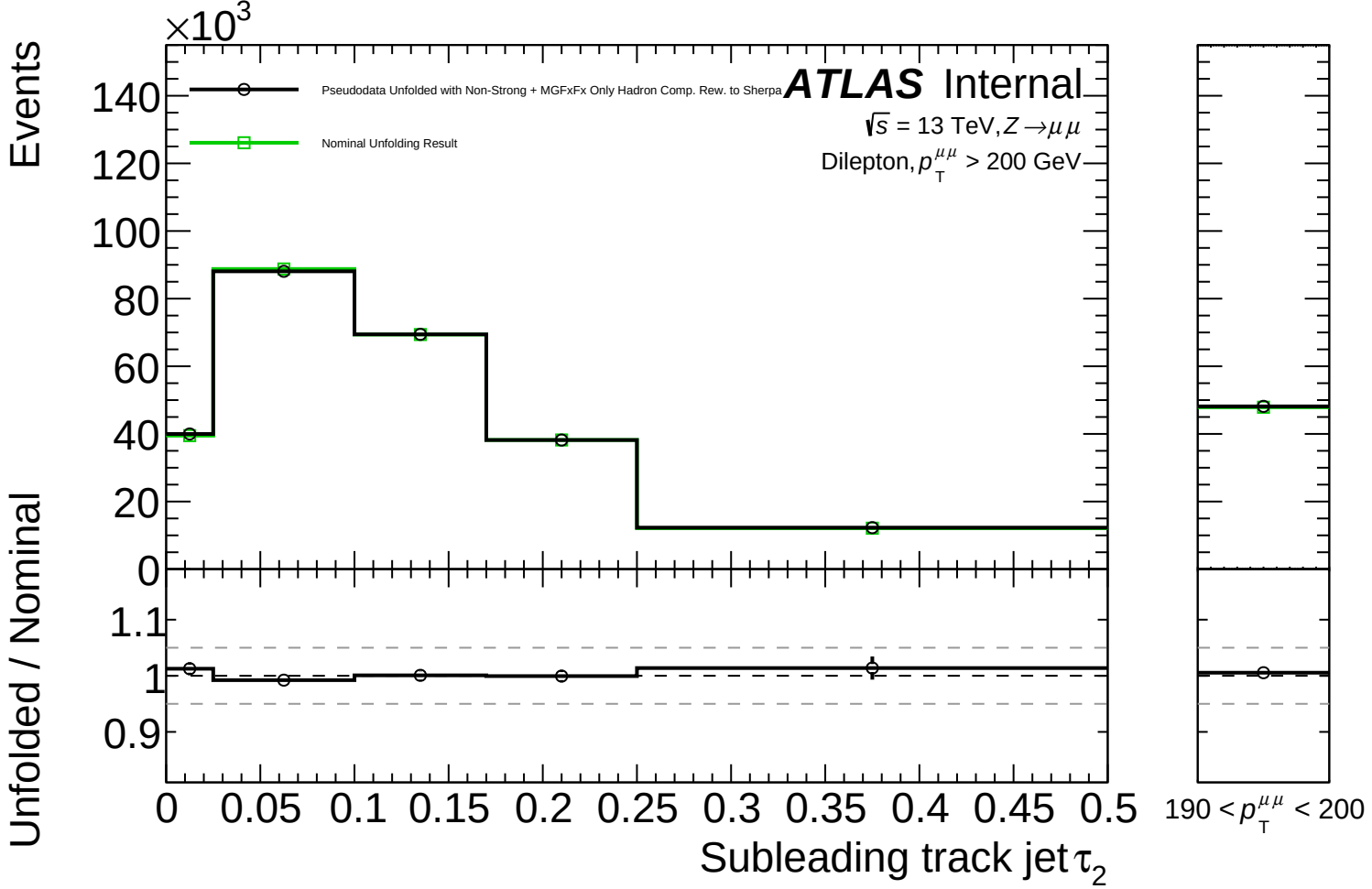


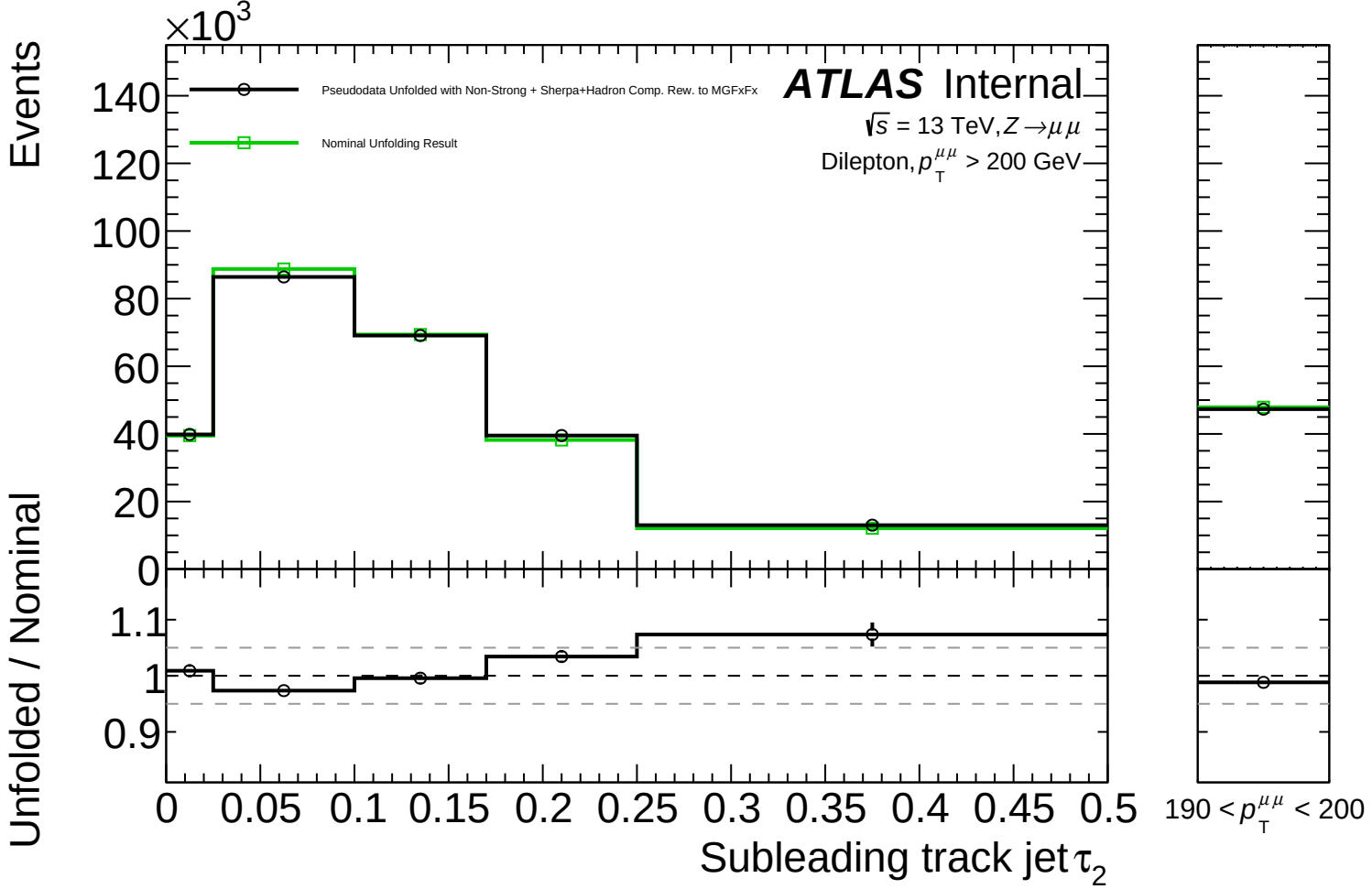


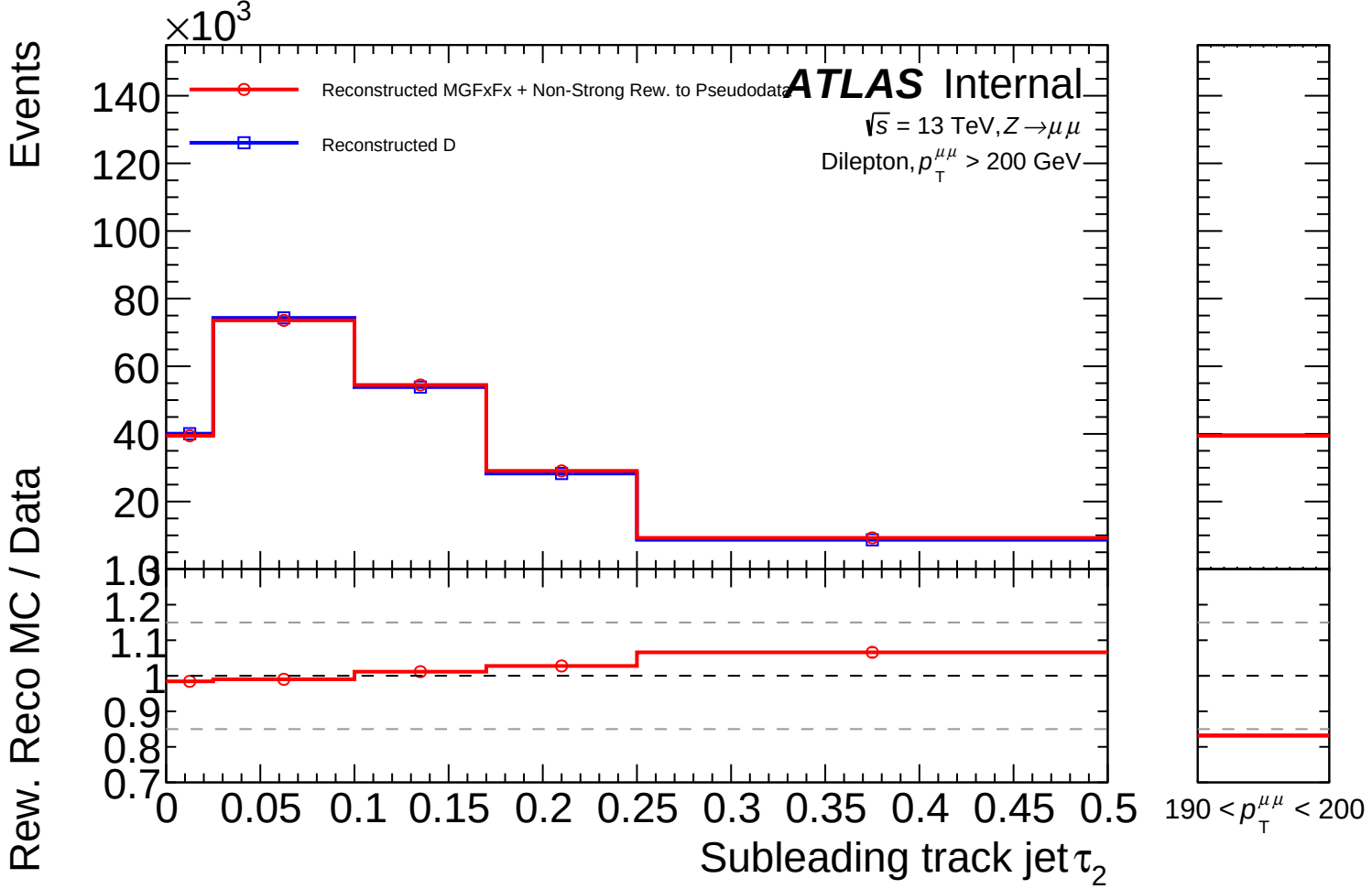


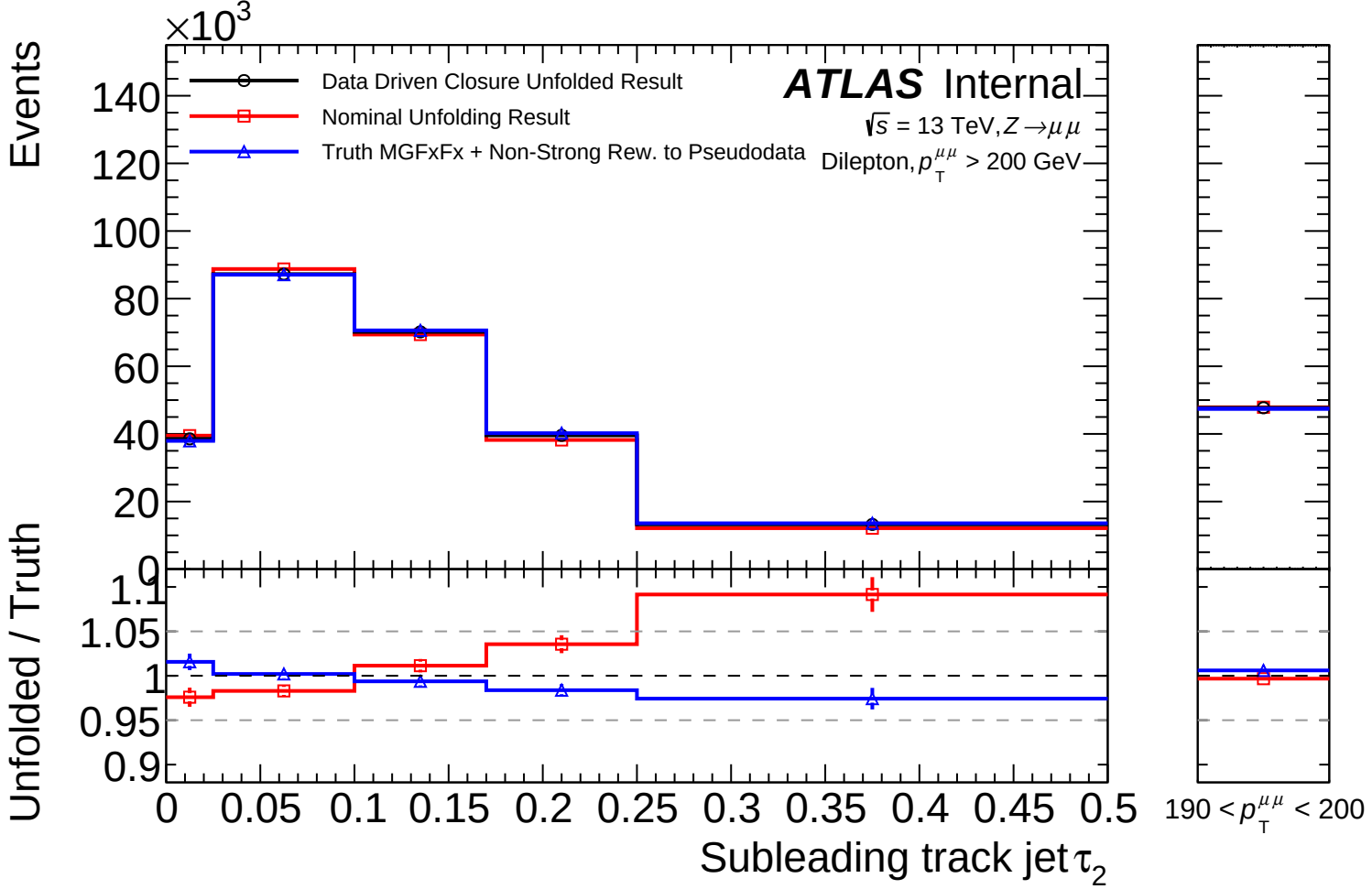




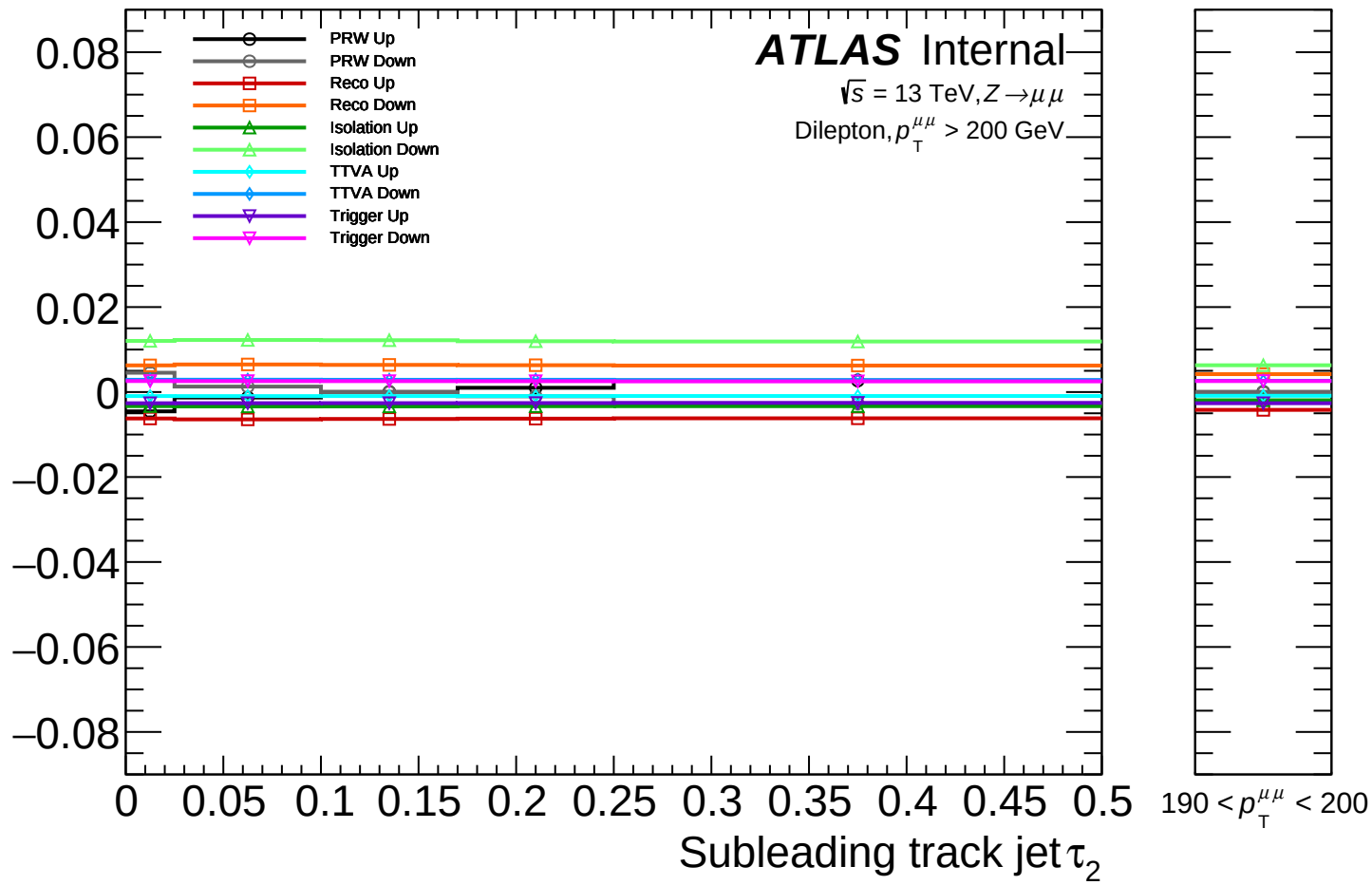




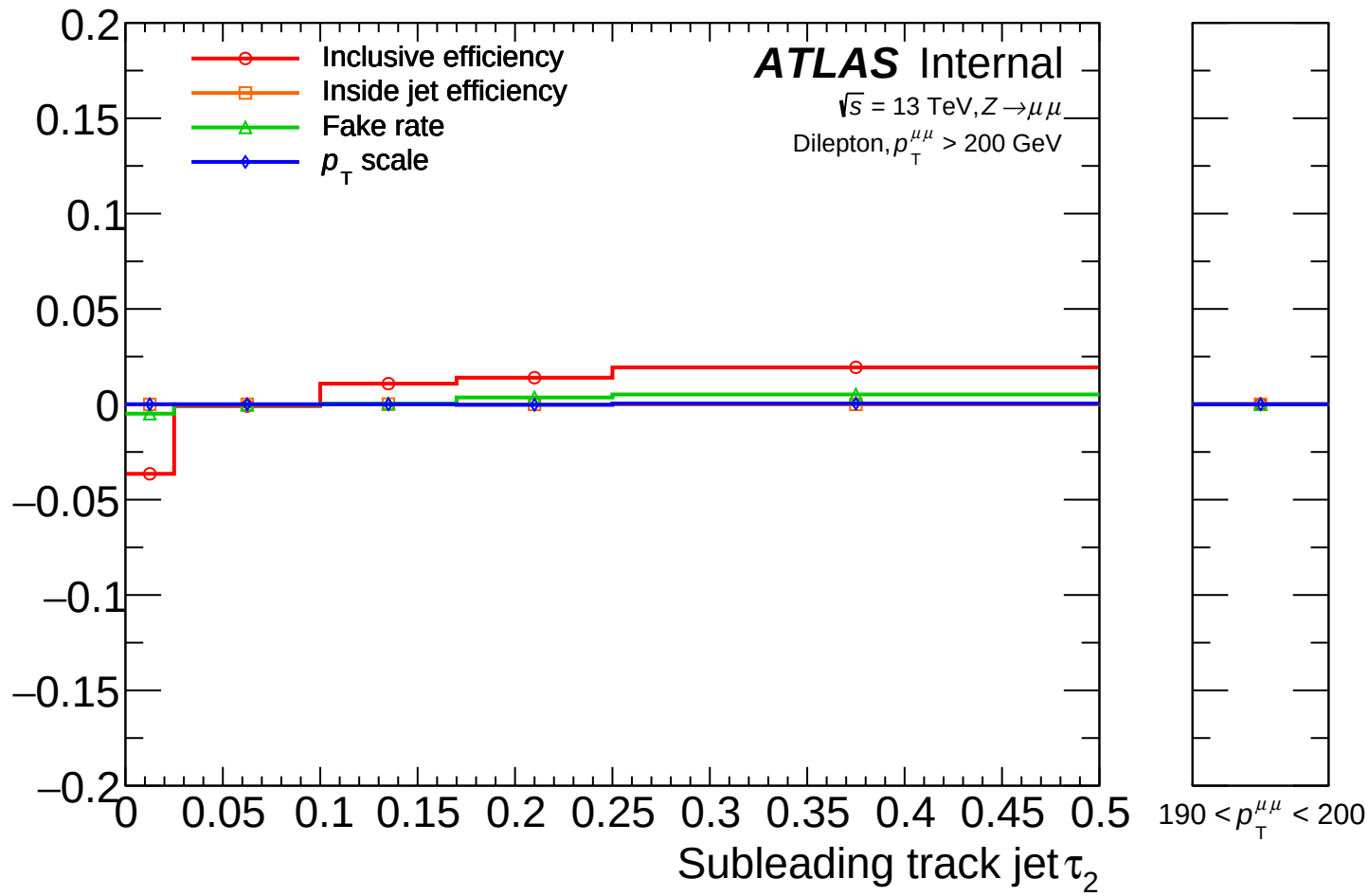




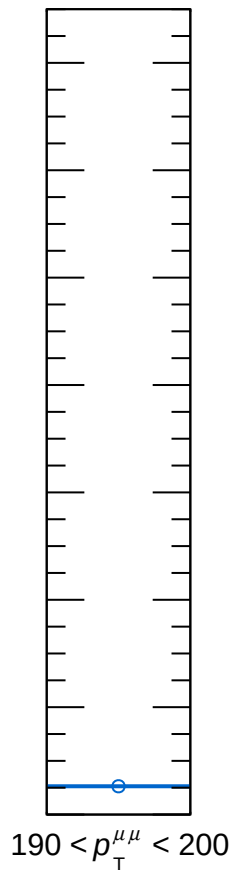
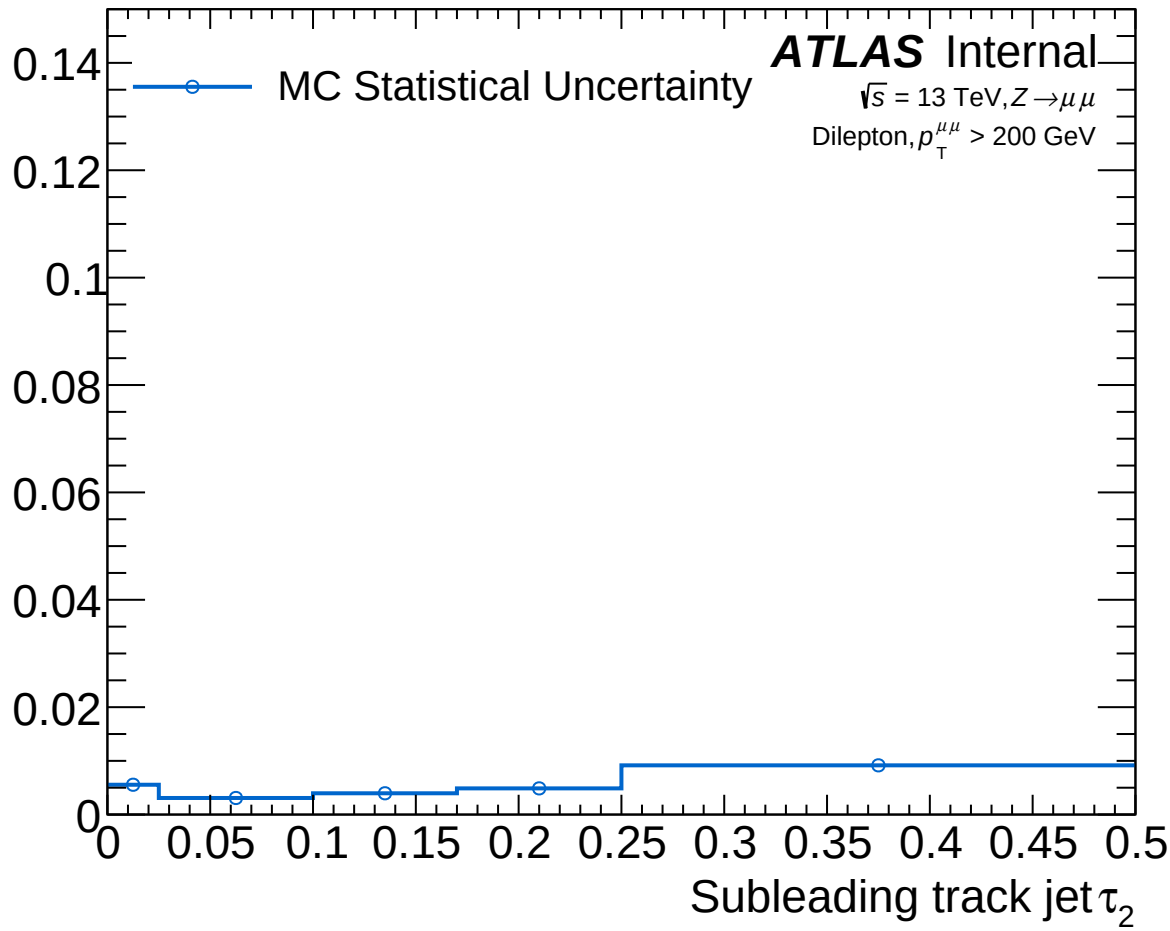
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



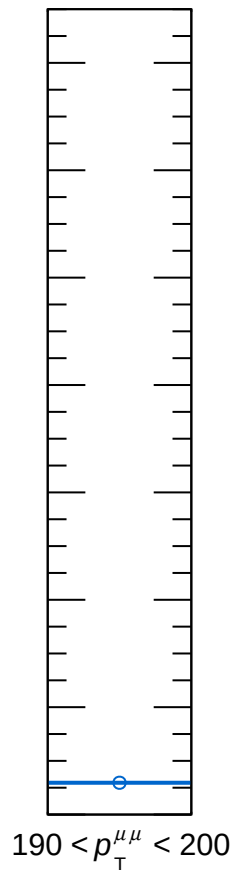
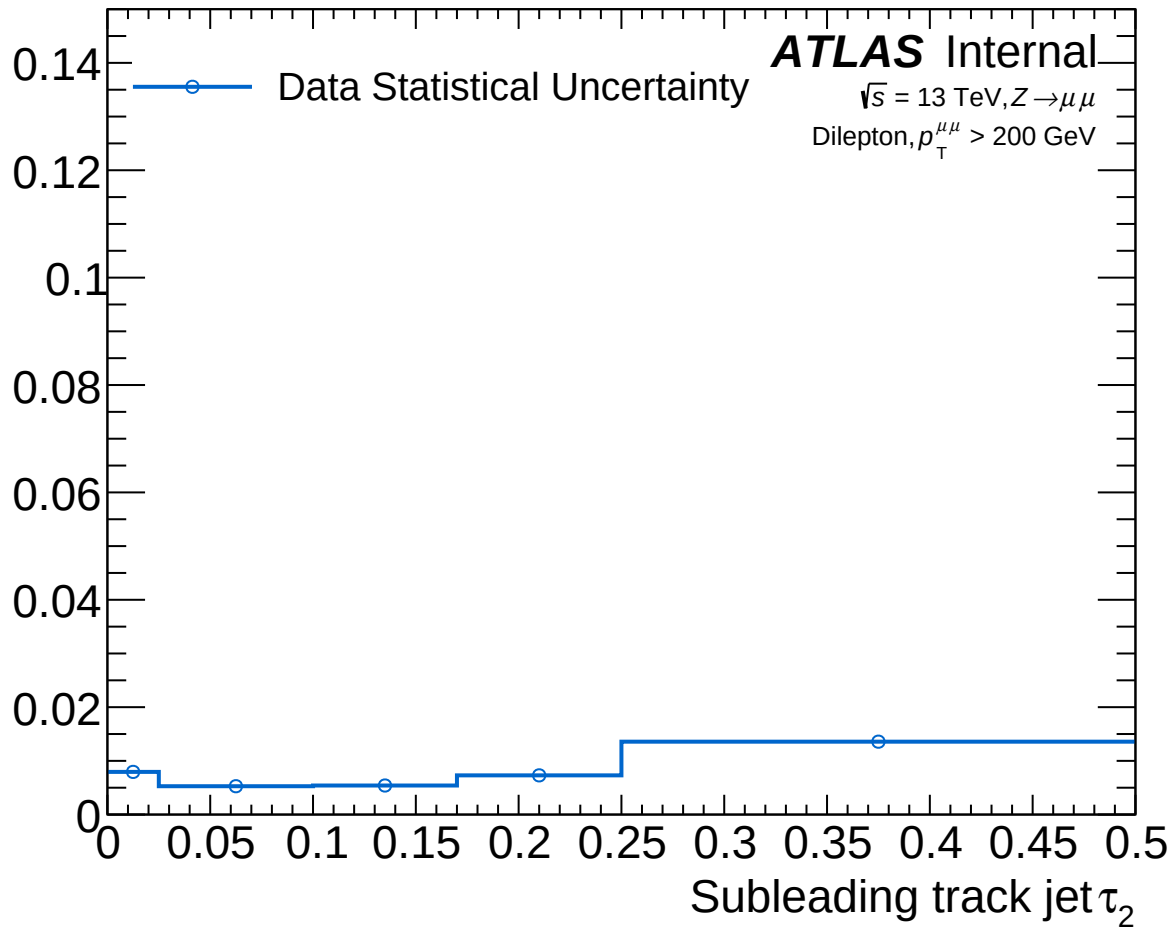
Fractional systematic impact

ATLAS Internal

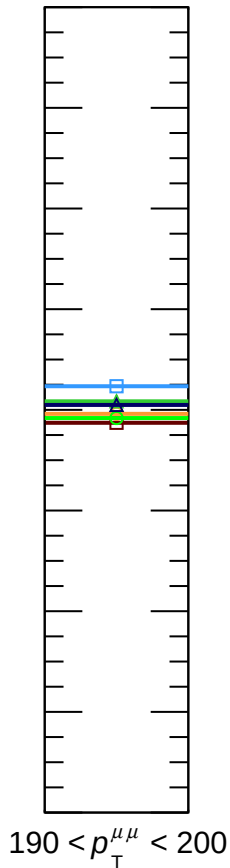
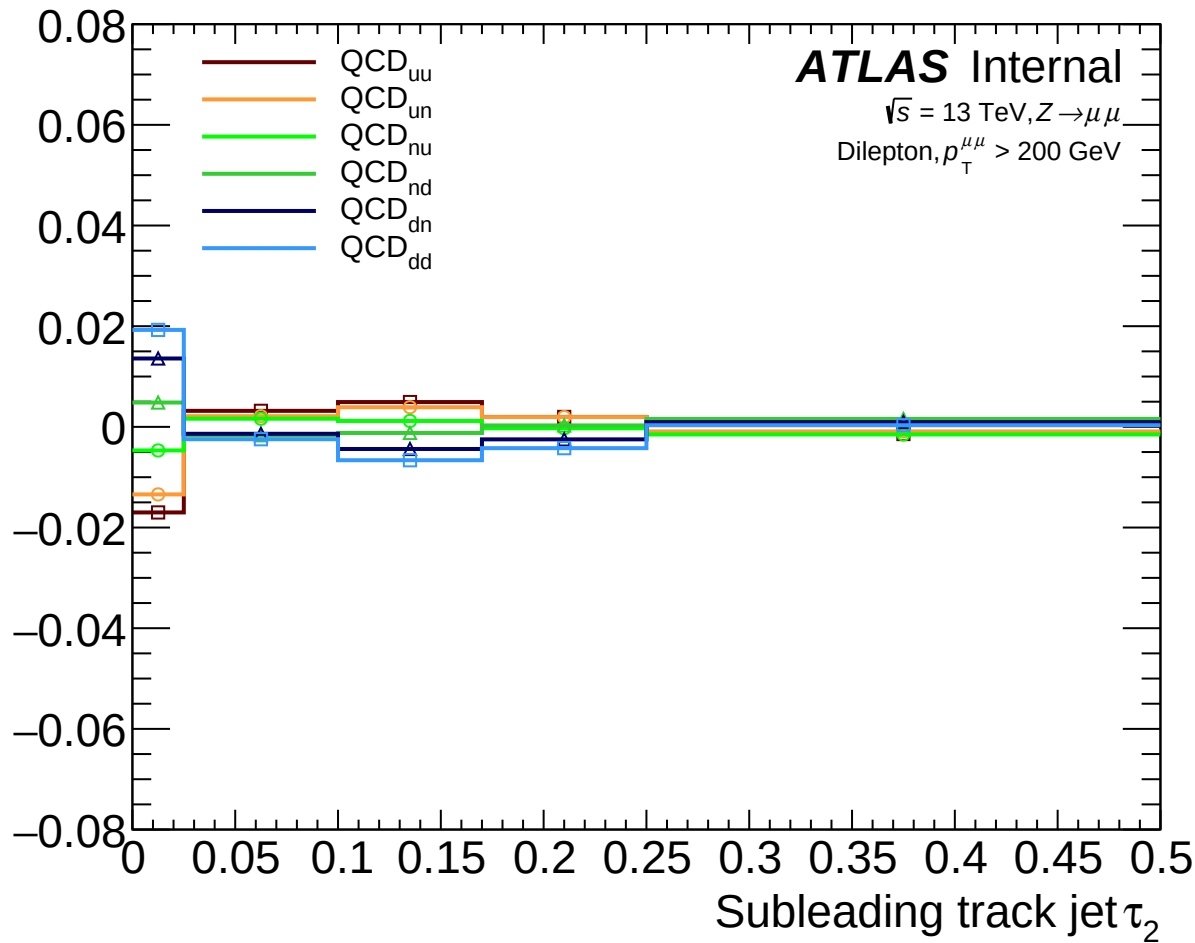
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

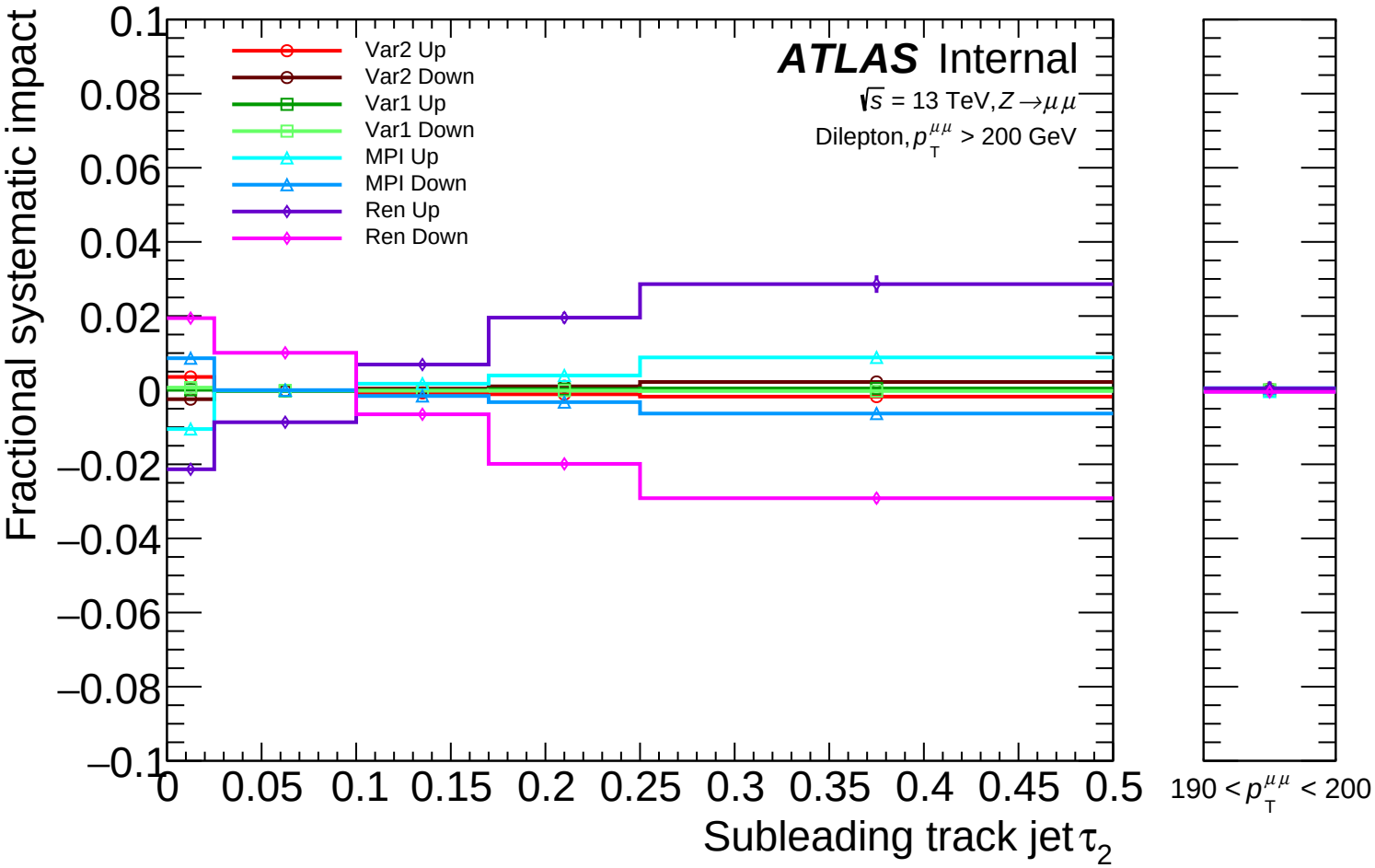
Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

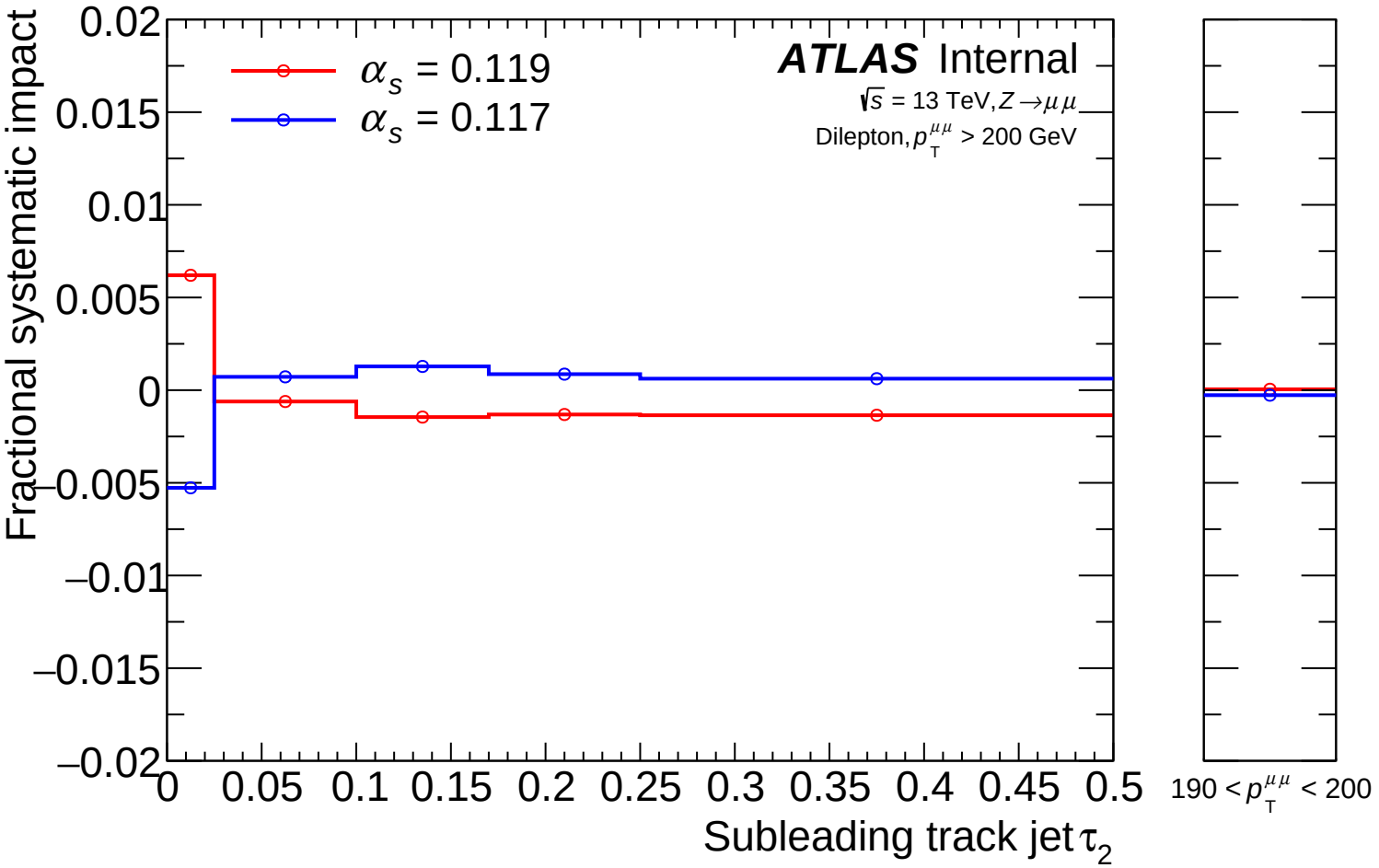
Data Statistical Uncertainty

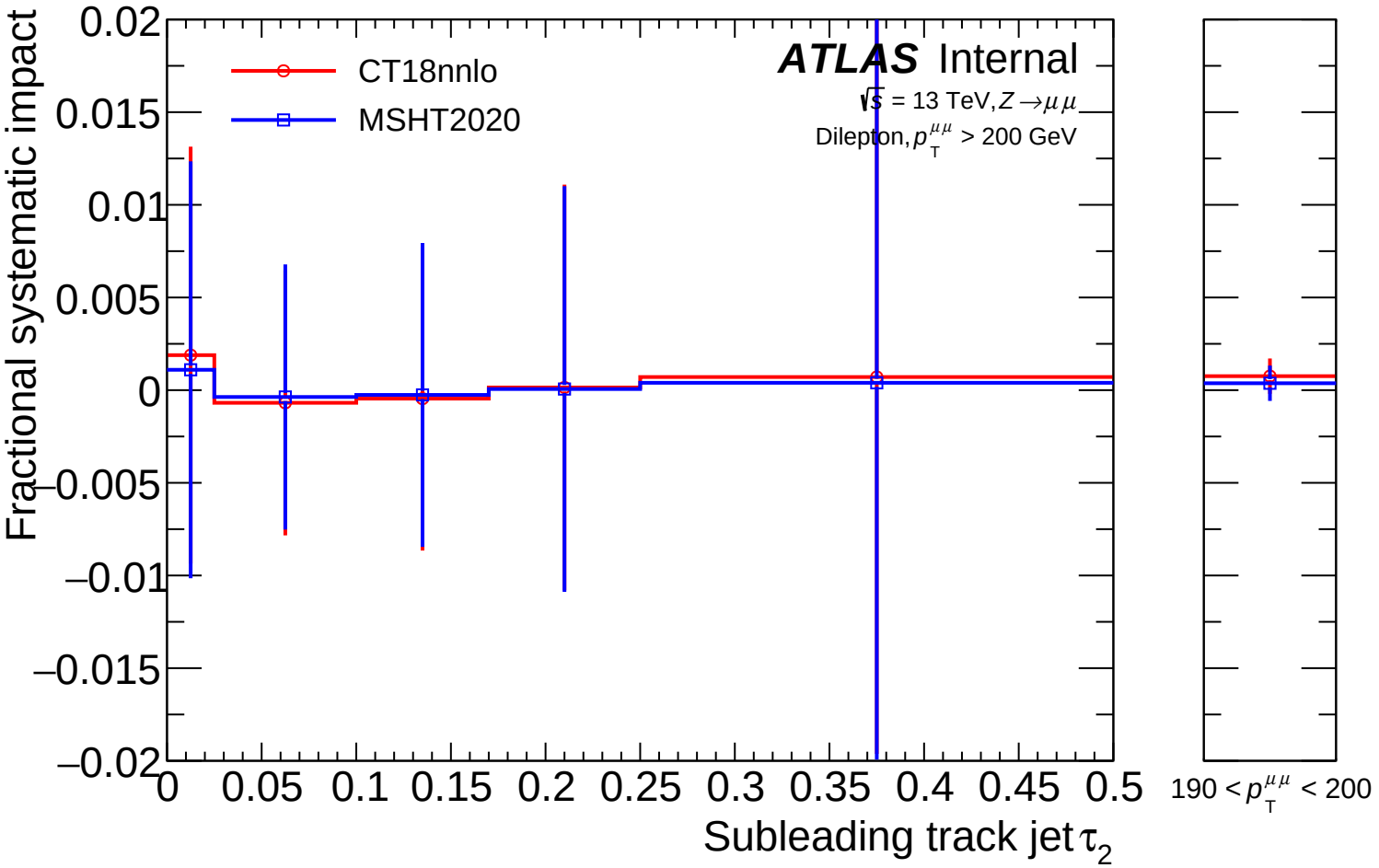


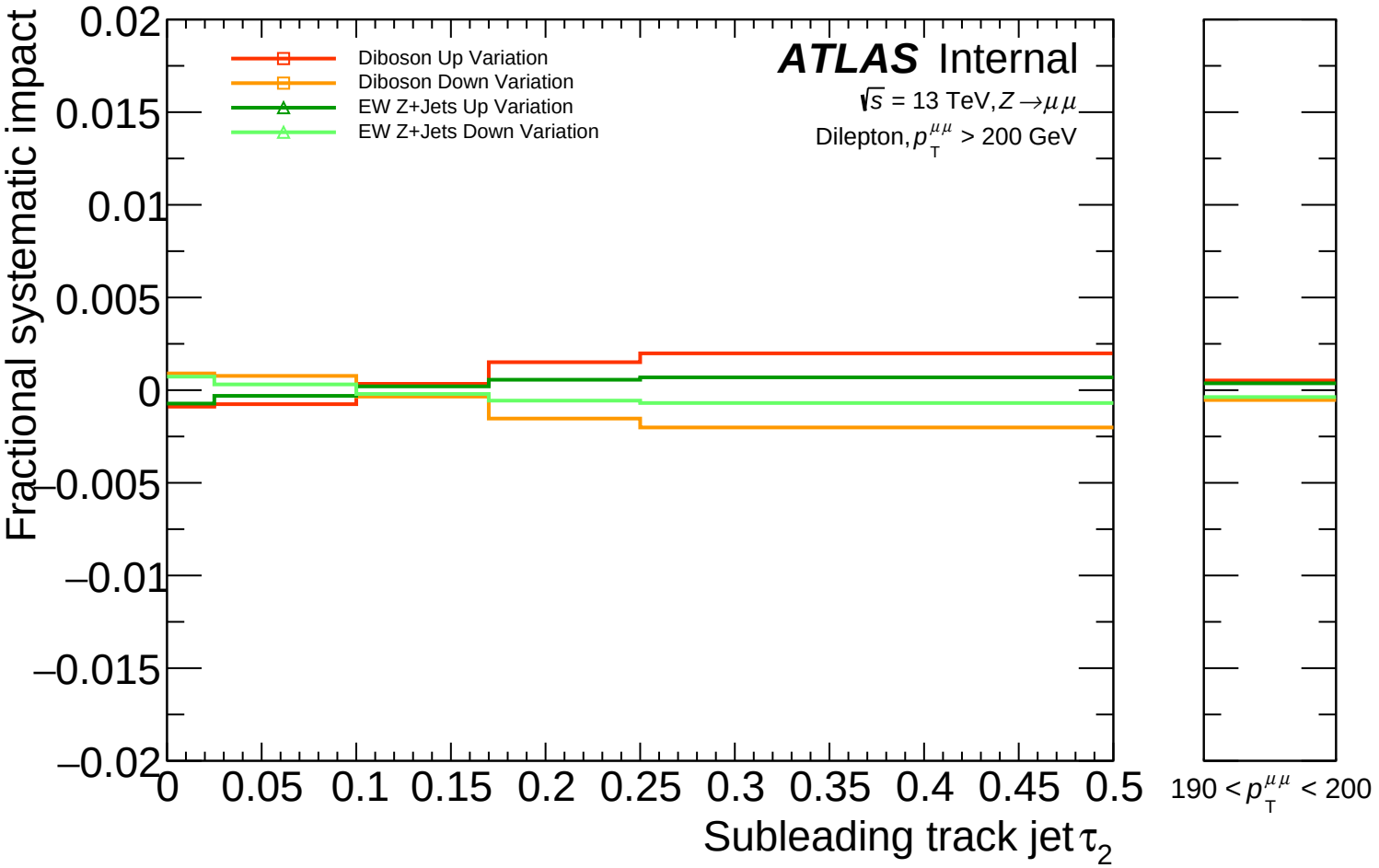
Fractional systematic impact









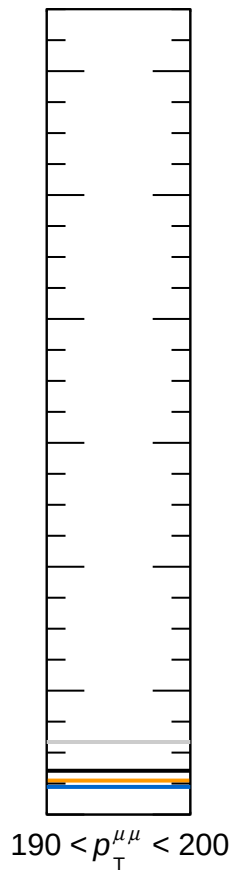
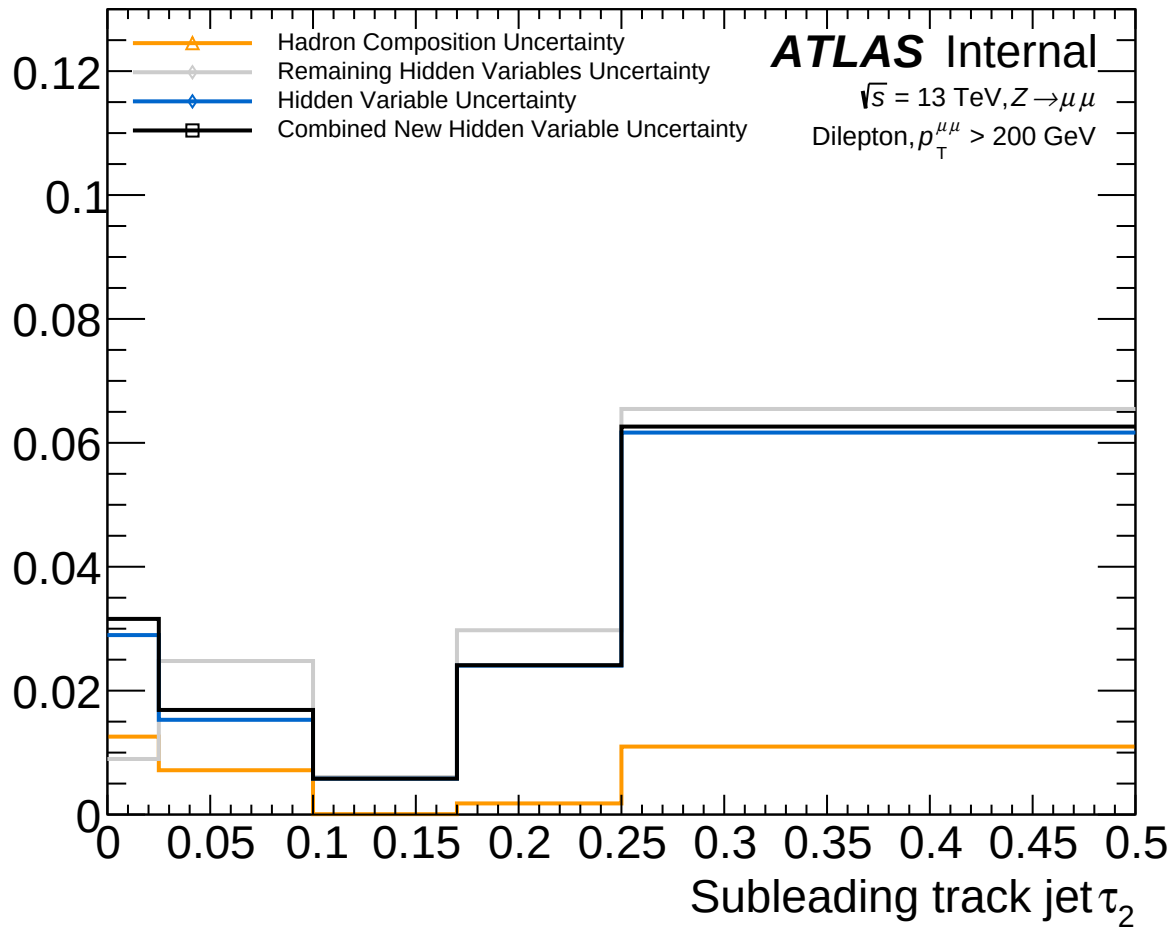


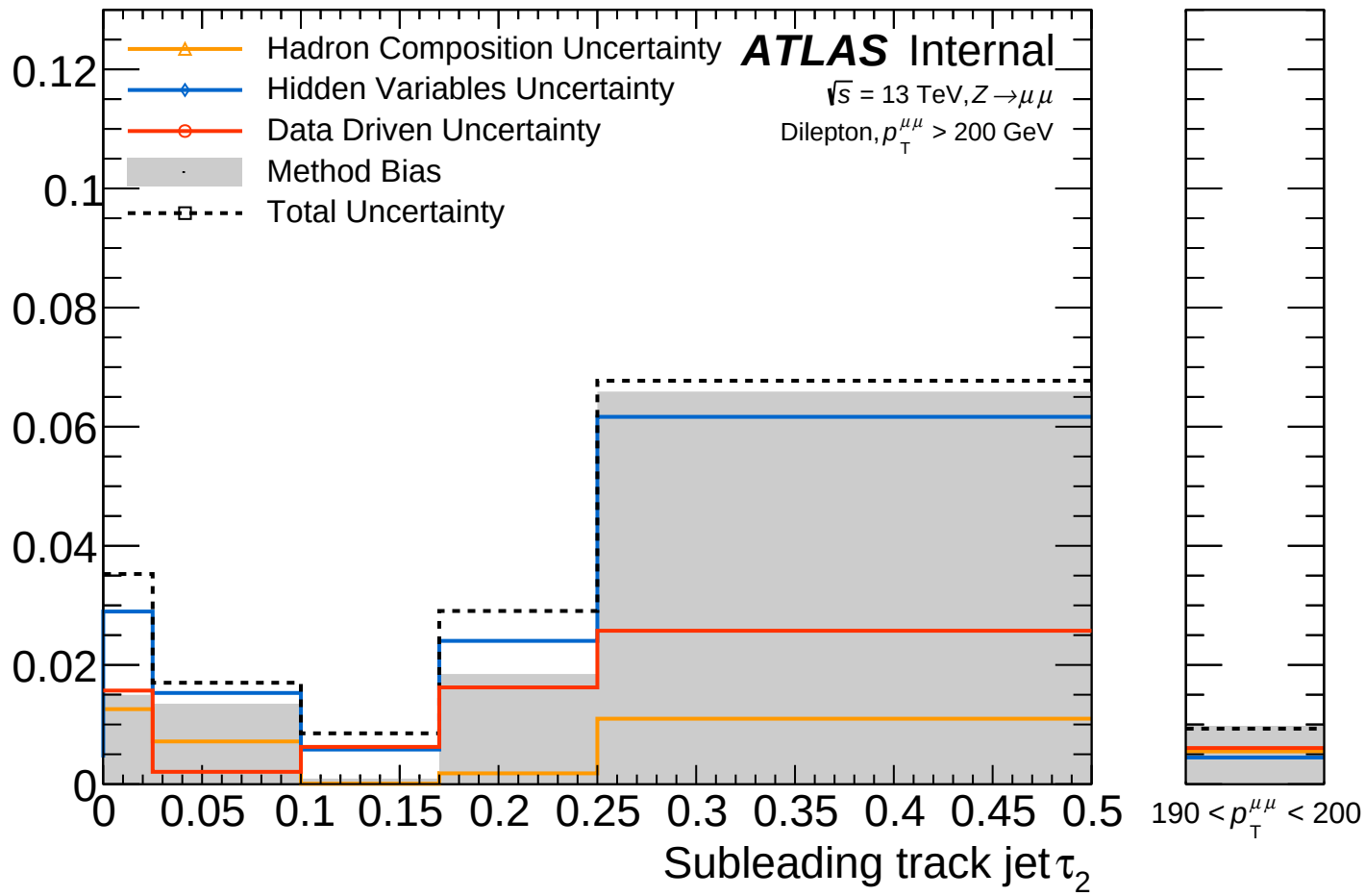
ATLAS Internal

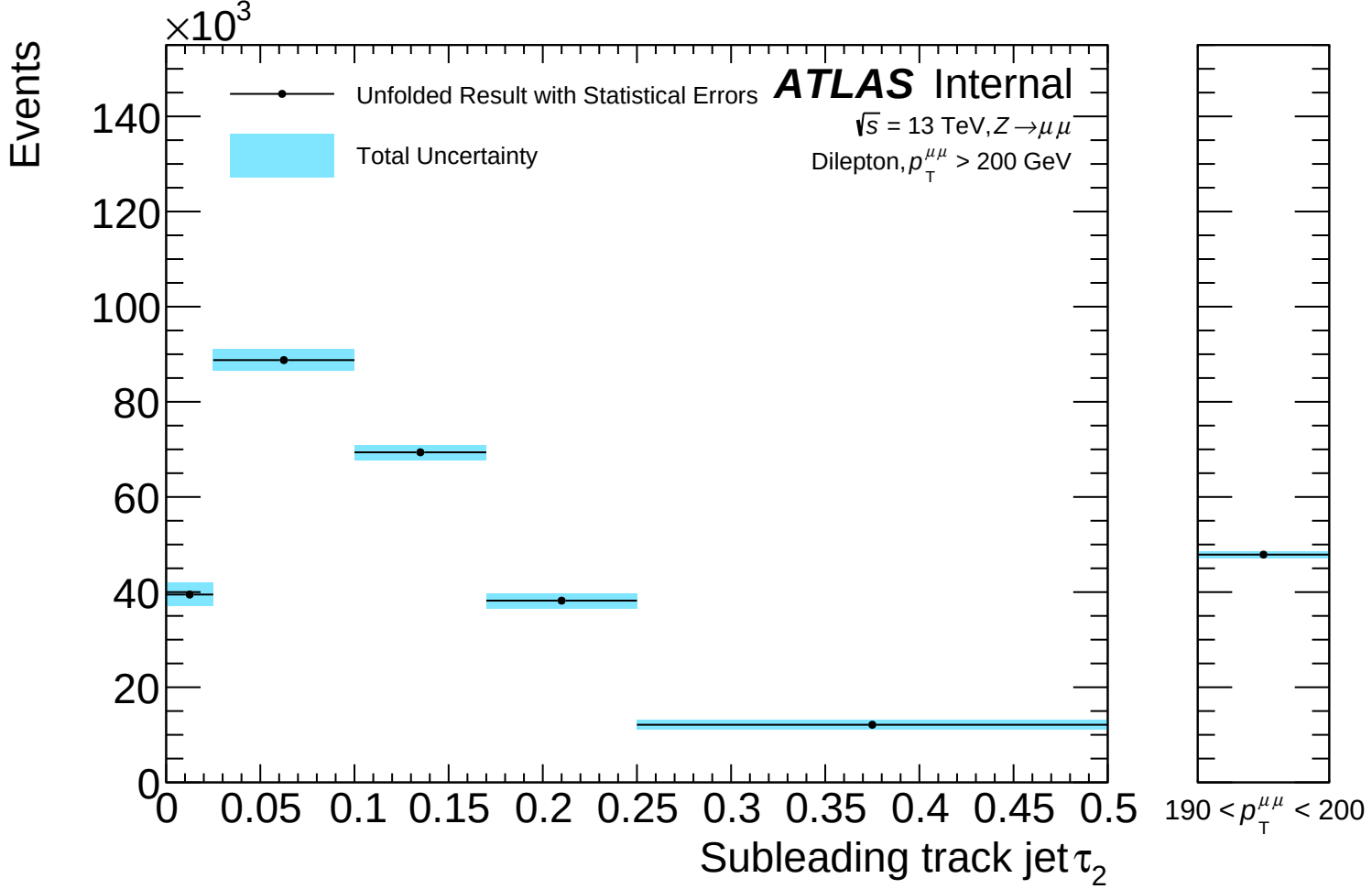
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

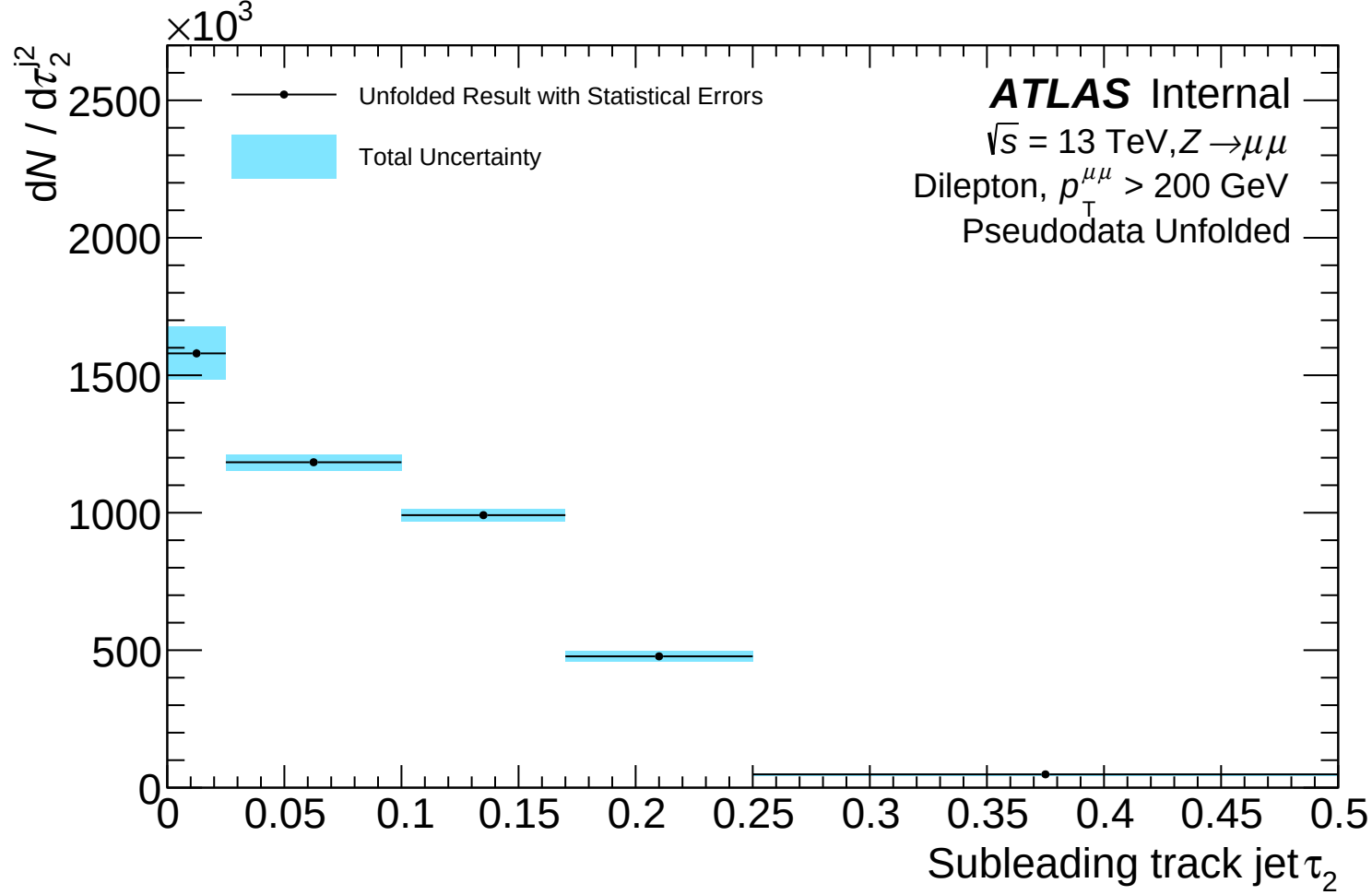
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

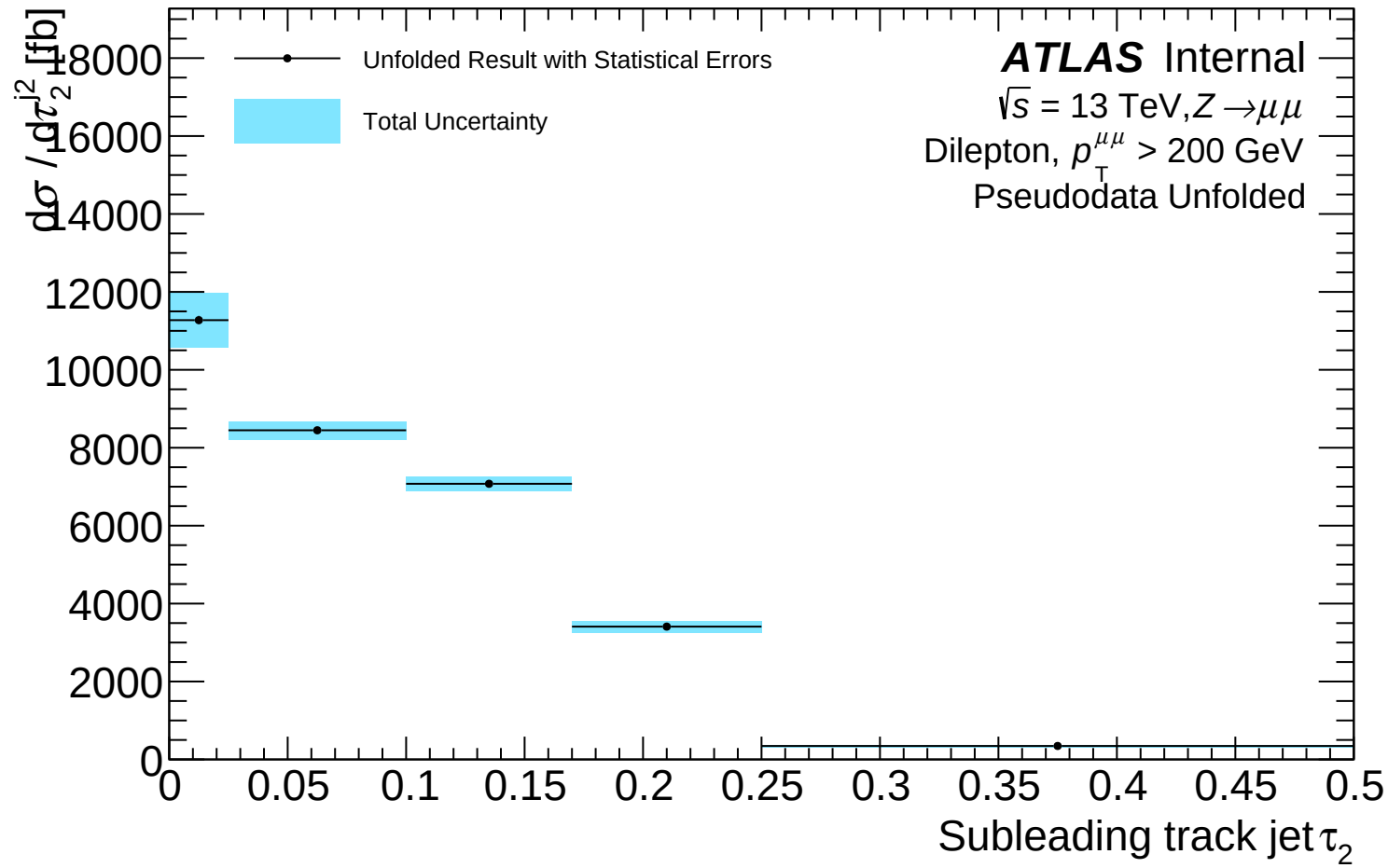
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty











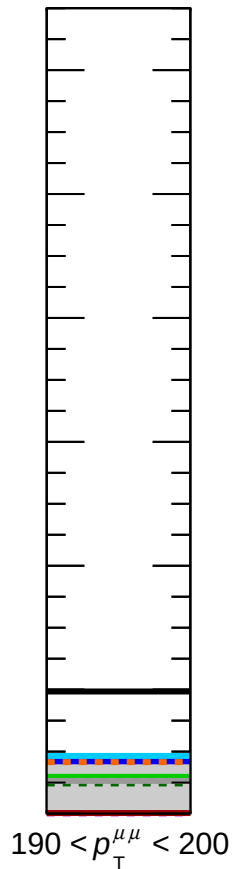
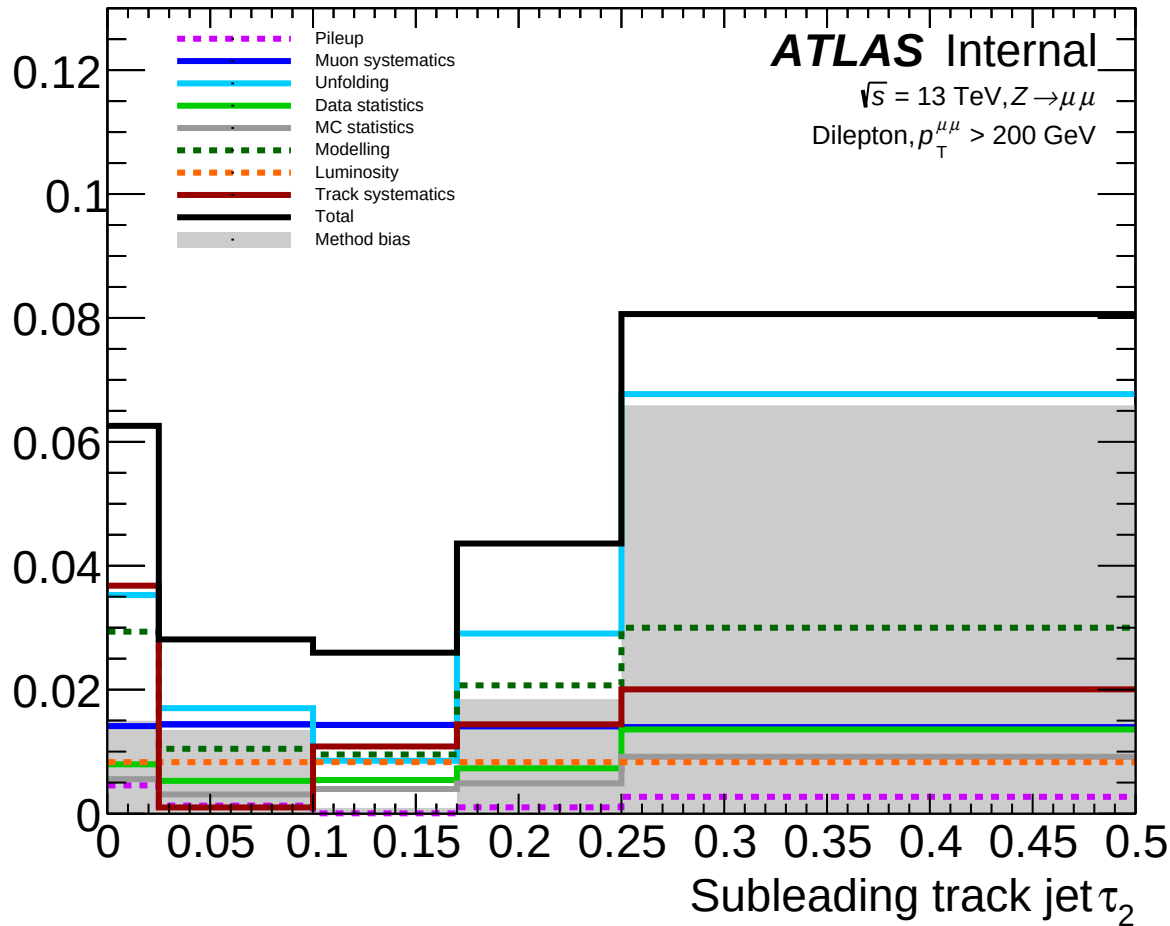
Fractional uncertainty

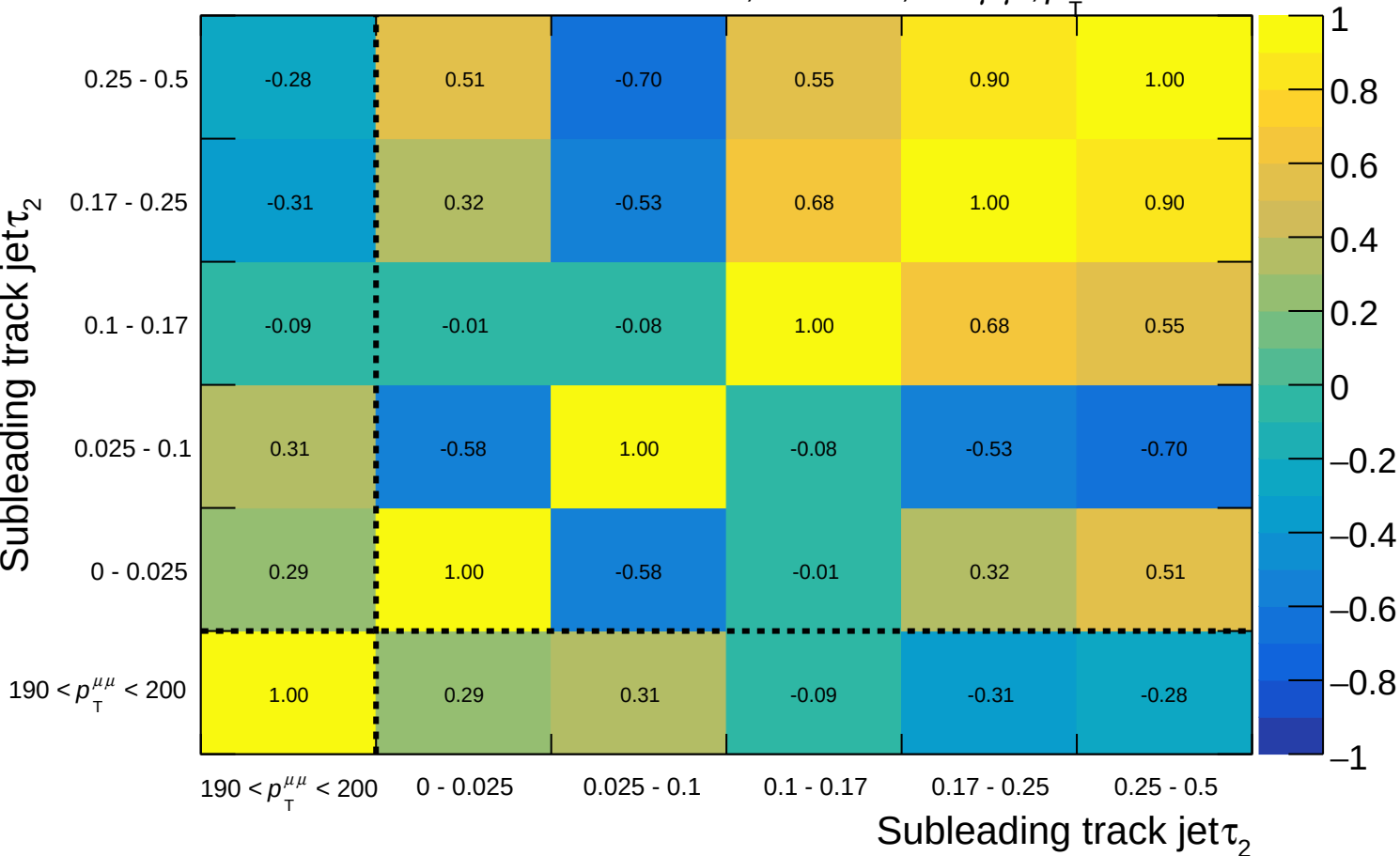
ATLAS Internal

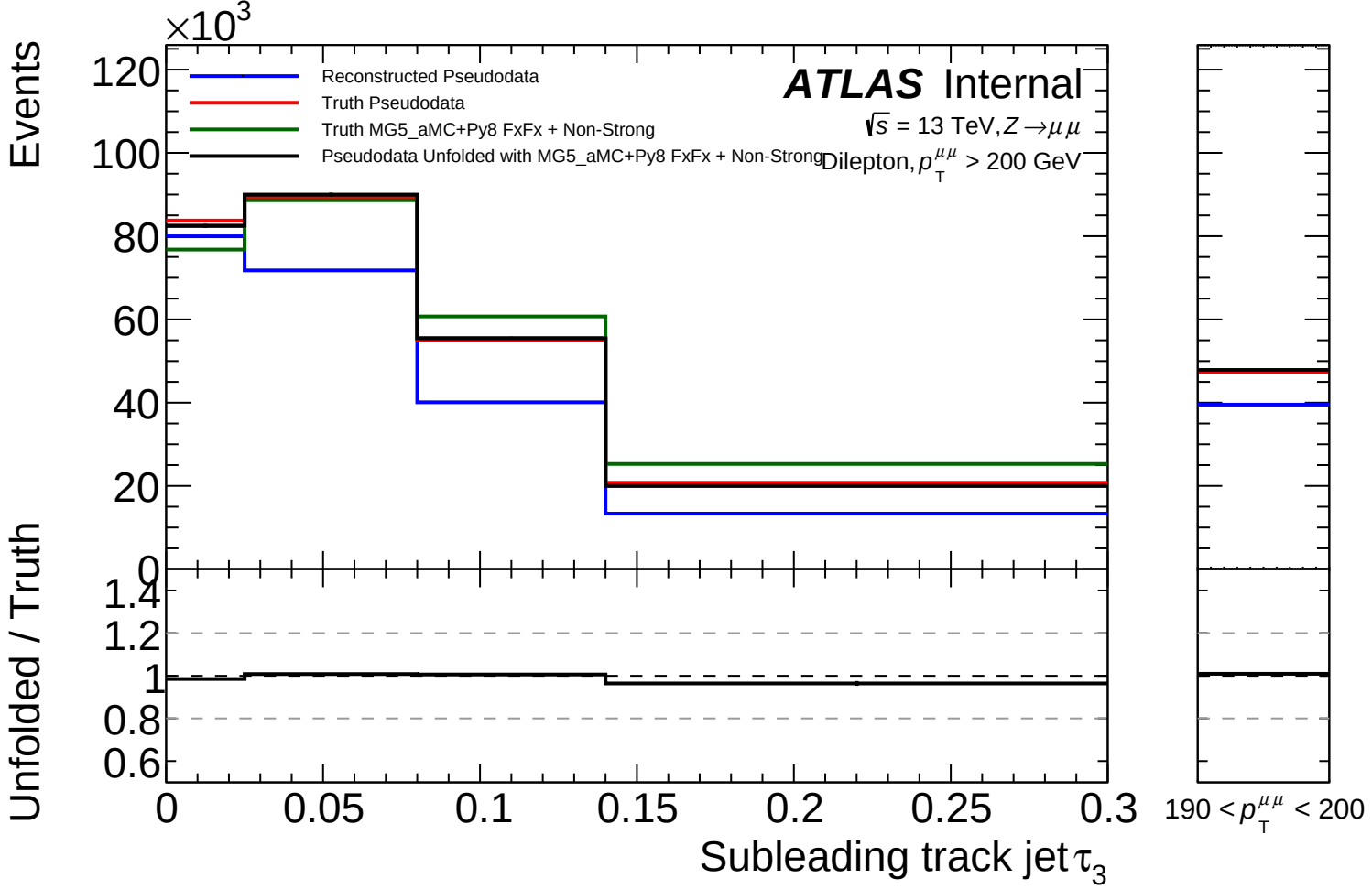
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

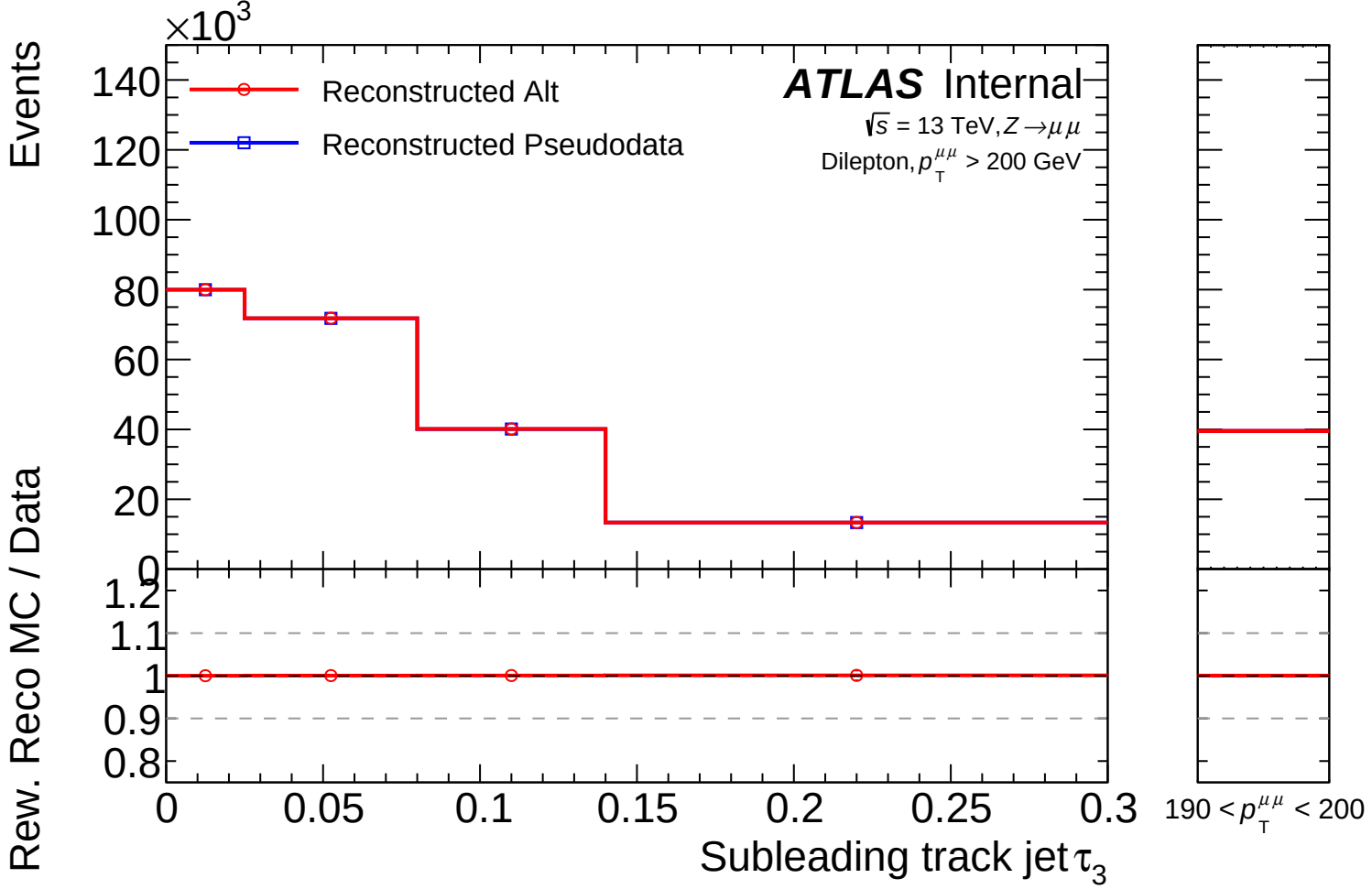
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

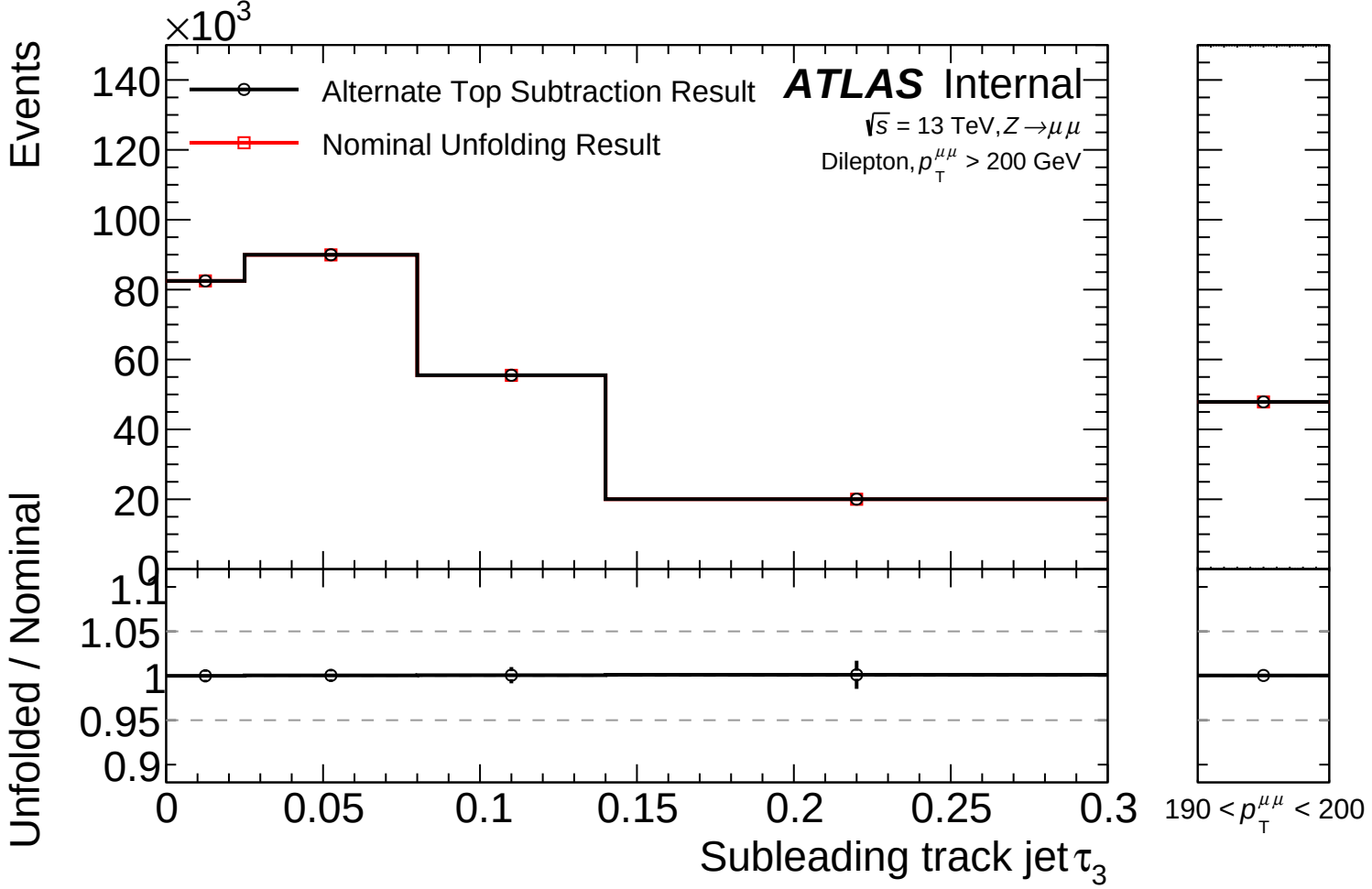
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_T^{\mu\mu} > 200 \text{ GeV}$ Subleading track jet τ_2 







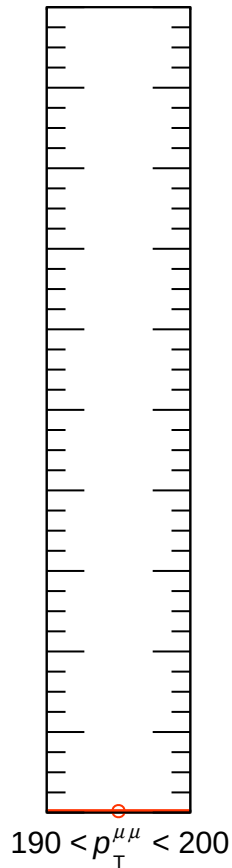
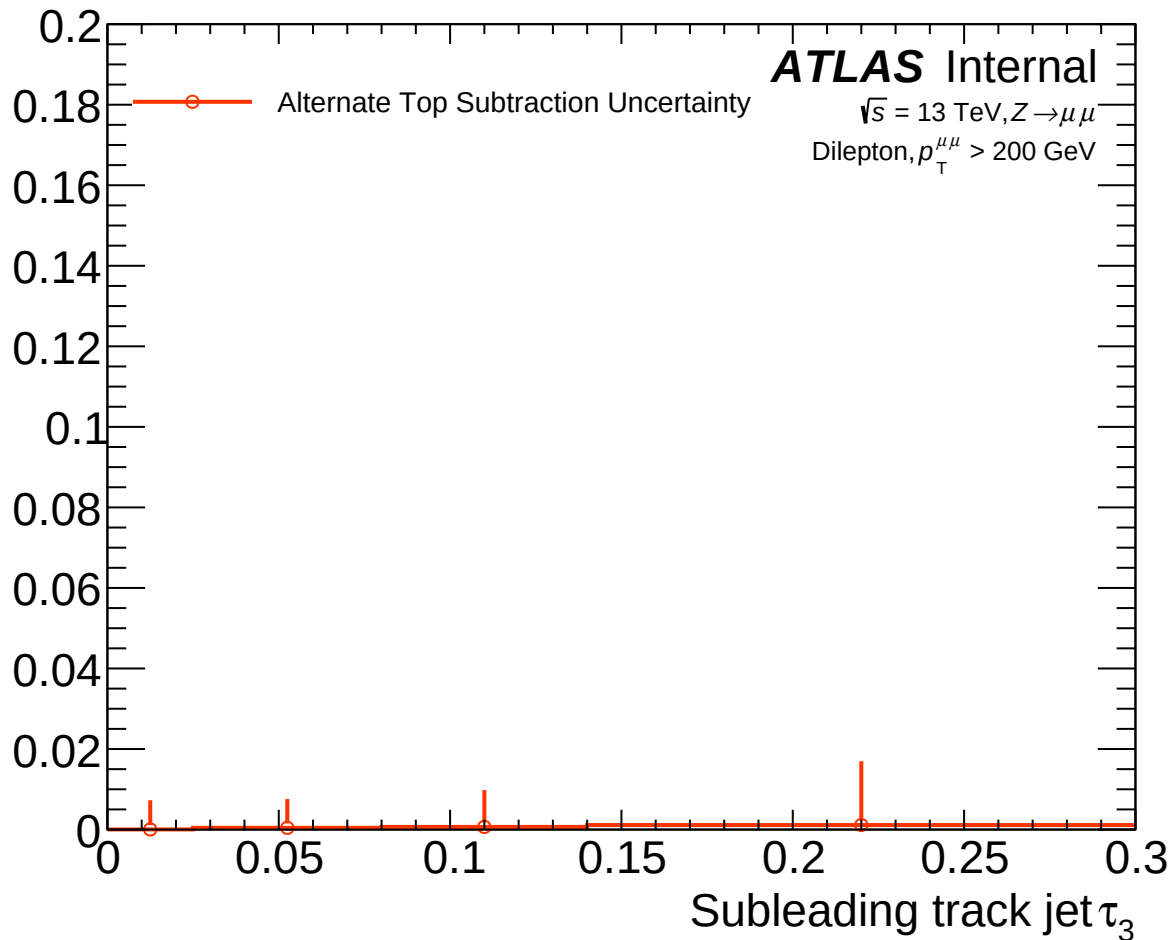
Fractional systematic impact

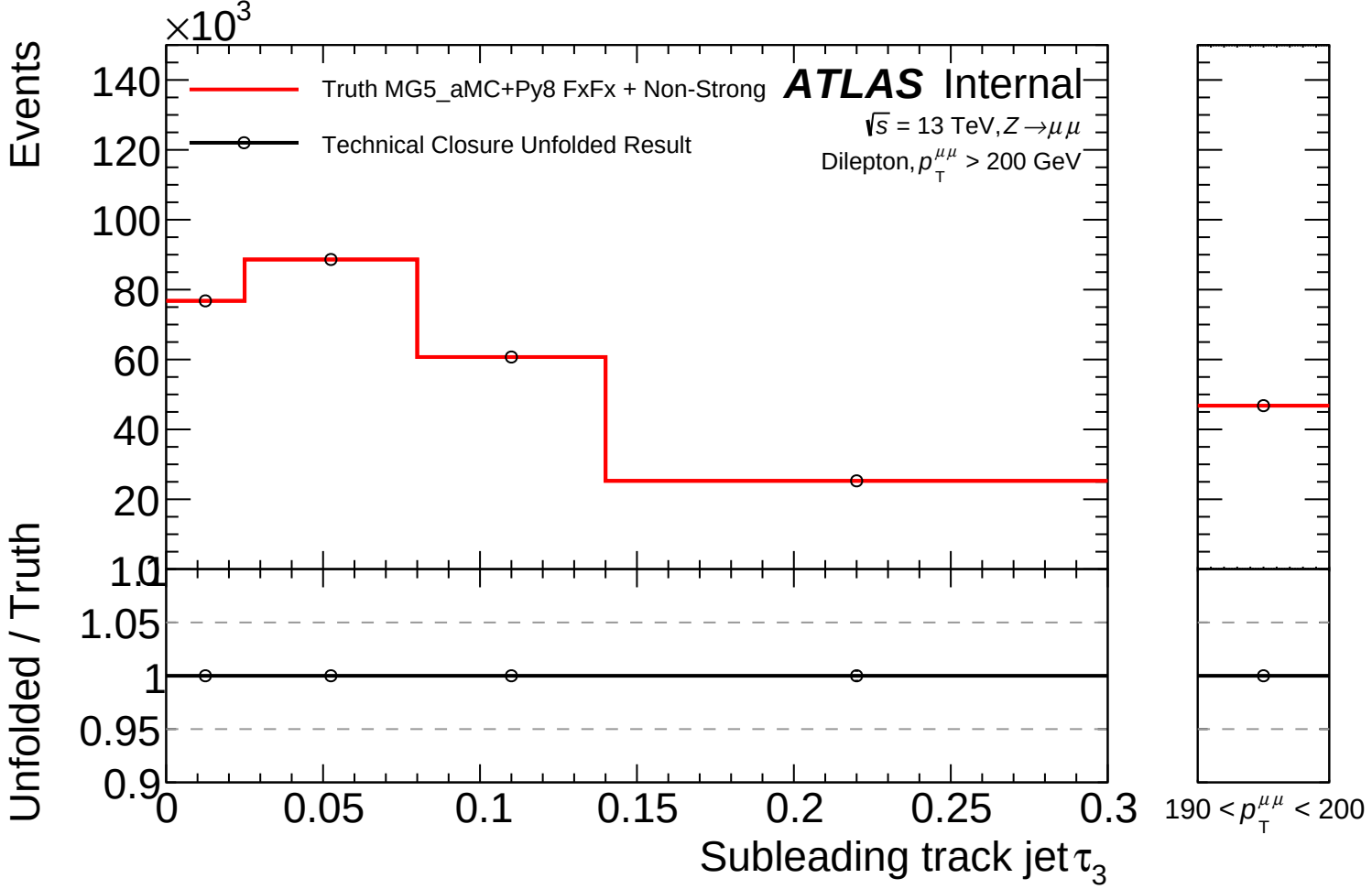
ATLAS Internal

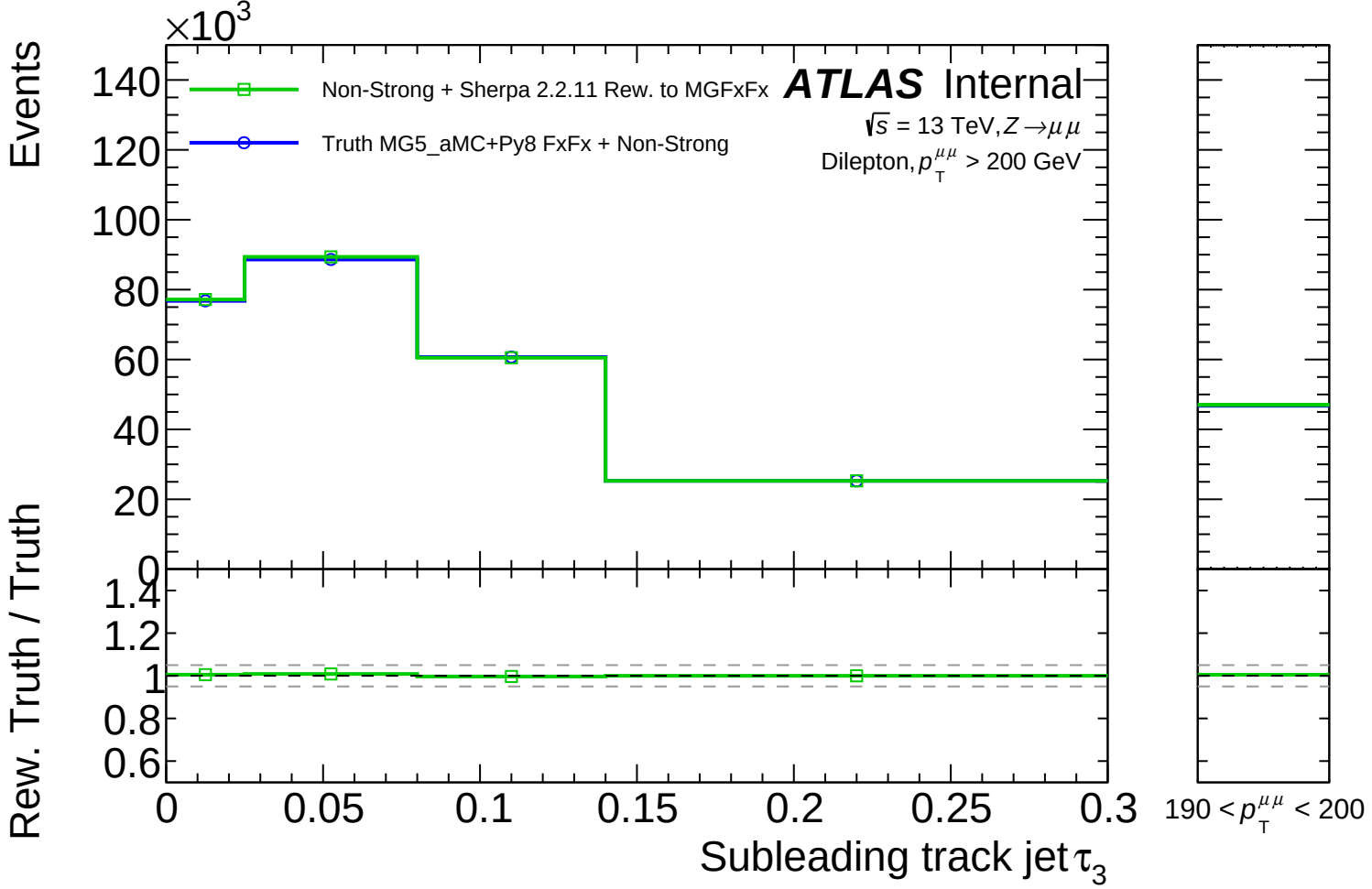
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

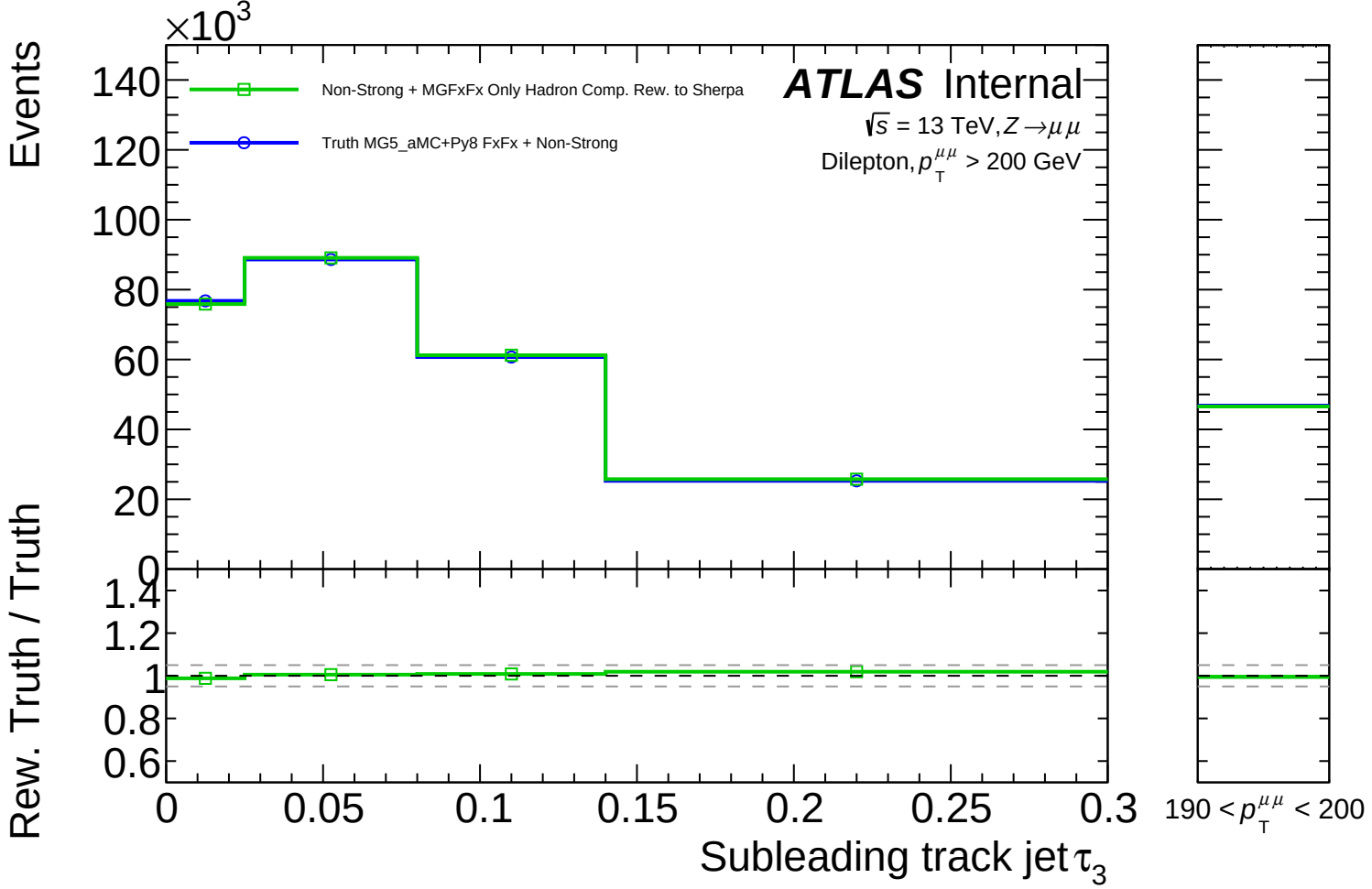
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

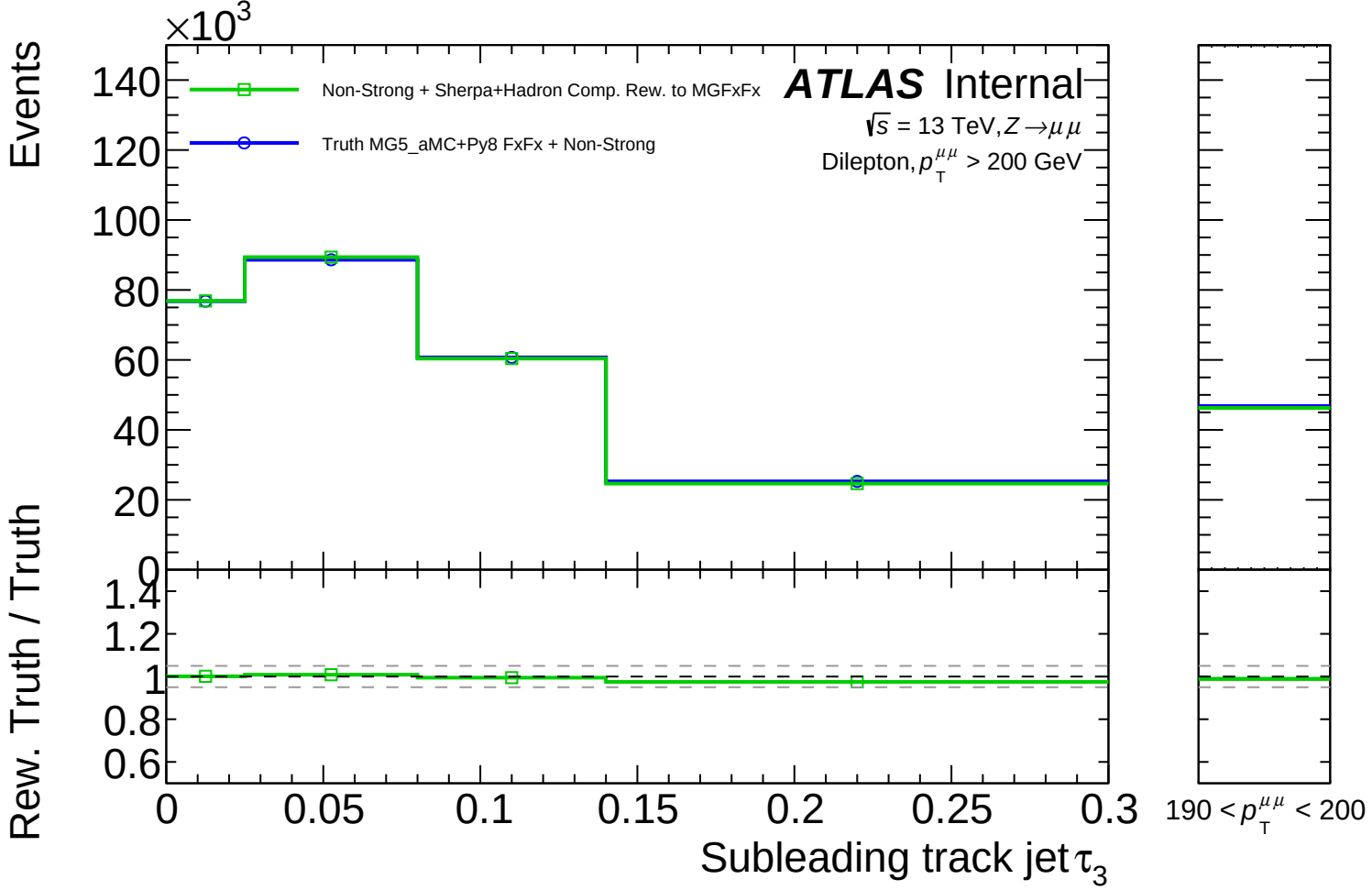
Alternate Top Subtraction Uncertainty

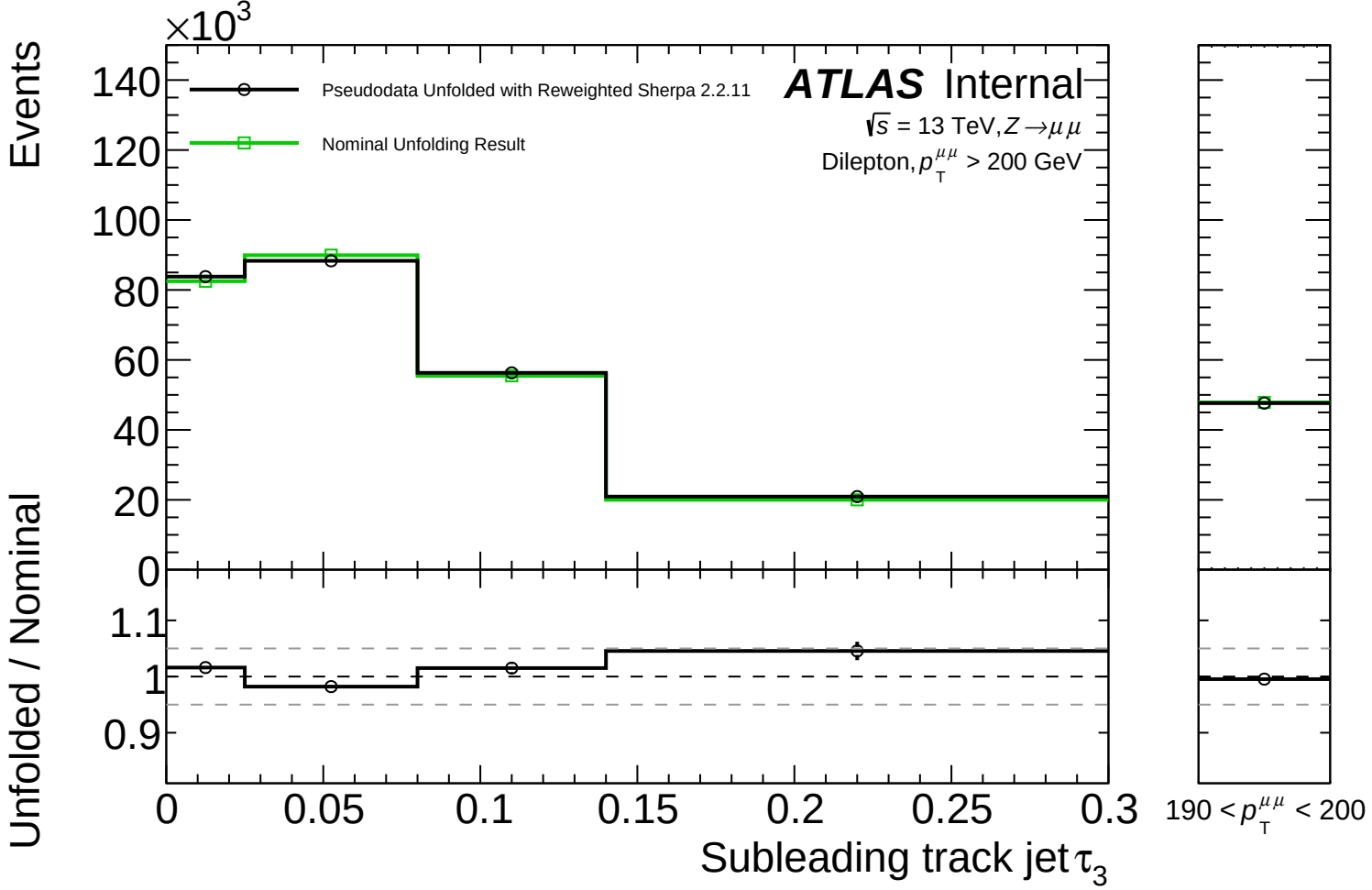


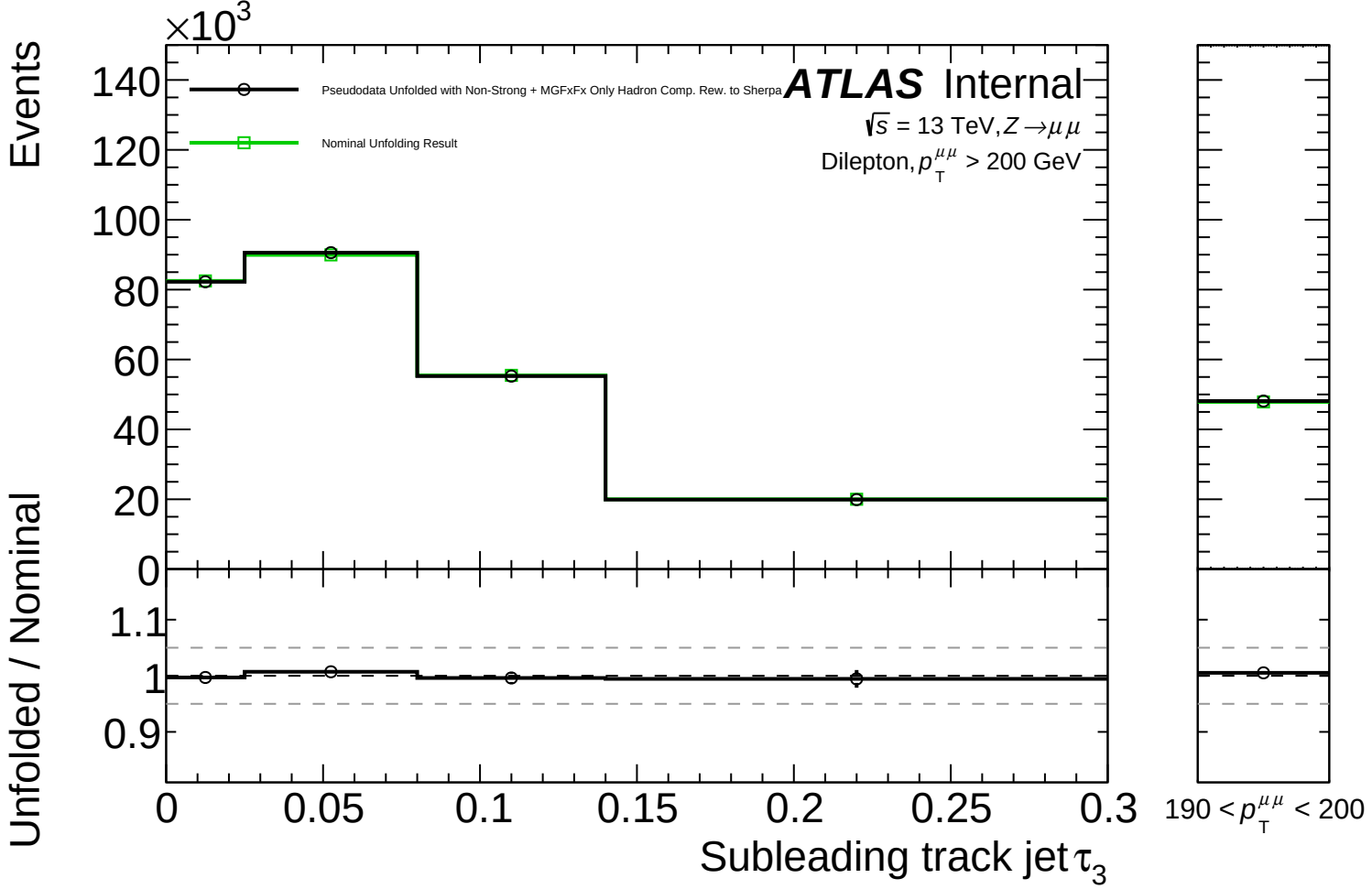


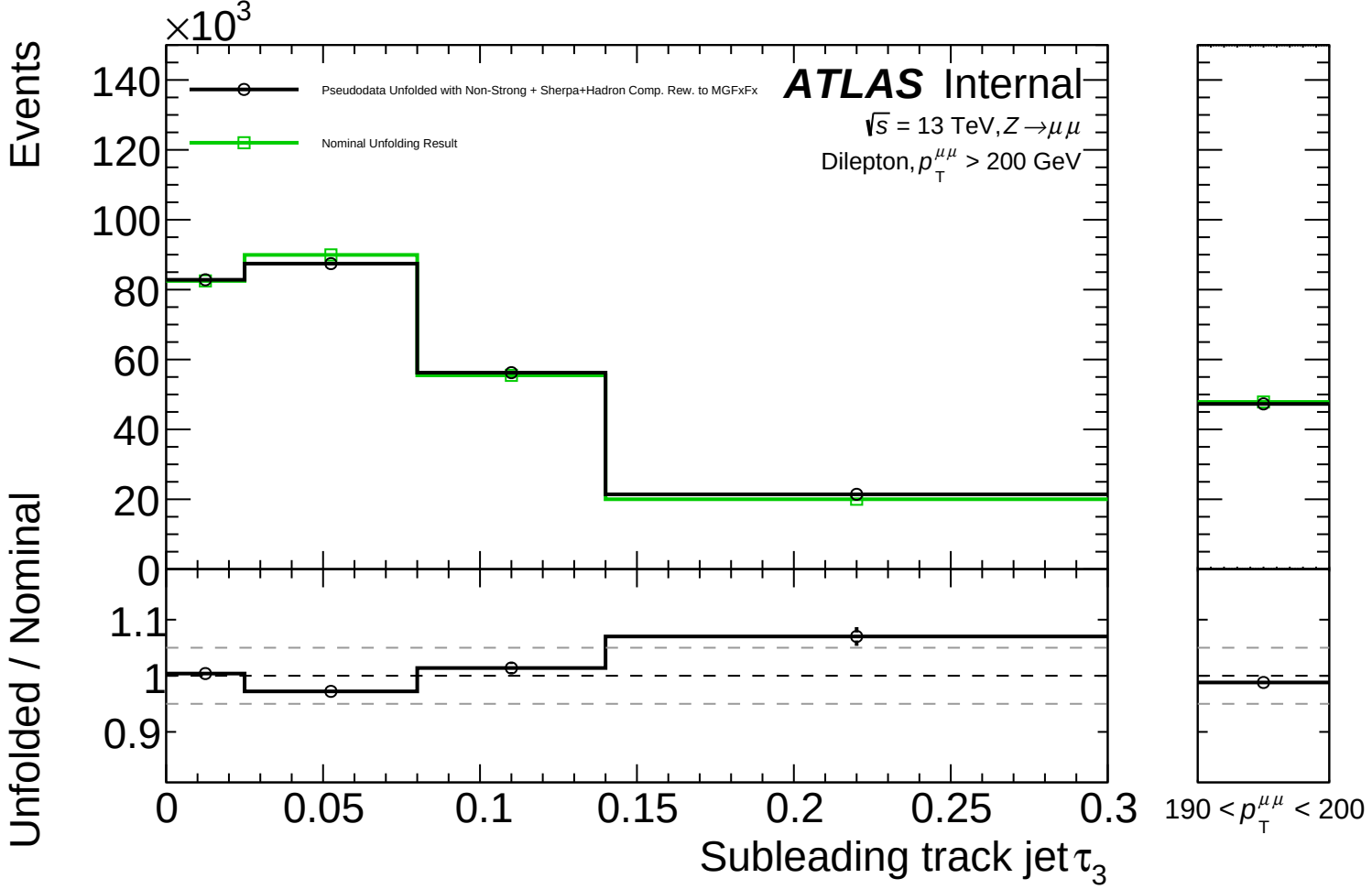


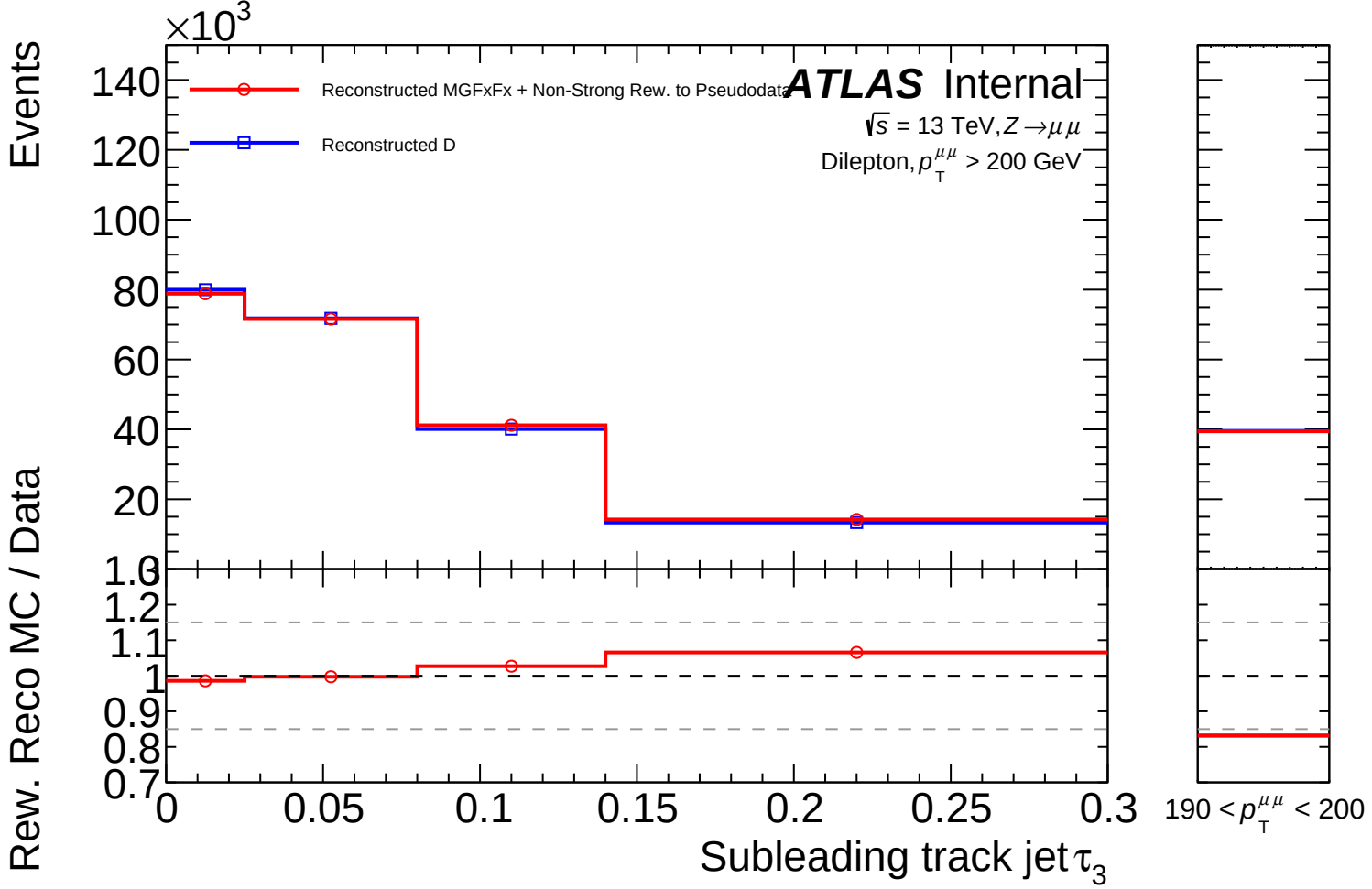


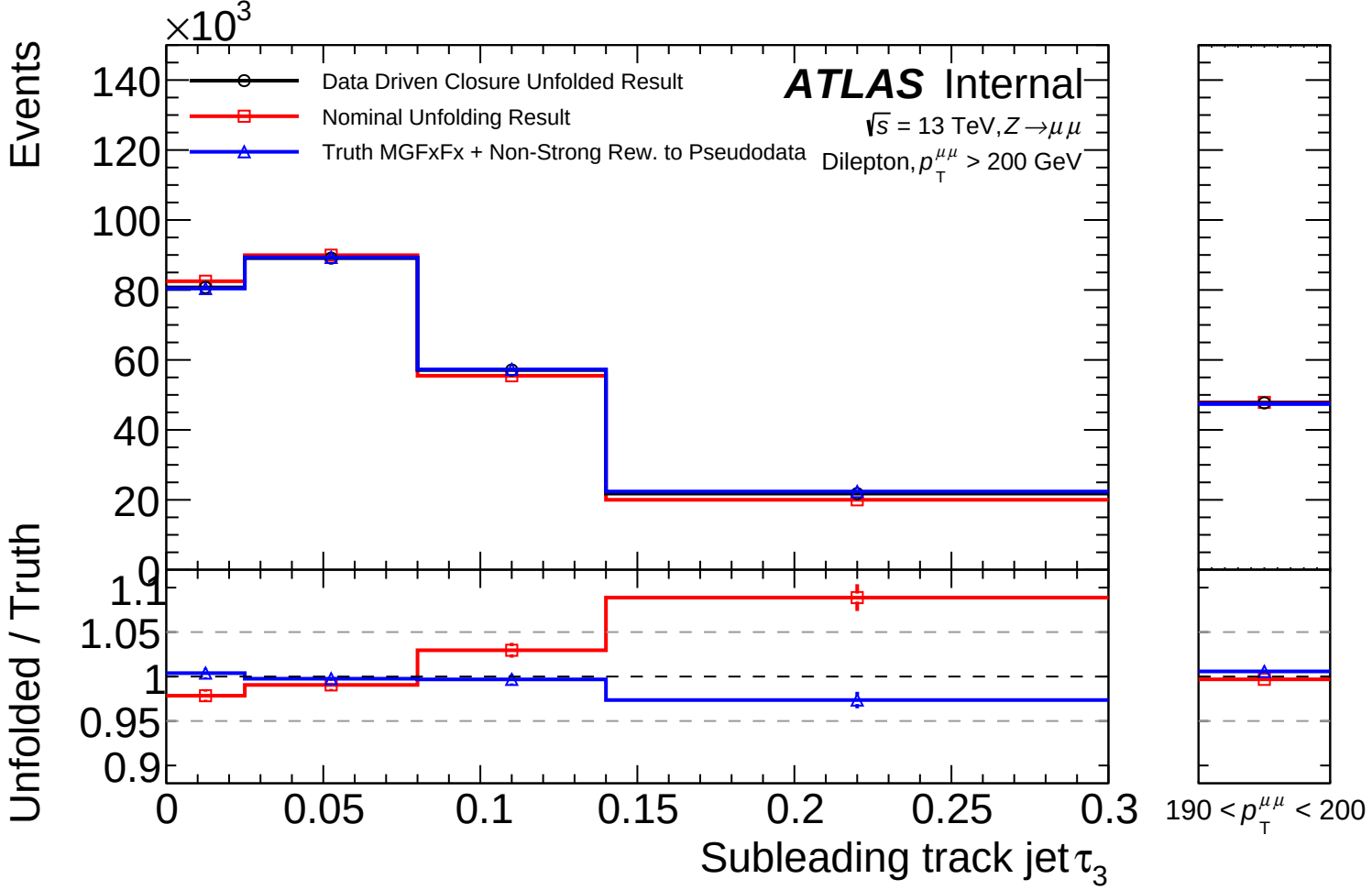




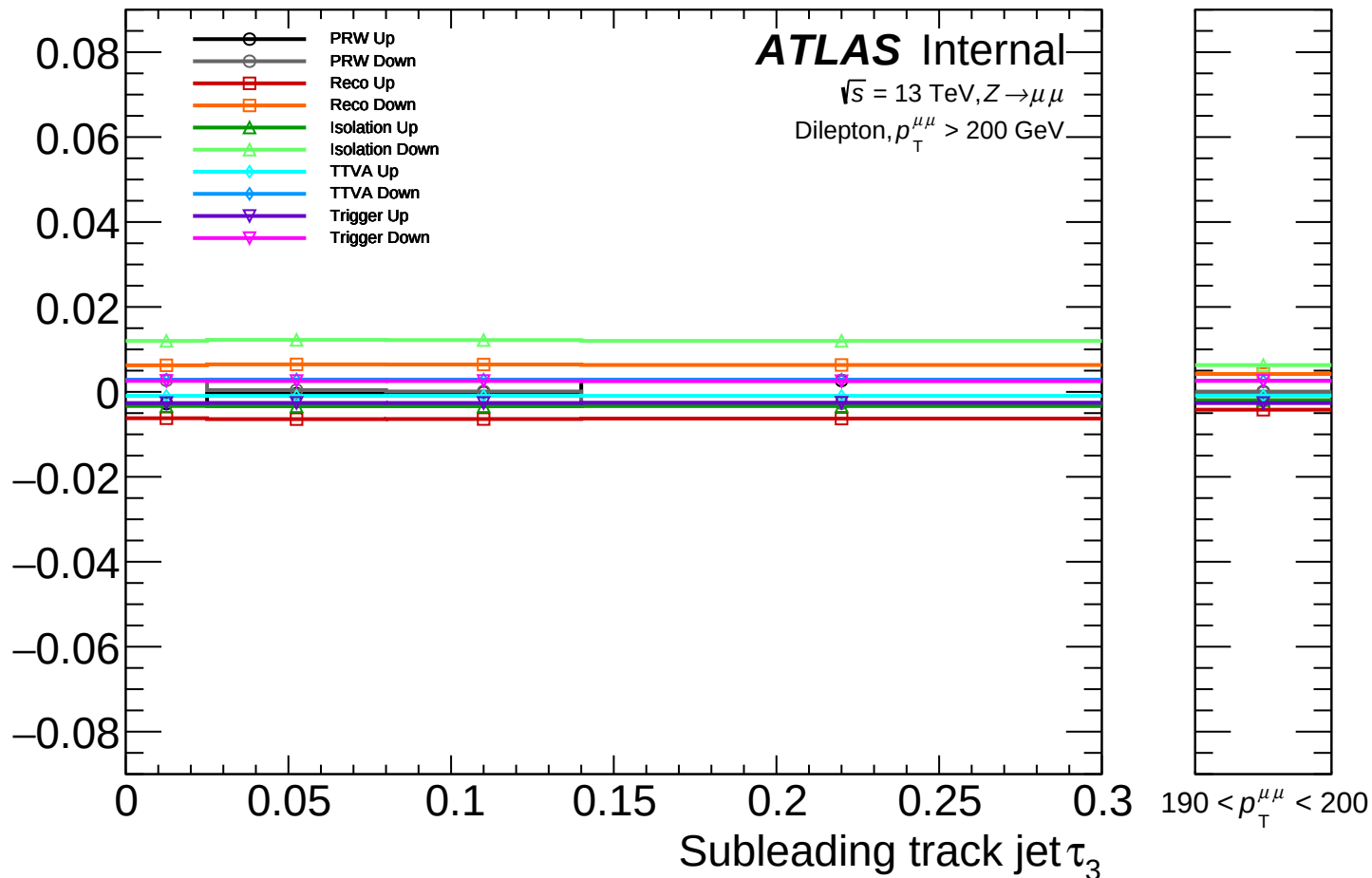




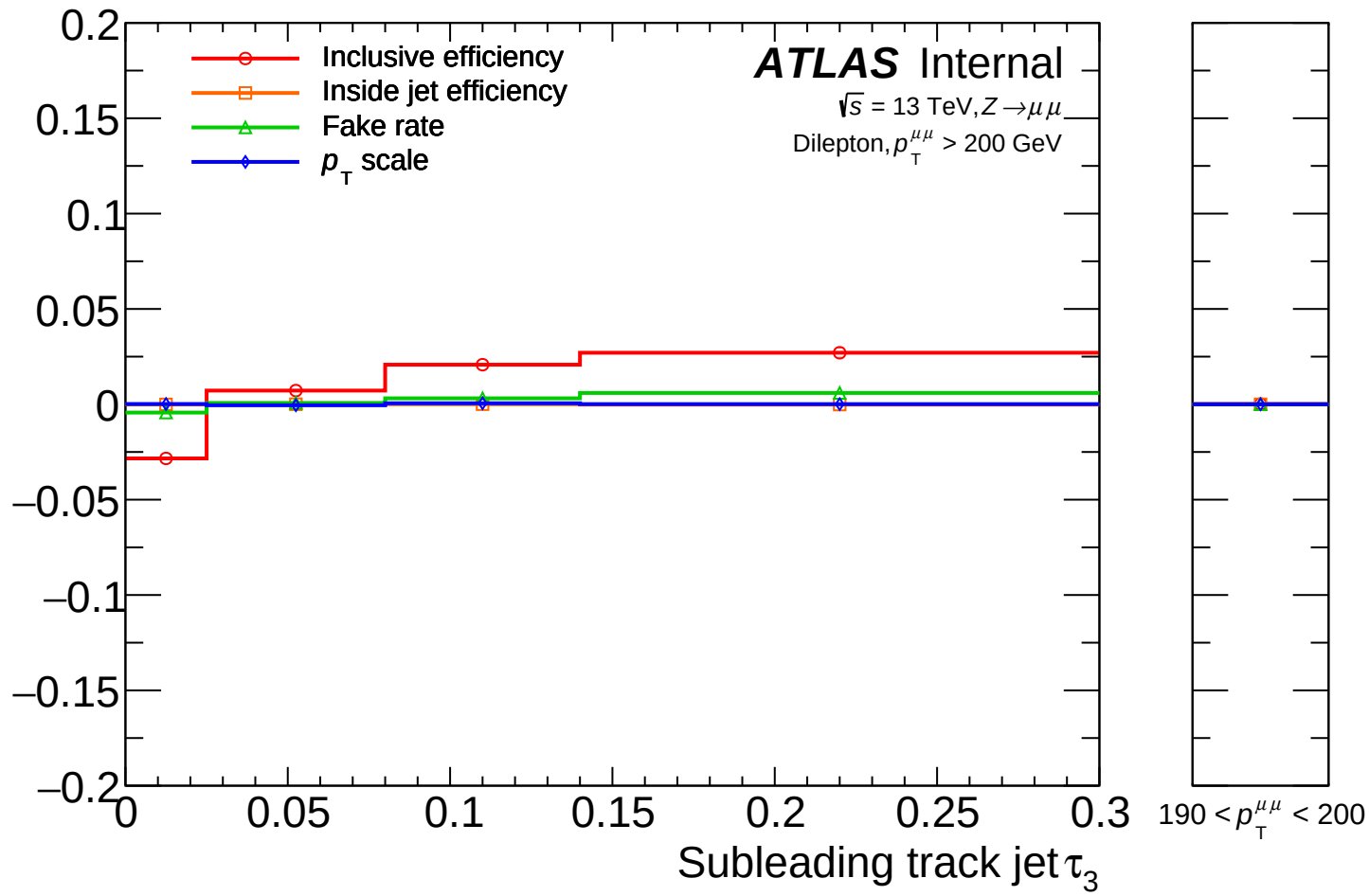




Fractional systematic impact



Fractional systematic impact



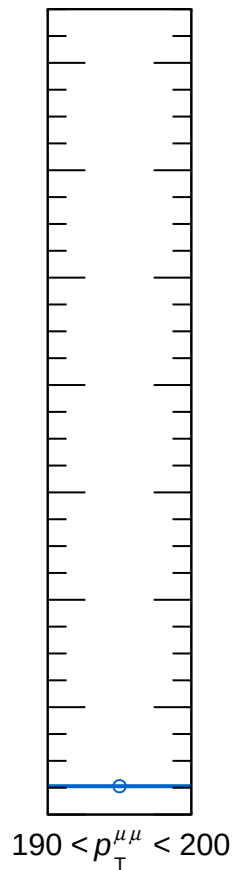
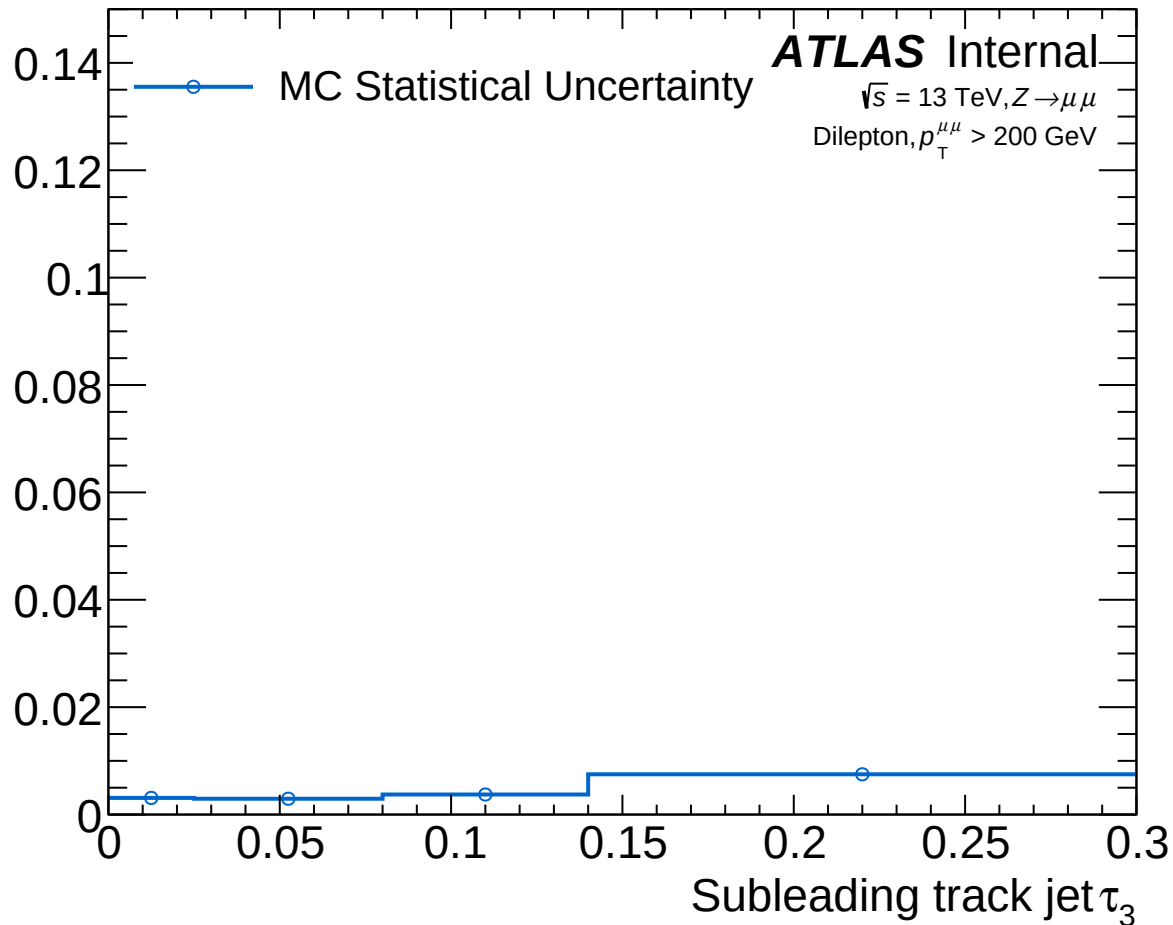
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



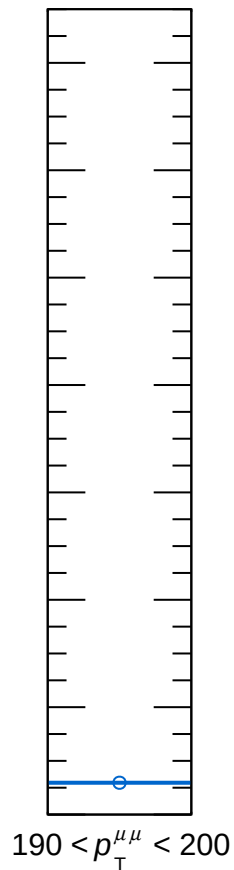
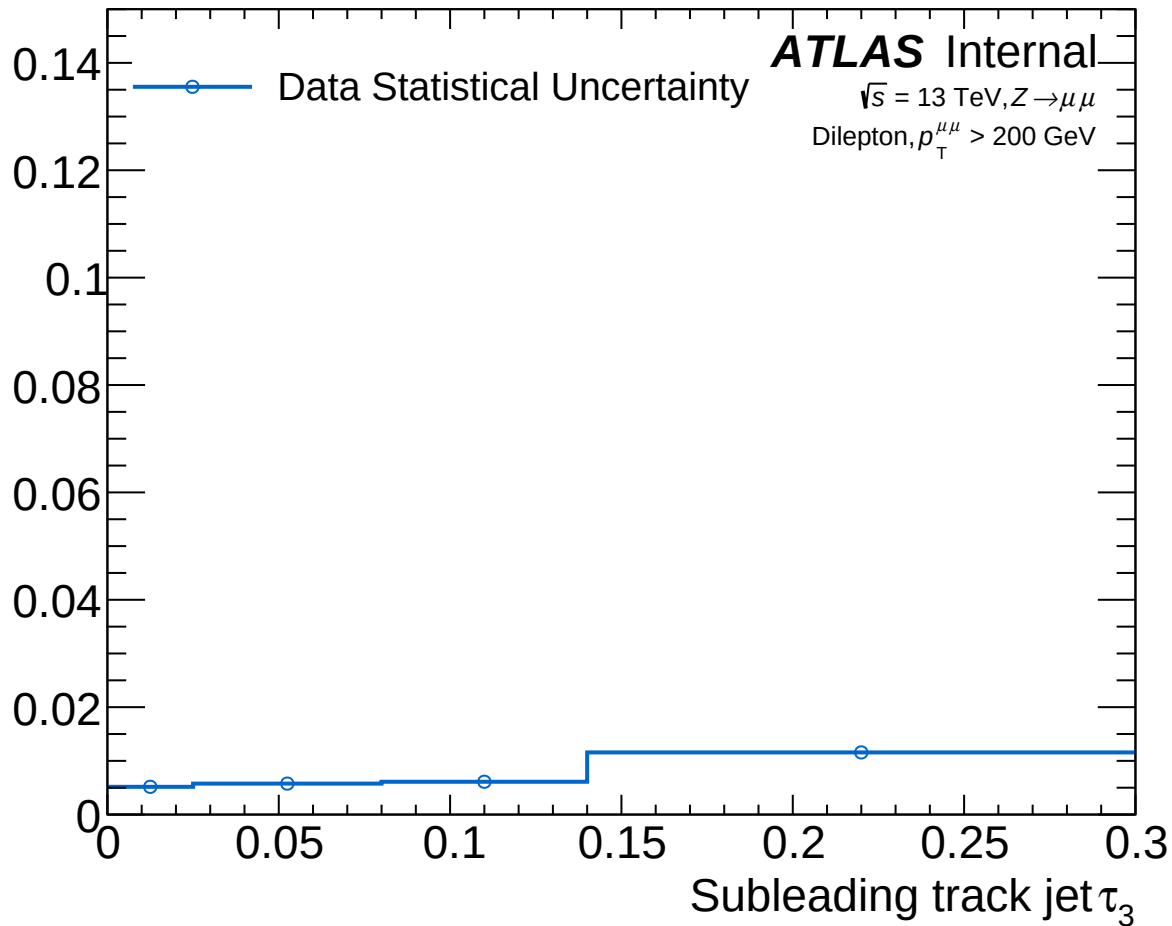
Fractional systematic impact

ATLAS Internal

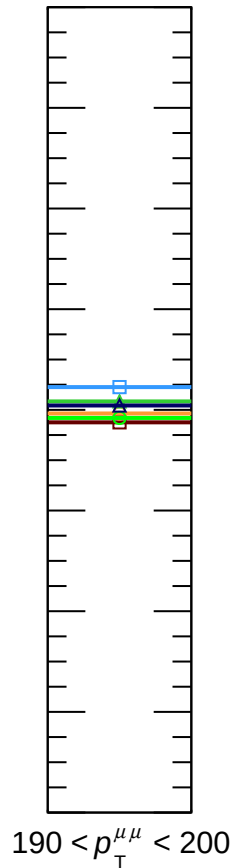
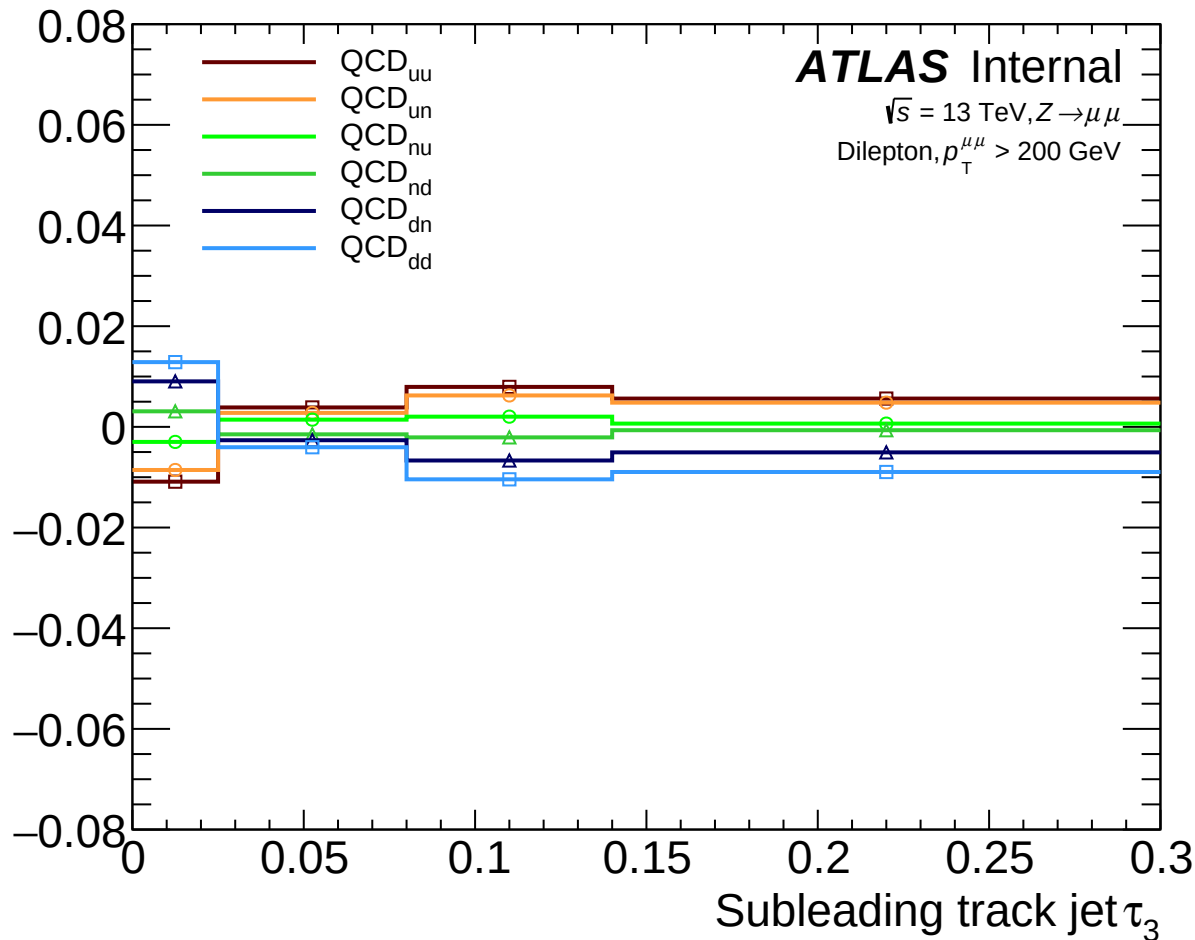
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

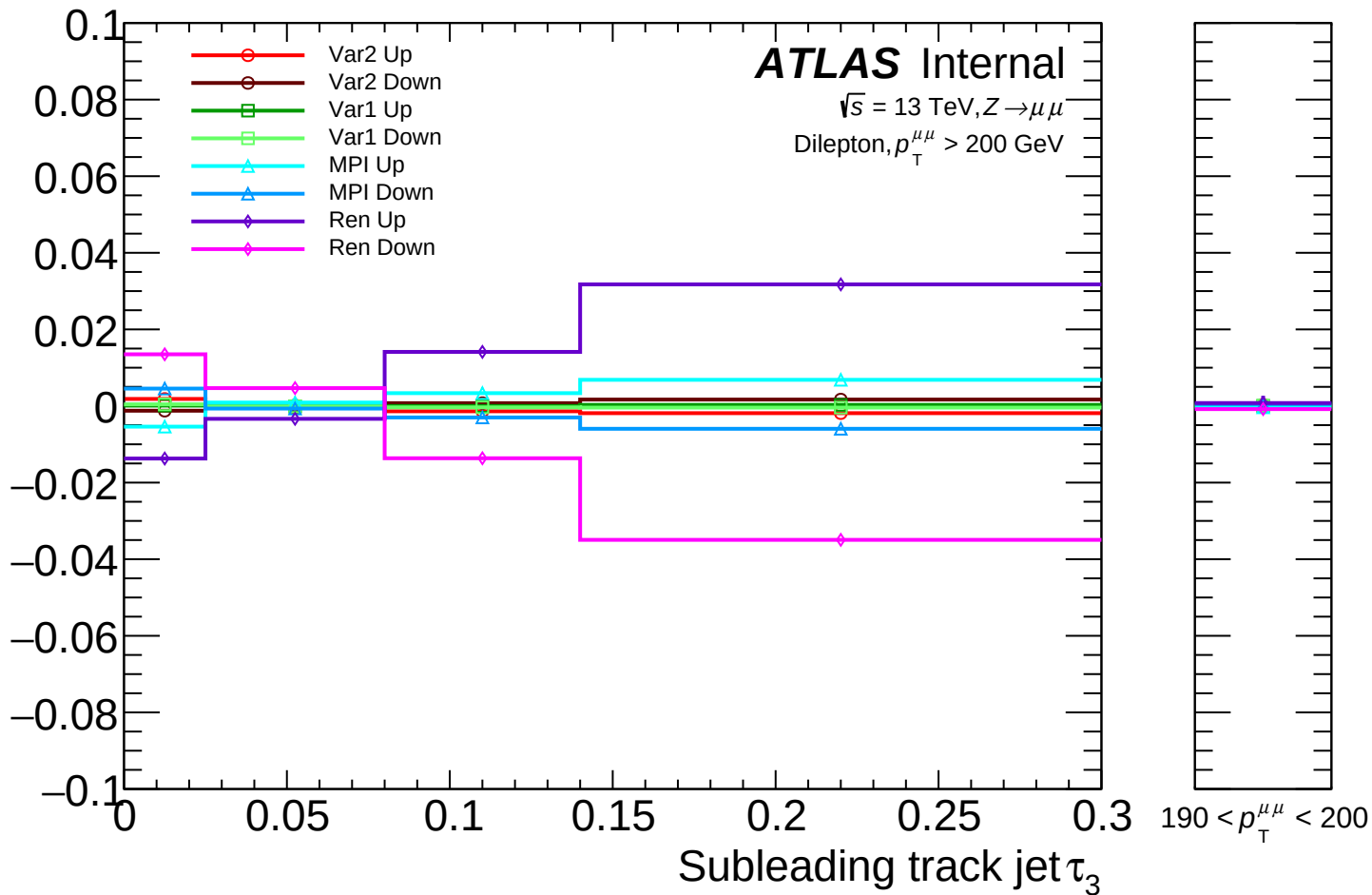
Data Statistical Uncertainty

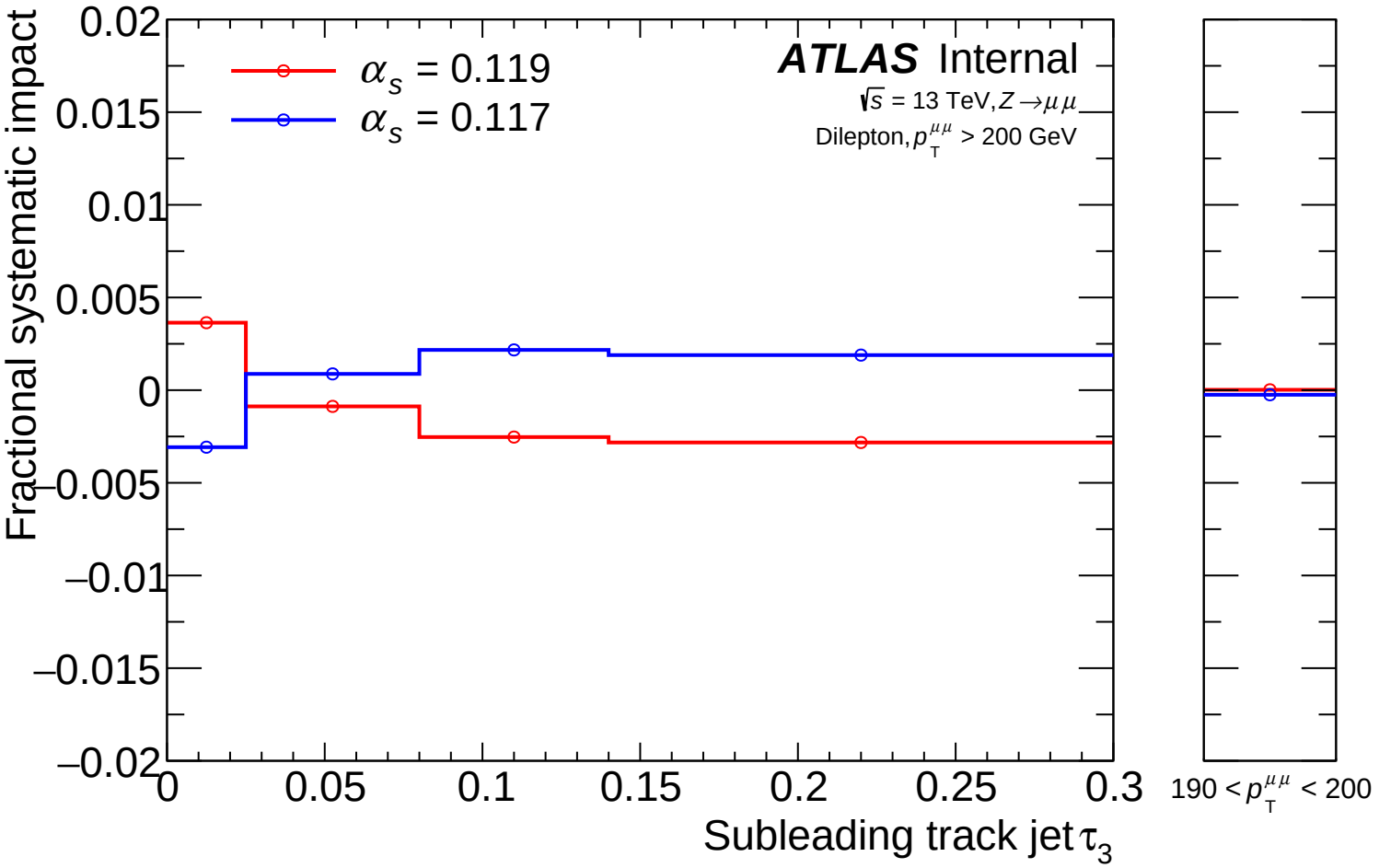


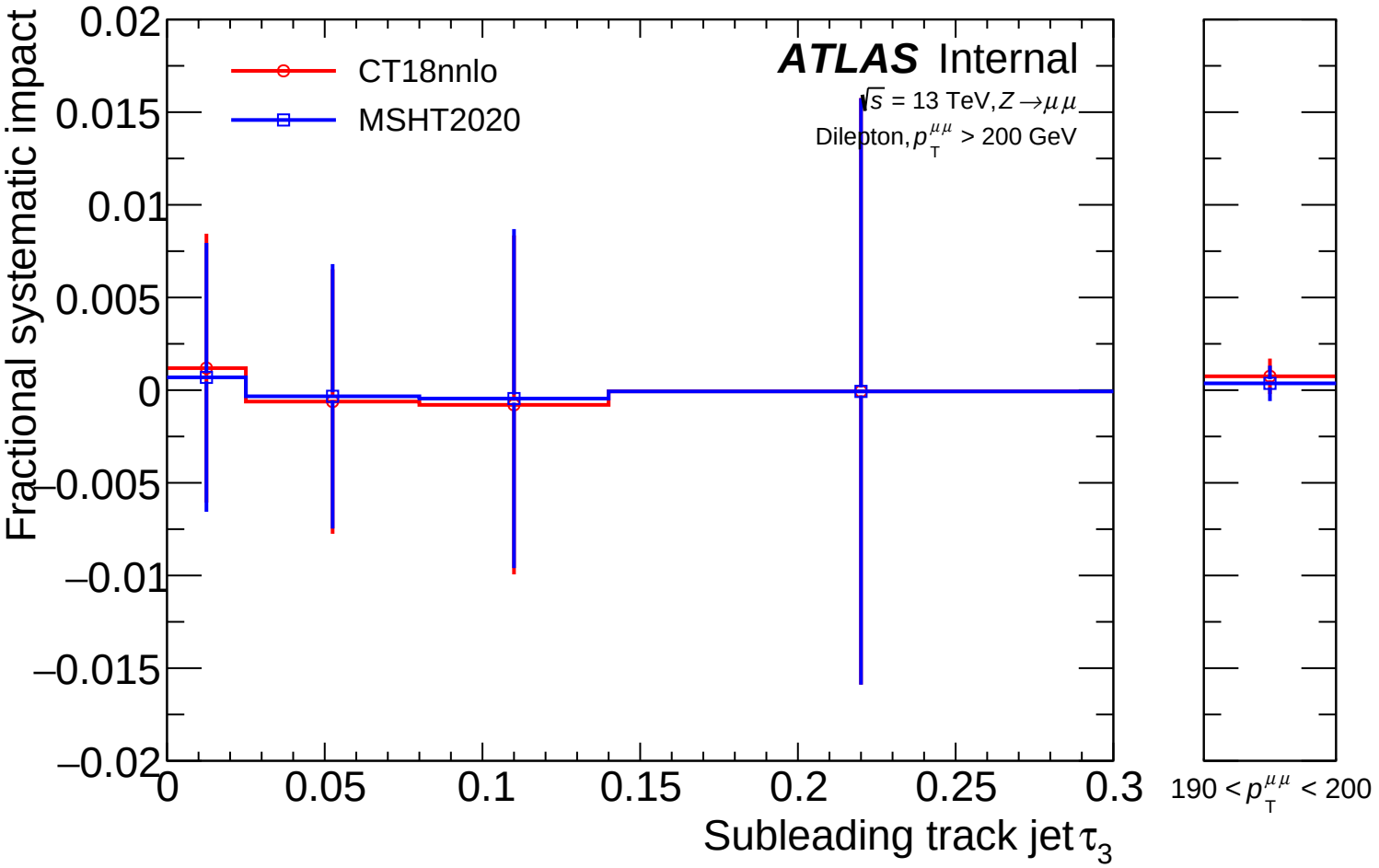
Fractional systematic impact



Fractional systematic impact







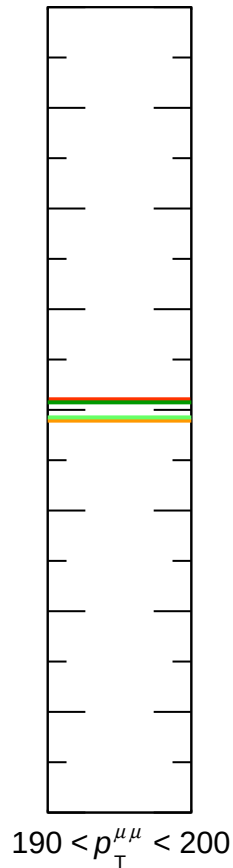
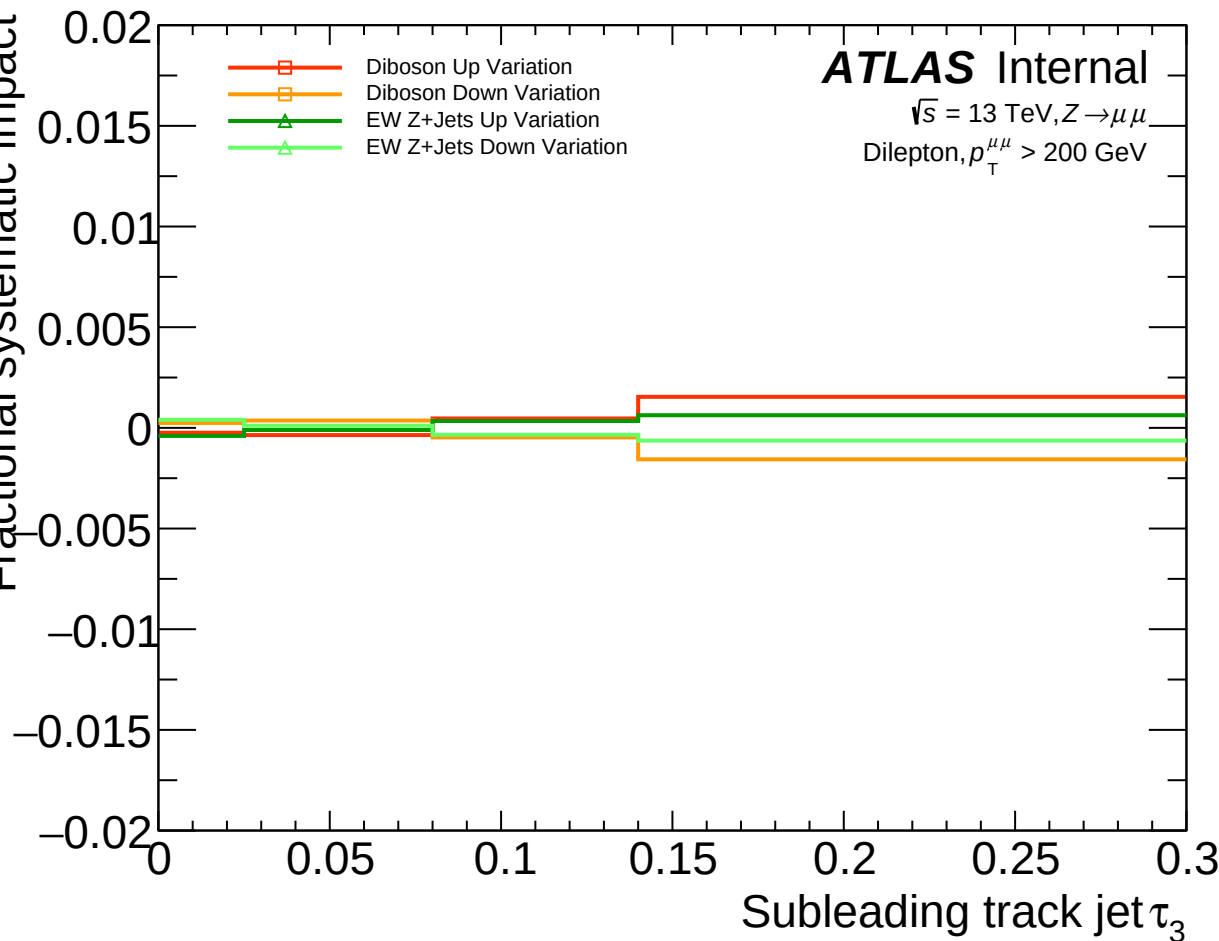
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

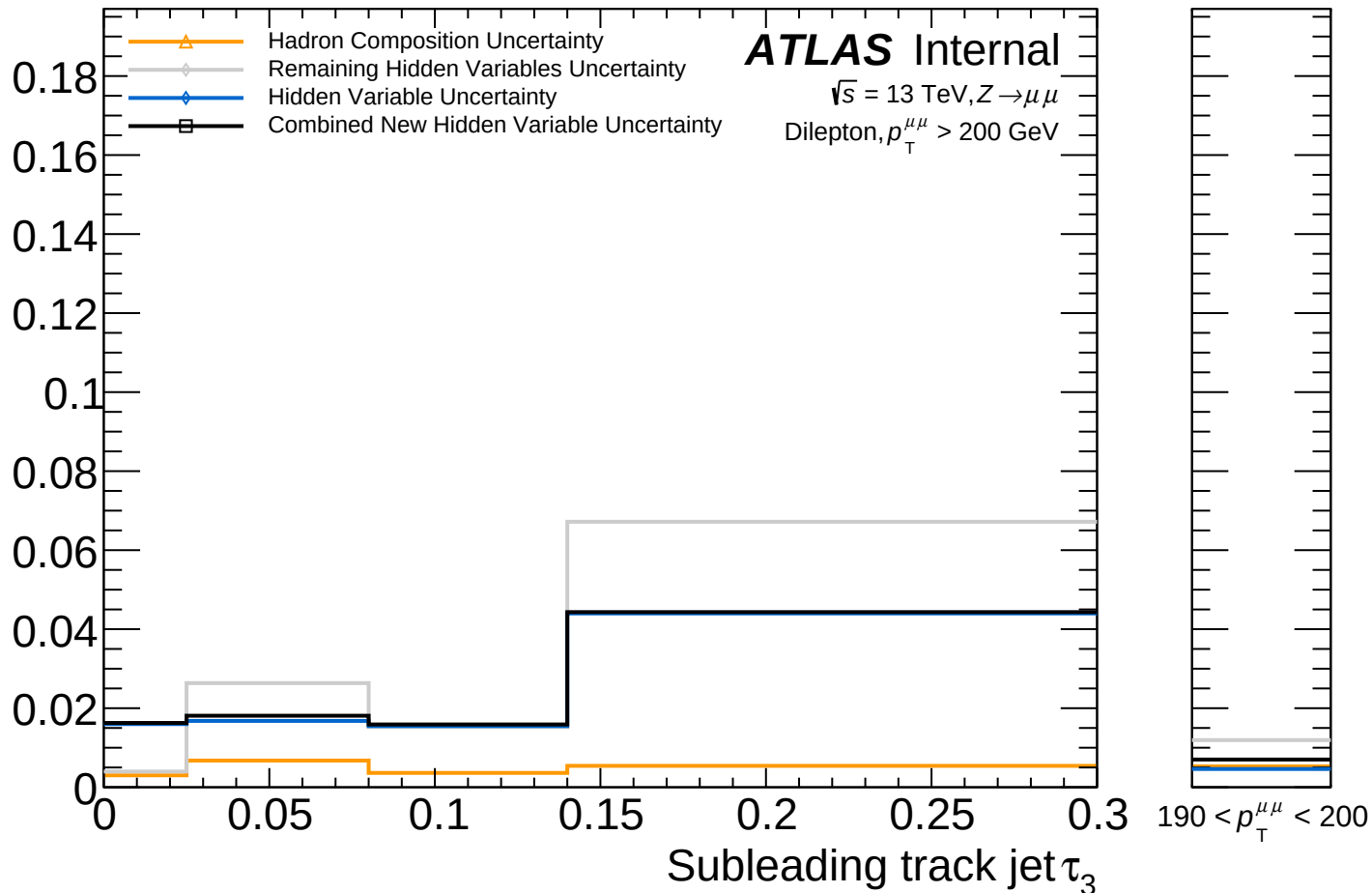


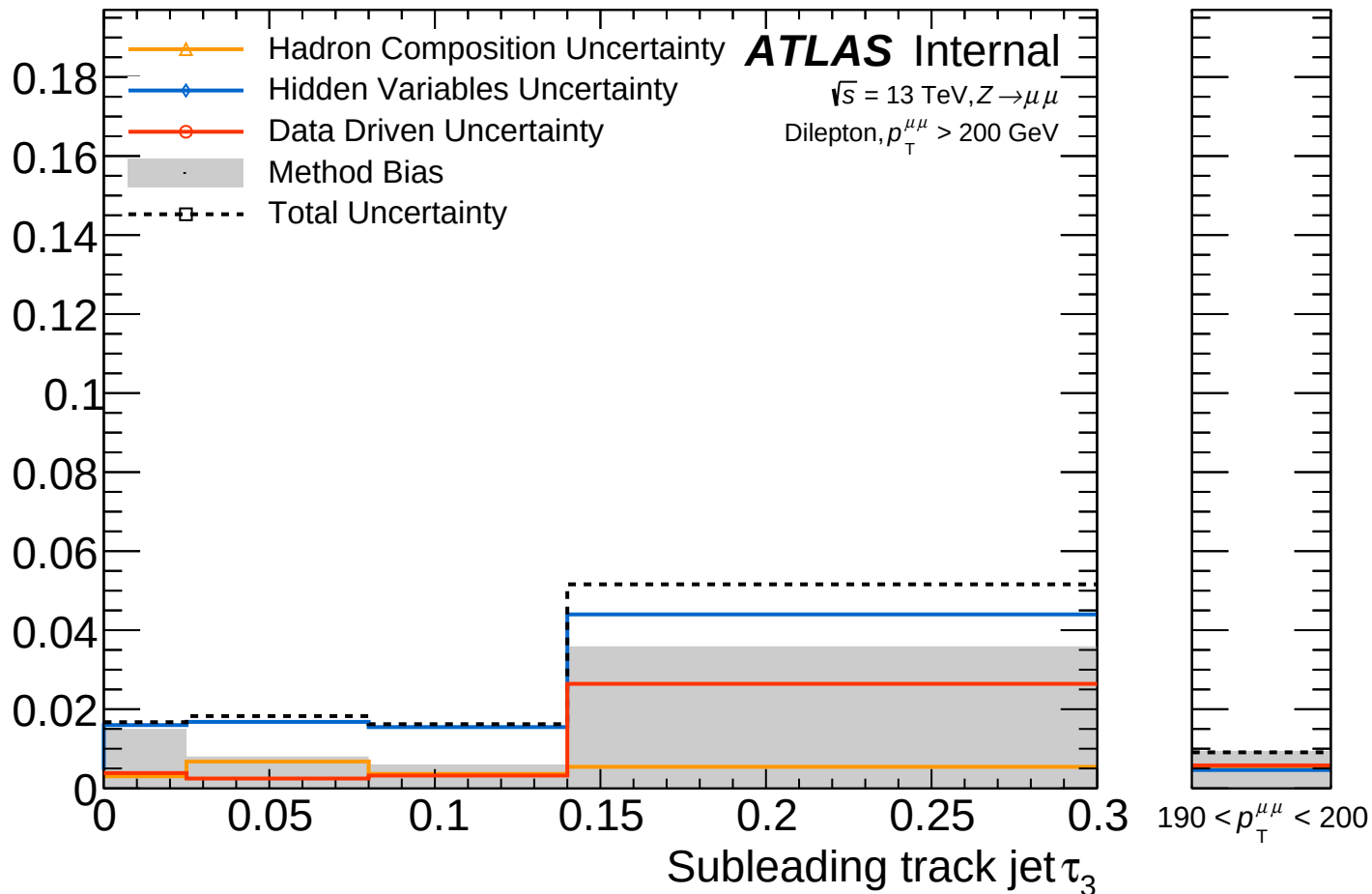
ATLAS Internal

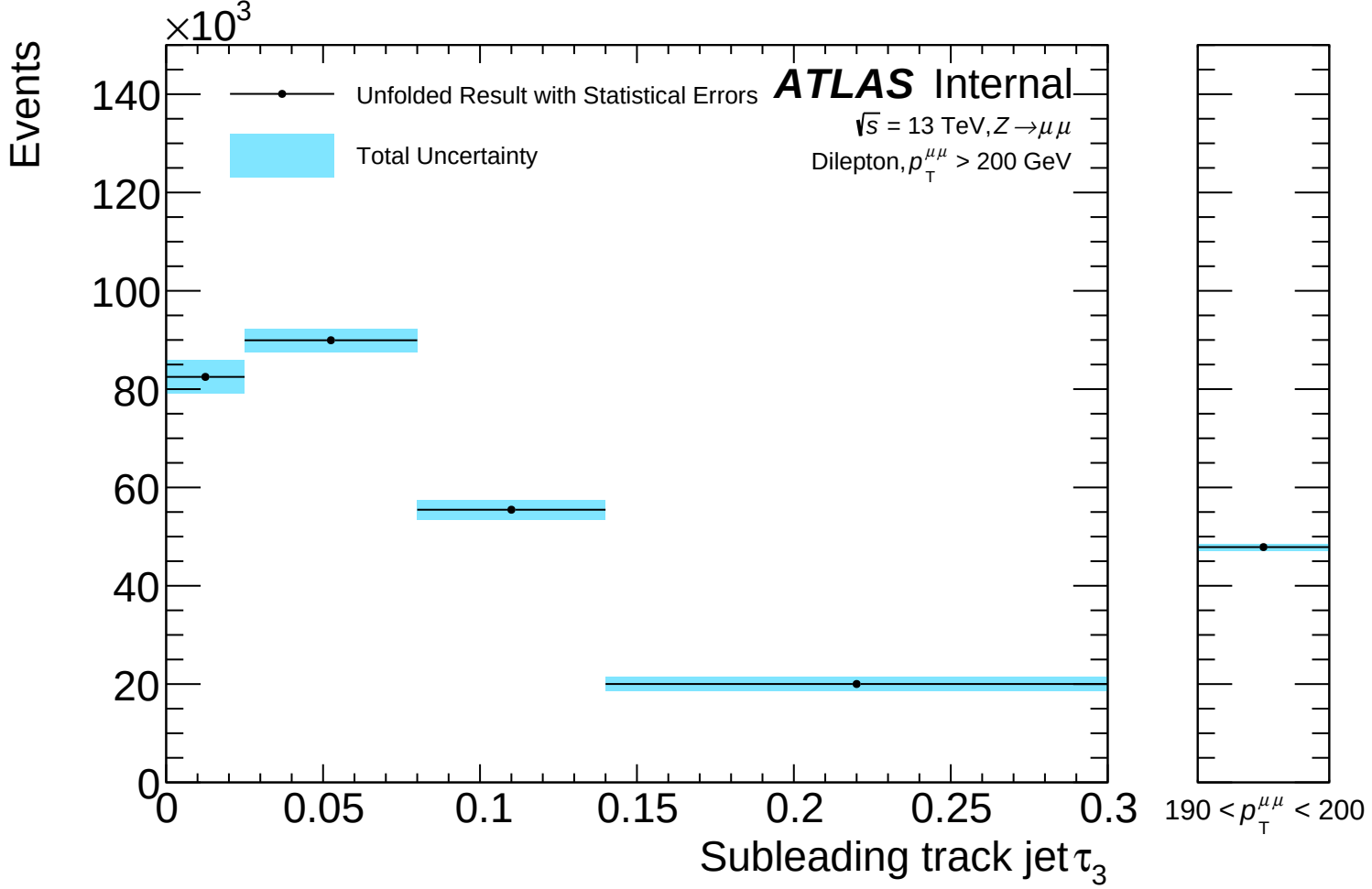
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

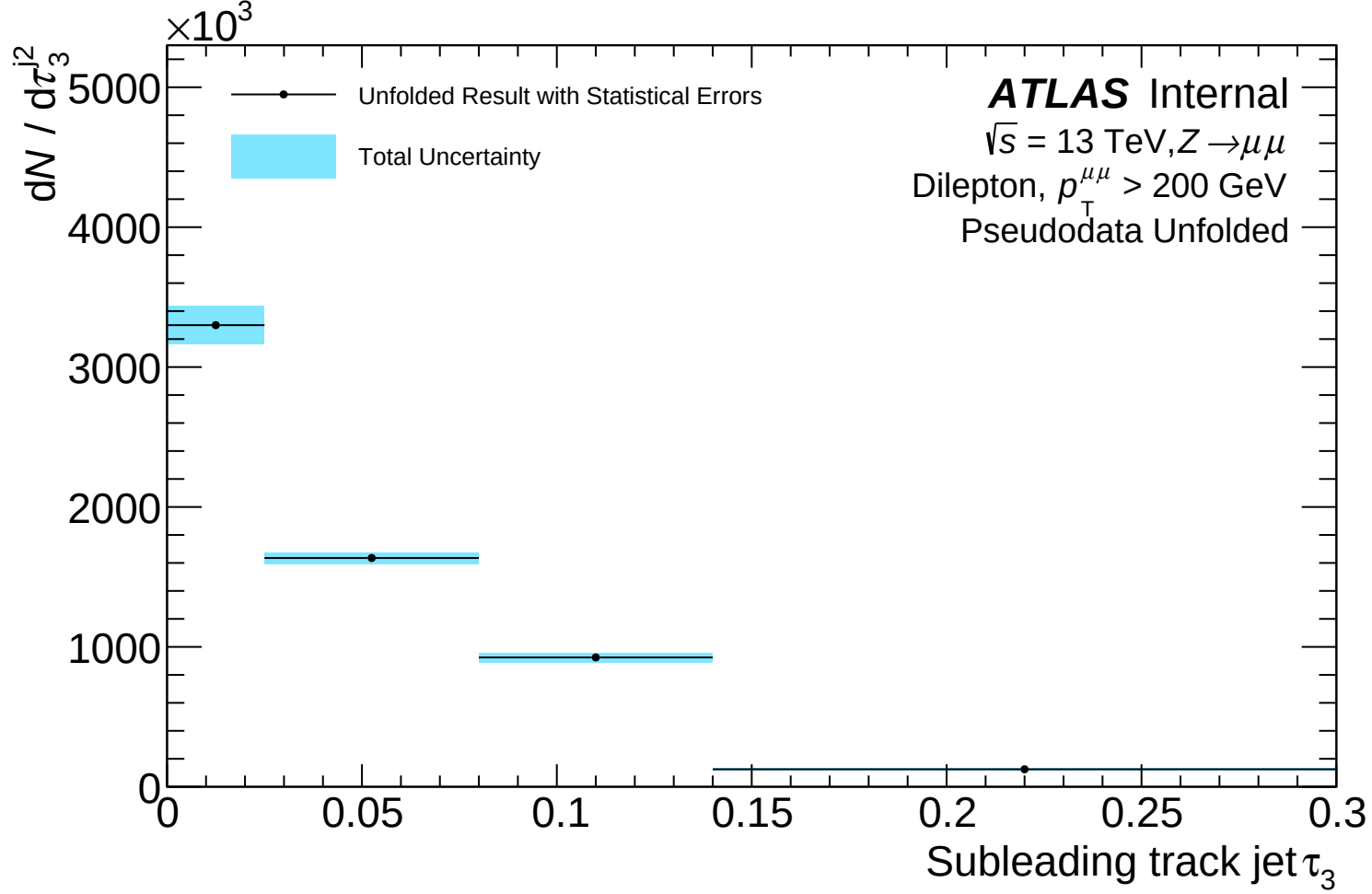
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

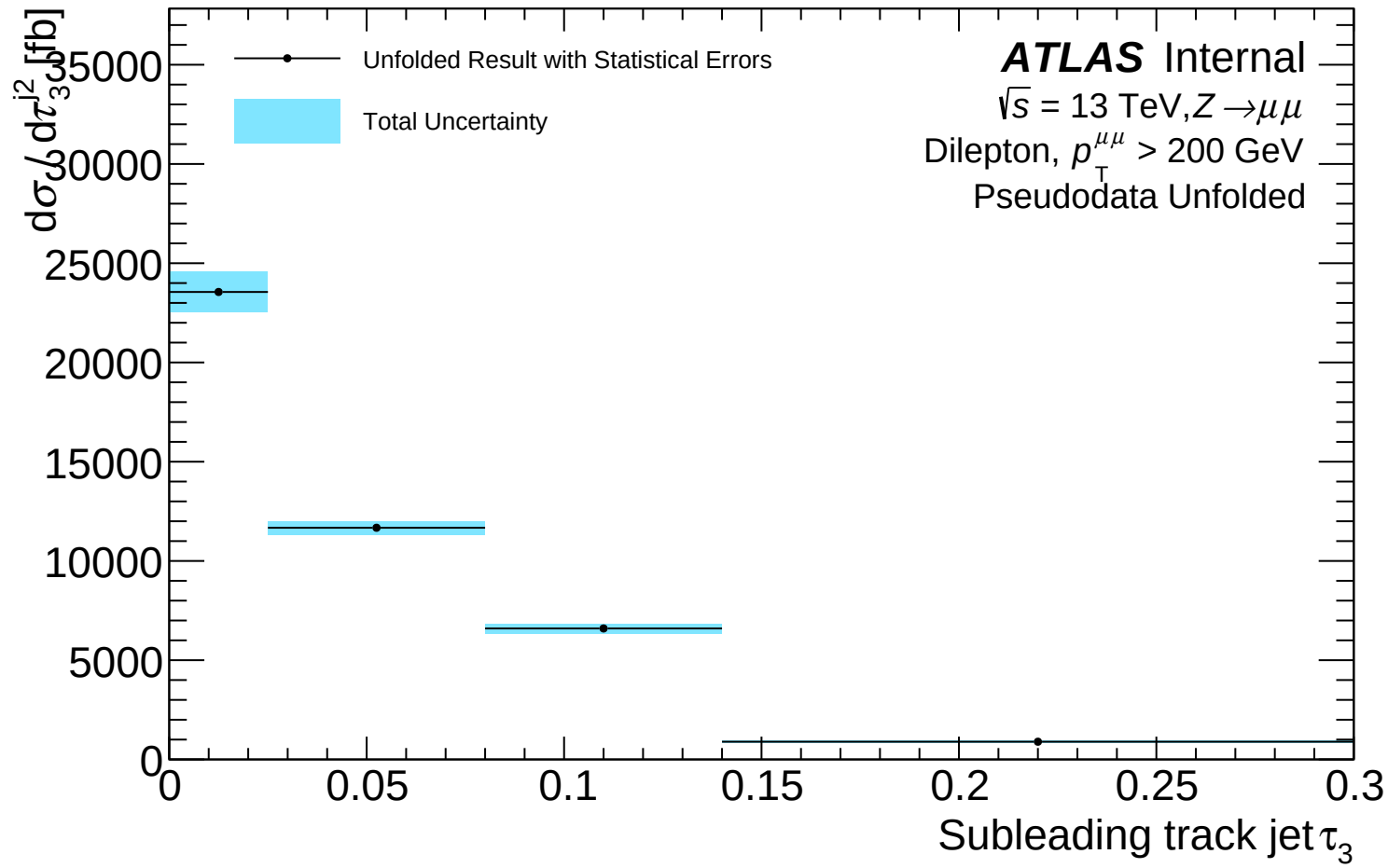
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty

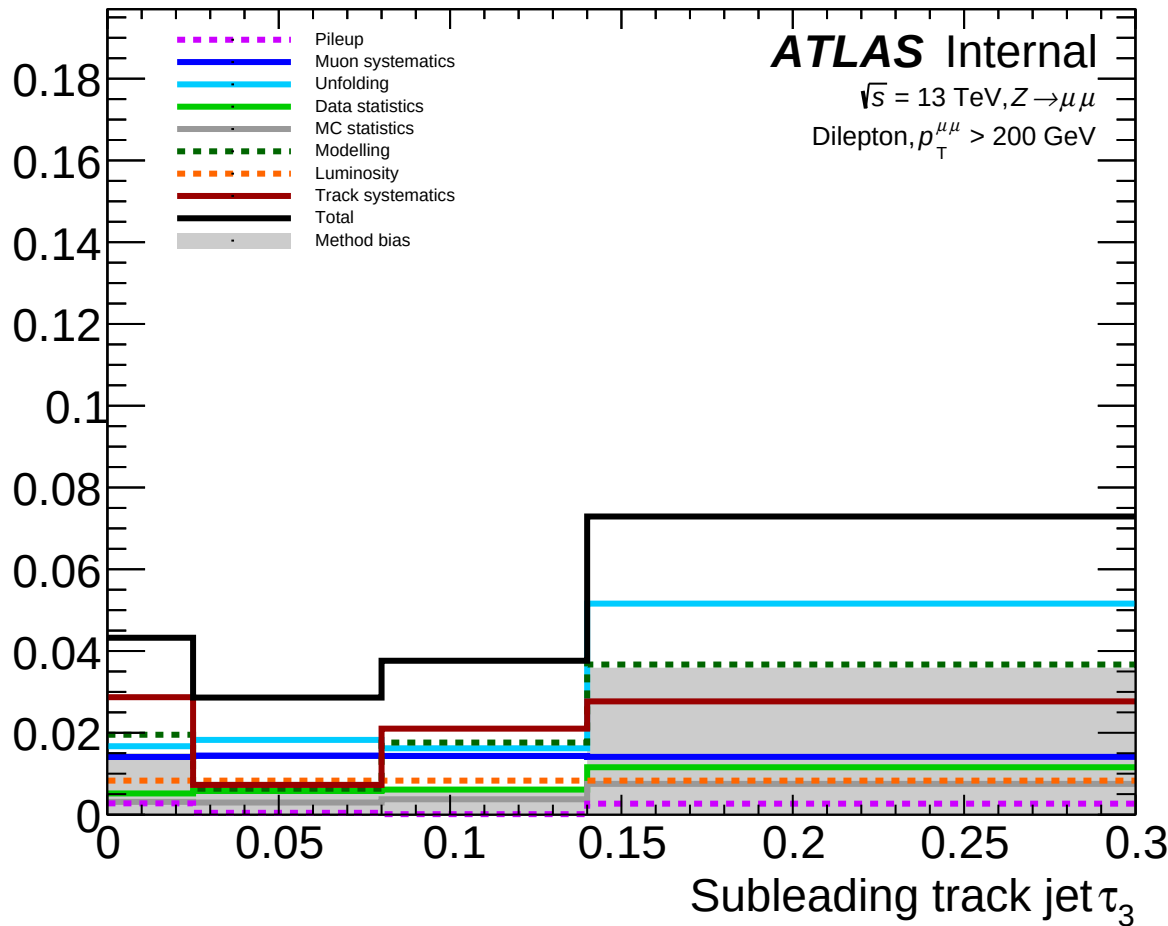


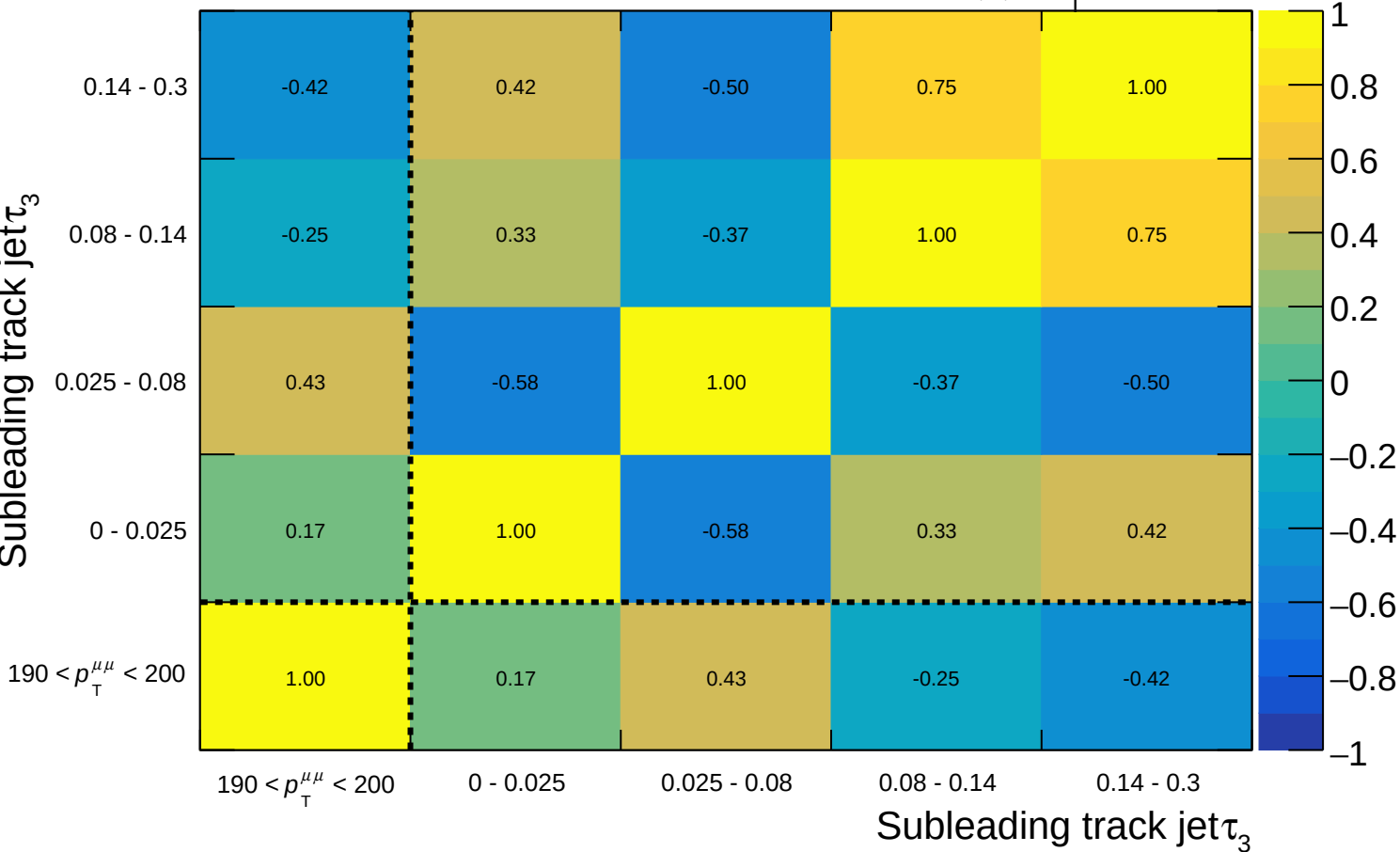


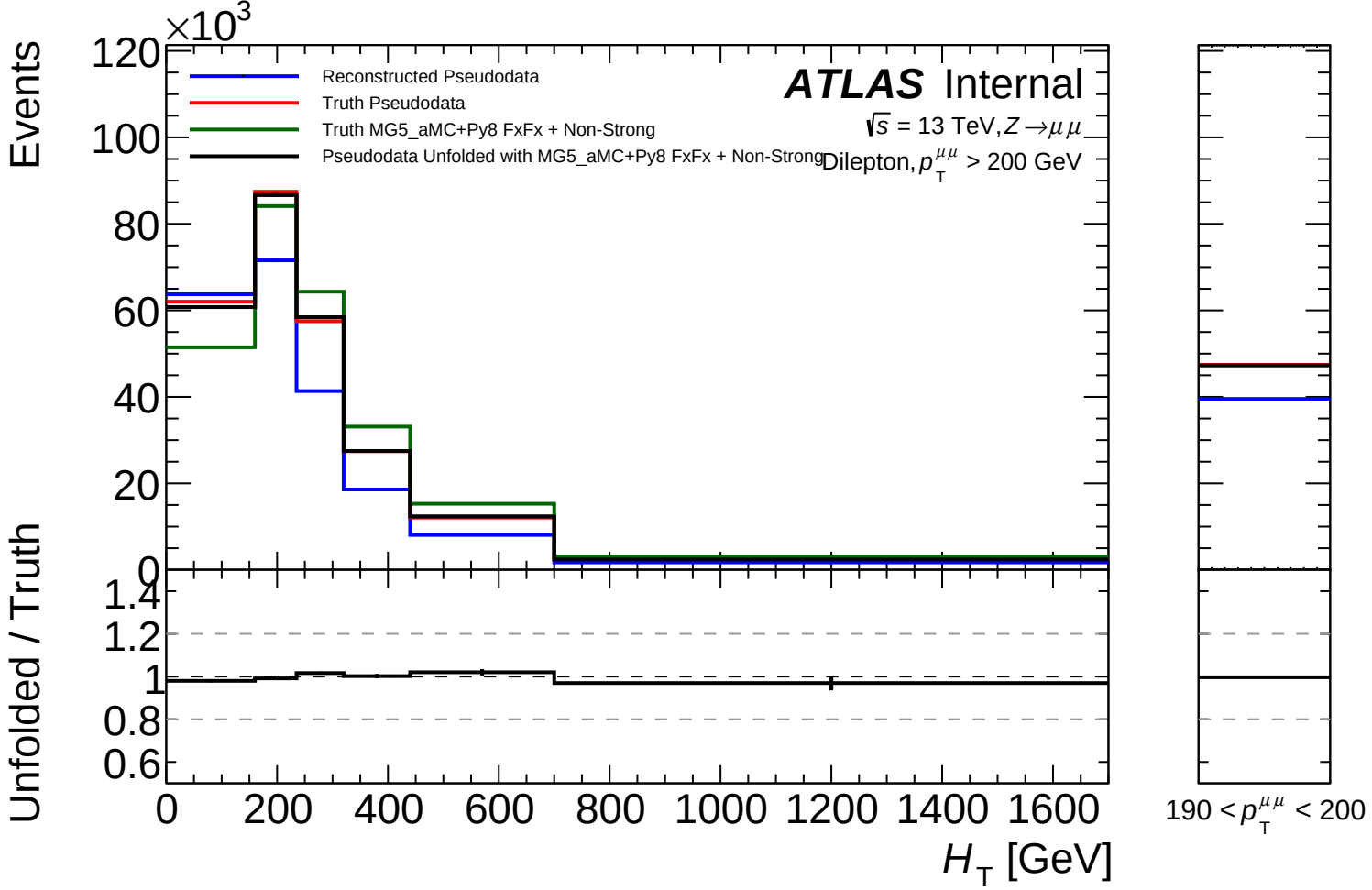


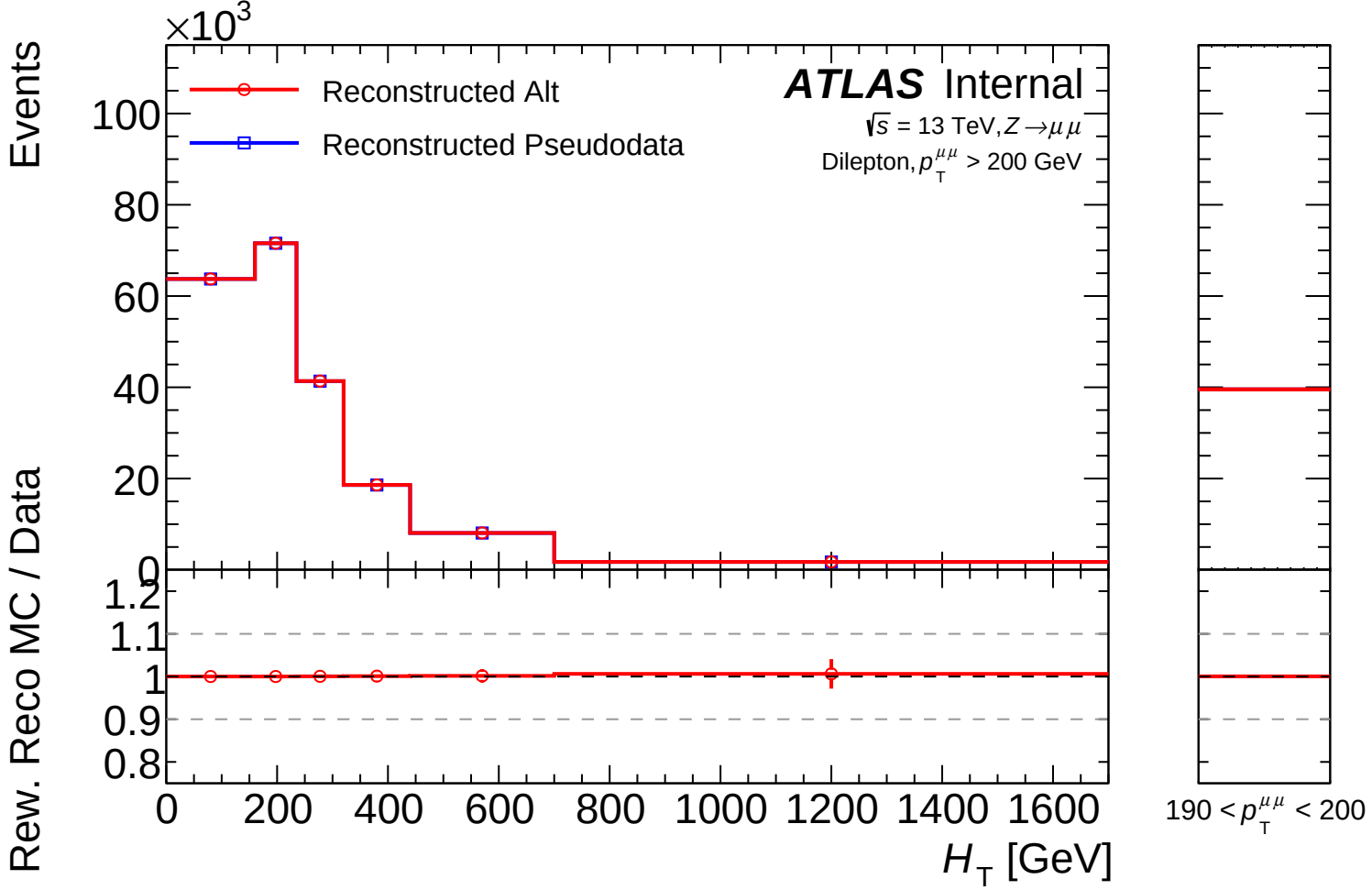


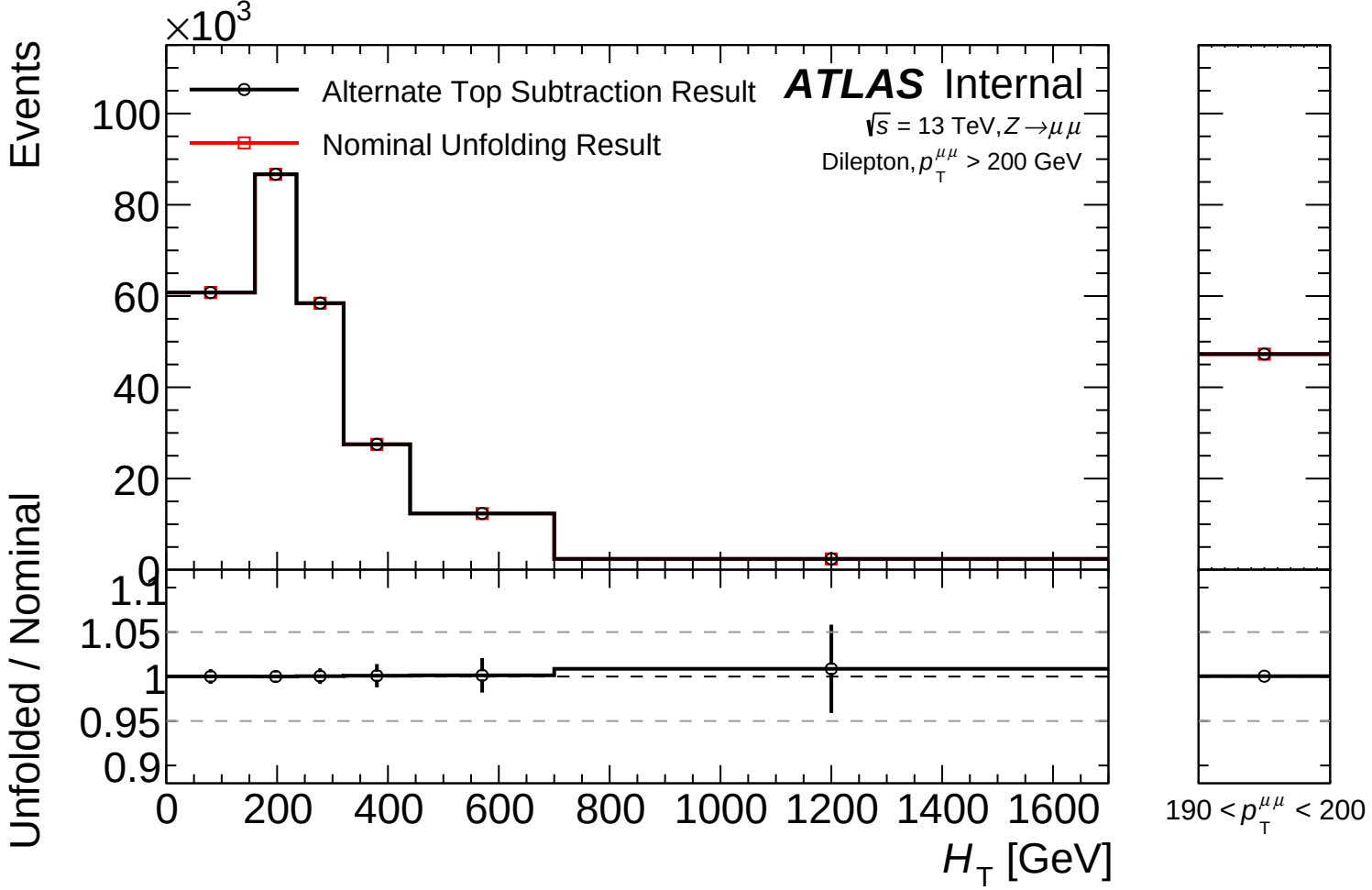




ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Subleading track jet τ_3 







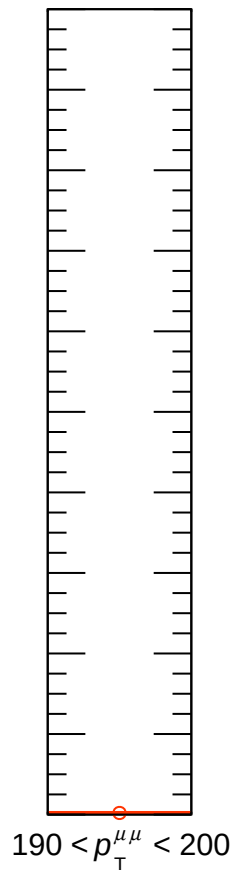
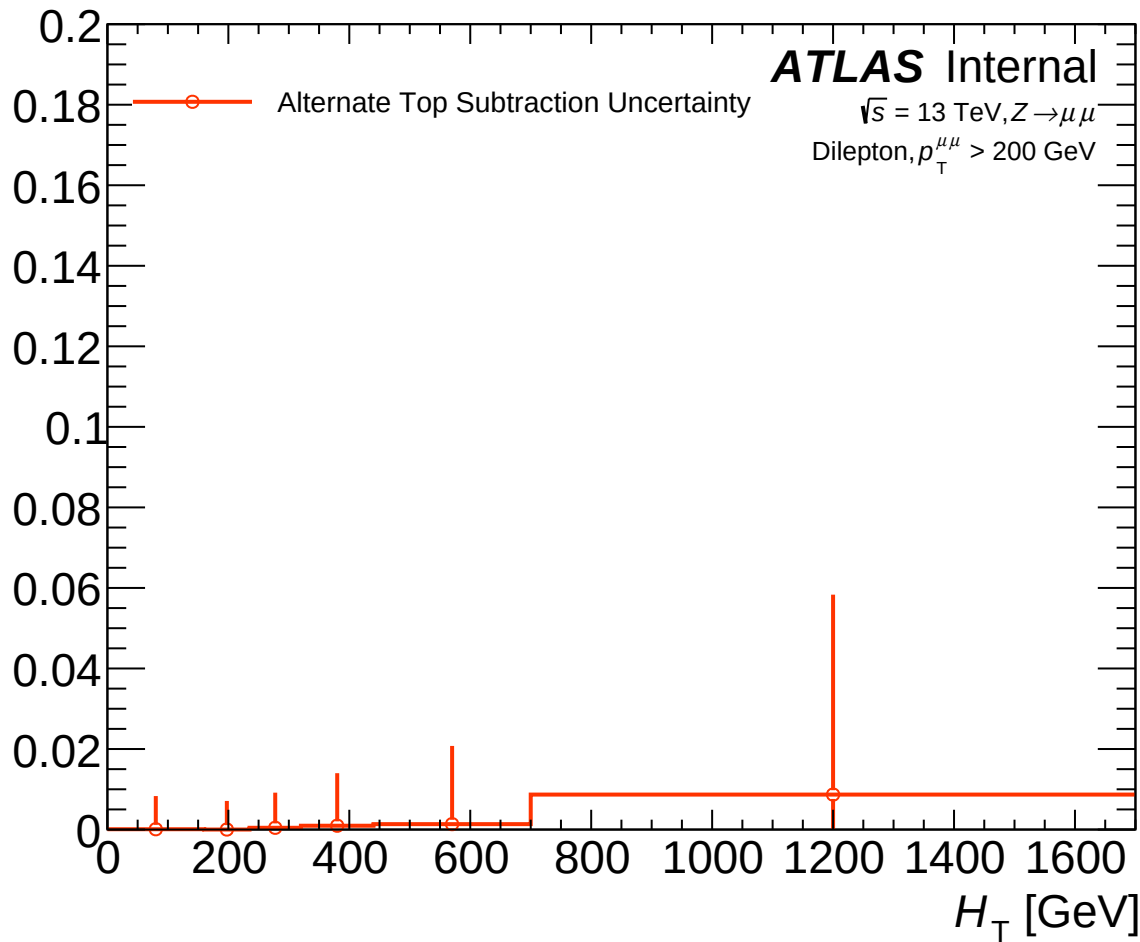
Fractional systematic impact

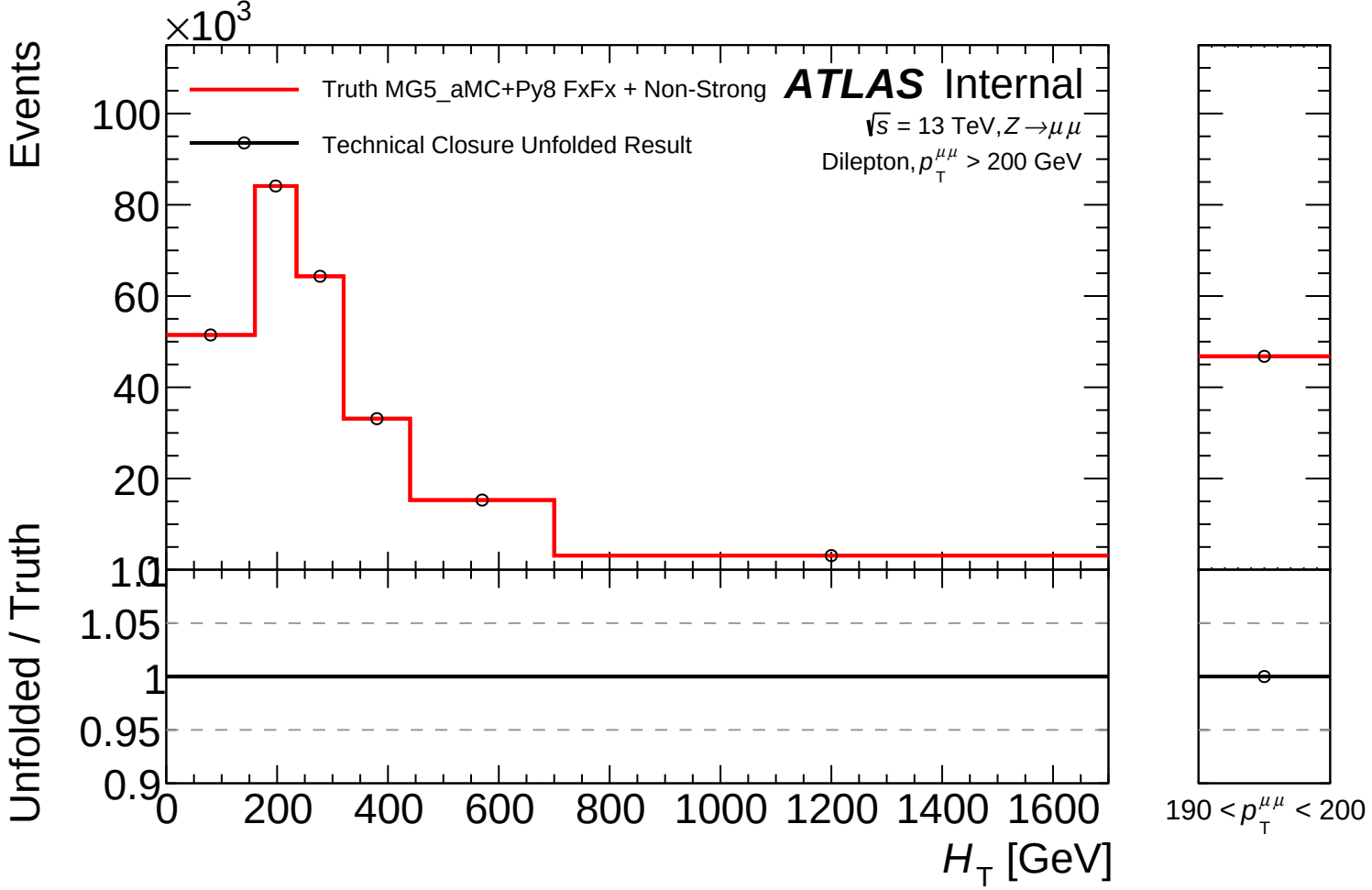
ATLAS Internal

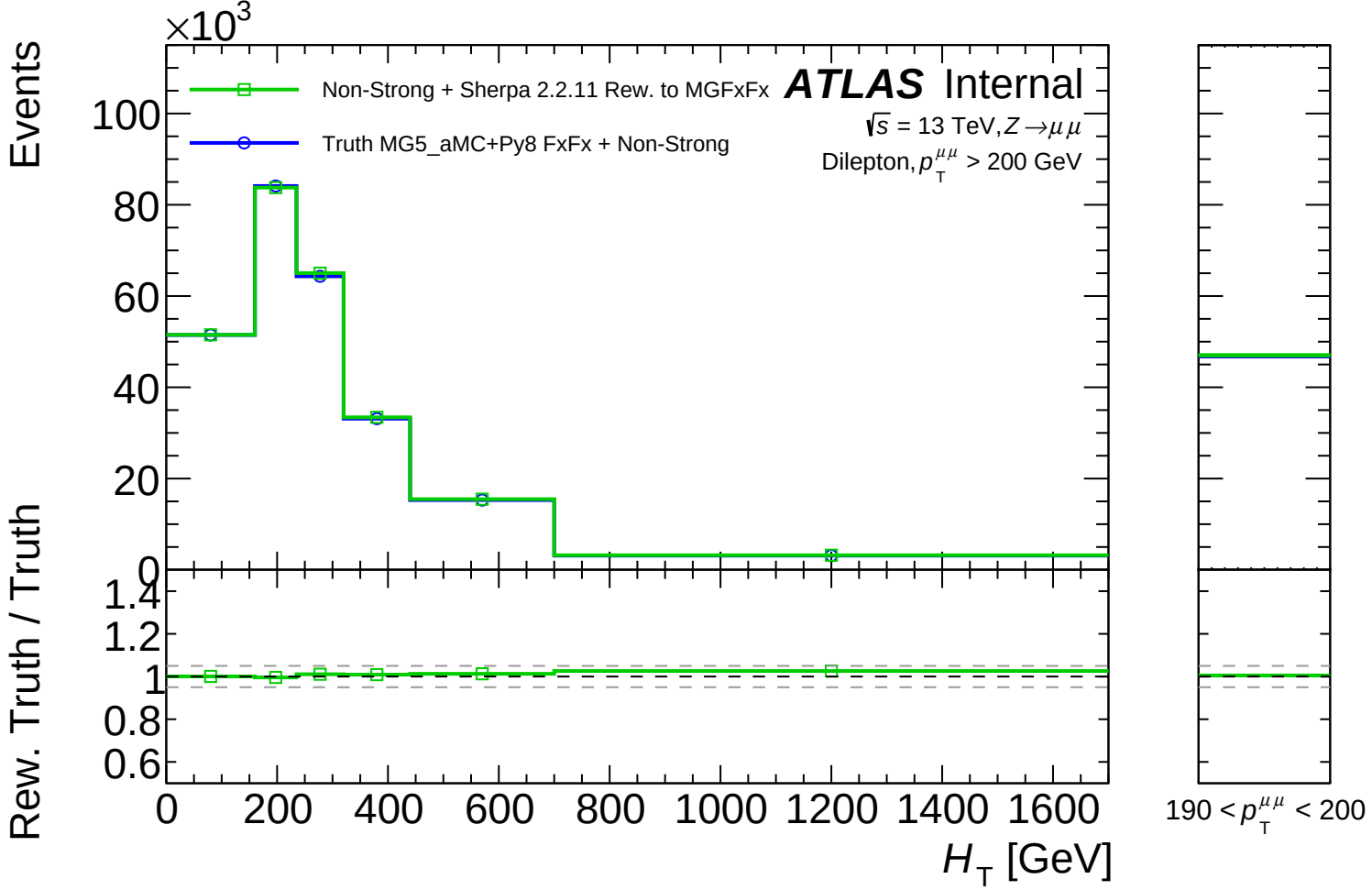
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

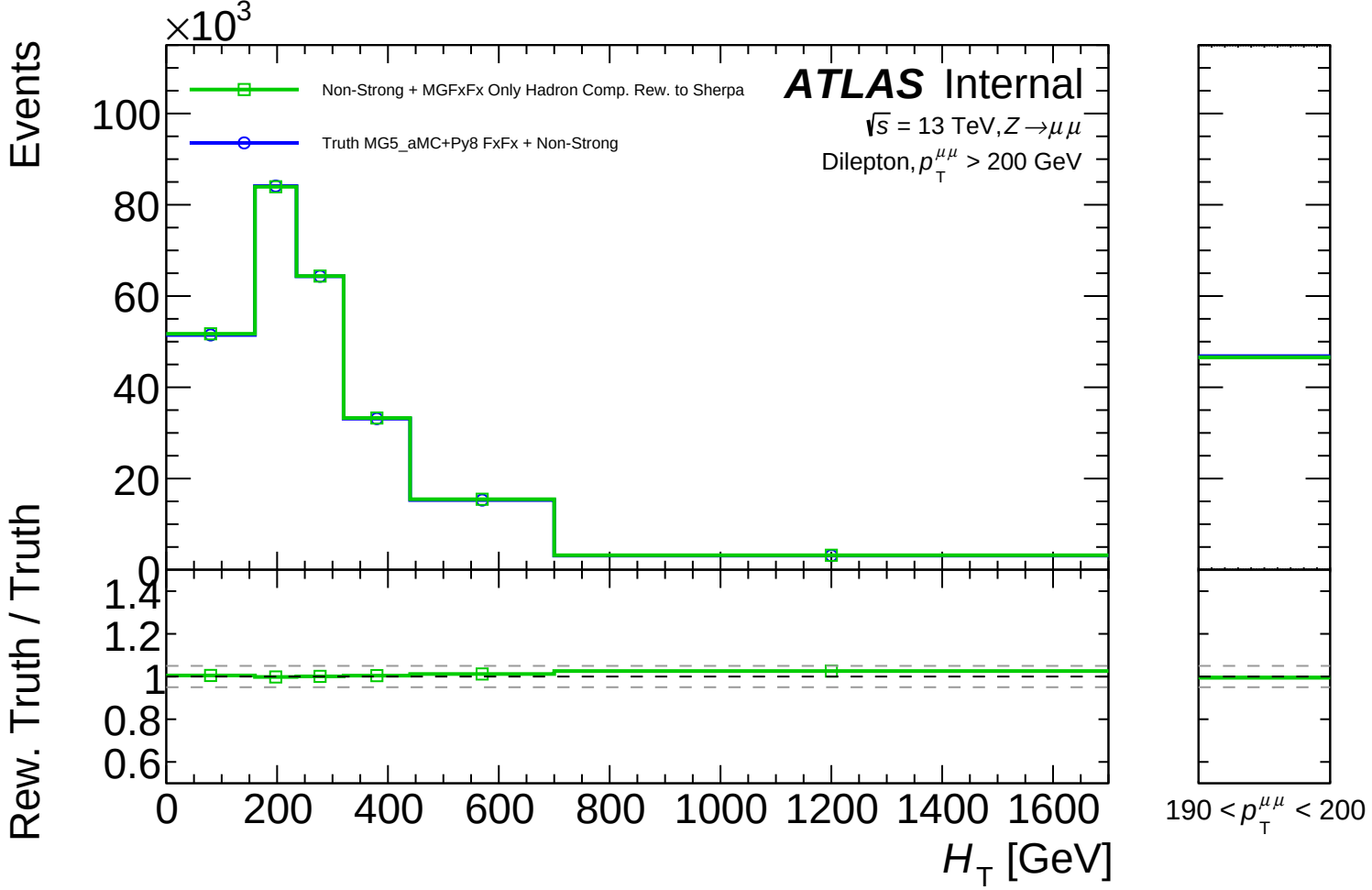
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

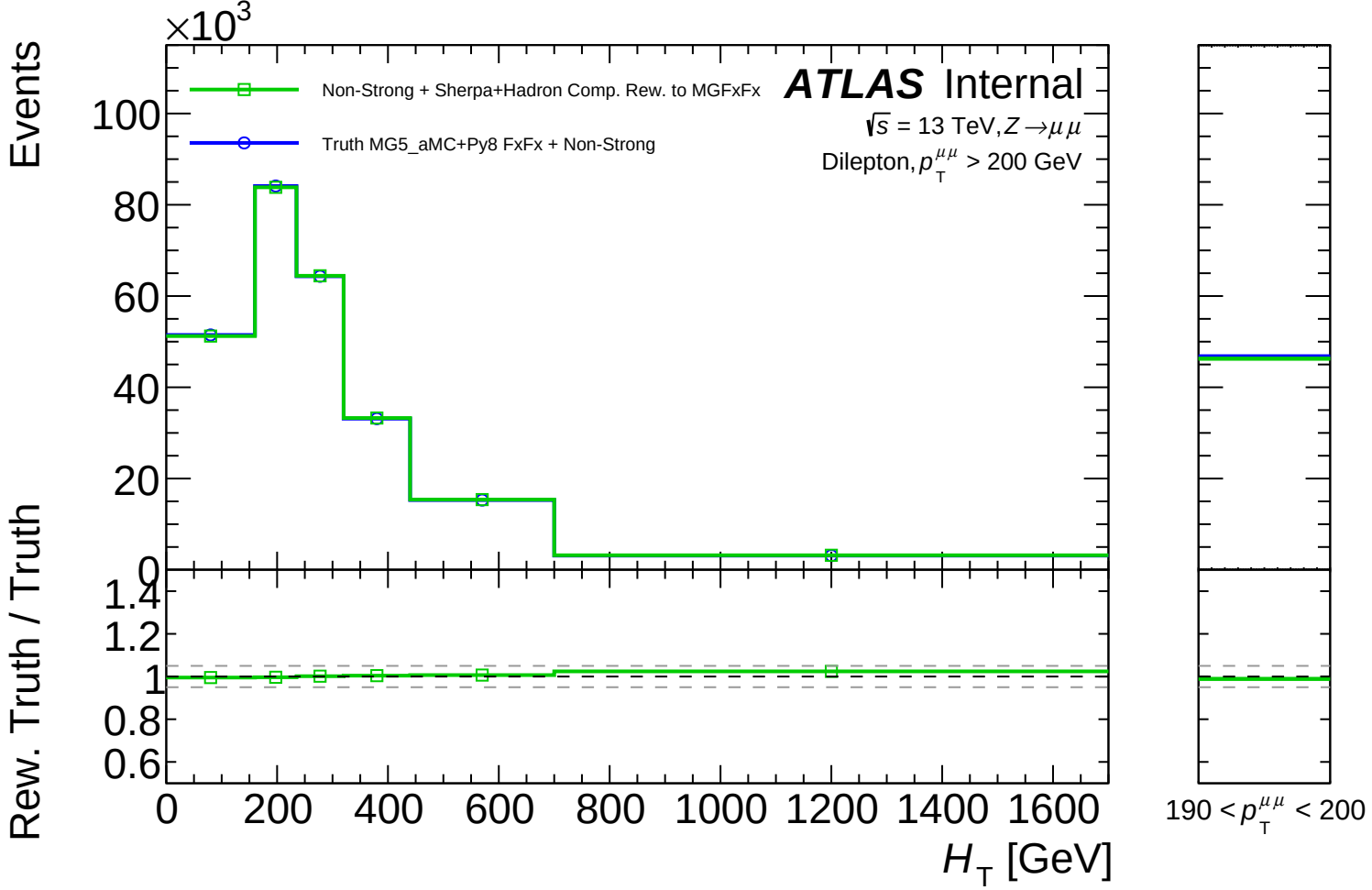
Alternate Top Subtraction Uncertainty

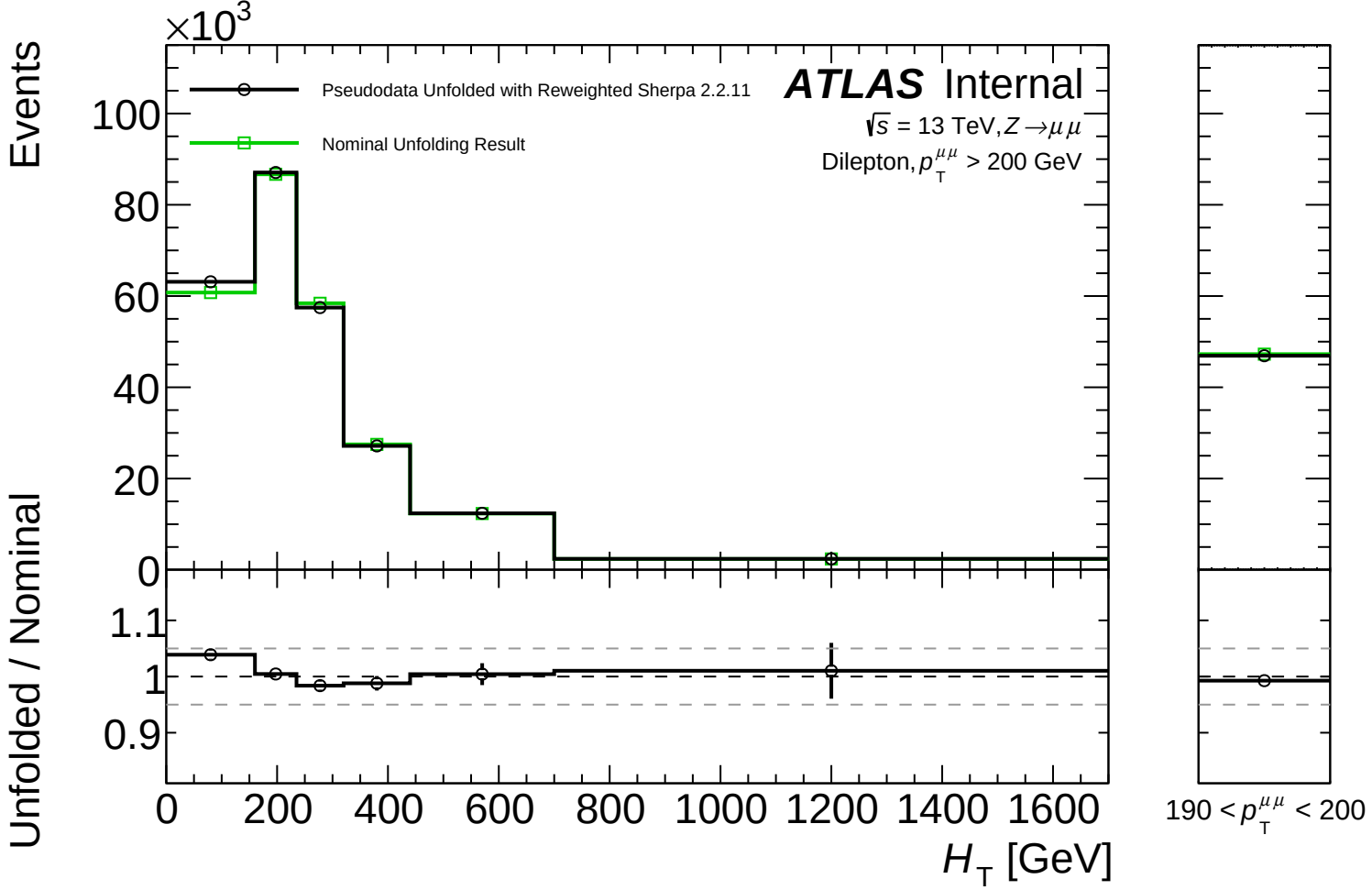


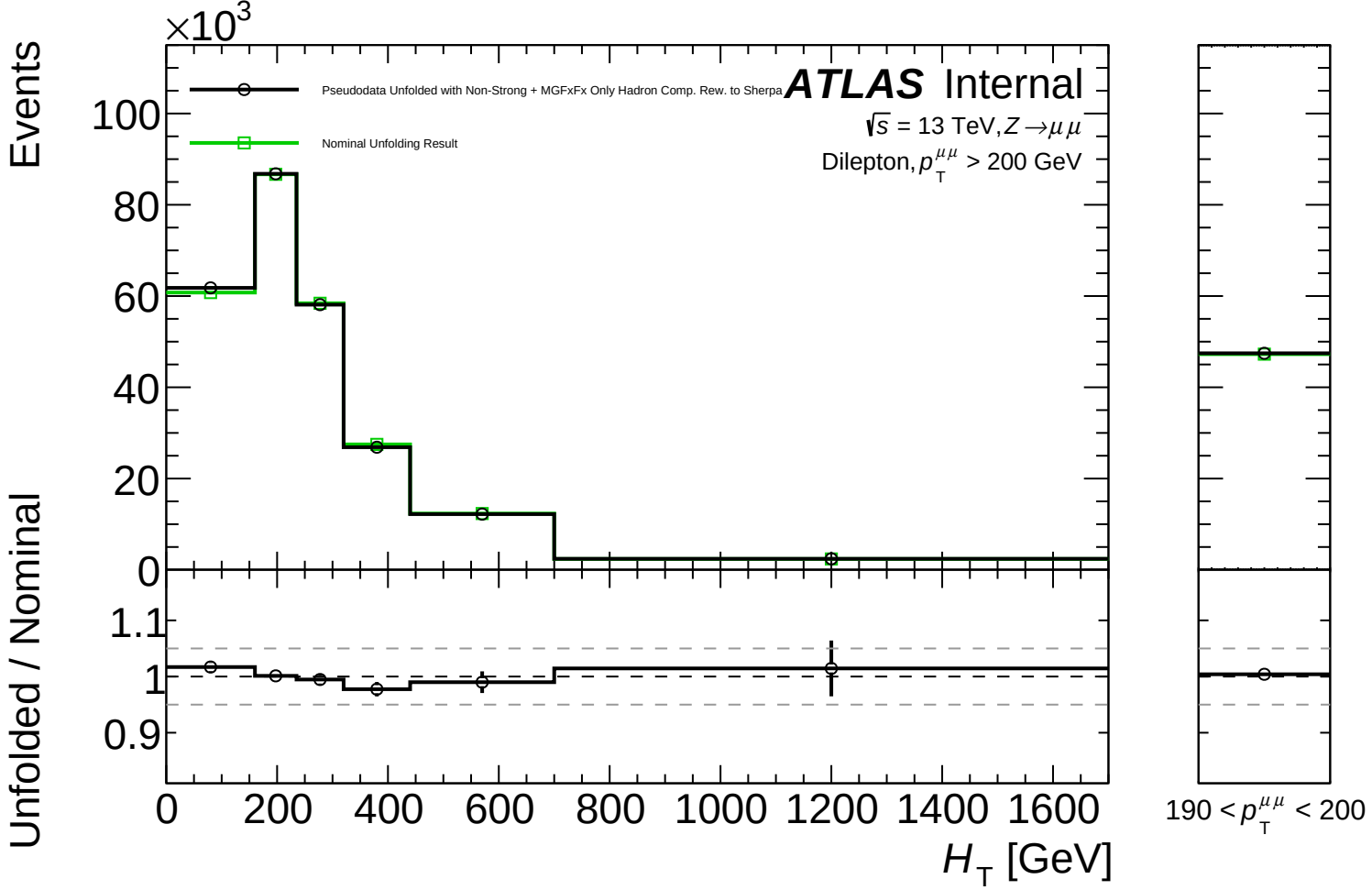


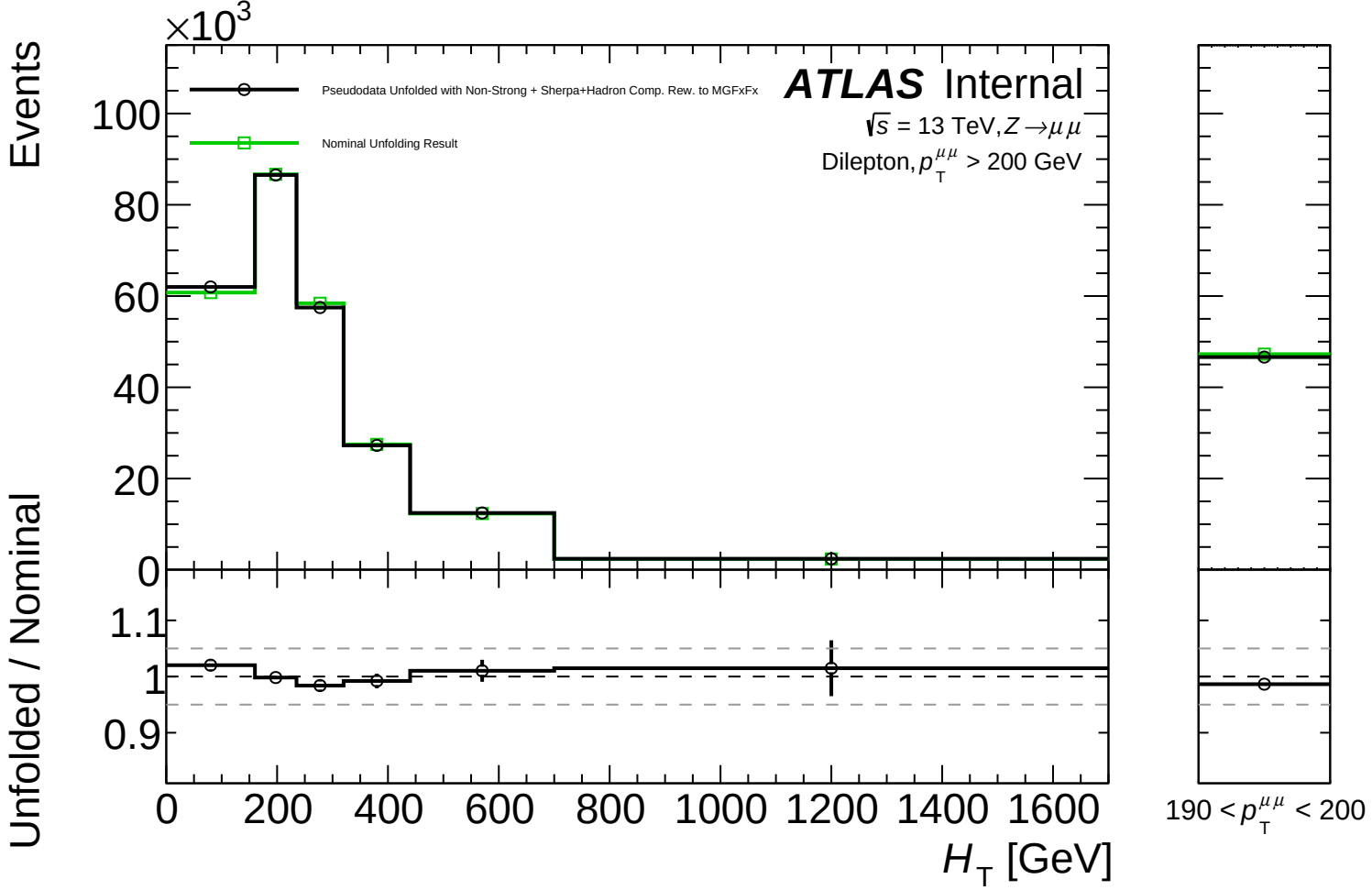






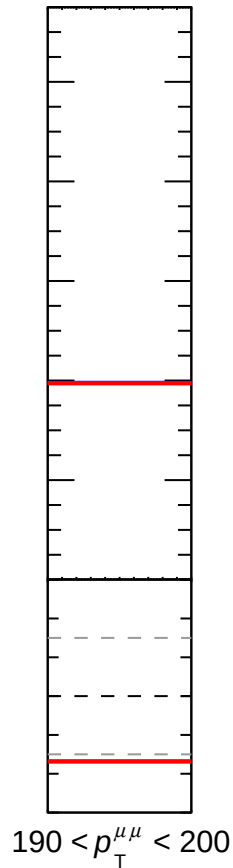
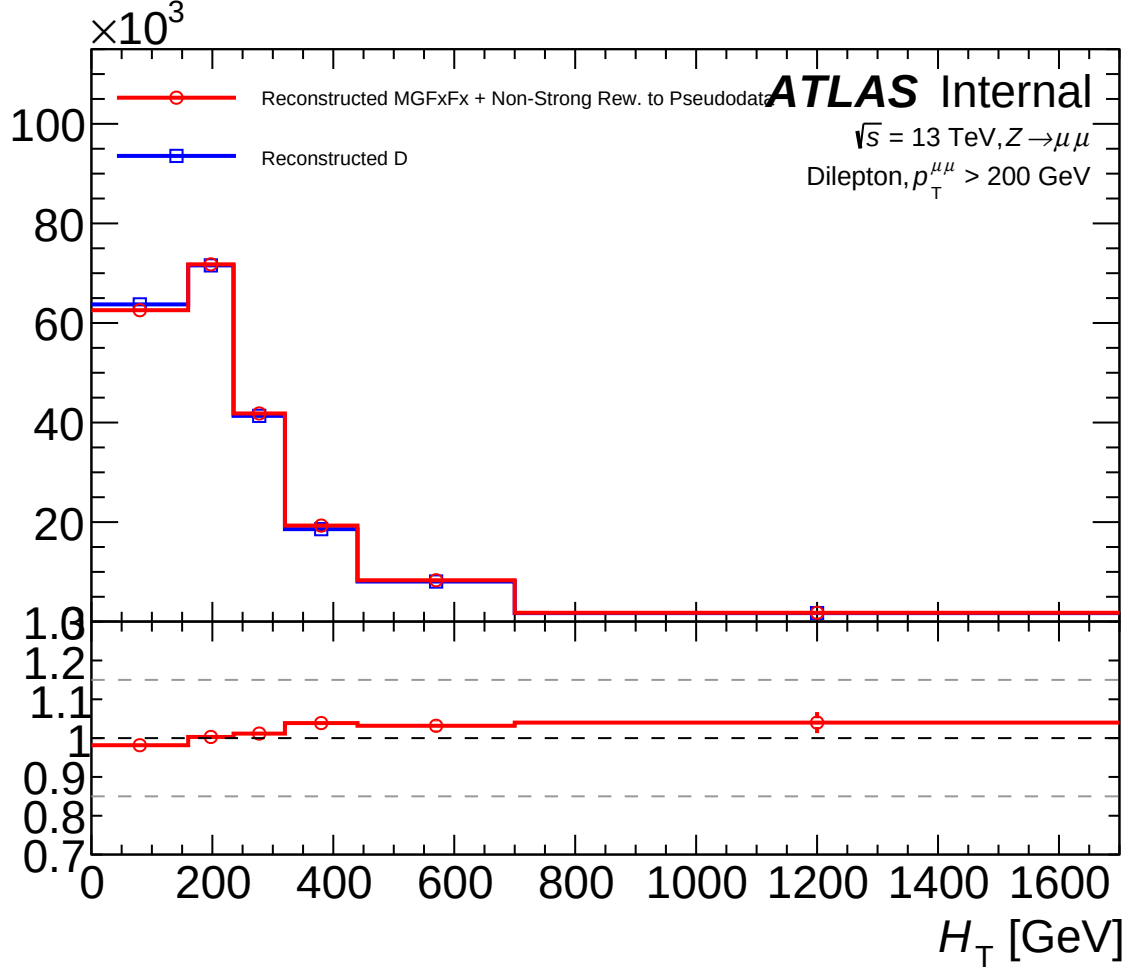


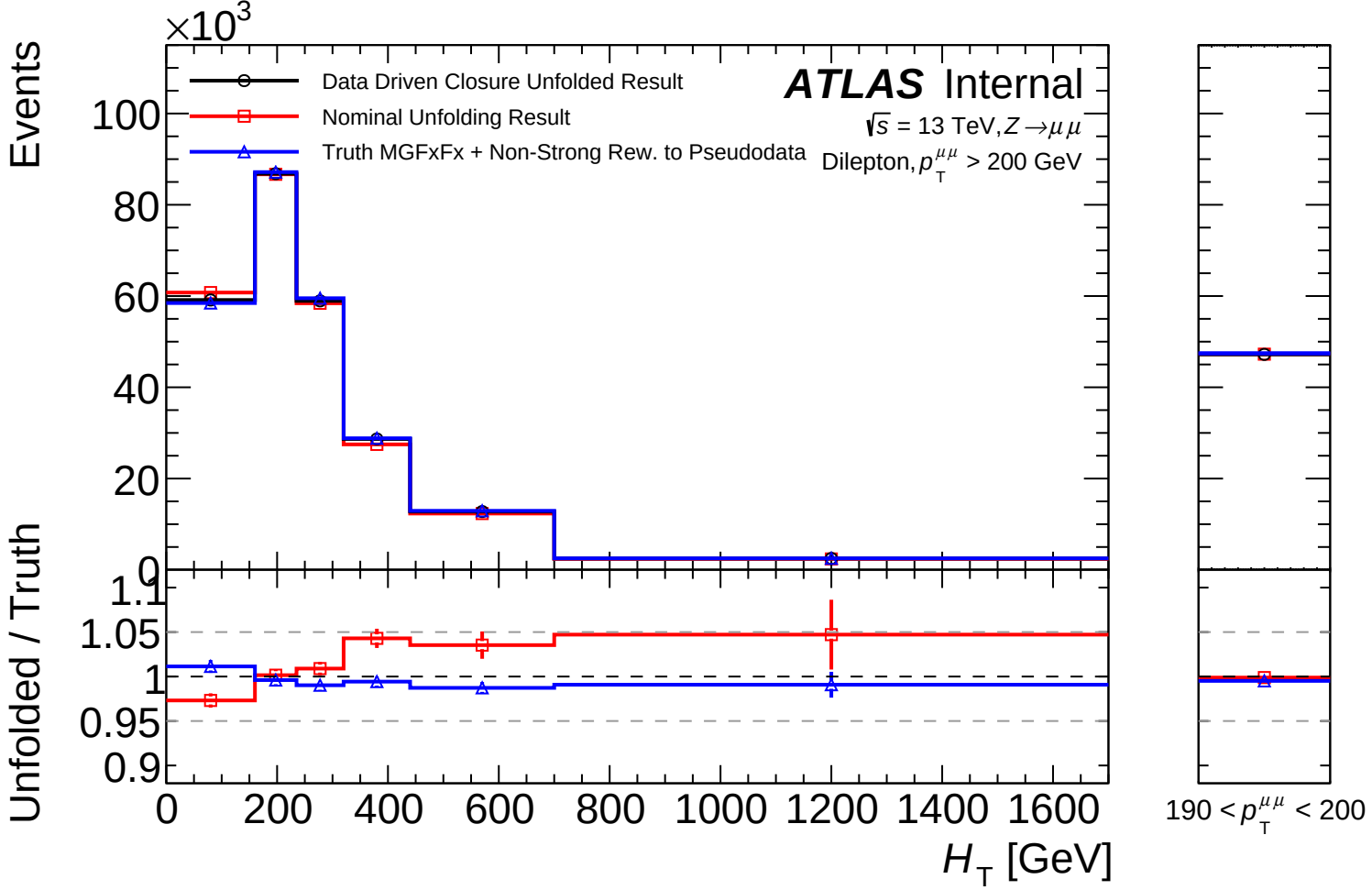




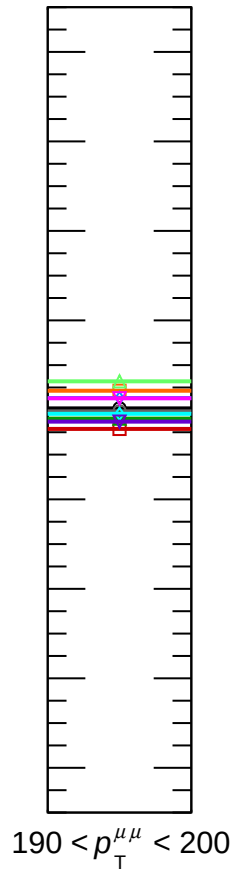
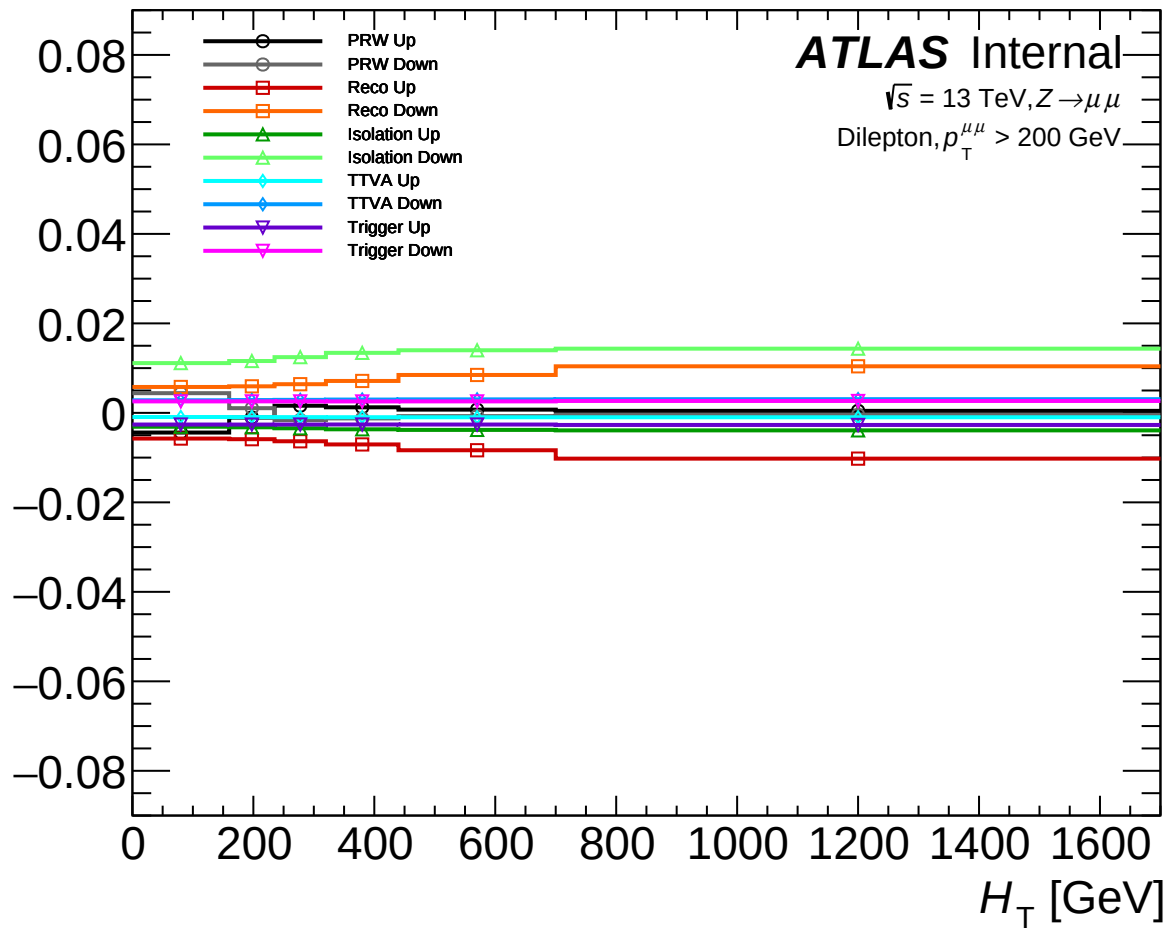
Events

Rew. Reco MC / Data

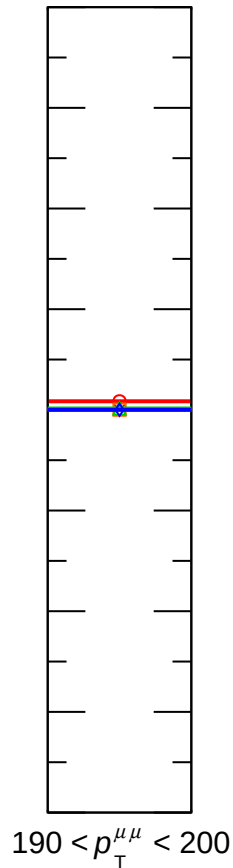
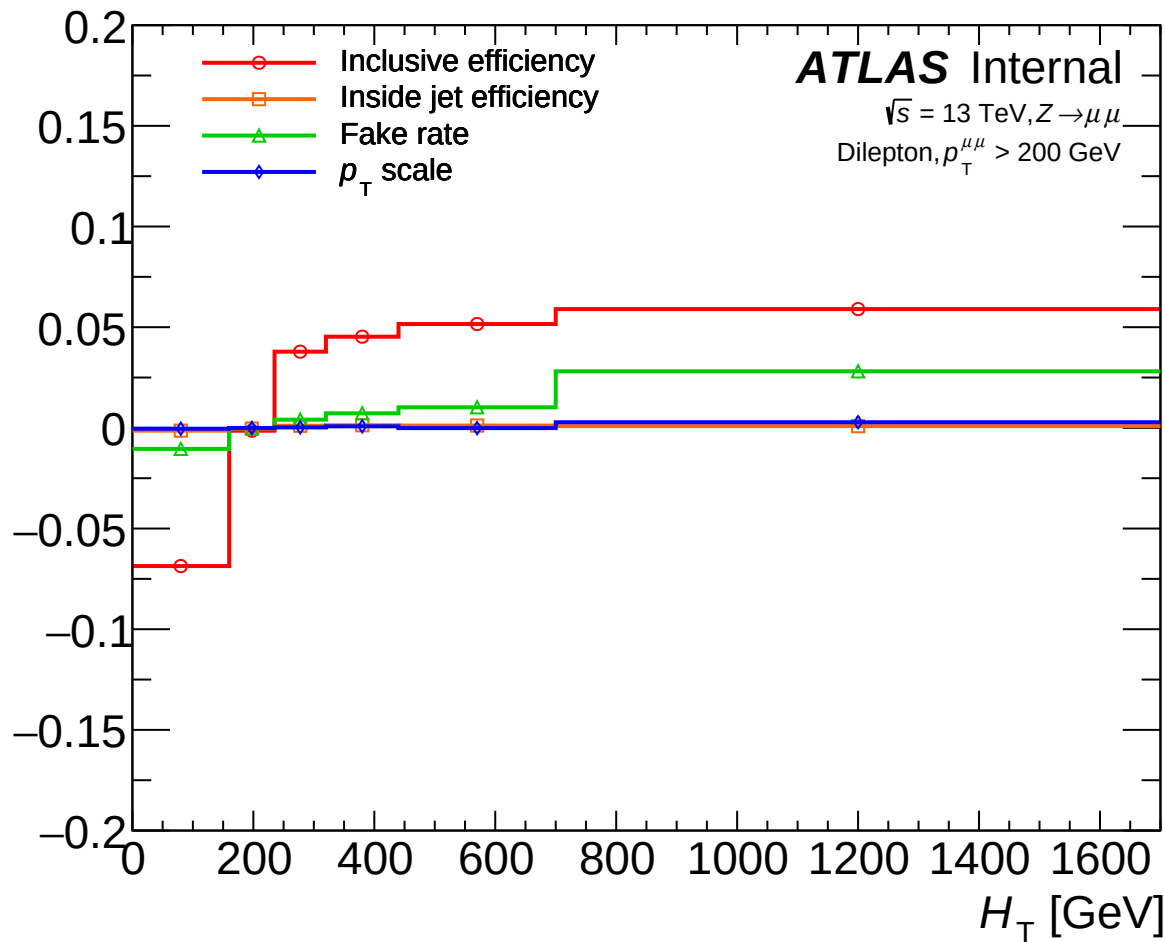




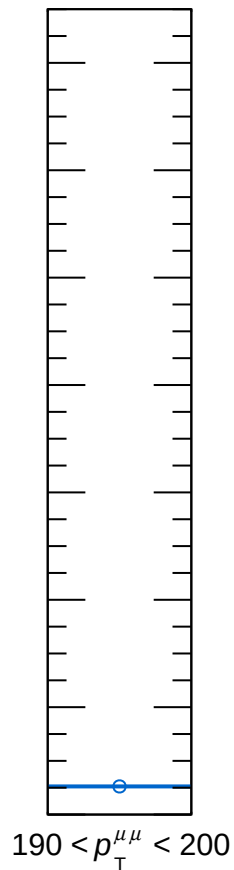
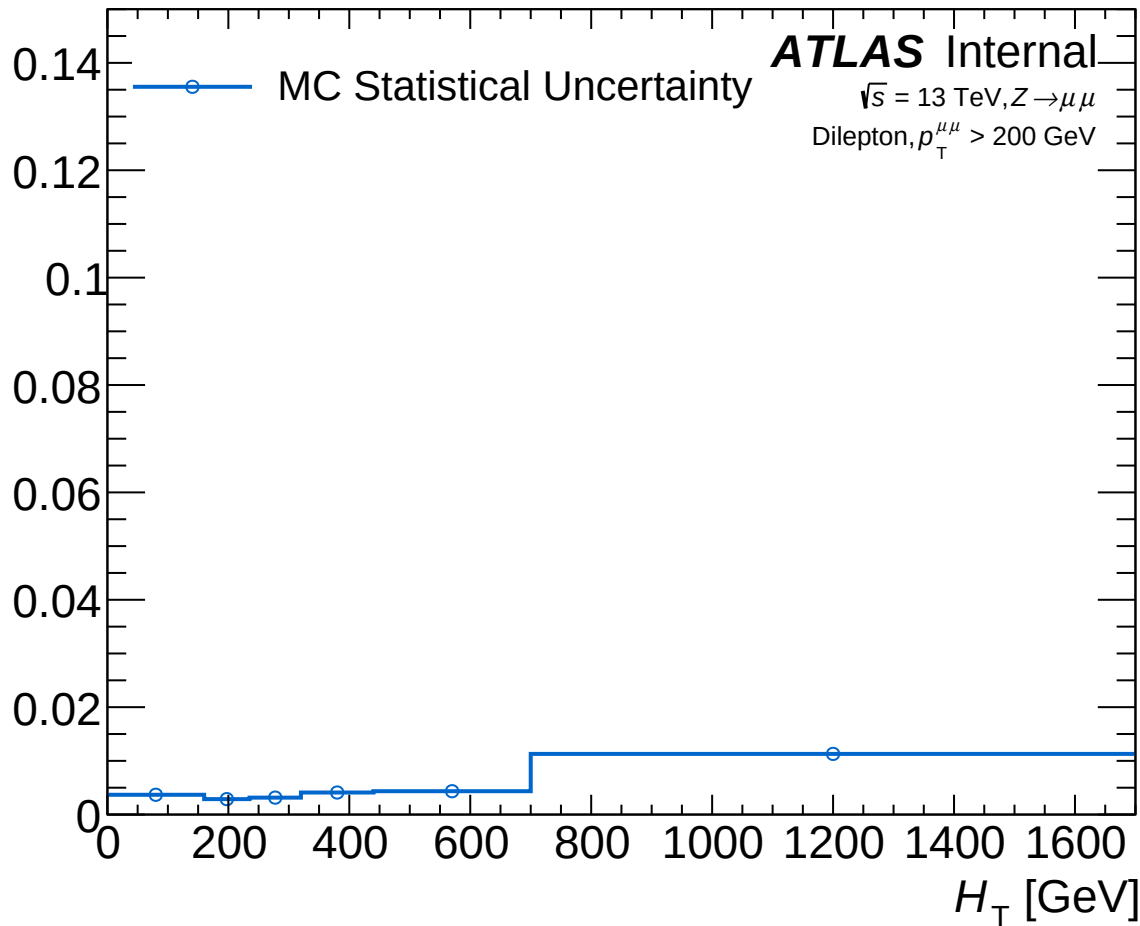
Fractional systematic impact



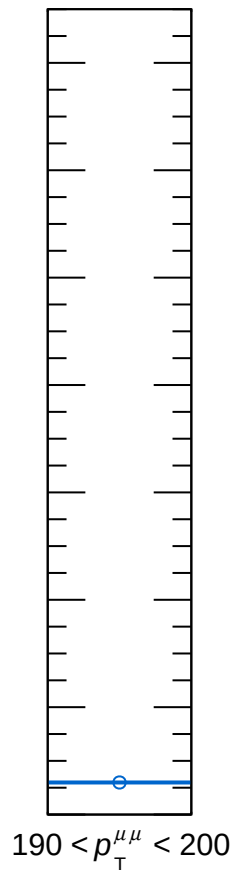
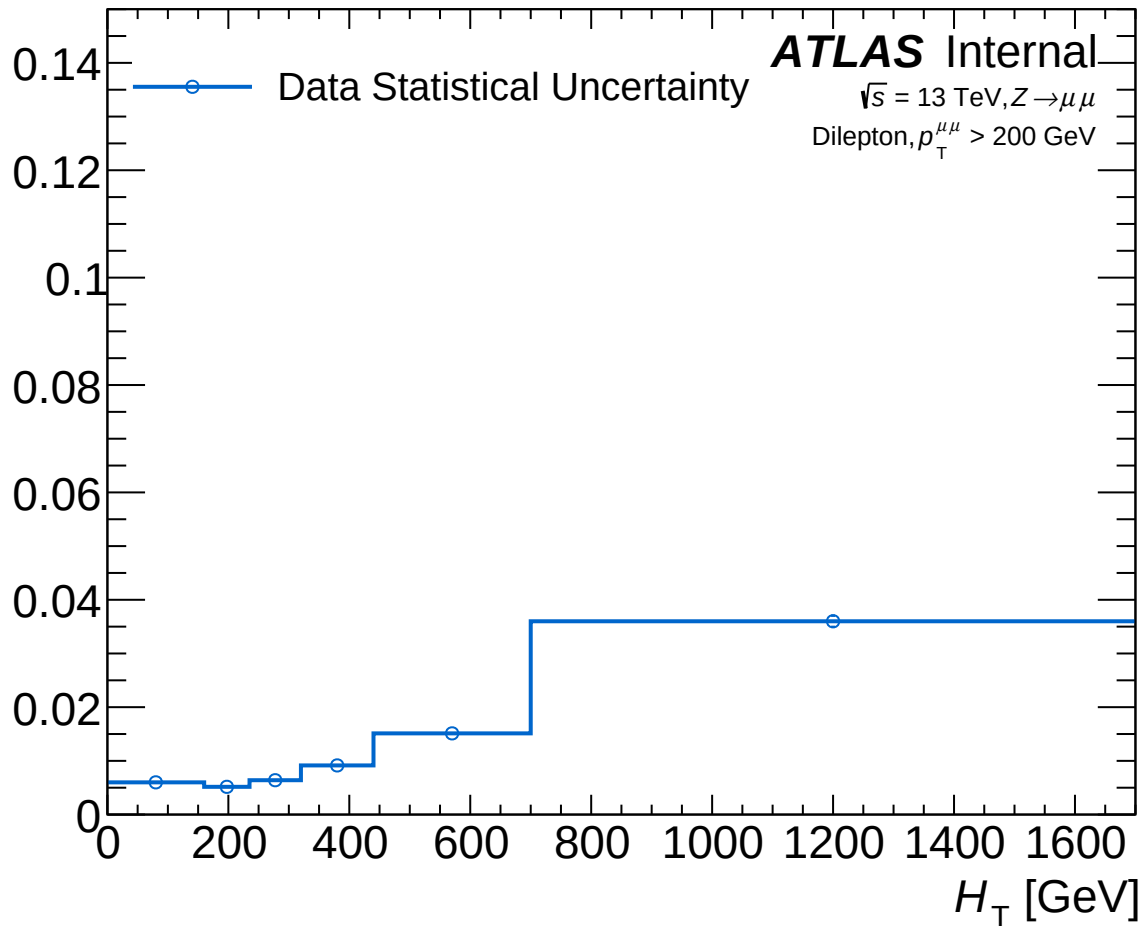
Fractional systematic impact



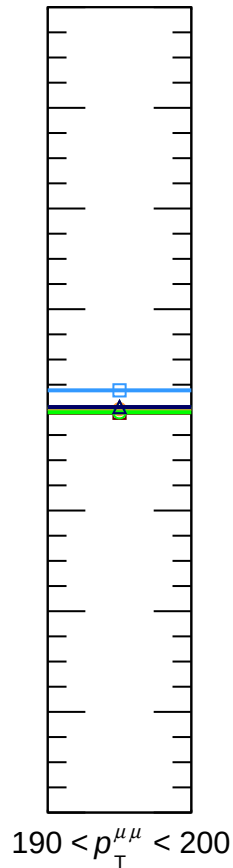
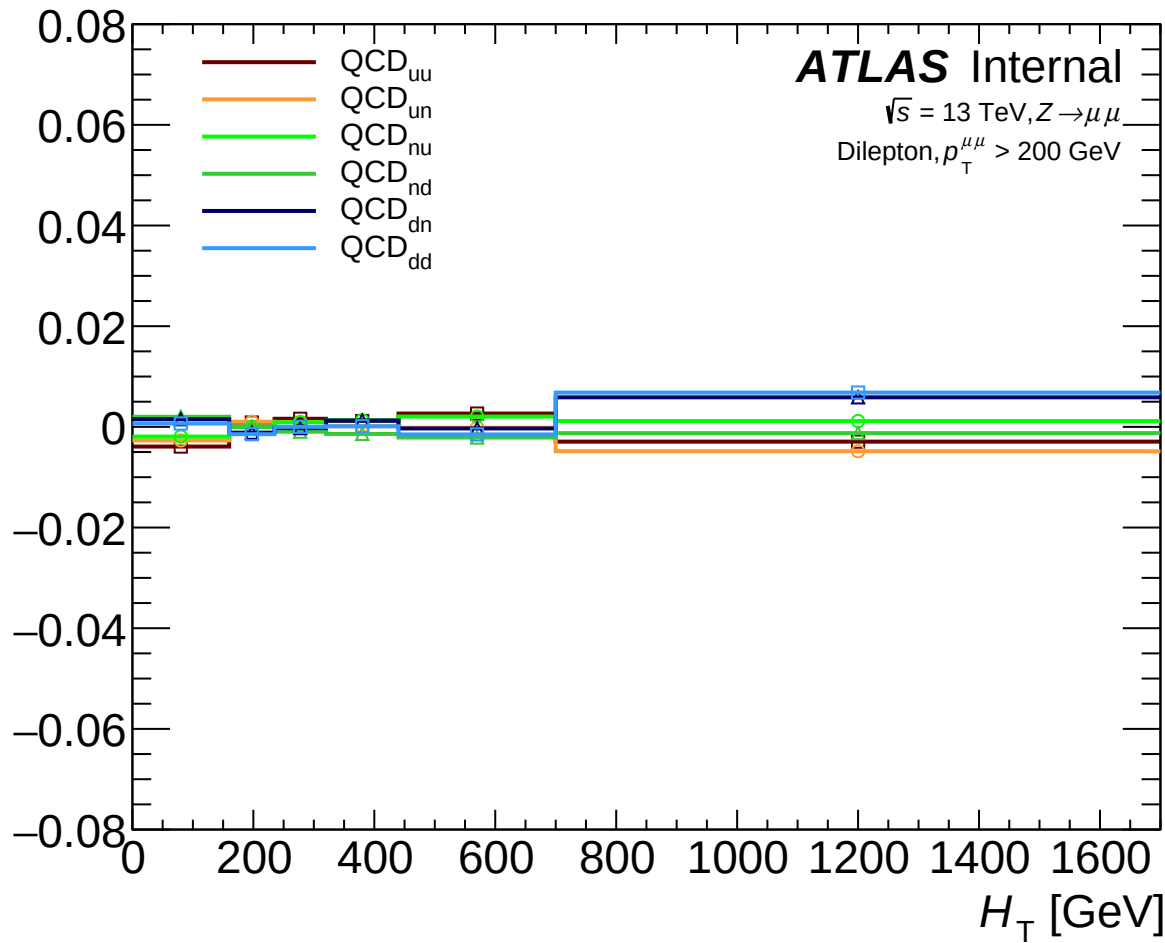
Fractional systematic impact



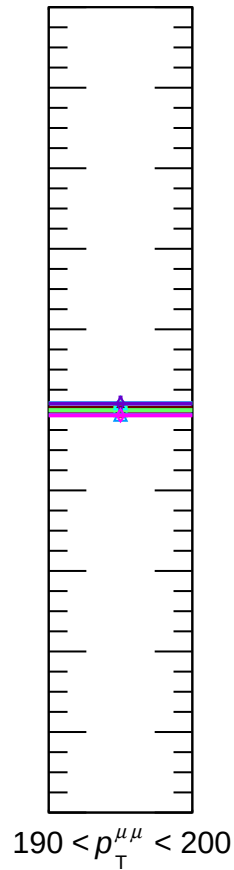
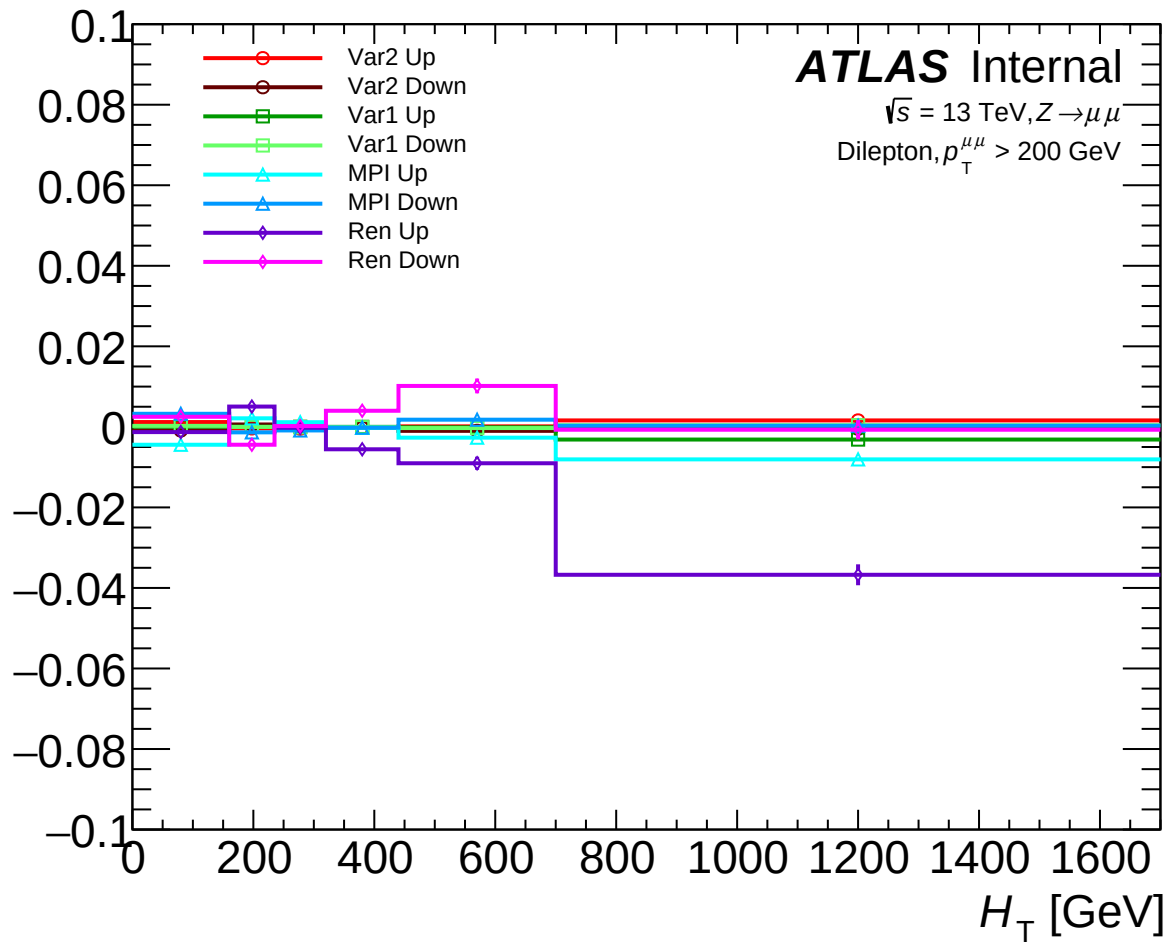
Fractional systematic impact



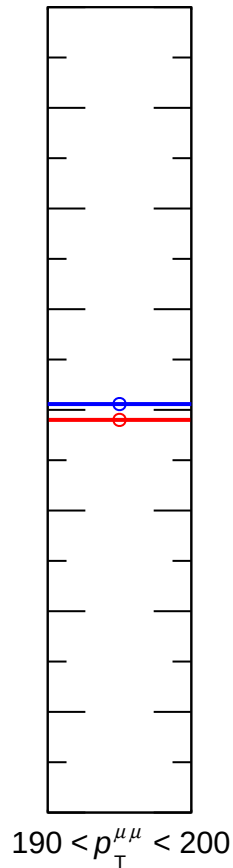
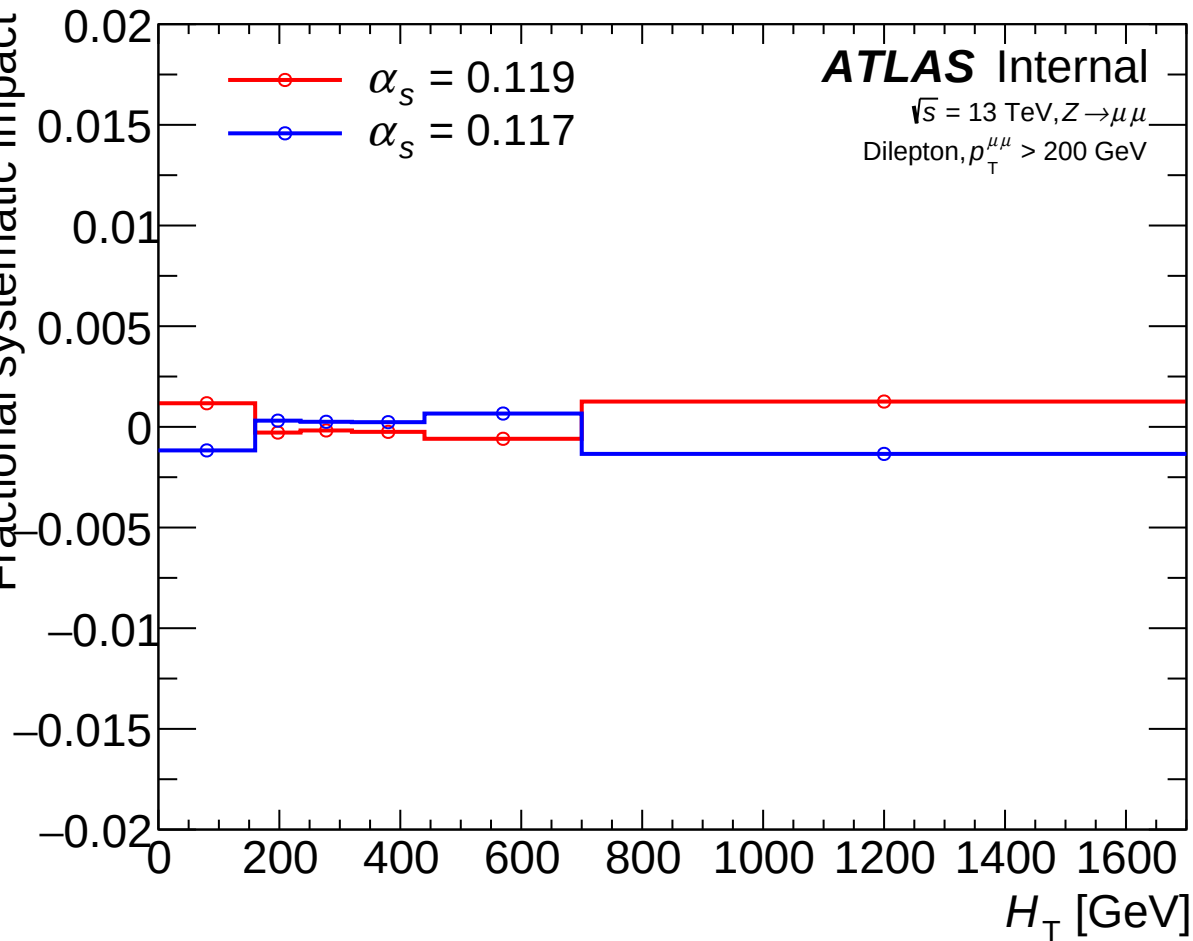
Fractional systematic impact



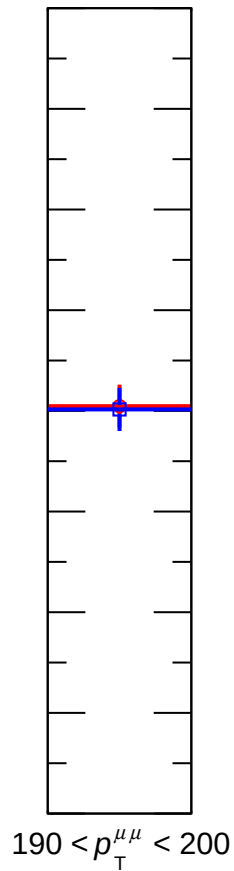
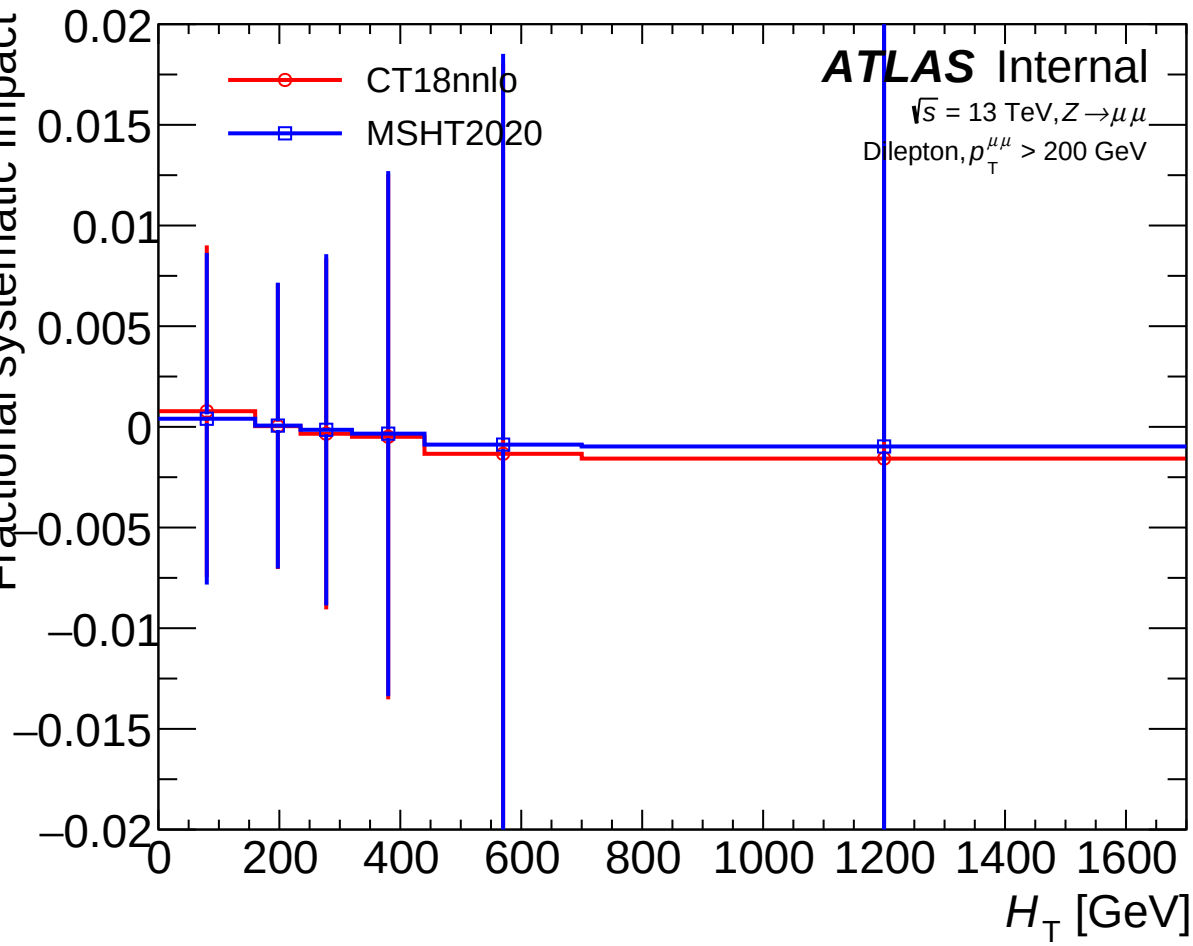
Fractional systematic impact



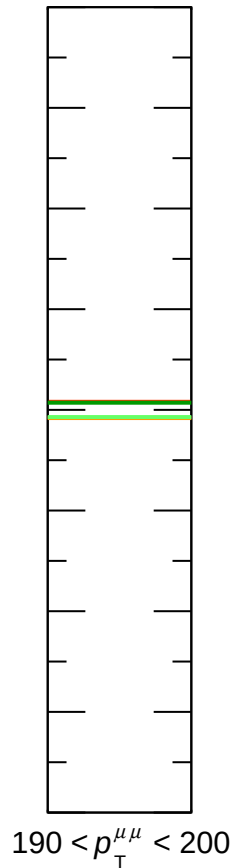
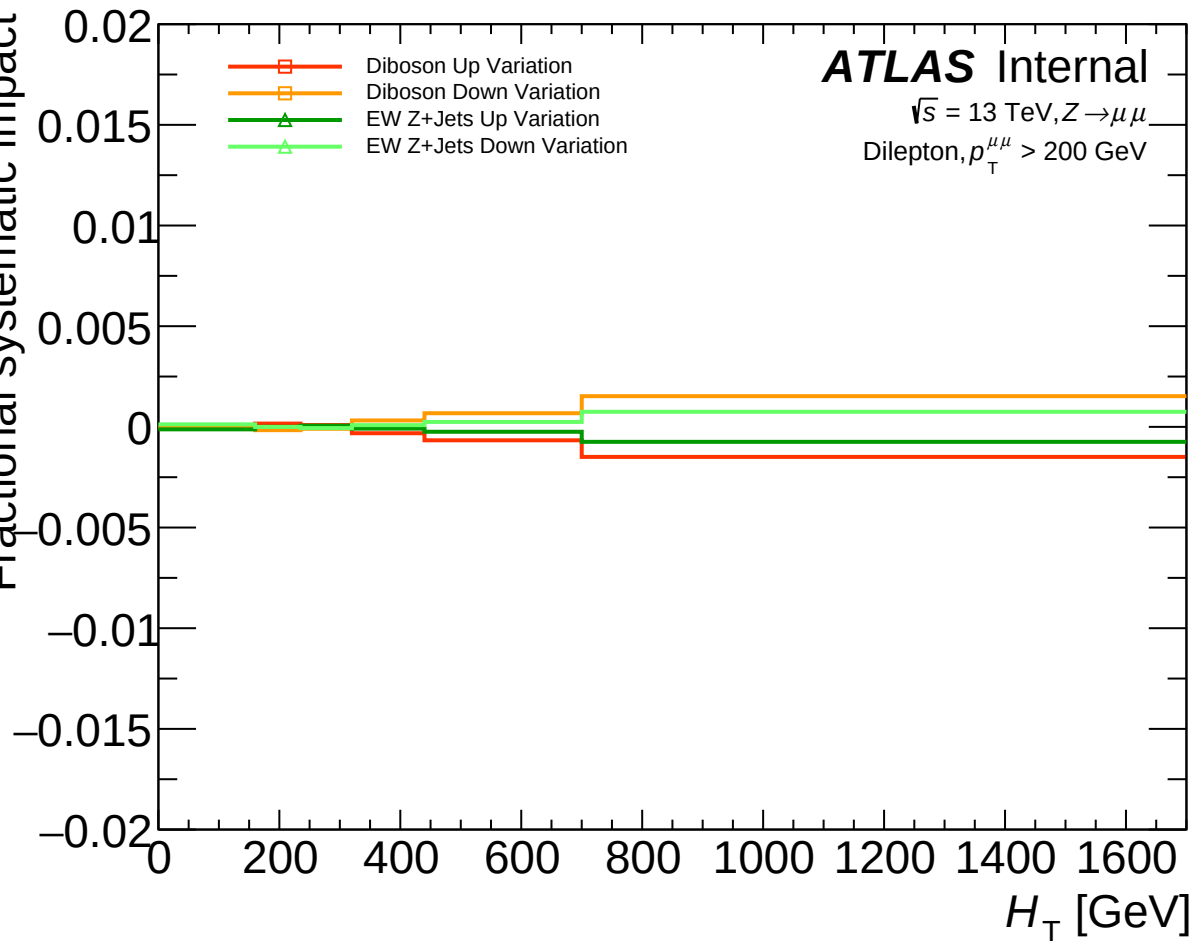
Fractional systematic impact



Fractional systematic impact



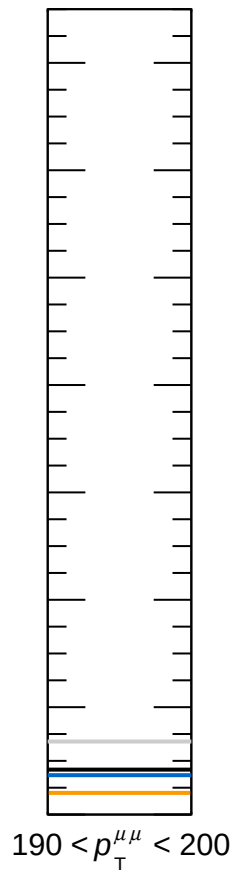
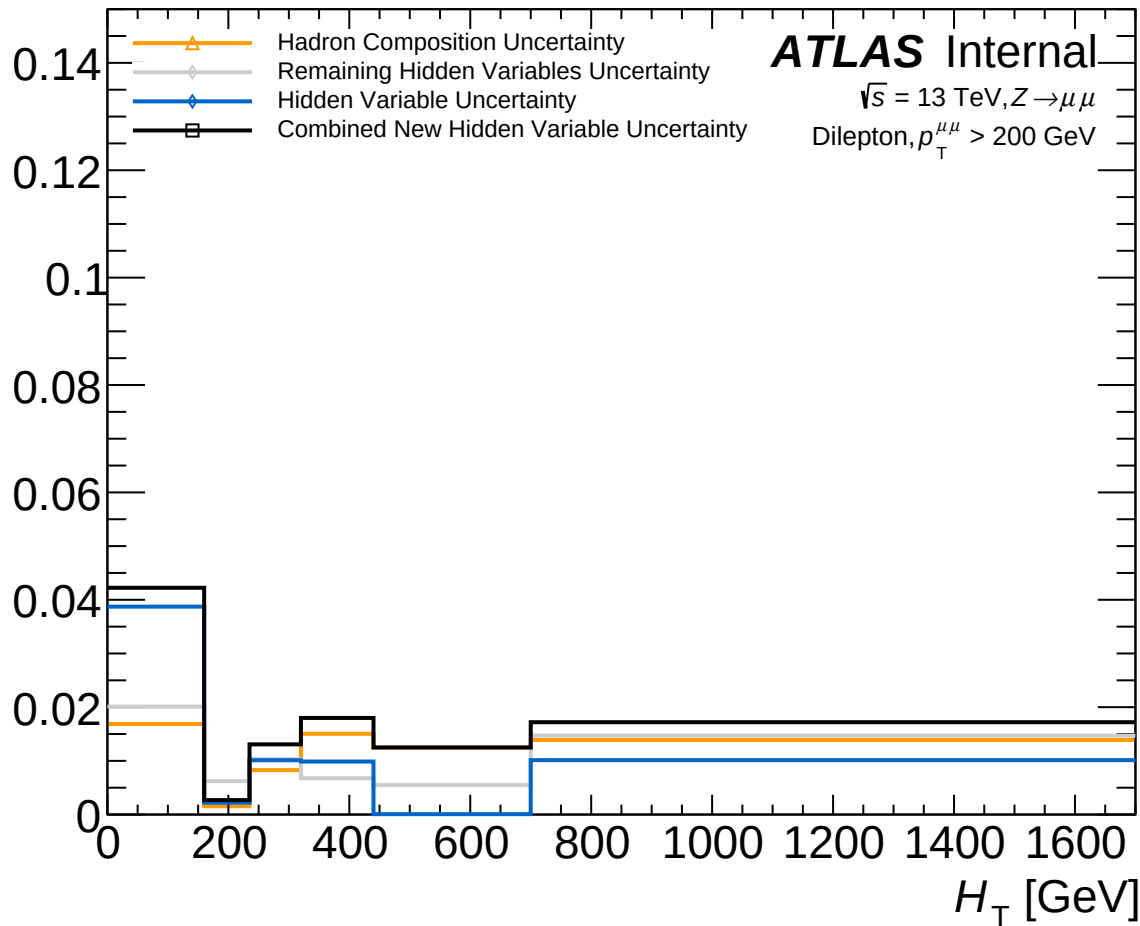
Fractional systematic impact

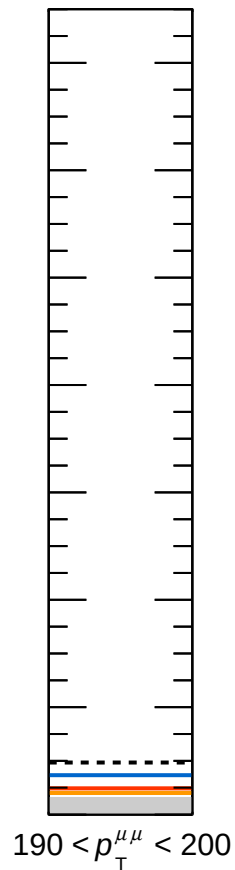
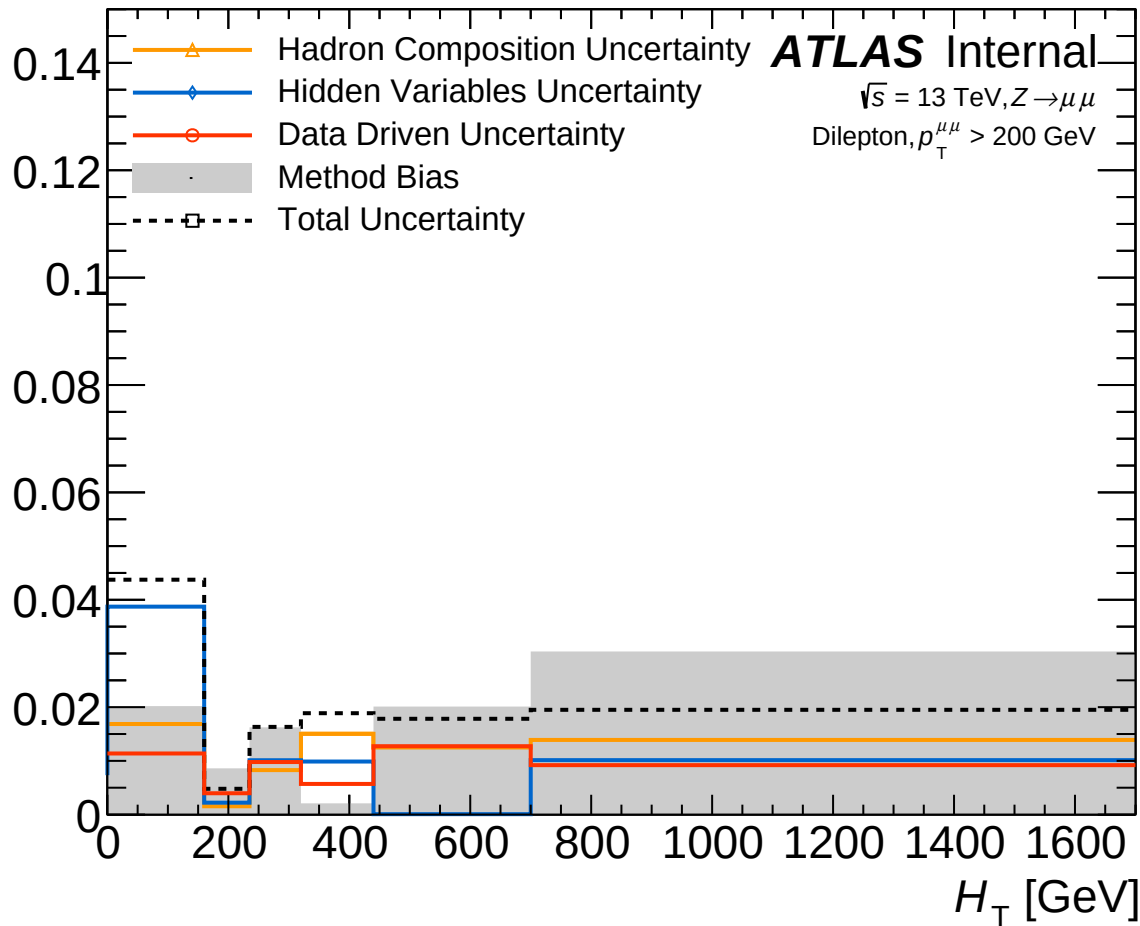


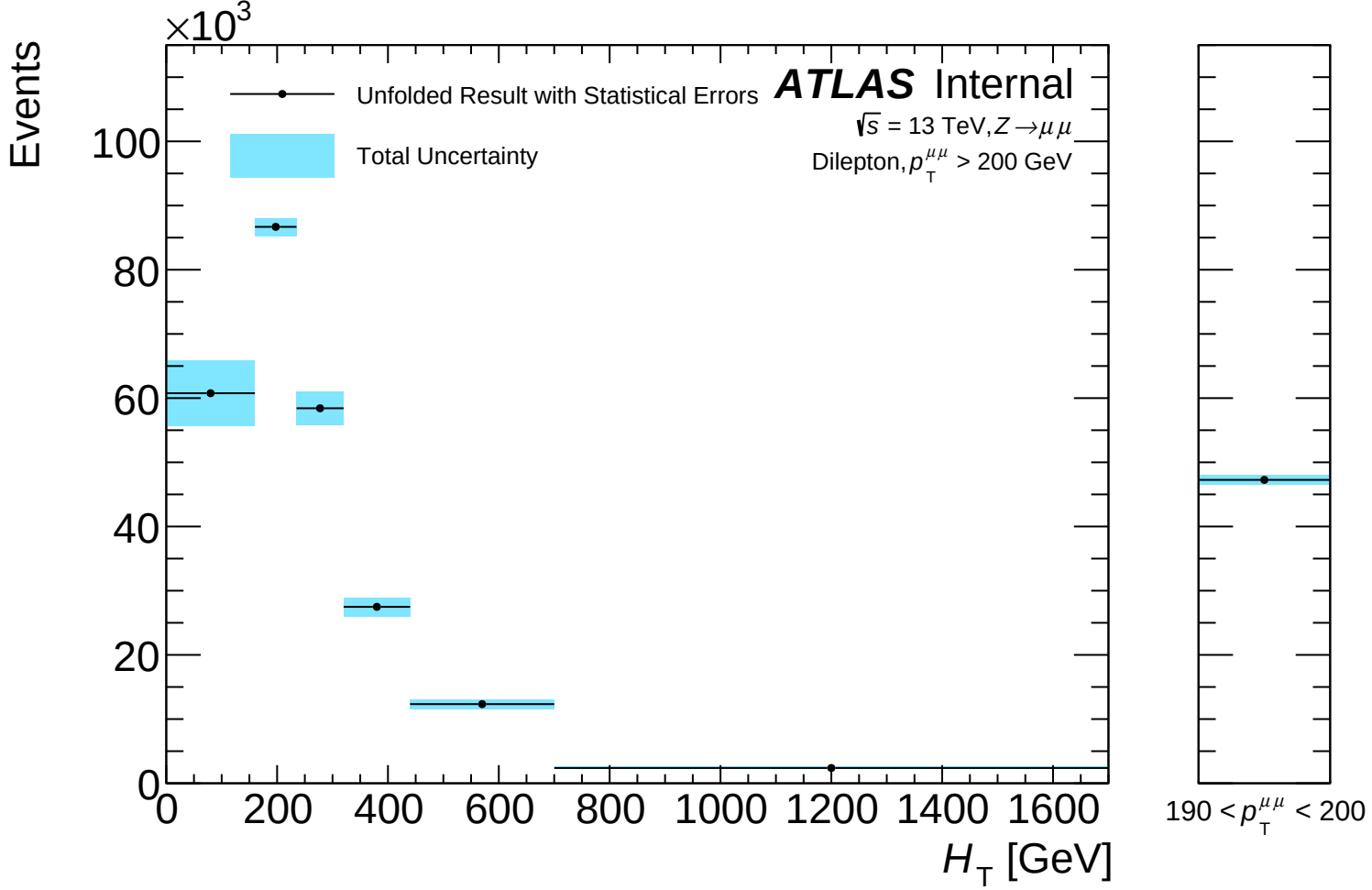
ATLAS Internal

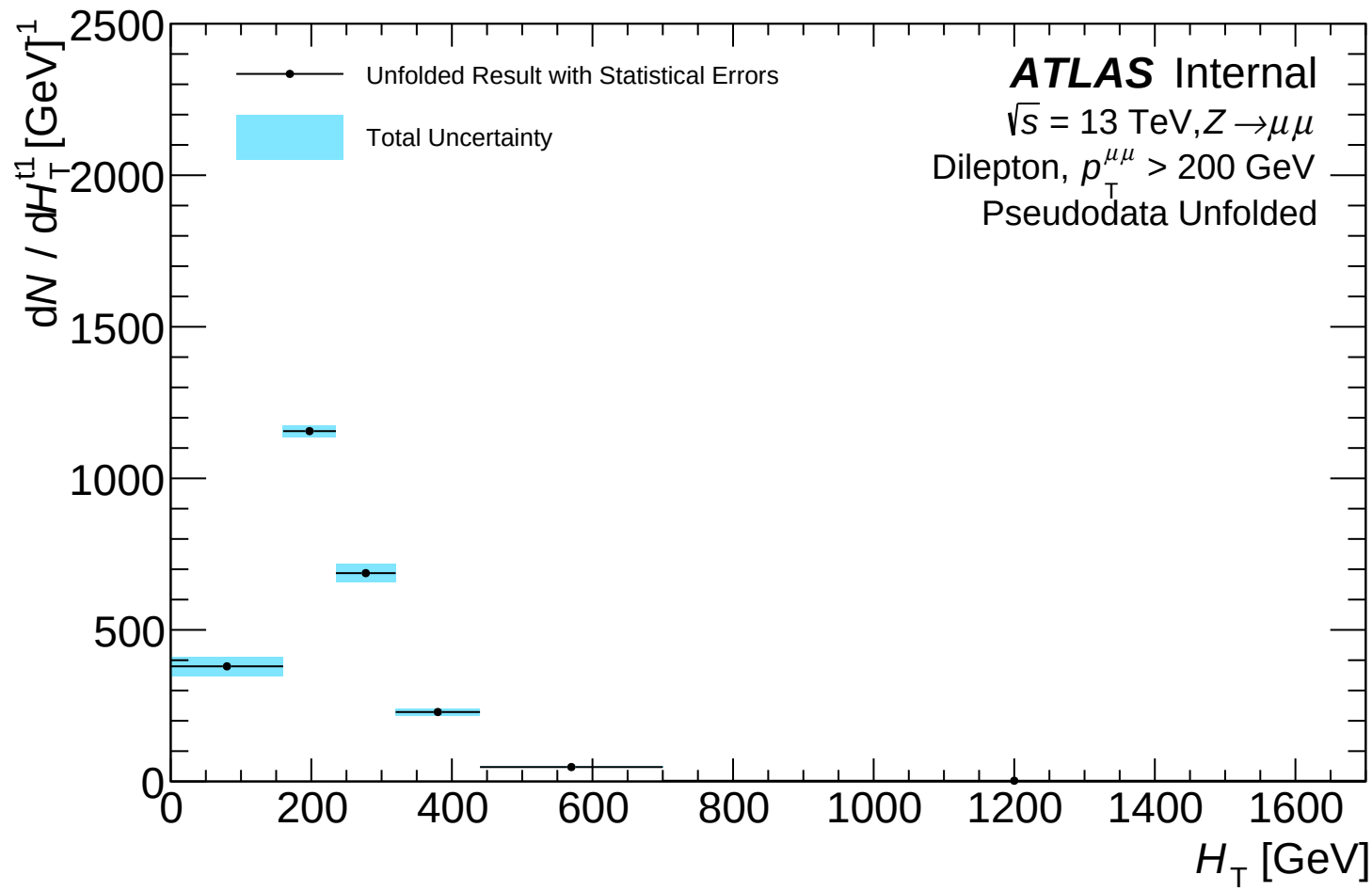
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

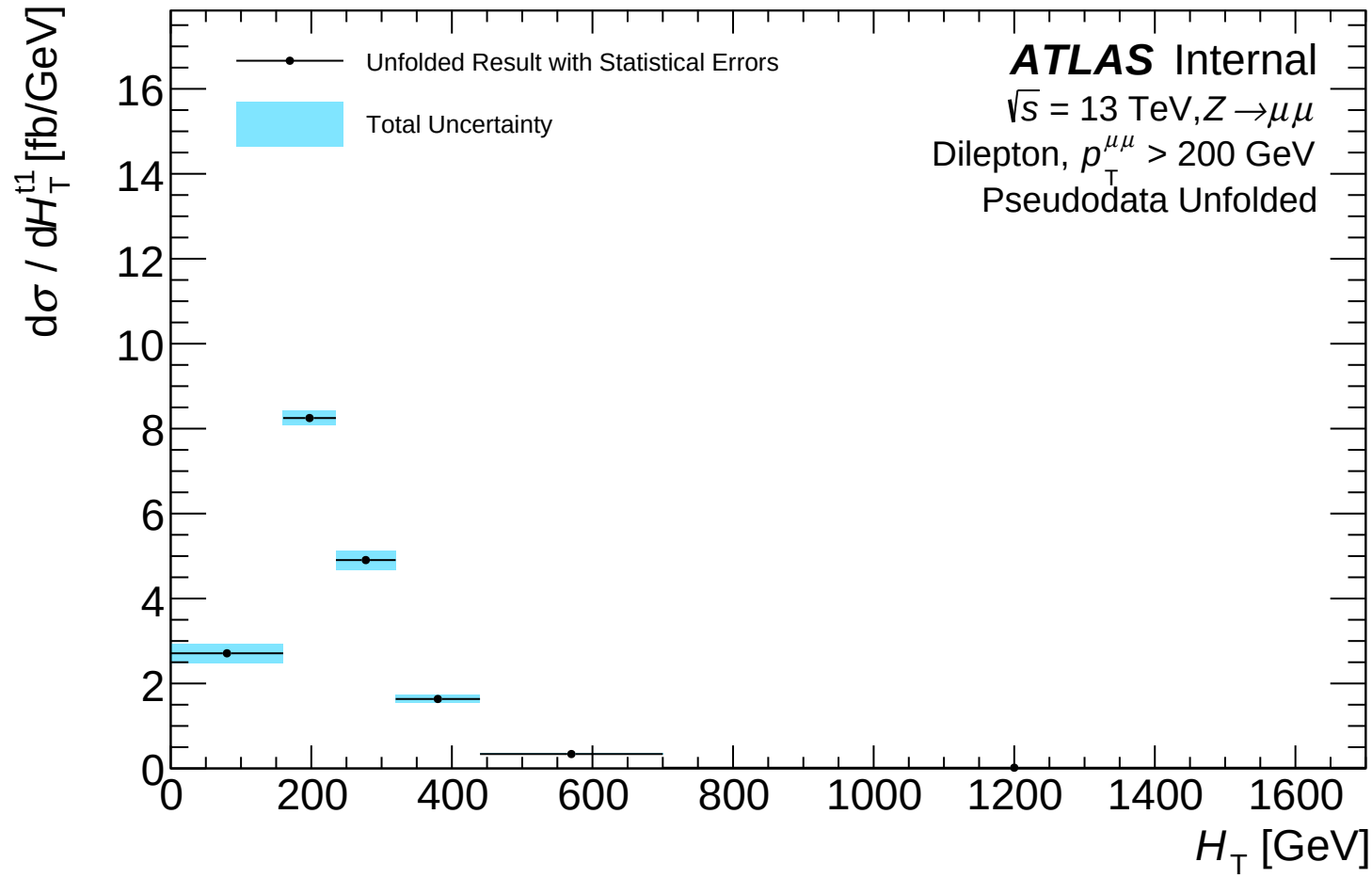
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$











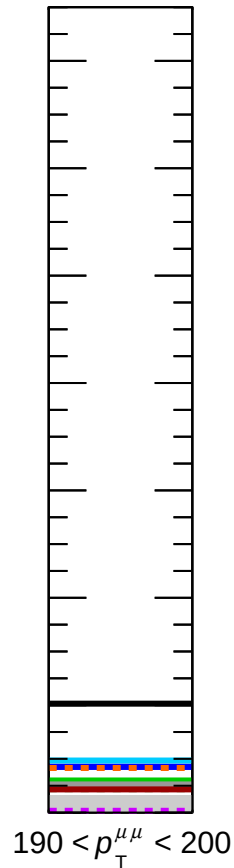
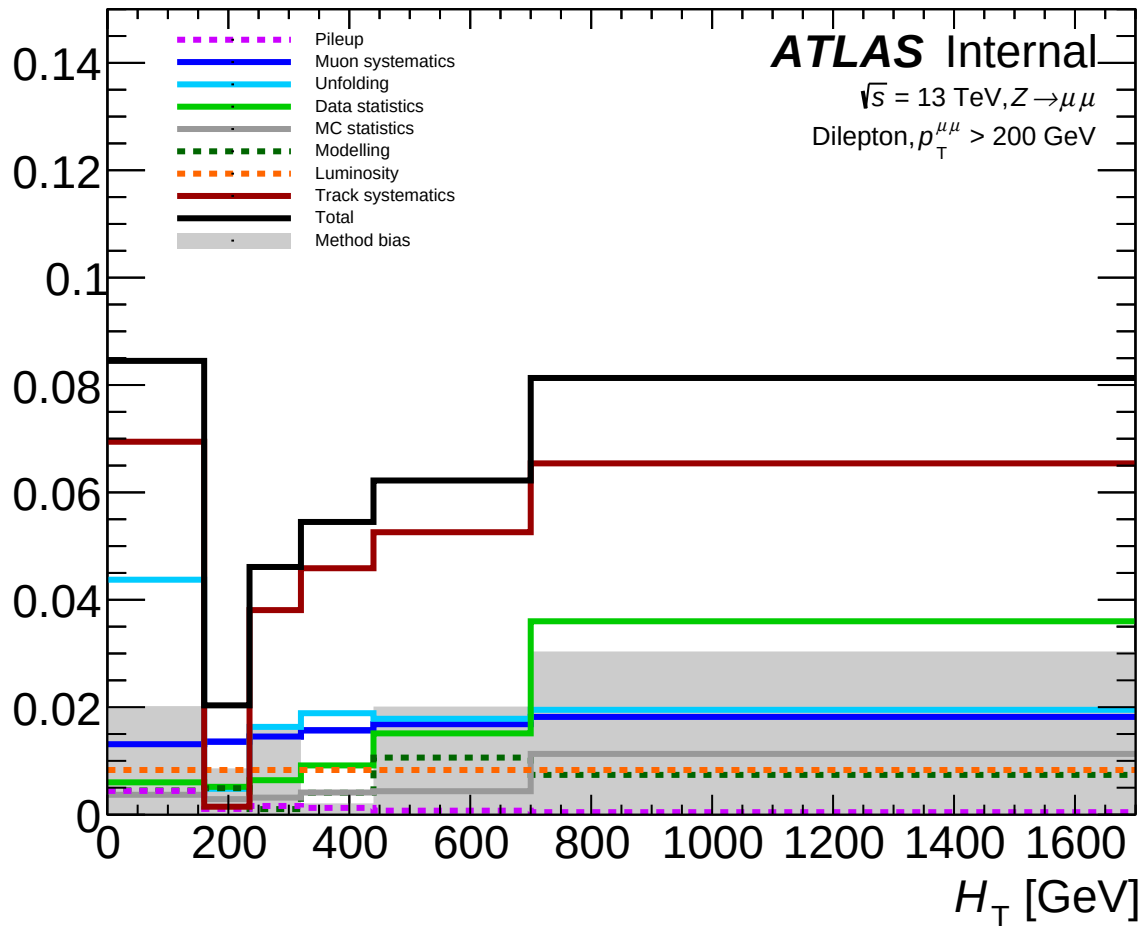
Fractional uncertainty

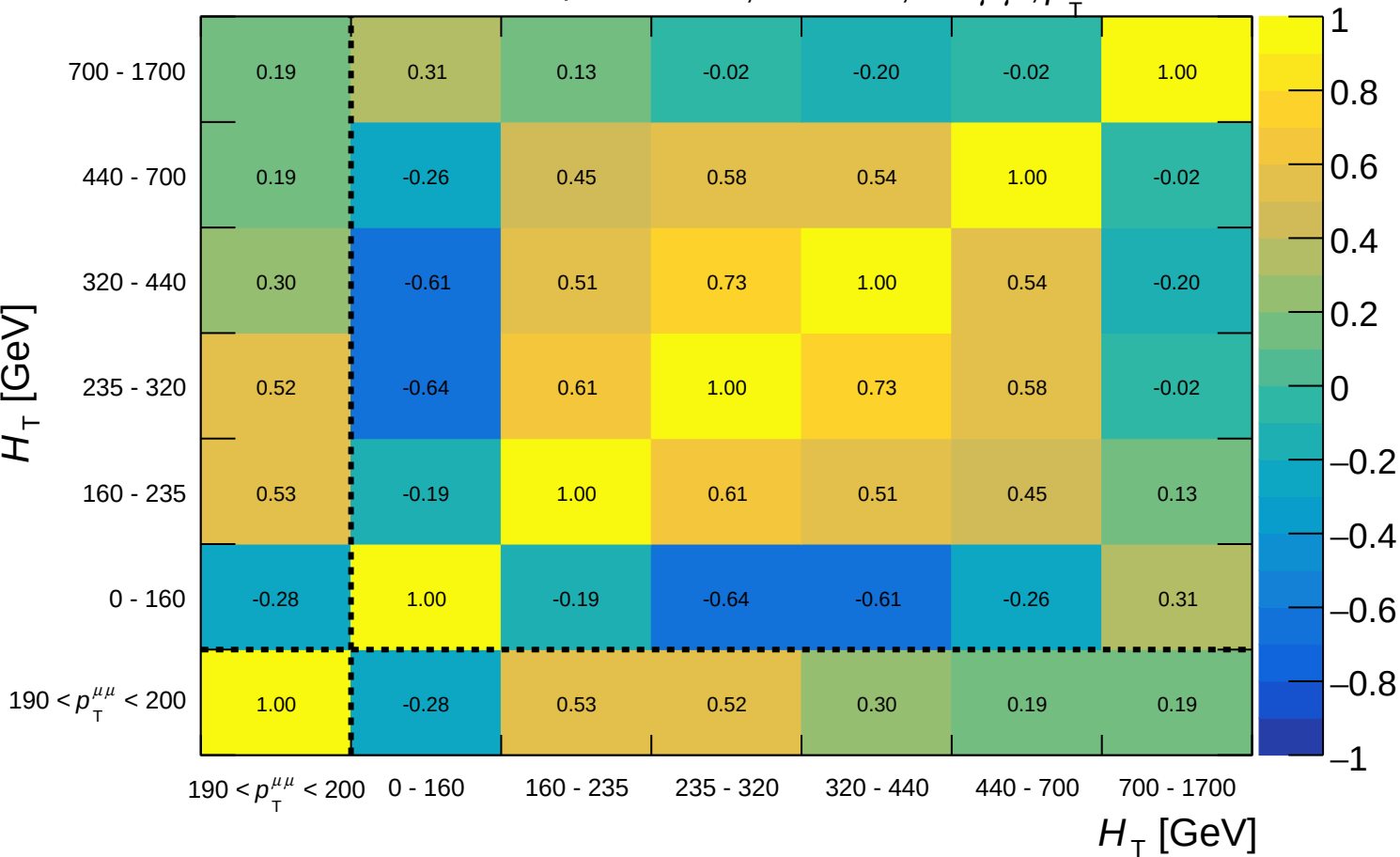
ATLAS Internal

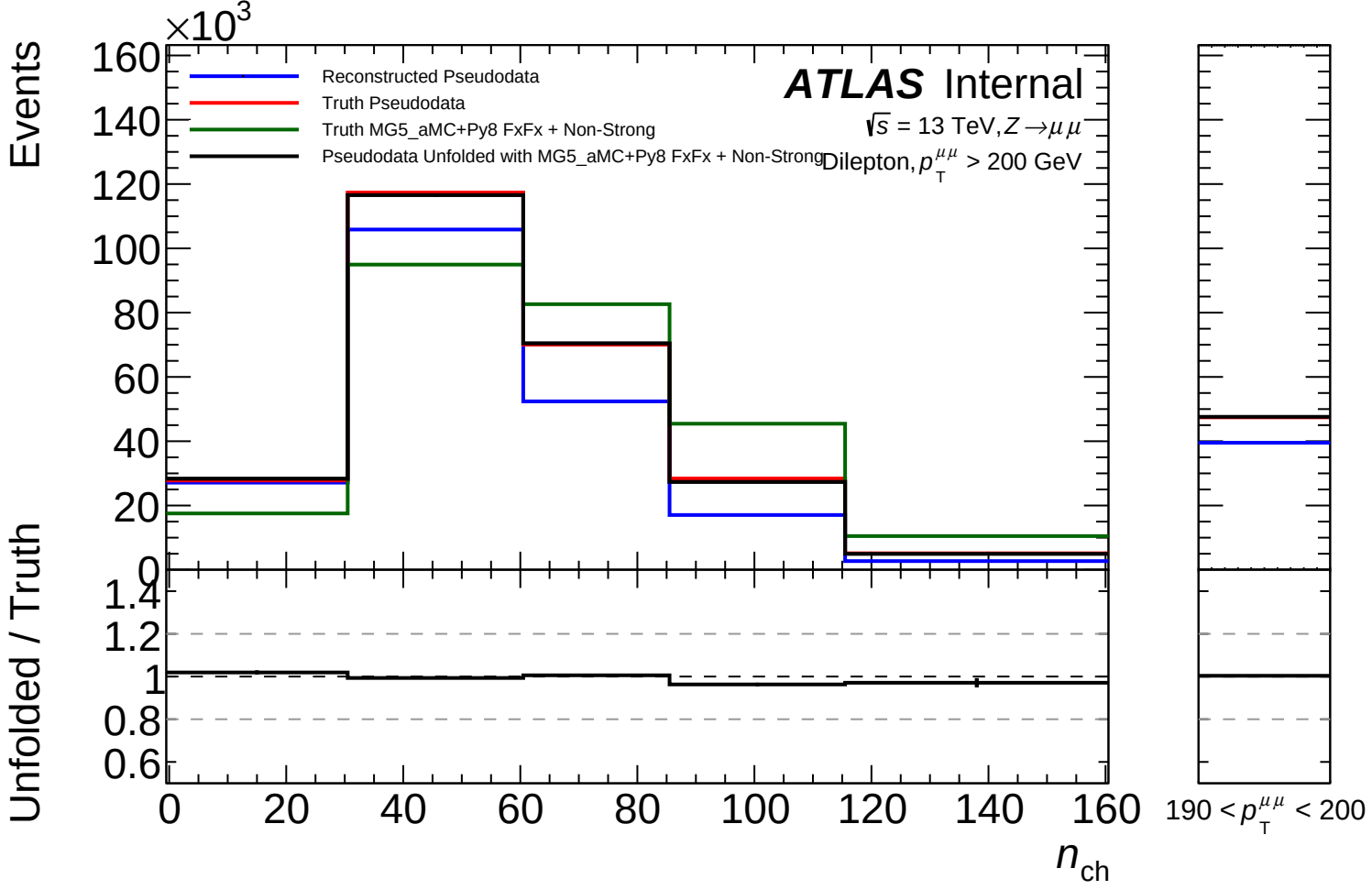
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

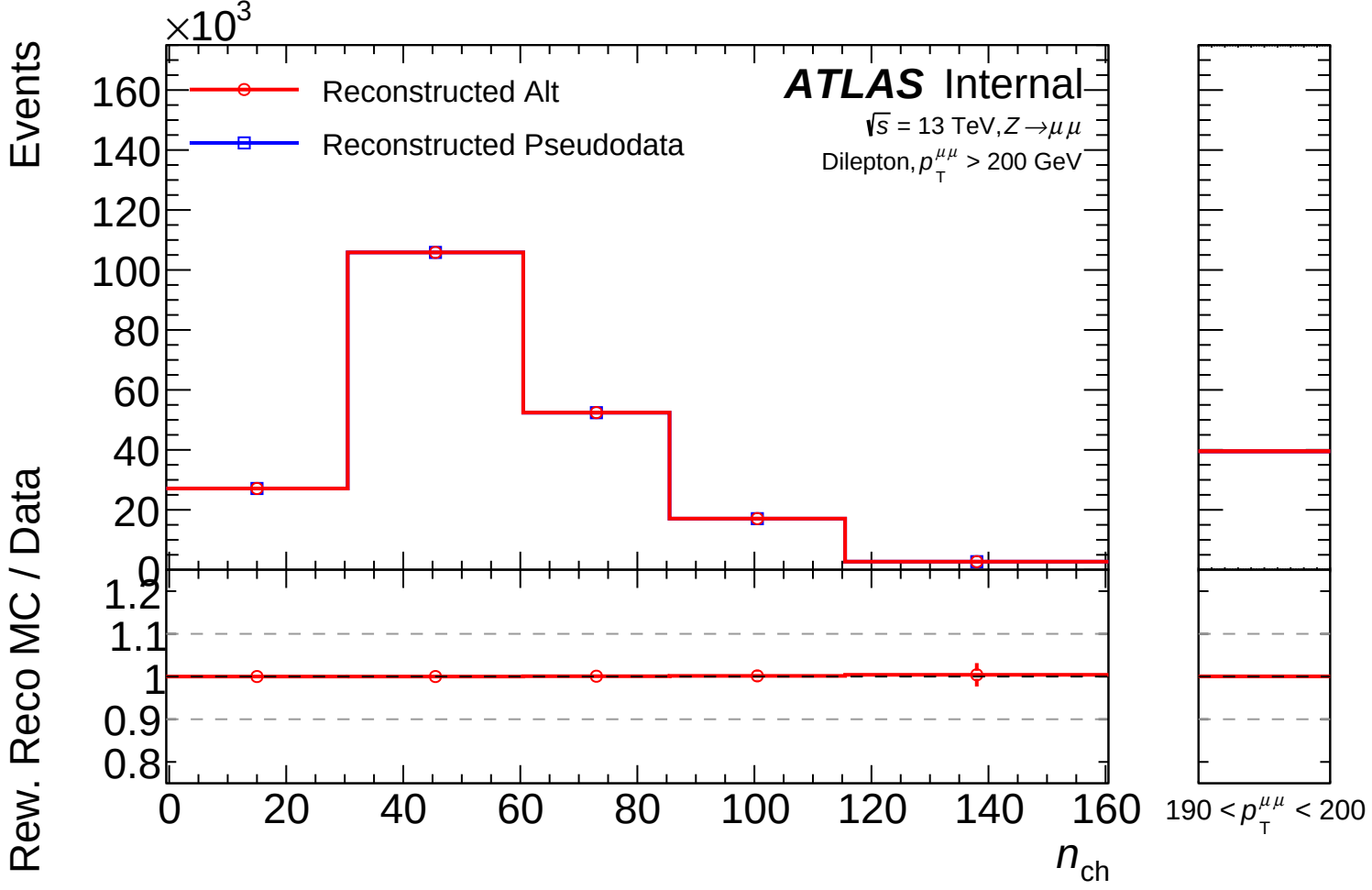
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

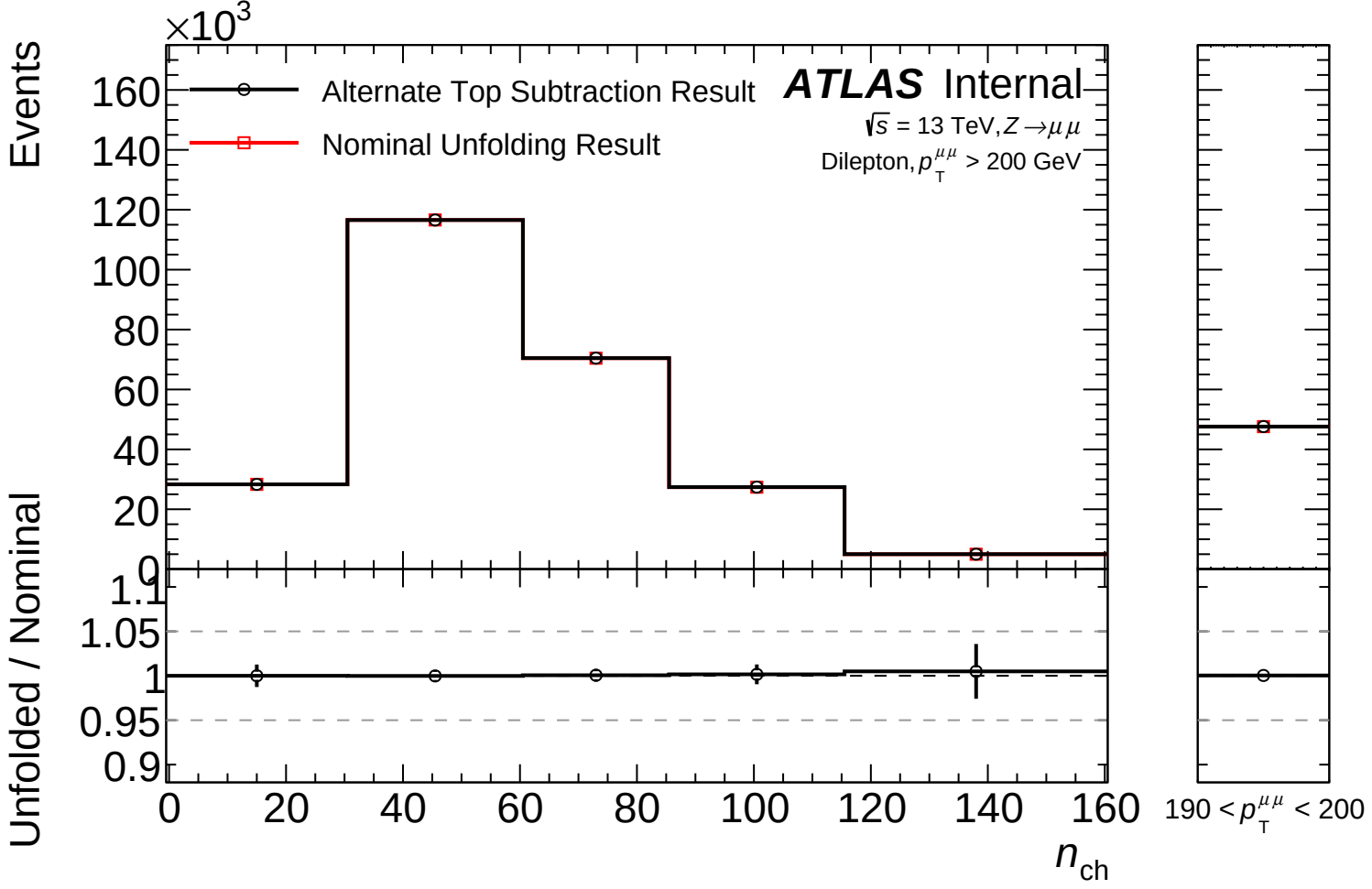
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $H_{\text{T}} [\text{GeV}]$ 







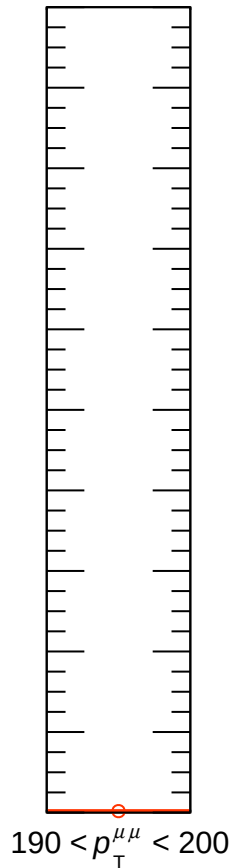
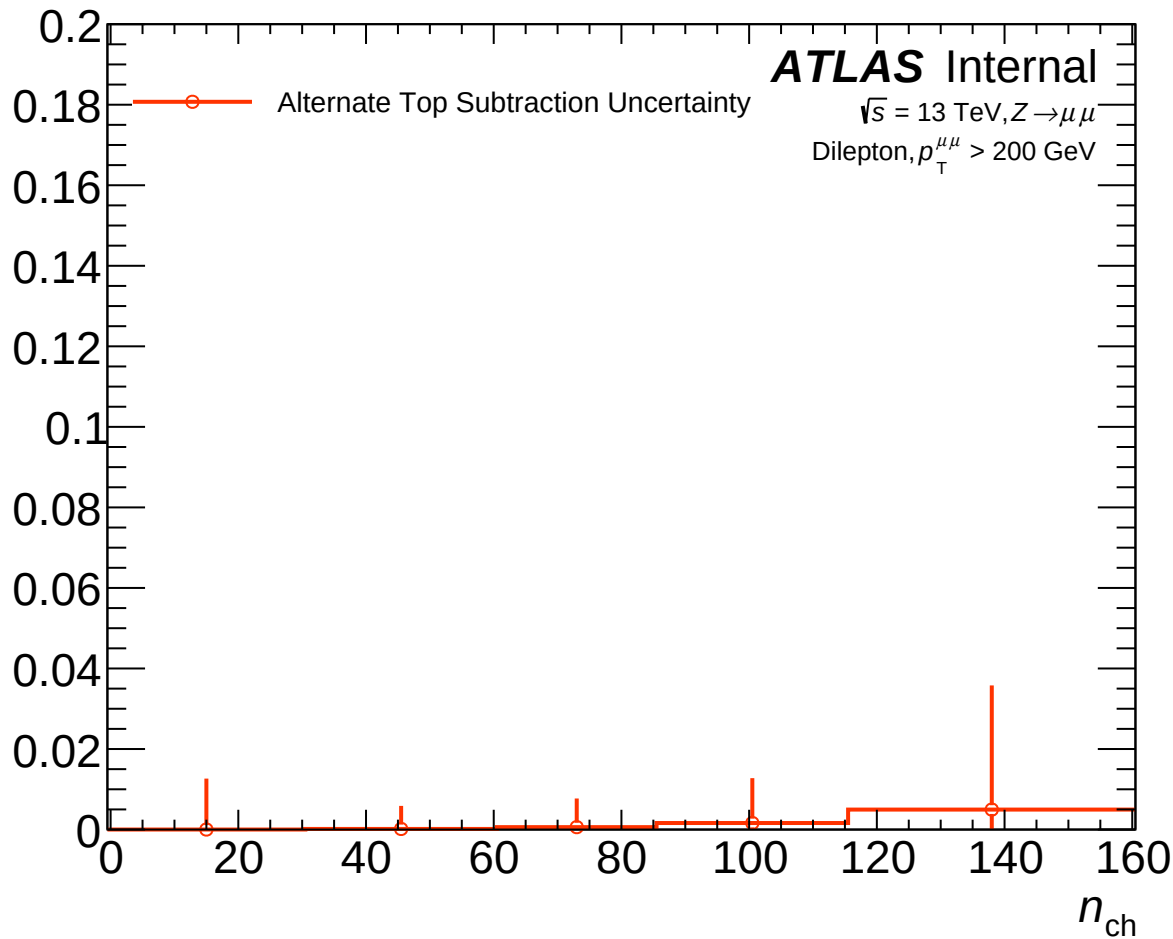
Fractional systematic impact

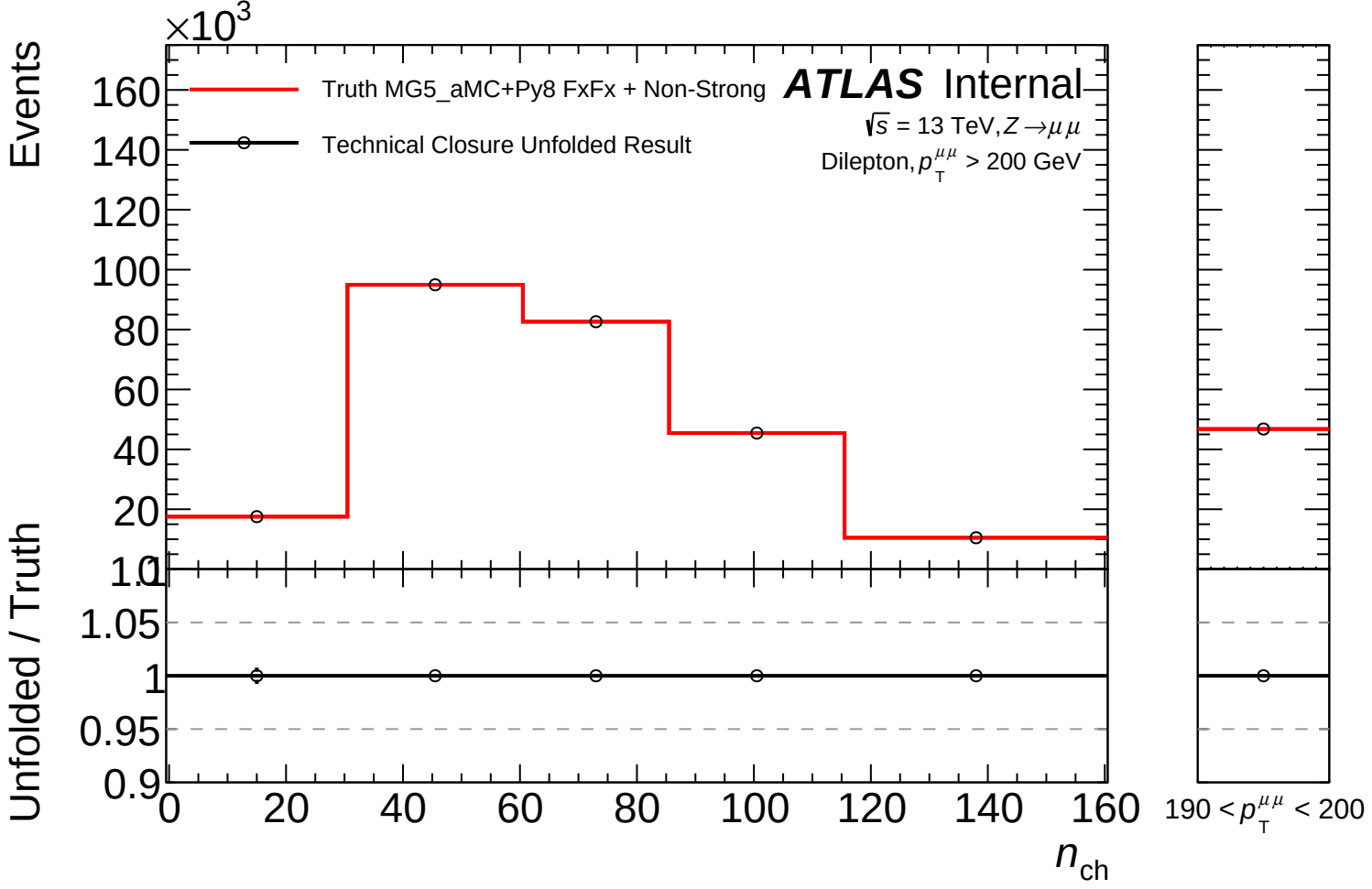
ATLAS Internal

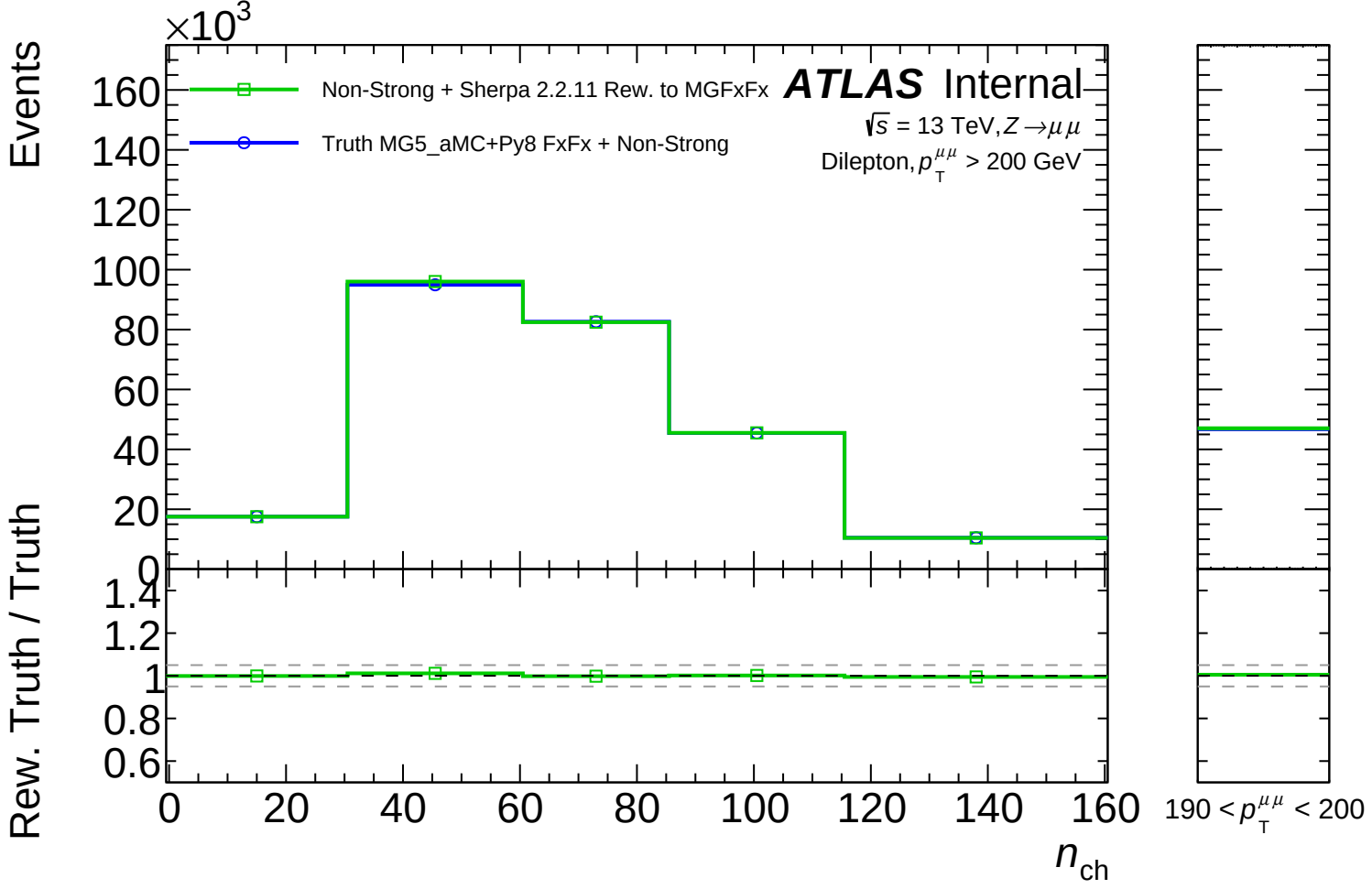
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

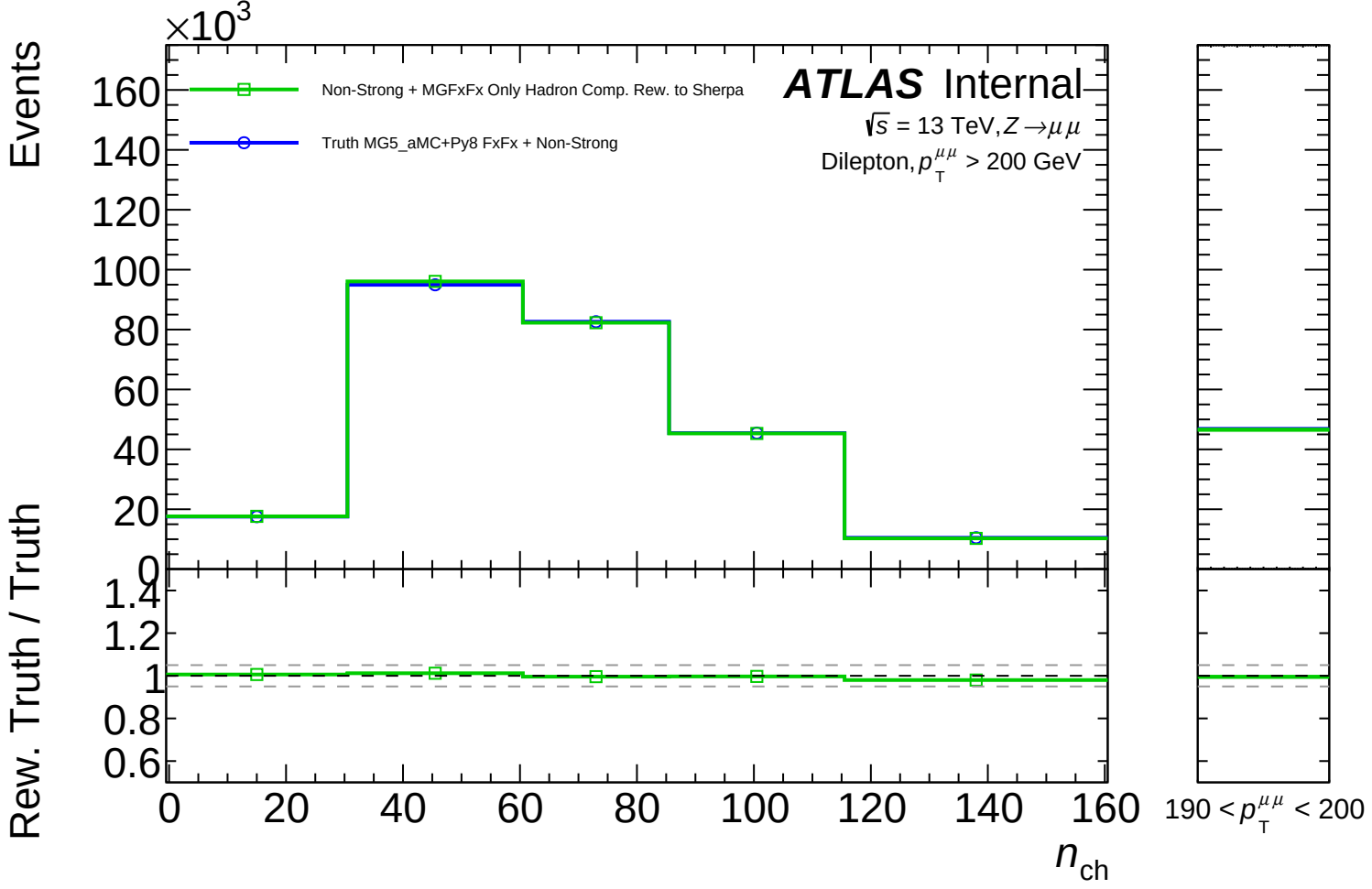
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

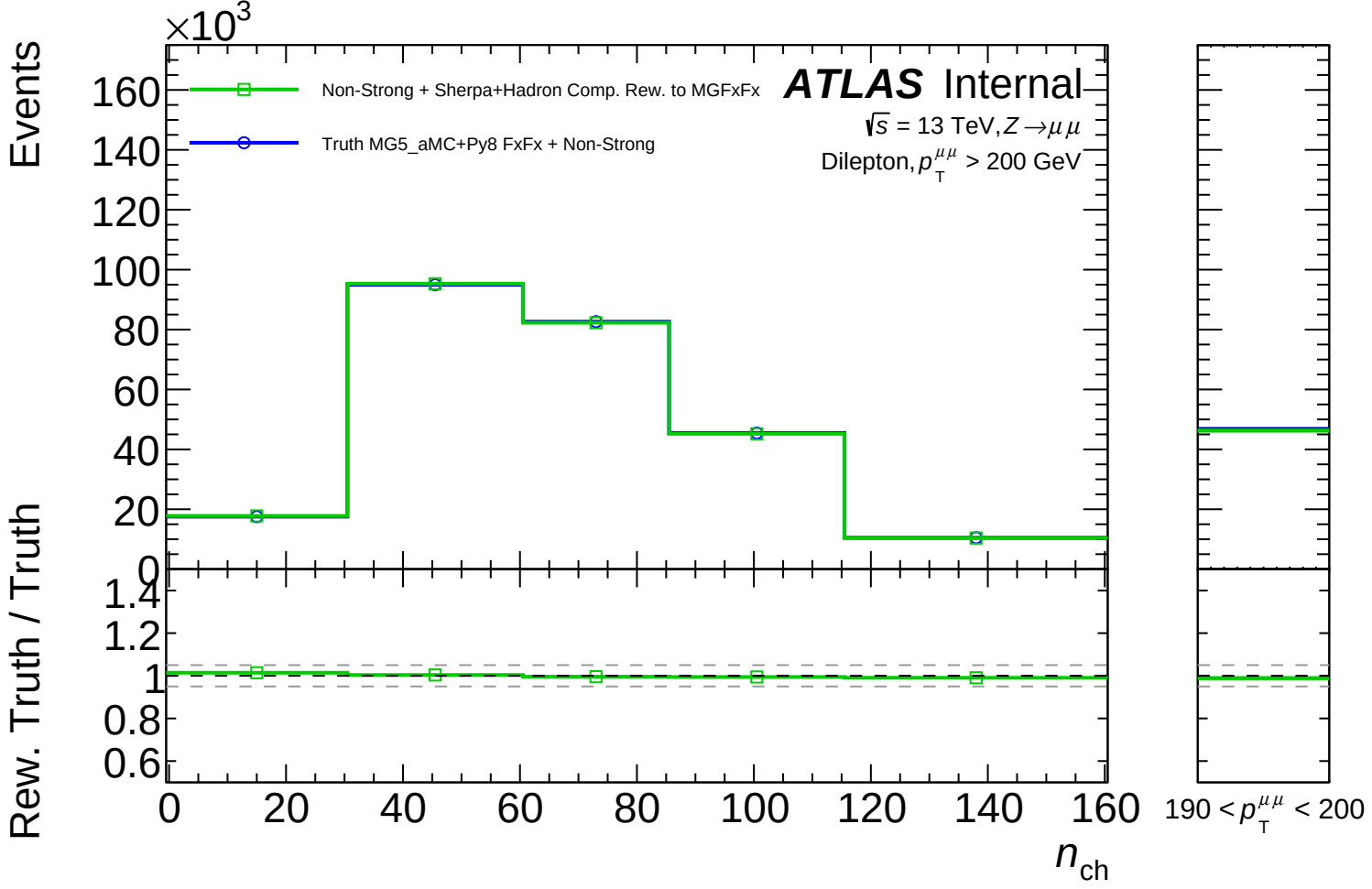
Alternate Top Subtraction Uncertainty

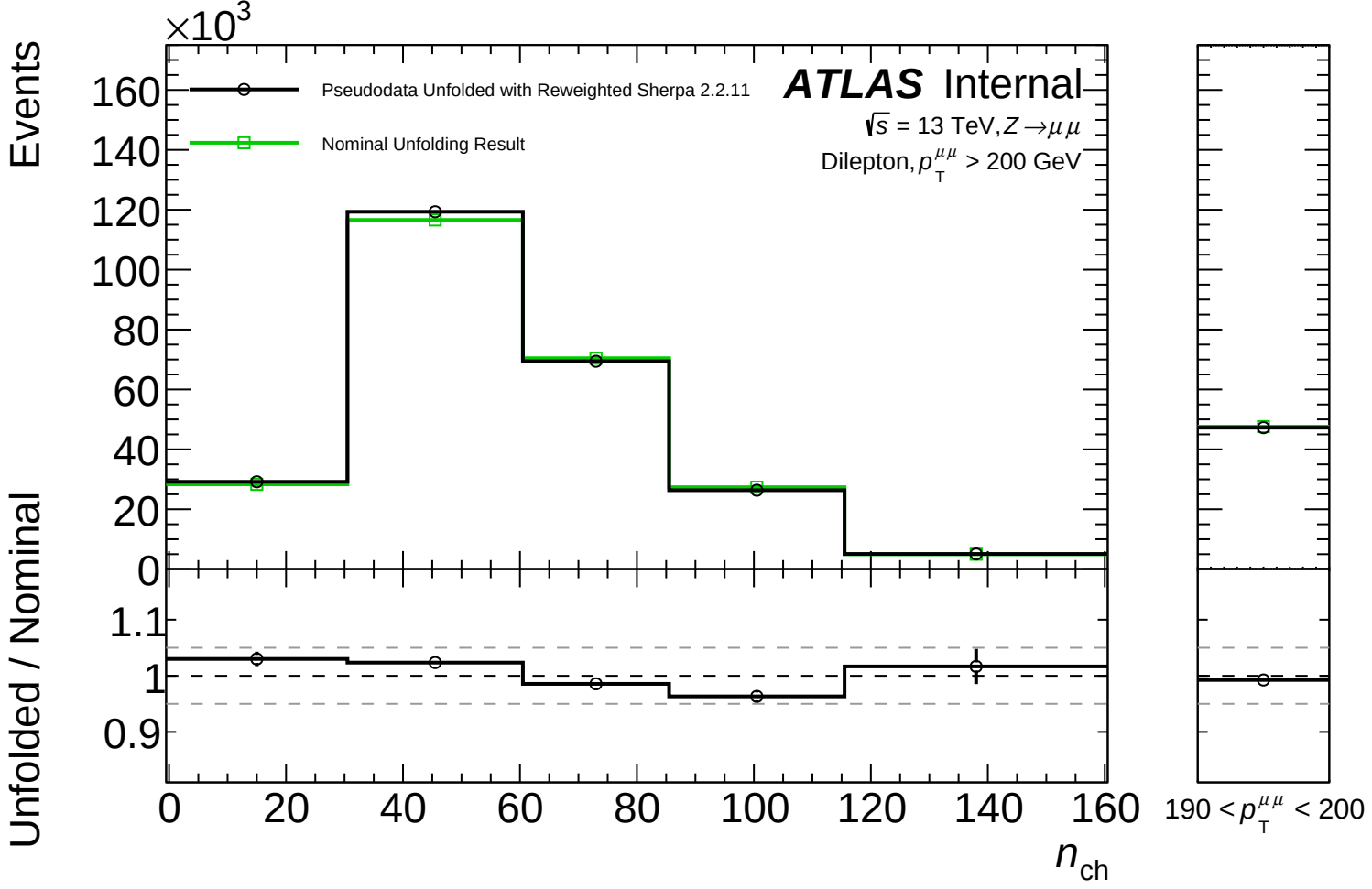


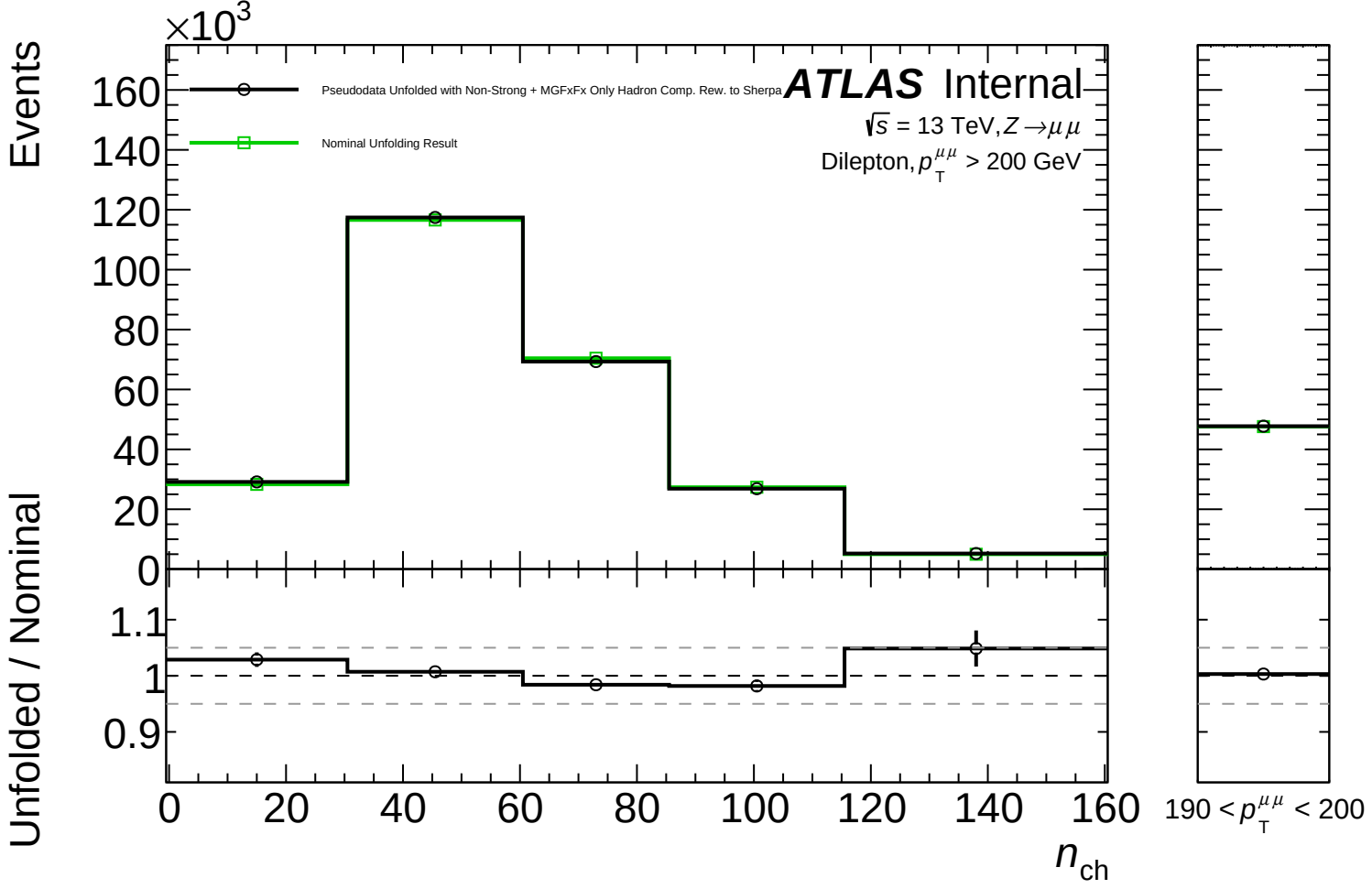


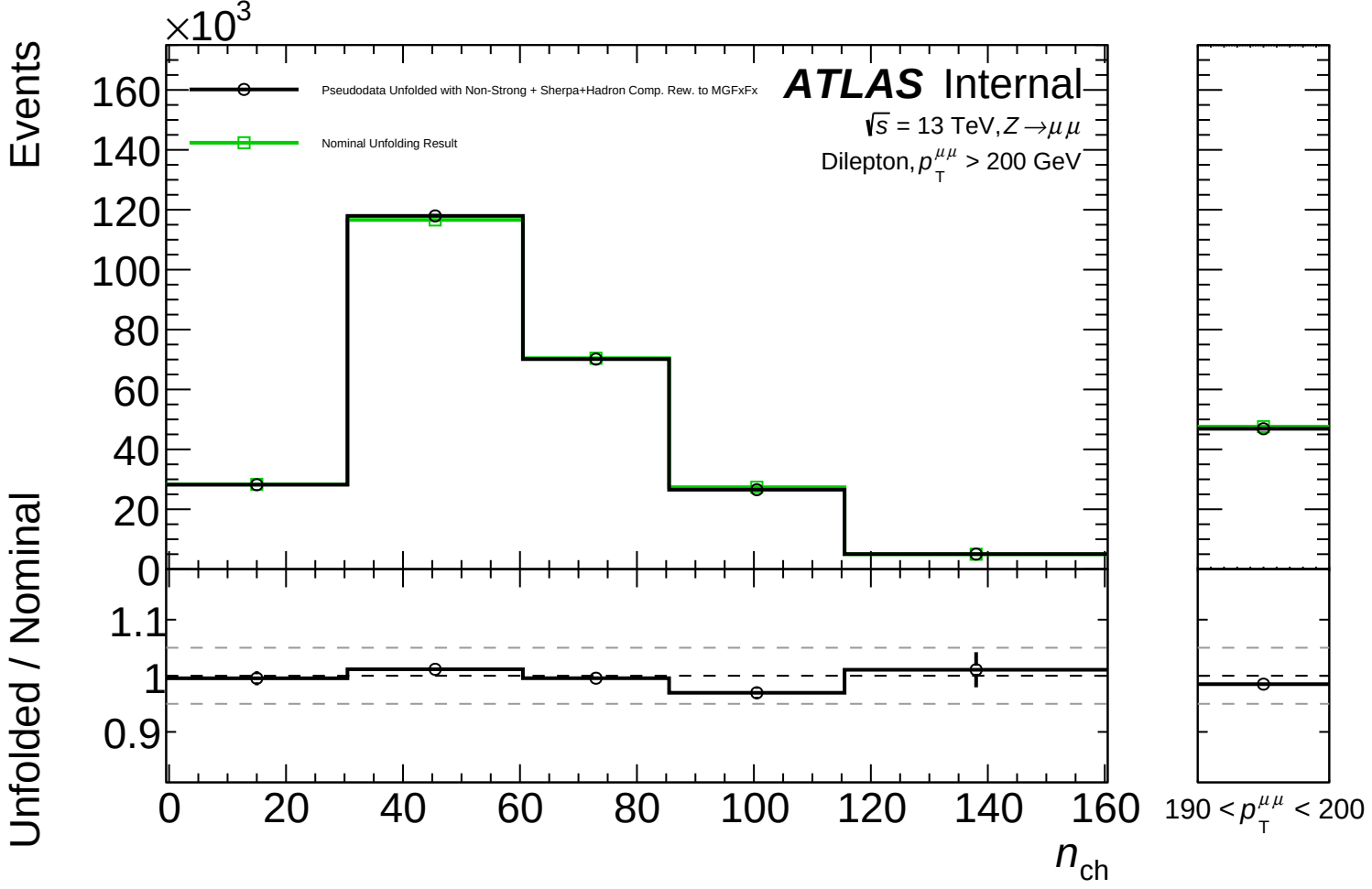


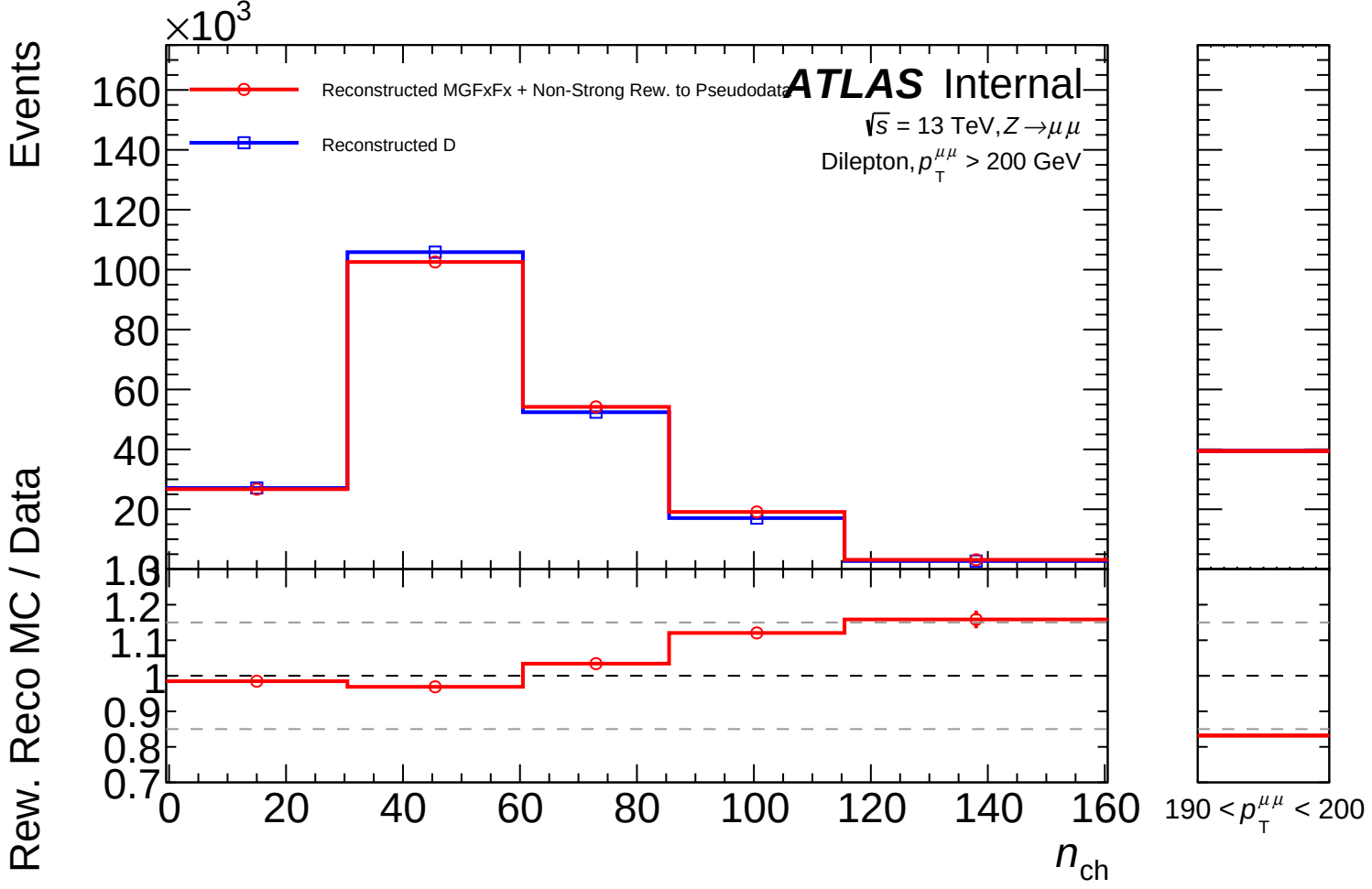


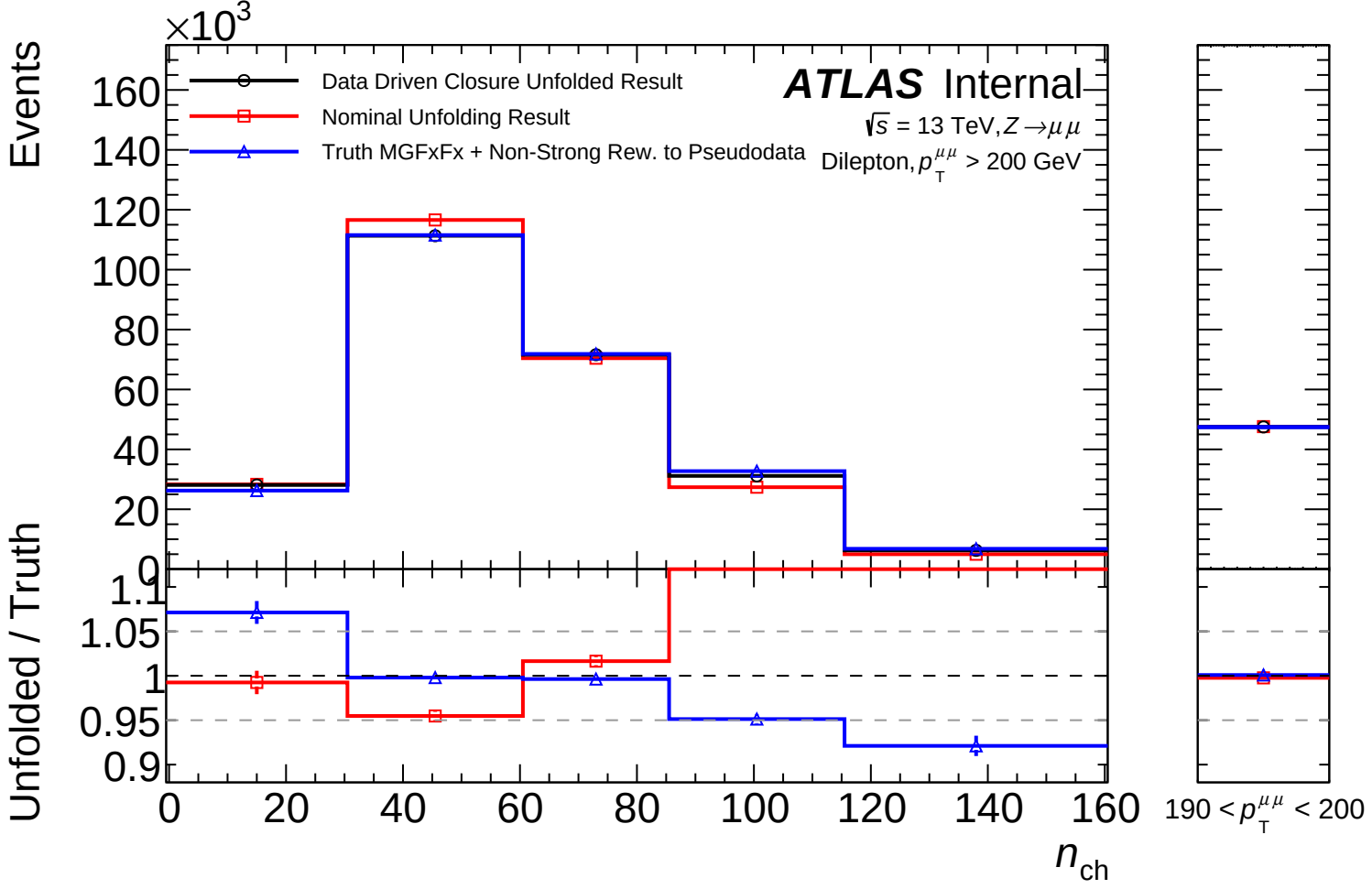




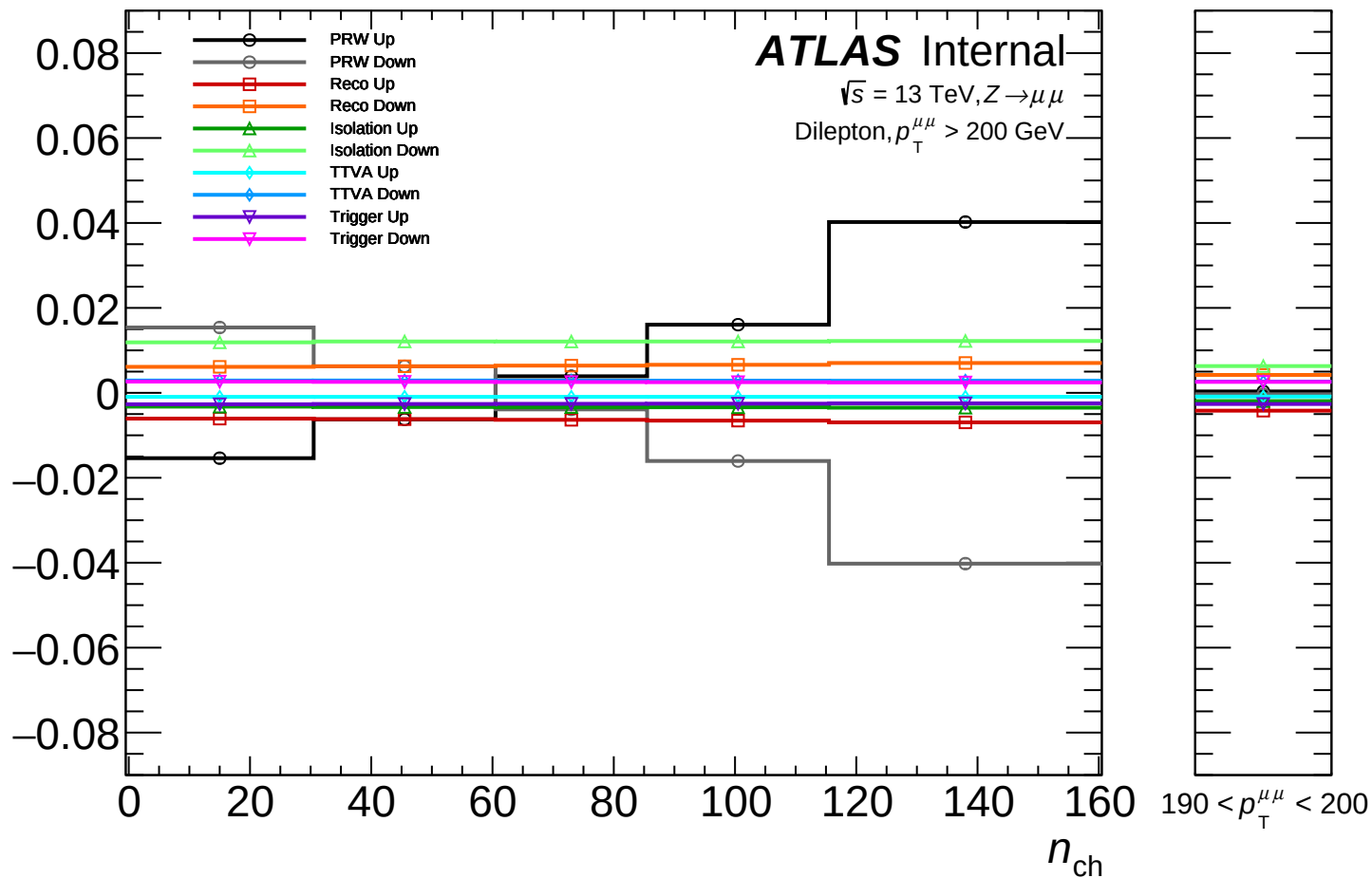




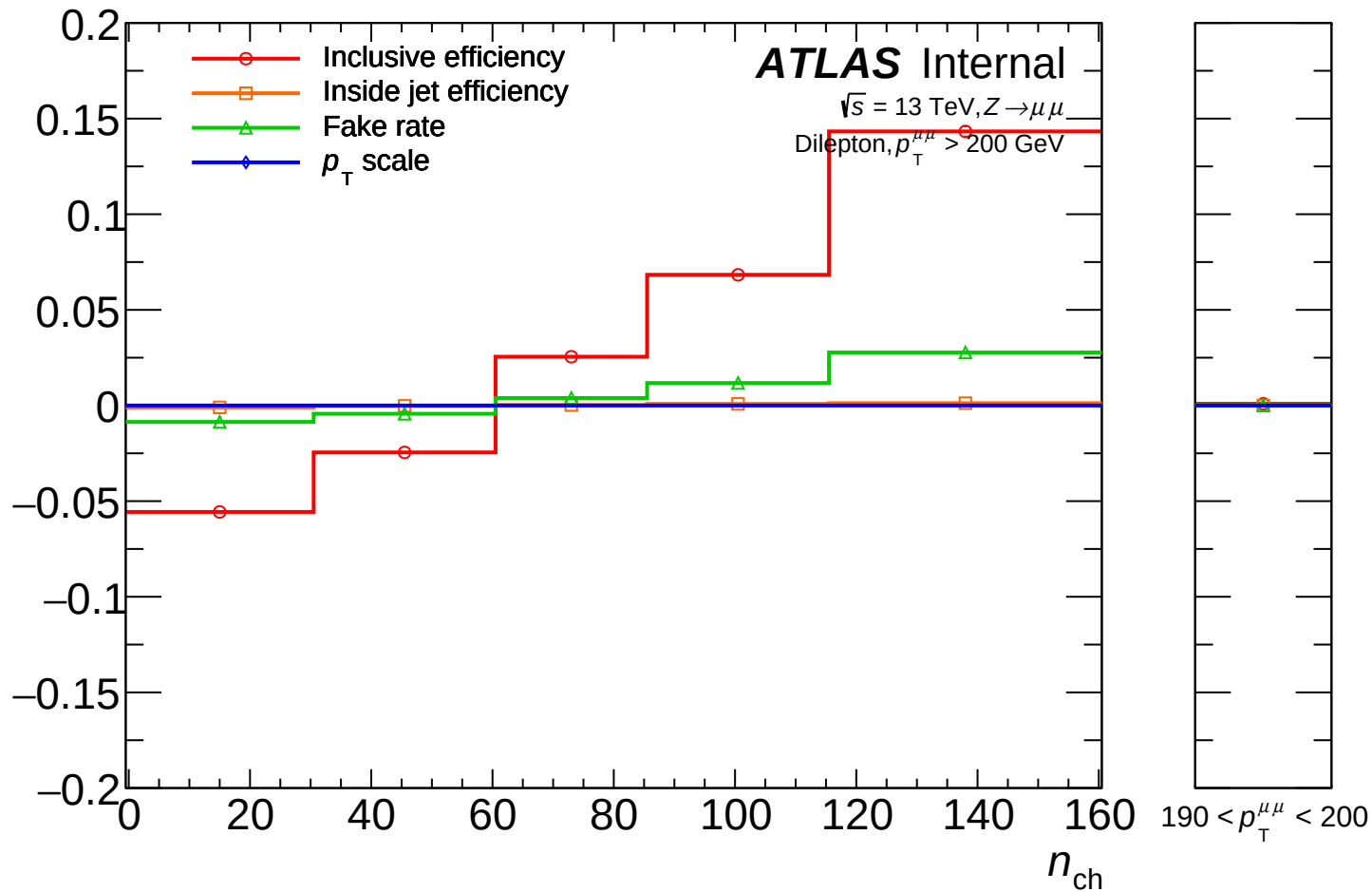




Fractional systematic impact



Fractional systematic impact



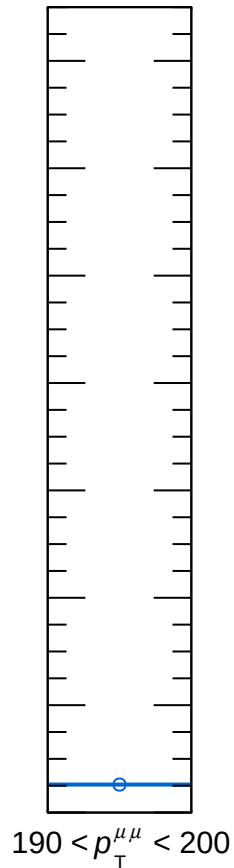
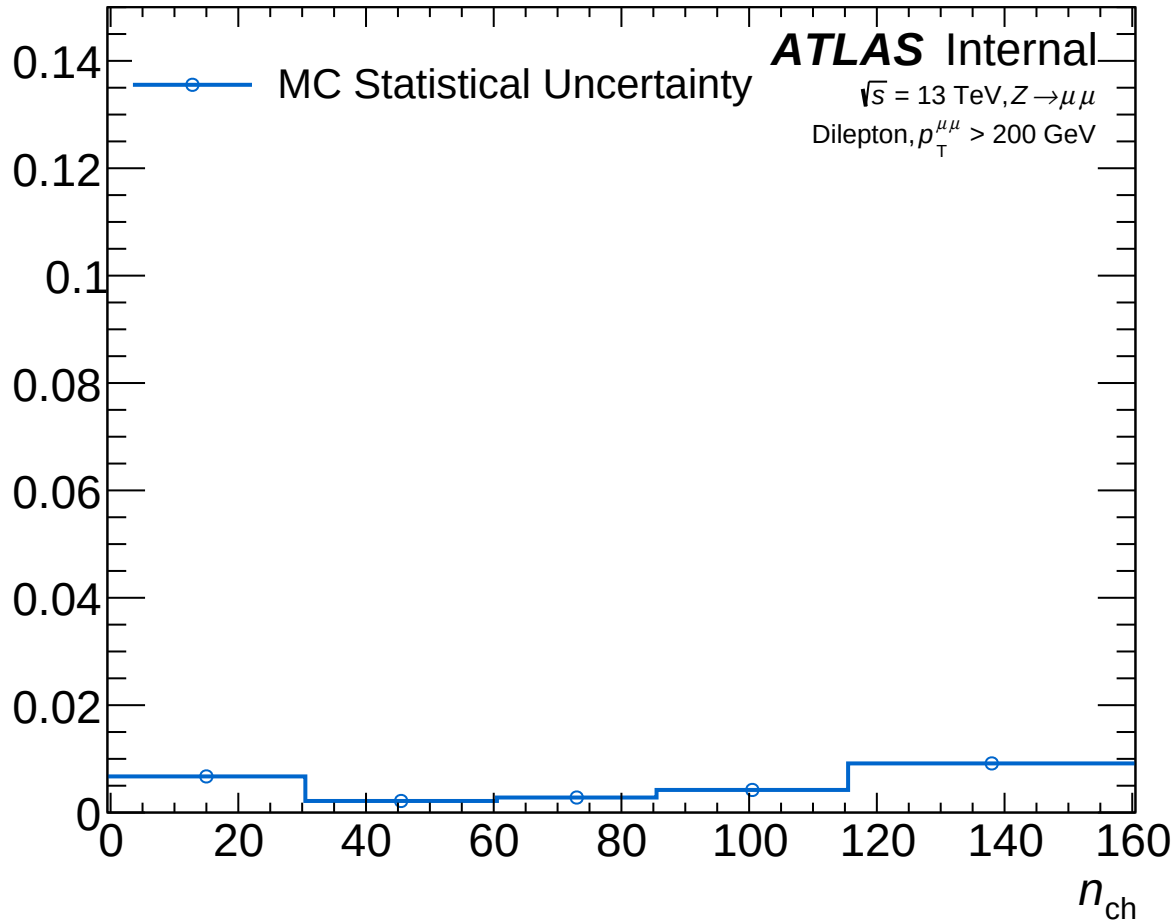
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



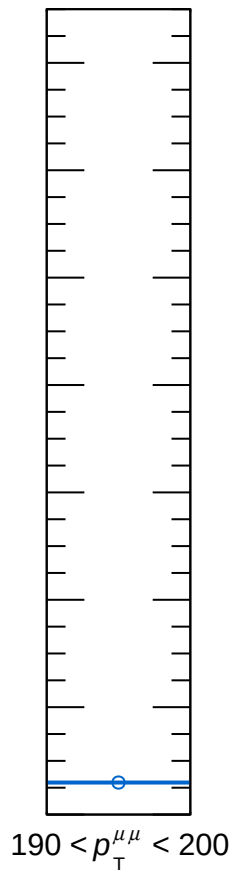
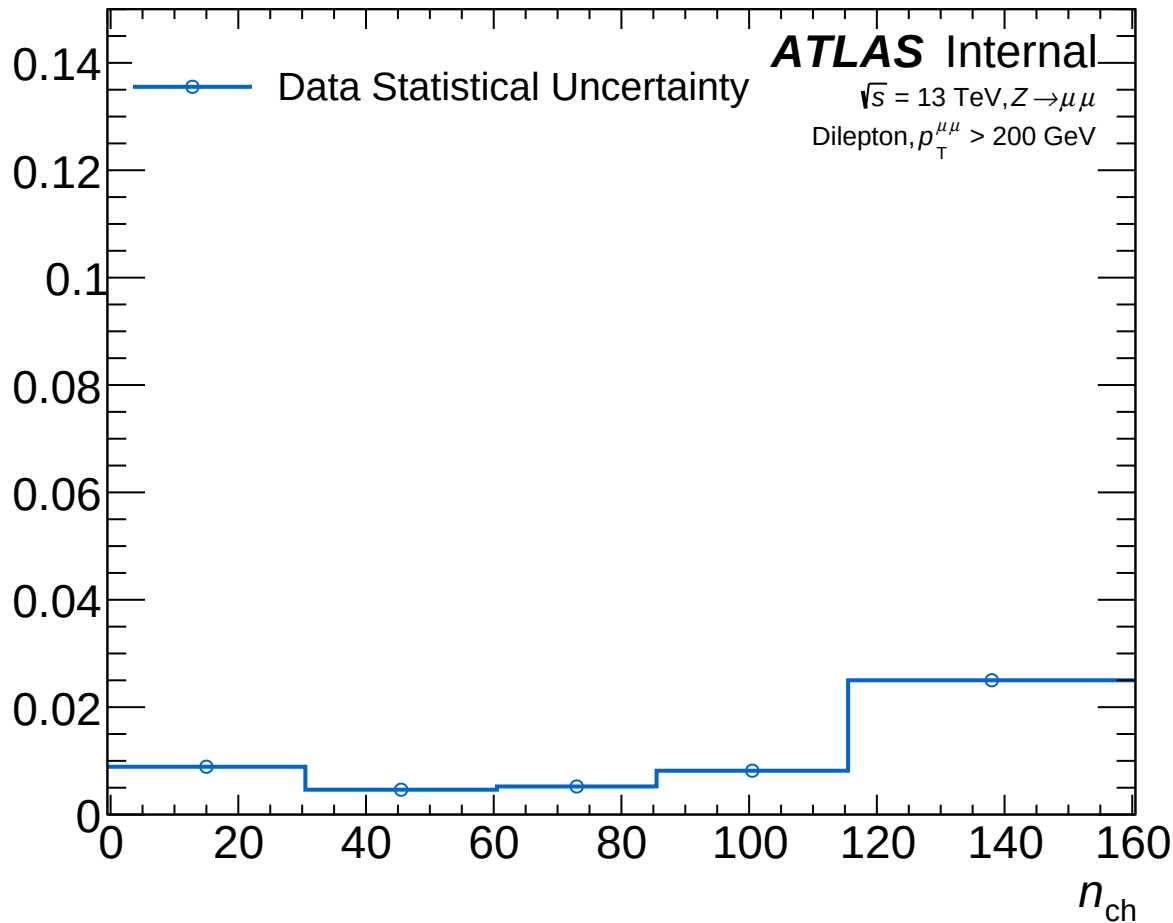
Fractional systematic impact

ATLAS Internal

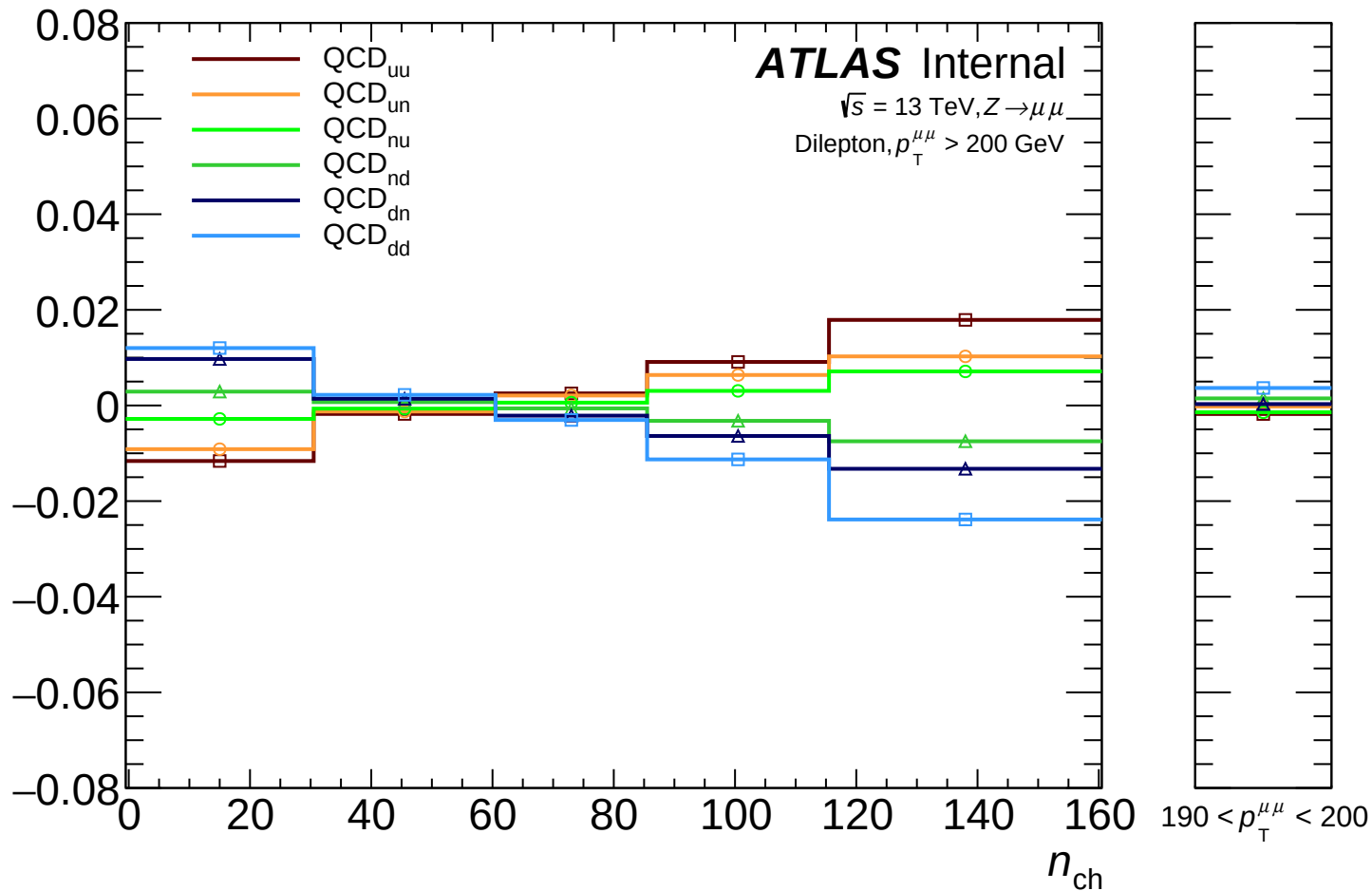
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

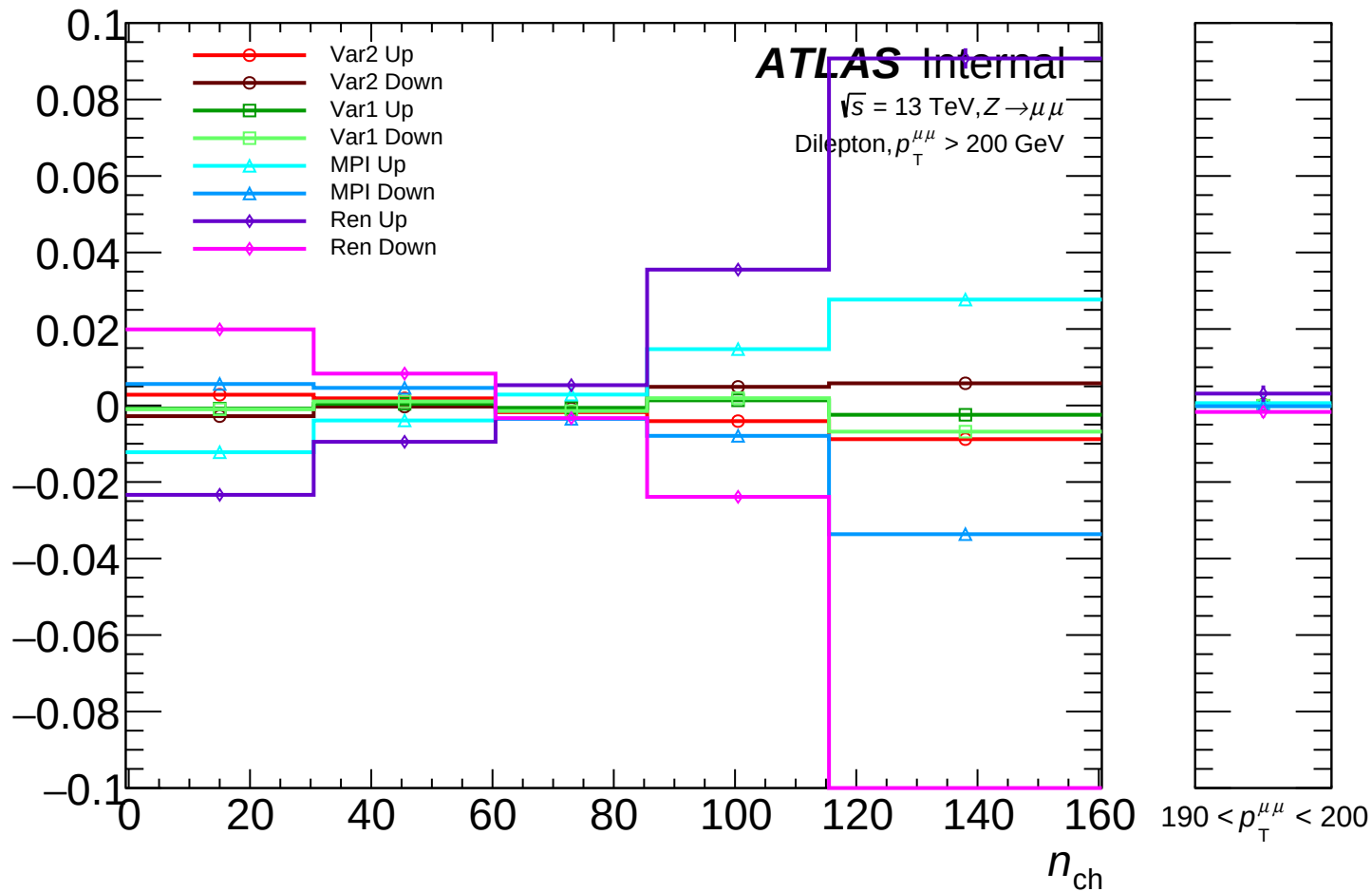
Data Statistical Uncertainty



Fractional systematic impact



Fractional systematic impact



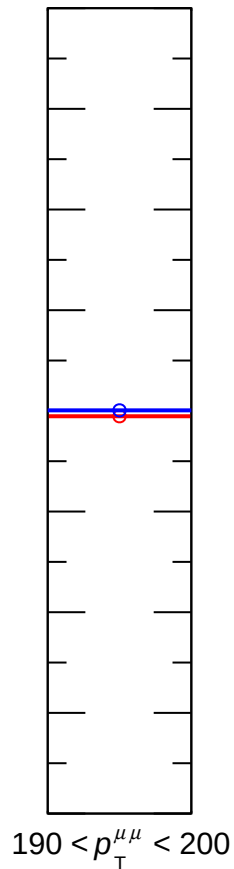
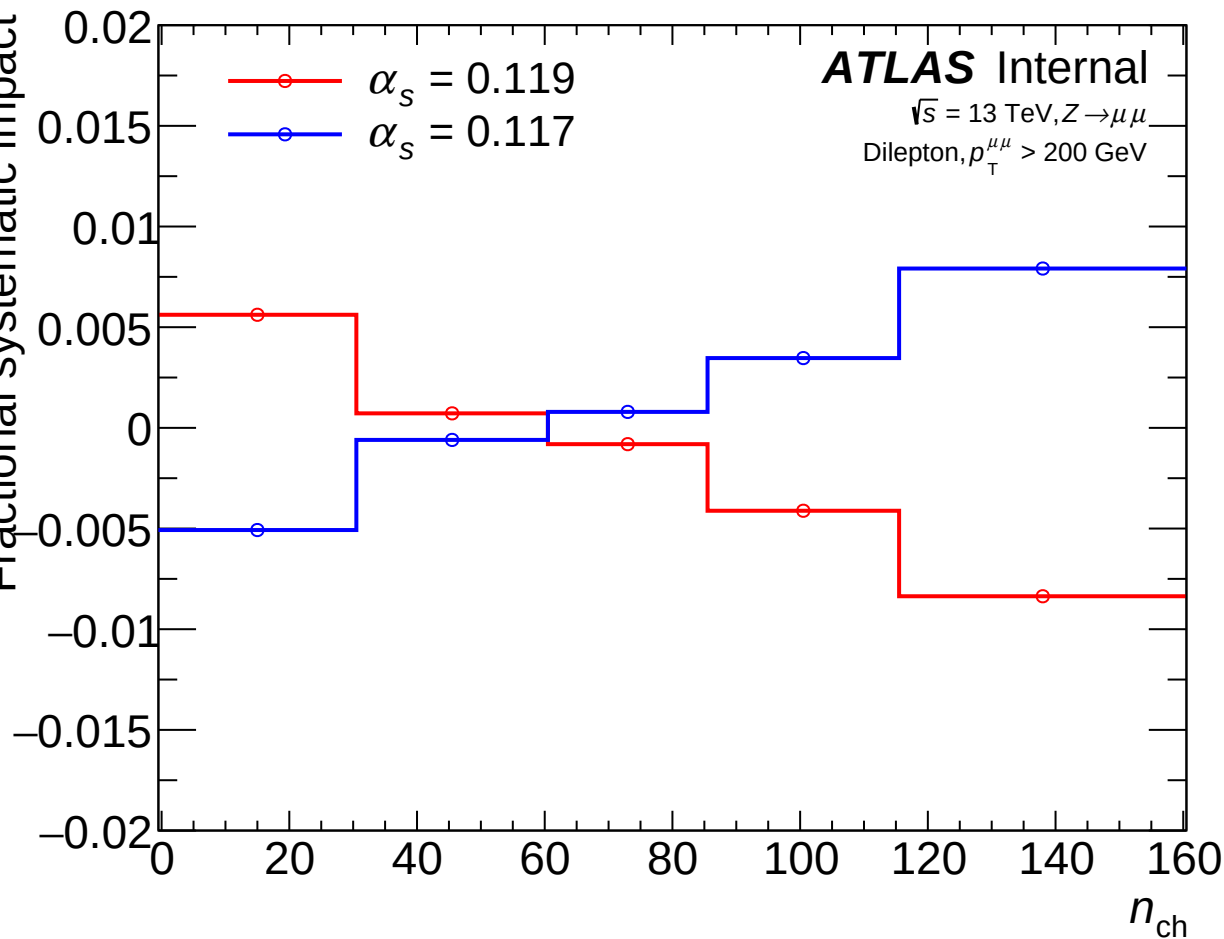
Fractional systematic impact

$\alpha_s = 0.119$
 $\alpha_s = 0.117$

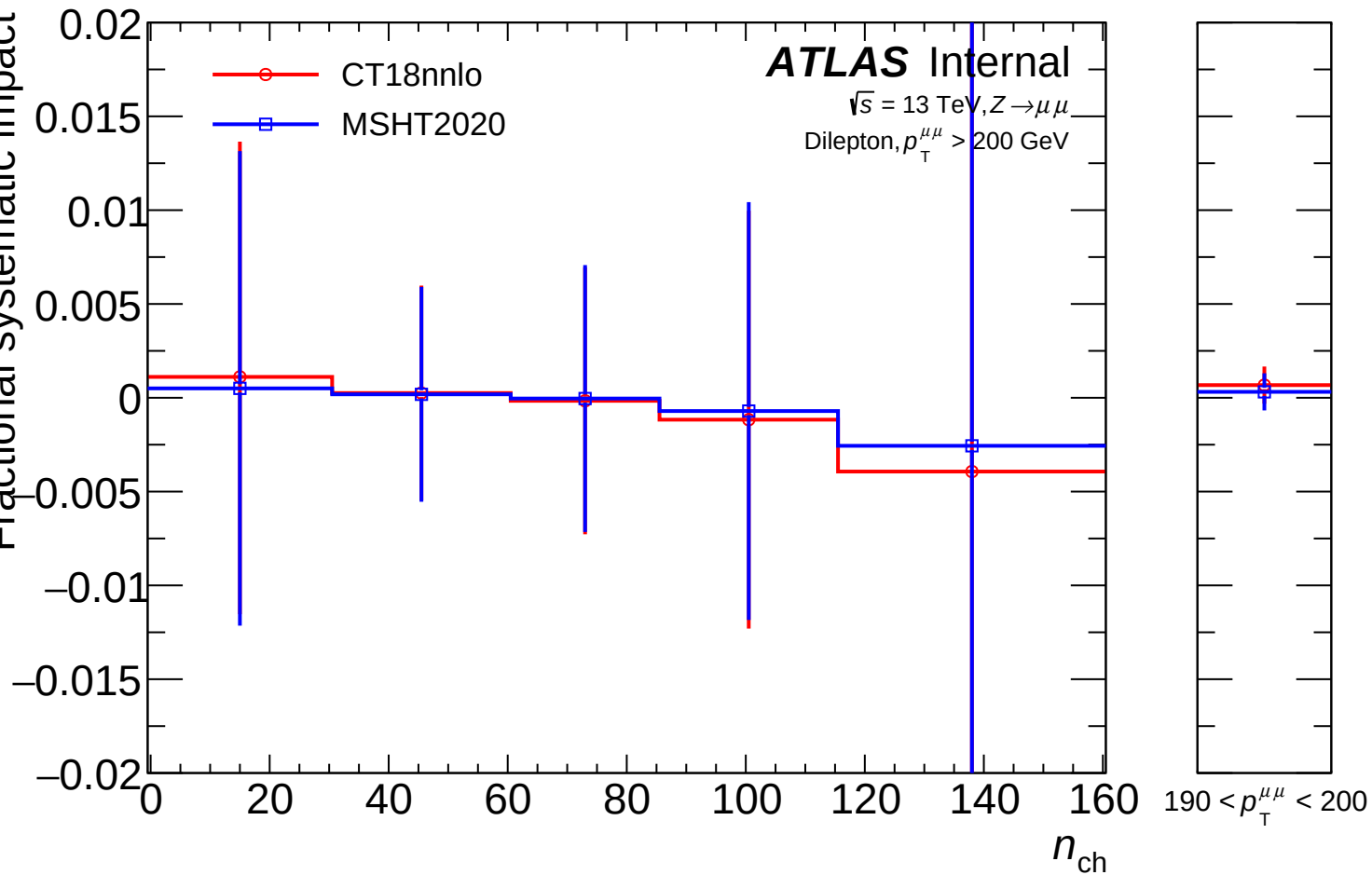
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$



Fractional systematic impact



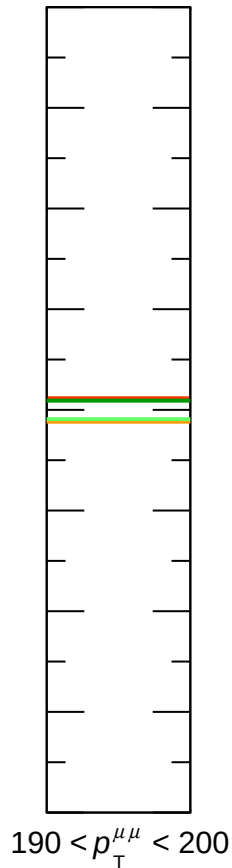
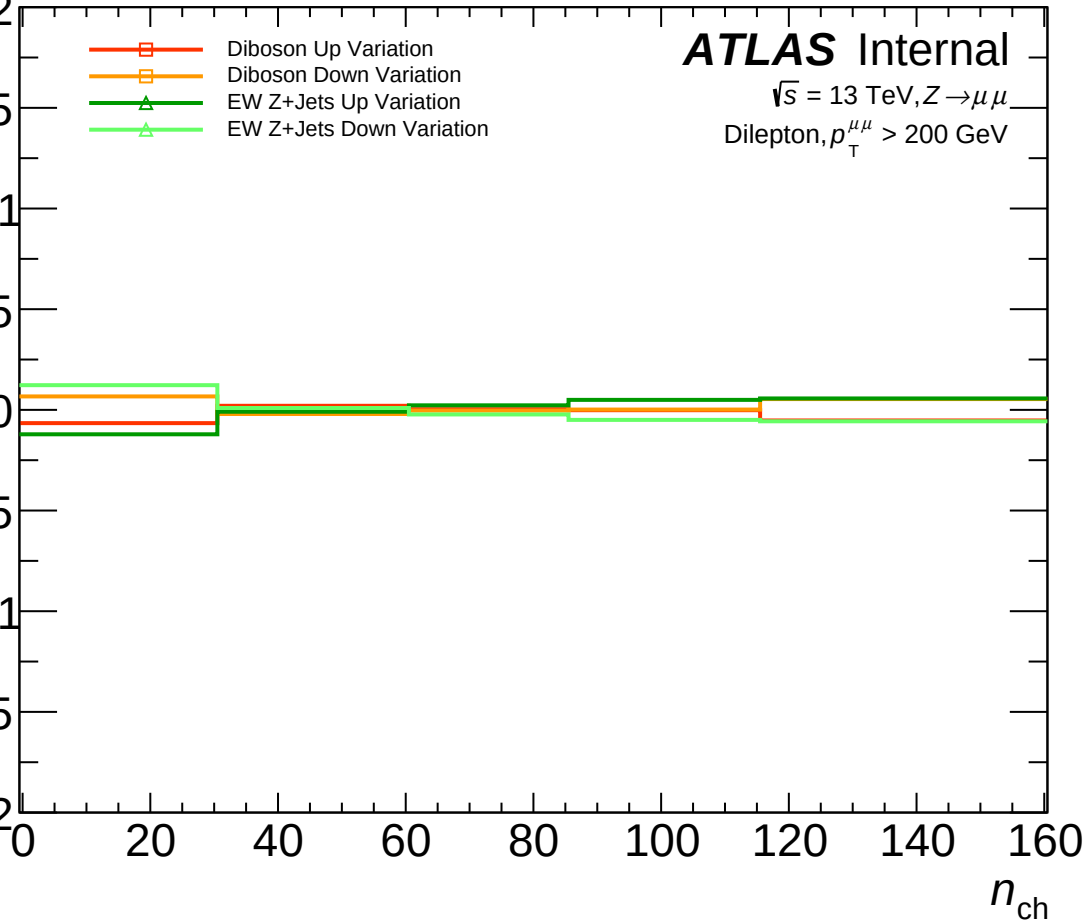
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

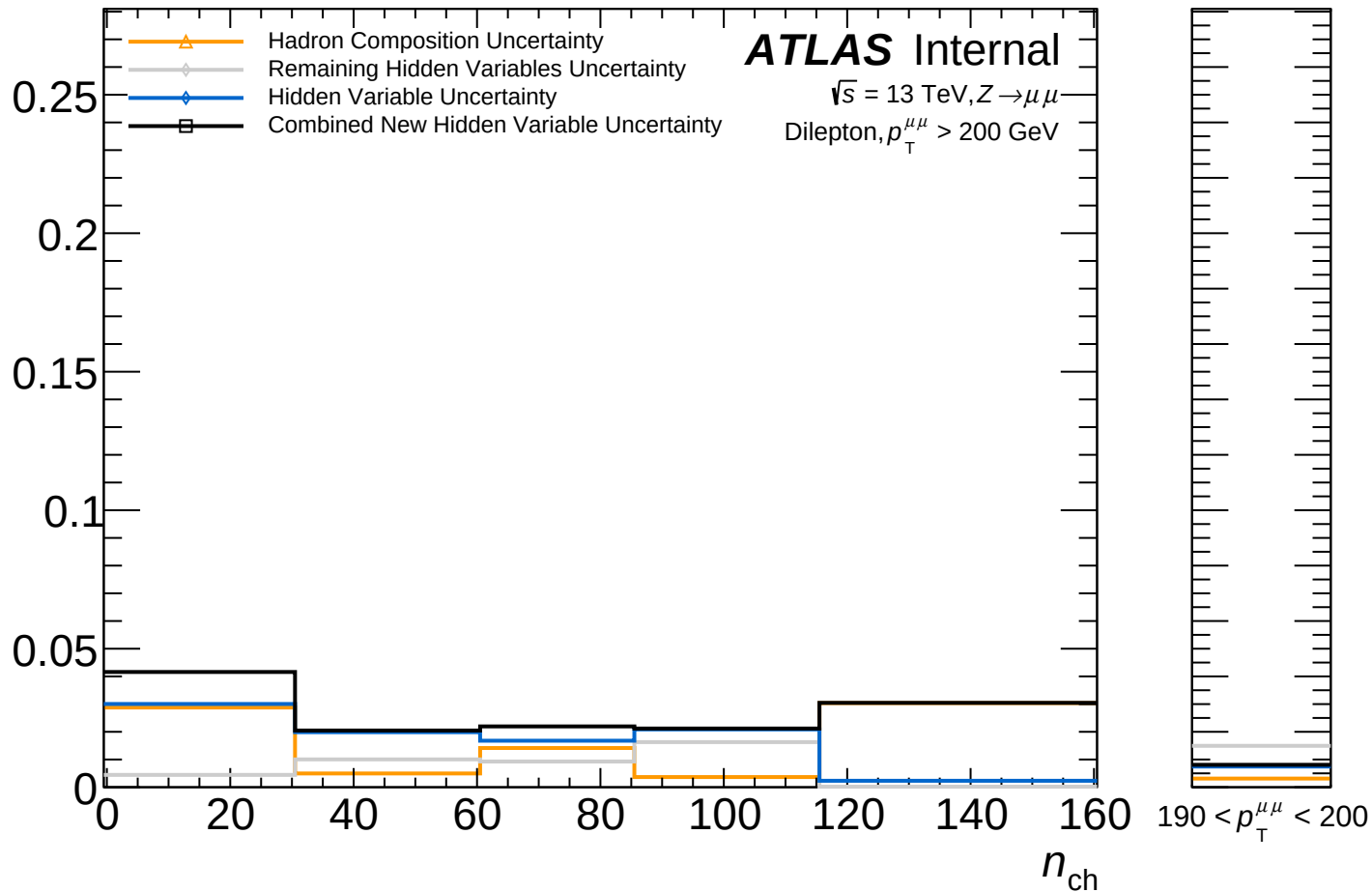


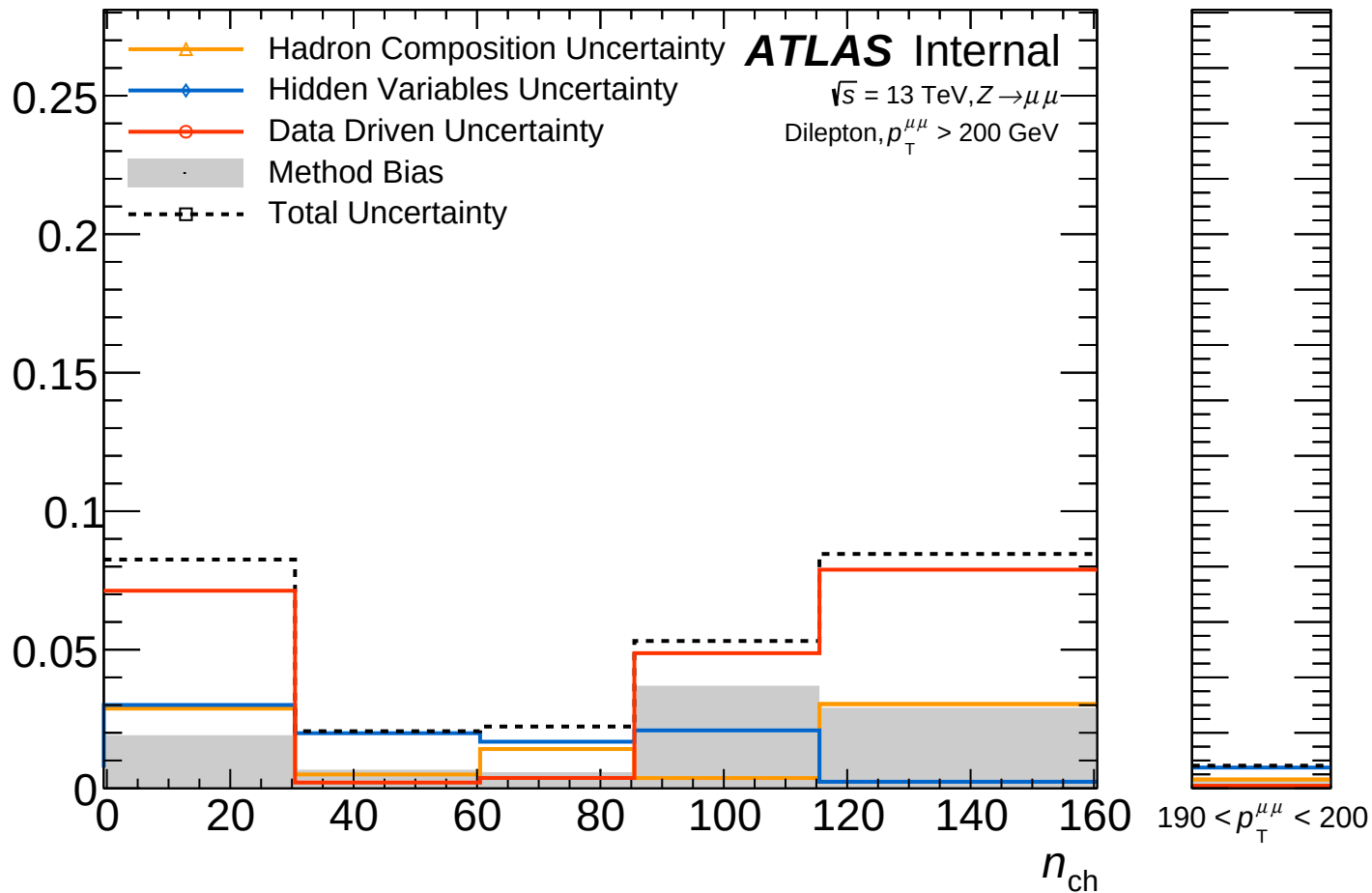
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty





Events

$\times 10^3$

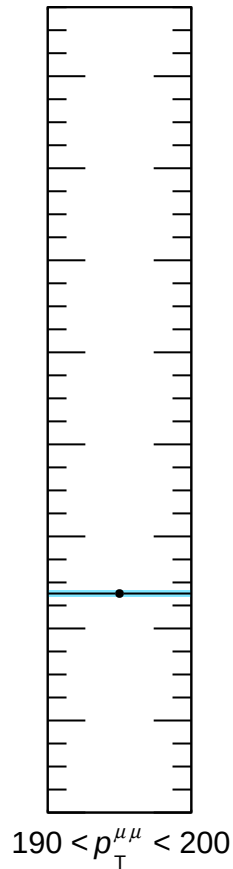
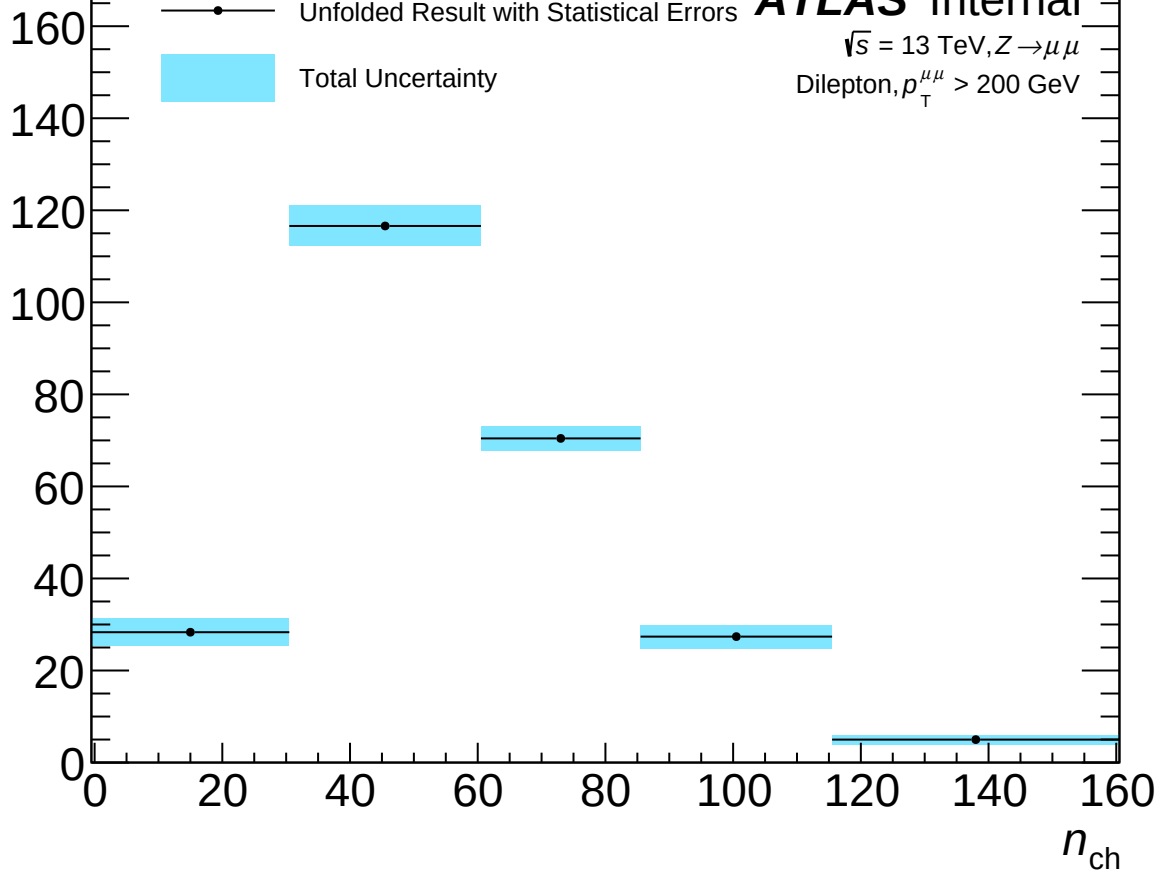
ATLAS Internal

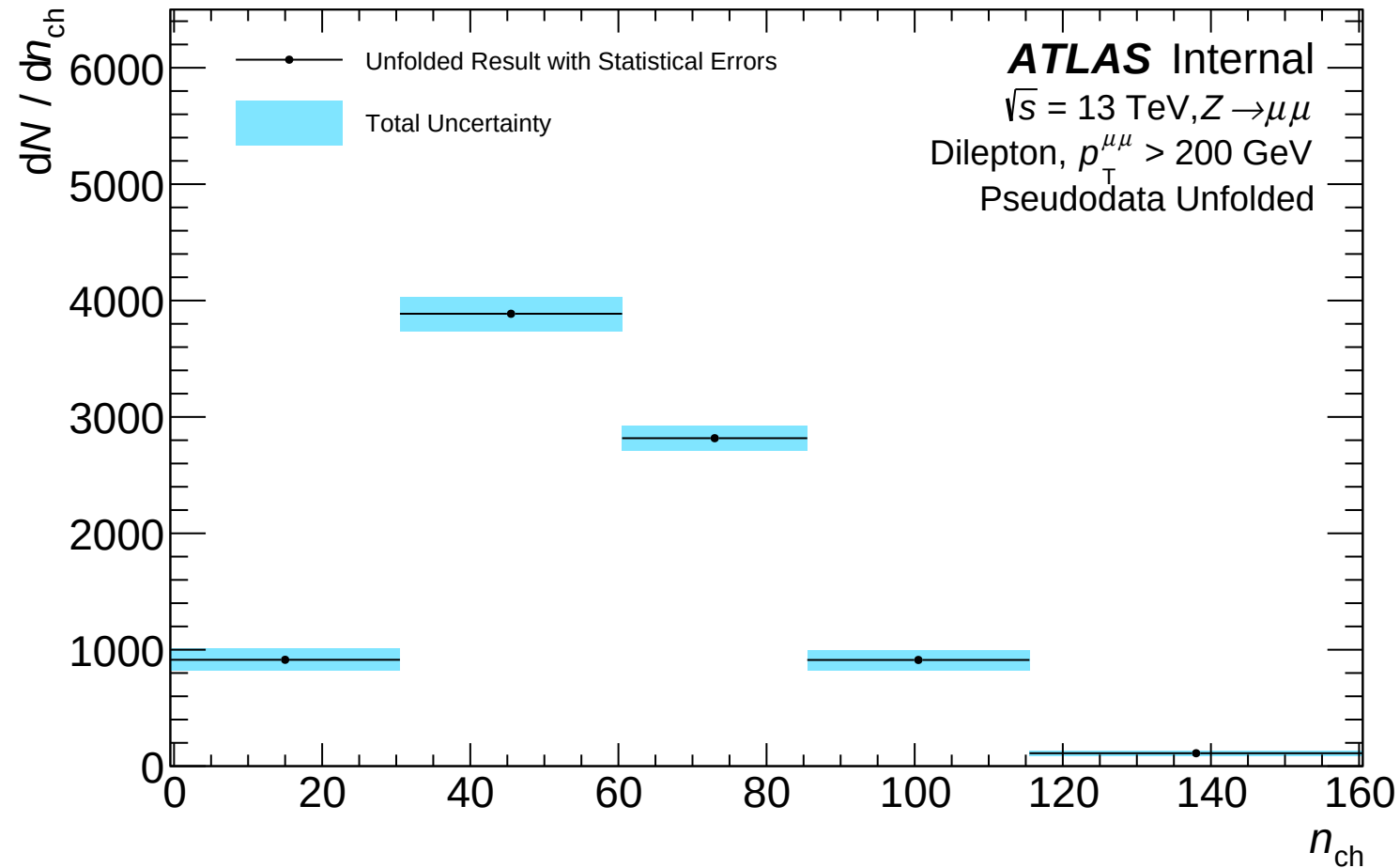
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

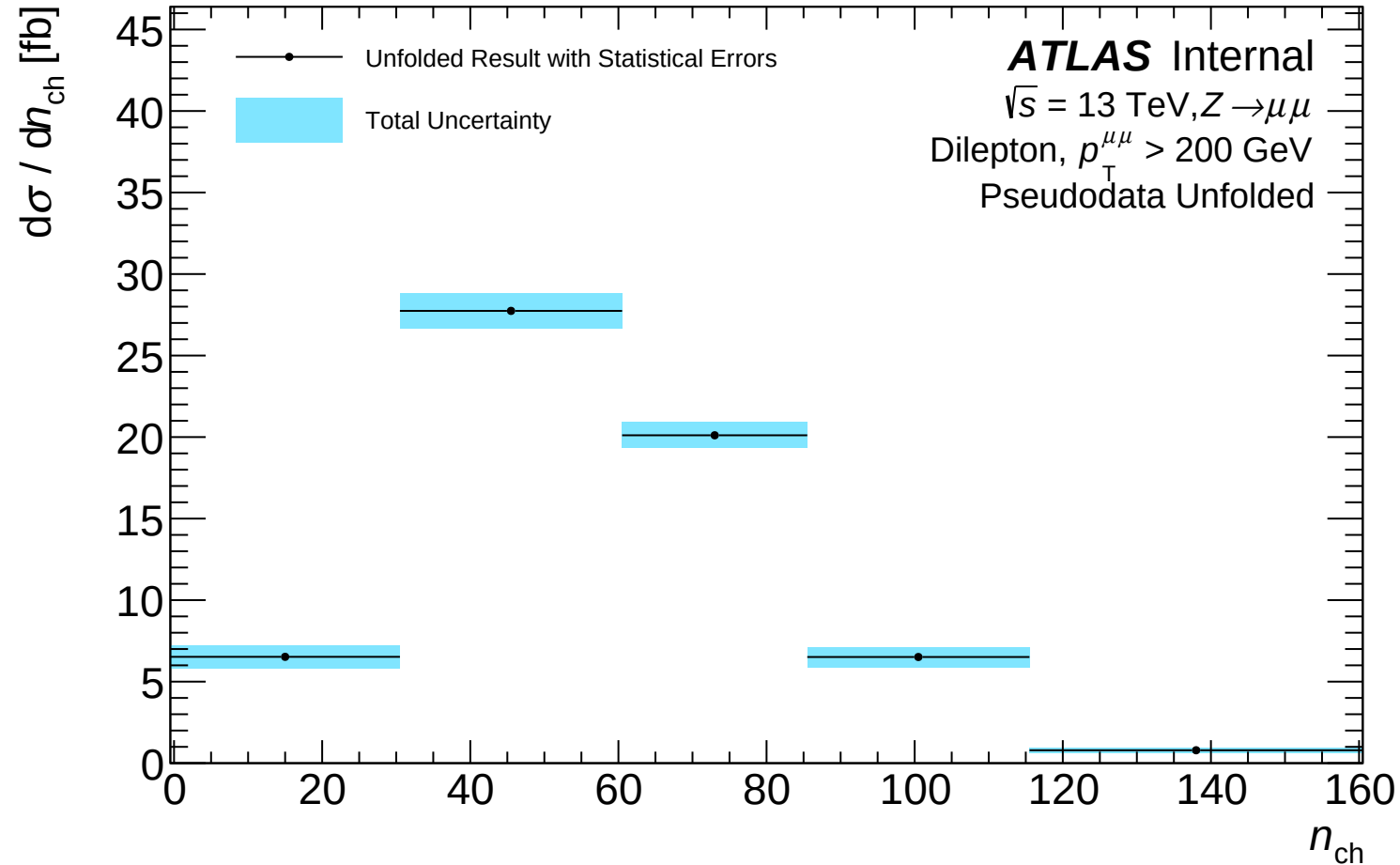
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

Unfolded Result with Statistical Errors

Total Uncertainty







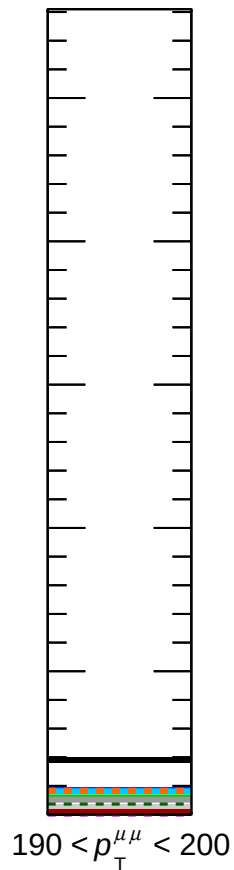
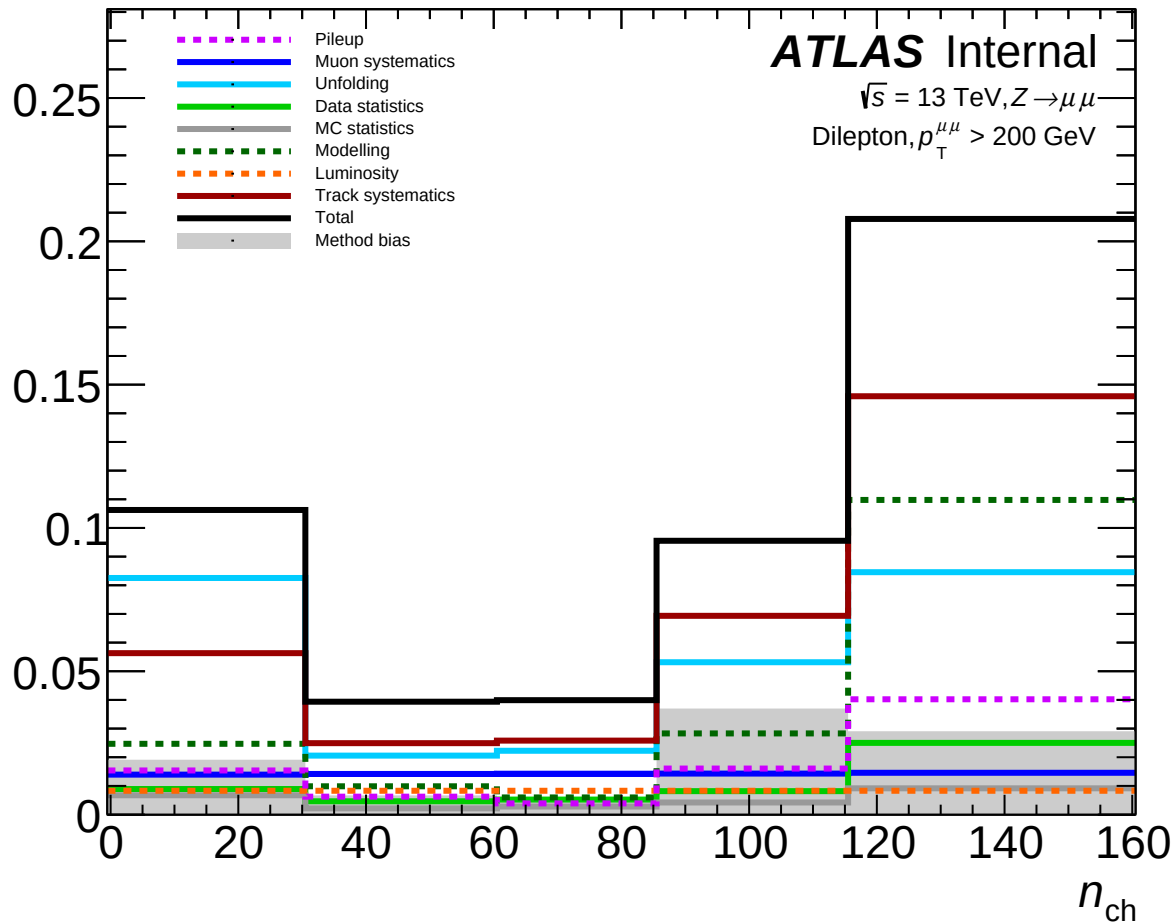
Fractional uncertainty

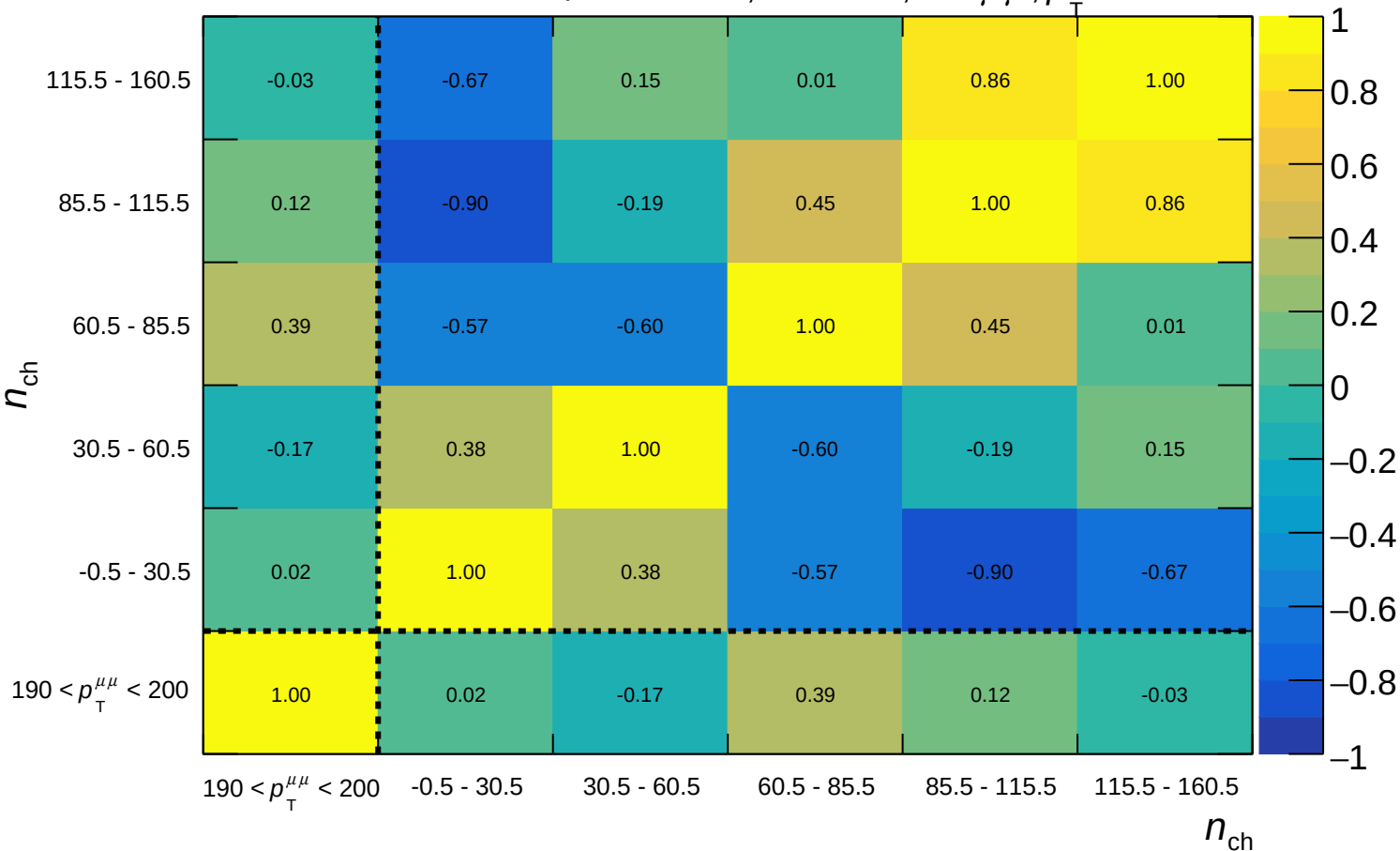
ATLAS Internal

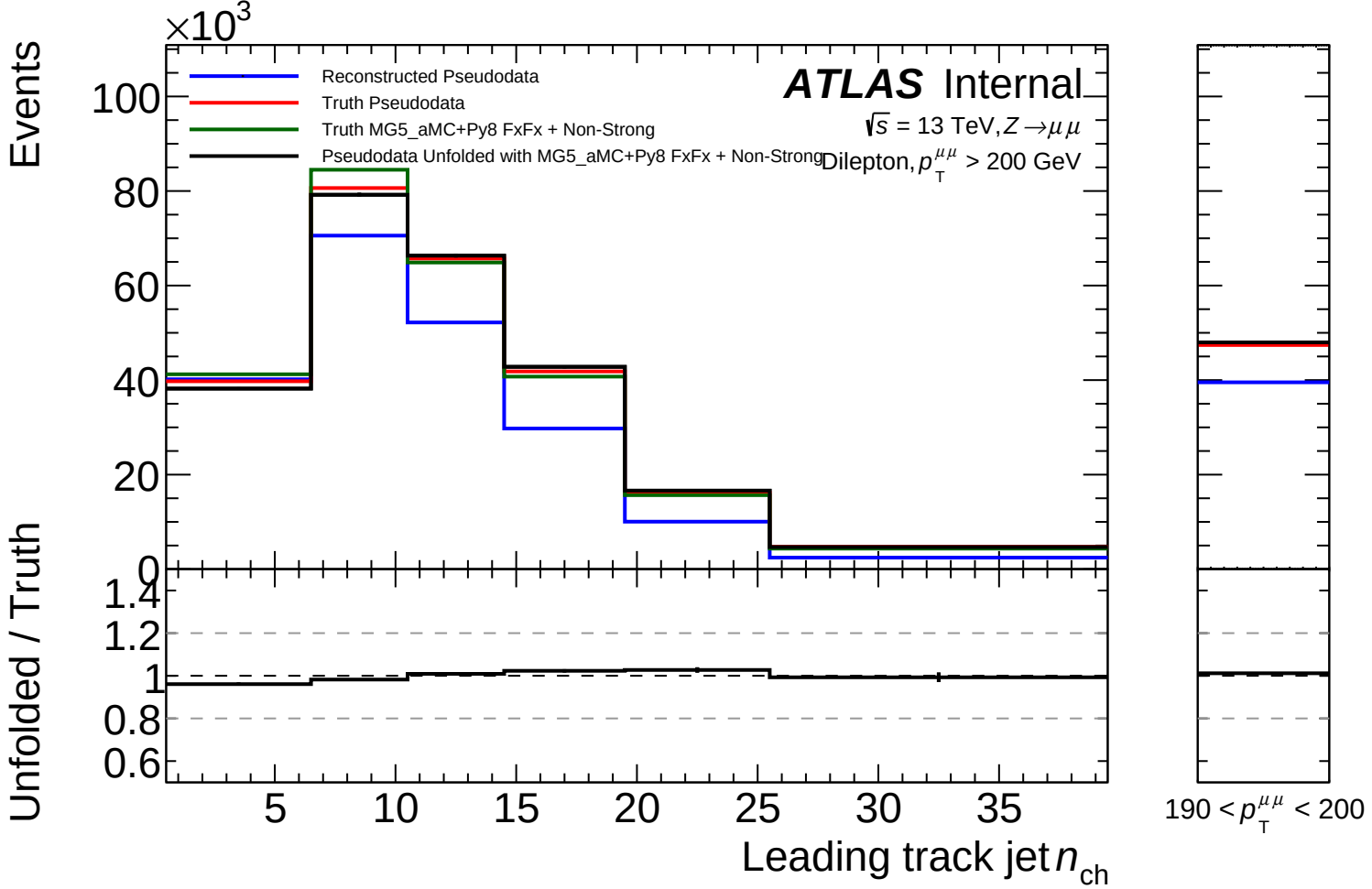
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

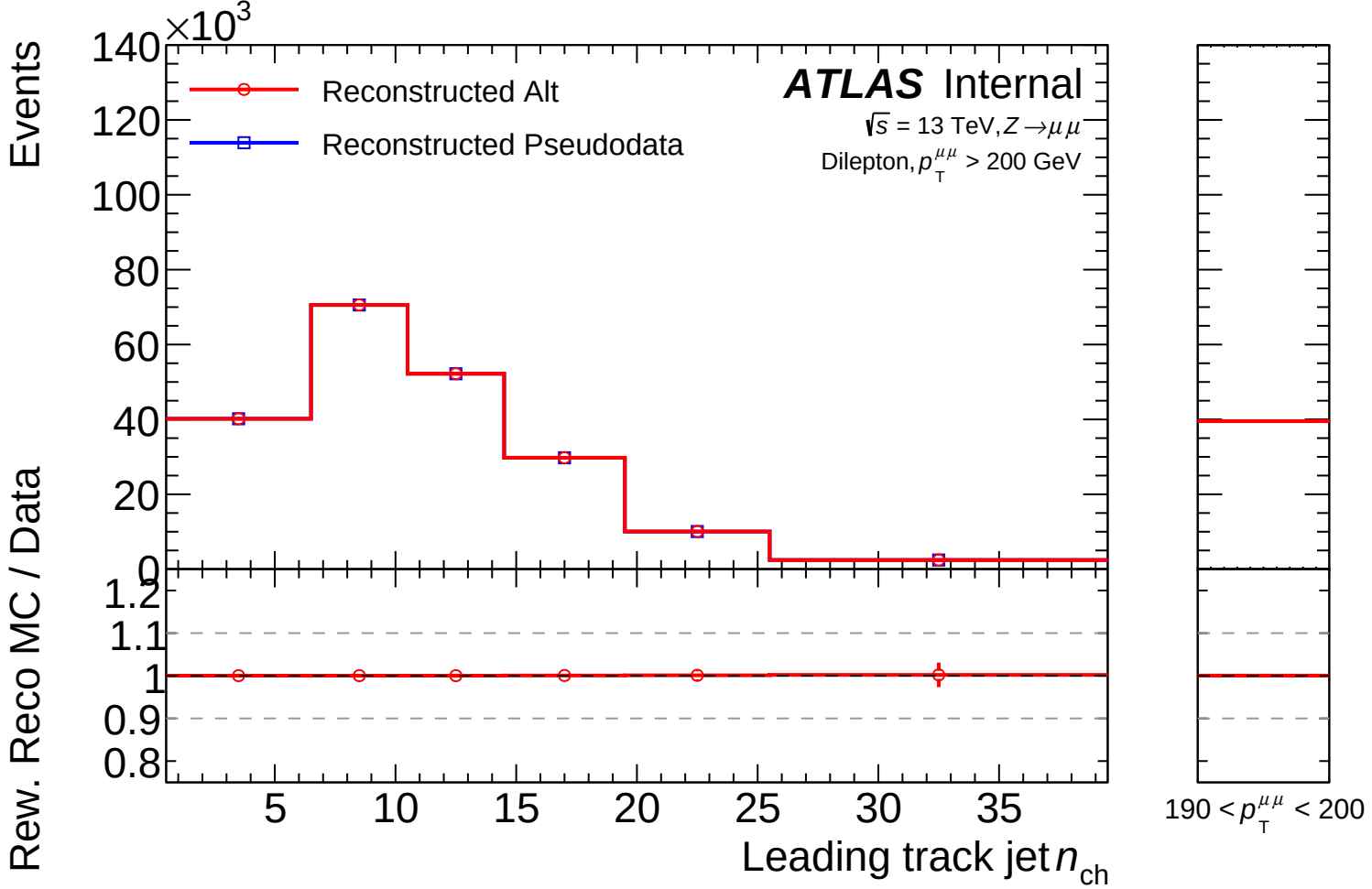
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

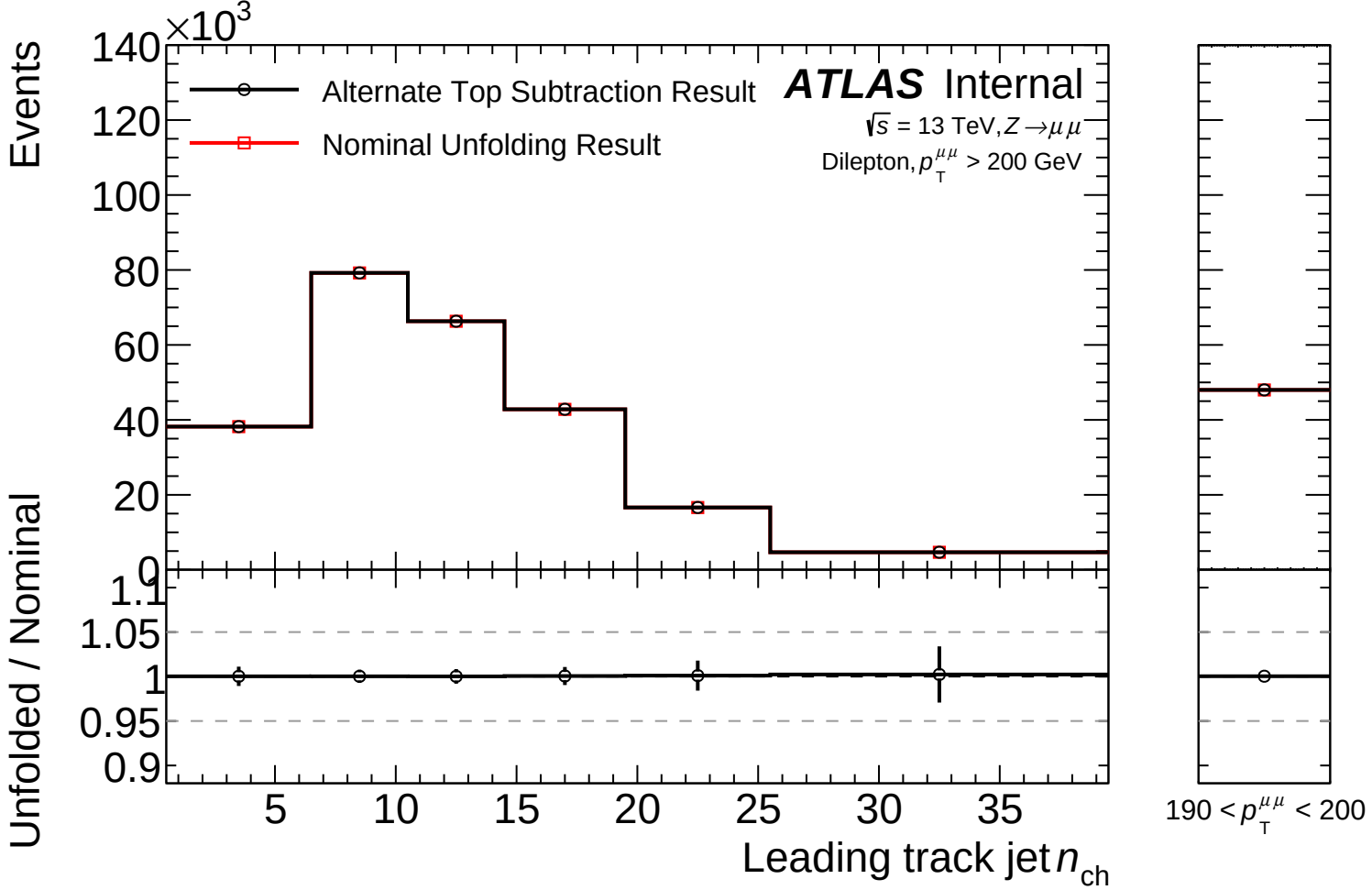
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ 







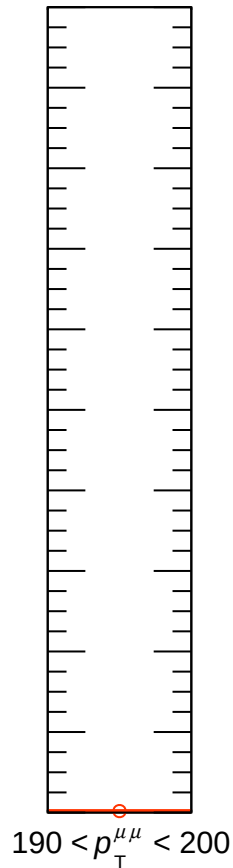
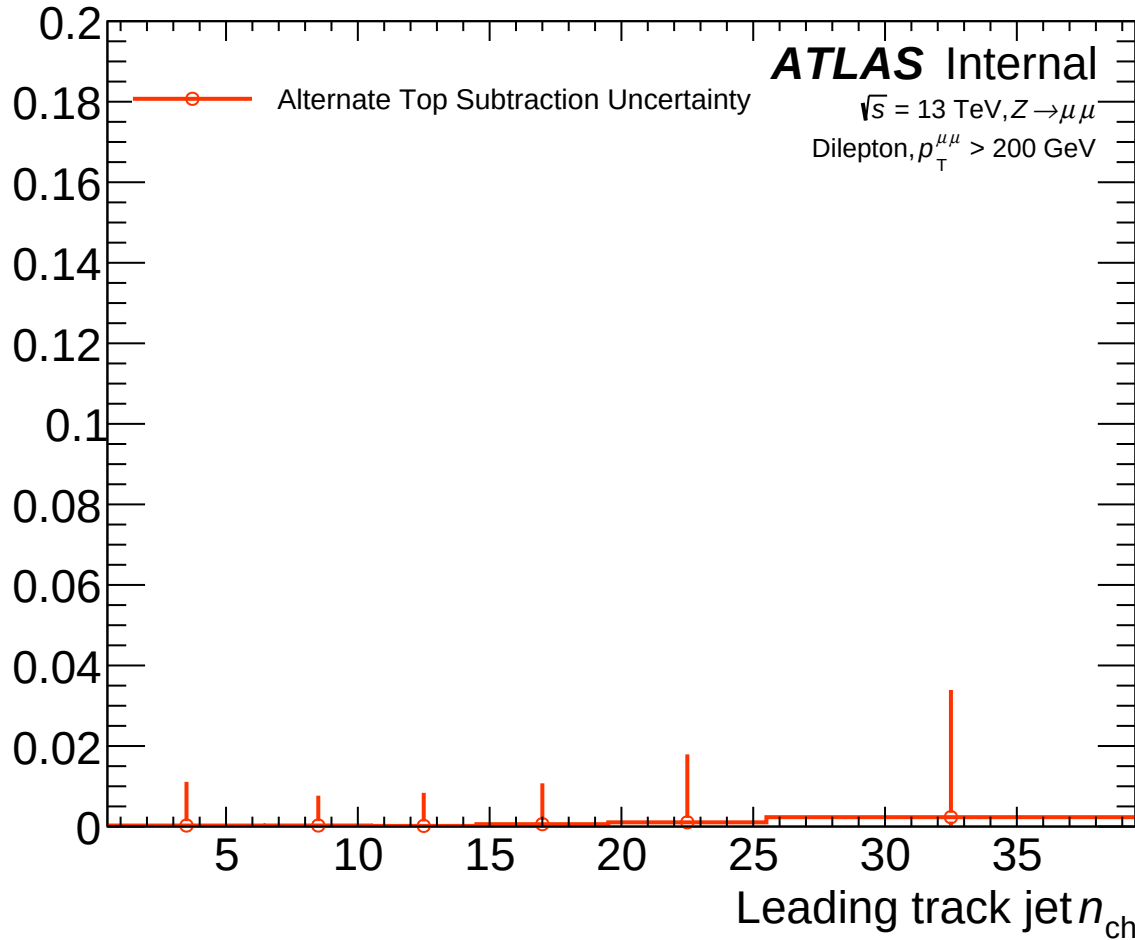
Fractional systematic impact

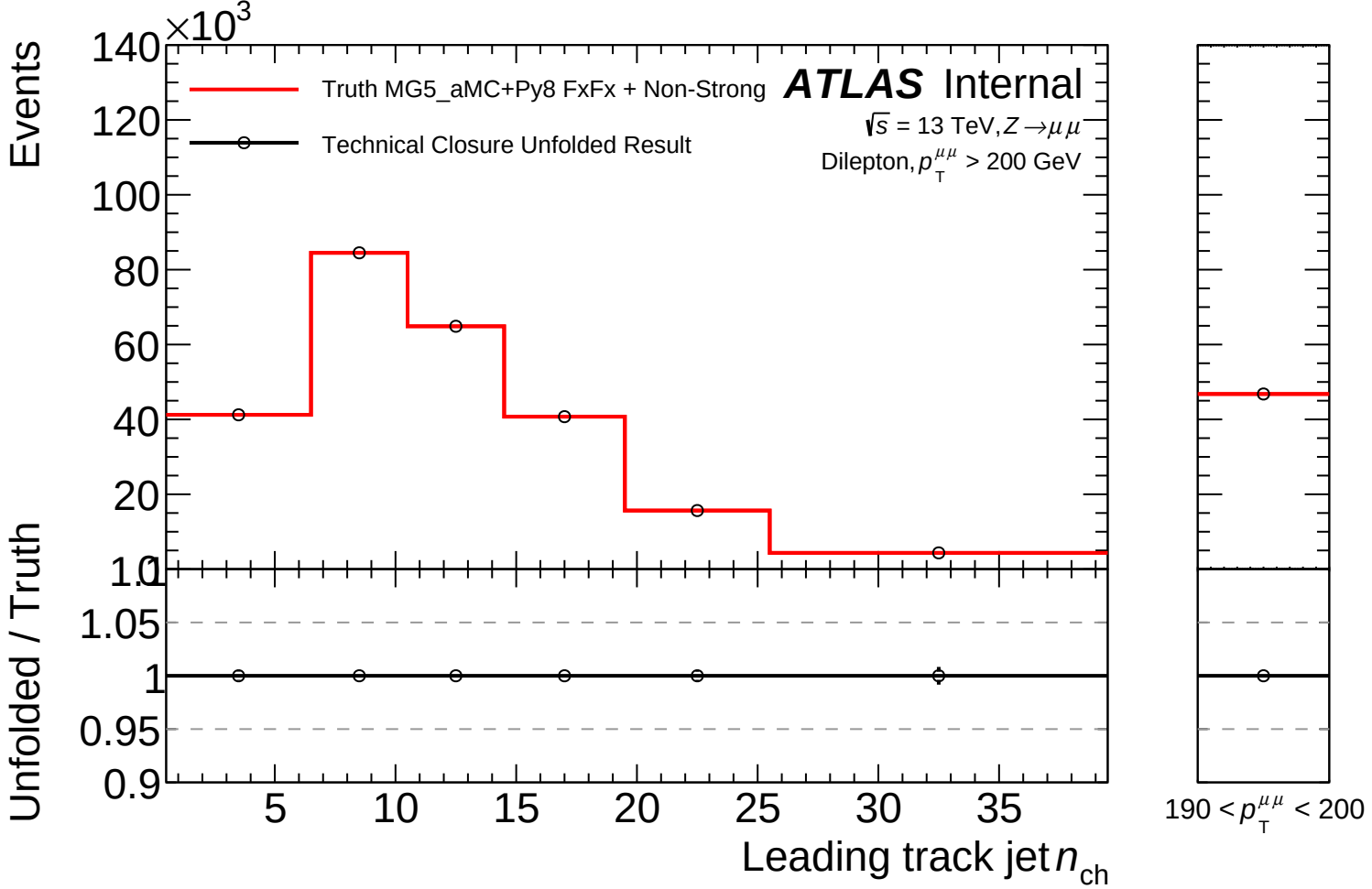
ATLAS Internal

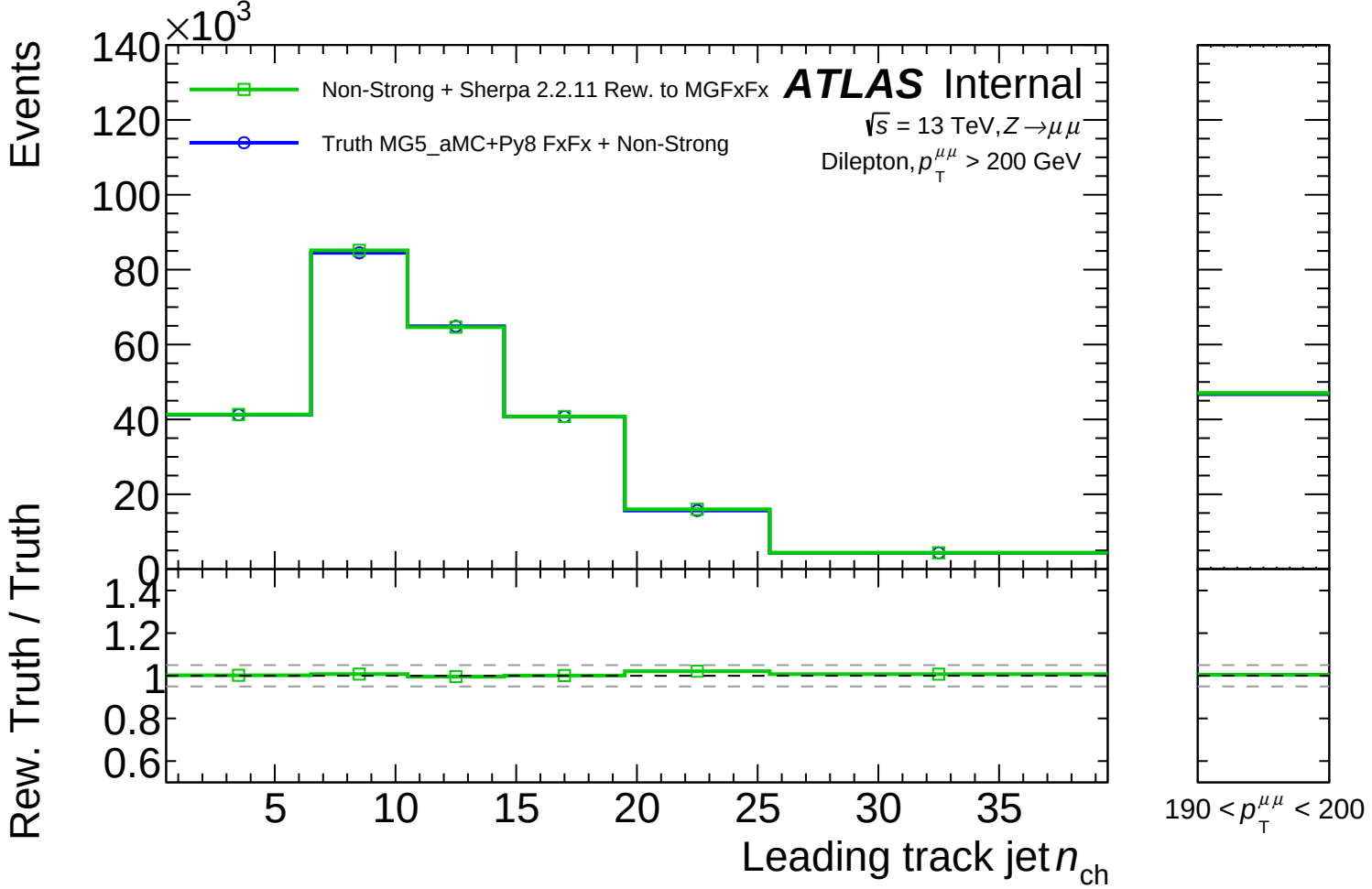
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

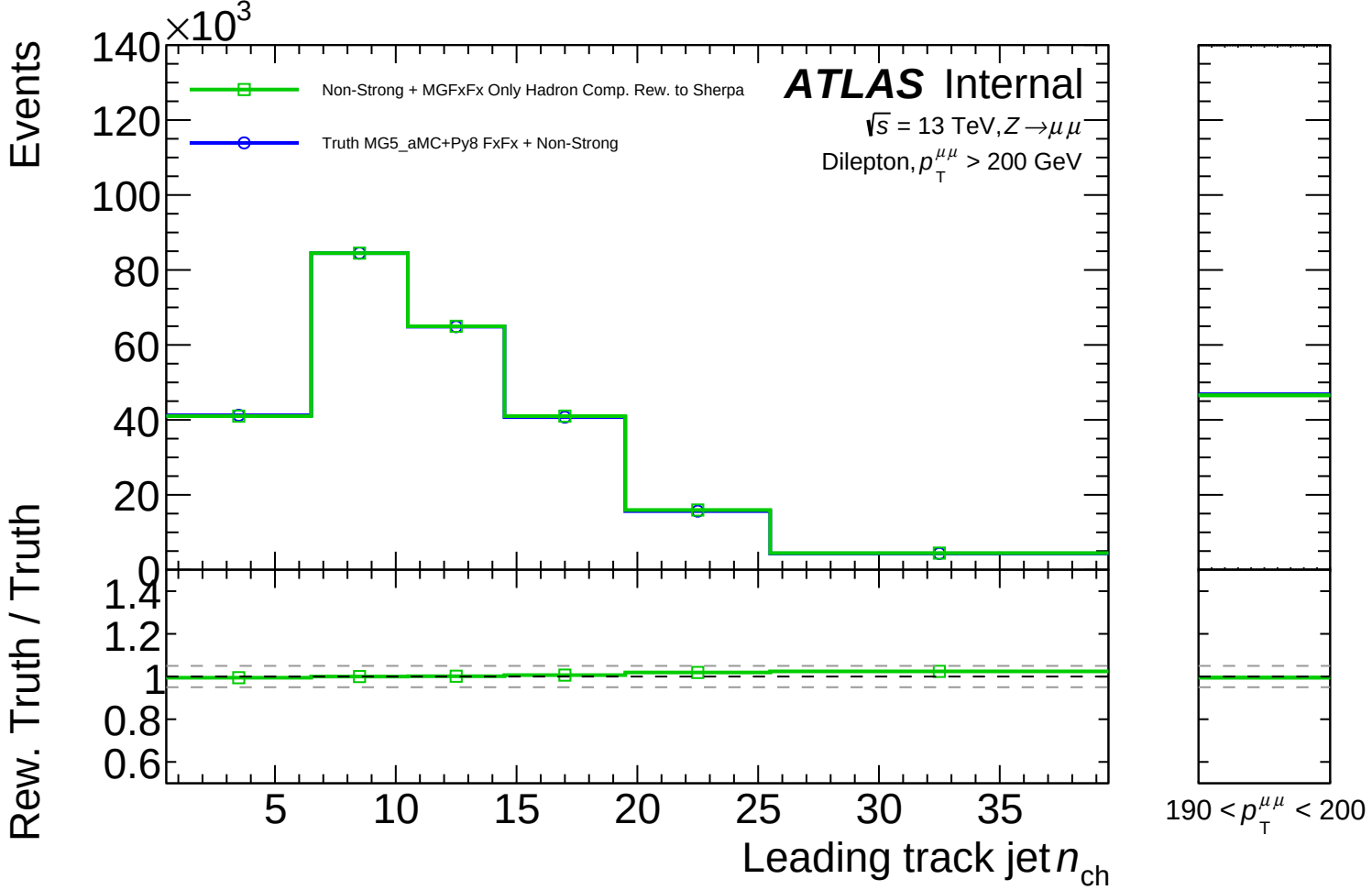
Dilepton, $p_T^{\mu\mu} > 200$ GeV

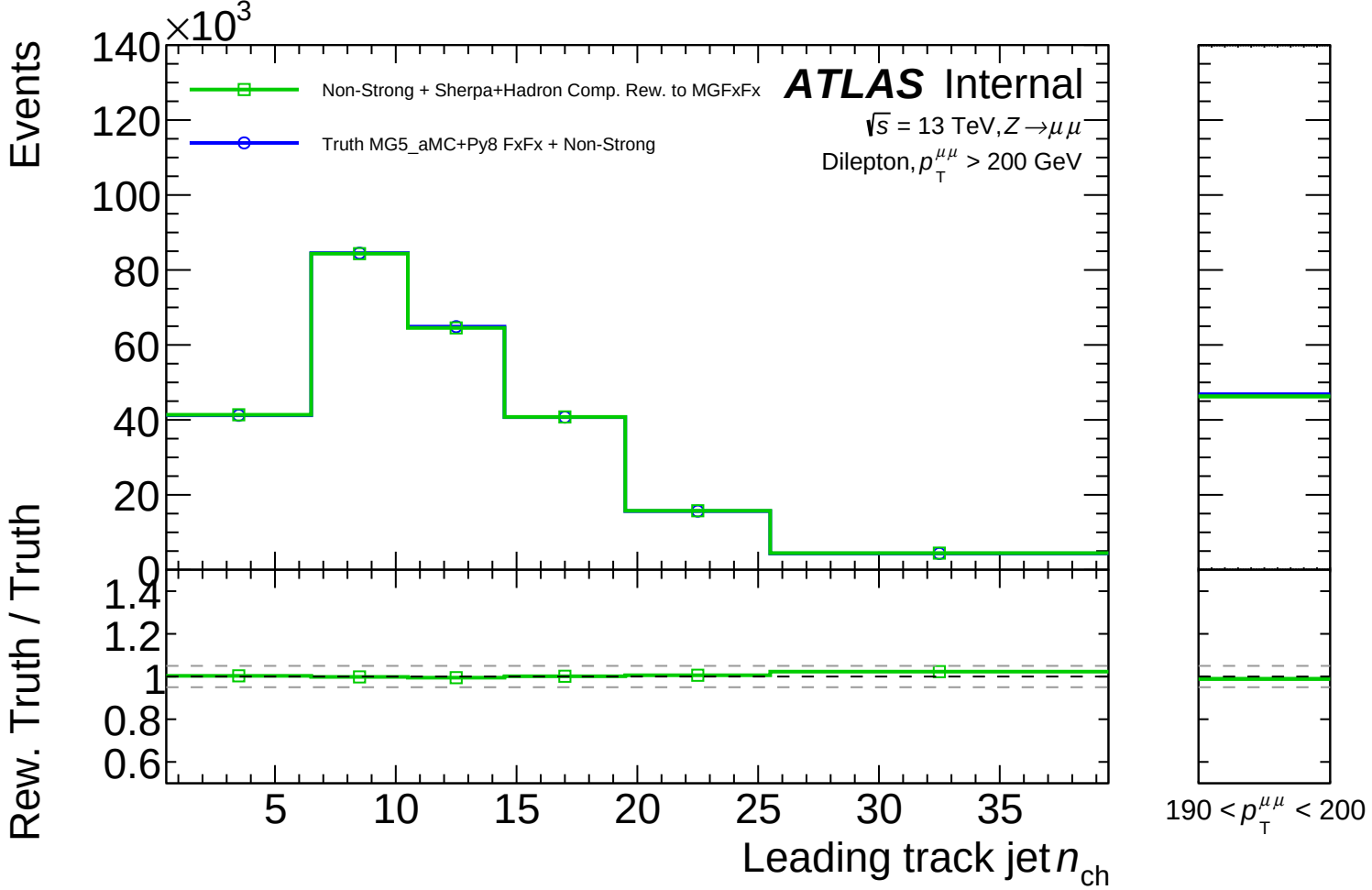
Alternate Top Subtraction Uncertainty

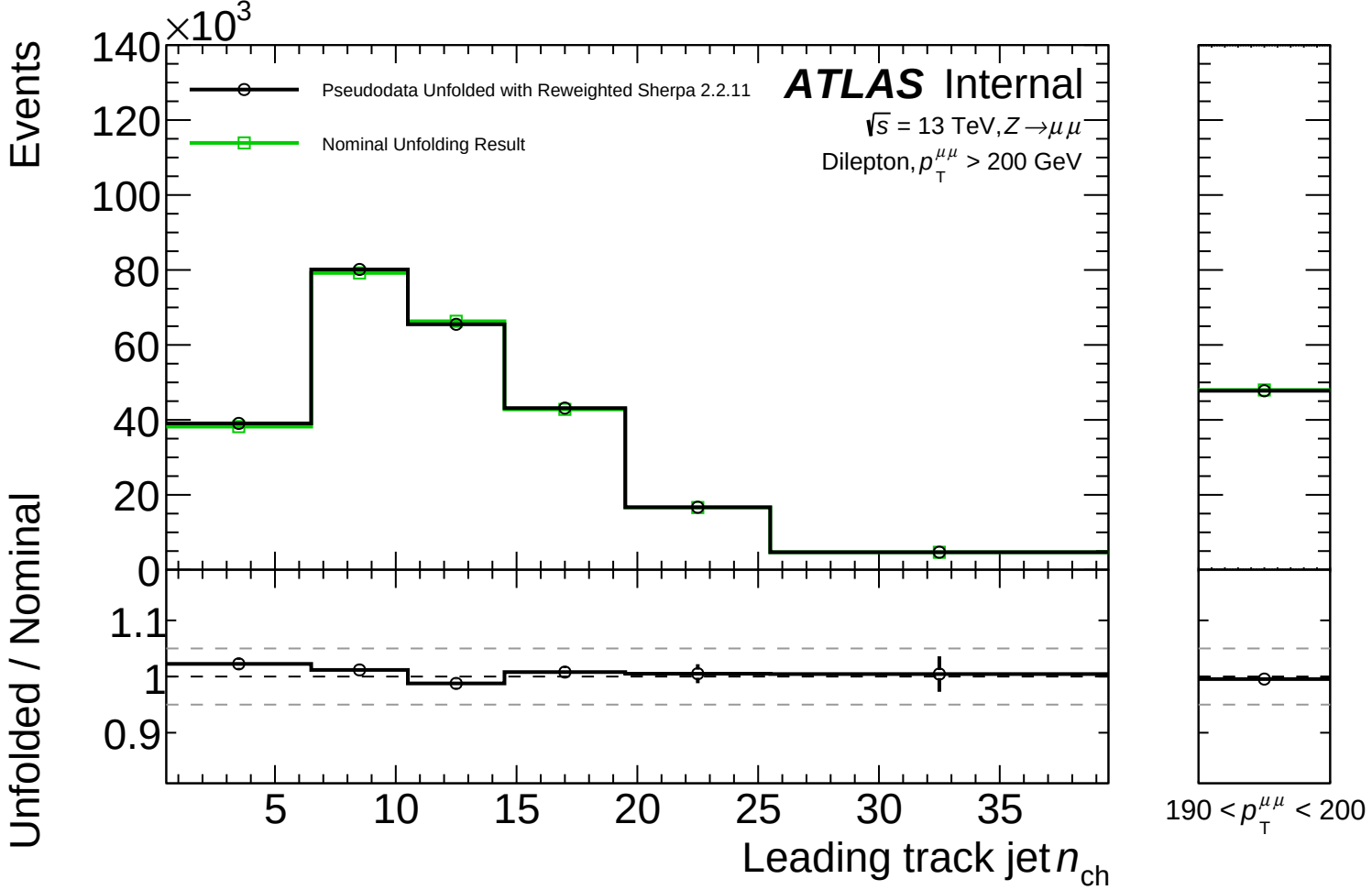


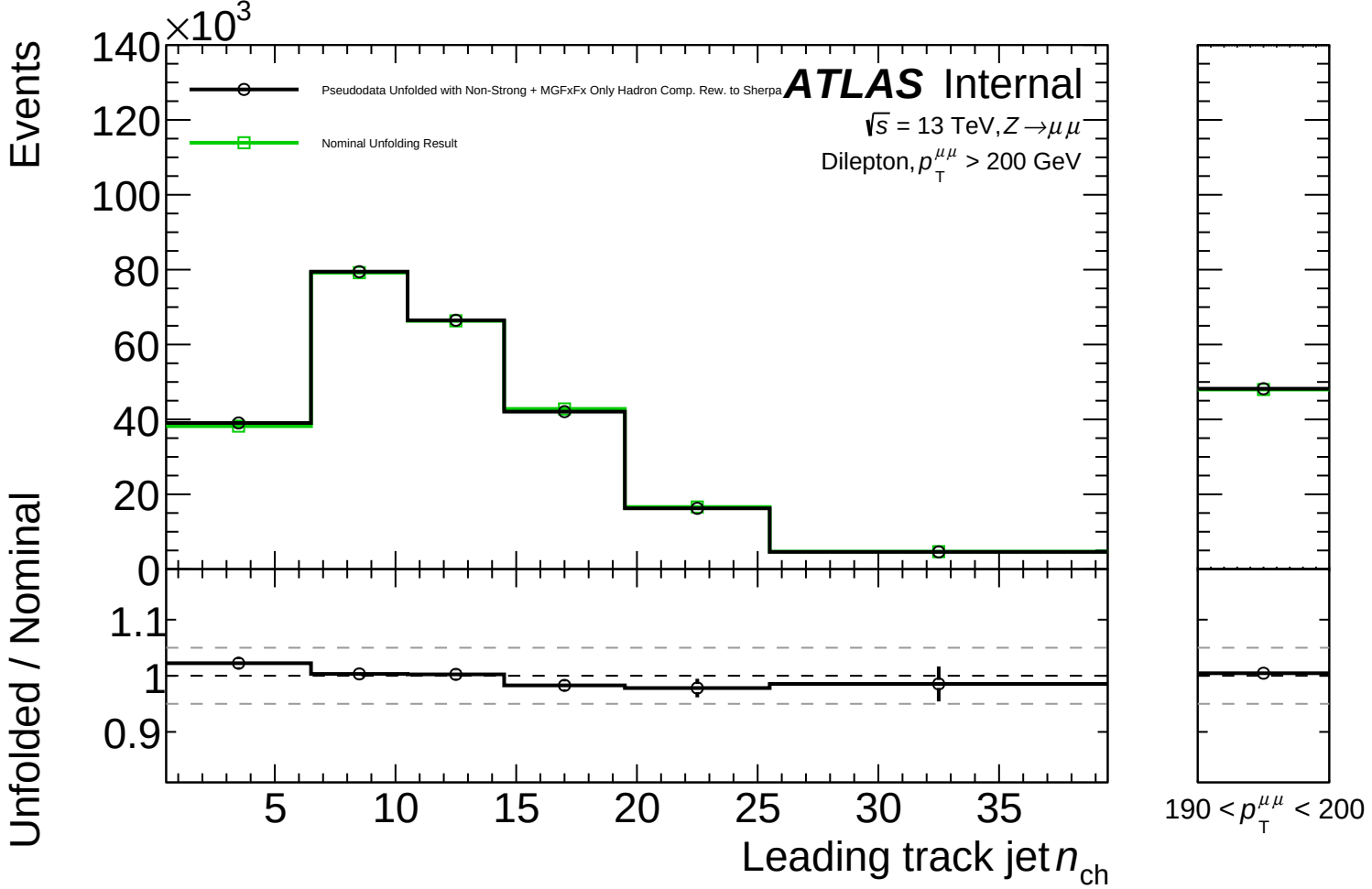


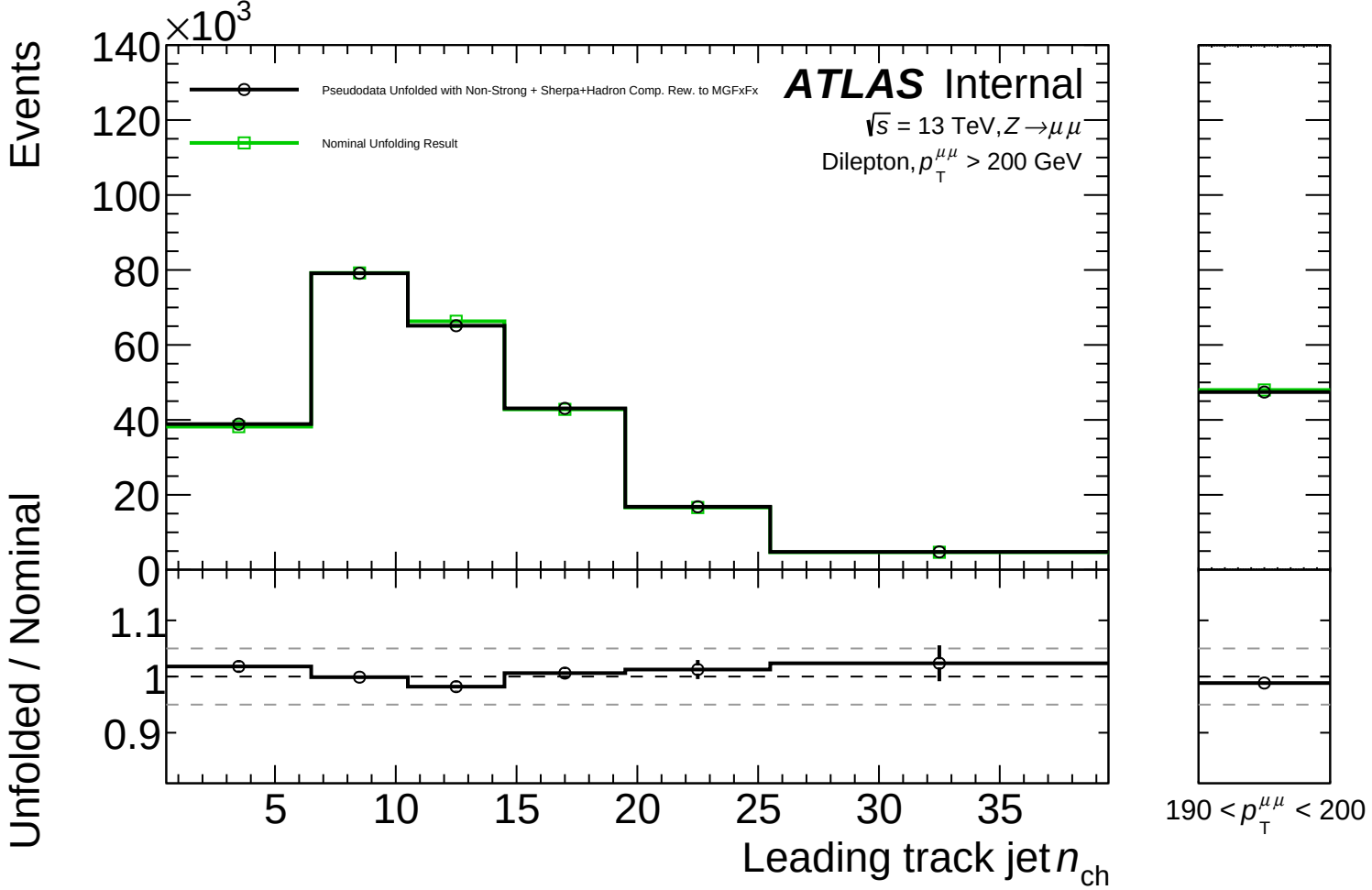


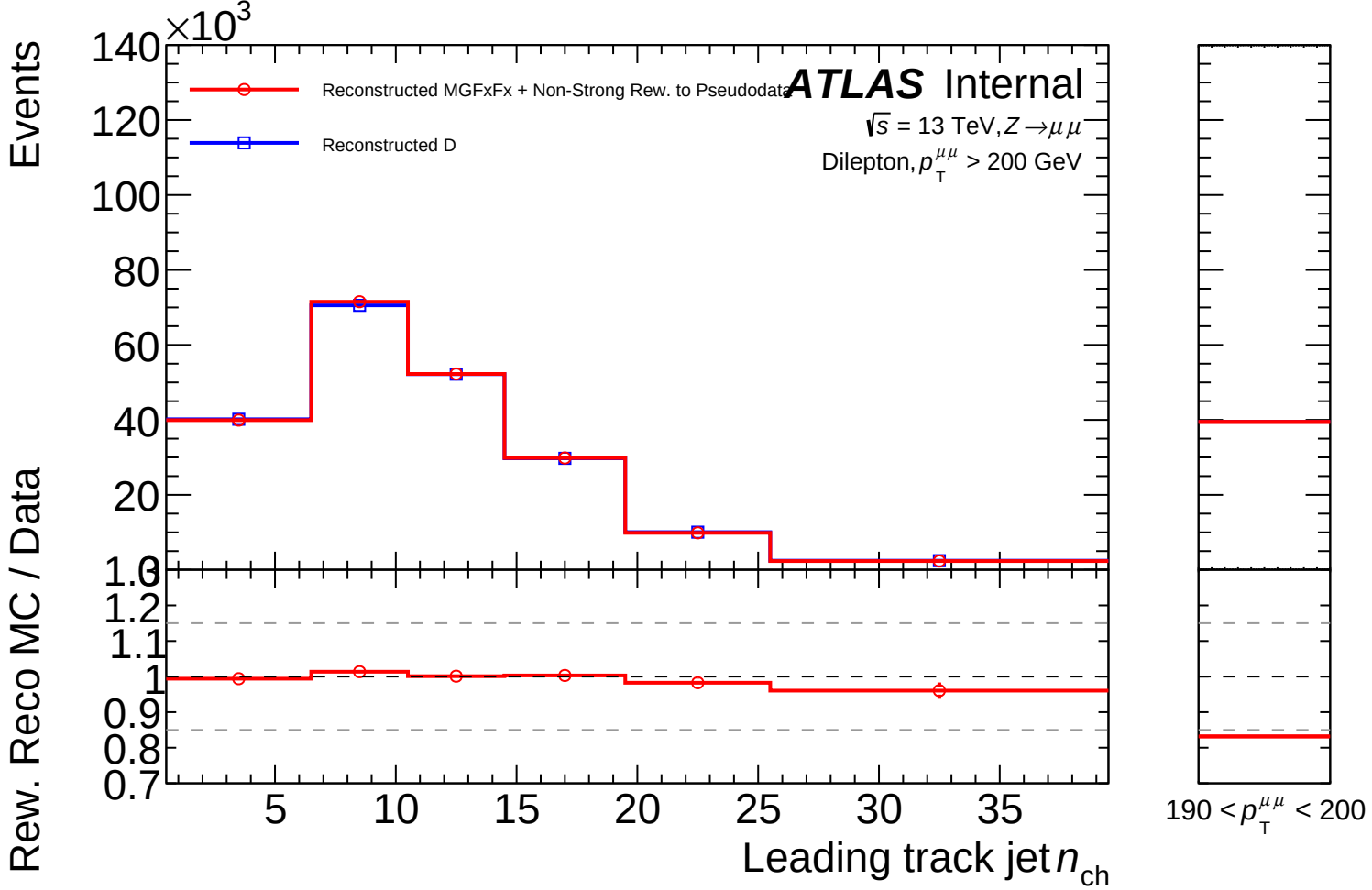


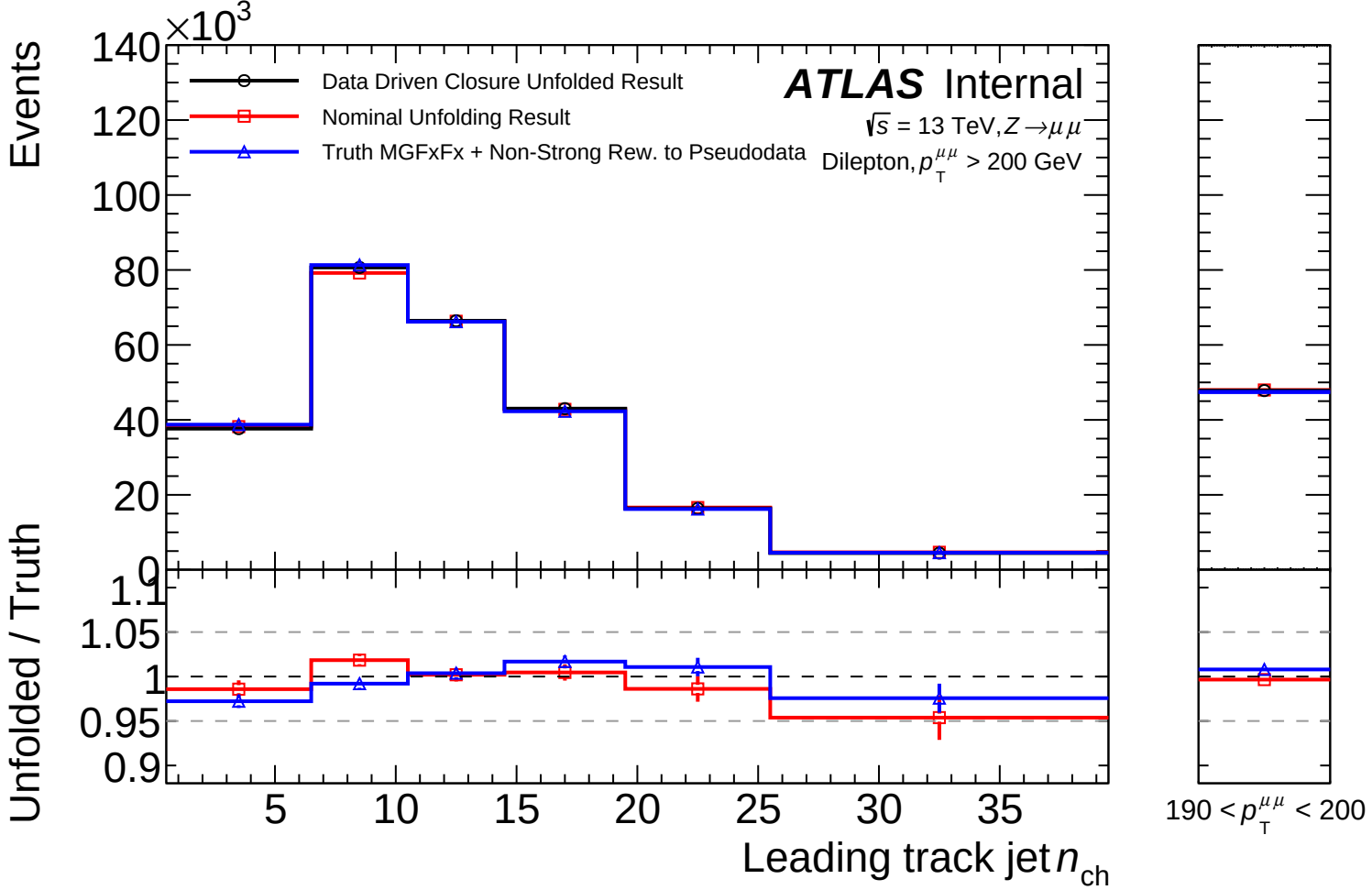




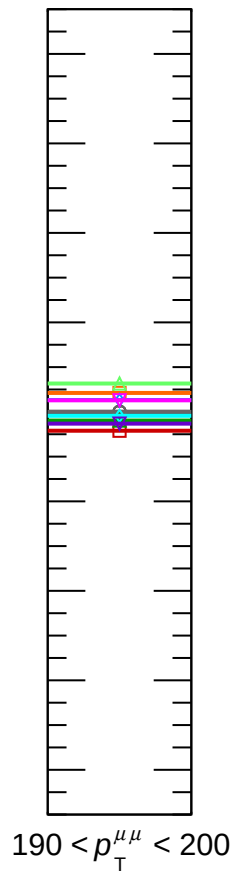
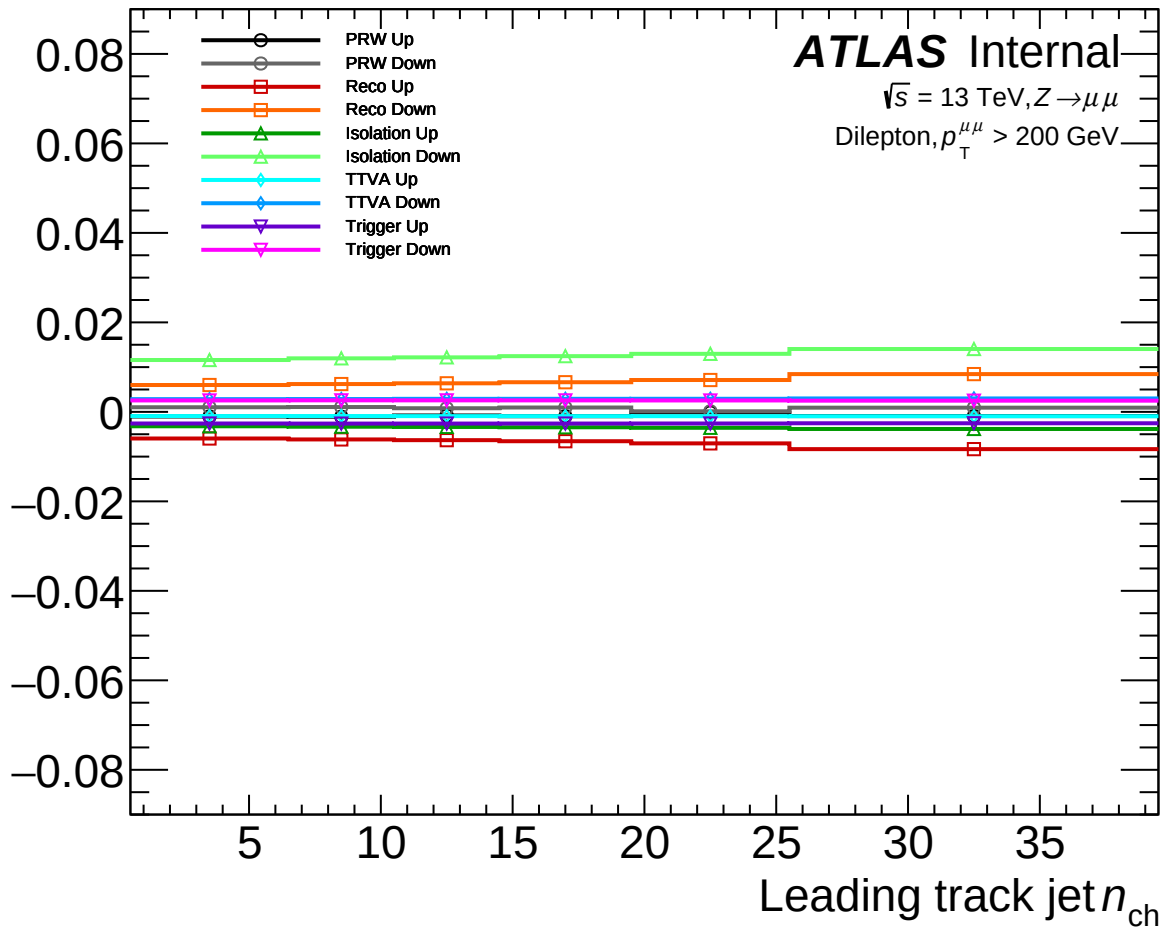




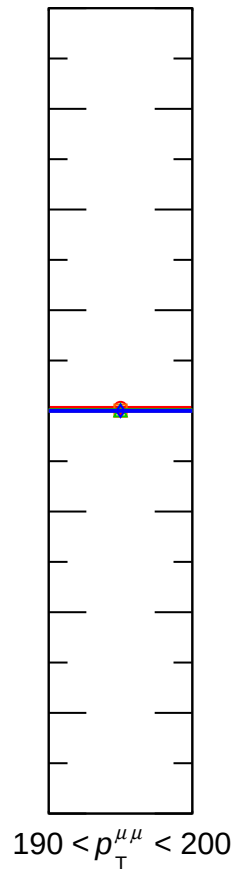
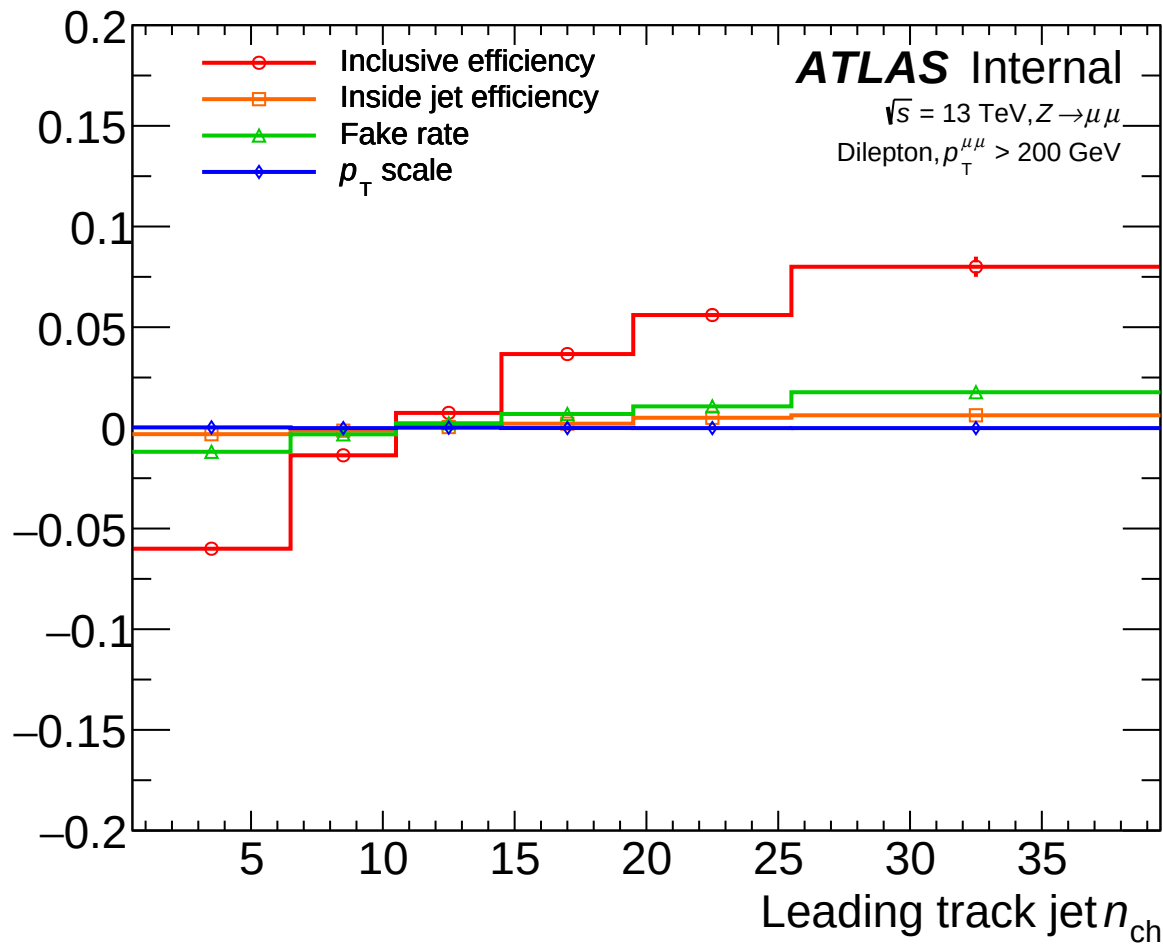




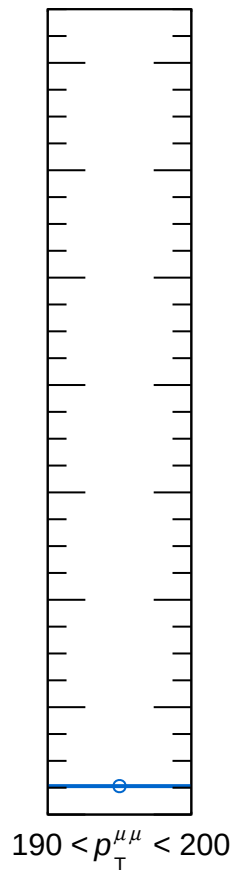
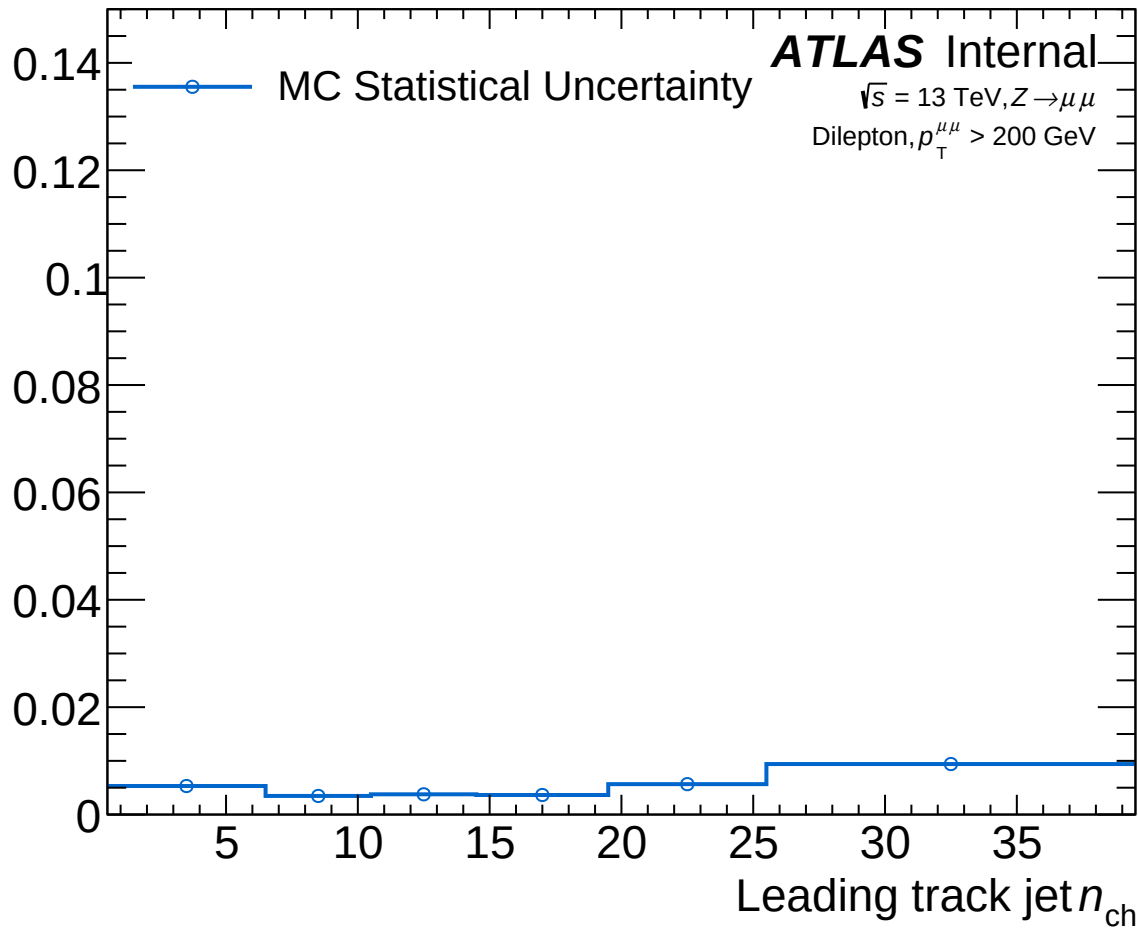
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



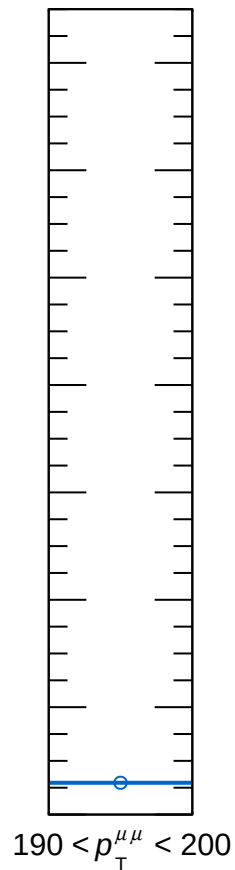
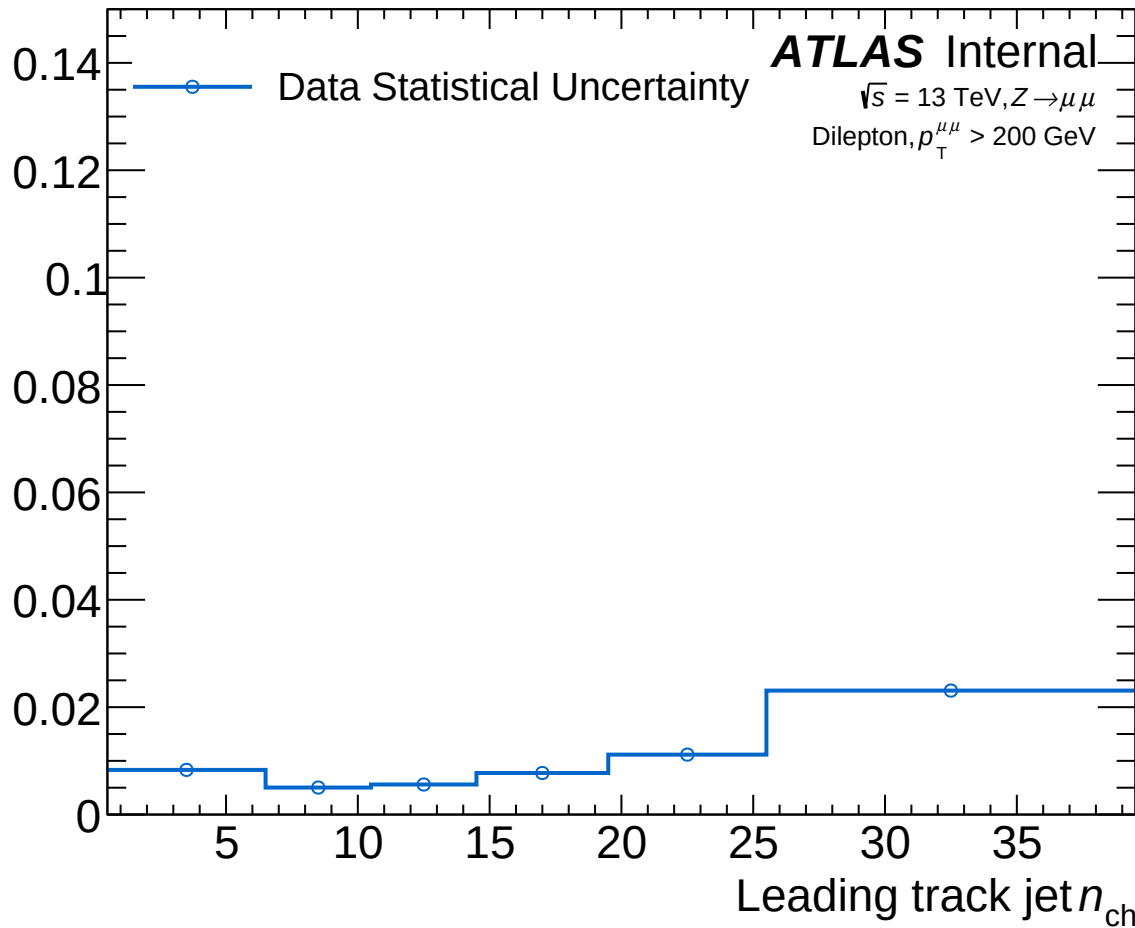
Fractional systematic impact

ATLAS Internal

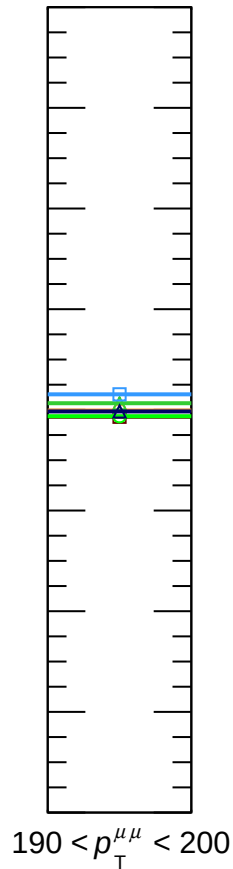
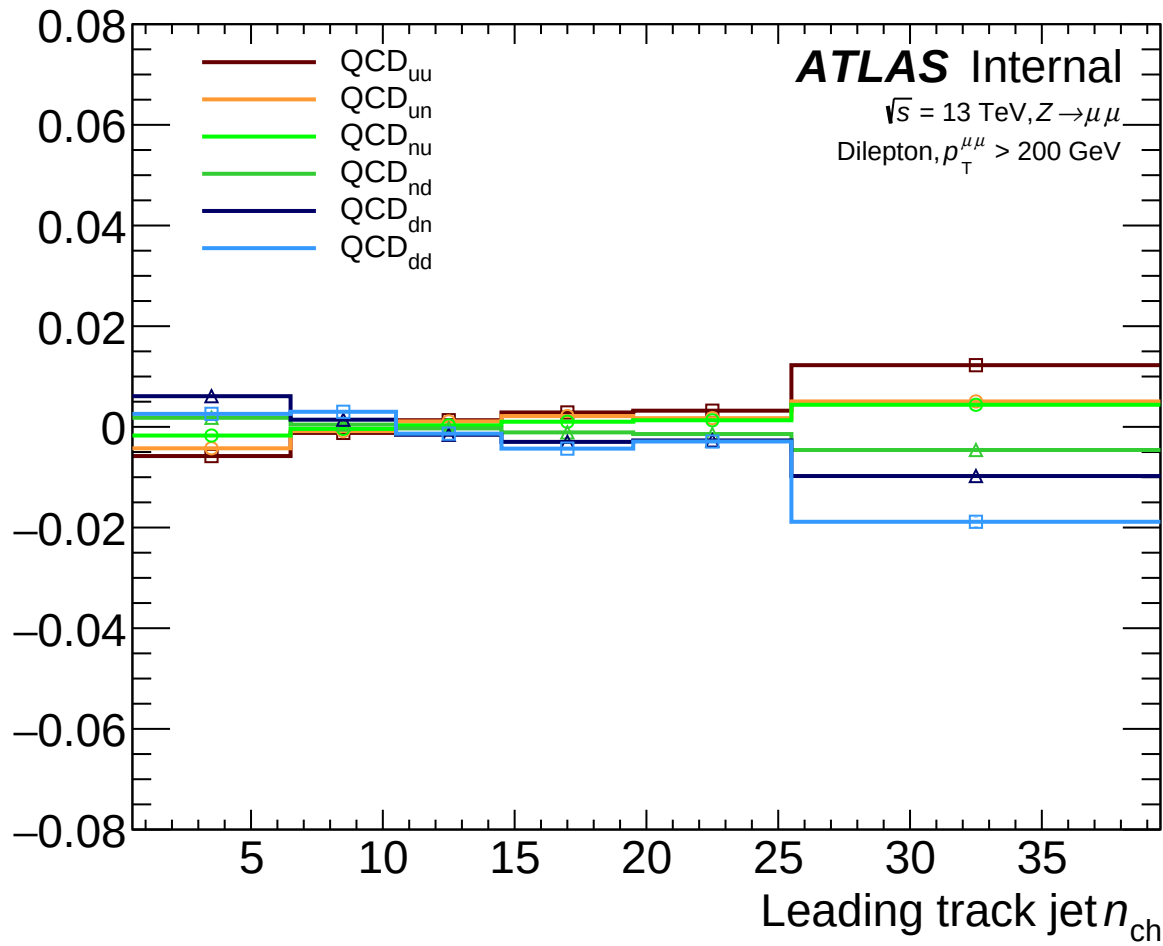
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

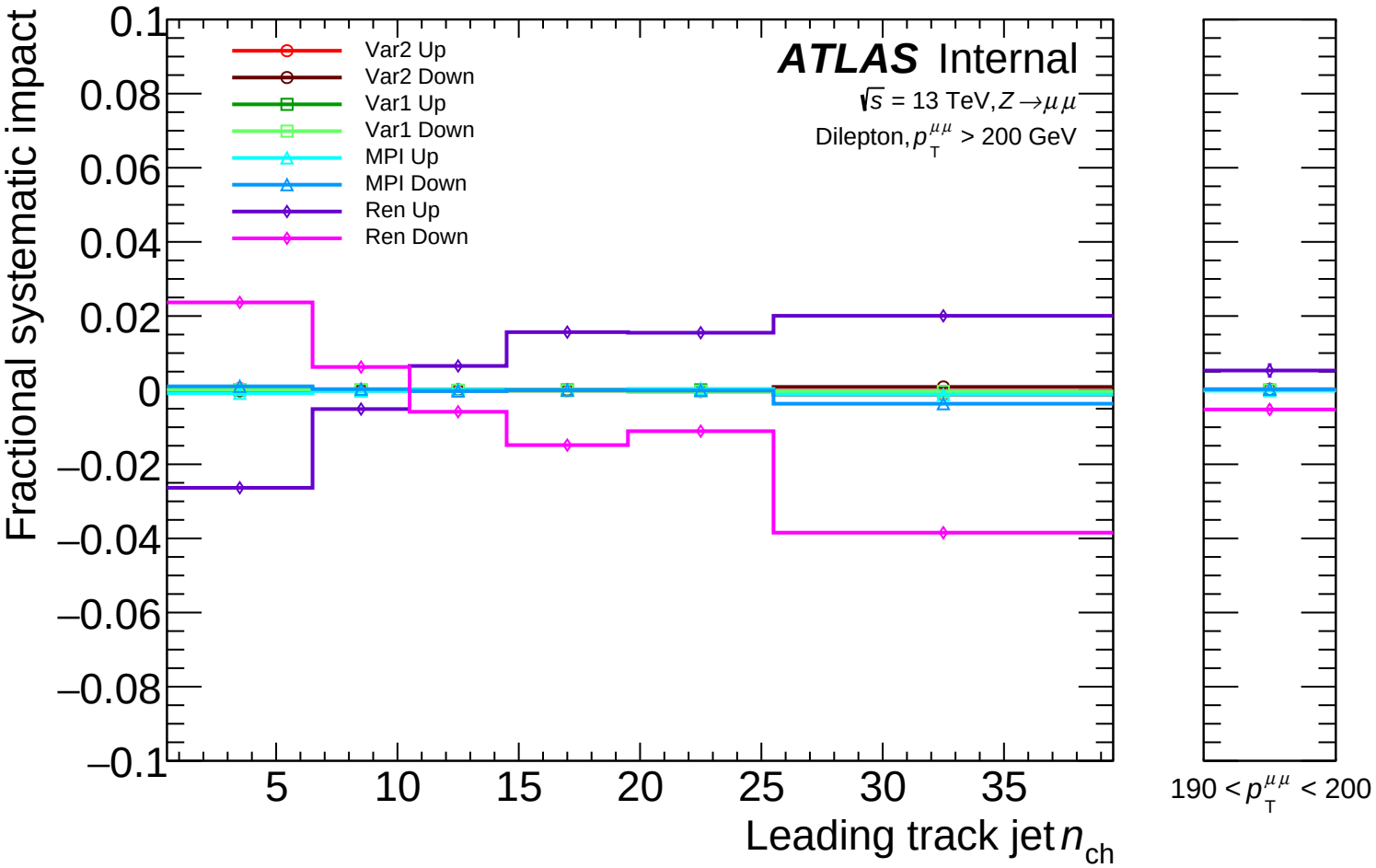
Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

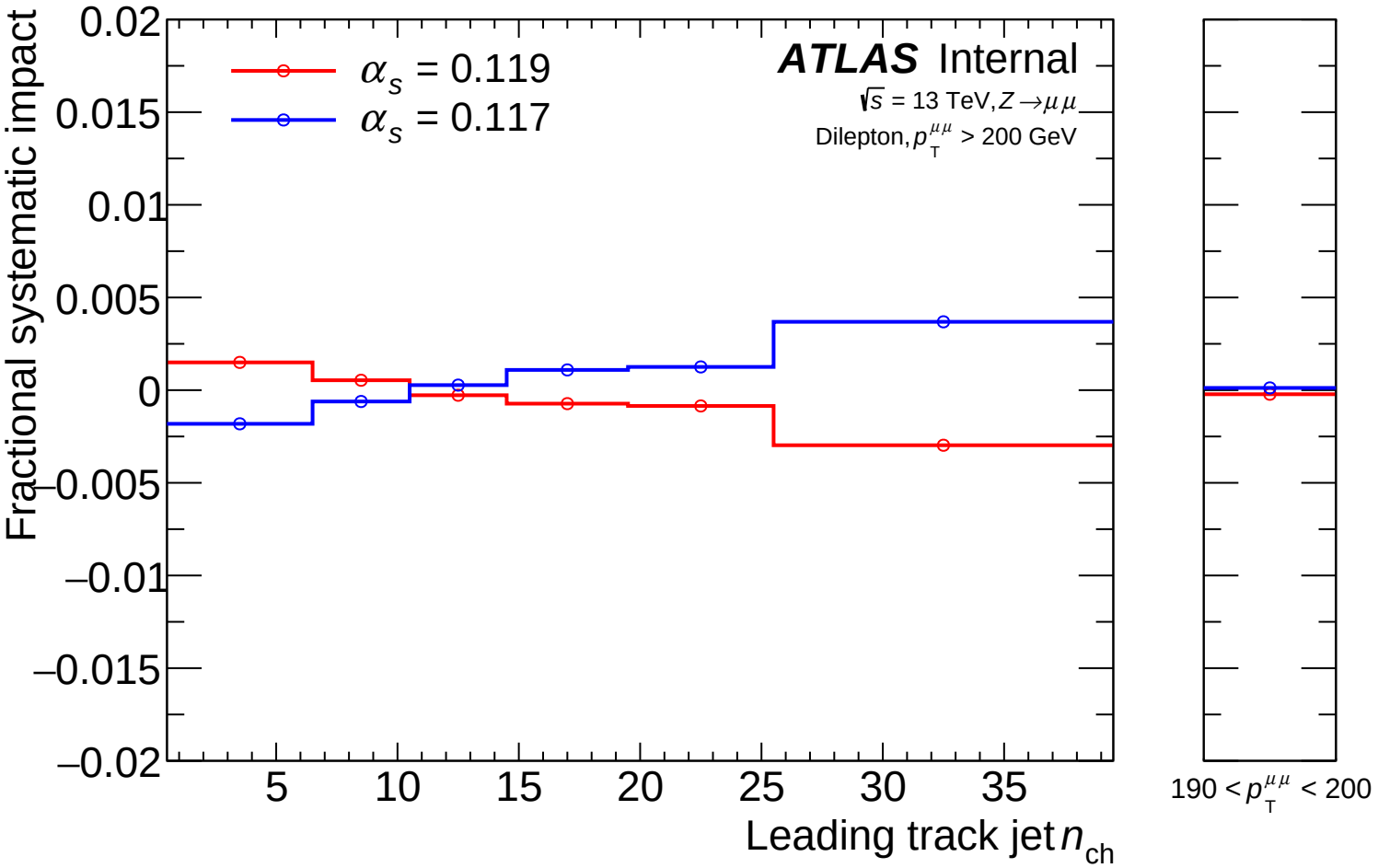
Data Statistical Uncertainty



Fractional systematic impact







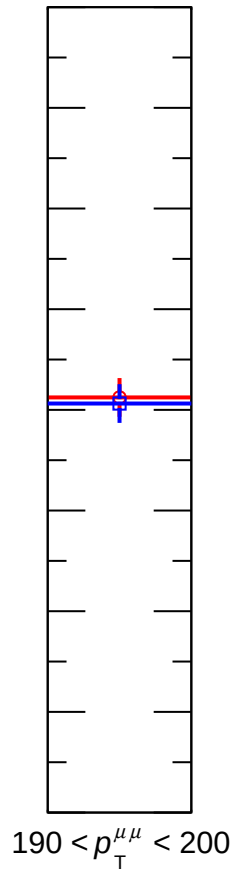
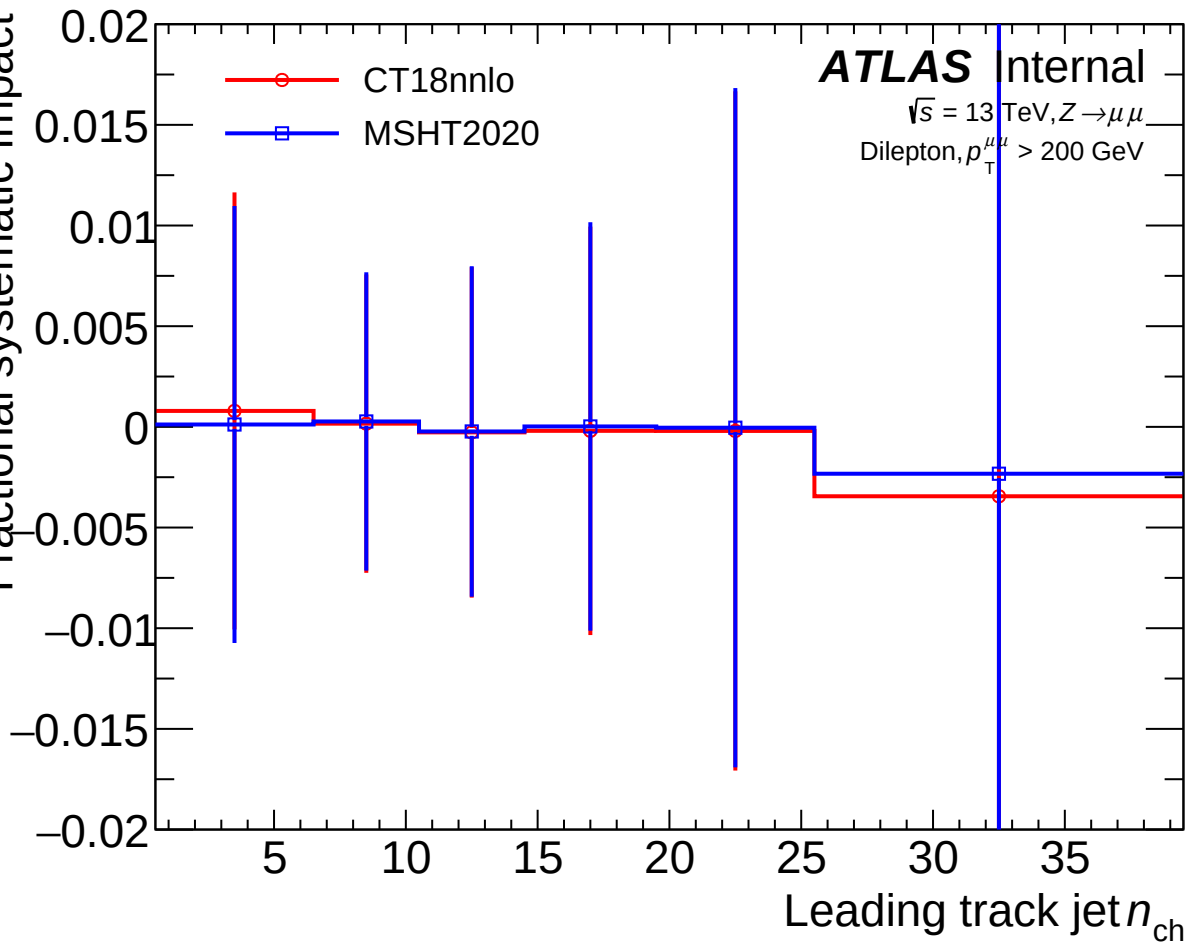
Fractional systematic impact

CT18nnlo
MSHT2020

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200$ GeV



Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

5

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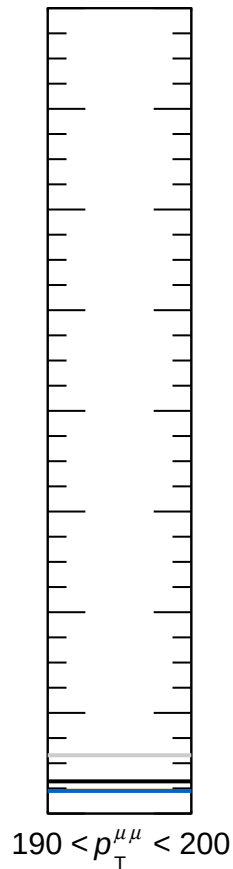
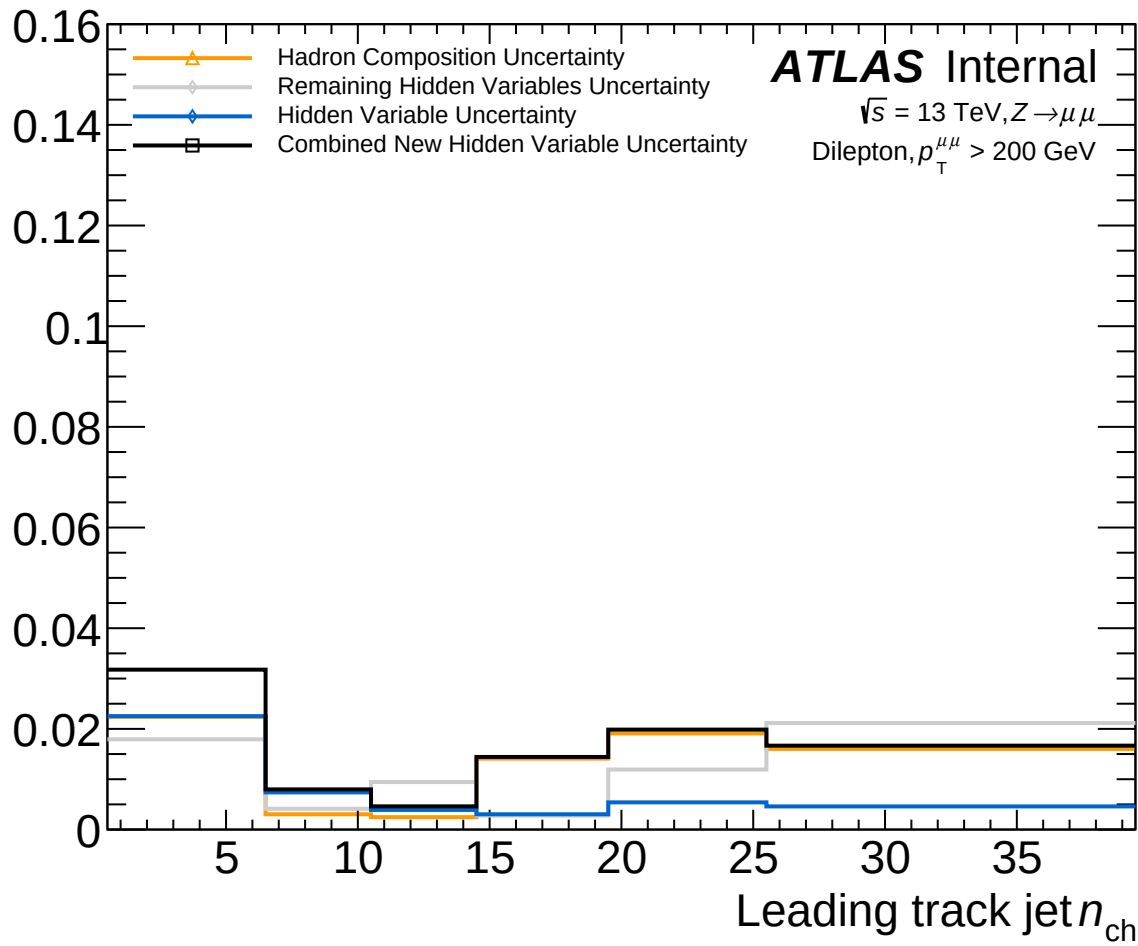
Leading track jet n_{ch}

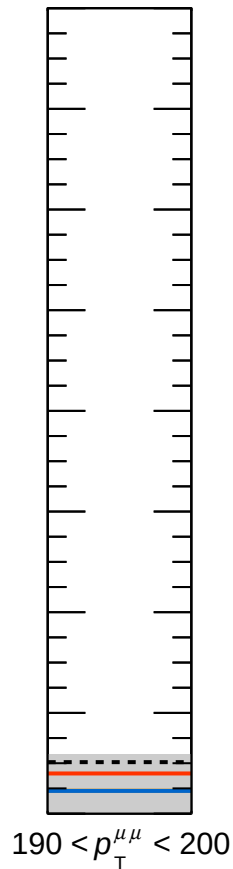
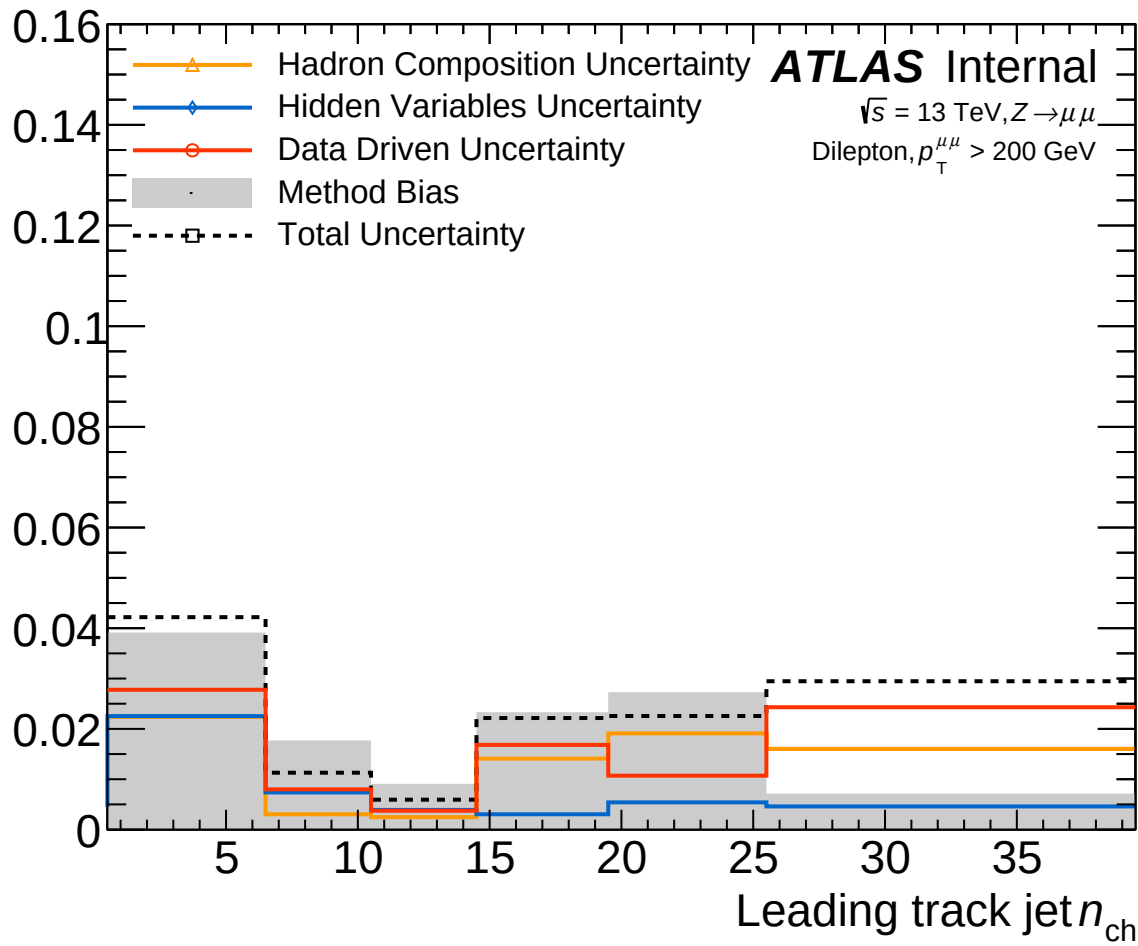
$190 < p_T^{\mu\mu} < 200$

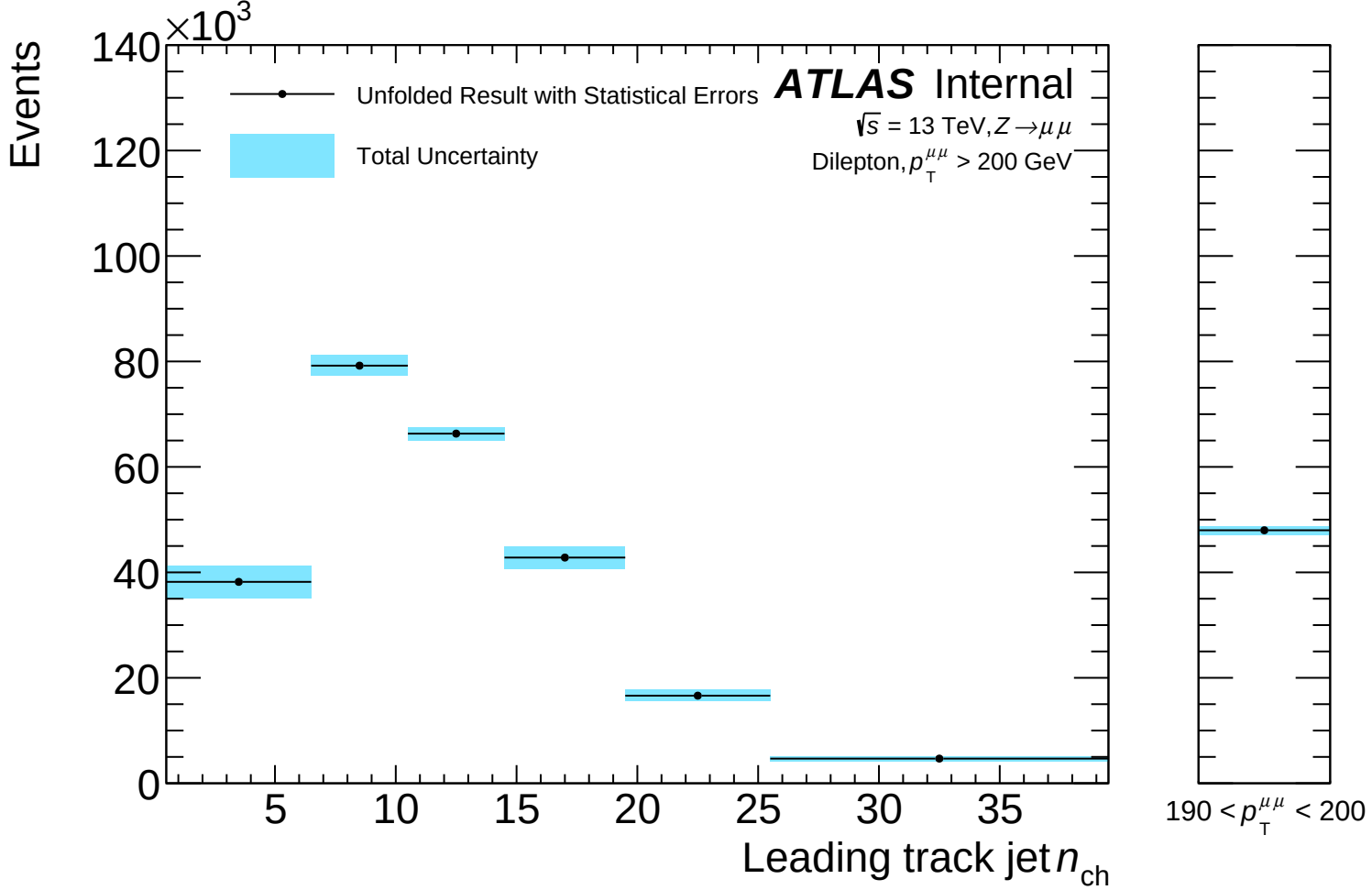
ATLAS Internal

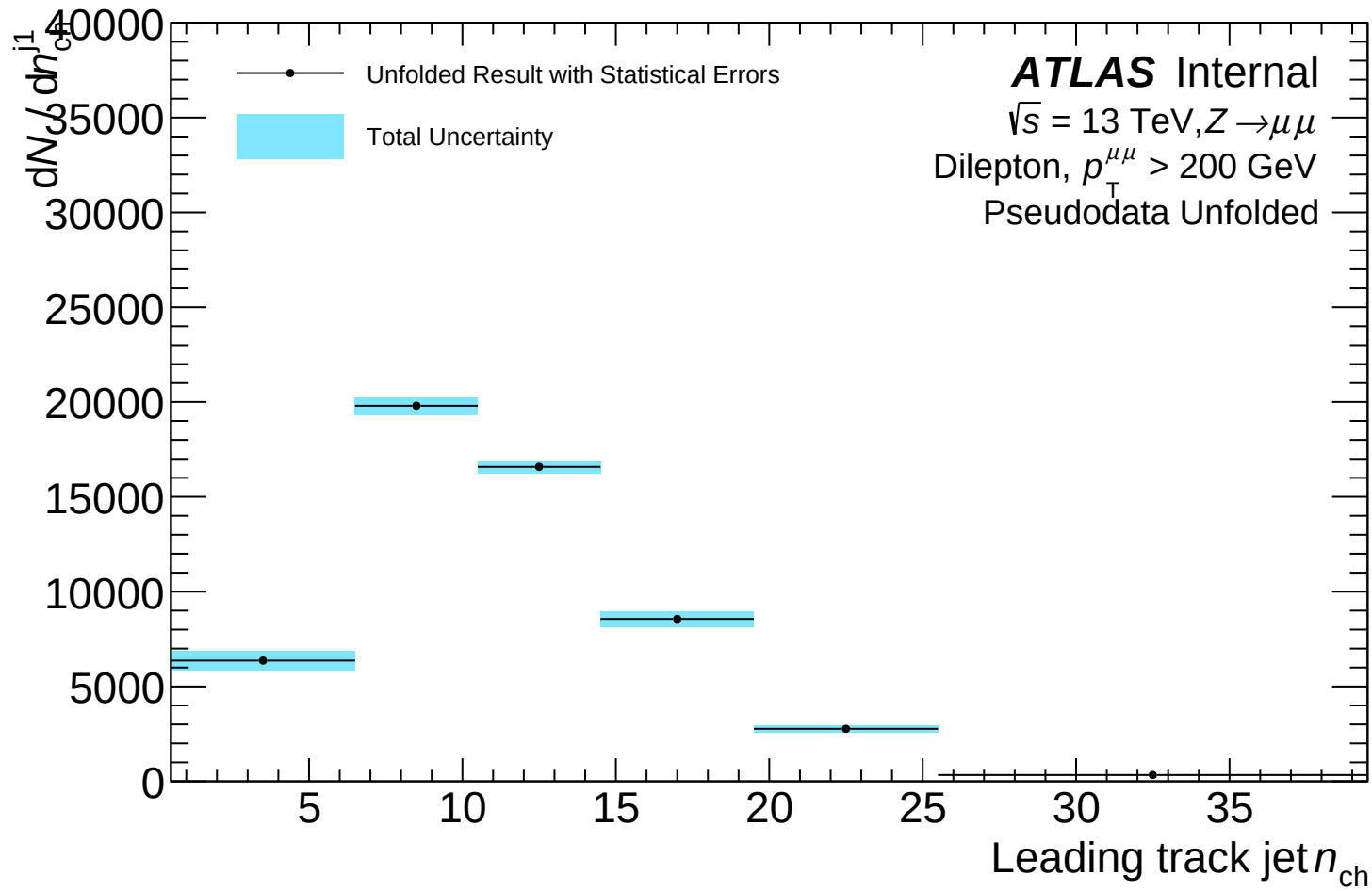
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

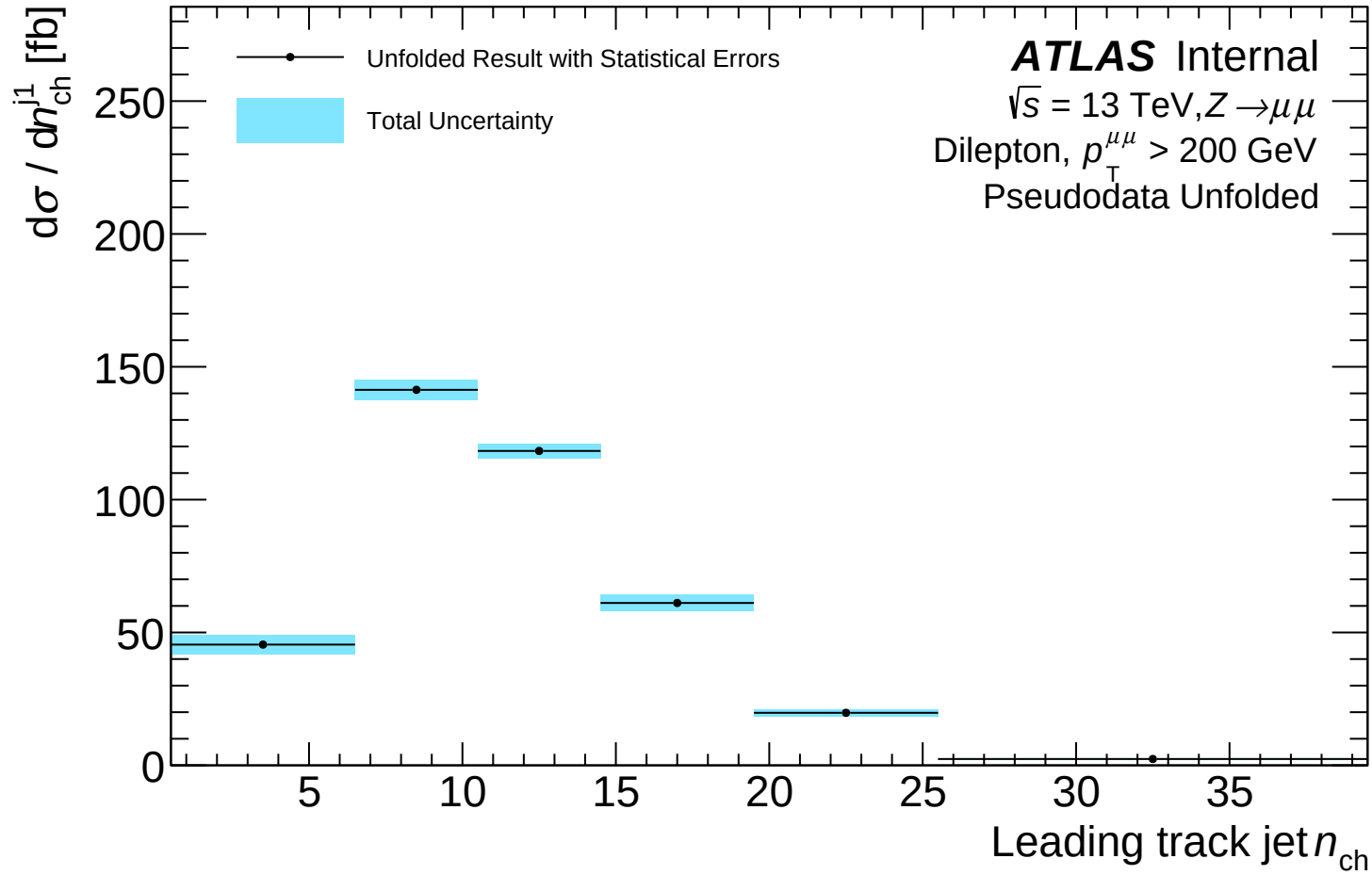
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$











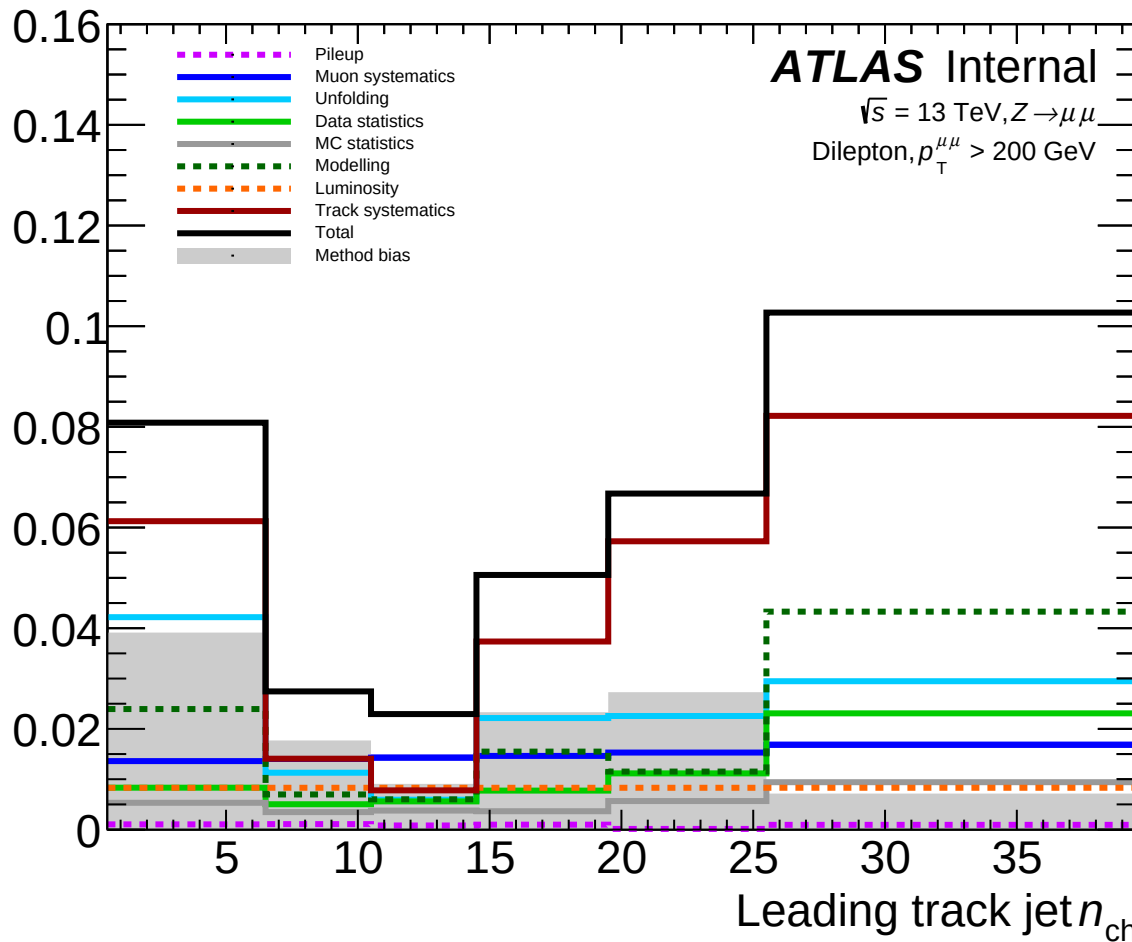
Fractional uncertainty

ATLAS Internal

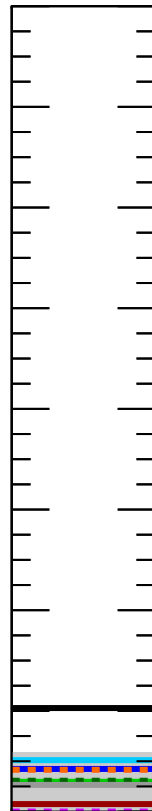
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

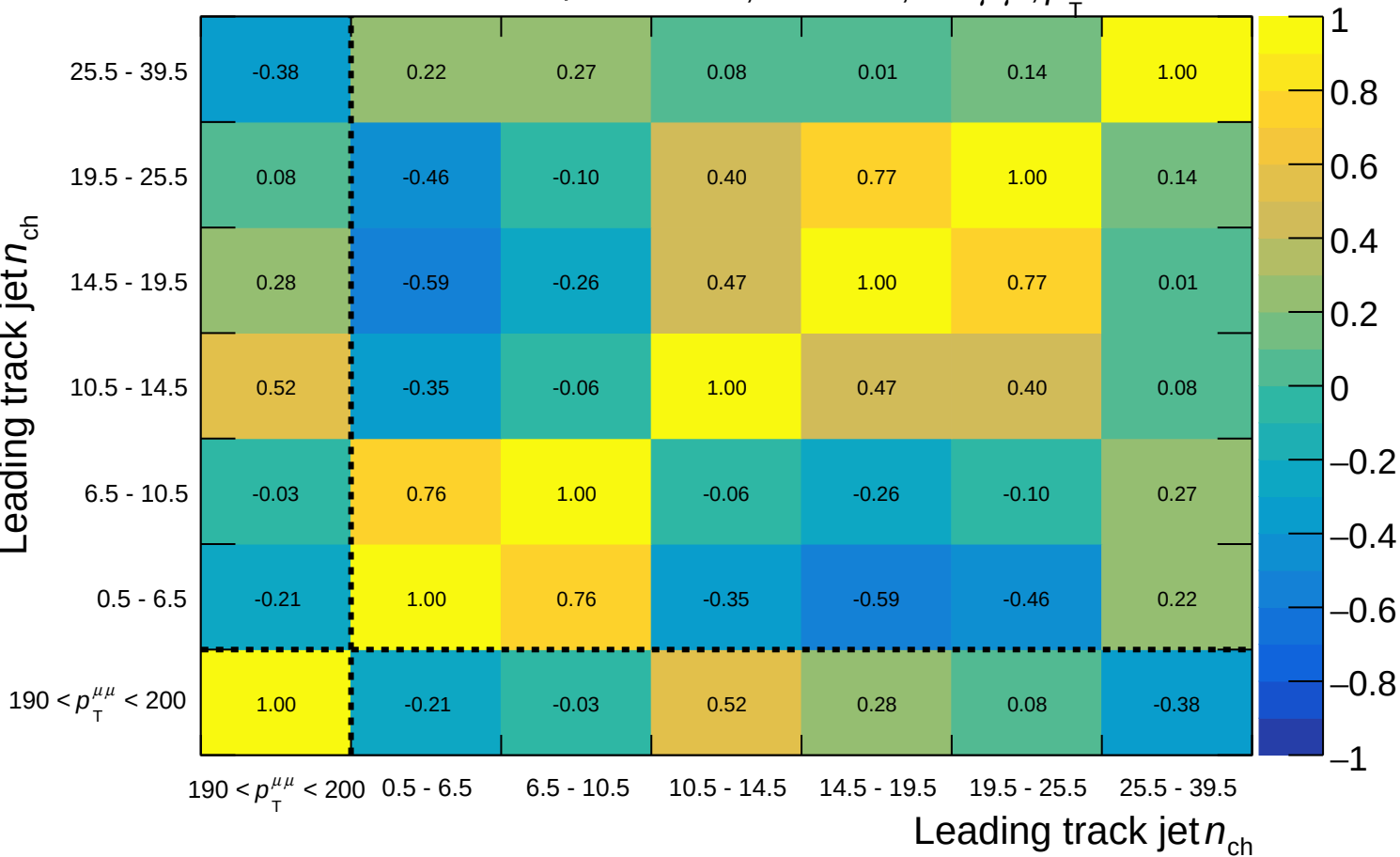
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

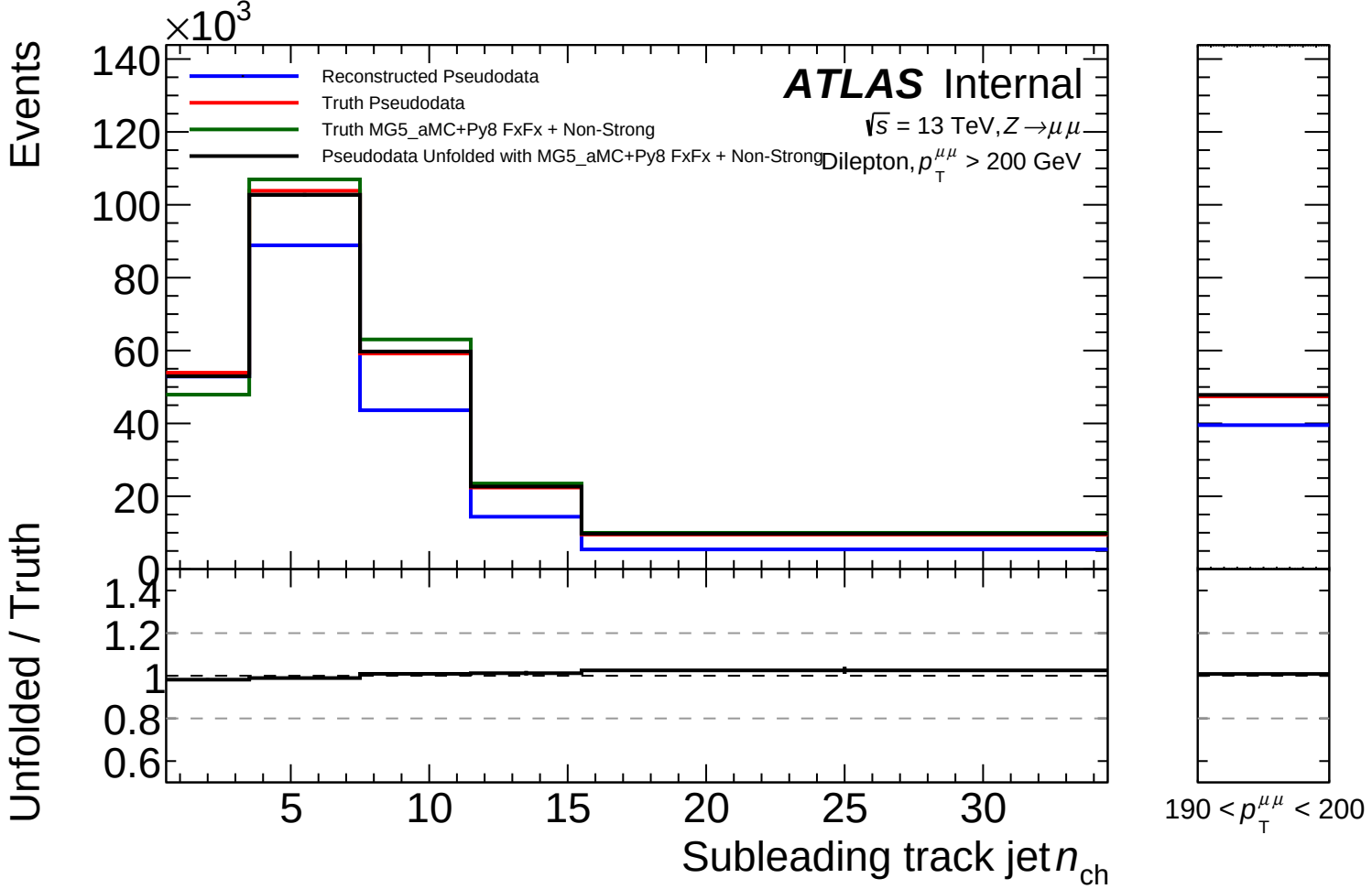
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

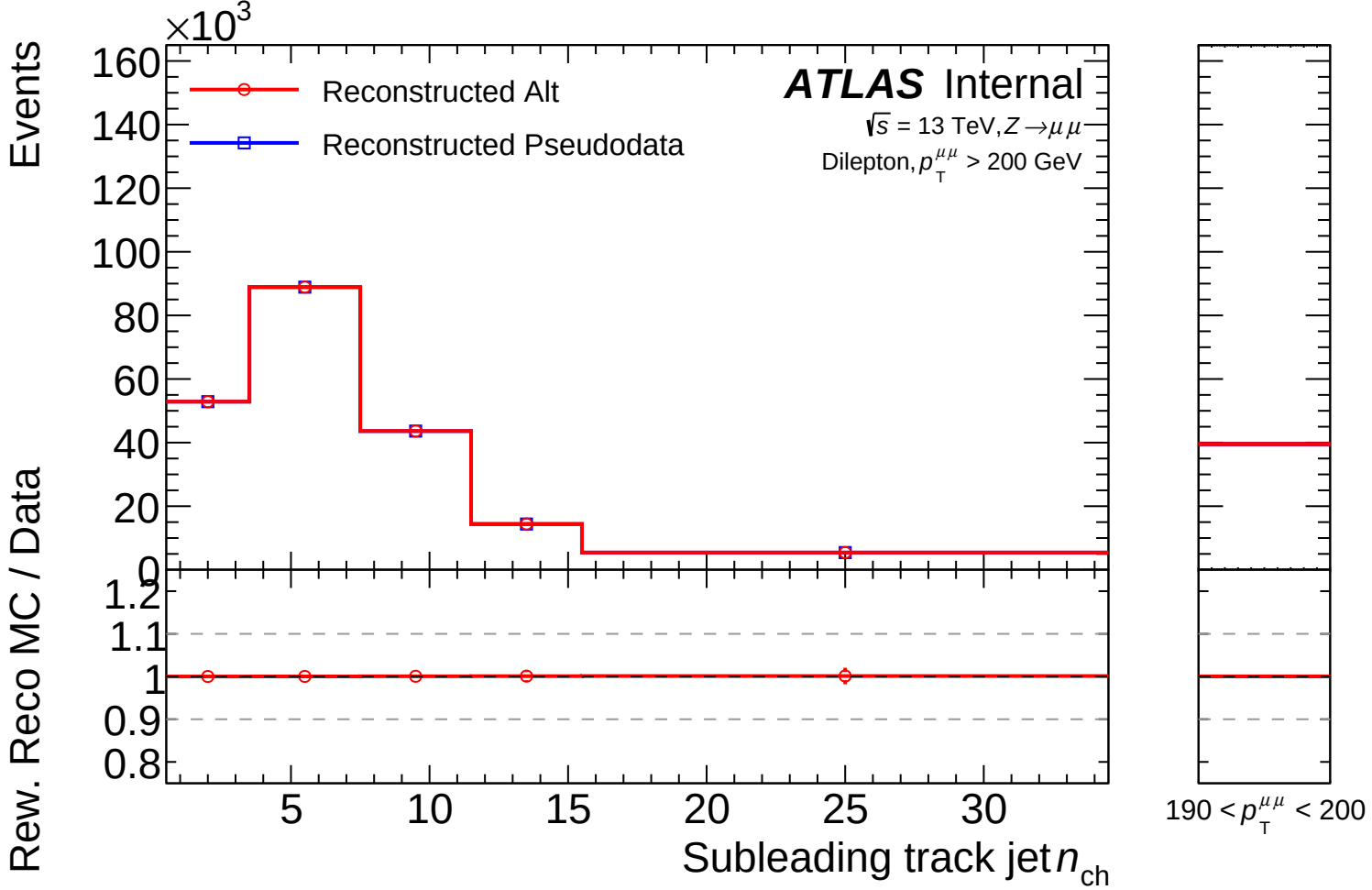


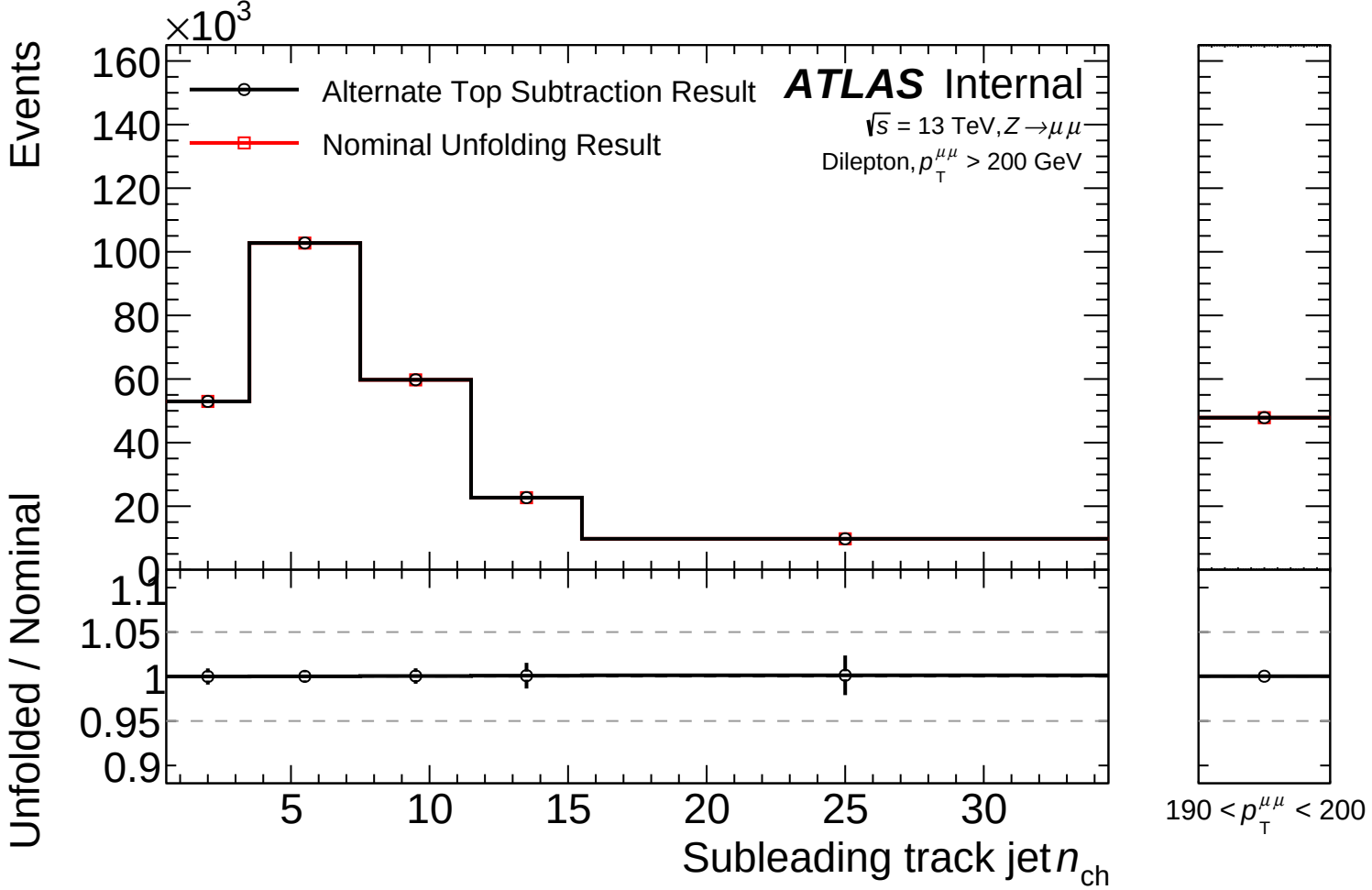
$190 < p_T^{\mu\mu} < 200$



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet n_{ch} 







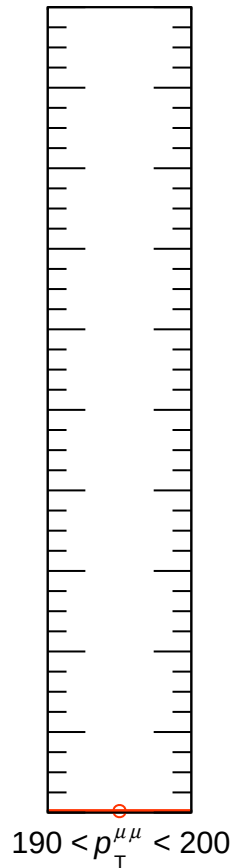
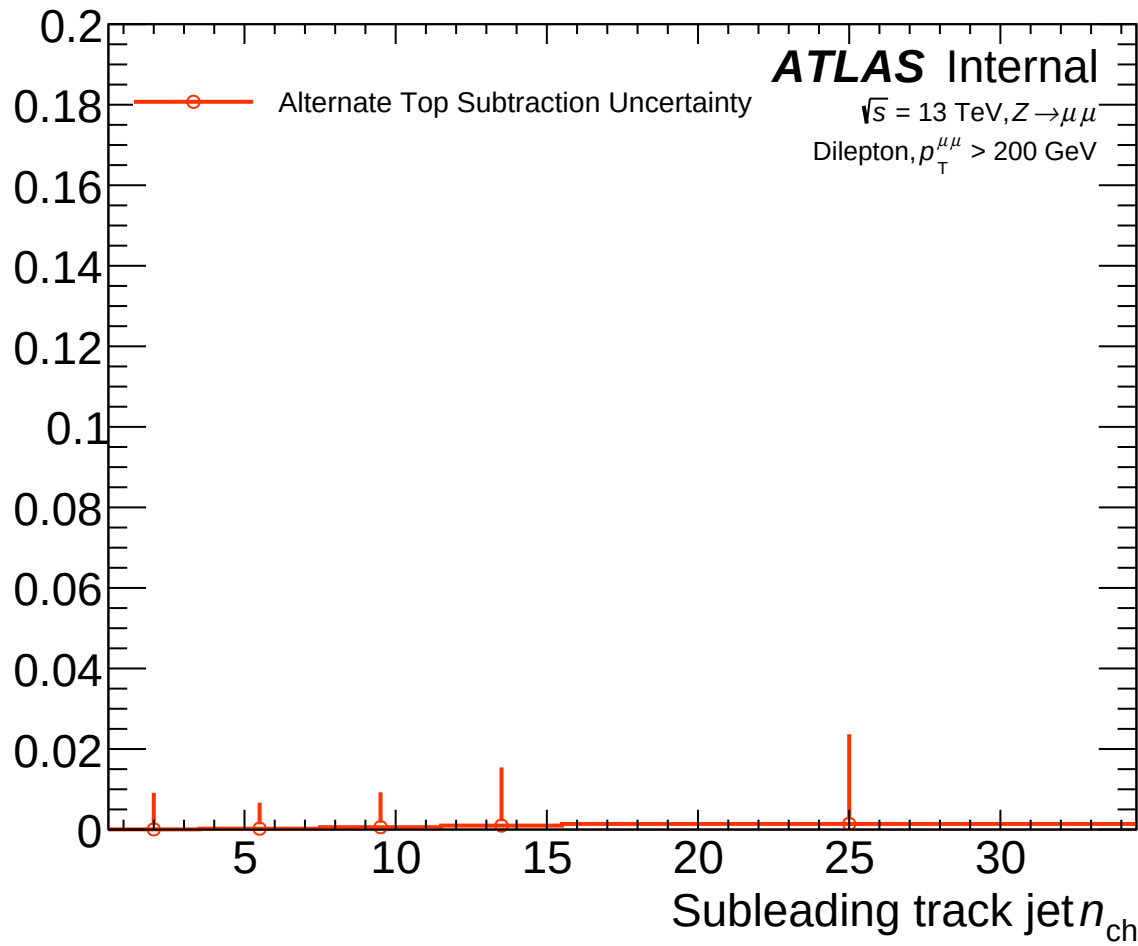
Fractional systematic impact

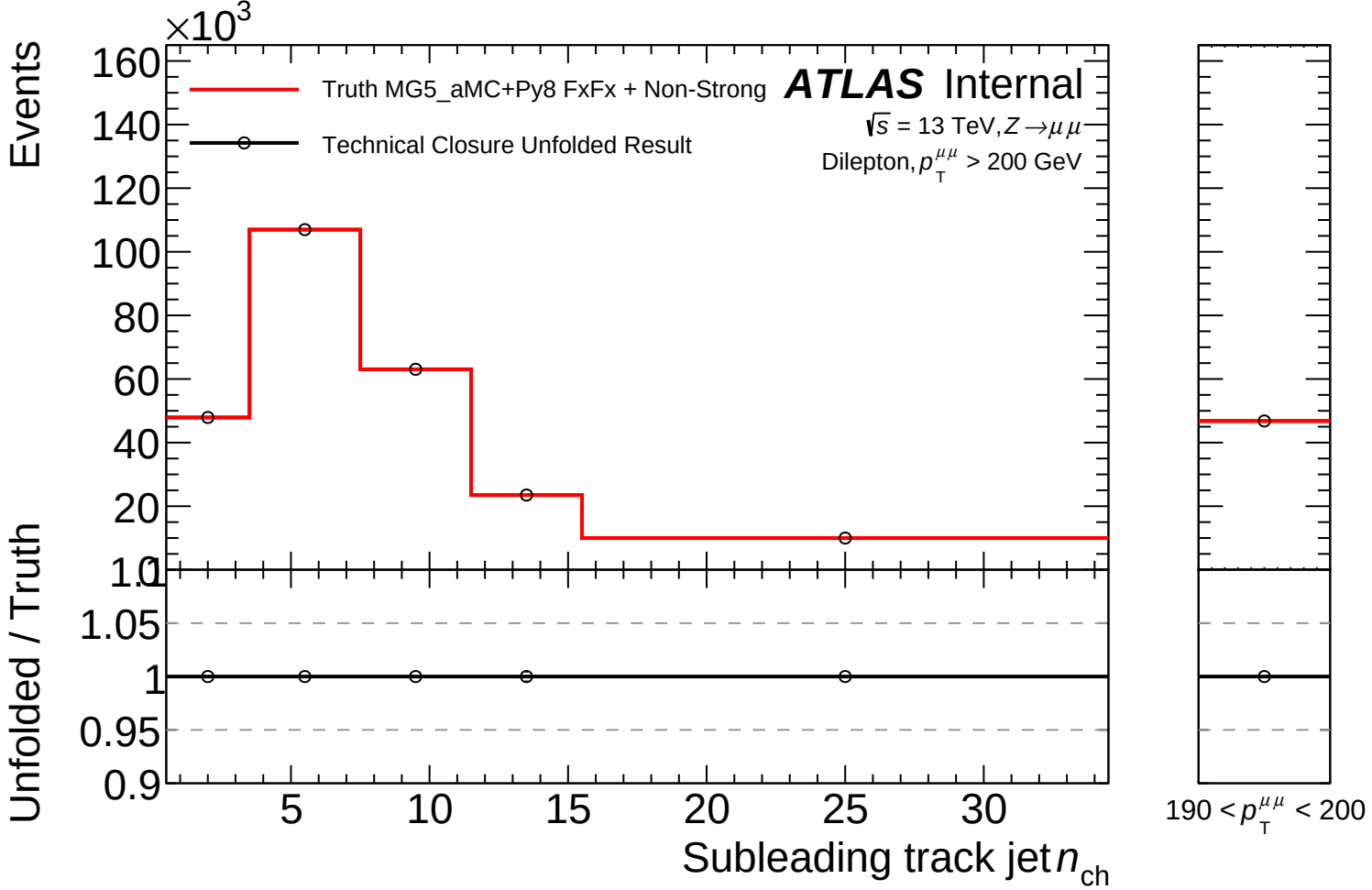
ATLAS Internal

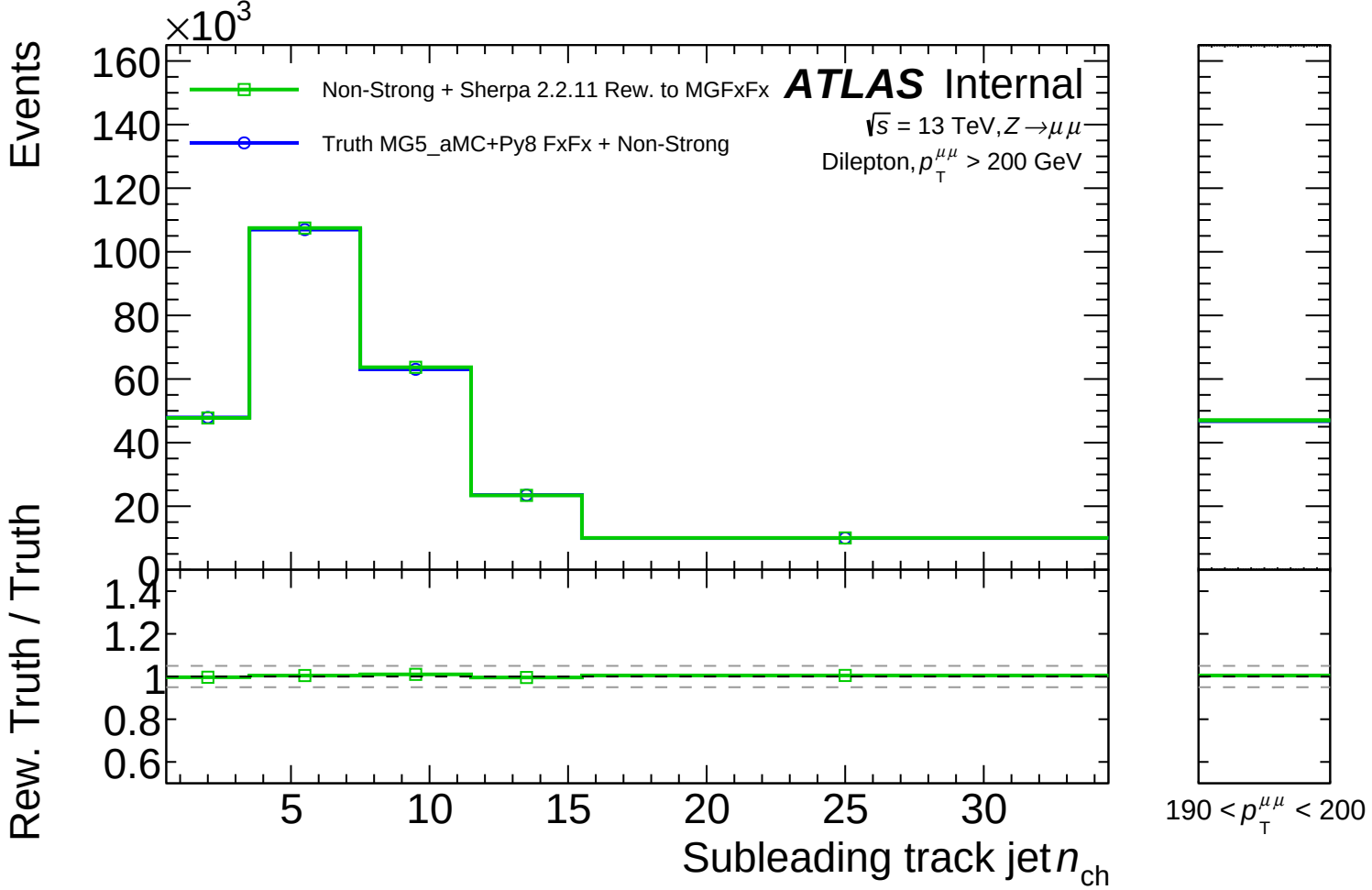
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

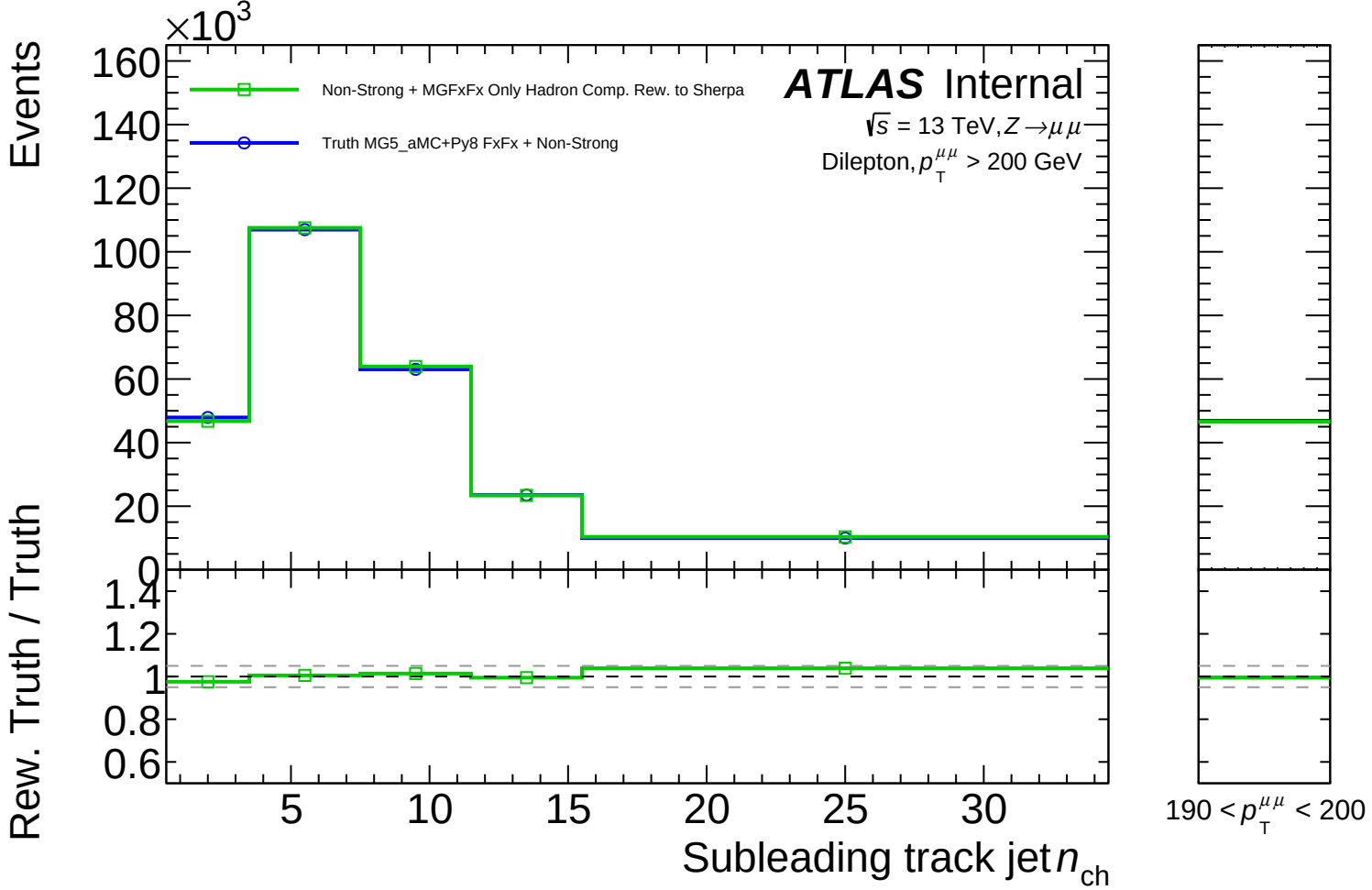
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

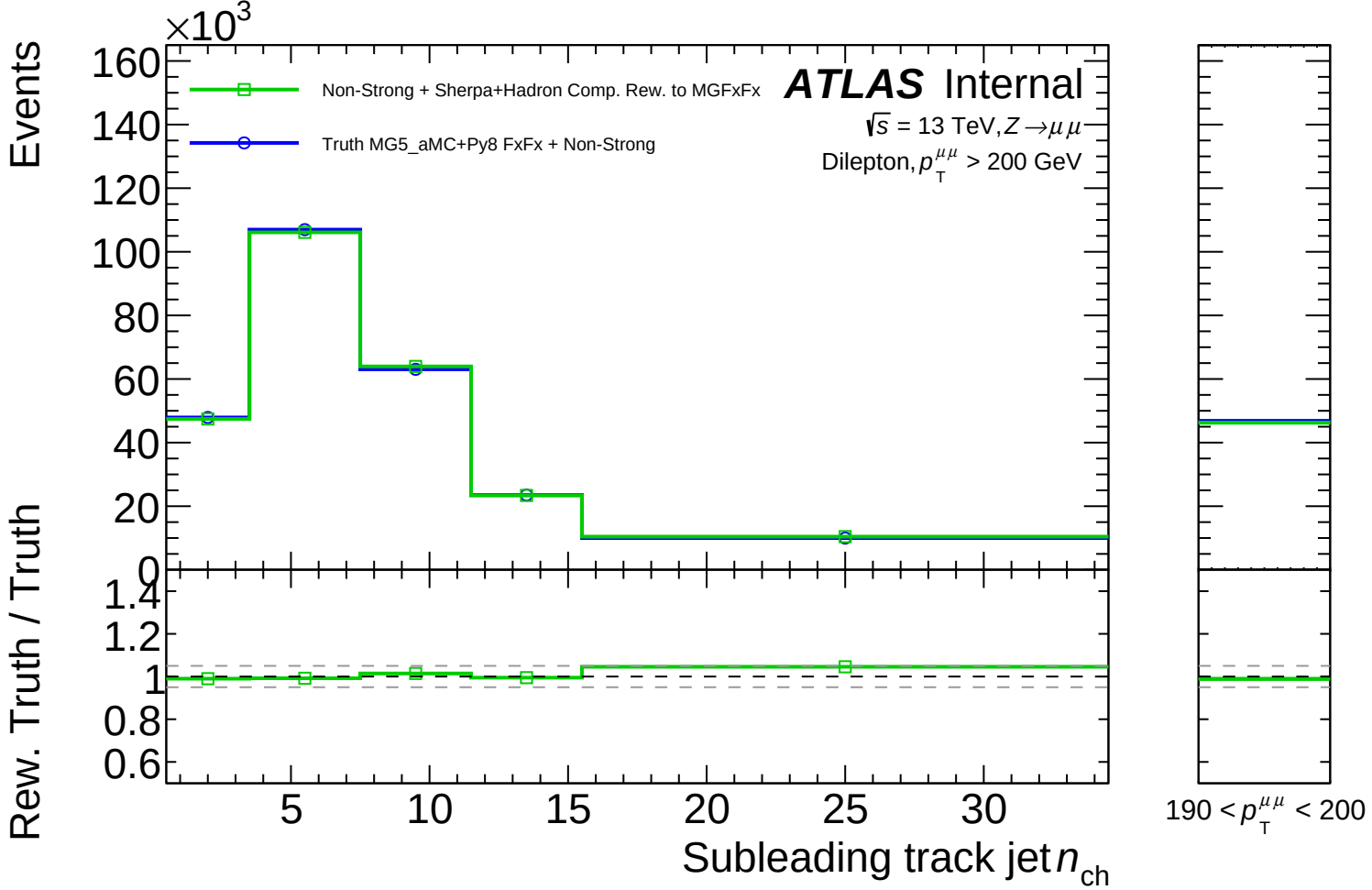
Alternate Top Subtraction Uncertainty

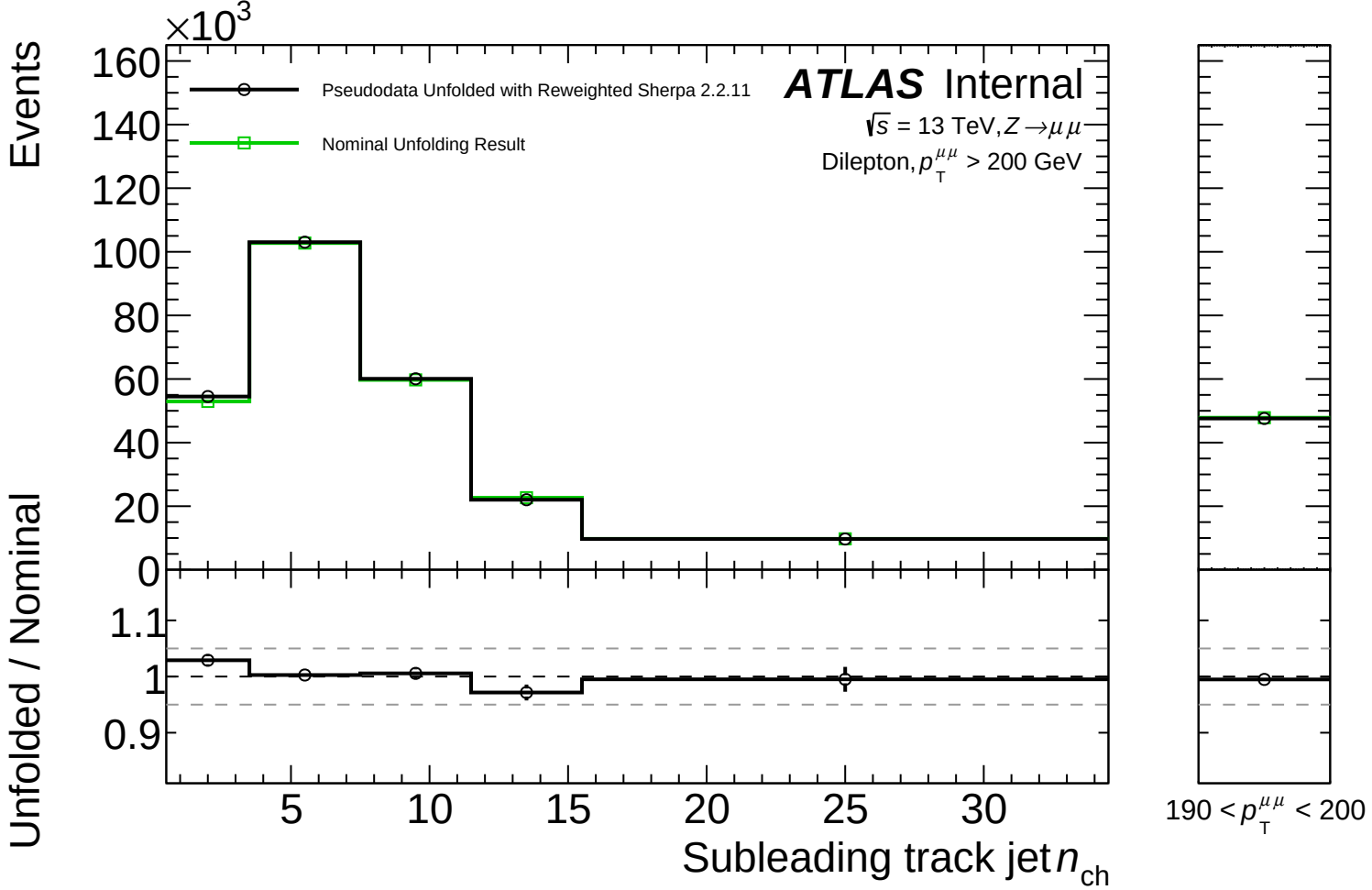






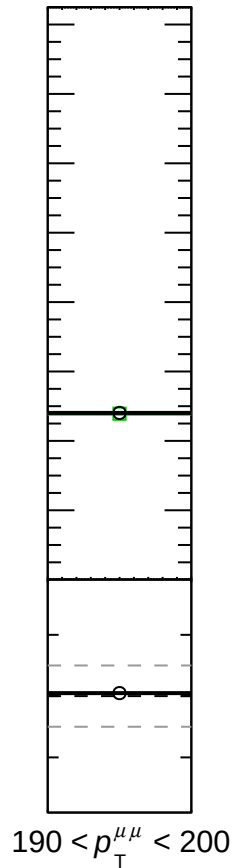
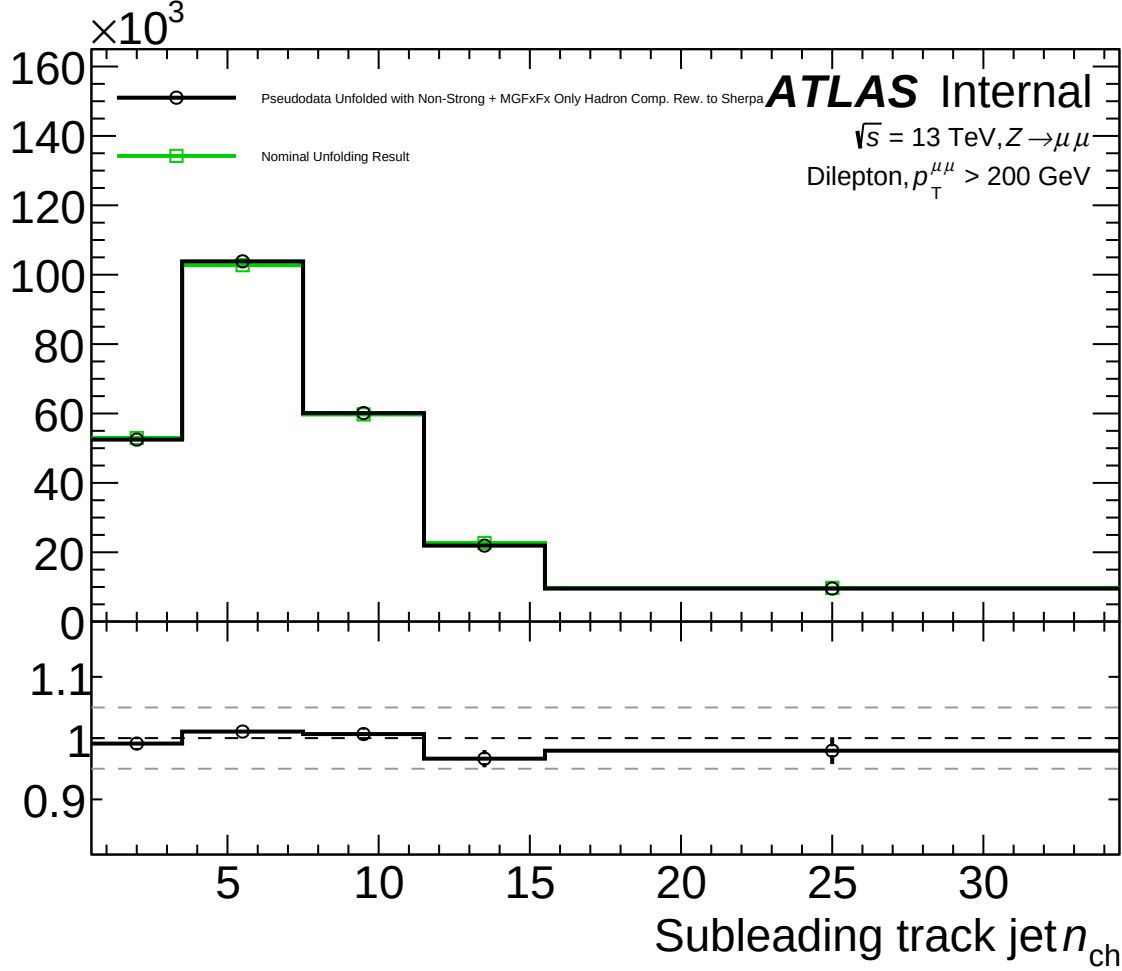


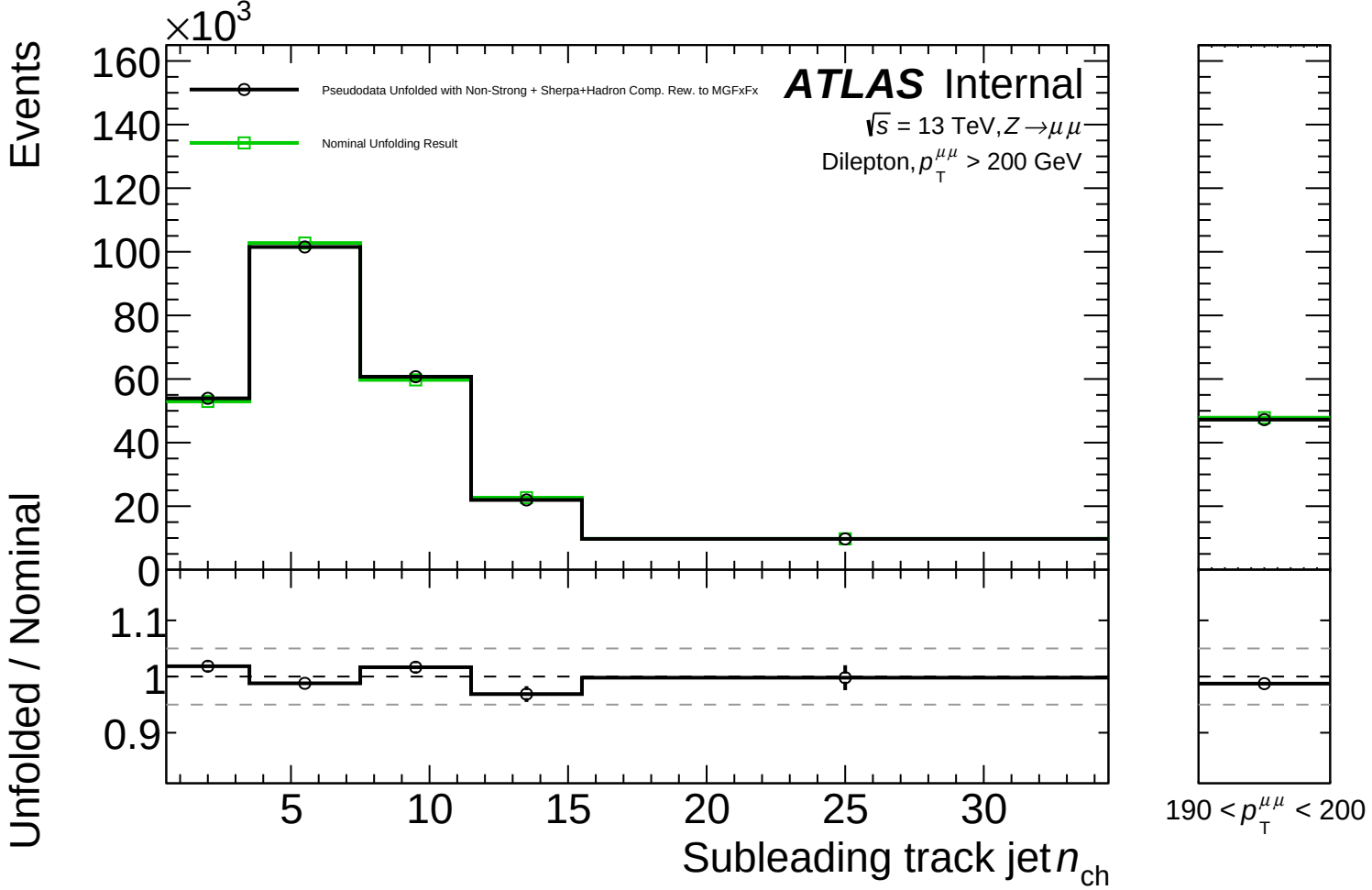


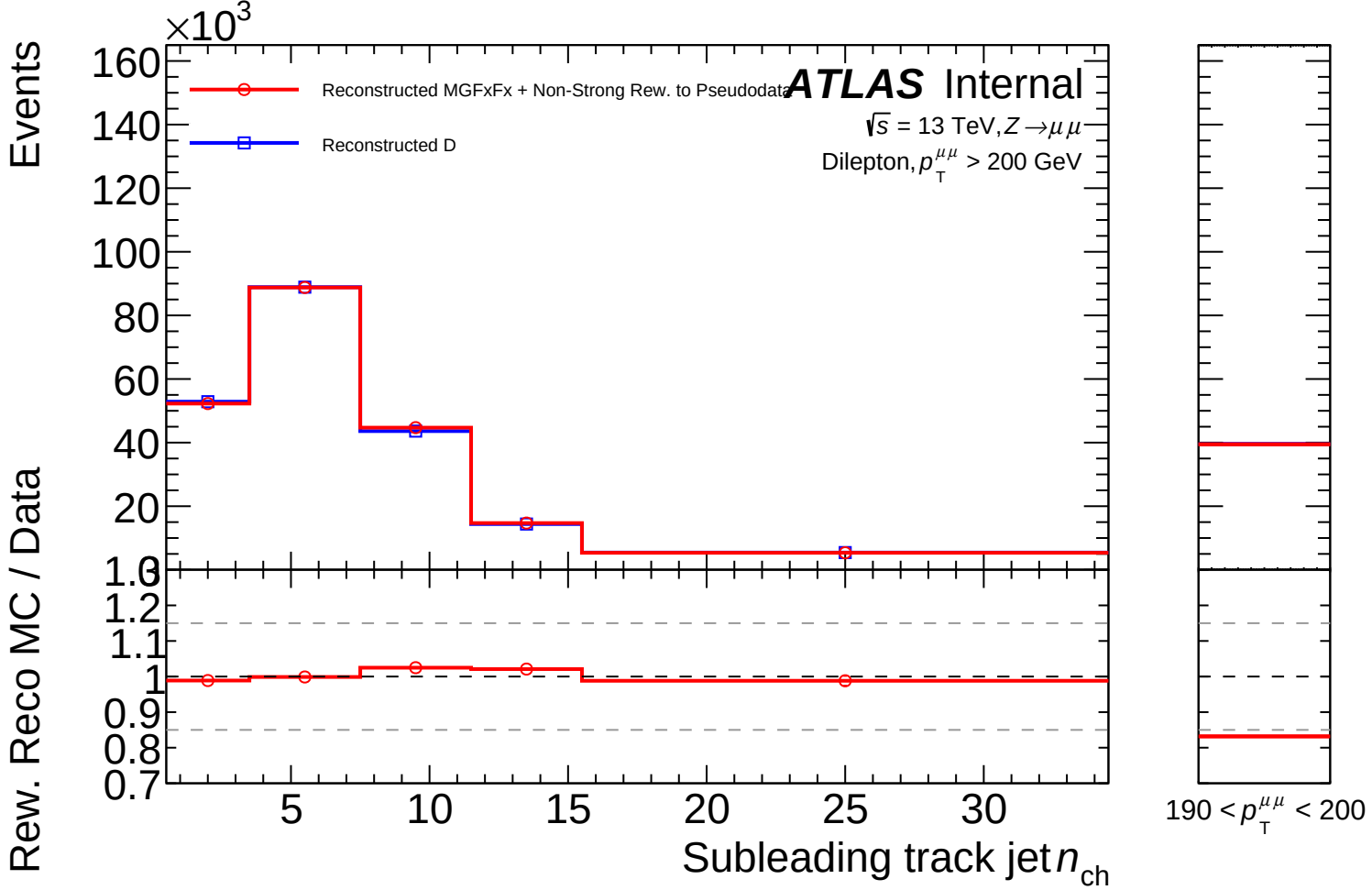


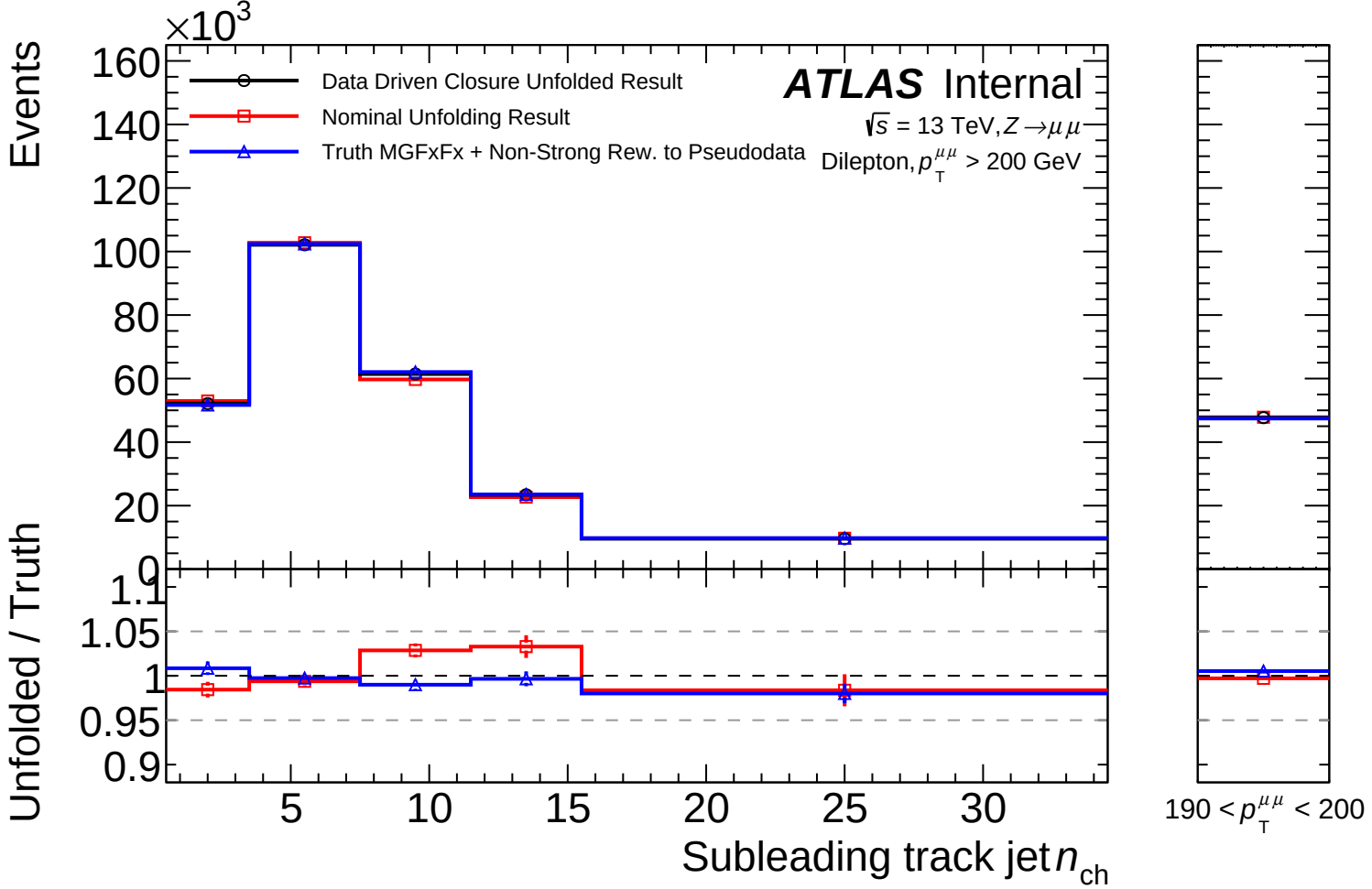
Events

Unfolded / Nominal

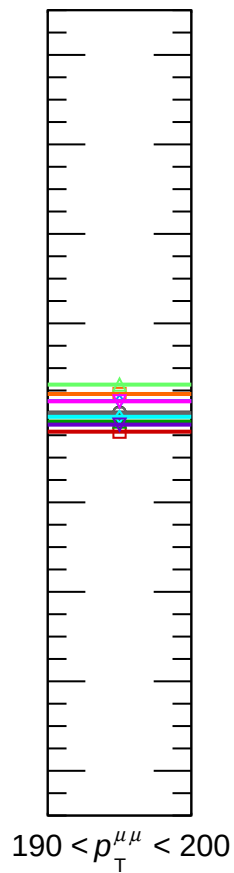
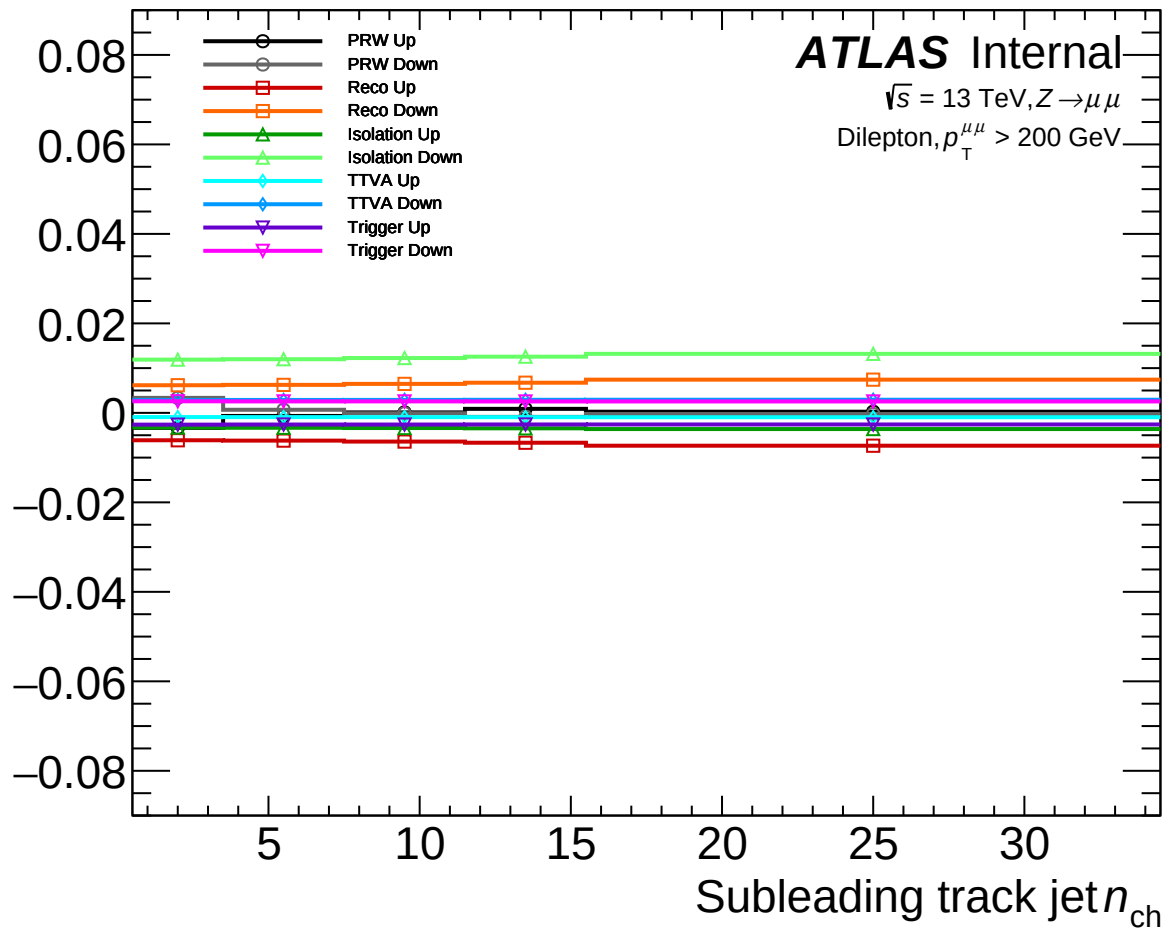




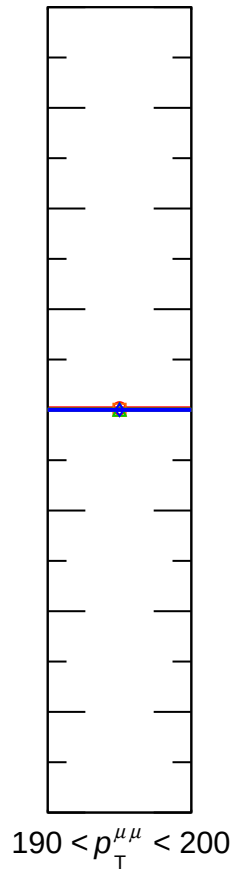
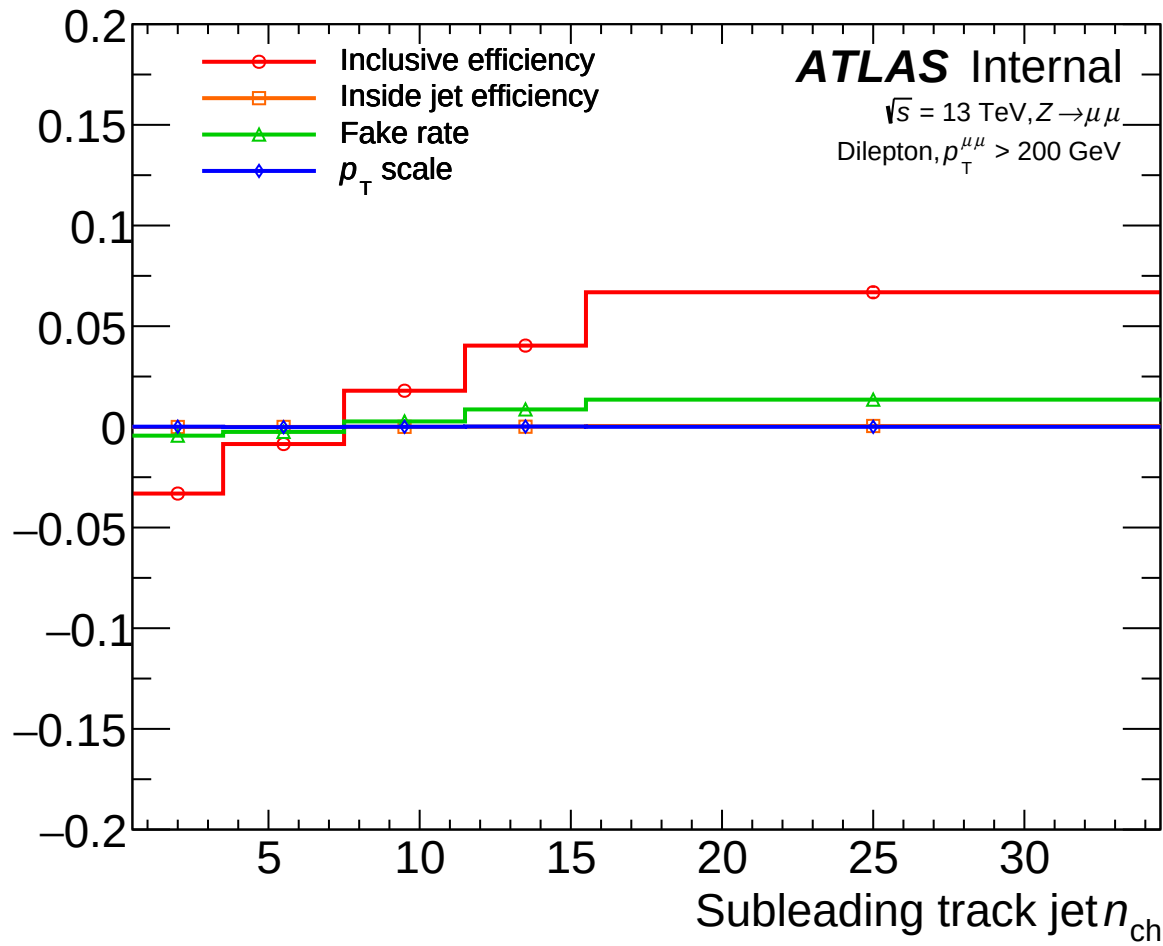




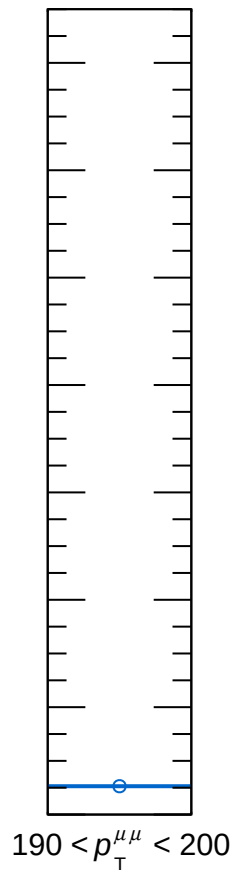
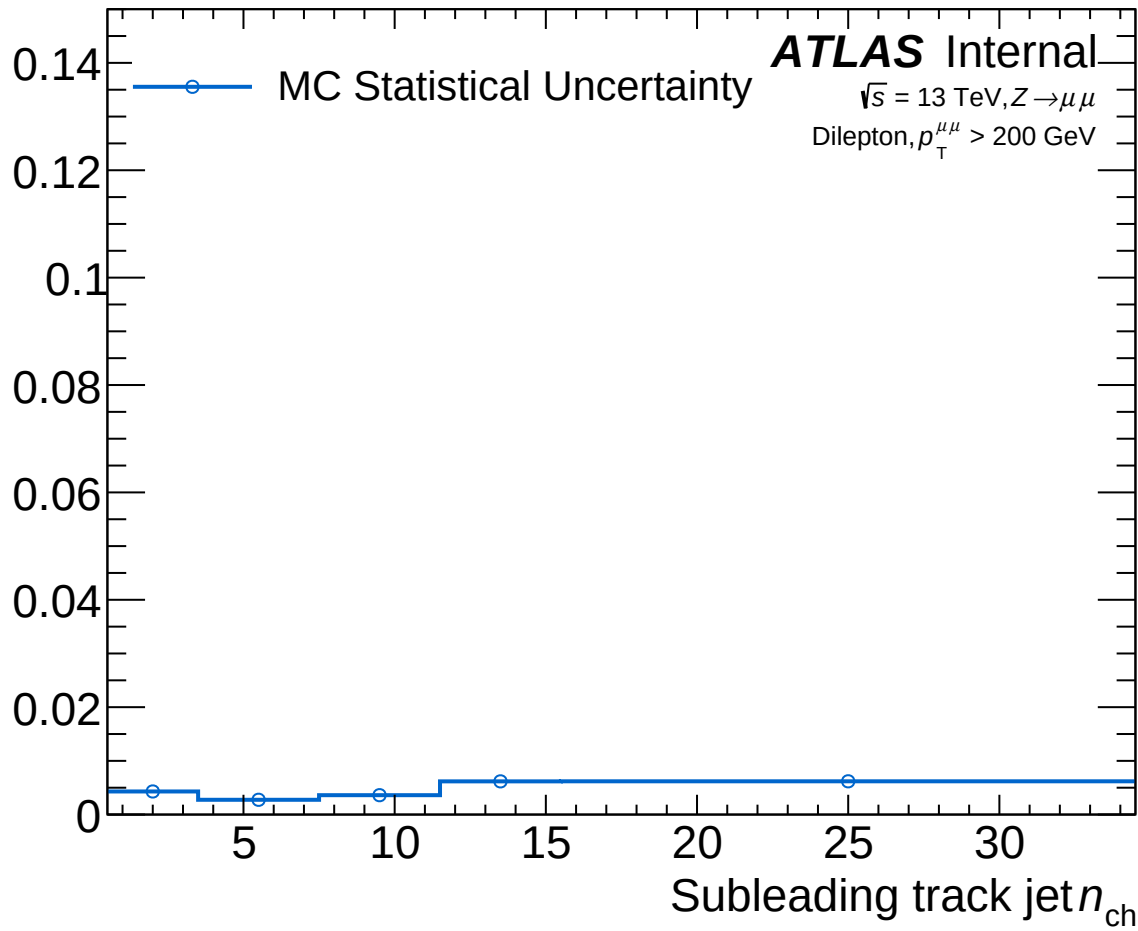
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



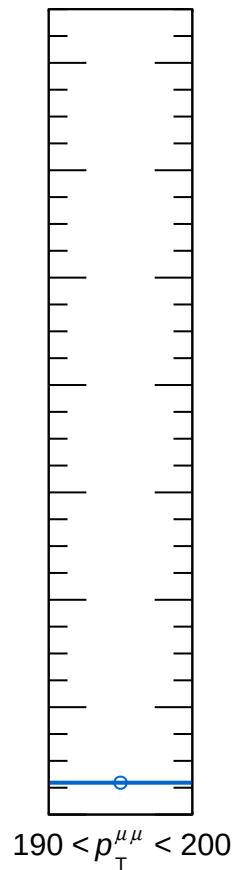
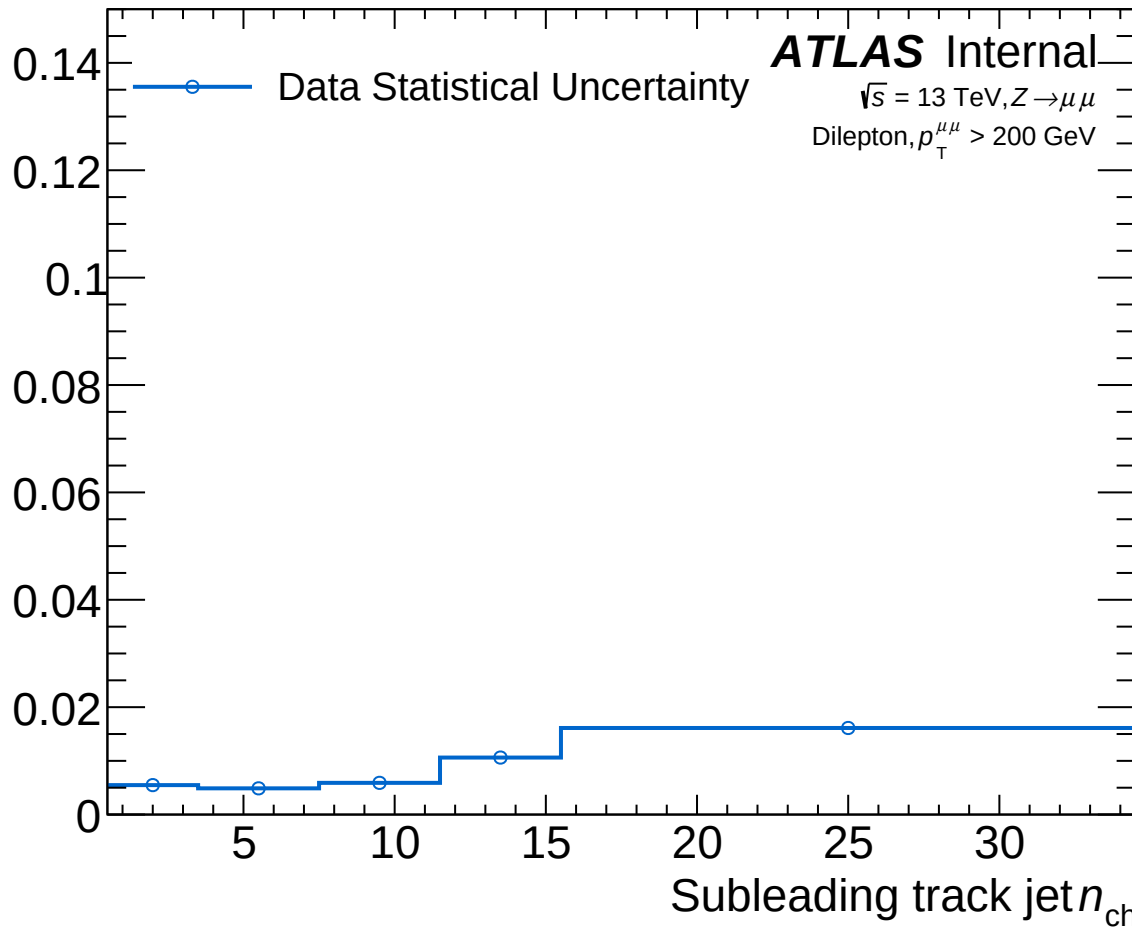
Fractional systematic impact

ATLAS Internal

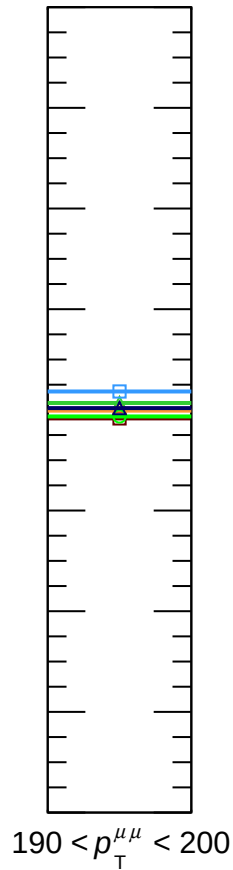
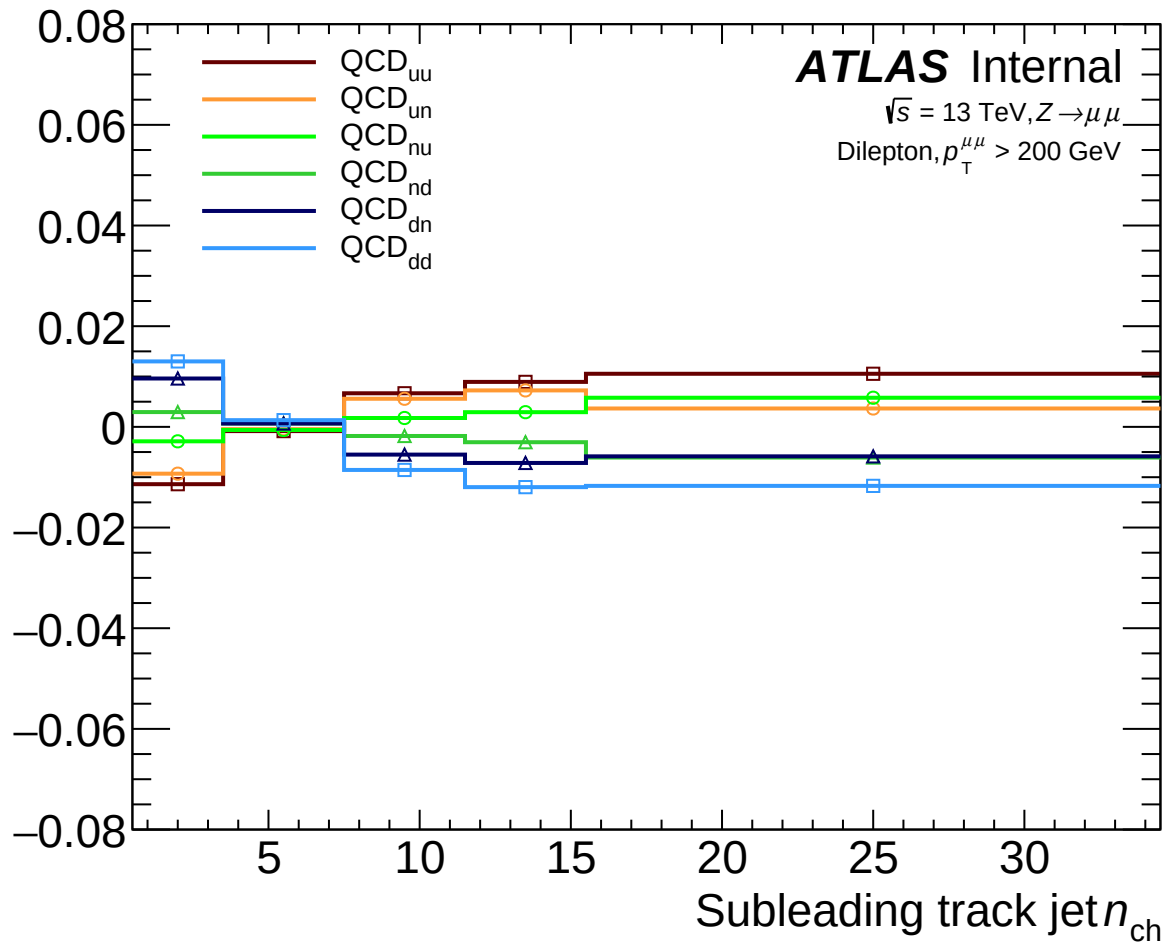
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

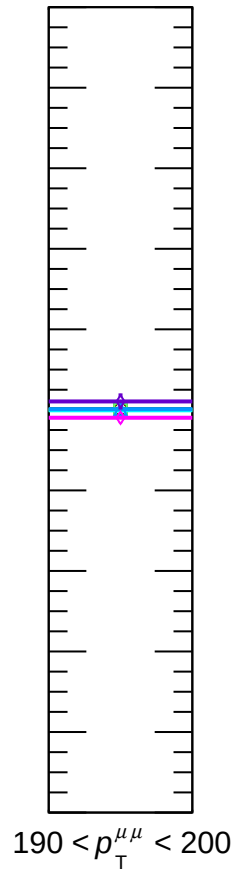
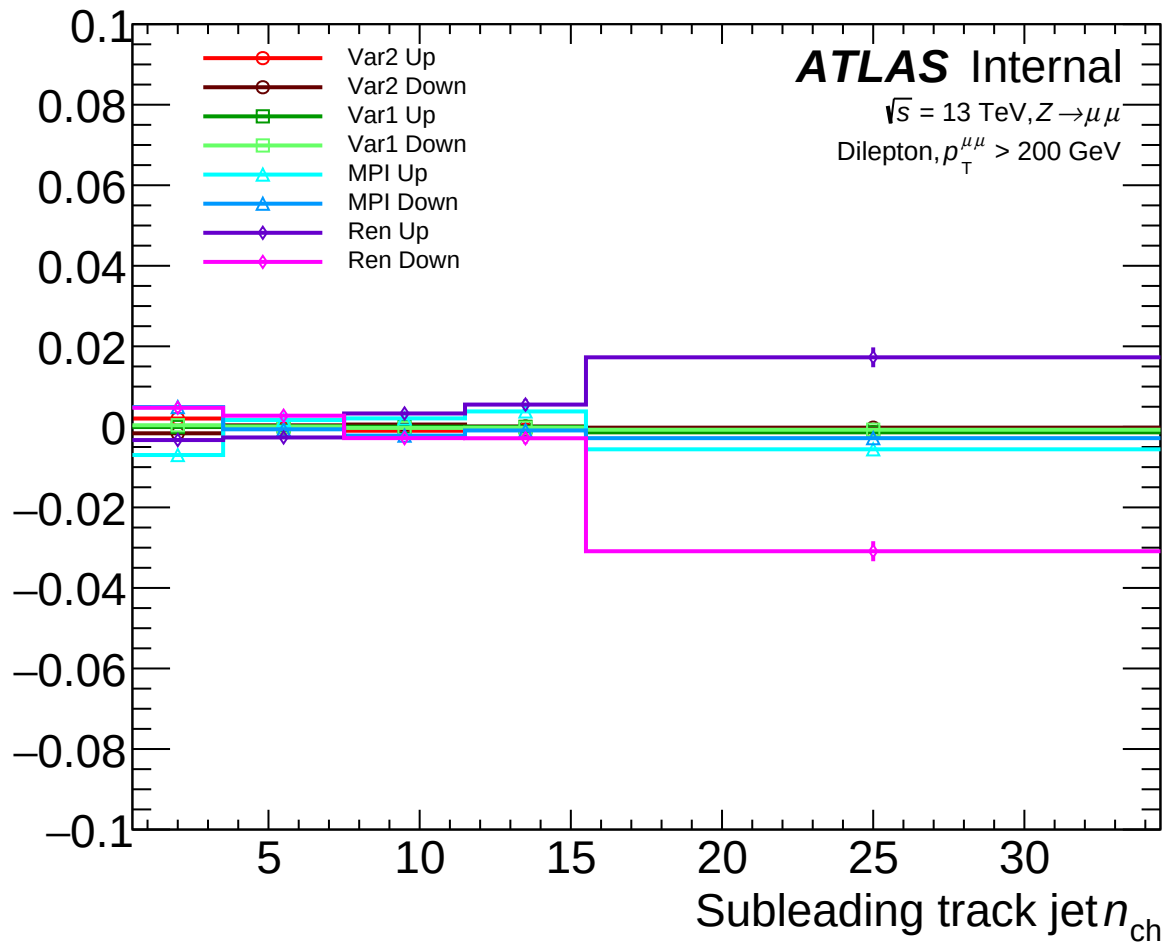
Data Statistical Uncertainty

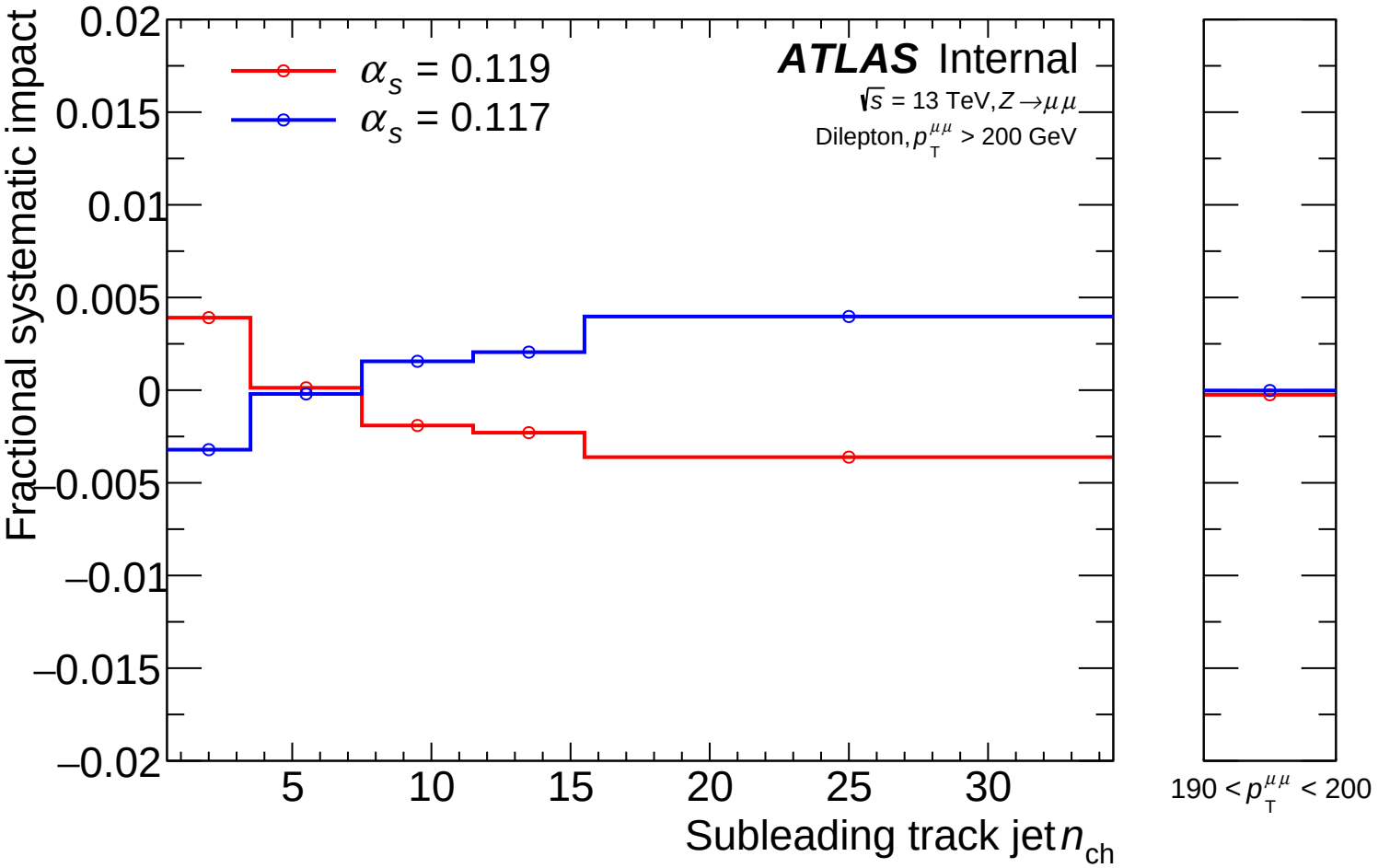


Fractional systematic impact

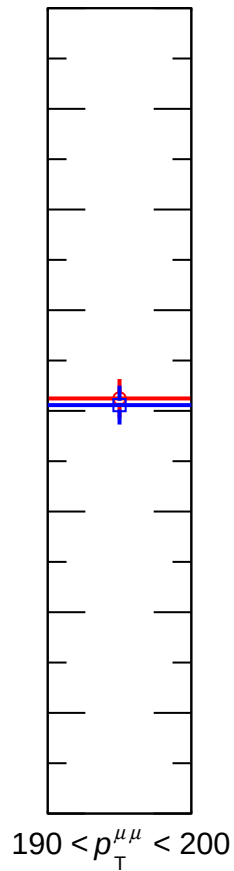
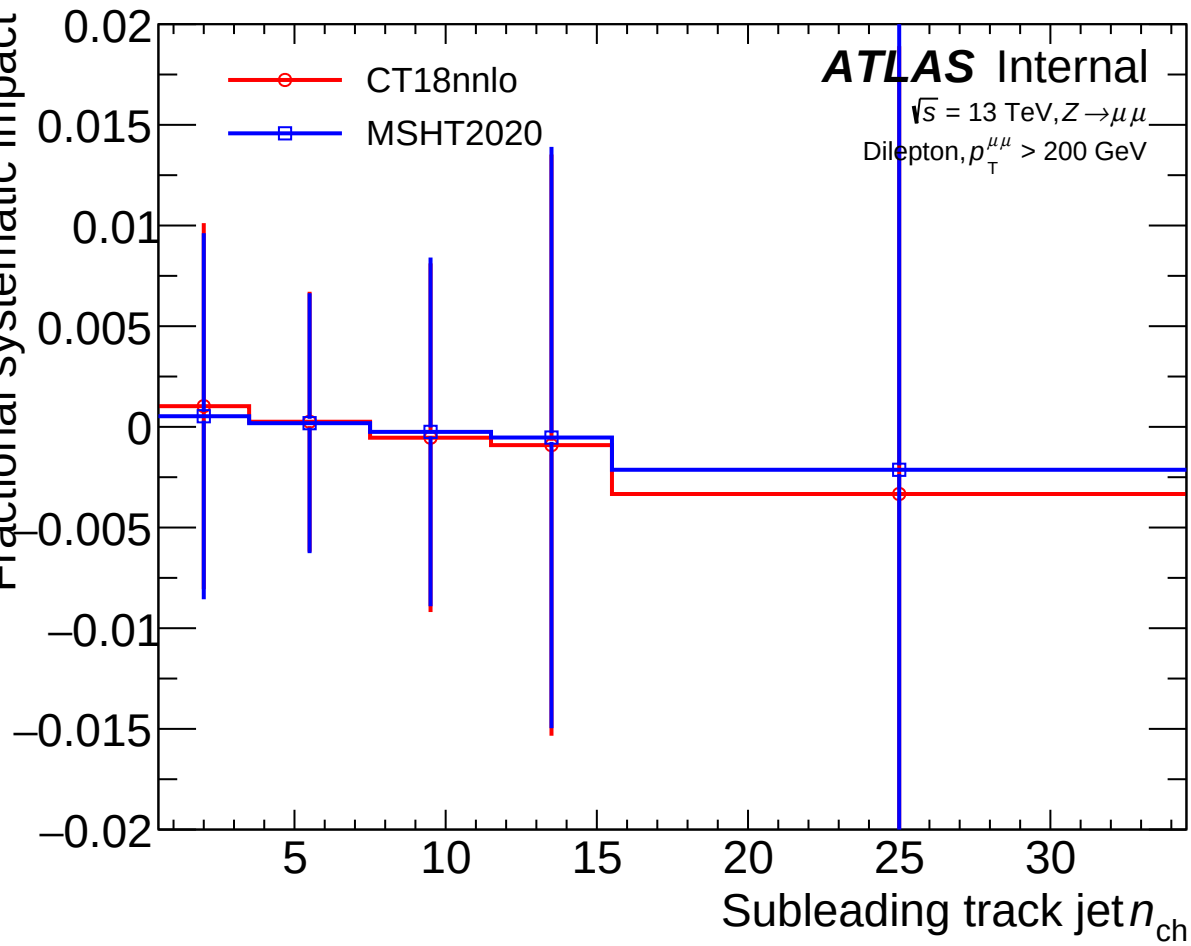


Fractional systematic impact





Fractional systematic impact



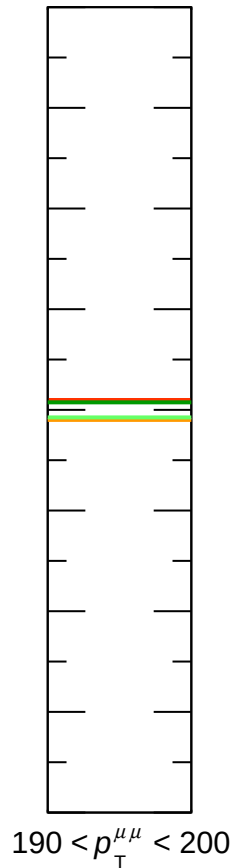
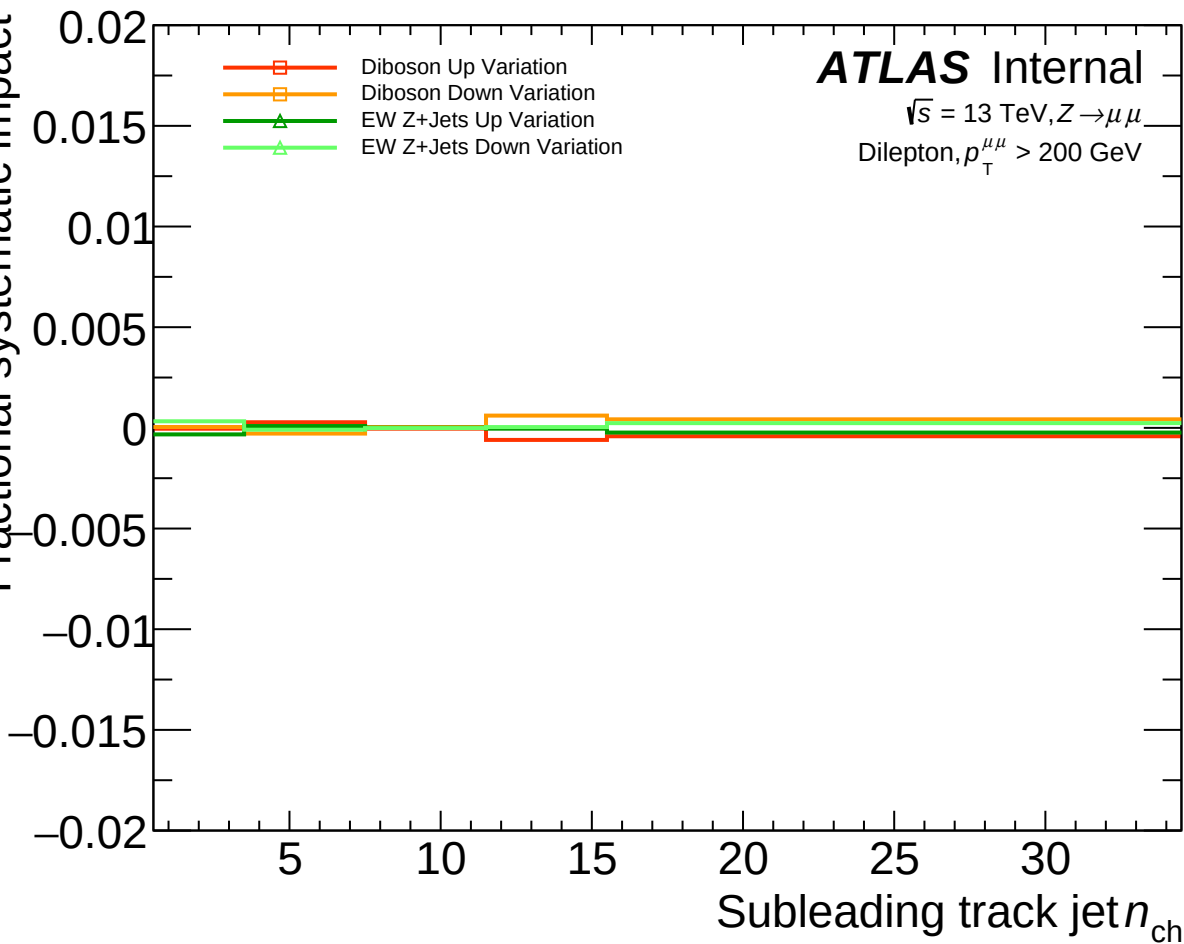
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

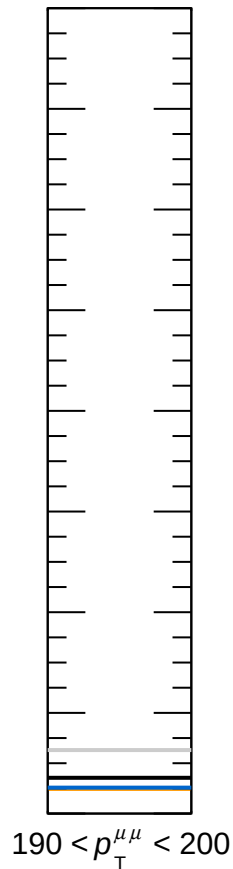
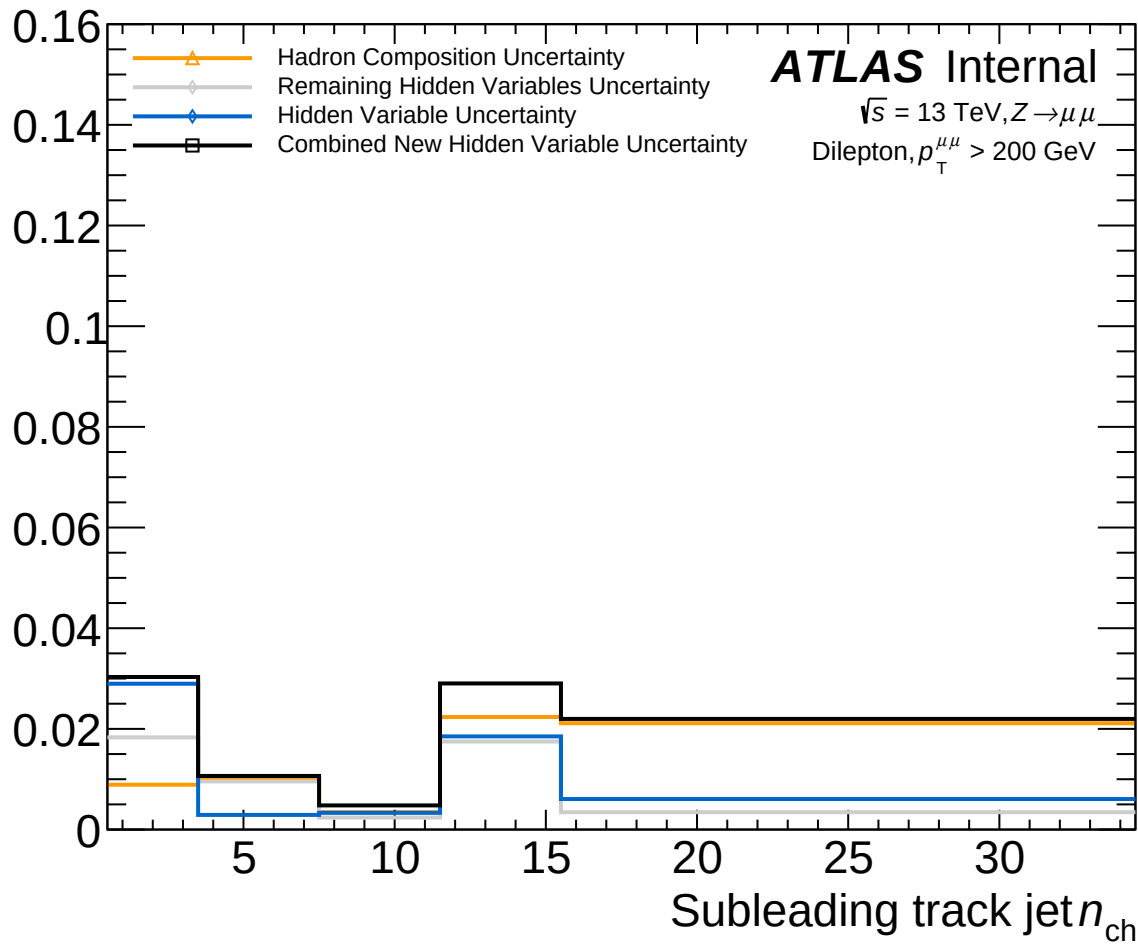
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

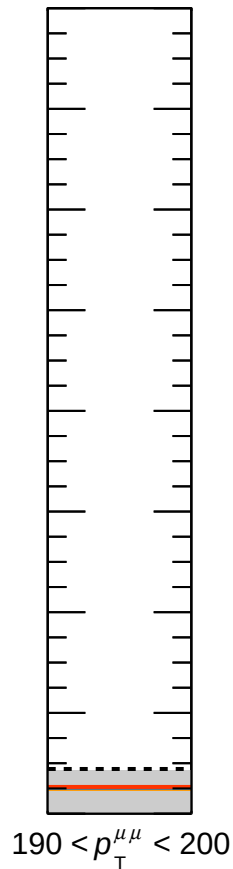
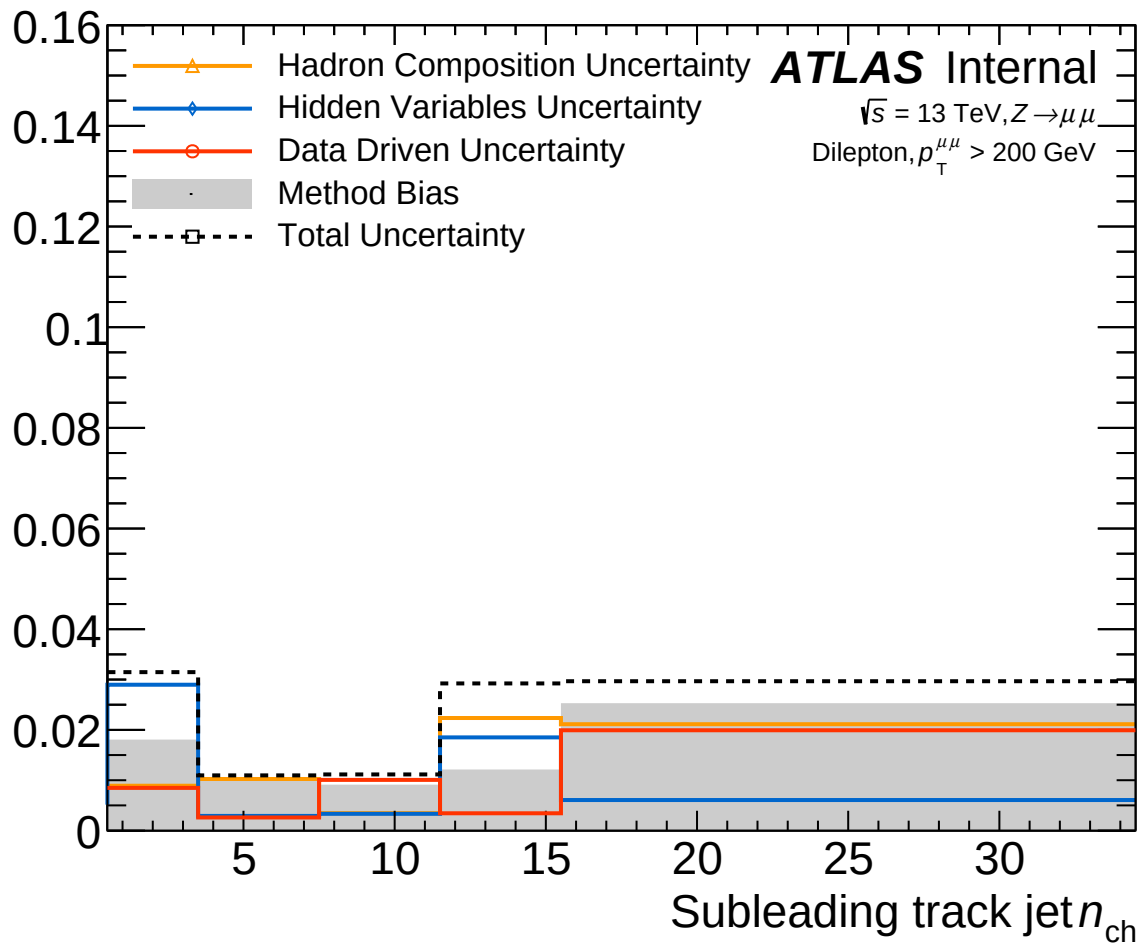


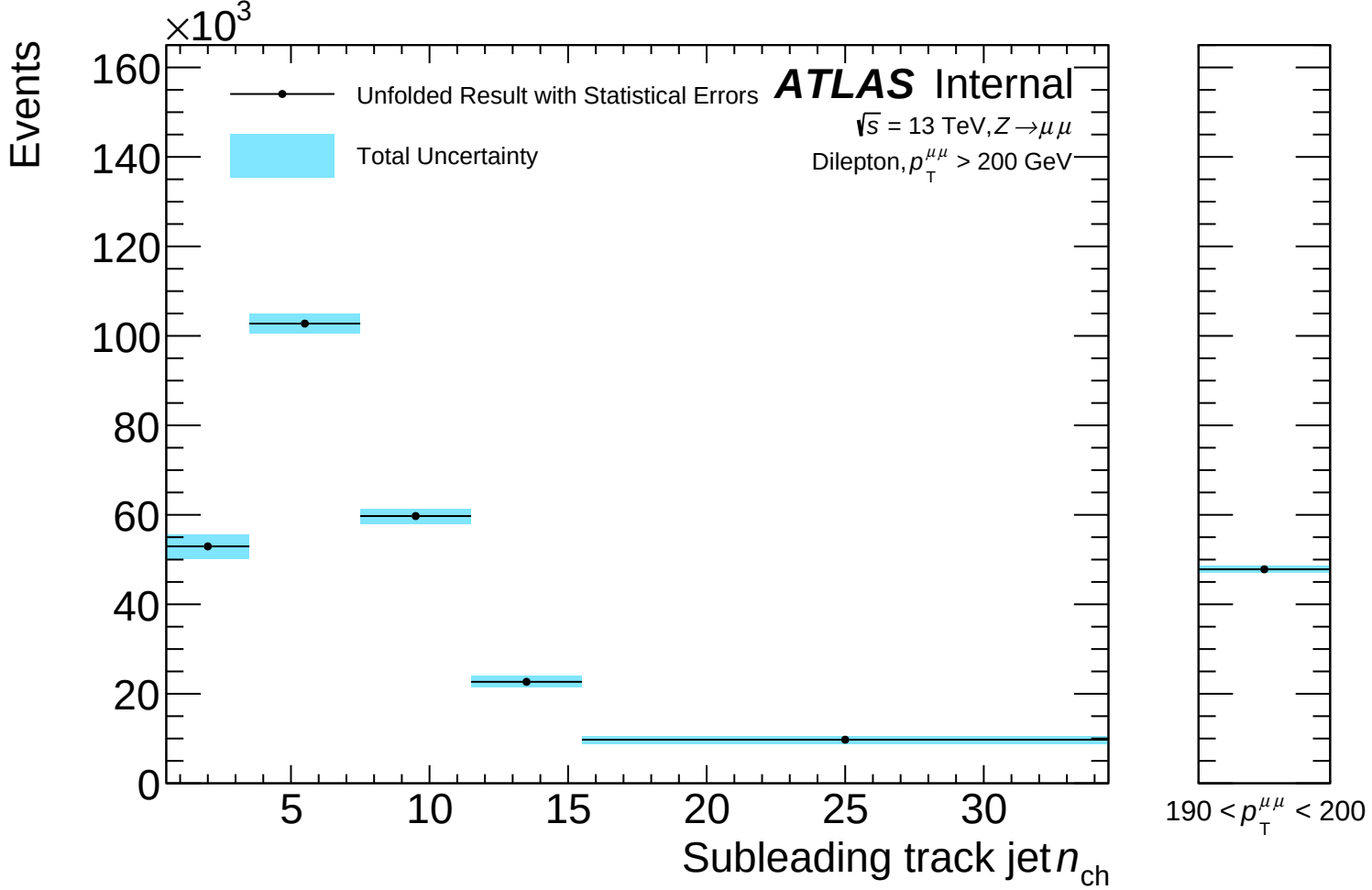
ATLAS Internal

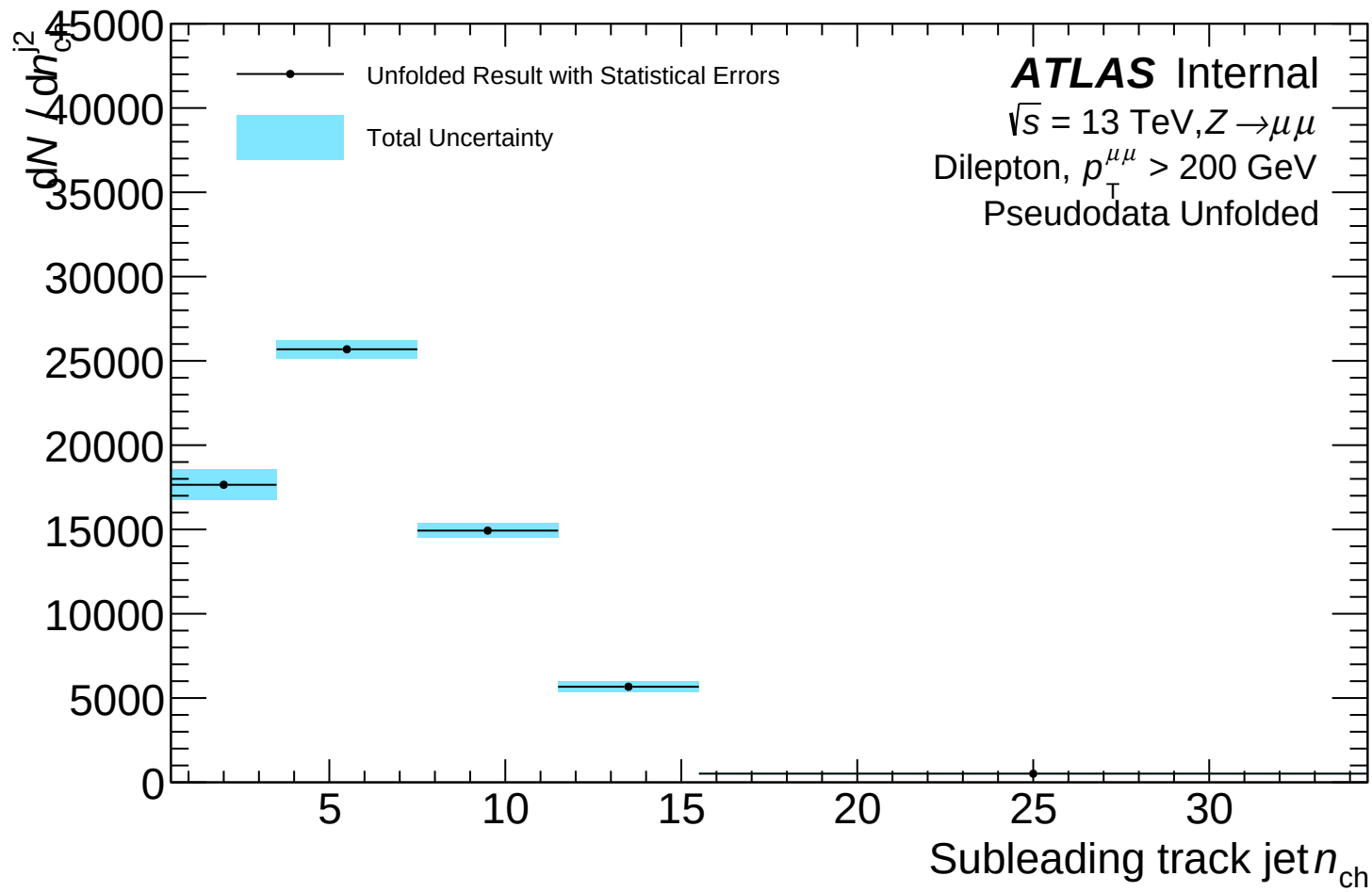
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

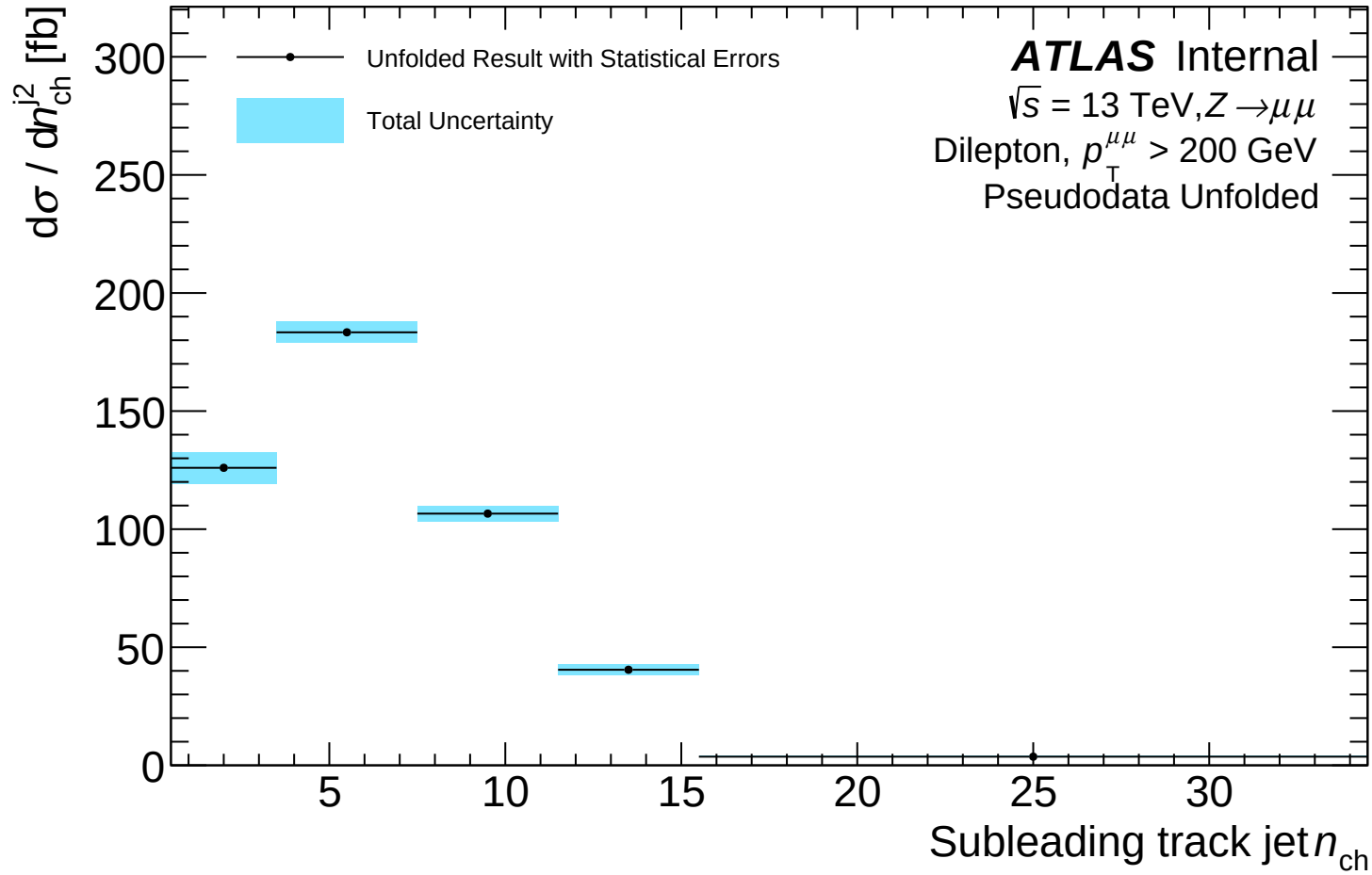
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$











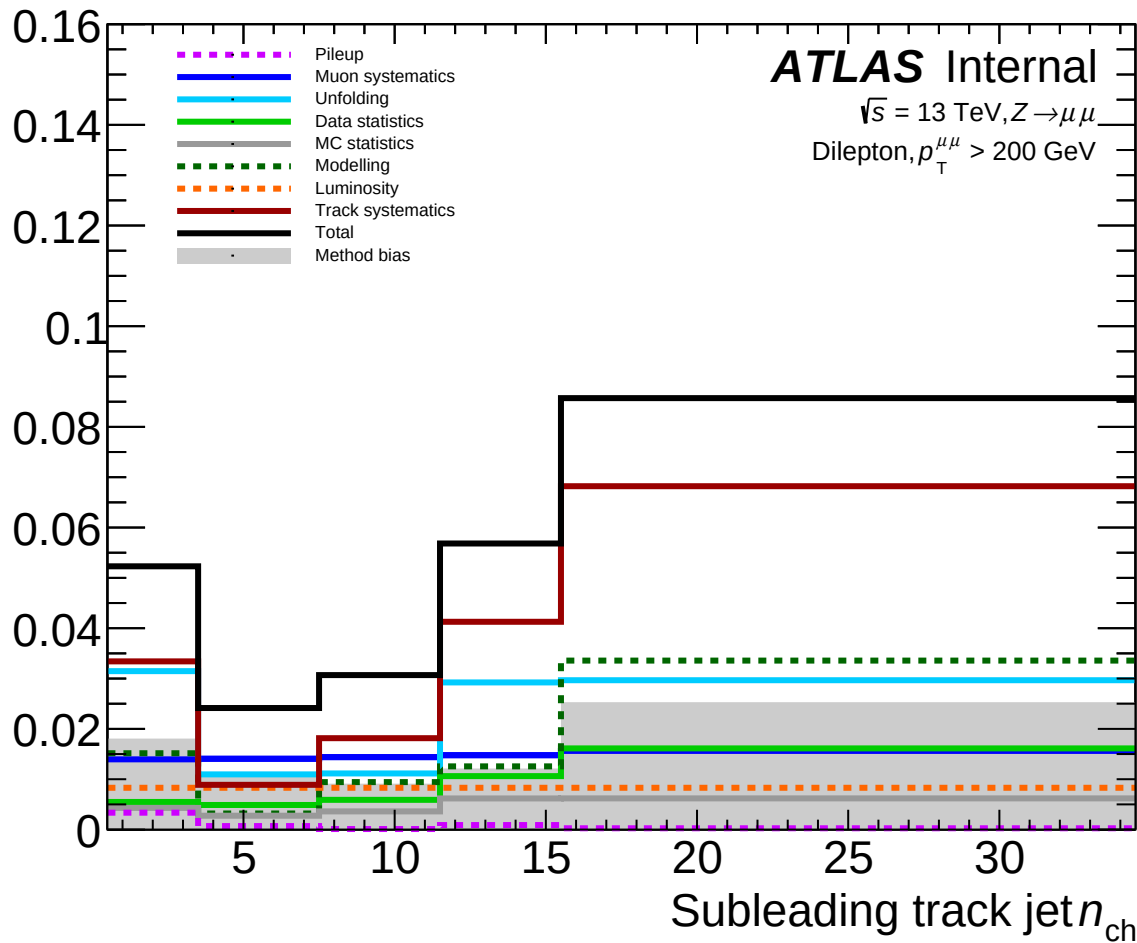
Fractional uncertainty

ATLAS Internal

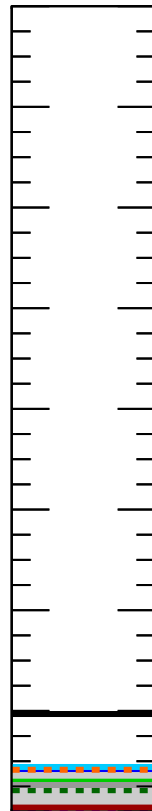
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

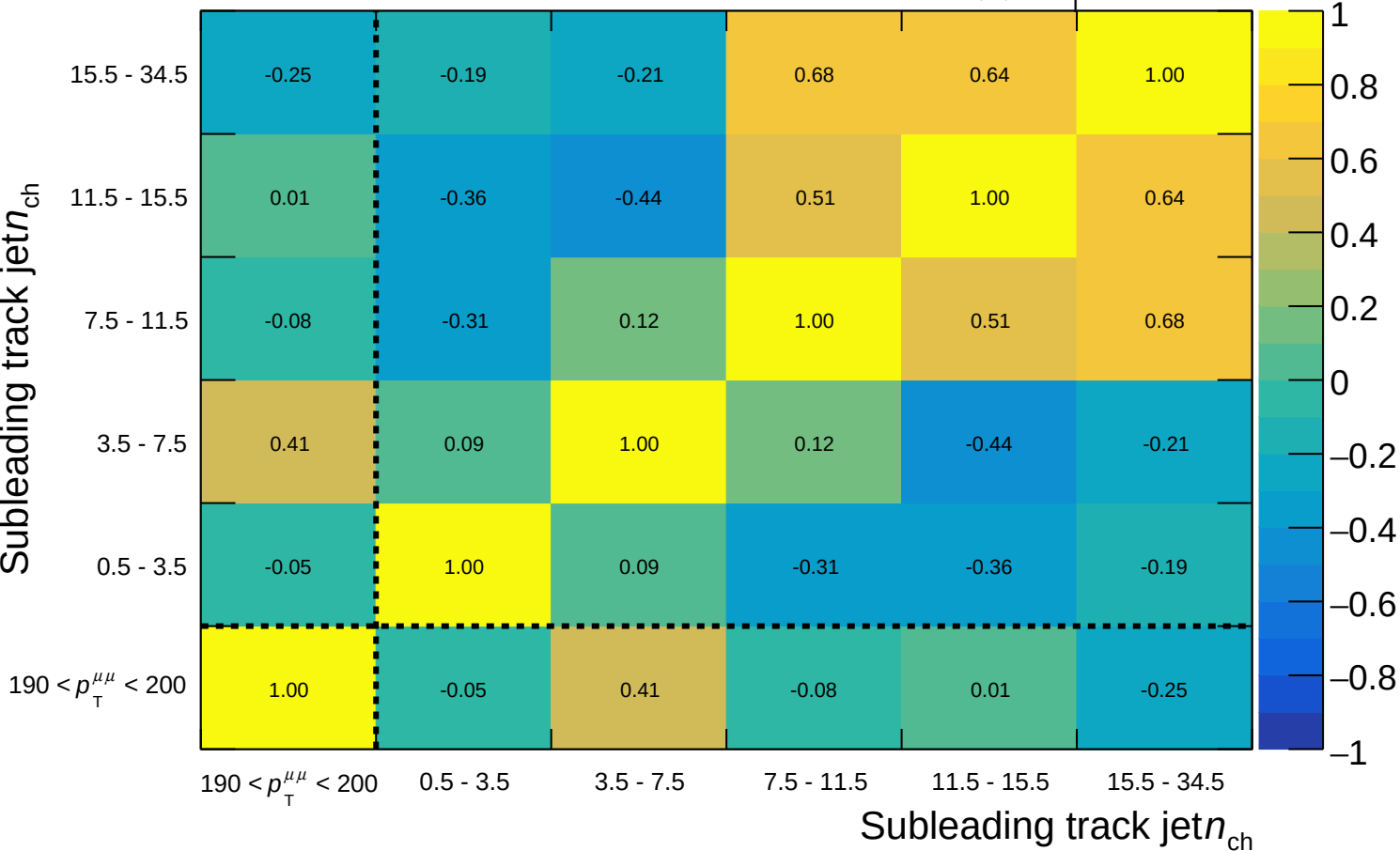
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

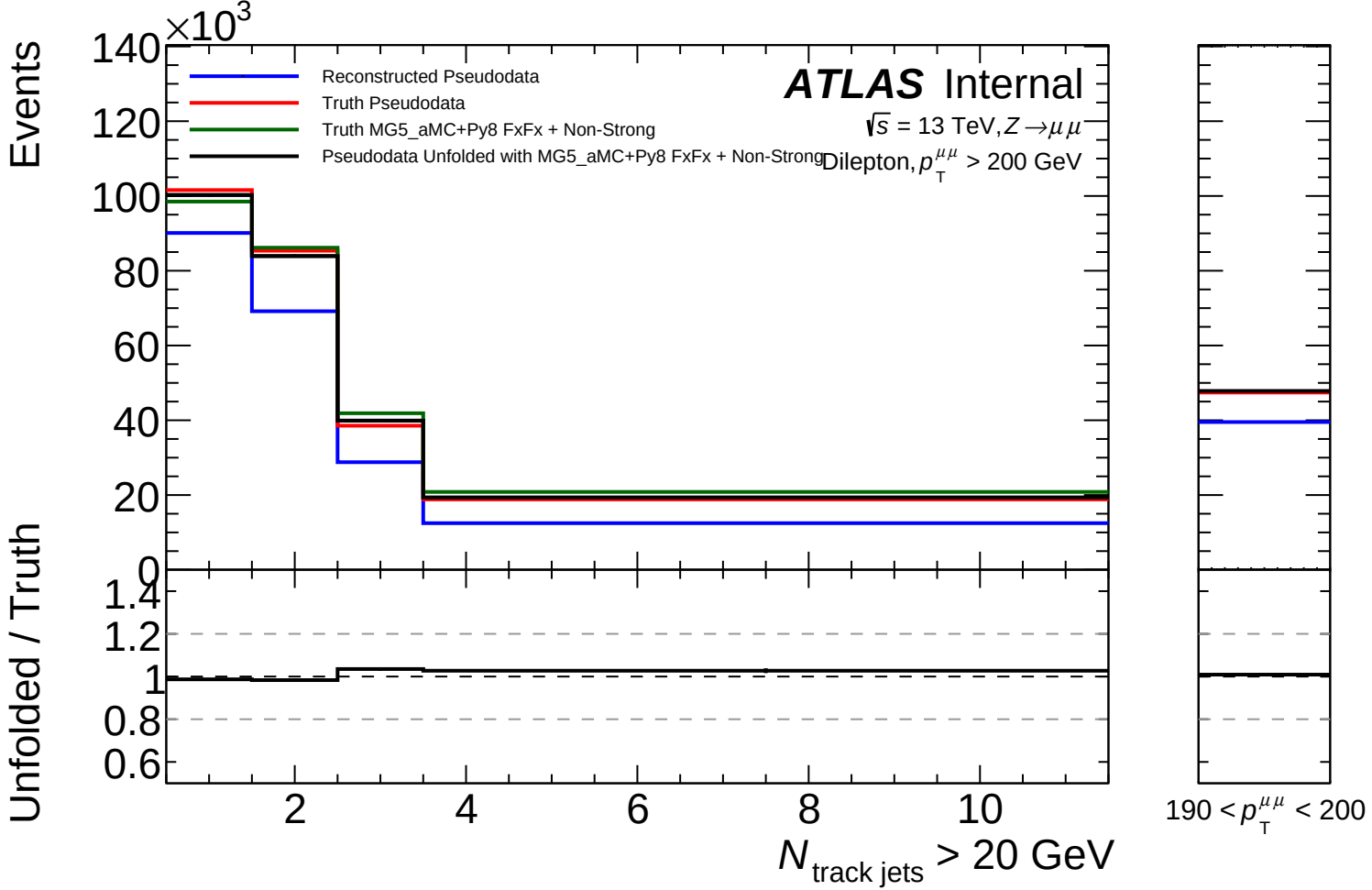
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

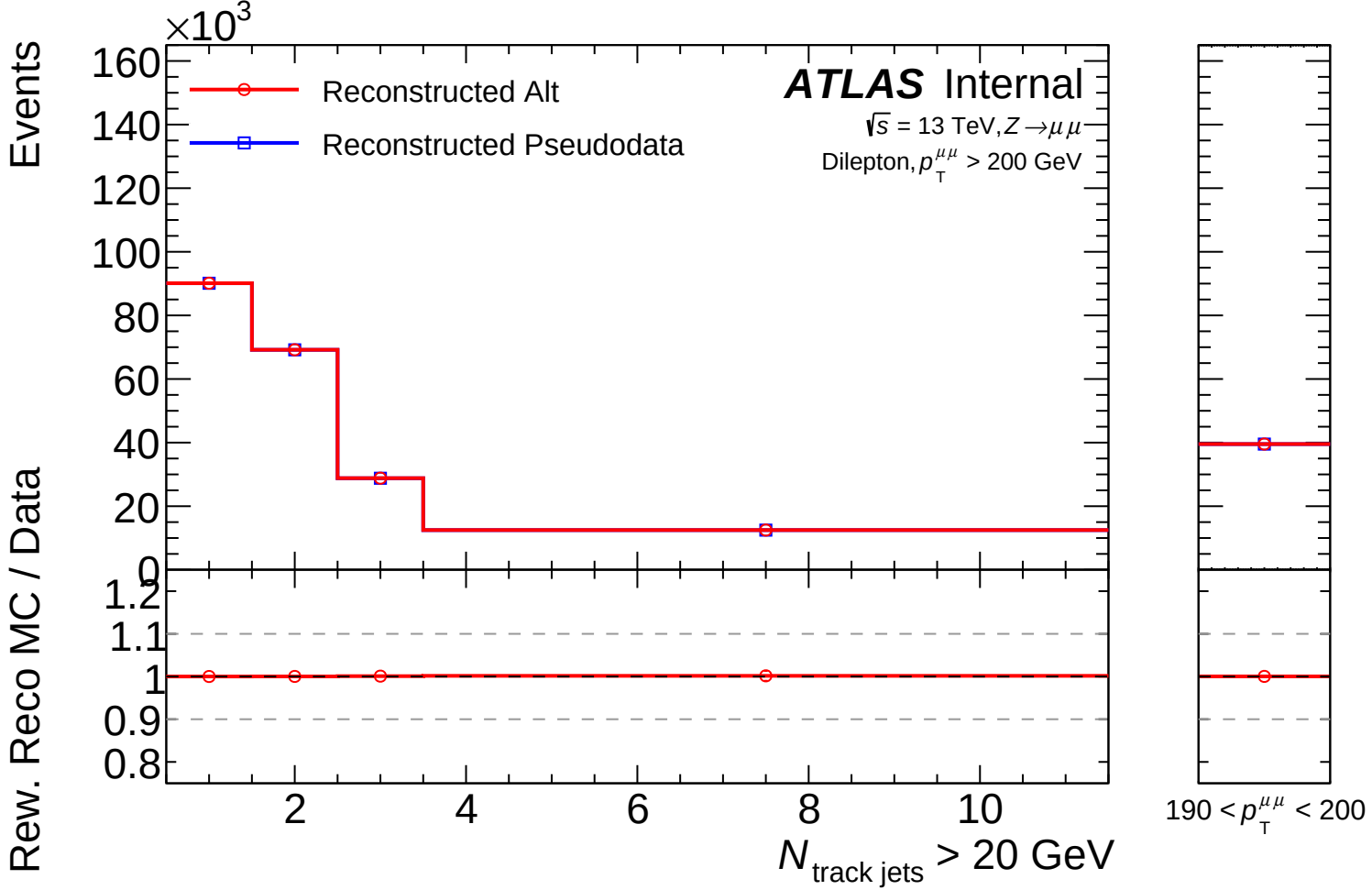


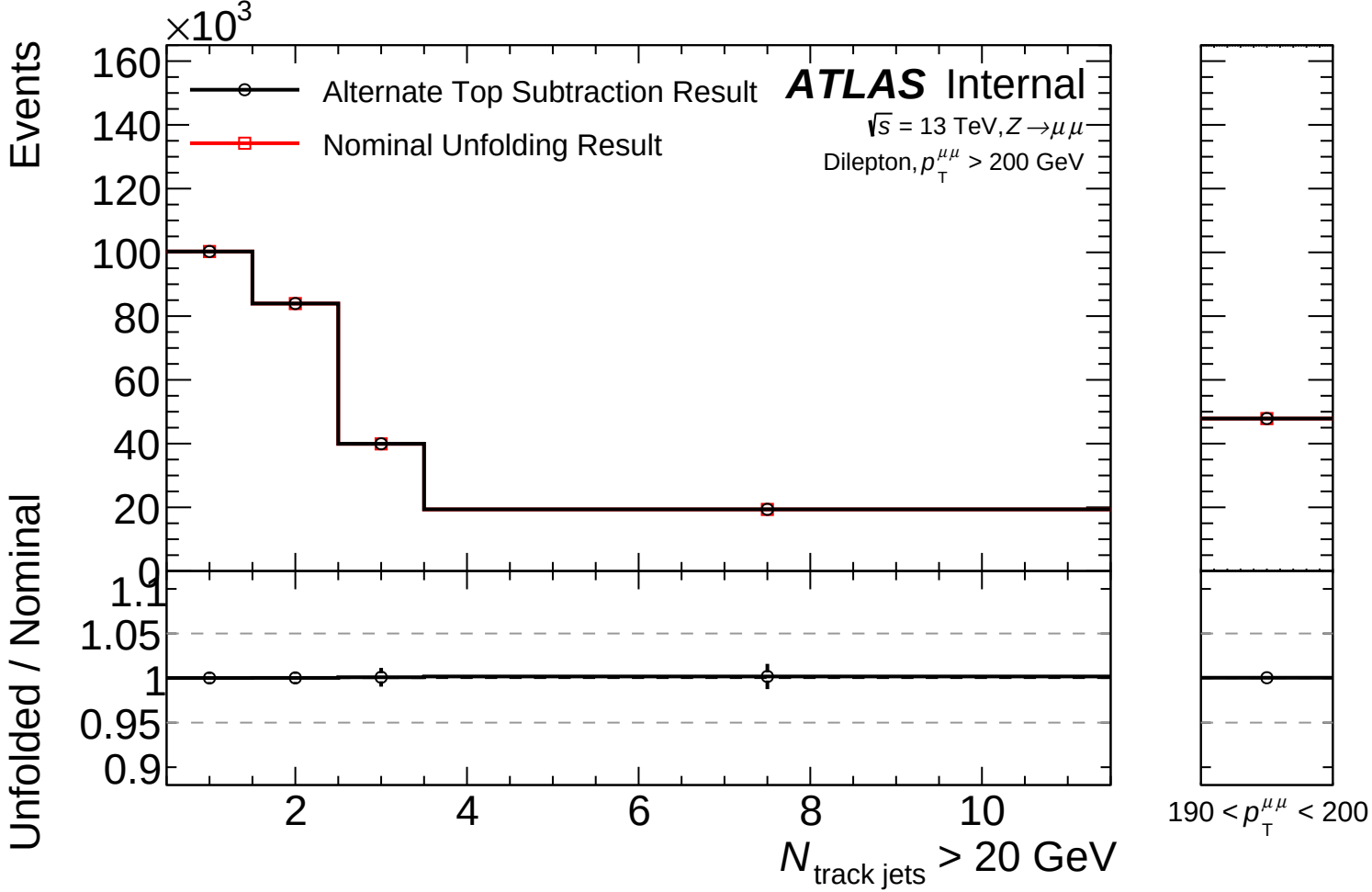
$190 < p_T^{\mu\mu} < 200$



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_T^{\mu\mu} > 200 \text{ GeV}$ Subleading track jet n_{ch} Subleading track jet n_{ch}







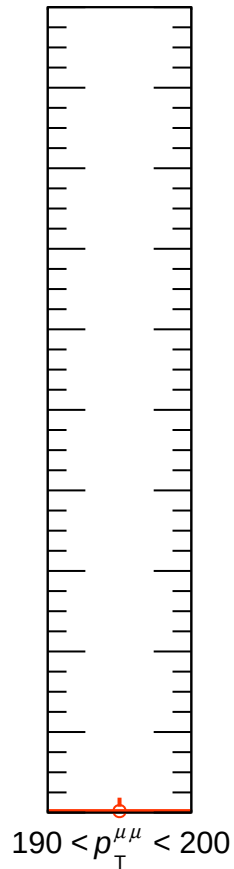
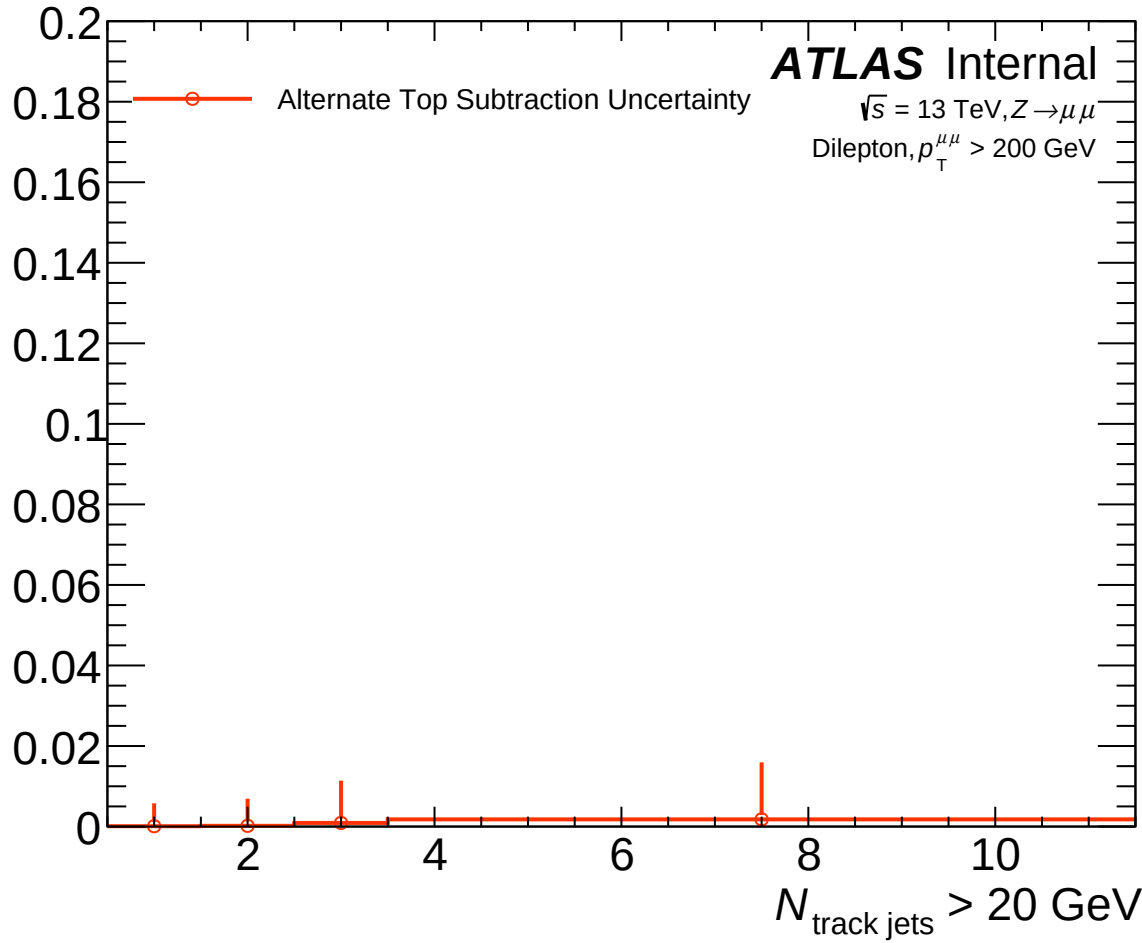
Fractional systematic impact

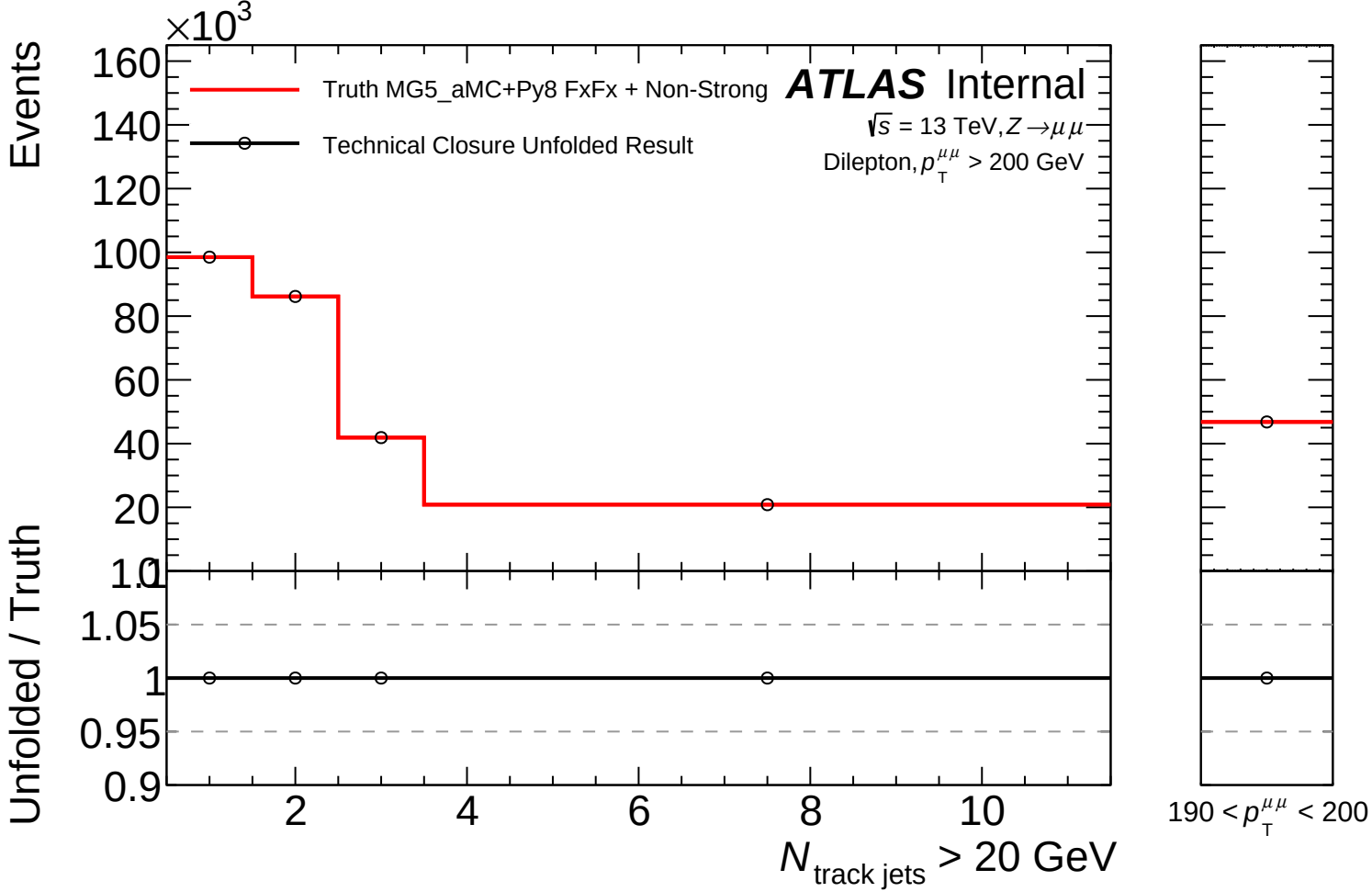
ATLAS Internal

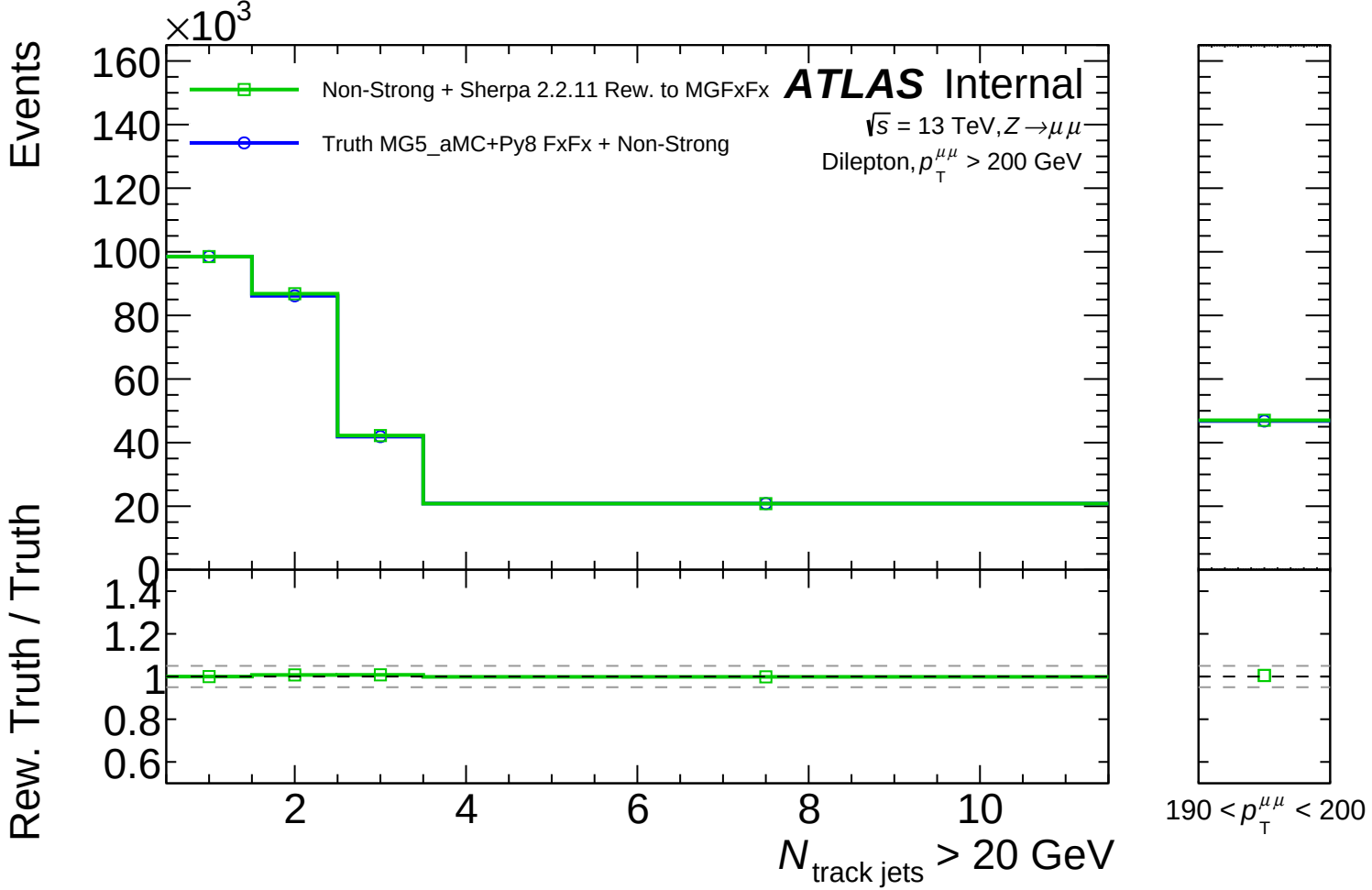
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

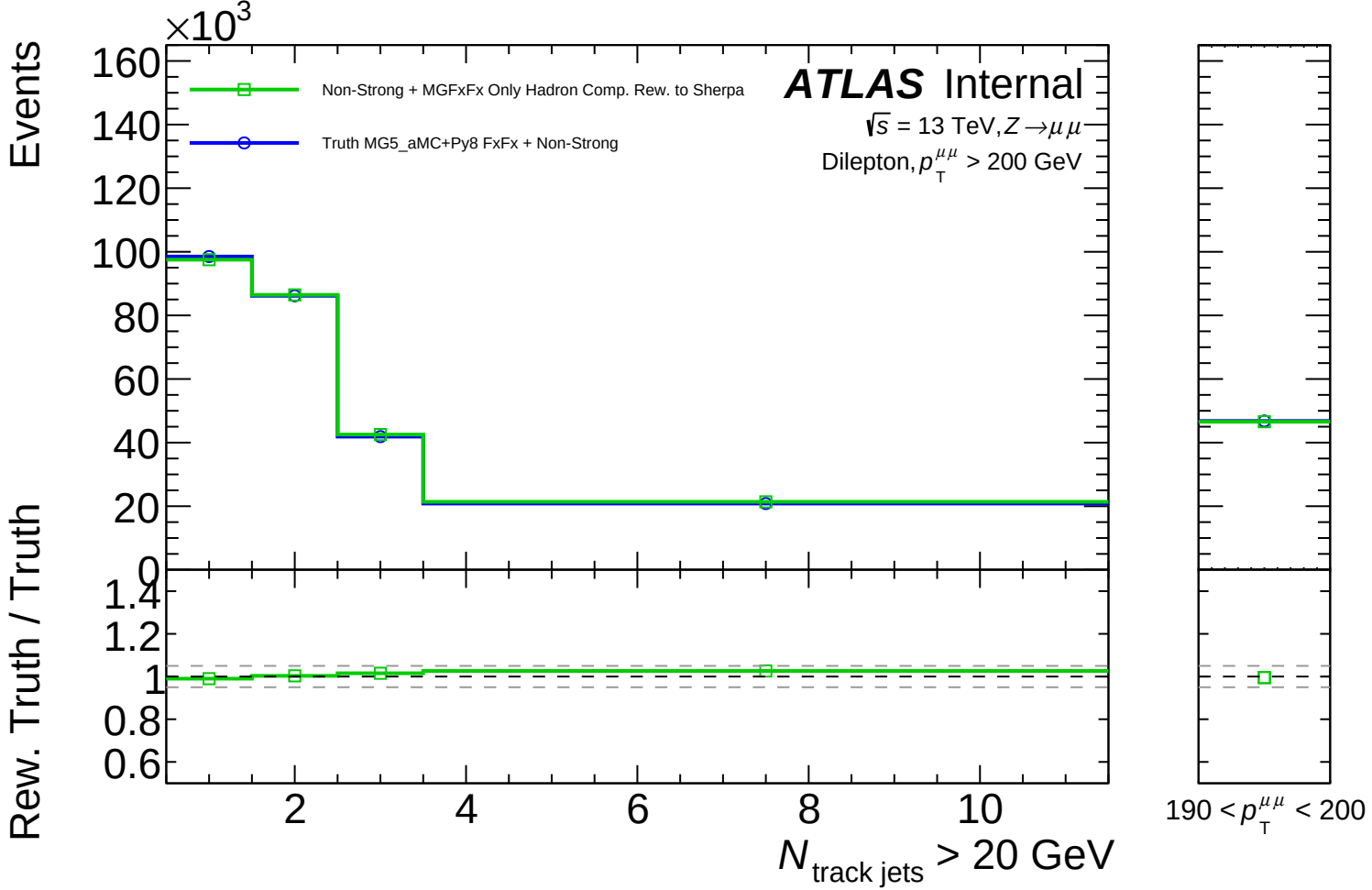
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

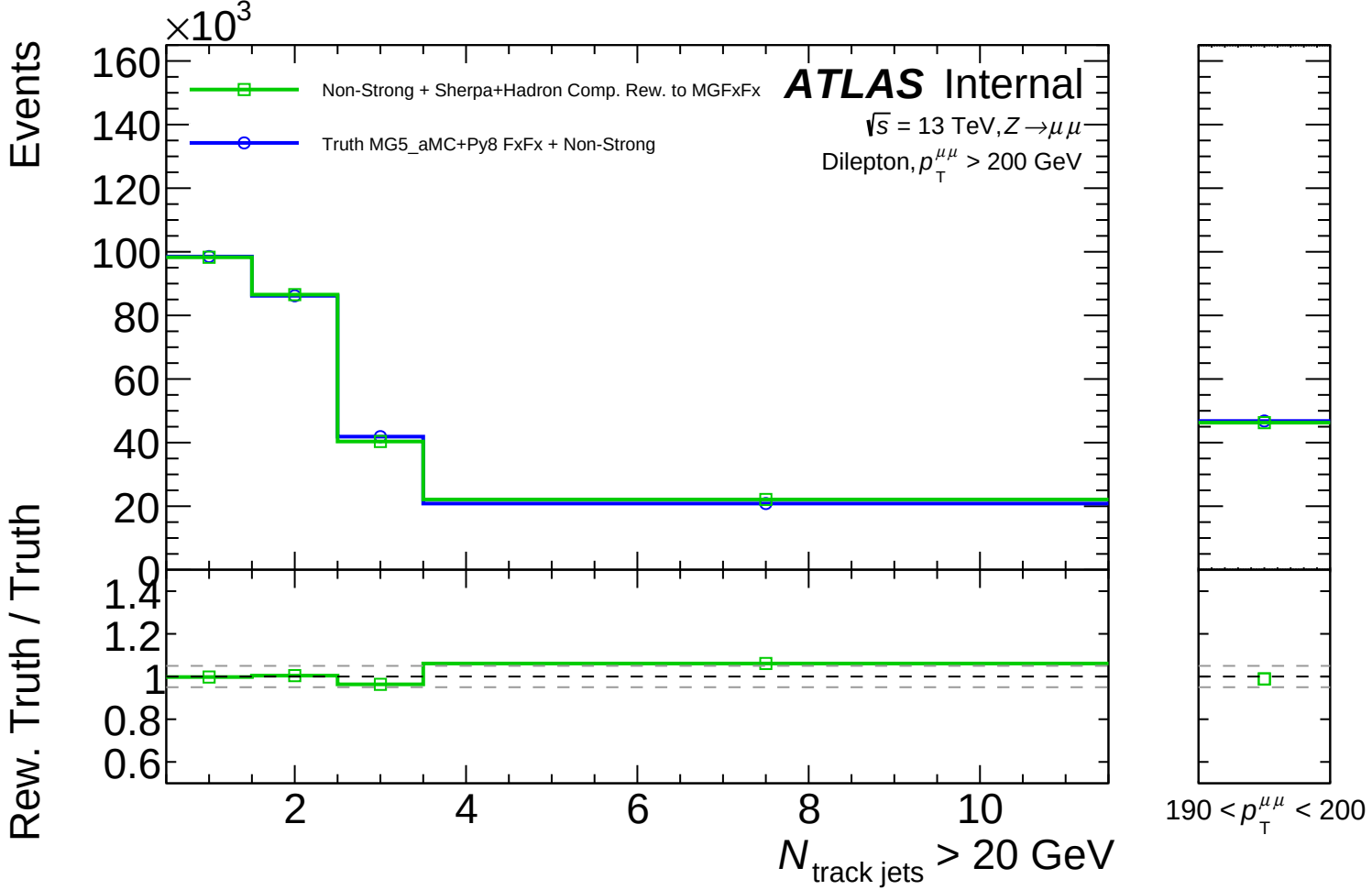
Alternate Top Subtraction Uncertainty

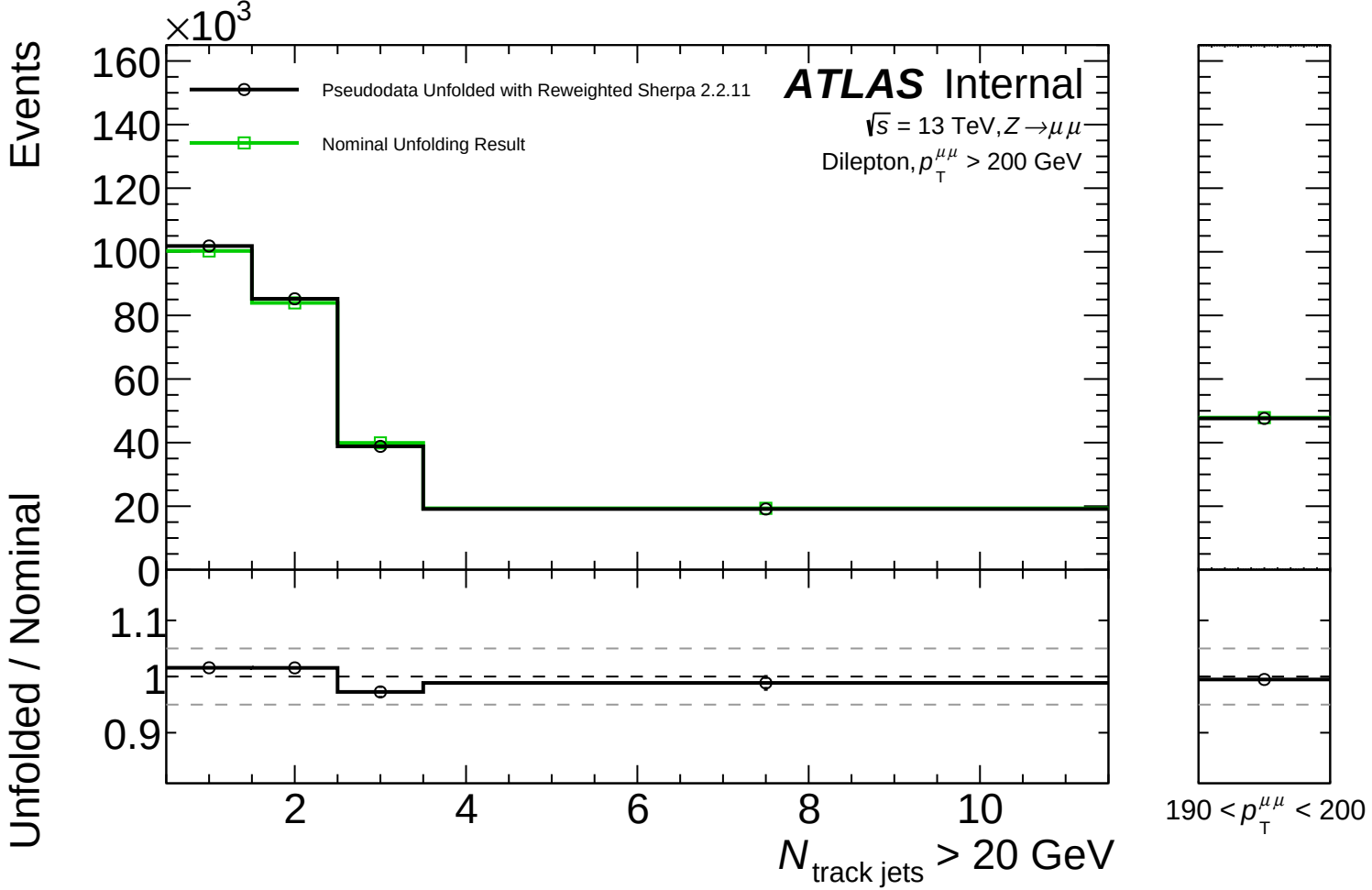


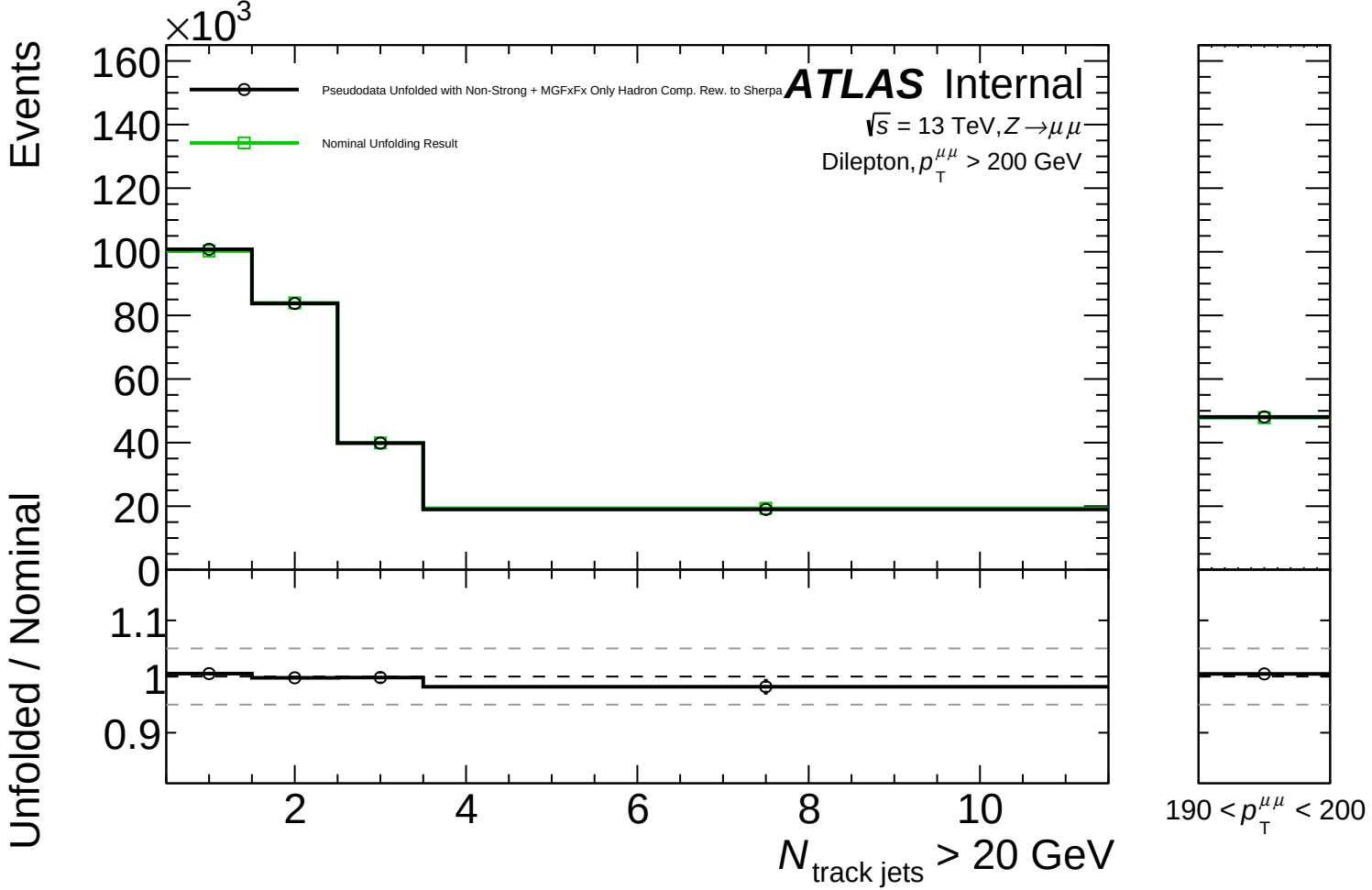






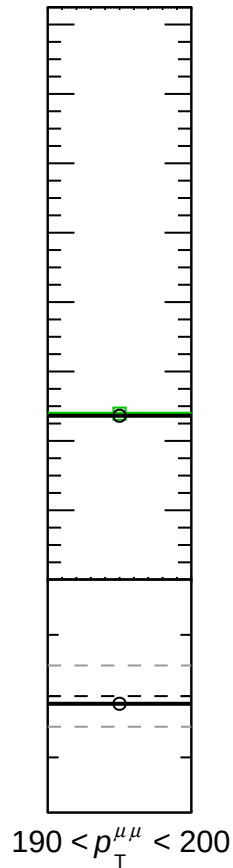
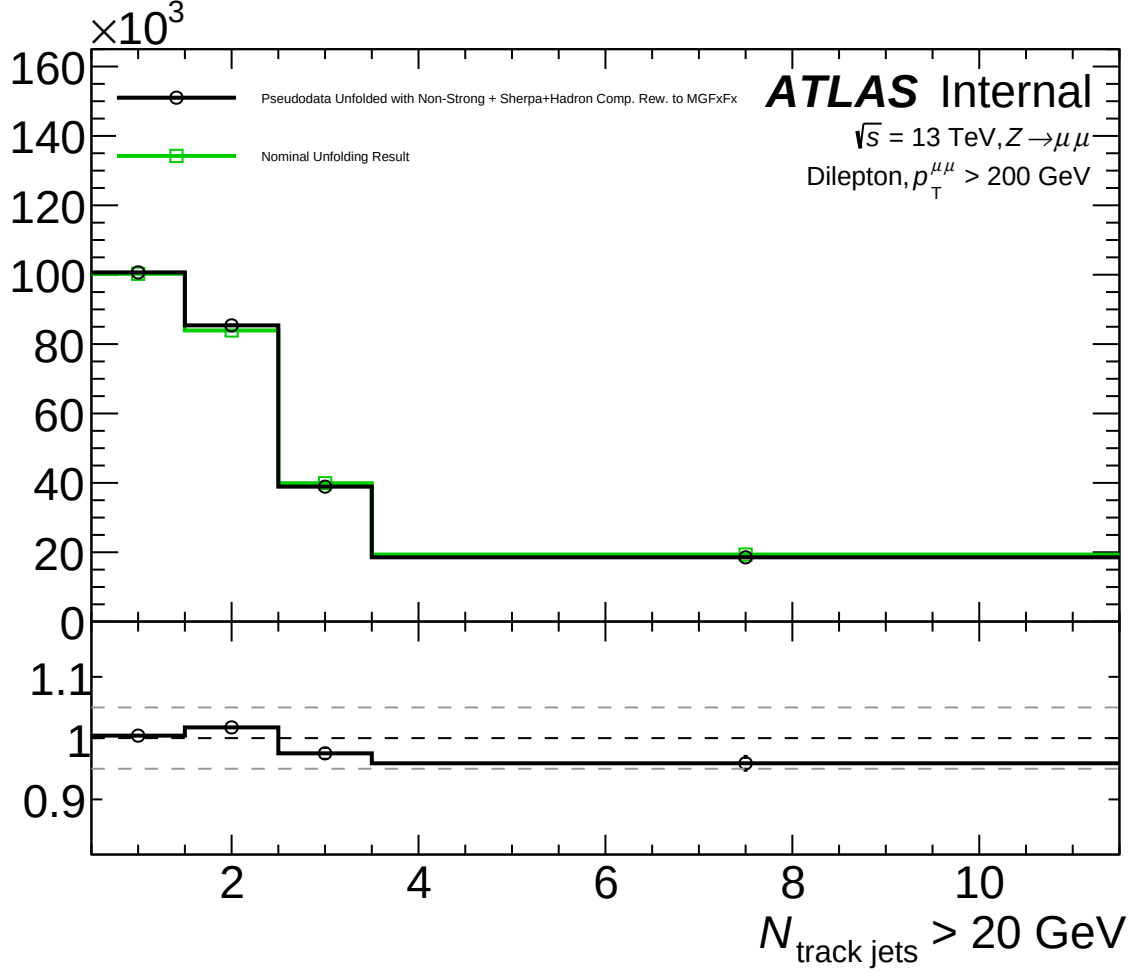






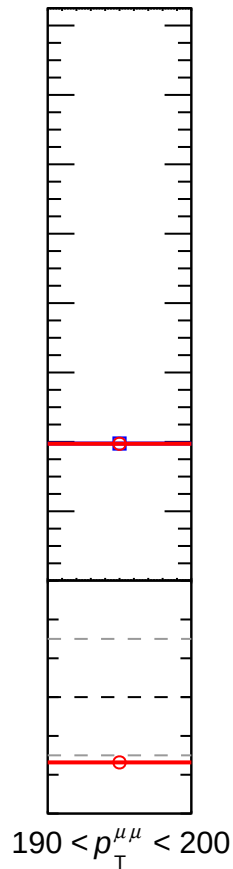
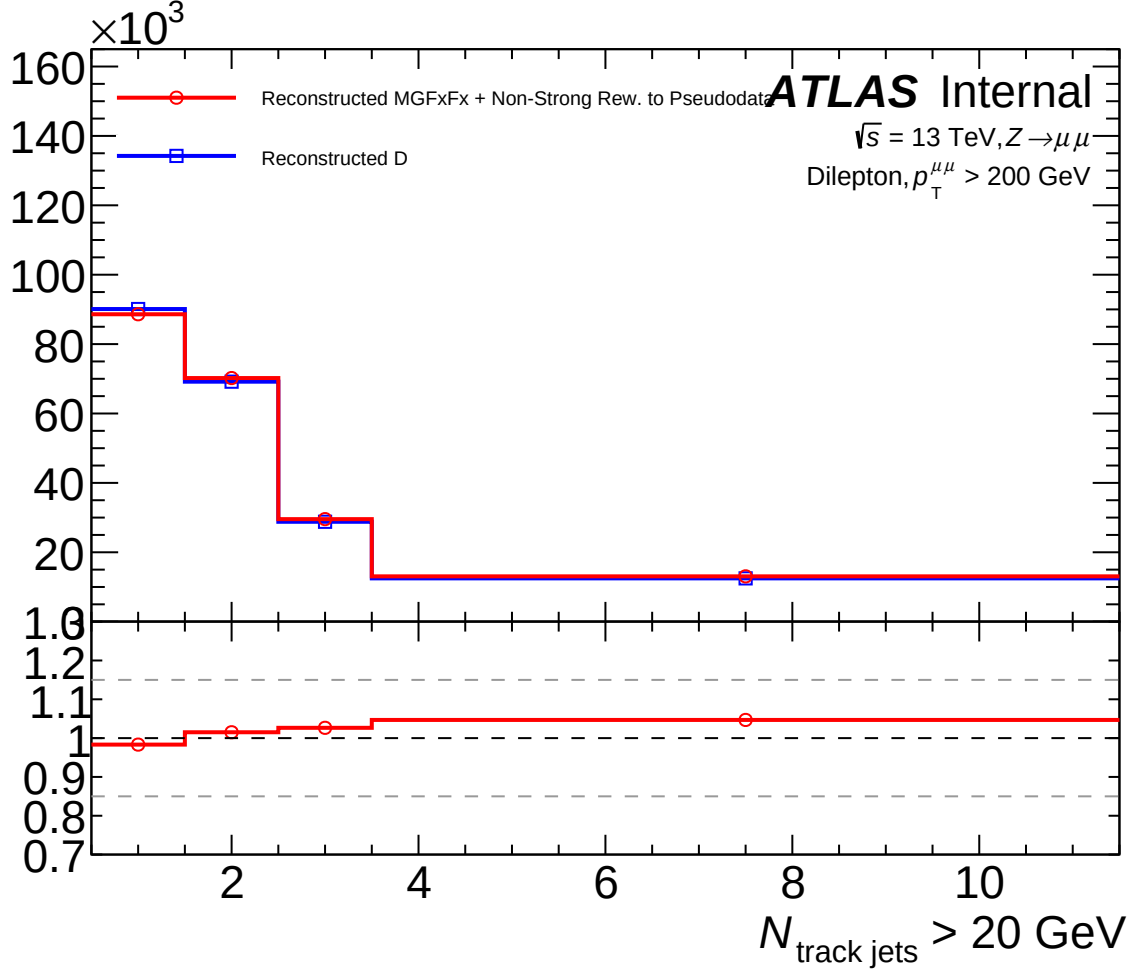
Events

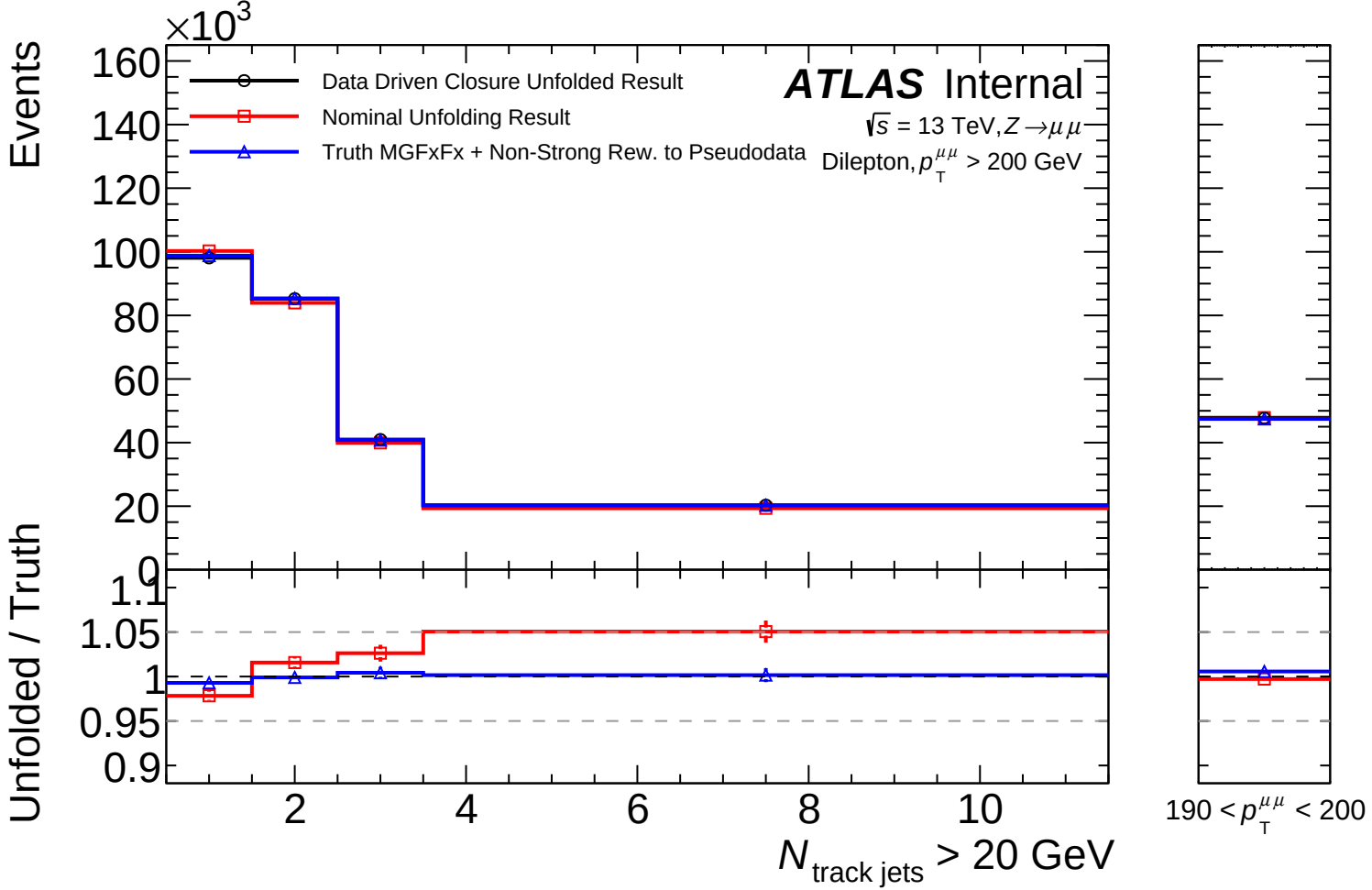
Unfolded / Nominal



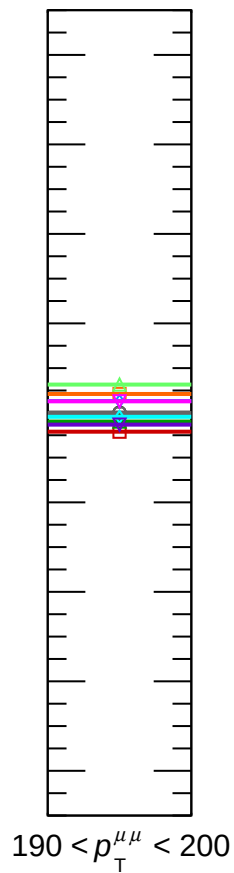
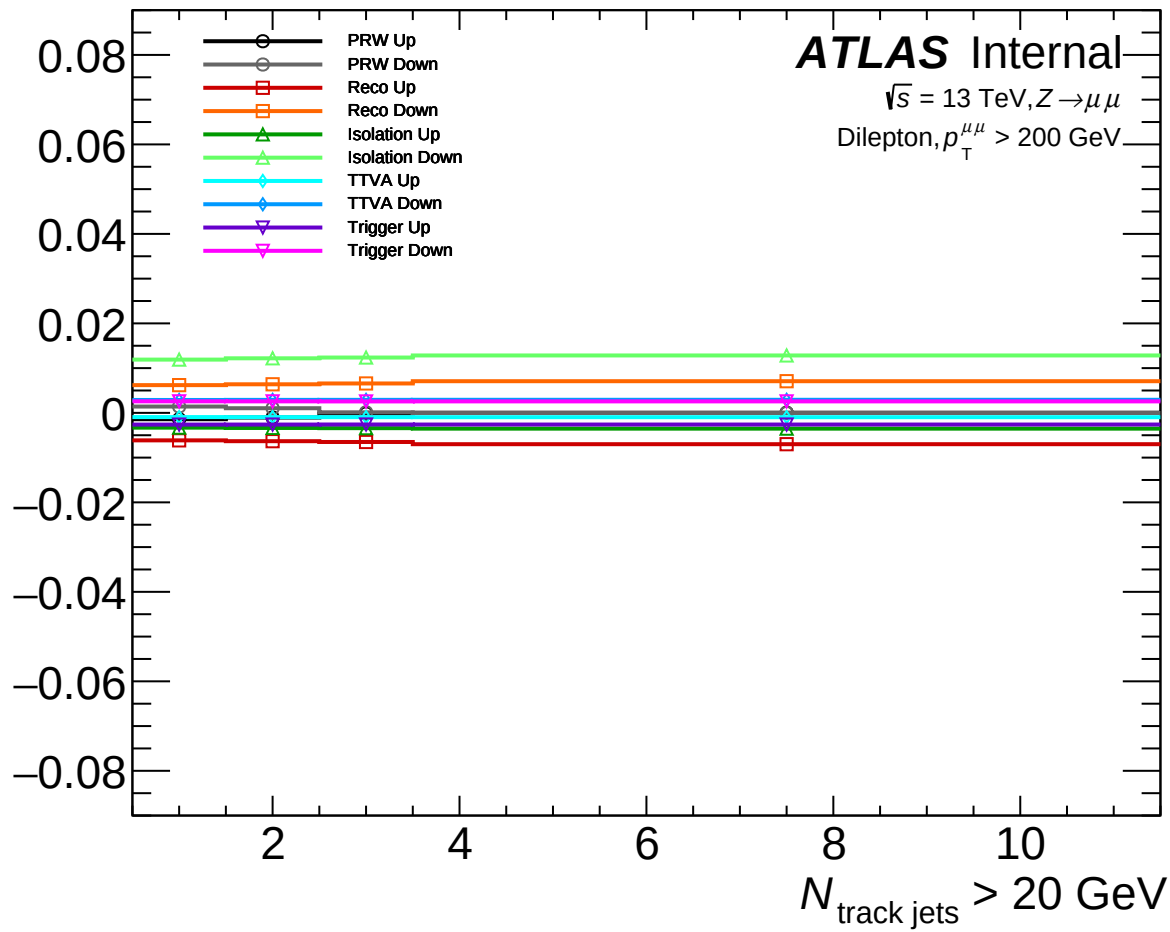
Events

Rew. Reco MC / Data

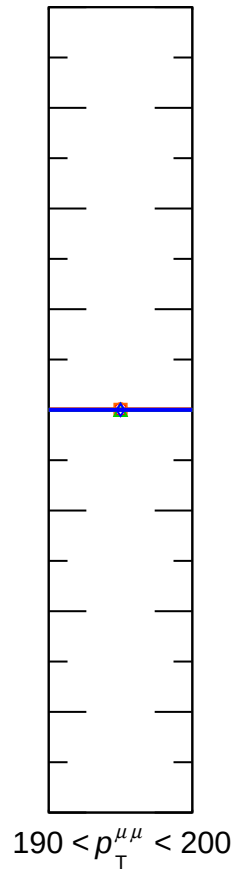
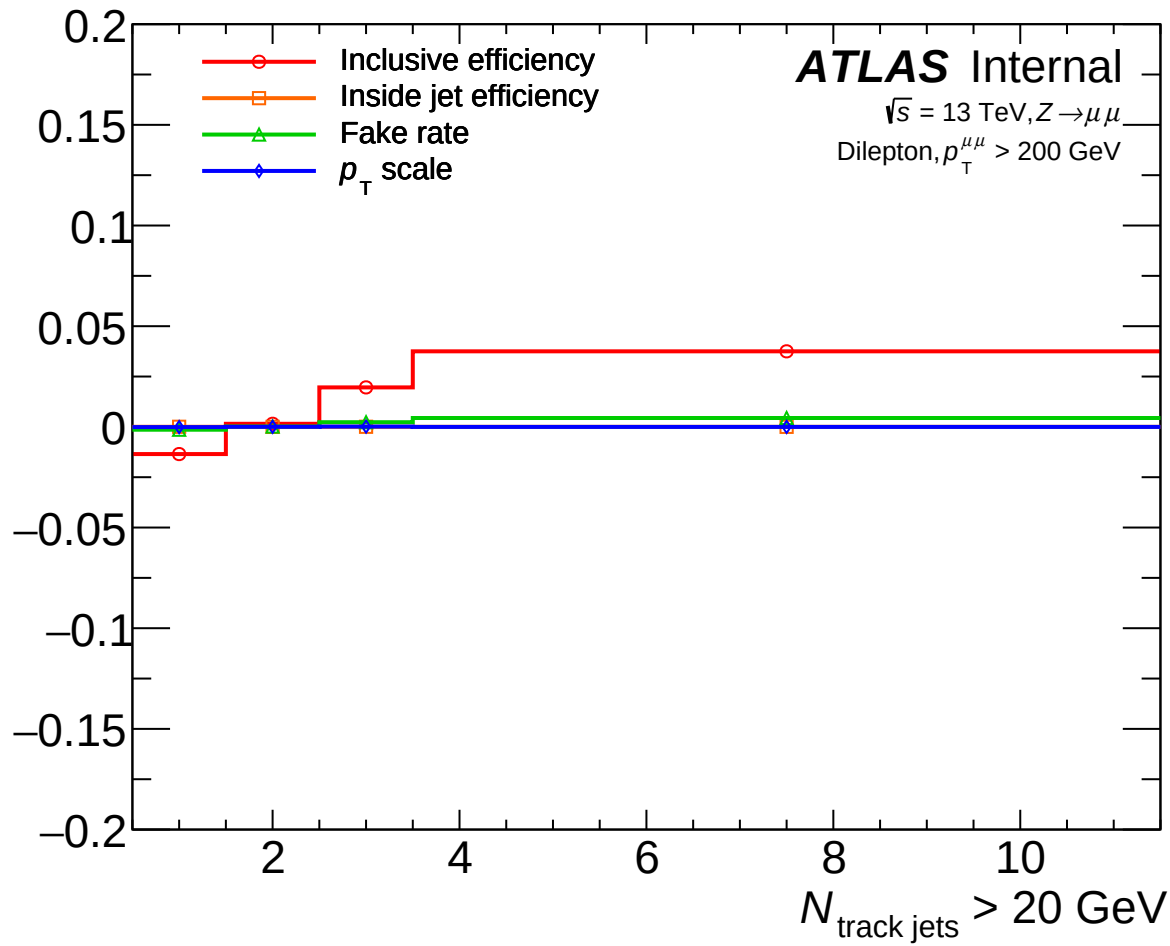




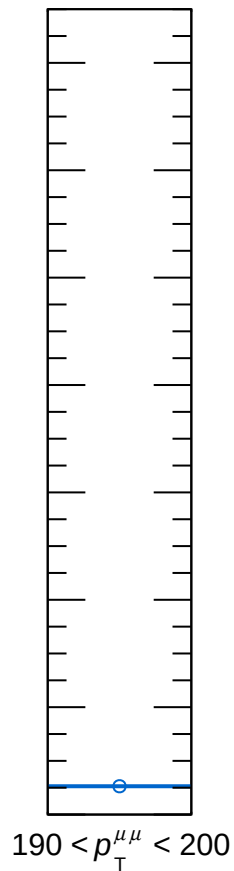
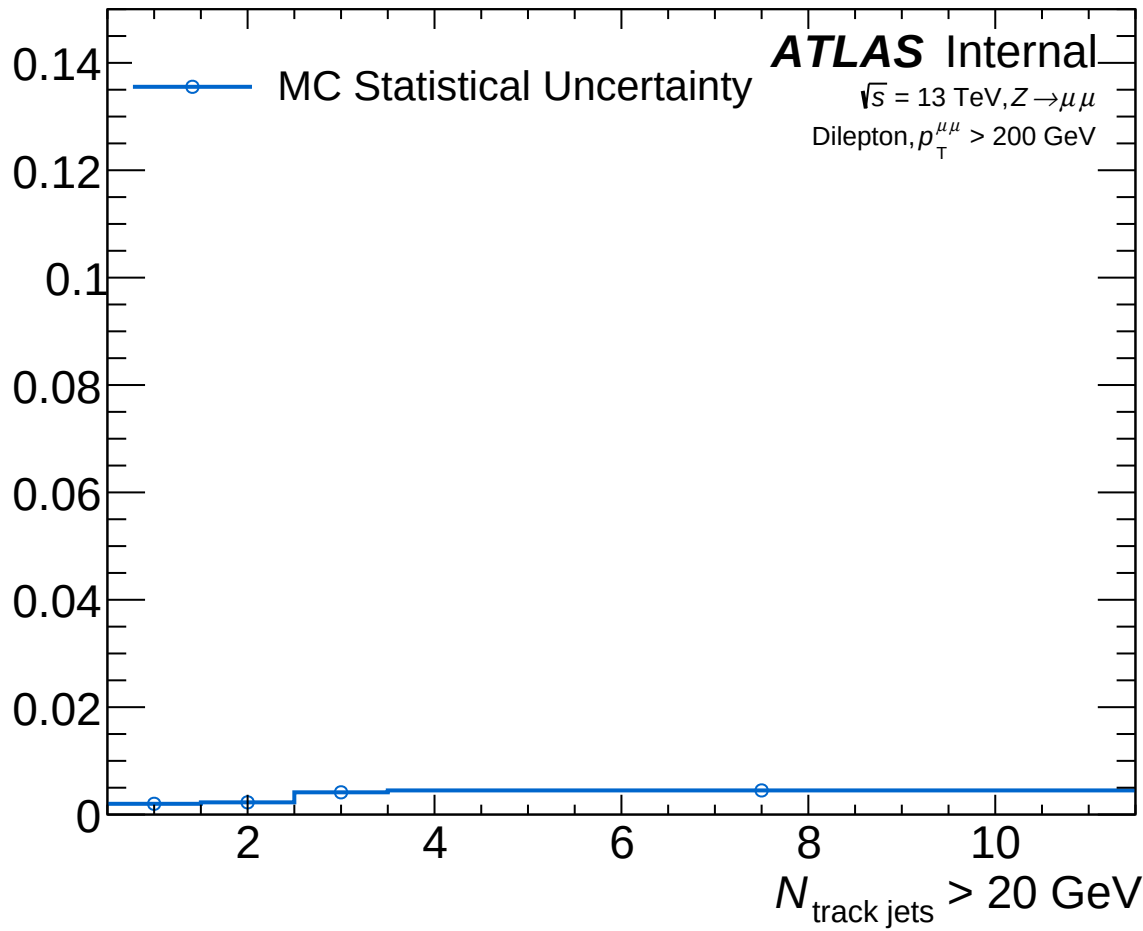
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



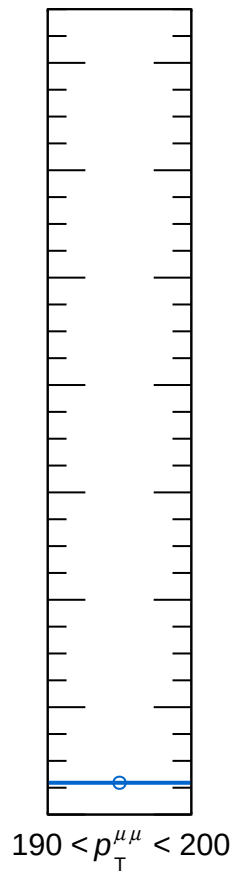
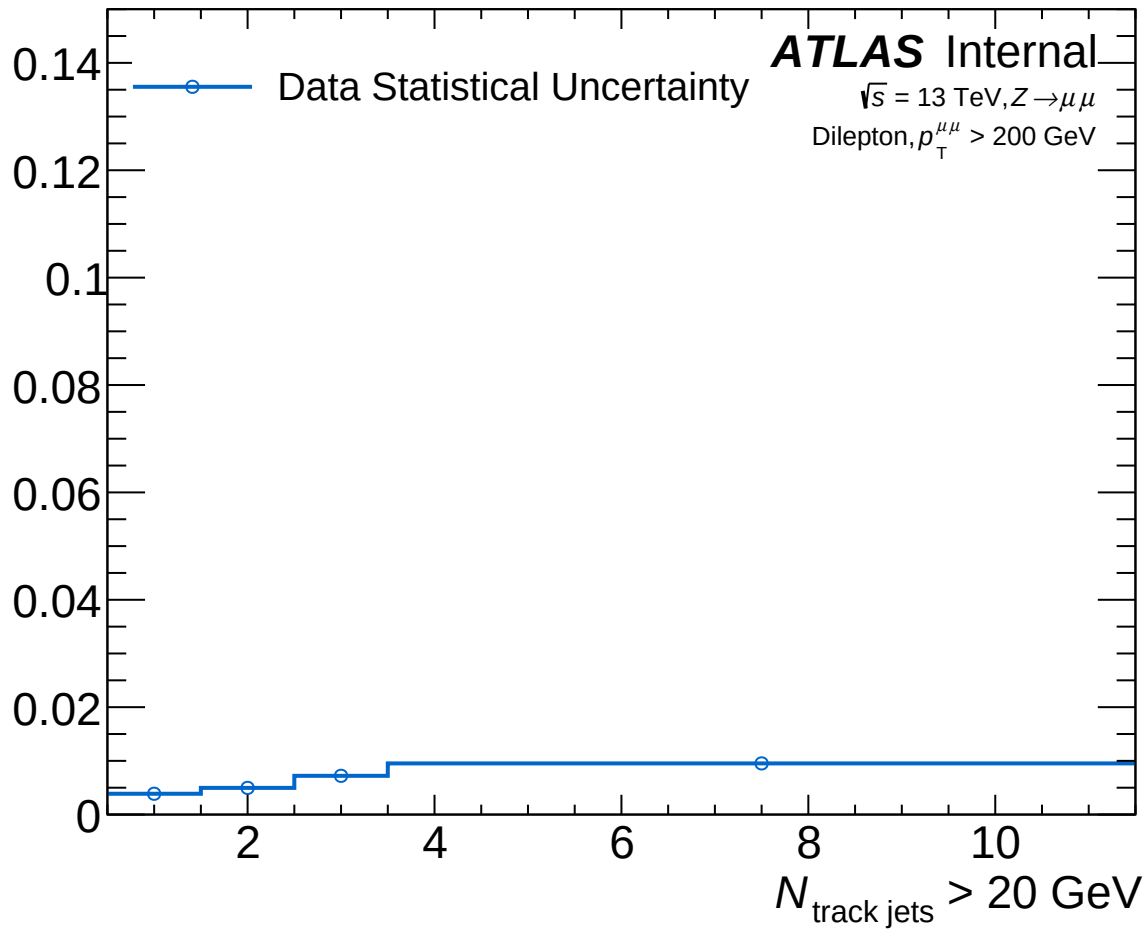
Fractional systematic impact

ATLAS Internal

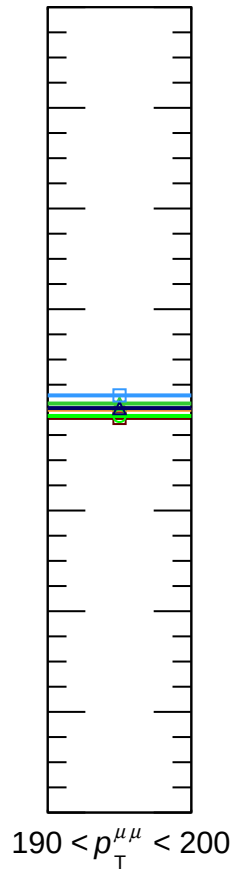
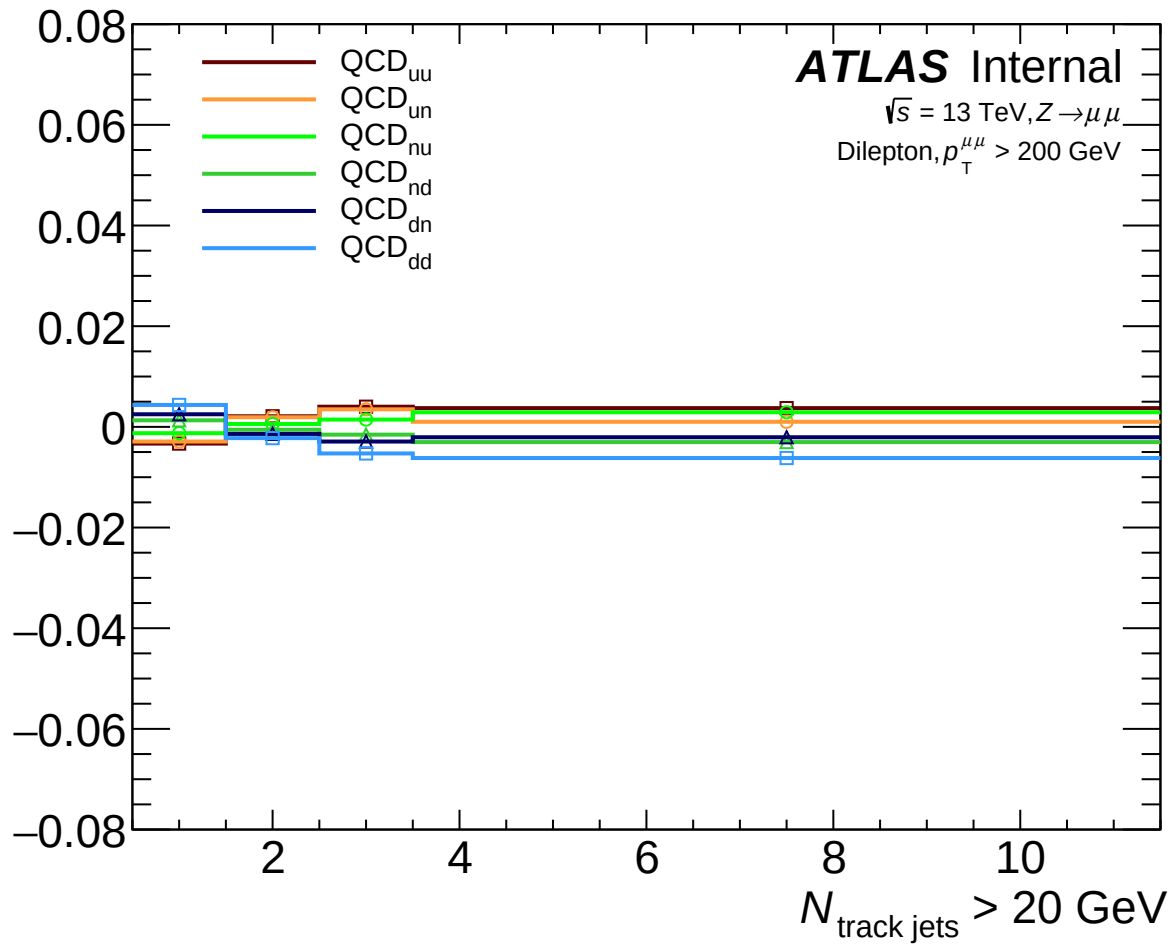
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

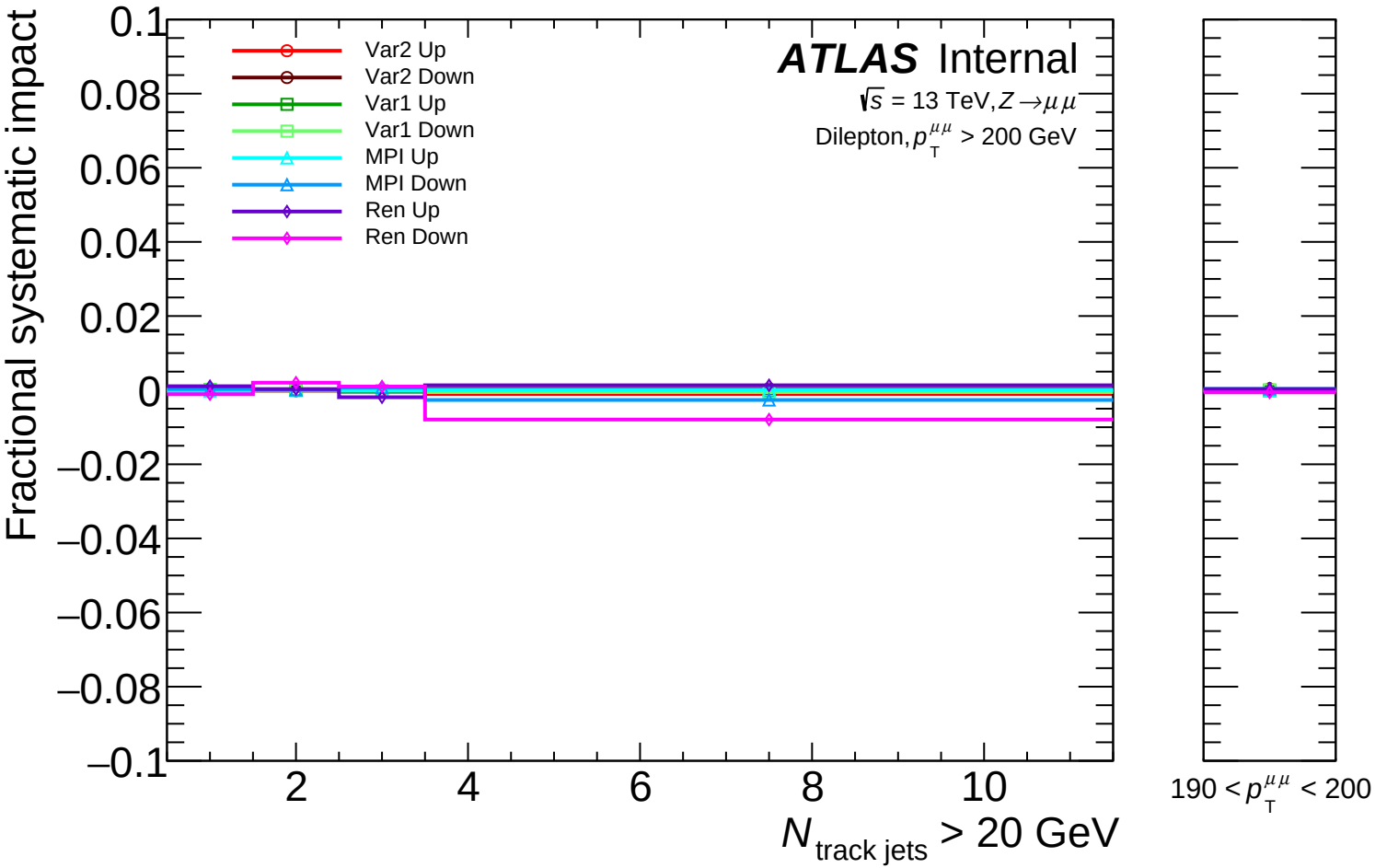
Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

Data Statistical Uncertainty



Fractional systematic impact





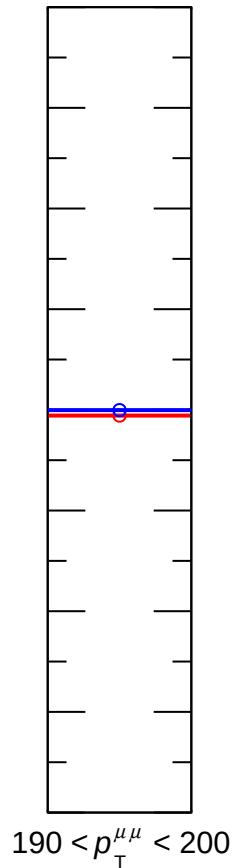
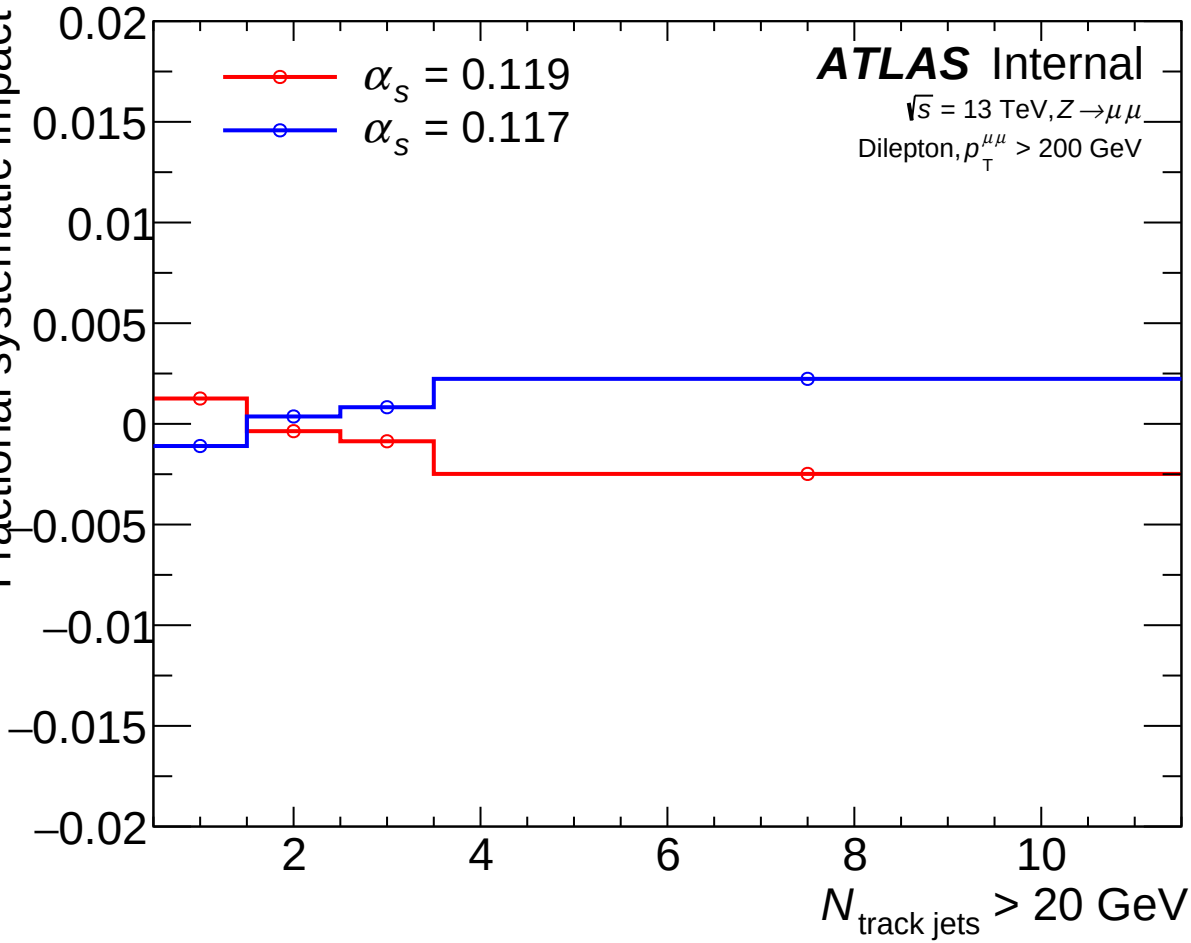
Fractional systematic impact

$\alpha_s = 0.119$
 $\alpha_s = 0.117$

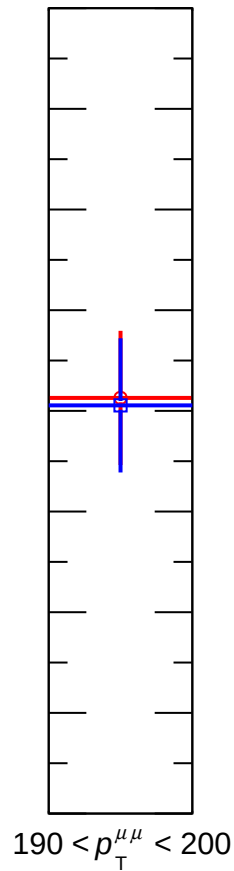
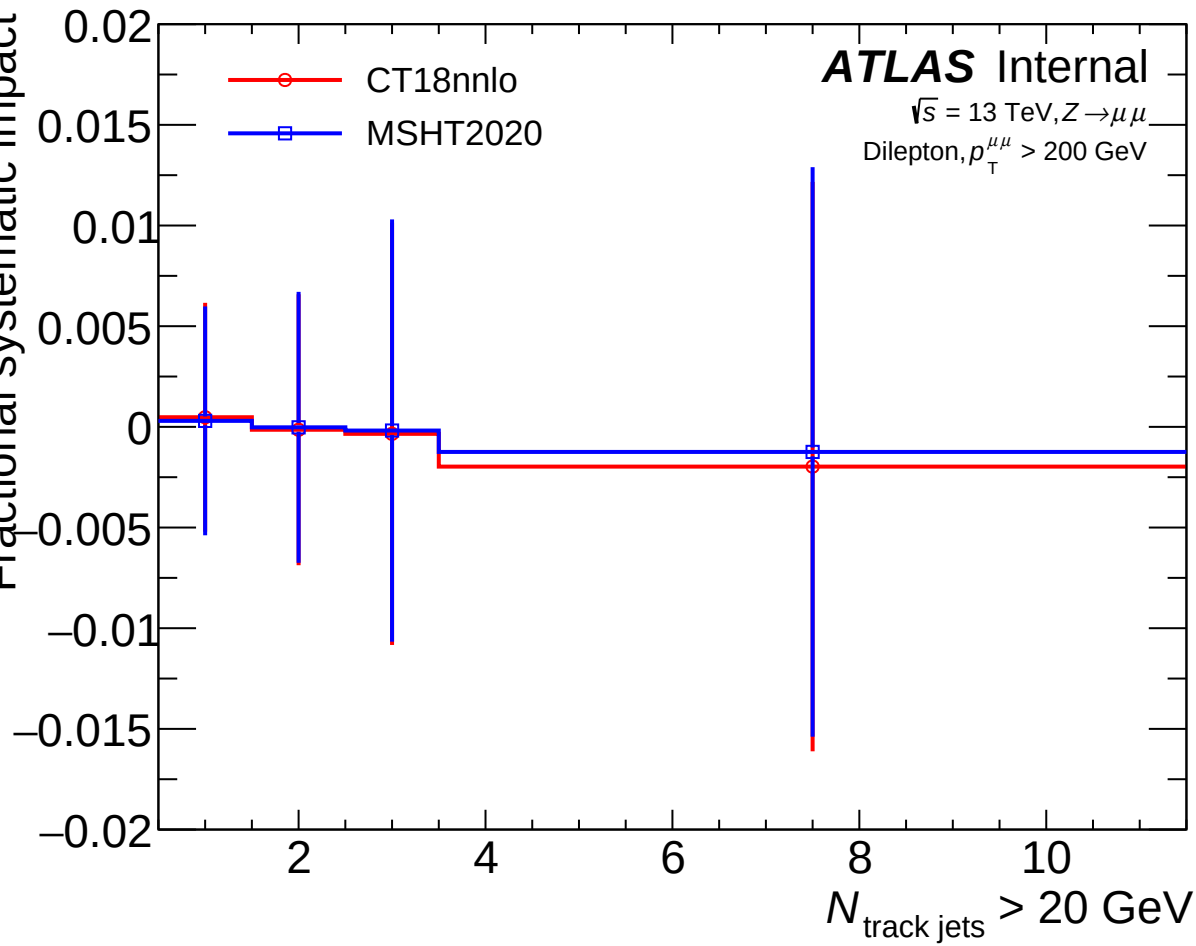
ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200$ GeV



Fractional systematic impact



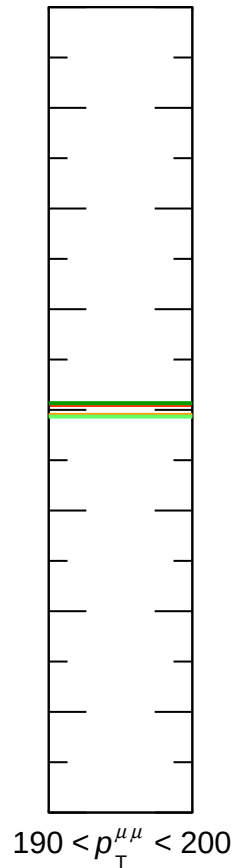
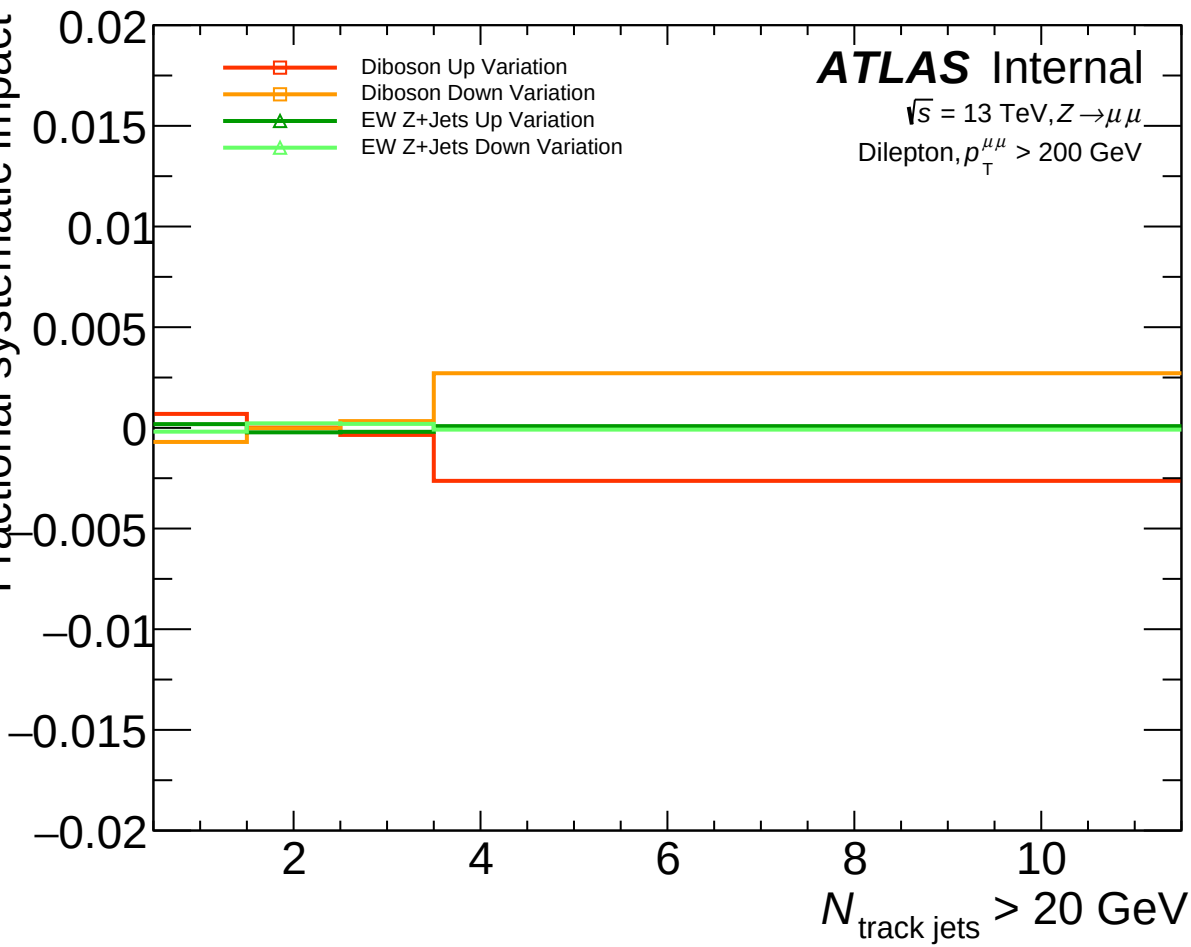
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

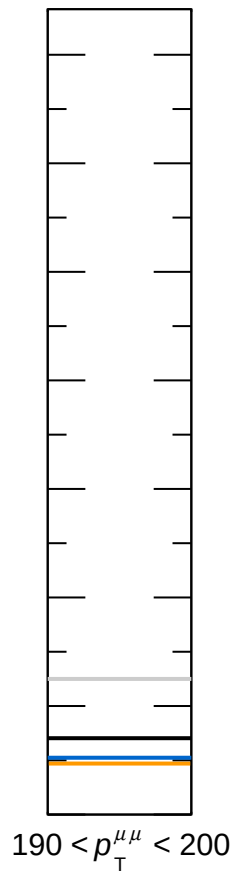
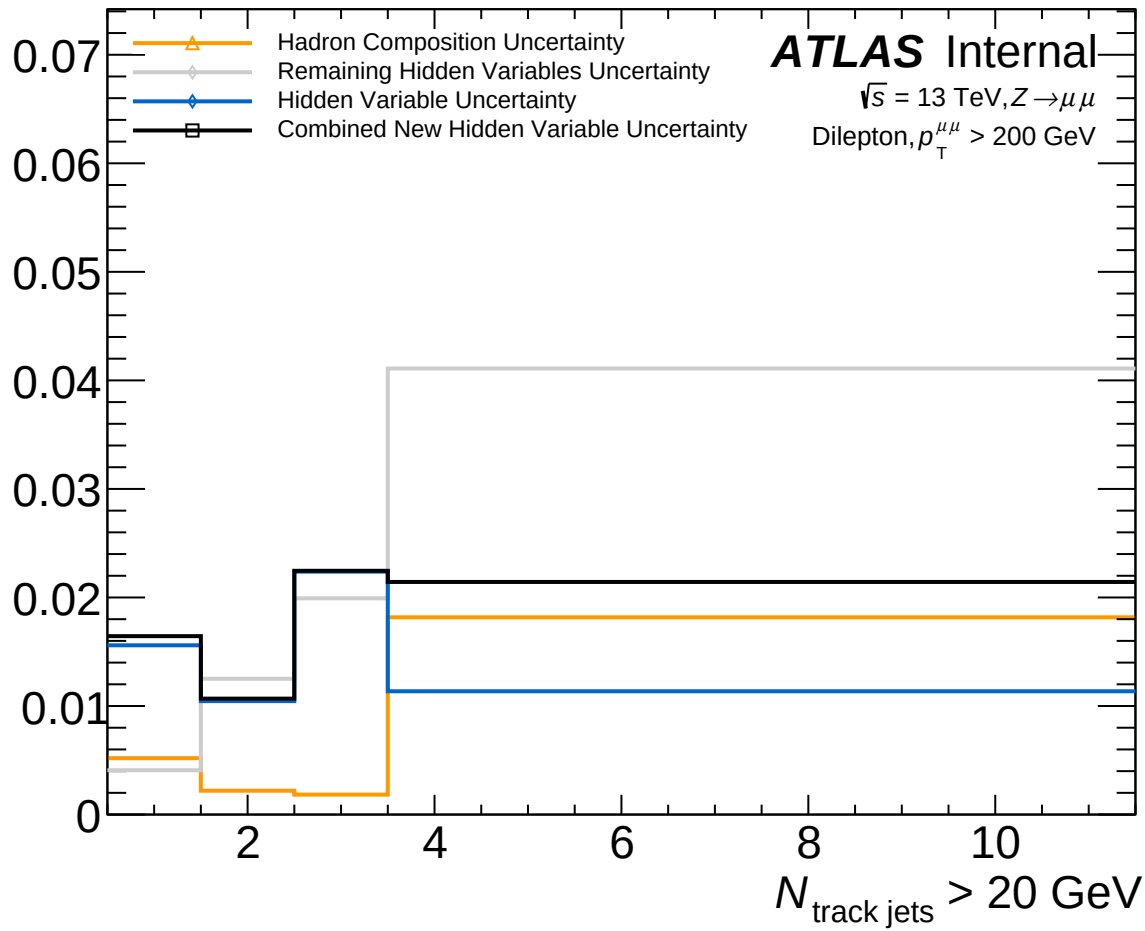
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

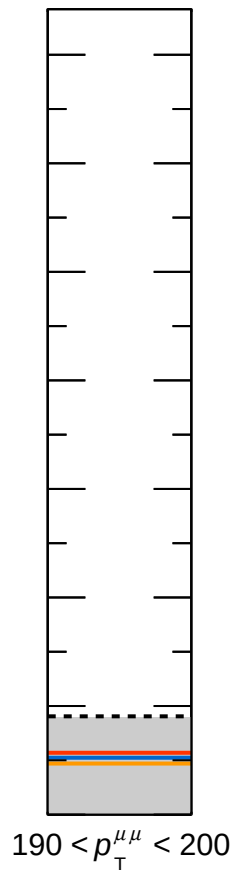
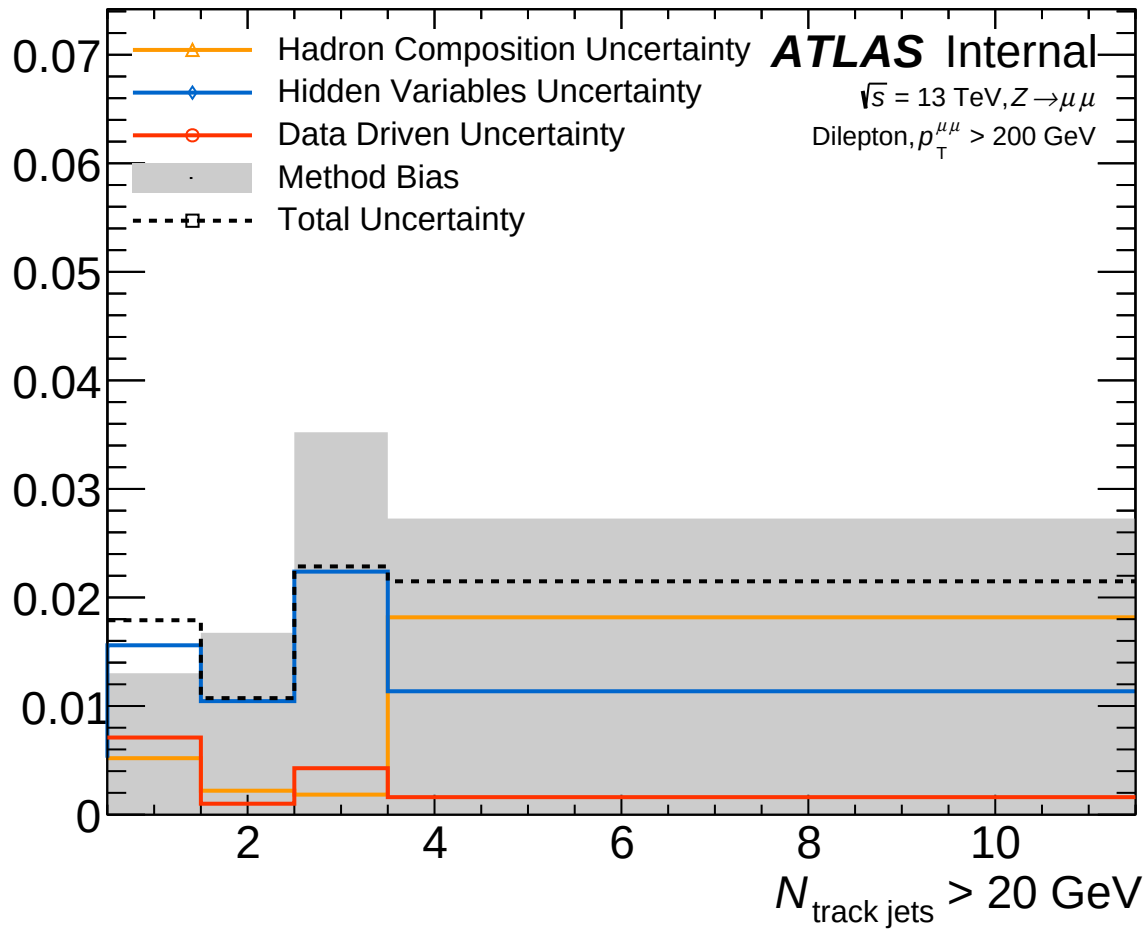


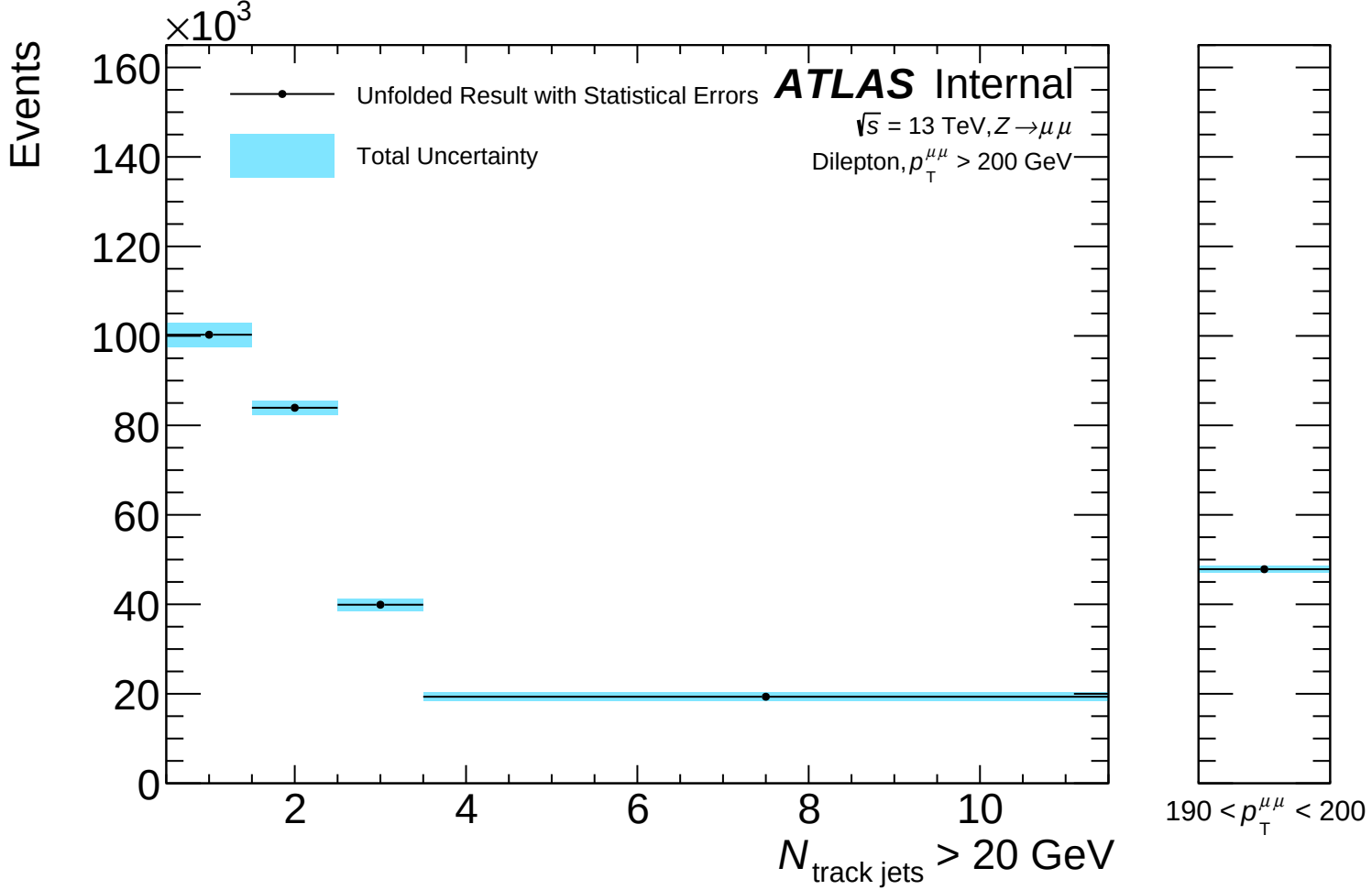
ATLAS Internal

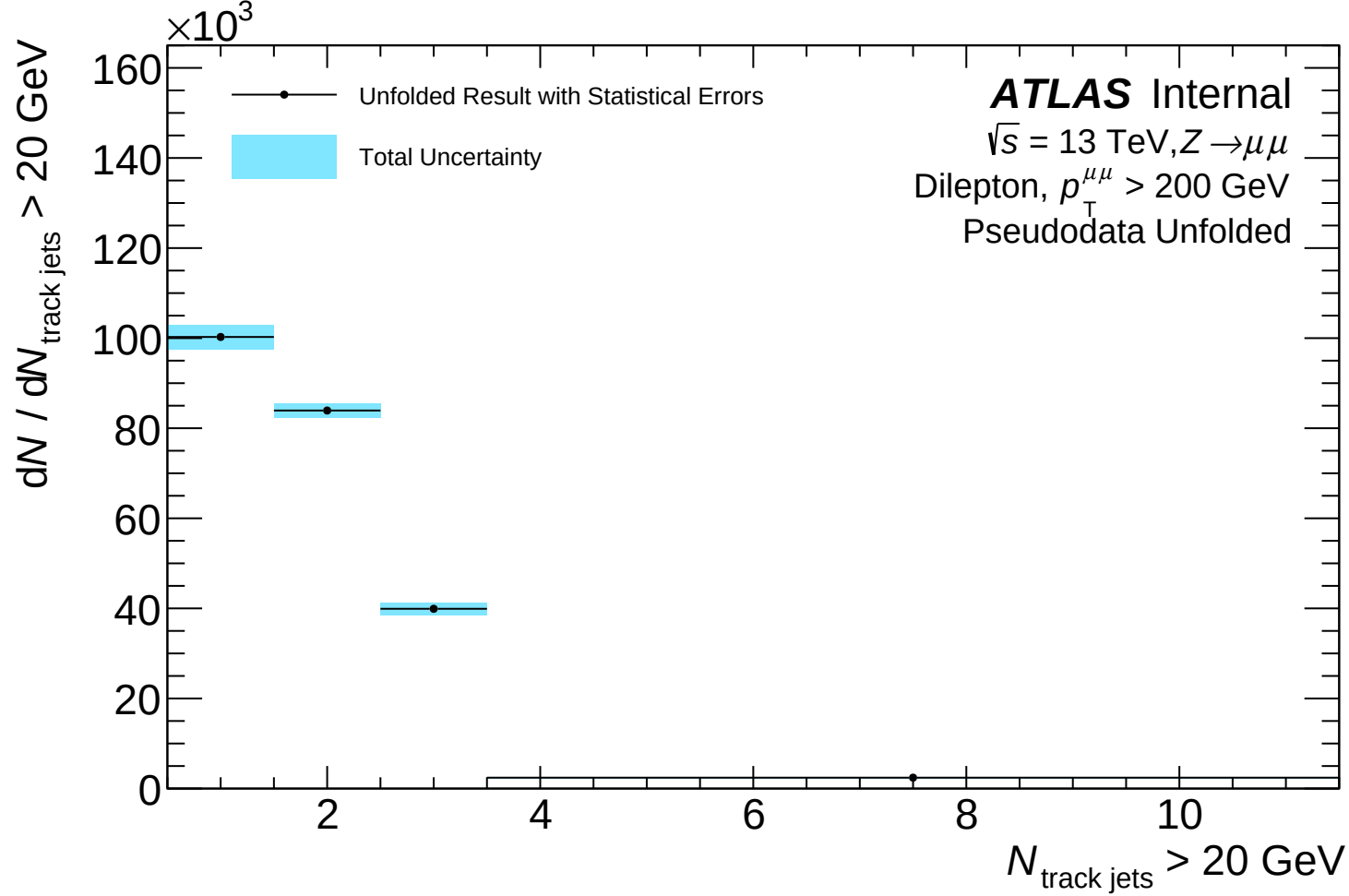
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

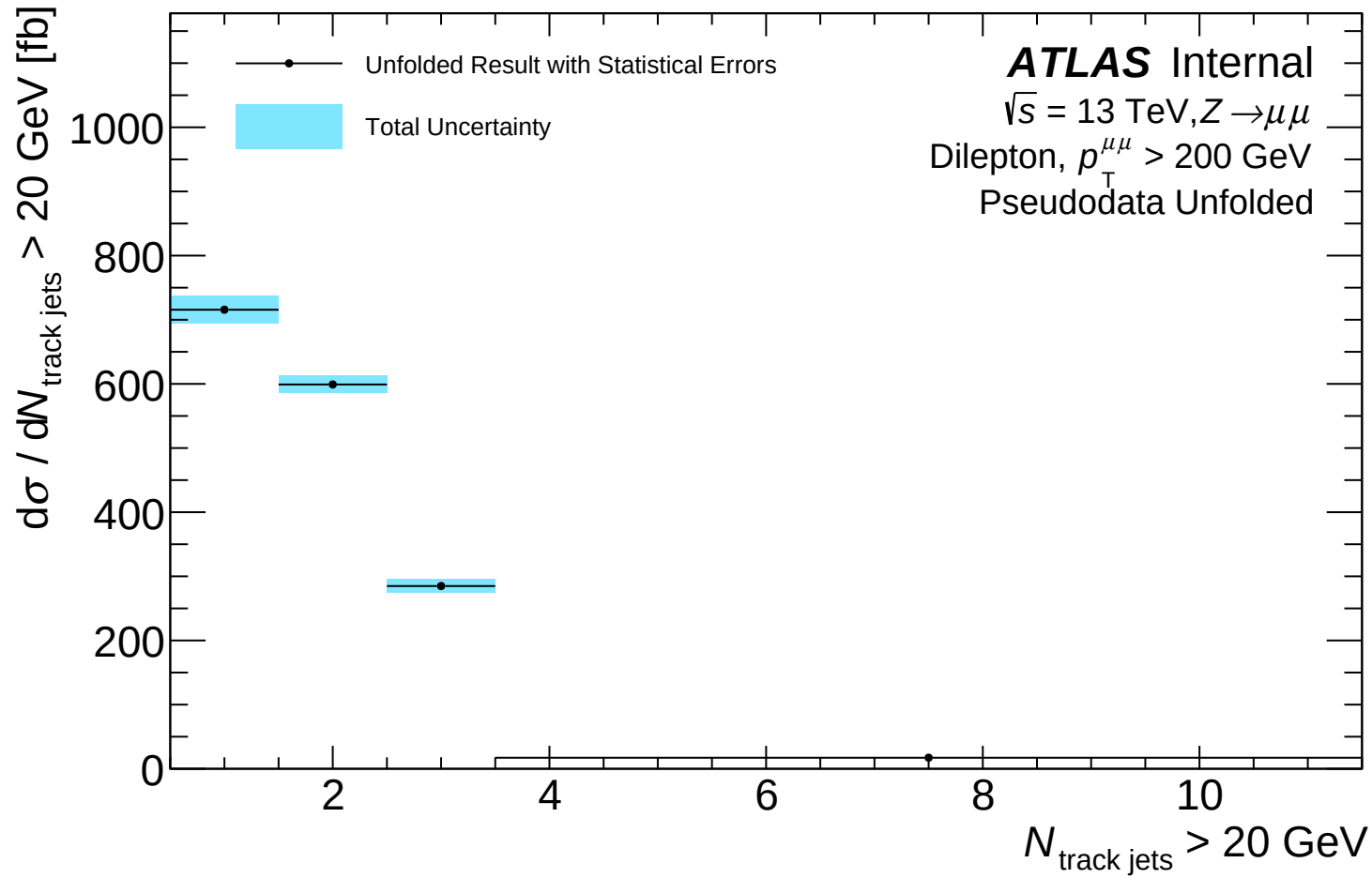
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$











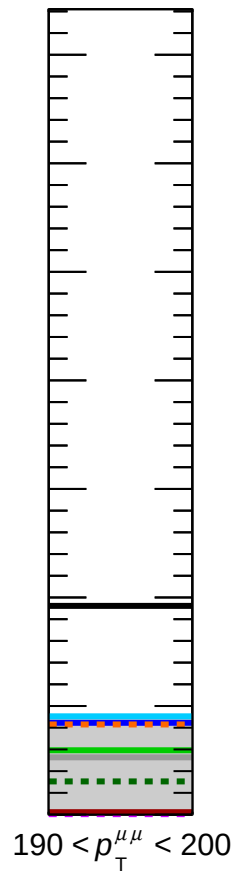
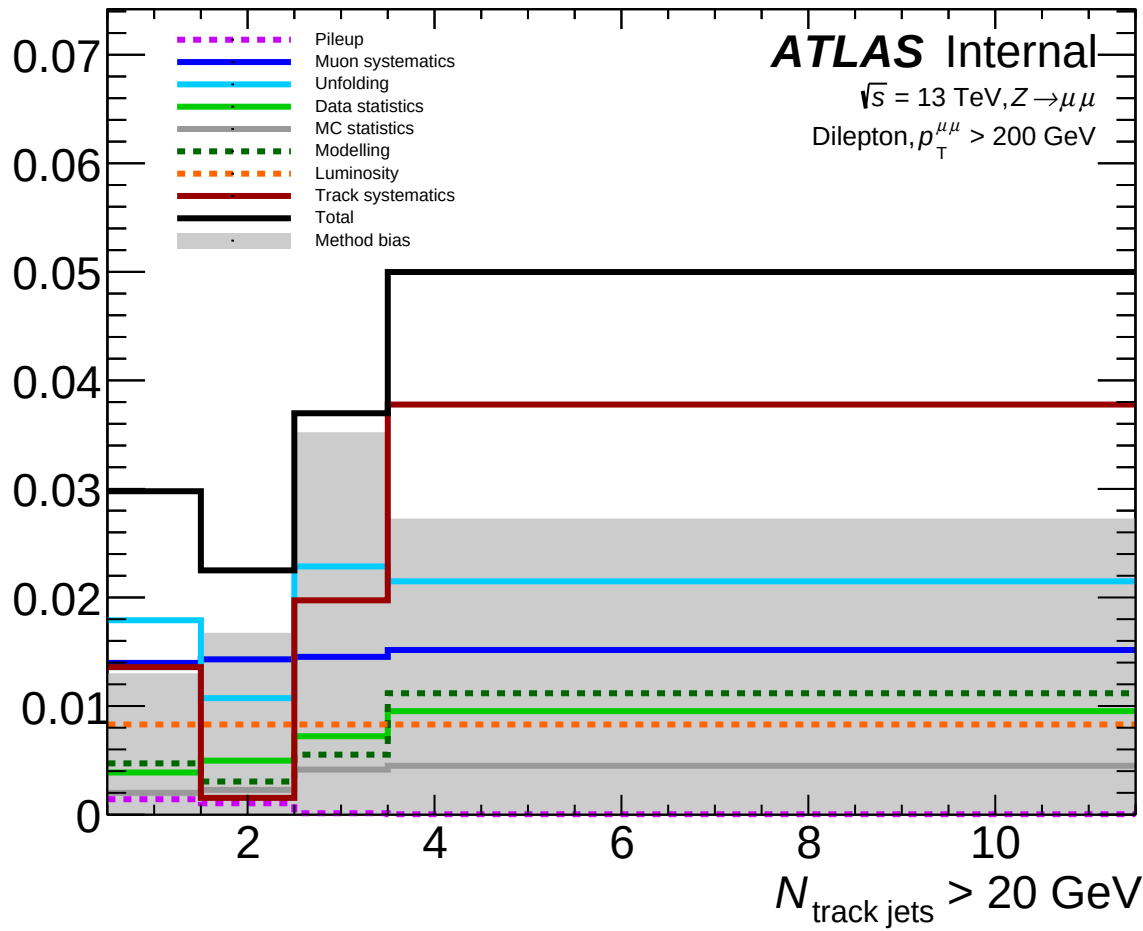
Fractional uncertainty

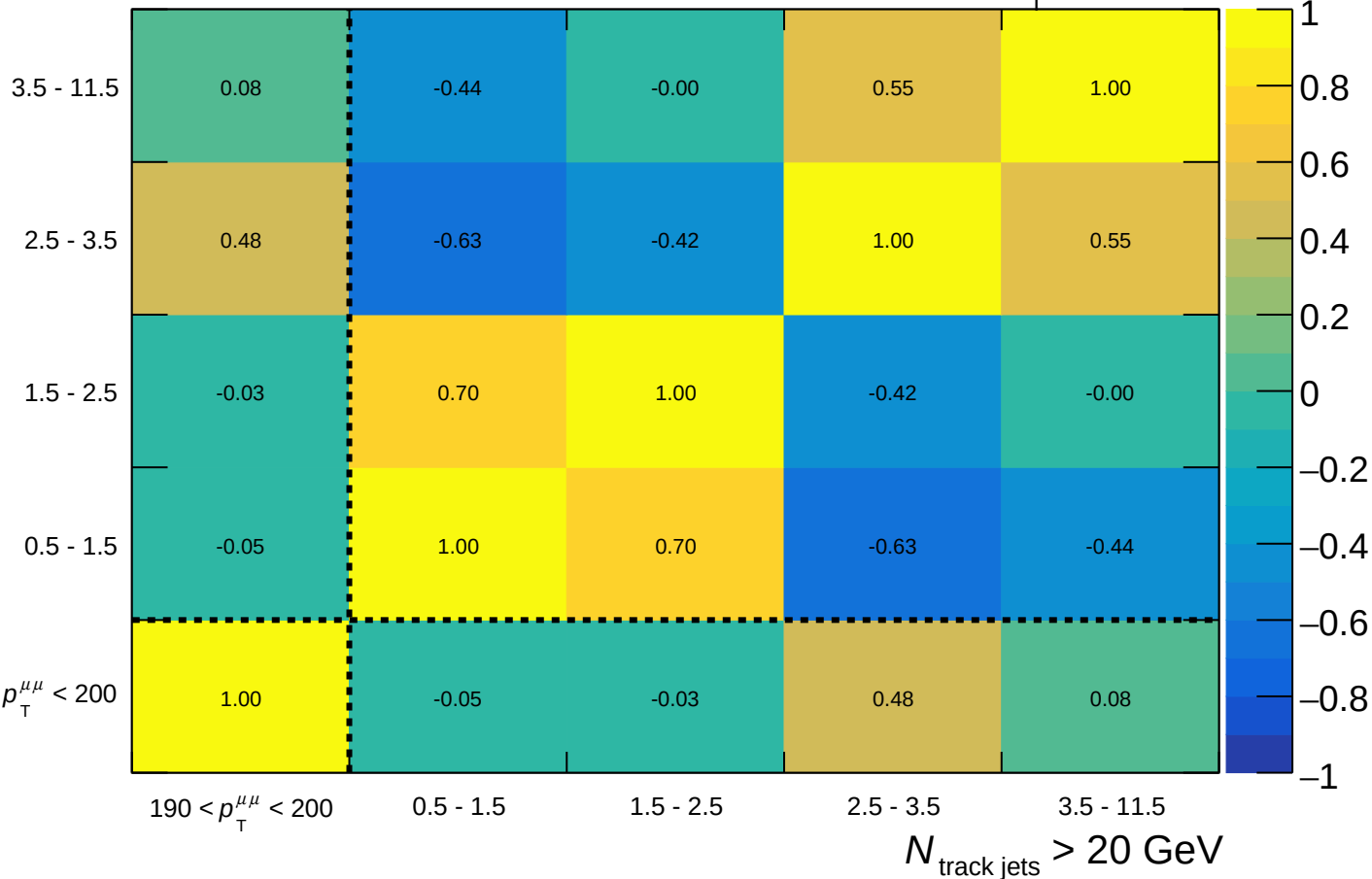
ATLAS Internal

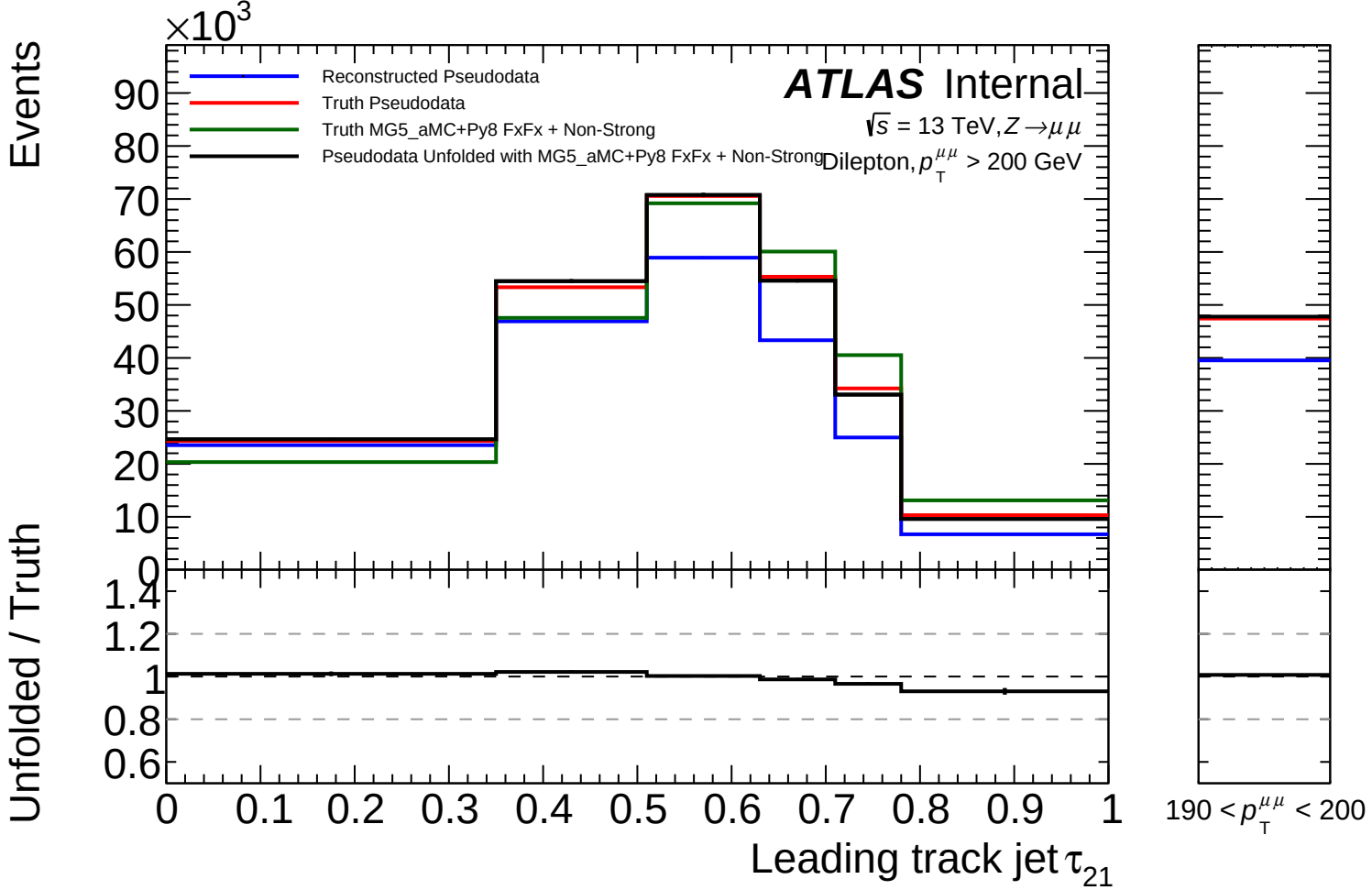
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

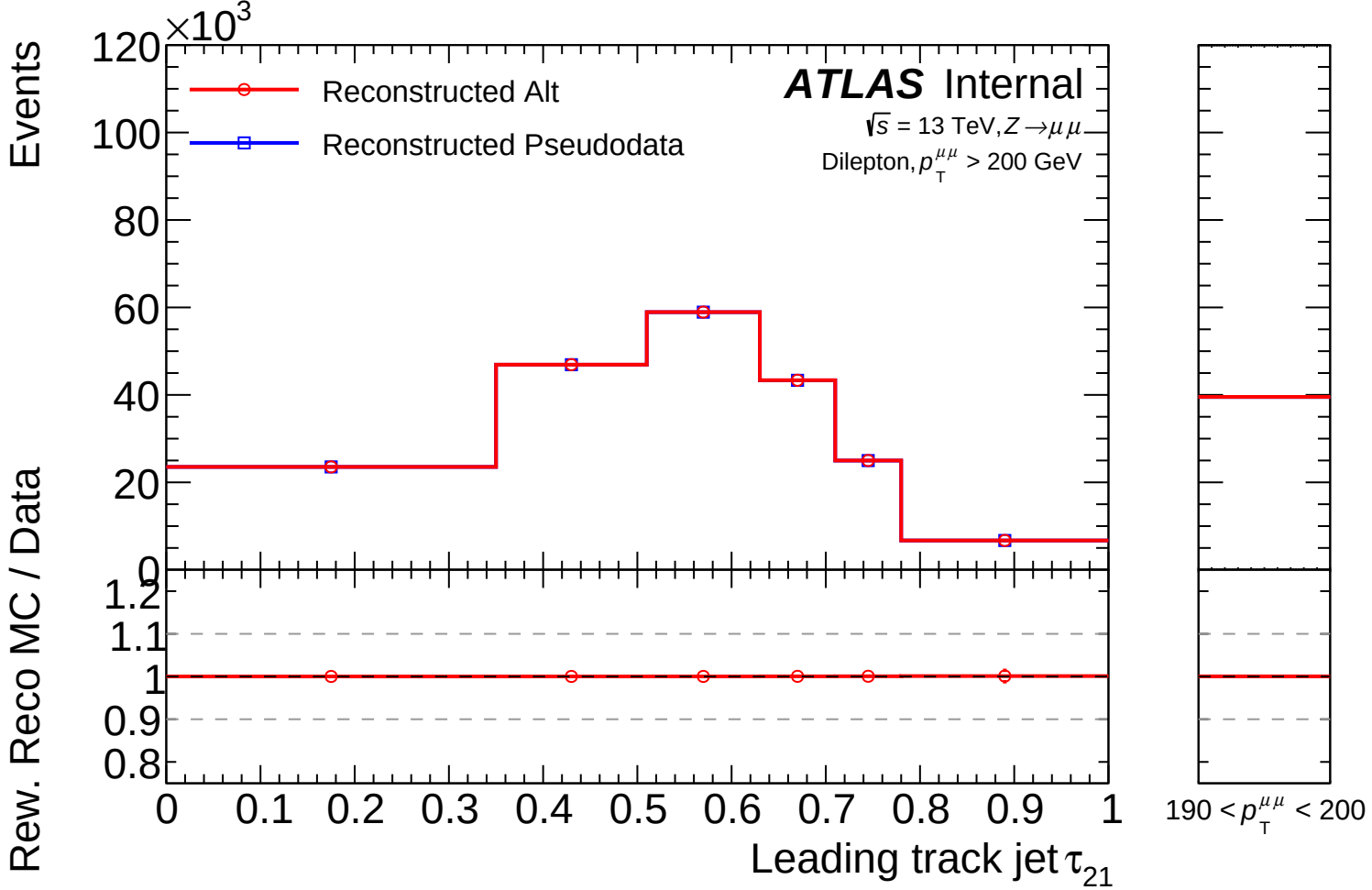
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

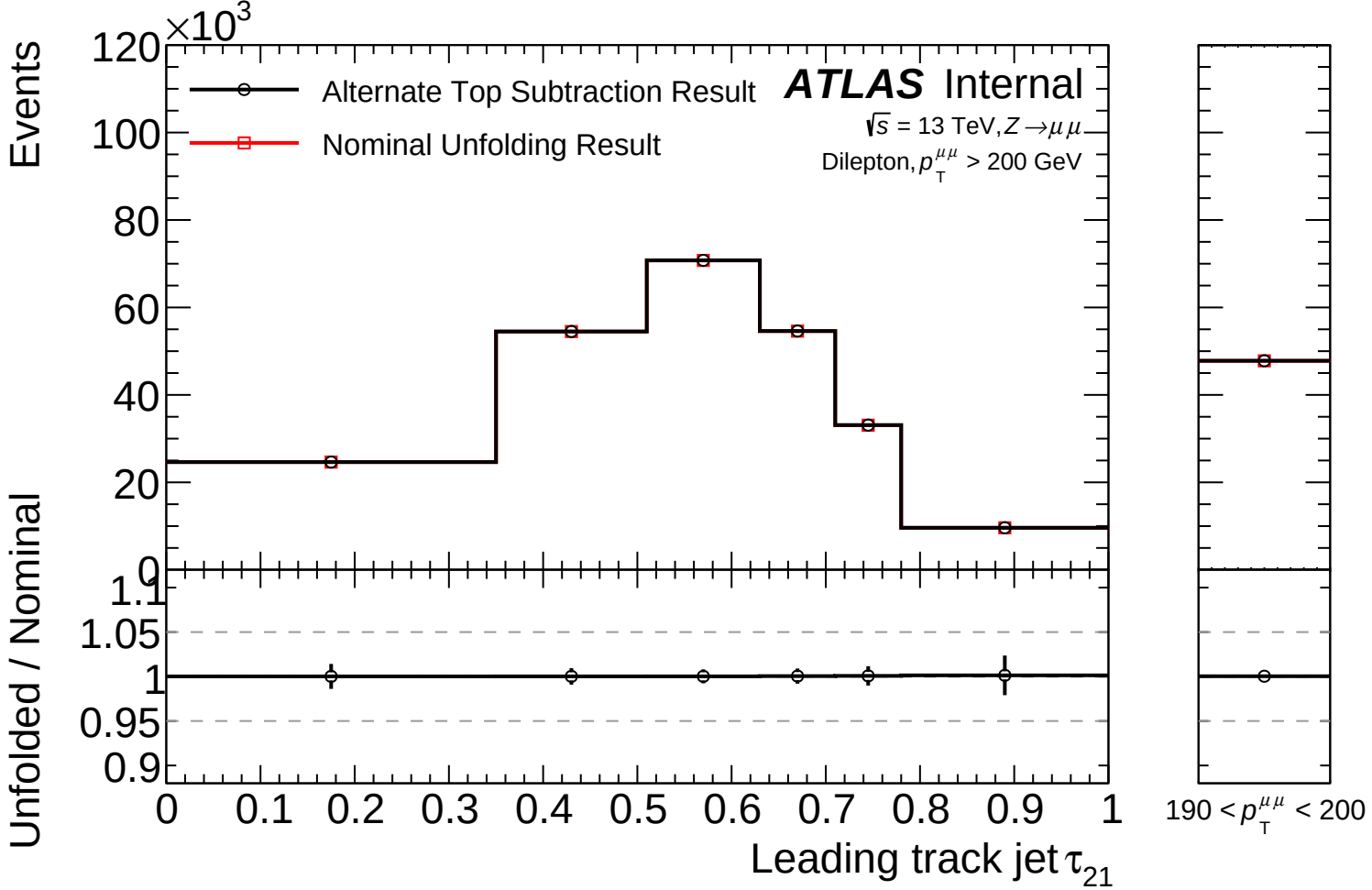
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $N_{\text{track jets}} > 20 \text{ GeV}$ 







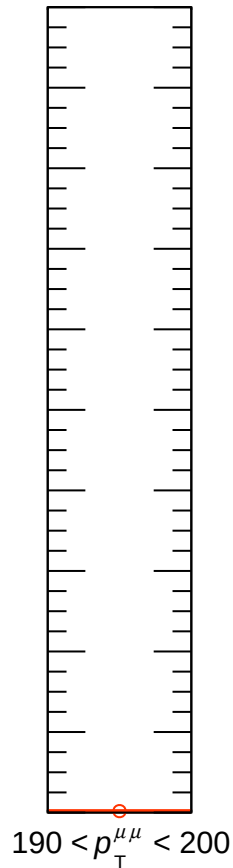
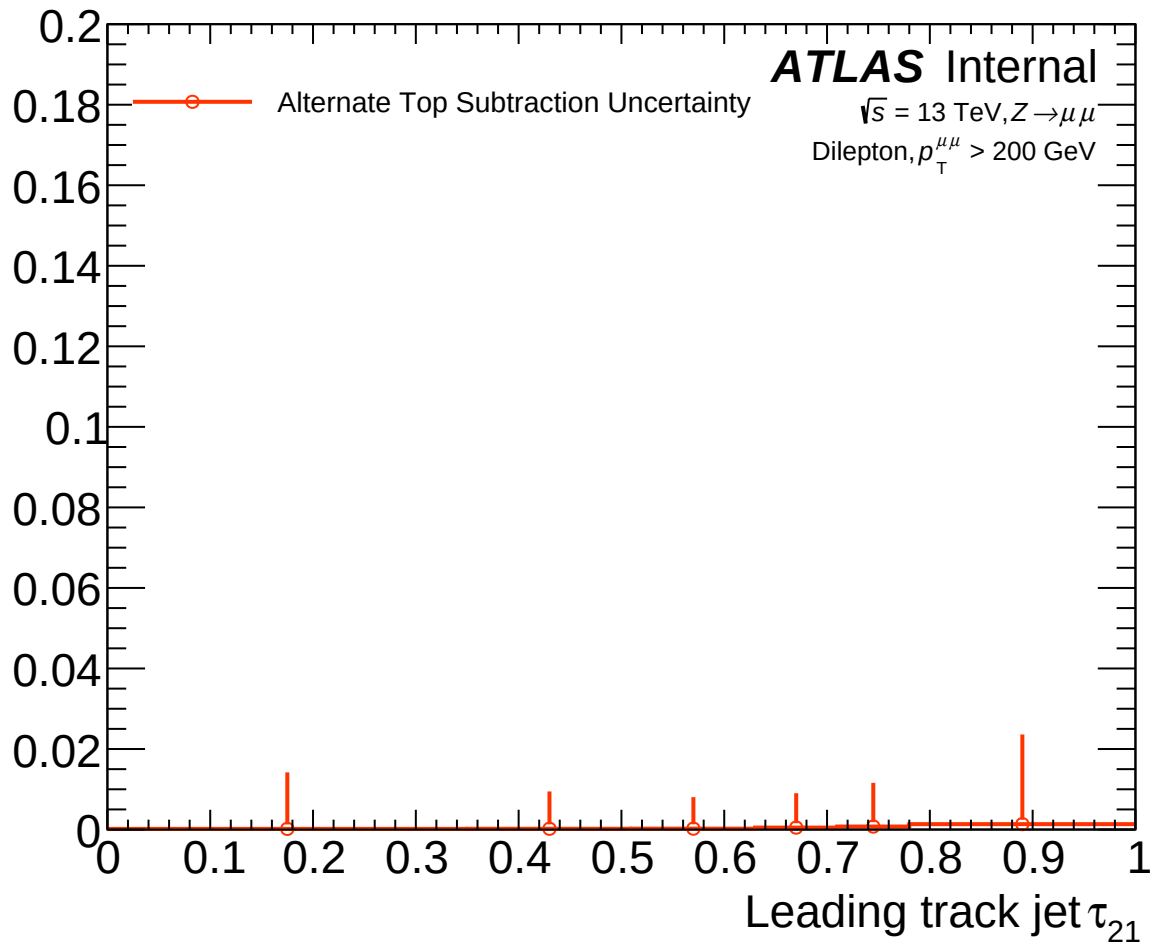
Fractional systematic impact

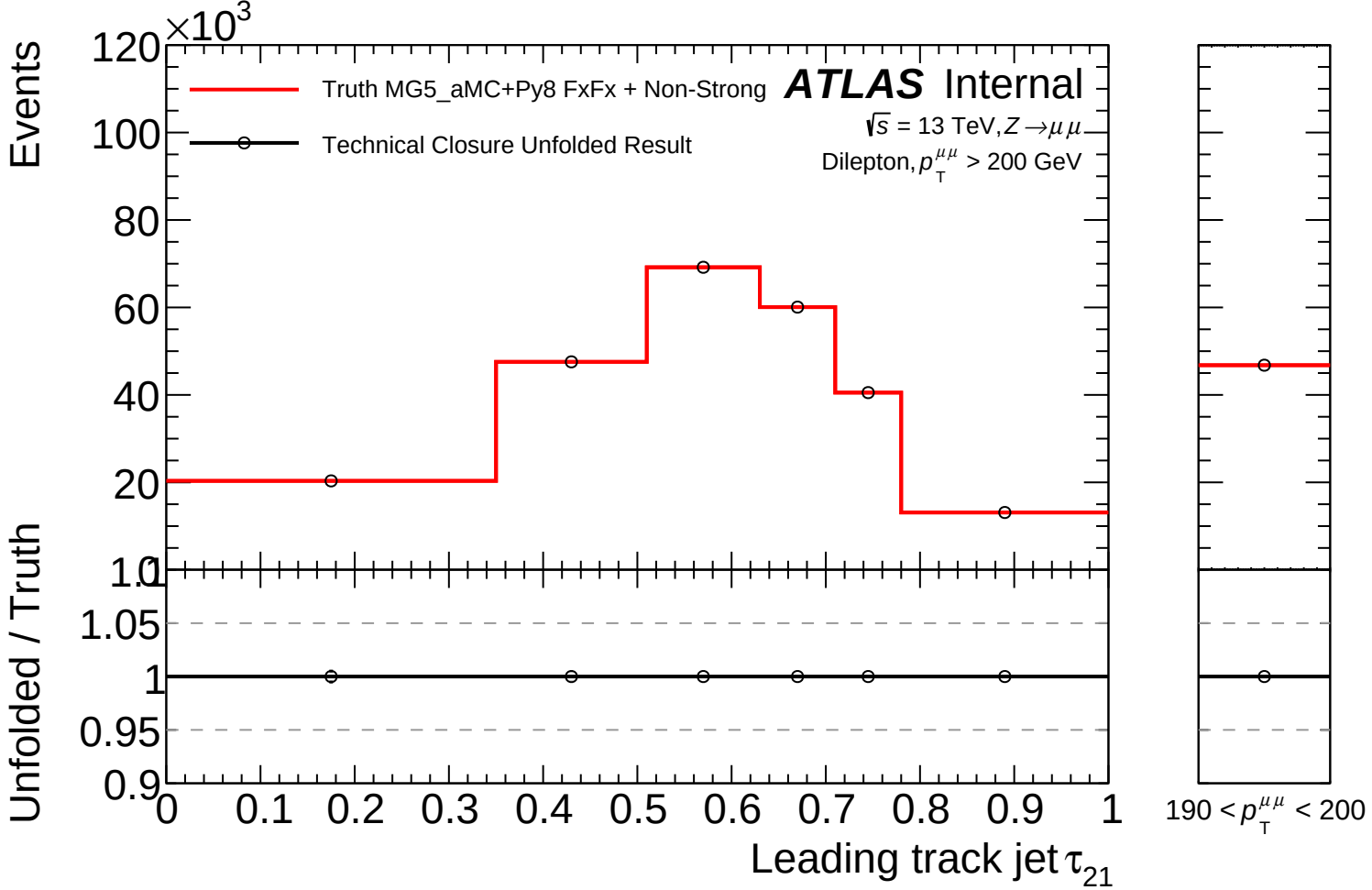
ATLAS Internal

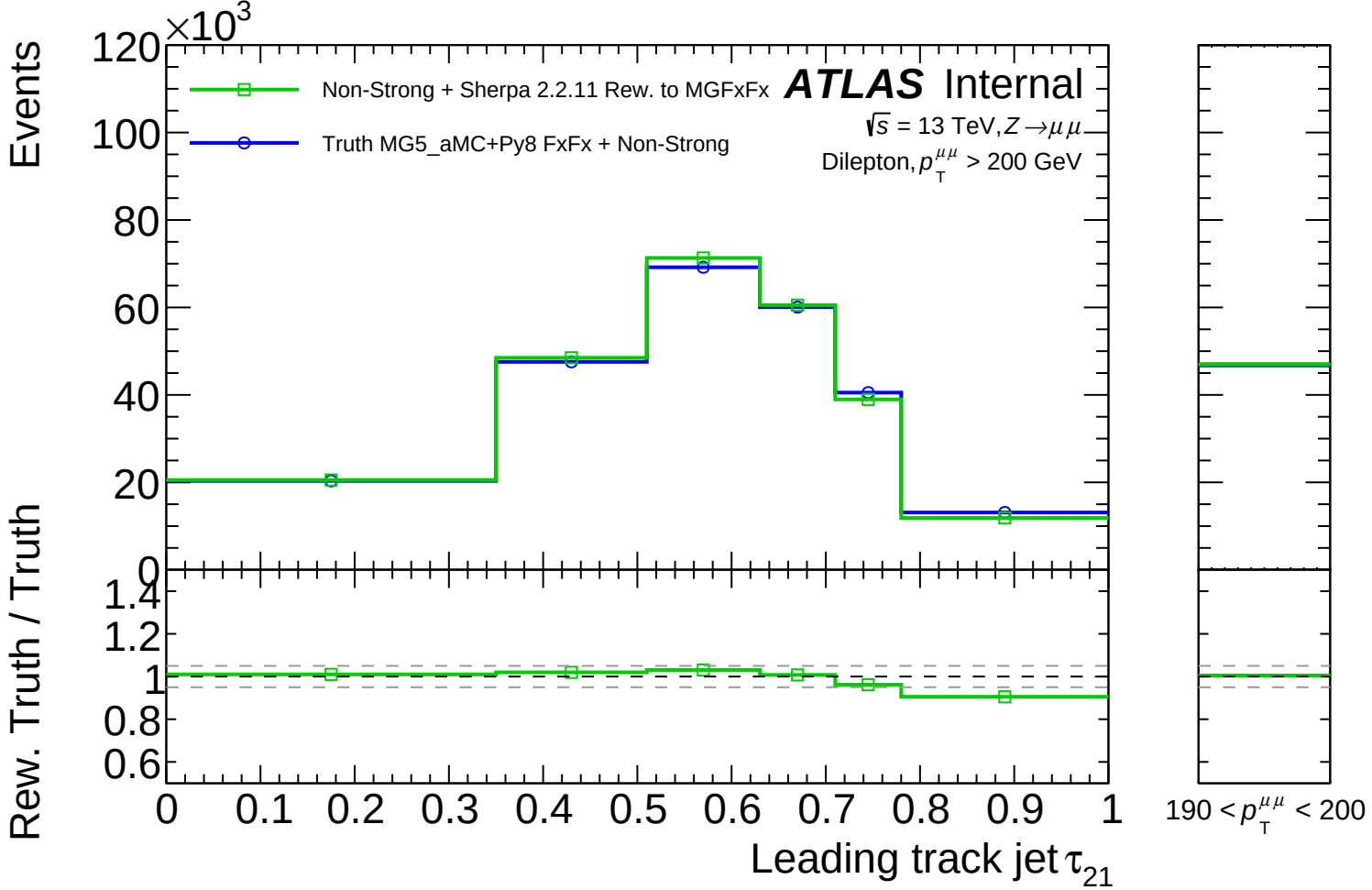
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

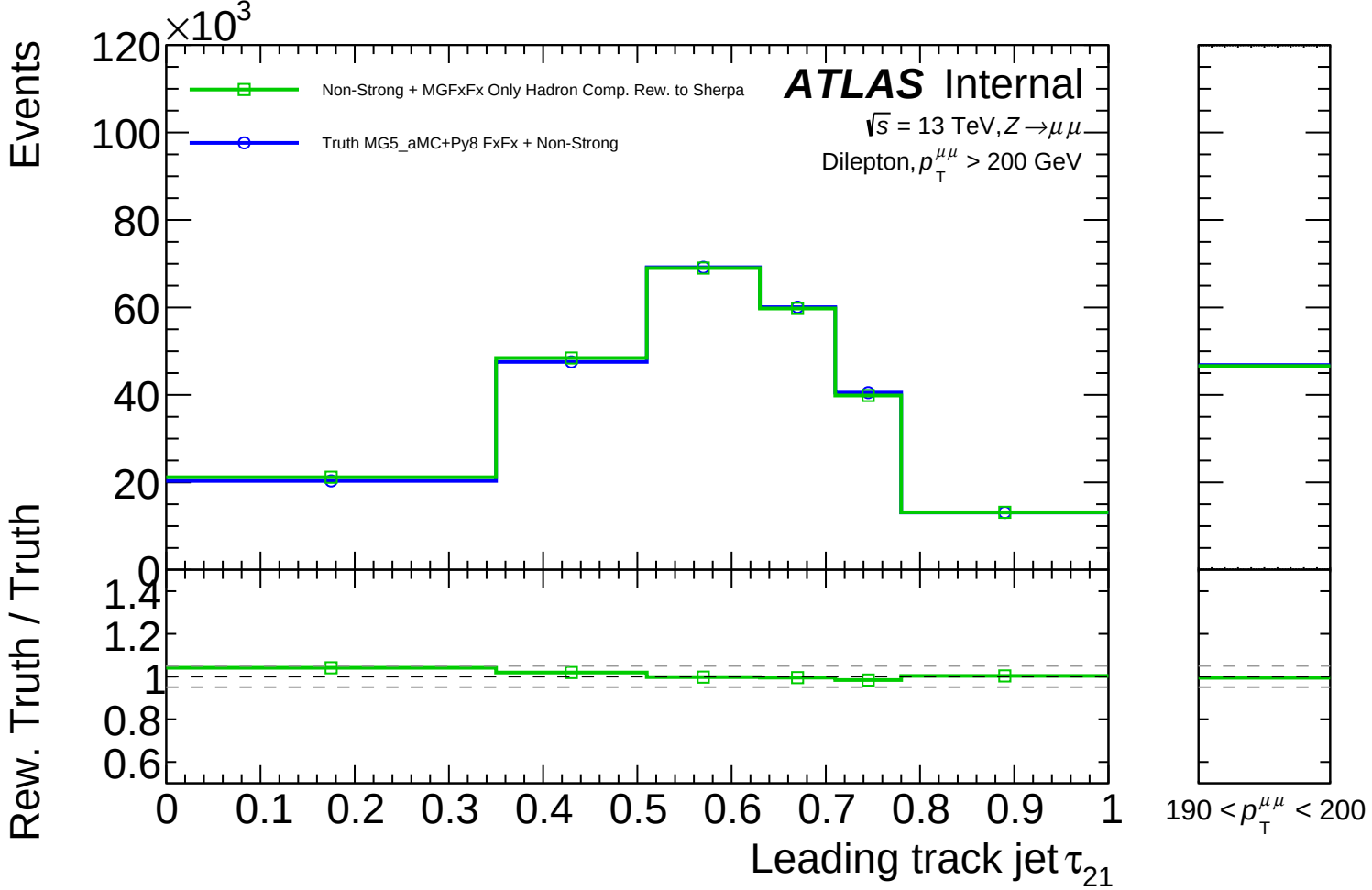
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

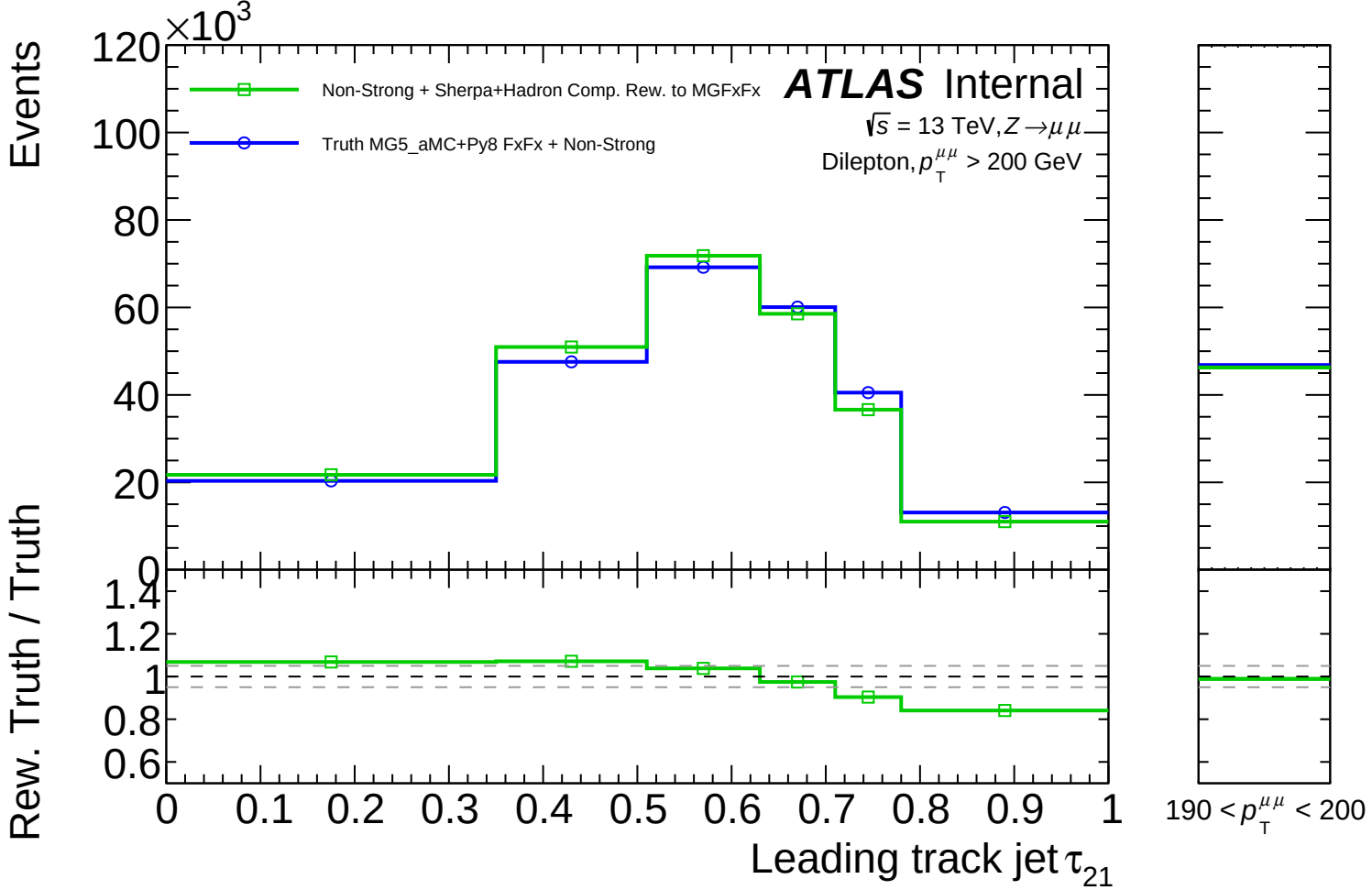
Alternate Top Subtraction Uncertainty

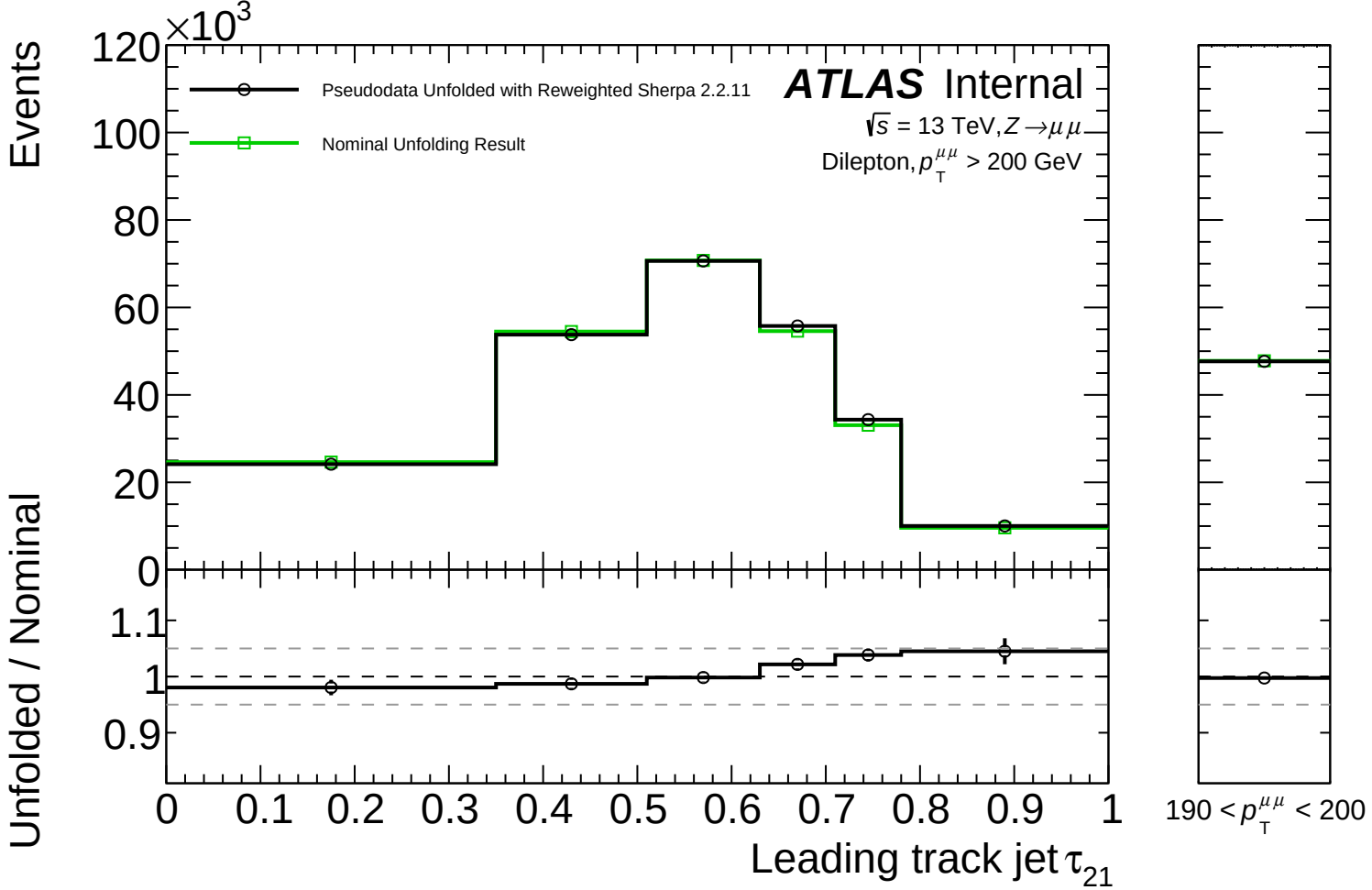


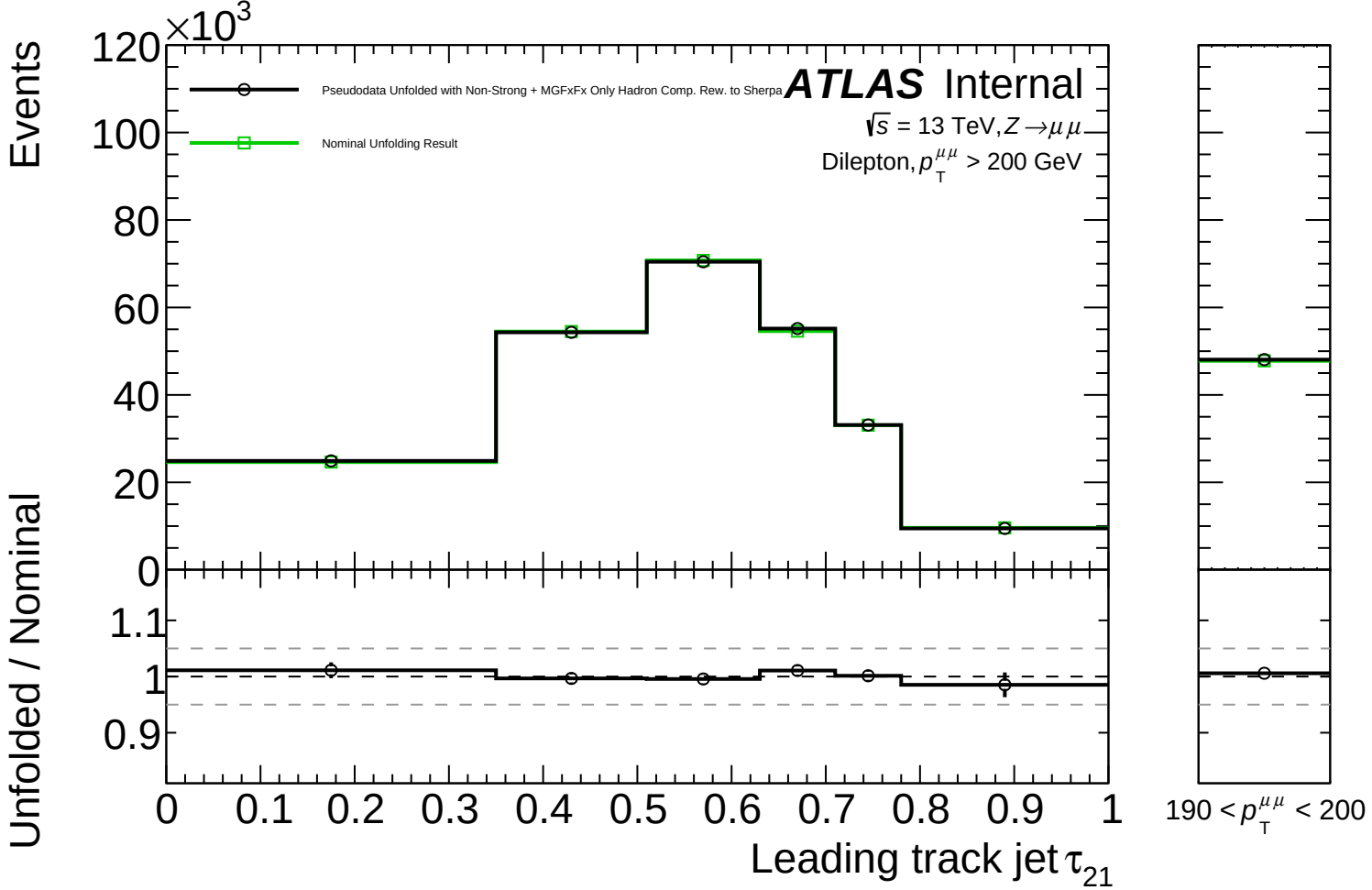


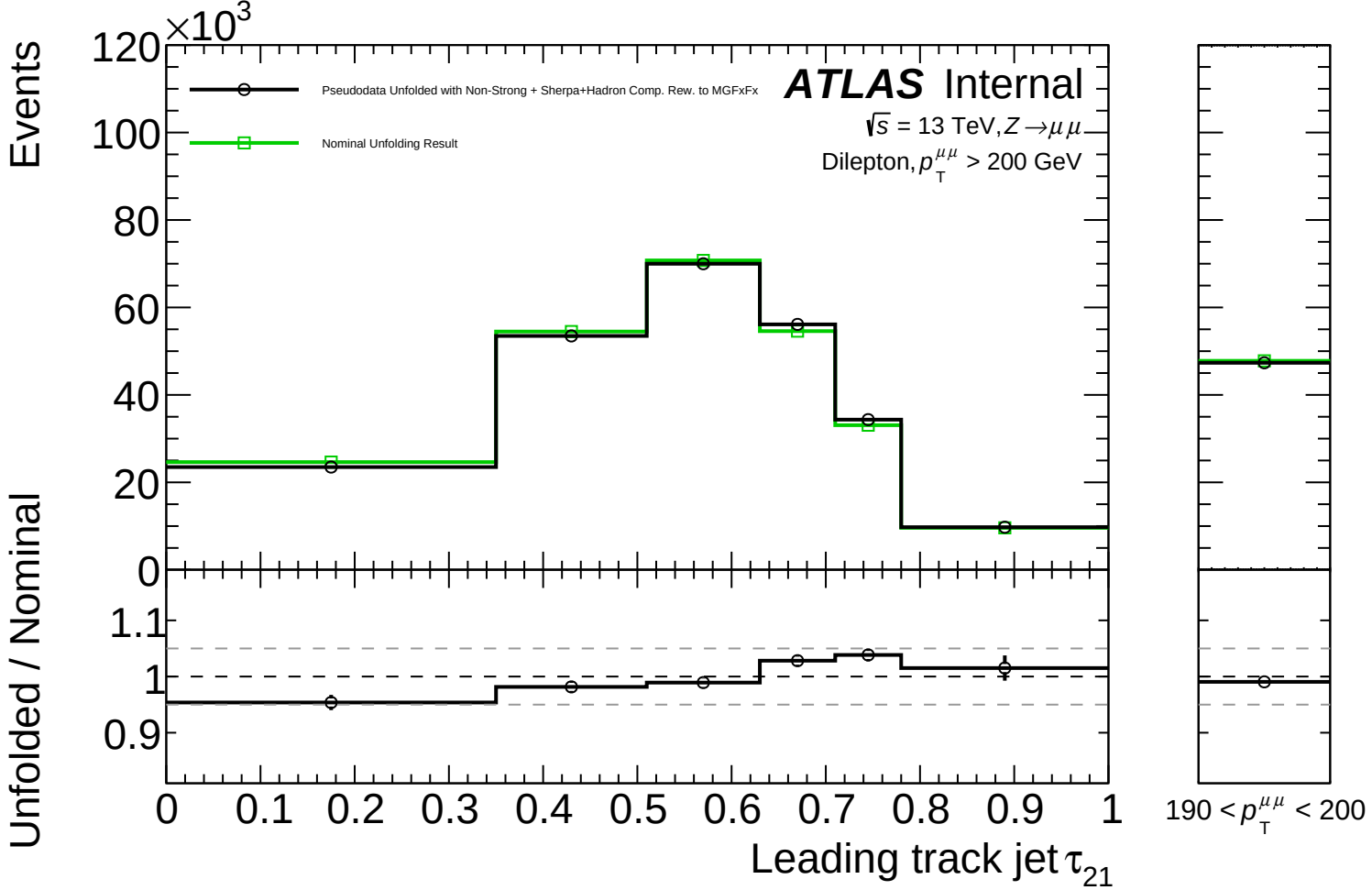


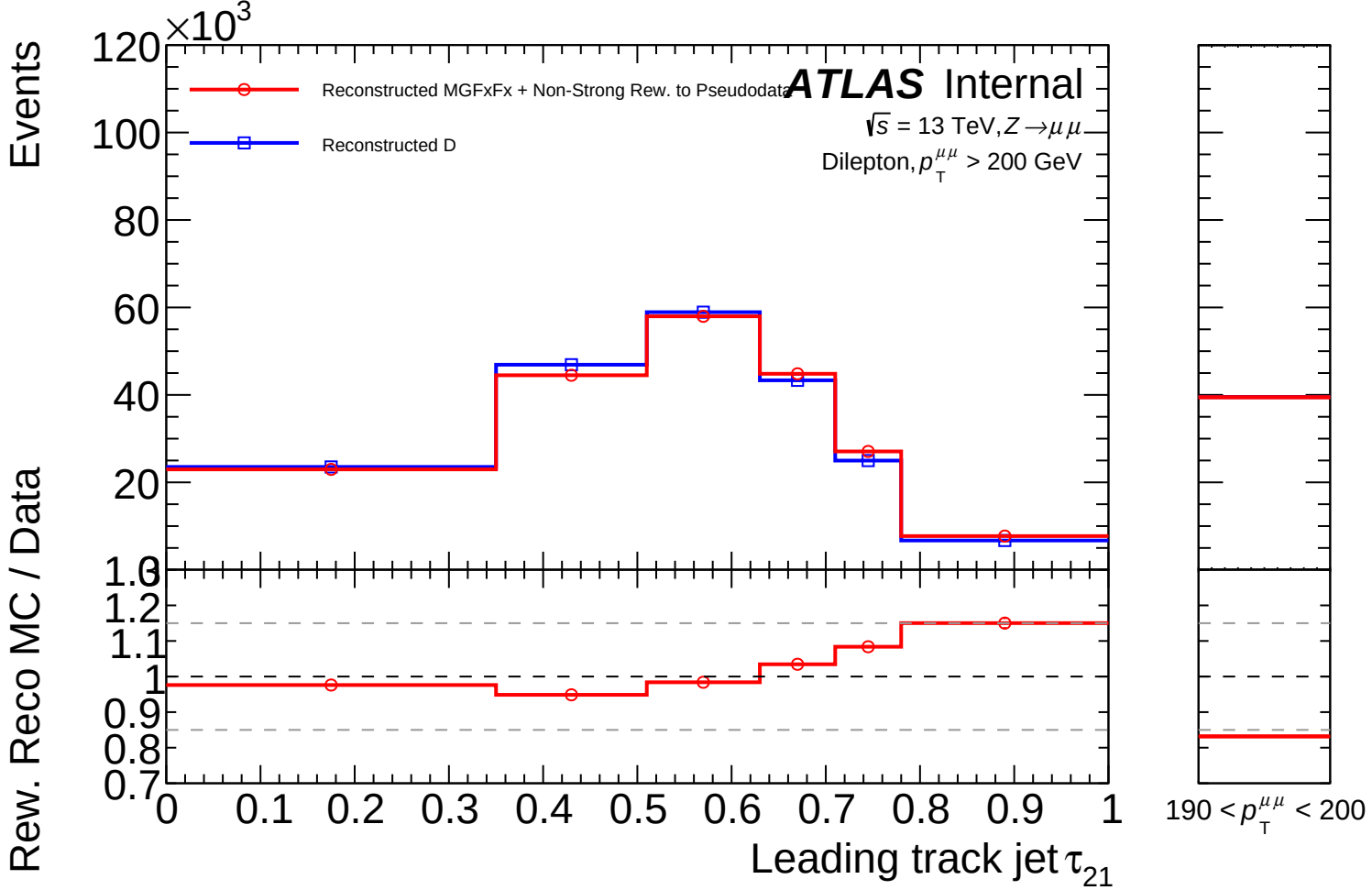


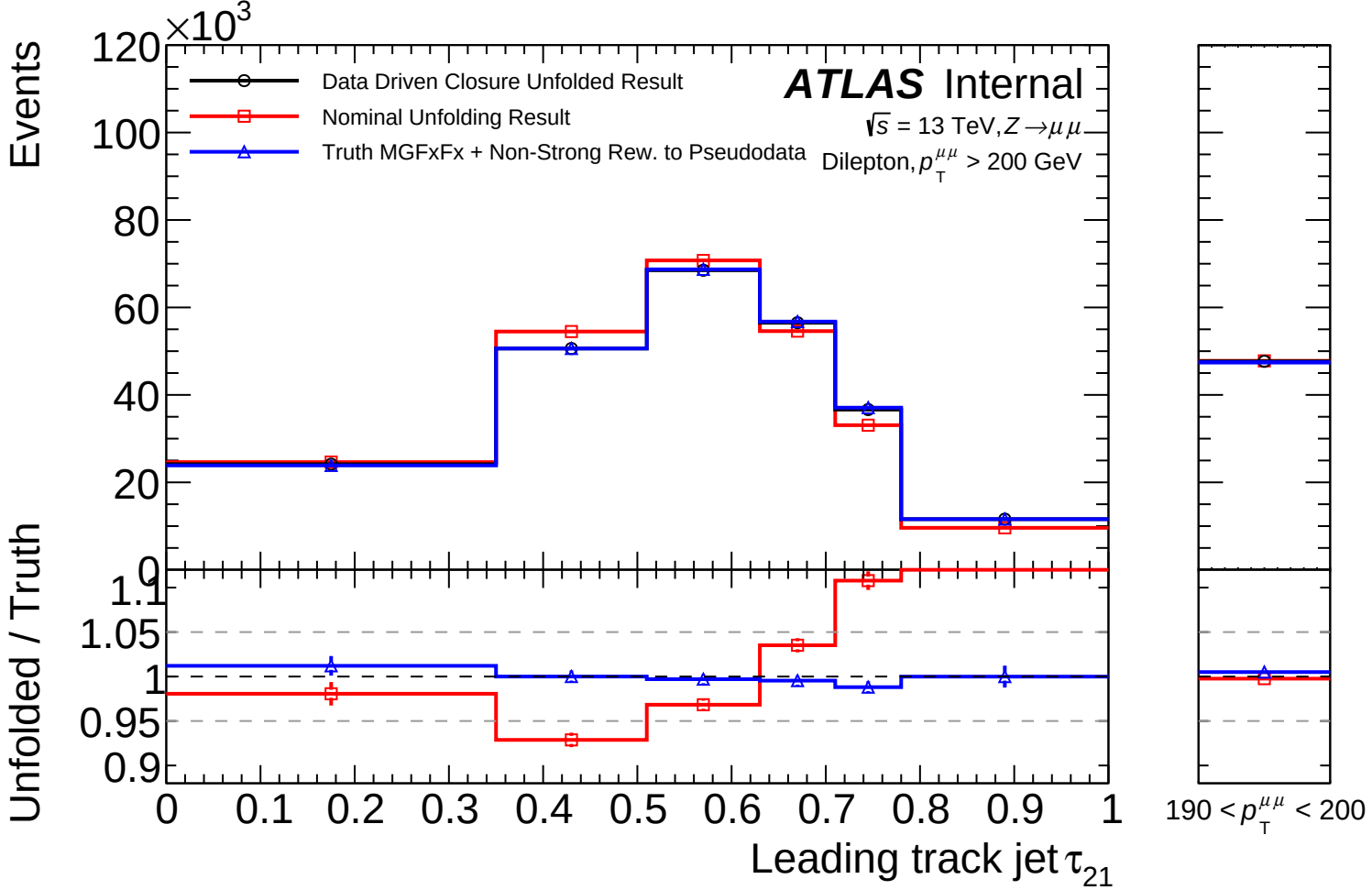




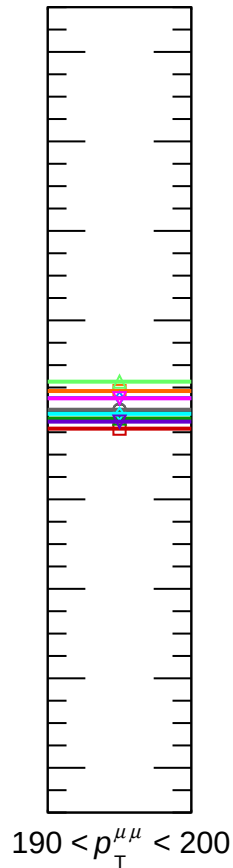
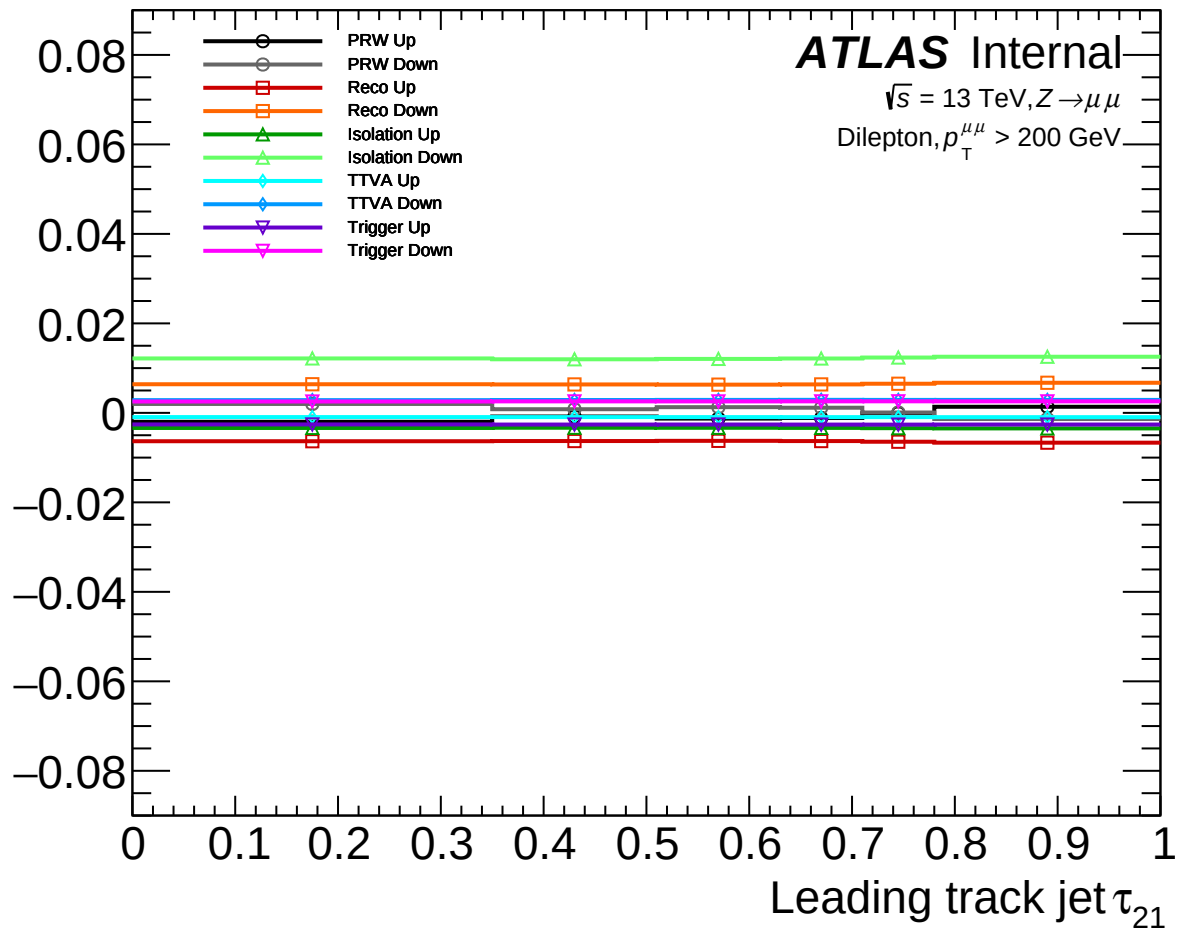




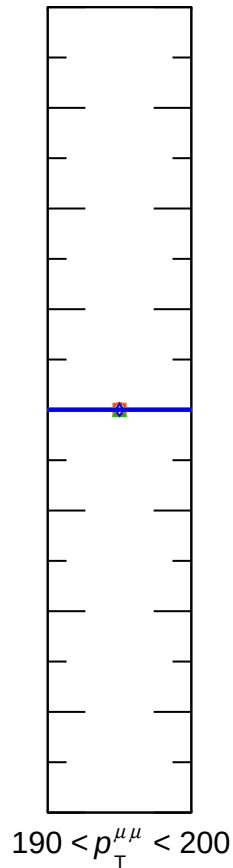
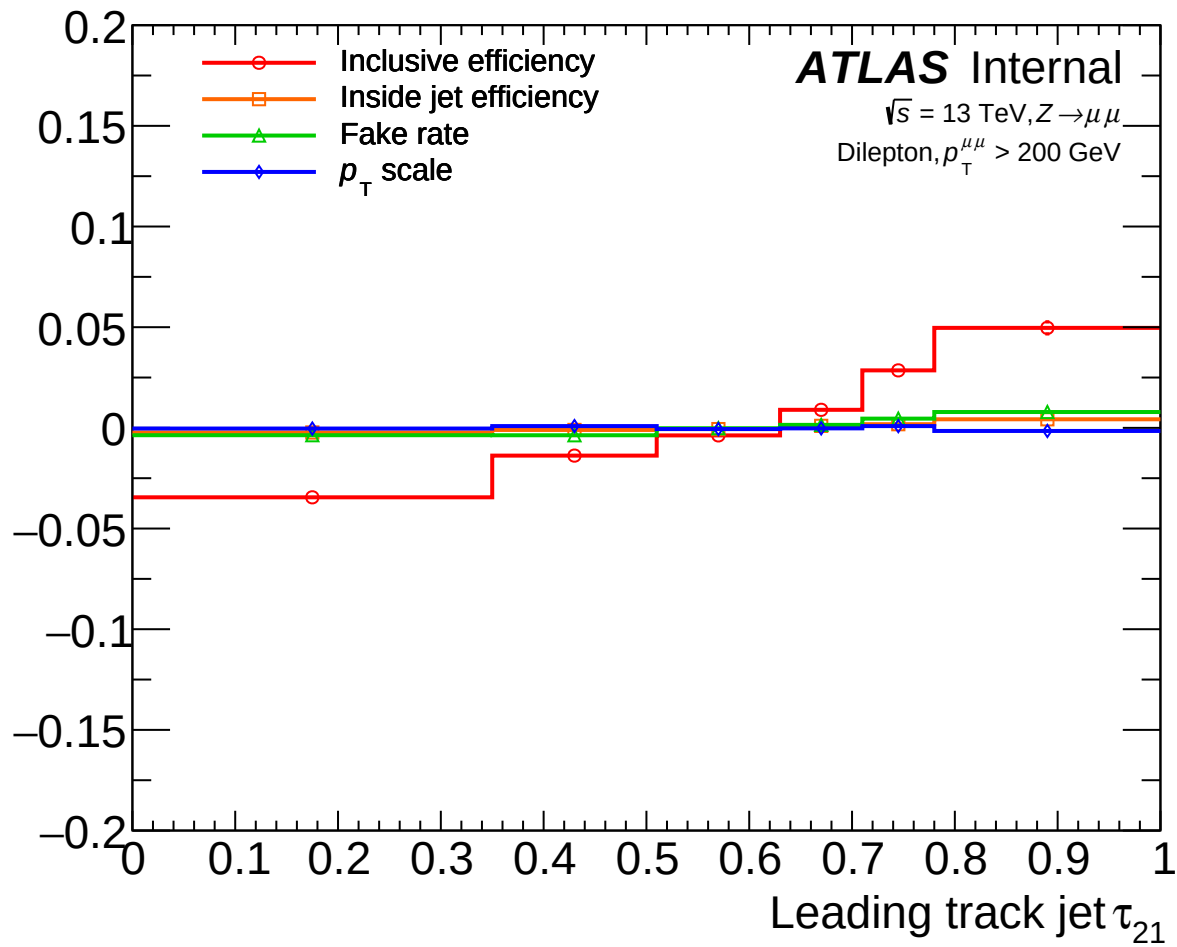




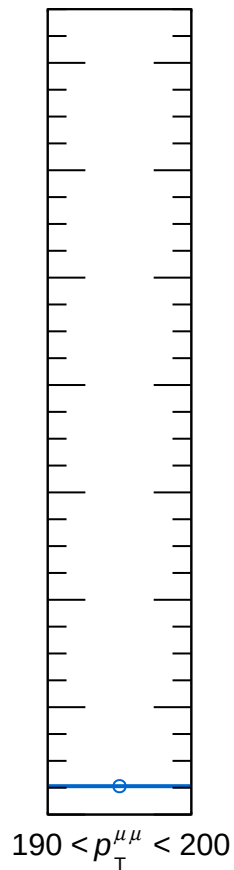
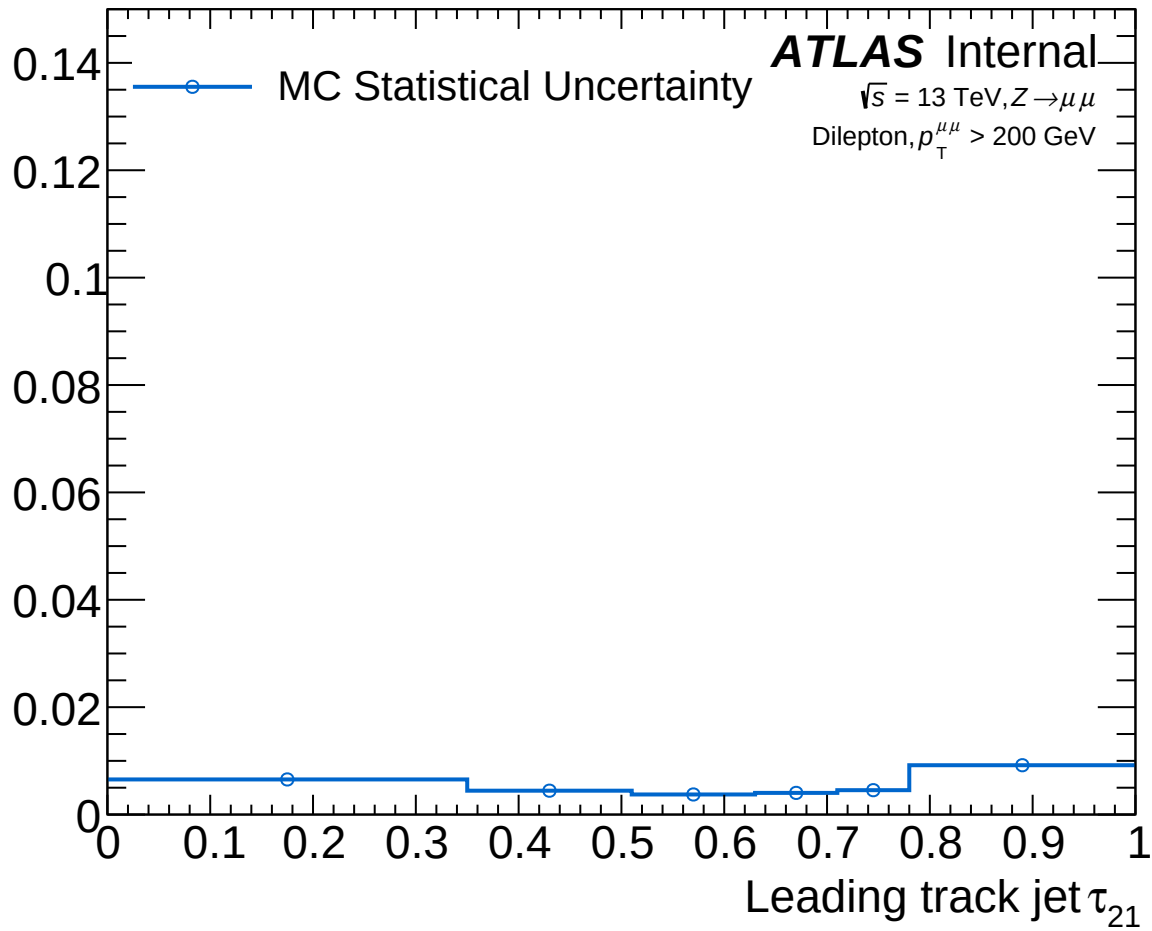
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



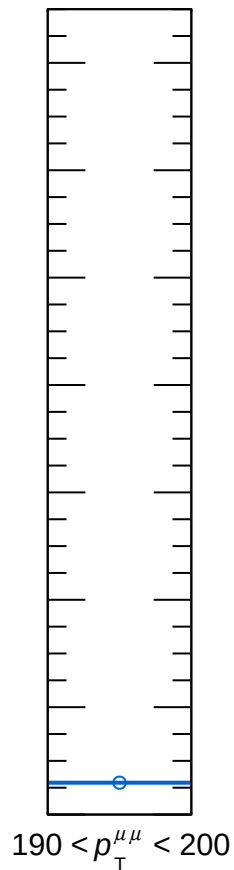
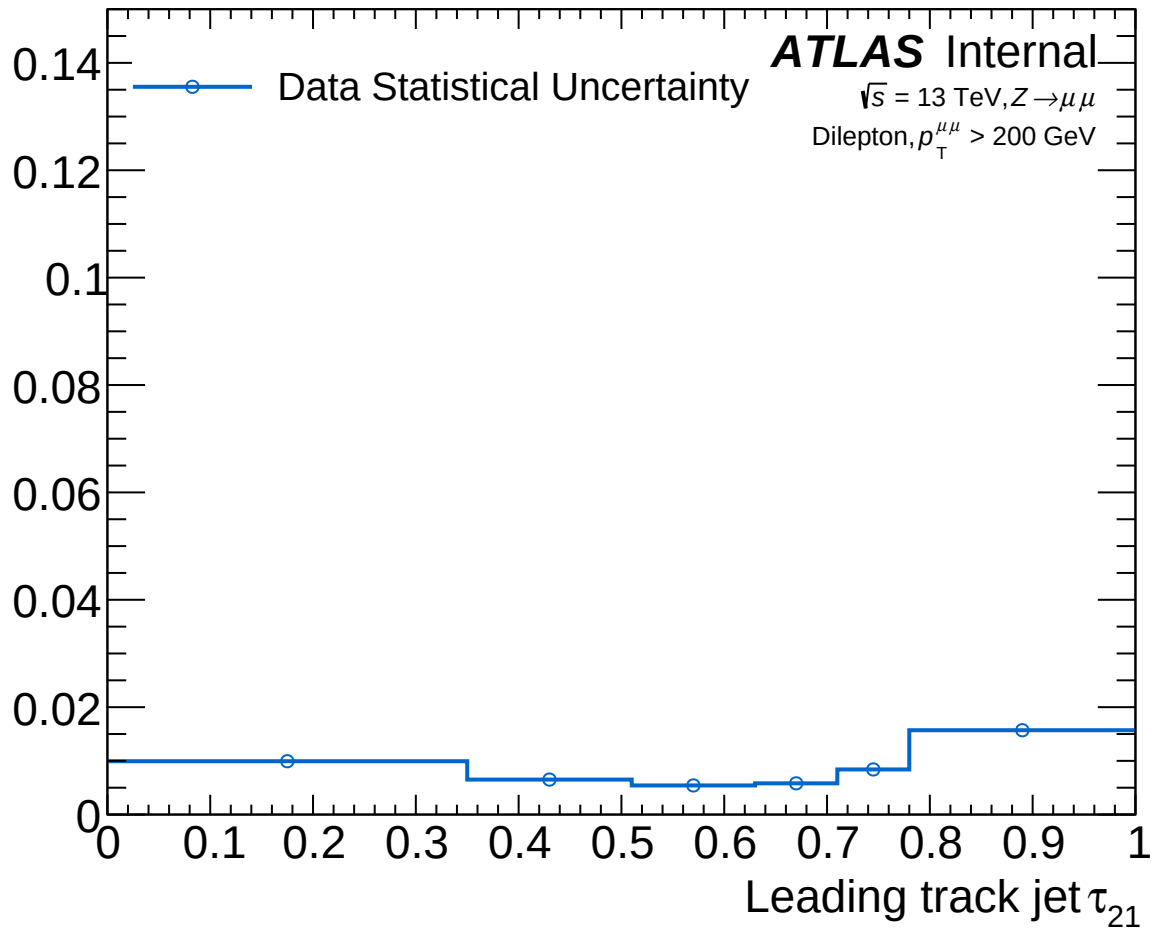
Fractional systematic impact

ATLAS Internal

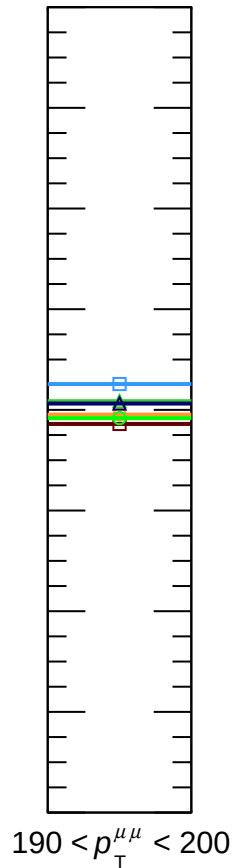
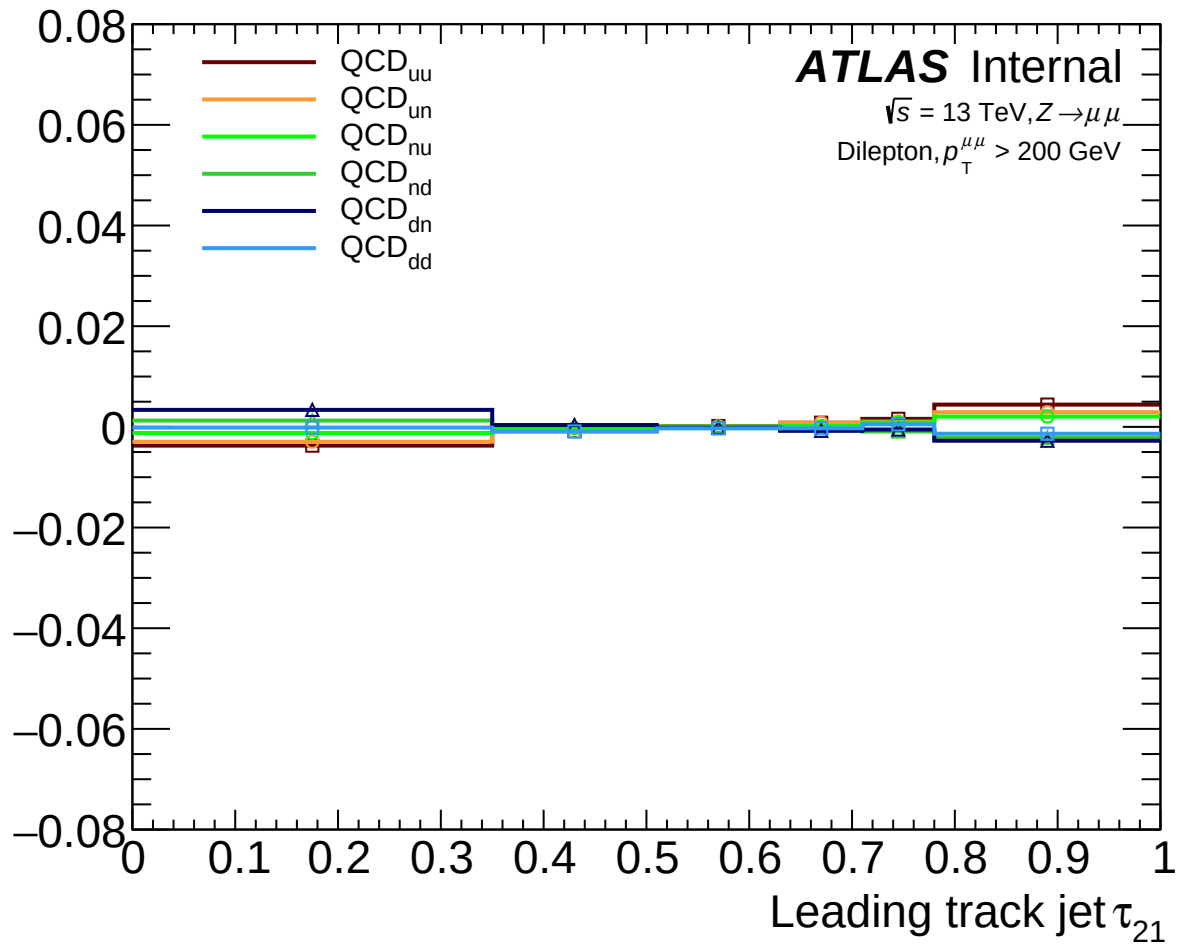
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

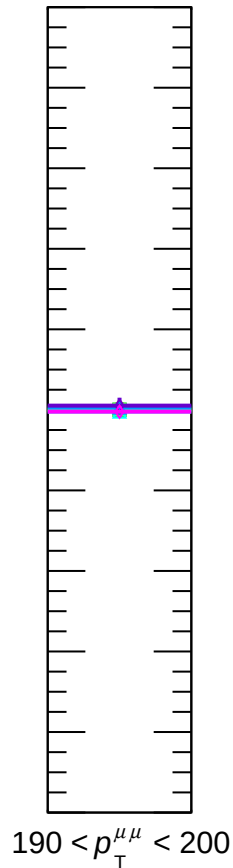
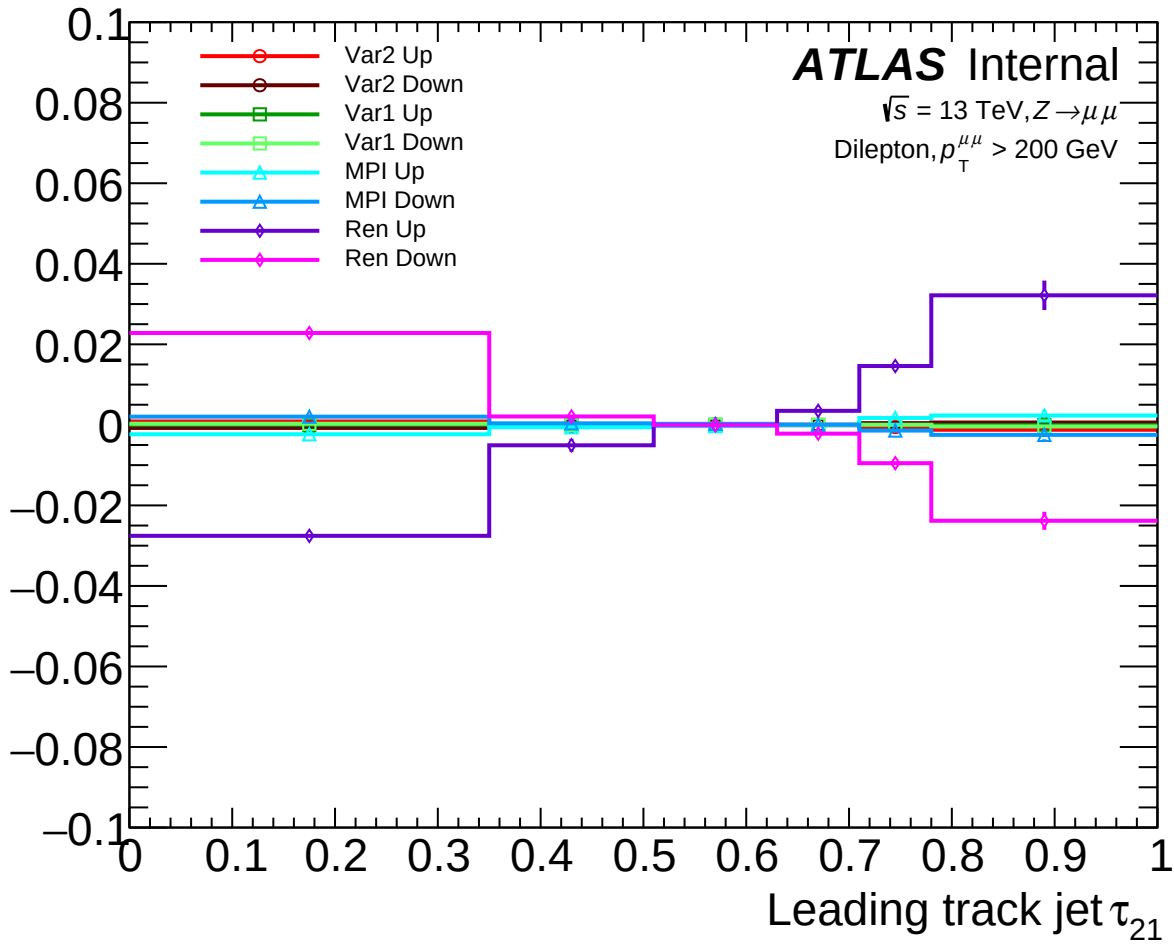
Data Statistical Uncertainty

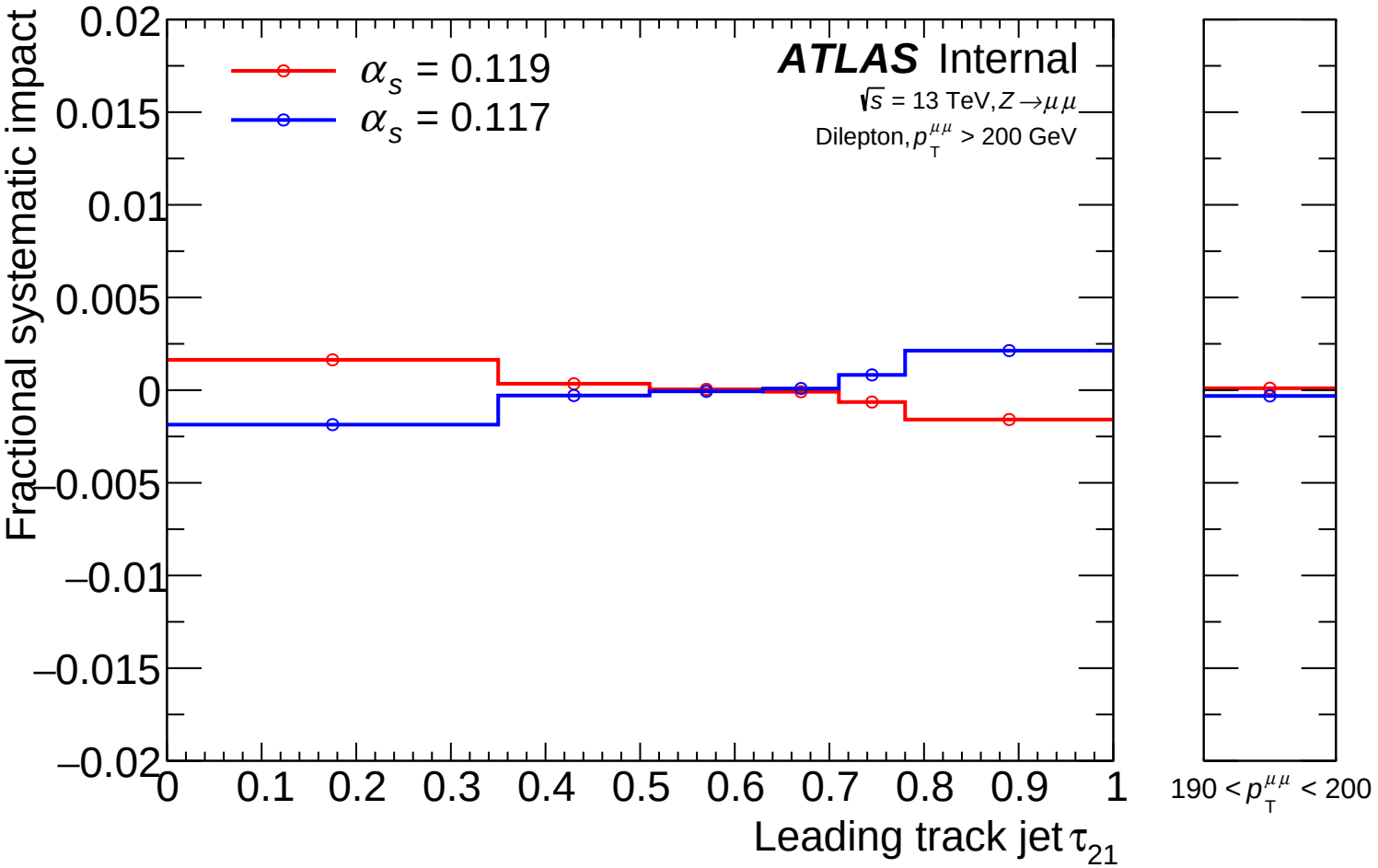


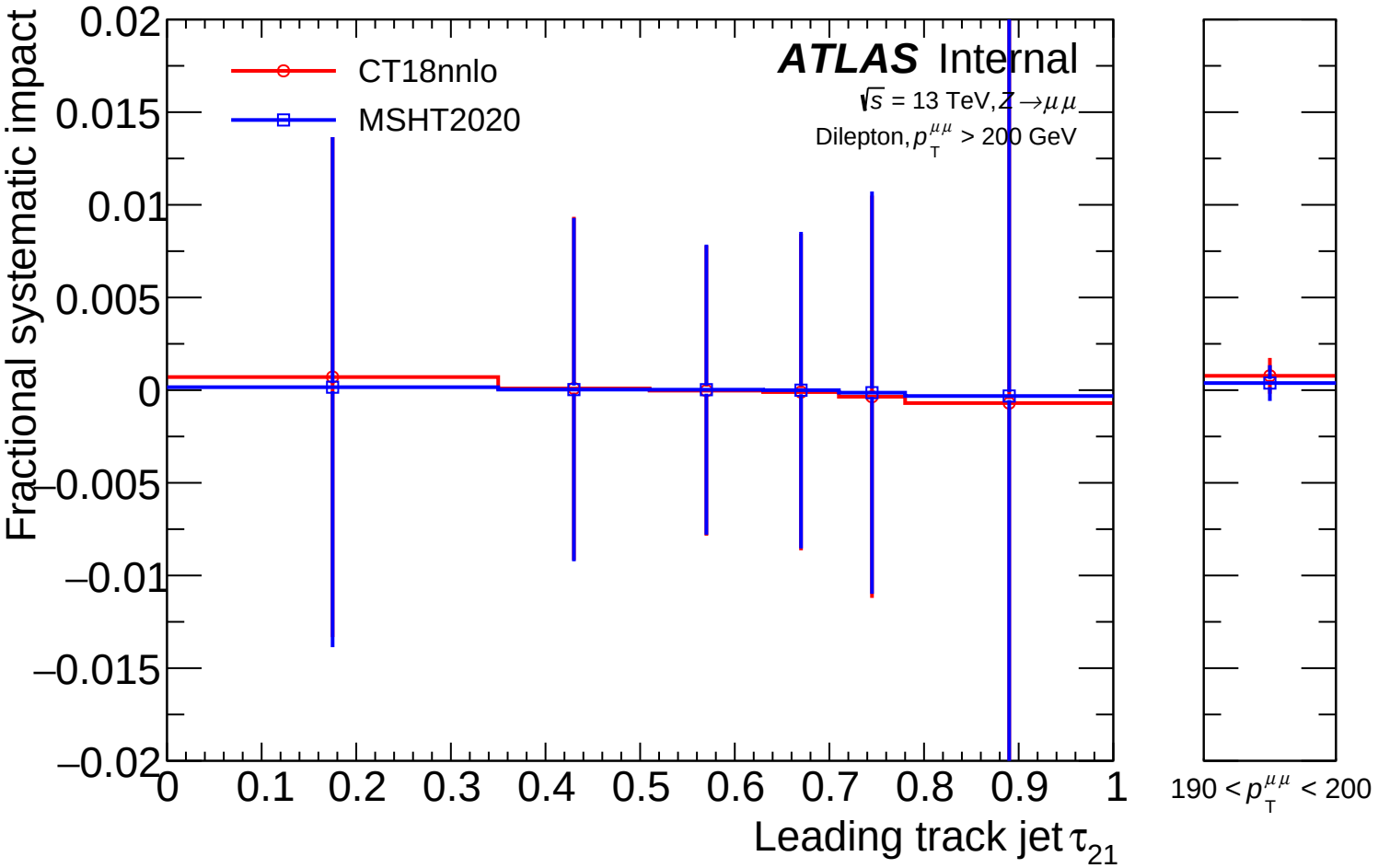
Fractional systematic impact



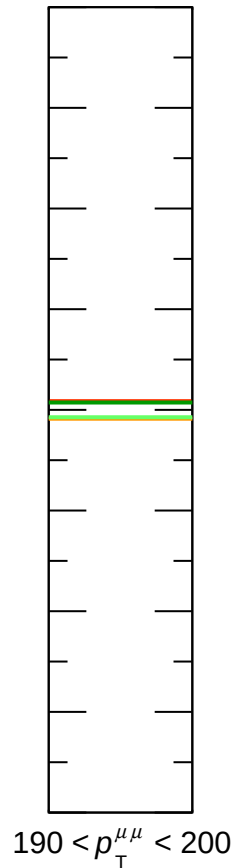
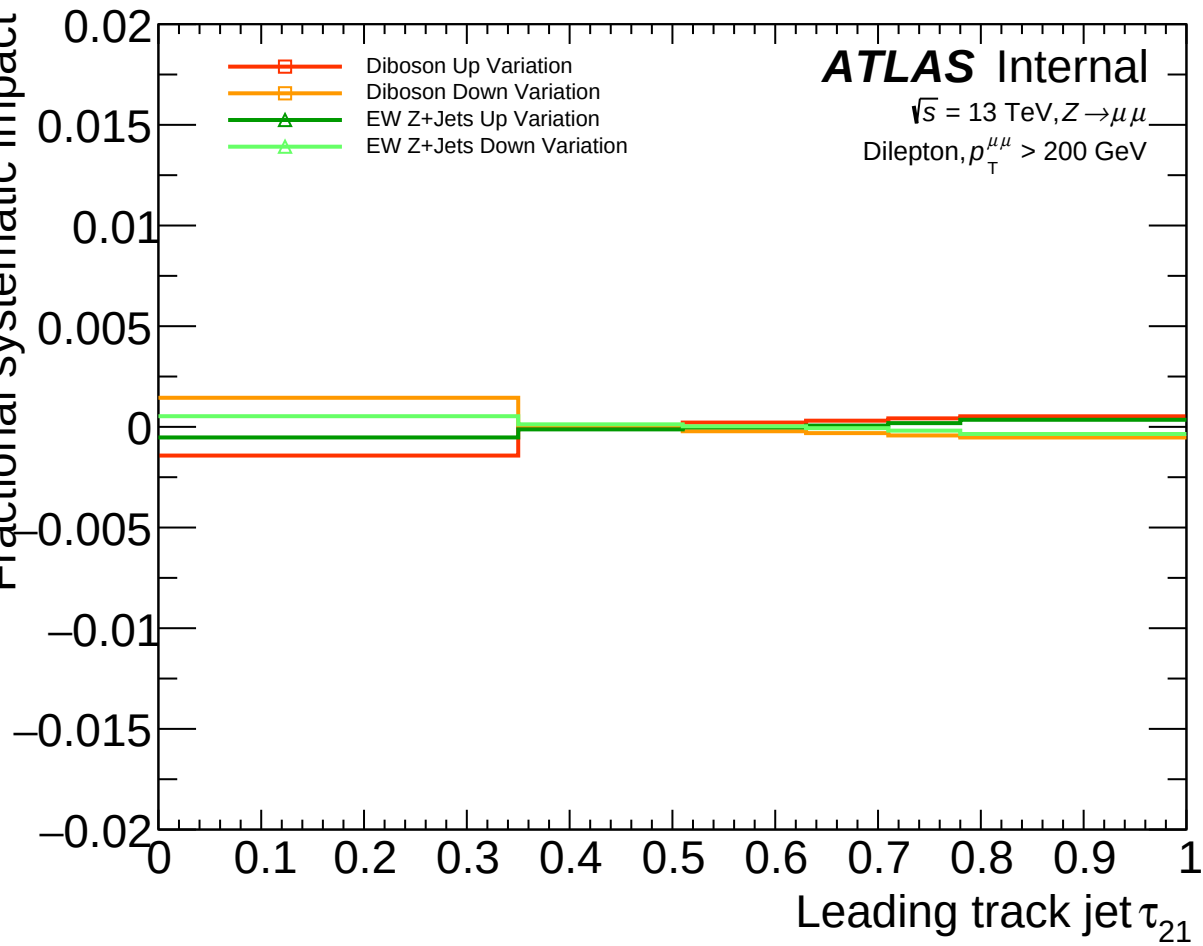
Fractional systematic impact







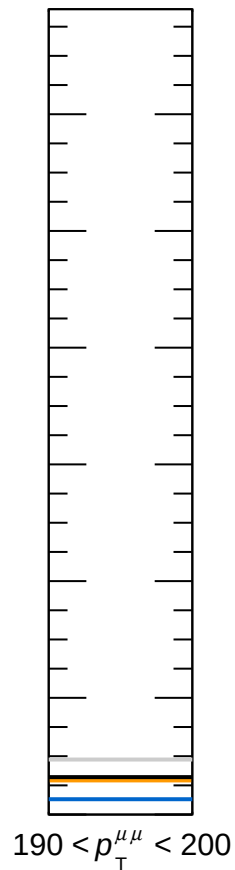
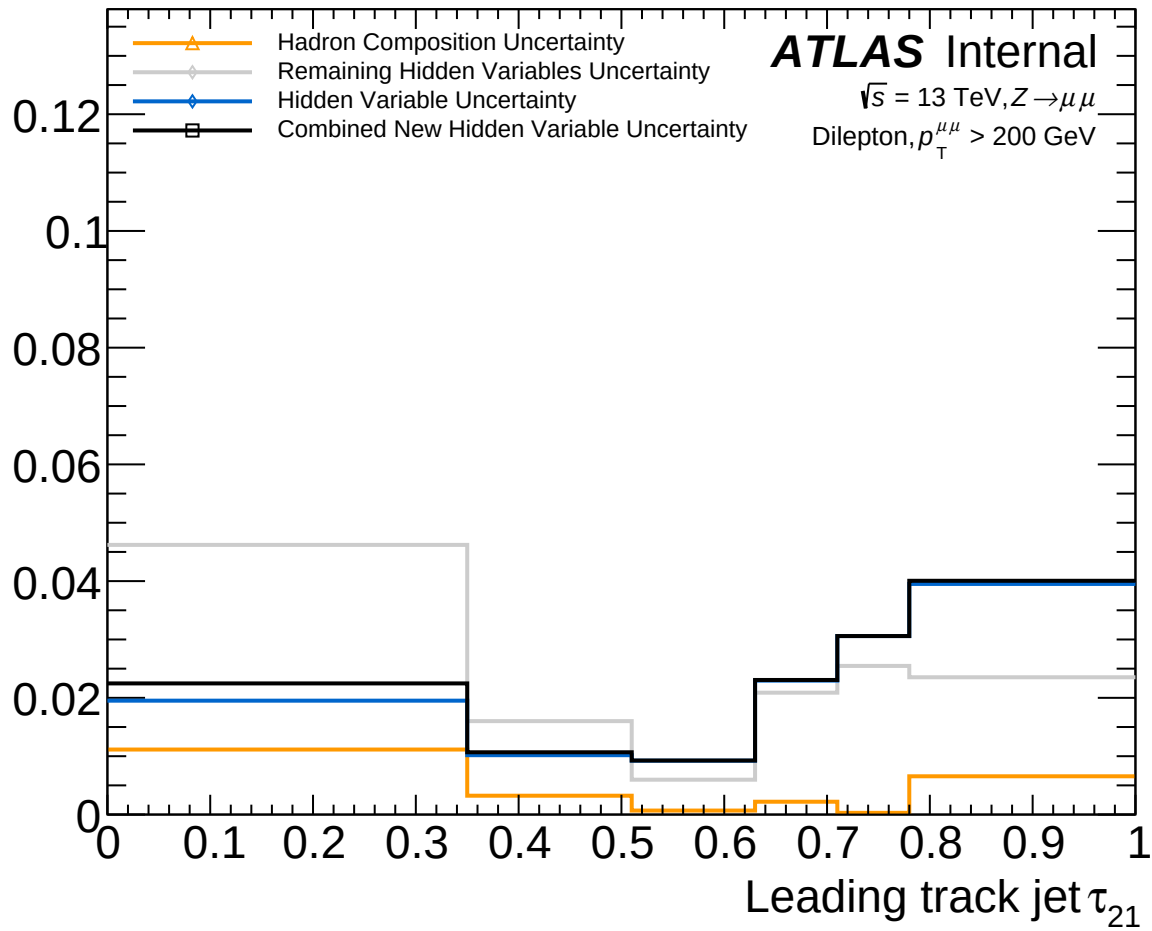
Fractional systematic impact

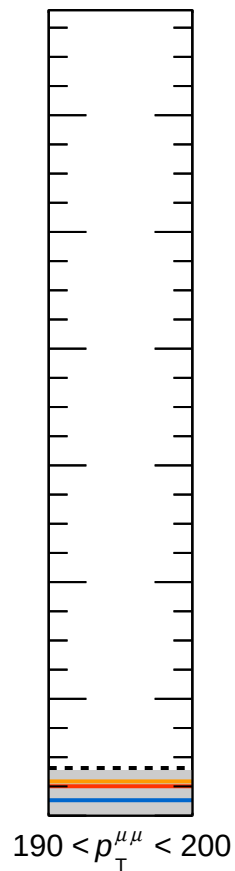
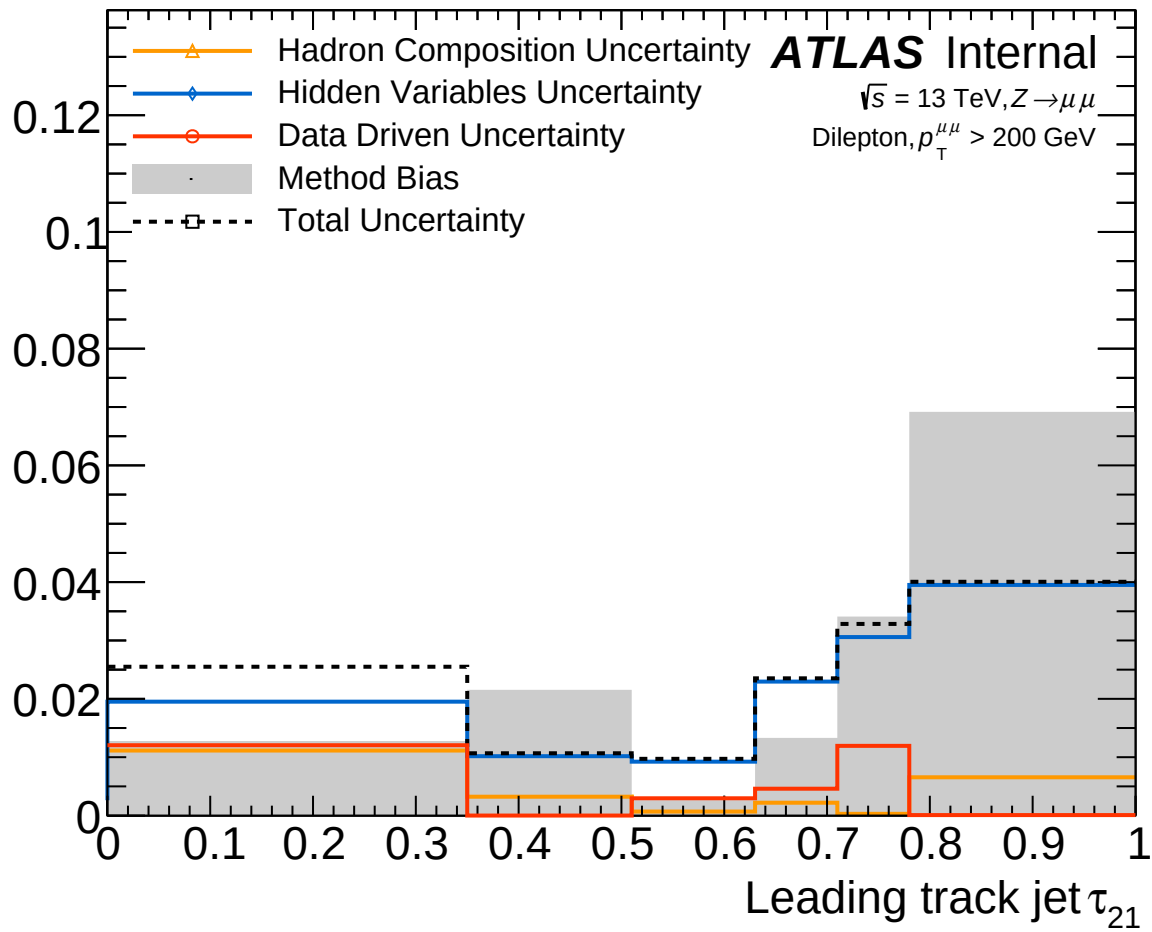


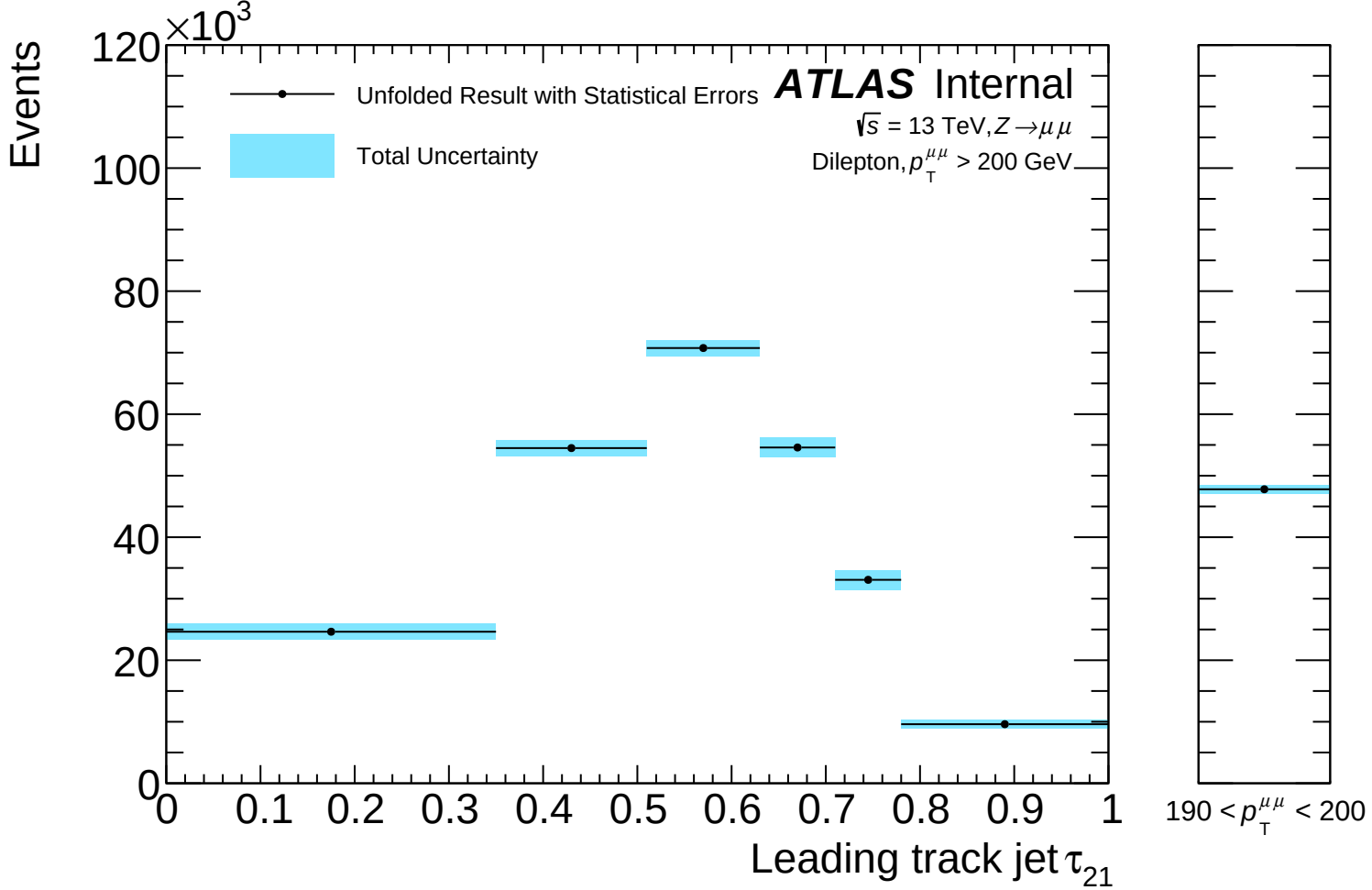
ATLAS Internal

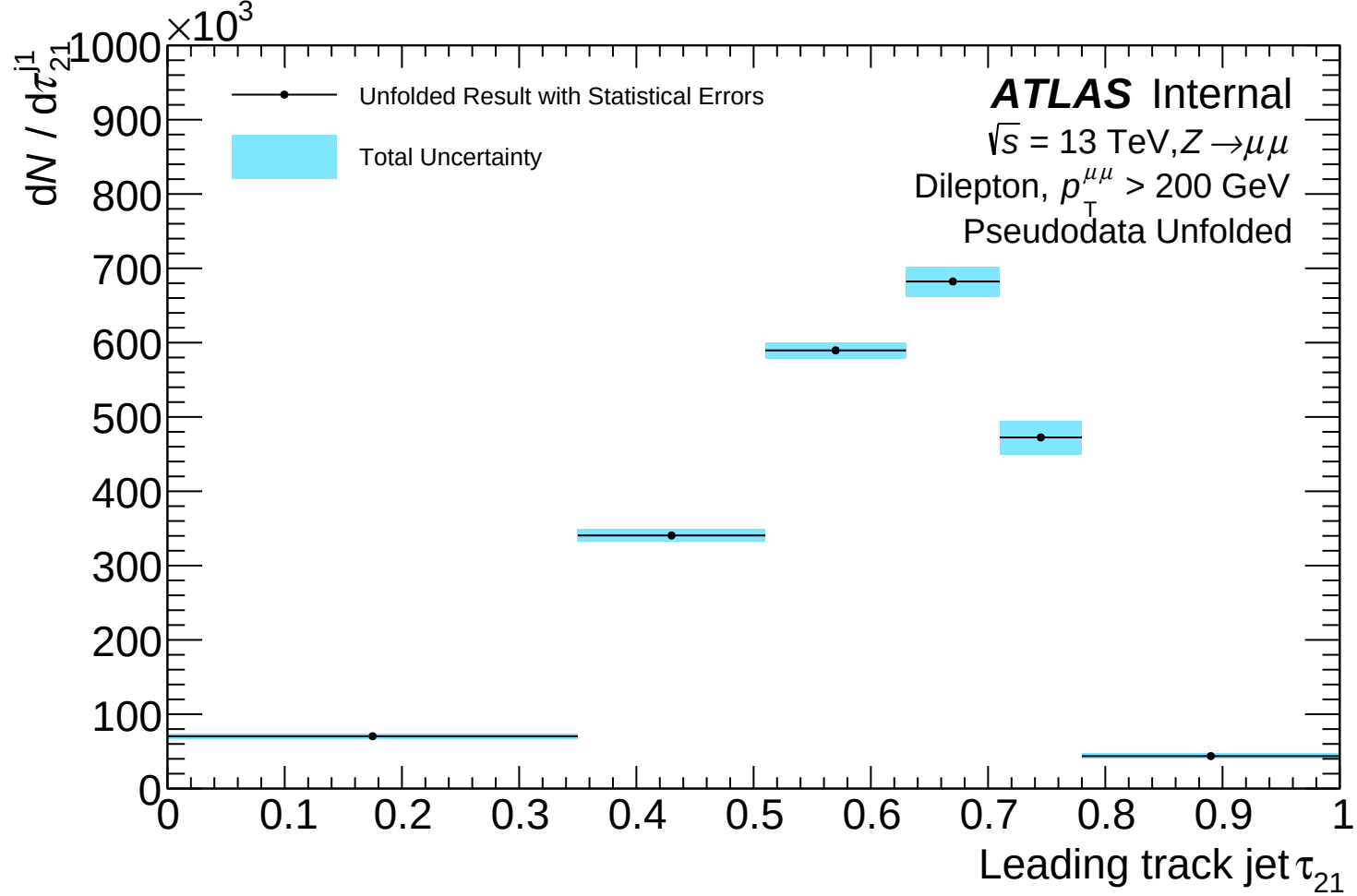
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

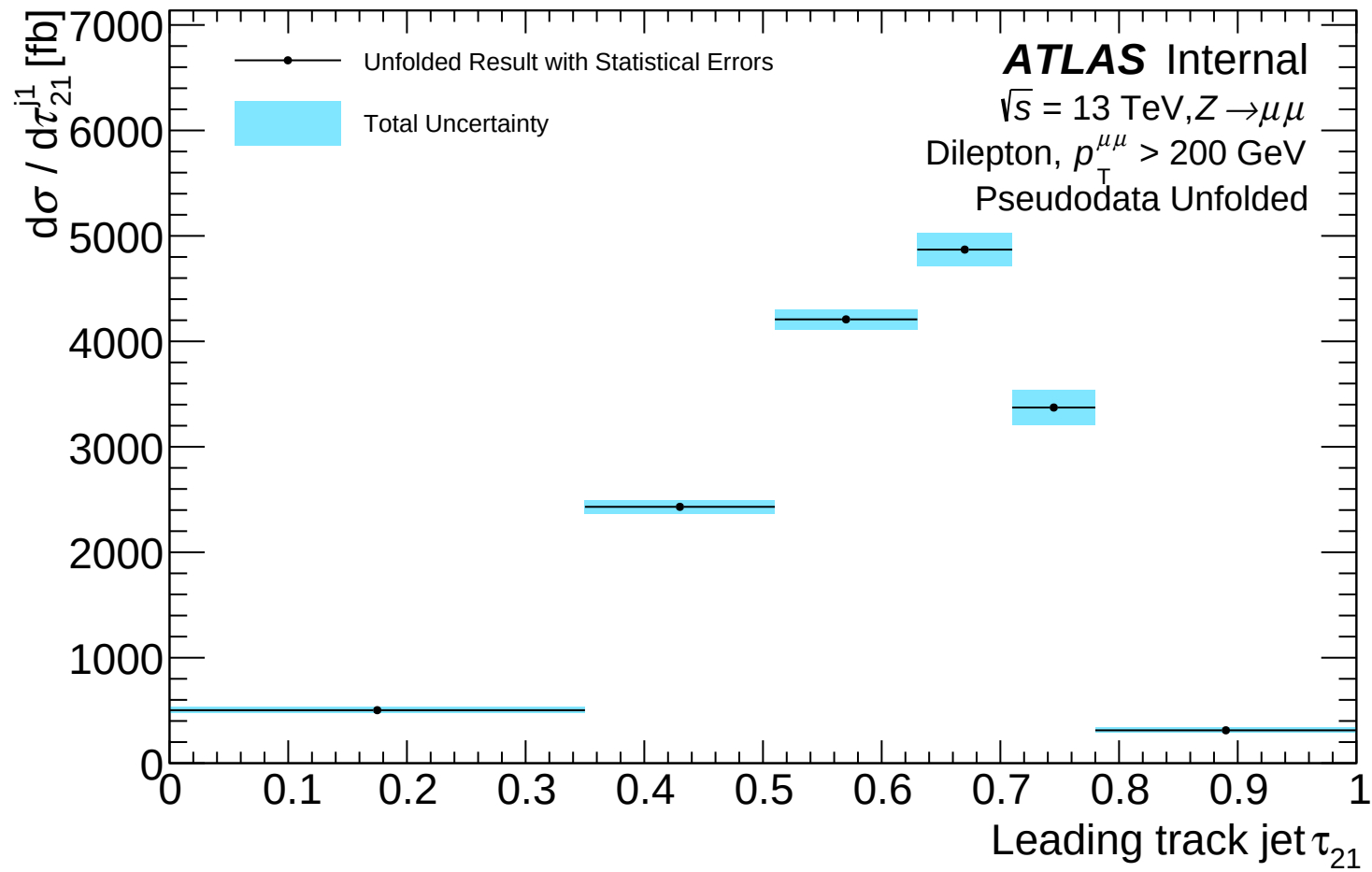
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$











Fractional uncertainty

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias

0.12

0.1

0.08

0.06

0.04

0.02

0

0

0.1 0.2 0.3

0.4 0.5

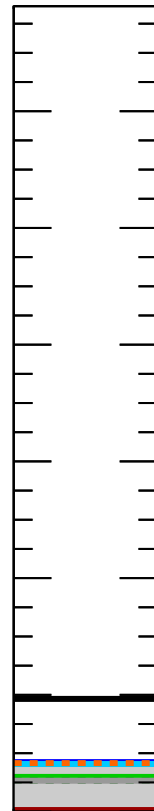
0.6 0.7

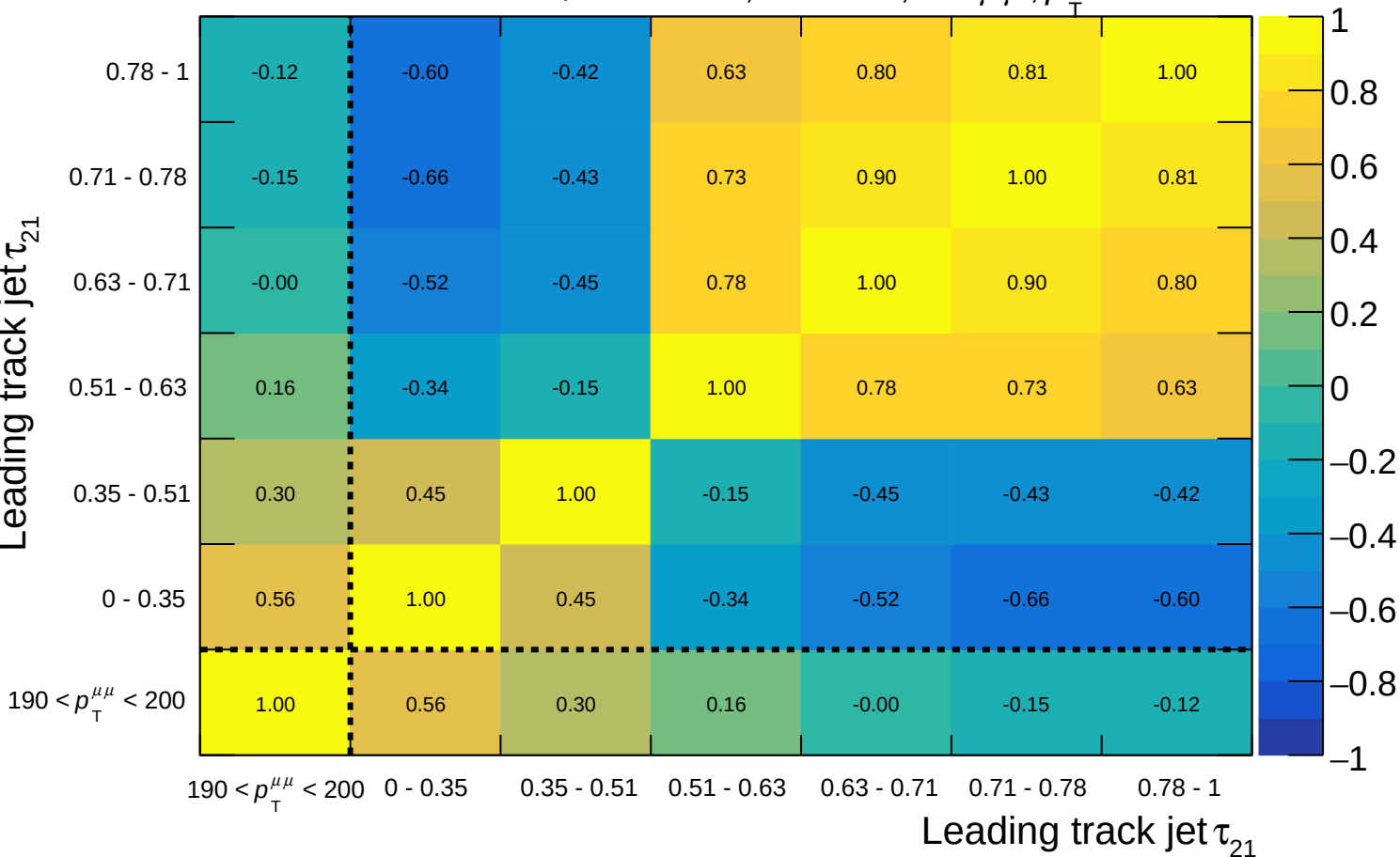
0.8 0.9

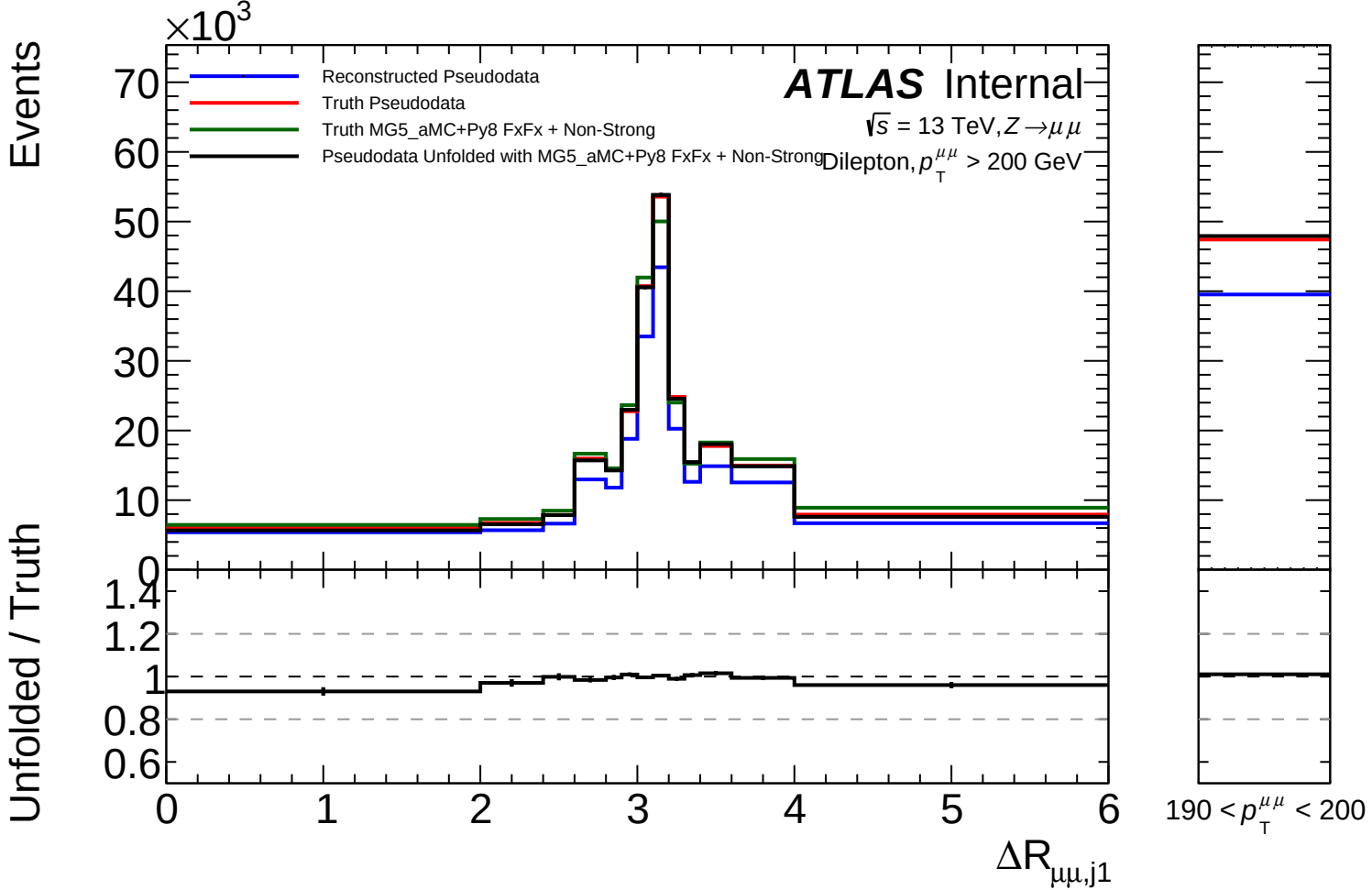
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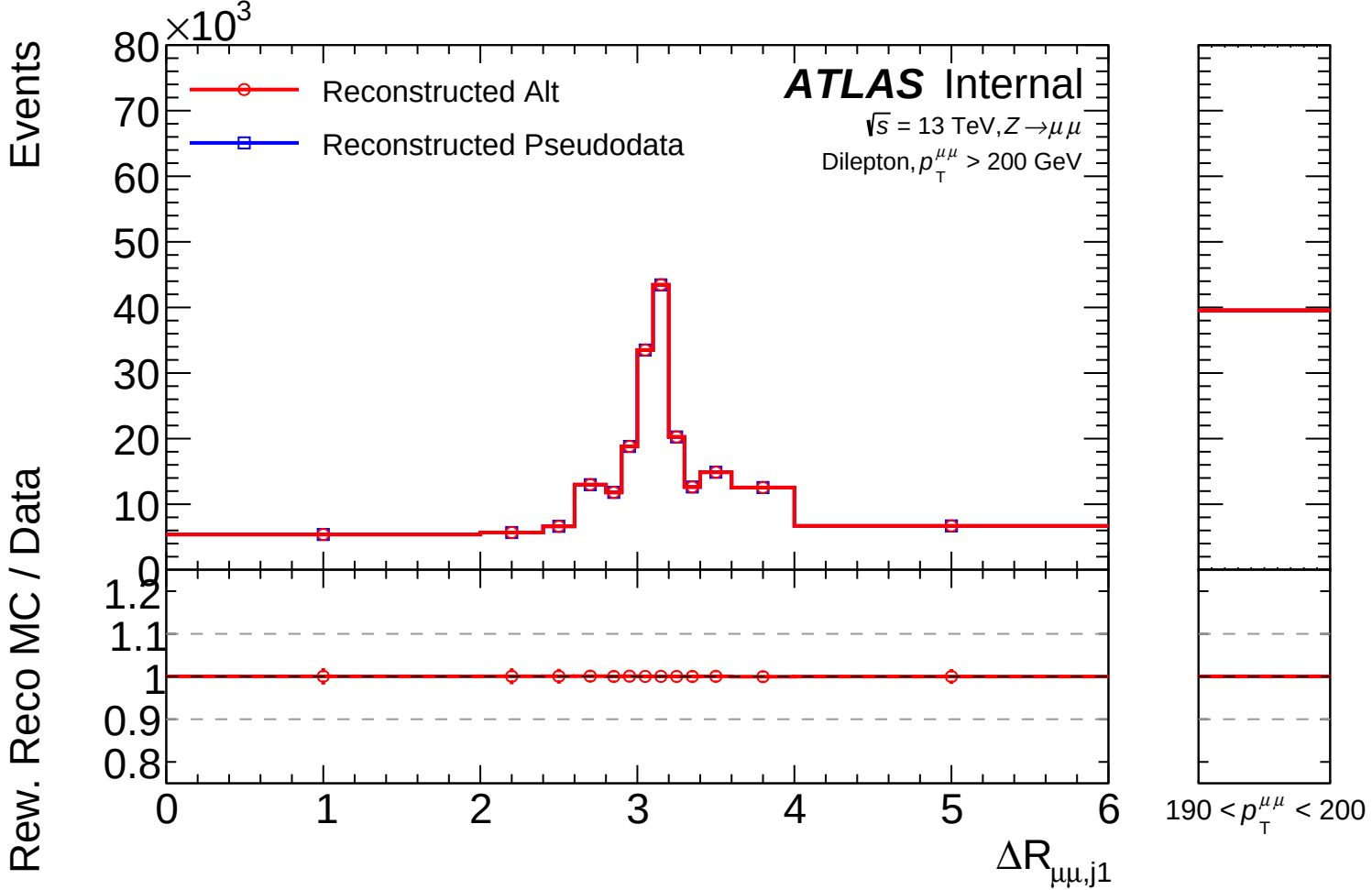
Leading track jet τ_{21}

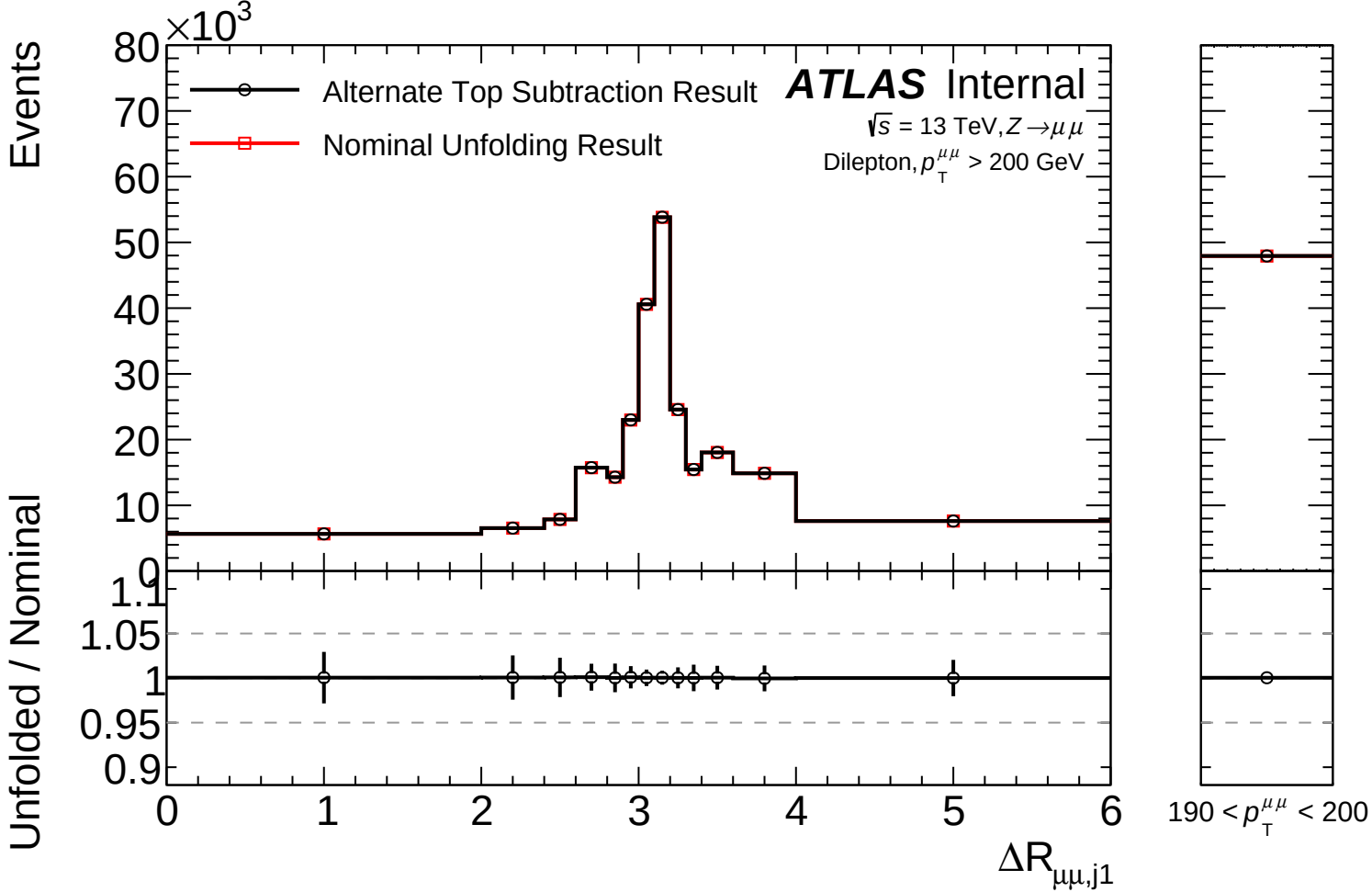
$190 < p_T^{\mu\mu} < 200$



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ Leading track jet τ_{21} 







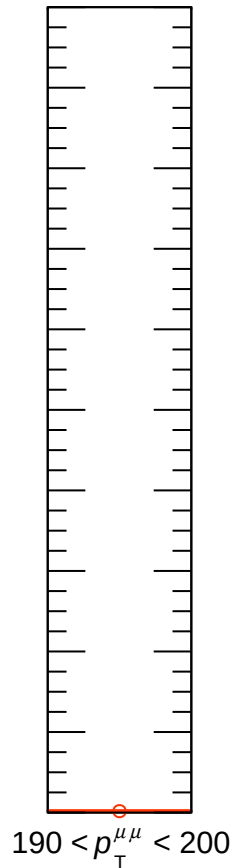
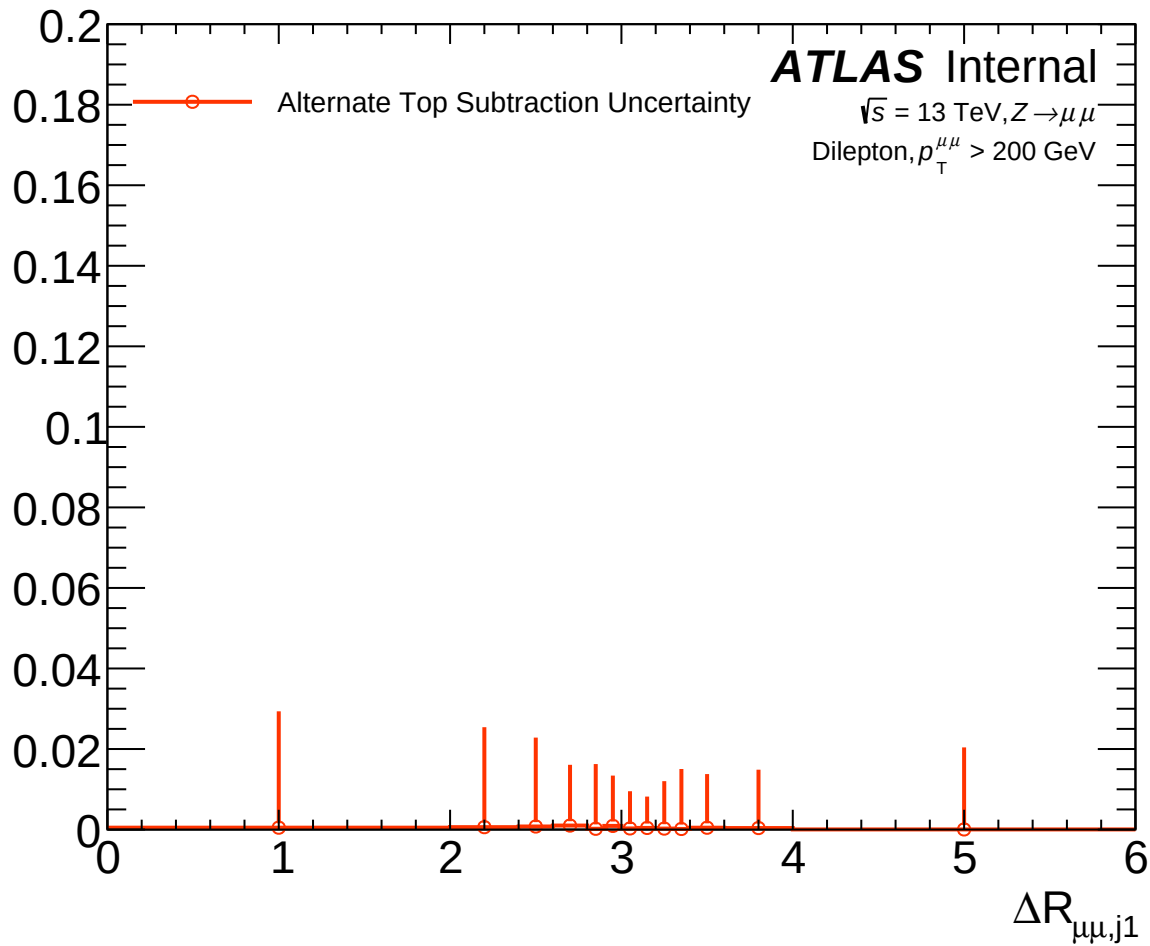
Fractional systematic impact

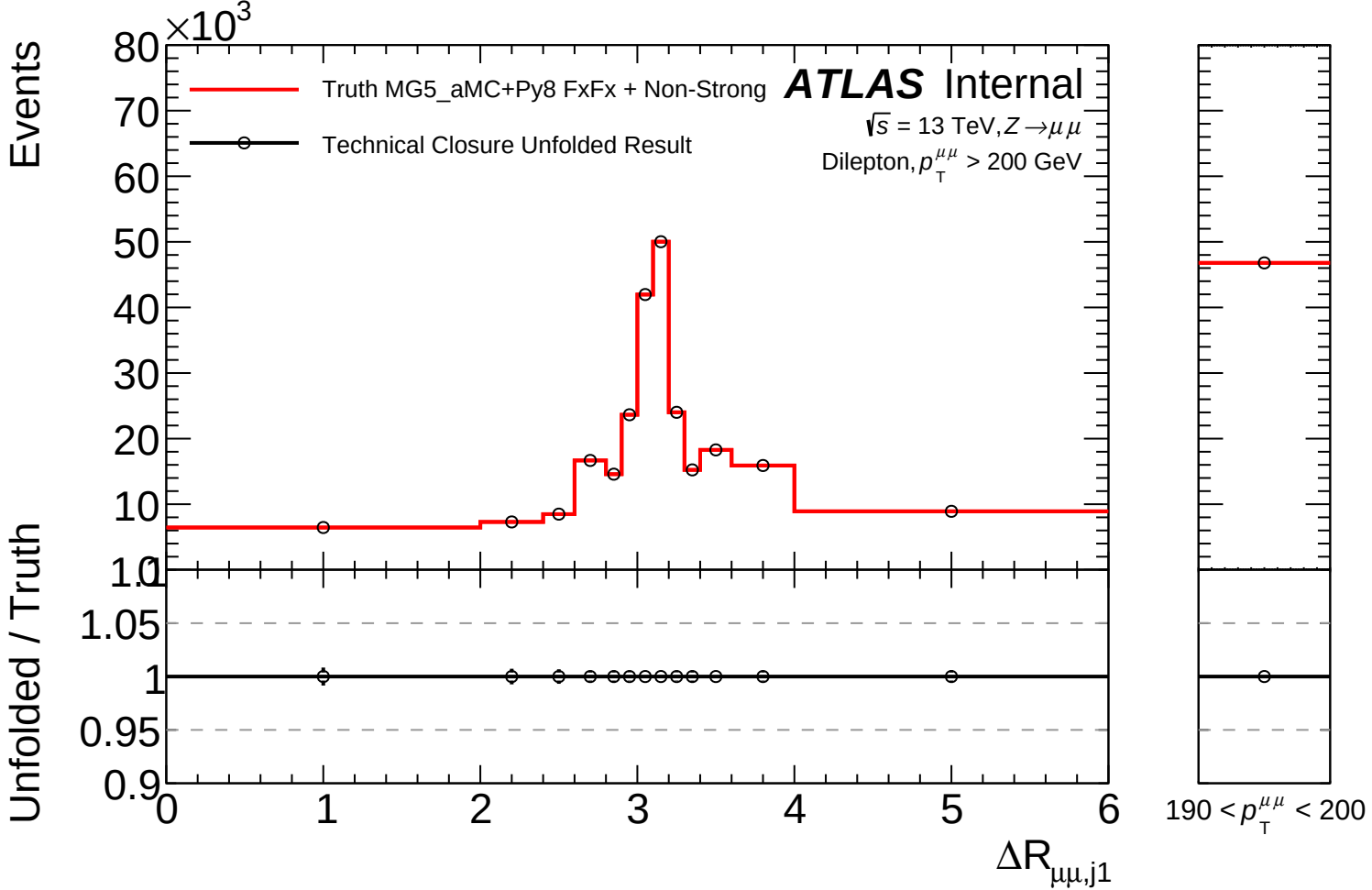
ATLAS Internal

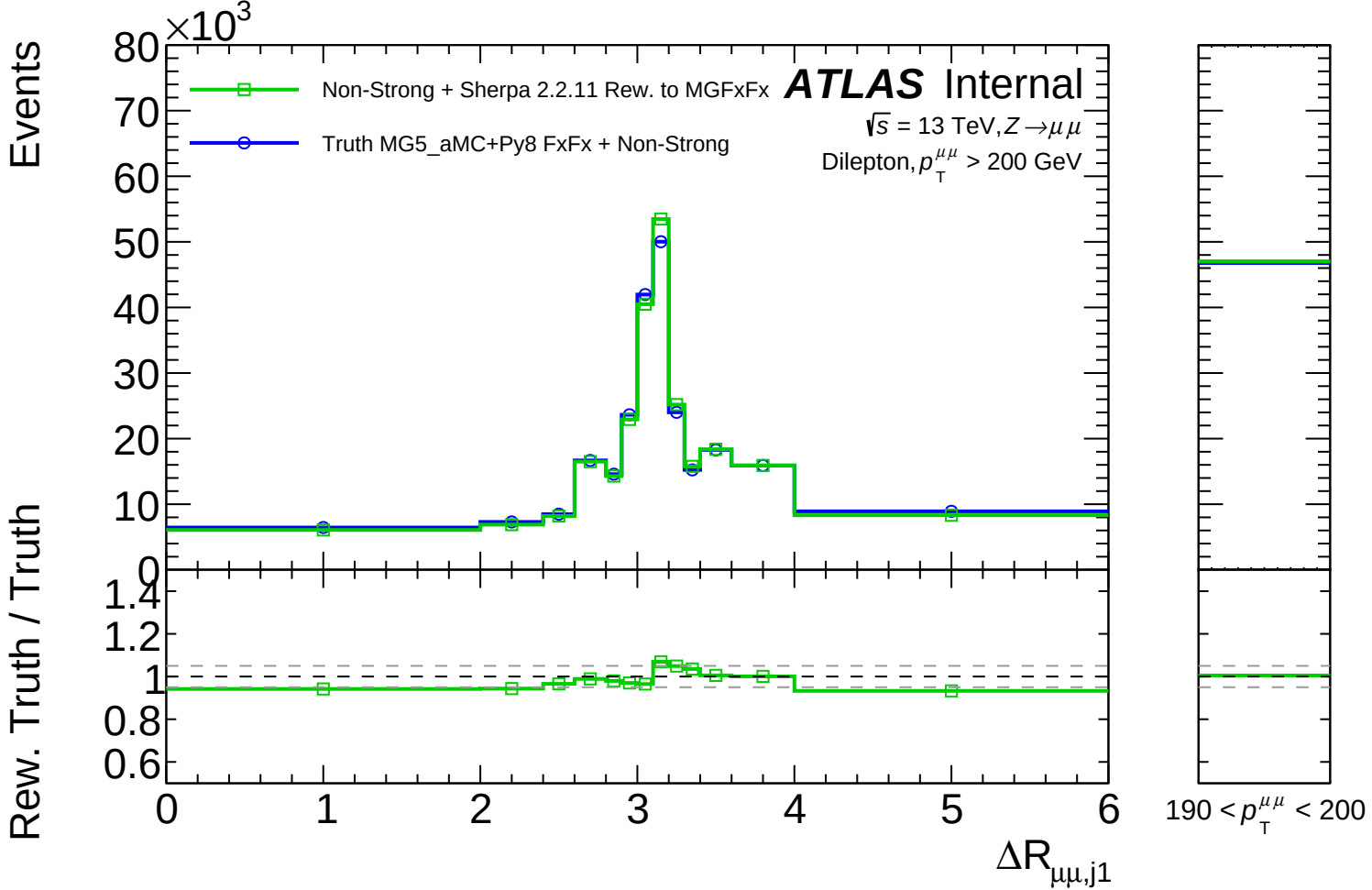
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

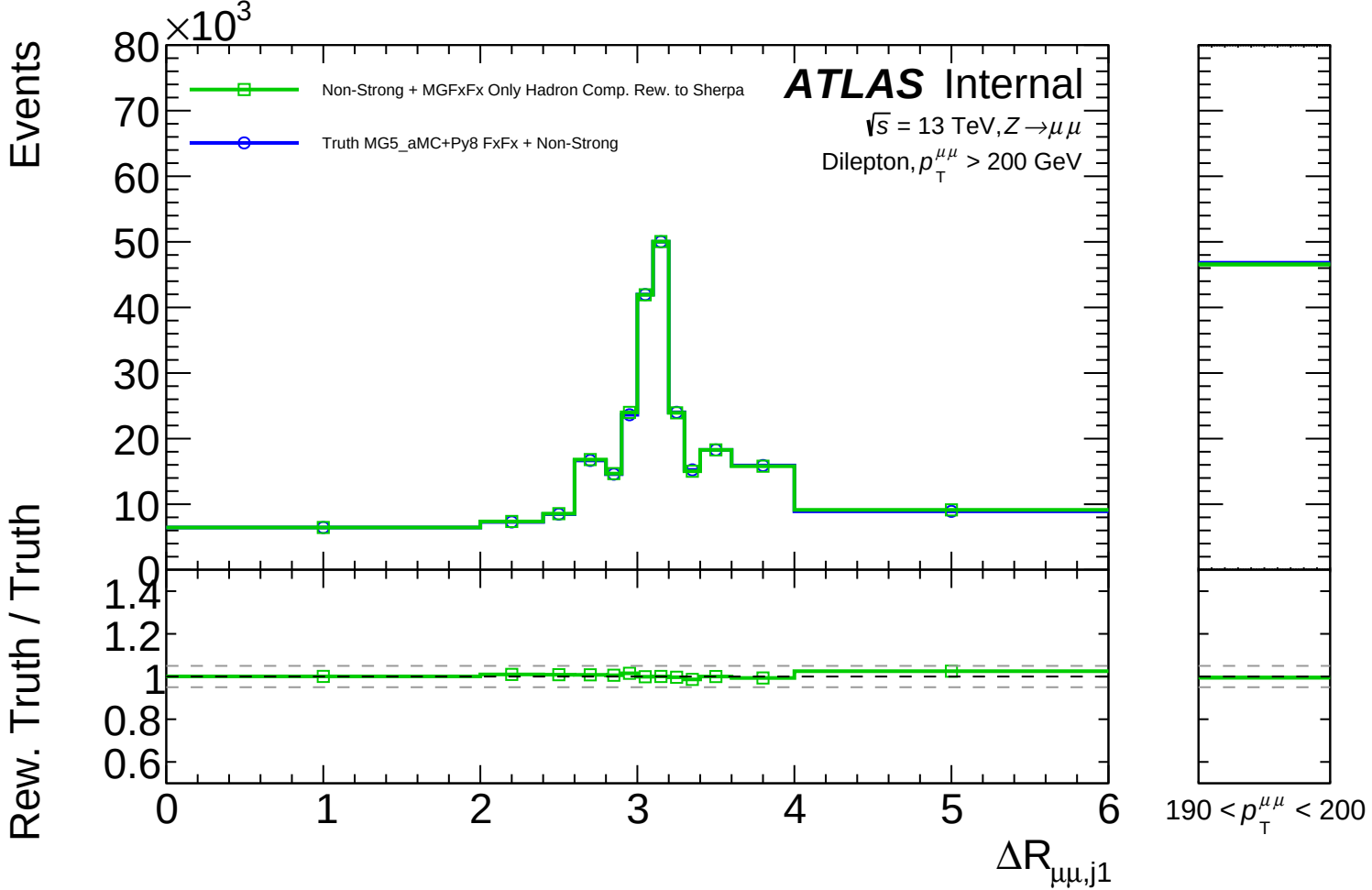
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

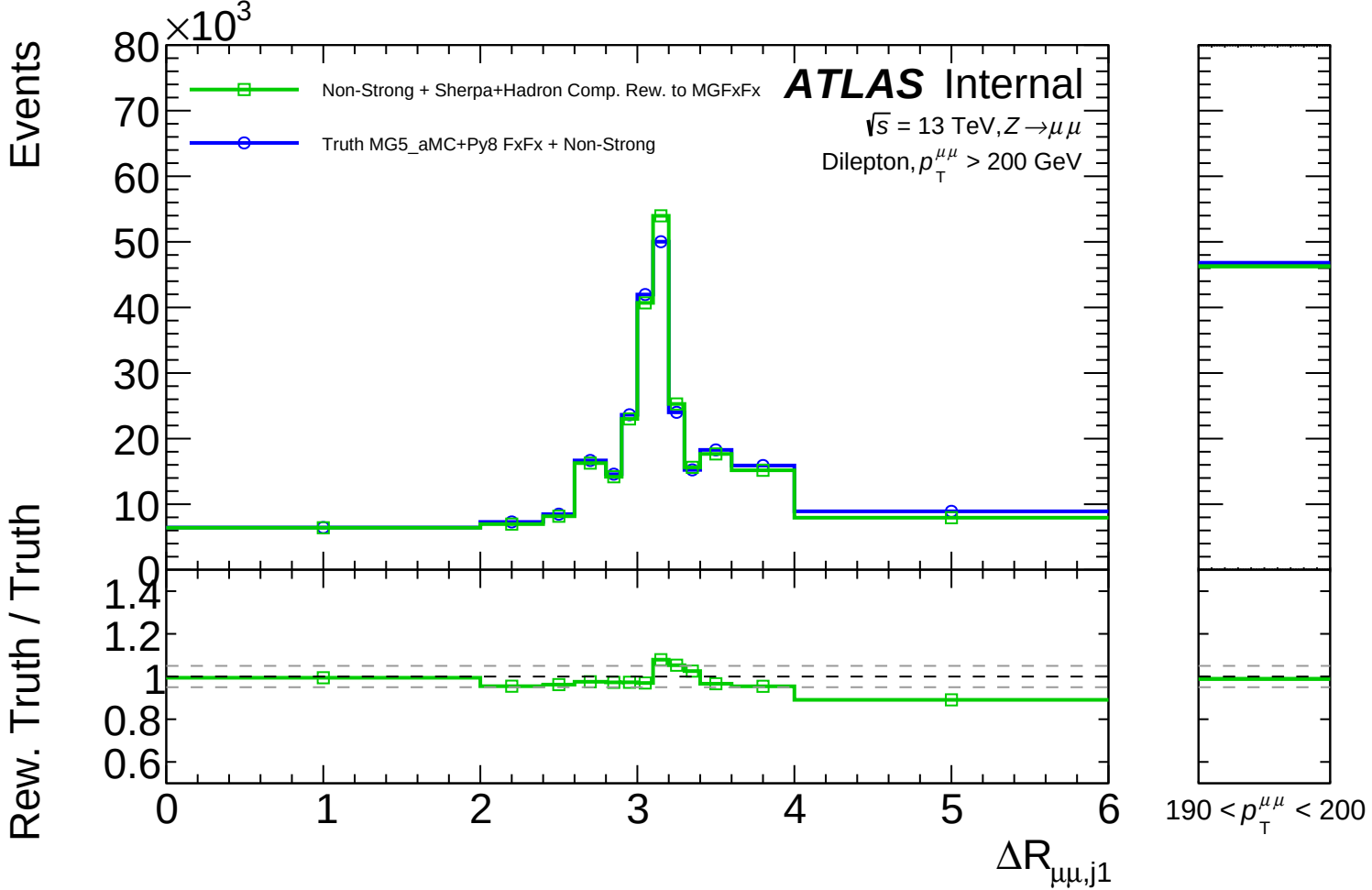
Alternate Top Subtraction Uncertainty

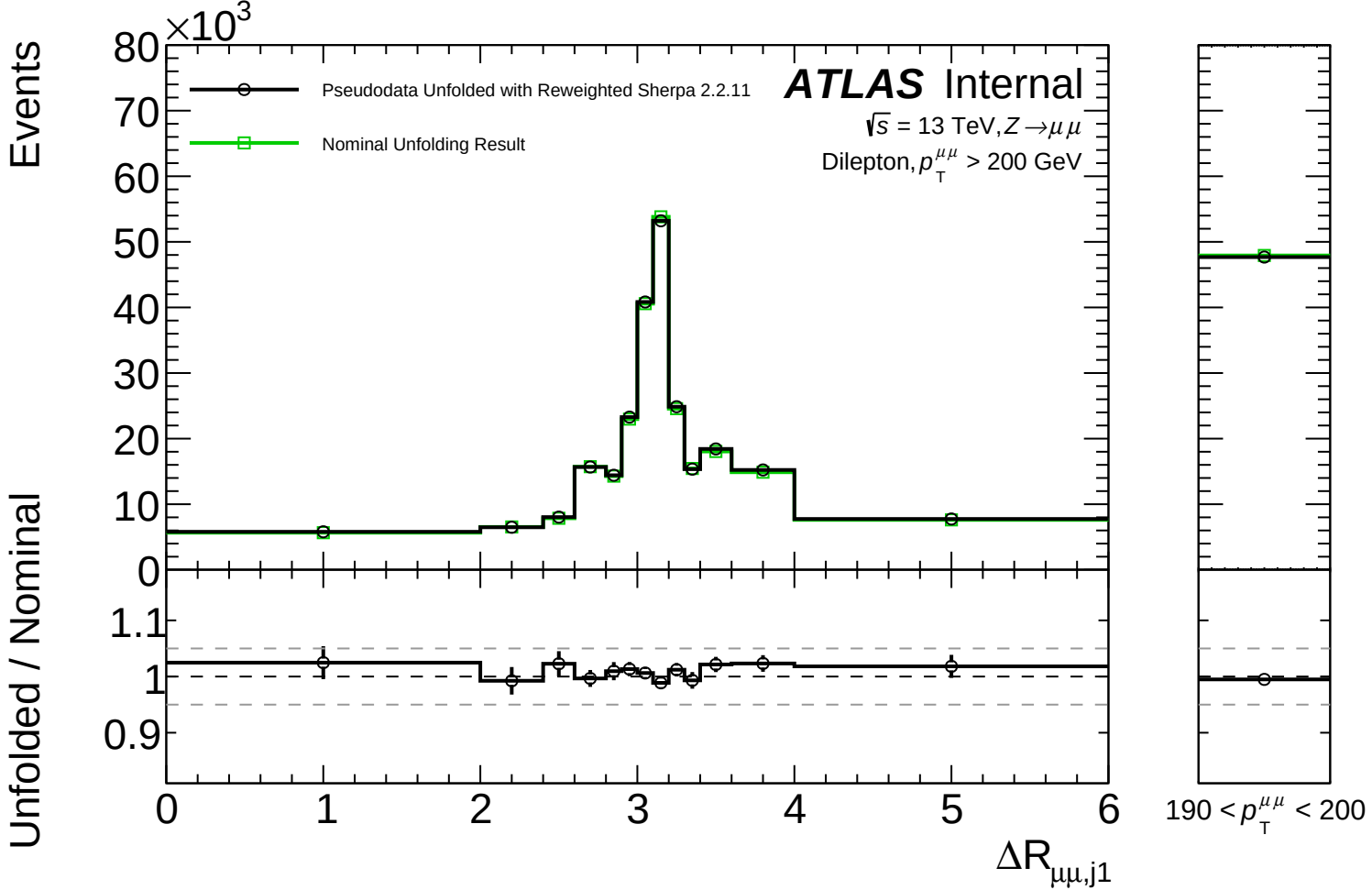


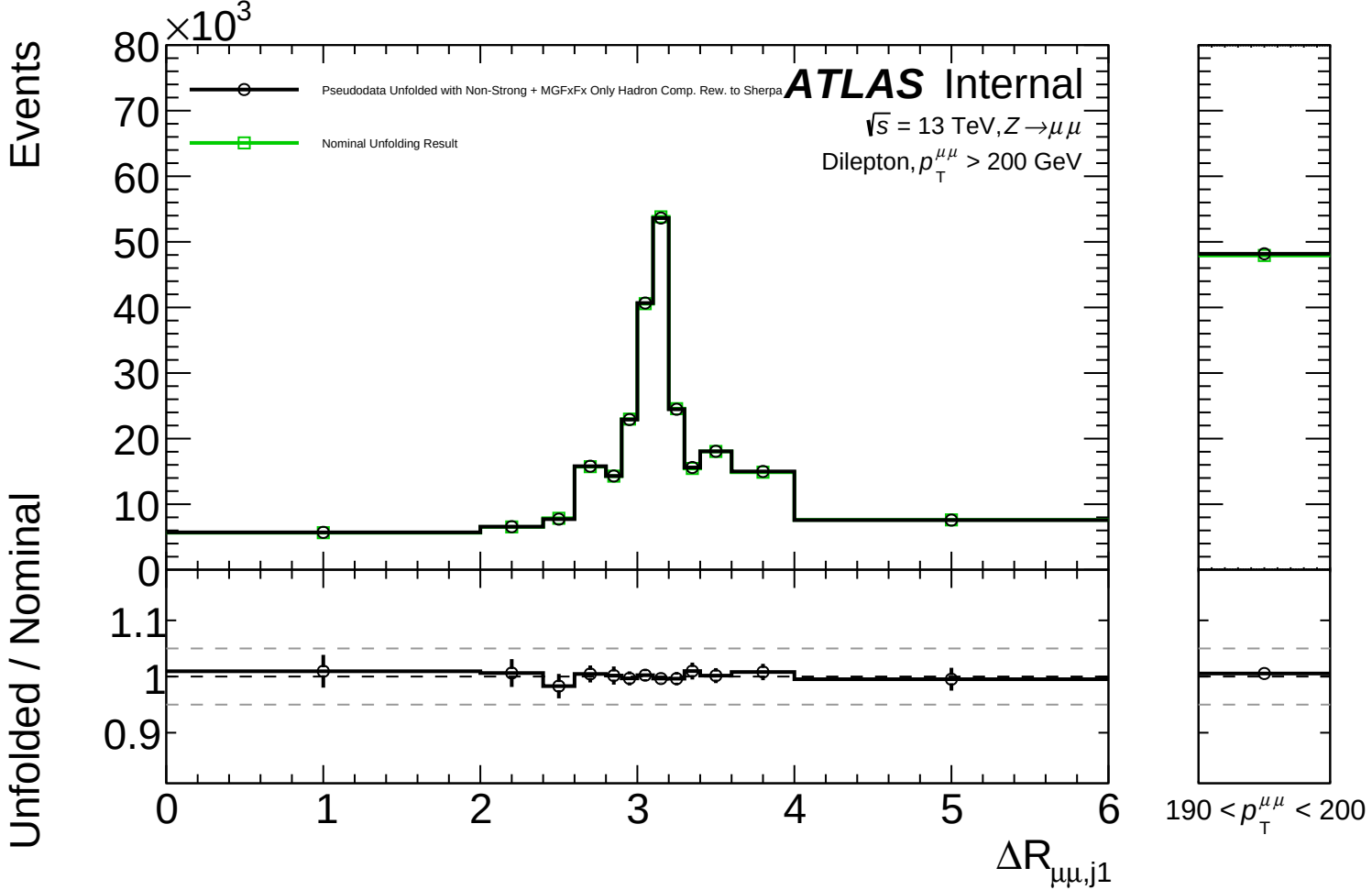






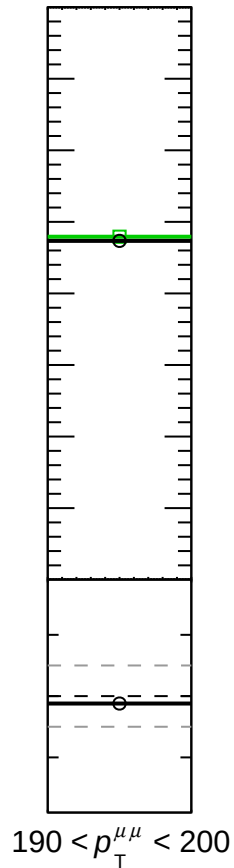
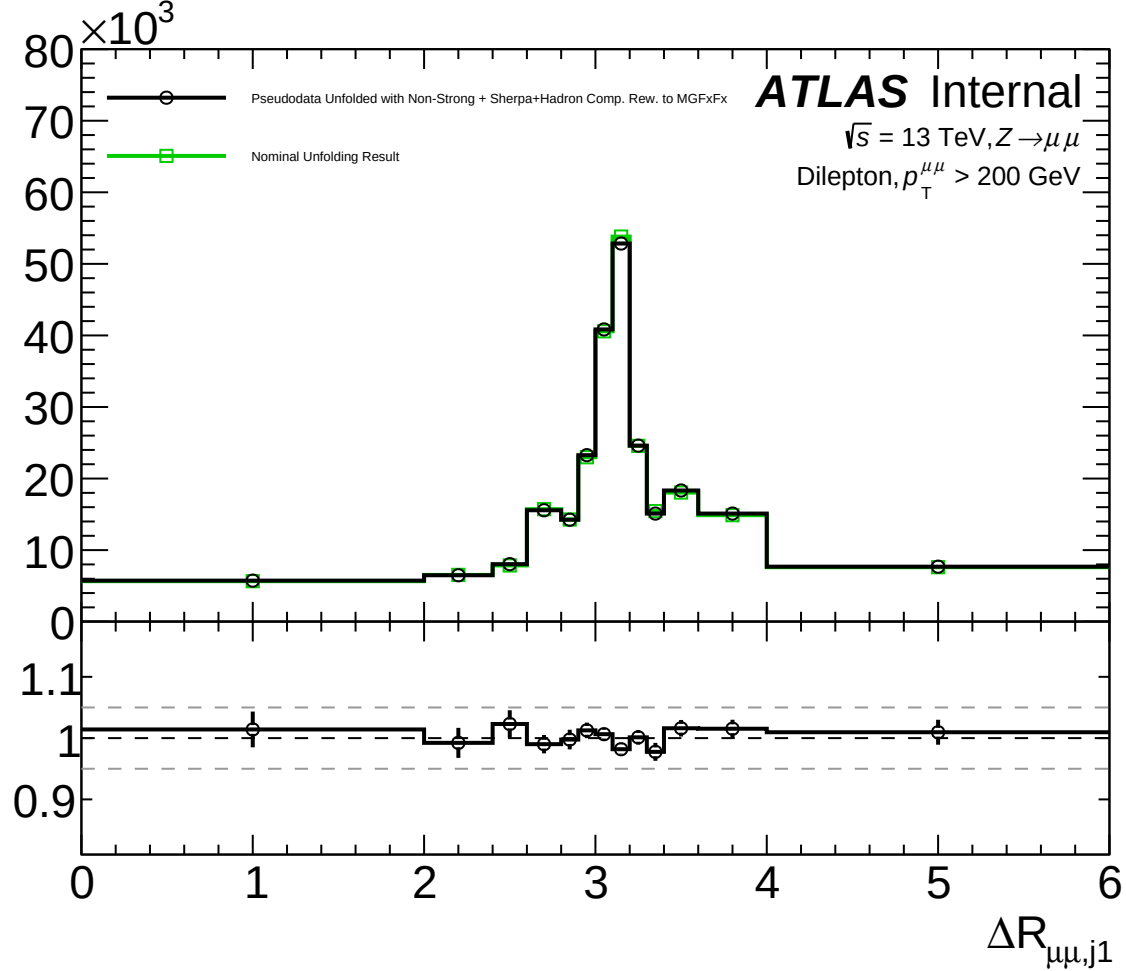






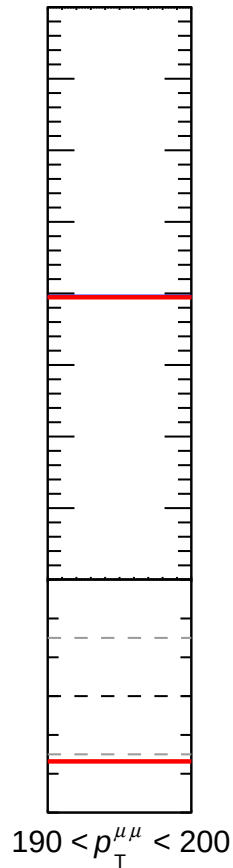
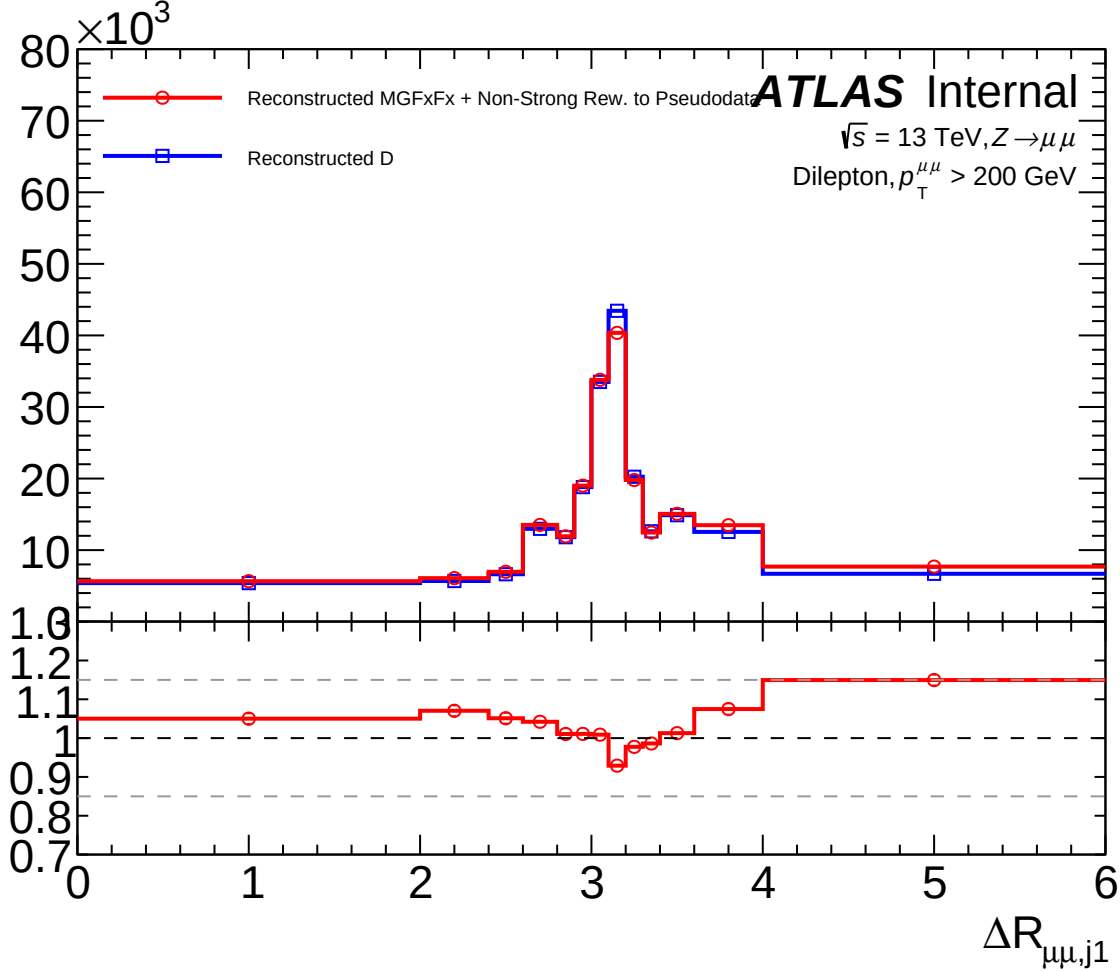
Events

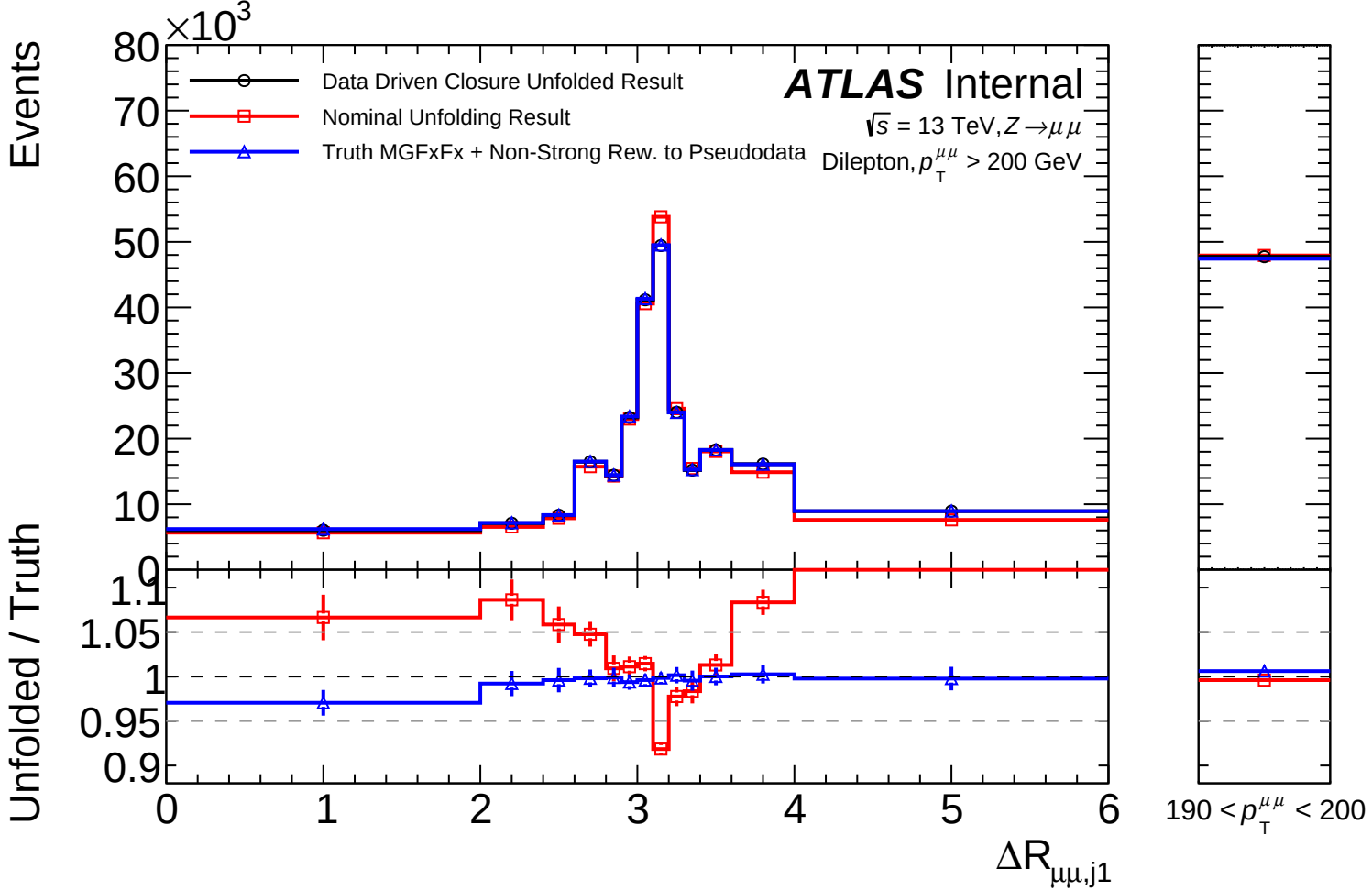
Unfolded / Nominal



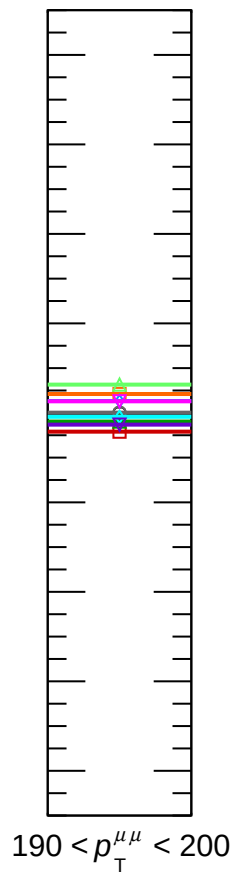
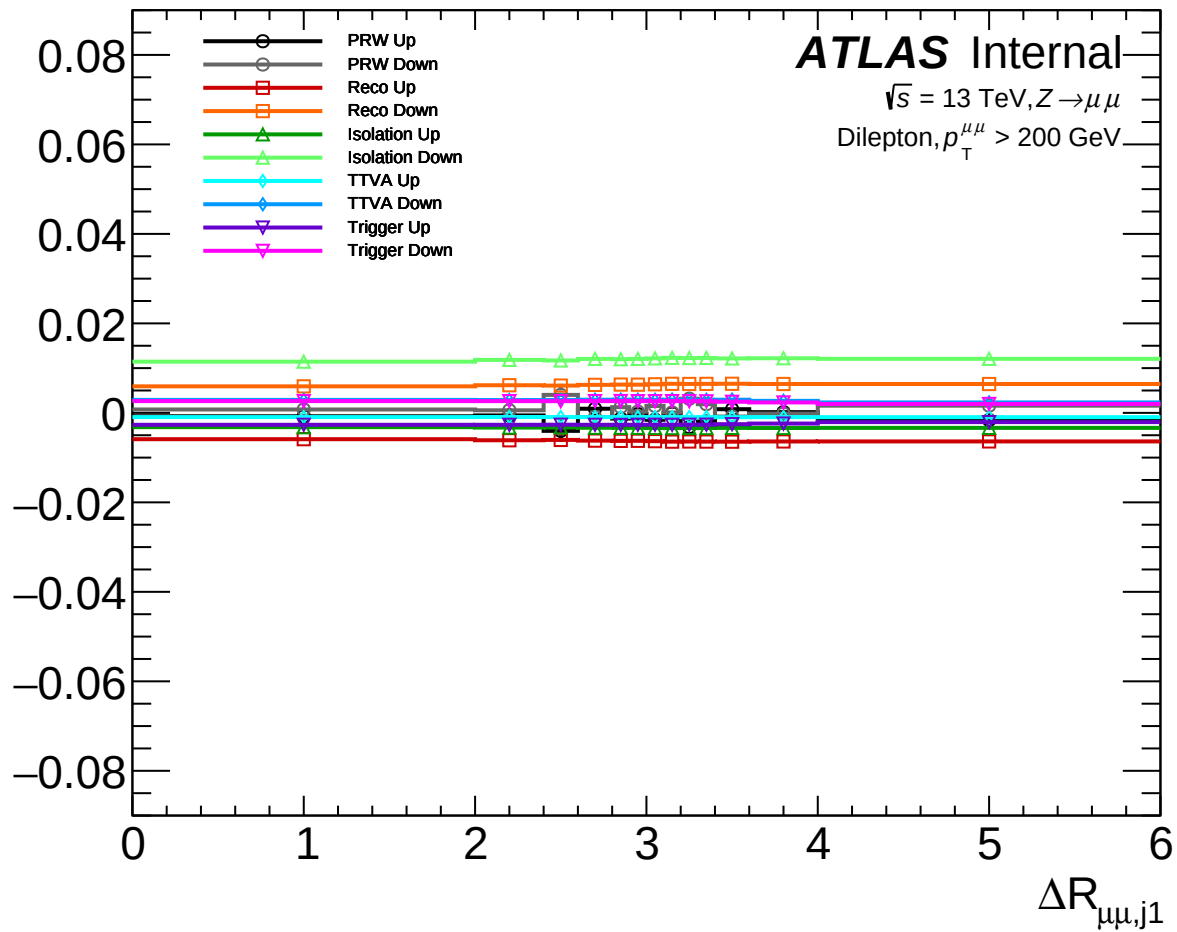
Events

Rew. Reco MC / Data

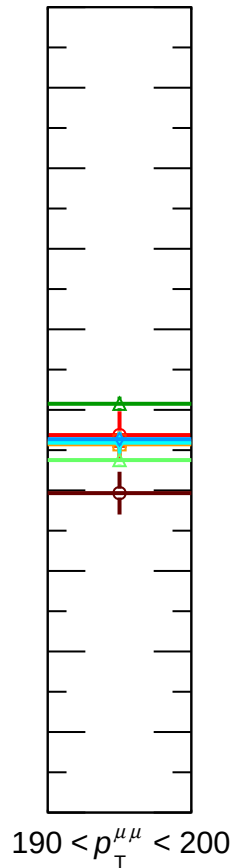
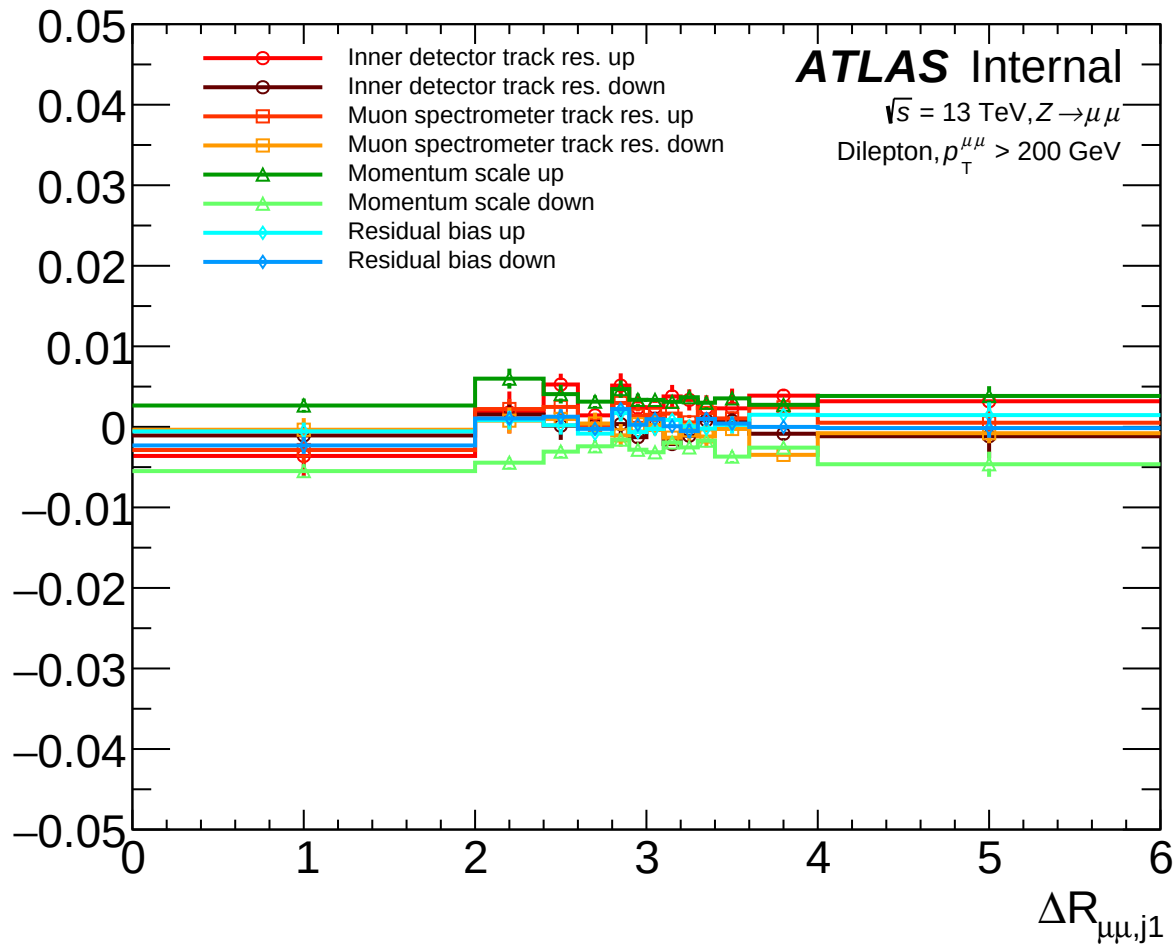




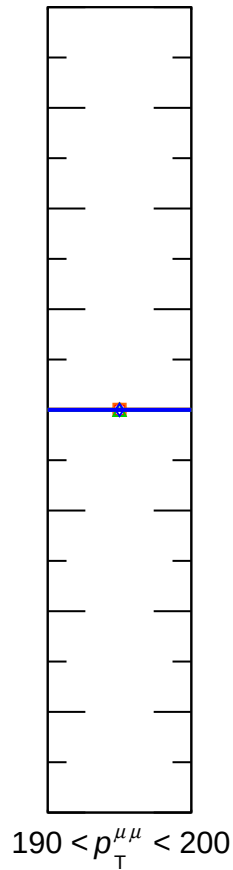
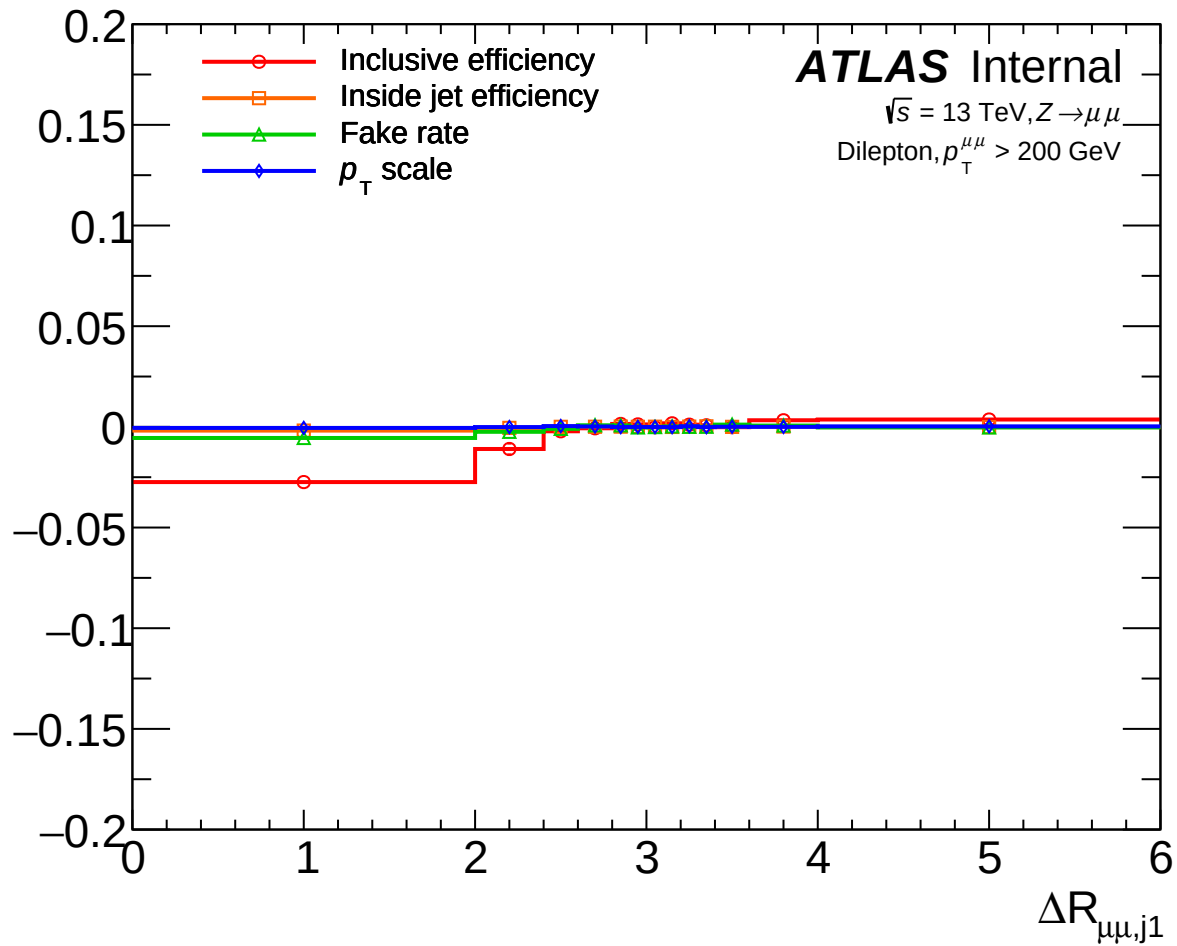
Fractional systematic impact



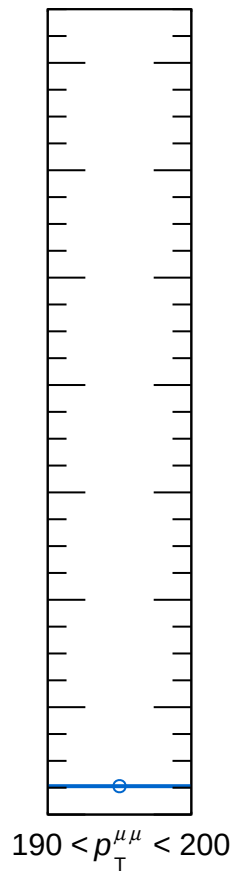
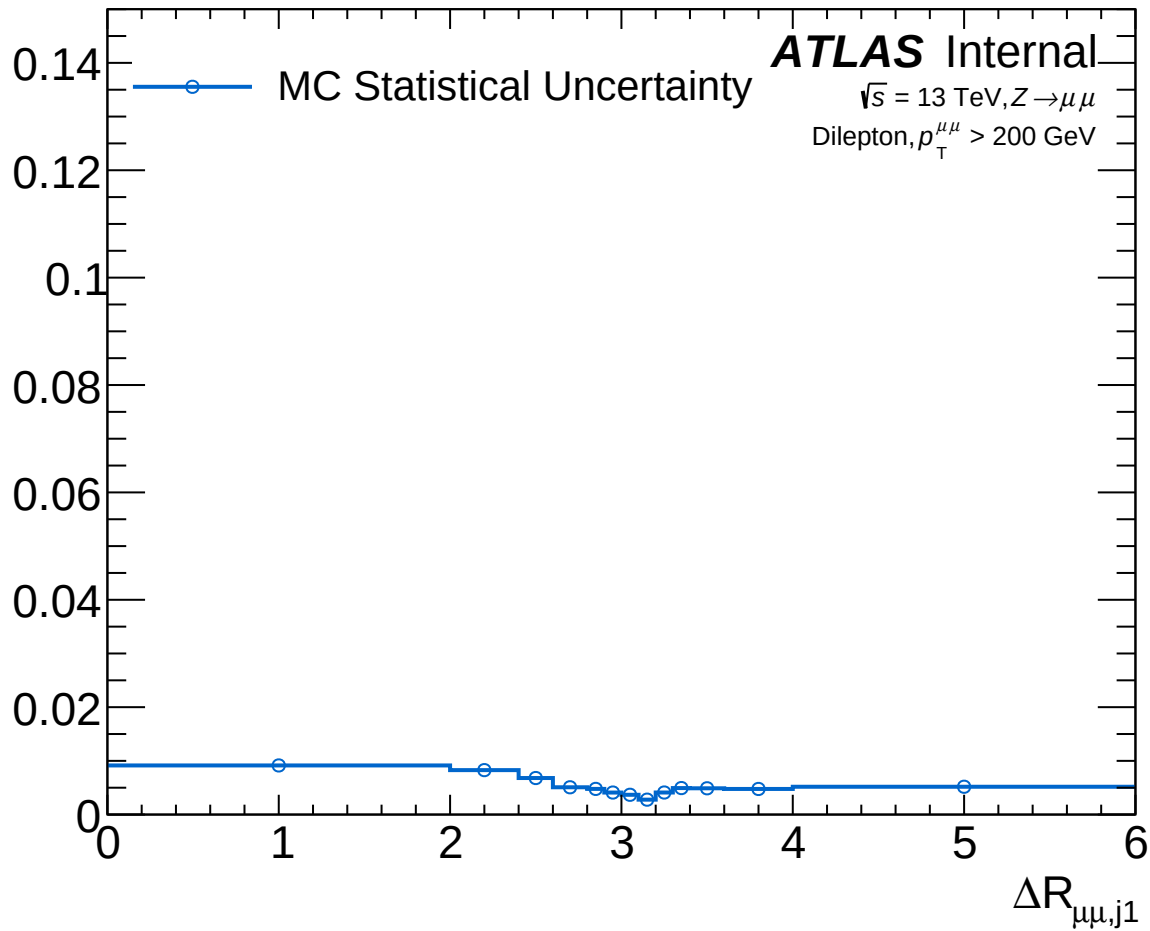
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



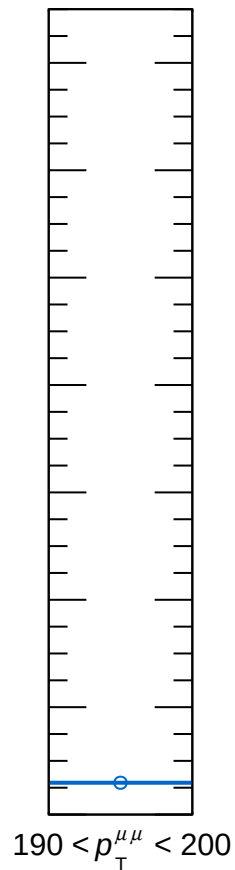
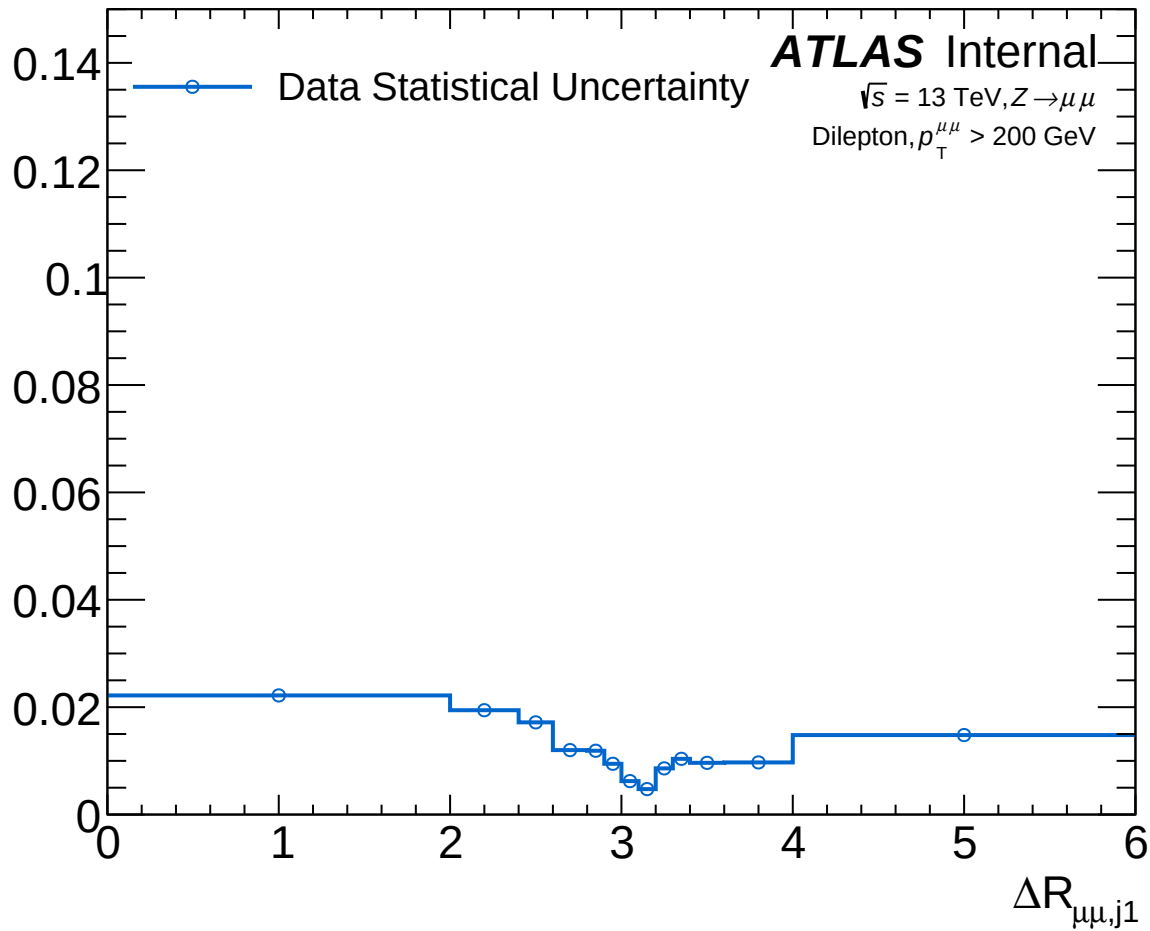
Fractional systematic impact

ATLAS Internal

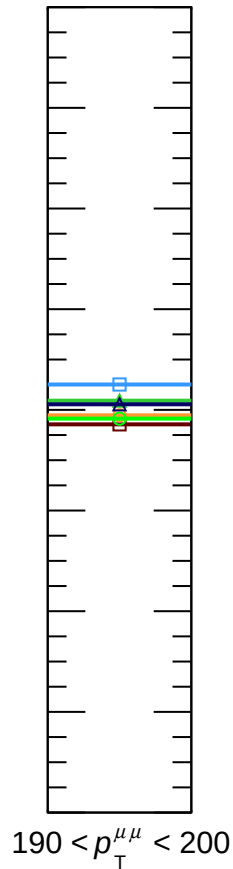
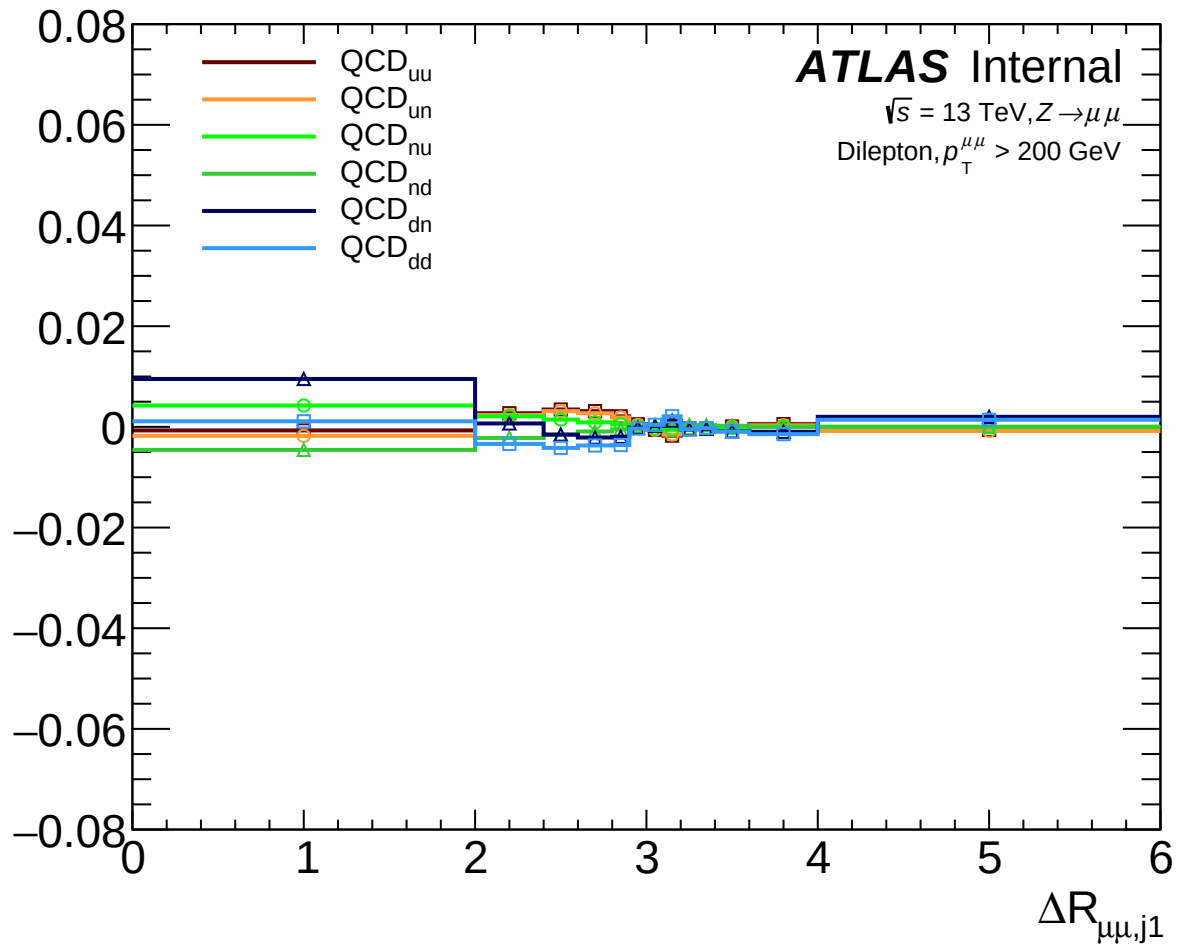
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

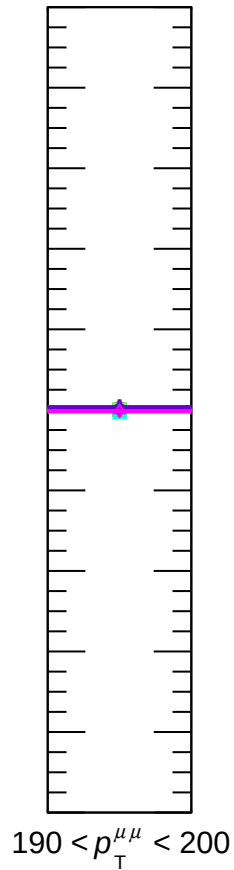
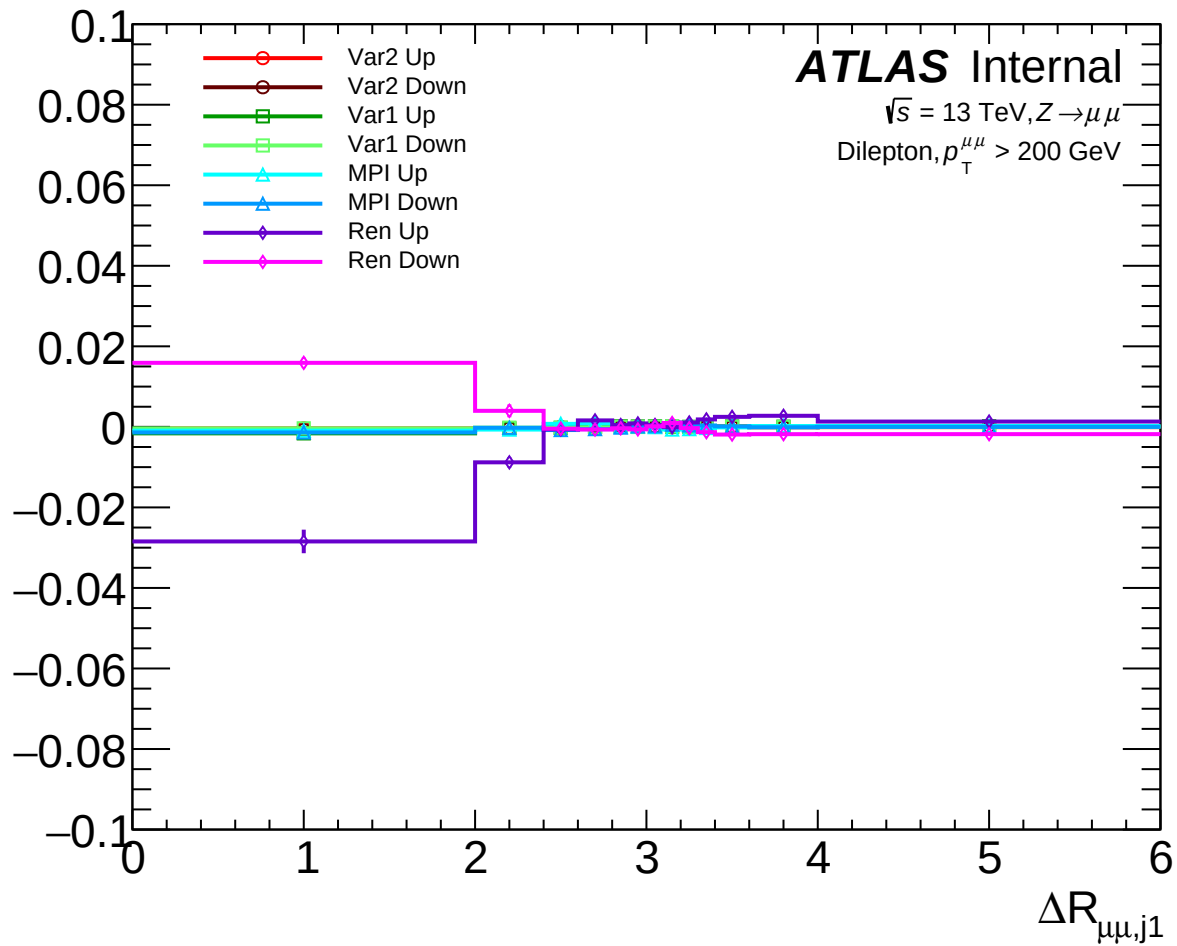
Data Statistical Uncertainty



Fractional systematic impact



Fractional systematic impact



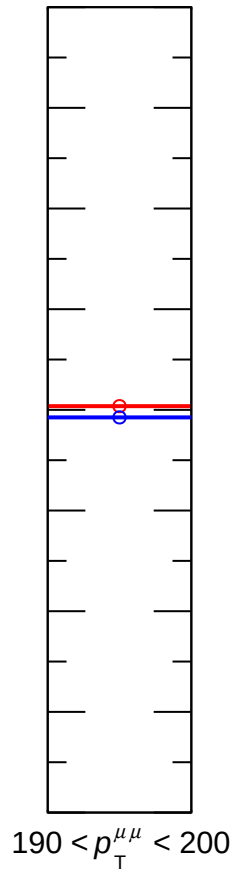
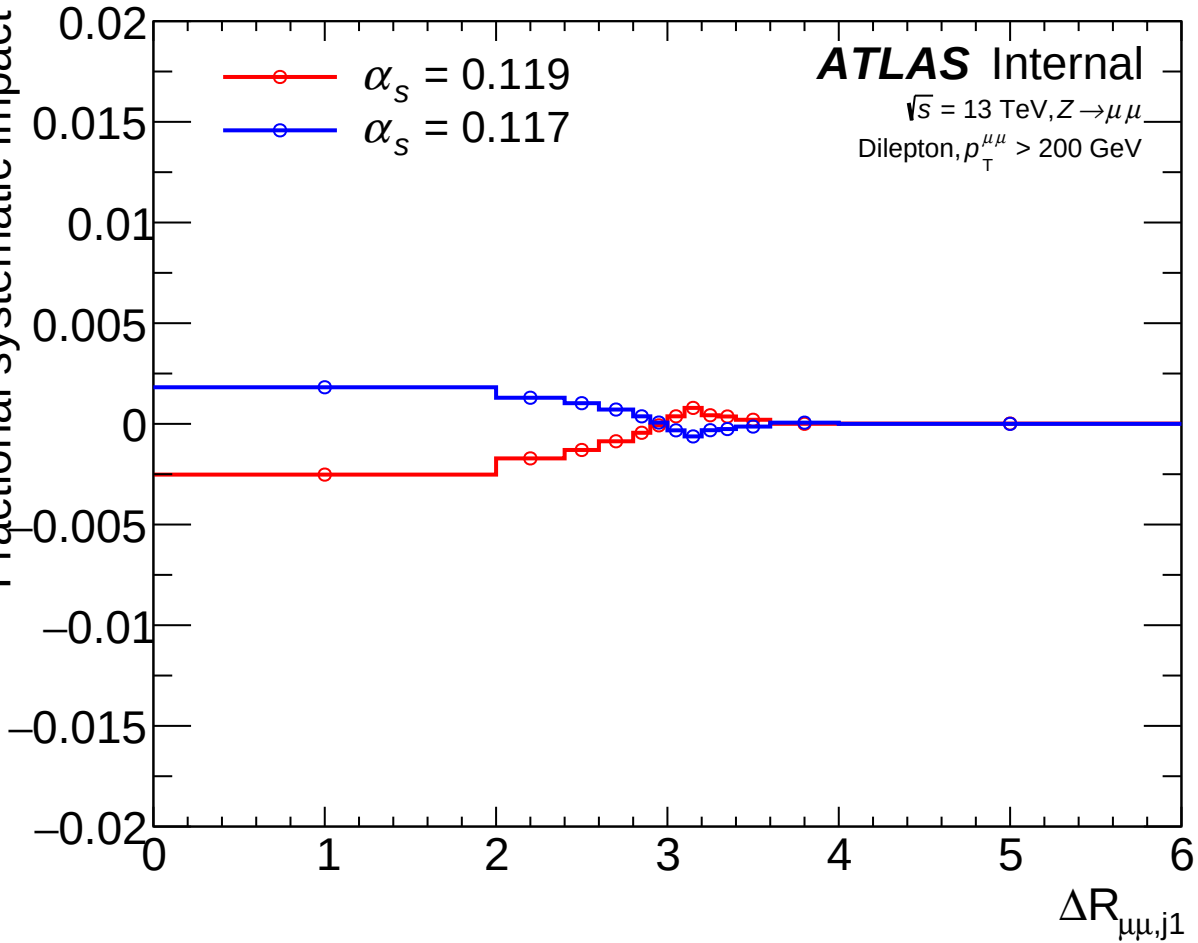
Fractional systematic impact

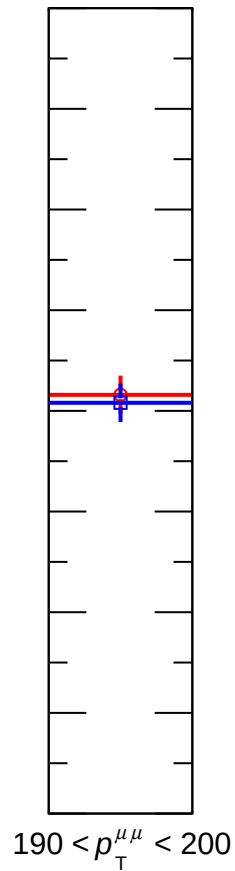
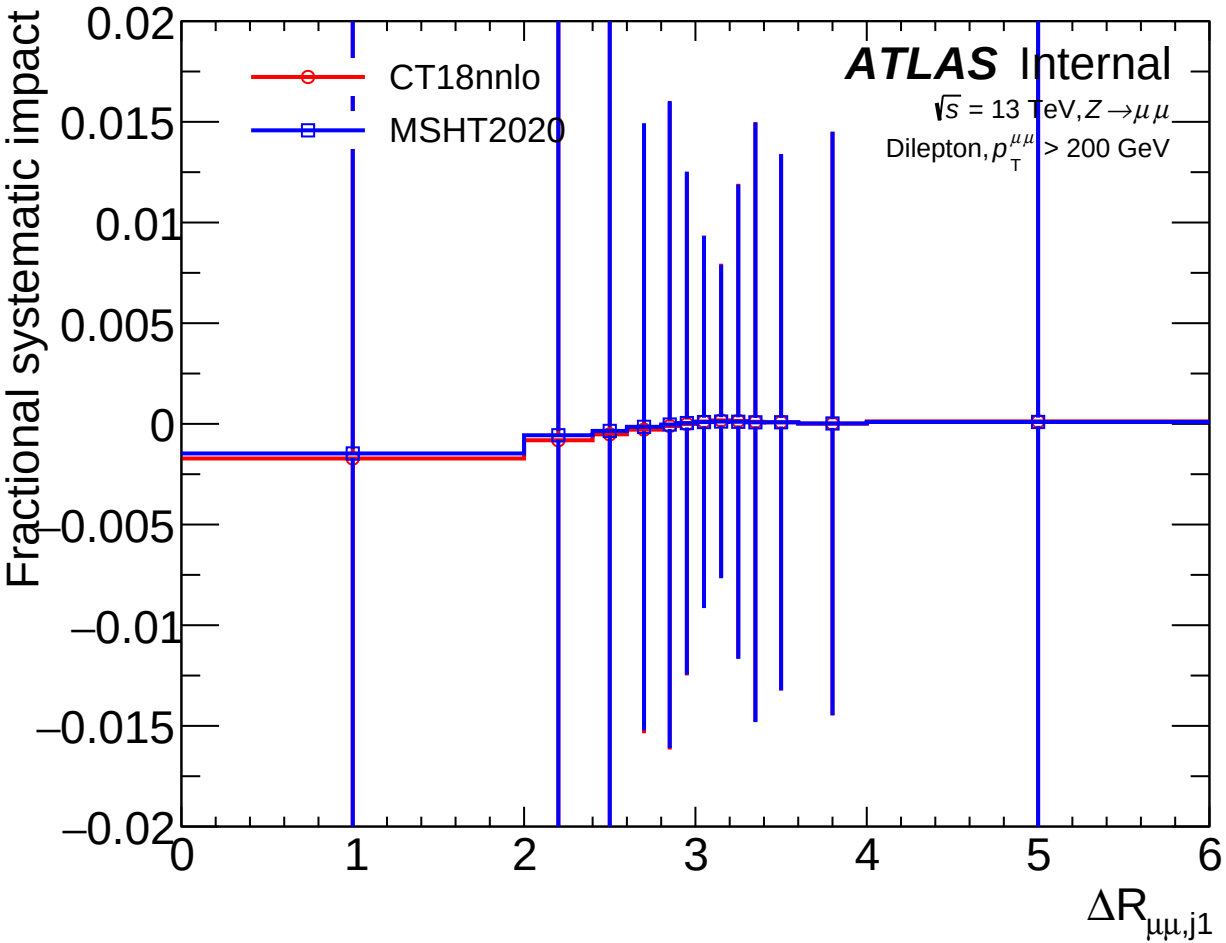
$\alpha_s = 0.119$
 $\alpha_s = 0.117$

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200$ GeV





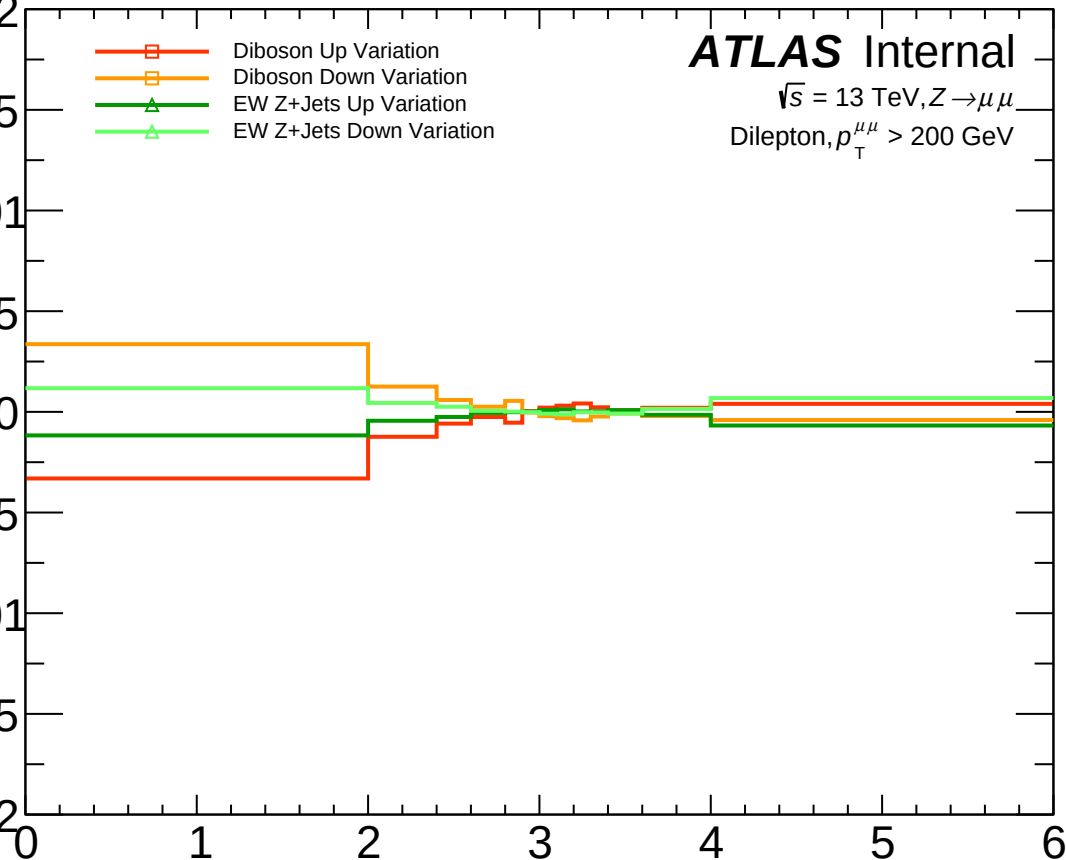
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

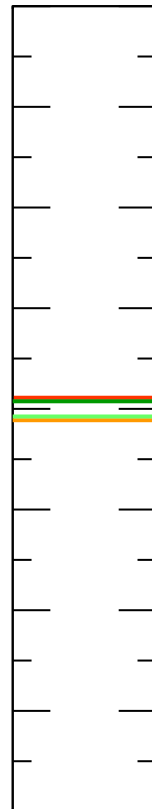
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$



$\Delta R_{\mu\mu, j1}$

$190 < p_T^{\mu\mu} < 200$

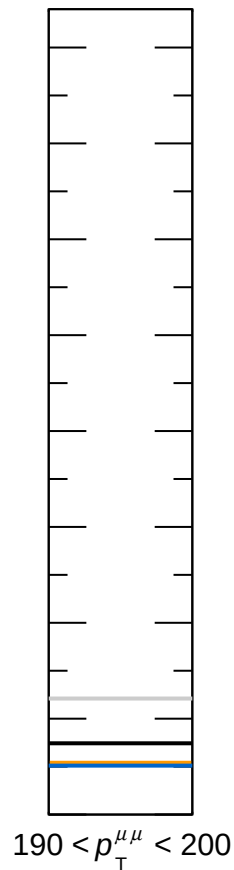
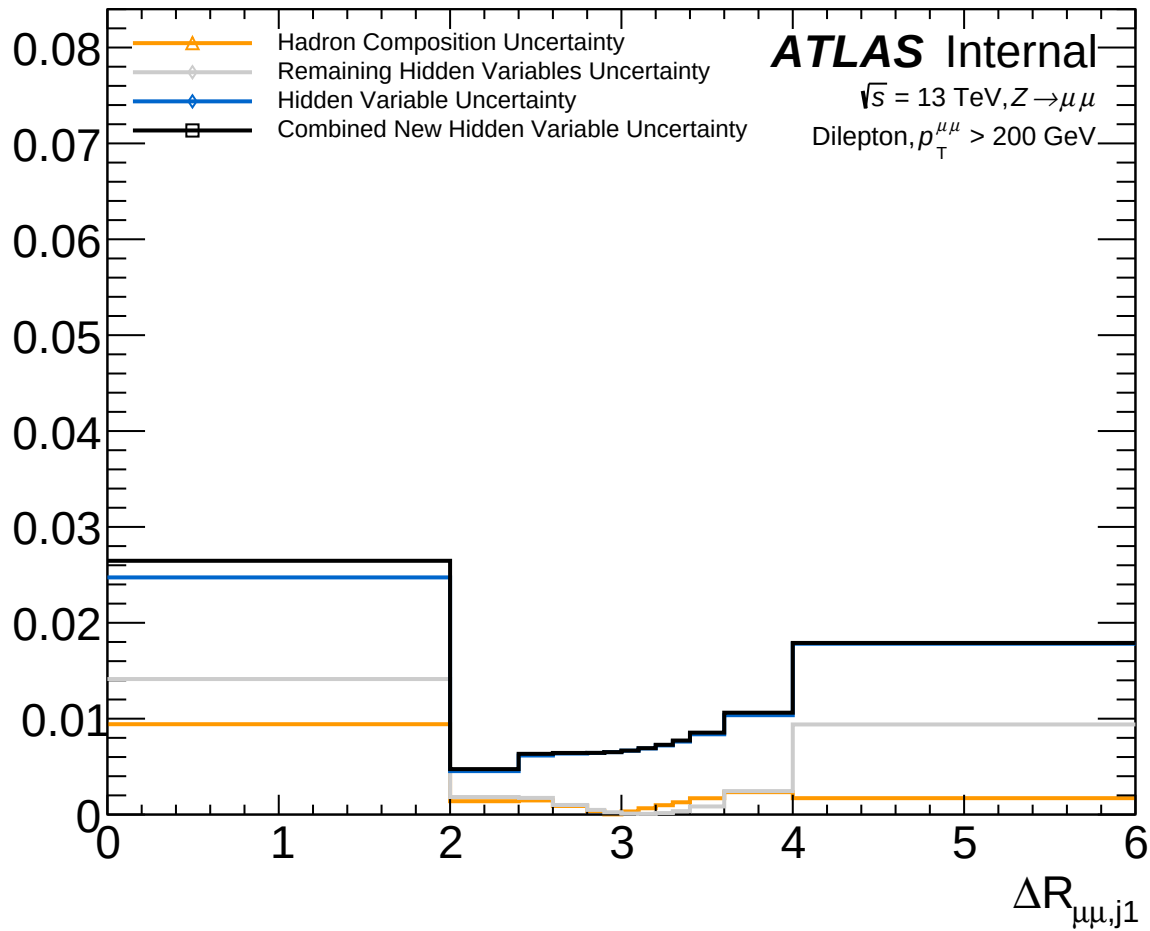


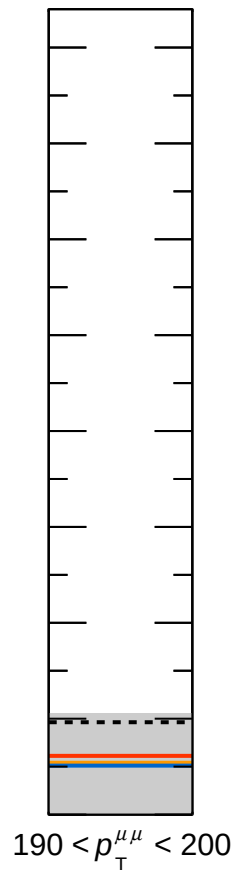
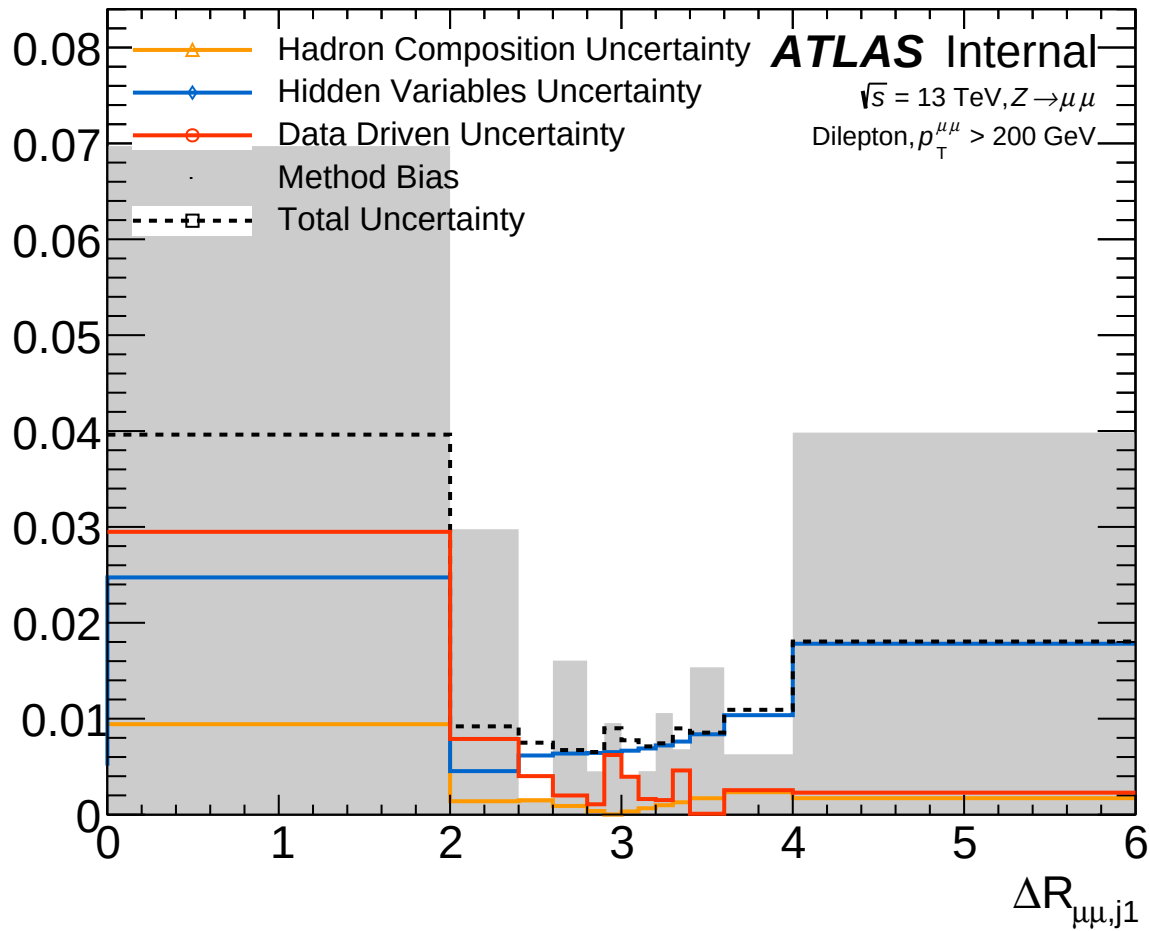
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

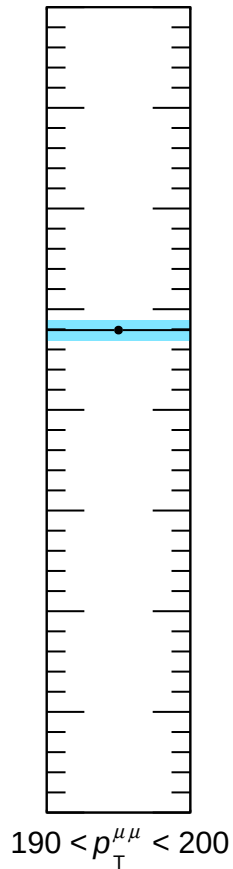
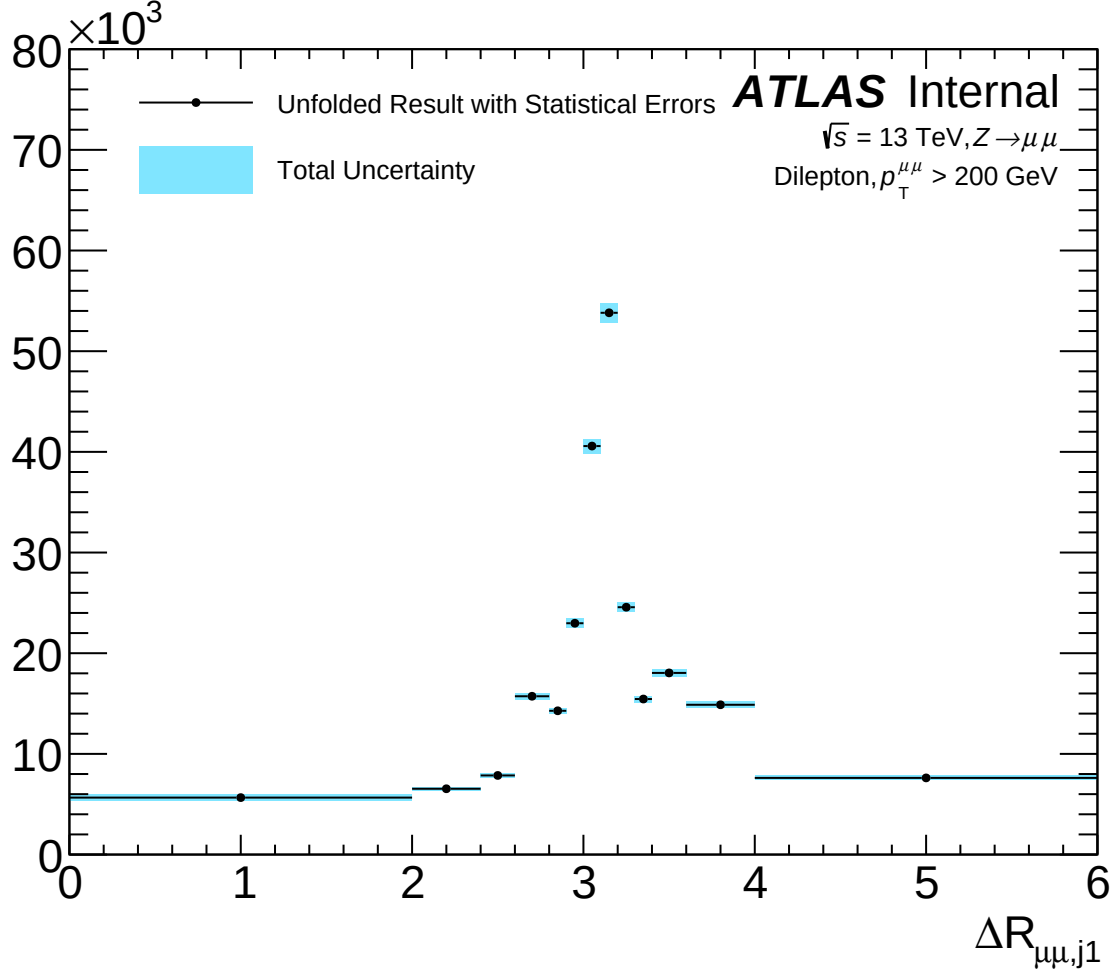
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

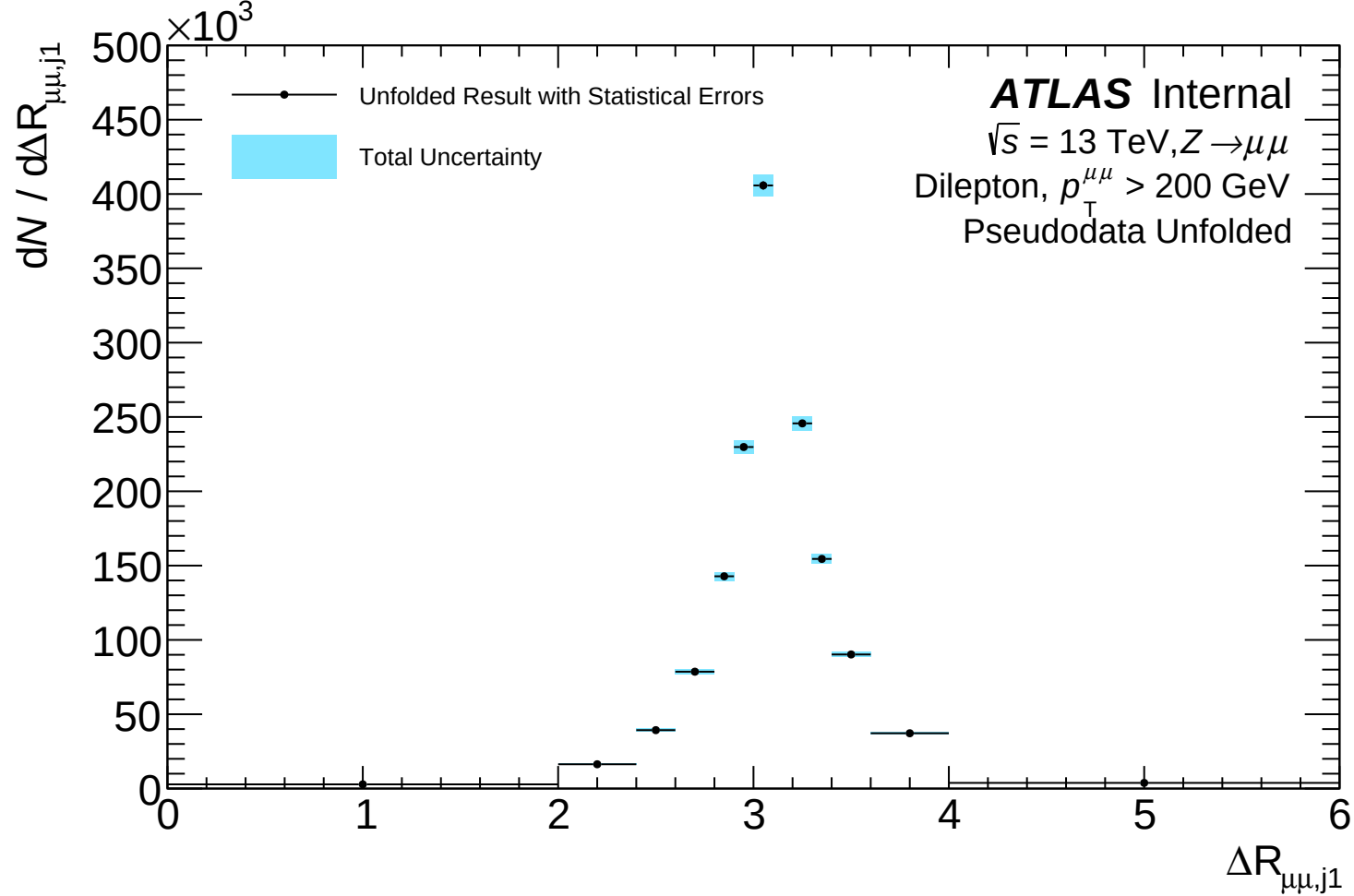
- Hadron Composition Uncertainty
- Remaining Hidden Variables Uncertainty
- Hidden Variable Uncertainty
- Combined New Hidden Variable Uncertainty

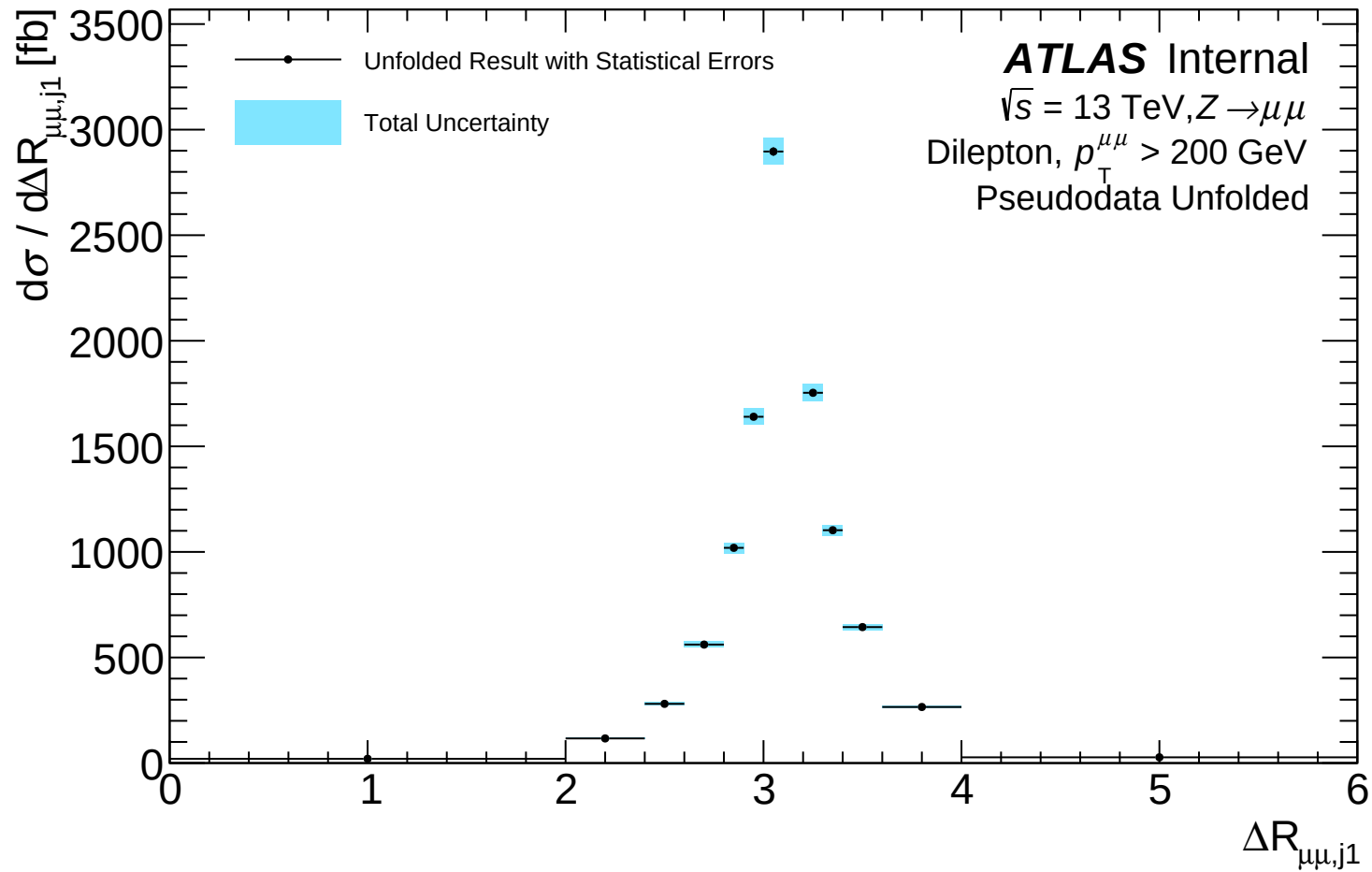




Events







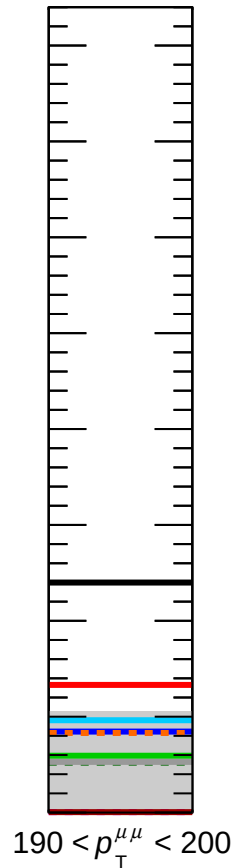
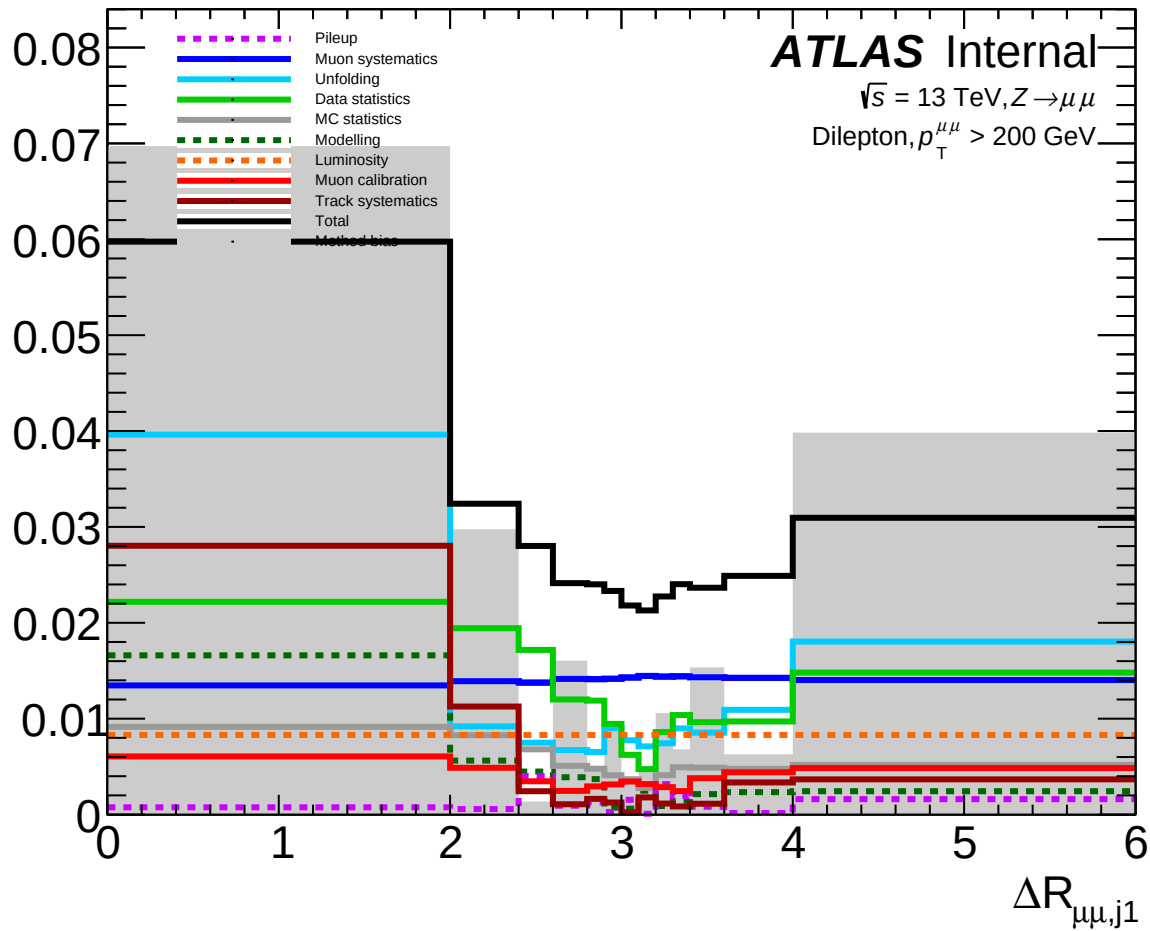
Fractional uncertainty

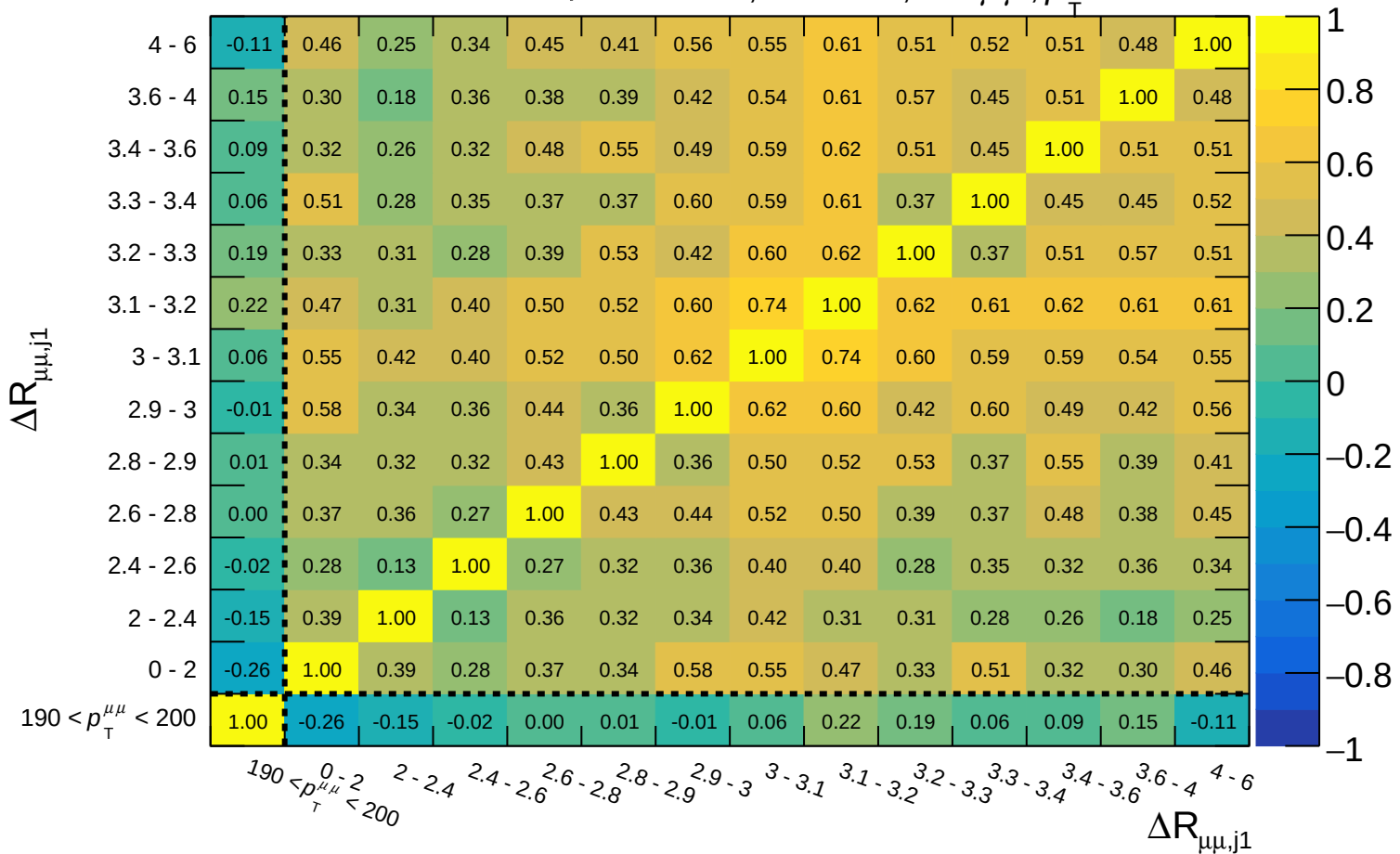
ATLAS Internal

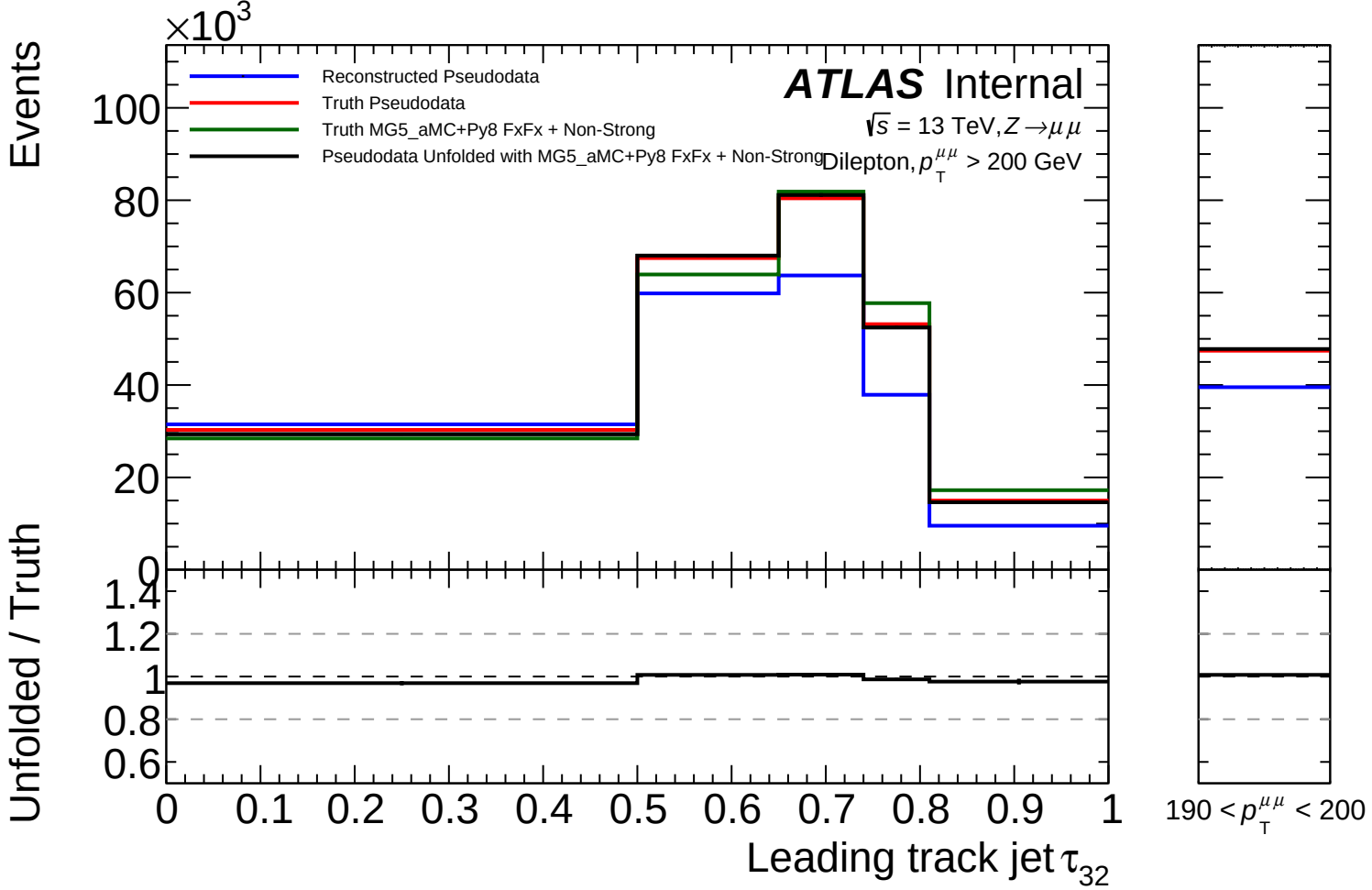
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

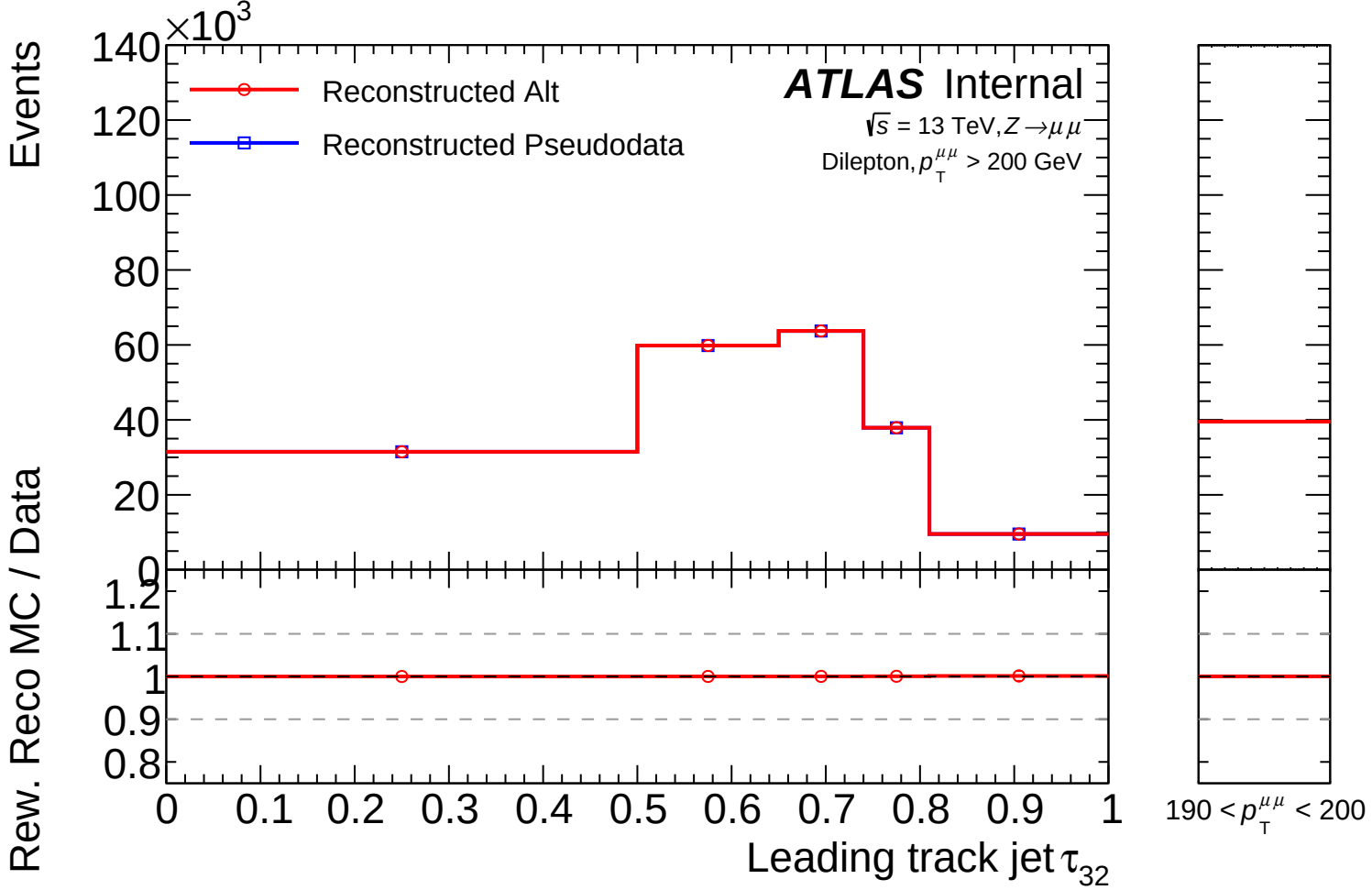
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

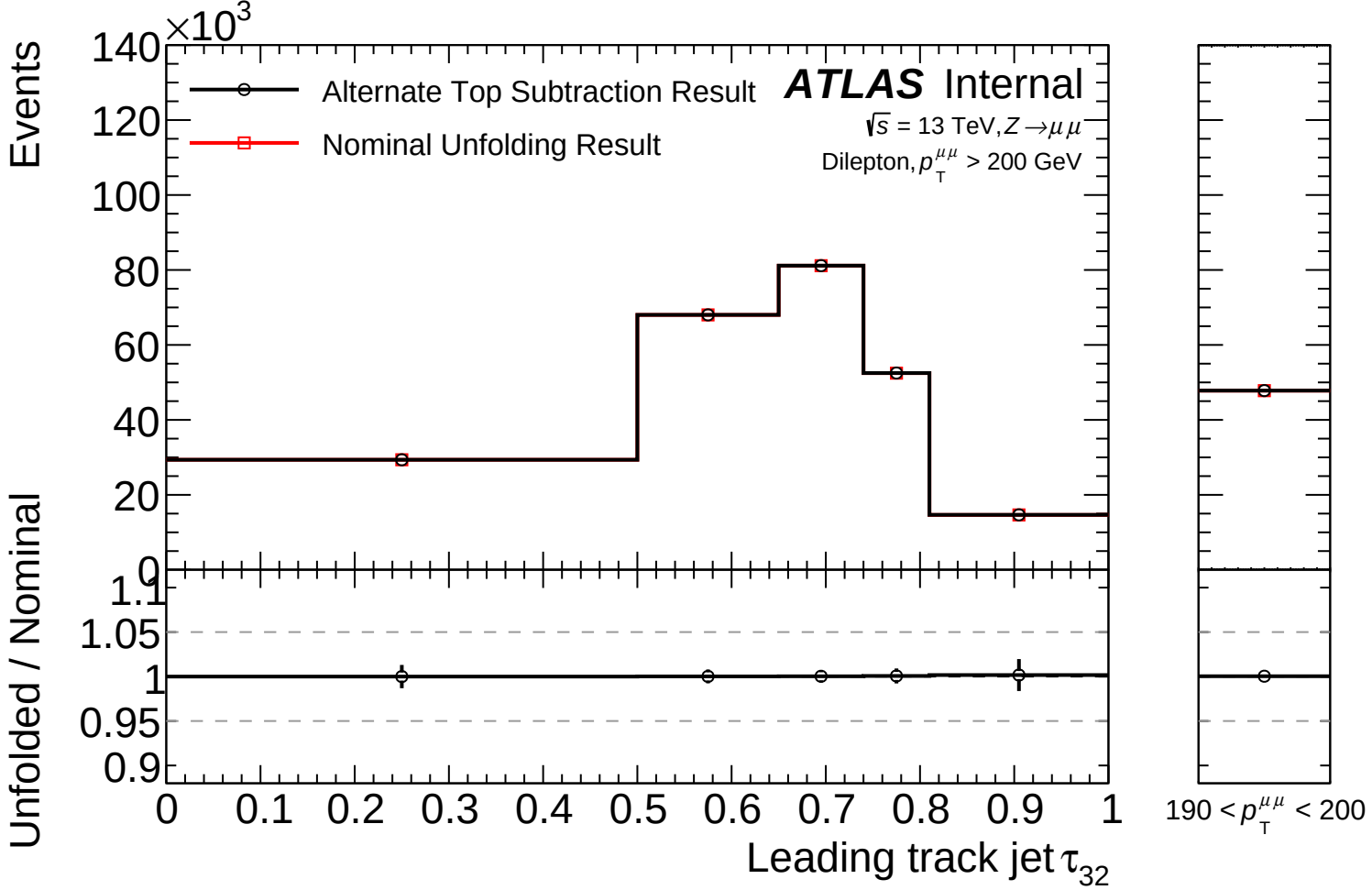
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Muon calibration
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ 







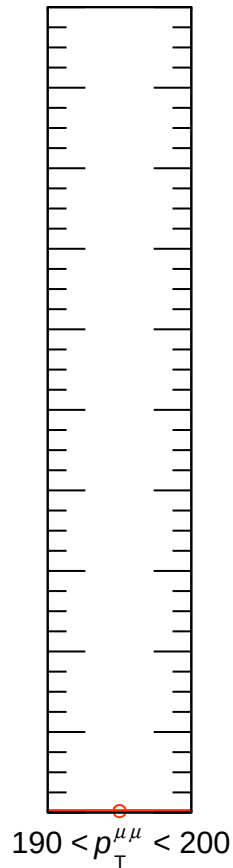
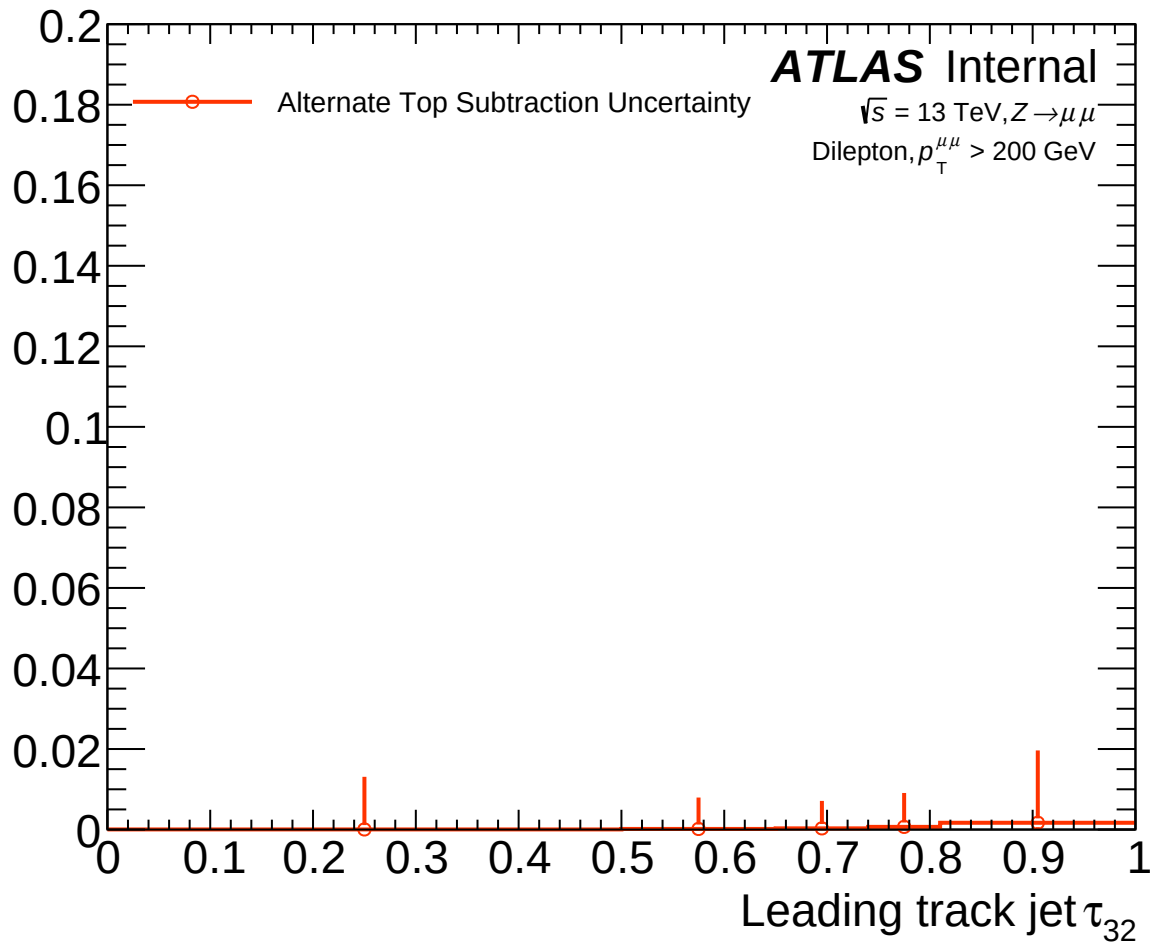
Fractional systematic impact

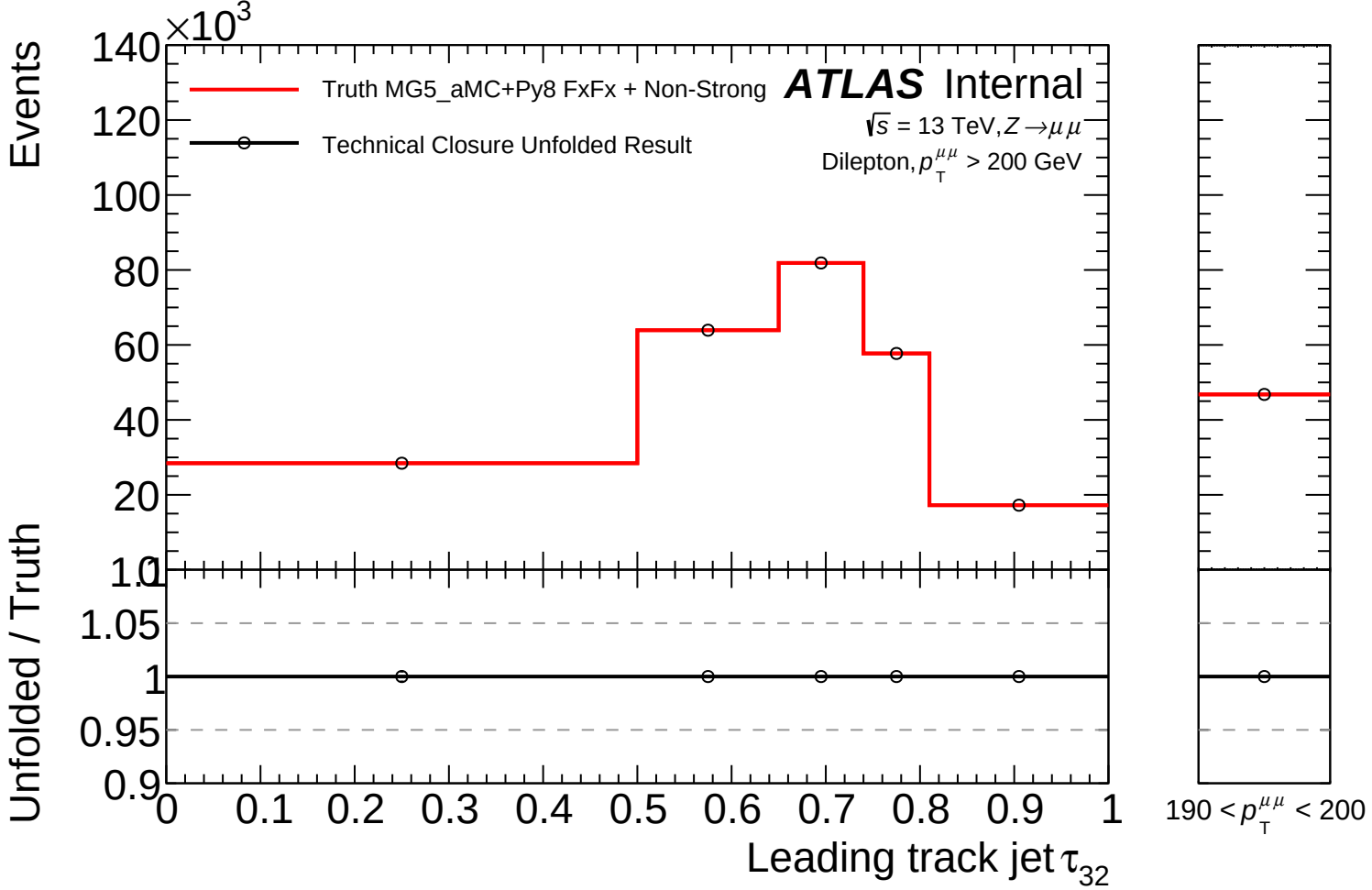
ATLAS Internal

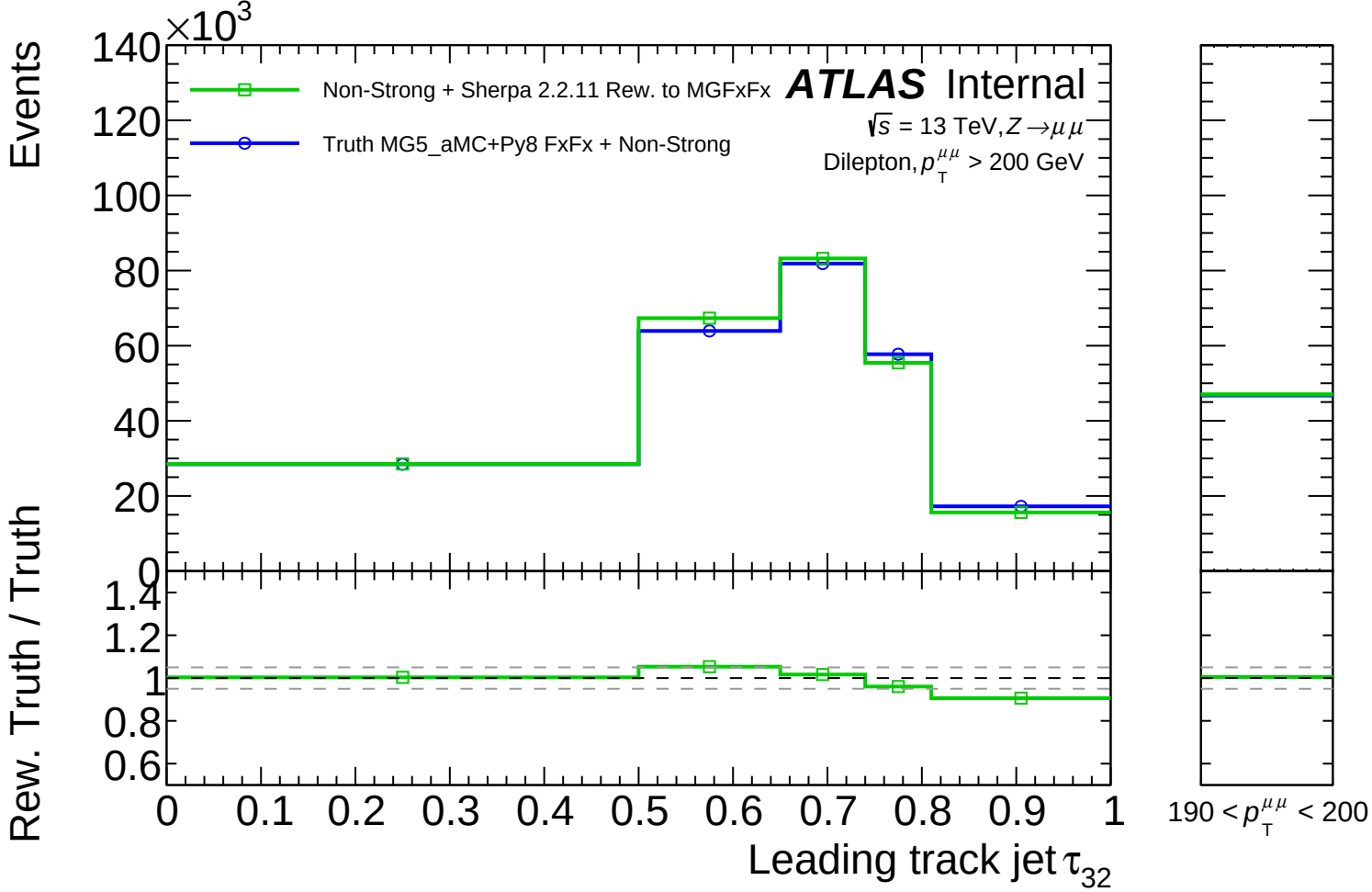
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

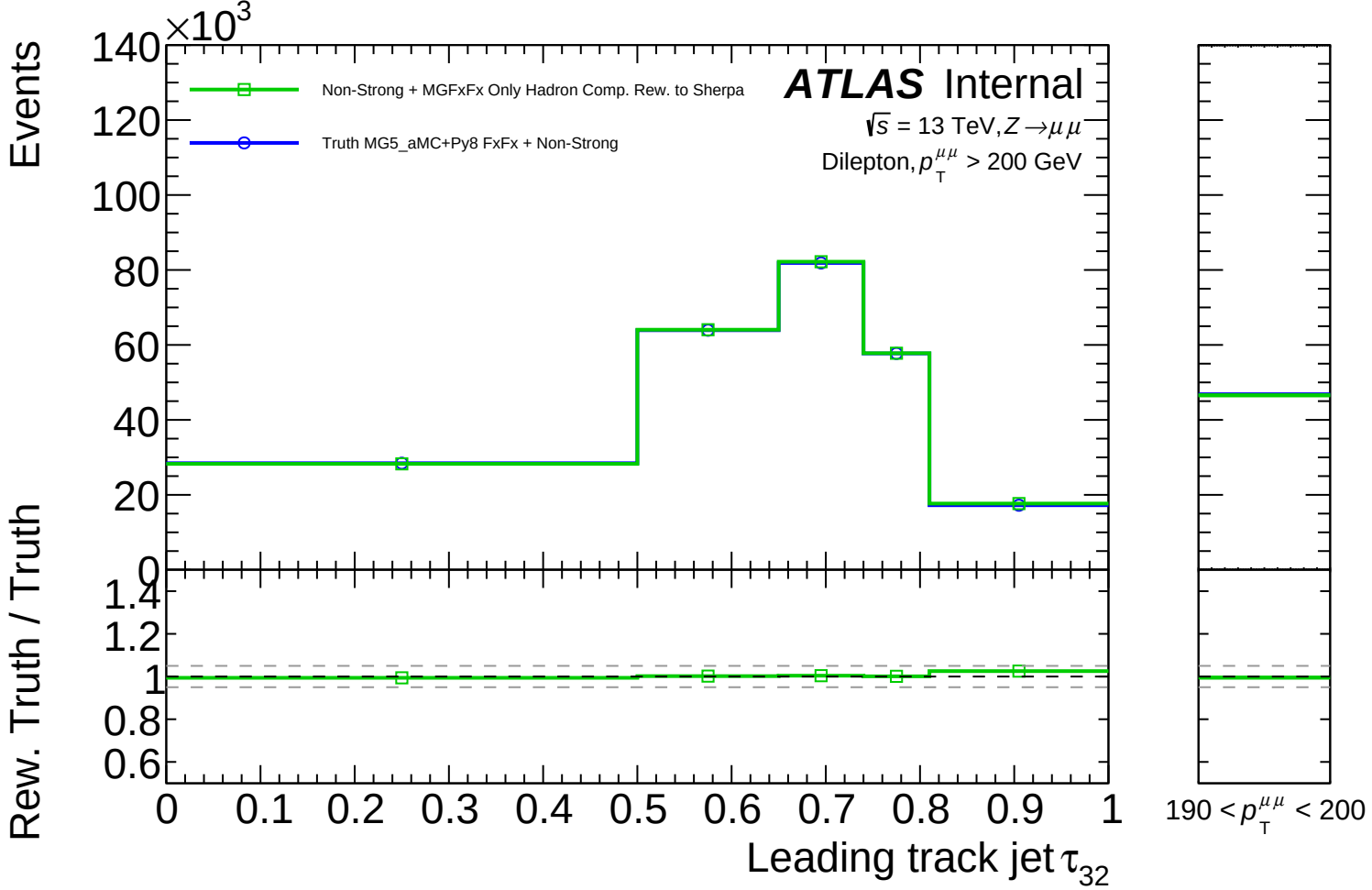
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

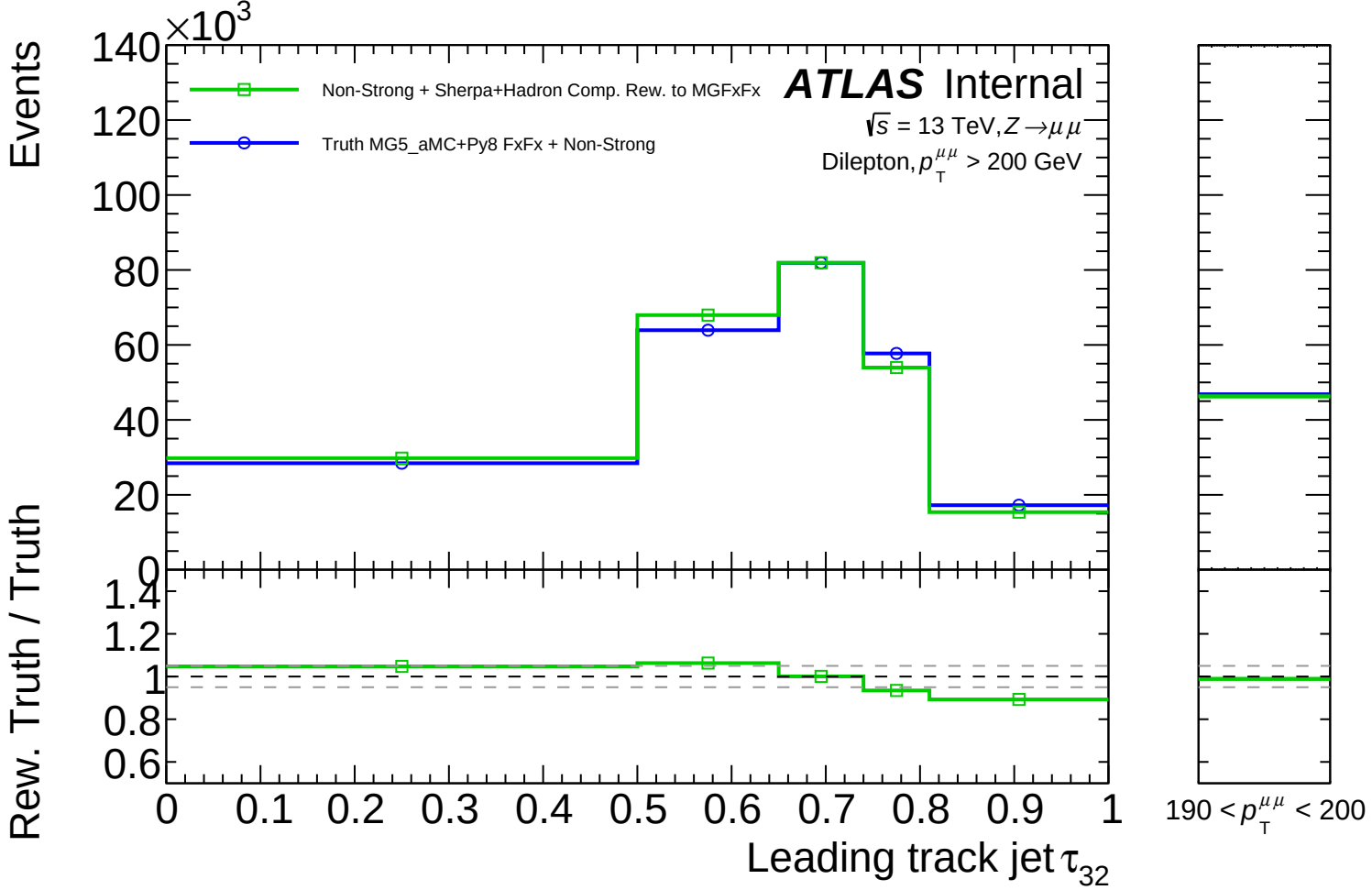
Alternate Top Subtraction Uncertainty

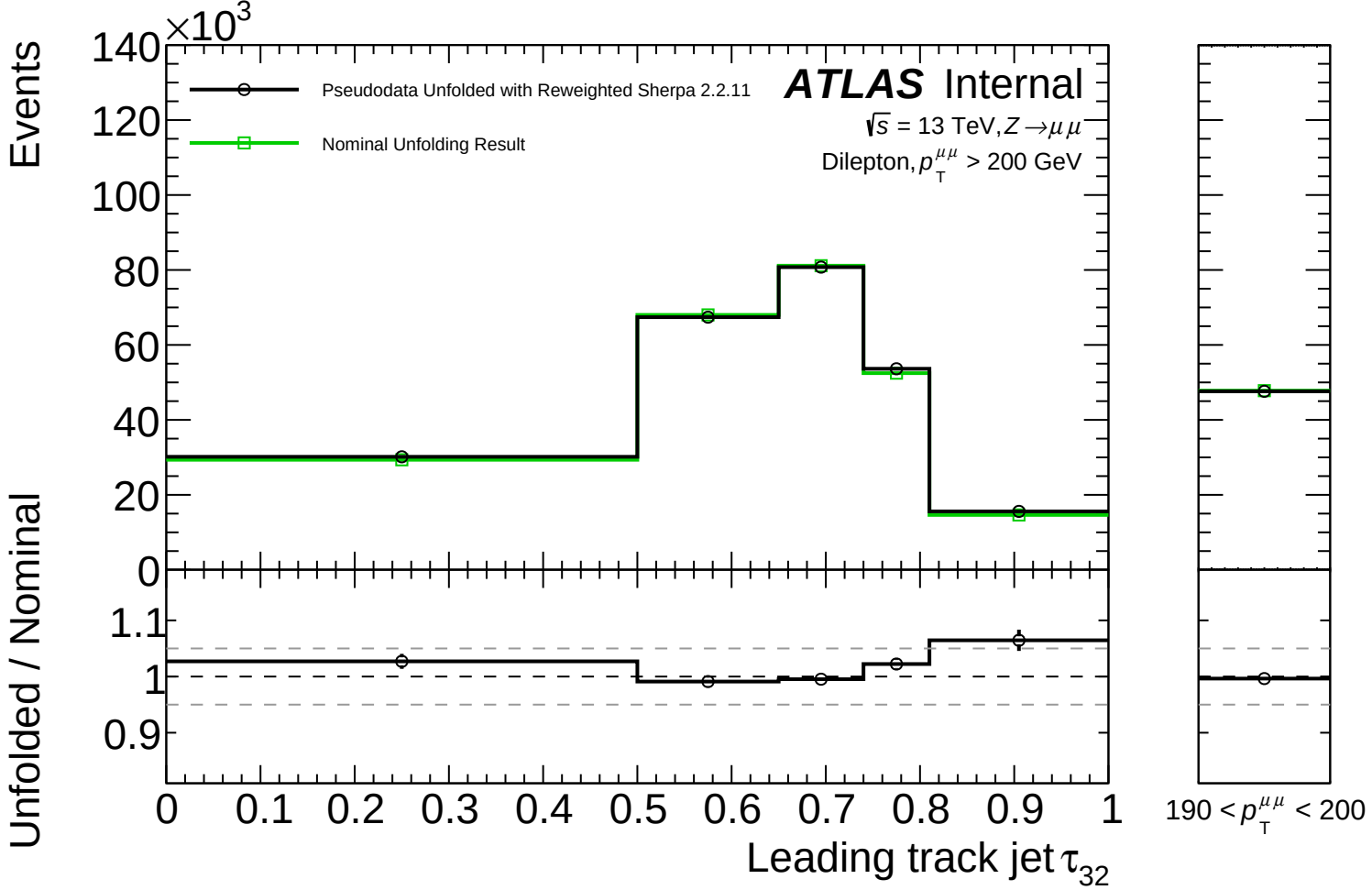


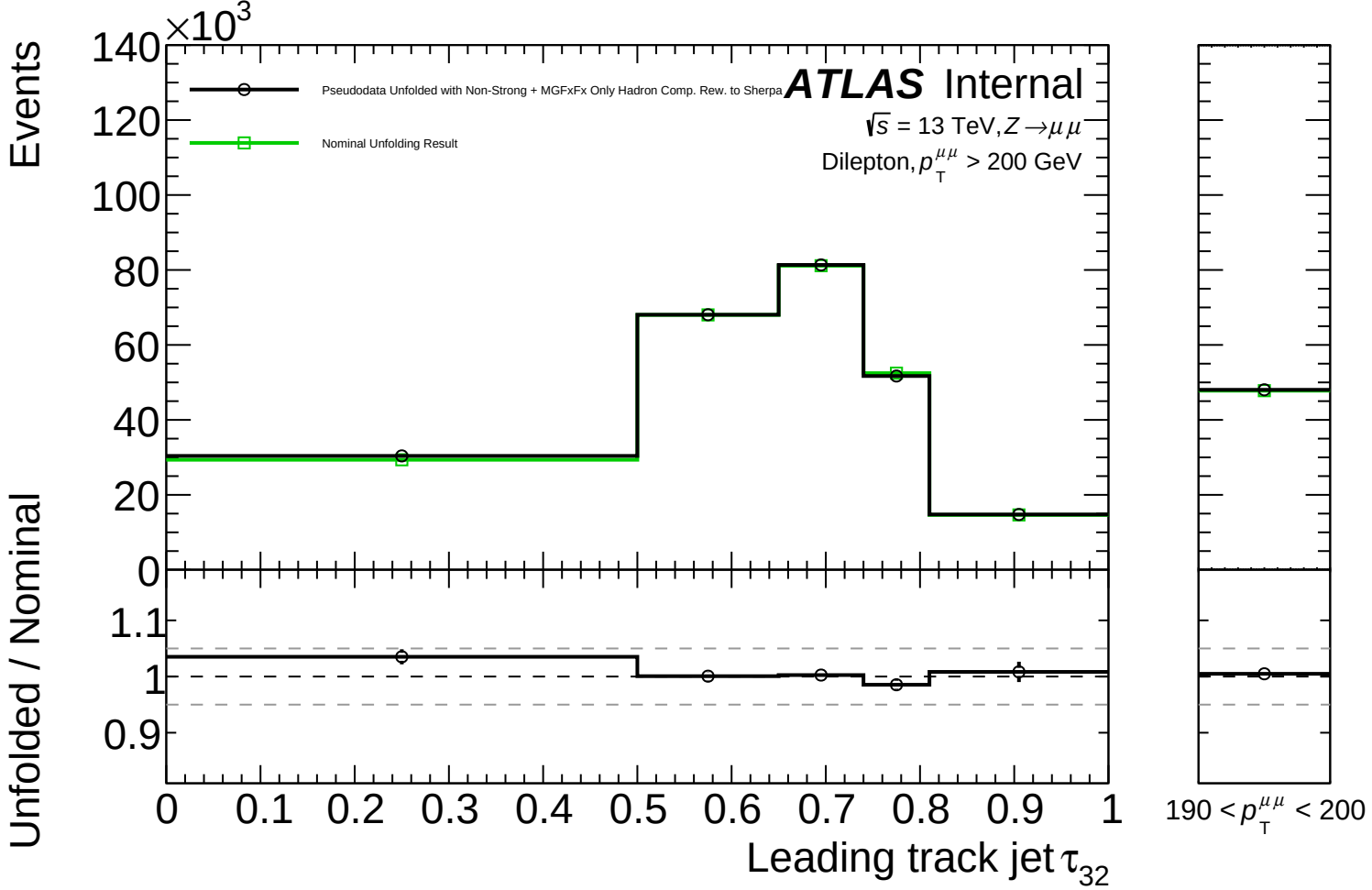


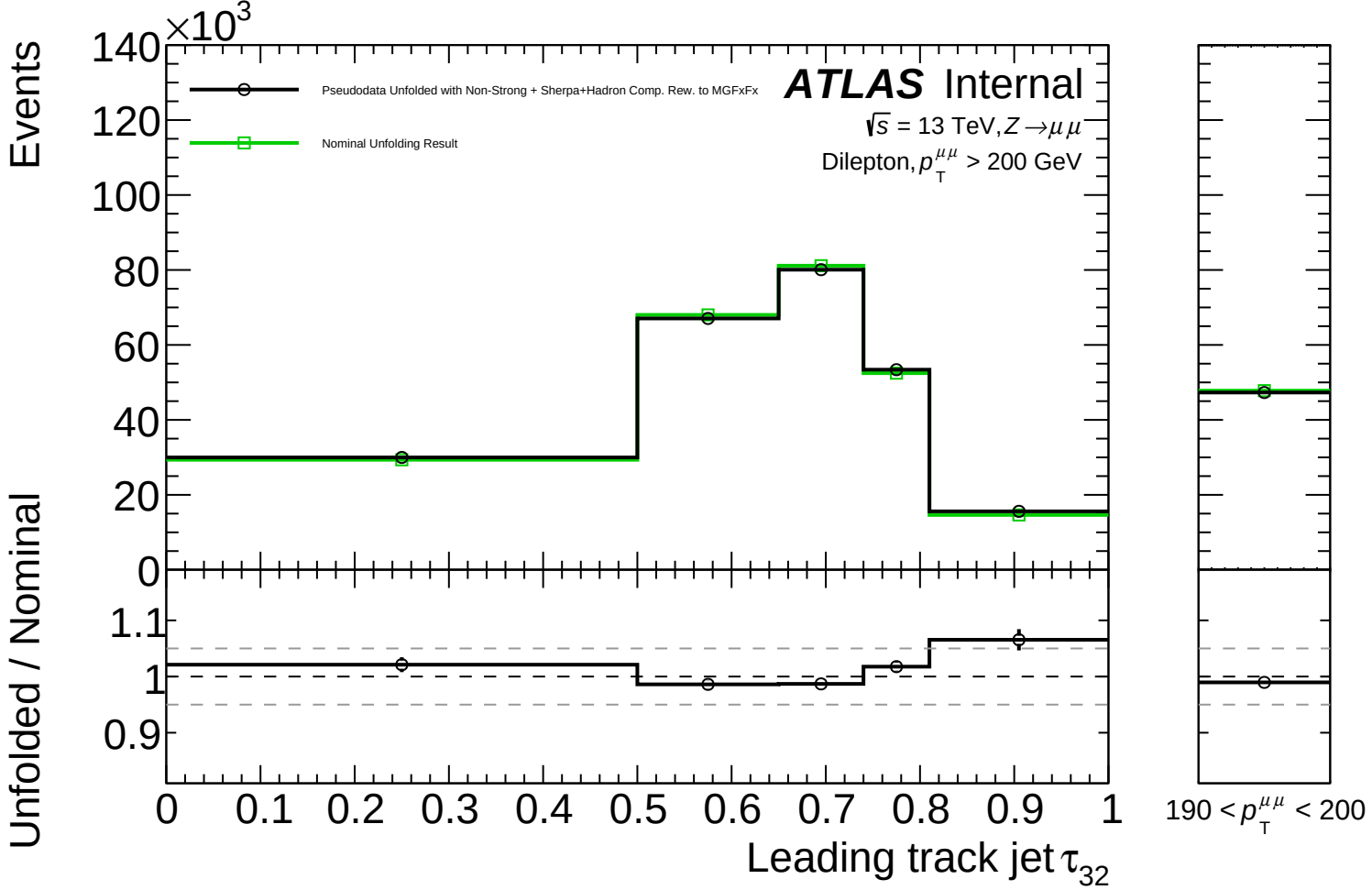


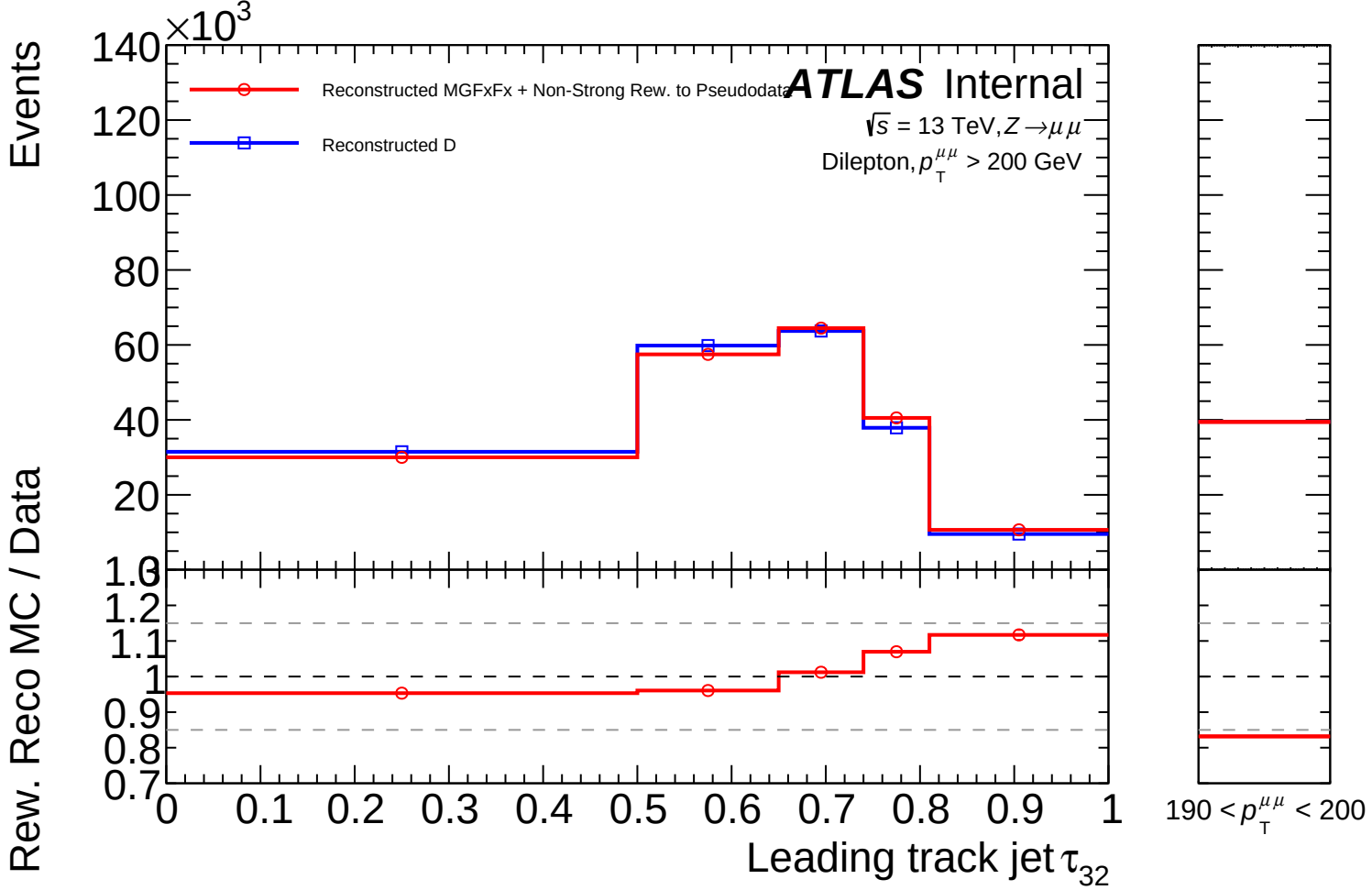


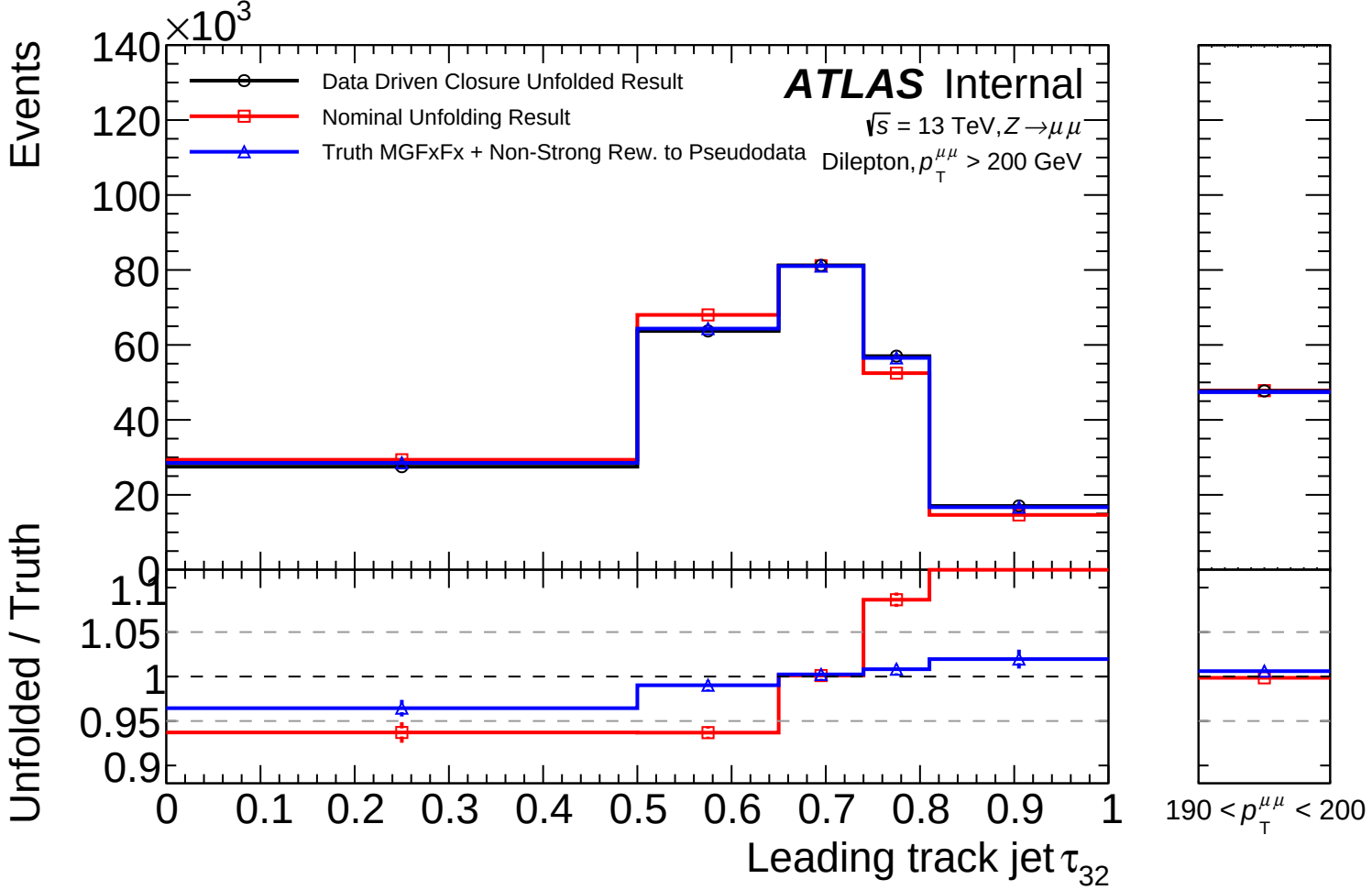




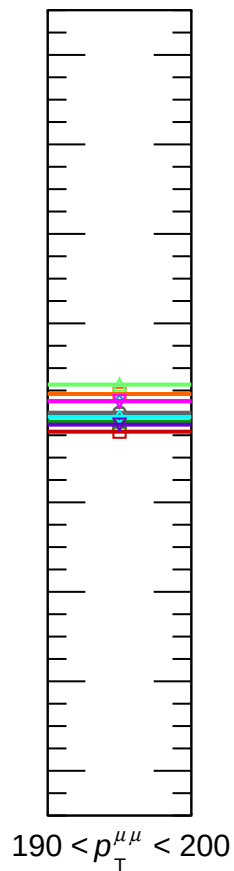
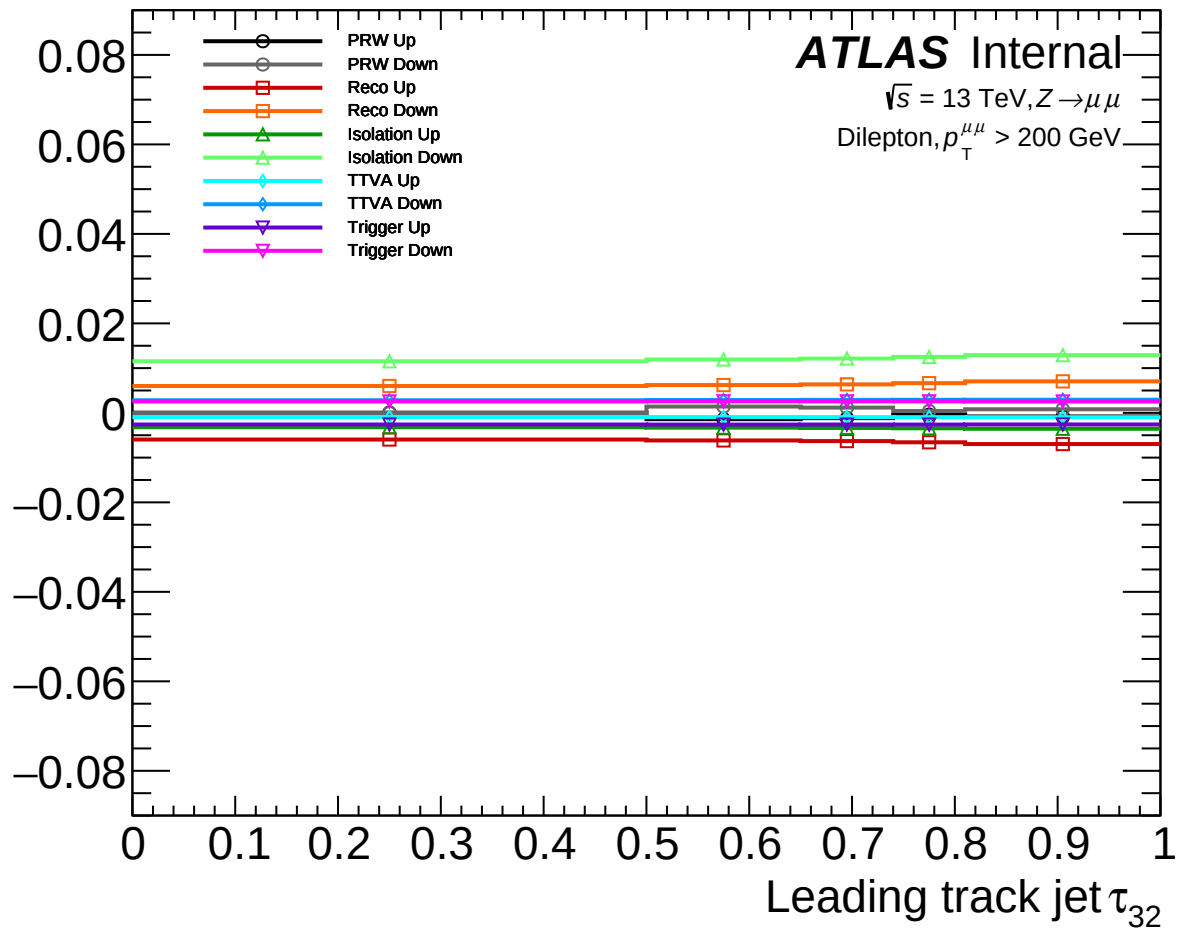




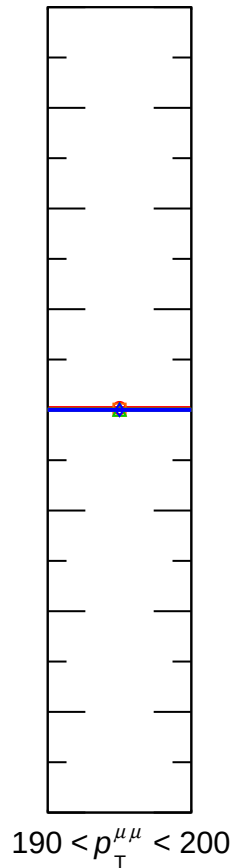
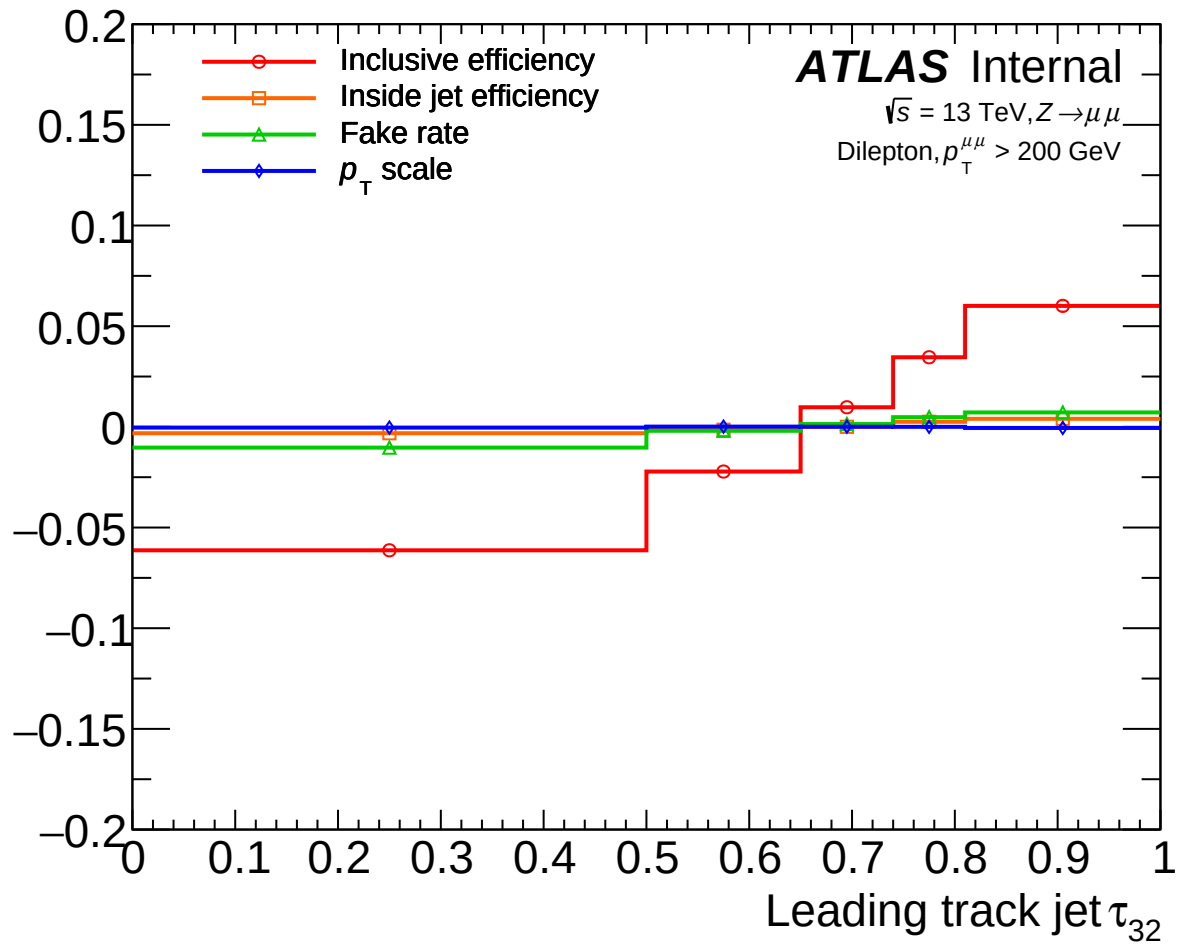




Fractional systematic impact



Fractional systematic impact



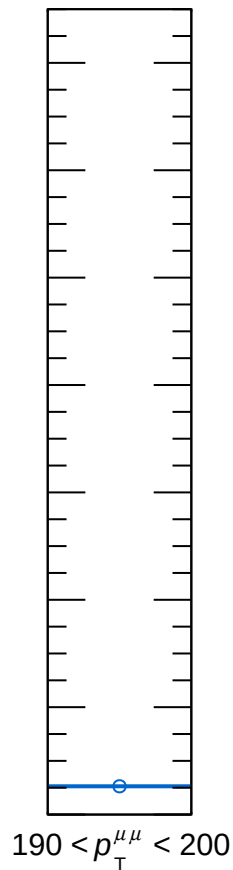
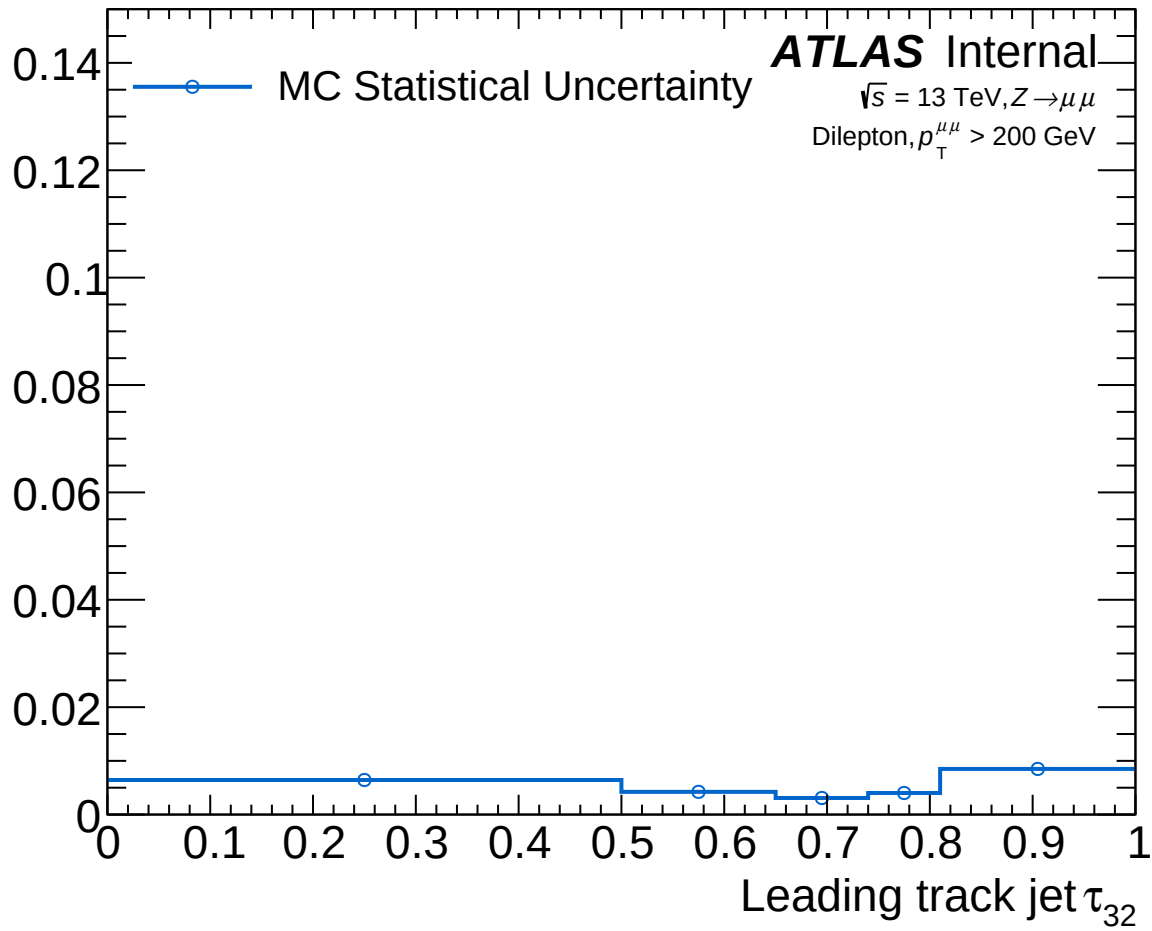
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



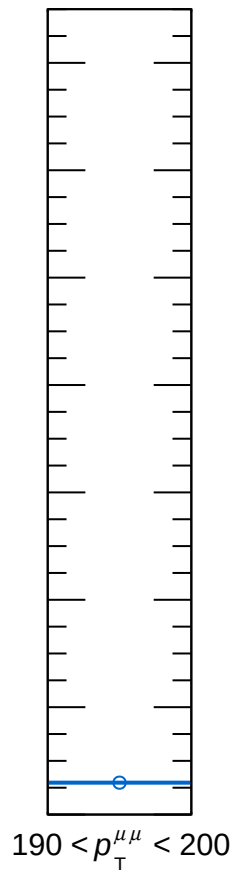
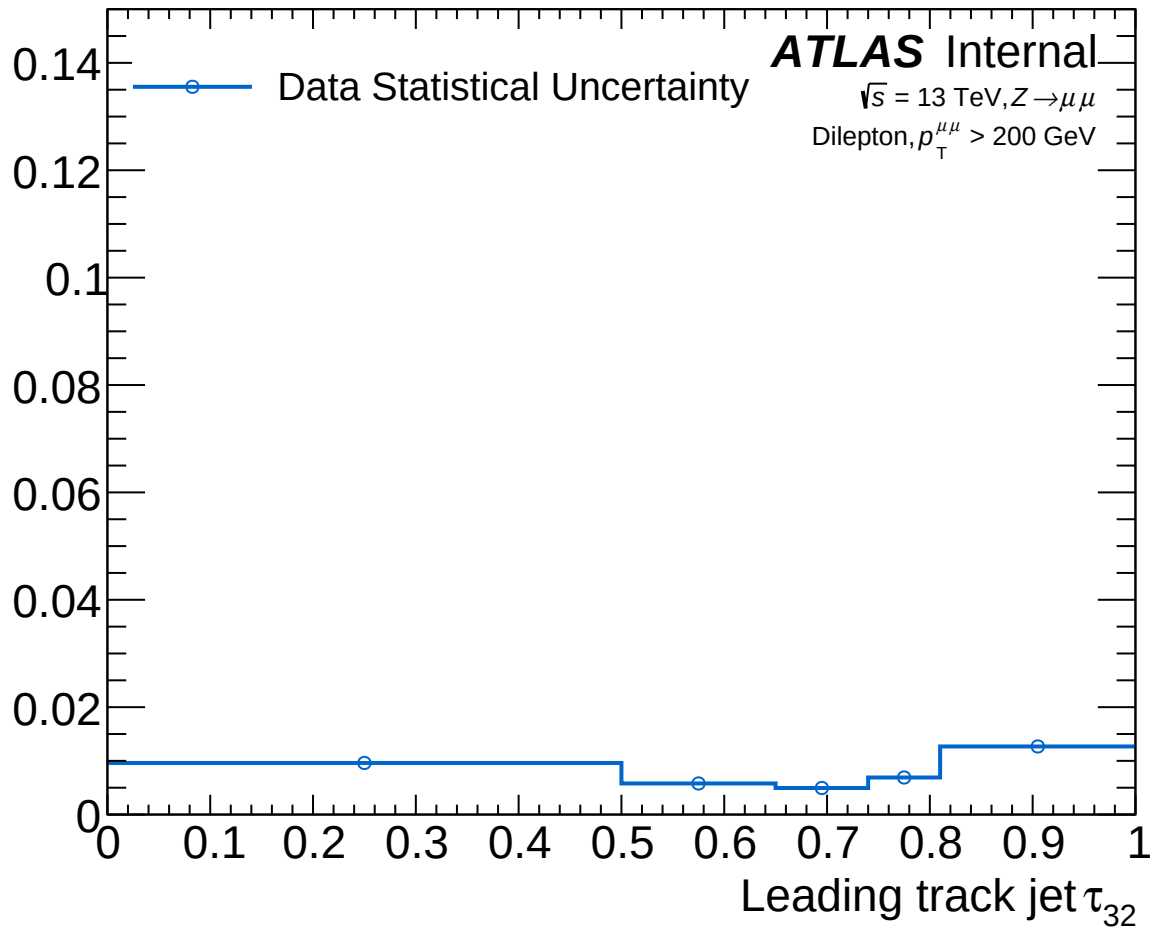
Fractional systematic impact

ATLAS Internal

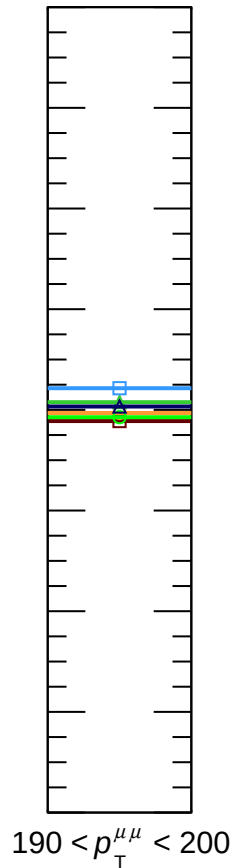
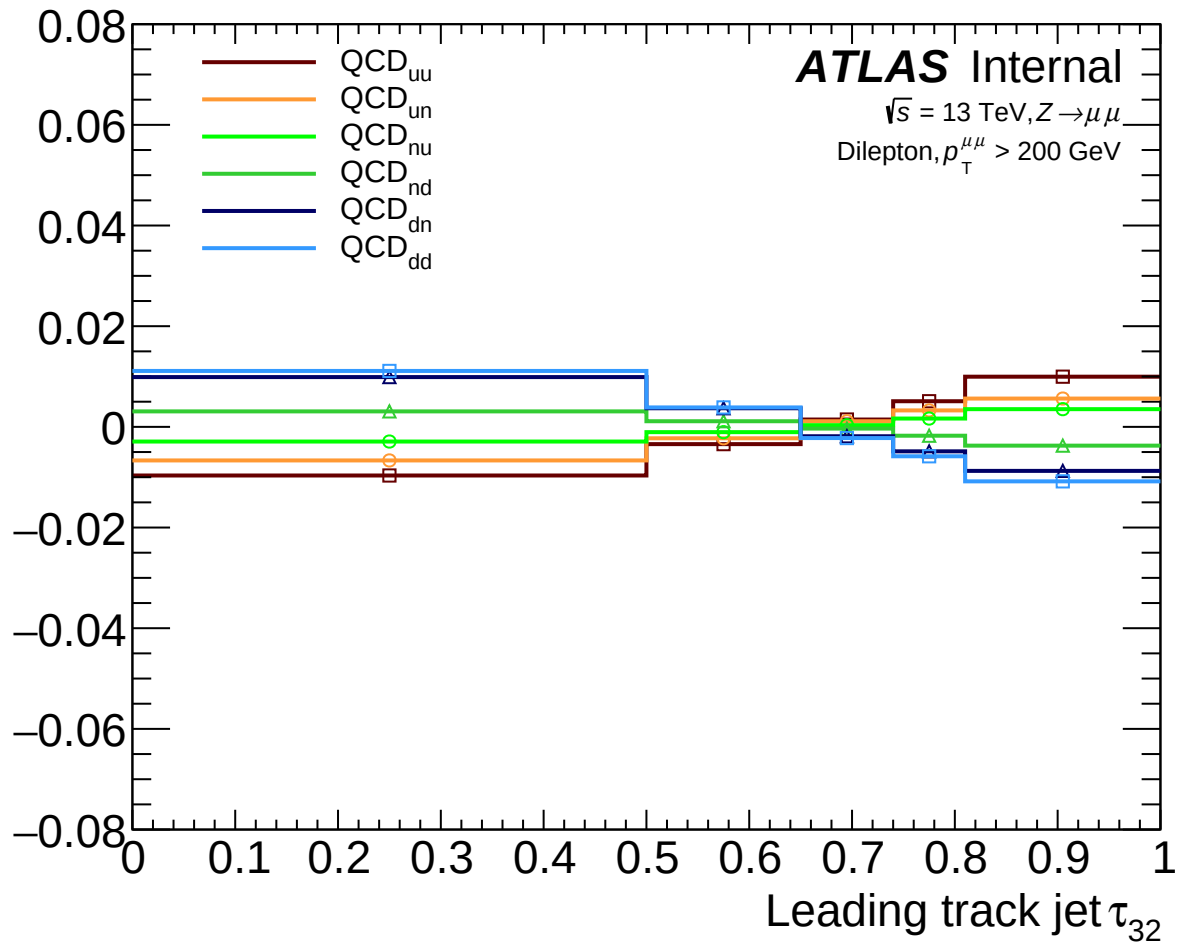
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

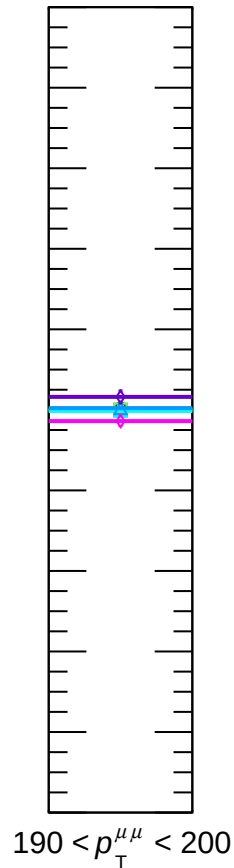
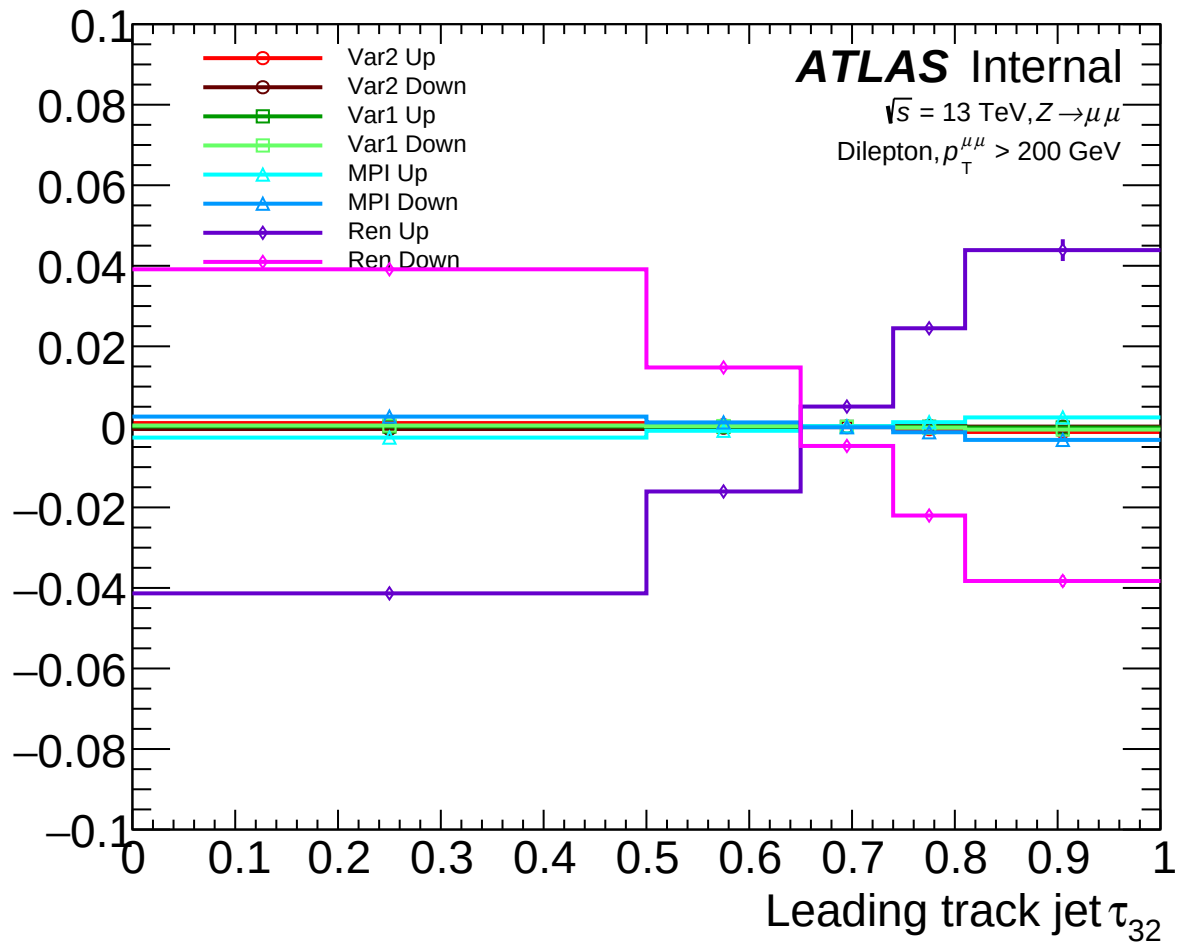
Data Statistical Uncertainty

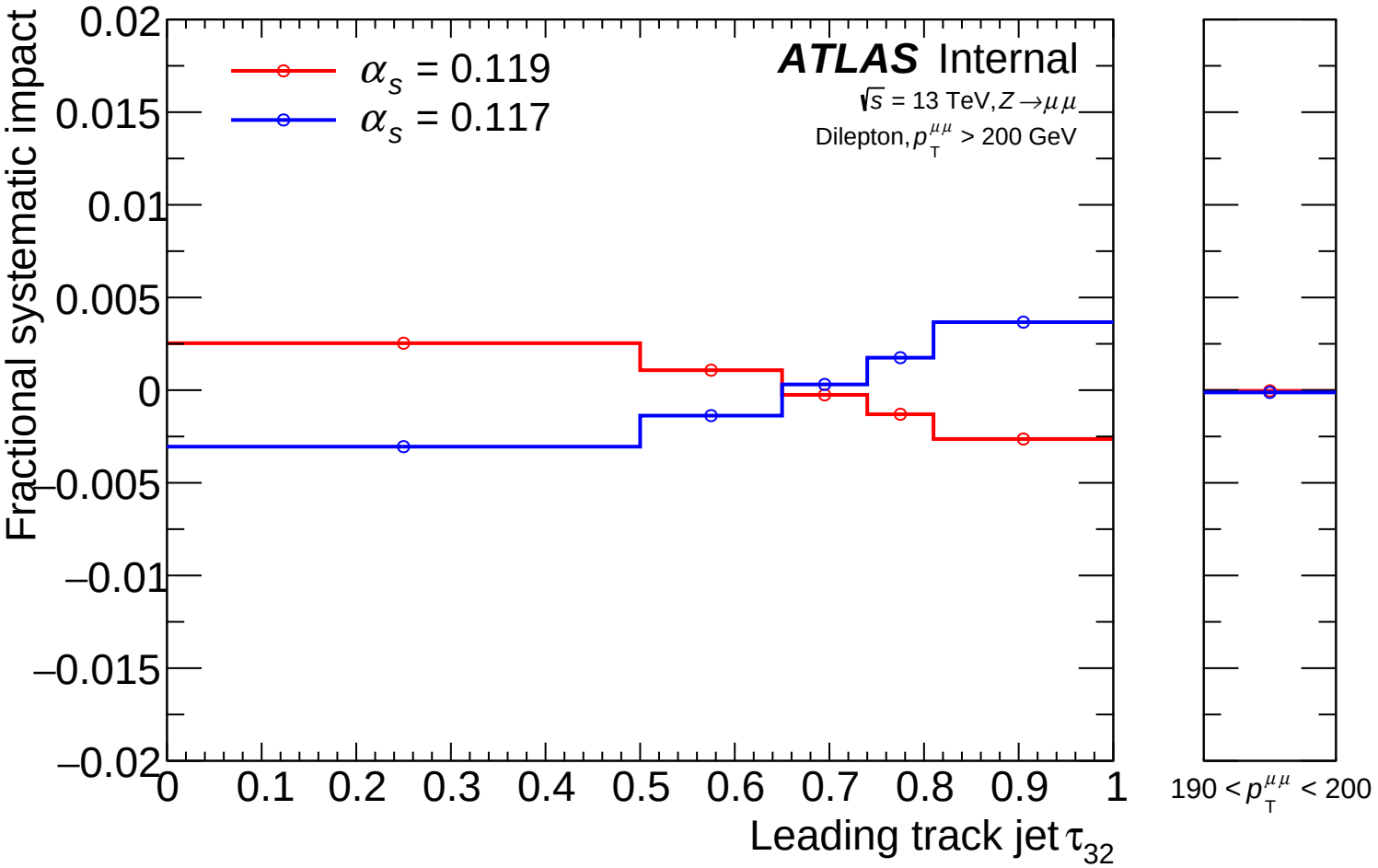


Fractional systematic impact



Fractional systematic impact





Fractional systematic impact

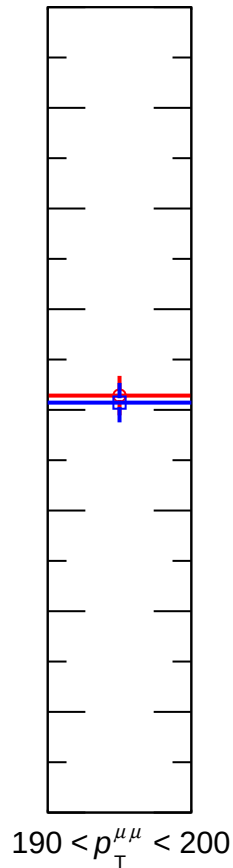
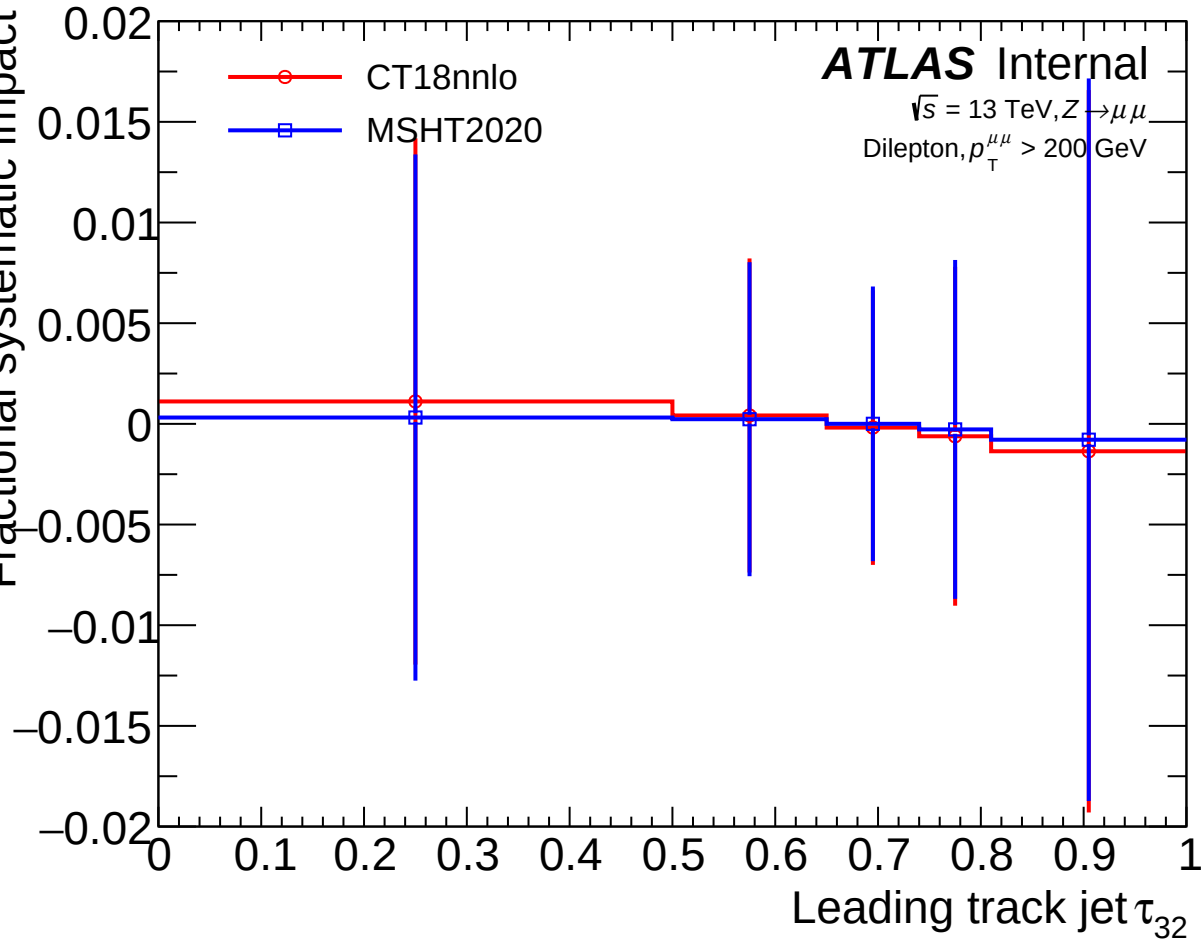
CT18nnlo

MSHT2020

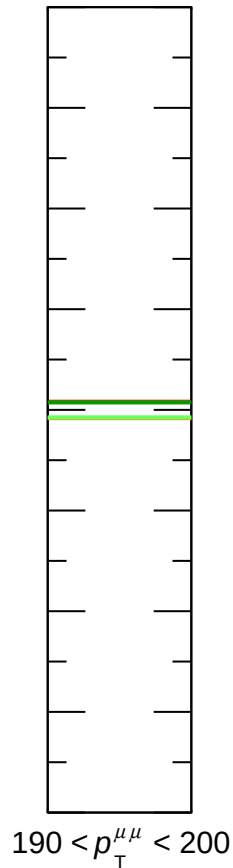
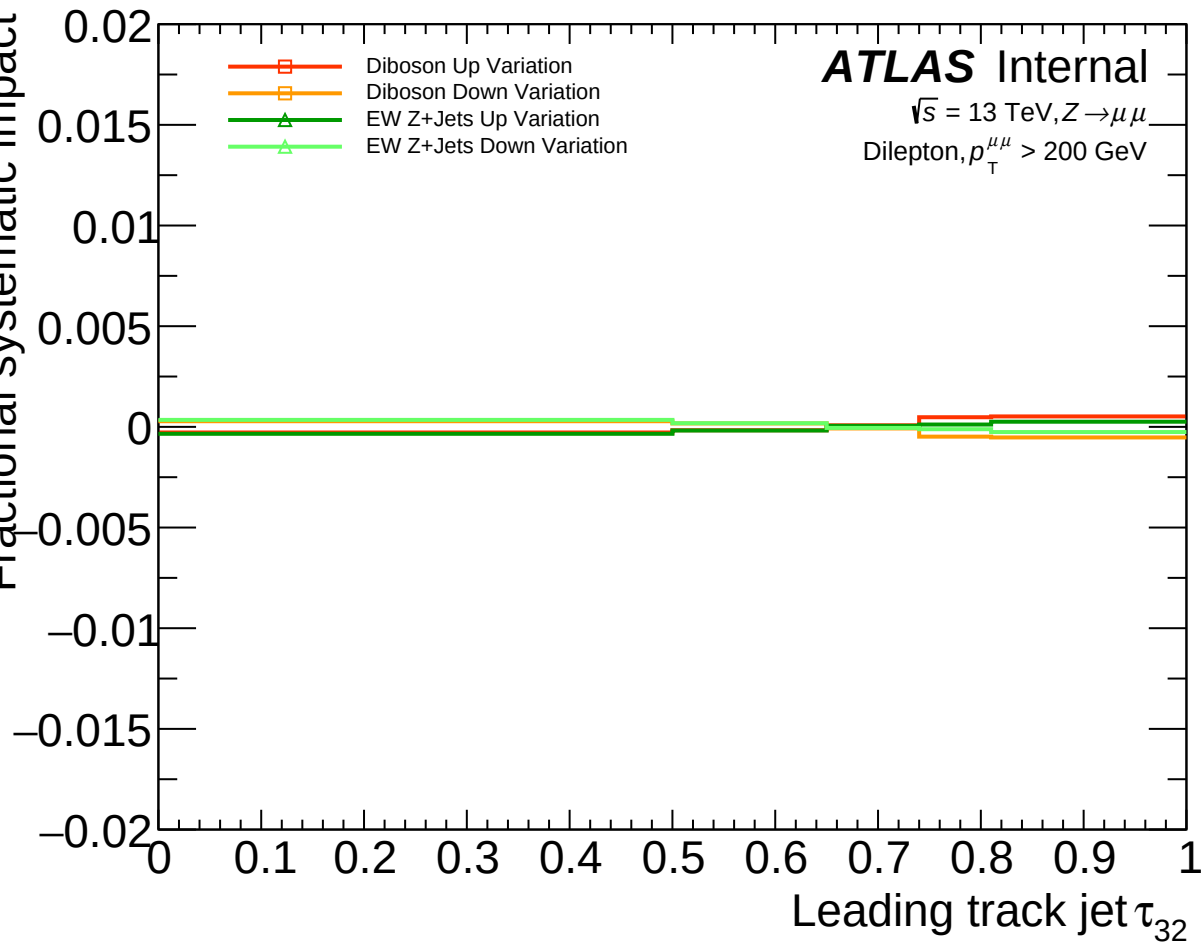
ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$



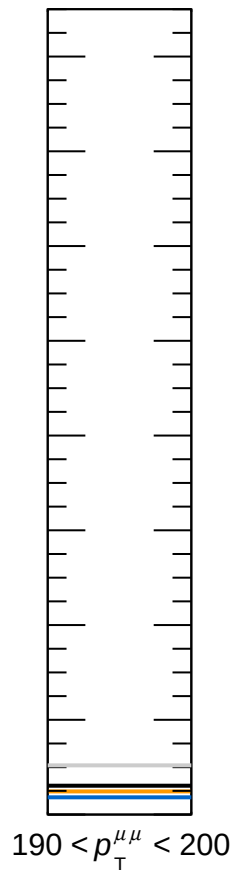
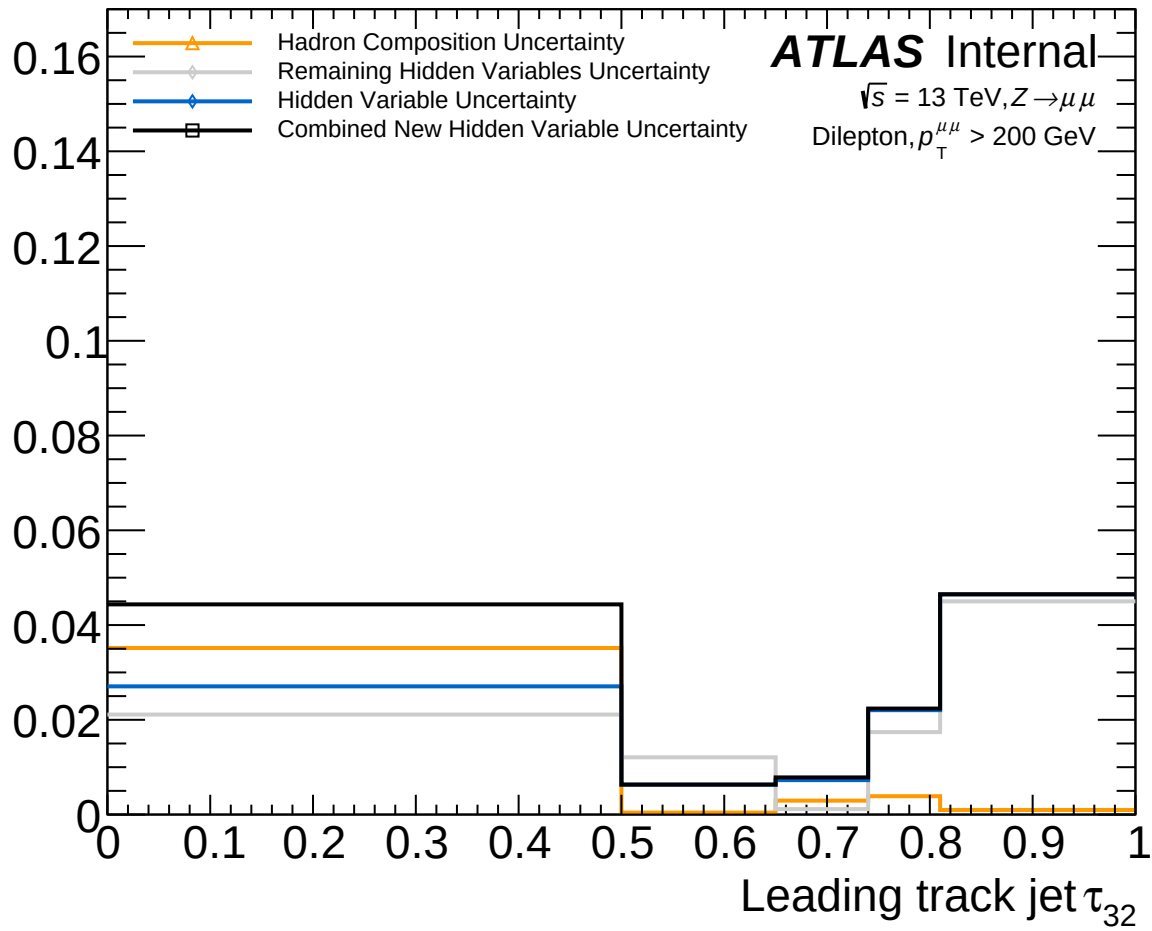
Fractional systematic impact

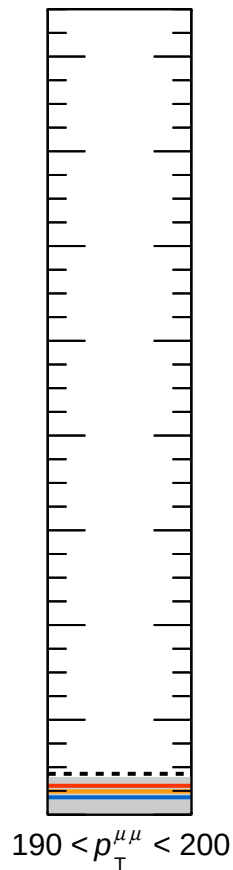
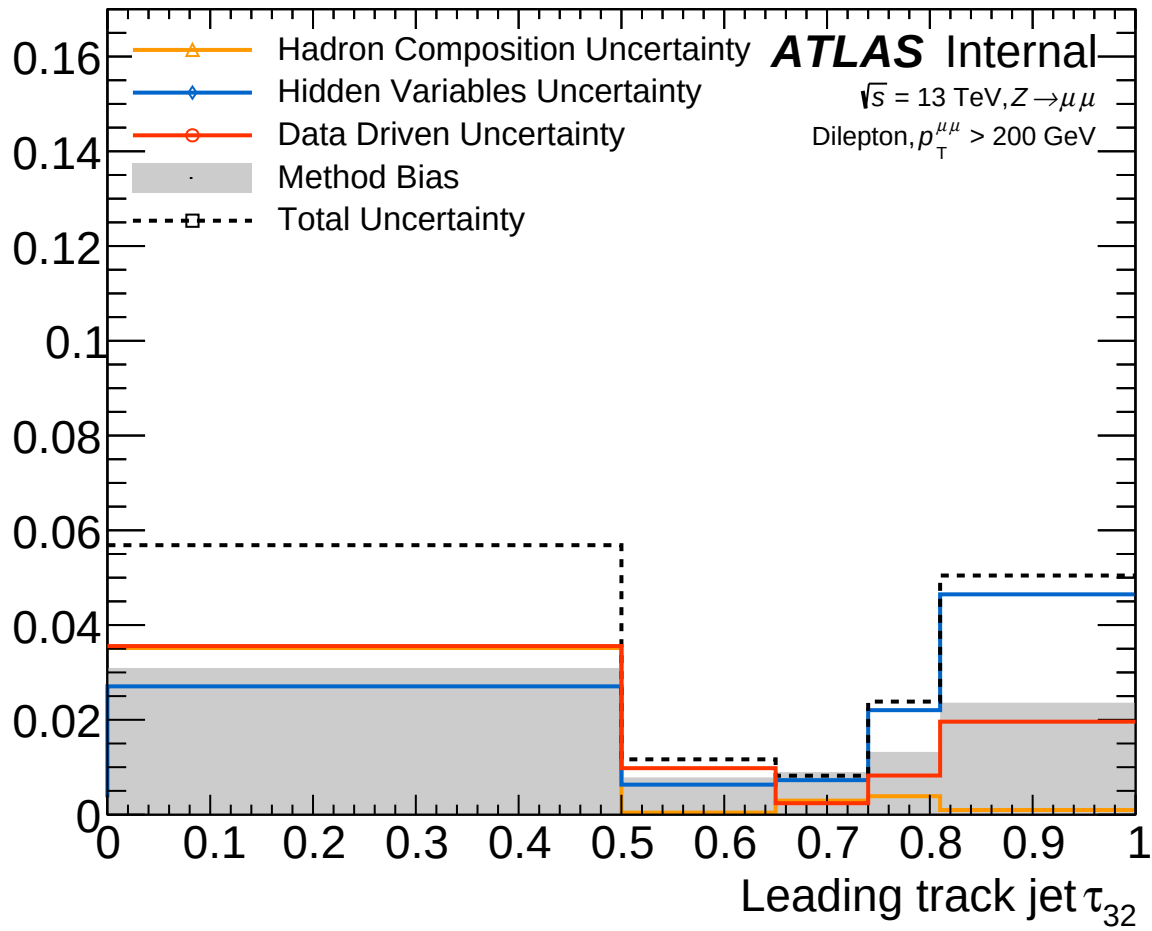


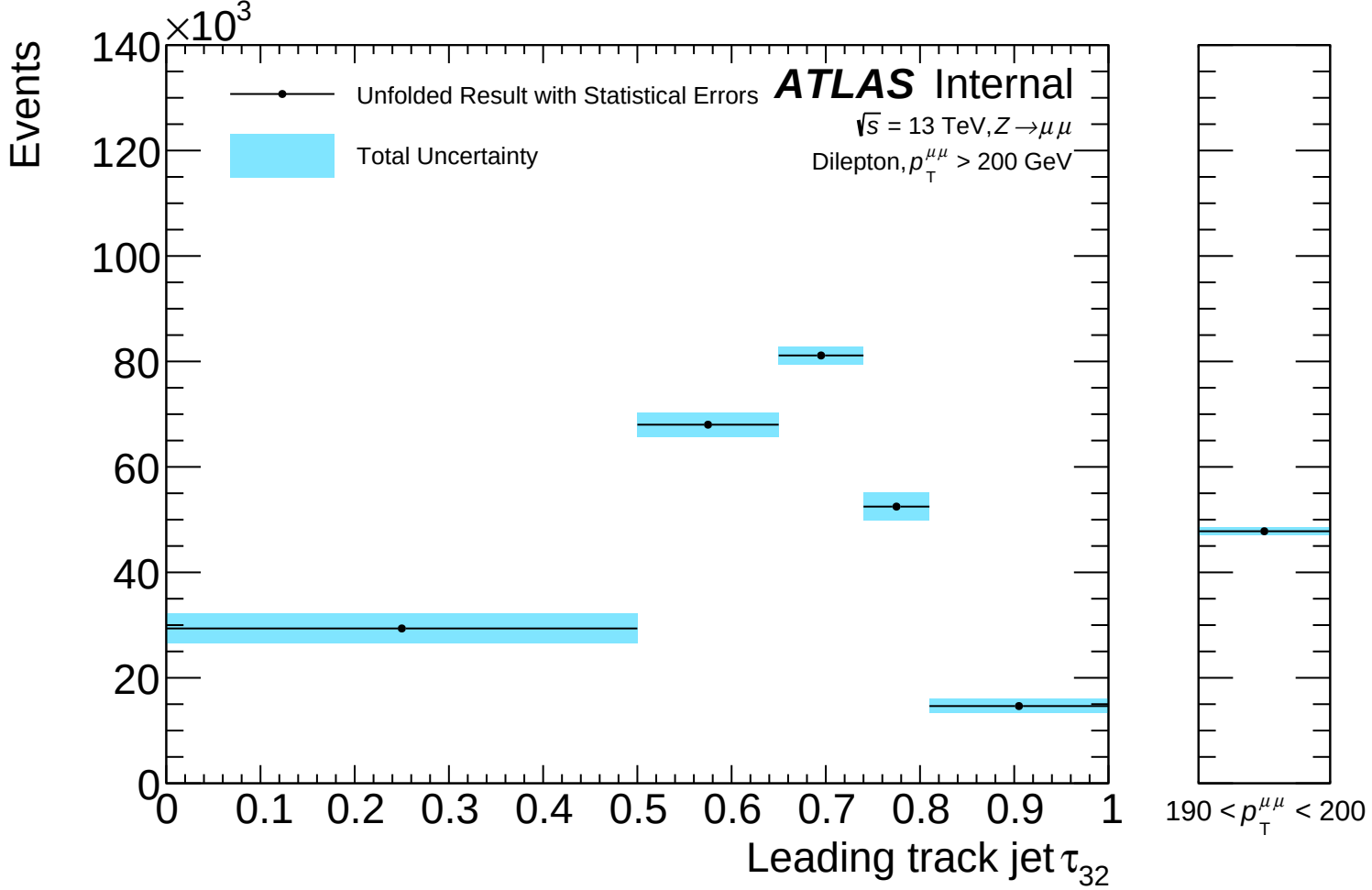
ATLAS Internal

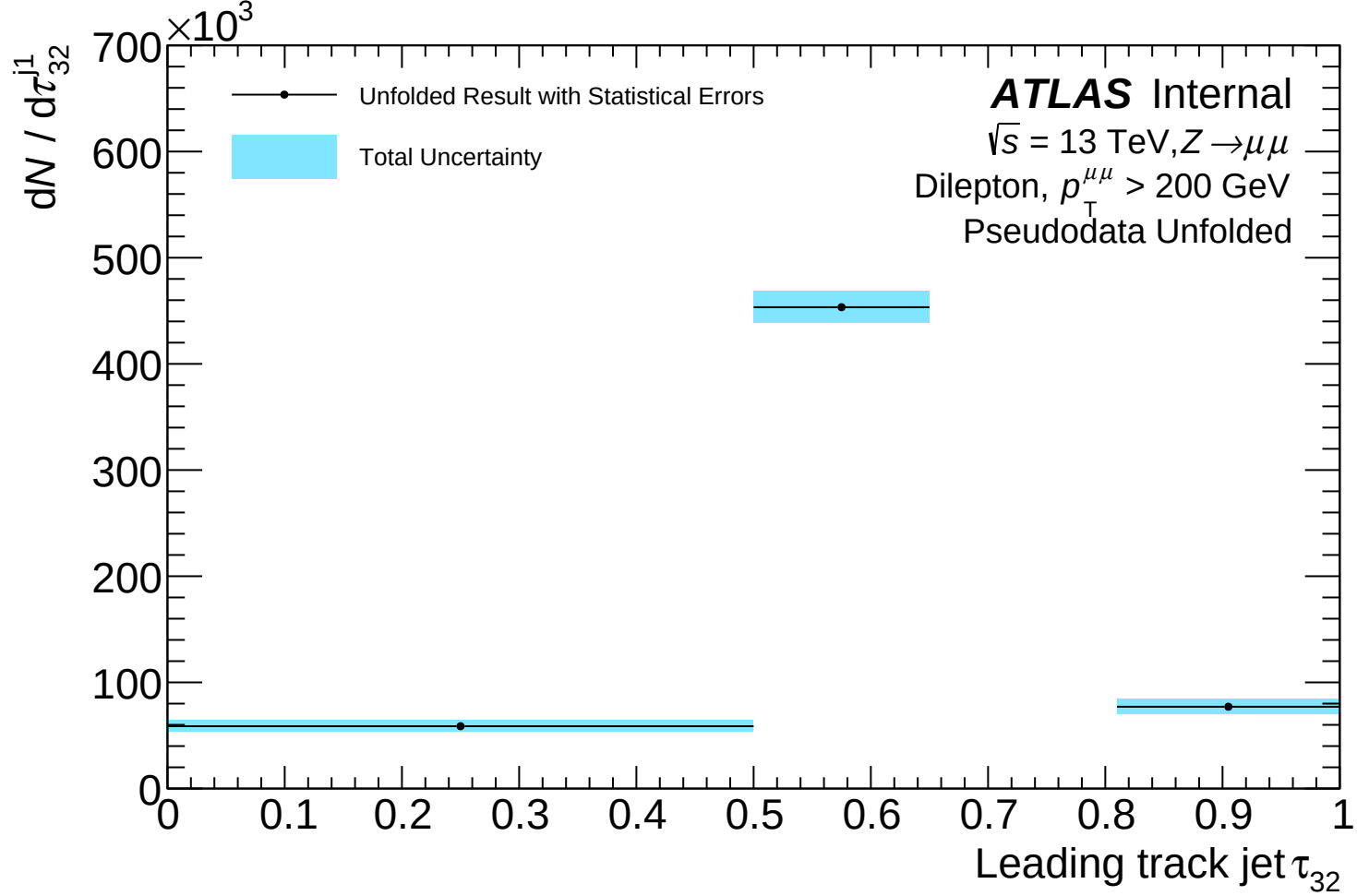
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

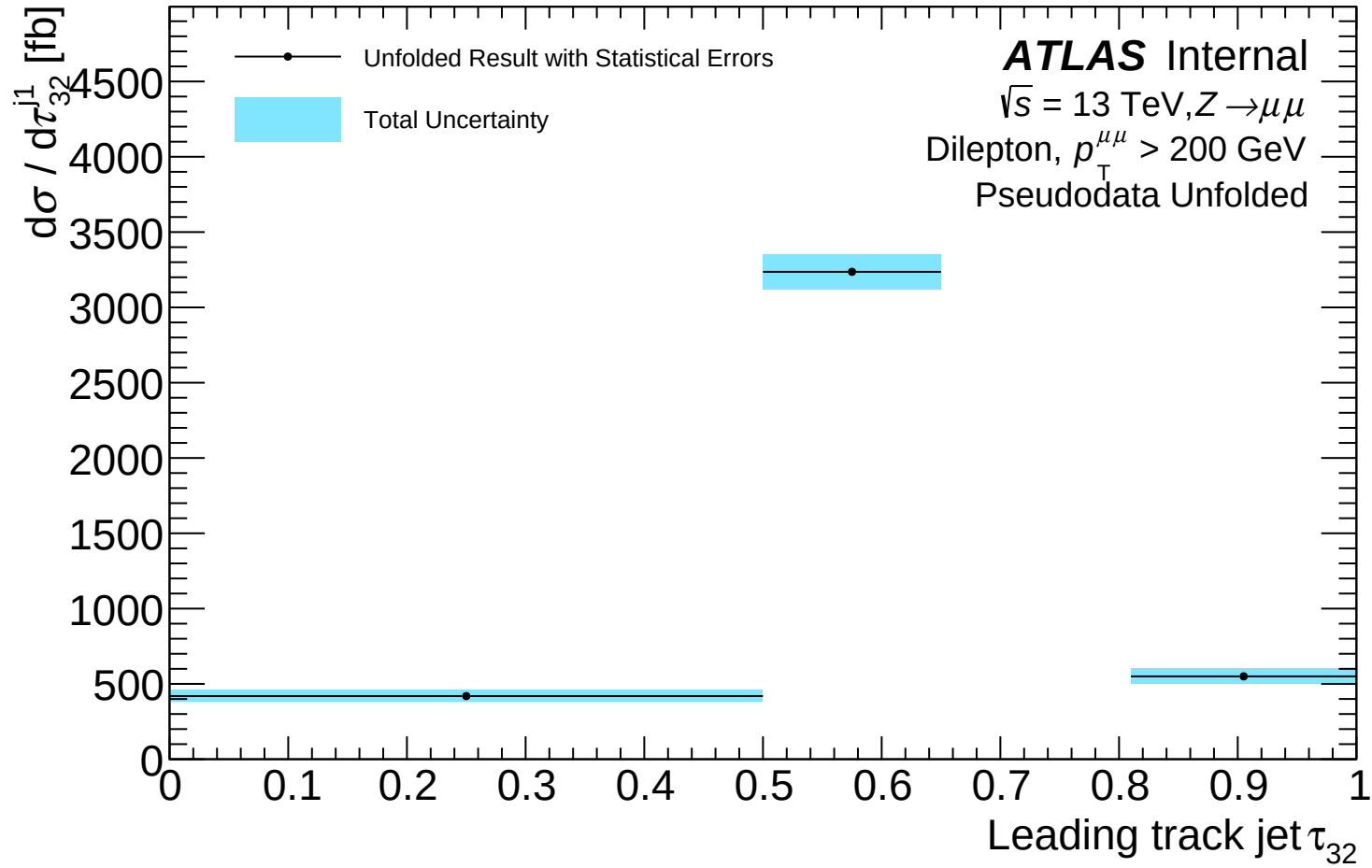
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$











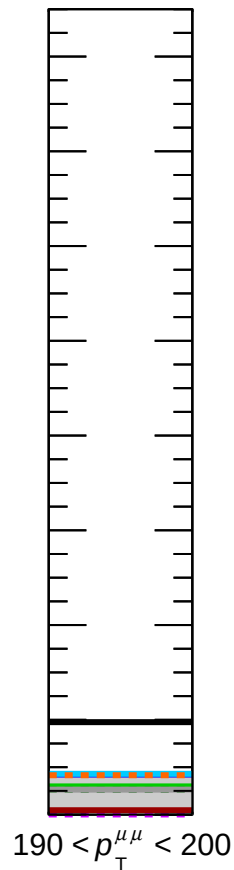
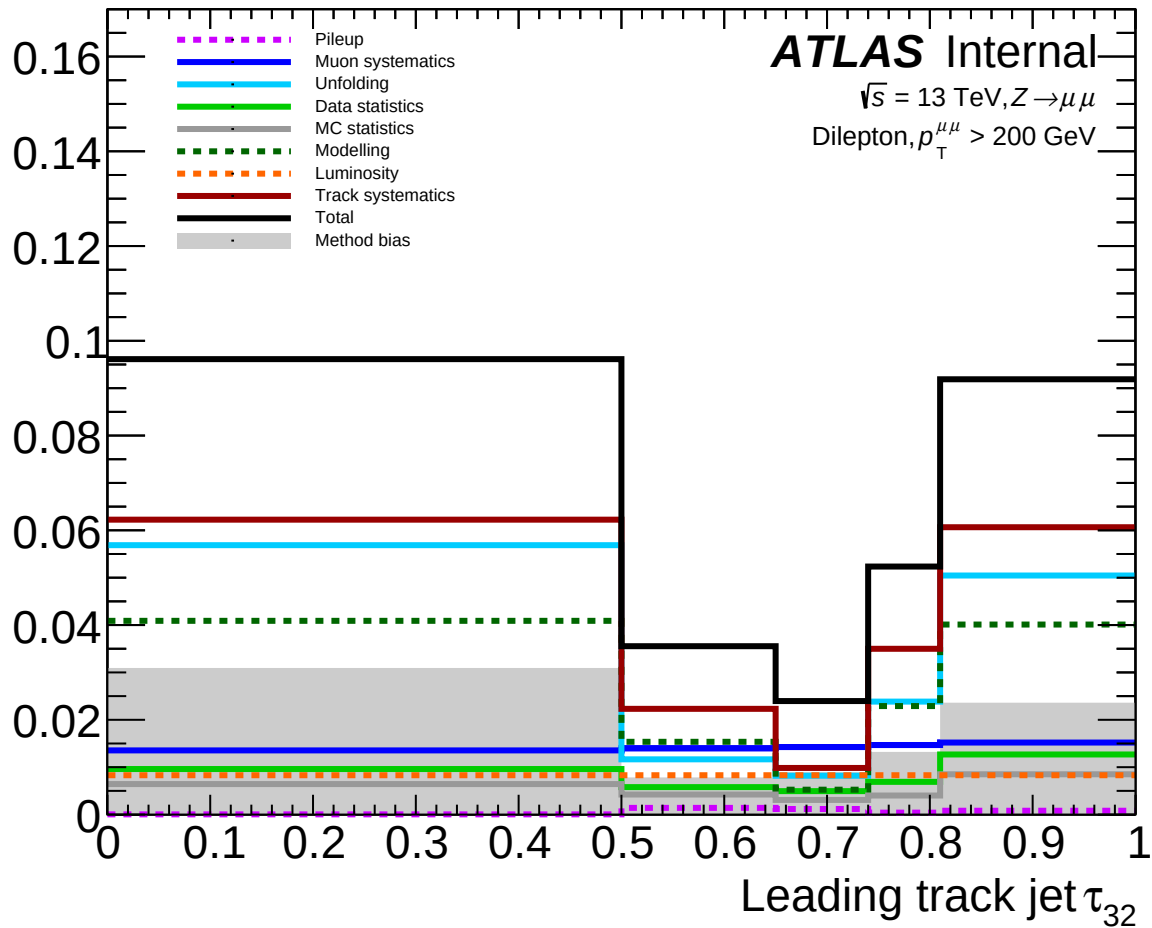
Fractional uncertainty

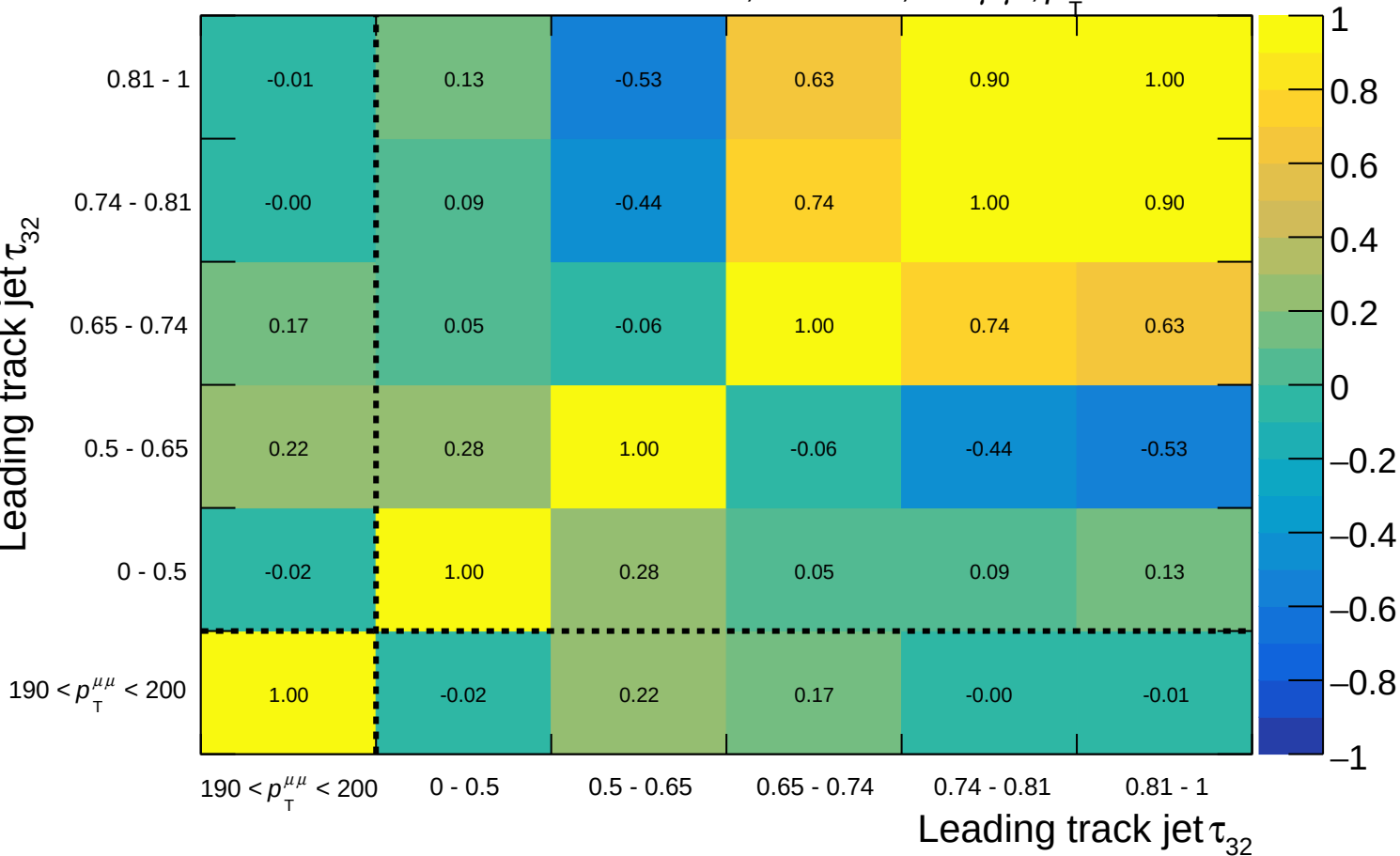
ATLAS Internal

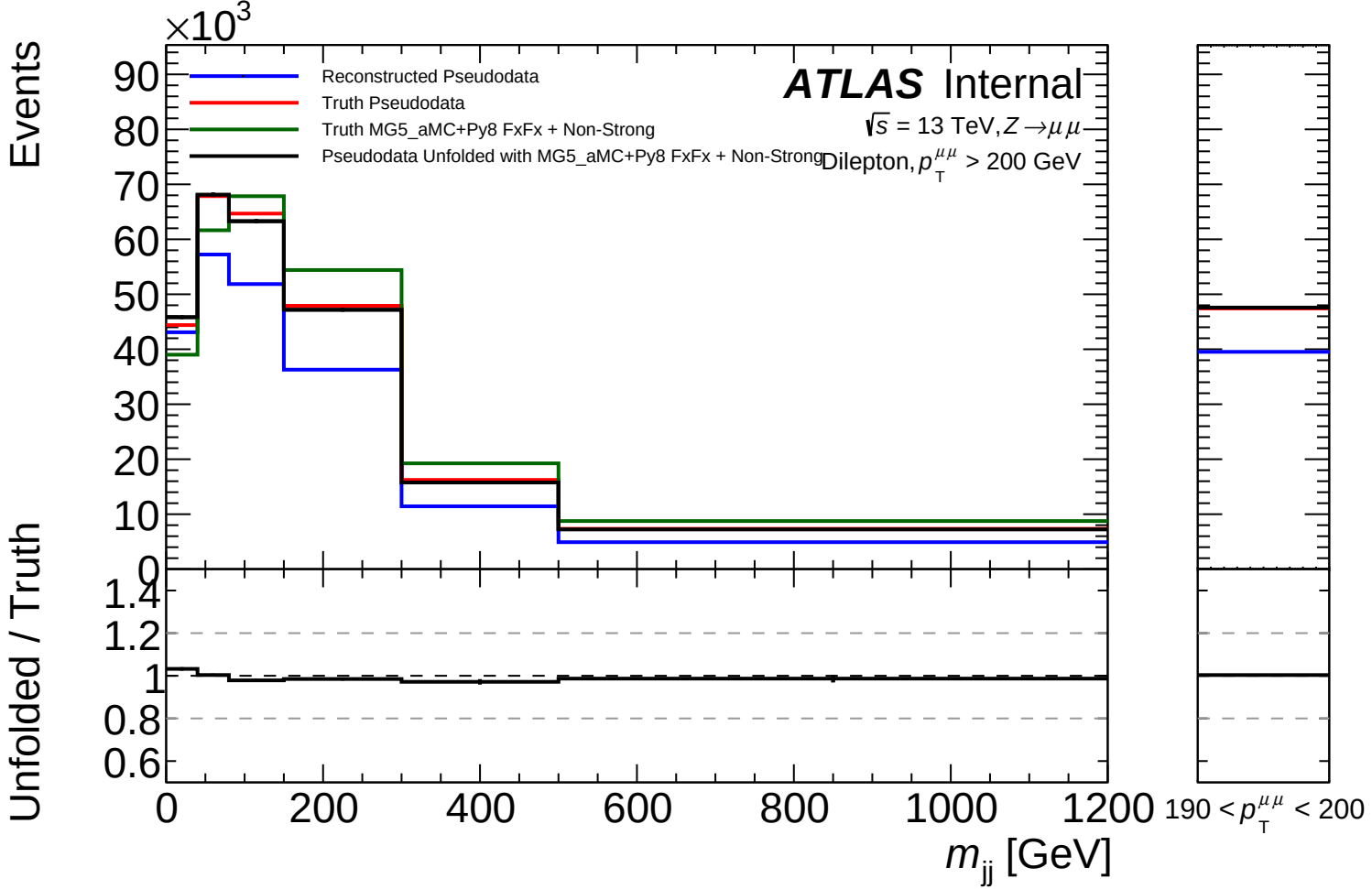
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

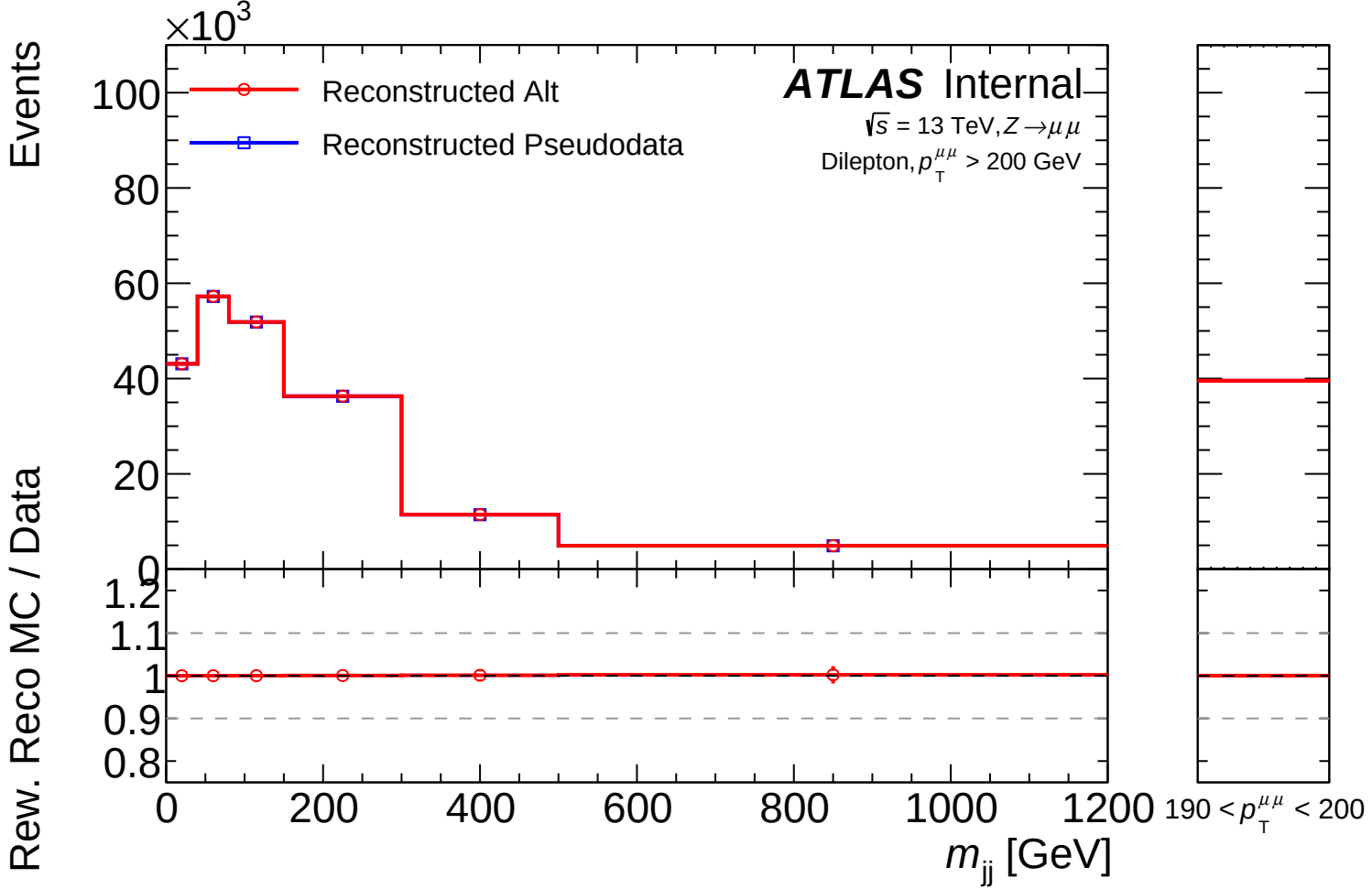
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

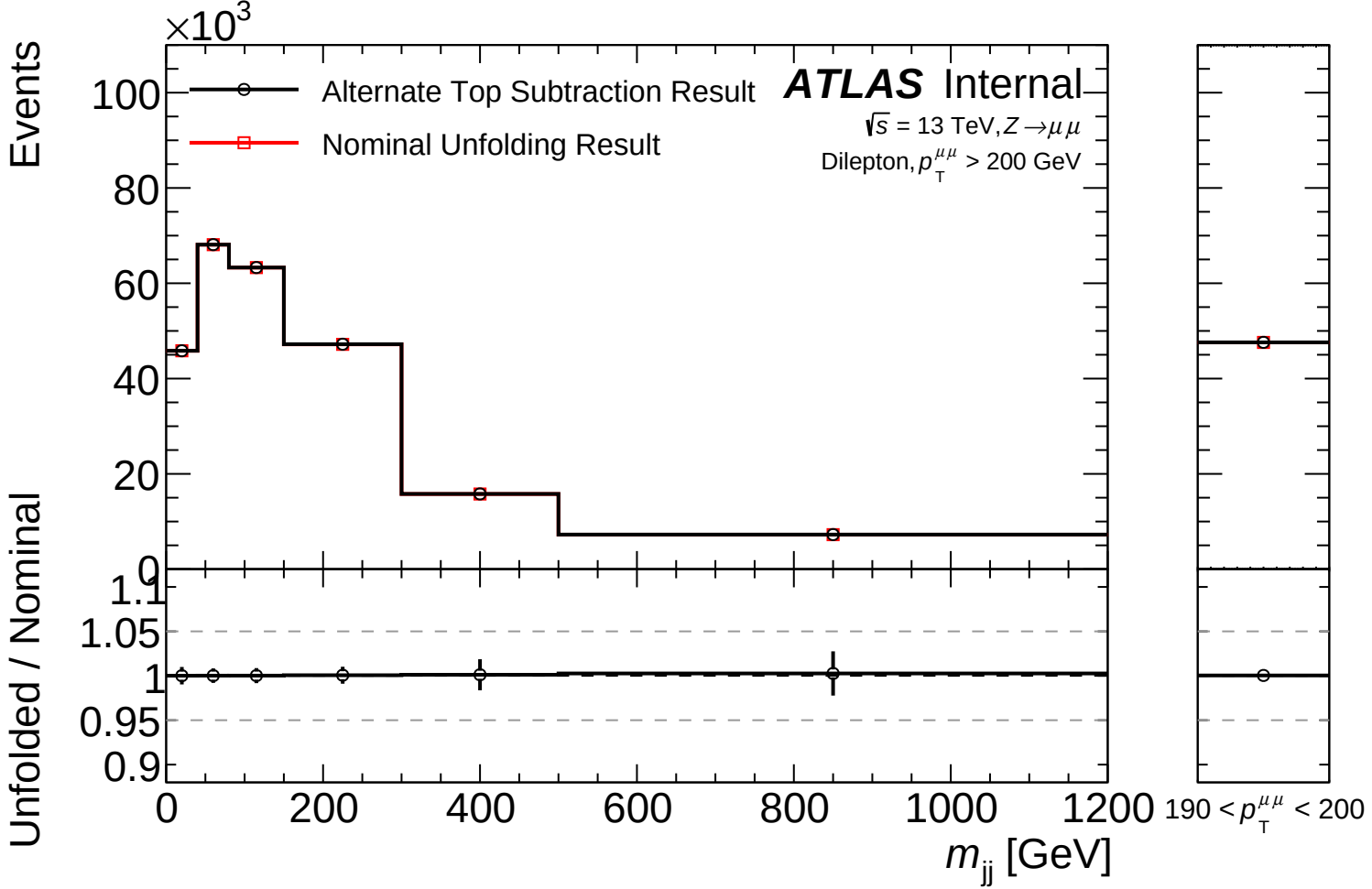
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_T^{\mu\mu} > 200 \text{ GeV}$ Leading track jet τ_{32} 







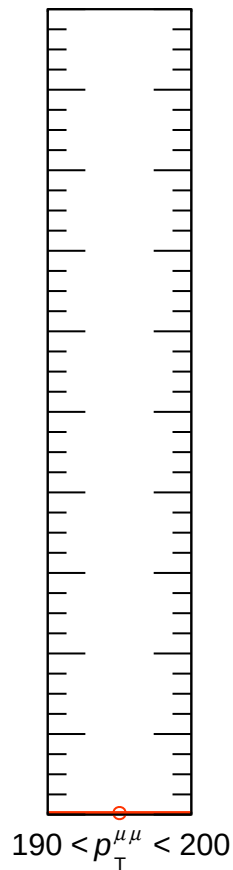
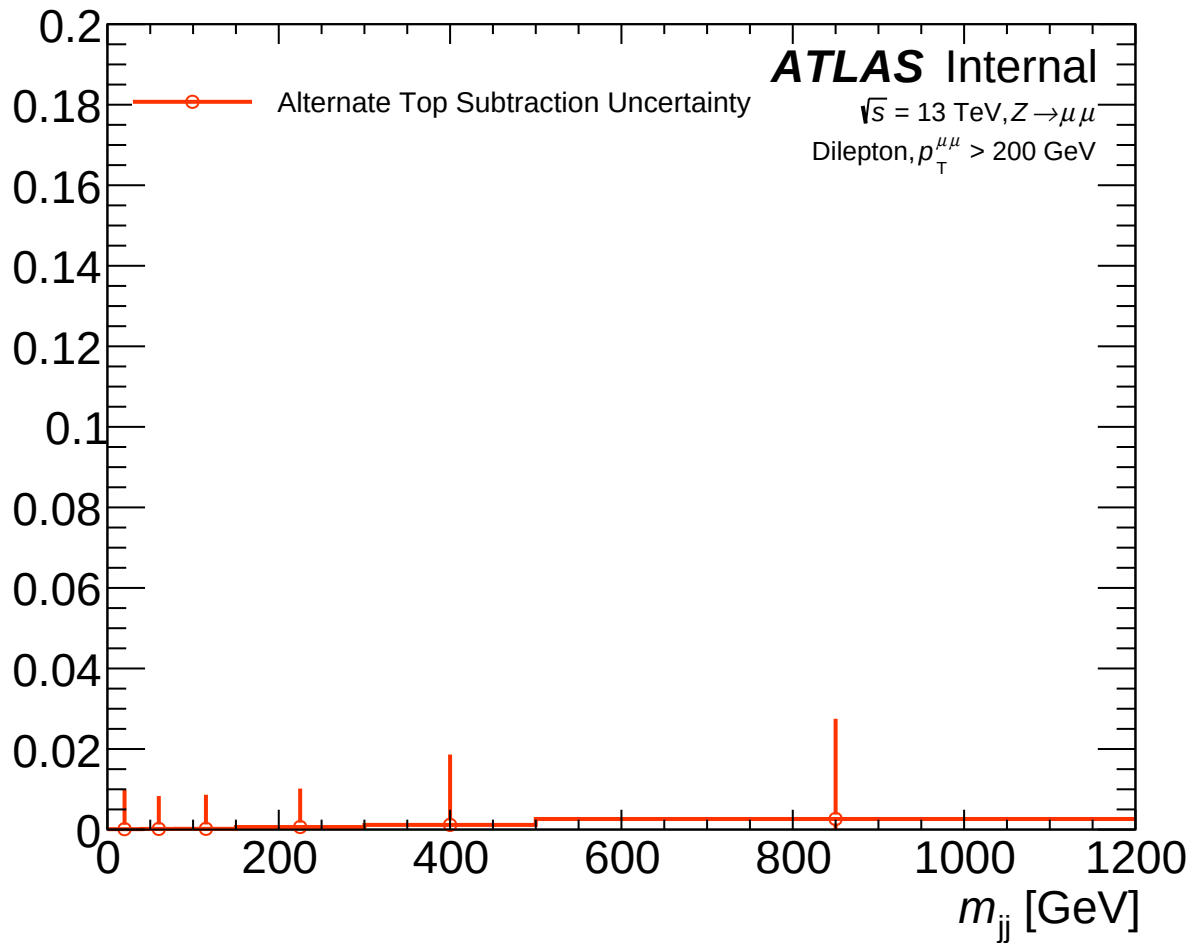
Fractional systematic impact

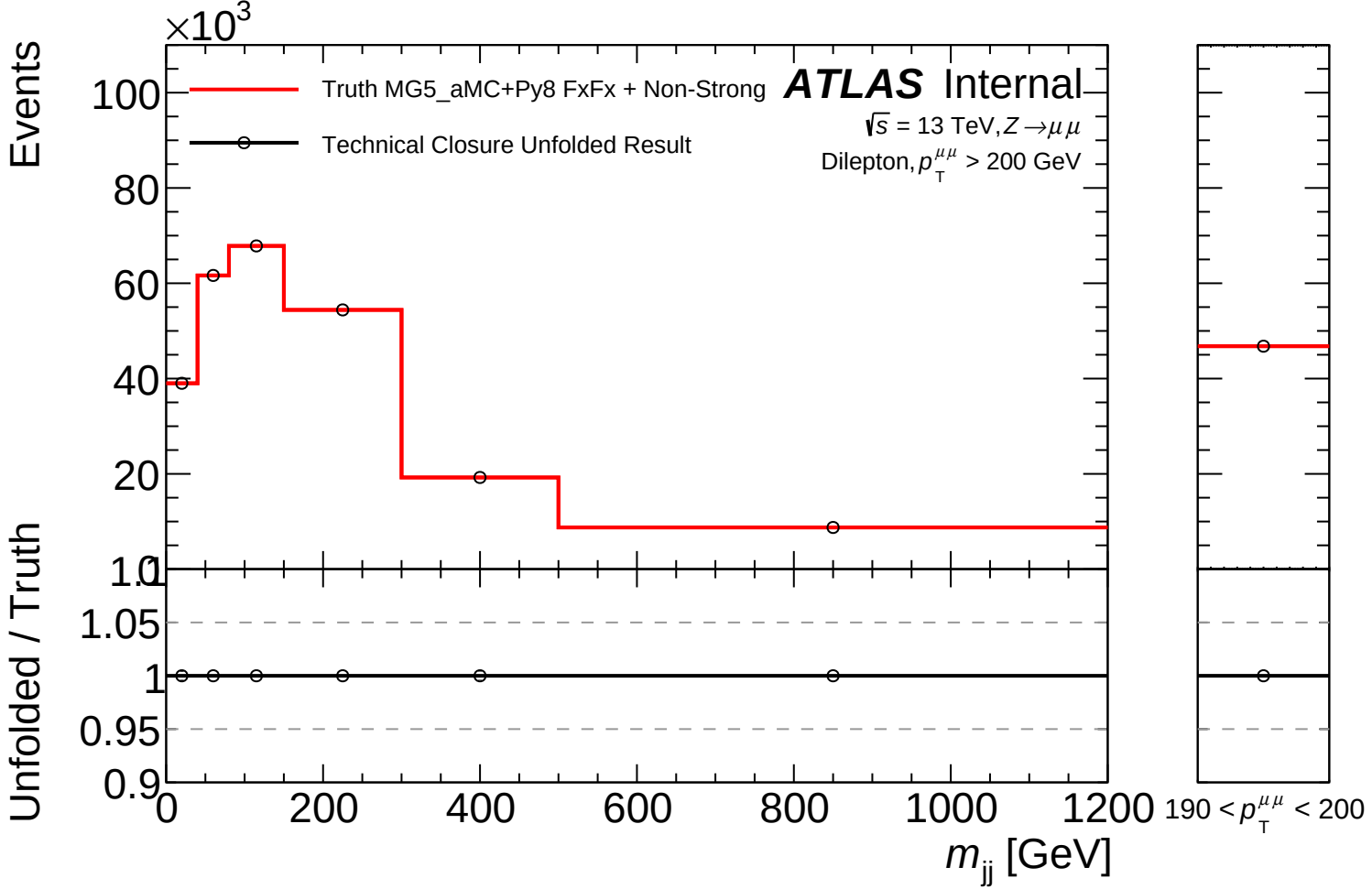
ATLAS Internal

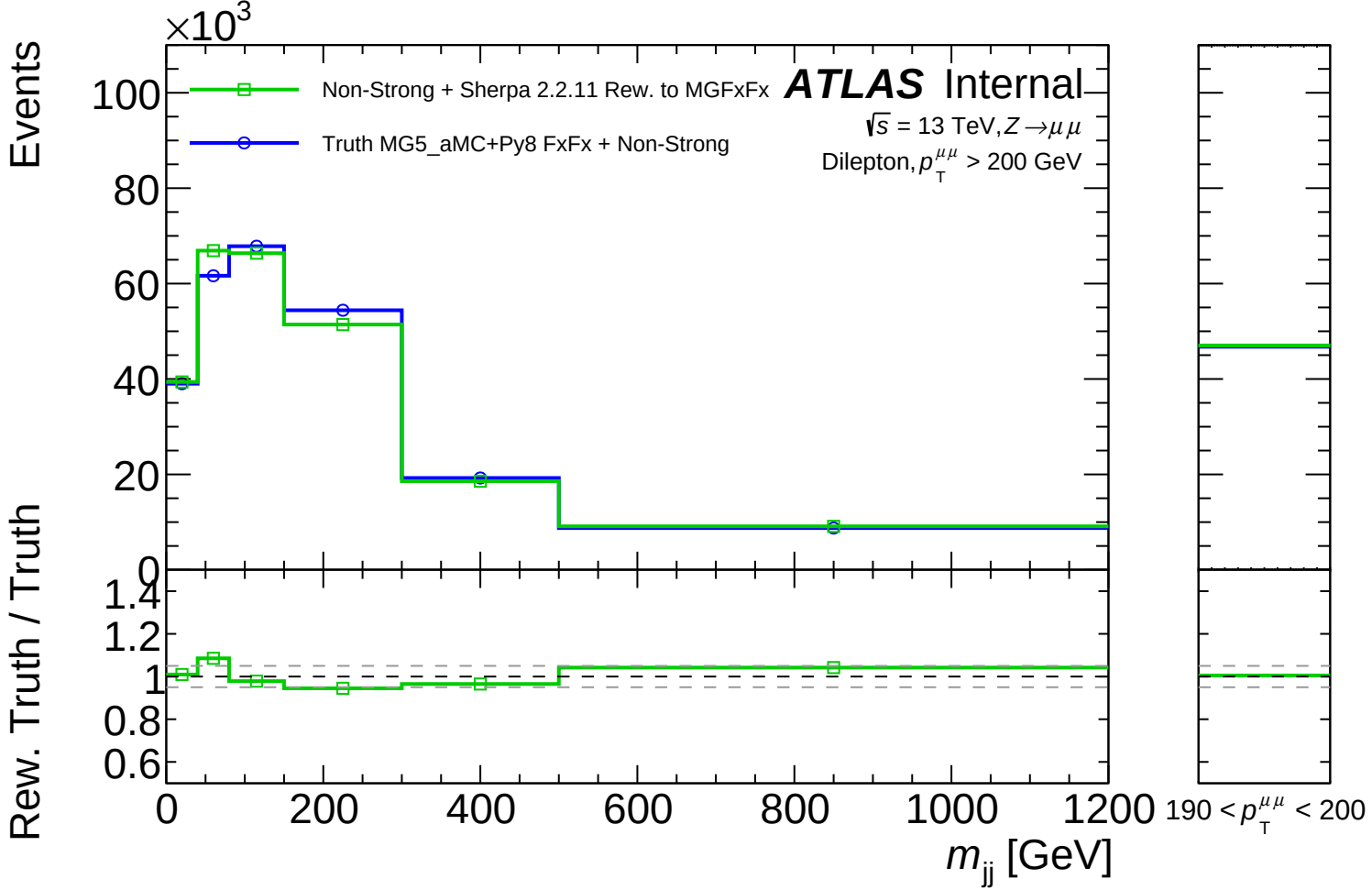
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

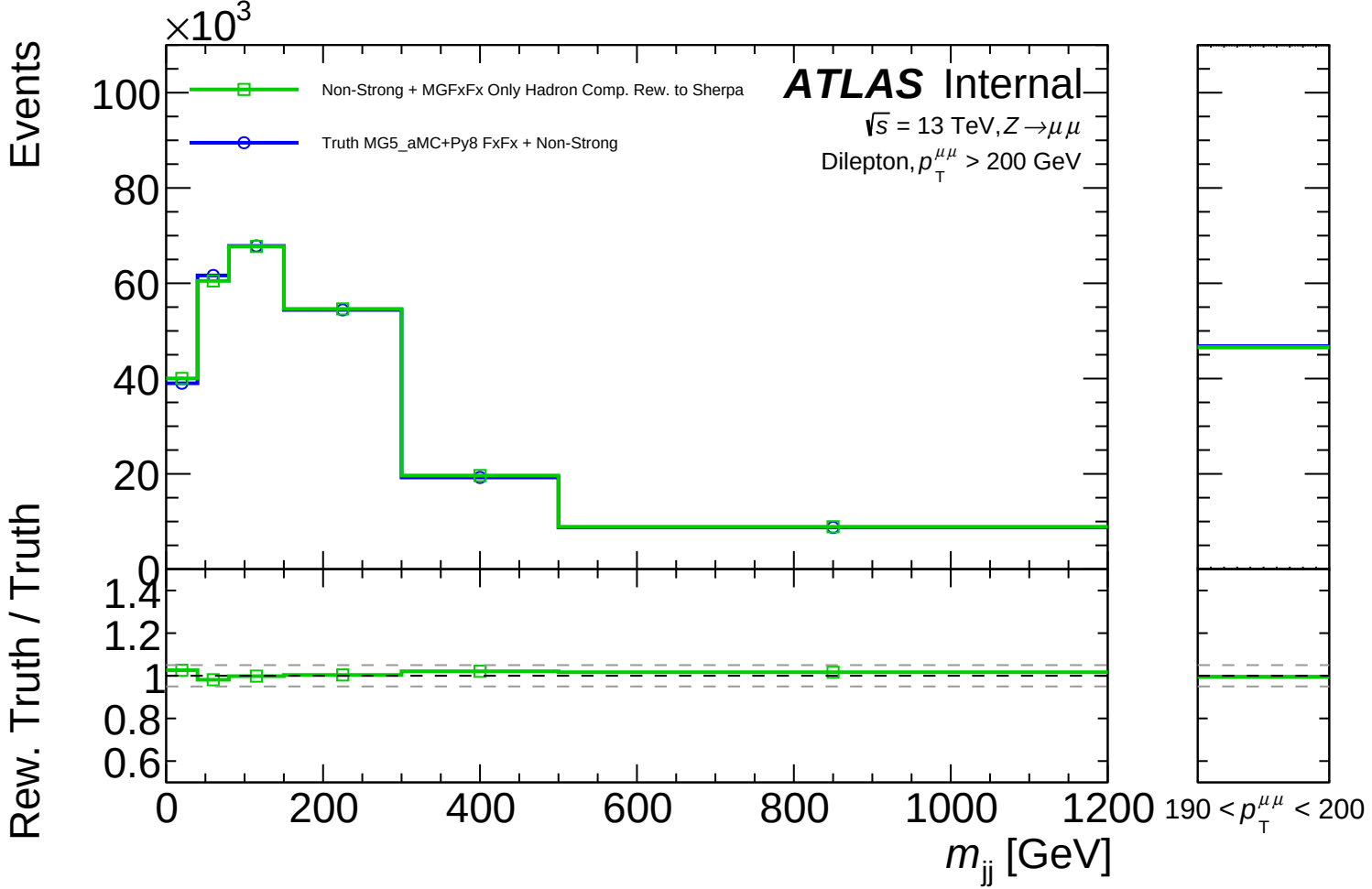
Dilepton, $p_T^{\mu\mu} > 200$ GeV

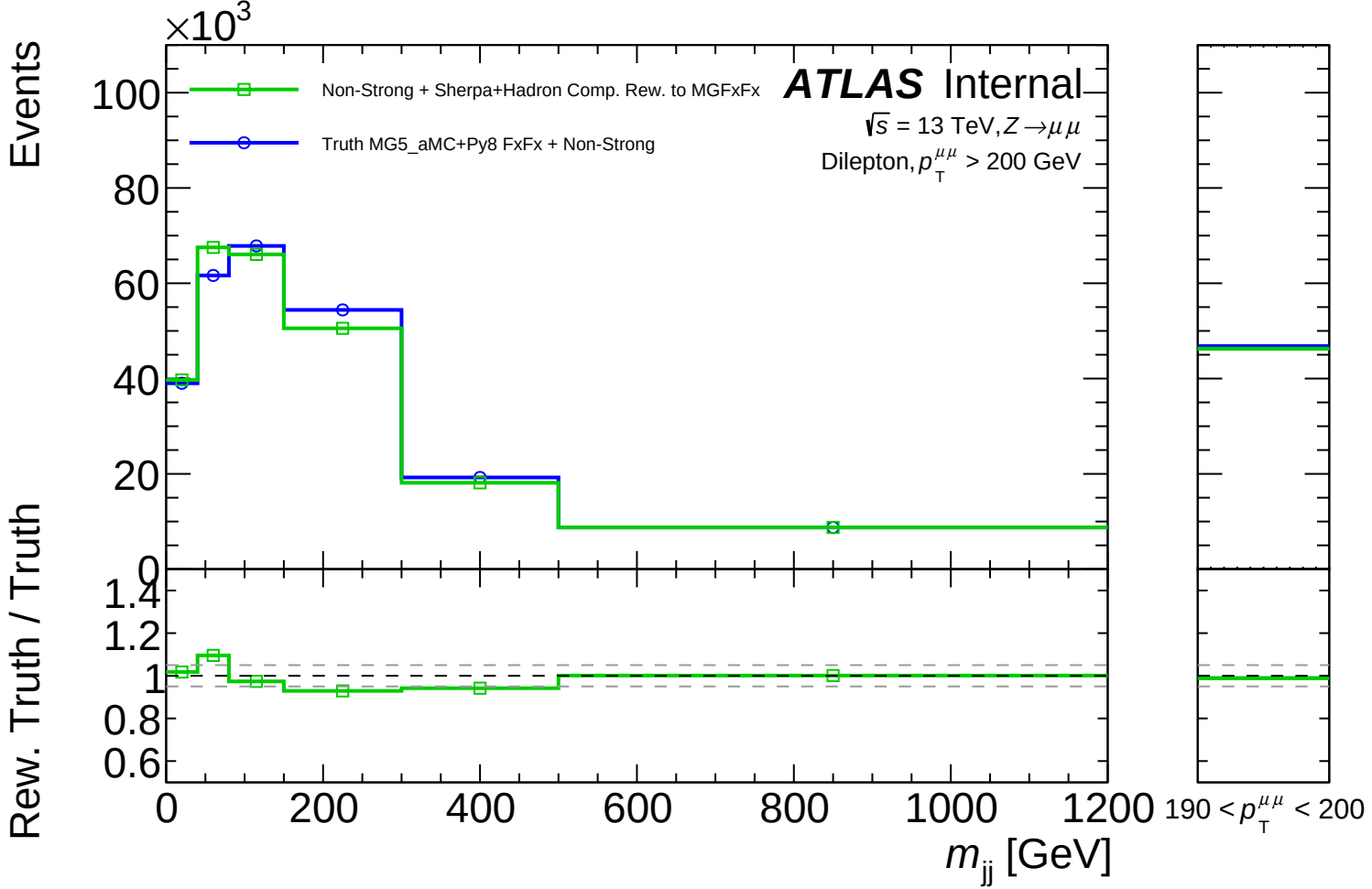
Alternate Top Subtraction Uncertainty

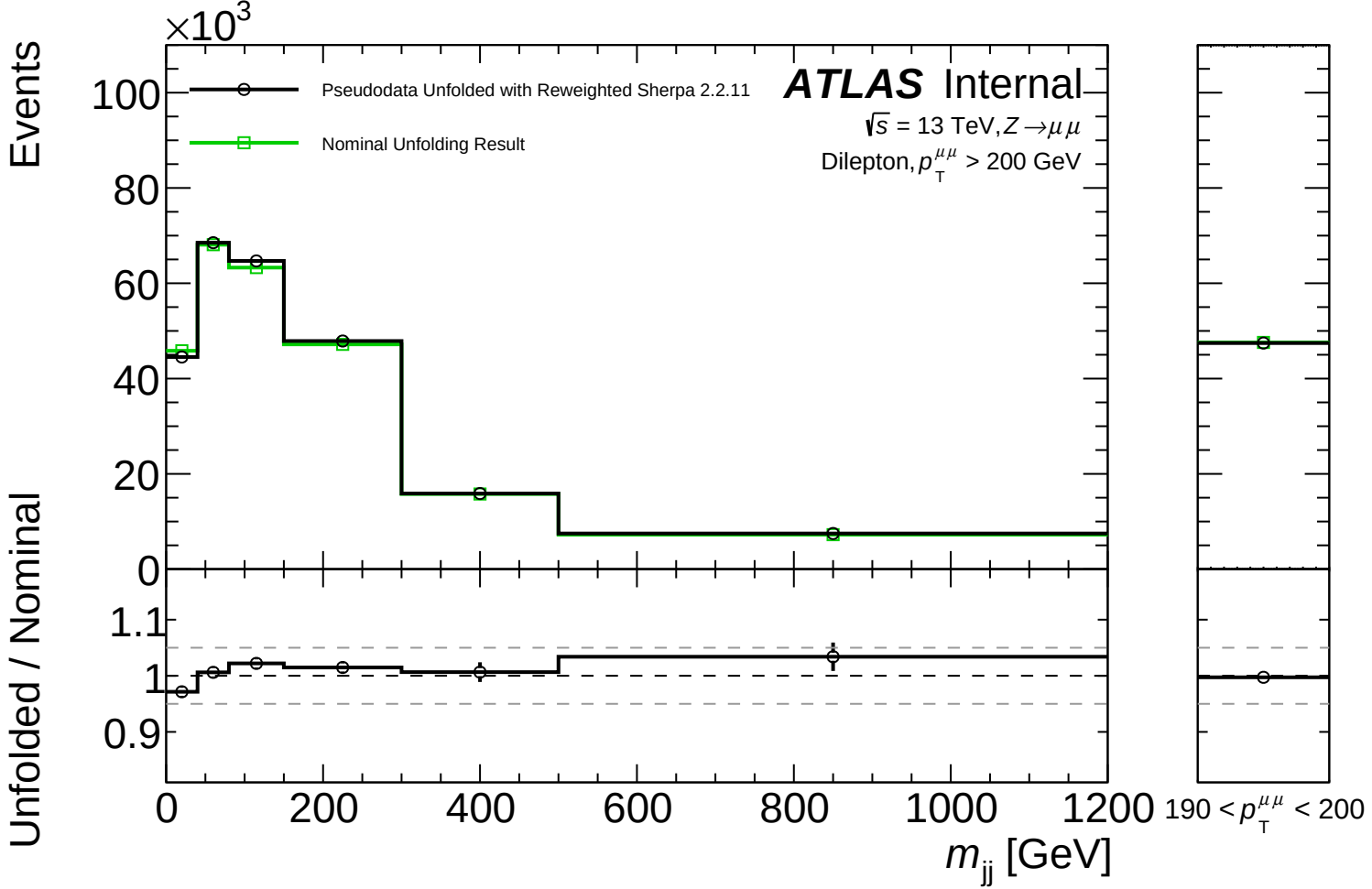


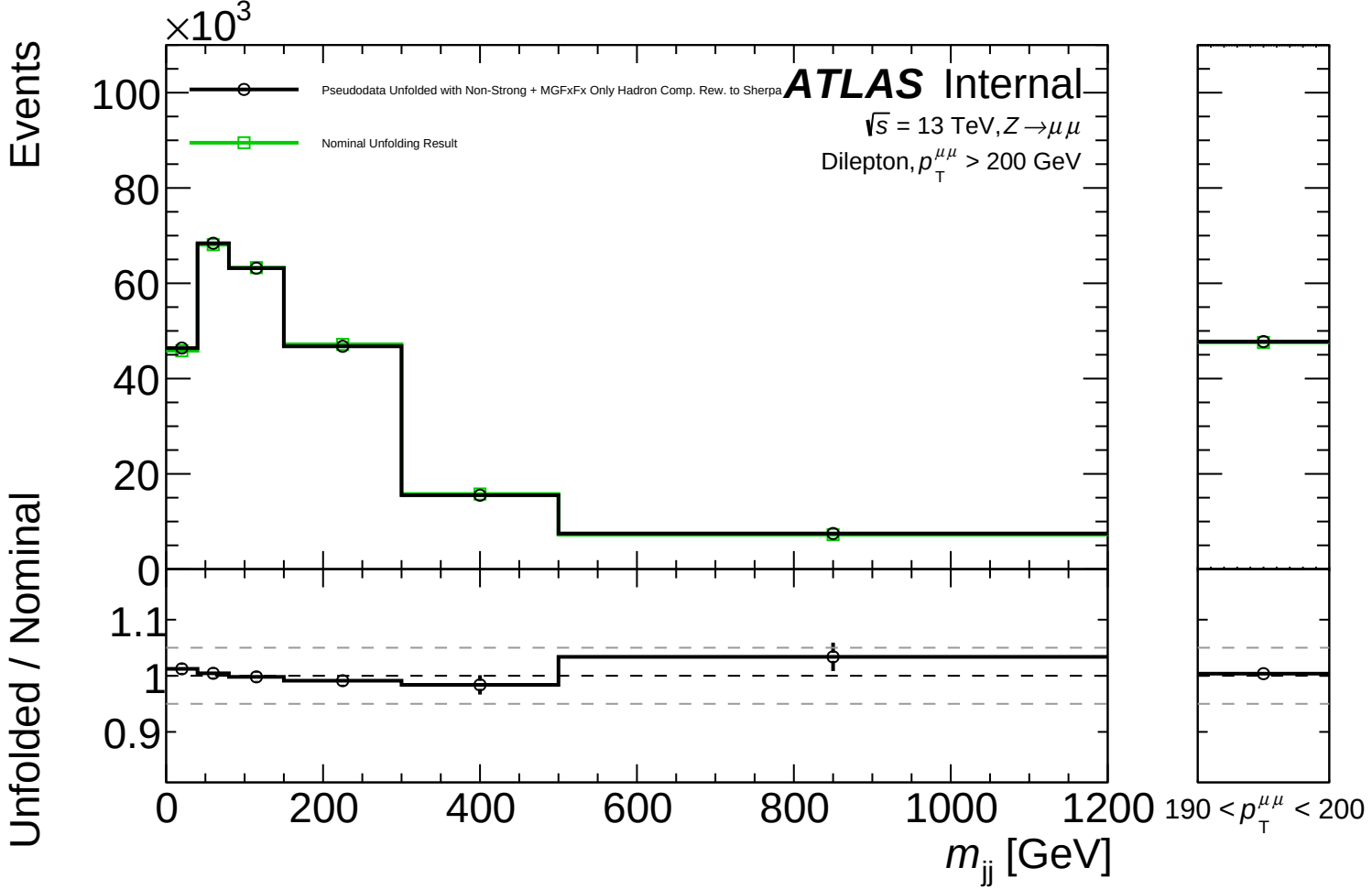


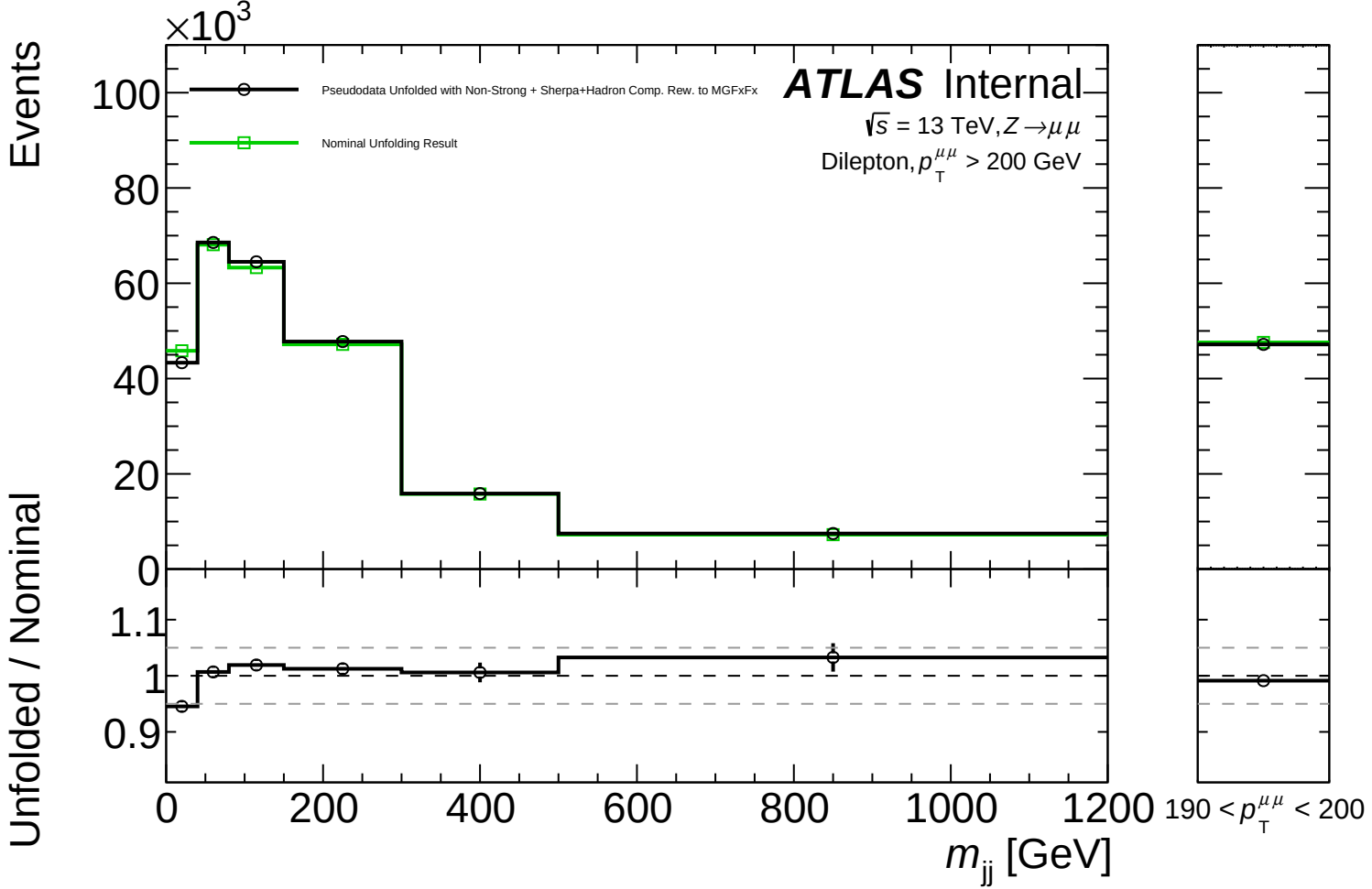


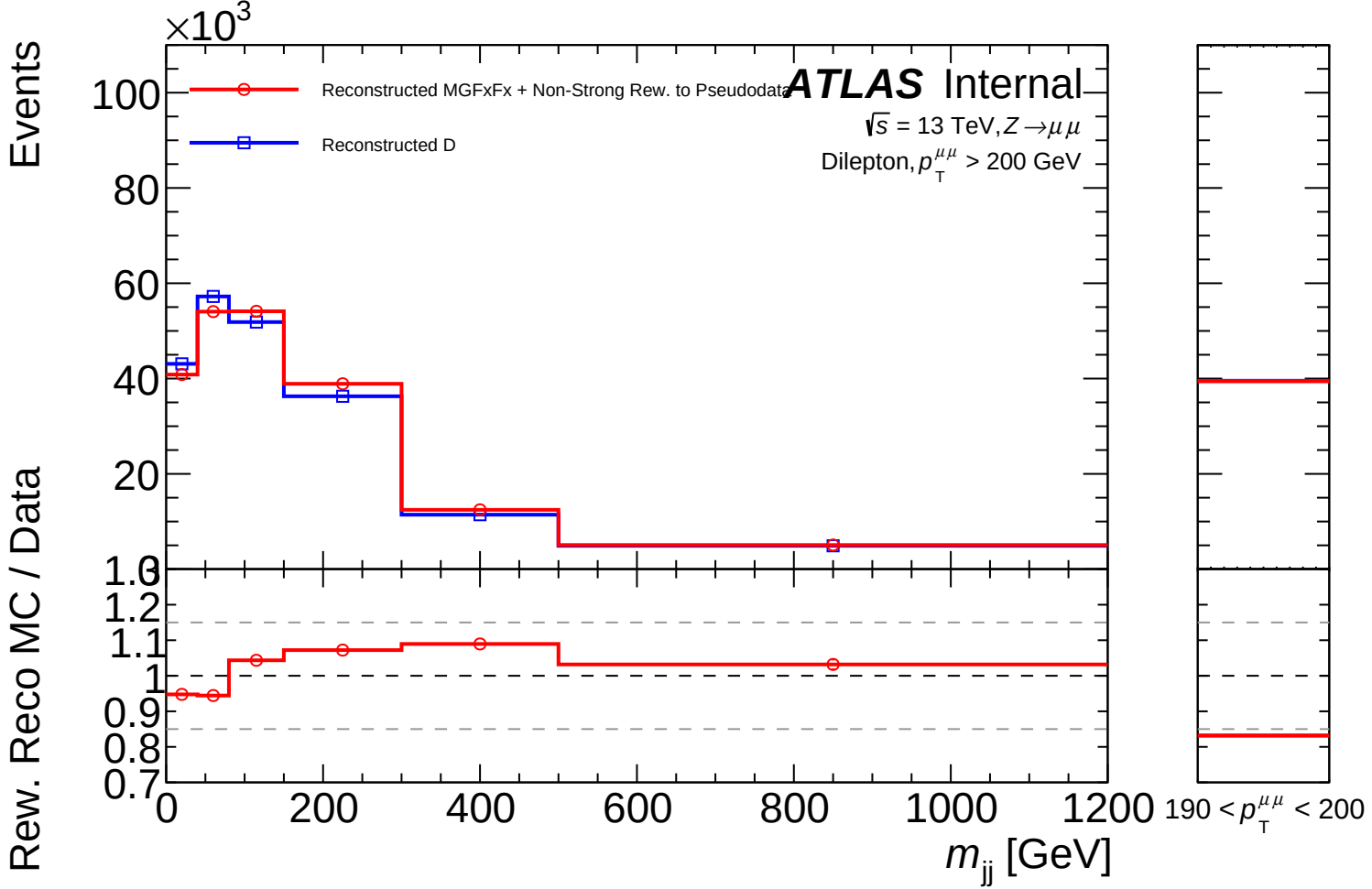


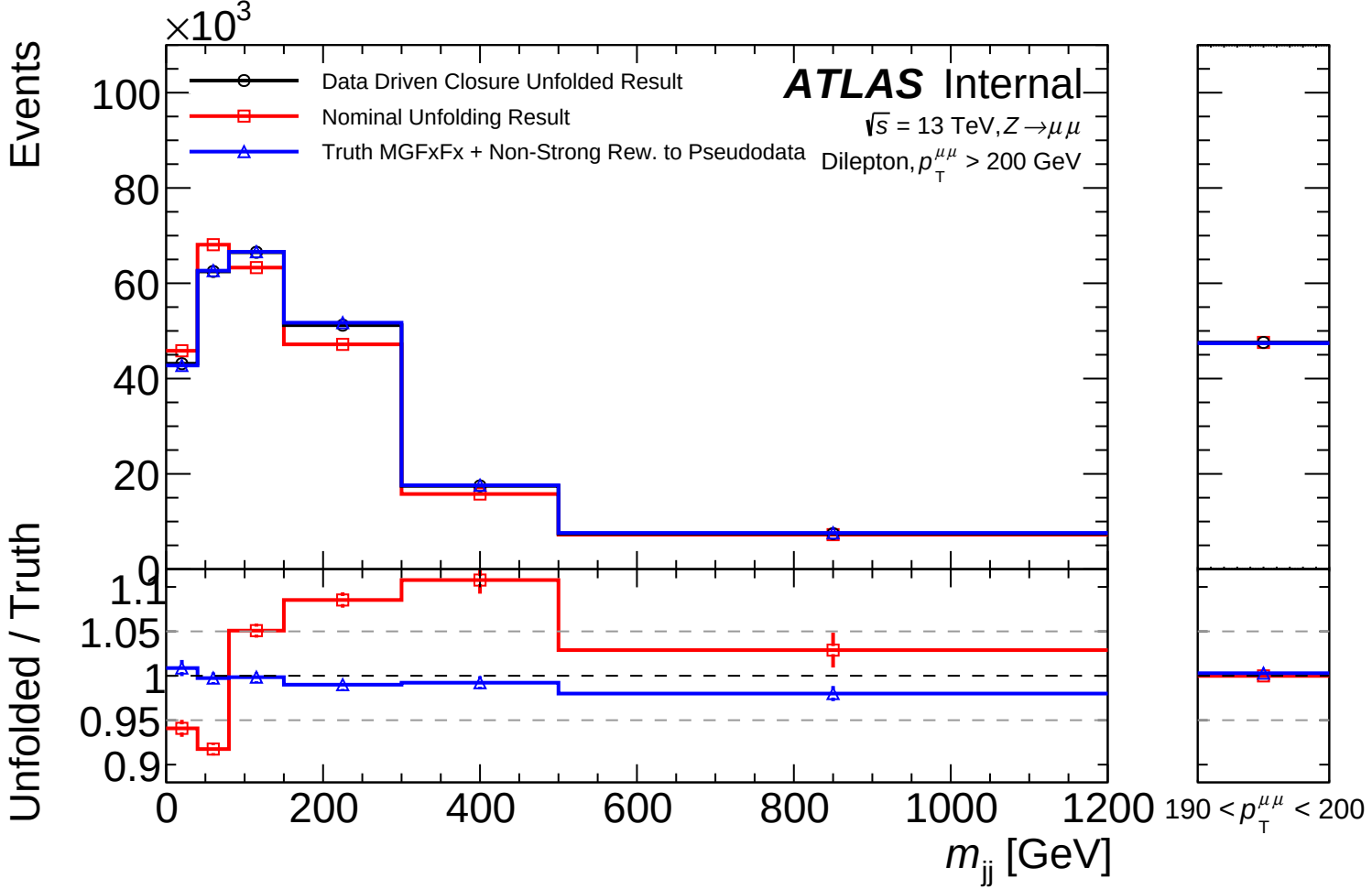




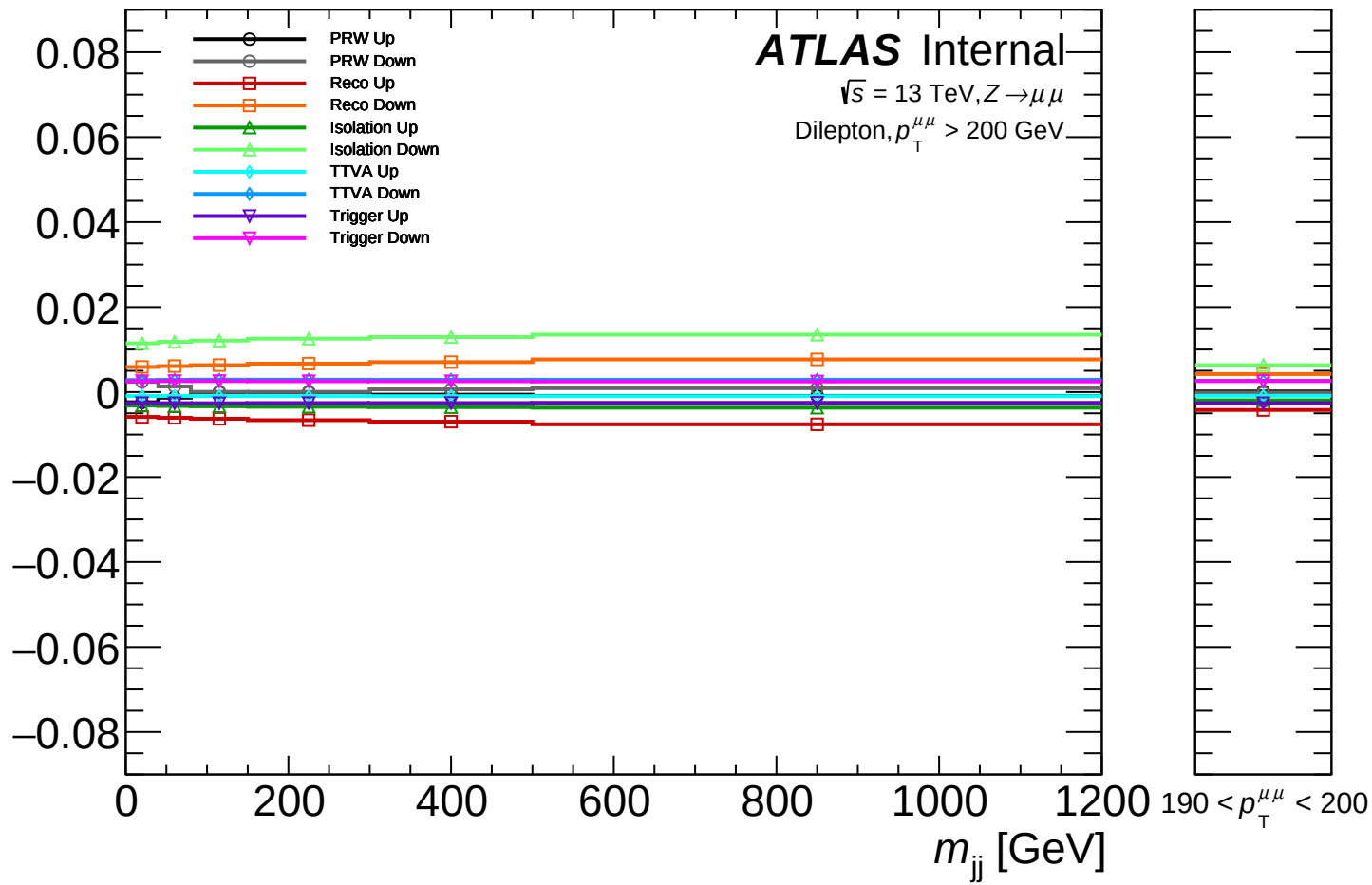




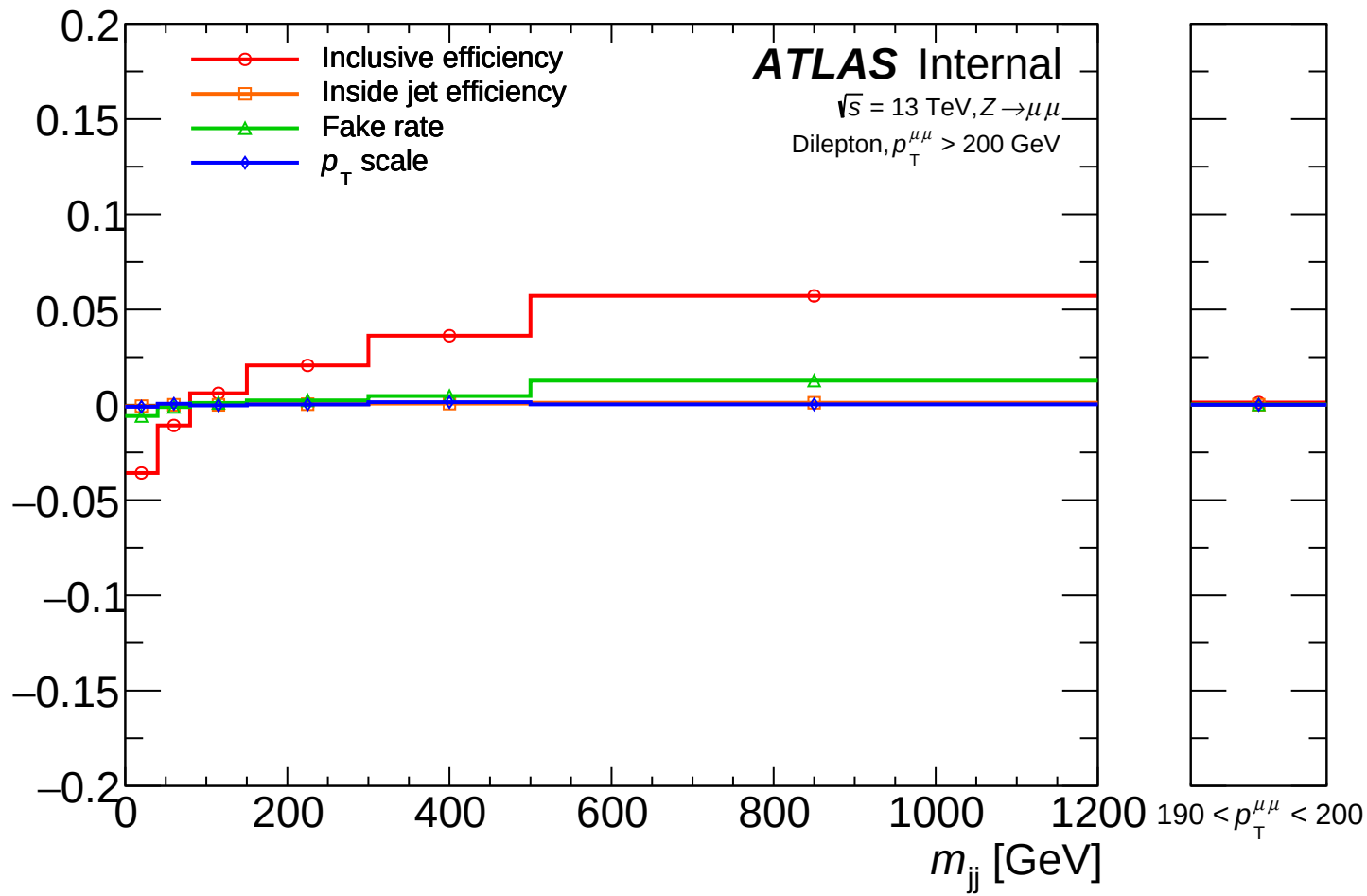




Fractional systematic impact



Fractional systematic impact



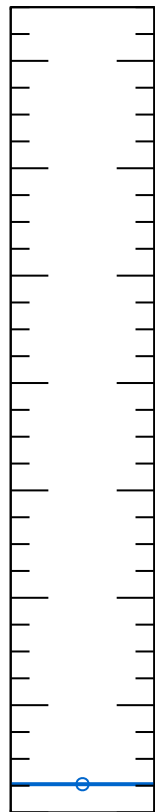
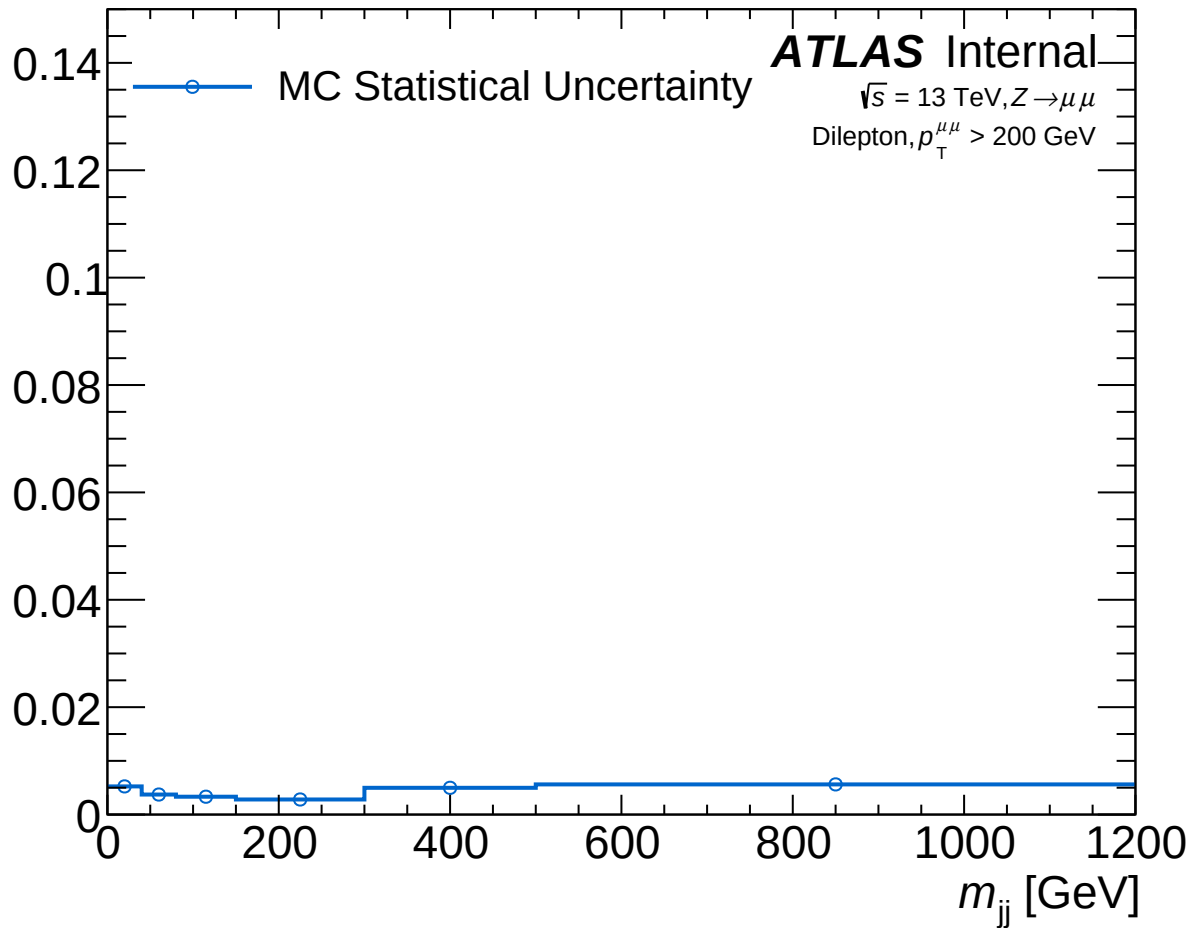
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

MC Statistical Uncertainty



$190 < p_{\text{T}}^{\mu\mu} < 200$

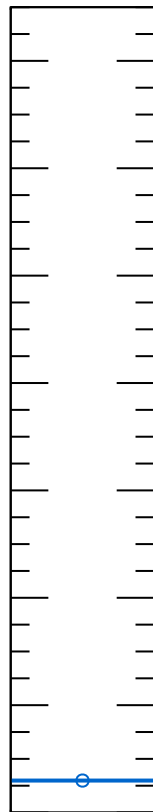
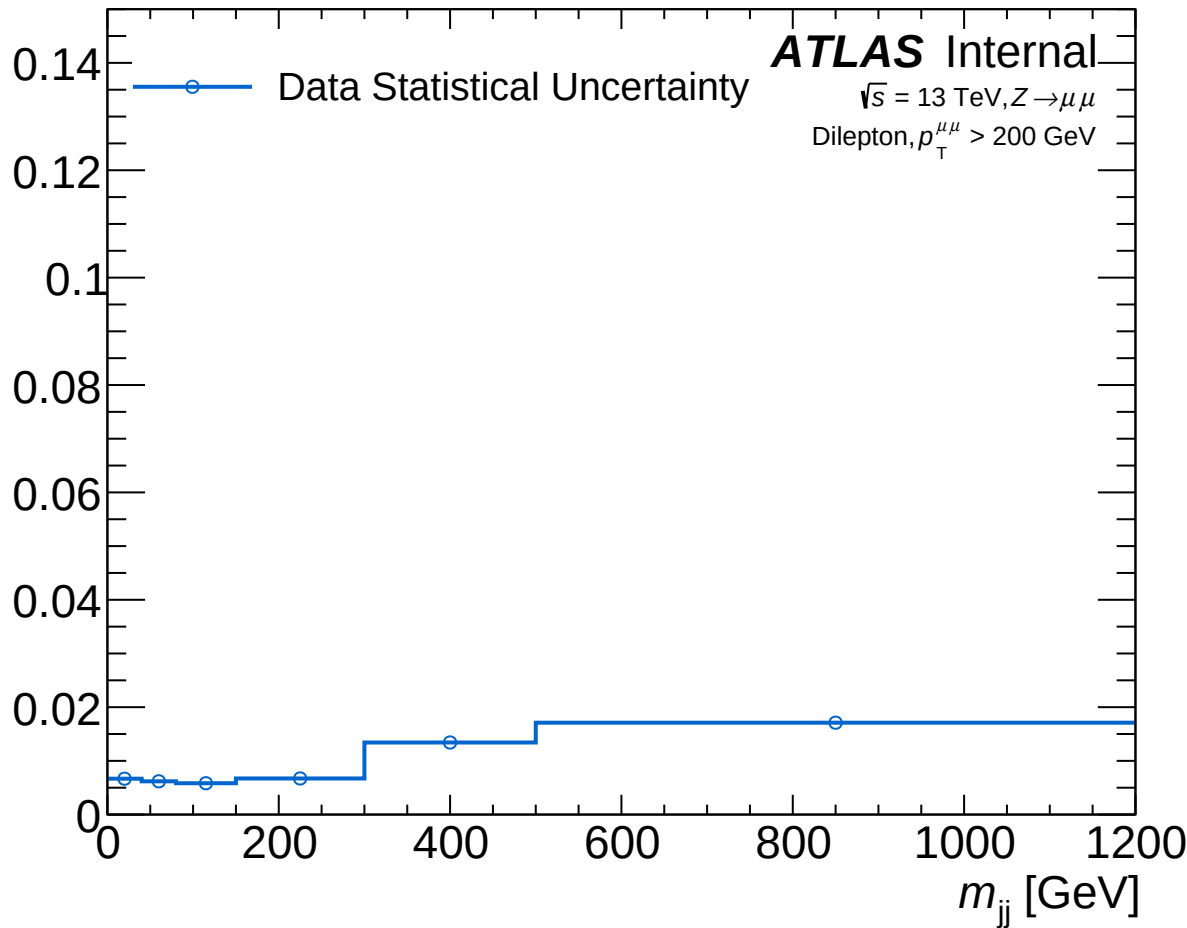
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

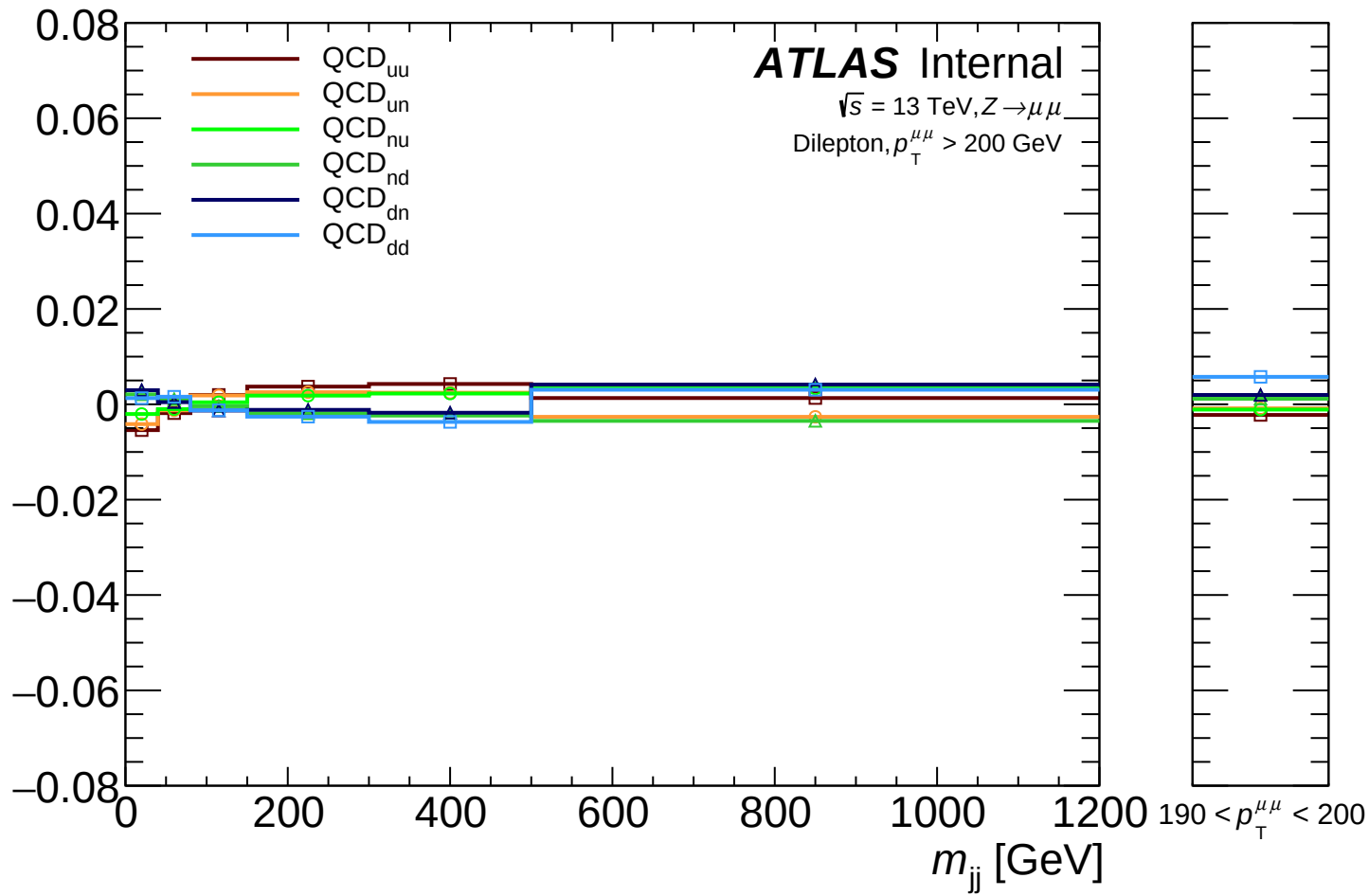
Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

Data Statistical Uncertainty

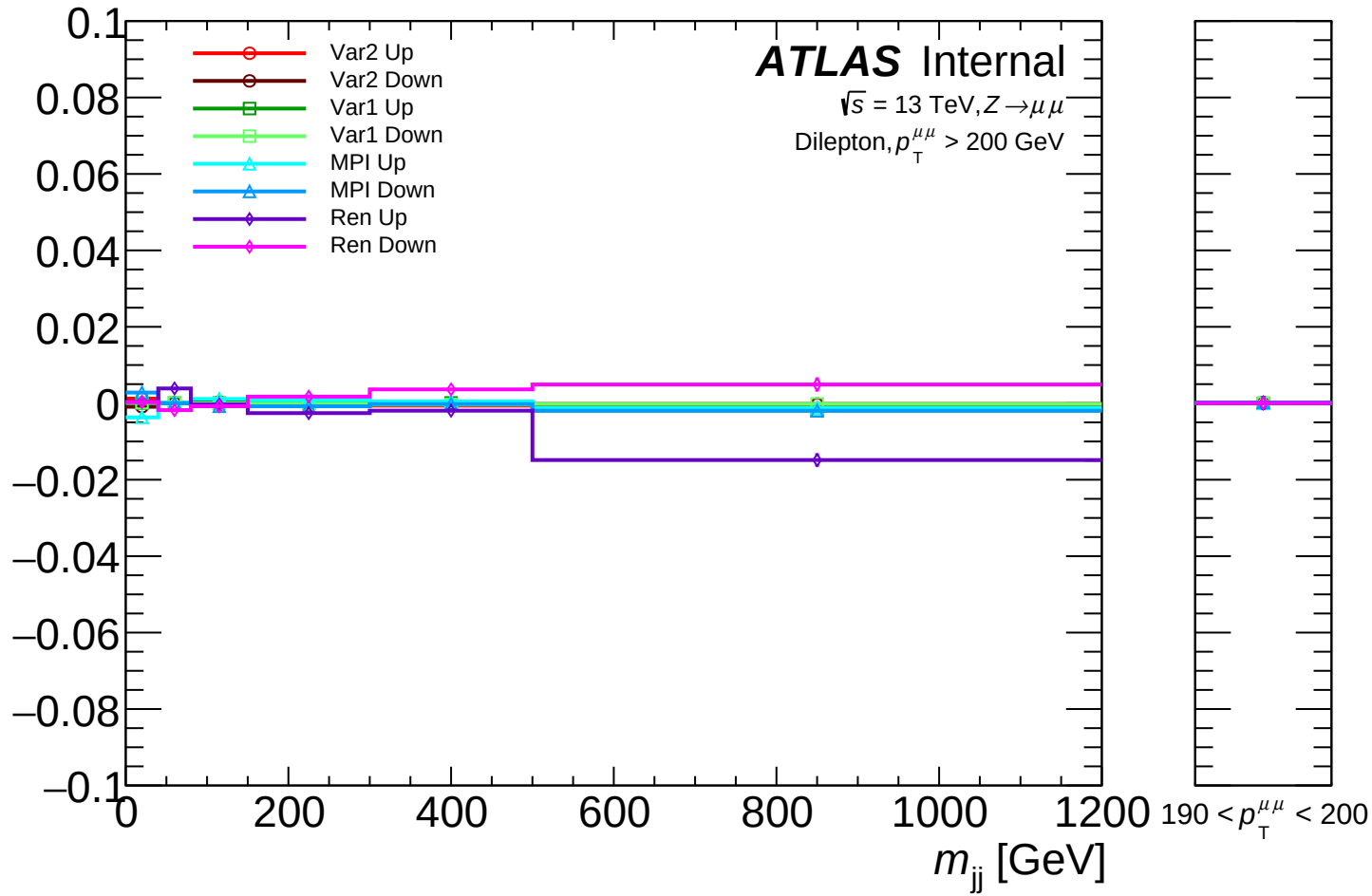


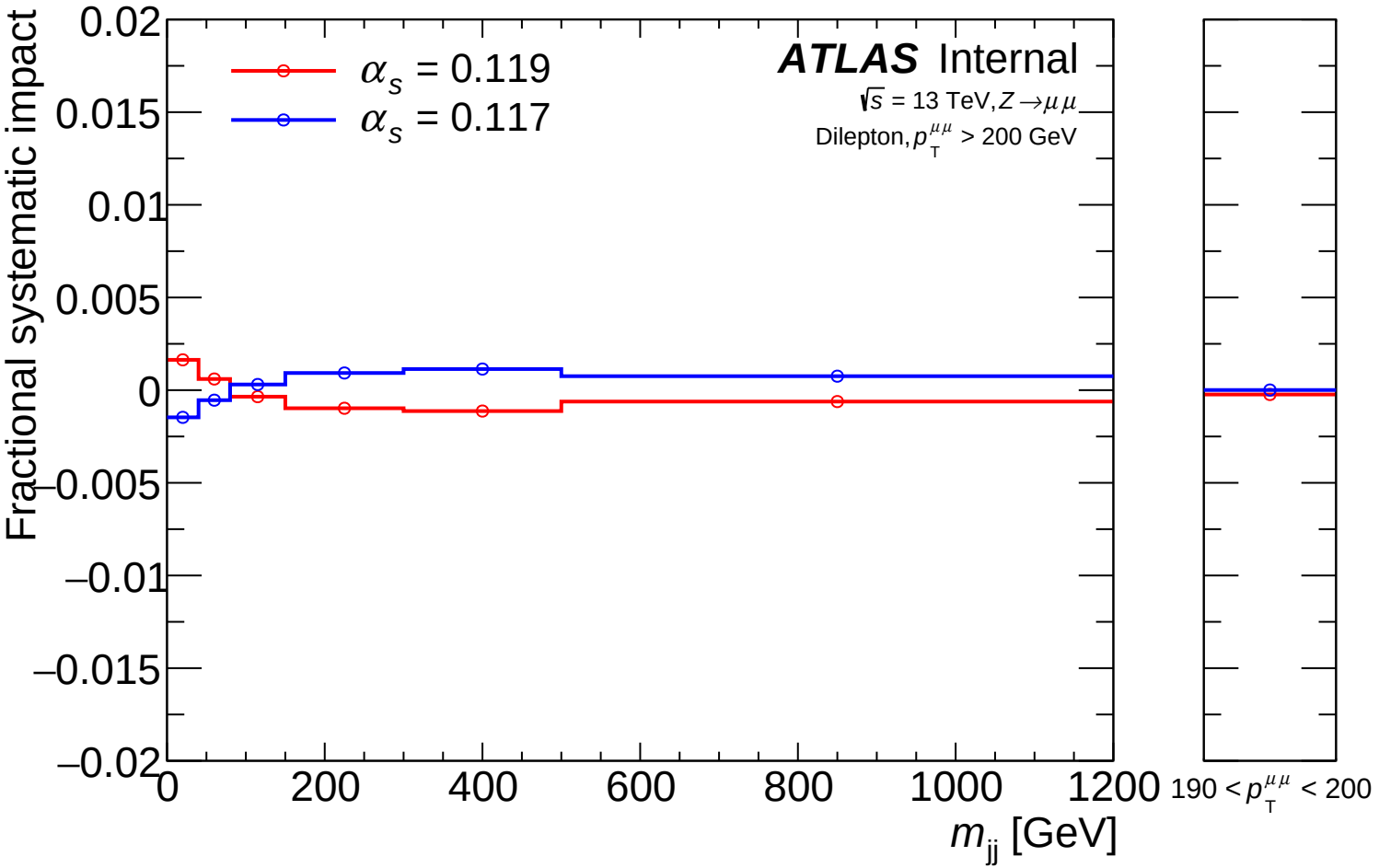
$190 < p_{\text{T}}^{\mu\mu} < 200$

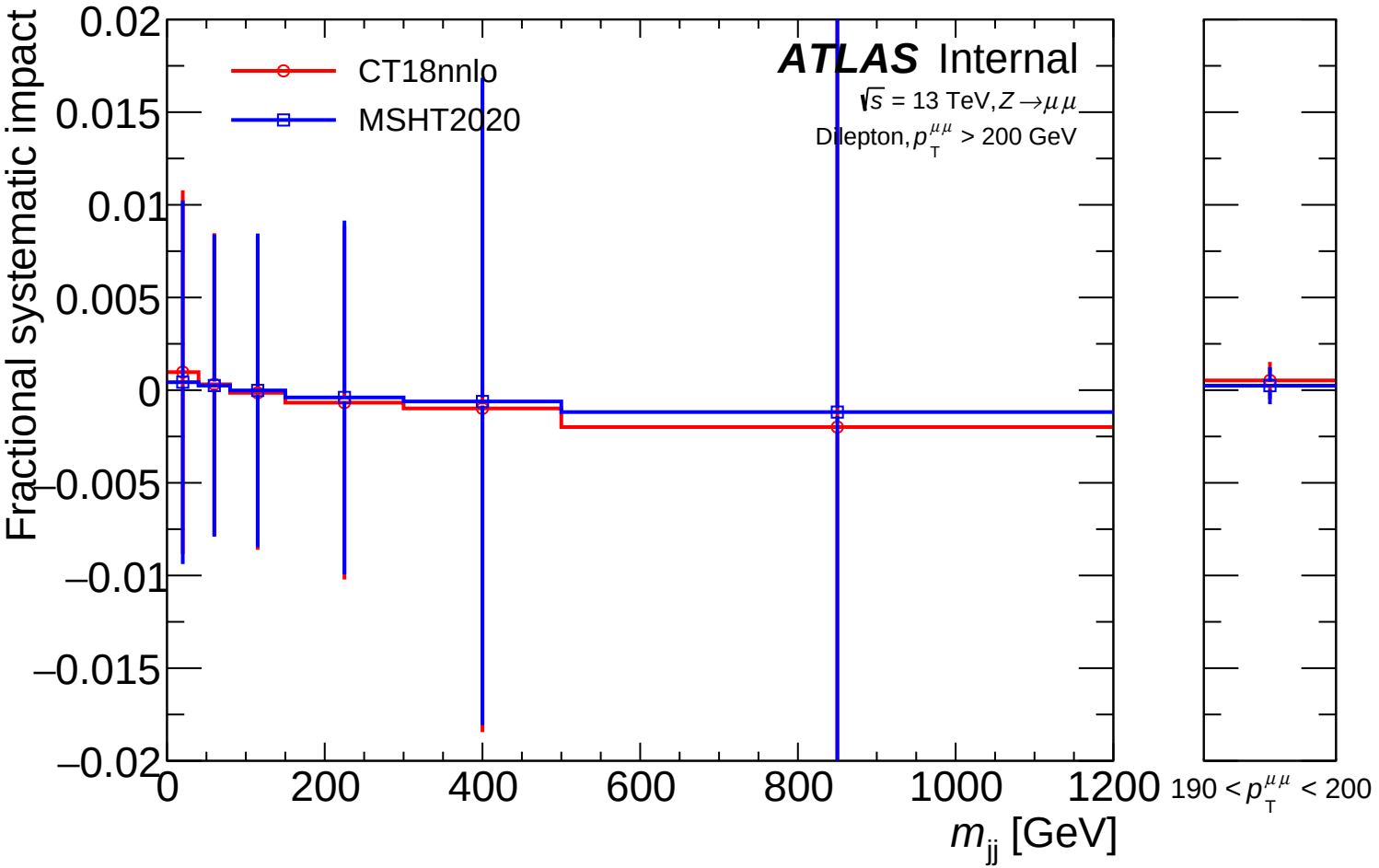
Fractional systematic impact



Fractional systematic impact







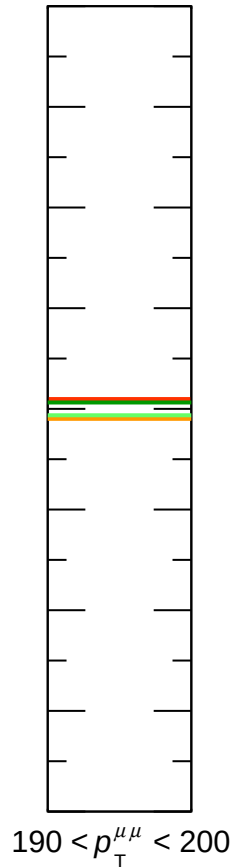
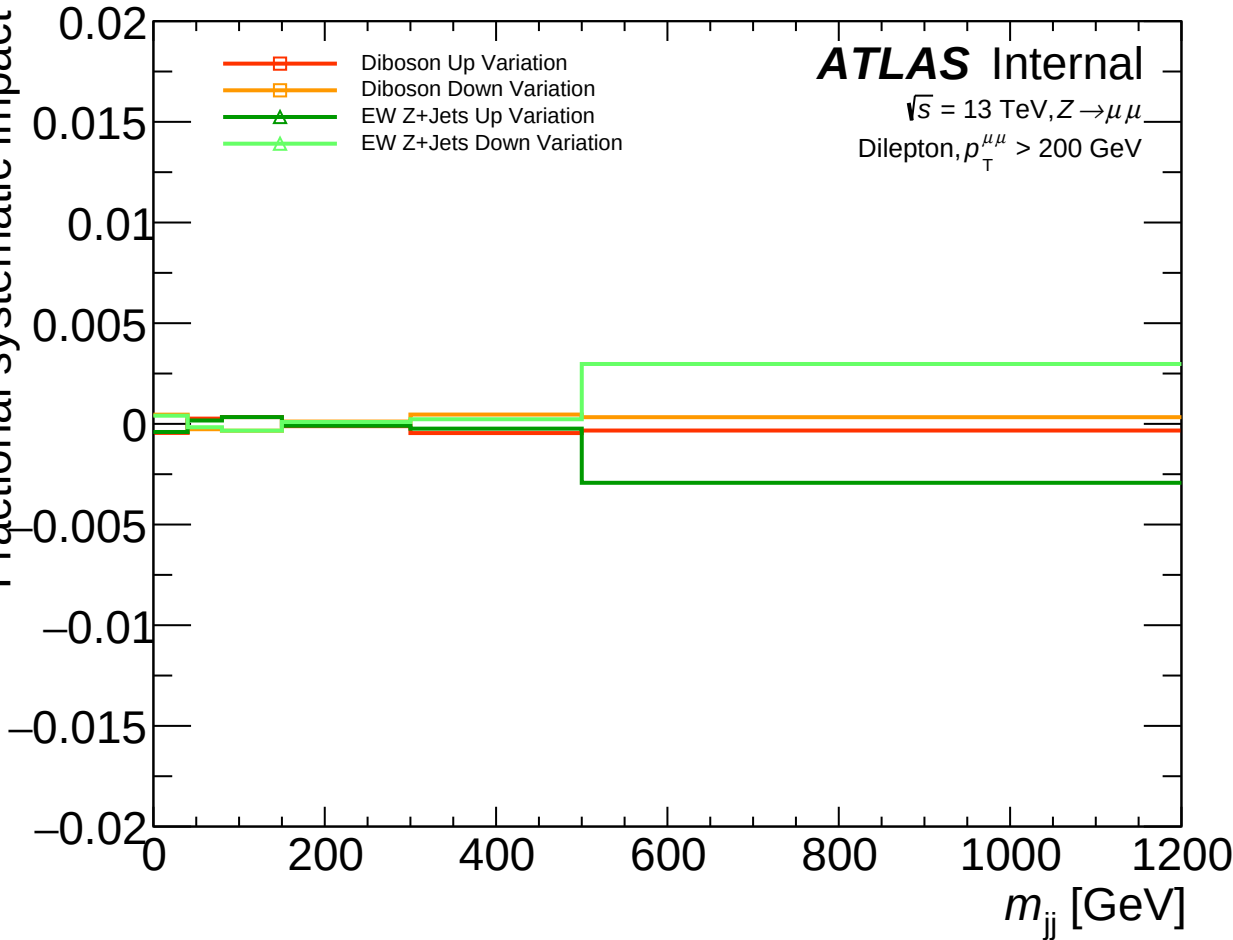
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

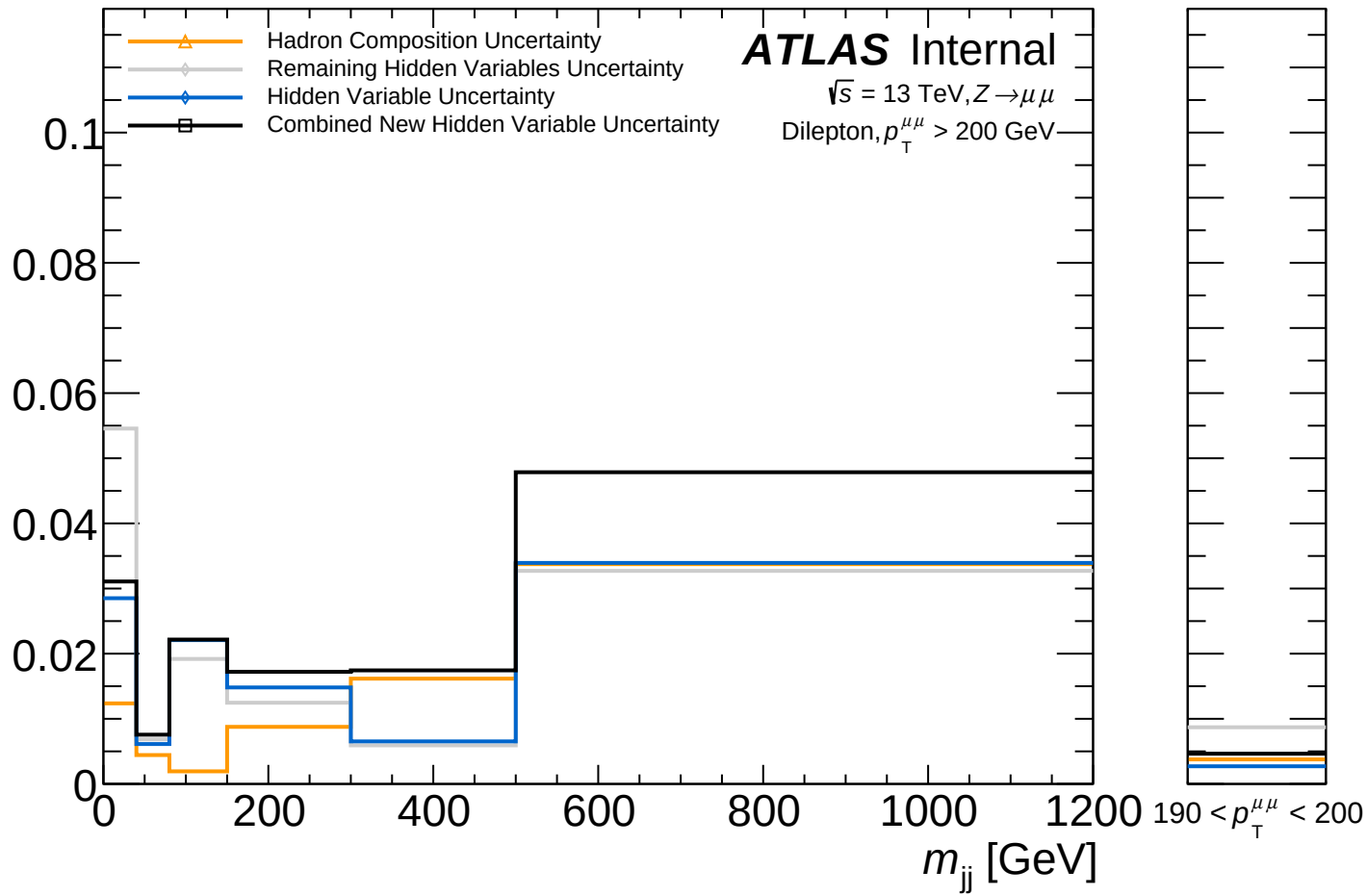
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

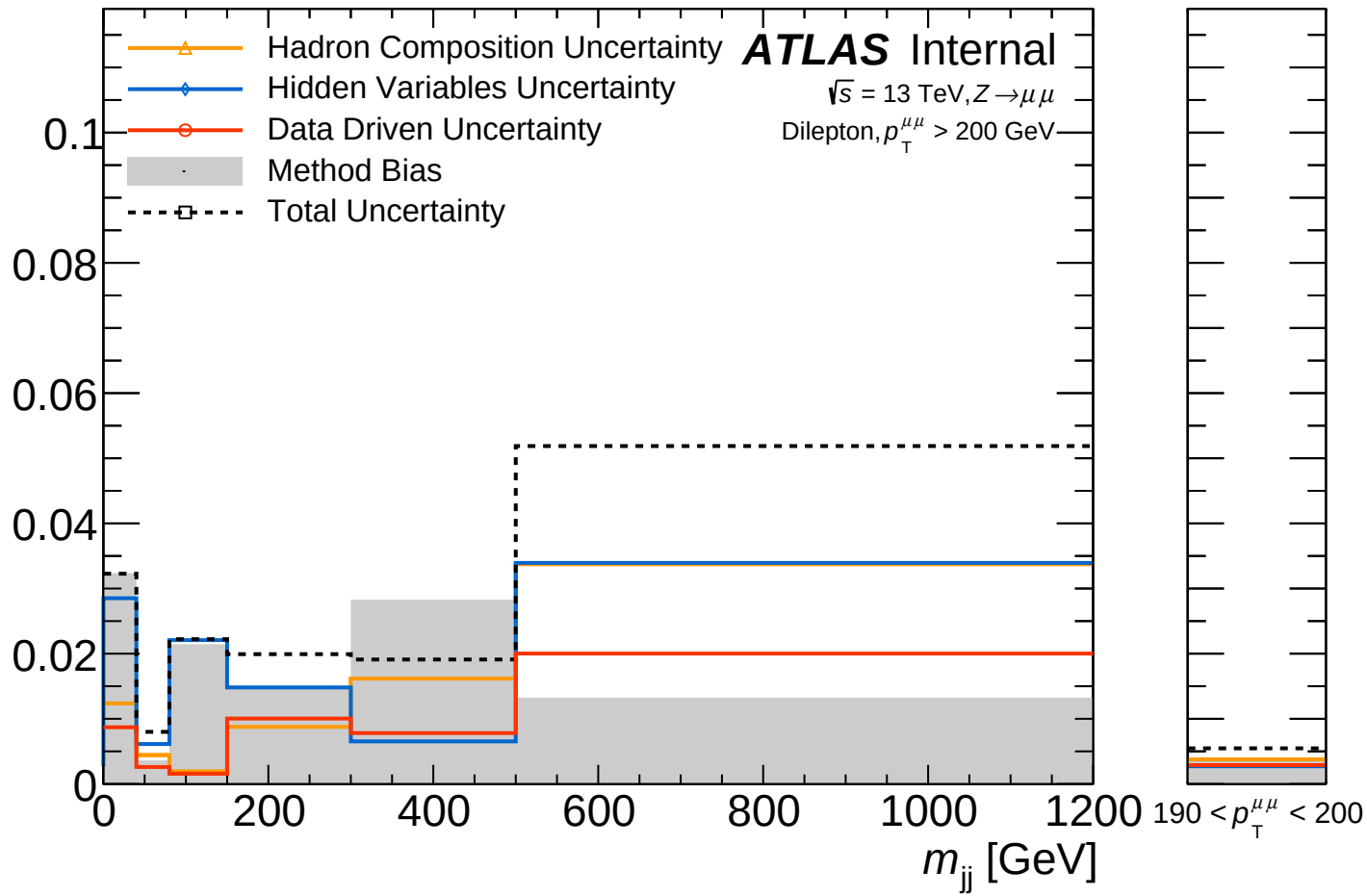


ATLAS Internal

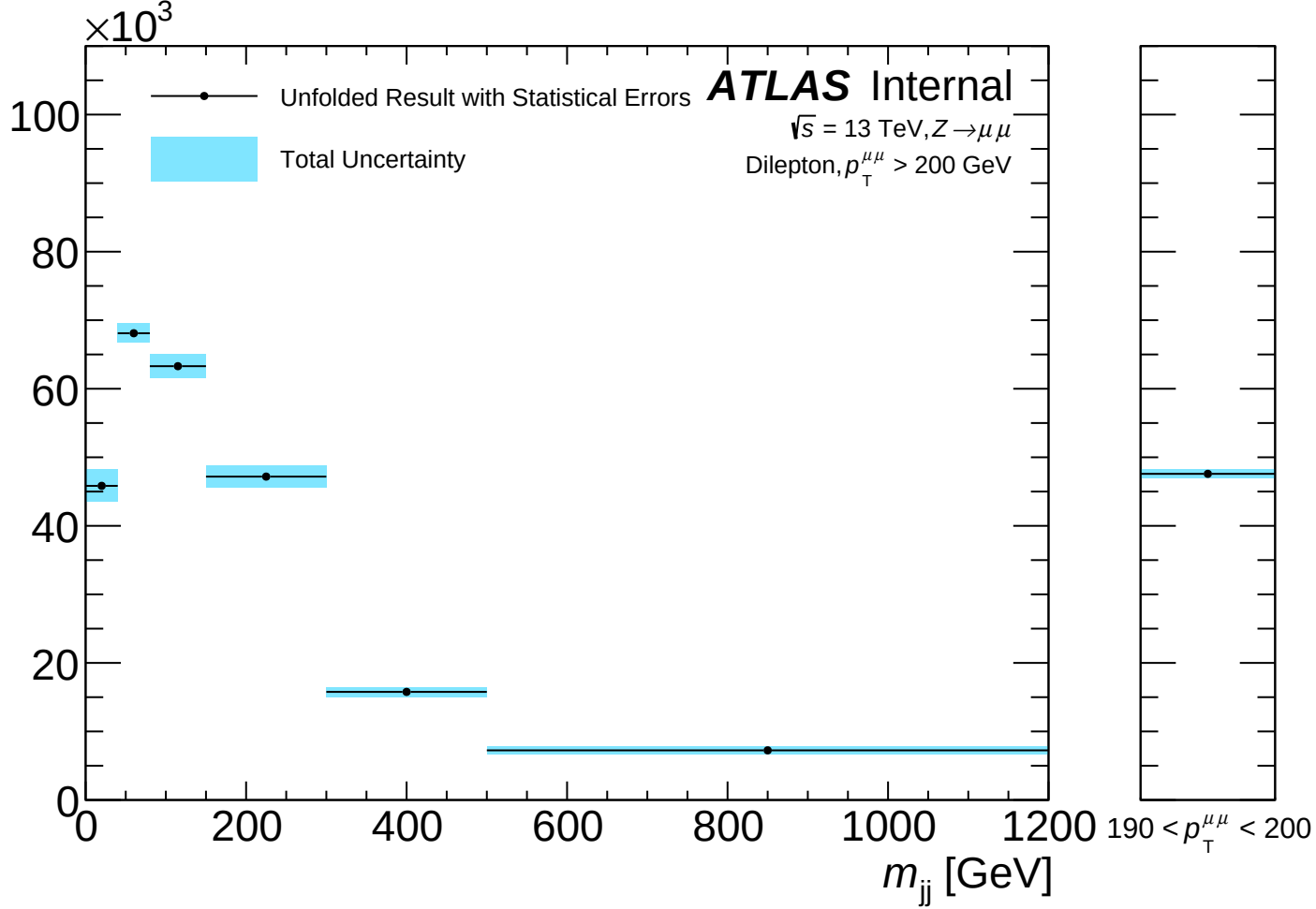
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

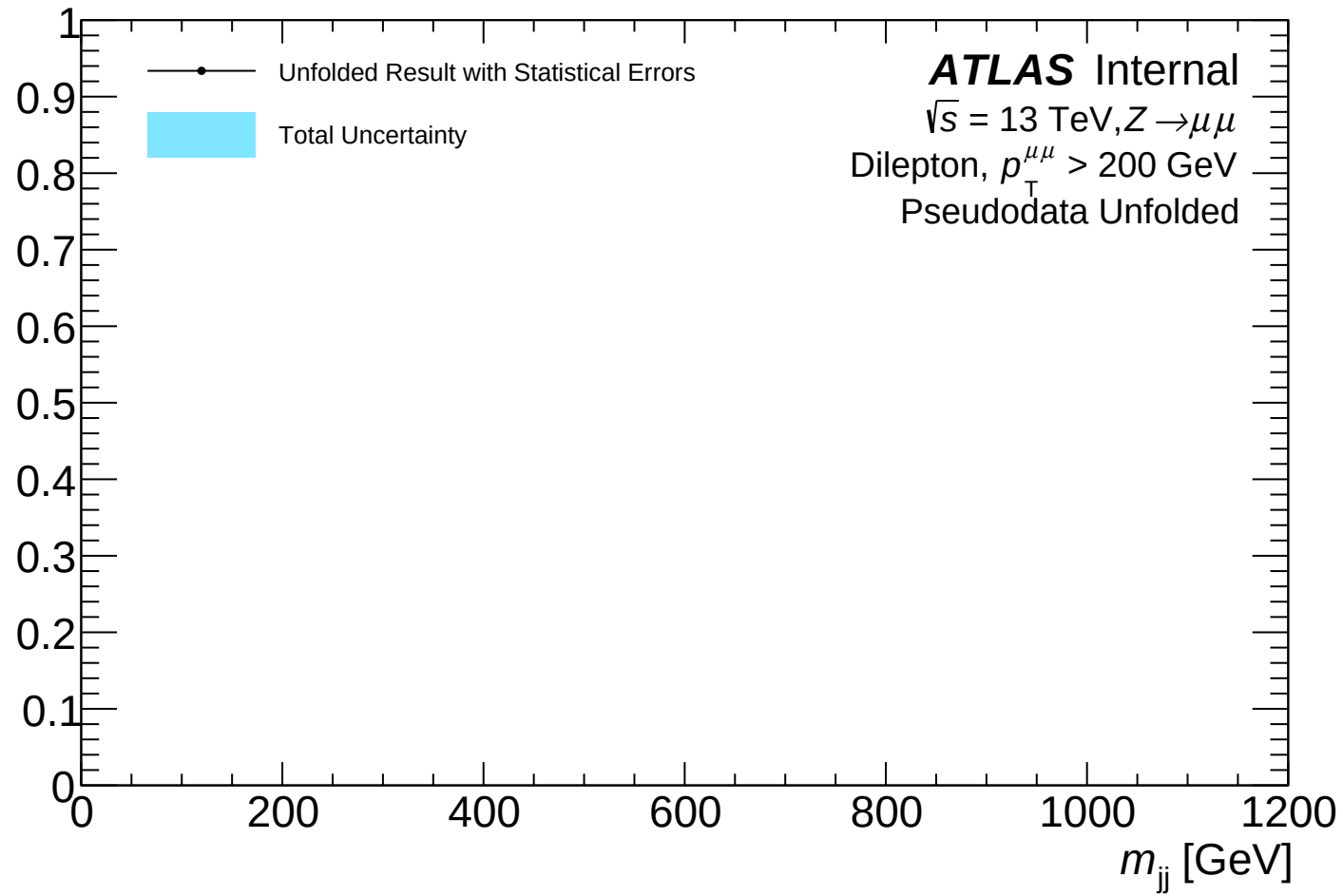
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

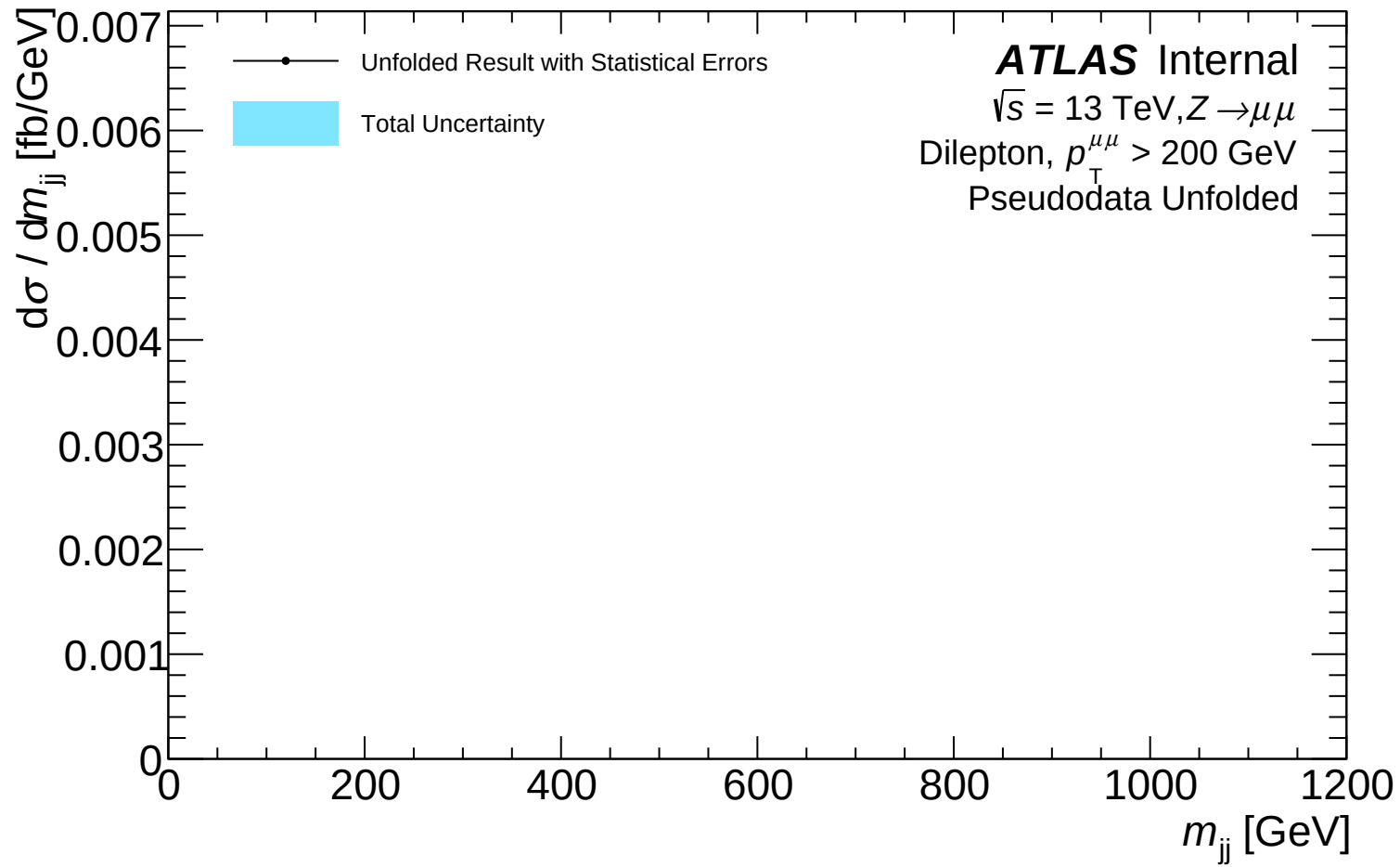




Events







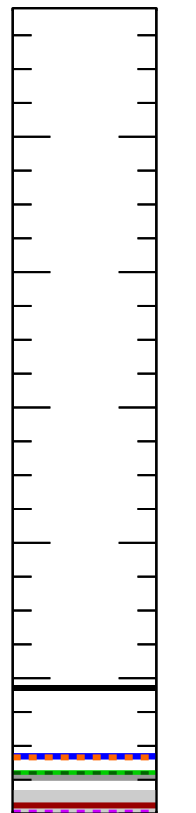
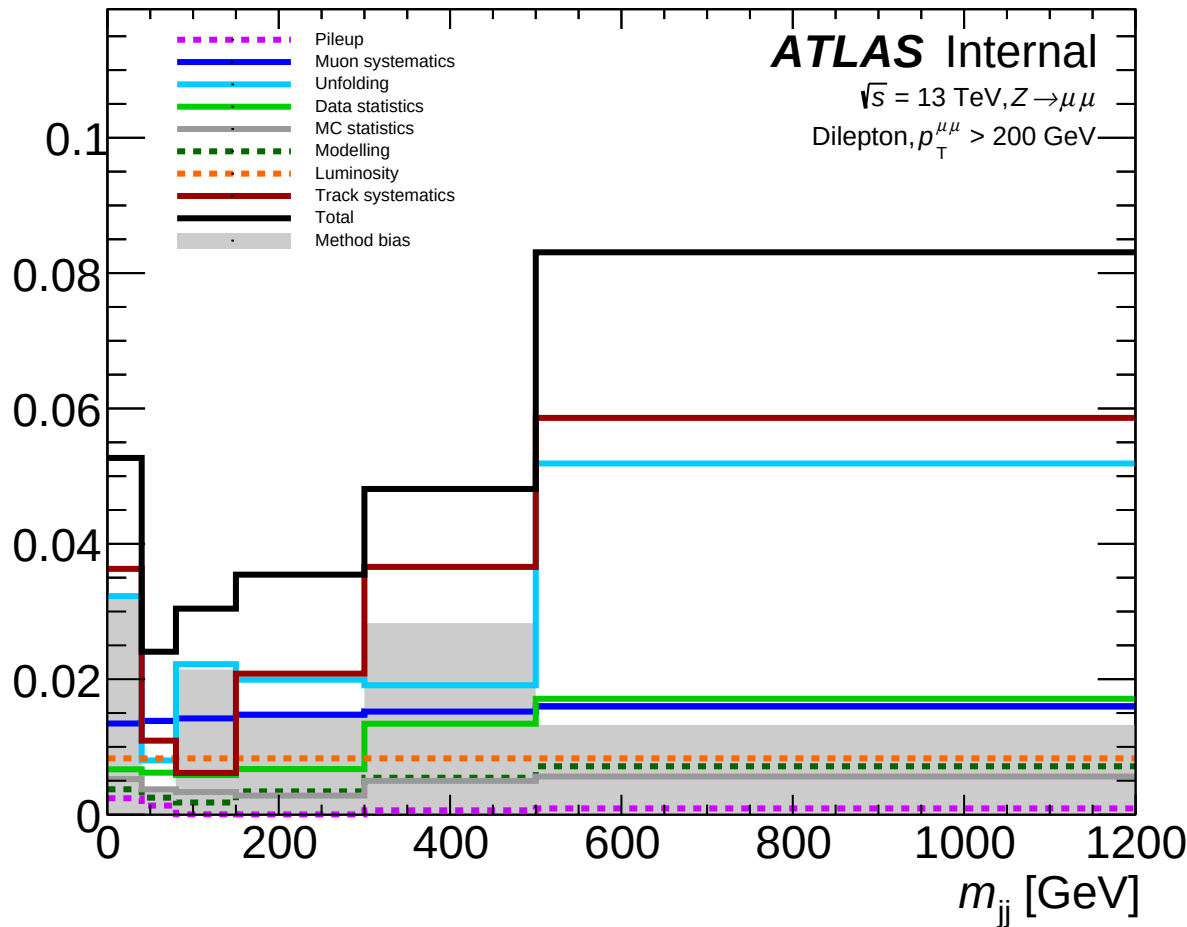
Fractional uncertainty

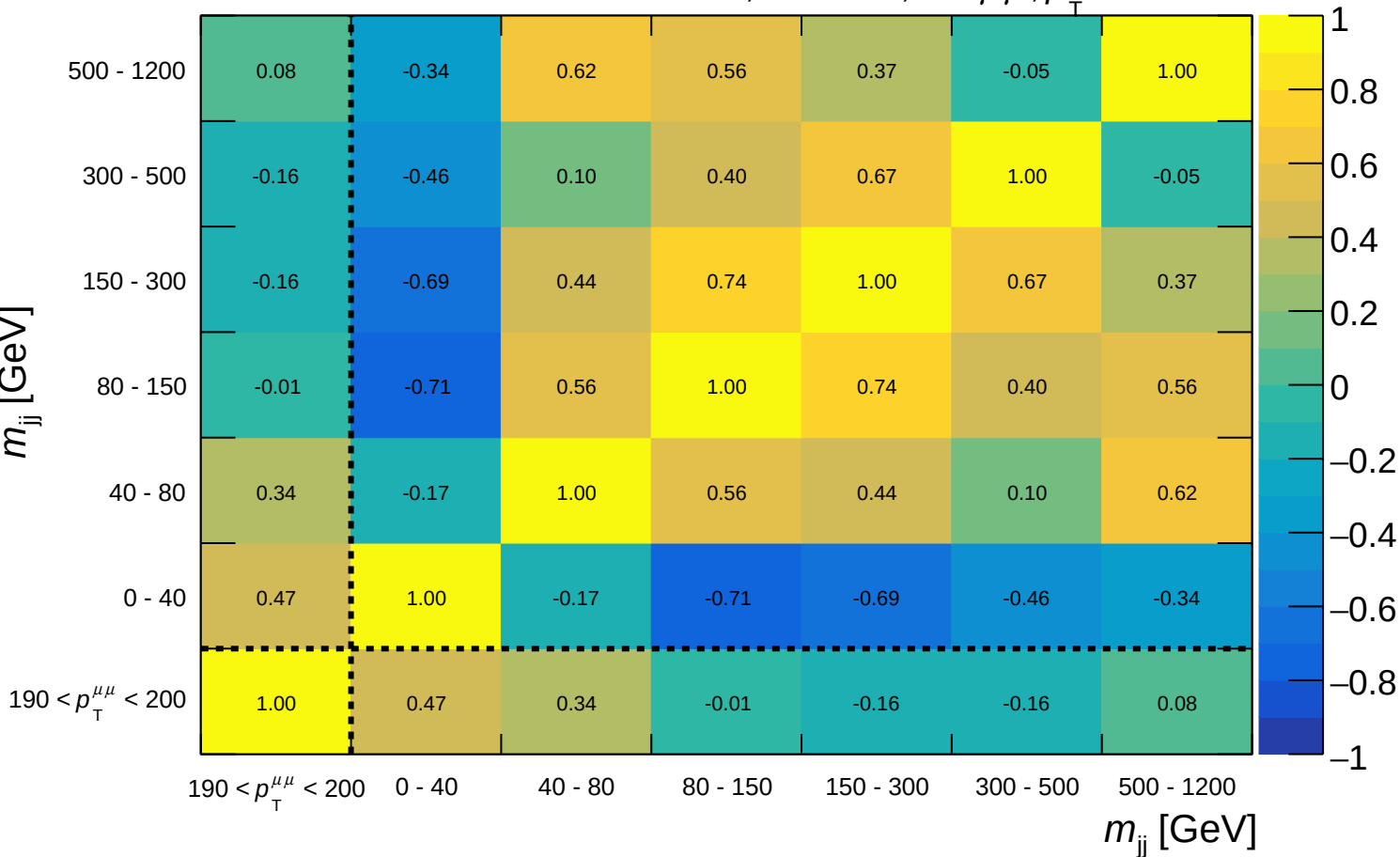
ATLAS Internal

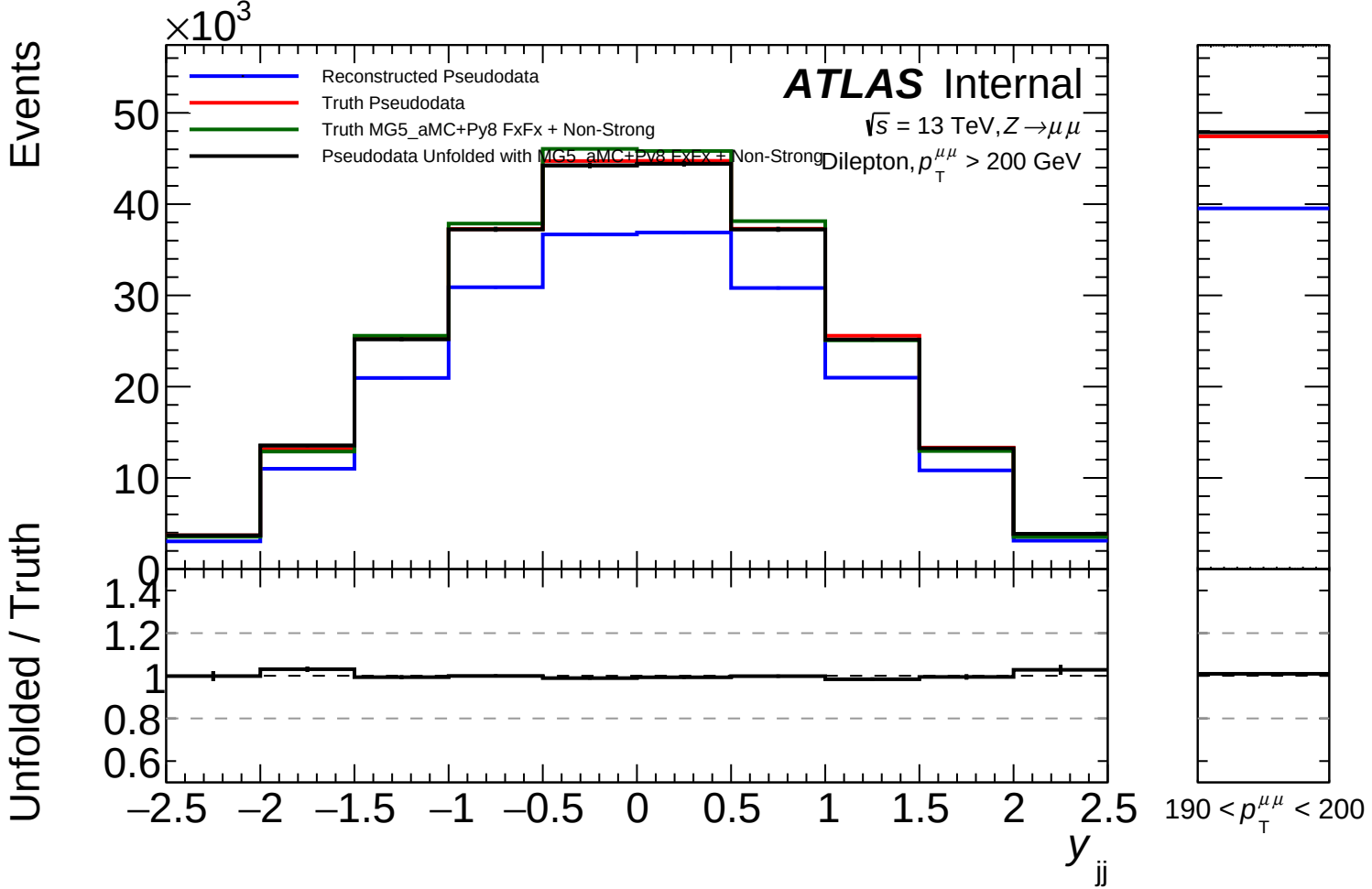
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

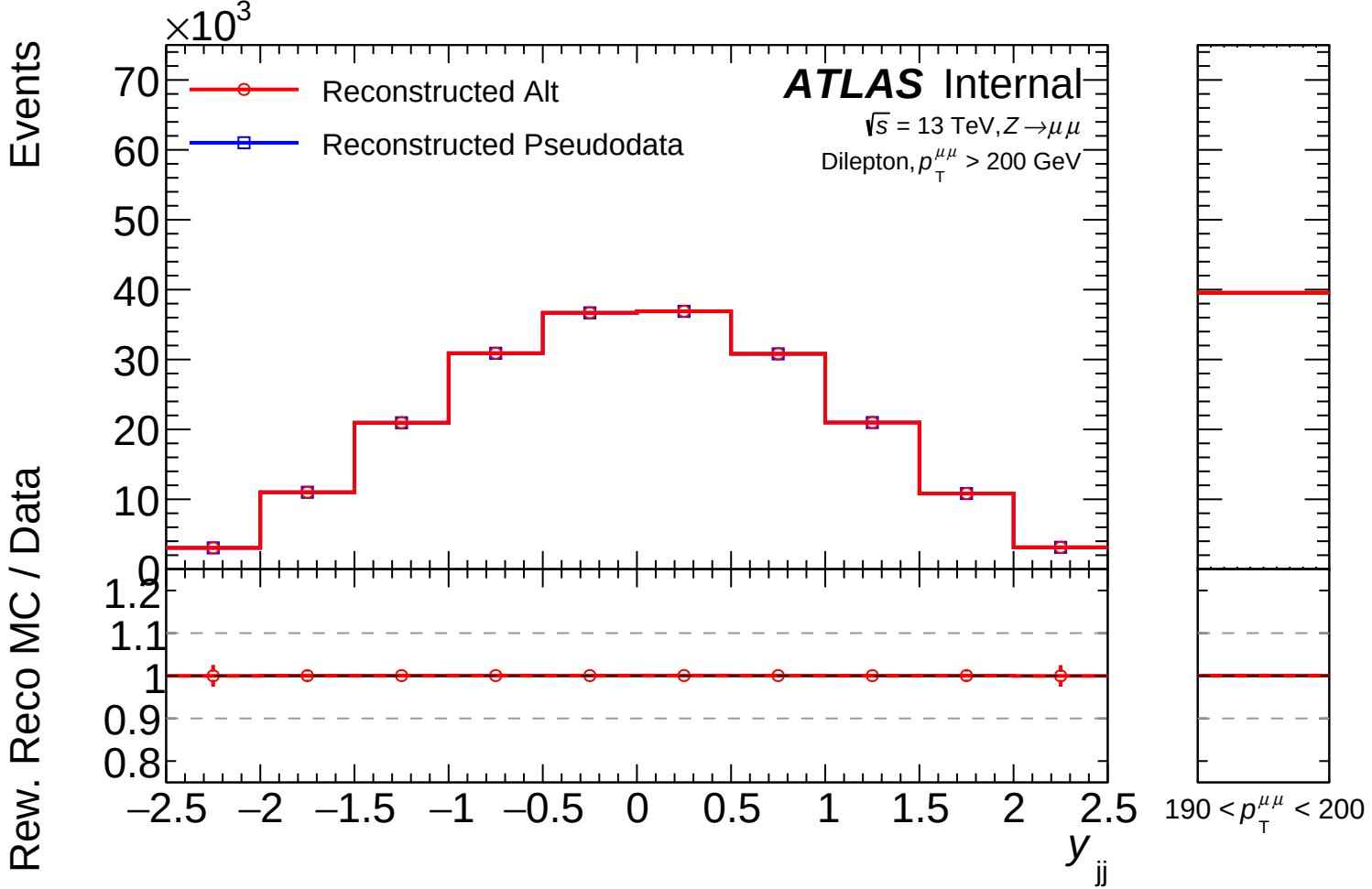
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

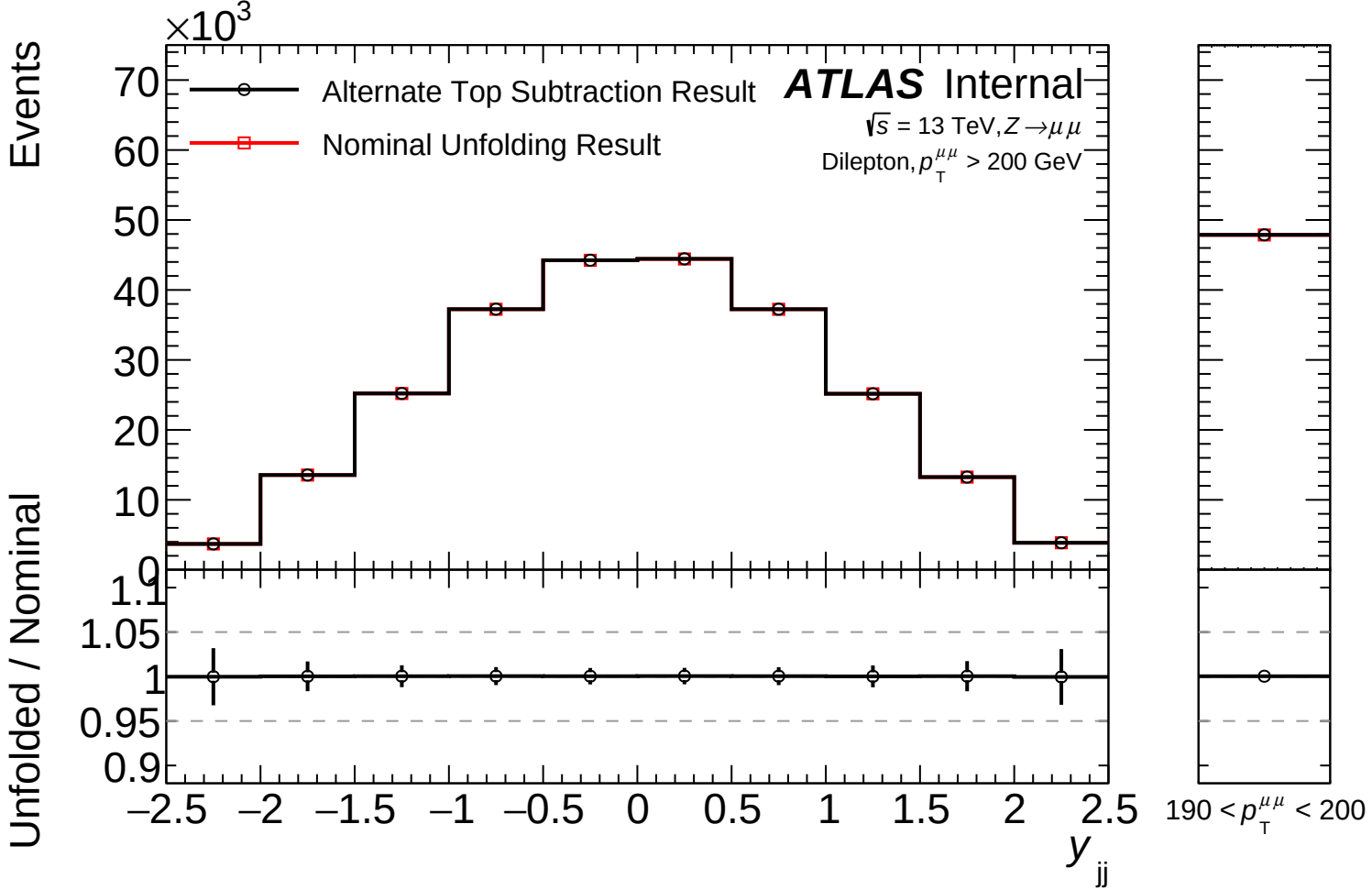
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $m_{\text{jj}} [\text{GeV}]$ 







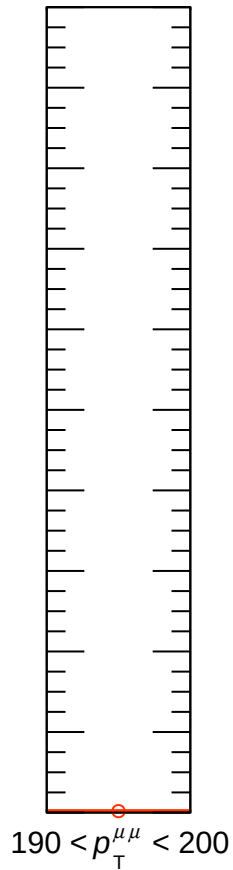
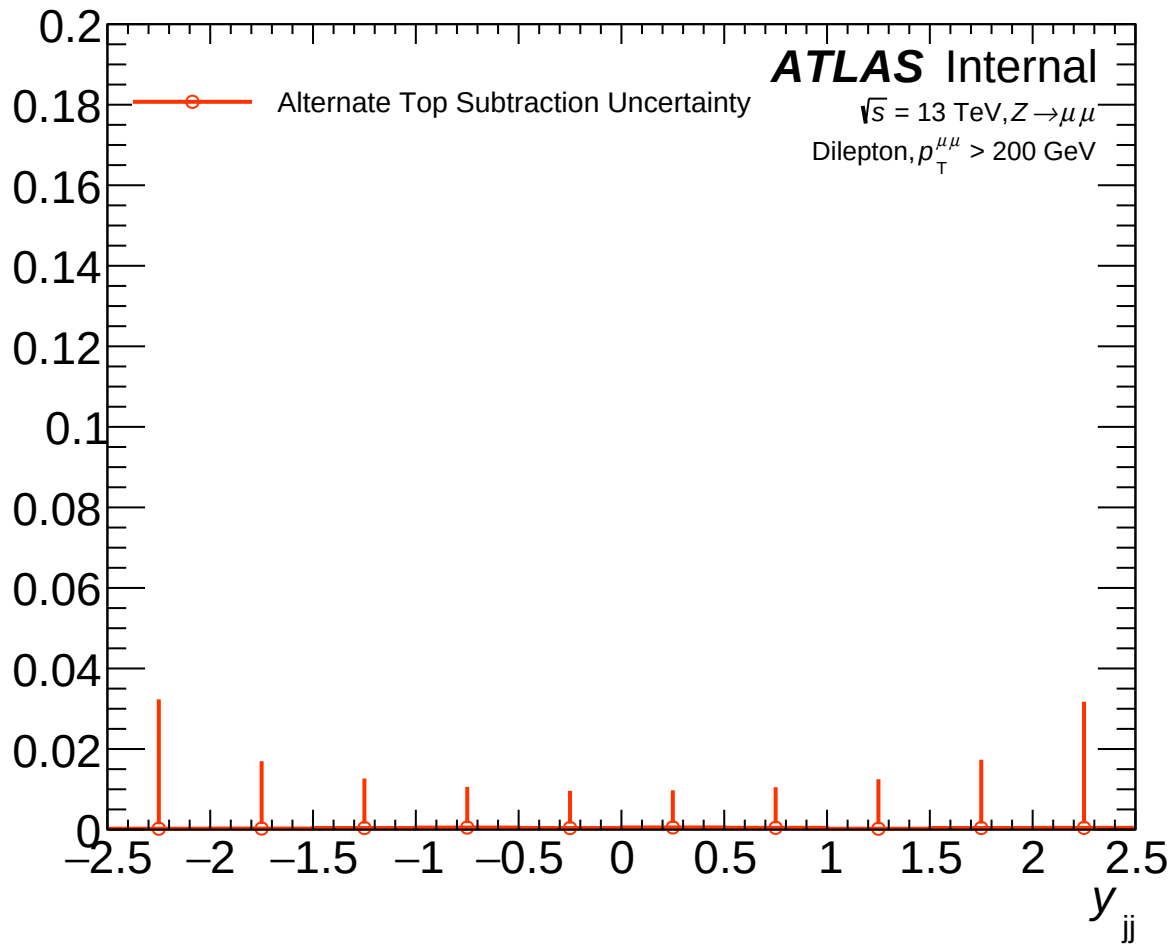
Fractional systematic impact

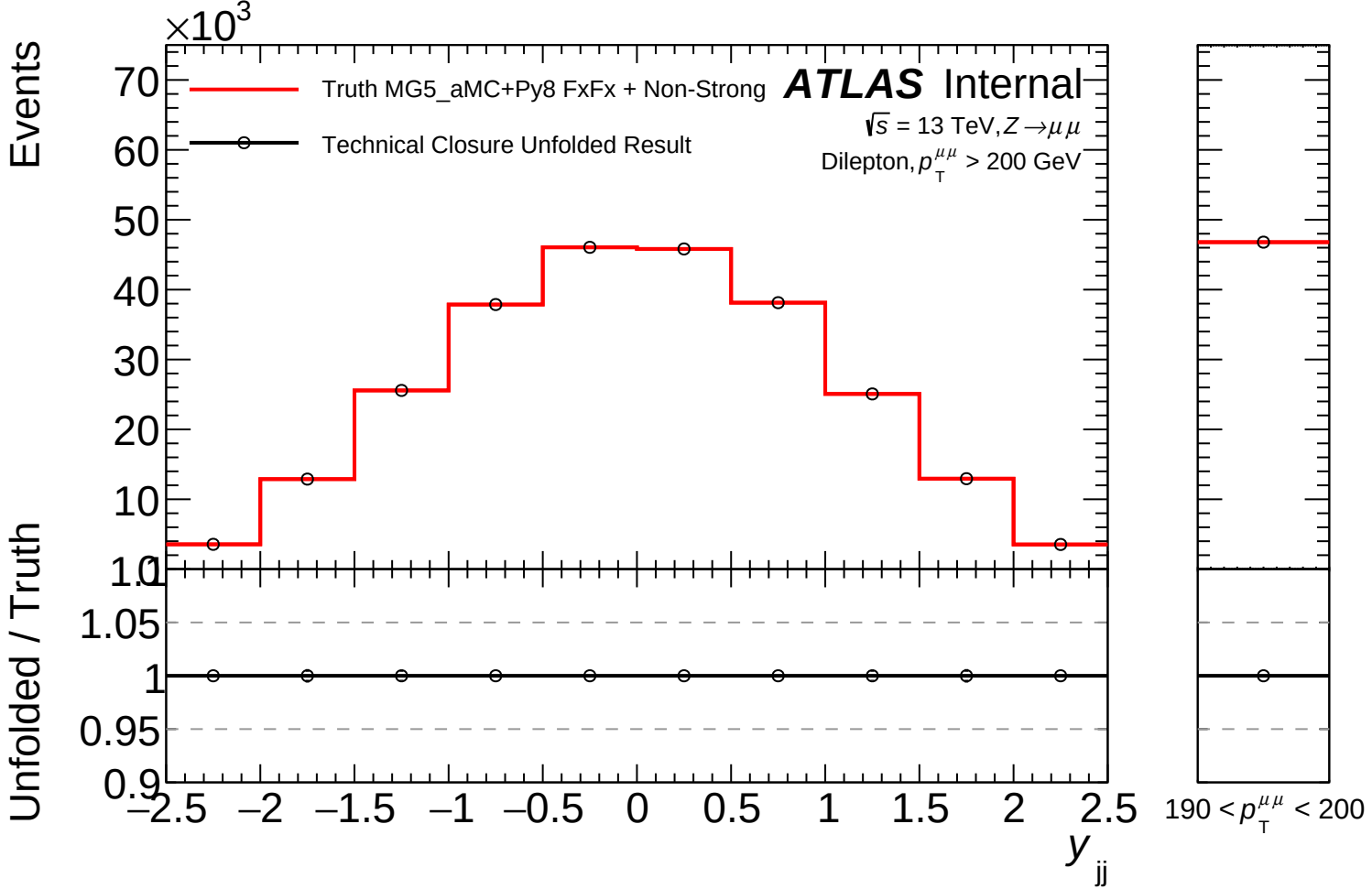
ATLAS Internal

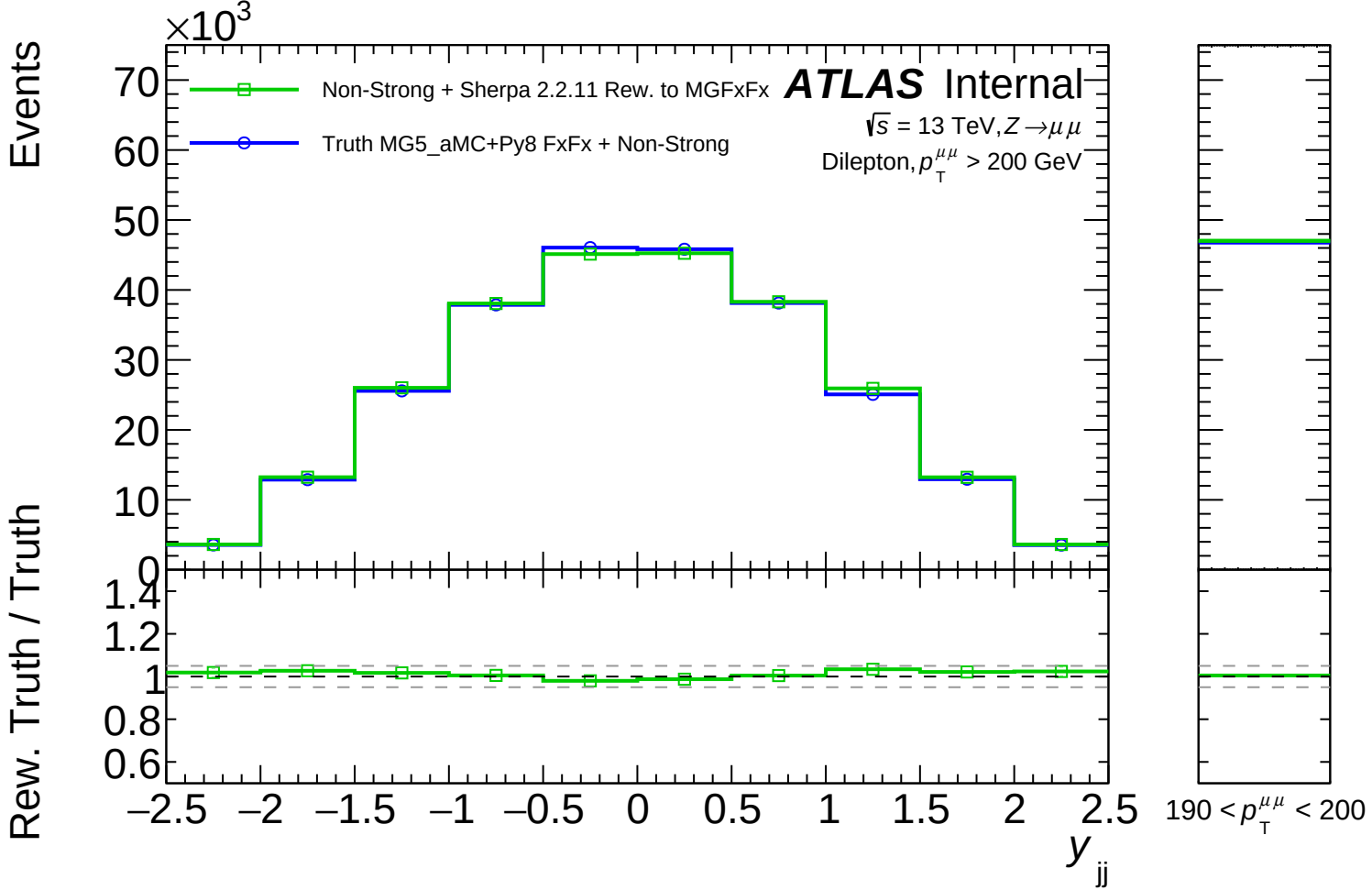
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

Alternate Top Subtraction Uncertainty

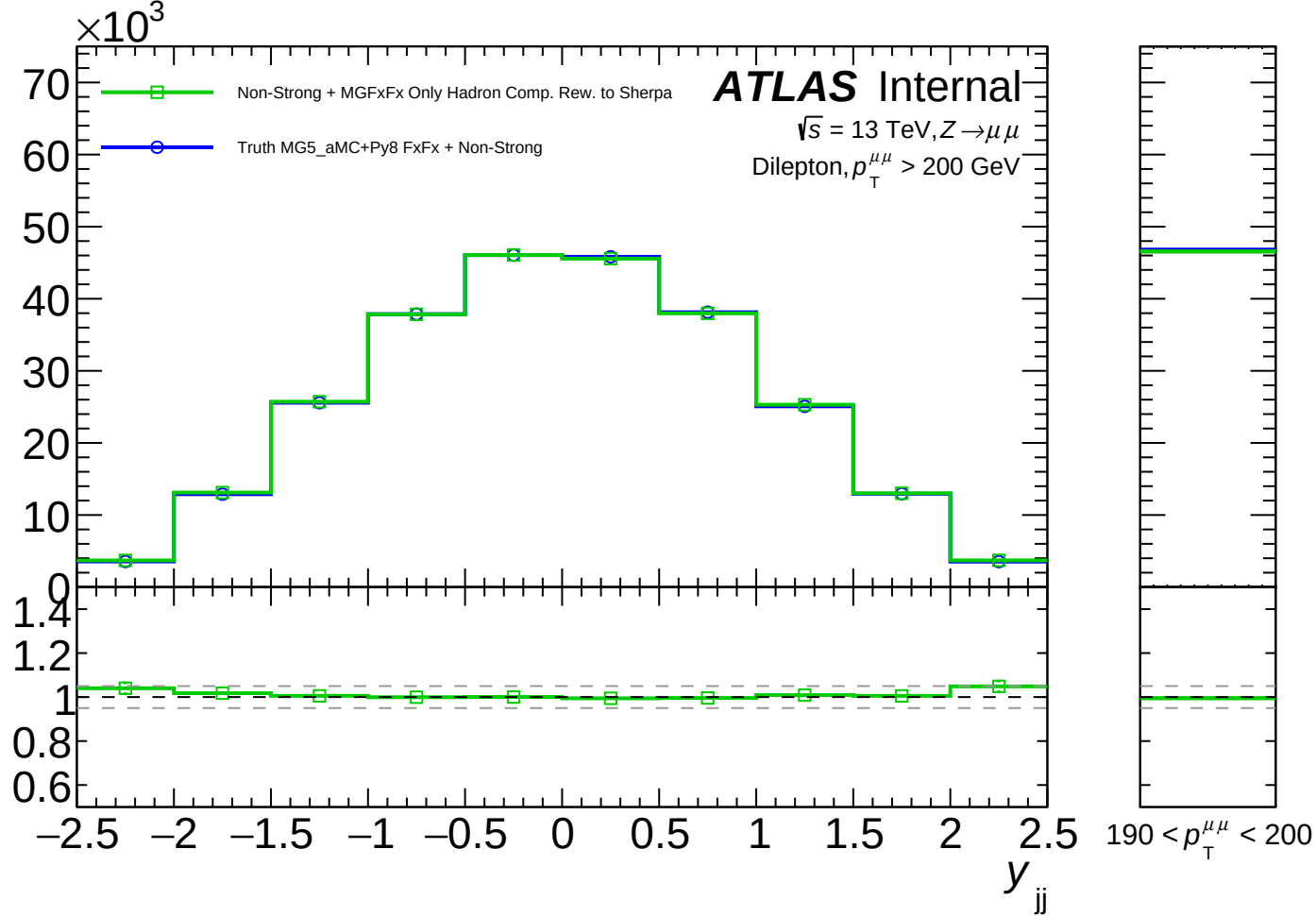






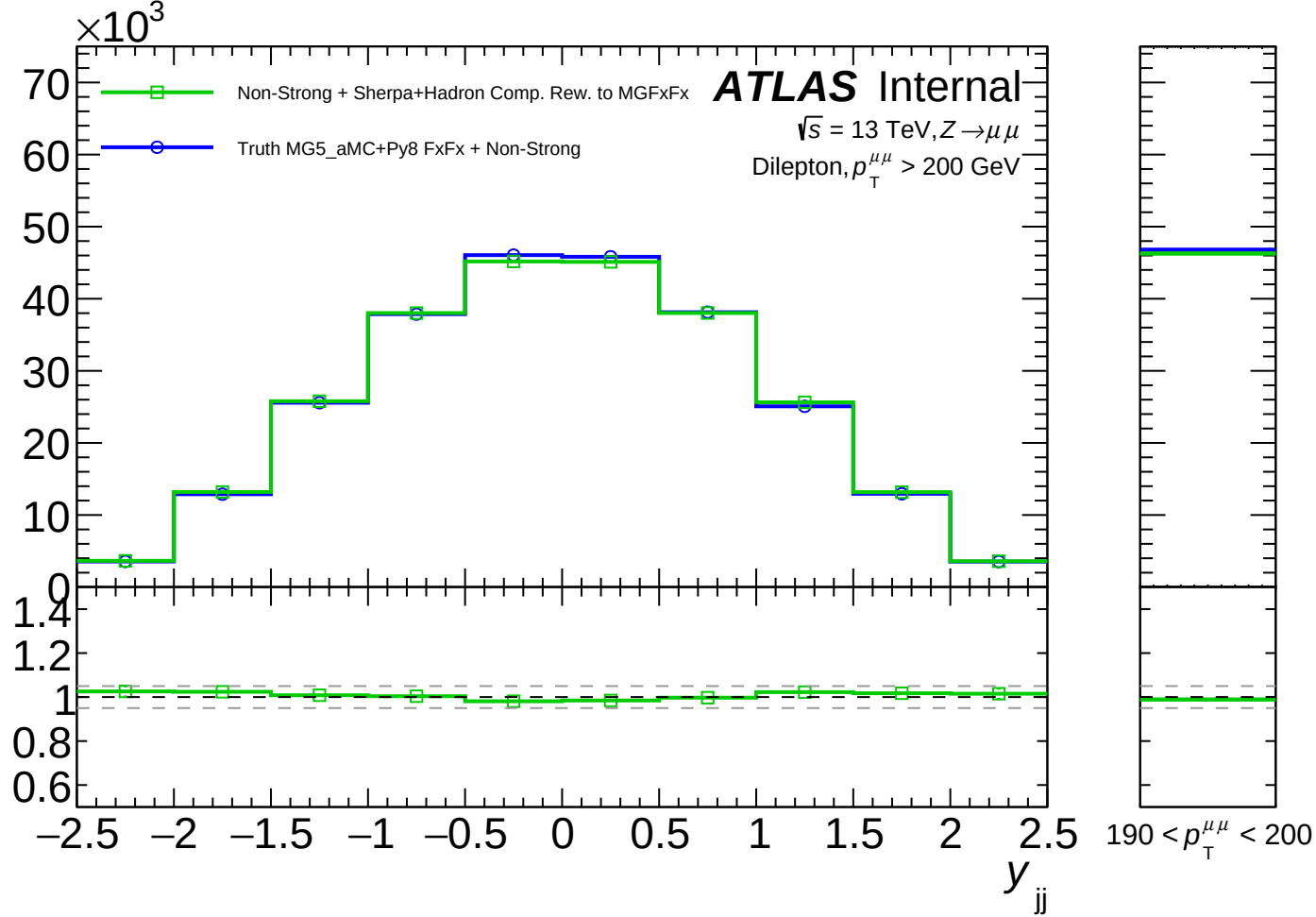
Events

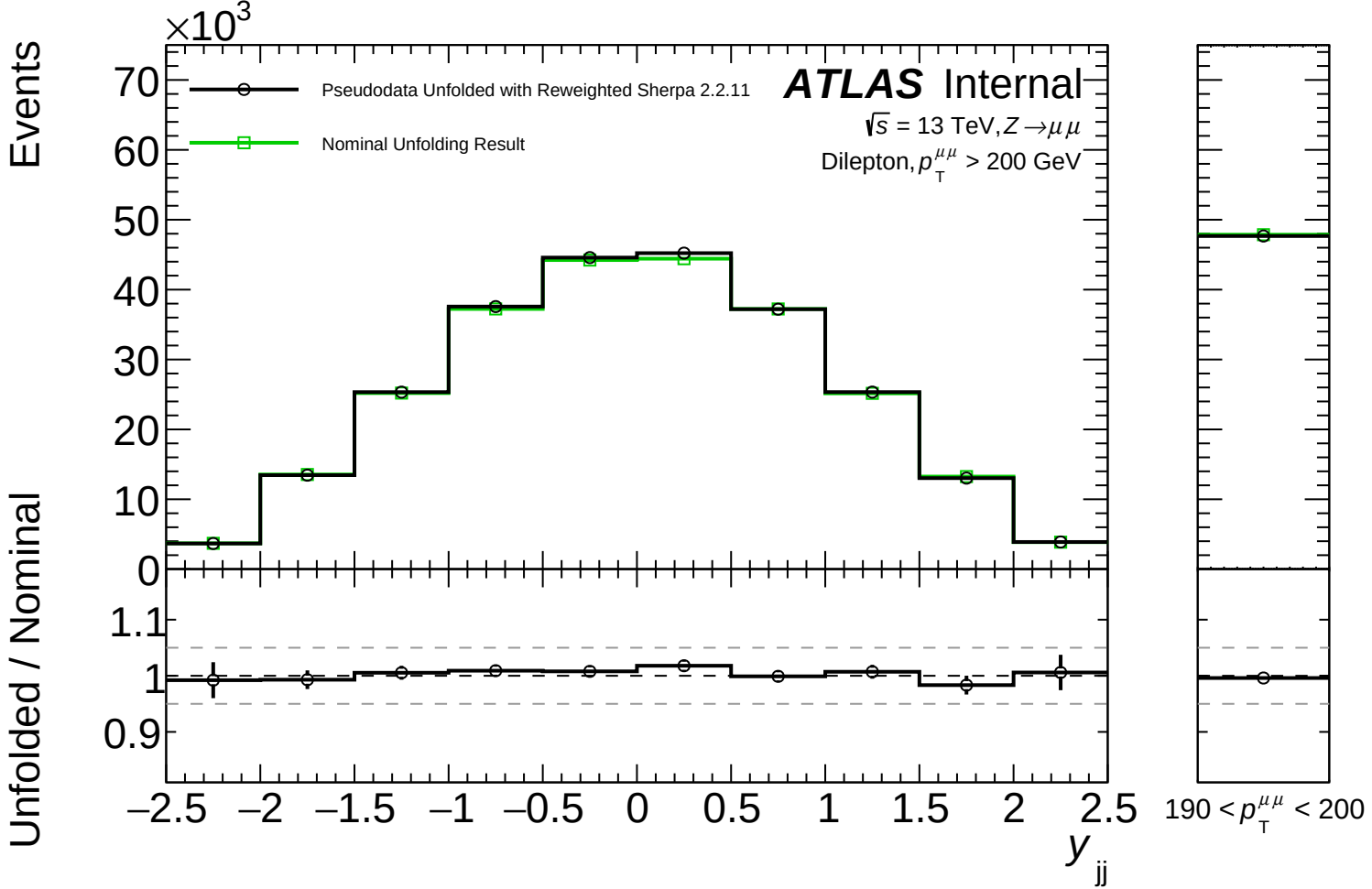
Rew. Truth / Truth

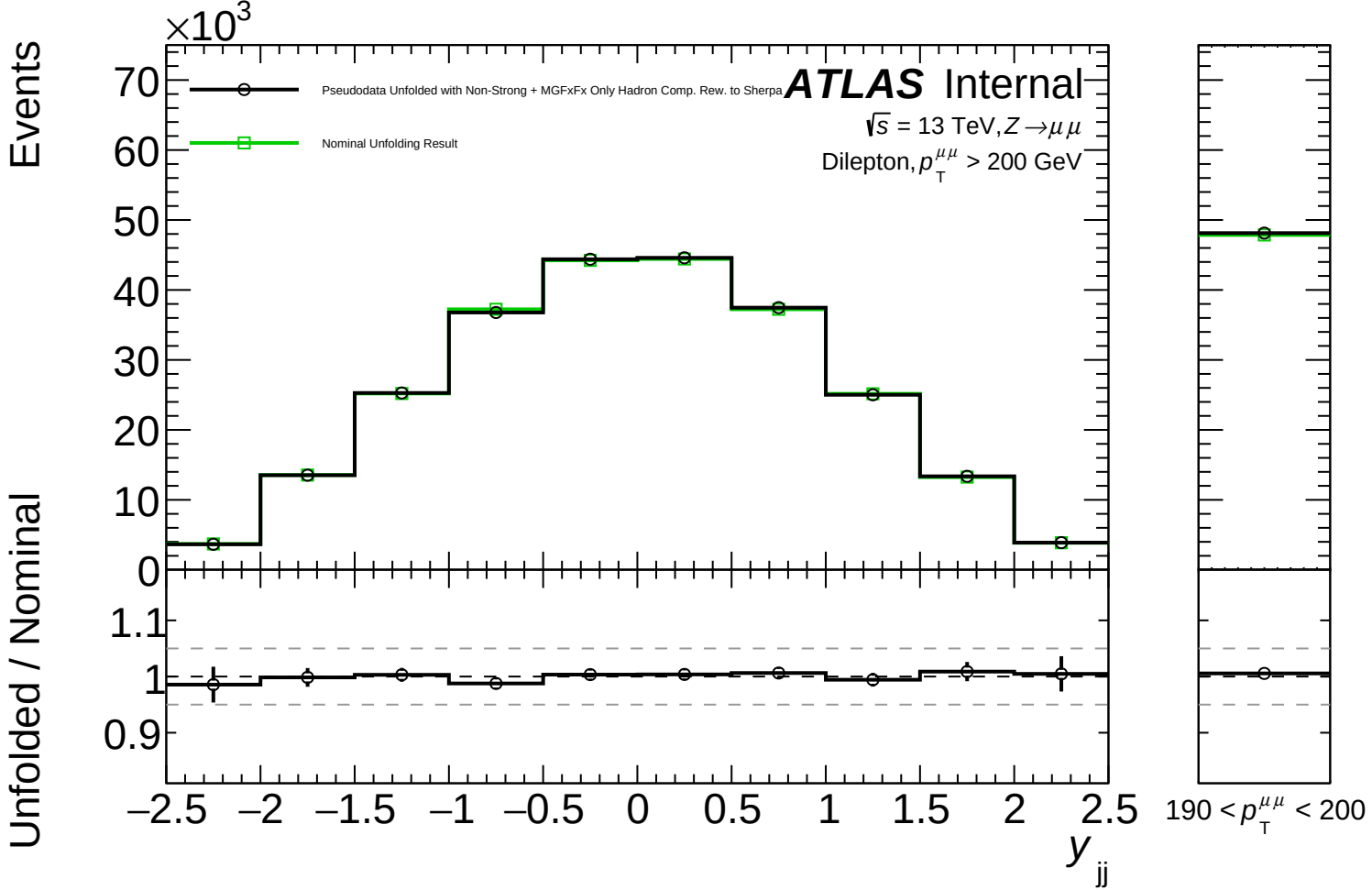


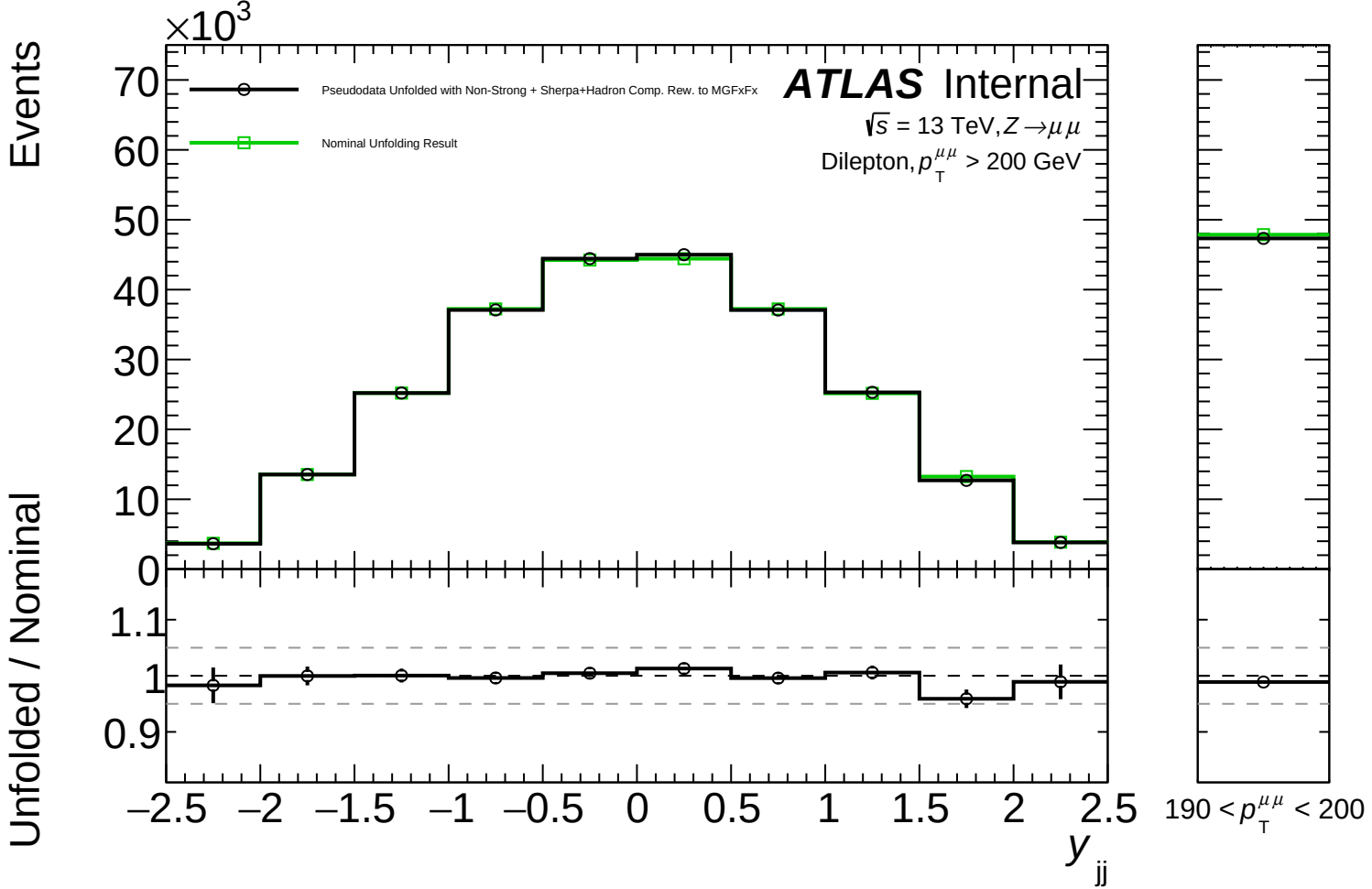
Events

Rew. Truth / Truth



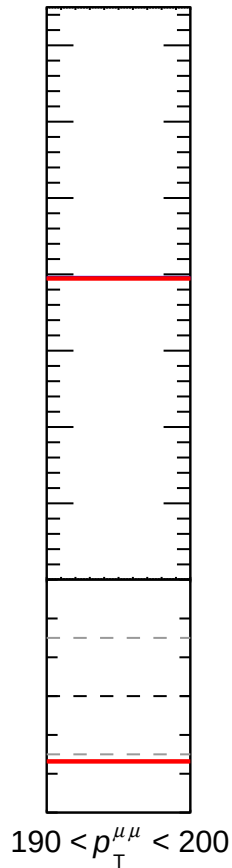
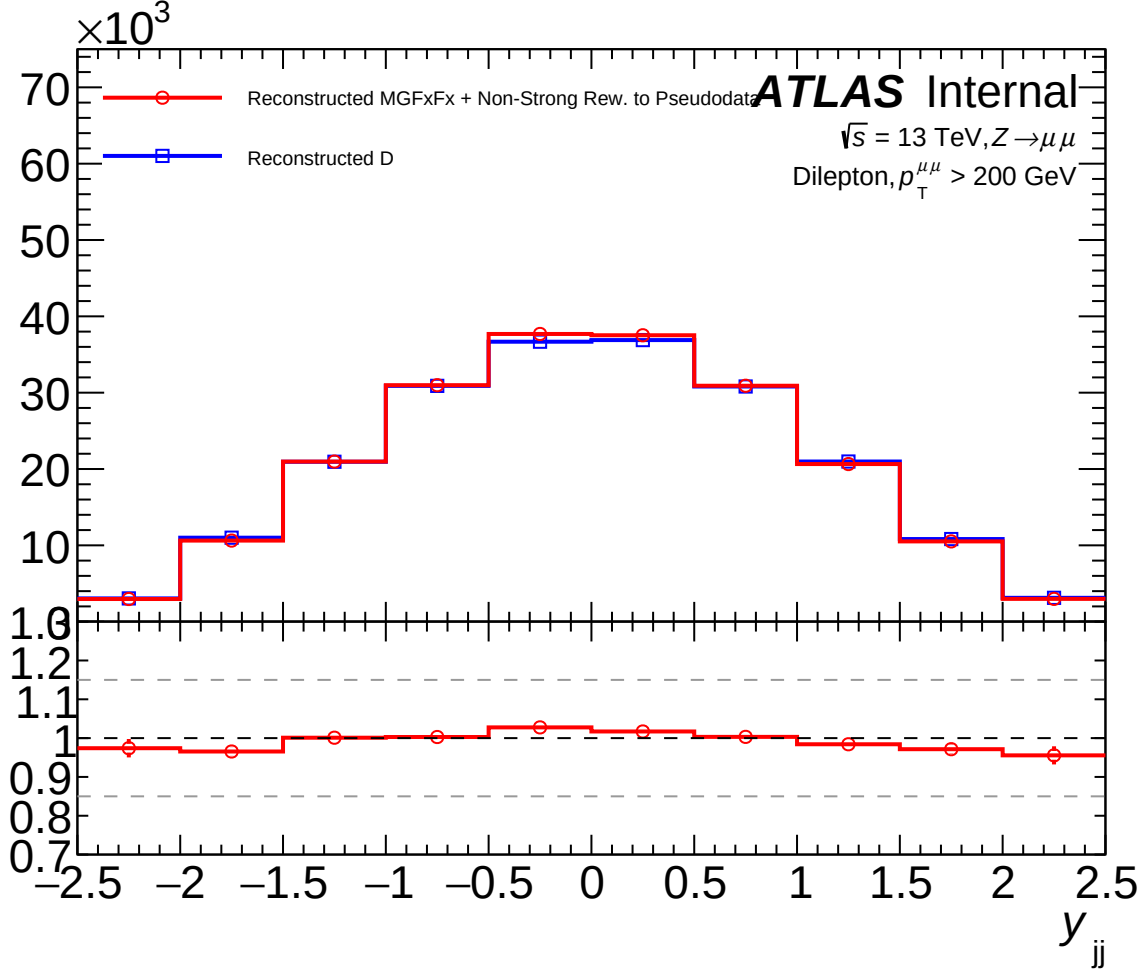


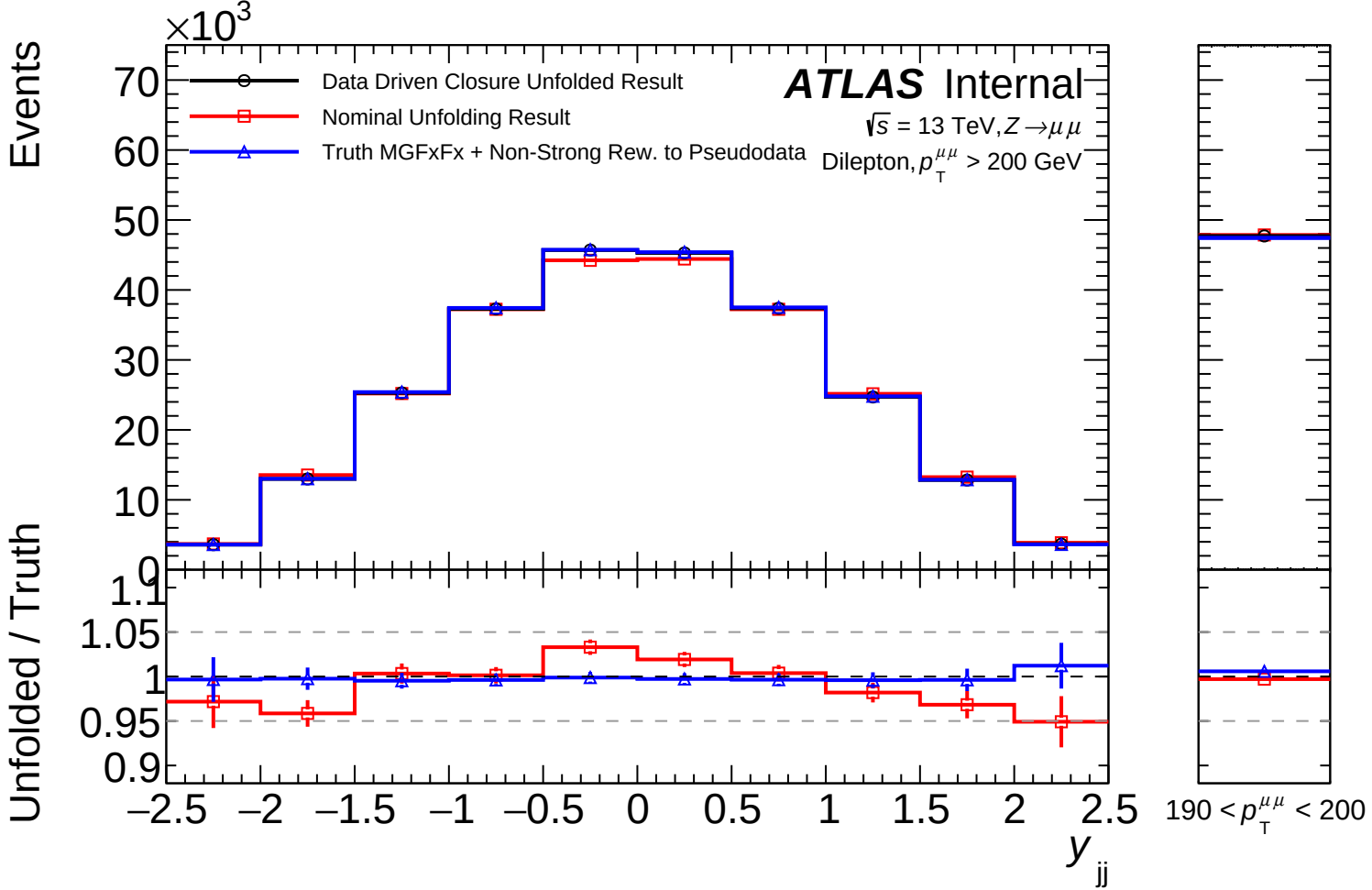




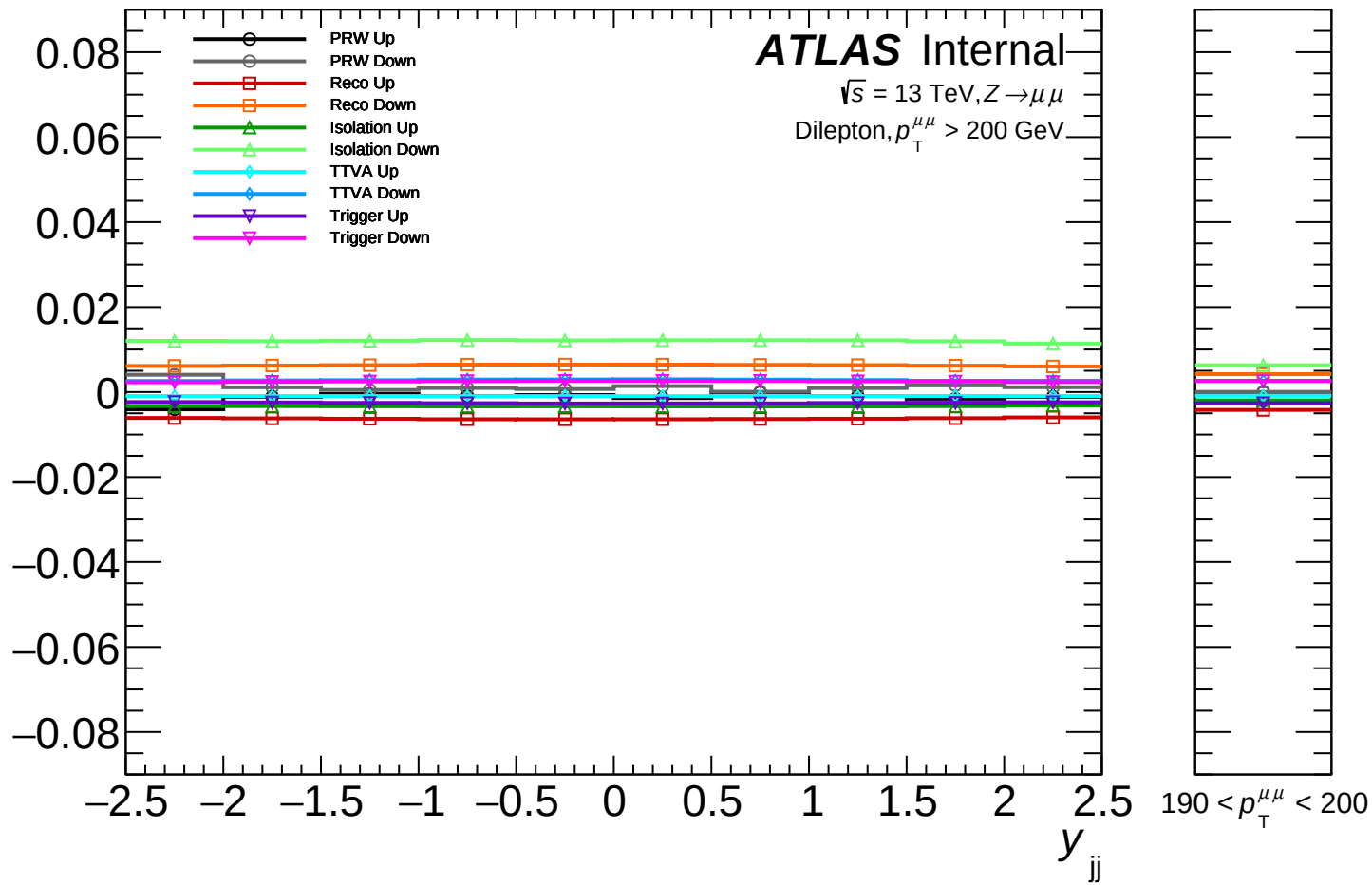
Events

Rew. Reco MC / Data

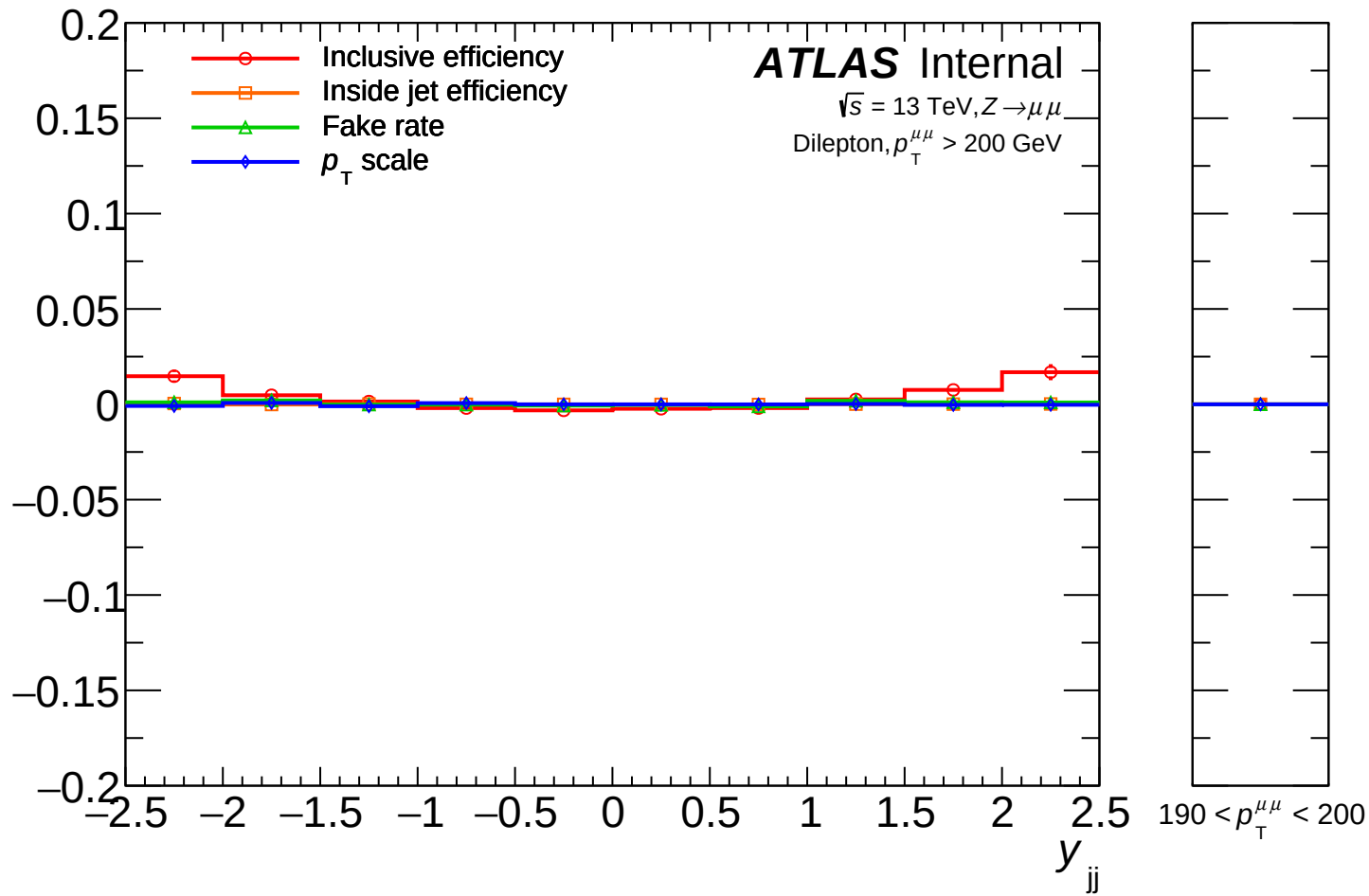




Fractional systematic impact



Fractional systematic impact



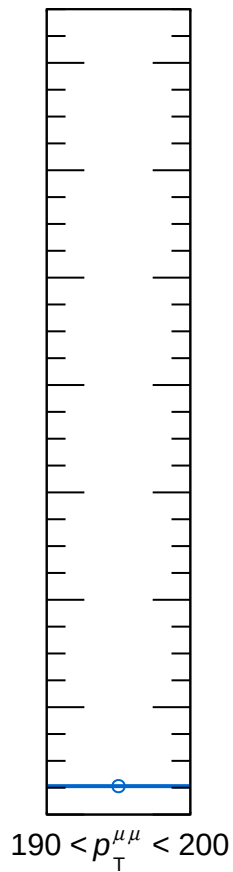
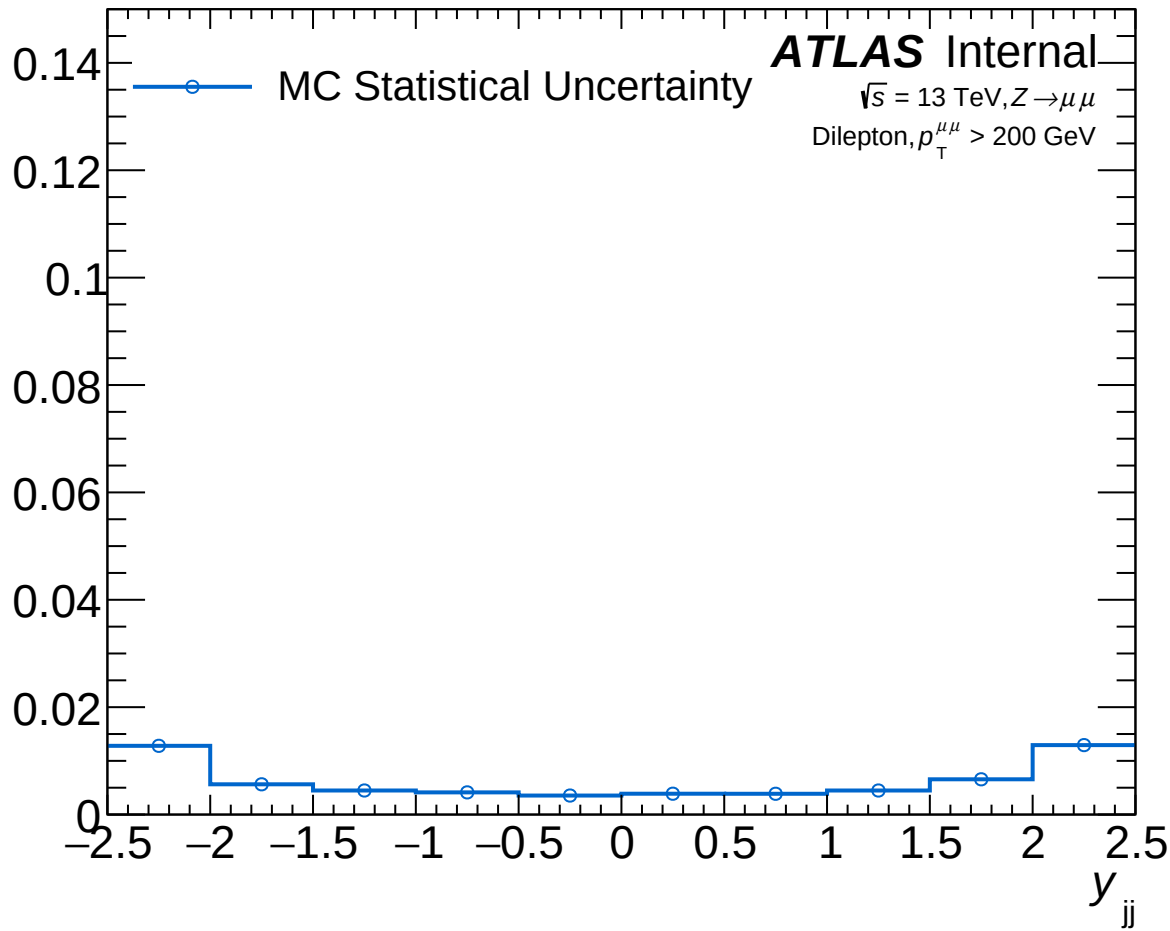
Fractional systematic impact

ATLAS Internal

$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

MC Statistical Uncertainty



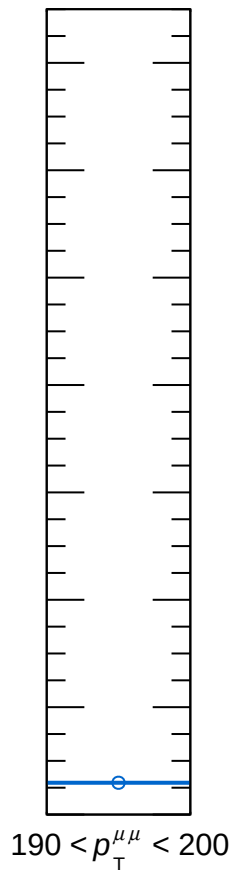
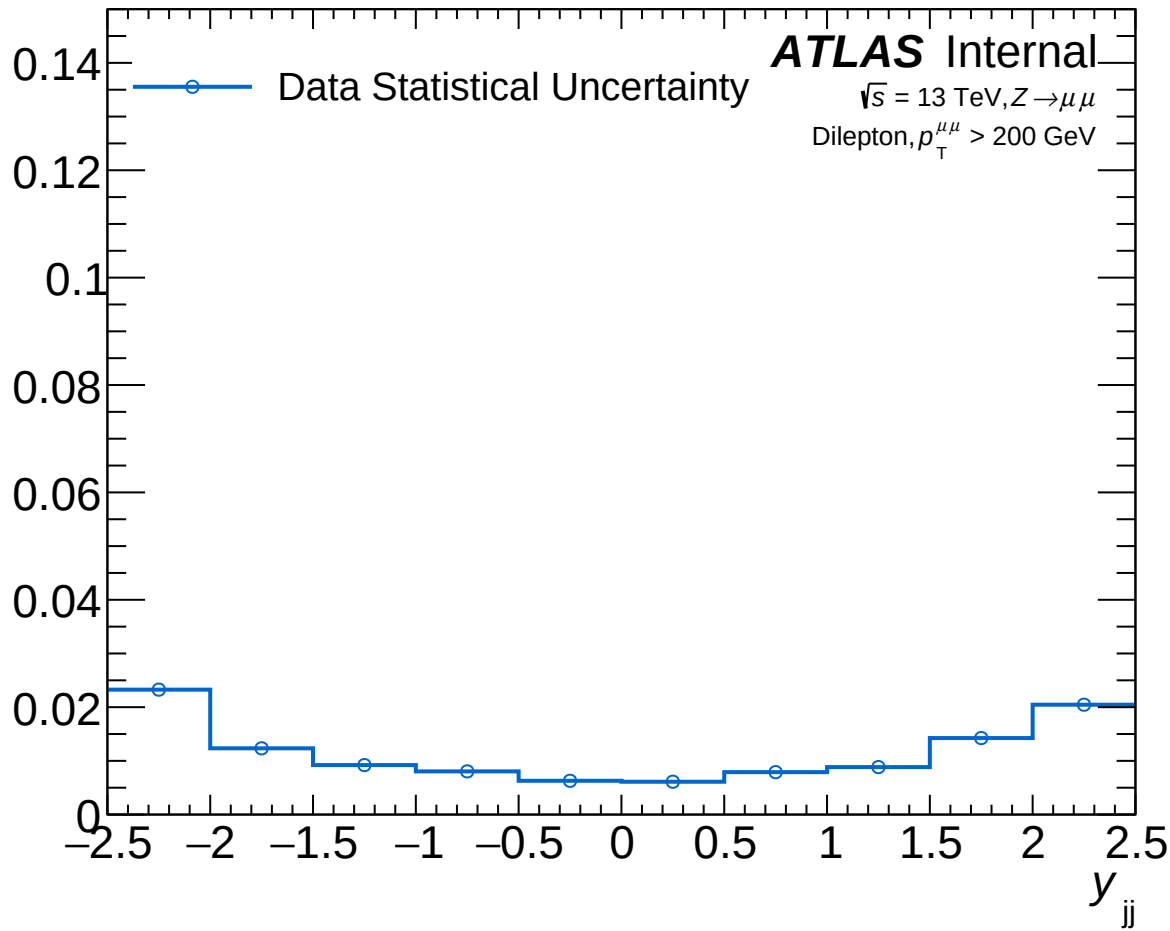
Fractional systematic impact

ATLAS Internal

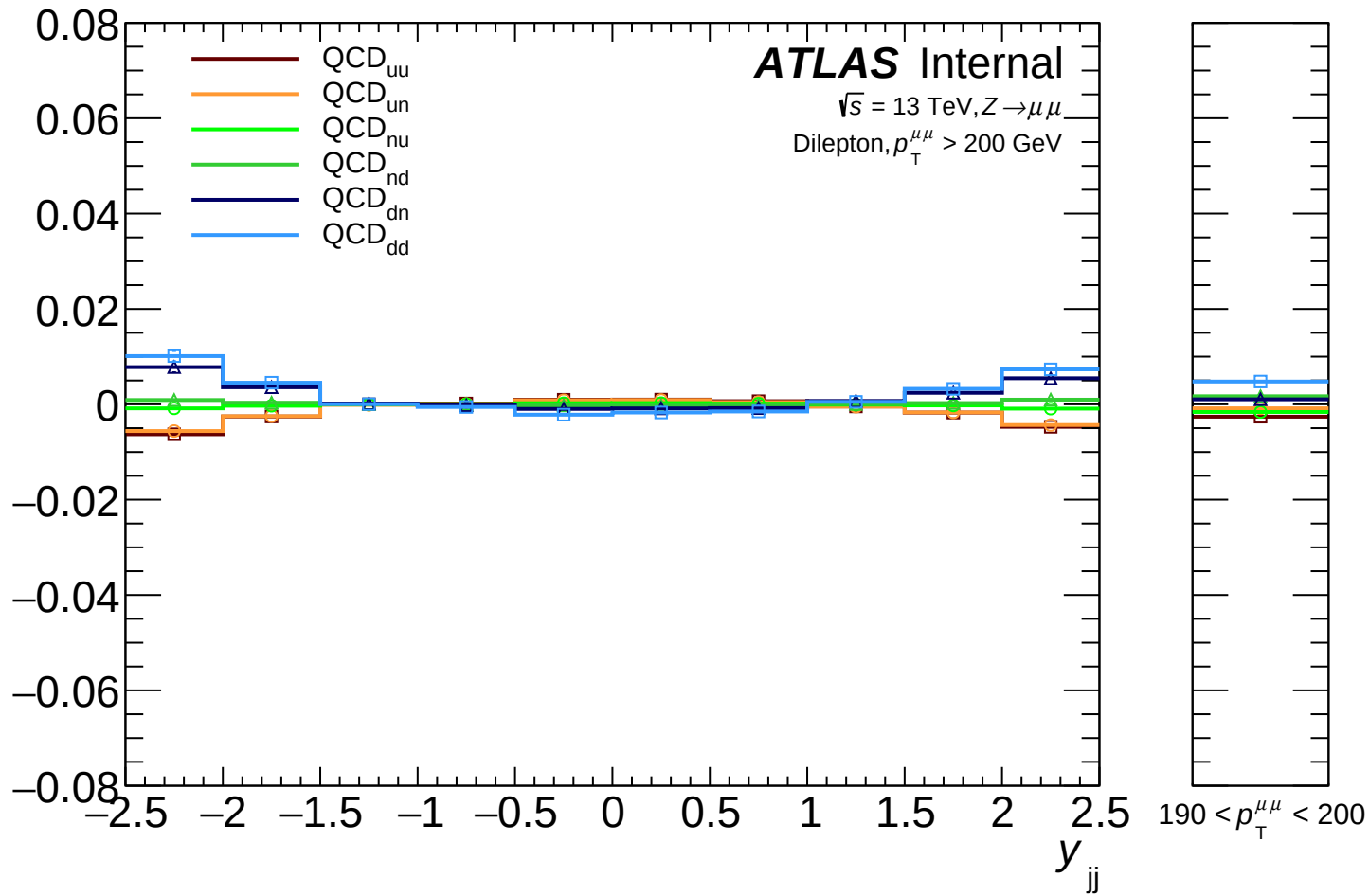
$\sqrt{s} = 13$ TeV, $Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200$ GeV

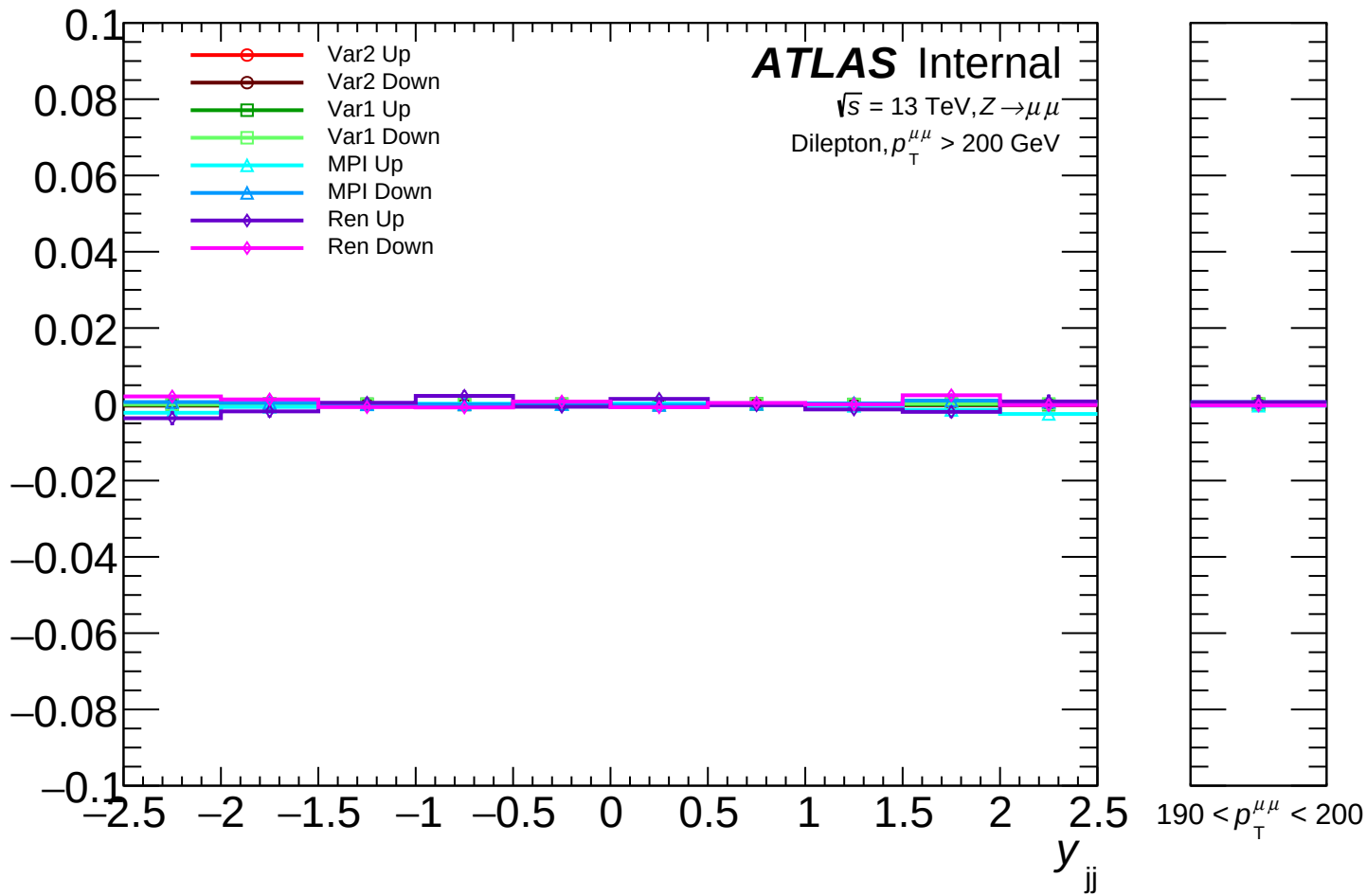
Data Statistical Uncertainty

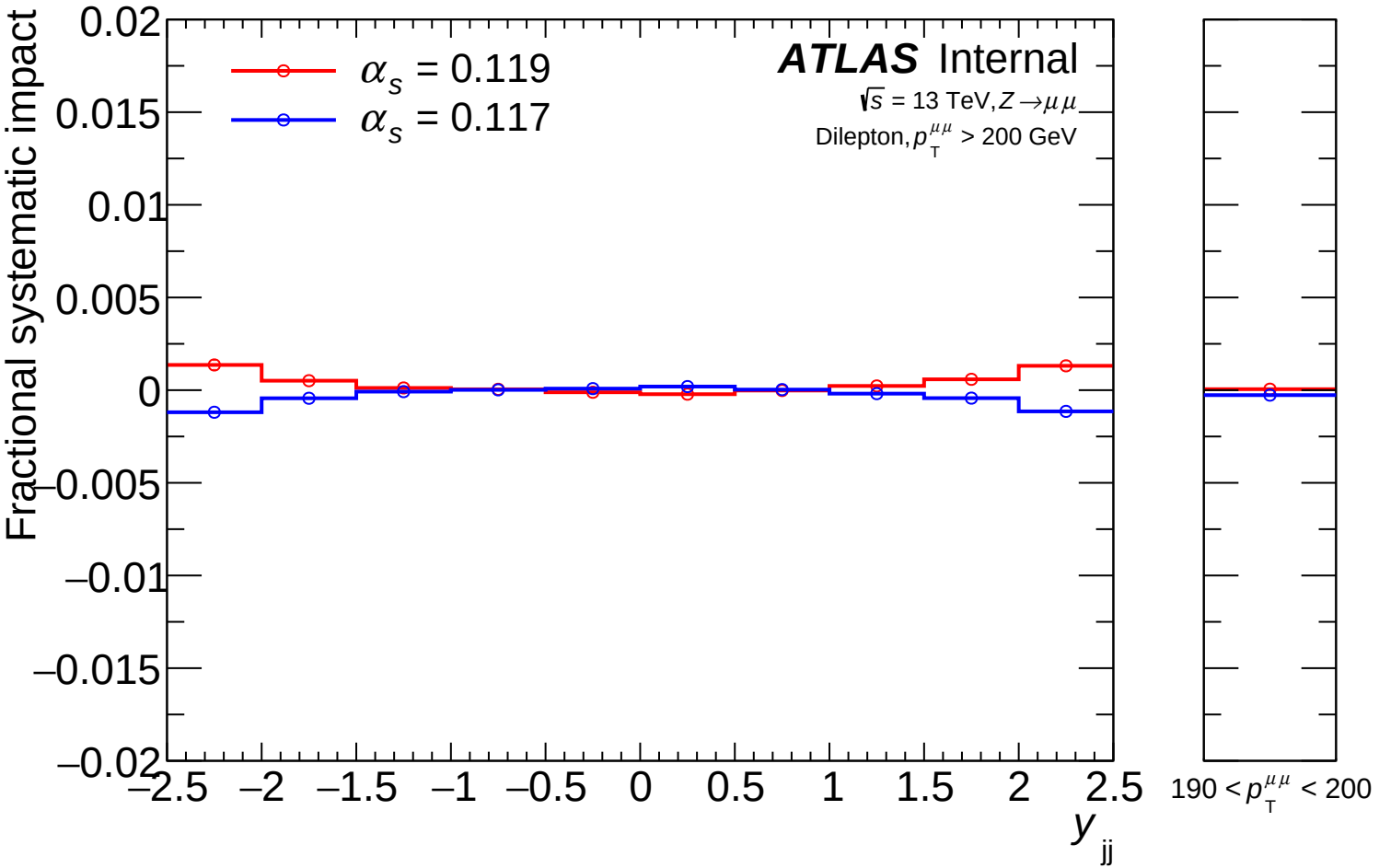


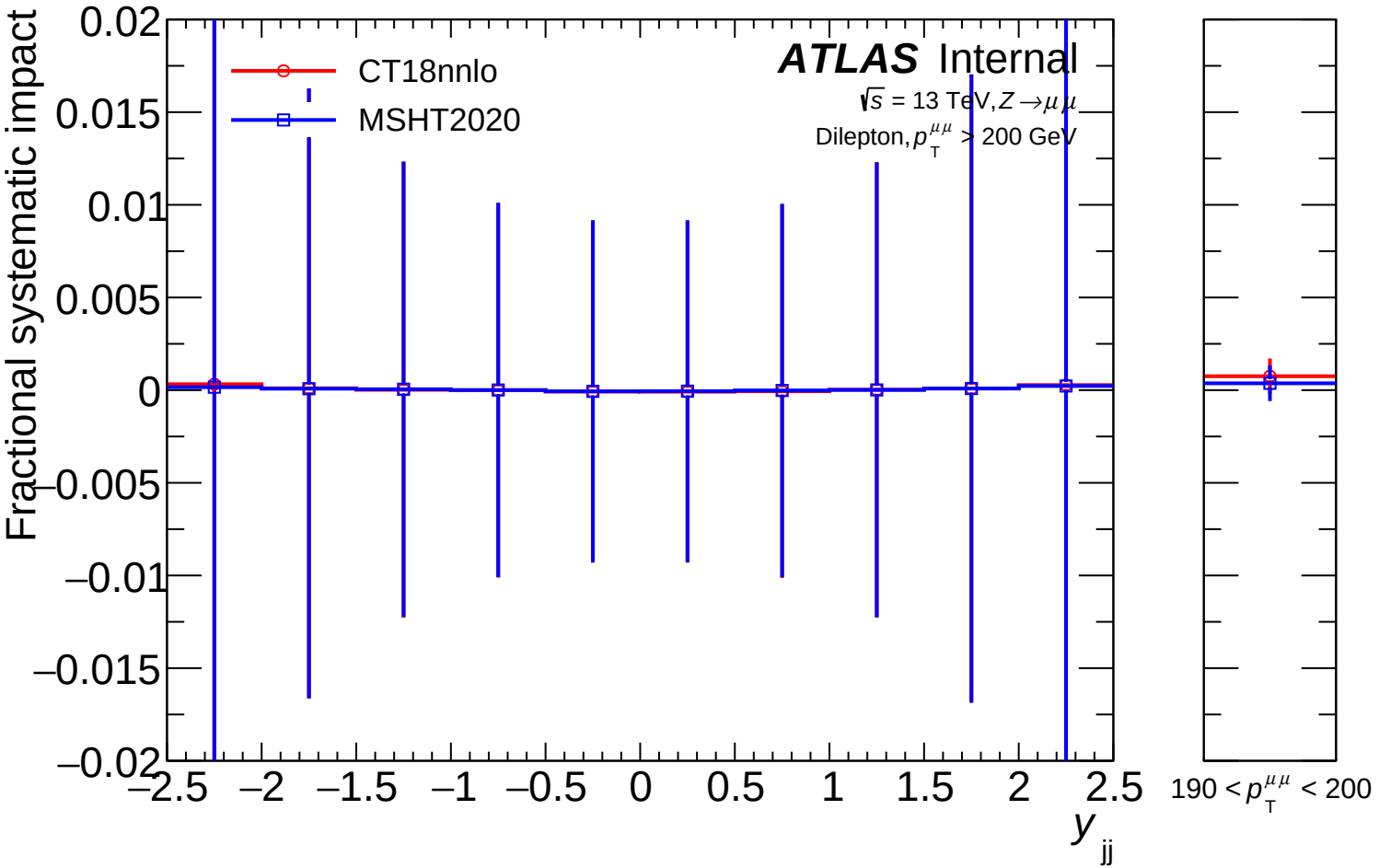
Fractional systematic impact



Fractional systematic impact







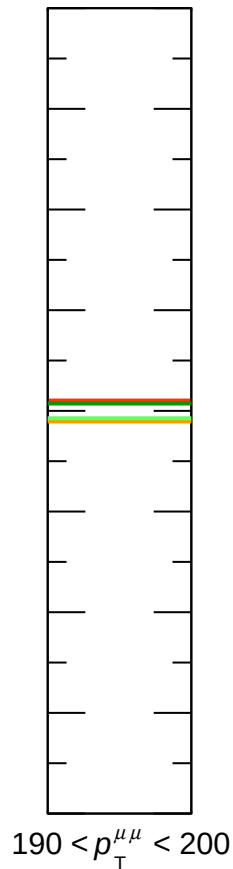
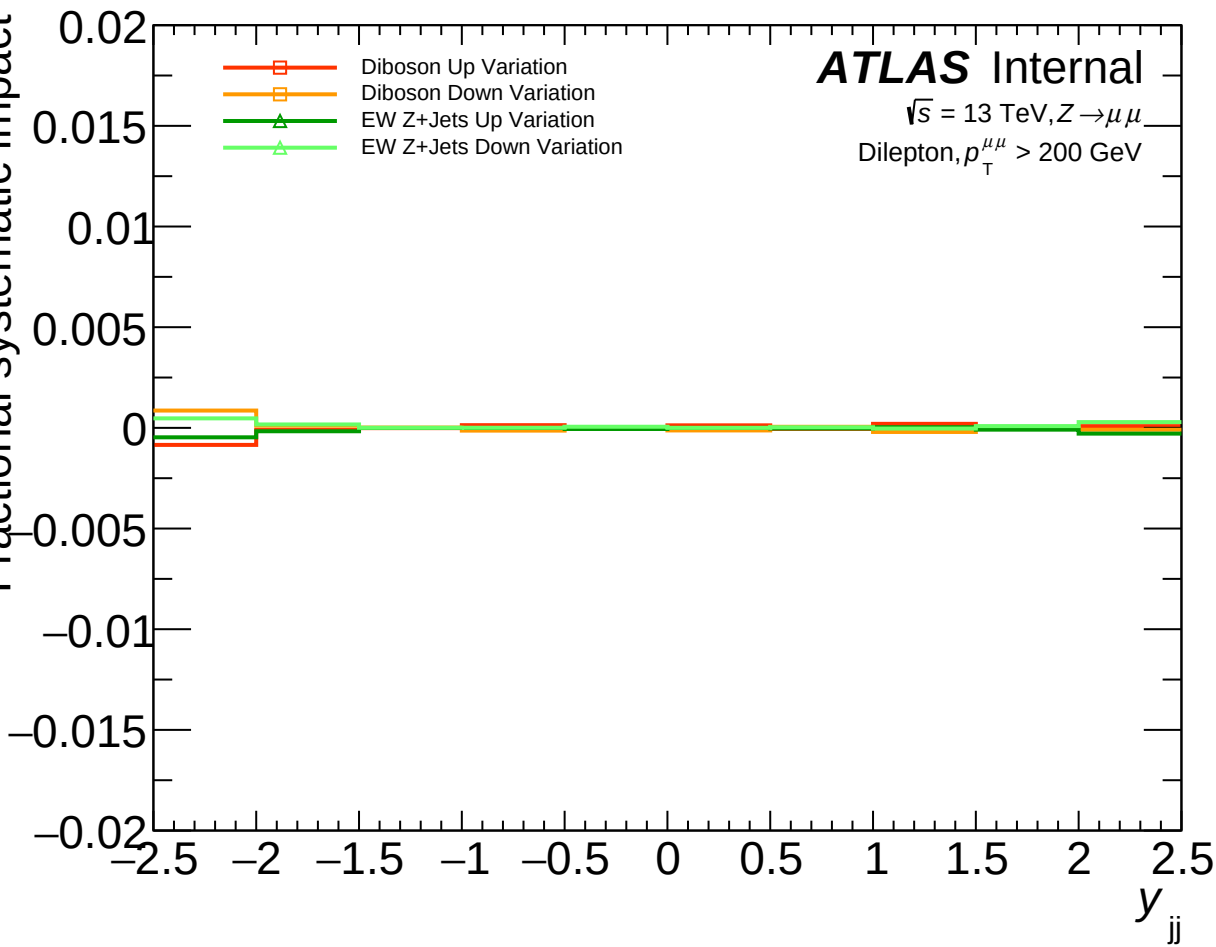
Fractional systematic impact

- Diboson Up Variation
- Diboson Down Variation
- EW Z+Jets Up Variation
- EW Z+Jets Down Variation

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

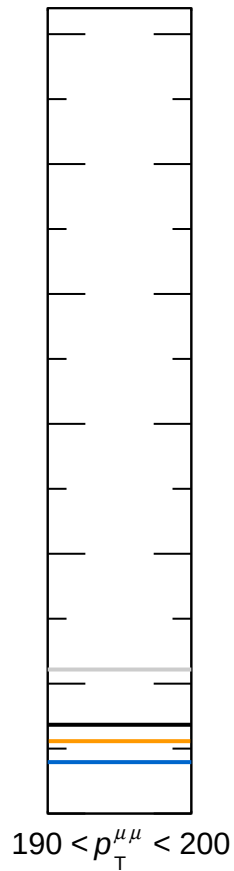
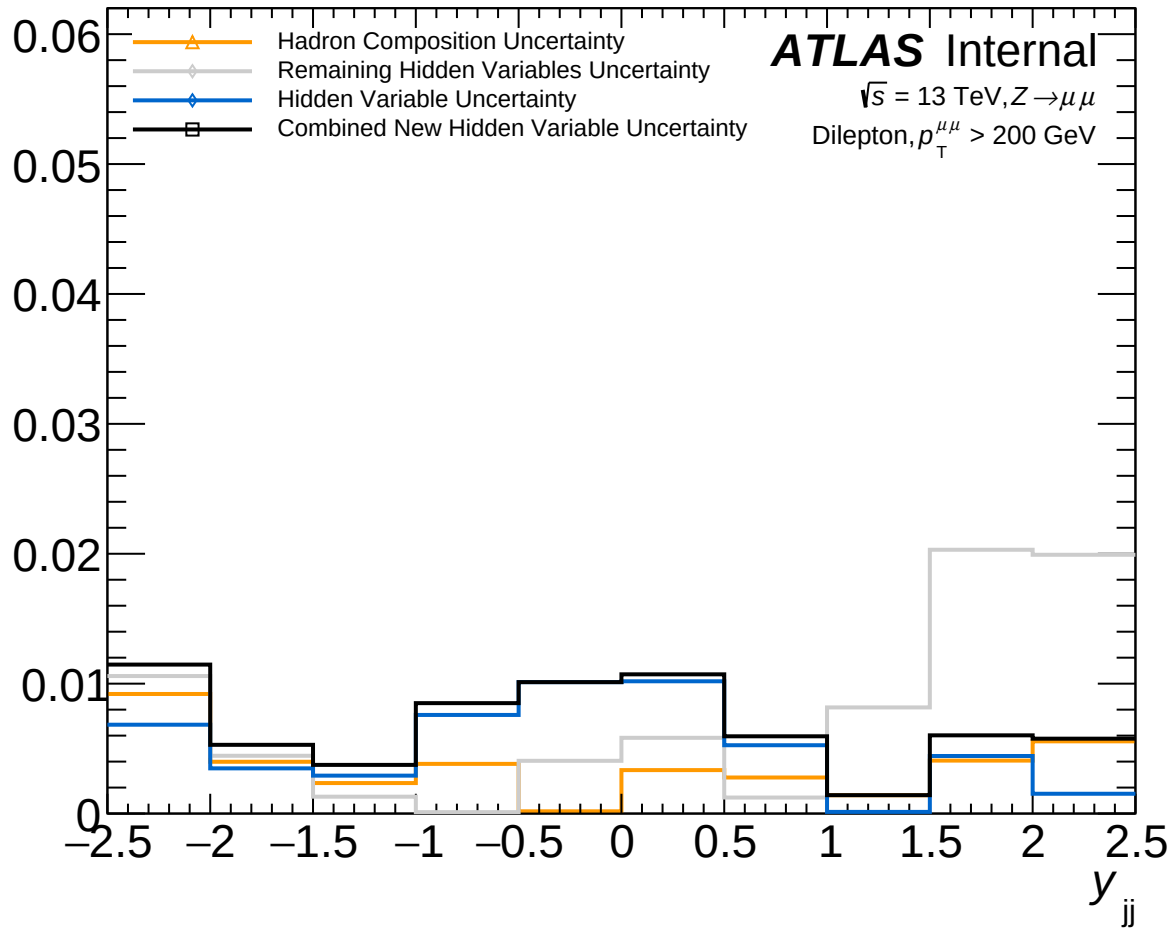
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

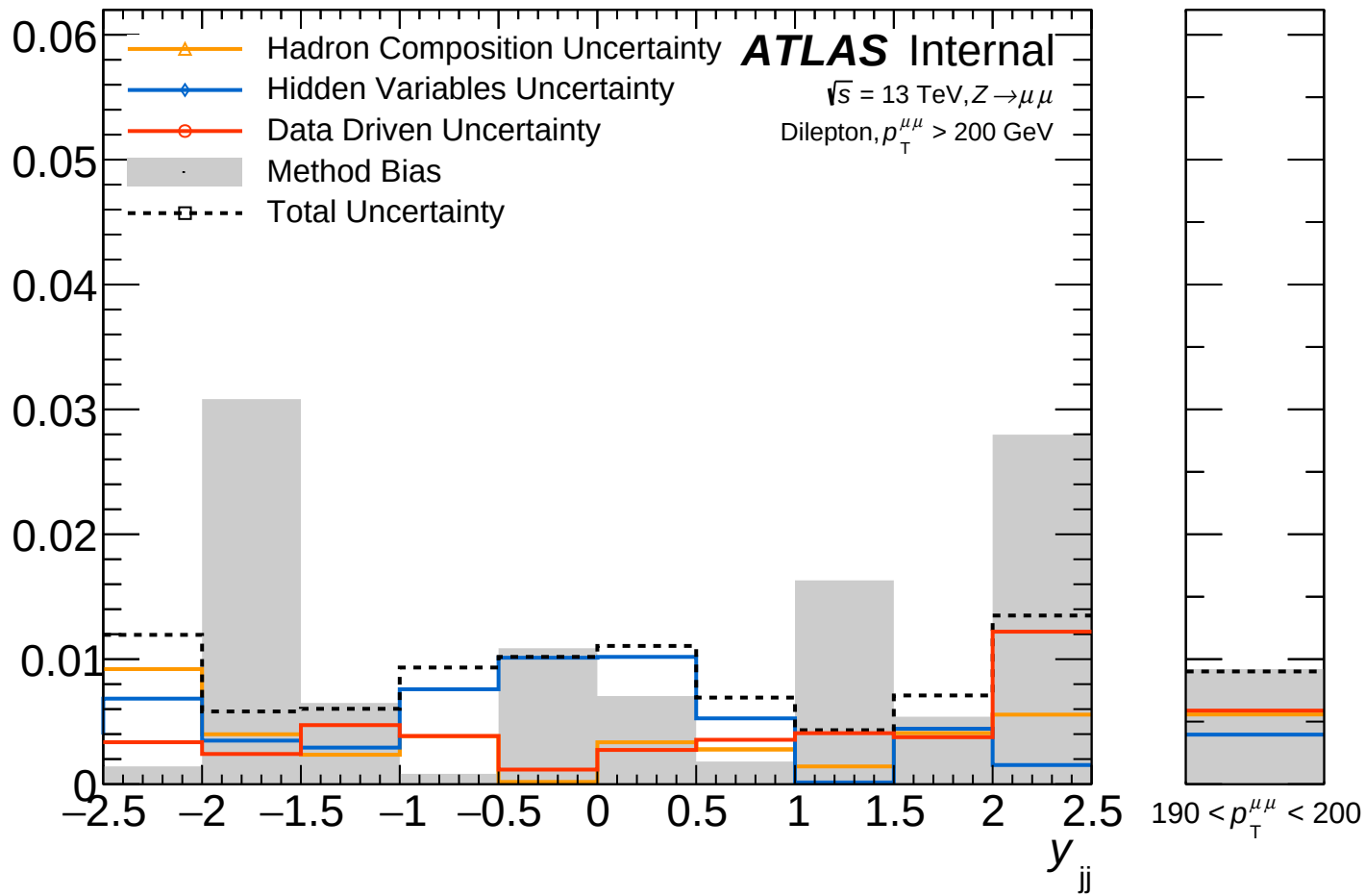


ATLAS Internal

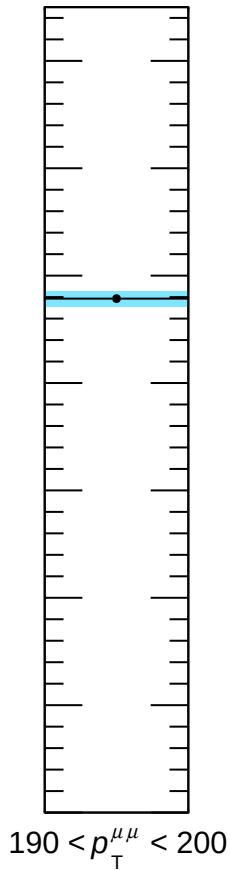
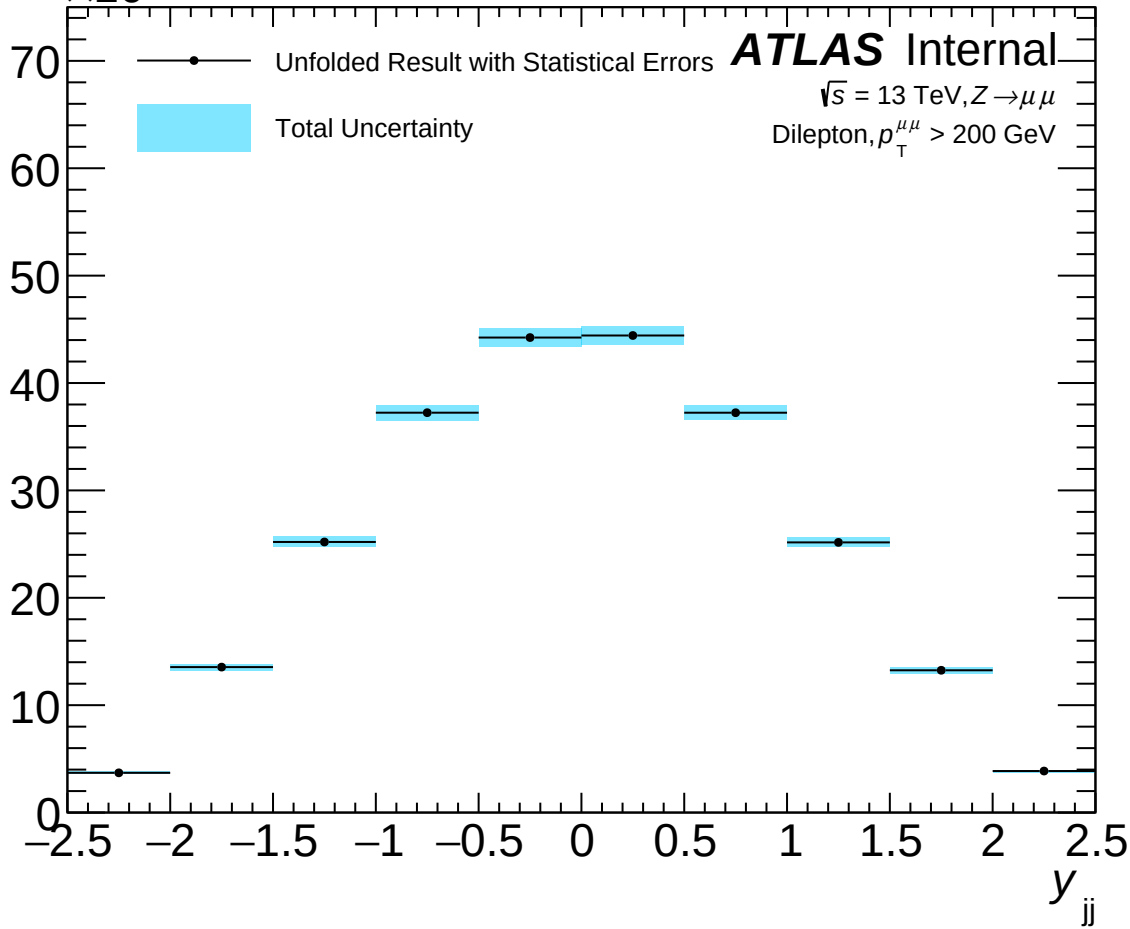
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

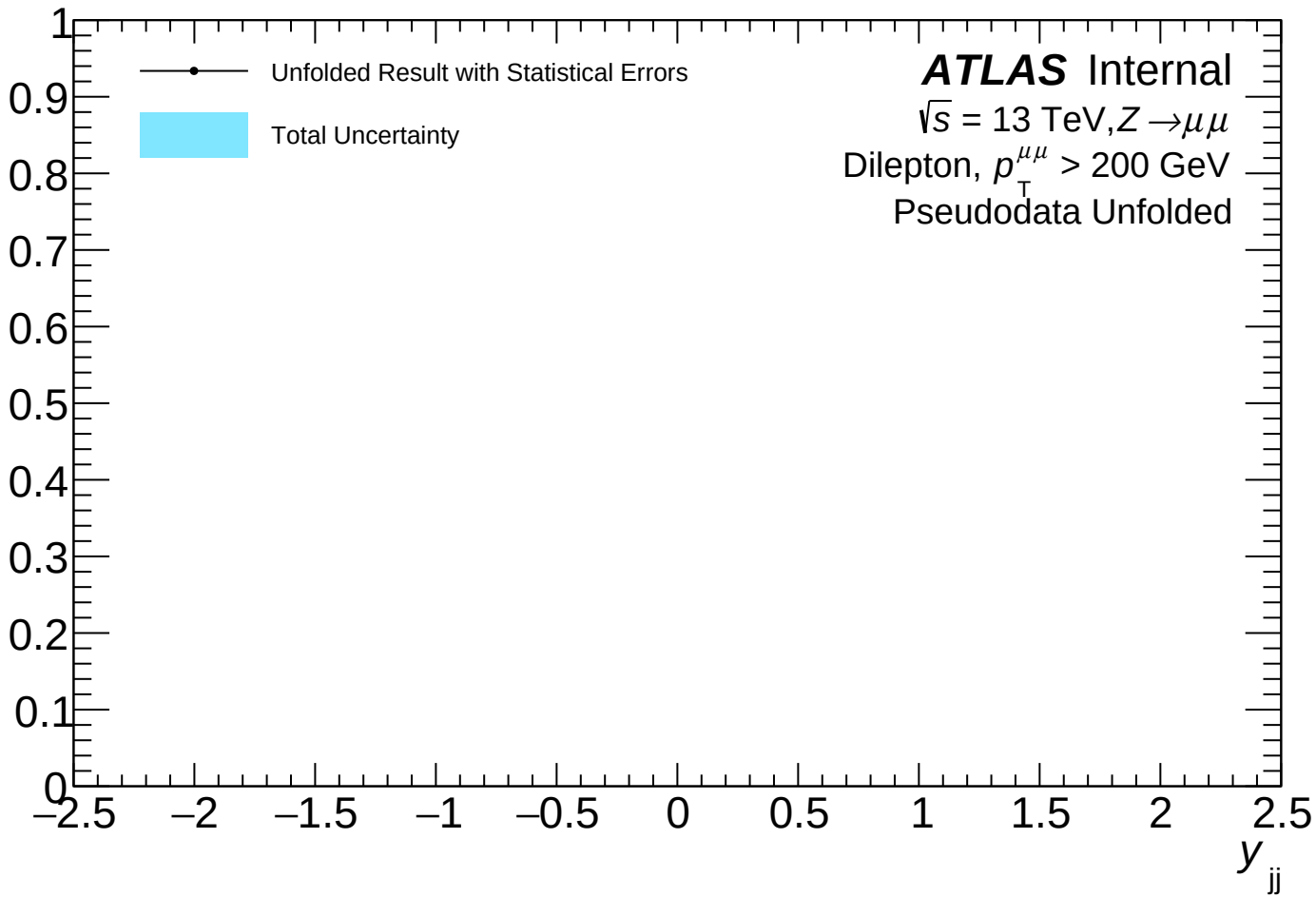
Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

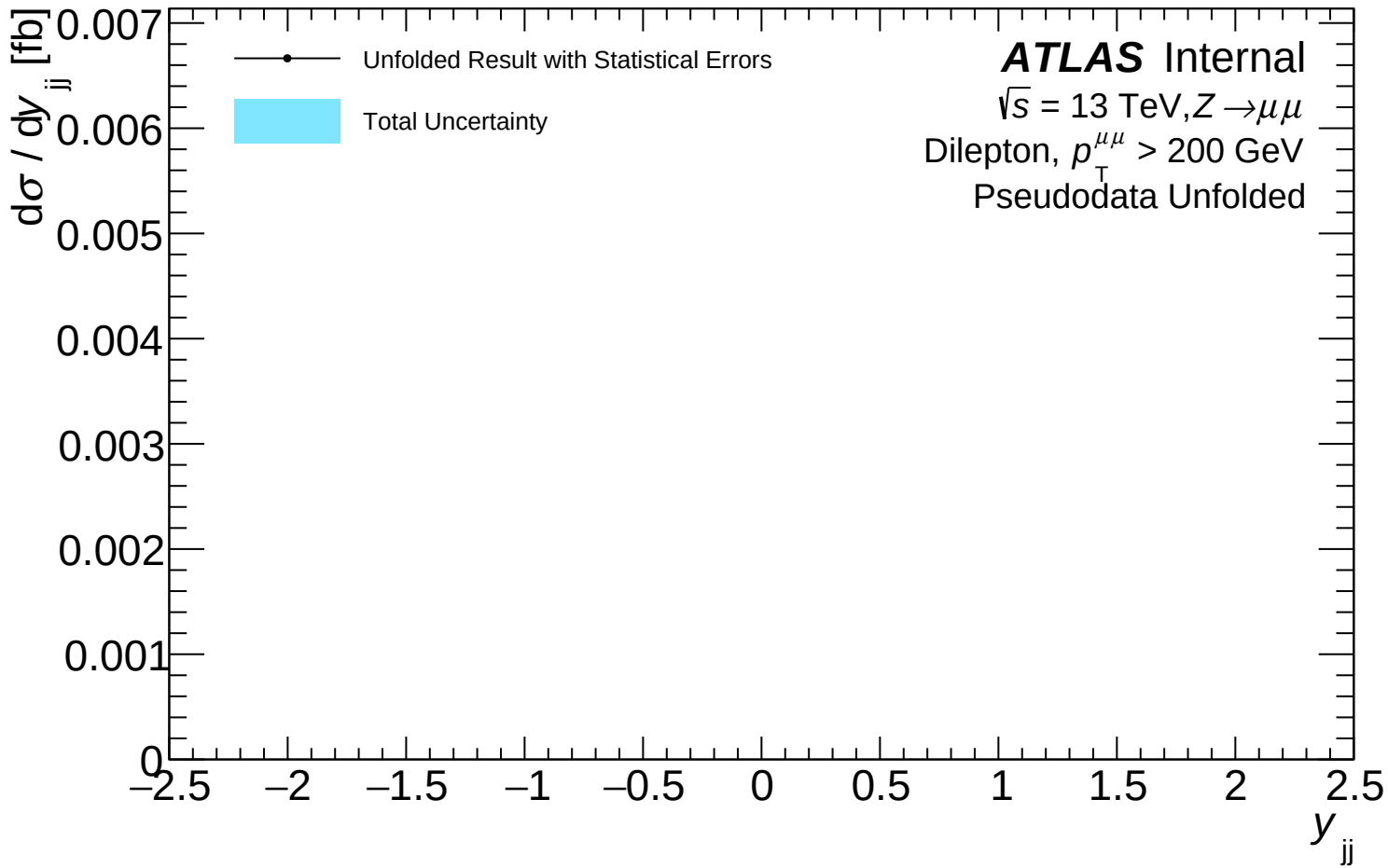




Events

 $\times 10^3$ 





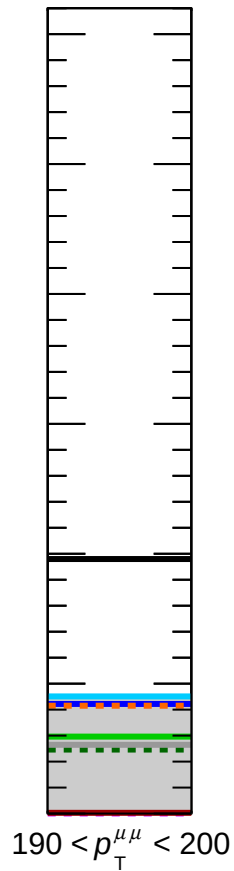
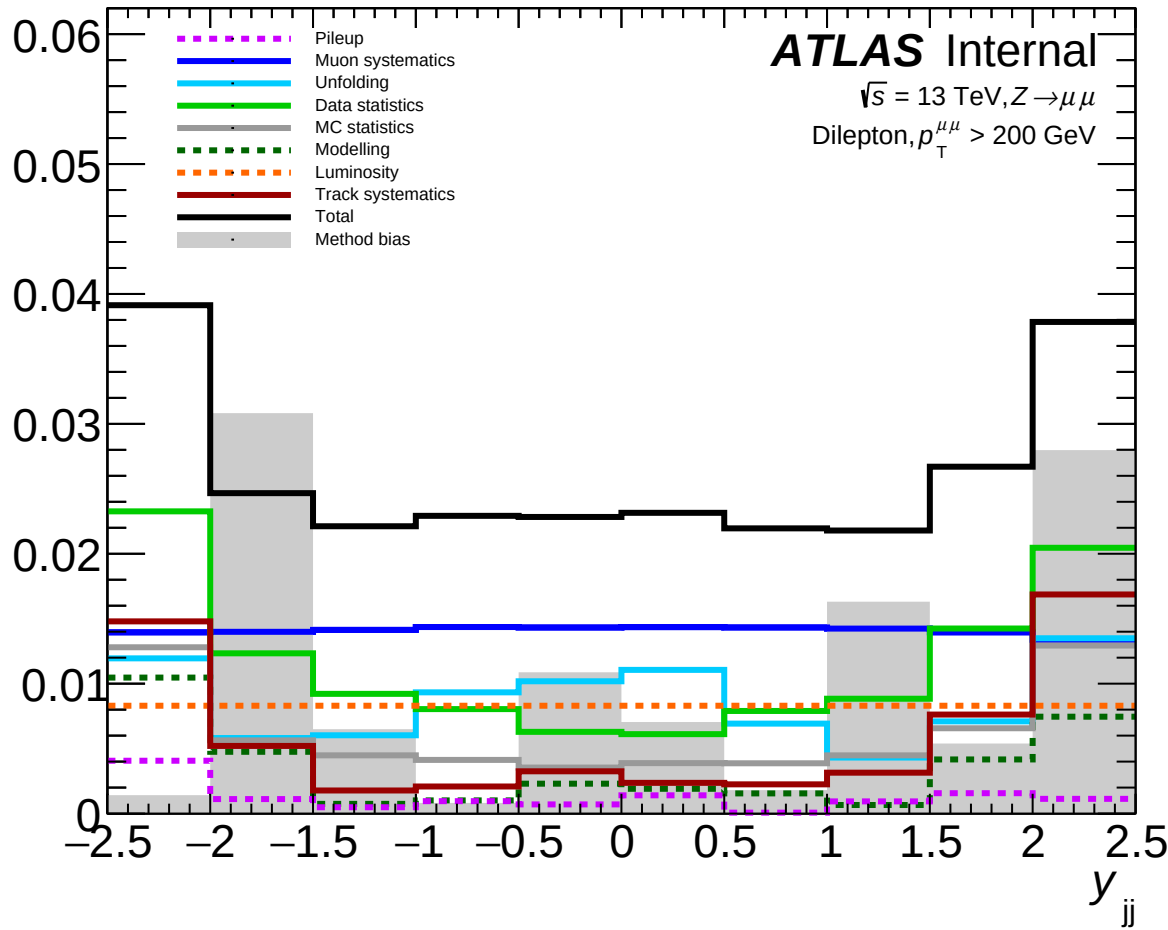
Fractional uncertainty

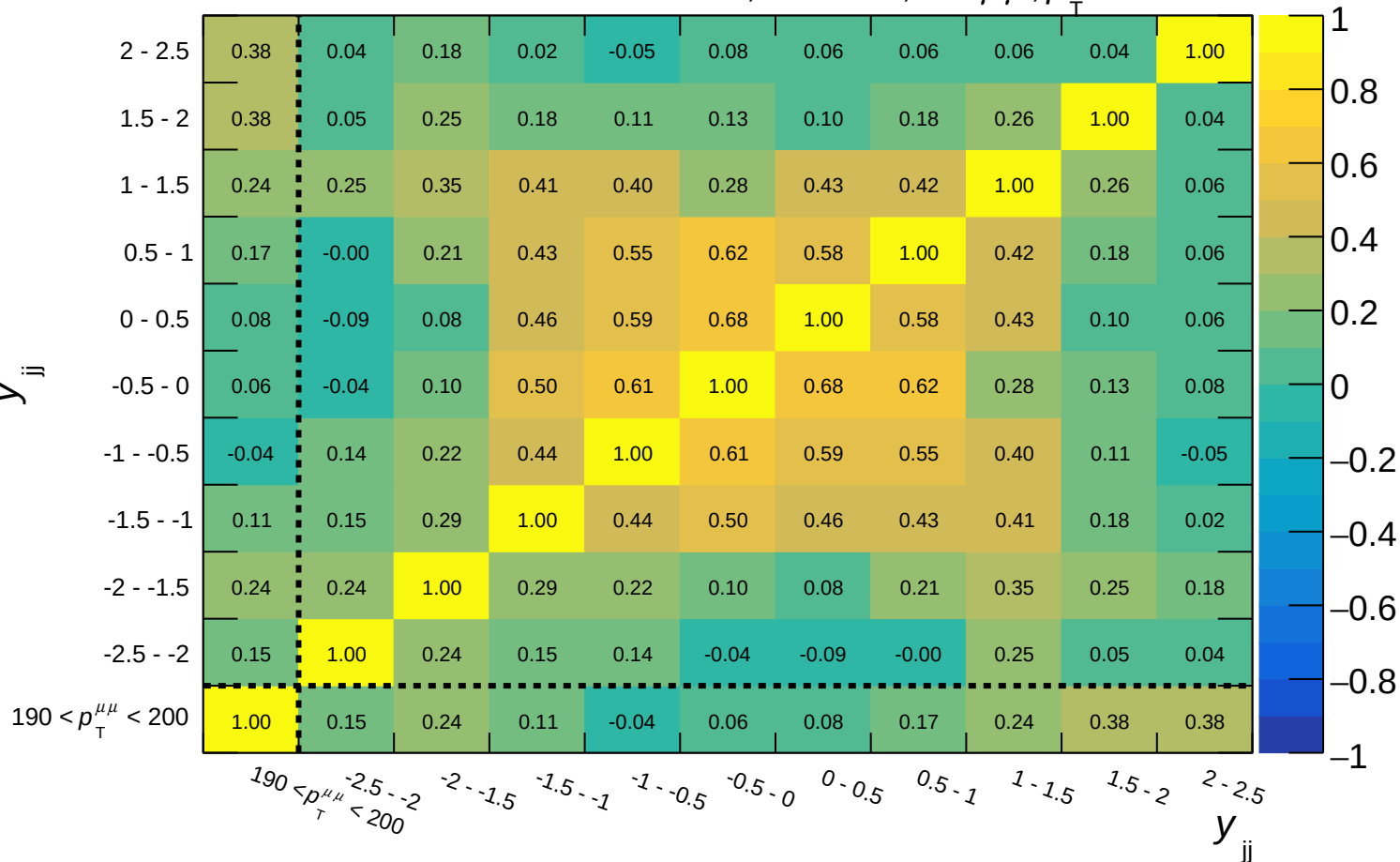
ATLAS Internal

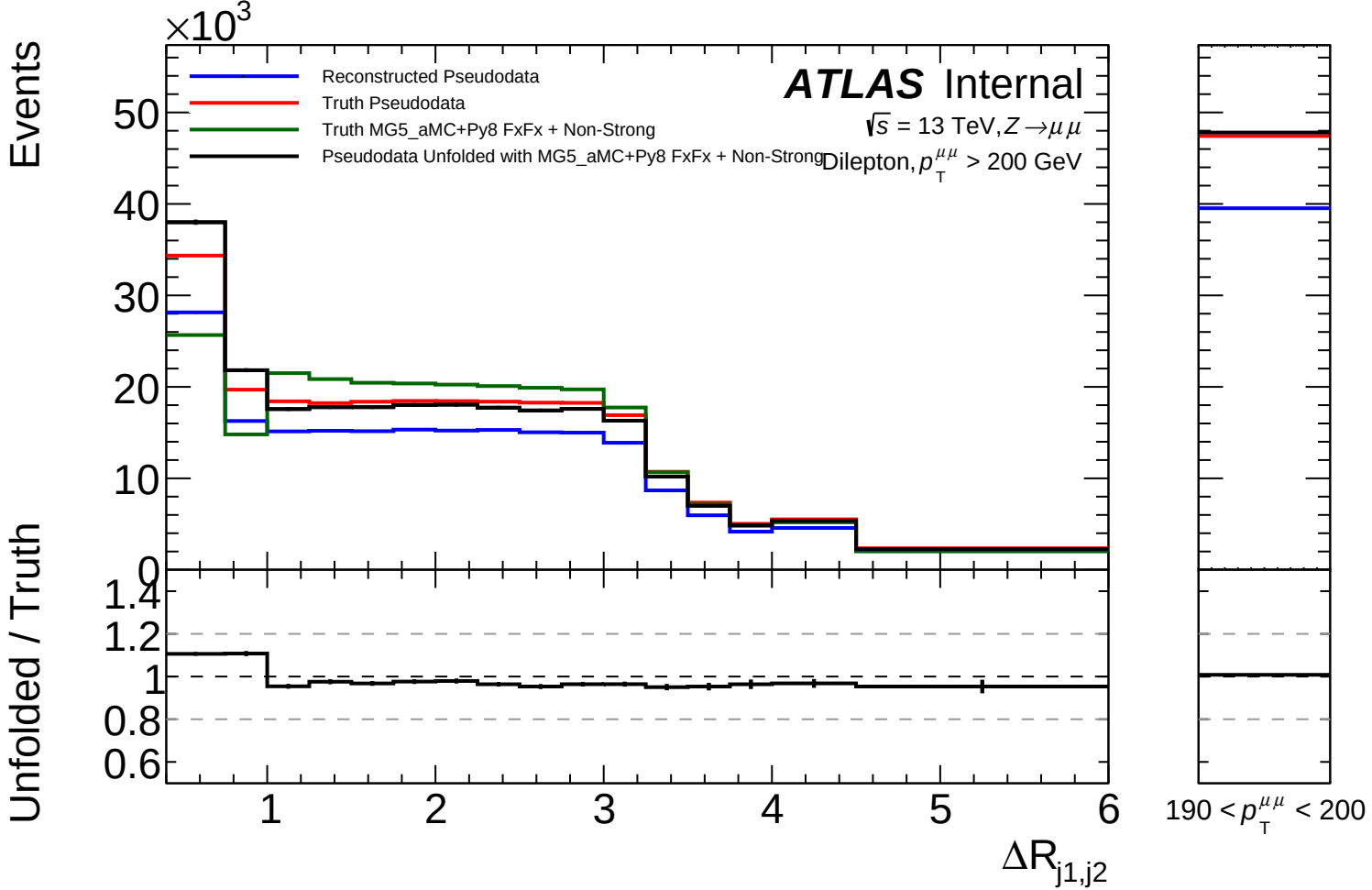
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

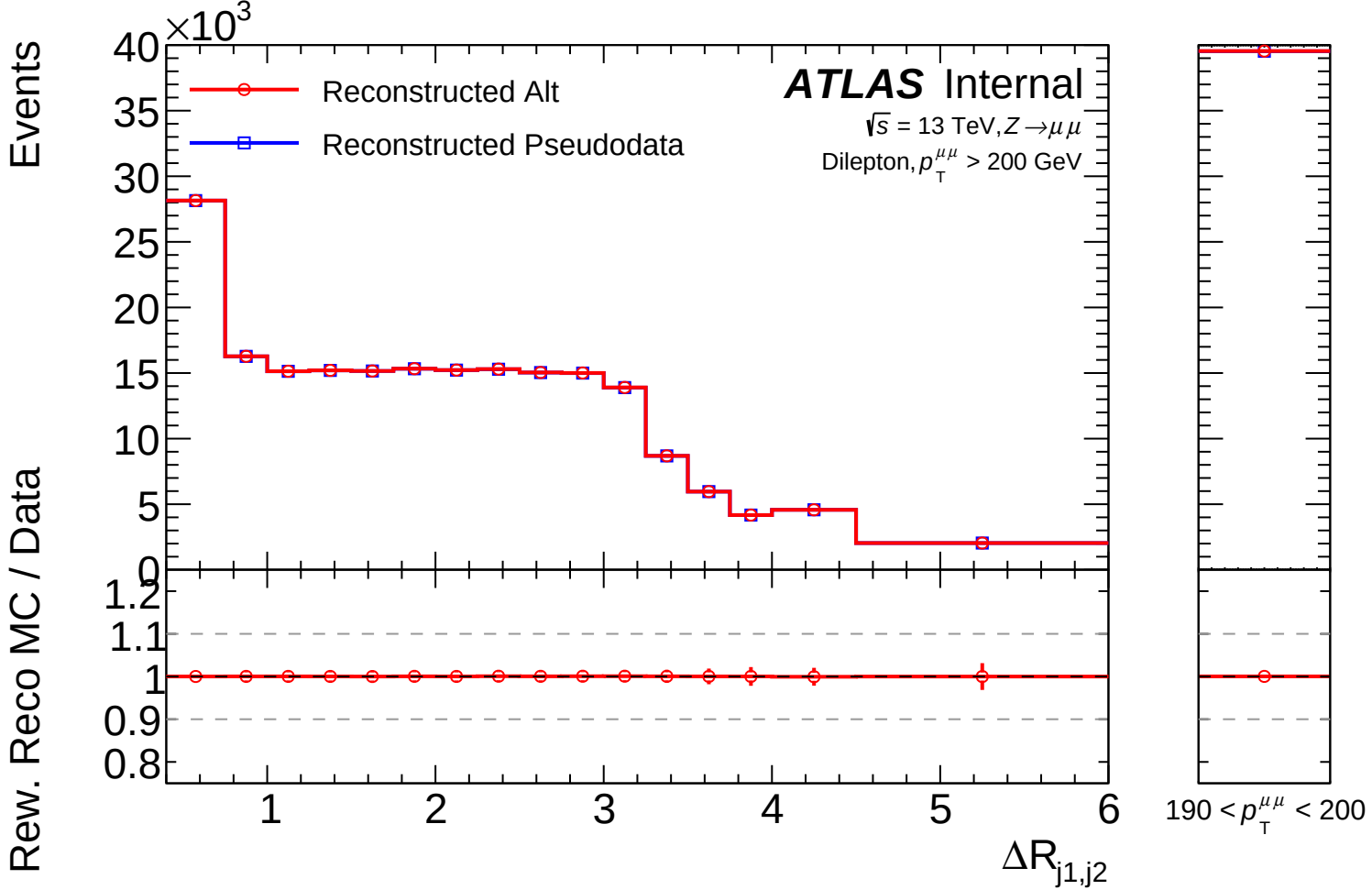
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

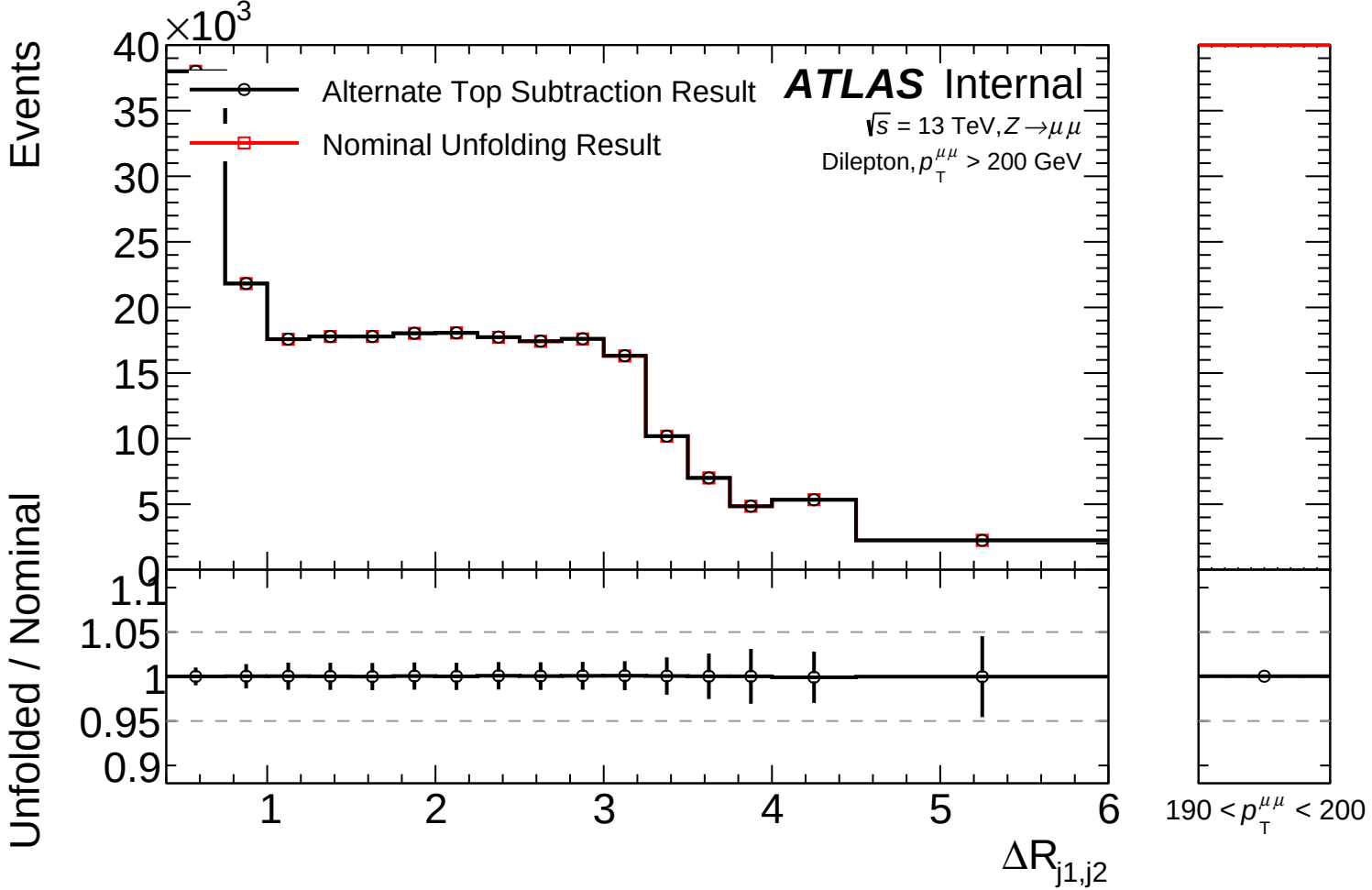
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ y_{ij} 







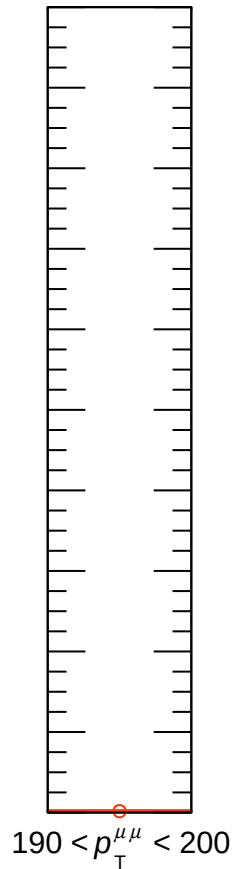
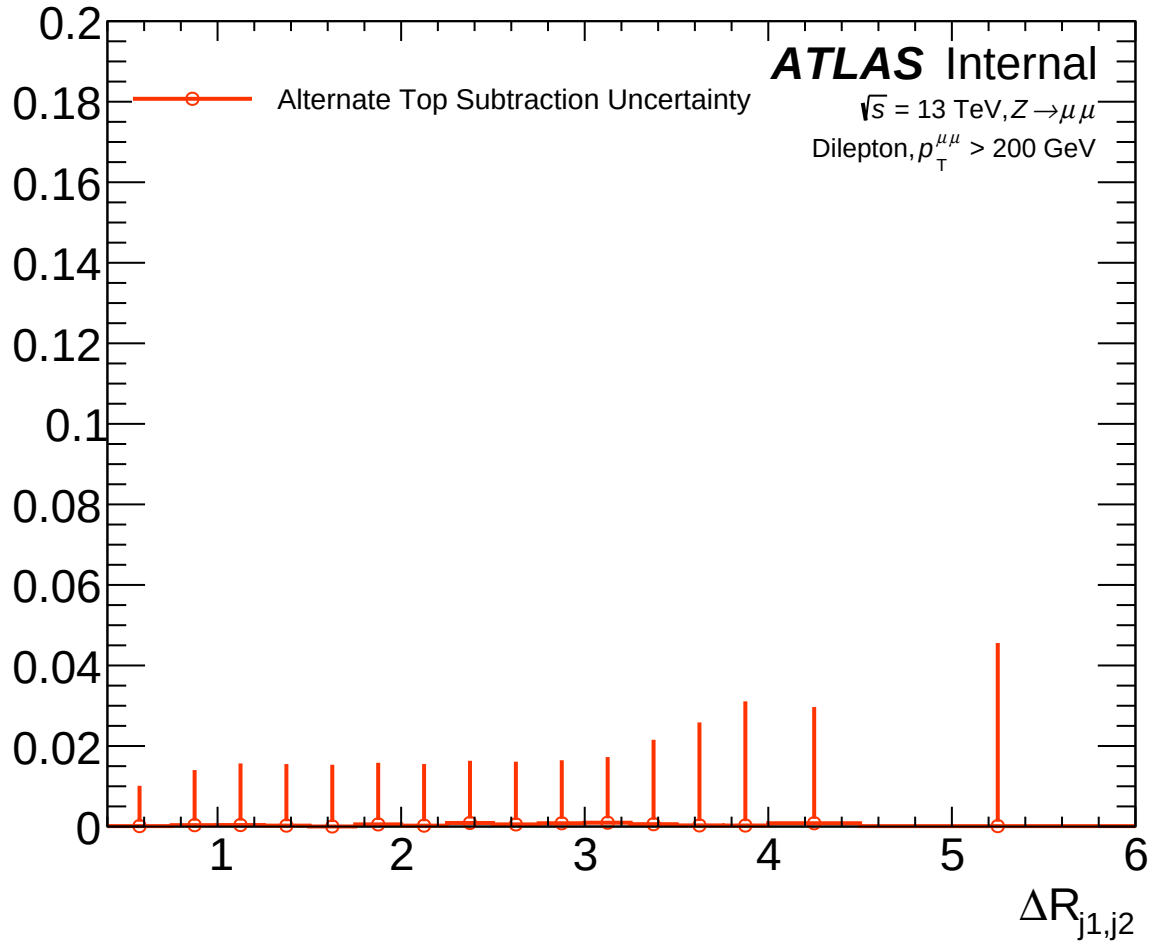
Fractional systematic impact

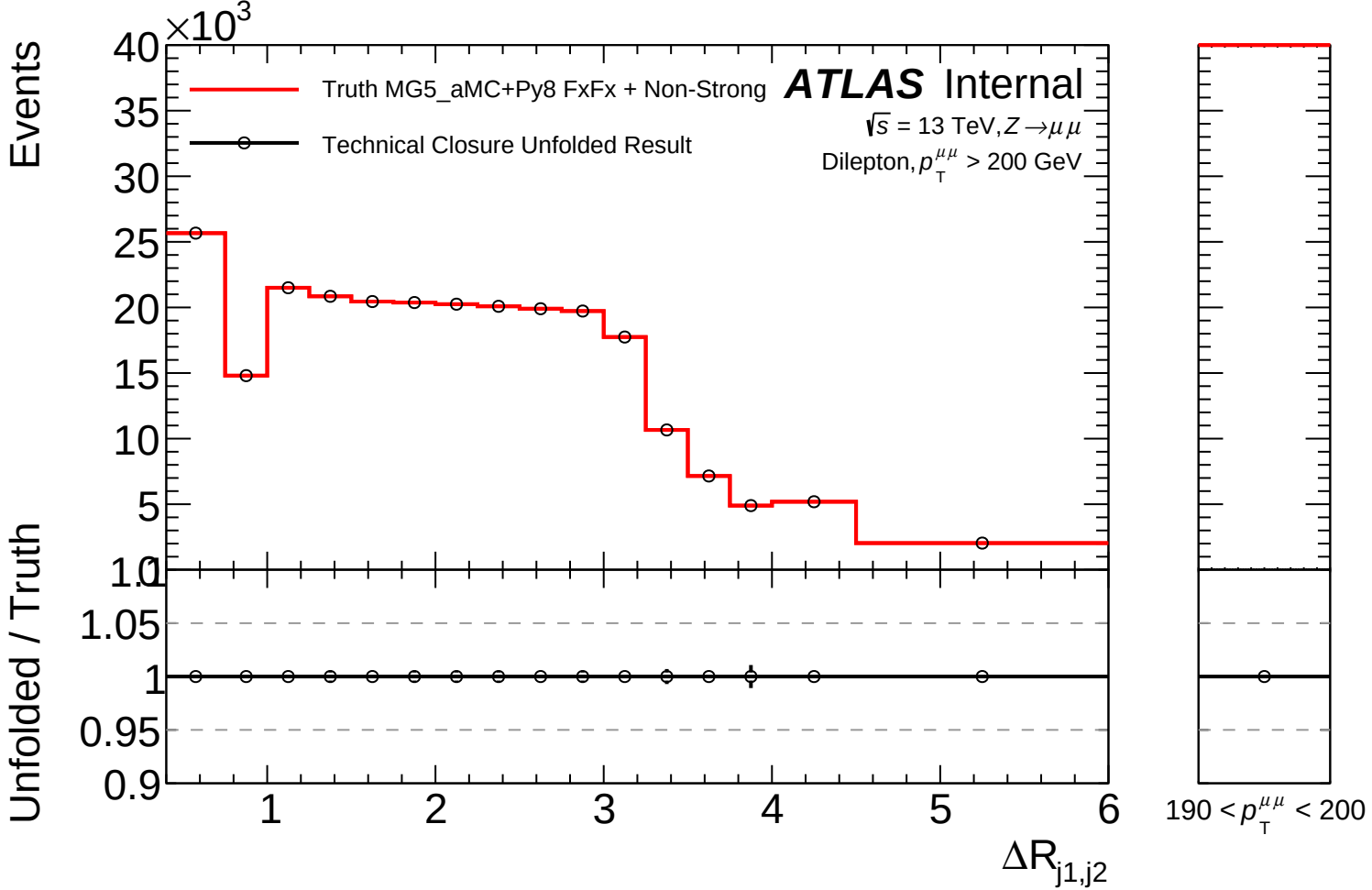
ATLAS Internal

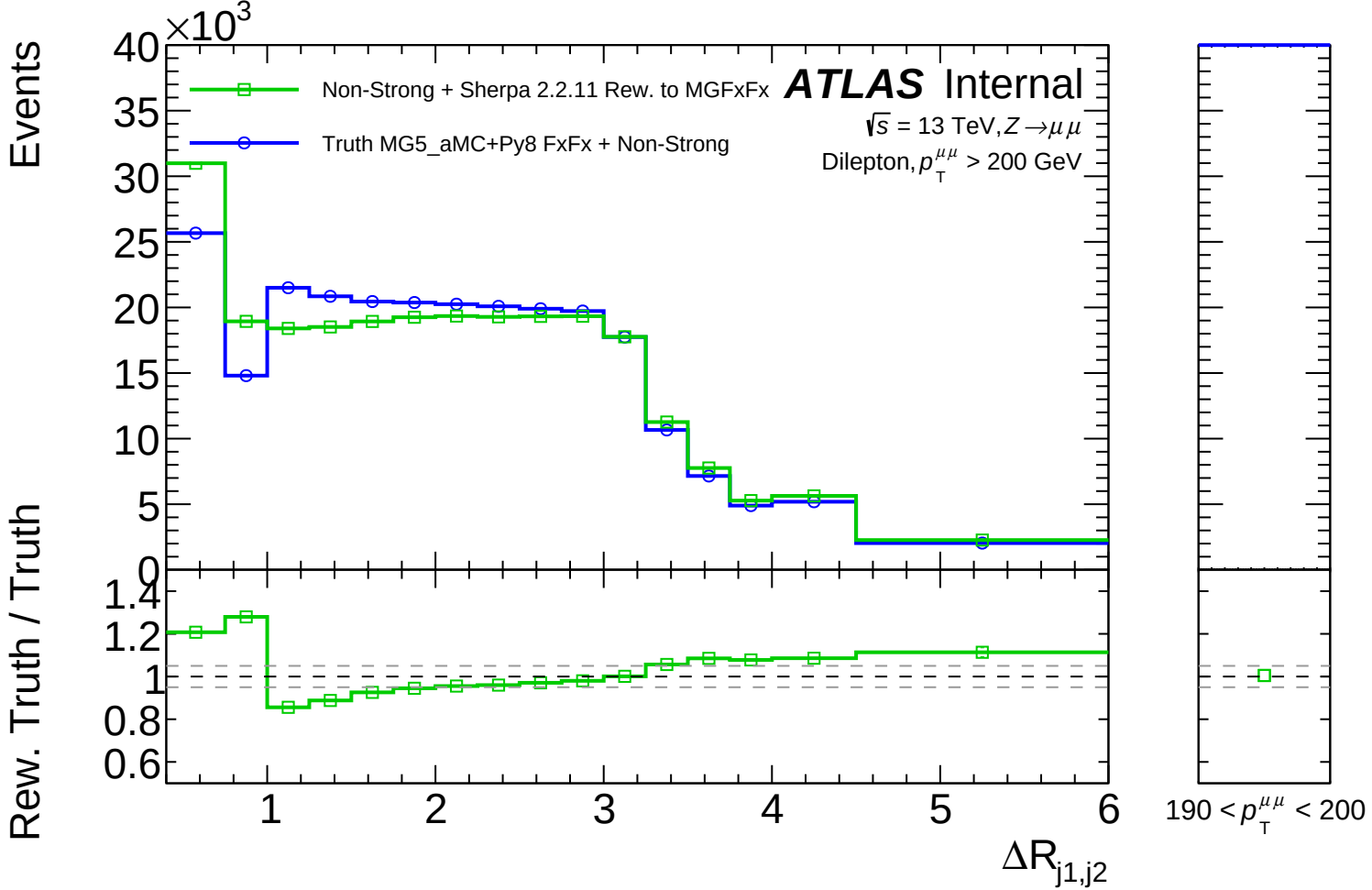
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

Alternate Top Subtraction Uncertainty

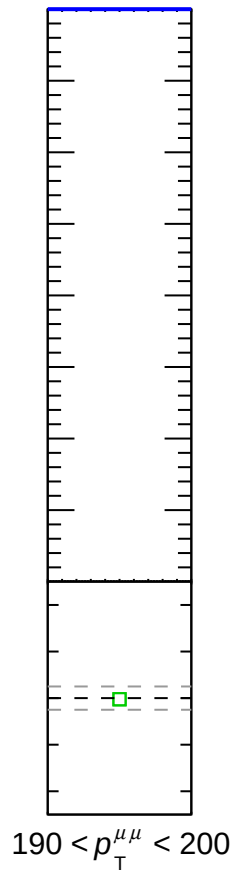
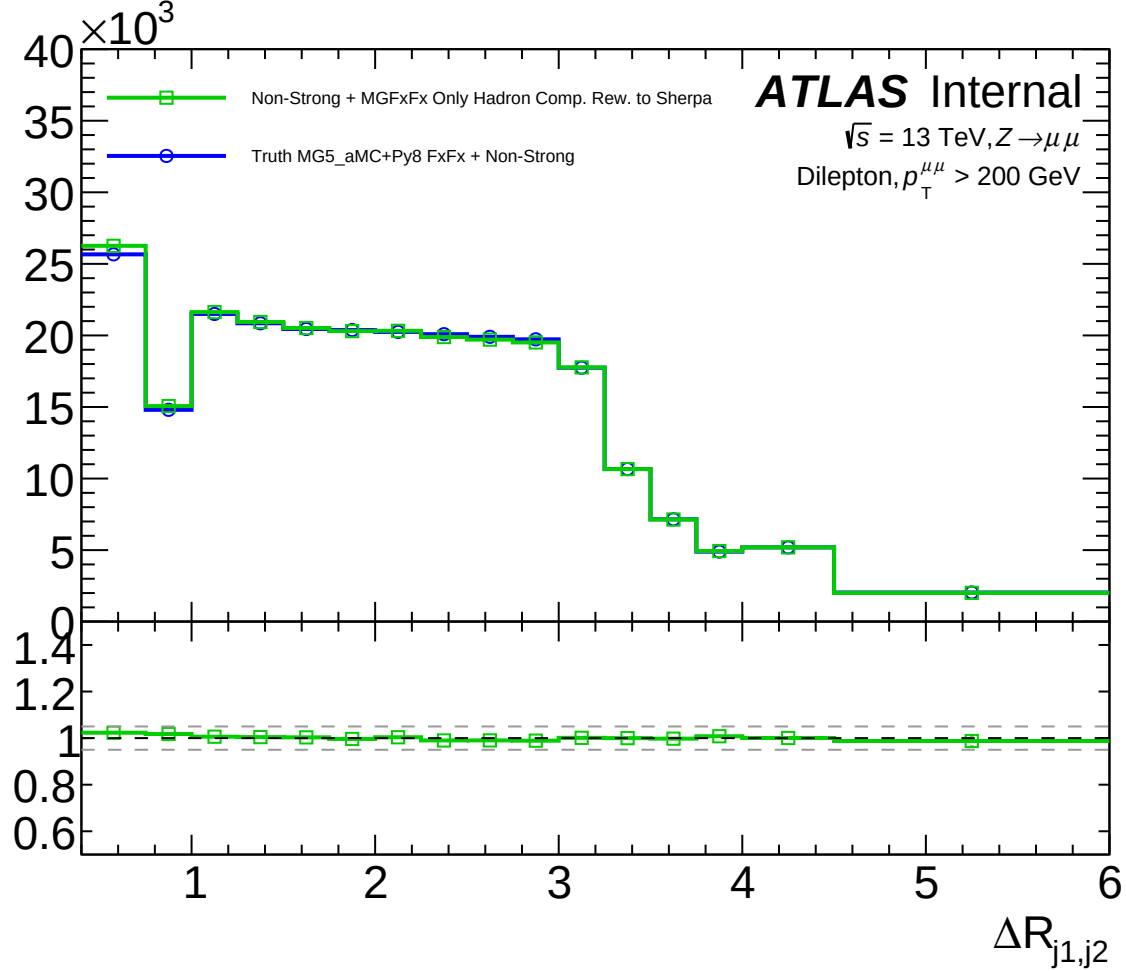


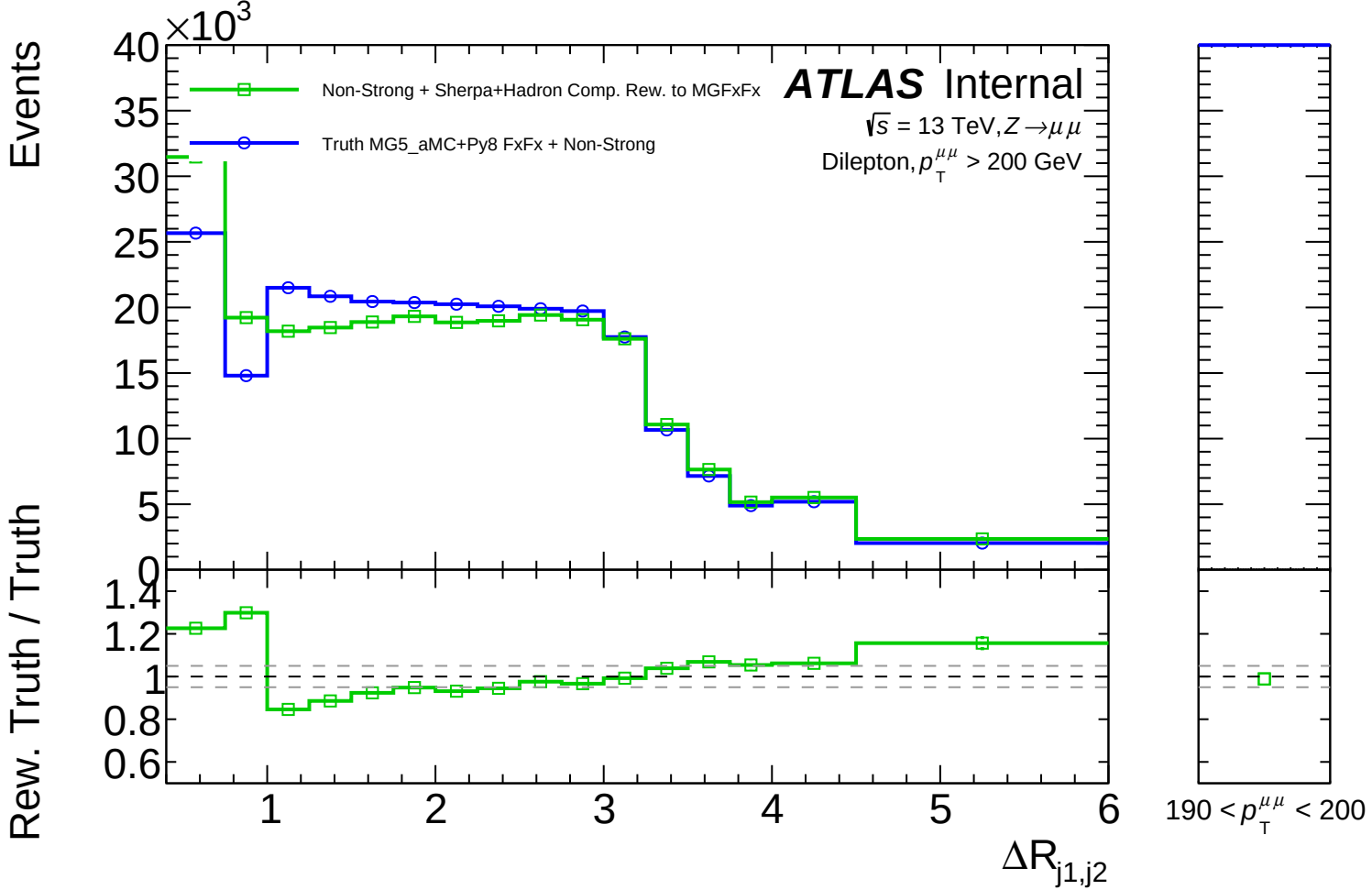


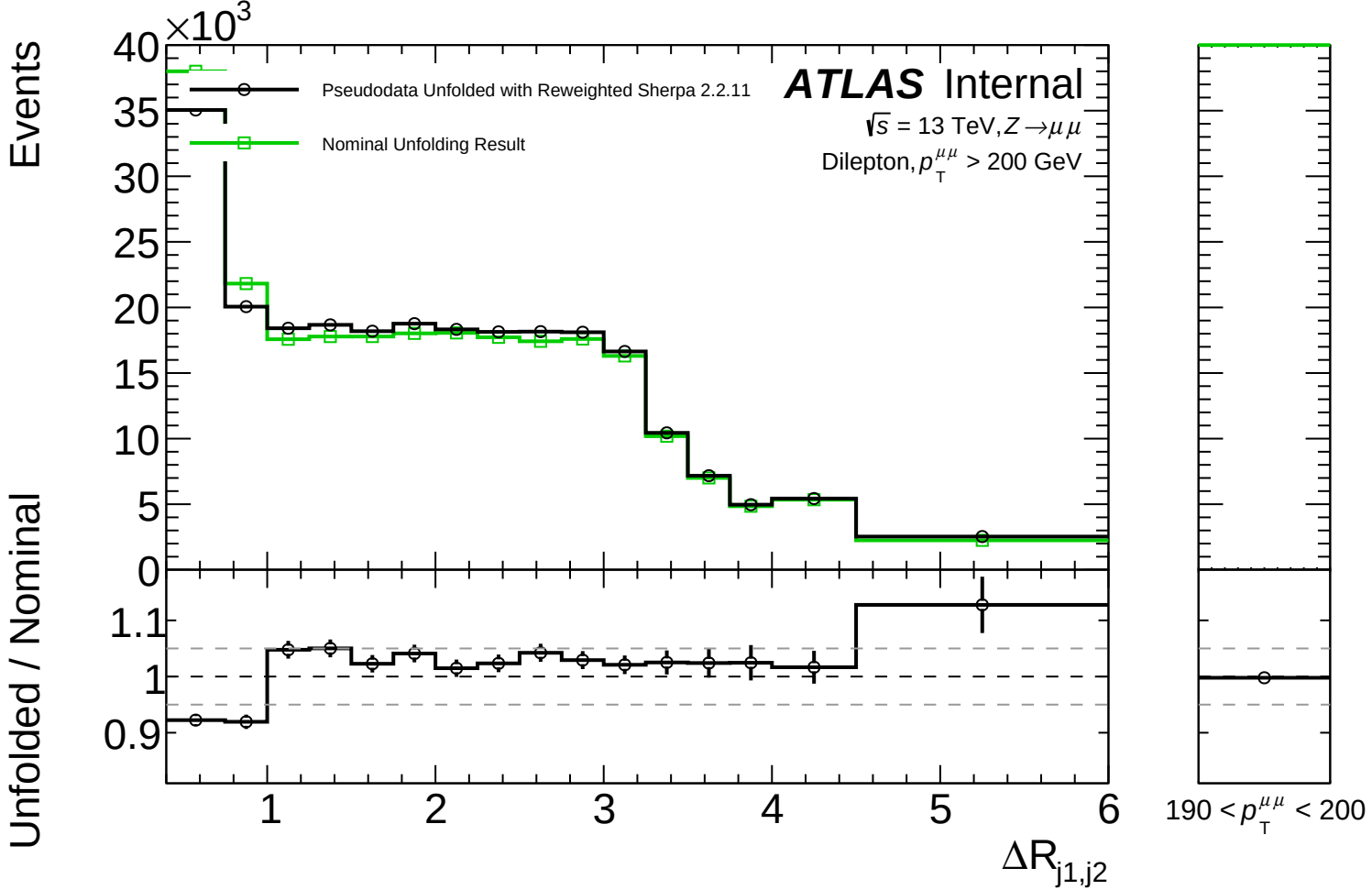


Events

Rew. Truth / Truth

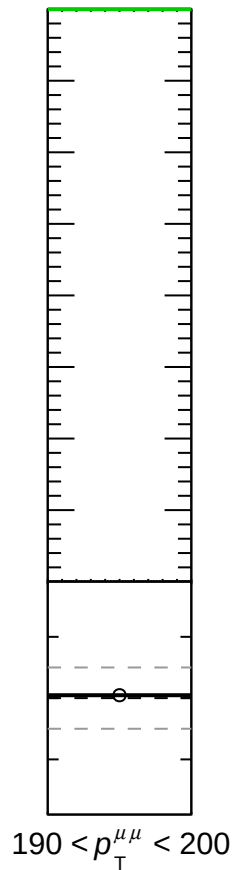
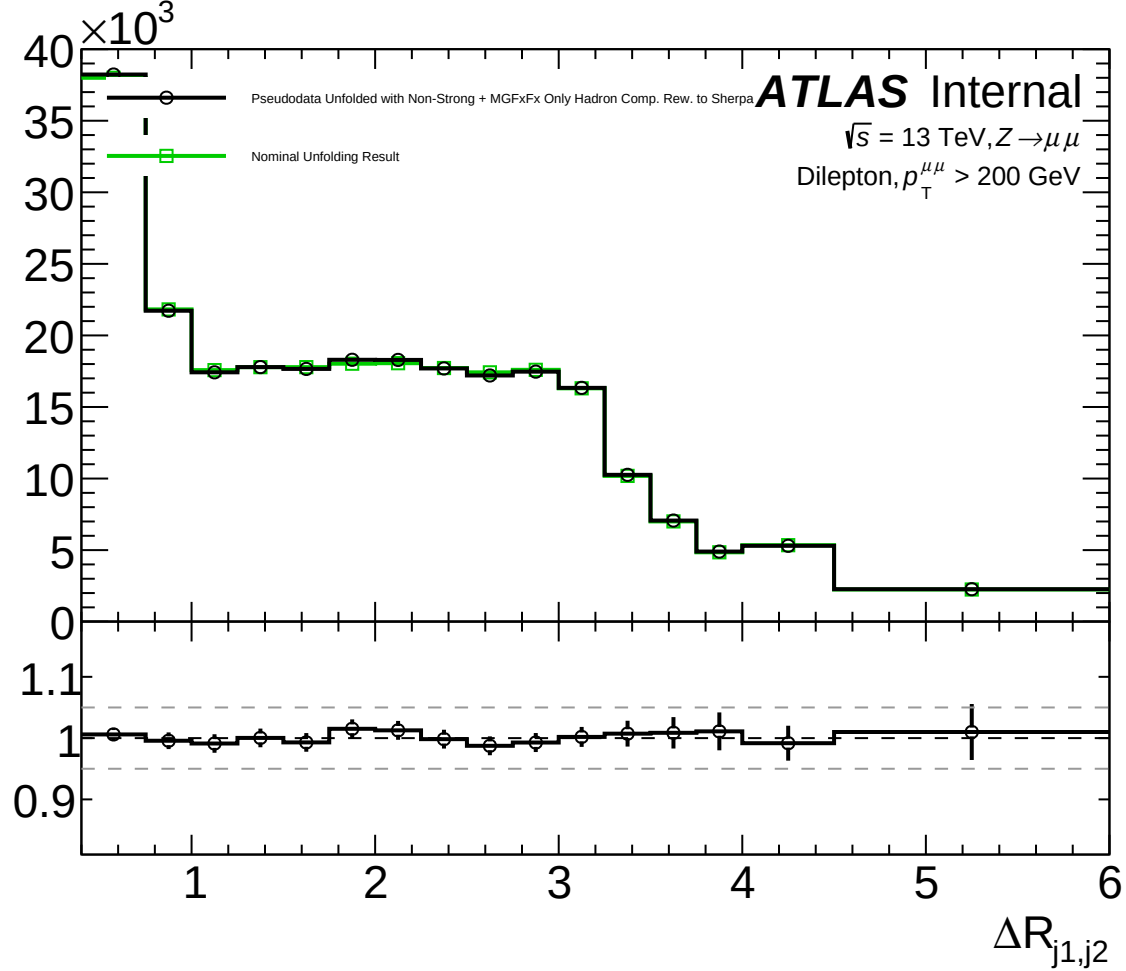


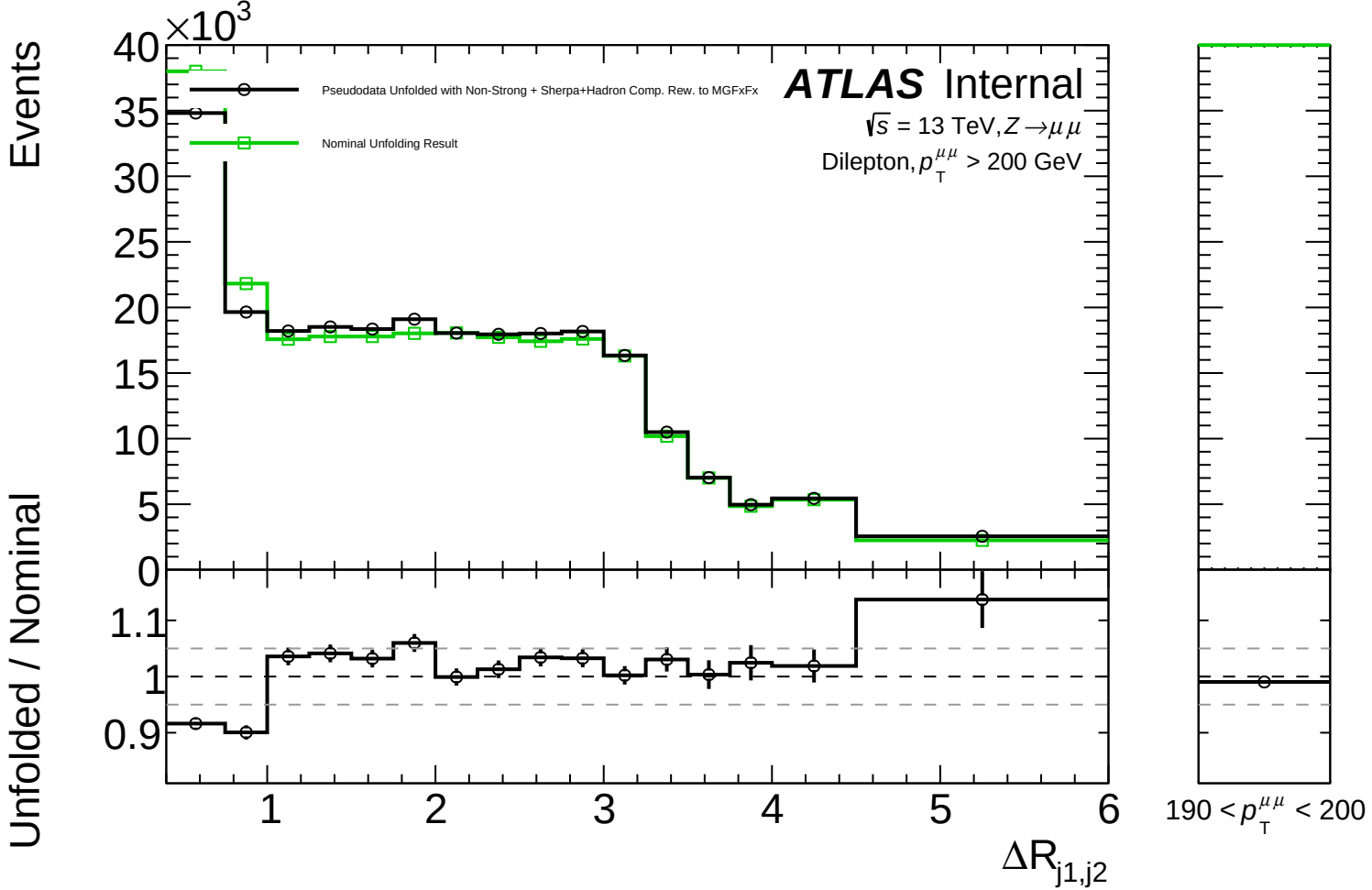




Events

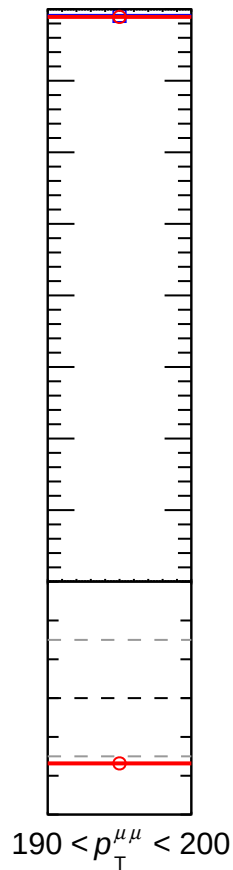
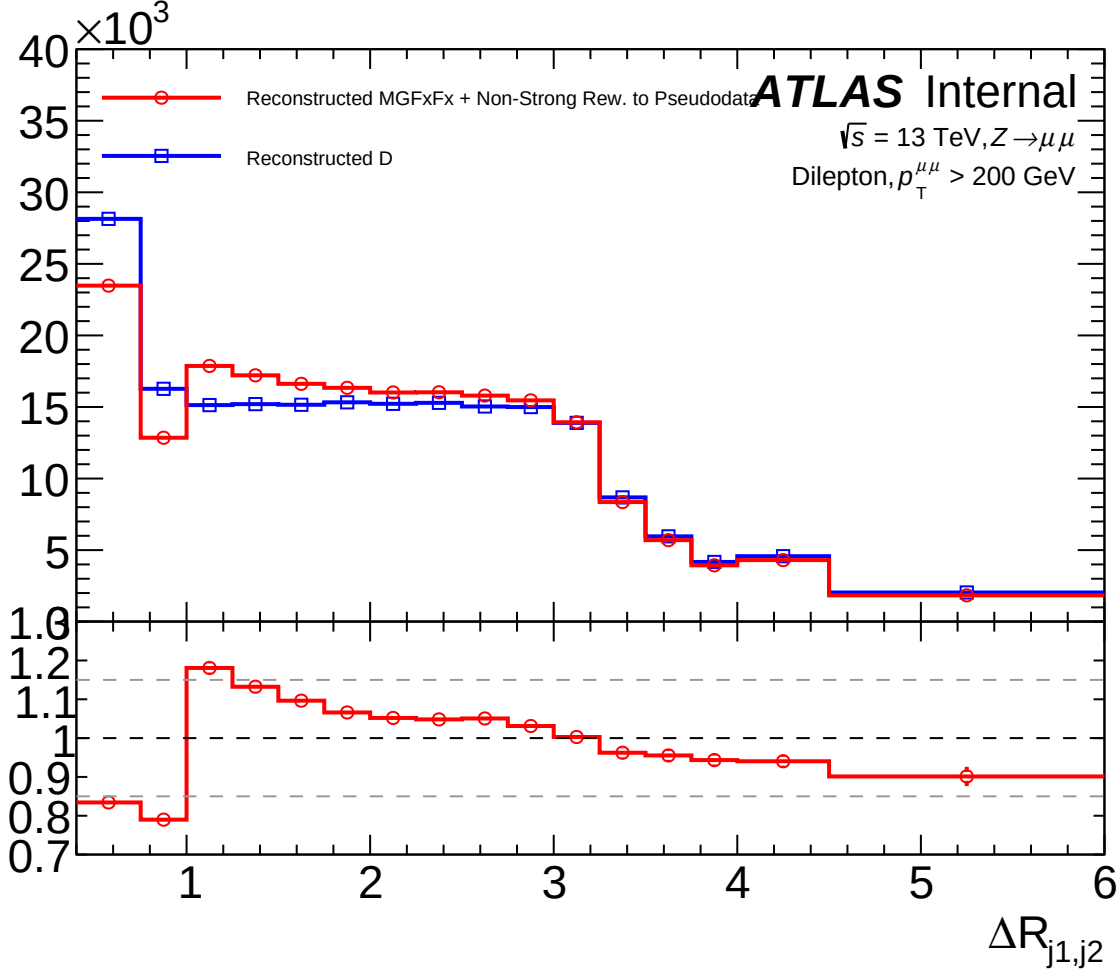
Unfolded / Nominal

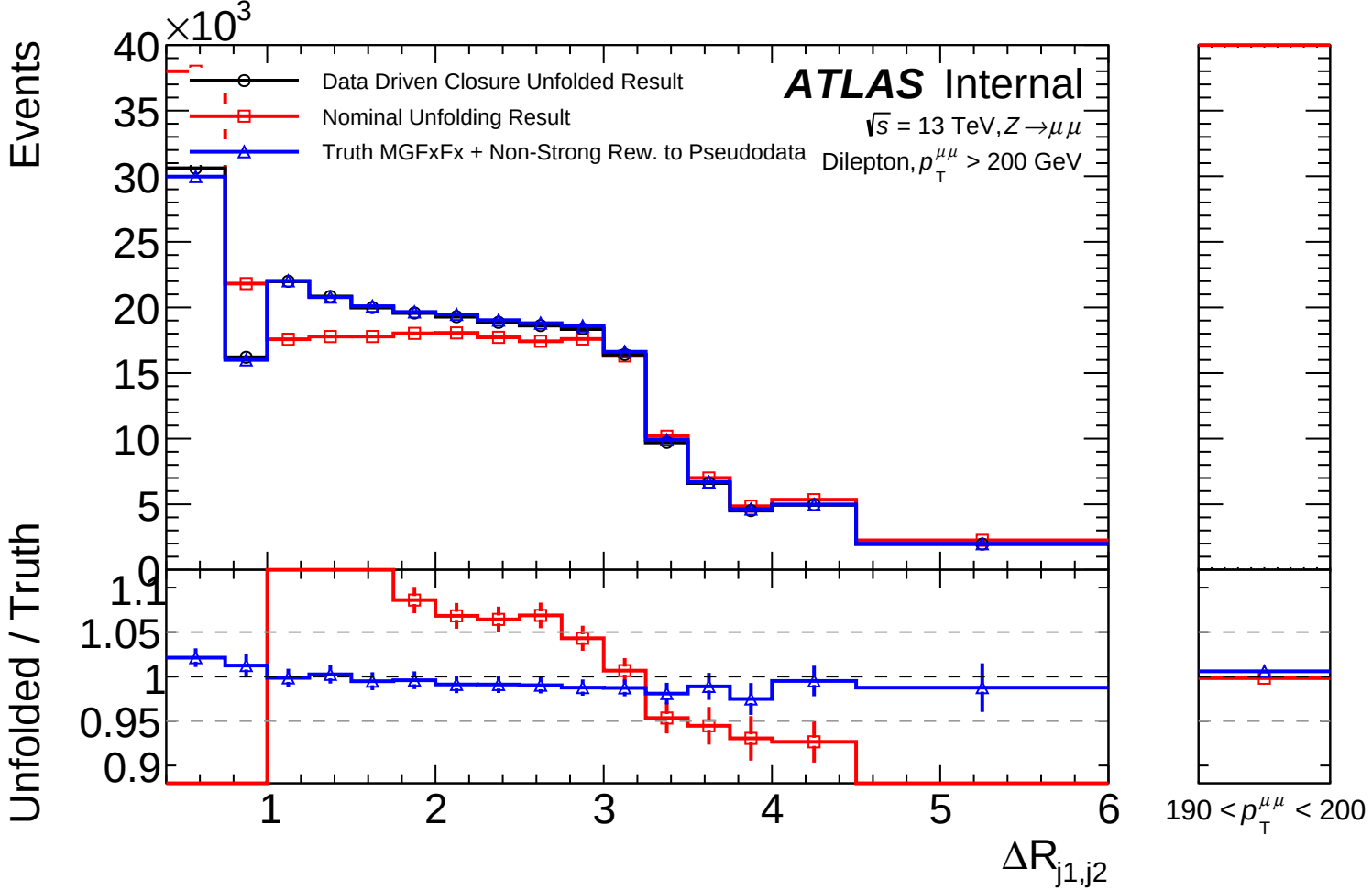




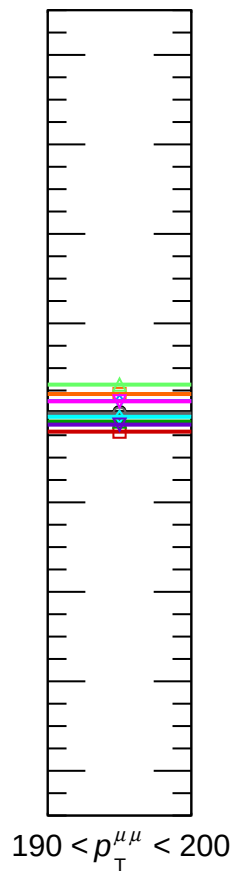
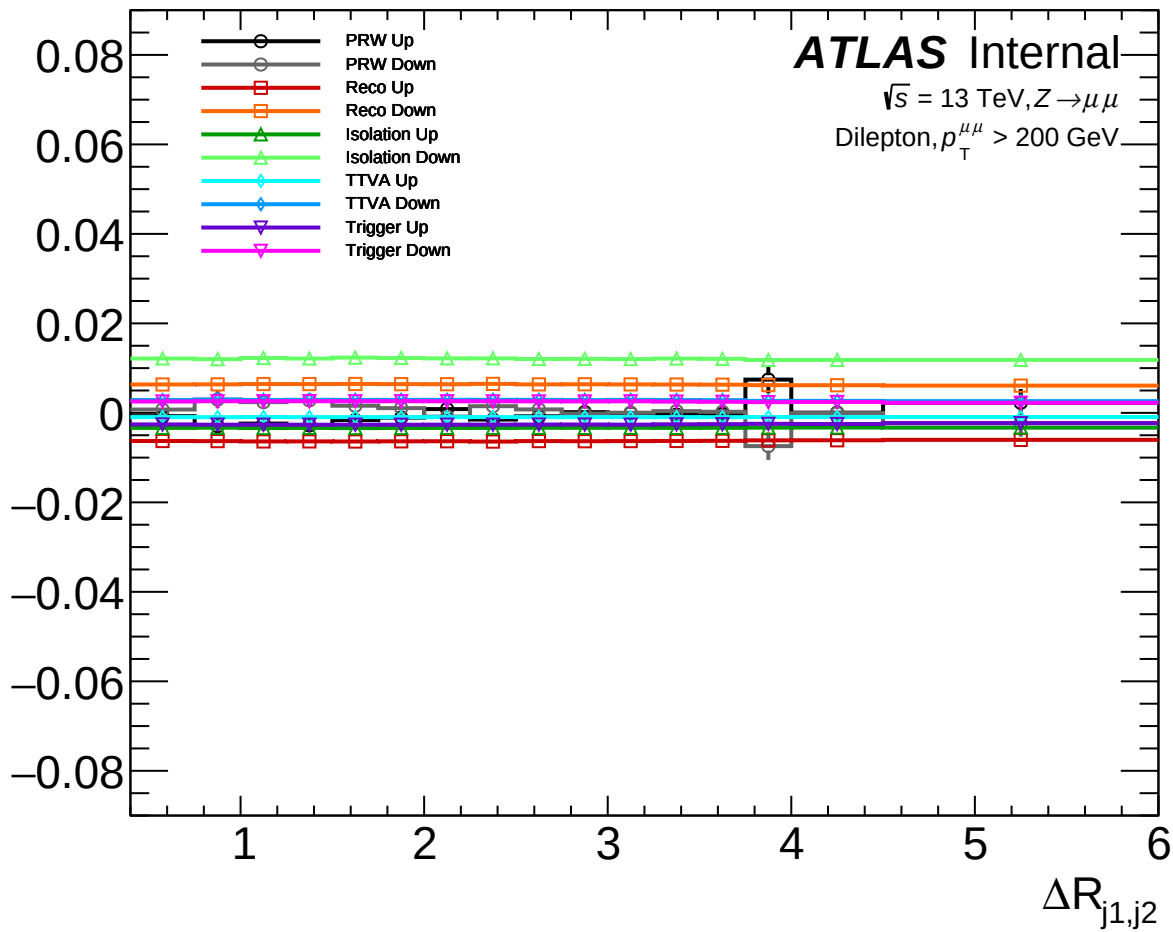
Events

Rew. Reco MC / Data

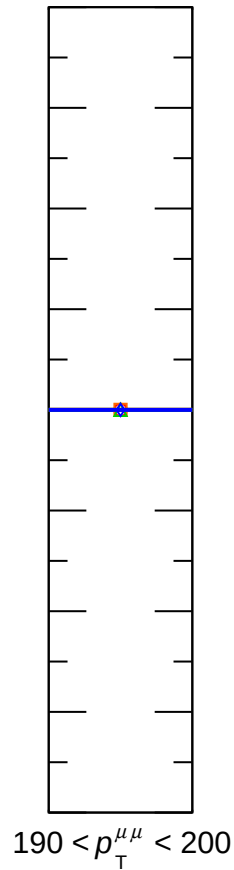
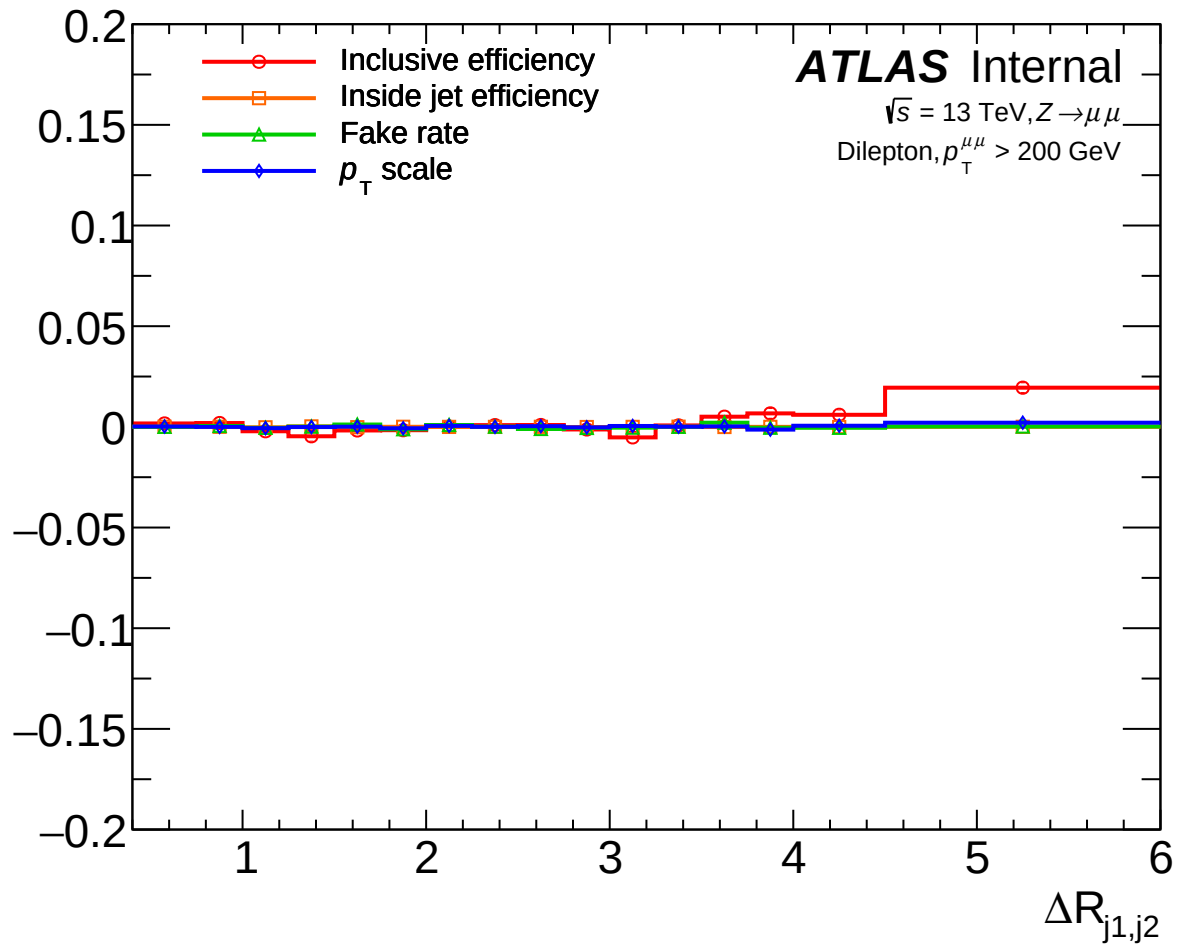




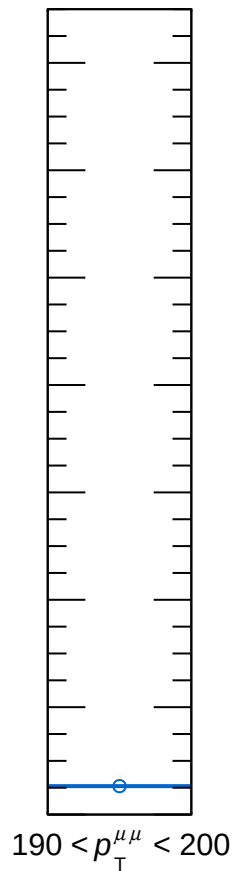
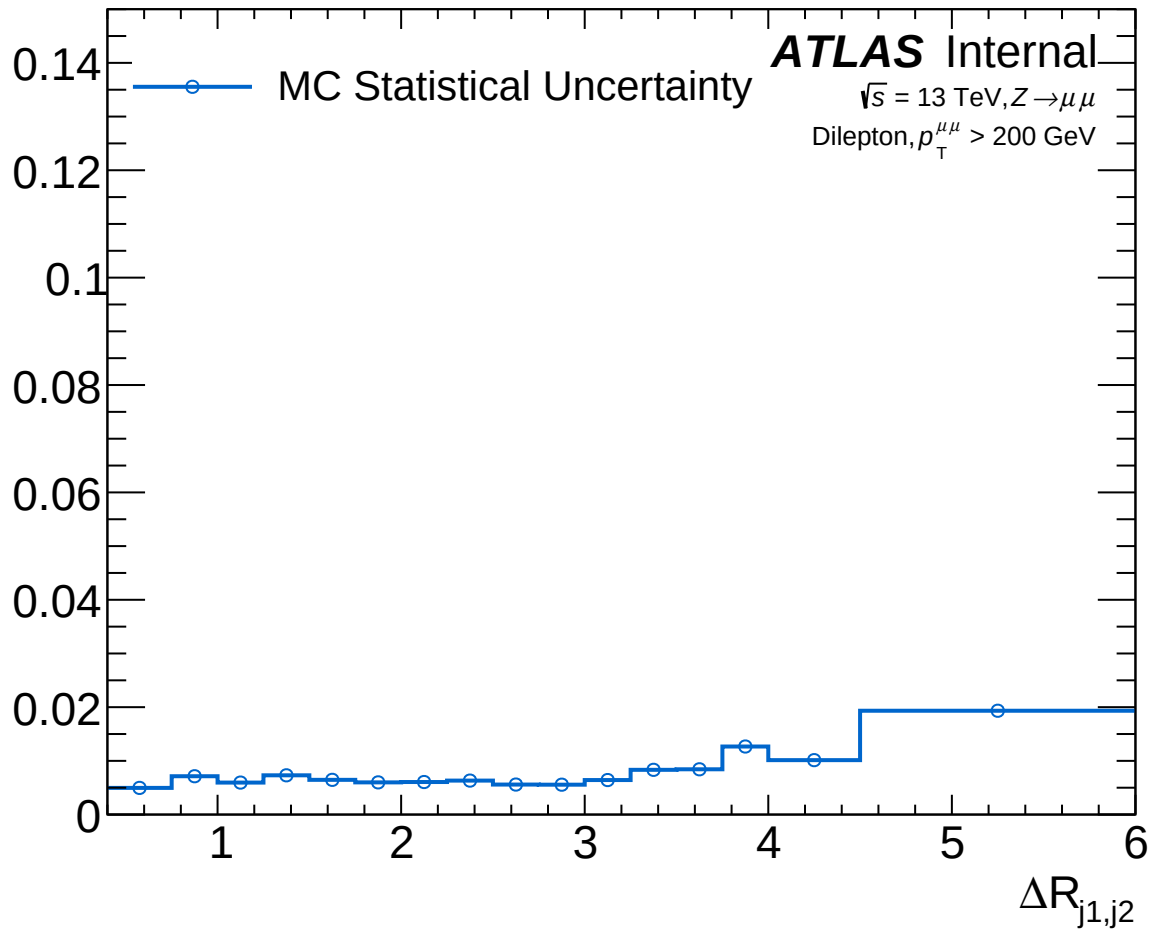
Fractional systematic impact



Fractional systematic impact



Fractional systematic impact



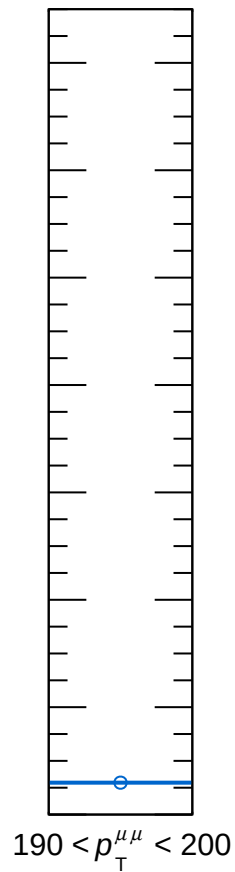
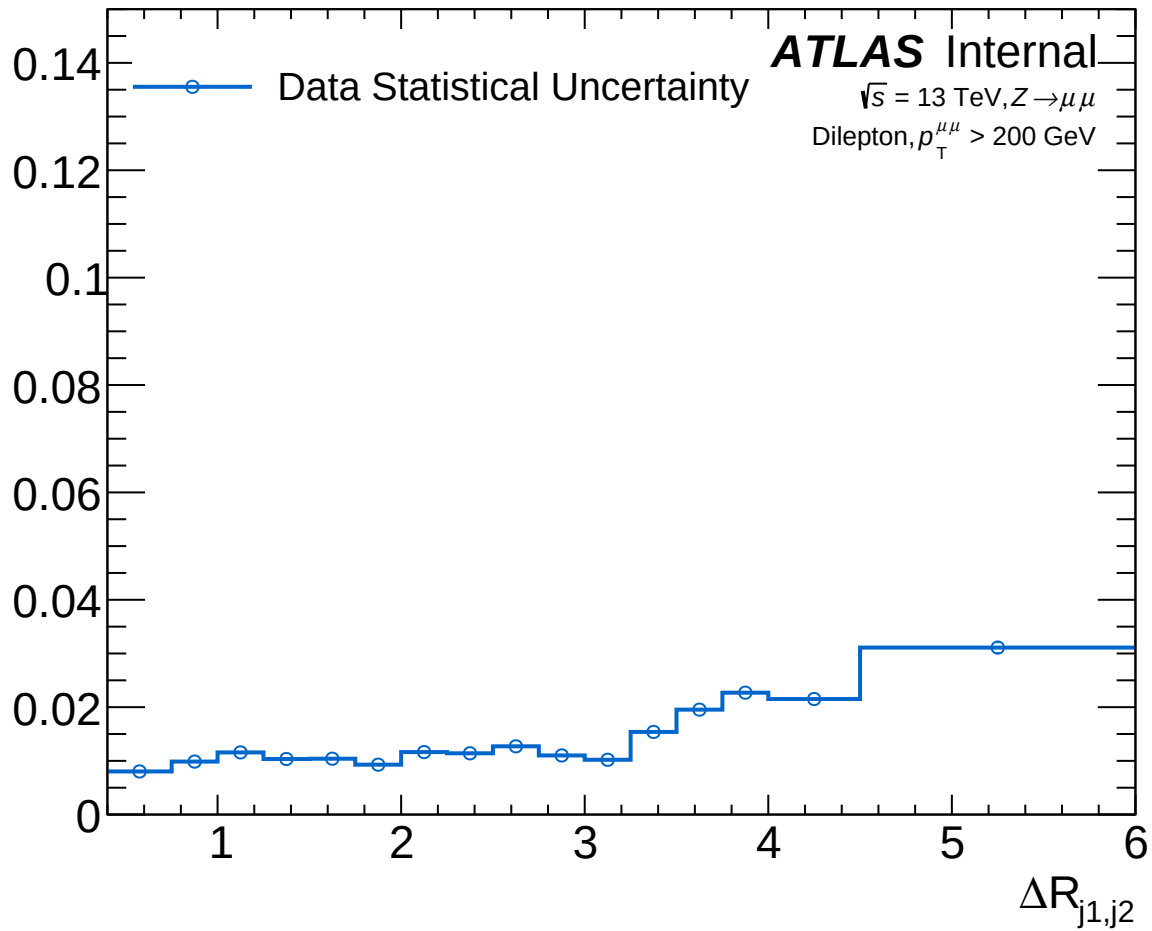
Fractional systematic impact

ATLAS Internal

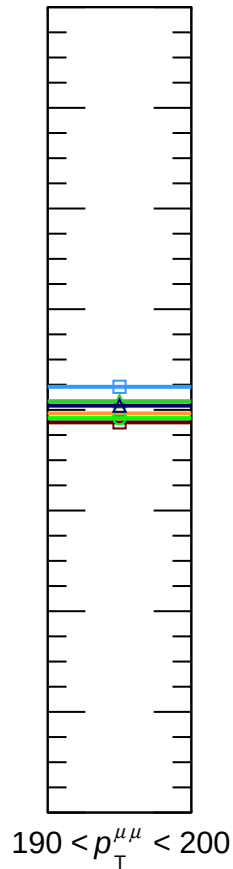
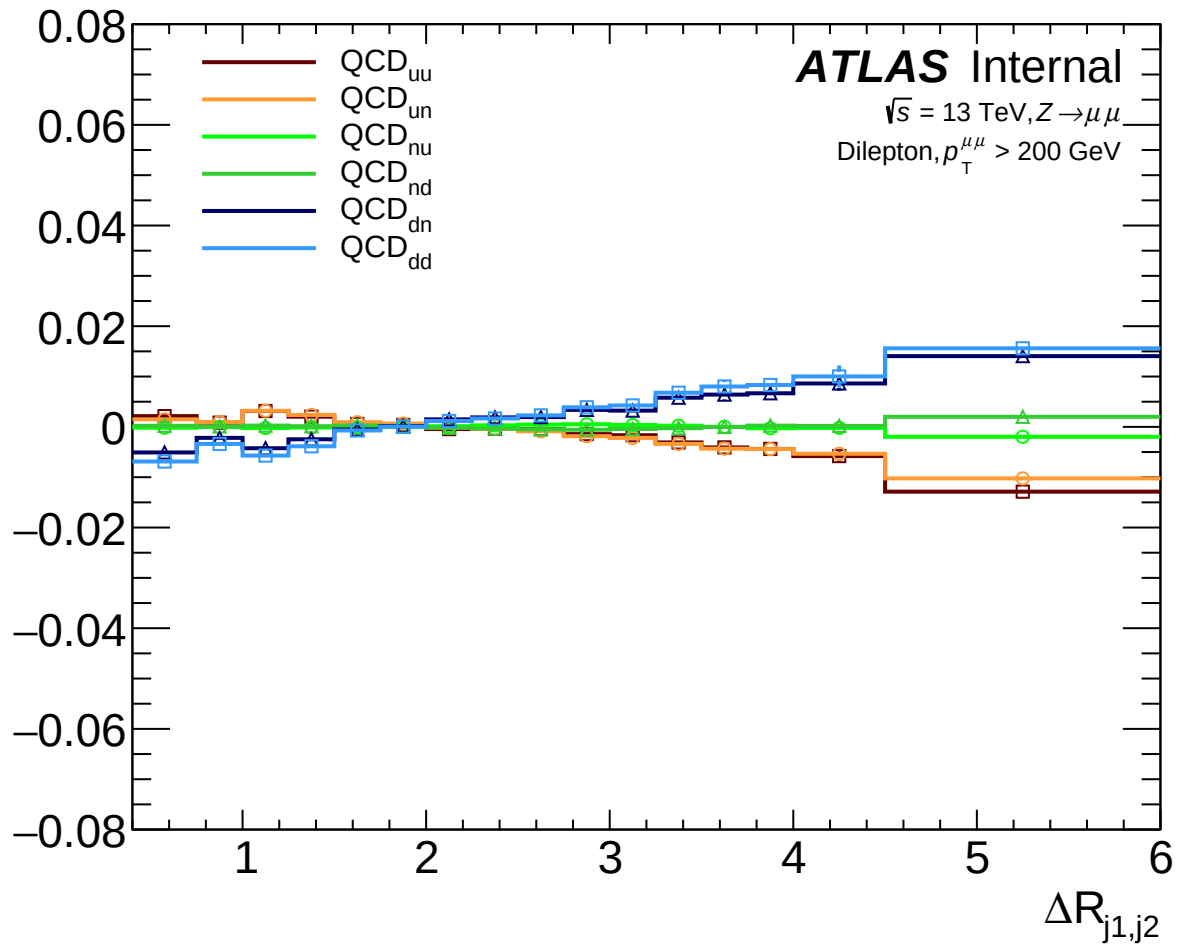
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

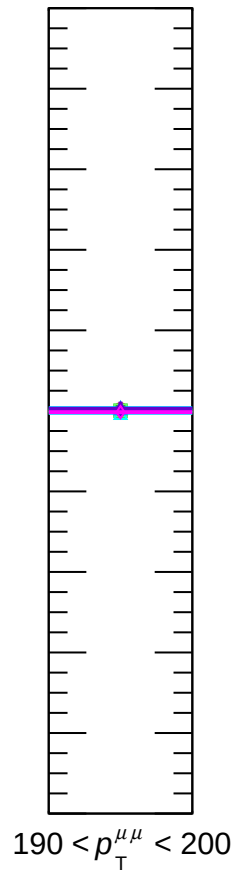
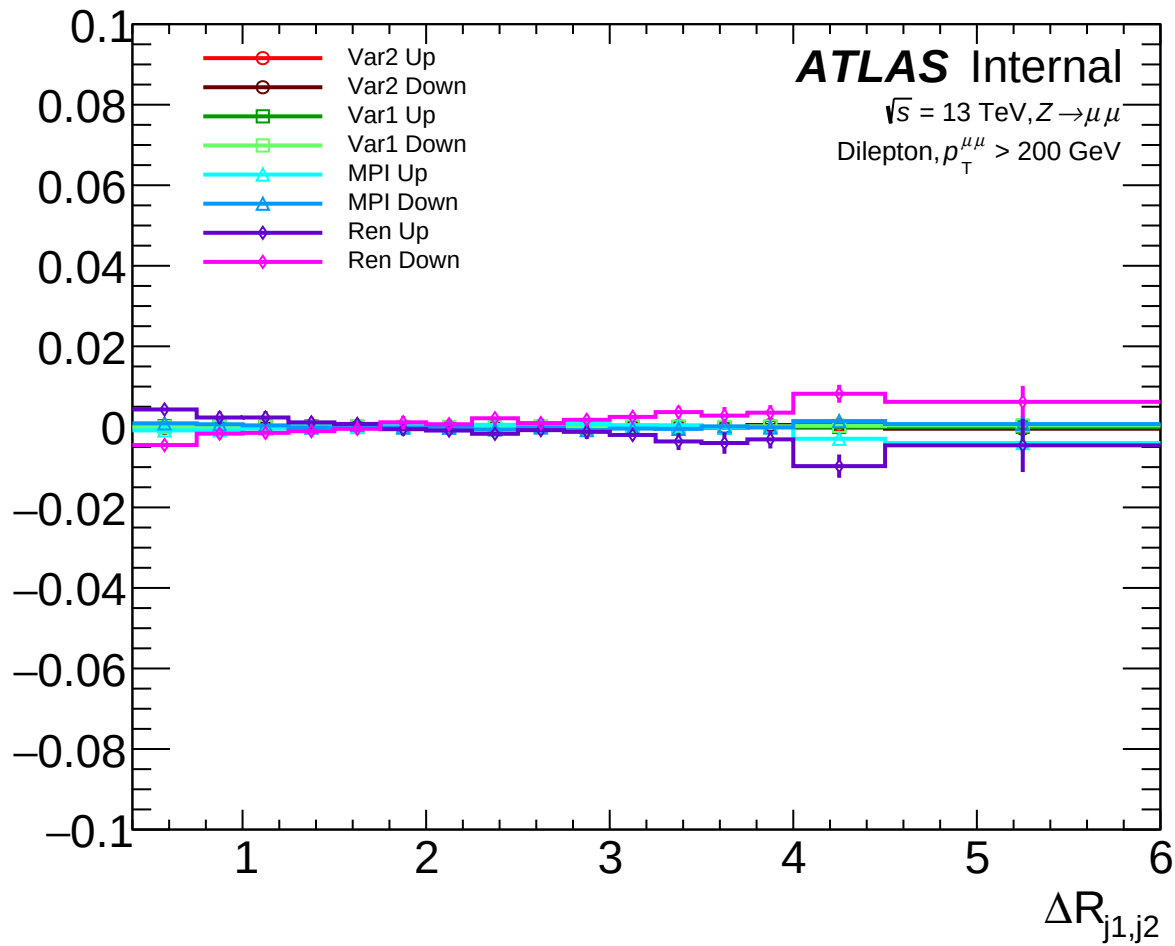
Data Statistical Uncertainty

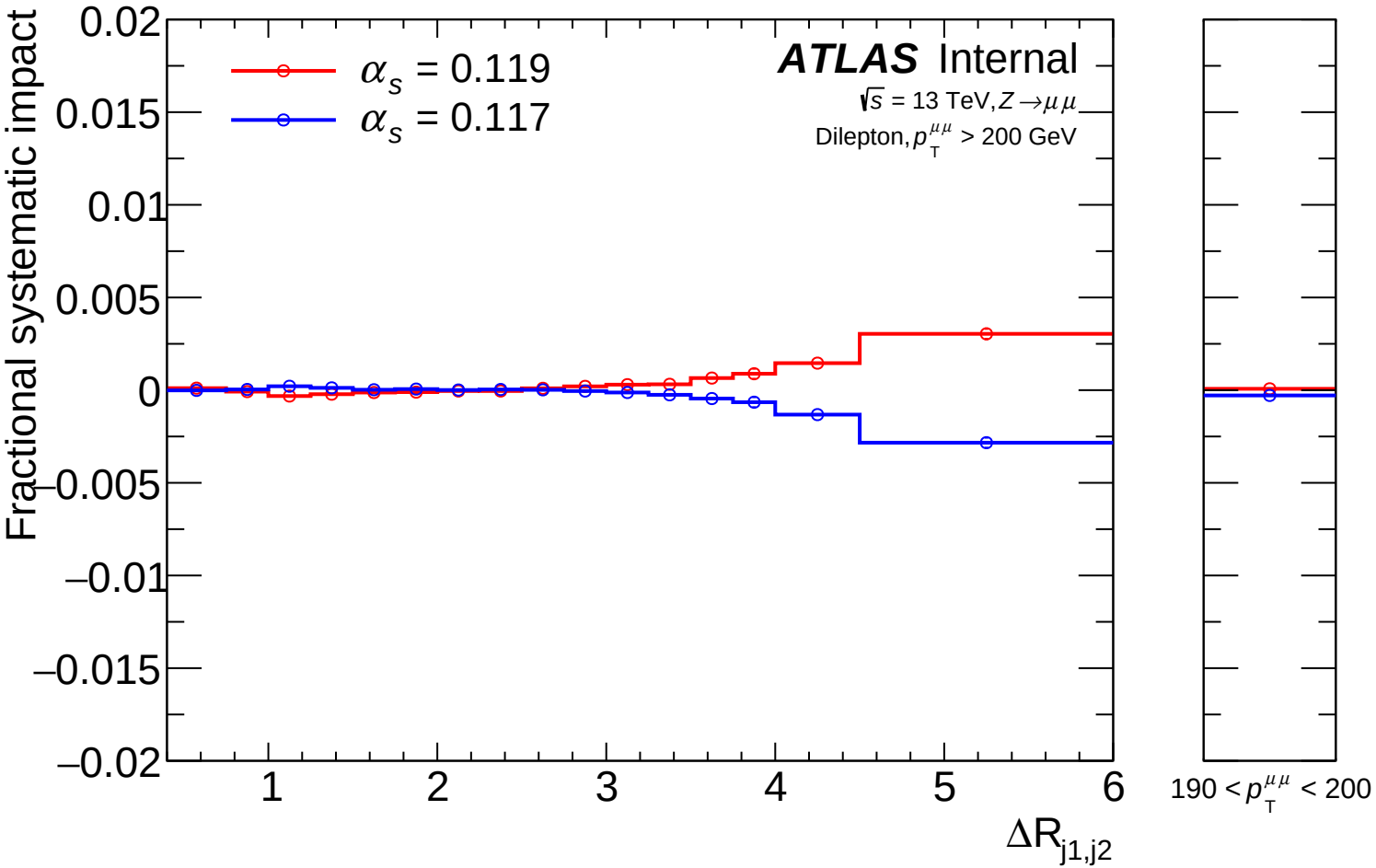


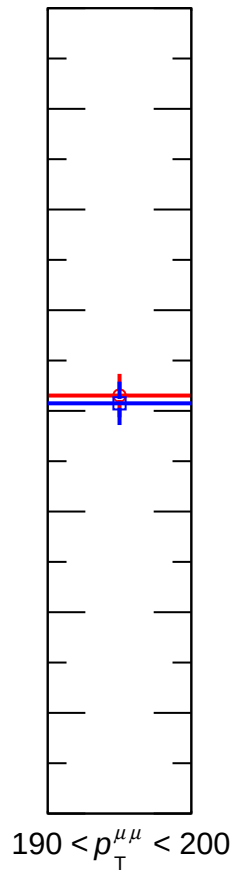
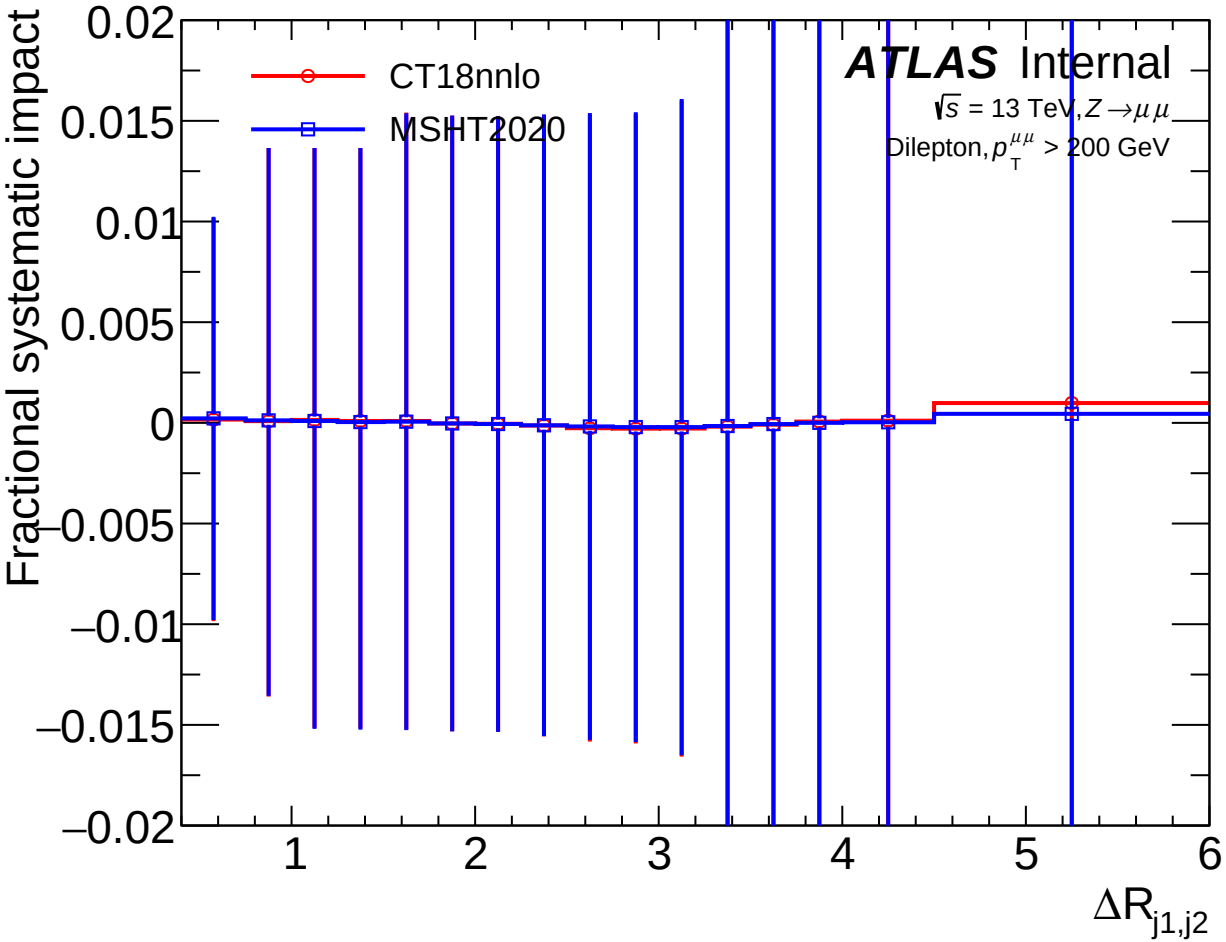
Fractional systematic impact



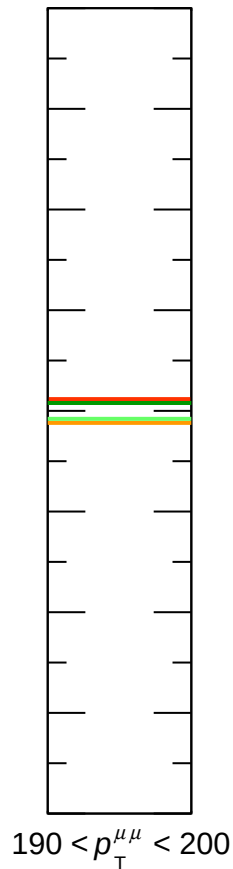
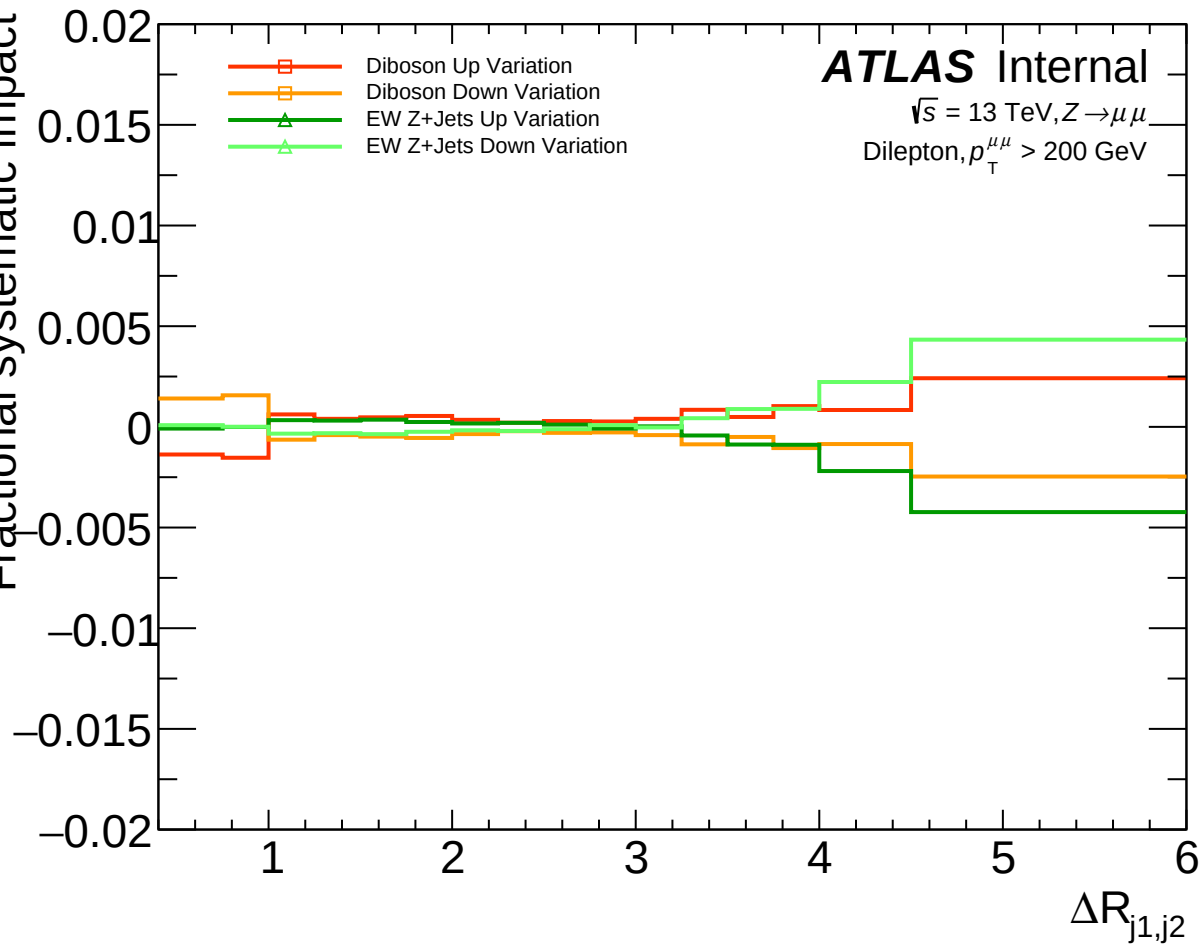
Fractional systematic impact







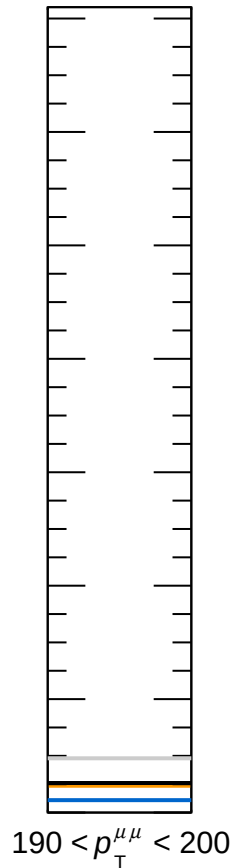
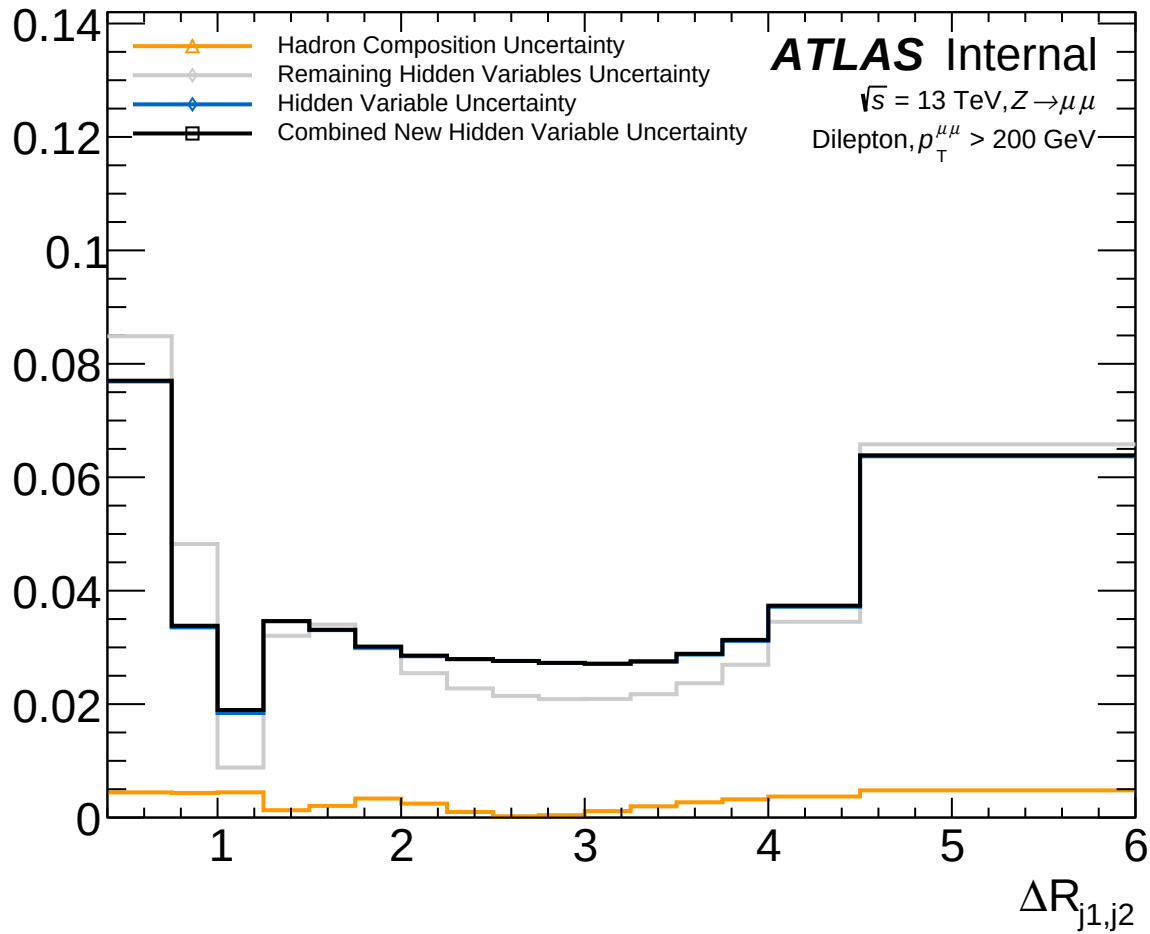
Fractional systematic impact

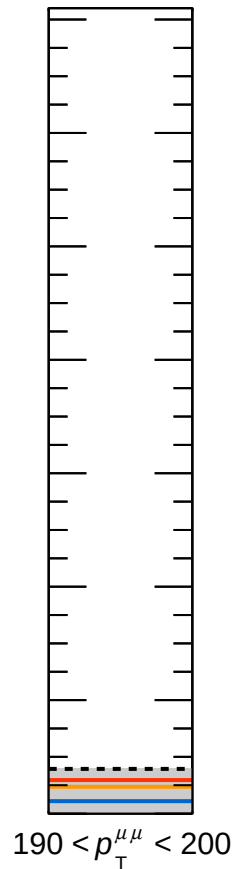
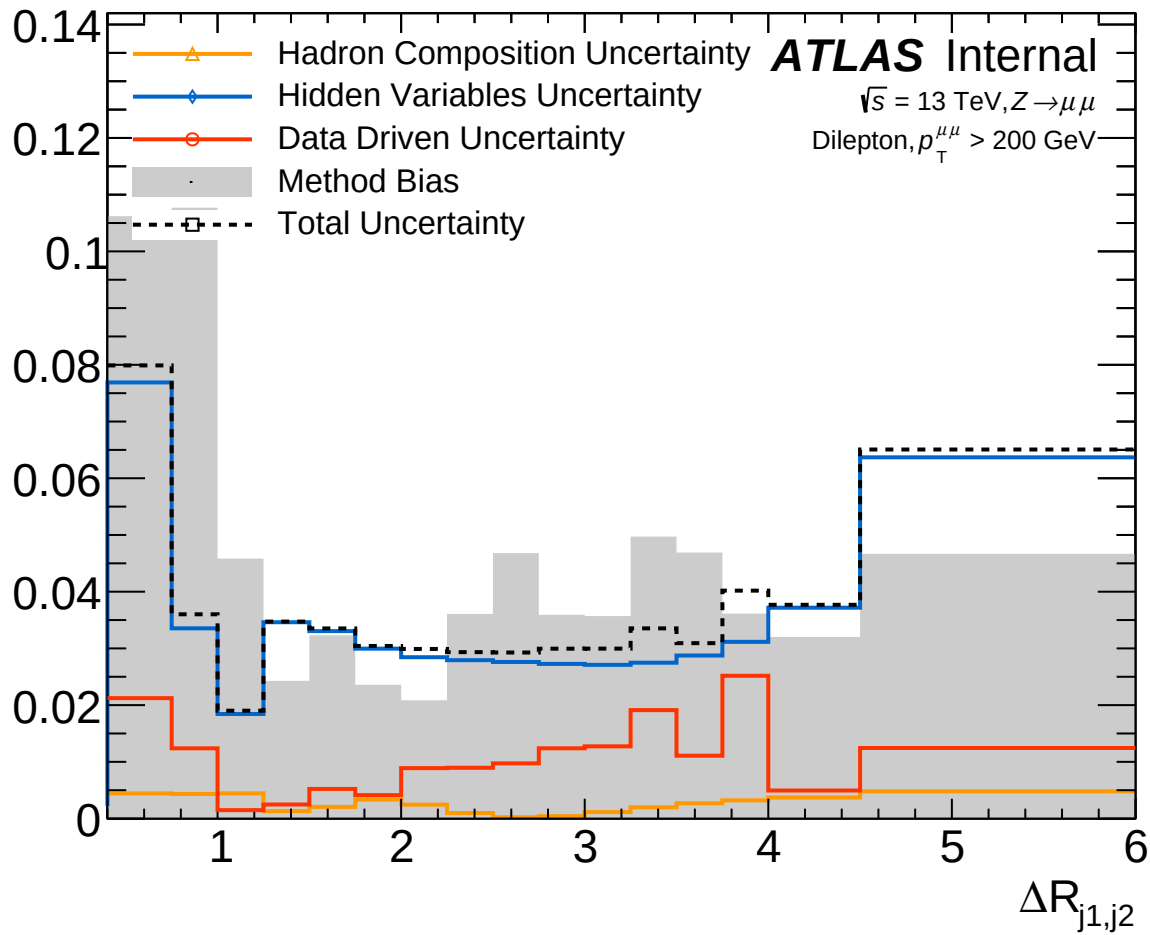


ATLAS Internal

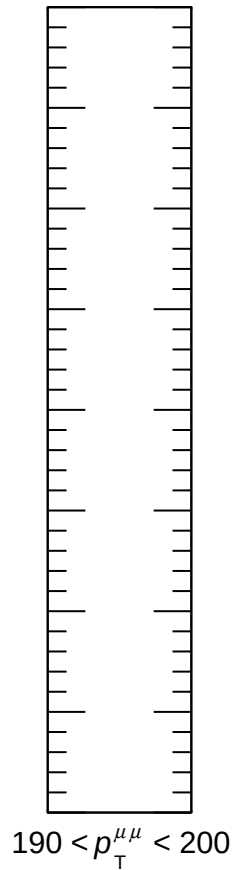
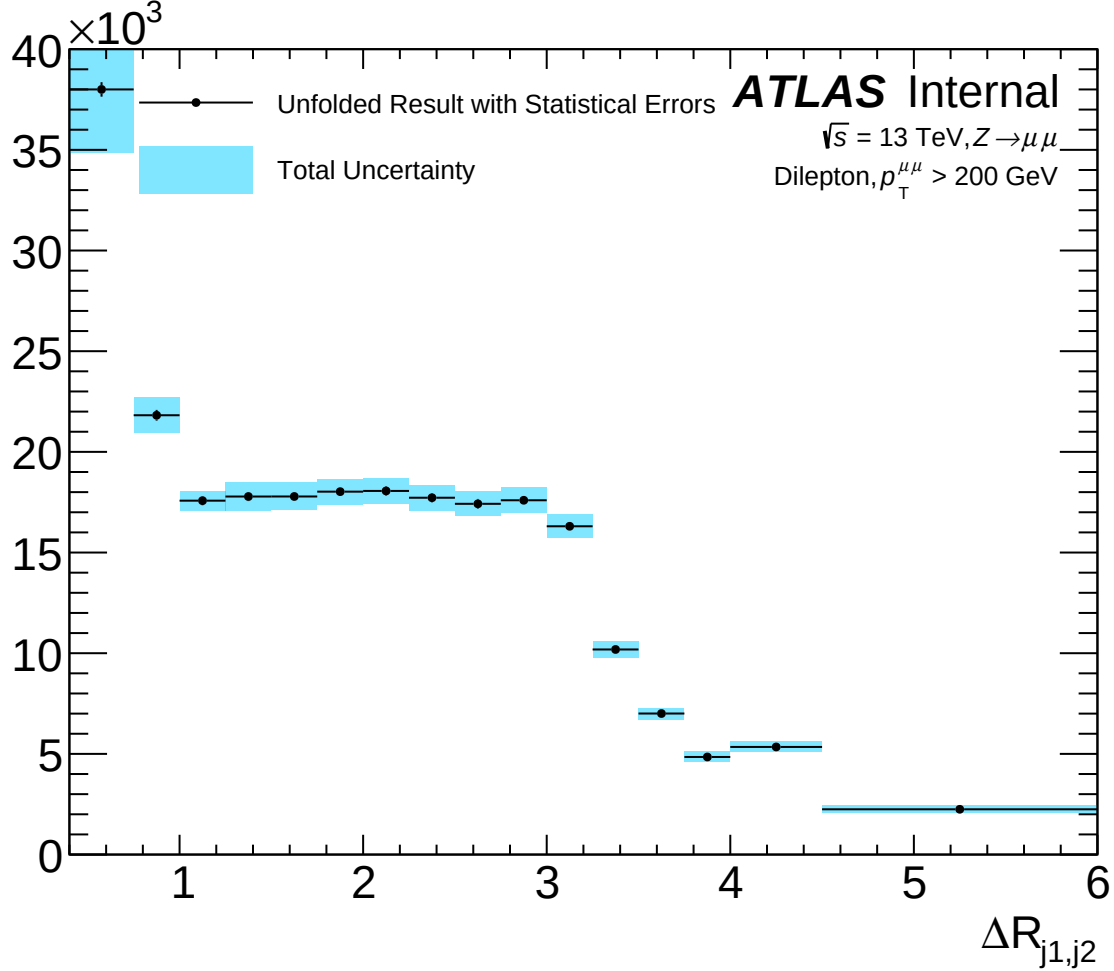
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

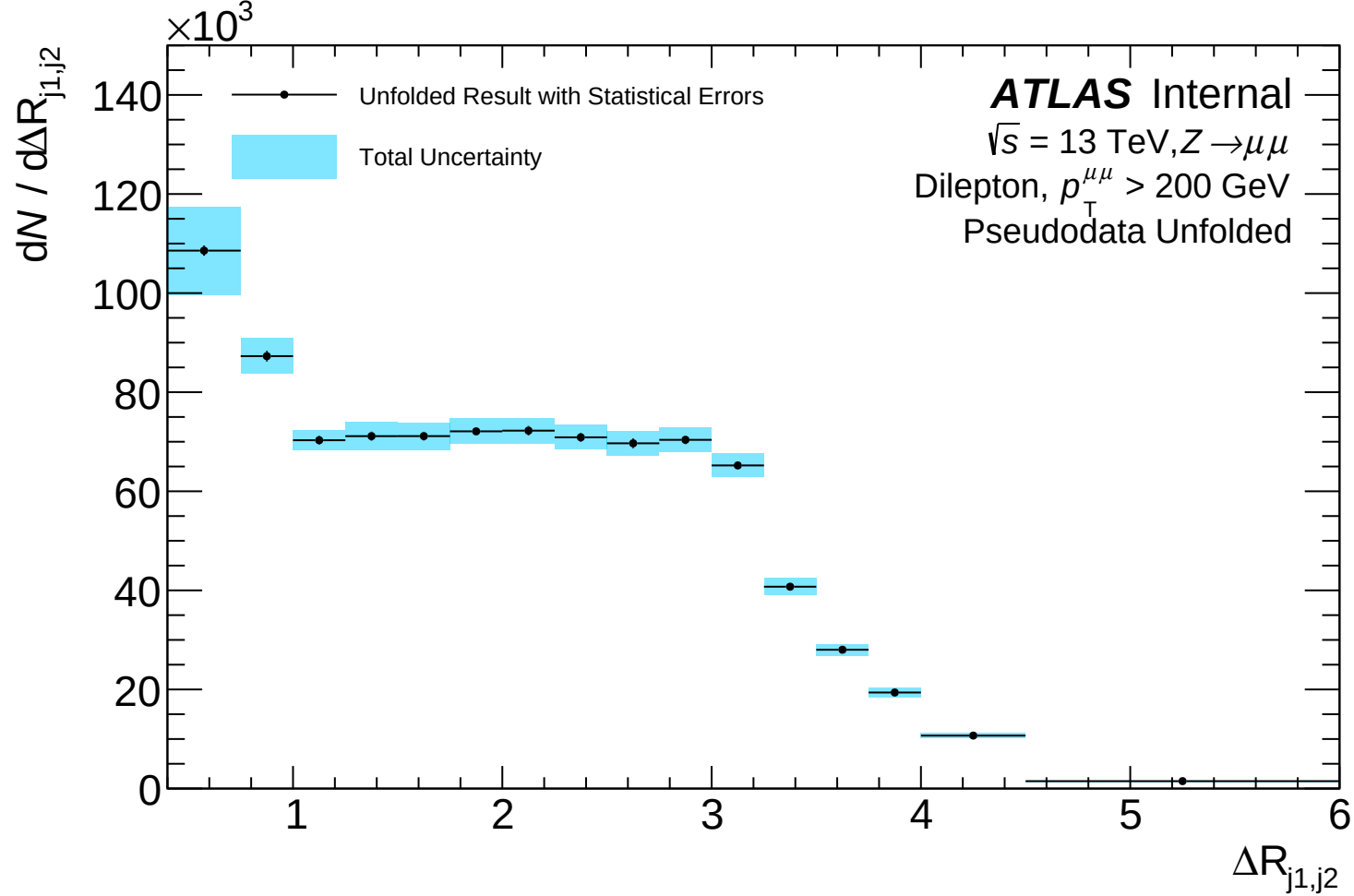
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

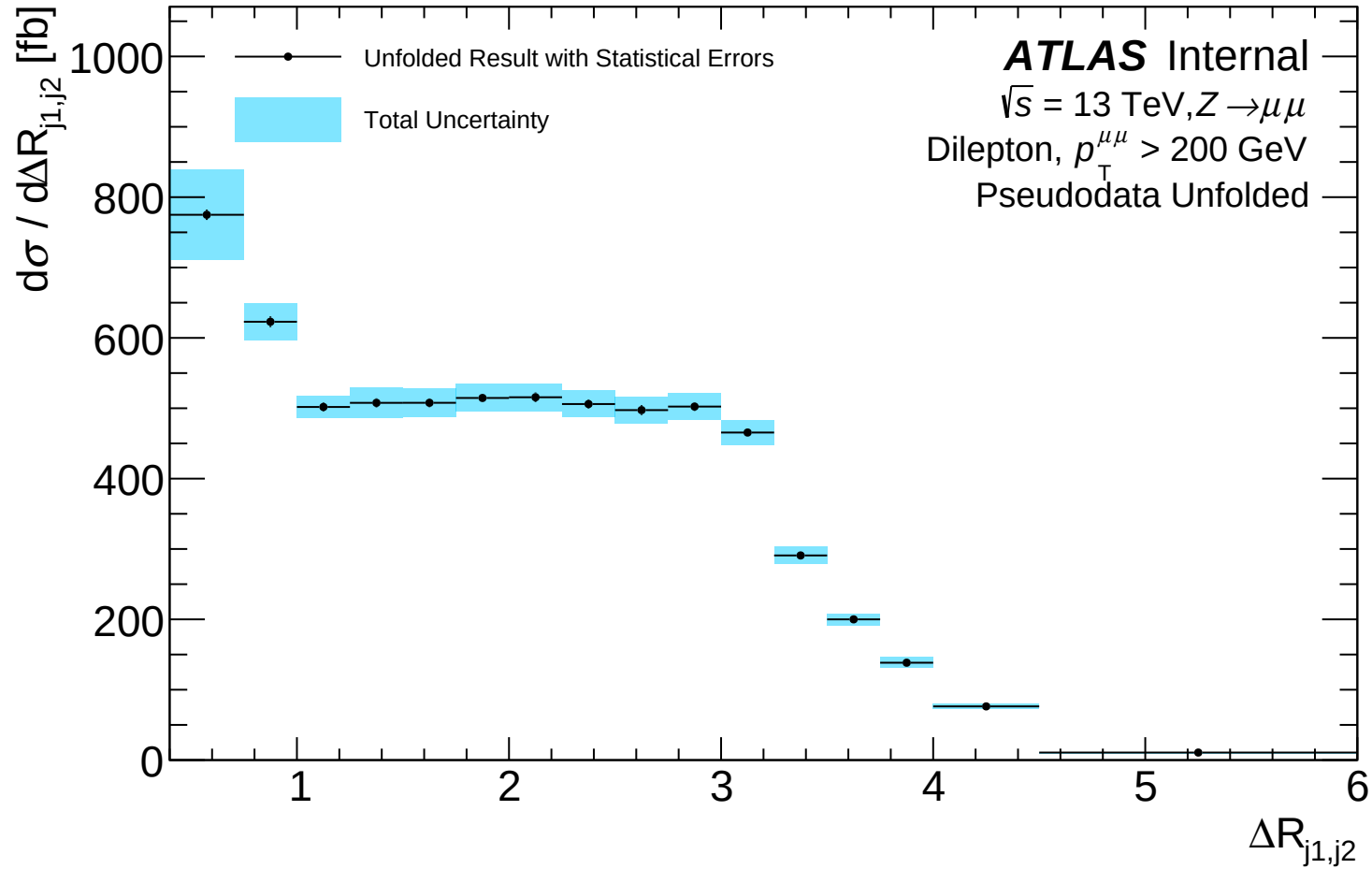




Events







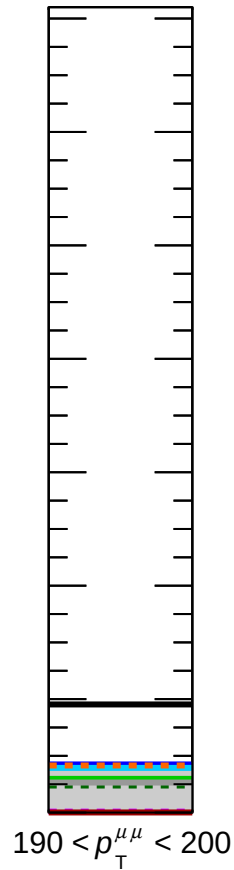
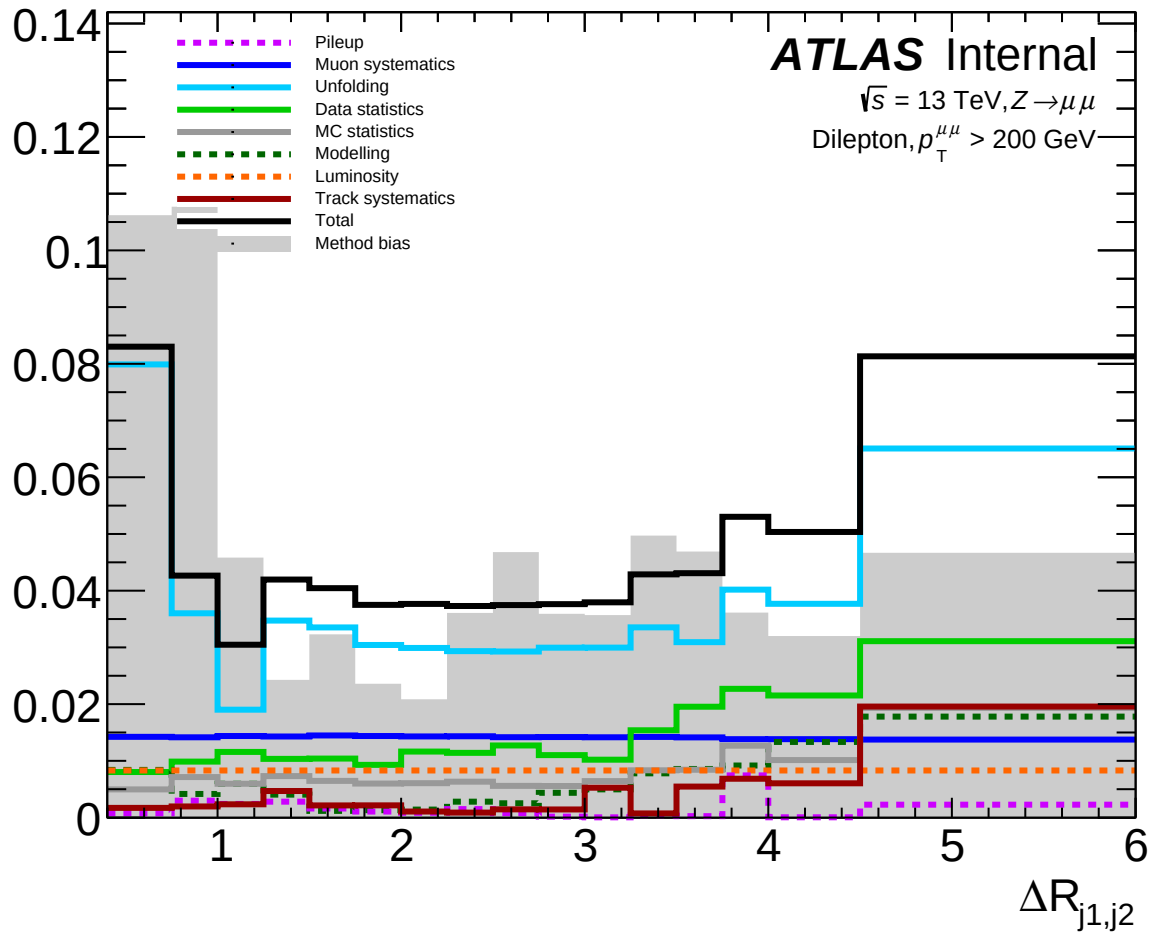
Fractional uncertainty

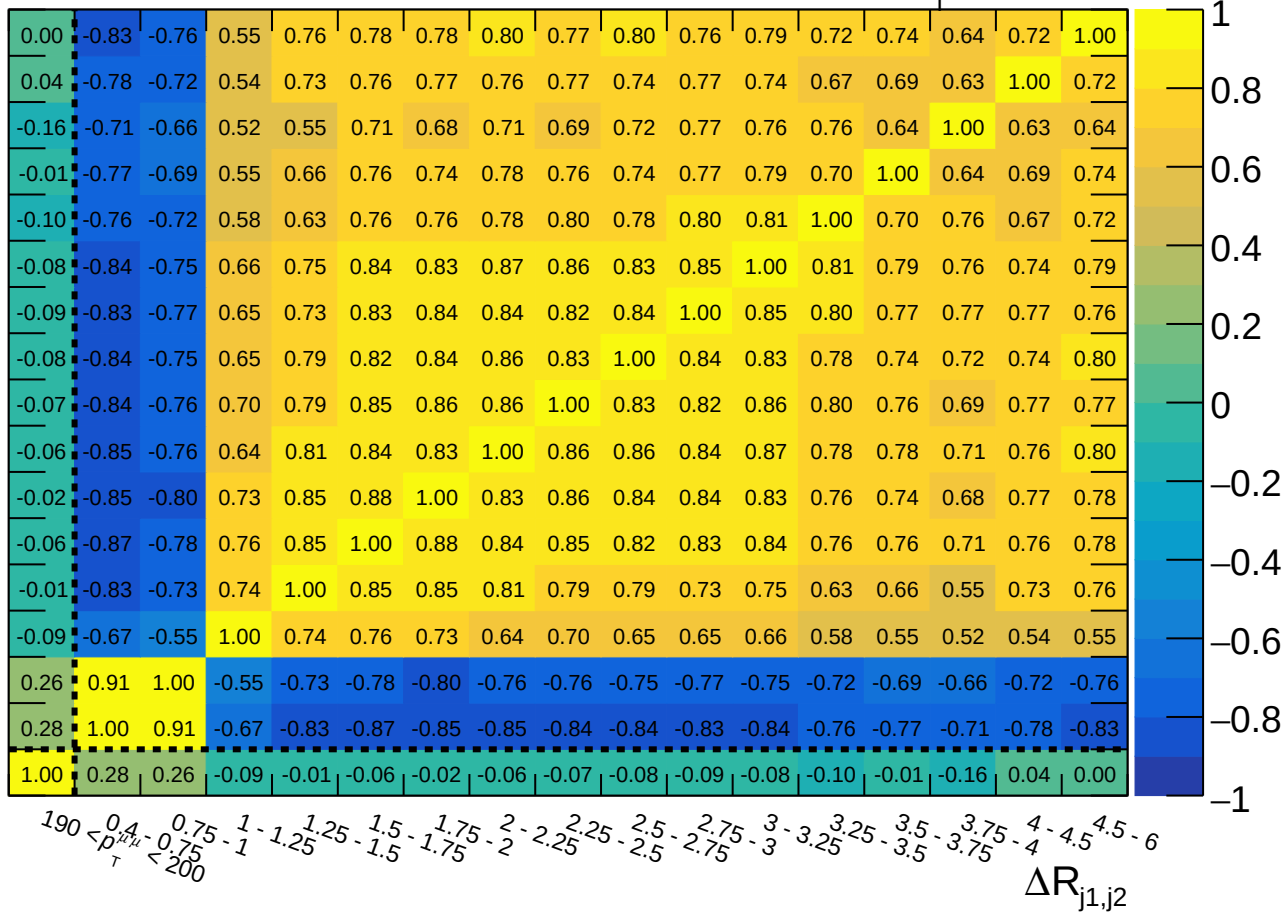
ATLAS Internal

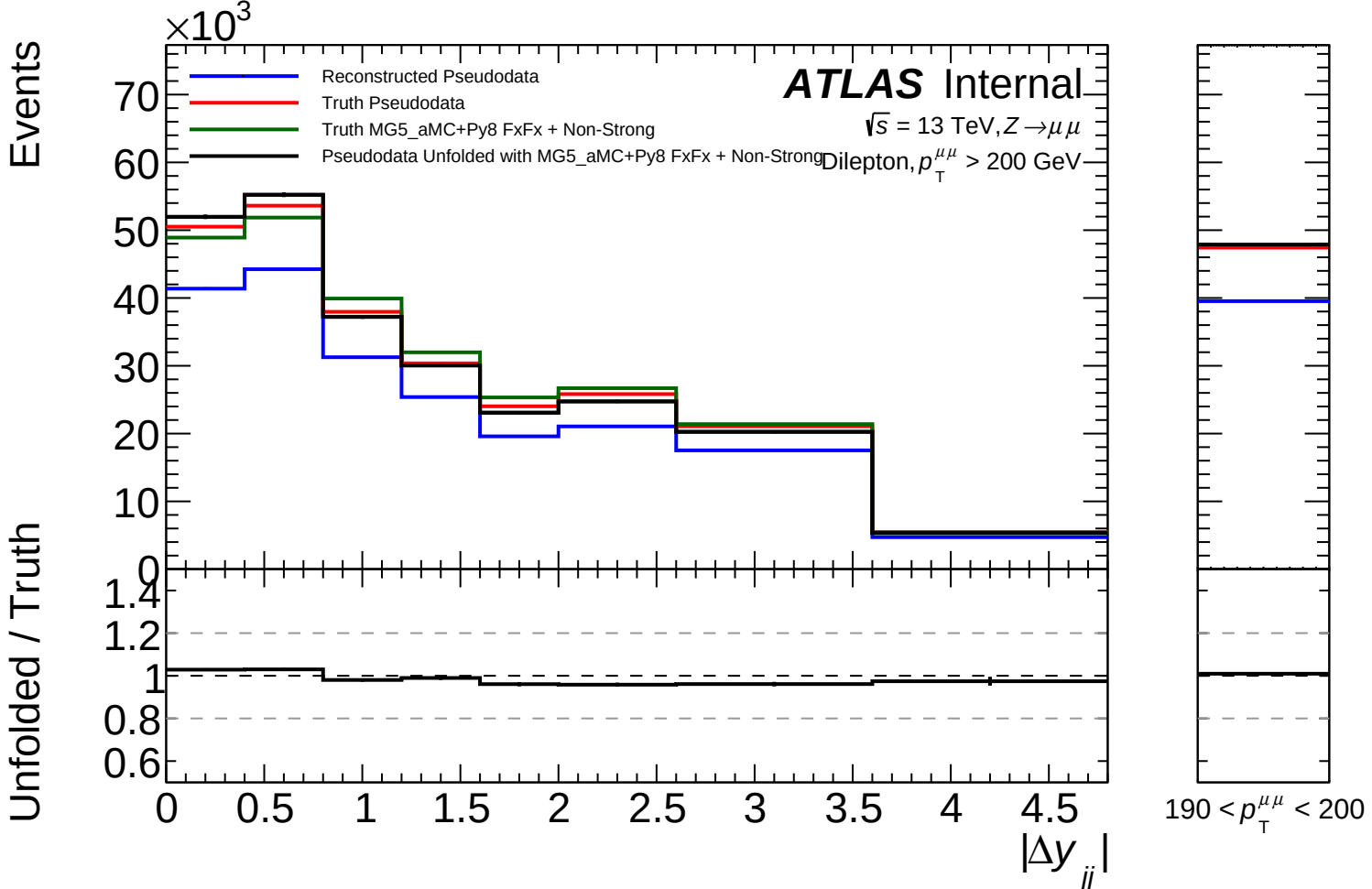
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

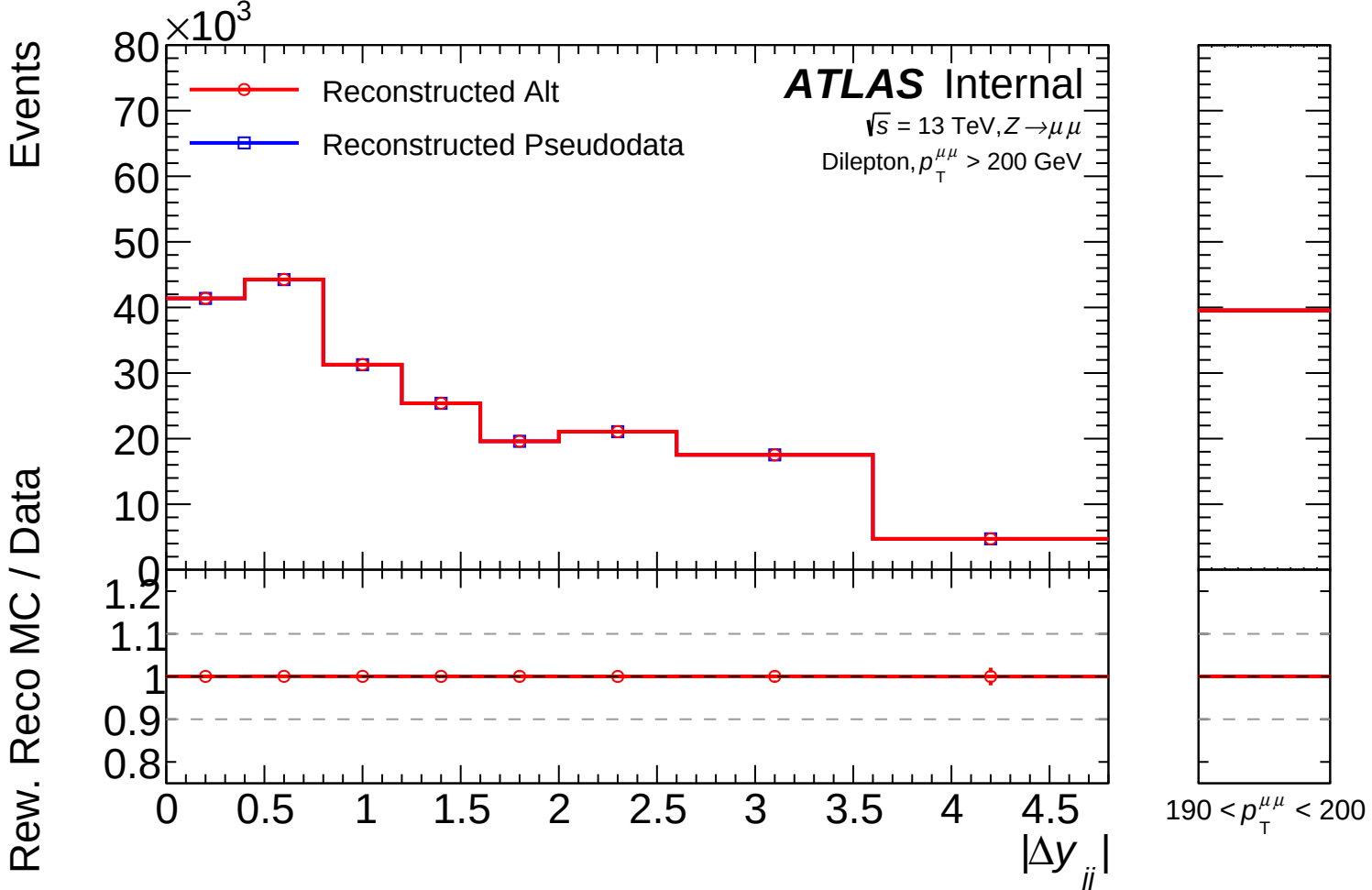
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

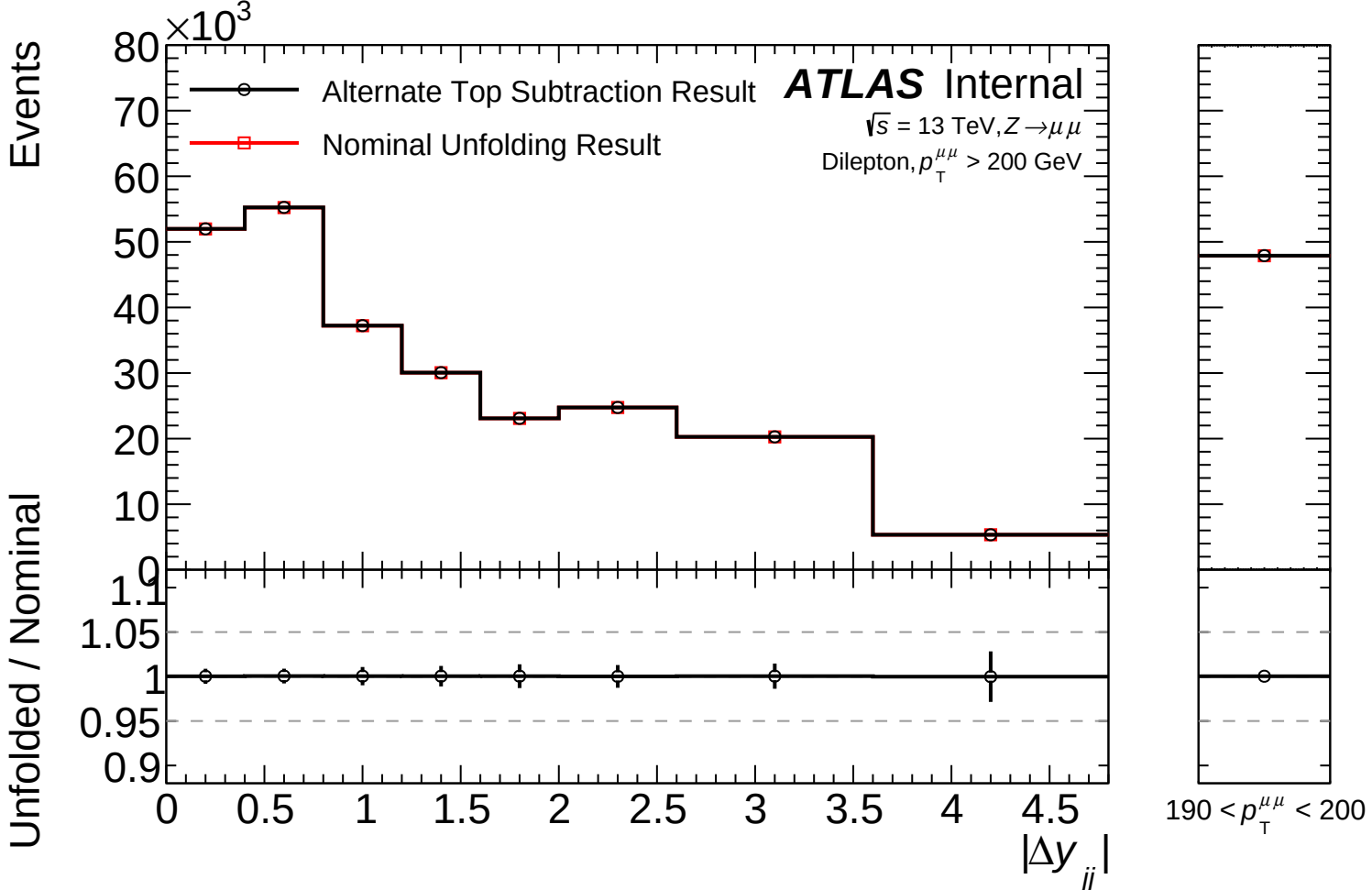
- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
- Modelling
- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ $\Delta R_{j1,j2}$ $190 < p_{\text{T}}^{\mu\mu} < 200$ 







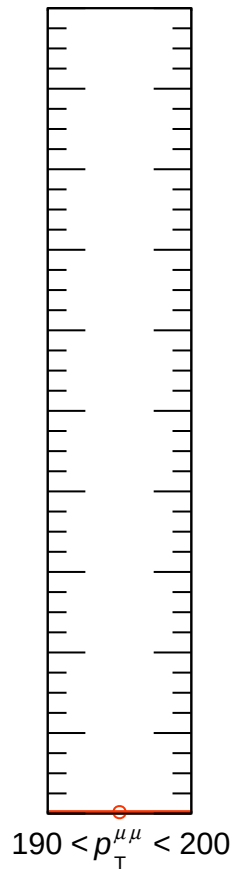
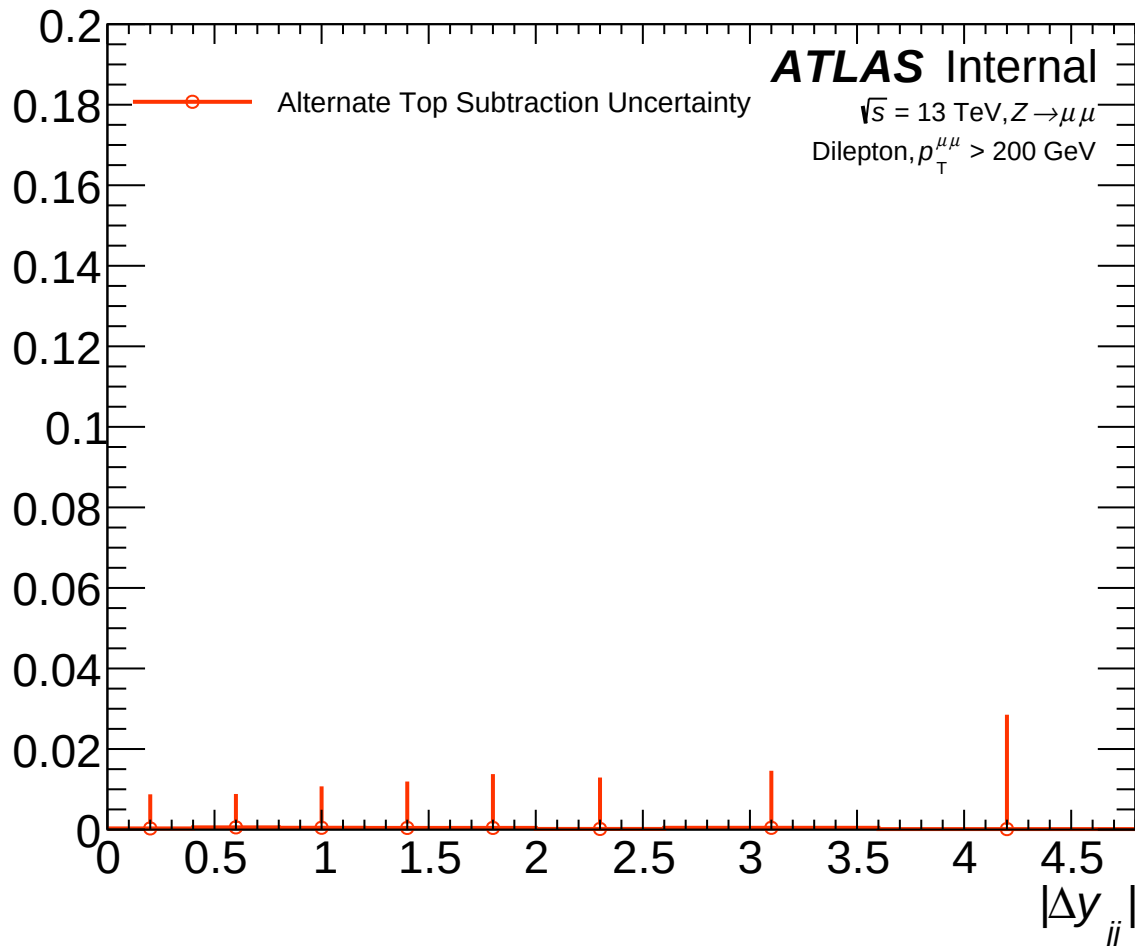
Fractional systematic impact

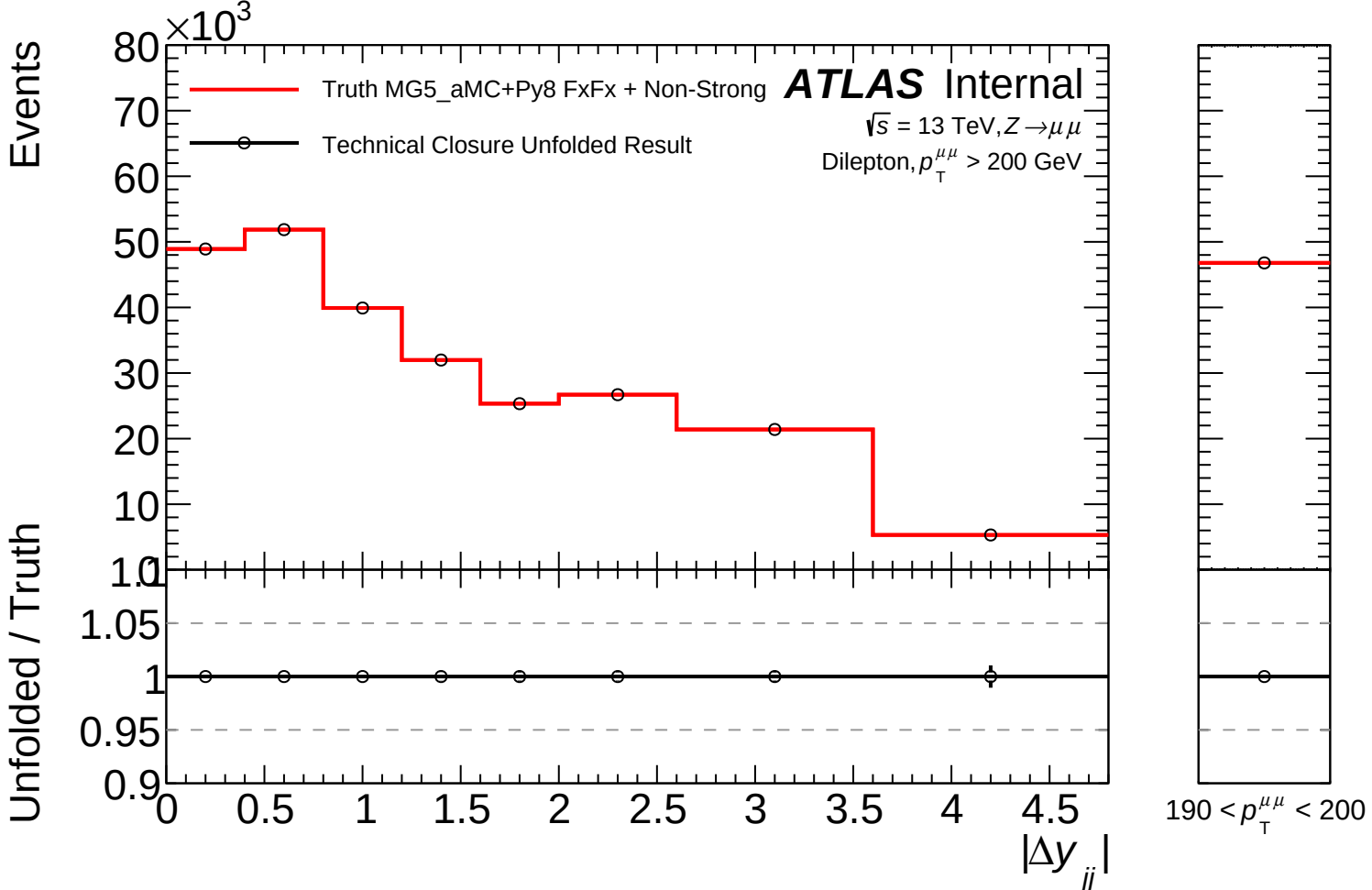
ATLAS Internal

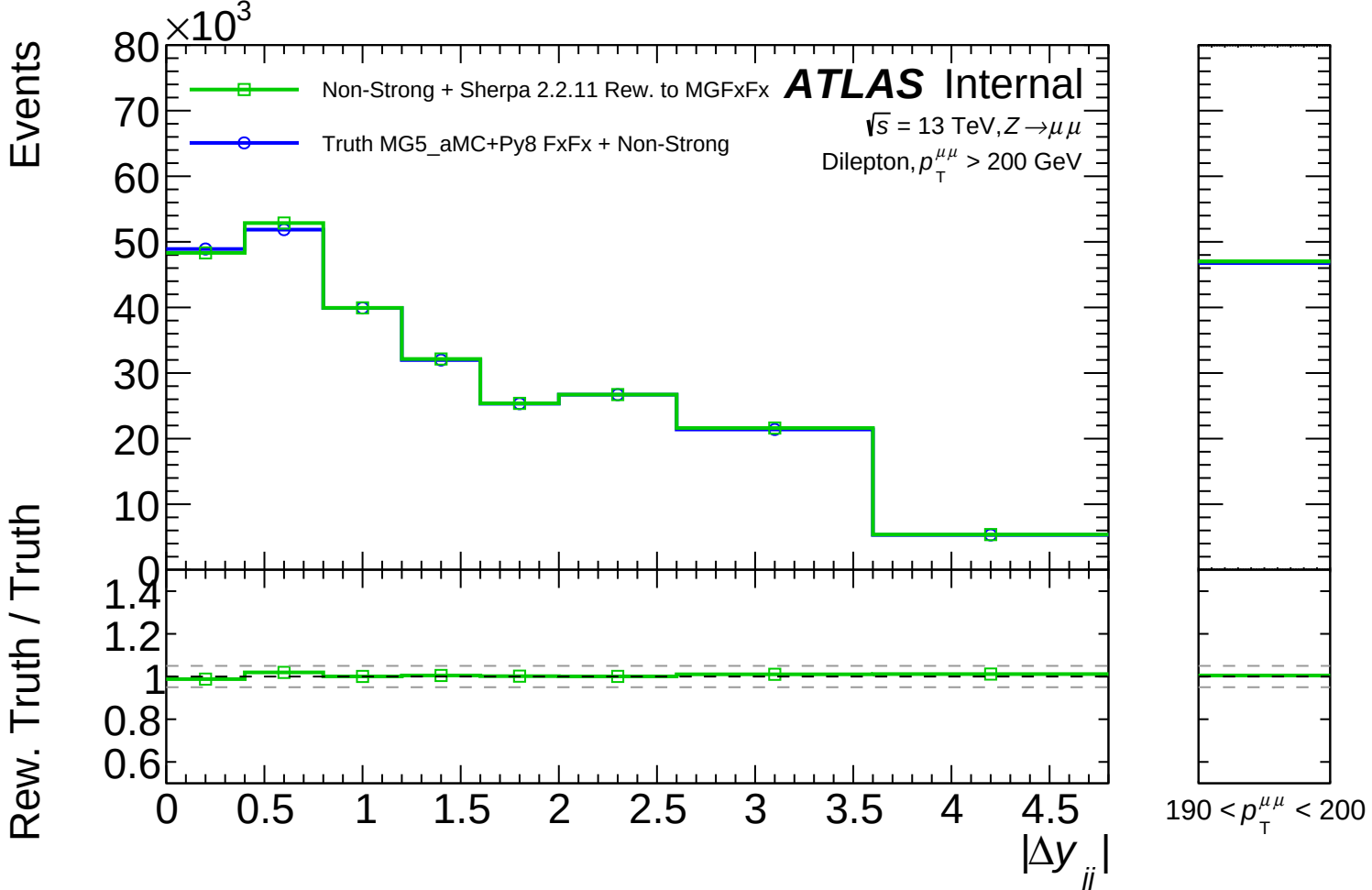
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

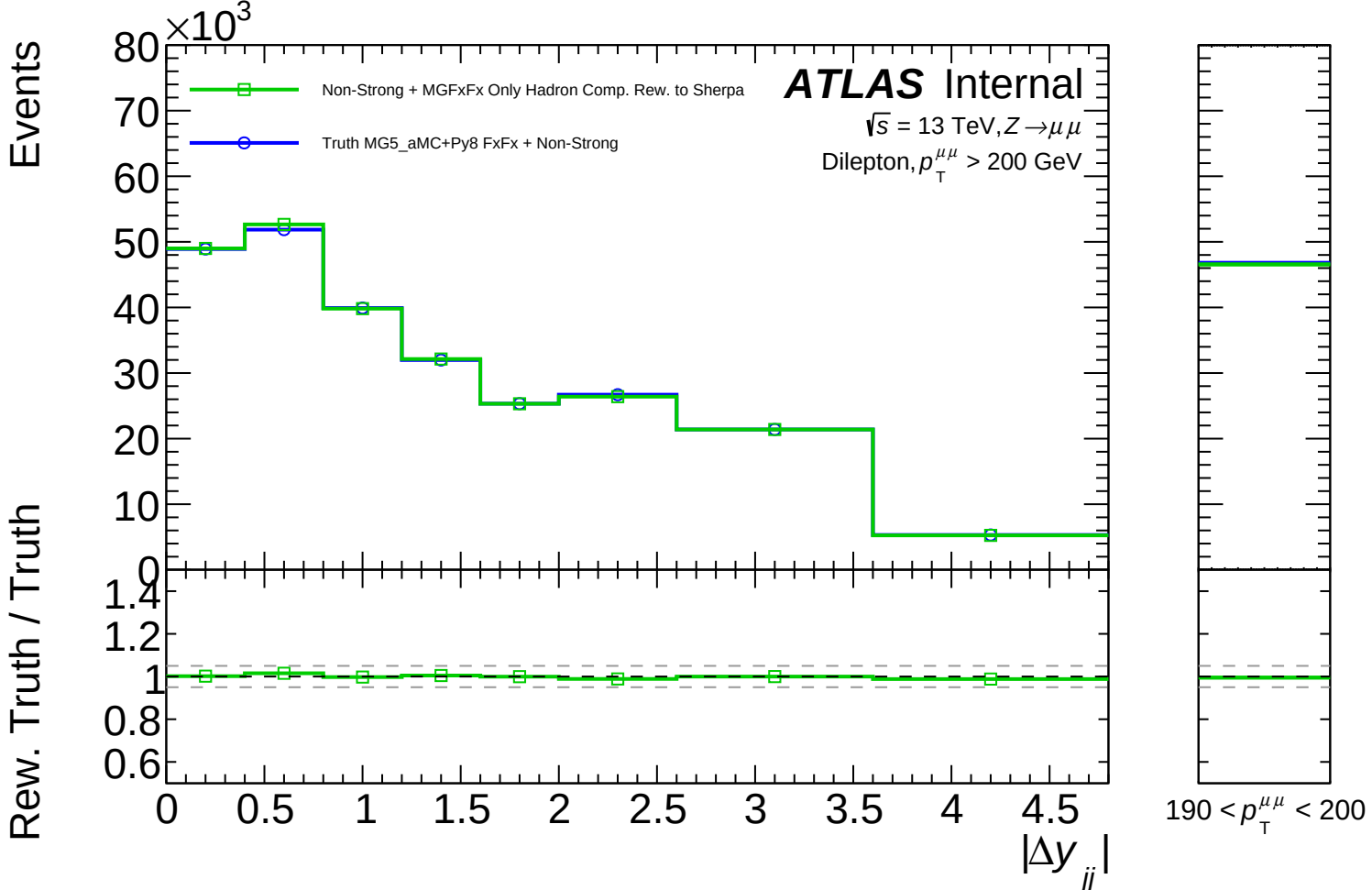
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

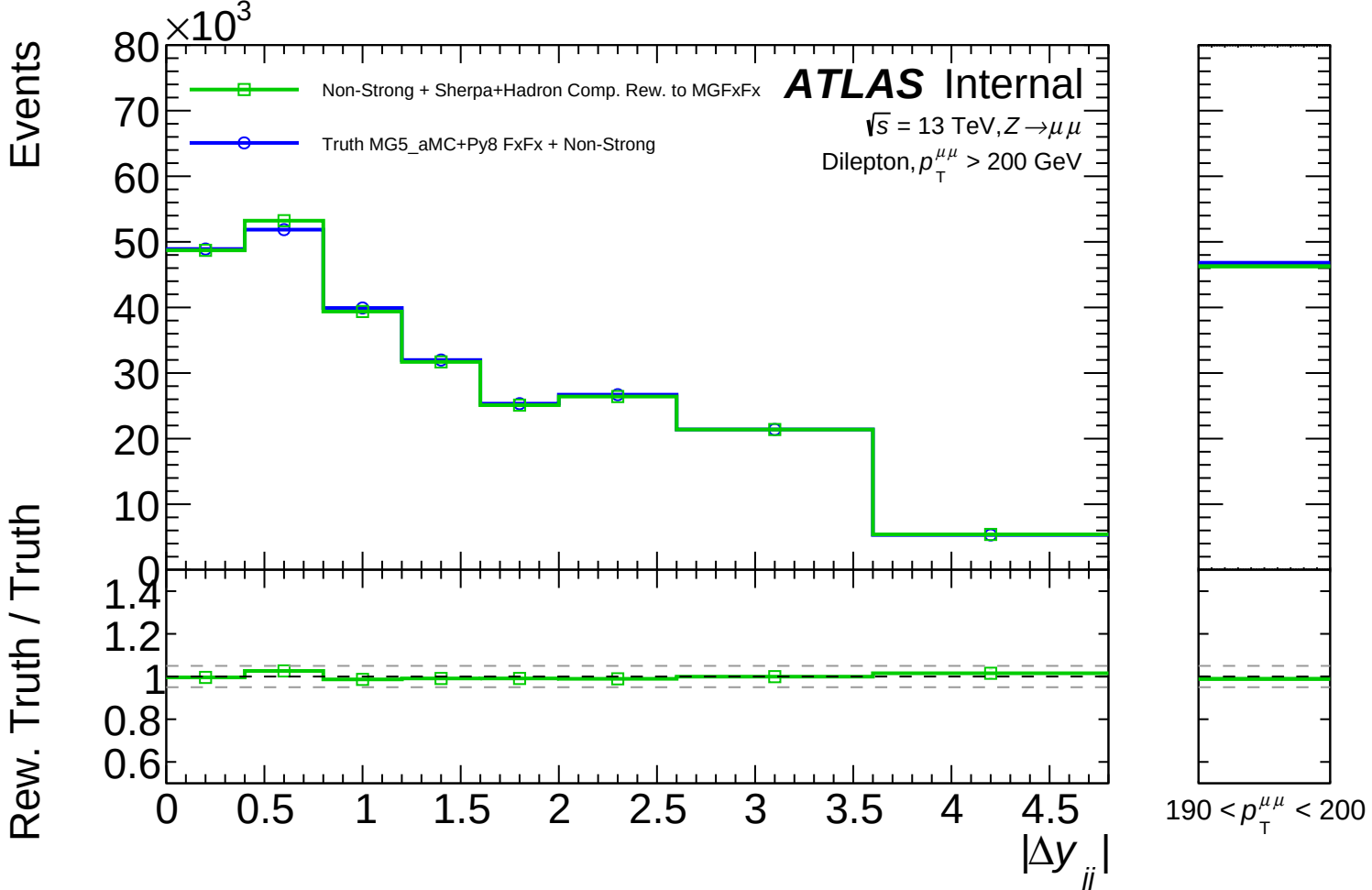
Alternate Top Subtraction Uncertainty

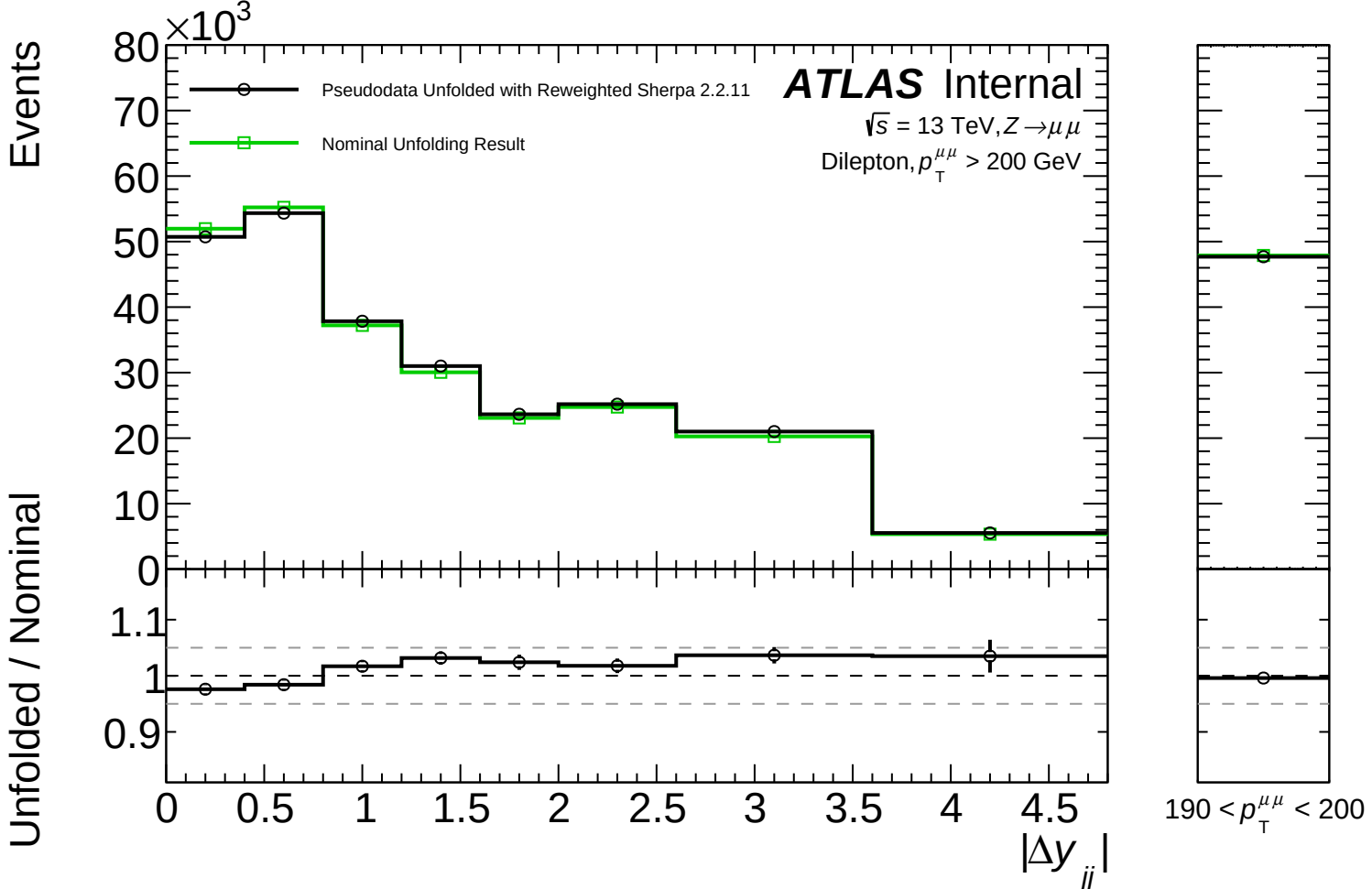


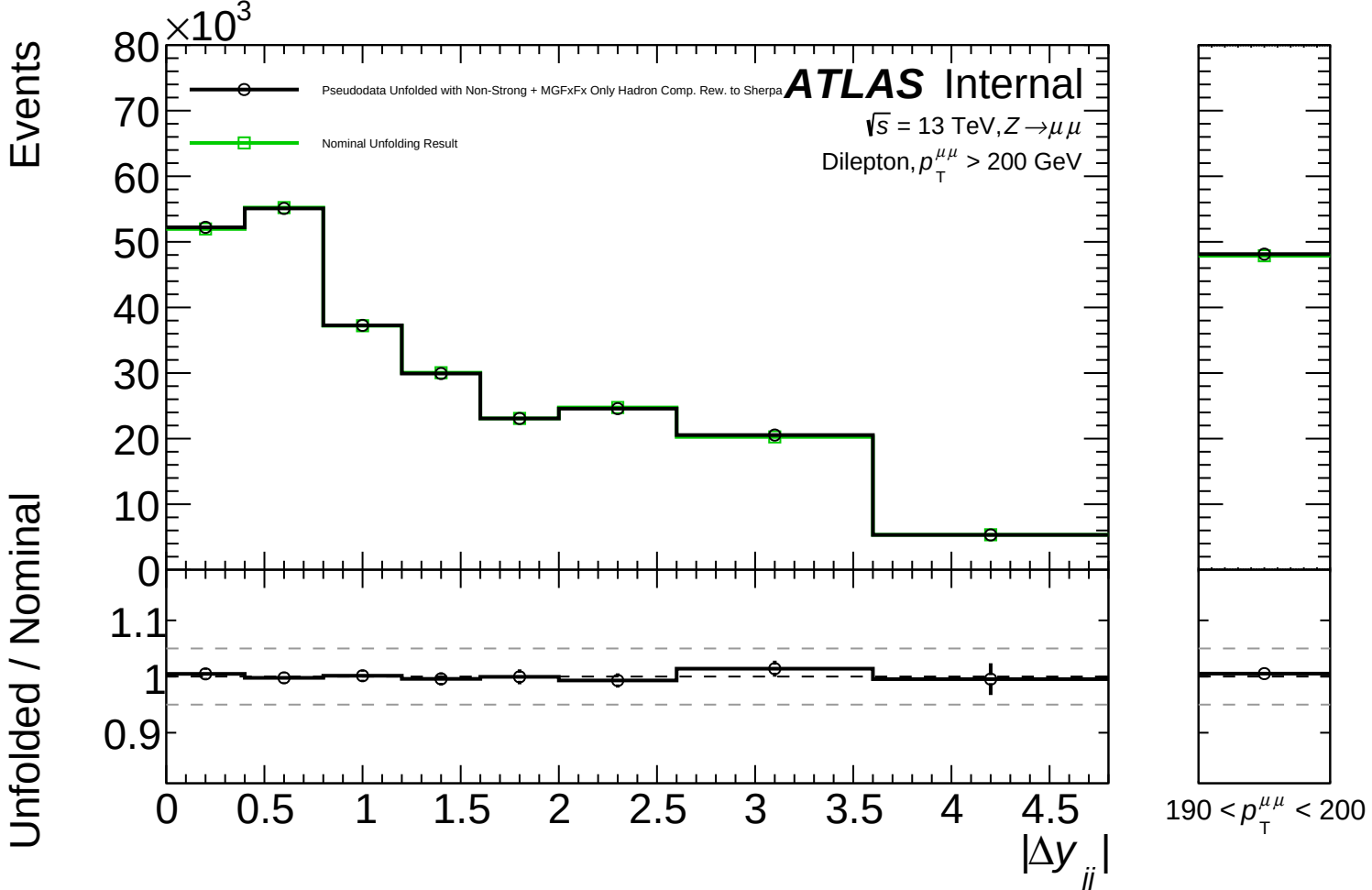


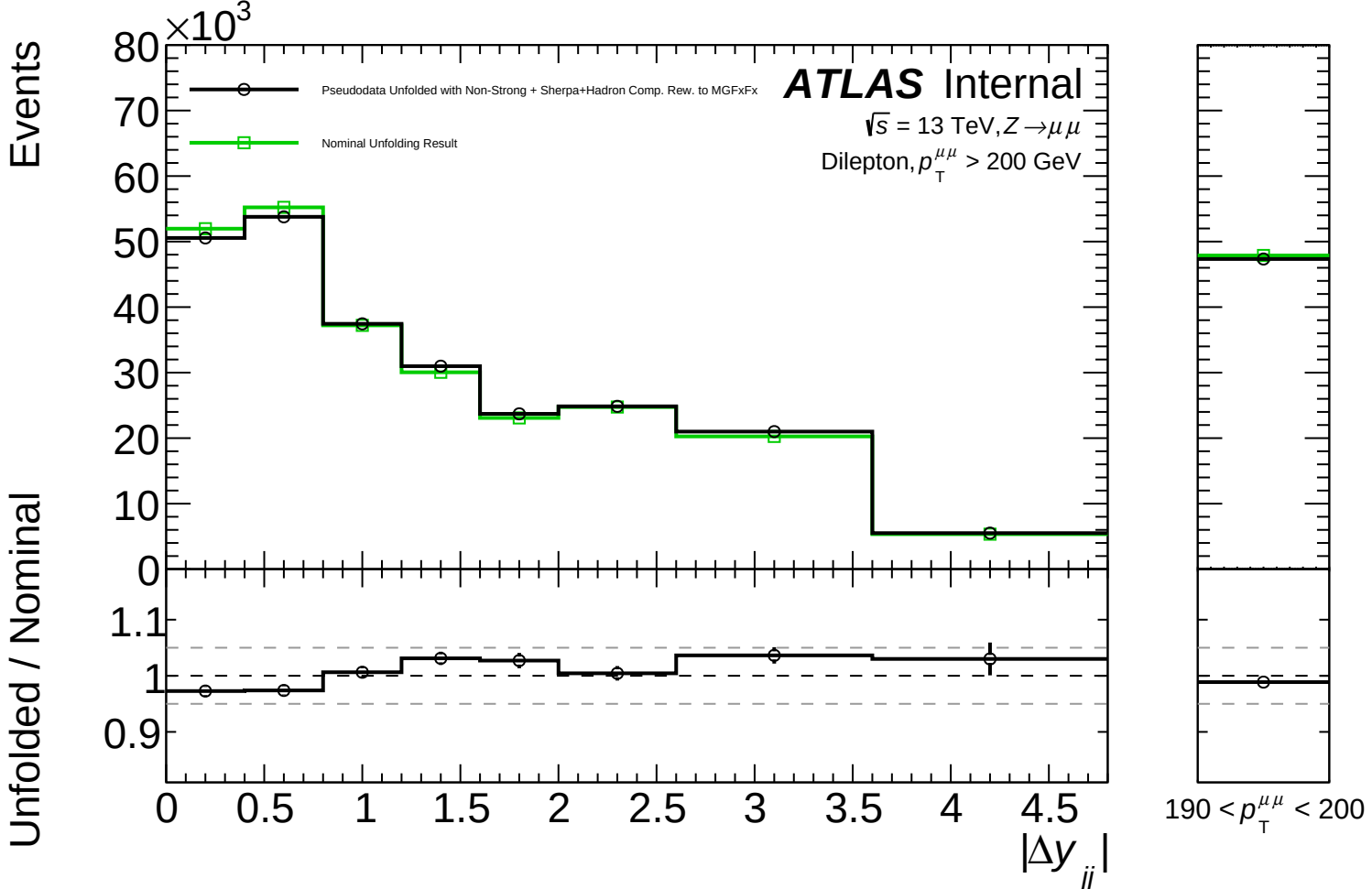


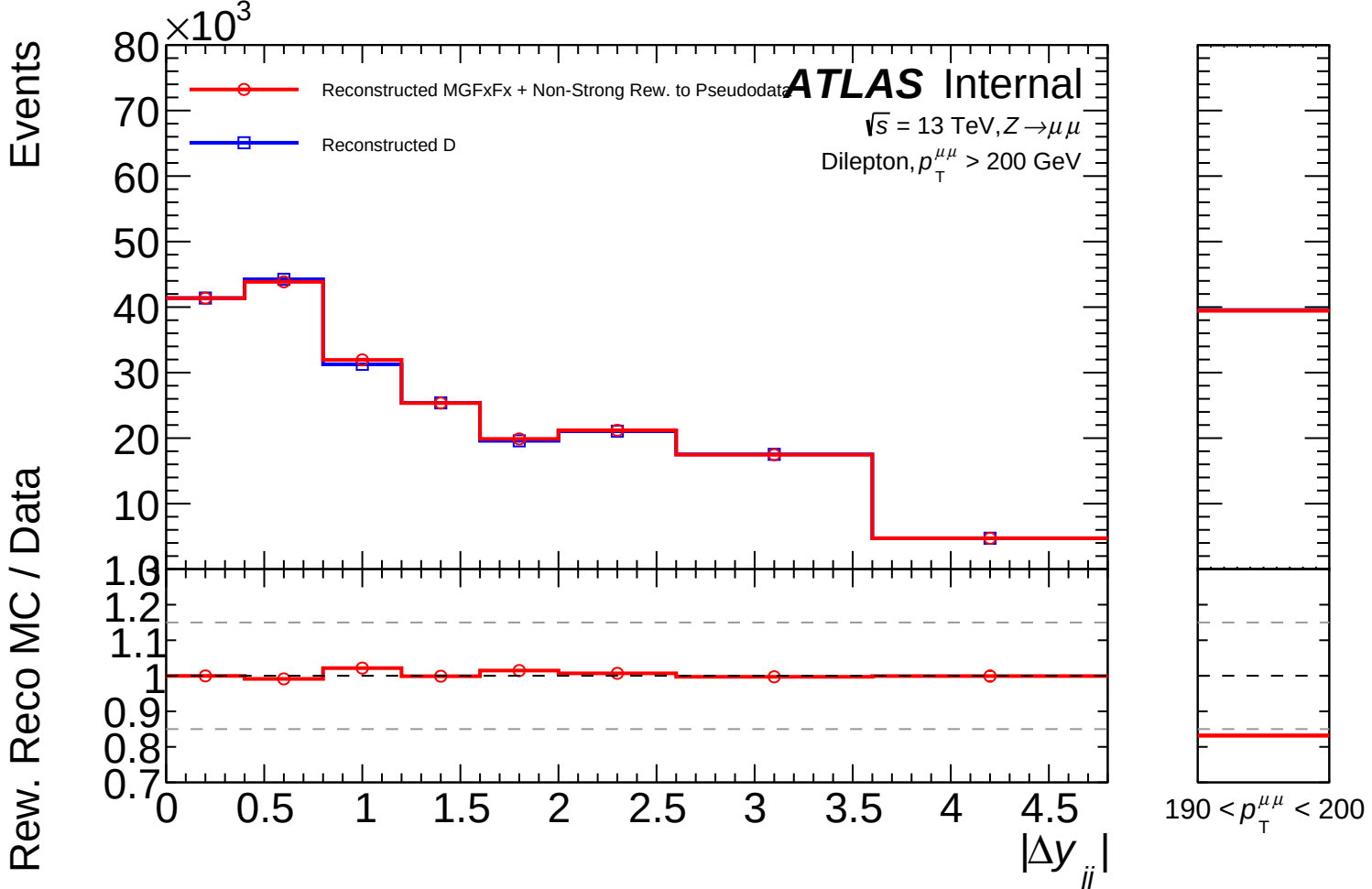


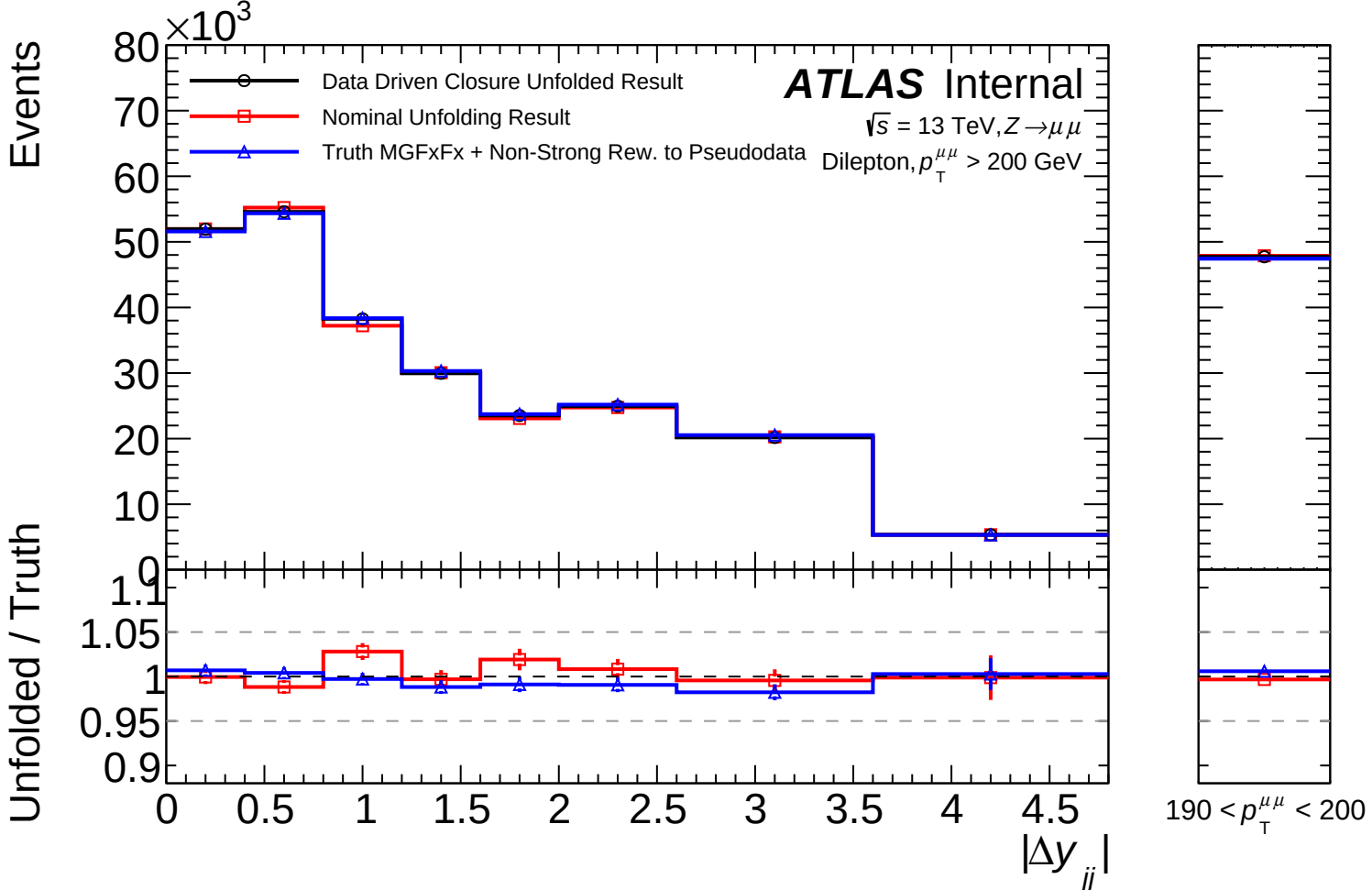




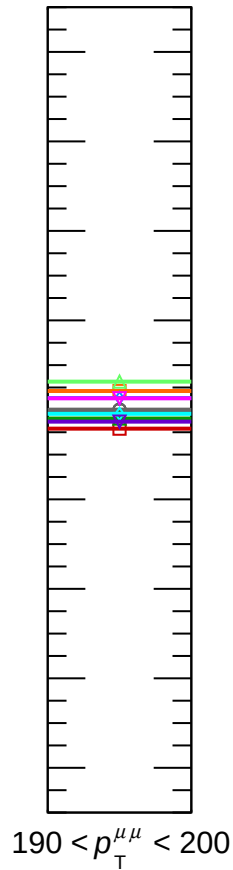
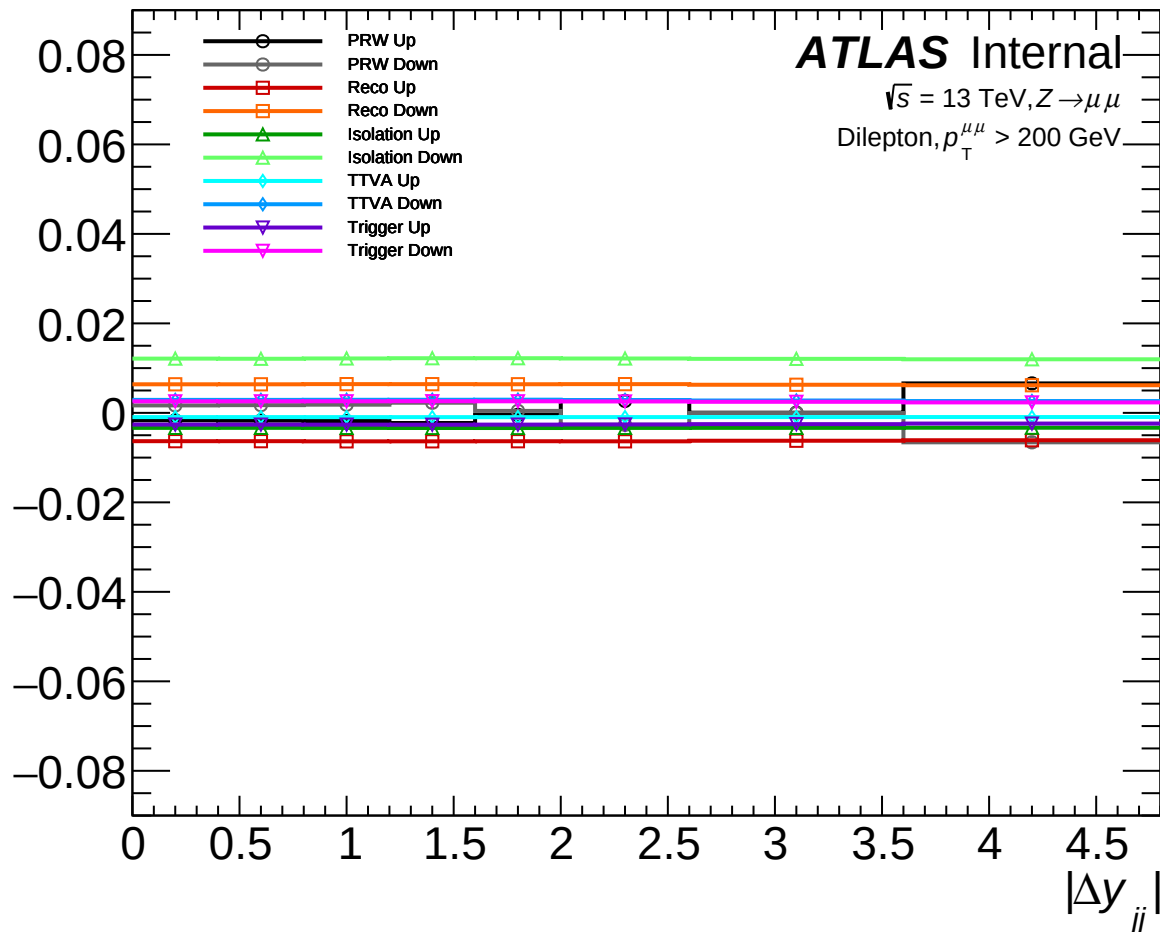




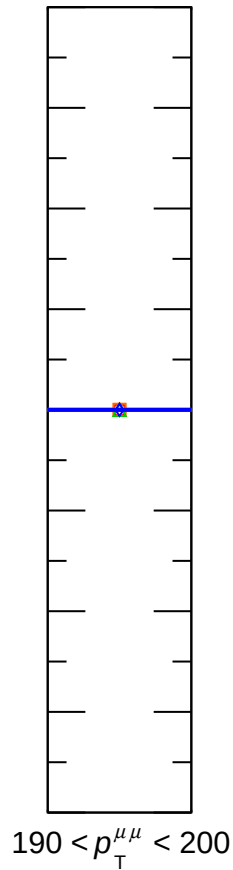
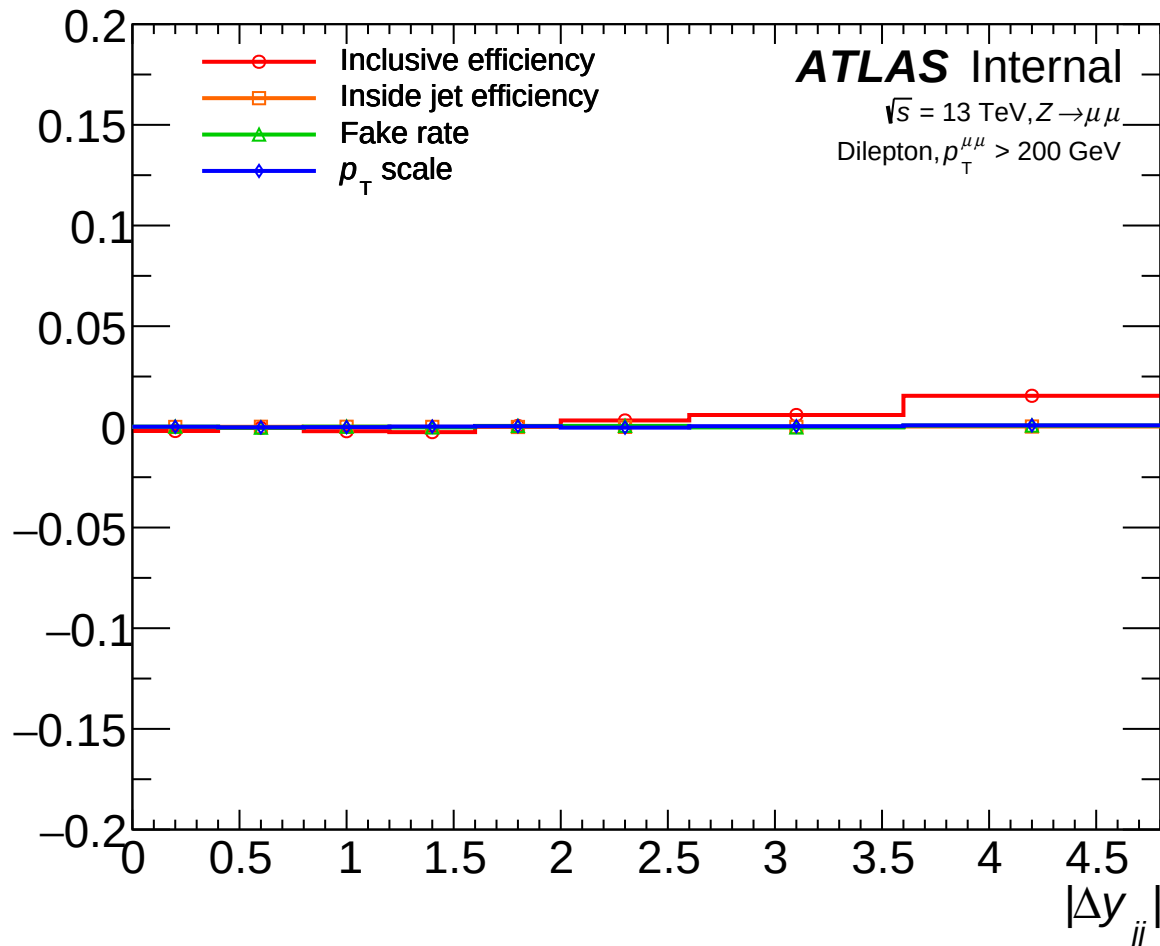




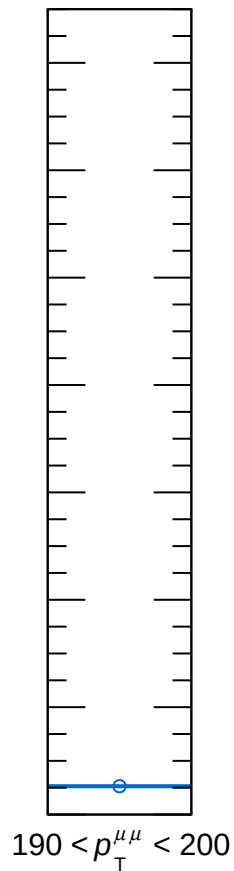
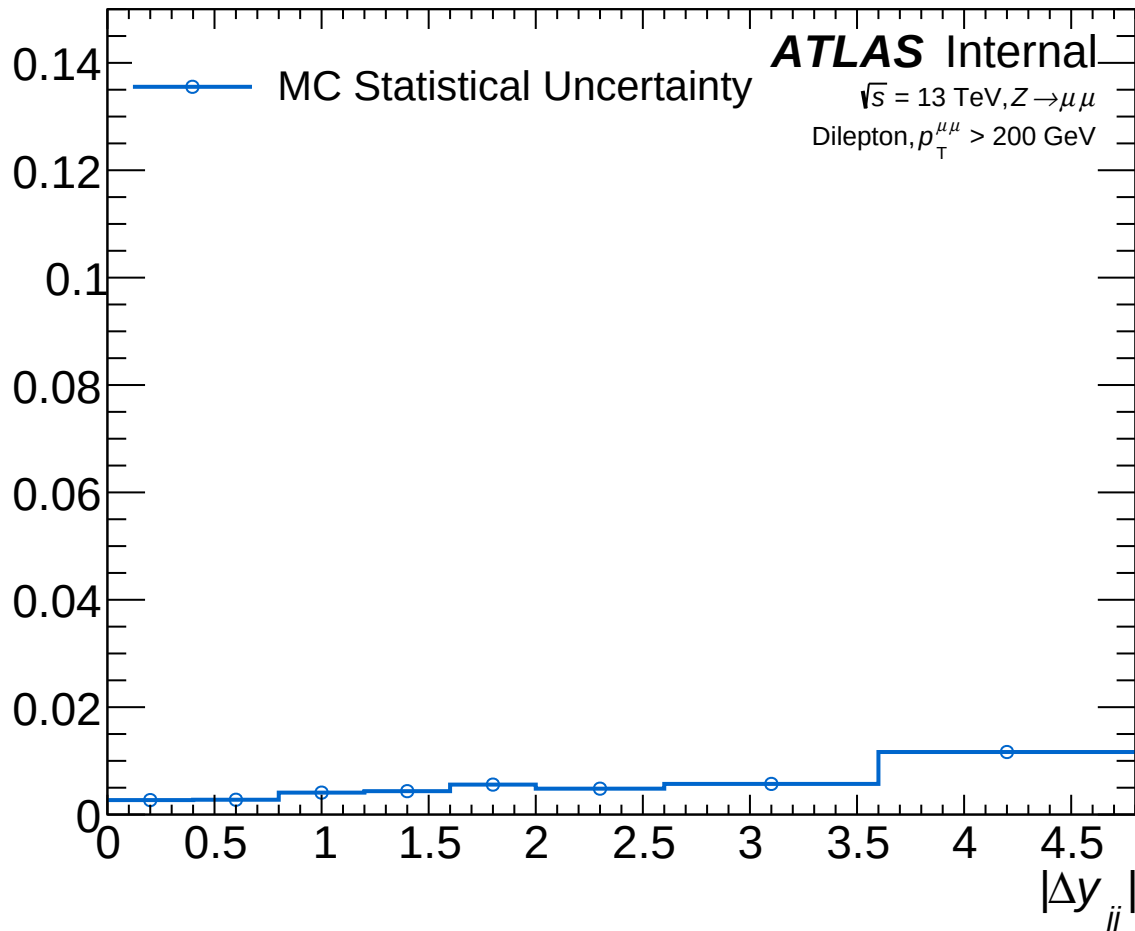
Fractional systematic impact



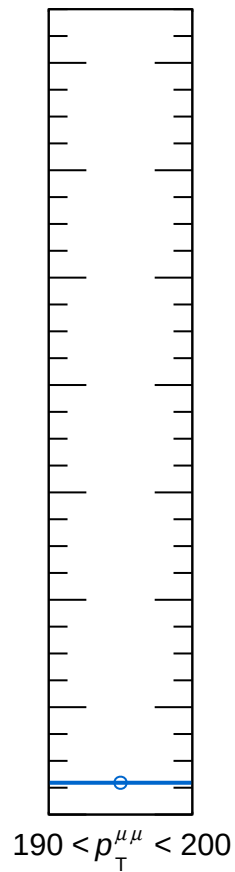
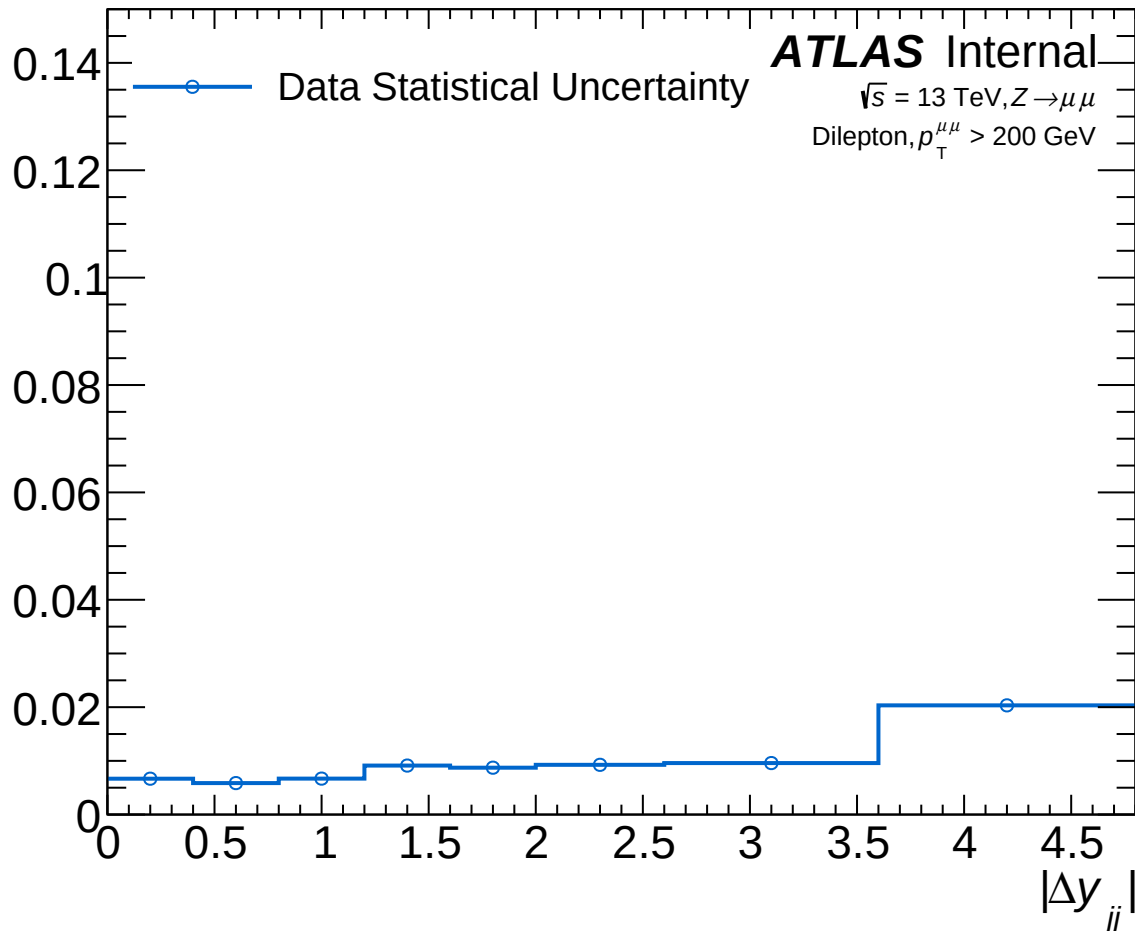
Fractional systematic impact



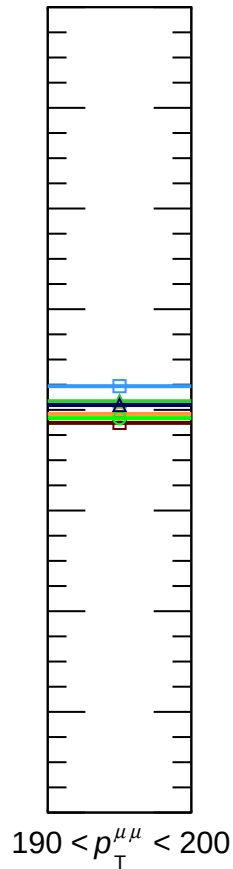
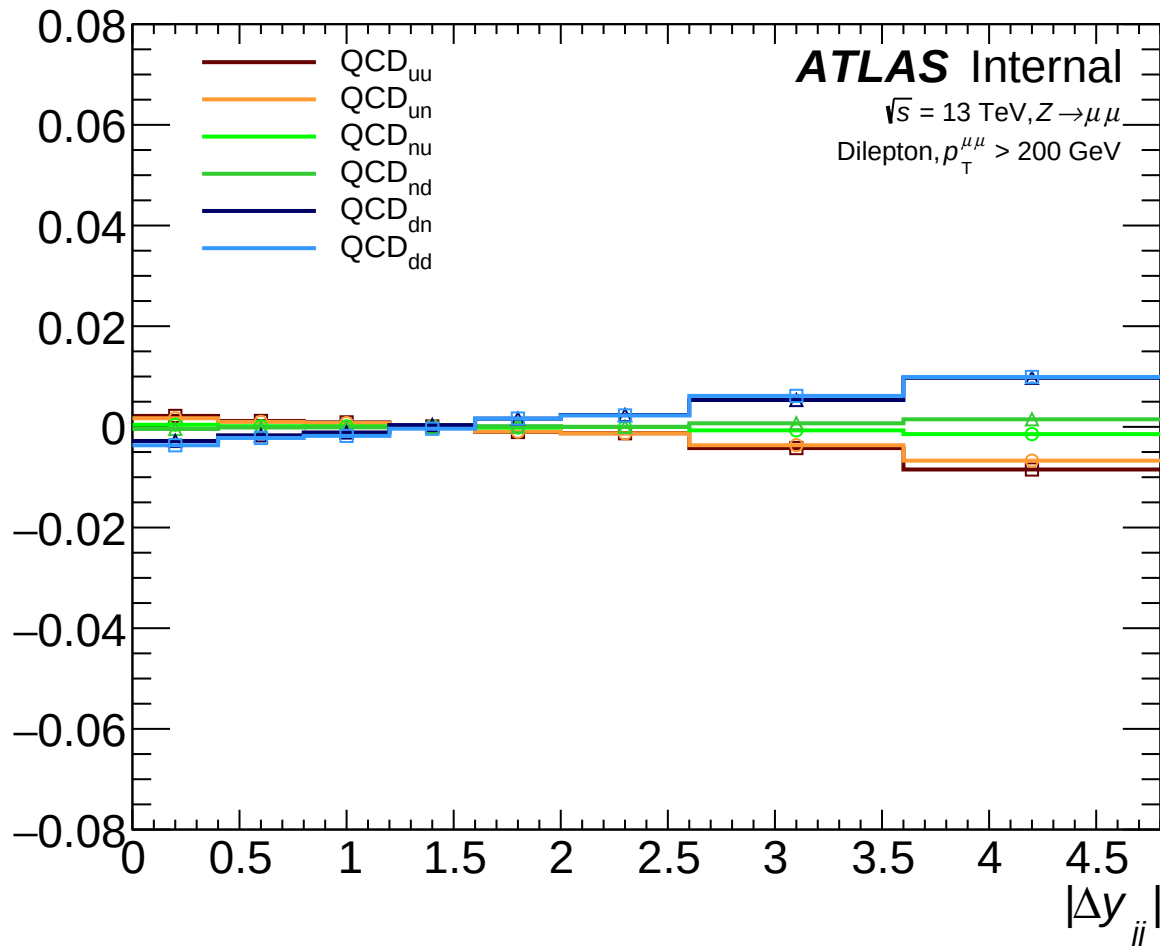
Fractional systematic impact



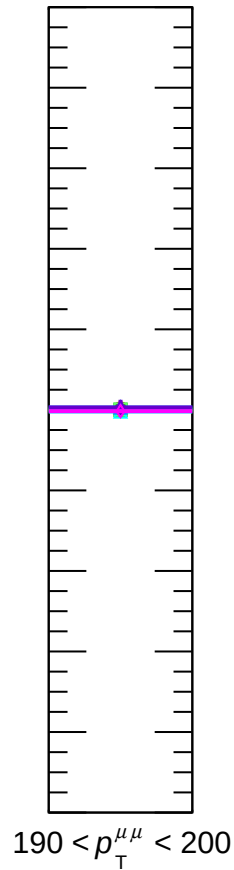
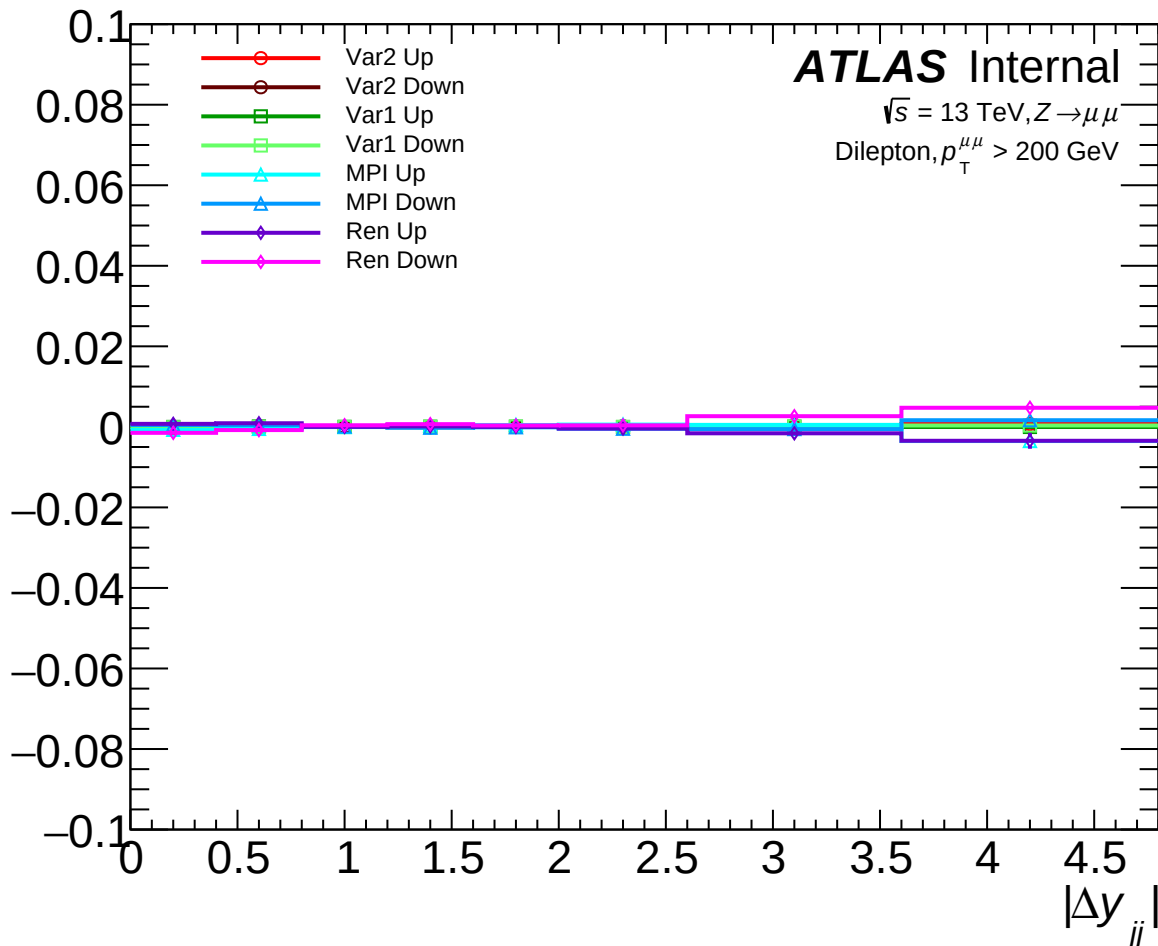
Fractional systematic impact

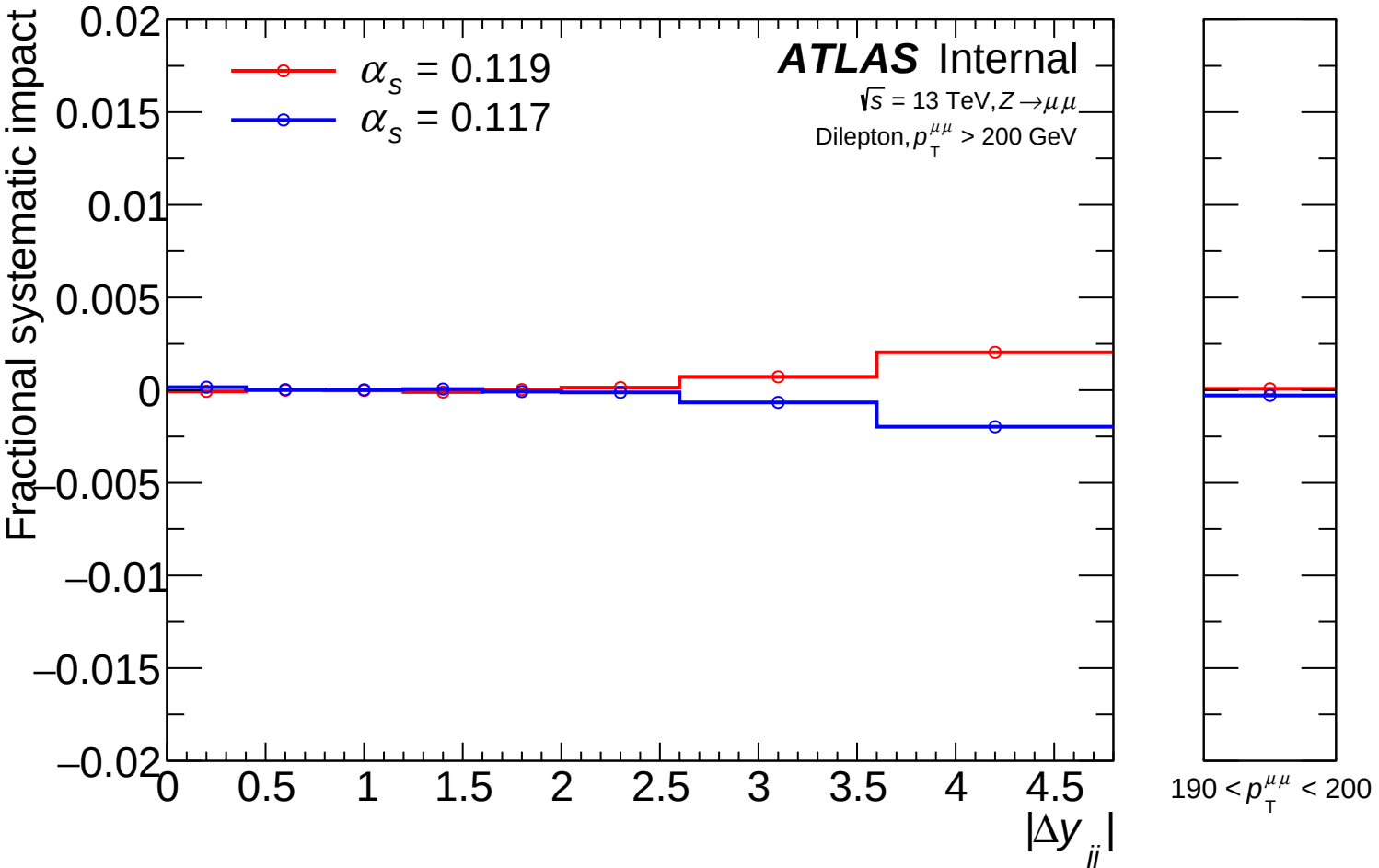


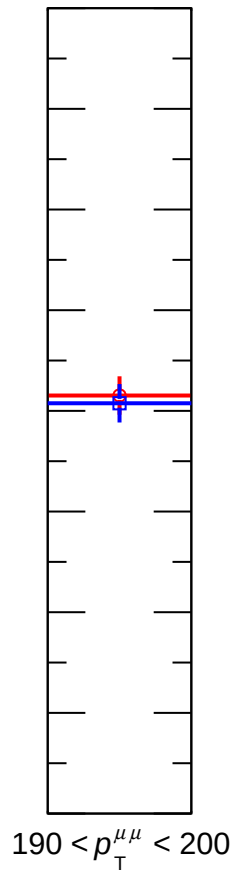
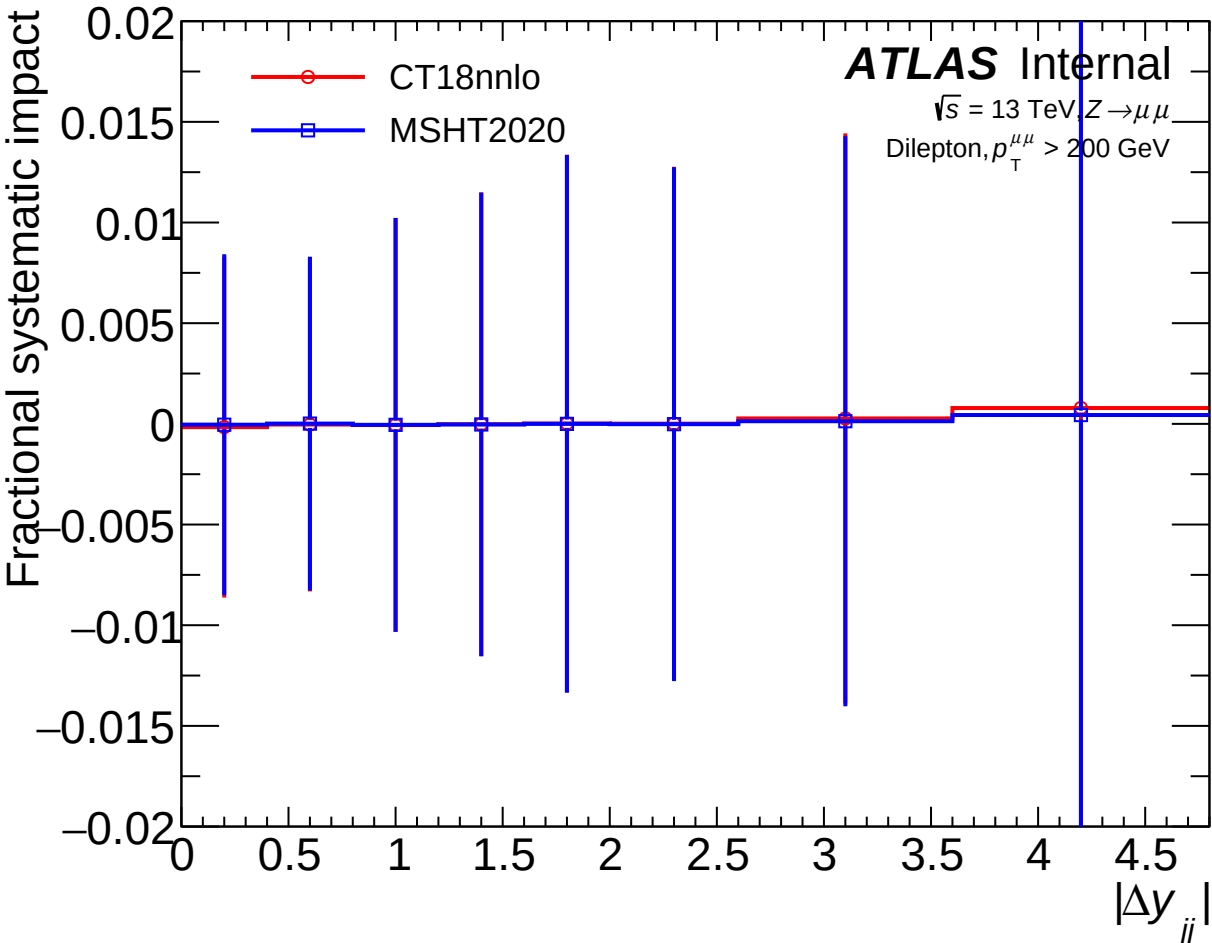
Fractional systematic impact



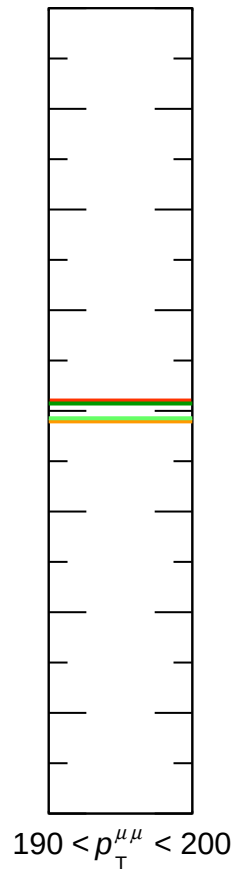
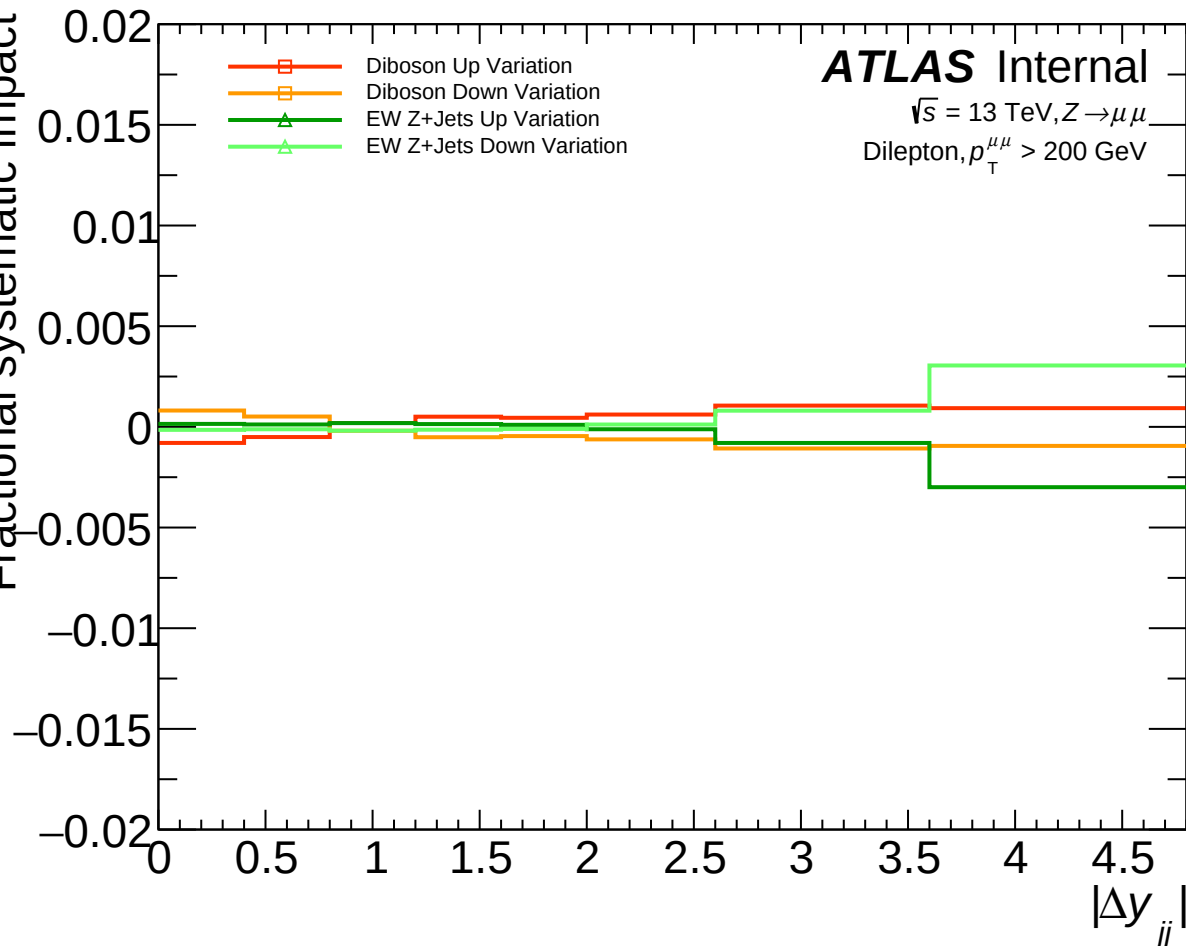
Fractional systematic impact







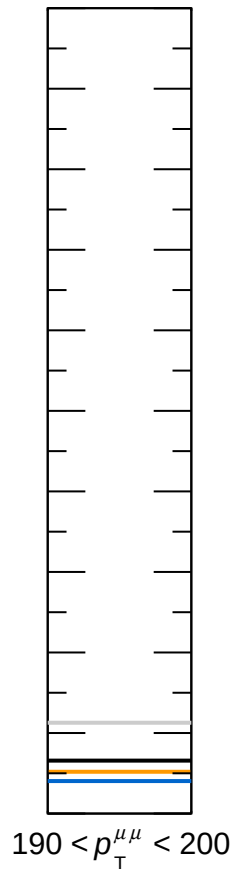
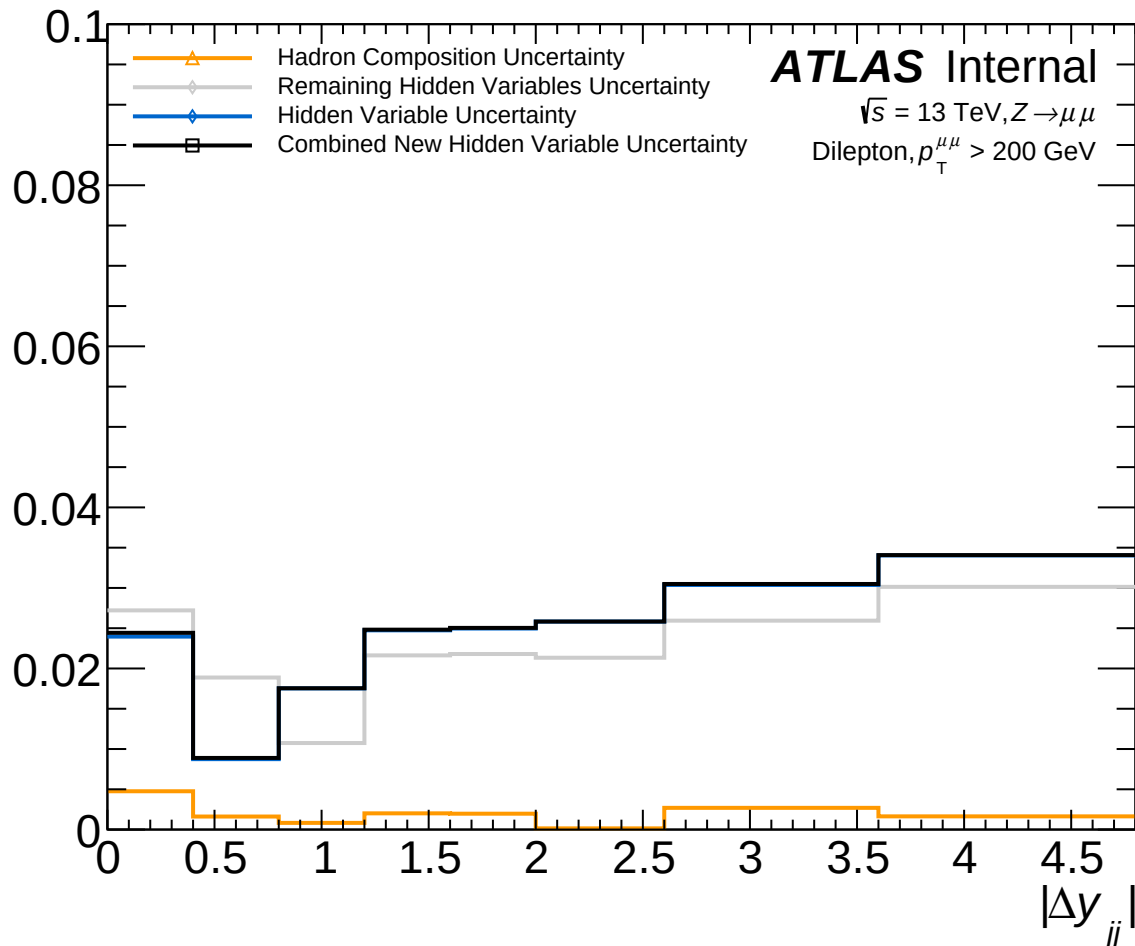
Fractional systematic impact

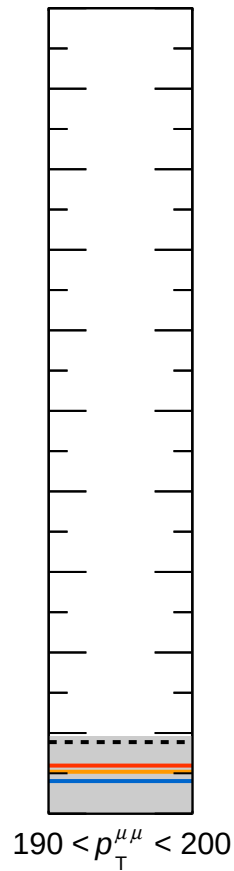
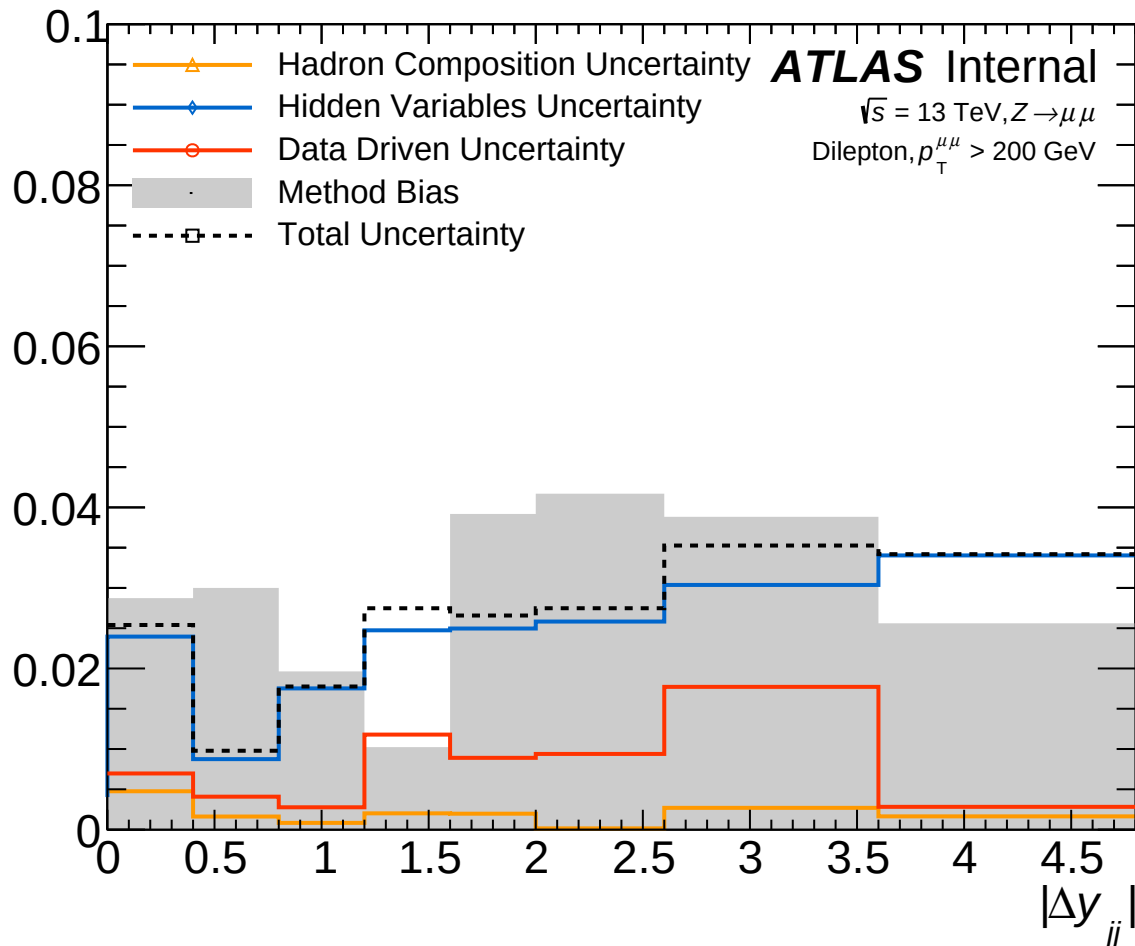


ATLAS Internal

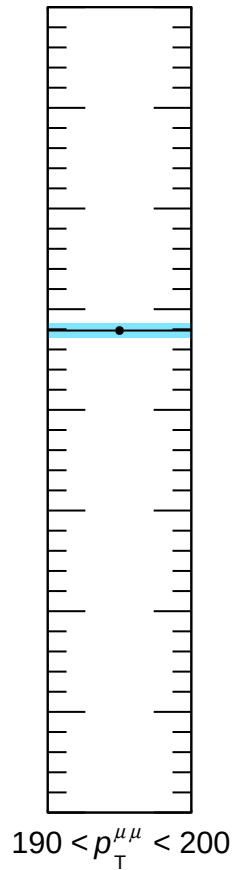
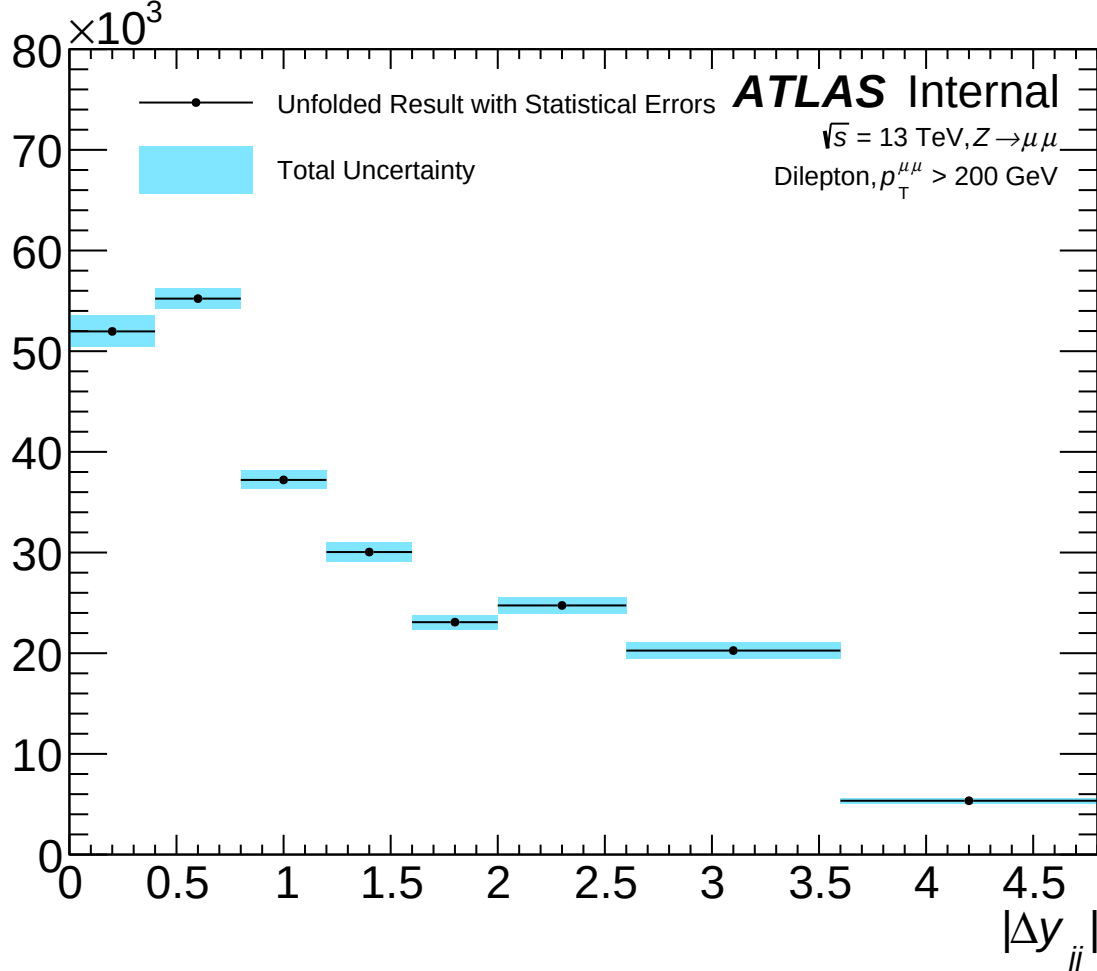
$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

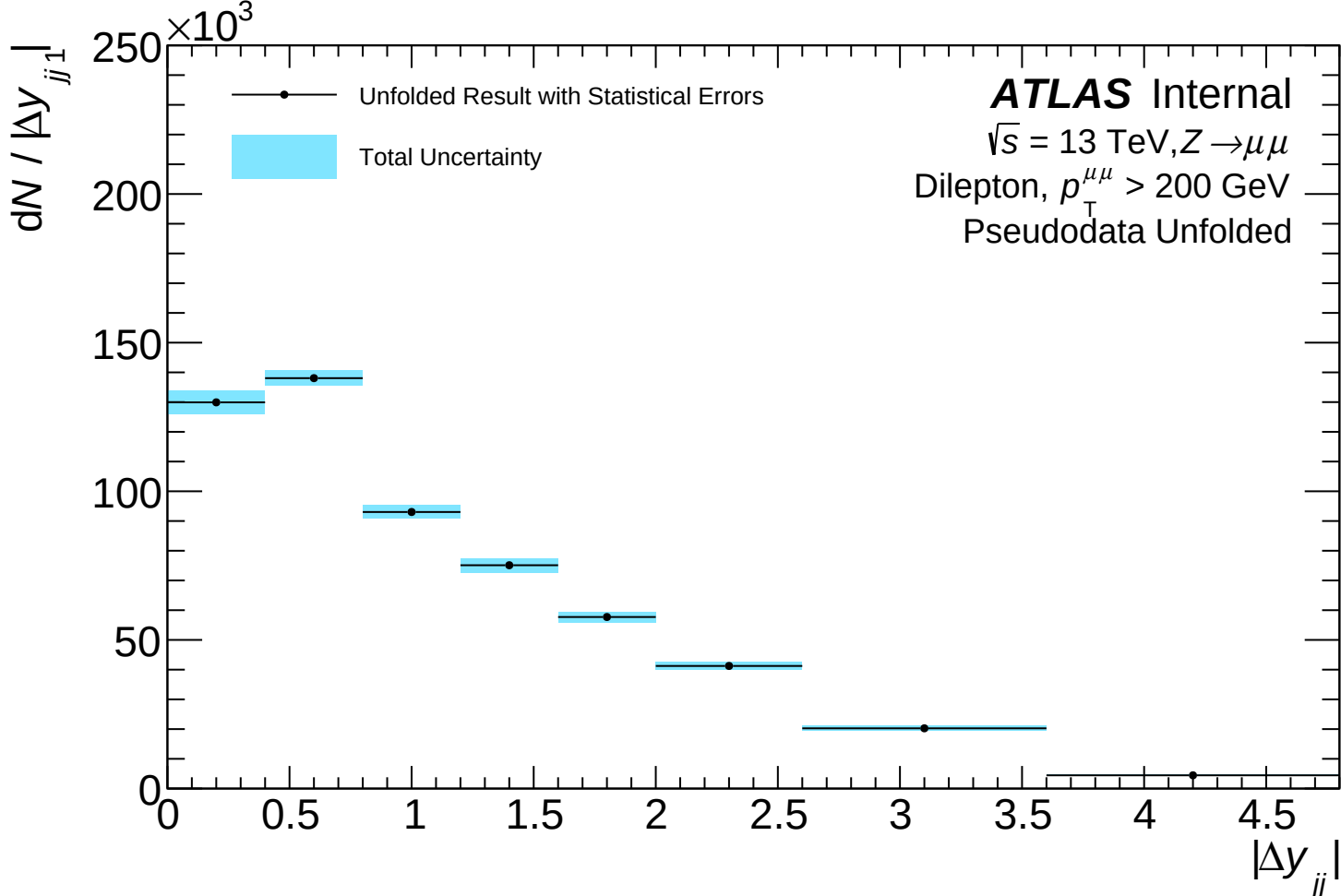
Dilepton, $p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$

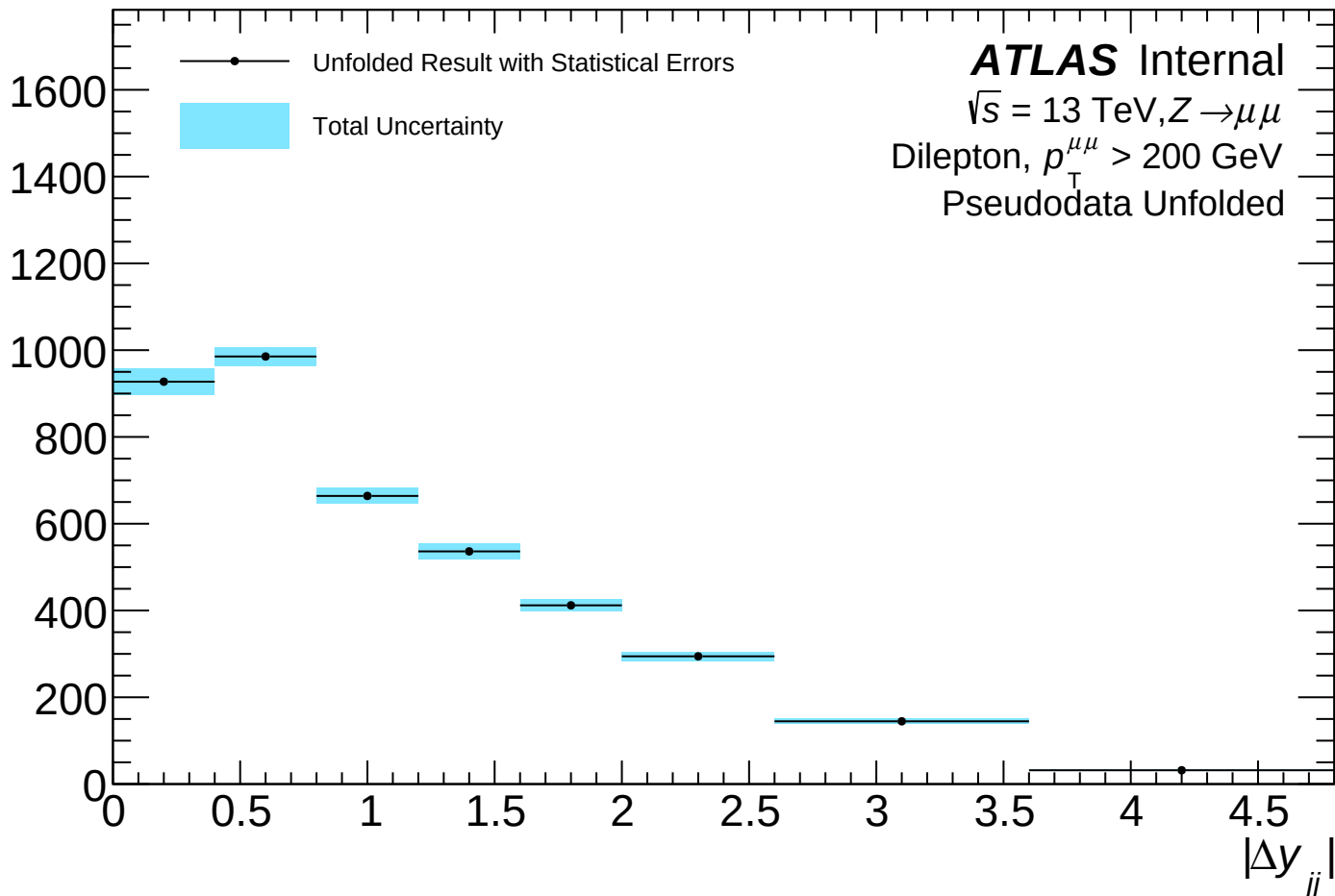




Events







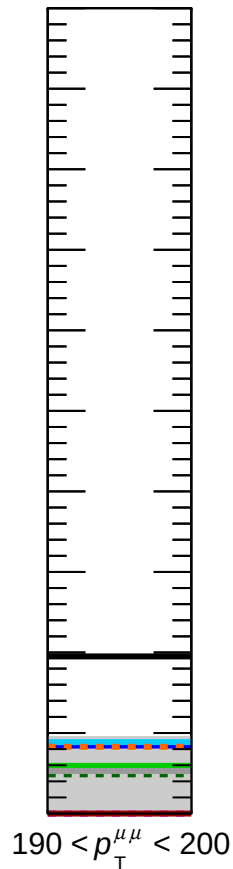
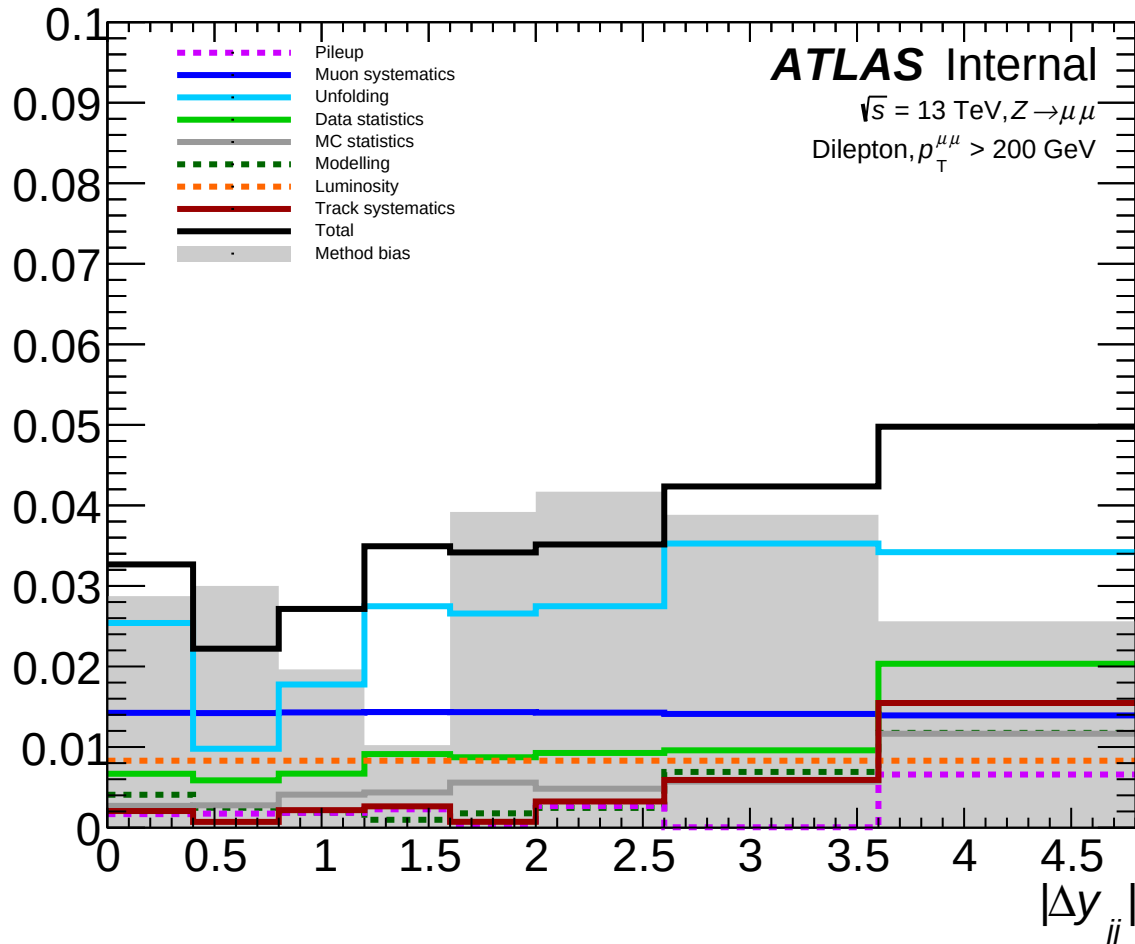
Fractional uncertainty

ATLAS Internal

$\sqrt{s} = 13 \text{ TeV}, Z \rightarrow \mu\mu$

Dilepton, $p_T^{\mu\mu} > 200 \text{ GeV}$

- Pileup
- Muon systematics
- Unfolding
- Data statistics
- MC statistics
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- Luminosity
- Track systematics
- Total
- Method bias



ATLAS Internal $\sqrt{s} = 13 \text{ TeV}, 140.1 \text{ fb}^{-1}, Z \rightarrow \mu\mu, p_{\text{T}}^{\mu\mu} > 200 \text{ GeV}$ 